Éléments de géométrie algébrique

A. Grothendieck and J. Dieudonné
Publications mathématiques de l'I.H.É.S

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What this is

This whole chapter is written by the translators.

This is a community translation of Alexander Grothendieck's and Jean Dieudonné's *Éléments de géométrie algébrique* (EGA). As it is a work in progress by multiple people, there will probably be a few mistakes—if you spot any then please do let us know[1].
To contribute, please visit

https://github.com/ryankeleti/ega.

On est désolés, Grothendieck.

In defense of a translation

From Wikipedia[2]:

> In January 2010, Grothendieck wrote the letter "Déclaration d'intention de non-publication" to Luc Illusie, claiming that all materials published in his absence have been published without his permission. He asks that none of his work be reproduced in whole or in part and that copies of this work be removed from libraries.[3] [...] This order may have been reversed later in 2010.[4]

It is a matter of often heated contention as to whether or not any translation of Grothendieck's work should take place, given his extremely explicit views on the matter. By no means do we mean to argue that somehow Grothendieck's wishes should be invalidated or ignored, nor do we wish to somehow twist his earlier words around in order to justify what we have done: we fully accept that he himself would probably have branded this project "an abomination". With this in mind, it remains to explain why we have gone ahead anyway.

First, and possibly foremost, it does not make sense (to us) for an individual to own the rights to knowledge. Arguments can be made about how the EGA is the product of years and years of intense work by Grothendieck (and Dieudonné, amongst others), and so *this* is something that he 'owns' and has full control over. Indeed, it is true that there are almost innumerably many sentences in these works that only Grothendieck himself could have engineered, but, in translation, we have never improved anything, but only (regrettably, but almost certainly) worsened. The work in these pages is that of Grothendieck; we have been not much more than typesetters and eager readers. However, there is some important point to be made about the fact that Grothendieck collaborated and worked with many other incredibly proficient mathematicians during the writing of this treatise; although it is impossible to pinpoint which parts exactly others may have contributed (and by no means do we wish to imply that any of this work is derivative or fraudulent in any way whatsoever—EGA was written *by Grothendieck*) it seems fair that, in some amount, there are bits of the EGAs that 'belong' to a broader collection of minds.

It is a very good idea here to repeat the oft-quoted aphorism: "the work here is not ours, but any mistakes are"—it is very understandable for an author to not want their name on something that they have not themselves written, or, at the very least, read. This may be, in part, a reason for Grothendieck's wishes, but that is pure speculation. Even so, we include this above disclaimer.

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[1] https://github.com/ryankeleti/ega/issues

[2] https://en.wikipedia.org/wiki/Alexander_Grothendieck#Retirement_into_reclusion_and_death

[3] Grothendieck's letter. *Secret Blogging Seminar.* 9 February 2010. Retrieved 3 September 2019.

[4] Réédition des SGA. Archived from the original on 29 June 2016. Retrieved 12 November 2013.

Secondly, then, we note that the French version of EGA is still entirely readily accessible. Anybody reading these copies who is not a native French speaker, will probably be translating at least some part of EGA into English in their head, or into their notebooks, as they read. This document is just the product of a few people doing exactly that, but then passing on their efforts to make things just that little bit easier for anyone else who follows.

Lastly, to quote another adage, "the guilty person is often the loudest". If it seems like we are over-eager to defend ourselves because we know that we are somehow in the wrong, it is because we are, at least partially. Working on this translation has meant going against Grothendieck's explicit requests, and for that we are sorry. We only hope that the freedom of knowledge is an excusable defense.

Notes from the translators

Grothendieck's writing style in EGA is quite particular, most notably for its long sentence structure. As translators, we have tried to give the best possible approximation of this style in English, resisting the temptation to "streamline" things in places where the language is more dense than usual.

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Any translations about which we are not entirely sure will be marked with a (?).

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Whenever a note is made by the translators, it will be prefaced by "[Trans.]".

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Along the margins we have provided the page numbers corresponding to the original text (of the first edition), as published by *Publications mathématiques de l'I.H.É.S.*, where the EGA were published as the following volumes:[5]

- EGA I (*tome 4, 1960*)
- EGA II (*tome 8, 1961*)
- EGA III, part 1 (*tome 11, 1961*)
- EGA III, part 2 (*tome 17, 1963*)
- EGA IV, part 1 (*tome 20, 1964*)
- EGA IV, part 2 (*tome 24, 1965*)
- EGA IV, part 3 (*tome 28, 1966*)
- EGA IV, part 4 (*tome 32, 1967*).

Due to EGA being a collection of volumes (one non-preliminary chapter, or part of a chapter, per volume), the page numbers reset at every new chapter. In addition, the preliminary section is stretched out over multiple volumes. To combat this, we label the pages as

$$\mathbf{X} \mid p,$$

referring to Chapter X, page p. For EGA III and IV, which are split across multiple chapters, we label the pages as

$$\mathbf{X}\text{-}n \mid p,$$

referring to Chapter X, part n, page p. In the case of the preliminaries (which are often collectively referred to as EGA 0), the preliminaries from Chapter Y are denoted as $\mathbf{0_Y}$.

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Later volumes (EGA II, III, and IV) include errata for earlier chapters. Where possible, we have used these to 'update' our translation, and entirely replace whatever mistakes might have been in the original copies of EGA I and II. If the change is minor (e.g. 'intersection' replacing 'inter-section') then we will not mention it; if it is anything more fundamental (e.g. X' replacing X) then we will include some margin note on the relevant line detailing the location of the erratum (e.g. $\mathbf{Err}_{\mathrm{II}}$ to denote that the correction is listed in the Errata section of EGA II).

[5] PDFs of which can be found online, hosted by the *Grothendieck circle*.

EGA IV SECTIONS

In EGA IV-1, the summary included a tentative list of section that EGA IV would contain. As EGA IV was written, §5, §7, and §§11–21 would contain different sections than initially envisaged. We include the original listing here:

§5. Dimension and depth for preschemes.
§7. Application to the relations between a local Noetherian ring and its completion. Excellent rings.
§11. Topological properties of finitely presented flat morphisms. Local flatness criteria.
§12. Study of fibres of finitely presented flat morphisms.
§13. Equidimensional morphisms.
§14. Universally open morphisms.
§15. Study of fibres of a universally open morphism.
§16. Differential invariants. Differentially smooth morphisms.
§17. Smooth morphisms, unramified morphisms, and étale morphisms.
§18. Supplement on étale morphisms. Henselian local rings.
§19. Regular immersions and transversely regular immersions.
§20. Hyperplane sections; generic projections.
§21. Infinitesimal extensions.

MATHEMATICAL WARNINGS

EGA uses *prescheme* for what is now usually called a scheme, and *scheme* for what is now usually called a separated scheme — we have decided to translate “literally” or “historically”, and thus continue to use the word *scheme* to mean separated scheme, and *prescheme* to mean scheme.

In some cases, we (the translators) have changed “$\to$” to “$\mapsto$” where appropriate.

Introduction

To Oscar Zariski and André Weil.

This memoir, and the many others will undoubtedly follow, are intended to form a treatise on the foundations of algebraic geometry. They do not, in principle, presume any particular knowledge of the subject, and it has even been recognised that such knowledge, despite its obvious advantages, could sometimes (because of the much-too-narrow interpretation—through the birational point of view—that it usually implies) be a hindrance to the one who wants to become familiar with the point of view and techniques presented here. However, we assume that the reader has a good knowledge of the following topics: **I** | 5

(a) *Commutative algebra*, as it is laid out, for example, in the volumes (in progress of being written) of the *Éléments* of N. Bourbaki (and, pending the publication of these volumes, in Samuel–Zariski [**SZ60**] and Samuel [**Sam53b**, **Sam53a**]).
(b) *Homological algebra*, for which we refer to Cartan–Eilenberg [**CE56**] (cited as (M)) and Godement [**God58**] (cited as (G)), as well as the recent article by A. Grothendieck [**Gro57**] (cited as (T)).
(c) *Sheaf theory*, where our main references will be (G) and (T); this theory provides the essential language for interpreting, in "geometric" terms, the essential notions of commutative algebra, and for "globalizing" them.
(d) Finally, it will be useful for the reader to have some familiarity with *functorial language*, which will be constantly used in this treatise, and for which the reader may consult (M), (G), and especially (T); the principles of this language and the main results of the general theory of functors will be described in more detail in a book currently in preparation by the authors of this treatise.

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It is not the place, in this introduction, to give a more or less summary description from the "schemes" point of view in algebraic geometry, nor the long list of reasons which made its adoption necessary, and in particular the systematic acceptance of nilpotent elements in the local rings of "manifolds" that we consider (which necessarily shifts the idea of rational maps into the background, in favor of those of regular maps or "morphisms"). To be precise, this treatise aims to systematically develop the language of schemes, and will demonstrate, we hope, its necessity. Although it would be easy to do so, we will not try to give here an "intuitive" introduction to the notions developed in Chapter I. For the reader who would like to have a glimpse of the preliminary study of the subject matter of this treatise, we refer them to the conference by A. Grothendieck at the International Congress of Mathematicians in Edinburgh in 1958 [**Gro58**], and the exposé [**Gro**] of the same author. The work [**Ser55a**] (cited as (FAC)) of J.-P. Serre can also be considered as an intermediary exposition between the classical point of view and the schemes point of view in algebraic geometry, and, as such, its reading may be an excellent preparation for the reading of our *Éléments*. **I** | 6

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We give below the general outline planned for this treatise, subject to later modifications, especially concerning the later chapters.

Chapter I. — The language of schemes.
— II. — Elementary global study of some classes of morphisms.
— III. — Cohomology of algebraic coherent sheaves. Applications.
— IV. — Local study of morphisms.
— V. — Elementary procedures of constructing schemes.
— VI. — Descent. General method of constructing schemes.
— VII. — Group schemes, principal fibre bundles.
— VIII. — Differential study of fibre bundles.
— IX. — The fundamental group.
— X. — Residues and duality.
— XI. — Theories of intersection, Chern classes, Riemann–Roch theorem.
— XII. — Abelian schemes and Picard schemes.
— XIII. — Weil cohomology.

In principle, all chapters are considered open to changes, and supplementary sections could always be added later; such sections would appear in separate fascicles in order to minimize the inconveniences accompanying whatever mode of publication adopted. When the writing of such a section is foreseen or in progress during the publication of a chapter, it will be mentioned in the summary of the chapter in question, even if, owing to certain orders of urgency, its actual publication clearly ought to have been later. For the convenience of the reader, we give in "Chapter 0" the necessary tools in commutative algebra, homological algebra, and sheaf theory, that will be used throughout this treatise, that are more or less well known but for which it was not possible to give convenient references. It is recommended for the reader to not read Chapter 0 except whilst reading the actual treatise, when the results to which we refer seem unfamiliar. Besides, we think that in this **I** | 7
way, the reading of this treatise could be a good method for the beginner to familiarize themselves with commutative algebra and homological algebra, whose study, when not accompanied with tangible applications, is considered tedious, or even depressing, by many.

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It is outside of our capabilities to give a historic overview, or even a summary thereof, of the ideas and results described herein. The text will contain only those references considered particularly useful for comprehension, and we indicate the origin of only the most important results. Formally, at least, the subjects discussed in our work are reasonably new, which explains the scarcity of references made to the fathers of algebraic geometry from the 19th to the beginning of the 20th century, whose works we know only by hear-say. It is suitable, however, to say some words here about the works which have most directly influenced the authors and contributed to the development of scheme-theoretic point of view. We absolutely must mention the fundamental work (FAC) of J.-P. Serre first, which has served as an introduction to algebraic geometry for more that one young student (the author of this treatise being one), deterred by the dryness of the classic *Foundations* of A. Weil [**Wei46**]. It is there that it is shown, for the first time, that the "Zariski topology" of an "abstract" algebraic variety is perfectly suited to applying certain techniques from algebraic topology, and notably to be able to define a cohomology theory. Further, the definition of an algebraic variety given therein is that which translates most naturally to the idea that we develop here[6]. Serre himself had incidentally noted that the cohomology theory of affine algebraic varieties could be translated without difficulty by replacing the affine algebras over a field by arbitrary commutative rings. Chapters I and II of this treatise, and the first two paragraphs of Chapter III, can thus be considered, for the most part, as easy translations, to this bigger framework, of the principal results of (FAC) and a later article

[6]Just as J.-P. Serre informed us, it is right to note that the idea of defining the structure of a manifold by the data of a sheaf of rings is due to H. Cartan, who took this idea as the starting point of his theory of analytic spaces. Of course, just as in algebraic geometry, it would be important in "analytic geometry" to allow the use of nilpotent elements in local rings of analytic spaces. This extension of the definition of H. Cartan and J.-P. Serre has recently been broached by H. Grauert [**Gra60**], and there is room to hope that a systematic report of analytic geometry in this setting will soon see the light of day. It is also evident that the ideas and techniques developed in this treatise retain a sense of analytic geometry, even though one must expect more considerable technical difficulties in this latter theory. We can foresee that algebraic geometry, by the simplicity of its methods, will be able to serve as a sort of formal model for future developments in the theory of analytic spaces.

of the same author [**Ser57**]. We have also vastly profited from the *Séminaire de géométrie algébrique* de C. Chevalley [**CC**]; in particular, the systematic usage of "constructible sets" introduced by him has turned out to be extremely useful in the theory of schemes (cf. Chapter IV). We have also borrowed from him the study of morphisms from the point of view of dimension (Chapter IV), that **I** | 8
translates with negligible change to the framework of schemes. It also merits noting that the idea of "schemes of local rings", introduced by Chevalley, naturally lends itself to being extended to algebraic geometry (not having, however, all the flexibility and generality that we intend to give it here); for the connections between this idea and our theory, see Chapter I, §8. One such extension has been developed by M. Nagata in a series of memoirs [**Nag58a**], which contain many special results concerning algebraic geometry over Dedekind rings[7].

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It goes without saying that a book on algebraic geometry, and especially a book dealing with the fundamentals, is of course influenced, if only by proxy, by mathematicians such as O. Zariski and A. Weil. In particular, the *Théorie des fonctions holomorphes* by Zariski [**Zar51**], reasonably flexible thanks to the cohomological methods and an existence theorem (Chapter III, §§4 and 5), is (along with the method of descent described in Chapter VI) one of the principal tools used in this treatise, and it seems to us one of the most powerful at our disposal in algebraic geometry.

The general technique in which it is employed can be sketched as follows (a typical example of which will be given in Chapter XI, in the study of the fundamental group). We have a proper morphism (Chapter II) $f : X \to Y$ of algebraic varieties (or, more generally, of schemes) that we wish to study on the neighbourhood of a point $y \in Y$, with the aim of resolving a problem P relative to a neighbourhood of y. We proceed step by step:

1st We can suppose that Y is affine, so that X becomes a scheme defined on the affine ring A of Y, and we can even replace A by the local ring of y. This reduction is always easy in practice (Chapter V) and brings us to the case where A is a *local* ring.

2nd We study the problem in question when A is a local *Artinian* ring. So that the problem P still makes sense when A is not assumed to be integral, we sometimes have to reformulate P, and it appears that we often obtain a better understanding of the problem in doing so, in an "infinitesimal" way.

3rd The theory of formal schemes (Chapter III, §§3, 4, and 5) lets us pass from the case of an Artinian ring to a *complete local ring*.

4th Finally, if A is an arbitrary local ring, considering "multiform (?) sections" of suitable schemes over X, approximating a given "formal" section (Chapter IV), will let us pass, by extension of scalars, to the completion of A, from a known result (about the scheme **I** | 9
induced by X by extension of scalars to the completion of A) to an analogous result for a finite simple (e.g. unramified) extension of A.

This sketch shows the importance of the systematic study of schemes defined over an Artinian ring A. The point of view of Serre in his formulation of the theory of local class fields, and the recent works of Greenberg, seem to suggest that such a study could be undertaken by functorially attaching, to some such scheme X, a scheme X' over the residue field k of A (assumed perfect) of dimension equal (in nice cases) to $n \dim X$, where n is the height of A.

As for the influence of A. Weil, it suffices to say that it is the need to develop the tools necessary to formulate, with full generality, the definition of "Weil cohomology", and to tackle the proof[8] of all the formal properties necessary to establish the famous conjectures in Diophantine geometry [**Wei49**], that has been one of the principal motivations for the writing of this treatise, as well as the desire to find the natural setting of the usual ideas and methods of algebraic geometry, and to give the authors the chance to understand said ideas and methods.

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[7] Among the works that come close to our point of view of algebraic geometry, we pick out the work of E. Kähler [**Käh58**] and a recent note of Chow and Igusa [**CI58**], which go back over certain results of (FAC) in the context of Nagata–Chevalley theory, as well as giving a Künneth formula.

[8] To avoid any misunderstanding, we point out that this task has barely been undertaken at the moment of writing this introduction, and still hasn't led to the proof of the Weil conjectures.

Finally, we believe it useful to warn the reader that they, as did all the authors themselves, will almost certainly have difficulty before becoming accustomed to the language of schemes, and to convince themselves that the usual constructions that suggest geometric intuition can be translated, in essentially only one sensible way, to this language. As in many parts of modern mathematics, the first intuition seems further and further away, in appearance, from the correct language needed to express the mathematics in question with complete precision and the desired level of generality. In practice, the psychological difficulty comes from the need to replicate some familiar set-theoretic constructions to a category that is already quite different from that of sets (the category of preschemes, or the category of preschemes over a given prescheme): Cartesian products, group laws, ring laws, module laws, fibre bundles, principal homogeneous fibre bundles, etc. It will most likely be difficult for the mathematician, in the future, to shy away from this new effort of abstraction (maybe rather negligible, on the whole, in comparison with that supplied by our fathers) to familiarize themselves with the theory of sets.

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The references are given following the numerical system; for example, in **III**, 4.9.3, the **III** indicates the volume, the 4 the chapter, the 9 the section, and the 3 the paragraph.[9] If we reference a volume from within itself then we omit the volume number.

[Trans] Page 10 in the original is left blank. **I** | 10

[9] *[Trans] This is not a direct translation of the original, but instead uses the language more familiar to modern book (and LaTeX document) layouts.*

CHAPTER 0

Preliminaries (EGA 0)

§1. Rings of fractions

1.0. Rings and Algebras $\mathbf{0_I}$ | 11

(1.0.1). All the rings considered in this treatise will have a *unit element*; all the modules over such a ring will be assumed to be *unitary*; the ring homomorphisms will always be assumed to *send the unit element to the unit element*; unless otherwise stated, a subring of a ring A will be assumed to *contain the unit element of* A. We will focus in particular on *commutative* rings, and when we speak of a ring without specifying any details, it will be implied that it is commutative. If A is a not-necessarily-commutative ring, by A-module we will we mean a *left* module unless stated otherwise.

(1.0.2). Let A and B be not-necessarily-commutative rings and $\phi : A \to B$ a homomorphism. Any left (resp. right) B-module M can be provided with a left (resp. right) A-module structure by $a \cdot m = \phi(a) \cdot m$ (resp. $m \cdot a = m \cdot \phi(a)$); when it will be necessary to distinguish M as an A-module or a B-module, we will denote by $M_{[\phi]}$ the left (resp. right) A-module defined as such. If L is an A-module, then a homomorphism $u : L \to M_{[\phi]}$ is a homomorphism of abelian groups such that $u(a \cdot x) = \phi(a) \cdot u(x)$ for $a \in A$, $x \in L$; we will also say that it is a *ϕ-homomorphism* $L \to M$, and that the pair (ϕ, u) (or, by abuse of language, u) is a *di-homomorphism* from (A, L) to (B, M). The pairs (A, L) consisting of a ring A and an A-module L thus form a *category* whose morphisms are di-homomorphisms.

(1.0.3). Under the hypotheses of (1.0.2), if $\mathfrak{J}$ is a left (resp. right) ideal of A, we denote by $B\mathfrak{J}$ (resp. $\mathfrak{J}B$) the left (resp. right) ideal $B\phi(\mathfrak{J})$ (resp. $\phi(\mathfrak{J})B$) of B generated by $\phi(\mathfrak{J})$; it is also the image of the canonical homomorphism $B \otimes_A \mathfrak{J} \to B$ (resp. $\mathfrak{J} \otimes_A B \to B$) of left (resp. right) B-modules.

(1.0.4). If A is a (commutative) ring, and B a not-necessarily-commutative ring, then the data of a structure of an *A-algebra* on B is equivalent to the data of a ring homomorphism $\phi : A \to B$ such that $\phi(A)$ is contained in the center of B. For all ideals $\mathfrak{J}$ of A, $\mathfrak{J}B = B\mathfrak{J}$ is then a two-sided ideal of B, and for every B-module M, $\mathfrak{J}M$ is then a B-module equal to $(B\mathfrak{J})M$.

(1.0.5). We will not dwell much on the notions of *modules of finite type* and (commutative) *algebras of finite type*; to say that an A-module M is of finite type means that there exists an exact sequence $\mathbf{0_I}$ | 12
$A^p \to M \to 0$. We say that an A-module M admits a *finite presentation* if it is isomorphic to the cokernel of a homomorphism $A^p \to A^q$, or, in other words, if there exists an exact sequence $A^p \to A^q \to M \to 0$. We note that for a *Noetherian* ring A, every A-module of finite type admits a finite presentation.

Let us recall that an A-algebra B is said to be *integral* over A if every element in B is a root in B of a monic polynomial with coefficients in A; equivalently, if every element of B is contained in a subalgebra of B which is an *A-module of finite type*. When this is so, and if B is commutative, the subalgebra of B generated by a finite subset of B is an A-module of finite type; for a commutative algebra B to be integral and of finite type over A, it is necessary and sufficient that B be an A-module of finite type; we also say that B is an *integral A-algebra of finite type* (or simply *finite*, if there is no chance of confusion). It should be noted that in these definitions, it is not assumed that the homomorphism $A \to B$ defining the A-algebra structure is injective.

(1.0.6). An *integral* ring (or an *integral domain*) is a ring in which the product of a finite family of elements $\neq 0$ is $\neq 0$; equivalently, in such a ring, we have $0 \neq 1$, and the product of two elements $\neq 0$ is $\neq 0$. A *prime* ideal of a ring A is an ideal $\mathfrak{p}$ such that $A/\mathfrak{p}$ is integral; this implies that $\mathfrak{p} \neq A$. For a ring A to have at least one prime ideal, it is necessary and sufficient that $A \neq \{0\}$.

(1.0.7). A *local* ring is a ring A in which there exists a unique maximal ideal, which is thus the complement of the invertible elements, and contains all the ideals $\neq A$. If A and B are local rings, and $\mathfrak{m}$ and $\mathfrak{n}$ their respective maximal ideals, then we say that a homomorphism $\phi : A \to B$ is *local* if $\phi(\mathfrak{m}) \subset \mathfrak{n}$ (or, equivalently, if $\phi^{-1}(\mathfrak{n}) = \mathfrak{m}$). By passing to quotients, such a homomorphism then defines a monomorphism from the residue field $A/\mathfrak{m}$ to the residue field $B/\mathfrak{n}$. The composition of any two local homomorphisms is a local homomorphism.

1.1. Radical of an ideal. Nilradical and radical of a ring

(1.1.1). Let $\mathfrak{a}$ be an ideal of a ring A; the *radical* of $\mathfrak{a}$, denoted by $\mathfrak{r}(\mathfrak{a})$, is the set of $x \in A$ such that $x^n \in \mathfrak{a}$ for an integer $n > 0$; it is an ideal containing $\mathfrak{a}$. We have $\mathfrak{r}(\mathfrak{r}(\mathfrak{a})) = \mathfrak{r}(\mathfrak{a})$; the relation $\mathfrak{a} \subset \mathfrak{b}$ implies $\mathfrak{r}(\mathfrak{a}) \subset \mathfrak{r}(\mathfrak{b})$; the radical of a finite intersection of ideals is the intersection of their radicals. If ϕ is a homomorphism from another ring A' to A, then we have $\mathfrak{r}(\phi^{-1}(\mathfrak{a})) = \phi^{-1}(\mathfrak{r}(\mathfrak{a}))$ for any ideal $\mathfrak{a} \subset A$. For an ideal to be the radical of an ideal, it is necessary and sufficient that it be an intersection of prime ideals. The radical of an ideal $\mathfrak{a}$ is the intersection of the *minimal* prime ideals which contain $\mathfrak{a}$; if A is Noetherian, there are finitely many of these minimal prime ideals.

The radical of the ideal (0) is also called the *nilradical* of A; it is the set $\mathfrak{N}$ of the nilpotent elements of A. We say that the ring A is *reduced* if $\mathfrak{N} = (0)$; for every ring A, the quotient $A/\mathfrak{N}$ of A by its nilradical is a reduced ring.

(1.1.2). Recall that the *radical* $\mathfrak{r}(A)$ of a (not-necessarily-commutative) ring A is the intersection of the maximal left ideals of A (and also the intersection of maximal right ideals). The radical of $A/\mathfrak{r}(A)$ is (0).

1.2. Modules and rings of fractions $\mathbf{0_I}$ | 13

(1.2.1). We say that a subset S of a ring A is *multiplicative* if $1 \in S$ and the product of two elements of S is in S. The examples which will be the most important in what follows are: 1st, the set S_f of powers f^n ($n \geqslant 0$) of an element $f \in A$; and 2nd, the complement $A - \mathfrak{p}$ of a *prime* ideal $\mathfrak{p}$ of A.

(1.2.2). Let S be a multiplicative subset of a ring A, and M an A-module; on the set $M \times S$, the relation between pairs (m_1, s_1) and (m_2, s_2):

"there exists an $s \in S$ such that $s(s_1 m_2 - s_2 m_1) = 0$"

is an equivalence relation. We denote by $S^{-1}M$ the quotient set of $M \times S$ by this relation, and by m/s the canonical image of the pair (m, s) in $S^{-1}M$; we call $i_M^S : m \mapsto m/1$ (also denoted i^S) the *canonical* map from M to $S^{-1}M$. This map is, in general, neither injective nor surjective; its kernel is the set of $m \in M$ such that there exists an $s \in S$ for which $sm = 0$.

On $S^{-1}M$ we define an additive group law by setting

$$(m_1/s_1) + (m_2/s_2) = (s_2 m_1 + s_1 m_2)/(s_1 s_2)$$

(one can check that it is independent of the choice of representative of the elements of $S^{-1}M$, which are equivalence classes). On $S^{-1}A$ we further define a multiplicative law by setting $(a_1/s_1)(a_2/s_2) = (a_1 a_2)/(s_1 s_2)$, and finally an exterior law on $S^{-1}M$, acted on by the set of elements of $S^{-1}A$, by setting $(a/s)(m/s') = (am)/(ss')$. It can then be shown that $S^{-1}A$ is endowed with a ring structure (called *the ring of fractions of A with denominators in S*) and $S^{-1}M$ with the structure of an $S^{-1}A$-module (called *the module of fractions of M with denominators in S*); for all $s \in S$, $s/1$ is invertible in $S^{-1}A$, its inverse being $1/s$. The canonical map i_A^S (resp. i_M^S) is a ring homomorphism (resp. a homomorphism of A-modules, $S^{-1}M$ being considered as an A-module by means of the homomorphism $i_A^S : A \to S^{-1}A$).

(1.2.3). If $S_f = \{f^n\}_{n \geqslant 0}$ for a $f \in A$, we write A_f and M_f instead of $S_f^{-1}A$ and $S_f^{-1}M$; when A_f is considered as algebra over A, we can write $A_f = A[1/f]$. A_f is isomorphic to the quotient algebra $A[T]/(fT - 1)A[T]$. When $f = 1$, A_f and M_f are canonically identified with A and M; if f is nilpotent, then A_f and M_f are 0.

When $S = A - \mathfrak{p}$, with $\mathfrak{p}$ a prime ideal of A, we write $A_\mathfrak{p}$ and $M_\mathfrak{p}$ instead of $S^{-1}A$ and $S^{-1}M$; $A_\mathfrak{p}$ is a *local ring* whose maximal ideal $\mathfrak{q}$ is generated by $i_A^S(\mathfrak{p})$, and we have $(i_A^S)^{-1}(\mathfrak{q}) = \mathfrak{p}$; by passing to quotients, i_A^S gives a monomorphism from the integral ring $A/\mathfrak{p}$ to the field $A_\mathfrak{p}/\mathfrak{q}$, which can be identified with the field of fractions of $A/\mathfrak{p}$.

(1.2.4). The ring of fractions $S^{-1}A$ and the canonical homomorphism i_A^S are a solution to a *universal mapping problem*: any homomorphism u from A to a ring B such that $u(S)$ is composed of *invertible* elements in B factors uniquely as

$$u : A \xrightarrow{i_A^S} S^{-1}A \xrightarrow{u^*} B$$

where u^* is a ring homomorphism. Under the same hypotheses, let M be an A-module, N a B-module, and $v : M \to N$ a homomorphism of A-modules (for the B-module structure on N defined by $u : A \to B$); then v factors uniquely as $\mathbf{0_I}$ | 14

$$v : M \xrightarrow{i_M^S} S^{-1}M \xrightarrow{v^*} N$$

where v^* is a homomorphism of $S^{-1}A$-modules (for the $S^{-1}A$-module structure on N defined by u^*).

(1.2.5). We define a canonical isomorphism $S^{-1}A \otimes_A M \simeq S^{-1}M$ of $S^{-1}A$-modules, sending the element $(a/s) \otimes m$ to the element $(am)/s$, with the inverse isomorphism sending m/s to $(1/s) \otimes m$.

(1.2.6). For every ideal $\mathfrak{a}'$ of $S^{-1}A$, $\mathfrak{a} = (i_A^S)^{-1}(\mathfrak{a}')$ is an ideal of A, and $\mathfrak{a}'$ is the ideal of $S^{-1}A$ generated by $i_A^S(\mathfrak{a})$, which can be identified with $S^{-1}\mathfrak{a}$ (1.3.2). The map $\mathfrak{p}' \mapsto (i_A^S)^{-1}(\mathfrak{p}')$ is an isomorphism, for the structure given by ordering, from the set of *prime* ideals of $S^{-1}A$ to the set of prime ideals $\mathfrak{p}$ of A such that $\mathfrak{p} \cap S = \varnothing$. In addition, the local rings $A_\mathfrak{p}$ and $(S^{-1}A)_{S^{-1}\mathfrak{p}}$ are canonically isomorphic (1.5.1).

(1.2.7). When A is an *integral* ring, for which K denotes its field of fractions, the canonical map $i_A^S : A \to S^{-1}A$ is injective for any multiplicative subset S not containing 0, and $S^{-1}A$ is then canonically identified with a subring of K containing A. In particular, for every prime ideal $\mathfrak{p}$ of A, $A_\mathfrak{p}$ is a local ring containing A, with maximal ideal $\mathfrak{p}A_\mathfrak{p}$, and we have $\mathfrak{p}A_\mathfrak{p} \cap A = \mathfrak{p}$.

(1.2.8). If A is a *reduced* ring (1.1.1), so is $S^{-1}A$: indeed, if $(x/s)^n = 0$ for $x \in A$, $s \in S$, then this means that there exists an $s' \in S$ such that $s'x^n = 0$, hence $(s'x)^n = 0$, which, by hypothesis, implies $s'x = 0$, so $x/s = 0$.

1.3. Functorial properties

(1.3.1). Let M and N be A-modules, and u an A-homomorphism $M \to N$. If S is a multiplicative subset of A, we define a $S^{-1}A$-homomorphism $S^{-1}M \to S^{-1}N$, denoted by $S^{-1}u$, by setting $S^{-1}u(m/s) = u(m)/s$; if $S^{-1}M$ and $S^{-1}N$ are canonically identified with $S^{-1}A \otimes_A M$ and $S^{-1}A \otimes_A N$ (1.2.5), then $S^{-1}u$ is considered as $1 \otimes u$. If P is a third A-module, and v an A-homomorphism $N \to P$, we have $S^{-1}(v \circ u) = (S^{-1}v) \circ (S^{-1}u)$; in other words, $S^{-1}M$ is a *covariant functor in* M, from the category of A-modules to that of $S^{-1}A$-modules (A and S being fixed).

(1.3.2). The functor $S^{-1}M$ is *exact*; in other words, if the sequence

$$M \xrightarrow{u} N \xrightarrow{v} P$$

is exact, then so is the sequence

$$S^{-1}M \xrightarrow{S^{-1}u} S^{-1}N \xrightarrow{S^{-1}v} S^{-1}P.$$

In particular, if $u : M \to N$ is injective (resp. surjective), the same is true for $S^{-1}u$; if N and P are submodules of M, $S^{-1}N$ and $S^{-1}P$ are canonically identified with submodules of $S^{-1}M$, and we have $\mathbf{0_I}$ | 15

$$S^{-1}(N+P) = S^{-1}N + S^{-1}P \text{ and } S^{-1}(N \cap P) = (S^{-1}N) \cap (S^{-1}P).$$

(1.3.3). Let $(M_\alpha, \phi_{\beta\alpha})$ be an inductive system of A-modules; then $(S^{-1}M_\alpha, S^{-1}\phi_{\beta\alpha})$ is an inductive system of $S^{-1}A$-modules. Expressing the $S^{-1}M_\alpha$ and $S^{-1}\phi_{\beta\alpha}$ as tensor products ((1.2.5) and (1.3.1)), it follows from the permutability of the tensor product and inductive limit operations that we have a canonical isomorphism

$$S^{-1} \varinjlim M_\alpha \simeq \varinjlim S^{-1}M_\alpha$$

which is we can further express by saying that the functor $S^{-1}M$ (in M) *commutes with inductive limits*.

(1.3.4). Let M and N be A-modules; there is a canonical *functorial* (in M and N) isomorphism

$$(S^{-1}M) \otimes_{S^{-1}A} (S^{-1}N) \simeq S^{-1}(M \otimes_A N)$$

which sends $(m/s) \otimes (n/t)$ to $(m \otimes n)/st$.

(1.3.5). We also have a *functorial* (in M and N) homomorphism

$$S^{-1}\operatorname{Hom}_A(M, N) \longrightarrow \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$$

which sends u/s to the homomorphism $m/t \mapsto u(m)/st$. When M has a finite presentation, the above homomorphism is an *isomorphism*: it is immediate when M is of the form A^r, and we pass to the general case by starting with the exact sequence $A^p \to A^q \to M \to 0$ and using the exactness of the functor $S^{-1}M$ and the left-exactness of the functor $\operatorname{Hom}_A(M, N)$ in M. Note that this is always the case when A is *Noetherian* and the A-module M is *of finite type*.

1.4. Change of multiplicative subset

(1.4.1). Let S and T be multiplicative subsets of a ring A such that $S \subset T$; there exists a canonical homomorphism $\rho_A^{T,S}$ (or simply $\rho^{T,S}$) from $S^{-1}A$ to $T^{-1}A$, sending the element denoted a/s of $S^{-1}A$ to the element denoted a/s in $T^{-1}A$; we have $i_A^T = \rho_A^{T,S} \circ i_A^S$. For every A-module M, there exists, in the same way, an $S^{-1}A$-linear map from $S^{-1}M$ to $T^{-1}M$ (the latter considered as an $S^{-1}A$-module by the homomorphism $\rho_A^{T,S}$), which sends the element m/s of $S^{-1}M$ to the element m/s of $T^{-1}M$; we denote this map by $\rho_M^{T,S}$, or simply $\rho^{T,S}$, and we still have $i_M^T = \rho_M^{T,S} \circ i_M^S$; by the canonical identification (1.2.5), $\rho_M^{T,S}$ is identified with $\rho_A^{T,S} \otimes 1$. The homomorphism $\rho_M^{T,S}$ is a *functorial morphism* (or natural transformation) from the functor $S^{-1}M$ to the functor $T^{-1}M$, in other words, the diagram

$$\begin{tikzcd}
S^{-1}M \arrow[r, "S^{-1}u"] \arrow[d, "\rho_M^{T,S}"'] & S^{-1}N \arrow[d, "\rho_N^{T,S}"] \\
T^{-1}M \arrow[r, "T^{-1}u"] & T^{-1}N
\end{tikzcd}$$

is commutative, for every homomorphism $u : M \to N$; $T^{-1}u$ is entirely determined by $S^{-1}u$, since, for $m \in M$ and $t \in T$, we have $\mathbf{0_I}$ | 16

$$(T^{-1}u)(m/t) = (t/1)^{-1}\rho^{T,S}((S^{-1}u)(m/1)).$$

(1.4.2). With the same notation, for A-modules M and N, the diagrams (cf. (1.3.4) and (1.3.5))

$$\begin{tikzcd}
(S^{-1}M) \otimes_{S^{-1}A} (S^{-1}N) \arrow[r, "\sim"] \arrow[d] & S^{-1}(M \otimes_A N) \arrow[d] \\
(T^{-1}M) \otimes_{T^{-1}A} (T^{-1}N) \arrow[r, "\sim"] & T^{-1}(M \otimes_A N)
\end{tikzcd}$$

$$\begin{tikzcd}
S^{-1}\operatorname{Hom}_A(M, N) \arrow[r] \arrow[d] & \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N) \arrow[d] \\
T^{-1}\operatorname{Hom}_A(M, N) \arrow[r] & \operatorname{Hom}_{T^{-1}A}(T^{-1}M, T^{-1}N)
\end{tikzcd}$$

are commutative.

(1.4.3). There is an important case, in which the homomorphism $\rho^{T,S}$ is *bijective*, when we then know that every element of T is a divisor of an element of S; we then identify the modules $S^{-1}M$ and $T^{-1}M$ via $\rho^{T,S}$. We say that S is *saturated* if every divisor in A of an element of S is in S; by replacing S with the set T of all the divisors of the elements of S (a set which is multiplicative and saturated), we see that we can always, if we wish, consider only modules of fractions $S^{-1}M$, where S is saturated.

(1.4.4). If S, T, and U are three multiplicative subsets of A such that $S \subset T \subset U$, then we have

$$\rho^{U,S} = \rho^{U,T} \circ \rho^{T,S}.$$

(1.4.5). Consider an *increasing filtered family* (S_α) of multiplicative subsets of A (we write $\alpha \leqslant \beta$ for $S_\alpha \subset S_\beta$), and let S be the multiplicative subset $\bigcup_\alpha S_\alpha$; let us put $\rho_{\beta\alpha} = \rho_A^{S_\beta,S_\alpha}$ for $\alpha \leqslant \beta$; according to (1.4.4), the homomorphisms $\rho_{\beta\alpha}$ define a ring A' as the *inductive limit* of the inductive system of rings $(S_\alpha^{-1}A, \rho_{\beta\alpha})$. Let ρ_α be the canonical map $S_\alpha^{-1}A \to A'$, and let $\phi_\alpha = \rho_A^{S,S_\alpha}$; as $\phi_\alpha = \phi_\beta \circ \rho_{\beta\alpha}$ for $\alpha \leqslant \beta$ according to (1.4.4), we can uniquely define a homomorphism $\phi : A' \to S^{-1}A$ such that the diagram

$$\begin{tikzcd}
 & S_\alpha^{-1}A \arrow[d, "\rho_{\beta\alpha}"] \arrow[ddl, "\rho_\alpha"'] \arrow[ddr, "\phi_\alpha"] & \\
 & S_\beta^{-1}A \arrow[dl, "\rho_\beta"] \arrow[dr, "\phi_\beta"'] & \\
A' \arrow[rr, "\phi"] & & S^{-1}A
\end{tikzcd} \qquad (\alpha \leqslant \beta)$$

is commutative. In fact, ϕ is an *isomorphism*; it is indeed immediate by construction that ϕ is surjective. On the other hand, if $\rho_\alpha(a/s_\alpha) \in A'$ is such that $\phi(\rho_\alpha(a/s_\alpha)) = 0$, then this means that $a/s_\alpha = 0$ in $S^{-1}A$, that is to say that there exists an $s \in S$ such that $sa = 0$; but there is a $\beta \geqslant \alpha$ such that $s \in S_\beta$, and consequently, as $\rho_\alpha(a/s_\alpha) = \rho_\beta(sa/ss_\alpha) = 0$, we find that ϕ is injective. The case for an A-module M is treated likewise, and we have thus defined canonical isomorphisms

$$\varinjlim S_\alpha^{-1}A \simeq (\varinjlim S_\alpha)^{-1}A, \ \varinjlim S_\alpha^{-1}M \simeq (\varinjlim S_\alpha)^{-1}M,$$

the second being *functorial* in M.

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(1.4.6). Let S_1 and S_2 be multiplicative subsets of A; then S_1S_2 is also a multiplicative subset of A. Let us denote by S_2' the canonical image of S_2 in the ring $S_1^{-1}A$, which is a multiplicative subset of this ring. For every A-module M there is then a functorial isomorphism

$$S_2'^{-1}(S_1^{-1}M) \simeq (S_1S_2)^{-1}M$$

which sends $(m/s_1)/(s_2/1)$ to the element $m/(s_1s_2)$.

1.5. Change of ring

(1.5.1). Let A and A' be rings, ϕ a homomorphism $A' \to A$, and S (resp. S') a multiplicative subset of A (resp. A'), such that $\phi(S') \subset S$; the composition homomorphism $A' \xrightarrow{\phi} A \to S^{-1}A$ factors as

$$A' \longrightarrow S'^{-1}A' \xrightarrow{\phi^{S'}} S^{-1}A,$$

by (1.2.4); where $\phi^{S'}(a'/s') = \phi(a')/\phi(s')$. If $A = \phi(A')$ and $S = \phi(S')$, then $\phi^{S'}$ is *surjective*. If $A' = A$ and ϕ is the identity, then $\phi^{S'}$ is exactly the homomorphism $\rho_A^{S,S'}$ defined in (1.4.1).

(1.5.2). Under the hypotheses of (1.5.1), let M be an A-module. There exists a canonical functorial morphism

$$\sigma : S'^{-1}(M_{[\phi]}) \longrightarrow (S^{-1}M)_{[\phi^{S'}]}$$

of $S'^{-1}A'$-modules, sending each element m/s' of $S'^{-1}(M_{[\phi]})$ to the element $m/\phi(s')$ of $(S^{-1}M)_{[\phi^{S'}]}$; indeed, we immediately see that this definition does not depend on the representative m/s' of the element in question. When $S = \phi(S')$, the homomorphism σ is *bijective*. When $A' = A$ and ϕ is the identity, σ is none other than the homomorphism $\rho_M^{S,S'}$ defined in (1.4.1).

When, in particular, we take $M = A$ the homomorphism ϕ defines an A'-algebra structure on A; $S'^{-1}(A_{[\phi]})$ is then endowed with a ring structure, with which it can be identified with $(\phi(S'))^{-1}A$, and the homomorphism $\sigma : S'^{-1}(A_{[\phi]}) \to S^{-1}A$ is a homomorphism of $S'^{-1}A'$-algebras.

(1.5.3). Let M and N be A-modules; by composing the homomorphisms defined in (1.3.4) and (1.5.2), we obtain a homomorphism

$$(S^{-1}M \otimes_{S^{-1}A} S^{-1}N)_{[\phi^{S'}]} \longleftarrow S'^{-1}((M \otimes N)_{[\phi]})$$

which is an isomorphism when $\phi(S') = S$. Similarly, by composing the homomorphisms in (1.3.5) and (1.5.2), we obtain a homomorphism

$$S'^{-1}((\mathrm{Hom}_A(M,N))_{[\phi]}) \longrightarrow (\mathrm{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N))_{[\phi^{S'}]}$$

which is an isomorphism when $\phi(S') = S$ and M admits a finite presentation.

(1.5.4). Let us now consider an A'-module N', and form the tensor product $N' \otimes_{A'} A_{[\phi]}$, which can be considered as an A-module by setting $a \cdot (n' \otimes b) = n' \otimes (ab)$. There is a functorial isomorphism of $S^{-1}A$-modules

$$\tau : (S'^{-1}N') \otimes_{S'^{-1}A'} (S^{-1}A)_{[\phi^{S'}]} \simeq S^{-1}(N' \otimes_{A'} A_{[\phi]})$$

which sends the element $(n'/s') \otimes (a/s)$ to the element $(n' \otimes a)/(\phi(s')s)$; indeed, we can show that $\mathbf{0_I}$ | 18
when we replace n'/s' (resp. a/s) by another expression of the same element, $(n' \otimes a)/(\phi(s')s)$ does not change; on the other hand, we can define a homomorphism inverse to τ by sending $(n' \otimes a)/s$ to the element $(n'/1) \otimes (a/s)$: we use the fact that $S^{-1}(N' \otimes_{A'} A_{[\phi]})$ is canonically isomorphic to $(N' \otimes_{A'} A_{[\phi]}) \otimes_A S^{-1}A$ (1.2.5), so also to $N' \otimes_{A'} (S^{-1}A)_{[\psi]}$, where we denote by ψ the composite homomorphism $a' \mapsto \phi(a')/1$ from A' to $S^{-1}A$.

(1.5.5). If M' and N' are A'-modules, then by composing the isomorphisms (1.3.4) and (1.5.4), we obtain an isomorphism

$$S'^{-1}M' \otimes_{S'^{-1}A'} S'^{-1}N' \otimes_{S'^{-1}A'} S^{-1}A \simeq S^{-1}(M' \otimes_{A'} N' \otimes_{A'} A).$$

Likewise, if M' admits a finite presentation, we have by (1.3.5) and (1.5.4) an isomorphism

$$\mathrm{Hom}_{S'^{-1}A'}(S'^{-1}M', S'^{-1}N') \otimes_{S'^{-1}A'} S^{-1}A \simeq S^{-1}(\mathrm{Hom}_{A'}(M',N') \otimes_{A'} A).$$

(1.5.6). Under the hypotheses of (1.5.1), let T (resp. T') be another multiplicative subset of A (resp. A') such that $S \subset T$ (resp. $S' \subset T'$) and $\phi(T') \subset T$. Then the diagram

$$\begin{tikzcd} S'^{-1}A' \arrow[r, "\phi^{S'}"] \arrow[d, "\rho^{T',S'}"'] & S^{-1}A \arrow[d, "\rho^{T,S}"] \\ T'^{-1}A' \arrow[r, "\phi^{T'}"] & T^{-1}A \end{tikzcd}$$

is commutative. If M is an A-module, then the diagram

$$\begin{tikzcd} S'^{-1}(M_{[\phi]}) \arrow[r, "\sigma"] \arrow[d, "\rho^{T',S'}"'] & (S^{-1}M)_{[\phi^{S'}]} \arrow[d, "\rho^{T,S}"] \\ T'^{-1}(M_{[\phi]}) \arrow[r, "\sigma"] & (T^{-1}M)_{[\phi^{T'}]} \end{tikzcd}$$

is commutative. Finally, if N' is an A'-module, then the diagram

$$\begin{tikzcd} (S'^{-1}N') \otimes_{S'^{-1}A'} (S^{-1}A)_{[\phi^{S'}]} \arrow[r, "\sim", "\tau"'] \arrow[d] & S^{-1}(N' \otimes_{A'} A_{[\phi]}) \arrow[d, "\rho^{T,S}"] \\ (T'^{-1}N') \otimes_{T'^{-1}A'} (T^{-1}A)_{[\phi^{T'}]} \arrow[r, "\sim", "\tau"'] & T^{-1}(N' \otimes_{A'} A_{[\phi]}) \end{tikzcd}$$

is commutative, the left vertical arrow obtained by applying $\rho_{N'}^{T',S'}$ to $S'^{-1}N'$ and $\rho_A^{T,S}$ to $S^{-1}A$.

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(1.5.7). Let A'' be a third ring, $\phi' : A'' \to A'$ a ring homomorphism, and S'' a multiplicative subset of A'' such that $\phi'(S'') \subset S'$. Let $\phi'' = \phi \circ \phi'$; then we have

$$\phi''^{S''} = \phi^{S'} \circ \phi'^{S''}.$$

Let M be an A-module; evidently we have $M_{[\phi'']} = (M_{[\phi]})_{[\phi']}$; if σ' and σ'' are the homomorphisms defined by ϕ' and ϕ'' in the same way as how σ is defined in (1.5.2) by ϕ, then we have the transitivity formula

$$\sigma'' = \sigma \circ \sigma'.$$

Finally, let N'' be an A''-module; the A-module $N'' \otimes_{A''} A_{[\phi'']}$ is canonically identified with $(N'' \otimes_{A''} A'_{[\phi']}) \otimes_{A'} A_{[\phi]}$, and likewise the $S^{-1}A$-module $(S''^{-1}N'') \otimes_{S''^{-1}A''} (S^{-1}A)_{[\phi''^{S''}]}$ is canonically identified with $((S''^{-1}N'') \otimes_{S''^{-1}A''} (S'^{-1}A')_{[\phi'^{S''}]}) \otimes_{S'^{-1}A'} (S^{-1}A)_{[\phi^{S'}]}$. With these identifications, if τ' and τ'' are the isomorphisms defined by ϕ' and ϕ'' in the same way as how τ is defined in (1.5.4) by ϕ, then we have the transitivity formula

$$\tau'' = \tau \circ (\tau' \otimes 1).$$

(1.5.8). Let A be a subring of a ring B; for every *minimal* prime ideal $\mathfrak{p}$ of A, there exists a minimal prime ideal $\mathfrak{q}$ of B such that $\mathfrak{p} = A \cap \mathfrak{q}$. Indeed, $A_{\mathfrak{p}}$ is a subring of $B_{\mathfrak{p}}$ (1.3.2) and has *a single prime ideal* $\mathfrak{p}'$ (1.2.6); since $B_{\mathfrak{p}}$ is not 0, it has at least one prime ideal $\mathfrak{q}'$ and we necessarily have $\mathfrak{q}' \cap A_{\mathfrak{p}} = \mathfrak{p}'$; the prime ideal $\mathfrak{q}_1$ of B, the inverse image of $\mathfrak{q}'$, is thus such that $\mathfrak{q}_1 \cap A = \mathfrak{p}$, and *a fortiori* we have $\mathfrak{q} \cap A = \mathfrak{p}$ for every minimal prime ideal $\mathfrak{q}$ of B contained in $\mathfrak{q}_1$.

1.6. Identification of the module M_f as an inductive limit

(1.6.1). Let M be an A-module and f an element of A. Consider a sequence (M_n) of A-modules, all identical to M, and for each pair of integers $m \leqslant n$, let ϕ_{nm} be the homomorphism $z \mapsto f^{n-m}z$ from M_m to M_n; it is immediate that $((M_n),(\phi_{nm}))$ is an *inductive system* of A-modules; let $N = \varinjlim M_n$ be the inductive limit of this system. We define a canonical *functorial* A-isomorphism from N to M_f. For this, let us note that, for all n, $\theta_n : z \mapsto z/f^n$ is an A-homomorphism from $M = M_n$ to M_f, and it follows from the definitions that we have $\theta_n \circ \phi_{nm} = \theta_m$ for $m \leqslant n$. As a result, there exists an A-homomorphism $\theta : N \to M_f$ such that, if ϕ_n denotes the canonical homomorphism $M_n \to N$, then we have $\theta_n = \theta \circ \phi_n$ for all n. Since, by hypothesis, every element of M_f is of the form z/f^n for at least one n, it is clear that θ is surjective. On the other hand, if $\theta(\phi_n(z)) = 0$, or, in other words, if $z/f^n = 0$, then there exists an integer $k > 0$ such that $f^k z = 0$, so $\phi_{n+k,n}(z) = 0$, which gives $\phi_n(z) = 0$. We can therefore identify M_f with $\varinjlim M_n$ via θ.

(1.6.2). Now write $M_{f,n}$, ϕ^f_{nm}, and ϕ^f_n instead of M_n, ϕ_{nm}, and ϕ_n. Let g be another element of A. Since f^n divides $f^n g^n$, we have a functorial homomorphism

$$\rho_{fg,f} : M_f \longrightarrow M_{fg} \quad ((1.4.1) \text{ and } (1.4.3));$$

if we identify M_f and M_{fg} with $\varinjlim M_{f,n}$ and $\varinjlim M_{fg,n}$ respectively, then $\rho_{fg,f}$ identifies with the $\mathbf{0_I}$ | 20
inductive limit of the maps $\rho^n_{fg,f} : M_{f,n} \to M_{fg,n}$, defined by $\rho^n_{fg,f}(z) = g^n z$. Indeed, this follows immediately from the commutativity of the diagram

$$\begin{tikzcd} M_{f,n} \arrow[r, "\rho^n_{fg,f}"] \arrow[d, "\phi^f_n"'] & M_{fg,n} \arrow[d, "\phi^{fg}_n"] \\ M_f \arrow[r, "\rho_{fg,f}"] & M_{fg}. \end{tikzcd}$$

1.7. Support of a module

(1.7.1). Given an A-module M, we define the *support* of M, denoted by $\operatorname{Supp}(M)$, to be the set of prime ideals $\mathfrak{p}$ of A such that $M_{\mathfrak{p}} \neq 0$. For it to be the case that $M = 0$, it is necessary and sufficient that $\operatorname{Supp}(M) = \varnothing$, because if $M_{\mathfrak{p}} = 0$ for all $\mathfrak{p}$, then the annihilator of an element $x \in M$ cannot be contained in any prime ideal of A, and so is the whole of A.

(1.7.2). If $0 \to N \to M \to P \to 0$ is an exact sequence of A-modules, then we have

$$\operatorname{Supp}(M) = \operatorname{Supp}(N) \cup \operatorname{Supp}(P)$$

because, for every prime ideal $\mathfrak{p}$ of A, the sequence $0 \to N_{\mathfrak{p}} \to M_{\mathfrak{p}} \to P_{\mathfrak{p}} \to 0$ is exact (1.3.2) and in order that $M_{\mathfrak{p}} = 0$, it is necessary and sufficient that $N_{\mathfrak{p}} = P_{\mathfrak{p}} = 0$.

(1.7.3). If M is the sum of a family (M_λ) of submodules, then $M_{\mathfrak{p}}$ is the sum of the $(M_\lambda)_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A ((1.3.3) and (1.3.2)), so $\operatorname{Supp}(M) = \bigcup_\lambda \operatorname{Supp}(M_\lambda)$.

(1.7.4). If M is an A-module *of finite type*, then $\operatorname{Supp}(M)$ is the set of prime ideals *containing the annihilator of* M. Indeed, if M is cyclic and generated by x, then to say that $M_{\mathfrak{p}} = 0$ is to say that that there exists an $s \notin \mathfrak{p}$ such that $s \cdot x = 0$, and thus that $\mathfrak{p}$ does not contain the annihilator of x. Now if M admits a finite system $(x_i)_{1 \leqslant i \leqslant n}$ of generators, and if $\mathfrak{a}_i$ is the annihilator of x_i, then it follows from (1.7.3) that $\operatorname{Supp}(M)$ is the set of the $\mathfrak{p}$ containing one of the $\mathfrak{a}_i$, or equivalently, the set of the $\mathfrak{p}$ containing $\mathfrak{a} = \bigcap_i \mathfrak{a}_i$, which is the annihilator of M.

(1.7.5). If M and N are two A-modules *of finite type*, then we have

$$\operatorname{Supp}(M \otimes_A N) = \operatorname{Supp}(M) \cap \operatorname{Supp}(N).$$

It is a question of seeing that, if $\mathfrak{p}$ is a prime ideal of A, then the condition $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \neq 0$ is equivalent to "$M_{\mathfrak{p}} \neq 0$ and $N_{\mathfrak{p}} \neq 0$" (taking (1.3.4) into account). In other words, it is a question of seeing that, if P and Q are modules of finite type over a *local* ring $B \neq 0$, then $P \otimes_B Q \neq 0$. Let $\mathfrak{m}$ be the maximal ideal of B. By Nakayama's Lemma, the vector spaces $P/\mathfrak{m}P$ and $Q/\mathfrak{m}Q$ are not 0, and so it is the same for the tensor product $(P/\mathfrak{m}P) \otimes_{B/\mathfrak{m}} (Q/\mathfrak{m}Q) = (P \otimes_B Q) \otimes_B (B/\mathfrak{m})$, whence the conclusion.

In particular, if M is an A-module of finite type, and $\mathfrak{a}$ an ideal of A, then $\operatorname{Supp}(M/\mathfrak{a}M)$ is the set of prime ideals containing both $\mathfrak{a}$ and the annihilator $\mathfrak{n}$ of M (1.7.4), that is, the set of prime ideals containing $\mathfrak{a} + \mathfrak{n}$.

§2. Irreducible spaces. Noetherian spaces

2.1. Irreducible spaces

$\mathbf{0_I}$ | 21

(2.1.1). We say that a topological space X is *irreducible* if it is nonempty and if it is not a union of two distinct closed subspaces of X. It is equivalent to say that $X \neq \varnothing$ and the intersection of two nonempty open sets (and consequently of a finite number of open sets) of X is nonempty, or that every nonempty open set is everywhere dense, or that any closed set is *rare*[1], or, lastly, that all open sets of X are *connected*.

(2.1.2). For a subspace Y of a topological space X to be irreducible, it is necessary and sufficient that its closure $\overline{Y}$ be irreducible. In particular, any subspace which is the closure $\overline{\{x\}}$ of a singleton is irreducible; we will express the relation $y \in \overline{\{x\}}$ (equivalent to $\overline{\{y\}} \subset \overline{\{x\}}$) by saying that y is a *specialization of* x or that x is a *generalization of* y. When there exists, in an irreducible space X, a point x such that $X = \overline{\{x\}}$, we will say that x is a *generic point of* X. Any nonempty open subset of X then contains x, and any subspace containing x has x as a generic point.

(2.1.3). Recall that a *Kolmogoroff space* is a topological space X satisfying the axiom of separation:

(T_0) If $x \neq y$ are any two points of X, there is an open set containing one of the points x and y, but not the other.

---

[1] [Trans] also known as *nowhere dense*.

If an irreducible Kolmogoroff space admits a generic point, it admits *exactly* one, since a nonempty open set contains any generic point.

Recall that a topological space X is said to be *quasi-compact* if, from any collection of open sets of X, one can extract a finite cover of X (or, equivalently, if any decreasing filtered family of nonempty closed sets has a nonempty intersection). If X is a quasi-compact space, then any nonempty closed subset A of X contains a *minimal* nonempty closed set M, because the set of nonempty closed subsets of A is inductive under the relation $\supset$; if, in addition, X is a Kolmogoroff space, M is necessarily a single point (or, as we say by abuse of language, is a *closed point*).

(2.1.4). In an irreducible space X, every nonempty open subspace U is irreducible, and if X admits a generic point x, x is also a generic point of U.

To prove this, let (U_α) be a cover (whose set of indices is nonempty) of a topological space X, consisting of nonempty open sets; if X is irreducible, it is necessary and sufficient that U_α is irreducible for all α, and that $U_\alpha \cap U_\beta \neq \varnothing$ for any α, β. The condition is clearly necessary; to see that it is sufficient, it suffices to prove that if V is a nonempty open subset of X, then $V \cap U_\alpha$ is nonempty for all α, since then $V \cap U_\alpha$ is dense in U_α for all α, and consequently V is dense in X. Now there is at least one index γ such that $V \cap U_\gamma \neq \varnothing$, so $V \cap U_\gamma$ is dense in U_γ, and as for all α, $U_\alpha \cap V_\alpha \neq \varnothing$, we also have $V \cap U_\alpha \cap U_\gamma \neq \varnothing$.

$0_{\mathbf{I}}$ | 22

(2.1.5). Let X be an irreducible space, and f a continuous map from X into a topological space Y. Then $f(X)$ is irreducible, and if x is a generic point of X, then $f(x)$ is a generic point of $f(X)$ and hence also of $\overline{f(X)}$. In particular, if, in addition, Y is irreducible and with a single generic point y, then for $f(X)$ to be everywhere dense, it is necessary and sufficient that $f(x) = y$.

(2.1.6). Any irreducible subspace of a topological space X is contained in a maximal irreducible subspace, which is necessarily closed. Maximal irreducible subspaces of X are called the *irreducible components* of X. If Z_1 and Z_2 are two irreducible components distinct from the space X, then $Z_1 \cap Z_2$ is a closed *rare* set in each of the subspaces Z_1, Z_2; in particular, if an irreducible component of X admits a generic point (2.1.2), such a point cannot belong to any other irreducible component. If X has only a *finite* number of irreducible components Z_i ($1 \leqslant i \leqslant n$), and if, for each i, we put $U_i = \bigcup_{j \neq i} Z_j$, then the U_i are open, irreducible, disjoint, and their union is dense in X. Let U be an open subset of a topological space X. If Z is an irreducible subset of X that intersects U, then $Z \cap U$ is open and dense in Z, thus irreducible; conversely, for any irreducible closed subset Y of U, the closure $\overline{Y}$ of Y in X is irreducible and $\overline{Y} \cap U = Y$. We conclude that there is a *bijective correspondence* between the irreducible components of U and the irreducible components of X which intersect U. **$\mathrm{Err}_{\mathrm{II}}$**

(2.1.7). If a topological space X is a union of a *finite* number of irreducible closed subspaces Y_i, then the irreducible components of X are the maximal elements of the set of the Y_i, because if Z is an irreducible closed subset of X, then Z is the union of the $Z \cap Y_i$, from which one sees that Z must be contained in one of the Y_i. Let Y be a subspace of a topological space X, and suppose that Y has only a finite number of irreducible components Y_i, ($1 \leqslant i \leqslant n$); then the closures $\overline{Y_i}$ in X are the irreducible components of Y.

(2.1.8). Let Y be an irreducible space admitting a single generic point y. Let X be a topological space, and f a continuous map from X to Y. Then, for any irreducible component Z of X intersecting $f^{-1}(y)$, $f(Z)$ is dense in Y. The converse is not necessarily true; however, if Z has a generic point z, and if $f(Z)$ is dense in Y, then we must have $f(z) = y$ (2.1.5); in addition, $Z \cap f^{-1}(y)$ is then the closure of $\{z\}$ in $f^{-1}(y)$ and is therefore irreducible, and as an irreducible subset of $f^{-1}(y)$ containing z is necessarily contained in Z (2.1.6), z is a generic point of $Z \cap f^{-1}(y)$. As any irreducible component of $f^{-1}(y)$ is contained in an irreducible component of X, we see that, if any irreducible component Z of X intersecting $f^{-1}(y)$ admits a generic point, then there is a *bijective correspondence* between all these components and all the irreducible components $Z \cap f^{-1}(y)$ of $f^{-1}(y)$, the generic points of Z being identical to those of $Z \cap f^{-1}(y)$.

2.2. Noetherian spaces

$\mathbf{0_I}$ | 23

(2.2.1). We say that a topological space X is *Noetherian* if the set of open subsets of X satisfies the *maximal* condition, or, equivalently, if the set of closed subsets of X satisfies the *minimal* condition. We say that X is *locally Noetherian* if each $x \in X$ admits a neighbourhood which is a Noetherian subspace.

(2.2.2). Let E be an ordered set satisfying the *minimal* condition, and let $\mathbf{P}$ be a property of the elements of E subject to the following condition: if $a \in E$ is such that for any $x < a$, $\mathbf{P}(x)$ is true, then $\mathbf{P}(a)$ is true. Under these conditions, $\mathbf{P}(x)$ *is true for all* $x \in E$ ("principle of Noetherian induction"). Indeed, let F be the set of $x \in E$ for which $\mathbf{P}(x)$ is false; if F were not empty, it would have a minimal element a, and as then $\mathbf{P}(x)$ is true for all $x < a$, $\mathbf{P}(a)$ would be true, which is a contradiction.

We will apply this principle in particular when E is a *set of closed subsets of a Noetherian space*.

(2.2.3). Any subspace of a Noetherian space is Noetherian. Conversely, any topological space that is a finite union of Noetherian subspaces is Noetherian.

(2.2.4). Any Noetherian space is quasi-compact; conversely, any topological space in which all open sets are quasi-compact is Noetherian.

(2.2.5). A Noetherian space has only a *finite* number of irreducible components, as we see by Noetherian induction.

§3. SUPPLEMENT ON SHEAVES

3.1. Sheaves with values in a category

(3.1.1). Let $\mathtt{C}$ be a category, $(A_\alpha)_{\alpha\in I}$, $(A_{\alpha\beta})_{(\alpha,\beta)\in I\times I}$ two families of objects of $\mathtt{C}$ such that $A_{\beta\alpha} = A_{\alpha\beta}$, and $(\rho_{\alpha\beta})_{(\alpha,\beta)\in I\times I}$ a family of morphisms $\rho_{\alpha\beta} : A_\alpha \to A_{\alpha\beta}$. We say that a pair consisting of an object A of $\mathtt{C}$ and a family of morphisms $\rho_\alpha : A \to A_\alpha$ is a *solution to the universal problem* defined by the data of the families (A_α), $(A_{\alpha\beta})$, and $(\rho_{\alpha\beta})$ if, for every object B of $\mathtt{C}$, the map which sends $f \in \operatorname{Hom}(B, A)$ to the family $(\rho_\alpha \circ f) \in \Pi_\alpha \operatorname{Hom}(B, A_\alpha)$ is a *bijection* of $\operatorname{Hom}(B, A)$ to the set of all (f_α) such that $\rho_{\alpha\beta} \circ f_\alpha = \rho_{\beta\alpha} \circ f_\beta$ for any pair of indices (α, β). If such a solution exists, it is unique up to an isomorphism.

(3.1.2). We will not recall the definition of a *presheaf* $U \mapsto \mathscr{F}(U)$ on a topological space X with values in a category $\mathtt{C}$ (G, I, 1.9); we say that such a presheaf is a *sheaf with values in* $\mathtt{C}$ if it satisfies the following axiom:

(F) *For any covering* (U_α) *of an open set* U *of* X *by open sets* U_α *contained in* U, *if we denote by* ρ_α (resp. $\rho_{\alpha\beta}$) *the restriction morphism*

$$\mathscr{F}(U) \longrightarrow \mathscr{F}(U_\alpha) \quad (\text{resp. } \mathscr{F}(U_\alpha) \longrightarrow \mathscr{F}(U_\alpha \cap U_\beta)),$$

the pair formed by $\mathscr{F}(U)$ *and the family* (ρ_α) *are a solution to the universal problem for* $(\mathscr{F}(U_\alpha))$, $(\mathscr{F}(U_\alpha \cap U_\beta))$, *and* $(\rho_{\alpha\beta})$ *in* (3.1.1)[2]. $\mathbf{0_I}$ | 24

Equivalently, we can say that, for each object T of $\mathtt{C}$, that the family $U \mapsto \operatorname{Hom}(T, \mathscr{F}(U))$ is a *sheaf of sets*.

(3.1.3). Assume that $\mathtt{C}$ is the category defined by a "type of structure with morphisms" Σ, the objects of $\mathtt{C}$ being the sets with structures of type Σ and morphisms those of Σ. Suppose that the category $\mathtt{C}$ also satisfies the following condition:

(E) If $(A, (\rho_\alpha))$ is a solution of a universal mapping problem *in the category* $\mathtt{C}$ for families (A_α), $(A_{\alpha\beta})$, $(\rho_{\alpha\beta})$, then it is also a solution of the universal mapping problem for the same families *in the category of sets* (that is, when we consider A, A_α, and $A_{\alpha\beta}$ as sets, ρ_α and $\rho_{\alpha\beta}$ as functions)[3].

[2]This is a special case of the more general notion of a (non-filtered) *projective limit* (*see* (T, I, 1.8) and the book in preparation announced in the introduction).

[3]It can be proved that it also means that the canonical functor $\mathtt{C} \to \mathtt{Set}$ *commutes with projective limits* (not necessarily filtered).

Under these conditions, the condition (F) gives that, when considered as a presheaf *of sets*, $U \mapsto \mathscr{F}(U)$ is a *sheaf*. In addition, for a map $u : T \to \mathscr{F}(U)$ to be a morphism of $\mathtt{C}$, it is necessary and sufficient, according to (F), that each map $\rho_\alpha \circ u$ is a morphism $T \to \mathscr{F}(U_\alpha)$, which means that the structure of type Σ on $\mathscr{F}(U)$ is the *initial structure* for the morphisms ρ_α. Conversely, suppose a presheaf $U \mapsto \mathscr{F}(U)$ on X, with values in $\mathtt{C}$, is a *sheaf of sets* and satisfies the previous condition; it is then clear that it satisfies (F), so it is a *sheaf with values in* $\mathtt{C}$.

(3.1.4). When Σ is a type of a group or ring structure, the fact that the presheaf $U \mapsto \mathscr{F}(U)$ with values in $\mathtt{C}$ is a sheaf of *sets* implies *ipso facto* that it is a sheaf with values in $\mathtt{C}$ (in other words, a sheaf of groups or rings within the meaning of (G))[4]. But it is not the same when, for example, $\mathtt{C}$ is the category of *topological rings* (with morphisms as continuous homomorphisms): a sheaf with values in $\mathtt{C}$ is a sheaf of rings $U \mapsto \mathscr{F}(U)$ such that for any open U and any covering of U by open sets $U_\alpha \subset U$, the topology of the ring $\mathscr{F}(U)$ is to be *the least fine* making the homomorphisms $\mathscr{F}(U) \to \mathscr{F}(U_\alpha)$ continuous. We will say in this case that $U \mapsto \mathscr{F}(U)$, considered as a sheaf of rings (without a topology), is *underlying* the sheaf of topological rings $U \mapsto \mathscr{F}(U)$. Morphisms $u_V : \mathscr{F}(V) \to \mathscr{G}(V)$ (V an arbitrary open subset of X) of sheaves of topological rings are therefore homomorphisms of the underlying sheaves of rings, such that u_V is *continuous* for all open $V \subset X$; to distinguish them from any homomorphisms of the sheaves of the underlying rings, we will call them continuous homomorphisms of sheaves of topological rings. We have similar definitions and conventions for sheaves of topological spaces or topological groups.

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(3.1.5). It is clear that for any category $\mathtt{C}$, if there is a presheaf (respectively a sheaf) $\mathscr{F}$ on X with values in $\mathtt{C}$ and U is an open set of X, the $\mathscr{F}(V)$ for open $V \subset U$ constitute a presheaf (or a sheaf) with values in $\mathtt{C}$, which we call the presheaf (or sheaf) *induced* by $\mathscr{F}$ on U and denote it by $\mathscr{F}|U$.

For any morphism $u : \mathscr{F} \to \mathscr{G}$ of presheaves on X with values in $\mathtt{C}$, we denote by $u|U$ the morphism $\mathscr{F}|U \to \mathscr{G}|U$ consisting of the u_V for $V \subset U$.

(3.1.6). Suppose now that the category $\mathtt{C}$ admits *inductive limits* (T, 1.8); then, for any presheaf (and in particular any sheaf) $\mathscr{F}$ on X with values in $\mathtt{C}$ and each $x \in X$, we can define the *stalk* $\mathscr{F}_x$ as the object of $\mathtt{C}$ defined by the inductive limit of the $\mathscr{F}(U)$ with respect to the filtered set (for $\supset$) of the open neighbourhoods U of x in X, and the morphisms $\rho_U^V : \mathscr{F}(V) \to \mathscr{F}(U)$. If $u : \mathscr{F} \to \mathscr{G}$ is a morphism of presheaves with values in $\mathtt{C}$, we define for each $x \in X$ the morphism $u_x : \mathscr{F}_x \to \mathscr{G}_x$ as the inductive limit of $u_U : \mathscr{F}(U) \to \mathscr{G}(U)$ with respect to all open neighbourhoods of x; we thus define $\mathscr{F}_x$ as a covariant functor in $\mathscr{F}$, with values in $\mathtt{C}$, for all $x \in X$.

When $\mathtt{C}$ is further defined by a kind of structure with morphisms Σ, we call *sections over* U of a *sheaf* $\mathscr{F}$ with values in $\mathtt{C}$ the elements of $\mathscr{F}(U)$, and we write $\Gamma(U, \mathscr{F})$ instead of $\mathscr{F}(U)$; for $s \in \Gamma(U, \mathscr{F})$, V an open set contained in U, we write $s|V$ instead of $\rho_V^U(s)$; for all $x \in U$, the canonical image of s in $\mathscr{F}_x$ is the *germ* of s at the point x, denoted by s_x (*we will never replace the notation* $s(x)$ *in this sense*, this notation being reserved for another notion relating to sheaves which will be considered in this treatise (5.5.1)).

If then $u : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves with values in $\mathtt{C}$, we will write $u(s)$ instead of $u_V(s)$ for all $s \in \Gamma(V, \mathscr{F})$. $\mathbf{Err_{II}}$

If $\mathscr{F}$ is a sheaf of commutative groups, or rings, or modules, we say that the set of $x \in X$ such that $\mathscr{F}_x \neq \{0\}$ is the *support* of $\mathscr{F}$, denoted $\operatorname{Supp}(\mathscr{F})$; this set is not necessarily closed in X.

When $\mathtt{C}$ is defined by a type of structure with morphisms, *we systematically refrain from using the point of view of "étalé spaces"* in terms of relating to sheaves with values in $\mathtt{C}$; in other words, we will never consider a sheaf as a topological space (nor even as the whole union of its stalks), and we will not consider also a morphism $u : \mathscr{F} \to \mathscr{G}$ of such sheaves on X as a continuous map of topological spaces.

[4] This is because in the category $\mathtt{C}$, any morphism that is a *bijection* (as a map of sets) is an *isomorphism*. This is no longer true when $\mathtt{C}$ is the category of topological spaces, for example.

3.2. Presheaves on an open basis

(3.2.1). We will restrict to the following categories $\mathtt{C}$ admitting *projective limits* (generalized, that is, corresponding to not necessarily filtered preordered sets, cf. (T, 1.8)). Let X be a topological space, $\mathfrak{B}$ an open basis for the topology of X. We will call a *presheaf on* $\mathfrak{B}$*, with values in* $\mathtt{C}$, a family of objects $\mathscr{F}(U) \in \mathtt{C}$, corresponding to each $U \in \mathfrak{B}$, and a family of morphisms $\rho_U^V : \mathscr{F}(V) \to \mathscr{F}(U)$ defined for any pair (U, V) of elements of $\mathfrak{B}$ such that $U \subset V$, with the conditions $\rho_U^U =$ identity and $\mathbf{0_I}$ | 26 $\rho_U^W = \rho_U^V \circ \rho_V^W$ if U, V, W in $\mathfrak{B}$ are such that $U \subset V \subset W$. We can associate a *presheaf with values in* $\mathtt{C}$: $U \mapsto \mathscr{F}(U)$ in the ordinary sense, taking for all open U, $\mathscr{F}'(U) = \varprojlim \mathscr{F}(V)$, where V runs through the ordered set (for $\subset$, *not filtered* in general) of $V \in \mathfrak{B}$ sets such that $V \subset U$, since the (V) form a projective system for the ρ_V^W ($V \subset W \subset U$, $V \in \mathfrak{B}$, $W \in \mathfrak{B}$). Indeed, if U, U' are two open sets of X such that $U \subset U'$, we define ${\rho'}_U^{U'}$ as the projective limit (for $V \subset U$) of the canonical morphisms $\mathscr{F}'(U') \to \mathscr{F}(V)$, in other words the unique morphism $\mathscr{F}'(U') \to \mathscr{F}'(U)$, which, when composed with the canonical morphisms $\mathscr{F}'(U) \to \mathscr{F}(V)$, gives the canonical morphisms $\mathscr{F}'(U') \to \mathscr{F}(V)$; the verification of the transitivity of ${\rho'}_U^{U'}$ is then immediate. Moreover, if $U \in \mathfrak{B}$, the canonical morphism $\mathscr{F}'(U) \to \mathscr{F}(U)$ is an isomorphism, allowing us to identify these two objects[5].

(3.2.2). For the presheaf $\mathscr{F}'$ thus defined to be a *sheaf*, it is necessary and sufficient that the presheaf $\mathscr{F}$ on $\mathfrak{B}$ satisfies the condition:

(F_0) *For any covering* (U_α) *of* $U \in \mathfrak{B}$ *by sets* $U_\alpha \in \mathfrak{B}$ *contained in* U*, and for any object* $T \in \mathtt{C}$*, the map which sends* $f \in \mathrm{Hom}(T, \mathscr{F}(U))$ *to the family* $(\rho_{U_\alpha}^U \circ f) \in \Pi_\alpha \mathrm{Hom}(T, \mathscr{F}(U_\alpha))$ *is a bijection from* $\mathrm{Hom}(T, \mathscr{F}(U))$ *to the set of all* (f_α) *such that* $\rho_V^{U_\alpha} \circ f_\alpha = \rho_V^{U_\beta} \circ f_\beta$ *for any pair of indices* (α, β) *and any* $V \in \mathfrak{B}$ *such that* $V \subset U_\alpha \cap U_\beta$[6].

The condition is obviously necessary. To show that it is sufficient, consider first a second basis $\mathfrak{B}'$ of the topology of X, *contained in* $\mathfrak{B}$, and show that if $\mathscr{F}''$ denotes the presheaf induced by the subfamily $(\mathscr{F}(V))_{V \in \mathfrak{B}'}$, $\mathscr{F}''$ is *canonically isomorphic* to $\mathscr{F}'$. Indeed, first the projective limit (for $V \in \mathfrak{B}'$, $V \subset U$) of the canonical morphisms $\mathscr{F}'(U) \to \mathscr{F}(V)$ is a morphism $\mathscr{F}'(U) \to \mathscr{F}''(U)$ for all open U. If $U \in \mathfrak{B}$, this morphism is an isomorphism, because by hypothesis the canonical morphisms $\mathscr{F}''(U) \to \mathscr{F}(V)$ for $V \in \mathfrak{B}'$, $V \subset U$, factorize as $\mathscr{F}''(U) \to \mathscr{F}(U) \to \mathscr{F}(V)$, and it is immediate to see that the composition of morphisms $\mathscr{F}(U) \to \mathscr{F}''(U)$ and $\mathscr{F}''(U) \to \mathscr{F}(U)$ thus defined are the identities. This being so, for all open U, the morphisms $\mathscr{F}''(U) \to \mathscr{F}''(W) = \mathscr{F}(W)$ for $W \in \mathfrak{B}$ and $W \subset U$ satisfy the conditions characterizing the projective limit of $\mathscr{F}(W)$ ($W \in \mathfrak{B}$, $W \subset U$), which proves our assertion given the uniqueness of a projective limit up to isomorphism.

This being so, let U be any open set of X, (U_α) a covering of U by the open sets contained in U, and $\mathfrak{B}'$ the subfamily of $\mathfrak{B}$ formed by the sets of $\mathfrak{B}$ contained in at least one U_α; it is clear that $\mathbf{0_I}$ | 27 $\mathfrak{B}'$ is still a basis of the topology of U, so $\mathscr{F}'(U)$ (resp. $\mathscr{F}''(U_\alpha)$) is the projective limit of $\mathscr{F}(V)$ for $V \in \mathfrak{B}'$ and $V \subset U$ (resp., $V \subset U_\alpha$), the axiom (F) is then immediately verified by virtue of the definition of the projective limit.

When (F_0) is satisfied, we will say by abuse of language that the presheaf $\mathscr{F}$ on the basis $\mathfrak{B}$ is a sheaf.

(3.2.3). Let $\mathscr{F}, \mathscr{G}$ be two presheaves on a basis $\mathfrak{B}$, with values in $\mathtt{C}$; we define a *morphism* $u : \mathscr{F} \to \mathscr{G}$ as a family $(u_V)_{V \in \mathfrak{B}}$ of morphisms $u_V : \mathscr{F}(V) \to \mathscr{G}(V)$ satisfying the usual compatibility conditions with the restriction morphisms ρ_V^W. With the notation of (3.2.1), we have a morphism $u' : \mathscr{F}' \to \mathscr{G}'$ of (ordinary) presheaves by taking for u'_U the projective limit of the u_V for $V \in \mathfrak{B}$ and $V \subset U$; the

[5] If X is a *Noetherian* space, we can still define $\mathscr{F}'(U)$ and show that it is a presheaf (in the ordinary sense) when one supposes only that $\mathtt{C}$ admits projective limits for *finite* projective systems. Indeed, if U is any open set of X, there is a *finite* covering (V_i) of U consisting of sets of $\mathfrak{B}$; for every couple (i, j) of indices, let (V_{ijk}) be a finite covering of $V_i \cap V_j$ formed by sets of $\mathfrak{B}$. Let I be the set of i and triples (i, j, k), ordered only by the relations $i > (i, j, k)$, $j > (i, j, k)$; we then take $\mathscr{F}'(U)$ to be the projective limit of the system of $\mathscr{F}(V_i)$ and $\mathscr{F}(V_{ijk})$; it is easy to verify that this does not depend on the coverings (V_i) and (V_{ijk}) and that $U \mapsto \mathscr{F}'(U)$ is a presheaf.

[6] It also means that the pair formed by $\mathscr{F}(U)$ and the $\rho_\alpha = \rho_{U_\alpha}^U$ is a *solution to the universal problem* defined in (3.1.1) by the data of $A_\alpha = \mathscr{F}(U_\alpha)$, $A_{\alpha\beta} = \Pi\mathscr{F}(V)$ (for $V \in \mathfrak{B}$ such that $V \subset U_\alpha \cap U_\beta$) and $\rho_{\alpha\beta} = (\rho''_V) : \mathscr{F}(U_\alpha) \to \Pi\mathscr{F}(V)$ defined by the condition that for $V \in \mathfrak{B}$, $V' \in \mathfrak{B}$, $W \in \mathfrak{B}$, $V \cup V' \subset U_\alpha \cap U_\beta$, $W \subset V \cap V'$, $\rho_W^V \circ \rho''_V = \rho_W^{V'} \circ \rho''_{V'}$.

verification of the compatibility conditions with the $\rho'^{U'}_{U}$ follows from the functorial properties of the projective limit.

(3.2.4). If the category $\mathtt{C}$ admits inductive limits, and if $\mathscr{F}$ is a presheaf on the basis $\mathfrak{B}$, with values in $\mathtt{C}$, for each $x \in X$ the neighbourhoods of x belonging to $\mathfrak{B}$ form a cofinal set (for $\supset$) in the set of neighbourhoods of x, therefore, if $\mathscr{F}'$ is the (ordinary) presheaf corresponding to $\mathscr{F}$, the stalk $\mathscr{F}'_x$ is equal to $\varinjlim_{\mathfrak{B}} \mathscr{F}(V)$ over the set of $V \in \mathfrak{B}$ containing x. If $u : \mathscr{F} \to \mathscr{G}$ is morphism of presheaves on $\mathfrak{B}$ with values in $\mathtt{C}$, $u' : \mathscr{F}' \to \mathscr{G}'$ the corresponding morphism of ordinary presheaves, u'_x is likewise the inductive limit of the morphisms $u_V : \mathscr{F}(V) \to \mathscr{G}(V)$ for $V \in \mathfrak{B}$, $x \in V$.

(3.2.5). We return to the general conditions of (3.2.1). If $\mathscr{F}$ is an ordinary *sheaf* with values in $\mathtt{C}$, $\mathscr{F}_1$ the sheaf *on* $\mathfrak{B}$ obtained by the restriction of $\mathscr{F}$ to $\mathfrak{B}$, then the ordinary sheaf $\mathscr{F}'_1$ obtained from $\mathscr{F}_1$ by the procedure of (3.2.1) is canonically isomorphic to $\mathscr{F}$, by virtue of the condition (F) and the uniqueness properties of the projective limit. We identify the ordinary sheaf $\mathscr{F}$ with $\mathscr{F}'_1$.

If $\mathscr{G}$ is a second (ordinary) sheaf on X with values in $\mathtt{C}$, and $u : \mathscr{F} \to \mathscr{G}$ a morphism, the preceding remark shows that the data of the $u_V : \mathscr{F}(V) \to \mathscr{G}(V)$ *for only the* $V \in \mathfrak{B}$ completely determines u; conversely, it is sufficient, the u_V being given for $V \in \mathfrak{B}$, to verify the commutative diagram with the restriction morphisms ρ^W_V for $V \in \mathfrak{B}$, $W \in \mathfrak{B}$, and $V \subset W$, for there to exist a unique morphism u' from $\mathscr{F}$ to $\mathscr{G}$ such that $u'_V = u_V$ for each $V \in \mathfrak{B}$ (3.2.3).

(3.2.6). Suppose that $\mathtt{C}$ admits projective limits. Then the category of *sheaves on X with values in* $\mathtt{C}$ admits *projective limits*; if $(\mathscr{F}_\lambda)$ is a projective system of sheaves on X with values in $\mathtt{C}$, the $\mathscr{F}(U) = \varprojlim_\lambda \mathscr{F}_\lambda(U)$ indeed define a presheaf with values in $\mathtt{C}$, and the verification of the axiom (F) follows from the transitivity of projective limits; the fact that $\mathscr{F}$ is then the projective limit of the $\mathscr{F}_\lambda$ is immediate.

When $\mathtt{C}$ is the category of sets, for each projective system $(\mathscr{H}_\lambda)$ such that $\mathscr{H}_\lambda$ is a *subsheaf* of $\mathscr{F}_\lambda$ $\mathbf{0_I}$ | 28
for each λ, $\varprojlim_\lambda \mathscr{H}_\lambda$ canonically identifies with a *subsheaf* of $\varprojlim_\lambda \mathscr{F}_\lambda$. If $\mathtt{C}$ is the category of abelian groups, the covariant functor $\varprojlim_\lambda \mathscr{F}_\lambda$ is *additive* and *left exact*.

3.3. Gluing sheaves

(3.3.1). Suppose still that the category $\mathtt{C}$ admits (generalized) projective limits. Let X be a topological space, $\mathfrak{U} = (U_\lambda)_{\lambda \in L}$ an open cover of X, and for each $\lambda \in L$, let $\mathscr{F}_\lambda$ be a sheaf on U_λ, with values in $\mathtt{C}$; for each pair of indices (λ, μ), suppose that we are given an *isomorphism* $\theta_{\lambda\mu} : \mathscr{F}_\mu|(U_\lambda \cap U_\mu) \simeq \mathscr{F}|(U_\lambda \cap U_\mu)$; in addition, suppose that for each triple (λ, μ, ν), if we denote by $\theta'_{\lambda\mu}, \theta'_{\mu\nu}, \theta'_{\lambda\nu}$ the restrictions of $\theta_{\lambda\mu}, \theta_{\mu\nu}, \theta_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$, then we have $\theta'_{\lambda\nu} = \theta'_{\lambda\mu} \circ \theta'_{\mu\nu}$ (*gluing condition* for the $\theta_{\lambda\mu}$). Then there exists a sheaf $\mathscr{F}$ on X, with values in $\mathtt{C}$, and for each λ an isomorphism $\eta_\lambda : \mathscr{F}|U_\lambda \simeq \mathscr{F}_\lambda$ such that, for each pair (λ, μ), if we denote by η'_λ and η'_μ the restrictions of η_λ and η_μ to $U_\lambda \cap U_\mu$, then we have $\theta_{\lambda\mu} = \eta'_\lambda \circ \eta'^{-1}_\mu$; in addition, $\mathscr{F}$ and the η_λ are determined up to unique isomorphism by these conditions. The uniqueness indeed follows immediately from (3.2.5). To establish the existence of $\mathscr{F}$, denote by $\mathfrak{B}$ the open basis consisting of the open sets contained in at least one U_λ, and for each $U \in \mathfrak{B}$, choose (by the Hilbert function τ) one of the $\mathscr{F}_\lambda(U)$ for one of the λ such that $U \subset U_\lambda$; if we denote this object by $\mathscr{F}(U)$, the ρ^V_U for $U \subset V$, $U \in \mathfrak{B}$, $V \in \mathfrak{B}$ are defined in an evident way (by means of the $\theta_{\lambda\mu}$), and the transitivity conditions is a consequence of the gluing condition; in addition, the verification of (F_0) is immediate, so the presheaf on $\mathfrak{B}$ thus clearly defines a sheaf, and we deduce by the general procedure (3.2.1) an (ordinary) sheaf still denoted $\mathscr{F}$ and which answers the question. We say that $\mathscr{F}$ is obtained by *gluing the $\mathscr{F}_\lambda$ by means of the* $\theta_{\lambda\mu}$ and we usually identify the $\mathscr{F}_\lambda$ and $\mathscr{F}|U_\lambda$ by means of the η_λ.

It is clear that each sheaf $\mathscr{F}$ on X with values in $\mathtt{C}$ can be considered as being obtained by the gluing of the sheaves $\mathscr{F}_\lambda = \mathscr{F}|U_\lambda$ (where (U_λ) is an arbitrary open cover of X), by means of the isomorphisms $\theta_{\lambda\mu}$ reduced to the identity.

(3.3.2). With the same notation, let $\mathscr{G}_\lambda$ be a second sheaf on U_λ (for each $\lambda \in L$) with values in $\mathtt{C}$, and for each pair (λ, μ) let us be given an isomorphism $\omega_{\lambda\mu} : \mathscr{G}_\mu|(U_\lambda \cap U_\mu) \simeq \mathscr{G}_\lambda|(U_\lambda \cap U_\mu)$, these isomorphisms satisfying the gluing condition. Finally, suppose that we are given for each λ a morphism $u_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$, and that the diagrams

(3.3.2.1)
$$\begin{tikzcd} \mathscr{F}_\mu|(U_\lambda \cap U_\mu) \arrow[r, "u_\mu"] \arrow[d] & \mathscr{G}_\mu|(U_\lambda \cap U_\mu) \arrow[d] \\ \mathscr{F}_\lambda|(U_\lambda \cap U_\mu) \arrow[r, "u_\lambda"] & \mathscr{G}_\lambda|(U_\lambda \cap U_\mu) \end{tikzcd}$$

are commutative. Then, if $\mathscr{G}$ is obtained by gluing the $\mathscr{G}_\lambda$ by means of the $\omega_{\lambda\mu}$, there exists a unique morphism $u : \mathscr{F} \to \mathscr{G}$ such that the diagrams $\mathbf{0_I}$ | 29

$$\begin{tikzcd} \mathscr{F}|U_\lambda \arrow[r, "u|U_\lambda"] \arrow[d] & \mathscr{G}|U_\lambda \arrow[d] \\ \mathscr{F}_\lambda \arrow[r, "u_\lambda"] & \mathscr{G}_\lambda \end{tikzcd}$$

are commutative; this follows immediately from (3.2.3). The correspondence between the family (u_λ) and u is in a functorial bijection with the subset of $\Pi_\lambda \operatorname{Hom}(\mathscr{F}_\lambda, \mathscr{G}_\lambda)$ satisfying the conditions (3.3.2.1) on $\operatorname{Hom}(\mathscr{F}, \mathscr{G})$.

(3.3.3). With the notation of (3.3.1), let V be an open set of X; it is immediate that the restrictions to $V \cap U_\lambda \cap U_\mu$ of the $\theta_{\lambda\mu}$ satisfy the gluing condition for the induced sheaves $\mathscr{F}_\lambda|(V \cap U_\lambda)$ and that the sheaves on V obtained by gluing the latter identifies canonically with $\mathscr{F}|V$.

3.4. Direct images of presheaves

(3.4.1). Let X, Y be two topological spaces, $\psi : X \to Y$ a continuous map. Let $\mathscr{F}$ be a presheaf on X with values in a category $\mathtt{C}$; for each open $U \subset Y$, let $\mathscr{G}(U) = \mathscr{F}(\psi^{-1}(U))$, and if U, V are two open subsets of Y such that $U \subset V$, let ρ_U^V be the morphism $\mathscr{F}(\psi^{-1}(V)) \to \mathscr{F}(\psi^{-1}(U))$; it is immediate that the $\mathscr{G}(U)$ and the ρ_U^V define a *presheaf* on Y with values in $\mathtt{C}$, that we call the *direct image of $\mathscr{F}$ by ψ* and we denote it by $\psi_*(\mathscr{F})$. If $\mathscr{F}$ is a sheaf, we immediately verify the axiom (F) for the presheaf $\mathscr{G}$, so $\psi_*(\mathscr{F})$ is a sheaf.

(3.4.2). Let $\mathscr{F}_1$, $\mathscr{F}_2$ be two presheaves of X with values in $\mathtt{C}$, and let $u : \mathscr{F}_1 \to \mathscr{F}_2$ be a morphism. When U varies over the set of open subsets of Y, the family of morphisms $u_{\psi^{-1}(U)} : \mathscr{F}_1(\psi^{-1}(U)) \to \mathscr{F}_2(\psi^{-1}(U))$ satisfies the compatibility conditions with the restriction morphisms, and as a result defines a morphism $\psi_*(u) : \psi_*(\mathscr{F}_1) \to \psi_*(\mathscr{F}_2)$. If $v : \mathscr{F}_2 \to \mathscr{F}_3$ is a morphism from $\mathscr{F}_2$ to a third presheaf on X with values in $\mathtt{C}$, we have $\psi_*(v \circ u) = \psi_*(v) \circ \psi_*(u)$; in other words, $\psi_*(\mathscr{F})$ is a *covariant functor* in $\mathscr{F}$, from the category of presheaves (resp. sheaves) on X with values in $\mathtt{C}$, to that of presheaves (resp. sheaves) on Y with values in $\mathtt{C}$.

(3.4.3). Let Z be a third topological space, $\psi' : Y \to Z$ a continuous map, and let $\psi'' = \psi' \circ \psi$. It is clear that we have $\psi''_*(\mathscr{F}) = \psi'_*(\psi_*(\mathscr{F}))$ for each presheaf $\mathscr{F}$ on X with values in $\mathtt{C}$; in addition, for each morphism $u : \mathscr{F} \to \mathscr{G}$ of such presheaves, we have $\psi''_*(u) = \psi'_*(\psi_*(u))$. In other words, ψ''_* is the *composition* of the functors ψ'_* and ψ_*, and this can be written as

$$(\psi' \circ \psi)_* = \psi'_* \circ \psi_*.$$

In addition, for each open set U of Y, the image under the restriction $\psi|\psi^{-1}(U)$ of the induced presheaf $\mathscr{F}|\psi^{-1}(U)$ is none other than the induced presheaf $\psi_*(\mathscr{F})|U$.

(3.4.4). Suppose that the category $\mathtt{C}$ admits inductive limits, and let $\mathscr{F}$ be a presheaf on X with values in $\mathtt{C}$; for all $x \in X$, the morphisms $\Gamma(\psi^{-1}(U), \mathscr{F}) \to \mathscr{F}_x$ (U an open neighbourhood of $\psi(x)$ in Y) form an inductive limit, which gives by passing to the limit a morphism $\psi_x : (\psi_*(\mathscr{F}))_{\psi(x)} \to \mathscr{F}_x$ of $\mathbf{0_I}$ | 30
the stalks; in general, these morphisms are *neither injective or surjective*. It is functorial; indeed, if

$u : \mathscr{F}_1 \to \mathscr{F}_2$ is a morphism of presheaves on X with values in $\mathtt{C}$, the diagram

$$\begin{tikzcd} (\psi_*(\mathscr{F}_1))_{\psi(x)} \arrow[r, "\psi_x"] \arrow[d, "(\psi_*(u))_{\psi(x)}"'] & (\mathscr{F}_1)_x \arrow[d, "u_x"] \\ (\psi_*(\mathscr{F}_2))_{\psi(x)} \arrow[r, "\psi_x"] & (\mathscr{F}_2)_x \end{tikzcd}$$

is commutative. If Z is a third topological space, $\psi' : Y \to Z$ a continuous map, and $\psi'' = \psi' \circ \psi$, then we have $\psi''_x = \psi_x \circ \psi'_{\psi(x)}$ for $x \in X$.

(3.4.5). Under the hypotheses of (3.4.4), suppose in addition that ψ is a *homeomorphism* from X to the subspace $\psi(X)$ of Y. Then, for each $x \in X$, ψ_x is an *isomorphism*. This applies in particular to the canonical injection j of a subset X of Y into Y.

(3.4.6). Suppose that $\mathtt{C}$ be the category of groups, or of rings, etc. If $\mathscr{F}$ is a sheaf on X with values in $\mathtt{C}$, of support S, and if $y \notin \overline{\psi(S)}$, then it follows from the definition of $\psi_*(\mathscr{F})$ that $(\psi_*(\mathscr{F}))_y = \{0\}$, or in other words, that the support of $\psi_*(\mathscr{F})$ is contained in $\overline{\psi(S)}$; but it is not necessarily contained in $\psi(S)$. Under the same hypotheses, if j is the canonical injection of a subset X of Y into Y, the sheaf $j_*(\mathscr{F})$ induces $\mathscr{F}$ on X; if moreover X is *closed* in Y, $j_*(\mathscr{F})$ is the sheaf on Y which induces $\mathscr{F}$ on X and 0 on $Y - X$ (G, II, 2.9.2), but it is in general distinct from the latter when we suppose that X is locally closed but not closed.

3.5. Inverse images of presheaves

(3.5.1). Under the hypotheses of (3.4.1), if $\mathscr{F}$ (resp. $\mathscr{G}$) is a presheaf on X (resp. Y) with values in $\mathtt{C}$, then each morphism $u : \mathscr{G} \to \psi_*(\mathscr{F})$ of presheaves on Y is called a *ϕ-morphism* from $\mathscr{G}$ to $\mathscr{F}$, and we denote it also by $\mathscr{G} \to \mathscr{F}$. We denote also by $\mathrm{Hom}_\phi(\mathscr{G}, \mathscr{F})$ the set of $\mathrm{Hom}_Y(\mathscr{G}, \psi_*(\mathscr{F}))$ the ψ-morphisms from $\mathscr{G}$ to $\mathscr{F}$. For each pair (U, V), where U is an open set of X, V an open set of Y such that $\psi(U) \subset V$, we have a morphism $u_{U,V} : \mathscr{G}(V) \to \mathscr{F}(U)$ by composing the restriction morphism $\mathscr{F}(\psi^{-1}(V)) \to \mathscr{F}(U)$ and the morphism $u_V : \mathscr{G}(V) \to \psi_*(\mathscr{F})(V) = \mathscr{F}(\psi^{-1}(V))$; it is immediate that these morphisms render commutative the diagrams

(3.5.1.1)
$$\begin{tikzcd} \mathscr{G}(V) \arrow[r, "u_{U,V}"] \arrow[d] & \mathscr{F}(U) \arrow[d] \\ \mathscr{G}(V') \arrow[r, "u_{U',V'}"] & \mathscr{F}(U') \end{tikzcd}$$

for $U' \subset U$, $V' \subset V$, $\psi(U') \subset V'$. Conversely, the data of a family $(u_{U,V})$ of morphisms rendering commutative the diagrams (3.5.1.1) define a ψ-morphism u, since it suffices to take $u_V = u_{\psi^{-1}(V),V}$.

If the category $\mathtt{C}$ admits (generalized) projective limits, and if $\mathfrak{B}$, $\mathfrak{B}'$ are bases for the topologies of X and Y respectively, to define a ψ-morphism u of *sheaves*, we can restrict to giving the $u_{U,V}$ for $U \in \mathfrak{B}$, $V \in \mathfrak{B}'$, and $\psi(U) \subset V$, satisfying the compatibility conditions of (3.5.1.1) for U, U' in $\mathfrak{B}$ and V, V' in $\mathfrak{B}'$; it indeed suffices to define u_W, for each open $W \subset Y$, as the projective limit of the $u_{U,V}$ for $V \in \mathfrak{B}'$ and $V \subset W$, $U \in \mathfrak{B}$ and $\psi(U) \subset V$. $\mathbf{0_I}$ | 31

When the category $\mathtt{C}$ admits inductive limits, we have, for each $x \in X$, a morphism $\mathscr{G}(V) \to \mathscr{F}(\psi^{-1}(V)) \to \mathscr{F}_x$, for each open neighbourhood V of $\psi(x)$ in Y, and these morphisms form an inductive system which gives by passing to the limit a morphism $\mathscr{G}_{\psi(x)} \to \mathscr{F}_x$.

(3.5.2). Under the hypotheses of (3.4.3), let $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ be presheaves with values in $\mathtt{C}$ on X, Y, Z respectively, and let $u : \mathscr{G} \to \psi_*(\mathscr{F})$, $v : \mathscr{H} \to \psi'_*(\mathscr{G})$ be a ψ-morphism and a ψ'-morphism respectively. We obtain a ψ''-morphism $w : \mathscr{H} \xrightarrow{v} \psi'_*(\mathscr{G}) \xrightarrow{\psi'_*(u)} \psi'_*(\psi_*(\mathscr{F})) = \psi''_*(\mathscr{F})$, that we call, by definition, the *composition* of u and v. We can therefore consider the pairs $(X, \mathscr{F})$ consisting of a topological space X and a presheaf $\mathscr{F}$ on X (with values in $\mathtt{C}$) as forming a *category*, the morphisms being the pairs $(\psi, \theta) : (X, \mathscr{F}) \to (Y, \mathscr{G})$ consisting of a continuous map $\psi : X \to Y$ and of a ψ-morphism $\theta : \mathscr{G} \to \mathscr{F}$.

(3.5.3). Let $\psi : X \to Y$ be a continuous map, $\mathscr{G}$ a *presheaf* on Y with values in $\mathtt{C}$. We call the *inverse image of $\mathscr{G}$ under ψ* the pair $(\mathscr{G}', \rho)$, where $\mathscr{G}'$ is a *sheaf* on X with values in $\mathtt{C}$, and $\rho : \mathscr{G} \to \mathscr{G}'$ a ψ-morphism (in other words a homomorphism $\mathscr{G} \to \psi_*(\mathscr{G}')$) such that, for each *sheaf* $\mathscr{F}$ on X with values in $\mathtt{C}$, the map

$$\mathrm{Hom}_X(\mathscr{G}', \mathscr{F}) \longrightarrow \mathrm{Hom}_\psi(\mathscr{G}, \mathscr{F}) \longrightarrow \mathrm{Hom}_Y(\mathscr{G}, \psi_*(\mathscr{F})) \tag{3.5.3.1}$$

sending v to $\psi_*(v) \circ \rho$, is a *bijection*; this map, being functorial in $\mathscr{F}$, then defines an isomorphism of functors in $\mathscr{F}$. The pair $(\mathscr{G}', \rho)$ is the solution of a universal problem, and we say it is *determined up to unique isomorphism* when it exists. We then write $\mathscr{G}' = \psi^*(\mathscr{G})$, $\rho = \rho_{\mathscr{G}}$, and by abuse of language, we say that $\psi^*(\mathscr{G})$ is *the inverse image sheaf* of $\mathscr{G}$ under ψ, and we agree that $\psi^*(\mathscr{G})$ is considered as equipped with a *canonical ψ-morphism* $\rho_{\mathscr{G}} : \mathscr{G} \to \psi^*(\mathscr{G})$, that is to say the *canonical homomorphism* of presheaves on Y:

$$\rho_{\mathscr{G}} : \mathscr{G} \longrightarrow \psi_*(\psi^*(\mathscr{G})). \tag{3.5.3.2}$$

For each homomorphism $v : \psi^*(\mathscr{G}) \to \mathscr{F}$ (where $\mathscr{F}$ is a sheaf on X with values in $\mathtt{C}$), we put $v^\flat = \psi_*(v) \circ \rho_{\mathscr{G}} : \mathscr{G} \to \psi_*(\mathscr{F})$. By definition, *each* morphism of presheaves $u : \mathscr{G} \to \psi_*(\mathscr{F})$ is of the form $v^\flat$ for a unique v, which we will denote $u^\sharp$. In other words, each morphism $u : \mathscr{G} \to \psi_*(\mathscr{F})$ of presheaves factorizes in a unique way as

$$u : \mathscr{G} \xrightarrow{\rho_{\mathscr{G}}} \psi_*(\psi^*(\mathscr{G})) \xrightarrow{\psi_*(u^\sharp)} \psi_*(\mathscr{F}). \tag{3.5.3.3}$$

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(3.5.4). Suppose now that the category $\mathtt{C}$ be such[7] that *each* presheaf $\mathscr{F}$ on Y with values in $\mathtt{C}$ admits an inverse image under ψ, and we denote it by $\psi^*(\mathscr{G})$.

We will see that we can define $\psi^*(\mathscr{G})$ as a *covariant functor* in $\mathscr{G}$, from the category of presheaves on Y with values in $\mathtt{C}$, to that of sheaves on X with values in $\mathtt{C}$, in such a way that the isomorphism $v \mapsto v^\flat$ is an *isomorphism of bifunctors*

$$\mathrm{Hom}_X(\psi^*(\mathscr{G}), \mathscr{F}) \simeq \mathrm{Hom}_Y(\mathscr{G}, \psi_*(\mathscr{F})) \tag{3.5.4.1}$$

in $\mathscr{G}$ and $\mathscr{F}$.

Indeed, for each morphism $w : \mathscr{G}_1 \to \mathscr{G}_2$ of presheaves on Y with values in $\mathtt{C}$, consider the composite morphism $\mathscr{G}_1 \xrightarrow{w} \mathscr{G}_2 \xrightarrow{\rho_{\mathscr{G}_2}} \psi_*(\psi^*(\mathscr{G}_2))$; to it corresponds a morphism $(\rho_{\mathscr{G}_2} \circ w)^\sharp : \psi^*(\mathscr{G}_1) \to \psi^*(\mathscr{G}_2)$, that we denote by $\psi^*(w)$. We therefore have, according to (3.5.3.3),

$$\psi_*(\psi^*(w)) \circ \rho_{\mathscr{G}_1} = \rho_{\mathscr{G}_2} \circ w. \tag{3.5.4.2}$$

For each morphism $u : \mathscr{G}_2 \to \psi_*(\mathscr{F})$, where $\mathscr{F}$ is a sheaf on X with values in $\mathtt{C}$, we have, according to (3.5.3.3), (3.5.4.2), and the definition of $u^\flat$, that

$$(u^\sharp \circ \psi^*(w))^\flat = \psi_*(u^\sharp) \circ \psi_*(\psi^*(w)) \circ \rho_{\mathscr{G}_1} = \psi_*(u^\sharp) \circ \rho_{\mathscr{G}_2} \circ w = u \circ w$$

where again

$$(u \circ w)^\sharp = u^\sharp \circ \psi^*(w). \tag{3.5.4.3}$$

If we take in particular for u a morphism $\mathscr{G}_2 \xrightarrow{w'} \mathscr{G}_3 \xrightarrow{\rho_{\mathscr{G}_3}} \psi_*(\psi^*(\mathscr{G}_3))$, it becomes $\psi^*(w' \circ w) = (\rho_{\mathscr{G}_3} \circ w' \circ w)^\sharp = (\rho_{\mathscr{G}_3} \circ w')^\sharp \circ \psi^*(w) = \psi^*(w') \circ \psi^*(w)$, hence our assertion.

Finally, for each sheaf $\mathscr{F}$ on X with values in $\mathtt{C}$, let $i_{\mathscr{F}}$ be the identity morphism of $\psi_*(\mathscr{F})$ and denote by

$$\sigma_{\mathscr{F}} : \psi^*(\psi_*(\mathscr{F})) \longrightarrow \mathscr{F}$$

the morphism $(i_{\mathscr{F}})^\sharp$; the formula (3.5.4.3) gives in particular the factorization

$$u^\sharp : \psi^*(\mathscr{G}) \xrightarrow{\psi^*(u)} \psi^*(\psi_*(\mathscr{F})) \xrightarrow{\sigma_{\mathscr{F}}} \mathscr{F} \tag{3.5.4.4}$$

for each morphism $u : \mathscr{G} \to \psi_*(\mathscr{F})$. We say that the morphism $\sigma_{\mathscr{F}}$ is *canonical*.

[7] In the book mentioned in the introduction, we will give very general conditions on the category $\mathtt{C}$ ensuring the existence of inverse images of presheaves with values in $\mathtt{C}$.

(3.5.5). Let $\psi' : Y \to Z$ be a continuous map, and suppose that each presheaf $\mathscr{H}$ on Z with values in $\mathtt{C}$ admits an inverse image $\psi'^*(\mathscr{H})$ under ψ'. Then (with the hypotheses of (3.5.4)) each presheaf $\mathscr{H}$ on Z with values in $\mathtt{C}$ admits an inverse image under $\psi'' = \psi' \circ \psi$ and we have a canonical functorial isomorphism

$$\psi''^*(\mathscr{H}) \simeq \psi^*(\psi'^*(\mathscr{H})). \tag{3.5.5.1}$$

This indeed follows immediately from the definitions, taking into account that $\psi''_* = \psi'_* \circ \psi_*$. In addition, if $u : \mathscr{G} \to \psi_*(\mathscr{F})$ is a ψ-morphism, $v : \mathscr{H} \to \psi'_*(\mathscr{G})$ a ψ'-morphism, and $w = \psi'_*(u) \circ v$ their composition (3.5.2), then we have immediately that $w^\sharp$ is the composite morphism $\mathbf{0_I} \mid 33$

$$w^\sharp : \psi^*(\psi'^*(\mathscr{H})) \xrightarrow{\psi^*(v^\sharp)} \psi^*(\mathscr{G}) \xrightarrow{u^\sharp} \mathscr{F}.$$

(3.5.6). We take in particular for ψ the identity map $1_X : X \to X$. Then if the inverse image under ψ of a presheaf $\mathscr{F}$ on X with values in $\mathtt{C}$ exists, we say that this inverse image is the *sheaf associated to the presheaf* $\mathscr{F}$. Each morphism $u : \mathscr{F} \to \mathscr{F}'$ from $\mathscr{F}$ to a *sheaf* $\mathscr{F}'$ with values in $\mathtt{C}$ factorizes in a unique way as $\mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} 1_X^*(\mathscr{F}) \xrightarrow{u^\sharp} \mathscr{F}'$.

3.6. Simple and locally simple sheaves

(3.6.1). We say that a *presheaf* $\mathscr{F}$ on X, with values in $\mathtt{C}$, is *constant* if the canonical morphisms $\mathscr{F}(X) \to \mathscr{F}(U)$ are *isomorphisms* for each nonempty open $U \subset X$; we note that $\mathscr{F}$ is not necessarily a sheaf. We say that a *sheaf* is *simple* if it is the associated sheaf (3.5.6) of a constant presheaf. We say that a sheaf $\mathscr{F}$ is *locally simple* if each $x \in X$ admits an open neighbourhood U such that $\mathscr{F}|U$ is simple.

(3.6.2). Suppose that X is *irreducible* (2.1.1); then the following properties are equivalent:

(a) *$\mathscr{F}$ is a constant presheaf on X;*
(b) *$\mathscr{F}$ is a simple sheaf on X;*
(c) *$\mathscr{F}$ is a locally simple sheaf on X.*

Indeed, let $\mathscr{F}$ be a constant presheaf on X; if U, V are two nonempty open sets in X, then $U \cap V$ is nonempty, so $\mathscr{F}(X) \to \mathscr{F}(U) \to \mathscr{F}(U \cap V)$ and $\mathscr{F}(X) \to \mathscr{F}(U)$ being isomorphisms, the same is true for $\mathscr{F}(U) \to \mathscr{F}(U \cap V)$, and similarly, $\mathscr{F}(V) \to \mathscr{F}(U \cap V)$ is an isomorphism. We therefore conclude immediately that the axiom (F) of (3.1.2) is clearly satisfied, $\mathscr{F}$ is isomorphic to its associated sheaf, and as a result (a) implies (b).

Now let (U_α) be an open cover of X by nonempty open sets and let $\mathscr{F}$ be a sheaf on X such that $\mathscr{F}|U_\alpha$ is simple for each α; as U_α is irreducible, $\mathscr{F}|U_\alpha$ is a constant presheaf according to the above. As $U_\alpha \cap U_\beta$ is not empty, $\mathscr{F}(U_\alpha) \to \mathscr{F}(U_\alpha \cap U_\beta)$ and $\mathscr{F}(U_\beta) \to \mathscr{F}(U_\alpha \cap U_\beta)$ are isomorphisms, hence we have a canonical isomorphism $\theta_{\alpha\beta} : \mathscr{F}(U_\alpha) \to \mathscr{F}(U_\beta)$ for each pair of indices. But then if we apply the condition (F) for $U = X$, we see that for each index α_0, $\mathscr{F}(U_{\alpha_0})$ and the $\theta_{\alpha_0\alpha}$ are solutions to the universal problem, which (according to the uniqueness) implies that $\mathscr{F}(X) \to \mathscr{F}(U_{\alpha_0})$ is an isomorphism, and hence proves that (c) implies (a).

3.7. Inverse images of presheaves of groups or rings $\mathbf{0_I} \mid 34$

(3.7.1). We will show that when we take $\mathtt{C}$ to be the category of sets, the inverse image under ψ for each presheaf $\mathscr{G}$ with values in $\mathtt{C}$ *always exists* (the notation and hypotheses on X, Y, ψ being that of (3.5.3)). Indeed, for each open $U \subset X$, define $\mathscr{G}'(U)$ as follows: an element s' of $\mathscr{G}'(U)$ is a family $(s'_x)_{x \in U}$, where $s'_x \in \mathscr{G}_{\psi(x)}$ for each $x \in U$, and where, for each $x \in U$, the following condition is satisfied: there exists an open neighbourhood V of $\psi(x)$ in Y, a neighbourhood $W \subset \psi^{-1}(V) \cap U$ of x, and an element $s \in \mathscr{G}(V)$ such that $s'_z = s_{\psi(x)}$ for all $z \in W$. We verify immediately that $U \mapsto \mathscr{G}'(U)$ clearly satisfies the axioms of a *sheaf*.

Now let $\mathscr{F}$ be a sheaf of sets on X, and let $u : \mathscr{G} \to \psi_*(\mathscr{F})$, $v : \mathscr{G}' \to \mathscr{F}$ be morphisms. We define $u^\sharp$ and $v^\flat$ in the following manner: if s' is a section of $\mathscr{G}'$ over a neighbourhood U of $x \in X$ and if V is an open neighbourhood of $\psi(x)$ and $s \in \mathscr{G}(V)$ such that we have $s'_z = s_{\psi(x)}$ for z in a neighbourhood of x contained in $\psi^{-1}(V) \cap U$, we take $u^\sharp_x(s'_x) = u_{\psi(x)}(s_{\psi(x)})$. Similarly, if $s \in \mathscr{G}(V)$ (V open in Y), $v^\flat(s)$ is the section of $\mathscr{F}$ over $\psi^{-1}(V)$, the image under v of the section s' of $\mathscr{G}'$ such that $s'_x = s_{\psi(x)}$

for all $x \in \psi^{-1}(V)$. In addition, the canonical homomorphism (3.5.3) $\rho : \mathscr{G} \to \psi_*(\psi^*(\mathscr{G}))$ is defined in the following manner: for each open $V \subset Y$ and each section $s \in \Gamma(V, \mathscr{G})$, $\rho(s)$ is the section $(s_{\psi(x)})_{x \in \psi^{-1}(V)}$ of $\psi^*(\mathscr{G})$ over $\psi^{-1}(V)$. The verification of the relations $(u^\sharp)^\flat = u$, $(v^\flat)^\sharp = v$, and $v^\flat = \psi_*(v) \circ \rho$ is immediate, and proves our assertion.

We check that, if $w : \mathscr{G}_1 \to \mathscr{G}_2$ is a homomorphism of sheaves of sets on Y, $\psi^*(w)$ is expressed in the following manner: if $s' = (s'_x)_{x \in U}$ is a section of $\psi^*(\mathscr{G}_1)$ over an open set U of X, then $(\psi^*(w))(s')$ is the family $(w_{\psi(x)}(s'_x))_{x \in U}$. Finally, it is immediate that for each open set V of Y, the inverse image of $\mathscr{G}|V$ under the restriction of ψ to $\psi^{-1}(V)$ is identical to the induced sheaf $\psi^*(\mathscr{G})|\psi^{-1}(V)$.

When ψ is the identity 1_X, we recover the definition of a sheaf of sets associated to a presheaf (G, II, 1.2). The above considerations apply without change when $\mathtt{C}$ is the category of groups or of rings (not necessarily commutative).

When X is any subset of a topological space Y, and j the canonical injection $X \to Y$, for each sheaf $\mathscr{G}$ on Y with values in a category $\mathtt{C}$, we call the *induced* sheaf of X by $\mathscr{G}$ the inverse image $j^*(\mathscr{G})$ (whenever it exists); for the sheaves of sets (or of groups, or of rings) we recover the usual definition (G, II, 1.5).

(3.7.2). Keeping the notation and hypotheses of (3.5.3), suppose that $\mathscr{G}$ is a *sheaf* of groups (resp. of rings) on Y. The definition of sections of $\psi^*(\mathscr{G})$ (3.7.1) shows (taking into account (3.4.4)) that the homomorphism of stalks $\psi_x \circ \rho_{\psi(x)} : \mathscr{G}_{\psi(x)} \to (\psi^*(\mathscr{G}))_x$ is a *functorial isomorphism in* $\mathscr{G}$, that identifies the two stalks; with this identification, $u_x^\sharp$ is identical to the homomorphism defined in (3.5.1), and in particular, we have $\mathrm{Supp}(\psi^*(\mathscr{G})) = \psi^{-1}(\mathrm{Supp}(\mathscr{G}))$.

An immediate consequence of this result is that *the functor* $\psi^*(\mathscr{G})$ *is exact in* $\mathscr{G}$ on the abelian category of sheaves of abelian groups.

3.8. Sheaves on pseudo-discrete spaces $\mathbf{0}_{\mathbf{I}}$ | 35

(3.8.1). Let X be a topological space whose topology admits a basis $\mathfrak{B}$ consisting of open *quasi-compact* subsets. Let $\mathscr{F}$ be a *sheaf of sets* on X; if we equip each of the $\mathscr{F}(U)$ with the *discrete* topology, $U \mapsto \mathscr{F}(U)$ is a *presheaf of topological spaces*. We will see that there exists a *sheaf of topological spaces* $\mathscr{F}'$ *associated to* $\mathscr{F}$ (3.5.6) such that $\Gamma(U, \mathscr{F}')$ is the discrete space $\mathscr{F}(U)$ for each open *quasi-compact* subsets U. It will suffice to show that the presheaf $U \mapsto \mathscr{F}(U)$ of discrete topological spaces *on* $\mathfrak{B}$ satisfy the condition (F_0) of (3.2.2), and more generally that if U is an open quasi-compact subset and if (U_α) is a cover of U by sets of $\mathfrak{B}$, then the least fine topology $\mathscr{T}$ on $\Gamma(U, \mathscr{F})$ making the maps $\Gamma(U, \mathscr{F}) \to \Gamma(U_\alpha, \mathscr{F})$ continuous is the *discrete* topology. There exists a finite number of indices α_i such that $U = \bigcup_i U_{\alpha_i}$. Let $s \in \Gamma(U, \mathscr{F})$ and let s_i be its image in $\Gamma(U_{\alpha_i}, \mathscr{F})$; the intersection of the inverse images of the sets $\{s_i\}$ is by definition a neighbourhood of s for $\mathscr{T}$; but since $\mathscr{F}$ is a sheaf of sets and the U_{α_i} cover U, this intersection is reduced to s, hence our assertion.

We note that if U is an open non-quasi-compact subset of X, the topological space $\Gamma(U, \mathscr{F}')$ still has $\Gamma(U, \mathscr{F})$ as the underlying set, but the topology is not discrete in general: it is the least fine making the maps $\Gamma(U, \mathscr{F}) \to \Gamma(V, \mathscr{F})$ continuous, for $V \in \mathfrak{B}$ and $V \subset U$ (the $\Gamma(V, \mathscr{F})$ being discrete).

The above considerations apply without modification to sheaves of groups or of rings (not necessarily commutative), and associate to them sheaves of *topological groups* or *topological rings*, respectively. To summarize, we say that the sheaf $\mathscr{F}'$ is the *pseudo-discrete* sheaf of *spaces* (resp. *groups*, *rings*) associated to a sheaf of sets (resp. groups, rings) $\mathscr{F}$.

(3.8.2). Let $\mathscr{F}, \mathscr{G}$ be two sheaves of sets (resp. groups, rings) on X, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism. Then u is thus a *continuous* homomorphism $\mathscr{F}' \to \mathscr{G}'$, if we denote by $\mathscr{F}'$ and $\mathscr{G}'$ the pseudo-discrete sheaves associated to $\mathscr{F}$ and $\mathscr{G}$; this follows indeed from (3.2.5).

(3.8.3). Let $\mathscr{F}$ be a sheaf of sets, $\mathscr{H}$ a subsheaf of $\mathscr{F}$, $\mathscr{F}'$ and $\mathscr{H}'$ the pseudo-discrete sheaves associated to $\mathscr{F}$ and $\mathscr{H}$ respectively. Then, for each open $U \subset X$, $\Gamma(U, \mathscr{H}')$ is *closed* in $\Gamma(U, \mathscr{F}')$: indeed, it is the intersection of the inverse images of the $\Gamma(V, \mathscr{H})$ (for $V \in \mathfrak{B}$, $V \subset U$) under the continuous maps $\Gamma(U, \mathscr{F}) \to \Gamma(V, \mathscr{F})$, and $\Gamma(V, \mathscr{H})$ is closed in the discrete space $\Gamma(V, \mathscr{F})$.

§4. Ringed spaces

4.1. Ringed spaces, sheaves of $\mathscr{A}$-modules, $\mathscr{A}$-algebras

(4.1.1). A *ringed space* (resp. topologically ringed space) is a pair $(X, \mathscr{A})$ consisting of a topological space X and a sheaf of (not necessarily commutative) rings (resp. of a sheaf of topological rings) $\mathscr{A}$ on X; we say that X is the *underlying* topological space of the ringed space $(X, \mathscr{A})$, and $\mathscr{A}$ the *structure sheaf*. The latter is denoted $\mathscr{O}_X$, and its stalk at a point $x \in X$ is denoted $\mathscr{O}_{X,x}$ or simply $\mathscr{O}_x$ when there is no chance of confusion.

We denote by 1 or e the *unit section* of $\mathscr{O}_X$ over X (the unit element of $\Gamma(X, \mathscr{O}_X)$).

As in this treatise we will have to consider in particular sheaves of *commutative* rings, it will be understood, when we speak of a ringed space $(X, \mathscr{A})$ without specification, that $\mathscr{A}$ is a sheaf of commutative rings.

The ringed spaces with not-necessarily-commutative structure sheaves (resp. the topologically ringed spaces) form a *category*, where we define a *morphism* $(X, \mathscr{A}) \to (Y, \mathscr{B})$ as a couple $(\psi, \theta) = \Psi$ consisting of a continuous map $\psi : X \to Y$ and a ψ-*morphism* $\theta : \mathscr{B} \to \mathscr{A}$ (3.5.1) of sheaves of **Err**$_{\mathrm{II}}$ rings (resp. of sheaves of topological rings); the *composition* of a second morphism $\Psi' = (\psi', \theta') : (Y, \mathscr{B}) \to (Z, \mathscr{C})$ and of Ψ, denoted $\Psi'' = \Psi' \circ \Psi$, is the morphism (ψ'', θ'') where $\psi'' = \psi' \circ \psi$, and θ'' is the composition of θ and θ' (equal to $\psi'_*(\theta) \circ \theta'$, cf. (3.5.2)). For ringed spaces, remember that we then have $\theta''^\sharp = \theta^\sharp \circ \psi^*(\theta'^\sharp)$ (3.5.5); therefore if $\theta'^\sharp$ and $\theta^\sharp$ are *injective* (resp. *surjective*), then the same is true of $\theta''^\sharp$, taking into account that $\psi_x \circ \rho_{\psi(x)}$ is an isomorphism for all $x \in X$ (3.7.2). We verify immediately, thanks to the above, that when ψ is an *injective* continuous map and when $\theta^\sharp$ is a *surjective* homomophism of sheaves of rings, the morphism (ψ, θ) is a *monomorphism* (T, 1.1) in the category of ringed spaces.

By abuse of language, we will often replace ψ by Ψ in notation, for example in writing $\Psi^{-1}(U)$ in place of $\psi^{-1}(U)$ for a subset U of Y, when the is no risk of confusion.

(4.1.2). For each subset M of X, the pair $(M, \mathscr{A}|M)$ is evidently a ringed space, said to be *induced* on M by the ringed space $(X, \mathscr{A})$ (and is still called the *restriction* of $(X, \mathscr{A})$ to M). If j is the canonical injection $M \to X$ and ω is the identity map of $\mathscr{A}|M$, $(j, \omega^\flat)$ is a monomorphism $(M, \mathscr{A}|M) \to (X, \mathscr{A})$ of ringed spaces, called the *canonical injection*. The composition of a morphism $\Psi : (X, \mathscr{A}) \to (Y, \mathscr{B})$ and this injection is called the *restriction* of Ψ to M.

(4.1.3). We will not revisit the definitions of $\mathscr{A}$-*modules* or *algebraic sheaves* on a ringed space $(X, \mathscr{A})$ (G, II, 2.2); when $\mathscr{A}$ is a sheaf of not necessarily commutative rings, by $\mathscr{A}$-module we will always mean "left $\mathscr{A}$-module" unless expressly stated otherwise. The $\mathscr{A}$-submodules of $\mathscr{A}$ will be called *sheaves of ideals* (left, right, or two-sided) in $\mathscr{A}$ or $\mathscr{A}$-*ideals*.

When $\mathscr{A}$ is a sheaf of commutative rings, and in the definition of $\mathscr{A}$-modules, we replace everywhere the *module* structure by that of an *algebra*, we obtain the definition of an $\mathscr{A}$-*algebra* on X. It is the same to say that an $\mathscr{A}$-algebra (not necessarily commutative) is a $\mathscr{A}$-module $\mathscr{C}$, given with a homomorphism of $\mathscr{A}$-modules $\phi : \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \to \mathscr{C}$ and a section e over X, such that: 1st the diagram

$$\begin{tikzcd} \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes\phi"'] & \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \arrow[d, "\phi"] \\ \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \arrow[r, "\phi"] & \mathscr{C} \end{tikzcd}$$

is commutative; 2nd for each open $U \subset X$ and each section $s \in \Gamma(U, \mathscr{C})$, we have $\phi((e|U) \otimes s) = \phi(s \otimes (e|U)) = s$. We say that $\mathscr{C}$ is a commutative $\mathscr{A}$-algebra if the diagram

$$\begin{tikzcd} \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \arrow[rr, "\sigma"] \arrow[dr, "\phi"'] & & \mathscr{C} \otimes_{\mathscr{A}} \mathscr{C} \arrow[dl, "\phi"] \\ & \mathscr{C} & \end{tikzcd}$$

is commutative, σ denoting the canonical symmetry (twist) map of the tensor product $\mathscr{C} \otimes_{\mathscr{A}} \mathscr{C}$.

The homomorphisms of $\mathscr{A}$-algebras are also defined as the homomorphisms of $\mathscr{A}$-modules in (G, II, 2.2), but naturally no longer form an abelian group.

If $\mathscr{M}$ is an $\mathscr{A}$-submodule of an $\mathscr{A}$-algebra $\mathscr{C}$, the $\mathscr{A}$-*subalgebra of* $\mathscr{C}$ *generated by* $\mathscr{M}$ is the sum of the images of the homomorphisms $\bigotimes^n \mathscr{M} \to \mathscr{C}$ (for each $n \geqslant 0$). This is also the sheaf associated to the presheaf $U \mapsto \mathscr{B}(U)$ of algebras, $\mathscr{B}(U)$ being the subalgebra of $\Gamma(U, \mathscr{C})$ generated by the submodule $\Gamma(U, \mathscr{M})$.

(4.1.4). We say that a sheaf of rings $\mathscr{A}$ on a topological space X is *reduced at a point x in X* if the stalk $\mathscr{A}_x$ is a *reduced* ring (1.1.1); we say that $\mathscr{A}$ is *reduced* if it is reduced at all points of X. Recall that a ring A is called *regular* if each of the local rings $A_{\mathfrak{p}}$ (where $\mathfrak{p}$ varies over the set of prime ideals of A) is a regular local ring; we will say that a sheaf of rings $\mathscr{A}$ on X is *regular at a point x* (resp. *regular*) if the stalk $\mathscr{A}_x$ is a regular ring (resp. if $\mathscr{A}$ is regular at each point). Finally, we will say that a sheaf of rings $\mathscr{A}$ on X is *normal at a point x* (resp. *normal*) if the stalk $\mathscr{A}_x$ is an *integral and integrally closed* ring (resp. if $\mathscr{A}$ is normal at each point). We will say that a ringed space $(X, \mathscr{A})$ has any of these preceeding properties if the sheaf of rings $\mathscr{A}$ has that property.

A *graded* sheaf of rings $\mathscr{A}$ is by definition a sheaf of rings that is the direct sum (G, II, 2,7) of a family $(\mathscr{A}_n)_{n \in \mathbf{Z}}$ of sheaves of abelian groups with the conditions $\mathscr{A}_m \mathscr{A}_n \subset \mathscr{A}_{m+n}$; a *graded $\mathscr{A}$-module* is an $\mathscr{A}$-module $\mathscr{F}$ that is the direct sum of a family $(\mathscr{F}_n)_{n \in \mathbf{Z}}$ of sheaves of abelian groups, satisfying the conditions $\mathscr{A}_m \mathscr{F}_n \subset \mathscr{F}_{m+n}$. It is equivalent to say that $(\mathscr{A}_m)_x (\mathscr{A}_n)_x \subset (\mathscr{A}_{m+n})_x$ (resp. $(\mathscr{A}_m)_x (\mathscr{F}_n)_x \subset (\mathscr{F}_{m+n})_x$) for each point x.

(4.1.5). Given a ringed space $(X, \mathscr{A})$ (not necessarily commutative), we will not recall here the definitions of the bifunctors $\mathscr{F} \otimes_{\mathscr{A}} \mathscr{G}$, $\mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{F})$, and $\operatorname{Hom}_{\mathscr{A}}(\mathscr{F}, \mathscr{G})$ (G, II, 2.8 and 2.2) in the categories of left or right (depending on the case) $\mathscr{A}$-modules, with values in the category of sheaves of abelian groups (or more generally of $\mathscr{C}$-modules, if $\mathscr{C}$ is the center of $\mathscr{A}$). The stalk $(\mathscr{F} \otimes_{\mathscr{A}} \mathscr{G})_x$ for each point $x \in X$ canonically identifies with $\mathscr{F}_x \otimes_{\mathscr{A}_x} \mathscr{G}_x$ and we define a canonical and functorial homomorphism $(\mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}))_x \to \operatorname{Hom}_{\mathscr{A}_x}(\mathscr{F}_x, \mathscr{G}_x)$ which is in general neither injective nor surjective. The bifunctors considered above are additive and in particular, commute with finite direct limits; $\mathscr{F} \otimes_{\mathscr{A}} \mathscr{G}$ is right exact in $\mathscr{F}$ and in $\mathscr{G}$, commutes with inductive limits, and $\mathscr{A} \otimes_{\mathscr{A}} \mathscr{G}$ (resp. $\mathscr{F} \otimes_{\mathscr{A}} \mathscr{A}$) canonically identifies with $\mathscr{G}$ (resp. $\mathscr{F}$). The functors $\mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G})$ and $\operatorname{Hom}_{\mathscr{A}}(\mathscr{F}, \mathscr{G})$ are *left exact* in $\mathscr{F}$ and $\mathscr{G}$; more precisely, if we have an exact sequence of the form $0 \to \mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$, the sequence

$$0 \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}') \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}) \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}'')$$

is exact, and if we have an exact sequence of the form $\mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, the sequence

$$0 \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}'', \mathscr{G}) \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}) \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{F}', \mathscr{G})$$

is exact, with the analogous properties for the functor Hom. In addiiton, $\mathscr{H}om_{\mathscr{A}}(\mathscr{A}, \mathscr{G})$ canonically identifies with $\mathscr{G}$; finally, for each open $U \subset X$, we have

$$\Gamma(U, \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G}) = \operatorname{Hom}_{\mathscr{A}|U}(\mathscr{F}|U, \mathscr{G}|U).$$

For each left (resp. right) $\mathscr{A}$-module, we define the *dual* of $\mathscr{F}$ and denote it by $\mathscr{F}^\vee$ the right (resp. left) $\mathscr{A}$-module $\mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{A})$.

Finally, if $\mathscr{A}$ is a sheaf of commutative rings, $\mathscr{F}$ an $\mathscr{A}$-module, $U \mapsto \wedge^p \Gamma(U, \mathscr{F})$ is a presheaf whose associated sheaf is an $\mathscr{A}$-module denoted $\wedge^p \mathscr{F}$ and is called the *p-th exterior power of $\mathscr{F}$*; we verify easily that the canonical map of the presheaf $U \mapsto \wedge^p \Gamma(U, \mathscr{F})$ to the associated sheaf $\wedge^p \mathscr{F}$ is *injective*, and for each $x \in X$, $(\wedge^p \mathscr{F})_x = \wedge^p(\mathscr{F}_x)$. It is clear that $\wedge^p \mathscr{F}$ is a covariant functor in $\mathscr{F}$.

(4.1.6). Suppose that $\mathscr{A}$ is a sheaf of not-necessarily-commutative rings, $\mathscr{J}$ a left sheaf of ideals of $\mathscr{A}$, $\mathscr{F}$ an left $\mathscr{A}$-module; we then denote by $\mathscr{J}\mathscr{F}$ the $\mathscr{A}$-submodule of $\mathscr{F}$, the image of $\mathscr{J} \otimes_{\mathbf{Z}} \mathscr{F}$ (where $\mathbf{Z}$ is the sheaf associated to the constant presheaf $U \mapsto \mathbf{Z}$) under the canonical map $\mathscr{J} \otimes_{\mathbf{Z}} \mathscr{F} \to \mathscr{F}$; it is clear that for each $x \in X$, we have $(\mathscr{J}\mathscr{F})_x = \mathscr{J}_x \mathscr{F}_x$. When $\mathscr{A}$ is commutative, $\mathscr{J}\mathscr{F}$ is also the canonical image of $\mathscr{J} \otimes_{\mathscr{A}} \mathscr{F} \to \mathscr{F}$. It is immediate that $\mathscr{J}\mathscr{F}$ is also the $\mathscr{A}$-module associated to the presheaf $U \mapsto \Gamma(U, \mathscr{J})\Gamma(U, \mathscr{F})$. If $\mathscr{J}_1$, $\mathscr{J}_2$ are two left sheaves of ideals of $\mathscr{A}$, we have $\mathscr{J}_1(\mathscr{J}_2\mathscr{F}) = (\mathscr{J}_1\mathscr{J}_2)\mathscr{F}$.

(4.1.7). Let $(X_\lambda, \mathscr{A}_\lambda)_{\lambda \in L}$ be a family of ringed spaces; for each couple (λ, μ), suppose we are given an open subset $V_{\lambda\mu}$ of X_λ, and an isomorphism of ringed spaces $\phi_{\lambda\mu} : (V_{\mu\lambda}, \mathscr{A}_\mu | V_{\lambda\mu}) \simeq (V_{\lambda\mu}, \mathscr{A}_\lambda | V_{\lambda\mu})$, with $V_{\lambda\lambda} = X_\lambda$, $\phi_{\lambda\lambda}$ being the identity. Furthermore, suppose that, for each triple (λ, μ, ν), if we denote by $\phi'_{\mu\lambda}$ the restriction of $\phi_{\mu\lambda}$ to $V_{\lambda\mu} \cap V_{\lambda\nu}$, $\phi'_{\mu\lambda}$ is an isomorphism from $(V_{\lambda\mu} \cap V_{\lambda\nu}, \mathscr{A}_\lambda | (V_{\lambda\mu} \cap V_{\lambda\nu}))$ to $(V_{\mu\nu} \cap V_{\mu\lambda}, \mathscr{A}_\mu | (V_{\mu\nu} \cap V_{\mu\lambda}))$ and that we have $\phi'_{\lambda\nu} = \phi'_{\lambda\mu} \circ \phi'_{\mu\nu}$ (*gluing condition* for the $\phi_{\lambda\mu}$).

We can first consider the topological space obtained by gluing (by means of the $\phi_{\lambda\mu}$) of the X_λ along $\mathbf{0_I}$ | 39
the $V_{\lambda\mu}$; if we identify X_λ with the corresponding open subset X'_λ in X, the hypotheses imply that the three sets $V_{\lambda\mu} \cap V_{\lambda\nu}$, $V_{\mu\nu} \cap V_{\mu\lambda}$, $V_{\nu\lambda} \cap V_{\nu\mu}$ identify with $X'_\lambda \cap X'_\mu \cap X'_\nu$. We can also transport to X'_λ the ringed space structure of X_λ, and if $\mathscr{A}'_\lambda$ are the transported sheaves of rings corresponding to the $\mathscr{A}_\lambda$, the $\mathscr{A}'_\lambda$ satisfy the gluing condition (3.3.1) and therefore define a sheaf of rings $\mathscr{A}$ on X; we say that $(X, \mathscr{A})$ is the ringed space obtained by *gluing the* $(X_\lambda, \mathscr{A}_\lambda)$ *along the* $V_{\lambda\mu}$, by means of the $\phi_{\lambda\mu}$.

4.2. Direct image of an $\mathscr{A}$-module

(4.2.1). Let $(X, \mathscr{A})$, $(Y, \mathscr{B})$ be two ringed spaces, $\Psi = (\psi, \theta)$ a morphism $(X, \mathscr{A}) \to (Y, \mathscr{B})$; $\psi_*(\mathscr{A})$ is then a sheaf of rings on Y, and θ a homomorphism $\mathscr{B} \to \psi_*(\mathscr{A})$ of sheaves of rings. Then let $\mathscr{F}$ be an $\mathscr{A}$-module; the direct image $\psi_*(\mathscr{F})$ is a sheaf of abelian groups on Y. In addition, for each open $U \subset Y$,

$$\Gamma(U, \psi_*(\mathscr{F})) = \Gamma(\psi^{-1}(U), \mathscr{F})$$

is equipped with the structure of a module over the ring $\Gamma(U, \psi_*(\mathscr{A})) = \Gamma(\psi^{-1}(U), \mathscr{A})$; the bilinear maps which define these structures are compatible with the restriction operations, defining on $\psi_*(\mathscr{F})$ the structure of a $\psi_*(\mathscr{A})$-module. The homomorphism $\theta : \mathscr{B} \to \psi_*(\mathscr{A})$ then defines also on $\psi_*(\mathscr{F})$ a *$\mathscr{B}$-module* structure; we say that this $\mathscr{B}$-module is the *direct image of $\mathscr{F}$ under the morphism* Ψ, and we denote it by $\Psi_*(\mathscr{F})$. If $\mathscr{F}_1$, $\mathscr{F}_2$ are two $\mathscr{A}$-modules over X and u an $\mathscr{A}$-homomorphism $\mathscr{F}_1 \to \mathscr{F}_2$, it is immediate (by considering the sections over the open subsets of Y) that $\psi_*(u)$ is a $\psi_*(\mathscr{A})$-homomorphism $\psi_*(\mathscr{F}_1) \to \psi_*(\mathscr{F}_2)$, and *a fortiori* a $\mathscr{B}$-homomorphism $\Psi_*(\mathscr{F}_1) \to \Psi_*(\mathscr{F}_2)$; as a $\mathscr{B}$-homomorphism, we denote it by $\Psi_*(u)$. So we see that Ψ_* is a *covariant functor* from the category of $\mathscr{A}$-modules to that of $\mathscr{B}$-modules. In addition, it is immediate that this functor is *left exact* (G, II, 2.12).

On $\psi_*(\mathscr{A})$, the structure of a $\mathscr{B}$-module and the structure of a sheaf of rings define a $\mathscr{B}$-algebra structure; we denote by $\Psi_*(\mathscr{A})$ this $\mathscr{B}$-algebra.

(4.2.2). Let $\mathscr{M}$, $\mathscr{N}$ be two $\mathscr{A}$-modules. For each open set U of Y, we have a canonical map

$$\Gamma(\psi^{-1}(U), \mathscr{M}) \times \Gamma(\psi^{-1}(U), \mathscr{N}) \longrightarrow \Gamma(\psi^{-1}(U), \mathscr{M} \otimes_{\mathscr{A}} \mathscr{N})$$

which is bilinear over the ring $\Gamma(\psi^{-1}(U), \mathscr{A}) = \Gamma(U, \psi_*(\mathscr{A}))$, and *a fortiori* over $\Gamma(U, \mathscr{B})$; it therefore defines a homomorphism

$$\Gamma(U, \Psi_*(\mathscr{M})) \otimes_{\Gamma(U,\mathscr{B})} \Gamma(U, \Psi_*(\mathscr{N})) \longrightarrow \Gamma(U, \Psi_*(\mathscr{M} \otimes_{\mathscr{A}} \mathscr{N}))$$

and as we check immediately that these homomorphisms are compatible with the restriction operations, they give a canonical functorial homomorphism of $\mathscr{B}$-modules

(4.2.2.1)
$$\Psi_*(\mathscr{M}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{N}) \longrightarrow \Psi_*(\mathscr{M} \otimes_{\mathscr{A}} \mathscr{N})$$

which is in general neither injective nor surjective. If $\mathscr{P}$ is a third $\mathscr{A}$-module, we check immediately $\mathbf{0_I}$ | 40
that the diagram

(4.2.2.2)
$$\begin{tikzcd} \Psi_*(\mathscr{M}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{N}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{P}) \arrow[r] \arrow[d] & \Psi_*(\mathscr{M} \otimes_{\mathscr{A}} \mathscr{N}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{P}) \arrow[d] \\ \Psi_*(\mathscr{M}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{N} \otimes_{\mathscr{A}} \mathscr{P}) \arrow[r] & \Psi_*(\mathscr{M} \otimes_{\mathscr{A}} \mathscr{N} \otimes_{\mathscr{A}} \mathscr{P}) \end{tikzcd}$$

is commutative.

(4.2.3). Let $\mathscr{M}$, $\mathscr{N}$ be two $\mathscr{A}$-modules. For each open $U \subset Y$, we have by definition that $\Gamma(\psi^{-1}(U), \mathscr{H}om_{\mathscr{A}}(\mathscr{M}, \mathscr{N})) = \operatorname{Hom}_{\mathscr{A}|V}(\mathscr{M}|V, \mathscr{N}|V)$, where we put $V = \psi^{-1}(U)$; the map $u \mapsto \Psi_*(u)$ is a homomorphism

$$\operatorname{Hom}_{\mathscr{A}|V}(\mathscr{M}|V, \mathscr{N}|V) \longrightarrow \operatorname{Hom}_{\mathscr{B}|U}(\Psi_*(\mathscr{M})|U, \Psi_*(\mathscr{N})|U)$$

on the $\Gamma(U, \mathscr{B})$-module structures; these homomorphisms are compatible with the restriction operations, hence they define a canonical functorial homomorphism of $\mathscr{B}$-modules

(4.2.3.1)
$$\Psi_*(\mathscr{H}om_{\mathscr{A}}(\mathscr{M}, \mathscr{N})) \longrightarrow \mathscr{H}om_{\mathscr{B}}(\Psi_*(\mathscr{M}), \Psi_*(\mathscr{N})).$$

(4.2.4). If $\mathscr{C}$ is an $\mathscr{A}$-algebra, the composite homomorphism

$$\Psi_*(\mathscr{C}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{C}) \longrightarrow \Psi_*(\mathscr{C} \otimes_{\mathscr{A}} \mathscr{C}) \longrightarrow \Psi_*(\mathscr{C})$$

defines on $\Psi_*(\mathscr{C})$ the structure of a *$\mathscr{B}$-algebra*, as a result of (4.2.2.2). We see similarly that if $\mathscr{M}$ is a $\mathscr{C}$-module, $\Psi_*(\mathscr{M})$ is canonically equipped with the structure of a $\Psi_*(\mathscr{C})$-module.

(4.2.5). Consider in particular the case where X is a *closed* subspace of Y and where ψ is the canonical injection $j : X \to Y$. If $\mathscr{B}' = \mathscr{B}|X = j^*(\mathscr{B})$ is the restriction of the sheaf of rings $\mathscr{B}$ to X, an $\mathscr{A}$-module $\mathscr{M}$ can be considered as a $\mathscr{B}'$-module by means of the homomorphism $\theta^\sharp : \mathscr{B}' \to \mathscr{A}$; then $\Psi_*(\mathscr{M})$ is the $\mathscr{B}$-module which induces $\mathscr{M}$ on X and 0 elsewhere. If $\mathscr{N}$ is a second $\mathscr{A}$-module, $\Psi_*(\mathscr{M}) \otimes_{\mathscr{B}} \Psi_*(\mathscr{N})$ canonically identifies with $\Psi_*(\mathscr{M} \otimes_{\mathscr{B}'} \mathscr{N})$ and $\mathscr{H}om_{\mathscr{B}}(\Psi_*(\mathscr{M}), \Psi_*(\mathscr{N}))$ with $\Psi_*(\mathscr{H}om_{\mathscr{B}'}(\mathscr{M}, \mathscr{N}))$.

(4.2.6). Let $(Z, \mathscr{C})$ be a third ringed space, $\Psi' = (\psi', \theta')$ a morphism $(Y, \mathscr{B}) \to (Z, \mathscr{C})$; if Ψ'' is the composite morphism $\Psi' \circ \Psi$, it is clear that we have $\Psi''_* = \Psi'_* \circ \Psi_*$.

4.3. Inverse image of an $\mathscr{A}$-module

(4.3.1). The hypotheses and notation being the same as (4.2.1), let $\mathscr{G}$ be a $\mathscr{B}$-module and $\psi^*(\mathscr{G})$ the inverse image (3.7.1) which is therefore a sheaf of abelian groups on X. The definition of sections of $\psi^*(\mathscr{G})$ and of $\psi^*(\mathscr{B})$ (3.7.1) shows that $\psi^*(\mathscr{G})$ is canonically equipped with a $\psi^*(\mathscr{B})$-module structure. On the other hand, the homomorphism $\theta^\sharp : \psi^*(\mathscr{B}) \to \mathscr{A}$ endows $\mathscr{A}$ with the a $\psi^*(\mathscr{B})$-module structure, which we denote by $\mathscr{A}_{[\theta]}$ when necessary to avoid confusion; the tensor product $\psi^*(\mathscr{G}) \otimes_{\psi^*(\mathscr{B})} \mathscr{A}_{[\theta]}$ is then equipped with an $\mathscr{A}$-module structure. We say that this $\mathscr{A}$-module is *the inverse image of $\mathscr{G}$ under the morphism* Ψ and we denote it by $\Psi^*(\mathscr{G})$. If $\mathscr{G}_1, \mathscr{G}_2$ are two $\mathscr{B}$-modules over Y, v a $\mathscr{B}$-homomorphism $\mathscr{G}_1 \to \mathscr{G}_2$, then $\psi^*(v)$, as we check immediately, is a $\psi^*(\mathscr{B})$-homomorphism from $\psi^*(\mathscr{G}_1)$ to $\psi^*(\mathscr{G}_2)$; as a result $\psi^*(v) \otimes 1$ is an $\mathscr{A}$-homomorphism $\Psi^*(\mathscr{G}_1) \to \Psi^*(\mathscr{G}_2)$, which we denote by $\Psi^*(v)$. So we define Ψ^* as a *covariant functor* from the category of $\mathscr{B}$-modules to that of $\mathscr{A}$-modules. Here, this functor (contrary to ψ^*) is no longer exact in general, but only *right exact*, the tensorization by $\mathscr{A}$ being a right exact functor to the category of $\psi^*(\mathscr{B})$-modules. $\mathbf{0_I}$ | 41

For each $x \in X$, we have $(\Psi^*(\mathscr{G}))_x = \mathscr{G}_{\psi(x)} \otimes_{\mathscr{B}_{\psi(x)}} \mathscr{A}_x$, according to (3.7.2). The support of $\Psi^*(\mathscr{G})$ is thus contained in $\psi^{-1}(\mathrm{Supp}(\mathscr{G}))$.

(4.3.2). Let $(\mathscr{G}_\lambda)$ be an inductive system of $\mathscr{B}$-modules, and let $\mathscr{G} = \varinjlim \mathscr{G}_\lambda$ be its inductive limit. The canonical homomorphisms $\mathscr{G}_\lambda \to \mathscr{G}$ define the $\psi^*(\mathscr{B})$-homomorphisms $\psi^*(\mathscr{G}_\lambda) \to \psi^*(\mathscr{G})$, which give a canonical homomorphism $\varinjlim \psi^*(\mathscr{G}_\lambda) \to \psi^*(\mathscr{G})$. As the stalk at a point of an inductive limit of sheaves is the inductive limit of the stalks at the same point (G, II, 1.11), the preceding canonical homomorphism is *bijective* (3.7.2). In addition, the tensor product commutes with inductive limits of sheaves, and we thus have a *canonical functorial isomorphism* $\varinjlim \Psi^*(\mathscr{G}_\lambda) \simeq \Psi^*(\varinjlim \mathscr{G}_\lambda)$ of $\mathscr{A}$-modules.

On the other hand, for a finite direct sum $\bigoplus_i \mathscr{G}_i$ of $\mathscr{B}$-modules, it is clear that $\psi^*(\bigoplus_i \mathscr{G}_i) = \bigoplus_i \psi^*(\mathscr{G}_i)$, therefore, by tensoring with $\mathscr{A}_{[\theta]}$,

$$\Psi^*\Big(\bigoplus_i \mathscr{G}_i\Big) = \bigoplus_i \Psi^*(\mathscr{G}_i). \tag{4.3.2.1}$$

By passing to the inductive limit, we deduce, in light of the above, that the above equality is still true for *any* direct sum.

(4.3.3). Let $\mathscr{G}_1$, $\mathscr{G}_2$ be two $\mathscr{B}$-modules; from the definition of the inverse images of sheaves of abelian groups (3.7.1), we obtain immediately a canonical homomorphism $\psi^*(\mathscr{G}_1) \otimes_{\psi^*(\mathscr{B})} \psi^*(\mathscr{G}_2) \to \psi^*(\mathscr{G}_1 \otimes_{\mathscr{B}} \mathscr{G}_2)$ of $\psi^*(\mathscr{B})$-modules, and the stalk at a point of a tensor product of sheaves being the tensor product of the stalks at this point (G, II, 2.8), we deduce from (3.7.2) that the above homomorphism is in fact a *isomorphism*. By tensoring with $\mathscr{A}$, we obtain a *canonical functorial isomorphism*

$$\Psi^*(\mathscr{G}_1) \otimes_{\mathscr{A}} \Psi^*(\mathscr{G}_2) \simeq \Psi^*(\mathscr{G}_1 \otimes_{\mathscr{B}} \mathscr{G}_2). \tag{4.3.3.1}$$

(4.3.4). Let $\mathscr{C}$ be a $\mathscr{B}$-algebra; the data of the algebra structure on $\mathscr{C}$ is the same as the data of a $\mathscr{B}$-homomorphism $\mathscr{C} \otimes_{\mathscr{B}} \mathscr{C} \to \mathscr{C}$ satisfying the associativity and commutativity conditions (conditions which are checked stalk-wise); the above isomorphism allows us to consider this homomorphism as a homomorphism of $\mathscr{A}$-modules $\Psi^*(\mathscr{C}) \otimes_{\mathscr{A}} \Psi^*(\mathscr{C}) \to \Psi^*(\mathscr{C})$ satisfying the same conditions, so $\Psi^*(\mathscr{C})$ is thus equipped with an $\mathscr{A}$-algebra structure. In particular, it follows immediately from the definitions that the $\mathscr{A}$-algebra $\Psi^*(\mathscr{B})$ is *equal to* $\mathscr{A}$ (up to a canonical isomorphism).

Similarly, if $\mathscr{M}$ is a $\mathscr{C}$-module, the data of this module structure is the same as that of a $\mathscr{B}$-homomorphism $\mathscr{C} \otimes_{\mathscr{B}} \mathscr{M} \to \mathscr{M}$ satisfying the associativity condition; hence we give a $\Psi^*(\mathscr{C})$-module structure on $\Psi^*(\mathscr{M})$. $\mathbf{0_I}$ | 42

(4.3.5). Let $\mathscr{J}$ be a sheaf of ideals of $\mathscr{B}$; as the functor ψ^* is exact, the $\psi^*(\mathscr{B})$-module $\psi^*(\mathscr{J})$ canonically identifies with a sheaf of ideals of $\psi^*(\mathscr{B})$; the canonical injection $\psi^*(\mathscr{J}) \to \psi^*(\mathscr{B})$ then gives a homomorphism of $\mathscr{A}$-modules $\Psi^*(\mathscr{J}) = \psi^*(\mathscr{J}) \otimes_{\psi^*(\mathscr{B})} \mathscr{A}_{[\theta]} \to \mathscr{A}$; we denote by $\Psi^*(\mathscr{J})\mathscr{A}$, or $\mathscr{J}\mathscr{A}$ if there is no fear of confusion, the image of $\Psi^*(\mathscr{J})$ under this homomorphism. So we have by definition $\mathscr{J}\mathscr{A} = \theta^\sharp(\psi^*(\mathscr{J}))\mathscr{A}$ and in particular, for each $x \in X$, $(\mathscr{J}\mathscr{A})_x = \theta_x^\sharp(\mathscr{J}_{\psi(x)})\mathscr{A}_x$, taking into account the canonical identification between the stalks of $\psi^*(\mathscr{J})$ and those of $\mathscr{J}$ (3.7.2). If $\mathscr{J}_1$, $\mathscr{J}_2$ are two sheaves of ideals of $\mathscr{B}$, then we have $(\mathscr{J}_1\mathscr{J}_2)\mathscr{A} = \mathscr{J}_1(\mathscr{J}_2\mathscr{A}) = (\mathscr{J}_1\mathscr{A})(\mathscr{J}_2\mathscr{A})$.

If $\mathscr{F}$ is an $\mathscr{A}$-module, we set $\mathscr{J}\mathscr{F} = (\mathscr{J}\mathscr{A})\mathscr{F}$.

(4.3.6). Let $(Z, \mathscr{C})$ be a third ringed space, $\Psi' = (\psi', \theta')$ a morphism $(Y, \mathscr{B}) \to (Z, \mathscr{C})$; if Ψ'' is the composite morphism $\Psi' \circ \Psi$, it follows from the definition (4.3.1) and from (4.3.3.1) that we have $\Psi''^* = \Psi^* \circ \Psi'^*$.

4.4. Relation between direct and inverse images

(4.4.1). The hypotheses and notation being the same as in (4.2.1), let $\mathscr{G}$ be a $\mathscr{B}$-module. By definition, a homomorphism $u : \mathscr{G} \to \Psi_*(\mathscr{F})$ of $\mathscr{B}$-modules is still called a Ψ*-morphisms from* $\mathscr{G}$ *to* $\mathscr{F}$, or simply a *homomorphism from* $\mathscr{G}$ *to* $\mathscr{F}$ and we write it as $u : \mathscr{G} \to \mathscr{F}$ when no confusion will occur. To give such a homomorphism is the same as giving, for each pair (U, V) where U is an open set of X, V an open set of Y such that $\psi(U) \subset V$, a *homomorphism* $u_{U,V} : \Gamma(V, \mathscr{G}) \to \Gamma(U, \mathscr{F})$ *of* $\Gamma(V, \mathscr{B})$-*modules*, $\Gamma(U, \mathscr{F})$ being considered as a $\Gamma(V, \mathscr{B})$-module by means of the ring homomorphism $\theta_{U,V} : \Gamma(V, \mathscr{B}) \to \Gamma(U, \mathscr{A})$; the $u_{U,V}$ must in addition render commutative the diagrams (3.5.1.1). It suffices, moreover, to define u by the data of the $u_{U,V}$ when U (resp. V) varies over a basis $\mathfrak{B}$ (resp. $\mathfrak{B}'$) for the topology of X (resp. Y) and to check the commutativity of (3.5.1.1) for these restrictions.

(4.4.2). Under the hypotheses of (4.2.1) and (4.2.6), let $\mathscr{H}$ be a $\mathscr{C}$-module, $v : \mathscr{H} \to \Psi'_*(\mathscr{G})$ a Ψ'-morphism; then $w : \mathscr{H} \xrightarrow{v} \Psi'_*(\mathscr{G}) \xrightarrow{\Psi'_*(u)} \Psi'_*(\Psi_*(\mathscr{F}))$ is a Ψ''-morphism which we call the *composition* of u and v.

(4.4.3). We will now see that we can define a canonical *isomorphism* of *bifunctors* in $\mathscr{F}$ and $\mathscr{G}$

$$\mathrm{Hom}_{\mathscr{A}}(\Psi^*(\mathscr{G}), \mathscr{F}) \simeq \mathrm{Hom}_{\mathscr{B}}(\mathscr{G}, \Psi_*(\mathscr{F})) \tag{4.4.3.1}$$

which we denote by $v \mapsto v_\theta^\flat$ (or simply $v \mapsto v^\flat$ if there is no chance of confusion); we denote by $u \mapsto u_\theta^\sharp$, or $u \mapsto u^\sharp$, the inverse isomorphism. This definition is the following: by composing $v : \Psi^*(\mathscr{G}) \to \mathscr{F}$ with the canonical map $\psi^*(\mathscr{G}) \to \Psi^*(\mathscr{G})$, we obtain a homomorphism of sheaves of groups $v' : \psi^*(\mathscr{G}) \to \mathscr{F}$, which is also a homomorphism of $\psi^*(\mathscr{B})$-modules. We obtain (3.7.1) a homomorphism $v'^\flat : \mathscr{G} \to \psi_*(\mathscr{F}) = \Psi_*(\mathscr{F})$, which is also a homomorphism of $\mathscr{B}$-modules as we check easily; we take $v_\theta^\flat = v'^\flat$. Similarly, for $u : \mathscr{G} \to \Psi_*(\mathscr{F})$, which is a homomorphism of $\mathscr{B}$-modules, we obtain (3.7.1) a homomorphism $u^\sharp : \psi^*(\mathscr{G}) \to \mathscr{F}$ of $\psi^*(\mathscr{B})$-modules, hence by tensoring with $\mathscr{A}$ we have a homomorphism of $\mathscr{A}$-modules $\Psi^*(\mathscr{G}) \to \mathscr{F}$, which we denote by $u_\theta^\sharp$. It is immediate to check that $(u_\theta^\sharp)_\theta^\flat = u$ and $(v_\theta^\flat)_\theta^\sharp = v$, so we have established the functorial nature in $\mathscr{F}$ of the isomorphism $v \mapsto v_\theta^\flat$. The functorial nature in $\mathscr{G}$ of $u \mapsto u_\theta^\sharp$ is then formally shown as in (3.5.4) (reasoning that would also prove the functorial nature of Ψ^* established in (4.3.1) directly). $\mathbf{0_I}$ | 43

If we take for v the identity homomorphism of $\Psi^*(\mathscr{B})$, $v_\theta^\flat$ is a homomorphism

$$\rho_{\mathscr{G}} : \mathscr{G} \longrightarrow \Psi_*(\Psi^*(\mathscr{G})); \tag{4.4.3.2}$$

if we take for u the identity homomorphism of $\Psi_*(\mathscr{F})$, $u_\theta^\sharp$ is a homomorphism

$$\sigma_{\mathscr{F}} : \Psi^*(\Psi_*(\mathscr{F})) \longrightarrow \mathscr{F}; \tag{4.4.3.3}$$

these homomorphisms will be called *canonical*. They are in general neither injective or surjective. We have canonical factorizations analogous to (3.5.3.3) and (3.5.4.4).

We note that if s is a section of $\mathscr{G}$ over an open set V of Y, $\rho_{\mathscr{G}}(s)$ is the section $s' \otimes 1$ of $\Psi^*(\mathscr{G})$ over $\psi^{-1}(V)$, s' being such that $s'_x = s_{\psi(x)}$ for all $x \in \psi^{-1}(V)$. We also note that if $u : \mathscr{G} \to \psi_*(\mathscr{F})$ is a homomorphism, it defines for all $x \in X$ a homomorphism $u_x : \mathscr{G}_{\psi(x)} \to \mathscr{F}_x$ on the stalks, obtained by composing $(u^\sharp)_x : (\Psi^*(\mathscr{G}))_x \to \mathscr{F}_x$ and the canonical homomorphism $s_x \mapsto s_x \otimes 1$ from $\mathscr{G}_{\psi(x)}$ to $(\Psi^*(\mathscr{G}))_x = \mathscr{G}_{\psi(x)} \otimes_{\mathscr{B}_{\psi(x)}} \mathscr{A}_x$. The homomorphism u_x is obtained also by passing to the inductive limit relative to the homomorphisms $\Gamma(V, \mathscr{G}) \xrightarrow{u} \Gamma(\psi^{-1}(V), \mathscr{F}) \to \mathscr{F}_x$, where V varies over the neighbourhoods of $\psi(x)$.

(4.4.4). Let $\mathscr{F}_1, \mathscr{F}_2$ be $\mathscr{A}$-modules, $\mathscr{G}_1, \mathscr{G}_2$ be $\mathscr{B}$-modules, u_i $(i = 1, 2)$ a homomorphism from $\mathscr{G}_i$ to $\mathscr{F}_i$. We denote by $u_1 \otimes u_2$ the homomorphism $u : \mathscr{G}_1 \otimes_{\mathscr{B}} \mathscr{G}_2 \to \mathscr{F}_1 \otimes_{\mathscr{A}} \mathscr{F}_2$ such that $u^\sharp = (u_1)^\sharp \otimes (u_2)^\sharp$ (taking into account (4.3.3.1)); we check that u is also the composition $\mathscr{G}_1 \otimes_{\mathscr{B}} \mathscr{G}_2 \to \Psi_*(\mathscr{F}_1) \otimes_{\mathscr{B}} \Psi_*(\mathscr{F}_2) \to \Psi_*(\mathscr{F}_1 \otimes_{\mathscr{A}} \mathscr{F}_2)$, where the first arrow is the ordinary tensor product $u_1 \otimes_{\mathscr{B}} u_2$ and the second is the canonical homomorphism (4.2.2.1).

(4.4.5). Let $(\mathscr{G}_\lambda)_{\lambda \in L}$ be an inductive system of $\mathscr{B}$-modules, and, for each $\lambda \in L$, let u_λ be a homomorphism $\mathscr{G}_\lambda \to \Psi_*(\mathscr{F})$, form an inductive limit; we put $\mathscr{G} = \varinjlim \mathscr{G}_\lambda$ and $u = \varinjlim u_\lambda$; then the $(u_\lambda)^\sharp$ form an inductive system of homomorphisms $\Psi^*(\mathscr{G}_\lambda) \to \mathscr{F}$, and the inductive limit of this system is none other than $u^\sharp$.

(4.4.6). Let $\mathscr{M}$, $\mathscr{N}$ be two $\mathscr{B}$-modules, V an open set of Y, $U = \psi^{-1}(V)$; the map $v \mapsto \Psi^*(v)$ is a homomorphism

$$\mathrm{Hom}_{\mathscr{B}|V}(\mathscr{M}|V, \mathscr{N}|V) \longrightarrow \mathrm{Hom}_{\mathscr{A}|U}(\Psi^*(\mathscr{M})|U, \Psi^*(\mathscr{N})|U)$$

for the $\Gamma(V, \mathscr{B})$-module structures ($\mathrm{Hom}_{\mathscr{A}|U}(\Psi^*(\mathscr{M})|U, \Psi^*(\mathscr{N})|U)$ is normaly equipped with the a $\Gamma(U, \psi^*(\mathscr{B}))$-module structure, and thanks to the canonical homomorphism (3.7.2) $\Gamma(V, \mathscr{B}) \to$ **0I** | 44
$\Gamma(U, \psi^*(\mathscr{B}))$, it is also a $\Gamma(V, \mathscr{B})$-module). We see immediately that these homomorphisms are compatible with the restriction morphisms, and as a result define a canonical functorial homomorphism

$$\gamma : \mathscr{H}om_{\mathscr{B}}(\mathscr{M}, \mathscr{N}) \longrightarrow \Psi_*(\mathscr{H}om_{\mathscr{A}}(\Psi^*(\mathscr{M}), \Psi^*(\mathscr{N}));$$

it also corresponds to this homomorphism the homomorphism

$$\gamma^\sharp : \Psi^*(\mathscr{H}om_{\mathscr{B}}(\mathscr{M}, \mathscr{N})) \longrightarrow \mathscr{H}om_{\mathscr{A}}(\Psi^*(\mathscr{M}), \Psi^*(\mathscr{N}))$$

and these canonical morphisms are functorial in $\mathscr{M}$ and $\mathscr{N}$.

(4.4.7). Suppose that $\mathscr{F}$ (resp. $\mathscr{G}$) is an $\mathscr{A}$-algebra (resp. a $\mathscr{B}$-algebra). If $u : \mathscr{G} \to \Psi_*(\mathscr{F})$ is a homomorphism of $\mathscr{B}$-algebras, $u^\sharp$ is a homomorphism $\Psi^*(\mathscr{G}) \to \mathscr{F}$ of $\mathscr{A}$-algebras; this follows from the commutativity of the diagram

$$\begin{tikzcd} \mathscr{G} \otimes_{\mathscr{B}} \mathscr{G} \arrow[r] \arrow[d] & \mathscr{G} \arrow[d, "u"] \\ \Psi_*(\mathscr{F} \otimes_{\mathscr{A}} \mathscr{F}) \arrow[r] & \Psi_*(\mathscr{F}) \end{tikzcd}$$

and from (4.4.4). Similarly, if $v : \Psi^*(\mathscr{G}) \to \mathscr{F}$ is a homomorphism of $\mathscr{A}$-algebras, $v^\flat : \mathscr{G} \to \Psi_*(\mathscr{F})$ is a homomorphism of $\mathscr{B}$-algebras.

(4.4.8). Let $(Z, \mathscr{C})$ be a third ringed space, $\Psi' = (\psi', \theta')$ a morphism $(Y, \mathscr{B}) \to (Z, \mathscr{C})$, and $\Psi'' : (X, \mathscr{A}) \to (Z, \mathscr{C})$ the composite morphism $\Psi' \circ \Psi$. Let $\mathscr{H}$ be a $\mathscr{C}$-module, u' a homomorphism from $\mathscr{H}$ to $\mathscr{G}$; the composition $v'' = v \circ v'$ is by definition the homomorphism from $\mathscr{H}$ to $\mathscr{F}$ defined by $\mathscr{H} \xrightarrow{v'} \Psi'_*(\mathscr{G}) \xrightarrow{\Psi'_*(v)} \Psi'_*(\Psi_*(\mathscr{F}))$; we check that $v''^\sharp$ is the homomorphism

$$\Psi^*(\Psi'^*(\mathscr{H})) \xrightarrow{\Psi^*(v'^\sharp)} \Psi^*(\mathscr{G}) \xrightarrow{v^\sharp} \mathscr{F}.$$

§5. Quasi-coherent and coherent sheaves

5.1. Quasi-coherent sheaves

(5.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space, $\mathscr{F}$ an $\mathscr{O}_X$-module. The data of a homomorphism $u : \mathscr{O}_X \to \mathscr{F}$ of $\mathscr{O}_X$-modules is equivalent to that of the section $s = u(1) \in \Gamma(X, \mathscr{F})$. Indeed, when s is given, for each section $t \in \Gamma(U, \mathscr{O}_X)$, we necessarily have $u(t) = t \cdot (s|U)$; we say that u is *defined by the section* s. If now I is any set of indices, consider the direct sum sheaf $\mathscr{O}_X^{(I)}$, and for each $i \in I$, let h_i be the canonical injection of the i-th factor into $\mathscr{O}_X^{(I)}$; we know that $u \mapsto (u \circ h_i)$ is an isomorphism from $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X^{(I)}, \mathscr{F})$ to the product $(\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{F}))^I$. So there is a canonical one-to-one correspondence between the homomorphisms $u : \mathscr{O}_X^{(I)} \to \mathscr{F}$ and the *families of sections* $(s_i)_{i \in I}$ *of* $\mathscr{F}$ *over* X. The homomorphism u corresponding to (s_i) sends an element $(a_i) \in (\Gamma(U, \mathscr{O}_X))^{(I)}$ to $\sum_{i \in I} a_i \cdot (s_i|U)$.

We say that $\mathscr{F}$ is *generated by the family* (s_i) if the homomorphism $\mathscr{O}_X^{(I)} \to \mathscr{F}$ defined for each family is *surjective* (in other words, if, for each $x \in X$, $\mathscr{F}_x$ is an $\mathscr{O}_x$-module generated by the $(s_i)_x$). **0_{I}** | 45
We say that $\mathscr{F}$ is *generated by its sections over* X if it is generated by the family of all these sections (or by a subfamily), in other words, if there exists a surjective homomorphism $\mathscr{O}_X^{(I)} \to \mathscr{F}$ for a suitable I.

We note that an $\mathscr{O}_X$-module $\mathscr{F}$ can be such that there exists a point $x_0 \in X$ for which $\mathscr{F}|U$ is not generated by its sections over U, *regardless of the choice of neighbourhood U of x_0*: it suffices to take $X = \mathbf{R}$, for $\mathscr{O}_X$ the simple sheaf $\mathbf{Z}$, for $\mathscr{F}$ the algebraic subsheaf of $\mathscr{O}_X$ such that $\mathscr{F}_0 = \{0\}$, $\mathscr{F}_x = \mathbf{Z}$ for $x \neq 0$, and finally $x_0 = 0$: the only section of $\mathscr{F}|U$ over U is 0 for a neighbourhood U of 0.

(5.1.2). Let $f : X \to Y$ be a morphism of ringed spaces. If $\mathscr{F}$ is an $\mathscr{O}_X$-module generated by its sections over X, then the canonical homomorphism $f^*(f_*(\mathscr{F})) \to \mathscr{F}$ (4.4.3.3) is *surjective*; indeed, with the notation of (5.1.1), $s_i \otimes 1$ is a section of $f^*(f_*(\mathscr{F}))$ over X, and its image in $\mathscr{F}$ is s_i. The example in (5.1.1) where f is the identity shows that the inverse of this proposition is false in general.

If $\mathscr{G}$ is an $\mathscr{O}_Y$-module generated by its sections over Y, then $f^*(\mathscr{G})$ is generated by its sections over X, since f^* is a right exact functor. **$\mathrm{Err}_{\mathrm{II}}$**

(5.1.3). We say that an $\mathscr{O}_X$-module $\mathscr{F}$ is *quasi-coherent* if for each $x \in X$ there is an open neighbourhood U of x such that $\mathscr{F}|U$ is isomorphic to the *cokernel* of a homomorphism of the form $\mathscr{O}_X^{(I)}|U \to \mathscr{O}_X^{(J)}|U$, where I and J are sets of arbitrary indices. It is clear that $\mathscr{O}_X$ is itself a quasi-coherent $\mathscr{O}_X$-module, and that any direct sum of quasi-coherent $\mathscr{O}_X$-modules is again a quasi-coherent $\mathscr{O}_X$-module. We say that an *$\mathscr{O}_X$-algebra $\mathscr{A}$* is *quasi-coherent* if it is quasi-coherent as an $\mathscr{O}_X$-module.

(5.1.4). Let $f : X \to Y$ be a morphism of ringed spaces. If $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_Y$-module, then $f^*(\mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module. Indeed, for each $x \in X$, there is an open neighbourhood V of $f(x)$ in Y such that $\mathscr{G}|V$ is the cokernel of a homomorphism $\mathscr{O}_Y^{(I)}|V \to \mathscr{O}_Y^{(J)}|V$. If $U = f^{-1}(V)$, and if f_U is the restriction of f to U, then we have $f^*(\mathscr{G})|U = f_U^*(\mathscr{G}|V)$; as f_U^* is right exact and commutes with direct sums, $f_U^*(\mathscr{G}|V)$ is the cokernel of a homomorphism $\mathscr{O}_X^{(I)}|U \to \mathscr{O}_X^{(J)}|U$.

5.2. Sheaves of finite type

(5.2.1). We say that an $\mathscr{O}_X$-module $\mathscr{F}$ is *of finite type* if for each $x \in X$ there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is generated by a *finite* family of sections over U, or if it is isomorphic to a sheaf quotient of a sheaf of the form $(\mathscr{O}_X|U)^p$ where p is finite. Each sheaf quotient of a sheaf of finite type is again a sheaf of finite type, as well as each finite direct sum and each finite tensor product of sheaves of finite type. An $\mathscr{O}_X$-module of finite type is not necessarily quasi-coherent, as we can see for the $\mathscr{O}_X$-module $\mathscr{O}_X/\mathscr{F}$, where $\mathscr{F}$ is the example in (5.1.1). If $\mathscr{F}$ is of finite type, then $\mathscr{F}_x$ is an $\mathscr{O}_x$-module of finite type for each $x \in X$, but the example in (5.1.1) shows that this condition is necessary but not sufficient in general.

(5.2.2). Let $\mathscr{F}$ be an $\mathscr{O}_X$-module *of finite type*. If s_i $(1 \leqslant i \leqslant n)$ are the sections of $\mathscr{F}$ over an open neighbourhood U of a point $x \in X$ and the $(s_i)_x$ generate $\mathscr{F}_x$, then there exists an open neighbourhood $V \subset U$ of x such that the $(s_i)_y$ generate $\mathscr{F}_y$ for all $y \in Y$ (FAC, I, 2, 12, prop. 1). In particular, we conclude that the support of $\mathscr{F}$ is *closed*.

Similarly, if $u : \mathscr{F} \to \mathscr{G}$ is a homomorphism such that $u_x = 0$, then there exists a neighbourhood U of x such that $u_y = 0$ for all $y \in U$. **0_I** | 46

(5.2.3). Suppose that X is *quasi-compact*, and let $\mathscr{F}$ and $\mathscr{G}$ be two $\mathscr{O}_X$-modules such that $\mathscr{G}$ is *of finite type*, $u : \mathscr{F} \to \mathscr{G}$ a *surjective* homomorphism. In addition, suppose that $\mathscr{F}$ is the inductive limit of an inductive system $(\mathscr{F}_\lambda)$ of $\mathscr{O}_X$-modules. Then there exists an index μ such that the homomorphism $\mathscr{F}_\mu \to \mathscr{G}$ is *surjective*. Indeed, for each $x \in X$, there exists a finite system of sections s_i of $\mathscr{G}$ over an open neighbourhood $U(x)$ of x such that the $(s_i)_y$ generate $\mathscr{G}_y$ for all $y \in U(x)$; there is then an open neighbourhood $V(x) \subset U(x)$ of x and n sections t_i of $\mathscr{F}$ over $V(x)$ such that $s_i|V(s) = u(t_i)$ for all i; we can also suppose that the t_i are the canonical images of sections of a similar sheaf $\mathscr{F}_{\lambda(x)}$ over $V(x)$. We then cover X with a finite number of neighbourhoods $V(x_k)$, and let μ be the maximal index of the $\lambda(x_k)$; it is clear that this index gives the answer.

Suppose still that X is quasi-compact, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module of finite type generated by its sections over X (5.1.1); then $\mathscr{F}$ is generated by a *finite* subfamily of these sections: indeed, it suffices to cover X by a finite number of open neighbourhoods U_k such that, for each k, there is a finite number of sections s_{ik} of $\mathscr{F}$ over X whose restrictions to U_k generate $\mathscr{F}|U_k$; it is clear that the s_{ik} then generate $\mathscr{F}$.

(5.2.4). Let $f : X \to Y$ be a morphism of ringed spaces. If $\mathscr{G}$ is an $\mathscr{O}_Y$-module of finite type, then $f^*(\mathscr{G})$ is an $\mathscr{O}_X$-module of finite type. Indeed, for each $x \in X$, there is an open neighbourhood V of $f(x)$ in Y and a surjective homomorphism $v : \mathscr{O}_Y^p|V \to \mathscr{G}|V$. If $U = f^{-1}(V)$ and if f_U is the restriction of f to U, then we have $f^*(\mathscr{G})|U = f_U^*(\mathscr{G}|V)$; since f_U^* is right exact (4.3.1) and commutes with direct sums (4.3.2), $f_U^*(v)$ is a surjective homomorphism $\mathscr{O}_X^p|U \to f^*(\mathscr{G})|U$. **Err_II**

(5.2.5). We say that an $\mathscr{O}_X$-module $\mathscr{F}$ *admits a finite presentation* if for each $x \in X$ there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is isomorphic to a *cokernel of a* $(\mathscr{O}_X|U)$*-homomorphism* $\mathscr{O}_X^p|U \to \mathscr{O}_X^q|U$, p and q being two integers > 0. Such an $\mathscr{O}_X$-module is therefore of finite type and quasi-coherent. If $f : X \to Y$ is a morphism of ringed spaces, and if $\mathscr{G}$ is an $\mathscr{O}_Y$-module admitting a finite presentation, then $f^*(\mathscr{G})$ admits a finite presentation, as shown in the argument of (5.1.4).

(5.2.6). Let $\mathscr{F}$ be an $\mathscr{O}_X$-module admitting a finite presentation (5.2.5); then, for each $\mathscr{O}_X$-module $\mathscr{H}$, the canonical functorial homomorphism

$$(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{H}))_x \longrightarrow \mathrm{Hom}_{\mathscr{O}_x}(\mathscr{F}_x, \mathscr{H}_x)$$

is *bijective* (T, 4.1.1).

(5.2.7). Let $\mathscr{F}$ and $\mathscr{G}$ be two $\mathscr{O}_X$-modules admitting a finite presentation. If for some $x \in X$, $\mathscr{F}_x$ and $\mathscr{G}_x$ are *isomorphic* as $\mathscr{O}_x$-modules, then there exists an open neighbourhood U of x such that $\mathscr{F}|U$ and $\mathscr{G}|U$ are *isomorphic*. Indeed, if $\phi : \mathscr{F}_x \to \mathscr{G}_x$ and $\psi : \mathscr{G}_x \to \mathscr{F}_x$ are an isomorphism and its inverse isomorphism, then there exists, according to (5.2.6), an open neighbourhood V of x and a section u (resp. v) of $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ (resp. $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{F})$) over V such that $u_x = \phi$ (resp. $v_x = \psi$). As $(u \circ v)_x$ and $(v \circ u)_x$ are the identity automorphisms, there exists an open neighbourhood $U \subset V$ of x such that $(u \circ v)|U$ and $(v \circ u)|U$ are the identity automorphisms, hence the proposition. **0_I** | 47

5.3. Coherent sheaves

(5.3.1). We say that an $\mathscr{O}_X$-module $\mathscr{F}$ is *coherent* if it satisfies the two following conditions:

(a) $\mathscr{F}$ is of finite type.
(b) for each open $U \subset X$, integer $n > 0$, and homomorphism $u : \mathscr{O}_X^n|U \to \mathscr{F}|U$, the kernel of u is of finite type.

We note that these two conditions are of a *local* nature.

For most of the proofs of the properties of coherent sheaves in what follows, cf. (FAC, I, 2).

(5.3.2). Each coherent $\mathscr{O}_X$-module admits a finite presentation (5.2.5); the inverse is not necessarily true, since $\mathscr{O}_X$ itself is not necessarily a coherent $\mathscr{O}_X$-module.

Each $\mathscr{O}_X$-submodule *of finite type* of a coherent $\mathscr{O}_X$-module is coherent; each *finite* direct sum of coherent $\mathscr{O}_X$-modules is a coherent $\mathscr{O}_X$-module.

(5.3.3). If $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of $\mathscr{O}_X$-modules and if two of these $\mathscr{O}_X$-modules are coherent, then so is the third.

(5.3.4). If $\mathscr{F}$ and $\mathscr{G}$ are two coherent $\mathscr{O}_X$-modules, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism, then $\operatorname{Im}(u)$, $\operatorname{Ker}(u)$, and $\operatorname{Coker}(u)$ are coherent $\mathscr{O}_X$-modules. In particular, if $\mathscr{F}$ and $\mathscr{G}$ are $\mathscr{O}_X$-submodules of a coherent $\mathscr{O}_X$-module, then $\mathscr{F} + \mathscr{G}$ and $\mathscr{F} \cap \mathscr{G}$ are coherent.

If $\mathscr{A} \to \mathscr{B} \to \mathscr{C} \to \mathscr{D} \to \mathscr{E}$ is an exact sequence of $\mathscr{O}_X$-modules, and if $\mathscr{A}$, $\mathscr{B}$, $\mathscr{D}$, $\mathscr{E}$ are coherent, then $\mathscr{C}$ is coherent. **Err$_{\text{II}}$**

(5.3.5). If $\mathscr{F}$ and $\mathscr{G}$ are two coherent $\mathscr{O}_X$-modules, then so are $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}$ are $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$.

(5.3.6). Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, $\mathscr{J}$ a coherent sheaf of ideals of $\mathscr{O}_X$. Then the $\mathscr{O}_X$-module $\mathscr{J}\mathscr{F}$ is coherent, as the image of $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{F}$ under the canonical homomorphism $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{F}$ ((5.3.4) and (5.3.5)).

(5.3.7). We say that an $\mathscr{O}_X$-algebra $\mathscr{A}$ is *coherent* if it is coherent as an $\mathscr{O}_X$-module. In particular, $\mathscr{O}_X$ is a *coherent sheaf of rings* if and only if for each open $U \subset X$ and each homomorphism of the form $u : \mathscr{O}_X^p|U \to \mathscr{O}_X|U$, the kernel of u is an $(\mathscr{O}_X|U)$-module of finite type.

If $\mathscr{O}_X$ is a coherent sheaf of rings, then each $\mathscr{O}_X$-module $\mathscr{F}$ admitting a finite presentation (5.2.5) is coherent, according to (5.3.4).

The *annihilator* of an $\mathscr{O}_X$-module $\mathscr{F}$ is the kernel $\mathscr{J}$ of the canonical homomorphism $\mathscr{O}_X \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{F})$ which sends each section $s \in \Gamma(U, \mathscr{O}_X)$ to the multiplication by s map in $\operatorname{Hom}(\mathscr{F}|U, \mathscr{F}|U)$; if $\mathscr{O}_X$ is coherent and if $\mathscr{F}$ is a coherent $\mathscr{O}_X$-module, then $\mathscr{J}$ is coherent ((5.3.4) and (5.3.5)) and for each $x \in X$, $\mathscr{J}_x$ is the annihilator of $\mathscr{F}_x$ (5.2.6).

(5.3.8). Suppose that $\mathscr{O}_X$ is coherent; let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, x a point of X, M a submodule of finite type of $\mathscr{F}_x$; then there exists an open neighbourhood U of x and a coherent $(\mathscr{O}_X|U)$-submodule $\mathscr{G}$ of $\mathscr{F}|U$ such that $\mathscr{G}_x = M$ (T, 4.1, Lemma 1). **$\mathbf{0_I}$ | 48**

This result, along with the properties of the $\mathscr{O}_X$-submodules of a coherent $\mathscr{O}_X$-module, impose the necessary conditions on the rings $\mathscr{O}_x$ such that $\mathscr{O}_X$ is coherent. For example (5.3.4), the intersection of two ideals of finite type of $\mathscr{O}_x$ must still be an ideal of finite type.

(5.3.9). Suppose that $\mathscr{O}_X$ is coherent, and let M be an $\mathscr{O}_x$-module admitting a finite presentation, therefore isomorphic to a cokernel of a homomorphism $\phi : \mathscr{O}_x^p \to \mathscr{O}_x^q$; then there exists an open neighbourhood U of X and a coherent $(\mathscr{O}_X|U)$-module $\mathscr{F}$ such that $\mathscr{F}_x$ is isomorphic to M. Indeed, according to (5.2.6), there exists a section u of $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{O}_X^p, \mathscr{O}_X^q)$ over an open neighbourhood U of x such that $u_x = \phi$; the cokernel $\mathscr{F}$ of the homomorphism $u : \mathscr{O}_X^p|U \to \mathscr{O}_X^q|U$ gives the answer (5.3.4).

(5.3.10). Suppose that $\mathscr{O}_X$ is coherent, and let $\mathscr{J}$ be a coherent sheaf of ideals of $\mathscr{O}_X$. For a $(\mathscr{O}_X/\mathscr{J})$-module $\mathscr{F}$ to be coherent, it is necessary and sufficient for it to be coherent as an $\mathscr{O}_X$-module. In particular, $\mathscr{O}_X/\mathscr{J}$ is a coherent sheaf of rings.

(5.3.11). Let $f : X \to Y$ be a morphism of ringed spaces, and suppose that $\mathscr{O}_X$ is coherent; then, for each coherent $\mathscr{O}_Y$-module $\mathscr{G}$, $f^*(\mathscr{G})$ is a coherent $\mathscr{O}_X$-module. Indeed, with the notation of (5.2.4), we can assume that $\mathscr{G}|V$ is the cokernel of a homomorphism $v : \mathscr{O}_Y^q|V \to \mathscr{O}_Y^p|V$; as f_U^* is right exact, $f^*(\mathscr{G})|U = f_U^*(\mathscr{G}|V)$ is the cokernel of the homomorphism $f_U^*(v) : \mathscr{O}_X^q|U \to \mathscr{O}_X^p|U$, hence our assertion.

(5.3.12). Let Y be a closed subset of X, $j : Y \to X$ the canonical injection, $\mathscr{O}_Y$ a sheaf of rings on Y, and set $\mathscr{O}_X = j_*(\mathscr{O}_Y)$. For an $\mathscr{O}_Y$-module $\mathscr{G}$ to be of finite type (resp. quasi-coherent, coherent), it is necessary and sufficient for $j_*(\mathscr{G})$ to be an $\mathscr{O}_X$-module of finite type (resp. quasi-coherent, coherent).

5.4. Locally free sheaves

(5.4.1). Let X be a ringed space. We say that an $\mathscr{O}_X$-module $\mathscr{F}$ is *locally free* if for each $x \in X$ there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is isomorphic to a $(\mathscr{O}_X|U)$-module of the form $\mathscr{O}_X^{(I)}|U$, where I can depend on U. If for each U, I is finite, then we say that $\mathscr{F}$ is *of finite rank*; if for each U, I has the same finite number of elements n, we say that $\mathscr{F}$ is *of rank n*. A locally free $\mathscr{O}_X$-module of rank 1 is called *invertible* (cf. (5.4.3)). If $\mathscr{F}$ is a locally free $\mathscr{O}_X$-module of finite rank, then for each $x \in X$, $\mathscr{F}_x$ is a free $\mathscr{O}_x$-module of finite rank $n(x)$, and there exists a neighbourhood U of x such that $\mathscr{F}|U$, is of rank $n(x)$; if X is connected, then $n(x)$ is *constant*.

It is clear that each locally free sheaf is quasi-coherent, and if $\mathscr{O}_X$ is a coherent sheaf of rings, then each locally free $\mathscr{O}_X$-module of finite rank is coherent.

If $\mathscr{L}$ is locally free, then $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{F}$ is an *exact* functor in $\mathscr{F}$ to the category of $\mathscr{O}_X$-modules.

We will mostly consider locally free $\mathscr{O}_X$-modules of finite rank, and when we speak of locally free sheaves without specifying, it will be understood that they are of *finite rank*. $\mathbf{0_I}$ | 49

Suppose that $\mathscr{O}_X$ is *coherent*, and let $\mathscr{F}$ be a *coherent* $\mathscr{O}_X$-module. Then, if at a point $x \in X$, $\mathscr{F}_x$ is an $\mathscr{O}_x$-module *free of rank n*, there exists a neighbourhood U of x such that $\mathscr{F}|U$ is *locally free of rank n*; in fact, $\mathscr{F}_x$ is then isomorphic to $\mathscr{O}_x^n$, and the proposition follows from (5.2.7). **Err$_{\text{II}}$**

(5.4.2). If $\mathscr{L}$, $\mathscr{F}$ are two $\mathscr{O}_X$-modules, we have a canonical functorial homomorphism

$$\mathscr{L}^\vee \otimes_{\mathscr{O}_X} \mathscr{F} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X) \otimes_{\mathscr{O}_X} \mathscr{F} \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{F}) \tag{5.4.2.1}$$

defined in the following way: for each open set U, send any pair (u, t), where $u \in \Gamma(U, \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X)) = \operatorname{Hom}(\mathscr{L}|U, \mathscr{O}_X|U)$ and $t \in \Gamma(U, \mathscr{F})$, to the element of $\operatorname{Hom}(\mathscr{L}|U, \mathscr{F}|U)$ which, for each $x \in U$, sends $s_x \in \mathscr{L}_x$ to the element $u_x(s_x)t_x$ of $\mathscr{F}_x$. If $\mathscr{L}$ is *locally free of finite rank*, then this homomorphism is *bijective*; the property being local, we can in fact reduce to the case where $\mathscr{L} = \mathscr{O}_X^n$; as for each $\mathscr{O}_X$-module $\mathscr{G}$, $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{O}_X^n, \mathscr{G})$ is canonically isomorphic to $\mathscr{G}^n$, we have reduced to the case $\mathscr{L} = \mathscr{O}_X$, which is immediate.

(5.4.3). If $\mathscr{L}$ is invertible, then so is its dual $\mathscr{L}^\vee = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X)$, since we can immediately reduce (as the question is local) to the case $\mathscr{L} = \mathscr{O}_X$. In addition, we have a canonical isomorphism

$$\mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X) \otimes_{\mathscr{O}_X} \mathscr{L} \simeq \mathscr{O}_X \tag{5.4.3.1}$$

as, according to (5.3.2), it suffices to define a canonical isomorphism $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{L}) \simeq \mathscr{O}_X$. For *each* $\mathscr{O}_X$-module $\mathscr{F}$, we have a canonical homomorphism $\mathscr{O}_X \simeq \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{F})$ (5.3.7). It remains to prove that if $\mathscr{F} = \mathscr{L}$ is invertible, then this homomorphism is bijective, and as the question is local, it reduces to the case $\mathscr{L} = \mathscr{O}_X$, which is immediate.

Due to the above, we put $\mathscr{L}^{-1} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X)$, and we say that $\mathscr{L}^{-1}$ is the *inverse* of $\mathscr{L}$. The terminology "invertible sheaf" can be justified in the following way when X is a point and $\mathscr{O}_X$ is a *local* ring A with maximal ideal $\mathfrak{m}$; if M and M' are two A-modules (M being of finite type) such that $M \otimes_A M'$ is isomorphic to A, then as $(A/\mathfrak{m}) \otimes_A (M \otimes_A M')$ identifies with $(M/\mathfrak{m}M) \otimes_{A/\mathfrak{m}} (M'/\mathfrak{m}M')$, this latter tensor product of vector spaces over the field $A/\mathfrak{m}$ is isomorphic to $A/\mathfrak{m}$, which requires $M/\mathfrak{m}M$ and $M'/\mathfrak{m}M'$ to be of dimension 1. For each element $z \in M$ not in $\mathfrak{m}M$, we have $M = Az + \mathfrak{m}M$, which implies that $M = Az$ according to Nakayama's Lemma, M being of finite type. Moreover, as the annihilator of z kills $M \otimes_A M'$, which is isomorphic to A, this annihilator is $\{0\}$, and as a result M is *isomorphic to A*. In the general case, this shows that $\mathscr{L}$ is an $\mathscr{O}_X$-module of finite type, such that there exists an $\mathscr{O}_X$-module $\mathscr{F}$ for which $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{F}$ is isomorphic to $\mathscr{O}_X$, and if in addition the rings $\mathscr{O}_x$ are local rings, then $\mathscr{L}_x$ is an $\mathscr{O}_x$-module isomorphic to $\mathscr{O}_x$ for each $x \in X$. If $\mathscr{O}_X$ and $\mathscr{L}$ are assumed to be *coherent*, then we conclude that $\mathscr{L}$ is invertible according to (5.2.7).

(5.4.4). If $\mathscr{L}$ and $\mathscr{L}'$ are two invertible $\mathscr{O}_X$-modules, then so is $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}'$, since the question is local, we can assume that $\mathscr{L} = \mathscr{O}_X$, and the result is then trivial. For each integer $n \geqslant 1$, we denote by $\mathscr{L}^{\otimes n}$ the tensor product of n copies of the sheaf $\mathscr{L}$; we set by convention $\mathscr{L}^{\otimes 0} = \mathscr{O}_X$, and for $n \geqslant 1$, $\mathscr{L}^{\otimes(-n)} = (\mathscr{L}^{-1})^{\otimes n}$. With these notation, there is then a *canonical functorial isomorphism* $\mathbf{0_I}$ | 50

$$\mathscr{L}^{\otimes m} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n} \simeq \mathscr{L}^{\otimes(n+m)} \tag{5.4.4.1}$$

for any rational integers m and n: indeed, by definition, we immediately reduce to the case where $m = -1$, $n = 1$, and the isomorphism in question is then that defined in (5.4.3).

(5.4.5). Let $f : Y \to X$ be a morphism of ringed spaces. If $\mathscr{L}$ is a locally free (resp. invertible) $\mathscr{O}_X$-module, then $f^*(\mathscr{L})$ is a locally free (resp. invertible) $\mathscr{O}_Y$-module: this follows immediately from that the inverse images of two locally isomorphic $\mathscr{O}_X$-modules are locally isomorphic, that f^* commutes with finite direct sums, and that $f^*(\mathscr{O}_X) = \mathscr{O}_Y$ (4.3.4). In addition, we know that we have a canonical functorial homomorphism $f^*(\mathscr{L}^\vee) \to (f^*(\mathscr{L}))^\vee$ (4.4.6), and when $\mathscr{L}$ is locally free, this homomorphism is *bijective*: indeed, we again reduce to the case where $\mathscr{L} = \mathscr{O}_X$ which is trivial. We conclude that if $\mathscr{L}$ is invertible, then $f^*(\mathscr{L}^{\otimes n})$ canonically identifies with $(f^*(\mathscr{L}))^{\otimes n}$ for each rational integer n.

(5.4.6). Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module; we denote by $\Gamma_\bullet(X, \mathscr{L})$ or simply $\Gamma_\bullet(\mathscr{L})$ the abelian group direct sum $\bigoplus_{n\in\mathbf{Z}} \Gamma(X, \mathscr{L}^{\otimes n})$; we equip it with the structure of a *graded ring*, by corresponding to a pair (s_n, s_m), where $s_n \in \Gamma(X, \mathscr{L}^{\otimes n})$, $s_m \in \Gamma(X, \mathscr{L}^{\otimes m})$, the section of $\mathscr{L}^{\otimes(n+m)}$ over X which corresponds canonically (5.4.4.1) to the section $s_n \otimes s_m$ of $\mathscr{L}^{\otimes n} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes m}$; the associativity of this multiplication is verified in an immediate way. It is clear that $\Gamma_\bullet(X, \mathscr{L})$ is a covariant functor in $\mathscr{L}$, with values in the category of graded rings.

If now $\mathscr{F}$ is any $\mathscr{O}_X$-module, then we set

$$\Gamma_\bullet(\mathscr{L}, \mathscr{F}) = \bigoplus_{n\in\mathbf{Z}} \Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}).$$

We equip this abelian group with the structure of a *graded module* over the graded ring $\Gamma_\bullet(\mathscr{L})$ in the following way: to a pair (s_n, u_m), where $s_n \in \Gamma(X, \mathscr{L}^{\otimes n})$ and $u_m \in \Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes m})$, we associate the section of $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes(m+n)}$ which canonically corresponds (5.4.4.1) to $s_n \otimes u_m$; the verification of the module axioms are immediate. For X and $\mathscr{L}$ fixed, $\Gamma_\bullet(\mathscr{L}, \mathscr{F})$ is a covariant functor in $\mathscr{F}$ with values in the category of graded $\Gamma_\bullet(\mathscr{L})$-modules; for X and $\mathscr{F}$ fixed, it is a covariant functor in $\mathscr{L}$ with values in the category of abelian groups.

If $f : Y \to X$ is a morphism of ringed spaces, the canonical homomorphism (4.4.3.2) $\rho : \mathscr{L}^{\otimes n} \to f_*(f^*(\mathscr{L}^{\otimes n}))$ defines a homomorphism of abelian groups $\Gamma(X, \mathscr{L}^{\otimes n}) \to \Gamma(Y, f^*(\mathscr{L}^{\otimes n}))$, and as $f^*(\mathscr{L}^{\otimes n}) = (f^*(\mathscr{L}))^{\otimes n})$, it follows from the definitions of the canonical homomorphisms (4.4.3.2) and (5.4.4.1) that the above homomorphisms define a *functorial homomorphism of graded rings* $\Gamma_\bullet(\mathscr{L}) \to \Gamma_\bullet(f^*(\mathscr{L}))$. The same canonical homomorphism (4.4.3) similarly defines a homomorphism of abelian groups $\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}) \to \Gamma(Y, f^*(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}))$, and as

$$f^*(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}) = f^*(\mathscr{F}) \otimes_{\mathscr{O}_Y} (f^*(\mathscr{L}))^{\otimes n} \quad (4.3.3.1),$$

these homomorphism (for n variable) define a *di-homomorphism of graded modules* $\Gamma_\bullet(\mathscr{L}, \mathscr{F}) \to \Gamma_\bullet(f^*(\mathscr{L}), f^*(\mathscr{F}))$. $\mathbf{0_I}$ | 51

(5.4.7). One can show that there exists a *set* $\mathfrak{M}$ (also denoted $\mathfrak{M}(X)$) of invertible $\mathscr{O}_X$-modules such that each invertible $\mathscr{O}_X$-module is isomorphic to a unique element of $\mathfrak{M}$ [8]; we define on $\mathfrak{M}$ a composition law by sending two elements $\mathscr{L}$ and $\mathscr{L}'$ of $\mathfrak{M}$ to the unique element of $\mathfrak{M}$ isomorphic to $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}'$. With this composition law, $\mathfrak{M}$ *is a group isomorphic to the cohomology group* $\mathrm{H}^1(X, \mathscr{O}_X^*)$, where $\mathscr{O}_X^*$ is the subsheaf of $\mathscr{O}_X$ such that $\Gamma(U, \mathscr{O}_X^*)$ is the group of invertible elements of the ring $\Gamma(U, \mathscr{O}_X)$ for each open $U \subset X$ ($\mathscr{O}_X^*$ is therefore a sheaf of *multiplicative* abelian groups).

We will note that for all open $U \subset X$, the group of sections $\Gamma(U, \mathscr{O}_X^*)$ canonically identifies with the *automorphism group* of the $(\mathscr{O}_X|U)$-module $\mathscr{O}_X|U$, the identification sending a section ε of $\mathscr{O}_X^*$ over U to the automorphism u of $\mathscr{O}_X|U$ such that $u_x(s_x) = \varepsilon_x s_x$ for all $x \in X$ and all $s_x \in \mathscr{O}_x$. Then let $\mathfrak{U} = (U_\lambda)$ be an open cover of X; the data, for each pair of indices (λ, μ), of an automorphism $\theta_{\lambda\mu}$ of $\mathscr{O}_X|(U_\lambda \cap U_\mu)$ is the same as giving a 1-*cochain* of the cover $\mathfrak{U}$, with values in $\mathscr{O}_X^*$, and say that the $\theta_{\lambda\mu}$ satisfy the gluing condition (3.3.1), meaning that the corresponding cochain is a *cocycle*. Similarly, the data, for each λ, of an automorphism ω_λ of $\mathscr{O}_X|U_\lambda$ is the same as the data of a 0-cochain of the cover $\mathfrak{U}$, with values in $\mathscr{O}_X^*$, and its *coboundary* corresponds to the family of automorphisms $(\omega_\lambda|U_\lambda \cap U_\mu) \circ (\omega_\mu|U_\lambda \cap U_\mu)^{-1}$. We can send each 1-cocycle of $\mathfrak{U}$ with values in $\mathscr{O}_X^*$ to the element of $\mathfrak{M}$ isomorphic to an invertible $\mathscr{O}_X$-module obtained by gluing with respect to the family of

[8] See the book in preparation cited in the introduction.

automorphisms $(\theta_{\lambda\mu})$ corresponding to this cocycle, and to two cohomologous coycles correspond two equal elements of $\mathfrak{M}$ (3.3.2); in other words, we thus define a map $\phi_{\mathfrak{U}} : \mathrm{H}^1(\mathfrak{U}, \mathscr{O}_X^*) \to \mathfrak{M}$. In addition, if $\mathfrak{B}$ is a second open cover of X, finer than $\mathfrak{U}$, then the diagram

$$\begin{tikzcd}
\mathrm{H}^1(\mathfrak{U}, \mathscr{O}_X^*) \arrow[dd] \arrow[dr, "\phi_{\mathfrak{U}}"] & \\
& \mathfrak{M} \\
\mathrm{H}^1(\mathfrak{B}, \mathscr{O}_X^*) \arrow[ur, "\phi_{\mathfrak{B}}"'] &
\end{tikzcd}$$

where the vertical arrow is the canonical homomorphism (G, II, 5.7), is commutative, as a result of (3.3.3). By passing to the inductive limit, we therefore obtain a map $\mathrm{H}^1(X, \mathscr{O}_X^*) \to \mathfrak{M}$, the Čech cohomology group $\check{\mathrm{H}}^1(X, \mathscr{O}_X^*)$ identifying as we know with the first cohomology group $\mathrm{H}^1(X, \mathscr{O}_X^*)$ (G, II, 5.9, Cor. of Thm. 5.9.1). This map is *surjective*: indeed, by definition, for each invertible $\mathscr{O}_X$-module $\mathscr{L}$, there is an open cover (U_λ) of X such that $\mathscr{L}$ is obtained by gluing the sheaves $\mathscr{O}_X|U_\lambda$ (3.3.1). It is also *injective*, since it suffices to prove for the maps $\mathrm{H}^1(\mathfrak{U}, \mathscr{O}_X) \to \mathfrak{M}$, and this follows from (3.3.2). It remains to show that the bijection thus defined is a group homomorphism. $\mathbf{0_I}$ | 52
Given two invertible $\mathscr{O}_X$-modules $\mathscr{L}$ and $\mathscr{L}'$, there is an open cover (U_λ) such that $\mathscr{L}|U_\lambda$ and $\mathscr{L}'|U_\lambda$ are isomorphic to $\mathscr{O}_X|U_\lambda$ for each λ; so there is for each index λ an element a_λ (resp. a'_λ) of $\Gamma(U_\lambda, \mathscr{L})$ (resp. $\Gamma(U_\lambda, \mathscr{L}')$) such that the elements of $\Gamma(U_\lambda, \mathscr{L})$ (resp. $\Gamma(U_\lambda, \mathscr{L}')$) are the $s_\lambda \cdot a_\lambda$ (resp. $s_\lambda \cdot a'_\lambda$), where s_λ varies over $\Gamma(U_\lambda, \mathscr{O}_X)$. The corresponding cocycles $(\varepsilon_{\lambda\mu})$, $(\varepsilon'_{\lambda\mu})$ are such that $s_\lambda \cdot a_\lambda = s_\mu \cdot a_\mu$ (resp. $s_\lambda \cdot a'_\lambda = s_\mu \cdot a'_\mu$) over $U_\lambda \cap U_\mu$ is equivalent to $s_\lambda = \varepsilon_{\lambda\mu} s_\mu$ (resp. $s_\lambda = \varepsilon'_{\lambda\mu} s_\mu$) over $U_\lambda \cap U_\mu$. As the sections of $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}'$ over U_λ are the finite sums of the $s_\lambda s'_\lambda \cdot (a_\lambda \otimes a'_\lambda)$ where s_λ and s'_λ vary over $\Gamma(U_\lambda, \mathscr{O}_X)$, it is clear that the cocycle $(\varepsilon_{\lambda\mu}, \varepsilon'_{\lambda\mu})$ corresponds to $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}'$, which finishes the proof. [9]

(5.4.8). Let $f = (\psi, \omega)$ be a morphism $Y \to X$ of ringed spaces. The functor $f^*(\mathscr{L})$ to the category of free $\mathscr{O}_X$-modules defines a map (which we still denote f^* by abuse of language) from the set $\mathfrak{M}(X)$ to the set $\mathfrak{M}(Y)$. Second, we have a canonical homomorphism (T, 3.2.2)

$$\mathrm{H}^1(X, \mathscr{O}_X^*) \longrightarrow \mathrm{H}^1(Y, \mathscr{O}_Y^*). \tag{5.4.8.1}$$

When we canonically identify (5.4.7) $\mathfrak{M}(X)$ and $\mathrm{H}^1(X, \mathscr{O}_X^*)$ (resp. $\mathfrak{M}(Y)$ and $\mathrm{H}^1(Y, \mathscr{O}_Y^*)$), the homomorphism (5.4.8.1) *identifies with the map* f^*. Indeed, if $\mathscr{L}$ comes from a cocycle $(\varepsilon_{\lambda\mu})$ corresponding to an open cover (U_λ) of X, then it suffices to show that $f^*(\mathscr{L})$ comes from a cocycle whose cohomology class is the image under (5.4.8.1) of $(\varepsilon_{\lambda\mu})$. If $\theta_{\lambda\mu}$ is the automorphism of $\mathscr{O}_X|(U_\lambda \cap U_\mu)$ which corresponds to $\varepsilon_{\lambda\mu}$, then it is clear that $f^*(\mathscr{L})$ is obtained by gluing the $\mathscr{O}_Y|\psi^{-1}(U_\lambda)$ by means of the automorphisms $f^*(\theta_{\lambda\mu})$, and it then suffices to check that these latter automorphisms corresponds to the cocycle $(\omega^\sharp(\varepsilon_{\lambda\mu}))$, which follows immediately from the definitions (we can identify $\varepsilon_{\lambda\mu}$ with its canonical image under ρ (3.7.2), a section of $\psi^*(\mathscr{O}_X^*)$ over $\psi^{-1}(U_\lambda \cap U_\mu)$).

(5.4.9). Let $\mathscr{E}$ and $\mathscr{F}$ be two $\mathscr{O}_X$-modules, $\mathscr{F}$ assumed to be *locally free*, and let $\mathscr{G}$ be an $\mathscr{O}_X$*-module extension of* $\mathscr{F}$ *by* $\mathscr{E}$, in other words there exists an exact sequence $0 \to \mathscr{E} \xrightarrow{i} \mathscr{G} \xrightarrow{p} \mathscr{F} \to 0$. Then, for each $x \in X$, there exists an open neighbourhood U of x such that $\mathscr{G}|U$ is isomorphic to the *direct sum* $\mathscr{E}|U \oplus \mathscr{F}|U$. We can reduce to the case where $\mathscr{F} = \mathscr{O}_X^n$; let e_i $(1 \leqslant i \leqslant n)$ be the canonical sections (5.5.5) of $\mathscr{O}_X^n$; there then exists an open neighbourhood U of x and n sections s_i of $\mathscr{G}$ over U such that $p(s_i|U) = e_i|U$ for $1 \leqslant i \leqslant n$. That being so, let f be the homomorphism $\mathscr{F}|U \to \mathscr{G}|U$ defined by the sections $s_i|U$ (5.1.1). It is immediate that for each open $V \subset U$, and each section $s \in \Gamma(V, \mathscr{G})$ we have $s - f(p(s)) \in \Gamma(V, \mathscr{E})$, hence our assertion.

(5.4.10). Let $f : X \to Y$ be a morphism of ringed spaces, $\mathscr{F}$ an $\mathscr{O}_X$-module, and $\mathscr{L}$ a locally free $\mathscr{O}_Y$-module of finite rank. Then there exists a canonical isomorphism

$$f_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{L} \simeq f_*(\mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{L})). \tag{5.4.10.1}$$

[9]For a general form of this result, see the book cited in the note on p. 51.

Indeed, for each $\mathscr{O}_Y$-module $\mathscr{L}$, we have a canonical homomorphism **0_I** | 53

$$f_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{L} \xrightarrow{1\otimes\rho} f_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} f_*(f^*(\mathscr{L})) \xrightarrow{\alpha} f_*(\mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{L})),$$

ρ the homomorphism (4.4.3.2) and α the homomorphism (4.2.2.1). To show that when $\mathscr{L}$ is locally free, this homomorphism is bijective, it suffices, since the questions is local, to consider the case where $\mathscr{L} = \mathscr{O}_X^n$; in addition, f_* and f^* commute with finite direct sums, so we can assume $n = 1$, and in this case the proposition follows immediately from the definitions and from the relation $f^*(\mathscr{O}_Y) = \mathscr{O}_X$.

5.5. Sheaves on a locally ringed space

(5.5.1). We say that a ringed space $(X, \mathscr{O}_X)$ is a *locally ringed space* if for each $x \in X$, $\mathscr{O}_x$ is a local ring; these ringed spaces will be by far the most frequent ringed spaces that we will consider in this work. We then denote by $\mathfrak{m}_x$ the *maximal ideal* of $\mathscr{O}_x$, by $k(x)$ the *residue field* $\mathscr{O}_x/\mathfrak{m}_x$; for each $\mathscr{O}_X$-module $\mathscr{F}$, each open set U of X, each point $x \in U$, and each section $f \in \Gamma(U, \mathscr{F})$, we denote by $f(x)$ the *class* of the germ $f_x \in \mathscr{F}_x$ mod. $\mathfrak{m}_x\mathscr{F}_x$, and we say that this is the *value* of f at the point x. The relation $f(x) = 0$ then means that $f_x \in \mathfrak{m}_x\mathscr{F}_x$; when this is so, we say (by abuse of language) that f *is zero at* x. We will take care not to confuse this relation with $f_x = 0$.

(5.5.2). Let X be a locally ringed space, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and f a section of $\mathscr{L}$ over X. There is then an *equivalence* between the three following properties for a point $x \in X$:

(a) *f_x is a generator of $\mathscr{L}_x$;*
(b) $f_x \notin \mathfrak{m}_x\mathscr{L}_x$ (in other words, $f(x) \neq 0$);
(c) *there exists a section g of $\mathscr{L}^{-1}$ over an open neighbourhood V of x such that the canonical image of $f \otimes g$ in $\Gamma(V, \mathscr{O}_X)$* (5.4.3) *is the unit section.*

Indeed, since the questions is local, we can reduce to the case where $\mathscr{L} = \mathscr{O}_X$; the equivalence of (a) and (b) are then evident, and it is clear that (c) implies (b). Conversely, if $f_x \notin \mathfrak{m}_x$, then f_x is invertible in $\mathscr{O}_x$, say $f_x g_x = 1_x$. By definition of germs of sections, this means that there exists a neighbourhood V of x and a section g of $\mathscr{O}_X$ over V such that $fg = 1$ in V, hence (c).

It follows immediately from the condition (c) that the set X_f of x satisfying the equivalent conditions (a), (b), (c) is *open* in X; following the terminology introduced in (5.5.1), this is the set of the x for which f *does not vanish*.

(5.5.3). Under the hypotheses of (5.5.2), let $\mathscr{L}'$ be a second invertible $\mathscr{O}_X$-module; then, if $f \in \Gamma(X, \mathscr{L})$, $g \in \Gamma(X, \mathscr{L}')$, we have

$$X_f \cap X_g = X_{f\otimes g}.$$

We can in fact reduce immediately to the case where $\mathscr{L} = \mathscr{L}' = \mathscr{O}_X$ (since the questions is local); as $f \otimes g$ then canonically identifies with the product fg, the proposition is evident.

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(5.5.4). Let $\mathscr{F}$ be a locally free $\mathscr{O}_X$ of rank n; it is immediate that $\wedge^p\mathscr{F}$ is a locally free $\mathscr{O}_X$-module of rank $\binom{n}{p}$ if $p \leqslant n$ and 0 if $p > n$, since the question is local and we can reduce to the case where $\mathscr{F} = \mathscr{O}_X^n$; in addition, for each $x \in X$, $(\wedge^p\mathscr{F})_x/\mathfrak{m}_x(\wedge^p\mathscr{F})_x$ is a vector space of dimension $\binom{n}{p}$ over $k(x)$, which canonically identifies with $\wedge^p(\mathscr{F}_x/\mathfrak{m}_x\mathscr{F}_x)$. Let $s_1, \ldots, s_p$ be the sections of $\mathscr{F}$ over an open subset U of X, and let $s = s_1 \wedge \cdots \wedge s_p$, which is a section of $\wedge^p\mathscr{F}$ over U (4.1.5); we have $s(x) = s_1(x) \wedge \cdots \wedge s_p(x)$, and as a result, we say that the $s_1(x), \ldots, s_p(x)$ are *linearly dependent* means $s(x) = 0$. We conclude that the *set of the $x \in X$ such that $s_1(x), \ldots, s_p(x)$ are linearly independent is open in* X: it suffices in fact, by reducing to the case where $\mathscr{F} = \mathscr{O}_X^n$, to apply (5.5.2) to the section image of s under one of the projections of $\wedge^p\mathscr{F} = \mathscr{O}_X^{\binom{n}{p}}$ to the $\binom{n}{p}$ factors.

In particular, if $s_1, \ldots, s_n$ are n sections of $\mathscr{F}$ over U such that $s_1(x), \ldots, s_n(x)$ are linearly independent for each point $x \in U$, then the homomorphism $u : \mathscr{O}_X^n|U \to \mathscr{F}|U$ defined by the s_i (5.1.1) is an *isomorphism*: indeed, we can restrict to the case where $\mathscr{F} = \mathscr{O}_X^n$ and where we canonically identify $\wedge^n\mathscr{F}$ and $\mathscr{O}_X$; $s = s_1 \wedge \cdots \wedge s_n$ is then an *invertible* section of $\mathscr{O}_X$ over U, and we define an inverse homomorphism for u by means of the Cramer formulas.

(5.5.5). Let $\mathscr{E}$ and $\mathscr{F}$ be two locally free $\mathscr{O}_X$-modules (of finite rank), and let $u : \mathscr{E} \to \mathscr{F}$ be a homomorphism. For there to exist a neighbourhood U of $x \in X$ such that $u|U$ is *injective* and that $\mathscr{F}|U$ is *the direct sum of the* $u(\mathscr{E})|U$ *and of a locally free* $(\mathscr{O}_X|U)$*-submodule* $\mathscr{G}$, it is necessary and sufficient that $u_x : \mathscr{E}_x \to \mathscr{F}_x$ gives, by passing to quotients, an *injective* homomorphism of vector spaces $\mathscr{E}_x/\mathfrak{m}_x\mathscr{E}_x \to \mathscr{F}_x/\mathfrak{m}_x\mathscr{F}_x$. The condition is indeed *necessary*, since $\mathscr{F}_x$ is then the direct sum of the free $\mathscr{O}_x$-modules $u_x(\mathscr{E}_x)$ and $\mathscr{G}_x$, so $\mathscr{F}_x/\mathfrak{m}_x\mathscr{F}_x$ is the direct sum of $u_x(\mathscr{E}_x)/\mathfrak{m}_x u_x(\mathscr{E}_x)$ and of $\mathscr{G}_x/\mathfrak{m}_x\mathscr{G}_x$. The condition is *sufficient*, since we can reduce to the case where $\mathscr{E} = \mathscr{O}_X^m$; let $s_1, \ldots, s_m$ be the images under u of the sections e_i of $\mathscr{O}_X^m$ such that $(e_i)_y$ is equal to the i-th element of the canonical basis of $\mathscr{O}_y^m$ for each $y \in Y$ (*canonical sections* of $\mathscr{O}_X^m$); by hypothesis, the $s_1(x), \ldots, s_m(x)$ are linearly independent, so if $\mathscr{F}$ is of rank n, then there exist $n-m$ sections $s_{m+1}, \ldots, s_n$ of $\mathscr{F}$ over a neighbourhood V of x such that the $s_i(x)$ $(1 \leqslant i \leqslant n)$ form a basis for $\mathscr{F}_x/\mathfrak{m}_x\mathscr{F}_x$. There then exists (5.5.4) a neighbourhood $U \subset V$ of x such that the $s_i(y)$ $(1 \leqslant i \leqslant n)$ form a basis for $\mathscr{F}_y/\mathfrak{m}_y\mathscr{F}_y$ for each $y \in V$, and we conclude (5.5.4) that there is an isomorphism from $\mathscr{F}|U$ to $\mathscr{O}_X^n|U$, sending the $s_i|U$ $(1 \leqslant i \leqslant m)$ to the $e_i|U$, which finishes the proof.

§6. Flatness

(6.0). The notion of flatness is due to J.-P. Serre [**Ser56**]; in the following, we omit the proofs of the results which are presented in the *Algèbre commutative* of N. Bourbaki, to which we refer the reader. We assume that all rings are commutative.[10]

If M, N are two A-modules, M' (resp. N') a submodule of M (resp. N), we denote by $\operatorname{Im}(M' \otimes_A N')$ the submodule of $M \otimes_A N$, the image under the canonical map $M' \otimes_A N' \to M \otimes_A N$. **$0_\mathrm{I}$** | 55

6.1. Flat modules

(6.1.1). Let M be an A-module. The following conditions are equivalent:

(a) The functor $M \otimes_A N$ is exact in N on the category of A-modules;
(b) $\operatorname{Tor}_i^A(M, N) = 0$ for each $i > 0$ and for each A-module N;
(c) $\operatorname{Tor}_1^A(M, N) = 0$ for each A-module N.

When M satisfies these conditions, we say that M is a *flat* A*-module*. It is clear that each free A-module is flat.

For M to be a flat A-module, it suffices that for each ideal $\mathfrak{J}$ of A, *of finite type*, the canonical map $M \otimes_A \mathfrak{J} \to M \otimes_A A = M$ is *injective*.

(6.1.2). Each inductive limit of flat A-modules is a flat A-module. For a direct sum $\bigoplus_{\lambda \in L} M_\lambda$ of A-modules to be a flat A-modules, it is necessary and sufficient that each of the A-modules M_λ is flat. In particular, every projective A-module is flat.

Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules, such that M'' is *flat*. Then, for each A-module N, the sequence

$$0 \longrightarrow M' \otimes_A N \longrightarrow M \otimes_A N \longrightarrow M'' \otimes_A N \longrightarrow 0$$

is exact. In addition, for M to be flat, is it necessary and sufficient that M' is (but it can be that M and M' are flat without $M'' = M/M'$ being so).

(6.1.3). Let M be a flat A-module, N any A-module; for two submodules N' N'' of N, we then have

$$\operatorname{Im}(M \otimes_A (N' + N'')) = \operatorname{Im}(M \otimes_A N') + \operatorname{Im}(M \otimes_A N''),$$
$$\operatorname{Im}(M \otimes_A (N' \cap N'')) = \operatorname{Im}(M \otimes_A N') \cap \operatorname{Im}(M \otimes_A N'')$$

(images taken in $M \otimes_A N$).

(6.1.4). Let M and N be two A-modules, M' (resp. N') a submodule of M (resp. N), and suppose that one of the modules M/M', N/N' is flat. Then we have $\operatorname{Im}(M' \otimes_A N') = \operatorname{Im}(M' \otimes_A N) \cap (M \otimes_A N')$ (images in $M \otimes_A N$). In particular, if $\mathfrak{J}$ is an ideal of A and if M/M' is flat, then we have $\mathfrak{J}M' = M' \cap \mathfrak{J}M$.

[10] See the exposé cited of N. Bourbaki for the generalization from most of the results to the noncommutative case.

6.2. Change of ring When an additive group M is equipped with multiple modules structures relative to the rings A, B, ..., we say that M is flat as an A-module, B-module, ..., we sometimes also say that M is *A-flat*, *B-flat*, ...

(6.2.1). Let A and B be two rings, M an A-module, N an (A, B)-bimodule. If M is flat and if N is B-flat, then $M \otimes_A N$ is B-flat. In particular, if M and N are two flat A-modules, then $M \otimes_A N$ is a flat A-module. If B is an A-algebra and if M is a flat A-module, then the B-module $M_{(B)} = M \otimes_A B$ $\mathbf{0_I}$ | 56
is flat. Finally, if B is an A-algebra which is flat as an A-module, and if N is a flat B-module, then N is also A-flat.

(6.2.2). Let A be a ring, B an A-algebra which is flat as an A-module. Let M, N be two A-modules, such that M admits a finite presentation; then the canonical homomorphism

$$\operatorname{Hom}_A(M, N) \otimes_A B \longrightarrow \operatorname{Hom}_B(M \otimes_A B, N \otimes_A B) \tag{6.2.2.1}$$

(sending $u \otimes b$ to the homomorphism $m \otimes b' \mapsto u(m) \otimes b'b$) is an isomorphism.

(6.2.3). Let $(A_\lambda, \phi_{\mu\lambda})$ be a filtered inductive system of rings; let $A = \varinjlim A_\lambda$. On the other hand, for each λ, let M_λ be an A_λ-module, and for $\lambda \leqslant \mu$ let $\theta_{\mu\lambda} : M_\lambda \to M_\mu$ be a $\phi_{\mu\lambda}$-homomorphism, such that $(M_\lambda, \theta_{\mu\lambda})$ is an inductive system; $M = \varinjlim M_\lambda$ is then an A-module. This being so, if for each λ, M_λ is a *flat* A_λ-module, then M is a *flat* A-module. Indeed, let $\mathfrak{J}$ be an ideal *of finite type* of A; by definition of the inductive limit, there exists an index λ and an ideal $\mathfrak{J}_\lambda$ of A_λ such that $\mathfrak{J} = \mathfrak{J}_\lambda A$. If we put $\mathfrak{J}'_\mu = \mathfrak{J}_\lambda A_\mu$ for $\mu \geqslant \lambda$, we also have $\mathfrak{J} = \varinjlim \mathfrak{J}'_\mu$ (where μ varies over the indices $\geqslant \lambda$), hence (the functor $\varinjlim$ being exact and commuting with tensor products)

$$M \otimes_A \mathfrak{J} = \varinjlim (M_\mu \otimes_{A_\mu} \mathfrak{J}'_\mu) = \varinjlim \mathfrak{J}'_\mu M_\mu = \mathfrak{J} M.$$

6.3. Local nature of flatness

(6.3.1). If A is a ring, S a multiplicative subset of A, $S^{-1}A$ is a *flat A-module*. Indeed, for each A-module N, $N \otimes_A S^{-1}A$ identifies with $S^{-1}N$ (1.2.5) and we know (1.3.2) that $S^{-1}N$ is an exact functor in N.

If now M is a flat A-module, $S^{-1}M = M \otimes_A S^{-1}A$ is a flat $S^{-1}A$-module (6.2.1), so it is also A-flat according to the above and from (6.2.1). In particular, if P is an $S^{-1}A$-module, we can consider it as an A-module isomorphic to $S^{-1}P$; for P to be A-flat, it is necessary and sufficient that it is $S^{-1}A$-flat.

(6.3.2). Let A be a ring, B an A-algebra, and T a multiplicative subset of B. If P is a B-module which is *A-flat*, $T^{-1}P$ is *A-flat*. Indeed, for each A-module N, we have $(T^{-1}P) \otimes_A N = (T^{-1}B \otimes_B P) \otimes_A N$ $= T^{-1}B \otimes_B (P \otimes_A N) = T^{-1}(P \otimes_A N)$; $T^{-1}(P \otimes_A N)$ is an exact functor in N, being the composition of the two exact functors $P \otimes_A N$ (in N) and $T^{-1}Q$ (in Q). If S is a multiplicative subset of A such that its image in B is *contained in* T, then $T^{-1}P$ is equal to $S^{-1}(T^{-1}P)$, so it is also $S^{-1}A$-flat according to (6.3.1).

(6.3.3). Let $\phi : A \to B$ be a ring homomorphism, M a B-module. The following properties are equivalent:

(a) M is a flat A-module.
(b) For each maximal ideal $\mathfrak{n}$ of B, $M_\mathfrak{n}$ is a flat A-module.
(c) For each maximal ideal $\mathfrak{n}$ of B, by setting $\mathfrak{m} = \phi^{-1}(\mathfrak{n})$, $M_\mathfrak{n}$ is a flat $A_\mathfrak{m}$-module.

Indeed, as $M_\mathfrak{n} = (M_\mathfrak{n})_\mathfrak{m}$, the equivalence of (b) and (c) follows from (6.3.1), and the fact that (a) implies (b) is a particular case of (6.3.2). It remains to see that (b) implies (a), that is $\mathbf{0_I}$ | 57
to say, that for each injective homomorhism $u : N' \to N$ of A-modules, the homomorphism $v = 1 \otimes u : M \otimes_A N' \to M \otimes_A N$ is injective. We have that v is also a homomorphism of B-modules, and we know that for it to be injective, it suffices that for each maximal ideal $\mathfrak{n}$ of B, $v_\mathfrak{n} : (M \otimes_A N')_\mathfrak{n} \to (M \otimes_A N)_\mathfrak{n}$ is injective. But as

$$(M \otimes_A N)_\mathfrak{n} = B_\mathfrak{n} \otimes_B (M \otimes_A N) = M_\mathfrak{n} \otimes_A N,$$

$v_\mathfrak{n}$ is none other that the homomorphism $1 \otimes u : M_\mathfrak{n} \otimes_A N' \to M_\mathfrak{n} \otimes_A N$, which is injective since $M_\mathfrak{n}$ is A-flat.

In particular (by taking $B = A$), for an A-module M to be flat, it is necessary and sufficient that $M_{\mathfrak{m}}$ is $A_{\mathfrak{m}}$-flat for each maximal ideal $\mathfrak{m}$ of A.

(6.3.4). Let M be an A-module; if M is flat, and if $f \in A$ does not divide 0 in A, f does not kill any element $\neq 0$ in M, since the homomorphism $m \mapsto f \cdot m$ is expressed as $1 \otimes u$, where u is the multiplication $a \mapsto f \cdot a$ on A and M is identified with $M \otimes_A A$; if u is injective, it is the same for $1 \otimes u$ since M is flat. In particular, if A is *integral*, M is *torsion-free*.

Conversely, suppose that A is integral, M is torsion-free, and suppose that for each maximal ideal $\mathfrak{m}$ of A, $A_{\mathfrak{m}}$ is a *discrete valuation ring*; then M is *A-flat*. Indeed, it suffices (6.3.3) to prove that $M_{\mathfrak{m}}$ is $A_{\mathfrak{m}}$-flat, and we can therefore suppose that A is already a discrete valuation ring. But as M is the inductive limit of its submodules of finite type, and these latter submodules are torsion-free, we can in addition reduce to the case where M is of finite type (6.1.2). The proposition follows in this case from that M is a free A-module.

In particular, if A is an *integral* ring, $\phi : A \to B$ a ring homomorphism making B a *flat* A-module and $\neq \{0\}$, ϕ is necessarily *injective*. Conversely, if B is integral, A a subring of B, and if for each maximal ideal $\mathfrak{m}$ of A, $A_{\mathfrak{m}}$ is a discrete valuation ring, then B is *A-flat*.

6.4. Faithfully flat modules

(6.4.1). For an A-module M, the following four properties are equivalent:

(a) For a sequence $N' \to N \to N''$ of A-modules to be exact, it is necessary and sufficient that the sequence $M \otimes_A N' \to M \otimes_A N \to M \otimes_A N''$ is exact;
(b) M is flat for each A-module N, the relation $M \otimes_A N = 0$ implies $N = 0$;
(c) M is flat for each homomorphism $v : N \to N'$ of A-modules, the relation $1_M \otimes v = 0$, 1_M being the identity automorphism of M;
(d) M is flat for each maximal ideal $\mathfrak{m}$ of A, $\mathfrak{m}M \neq M$.

When M satisfies these conditions, we say that M is a *faithfully flat* A-module; M is then necessarily a *faithful* module. In addition, if $u : N \to N'$ is a homomorphism of A-modules, then for u to be injective (resp. surjective, bijective), it is necessary and sufficient that $1 \otimes u : M \otimes_A N \to M \otimes_A N'$ is so.

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(6.4.2). A free module $\neq \{0\}$ is faithfully flat; it is the same for the direct sum of a flat module and a faithfully flat module. If S is a multiplicative subset of A, then $S^{-1}A$ is a faithfully flat A-module if S consists of invertible elements (so $S^{-1}A = A$).

(6.4.3). Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules; if M' and M'' are flat, and if one of the two is faithfully flat, then M is also faithfully flat.

(6.4.4). Let A and B be two rings, M an A-module, N an (A, B)-bimodule. If M is faithfully flat and if N is a faithfully flat B-module, then $M \otimes_A N$ is a faithfully flat B-module. In particular, if M and N are two faithfully flat A-modules, then so is $M \otimes_A N$. If B is an A-algebra and if M is a faithfully flat A-module, the B-module $M_{(B)}$ is faithfully flat.

(6.4.5). If M is a faithfully flat A-modules and if S is a multiplicative subset of A, $S^{-1}M$ is a faithfully flat $S^{-1}A$-module, since $S^{-1}M = M \otimes_A (S^{-1}A)$ (6.4.4). Conversely, if for each maximal ideal $\mathfrak{m}$ of A, $M_{\mathfrak{m}}$ is a faithfully flat $A_{\mathfrak{m}}$-module, then M is a faithfully flat A-module, since M is A-flat (6.3.3), and we have

$$M_{\mathfrak{m}}/\mathfrak{m}M_{\mathfrak{m}} = (M \otimes_A A_{\mathfrak{m}}) \otimes_{A_{\mathfrak{m}}} (A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}) = M \otimes_A (A/\mathfrak{m}) = M/\mathfrak{m}M,$$

so the hypotheses imply that $M/\mathfrak{m}M \neq 0$ for each maximal ideal $\mathfrak{m}$ of A, which proves our assertion (6.4.1).

6.5. Restriction of scalars

(6.5.1). Let A be a ring, $\phi : A \to B$ a ring homomorphism making B an A-algebra. Suppose that there exists a B-module N which is a *faithfully flat* A-module. Then, for each A-module M, the homomorphism $x \mapsto 1 \otimes x$ from M to $B \otimes_A M = M_{(B)}$ is *injective*. In particular, ϕ is injective; for each ideal $\mathfrak{a}$ of A, we have $\phi^{-1}(\mathfrak{a}B) = \mathfrak{a}$; for each maximal (resp. prime) ideal $\mathfrak{m}$ of A, there exists a maximal (resp. prime) ideal $\mathfrak{n}$ of B such that $\phi^{-1}(\mathfrak{n}) = \mathfrak{m}$.

(6.5.2). When the conditions of (6.5.1) are satisfied, we identify A with the subring of B by ϕ and more generally, for each A-module M, we identify M with an A-submodule of $M_{(B)}$. We note that if B is also *Noetherian*, then so is A, since the map $\mathfrak{a} \mapsto \mathfrak{a}B$ is an increasing injection from the set of ideals of A to the set of ideals of B; the existence of an infinite strictly increasing sequence of ideals of A thus implies the existence of an analogous sequence of ideals of B.

6.6. Faithfully flat rings

(6.6.1). Let $\phi : A \to B$ be a ring homomorphism making B an A-algebra. The following five properties are equivalent:

(a) B is a faithfully flat A-module (in other words, $M_{(B)}$ is an *exact* and *faithful* functor in M).
(b) The homomorphism ϕ is injective and the A-module $B/\phi(A)$ is flat. $\mathbf{0_I}$ | 59
(c) The A-module B is flat (in other words, the functor $M_{(B)}$ is *exact*), and for each A-module M. the homomorphism $x \mapsto 1 \otimes x$ from M to $M_{(B)}$ is injective.
(d) The A-module B is flat and for each ideal $\mathfrak{a}$ of A, we have $\phi^{-1}(\mathfrak{a}B) = \mathfrak{a}$.
(e) The A-module B is flat and for each maximal ideal $\mathfrak{m}$ of A, there exists a maximal ideal $\mathfrak{n}$ of B such that $\phi^{-1}(\mathfrak{n}) = \mathfrak{m}$.

When these conditions are satisfied, we identify A with a subring of B.

(6.6.2). Let A be a *local* ring, $\mathfrak{m}$ its maximal ideal, and B an A-algebra such that $\mathfrak{m}B \neq B$ (which is so when for example B is a local ring and $A \to B$ is a *local* homomorphism). If B is a *flat* A-module, B is a *faithfully flat* A-module. Indeed, this follows from (6.4.1, (d)). Under the indicated conditions, we thus see that if B is Noetherian, then so too is A (6.5.2). $\mathbf{Err_{II}}$

(6.6.3). Let B be an A-algebra which is a faithfully flat A-module. For each A-module M and each A-submodule M' of M, we have (by identifying M with an A-submodule of $M_{(B)}$) $M' = M \cap M'_{(B)}$. For M to be a flat (resp. faithfully flat) A-module, it is necessary and sufficient that $M_{(B)}$ is a flat (resp. faithfully flat) B-module.

(6.6.4). Let B be an A-algebra, N a faithfully flat B-module. For B to be a flat (resp. faithfully flat) A-module, it is necessary and sufficient that N is.

In particular, let C be a B-algebra; if the ring C is faithfully flat over B and B is faithfully flat over A, then C is faithfully flat over A; if C is faithfully flat over B and over A, then B is faithfully flat over A.

6.7. Flat morphisms of ringed spaces

(6.7.1). Let $f : X \to Y$ be a morphism of ringed spaces, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module. We say that $\mathscr{F}$ is *f-flat* (or *Y-flat* when there is no chance of confusion with f) *at a point* $x \in X$ if $\mathscr{F}_x$ is a flat $\mathscr{O}_{f(x)}$-module; we say that $\mathscr{F}$ is *f-flat over* $y \in Y$ if $\mathscr{F}$ is f-flat for all the points $x \in f^{-1}(y)$; we say that $\mathscr{F}$ is *f-flat* if $\mathscr{F}$ is f-flat at all the points of X. We say that the morphism f is *flat at* $x \in X$ (resp. *flat over* $y \in Y$, resp. *flat*) if $\mathscr{O}_X$ is f-flat at x (resp. f-flat over y, resp. f-flat). If f is a flat morphism, we then say that X is *flat over* Y, or *Y-flat*. $\mathbf{Err_{II}}$

(6.7.2). With the notation of (6.7.1), if $\mathscr{F}$ is f-flat at x, for each open neighbourhood U of $y = f(x)$, the functor $(f^*(\mathscr{G}) \otimes_{\mathscr{O}_X} \mathscr{F})_x$ in $\mathscr{G}$ is *exact* on the category of $(\mathscr{O}_Y|U)$-modules; indeed, this stalk canonically identifies with $\mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$, and our assertion follows from the definition. In particular, if f is a *flat* morphism, the functor f^* is *exact* on the category of $\mathscr{O}_Y$-modules.

(6.7.3). Conversely, suppose the sheaf of rings $\mathscr{O}_Y$ is *coherent*, and suppose that for *each* open neighbourhood U of y, the functor $(f^*(\mathscr{G}) \otimes_{\mathscr{O}_X} \mathscr{F})_x$ is exact in $\mathscr{G}$ on the category of *coherent* $(\mathscr{O}_Y|U)$-modules. Then $\mathscr{F}$ is *f-flat at x*. In fact, it suffices to prove that for each ideal of finite type $\mathfrak{J}$ of $\mathscr{O}_y$, the canonical homomorphism $\mathfrak{J} \otimes_{\mathscr{O}_y} \mathscr{F}_x \to \mathscr{F}_x$ is injective (6.1.1). We know (5.3.8) that there then exists an open neighbourhood U of y and a coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Y|U$ such that $\mathscr{J}_y = \mathfrak{J}$, hence the conclusion. $\mathbf{0_I}$ | 60

(6.7.4). The results of (6.1) for flat modules are immediately translated into propositions for sheaves with are f-flat at a point:

If $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is an exact sequence of $\mathscr{O}_X$-modules and if $\mathscr{F}''$ is f-flat at a point $x \in X$, then, for each open neighbourhood U of $y = f(x)$ and each $(\mathscr{O}_Y|U)$-module $\mathscr{G}$, the sequence

$$0 \longrightarrow (f^*(\mathscr{G}) \otimes_{\mathscr{O}_X} \mathscr{F}')_x \longrightarrow (f^*(\mathscr{G}) \otimes_{\mathscr{O}_X} \mathscr{F})_x \longrightarrow (f^*(\mathscr{G}) \otimes_{\mathscr{O}_X} \mathscr{F}'')_x \longrightarrow 0$$

is exact. For $\mathscr{F}$ to be f-flat at x, it is necessary and sufficient that $\mathscr{F}'$ is. We have similar statements for the corresponding notions of a f-flat $\mathscr{O}_X$-modules over $y \in Y$, or of a f-flat $\mathscr{O}_X$-module.

(6.7.5). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of ringed spaces; let $x \in X$, $y = f(x)$, and $\mathscr{F}$ be an $\mathscr{O}_X$-module. If $\mathscr{F}$ is f-flat at the point x and if the morphism g is flat at the point y, then $\mathscr{F}$ is $(g \circ f)$-flat at x (6.2.1). In particular, if f and g are flat morphisms, then $g \circ f$ is flat.

(6.7.6). Let X, Y be two ringed spaces, $f : X \to Y$ a *flat* morphism. Then the canonical homomorphism of bifunctors (4.4.6)

$$f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F},\mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{G})) \tag{6.7.6.1}$$

is an *isomorphism* when $\mathscr{F}$ admits a *finite presentation* (5.2.5).

Indeed, since the questions is local, we can assume that there exists an exact sequence $\mathscr{O}_Y^m \to \mathscr{O}_Y^n \to \mathscr{F} \to 0$. The two sides of (6.7.6.1) are right exact functors in $\mathscr{F}$ according to the hypotheses on f; we then have reduced to proving the proposition in the case where $\mathscr{F} = \mathscr{O}_Y$, in which the result is trivial.

(6.7.8). We say that a morphism $f : X \to Y$ of ringed spaces is *faithfully flat* if f is *surjective* and if, for each $x \in X$, $\mathscr{O}_x$ is a *faithfully flat* $\mathscr{O}_{f(x)}$-module. When X and Y are locally ringed spaces (5.5.1), it is equivalent to say that the morphism f is *surjective* and *flat* (6.6.2). When f is faithfully flat, f^* is an *exact* and *faithful* functor on the category of $\mathscr{O}_Y$-modules (6.6.1, a), and for an $\mathscr{O}_Y$-module $\mathscr{G}$ to be Y-flat, it is necessary and sufficient that $f^*(\mathscr{G})$ is (6.6.3).

§7. Adic rings

7.1. Admissible rings

(7.1.1). Recall that in a topological ring A (not necessarily separated), we say that an element x is *topologically nilpotent* if 0 is *a* limit of the sequence $(x^n)_{n\geqslant 0}$. We say that a topological ring A is *linearly topologized* if there exists a fundamental system of neighbourhoods of 0 in A of (necessarily *open*) *ideals*.

Definition (7.1.2). — In a linearly topologized ring A, we say that an ideal $\mathfrak{J}$ is an *ideal of definition* if $\mathfrak{J}$ is open and if, for each neighbourhood V of 0, there exists a integer $n > 0$ such that $\mathfrak{J}^n \subset V$ *(which we express, by abuse of language, by saying that the sequence* $(\mathfrak{J}^n)$ tends to 0*).* We say that a linearly topologized ring A is *preadmissible* if there exists in A an ideal of definition; we say that A is *admissible* if it is preadmissible and if in addition it is separated and complete. $\mathbf{0_I}$ | 61

It is clear that if $\mathfrak{J}$ is an ideal of definition, $\mathfrak{L}$ an open ideal of A, then $\mathfrak{J} \cap \mathfrak{L}$ is also an ideal of definition; the ideals of definition of a preadmissible ring A thus form a *fundamental system of neighbourhoods of* 0.

Lemma (7.1.3). — *Let A be a linearly topologized ring.*

(i) *For $x \in A$ to be topologically nilpotent, it is necessary and sufficient that for each open ideal $\mathfrak{J}$ of A, the canonical image of x in $A/\mathfrak{J}$ is nilpotent. The set $\mathfrak{T}$ of topologically nilpotent elements of A is an ideal.*

(ii) *Suppose that in addition A is preadmissible, and let $\mathfrak{J}$ be an ideal of definition for A. For $x \in A$ to be topologically nilpotent, it is necessary and sufficient that its canonical image in $A/\mathfrak{J}$ is nilpotent; the ideal $\mathfrak{T}$ is the inverse image of the nilradical of $A/\mathfrak{J}$ and is thus open.*

PROOF. (i) follows immediately from the definitions. To prove (ii), it suffices to note that for each neighbourhood V of 0 in A, there exists an $n > 0$ such that $\mathfrak{J}^n \subset V$; if $x \in A$ is such that $x^m \in \mathfrak{J}$, we have $x^{mq} \in V$ for $q \geqslant n$, so x is topologically nilpotent. □

Proposition (7.1.4). — *Let A be a preadmissible ring, $\mathfrak{J}$ an ideal of definition for A.*

(i) *For an ideal $\mathfrak{J}'$ of A to be contained in an ideal of definition, it is necessary and sufficient that there exists an integer $n > 0$ such that $\mathfrak{J}'^n \subset \mathfrak{J}$.*
(ii) *For an $x \in A$ to be contained in an ideal of definition, it is necessary and sufficient that it is topologically nilpotent.*

PROOF.

(i) If $\mathfrak{J}'^n \subset \mathfrak{J}$, then for each open neighbourhood V of 0 in A, there exists an m such that $\mathfrak{J}^m \subset V$, thus $\mathfrak{J}'^{mn} \subset V$.
(ii) The condition is evidently necessary; it is sufficient, since if it satisfied, then there exists an n such that $x^n \in \mathfrak{J}$, so $\mathfrak{J}' = \mathfrak{J} + Ax$ is an ideal of definition, because it is open, and $\mathfrak{J}'^n \subset \mathfrak{J}$. □

Corollary (7.1.5). — *In a preadmissible ring A, an open prime ideal contains all the ideals of definition.*

Corollary (7.1.6). — *The notation and hypotheses being that of (7.1.4), the following properties of an ideal $\mathfrak{J}_0$ of A are equivalent:*

(a) *$\mathfrak{J}_0$ is the largest ideal of definition of A;*
(b) *$\mathfrak{J}_0$ is a maximal ideal of definition;*
(c) *$\mathfrak{J}_0$ is an ideal of definition such that the ring $A/\mathfrak{J}_0$ is reduced.*

For there to exist an ideal $\mathfrak{J}_0$ to have these properties, it is necessary and sufficient that the nilradical of $A/\mathfrak{J}$ to be nilpotent; $\mathfrak{J}_0$ is then equal to the ideal $\mathfrak{T}$ of topologically nilpotent elements of A.

PROOF. It is clear that (a) implies (b), and (b) implies (c) according to (7.1.4, ii), and (7.1.3, ii); for the same reason, (c) implies (a). The latter assertion follows from (7.1.4, i) and (7.1.3, ii). □

When $\mathfrak{T}/\mathfrak{J}$, the nilradical of $A/\mathfrak{J}$, is nilpotent, and we denote by A_{red} the (reduced) quotient ring $A/\mathfrak{T}$.

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Corollary (7.1.7). — *A preadmissible Noetherian ring admits a largest ideal of definition.*

Corollary (7.1.8). — *If a preadmissible ring A is such that, for an ideal of definition $\mathfrak{J}$, the powers $\mathfrak{J}^n$ ($n > 0$) form a fundamental system of neighbourhoods of 0, it is the same for the powers $\mathfrak{J}'^n$ for each ideal of definition $\mathfrak{J}'$ of A.*

Definition (7.1.9). — We say that a preadmissible ring A is *preadic* if there exists an ideal of definition $\mathfrak{J}$ for A such that the $\mathfrak{J}^n$ form a fundamental system of neighbourhoods of 0 in A *(or equivalently, such that the $\mathfrak{J}^n$ are* open*)*. We call a ring *adic* if it is a separated and complete preadic ring.

If $\mathfrak{J}$ is an ideal of definition for a preadic (resp. adic) ring A, we say that A is a $\mathfrak{J}$-*preadic* (resp. $\mathfrak{J}$-*adic*) ring, and that its topology is the $\mathfrak{J}$-*preadic* (resp. $\mathfrak{J}$-*adic*) topology. More generally, if M is an A-module, the topology on M having for a fundamental system of neighbourhoods of 0 the submodules $\mathfrak{J}^n M$ is called the $\mathfrak{J}$-*preadic* (resp. $\mathfrak{J}$-*adic*) topology. According to (7.1.8), these topologies are independent of the choice of the ideal of definition $\mathfrak{J}$.

Proposition (7.1.10). — *Let A be an admissible ring, $\mathfrak{J}$ an ideal of definition for A. Then $\mathfrak{J}$ is contained in the radical of A.*

This statement is equivalent to any of the following corollaries:

Corollary (7.1.11). — *For each $x \in \mathfrak{J}$, $1 + x$ is invertible in A.*

Corollary (7.1.12). — *For $f \in A$ to be invertible in A, it is necessary and sufficient that its canonical image in $A/\mathfrak{J}$ is invertible in $A/\mathfrak{J}$.*

Corollary (7.1.13). — *For each A-module M of finite type, the relation $M = \mathfrak{J}M$ (equivalent to $M \otimes_A (A/\mathfrak{J}) = 0$) implies that $M = 0$.*

Corollary (7.1.14). — *Let $u : M \to N$ be a homomorphism of A-modules, N being of finite type; for u to be surjective, it is necessary and sufficient that $u \otimes 1 : M \otimes_A (A/\mathfrak{J}) \to N \otimes_A (A/\mathfrak{J})$ is.*

Proof. The equivalence of (7.1.10) and (7.1.11) follows from Bourbaki, *Alg.*, chap. VIII, §6, no. 3, th. 1, and the equivalence of (7.1.10) and (7.1.10) and (7.1.13) follows from *loc. cit.*, th. 2; the fact that (7.1.10) implies (7.1.14) follows from *loc. cit.*, cor. 4 of the prop. 6; on the other hand, (7.1.14) implies (7.1.13) by applying the zero homomorphism. Finally, (7.1.10) implies that if f is invertible in $A/\mathfrak{J}$, then f is not contained in any maximal ideal of A, thus f is invertible in A, in other words, (7.1.10) implies (7.1.12); conversely, (7.1.12) implies (7.1.11).

It therefore remains to prove (7.1.11). Now as A is separated and complete, and the sequence $(\mathfrak{J}^n)$ tends to 0, it is immediate that the series $\sum_{n=0}^{\infty}(-1)^n x^n$ is convergent in A, and that if y is its sum, then we have $y(1+x) = 1$. □

7.2. Adic rings and projective limits

(7.2.1). Each projective limit of *discrete* rings is evidently a linearly topologized ring, separated and compact. Conversely, let A be a linearly topologized ring, and let $(\mathfrak{J}_\lambda)$ be a fundamental system of open neighbourhoods of 0 in A consisting of ideals. The canonical maps $\phi_\lambda : A \to A/\mathfrak{J}_\lambda$ form a projective system of continuous representations and therefore define a continuous representation $\phi : A \to \varprojlim A/\mathfrak{J}_\lambda$; if A is *separated*, then ϕ is a topological isomorphism from A to an everywhere-dense subring of $\varprojlim A/\mathfrak{J}_\lambda$; if in addition A is *complete*, then ϕ is a topological isomorphism from A to $\varprojlim A/\mathfrak{J}_\lambda$. $\mathbf{0_I}$ | 63

Lemma (7.2.2). — *For a linearly topologized ring to be admissible, it is necessary and sufficient that it is isomorphic to a projective limit $A = \varprojlim A_\lambda$, where $(A_\lambda, u_{\lambda\mu})$ is a projective limit of discrete rings having for the set of indices a filtered ordered (by $\leqslant$) L which admits a smallest element denoted 0 and satisfies the following conditions: 1st. the $u_\lambda : A \to A_\lambda$ are sujective; 2nd. the kernel $\mathfrak{J}_\lambda$ of $u_{0\lambda} : A_\lambda \to A_0$ is nilpotent. When this is so, the kernel $\mathfrak{J}$ of $u_0 : A \to A_0$ is equal to $\varprojlim \mathfrak{J}_\lambda$.*

Proof. The necessity of the condition follows from (7.2.1), by choosing $(\mathfrak{J}_\lambda)$ to be a fundamental system of neighbourhoods of 0 consisting of ideals of definitions contained in an ideal of definition $\mathfrak{J}_0$ and by applying (7.1.4, i). The converse follows from the definition of the projective limit and from (7.2.1), and the latter assertion is immediate. □

(7.2.3). Let A be an *admissible* topological ring, $\mathfrak{J}$ an ideal of A contained in an ideal of definition (in other words (7.1.4) such that $(\mathfrak{J}^n)$ tends to 0); we can consider on A the ring topology having for a fundamental system of neighbourhoods of 0 the powers $\mathfrak{J}^n$ ($n > 0$); we call again this the $\mathfrak{J}$-*preadic* topology. The hypothesis that A is admissible implies that $\bigcup_n \mathfrak{J}^n = 0$, therefore the $\mathfrak{J}$-preadic topology on A is *separated*; let $\widehat{A} = \varprojlim A/\mathfrak{J}^n$ be the completion of A for this topology (where the $A/\mathfrak{J}^n$ are equipped with the discrete topology), and denote by u the (not necessarily continuous) ring homomorphism $A \to \widehat{A}$, the projective limit of the sequence of homomorphisms $u_n : A \to A/\mathfrak{J}^n$. On the other hand, the $\mathfrak{J}$-preadic topology on A is finer than the given topology $\mathscr{T}$ on A; as A is separated and complete for $\mathscr{T}$, we can extend by continuity the identity map of A (equipped with the $\mathfrak{J}$-preadic topology) to A equipped with $\mathscr{T}$; this gives a continuous representation $v : \widehat{A} \to A$.

Proposition (7.2.4). — *If A is an admissible ring and $\mathfrak{J}$ is contained in an ideal of definition of A, then A is separated and complete for the $\mathfrak{J}$-preadic topology.*

Proof. With the notation of (7.2.3), it is immediate that $v \circ u$ is the identity map of A. On the other hand, $u_n \circ v : \widehat{A} \to A/\mathfrak{J}^n$ is the extension by continuity (for the $\mathfrak{J}$-preadic topology on A and the discrete topology on $A/\mathfrak{J}^n$) of the canonical map u_n; in other words, it is the canonical map from $\widehat{A} = \varprojlim_k A/\mathfrak{J}^k$ to $A/\mathfrak{J}^n$; $u \circ v$ is therefore the projective limit of this sequence of maps, which is by definition the identity map of $\widehat{A}$; this proves the proposition. □

Corollary (7.2.5). — *Under the hypotheses of (7.2.3), the following conditions are equivalent:*

(a) *the homomorphism u is continuous;* $\mathbf{0_I}$ | 64
(b) *the homomorphism v is bicontinous;*
(c) *A is a $\mathfrak{J}$-adic ring.*

Corollary (7.2.6). — *Let A be an admissible ring A, $\mathfrak{J}$ an ideal of definition for A. For A to be Noetherian, it is necessary and sufficient for $A/\mathfrak{J}$ to be Noetherian and for $\mathfrak{J}/\mathfrak{J}^2$ to be an $A/\mathfrak{J}$-module of finite type.*

These conditions are evidently necessary. Conversely, suppose the conditions are satisfied; as according to (7.2.4) A is complete for the $\mathfrak{J}$-preadic topology, for it to be Noetherian, it is necessary and sufficient that the associated graded ring $\operatorname{grad}(A)$ (for the filtration on the $\mathfrak{J}^n$) is ([**CC**, p .18–07, th. 4]). Now, let $a_1, \ldots, a_n$ be the elements of $\mathfrak{J}$ whose classes mod. $\mathfrak{J}^2$ are the generators of $\mathfrak{J}/\mathfrak{J}^2$ as a $A/\mathfrak{J}$-module. It is immediate by induction that the classes mod. $\mathfrak{J}^{m+1}$ of the monomials of total degree m in the a_i ($1 \leqslant i \leqslant n$) form a system of generators for the $A/\mathfrak{J}$-module $\mathfrak{J}^m/\mathfrak{J}^{m+1}$. We conclude that $\operatorname{grad}(A)$ is a ring isomorphic to a quotient of $(A/\mathfrak{J})[T_1, \ldots, T_n]$ (T_i indeterminates), which finishes the proof.

Proposition (7.2.7). — *Let (A_i, u_{ij}) be a projective system ($i \in \mathbf{N}$) of discrete rings, and for each integer i, let $\mathfrak{J}_i$ be the kernel in A_i of the homomorphism $u_{0i} : A_i \to A_0$. We suppose that:*

(a) *For $i \leqslant j$, u_{ij} is surjective and its kernel is $\mathfrak{J}_j^{i+1}$* (therefore A_i is isomorphic to $A_j/\mathfrak{J}_j^{i+1}$).

(b) *$\mathfrak{J}_1/\mathfrak{J}_1^2$ ($= \mathfrak{J}_1$) is a module of finite type over $A_0 = A_1/\mathfrak{J}_1$.*

Let $A = \varprojlim_i A_i$, and for each integer $n \geqslant 0$, let u_n be the canonical homomorphism $A \to A_n$, and let $\mathfrak{J}^{(n+1)} \subset A$ be its kernel. Then we have these conditions:

(i) *A is an adic ring, having $\mathfrak{J} = \mathfrak{J}^{(1)}$ for an ideal of definition.*

(ii) *We have $\mathfrak{J}^{(n)} = \mathfrak{J}^n$ for each $n \geqslant 1$.*

(iii) *$\mathfrak{J}/\mathfrak{J}^2$ is isomorphic to $\mathfrak{J}_1 = \mathfrak{J}_1/\mathfrak{J}_1^2$, and as a result is a module of finite type over $A_0 = A/\mathfrak{J}$.*

Proof. The hypothesis of surjectivity on the u_{ij} implies that u_n is surjective; in addition, the hypothesis (a) implies that $\mathfrak{J}_j^{j+1} = 0$, therefore A is an admissible ring (7.2.2); by definition, the $\mathfrak{J}^{(n)}$ form a fundamental system of neighbourhoods of 0 in A, so (ii) implies (i). In addition, we have $\mathfrak{J} = \varprojlim_i \mathfrak{J}_i$ and the maps $\mathfrak{J} \to \mathfrak{J}_i$ are surjective, so (ii) implies (iii), and it remains to prove (ii). By definition, $\mathfrak{J}^{(n)}$ consists of the elements $(x_k)_{k \geqslant 0}$ of A such that $x_k = 0$ for $k < n$, therefore $\mathfrak{J}^{(n)}\mathfrak{J}^{(m)} \subset \mathfrak{J}^{(n+m)}$, in other words the $\mathfrak{J}^{(n)}$ form a *filtration* of A. On the other hand, $\mathfrak{J}^{(n)}/\mathfrak{J}^{(n+1)}$ is isomorphic to the projection from $\mathfrak{J}^{(n)}$ to A_n; as $\mathfrak{J}^{(n)} = \varprojlim_{i > n} \mathfrak{J}_i^n$, this projection is none other than $\mathfrak{J}_n^n$, which is a module over $A_0 = A_n/\mathfrak{J}_n$. Now let $a_j = (a_{jk})_{k \geqslant 0}$ be r elements of $\mathfrak{J} = \mathfrak{J}^{(1)}$ such that $a_{11}, \ldots, a_{r1}$ form a system of generators for $\mathfrak{J}_1$ over A_0; we will see that the set S_n of monomials of total degree n and the a_j generate the ideal $\mathfrak{J}^{(n)}$ of A. As $\mathfrak{J}_i^{i+1} = 0$, it is clear first of all that $S_n \subset \mathfrak{J}^{(n)}$; since A is complete for the filtration $(\mathfrak{J}^{(m)})$, it suffices to prove that the set $\overline{S}_n$ of classes mod. $\mathfrak{J}^{(n+1)}$ of elements of S_n generate the graded module $\operatorname{grad}(\mathfrak{J}^{(n)})$ over the graded ring $\operatorname{grad}(A)$ for the above filtration ([**CC**, p. 18–06, lemme]); according to the definition of the multiplication on $\operatorname{grad}(A)$, it $\mathbf{0_I}$ | 65
suffices to prove that for each m, $\overline{S}_m$ is a system of generators for the A_0-module $\mathfrak{J}^{(m)}/\mathfrak{J}^{(m+1)}$, or that $\mathfrak{J}_m^m$ is generated by the monomials of degree m in the a_{jm} ($1 \leqslant j \leqslant r$). For this, it remains to show that $\mathfrak{J}_m$ is generated (as an A_m-module) by the monomials of degree $\leqslant m$ with respect to to a_{jm}; the proposition being evident by definition for $m = 1$, we argue by induction on m, and let $\mathfrak{J}'_m$ be the A_m-submodule of $\mathfrak{J}_m$ generated by these monomials. The relation $\mathfrak{J}_{m-1} = \mathfrak{J}_m/\mathfrak{J}_m^m$ and the induction hypothesis prove that $\mathfrak{J}_m = \mathfrak{J}'_m + \mathfrak{J}_m^m$, hence, since $\mathfrak{J}_m^{m+1} = 0$, we have $\mathfrak{J}_m^m = \mathfrak{J}'^{\,m}_m$, and finally $\mathfrak{J}_m = \mathfrak{J}'_m$. □

Corollary (7.2.8). — *Under the conditions of Proposition (7.2.7), for A to be Noetherian, it is necessary and sufficient that A_0 is.*

Proof. This follows immediately from Corollary (7.2.6). □

Proposition (7.2.9). — *Suppose the hypotheses of Proposition (7.2.7): for each integer i, let M_i be an A_i-module, and for $i \leqslant j$, let $v_{ij} : M_j \to M_i$ be a u_{ij}-homomorphism, such that (M_i, v_{ij}) is a projective system. In addition, suppose that M_0 is an A_0-module of finite type and that the v_{ij} are surjective with kernel $\mathfrak{J}_j^{i+1}M_j$. Then $M = \varprojlim M_i$ is an A-module of finite type, and the kernel of the surjective u_n-homomorphism $v_n : M \to M_n$ is $\mathfrak{J}^{n+1}M$* (such that M_n identifies with $M/\mathfrak{J}^{n+1}M = M \otimes_A (A/\mathfrak{J}^{n+1})$).

PROOF. Let $z_h = (z_{hk})_{k\geqslant 0}$ be a system of s elements of M such that the z_{h0} ($1 \leqslant h \leqslant s$) forms a system of generators for M_0; we will show that the z_h generate the A-module M. The A-module M is separated and complete for the filtration by the $M^{(n)}$, where $M^{(n)}$ is the set of $y = (y_k)_{k\geqslant 0}$ in M such that $y_k = 0$ for $k < n$; it is clear that we have $\mathfrak{J}^{(n)}M \subset M^{(n)}$ and that $M^{(n)}/M^{(n+1)} = \mathfrak{J}_n^n M_n$. We therefore have reduced to showing that the classes of the z_h modulo $M^{(0)}$ generate the graded module $\operatorname{grad}(M)$ (by the above filtration) over the graded ring $\operatorname{grad}(A)$ [**CC**, p. 18–06, lemme]; for this, we observe that it suffices to prove that the z_{hn} ($1 \leqslant h \leqslant s$) generate the A_n-module M_n. We argue by induction on n, the proposition being evident by definition for $n = 0$; the relation $M_{n-1} = M_n/\mathfrak{J}_n^n M_n$ and the induction hypothesis show that if M_n' is the submodule of M_n generated by the z_{hn}, we have that $M_n = M_n' + \mathfrak{J}_n^n M_n$, and as $\mathfrak{J}_n$ is nilpotent, this implies that $M_n = M_n'$. Similarly, passing to the associated graded modules shows that the canonical map from $\mathfrak{J}^{(n)}$ to $M^{(n)}$ is surjective (thus bijection), in other words that $\mathfrak{J}^{(n)}M = \mathfrak{J}^n M$ is the kernel of $M \to M_{n-1}$. ☐ **Err$_{\text{II}}$**

Corollary (7.2.10). — *Let (N_i, w_{ij}) be a second projective system of A_i-modules satisfying the conditions of Proposition (7.2.9), and let $N = \varprojlim N_i$. There is a bijective correspondence between the projective systems (h_i) of A_i-homomorphisms $h_i : M_i \to N_i$ and the homomorphisms of A-modules $h : M \to N$ (which is necessarily continuous for the $\mathfrak{J}$-adic topologies).*

PROOF. It is clear that if $h : M \to N$ is an A-homomorphism, then we have $h(\mathfrak{J}^n M) \subset \mathfrak{J}^n N$, hence the continuity of h; by passing to quotients, there corresponds to h a projective system of A_i-homomorphisms $h_i : M_i \to N_i$, whose projective limit is h, hence the corollary. ☐

Remark (7.2.11). — Let A be an adic ring with an ideal of definition $\mathfrak{J}$ such that $\mathfrak{J}/\mathfrak{J}^2$ is an $A/\mathfrak{J}$-module of finite type; it is clear that the $A_i = A/\mathfrak{J}^{i+1}$ satisfy the conditions of Proposition (7.2.7); as **0$_{\text{I}}$ | 66** A is the projective limit of the A_i, we see that Proposition (7.2.7) gives the description of *all* the adic rings of the type considered (and in particular of all the *adic Noetherian* rings).

Example (7.2.12). — Let B be a ring, $\mathfrak{J}$ an ideal of B such that $\mathfrak{J}/\mathfrak{J}^2$ is a module of finite type over $B/\mathfrak{J}$ (or over B, equivalently); set $A = \varprojlim_n B/\mathfrak{J}^{n+1}$; A is the separated completion of B equipped with the $\mathfrak{J}$-preadic topology. If $A_n = B/\mathfrak{J}^{n+1}$, then it is immediate that the A_n satisfy the conditions of Proposition (7.2.7); therefore A is an adic ring and if $\overline{\mathfrak{J}}$ is the closure in A of the canonical image of $\mathfrak{J}$, then $\overline{\mathfrak{J}}$ is an ideal of definition for A, $\overline{\mathfrak{J}^n}$ is the closure of the canonical image of $\mathfrak{J}^n$, $A/\overline{\mathfrak{J}^n}$ identifies with $B/\mathfrak{J}^n$ and $\overline{\mathfrak{J}}/\overline{\mathfrak{J}}^2$ is isomorphic to $\mathfrak{J}/\mathfrak{J}^2$ as an $A/\overline{\mathfrak{J}}$-module. Similarly, if N is such that $N/\mathfrak{J}N$ is a B-module of finite type, and if we set $M_i = N/\mathfrak{J}^{i+1}N$, then $M = \varprojlim M_i$ is an A-module of finite type, isomorphic to the separated completion of N for the $\mathfrak{J}$-preadic topology, and $\overline{\mathfrak{J}^n}M$ identifies with the closure of the canonical image of $\mathfrak{J}^n N$, and $M/\overline{\mathfrak{J}^n}M$ identifies with $N/\mathfrak{J}^n N$.

7.3. Preadic Noetherian rings

(7.3.1). Let A be a ring, $\mathfrak{J}$ an ideal of A, and M an A-module; we denote by $\widehat{A} = \varprojlim_n A/\mathfrak{J}^n$ (resp. $\widehat{M} = \varprojlim_n M/\mathfrak{J}^n M$) the separated completion of A (resp. M) for the $\mathfrak{J}$-preadic topology. Let $M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$ be an exact sequence of A-modules; as $M/\mathfrak{J}^n M = M \otimes_A (A/\mathfrak{J}^n)$, the sequence

$$M'/\mathfrak{J}^n M' \xrightarrow{u_n} M/\mathfrak{J}^n M \xrightarrow{v_n} M''/\mathfrak{J}^n M'' \longrightarrow 0$$

is exact for each n. In addition, as $v(\mathfrak{J}^n M) = \mathfrak{J}^n v(M) = \mathfrak{J}^n M''$, $\widehat{v} = \varprojlim v_n$ is surjective (Bourbaki, *Top. gén.*, Chap. IX, 2nd ed., p. 60, Cor. 2). On the other hand, if $z = (z_k)$ is an element of the kernel of $\widehat{v}$, then for each integer k, there exists a $z_k' \in M'/\mathfrak{J}^k M'$ such that $u_k(z_k') = z_k$; we conclude that there exists a $z' = (z_n') \in \widehat{M'}$ such that the first k components of $\widehat{u}(z')$ coincide with the z; in other words, the image of $\widehat{M'}$ under $\widehat{u}$ is *dense* in the kernel of $\widehat{v}$.

If we suppose that A is *Noetherian*, then so is $\widehat{A}$, according to (7.2.12), $\mathfrak{J}/\mathfrak{J}^2$ is then an A-module of finite type. In addition, we have the following theorem.

Theorem (7.3.2). — (Krull's Theorem). *Let A be a* Noetherian *ring, $\mathfrak{J}$ an ideal of A, M an A-module* of finite type, *and M' a submodule of M; then the induced topology on M' by the $\mathfrak{J}$-preadic topology of M is identical to the $\mathfrak{J}$-preadic topology of M'.*

This follows immediately from

Lemma (7.3.2.1). — (Artin–Rees Lemma). *Under the hypotheses of (7.3.2), there exists an integer p such that, for $n \geqslant p$, we have*

$$M' \cap \mathfrak{J}^n M = \mathfrak{J}^{n-p}(M' \cap \mathfrak{J}^p M).$$

For the proof, see [**CC**, p. 2–04].

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Corollary (7.3.3). — *Under the hypotheses of (7.3.2), the canonical map $M \otimes_A \widehat{A} \to \widehat{M}$ is bijective, and the functor $M \otimes_A \widehat{A}$ is exact in M on the category of A-modules of finite type; as a result, the separated $\mathfrak{J}$-adic completion $\widehat{A}$ is a flat A-module (6.1.1).*

PROOF. We first note that $\widehat{M}$ is an *exact* functor in M on the category of A-modules of finite type. Indeed, let $0 \to M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$ be an exact sequence; we have seen that $\widehat{v} : \widehat{M} \to \widehat{M''}$ is surjective (7.3.1); on the other hand, if i is the canonical homomorphism $M \to \widehat{M}$, it follows from Krull's Theorem (7.3.2) that the closure of $i(u(M'))$ in $\widehat{M}$ identifies with the separated completion of M' for the $\mathfrak{J}$-preadic topology; thus $\widehat{u}$ is injective, and according to (7.3.1), the image of $\widehat{u}$ is equal to the kernel of $\widehat{v}$.

This being so, the canonical map $M \otimes_A \widehat{A} \to \widehat{M}$ is obtained by passing to the projective limit of the maps $M \otimes_A \widehat{A} \to M \otimes_A (A/\mathfrak{J}^n) = M/\mathfrak{J}^n M$. It is clear that this map is bijective when M is of the form A^p. If M is an A-module of finite type, then we have an exact sequence $A^p \to A^q \to M \to 0$, hence, by virtue of the *right* exactness of the functors $M \otimes_A \widehat{A}$ and $\widehat{M}$ (in M) on the category of A-modules of finite type, we have the commutative diagram

$$\begin{tikzcd} A^p \otimes_A \widehat{A} \arrow[r] \arrow[d] & A^q \otimes_A \widehat{A} \arrow[r] \arrow[d] & M \otimes_A \widehat{A} \arrow[r] \arrow[d] & 0 \\ \widehat{A^p} \arrow[r] & \widehat{A^q} \arrow[r] & \widehat{M} \arrow[r] & 0, \end{tikzcd}$$

where the two rows are exact and the first two vertical arrows are isomorphisms; this immediately finishes the proof. □

Corollary (7.3.4). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A, M and N two A-modules of finite type; we have the canonical functorial isomorphisms*

$$(M \otimes_A N)^\wedge \simeq \widehat{M} \otimes_{\widehat{A}} \widehat{N}, \ (\operatorname{Hom}_A(M,N))^\wedge \simeq \operatorname{Hom}_{\widehat{A}}(\widehat{M}, \widehat{N}).$$

PROOF. This follows from Corollary (7.3.3), (6.2.1), and (6.2.2). □

Corollary (7.3.5). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A. The following conditions are equivalent:*

(a) *$\mathfrak{J}$ is contained in the radical of A.*
(b) *$\widehat{A}$ is a faithfully flat A-module (6.4.1).*
(c) *Each A-module of finite type is separated for the $\mathfrak{J}$-preadic topology.*
(d) *Each submodule of an A-module of finite type is closed for the $\mathfrak{J}$-preadic topology.*

PROOF. As $\widehat{A}$ is a flat A-module, the conditions (b) and (c) are equivalent, since (b) is equivalent to saying that if M is an A-module of finite type, then the canonical map $M \to \widehat{M} = M \otimes_A \widehat{A}$ is injective (6.6.1, c). It is immediate that (c) implies (d), since if N is a submodule of an A-module M of finite type, then M/N is separated for the $\mathfrak{J}$-preadic topology, so N is closed in M. We show that (d) implies (a): if $\mathfrak{m}$ is a maximal ideal of A, then $\mathfrak{m}$ is closed in A for the $\mathfrak{J}$-preadic topology, so $\mathfrak{m} = \bigcap_{p \geqslant 0}(\mathfrak{m} + \mathfrak{J}^p)$, and as $\mathfrak{m} + \mathfrak{J}^p$ is necessarily equal to A or to $\mathfrak{m}$, we have that $\mathfrak{m} + \mathfrak{J}^p = \mathfrak{m}$ for large enough p, hence $\mathfrak{J}^p \subset \mathfrak{m}$, and $\mathfrak{J} \subset \mathfrak{m}$ when $\mathfrak{m}$ is prime. Finally, (a) implies (b): indeed, let $\mathbf{0_I}$ | 68
P be the closure of $\{0\}$ in an A-module M of finite type, for the $\mathfrak{J}$-preadic topology; according to Krull's Theorem (7.3.2), the topology induced on P by the $\mathfrak{J}$-preadic topology of M is the $\mathfrak{J}$-preadic topology of P, so $\mathfrak{J}P = P$; as P is of finite type, it follows from Nakayama's Lemma that $P = 0$ ($\mathfrak{J}$ being contained in the radical of A). □

We note that the conditions of Corollary (7.3.5) are satisfied when A is a *local Noetherian ring* and $\mathfrak{J} \neq A$ is any ideal of A.

Corollary (7.3.6). — *If A is a $\mathfrak{J}$-preadic Noetherian ring, then each A-module of finite type is separated and complete for the $\mathfrak{J}$-preadic topology.*

PROOF. As we then have $\widehat{A} = A$, this follows immediately from Corollary (7.3.3). □

We conclude that Proposition (7.2.9) gives the description of *all* the modules of finite type over an adic Noetherian ring.

Corollary (7.3.7). — *Under the hypotheses of (7.3.2), the kernel of the canonical map $M \to \widehat{M} = M \otimes_A \widehat{A}$ is the set of the $x \in M$ killed by an element of $1 + \mathfrak{J}$.*

PROOF. For each $x \in M$ in this kernel, it is necessary and sufficient that the separated completion of the submodule Ax is 0 (by Krull's Theorem (7.3.2)), in other words, that $x \in \mathfrak{J}x$. □

7.4. Quasi-finite modules over local rings

Definition (7.4.1). — Given a local ring A, with maximal ideal $\mathfrak{m}$, we say that an A-module M is quasi-finite (over A) if $M/\mathfrak{m}M$ is of finite rank over the residue field $k = A/\mathfrak{m}$.

When A is *Noetherian*, the separated completion $\widehat{M}$ of M for the $\mathfrak{m}$-preadic topology is then an *$\widehat{A}$-module of finite type*; indeed, as $\mathfrak{m}/\mathfrak{m}^2$ is then an A-module of finite type, this follows from Example (7.2.12) and from the hypothesis on $M/\mathfrak{m}M$.

In particular, if we suppose that in addition A is *complete* and M is *separated* for the $\mathfrak{m}$-preadic topology (in other words, $\bigcap_n \mathfrak{m}^n M = 0$), then M is also an *A-module of finite type*: indeed, $\widehat{M}$ is then an A-module of finite type, and as M identifies with a submodule of $\widehat{M}$, M is also of finite type (and is indeed identical to its completion according to Corollary (7.3.6)).

Proposition (7.4.2). — *Let A, B be two local rings, $\mathfrak{m}$, $\mathfrak{n}$ their maximal ideals, and suppose that B is Noetherian. Let $\phi : A \to B$ be a local homomorphism, M a B-module of finite type. If M is a quasi-finite A-module, then the $\mathfrak{m}$-preadic and $\mathfrak{n}$-preadic topologies on M are identical, thus separated.*

PROOF. We note that by hypothesis $M/\mathfrak{m}M$ is of *finite length* as an A-module, thus also *a fortiori* as a B-module. We conclude that $\mathfrak{n}$ is the *unique prime ideal* of B containing the annihilator of $M/\mathfrak{m}M$: indeed, we immediately reduce (according to (1.7.4) and (1.7.2)) to the case where $M/\mathfrak{m}M$ is *simple*, thus necessarily isomorphic to $B/\mathfrak{n}$, and our assertion is evident in this case. On the other hand, as M is a B-module of finite type, the prime ideals which contain the annihilator of $M/\mathfrak{m}M$ are those which contain $\mathfrak{m}B + \mathfrak{b}$, where we denote by $\mathfrak{b}$ the annihilator of the B-module M (1.7.5). As B is Noetherian, we conclude ([Sam53b, p. 127, Cor. 4]) that $\mathfrak{m}B + \mathfrak{b}$ is an ideal of definition for B, in $\mathbf{0_I}$ | 69
other words there exists a $k > 0$ such that $\mathfrak{n}^k \subset \mathfrak{m}B + \mathfrak{b} \subset \mathfrak{n}$; as a result, for each $h > 0$, we have

$$\mathfrak{n}^{hk} \subset (\mathfrak{m}B + \mathfrak{b})^h M = \mathfrak{m}^h M \subset \mathfrak{n}^h M,$$

which proves that the $\mathfrak{m}$-preadic and $\mathfrak{n}$-preadic topologies on M are the same; the second is separated according to Corollary (7.3.5). □

Corollary (7.4.3). — *Under the hypotheses of Proposition (7.2.4), if in addition A is Noetherian and complete for the $\mathfrak{m}$-preadic topology, then M is an A-module of finite type.*

PROOF. Indeed, M is then separated for the $\mathfrak{m}$-preadic topology, and our assertion follows from the remark after Definition (7.4.1). □

(7.4.4). The most important case of Proposition (7.4.2) is when B is a quasi-finite A-module, i.e., $B/\mathfrak{m}B$ is an *algebra of finite rank over $k = A/\mathfrak{m}$*; furthermore, this condition can be broken down into the combination of the following:

(i) *$\mathfrak{m}B$ is an ideal of definition for B;*
(ii) *$B/\mathfrak{n}$ is an extension of finite rank of the field $A/\mathfrak{m}$.*

When this is so, every B-module of finite type is evidently a quasi-finite A-module.

Corollary (7.4.5). — *Under the hypotheses of Proposition (7.4.2), if $\mathfrak{b}$ is the annihilator of the B-module M, then $B/\mathfrak{b}$ is a quasi-finite A-module.*

PROOF. Suppose $M \neq 0$ (otherwise the corollary is evident). We can consider M as a module over the local Noetherian ring $B/\mathfrak{b}$; its annihilator then being 0, the proof of Proposition (7.4.2) shows that $\mathfrak{m}(B/\mathfrak{b})$ is an ideal of definition for $B/\mathfrak{b}$. On the other hand, $M/\mathfrak{n}M$ is a vector space of finite rank over $A/\mathfrak{m}$, being a quotient of $M/\mathfrak{m}M$, which is by hypothesis of finite rank over $A/\mathfrak{m}$; as $M \neq 0$, we have $M \neq \mathfrak{n}M$ by virtue of Nakayama's Lemma; as $M/\mathfrak{n}M$ is a vector space $\neq 0$ over $B/\mathfrak{n}$, the fact that it is of finite rank over $A/\mathfrak{m}$ implies that $B/\mathfrak{n}$ is also of finite rank over $A/\mathfrak{m}$; the conclusion follows from (7.4.4) applied to the ring $B/\mathfrak{b}$. □

7.5. Rings of restricted formal series

(7.5.1). Let A be a topological ring, linearly topologized, separated and complete; let $(\mathfrak{J}_\lambda)$ be a fundamental system of neighbourhoods of 0 in A consisting of (open) ideals, such that A canonically identifies with $\varprojlim A/\mathfrak{J}_\lambda$ (7.2.1). For each λ, let $B_\lambda = (A/\mathfrak{J}_\lambda)[T_1, \ldots, T_r]$, where the T_i are indeterminates; it is clear that the B_λ form a projective system of discrete rings. We set $\varprojlim B_\lambda = A\{T_1, \ldots, T_r\}$, and we will see that this topological ring is independent of the fundamental system of ideals $(\mathfrak{J}_\lambda)$ considered. More precisely, let A' be the subring of the ring of formal series $A[\![T_1, \ldots, T_r]\!]$ consisting of formal series $\sum_\alpha c_\alpha T^\alpha$ (with $\alpha = (\alpha_1, \ldots, \alpha_r) \in \mathbf{N}^r$) such that $\lim c_\alpha = 0$ (according to the filter by compliments of finite subsets of $\mathbf{N}^r$); we say that these series are the *restricted* formal series in the T_i, with coefficients in A. For each neighbourhood V of 0 in A, let V' be the set of $x = \sum_\alpha c_\alpha T^\alpha \in A'$ $\mathbf{0_I}$ | 70
such that $c_\alpha \in V$ for all α. We verify immediately that the V' form a fundamental system of neighbourhoods of 0 defining on A' a *separated* ring topology; we will canonically define a *topological isomorphism* from the ring $A\{T_1, \ldots, T_r\}$ to A'. For each $\alpha \in \mathbf{N}^r$ and each λ, let $\phi_{\lambda,\alpha}$ be the map from $(A/\mathfrak{J}_\lambda)[T_1, \ldots, T_r]$ to $A/\mathfrak{J}\lambda$ which sends each polynomial in the first ring to coefficient of T^α in that polynomial. It is clear that the $\phi_{\lambda,\alpha}$ form a projective system of homomorphisms of $A/\mathfrak{J}_\lambda$-modules, so their projective limit is a continuous homomorphism $\phi_\alpha : A\{T_1, \ldots, T_r\} \to A$; we will see that, for each $y \in A\{T_1, \ldots, T_r\}$, the formal series $\sum_\alpha \phi_\alpha(y)T^\alpha$ is *restricted*. Indeed, if y_λ is the component of y in B_λ, and if we denote by H_λ the finite set of the $\alpha \in \mathbf{N}^r$ for which the coefficients of the polynomial y_λ are nonzero, then we have $\phi_{\lambda,\alpha}(y_\mu) \in \mathfrak{J}_\lambda$ for $\mathfrak{J}_\mu \subset \mathfrak{J}_\lambda$ and $\alpha \notin H_\lambda$, and by passing to the limit, $\phi_\alpha(y) \in \mathfrak{J}_\lambda$ for $\alpha \notin H_\lambda$. We thus define a ring homomorphism $\phi : A\{T_1, \ldots, T_r\} \to A'$ by setting $\phi(y) = \sum_\alpha \phi_\alpha(y)T^\alpha$, and it is immediate that ϕ is continuous. Conversely, if θ_λ is the canonical homomorphism $A \to A/\mathfrak{J}_\lambda$, then for each element $z = \sum_\alpha c_\alpha T^\alpha \in A'$ and each λ, there are only a finite number of indices α such that $\theta_\lambda(c_\alpha) \neq 0$, and as a result $\psi_\lambda(z) = \sum_\alpha \theta_\lambda(c_\alpha)T^\alpha$ is in B_λ; the ψ_λ are continuous and form a projective system of homomorphisms whose projective limit is a continuous homomorphism $\psi : A' \to A\{T_1, \ldots, T_r\}$; is remains to verify that $\phi \circ \psi$ and $\psi \circ \phi$ are the identity automorphisms, which is immediate.

(7.5.2). We identify $A\{T_1, \ldots, T_r\}$ with the ring A' of restricted formal series by means of the isomorphisms defined in (7.5.1). The canonical isomorphisms

$$((A/\mathfrak{J}_\lambda)[T_1, \ldots, T_r])[T_{r+1}, \ldots, T_s] \simeq (A/\mathfrak{J}_\lambda)[T_1, \ldots, T_s]$$

define, by passing to the projective limit, a canonical isomorphism

$$(A\{T_1, \ldots, T_r\})\{T_{r+1}, \ldots, T_s\} \simeq A\{T_1, \ldots, T_s\}.$$

(7.5.3). For every continuous homomorphism $u : A \to B$ from A to a linearly topologized ring B, separated and complete, and each system $(b_1, \ldots, b_r)$ of r elements of B, there exists *a unique continuous homomorphism* $\overline{u} : A\{T_1, \ldots, T_r\} \to B$, such that $\overline{u}(a) = u(a)$ for all $a \in A$ and $\overline{u}(T_j) = b_j$ for $1 \leqslant j \leqslant r$. It suffices to set

$$\overline{u}\Big(\sum_\alpha c_\alpha T^\alpha\Big) = \sum_\alpha u(c_\alpha) b_1^{\alpha_1} \cdots b_r^{\alpha_r};$$

the verification of the fact that the family $(u(c_\alpha)b_1^{\alpha_1} \cdots b_r^{\alpha_r})$ is summable in B and that $\overline{u}$ is continuous are immediate and left to the reader. We note that this property (for arbitrary B and b_j) *characterize* the topological ring $A\{T_1, \ldots, T_r\}$ up to unique isomorphism.

Proposition (7.5.4). —

(i) *If A is an admissible ring, then so is $A' = A\{T_1, \ldots, T_r\}$.*

(ii) *Let A be an adic ring, $\mathfrak{J}$ an ideal of definition for A such that $\mathfrak{J}/\mathfrak{J}^2$ is of finite type over $A/\mathfrak{J}$. If we set $\mathfrak{J}' = \mathfrak{J}A'$, then A' is also a $\mathfrak{J}'$-adic ring, and $\mathfrak{J}'/\mathfrak{J}'^2$ is of finite type over $A'/\mathfrak{J}'$. If in addition A is Noetherian, then so is A'.* $\mathbf{0_I}$ | 71

PROOF.

(i) If $\mathfrak{J}$ is an ideal of A, $\mathfrak{J}'$ the ideal of A' consisting of the $\sum_\alpha c_\alpha T^\alpha$ such that $c_\alpha \in \mathfrak{J}$ for all α, then $(\mathfrak{J}')^n \subset (\mathfrak{J}^n)'$; if $\mathfrak{J}$ is an ideal of definition for A, then $\mathfrak{J}'$ is also an ideal of definition for A'.

(ii) Set $A_i = A/\mathfrak{J}^{i+1}$, and for $i \leqslant j$, let u_{ij} be the canonical homomorphism $A/\mathfrak{J}^{j+1} \to A/\mathfrak{J}^{i+1}$; set $A'_i = A_i[T_1, \ldots, T_r]$, and let u'_{ij} be the homomorphism $A'_j \to A'_i$ ($i \leqslant j$) obtained by applying u_{ij} to the coefficients of the polynomials in A'_j. We will show that the projective system (A'_i, u'_{ij}) satisfies the conditions of Proposition (7.2.7); as $\mathfrak{J}'$ is the kernel of $A' \to A'_0$, this proves the first assertion of (ii). It is clear that the u'_{ij} are surjective; the kernel $\mathfrak{J}'_i$ of u_{0i} is the set of polynomials in $A_i[T_1, \ldots, T_r]$ whose coefficients are in $\mathfrak{J}/\mathfrak{J}^{i+1}$; in particular, $\mathfrak{J}'_1$ is the set of polynomials in $A_1[T_1, \ldots, T_r]$ whose coefficients are in $\mathfrak{J}/\mathfrak{J}^2$. As $\mathfrak{J}/\mathfrak{J}^2$ is of finite type over $A_1 = A/\mathfrak{J}^2$, we see that $\mathfrak{J}'_1/{\mathfrak{J}'_1}^2$ is a module of finite type over A'_1 (or equivalently, over $A'_0 = A'_1/\mathfrak{J}'_1$). We will now show that the kernel of u_{ij} is ${\mathfrak{J}'_j}^{i+1}$. It is evident that ${\mathfrak{J}'_j}^{i+1}$ is contained in this kernel. On the other hand, let $a_1, \ldots, a_m$ be the elements of $\mathfrak{J}$ whose classes mod $\mathfrak{J}^2$ generate $\mathfrak{J}/\mathfrak{J}^2$; we verify immediately that the classes mod $\mathfrak{J}^{j+1}$ of monomials of degree $\leqslant j$ in the a_k ($1 \leqslant k \leqslant m$) generate $\mathfrak{J}/\mathfrak{J}^{j+1}$, and the classes of monomials of degree $> i$ and $\leqslant j$ generate $\mathfrak{J}^{i+1}/\mathfrak{J}^{j+1}$; a monomial in the T_k having such an element for a coefficient is thus a product of $i+1$ elements of $\mathfrak{J}'_i$, which establishes our assertion. Finally, if A is Noetherian, then so is $A'/\mathfrak{J}' = (A/\mathfrak{J})[T_1, \ldots, T_r]$, hence A' is Notherian (7.2.8).

□

Proposition (7.5.5). — *Let A be a Notherian $\mathfrak{J}$-adic ring, B an admissible topological ring, $\phi : A \to B$ a continuous homomorphism, making B and A-algebra. The following conditions are equivalent:*

(a) *B is Noetherian and $\mathfrak{J}B$-adic, and $B/\mathfrak{J}B$ is an algebra of finite type over $A/\mathfrak{J}$.*

(b) *B is topologically A-isomorphic to $\varprojlim B_n$, where $B_n = B_m/\mathfrak{J}^{n+1}B_m$ for $m \geqslant n$, and B_1 is an algebra of finite type over $A_1 = A/\mathfrak{J}^2$.*

(c) *B is topologically A-isomorphic to a quotient of an algebra of the form $A\{T_1, \ldots, T_r\}$ by an ideal (necessarily closed according to Corollary (7.3.6) and Proposition (7.5.4, ii)).*

PROOF. As A is Noetherian, so is $A' = A\{T_1, \ldots, T_r\}$ (7.5.4), so (c) implies that B is Noetherian; as $\mathfrak{J}' = \mathfrak{J}A'$ is an open neighbourhood of 0 in A' such that the $\mathfrak{J}'^n$ form a fundamental system of neighbourhoods of 0, the images $\mathfrak{J}^n B$ of the $\mathfrak{J}'^n$ form a fundamental system of neighbourhoods of 0 in B, and as B is separated and complete, B is a $\mathfrak{J}B$-adic ring. Finally, $B/\mathfrak{J}B$ is an algebra (over $(A/\mathfrak{J})$ quotient of $A'/\mathfrak{J}A' = (A/\mathfrak{J})[T_1, \ldots, T_r]$, so it is of finite type, which proves that (c) implies (a).

If B is $\mathfrak{J}B$-adic and Noetherian, then B is isomorphic to $\varprojlim B_n$, where $B_n = B/\mathfrak{J}^{n+1}B$ (7.2.11), and $\mathfrak{J}B/\mathfrak{J}^2B$ is a module of finite type over $B/\mathfrak{J}B$. Let $(a_j)_{1 \leqslant j \leqslant s}$ be a system of generators for the $B/\mathfrak{J}B$-module $\mathfrak{J}B/\mathfrak{J}^2B$, and let $(c_i)_{1 \leqslant i \leqslant r}$ be a system of elements of $B/\mathfrak{J}^2B$ such that the classes mod $\mathfrak{J}B/\mathfrak{J}^2B$ form a system of generators for the $A/\mathfrak{J}$-algebra $B/\mathfrak{J}B$; we see immediately that the $c_i a_j$ form a system of generators for the $A/\mathfrak{J}^2$-algebra $B/\mathfrak{J}^2B$, hence (a) implies (b). $\mathbf{0_I}$ | 72

It remains to prove that (b) implies (c). The hypotheses imply that B_1 is a Noetherian ring, and as $B_1 = B_2/\mathfrak{J}^2B_2$, we have $\mathfrak{J}^2B_1 = 0$, hence $\mathfrak{J}B_1 = \mathfrak{J}B_1/\mathfrak{J}^2B_1$ is a B_0-module of finite type. The conditions of Proposition (7.2.7) are thus satisfied by the projective system (B_n) and B is a $\mathfrak{J}B$-adic ring. Let $(c_i)_{1 \leqslant i \leqslant r}$ be a finite system of elements of B whose classes mod $\mathfrak{J}B$ generate the $A/\mathfrak{J}$-algebra $B/\mathfrak{J}B$, and whose linear combinations with coefficients in $\mathfrak{J}$ are such that their classes mod $\mathfrak{J}^2B$ generate the B_0-module $\mathfrak{J}B/\mathfrak{J}^2B$. There exists a continuous A-homomorphism u from $A' = A\{T_1, \ldots, T_r\}$ to B which reduces to ϕ on A and is such that $u(T_i) = c_i$ for $1 \leqslant i \leqslant r$ (7.5.3); if we prove that u is *surjective*, then (c) will be established, since from $u(A') = B$ we deduce that $u(\mathfrak{J}^n A') = \mathfrak{J}^n B$, in other words that u is a strict morphism of topological rings and B is this isomorphic to a quotient of A' by a closed ideal. As B is complete for the $\mathfrak{J}B$-adic topology, it suffices ([**CC**, p. 18–07]) to show that the homomorphism $\mathrm{grad}(A') \to \mathrm{grad}(B)$, induced canonically by u for the $\mathfrak{J}$-adic filtrations on A' and B, is surjective. But by definition, the homomorphisms $A'/\mathfrak{J}A' \to B/\mathfrak{J}B$ and $\mathfrak{J}A'/\mathfrak{J}^2A' \to \mathfrak{J}B/\mathfrak{J}^2B$ induced by u are surjective; by induction on n, we immediately deduce that so is $\mathfrak{J}A'/\mathfrak{J}^nA' \to \mathfrak{J}B/\mathfrak{J}^nB$, and *a fortiori* so is $\mathfrak{J}^nA'/\mathfrak{J}^{n+1}A' \to \mathfrak{J}^nB/\mathfrak{J}^{n+1}B$, which finishes the proof. □

7.6. Completed rings of fractions

(7.6.1). Let A be a linearly topologized ring, $(\mathfrak{J}_\lambda)$ a fundamental system of neighbourhoods of 0 in A consisting of ideals, S a multiplicative subset of A. Let u_λ be the canonical homomorphism $A \to A_\lambda = A/\mathfrak{J}_\lambda$, and for $\mathfrak{J}_\mu \subset \mathfrak{J}_\lambda$, let $u_{\lambda\mu}$ be the canonical homomorphism $A_\mu \to A_\lambda$. Set $S_\lambda = u_\lambda(S)$, so that $u_{\lambda\mu}(S_\mu) = S_\lambda$. The $u_{\lambda\mu}$ canonically induce surjective homomorphisms $S_\mu^{-1}A_\mu \to S_\lambda^{-1}A_\lambda$, for which these rings form a projective system; we denote by $A\{S^{-1}\}$ the projective limit of this system. This definition does not depend on the fundamental system of neighbourhoods $(\mathfrak{J}_\lambda)$ chosen; indeed:

Proposition (7.6.2). — *The ring $A\{S^{-1}\}$ is topologically isomorphic to the separated completion of the ring $S^{-1}A$ for the topology which has a fundamental system of neighbourhoods of* 0 *consisting of the $S^{-1}\mathfrak{J}_\lambda$.*

PROOF. If v_λ is the canonical homomorphism $S^{-1}A \to S_\lambda^{-1}A_\lambda$ induced by u_λ, then the kernel of v_λ is surjective, hence the proposition (7.2.1). □

Corollary (7.6.3). — *If S' is the canonical image of S in the separated completion $\widehat{A}$ of A, then $A\{S^{-1}\}$ canonically identifies with $\widehat{A}\{S'^{-1}\}$.*

We note that if A is separated and complete, then it is not necessarily the same for $S^{-1}A$ with the topology defined by the $S^{-1}\mathfrak{J}_\lambda$, as we see for example by taking S to be the set of the f^n $(n \geqslant 0)$, where f is topologically nilpotent but not nilpotent: indeed, $S^{-1}A$ is not 0 and on the other hand, for each λ there exists an n such that $f^n \in \mathfrak{J}_\lambda$, so $1 = f^n/f^n \in S^{-1}\mathfrak{J}_\lambda$ and $S^{-1}\mathfrak{J}_\lambda = S^{-1}A$.

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Corollary (7.6.4). — *If, in A,* 0 *does not belong to S, then the ring $A\{S^{-1}\}$ is not* 0*.*

PROOF. Indeed, 0 does not belong to $\{1\}$ in the ring $S^{-1}A$; otherwise, we would have that $1 \in S^{-1}\mathfrak{J}_\lambda$ for each open ideal $\mathfrak{J}_\lambda$ of A, and it follows that $\mathfrak{J}_\lambda \cap S \neq \varnothing$ for all λ, contradicting the hypothesis. □

(7.6.5). We say that $A\{S^{-1}\}$ is the *completed ring of fractions* of A with denominators in S. With the above notation, it is clear that the inverse image of $S^{-1}\mathfrak{J}_\lambda$ in A contains $\mathfrak{J}_\lambda$, hence the canonical map $A \to S^{-1}A$ is continuous, and if we compose it with the canonical map $S^{-1}A \to A\{S^{-1}\}$, we obtain a canonical continuous homomorphism $A \to A\{S^{-1}\}$, the projective limit of the homomorphisms $A \to S_\lambda^{-1}A_\lambda$.

(7.6.6). The couple consisting of $A\{S^{-1}\}$ and the canonical homomorphism $A \to A\{S^{-1}\}$ are characterized by the following *universal property*: every continuous homomorphism u from A to a linearly topologized ring B, separated and complete, such that $u(S)$ consists of the invertible elements of B, uniquely factorizes as $A \to A\{S^{-1}\} \xrightarrow{u'} B$, where u' is continuous. Indeed, u uniquely factorizes as $A \to S^{-1}A \xrightarrow{v'} B$; as for each open ideal $\mathfrak{K}$ of B we have that $u^{-1}(\mathfrak{K})$ contains a $\mathfrak{J}_\lambda$, $v'^{-1}(\mathfrak{K})$ necessarily contains $S^{-1}\mathfrak{J}_\lambda$, so v' is continuous; since B is separated and complete, v' uniquely factorizes as $S^{-1}A \to A\{S^{-1}\} \xrightarrow{u'} B$, where u' is continuous; hence our assertion.

(7.6.7). Let B be a second linearly topologized ring, T a multiplicative subset of B, $\phi : A \to B$ a continuous homomorphism such that $\phi(S) \subset T$. According to the above, the continuous homomorphism $A \xrightarrow{\phi} B \to B\{T^{-1}\}$ uniquely factorizes as $A \to A\{S^{-1}\} \xrightarrow{\phi'} B\{T^{-1}\}$, where ϕ' is continuous. In particular, if $B = A$ and if ϕ is the identity, we see that for $S \subset T$ we have a continuous homomorphism $\rho^{T,S} : A\{S^{-1}\} \to A\{T^{-1}\}$ obtained by passing to the separated completion from $S^{-1}A \to T^{-1}A$; if U is a third multiplicative subset of A such that $S \subset T \subset U$, then we have $\rho^{U,S} = \rho^{U,T} \circ \rho^{T,S}$.

(7.6.8). Let S_1, S_2 be two multiplicative subsets of A, and let S'_2 be the canonical image of S_2 in $A\{S_1^{-1}\}$; we then have a canonical topological isomorphism $A\{(S_1S_2)^{-1}\} \simeq A\{S_1^{-1}\}\{S_2'^{-1}\}$, as we see from the canonical isomorphism $(S_1S_2)^{-1}A \simeq S_2''^{-1}(S_1^{-1}A)$ (where S''_2 is the canonical image of S_2 in $S_1^{-1}A$), which is bicontinuous.

(7.6.9). Let $\mathfrak{a}$ be an *open* ideal of A; we can assume that $\mathfrak{J}_\lambda \subset \mathfrak{a}$ for all λ, and as a result $S^{-1}\mathfrak{J}_\lambda \subset S^{-1}\mathfrak{a}$ in the ring $S^{-1}A$, in other words, $S^{-1}\mathfrak{a}$ is an *open* ideal of $S^{-1}A$; we denote by $\mathfrak{a}\{S^{-1}\}$ its separated completion, equal to $\varprojlim(S^{-1}\mathfrak{a}/S^{-1}\mathfrak{J}_\lambda)$, which is an *open* ideal of $A\{S^{-1}\}$, isomorphic to the closure of the canonical image of $S^{-1}\mathfrak{a}$. In addition, *the discrete ring $A\{S^{-1}\}/\mathfrak{a}\{S^{-1}\}$ is canonically isomorphic to $S^{-1}A/S^{-1}\mathfrak{a} = S^{-1}(A/\mathfrak{a})$*. Conversely, if $\mathfrak{a}'$ is an open ideal of $A\{S^{-1}\}$, then $\mathfrak{a}'$ contains an ideal of the form $\mathfrak{J}_\lambda\{S^{-1}\}$ which is the inverse image of an ideal of $S^{-1}A/S^{-1}\mathfrak{J}_\lambda$, which is necessarily (1.2.6) of the form $S^{-1}\mathfrak{a}$, where $\mathfrak{a} \supset \mathfrak{J}_\lambda$. We conclude that $\mathfrak{a}' = \mathfrak{a}\{S^{-1}\}$. In particular (1.2.6):

Proposition (7.6.10). — *The map $\mathfrak{p} \mapsto \mathfrak{p}\{S^{-1}\}$ is an increasing bijection from the set of* open prime *ideals $\mathfrak{p}$ of A such that $\mathfrak{p} \cap S = \varnothing$ to the set of* open prime *ideals of $A\{S^{-1}\}$; in addition, the field of fractions of $A\{S^{-1}\}/\mathfrak{p}\{S^{-1}\}$ is canonically isomorphic to that of $A/\mathfrak{p}$.* $\mathbf{0_I}$ | 74

Proposition (7.6.11). —

(i) *If A is an admissible ring, then so is $A' = A\{S^{-1}\}$, and for every ideal of definition $\mathfrak{J}$ for A, $\mathfrak{J}' = \mathfrak{J}\{S^{-1}\}$ is an ideal of definition for A'.*
(ii) *Let A be an adic ring, $\mathfrak{J}$ an ideal of definition for A such that $\mathfrak{J}/\mathfrak{J}^2$ is of finite type over $A/\mathfrak{J}$; then A' is a $\mathfrak{J}'$-adic ring and $\mathfrak{J}'/\mathfrak{J}'^2$ is of finite type over $A'/\mathfrak{J}'$. If in addition A is Noetherian, then so is A'.*

PROOF.

(i) If $\mathfrak{J}$ is an ideal of definition for A, then it is clear that $S^{-1}\mathfrak{J}$ is an ideal of definition for the topological ring $S^{-1}A$, since we have $(S^{-1}\mathfrak{J})^n = S^{-1}\mathfrak{J}^n$. Let A'' be the separated ring associated to $S^{-1}A$, $\mathfrak{J}''$ the image of $S^{-1}\mathfrak{J}$ in A''; the image of $S^{-1}\mathfrak{J}^n$ is $\mathfrak{J}''^n$, so $\mathfrak{J}''^n$ tends to 0 in A''; as $\mathfrak{J}'$ is the closure of $\mathfrak{J}''$ in A', $\mathfrak{J}'^n$ is contained in the closure of $\mathfrak{J}''^n$, hence tends to 0 in A'.
(ii) Set $A_i = A/\mathfrak{J}^{i+1}$, and for $i \leqslant j$, let u_{ij} be the canonical homomorphism $A/\mathfrak{J}^{j+1} \to A/\mathfrak{J}^{i+1}$; let S_i be the canonical image of S in A_i, and set $A'_i = S_i^{-1}A_i$; finally, let $u'_{ij} : A'_j \to A'_i$ be the homomorphism canonically induced by u_{ij}. We show that the projective system (A'_i, u'_{ij}) satisfies the conditions of Proposition (7.2.7): it is clear that the u'_{ij} are surjective; on the other hand, the kernel of u'_{ij} is $S_j^{-1}(\mathfrak{J}^{i+1}/\mathfrak{J}^{j+1})$ (1.3.2), equal to $\mathfrak{J}_j'^{i+1}$, where $\mathfrak{J}'_j = S_j^{-1}(\mathfrak{J}/\mathfrak{J}^{j+1})$; finally, $\mathfrak{J}'_1/\mathfrak{J}_1'^2 = S_1^{-1}(\mathfrak{J}/\mathfrak{J}^2)$, and as $\mathfrak{J}/\mathfrak{J}^2$ is of finite type over $A/\mathfrak{J}^2$, $\mathfrak{J}'_1/\mathfrak{J}_1'^2$ is of finite type over A'_1. Finally, if A is Noetherian, then so is $A'_0 = S_0^{-1}(A/\mathfrak{J})$, which finishes the proof (7.2.8).

□

Corollary (7.6.12). — *Under the hypotheses of Proposition (7.6.11, ii), we have $(\mathfrak{J}\{S^{-1}\})^n = \mathfrak{J}^n\{S^{-1}\}$.*

PROOF. This follows from Proposition (7.2.7) and the proof of Proposition (7.6.11). □

Proposition (7.6.13). — *Let A be an adic Noetherian ring, S a multiplicative subset of A; then $A\{S^{-1}\}$ is a flat A-module.*

PROOF. If $\mathfrak{J}$ is an ideal of definition for A, then $A\{S^{-1}\}$ is the separated completion of the Noetherian ring $S^{-1}A$ equipped with the $S{-}1\mathfrak{J}$-preadic topology; as a result (7.3.3) $A\{S^{-1}\}$ is a flat $S^{-1}A$-module; as $S^{-1}A$ is a flat A-module (6.3.1), the proposition follows from the transitivity of flatness (6.2.1). □

Corollary (7.6.14). — *Under the hypotheses of Proposition (7.6.13), let $S' \subset S$ be a second multiplicative subset of A; then $A\{S^{-1}\}$ is a flat $A\{S'^{-1}\}$-module.*

PROOF. By (7.6.8), $A\{S^{-1}\}$ canonically identifies with $A\{S'^{-1}\}\{S_0^{-1}\}$, where S_0 is the canonical image of S in $A\{S'^{-1}\}$, and $A\{S'^{-1}\}$ is Noetherian (7.6.11). □

(7.6.15). For each element f of a linearly topologized ring A, we denote by $A_{\{f\}}$ the completed ring of fractions $A\{S_f^{-1}\}$, where S_f is the multiplicative set of the f^n ($n \geqslant 0$); for each open ideal $\mathfrak{a}$ of A, we write $\mathfrak{a}_{\{f\}}$ for $\mathfrak{a}\{S_f^{-1}\}$. If g is a second element of A, then we have a canonical continuous homomorphism $A_{\{f\}} \to A_{\{fg\}}$ (7.6.7). When f varies over a multiplicative subset S of A, the $A_{\{f\}}$ form a filtered inductive system with the above homomorphisms; we set $A_{\{S\}} = \varinjlim_{f\in S} A_{\{f\}}$. For every $f \in S$, we have a homomorphism $A_{\{f\}} \to A\{S^{-1}\}$ (7.6.7), and these homomorphisms form an inductive system; by passing to the inductive limit, they thus define a canonical homomorphism $A_{\{S\}} \to A\{S^{-1}\}$. $\mathbf{0_I}$ | 75

Proposition (7.6.16). — *If A is a Noetherian ring, then $A\{S^{-1}\}$ is a flat module over $A_{\{S\}}$.*

PROOF. By (7.6.14), $A\{S^{-1}\}$ is flat for each of the rings $A_{\{f\}}$ for $f \in S$, and the conclusion follows from (6.2.3). □

Proposition (7.6.17). — *Let $\mathfrak{p}$ be an open prime ideal in an admissible ring A, and let $S = A - \mathfrak{p}$. Then the rings $A\{S^{-1}\}$ and $A_{\{S\}}$ are local rings, the canonical homomorphism $A_{\{S\}} \to A\{S^{-1}\}$ is local, and the residue fields of $A_{\{S\}}$ and $A\{S^{-1}\}$ are canonically isomorphic to the field of fractions of $A/\mathfrak{p}$.*

PROOF. Let $\mathfrak{J} \subset \mathfrak{p}$ be an ideal of definition for A; we have $S^{-1}\mathfrak{J} \subset S^{-1}\mathfrak{p} = \mathfrak{p}A_{\mathfrak{p}}$, so $A_{\mathfrak{p}}/S^{-1}\mathfrak{J}$ is a local ring; we conclude from Corollary (7.1.12), (7.6.9), and Proposition (7.6.11, i) that $A\{S^{-1}\}$ is a local ring. Set $\mathfrak{m} = \varinjlim_{f\in S} \mathfrak{p}_{\{f\}}$, which is an ideal in $A_{\{S\}}$; we will see that each element in $A_{\{S\}}$ not in $\mathfrak{m}$ is invertible. Indeed, such an element is the image in $A_{\{S\}}$ of an element $z \in A_{\{f\}}$ not in $\mathfrak{p}_{\{f\}}$, for an $f \in S$; its canonical image z_0 in $A_{\{f\}}/\mathfrak{J}_{\{f\}} = S_f^{-1}(A/\mathfrak{J})$ therefore is not in $S_f^{-1}(\mathfrak{p}/\mathfrak{J})$ (7.6.9), which means that $z_0 = \overline{x}/\overline{f}^k$, where $x \notin \mathfrak{p}$ and $\overline{x}$, $\overline{f}$ are the classes of x, f mod $\mathfrak{J}$. As $x \in S$, we have $g = xf \in S$, and in $S_g^{-1}A$, the canonical image $y_0 = x^{k+1}/g^k$ of $x/f^k \in S_f^{-1}A$ admits an inverse $x^{k-1}f^{2k}/g^k$. This implies *a fortiori* that the image of y_0 in $S_g^{-1}A/S_g^{-1}\mathfrak{J}$ is invertible, so ((7.6.9) and Corollary (7.1.12)) the canonical image y of z in $A_{\{g\}}$ is invertible; the image of z in $A_{\{S\}}$ (equal to that of y) is as a result invertible. We thus see that $A_{\{S\}}$ a local ring with maximal ideal $\mathfrak{m}$; in addition, the image of $\mathfrak{p}_{\{f\}}$ in $A\{S^{-1}\}$ is contained in the maximal ideal $\mathfrak{p}\{S^{-1}\}$ of this ring; *a fortiori*, the image of $\mathfrak{m}$ in $A\{S^{-1}\}$ is contained in $\mathfrak{p}\{S^{-1}\}$, so the canonical homomorphism $A_{\{S\}} \to A\{S^{-1}\}$ is local. Finally, as each element of $A\{S^{-1}\}/\mathfrak{p}\{S^{-1}\}$ is the image of an element in the ring $S_f^{-1}A$ for a suitable $f \in S$, the homomorphism $A_{\{S\}} \to A\{S^{-1}\}/\mathfrak{p}\{S^{-1}\}$ is surjective, and gives an isomorphism of the residue fields by passing to quotients. □

Corollary (7.6.18). — *Under the hypotheses of Proposition (7.6.17), if we suppose also that A is an adic Noetherian ring, then the local rings $A\{S^{-1}\}$ and $A_{\{S\}}$ are Noetherian, and $A\{S^{-1}\}$ is a faithfully flat $A_{\{S\}}$-module.*

PROOF. We know from before (7.6.11, ii) that $A\{S^{-1}\}$ is Noetherian and $A_{\{S\}}$-flat (7.6.16); as the homomorphism $A_{\{S\}} \to A\{S^{-1}\}$ is local, we conclude that $A\{S^{-1}\}$ is a faithfully flat $A_{\{S\}}$-module (6.6.2), and as a result that $A_{\{S\}}$ is Noetherian (6.5.2). □

7.7. Completed tensor products

(7.7.1). Let A be a linearly topologized ring, M, N two linearly topologized A-modules. Let $\mathfrak{J}$, V, W be open neighbourhoods of 0 in A, M, N respectively, which are A-modules, and such that $\mathfrak{J}M \subset V$, $\mathfrak{J}N \subset W$, so that M/V and N/W can be considered as $A/\mathfrak{J}$-modules. When $\mathfrak{J}$, V, W vary over the systems of open neighbourhoods satisfying these properties, it is immediate that the modules $(M/V) \otimes_{A/\mathfrak{J}} (N/W)$ form a projective system of modules over the projective system of rings $A/\mathfrak{J}$; $\mathbf{0_I}$ | 76
by passing to the projective limit, we thus obtain a module over the separated completion $\widehat{A}$ of A, which we call the *completed tensor product* of M and N and denote by $(M \otimes_A N)^\wedge$. If we have that M/V is canonically isomorphic to $\widehat{M}/\overline{V}$, where $\widehat{M}$ is the separated completion of M and $\overline{V}$ the closure in $\widehat{M}$ of the image of V, then we see that the completed tensor product $(M \otimes_A N)^\wedge$ canonically identifies with $(\widehat{M} \otimes_{\widehat{A}} \widehat{N})^\wedge$, which we denote by $\widehat{M} \widehat{\otimes}_{\widehat{A}} \widehat{N}$.

(7.7.2). With the above notation, the tensor products $(M/V) \otimes_A (N/W)$ and $(M/V) \otimes_{A/\mathfrak{J}} (N/W)$ identify canonically; they identify with $(M \otimes_A N)/(\mathrm{Im}(V \otimes_A N) + \mathrm{Im}(M \otimes_A W))$. We conclude that $(M \otimes_A N)^\wedge$ is *the separated completion of the A-module $M \otimes_A N$, equipped with the topology for which the submodules*

$$\mathrm{Im}(V \otimes_A N) + \mathrm{Im}(M \otimes_A W)$$

form a fundamental system of neighbourhoods of 0 (V and W varying over the set of open submodules of M and N respectively); we say for brevity that this topology is the *tensor product* of the given topologies on M and N.

(7.7.3). Let M', N' be two linearly topologized A-modules, $u : M \to M'$, $v : N \to N'$ two continuous homomorphisms; it is immediate that $u \otimes v$ is continuous for the tensor product topologies on $M \otimes_A N$ and $M' \otimes_A N'$ respectively; by passing to the separated completions, we obtain a continuous homomorphism $(M \otimes_A N)^\wedge \to (M' \otimes_A N')^\wedge$, which we denote by $u \widehat{\otimes} v$; $(M \otimes_A N)^\wedge$ is thus a *bifunctor* in M and N on the category of linearly topologized A-modules.

(7.7.4). We similarly define the completed tensor product of any finite number of linearly topologized A-modules; it is immediate that this product has the usual properties of associativity and commutativity.

(7.7.5). If B, C are two linearly topologized A-algebras, then the tensor product topology on $B \otimes_A C$ has for a fundamental system of neighbourhoods of 0 the *ideals* $\mathrm{Im}(\mathfrak{K} \otimes_A C) + \mathrm{Im}(B \otimes_A \mathfrak{L})$ of the algebra $B \otimes_A C$, $\mathfrak{K}$ (resp. $\mathfrak{L}$) varying over the set of open ideals of B (resp. C). As a result, $(B \otimes_A C)^\wedge$ is equipped with the structure of a *topological $\widehat{A}$-algebra*, the projective limit of the projective system of $A/\mathfrak{J}$-algebras $(B/\mathfrak{K}) \otimes_{A/\mathfrak{J}} (C/\mathfrak{L})$ ($\mathfrak{J}$ the open ideal of A such that $\mathfrak{J}B \subset \mathfrak{K}$, $\mathfrak{J}C \subset \mathfrak{L}$; it always exists). We say that this algebra is the *completed tensor product* of the algebras B and C.

(7.7.6). The A-algebra homomorphisms $b \mapsto b \otimes 1$, $c \mapsto 1 \otimes c$ from B and C to $B \otimes_A C$ are continuous when we equip the latter algebra with the tensor product topology; by composing with the canonical homomorphism from $B \otimes_A C$ to its separated completion, they give *canonical homomorphisms* $\rho : B \to (B \otimes_A C)^\wedge$, $\sigma : C \to (B \otimes_A C)^\wedge$. The algebra $(B \otimes_A C)^\wedge$ and the homomorphisms ρ and σ have in addition the following *universal property*: for every linearly topologized A-algebra D, separated and complete, and each pair of continuous A-homomorphisms $u : B \to D$, $v : C \to D$, there exists a unique continuous A-homomorphism $w : (B \otimes_A C)^\wedge \to D$ such that $u = w \circ \rho$ and $v = w \circ \sigma$. Indeed, there already exists a unique A-homomorphism $w_0 : B \otimes_A C \to D$ such that $u(b) = w_0(b \otimes 1)$ and $v(c) = w_0(1 \otimes c)$, and it remains to prove that w_0 is *continuous*, since it then gives a continuous homomorphism w by passing to the separated completion. If $\mathfrak{M}$ is an open ideal $\mathbf{0_I}$ | 77
of D, then there exists by hypothesis open ideals $\mathfrak{K} \subset B$, $\mathfrak{L} \subset C$ such that $u(\mathfrak{K}) \subset \mathfrak{M}$, $v(\mathfrak{L}) \subset \mathfrak{M}$; the image under w_0 of $\mathrm{Im}(\mathfrak{K} \otimes_A C) + \mathrm{Im}(B \otimes_A \mathfrak{L})$ is again contained in $\mathfrak{M}$, hence our assertion.

Proposition (7.7.7). — *If B and C are two preadmissible A-algebras, then $(B \otimes_A C)^\wedge$ is admissible, and if $\mathfrak{K}$ (resp. $\mathfrak{L}$) is an ideal of definition for B (resp. C), then the closure in $(B \otimes_A C)^\wedge$ of the canonical image of $\mathfrak{H} = \mathrm{Im}(\mathfrak{K} \otimes_A C) + \mathrm{Im}(B \otimes_A \mathfrak{L})$ is an ideal of definition.*

PROOF. It suffices to show that $\mathfrak{H}^n$ tends to 0 for the tensor product topology, which follows immediately from the inclusion

$$\mathfrak{H}^{2n} \subset \mathrm{Im}(\mathfrak{K}^n \otimes_A C) + \mathrm{Im}(B \otimes_A \mathfrak{L}^n).$$

□

Proposition (7.7.8). — *Let A be a preadic ring, $\mathfrak{J}$ an ideal of definition for A, M an A-module* of finite type, *equipped with the $\mathfrak{J}$-preadic topology. For every topological adic Noetherian A-algebra B, $B \otimes_A M$ identifies with the completed tensor product $(B \otimes_A M)^\wedge$.*

Proof. If $\mathfrak{K}$ is an ideal of definition for B, there exists by hypothesis an integer m such that $\mathfrak{J}^m B \subset \mathfrak{K}$, so $\operatorname{Im}(B \otimes_A \mathfrak{J}^{nm} M) = \operatorname{Im}(\mathfrak{J}^{nm} B \otimes_A M) \subset \operatorname{Im}(\mathfrak{K}^n B \otimes_A M) = \mathfrak{K}^n (B \otimes_A M)$; we conclude that over $B \otimes_A M$, the tensor products of the topologies of B and M is the $\mathfrak{K}$-preadic topology. As $B \otimes_A M$ is a B-module of finite type, the proposition follows from Corollary (7.3.6). □

7.8. Topologies on modules of homomorphisms

(7.8.1). Let A be a *Noetherian $\mathfrak{J}$-adic* ring, M and N two A-modules of finite type, equipped with the $\mathfrak{J}$-preadic topology; we know (7.3.6) that they are separated and complete; in addition, every A-homomorphism $M \to N$ is automatically continuous, and the A-module $\operatorname{Hom}_A(M, N)$ is of finite type. For every integer $i \geqslant 0$, set $A_i = A/\mathfrak{J}^{i+1}$, $M_i = M/\mathfrak{J}^{i+1}M$, $N_i = N/\mathfrak{J}^{i+1}N$; for $i \leqslant j$, every homomorphism $u_j : M_j \to N_j$ maps $\mathfrak{J}^{i+1}M_j$ to $\mathfrak{J}^{i+1}N_j$, thus giving by passage to quotients a homomorphism $u_i : M_i \to N_i$, which defines a canonical homomorphism $\operatorname{Hom}_{A_j}(M_j, N_j) \to \operatorname{Hom}_{A_i}(M_i, N_i)$; in addition, the $\operatorname{Hom}_{A_i}(M_i, N_i)$ form a *projective system* for these homomorphisms, and it follows from Corollary (7.2.10) that there is a canonical isomorphism $\phi : \operatorname{Hom}_A(M, N) \to \varprojlim_i \operatorname{Hom}_{A_i}(M_i, N_i)$. In addition:

Proposition (7.8.2). — *If M and N are modules of finite type over a $\mathfrak{J}$-adic Noetherian ring A, then the submodules* $\operatorname{Hom}_A(M, \mathfrak{J}^{i+1}N)$ *form a fundamental system of neighbourhoods of* 0 *in* $\operatorname{Hom}_A(M, N)$ *for the $\mathfrak{J}$-adic topology, and the canonical isomorphism ϕ* : $\operatorname{Hom}_A(M, N) \to \varprojlim_i \operatorname{Hom}_{A_i}(M_i, N_i)$ *is a topological isomorphism.*

Proof. We can consider M as the quotient of a free A-module L of finite type, and as a result identify $\operatorname{Hom}_A(M, N)$ as a submodule of $\operatorname{Hom}_A(L, N)$; in this identification, $\operatorname{Hom}_A(M, \mathfrak{J}^{i+1}N)$ is the intersection of $\operatorname{Hom}_A(M, N)$ and $\operatorname{Hom}_A(L, \mathfrak{J}^{i+1}N)$; as the induced topology on $\operatorname{Hom}_A(M, N)$ by the $\mathfrak{J}$-adic topology of $\operatorname{Hom}_A(L, N)$ is the $\mathfrak{J}$-adic (7.3.2), we have reduced to proving the first assertion for $M = L = A^m$; but then $\operatorname{Hom}_A(L, N) = N^m$, $\operatorname{Hom}_A(L, \mathfrak{J}^{i+1}N) = (\mathfrak{J}^{i+1}N)^m = \mathfrak{J}^{i+1}N^m$ and the $\mathbf{0_I}$ | 78
result is evident. To establish the second assertion, we note that the image of $\operatorname{Hom}_A(M, \mathfrak{J}^{i+1}N)$ in $\operatorname{Hom}_{A_j}(M_j, N_j)$ is zero for $j \leqslant i$, hence ϕ is continuous; conversely, the inverse image in $\operatorname{Hom}_A(M, N)$ of 0 of $\operatorname{Hom}_{A_i}(M_i, N_i)$ is $\operatorname{Hom}_A(M, \mathfrak{J}^{i+1}N)$, so ϕ is bicontinuous. □

If we only suppose that A is a *Noetherian $\mathfrak{J}$-preadic* ring, M and N two A-modules of finite type, separated for the $\mathfrak{J}$-preadic topology, then the following proof shows that the first assertion of Proposition (7.8.2) remains valid, and that ϕ is a topological isomorphism from $\operatorname{Hom}_A(M, N)$ to a submodule of $\varprojlim_i \operatorname{Hom}_{A_i}(M_i, N_i)$.

Proposition (7.8.3). — *Under the hypotheses of Proposition (7.8.2), the set of injective (resp. surjective, bijective) homomorphisms from M to N is an open subset of* $\operatorname{Hom}_A(M, N)$.

Proof. According to Corollaries (7.3.5) and (7.1.14), for u to be injective, it is necessary and sufficient that the corresponding homomorphism $u_0 : M/\mathfrak{J}M \to N/\mathfrak{J}N$ is, and the set of surjective homomorphisms from M to N is thus the inverse image under the continuous map $\operatorname{Hom}_A(M, N) \to \operatorname{Hom}_{A_0}(M_0, N_0)$ of a subset of a discrete space. We now show that the set of injective homomorphisms is open; let v be such a homomorphism and set $M' = v(M)$; by the Artin–Rees Lemma (7.3.2.1), there exists an integer $k \geqslant 0$ such that $M' \cap \mathfrak{J}^{m+k}N \subset \mathfrak{J}^m M'$ for all $m > 0$; we will see that for all $w \in \mathfrak{J}^{k+1} \operatorname{Hom}_A(M, N)$, $u = v + w$ is injective, which will finish the proof. Indeed, let $x \in M$ be such that $u(x) = 0$; we prove that for every $i \geqslant 0$ the relation $x \in \mathfrak{J}^i M$ implies that $x \in \mathfrak{J}^{i+1}M$; this follows from $x \in \bigcap_{i \geqslant 0} \mathfrak{J}^i M = (0)$. Indeed, we then have $w(x) \in \mathfrak{J}^{i+k+1}N$, and as a result $w(x) = -v(x) \in M'$, so $v(x) \in M' \cap \mathfrak{J}^{i+k+1}N \subset \mathfrak{J}^{i+1}M'$, and as v is an isomorphism from M to M', $x \in \mathfrak{J}^{i+1}M$; q.e.d. □

§8. Representable functors

8.1. Representable functors $\mathbf{0}_{\mathbf{III}}$ | 5

(8.1.1). We denote by $\mathtt{Set}$ the category of sets. Let $\mathtt{C}$ be a category; for two objects X, Y of $\mathtt{C}$, we set $h_X(Y) = \mathrm{Hom}(Y, X)$; for each morphism $u : Y \to Y'$ in $\mathtt{C}$, we denote by $h_X(u)$ the map $v \mapsto vu$ from $\mathrm{Hom}(Y', X)$ to $\mathrm{Hom}(Y, X)$. It is immediate that with these definitions, $h_X : \mathtt{C} \to \mathtt{Set}$ is a *contravariant functor*, i.e., an object of the category $\mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set})$, of covariant functors from the category $\mathtt{C}^{\mathrm{op}}$ (the dual of the category $\mathtt{C}$) to the category $\mathtt{Set}$ (T, 1.7, (d) and [Car]).

(8.1.2). Now let $w : X \to X'$ be a morphism in $\mathtt{C}$; for each $Y \in \mathtt{C}$ and each $v \in \mathrm{Hom}(Y, X) = h_X(Y)$, we have $wv \in \mathrm{Hom}(Y, X') = h_{X'}(Y)$; we denote by $h_w(Y)$ the map $v \mapsto wv$ from $h_X(Y)$ to $h_{X'}(Y)$. It is immediate that for each morphism $u : Y \to Y'$ in $\mathtt{C}$, the diagram

$$\begin{tikzcd} h_X(Y') \arrow[r, "h_X(u)"] \arrow[d, "h_w(Y')"'] & h_X(Y) \arrow[d, "h_w(Y)"] \\ h_{X'}(Y') \arrow[r, "h_{X'}(u)"] & h_{X'}(Y) \end{tikzcd}$$

is commutative; in other words, h_w is a *natural transformation (or functorial morphism)* $h_X \to h_{X'}$ (T, 1.2), also a morphism in the category $\mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set})$ (T, 1.7, (d)). The definitions of h_X and of h_w therefore constitute the definition of a *canonical covariant functor*

(8.1.2.1) $$h : \mathtt{C} \longrightarrow \mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set}), \quad X \longmapsto h_X.$$

(8.1.3). Let X be an object in $\mathtt{C}$, F a contravariant functor from $\mathtt{C}$ to $\mathtt{Set}$ (an object of $\mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set})$). Let $g : h_X \to F$ be a *natural transformation*: for all $Y \in \mathtt{C}$, $g(Y)$ is thus a map $h_X(Y) \to F(Y)$ such that $\mathbf{0}_{\mathbf{III}}$ | 6
for each morphism $u : Y \to Y'$ in $\mathtt{C}$, the diagram

(8.1.3.1) $$\begin{tikzcd} h_X(Y') \arrow[r, "h_X(u)"] \arrow[d, "g(Y')"'] & h_X(Y) \arrow[d, "g(Y)"] \\ F(Y') \arrow[r, "F(u)"] & F(Y) \end{tikzcd}$$

is commutative. In particular, we have a map $g(X) : h_X(X) = \mathrm{Hom}(X, X) \to F(X)$, hence an element

(8.1.3.2) $$\alpha(g) = (g(X))(1_X) \in F(X)$$

and as a result a canonical map

(8.1.3.3) $$\alpha : \mathrm{Hom}(h_X, F) \longrightarrow F(X).$$

Conversely, consider an element $\xi \in F(X)$; for each morphism $v : Y \to X$ in $\mathtt{C}$, $F(v)$ is a map $F(X) \to F(Y)$; consider the map

(8.1.3.4) $$v \longmapsto (F(v))(\xi)$$

from $h_X(Y)$ to $F(Y)$; if we denote by $(\beta(\xi))(Y)$ this map,

(8.1.3.5) $$\beta(\xi) : h_X \longrightarrow F$$

is a *natural transformation*, since for each morphism $u : Y \to Y'$ in $\mathtt{C}$ we have $(F(vu))(\xi) = (F(v) \circ F(u))(\xi)$, which makes (8.1.3.1) commutative for $g = \beta(\xi)$. We have thus defined a canonical map

(8.1.3.6) $$\beta : F(X) \longrightarrow \mathrm{Hom}(h_X, F).$$

Proposition (8.1.4). — *The maps α and β are the inverse bijections of each other.*

PROOF. We calculate $\alpha(\beta(\xi))$ for $\xi \in F(X)$; for each $Y \in \mathtt{C}$, $(\beta(\xi))(Y)$ is a map $g_1(Y) : v \mapsto (F(v))(\xi)$ from $h_X(Y)$ to $F(Y)$. We thus have

$$\alpha(\beta(\xi)) = (g_1(X))(1_X) = (F(1_X))(\xi) = 1_{F(X)}(\xi) = \xi.$$

We now calculate $\beta(\alpha(g))$ for $g \in \mathrm{Hom}(h_X, F)$; for each $Y \in \mathtt{C}$, $(\beta(\alpha(g)))(Y)$ is the map $v \mapsto (F(v))((g(X))(1_X))$; according to the commutativity of (8.1.3.1), this map is none other than $v \mapsto$

$(g(Y))((h_X(v))(1_X)) = (g(Y))(v)$ by definition of $h_X(v)$, in other words, it is equal to $g(Y)$, which finishes the proof. □

(8.1.5). Recall that a *subcategory* $\mathtt{C}'$ of a category $\mathtt{C}$ is defined by the condition that its objects are objects of $\mathtt{C}$, and that if X', Y' are two objects of $\mathtt{C}'$, then the set $\mathrm{Hom}_{\mathtt{C}'}(X', Y')$ of morphisms $X' \to Y'$ *in* $\mathtt{C}'$ is a subset of the set $\mathrm{Hom}_{\mathtt{C}}(X', Y')$ of morphisms $X' \to Y'$ *in* $\mathtt{C}$, the canonical map of "composition of morphisms"

$$\mathrm{Hom}_{\mathtt{C}'}(X', Y') \times \mathrm{Hom}_{\mathtt{C}'}(Y', Z') \longrightarrow \mathrm{Hom}_{\mathtt{C}'}(X', Z')$$

being the restriction of the canonical map $\mathbf{0}_{\mathrm{III}}$ | 7

$$\mathrm{Hom}_{\mathtt{C}}(X', Y') \times \mathrm{Hom}_{\mathtt{C}}(Y', Z') \longrightarrow \mathrm{Hom}_{\mathtt{C}}(X', Z').$$

We say that $\mathtt{C}'$ is a *full* subcategory of $\mathtt{C}$ if $\mathrm{Hom}_{\mathtt{C}'}(X', Y') = \mathrm{Hom}_{\mathtt{C}}(X', Y')$ for every pair of objects in $\mathtt{C}'$. The subcategory $\mathtt{C}''$ of $\mathtt{C}$ consisting of the objects of $\mathtt{C}$ isomorphic to objects of $\mathtt{C}'$ is then again a full subcategory of $\mathtt{C}$, *equivalent* (T, 1.2) to $\mathtt{C}'$ as we verify easily.

A covariant functor $F : \mathtt{C}_1 \to \mathtt{C}_2$ is called *fully faithful* if for every pair of objects X_1, Y_1 of $\mathtt{C}_1$, the map $u \mapsto F(u)$ from $\mathrm{Hom}(X_1, Y_1)$ to $\mathrm{Hom}(F(X_1), F(Y_1))$ is *bijective*; this implies that the subcategory $F(\mathtt{C}_1)$ of $\mathtt{C}_2$ is *full*. In addition, if two objects X_1, X_1' have the same image X_2, then there exists a unique isomorphism $u : X_1 \to X_1'$ such that $F(u) = 1_{X_1}$. For each object X_2 of $F(\mathtt{C}_1)$, let $G(X_2)$ be one of the objects X_1 of $\mathtt{C}_1$ such that $F(X_1) = X_2$ (G is defined by means of the axiom of choice); for each morphism $v : X_2 \to Y_2$ in $F(\mathtt{C}_1)$, $G(v)$ will be the unique morphism $u : G(X_2) \to G(Y_2)$ such that $F(u) = v$; G is then a *functor* from $F(\mathtt{C}_1)$ to $\mathtt{C}_1$; FG is the identity functor on $F(\mathtt{C}_1)$, and the above shows that there exists an isomorphism of functors $\phi : 1_{\mathtt{C}_1} \to GF$ such that F, G, ϕ, and the identity $1_{F(\mathtt{C}_1)} \to FG$ defines an *equivalence* between the category $\mathtt{C}_1$ and the full subcategory $F(\mathtt{C}_1)$ of $\mathtt{C}_2$ (T, 1.2).

(8.1.6). We apply Proposition (8.1.4) to the case where F is $h_{X'}$, X' being any object of $\mathtt{C}$; the map $\beta : \mathrm{Hom}(X, X') \to \mathrm{Hom}(h_X, h_{X'})$ is none other than the map $w \mapsto h_w$ defined in (8.1.2); this map being *bijective*, we see with the terminology of (8.1.5) that:

Proposition (8.1.7). — *The canonical functor $h : \mathtt{C} \to \mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set})$ is fully faithful.*

(8.1.8). Let F be a contravariant functor from $\mathtt{C}$ to $\mathtt{Set}$; we say that F is *representable* if there exists an object $X \in \mathtt{C}$ such that F is *isomorphic* to h_X; it follows from Proposition (8.1.7) that the data of an $X \in \mathtt{C}$ and an isomorphism of functors $g : h_X \to F$ determines X up to unique isomorphism. Proposition (8.1.7) then implies that h defines an *equivalence* between $\mathtt{C}$ and the full subcategory of $\mathtt{Hom}(\mathtt{C}^{\mathrm{op}}, \mathtt{Set})$ consisting of the *contravariant representable functors*. It follows from Proposition (8.1.4) that the data of a natural transformation $g : h_X \to F$ is equivalent to that of an element $\xi \in F(X)$; to say that g is an *isomorphism* is equivalent to the following condition on ξ: *for every object Y of $\mathtt{C}$ the map $v \mapsto (F(v))(\xi)$ from $\mathrm{Hom}(Y, X)$ to $F(Y)$ is bijective*. When ξ satisfies this condition, we say that the pair (X, ξ) *represents* the representable functor F, By abuse of language, we also say that the object $X \in \mathtt{C}$ represents F if there exists a $\xi \in F(X)$ such that (X, ξ) represents F, in other words if h_X is isomorphic to F.

Let F, F' be two contravariant representable functors from $\mathtt{C}$ to $\mathtt{Set}$, $h_X \to F$ and $h_{X'} \to F'$ two isomorphisms of functors. Then it follows from (8.1.6) that there is a canonical bijective correspondence between $\mathrm{Hom}(X, X')$ and the set $\mathrm{Hom}(F, F')$ of natural transformations $F \to F'$.

(8.1.9). *Example I. Projective limits*. The notion of a contravariant representable functor covers in particular the "dual" notion of the usual notion of a "solution to a universal problem". More $\mathbf{0}_{\mathrm{III}}$ | 8 generally, we will see that the notion of the *projective limit* is a special case of the notion of a representable functor. Recall that in a category $\mathtt{C}$, we define a *projective system* by the data of a preordered set I, a family $(A_\alpha)_{\alpha \in I}$ of objects of $\mathtt{C}$, and for every pair of indices (α, β) such that $\alpha \leqslant \beta$, a morphism $u_{\alpha\beta} : A_\beta \to A_\alpha$. A *projective limit* of this system in $\mathtt{C}$ consists of an object B of $\mathtt{C}$ (denoted $\varprojlim A_\alpha$), and for each $\alpha \in I$, a morphism $u_\alpha : B \to A_\alpha$ such that: 1st. $u_\alpha = u_{\alpha\beta} u_\beta$ for $\alpha \leqslant \beta$; 2nd. for every object X of $\mathtt{C}$ and every family $(v_\alpha)_{\alpha \in I}$ of morphisms $v_\alpha : X \to A_\alpha$ such that $v_\alpha = u_{\alpha\beta} v_\beta$ for $\alpha \leqslant \beta$, there exists a unique morphism $v : X \to B$ (denoted $\varprojlim v_\alpha$) such that $v_\alpha = u_\alpha v$ for all $\alpha \in I$ (T, 1.8). This can be interpreted in the following way: the $u_{\alpha\beta}$ canonically define maps

$$\overline{u}_{\alpha\beta} : \mathrm{Hom}(X, A_\beta) \longrightarrow \mathrm{Hom}(X, A_\alpha)$$

which define a *projective system* of sets $(\mathrm{Hom}(X, A_\alpha), \overline{u}_{\alpha\beta})$, and (v_α) is by definition an element of the set $\varprojlim \mathrm{Hom}(X, A_\alpha)$; it is clear that $X \mapsto \varprojlim \mathrm{Hom}(X, A_\alpha)$ is a *contravariant functor* from $\mathtt{C}$ to $\mathtt{Set}$, and the existence of the projective limit B is equivalent to saying that $(v_\alpha) \mapsto \varprojlim v_\alpha$ is an *isomorphism* of functors in X

(8.1.9.1) $$\varprojlim \mathrm{Hom}(X, A_\alpha) \simeq \mathrm{Hom}(X, B),$$

in other words, that the functor $X \mapsto \varprojlim \mathrm{Hom}(X, A_\alpha)$ is *representable*.

(8.1.10). *Example II. Final objects*. Let $\mathtt{C}$ be a category, $\{a\}$ a singleton set. Consider the contravariant functor $F : \mathtt{C} \to \mathtt{Set}$ which sends every object X of $\mathtt{C}$ to the set $\{a\}$, and every morphism $X \to X'$ in $\mathtt{C}$ to the unique map $\{a\} \to \{a\}$. To say that this functor is *representable* means that there exists an object $e \in \mathtt{C}$ such that for every $Y \in \mathtt{C}$, $\mathrm{Hom}(Y, e) = h_e(Y)$ is a *singleton set*; we say that e is an *final object* of $\mathtt{C}$, and it is clear that two final objects of $\mathtt{C}$ are isomorphic (which allows us to define, in general with the axiom of choice, *one* final object of $\mathtt{C}$ which we then denote $e_{\mathtt{C}}$). For example, in the category $\mathtt{Set}$, the final objects are the singleton sets; in the category of *augmented algebras* over a field K (where the morphisms are the algebra homomorphisms compatible with the augmentation), K is a final object; in the category of *S-preschemes* (**I**, 2.5.1), S is a final object.

(8.1.11). For two objects X and Y of a category $\mathtt{C}$, set $h'_X(Y) = \mathrm{Hom}(X, Y)$, and for every morphism $u : Y \to Y'$, let $h_X(u)$ be the map $v \mapsto vu$ from $\mathrm{Hom}(X, Y)$ to $\mathrm{Hom}(X, Y')$; h'_X is then a *covariant functor* $\mathtt{C} \to \mathtt{Set}$, so we deduce as in (8.1.2) the definition of a canonical covariant functor $h' : \mathtt{C}^{\mathtt{op}} \to \mathtt{Hom}(\mathtt{C}, \mathtt{Set})$; a *covariant* functor F from $\mathtt{C}$ to $\mathtt{Set}$, in other words an object of $\mathtt{Hom}(\mathtt{C}, \mathtt{Set})$, is then *representable* if there exists an object $X \in \mathtt{C}$ (necessarily unique up to unique isomorphism) such that F is *isomorphic to h'_X*; we leave it to the reader to develop the "dual" notions of the above, which this time cover the notion of an *inductive limit*, and in particular the usual notion of a "solution to a universal problem".

8.2. Algebraic structures in categories

(8.2.1). Given two contravariant functors F and F' from $\mathtt{C}$ to $\mathtt{Set}$, recall that for every object $Y \in \mathtt{C}$, $\mathbf{0_{III}}$ | 9
we set $(F \times F')(Y) = F(Y) \times F'(Y)$, and for every morphism $u : Y \to Y'$ in $\mathtt{C}$, we set $(F \times F')(u) = F(u) \times F'(u)$, which is the map $(t, t') \mapsto (F(u)(t), F'(u)(t'))$ from $F(Y') \times F'(Y')$ to $F(Y) \times F'(Y)$; $F \times F' : \mathtt{C} \to \mathtt{Set}$ is thus a *contravariant functor* (which is none other than the *product* of the objects F and F' in the category $\mathtt{Hom}(\mathtt{C}^{\mathtt{op}}, \mathtt{Set})$). Given an object $X \in \mathtt{C}$, we call an *internal composition law* on X a *natural transformation*

(8.2.1.1) $$\gamma_X : h_X \times h_X \longrightarrow h_X.$$

In other words (T, 1.2), for every object $Y \in \mathtt{C}$, $\gamma_X(Y)$ is a map $h_X(Y) \times h_X(Y) \to h_X(Y)$ (thus by definition an *internal composition law* on the set $h_X(Y)$) with the condition that for every morphism $u : Y \to Y'$ in $\mathtt{C}$, the diagram

$$\begin{tikzcd} h_X(Y') \times h_X(Y') \arrow[r, "h_X(u) \times h_X(u)"] \arrow[d, "\gamma_X(Y')"'] & h_X(Y) \times h_X(Y) \arrow[d, "\gamma_X(Y)"] \\ h_X(Y') \arrow[r, "h_X(u)"] & h_X(Y) \end{tikzcd}$$

is commutative; this implies that for the composition laws $\gamma_X(Y)$ and $\gamma_X(Y')$, $h_X(u)$ is a *homomorphism* from $h_X(Y')$ to $h_X(Y)$.

In a similar way, given two objects Z and X of $\mathtt{C}$, we call an *external composition law on X, with Z as its domain of operators* a natural transformation

(8.2.1.2) $$\omega_{X,Z} : h_Z \times h_X \longrightarrow h_X.$$

We see as above that for every $Y \in \mathtt{C}$, $\omega_{X,Z}(Y)$ is an external composition law on $h_X(Y)$, with $h_Z(Y)$ as its domain of operators and such that for every morphism $u : Y \to Y'$, $h_X(u)$ and $h_Z(u)$ form a *di-homomorphism* from $(h_Z(Y'), h_X(Y'))$ to $(h_Z(Y), h_X(Y))$.

(8.2.2). Let X' be a second object of $\mathtt{C}$, and suppose we are given an internal composition law $\gamma_{X'}$ on X'; we say that a morphism $w : X \to X'$ in $\mathtt{C}$ is a *homomorphism* for the composition laws if for every $Y \in \mathtt{C}$, $h_w(Y) : h_X(Y) \to h_{X'}(Y)$ is a *homomorphism* for the composition laws $\gamma_X(Y)$ and $\gamma_{X'}(Y)$. If X'' is a third object of $\mathtt{C}$ equipped with an internal composition law $\gamma_{X''}$ and $w' : X' \to X''$ is a morphism $\mathbf{0_{III}}$ | 10
in $\mathtt{C}$ which is a homomorphism for $\gamma_{X'}$ and $\gamma_{X''}$, then it is clear that the morphism $w'w : X \to X''$ is a homomorphism for the composition laws γ_X and $\gamma_{X''}$. An isomorphism $w : X \simeq X'$ in $\mathtt{C}$ is called an *isomorphism for the composition laws γ_X and $\gamma_{X'}$* if w is a homomorphism for these composition laws, and if its inverse morphism w^{-1} is a homomorphism for the composition laws $\gamma_{X'}$ and γ_X.

We define in a similar way the *di-homomorphisms* for pairs of objects of $\mathtt{C}$ equipped with external composition laws.

(8.2.3). When an internal composition law γ_X on an object $X \in \mathtt{C}$ is such that $\gamma_X(Y)$ is a *group law* on $h_X(Y)$ for *every* $Y \in \mathtt{C}$, we say that X, equipped with this law, is a $\mathtt{C}$-*group* or a *group object in* $\mathtt{C}$. We similarly define $\mathtt{C}$-*rings*, $\mathtt{C}$-*modules*, etc.

(8.2.4). Suppose that the *product* $X \times X$ of an object $X \in \mathtt{C}$ by itself exists in $\mathtt{C}$; by definition, we then have $h_{X\times X} = h_X \times h_X$ up to canonical isomorphism, since it is a particular case of the projective limit (8.1.9); an internal composition law on X can thus be considered as a functorial morphism $\gamma_X : h_{X\times X} \to h_X$, and thus canonically determine (8.1.6) an element $c_X \in \operatorname{Hom}(X \times X, X)$ such that $h_{c_X} = \gamma_X$; in this case, the data of an internal composition law on X is equivalent to the data of a morphism $X \times X \to X$; when $\mathtt{C}$ is the category $\mathtt{Set}$, we recover the classical notion of an internal composition law on a set. We have an analogous result for an external composition law when the product $Z \times X$ exists in $\mathtt{C}$.

(8.2.5). With the above notation, suppose that in addition $X \times X \times X$ exists in $\mathtt{C}$; the characterization of the product as an object representing a functor (8.1.9) implies the existence of canonical isomorphisms

$$(X \times X) \times X \simeq X \times X \times X \simeq X \times (X \times X);$$

if we canonically identity $X \times X \times X$ with $(X \times X) \times X$, then the map $\gamma_X(Y) \times 1_{h_X(Y)}$ identifies with $h_{c_X \times 1_X}(Y)$ for all $Y \in \mathtt{C}$. As a result, it is equivalent to say that for every $Y \in \mathtt{C}$, the internal law $\gamma_X(Y)$ is associative, or that the diagram of maps

$$\begin{tikzcd}
h_X(Y) \times h_X(Y) \times h_X(Y) \arrow[r, "\gamma_X(Y)\times 1"] \arrow[d, "1\times\gamma_X(Y)"'] & h_X(Y) \times h_X(Y) \arrow[d, "\gamma_X(Y)"] \\
h_X(Y) \times h_X(Y) \arrow[r, "\gamma_X(Y)"] & h_X(Y)
\end{tikzcd}$$

is commutative, or that the diagram of morphisms $\mathbf{0_{III}}$ | 11

$$\begin{tikzcd}
X \times X \times X \arrow[r, "c_X\times 1_X"] \arrow[d, "1_X\times c_X"'] & X \times X \arrow[d, "c_X"] \\
X \times X \arrow[r, "c_X"] & X
\end{tikzcd}$$

is commutative.

(8.2.6). Under the hypotheses of (8.2.5), if we want to express, for every $Y \in \mathtt{C}$, the internal law $\gamma_X(Y)$ as a *group law*, then it is first necessary that it is associative, and second that there exists a map $\alpha_X(Y) : h_X(Y) \to h_X(Y)$ having the properties of the *inverse* operation of a group; as for every morphism $u : Y \to Y'$ in $\mathtt{C}$, we have seen that $h_X(u)$ must be a group homomorphism $h_X(Y') \to h_X(Y)$, we first see that $\alpha_X : h_X \to h_X$ must be a *natural transformation*. On the other hand, one can express the characteristic properties of the inverse $s \mapsto s^{-1}$ of a group G without involving the identity element: it suffices to check that the two composite maps

$$(s,t) \longmapsto (s,s^{-1},t) \longmapsto (s,s^{-1}t) \longmapsto s(s^{-1}t),$$

$$(s,t) \longmapsto (s,s^{-1},t) \longmapsto (s,ts^{-1}) \longmapsto (ts^{-1})s$$

are equal to the second projection $(s,t) \mapsto t$ from $G \times G$ to G. By (8.1.3), we have $\alpha_X = h_{a_X}$, where $a_X \in \mathrm{Hom}(X,X)$; the first condition above then expresses that the composite morphism

$$X \times X \xrightarrow{(1_X,a_X)\times 1_X} X \times X \times X \xrightarrow{1_X \times c_X} X \times X \xrightarrow{c_X} X$$

is the second projection $X \times X \to X$ in $\mathtt{C}$, and the second condition is similar.

(8.2.7). Now suppose that there exists a *final object* e (8.1.10) in $\mathtt{C}$. Let us always assume that $\gamma_X(Y)$ is a group law on $h_X(Y)$ for every $Y \in \mathtt{C}$, and denote by $\eta_X(Y)$ the identity element of $\gamma_X(Y)$. As, for every morphism $u : Y \to Y'$ in $\mathtt{C}$, $h_X(u)$ is a group homomorphism, we have $\eta_X(Y) = (h_X(u))(\eta_X(Y'))$; taking in particular $Y' = e$, in which case u is the unique element ε of $\mathrm{Hom}(Y,e)$, we see that the element $\eta_X(e)$ completely determines $\eta_X(Y)$ for every $Y \in \mathtt{C}$. Set $e_X = \eta_X(X)$, the identity element of the group $h_X(X) = \mathrm{Hom}(X,X)$; the commutativity of the diagram

$$\begin{tikzcd} h_X(e) \arrow[r, "h_X(\varepsilon)"] \arrow[d, "h_{e_X}(e)"'] & h_X(Y) \arrow[d, "h_{e_X}(Y)"] \\ h_X(e) \arrow[r, "h_X(\varepsilon)"] & h_X(Y) \end{tikzcd}$$

(cf. (8.1.2)) shows that, on the set $h_X(Y)$, the map $h_{e_X}(Y)$ is none other than $s \mapsto \eta_X(Y)$ sending every element to the identity element. We then verify that the fact that $\eta_X(Y)$ is the identity element of $\gamma_X(Y)$ for every $Y \in \mathtt{C}$ is equivalent to saying that the composite morphism $\mathbf{0}_{\mathbf{III}}$ | 12

$$X \xrightarrow{(1_X,1_X)} X \times X \xrightarrow{1_X \times e_X} X \times X \xrightarrow{c_X} X,$$

and the analog in which we swap 1_X and e_X, are both *equal* to 1_X.

(8.2.8). One could of course easily extend the examples of algebraic structures in categories. The example of groups was treated with enough detail, but latter on we will usually leave it to the reader to develop analogous notions for the examples of algebraic structures we will encounter.

§9. CONSTRUCTIBLE SETS

9.1. Constructible sets

Definition (9.1.1). — We say that a continuous map $f : X \to Y$ is *quasi-compact* if for every quasi-compact open subset U of Y, $f^{-1}(U)$ is quasi-compact. We say that a subset Z of a topological space X is *retrocompact* in X if the canonical injection $Z \to X$ is quasi-compact, in other words, if for every quasi-compact open subset U of X, $U \cap Z$ is quasi-compact.

A *closed* subset of X is retrocompact in X, but a quasi-compact subset of X is not necessarily retrocompact in X. If X is quasi-compact, every retrocompact open subset of X is quasi-compact. It is clear that every *finite* union of retrocompact sets in X is retrocompact in X, as every finite union of quasi-compact sets is quasi-compact. Every finite intersection of retrocompact open sets in X is a retrocompact open set in X. In a *locally Noetherian* space X, every quasi-compact set is a Noetherian subspace, and as a result *every subset* of X is retrocompact in X.

Definition (9.1.2). — Given a topological space X, we say that a subset of X is *constructible* if it belongs to the smallest set of subsets $\mathfrak{F}$ of X containing all the retrocompact open subsets of $\mathfrak{F}$ and is stable under finite intersections and complements (which implies that $\mathfrak{F}$ is also stable under finite unions).

Proposition (9.1.3). — *For a subset of X to be constructible, it is necessary and sufficient for it to be a finite union of sets of the form $U \cap \complement V$, where U and V are retrocompact open sets in X.*

PROOF. It is clear that the condition is sufficient. To see that it is necessary, consider the set $\mathfrak{G}$ of finite unions of sets of the form $U \cap \complement V$, where U and V are retrocompact open sets in X; it suffices to see that every complement of a set in $\mathfrak{G}$ is in $\mathfrak{G}$. Let $Z = \bigcup_{i \in I}(U_i \cap \complement V_i)$, where I is finite, U_i and V_i retrocompact open sets in X; we have $\complement Z = \bigcap_{i \in I}(V_i \cup \complement U_i)$, so Z is a finite union of sets which are intersections of a certain number of the V_i and of a certain number of the $\complement U_i$, thus of the form $V \cap \complement U$, where U is the union of a certain number of the U_i and V is the intersection of a certain $\mathbf{0}_{\mathbf{III}}$ | 13
number of the V_i; but we have noted above that finite unions and intersections of retrocompact open sets in X are retrocompact open sets in X, hence the conclusion. □

Corollary (9.1.4). — *Every constructible subset of X is retrocompact in X.*

PROOF. It suffices to show that if U and V are retrocompact open sets in X, then $U \cap \complement V$ is retrocompact in X; if W is a quasi-compact open set in X, then $W \cap U \cap \complement V$ is closed in the quasi-compact space $W \cap U$, hence it is quasi-compact. □

In particular:

Corollary (9.1.5). — *For an open subset U of X to be constructible, it is necessary and sufficient for it to be retrocompact in X. For a closed subset F of X to be constructible, it is necessary and sufficient for the open set $\complement F$ to be retrocompact.*

(9.1.6). An important case is when every quasi-compact open subset of X is retrocompact, in other words, when the intersection of two quasi-compact open subsets of X is quasi-compact (cf. (**I**, 5.5.6)). When X is also quasi-compact, this implies that the retrocompact open subsets of X are identical to the quasi-compact open subsets of X, and the constructible subsets of X are finite unions of sets of the form $U \cap \complement V$, where U and V are quasi-compact open sets.

Corollary (9.1.7). — *For a subset of a Noetherian space to be constructible, it is necessary and sufficient for it to be a finite union of locally closed subsets of X.*

Proposition (9.1.8). — *Let X be a topological space, U an open subset of X.*

(i) *If T is a constructible subset of X, then $T \cap U$ is a constructible subset of U.*
(ii) *In addition, suppose that U is retrocompact in X. For a subset Z of U to be constructible in X, it is necessary and sufficient for it to be constructible in U.*

PROOF.

(i) Using Proposition (9.1.3), we reduce to showing that if T is a retrocompact open set in X, then $T \cap U$ is a retrocompact open set in U, in other words, for every quasi-compact open $W \subset U$, $T \cap U \cap W = T \cap W$ is quasi-compact, which immediately follows from the hypothesis.
(ii) The condition is necessary by (i), so it remains to show that it is sufficient. By Proposition (9.1.3), it suffices to consider the case where Z is a retrocompact open set *in U*, because it will then follow that $U - Z$ is constructible in X, and if Z and Z' are two retrocompact opens *in U*, then $Z \cap (U - Z')$ will be constructible in X. If W is a quasi-compact open set in X and Z a retrocompact open set in U, then we have $Z \cap W = Z \cap (W \cap U)$, and by hypothesis $W \cap U$ is a quasi-compact open set in U; so $W \cap Z$ is quasi-compact, and as a result Z is a retrocompact open set in X, and *a fortiori* constructible in X.

□

Corollary (9.1.9). — *Let X be a topological space, $(U_i)_{i \in I}$ a finite cover of X by retrocompact open sets in X. For a subset Z of X to be constructible in X, it is necessary and sufficient for $Z \cap U_i$ to be constructible in U_i for all $i \in I$.*

(9.1.10). In particular, suppose that X is quasi-compact and every point of X admits a fundamental $\mathbf{0}_{\mathbf{III}}$ | 14
system of retrocompact open neighbourhoods in X (and *a fortiori* quasi-compact); then the condition for a subset Z of X to be constructible in X is of a *local* nature, in other words, it is necessary and sufficient that for every $x \in X$, there exists an open neighbourhood V of x such that $V \cap Z$ is constructible in V. Indeed, if this condition is satisfied, then there exists for every $x \in X$ an open neighbourhood V of x which is *retrocompact in X* and such that $V \cap Z$ is constructible in V, by the

hypotheses on X and by Proposition (9.1.8, i); it then suffices to cover X by a finite number of these neighbourhoods and to apply Corollary (9.1.9).

Definition (9.1.11). — Let X be a topological space. We say that a subset T of X is *locally constructible* in X if for every $x \in X$ there exists an open neighbourhood V of x such that $T \cap V$ is constructible in V.

It follows from Proposition (9.1.8, i) that if V is such that $V \cap T$ is constructible in V, then for every open $W \subset V$, $W \cap T$ is constructible in W. If T is locally constructible in X, then for every open set U in X, $T \cap U$ is locally constructible in U, as a result of the above remark. The same remark shows that the set of locally constructible subsets of X is stable under finite unions and finite intersections; on the other hand, it is clear that it is also stable under taking complements.

Proposition (9.1.12). — *Let X be a topological space. Every constructible set in X is locally constructible in X. The converse is true if X is quasi-compact and if its topology admits a basis formed by the retrocompact sets in X.*

PROOF. The first assertion follows from Definition (9.1.11) and the second from (9.1.10). □

Corollary (9.1.13). — *Let X be a topological space whose topology admits a basis formed by the retrocompact sets in X. Then every locally constructible subset T of X is retrocompact in X.*

PROOF. Let U be a quasi-compact open set in X; $T \cap U$ is locally constructible in U, hence constructible in U by Proposition (9.1.12), and as a result quasi-compact by Corollary (9.1.4). □

9.2. Constructible subsets of Noetherian spaces

(9.2.1). We have seen (9.1.7) that in a Noetherian space X, the constructible subsets of X are the *finite unions of locally closed subsets of X*.

The inverse image of a constructible set in X by a continuous map from a Noetherian space X' to X is constructible in X'. If Y is a constructible subset of a Noetherian space X, then the subsets of Y are constructible as subspaces of Y and are identical to those which are constructible as subspaces of X.

Proposition (9.2.2). — *Let X be an irreducible Noetherian space, E a constructible subset of X. For E to be everywhere dense in X, it is necessary and sufficient for E to contain a nonempty open subset of X.*

PROOF. The condition is evidently sufficient, as every nonempty open set is dense in X. Conversely, let $E = \bigcup_{i=1}^{n}(U_i \cap F_i)$ be a constructible subset of X, the U_i being nonempty open sets and the F_i closed in X; we then have $\overline{E} \subset \bigcup_i F_i$. As a result, if $\overline{E} = X$, then X is equal to one of the F_i, hence $E \supset U_i$, which finishes the proof. □

When X admits a generic point x (**0**, 2.1.2), the condition of Proposition (9.2.2) is equivalent to the relation $x \in E$.

Proposition (9.2.3). — *Let X be a Noetherian space. For a subset E of X to be constructible, it is necessary and sufficient that, for every irreducible closed subset Y of X, $E \cap Y$ is rare in Y or contains a nonempty open subset of Y.*

PROOF. The necessity of the condition follows from the fact that $E \cap Y$ must be a constructible subset of Y and from Proposition (9.2.2), since a nondense subset of Y is necessarily rare in the irreducible space Y (**0**, 2.1.1). To prove that the condition is sufficient, apply the principle of Noetherian induction (**0**, 2.2.2) to the set $\mathfrak{F}$ of closed subsets Y of X such that $Y \cap E$ is constructible (with respect to Y or with respect to X, which are equivalent): we can thus assume that for every closed subset $Y \neq X$ of X, $E \cap Y$ is constructible. First suppose that X is not irreducible, and let X_i $(1 \leqslant i \leqslant m)$ are its irreducible components, necessarily of finite number (**0**, 2.2.5); by hypothesis the $E \cap X_i$ are constructible, hence their union E is as well. Suppose now that X is irreducible; then by hypothesis, if E is rare, then $\overline{E} \neq X$ and $E = E \cap \overline{E}$ is constructible; if E contains a nonempty open subset U of X, then it is the union of U and $E \cap (X - U)$; but $X - U$ is a closed set distinct from X, so $E \cap (X - U)$ is constructible; as a result, E is itself constructible, which finishes the proof. □

Corollary (9.2.4). — *Let X be a Noetherian space, (E_α) an increasing filtered family of constructible subsets of X, such that*

(1st) *X is the union of the family (E_α).*

(2nd) *Every irreducible closed subset of X is contained in the closure of one of the E_α.*

Then there exists an index α such that $X = E_\alpha$.

When every irreducible closed subset of X admits a generic point, the hypothesis (1st) *can be omitted.*

Proof. We apply the principle of Noetherian induction (**0**, 2.2.2) to the set $\mathfrak{M}$ of closed subsets of X contained in at least one of the E_α; we can thus suppose that every closed subset $Y \neq X$ of X is contained in one of the E_α. The proposition is evident if X is not irreducible, because each of the irreducible components X_i of X ($1 \leqslant i \leqslant m$) is contained in an E_{α_i}, and there exists an E_α containing all of the E_{α_i}. Now suppose that X is irreducible. By hypothesis, there exists a β such that $X = \overline{E_\beta}$, so (9.2.2) E_β contains a nonempty open subset U of X. But then the closed set $X - U$ is contained in an E_γ, and it suffices to take an E_α containing E_β and E_γ. When every irreducible closed subset Y of X admits a generic point y, there exists α such that $y \in E_\alpha$, so $Y = \overline{\{y\}} \subset \overline{E_\alpha}$, and condition (2nd) is therefore a consequence of (1st). □ $\mathbf{0}_{\mathbf{III}}$ | 16

Proposition (9.2.5). — *Let X be a Noetherian space, x a point of X, and E a constructible subset of X. For E to be a neighbourhood of x, it is necessary and sufficient that for every irreducible closed subset Y of X containing x, $E \cap Y$ is dense in Y (if there exists a generic point y of Y, this also implies (9.2.2) that $y \in E$).*

Proof. The condition is evidently necessary; we will prove that it is sufficient. Applying the principle of Noetherian induction to the set $\mathfrak{M}$ of closed subsets Y of X containing x and such that $E \cap Y$ is a neighbourhood of x in Y, we can assume that every closed subset $Y \neq X$ of X containing x belongs to $\mathfrak{M}$. If X is not irreducible, then each of the irreducible components X_i of X containing x are distinct from X, hence $E \cap X_i$ is a neighbourhood of x with respect to X_i; as a result, E is a neighbourhood of x in the union of the irreducible components of X containing x, and as this union is a neighbourhood of x in X, so is E. If X is irreducible, then E is dense in X by hypothesis, so it contains a nonempty open subset U of X (9.2.2); the proposition is then evident if $x \in U$; otherwise, x is by hypothesis inside $E \cap (X - U)$ with respect to $X - U$, so the closure of $X - E$ in X does not contain x, and the complement of this closure is a neighbourhood of x contained in E, which finishes the proof. □

Corollary (9.2.6). — *Let X be a Noetherian space, E a subset of X. For E to be an open set in X, it is necessary and sufficient that for every irreducible closed subset Y of X intersecting E, $E \cap Y$ contains a nonempty open subset of Y.*

Proof. The condition is evidently necessary; conversely, if it is satisfied, then it implies that E is constructible by Proposition (9.2.3). In addition, Proposition (9.2.5) shows that E is then a neighbourhood of each of its points, hence the conclusion. □

9.3. Constructible functions

Definition (9.3.1). — Let h be a map from a topological space X to a set T. We say that h is *constructible* if $h^{-1}(t)$ is constructible for every $t \in T$, and empty except for finitely many values of t; then for every subset S of T, $h^{-1}(S)$ is constructible. We say that h is *locally constructible* if every $x \in X$ has an open neighbourhood V such that $h|V$ is constructible.

Every constructible function is locally constructible; the converse is true when X is quasi-compact and admits a basis formed by the retrocompact open sets in X (in particular, when X is Noetherian).

Proposition (9.3.2). — *Let h be a map from a Noetherian space X to a set T. For h to be constructible, it is necessary and sufficient that for every irreducible closed subset Y of X, there exists a nonempty subset U of Y, open with respect to Y, in which h is constant.*

Proof. The condition is necessary: indeed, by hypothesis, h does not take finitely many values t_i on Y, and each of the sets $h^{-1}(t_i) \cap Y$ is constructible in Y (9.2.1); as they can not all be rare subsets of the space Y, at least one of them contains a nonempty open set (9.2.3).

To see that the condition is sufficient, we apply the principle of Noetherian induction on the set $\mathfrak{M}$ of closed subsets Y of X such that the restriction $h|Y$ is constructible; we can thus assume that for every closed subset $Y \neq X$ of X, $h|Y$ is constructible. If X is not irreducible, then the restriction of h to each of the (finitely many) irreducible components X_i of X is constructible, and it then follows immediately from Definition (9.3.1) that h is constructible. If X is irreducible, then there exists by hypothesis a nonempty open subset U of X on which h is constant; on the other hand, the restriction of h to $X - U$ is constructible by hypothesis, and it follows immediately that h is constructible. □ **0**$_{\mathbf{III}}$ | 17

Corollary (9.3.3). — *Let X be a Noetherian space in which every irreducible closed subset admits a generic point. If h is a map from X to a set T such that, for every $t \in T$, $h^{-1}(t)$ is constructible, then h is constructible.*

Proof. If Y is an irreducible closed subset of X and y its generic point, then $Y \cap h^{-1}(h(y))$ is constructible and contains y, hence (9.2.2) this set contains a nonempty open subset of Y, and it suffices to apply Proposition (9.3.2). □

Proposition (9.3.4). — *Let X be a Noetherian space in which every irreducible closed subset admits a generic point, h a constructible map from X to an ordered set. For h to be upper semi-continuous on X, it is necessary and sufficient that for every $x \in X$ and every specialization* (**0**, 2.1.2) *x' of x, we have $h(x') \leqslant h(x)$.*

Proof. The function h does not take a finite number of values; therefore, to say that it is upper semi-continuous means that for every $x \in X$, the set E of the $y \in X$ such that $h(y) \leqslant h(x)$ is a neighbourhood of x. By hypothesis, E is a constructible subset of X; on the other hand,to say that an irreducible closed subset Y of X contains x means that its generic point y is a specialization of x; the conclusion then follows from Proposition (9.2.5). □

§10. Supplement on flat modules

For any proofs missing in (10.1) and (10.2), we refer the reader to Bourbaki, *Alg. comm.*, chap. II and III.

10.1. Relations between flat modules and free modules

(10.1.1). Let A be a ring, $\mathfrak{J}$ an ideal of A, and M an A-module; for every integer $p \geqslant 0$, we have a canonical homomorphism of $(A/\mathfrak{J})$-modules

(10.1.1.1) $$\phi_p : (M/\mathfrak{J}M) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^p/\mathfrak{J}^{p+1}) \longrightarrow \mathfrak{J}^p M/\mathfrak{J}^{p+1}M,$$

which is evidently *surjective*. We denote by $\mathrm{gr}(A) = \oplus_{p\geqslant 0}\mathfrak{J}^p/\mathfrak{J}^{p+1}$ the graded ring associated to A filtered by the $\mathfrak{J}^p$, and by $\mathrm{gr}(M) = \oplus_{p\geqslant 0}\mathfrak{J}^p M/\mathfrak{J}^{p+1}M$ the graded $\mathrm{gr}(A)$-module associated to M filtered by the $\mathfrak{J}^p M$; we then have $\mathrm{gr}_p(A) = \mathfrak{J}^p/\mathfrak{J}^{p+1}$, and $\mathrm{gr}_p(M) = \mathfrak{J}^p M/\mathfrak{J}^{p+1}M$; the ϕ define a *surjective* homomorphism of graded $\mathrm{gr}(A)$-modules

(10.1.1.2) $$\phi : \mathrm{gr}_0(M) \otimes_{\mathrm{gr}_0(A)} \mathrm{gr}(A) \longrightarrow \mathrm{gr}(M).$$

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(10.1.2). Suppose that *one* of the following hypotheses is satisfied:

(i) $\mathfrak{J}$ is nilpotent;
(ii) A is Noetherian, $\mathfrak{J}$ is contained in the radical of A, and M is of finite type.

Then the following properties are equivalent.

(a) M is a free A-module.
(b) $M/\mathfrak{J}M = N \otimes_A (A/\mathfrak{J})$ is a free $(A/\mathfrak{J})$-module, and $\mathrm{Tor}_1^A(M, A/\mathfrak{J}) = 0$.
(c) $M/\mathfrak{J}M$ is a free $(A/\mathfrak{J})$-module, and the canonical homomorphism (10.1.1.2) is injective (and thus bijective).

(10.1.3). Suppose that $A/\mathfrak{J}$ is a *field* (in other words, that $\mathfrak{J}$ is maximal), and that one of the hypotheses, (i) and (ii), of (10.1.2) is satisfied. Then the following properties are equivalent.

(a) M is a free A-module.
(b) M is a projective A-module.
(c) M is a flat A-module.

(d) $\mathrm{Tor}_1^A(M, A/\mathfrak{J}) = 0$.
(e) The canonical homomorphism (10.1.1.2) is bijective.

This result can be applied, in particular, to the following two cases:

(i) M is an *arbitrary* module, over a local ring A whose maximal ideal $\mathfrak{J}$ is *nilpotent* (for example, a local Artinian ring);
(ii) M is a module *of finite type* over a *local Noetherian* ring.

10.2. Local flatness criteria

(10.2.1). With the hypotheses and notation of (10.1.1), consider the following conditions.

(a) M is a flat A-module.
(b) $M/\mathfrak{J}M$ is a flat $(A/\mathfrak{J})$-module, and $\mathrm{Tor}_1^A(M, A/\mathfrak{J}) = 0$.
(c) $M/\mathfrak{J}M$ is a flat $(A/\mathfrak{J})$-module, and the canonical homomorphism (10.1.1.2) is bijective.
(d) For all $n \geqslant 1$, $M/\mathfrak{J}^n M$ is a flat $(A/\mathfrak{J}^n)$-module.

Then we have the implications

$$(a) \implies (b) \implies (c) \implies (d),$$

and, if $\mathfrak{J}$ is *nilpotent*, then the four conditions are *equivalent*. This is also the case if A is Noetherian and M is *ideally separated*, that is to say, for every ideal $\mathfrak{a}$ of A, the A-module $\mathfrak{a} \otimes_A M$ is *separated* for the $\mathfrak{J}$-preadic topology.

(10.2.2). Let A be a Noetherian ring, B a commutative Noetherian A-algebra, $\mathfrak{J}$ an ideal of A such that $\mathfrak{J}B$ is contained in the radical of B, and M a B-*module* of finite type. Then, when M is considered as an A-*module*, the four conditions of (10.2.1) are *equivalent*. This result applies first and foremost in $\mathbf{0}_{\mathbf{III\text{-}1}}$ | 10
the case where A and B are *local* Noetherian rings, with the homomorphism $A \to B$ being a *local* homomorphism. More specifically, if $\mathfrak{J}$ is then the *maximal* ideal of A, we can, in conditions *(b)* and *(c)*, remove the hypothesis that $M/\mathfrak{J}M$ is flat, since it is automatically satisfied, and condition *(d)* implies that the modules $M/\mathfrak{J}^n M$ are *free* over the $A/\mathfrak{J}^n$.

(10.2.3). With the hypotheses on A, B, $\mathfrak{J}$, and M from the start of (10.2.2), let $\widehat{A}$ be the separated completion of A for the $\mathfrak{J}$-preadic topology, and $\widehat{M}$ the separated completion of M for the $\mathfrak{J}B$-preadic topology. Then, for M to be a flat A-module, it is necessary and sufficient for $\widehat{M}$ to be a flat $\widehat{A}$-module.

(10.2.4). Let $\rho : A \to B$ be a local homomorphism of local Noetherian rings, k the residue field of A, and M and N both B-modules of finite type, with N assumed to be A-*flat*. Let $u : M \to N$ be a B-homomorphism. Then the following conditions are equivalent.

(a) u is injective, and $\mathrm{Coker}(u)$ is a flat A-module.
(b) $u \otimes 1 : M \otimes_A k \to N \otimes_A k$ is injective.

(10.2.5). Let $\rho : A \to B$ and $\sigma : B \to C$ be local homomorphisms of local Noetherian rings, k the residue field of A, and M a C-module of finite type. Suppose that B is a *flat* A-module. Then the following conditions are equivalent.

(a) M is a flat B-module.
(b) M is a flat A-module, and $M \otimes_A k$ is a flat $(B \otimes_A k)$-module.

Proposition (10.2.6). — *Let A and B be local Noetherian rings, $\rho : A \to B$ a local homomorphism, $\mathfrak{J}$ an ideal of B contained in the maximal ideal, and M a B-module of finite type. Suppose that, for all $n \geqslant 0$, $M_n = M/\mathfrak{J}^{n+1}M$ is a flat A-module. Then M is a flat A-module.*

PROOF. We have to prove that, for every injective homomorphism $u : N' \to N$ of A-modules of finite type, $v = 1 \otimes u : M \otimes_A N' \to M \otimes_A N$ is injective. But $M \otimes_A N'$ and $M \otimes_A N$ are B-modules of finite type, and thus separated for the $\mathfrak{J}$-preadic topology ($\mathbf{0}_{\mathbf{I}}$, 7.3.5); it thus suffices to prove that the homomorphism $\widehat{v} : \widehat{M \otimes_A N'} \to \widehat{M \otimes_A N}$ of the separated completions is injective. But $\widehat{v} = \varprojlim v_n$, where v_n is the homomorphism $1 \otimes u : M_n \otimes_A N' \to M_n \otimes_A N$; since, by hypothesis, M_n is A-flat, v_n is injective for all n, and thus so too is v, because the functor $\varprojlim$ is left exact. □

Corollary (10.2.7). — *Let A be a Noetherian ring, B a local Noetherian ring, $\rho : A \to B$ a homomorphism, f an element of the maximal ideal of B, and M a B-module of finite type. Suppose that the homothety $f_M : x \to fx$ on M is injective, and that M/fM is a flat A-module. Then M is a flat A-module.*

PROOF. Let $M_i = f^i M$ for $i \geqslant 0$; since f_M is injective, M_i/M_{i+1} is isomorphic to M/fM, and thus A-flat for all $i \geqslant 0$; the exact sequence

$$0 \longrightarrow M_i/M_{i+1} \longrightarrow M/M_{i+1} \longrightarrow M/M_i \longrightarrow 0$$

gives us, by induction on i, that M/M_i is *A-flat* for all $i \geqslant 0$ ($\mathbf{0_I}$, 6.1.2); we can thus apply (10.2.6). We can also argue directly as follows: for every A-module N of finite type, $M \otimes_A N$ is a B-module of finite type; since f belongs to the radical $\mathfrak{n}$ of B, the (f)-adic topology on $M \otimes_A N$ is finer than the $\mathfrak{u}$-adic topology, and we know that the latter is *separated* ($\mathbf{0_I}$,**0.7.3.5**,) Now, since M/M_i is A-flat, we $\mathbf{0_{III\text{-}1}}$ | 20
have that

$$f^i(M \otimes_A N) = \operatorname{Im}(M_i \otimes_A N \longrightarrow M \otimes_A N) = \operatorname{Ker}(M \otimes_A N \longrightarrow (M/M_i) \otimes_A N)$$

by ($\mathbf{0_I}$, 6.1.2). So let N be an A-module of finite type, and N' a submodule of N, with canonical injection $j : N' \to N$; in the commutative diagram

$$\begin{tikzcd} M \otimes_A N' \arrow[r] \arrow[d, "1_M \otimes j"'] & (M/M_i) \otimes_A N' \arrow[d, "1_{M/M_i} \otimes j"] \\ M \otimes_A N \arrow[r] & (M/M_i) \otimes_A N \end{tikzcd}$$

$1_{M/M_i} \otimes j$ is injective, because M/M_i is A-flat; we thus conclude that

$$\operatorname{Ker}(M \otimes_A N' \longrightarrow M \otimes_A N) \subset \operatorname{Ker}(M \otimes_A N' \longrightarrow (M/M_i) \otimes_A N')$$

for any i; since the intersection (over i) of the latter kernel is 0, as we saw above, so too is the intersection (over i) of the former, and so M is A-flat. □

Proposition (10.2.8). — *Let A be a reduced Noetherian ring, and M an A-module of finite type. Suppose that, for every A-algebra B (which is then a discrete valuation ring), $M \otimes_A B$ is a flat B-module (and thus free* (10.1.3)*). Then M is a flat A-module.*

PROOF. We know that, for M to be flat, it is necessary and suffices for $M_{\mathfrak{m}}$ to be a flat $A_{\mathfrak{m}}$-module for every maximal ideal $\mathfrak{m}$ of A ($\mathbf{0_I}$, 6.3.3); we can thus restrict to the case where A is *local* ($\mathbf{0_I}$, 1.2.8). So let $\mathfrak{m}$ be the maximal ideal of A, $\mathfrak{p}_i$ ($1 \leqslant i \leqslant r$) the minimal prime ideals of A, and k the residue field $A/\mathfrak{m}$. We know (**II**, 7.1.7) that there exists, for each i, a discrete valuation ring B_i that has the same field of fractions K_i as the integral ring $A/\mathfrak{p}_i$, and that, further, dominates $A/\mathfrak{p}_i$. Let $M_i = M \otimes_A B_i$. By hypothesis, M_i is free over B_i, and so, denoting by k_i the residue field of B_i, we have

$$\operatorname{rg}_{k_i}(M_i \otimes_{B_i} k_i) = \operatorname{rg}_{K_i}(M_i \otimes_{B_i} K_i). \tag{10.2.8.1}$$

But it is clear that the composite homomorphism $A \to A/\mathfrak{p}_i \to B_i$ is local, and so k is an extension of k_i, and that we have $M_i \otimes_{B_i} k_i = M \otimes_A k_i = (M \otimes_A k) \otimes_k k_i$, and also that $M_i \otimes_{B_i} K_i = M \otimes_A K_i$. Equation (10.2.8.1) thus implies that

$$\operatorname{rg}_k(M \otimes_A k) = \operatorname{rg}_{K_i}(M \otimes_A K_i) \quad \text{for } 1 \leqslant i \leqslant r$$

and since A is reduced, we know that this condition implies that M is a *free* A-module (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 7). □

10.3. Existence of flat extensions of local rings

Proposition (10.3.1). — *Let A be a local Noetherian ring, with maximal ideal $\mathfrak{J}$, and residue field $k = A/\mathfrak{J}$. Let K be a field extension of k. Then there exists a local homomorphism from A to a local Noetherian ring B, such that $B/\mathfrak{J}B$ is k-isomorphic to K, and such that B is a flat A-module.*

The rest of this section is devoted to proving this proposition, step-by-step.

(10.3.1.1). First suppose that $K = k(T)$, where T is an indeterminate. In the ring of polynomials $A' = A[T]$, consider the prime ideal $\mathfrak{J}' = \mathfrak{J}A$, consisting of the polynomials that have coefficients $\mathbf{0}_{\mathbf{III\text{-}1}}$ | 21 in the ideal $\mathfrak{J}$; it is clear that $A'/\mathfrak{J}'$ is canonically isomorphic to $k[T]$. We will show that the ring of fractions $B = A'_{\mathfrak{J}}$ is that for which we are searching (that is, a ring which satisfies the conditions of the conclusion of the proposition); it is clearly a local Noetherian ring, with maximal ideal $\mathfrak{L} = \mathfrak{J}B$. Further, $B/\mathfrak{L} = (A'/\mathfrak{J}')_{\mathfrak{J}'} = (k[T])_{\mathfrak{J}'}$ is exactly the field of fractions K of $k[T]$. Finally, B is a flat A'-module, and A' is a free A-module, so B is a flat A-module ($\mathbf{0}_{\mathbf{I}}$, 6.2.1).

(10.3.1.2). Now suppose that $K = k(t) = k[t]$, where t is algebraic over k; let $f \in k[T]$ be the minimal polynomial of t; there exists a monic polynomial $F \in A[T]$ whose canonical image in $k[T]$ is f. So let $A' = A[T]$, and let $\mathfrak{J}'$ be the ideal $\mathfrak{J}A' + (F)$ in A'. We will see that the quotient ring $B = A'/(F)$ is that for which we are searching. First of all, it is clear that B is a *free* A-module, and thus flat. The ring $A'/\mathfrak{J}'$ is isomorphic to

$$(A'/\mathfrak{J}A')/\big((\mathfrak{J}A' + (F))/\mathfrak{J}A'\big) = k[T]/(f) = K;$$

the image $\mathfrak{L}$ of $\mathfrak{J}'$ in B is thus maximal, and we evidently have that $\mathfrak{L} = \mathfrak{J}B$. Finally, B is a semi-local ring, because it is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 3 of prop. 9), and its maximal ideals are in bijective correspondence with those of $B/\mathfrak{J}B$ ([**SZ60**, vol. I, p. 259]); the previous arguments then prove that B is a local ring.

Lemma (10.3.1.3). — *Let $(A_\lambda, f_{\mu\lambda})$ be a filtered inductive system of local rings, such that the $f_{\mu\lambda}$ are local homomorphisms; let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ, and let $K_\lambda = A_\lambda/\mathfrak{m}_\lambda$. Then $A' = \varinjlim A_\lambda$ is a local ring, with maximal ideal $\mathfrak{m} = \varinjlim \mathfrak{m}_\lambda$, and residue field $K = \varinjlim K_\lambda$. Further, if $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$ with $\lambda < \mu$, then we have $\mathfrak{m}' = \mathfrak{m}_\lambda A'$ for all λ. If, further, for $\lambda < \mu$, A_μ is a flat A_λ-module, and if all the A_λ are Noetherian, then A' is a flat Noetherian A_λ-modules for all λ.*

Proof. Since, by hypothesis, $(f_\mu\lambda)(\mathfrak{m}_\lambda) \subset \mathfrak{m}_\mu$ for $\lambda < \mu$, the $\mathfrak{m}_\lambda$ form an inductive system, and its limit $\mathfrak{m}'$ is evidently an ideal of A'. Further, if $x' \notin \mathfrak{m}'$, there exists a λ such that $x' = f_\lambda(x_\lambda)$ for some $x_\lambda \in A_\lambda$ (where $f_\lambda : A_\lambda \to A'$ denotes the canonical homomorphism); because $x' \notin \mathfrak{m}'$, we necessarily have that $x_\lambda \notin \mathfrak{m}_\lambda$, and so x_λ admits an inverse y_λ in A_λ, and $y' = f_\lambda(y_\lambda)$ is the inverse of x' in A', which proves that A' is a local ring with maximal ideal $\mathfrak{m}'$; the claim about K follows immediately from the fact that $\varinjlim$ is an exact functor. The hypothesis that $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$ implies that the canonical map $\mathfrak{m}_\lambda \otimes_{A_\lambda} A_\mu \to \mathfrak{m}_\mu$ is surjective; the equality $\mathfrak{m}' = \mathfrak{m}_\lambda A'$ then follows from, again, the fact that the functor $\varinjlim$ is exact and commutes with the tensor product.

Now suppose that, for $\lambda < \mu$, we have $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$, and that A_μ is a flat A_λ-module. Then A' is a flat A_λ-module for all λ, by ($\mathbf{0}_{\mathbf{I}}$, 6.2.3); since A' and A_λ are local rings, and since $\mathfrak{m}' = \mathfrak{m}_\lambda A'$, A' is even a *faithfully flat* A_λ-module ($\mathbf{0}_{\mathbf{I}}$, 6.6.2). Finally, suppose further that the A_λ are *Noetherian*; the $\mathfrak{m}_\lambda$-adic topologies are then separated ($\mathbf{0}_{\mathbf{I}}$, 7.3.5); we now show that, from this, it follows that the $\mathfrak{m}'$-adic topology on A' is *separated*. Indeed, if $x' \in A'$ belongs to all the $\mathfrak{m}'^n$ ($n \geqslant 0$), then it is the image of some $x_\mu \in A_\mu$ for a specific index μ, and since the inverse image in A_μ of $\mathfrak{m}'^n = \mathfrak{m}_\mu^n A'$ is $\mathfrak{m}_\mu^n$ ($\mathbf{0}_{\mathbf{I}}$, 6.6.1), x_μ belongs to all the $\mathfrak{m}_\mu^n$, so $x_\mu = 0$, by hypothesis, and so $x' = 0$. Denote by $\widehat{A}'$ the completion of A' for the $\mathfrak{m}'$-adic topology; the above shows that we have $A' \subset \widehat{A}'$. We will now show that $\widehat{A}'$ is *Noetherian* and A_λ-*flat* for all λ; from this, it will follow that $\widehat{A}'$ is A'-flat ($\mathbf{0}_{\mathbf{I}}$, 6.2.3), $\mathbf{0}_{\mathbf{III\text{-}1}}$ | 22 and since $\mathfrak{m}'\widehat{A}' \neq \widehat{A}'$, that $\widehat{A}'$ is a faithfully flat A'-module ($\mathbf{0}_{\mathbf{I}}$, 6.6.2), whence the final conclusion that A' is *Noetherian* ($\mathbf{0}_{\mathbf{I}}$, 6.5.2), which will finish the proof of the lemma.

We have $\widehat{A}' = \varprojlim_n A'/\mathfrak{m}'^n$; by the fact that A' is A_λ-flat, we have that

$$\mathfrak{m}'^n/\mathfrak{m}'^{n+1} = (\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}) \otimes_{A_\lambda} A' = (\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}) \otimes_{K_\lambda} (K_\lambda \otimes_{A_\lambda} A') = (\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}) \otimes_{K_\lambda} K;$$

since $\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}$ is a K_λ-vector space of finite dimension, $\mathfrak{m}_\lambda'^n/\mathfrak{m}_\lambda'^{n+1}$ is a K-vector space of finite dimension for all $n \geqslant 0$. It thus follows from ($\mathbf{0_I}$, 7.2.12) and ($\mathbf{0_I}$, 7.2.8) that $\widehat{A}'$ is *Noetherian*. We further know that the maximal ideal of $\widehat{A}'$ is $\mathfrak{m}'A'$, and that $\widehat{A}'/\mathfrak{m}'^n\widehat{A}'$ is isomorphic to $A'/\mathfrak{m}'^n$; since $A'/\mathfrak{m}'^n = (A_\lambda/\mathfrak{m}_\lambda^n)\otimes_{A_\lambda} A'$, we see that $A'/\mathfrak{m}'^n$ is a flat $(A_\lambda/\mathfrak{m}_\lambda^n)$-module ($\mathbf{0_I}$, 6.2.1); criterion (10.2.2) is thus applicable to the Noetherian A_λ-algebra $\widehat{A}'$, and shows that $\widehat{A}'$ is A_λ-*flat*. □

(10.3.1.4). We now treat the general case. There exists an ordinal γ and, for every ordinal $\lambda \leqslant \gamma$, a subfield k_λ of K that contains k, such that (i) for all $\lambda < \gamma$, $k_{\lambda+1}$ is an extension of k_λ generated by a single element; (ii) for every limit ordinal μ, $k_\mu = \bigcup_{\lambda<\mu} k_\lambda$; and (iii) $K = k_\gamma$. In fact, it suffices to consider a bijection $\xi \mapsto t_\xi$ from the set of ordinals $\xi \leqslant \beta$ (for some suitable β) to K, and to define k_λ by transfinite induction (for $\lambda \leqslant \beta$) as the union of the k_μ for $\mu < \lambda$ if λ is a limit ordinal, and as $k_\nu(t_\xi)$ if $\lambda = \nu+1$, where ξ is the smallest ordinal such that $t_\xi \notin k_\nu$; γ is then, by definition, the smallest ordinal $\leqslant \beta$ such that $k_\gamma = K$.

With this in mind, we will define, by transfinite induction, a family of local Noetherian rings A_λ for $\lambda \leqslant \gamma$, and local homomorphisms $f_{\mu\lambda} : A_\lambda \to A_\mu$ for $\lambda \leqslant \mu$, satisfying the following conditions:

(i) $(A_\lambda, f_{\mu\lambda})$ is an inductive system, and $A_0 = A$;
(ii) for all λ, we have a k-isomorphism $A_\lambda/\mathfrak{J}A_\lambda \simeq k_\lambda$;
(iii) for $\lambda \leqslant \mu$, A_μ is a flat A_λ-module.

So suppose that the A_λ and the $f_{\mu\lambda}$ are defined for $\lambda < \mu < \xi$, and suppose, first of all, that $\xi = \zeta + 1$, so that $k_\xi = k_\zeta(t)$. If t is transcendental over k_ζ, we define A_ζ, following the procedure of (10.3.1.1), to be equal to $(A_\zeta[t])_{\mathfrak{J}A_{\zeta[t]}}$; the canonical map is $f_{\zeta\xi}$, and, for $\lambda < \zeta$, we take $f_{\xi\lambda} = f_{\xi\zeta}\circ f_{\zeta\lambda}$; the verification of conditions (i) to (iii) is then immediate, given that what we have shown in (10.3.1.1). So now suppose that t is algebraic, and let h be its minimal polynomial in $k_\zeta[T]$, and H a monic polynomial in $A_\zeta[T]$ whose image in $k_\zeta[T]$ is h; we then take A_ξ to be equal to $A_\zeta[T](H)$, with the $f_{\xi\lambda}$ being defined as before; the verification of conditions (i) to (iii) then follows from what we have shown in (10.3.1.2).

Now suppose that ξ has no predecessor; we then take A_ξ to be the inductive limit of the inductive system of local rings $(A_\lambda, f_{\mu\lambda})$ for $\lambda < \xi$; we define $f_{\xi\lambda}$ as the canonical map for $\lambda < \xi$. The fact that A_ξ is local and Noetherian, that the $f_{\xi\lambda}$ are local homomorphism, and that conditions (i) to (iii) are satisfied for $\lambda \leqslant \xi$ then follows from the induction hypothesis, and from Lemma (10.3.1.3). $\mathbf{0_{III\text{-}1}}$ | 23
With this construction, it is clear that the ring $B = A_\gamma$ satisfies the conditions of (10.3.1). □

We note that, by (10.2.1, c), we have a canonical isomorphism

$$\mathrm{gr}(A)\otimes_k K \xrightarrow{\sim} \mathrm{gr}(B). \tag{10.3.1.5}$$

We can also replace B by its $\mathfrak{J}B$-adic completion $\widehat{B}$ without changing the conclusions of (10.3.1), because $\widehat{B}$ is a flat B-module ($\mathbf{0_I}$, 7.3.3), and thus a flat A-module ($\mathbf{0_I}$, 6.2.1).

We have also shown the following:

Corollary (10.3.2). — *If K is an extension of finite degree, then we can assume that B is a finite A-algebra.*

§11. Supplement on homological algebra

11.1. Review of spectral sequences

(11.1.1). In the following, we use a more general notion of a spectral sequence than that defined in (T, 2.4); keeping the notation of (T, 2.4), we call a *spectral sequence* in an abelian category $\mathtt{C}$ a system E consisting of the following parts:

(a) A family (E_r^{pq}) of objects of $\mathtt{C}$ defined for $p,q \in \mathbf{Z}$ and $r \geqslant 2$.
(b) A family of morphisms $d_r^{pq} : E_r^{pq} \to E_r^{p+r,q-r+1}$ such that $d_r^{p+r,q-r+1}d_r^{pq} = 0$. We set $Z_{r+1}(E_r^{pq}) = \mathrm{Ker}(d_r^{pq})$ and $B_{r+1}(E_r^{pq}) = \mathrm{Im}(d_r^{p+r,q-r+1})$, so that
$$B_{r+1}(E_r^{pq}) \subset Z_{r+1}(E_r^{pq}) \subset E_r^{pq}.$$
(c) A family of isomorphisms $\alpha_r^{pq} : Z_{r+1}(E_r^{pq})/B_{r+1}(E_r^{pq}) \simeq E_{r+1}^{pq}$.
We then define for $k \geqslant r+1$, by induction on k, the subobjects $B_k(E_r^{pq})$ and $Z_k(E_r^{pq})$ as the inverse images, under the canonical morphism $E_r^{pq} \to E_r^{pq}/B_{r+1}(E_r^{pq})$ of the subobjects

of this quotient identified via α_r^{pq} with the subobjects $B_k(E_{r+1}^{pq})$ and $Z_k(E_{r+1}^{pq})$ respectively. It is clear that we then have, up to isomorphism,

(11.1.1.1)
$$Z_k(E_r^{pq})/B_k(E_r^{pq}) = E_k^{pq} \text{ for } k \geqslant r+1,$$

and, if we set $B_r(E_r^{pq}) = 0$ and $Z_r(E_r^{pq}) = E_r^{pq}$, then we have the inclusion relations

(11.1.1.1.2)
$$0 = B_r(E_r^{pq}) \subset B_{r+1}(E_r^{pq}) \subset B_{r+2}(E_r^{pq}) \subset \cdots \subset Z_{r+2}(E_r^{pq}) \subset Z_{r+1}(E_r^{pq}) \subset Z_r(E_r^{pq}) = E_r^{pq}.$$

The other parts of the data of E are then:

(d) Two subobjects $B_\infty(E_2^{pq})$ and $Z_\infty(E_2^{pq})$ of E_2^{pq} such that we have $B_\infty(E_2^{pq}) \subset Z_\infty(E_2^{pq})$ and, for every $k \geqslant 2$,
$$B_k(E_2^{pq}) \subset B_\infty(E_2^{pq}) \text{ and } Z_\infty(E_2^{pq}) \subset Z_k(E_2^{pq}).$$

We set

(11.1.1.1.3)
$$E_\infty^{pq} = Z_\infty(E_2^{pq})/B_\infty(E_2^{pq}).$$

(e) A family (E^n) of objects of $\mathtt{C}$, each equipped with a *decreasing filtration* $(F^p(E^n))_{p\in\mathbf{Z}}$. As **$0_{\mathbf{III}}$** | 24
usual, we denote by $\mathrm{gr}(E^n)$ the graded object associated to the filtered object E^n, the direct sum of the $\mathrm{gr}_p(E^n) = F^p(E^n)/F^{p+1}(E^n)$.

(f) For every pair $(p,q) \in \mathbf{Z}\times\mathbf{Z}$, an isomorphism $\beta^{pq} : E_\infty^{pq} \simeq \mathrm{gr}_p(E^{p+q})$.

The family (E^n), without the filtrations, is called the *abutment* (or *limit*) of the spectral sequence E.

Suppose that the category $\mathtt{C}$ admits infinite direct sums, or that for every $r \geqslant 2$ and every $n \in \mathbf{Z}$, there are finitely many pairs (p,q) such that $p+q=n$ and $E_r^{pq} \neq 0$ (it suffices for it to hold for $r=2$). Then the $E_r^{(n)} = \sum_{p+q=n} E_r^{pq}$ are defined, and we if denote by $d_r^{(n)}$ the morphism $E_r^{(n)} \to E_r^{(n+1)}$ whose restriction to E_r^{pq} is d_r^{pq} for every pair (p,q) such that $p+q=n$, then $d_r^{(n+1)} \circ d_r^{(n)} = 0$, in other words, $(E_r^{(n)})_{n\in\mathbf{Z}}$ is a *complex* $E_r^{(\bullet)}$ in $\mathtt{C}$, with differentials of degree $+1$, and it follows from (c) that $\mathrm{H}^n(E_n^{(\bullet)})$ is *isomorphic to* $E_{r+1}^{(n)}$ for every $r \geqslant 2$.

(11.1.2). A *morphism* $u : E \to E'$ from a spectral sequence E to a spectral sequence $E' = (E_r'^{pq}, E'^n)$ consists of systems of morphisms $u_r^{pq} : E_r^{pq} \to E_r'^{pq}$ and $u^n : E^n \to E'^n$, the u^n compatible with the filtrations on E^n and E'^n, and the diagrams

$$\begin{tikzcd}
E_r^{pq} \arrow[r, "d_r^{pq}"] \arrow[d, "u_r^{pq}"'] & E_r^{p+r,q-r+1} \arrow[d, "u_r^{p+r,q-r+1}"] \\
E_r'^{pq} \arrow[r, "d_r'^{pq}"] & E_r''^{p+r,q-r+1}
\end{tikzcd}$$

being commutative; in addition, by passing to quotients, u_r^{pq} gives a morphism $\overline{u}_r^{pq} : Z_{r+1}(Z_r^{pq})/B_{r+1}(E_r^{pq}) \to Z_{r+1}(E_r'^{pq})/B_{r+1}(E_r'^{pq})$ and we must have $\alpha_r'^{pq} \circ \overline{u}_r^{pq} = u_{r+1}^{pq} \circ \alpha_r^{pq}$; finally, we must have $u_2^{pq}(B_\infty(E_2^{pq})) \subset B_\infty(E_2'^{pq})$ and $u_2^{pq}(Z_\infty(E_2^{pq})) \subset Z_\infty(E_2'^{pq})$; by passing to quotients, u_2^{pq} then gives a morphism $u_\infty'^{pq} : E_\infty^{pq} \to E_\infty'^{pq}$, and the diagram

$$\begin{tikzcd}
E_\infty^{pq} \arrow[r, "u_\infty'^{pq}"] \arrow[d, "\beta^{pq}"'] & E_\infty'^{pq} \arrow[d, "\beta'^{pq}"] \\
\mathrm{gr}_p(E^{p+q}) \arrow[r, "\mathrm{gr}_p(u^{p+q})"] & \mathrm{gr}_p(E'^{p+q})
\end{tikzcd}$$

must be commutative.

The above definitions show, by induction on r, that if the u_2^{pq} are *isomorphisms*, then so are the u_r^{pq} for $r \geqslant 2$; *if in addition we know that $u_2^{pq}(B_\infty(E_2^{pq})) = B_\infty(E_2'^{pq})$ and $u_2^{pq}(Z_\infty(E_2^{pq})) = Z_\infty(E_2'^{pq})$ and the u^n are isomorphisms*, then we can conclude that u is an *isomorphism*.

(11.1.3). Recall that if $(F^p(X))_{p\in\mathbf{Z}}$ is a (decreasing) *filtration* of an object $X \in \mathtt{C}$, then we say that this filtration is *separated* if $\inf(F^p(X)) = 0$, *discrete* if there exists a p such that $F^p(X) = 0$, *exhaustive* (or *coseparated*) if $\sup(F^p(X)) = X$, *codiscrete* if there exists a p such that $F^p(X) = X$.

We say that a spectral sequence $E = (E_r^{pq}, E^n)$ is *weakly convergent* if we have $B_\infty(E_2^{pq}) = \sup_k(B_k(E_2^{pq}))$ and $Z_\infty(E_2^{pq}) = \inf_k(Z_k(E_2^{pq}))$ (in other words, the objects of $B_\infty(E_2^{pq})$ and $Z_\infty(E_2^{pq})$ are determined from the data of (a) and (c) of the spectral sequence E). We say that the spectral sequence E is *regular* if it is weakly convergent and if, in addition:

(1st) For every pair (p,q), the decreasing sequence $(Z_k(E_2^{pq}))_{k\geqslant 2}$ is *stable*; the hypothesis that E is weakly convergent then implies that $Z_\infty(E_2^{pq}) = Z_k(E_2^{pq})$ for k large enough (depending on p and q).

(2nd) For every n, the filtration $(F^p(E^n))_{p\in\mathbf{Z}}$ of E^n is *discrete* and *exhaustive*.

We say that the spectral sequence E is *coregular* if it is weakly convergent and if, in addition:

(3rd) For every pair (p,q), the increasing sequence $(B_k(E_2^{pq}))_{k\geqslant 2}$ is *stable*, which implies that $B_\infty(E_2^{pq}) = B_k(E_2^{pq})$, and as a result, $E_\infty^{pq} = \inf E_k^{pq}$.

(4th) For every n, the filtration of E^n is *codiscrete*.

Finally, we say that E is *biregular* if it is both regular and coregular, in other words if we have the following conditions:

(a) For every pair (p,q), the sequences $(B_k(E_2^{pq}))_{k\geqslant 2}$ and $(Z_k(E_2^{pq}))_{k\geqslant 2}$ are *stable* and we have $B_\infty(E_2^{pq}) = B_k(E_2^{pq})$ and $Z_\infty(E_2^{pq}) = Z_k(E_2^{pq})$ for k large enough (which implies that $E_\infty^{pq} = E_k^{pq}$).

(b) For every n, the filatration $(F^p(E^n))_{p\in\mathbf{Z}}$ is *discrete* and *codiscrete* (which we also call *finite*).

The spectral sequences defined in (T, 2.4) are thus biregular spectral sequences.

(11.1.4). Suppose that in the category $\mathtt{C}$, filtered inductive limits exist and the functor $\varinjlim$ is *exact* (which is equivalent to saying that the axiom (AB 5) of (T, 1.5) is satisfied (cf. T, 1.8)). The condition that the filtration $(F^p(X))_{p\in\mathbf{Z}}$ of an object $X \in \mathtt{C}$ is exhaustive is then expressed as $\varinjlim_{p\to-\infty} F^p(X) = X$. If a spectral sequence E is weakly convergent, then we have $B_\infty(E_2^{pq}) = \varinjlim_{k\to\infty} B_k(E_2^{pq})$; if in addition $u : E \to E'$ is a morphism from E to a weakly convergent spectral sequence E' in $\mathtt{C}$, then we have $u_2^{pq}(B_\infty(E_2^{pq})) = B_\infty(E_2'^{pq})$, by the exactness of $\varinjlim$. In addition:

Proposition (11.1.5). — *Let $\mathtt{C}$ be an abelian category in which filtered inductive limits are exact, E and E' two regular spectral sequences in $\mathtt{C}$, $u : E \to E'$ a morphism of spectral sequences. If the u_2^{pq} are isomorphisms, then so is u.*

Proof. We already know (11.1.2) that the u_r^{pq} are isomorphisms and that

$$u_2^{pq}(B_\infty(E_2^{pq})) = B_\infty(E_2'^{pq});$$

the hypothesis that E and E' are regular also implies that $u_2^{pq}(Z_\infty(E_2^{pq})) = Z_\infty(E_2'^{pq})$, and as u_2^{pq} is an isomorphism, so is $u_\infty'^{pq}$; we thus conclude that $\mathrm{gr}_p(u^{p+q})$ is also an isomorphism. But as the filtrations of the E^n and the E'^n are discrete and exhaustive, this implies that the u^n are also isomorphisms (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, th. 1). $\mathbf{0_{III}}$ | 26 □

(11.1.6). It follows from (11.1.1.2) and the definition (11.1.1.3) that if, for a spectral sequence E, we have $E_r^{pq} = 0$, then we have $E_k^{pq} = 0$ for $k \geqslant r$ and $E_\infty^{pq} = 0$. We say that a spectral sequence *degenerates* if there exists an integer $r \geqslant 2$ and, for every integer $n \in \mathbf{Z}$, an integer $q(n)$ such that $E_r^{n-q,q} = 0$ *for every* $q \neq q(n)$. We first deduce from the previous remark that we also have $E_k^{n-q,q} = 0$ for $k \geqslant r$ (including $k = \infty$) and $q \neq q(n)$. In addition, the definition of E_{r+1}^{pq} shows that we have $E_{r+1}^{n-q(n),q(n)} = E_r^{n-q(n),q(n)}$; if E is *weakly convergent*, then we also have $E_\infty^{n-q(n),q(n)} = E_r^{n-q(n),q(n)}$; in other words, for every $n \in \mathbf{Z}$, $\mathrm{gr}_p(E^n) = 0$ for $p \neq q(n)$ and $\mathrm{gr}_{q(n)}(E^n) = E_r^{n-q(n),q(n)}$. If in addition the filtration of E^n is *discrete* and *exhaustive*, then the spectral sequence E is *regular*, and we have $E^n = E_r^{n-q(n),q(n)}$ up to unique isomorphism.

(11.1.7). Suppose that filtered inductive limits exist and are exact in the category $\mathtt{C}$, and let $(E_\lambda, u_{\mu\lambda})$ be an inductive system (over a filtered set of indices) of spectral sequeneces in $\mathtt{C}$. Then the *inductive limit* of this inductive system exists in the additive category of spectral sequences of objects of $\mathtt{C}$: to see this, it suffices to define E_r^{pq}, d_r^{pq}, α_r^{pq}, $B_\infty(E_2^{pq})$, $Z_\infty(E_2^{pq})$, E^n, $F^p(E^n)$, and β^{pq} as the respective inductive limits of the $E_{r,\lambda}^{pq}$, $d_{r,\lambda}^{pq}$, $\alpha_{r,\lambda}^{pq}$, $B_\infty(E_{2,\lambda}^{pq})$, $Z_\infty(E_{2,\lambda}^{pq})$, E_λ^n, $F^p(E_\lambda^n)$, and β_λ^{pq}; the verification of the conditions of (11.1.1) follows from the exactness of the functor $\varinjlim$ on $\mathtt{C}$.

Remark (11.1.8). — Suppose that the category $\mathtt{C}$ is the category of A-modules over a *Noetherian* ring A (resp. a ring A). Then the definitions of (11.1.1) show that if, for a given r, the E_r^{pq} are A-modules *of finite type* (resp. *of finite length*), then so are each of the modules E_s^{pq} for $s \geqslant r$, hence so is E_∞^{pq}. If in addition the filtration of the abutment/limit (E^n) is *discrete* or *codiscrete* for all n, then we conclude that each of the E^n is also an A-module *of finite type* (resp. *of finite length*).

(11.1.9). We will have to consider conditions which ensure that a spectral sequence E is biregular is a "uniform" way in $p+q=n$. We will then use the following lemma:

Lemma (11.1.10). — *Let (E_r^{pq}) be a family of objects of $\mathtt{C}$ related by the data of* (a), (b), *and* (c) *of* (11.1.1). *For a fixed integer n, the following properties are equivalent:*

(a) *There exists an integer $r(n)$ such that for $r \geqslant r(n)$, $p+q=n$ or $p+q=n-1$, the morphisms d_r^{pq} are zero.*
(b) *There exists an integer $r(n)$ such that for $p+q=n$ or $p+q=n+1$, we have $B_r(E_2^{pq}) = B_s(E_2^{pq})$ for $s \geqslant r \geqslant r(n)$.*
(c) *There exists an integer $r(n)$ such that for $p+q=n$ or $p+q=n-1$, we have $Z_r(E_2^{pq}) = Z_s(E_2^{pq})$ for $s \geqslant r \geqslant r(n)$.*
(d) *There exists an integer $r(n)$ such that for $p+q=n$, we have $B_r(E_2^{pq}) = B_s(E_2^{pq})$ and $Z_r(E_2^{pq}) = Z_s(E_2^{pq})$ for $s \geqslant r \geqslant r(n)$.*

PROOF. According to the conditions (a), (b), and (c) of (11.1.1), we have that saying $Z_{r+1}(E_2^{pq}) = Z_r(E_2^{pq})$ is equivalent to saying that $d_r^{pq}=0$ and that saying $B_r(E_2^{p+r,q-r+1}) = B_{r+1}(E_2^{p+r,q-r+1})$ is equivalent to saying that $d_r^{pq}=0$; the lemma immediately follows from this remark. □

11.2. The spectral sequence of a filtered complex

(11.2.1). Given an abelian category $\mathtt{C}$, we will agree to denote by notation such as $K^\bullet$ the *complexes* $(K^i)_{i\in\mathbf{Z}}$ of objects of $\mathtt{C}$ whose differential is of degree $+1$, and by the notation such as $K_\bullet$ the complexes $(K_i)_{i\in\mathbf{Z}}$ of objects of $\mathtt{C}$ whose differential is of degree -1. To each complex $K^\bullet = (K^i)$ whose differential d is of degree $+1$, we can associate a complex $K'_\bullet = (K'_i)$ by setting $K'_i = K^{-i}$, the differential $K'_i \to K'_{i-1}$ being the operator $d : K^{-i} \to K^{-i-1}$; and *vice versa*, which, depending on the circumstances, will allow one to consider either one of the types of complexes and translate any result from one type into results for the other. We similarly denote by notation such as $K^{\bullet\bullet} = (K^{ij})$ (resp. $K_{\bullet\bullet} = (K_{ij})$) the *bicomplexes* (or *double complexes*) of objects of $\mathtt{C}$ in which the *two* differentials are of degree $+1$ (resp. -1); we can still pass from one type to the other by changing the signs of the indices, and we have similar notation and remarks for any multicomplexes. The notation $K^\bullet$ and $K_\bullet$ will also be used for $\mathbf{Z}$-*graded objects* of $\mathtt{C}$, which are not necessarily complexes (they can be considered as such for the *zero* differentials); for example, we write $\mathrm{H}^\bullet(K^\bullet) = (\mathrm{H}^i(K^\bullet))_{i\in\mathbf{Z}}$ for the *cohomology* of a complex $K^\bullet$ whose differential is of degree $+1$, and $H_\bullet(K_\bullet) = (\mathrm{H}_i(K_\bullet))_{i\in\mathbf{Z}}$ for the *homology* of a complex $K_\bullet$ whose differential is of degree -1; when we pass from $K^\bullet$ to $K'_\bullet$ by the method described above, we have $\mathrm{H}_i(K'_\bullet) = \mathrm{H}^{-i}(K^\bullet)$.

Recall in this case that for a complex $K^\bullet$ (resp. $K_\bullet$), we will write in general $Z^i(K^\bullet) = \mathrm{Ker}(K^i \to K^{i+1})$ ("object of cocycles") and $B^i(K^\bullet) = \mathrm{Im}(K^{i-1} \to K^i)$ ("object of coboundaries") (resp. $Z_i(K_\bullet) = \mathrm{Ker}(K_i \to K_{i-1})$ ("object of cycles") and $B_i(K_\bullet) = \mathrm{Im}(K_{i+1} \to K_i)$ ("object of boundaries")) so that $\mathrm{H}^i(K^\bullet) = Z^i(K^\bullet)/B^i(K^\bullet)$ (resp. $\mathrm{H}_i(K_\bullet) = Z_i(K_\bullet)/B_i(K_\bullet)$).

If $K^\bullet = (K^i)$ (resp. $K_\bullet = (K_i)$) is a complex in $\mathtt{C}$ and $T : \mathtt{C} \to \mathtt{C}'$ a functor from $\mathtt{C}$ to an abelian category $\mathtt{C}'$, then we denote by $T(K^\bullet)$ (resp. $T(K_\bullet)$) the complex $(T(K^i))$ (resp. $(T(K_i))$) in $\mathtt{C}'$.

We will not review the definition of the ∂-*functors* (T, 2.1), except to note that we *also* say ∂-functor in place of ∂^*-functor when the morphism ∂ decreases the degree of a unit, the context clarifying this point if there is cause for confusion.

Finally, we say that a *graded object* $(A_i)_{i\in\mathbf{Z}}$ of $\mathtt{C}$ is *bounded below* (resp. *above*) if there exists an i_0 such that $A_i = 0$ for $i < i_0$ (resp. $i > i_0$).

(11.2.2). Let $K^\bullet$ be a complex in $\mathtt{C}$ whose differential d is of degree $+1$, and suppose it is equipped with a *filtration* $F(K^\bullet) = (F^p(K^\bullet))_{p\in\mathbf{Z}}$ consisting of *graded* subobjects of $K^\bullet$, in other words, $F^p(K^\bullet) =$ $\mathbf{0_{III}}$ | 28
$(K^i \cap F^p(K^\bullet))_{i\in\mathbf{Z}}$; in addition, we assume that $d(F^p(K^\bullet)) \subset F^p(K^\bullet)$ for every $p \in \mathbf{Z}$. Let us quickly recall how one *functorially* defines a spectral sequence $E(K^\bullet)$ from $K^\bullet$ (M, XV, 4 and G, I, 4.3). For $r \geqslant 2$, the canonical morphism $F^p(K^\bullet)/F^{p+r}(K^\bullet) \to F^p(K^\bullet)/F^{p+1}(K^\bullet)$ defines a morphism in cohomology

$$\mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+r}(K^\bullet)) \longrightarrow \mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+1}(K^\bullet)).$$

We denote by $Z_r^{pq}(K^\bullet)$ the image of this morphism. Similarly, from the exact sequence

$$0 \longrightarrow F^p(K^\bullet)/F^{p+1}(K^\bullet) \longrightarrow F^{p-r+1}(K^\bullet)/F^{p+1}(K^\bullet) \longrightarrow F^{p-r+1}(K^\bullet)/F^p(K^\bullet) \longrightarrow 0,$$

we deduce from the exact sequence in cohomology a morphism

$$\mathrm{H}^{p+q-1}(F^{p-r+1}(K^\bullet)/F^p(K^\bullet)) \longrightarrow \mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+1}(K^\bullet)),$$

and we denote by $B_r^{pq}(K^\bullet)$ the image of this morphism; we show that $B_r^{pq}(K^\bullet) \subset Z_r^{pq}(K^\bullet)$ and we take $E_r^{pq}(K^\bullet) = Z_r^{pq}(K^\bullet)/B_r^{pq}(K^\bullet)$; we will not specify the definition of the d_r^{pq} or the α_r^{pq}.

We note here that all the $Z_r^{pq}(K^\bullet)$ and $B_r^{pq}(K^\bullet)$, for p and q fixed, are subobjects of the same object $\mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+1}(K^\bullet))$, which we denote by $Z_1^{pq}(K^\bullet)$; we set $B_1^{pq}(K^\bullet) = 0$, so that the above definitions of $Z_r^{pq}(K^\bullet)$ and $B_r^{pq}(K^\bullet)$ also work for $r = 1$; we still set $E_1^{pq}(K^\bullet) = Z_1^{pq}(K^\bullet)$. We define d_1^{pq} and α_1^{pq} such that the conditions of (11.1.1) are satisfied for $r = 1$. On the other hand, we define the subobjects $Z_\infty^{pq}(K^\bullet)$ as the image of the morphism

$$\mathrm{H}^{p+q}(F^p(K^\bullet)) \longrightarrow \mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+1}(K^\bullet)) = E_1^{pq}(K^\bullet),$$

and $B_\infty^{pq}(K^\bullet)$ as the image of the morphism

$$\mathrm{H}^{p+q-1}(K^\bullet/F^p(K^\bullet)) \longrightarrow \mathrm{H}^{p+q}(F^p(K^\bullet)/F^{p+1}(K^\bullet)) = E_1^{pq}(K^\bullet),$$

induced as above from the exact sequence in cohomology. We set $Z_\infty(E_2^{pq}(K^\bullet))$ and $B_\infty(E_2^{pq}(K^\bullet))$ to be the canonical images of $E_2^{pq}(K^\bullet)$ in $Z_\infty^{pq}(K^\bullet)$ and $B_\infty^{pq}(K^\bullet)$.

Finally, we denote by $F^p(\mathrm{H}^n(K^\bullet))$ the image in $\mathrm{H}^n(K^\bullet)$ of the morphism $\mathrm{H}^n(F^p(K^\bullet)) \to \mathrm{H}^n(K^\bullet)$ induced from the canonical injection $F^p(K^\bullet) \to K^\bullet$; by the exact sequence in cohomology, this is also the kernel of the morphism $\mathrm{H}^n(K^\bullet) \to \mathrm{H}^n(K^\bullet/F^p(K^\bullet))$. This defines a filtration on $E^n(K^\bullet) = \mathrm{H}^n(K^\bullet)$; we will not give here the definition of the isomorphisms β^{pq}.

(11.2.3). The *functorial* nature of $E(K^\bullet)$ is understood in the following way: given two *filtered* complexes $K^\bullet$ and $K'^\bullet$ in $\mathtt{C}$ and a morphism of complexes $u : K^\bullet \to K'^\bullet$ that is *compatible with the filtrations*, we induce in an evident way the morphisms u_r^{pq} (for $r \geqslant 1$) and u^n, and we show that these morphisms are compatible with the d_r^{pq}, α_r^{pq}, and β^{pq} in the sense of (11.1.2), and thus given a well-defined morphism $E(u) : E(K^\bullet) \to E(K'^\bullet)$ of spectral sequences. In addition, we show that if u and v are morphisms $K^\bullet \to K'^\bullet$ of the above type, *homotopic in degree* $\leqslant k$, then $u_r^{pq} = v_r^{pq}$ for $r > k$ and $u^n = v^n$ for all n (M, XV, 3.1).

(11.2.4). Suppose that filtered inductive limits in $\mathtt{C}$ are exact. Then if the filtration $(F^p(K^\bullet))$ of $\mathbf{0_{III}}$ | 29
$K^\bullet$ is *exhaustive*, then so is the filatration $(F^p(\mathrm{H}^n(K^\bullet)))$ for all n, since by hypothesis we have $K^\bullet = \varinjlim_{p\to-\infty} F^p(K^\bullet)$ and since the hypothesis on $\mathtt{C}$ implies that cohomology commutes with inductive limits. In addition, for the same reason, we have $B_\infty(E_2^{pq}(K^\bullet)) = \sup_k B_k(E_2^{pq}(K^\bullet))$. We say that the filtration $(F^p(K^\bullet))$ of $K^\bullet$ is *regular* if for every n there exists an integer $u(n)$ such that $\mathrm{H}^n(F^p(K^\bullet)) = 0$ for $p > u(n)$. This is particularly the case when the filtration of $K^\bullet$ is *discrete*. When the filtration of $K^\bullet$ is regular and exhaustive, and filtered inductive limits are exact in $\mathtt{C}$, we have (M, XV, 4) that the spectral sequence $E(K^\bullet)$ is *regular*.

11.3. The spectral sequences of a bicomplex

(11.3.1). With regard the conventions for bicomplexes, we follow those of (T, 2.4) rather than those of (M), the two differentials d' and d'' (of degree $+1$) of such a bicomplex $K^{\bullet\bullet} = (K^{ij})$ thus being assumed to be *permutable*. Suppose that *one* of the following two conditions is satisfied:

(1) *Infinite direct sums* exist in $\mathtt{C}$;
(2) For all $n \in \mathbf{Z}$, there is only a *finite* number of pairs (p,q) such that $p+q=n$ and $K^{pq} \neq 0$.

Then the bicomplex $K^{\bullet\bullet}$ defines a (simple) *complex* $(K'^n)_{n\in\mathbf{Z}}$ with $K'^n = \sum_{i+j=n} K^{ij}$, the differential d (of degree $+1$) of this complex being given by $dx = d'x + (-1)^i d''x$ for $x \in K^{ij}$. When we later speak of the spectral sequence of a (simple) *complex* that is *defined by a bicomplex* $K^{\bullet\bullet}$, it will always be understood that one of the above conditions is satisfied. We adopt the analogous conventions for multicomplexes.

We denote by $K^{i,\bullet}$ (resp. $K^{\bullet,j}$) the simple complex $(K^{ij})_{j\in\mathbf{Z}}$ (resp. $(K^{ij})_{i\in\mathbf{Z}}$), by $\mathrm{Z}^p_{\mathrm{II}}(K^{i,\bullet})$, $B^p_{\mathrm{II}}(K^{i,\bullet})$, $\mathrm{H}^p_{\mathrm{II}}(K^{i,\bullet})$ (resp. $\mathrm{Z}^p_{\mathrm{I}}(K^{\bullet,j})$, $B^p_{\mathrm{I}}(K^{\bullet,j})$, $\mathrm{H}^p_{\mathrm{I}}(K^{\bullet,j})$) its p objects of cocycles, of coboundaries, and of cohomology, respectively; the differential $d' : K^{i,\bullet} \to K^{i+1,\bullet}$ is a morphism of complexes, which thus gives an operator on the cocycles, coboundaries, and cohomology,

$$d' : \mathrm{Z}^p_{\mathrm{II}}(K^{i,\bullet}) \longrightarrow \mathrm{Z}^p_{\mathrm{II}}(K^{i+1,\bullet})$$
$$d' : B^p_{\mathrm{II}}(K^{i,\bullet}) \longrightarrow B^p_{\mathrm{II}}(K^{i+1,\bullet})$$
$$d' : \mathrm{H}^p_{\mathrm{II}}(K^{i,\bullet}) \longrightarrow \mathrm{H}^p_{\mathrm{II}}(K^{i+1,\bullet})$$

and it is clear that for these operators, $(\mathrm{Z}^p_{\mathrm{II}}(K^{i,\bullet}))_{i\in\mathbf{Z}}$, $(B^p_{\mathrm{II}}(K^{i,\bullet}))_{i\in\mathbf{Z}}$, and $(\mathrm{H}^p_{\mathrm{II}}(K^{i,\bullet}))_{i\in\mathbf{Z}}$ are complexes; we denote the complex $(\mathrm{H}^p_{\mathrm{II}}(K^{i,\bullet}))_{i\in\mathbf{Z}}$ by $\mathrm{H}^p_{\mathrm{II}}(K^{\bullet\bullet})$, and its q objects of cocycles, coboundaries, and cohomology by $\mathrm{Z}^q_{\mathrm{I}}(\mathrm{H}^p_{\mathrm{II}}(K^{\bullet\bullet}))$, $B^q_{\mathrm{I}}(\mathrm{H}^p_{\mathrm{II}}(K^{\bullet\bullet}))$, and $\mathrm{H}^q_{\mathrm{I}}(\mathrm{H}^p_{\mathrm{II}}(K^{\bullet\bullet}))$. We similarly define the complexes $\mathrm{H}^p_{\mathrm{I}}(K^{\bullet\bullet})$ and their cohomology objects $\mathrm{H}^q_{\mathrm{II}}(\mathrm{H}^p_{\mathrm{I}}(K^{\bullet\bullet}))$. Recall, however, that $\mathrm{H}^n(K^{\bullet\bullet})$ denotes the n object of the cohomology of the *(simple)* complex defined by $K^{\bullet\bullet}$.

(11.3.2). On the complex defined by a bicomplex $K^{\bullet\bullet}$, we can consider two canonical filtrations, $(F^p_{\mathrm{I}}(K^{\bullet\bullet}))$ and $(F^p_{\mathrm{II}}(K^{\bullet\bullet}))$, given by

(11.3.2.1) $$F^p_{\mathrm{I}}(K^{\bullet\bullet}) = \left(\sum_{i+j=n, i\geqslant p} K^{ij}\right)_{n\in\mathbf{Z}} \quad\text{and}\quad F^p_{\mathrm{II}}(K^{\bullet\bullet}) = \left(\sum_{i+j=n, j\geqslant p} K^{ij}\right)_{n\in\mathbf{Z}}$$

which, by definition, are graded subobjects of the (simple) complex defined by $K^{\bullet\bullet}$, and thus make $\mathbf{0_{III}}$ | 30
this complex a filtered complex; moreover, is is clear that these filtrations are *exhaustive* and *separated*.

There corresponds to each of these filtrations a spectral sequence (11.2.2); we denote by $'E(K^{\bullet\bullet})$ and $''E(K^{\bullet\bullet})$ the spectral sequences corresponding to $(F^p_{\mathrm{I}}(K^{\bullet\bullet}))$ and $(F^p_{\mathrm{II}}(K^{\bullet\bullet}))$ respectively, called the *spectral sequence of the bicomplex* $K^{\bullet\bullet}$, and both having as their abutment the cohomology $(\mathrm{H}^n(K^{\bullet\bullet}))$. We can further show (M, XV, 6) that we have

(11.3.2.2) $$'E^{pq}_2(K^{\bullet\bullet}) = \mathrm{H}^p_{\mathrm{I}}(\mathrm{H}^p_{\mathrm{II}}(K^{\bullet\bullet})) \quad\text{and}\quad ''E^{pq}_2(K^{\bullet\bullet})) = \mathrm{H}^p_{\mathrm{II}}(\mathrm{H}^p_{\mathrm{I}}(K^{\bullet\bullet})).$$

Every morphism $u : K^{\bullet\bullet} \to K'^{\bullet\bullet}$ of bicomplexes is *ipso facto* compatible with the filtrations of the same type of $K^{\bullet\bullet}$ and $K'^{\bullet\bullet}$, thus defines a morphism for each of the two spectral sequences; in addition, two *homotopic* morphisms define a homotopy *of order* $\leqslant 1$ of the corresponding filtered (simple) complexes, thus the *same* morphism for each of the two spectral sequences (M, XV, 6.1).

Proposition (11.3.3). — *Let $K^{\bullet\bullet} = (K^{ij})$ be a bicomplex in an abelian category $\mathtt{C}$.*

(i) *If there exist i_0 and j_0 such that $K^{ij} = 0$ for $i < i_0$ or $j < j_0$ (resp. $i > i_0$ or $j > j_0$), then the two spectral sequences $'E(K^{\bullet\bullet})$ and $''E(K^{\bullet\bullet})$ are biregular.*
(ii) *If there exist i_0 and i_1 such that $K^{ij} = 0$ for $i < i_0$ or $i > i_1$ (resp. if there exist j_0 and j_1 such that $K^{ij} = 0$ for $j < j_0$ or $j > j_1$), then the two spectral sequences $'E(K^{\bullet\bullet})$ and $''E(K^{\bullet\bullet})$ are biregular.*
(iii) *If there exists i_0 such that $K^{ij} = 0$ for $i > i_0$ (resp. if there exists j_0 such that $K^{ij} = 0$ for $j < j_0$), then the spectral sequence $'E(K^{\bullet\bullet})$ is regular.*
(iv) *If there exists i_0 such that $K^{ij} = 0$ for $i < i_0$ (resp. if there exists j_0 such that $K^{ij} = 0$ for $j > j_0$), then the spectral sequence $''E(K^{\bullet\bullet})$ is regular.*

PROOF. The proposition follows immediately from the definitions (11.1.3) and from (11.2.4), as well as from the following observations relating to the filtration F_{I} (and similar observations that we can deduce for F_{II} by exchange the roles of the two indices in $K^{\bullet\bullet}$):

1° If there exists i_0 such that $K^{ij} = 0$ for $i > i_0$, then the filtration $F_{\mathrm{I}}(K^{\bullet\bullet})$ is *discrete*.

2° If there exists i_0 such that $K^{ij} = 0$ for $i < i_0$, then the filtration $F_{\mathrm{I}}(K^{\bullet\bullet})$ is *co-discrete*. We can then immediate deduce that it is the same for the corresponding filtration $F_{\mathrm{I}}(\mathrm{H}^n(K^{\bullet\bullet}))$ for all n. Furthermore, the definition of the B_r^{pq} corresponding to the filtration $F_{\mathrm{I}}(K^{\bullet\bullet})$ (11.2.2) shows that for any pair (p,q), the sequence $(B_r^{pq})_{r\leqslant 2}$ is stable.

3° If there exists j_0 such that $K^{ij} = 0$ for $j < j_0$, then we have
$$F_{\mathrm{I}}^{p+r}(K^{\bullet\bullet}) \cap (\sum_{i+j=n} K^{ij}) = 0$$
as soon as $p + r + j_0 > n$, so $Z_r^{pq} = Z_\infty(E_2^{pq})$ for $r > q - j_0 + 1$. On the other hand, $\mathrm{H}^n(F_{\mathrm{I}}^p(K^{\bullet\bullet})) = 0$ for $p > n - j_0 + 1$.

4° If there exists j_0 such that $K^{ij} = 0$ for $j > j_0$, then we have
$$F_{\mathrm{I}}^{p-r+1}(K^{\bullet\bullet}) \cap (\sum_{i+j=n} K^{ij}) = \sum_{i+j=n} K^{ij}$$
as soon as $p - r + 1 + j_0 < n$, so $B_r^{pq} = B_\infty(E_2^{pq})$ for $r < j_0 - q + 1$. On the other hand, $\mathrm{H}^n(F_{\mathrm{I}}^p(K^{\bullet\bullet})) = \mathrm{H}^n(K^{\bullet\bullet})$ for $p + j_0 < n - 1$.

□

(11.3.4). Suppose that the bicomplex $K^{\bullet\bullet} = (K^{ij})$ is such that $K^{ij} = 0$ for $i < 0$ or $j < 0$. We know that we can define, for all $p \in \mathbf{Z}$, a canonical "edge-homomorphism". $\mathbf{0_{III}}$ | 31

(11.3.4.1)
$${}'E_2^{p0}(K^{\bullet\bullet}) \to \mathrm{H}^p(K^{\bullet\bullet})$$

(M, XV, 6). Recall that this is due to, on the one hand, the spectral sequence ${}'E(K^{\bullet\bullet})$, since $Z_r^{p0} = Z_{\mathrm{I}}^p(Z_{\mathrm{II}}^0(K^{\bullet\bullet}))$ for $2 \leqslant r \leqslant +\infty$, and on the other hand, that $\mathrm{H}^p(F_{\mathrm{I}}^{p+1}(K^{\bullet\bullet})) = 0$, so that the isomorphism $\beta^{p0} : {}'E_\infty^{p0} \xrightarrow{\sim} \mathrm{H}^p(F_1^p)/\mathrm{H}^p(F_1^{p+1})$ gives a homomorphism ${}'E_\infty^{p0} \to \mathrm{H}^p(F_1^p(K^{\bullet\bullet})) \to \mathrm{H}^p(K^{\bullet\bullet})$. The equality of all the Z_r^{p0} allows us to define a canonical homomorphism ${}'E_r^{p0} \to' E_s^{p0}$ for $r \leqslant s$, and, in particular, a homomorphism ${}'E_2^{p0} \to' E_\infty^{p0}$, from the composition of the edge-homomorphism ${}'E_2^{p0}(K^{\bullet\bullet}) \to \mathrm{H}^p(K^{\bullet\bullet})$. Furthermore, we can immediately verify that, in the class mod B_2^{p0} of an element $z \in \mathbf{Z}_{\mathrm{II}}^0(K^{\bullet\bullet}) \subset K^{p0}$ such that $d'z = 0$, the edge-homomorphism thus defined associates, to the class of mod B_∞^{p0} in ${}'E_\infty^{p0}$, the cohomology class of z in $H^p(K^{\bullet\bullet})$. We therefore finally see that the *edge-homomorphism* (11.3.4.1) *comes from, by passing to cohomology, the canonical injection* $Z_{\mathrm{II}}^0(K^{\bullet\bullet}) \to K^{\bullet\bullet}$ (where $K^{\bullet\bullet}$ is considered as simple complex). We naturally interpret the edge-homomorphism in the same way

(11.3.4.2)
$${}''E_2^{p0}(K^{\bullet\bullet}) \to \mathrm{H}^p(K^{\bullet\bullet})$$

as coming from the canonical injection $Z_{\mathrm{I}}^0(K^{\bullet\bullet}) \to K^{\bullet\bullet}$.

(11.3.5). Now let $K_{\bullet\bullet} = (K_{ij})$ be a bicomplex in $\mathtt{C}$ whose two differential operators are of degrees -1. We then write $K_{i,\bullet}$ (resp. $K_{\bullet,j}$) to mean the simple complex $(K_{ij})_{j\in\mathbf{Z}}$ (resp. $(K_{ij})_{i\in\mathbf{Z}}$), $\mathrm{H}_p^{\mathrm{II}}(K_{i,\bullet})$ (resp. $\mathrm{H}_p^{\mathrm{I}}(K_{\bullet,j})$) to mean the p-th homology object, $\mathrm{H}_p^{\mathrm{II}}(K_{\bullet\bullet})$ (resp. $\mathrm{H}_p^{\mathrm{I}}(K_{\bullet\bullet})$) to mean the complex $(\mathrm{H}_p^{\mathrm{II}}(K_{i,\bullet}))_{i\in\mathbf{Z}}$ (resp. $(\mathrm{H}_p^{\mathrm{I}}(K_{\bullet,j}))_{j\in\mathbf{Z}}$), and $\mathrm{H}_q^{\mathrm{I}}(\mathrm{H}_p^{\mathrm{II}}(K_{\bullet\bullet}))$ (resp. $\mathrm{H}_q^{\mathrm{II}}(\mathrm{H}_p^{\mathrm{I}}(K_{\bullet\bullet}))$) to mean the q-th homology object; we use analogous notation for the objects of cycles and objects of boundaries; finally, we will denote by $\mathrm{H}_n(K_{\bullet\bullet})$ the n-th homology object (when it exists) of the simple complex (with a differential operator of degree -1) defined by $K_{\bullet\bullet}$. Let $K'^{\bullet\bullet} = (K'^{ij})$ with $K'^{\bullet\bullet} = (K_{-i,-j})$ be the bicomplex with differential operators of degrees $+1$ associated to $K_{\bullet\bullet}$. By definition, the *spectral sequences* of $K_{\bullet\bullet}$ are those of $K'^{\bullet\bullet}$, that we write as ${}'E(K_{\bullet\bullet})$ and ${}''E(K_{\bullet\bullet})$, where, however, we change the notation, by putting
$${}'E_{pq}^r(K_{\bullet\bullet}) =' E_r^{-p,-q}(K'^{\bullet\bullet}) \quad \text{and} \quad {}''E_{pq}^r(K_{\bullet\bullet}) ='' E_r^{-p,-q}(K'^{\bullet\bullet}),$$

for $2 \leqslant r \leqslant \infty$. With this notation, we have

$$'E^2_{pq}(K_{\bullet\bullet}) = \mathrm{H}^{\mathrm{I}}_p(\mathrm{H}^{\mathrm{II}}_q(K_{\bullet\bullet})) \quad \text{and} \quad 'E^2_{pq}(K_{\bullet\bullet}) = \mathrm{H}^{\mathrm{II}}_q(\mathrm{H}^{\mathrm{I}}_p(K_{\bullet\bullet})).$$

To avoid sign errors, it will be preferable in general to return to the complex $K'^{\bullet\bullet}$ for the relations between these spectral sequences and their abutments. We note, however, the criteria corresponding to (11.3.3):

(11.3.6). The spectral sequences $'E(K_{\bullet\bullet})$ and $''E(K_{\bullet\bullet})$ are *biregular* in the following cases: $\mathbf{0}_{\mathbf{III}}$ | 32

(a) There exist i_0 and j_0 such that $K_{ij} = 0$ for $i > i_0$ and $j > j_0$ (resp. for $i < i_0$ and $j < j_0$);
(b) There exist i_0 and i_1 such that $K_{ij} = 0$ for $i < i_0$ and $i > i_1$;
(c) There exist j_0 and j_1 such that $K_{ij} = 0$ for $j < j_0$ and $j > j_1$.

The sequence $'E(K_{\bullet\bullet})$ is *regular* if there exists i_0 such that $K_{ij} = 0$ for $i < i_0$ and $K_{ij} = 0$ for $j > j_0$.

The sequence $''E(K_{\bullet\bullet})$ is *regular* if there exists i_0 such that $K_{ij} = 0$ for $i > i_0$ and $K_{ij} = 0$ for $j < j_0$.

11.4. Hypercohomology of a functor with respect to a complex $K^\bullet$

(11.4.1). Let C be an abelian category. Recall that we defined the *right resolution* (or *cohomological resolution*) of a object A in C to be a complex of objects in C, whose differential operator is of degree $+1$,

$$0 \to L^0 \to L^1 \to L^2 \ldots$$

endowed with a morphism $\varepsilon : A \to L^0$, called the *augmentation* of the resolution, which we can consider as a morphism of complexes

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\varepsilon"'] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \ldots \\ 0 \arrow[r] & L^0 \arrow[r] & L^1 \arrow[r] & L^2 \arrow[r] & \ldots \end{tikzcd}$$

such that the sequence

$$0 \to A \xrightarrow{\varepsilon} L^0 \to L^1 \ldots$$

is exact. Similarly, a *left resolution* (or *homological resolution*) of A is a complex $0 \leftarrow L_0 \leftarrow L_1 \leftarrow L_2 \ldots$ of objects in C, whose differential operator is of degree -1, endowed with an *augmentation* $L_0 \xrightarrow{\varepsilon} A$ such that the sequence

$$0 \leftarrow A \xleftarrow{\varepsilon} L_0 \leftarrow L_1 \ldots$$

is exact.

When the right resolution $(L_i)_i \geqslant 0$ of a object A is such that $L_i = 0$ for $i \geqslant n+1$, we say that this resolution is *of length* $\leqslant n$. We define a left resolution of length $\leqslant n$ similarly. A resolution which is of of length $\leqslant n$ for some integer n is said to be *finite*.

A resolution of A is called *projective* (resp. *injective*) if the objects of C, apart from A, of which it is composed are *projective* (resp. *injective*). When C is the category of (left, say) modules over a ring, we say that a resolution of A is *flat* (resp. *free*) when the modules, apart from A, of which it is composed are *flat* (resp. *free*).

(11.4.2). Let $K^\bullet = (K^i)_{i\in\mathbf{Z}}$ be a complex of objects of C, whose differential operator is of degree $+1$. We define the *right Cartan–Eilenberg resolution* of $K^\bullet$ to be a pair consisting of a bicomplex $L^{\bullet\bullet} = (L^{ij})$ with differential operator of degree $+1$ and such that $L^{ij} = 0$ for $j \leqslant 0$, and a morphism of simple complexes $\varepsilon : K^\bullet \to L^{\bullet,0}$, such that the following condition are fulfilled: $\mathbf{0}_{\mathbf{III}}$ | 33

(i) For each index i, the sequences

$$\begin{aligned} 0 \to &K^i \xrightarrow{\varepsilon} L^{i0} \to L^{i1} \to \ldots \\ 0 \to &B^i(K^\bullet) \xrightarrow{\varepsilon} B^i_{\mathrm{I}}(L^{\bullet,0}) \to B^i_{\mathrm{I}}(L^{\bullet,1}) \to \ldots \\ 0 \to &Z^i(K^\bullet) \xrightarrow{\varepsilon} Z^i_{\mathrm{I}}(L^{\bullet,0}) \to Z^i_{\mathrm{I}}(L^{\bullet,1}) \to \ldots \\ 0 \to &\mathrm{H}^i(K^\bullet) \xrightarrow{\varepsilon} \mathrm{H}^i_{\mathrm{I}}(L^{\bullet,0}) \to \mathrm{H}^i_{\mathrm{I}}(L^{\bullet,1}) \to \ldots \end{aligned}$$

are exact, or, in other words, $(L^{i,\bullet})$, $(B^i_{\mathrm{I}}(L^{\bullet\bullet}))$, $(Z^i_{\mathrm{I}}(L^{\bullet\bullet}))$ and $(\mathrm{H}^i_{\mathrm{I}}(L^{\bullet\bullet}))$ are *resolutions* of $K^i, B^i(K^\bullet), Z^i(K^\bullet)$ and $\mathrm{H}^i(K^\bullet)$ (respectively).

(ii) For each j, the simple complex $L^{\bullet,j}$ is *split*, or, in other words, the exact sequences

$$0 \to B^i_{\mathrm{I}}(L^{\bullet,j}) \to Z^i_{\mathrm{I}}(L^{\bullet,j}) \to \mathrm{H}^i_{\mathrm{I}}(L^{\bullet,j}) \to 0 \tag{11.4.2.1}$$

$$0 \to Z^i_{\mathrm{I}}(L^{\bullet,j}) \to L^{ij} \to B^{i+1}_{\mathrm{I}}(L^{\bullet,j}) \to 0 \tag{11.4.2.2}$$

are *split*.

We can prove (M, XVII, 1.2) that if every object of $\mathtt{C}$ is subobjects of an injective object, then every complex $K^\bullet$ of $\mathtt{C}$ admits an *injective* Cartan–Eilenberg resolution, i.e. consisting of injective objects of L^{ij} (the condition (ii) above then implying that the $(B^i_{\mathrm{I}}(L^{\bullet,j}))$, $(Z^i_{\mathrm{I}}(L^{\bullet,j}))$ and $(\mathrm{H}^i_{\mathrm{I}}(L^{\bullet,j}))$ also consist of injective objects). Furthermore, for any morphsim $f : K^\bullet \to K'^\bullet$ of complexes of $\mathtt{C}$, any Cartan–Eilenberg resolution $L^{\bullet\bullet}$ of $K^\bullet$, and any *injective* Cartan–Eilenberg resolution $L'^{\bullet\bullet}$ of $K'^\bullet$, there exists a morphism of bicomplexes $F : L^{\bullet\bullet} \to L'^{\bullet\bullet}$ that is compatible with f and its augmentation, and if f and g are homotopic morphisms from $K^\bullet$ to $K'^\bullet$, then the corresponding morphisms from $L^{\bullet\bullet}$ to $L'^{\bullet\bullet}$ are also homotopic *(loc. cit.)*.

When $K^\bullet$ is *bounded below* (resp. *bounded above*), we can take $L^{\bullet\bullet}$ to be such that $L^{ij} = 0$ for $i < i_0$ (resp. $i > i_0$) if $K^i = 0$ for $i < i_0$ (resp. $i > i_0$) (M, XVII, 1.3).

Suppose, on the other hand, that there exists a integer n such that every object of $\mathtt{C}$ admits a *injective resolution of length* $\leqslant n$; then we can assume that we have $L^{ij} = 0$ for $j > n$ (M, XVII, 1.4).

(11.4.3). Now let T be an *additive covariant functor* from $\mathtt{C}$ to an abelian category $\mathtt{C}'$. Given a complex $K^\bullet$ of $\mathtt{C}$ and an *injective* Cartan–Eilenberg resolution of $L^{\bullet\bullet}$ of $K^\bullet$, suppose that the (simple) complex defined by the bicomplex $T(L^{\bullet\bullet})$ exists (11.3.1); then the two spectral sequences $'E(T(L^{\bullet\bullet}))$ and $''E(T(L^{\bullet\bullet}))$ of this bicomplex are called the *hypercohomology spectral sequences* of T with respect to $K^\bullet$; by (11.4.2) and (11.4.3), they effectively only depend on the complex $K^\bullet$ effectively, not on the choice of the injective Cartan–Eilenberg resolution $L^{\bullet\bullet}$; furthermore, they depend *functorially* on $K^\bullet$. They have the same abutment $\mathrm{H}^\bullet(T(L^{\bullet\bullet}))$, also called the *hypercohomology* of T with respect to $K^\bullet$, and denoted by $\mathrm{R}^\bullet T(K^\bullet)$. We can show that the E_2 terms of the two spectral sequences above are given by

$$'E_2^{pq} = H^p(\mathrm{R}^q T(K^\bullet)) \tag{11.4.3.1}$$

$$''E_2^{pq} = \mathrm{R}^p T(H^q(K^\bullet)) \tag{11.4.3.2}$$

where $\mathrm{R}^p T$ denotes, as usual, the p-th *derived functor* of T for $p \in \mathbf{Z}$; $\mathrm{R}^q T(K^\bullet)$ denotes the complexes $(\mathrm{R}^q T(K^i))_{i \in \mathbf{Z}}$. Unless explicitly otherwise mentioned, we will henceforth assume that every object of $\mathtt{C}$ is a subobject of an injective object of $\mathtt{C}$, so that injective Cartan–Eilenberg resolutions exist for any complex of $\mathtt{C}$. Since $L^{ij} = 0$ for $j < 0$, the criteria of (11.3.3) show that the *two* hypercohomology spectral sequences of T with respect to $K^\bullet$ exist and are *biregular* in the each of the following two cases: $\mathbf{0}_{\mathbf{III}}$ | 34

(1) $K^\bullet$ is bounded below;
(2) Every object of $\mathtt{C}$ admits an injective resolution of length at most n, for some integer n independent of the object in question.

Indeed, we can suppose, in the first case, that (11.4.2) there exists i_0 such that $L^{ij} = 0$ for $i < i_0$, and, in the second case, that there exists j_1 such that $L^{ij} = 0$ for $j > j_1$; in each of these two cases, it is furthermore clear that, for any given n, there exist only a finite number of pairs (i, j) such that $L^{ij} \neq 0$ and $i + j = n$, which proves our claims.

If we assume that filtrant inductive limits exist in $\mathtt{C}'$ and are exact (which implies, in particular, the existence of infinite direct sums in $\mathtt{C}'$), then the complex defined by the bicomplex $T(L^{\bullet\bullet})$ exists, and criterion (11.3.3) shows that the sequence $'E(T(L^{\bullet\bullet}))$ is always *regular*.

If $K^\bullet$ is a complex such that all the K^i are zero except for a single K^{i_0}, then $\mathrm{R}^n T(K^\bullet)$ is isomorphic to $\mathrm{R}^{n-i_0} T(K^\bullet)$, as follows immediately from the definitions by taking a Cartan–Eilenberg resolution $L^{\bullet\bullet}$ such that $L^{ij} = 0$ for $i \neq i_0$.

If $K^\bullet$ and $K'^\bullet$ are two complexes of $\mathtt{C}$, and f and g are homotopic morphism from $K^\bullet$ to $K'^\bullet$, then the morphisms $\mathrm{R}^\bullet T(K^\bullet) \to \mathrm{R}^\bullet T(K'^\bullet)$ induced by f and g are identical, and the same is true for the morphisms of the cohomology spectral sequences.

Proposition (11.4.5). — *Suppose that filtrant inductive limits exist in $\mathtt{C}'$ and are exact. If $\mathrm{R}^n T(K^i) = 0$ for all $n > 0$ and all $i \in \mathbf{Z}$, then we have functorial isomorphisms*

$$\mathrm{R}^i T(K^\bullet) \xrightarrow{\sim} \mathrm{H}^i(T(K^\bullet)) \tag{11.4.5.1}$$

for all $i \in \mathbf{Z}$.

Proof. Indeed, the only non-zero E_2 terms of the first spectral sequence (11.4.3.1) are then ${}'E_2^{p0} = \mathrm{H}^p(T(K^\bullet))$; in other words, this sequence is *degenerate*; since it is regular (11.4.4), the conclusion follows from (11.1.6). □

(11.4.6). Now consider, for example, a covariant bifunctor $(M,N) \to T(M,N)$ from $\mathtt{C} \times \mathtt{C}$ to $\mathtt{C}''$, where $\mathtt{C}$, $\mathtt{C}'$, and $\mathtt{C}''$ are three abelian categories; we assume, for simplicity, that T is additive in each of its arguments, and furthermore that every object of $\mathtt{C}$ and every object of $\mathtt{C}'$ are subobjects of an injective object, and that filtrant inductive limits exist in $\mathtt{C}$ and are exact. We then define the *hypercohomology* of T with respect to the complexes $K^\bullet$ and $K'^\bullet$ of $\mathtt{C}$ and $\mathtt{C}'$ (respectively), with differential operators of degree $+1$, by taking $K^\bullet$ (resp. $K'^\bullet$) to be an injective Cartan–Eilenberg resolution $L^{\bullet\bullet}$ (resp. $L'^{\bullet\bullet}$); then $T(L^{\bullet\bullet}, L'^{\bullet\bullet})$ is a quadricomplex of $\mathtt{C}''$, which we consider as a *bicomplex* of $\mathtt{C}'$, with the degree of $T(L^{ij}, L'^{hk})$ being the integers $i+h$ and $j+k$. The *hypercohomology* of T with respect to $K^\bullet$ and $K'^\bullet$ is by definition the cohomology $\mathrm{H}^\bullet(T(L^{\bullet\bullet}, L'^{\bullet\bullet}))$ of this bicomplex (in other words, that of the associated simple complex) and is denoted by $\mathrm{R}^\bullet T(K^\bullet, K'^\bullet)$; it is the abutment of two spectral sequences whose E_2 terms are given by **0**$_{\mathbf{III}}$ | 35

$${}'E_2^{pq} = \mathrm{H}^p(\mathrm{R}^q T(K^\bullet, K'^\bullet)) \tag{11.4.6.1}$$

$${}''E_2^{pq} = \sum_{q'+q''=q} \mathrm{R}^p T(\mathrm{H}^{q'} K^\bullet, \mathrm{H}^{q''} K'^\bullet) \tag{11.4.6.2}$$

(cf. M, XVII, 2).

Here $\mathrm{R}^q T(K^\bullet, K'^\bullet)$ is the bicomplex $(\mathrm{R}^q T(K^i, K'^j))_{(i,j)\in\mathbf{Z}\times\mathbf{Z}}$, and the right-hand side of (11.4.6.1) is its cohomology when we consider it as a simple complex.

Furthermore, the first spectral sequence is always regular, and the two spectral sequences are biregular when there exists n such that every object of $\mathtt{C}$ and every object of $\mathtt{C}'$ admit an injective resolution of length $\leqslant n$, or when $K^\bullet$ and $K'^\bullet$ are bounded below; in the latter case, we can furthermore omit the hypothesis that inductive limits exist in $\mathtt{C}$ and $\mathtt{C}'$.

If $K^\bullet$ and $K'^\bullet$ are two other complexes of $\mathtt{C}$ and $\mathtt{C}'$ (respectively), f and g *homotopic* morphisms from $K^\bullet$ to $K_1^\bullet$, and f' and g' *homotopic* morphisms from $K'^\bullet$ to $K_1'^\bullet$, then the morphisms $\mathrm{R}^\bullet T(K^\bullet, K'^\bullet) \to \mathrm{R}^\bullet T(K_1^\bullet, K_1'^\bullet)$ induced by f and f' on the one hand, and by g and g' on the other hand, are identical, and the same is true for the morphisms of the hypercohomology spectral sequences.

We can generalize easily to any additive covariant multifunctor.

Proposition (11.4.7). — *Suppose that for any injective object $\mathtt{I}$ of $\mathtt{C}$ (resp. $\mathtt{I}'$ of $\mathtt{C}'$), $\mathtt{A}' \mapsto T(\mathtt{I}, \mathtt{A}')$ (resp. $\mathtt{A} \mapsto T(\mathtt{A}, \mathtt{I}')$) is an exact functor. Then, with the notation of (11.4.6), we have a canonical isomorphism*

$$\mathrm{R}^\bullet T(K^\bullet, K'^\bullet) \xrightarrow{\sim} \mathrm{H}^\bullet(T(L^{\bullet\bullet}, K'^\bullet)) \xrightarrow{\sim} \mathrm{H}^\bullet(T(K^\bullet, L'^{\bullet\bullet}))$$

where the last two terms are the cohomology of simple complexes defined by the tricomplexes $T(L^{\bullet\bullet}, K'^\bullet)$ and $T(K^\bullet, L^{\bullet\bullet})$ (respectively).

Proof. Let us define, for example, the first of these isomorphisms. The quadricomplex $T(L^{\bullet\bullet}, L'^{\bullet\bullet})$ can be considered as a bicomplex, where the degrees of $T(L^{ij}, L'^{hk})$ are the integers $i+j$ and $h+k$. Since, for each h, $L'^{h,\bullet}$ is a *resolution* of K'^h, we have, for this bicomplex, by virtue of the hypotheses on T, $\mathrm{H}^q_{\mathrm{II}}(T(L^{\bullet\bullet}, L'^{\bullet\bullet})) = 0$ for $q \neq 0$, and $\mathrm{H}^0_{\mathrm{II}}(T(L^{\bullet\bullet}, L'^{\bullet\bullet})) = T(L^{\bullet\bullet}, K'^\bullet)$; the first spectral sequence of this bicomplex is thus *degenerate*; since $L'^{hk} = 0$ for $k < 0$, this sequence is also regular (11.3.3), and the conclusion then follows from (11.1.6). □

We have similar results for a covariant multifunctor of any number n of arguments: in the calculation of the hypercohomology, it is not necessary to replace *all* the complexes by a Cartan–Eilenberg resolution, but only $n-1$ of them, provided that, when we fix $n-1$ arbitrary arguments by taking them to be *injective* objects, the covariant functor in the remaining argument is *exact*.

11.5. Passage to the inductive limit in the hypercohomology $\mathbf{0}_{\mathbf{III}}$ | 36

11.6. Hypercohomology of a functor with respect to a complex $K_\bullet$

11.7. Hypercohomology of a functor with respect to a bicomplex $K_{\bullet\bullet}$

11.8. Supplement on the cohomology of simplicial complexes

11.9. A lemma on the complexes of finite type

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§12. Supplement on sheaf cohomology

12.1. Cohomology of sheaves of modules on ringed spaces

§13. Projective limits in homological algebra

§14. Combinatorial dimension of a topological space

Summary

§14. Combinatorial dimension of a topological space. **IV-1** | 5
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§23. Japanese rings.

Almost all of the preceding sections have been focused on the exposition of ideas of commutative algebra that will be used throughout Chapter IV. Even though a large amount of these ideas already appear in multiple works ([**CC**], [**Sam53a**], [**SZ60**], [**Ser55b**], [**Nag62**]), we thought that it would be more practical for the reader to have a coherent, vaguely independent exposition. Together with §§5, 6, and 7 of Chapter IV (where we use the language of schemes), these sections constitute, in the middle of our treatise, a miniature special treatise, somewhat independent of Chapters I to III, and one that aims to present, in a coherent manner, the properties of rings that "behave well" with respect to operations such as completion, or integral closure, by systematically associating these properties to more general ideas.[11]

[11] The majority of properties which we discuss were discovered by Chevalley, Zariski, Nagata, and Serre. The method used here was first developed in the autumn of 1961, in a course taught at Harvard University by A. Grothendieck.

14.1. Combinatorial dimension of a topological space

$\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 6

(14.1.1). Let I be an ordered set; a *chain* of elements of I is, by definition, a strictly-increasing finite sequence $i_0 < i_1 < \cdots < i_n$ of elements of I ($n \geqslant 0$); by definition, the *length* of this chain is n. If X is a topological space, the set of its *irreducible closed* subsets is ordered by inclusion, and so we have the notion of a *chain* of irreducible closed subsets of X.

Definition (14.1.2). — Let X be a topological space. We define the combinatorial dimension of X (or simply the dimension of X, if there is no risk of confusion), denoted by $\dim c(X)$ (or simply $\dim(X)$), to be the upper bound of lengths of chains of irreducible closed subsets of X. For all $x \in X$, we define the combinatorial dimension of X at x (or simply the dimension of X at x), denoted by $\dim_x(X)$, to be the number $\inf_U(\dim(U))$, where U varies over the open neighbourhoods of x in X.

It follows from this definition that we have

$$\dim(\varnothing) = -\infty$$

(the upper bound in $\overline{\mathbf{R}}$ of the empty set being $-\infty$). If (X_α) is the family of irreducible components of X, then we have

$$\dim(X) = \sup_\alpha \dim(X_\alpha), \tag{14.1.2.1}$$

because every chain of irreducible closed subsets of X is, by definition, contained in some irreducible component of X, and, conversely, the irreducible components are closed in X, so every irreducible closed subset of an X_α is a irreducible closed subset of X.

Definition (14.1.3). — We say that a topological space X is equidimensional if all its irreducible components have the same dimension (which is thus equal to $\dim(X)$, by (14.1.2.1)).

Proposition (14.1.4). —

(i) *For every closed subset Y of a topological space X, we have* $\dim(Y) \leqslant \dim(X)$.
(ii) *If a topological space X is a finite union of closed subsets X_i, then we have* $\dim(X) = \sup_i \dim(X_i)$.

PROOF. For every irreducible closed subset Z of Y, the closure $\overline{Z}$ of Z in X is irreducible ($\mathbf{0}_\mathbf{I}$, 2.1.2), and $\overline{Z} \cap Y = Z$, whence (i). Now, if $X = \bigcup_{i=1}^n X_i$, where the X_i are closed, then every irreducible closed subset of X is contained in one of the X_i ($\mathbf{0}_\mathbf{I}$, 2.1.1), and so every chain of irreducible closed subsets of X is contained in one of the X_i, whence (ii). □

From (14.1.4, i), we see that, for all $x \in X$, we can also write

$$\dim_x(X) = \lim_U \dim(U), \tag{14.1.4.1}$$

where the limit is taken over the downward-directed set of open neighbourhoods of x in X.

$\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 7

Corollary (14.1.5). — *Let X be a topological space, x a point of X, U a neighbourhood of x, and Y_i ($1 \leqslant i \leqslant n$) closed subsets of U such that, for all i, $x \in Y_i$, and such that U is the union of the Y_i. Then we have*

$$\dim_x(X) = \sup_i(\dim_x(Y_i)). \tag{14.1.5.1}$$

PROOF. It follows from (14.1.4, ii) that we have $\dim_x(X) = \inf_V(\sup_i(\dim(Y_i \cap V)))$, where V ranges over the set of open neighbourhoods of x that are contained in U; similarly, we have $\dim_x(Y_i) = \inf_V(\dim(Y_i \cap V))$ for all i. The corollary is thus evident if

$$\sup_i(\dim_x(Y_i)) = +\infty;$$

if this were not the case, then there would be an open neighbourhood $V_0 \subset U$ of x such that $\dim(Y_i \cap V) = \dim_x(Y_i)$ for $1 \leqslant i \leqslant n$ and for all $V \subset V_0$, whence the conclusion. □

Proposition (14.1.6). — *For every topological space X, we have* $\dim(X) = \sup_{x \in X} \dim_x(X)$.

PROOF. It follows from Definition (14.1.2) and Proposition (14.1.4) that $\dim_x(X) \leqslant \dim(X)$ for all $x \in X$. Now, let $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ be a chain of irreducible closed subsets of X, and let $x \in Z_0$; for every open subset $U \subset X$ that contains x, $U \cap Z_i$ is irreducible ($\mathbf{0_I}$, 2.1.6) and closed in U, and since we have $\overline{U \cap Z_i} = Z_i$ in X, the $U \cap Z_i$ are pairwise distinct; thus $\dim(U) \geqslant n$, which finishes the proof. □

Corollary (14.1.7). — *If (X_α) is an open, or closed and locally finite, cover of X, then $\dim(X) = \sup_\alpha(\dim(X_\alpha))$.*

PROOF. If X_α is a neighbourhood of $x \in X$, then $\dim_x(X) \leqslant \dim(X_\alpha)$, whence the claim for open covers. On the other hand, if the X_α are closed, and U is a neighbourhood of $x \in X$ which meets only finitely many of the X_α, then

$$\dim_x(X) \leqslant \dim(U) = \sup_\alpha(\dim(U \cap X_\alpha)) \leqslant \sup_\alpha(\dim(X_\alpha))$$

by (14.1.4), whence the other claim. □

Corollary (14.1.8). — *Let X be a Noetherian Kolmogoroff space ($\mathbf{0_I}$, 2.1.3), and F the set of closed points of X. Then $\dim(X) = \sup_{x \in F} \dim_x(X)$.*

PROOF. With the notation from the proof of (14.1.6), it suffices to note that there exists a closed point in Z_0 ($\mathbf{0_I}$, 2.1.3). □

Proposition (14.1.9). — *Let X be a nonempty Noetherian Kolmogoroff space. To have $\dim(X) = 0$, it is necessary and sufficient for X to be finite and discrete.*

PROOF. If a space X is separated (and *a fortiori* if X is a discrete space), then all the irreducible closed subsets of X are single points, and so $\dim(X) = 0$. Conversely, suppose that X is a Noetherian Kolmogoroff space such that $\dim(X) = 0$; since every irreducible component of X contains a closed point ($\mathbf{0_I}$, 2.1.3), it must be exactly this single point. Since X has only a finite number of irreducible components, it is thus finite and discrete. □

Corollary (14.1.10). — *Let X be a Noetherian Kolmogoroff space. For a point $x \in X$ to be isolated, it is necessary and sufficient to have $\dim_x(X) = 0$.*

PROOF. The condition is clearly necessary (without any hypotheses on X). It is also sufficient, because it implies that $\dim(U) = 0$ for any open neighbourhood U of x, and since U is a Noetherian Kolmogoroff space, U is finite and discrete. □ $\mathbf{0_{IV\text{-}1}}$ | 8

Proposition (14.1.11). — *The function $x \mapsto \dim_x(X)$ is upper semi-continuous on X.*

PROOF. It is clear that this function is upper semi-continuous at every point where its value is $+\infty$. So suppose that $\dim_x(X) = n < +\infty$; then Equation (14.1.4.1) shows that there exists an open neighbourhood U_0 of x such that $\dim(U) = n$ for every open neighbourhood $U \subset U_0$ of x. So, for all $y \in U_0$ and every open neighbourhood $V \subset U_0$ of y, we have $\dim(V) \leqslant \dim(U_0) = n$ (14.1.4); we thus deduce from (14.1.4.1) that $\dim_y(X) \leqslant n$. □

Remark (14.1.12). — If X and Y are topological spaces, and $f : X \to Y$ a continuous map, then it can be the case that $\dim(f(X)) > \dim(X)$; we obtain such an example by taking X to be a discrete space with 2 points, a and b, and Y to be the set $\{a, b\}$ endowed with the topology for which[12] the closed sets are $\varnothing$, $\{a\}$, and $\{a, b\}$; if $f : X \to Y$ is the identity map, then $\dim(Y) = 1$ and $\dim(X) = 0$. We note that Y is the spectrum of a discrete valuation ring A, of which a is the unique closed point, and b the generic point; if K and k are the field of fractions and the residue field of A (respectively), then X is the spectrum of the ring $k \times K$, and f is the continuous map corresponding to the homomorphism $(\phi, \psi) : A \to k \times K$, where $\phi : A \to k$ and $\psi : A \to K$ are the canonical homomorphism (cf. (**IV**, 5.4.3)).

[12] *[Trans.] This is now often referred to as the* Sierpiński space, *or the* connected two-point set.

14.2. Codimension of a closed subset

Definition (14.2.1). — Given an irreducible closed subset Y of a topological space X, we define the combinatorial codimension (or simply codimension) of Y in X, denoted by $\operatorname{codim}(Y, X)$, to be the upper bound of the lengths of chains of irreducible closed subsets of X of which Y is the smallest element. If Y is an arbitrary closed subset of X, then we define the codimension of Y in X, again denoted by $\operatorname{codim}(Y, X)$, to be the lower bound of the codimensions in X of the irreducible components of Y. We say that X is equicodimensional if all the minimal irreducible closed subsets of X has the same codimension in X.

It follows from this definition that $\operatorname{codim}(\varnothing, X) = +\infty$, since the lower bound of the empty set of $\overline{\mathbf{R}}$ is $+\infty$. If Y is closed in X, and if (X_α) (resp. (Y_α)) is the family of irreducible components of X (resp. Y), then every Y_β is contained in some X_α, and, more generally, every chain of irreducible closed subsets of X of which Y_β is the smallest element is formed of subsets of some X_α; we thus have

(14.2.1.1) $$\operatorname{codim}(Y, X) = \inf_\beta(\sup_\alpha(\operatorname{codim}(Y_\beta, X_\alpha))),$$

where, for every β, α ranges over the set of indices such that $Y_\beta \subset X_\alpha$.

Proposition (14.2.2). — *Let X be a topological space.*

(i) *If Φ is the set of irreducible closed subsets of X, then*

(14.2.2.1) $$\dim(X) = \sup_{Y \in \Phi}(\operatorname{codim}(Y, X)).$$

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(ii) *For every nonempty closed subset Y of X, we have*

(14.2.2.2) $$\dim(Y) + \operatorname{codim}(Y, X) \leqslant \dim(X).$$

(iii) *If Y, Z, and T are closed subsets of X such that $Y \subset Z \subset T$, then*

(14.2.2.3) $$\operatorname{codim}(Y, Z) + \operatorname{codim}(Z, T) \leqslant \operatorname{codim}(Y, T).$$

(iv) *For a closed subset Y of X to be such that* $\operatorname{codim}(Y, X) = 0$, *it is necessary and sufficient for Y to contain an irreducible component of X.*

PROOF. Claims (i) and (iv) are immediate consequences of Definition (14.2.1). To show (ii), we can restrict to the case where Y is irreducible, and then the equation follows from Definitions (14.1.1) and (14.2.1). Finally, to show (iii), we can, by Definition (14.2.1), first restrict to the case where Y is irreducible; then $\operatorname{codim}(Y, Z) = \sup_\alpha(\operatorname{codim}(Y, Z_\alpha))$ for the irreducible components Z_α of Z that contain Y; it is clear that $\operatorname{codim}(Y, T) \geqslant \operatorname{codim}(Y, Z)$, so the inequality is true if $\operatorname{codim}(Y, Z) = +\infty$; but if this were not the case, then there would exist some α such that $\operatorname{codim}(Y, Z) = \operatorname{codim}(Y, Z_\alpha)$, and by (14.2.1), we can restrict to the case where Z itself is irreducible; but then the inequality in (14.2.2.3) is an evident consequence of Definition (14.2.1). □

Proposition (14.2.3). — *Let X be a topological space, and Y a closed subset of X. For every open subset U of X, we have*

(14.2.3.1) $$\operatorname{codim}(Y \cap U, U) \geqslant \operatorname{codim}(Y, X).$$

Furthermore, for this inequality (14.2.3.1) *to be an equality, it is necessary and sufficient to have* $\operatorname{codim}(Y, X) = \inf_\alpha(\operatorname{codim}(Y_\alpha, X))$, *where (Y_α) is the family of irreducible components of Y that meet U.*

PROOF. We know ($\mathbf{0}_\mathbf{I}$, 2.1.6) that $Z \mapsto \overline{Z}$ is a bijection from the set of irreducible closed subsets of U to the set of irreducible closed subsets of X that meet U, and, in particular, induces a correspondence between the irreducible components of $Y \cap U$ and the irreducible components of Y that meet U; if Y_α is one of the latter such components, then we have $\operatorname{codim}(Y_\alpha, X) = \operatorname{codim}(Y_\alpha \cap U, U)$, and the proposition then follows from Definition (14.2.1). □

Definition (14.2.4). — Let X be a topological space, Y a closed subset of X, and x a point of X. We define the codimension of Y in X at the point x, denoted by $\operatorname{codim}_x(Y, X)$, to be the number $\sup_U(\operatorname{codim}(Y \cap U, U))$, where U ranges over the set of open neighbourhoods of x in X.

By (14.2.3), we can also write

(14.2.4.1) $$\operatorname{codim}_x(Y,X) = \lim_{U}(\operatorname{codim}(Y \cap U, U)),$$

where the limit is taken over the downward-directed set of open neighbourhoods of x in X. We note that we have

$$\operatorname{codim}_x(Y,X) = +\infty \text{ if } x \in X - Y.$$

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Proposition (14.2.5). — *If $(Y_i)_{1\leqslant i\leqslant n}$ is a finite family of closed subsets of a topological space X, and Y is the union of this family, then*

(14.2.5.1) $$\operatorname{codim}(Y,X) = \inf_i(\operatorname{codim}(Y_i,X)).$$

PROOF. Every irreducible component of one of the Y_i is contained in an irreducible component of Y, and, conversely, every irreducible component of Y is also an irreducible component of ome of the Y_i ($\mathbf{0}_\mathbf{I}$, 2.1.1); the conclusion then follows from Definition (14.2.1) and the inequality in (14.2.2.3). □

Corollary. — *Let X be a topological space, and Y a locally-Noetherian closed subspace of X.*

(i) *For all $x \in X$, there exists only a finite number of irreducible components Y_i ($1 \leqslant i \leqslant n$) of Y that contain x, and we have $\operatorname{codim}_x(Y,X) = \inf_i(\operatorname{codim}(Y_i,X))$.*
(ii) *The function $x \mapsto \operatorname{codim}_x(Y,X)$ is lower semi-continuous on X.*

PROOF. By hypothesis, there exists an open neighbourhood U_0 of x in X such that $Y \cap U_0$ is Noetherian, and so U_0 has only a finite number of irreducible components, which are the intersections of U_0 with the irreducible components of Y; *a fortiori* there are only a finite number of irreducible components Y_i ($1 \leqslant i \leqslant n$) of Y that contain x, and we can, by replacing U_0 with an open neighbourhood $U \subset U_0$ of x that doesn't meet any of the Y_j that do not contain x, assume that the $Y_i \cap U$ are the irreducible components of $Y \cap U$; for every open neighbourhood $V \subset U$ of x in X, the $Y_i \cap V$ are thus the irreducible components of $Y \cap V$, and (14.2.3) then shows that $\operatorname{codim}(Y_i,X) = \operatorname{codim}(Y_i \cap V, V)$, which proves (i). Further, for every $x' \in U$, the irreducible components of Y that contain x' are certain Y_i, and so $\operatorname{codim}_{x'}(Y,X) \geqslant \operatorname{codim}_x(Y,X)$, which proves (ii). □

14.3. The chain condition

(14.3.1). In a topological space X, we say that a chain $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ of irreducible closed subsets if *saturated* if there does not exist an irreducible closed subset Z', distinct from each of the Z_i, such that $Z_k \subset Z' \subset Z_{k+1}$ for any k.

Proposition (14.3.2). — *Let X be a topological space such that, for any two irreducible closed subsets Y and Z of X with $Y \subset Z$, we have $\operatorname{codim}(Y,Z) < +\infty$. The following two conditions are equivalent.*

(a) *Any two saturated chains of closed irreducible subsets of X that have the same first and last elements as one another have the same length.*
(b) *If Y, Z, and T are irreducible closed subsets of X such that $Y \subset Z \subset T$, then*

(14.3.2.1) $$\operatorname{codim}(Y,T) = \operatorname{codim}(Y,Z) + \operatorname{codim}(Z,T).$$

PROOF. It is immediate that (a) implies (b). Conversely, suppose that (b) is satisfied, and we will show that if we have two saturated chains with the same first and last elements as one another, of lengths m and $n \leqslant m$ (respectively), then $m = n$. We proceed by induction on n, with the proposition being clear for $n = 1$. So suppose that $1 < n < m$, and let $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ be a saturated chain such that there exists another saturated chain, with first element Z_0 and last element Z_n, of length m. Since $\operatorname{codim}(Z_0,Z_n) \geqslant m > n$, and $\operatorname{codim}(Z_0,Z_1) = 1$, it follows from *(b)* that $\operatorname{codim}(Z_1,Z_n) = \operatorname{codim}(Z_0,Z_n) - 1 > n-1$, which contradicts our induction hypothesis. □ $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 11

When the conditions of (14.3.2) are satisfied, we say that X satisfies the *chain condition*, or that it is a *catenary space*. It is clear that every closed subspace of a catenary space is catenary.

Proposition (14.3.3). — *Let X be a Noetherian Kolmogoroff space of finite dimension. The following conditions are equivalent.*

(a) *Any two maximal chains of irreducible closed subsets of X have the same length.*
(b) *X is equidimensional, equicodimensional, and catenary.*
(c) *X is equidimensional, and, for any irreducible closed subsets Y and Z of X with $Y \subset Z$, we have*

$$\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z). \tag{14.3.3.1}$$

(d) *X is equicodimensional, and, for any irreducible closed subsets Y and Z of X with $Y \subset Z$, we have*

$$\operatorname{codim}(Y, Z) = \operatorname{codim}(Y, Z) + \operatorname{codim}(Z, X). \tag{14.3.3.2}$$

PROOF. The hypotheses on X imply that the first and last elements of a maximal chain of irreducible closed subsets of X are necessarily a closed point and an irreducible component of X (respectively) ($\mathbf{0_I}$, 2.1.3); further, every saturated chain with first element Y and last element Z (thus $Y \subset Z$) is contained in a maximal chain whose elements differ from those of the given chain, or are contained in Y, or contain Z. These remarks immediately establish the equivalence between (a) and (b), and also show that if (a) is satisfied, then we have, for every irreducible closed subset Y to X,

$$\dim(Y) + \operatorname{codim}(Y, X) = \dim(X); \tag{14.3.3.3}$$

from (14.3.2.1), we immediately deduce (14.3.3.1) and (14.3.3.2) from (14.3.3.3). Conversely, (14.3.3.1) implies (14.3.2.1), and so (14.3.3.1) implies the chain condition, by (14.3.2); further, by applying (14.3.3.1) to the case where Y is a single closed point x of X, and Z is an irreducible component of X, we get that $\operatorname{codim}(\{x\}, X) = \dim(Z)$; we thus conclude that (c) implies (b). Similarly, (14.3.3.2) implies (14.3.2.1), and thus the chain condition; further, with the same choice of Y and Z as above, (14.3.3.2) again implies that $\operatorname{codim}(\{x\}, X) = \dim(Z)$, and so (since every irreducible component of X contains a closed point, by ($\mathbf{0_I}$, 2.1.3)), (d) implies (b). □

We say that a Noetherian Kolmogoroff space is *biequidimensional* if it is of *finite dimension* and if it verifies any of the (equivalent) conditions of (14.3.3).

Corollary (14.3.4). — *Let X be a biequidimensional Noetherian Kolmogoroff space; then, for every closed point x of X, and every irreducible component Z of X, we have*

$$\dim(X) = \dim(Z) = \operatorname{codim}(\{x\}, X) = \dim_x(X). \tag{14.3.4.1}$$

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PROOF. The last equality follows from the fact that, if $Y_0 = \{x\} \subset Y_1 \subset \cdots \subset Y_m$ is a maximal chain of irreducible closed subsets of X, and U an open neighbourhood of x, then the $U \cap Y_i$ are pairwise disjoint irreducible closed subsets of U (because $\overline{U \cap Y_i} = Y_i$), whence $\dim(U) = \dim(X)$, by (14.1.4). □

Corollary (14.3.5). — *Let X be a Noetherian Kolmogoroff space; if X is biequidimensional, then so is every union of irreducible components of X, and every irreducible closed subset of X. In addition, for every closed subset Y of X, we have*

$$\dim(Y) + \operatorname{codim}(Y, X) = \dim(X). \tag{14.3.5.1}$$

PROOF. Every chain of irreducible closed subsets of X is contained in an irreducible component of X, and so the first claim follows immediately from (14.3.3). Further, if X' is an irreducible closed subset of X, then X' trivially satisfies the conditions of (14.3.3, c), whence the second claim.

Finally, to show (14.3.5.1), note that we have seen, in the proof of (14.3.3), that this equation is true whenever Y is irreducible; if Y_i ($1 \leqslant i \leqslant m$) are the irreducible components of Y, then the Y_i for which $\dim(Y_i)$ is the largest are also those for which $\operatorname{codim}(Y_i, X)$ is the smallest; so (14.3.5.1) follows from the definitions of $\dim(Y)$ and $\operatorname{codim}(Y, X)$. □

Remark (14.3.6). — The reader will note that the proof of (14.3.2) applies to any ordered set, and the fact that we are working with the example of a set of irreducible closed subsets of a topological space is not used at all in the proof. It is the same in the proof of (14.3.3), which holds, more generally, for any ordered set E such that, for all $x \in E$, there exists some $z \leqslant x$ which is *minimal* in E, and such that the length of chains of elements of E is bounded.

§15. M-REGULAR SEQUENCES AND $\mathscr{F}$-REGULAR SEQUENCES

15.1. M-regular sequences and M-quasi-regular sequences

(15.1.1). Let A be a ring, N an A-module; in what follows, we will write $N[T_1, \ldots, T_n]$, for short, for the graded A-module $N \otimes_A A[T_1, \ldots, T_n]$ (T_i indeterminates), where N is graded by the trivial graduation and $A[T_1, \ldots, T_n]$ by the total degree. $\mathbf{0_{IV\text{-}1}}$ | 12

Let M be an A-module, f_i ($1 \leqslant i \leqslant n$) elements of A, $\mathfrak{J} = \sum_{i=1}^{n} f_i A$ the ideal of A generated by the f_i, and let us equip M with the $\mathfrak{J}$-preadic filtration; we then define a *surjective graded homomorphism of graded A-modules*, of degree 0,

$$\varphi : (\mathrm{gr}_0(M))[T_1, \ldots, T_n] \longrightarrow \mathrm{gr}_\bullet(M) \tag{15.1.1.1}$$

in the following way: if $\alpha \in A/\mathfrak{J}$ is the class of an element $a \in A$, $\xi \in M_k/M_{k+1}$ the class mod $\mathfrak{J}M$ of an element $x \in M$, to the couple formed by ξ and a polynomial $P[T_1, \ldots, T_n]$ homogeneous of degree k, we make correspond the class mod $\mathfrak{J}^{k+1}M$ of $P(f_1, \ldots, f_n)x$; it is immediate that this class depends only on ξ and on P, and if S_k is the set of homogeneous polynomials of degree k in $A[T_1, \ldots, T_n]$, we have thus defined an A-linear map $\mathbf{0_{IV\text{-}1}}$ | 13

$$\varphi_k : \mathrm{gr}_0(M) \otimes_A S_k \longrightarrow \mathrm{gr}_k(M),$$

which is surjective by definition of $\mathfrak{J}^k M$.

We will say in what follows that an element $f \in A$ is *not a zero divisor in an A-module M* if the homothety $f_M : x \mapsto fx$ of M is injective.

Lemma (15.1.2). — *Let M be an A-module, f an element of A, and let us equip M with the (fA)-preadic filtration. If f is not a zero divisor in M, the canonical homomorphism defined in (15.1.1):*

$$\varphi : (M/fM)[T] \longrightarrow \mathrm{gr}_\bullet(M) \tag{15.1.2.1}$$

is bijective. The converse is true if M is separated for the (fA)-preadic topology.

Proof. We have here $\mathrm{gr}_k(M) = f^k M / f^{k+1} M$; to say that φ is injective means that for all $x \in M$ and all k, the relation $f^k x \in f^{k+1} M$ implies $x \in fM$; it is clear that this is indeed the case if f is not a zero divisor in M. Conversely, note that the homothety $f_M : x \mapsto fx$ defines, by passing to quotients, homomorphisms $\mathrm{gr}_k(M) \to \mathrm{gr}_{k+1}(M)$, hence a graded homomorphism g of degree 1 of $\mathrm{gr}_\bullet(M)$ into itself; and to say that φ is injective also means that g is injective. If we further suppose the filtration of M separated, it follows that f_M is injective (Bourbaki, *Alg. comm.*, chap. III, § 2, no. 8, cor. 1 of th. 1). □

Corollary (15.1.3). — *Let A be a noetherian ring, M a finitely generated A-module. For f to be not a zero divisor in M, it is necessary and sufficient that φ be bijective.*

Proof. We need only show that the condition is sufficient; to prove that f_M is injective, it suffices to show that for every prime ideal $\mathfrak{p}$ of A, the homothety of ratio $f' = f/1 \in A' = A_{\mathfrak{p}}$ in the $A_{\mathfrak{p}}$-module $M_{\mathfrak{p}} = M'$ is injective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 4, th. 1). Now $f'^k M' / f'^{k+1} M' = (f^k M / f^{k+1} M) \otimes_A A'$, and it is immediate that the canonical homomorphism $\varphi' : (M'/f'M')[T] \to \mathrm{gr}_\bullet(M')$ is equal to $\varphi \otimes 1_{A'}$; if φ is injective, so is φ' ($\mathbf{0_I}$, 1.3.2). We are thus reduced to the case where A is a noetherian *local* ring, whose maximal ideal we denote by $\mathfrak{m}$. If $f \notin \mathfrak{m}$, then f is invertible and there is nothing to prove. If $f \in \mathfrak{m}$, we know that M is separated for the (fA)-preadic topology ($\mathbf{0_I}$, 7.3.5), and it suffices therefore to apply (15.1.2). □

Definition (15.1.4). — Let A be a ring, M an A-module. We say that an element $f \in A$ is *M-regular* (resp. *M-quasi-regular*) if f is not a zero divisor in M (resp. if the homomorphism (15.1.2.1) is bijective).

We have thus just shown that every M-regular element is M-quasi-regular, and that the converse is true if A is noetherian and M a finitely generated A-module. When M is no longer assumed to be finitely generated, the converse can fail: for example, if A is a principal ideal domain that is not a field, K its field of fractions, and M the A-module K/A, we have $fM = M$ for all $f \neq 0$ in A, hence both sides of (15.1.2.1) reduce to 0, but f is then a zero divisor in M.

(15.1.5). Suppose now that M is equipped with a *filtration* (M_k) such that $M_0 = M$, and denote by $\mathrm{gr}_\bullet(M) = \bigoplus_{k\geqslant 0} M_k/M_{k+1}$ the associated graded A-module. The notations being those of (15.1.1), set, for all $k \geqslant 0$, $\mathbf{0_{IV\text{-}1}}$ | 14

(15.1.5.1) $$M'_k = M_k + \mathfrak{J}M_{k-1} + \cdots + \mathfrak{J}^{k-1}M_1 + \mathfrak{J}^k M_0.$$

It is clear that (M'_k) is a filtration on M. When $M_k = \mathfrak{Q}^k M$, where $\mathfrak{Q}$ is an ideal of A, the filtration (M'_k) is none other than the $(\mathfrak{J}+\mathfrak{Q})$-preadic filtration. We will again write $(\mathrm{gr}_\bullet(M)\otimes_A (A/\mathfrak{J}))[T_1,\dots,T_n]$ for the graded tensor product of graded A-modules (**II**, 2.1.2) $(\mathrm{gr}_\bullet(M)\otimes_A (A/\mathfrak{J}))\otimes_A A[T_1,\dots,T_n]$. We define a *surjective graded homomorphism of graded A-modules*, of degree 0,

(15.1.5.2) $$\psi : (\mathrm{gr}_\bullet(M)\otimes_A (A/\mathfrak{J}))[T_1,\dots,T_n] \longrightarrow \mathrm{gr}'_\bullet(M)$$

in the following way: if $\alpha \in A/\mathfrak{J}$ is the class of an element $a \in A$, $\xi \in M_k/M_{k+1}$ the class of an element $x \in M_k$, and $P[T_1,\dots,T_n]$ a polynomial homogeneous of degree h, we make correspond to the triplet (α,ξ,P) the class mod M'_{k+h+1} of $P(f_1,\dots,f_n)ax$; one verifies immediately that this element depends only on ξ, α, and P, and this defines the surjective graded homomorphism (15.1.5.2), by virtue of the definition (15.1.5.1). One recovers the homomorphism (15.1.1.1) by considering the filtration on M such that $M_1 = 0$.

Lemma (15.1.6). — *We keep the notations of* (15.1.5) *and suppose $n = 1$, setting $f = f_1$. If f is $\mathrm{gr}_\bullet(M)$-regular, the homomorphism* (15.1.5.2) *is bijective.*

Proof. Let Q_k (resp. Q'_k) be the sub-A-module of elements of degree k in the first (resp. second) member of (15.1.5.2). We equip Q_k with the filtration

(15.1.6.1) $$(Q_k)_i = \sum_{j\leqslant k-i} (\mathrm{gr}_{k-j}(M)\otimes_A (A/fA))T^j$$

so that $(Q_k)_0 = Q_k$ and $(Q_k)_{k+1} = 0$, and for this filtration, one has

$$\mathrm{gr}_i(Q_k) = (\mathrm{gr}_i(M)\otimes_A (A/fA))T^{k-i}.$$

We equip Q'_k with the filtration formed by $\psi((Q_k)_i) = (Q'_k)_i$. As these filtrations are *finite*, it suffices to prove that the homomorphisms $\psi_{ki} : \mathrm{gr}_i(Q_k) \to \mathrm{gr}_i(Q'_k)$ induced by ψ_k are injective (Bourbaki, *Alg. comm.*, chap. III, § 3, no. 8, cor. 1 of th. 1). Now $\mathrm{gr}_i(Q_k) = ((M_i/M_{i+1})\otimes_A (A/fA))T^{k-i}$; on the other hand, $(Q'_k)_{i+1}$ is the image of

$$M_k + fM_{k-1} + \cdots + f^{k-i-1}M_{i+1}$$

in M'_k/M'_{k+1}. We are thus led to write that, for $x \in M_i$, the relation $\mathbf{0_{IV\text{-}1}}$ | 15

(15.1.6.2) $$f^{k-i}x \in M_k + fM_{k-1} + \cdots + f^{k-i-1}M_{i+1} + M'_{k+1}$$

implies $x \in fM_i + M_{i+1}$. Now, the second member of (15.1.6.2) is contained in $M_{i+1} + M'_{k+1}$, and by virtue of (15.1.5.1), M'_{k+1} is contained in $M_{i+1} + f^{k+1-i}M$. The hypothesis on f implies that f is not a zero divisor in M/M_h for all h (Bourbaki, *loc. cit.*). The relation $f^{k-i}x \in f^{k+1-i}M + M_{i+1}$ thus implies (reasoning by induction) $x \in fM_i + M_{i+1}$. □

Definition (15.1.7). — Let $(f_i)_{1\leqslant i\leqslant n}$ be a sequence of elements of A. We say that it is *M-regular* if, for $1 \leqslant i \leqslant n$, f_i is $(M/(\sum_{1\leqslant j\leqslant i-1} f_jM))$-regular. We say that the sequence (f_i) is *M-quasi-regular* if the canonical homomorphism φ defined in (15.1.1.1) is bijective.

When $M = A$, we say simply "regular sequence" and "quasi-regular sequence".

Proposition (15.1.8). — *The notations being those of* (15.1.5)*, suppose that the sequence $(f_i)_{1\leqslant i\leqslant n}$ is $\mathrm{gr}_\bullet(M)$-regular. Then the canonical homomorphism* (15.1.5.2) *is bijective.*

Proof. The proposition is none other than (15.1.6) for $n = 1$; let us reason by induction on n. Set $\mathfrak{J}'' = \sum_{i=1}^{n-1} f_iA$, so that $\mathfrak{J} = \mathfrak{J}'' + f_nA$, and let

$$M''_k = M_k + \mathfrak{J}''M_{k-1} + \cdots + \mathfrak{J}''^{k-1}M_1 + \mathfrak{J}''^k M_0;$$

one can then write

$$M'_k = M''_k + f_nM''_{k-1} + \cdots + f_n^{k-1}M''_1 + f_n^k M''_0.$$

Let us write $\mathrm{gr}''_\bullet(M)$ for the graded A-module associated to M filtered by the filtration (M''_k). One can write, up to canonical isomorphisms,

$$(\mathrm{gr}_\bullet(M) \otimes (A/\mathfrak{J})) \otimes_A A[T_1, \ldots, T_n] = (\mathrm{gr}_\bullet(M) \otimes (A/\mathfrak{J}''))[T_1, \ldots, T_{n-1}] \otimes (A/f_n A) \otimes_A A[T_n]$$

and by (15.1.5.2), ψ factorizes as

$$(\mathrm{gr}''_\bullet(M) \otimes_A (A/f_n A))[T_n] \longrightarrow \mathrm{gr}'_\bullet(M) \tag{15.1.8.1}$$

and

$$((\mathrm{gr}''_0(M))[T_1, \ldots, T_{n-1}]) \otimes (A/f_n A) \otimes_A A[T_n] \xrightarrow{\psi' \otimes 1} \mathrm{gr}''_\bullet(M) \otimes (A/f_n A) \otimes_A A[T_n]$$

where ψ' is the map (15.1.5.2) in which one replaces n by $n-1$, $\mathfrak{J}$ by $\mathfrak{J}''$, and $\mathrm{gr}'_\bullet(M)$ by $\mathrm{gr}''_\bullet(M)$. The induction hypothesis implies that ψ' is bijective, and it remains to see, by applying (15.1.6), that it suffices to show that f_n is $\mathrm{gr}''_\bullet(M)$-regular. But $\mathrm{gr}''_\bullet(M)$ identifies with $(\mathrm{gr}_\bullet(M) \otimes (A/\mathfrak{J}''))[T_1, \ldots, T_{n-1}]$; by hypothesis, the homothety of ratio f_n in $\mathrm{gr}_\bullet(M) \otimes (A/\mathfrak{J}'')$ is bijective; since $A[T_1, \ldots, T_{n-1}]$ is a free A-module, it follows that f_n is $\mathrm{gr}''_\bullet(M)$-regular. □

Proposition (15.1.9). — *Let M be an A-module, $(f_i)_{1 \leqslant i \leqslant n}$ a sequence of elements of A. If the sequence (f_i) is M-regular, it is M-quasi-regular. The converse is true when one supposes in addition that the A-modules M, $M/f_1M, \ldots, M/(\sum_{1 \leqslant i \leqslant n-1} f_i M)$ are separated for the $\mathfrak{J}$-preadic topology.*

PROOF. The first assertion follows from (15.1.8), taking $M_1 = 0$, and by (15.1.5.2), φ factorizes as **$0_{\text{IV-1}}$** | 16

$$(\mathrm{gr}''_\bullet(M) \otimes (A/f_n A))[T_n] \longrightarrow \mathrm{gr}_\bullet(M) \tag{15.1.9.1}$$

and $\varphi' \otimes 1$, where φ' is the canonical homomorphism

$$(\mathrm{gr}''_0(M))[T_1, \ldots, T_{n-1}] \longrightarrow \mathrm{gr}''_\bullet(M).$$

As these two homomorphisms are surjective, they are both bijective if φ is supposed bijective, hence φ' is necessarily injective, and thus bijective since it is surjective. One can then reason by induction on n (the case $n = 1$ follows from (15.1.2)), since on M, $M/f_1M, \ldots, M/(\sum_{j=1}^{n-2} f_j M)$ the $\mathfrak{J}''$-preadic topology is finer than the $\mathfrak{J}$-preadic topology, and is thus separated. One sees therefore that the sequence $(f_i)_{1 \leqslant i \leqslant n-1}$ is M-regular. On the other hand, let us show that if $y \in M/\mathfrak{J}''M$ is such that $f_n^k y \in f_n^{k+1}(M/\mathfrak{J}''M)$, then $y \in f_n(M/\mathfrak{J}''M)$. Indeed, if x is a representative of y, let us show that $f_n^k x \in \mathfrak{J}^{k+1}M$; our assertion will then follow from the fact that (15.1.9.1) is injective. The hypothesis implies $f_n^k x \in f_n^{k+1}M + \mathfrak{J}''M \subset \mathfrak{J}M$, whence, since φ is injective, $f_n^{k-1}x \in \mathfrak{J}^k M$; by induction, one thus obtains $x \in \mathfrak{J}M$, and finally $f_n^k x \in \mathfrak{J}^{k+1}M$. We finish the reasoning as in (15.1.2): on $M/\mathfrak{J}''M$, the $\mathfrak{J}$-preadic filtration is identical to the (f_nA)-preadic filtration, which is separated by hypothesis, and one concludes that the homothety of ratio f_n in $M/\mathfrak{J}''M$ is injective, which finishes showing that the sequence $(f_i)_{1 \leqslant i \leqslant n}$ is M-regular. □

Corollary (15.1.10). — *With the notations of (15.1.9), suppose that for every finitely generated A-module N, $M \otimes_A N$ is separated for the $\mathfrak{J}$-preadic topology. Then, for the sequence (f_i) to be M-regular, it is necessary and sufficient that it be M-quasi-regular.*

One will note that the hypothesis of (15.1.10) implies that, if the sequence $(f_i)_{1 \leqslant i \leqslant n}$ is M-regular, so is the sequence $(f_{\sigma(i)})$ for every permutation σ of the interval $[1, n]$.

Corollary (15.1.11). — *Let A be a noetherian ring, M a finitely generated A-module, $(f_i)_{1 \leqslant i \leqslant n}$ a finite sequence of elements contained in the radical of A. Then, for the sequence (f_i) to be M-regular, it is necessary and sufficient that it be M-quasi-regular.*

PROOF. This is a particular case of (15.1.10), since every finitely generated A-module is then separated for the $\mathfrak{J}$-preadic topology ($\mathbf{0_I}$, 7.3.5). □

Remarks (15.1.12). —

(i) When A is a noetherian local ring, M a finitely generated A-module, and when the f_i are contained in the maximal ideal of A, the notion of M-regular sequence was introduced by J.-P. Serre under the name of *M-suite* [**?**].

(ii) Even if A is noetherian and $M = A$, a sequence (f, g) of two elements of A that is quasi-regular is not necessarily regular. For example, the sequence $(0, 1)$ is quasi-regular, since then $\mathfrak{J} = fA + gA = A$, hence $\mathrm{gr}_\bullet(A) = 0$; but if $A \neq 0$ it is not regular, since 0 is not A-regular. One will note however that in this case the sequence $(1, 0)$ is regular, since 1 is not a zero divisor in A, and since $A/A = 0$, 0 is not a zero divisor in this module (cf. (15.1.1)).

(iii) In (15.1.6), one cannot in general replace the hypothesis that f is $\mathrm{gr}_\bullet(M)$-regular by the hypothesis that it is $\mathrm{gr}_\bullet(M)$-quasi-regular. To see this, let us take the example (15.1.4) where A is a principal ideal domain with field of fractions $K \neq A$, and let us take $M = K$, the filtration (M_k) of M being defined by $M_0 = M = K$, $M_1 = A$, $M_k = 0$ for $k \geqslant 2$, whence $\mathrm{gr}_0(M) = K/A$, $\mathrm{gr}_1(M) = A$, $\mathrm{gr}_k(M) = 0$ for $k \geqslant 2$. If $f \neq 0$ in A, one sees that f is not $\mathrm{gr}_\bullet(M)$-regular, but since $f^k \,\mathrm{gr}_\bullet(M)/f^{k+1}\,\mathrm{gr}_\bullet(M) = f^k A/f^{k+1}A$, it is immediate that the homomorphism (15.1.2.1) (where M is replaced by $\mathrm{gr}_\bullet(M)$) is injective, hence f is $\mathrm{gr}_\bullet(M)$-quasi-regular. But with the notations of (15.1.6), one then has $M'_k = K$ for all $k \geqslant 0$, hence $\mathrm{gr}'_\bullet(M) = 0$, while the first member of (15.1.5.2) is not reduced to 0. $\mathbf{0_{IV\text{-}1}}$ | 17

Proposition (15.1.13). — *Let A be a ring, M, N two A-modules, $(f_i)_{1 \leqslant i \leqslant n}$ a sequence of elements of A. If the sequence (f_i) is M-regular and if N is a flat A-module, then the sequence (f_i) is $(M \otimes_A N)$-regular. The converse is true if N is a faithfully flat A-module.*

Proof. Indeed (without any hypothesis on N), $(M \otimes_A N)/(\sum_{j=1}^{i-1} f_j(M \otimes_A N))$ is isomorphic to $(M/(\sum_{j=1}^{i-1} f_j M)) \otimes_A N$; it suffices then to apply ($\mathbf{0_I}$, 6.1.1) and ($\mathbf{0_I}$, 6.4.1) to the homothety of ratio f_i in $M/(\sum_{j=1}^{i-1} f_j M)$. □

Corollary (15.1.14). — *Let A be a ring, M an A-module, $(f_i)_{1 \leqslant i \leqslant n}$ a finite sequence of elements of A. Let $\rho : A \to B$ be a ring homomorphism, $M' = M_{(B)} = M \otimes_A B$, $f'_i = \rho(f_i)$ for $1 \leqslant i \leqslant n$. If B is a flat A-module and if the sequence (f_i) is M-regular, then the sequence (f'_i) is M'-regular; the converse is true if B is a faithfully flat A-module.*

Proof. Indeed, $M'/(\sum_{j=1}^{i-1} f'_j M')$ identifies with $(M/(\sum_{j=1}^{i-1} f_j M)) \otimes_A B$, and the homothety of ratio f'_i in the B-module $M'/(\sum_{j=1}^{i-1} f'_j M')$ with the homothety of ratio f_i in the A-module $(M/(\sum_{j=1}^{i-1} f_j M)) \otimes_A B$; the assertion thus follows from (15.1.13). □

Proposition (15.1.15). — *Let A be a ring, B a commutative A-algebra, $(f_i)_{1 \leqslant i \leqslant n}$ a finite sequence of elements of B, M a B-module. Suppose that the sequence (f_i) is M-regular and that each of the A-modules $M/(\sum_{j=1}^{i-1} f_j M)$ $(1 \leqslant i \leqslant n)$ is flat. Then, for every ring homomorphism $\rho : A \to A'$, setting $B' = B \otimes_A A'$, $M' = M \otimes_A A'$, and $f'_i = f_i \otimes 1$ $(1 \leqslant i \leqslant n)$, the sequence (f'_i) is M'-regular and the A'-modules $M'/(\sum_{j=1}^{i-1} f'_j M')$ $(1 \leqslant i \leqslant n)$ are flat.*

Proof. Consider the exact sequence (by hypothesis) $\mathbf{0_{IV\text{-}1}}$ | 18

$$0 \longrightarrow M \xrightarrow{h_1} M \longrightarrow M/f_1 M \longrightarrow 0$$

(the arrow $M \xrightarrow{h_1} M$ being the homothety of ratio f_1 in M). The hypothesis that M/f_1M is A-flat implies that the sequence

$$0 \longrightarrow M \otimes_A A' \xrightarrow{h_1 \otimes 1} M \otimes_A A' \longrightarrow (M/f_1 M) \otimes_A A' \longrightarrow 0$$

is exact ($\mathbf{0_I}$, 6.1.2). This shows that the homothety of ratio f'_1 in M' is injective.

Set then, for $1 \leqslant i \leqslant n-1$, $M_i = M/(\sum_{j=1}^{i} f_j M)$, $M'_i = M'/(\sum_{j=1}^{i} f'_j M')$; we have $M'_i = M_i \otimes_A A'$, $M_i/f_{i+1}M_i = M_{i+1}$, $M'_i/f'_{i+1}M'_i = M'_{i+1}$; it suffices to replace M and f_1 by M_i and f_{i+1} to conclude that the homothety of ratio f'_{i+1} in M'_i is injective, by reason of the flatness of $M_i/f_{i+1}M_i$. The last assertion follows from ($\mathbf{0_I}$, 6.2.1). □

Proposition (15.1.16). — *Let A, B be two noetherian local rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, M a finitely generated B-module. Let $(f_i)_{1 \leqslant i \leqslant n}$ be a sequence of elements of the maximal ideal of B, and for each i, let g_i be the image of f_i in $B \otimes_A k$. The following conditions are equivalent:*

a) *The sequence (f_i) is M-regular and the quotients $M_i = M/(\sum_{j=1}^{i} f_j M)$ are flat A-modules for $1 \leqslant i \leqslant n$.*
b) *The sequence (f_i) is M-regular and the A-module M_n is flat.*
c) *The A-module M is flat and the sequence $(g_i)_{1\leqslant i\leqslant n}$ is $(M \otimes_A k)$-regular.*
d) *The A-module M is flat, and for every ring homomorphism $\rho : A \to A'$, if we set $B' = B \otimes_A A'$, $M' = M \otimes_A A'$, $f'_i = f_i \otimes 1$ $(1 \leqslant i \leqslant n)$, the sequence (f'_i) is M'-regular.*

Proof. It is clear that a) implies b); to show that b) implies d), it suffices, by virtue of (15.1.15), to show that b) implies that the M_i are flat A-modules for $0 \leqslant i \leqslant n-1$ (setting $M_0 = M$). As $M_{i+1} = M_i/f_{i+1}M_i$, we are reduced, by descending induction, to proving that if h is an element of the maximal ideal of B such that the homothety of ratio h in M is injective and M/hM is a flat A-module, then M is a flat A-module; but this is none other than ($\mathbf{0_{III}}$, 10.2.7). It is clear that c) is a particular case of d) obtained by taking $A' = k$. It thus remains to prove that c) implies a). As M is a flat A-module and a finitely generated B-module, it follows from c) and from ($\mathbf{0_{III}}$, 10.2.4) that the homothety of ratio f_1 in M is injective and that its cokernel $M/f_1M = M_1$ is a flat A-module. Let us reason by induction on i and suppose shown that M_i is a flat A-module; as the homothety of ratio g_{i+1} in $M_i \otimes_A k$ is injective by hypothesis, it follows again from ($\mathbf{0_{III}}$, 10.2.4) that the homothety of ratio f_{i+1} in M_i is injective and that $M_i/f_{i+1}M_i = M_{i+1}$ is a flat A-module, which finishes the proof. One will note that the proof that c) implies a) does not use the fact that the f_i belong to the maximal ideal of B. □

Corollary (15.1.17). — *The hypotheses on A and B being those of (15.1.16), let M be a finitely generated B-module, flat over A, R a sub-B-module of M such that $N = M/R$ is a flat A-module. Let $(f_i)_{1\leqslant i\leqslant n}$ be a sequence of elements of the maximal ideal of B having the following properties:*

(i) *$f_iM \subset R$ for $1 \leqslant i \leqslant n$.*
(ii) *If g_i is the image of f_i in $B \otimes_A k$, the sequence $(g_i)_{1\leqslant i\leqslant n}$ is $(M \otimes_A k)$-regular.*
(iii) *The sum of the sub-$(B \otimes_A k)$-modules $g_i(M \otimes_A k)$ of $M \otimes_A k$ is the image of $R \otimes_A k$ in $M \otimes_A k$ (equal to the kernel of the canonical homomorphism $M \otimes_A k \to N \otimes_A k$).*

Then the sequence (f_i) is M-regular and $R = \sum_{i=1}^{n} f_iM$.

Proof. As N is a flat A-module, it follows from the exactness of the sequence $0 \to R \to M \to N \to 0$ that the sequence $0 \to R \otimes_A k \to M \otimes_A k \to N \otimes_A k \to 0$ is exact ($\mathbf{0_{I}}$, 6.1.2); we thus have $R \otimes_A k = \sum g_i(M \otimes_A k)$. As M is a finitely generated B-module and B is noetherian, R is also a finitely generated B-module, whence $R = \sum f_iM$ by Nakayama's lemma, since the f_i belong to the maximal ideal of B. □ $\mathbf{0_{IV\text{-}1}}$ | 19

Lemma (15.1.18). — *Let A be a ring, M an A-module equipped with a separated and exhaustive filtration $(M_k)_{k\geqslant 0}$, and let $f \in A$ be a $\operatorname{gr}_\bullet(M)$-regular element. Then:*

(i) *f is M-regular.*
(ii) *For all k, one has $M_k \cap fM = fM_k$, and the canonical image M'_k of M_k in $M' = M/fM$ is thus canonically isomorphic to M_k/fM_k.*
(iii) *The sequence*

$$0 \longrightarrow M_{k+1}/fM_{k+1} \longrightarrow M_k/fM_k \longrightarrow \operatorname{gr}_k(M)/f \cdot \operatorname{gr}_k(M) \longrightarrow 0$$

is exact, and if one equips M' with the quotient filtration (M'_k), $\operatorname{gr}_\bullet(M')$ is thus canonically isomorphic to $\operatorname{gr}_\bullet(M)/f \cdot \operatorname{gr}_\bullet(M)$.

Proof. Assertion (i) follows from the hypothesis on f and from Bourbaki, *Alg. comm.*, chap. III, § 2, no. 8, cor. 1 of th. 1. As $\operatorname{gr}_k(M/M_k)$ is a direct factor of $\operatorname{gr}_k(M)$, f is $\operatorname{gr}_\bullet(M/M_k)$-regular by (i), which establishes (ii). To prove (iii), it suffices to see that the map $M_{k+1}/fM_{k+1} \to M_k/fM_k$ is injective: now, if $x \in M_{k+1}$ is such that $x = fy$ for $y \in M$, it follows from (ii) that $x \in fM_{k+1}$, which finishes proving the lemma. □

Proposition (15.1.19). — *Let A be a ring, M an A-module equipped with a filtration $(M_k)_{k\geqslant 0}$ such that $M_0 = M$, $(f_i)_{1\leqslant i\leqslant n}$ a $\operatorname{gr}_\bullet(M)$-regular sequence of elements of A. Suppose in addition that, for $0 \leqslant i \leqslant n-1$, the quotient filtration of (M_k) on $M/(\sum_{j\leqslant i} f_jM)$ is separated. Then the sequence (f_i) is M-regular, and if* *one sets $M' = M/(\sum_{i\leqslant n} f_i M)$ and equips M' with the quotient filtration of (M_k), $\mathrm{gr}_\bullet(M')$ is canonically isomorphic to $\mathrm{gr}_\bullet(M)/(\sum_{i\leqslant n} f_i\,\mathrm{gr}_\bullet(M))$.*

PROOF. For $n = 1$, the proposition follows from the lemma (15.1.18). Let us reason by induction on n, setting $N = M/(\sum_{i\leqslant n-1} f_i M)$ and denoting by (N_k) the quotient filtration of (M_k) on N, which is separated by hypothesis; moreover, $\mathrm{gr}_\bullet(N)$ is isomorphic to

$$\mathrm{gr}_\bullet(M)/\Big(\sum_{i\leqslant n-1} f_i\,\mathrm{gr}_\bullet(M)\Big).$$

By hypothesis, f_n is thus $\mathrm{gr}_\bullet(N)$-regular, and it follows from the lemma (15.1.18) applied to N that f_n is N-regular and $\mathrm{gr}_\bullet(M')$ is isomorphic to $\mathrm{gr}_\bullet(N)/f_n\,\mathrm{gr}_\bullet(N)$, which proves the proposition. □

The proposition (15.1.19) will apply in particular for filtrations satisfying one or the other of the following hypotheses:

1° The filtration (M_k) is finite and separated (since this implies $M_k = 0$ for k large enough);

2° A is a noetherian ring, $\mathfrak{J}$ an ideal contained in the radical of A, M a finitely generated A-module, and (M_k) the $\mathfrak{J}$-preadic filtration ($\mathbf{0_I}$, 7.3.5).

Corollary (15.1.20). — *Let A be a ring, M an A-module, $(f_i)_{1\leqslant i\leqslant n}$ an M-regular sequence of elements of A. Then, for any integers $\nu_i \geqslant 1$, the sequence $(f_i^{\nu_i})$ is M-regular, and $M/(\sum_{i\leqslant n} f_i^{\nu_i} M)$ has a composition series whose quotients are isomorphic to $M/(\sum_{i\leqslant n} f_i M)$.* $\mathbf{0_{IV\text{-}1}}$ | 20

PROOF. Let us reason by induction on n; for $n = 1$ the assertions are immediate since $f^j M/f^{j+1}M$ is isomorphic to M/fM, the homothety of ratio f in M being injective by hypothesis. Set then $N = M/(\sum_{i\leqslant n-1} f_i^{\nu_i} M)$; by the induction hypothesis, N admits a finite filtration (N_k) for which $\mathrm{gr}_k(N)$ are isomorphic to $M/(\sum_{i\leqslant n} f_i M)$; by hypothesis f_n is thus $\mathrm{gr}_\bullet(N)$-regular; it follows by (15.1.19) that f_n is N-regular. Applying the case $n = 1$ proved at the start, one concludes that $f_n^{\nu_n}$ is N-regular and that $N/f_n^{\nu_n}N$ admits a composition series whose quotients are isomorphic to N/f_nN; but for the filtration of N/f_nN given by the quotient filtration of (N_k), $\mathrm{gr}_k(N/f_nN)$ is isomorphic to $\mathrm{gr}_k(N)/f_n\,\mathrm{gr}_k(N)$ by (15.1.19) for all k, hence to $M/(\sum_{i\leqslant n} f_i M)$, which proves the corollary. □

Proposition (15.1.21). — *Let A, B be two noetherian local rings, $\varphi : A \to B$ a local homomorphism, M a finitely generated B-module, $(f_i)_{1\leqslant i\leqslant n}$ an A-regular sequence of elements of the maximal ideal of A. The following properties are equivalent:*

a) M is a flat A-module.

b) If $\mathfrak{J}$ is the ideal $\sum_{i\leqslant n} f_i A$, $M/\mathfrak{J}M$ is a flat $A/\mathfrak{J}$-module and the sequence (f_i) is M-regular.

PROOF. The fact that a) implies b) follows from (15.1.13) and from ($\mathbf{0_{III}}$, 10.2.2). Conversely, if the sequence (f_i) is M-regular, it is M-quasi-regular (15.1.9), hence the homomorphism (15.1.1.1) is bijective. But as the sequence $(f_i)_{1\leqslant i\leqslant n}$ is also supposed A-regular, it is A-quasi-regular, and the corresponding canonical homomorphism $(A/\mathfrak{J})[T_1,\dots,T_n] \to \mathrm{gr}_\bullet(A)$ is bijective. One concludes that the canonical morphism ($\mathbf{0_{III}}$, 10.1.2)

$$\mathrm{gr}_0(M) \otimes_{A/\mathfrak{J}} \mathrm{gr}_\bullet(A) \longrightarrow \mathrm{gr}_\bullet(M)$$

is bijective. The conclusion then follows from ($\mathbf{0_{III}}$, 10.2.2), the f_i being in the maximal ideal of A and the homomorphism φ being local. □

15.2. $\mathscr{F}$-regular sequences

(15.2.1). Let X be a ringed space, $\mathscr{F}$ an $\mathscr{O}_X$-Module, f_i $(1 \leqslant i \leqslant n)$ a sequence of sections of $\mathscr{O}_X$ over X, $\mathscr{J}$ the Ideal of $\mathscr{O}_X$ generated by the f_i (image of $\mathscr{O}_X^n$ by the homomorphism $\mathscr{O}_X^n \to \mathscr{O}_X$ defined canonically by the f_i ($\mathbf{0_I}$, 5.1.1)). The sub-$\mathscr{O}_X$-Modules $\mathscr{J}^k\mathscr{F}$ define a *filtration* on $\mathscr{F}$, and one sets further $\mathrm{gr}_k(\mathscr{F}) = \mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}$; if S_k is the set of homogeneous polynomials of total degree k in the ring $\mathbf{Z}[T_1, \ldots, T_n]$, S_k can be considered as a simple sheaf on X, and one defines as in (15.1.1) a canonical homomorphism $\mathbf{0_{IV\text{-}1}}$ | 21

$$\varphi_k : \mathrm{gr}_0(\mathscr{F}) \otimes_{\mathbf{Z}} S_k \longrightarrow \mathrm{gr}_k(\mathscr{F}) \tag{15.2.1.1}$$

in the following way: for every open set U of X (or of a base of the topology of X), one considers the homomorphism (15.1.1.1)

$$\varphi_{k,U} : (\mathrm{gr}_0(\Gamma(U, \mathscr{F}))) \otimes_{\mathbf{Z}} S_k \longrightarrow \mathrm{gr}_k(\Gamma(U, \mathscr{F}))$$

defined by the n sections $f_i|U \in \Gamma(U, \mathscr{O}_X)$, the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathscr{F})$ being filtered by the sub-modules $(\Gamma(U, \mathscr{J}))^k\Gamma(U, \mathscr{F})$. The $\mathscr{O}_X$-Module $\mathrm{gr}_k(\mathscr{F})$ being associated to the presheaf $U \mapsto \mathrm{gr}_k(\Gamma(U, \mathscr{F}))$ for all k, the $\varphi_{k,U}$ define a homomorphism of (15.2.1.1). By virtue of the exactness properties of inductive limits of modules and their commutation with tensor products, it follows from the above that for all $x \in X$, the homomorphism of the stalks of the two members of (15.2.1.1) at the point x identifies with the homomorphism of (15.1.1)

$$\mathrm{gr}_0(\mathscr{F}_x) \otimes_{\mathbf{Z}} S_k \longrightarrow \mathrm{gr}_k(\mathscr{F}_x) \tag{15.2.1.2}$$

defined by the sequence $((f_i)_x)_{1 \leqslant i \leqslant n}$ of elements of $\mathscr{O}_x$.

(15.2.2). The notations being those of (15.2.1), we say that the sequence $(f_i)_{1 \leqslant i \leqslant n}$ is *$\mathscr{F}$-regular* if, denoting by $f_j\mathscr{F}$ the sub-$\mathscr{O}_X$-Module of $\mathscr{F}$, the image of the endomorphism of $\mathscr{F}$ defined by f_j, the endomorphism of $\mathscr{F}/(\sum_{j=1}^{i-1} f_j\mathscr{F})$ defined by f_i is *injective* for $1 \leqslant i \leqslant n$; we say that the sequence (f_i) is *$\mathscr{F}$-quasi-regular* if the homomorphisms (15.2.1.1) are injective. It follows from these definitions and the remarks of (15.2.1) that for the sequence (f_i) to be $\mathscr{F}$-regular (resp. $\mathscr{F}$-quasi-regular), it is necessary and sufficient that, for all $x \in X$, the sequence $((f_i)_x)_{1 \leqslant i \leqslant n}$ be $\mathscr{F}_x$-regular (resp. $\mathscr{F}_x$-quasi-regular). Every $\mathscr{F}$-regular sequence is thus $\mathscr{F}$-quasi-regular (15.1.9); the converse is true if $\mathscr{F}$ is an $\mathscr{O}_X$-Module *of finite type*, if the $\mathscr{O}_x$ are *noetherian* rings for all $x \in X$, and if moreover the $(f_i)_x$ belong to the *radical* of $\mathscr{O}_x$ for all $x \in X$ (15.1.11).

If $X = \mathrm{Spec}(A)$ is an affine scheme, and $\mathscr{F} = \widetilde{M}$, it follows from (**I**, 1.3) that for the sequence (f_i) to be $\mathscr{F}$-regular (resp. $\mathscr{F}$-quasi-regular), it is necessary and sufficient that it be M-regular (resp. M-quasi-regular).

When $\mathscr{F} = \mathscr{O}_X$, one omits the mention of $\mathscr{F}$ in the preceding definitions.

Remark (15.2.3). — When $X = \mathrm{Spec}(A)$ and $\mathfrak{J} = \sum_{i=1}^n f_iA$, the sequence (f_j) is quasi-regular for *every* A-module M in every open set U not meeting $V(\mathfrak{J})$, since for every point $x \in U$, $\mathfrak{J}_x = \mathscr{O}_x$ and the two members of (15.2.1.2) are zero. The notion of $\widetilde{M}$-regular sequence is thus of interest only in a neighbourhood of the points of $V(\mathfrak{J})$.

Proposition (15.2.4). — *Let X be a ringed space, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, x a point of X, $(f_i)_{1 \leqslant i \leqslant n}$ a sequence of elements of $\Gamma(X, \mathscr{O}_X)$. If the sequence $((f_i)_x)_{1 \leqslant i \leqslant n}$ is $\mathscr{F}_x$-regular, there exists an open neighbourhood U of x such that the sequence $(f_i|U)_{1 \leqslant i \leqslant n}$ is $(\mathscr{F}|U)$-regular.*

PROOF. Indeed, for $1 \leqslant i \leqslant n$, the $\mathscr{O}_X$-Module $\mathscr{F}/(\sum_{j=1}^{i-1} f_j\mathscr{F})$ is coherent ($\mathbf{0_I}$, 5.3.4), hence so is the kernel of the endomorphism of this $\mathscr{O}_X$-Module defined by f_i ($\mathbf{0_I}$, 5.3.4); as the stalk of this kernel at the point x is zero, the same holds for its restriction to a neighbourhood of x ($\mathbf{0_I}$, 5.2.2). □

Proposition (15.2.5). — *Let $u = (\psi, \theta) : X \to Y$ be a flat morphism of ringed spaces ($\mathbf{0_I}$, 6.7.1); let $\mathscr{F}$ be an $\mathscr{O}_Y$-Module, $\mathscr{G} = u^*(\mathscr{F})$, $(f_i)_{1 \leqslant i \leqslant n}$ a sequence of elements of $\Gamma(Y, \mathscr{O}_Y)$, $(g_i)_{1 \leqslant i \leqslant n}$ the sequence of their images $g_i = \theta(f_i) \in \Gamma(Y, u_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$. If the sequence (f_i) is $\mathscr{F}$-regular, the sequence (g_i) is $\mathscr{G}$-regular. The converse is true if u is a faithfully flat morphism ($\mathbf{0_I}$, 6.7.8).* $\mathbf{0_{IV\text{-}1}}$ | 22

PROOF. Let x be a point of X, $y = u(x) = \psi(x)$; we have

$$\mathscr{G}_x = \psi_x^*(\mathscr{F}_y) \otimes_{\psi_x^*(\mathscr{O}_y)} \mathscr{O}_x, \qquad \text{and} \qquad (g_i)_x = \theta_x^*(\psi_x^*((f_i)_y));$$

as $\theta_x^* : \psi_x^*(\mathscr{O}_y) \to \mathscr{O}_x$ makes $\mathscr{O}_x$ a flat $\psi_x^*(\mathscr{O}_y)$-module, and since the functor ψ^* is exact, the first assertion follows from the corresponding assertion of (15.1.14); the same holds for the second, taking into account the fact that u being faithfully flat means that ψ is surjective and that θ_x^* makes $\mathscr{O}_x$ a faithfully flat $\psi_x^*(\mathscr{O}_y)$-module. ☐

§16. Dimension and depth of Noetherian local rings

16.1. Dimension of a ring

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(16.1.1). We call the *dimension* (or *Krull dimension*) of a ring A and denote by dim A the (combinatorial) dimension of its spectrum Spec(A) (14.2.1); as the irreducible closed subsets of Spec(A) are the $V(\mathfrak{p})$, where $\mathfrak{p}$ is a prime ideal of A (Bourbaki, *Alg. comm.*, chap. II, § 4, n° 3, cor. 1 of prop. 14), it amounts to saying that dim(A) is *the upper bound of the lengths of chains* (14.1.1)

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_n$$

of *prime ideals* of A. If $A \neq 0$, we have $\dim A \geqslant 0$ (in particular an artinian ring $\neq 0$ is of dimension 0 (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, prop. 7); a Dedekind ring that is not a field (in particular $\mathbf{Z}$ or a polynomial ring $k[T]$ over a field k) is of dimension 1 (Bourbaki, *Alg. comm.*, chap. VII, § 2). A Noetherian ring is not necessarily of finite dimension [30]. If $(\mathfrak{p}_\alpha)$ is the family of minimal prime ideals of A, we have (14.1.2.1)

$$\dim(A) = \sup_\alpha(\dim(A/\mathfrak{p}_\alpha)). \tag{16.1.1.1}$$

(16.1.2). For every ideal $\mathfrak{a}$ of A, the prime ideals of $A/\mathfrak{a}$ correspond bijectively to the prime ideals of A containing $\mathfrak{a}$, hence

$$\dim(A/\mathfrak{a}) \leqslant \dim(A). \tag{16.1.2.1}$$

If in addition $\mathfrak{a}$ *is not contained in any minimal prime ideal* $\mathfrak{p}_\alpha$ of A, every chain of prime ideals of A containing $\mathfrak{a}$ can be completed by one of the $\mathfrak{p}_\alpha$, hence

$$1 + \dim(A/\mathfrak{a}) \leqslant \dim(A). \tag{16.1.2.2}$$

(16.1.3). If S is a multiplicative subset of A, the prime ideals of $S^{-1}A$ correspond bijectively to the prime ideals of A not meeting S, hence

$$\dim(S^{-1}A) \leqslant \dim(A). \tag{16.1.3.1}$$

For every prime ideal $\mathfrak{p}$ of A, we have, by definition (14.2.1)

$$\dim(A_\mathfrak{p}) = \operatorname{codim}(V(\mathfrak{p}), \operatorname{Spec}(A)). \tag{16.1.3.2}$$

This number is also called the *height* of the prime ideal $\mathfrak{p}$ and denoted $\operatorname{ht}(\mathfrak{p})$. More generally, for every ideal $\mathfrak{a}$ of A, we call the *height* of $\mathfrak{a}$ the number

$$\operatorname{ht}(\mathfrak{a}) = \operatorname{codim}(V(\mathfrak{a}), \operatorname{Spec}(A)). \tag{16.1.3.3}$$

If $(\mathfrak{p}_\lambda)$ is the family of minimal prime ideals among those containing $\mathfrak{a}$, we have then (14.2.1)

$$\operatorname{ht}(\mathfrak{a}) = \inf_\lambda(\operatorname{ht}(\mathfrak{p}_\lambda)). \tag{16.1.3.4}$$

We evidently have

$$\dim(A) = \sup_\mathfrak{m}(\dim(A_\mathfrak{m})) \tag{16.1.3.5}$$

where $\mathfrak{m}$ ranges over the set of maximal ideals of A.

(16.1.4). For every prime ideal $\mathfrak{p}$ of A, we have (14.2.2)

$$\dim(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) \leqslant \dim A. \tag{16.1.4.1}$$

We say that A is *catenary* (resp. *equidimensional*, *equicodimensional*, *biequidimensional*) when Spec(A) is catenary (resp. equidimensional, equicodimensional, biequidimensional). When A is Noetherian and biequidimensional, the two members of (16.1.4.1) are equal for every prime ideal $\mathfrak{p}$ of A (14.3.5).

For every ideal $\mathfrak{a}$ of A, the prime ideals of $A/\mathfrak{a}$ correspond bijectively to the prime ideals of A containing $\mathfrak{a}$, hence, *if* A *is catenary, so is* $A/\mathfrak{a}$. Similarly, for every multiplicative subset S of A, the prime ideals of $S^{-1}A$ correspond bijectively to the prime ideals of A not meeting S, hence *if* A *is catenary, so is* $S^{-1}A$.

Furthermore, as the prime ideals of A contained in a prime ideal $\mathfrak{p}$ correspond bijectively to the prime ideals of $A_\mathfrak{p}$, for A to be catenary it suffices that $A_\mathfrak{p}$ be so for every $\mathfrak{p}$. To say that A is catenary means ((14.3.2) and (16.1.3.2)) that for every pair of prime ideals $\mathfrak{p}$, $\mathfrak{q}$ of A such that $\mathfrak{q} \subset \mathfrak{p}$, we have

(16.1.4.2) $$\dim(A_\mathfrak{q}) + \dim(A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}) = \dim(A_\mathfrak{p}).$$

If A is biequidimensional, it is the same for $A_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A.

Proposition (16.1.5). — *Let* $\rho : A \to B$ *be a ring homomorphism making* B *an integral* A*-algebra. Then* $\dim(B) \leqslant \dim(A)$*; if in addition* ρ *is injective, we have* $\dim(B) = \dim(A)$.

PROOF. If $\mathfrak{n}$ is the kernel of ρ, $\rho(A)$ is isomorphic to $A/\mathfrak{n}$, hence $\dim \rho(A) \leqslant \dim(A)$ by (16.1.2.1), and since B is integral over $\rho(A)$, one can reduce to proving the second assertion. If $\mathfrak{p}'_0 \subset \mathfrak{p}'_1 \subset \ldots \subset \mathfrak{p}'_n$ is a chain of prime ideals of B, the $\mathfrak{p}'_i \cap A$ are then distinct prime ideals in A (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, cor. 1 of prop. 1), hence they form a chain of prime ideals. Conversely, for every chain $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_m$ of prime ideals of A, we know that for each i there exists a prime ideal $\mathfrak{p}'_i$ of B such that $\mathfrak{p}'_i \cap A = \mathfrak{p}_i$ and $\mathfrak{p}'_i \subset \mathfrak{p}'_{i+1}$ for $0 \leqslant i \leqslant m-1$ (*loc. cit.*, cor. 2 of th. 1). Hence the conclusion. $\square$ $\mathbf{0_{IV\text{-}1}}$ | 24

Proposition (16.1.6). — *Let* B *be an integral ring,* A *a subring local to* B*,* $\mathfrak{m}$ *its maximal ideal. Suppose that* A *is integrally closed and that* B *is integral over* A*; then, for every maximal ideal* $\mathfrak{n}$ *of* B*, we have* $\dim(B_\mathfrak{n}) = \dim(A)$.

PROOF. The first part of the reasoning of (16.1.5) shows that $\dim(B_\mathfrak{n}) \leqslant \dim(A)$. Conversely, consider a chain of prime ideals $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_k = \mathfrak{m}$ of A; we know that $\mathfrak{n} \cap A = \mathfrak{m}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, prop. 1) and by virtue of the hypotheses made, one can apply the second theorem of Cohen-Seidenberg (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 4, th. 3), which proves the existence of a chain of prime ideals $\mathfrak{p}'_0 \subset \mathfrak{p}'_1 \subset \ldots \subset \mathfrak{p}'_k = \mathfrak{n}$ of B such that $\mathfrak{p}'_i \cap A = \mathfrak{p}_i$ for each i; from this one deduces immediately $\dim(B_\mathfrak{n}) \geqslant \dim(A)$, whence the proposition. $\square$

(16.1.7). Let M be an A-module; we call the *dimension* of M and denote by $\dim_A(M)$ or $\dim(M)$ the dimension of the ring $A/\operatorname{Ann}(M)$ (isomorphic to the ring of homotheties A_M of M). We thus have $\dim(M) = \dim(M)$ for every $k > 0$. If M is *of finite type*, the prime ideals of A containing $\operatorname{Ann}(M)$ are exactly those belonging to $\operatorname{Supp}(M)$ ($\mathbf{0}_\mathrm{I}$, 1.7.4) and the latter is closed in $\operatorname{Spec}(A)$, hence we then have

(16.1.7.1) $$\dim(M) = \dim(\operatorname{Supp}(M)) \leqslant \dim(A).$$

If M, N are two A-modules such that $N \subset M$, we have

(16.1.7.2) $$\dim(N) \leqslant \dim(M), \qquad \dim(M/N) \leqslant \dim(M).$$

If S is a multiplicative subset of A and if M is of finite type, we have

(16.1.7.3) $$\dim_{S^{-1}A}(S^{-1}M) \leqslant \dim_A(M).$$

Furthermore, every maximal chain of prime ideals containing $\operatorname{Ann}(M)$ terminates at a maximal ideal $\mathfrak{m}$, hence its length is equal to the length of the chain of corresponding prime ideals of $A_\mathfrak{m}$ (which contain $(\operatorname{Ann}(M))_\mathfrak{m}$); from this remark and from (16.1.7.3), one draws

(16.1.7.4) $$\dim_A(M) = \sup_\mathfrak{m}(\dim_{A_\mathfrak{m}}(M_\mathfrak{m}))$$

where $\mathfrak{m}$ ranges over the set of maximal ideals of A. We say that M is *catenary* (resp. *equidimensional*, *equicodimensional*, *biequidimensional*) when $A/\operatorname{Ann}(M)$ is.

Proposition (16.1.8). — *Let* A *be a ring,* M *an* A*-module of finite type,* $\mathfrak{p}$ *a prime ideal of* A*. We have*

(16.1.8.1) $$\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim_A(M/\mathfrak{p}M) \leqslant \dim_A(M).$$

Proof. We have $\mathrm{Ann}(M_{\mathfrak{p}}) = (\mathrm{Ann}(M))_{\mathfrak{p}}$, hence the prime ideals of $A_{\mathfrak{p}}$ containing $\mathrm{Ann}(M_{\mathfrak{p}})$ correspond bijectively to the prime ideals of A containing $\mathrm{Ann}(M)$ and contained in $\mathfrak{p}$. On the other hand, if $\mathrm{Ann}(M) = \mathfrak{n}$, $V(\mathrm{Ann}(M/\mathfrak{p}M)) = V(\mathfrak{p}+\mathfrak{n})$ ($\mathbf{0}_{\mathrm{I}}$, 1.7.5), hence the prime ideals of A containing $\mathrm{Ann}(M/\mathfrak{p}M)$ contain $\mathfrak{p}+\mathfrak{n}$; whence the conclusion. We note in addition that $\mathbf{0}_{\textbf{IV-1}}$ | 25

(16.1.8.2) $$\dim_{A/\mathfrak{p}}(M/\mathfrak{p}M) = \dim_A(M/\mathfrak{p}M).$$

□

Proposition (16.1.9). — *Let* A *be a Noetherian ring,* $\rho : A \to B$ *a ring homomorphism,* M *a* B*-module such that* $M_{[\rho]}$ *is an* A*-module of finite type. Then*

(16.1.9.1) $$\dim_A(M_{[\rho]}) = \dim_B(M).$$

Proof. Indeed, the ring B_M of homotheties of the B-module M is a subring of the A-module $C = \mathrm{End}_A(M_{[\rho]})$, and C is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 1, lemma 2), hence it is the same for B_M; moreover the ring A_M of homotheties of the A-module $M_{[\rho]}$ is a subring of B_M, canonical image of A in C, hence B_M is finite over A_M; the conclusion then follows from (16.1.5). □

(16.1.10). Let A be a Noetherian ring; if M is an A-module of *finite length*, we know (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, prop. 7 and cor. 1 of prop. 7) that $\mathrm{Supp}(M)$ is a finite discrete set, hence $\dim(M) = 0$ if $M \neq 0$; conversely, if M is an A-module of finite type such that $\dim(M) = 0$, every point of $\mathrm{Supp}(M)$ is closed, otherwise said it is a maximal ideal of A, hence (*loc. cit.*, prop. 7) M is of finite length.

16.2. Dimension of a semi-local Noetherian ring

(16.2.1). Let A be a *semi-local Noetherian* ring, $\mathfrak{r}$ its radical; recall that an *ideal of definition* $\mathfrak{q}$ of A is an ideal such that $\mathfrak{q} \subset \mathfrak{r}$ and $\mathfrak{q}$ contains a power of $\mathfrak{r}$, or equivalently, that the only prime ideals containing $\mathfrak{q}$ are *maximal*, or that $A/\mathfrak{q}$ is an artinian ring. Let M be an A-module of finite type; for every integer $n > 0$, $M/\mathfrak{q}^n M$ is an A-module of *finite length*, equal to $\sum_{i=0}^{n-1} \mathrm{long}(\mathrm{gr}_i(M))$, where $\mathrm{gr}_i(M)$ is the $(A/\mathfrak{q})$-graded module associated to M equipped with the $\mathfrak{q}$-preadic filtration. We say that there exists a *polynomial* $Q(n)$ with rational coefficients such that $\mathrm{long}(\mathrm{gr}_n(M)) = Q(n)$ for n sufficiently large (**III**, 2.5.4); from this one deduces immediately that there exists a polynomial $P_{\mathfrak{q}}(M, n)$ in n, with rational coefficients, such that $\mathrm{long}(M/\mathfrak{q}^n M) = P_{\mathfrak{q}}(M, n)$ *for n sufficiently large*; it is clear that this polynomial is unique and that its leading coefficient is > 0 if $M \neq 0$; we call it the *Hilbert–Samuel polynomial* of M for the $\mathfrak{q}$-preadic filtration.

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Lemma (16.2.2.1). — *Let* (M_n) *be a* $\mathfrak{q}$*-good filtration of* M *(Bourbaki,* Alg. comm.*, chap. III, § 3, nº 1); then* $\mathrm{long}(M/M_n)$ *is equal, for n sufficiently large, to a polynomial in n whose degree and leading coefficient are the same as those of* $P_{\mathfrak{q}}(M, n)$*.*

Proof. Indeed, there exists n_0 such that $M_{n+1} = \mathfrak{q}M_n$ for $n \geqslant n_0$, whence the inclusions $\mathfrak{q}^{n+n_0}M \subset M_n \subset \mathfrak{q}^n M \subset M_{n_{n_0}}$ for n sufficiently large, which implies

$$P_{\mathfrak{q}}(M, n+n_0) \geqslant \mathrm{long}(M/M_n) \geqslant P_{\mathfrak{q}}(M, M_n)$$

hence the conclusion. □

If $\mathfrak{q}'$ is a second ideal of definition, one has $\mathfrak{q}' \supset \mathfrak{q}^m$ for some integer m, hence $P_{\mathfrak{q}'}(M, n) \leqslant P_{\mathfrak{q}}(M, mn)$; the *degree* of $P_{\mathfrak{q}}(M, n)$ therefore *does not depend on the ideal of definition* $\mathfrak{q}$; we denote it $d(M)$.

Lemma (16.2.2.2). — *Let* $0 \to M' \to M \to M'' \to 0$ *be a short exact sequence of* A*-modules of finite type. Then the polynomial* $P_{\mathfrak{q}}(M, n) - P_{\mathfrak{q}}(M', n) - P_{\mathfrak{q}}(M'', n)$ *has degree* $\leqslant d(M') - 1 \leqslant d(M) - 1$*.*

Proof. Indeed, the filtration of the $M'_n = M' \cap \mathfrak{q}^n M''$ is $\mathfrak{q}$-good (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 1, prop. 1); as $\mathrm{long}(M/\mathfrak{q}^n M) = \mathrm{long}(M''/\mathfrak{q}^n M'') + \mathrm{long}(M'/M'_n)$, the lemma follows immediately from (16.2.2.1) applied to M'. □

Theorem (16.2.3). — (Krull-Chevalley-Samuel). *Let* A *be a semi-local Noetherian ring,* $\mathfrak{r}$ *its radical,* $M \neq 0$ *an* A*-module of finite type,* $d(M)$ *the degree of the Hilbert–Samuel polynomial of* M *(for any ideal of definition of* A*); let* $s(M)$ *denote the lower bound of the number of elements* $x_1, \dots, x_n$ *of* $\mathfrak{r}$ *such that* $M/(x_1M + \dots + x_nM)$ *is of finite length. Then*

$$d(M) = s(M) = \dim(M).$$

The proof of (16.2.3) proceeds in three steps:

(16.2.3.2). $\dim(M) \leqslant d(M)$.

We argue by induction on $d(M)$, the case $d(M) = 0$ being trivial. Suppose thus $d(M) \geqslant 1$, and let $\mathfrak{p}_0 \in \mathrm{Supp}(M)$ such that $\dim(A/\mathfrak{p}_0) = \dim(M)$. This implies that $\mathfrak{p}_0$ is a *minimal* element of $\mathrm{Supp}(M)$, hence is associated to M (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 4, th. 2) and M therefore contains a submodule N isomorphic to $A/\mathfrak{p}_0$ (*loc. cit.*, § 1, n° 1); as $d(M) \geqslant d(N)$ by (16.2.2.2), it suffices to prove that $\dim(N) \leqslant d(N)$. Let then $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \dots \subset \mathfrak{p}_n$ be a chain of distinct prime ideals in A, and let us show that $n \leqslant d(N)$. This is evident if $n = 0$. Otherwise, there exists $x \in \mathfrak{p}_1 \cap \mathfrak{r}$ such that $x \notin \mathfrak{p}_0$, for the relation $\mathfrak{p}_0 \supset \mathfrak{p}_1 \cap \mathfrak{r}$ would imply $\mathfrak{p}_0 \supset \mathfrak{r}$ since $\mathfrak{p}_0$ does not contain $\mathfrak{p}_1$, hence $\mathfrak{p}_0$ would be maximal, which is absurd. We have $\mathfrak{p}_i \in \mathrm{Supp}(N/xN)$, for $i \geqslant 1$, hence $n - 1 \leqslant \dim(N/xN)$ (16.1.7). As ${}_xN = 0$ by the choice of x, we have $d(N/xN) \leqslant d(N) - 1$ by (16.2.3.1), (iii)). The induction hypothesis then implies $d(N/xN) = \dim(N/xN) \geqslant n - 1$, hence $n - 1 \leqslant d(N) - 1$.

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(16.2.3.3). $\dim(M) \leqslant s(M)$.

Let indeed $(x_i)_{1 \leqslant i \leqslant n}$ be a family of elements of $\mathfrak{r}$ such that, if we set $\mathfrak{a} = x_1A + \dots + x_nA$, $M/\mathfrak{a}M$ is of finite length. The ideals associated to $M/\mathfrak{a}M$ are then maximal (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, prop. 7) and since these are the only prime ideals containing $\mathfrak{q} = \mathfrak{a} + (\mathfrak{r} \cap \mathrm{Ann}(M))$, $A/\mathfrak{q}$ is artinian (*loc. cit.*, prop. 9) and $\mathfrak{q}$ is an ideal of definition of A. But as $\mathfrak{q}^nM/\mathfrak{q}^{n+1}M = \mathfrak{a}^nM/\mathfrak{a}^{n+1}M$, and if M is generated by r elements z_j, $\mathfrak{a}^nM/\mathfrak{a}^{n+1}M$ is an $(A/\mathfrak{q})$-module generated by the canonical images of $x_1^{\nu_1}x_2^{\nu_2}\cdots x_n^{\nu_n}z_j$ where $\sum \nu_i = m$, hence by $r\binom{m+n-1}{n-1}$ elements; its length is therefore $\leqslant ar\binom{m+n-1}{n-1}$, where a is a constant, and one deduces immediately that the degree of $P_{\mathfrak{q}}(M, n)$ is $\leqslant n$; whence $d(M) \leqslant s(M)$.

(16.2.3.4). $s(M) \leqslant \dim(M)$.

Let us note first that $n = \dim(M)$ is finite by (16.2.3.2). We argue by induction on n; the proposition is evident for $n = 0$, since M is then of finite length (16.1.10). Suppose $n \geqslant 1$, and let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant m)$ be the prime ideals of $\mathrm{Supp}(M)$ such that $\dim(A/\mathfrak{p}_i) = n$; the definition of the $\mathfrak{p}_i$ shows that they are minimal elements of $\mathrm{Supp}(M)$, hence finite in number (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 4, th. 2). As $n \geqslant 1$, the $\mathfrak{p}_i$ are not maximal, and there therefore exists $x \in \mathfrak{r}$ such that $x \notin \mathfrak{p}_i$ for every i (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2). By virtue of (16.2.3.1), we have $s(M/xM) \leqslant s(M) + 1$ and $\dim(M) \geqslant \dim(M/xM) + 1$; the induction hypothesis gives the conclusion.

Lemma (16.2.3.1). — *For every* $x \in \mathfrak{r}$*, let* ${}_xM$ *be the submodule of* M *of elements annihilated by* x*. Then:*

(i) $s(M) \leqslant s(M/xM) + 1$.

(ii) *Let* $\mathfrak{p}_i$ *be the prime ideals belonging to* $\mathrm{Supp}(M)$ *and such that*

$$\dim(A/\mathfrak{p}_i) = \dim(M) \quad (1 \leqslant i \leqslant m).$$

If $x \notin \mathfrak{p}_i$ *for every* i*, we have* $\dim(M/xM) \leqslant \dim(M) - 1$.

(iii) *If* $\mathfrak{q}$ *is an ideal of definition of* A*, the polynomial*

$$P_{\mathfrak{q}}({}_xM, n) - P_{\mathfrak{q}}(M/xM, n)$$

has degree $\leqslant d(M) - 1$.

Proof. The assertion (i) is trivial, since if $(x_i)_{1 \leqslant i \leqslant m}$ is such that for the module $N = M/xM$, $N/(x_1N + \dots + x_mN)$ is of finite length, this amounts to saying that $M/(xM + x_1M + \dots + x_mM)$ is isomorphic to this module.

For (ii), note that, since $\mathfrak{a}$ is the annihilator of M, the prime ideals containing the annihilator of M/xM are those containing $\mathfrak{a} + Ax$ ($\mathbf{0}_{\mathrm{I}}$, 1.7.5); none of the $\mathfrak{p}_i$ can therefore contain $\mathfrak{a} + Ax$ and by definition of $\dim(M)$ (16.1.7) every chain of prime ideals containing $\mathfrak{a} + Ax$ has therefore length $\leqslant \dim(M) - 1$.

Finally, to prove (iii), note that one has two short exact sequences

$$0 \longrightarrow {}_xM \longrightarrow M \xrightarrow{x} xM \longrightarrow 0, \qquad 0 \longrightarrow xM \longrightarrow M \longrightarrow M/xM \longrightarrow 0$$

and the assertion follows immediately from the lemma (16.2.2.2). □

Corollary (16.2.4). — *Under the hypotheses of (16.2.3), we have*

(16.2.4.1) $$\dim_{\widehat{A}}(\widehat{M}) = \dim_A(M).$$

Proof. Indeed, $M/\mathfrak{r}^n M$ and $\widehat{M}/\mathfrak{r}^n\widehat{M}$ are isomorphic, hence $d(\widehat{M}) = d(M)$. □

Corollary (16.2.5). — *The dimension of a semi-local Noetherian ring A is finite and equal to the minimum number of elements of the radical of A generating an ideal of definition.*

Proof. This is the equality $s(M) = \dim(M)$ for $M = A$. □

In particular:

Corollary (16.2.6). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$ its residue field; we have*

(16.2.6.1) $$\dim(A) \leqslant \mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2).$$

Proof. Indeed, we know that if the x_i ($1 \leqslant i \leqslant n$) are elements of $\mathfrak{m}$ whose classes mod. $\mathfrak{m}^2$ form a basis of the k-vector space $\mathfrak{m}/\mathfrak{m}^2$, the x_i generate $\mathfrak{m}$ (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 2, cor. 2 of prop. 4). □

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Proposition (16.2.7). — *Let k be a field, B a graded k-algebra of finite type generated by its elements of degree 1 and such that $B_0 = k$; let E be a graded B-module of finite type, such that* (**III**, 2.5.4) *for n sufficiently large, $\mathrm{rg}_k(E_n) = r(n)$ is of the form $\Phi(n) - \Phi(n-1)$, where Φ is a polynomial in n of degree d. Then there exist d elements homogeneous of degree $\geqslant 1$ in B, $t_1, \ldots, t_d$, of a family $(t_i)_{1\leqslant i\leqslant d}$ such that $E/(\sum_i^d t_iE)$ is of finite rank over k.*

Proof. Let $\mathfrak{a}$ be the maximal ideal $\bigoplus_{n\geqslant 1} B_n$ of B, and, for every n, let E'_n be the sub-B-module $\bigoplus_{h\geqslant n+1} E_h$ of E. Every element of $B - \mathfrak{a}$ determines an injective homothety in the B-module artinian E/E'_n, and this homothety is thus bijective (Bourbaki, *Alg.*, chap. VIII, § 2, nº 2, lemma 3), from which one deduces that $(E/E'_n)_{\mathfrak{q}}$ is a polynomial in n of degree d for n sufficiently large; as the residue field of $B_{\mathfrak{q}}$ is k, one deduces from the hypothesis that $\mathrm{long}_{B_{\mathfrak{q}}}((E/E'_{\mathfrak{q}})_{\mathfrak{q}})$ is a polynomial in n of degree d; moreover, E is a B-module of finite type and B is generated over k by its elements of degree 1 homogeneous, hence the filtration (E'_n) is $\mathfrak{a}$-good (Bourbaki, *Alg. comm.*, chap. III, § 1, nº 3, lemma 1), hence the filtration $(E'_n)_{\mathfrak{q}}$ on $E_{\mathfrak{q}}$ is $\mathfrak{a}B_{\mathfrak{q}}$-good. Since $B_{\mathfrak{q}}$ is a local Noetherian ring, one concludes (16.2.3) and from (16.2.3.1) that $\dim(E_{\mathfrak{q}}) = d$, the lemma being evident if $d = 0$. Suppose $d \geqslant 1$, and observe that the minimal prime ideals $\mathfrak{p}_i$ ($1 \leqslant i \leqslant r$) in $\mathrm{Ass}(E)$ are such that $\mathfrak{a}$ is distinct from the union of the $\mathfrak{p}_i$ since $d \geqslant 1$; there is therefore a *homogeneous* element $t_1 \in \mathfrak{a}$ not belonging to any of the $\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. III, § 1, nº 4, prop. 8). As the prime ideals $(\mathfrak{p}_i)_{\mathfrak{q}}$ are the minimal elements of $\mathrm{Ass}(E_{\mathfrak{q}})$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 2, prop. 5), we have $\dim(E_{\mathfrak{q}}/t_1E_{\mathfrak{q}}) = d - 1$ (16.3.4). It suffices then to apply the induction hypothesis to obtain the conclusion. □

16.3. Systems of parameters in a Noetherian local ring

Proposition (16.3.1). — *Let* A *be a semi-local Noetherian ring,* $X = \mathrm{Spec}(A)$, $\mathfrak{a}$ *an ideal of* A *distinct from* A *such that* $\mathrm{codim}(V(\mathfrak{a}), X) = r > 0$. *Then there exist r elements* $x_1, \ldots, x_r$ *of* $\mathfrak{a}$, *forming part of a system of parameters of* A, *such that*

$$\mathrm{codim}(V(\mathfrak{a}), V(Ax_1 + \ldots + Ax_r)) = 0.$$

PROOF. Let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ be the minimal prime ideals of the ring A, $\mathfrak{q}_j$ $(1 \leqslant j \leqslant n)$ the minimal prime ideals among those containing $\mathfrak{a}$; we have (14.2.1) $\mathrm{codim}(V(\mathfrak{a}), X) = \inf_j(\mathrm{codim}(V(\mathfrak{q}_j), X)) = \inf_j(\dim(A_{\mathfrak{q}_j/A_{\mathfrak{q}_j}}x))$ (16.1.3.2). The hypothesis $r > 0$ implies that $\mathfrak{a}$ is not contained in any of the $\mathfrak{p}_i$; hence there exists $x \in \mathfrak{a}$ not belonging to any of the $\mathfrak{p}_i$ (Bourbaki, *Alg.*, chap. II, § 1, n° 1, prop. 2). One has as above $\mathrm{codim}(V(\mathfrak{a}), V(Ax)) = \inf_j(\dim((A/Ax)_{\mathfrak{q}_j/A_{\mathfrak{q}_j}}x))$; but as $x/1$ is not in any of the minimal prime ideals of $A_{\mathfrak{q}_j}$, we have $\dim(A_{\mathfrak{q}_j}/A_{\mathfrak{q}_j}x) = \dim(A_{\mathfrak{q}_j}) - 1$ (16.3.4), whence $\mathrm{codim}(V(\mathfrak{a}), V(Ax)) = r - 1$. For the same reason, $\dim(A/Ax) = \dim(A) - 1$ (16.3.4). We now argue by induction on r; if $r > 1$, there exist $r - 1$ elements x'_i $(2 \leqslant i \leqslant r)$ of $A' = A/Ax$, forming part of a system of parameters of A'; for every i let x_i be an element of the class x'_i in A, with $x_i \in \mathfrak{a}$ for $2 \leqslant i \leqslant r$; it is immediate that $A/(Ax + Ax_2 + \ldots + Ax_r)$ is of finite length, and as $\dim(A) = s$, we see that $x_1 = x$ and the x_i for $i \geqslant 2$ satisfy the conditions of the statement. □

For $n = 1$, one obtains the special case: $\mathbf{0}_{\textbf{IV-1}}$ | 29

Corollary (16.3.2). — ("Hauptidealsatz"). *For a prime ideal in a Noetherian ring to be of height* $\leqslant 1$, *it is necessary and sufficient that it be minimal in the set of prime ideals containing a principal convex ideal.*

Corollary (16.3.3). — (Artin-Tate). *Let* A *be an integral Noetherian ring. The following conditions are equivalent:*

a) A *is a semi-local ring of dimension* $\leqslant 1$.
b) There exists $f \neq 0$ *in* A *such that* A_f *is a field.*

PROOF. The condition *a*) is equivalent to saying that A has only finitely many prime ideals $\mathfrak{m}_i \neq 0$, and that these ideals are all maximal (0 being a prime ideal). As the product of the $\mathfrak{m}_i$ is not zero, since A is integral, an element $f \neq 0$ of this product belongs to every maximal ideal, hence (0) is the only prime ideal of the integral ring A_f, otherwise said A_f is a field. (One notes that this part of the argument does not use the fact that A is Noetherian.) Let us now prove that *b*) implies *a*); the hypothesis *b*) means that every prime ideal $\mathfrak{p} \neq 0$ of A contains f. Let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ be the minimal prime ideals among those containing f (which are finite in number since A/fA is Noetherian); it suffices to prove that the $\mathfrak{p}_i$ are *maximal* ideals, since A is a field. Suppose then that there exists a maximal ideal $\mathfrak{m}$ containing one of the $\mathfrak{p}_i$ and distinct from the $\mathfrak{p}_i$; $\mathfrak{m}$ is therefore not contained in the union of the $\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2), otherwise said there exists $g \in \mathfrak{m}$ not belonging to any of the $\mathfrak{p}_i$. Let $\mathfrak{q}$ be a minimal prime ideal among those containing g; it follows from the "Hauptidealsatz" (16.3.2) that $\mathfrak{q}$ is of height 1, hence $\neq 0$; it therefore contains by assumption f, hence also one of the $\mathfrak{p}_i$, and as g does not belong to any of the $\mathfrak{p}_i$, $\mathfrak{q}$ is distinct from the $\mathfrak{p}_i$; it would thus be of height $\geqslant 2$, which is a contradiction. □

Proposition (16.3.4). — *Let* A *be a semi-local Noetherian ring,* $\mathfrak{r}$ *its radical,* M *an* A*-module of finite type,* $\mathfrak{p}_i$ *the elements of* $\mathrm{Supp}(M)$ *such that* $\dim(A/\mathfrak{p}_i) = \dim(M)$. *Then the* $\mathfrak{p}_i$ *are minimal elements of* $\mathrm{Ass}(M)$, *and* $\dim(M/\mathfrak{p}_i M) = \dim(M)$. *For every* $x \in \mathfrak{r}$, *we have*

(16.3.4.1) $$\dim(M/xM) \geqslant \dim(M) - 1.$$

Furthermore, for the two members of (16.3.4.1) *to be equal, it is necessary and sufficient that x not belong to any of the* $\mathfrak{p}_i$.

PROOF. The first assertion is immediate, since the $\mathfrak{p}_i$ are by definition the minimal elements of $\mathrm{Supp}(M)$, and they are also the minimal elements of $\mathrm{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 4, th. 2). Furthermore, $\mathrm{Supp}(M/\mathfrak{p}_i M)$ is the set of prime ideals of A containing $\mathrm{Ann}(M) + A\mathfrak{p}_i$; as these contain the $\mathfrak{p}_i$ as well, $\dim(M/\mathfrak{p}_i M) = \dim(M)$.

Set $\mathrm{N} = \mathrm{M}/x\mathrm{M}$. If $x_1, \ldots, x_n$ are elements of $\mathfrak{r}$ such that $\mathrm{M}/(x\mathrm{M} + x_1\mathrm{M} + \ldots + x_n\mathrm{M})$ is of finite length, we have $s(\mathrm{M}) \leqslant n+1$, hence $\dim(\mathrm{M}) \leqslant n+1$ by (16.2.3), and as one can choose them so that $n = s(\mathrm{N}) = \dim(\mathrm{N})$, we get (16.3.4.1).

The fact that the two members of (16.3.4.1) are equal when x does not belong to any of the $\mathfrak{p}_i$ follows from (16.2.3.1), (ii) and (16.2.3). Conversely, if $\dim(\mathrm{M}/x\mathrm{M}) = \dim(\mathrm{M}) - 1$, none of the $\mathfrak{p}_i$ can belong to $\mathrm{Supp}(\mathrm{M}/x\mathrm{M})$, and the prime ideals in the support of the latter are those containing $\mathrm{Ann}(\mathrm{M}) + \mathrm{A}x$; as they contain the $\mathfrak{p}_i$, they cannot contain x. □ $\mathbf{0_{IV\text{-}1}}$ | 30

Corollary (16.3.5). — *With the notations of (16.3.4), for every ideal* $\mathfrak{a} \subset \mathfrak{p}_i$, *we have* $\dim(\mathrm{M}/\mathfrak{a}\mathrm{M}) = \dim(\mathrm{M})$ *and if one sets* $\mathrm{N} = \mathrm{M}/\mathfrak{a}\mathrm{M}$, *the relation* $\dim(\mathrm{M}/x\mathrm{M}) = \dim(\mathrm{M}) - 1$ *implies* $\dim(\mathrm{N}/x\mathrm{N}) = \dim(\mathrm{N}) - 1$.

Proof. Indeed, $\mathrm{Supp}(\mathrm{N})$ is the set of prime ideals containing $\mathfrak{a} + \mathrm{Ann}(\mathrm{M})$, hence $\mathfrak{p}_i$ belongs to $\mathrm{Supp}(\mathrm{N})$ and $\dim(\mathrm{N}) \leqslant \dim(\mathrm{M}) = \dim(\mathrm{A}/\mathfrak{p}_i)$, hence $\dim(\mathrm{A}/\mathfrak{p}_i) = \dim(\mathrm{N})$; furthermore the prime ideals $\mathfrak{q} \in \mathrm{Supp}(\mathrm{N})$ such that $\dim(\mathrm{A}/\mathfrak{q}) = \dim(\mathrm{N})$ are certain of the $\mathfrak{p}_j$; if x does not belong to any of the $\mathfrak{p}_j$, we have then $\dim(\mathrm{N}/x\mathrm{N}) = \dim(\mathrm{N}) - 1$ by virtue of (16.3.4). □

Definition (16.3.6). — Let A be a semi-local Noetherian ring, $\mathfrak{r}$ its radical, M an A-module of finite type, and set $n = \dim(\mathrm{M})$. A *system of parameters* for M is any family $(x_i)_{1 \leqslant i \leqslant n}$ of n elements of $\mathfrak{r}$ such that $\mathrm{M}/(x_1\mathrm{M} + \ldots + x_n\mathrm{M})$ is of *finite length*.

We have seen (16.2.3) that such systems always exist. We have seen in the course of the proof of (16.2.3.3) that if $\mathfrak{J} = \sum_{i=1}^{n} x_i\mathrm{A}$ is such that $\mathrm{M}/\mathfrak{J}\mathrm{M}$ is of finite length, $\mathfrak{q} = \mathfrak{J} + \mathrm{Ann}(\mathrm{M})$ is an *ideal of definition* of A, and conversely, if $\mathrm{M}/\mathfrak{J}\mathrm{M}$ is of finite type over the artinian ring $\mathrm{A}/\mathfrak{q}$, hence is of finite length. It amounts to the same to say that $(x_i)_{1 \leqslant i \leqslant n}$ is a system of parameters *for* M or *for* $\mathrm{A}/\mathrm{Ann}(\mathrm{M})$ (or equivalently that their images in $\mathrm{A}/\mathrm{Ann}(\mathrm{M})$ form a system of parameters for this ring). If $(x_i)_{1 \leqslant i \leqslant n}$ is a system of parameters for M, so is (x_i^k) for every integer $k \geqslant 1$, since one can reduce, by virtue of what precedes, to the case $\mathrm{M} = \mathrm{A}$, and if $\mathfrak{J}$ is an ideal of definition of A, $\sum_i x_i^k\mathrm{A}$ contains $\mathfrak{J}^{kn}$, hence is also an ideal of definition.

Proposition (16.3.7). — *Let* A *be a semi-local Noetherian ring,* $\mathfrak{r}$ *its radical,* $x_1, \ldots, x_k$ *elements of* $\mathfrak{r}$, M *an* A*-module of finite type. Then*

(16.3.7.1) $$\dim(\mathrm{M}/(x_1\mathrm{M} + \ldots + x_k\mathrm{M})) \geqslant \dim(\mathrm{M}) - k.$$

Furthermore, for the two members of (16.3.7.1) to be equal, it is necessary and sufficient that there exist elements x_i $(k+1 \leqslant i \leqslant n = \dim(\mathrm{M}))$ *of* $\mathfrak{r}$ *such that* $(x_i)_{1 \leqslant i \leqslant n}$ *is a system of parameters for* M.

Proof. The inequality (16.3.7.1) follows from (16.3.4.1) by induction on k. If the two members of (16.3.7.1) are equal, and if $x_{k+1}, \ldots, x_n$ is a system of parameters for $\mathrm{M}/(x_1\mathrm{M} + \ldots + x_k\mathrm{M})$, it is clear that $\mathrm{M}/(x_1\mathrm{M} + \ldots + x_k\mathrm{M} + x_{k+1}\mathrm{M} + \ldots + x_n\mathrm{M})$ is of finite length, hence $(x_i)_{1 \leqslant i \leqslant n}$ is a system of parameters for M; conversely, if there exist x_i $(k+1 \leqslant i \leqslant n)$ having this property and if one sets $\mathrm{N} = \mathrm{M}/(x_1\mathrm{M} + \ldots + x_k\mathrm{M})$, it is clear that $\mathrm{N}/(x_{k+1}\mathrm{N} + \ldots + x_n\mathrm{N})$ is of finite length, hence $\dim(\mathrm{N}) \leqslant n - k$, and the two members are equal by virtue of (16.2.3). □

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Proposition (16.3.8). — *Let* A *be a semi-local Noetherian ring,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ *its spectrum,* $\mathfrak{a}$ *an ideal of* A *distinct from* A *such that* $\mathrm{codim}(\mathrm{V}(\mathfrak{a}), \mathrm{X}) = r > 0$. *Then there exist r elements* $x_1, \ldots, x_r$ *of* $\mathfrak{a}$, *forming part of a system of parameters of* A, *such that*

$$\mathrm{codim}(\mathrm{V}(\mathfrak{a}), \mathrm{V}(\mathrm{A}x_1 + \ldots + \mathrm{A}x_r)) = 0.$$

Proof. Let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant m)$ be the minimal prime ideals of the ring A, $\mathfrak{q}_j$ $(1 \leqslant j \leqslant n)$ the minimal prime ideals among those containing $\mathfrak{a}$; we have (14.2.1) $\mathrm{codim}(\mathrm{V}(\mathfrak{a}), \mathrm{X}) = \inf_j(\dim(\mathrm{A}_{\mathfrak{q}_j}))$ (16.1.3.2). The hypothesis $r > 0$ implies that $\mathfrak{a}$ is not contained in any of the $\mathfrak{p}_i$; hence there exists $x \in \mathfrak{a}$ not belonging to any of the $\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2). One has as above $\mathrm{codim}(\mathrm{V}(\mathfrak{a}), \mathrm{V}(\mathrm{A}x)) = \inf_j(\dim((\mathrm{A}/\mathrm{A}x)_{\mathfrak{q}_j}))$; but as $x/1$ is not in any of the minimal prime ideals of $\mathrm{A}_{\mathfrak{q}_j}$, $\dim(\mathrm{A}_{\mathfrak{q}_j}/\mathrm{A}_{\mathfrak{q}_j}x) = \dim(\mathrm{A}_{\mathfrak{q}_j}) - 1$ (16.3.4), whence $\mathrm{codim}(\mathrm{V}(\mathfrak{a}), \mathrm{V}(\mathrm{A}x)) = r - 1$. For the same reason, $\dim(\mathrm{A}/\mathrm{A}x) = \dim(\mathrm{A}) - 1$. We argue by induction on r; if $r > 1$, there exist $r - 1$ elements x_i' $(2 \leqslant i \leqslant r)$ of $\mathrm{A}' = \mathrm{A}/\mathrm{A}x$, forming part of a system of parameters of A' and such that $\operatorname{codim}(V(\mathfrak{a}/Ax), V(A'x'_2 + \ldots + A'x'_r)) = 0$. Let $(x'_i)_{r+1\leqslant i\leqslant s}$ be elements of A' with $x_i \in \mathfrak{a}$ for $2 \leqslant i \leqslant r$; it is immediate that $A/(Ax + Ax_2 + \ldots + Ax_s)$ is of finite length, and as $\dim(A) = s$, we see that $x_1 = x$ and the x_i for $i \geqslant 2$ satisfy the conditions of the statement. □

Proposition (16.3.9). — *Let* A, B *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, k *its residue field,* $\varphi : A \to B$ *a local homomorphism.*

(i) *We have*

(16.3.9.1)
$$\dim(B) \leqslant \dim(A) + \dim(B \otimes_A k).$$

(ii) *Let* $M \neq 0$ *be an* A*-module of finite type,* $N \neq 0$ *a* B*-module of finite type. We have*

(16.3.9.2)
$$\dim_B(M \otimes_A N) \leqslant \dim_A(M) + \dim_{B\otimes_A k}(N \otimes_A k).$$

PROOF. Let m be the dimension of A, $(s_i)_{1\leqslant i\leqslant m}$ a system of parameters of A (16.3.6), so that $\mathfrak{a} = \sum_i s_i A$, $A/\mathfrak{a}$ is an A-module of finite length. Then the artinian ring $A/\mathfrak{a}$ has nilpotent maximal ideal $\mathfrak{m}/\mathfrak{a}$; the ideal $\mathfrak{m}B/\mathfrak{a}B$ of $B/\mathfrak{a}B$, image of $(\mathfrak{m}/\mathfrak{a}) \otimes_A B$, is therefore also nilpotent, and hence $\dim(B/\mathfrak{a}B) = \dim(B/\mathfrak{m}B)$ (the spectra of these two rings having the same underlying space). On the other hand $B/\mathfrak{a}B = B/(s_1B + \ldots + s_mB)$, and the images of the s_i in B belong to the maximal ideal of B. Hence by (16.3.7), we have

$$\dim(B) \leqslant m + \dim(B \otimes_A k)$$

which is (16.3.9.1).

To prove (16.3.9.2), note that if $\mathfrak{r}$ (resp. $\mathfrak{s}$) is the annihilator of M (resp. N) in A (resp. B), we have $\dim_A(M) = \dim(A/\mathfrak{r})$, $\dim_B(N) = \dim(B/\mathfrak{s})$ by definition (16.1.7); on the other hand, $\operatorname{Supp}(M \otimes_A N)$ is a closed subset of $\operatorname{Spec}(B)$ that identifies (**I**, 9.1.13) to $\operatorname{Supp}(M) \times_{\operatorname{Spec}(A)} \operatorname{Supp}(N) = \operatorname{Spec}((B/\mathfrak{s}) \otimes_A (A/\mathfrak{r})) = \operatorname{Spec}(B/(\mathfrak{s} + \mathfrak{r}B))$. As $N \otimes_A k$ is a B-module of finite type, we have $\dim_{B\otimes_A k}(N \otimes_A k) = \dim_B(N \otimes_A k)$ (16.1.9), and $\operatorname{Supp}(N \otimes_A k)$ is equal to $\operatorname{Spec}(B/(\mathfrak{s} + \mathfrak{m}B))$ by the same reasoning as above. Applying (16.3.9.1) to $A/\mathfrak{r}$ and to $B/(\mathfrak{s} + \mathfrak{r}B)$, it comes $\mathbf{0_{IV\text{-}1}}$ | 32

$$\dim(B/(\mathfrak{s} + \mathfrak{r}B)) \leqslant \dim(A/\mathfrak{r}) + \dim(B/(\mathfrak{s} + \mathfrak{m}B))$$

which is nothing other than (16.3.9.2). □

Corollary (16.3.10). — *Under the hypotheses of* (16.3.9), *if* $\mathfrak{m}B$ *is an ideal of definition of* B, *we have* $\dim(B) \leqslant \dim(A)$*; if in addition* A *is integral and* $\dim(A) = \dim(B)$, φ *is injective.*

PROOF. The first assertion follows from (16.3.9) since then $\dim(B/\mathfrak{m}B) = 0$. One can moreover replace A by $\varphi(A)$, hence $\dim(B) \leqslant \dim(\varphi(A))$; if $\mathfrak{a} = \operatorname{Ker}(\varphi) \neq 0$, and if A is integral, we have $\dim(\varphi(A)) = \dim(A/\mathfrak{a}) < \dim(A)$ (16.1.2.2), hence one cannot have $\dim(A) = \dim(B)$ if $\mathfrak{a} = 0$. □

16.4. Depth and codepth

(16.4.0). In the 3rd part of chapter III, we shall develop the notion of depth from the cohomological point of view and in a more general framework.

Proposition (16.4.1). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* M *an* A*-module of finite type. Then every* M*-regular sequence* $(x_i)_{1\leqslant i\leqslant r}$ (15.1.7) *formed of elements of* $\mathfrak{m}$, *is part of a system of parameters for* M.

PROOF. We argue by induction on r; consider the A-module

$$N = M/(x_1M + \ldots + x_{r-1}M);$$

by the induction hypothesis $x_1, \ldots, x_{r-1}$ form part of a system of parameters for M, hence $\dim(N) = \dim(M) - (r-1)$ (16.3.7). As by hypothesis the homothety of ratio x_r in N is injective, x_r does not belong to any of the prime ideals associated to N, hence (16.3.4) $\dim(N/x_rN) = \dim(N) - 1$, which can also be written

$$\dim(M/(x_1M + \ldots + x_rM)) = \dim(M) - r;$$

the conclusion then follows from (16.3.7). □

For an M-regular sequence formed of elements of $\mathfrak{m}$, $M \neq 0$, there are at most $\dim(M)$ elements.

Lemma (16.4.2). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $k = \mathrm{A}/\mathfrak{m}$ *its residue field,* M *an* A*-module. For an* M*-regular sequence* $(x_i)_{1\leqslant i\leqslant n}$ *of elements of* $\mathfrak{m}$ *to be maximal, it is necessary and sufficient that* $\mathrm{Hom}_{\mathrm{A}}(k, \mathrm{M}/(x_1\mathrm{M} + \ldots + x_n\mathrm{M})) \neq 0$.

PROOF. To say that (x_i) is maximal means that, for no $x \in \mathfrak{m}$, the homothety of ratio x in $\mathrm{N} = \mathrm{M}/(x_1\mathrm{M} + \ldots + x_n\mathrm{M})$ is injective, or equivalently that $\mathfrak{m}$ is contained in the union of the prime ideals associated to N (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 1, cor. 2 of prop. 2), hence equal to one of them, since it is a maximal ideal (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2); otherwise said, there is an element $z \in \mathrm{N}$ whose annihilator is $\mathfrak{m}$, and hence the submodule Az of N is isomorphic to k; whence the lemma. ∎ **$0_{\text{IV-1}}$ | 33**

Lemma (16.4.3). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $k = \mathrm{A}/\mathfrak{m}$ *its residue field,* M *an* A*-module,* $(x_i)_{1\leqslant i\leqslant n}$ *an* M*-regular sequence of elements of* $\mathfrak{m}$*. Then the* A*-modules* $\mathrm{Hom}_{\mathrm{A}}(k, \mathrm{M}/(x_1\mathrm{M} + \ldots + x_n\mathrm{M}))$ *and* $\mathrm{Ext}^n_{\mathrm{A}}(k, \mathrm{M})$ *are isomorphic, and for* $n > 1$*, they are also isomorphic to* $\mathrm{Ext}^{n-1}_{\mathrm{A}}(k, \mathrm{M}/x_1\mathrm{M})$.

PROOF. We argue by induction on n, the proposition being evident for $n = 0$. Set $\mathrm{N} = \mathrm{M}/x_1\mathrm{M}$; the sequence $(x_i)_{2\leqslant i\leqslant n}$ is N-regular, hence

$$\mathrm{Hom}_{\mathrm{A}}(k, \mathrm{M}/(x_1\mathrm{M} + \ldots + x_n\mathrm{M})) = \mathrm{Hom}_{\mathrm{A}}(k, \mathrm{N}/(x_2\mathrm{N} + \ldots + x_n\mathrm{N}))$$

is isomorphic to $\mathrm{Ext}^{n-1}_{\mathrm{A}}(k, \mathrm{N})$ by the induction hypothesis. Consider then the short exact sequence $0 \to \mathrm{M} \xrightarrow{x_1} \mathrm{M} \to \mathrm{M}/x_1\mathrm{M} = \mathrm{N} \to 0$; one deduces the exact sequence of Ext

(16.4.3.1) $$\mathrm{Ext}^{n-1}_{\mathrm{A}}(k, \mathrm{M}) \longrightarrow \mathrm{Ext}^{n-1}_{\mathrm{A}}(k, \mathrm{N}) \longrightarrow \mathrm{Ext}^{n}_{\mathrm{A}}(k, \mathrm{M}) \xrightarrow{x_1} \mathrm{Ext}^{n}_{\mathrm{A}}(k, \mathrm{M}).$$

By the induction hypothesis, $\mathrm{Ext}^{n-1}_{\mathrm{A}}(k, \mathrm{M})$ is isomorphic to $\mathrm{Hom}_{\mathrm{A}}(k, \mathrm{M}/(x_1\mathrm{M} + \ldots + x_{n-1}\mathrm{M}))$, which is zero since $(x_i)_{1\leqslant i\leqslant n-1}$ is not a maximal M-regular sequence (16.4.2). On the other hand, as $x_1 \in \mathfrak{m}$, the homothety of ratio x_1 in the A-module $\mathrm{Hom}_{\mathrm{A}}(k, \mathrm{T})$ is zero for every A-module T, hence it is the same for every A-module $\mathrm{Ext}^n_{\mathrm{A}}(k, \mathrm{T})$; the assertions of the lemma then follow immediately from the exact sequence (16.4.3.1). ∎

Corollary (16.4.4). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $k = \mathrm{A}/\mathfrak{m}$ *its residue field,* M *an* A*-module of finite type* $\neq 0$*. All maximal* M*-regular sequences of elements of* $\mathfrak{m}$ *have the same number of elements, which is the smallest integer n such that* $\mathrm{Ext}^n_{\mathrm{A}}(k, \mathrm{M}) \neq 0$.

Definition (16.4.5). — Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, M an A-module of finite type. The *depth* of M is denoted $\mathrm{prof}_{\mathrm{A}}\,\mathrm{M}$ or $\mathrm{prof}(\mathrm{M})$, the integer defined as the upper bound of the lengths of M-regular sequences formed of elements of $\mathfrak{m}$. We thus have $\mathrm{prof}(\mathrm{M}) = +\infty$ if and only if $\mathrm{M} = 0$, and, by (16.4.1),

(16.4.5.1) $$\mathrm{prof}(\mathrm{M}) \leqslant \dim(\mathrm{M}) \quad \text{if} \quad \mathrm{M} \neq 0.$$

We evidently have $\mathrm{prof}(\mathrm{M}^k) = \mathrm{prof}(\mathrm{M})$ for every $k > 0$.

Proposition (16.4.6). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* M *an* A*-module of finite type.*

(i) *For* $\mathrm{prof}(\mathrm{M}) = 0$*, it is necessary and sufficient that* $\mathfrak{m} \in \mathrm{Ass}(\mathrm{M})$ *(or equivalently that there exist a submodule of* M *isomorphic to the residue field* $k = \mathrm{A}/\mathfrak{m}$ *of* A*).*

(ii) *For every* M*-regular element x belonging to the maximal ideal of* A*, we have*

(16.4.6.1) $$\mathrm{prof}(\mathrm{M}/x\mathrm{M}) = \mathrm{prof}(\mathrm{M}) - 1.$$

(iii) *If* $\mathrm{M} \neq 0$*, we have*

(16.4.6.2) $$\mathrm{prof}(\mathrm{M}) \leqslant \inf_{\mathfrak{p}\in\mathrm{Ass}(\mathrm{M})} \dim(\mathrm{A}/\mathfrak{p}) \leqslant \dim(\mathrm{M}).$$

(iv) *We have* $\mathrm{prof}_{\mathrm{A}}(\mathrm{M}) = \mathrm{prof}_{\widehat{\mathrm{A}}}(\widehat{\mathrm{M}})$.

PROOF. (i) To say $\mathrm{prof}(\mathrm{M}) = 0$ means that there is no $x \in \mathfrak{m}$ such that the homothety of ratio x in M is injective, that is, every $x \in \mathfrak{m}$ belongs to a prime ideal associated to M (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 1, cor. 2 of prop. 2); but $\mathfrak{m}$ cannot be the union of prime ideals associated to M unless it is equal to one of them (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2).

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(ii) If $(x_i)_{1 \leqslant i \leqslant m}$ is a $(\mathrm{M}/x\mathrm{M})$-regular maximal sequence of elements of $\mathfrak{m}$, the sequence formed of x and the x_i is M-regular, and it results from the definition that it is maximal when the first sequence is; whence the conclusion.

(iii) We argue by induction on $n = \mathrm{prof}(\mathrm{M})$, the assertion being evident for $n = 0$. We use the following lemma: □

Lemma (16.4.6.3). — *Let* A *be a Noetherian ring,* M *an* A*-module of finite type,* $t \in \mathrm{A}$ *an* M*-regular element. Then, for every* $\mathfrak{p} \in \mathrm{Ass}(\mathrm{M})$, *every* minimal *prime ideal among those containing* $\mathfrak{p} + \mathrm{A}t$ *belongs to* $\mathrm{Ass}(\mathrm{M}/t\mathrm{M})$.

PROOF OF THE LEMMA. We know that there exists a short exact sequence

$$0 \longrightarrow \mathrm{M}' \longrightarrow \mathrm{M} \longrightarrow \mathrm{M}'' \longrightarrow 0$$

such that $\mathrm{Ass}(\mathrm{M}') = \{\mathfrak{p}\}$, $\mathrm{Ass}(\mathrm{M}'') = \mathrm{Ass}(\mathrm{M}) - \{\mathfrak{p}\}$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 1, prop. 4); as t is M-regular, it is also M′-regular and M″-regular (*loc. cit.*, § 1, nº 1, cor. 2 of prop. 2). One deduces that the sequence

$$0 \longrightarrow \mathrm{M}'/t\mathrm{M}' \longrightarrow \mathrm{M}/t\mathrm{M} \longrightarrow \mathrm{M}''/t\mathrm{M}'' \longrightarrow 0$$

is exact (15.1.18). But the prime ideals of A containing $\mathfrak{p}$ are the points of $\mathrm{Supp}(\mathrm{M}')$ (*loc. cit.*, § 1, nº 3, prop. 7), hence the prime ideals containing $\mathfrak{p} + \mathrm{A}t$ are the points of the support of $\mathrm{M}'/t\mathrm{M}' = \mathrm{M}' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A})$ ($\mathbf{0}_{\mathrm{I}}$, 1.7.5); a minimal element of the set of these prime ideals belongs then to $\mathrm{Ass}(\mathrm{M}'/t\mathrm{M}')$ (Bourbaki, *loc. cit.*, § 1, nº 3, cor. 1 of prop. 7), and *a fortiori* to $\mathrm{Ass}(\mathrm{M}/t\mathrm{M})$. □

CONTINUATION OF THE PROOF OF (16.4.6). This lemma established, we return to the proof of (iii). Let $x \in \mathfrak{m}$ be an M-regular element, so that $\mathrm{prof}(\mathrm{M}/x\mathrm{M}) = n - 1$ by (ii); for every $\mathfrak{p} \in \mathrm{Ass}(\mathrm{M})$, it follows from the lemma that there is a prime ideal $\mathfrak{p}'$ associated to $\mathrm{M}/x\mathrm{M}$ and containing $\mathfrak{p} + x\mathrm{A}$; but as x is M-regular, we have $x \notin \mathfrak{p}$, whence $\mathfrak{p}' \neq \mathfrak{p}$ and hence $\dim(\mathrm{A}/\mathfrak{p}') \leqslant \dim(\mathrm{A}/\mathfrak{p}) - 1$. The induction hypothesis shows that $n - 1 \leqslant \dim(\mathrm{A}/\mathfrak{p}')$; one concludes that $n \leqslant \dim(\mathrm{A}/\mathfrak{p})$ for every $\mathfrak{p} \in \mathrm{Ass}(\mathrm{M})$.

(iv) One can restrict to the case $\mathrm{M} \neq 0$. Recall that $\widehat{\mathrm{M}} = \mathrm{M} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ is a canonical isomorphism ($\mathbf{0}_{\mathrm{I}}$, 7.3.3); as $\widehat{\mathrm{A}}$ is a flat A-module, there are canonical isomorphisms $\mathrm{Ext}^i_{\mathrm{A}}(k, \mathrm{M}) \otimes_{\mathrm{A}} \widehat{\mathrm{A}} \xrightarrow{\sim} \mathrm{Ext}^i_{\widehat{\mathrm{A}}}(k, \mathrm{M} \otimes_{\mathrm{A}} \widehat{\mathrm{A}})$ since $k = k \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ is a canonical isomorphism ($\mathbf{0}_{\mathrm{III}}$, 12.3.5). The assertion (iv) then follows from (16.4.4) and the fact that $\widehat{\mathrm{A}}$ is a faithfully flat A-module ($\mathbf{0}_{\mathrm{I}}$, 7.3.5). □

(16.4.6.4). One notes that if $\mathfrak{p}$ is a prime ideal of A, one can *also* have $\mathrm{prof}_{\mathrm{A}_{\mathfrak{p}}}(\mathrm{M}_{\mathfrak{p}}) < \mathrm{prof}_{\mathrm{A}}(\mathrm{M})$ and $\mathrm{prof}_{\mathrm{A}_{\mathfrak{p}}}(\mathrm{M}_{\mathfrak{p}}) > \mathrm{prof}_{\mathrm{A}}(\mathrm{M})$; for an example of the first case, it suffices to take A a local integral ring of dimension $\geqslant 1$; we have $\mathrm{prof}(\mathrm{A}) \geqslant 1$ but $\mathrm{prof}(\mathrm{K}) = 0$ for the field of fractions K of A. For an example of the second case, consider a Noetherian integral local ring A_0 of dimension $\geqslant 2$, with maximal ideal $\mathfrak{m}_0$ and residue field $k = \mathrm{A}_0/\mathfrak{m}_0$; let $\mathrm{A} = \mathrm{A}_0 \oplus \mathfrak{J}$ be the trivial extension of A_0 by the A_0-module k (so that the multiplication in A is given by $(x, z)(x', z') = (xx', xz' + x'z)$). It is clear that $\mathfrak{J}$ is the nilradical of A and $\mathrm{A}/\mathfrak{J}$ is isomorphic to A_0, so that every prime ideal $\mathfrak{p}$ of A is of the form $\mathfrak{p}_0 \oplus \mathfrak{J}$, where $\mathfrak{p}_0$ is a prime ideal of A_0; in particular $\mathfrak{m} = \mathfrak{m}_0 \oplus \mathfrak{J}$ is the only maximal ideal of A. As every element of $\mathfrak{m}_0$ annihilates $\mathfrak{J}$, we have $\mathrm{prof}(\mathrm{A}) = \mathrm{prof}(\mathrm{A}_{\mathfrak{m}}) = 0$; on the contrary, if $\mathfrak{p}_0 \neq \mathfrak{m}_0$, the $\mathfrak{p}_i$ are not maximal, hence $\mathrm{A}_{\mathfrak{p}}$ is canonically isomorphic to $(\mathrm{A}_0)_{\mathfrak{p}_0}$, and as A_0 is integral, $\mathrm{prof}((\mathrm{A}_0)_{\mathfrak{p}_0}) \geqslant 1$ if $\mathfrak{p}_0 \neq 0$.

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Corollary (16.4.7). — *For a local Noetherian reduced ring* A *to be of depth* 0, *it is necessary and sufficient that* A *be a field.*

PROOF. Indeed, as the intersection of the minimal prime ideals of A is 0, A has only one minimal prime ideal, hence $\mathfrak{m}$ is also the only maximal prime ideal; by (16.4.6), (i), if $\mathrm{prof}(\mathrm{A}) = 0$, $\mathfrak{m}$ is also the only prime ideal of A, and as A is reduced, $\mathfrak{m} = 0$. □

Proposition (16.4.8). — *Let* A, B *be two local Noetherian rings,* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* M *a* B*-module such that* $\mathrm{M}_{[\rho]}$ *is an* A*-module of finite type. Then*

(16.4.8.1) $$\mathrm{prof}_{\mathrm{A}}(\mathrm{M}_{[\rho]}) = \mathrm{prof}_{\mathrm{B}}(\mathrm{M}).$$

PROOF. One can restrict to the case $M \neq 0$; suppose $\mathrm{prof}_A(M_{[\rho]}) = n$. If $(x_i)_{1\leqslant i\leqslant n}$ is a $(M_{[\rho]})$-regular maximal sequence formed of elements of the maximal ideal of A, the $\rho(x_i)$ constitute an M-regular sequence formed of elements of the maximal ideal of B, and we have then, setting $N = M/(\rho(x_1)M + \ldots + \rho(x_n)M)$, $\mathrm{prof}_B(N) = \mathrm{prof}_B(M) - n$ (16.4.6), (ii)); similarly $N_{[\rho]} = M_{[\rho]}/(x_1 M_{[\rho]} + \ldots + x_n M_{[\rho]})$ and $\mathrm{prof}_A(N_{[\rho]}) = 0$; one is thus reduced to proving the proposition when $n = 0$. Let k be the residue field of A; the hypothesis $\mathrm{prof}_A(M_{[\rho]}) = 0$ implies that $P \neq 0$ (16.4.6), (i)); furthermore, P is an A-module of finite type (since $M_{[\rho]}$ is an A-module of finite type), hence a k-vector space of finite type, annihilated by the maximal ideal of A, and finally a module of *finite length*. Moreover, P is a submodule of the B-module M, and since it is of finite length as an A-module, it is *a fortiori* of finite length as a B-module. As it is $\neq 0$, it contains a simple sub-B-module, that is to say one isomorphic to the residue field of B; one concludes then by (16.4.6), (i)) that $\mathrm{prof}_B(M) = 0$. □

Definition (16.4.9). — Let A be a local Noetherian ring, M an A-module of finite type. If $M \neq 0$, we call the *codepth* of M and denote $\mathrm{coprof}_A(M)$ or $\mathrm{coprof}(M)$, the finite integer $\dim_A(M) - \mathrm{prof}_A(M) \geqslant 0$ (16.4.5.1). If $M = 0$, we set $\mathrm{coprof}(M) = 0$. We evidently have $\mathrm{coprof}(M^k) = \mathrm{coprof}(M)$ for every $k > 0$.

Proposition (16.4.10). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type.*

(i) *For every* M*-regular element* x *belonging to the maximal ideal of* A*, we have* $\mathrm{coprof}(M/xM) = \mathrm{coprof}(M)$.
(ii) *We have* $\mathrm{coprof}_A(M) = \mathrm{coprof}_{\widehat{A}}(\widehat{M})$.

PROOF. Indeed, (i) follows from (16.3.4) and (16.4.6), (ii)), and (ii) from (16.2.4) and (16.4.6), (iv)). □

We shall show later (**IV**, 6.11.5) that for every prime ideal $\mathfrak{p}$ of A, we have $\mathrm{coprof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \leqslant \mathrm{coprof}_A(M)$.

Proposition (16.4.11). — *Under the hypotheses of* (16.4.8)*, we have*

(16.4.11.1) $$\mathrm{coprof}_A(M_{[\rho]}) = \mathrm{coprof}_B(M).$$

PROOF. This follows from (16.1.9) and (16.4.8). □

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16.5. Cohen-Macaulay modules

Definition (16.5.1). — Let A be a local Noetherian ring. An A-module of finite type M is called a *Cohen-Macaulay module* if $\mathrm{coprof}(M) = 0$, that is, if $M = 0$ or $\dim(M) = \mathrm{prof}(M)$. We say that A is a *Cohen-Macaulay ring* if the A-module A is a Cohen-Macaulay module.

An A-module of finite length is a Cohen-Macaulay module since it is of dimension 0 (16.1.10). A local reduced ring of dimension 1 is a Cohen-Macaulay ring, since then its maximal ideal $\mathfrak{m}$ cannot be associated to A (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, prop. 10), hence $\mathrm{prof}(A) = 1$ by virtue of (16.4.6), (i)). If A is an *integral local integrally closed* ring of dimension $\geqslant 2$, we have $\mathrm{prof}(A) \geqslant 2$: indeed, if $x \neq 0$ is an element of $\mathfrak{m}$, we know (Bourbaki, *Alg. comm.*, chap. VII, § 1, nº 4, prop. 8) that the prime ideals associated to A/xA are of height 1, hence distinct of $\mathfrak{m}$ and their union is therefore distinct of $\mathfrak{m}$ (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2); there thus exist (x, y) sequences that are A-regular and formed of elements of $\mathfrak{m}$, whence our assertion. One concludes that an integral local integrally closed ring of dimension 2 is a Cohen-Macaulay ring.

Proposition (16.5.2). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type. For* M *to be a Cohen-Macaulay module, it is necessary and sufficient that* $\widehat{M}$ *be a* $\widehat{A}$*-module of Cohen-Macaulay.*

PROOF. This follows immediately from (16.4.10), (ii)). □

Proposition (16.5.3). — *Under the hypotheses of* (16.4.8)*, for* M *to be a* B*-module of Cohen-Macaulay, it is necessary and sufficient that* $M_{[\rho]}$ *be an* A*-module of Cohen-Macaulay.*

PROOF. This follows from (16.4.11). □

Proposition (16.5.4). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type* $\neq 0$*. If* M *is a Cohen-Macaulay module, we have* $\dim(A/\mathfrak{p}) = \dim(M)$ *for every prime ideal* $\mathfrak{p} \in \mathrm{Ass}(M)$*; in particular no prime ideal associated to* M *is immersed.*

PROOF. This follows immediately from the inequalities (16.4.6.2). □

One can also say that M (or $\mathrm{Supp}(M)$) is *equidimensional* (16.1.7).

Proposition (16.5.5). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* M *an* A*-module of finite type* $\neq 0$*,* x *an element of* $\mathfrak{m}$*. Suppose that* M *is a Cohen-Macaulay module; then the following conditions are equivalent:*

a) x *is* M*-regular;*
b) $\dim(M/xM) = \dim(M) - 1$*;*
c) x *does not belong to any* minimal *element of* $\mathrm{Supp}(M)$*.*

Furthermore, M/xM *is then a Cohen-Macaulay module.*

PROOF. We have seen (16.5.4) that all the $\mathfrak{p}$ in $\mathrm{Ass}(M)$ are minimal in $\mathrm{Supp}(M)$ and such that $\dim(A/\mathfrak{p}) = \dim(M)$; the equivalence of a) and c) then follows from Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 2 of prop. 2; the equivalence of b) and c) follows from (16.3.4). The fact that M/xM is a Cohen-Macaulay module follows finally from (16.4.10), (i)). □ **$0_{\text{IV-1}}$** | 37

Corollary (16.5.6). — *Under the hypotheses of* (16.5.5)*, let* $(x_i)_{1\leqslant i\leqslant r}$ *be a sequence of elements of* $\mathfrak{m}$*. The following conditions are equivalent:*

a) The sequence (x_i) *is* M*-regular.*
b) $\dim(M/(x_1M + \ldots + x_rM)) = \dim(M) - r$*.*
c) The x_i *form part of a system of parameters for* M*.*

Furthermore, when these conditions are satisfied, $N = M/(x_1M + \ldots + x_rM)$ *is a Cohen-Macaulay module, hence for every prime ideal* $\mathfrak{p} \in \mathrm{Ass}(N)$*, we have* $\dim(A/\mathfrak{p}) = \dim(M) - r$*.*

PROOF. We already know that b) and c) are equivalent (16.3.6) and that a) implies c) (16.4.1) without hypothesis on M and to prove the last assertion of the statement, we argue by induction on r, the assertion being trivial for $r = 0$. Since x_1 is M-regular, $\dim(M/x_1M) = \dim(M) - 1$ (16.3.6), hence x_1 is M-regular and M/x_1M is Cohen-Macaulay by (16.5.5). Since the sequence $(x_i)_{2\leqslant i\leqslant r}$ forms part of a system of parameters for $P = M/x_1M$, the induction hypothesis shows that $(x_i)_{2\leqslant i\leqslant r}$ is a P-regular sequence and that $P/(x_2P + \ldots + x_rP) = N$ is a Cohen-Macaulay module; furthermore $(x_i)_{1\leqslant i\leqslant r}$ is M-regular. □

Remark (16.5.7). — It follows from (16.5.6) that there is an identity between *systems of parameters* for M and the maximal M-regular sequences of elements of $\mathfrak{m}$ when M is a Cohen-Macaulay module. Conversely, it follows from (16.4.1) that if there is any M-regular sequence that is a system of parameters for M, then M is a Cohen-Macaulay module. The last property stated in (16.5.6) also *characterizes* Cohen-Macaulay modules:

Proposition (16.5.8). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type. Suppose that for every sequence* $(x_i)_{1\leqslant i\leqslant r}$ *of elements of* A *forming part of a system of parameters for* M*, and for every prime ideal* $\mathfrak{p}$ *associated to* $N = M/(x_1M + \ldots + x_rM)$*, we have* $\dim(A/\mathfrak{p}) = \dim(N)$*. Then* M *is a Cohen-Macaulay module.*

PROOF. Let $(y_i)_{1\leqslant i\leqslant n}$ be a system of parameters for M; it suffices to show that the sequence (y_i) is M-regular. We argue by induction on $n = \dim(M)$, the proposition being trivial for $n = 0$. By hypothesis (applied with $r = 0$) the prime ideals $\mathfrak{p}_j$ associated to M are all such that $\dim(A/\mathfrak{p}_j) = \dim(M)$; as y_1 forms part of a system of parameters, $\dim(M/y_1M) = \dim(M) - 1$ (16.3.7) and (16.3.4), hence y_1 does not belong to any of the $\mathfrak{p}_j$, hence is M-regular. Moreover $N/y_1N = M/y_1M$, and as $y_1 \in \mathfrak{p}$, the hypothesis $\dim(A/\mathfrak{p}) = \dim(M)$ implies that $\dim(M/y_1M) = \dim(M) - 1$; one can therefore apply the induction hypothesis; applying the induction hypothesis to P, one sees that $(y_i)_{2\leqslant i\leqslant n}$ is a P-regular sequence, which proves the proposition. □

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Proposition (16.5.9). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type. Suppose that* M *is a Cohen-Macaulay module. Let* $\mathfrak{p} \in \operatorname{Supp}(M)$*, and set* $r = \dim(M) - \dim(M/\mathfrak{p}M)$*. Then there exists an* M*-regular sequence* $(x_i)_{1 \leqslant i \leqslant r}$*, formed of elements of* $\mathfrak{p}$*; for every such sequence, we have*

$$\dim(M/(x_1M + \ldots + x_rM)) = \dim(M/\mathfrak{p}M) = \dim(A/\mathfrak{p}),$$

and $\mathfrak{p}$ *is a* minimal *element of* $\operatorname{Ass}(M/(x_1M + \ldots + x_rM))$.

Proof. Let us first prove the existence of the x_i; there is nothing to prove for $r = 0$. For $r > 0$, we proceed by induction. As for every prime ideal $\mathfrak{q}_i$ associated to M we have $\dim(A/\mathfrak{q}_i) = \dim(M)$ (16.5.4), the hypothesis $\dim(M/\mathfrak{p}M) < \dim(M)$ implies that the minimal prime ideals among those containing $\operatorname{Ann}(M)$ (16.1.2.2) and (16.1.7)); one concludes that $\mathfrak{p}$ is not contained in any of the $\mathfrak{q}_i$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2), and hence there exists $x_1 \in \mathfrak{p}$ that is M-regular (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 2 of prop. 2). On deduces then from (16.5.5) that $N = M/x_1M$ is a Cohen-Macaulay module, of dimension $\dim(M) - 1$; as $x_1 \in \mathfrak{p}$, moreover $N/\mathfrak{p}N = M/\mathfrak{p}M$, whence $\dim(N) - \dim(N/\mathfrak{p}N) = r - 1$; one can therefore apply the induction hypothesis to N, and if $(x_i)_{2 \leqslant i \leqslant r}$ is an N-regular sequence formed of elements of $\mathfrak{p}$, it is clear that $(x_i)_{1 \leqslant i \leqslant r}$ is an M-regular sequence formed of elements of $\mathfrak{p}$.

Now if one sets $P = M/(x_1M + \ldots + x_rM)$, we have $P/\mathfrak{p}P = M/\mathfrak{p}M$ since the x_i are in $\mathfrak{p}$, and $\dim(P) = \dim(P/\mathfrak{p}P)$ by (16.5.6). As $A_\mathfrak{p}$ is regular, and the hypothesis a) implies that $B_\mathfrak{p} = A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$ is also regular, the prime ideals $\mathfrak{q}_i$ associated to P satisfy $\dim(A/\mathfrak{q}_i) = \dim(P)$ (16.5.4). As $\mathfrak{p} \in \operatorname{Supp}(P)$ ($\mathbf{0}_\mathrm{I}$, 1.7.5), $\mathfrak{p}$ contains one of the $\mathfrak{q}'_i \in \operatorname{Ass}(P)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 4, th. 2), and as $\dim(P) = \dim(A/\mathfrak{p}) \leqslant \dim(A/\mathfrak{q}'_i) = \dim(P)$, one necessarily has $\mathfrak{p} = \mathfrak{q}'_i$, hence the result. □

Corollary (16.5.10). — *Under the same hypotheses as in* (16.5.9)*:*

(i) $M_\mathfrak{p}$ *is an* $A_\mathfrak{p}$*-module of Cohen-Macaulay.*
(ii) *We have*

(16.5.10.1) $$\dim(M) = \dim_A(M/\mathfrak{p}M) + \dim_{A_\mathfrak{p}}(M_\mathfrak{p}).$$

Proof. With the notations of (16.5.9), let x'_i be the canonical images of the x_i in $A_\mathfrak{p}$ $(1 \leqslant i \leqslant r)$; as the x'_i belong to the maximal ideal of $A_\mathfrak{p}$ and form a $M_\mathfrak{p}$-regular sequence by flatness (15.1.14), we have

$$\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \geqslant r = \dim(M) - \dim(M/\mathfrak{p}M) \geqslant \dim_{A_\mathfrak{p}}(M_\mathfrak{p})$$

(by virtue of (16.1.8.1)). But taking into account (16.4.5.1), the three terms of this inequality are necessarily equal, whence the corollary. □

Corollary (16.5.11). — *Let* A *be a local Noetherian ring; suppose that there exists an* A*-module of finite type* M *with support* $\operatorname{Spec}(A)$*, that is a Cohen-Macaulay module. Then, for every prime ideal* $\mathfrak{p}$ *of* A*, we have*

(16.5.11.1) $$\dim(A) = \dim(A/\mathfrak{p}) + \dim(A_\mathfrak{p}).$$

Proof. Indeed, $\operatorname{Supp}(M/\mathfrak{p}M) = \operatorname{Spec}(A/\mathfrak{p})$ ($\mathbf{0}_\mathrm{I}$, 1.7.5) and $\operatorname{Supp}(M_\mathfrak{p}) = \operatorname{Spec}(A_\mathfrak{p})$; the relation (16.5.10.1) is thus a special case of (16.5.10.1). □

Corollary (16.5.12). — *Every quotient ring of a local Noetherian ring* A *satisfying the conditions of* (16.5.11) *(in particular every quotient ring of a local Cohen-Macaulay ring) is* catenary.

Proof. It suffices evidently to prove that A itself is catenary, that is to say for two prime ideals $\mathfrak{p} \subset \mathfrak{q}$ of A, we have (16.1.4.2)

$$\dim(A_\mathfrak{q}) = \dim(A_\mathfrak{q}/\mathfrak{p}A_\mathfrak{q}) + \dim(A_\mathfrak{p}).$$

But, by virtue of (16.5.10), (i)) $A_\mathfrak{q}$ satisfies the same hypotheses as A, and the preceding relation is then none other than (16.5.11.1) applied to $A_\mathfrak{q}$ and to the prime ideal $\mathfrak{p}A_\mathfrak{q}$ of $A_\mathfrak{q}$. □

(16.5.13). Let A be a Noetherian ring, M an A-module of finite type. We say that M is a *Cohen-Macaulay* A-*module* if, for every prime ideal $\mathfrak{p}$ of A, $M_\mathfrak{p}$ is an $A_\mathfrak{p}$-module of Cohen-Macaulay; by virtue of (16.5.10) this definition coincides with (16.5.1) when A is *local*. We say that A is a *Cohen-Macaulay ring* if all the $A_\mathfrak{p}$ are. It is clear that if M (resp. A) is a Cohen-Macaulay A-module (resp. a Cohen-Macaulay ring), $S^{-1}M$ is a Cohen-Macaulay $S^{-1}A$-module (resp. $S^{-1}A$ is a Cohen-Macaulay ring) for every multiplicative subset S of A.

§17. Regular rings

17.1. Definition of regular rings

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Proposition (17.1.1). — *Let A be a Noetherian local ring of dimension n, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$ its residue field. The following conditions are equivalent:*

a) *The canonical surjective homomorphism of graded k-modules (15.1.1)*

$$\varphi : \mathbf{S}^*_k(\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \mathrm{gr}^*_\mathfrak{m}(A)$$

(where the second term is the graded k-module associated to A equipped with the $\mathfrak{m}$-preadic filtration) is bijective.

b) *We have $\mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2) = n$.*

c) *The ideal $\mathfrak{m}$ admits a system of generators of n elements.*

d) *The ideal $\mathfrak{m}$ admits a system of generators that is an A-regular sequence.*

Proof. By Nakayama's lemma, $\mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2)$ is the smallest number of elements of a system of generators of $\mathfrak{m}$, so b) and c) are equivalent. On the other hand, if $(x_i)_{1\leqslant i\leqslant r}$ is a system of generators of $\mathfrak{m}$ whose classes $\overline{x}_i$ mod. $\mathfrak{m}^2$ form a basis of the k-vector space $\mathfrak{m}/\mathfrak{m}^2$, $\mathbf{S}^*(\mathfrak{m}/\mathfrak{m}^2)$ is isomorphic to the polynomial ring $k[\mathrm{T}_1,\ldots,\mathrm{T}_r]$; taking into account that every A-module of finite type is separated $\mathbf{0}_{\textbf{IV-1}}$ | 40
for the $\mathfrak{m}$-preadic filtration ($\mathbf{0}_\mathbf{I}$, 7.3.5), it follows from (15.1.9) that conditions a) and d) are equivalent; moreover, since every A-regular sequence of elements of $\mathfrak{m}$ has at most n elements (16.4.1), we see that d) implies c). It remains to prove that c) implies a).

Let us set, for brevity, $\mathrm{S} = \mathbf{S}^*(\mathfrak{m}/\mathfrak{m}^2) = k[\mathrm{T}_1,\ldots,\mathrm{T}_n]$, $\mathrm{G} = \mathrm{gr}^*_\mathfrak{m}(A)$, and consider the exact sequence $0 \to \mathfrak{J} \to \mathrm{S} \xrightarrow{\varphi} \mathrm{G} \to 0$, where the kernel $\mathfrak{J}$ of φ is thus a graded ideal of S. For every integer $s > 0$, we thus have

$$\binom{s+n-1}{n-1} = \mathrm{long}(\mathrm{S}_s) = \mathrm{long}(\mathrm{G}_s) + \mathrm{long}(\mathfrak{J}_s). \tag{17.1.1.1}$$

Suppose $\mathfrak{J} \neq 0$, so that there exists a homogeneous element $u \in \mathfrak{J}$ of degree $h > 0$; $\mathfrak{J}_s$ contains $u\mathrm{S}_{s-h}$ for $s \geqslant h$, and since $u \neq 0$ and S is integral, $u\mathrm{S}_{s-h}$ is isomorphic to S_{s-h} as a k-vector space; we would thus have $\mathrm{long}(\mathfrak{J}_s) \geqslant \mathrm{long}(\mathrm{S}_{s-h}) = \binom{s-h+n-1}{n-1}$, whence for $s \geqslant h$,

$$\mathrm{long}(\mathrm{G}_s) \leqslant \binom{s+n-1}{n-1} - \binom{s-h+n-1}{n-1}. \tag{17.1.1.2}$$

But the second term of (17.1.1.2) is a polynomial in s of degree $\leqslant n-2$; since $\mathrm{G}_s = \mathfrak{m}^s/\mathfrak{m}^{s+1}$, and for s sufficiently large, $\mathrm{long}(\mathrm{G}_s)$ is a polynomial in s of degree exactly equal to $n-1$ whose leading coefficient is > 0 (16.2.1), which contradicts the inequality (17.1.1.2). □

Definition (17.1.2). — We say that a Noetherian local ring that satisfies the equivalent conditions of (17.1.1) is *regular*.

Corollary (17.1.3). — *A regular local ring is integral, integrally closed, and a Cohen-Macaulay ring.*

Proof. The last assertion follows from (17.1.1, d)); on the other hand, if A is regular, $\mathrm{gr}^*_\mathfrak{m}(A)$ is integral and completely integrally closed, being isomorphic to $k[\mathrm{T}_1,\ldots,\mathrm{T}_n]$; we conclude that A also possesses these two properties (Bourbaki, *Alg. comm.*, chap. III, §2, n° 3, cor. of prop. 1 and chap. V, §1, n° 5, prop. 15). □

Example (17.1.4). — (i) A regular local ring of dimension 0, being integral by (17.1.3), is necessarily a *field*, and conversely.

(ii) For a Noetherian local ring of dimension 1 to be regular, it is necessary and sufficient that it be a *discrete valuation ring*: indeed, to say that a Noetherian local ring of dimension 1 is regular means that its maximal ideal is *principal*, and the conclusion follows from Bourbaki, *Alg. comm.*, chap. VI, §3, n° 6, prop. 9.

(iii) Let k be a field; the formal power series ring $A = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$ is a *regular ring of dimension* n; indeed, it is clear that the T_i $(1 \leqslant i \leqslant n)$ generate the maximal ideal $\mathfrak{m}$ of A and form an A-regular sequence, since $A/(\mathrm{T}_1 A + \cdots + \mathrm{T}_i A)$ is isomorphic to $k[[\mathrm{T}_{i+1}, \ldots, \mathrm{T}_n]]$.

Proposition (17.1.5). — *For a Noetherian local ring A to be regular, it is necessary and sufficient that its completion $\widehat{A}$ be regular.*

PROOF. Indeed, the maximal ideal of $\widehat{A}$ is $\mathfrak{m}\widehat{A}$ and we know that $\mathfrak{m}^h/\mathfrak{m}^{h+1}$ and $(\mathfrak{m}\widehat{A})^h/(\mathfrak{m}\widehat{A})^{h+1}$ are isomorphic k-vector spaces; condition a) of (17.1.1) is thus the same for A and $\widehat{A}$. □ $\mathbf{0_{IV\text{-}1}}$ | 41

Definition (17.1.6). — Given a Noetherian local ring A, with maximal ideal $\mathfrak{m}$, we call a *regular system of parameters* of A a system of parameters for A (16.3.5) that generates $\mathfrak{m}$.

The existence of such a system is equivalent to the fact that A is regular (17.1.1); if $(x_i)_{1 \leqslant i \leqslant n}$ is a regular system of parameters, it is a minimal system of generators of $\mathfrak{m}$, whose classes mod. $\mathfrak{m}^2$ form a basis of $\mathfrak{m}/\mathfrak{m}^2$ over k; by virtue of (17.1.1, a)) and (15.1.9), (x_i) is a maximal A-regular sequence (whence the terminology).

One should note however that, in a regular ring, a system of parameters for A that is also an A-regular sequence is *not* necessarily a regular system of parameters, as shown by the example of k-th powers of a regular system of parameters (15.1.20).

Proposition (17.1.7). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$ its residue field, $(x_i)_{1 \leqslant i \leqslant r}$ a sequence of elements of $\mathfrak{m}$, $\mathfrak{J}$ the ideal $x_1 A + \cdots + x_r A$. The following conditions are equivalent:*

a) *A is regular and the x_i are part of a regular system of parameters of A.*
a') *A is regular and the classes $\overline{x}_i$ of the x_i mod. $\mathfrak{m}^2$ are linearly independent over k.*
b) *The x_i are part of a system of parameters for A and $A/\mathfrak{J}$ is a regular ring.*

Moreover, when these conditions are satisfied, $\mathfrak{J}$ is a prime ideal.

PROOF. The last assertion follows trivially from the fact that $A/\mathfrak{J}$, being regular, is integral (17.1.3). Set $n = \dim(A)$. The equivalence of a) and a') is immediate; indeed, the classes mod. $\mathfrak{m}^2$ of the elements of a regular system of parameters form a basis of $\mathfrak{m}/\mathfrak{m}^2$ over k, and conversely, if the $\overline{x}_i$ $(1 \leqslant i \leqslant r)$ are linearly independent over k, one can find $n-r$ elements $x_i \in \mathfrak{m}$ $(r+1 \leqslant i \leqslant n)$ such that the $\overline{x}_i$ $(1 \leqslant i \leqslant n)$ form a basis of $\mathfrak{m}/\mathfrak{m}^2$, so $(x_i)_{1 \leqslant i \leqslant n}$ is a regular system of parameters. To see that a) implies b), note that since the x_i are part of a system of parameters of A, we have $\dim(A/\mathfrak{J}) = n - r$ (16.3.6); let $\mathfrak{n} = \mathfrak{m}/\mathfrak{J}$ be the ideal of $A/\mathfrak{J}$. We have an exact sequence

(17.1.7.1) $$0 \longrightarrow (\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{m}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{n}/\mathfrak{n}^2 \longrightarrow 0$$

since $\mathfrak{n}^2 = (\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{J}$; hypothesis a) implies that $\mathrm{rg}_k((\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{m}^2) = r$, so $\mathrm{rg}_k(\mathfrak{n}/\mathfrak{n}^2) = n - r = \dim(A/\mathfrak{J})$, which proves b) by virtue of (17.1.1, b)).

Conversely, to prove that b) implies a), note that the fact that the x_i are part of a system of parameters implies $\dim(A/\mathfrak{J}) = n - r$ (16.3.6); the hypothesis that $A/\mathfrak{J}$ is regular implies on the other hand $\mathrm{rg}_k(\mathfrak{n}/\mathfrak{n}^2) = n - r$ (17.1.1); moreover we evidently have $\mathrm{rg}_k((\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{m}^2) \leqslant r$, so from (17.1.7.1) we obtain $\mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2) \leqslant r + (n - r) = n$; thus A is regular by (16.2.6) and (17.1.1). Furthermore the relation $\mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2) = n$ then implies $\mathrm{rg}_k((\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{m}^2) = r$, and consequently the $\overline{x}_i$ are linearly independent over k. □

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Corollary (17.1.8). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, t an element of $\mathfrak{m}$. The following conditions are equivalent:*

a) *A/tA is regular and t is a non-zero-divisor in A.*
b) *A is regular and $t \notin \mathfrak{m}^2$.*

PROOF. This is the special case $r = 1$ of (17.1.7) (taking into account that an element of $\mathfrak{m}$ that is a non-zero-divisor is part of a system of parameters (16.4.1)). □

Corollary (17.1.9). — *Let A be a regular local ring, $\mathfrak{m}$ its maximal ideal, $\mathfrak{J}$ an ideal of A contained in $\mathfrak{m}$. The following conditions are equivalent:*

a) *The ring $A/\mathfrak{J}$ is regular.*
b) *$\mathfrak{J}$ is generated by a sequence $(x_i)_{1\leqslant i\leqslant r}$ forming part of a regular system of parameters of A.*

PROOF. We have already seen (17.1.7) that b) implies a). Conversely, suppose that $A/\mathfrak{J}$ is regular (which implies that $\mathfrak{J}$ is prime) and let $n = \dim(A)$, $n - r = \dim(A/\mathfrak{J})$; with the notation of (17.1.7), the exact sequence (17.1.7.1) gives $\mathrm{rg}_k((\mathfrak{m}^2 + \mathfrak{J})/\mathfrak{m}^2) = r$, so there exists a sequence $(x_i)_{1\leqslant i\leqslant r}$ of elements of $\mathfrak{J}$ forming part of a regular system of parameters of A. Set $\mathfrak{J}' = x_1A + \cdots + x_rA$; it follows from (17.1.7) that $\mathfrak{J}'$ is a prime ideal of A and that $A/\mathfrak{J}'$ is regular and of dimension $n - r$; but since $A/\mathfrak{J} = (A/\mathfrak{J}')/(\mathfrak{J}/\mathfrak{J}')$ and $\mathfrak{J}/\mathfrak{J}'$ is prime in the integral ring $A/\mathfrak{J}'$, the dimensions of $A/\mathfrak{J}$ and $A/\mathfrak{J}'$ can only be equal if $\mathfrak{J} = \mathfrak{J}'$ (16.1.2.2). □

17.2. Reminders on projective and injective dimension of modules

(17.2.1). Let A be a ring, M an A-module. Recall (M, VI, 2) that we call *projective dimension* (resp. *injective dimension*) of M and denote $\dim.\mathrm{proj}(M)$ or $\dim.\mathrm{proj}_A(M)$ (resp. $\dim.\mathrm{inj}(M)$ or $\dim.\mathrm{inj}_A(M)$) the smallest n (integer or equal to $+\infty$) such that there exists a left projective resolution (resp. a right injective resolution) of M of length n. It is equivalent (loc. cit.) to say that n is the smallest number such that for every A-module N, we have $\mathrm{Ext}^i_A(M,N) = 0$ (resp. $\mathrm{Ext}^i_A(N,M) = 0$) for all $i > n$, or only for $i = n+1$. For every direct factor M' of M, we thus have $\dim.\mathrm{proj}(M') \leqslant \dim.\mathrm{proj}(M)$ since $\mathrm{Ext}^i_A(M',N)$ is a direct factor of $\mathrm{Ext}^i_A(M,N)$; similarly $\dim.\mathrm{inj}(M') \leqslant \dim.\mathrm{inj}(M)$. A condition equivalent to $\dim.\mathrm{proj}(M) = n$ (resp. $\dim.\mathrm{inj}(M) = n$) is that n is the smallest number such that, for every exact sequence

$$0 \longrightarrow \mathrm{R} \longrightarrow \mathrm{P}_{n-1} \longrightarrow \cdots \longrightarrow \mathrm{P}_0 \longrightarrow M \longrightarrow 0$$

(resp. $0 \to M \to \mathrm{Q}_0 \to \cdots \to \mathrm{Q}_{n-1} \to \mathrm{C} \to 0$), where all the P_i for $0 \leqslant i < n$ are projective (resp. all the Q_i for $0 \leqslant i < n$ are injective), R is projective (resp. C is injective).

Remark (17.2.2). —
(i) To say that M is a projective (resp. injective) A-module is therefore equivalent to saying that $\dim.\mathrm{proj}(M) = 0$ (resp. $\dim.\mathrm{inj}(M) = 0$).
(ii) Let T be a covariant additive functor from the category of A-modules to an abelian category. It follows immediately from the definitions of (17.2.1) and those of derived functors that if $\dim.\mathrm{proj}(M) \leqslant n$ (resp. $\dim.\mathrm{inj}(M) \leqslant n$), we have $\mathrm{L}^i\mathrm{T}(M) = 0$ (resp. $\mathrm{R}^i\mathrm{T}(M) = 0$) for all $i > n$. $\mathbf{0}_{\textbf{IV-1}}$ | 43
(iii) If we suppose A Noetherian and M of finite type, the last interpretation of the projective dimension given in (17.2.1) shows that if $\dim.\mathrm{proj}(M) = n$, M admits a left resolution of length n formed of projective modules *of finite type*. If moreover A is a Noetherian *local* ring, M admits a resolution of length n by *free* modules of finite type, a projective module of finite type then being free (Bourbaki, *Alg. comm.*, chap. II, §3, n° 2, cor. 2 of prop. 5).

Lemma (17.2.3). — *Let A be a ring, M an A-module. For $\dim.\mathrm{inj}(M) \leqslant n$, it is necessary and sufficient that for every monogenic A-module N, we have $\mathrm{Ext}^{n+1}_A(N,M) = 0$.*

PROOF. With the notation of (17.2.1), it suffices to prove that C is injective; for every A-module N, $\mathrm{Ext}^{n+1}_A(N,M)$ is isomorphic to $\mathrm{Ext}^1_A(N,\mathrm{C})$ (M, V, 7), so we have $\mathrm{Ext}^1_A(N,\mathrm{C}) = 0$ for every module N of the form $A/\mathfrak{J}$, where $\mathfrak{J}$ is any ideal of A. The exact sequence of Ext then shows that the canonical homomorphism $\mathrm{Hom}(A,\mathrm{C}) \to \mathrm{Hom}(\mathfrak{J},\mathrm{C})$ is surjective for every ideal $\mathfrak{J}$ of A, which implies that C is an injective A-module (M, I, 3.2). □

Lemma (17.2.4). — *Let A be a Noetherian local ring, M an A-module of finite type. For $\dim.\mathrm{proj}(M) \leqslant n$, it is necessary and sufficient that for every monogenic A-module N, we have $\mathrm{Ext}^{n+1}_A(M,N) = 0$.*

PROOF. We know in fact (M, VI, 2.5) that the condition $\dim.\mathrm{proj}(M) \leqslant n$ is equivalent to $\mathrm{Ext}^{n+1}_A(M,N) = 0$ for every A-module N of finite type; to see that the condition of the statement is also sufficient, we argue by induction on the number m of generators of N: there is a submodule N_1 of N generated by $m-1$ elements and such that $\mathrm{N}_2 = N/\mathrm{N}_1$ is monogenic; from the exact sequence $0 \to \mathrm{N}_1 \to N \to \mathrm{N}_2 \to 0$, we then deduce the exact sequence $\mathrm{Ext}^{n+1}_A(M,\mathrm{N}_1) \to \mathrm{Ext}^{n+1}_A(M,N) \to$

$\operatorname{Ext}_A^{n+1}(M, \mathrm{N}_2)$ and the induction hypothesis shows that the condition of the statement indeed implies $\operatorname{Ext}_A^{n+1}(M,N) = 0$. □

Corollary (17.2.5). — *Let A be a Noetherian ring, M an A-module. We have*

(17.2.5.1) $$\operatorname{dim.inj}_A(M) = \sup_{\mathfrak{m}} \big(\operatorname{dim.inj}_{A_\mathfrak{m}}(M_\mathfrak{m})\big)$$

and if M is of finite type

(17.2.5.2) $$\operatorname{dim.proj}_A(M) = \sup_{\mathfrak{m}} \big(\operatorname{dim.proj}_{A_\mathfrak{m}}(M_\mathfrak{m})\big)$$

where $\mathfrak{m}$ ranges over the set of prime ideals (or the set of maximal ideals) of A.

PROOF. Indeed, if N is an A-module of finite type (thus of finite presentation), we have, for every multiplicative subset S of A and for all $i \geqslant 0$ $\mathbf{0}_{\textbf{IV-1}}$ | 44

$$S^{-1}\operatorname{Ext}_A^i(N,M) = \operatorname{Ext}^i_{S^{-1}A}(S^{-1}N, S^{-1}M)$$

by flatness, considering a free resolution of M and using the fact that the preceding relation is true for $i = 0$ (Bourbaki, *Alg. comm.*, chap. II, §2, nº 7, prop. 19). In particular $\operatorname{Ext}^i_{A_\mathfrak{m}}(A_\mathfrak{m}/\mathfrak{J}A_\mathfrak{m}, M_\mathfrak{m}) = (\operatorname{Ext}^i_A(A/\mathfrak{J}, M))_\mathfrak{m}$ for every prime ideal $\mathfrak{m}$ and every ideal $\mathfrak{J}$ of A; taking into account (17.2.3) and the fact that every ideal of $A_\mathfrak{m}$ is of the form $\mathfrak{J}A_\mathfrak{m}$ for a suitable ideal $\mathfrak{J}$ of A, we deduce formula (17.2.5.1) from what precedes and from Bourbaki, *Alg. comm.*, chap. II, §3, nº 3, th. 1. We proceed similarly for (17.2.5.2) using this time (17.2.4) and interchanging the roles of M and N. □

For Noetherian rings and modules of finite type, the study of projective or injective dimension is thus reduced by (17.2.5) to the case of local rings. We then have the

Lemma (17.2.6). — *Let A be a Noetherian local ring, k its residue field, M an A-module of finite type. For* $\operatorname{dim.proj}(M) \leqslant n$, *it is necessary and sufficient that* $\operatorname{Tor}_i^A(M,k) = 0$ *for $i > n$, and it suffices that* $\operatorname{Tor}_{n+1}^A(M,k) = 0$.

PROOF. The necessity is a special case of the remark (17.2.2, (ii)), applied to the covariant functor $M \mapsto k \otimes_A M$. To prove the sufficiency, it suffices, with the notation of (17.2.1), to establish that R is projective when the P_i are assumed of finite type; now $\operatorname{Tor}^A_{n+1}(M,k)$ is isomorphic to $\operatorname{Tor}^A_1(\mathrm{R},k)$ (M, V, 7); and we know that, since R is of finite type, the condition $\operatorname{Tor}_1^A(\mathrm{R},k) = 0$ implies that R is free (Bourbaki, *Alg. comm.*, chap. II, §3, nº 2, cor. 2 of prop. 5). □

Corollary (17.2.7). — *Under the hypotheses of (17.2.6), let x be an M-regular element of A belonging to the maximal ideal of A; then we have*

(17.2.7.1) $$\operatorname{dim.proj}(M/xM) = \operatorname{dim.proj}(M) + 1.$$

PROOF. Indeed, from the exact sequence $0 \to M \xrightarrow{x} M \to M/xM \to 0$ defined by multiplication by x in M, we deduce by the exact sequence of Tor:

$$\operatorname{Tor}_i^A(M,k) \xrightarrow{x} \operatorname{Tor}_i^A(M/xM,k) \longrightarrow \operatorname{Tor}_{i-1}^A(M,k) \xrightarrow{x} \operatorname{Tor}_{i-1}^A(M,k)$$

for all $i \geqslant 1$. Now, for every A-module N, the homothety of ratio x in $M \otimes_A N$ arises equally well from the homothety of ratio x in M as from the homothety of ratio x in N; we conclude immediately, since the homothety of ratio x in k is zero by definition, that for every i, the homomorphism $\operatorname{Tor}_i^A(M,k) \xrightarrow{x} \operatorname{Tor}_i^A(M,k)$ is zero; in other words, we have the exact sequence

$$0 \longrightarrow \operatorname{Tor}_i^A(M,k) \longrightarrow \operatorname{Tor}_i^A(M/xM,k) \longrightarrow \operatorname{Tor}_{i-1}^A(M,k) \longrightarrow 0.$$

If $\operatorname{dim.proj}(M) = n$, we have $\operatorname{Tor}_n^A(M,k) \neq 0$ and $\operatorname{Tor}^A_{n+1}(M,k) = \operatorname{Tor}^A_{n+2}(M,k) = 0$ by (17.2.6). It thus follows from what precedes that $\operatorname{Tor}^A_{n+1}(M/xM,k) \neq 0$ and $\operatorname{Tor}^A_{n+2}(M/xM,k) = 0$, so $\operatorname{dim.proj}(M/xM) = n+1$ by (17.2.6). □

Proposition (17.2.8). — (M. Auslander) *Let A be a ring. The following conditions are equivalent:*

a) *Every A-module is of projective dimension $\leqslant n$.*
a') *Every A-module of finite type is of projective dimension $\leqslant n$.*
b) *Every A-module is of injective dimension $\leqslant n$.*

c) *For every pair of A-modules M, N we have* $\operatorname{Ext}_A^{n+1}(M,N)=0$.
c') *For every pair of A-modules M, N such that M is of finite type (or only monogenic), we have* $\operatorname{Ext}_A^{n+1}(M,N)=0$.

PROOF. This follows immediately from (17.2.1) and (17.2.3). □

The smallest number n (integer or $+\infty$) for which the equivalent conditions of (17.2.8) are satisfied is called the *global cohomological dimension* (or simply *cohomological dimension*) of A and denoted $\dim.\operatorname{coh}(A)$. $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 45

Proposition (17.2.9). — *Let A be a Noetherian ring. The following conditions are equivalent:*

a) $\dim.\operatorname{coh}(A)\leqslant n$.
b) *Every A-module of finite type is of injective dimension $\leqslant n$.*
c) *For every pair of A-modules of finite type, M, N, we have* $\operatorname{Ext}_A^{n+1}(M,N)=0$.

PROOF. This follows immediately from the definition and from (17.2.4). □

Corollary (17.2.10). — *If A is a Noetherian ring, we have*

(17.2.10.1) $$\dim.\operatorname{coh}(A)=\sup_{\mathfrak{m}}\big(\dim.\operatorname{coh}(A_{\mathfrak{m}})\big)$$

where $\mathfrak{m}$ ranges over the spectrum of A (or the set of maximal ideals of A).

PROOF. This follows immediately from (17.2.9) and (17.2.5). □

Proposition (17.2.11). — *Let A be a Noetherian local ring, k its residue field. For* $\dim.\operatorname{coh}(A)\leqslant n$, *it is necessary that* $\operatorname{Tor}_i^A(k,k)=0$ *for $i>n$, and it suffices that* $\operatorname{Tor}_{n+1}^A(k,k)=0$.

PROOF. Taking into account (17.2.6), it suffices to prove that the relations $\dim.\operatorname{coh}(A)\leqslant n$ and $\dim.\operatorname{proj}_A(k)\leqslant n$ are equivalent. It is clear that the first implies the second by definition. Conversely, if $\dim.\operatorname{proj}_A(k)\leqslant n$, we have $\operatorname{Tor}_i^A(M,k)=0$ for $i>n$ and for every A-module M by (17.2.2, (ii)); thus $\dim.\operatorname{proj}(M)\leqslant n$, which proves the proposition by (17.2.8). □

Corollary (17.2.12). — *Let A be a Noetherian ring. For* $\dim.\operatorname{coh}(A)\leqslant n$, *it is necessary that, for every maximal ideal $\mathfrak{m}$ of A, we have* $\operatorname{Tor}_i^{A_{\mathfrak{m}}}(A/\mathfrak{m},A/\mathfrak{m})=0$ *for $i>n$, and it suffices that these relations be satisfied for $i=n+1$.*

PROOF. This follows immediately from (17.2.11) and (17.2.10). □

Proposition (17.2.13). — *Let A, B be two Noetherian local rings, $\rho: A\to B$ a local homomorphism making B a flat A-module. Then we have*

(17.2.13.1) $$\dim.\operatorname{coh}(A)\leqslant\dim.\operatorname{coh}(B).$$

PROOF. Suppose that $\dim.\operatorname{coh}(B)=n$ is finite; it suffices to prove that for every pair (M,N) of A-modules of finite type, we have $\operatorname{Tor}_i^A(M,N)=0$ for $i>n$ (17.2.11). Since B is a faithfully flat A-module ($\mathbf{0}_{\mathbf{I}}$, 6.6.2), it amounts to the same ($\mathbf{0}_{\mathbf{I}}$, 6.4.1) to show that

(17.2.13.2) $$\operatorname{Tor}_i^A(M,N)\otimes_A B=0.$$

Now, if $\mathrm{L}_\bullet=(\mathrm{L}_j)$ is a right resolution of M by free A-modules, it follows from the fact that B is a flat A-module that $\mathrm{L}_\bullet\otimes_A B=(\mathrm{L}_j\otimes_A B)$ is a right resolution of the B-module $M\otimes_A B$ by free B-modules; moreover, we have $(\mathrm{L}_j\otimes_A B)\otimes_B(N\otimes_A B)=(\mathrm{L}_j\otimes_A N)\otimes_A B$, whence we conclude immediately that the first term of (17.2.13.2) is equal to $\operatorname{Tor}_i^B(M\otimes_A B,N\otimes_A B)$; the hypothesis on B implies that this B-module is zero for $i>n$ (17.2.2, (ii)), whence the conclusion. □ $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 46

(17.2.14). Let $(X,\mathscr{O}_X)$ be a ringed space, $\mathscr{F}$ an $\mathscr{O}_X$-Module; we call the *pointwise projective dimension* (resp. *injective dimension*) of $\mathscr{F}$ and denote $\dim.\operatorname{proj}(\mathscr{F})$ (resp. $\dim.\operatorname{inj}(\mathscr{F})$) the number $\sup_{x\in X}(\dim.\operatorname{proj}(\mathscr{F}_x))$ (resp. $\sup_{x\in X}(\dim.\operatorname{inj}(\mathscr{F}_x))$). We call the *pointwise cohomological dimension* of X and denote $\dim.\operatorname{coh}(\mathscr{O}_x)$ the number $\sup_{x\in X}(\dim.\operatorname{coh}(\mathscr{O}_x))$. It follows from (17.2.5) and (17.2.10) that when A is a Noetherian ring, M an A-module of finite type and $X=\operatorname{Spec}(A)$, we have $\dim.\operatorname{proj}(\widetilde{M})=\dim.\operatorname{proj}(M)$, $\dim.\operatorname{inj}(\widetilde{M})=\dim.\operatorname{inj}(M)$ and $\dim.\operatorname{coh}(X)=\dim.\operatorname{coh}(A)$. We call the *projective dimension* (resp. *injective dimension*) of $\mathscr{F}$ at a point $x\in X$ the projective (resp. injective) dimension of $\mathscr{F}_x$, and the *cohomological dimension of X at a point x* the cohomological dimension of $\mathscr{O}_x$.

Proposition (17.2.15). — *Let X, Y be two ringed spaces in Noetherian local rings, $f : X \to Y$ a flat morphism. If $\dim.\mathrm{coh}(X) \leqslant n$, then Y is of cohomological dimension $\leqslant n$ at every point of $f(X)$.*

PROOF. This follows immediately from (17.2.13). □

17.3. Cohomological theory of regular rings

Theorem (17.3.1). — (Hilbert-Serre) *Let A be a Noetherian local ring. For A to be of finite cohomological dimension, it is necessary and sufficient that A be regular; we then have*

$$\dim.\mathrm{coh}(A) = \dim(A). \tag{17.3.1.1}$$

PROOF. Suppose A regular; let $\mathfrak{m}$ be its maximal ideal, $\mathbf{x} = (x_i)_{1 \leqslant i \leqslant n}$ a regular system of parameters of A (17.1.6); consider the Koszul complex $\mathrm{K}_\bullet(\mathbf{x})$ (**III**, 1.1.1), which is formed of free A-modules, with $\mathrm{K}_i(\mathbf{x}) = 0$ for $i > n$; since (x_i) is a regular sequence, we have $\mathrm{H}_i(\mathrm{K}_\bullet(\mathbf{x})) = 0$ for $i > 0$ (**III**, 1.1.4 and 1.1.3.3) and $\mathrm{H}_0(\mathrm{K}_\bullet(\mathbf{x})) = A/(x_1A + \cdots + x_nA) = A/\mathfrak{m} = k$; the $\mathrm{K}_i(\mathbf{x})$ thus form a free resolution of k of length n. Now, the fact that the x_i belong to $\mathfrak{m}$ immediately implies that in the complex $\mathrm{K}_\bullet(\mathbf{x}, k) = \mathrm{K}_\bullet(\mathbf{x}) \otimes_A k$, the boundary operator is zero in all dimensions, so that we have, by definition, $\mathrm{Tor}_i^A(k,k) = \bigwedge^i(k^n)$; the equality (17.3.1.1) thus follows immediately from (17.2.11) (this result is essentially the "syzygy theorem" of Hilbert).

Let us now show that if A is a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, and if $\dim.\mathrm{proj}_A(\mathfrak{m})$ is *finite*, then A is regular, which will finish proving (17.3.1). We proceed by induction on $n = \mathrm{rg}_k(\mathfrak{m}/\mathfrak{m}^2)$. For $n = 0$, we have $\mathfrak{m} = 0$ and the assertion is trivial. □

Lemma (17.3.1.2). — (Nagata) *If every element of $\mathfrak{m} - \mathfrak{m}^2$ is a zero divisor in A, there exists $c \neq 0$ in A such that $c\mathfrak{m} = 0$ (in other words $\mathfrak{m} \in \mathrm{Ass}(A)$).*

PROOF. We may restrict to the case $\mathfrak{m} \neq 0$, hence $\mathfrak{m} \neq \mathfrak{m}^2$. The hypothesis implies that $\mathfrak{m} - \mathfrak{m}^2$ $\mathbf{0_{IV\text{-}1}}$ | 47
is contained in the union of the ideals $\mathfrak{p}_i$ of $\mathrm{Ass}(A)$ (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, cor. 3 of prop. 2); thus $\mathfrak{m}$ is contained in the union of $\mathfrak{m}^2$ and the $\mathfrak{p}_i$ and since $\mathfrak{m} \neq \mathfrak{m}^2$ (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2); $\mathfrak{m}$ being maximal, this proves the lemma. □

Lemma (17.3.1.3). — *If $a \in \mathfrak{m} - \mathfrak{m}^2$, $\mathfrak{m}/Aa$ is isomorphic to a direct factor of $\mathfrak{m}/a\mathfrak{m}$.*

PROOF. There exists a minimal system of generators of $\mathfrak{m}$ containing a (whose images in $\mathfrak{m}/\mathfrak{m}^2$ form a basis of this k-vector space); let $\mathfrak{b}$ be the ideal generated by the elements of this system other than a. Since the relation $xa \in \mathfrak{b}$ implies $x \in \mathfrak{m}$ (considering the image of xa in $\mathfrak{m}/\mathfrak{m}^2$), we have $\mathfrak{b} \cap Aa \subset a\mathfrak{m}$ whence a homomorphism $\mathfrak{b}/(\mathfrak{b} \cap Aa) \to \mathfrak{m}/a\mathfrak{m}$ deduced from the injection $\mathfrak{b} \to \mathfrak{m}$ by passing to quotients, and which is injective. Moreover, $\mathfrak{b} + Aa = \mathfrak{m}$, and the composed homomorphism

$$\mathfrak{m}/Aa = (\mathfrak{b} + Aa)/Aa \xrightarrow{\sim} \mathfrak{b}/(\mathfrak{b} \cap Aa) \longrightarrow \mathfrak{m}/a\mathfrak{m} \longrightarrow \mathfrak{m}/Aa$$

is the identity; whence the lemma. □

Lemma (17.3.1.4). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, E an A-module of finite type and of finite projective dimension. If $a \in \mathfrak{m}$ is A-regular and E-regular, then E/aE is an (A/aA)-module of finite projective dimension, at most equal to $\dim.\mathrm{proj}_A(E)$.*

PROOF. We argue by induction on $h = \dim.\mathrm{proj}_A(E)$, the case $h = 0$ being trivial since then E is a projective A-module, hence E/aE is a projective (A/aA)-module. There exists an exact sequence

$$0 \longrightarrow N \longrightarrow L \longrightarrow E \longrightarrow 0$$

where L is free and $\dim.\mathrm{proj}_A(N) = h - 1$ (17.2.2, (iii)), N being of finite type. Moreover, the sequence

$$0 \longrightarrow N/aN \longrightarrow L/aL \longrightarrow E/aE \longrightarrow 0$$

is exact (15.1.18). Since a is A-regular, it is also N-regular since L is free; the induction hypothesis $\mathbf{0_{IV\text{-}1}}$ | 48
implies that N/aN is an (A/aA)-module of projective dimension $\leqslant h - 1$, and since L/aL is a free (A/aA)-module, E/aE is an (A/aA)-module of projective dimension $\leqslant h$. □

Let us now examine two cases:

I. — Suppose first that every element of $\mathfrak{m} - \mathfrak{m}^2$ is a zero divisor in A, in which case (17.3.1.2) there exists $c \neq 0$ in A such that $c\mathfrak{m} = 0$. Let us show then that $\mathfrak{m} = 0$. If this were not the case, note first that $\mathfrak{m}$ could not be a projective A-module, since it would be free (Bourbaki, *Alg. comm.*, chap. II, §3, n° 2, cor. 2 of prop. 5), which contradicts the relation $c\mathfrak{m} = 0$. We would thus have $n = \dim.\mathrm{coh}(A) \geqslant 1$. Since $\mathfrak{m} \in \mathrm{Ass}(A)$, the exact sequence of Tor would give the exact sequence $0 \to \mathrm{Tor}^A_{n+1}(E,k) \to \mathrm{Tor}^A_n(k,k) \to 0$; but we have $\mathrm{Tor}^A_{n+1}(E,k) = 0$ (17.2.2, (ii)) and $\mathrm{Tor}^A_n(k,k) \neq 0$ (17.2.11) and we have reached a contradiction.

II. — We can thus restrict to the case where there exists $a \in \mathfrak{m} - \mathfrak{m}^2$ that is A-regular, and consequently also $\mathfrak{m}$-regular. Consider the ring $A' = A/aA$ and its maximal ideal $\mathfrak{m}' = \mathfrak{m}/aA$; it is clear that $\mathrm{rg}_k(\mathfrak{m}'/\mathfrak{m}'^2) = n - 1$. By (17.3.1.4), $\mathfrak{m}/a\mathfrak{m}$ is an A'-module of finite projective dimension, hence so is $\mathfrak{m}' = \mathfrak{m}/aA$, which is a direct factor of it (17.3.1.3) and (17.2.1). The induction hypothesis implies therefore that $A' = A/aA$ is regular, which proves by (17.1.8) that A is regular.

Corollary (17.3.2). — *If A is a regular local ring, $A_\mathfrak{p}$ is regular for every prime ideal $\mathfrak{p}$ of A.*

PROOF. Indeed, we have seen (17.2.10) that $\dim.\mathrm{coh}(A_\mathfrak{p}) \leqslant \dim.\mathrm{coh}(A)$, so the conclusion follows from (17.3.1). □

Proposition (17.3.3). — *Let A, B be two Noetherian local rings, $\rho : A \to B$ a local homomorphism, $\mathfrak{m}$ (resp. $\mathfrak{n}$) the maximal ideal of A (resp. B), $k = A/\mathfrak{m}$ (resp. $k' = B/\mathfrak{n}$) the residue field of A (resp. B). We thus have a canonical homomorphism of k'-vector spaces*

(17.3.3.1) $$\psi : (\mathfrak{m}/\mathfrak{m}^2) \otimes_k k' \longrightarrow \mathfrak{n}/\mathfrak{n}^2.$$

(i) *If B is regular and B is a flat A-module, A is regular.*
(ii) *The following conditions are equivalent:*
  a) *B is regular and the homomorphism ψ (17.3.3.1) is injective.*
  b) *A and B are regular, and for a regular system of parameters $(x_i)_{1 \leqslant i \leqslant m}$ of A, the $\rho(x_i)$ are part of a regular system of parameters of B (in which case this property holds for every regular system of parameters of A).*
  c) *B and $B \otimes_A k$ are regular, and B is a flat A-module.*
  d) *A and $B \otimes_A k$ are regular, and B is a flat A-module.*
  e) *A and $B \otimes_A k$ are regular, and we have*

(17.3.3.2) $$\dim(B) = \dim(A) + \dim(B \otimes_A k).$$

PROOF. (i) We have $\dim.\mathrm{coh}(A) \leqslant \dim.\mathrm{coh}(B)$ by (17.2.13), so it suffices to apply (17.3.1). $\mathbf{0}_{\text{IV-1}}$ | 49

(ii) When A is assumed regular and (x_i) is a regular system of parameters of A, saying that B is a flat A-module is equivalent, by (15.1.21), to saying that the sequence of $y_i = \rho(x_i)$ is B-regular (since $B \otimes_A k$ is a flat k-module). On the other hand, since $\dim(A) = m$ and the sequence (y_i) generates the ideal $\mathfrak{m}B$, the relation (17.3.3.2), which can also be written $\dim(B/\mathfrak{m}B) = \dim(B) - m$, is equivalent to saying that the sequence $(y_i)_{1 \leqslant i \leqslant m}$ is part of a system of parameters of B (16.3.7). We thus see that conditions b), d) and e) are respectively equivalent to the following:

b') A and B are regular and $(y_i)_{1 \leqslant i \leqslant m}$ is part of a regular system of parameters of B.
d') A is regular, $B/(\sum y_i B)$ is regular and the sequence (y_i) is B-regular.
e') A is regular, $B/(\sum y_i B)$ is regular, (y_i) is part of a system of parameters of B.

Now, b') and e') are equivalent by (17.1.7), and since d') implies e') and is implied by b') (17.1.7), it is equivalent to both. The conjunction of b) and d) trivially implies e), and by (i), implies d); we have thus proved the equivalence of b), c), d) and e). Moreover, it is clear that b) implies a), the classes of the x_i in $\mathfrak{m}/\mathfrak{m}^2$ then forming a basis of this k-vector space and the classes of the y_i in $\mathfrak{n}/\mathfrak{n}^2$ a free system of this k'-vector space. It thus remains to prove that a) implies e). Set $V = \mathfrak{m}/\mathfrak{m}^2$, $W = \mathfrak{n}/\mathfrak{n}^2$; it is immediate that $\psi(V \otimes_k k') = (\mathfrak{n}^2 + \mathfrak{m}B)/\mathfrak{n}^2$. On the one hand, by (16.3.9)

(17.3.3.3) $$\dim(B) \leqslant \dim(A) + \dim(B \otimes_A k);$$

secondly (16.2.6), we have

(17.3.3.4) $$\dim(A) \leqslant \mathrm{rg}_k(V \otimes_k k').$$

Finally, in the local ring $B \otimes_A k = B/\mathfrak{m}B$, $\mathfrak{n}' = \mathfrak{n}/\mathfrak{m}B$ is the maximal ideal, k' the residue field, and $\mathfrak{n}'/\mathfrak{n}'^2$ is isomorphic to $\mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B) = W/\psi(V \otimes_k k')$; we thus have, by (16.2.6)

(17.3.3.5) $$\dim(B \otimes_A k) \leqslant \mathrm{rg}_{k'} W - \mathrm{rg}_{k'} \psi(V \otimes_k k').$$

Finally, since B is assumed regular, we have $\dim(B) = \mathrm{rg}_{k'} W$ (17.1.1); we thus conclude from (17.3.3.3), (17.3.3.4) and (17.3.3.5) that $\mathbf{0_{IV\text{-}1}}$ | 50

$$\mathrm{rg}_{k'} W \leqslant \dim(A) + \dim(B \otimes_A k) \leqslant \mathrm{rg}_{k'} W + \mathrm{rg}_{k'}(V \otimes_k k') - \mathrm{rg}_{k'} \psi(V \otimes_k k')$$

and saying that the two extreme terms of this inequality are equal means that ψ is injective. Condition a) thus necessarily implies that in each of the relations (17.3.3.3), (17.3.3.4) and (17.3.3.5), the two sides are equal; now, equality in (17.3.3.4) (resp. (17.3.3.5)) means that A (resp. $B \otimes_A k$) is regular (17.1.1); we have thus proved that a) implies e). □

Proposition (17.3.4). — *Let A be a regular local ring of dimension n, $\mathfrak{m}$ its maximal ideal. For every non-zero A-module of finite type M, we have*

(17.3.4.1) $$\mathrm{prof}(M) + \dim.\mathrm{proj}(M) = n.$$

PROOF. We argue by induction on $m = \mathrm{prof}(M)$. If $m = 0$, we know (16.4.6, (i)) that there exists a sub-A-module N of M isomorphic to $k = A/\mathfrak{m}$; applying the Tor exact sequence to the exact sequence $0 \to N \to M \to M/N \to 0$, we obtain an exact sequence

$$\mathrm{Tor}^A_{n+1}(M/N, k) \longrightarrow \mathrm{Tor}^A_n(N, k) \longrightarrow \mathrm{Tor}^A_n(M, k),$$

and the hypothesis that A is regular implies $\mathrm{Tor}^A_{n+1}(M/N, k) = 0$ ((17.3.1.1) and (17.2.6)), so $\mathrm{Tor}^A_n(k, k)$ is isomorphic to a submodule of $\mathrm{Tor}^A_n(M, k)$; applying again (17.3.1.1) and (17.2.6), we see that $\mathrm{Tor}^A_n(M, k) \neq 0$, so $\dim.\mathrm{proj}(M) \geqslant n$ by (17.2.6); but since $\dim.\mathrm{proj}(M) \leqslant n$ by (17.3.1.1), we indeed have $\dim.\mathrm{proj}(M) = n$. Suppose now $m > 0$, and let x be an M-regular element belonging to $\mathfrak{m}$; we then know (16.4.6, (i)) that $\mathrm{prof}(M/xM) = m - 1$, and on the other hand (17.2.7) $\dim.\mathrm{proj}(M/xM) = \dim.\mathrm{proj}(M) + 1$; the induction hypothesis immediately proves the relation (17.3.4.1). □

Corollary (17.3.5). —
(i) *Let A be a regular local ring of dimension n; for an A-module $M \neq 0$ of finite type to be free, it is necessary and sufficient that M be a Cohen-Macaulay A-module of dimension n.*
(ii) *Let A be a regular local ring, B a local ring, $\rho : A \to B$ a local homomorphism making B a free A-module. For B to be a free Cohen-Macaulay A-module, it is necessary and sufficient that B be a Cohen-Macaulay ring and that ρ be injective (or, equivalently* (16.1.5), *that* $\dim(B) = \dim(A)$*).*

PROOF. (i) This follows from (17.3.4.1) and from the fact that for an A-module of finite type, it is equivalent to say that this module is projective or free (Bourbaki, *Alg. comm.*, chap. II, §3, nº 2, cor. 2 of prop. 5); the free A-modules of finite type M are thus characterised by the relation $\dim.\mathrm{proj}(M) = 0$ (17.2.2, (i)).

(ii) To say that B is a Cohen-Macaulay ring is equivalent to saying that B is a Cohen-Macaulay A-module (16.5.3), so it suffices to apply (i), since $\dim B = \dim A$ (16.1.5). □

(17.3.6). We say that a Noetherian ring A is *regular* if for every prime ideal $\mathfrak{p}$ of A, the local ring $A_\mathfrak{p}$ is regular; when A is itself a local ring, it follows from (17.3.2) that this definition is equivalent to that of (17.1.2). For A to be regular, it suffices, by (17.3.2), that $A_\mathfrak{m}$ be regular for every *maximal* ideal $\mathfrak{m}$ of A. Moreover, it follows immediately from this definition that for every multiplicative subset S of A, $S^{-1}A$ is regular.

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Proposition (17.3.7). — *If A is a regular Noetherian ring, every polynomial ring $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is regular.*

PROOF. It clearly suffices to prove that the polynomial ring $B = A[\mathrm{T}]$ is regular; since for every prime ideal $\mathfrak{q}$ of B, $B_\mathfrak{q}$ is a flat $A_\mathfrak{p}$-module, where $\mathfrak{p} = \mathfrak{q} \cap A$ ($\mathbf{0_I}$, 6.3.1); it thus suffices, by (17.1.10), to prove that $B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}$ is regular, and since this ring is a local ring at a prime ideal of $A_\mathfrak{p}[\mathrm{T}]/\mathfrak{p}A_\mathfrak{p}[\mathrm{T}]$ (where k is the residue field $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$), it suffices to prove that $k[\mathrm{T}]$ is regular; now, $\mathrm{C} = k[\mathrm{T}]$ being principal, the local rings of C are discrete valuation rings or a field (for the ideal (0)), thus regular (17.1.4), which concludes the proof. ☐

Corollary (17.3.8). — *If A is a regular Noetherian ring, every formal power series ring $A[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ is regular.*

PROOF. Let $\mathfrak{J}$ be the ideal generated by the T_i in the polynomial ring $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$; since the latter is regular by (17.3.7), and $A[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ is the completion of $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ for the $\mathfrak{J}$-preadic topology (Bourbaki, *Alg. comm.*, chap. III, §2, nº 12, *Example* 1), the conclusion follows from the following lemma. ☐

Lemma (17.3.8.1). — *Let A be a regular ring, $\mathfrak{J}$ an ideal of A, $\widehat{A}$ the separated completion of A for the $\mathfrak{J}$-preadic topology. Then $\widehat{A}$ is regular.*

PROOF. It suffices in fact (17.3.6) to see that for every maximal ideal $\mathfrak{n}$ of $\widehat{A}$, the local ring $\widehat{A}_{\mathfrak{n}}$ is regular; we know (Bourbaki, *Alg. comm.*, chap. III, §3, nº 4, prop. 8) that $\mathfrak{n}$ is of the form $\mathfrak{m}\widehat{A}$, where $\mathfrak{m}$ is a maximal ideal of A containing $\mathfrak{J}$, and that the canonical homomorphism $A \to \widehat{A}$ gives an injective homomorphism $A_{\mathfrak{m}} \to \widehat{A}_{\mathfrak{n}}$, such that the $\mathfrak{n}\widehat{A}_{\mathfrak{n}}$-preadic topology of $\widehat{A}_{\mathfrak{n}}$ induces on $A_{\mathfrak{m}}$ the $\mathfrak{m}A_{\mathfrak{m}}$-preadic topology, and that $A_{\mathfrak{m}}$ is dense in $\widehat{A}_{\mathfrak{n}}$ for the $\mathfrak{n}\widehat{A}_{\mathfrak{n}}$-preadic topology; consequently the completions of the Noetherian local rings $A_{\mathfrak{m}}$ and $\widehat{A}_{\mathfrak{n}}$ are canonically isomorphic. Now, by hypothesis $A_{\mathfrak{m}}$ is regular, hence so is its completion (17.1.5) and since the completion of $\widehat{A}_{\mathfrak{n}}$ is regular, it follows that $\widehat{A}_{\mathfrak{n}}$ is regular by (17.1.5). ☐

Corollary (17.3.9). — *Let A be a Noetherian ring, quotient of a regular Noetherian ring B. If C is an A-algebra of finite type, every ring of fractions $S^{-1}C$ of C is a quotient of a regular ring.*

PROOF. Indeed, C is a quotient of a polynomial ring $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$; since $A = B/\mathfrak{J}$, where $\mathfrak{J}$ is an ideal of B, we have $A[\mathrm{T}_1, \dots, \mathrm{T}_n] = B[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathfrak{J}B[\mathrm{T}_1, \dots, \mathrm{T}_n]$; by (17.3.7), we may restrict to the case where C is a quotient of a regular ring B'; but if S' is the inverse image of S in B', S' is a multiplicative subset of B' and C is a quotient ring of $S'^{-1}B'$, so that ultimately everything reduces to seeing that when C is regular, $S^{-1}C$ is also regular, which we saw in (17.3.6). ☐

The importance of quotients of regular rings comes among other things from the property that they are *catenary* rings (16.5.12).

All the important rings in the applications to algebraic geometry are quotients of regular rings.

§18. SUPPLEMENT ON EXTENSIONS OF ALGEBRAS

This paragraph assembles a certain number of functorial constructions on rings, which will be $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 51
used repeatedly in §§ 19 and 20; it contains no non-trivial result.

18.1. Inverse images of augmented rings

(18.1.1). Given a ring A (not necessarily commutative), the category of *A-rings* has for objects the couples (B, ρ) formed of a ring B and a homomorphism of rings $\rho : A \to B$, for morphisms (also called *A-homomorphisms*) the homomorphisms of rings $u : B \to C$ such that, if $\rho : A \to B$ and $\sigma : A \to C$ are the homomorphisms (called *structural*) defining the structure of A-ring on B and C respectively, the diagram

(18.1.1.1)
$$\begin{tikzcd} B \arrow[r, "u"] & C \\ & A \arrow[ul, "\rho"] \arrow[u, "\sigma"'] \end{tikzcd}$$

is commutative. The *kernel* $\mathfrak{J}$ of u is a bilateral ideal that is an *A-bimodule* (for ρ).

If $f : A' \to A$ is a homomorphism of rings, for every A-ring (B, ρ), $(B, \rho \circ f)$ is an A'-ring, and if $u : B \to C$ is an A-homomorphism of the A-ring (B, ρ) in the A-ring (C, σ), it is also an A'-homomorphism of the A'-ring $(B, \rho \circ f)$ in the A'-ring $(C, \sigma \circ f)$; one thus defines a functor from the category of A-rings into that of A'-rings.

(18.1.2). Let B, E, F be three A-rings, $f : E \to B$, $g : F \to B$ two A-homomorphisms. Recall that one calls *fibre product* of E and F over B (for the homomorphisms f and g) and one denotes $E \times_B F$ the sub-ring of the product $E \times F$ formed of the couples (x, y) such that $f(x) = g(y)$; the restrictions $p_1 : G \to E$, $p_2 : G \to F$ of the projections pr_1 and pr_2 to G are still called the *canonical projections*; the structure of A-ring of G is defined by the homomorphism $\alpha \mapsto (\rho(\alpha), \sigma(\alpha))$ (where $\rho : A \to E$ and $\sigma : A \to F$ are the structural homomorphisms), which effectively applies A into G by virtue of (18.1.1). The characteristic property of $E \times_B F$ is that, for every couple of A-homomorphisms $u : C \to E$, $v : C \to F$ such that $f \circ u = g \circ v$, there exists one and only one A-homomorphism $w : C \to G$ such that $u = p_1 \circ w$, $v = p_2 \circ w$. One can still say that $E \times_B F$ is the *projective limit* of the projective system formed by B, E, F and the A-homomorphisms f, g, in the category of A-rings ($\mathbf{0_{III}}$, 8.1.9).

(18.1.3). Let $\mathfrak{J}$ be the bilateral ideal of E, kernel of f; it is immediate that the kernel $\mathfrak{J}'$ of $p_2 : G \to F$ is the ideal formed of the elements $(x, 0)$ where $x \in \mathfrak{J}$; the restriction $i_1 : \mathfrak{J}' \to \mathfrak{J}$ of p_1 is an isomorphism of A-bimodules, and also an isomorphism of A-bimodules. Similarly, if $\mathfrak{K}$ is the kernel of g, the kernel $\mathfrak{K}'$ of $p_1 : G \to E$ is the ideal of the elements $(0, y)$ where $y \in K$, and the restriction $i_2 : \mathfrak{K}' \to \mathfrak{K}$ of p_2 is an isomorphism (in the same sense). Finally, it is clear that the kernel of the A-homomorphism $q = f \circ p_1 = g \circ p_2$ of $E \times_B F$ into B is $\mathfrak{J} \times \mathfrak{K} = \mathfrak{J}' \oplus \mathfrak{K}'$, so that one has a commutative diagram

(18.1.3.1)
$$\begin{tikzcd}
 & 0 & 0 & 0 & \\
 & \mathfrak{J}' \oplus \mathfrak{K}' \arrow[u] & & & \\
0 \arrow[r] & \mathfrak{J}' \arrow[r] \arrow[u, hook] & G \arrow[r, "p_2"] & F \arrow[r] & 0 \\
 & & 0 \arrow[r] & \mathfrak{J} \arrow[u, hook] & \\
0 \arrow[r] & \mathfrak{J} \arrow[r] & E \arrow[r, "f"] & B \arrow[r] & 0
\end{tikzcd}$$

The definitions and results of (18.1.2) and (18.1.3) extend immediately to the fibre product of any *family* $(E_\lambda)_{\lambda \in I}$ of A-rings defined by a family of A-homomorphisms $f_\lambda : E_\lambda \to B$. We leave the formulation of these results to the reader.

(18.1.4). Conforming to the terminology of (M, VIII), we call *A-ring augmented over B* an A-ring E equipped with an A-homomorphism *surjective* (called *augmentation* of E), $f : E \to B$; the kernel $\mathfrak{J}$ of f is called the *ideal of augmentation*. One says that the A-augmented ring E is *trivial* if there exists an A-homomorphism $s : B \to E$ that is *inverse to the right* of the augmentation $f : E \to B$ (in other words $f \circ s = 1_B$). The exact sequence of A-bimodules

$$0 \longrightarrow \mathfrak{J} \longrightarrow E \xrightarrow{f} B \longrightarrow 0$$

is then *split*; in other words, one can identify the A-bimodule E with $B \times \mathfrak{J}$, and the multiplication in $\mathbf{0_{IV\text{-}1}}$ | 53
E is then given by

$$(b, z)(b', z') = (bb', bz' + zb' + zz'),$$

$\mathfrak{J}$ being considered as a B-bimodule by means of $s : B \to E$.

(18.1.5). With the notations of (18.1.2), suppose that the A-homomorphism f makes E into an A-ring *augmented over B*. Then it is clear that $p_2 : G \to F$ is also surjective, in other words G is given a structure of A-ring *augmented over F*, that one calls the *inverse image* by $g : F \to B$ of the augmented A-ring E.

Proposition (18.1.6). — *For the inverse image $G = E \times_B F$ by $g : F \to B$ of the augmented A-ring E to be a trivial augmented A-ring, it is necessary and sufficient that there exist an A-homomorphism $u : F \to E$ making the diagram*

$$\begin{tikzcd} & F \arrow[dl, "u"'] \arrow[d, "g"] \\ E \arrow[r, "f"'] & B \end{tikzcd}$$

commutative.

PROOF. The condition is evidently necessary, taking $u = p_1 \circ s$, where $s : F \to G$ is an A-homomorphism inverse to the right of p_2. Inversely, if there exists such an A-homomorphism u, the existence of the inverse to the right s of p_2 results from the universal property of the fibre product (18.1.2) applied to the A-homomorphisms $u : F \to E$ and $1_F : F \to F$.

This result implies in particular that if E is a trivial augmented A-ring, so is every inverse image. □

(18.1.7). Let us take up again the situation described in (18.1.2) and (18.1.3); on $\mathfrak{N}$ one has evidently a structure of *G-bimodule*, coming from its structure of F-bimodule and the homomorphism of rings $p_2 : G \to F$. As i_2 is bijective one has moreover an injection $\theta = j \circ i_2^{-1} : \mathfrak{N} \to G$, which is a homomorphism of *G-bimodules*. We shall now see inversely that, when one supposes moreover that f and g are *surjective*, in other words that E and F are A-rings *augmented* over B, the data of such a homomorphism θ allows us to reconstruct the augmented ring E (over B) from the augmented ring G (over F).

More precisely, let us give ourselves an A-ring F augmented over B and an A-ring G augmented over F, the ideals of augmentation being denoted $\mathfrak{N}$ and $\mathfrak{J}'$ respectively:

(18.1.7.1)
$$\begin{tikzcd} & 0 \arrow[d] & & & \\ & \mathfrak{N} \arrow[d] \arrow[dl, bend right, "\theta"] & & & \\ 0 \arrow[r] & \mathfrak{J}' \arrow[r] & G \arrow[r, "h"] & F \arrow[r] & 0 \\ & & & B \arrow[u, "g'"'] \arrow[d] & \\ & & & 0 & \end{tikzcd}$$

The homomorphism h defines on all the ideals of F (and in particular on $\mathfrak{N}$) a structure of G-bimodule. Let $\theta : \mathfrak{N} \to G$ be a homomorphism of G-bimodules making the diagram (18.1.7.1) commutative; this implies that θ is *injective* and that $\mathfrak{J}' \cap \theta(\mathfrak{N}) = 0$; as $\theta(\mathfrak{N}) = \mathfrak{N}$, the homomorphism $g \circ h : G \to B$ composed is surjective; moreover the image $\mathfrak{J}$ of $\mathfrak{J}'$ in $E = G/\theta(\mathfrak{N})$, and the restriction of q to $\mathfrak{J}'$ the homomorphism canonique $q : G \to E$ is injective. One can therefore form the *fibre product* $G' = E \times_B F$, and as the two A-homomorphisms $q : G \to E$ and $h : G \to F$ are such that $f \circ q = g \circ h$, they define a unique A-homomorphism $u : G' \to G$ by the universal property of G' (18.1.2); we shall now see that u is *bijective*. It suffices to show that u is compatible with the finite filtrations on G' and G formed respectively by G' and $\mathfrak{J}' = \mathrm{Ker}(G' \to F)$, and by G and $\mathfrak{J}'$; moreover, on has seen (18.1.3) that $\mathrm{gr}_0\, u : F \to F$ is the identity and the assertion (ii) in turn results from $\mathrm{gr}_1\, u : \mathfrak{J}'' \to \mathfrak{J}'$ is bijective, hence u itself is bijective (Bourbaki, *Alg. comm.*, chap. III, § 2, no. 8, cor. 3 of th. 1).

18.2. Extensions of a ring by a bimodule

(18.2.1). Let E be an A-ring augmented over B, $f : E \to B$ the augmentation, $\mathfrak{J} = \operatorname{Ker}(f)$ the ideal of augmentation. If $\mathfrak{J}^2 = 0$, $\mathfrak{J}$ is not only an E-bimodule but also a B-*bimodule* since B is isomorphic to $E/\mathfrak{J}$; in a precise manner, every $b \in B$ is of the form $f(x)$ with $x \in E$, and if z, z' are in $\mathfrak{J}$, one has $(x+z)z' = xz'$ (resp. $z'(x+z) = z'x$) so that the value of xz' (resp. $z'x$) does not depend on the element $x \in f^{-1}(b)$, and one can write bz' (resp. $z'b$) which defines the structure of B-bimodule considered. Inversely, if $\mathfrak{J}$ is equipped with a structure of B-bimodule such that $xz' = f(x)z'$ and $z'x = z'f(x)$ for $x \in E$ and $z' \in \mathfrak{J}$, it is clear that $\mathfrak{J}^2 = 0$ for this bimodule structure.

(18.2.2). One calls *A-extension of an A-ring B by a B-bimodule L* an *exact* sequence of homomorphisms of A-bimodules

$$0 \longrightarrow L \xrightarrow{j} E \xrightarrow{f} B \longrightarrow 0$$

where E is an A-ring, f an A-homomorphism of rings and one has, for $x \in E$ and $z \in L$,

$$j(f(x)z) = xj(z), \quad j(zf(x)) = j(z)x$$

hence (18.2.1) $j(L)$ is a bilateral ideal of square zero of E. By abuse of language one also says that E is an extension of B by L. One says that two A-extensions E, E' of B by L are *A-equivalent* if there exists an isomorphism of A-rings $u : E \xrightarrow{\sim} E'$ making the diagram

(18.2.2.1)
$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r] & E \arrow[r] \arrow[d, "u"] & B \arrow[r] & 0 \\
0 \arrow[r] & L \arrow[r] & E' \arrow[r] & B \arrow[r] & 0
\end{tikzcd}$$

commutative.

(18.2.3). One says that an A-extension E of B by a B-bimodule L is *A-trivial* if E is a trivial augmented A-ring (over B) (18.1.4). One defines on the A-bimodule product $B \times L$ a structure of A-ring by setting $(x,s)(y,t) = (xy, xt + sy)$, as one easily checks, and the canonical maps $j : L \to B \times L$ and $f : B \times L \to B$ define an A-extension of B by L that is A-trivial, the canonical map $g : B \to B \times L$ being an A-homomorphism inverse to the right of f. One says this extension is the *trivial type extension* of B by L and one denotes it $D_B(L)$; it is immediate that every A-trivial A-extension of B by a bimodule is A-equivalent to $D_B(L)$. $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 55

One will note that every extension of the A-ring A itself by a bimodule is necessarily A-trivial.

(18.2.4). Given two A-extensions $L \xrightarrow{j} E \xrightarrow{f} B$, $L' \xrightarrow{j'} E' \xrightarrow{f'} B'$, a *morphism* of the first into the second is by definition a triplet of homomorphisms of A-bimodules (u, v, w) such that the diagram

(18.2.4.1)
$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r, "j"] \arrow[d, "w"'] & E \arrow[r, "f"] \arrow[d, "u"'] & B \arrow[r] \arrow[d, "v"'] & 0 \\
0 \arrow[r] & L' \arrow[r, "j'"'] & E' \arrow[r, "f'"'] & B' \arrow[r] & 0
\end{tikzcd}$$

is commutative, u and v being A-homomorphisms of *rings* and w being such that $w(bz) = v(b)w(z)$ and $w(zb) = w(z)v(b)$ for $z \in L$ and $b \in B$ (in other words, the couple (v, w) constitutes a *di-homomorphism* of the B-bimodule L into the B'-bimodule L'); it is clear that if (u', v', w') is a morphism from $L' \to E' \to B'$ into an A-extension $L'' \to E'' \to B''$, $(u' \circ u, v' \circ v, w' \circ w)$ is still a morphism, which justifies the terminology.

The consideration of the two commutative squares of the diagram (18.2.4.1) will lead us to two operations on extensions of A-rings.

(18.2.5). In the first place, consider an A-extension E' of B' by L'

$$0 \longrightarrow L' \xrightarrow{j'} E' \xrightarrow{f'} B' \longrightarrow 0$$

and an A-homomorphism of rings $v : B \to B'$, and let $F = E' \times_{B'} B$ be the *inverse image* of E' (18.1.5), so that one has a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & L_0 \arrow[r] \arrow[d, "i"] & F \arrow[r, "p_2"] \arrow[d, "p_1"] & B \arrow[r] \arrow[d, "v"] & 0 \\
0 \arrow[r] & L' \arrow[r, "j'"'] & E' \arrow[r, "f'"'] & B' \arrow[r] & 0
\end{tikzcd}$$

where the two lines are exact, p_1 and p_2 are canonical homomorphisms and i is bijective; one sees from the definition (18.1.2) and (18.1.3) that $L_0^2 = 0$, so that one can consider F as an A-extension of B by L' (where L' being naturally considered as a B-bimodule by means of the homomorphism of rings v). The functorial character of the fibre product with respect to each of the factors shows moreover that if one has a morphism of two extensions of B'

$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r] \arrow[d, "1_L"] & E_1 \arrow[r] \arrow[d, "g"] & B_1 \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & L \arrow[r] & E_2 \arrow[r] & B_2 \arrow[r] & 0
\end{tikzcd}$$

one deduces from it canonically a morphism of inverse image extensions

$$\begin{tikzcd}
0 \arrow[r] & L' \arrow[r] \arrow[d, "1_{L'}"] & E_1 \oplus_L L' \arrow[r] \arrow[d, "g \oplus 1_{L'}"] & B_1 \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & L' \arrow[r] & E_2 \oplus_L L' \arrow[r] & B_2 \arrow[r] & 0
\end{tikzcd}$$

In particular, if E_1 and E_2 are A-equivalent A-extensions of B by L, their inverse image extensions of B by L' obtained by w are A-equivalent.

When a morphism (18.2.4.1) of A-extensions factors through the inverse image of E' by the homomorphism $w : L \to L'$ (where L' is considered as B-bimodule by means of the homomorphism of rings $v : B \to B'$): in fact, the definition of the amalgamated sum shows that there exists a unique A-homomorphism $u_0 : E \to F = E' \times_{B'} B$ making the diagram

$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r, "j"] \arrow[d, "w_0"] & E \arrow[r, "f"] \arrow[d, "u_0"] & B \arrow[r] \arrow[d, "1_B"] & 0 \\
0 \arrow[r] & L'_0 \arrow[r] \arrow[d, "i"] & F \arrow[r, "p_2"] \arrow[d, "p_1"] & B \arrow[r] \arrow[d, "v"] & 0 \\
0 \arrow[r] & L' \arrow[r, "j'"'] & E' \arrow[r, "f'"'] & B' \arrow[r] & 0
\end{tikzcd}$$

commutative, with $u = u_0 \circ j_1$, $w = j_0 \circ w_0$, j_1 and j_2 being the canonical homomorphisms; one verifies immediately that u_0 is also a homomorphism of rings.

(18.2.6). Let us study the inverse image of extensions by surjective A-homomorphisms. Consider a surjective A-homomorphism $v : B \to B'$, an A-extension F of B by a B-bimodule L, and the bilateral ideal $\mathfrak{N}$ of B, kernel of v (which can be considered as an F-bimodule by means of the augmentation $F \to B$); one has seen (18.1.7) that every homomorphism of F-bimodules $\theta : \mathfrak{N} \to F$ making the diagram

$$\begin{tikzcd}
& \mathfrak{N} \arrow[d] \arrow[dl, bend right, "\theta"'] \\
F \arrow[r] & B
\end{tikzcd}$$

commutative determines an extension E' of $B' = B/\mathfrak{N}$ by L whose inverse image by $v : B \to B'$ is A-equivalent to F, and that every A-extension E of B' by L having this last property is obtained in this way (up to an A-equivalence close). Moreover, for two homomorphisms θ_1, θ_2 of F-bimodules of $\mathfrak{R}$ into F to give two A-equivalent A-extensions of B by L, it is necessary and sufficient that there exist an A-*equivalence* u of the A-extension F onto itself such that $\theta_2 = u \circ \theta_1$.

(18.2.7). Consider now the left square of (18.2.4.1), and recall first the notion of *amalgamated sum* in the category of A-bimodules: given three A-bimodules X, Y, Z and two A-homomorphisms $f : X \to Y$, $g : X \to Z$, the amalgamated sum $Y \oplus_X Z$ is the *inductive limit* of the inductive system formed by X, Y, Z and the A-homomorphisms f, g in the category of A-bimodules ($\mathbf{0_{III}}$, 8.1.11). One defines this A-bimodule as quotient of the product $Y \times Z$ by the sub-A-bimodule M image of X by the homomorphism $x \mapsto (f(x), -g(x))$. Its characteristic property is that, for every couple of homomorphisms of A-bimodules $u : Y \to T$, $v : Z \to T$ such that $u \circ f = v \circ g$, there exists a unique homomorphism $w : Y \oplus_X Z \to T$ such that $u = w \circ j_1$ and $v = w \circ j_2$, where $j_1 : Y \to Y \oplus_X Z$ and $j_2 : Z \to Y \oplus_X Z$ are the canonical maps.

(18.2.8). Consider now an A-extension E of B by a B-bimodule L:

$$0 \longrightarrow L \xrightarrow{j} E \xrightarrow{f} B \longrightarrow 0$$

and let moreover $w : L \to L'$ be a homomorphism of B-*bimodules*. Let H be the A-bimodule amalgamated sum $E \oplus_L L'$; let us show how one can equip it with a structure of A-*ring* and define an A-*extension*

$$0 \longrightarrow L' \longrightarrow H \longrightarrow B \longrightarrow 0.$$

Let us note for this that L' is equipped with a structure of E-*bimodule* by means of the homomorphism of augmentation $E \to B$; one can therefore form the A-extension of trivial type $G = D_B(L')$ (18.2.3). Consider then the map $\theta : z \mapsto (j(z), -w(z))$ from L into G; it is a homomorphism of G-*bimodules* (L being considered as a G-bimodule by means of the canonical homomorphism $\rho : G \to E$). Indeed, for $(x, z') \in G$ and $z \in L$, one has $(j(z), -w(z))(x, z') = (j(z)x, j(z)z' - w(z)f(x))$; or $j(z)z' = f(j(z))z' = 0$ by definition of the structure of E-bimodule on L', and $w(z)f(x) = w(zf(x))$ by definition of the structure of B-bimodule on L'; one thus verifies that θ is also a homomorphism of G-module on the left. One can then apply the result of (18.1.7) to the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & & & \\
 & L \arrow[d] \arrow[dl, bend right, "\theta"] & & & \\
0 \arrow[r] & L' \arrow[r] & G \arrow[r, "p"] & E \arrow[r] & 0 \\
 & & & B \arrow[u, "f"'] \arrow[d] & \\
 & & & 0 &
\end{tikzcd}$$

the result of (18.1.7). As $H = G/\theta(L)$ by definition, our assertion is an immediate consequence of (18.1.7).

One says that the A-extension H of B by L' is *deduced* of E *by the homomorphism* $w : L \to L'$. The functorial character of the amalgamated sum in each of the summands shows moreover that if one has a morphism of extensions

$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r] \arrow[d, "1_L"] & E_1 \arrow[r] \arrow[d, "g"] & B_1 \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & L \arrow[r] & E_2 \arrow[r] & B_2 \arrow[r] & 0
\end{tikzcd}$$

one deduces from it canonically a morphism of deduced extensions

$$\begin{tikzcd} 0 \arrow[r] & L' \arrow[r] \arrow[d, "1_{L'}"] & E_1 \oplus_L L' \arrow[r] \arrow[d] & B_1 \arrow[r] \arrow[d, "h"] & 0 \\ 0 \arrow[r] & L' \arrow[r] & E_2 \oplus_L L' \arrow[r] & B_2 \arrow[r] & 0 \end{tikzcd}$$

In particular, if E_1 and E_2 are A-equivalent A-extensions of B by L, the extensions of B by L' obtained by w are A-equivalent.

When one has a morphism (18.2.4.1) of A-extensions, it factors through the A-extension H of B by L' deduced from E by the homomorphism $w : L \to L'$ (where L' being considered as B-bimodule by means of the homomorphism of rings $v : B \to B'$): in fact, the definition of the amalgamated sum shows that there exists a unique A-homomorphism $u_0 : H \to E'$ of A-bimodules, making the diagram

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r, "j"] \arrow[d, "w_0"] & E \arrow[r, "f"] \arrow[d, "j_1"] & B \arrow[r] \arrow[d, "1_B"] & 0 \\ 0 \arrow[r] & L'_0 \arrow[r] \arrow[d, "j_e"] & H \arrow[r, "f_e"] \arrow[d, "u_e"] & B \arrow[r] \arrow[d, "v"] & 0 \\ 0 \arrow[r] & L' \arrow[r, "j'"'] & E' \arrow[r, "f'"'] & B' \arrow[r] & 0 \end{tikzcd}$$

commutative, with $u = u_0 \circ j_1$, $w = j_0 \circ w_0$, j_1 and j_2 being the canonical homomorphisms; one verifies immediately that u_0 is also a homomorphism of rings.

Let us note moreover the functorial properties relative to trivial extensions:

Proposition (18.2.9). — *Let B, B' be two A-rings, L a B-bimodule, L' a B'-bimodule, $v : B \to B'$ an A-homomorphism of rings, $w : L \to L'$ a homomorphism of A-bimodules such that (v, w) is a di-homomorphism of bimodules. Then there exists a unique A-homomorphism of rings $u : D_B(L) \to D_{B'}(L')$ making commutative the diagrams*

$$\begin{tikzcd} D_B(L) \arrow[r, "u"] & D_{B'}(L') \\ B \arrow[u] \arrow[r, "v"'] & B' \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} D_B(L) \arrow[r, "u"] & D_{B'}(L') \\ L \arrow[u] \arrow[r, "w"'] & L' \arrow[u] \end{tikzcd}$$

where the vertical arrows are the canonical injections.

PROOF. Indeed, u cannot be other than the map $(x, s) \mapsto (v(x), w(s))$ and it remains to verify that it is an A-homomorphism of rings, which results trivially from the definition of the multiplication (18.2.3). □

Proposition (18.2.10). — *Let B be an A-ring, L a B-bimodule, E an A-extension of B by L. One defines a bijective map of the set G of homomorphisms of B-rings of E onto the set G' of A-equivalences of $D_B(L)$ onto E by associating to every $g \in G$ the A-equivalence $g' : (x, s) \mapsto g(x) + s$; the inverse map associates to every $g' \in G'$ the B-homomorphism inverse to the right of the augmentation $E \to B$.*

PROOF. This results immediately from the definitions. □

18.3. The group of classes of A-extensions

(18.3.1). Consider an A-ring B and a B-bimodule L fixed; then the relation "E and E' are A-equivalent" between A-extensions E, E' of B by L is an equivalence relation and for this relation one can speak of the *set* of classes of A-extensions of B by L. It suffices to see this to remark that, for every $x \in B$, c_x is an element of E whose image in B is x, every $z \in E$ can be written in one and only one way in the form $c_x + t$, where $t \in L$, and one can write $c_{x+y} = c_x + c_y + \varphi(x,y)$, $c_{xy} = c_x c_y + \psi(x,y)$, where $\varphi(x,y)$ and $\psi(x,y)$ are elements of L, the maps φ and ψ from $B \times B$ into L being the expressions of the fact that E is an A-ring, which is pointless to write here. Every A-extension of B by L is therefore A-equivalent to an A-extension whose underlying set is $B \times L$, hence our conclusion.

(18.3.2). The A-ring B being fixed, denote provisionally by $T(L)$ the set of classes of A-extensions of B by L. For every B-homomorphism of B-bimodules $w : L \to L'$, one defines canonically an application $T(w) : T(L) \to T(L')$ by associating to the class of an A-extension E of B by L the class of the A-extension $E \oplus_L L'$ deduced of E by w, by virtue of (18.2.8). If $w' : L' \to L''$ is a second homomorphism of B-bimodules, one has also

(18.3.2.1) $$T(w' \circ w) = T(w') \circ T(w).$$

Indeed, one knows that there exists a canonical isomorphism of A-bimodules

$$E \oplus_L L'' \xrightarrow{\sim} (E \oplus_L L') \oplus_{L'} L''$$

by virtue of the general properties of inductive limits (cf. for example (**I**, 3.3.9)), and it is immediate to verify that it is an A-equivalence of A-extensions, hence our assertion. One will identify canonically $T(L)$ to $\prod_\alpha T(L_\alpha)$ by the preceding application. One verifies also immediately that if $(L'_\alpha)_{\alpha \in I}$ is a second family of B-bimodules and, for each α, $w_\alpha : L_\alpha \to L'_\alpha$ a homomorphism of B-bimodules, then, setting $w = \prod_\alpha w_\alpha : L \to L' = \prod_\alpha L'_\alpha$, $T(w)$ identifies with $\prod_\alpha T(w_\alpha)$ when one makes the preceding identification.

(18.3.3). Consider now a family $(L_\alpha)_{\alpha \in I}$ of B-bimodules and their product $L = \prod_\alpha L_\alpha$; the projections $\mathrm{pr}_\alpha : L \to L_\alpha$ define applications

$$T(\mathrm{pr}_\alpha) : T(L) \longrightarrow T(L_\alpha),$$

hence a canonical application

(18.3.3.1) $$\prod_\alpha T(\mathrm{pr}_\alpha) : T(L) \longrightarrow \prod_{\alpha \in I} T(L_\alpha).$$

We shall see that this application is *bijective*. Indeed, for every $\alpha \in I$, let E_α be an A-extension of B by L_α, and let F be the *fibre product* of the E_α on B (18.1.3); it is immediate that if one replaces each E_α by an A-equivalent A-extension E'_α, the fibre product F' of the E'_α on B is A-equivalent to F. One has thus defined an application $\prod_\alpha T(L_\alpha) \to T(L)$ and it is clear that this application is the reciprocal of (18.3.3.1), hence our assertion. One will identify canonically $T(L)$ with $\prod_\alpha T(L_\alpha)$ by the preceding application. One verifies also immediately that if $(L'_\alpha)_{\alpha \in I}$ is a second family of B-bimodules and, for each α, $w_\alpha : L_\alpha \to L'_\alpha$ a homomorphism of B-bimodules, then, setting $w = \prod_\alpha w_\alpha : L \to L' = \prod_\alpha L'_\alpha$, $T(w)$ identifies with $\prod_\alpha T(w_\alpha)$ when one makes the preceding identification.

(18.3.4). Since, for a B-bimodule L, the addition is a homomorphism $s : L \times L \to L$ of B-*bimodules*, and it is also the symmetry $t : L \to L$ of the additive law of L. One deduces from them a law of composition

$$T(s) : T(L) \times T(L) \longrightarrow T(L)$$

in $T(L)$ by virtue of (18.3.3), and this law is a law of *commutative group* of which $T(t)$ is the symmetry, as it results from the definition of an object in groups by means of commutative diagrams ($\mathbf{0_{III}}$, 8.2.5 and 8.2.6). We shall designate by $\mathrm{Exan}_A(B, L)$ the commutative group thus defined and we say it is the *group of classes of A-extensions of B by L*.

(18.3.5). Denote by $\boldsymbol{K}$ the category whose objects are the triplets (A, B, L) where A is a ring, B an A-ring and L a B-bimodule; the *morphisms* of this category are the triplets (u, v, w) where $u : A' \to A$ $\mathbf{0_{IV\text{-}1}}$ | 61 and $v : B' \to B$ are two homomorphisms of rings making commutative the left square of the diagram

$$\begin{tikzcd} A \arrow[r] & B & L \\ A' \arrow[u, "u"] \arrow[r] & B' \arrow[u, "v"'] & L' \arrow[u, "w"'] \end{tikzcd}$$

where the horizontal arrows are the structural homomorphisms; finally $w : L \to L'$ is a homomorphism of commutative groups such that one has $w(v(b')z) = b'w(z)$ and $w(zv(b')) = w(z)b'$ for all $z \in L$ and $b' \in B'$ (in other words, w is an homomorphism of B'-bimodules when L is equipped with its structure of B'-bimodule defined by v). The composition of morphisms is defined by $(u', v', w') \circ (u, v, w) = (u \circ u', v \circ v', w' \circ w)$, which justifies immediately. We propose to show that

(18.3.5.1) $$(A, B, L) \longmapsto \mathrm{Exan}_A(B, L)$$

is a *covariant functor* from the category $\boldsymbol{K}$ into the category $\boldsymbol{Ab}$ of commutative groups. It is therefore a matter of defining for every triplet (u, v, w) as above, a homomorphism of commutative groups

$$(u, v, w)_* : \mathrm{Exan}_A(B, L) \longrightarrow \mathrm{Exan}_{A'}(B', L').$$

By virtue of the definition of morphisms in $\boldsymbol{K}$, one can write

$$(u, v, w) = (1_{A'}, 1_{B'}, w) \circ (1_{A'}, v, 1_L) \circ (u, 1_B, 1_L)$$

where, in the first factor, L is equipped with its structure of B'-bimodule defined by v; we shall therefore define $(u, v, w)_*$ when two of the three homomorphisms u, v, w are reduced to the identity.

(18.3.6). We take first $(1_A, 1_B, w)_*$ the application

(18.3.6.1) $$w_* : \mathrm{Exan}_A(B, L) \longrightarrow \mathrm{Exan}_A(B, L')$$

denoted $T(w)$ in (18.3.2); it is immediate to verify that it is a homomorphism of groups, this property expressing the commutativity of diagrams, transformed by T of analogous diagrams for L and L'.

The application $(1_A, v, 1_L)_*$ is the application

(18.3.6.2) $$v^* : \mathrm{Exan}_A(B, L) \longrightarrow \mathrm{Exan}_A(B', L)$$

defined in the following way: if E is an A-extension of B by L, one has seen that $E \times_B B'$ is an A-extension of B' by L (18.2.5) and that if one replaces E by an A-equivalent A-extension E', $E' \times_B B'$ is A-equivalent to $E \times_B B'$; the image by v^* of the class of E is the class of $E \times_B B'$. One verifies immediately that if $w : L \to L'$ is a homomorphism of B-bimodules, the diagram

(18.3.6.3) $$\begin{tikzcd} \mathrm{Exan}_A(B, L) \arrow[r, "v^*"] \arrow[d, "w_*"'] & \mathrm{Exan}_{A'}(B, L) \arrow[d, "w_*"] \\ \mathrm{Exan}_A(B, L') \arrow[r, "v^*"'] & \mathrm{Exan}_{A'}(B, L') \end{tikzcd}$$

is commutative, L and L' being considered as B'-bimodules by means of v in the right column. Replacing L and L' by $L \times L$ and L, and w by the addition s in L, one concludes as above that v^* is indeed a *homomorphism of groups*.

(18.3.6.4). Finally, the application $(u, 1_B, 1_L)_*$ is the application

(1) $$u^* : \mathrm{Exan}_A(B, L) \longrightarrow \mathrm{Exan}_{A'}(B, L)$$

which is obtained by associating to an A-extension E of B by L the ring E considered as an A'-*ring* by means of u (18.1.1), which is evidently also an A'-extension of B by L, B being also considered as A'-ring by means of u; it is clear that an A-equivalence is also an A'-equivalence, whence the application (18.3.6.4), which, for every homomorphism $w : L \to L'$ of B-bimodules, also makes commutative the diagram analogous to that

$$\begin{tikzcd} \mathrm{Exan}_A(B,L) \arrow[r, "u^*"] \arrow[d, "w_*"'] & \mathrm{Exan}_{A'}(B,L) \arrow[d, "w_*"] \\ \mathrm{Exan}_A(B,L') \arrow[r, "u^*"'] & \mathrm{Exan}_{A'}(B,L') \end{tikzcd}$$

One deduces as above that u^* is indeed a homomorphism of groups.

This being so, one sets $(u,v,w)_* = (1_{A'}, 1_{B'}, w)_* \circ (1_{A'}, v, 1_L)_* \circ (u, 1_B, 1_L)_*$ and one verifies easily, by reason of the commutativity of the diagrams (18.3.6.3) and (18.3.6.5), that one has $(u \circ u', v \circ v', w' \circ w)_* = (u', v', w')_* \circ (u,v,w)_*$, which finishes the proof that (18.3.5.1) is a functor.

The existence of the homomorphism of groups (18.3.6.1) shows in particular that if E is a trivial A-extension of B by L, the extension $E \oplus_L L'$ of B by L' defined in (18.2.8) is also trivial, which one moreover verifies directly without difficulty.

On the other hand, for every element z of the *centre* Z of B, the homothety $h_z : y \mapsto yz = zy$ is an endomorphism of the B-bimodule L, hence $(h_z)_*$ is an endomorphism of the commutative group $\mathrm{Exan}_A(B,L)$, and by functoriality, these endomorphisms define on $\mathrm{Exan}_A(B,L)$ a structure of *Z-module*.

(18.3.7). Let A, A' be two rings, $u : A' \to A$ a homomorphism, B an A-ring and L a B-bimodule. The *kernel* of the homomorphism of groups

$$u^* : \mathrm{Exan}_A(B,L) \longrightarrow \mathrm{Exan}_{A'}(B,L)$$

is formed by definition of classes of A-extensions of B by L that are A'-trivial when one considers them as A'-extensions by means of u. One denotes this kernel by the notation $\mathrm{Exan}_{A/A'}(B,L)$ when this causes no confusion.

If Λ is a ring, and if one has a commutative diagram of Λ-*homomorphisms* of Λ-rings

(18.3.7.1)
$$\begin{tikzcd} B' \arrow[r] & B \\ A' \arrow[r] \arrow[u] & A \arrow[u] \end{tikzcd}$$

one deduces from it canonically homomorphisms $\mathbf{0}_{\textbf{IV-1}}$ | 63

(18.3.7.2)
$$\mathrm{Exan}_{A/\Lambda}(B,L) \longrightarrow \mathrm{Exan}_{A'/\Lambda}(B,L) \longrightarrow \mathrm{Exan}_{A'/\Lambda}(B',L)$$

which come from those of (18.3.7.1) by functoriality.

Proposition (18.3.8). — *Let B be a ring, $\mathfrak{J}$ a bilateral ideal of B, $C = B/\mathfrak{J}$ the quotient ring canonically equipped with a structure of C-bimodule. For every C-bimodule L, the group $\mathrm{Hom}_C(\mathfrak{J}/\mathfrak{J}^2, L)$ of the additive group of homomorphisms of C-bimodules of $\mathfrak{J}/\mathfrak{J}^2$ into L. One defines then an isomorphism of commutative groups*

(18.3.8.1)
$$\eta_L : \mathrm{Hom}_C(\mathfrak{J}/\mathfrak{J}^2, L) \xrightarrow{\sim} \mathrm{Exan}_B(C,L)$$

by associating to every C-homomorphism $w : \mathfrak{J}/\mathfrak{J}^2 \to L$ (which is a fortiori *a B-homomorphism) the class of the extension $(B/\mathfrak{J}^2) \oplus_{(\mathfrak{J}/\mathfrak{J}^2)} L$ deduced of the extension $B/\mathfrak{J}^2$ of C by $\mathfrak{J}/\mathfrak{J}^2$ by means of w; the reciprocal isomorphism associates to a class of B-extension E of C by L the homomorphism w such that the composed $\mathfrak{J} \to \mathfrak{J}/\mathfrak{J}^2 \xrightarrow{w} L$ is the restriction to $\mathfrak{J}$ of the structural homomorphism $B \to E$.*

Let $F = B/\mathfrak{J}^2$, which is a B-extension of C by $\mathfrak{J}/\mathfrak{J}^2$. For every B-extension $0 \to L \xrightarrow{j} E \xrightarrow{p} C \to 0$ of C by L, the structural homomorphism $f : B \to E$ is such that the composed $B \xrightarrow{f} E \xrightarrow{p} C$ is the canonical homomorphism $B \to B/\mathfrak{J} = C$. As the image of $\mathfrak{J}$ by $p \circ f$ is zero, $f(\mathfrak{J})$ is contained in the kernel $j(L)$, and as $j(L)$ is of square zero, $f(\mathfrak{J}^2) = 0$; hence f factors as $B \to B/\mathfrak{J}^2 = F \xrightarrow{u} E$, and if w is the restriction of u

to $\mathfrak{J}/\mathfrak{J}^2$, one has a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{J}/\mathfrak{J}^2 \arrow[r] \arrow[d, "w"] & F \arrow[r, "g"] \arrow[d, "u"] & C \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & L \arrow[r, "j"'] & E \arrow[r, "p"'] & C \arrow[r] & 0
\end{tikzcd}$$

in other words $(u, 1_C, w)$ is a morphism of extensions (18.2.4). One deduces from it a morphism of extensions

$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r] \arrow[d, equal] & E' \arrow[r] \arrow[d, "u'"] & C \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & L \arrow[r, "j"'] & E \arrow[r, "p"'] & C \arrow[r] & 0
\end{tikzcd}$$

*it remains to prove that u' is a B-*equivalence*, which establishes η_L is bijective. It suffices to prove (18.1.7) that the inverse image G of the extension E of C by F is F-*trivial*, which is evident since g factors as $F \xrightarrow{u} E \xrightarrow{p} C$ (18.1.6).*

It remains to see that η_L is a homomorphism of groups*; or, for every B-homomorphism $h : L \to L'$, it is immediate that the diagram*

$$\begin{tikzcd}
\mathrm{Hom}_C(\mathfrak{J}/\mathfrak{J}^2, L) \arrow[r, "\eta_L"] \arrow[d, "\mathrm{Hom}(1,h)"] & \mathrm{Exan}_B(C, L) \arrow[d, "h_*"] \\
\mathrm{Hom}_C(\mathfrak{J}/\mathfrak{J}^2, L') \arrow[r, "\eta_{L'}"'] & \mathrm{Exan}_B(C, L')
\end{tikzcd}$$

is commutative. It suffices to apply this remark to the homomorphism $L \times L \to L$ defining the addition to conclude.

18.4. Extensions of algebras

(18.4.1). Let A be a *commutative* ring. The category of A-*algebras* is then a full sub-category of that of A-rings, characterized by the fact that the structural homomorphisms $\rho : A \to B$ are *central*. $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 64

If B is an A-algebra, L a B-bimodule, it is immediate that the A-extension of trivial type $D_B(L)$ is an A-algebra. Similarly, in the construction of (18.2.5) (resp. of (18.2.8)), if B, B' and E (resp. B and E) are A-algebras, it is the same for $E' \times_{B'} B$ (resp. for $E \oplus_L L'$). Finally, if B is an A-algebra and E an A-extension of B by a B-bimodule L that is an A-algebra, every A-extension E' of B by L, *A-equivalent* to E, is also an A-algebra. One deduces then immediately from the definition (18.3.4) that the classes of A-extensions equivalent to *A-algebras* form a *sub-group*, denoted $\mathrm{Exal}_A(B, L)$, of $\mathrm{Exan}_A(B, L)$. Let $\boldsymbol{K}'$ be the full sub-category of $\boldsymbol{K}$ defined in (18.3.5), whose objects (A, B, L) are such that A is commutative and B an A-algebra. Then what precedes shows that

$$(A, B, L) \longmapsto \mathrm{Exal}_A(B, L)$$

is a *covariant functor* of $\boldsymbol{K}'$ in $\boldsymbol{Ab}$. The results of (18.3.7) are unchanged when one replaces Exan everywhere by Exal.

(18.4.2). Suppose always A commutative. The remarks of (18.4.1) remain valid when one replaces “algebra” by “commutative algebra” and “bimodule” by “module”. If B is a commutative A-algebra and L a B-module, the classes of A-extensions equivalent to A-rings that are commutative *A-algebras* (which amounts to the same, commutative *A-rings*) form a *sub-group* of $\mathrm{Exal}_A(B, L)$, denoted $\mathrm{Exalcom}_A(B, L)$. If $\boldsymbol{K}''$ is the full sub-category of $\boldsymbol{K}'$ where A and B are commutative and L a B-module,

$$(A, B, L) \longmapsto \mathrm{Exalcom}_A(B, L)$$

is still a *covariant functor* of $\boldsymbol{K}''$ in $\boldsymbol{Ab}$. One can also replace Exan by Exalcom in (18.3.7). Finally, if in (18.3.8), one supposes B to be a commutative ring and L a C-*module*, the same reasoning gives a canonical isomorphism

$$\mathrm{Hom}_C(\mathfrak{J}/\mathfrak{J}^2, L) \xrightarrow{\sim} \mathrm{Exalcom}_B(C, L) \qquad (18.4.2.1)$$

where the first member is the group of homomorphisms of *C-modules*.

(18.4.3). Let A be a commutative ring. An important case of extensions of A-algebras is formed of A-algebras E, extensions of an A-algebra B by a B-bimodule L such that E, as A-*module*, is the *trivial* extension of the A-module B by the A-module L; in other words, the exact sequence of A-*modules* $0 \to E \to L \to B \to 0$ is *split*. One says then that E is a *Hochschild extension* of B by L. This will always be the case when B is an A-module *projective*, and in particular when A is a commutative *field*. As an A-module, one can identify E with $B \times L$, the multiplication in E being given by $(x,s)(y,t) = (xy, xt + sy + f(x,y))$ with $f(x,y) \in L$. If one writes that this multiplication defines on $B \times L$ a structure of A-algebra, one finds (M, XIV, 2) that f must be an A-*bilinear* map of $B \times B$ into L, such that

$$f(xy,z) + f(x,y)z = xf(y,z) + f(x,yz) \tag{18.4.3.1}$$

in other words, f is a 2-*cocycle* on B with values in L, in the sense of Hochschild; for the extension E to be A-trivial, it is necessary and sufficient that

$$f(x,y) = xg(y) - g(xy) + g(x)y \tag{18.4.3.2}$$

where g is an A-*linear* map of B into L, in other words f must be a 2-*cobord* in the sense of Hochschild. One deduces from it that the classes of Hochschild extensions of B by L form a sub-group of $\mathrm{Exal}_A(B,L)$, isomorphic to the *Hochschild cohomology group* $\mathrm{H}^2_A(B,L)$.

If B is a commutative A-algebra, and L a B-module, the commutative Hochschild extensions of B by L correspond to the *symmetric* 2-cocycles, i.e. those such that $f(y,x) = f(x,y)$. The classes of commutative Hochschild extensions of B by L form therefore a sub-group of $\mathrm{Exalcom}_A(B,L)$, isomorphic to the sub-group of $\mathrm{H}^2_A(B,L)$ image of the group of symmetric 2-cocycles, which we shall denote $\mathrm{H}^2_A(B,L)^s$.

(18.4.4). Let us restrict to the case where A and B are *commutative* and L a B-module, and recall in this case the equivalent definition of the Hochschild cohomology groups $\mathrm{H}^i_A(B,L)$ (M, IX, 6). One considers a complex $\mathrm{P}_\bullet = (\mathrm{P}_n)_{n\geqslant 0}$ of B-modules, where $\mathrm{P}_n = B^{\otimes(n+1)} = B \otimes_A B \otimes \cdots \otimes_A B$ ($n+1$ times), the structure of B-module being defined by $x(y_1 \otimes y_2 \otimes \cdots \otimes y_{n+1}) = (xy_1) \otimes y_2 \otimes \cdots \otimes y_{n+1}$; the derivation, of degree -1, $d_n : \mathrm{P}_n \to \mathrm{P}_{n-1}$, is defined by

$$\begin{aligned} d_n(x_1 \otimes x_2 \otimes \cdots \otimes x_{n+1}) = (x_1x_2) \otimes x_3 \otimes \cdots \otimes x_{n+1} - x_1 \otimes (x_2x_3) \otimes \cdots \otimes x_{n+1} + \cdots \\ + (-1)^{n-1} x_1 \otimes x_2 \otimes \cdots \otimes (x_n x_{n+1}) + (-1)^n (x_{n+1}x_1) \otimes x_2 \otimes \cdots \otimes x_n \end{aligned}$$

which is B-linear since B is commutative.

A 2-cocycle of this complex with values in L is a B-linear map h of $B \otimes_A B \otimes_A B$ into L such that $h(d_3(x \otimes y \otimes z \otimes t)) = 0$; but as $h(x \otimes y \otimes z) = xh(1 \otimes y \otimes z)$, the cocycle h is determined by the A-bilinear map $(y,z) \mapsto f(y,z) = h(1 \otimes y \otimes z)$ of B into L; one obtains $h'(d_2(x \otimes y \otimes z)) = 0$, which gives back the condition (18.4.3.1). Similarly, a 2-cobord will be a map of the form $x \otimes y \otimes z \mapsto h'(d_2(x \otimes y \otimes z))$, where $h' : B \otimes_A B \to L$ is linear, and here h' is determined by the A-linear map $g : y \mapsto h'(1 \otimes y)$ of B into L; one obtains then $h'(d_2(1 \otimes y \otimes z)) = xg(y) - g(xy) + yg(x)$, which gives back (18.4.3.2). One proceeds similarly for the 1-cocycles and one sees that one has

$$\mathrm{H}^i_A(B,L) = \mathrm{H}^i(\mathrm{Hom}_B(\mathrm{P}_\bullet, L)). \tag{18.4.4.1}$$

(18.4.5). Under the conditions of (18.4.4), one can reinterpret in the same way the group $\mathrm{H}^2_A(B,L)^s$ (18.4.3). For this, modify the complex $\mathrm{P}_\bullet$ in degree 3 considering a new complex

$$\mathrm{P}'_\bullet : \mathrm{P}'_3 \xrightarrow{d'_3} \mathrm{P}_2 \xrightarrow{d_2} \mathrm{P}_1;$$

we take $\mathrm{P}'_3 = \mathrm{P}_3 \oplus (B \otimes_A B \otimes_A B)$, d'_3 coinciding with d_3 on P_3, and being given by

$$d'_3(x \otimes y \otimes z) = x \otimes y \otimes z - x \otimes z \otimes y.$$

The relation $d_2 d'_3 = 0$ results from the above with the commutativity of B. With the notations introduced above, a 2-cocycle of $\mathrm{P}'_\bullet$ corresponds now to an A-bilinear map f of $B \times B$ into L that is *symmetric* and verifies (18.4.3.1); consequently, one has $\mathrm{H}^2_A(B,L)^s = \mathrm{H}^2(\mathrm{Hom}_B(\mathrm{P}'_\bullet, L))$.

(18.4.6). In the particular case where one considers a commutative *field* k, an extension K of k, one has $\mathrm{Ext}^1_K(M, L) = 0$ whatever the K-vector spaces L, M and consequently (M, VI, 3.3 *a*)), one has a canonical isomorphism

(18.4.6.1) $$\mathrm{H}^1_k(K, L) \simeq \mathrm{Hom}_K(\mathrm{H}_1(\mathrm{P}_\bullet), L)$$

and similarly

(18.4.6.2) $$\mathrm{H}^2_k(K, L)^s \simeq \mathrm{Hom}_K(\mathrm{H}_2(\mathrm{P}'_\bullet), L).$$

18.5. Case of topological rings

(18.5.1). Let B be a *topological* ring whose topology is *linear*, $\rho : A \to B$ a continuous homomorphism, L a B-bimodule topological, and suppose that there exists a bilateral ideal *open* $\mathfrak{R}_0$ of B such that $\mathfrak{R}_0.L = L.\mathfrak{R}_0 = 0$, so that L can be considered as a $(B/\mathfrak{R})$-bimodule for every bilateral ideal open $\mathbf{0_{IV\text{-}1}}$ | 67
$\mathfrak{R} \subset \mathfrak{R}_0$. Let then $\mathfrak{R}$ be a bilateral ideal open of A and $\mathfrak{J}$ a bilateral ideal open of B such that $\rho(\mathfrak{J}) \subset \mathfrak{R}$, so that $B/\mathfrak{R}$ can be considered as an $(A/\mathfrak{J})$-ring discretely; the preceding remark proves that the group $\mathrm{Exan}_{A/\mathfrak{J}}(B/\mathfrak{R}, L)$ is defined; moreover, if $\mathfrak{R}' \subset \mathfrak{R}$, $\mathfrak{J}' \subset \mathfrak{J}$ are two bilateral ideals open of A and B respectively such that $\rho(\mathfrak{J}') \subset \mathfrak{R}'$, one has by (18.3.5.1) a canonical homomorphism

(18.5.1.1) $$\mathrm{Exan}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \longrightarrow \mathrm{Exan}_{A/\mathfrak{J}'}(B/\mathfrak{R}', L).$$

The set of couples of bilateral ideals open $(\mathfrak{J}, \mathfrak{R})$ such that $\rho(\mathfrak{J}) \subset \mathfrak{R} \subset \mathfrak{R}_0$ is evidently ordered to the right for the relation "$\mathfrak{J} \supset \mathfrak{J}'$ and $\mathfrak{R} \supset \mathfrak{R}'$" and the applications (18.5.1.1) define an *inductive system* of additive groups having this set for set of indices. One sets, by abuse of notation (since it is no longer a matter of a group of extensions in bijective natural correspondence with a set of extensions)

(18.5.1.2) $$\mathrm{Exantop}_A(B, L) = \varinjlim \mathrm{Exan}_{A/\mathfrak{J}}(B/\mathfrak{R}, L).$$

To say that the second member of (18.5.1.2) is *zero* means therefore that, for every couple of bilateral ideals open $\mathfrak{J} \subset A$, $\mathfrak{R} \subset B$ such that $\rho(\mathfrak{J}) \subset \mathfrak{R} \subset \mathfrak{R}_0$ and every $(A/\mathfrak{J})$-extension E of $B/\mathfrak{R}$ by L, there exist two open ideals $\mathfrak{J}' \subset \mathfrak{J}$, $\mathfrak{R}' \subset \mathfrak{R}$ such that $\rho(\mathfrak{J}') \subset \mathfrak{R}'$ and the inverse image of E by the homomorphism $B/\mathfrak{R}' \to B/\mathfrak{R}$ is *trivial*.

We leave to the reader the analogous definition of inductive limits $\mathrm{Exaltop}_A(B, L)$ from $\mathrm{Exal}_{A/\mathfrak{J}}(B/\mathfrak{R}, L)$ and of $\mathrm{Exalcotop}_{A/\mathfrak{J}}(B/\mathfrak{R}, L)$ for the case where A is commutative and B an A-algebra (resp. a commutative A-algebra) topological.

(18.5.2). If one has a commutative diagram of continuous homomorphisms of rings

$$\begin{tikzcd} B' \arrow[r] & B \\ A' \arrow[u] \arrow[r] & A \arrow[u] \end{tikzcd}$$

one deduces from it canonically two homomorphisms of additive groups

$$\mathrm{Exantop}_A(B, L) \longrightarrow \mathrm{Exantop}_{A'}(B, L) \longrightarrow \mathrm{Exantop}_{A'}(B', L)$$

by passage to the inductive limit from (18.3.6.1).

By virtue of the exactness of the $\varinjlim$ functor in the category of commutative groups, the kernel of the homomorphism

$$\mathrm{Exantop}_A(B, L) \longrightarrow \mathrm{Exantop}_{A'}(B, L)$$

is the inductive limit of the kernels of the homomorphisms

$$\mathrm{Exan}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \longrightarrow \mathrm{Exan}_{A'/\mathfrak{J}'}(B/\mathfrak{R}', L)$$

where one took for $\mathfrak{J}'$ the inverse image of $\mathfrak{J}$; one denotes this kernel $\mathrm{Exantop}_{A/A'}(B, L)$. One defines similarly $\mathrm{Exaltop}_{A/A'}(B, L)$ and $\mathrm{Exalcotop}_{A/A'}(B, L)$. Finally, if one has an homomorphism

(18.5.3). Given a topological ring C and two C-bimodules topological M, N, one denotes $\mathrm{Hom.\ cont}_C(M, N)$ the additive group of C-homomorphisms *continuous* from M into N.

Lemma (18.5.3.1). — *Let C be a topological ring, E, L two C-bimodules topological; suppose that the topologies are linear, that L is discrete and annihilated by a bilateral ideal open of C. Then one has a canonical isomorphism*

$$\varinjlim \operatorname{Hom}_{C/\mathfrak{R}}(E/V, L) \xrightarrow{\sim} \operatorname{Hom.\ cont}_C(E, L) \tag{18.5.3.2}$$

where in the first member the inductive limit is taken over the ordered filtering set of the couples $(\mathfrak{R}, V)$ such that $\mathfrak{R}$ is a bilateral ideal open of C, V a sub-C-bimodule open of E, such that $\mathfrak{R}.L = L.\mathfrak{R} = 0$, $\mathfrak{R}.E \subset V$, $E.\mathfrak{R} \subset V$.

PROOF. As $C/\mathfrak{R}$ and E/V are discrete, one has homomorphisms canoniques $w_{\mathfrak{R},V} : \operatorname{Hom}_{C/\mathfrak{R}}(E/V, L) \to \operatorname{Hom.\ cont}_C(E, L)$ forming an inductive system, hence a homomorphism from the first member of (18.5.3.2) in $\operatorname{Hom.\ cont}_C(E, L)$. As the homomorphism $E/V' \to E/V$ is surjective for $V' \subset V$, it follows from the definition that the homomorphism $\operatorname{Hom}_{C/\mathfrak{R}}(E/V, L) \to \operatorname{Hom}_{C/\mathfrak{R}}(E/V', L)$ for $\mathfrak{R}' \subset \mathfrak{R}$ is injective. On the other hand, if u is a C-homomorphism continuous of E into L, its kernel is an open sub-bimodule V_0 of E, and $\mathfrak{R}_0.E \subset V_0$, $E.\mathfrak{R}_0 \subset V_0$, it is clear that u is the canonical image of a $(C/\mathfrak{R}_0)$-homomorphism of E/V_0 into L, hence (18.5.3.2) is surjective.

This being so, the proposition (18.3.8) generalizes as follows to topological rings: ☐

Proposition (18.5.4). — *Let B be a topological ring linearly topologized, $\mathfrak{J}$ a bilateral ideal of B, $C = B/\mathfrak{J}$ the quotient topological ring; $\mathfrak{J}/\mathfrak{J}^2$ (where $\mathfrak{J}$ is equipped with the topology induced by that of B and $\mathfrak{J}/\mathfrak{J}^2$ with the quotient topology of that of $\mathfrak{J}$) is then canonically equipped with a structure of C-bimodule topological. For every C-bimodule discrete L annihilated by a bilateral ideal open of C, there exists then a canonical isomorphism*

$$\operatorname{Hom.\ cont}_C(\mathfrak{J}/\mathfrak{J}^2, L) \xrightarrow{\sim} \operatorname{Exantop}_B(C, L). \tag{18.5.4.1}$$

PROOF. Indeed, for every open $\mathfrak{R}$ of B such that $(\mathfrak{J} + \mathfrak{R})/\mathfrak{J}$ annihilates L, one has, by (18.3.9), a canonical isomorphism

$$\operatorname{Hom}_{B/(\mathfrak{J}+\mathfrak{R})}((\mathfrak{J} + \mathfrak{R})/(\mathfrak{J}^2 + \mathfrak{R}), L) \xrightarrow{\sim} \operatorname{Exan}_{B/\mathfrak{R}}(B/(\mathfrak{J} + \mathfrak{R}), L)$$

and it suffices to pass to the inductive limit, taking into account (18.5.3.1). ☐

§19. FORMALLY SMOOTH ALGEBRAS AND COHEN RINGS

19.1. Introduction

(19.0.1). In Chapter IV, we will introduce and study, among other things, an important class of morphisms of preschemes, the *smooth* morphisms[13]. One of their fundamental properties (and which, together with a finiteness condition, characterizes them) is a property of *lifting of morphisms*: if $f : X \to Y$ is a smooth morphism, $g : Y' \to Y$ a morphism, then for every morphism $h : Y'' \to Y'$ making Y'' a prescheme "slightly different" from Y', every Y-morphism $Y'' \to X$ factorizes as $Y'' \xrightarrow{h} Y' \to X$. More precisely, let us restrict to the case where $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine; B is then called a *formally smooth* A-algebra if, for every A-algebra C and every nilpotent ideal $\mathfrak{J}$ of C, every A-homomorphism $B \to C/\mathfrak{J}$ *lifts to* $B \to C \to C/\mathfrak{J}$. In other terms, the map **$0_{\text{IV-1}}$ | 69**

$$\operatorname{Hom}_A(B, C) \longrightarrow \operatorname{Hom}_A(B, C/\mathfrak{J})$$

is *surjective*. In many applications, X will appear as a contravariant representable functor $Y' \mapsto F(Y')$ from the category of Y-preschemes to that of sets, so that ($\mathbf{0_{III}}$, 8.1.8) one will have $F(Y') = \operatorname{Hom}_Y(Y', X)$. In the affine case, setting $F^0(C) = F(\operatorname{Spec}(C))$, the verification of the fact that, under the conditions above, $F^0(C) \to F^0(C/\mathfrak{J})$ is *surjective* (which can be done even without knowing that F is representable) will establish that f is smooth, provided the additional finiteness condition is satisfied.

[13]Also called *simple* morphisms in certain recent works (inspired by the classical terminology of "simple point"); this terminology lends itself however to confusion, notably in the theory of algebraic groups.

(19.0.2). For *topological algebras* over a topological ring A, there is an analogous notion of *formally smooth algebra* that we will not make precise here (cf. (19.3.1)). The study of these notions is first done from an elementary point of view in § 19, then, with the aid of the properties of differentials that will be developed in §§ 20 and 21, more delicate theorems will be proved in § 22. Let us summarize here the main results on formally smooth algebras: $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 70

I. — Let A, B be two noetherian local rings, $\varphi : A \to B$ a local homomorphism, k the residue field of A, and let $B_0 = B \otimes_A k$; we equip A, B, and B_0 with their preadic topologies and k with the discrete topology. Then, for B to be a formally smooth A-algebra, it is necessary and sufficient that B be a *flat* A-module and that B_0 be a formally smooth k-algebra (19.7.1). This theorem thus reduces formal smoothness, for noetherian local rings, to the same question for noetherian local rings that are algebras over a field.

II. — Let k be a field, A a noetherian local ring that is a k-algebra. For A to be formally smooth, it is necessary and sufficient that A be *geometrically regular* over k, that is to say (cf. **IV**, 6.7.6), for every finite extension k' of k, the semi-local ring $A \otimes_k k'$ be regular (17.3.6). Moreover, if K is the residue field of A, $\widehat{A}$ is isomorphic to a power series ring $K[[T_1, \ldots, T_n]]$ (19.6.5). Moreover, the structure of k-algebra of A is entirely determined by the residue field K of A and the dimension of A; the latter can furthermore be arbitrary subject to the inequality $\dim(A) \geqslant \mathrm{rg}_k(\Gamma_{K/k})$, where $\Gamma_{K/k}$ is the "module of imperfection" of K (22.2.6).

In particular, for an extension K of k to be formally smooth, it is necessary and sufficient that K be a *separable* extension of k (19.6.1).

III. — Let A be a noetherian local ring, $\mathfrak{J}$ an ideal of A distinct from A, $A_0 = A/\mathfrak{J}$, B_0 a complete noetherian local ring, $A_0 \to B_0$ a homomorphism making B_0 a formally smooth A_0-algebra. Then there exists a complete noetherian local ring B, a local homomorphism $A \to B$ making B a flat A-module, and an A_0-isomorphism $B \otimes_A A_0 \xrightarrow{\sim} B_0$ (so that, by I), B is a formally smooth A-algebra); moreover B is determined by these properties up to isomorphism (19.7.2). This theorem contains in particular the *theorems of Cohen* on the structure of complete noetherian local rings (19.8), which will play an important role in §§ 6 and 7 of Chapter IV.

IV. — The interest in the study of formally smooth noetherian local rings over another comes from the following "pointwise" characterization of smooth morphisms: if X and Y are locally noetherian preschemes, $f : X \to Y$ a locally finitely presented morphism, then, for f to be smooth, it is necessary and sufficient that for every $x \in X$, the ring $\mathscr{O}_x$ be formally smooth over $\mathscr{O}_{f(x)}$. In particular, if $Y = \mathrm{Spec}(k)$, where k is a perfect field, to say that f is smooth is equivalent to saying that X is a *regular* prescheme.

V. — Finally, we will see in §§ 20, 21, and 22 that the notion of formally smooth algebra is introduced naturally in the theory of *Kähler differentials*, the two theories shedding light on each other.

(19.0.3). Throughout this section and the following ones, topological rings and modules will be assumed to be *linearly topologized* ($\mathbf{0}_\mathbf{I}$, 7.1.1); the topological rings considered will be assumed to be *commutative*, unless explicit mention to the contrary. Recall that if A and B are two topological rings, $\rho : A \to B$ a ring homomorphism defining on B a structure of A-algebra, we say that B is a *topological A-algebra* if ρ is continuous for the topologies considered.

For brevity, in a topological ring A (resp. a topological A-module M), we will say "fundamental system of open ideals (resp. submodules)" instead of "fundamental system of neighbourhoods of 0 consisting of ideals (resp. submodules)".

Given a topological ring A and a topological A-module M, the sets $\mathfrak{J}M$, where $\mathfrak{J}$ ranges over a fundamental system of open ideals in A, form a fundamental system of open submodules of a topology on M making M a topological A-module, said to be *derived* from the topology of A. $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 71

Let M be a topological A-module whose topology is *less fine* than the topology derived from that of A; then, if N is an open submodule of M, the discrete A-module M/N is annihilated by an open ideal $\mathfrak{R}$ of A, since by hypothesis there exists such an ideal $\mathfrak{R}$ with $\mathfrak{R} \cdot M \subset N$.

If M and N are two topological A-modules whose topologies are both derived from that of A, then every A-homomorphism $u : M \to N$ is *continuous*, since for every neighbourhood V

of 0 in N, there exists by hypothesis an open ideal $\mathfrak{J}$ of A such that $\mathfrak{J}N \subset V$, and one thus has $u(\mathfrak{J}M) \subset \mathfrak{J}N \subset V$.

When B is a topological A-algebra, the topology on B *derived* from that of A is *finer* than the topology given on B, since for every open ideal $\mathfrak{K}$ of B, there is by hypothesis an open ideal $\mathfrak{J}$ of A such that $\mathfrak{J}B \subset \mathfrak{K}$.

19.2. Epimorphisms and formal monomorphisms

Proposition (19.1.1). — *Let A be a topological ring, M, N two topological A-modules, (W_λ) a fundamental system of open submodules in N, $u : M \to N$ a continuous A-homomorphism, $\widehat{M}$, $\widehat{N}$ the separated completions of M and N, $\widehat{u} : \widehat{M} \to \widehat{N}$ the continuous extension of u to the separated completions.*

(i) The following conditions are equivalent:
- *a) $u(M)$ is dense in N.*
- *a′) $\widehat{u}(\widehat{M})$ is dense in $\widehat{N}$.*
- *a″) For all λ, the composed homomorphism $M \xrightarrow{u} N \to N/W_\lambda$ is surjective.*

(ii) The following conditions are equivalent:
- *b) The inverse image by u of the topology of N is equal to the topology of M.*
- *b′) $\widehat{u}$ is an isomorphism of the topological $\widehat{A}$-module $\widehat{M}$ onto the sub-$\widehat{A}$-module $\widehat{u}(\widehat{M})$ of $\widehat{N}$ (which is necessarily closed).*
- *b″) The $u^{-1}(W_\lambda)$ form a fundamental system of neighbourhoods of 0 in M.*

This follows immediately from the definition of separated completions, and from the fact that N/W_λ is discrete.

(19.1.2). When the equivalent conditions of (i) (resp. (ii)) in (19.1.1) are satisfied, we say that u is a *formal epimorphism* (resp. a *formal monomorphism*). We say that u is a *formal bimorphism* if it is both a formal epimorphism and a formal monomorphism; this amounts to the same, by virtue of (19.1.1), as saying that $\widehat{u}$ is an *isomorphism* of the topological $\widehat{A}$-module $\widehat{M}$ onto the topological $\widehat{A}$-module $\widehat{N}$.

Proposition (19.1.3). — *Let A be a topological ring, M, N two topological A-modules, (V_λ), (W_λ) two fundamental systems of open submodules in M and N respectively, having the same index set, and such that $u(V_\lambda) \subset W_\lambda$ for all λ; let $u_\lambda : M/V_\lambda \to N/W_\lambda$ be the homomorphism induced by u by passing to quotients. Then:*

- *(i) For u to be a formal epimorphism, it is necessary and sufficient that, for all λ, u_λ be surjective.*
- *(ii) For u to be a formal monomorphism, it is necessary and sufficient that, for all λ, there exist μ such that $V_\mu \subset V_\lambda$ and that $\mathrm{Ker}(u_\mu)$ be contained in the kernel V_λ/V_μ of the canonical map $M/V_\mu \to M/V_\lambda$.*
- *(iii) For u to be a formal bimorphism, it is necessary and sufficient that, for all λ, u_λ be surjective and that there exist μ such that $V_\mu \subset V_\lambda$ and that the canonical homomorphism $M/V_\mu \to M/V_\lambda$ be the unique factorization $M/V_\mu \xrightarrow{u_\mu} N/W_\mu \xrightarrow{h} M/V_\lambda$ (where $N/W_\mu \to M/V_\lambda$ is necessarily unique).*

PROOF. Assertion (i) is immediate; on the other hand, $\mathrm{Ker}(u_\mu) = u^{-1}(W_\mu)/V_\mu$, and the kernel of the canonical map $M/V_\mu \to M/V_\lambda$ is V_λ/V_μ; the condition of (ii) thus amounts to saying that $u^{-1}(W_\mu) \subset V_\lambda$, and assertion (ii) follows immediately; finally, when u_μ is surjective, to say that $\mathrm{Ker}(u_\mu)$ is contained in V_λ/V_μ, or to say that $M/V_\mu \to M/V_\lambda$ factorizes as $M/V_\mu \xrightarrow{u_\mu} N/W_\mu \to M/V_\lambda$, where v is a homomorphism $N/W_\mu \to M/V_\lambda$, is the same thing, the latter being necessarily an isomorphism. □

Corollary (19.1.4). — *Let A be a topological ring, M, N two topological A-modules whose topologies* $\mathbf{0_{IV\text{-}1}}$ | 72
are derived from that of A, $u : M \to N$ a formal epimorphism. Let $(\mathfrak{J}_\lambda)$ be a fundamental system of open ideals in A. For u to be a formal bimorphism, it is necessary and sufficient that for all λ, the homomorphism $u_\lambda : M/\mathfrak{J}_\lambda M \to N/\mathfrak{J}_\lambda N$ induced by u by passing to quotients be bijective.

PROOF. Indeed, $u(\mathfrak{J}_\lambda M) \subset \mathfrak{J}_\lambda N$ and one can apply the criterion (19.1.3), (iii); but if one has a factorization $M/\mathfrak{J}_\lambda M \xrightarrow{u_\lambda} N/\mathfrak{J}_\lambda N \xrightarrow{h} M/\mathfrak{J}_\lambda M$, the image of $\mathfrak{J}_\lambda M$ by u_μ is $\mathfrak{J}_\lambda N/\mathfrak{J}_\mu N$, hence the image of $\mathfrak{J}_\lambda N/\mathfrak{J}_\mu N$ by v is 0 and v factorizes as

$$N/\mathfrak{J}_\mu N \longrightarrow N/\mathfrak{J}_\lambda N \xrightarrow{v'} M/\mathfrak{J}_\lambda M;$$

but then the automorphism identical to that of $M/\mathfrak{J}_\lambda M$ factorizes as

$$M/\mathfrak{J}_\lambda M \xrightarrow{u_\lambda} N/\mathfrak{J}_\lambda N \xrightarrow{v'} M/\mathfrak{J}_\lambda M,$$

which shows that u_λ is injective, and as we already know it is surjective, this proves the corollary. □

Proposition (19.1.5). — *Let A be a topological ring, M, N two topological A-modules, N being projective, $u : M \to N$ an A-homomorphism. Let $\mathfrak{J}$ be an ideal of A; suppose one of the following conditions is verified:*

(i) $\mathfrak{J}$ is nilpotent.
(ii) $\mathfrak{J}$ is contained in the radical of A and M is of finite type.

Then, for u to be left invertible, it is necessary and sufficient that the homomorphism $u_0 : M/\mathfrak{J}M \to N/\mathfrak{J}N$ of $(A/\mathfrak{J})$-modules, induced by u by passing to quotients, be left invertible.

PROOF. The condition is obviously necessary. Let us prove it is sufficient. Let v_0 be a left inverse of u_0; the composed homomorphism $N \xrightarrow{u} M \to M/\mathfrak{J}M$ factorizes as $N \xrightarrow{\widetilde{u}} M/\mathfrak{J}M$ since N is projective; then $w = v_0 u$ is an endomorphism of M such that, for all n, the endomorphism $(wu)_n$ of $M_n = M/\mathfrak{J}^n M$ induced by wu by passing to quotients is the identity; this implies that $wu = 1_M$ since M is separated and complete. □

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Corollary (19.1.6). — *If there exists a continuous $\widehat{A}$-homomorphism $s : \widehat{N} \to \widehat{M}$ such that $s \circ \widehat{u} = 1_{\widehat{M}}$, then u is formally left invertible.*

PROOF. One will note that $\widehat{u}$ is then an isomorphism of $\widehat{M}$ onto a *direct summand* of $\widehat{N}$. To prove (19.1.6), it suffices to note that with the notations of (19.1.5), *a*), $v : M \to E$ extends to a $\widehat{A}$-homomorphism $\widehat{v} : \widehat{M} \to E$ since E is discrete; let $j : M \to \widehat{M}$ and $j' : N \to \widehat{N}$ be the canonical homomorphisms; then, setting $w = \widehat{v} \circ s \circ j'$, one has $w \circ u = \widehat{v} \circ s \circ j' \circ u = \widehat{v} \circ s \circ \widehat{u} \circ j = \widehat{v} \circ j = v$. □

Corollary (19.1.7). — *Suppose that the topologies of M and N are derived from that of A, and let $(\mathfrak{J}_\lambda)$ be a fundamental system of open ideals in A. For u to be formally left invertible, it is necessary and sufficient that, for all λ, the homomorphism $u_\lambda : M/\mathfrak{J}_\lambda M \to N/\mathfrak{J}_\lambda N$, induced by u by passing to quotients, be left invertible (in other words, be an isomorphism of $M/\mathfrak{J}_\lambda M$ onto a direct summand of $N/\mathfrak{J}_\lambda N$).*

Proposition (19.1.8). — *Let A be a topological ring admitting a fundamental system of open ideals which is a countable decreasing sequence $(\mathfrak{J}_n)$. Let M, N be two topological A-modules whose topologies are derived from those of A. Set $M_n = M/\mathfrak{J}_n M$, $N_n = N/\mathfrak{J}_n N$, and let $u_n : M_n \to N_n$ be the $(A/\mathfrak{J}_n)$-homomorphism induced by u by passing to quotients. Suppose that, for all n, $P_n = \mathrm{Coker}(u_n)$ is a $(A/\mathfrak{J}_n)$-module projective and that M is separated and complete. Then, for u to be formally left invertible (and u is then an isomorphism of M onto a direct summand of N).*

Proposition (19.1.9). — *Let A be a topological ring that is preadmissible ($\mathbf{0_I}$, 7.1.2), $\mathfrak{Q}$ an ideal of definition of A, $(\mathfrak{J}_\lambda)$ a fundamental system of open ideals in A. Let M, N be two topological A-modules whose topologies are derived from those of A, and such that for all λ, $N/\mathfrak{J}_\lambda N$ be an $(A/\mathfrak{J}_\lambda)$-module projective (cf. (19.2.3)). Let $u : M \to N$ be an A-homomorphism. Then the following conditions are equivalent:*

a) u is formally left invertible.
b) The homomorphism $u_0 : M/\mathfrak{Q}M \to N/\mathfrak{Q}N$ of $(A/\mathfrak{J})$-modules, induced by u by passing to quotients, is left invertible.

Proposition (19.1.10). — *Let A be a ring, M, N two A-modules, N being projective, $u : M \to N$ an A-homomorphism. Let $\mathfrak{J}$ be an ideal of A; suppose one of the following conditions is verified:* $\mathbf{0_{IV\text{-}1}}$ | 75

(i) $\mathfrak{J}$ is nilpotent.
(ii) $\mathfrak{J}$ is contained in the radical of A and M is of finite type.

Then, for u to be left invertible, it is necessary and sufficient that the homomorphism $u_0 : M/\mathfrak{J}M \to N/\mathfrak{J}N$ of $(A/\mathfrak{J})$-modules, induced by u by passing to quotients, be left invertible.

Corollary (19.1.11). — *Let A be a local ring, k its residue field, M a finitely generated A-module, N a projective A-module, $u : M \to N$ a homomorphism. For u to be left invertible, it is necessary and sufficient that there exist a system of generators $(x_i)_{1 \leqslant i \leqslant m}$ of M such that the images $x_i \otimes 1 : M \otimes_A k \to N \otimes_A k$ of the $x_i \otimes 1$ be linearly independent in $N \otimes_A k$; the x_i then form a base of M.* $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 76

Corollary (19.1.12). — *Let A be a ring, M a finitely generated A-module, N a projective A-module, $u : M \to N$ a homomorphism. For every prime ideal $\mathfrak{p} \in \operatorname{Spec}(A)$, the following conditions are equivalent:*

a) The homomorphism $u_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is left invertible.
b) The homomorphism $u \otimes 1 : M \otimes_A k(\mathfrak{p}) \to N \otimes_A k(\mathfrak{p})$ is injective (where $k(\mathfrak{p})$ denotes the residue field of A at $\mathfrak{p}$).
c) There exists a finite system of elements $x_i \in M$ ($1 \leqslant i \leqslant m$) such that the images of the x_i in $M_{\mathfrak{p}}$ generate $M_{\mathfrak{p}}$, and a system of m linear forms y_i^* on N ($1 \leqslant i \leqslant m$) such that $\det(\langle y_i^*, u(x_j)\rangle) \notin \mathfrak{p}$.
d) There exists $f \in A - \mathfrak{p}$ such that the homomorphism $u_f : M_f \to N_f$ is left invertible.

Moreover, the set of primes $\mathfrak{p} \in \operatorname{Spec}(A)$ satisfying these conditions is open in $\operatorname{Spec}(A)$.

Proposition (19.1.13). — *Suppose the hypotheses of (19.1.9) verified, and suppose in addition that $(\mathfrak{J}_\lambda)$ be a decreasing sequence $(\mathfrak{J}_n)$ and that M be separated and complete. Then the conditions a) and b) of (19.1.9) are also equivalent to:*

c) u is left invertible.

Proposition (19.1.14). — *Let A be a ring, M a finitely generated A-module, N a projective A-module, $u : M \to N$ an A-homomorphism.*

(i) For u to be left invertible, it is necessary and sufficient that, for every maximal ideal $\mathfrak{m}$ of A, $u_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ be left invertible.
(ii) Let A' be an A-algebra which is a faithfully flat A-module. For u to be left invertible, it is necessary and sufficient that $u \otimes 1 : M \otimes_A A' \to N \otimes_A A'$ be left invertible.

Remark (19.1.15). — The "dual" notion to that of formally left invertible homomorphism is that of *formally right invertible* homomorphism; a continuous A-homomorphism $u : M \to N$ satisfies, by definition, the following condition: for every open submodule W' of N, there exist an open submodule $V'' \subset V' \cap u^{-1}(W')$ of M, an open submodule $W'' \subset W'$ of N, and a homomorphism $h : N/W'' \to M/V'$ such that the canonical homomorphism $N/W'' \to N/W'$ factorizes as $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 78

$$N/W'' \xrightarrow{h} M/V' \xrightarrow{u'} N/W'$$

where u' is induced by u by passing to quotients.

19.3. Formally projective modules

Definition (19.2.1). — Let A be a topological ring, M a topological A-module. We say that M is *formally projective* if it satisfies the following condition: for every open ideal $\mathfrak{J}$ of A, every discrete $(A/\mathfrak{J})$-module P, every surjective A-homomorphism $v : M \to P \to Q$, and every continuous A-homomorphism $w : M \to P$ such that $v = u \circ w$.

(19.2.2). To verify the condition of (19.2.1), one can obviously restrict to the case where v is itself surjective (replacing Q by $v(M)$ and P by $u^{-1}(v(M))$) at the case where v is itself surjective; Q is then isomorphic to M/V, where V is an open submodule of M such that $\mathfrak{J}M \subset V$; the condition of (19.2.1) then amounts to saying that for every $(A/\mathfrak{J})$-module discrete P and every A-homomorphism $u : P \to M/V$, there exists an open submodule $V' \subset V$ and an A-homomorphism $w : M/V' \to P$, such that the canonical homomorphism $M/V' \to M/V$ factorizes as $M/V' \xrightarrow{w} P \xrightarrow{u} M/V$. One will note that it suffices to verify this condition when $\mathfrak{J}$ ranges over a fundamental system of neighbourhoods of 0 formed of ideals in A.

(19.2.3). Suppose that there exists a fundamental system $(\mathfrak{J}_\lambda)$ of open ideals in A and a fundamental system (V_λ) of open submodules in M, having the same index set, such that $\mathfrak{J}_\lambda \subset V_\lambda$ for all λ, and such that M/V_λ be an $(A/\mathfrak{J}_\lambda)$-module *projective*. Then M is formally projective; it suffices indeed, with the notations of (19.2.2), to take λ such that $\mathfrak{J}_\lambda \subset \mathfrak{J}$ and $V_\lambda \subset V$; as P is also an $(A/\mathfrak{J}_\lambda)$-module, the factorization of the canonical homomorphism $M/V_\lambda \to M/V$ into $M/V_\lambda \to P \to M/V$ results from what M/V_λ is supposed to be an $(A/\mathfrak{J}_\lambda)$-module projective.

When the stricter condition above is satisfied, we say that M is *strictly formally projective*.

Proposition (19.2.4). — *Let A be a topological ring, M a topological A-module whose topology is derived from that of A. For M to be formally projective, it is necessary and sufficient that it be strictly formally projective.*

Proposition (19.2.5). — *Let A be a topological ring, M a topological A-module.* $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 79

(i) *It suffices to remark that when $\mathfrak{J}$ (resp. V) ranges over the set of open ideals (resp. submodules) of A (resp. M), the separated completed $\widehat{\mathfrak{J}}$ (resp. $\widehat{V}$) ranges over the set of open ideals of $\widehat{A}$ (resp. open submodules of $\widehat{M}$), and $\widehat{A}/\widehat{\mathfrak{J}} = A/\mathfrak{J}$ (resp. $\widehat{M}/\widehat{V} = M/V$) up to canonical isomorphism; as the notion of formally projective module (resp. strictly formally projective) only involves the $A/\mathfrak{J}$ and the M/V, one deduces (i) immediately.*

(ii) *Suppose first that M is formally projective and set $M' = M \otimes_A A'$; let $\mathfrak{J}'$ be an open ideal of A', P', Q' two discrete $(A'/\mathfrak{J}')$-modules, $u' : P' \to Q'$ a surjective A'-homomorphism, $v' : M' \to Q'$ a continuous A'-homomorphism. There is an open ideal $\mathfrak{J}$ of A such that $\mathfrak{J}A' \subset \mathfrak{J}'$, hence P' and Q' are also discrete $(A/\mathfrak{J})$-modules; as M is formally projective, there exists an A-homomorphism $w : M \to P'$ such that $u' \circ w = v' \circ j$, where $j : M \to M'$ is the canonical homomorphism. The existence of a continuous A'-homomorphism $w' : M' \to P'$ such that $v' = u' \circ w'$ follows.*

19.4. Formally smooth algebras

Definition (19.3.1). — Let A be a topological ring, B a topological A-algebra. We say that B is a *formally smooth* A-algebra if, for every discrete topological A-algebra C, and every nilpotent ideal $\mathfrak{J}$ of C, every continuous A-homomorphism $u : B \to C/\mathfrak{J}$ *factorizes* as $B \xrightarrow{v} C \xrightarrow{\varphi} C/\mathfrak{J}$, where v is a continuous A-homomorphism and φ the canonical homomorphism.

The definition (19.3.1) amounts to saying that the following property holds: $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 80

(P) For every open ideal $\mathfrak{R}$ of the A-algebra B and every continuous A-homomorphism $u' : B/\mathfrak{R} \to C/\mathfrak{J}$ such that the homomorphism $B \to B/\mathfrak{R} \xrightarrow{u'} C/\mathfrak{J}$ is a continuous A-homomorphism, where v' is also an A-homomorphism.

Indeed, if $u : B \to C/\mathfrak{J}$ is a continuous A-homomorphism, its kernel $\mathfrak{R}$ is an open ideal of B, and if (P) is satisfied, it suffices to take u' and to take for v the composition $B \to B/\mathfrak{R} \xrightarrow{v'} C$ to satisfy the conditions of (19.3.1). Conversely, suppose B is a formally smooth A-algebra; let us give ourselves an open ideal $\mathfrak{R}$ of B and a continuous A-homomorphism $u' : B/\mathfrak{R} \to C/\mathfrak{J}$; if $v : B \to C$ is a continuous A-homomorphism such that u factorizes as $B \xrightarrow{v} C \to C/\mathfrak{J}$, the ideal $\mathfrak{R}' = \operatorname{Ker}(v) \cap \mathfrak{R}$ of B is open and contained in $\mathfrak{R}$; by taking v' to factorize as $B \to B/\mathfrak{R}' \xrightarrow{v'} C$, and v' clearly satisfies the condition (P).

Proposition (19.3.2). — *Let A be a discrete ring, V a projective A-module; the symmetric algebra $B = \mathrm{S}_A^\bullet(V)$, equipped with the discrete topology, is a formally smooth A-algebra.*

PROOF. Indeed, the notations C and $\mathfrak{J}$ having the same meaning as in (19.3.1), let $u : B \to C/\mathfrak{J}$; as V is projective, u_1, the restriction of u to V, lifts to a homomorphism of A-modules $v_1 : V \to C$ such that u_1 factorizes as $V \xrightarrow{v_1} C \xrightarrow{\varphi} C/\mathfrak{J}$; as V is projective, u_1 extends to a homomorphism of A-algebras $v : \mathrm{S}_A^\bullet(V) \to C$, and v coincides with u in V, hence $\varphi \circ v = u$, which proves the proposition. □

Corollary (19.3.3). — *If A is a discrete ring, every polynomial algebra $B = A[T_\alpha]_{\alpha \in I}$, equipped with the discrete topology, is a formally smooth A-algebra.*

Proposition (19.3.4). — *Let A be a topological ring, and let $B = A[[T_\alpha]]_{\alpha \in I}$ be a ring of formal power series $\sum c_\lambda T^\lambda$ (algebra large over A of the monoid $\mathbf{N}^{(I)}$, identified as a topological A-module with the product $A^{\mathbf{N}^{(I)}}$). If one equips B with the product topology, B is a formally smooth topological A-algebra.*

Proposition (19.3.5). — *Let A be a topological ring.*

(i) *A is a formally smooth A-algebra.*

(ii) *If B is a formally smooth A-algebra and C a formally smooth B-algebra, then C is a formally smooth A-algebra.*

(iii) *Let B be a formally smooth A-algebra, A' a topological A-algebra; then the topological A'-algebra $B \otimes_A A'$ ($\mathbf{0_I}$, 7.7.5 and 7.7.6) is formally smooth.*

(iv) *Let B be a topological A-algebra, S (resp. T) a multiplicative subset of A (resp. B) such that the canonical image of S in B is contained in T. If B is a formally smooth A-algebra, then $T^{-1}B$ is an $S^{-1}A$-algebra formally smooth.*

(v) *Let B_i ($1 \leqslant i \leqslant n$) be topological A-algebras. For $\prod_{i=1}^n B_i$ to be a formally smooth A-algebra, it is necessary and sufficient that each of the B_i be so.*

Proposition (19.3.6). — *Let A be a topological ring, B a topological A-algebra, $\widehat{A}$ and $\widehat{B}$ the separated completions of A and B, so that $\widehat{B}$ is a topological $\widehat{A}$-algebra. The following conditions are equivalent:* $\mathbf{0_{IV\text{-}1}}$ | 83

a) *B is a formally smooth A-algebra.*
b) *$\widehat{B}$ is a formally smooth A-algebra.*
c) *$\widehat{B}$ is a formally smooth $\widehat{A}$-algebra.*

PROOF. Of course, the structure of $\widehat{A}$-algebra on $\widehat{B}$ is defined by the homomorphism $\widehat{\varphi}$, where $\varphi : A \to B$ is the homomorphism defining the structure of A-algebra on B. As every discrete A-algebra C is separated and complete, it is also a $\widehat{A}$-algebra (by canonical prolongation of the homomorphism of A in C), and a continuous A-homomorphism of B in C gives by prolongation a canonical $\widehat{A}$-homomorphism (and *a fortiori* an A-homomorphism) of $\widehat{B}$ in C (in other terms, every continuous A-homomorphism $B \to C$ factorizes as $B \to \widehat{B} \to C$ uniquely). These remarks and the definition (19.3.1) immediately imply the proposition. □

Corollary (19.3.7). — *Under the hypotheses of (19.3.5), (iv)), the $A\{S^{-1}\}$-algebra topological $B\{T^{-1}\}$ is formally smooth.*

Proposition (19.3.8). — *Let A be a topological ring, B a topological A-algebra, and suppose that for every open ideal $\mathfrak{N}$ of B, $\mathfrak{N}^2$ is also open. Let A', B' be the rings A and B equipped with topologies finer than the initial topologies, and suppose that the canonical homomorphism $\varphi : A \to B$ is still a continuous homomorphism of A' in B'. Then, if B' is a formally smooth A'-algebra, B is a formally smooth A-algebra.*

PROOF. It suffices to apply the following lemma: □

Lemma (19.3.8.1). — *Let C be a discrete A-algebra, $\mathfrak{J}$ a nilpotent ideal of C. Suppose that the square of every open ideal of B is also open. Then, if a composed homomorphism $v : B \xrightarrow{u} C \to C/\mathfrak{J}$ is continuous, the homomorphism u is continuous.*

PROOF. Indeed, $\mathfrak{N} = u^{-1}(\mathfrak{J})$ is the kernel of v, and it is open by hypothesis; as $\mathfrak{J}^n = 0$, $\mathfrak{N}^n$ is contained in $\mathrm{Ker}(u)$; but by induction on h, it follows from the hypothesis that $\mathfrak{N}^{2^h}$ is also open, hence $\mathfrak{N}^n$ for all n, and consequently $\mathrm{Ker}(u)$ is also open, which proves our assertion. □

(19.3.9). We will now see that the property of being formally smooth implies properties of "lifting" of homomorphisms under more general conditions than those of the definition (19.3.1).

Proposition (19.3.10). — *Let A be a topological ring, B a formally smooth topological A-algebra. Let C be a topological A-algebra, $\mathfrak{J}$ an ideal of C, satisfying the following conditions:* $\mathbf{0_{IV\text{-}1}}$ | 84

1° *C is metrizable and complete.*
2° *$\mathfrak{J}$ is closed and the sequence $(\mathfrak{J}^n)$ tends to 0.*

Then, every continuous A-homomorphism $u : B \to C/\mathfrak{J}$ factorizes as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism.

PROOF. Let $(\mathfrak{Q}_n)$ be a decreasing sequence of ideals of C forming a fundamental system of neighbourhoods of 0 in this algebra; as there exists k such that $\mathfrak{J}^k \subset \mathfrak{Q}_n$, $(\mathfrak{J} + \mathfrak{Q}_n)/\mathfrak{Q}_n$ is nilpotent. For all n, let u_n be the continuous A-homomorphism $B \xrightarrow{u} C/\mathfrak{J} \to C/(\mathfrak{J} + \mathfrak{Q}_n) = (C/\mathfrak{Q}_n)/((\mathfrak{J} + \mathfrak{Q}_n)/\mathfrak{Q}_n)$; by hypothesis u_n factorizes as $B \xrightarrow{v_n} C/\mathfrak{Q}_n \xrightarrow{\varphi_n} C/(\mathfrak{J} + \mathfrak{Q}_n)$, where v_n is a continuous A-homomorphism. Let us show that one can choose step by step the v_n so that v_n factorizes as

$$B \xrightarrow{v_{n+1}} C/\mathfrak{Q}_{n+1} \longrightarrow C/\mathfrak{Q}_n$$

for all n (in other words, the v_n form a *projective system* of homomorphisms). This will follow from the following lemma: □

Lemma (19.3.10.1). — *Let B be a formally smooth A-algebra, E, E' two discrete topological A-algebras, $\mathfrak{N}$ (resp. $\mathfrak{N}'$) a nilpotent ideal of E (resp. E'), $f : E \to E'$ a surjective A-homomorphism such that $f(\mathfrak{N}) \subset \mathfrak{N}'$, $f' : E/\mathfrak{N} \to E'/\mathfrak{N}'$ the homomorphism induced by f by passing to quotients. Let $u : B \to E/\mathfrak{N}$ be a continuous A-homomorphism, u' the composed homomorphism $B \xrightarrow{u} E/\mathfrak{N} \xrightarrow{f'} E'/\mathfrak{N}'$, and let $v' : B \to E'$ be a continuous A-homomorphism such that u' factorizes as $B \xrightarrow{v'} E' \to E'/\mathfrak{N}'$. Then there exists a continuous A-homomorphism $v : B \to E$ such that v' factorizes as $B \xrightarrow{v} E \xrightarrow{f} E'$.*

Corollary (19.3.11). — *Let A be a topological ring, B a formally smooth topological A-algebra, C a noetherian complete local ring, $\mathfrak{J}$ an ideal distinct from C, $\varphi : A \to C$ a continuous homomorphism making C a topological A-algebra. Then every continuous A-homomorphism $u_0 : B \to C_0 = C/\mathfrak{J}$ factorizes as $B \xrightarrow{u} C \to C/\mathfrak{J}$, where u is a continuous A-homomorphism.* $\mathbf{0_{IV\text{-}1}}$ | 85

PROOF. All the conditions of (19.3.10) are indeed satisfied ($\mathbf{0_I}$, 7.3.5). ☐

Proposition (19.3.12). — *Let A be a topological ring, B a topological A-algebra, C a topological A-algebra, $\mathfrak{J}$ an ideal of C_λ, satisfying the following conditions:*

1° *There exists a fundamental system of open ideals $\mathfrak{Q}_\lambda$ of C, such that the $C_\lambda = C/\mathfrak{Q}_\lambda$ are artinian rings and that $C \xrightarrow{\sim} \varprojlim C_\lambda$.*
2° *$\mathfrak{J}$ is closed in C and topologically nilpotent.*
3° *The square of every open ideal of B is open.*

Under these conditions, every continuous A-homomorphism $u : B \to C/\mathfrak{J}$ factorizes as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism.

19.5. First criteria for formal smoothness

Proposition (19.4.1). — *Let A be a topological ring, B a topological A-algebra; suppose that there exist two decreasing filtering families $(\mathfrak{J}_\alpha)_{\alpha\in I}$, $(\mathfrak{N}_\alpha)_{\alpha\in I}$ of ideals of A and B respectively, such that:* $\mathbf{0_{IV\text{-}1}}$ | 86

1° *$(\mathfrak{J}_\alpha)$ tends to 0 in A and $(\mathfrak{N}_\alpha)$ tends to 0 in B;*
2° *for all $\alpha \in I$, we have $\mathfrak{J}_\alpha B \subset \mathfrak{N}_\alpha$ (so that $B/\mathfrak{N}_\alpha$ is an $(A/\mathfrak{J}_\alpha)$-topological algebra);*
3° *for all $\alpha \in I$, $B/\mathfrak{N}_\alpha$ is a formally smooth $(A/\mathfrak{J}_\alpha)$-algebra.*

Then B is a formally smooth A-algebra.

PROOF. Let C be a discrete A-algebra, $\mathfrak{L}$ a nilpotent ideal of C, and $\mathfrak{N}$ an open ideal of B; by hypothesis there exists $\alpha \in I$ such that $\mathfrak{N}_\alpha \subset \mathfrak{N}$, so $B/\mathfrak{N}$ is a quotient of $B/\mathfrak{N}_\alpha$. Every A-homomorphism $u' : B/\mathfrak{N} \to C/\mathfrak{L}$ is also an $(A/\mathfrak{J}_\alpha)$-homomorphism, and so it also lifts from $B/\mathfrak{N}_\alpha$; whence the conclusion, by virtue of the remark following (19.3.1). ☐

Corollary (19.4.2). — *Let A be a topological ring, B a topological A-algebra, $(\mathfrak{J}_\alpha)_{\alpha\in I}$ a decreasing filtering family of ideals of A tending to 0. For B to be a formally smooth A-algebra, it is necessary and sufficient that for all $\alpha \in I$, setting $A_\alpha = A/\mathfrak{J}_\alpha$, $B_\alpha = B/\mathfrak{J}_\alpha B$, B_α be a formally smooth A_α-algebra.*

PROOF. The condition is sufficient by (19.4.1), and it is also necessary by (19.3.5), (iii). ☐

Proposition (19.4.3). — *Let A be a topological ring, B a topological A-algebra. Suppose that for every discrete A-algebra C and every ideal $\mathfrak{J}$ of C such that $\mathfrak{J}^2 = 0$, every continuous A-homomorphism $u : B \to C/\mathfrak{J}$ factorizes as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism. Then B is a formally smooth A-algebra.*

PROOF. With the same notation, let $\mathfrak{N}$ be any nilpotent ideal of C, and consider a continuous A-homomorphism $u' : B \to C/\mathfrak{N}$. Suppose $\mathfrak{N}^m = 0$, and set $C_i = C/\mathfrak{N}^i$ for $1 \leqslant i \leqslant m$, so that $C_m = C$, that $\mathfrak{N}^{i-1}/\mathfrak{N}^i$ is an ideal of square zero in C_i, and $C_i = C_{i+1}/\mathfrak{J}_{i+1}$; the hypothesis then gives by induction on i the existence of continuous A-homomorphisms $u_i : B \to C_i$ such that $u_1 = u$ and that u_i factorizes as $B \xrightarrow{u_{i+1}} C_{i+1} \to C_{i+1}/\mathfrak{J}_{i+1} = C_i$; whence the conclusion. ☐

Proposition (19.4.4). — *Let A be a topological ring, B a topological A-algebra (commutative). For B to be a formally smooth A-algebra, it is necessary and sufficient that for every discrete topological B-module L, annihilated by an open ideal of B, we have (cf. (18.5.1))* $\mathbf{0_{IV\text{-}1}}$ | 87

$$\mathrm{Exalcotop}_A(B, L) = 0.$$

PROOF. Let $(\mathfrak{R}_\lambda)$ be a fundamental decreasing system of open ideals of B and set $B_\lambda = B/\mathfrak{R}_\lambda$. Consider a discrete topological A-algebra C and an ideal $\mathfrak{J}$ of C such that $\mathfrak{J}^2 = 0$, so that C is an A-extension of $C/\mathfrak{J}$ by $\mathfrak{J}$; suppose given an A-homomorphism $u : B_\lambda \to C/\mathfrak{J}$ and form the A-extension E_λ of B_λ by $\mathfrak{J}$, inverse image of C by the homomorphism u (18.2.5); it is a topological A-algebra for the discrete topology. If $\mathrm{Exalcotop}_A(B, L) = 0$, there exists by definition (18.5.1) a μ such that $\mathfrak{R}_\mu \subset \mathfrak{R}_\lambda$ and the inverse image extension E_μ is A-trivial; but this means (18.1.6) that there exists a continuous A-homomorphism $v : B_\mu \to E_\lambda$ such that the canonical homomorphism $B_\mu \to B_\lambda$ factorizes as $B_\mu \xrightarrow{v} E_\lambda \to B_\lambda$; we conclude immediately that $B_\mu \to B_\lambda \xrightarrow{u} C/\mathfrak{J}$ factorizes as $B_\mu \xrightarrow{v} E_\lambda \to C \to C/\mathfrak{J}$, and this proves, by virtue of (19.3.1) and (19.4.3), that B is a formally smooth A-algebra. The converse is immediate, applying (19.3.1) to the case where C is a topological A-algebra which is an A-extension of B_λ by L, and to the identical homomorphism $B_\lambda \to B_\lambda = C/L$. □

When A and B are discrete rings, the criterion of formal smoothness (19.4.4) reduces to

(19.4.4.1) $$\mathrm{Exalcom}_A(B, L) = 0 \qquad \text{for all } B\text{-modules } L;$$

in other words, *every commutative A-extension of B by a B-module must be A-trivial*.

Corollary (19.4.5). — *Let A be a topological ring (commutative), B a topological A-algebra (commutative).*

(i) *Suppose that B is a formally smooth A-algebra. Then, for every open ideal $\mathfrak{R}$ of B, every $(B/\mathfrak{R})$-module L, and every Hochschild A-bilinear symmetric 2-cocycle f of $(B/\mathfrak{R}) \times (B/\mathfrak{R})$ in L (18.4.3), there exists an open ideal $\mathfrak{R}' \subset \mathfrak{R}$ such that, if $\varphi : B/\mathfrak{R}' \to B/\mathfrak{R}$ is the canonical homomorphism, $f \circ (\varphi \times \varphi)$ is a Hochschild A-bilinear 2-coboundary of $(B/\mathfrak{R}') \times (B/\mathfrak{R}')$ in L.*

(ii) *If B is a formally projective A-module (19.2.1) and if the condition of* (i) *is satisfied, B is a formally smooth A-algebra.*

PROOF. (i) The 2-cocycle f defines an A-extension of Hochschild C of $B/\mathfrak{R}$ by L (18.4.3). Applying (19.3.1) to C, to the ideal of square zero L of C, and to the identical homomorphism $B/\mathfrak{R} \to B/\mathfrak{R} = C/L$, we deduce the condition (i) by virtue of (18.4.3).

(ii) Let us apply the criterion (19.4.3), considering an open ideal $\mathfrak{J}$ of A, an open ideal $\mathfrak{R}$ of B, and an $(A/\mathfrak{J})$-extension E of $B/\mathfrak{R}$ by L. As B is a formally projective A-module, the canonical continuous A-linear application $\rho : B \to B/\mathfrak{R}$ factorizes as $B \xrightarrow{w} E \to B/\mathfrak{R}$, where w is a continuous A-linear application (19.2.1), which itself factorizes as $B \to B/\mathfrak{R}' \xrightarrow{w'} E$ where $\mathfrak{R}'$ is an open ideal of B contained in $\mathfrak{R}$; replacing $\mathfrak{J}$ by a smaller ideal, one can suppose that $\mathfrak{J}B \subset \mathfrak{R}'$. Then the inverse image of the extension E of $B/\mathfrak{R}$ by L, via the canonical homomorphism $B/\mathfrak{R}' \to B/\mathfrak{R}$, is an $(A/\mathfrak{J})$-extension *of Hochschild* E' of $B/\mathfrak{R}'$ by L; applying (i) to a cocycle defining this extension (18.4.3), we conclude that there $\mathbf{0_{IV\text{-}1}}$ | 88
exists an open ideal $\mathfrak{R}'' \subset \mathfrak{R}'$ of B such that the inverse image E'' of E by $B/\mathfrak{R}'' \to B/\mathfrak{R}$ is A-trivial, Q.E.D.

□

Corollary (19.4.6). — *Let A be a topological ring, B a topological A-algebra which is a formally projective A-module. Let A' be an A-algebra equipped with the topology derived from that of A. Suppose furthermore that A' is a faithfully flat A-module, and that one of the following conditions is satisfied:*

1° *There exists a fundamental system $(\mathfrak{J}_\lambda)$ of open ideals of A and a fundamental system $(\mathfrak{M}_\lambda)$ of open ideals of B, with the same index set, such that for all λ we have $\mathfrak{J}_\lambda B \subset \mathfrak{M}_\lambda$ and $B/\mathfrak{M}_\lambda$ is a projective $(A/\mathfrak{J}_\lambda)$-module of finite type.*

2° *A' is a projective A-module of finite type.*

Then, for $B' = B \otimes_A A'$ (equipped with the tensor product topology) to be a formally smooth A'-algebra, it is necessary and sufficient that B be a formally smooth A-algebra.

PROOF. The sufficiency of the condition is contained in (19.3.5), (iii), without any additional hypothesis on A'. To prove the converse, we apply the criterion (19.4.5); under hypothesis 2°, we will again denote by $(\mathfrak{M}_\lambda)$ a fundamental system of open ideals of B, and for each λ, by $\mathfrak{J}_\lambda$ an open ideal of A such that $\mathfrak{J}_\lambda B \subset \mathfrak{M}_\lambda$; in both cases, we set $A_\lambda = A/\mathfrak{J}_\lambda$, $B_\lambda = B/\mathfrak{M}_\lambda$, $A'_\lambda = A_\lambda \otimes_A A'$, $B'_\lambda = B_\lambda \otimes_A A'$. Let f be a Hochschild A_λ-bilinear symmetric 2-cocycle of $B_\lambda \times B_\lambda$ in a B_λ-module

L; by extension of scalars, we deduce a Hochschild A'_λ-bilinear symmetric 2-cocycle of $B'_\lambda \times B'_\lambda$ in $L' = L \otimes_A A'$. As B' is by hypothesis a formally smooth A'-algebra, there exists by (19.4.5), (i), a μ such that $\mathfrak{M}_\mu \subset \mathfrak{M}_\lambda$ and such that, if $\varphi' : B'_\mu \to B'_\lambda$ is the canonical homomorphism, the image c' of $f' \circ (\varphi' \times \varphi')$ in the Hochschild group $\mathrm{H}^2_{A'_\mu}(B'_\mu, L')^s$ is zero; it is clear that if $\varphi : B_\mu \to B_\lambda$ is the canonical homomorphism, and c the class of $f \circ (\varphi \times \varphi)$ in the group $\mathrm{H}^2_{A_\mu}(B_\mu, L)^s$, then c' is the canonical image of c. Now, if $\mathrm{P}_\bullet$ is the complex relative to the rings A_μ and B_μ defined in (18.4.5), the analogous complex relative to the rings A'_μ and B'_μ is evidently $\mathrm{P}_\bullet \otimes_A A'$; under hypothesis 1°, the construction of $\mathrm{P}_\bullet$ shows that it is a projective A_μ-module of finite type. One concludes from Bourbaki, *Alg.*, chap. II, 3e éd., § 5, n° 3, prop. 7 that, under both hypotheses, one has up to canonical isomorphism; as A' is a flat A-module, one has by (18.4.5)

$$\mathrm{H}^2_{A'_\mu}(B'_\mu, L')^s = (\mathrm{H}^2_{A_\mu}(B_\mu, L)^s) \otimes_A A'$$

and one can therefore write $c' = c \otimes 1$; but as A' is a faithfully flat A-module, the hypothesis $c' = 0$ implies $c = 0$, which completes the proof. □

Proposition (19.4.7). — *Let A be a preadmissible topological ring, $\mathfrak{J}$ an ideal of definition of A ($\mathbf{0_I}$, 7.1.2), B a topological A-algebra which is a formally projective A-module. Consider the topological quotient rings $A_0 = A/\mathfrak{J}$, $B_0 = B/\mathfrak{J}B$; then, for B to be a formally smooth A-algebra, it is necessary and sufficient that B_0 be a formally smooth A_0-algebra.*

Proof. The necessity of the condition follows from (19.4.2). To see that it is sufficient, we first note that, by considering a fundamental system of open neighborhoods of 0 in A formed by ideals $\mathfrak{J}_\lambda$ contained in $\mathfrak{J}$, one can by virtue of (19.4.2) reduce to the case where A is discrete, since $B/\mathfrak{J}_\lambda B$ is a formally projective $(A/\mathfrak{J}_\lambda)$-module (19.2.5); by definition of a preadmissible ring, $\mathfrak{J}$ is then nilpotent. It furthermore suffices to prove the proposition when $\mathfrak{J}^2 = 0$, for if $\mathfrak{J}^m = 0$, one applies it successively to the rings $A_k = A/\mathfrak{J}^k$ and $B_k = B/\mathfrak{J}^k B$ and to the ideal $\mathfrak{J}^{k-1}/\mathfrak{J}^k$ of A_k for $k = 2, 3, \ldots, m$, noting (19.2.5) that $B_k = B \otimes_A A_k$ is a formally projective A_k-module. Let us apply the criterion (19.4.5), (ii), considering an open ideal $\mathfrak{R}$ of B and a Hochschild A-bilinear symmetric 2-cocycle f of $(B/\mathfrak{R}) \times (B/\mathfrak{R})$ in a $(B/\mathfrak{R})$-module L. Consider first the special case where $\mathfrak{J}L = 0$, so that L can also be viewed as a $(B_0/\mathfrak{R}B_0)$-module, and f factorizes as $\mathbf{0_{IV\text{-}1}}$ | 89

$$(B/\mathfrak{R}) \times (B/\mathfrak{R}) \longrightarrow (B_0/\mathfrak{R}B_0) \times (B_0/\mathfrak{R}B_0) \xrightarrow{f_0} L$$

where f_0 is a Hochschild bilinear symmetric 2-cocycle. By hypothesis, there exists an open ideal $\mathfrak{R}' \subset \mathfrak{R}$ in B and an A_0-linear application $g_0 : B_0/\mathfrak{R}'B_0 \to L$ such that $f_0(\varphi_0(x), \varphi_0(y)) = xg_0(y) - g_0(xy) + g_0(x)y$ for x, y in $B_0/\mathfrak{R}'B_0$, where $\varphi_0 : B_0/\mathfrak{R}'B_0 \to B_0/\mathfrak{R}B_0$ is the canonical homomorphism. One concludes immediately that the composed A-linear application $g : B/\mathfrak{R}' \to B_0/\mathfrak{R}'B_0 \xrightarrow{g_0} L$ is such that $dg = f \circ (\varphi \times \varphi)$, where $\varphi : B/\mathfrak{R}' \to B/\mathfrak{R}$ is the canonical homomorphism.

Let us now pass to the general case, and consider first the $(B/\mathfrak{R})$-module $L/\mathfrak{J}L = L'$, for which $\mathfrak{J}L' = 0$; if f' is the A-bilinear composed application $(B/\mathfrak{R}) \times (B/\mathfrak{R}) \xrightarrow{f} L \to L'$, f' is again a symmetric Hochschild 2-cocycle, and by virtue of what precedes, there exists an open ideal $\mathfrak{R}' \subset \mathfrak{R}$ in B and an A-linear application $g' : B/\mathfrak{R}' \to L'$ satisfying $dg' = f' \circ (\varphi' \times \varphi')$ for the canonical application $\varphi' : B/\mathfrak{R}' \to B/\mathfrak{R}$. As B is a formally projective A-module, there exists an A-linear application $g_1 : B/\mathfrak{R}'_1 \to L$ lifting $B/\mathfrak{R}'_1 \to B/\mathfrak{R}' \xrightarrow{g'} L'$. Consider then the Hochschild 2-cocycle

$$f_1(x, y) = f(\varphi_1(x), \varphi_1(y)) - xg_1(y) + g_1(xy) - g_1(x)y,$$

a symmetric A-bilinear application of $(B/\mathfrak{R}'_1) \times (B/\mathfrak{R}'_1)$ in L. The choice of g_1 implies that f_1 takes its values in $\mathfrak{J}L$. As $\mathfrak{J}(\mathfrak{J}L) = 0$, one can again apply the first case, so there exists an open ideal $\mathfrak{R}'' \subset \mathfrak{R}'_1$ and an A-linear application $g_2 : B/\mathfrak{R}'' \to \mathfrak{J}L$ such that

$$f_1(\varphi_2(x), \varphi_2(y)) = xg_2(y) - g_2(xy) + g_2(x)y$$

in $(B/\mathfrak{R}'') \times (B/\mathfrak{R}'')$, where $\varphi_2 : B/\mathfrak{R}'' \to B/\mathfrak{R}'_1$ is the canonical application; the A-linear application $g = g_2 + g_1 \circ \varphi_2 : B/\mathfrak{R}'' \to L$ therefore satisfies $dg = f \circ (\varphi \times \varphi)$ for the canonical application $\varphi : B/\mathfrak{R}'' \to B/\mathfrak{R}$. □

19.6. Formal smoothness and associated graded rings

(19.5.1). Let C be a topological ring (commutative), V a topological C-module, and consider the symmetric algebra $\mathbf{S}_C^*(V) = \bigoplus_n \mathbf{S}_C^n(V)$, which we will canonically equip with a *linear topology*, compatible with its C-algebra structure. For this, consider an open submodule U of V, and the *graded ideal* $U.\mathbf{S}_C^*(V)$ it generates in $\mathbf{S}_C^*(V)$; we take as fundamental system of neighborhoods of 0 in $\mathbf{S}_C^*(V)$ the set of sums $\mathfrak{a}.\mathbf{S}_C^*(V) + U.\mathbf{S}_C^*(V)$, where $\mathfrak{a}$ (resp. U) runs through a fundamental system of open ideals (resp. of open submodules) of C (resp. V). We note that if the topology of V is less fine than the topology derived from that of C, one can restrict to pairs $(\mathfrak{a}, U)$ such that $\mathfrak{a}V \subset U$, so $\mathfrak{a}.\mathbf{S}_C^*(V) + U.\mathbf{S}_C^*(V) = \mathfrak{a}.1_C + U.\mathbf{S}_C^*(V)$; in this case the topology induced on each $\mathbf{S}_C^n(V)$ for $n \geqslant 1$ has as fundamental system of neighborhoods of 0 the $U.\mathbf{S}_C^{n-1}(V)$, where U runs through the open submodules of V; in particular, on $V = \mathbf{S}_C^1(V)$, it coincides with the given topology (in general it is less fine than the latter). Furthermore, in all cases, the *topological algebra* $\mathbf{S}_C^*(V)$ thus defined is, for the categories of topological C-modules and of topological C-algebras, the solution of the same universal problem as for the categories of C-modules and C-algebras. Indeed, let E be a topological C-algebra, $u : V \to E$ a homomorphism of C-modules, $\mathbf{S}(u) = \widetilde{u}$ its canonical extension to $\mathbf{S}_C^*(V)$. Suppose u continuous; then, if $\mathfrak{b}$ is an open ideal of E, its inverse image $U = u^{-1}(\mathfrak{b})$ is an open C-submodule of V, and the image by $\widetilde{u}$ of $U.\mathbf{S}_C^*(V)$ is therefore contained in $\mathfrak{b}$; as moreover there exists an open ideal $\mathfrak{a}$ of C such that $\mathfrak{a}.E \subset \mathfrak{b}$, hence $\widetilde{u}(\mathfrak{a}.\mathbf{S}_C^*(V)) \subset \mathfrak{b}$, this proves that $\widetilde{u}$ is continuous. Conversely, if $\widetilde{u}$ is continuous, and if $\mathfrak{b}$ is an open ideal of E, there exists an open submodule U of V such that $\widetilde{u}(U.\mathbf{S}_C^*(V)) \subset \mathfrak{b}$, and in particular $\widetilde{u}(U.1_C) \subset \mathfrak{b}$, that is $u(U) \subset \mathfrak{b}$, so u is continuous. We also recall that one has a canonical isomorphism of topological C-modules (discrete) $\mathbf{0_{IV\text{-}1}}$ | 90

(19.5.1.1)
$$\mathbf{S}_C^n(V)/U.\mathbf{S}_C^{n-1}(V) \xrightarrow{\sim} \mathbf{S}_C^n(V/U).$$

(19.5.2). Let A be a topological ring, B a topological A-algebra (commutative), $\mathfrak{J}$ an ideal of B (not necessarily open or closed); in all that follows, we equip $\mathfrak{J}$ and the $\mathfrak{J}^n$ ($n \geqslant 2$) with the topology induced from that of B, the quotients $C = B/\mathfrak{J}$, $\mathfrak{J}^n/\mathfrak{J}^{n+1} = \mathrm{gr}_{\mathfrak{J}}^n(B)$ with the quotient topology, so that the $\mathfrak{J}^n/\mathfrak{J}^{n+1}$ are topological C-modules; the canonical injection $\mathfrak{J}/\mathfrak{J}^2 \to \mathrm{gr}_{\mathfrak{J}}(B)$ extends to a homomorphism (not topological at first) of C-algebras

$$\varphi : \mathbf{S}_C^*(\mathfrak{J}/\mathfrak{J}^2) \longrightarrow \mathrm{gr}_{\mathfrak{J}}(B)$$

which for each n gives a surjective homomorphism of C-modules

(19.5.2.1)
$$\varphi_n : \mathbf{S}_C^n(\mathfrak{J}/\mathfrak{J}^2) \longrightarrow \mathfrak{J}^n/\mathfrak{J}^{n+1} = \mathrm{gr}_{\mathfrak{J}}^n(B).$$

When $\mathbf{S}_C^n(\mathfrak{J}/\mathfrak{J}^2)$ is equipped with the topology defined in (19.5.1), the homomorphisms φ_n are continuous, by virtue of the universal property (19.5.1) of $\mathbf{S}_C^*(\mathfrak{J}/\mathfrak{J}^2)$ applied to the topological algebra $E = \mathrm{gr}_{\mathfrak{J}/\mathfrak{J}^n}(B/\mathfrak{J}^n)$ equipped with the product topology of those of the $\mathfrak{J}^k/\mathfrak{J}^{k+1}$, and to the canonical injection $\mathfrak{J}/\mathfrak{J}^2 \to E$. Note that here the topology of $\mathfrak{J}/\mathfrak{J}^2$ is less fine than that of C (this latter topology on $\mathfrak{J}/\mathfrak{J}^2$ being also the topology derived from that of B). $\mathbf{0_{IV\text{-}1}}$ | 91

Theorem (19.5.3). — *Let A be a topological ring, B a topological A-algebra, $\mathfrak{J}$ an ideal of B, $C = B/\mathfrak{J}$ the topological quotient A-algebra. Suppose that the discrete A-algebra C is formally smooth. Then:*

(i) *If B is a formally smooth A-algebra, $\mathfrak{J}/\mathfrak{J}^2$ is a topologically formally projective C-module (19.2.1).*

(ii) *If B is a formally smooth A-algebra and a preadmissible ring ($\mathbf{0_I}$, 7.1.2), the homomorphisms φ_n (19.5.2) are formal bimorphisms (19.1.2).*

(iii) *Conversely, suppose that B is preadmissible, that the sequence $(\mathfrak{J}^n)$ tends to 0 in B, that $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module, and that the φ_n are formal bimorphisms. Then B is a formally smooth A-algebra.*

The proof of this theorem is long and encumbered with technical details; we will therefore begin by proving a simpler corollary, where the guiding ideas appear more clearly; this corollary is moreover the particular case of the theorem (19.5.3) that will be most frequently used in the sequel.

Corollary (19.5.4). — *Let A be a topological ring, B a topological A-algebra, $\mathfrak{J}$ an ideal of B such that the topology of B is the $\mathfrak{J}$-preadic topology. Suppose that the discrete A-algebra $C = B/\mathfrak{J}$ is formally smooth. Then the three following conditions are equivalent:*

a) B is a formally smooth A-algebra.
b) $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module and the canonical homomorphism

$$\varphi : \mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) \longrightarrow \mathrm{gr}_{\mathfrak{J}}(B)$$

is bijective.
c) The separated completion $\widehat{B}$ of B is a topological $\widehat{A}$-algebra isomorphic to an $\widehat{A}$-algebra of the form B', where $B' = \mathbf{S}^*_C(V)$, V being a projective C-module and B' being equipped with the B'^{+}-preadic topology, where B'^{+} is the ideal of augmentation.

(19.5.4.1). Let us first prove that $a)$ implies that $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module. Let P and Q be two C-modules, $u : P \to Q$ a surjective C-homomorphism, and $v : \mathfrak{J}/\mathfrak{J}^2 \to Q$ a C-homomorphism. The ring $B/\mathfrak{J}^2$ being viewed as a B-extension of C by $\mathfrak{J}/\mathfrak{J}^2$, we deduce from v a B-extension $E = (B/\mathfrak{J}^2) \oplus_{(\mathfrak{J}/\mathfrak{J}^2)} Q$ of C by Q (18.2.8). As C is a discrete formally smooth A-algebra, the extension E is A-trivial (19.4.4) and can therefore be identified with $\mathrm{D}_C(Q)$ (18.2.3). The surjective homomorphism $u : P \to Q$ then canonically defines a surjective A-homomorphism $\psi : \mathrm{D}_C(P) \to \mathrm{D}_C(Q)$ (18.2.9) whose kernel is an ideal $\mathfrak{L}$ of the extension $E' = \mathrm{D}_C(P)$, contained in P, and *a fortiori* of square zero. Let $f : B \to E = E'/\mathfrak{L}$ be the homomorphism defining the B-algebra structure of E: as $\mathfrak{L}$ is of square zero and B is a formally smooth A-algebra, f factorizes as $B \xrightarrow{g} E' \to E'/\mathfrak{L}$, where g is a continuous A-homomorphism. The diagram $\mathbf{0_{IV\text{-}1}}$ | 92

$$\begin{tikzcd} E' \arrow[r, "\psi"] & E \\ B \arrow[u, "h'"] \arrow[r, "\theta"'] & B/\mathfrak{J}^2 \arrow[u, "v'"'] \end{tikzcd}$$

where v' is deduced from v (18.2.8), is commutative. Moreover, the image of $\mathfrak{J}$ by $\psi \circ h'$ is contained in Q, so the image of $\mathfrak{J}$ by h' is contained in P, and the image of $\mathfrak{J}^2$ by h' is zero. Restricting h' and θ to $\mathfrak{J}$, one obtains a commutative diagram

$$\begin{tikzcd} P \arrow[r, "u"] & Q \\ \mathfrak{J}/\mathfrak{J}^2 \arrow[u, "w"] \arrow[ur, "v"'] & \end{tikzcd}$$

which proves that the C-module $\mathfrak{J}/\mathfrak{J}^2$ is projective.

(19.5.4.2). Let us prove secondly that $a)$ implies that φ is bijective, which will complete the proof that $a)$ implies $b)$. Set $E_n = B/\mathfrak{J}^{n+1}$, and denote by F_n the quotient of $\mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2)$ by the $(n+1)$-th power of its ideal of augmentation. As $\mathfrak{J}^{n+1}$ is nilpotent in E_n and C is a discrete formally smooth A-algebra, the identity automorphism of C factorizes as $C \xrightarrow{f} E_n \to C = E_n/(\mathfrak{J}/\mathfrak{J}^{n+1})$ where f is an A-homomorphism; moreover, $\mathfrak{J}/\mathfrak{J}^2$ being a projective C-module, the identity of $\mathfrak{J}/\mathfrak{J}^2$ factorizes as $\mathfrak{J}/\mathfrak{J}^2 \to \mathfrak{J}/\mathfrak{J}^{n+1} \to \mathfrak{J}/\mathfrak{J}^2$, and from f and g one obtains canonically a homomorphism of C-algebras $\mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) \to E_n$ which annihilates on the $(n+1)$-th power of the ideal of augmentation, whence, passing to the quotient, a surjective homomorphism of algebras $v : F_n \to E_n$ such that $\mathrm{gr}^0(v)$ and $\mathrm{gr}^1(v)$ are the identity automorphisms of C and of $\mathfrak{J}/\mathfrak{J}^2$. By definition of the canonical homomorphism φ, we see that $\mathrm{gr}^j(v) = \varphi_j$ for all $j \leqslant n$. Note now that the kernel $\mathfrak{N}$ is a nilpotent ideal of F_n, so that E_n is identified with $F_n/\mathfrak{N}$. As B is a formally smooth A-algebra, the canonical A-homomorphism $p_n : B \to E_n = B/\mathfrak{J}^{n+1}$, which is continuous, factorizes as $B \xrightarrow{w} F_n \xrightarrow{v} E_n$, where w is a continuous A-homomorphism; as $\mathrm{gr}^0(v)$ is the identity, $w(\mathfrak{J})$ is contained in the ideal of augmentation of F_n, hence $w(\mathfrak{J}^{n+1}) = 0$, so w factorizes as $B \xrightarrow{p_n} E_n \xrightarrow{w'} F_n$, and the composite

$E_n \xrightarrow{w'} F_n \xrightarrow{v} E_n$ is the identity. Since $\mathrm{gr}^j(v) = \varphi_j$ for all $j \leqslant n$, making $j = 0$ and $j = 1$, one sees that φ_n is injective, which proves that $a)$ implies $b)$.

(19.5.4.3). Let us next prove that $b)$ implies $a)$. The same reasoning as at the beginning of (19.5.4.2) $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 93
proves the existence of a surjective A-homomorphism of algebras $v : F_n \to E_n$ such that $\mathrm{gr}^j(v) = \varphi_j$ for all j and the filtrations of F_n and E_n are finite; we conclude that v is *bijective* (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 of th. 1). Let now G be a discrete topological A-algebra, $\mathfrak{R}$ an ideal of square zero in G, $f : B \to G/\mathfrak{R}$ a continuous A-homomorphism of algebras. As G is discrete, there exists an integer m such that f annihilates on $\mathfrak{J}^m$, so f factorizes as $B \to E_n \xrightarrow{f_n} G/\mathfrak{R}$, where we take $n = 2m$. One has therefore by composition a continuous A-homomorphism $r : C \to F_n \xrightarrow{v} E_n \xrightarrow{f_n} G/\mathfrak{R}$, and as C is a discrete formally smooth A-algebra, r factorizes as $C \xrightarrow{r'} G \to G/\mathfrak{R}$, so G is thus equipped with a structure of C-algebra. On the other hand, the restriction to $\mathfrak{J}/\mathfrak{J}^2$ of the C-homomorphism $f'_n : F_n \xrightarrow{v} E_n \xrightarrow{f_n} G/\mathfrak{R}$ is a C-linear application $g : \mathfrak{J}/\mathfrak{J}^2 \to G/\mathfrak{R}$. As $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module, g factorizes as $\mathfrak{J}/\mathfrak{J}^2 \xrightarrow{h} G \to G/\mathfrak{R}$, where h is a C-linear application. By extension, one deduces a homomorphism of C-algebras $w : \mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) \to G$, and by construction every element of degree m has for image by w an element of $\mathfrak{R}$, so every element of degree $n = 2m$ has zero image since $\mathfrak{R}^2 = 0$; in other words, w factorizes as $\mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) \to F_n \xrightarrow{w'} G$. By construction, the composite $F_n \xrightarrow{w'} G \xrightarrow{\theta} G/\mathfrak{R}$ coincides with f'_n in C and in $\mathfrak{J}/\mathfrak{J}^2$, hence is equal to f'_n. Finally, the composite $B \to E_n \xrightarrow{w' \circ v^{-1}} G$ being equal to f, one sees that B is a formally smooth A-algebra.

(19.5.4.4). It remains to prove the equivalence of $a)$ and $c)$. In the first place, $c)$ implies that B' is a C-algebra formally smooth for the discrete topology (19.3.8); as C is a formally smooth A-algebra, B' is also a formally smooth A-algebra (19.3.6), hence B is a formally smooth A-algebra (19.3.6). It remains to see that $a)$ implies $c)$. Note first that C is a formally smooth A-algebra, the identical homomorphism $C \to B/\mathfrak{J}$ factorizes as $C \to \widehat{B} \to B/\mathfrak{J}$, where $C \to \widehat{B}$ is an A-homomorphism (19.3.10); hence $\widehat{B}$, and therefore all the $B/\mathfrak{J}^{n+1}$, are equipped with structures of C-algebras. On the other hand, as $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module by $b)$, the canonical injection $\mathfrak{J}/\mathfrak{J}^2 \to B/\mathfrak{J}^2$ allows one to form a projective system of C-homomorphisms $u_n : \mathfrak{J}/\mathfrak{J}^2 \to B/\mathfrak{J}^{n+1}$ for $n \geqslant 1$, hence also (by the universal property of the symmetric algebra) a projective system of homomorphisms of C-algebras $v_n : \mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) = B' \to B/\mathfrak{J}^{n+1}$; moreover it is clear that v_n annihilates on $(B'^{+})^{n+1}$ and as $\mathrm{gr}^0(v_n) = \varphi_0$ and $\mathrm{gr}^1(v_n) = \varphi_1$, one has $\mathrm{gr}^j(v_n) = \varphi_j$ for $0 \leqslant j \leqslant n$. As the φ_j are isomorphisms by $b)$, and the filtrations of $B'/B'^{+^{n+1}}$ and of $B/\mathfrak{J}^{n+1}$ are finite, one concludes that v_n is *bijective* for all n (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 of th. 1); whence $c)$ by passage to the projective $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 94
limit.

Remark (19.5.5). — Under the general hypotheses of (19.5.4), suppose furthermore that $\mathfrak{J}$ is a *maximal* ideal of B, so that $C = B/\mathfrak{J}$ is a field, and that $\mathfrak{J}/\mathfrak{J}^2$ is a vector space of *finite* dimension. Then the conditions $a)$, $b)$ and $c)$ of (19.5.4) are also equivalent to the following:

d) *Given* a polynomial algebra $E = C[T_1, \dots, T_n]$ over the field C, and denoting by $\mathfrak{n}$ the maximal ideal of E generated by the T_i ($1 \leqslant i \leqslant n$), then for every ideal $\mathfrak{b}$ of E containing a power $\mathfrak{n}^k$, every homomorphism $v : B \to E/\mathfrak{b}$ of A-algebras C-augmented factorizes as $B \xrightarrow{w} E/\mathfrak{b}$, where w is an A-homomorphism.

Indeed, $\mathfrak{J}/\mathfrak{J}^2$ being here a free C-module, it suffices to verify condition $d)$, where $\mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2)$ identifies with the polynomial algebra $E = C[T_1, \dots, T_n]$, where $n = \mathrm{rg}_C(\mathfrak{J}/\mathfrak{J}^2)$, and the ideal of augmentation of E is the ideal $\mathfrak{n}$ generated by the T_i. For every integer $k \geqslant 0$, the hypothesis $d)$ allows one to factorize the canonical homomorphism $B \to B/\mathfrak{J}^{k+1}$ into $B \xrightarrow{w} E/\mathfrak{n}^k \to E/\mathfrak{b}$, and $a)$ follows by (19.5.4.2).

(19.5.6). *Proof of the theorem* (19.5.3). — Let $(\mathfrak{b}_\alpha)$ be a fundamental system of decreasing open ideals of B. We set

$$B_\alpha = B/\mathfrak{b}_\alpha, \quad C_\alpha = B/(\mathfrak{b}_\alpha + \mathfrak{J}), \quad \mathfrak{J}_\alpha = (\mathfrak{b}_\alpha + \mathfrak{J})/\mathfrak{b}_\alpha,$$

so that $E_0 = C$; and we set

$$E_{\alpha,n} = B/(\mathfrak{b}_\alpha + \mathfrak{J}^{n+1}) = B_\alpha/\mathfrak{J}_\alpha^{n+1},$$

a discrete quotient ring.

(19.5.6.1). The C-modules $((\mathfrak{b}_\alpha \cap \mathfrak{J}^n) + \mathfrak{J}^{n+1})/\mathfrak{J}^{n+1}$ of $\mathfrak{J}^n/\mathfrak{J}^{n+1}$ form a fundamental system of open submodules of the topological C-module $\mathfrak{J}^n/\mathfrak{J}^{n+1}$; as $(\mathfrak{b}_\alpha \cap \mathfrak{J}^n) + \mathfrak{J}^{n+1} = \mathfrak{J}^n \cap (\mathfrak{b}_\alpha + \mathfrak{J}^{n+1})$, we have

$$\mathfrak{J}^n/((\mathfrak{b}_\alpha \cap \mathfrak{J}^n) + \mathfrak{J}^{n+1}) = \mathfrak{J}^n/(\mathfrak{J}^n \cap (\mathfrak{b}_\alpha + \mathfrak{J}^{n+1})) = (\mathfrak{b}_\alpha + \mathfrak{J}^n)/(\mathfrak{b}_\alpha + \mathfrak{J}^{n+1}) = \mathfrak{J}_\alpha^n/\mathfrak{J}_\alpha^{n+1}.$$

(19.5.6.2). For all $n \geqslant 0$, we set $E_n = B/\mathfrak{J}^{n+1}$, so that $E_0 = C$; and we set $\mathbf{0_{IV\text{-}1}}$ | 95

$$E_{\alpha,n} = B/(\mathfrak{b}_\alpha + \mathfrak{J}^{n+1}) = B_\alpha/\mathfrak{J}_\alpha^{n+1},$$

a discrete quotient ring. Likewise, in $\mathrm{gr}^n_{\mathfrak{J}}(B) = \mathfrak{J}^n/\mathfrak{J}^{n+1}$, we have seen that the $((\mathfrak{b}_\alpha \cap \mathfrak{J}^n) + \mathfrak{J}^{n+1})/\mathfrak{J}^{n+1}$ form a fundamental system of open submodules. Consider the symmetric algebra $\mathbf{S}^*_{C_\alpha}(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^2)$; we denote by $F_{\alpha,n}$ the quotient of $\mathbf{S}^*_{C_\alpha}(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^2)$ by the $(n+1)$-th power of its ideal of augmentation. For a fixed $n \geqslant 1$, it follows from (19.5.1.1) that the $\mathbf{S}^n_C(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^2)$ are the quotients of $\mathbf{S}^n_C(\mathfrak{J}/\mathfrak{J}^2)$ by a fundamental system of open submodules within this topological C-module.

To abbreviate the language, we will say that for $\alpha \leqslant \beta$, the canonical homomorphisms $B_\beta \to B_\alpha$, $C_\beta \to C_\alpha$, $E_{\beta,n} \to E_{\alpha,n}$, $F_{\beta,n} \to F_{\alpha,n}$, etc., are *homomorphisms of transition*.

Lemma (19.5.6.3). — *Suppose that C is a formally smooth A-algebra, that the A-algebra B is formally smooth, the ring B preadmissible, and the C-module $\mathfrak{J}/\mathfrak{J}^2$ formally projective. Then:*

(i) *For every α, there exists $\beta \geqslant \alpha$ and an A-homomorphism surjective of algebras*

$$v_{\alpha\beta} : F_{\beta,n} \longrightarrow E_{\alpha,n}$$

such that $\mathrm{gr}^0(v_{\alpha\beta}) : C_\beta \to C_\alpha$ *and* $\mathrm{gr}^1(v_{\alpha\beta}) : \mathfrak{J}_\beta/\mathfrak{J}_\beta^2 \to \mathfrak{J}_\alpha/\mathfrak{J}_\alpha^2$ *are the homomorphisms of transition;* 2° *the composite* $E_{\lambda,n} \xrightarrow{w_{\beta\lambda}} F_{\beta,n} \xrightarrow{v_{\alpha\beta}} E_{\alpha,n}$ *is the homomorphism of transition.*

(ii) *If β and $v_{\alpha\beta}$ satisfy the conditions of* (i)*, then for every $\gamma \geqslant \beta$, there exists $\delta \geqslant \gamma$ and an A-homomorphism surjective of algebras $v_{\gamma\delta} : F_{\delta,n} \to E_{\gamma,n}$ satisfying the conditions of* (i) *(for γ and δ) and making commutative the diagram*

$$\begin{tikzcd} F_{\beta,n} \arrow[r, "v_{\alpha\beta}"] & E_{\alpha,n} \\ F_{\delta,n} \arrow[u] \arrow[r, "v_{\gamma\delta}"'] & E_{\gamma,n} \arrow[u] \end{tikzcd}$$

where the vertical arrows are the homomorphisms of transition.

(19.5.6.4). $\mathbf{0_{IV\text{-}1}}$ | 96

(i) In the discrete topological A-algebra $E_{\alpha,n}$, the ideal $\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1}$ is nilpotent, and the isomorphism $C_\alpha \xrightarrow{\sim} E_{\alpha,n}/(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1})$ gives by composition a continuous A-homomorphism $C \to C_\alpha \to E_{\alpha,n}/(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1})$; as C is a formally smooth A-algebra, this homomorphism factorizes as $C \xrightarrow{f_\alpha} E_{\alpha,n} \to E_{\alpha,n}/(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1})$, where f_α is a continuous A-homomorphism; hence $\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1}$ becomes, by means of f_α, a topological C-module annihilated by an open ideal of C. The hypothesis that $\mathfrak{J}/\mathfrak{J}^2$ is a formally projective C-module then implies the existence of a continuous C-linear application $g_\alpha : \mathfrak{J}/\mathfrak{J}^2 \to \mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1}$ making commutative the diagram

$$\begin{tikzcd} \mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1} \arrow[r] & \mathfrak{J}_\alpha/\mathfrak{J}_\alpha^2 \\ & \mathfrak{J}/\mathfrak{J}^2 \arrow[ul, "g_\alpha"] \arrow[u] \end{tikzcd}$$

As f_α and g_α are continuous, there exists $\beta \geqslant \alpha$ such that these homomorphisms factorize through

$$C \longrightarrow C_\beta \xrightarrow{f'_{\alpha\beta}} E_{\alpha,n}$$

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \mathfrak{J}_\beta/\mathfrak{J}_\beta^2 \xrightarrow{g'_{\alpha\beta}} \mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1}$$

and from $f'_{\alpha\beta}$ and $g'_{\alpha\beta}$, one obtains a homomorphism of C_β-algebras

$$\mathbf{S}^*_{C_\beta}(\mathfrak{J}_\beta/\mathfrak{J}_\beta^2) \longrightarrow E_{\alpha,n}$$

which, since by definition $g'_{\alpha\beta}$ annihilates on the $(n+1)$-th power of the ideal of augmentation of $\mathbf{S}^*_{C_\beta}(\mathfrak{J}_\beta/\mathfrak{J}_\beta^2)$, passing to the quotient by this power, one obtains the desired homomorphism $v_{\alpha\beta}$, taking into account the definitions of f_α and g_α; the surjectivity of $v_{\alpha\beta}$ follows from that of the two homomorphisms $\mathrm{gr}^0(v_{\alpha\beta})$ and $\mathrm{gr}^1(v_{\alpha\beta})$, since this implies that $\mathrm{gr}(v_{\alpha\beta})$ is surjective (the graded algebra $\mathrm{gr}^*(E_{\alpha,n})$ being generated by $\mathrm{gr}^0(E_{\alpha,n})$ and $\mathrm{gr}^1(E_{\alpha,n})$), and as the filtrations considered are finite, one can apply Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 of th. 1.

(ii) The hypothesis that B is preadmissible means that one can suppose all the $\mathfrak{b}_\alpha$ contained in the same $\mathfrak{b}_{\alpha_0}$, whose powers tend to 0. This implies in particular that the kernel of the homomorphism of transition $E_{\gamma,n} \to E_{\alpha,n}$, equal to $(\mathfrak{b}_\gamma + \mathfrak{J}^{n+1})/(\mathfrak{b}_\alpha + \mathfrak{J}^{n+1})$, is *nilpotent*; applying the lemma (19.3.10.1), one sees that one can choose f_γ so that the diagram

$$\begin{tikzcd} & E_{\alpha,n} \\ C \arrow[ur, "f_\alpha"] \arrow[r, "f_\gamma"'] & E_{\gamma,n} \arrow[u] \end{tikzcd}$$

(where the vertical arrow is the homomorphism of transition) is commutative. The hypothesis that $\mathfrak{J}/\mathfrak{J}^2$ is a formally projective C-module furthermore allows one to choose $g_\gamma : \mathfrak{J}/\mathfrak{J}^2 \to \mathfrak{J}_\gamma/\mathfrak{J}_\gamma^{n+1}$ so that the appropriate diagram is commutative (it suffices to note that the image by (g_α, p_γ) of $\mathfrak{J}/\mathfrak{J}^2$ in the product module $(\mathfrak{J}_\alpha/\mathfrak{J}_\alpha^{n+1}) \times (\mathfrak{J}_\gamma/\mathfrak{J}_\gamma^2)$ is contained in the canonical image Q of this product module; the existence of g_γ and the relation $\beta \leqslant \gamma$ then suffices to obtain (19.5.6.4) and the commutativity of the diagram.

Lemma (19.5.6.5). — *Suppose that the A-algebras B and C are both formally smooth, and that B is preadmissible. For every system of two indices $\alpha \leqslant \beta$ and every homomorphism $v_{\alpha\beta} : F_{\beta,n} \to E_{\alpha,n}$ satisfying the conditions of* (19.5.6.3), (i), *there exists an index $\lambda \geqslant \beta$ and an A-homomorphism of algebras* $\mathbf{0_{IV\text{-}1}}$ | 97

$$w_{\beta\lambda} : E_{\lambda,n} \longrightarrow F_{\beta,n}$$

such that: 1° $\mathrm{gr}^0(w_{\beta\lambda}) : C_\lambda \to C_\beta$ *and* $\mathrm{gr}^1(w_{\beta\lambda}) : \mathfrak{J}_\lambda/\mathfrak{J}_\lambda^2 \to \mathfrak{J}_\beta/\mathfrak{J}_\beta^2$ *are the homomorphisms of transition;* 2° *the composite* $E_{\lambda,n} \xrightarrow{w_{\beta\lambda}} F_{\beta,n} \xrightarrow{v_{\alpha\beta}} E_{\alpha,n}$ *is the homomorphism of transition.*

PROOF. Let us apply the lemma (19.5.6.3), (ii) with $\gamma = \beta$, which gives a $\delta \geqslant \beta$ and a $v_{\beta\delta} : F_{\delta,n} \to E_{\beta,n}$. Recall that $v_{\beta\delta}$ is surjective; on the other hand, its kernel $\mathfrak{N}$ is nilpotent since B is preadmissible. As B is a formally smooth A-algebra, the A-homomorphism canonical $\rho_{\beta,n} : B \to E_{\beta,n}$, which is continuous, factorizes as $B \xrightarrow{w} F_{\beta,n} \xrightarrow{v_{\beta\delta}} E_{\beta,n}$, where w is a continuous A-homomorphism. As $F_{\beta,n}$ is discrete, there exists $\lambda \geqslant \delta$ such that w is zero on $\mathfrak{b}_\lambda$, so w factorizes as $B \to B/\mathfrak{b}_\lambda \xrightarrow{w'_\lambda} F_{\beta,n}$. Consider the composed homomorphism $u'_\lambda : B/\mathfrak{b}_\lambda \xrightarrow{w'_\lambda} F_{\beta,n} \xrightarrow{v_{\alpha\beta}} E_{\alpha,n}$, where $q_{\beta\delta}$ is the homomorphism of transition; as the composite $v_{\beta\delta} \circ w_{\beta\lambda}$ is the homomorphism of transition $C_\lambda \to C_\beta$, this A-homomorphism is the canonical homomorphism $B/\mathfrak{b}_\lambda \to B/(\mathfrak{b}_\alpha + \mathfrak{J}^{n+1}) = E_{\alpha,n}$. Moreover, as $\varphi_{\gamma\gamma,0}$ and $\varphi_{\gamma\gamma,1}$ are the homomorphisms of transition identities, $u^0_{\beta\gamma} : C_\gamma \to C_\beta$ and $u^1_{\beta\gamma} : \mathfrak{J}_\gamma/\mathfrak{J}_\gamma^2 \to \mathfrak{J}_\beta/\mathfrak{J}_\beta^2$

are the homomorphisms of transition. From w'_λ, noting that $\mathrm{gr}^0(q_{\beta\delta}) = \mathrm{gr}^0(v_{\beta\delta}) : C_\delta \to C_\beta$ is the homomorphism of transition, the composite $C_\lambda \to C_\delta \to C_\beta$ is the homomorphism of transition. Similarly, $\mathrm{gr}^1(v_{\beta\delta} \circ w_{\beta\lambda})$ is the homomorphism of transition, and the required properties follow. □

Lemma (19.5.6.7). — *Suppose that B is preadmissible, that $\mathfrak{J}/\mathfrak{J}^2$ is a formally projective C-module, that C is a formally smooth A-algebra, and that the φ_n are formal bimorphisms. Then, for every pair of indices α, β such that $\alpha \leqslant \beta$ and every homomorphism $v_{\alpha\beta} : F_{\beta,n} \to E_{\alpha,n}$ satisfying the conditions of* (19.5.6.3), (i), *there exists an index $\gamma \geqslant \beta$ such that, for every index $\mu \geqslant \gamma$, there exists a commutative diagram of A-homomorphisms* **$0_{\text{IV-1}}$ | 98**

$$\begin{array}{ccc} F_{\beta,n} & \xrightarrow{v_{\alpha\beta}} & E_{\alpha,n} \\ {\scriptstyle u_{\beta\mu}}\uparrow & & \uparrow{\scriptstyle p_{\alpha\mu}} \\ F_{\mu,n} & & E_{\mu,n} \end{array}$$

where $p_{\alpha\mu}$ is the homomorphism of transition.

(19.5.6.8). *Proof of* (19.5.3), (iii). — To show that φ_n is a formal bimorphism, we apply the criterion of (19.1.3) and the conditions of (19.5.6.5) being satisfied by hypothesis; let us determine, for every α, $v_{\alpha\beta}$ and $w_{\beta\lambda}$ satisfying the conclusions of this lemma. The homomorphism **$0_{\text{IV-1}}$ | 99**

$$\mathrm{gr}^n(v_{\alpha\beta}) : \mathrm{gr}^n(F_{\beta,n}) \longrightarrow \mathrm{gr}^n(E_{\alpha,n})$$

is none other than the homomorphism

$$\varphi_{\alpha\beta,n} : \mathbf{S}^n_{C_\beta}(\mathfrak{J}_\beta/\mathfrak{J}_\beta^2) \longrightarrow \mathfrak{J}_\alpha^n/\mathfrak{J}_\alpha^{n+1}$$

deduced from the canonical homomorphism φ_n (19.5.2.1) by passage to quotients; indeed, it follows from (19.5.6.3) and the definition of the canonical homomorphism φ that $\mathrm{gr}^j(v_{\alpha\beta})$ coincides with $\varphi_{\alpha\beta,j}$ for all j, since φ_n is surjective, c' is *a fortiori* a formal epimorphism; moreover, for $j \leqslant n$, as $\mathrm{gr}^j(v_{\alpha\beta}) = \varphi_j$ for all j, the homomorphism composed

$$\mathrm{gr}^j(F_{\lambda,n}) \xrightarrow{\varphi_{\lambda\lambda,j}} \mathrm{gr}^j(E_{\lambda,n}) \xrightarrow{\mathrm{gr}^j(w_{\beta\lambda})} \mathrm{gr}^j(F_{\beta,n})$$

is the homomorphism of transition, since this is true for $j = 0$ and $j = 1$ by (19.5.6.5), and as $\mathrm{gr}^*(E_{\lambda,n})$ is generated by $\mathrm{gr}^1(E_{\lambda,n})$; composing with the homomorphism of transition $\mathrm{gr}^n(F_{\beta,n}) \to \mathrm{gr}^n(F_{\alpha,n})$, one sees therefore, for $j = n$, that one has factorized the homomorphism of transition $\mathrm{gr}^n(F_{\lambda,n}) \to \mathrm{gr}^n(F_{\alpha,n})$ as

$$\mathrm{gr}^n(F_{\lambda,n}) \xrightarrow{\varphi_{\lambda\lambda,n}} \mathrm{gr}^n(E_{\lambda,n}) \longrightarrow \mathrm{gr}^n(F_{\alpha,n})$$

which is the condition of (19.1.3) for φ_n to be a formal bimorphism.

Let G be a discrete topological A-algebra, $\mathfrak{N}$ an ideal of square zero in G, $f : B \to G/\mathfrak{N}$ a continuous A-homomorphism of algebras. As $G/\mathfrak{N}$ is discrete, there exists an index α such that f annihilates on $\mathfrak{b}_\alpha$; moreover, as $\mathfrak{J}^m \subset \mathfrak{b}_\alpha$ for a sufficiently large m, f factorizes as

$$B \xrightarrow{p_\alpha} E_{\alpha,n} \xrightarrow{f_{\alpha,n}} G/\mathfrak{N}$$

where we take $n = 2m$. The hypotheses of the lemma (19.5.6.3) being satisfied, there exists a $\beta \geqslant \alpha$ and a $v_{\alpha\beta} : F_{\beta,n} \to E_{\alpha,n}$; thus the composite $f_{\alpha,n} \circ v_{\alpha\beta} : F_{\beta,n} \to G/\mathfrak{N}$ is a continuous A-homomorphism. Since $F_{\beta,n}$ is a C_β-algebra, hence in particular a discrete topological C-algebra, this gives by composition a continuous A-homomorphism $r : C \to G/\mathfrak{N}$. Since C is a formally smooth A-algebra, r factorizes as $C \xrightarrow{r'} G \to G/\mathfrak{N}$, and G is thereby equipped with the structure of a discrete topological C-algebra. On the other hand, by composition with the canonical homomorphism

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \mathfrak{J}_\beta/\mathfrak{J}_\beta^2 \longrightarrow F_{\beta,n},$$

one deduces from $f_{\alpha,n} \circ v_{\alpha\beta}$ a continuous C-homomorphism $g : \mathfrak{J}/\mathfrak{J}^2 \to G/\mathfrak{N}$. Since $\mathfrak{J}/\mathfrak{J}^2$ is a formally projective C-module, g factorizes as $\mathfrak{J}/\mathfrak{J}^2 \xrightarrow{h} G \to G/\mathfrak{N}$, where h is a continuous C-homomorphism. As G is discrete, there exists an index $\gamma \geqslant \beta$ such that h annihilates the image of $((\mathfrak{b}_\gamma \cap \mathfrak{J}) + \mathfrak{J}^2)/\mathfrak{J}^2$ and r' annihilates the image of $(\mathfrak{b}_\gamma + \mathfrak{J})/\mathfrak{J}$; therefore h factorizes as a C_γ-homomorphism $\mathfrak{J}_\gamma/\mathfrak{J}_\gamma^2 \to G$. By extension, this gives a homomorphism of C_γ-algebras

$$w_\gamma : \mathbf{S}^*_{C_\gamma}(\mathfrak{J}_\gamma/\mathfrak{J}_\gamma^2) \longrightarrow G.$$

By construction, every element of degree m has image in $\mathfrak{N}$ under w_γ, hence every element of degree $n = 2m$ has zero image, since $\mathfrak{N}^2 = 0$; in other words, w_γ factorizes as $F_{\gamma,n} \xrightarrow{w'_\gamma} G$. Applying the lemma (19.5.6.7) to $v_{\alpha\beta}$, there exists a $\delta \geqslant \gamma$ and a $u_{\gamma\delta} : E_{\delta,n} \to F_{\gamma,n}$, so that the composite $v_{\alpha\beta} \circ u_{\gamma\delta}$ is the homomorphism of transition. One obtains therefore finally a commutative diagram of continuous A-homomorphisms

$$\begin{tikzcd}
 & E_{\alpha,n} \arrow[r, "f_{\alpha,n}"] & G/\mathfrak{N} & \\
 & F_{\beta,n} \arrow[u, "v_{\alpha\beta}"] & & \\
B \arrow[r, "p_\delta"'] & E_{\delta,n} \arrow[u, "u_{\gamma\delta}"] & F_{\gamma,n} \arrow[l, "w_\gamma"'] \arrow[r, "w'_\gamma"'] & G
\end{tikzcd}$$

and the composite $f' : B \to G$ of the homomorphisms of the bottom line is such that f factorizes as $B \xrightarrow{f'} G \to G/\mathfrak{N}$, which proves that B is a formally smooth A-algebra. The proof of the theorem (19.5.3) is thus complete.

Corollary (19.5.7). — *Let A be a topological ring, B a topological A-algebra, $(\mathfrak{b}_\lambda)$ a fundamental system of open ideals of B, $\mathfrak{J}$ an ideal of B, and $C = B/\mathfrak{J}$ the topological quotient A-algebra. Set $C_\lambda = B/(\mathfrak{b}_\lambda + \mathfrak{J})$. Suppose that:*

1° *for all n, the topology induced on $\mathfrak{J}^n$ by that of B is also the topology of the B-module $\mathfrak{J}^n$ deduced from the topology of B (19.0.2); this condition is satisfied in particular if B is Noetherian and its topology is preadic ($\mathbf{0_I}$, 7.3.2);*

2° *C is a formally smooth A-algebra.*

Under these conditions:

(i) *If B is a formally smooth A-algebra, then, for all λ, $(\mathfrak{J}/\mathfrak{J}^2) \otimes_C C_\lambda$ is a projective C_λ-module.*

(ii) *If B is a formally smooth A-algebra and a preadmissible ring, then, for all λ, the canonical homomorphism*

(19.5.7.1) $$\varphi_\lambda = \varphi \otimes 1_{C_\lambda} : \mathbf{S}^*_{C_\lambda}((\mathfrak{J}/\mathfrak{J}^2) \otimes_C C_\lambda) \longrightarrow \mathrm{gr}_{\mathfrak{J}}(B) \otimes_C C_\lambda$$

is bijective.

(iii) *Conversely, suppose that B is preadmissible, that the sequence $(\mathfrak{J}^n)$ tends to 0, and that, for all λ, $(\mathfrak{J}/\mathfrak{J}^2) \otimes_C C_\lambda$ is a projective C_λ-module and the homomorphism (19.5.7.1) is bijective. Then B is a formally smooth A-algebra.*

PROOF. Indeed, the topology of $\mathfrak{J}/\mathfrak{J}^2$ and those of the $\mathbf{S}^n_C(\mathfrak{J}/\mathfrak{J}^2)$ are then also deduced from that of B (19.5.1); the conclusions are immediate consequences of (19.5.3) and of the criteria of (19.1.4) and (19.2.4) specialized to this type of topology. □

Remark (19.5.8). — Suppose that $\mathfrak{J}/\mathfrak{J}^2$ is a C-module of finite type and that, for the quotient topology inherited from B, C is a Zariski ring. Let $\mathfrak{r}$ be an ideal of definition of C, so that the $\mathfrak{r}^n$ form a fundamental system of neighborhoods of 0 in C. Then the conclusions of (i) and (ii) in (19.5.7) may be replaced by the following:

(i') $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module.

(ii′) The canonical homomorphism

$$\varphi : \mathbf{S}^*_C(\mathfrak{J}/\mathfrak{J}^2) \longrightarrow \mathrm{gr}_{\mathfrak{J}}(B)$$

is bijective.

Indeed, it is clear that (i′) implies the conclusion of (i) in (19.5.7). Conversely, if $(\mathfrak{J}/\mathfrak{J}^2) \otimes_C C_\lambda$ is a projective C_λ-module for all λ, then $(\mathfrak{J}/\mathfrak{J}^2) \otimes_C (C/\mathfrak{r}^n)$ is a projective, hence flat, $(C/\mathfrak{r}^n)$-module for every n; one concludes that $\mathfrak{J}/\mathfrak{J}^2$ is a flat C-module ($\mathbf{0}_{\mathbf{III}}$, 10.2.2), hence projective since it is finitely presented (Bourbaki, *Alg. comm.*, chap. II, §5, n° 2, cor. 2 of th. 1). On the other hand, the C-modules $\mathbf{S}^n_C(\mathfrak{J}/\mathfrak{J}^2)$ and $\mathrm{gr}^n_{\mathfrak{J}}(B)$ are of finite type, and when C is a Zariski ring it is equivalent to say that φ_n is bijective or that $\varphi_n \otimes 1_{C_\lambda}$ is bijective for all λ (Bourbaki, *Alg. comm.*, chap. III, §3, n° 5, prop. 9); therefore (ii) is equivalent to (ii′).

19.7. The case of algebras over a field

Theorem (19.6.1). — (Cohen). — *Let k be a field, K an extension of k, k and K being equipped with the discrete topologies. For K to be a formally smooth k-algebra, it is necessary and sufficient that K be a separable extension of k.* $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 100

Proof. The necessity of the condition will be established in (19.6.5.1) (and of course is not used until then); we will content ourselves here with proving that the condition is *sufficient*. We distinguish two cases:

I. — K is a separable extension of finite type of k. We know then (Bourbaki, *Alg.*, chap. V, § 9, n° 3, th. 2) that there exists a pure sub-extension $K' = k(T_1, \ldots, T_n)$ of K such that K is a finite separable algebraic extension of K'. In view of (19.3.5), (ii), one can therefore reduce to the case where $K = K'$, or to the case where K is algebraically finite over k. In the first case, $A = k[T_1, \ldots, T_n]$ is a formally smooth k-algebra (19.3.3), hence $k(T_1, \ldots, T_n) = S^{-1}A$ where $S = A - \{0\}$ is also formally smooth (19.3.5), (iv). In the second case, *all* Hochschild cohomology groups $\mathrm{H}^n_k(K, L)$ are zero for every (K, K)-bimodule L: indeed, if one considers the k-algebra tensor product $C = K \otimes_k K$, L is a C-module on the left and the cohomology group $\mathrm{H}^n_k(K, L)$ is equal to $\mathrm{Ext}^n_C(K, L)$, where K is also viewed as (K, K)-bimodule, hence as a left C-module (M, IX, 4). Now, as K is a finite separable $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 101
extension of k, we know that $K \otimes_k K$ is a direct product of extensions of k, one of which is K itself (Bourbaki, *Alg.*, chap. VIII, § 8, prop. 3); this implies that K is a *projective* C-module, whence our assertion. Every k-extension of K of zero kernel L is therefore k-trivial, and *a fortiori* all commutative k-extensions are trivial, whence the theorem in this case (19.4.4).

II. — General case. — With the notation of (18.4.5), it suffices to prove that $\mathrm{Hom}_K(\mathrm{H}_2(\mathrm{P}'_\bullet), L) = 0$ for *every* K-vector space L, which means that $\mathrm{H}_2(\mathrm{P}'_\bullet) = 0$. If K is a union of a filtering family of sub-extensions $K^{(\alpha)}$ of k, $\mathrm{P}'_\bullet$ is the *inductive limit* of the corresponding complexes $\mathrm{P}'^{(\alpha)}_\bullet$, since the $\varinjlim$ functor commutes with the tensor product in the category of k-modules; by the exactness of the $\varinjlim$ functor in this category, one has $\mathrm{H}_2(\mathrm{P}'_\bullet) = \varinjlim \mathrm{H}_2(\mathrm{P}'^{(\alpha)}_\bullet)$. As K is separable, it follows that the same holds for every sub-extension of K and that K is a union of sub-extensions of finite type, so the first part of the proof implies that K is a formally smooth k-algebra, Q.E.D. □

Corollary (19.6.2). — *Let A be a separated complete local ring, K its residue field, k a subfield of K such that K is a separable extension of k. Then there exists a subfield K' of A containing k, such that the restriction to K' of the canonical homomorphism $A \to K$ is an isomorphism of K' onto K.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of A. By the hypothesis and (19.6.1), K is a formally smooth k-algebra; applying (19.3.10) (replacing C by A and $\mathfrak{J}$ by $\mathfrak{m}$), one sees that the identity automorphism of $K = A/\mathfrak{m}$ factorizes as $K \xrightarrow{u} A \to A/\mathfrak{m}$, where u is a k-homomorphism, hence necessarily a k-isomorphism of K onto a subfield K' of A containing k. □

Corollary (19.6.3). — *Let A be a noetherian complete local ring, $\mathfrak{m}$ its maximal ideal, k its residue field. The following conditions are equivalent:*

a) A contains a subfield.
b) If p is the characteristic of k, one has $pA = 0$.
c) There exists a field K such that A is isomorphic to a quotient ring of a formal power series ring $B = K[[T_1, \ldots, T_n]]$.

When this is the case, one can suppose that A is isomorphic to $B/\mathfrak{b}$, where $\mathfrak{b}$ is contained in the square of the maximal ideal of B.

PROOF. It is immediate that $c)$ implies $a)$, since A is $\neq 0$ because it is a local ring; as K is a field and the composed homomorphism $K \xrightarrow{\varphi} B \to A$ (where φ is the canonical injection) is not zero, it is injective. To see that $a)$ implies $b)$, it suffices to remark that if K is a subfield of A, K is isomorphic to a subfield of k and has therefore the same characteristic p; as $p.1 = 0$ in K, hence in A, one has $pA = 0$. Reciprocally, $b)$ implies $a)$, for the composite of the canonical homomorphisms $\mathbf{Z} \xrightarrow{j} A \to k$ has kernel $p\mathbf{Z}$, and one has $j(p\mathbf{Z}) = 0$, hence $\mathrm{Ker}(j) = p\mathbf{Z}$ and A contains a ring A' isomorphic to $\mathbf{Z}/p\mathbf{Z}$ and not meeting $\mathfrak{m}$; if $p > 0$, A' is a field; if not, every element of A' being invertible in A, the field of fractions of A' (isomorphic to $\mathbf{Q}$) is also contained in A. **$\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$** | 102

Let us prove finally that $a)$ implies $c)$; the condition $a)$ implies that A contains a prime subfield k_0, isomorphic to the prime subfield of k, and whose k is therefore a separable extension. Applying (19.6.2), one sees that there exists an isomorphism f of k onto a subfield K of A. On the other hand, let $(x_i)_{1 \leqslant i \leqslant n}$ be a system of elements of $\mathfrak{m}$ such that the classes mod. $\mathfrak{m}^2$ of the x_i form a basis (over k) of $\mathfrak{m}/\mathfrak{m}^2$. Since A is complete, there exists a continuous ring homomorphism $u : k[[T_1, \ldots, T_n]] \to A$ such that u equals f on k and $u(T_i) = x_i$ for all i, and this homomorphism is surjective by virtue of the choice of the x_i (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 9, prop. 11). □

Theorem (19.6.4). — *Let k be a field, A a local noetherian k-algebra, $\mathfrak{m}$ its maximal ideal, $K = A/\mathfrak{m}$ its residue field, A being equipped with the $\mathfrak{m}$-preadic topology. Suppose that K is a separable extension of k. Then the following conditions are equivalent:*

- *a) A is a formally smooth k-algebra.*
- *b) The completion $\widehat{A}$ of A is k-isomorphic to a formal power series ring $K[[T_1, \ldots, T_n]]$ (where the k-algebra structure is defined by the composed homomorphism $k \xrightarrow{\varphi} K \xrightarrow{\psi} K[[T_1, \ldots, T_n]]$, where ψ is the canonical injection and φ the homomorphism defining the extension structure of K).*
- *b′) $\widehat{A}$ is a formal power series ring isomorphic to $K[[T_1, \ldots, T_n]]$.*
- *c) A is a regular local ring.*

PROOF. The fact that $a)$ implies $b)$ is a particular case of (19.5.4) (equivalence of $a)$ and $c)$), applied replacing k by A, B by A and $\mathfrak{J}$ by $\mathfrak{m}$, taking into account (19.6.1). It is clear that $b)$ implies $b')$ and $b')$ implies $c)$ by (17.1.1). Finally, $c)$ is verified: it follows from (17.1.1), $a)$ and from (19.5.4) (equivalence of $b)$ and $a)$), applied to $K = A/\mathfrak{m}$, replacing C by A, that A is a formally smooth k-algebra. □

Corollary (19.6.5). — *Let k be a field, A a local noetherian k-algebra, $\mathfrak{m}$ its maximal ideal, A being equipped with the $\mathfrak{m}$-preadic topology. Suppose that A is a formally smooth k-algebra; then A is* geometrically regular *over k, in other words (cf.* **IV***, 6.7.6), for every finite extension k' of k, the semi-local ring $A' = A \otimes_k k'$ is regular (17.3.6). Furthermore, if K is the residue field of A, $\widehat{A}$ is isomorphic to a formal power series ring $K[[T_1, \ldots, T_n]]$.* **$\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$** | 103

PROOF. As $\widehat{A}' = \widehat{A} \otimes_k k'$, it follows from (19.3.6) and (17.1.5) (applied to the maximal ideals of A') that one can reduce to the case where A is complete; then A' is a formally smooth k'-algebra (19.3.5), (iii) and is moreover a direct product of k'-local algebras, which are also formally smooth (19.3.5), (v). One is therefore reduced to proving that A is *regular*. Let k_0 be the prime subfield of k; as k is a separable extension of k_0, it is a formally smooth k_0-algebra (19.6.1), and by the hypothesis, so is A (19.3.5), (ii). As the residue field K of A is a separable extension of k_0, one can then apply (19.6.4) to A and k_0, whence the conclusion. □

Corollary (19.6.5.1). — *Let A be a noetherian local ring, k a subfield of A such that A is a formally smooth k-algebra (for its preadic topology). Then every field K such that $k \subset K \subset A$ is a separable extension of k.*

PROOF. Indeed, for every finite extension k' of k, the ring $K \otimes_k k'$ is identified with a sub-ring of $A' = A \otimes_k k'$; as A' is a regular ring, hence reduced, so is $K \otimes_k k'$, which proves that K is a separable extension of k (Bourbaki, *Alg.*, chap. VIII, § 7, n° 3, th. 1). □

Remark (19.6.5.2). — If K is a *non-separable* extension of k, the formal power series ring $K[[T_1, \dots, T_n]]$ is therefore *not* a formally smooth k-algebra (for the usual k-algebra structure recalled in (19.6.4)). On the other hand, there exist formally smooth k-algebras A which are complete noetherian local rings whose residue field K is a non-separable extension of k (for example the completions of the localizations of $B = k[T_1, \dots, T_n]$ at maximal ideals $\mathfrak{n}$ such that $B/\mathfrak{n}$ is a finite non-separable algebraic extension of k); such a ring cannot therefore be k-isomorphic to $K[[T_1, \dots, T_n]]$ even though it is *isomorphic* to it by virtue of (19.6.5).

One can in certain cases make the corollary (19.6.5) more precise:

Proposition (19.6.6). — *Let k be a field, A a local noetherian k-algebra, $\mathfrak{m}$ its maximal ideal, A being equipped with the $\mathfrak{m}$-preadic topology. Suppose that the residue field $K = A/\mathfrak{m}$ of A is such that there exists a finite radicial extension k_0 of k for which the residue field of the local ring $K \otimes_k k_0$ is a separable extension of k_0 (we will express this property later by saying that K is of* finite radicial multiplicity *over k (**IV**, 4.7.4) and we will see that this condition is always satisfied when K is an extension of finite type of k). Then the following conditions are equivalent:*

- a) *A is a formally smooth k-algebra.*
- b) *There exists a finite radicial extension k' of k such that $\widehat{A} \otimes_k k'$ is k'-isomorphic to a formal power series algebra $K'[[T_1, \dots, T_n]]$, where K' is a separable extension of k'.*
- b′) *There exists a finite extension k' of k such that the semi-local ring $\widehat{A} \otimes_k k'$ has at least one local component which is k'-isomorphic to a formal power series algebra $K'[[T_1, \dots, T_n]]$, where K' is a separable extension of k'.*
- c) *A is geometrically regular over k (19.6.5).*

PROOF. Note first that if k' is a finite radicial extension of k, there is only a single maximal ideal **$0_{\text{IV-1}}$** | 104
of $A' = A \otimes_k k'$ above $\mathfrak{m}$, formed of elements whose p^h-th power (p being the characteristic of k) is in $\mathfrak{m}$ for a suitable h (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 3, lemma 4); A' is therefore a local ring, and so is its completion $\widehat{A}'$. We recall on the other hand that if K is a separable extension of k, then, for every finite extension k'' of k, $K \otimes_k k''$ is a direct product of fields (Bourbaki, *Alg.*, chap. VIII, § 7, n° 3, cor. 1 of th. 1), and therefore $\mathfrak{m} \otimes_k k''$ is the radical of $A \otimes_k k''$, and the residue fields at the maximal ideals of $A \otimes_k k''$ are the field components of $K \otimes_k k''$; furthermore, these fields are separable extensions of k'' since for every extension k_1 of k'', $K \otimes_k k_1 = (K \otimes_k k'') \otimes_{k''} k_1$ is without radical (*loc. cit.*, th. 1). Applying these remarks to the field K of finite radicial multiplicity over k, one sees that for *every* finite extension k'' of k_0, $(K \otimes_k k'')_{\text{red}}$ is separable over k''.

Let us pass to the proof of (19.6.6). It is trivial that b) implies b'). If one sets $B' = K'[[T_1, \dots, T_n]]$, the hypothesis that K' is separable over k' entails, by the remarks of the beginning, that for every finite extension k'' of k', the local components of the semi-local complete ring $B' \otimes_{k'} k''$ (equal to formal power series algebras $(K' \otimes_{k'} k'')[[T_1, \dots, T_n]]$) are the field components of $K' \otimes_{k'} k''$, and the K''_i are separable over k'_0; since each of these local rings is k'_0-isomorphic to a formal power series algebra $K''_i[[T_1, \dots, T_n]]$, where K' is separable, one sees that the hypothesis b') is still verified when one replaces k' by any of its finite extensions. In particular, one can suppose k' *quasi-Galois*, hence a Galois extension of a radicial extension k'_0 of k; $A' = \widehat{A} \otimes_k k'$ can then be written $A'_0 = \widehat{A} \otimes_k k'_0$, where $A'_0 = \widehat{A} \otimes_k k'_0$ is a complete local ring. One of the local direct components of A', viewed as k'_0-algebras, is k'_0-isomorphic to a formal power series ring $K'_0[[T_1, \dots, T_n]]$, where K'_0 is separable over k'_0; in view of (19.6.4), A'_0 is therefore a formally smooth k'_0-algebra. But as k'_0 is a k-module which is faithfully flat, projective and of finite type, one concludes by (19.4.7) that $\widehat{A}$ is a formally smooth k-algebra.

On the other hand, that a) implies c) has already been seen (19.6.5); it remains therefore to verify that c) implies b). If c) is verified, then, for every finite radicial extension k' of k containing k_0, $A \otimes_k k'$ is a local regular ring, whose residue field K' is a separable extension of k'; it follows therefore from (19.6.4) that its completion $\widehat{A} \otimes_k k'$ is k'-isomorphic to a formal power series ring $K'[[T_1, \dots, T_n]]$. □

Remarks (19.6.7). — (i) We note that the hypothesis that K is of finite radicial multiplicity over k was only used in the proof of the implication b') $\Rightarrow$ a).

(ii) We will prove later (22.5.8) that $a)$ and $c)$ are equivalent, *without* any hypothesis on the extension K of k.

19.8. The case of local homomorphisms; existence and uniqueness theorems In this section, whenever a semi-local ring is considered as a topological ring, it is always understood that it is endowed with its $\mathfrak{r}$-preadic topology, where $\mathfrak{r}$ is its radical. Every local homomorphism of local rings is thus automatically continuous.

Theorem (19.7.1). — *Let A, B be two local noetherian rings, $\mathfrak{m}$, $\mathfrak{n}$ their respective maximal ideals, $k = A/\mathfrak{m}$ the residue field of A; suppose that A and B are equipped respectively with the $\mathfrak{m}$-preadic and $\mathfrak{n}$-preadic topologies. Let $\varphi : A \to B$ be a local homomorphism, and set $B_0 = B \otimes_A k$. The following properties are equivalent:*

a) *B is a formally smooth A-algebra.*
b) *B is a flat A-module and $B_0 = B/\mathfrak{m}B$ (equipped with the quotient topology) is a formally smooth k-algebra.*

PROOF. The proof is carried out in several steps.

(19.7.1.1) Let us first show that $b)$ implies $a)$; we will apply the criterion (19.4.7), with $\mathfrak{J} = \mathfrak{m}$; by virtue of the second hypothesis in $b)$, it suffices to show that B is a *formally projective* A-module. The hypothesis implies that for all $h > 0$, $B/\mathfrak{m}^h B$ is a flat $(A/\mathfrak{m}^h)$-module ($\mathbf{0_{III}}$, 10.2.1); as the $\mathfrak{m}^h$ form a fundamental system of neighbourhoods of 0 in A and $(B/\mathfrak{m}^h B) \otimes_{A/\mathfrak{m}^h} k = B_0$, we can replace A and B by $A/\mathfrak{m}^h$ and $B/\mathfrak{m}^h B$ respectively, and thus suppose that A is *artinian* (hence discrete). As B_0 is a k-algebra formally smooth, it is a regular ring (19.6.5); let $(x_i^0)_{1 \leqslant i \leqslant n}$ be a regular system of parameters for B_0 (17.1.6), and for each i, let $x_i \in B$ be such that x_i^0 is its image in $B_0 = B/\mathfrak{m}B$; as the $\mathbf{0_{IV\text{-}1}}$ | 105
x_i^0 generate the maximal ideal $\mathfrak{n}_0 = \mathfrak{n}/\mathfrak{m}B$ of B_0, the ideals $\mathfrak{Q}_m = \sum_{i=1}^n (x_i^0)^m B_0$ (for $m > 0$) form a fundamental system of neighbourhoods of 0 in B_0, since $\mathfrak{Q}_m$ is evidently contained in $\mathfrak{n}_0^m$, and $\mathfrak{n}_0^m$ contains $\mathfrak{Q}_m^m$. Set $\mathfrak{J}_m = \sum_{i=1}^n x_i^m B$ for all $m > 0$; it is clear that $\mathfrak{n} = \mathfrak{J}_1 + \mathfrak{m}B$; as there exists $h > 0$ such that $\mathfrak{m}^h = 0$, one has $\mathfrak{n}^m \subset \mathfrak{J}_1^{m-h} \subset \mathfrak{n}^{m-h}$ for $m > h$, and as we have seen that $\mathfrak{J}_m \supset \mathfrak{J}_1^{mn}$, the $\mathfrak{J}_m$ form a fundamental system of neighbourhoods of 0 in B. The whole argument then amounts to proving that the $B/\mathfrak{J}^m$ are *free* A-modules, and it amounts to the same to verify that these are *flat* A-modules ($\mathbf{0_{III}}$, 10.1.3). Now, the hypothesis that (x_i^0) is a B_0-regular sequence of elements of the maximal ideal of B_0 implies the same property for the sequence $(x_i^0)^m$ $(1 \leqslant i \leqslant n)$ for all $m > 0$ (15.1.20); the conclusion thus follows from (15.1.16), $b)$ and $c)$). ☐

Lemma (19.7.1.2). — *Let A be a topological ring, B, C two topological A-algebras that are local noetherian rings. Suppose further that C is complete and that the residue field $B/\mathfrak{m}$ of B is an A-module of finite type. Let E be the completed tensor product $B \widehat{\otimes}_A C$. Then:*

(i) *E is a complete semi-local noetherian ring.*
(ii) *The ideal $\mathfrak{m}E$ is contained in the radical of E, and for all $h > 0$, $E/\mathfrak{m}^h E$ is isomorphic to $(B/\mathfrak{m}^h) \otimes_A C$.*
(iii) *If C is a flat A-module, E is a flat B-module.*

PROOF. By definition, E is the separated completion of the tensor product $B \otimes_A C$ for the topology defined by the ideals $\operatorname{Im}(\mathfrak{m}^i \otimes_A C) + \operatorname{Im}(B \otimes_A \mathfrak{n}')$ ($\mathbf{0_I}$, 7.7.5). If one sets $\mathfrak{r} = \operatorname{Im}(\mathfrak{m} \otimes_A C) + \operatorname{Im}(B \otimes_A \mathfrak{n})$, one has $\mathfrak{r}^{2i} \subset \operatorname{Im}(\mathfrak{m}^i \otimes_A C) + \operatorname{Im}(B \otimes_A \mathfrak{n}^i) \subset \mathfrak{r}^i$, so E is also the separated completion of $B \otimes_A C$ for the $\mathfrak{r}$-preadic topology. By hypothesis, $(B \otimes_A C)/\mathfrak{r} = (B/\mathfrak{m}) \otimes_A (C/\mathfrak{n})$ is a $(C/\mathfrak{n})$-module of finite type, hence an artinian ring; moreover, $\mathfrak{r}/\mathfrak{r}^2$, being a quotient of $((\mathfrak{m}/\mathfrak{m}^2) \otimes_A C) \oplus (B \otimes_A (\mathfrak{n}/\mathfrak{n}^2))$, is a $(B \otimes_A C)$-module of finite type; applying ($\mathbf{0_I}$, 7.2.11), one sees that E is a *noetherian* ring; moreover, $E/\mathfrak{r}E$, being artinian, E is *semi-local*. Note now that E, which is isomorphic to $\varprojlim((B/\mathfrak{m}^i) \otimes_A (C/\mathfrak{n}^j))$, is also isomorphic, by the theorem of the double projective $\mathbf{0_{IV\text{-}1}}$ | 106
limit, to $\varprojlim_i(\varprojlim_j((B/\mathfrak{m}^i) \otimes_A (C/\mathfrak{n}^j)))$; but $\varprojlim_j((B/\mathfrak{m}^i) \otimes_A (C/\mathfrak{n}^j))$ is the separated completion of $(B/\mathfrak{m}^i) \otimes_A C$, and as C is complete, this is none other than $(B/\mathfrak{m}^i) \otimes_A C$ itself, since $B/\mathfrak{m}^i$ is an A-module of finite type (as $\mathfrak{m}/\mathfrak{m}^i$ is a $(B/\mathfrak{m})$-module of finite type ($\mathbf{0_I}$, 7.3.6)). One therefore has $E = \varprojlim_i((B/\mathfrak{m}^i) \otimes_A C)$. For every integer $h > 0$, $\mathfrak{m}^h E$, being an ideal of E, is closed in E ($\mathbf{0_I}$, 7.3.5), hence complete, and on the other hand it is evidently dense in $\varprojlim_i(\operatorname{Im}(\mathfrak{m}^h/\mathfrak{m}^{h+i}) \otimes_A C))$,

hence equal to this projective limit. Moreover, all the projective systems considered are defined by *surjective* homomorphisms; it follows therefore from ($\mathbf{0_{III}}$, 13.2.2) that $E/\mathfrak{m}^h E$ is isomorphic to $(B/\mathfrak{m}^h) \otimes_A C = (B \otimes_A C)/\operatorname{Im}(\mathfrak{m}^h \otimes_A C)$. In particular $\operatorname{Im}(\mathfrak{m} \otimes_A C) \subset \mathfrak{r}$, which shows that $\mathfrak{m}E$ is contained in $\mathfrak{r}E$, hence in the radical of E. Finally, the hypothesis that C is a flat A-module implies that $(B/\mathfrak{m}^h) \otimes_A C = E/\mathfrak{m}^h E$ is a flat $(B/\mathfrak{m}^h)$-module for all $h > 0$: since B and E are noetherian and $\mathfrak{m}E$ is contained in the radical of E, it follows from ($\mathbf{0_{III}}$, 10.2.2) that E is a flat B-module. □

Lemma (19.7.1.3). — *Let A be a local noetherian ring, $\mathfrak{m}$ its maximal ideal, k its residue field, B_0 a local noetherian k-algebra; suppose that B_0 is complete and regular. Then there exists a topological A-algebra B that is a complete local noetherian ring, a flat A-module, and such that B_0 is k-isomorphic to $B \otimes_A k = B/\mathfrak{m}B$.*

PROOF. As $\widehat{A}$ is flat over A and has the same residue field, we can restrict to the case where A is *complete*.

Let K be the residue field of B_0, and let us distinguish two cases:

Case I: K is a *separable* extension of k. By virtue of (19.6.4), B_0 is k-isomorphic to a formal power series ring $K[[T_1, \dots, T_n]]$. When $B_0 = K$, the lemma has already been proved ($\mathbf{0_{III}}$, 10.3.1); let C be a complete local noetherian ring that is a flat A-module and such that $C \otimes_A k$ is isomorphic to K. For $n \geqslant 1$, it suffices to take (using the preceding notation) $B = C[[T_1, \dots, T_n]]$; indeed it is known (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 4, cor. 3 of th. 1) that B is a flat C-module, hence also a flat A-module, and on the other hand, it is immediate that $C[[T_1, \dots, T_n]] \otimes_A k$ is isomorphic to $(C/\mathfrak{m}C)[[T_1, \dots, T_n]] = B_0$.

Case II: K is of *characteristic* $p > 0$, and is therefore of the same characteristic as k. Denote by $\mathbf{F}_p$ the prime field, and by $W(P)$ the complete local ring of p-adic numbers $\mathbf{Z}_p$, which is a complete discrete valuation ring (hence *regular*), with P as residue field. Let us first show that there exists a continuous ring homomorphism $W(P) \to A$, thus making A into a $W(P)$-*topological algebra*. Indeed, if $j : \mathbf{Z} \to A$ is the canonical homomorphism, one has $j(p\mathbf{Z}) \subset \mathfrak{m}$ by hypothesis, whence $j^{-1}(\mathfrak{m}) = p\mathbf{Z}$, and j therefore factors as $\mathbf{Z} \to \mathbf{Z}_{p\mathbf{Z}} \xrightarrow{f} A$, where f is a local homomorphism, hence continuous, which (since A is complete) extends by continuity to the desired homomorphism $W(P) \to A$. $\mathbf{0_{IV\text{-}1}}$ | 107

As k is a separable extension of P, case I) shows that there is a local homomorphism $W(P) \to W(k)$, where $W(k)$ is a complete local noetherian ring and a flat $W(P)$-module, such that $W(k) \otimes_{W(P)} P$ is isomorphic to k. Moreover, as the uniformiser p of $W(P)$ is a $W(k)$-regular element by flatness ($\mathbf{0_I}$, 6.3.4), and as $W(k)/pW(k) = k$, the maximal ideal of $W(k)$ is $pW(k)$, which implies that this ring is a *complete discrete valuation ring* (Bourbaki, *Alg. comm.*, chap. VI, § 3, n° 5, prop. 9). By (19.7.1.1) (as k is separable over P, hence a formally smooth P-algebra (19.6.1)), one sees that $W(k)$ is a $W(P)$-algebra *formally smooth*. The continuous $W(P)$-homomorphism $W(k) \to k$ therefore factors as $W(k) \to A \to k$ (19.3.11), which allows us to consider A as a $W(k)$-topological algebra. Applying now case I) to B_0 considered as a P-algebra and to $W(P)$, one sees that there exists a $W(P)$-algebra B_P which is a complete local noetherian ring, a flat $W(P)$-module, and such that $B_P \otimes_{W(P)} P$ is P-isomorphic to B_0. Using again the fact that $W(k)$ is a $W(P)$-algebra formally smooth, one sees by (19.3.11) that the homomorphism $W(k) \to k \to B_0$ factors as $W(k) \to B_P \to B_0$; moreover, as $k = W(k)/pW(k)$, one has $B_P \otimes_{W(k)} k = B_P/pB_P = B_P \otimes_{W(P)} P = B_0$. Let us show that B_P is a flat $W(k)$-module; it suffices to verify that B_P is a $W(k)$-module *without torsion* ($\mathbf{0_I}$, 6.3.4), or equivalently that p is a B_P-regular element, which follows from the fact that B_P is a flat $W(P)$-module ($\mathbf{0_I}$, 6.3.4).

Set now
$$B = B_P \widehat{\otimes}_{W(k)} A$$
and note that the residue field of A being equal to that of $W(k)$, is *a fortiori* a $W(k)$-module of finite type. It follows first from (19.7.1.2) that B is a complete semi-local noetherian ring, $\mathfrak{m}B$ is contained in the radical of B; moreover $B/\mathfrak{m}B$ is k-isomorphic to $B_P \otimes_{W(k)} k = B_0$, hence B is in fact a *local* ring. As B_P is a flat $W(k)$-module, (19.7.1.2) shows finally that B is a flat A-module. Q.E.D. □

Lemma (19.7.1.4). — *Let A be a ring, $\mathfrak{J}$ an ideal of A, M, N two A-modules separated for the $\mathfrak{J}$-preadic topology. Suppose further that N is complete for the $\mathfrak{J}$-preadic topology and that M is a flat A-module. Let $u : N \to M$ be an A-homomorphism; if $u \otimes 1 : N \otimes_A (A/\mathfrak{J}) \to M \otimes_A (A/\mathfrak{J})$ is bijective, then u is bijective.* $\mathbf{0_{IV\text{-}1}}$ | 108

PROOF. The graded modules being taken relative to the $\mathfrak{J}$-preadic filtrations, it follows from the hypotheses on M and N relative to the $\mathfrak{J}$-preadic topologies that it suffices to prove that $\mathrm{gr}(u) : \mathrm{gr}_\bullet(N) \to \mathrm{gr}_\bullet(M)$ is *bijective* (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 of th. 1). Now, there is a commutative diagram

$$\begin{tikzcd}
\mathrm{gr}_0(N) \otimes_{A_0} \mathrm{gr}_\bullet(A) \arrow[r, "\mathrm{gr}_0(u)\otimes 1"] \arrow[d, "\varphi_N"'] & \mathrm{gr}_0(M) \otimes_{A_0} \mathrm{gr}_\bullet(A) \arrow[d, "\varphi_M"] \\
\mathrm{gr}_\bullet(N) \arrow[r, "\mathrm{gr}(u)"] & \mathrm{gr}_\bullet(M)
\end{tikzcd}$$

where $A_0 = A/\mathfrak{J} = \mathrm{gr}_0(A)$, and φ_M and φ_N are the canonical maps ($\mathbf{0_{III}}$, 10.1.2). By hypothesis, $\mathrm{gr}_0(u)$ is bijective as well as φ_M ($\mathbf{0_{III}}$, 10.2.1), and φ_N is surjective; one deduces first that $\mathrm{gr}_0(u) \otimes 1$ is bijective, then that φ_N is injective, hence bijective, and finally that $\mathrm{gr}(u)$ is bijective. □

Lemma (19.7.1.5). — *Let A be a noetherian ring, $\mathfrak{J}$ an ideal of A, B, B' two A-algebras that are local noetherian rings, the homomorphisms $A \to B$, $A \to B'$ being continuous for the $\mathfrak{J}$-preadic topology on A. Suppose that:*

1° *B and B' are complete for the $\mathfrak{J}$-preadic topologies;*
2° *B is a formally smooth A-algebra;*
3° *B' is a flat A-module.*

Set $A_0 = A/\mathfrak{J}$, and let $u_0 : B \otimes_A A_0 \to B' \otimes_A A_0$ be an A_0-isomorphism; then there exists an A-isomorphism $u : B \to B'$ such that $u_0 = u \otimes 1$ (which implies that B' is also a formally smooth A-algebra and B is a flat A-module).

PROOF. Set $B_0 = B \otimes_A A_0$, $B'_0 = B' \otimes_A A_0$. Note that if $\mathfrak{m}$ and $\mathfrak{m}'$ are the maximal ideals of B and B', the $\mathfrak{J}$-preadic topologies on B and B' are separated since $\mathfrak{J}B \subset \mathfrak{m}$ and $\mathfrak{J}B' \subset \mathfrak{m}'$; moreover, as $\mathfrak{J}B'$ is closed in B' for the $\mathfrak{J}$-preadic topology, the composed homomorphism $B \to B_0 \xrightarrow{u_0} B'_0$, which is continuous for the $\mathfrak{J}$-preadic topologies, factors as $B \xrightarrow{\bar{u}} B'$, where $\bar{u}$ is a continuous A-homomorphism (19.3.10). One has $\bar{u}_0 = u \otimes 1$, and the hypothesis that u_0 is bijective implies that $\bar{u}$ is bijective, by virtue of (19.7.1.4). □

(19.7.1.6) *End of the proof of* (19.7.1). — To finish proving (19.7.1), it remains to show that *a*) implies *b*); it is already known that *a*) implies that B_0 is a k-algebra formally smooth (19.3.5), (iii), hence it suffices to prove that B is a flat A-module. It amounts to the same to establish that $\widehat{B}$ is a flat $\widehat{A}$-module (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 4), and one knows that $\widehat{B}$ is an $\widehat{A}$-algebra formally smooth (19.3.6); one can therefore restrict to the case where A and B are *complete*. As B_0 is a k-algebra formally smooth, it is a regular ring (19.6.5) and complete ($\mathbf{0_I}$, 6.3.5); applying (19.7.1.3), one sees that there exists a complete local noetherian ring B' that is a flat A-module and a k-isomorphism $B_0 \simeq B'_0 = B' \otimes_A k$. It then suffices to apply (19.7.1.5) (taking $\mathfrak{J}$ to be the maximal ideal of A) to obtain that B is A-isomorphic to B', hence is a flat A-module. Q.E.D.

Theorem (19.7.2). — *Let A be a local noetherian ring, $\mathfrak{J}$ an ideal contained in the maximal ideal of A,* $\mathbf{0_{IV\text{-}1}}$ | 109
$A_0 = A/\mathfrak{J}$, B_0 a complete local noetherian ring, $A_0 \to B_0$ a local homomorphism making B_0 into a formally smooth A_0-algebra. Then there exists a complete local noetherian ring B, a local homomorphism $A \to B$ making B into a flat A-module, and an A_0-isomorphism $u : B \otimes_A A_0 \xrightarrow{\sim} B_0$. If (B', u') is a couple satisfying the same conditions as (B, u), there exists an A-isomorphism $v : B \xrightarrow{\sim} B'$ making the diagram

$$\begin{tikzcd}
B \otimes_A A_0 \arrow[rr, "v\otimes 1"] \arrow[dr, "u"'] & & B' \otimes_A A_0 \arrow[dl, "u'"] \\
& B_0 &
\end{tikzcd}$$

commutative.

PROOF. Let $\mathfrak{m}$ be the maximal ideal of A, so that $\mathfrak{m}_0 = \mathfrak{m}/\mathfrak{J}$ is the maximal ideal of A_0, A and A_0 having the same residue field k. Set $B_{00} = B_0 \otimes_{A_0} k$; as B_{00} is a formally smooth k-algebra (19.3.5), (iii), it is a regular local ring (19.6.5); applying (19.7.1.3), one sees that there exists a topological A-algebra B which is a complete local noetherian ring, a flat A-module, and for which one has a k-isomorphism $u_{00} : B \otimes_A k \xrightarrow{\sim} B_{00}$. Note that by virtue of (19.7.1), B is a formally smooth A-algebra, hence $B \otimes_A A_0 = B/\mathfrak{J}B$ is a formally smooth A_0-algebra (19.3.5), (iii) and a complete local noetherian ring; moreover, B_0 is a flat A_0-module by virtue of the hypothesis and of (19.7.1); as one has a k-isomorphism $u_{00} : B \otimes_A k = (B \otimes_A A_0) \otimes_{A_0} k \xrightarrow{\sim} B_0 \otimes_{A_0} k = B_{00}$, one deduces from (19.7.1.5), applied by replacing A by A_0 and $\mathfrak{J}$ by $\mathfrak{m}_0$, that there exists an A_0-isomorphism $u : B \otimes_A A_0 \xrightarrow{\sim} B_0$ such that $u_{00} = u \otimes 1$. As for the uniqueness assertion, note that the ideals $\mathfrak{J}^h B$ (resp. $\mathfrak{J}^h B'$) are closed in B (resp. B') ($\mathbf{0_I}$, 7.3.5), hence B and B' are separated and complete for the $\mathfrak{J}$-preadic topologies (Bourbaki, *Top. gén.*, chap. III, 3ᵉ éd., § 3, n° 5, cor. 2 of prop. 9); one has by hypothesis an A_0-isomorphism $v_0 : B \otimes_A A_0 \xrightarrow{\sim} B' \otimes_A A_0$ such that $u' v_0 = u$; as B is a formally smooth A-algebra and B' a flat A-module, one can apply (19.7.1.5), whence the existence of the A-isomorphism v answering the question. □

Remarks (19.7.3). — (i) We note that the uniqueness assertion in (19.7.2) is still valid if one only supposes that B and B' are complete *for the $\mathfrak{J}$-preadic topologies*. We do not know whether one can similarly improve the existence assertion, that is, whether one can dispense with assuming the local ring B_0 to be complete (for its $\mathfrak{n}_0$-preadic topology, where $\mathfrak{n}_0$ denotes its maximal ideal) by only requiring that B be complete for the $\mathfrak{J}$-preadic topology. When A_0 is complete for the $\mathfrak{m}$-preadic topology, one can see that this problem reduces to the following: if B_0 is a regular local noetherian ring (not necessarily complete) containing the prime field $\mathbf{F}_p = \mathbf{Z}/p\mathbf{Z}$, does there exist for every $n > 1$ a flat $(\mathbf{Z}/p^n\mathbf{Z})$-algebra B such that B/pB is isomorphic to B_0?

(ii) We note that in general, the isomorphism v whose existence is asserted in (19.7.2) is *not unique* (cf. (19.8.7)).

19.9. Cohen algebras and Cohen rings; application to the structure of complete local rings
The results of this section are immediate applications of the theorems of (19.7), but deserve to be stated explicitly because of their practical importance.

Definition (19.8.1). — Let A, B be two local noetherian rings, $\mathfrak{m}$ the maximal ideal of A, $k = A/\mathfrak{m}$ its residue field, $\varphi : A \to B$ a local homomorphism, making B into an A-algebra. We say that B is a *Cohen A-algebra* if it satisfies the following conditions:

(i) B is a complete ring.
(ii) B is a flat A-module.
(iii) $B \otimes_A k$ is a field (in other words, $\mathfrak{m}B$ is the maximal ideal of B) which is a separable extension of k.

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Theorem (19.8.2). — *Let A be a local noetherian ring, k its residue field.*

(i) *If B is a Cohen A-algebra, B is a formally smooth A-algebra. For every complete local noetherian ring C, every local homomorphism $A \to C$ and every ideal $\mathfrak{J} \neq C$ in C, every A-homomorphism $B \to C/\mathfrak{J}$ therefore factors as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is an A-homomorphism (necessarily local).*
(ii) *For every field K, separable extension of k, there exists a Cohen A-algebra B such that $B \otimes_A k$ is k-isomorphic to K, and such an A-algebra is unique up to isomorphism.*

PROOF. As K is a formally smooth k-algebra (19.6.1), the assertion (i) follows from (19.7.1). To prove (ii), one can restrict to the case where A is *complete*, for it amounts to the same to say that B is a flat A-module or a flat $\widehat{A}$-module ($\mathbf{0_{III}}$, 10.2.3), one has $\mathfrak{m}B = \mathfrak{m}\widehat{A}B$ and k is the residue field of $\widehat{A}$. It then suffices to apply (19.7.2) taking $\mathfrak{J} = \mathfrak{m}$ and $B_0 = K$ (and using (19.6.1)). □

Definition (19.8.3). — A *prime local ring* is a local ring of the form $\mathbf{Z}_{p\mathbf{Z}}$, where $p\mathbf{Z}$ is a prime ideal of $\mathbf{Z}$. A *complete prime local ring* is the completion of a prime local ring.

The prime local rings are therefore of two kinds:

1° Those corresponding to maximal ideals $p\mathbf{Z}$ where $p \neq 0$ is a prime number; $\mathbf{Z}_{p\mathbf{Z}}$ is a discrete valuation ring, whose completion is the ring of p-adic integers usually denoted $\mathbf{Z}_p$.[14]

2° For the prime ideal $p\mathbf{Z} = (0)$, $\mathbf{Z}_{p\mathbf{Z}}$ is the field of rational numbers $\mathbf{Q}$, which is identical to its completion (the topology being naturally the topology of a local noetherian ring, hence discrete here).

The terminology of (19.8.3), analogous to that of "prime fields", is justified in the same way: for every local ring A, consider the canonical homomorphism $\varphi : \mathbf{Z} \to A$, and let $p\mathbf{Z}$ be the inverse image under this homomorphism of the maximal ideal $\mathfrak{m}$ of A; $p\mathbf{Z}$ is a prime ideal of $\mathbf{Z}$ and the preceding homomorphism therefore factors as $\mathbf{Z} \to \mathbf{Z}_{p\mathbf{Z}} \xrightarrow{\psi} A$; moreover, as φ is the *unique* homomorphism of $\mathbf{Z}$ into A, p and ψ are determined in a unique way. In other words, for every local ring A, there is a unique homomorphism $\psi : P \to A$, where P is a prime local ring; if moreover A is separated and complete, one can extend this homomorphism by completion, and there is thus a unique homomorphism $\widehat{\psi} : \widehat{P} \to A$, where $\widehat{P}$ is a complete prime local ring. Moreover, ψ gives by passage to quotients a homomorphism of the residue field $\mathbf{F}_p$ if $p > 0$ (resp. $\mathbf{Q}$ if $p = 0$) into the residue field k of A, and p is therefore the *characteristic* of k.

If one takes in particular A to be a prime local ring (resp. complete prime local ring), one sees that there exists in such a ring only *a single endomorphism*, namely the identity.

Definition (19.8.4). — Let A be a local ring, $P \to A$ the unique homomorphism from a prime local ring P into A, p the characteristic of the residue fields of P and A. We say that A is a *Cohen ring* if it is a Cohen P-algebra, that is (19.8.1), if: **$0_{\text{IV-1}}$** | 111

1° A is noetherian and complete.

2° A is a flat P-module (which also amounts to saying that A is a flat $\widehat{P}$-module (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 4)).

3° A/pA is a field (necessarily separable over the residue field of P, this field being prime).

If $p = 0$, these conditions are equivalent to saying that A is a field of characteristic 0. If $p > 0$, one necessarily has $pA \neq 0$; condition 3° means that pA is the maximal ideal $\mathfrak{m}$ of A; condition 2° means that p is A-regular, since P is a discrete valuation ring ($\mathbf{0_I}$, 6.3.4). Hence A is a regular ring (17.1.1), *d*)) of dimension 1, and therefore a discrete valuation ring, complete by virtue of 1°; in summary:

Proposition (19.8.5). — *The Cohen rings are the fields of characteristic* 0 *and the complete discrete valuation rings whose residue field has characteristic $p > 0$, and whose maximal ideal is generated by $p \cdot 1$ (*1 *being the unity of the ring).*

We note that in the second case, $p \cdot 1 \neq 0$ since p is A-regular, hence one can identify $p \cdot 1$ with the integer p; the canonical homomorphism $\mathbf{Z}_p \to A$ is injective, and one identifies $p \cdot 1$ with the element p of $\mathbf{Z}_p$; one says in this case that A is a *p-ring* (or *Cohen p-ring*).

Theorem (19.8.6). — (Cohen). —

(i) *Let W be a Cohen ring, C a complete local noetherian ring, $\mathfrak{J}$ an ideal of C distinct from C. Then every local homomorphism $u : W \to C/\mathfrak{J}$ factors as $W \xrightarrow{v} C \to C/\mathfrak{J}$ where v is a local homomorphism.*

(ii) *Let K be a field. There exists a Cohen ring W whose residue field is isomorphic to K. If W' is a second Cohen ring, K' its residue field, every isomorphism $u : K \xrightarrow{\sim} K'$ arises by passage to quotients from an isomorphism $v : W \xrightarrow{\sim} W'$.*

PROOF. This is nothing other than (19.8.2) applied to the case where A is a prime local ring. □

Remarks (19.8.7). — (i) When K is of characteristic 0, part (ii) of (19.8.6) becomes trivial.

(ii) The homomorphism v of (19.8.6), (i) is *not* necessarily determined uniquely by u, as is shown already by the case where W is a field of characteristic 0, $\mathfrak{J}^2 = 0$ and u is an isomorphism (cf. (21.5.5)). Similarly, in (19.8.6), (ii), the isomorphism v is not necessarily determined uniquely by u (cf. (21.5.5)). However, when K is *perfect* and of characteristic

[14] This notation, now universally used, is in conflict with the notation A_f adopted in ($\mathbf{0_I}$, 1.2.3): with $A = \mathbf{Z}$ and $f = p$, A_f denotes the ring of rational numbers of the form k/p^n ($k \in \mathbf{Z}$, n an integer $\geqslant 0$); we will always avoid using the notation $\mathbf{Z}_p$ to denote this latter ring.

$p > 0$, one will see (21.5.5) that in (19.8.6), (ii) the isomorphism v is unique. One will also see later that in this case W identifies with the ring $W_\infty(K)$ of *Witt vectors of infinite length* over K.

(iii) In (19.8.6), (i), one can weaken the hypotheses on C by using (19.3.10) and (19.3.12).

Theorem (19.8.8). — (Cohen). — *Let A be a complete local noetherian ring, k its residue field.* $\mathbf{0}_{\textbf{IV-1}}$ | 112

(i) *There exists a Cohen ring W such that A is isomorphic to a quotient ring of a formal power series ring $W[[T_1, \ldots, T_n]]$ (and in particular A is isomorphic to a quotient of a complete regular local ring (17.3.8)). If A contains a field, it is isomorphic to a quotient ring of $k[[T_1, \ldots, T_n]]$.*

(ii) *Suppose moreover that A is an integral domain. Then there exists a subring B of A such that: 1° B is isomorphic to a formal power series ring over a ring C which is a field or a Cohen ring (which implies that B is a complete regular local ring (17.3.8)); 2° B has the same residue field as A and the injection $B \to A$ is a local homomorphism; 3° A is a finite B-algebra.*

PROOF. Let $\mathfrak{m}$ be the maximal ideal of A. There exists a Cohen ring W whose residue field is isomorphic to k (19.8.6), (ii); one therefore has a homomorphism $W \to A/\mathfrak{m}$, which then factors as $W \xrightarrow{u} A \to A/\mathfrak{m}$, where u is a local homomorphism (19.8.6), (i). For every finite family $(x_i)_{1\leqslant i\leqslant n}$ of elements of $\mathfrak{m}$, there then exists a local homomorphism $v : W[[T_1, \ldots, T_n]] \to A$ extending u and such that $v(T_i) = x_i$ for all i (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6). When A contains a field, it contains a prime field P, of which k is an extension (necessarily separable), and therefore A contains a field isomorphic to k (19.6.2); one can then replace W by k in the preceding definition of v.

(i) Let us first take the x_i to be a system of generators of $\mathfrak{m}$. As W has the same residue field as A, and the classes of the x_i generate the graded ring $\mathrm{gr}_\bullet(A)$ as a k-algebra, $\mathrm{gr}(v) : \mathrm{gr}_\bullet(W[[T_1, \ldots, T_n]]) \to \mathrm{gr}_\bullet(A)$ is surjective; one deduces that v is itself surjective (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 2 of th. 1). We recall that the case where A contains a field has already been seen and is included here only for convenience (19.6.3).

(ii) If A contains a field, it contains a field k' isomorphic to k as we have seen; one then considers a system of parameters $(y_j)_{1\leqslant j\leqslant m}$ of A (16.3.6), one takes $B = k[[T_1, \ldots, T_m]]$ and one considers the local homomorphism $w : B \to A$ which coincides in k with an isomorphism $k \to k'$ and is such that $w(T_j) = y_j$ for $1 \leqslant j \leqslant m$. If A does not contain a field, the unique homomorphism $\mathbf{Z}_{p\mathbf{Z}} \to A$ (19.8.3) is necessarily injective (for otherwise, as A is an integral domain, its kernel would be the maximal ideal of $\mathbf{Z}_{p\mathbf{Z}}$ and its image would be isomorphic to a field; moreover one has $p > 0$ by hypothesis, $\mathbf{Z}_{p\mathbf{Z}}$ being a field if $p = 0$); the element $p \cdot 1$ of A (identified with p) is a non-zero-divisor in A, and is contained in $\mathfrak{m}$, hence (16.3.4) and (16.3.7) there exist $m - 1$ other elements x_i ($2 \leqslant i \leqslant m$) of $\mathfrak{m}$ forming, with p, a system of parameters of A. The Cohen ring W considered at the beginning of the proof is then a discrete valuation ring whose residue field is isomorphic to k, in which p generates the maximal ideal (19.8.5), and the unique homomorphism $u : W \to A$ defined at the beginning maps p to itself. One then takes $B = W[[T_1, \ldots, T_{m-1}]]$ and considers the local homomorphism $w : B \to A$ which coincides with u in W and is such that $w(T_i) = x_i$ for $1 \leqslant i \leqslant m - 1$. In both cases, $\mathbf{0}_{\textbf{IV-1}}$ | 113
if $\mathfrak{n}$ is the maximal ideal of B, it is clear that $\mathfrak{n}A$ is an ideal of definition of A; as moreover $B/\mathfrak{n}$ and $A/\mathfrak{m}$ are isomorphic, A is a quasi-finite B-module ($\mathbf{0}_\mathbf{I}$, 7.4.4), hence a B-module of finite type since B is complete and A is separated for the $\mathfrak{n}$-preadic topologies ($\mathbf{0}_\mathbf{I}$, 7.4.1). On the other hand, in both cases one has $\dim(B) = \dim(A) = m$; in the first case, this follows from (17.1.4), (iii); in the second, one sees directly that p and the T_i ($1 \leqslant i \leqslant m - 1$) form a B-regular sequence generating $\mathfrak{n}$, or one can also use the fact that these elements generate $\mathfrak{n}$ and that $\dim(B) \geqslant \dim(A)$ by (16.3.10). As A and B are integral domains, one finally deduces from (16.3.10) that w is injective, which completes the proof. □

Corollary (19.8.9). — *Let A be a complete local noetherian integral domain containing a field k_0; let k be the residue field of A, and suppose that k is finite over k_0. Then, in the conclusion of (19.8.8), (ii), one can replace 1° and 2° by the condition that B is of the form $k_0[[T_1, \ldots, T_n]]$, the canonical injection $B \to A$ being a local k_0-homomorphism (for the usual k_0-algebra structure of B).*

PROOF. Indeed, taking up the proof of (19.8.8), (ii), one defines this time $w : k_0[[T_1, \ldots, T_n]] \to A$ as coinciding in k_0 with the identity and mapping T_j to y_j for $1 \leqslant j \leqslant n$. The hypothesis that k is of finite degree over k_0 again implies that A is a quasi-finite B-module, hence of finite type by ($\mathbf{0_I}$, 7.4.1), and one concludes as in (19.8.8). □

Corollary (19.8.10). — *Let A be a local artinian ring whose maximal ideal $\mathfrak{m}$ is of square zero; then there exists a regular local noetherian ring B, with maximal ideal $\mathfrak{n}$, such that A is isomorphic to $B/\mathfrak{n}^2$.*

PROOF. Let K be the residue field $A/\mathfrak{m}$ of A, and n the rank of $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{m}$ over K. If A contains a field, it follows from (19.6.3) that A is isomorphic to $B/\mathfrak{b}$, where $B = K[[T_1, \ldots, T_n]]$ and $\mathfrak{b}$ is contained in the square of the maximal ideal $\mathfrak{n}$ of B; but as $\mathrm{long}(B/\mathfrak{n}^2) = n + 1 = \mathrm{long}(A)$, one necessarily has $\mathfrak{b} = \mathfrak{n}^2$.

Suppose next that A does not contain a field; this implies that K is of characteristic $p > 0$ and that $p \cdot 1 \neq 0$ in A (19.6.3); hence $p \cdot 1$ is an element of $\mathfrak{m}$, and there are therefore $n - 1$ other elements x_i ($2 \leqslant i \leqslant n$) of $\mathfrak{m}$ forming with $p \cdot 1$ a basis of $\mathfrak{m}$ over K. Let W be a Cohen ring whose residue field is isomorphic to K; W is a discrete valuation ring in which p generates the maximal ideal; one has seen in the proof of (19.8.8) that there is a homomorphism $u : W \to A$ mapping p to itself and which by passage to quotients gives the identity on K. One takes $B = W[[T_2, \ldots, T_n]]$ and considers the local homomorphism $w : B \to A$ which coincides with u in W and is such that $w(T_i) = x_i$ for $2 \leqslant i \leqslant n$. It is clear that w is surjective and that its kernel $\mathfrak{b}$ is contained in the square of the maximal ideal $\mathfrak{n} = pB + BT_2 + \ldots + BT_n$ of B; as $\mathrm{long}(B/\mathfrak{n}^2) = n + 1 = \mathrm{long}(A)$, one again has $\mathfrak{b} = \mathfrak{n}^2$. □

Proposition (19.8.11). — *Let A be a local artinian ring, $\mathfrak{m}$ its maximal ideal, k its residue field. For A to be isomorphic to a quotient ring of a Cohen ring, it is necessary and sufficient that $\mathfrak{m}$ be generated by $p \cdot 1$, where p is the characteristic of k.*

PROOF. The condition is evidently necessary (19.8.5). To see that it is sufficient, one observes, as at the beginning of the proof of (19.8.8), that there exists a Cohen ring W whose residue field is isomorphic to k and a local homomorphism $u : W \to A$. Moreover, if one considers the composed homomorphism $\mathbf{Z}_{p\mathbf{Z}} \to W \xrightarrow{u} A$ (which is necessarily the unique homomorphism of $\mathbf{Z}_{p\mathbf{Z}}$ into A), one sees that the image under u of the element $p \cdot 1$ of W is the element $p \cdot 1$ of A; as the element $p \cdot 1$ of W generates the maximal ideal of this ring, one deduces immediately from the hypothesis that $\mathrm{gr}(u) : \mathrm{gr}_\bullet(W) \to \mathrm{gr}_\bullet(A)$ is surjective, and therefore so is u (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 2 of th. 1). □ $\mathbf{0_{IV\text{-}1}}$ | 114

19.10. Relatively formally smooth algebras

Definition (19.9.1). — Let Λ be a topological ring, A a topological Λ-algebra, B a topological A-algebra. We say that B is a *formally smooth A-algebra relative to* Λ if, for every discrete topological A-algebra C, and every nilpotent ideal $\mathfrak{J}$ of C, every continuous A-homomorphism $u_0 : B \to C/\mathfrak{J}$ which factors as $B \xrightarrow{u} C \to C/\mathfrak{J}$, where u is a continuous Λ-homomorphism, also factors as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism.

It follows from this definition that if B is a *formally smooth* A-algebra, then B is also formally smooth relative to Λ, for every structure of topological Λ-algebra defined on A (in other words, every continuous ring homomorphism $\Lambda \to A$).

Proposition (19.9.2). — *Let Λ be a topological ring, A a topological Λ-algebra.*

(i) *A is a Λ-algebra formally smooth relative to Λ.*
(ii) *If B is an A-algebra formally smooth relative to Λ and C is a B-algebra formally smooth relative to Λ, then C is an A-algebra formally smooth relative to Λ.*
(iii) *Let B be an A-algebra formally smooth relative to Λ, A' a topological A-algebra, then the topological A'-algebra $B \otimes_A A'$ is formally smooth relative to Λ.*
(iv) *Let B be a topological A-algebra, S (resp. T) a multiplicative subset of A (resp. B) such that the canonical image of S in B is contained in T. If B is an A-algebra formally smooth relative to Λ, then $T^{-1}B$ is an $S^{-1}A$-algebra formally smooth relative to Λ.*
(v) *Let B_i ($1 \leqslant i \leqslant n$) be topological A-algebras. For $\prod_{i=1}^n B_i$ to be an A-algebra formally smooth relative to Λ, it is necessary and sufficient that each of the B_i be so.*

PROOF. Assertion (i) is trivial, and the proof of the others closely follows the proofs of (19.3.5); it is therefore left to the reader. □

Corollary (19.9.3). — *Let A be a topological ring, A' and B two topological A-algebras. Then the topological A-algebra $B' = B \otimes_A A'$ is a B-algebra formally smooth relative to A.*

PROOF. This follows from (19.9.2), (i) and (iii). ☐

Proposition (19.9.4). — *Let Λ be a topological ring, A a topological Λ-algebra, B a topological A-algebra. The following conditions are equivalent:*

a) *B is an A-algebra formally smooth relative to Λ.*
b) *$\widehat{B}$ is an A-algebra formally smooth relative to Λ.*
c) *$\widehat{B}$ is an $\widehat{A}$-algebra formally smooth relative to Λ.*
d) *$\widehat{B}$ is an $\widehat{A}$-algebra formally smooth relative to $\widehat{\Lambda}$.*

PROOF. The proof is again left to the reader, modelled on that of (19.3.6). ☐

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(19.9.5). Similarly, the statement (19.3.8) is still valid (with the same proof) when one replaces "formally smooth" by "formally smooth relative to Λ". If in the statement of (19.3.10) one replaces "formally smooth" by "formally smooth relative to Λ", the conclusion is replaced by the following (the proof remaining essentially unchanged): *every continuous A-homomorphism $u_0 : B \to C/\mathfrak{J}$ which factors as $B \xrightarrow{u} C \to C/\mathfrak{J}$, where u is a continuous Λ-homomorphism, also factors as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism.*

(19.9.6). The criteria for formal smoothness (19.4.1) and (19.4.2) are also valid when one replaces "formally smooth" by "formally smooth relative to Λ", the proofs remaining practically unchanged.

Proposition (19.9.7). — *Let Λ be a topological ring, A a topological Λ-algebra, B a topological A-algebra. Suppose that for every discrete A-algebra C and every ideal $\mathfrak{J}$ of C such that $\mathfrak{J}^2 = 0$, every continuous A-homomorphism $u_0 : B \to C/\mathfrak{J}$ which factors as $B \xrightarrow{u} C \to C/\mathfrak{J}$, where u is a continuous Λ-homomorphism, also factors as $B \xrightarrow{v} C \to C/\mathfrak{J}$, where v is a continuous A-homomorphism. Then B is an A-algebra formally smooth relative to Λ.*

PROOF. The proof of (19.4.3) transcribes immediately. ☐

Proposition (19.9.8). — *Let Λ be a topological ring, A a topological Λ-algebra, B a topological A-algebra. For B to be an A-algebra formally smooth relative to Λ, it is necessary and sufficient that for every discrete topological B-module L, annihilated by an open ideal of B, one has (cf. 18.4.2)*

$$\mathrm{Exalcotop}_{A/\Lambda}(B, L) = 0.$$

PROOF. Using the notation of the proof of (19.4.4), it suffices to note here that one can suppose that the extension E_Λ of B_Λ is Λ-trivial; the rest of the proof is then unchanged. ☐

When Λ, A and B are discrete rings, the criterion (19.9.8) reduces to

(19.9.8.1) $$\mathrm{Exalcom}_{A/\Lambda}(B, L) = 0 \quad \text{for every } B\text{-module } L;$$

in other words, *every commutative A-extension of B by a B-module, which is also Λ-trivial, is also A-trivial.*

19.11. Formally unramified algebras and formally étale algebras

(19.10.1). We will also need two notions closely related to that of a formally smooth algebra.

Definition (19.10.2). — Let A be a topological ring, B a topological A-algebra. We say that B is a *formally unramified* A-algebra if, for every A-homomorphism $p : E \to C$ of discrete A-algebras, whose kernel is nilpotent, and every continuous A-homomorphism $u : B \to C$, there exists *at most one* continuous A-homomorphism $v : B \to E$ such that u factors as $B \xrightarrow{v} E \xrightarrow{p} C$. We say that B is a *formally étale* A-algebra if it is formally smooth and formally unramified, in other words, if, for every surjective A-homomorphism $p : E \to C$ of discrete A-algebras, whose kernel is nilpotent, and for every continuous A-homomorphism $u : B \to C$, there exists *one and only one* continuous A-homomorphism $v : B \to E$ such that u factors as $B \xrightarrow{v} E \xrightarrow{p} C$.

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Examples (19.10.3). — We restrict here to *discrete* rings.

(i) If the structural homomorphism $\rho : A \to B$ is *surjective*, B is a *formally unramified* A-algebra, for with the notation of (19.10.2), the composed homomorphism $A \xrightarrow{j} B \xrightarrow{v} E$ must be the structural homomorphism, hence v is unique (if it exists). On the other hand, B is not formally smooth unless ρ is *bijective*.

(ii) Let A be a ring, S a multiplicative subset of A; then $B = S^{-1}A$ is a *formally étale* A-algebra. Indeed, let $j : A \to B$ be the canonical homomorphism, $\rho : A \to E$ a ring homomorphism, $\mathfrak{J}$ a nilpotent ideal of E, $p : E \to C = E/\mathfrak{J}$ the canonical homomorphism. If $u : B \to C$ is an A-homomorphism, $u(s/1)$ is invertible in C for $s \in S$ since $s/1$ is invertible in B; but $u(s/1) = p(\rho(s))$, and as $\mathfrak{J}$ is nilpotent, the fact that the image of $\rho(s)$ in $E/\mathfrak{J}$ is invertible implies that $\rho(s)$ is itself invertible in E ($\mathbf{0_I}$, 7.1.12); one concludes that ρ factors in a unique way as $A \xrightarrow{j} B \xrightarrow{v} E$, and one evidently has $u \circ j = (p \circ v) \circ j$, whence $u = p \circ v$, which proves our assertion.

(iii) A finite separable extension k' of a field k is a formally smooth k-algebra (19.6.1); we will see later that for a *finite type* extension K of a field k to be a formally étale k-algebra, it is necessary and sufficient that it be formally unramified, and this is also equivalent to K being a finite and separable extension of k (21.7.4).

(iv) A polynomial algebra $A[X_\alpha]_{\alpha \in I}$ over a ring A is *formally smooth* (19.3.3) but *not formally étale* if I is non-empty, as follows immediately from the definitions.

Remark (19.10.4). — The same reasoning as in (19.4.3) shows that in order to verify that a topological A-algebra B is formally unramified, one can, in the definition (19.10.2), restrict to the case where the kernel of $p : E \to C$ is of *square zero*; one can moreover restrict to the case where $u(B) \subset p(E)$ (for otherwise there is no v factoring u) and, replacing C by $u(B)$ and E by $p^{-1}(u(B))$, suppose u and p *surjective*.

§20. Derivations and differentials

The notions introduced in this section will be taken up again in geometric form in chapter IV, $\mathbf{0_{IV\text{-}1}}$ | 116
§16, and will play an important role in the study of preschemes. Their importance in this chapter stems first from their relations with the notion of a formally smooth algebra, and in particular from the theorems (20.4.9), (20.5.7) and (20.5.12), which will be translated into geometric language in the section of chapter IV devoted to smooth morphisms, and are in constant use in the applications. On the other hand, the differential notions will serve to prove in §22 important criteria for regularity which will play an essential role in the detailed study of noetherian local rings made in §7 of chapter IV.

20.1. Derivations and algebra extensions $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 117

Proposition (20.1.1). — *Let A be a ring (not necessarily commutative), B, E, C A-rings, $p : E \to C$ an A-homomorphism whose kernel $\mathfrak{L}$ is of square zero, $u : B \to C$ an A-homomorphism. Suppose that there exists an A-homomorphism $v_0 : B \to E$ such that u factors as $B \xrightarrow{v_0} E \xrightarrow{p} C$. Then the set of A-homomorphisms $v : B \to E$ such that $u = p \circ v$ is identical to the set of maps $v_0 + D$, where $D : B \to \mathfrak{L}$ is a map satisfying the two following conditions:*

(i) *D is a homomorphism of A-bimodules.*
(ii) *For all f, g in B, one has*

$$D(fg) = f.D(g) + D(f).g. \tag{20.1.1.1}$$

Proof. To say that $p(v(f)) = p(v_0(f))$ for $f \in B$ means that $D(f) = v(f) - v_0(f)$ belongs to $\mathfrak{L}$; if one writes $v(fg) = v(f)v(g)$, one obtains the relation (20.1.1.1), $\mathfrak{L}$ being of square zero, and condition (i) follows from the fact that v and v_0 are A-homomorphisms.

If $\rho : A \to B$ is the structural homomorphism, one deduces from (20.1.1.1) that one has $D(\rho(a)f) = \rho(a)D(f) + D(\rho(a))f$ for all $a \in A$; but one must also have $D(\rho(a)f) = \rho(a)D(f)$ by (i), hence, setting $f = 1$, one obtains

$$D(\rho(a)) = 0 \qquad \text{for} \quad a \in A;$$

conversely, if D is null on $\rho(A)$ and satisfies (20.1.1.1), it also satisfies (i). □

Definition (20.1.2). — Given an A-ring B and a B-bimodule L, we call an *A-derivation* of B into L a map $D : B \to L$ satisfying conditions (i) and (ii) of (20.1.1).

It follows from (20.1.1) that the kernel of an A-derivation $D : B \to L$ is a sub-A-ring of B.

One sometimes calls a *derivation* of B into L a $\mathbf{Z}$-derivation, that is, an additive map of B into L satisfying (20.1.1.1); one then has $D(1) = 0$ for such a map. If B is an algebra over a *prime field* P, every $\mathbf{Z}$-derivation of B is a P-derivation: this is evident from the above if P is of characteristic > 0, and otherwise, the relation $n \cdot n^{-1} = 1$ for $n \in \mathbf{Z}$ gives

$$nD(n^{-1}) = 0$$

hence $D(n^{-1}) = 0$ since P is of characteristic 0.

It follows immediately from the definition (20.1.2) that if D, D' are two A-derivations of B into L, then so is $D - D'$. In other words, the set of A-derivations of B into L is endowed with a structure of *additive group*; we denote this group $\mathrm{Der}_A(B, L)$. If A is *commutative*, B a commutative *A-algebra*, and L a B-module, then, for every A-derivation D of B into L and every $a \in A$, aD is again an A-derivation of B into L, in other words $\mathrm{Der}_A(B, L)$ is then endowed with a structure of *A-module*.

Proposition (20.1.1) can be interpreted in the following way: $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 118

Corollary (20.1.3). — *Given two A-homomorphisms of A-rings $p : E \to C$, $u : B \to C$ where the first has kernel $\mathfrak{L}$ of square zero, the set of A-homomorphisms $v : B \to E$ such that $u = p \circ v$ is either empty or a principal homogeneous space for the group $\mathrm{Der}_A(B, \mathfrak{L})$.*

In particular:

Corollary (20.1.4). — *Let A be a ring, C an A-ring, L a C-bimodule, E a trivial A-extension of C by L, $p : E \to C$ the augmentation. The map which, to every derivation $D \in \mathrm{Der}_A(C, L)$, associates the map $x \mapsto x + D(p(x))$ is an isomorphism of the group $\mathrm{Der}_A(C, L)$ onto the group of A-equivalences of E with itself.*

Proof. Apply (20.1.1) for $B = E$, $v_0 = \mathrm{I}_E$: the set of A-homomorphisms $v : E \to E$ such that $p \circ v = p$ is identical to the set of maps $v = \mathrm{I}_E + D'$, where $D' \in \mathrm{Der}_A(E, L)$. To say that such an A-homomorphism v is an A-equivalence amounts to saying that $D'(x) = 0$ in L, or equivalently that D' factors as $E \xrightarrow{p} C \xrightarrow{D} L$, where D is an A-derivation. Whence the corollary. □

For trivial extensions, (20.1.3) gives:

Corollary (20.1.5). — *Let A be a ring, B, C two A-rings, L a C-bimodule, $u : B \to C$ an A-homomorphism; the map which, to every derivation $D \in \mathrm{Der}_A(B, L)$, associates the map $\alpha : x \mapsto (u(x), D(x))$ is a bijection onto the set of A-homomorphisms $v : B \to D_C(L)$ (cf. 18.2.3) such that u factors as $B \xrightarrow{v} D_C(L) \to C$.*

More particularly:

Corollary (20.1.6). — *Let A be a ring, B an A-ring, L a B-bimodule. If, to every derivation $D \in \mathrm{Der}_A(B, L)$, one associates: 1° the A-equivalence $(x, y) \mapsto (x, y + D(x))$ of the extension $D_B(L)$ with itself; 2° the A-homomorphism $x \mapsto (x, D(x))$ of B into $D_B(L)$, inverse on the right of the augmentation homomorphism $D_B(L) \to B$; one defines canonical bijections between:*

(i) *the set $\mathrm{Der}_A(B, L)$;*
(ii) *the set of A-equivalences of $D_B(L)$ with itself;*
(iii) *the set of A-homomorphisms inverse on the right of the augmentation homomorphism $D_B(L) \to B$.*

We note that the bijection thus established between (i) and (ii) respects the group structures, and that the bijection between (ii) and (iii) is none other than the bijection already defined in (18.3.8).

Corollary (20.1.7). — *Let A be a ring, C an A-ring, L a C-bimodule, E a trivial A-extension of C by L, $p : E \to C$ the augmentation. The set of A-derivations D of E into L such that $D(x) = x$ in L is identical to the set of maps $x \mapsto x - w(p(x))$, where w runs through the set of A-homomorphisms inverse on the right of p.*

Proof. It suffices to apply (20.1.1) for $B = E$, $v_0 = \mathrm{I}_E$; if $v = \mathrm{I}_E - D$, the condition $D(x) = x$ for $x \in L$ is equivalent to $v(x) = 0$ for $x \in L$, that is, v factors as $E \xrightarrow{p} C \xrightarrow{D} L$, where D is an A-derivation; the condition $p \circ v = p$ then gives $p \circ w = \mathrm{I}_C$, i.e., w is inverse on the right of p. □

20.2. Functorial properties of derivations $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 119

(20.2.1). Let A be a ring, B an A-ring, L a B-bimodule; if L' is a second B-bimodule and $w : L \to L'$ a homomorphism of B-bimodules, it is clear that the map $D \mapsto w \circ D$ is a homomorphism of additive groups

(20.2.1.1) $$w_0 : \mathrm{Der}_A(B, L) \longrightarrow \mathrm{Der}_A(B, L')$$

and if $w' : L' \to L''$ is a second homomorphism of B-bimodules, one has $(w' \circ w)_0 = w'_0 \circ w_0$. When A is commutative, B a commutative A-algebra and L a B-module, (20.2.1.1) is a homomorphism of A-*modules*.

In the second place, let B' be an A-ring, $v : B' \to B$ an A-homomorphism which makes L into a B'-bimodule; then the map $D \mapsto D \circ v$ is a homomorphism of additive groups,

(20.2.1.2) $$v^0 : \mathrm{Der}_A(B, L) \longrightarrow \mathrm{Der}_A(B', L)$$

as follows from (20.1.1.1); if $v' : B'' \to B'$ is a second A-homomorphism, one has $(v \circ v')^0 = v'^0 \circ v^0$. When A, B and B' are commutative and L a B-module, (20.2.1.2) is a homomorphism of A-modules.

Finally, let $u : A' \to A$ be a ring homomorphism making B into an A'-ring; every A-derivation $D \in \mathrm{Der}_A(B, L)$ is also an A'-derivation, whence a canonical injection of groups

(20.2.1.3) $$u^0 : \mathrm{Der}_A(B, L) \longrightarrow \mathrm{Der}_{A'}(B, L)$$

and if $u' : A'' \to A'$ is a second ring homomorphism, one has $(u \circ u')^0 = u^0 \circ u'^0$; when A, A' and B are commutative and L a B-module (20.2.1.3) is a *di*-homomorphism of modules (relative to u).

One can also say that

$$(A, B, L) \rightsquigarrow \mathrm{Der}_A(B, L)$$

is a *covariant functor* from the category $\boldsymbol{K}$ defined in (18.3.5) into the category $\boldsymbol{Ab}$ of commutative groups, by associating to every triplet (u, v, w) constituting a morphism of $\boldsymbol{K}$ the homomorphism $w_0 \circ v^0 \circ u^0$; the verification of the functoriality results from the commutativity of the diagrams

$$\begin{tikzcd} \mathrm{Der}_A(B, L) \arrow[r, "v^0"] \arrow[d, "w_0"'] & \mathrm{Der}_A(B', L) \arrow[d, "w_0"] \\ \mathrm{Der}_A(B, L') \arrow[r, "v^0"] & \mathrm{Der}_A(B', L') \end{tikzcd} \qquad \begin{tikzcd} \mathrm{Der}_A(B, L) \arrow[r, "u^0"] \arrow[d, "w_0"'] & \mathrm{Der}_{A'}(B, L) \arrow[d, "w_0"] \\ \mathrm{Der}_A(B, L') \arrow[r, "u^0"] & \mathrm{Der}_{A'}(B, L') \end{tikzcd}$$

for every homomorphism $w : L \to L'$ of B-bimodules.

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Theorem (20.2.2). — *Let $u : A \to B$, $v : B \to C$ be two ring homomorphisms, L a C-bimodule. There is a canonical exact sequence of commutative groups*

$$(20.2.2.1)\qquad 0 \longrightarrow \mathrm{Der}_B(C,L) \xrightarrow{u^0} \mathrm{Der}_A(C,L) \xrightarrow{v^0} \mathrm{Der}_A(B,L) \xrightarrow{\partial}$$
$$\longrightarrow \mathrm{Exan}_B(C,L) \xrightarrow{u^1} \mathrm{Exan}_A(C,L) \xrightarrow{v^1} \mathrm{Exan}_A(B,L)$$

where u^0, v^0 are the homomorphisms (20.2.1.3) *and* (20.2.1.2) *respectively, u^1, v^1 are the homomorphisms defined in (18.3.4.1) and (18.3.1) respectively, and where ∂ is defined as follows: for every derivation D of B into L, $\partial(D)$ is the class of the B-extension of C by L defined by the A-homomorphism $\alpha : x \mapsto (v(x), D(x))$* (*cf.* (20.1.5)). *Moreover, the exact sequence* (20.2.2.1) *is functorial in L* (*for the homomorphisms defined in* (20.2.1.1) *and* (18.1.1) *respectively*).

PROOF. As D is a derivation (and *a fortiori* a $\mathbf{Z}$-derivation) of B into L, the A-homomorphism $\alpha : x \mapsto (v(x), D(x))$ indeed defines on $D_C(L)$ a structure of B-extension, hence $\partial(D)$ is well defined (20.1.5). We must verify the exactness at the 5 places:

1) The exactness at $\mathrm{Der}_B(C,L)$ is trivial (cf. (20.2.1)).

2) By definition (20.2.1) the kernel of v^0 is the set of A-derivations of C which vanish on B, that is, those which are also B-derivations; whence the exactness at $\mathrm{Der}_A(C,L)$.

3) The kernel of ∂ is formed of the derivations $D \in \mathrm{Der}_A(B,L)$ for which the B-extension defined by $\alpha : x \mapsto (v(x), D(x))$ is B-trivial; this means (18.2.3) that there exists a B-homomorphism $z \mapsto (z, w(z))$ of C into $D_C(L)$ (the structure of B-ring of $D_C(L)$ being defined by α); but such a homomorphism, being *a fortiori* an A-homomorphism, is of the form $z \mapsto (z, D'(z))$ where $D' \in \mathrm{Der}_A(C,L)$ (20.1.6); and writing that this is a B-homomorphism, it comes

$$D(x)z + v(x)D'(z) = D'(v(x)z) = v(x)D'(z) + D'(v(x))z$$

for $x \in B$, $z \in C$, which gives $D'(v(x)) = D(x)$; the kernel of ∂ is therefore the image of v^0.

4) The kernel of u^1 is the set of classes of B-extensions of C by L which are A-trivial, hence (up to equivalence) of the form $D_C(L)$, where the structure of A-ring of $D_C(L)$ is defined by the homomorphism $t \mapsto (u(t), 0)$. Now, every structure of B-extension on $D_C(L)$ is defined by a homomorphism $\alpha : x \mapsto (v(x), D(x))$, where D is a $\mathbf{Z}$-derivation of B into L (20.1.5); to say that the structure of A-ring of this B-extension is deduced from its structure of B-ring by means of $u : A \to B$ means that $D(u(t)) = 0$ for $t \in A$, hence that D is an A-derivation, or equivalently that the class of the B-extension considered is of the form $\partial(D)$; whence the exactness at $\mathrm{Exan}_B(C,L)$.

5) The kernel of v^1 is the set of classes of A-extensions E of C by L which trivialise on $v(B)$, that is, for which there is an A-homomorphism $w : B \to E$ such that v factors as $B \xrightarrow{w} E \to C$; or an A-homomorphism defining on E a structure of B-extension whose class has for image by u^1 the class of the A-extension given; the converse being trivially true, this proves the exactness at $\mathrm{Exan}_A(C,L)$.

Finally, the functoriality in L follows trivially from the definitions. □

Corollary (20.2.3). — *Let A, B, C be three commutative rings, $u : A \to B$, $v : B \to C$ two ring homomorphisms, L a C-module. There is a canonical exact sequence of A-modules*

(20.2.3.1)

$$0 \longrightarrow \mathrm{Der}_B(C,L) \xrightarrow{u^0} \mathrm{Der}_A(C,L) \xrightarrow{v^0} \mathrm{Der}_A(B,L) \xrightarrow{\partial}$$
$$\longrightarrow \mathrm{Exalcom}_B(C,L) \xrightarrow{u^1} \mathrm{Exalcom}_A(C,L) \xrightarrow{v^1} \mathrm{Exalcom}_A(B,L)$$

functorial in L.

PROOF. The reasoning is the same as in (20.2.2) once one has verified that for every derivation $D \in \mathrm{Der}_A(B,L)$, $\partial(D)$ is indeed the class of a B-extension of C by L which is a commutative B-algebra; but this follows immediately from the commutativity of C and from the fact that L is a C-module. □

Corollary (20.2.4). — *Under the hypotheses of* (20.2.2) (*resp.* (20.2.3)), *there is a canonical exact sequence, functorial in* L,

$$0 \longrightarrow \mathrm{Der}_B(C,L) \xrightarrow{u^0} \mathrm{Der}_A(C,L) \xrightarrow{v^0} \mathrm{Der}_A(B,L) \xrightarrow{\partial} \mathrm{Exan}_{B/A}(C,L) \longrightarrow 0 \tag{20.2.4.1}$$

(*resp.*

$$0 \longrightarrow \mathrm{Der}_B(C,L) \xrightarrow{u^0} \mathrm{Der}_A(C,L) \xrightarrow{v^0} \mathrm{Der}_A(B,L) \xrightarrow{\partial} \mathrm{Exalcom}_{B/A}(C,L) \longrightarrow 0). \tag{20.2.4.2}$$

This follows from the definition of $\mathrm{Exan}_{B/A}(C,L)$ (*resp.* $\mathrm{Exalcom}_{B/A}(C,L)$) (*(18.3.7) and (18.4.2)*).

Remark (20.2.5). — Suppose that one has a commutative diagram of ring homomorphisms

$$\begin{tikzcd} A \arrow[r] & B \arrow[r] & C \\ A' \arrow[r] \arrow[u] & B' \arrow[r] \arrow[u] & C' \arrow[u] \end{tikzcd}$$

Then there is a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Der}_B(C,L) \arrow[r] \arrow[d] & \mathrm{Der}_A(C,L) \arrow[r] \arrow[d] & \mathrm{Der}_A(B,L) \arrow[r, "\partial"] \arrow[d] & \mathrm{Exan}_B(C,L) \arrow[r] \arrow[d] & \mathrm{Exan}_A(C,L) \arrow[r] \arrow[d] & \mathrm{Exan}_A(B,L) \arrow[d] \\ 0 \arrow[r] & \mathrm{Der}_{B'}(C',L) \arrow[r] & \mathrm{Der}_{A'}(C',L) \arrow[r] & \mathrm{Der}_{A'}(B',L) \arrow[r, "\partial"] & \mathrm{Exan}_{B'}(C',L) \arrow[r] & \mathrm{Exan}_{A'}(C',L) \arrow[r] & \mathrm{Exan}_{A'}(B',L) \end{tikzcd}$$

and similarly for the exact sequences (20.2.3.1), (20.2.4.1) and (20.2.4.2).

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(20.3.1). Given two topological rings A, B (linearly topologised, as always), we will denote by $\mathrm{Hom.\ cont}(A,B)$ the set of continuous homomorphisms from A to B. Given a topological ring A, if B and C are two topological A-rings, we denote $\mathrm{Hom.\ cont}_A(B,C)$ the set of continuous A-homomorphisms from B to C.

Let A be a topological ring, B a topological A-ring, L a topological B-bimodule; we denote $\mathrm{Dér.\ cont}_A(B,L)$ the set of *continuous* A-derivations of B into L; it is clear that this is a *subgroup* of $\mathrm{Der}_A(B,L)$ (and a sub-A-module when A and B are commutative and L a B-module).

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(20.3.2). It is immediate that the proposition (20.1.1) remains valid when one replaces "ring" by "topological ring" and "homomorphism" by "continuous homomorphism". Similarly, (20.1.3) and (20.1.4) remain valid by replacing Der by Dér. cont, "ring" (resp. "bimodule") by "topological ring" (resp. "topological bimodule"), "homomorphism" by "continuous homomorphism"; one must naturally suppose in (20.1.4) that p is continuous, and replace "A-equivalences" by "continuous A-equivalences". One has the same results for (20.1.5) and (20.1.6), provided one takes as topology on $D_C(L)$ (resp. $D_B(L)$) the product topology (on $C \times L$, resp. $B \times L$).

Proposition (20.3.3). — *Let A be a topological ring, B a topological A-ring, L a discrete topological B-bimodule annihilated by an open bilateral ideal of B. If in B the square of every open bilateral ideal is open, one has* $\mathrm{Dér.\ cont}_A(B,L) = \mathrm{Der}_A(B,L)$.

Proof. Indeed, if $\mathfrak{R}$ is an open bilateral ideal of B annihilating L, and D an A-derivation of B into L, one has $D(\mathfrak{R}^2) \subset \mathfrak{R}.D(\mathfrak{R}) + D(\mathfrak{R}).\mathfrak{R} = 0$ (20.1.1.1), hence D is continuous. □

(20.3.4). All the results of (20.2.1) remain valid when one replaces "ring" by "topological ring", "bimodule" by "topological bimodule", "homomorphism" by "continuous homomorphism" and Der by Dér. cont.

Proposition (20.3.5). — *Let A be a topological ring, B a topological A-ring, L a discrete topological B-bimodule annihilated by an open bilateral ideal of B. There is then a canonical isomorphism*

$$\varinjlim \mathrm{Der}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \xrightarrow{\sim} \mathrm{Dér.\ cont}_A(B,L) \tag{20.3.5.1}$$

where in the first member the inductive limit is taken over the ordered filtered set of couples $(\mathfrak{J},\mathfrak{R})$ of bilateral ideals such that $\mathfrak{R}.L = L.\mathfrak{R} = 0$, $\mathfrak{J}.B \subset \mathfrak{R}$, $B.\mathfrak{J} \subset \mathfrak{R}$.

PROOF. As $A/\mathfrak{J}$ and $B/\mathfrak{R}$ are discrete, one has canonical homomorphisms $w_{\mathfrak{J},\mathfrak{R}} : \mathrm{Der}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \to$ Dér. $\mathrm{cont}_A(B, L)$ forming an inductive system (20.3.4), whence the homomorphism (20.3.5.1) by passage to the inductive limit. As the homomorphism $B/\mathfrak{R}' \to B/\mathfrak{R}$ is surjective for $\mathfrak{R} \supset \mathfrak{R}'$, it follows immediately from the definition that the homomorphism $\mathrm{Der}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \to \mathrm{Der}_{A/\mathfrak{J}}(B/\mathfrak{R}', L)$ (with $\mathfrak{R}.L = L.\mathfrak{R} = 0$, $\mathfrak{J}.B \subset \mathfrak{R}$, $B.\mathfrak{J} \subset \mathfrak{R}'$) is injective, and it follows similarly that the homomorphism $\mathrm{Der}_{A/\mathfrak{J}}(B/\mathfrak{R}, L) \to \mathrm{Der}_{A/\mathfrak{J}'}(B/\mathfrak{R}, L)$ for $\mathfrak{J}' \subset \mathfrak{J}$ (20.2.1) is injective; one concludes that the homomorphism (20.3.5.1) is injective. On the other hand, if D is a continuous A-derivation of B into L, its kernel contains an open bilateral ideal $\mathfrak{R}_0$ of B, and if $\mathfrak{J}_0$ is an open bilateral ideal of A such that $\mathfrak{J}_0 B \subset \mathfrak{R}_0$ and $B\mathfrak{J}_0 \subset \mathfrak{R}_0$, it is clear that D is the canonical image of an $(A/\mathfrak{J}_0)$-derivation of $B/\mathfrak{R}_0$ into L, hence (20.3.5.1) is surjective. □

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Proposition (20.3.6). — *Let $u : A \to B$, $v : B \to C$ be two continuous homomorphisms of topological rings, L a discrete C-bimodule annihilated by an open bilateral ideal of C. There is a canonical exact sequence*

(20.3.6.1)
$$0 \longrightarrow \text{Dér. cont}_B(C, L) \xrightarrow{u^0} \text{Dér. cont}_A(C, L) \xrightarrow{v^0} \text{Dér. cont}_A(B, L) \xrightarrow{\partial}$$
$$\longrightarrow \mathrm{Exantop}_B(C, L) \xrightarrow{u^1} \mathrm{Exantop}_A(C, L) \xrightarrow{v^1} \mathrm{Exantop}_A(B, L)$$

where ∂ is defined by passage to the inductive limit from the homomorphism ∂ of (20.2.2.1); this exact sequence is functorial in L (in the category of discrete C-bimodules annihilated by open bilateral ideals).

PROOF. This follows from the exactness of the functor $\varinjlim$, from (20.2.2). □

Corollary (20.3.7). — *Let A, B, C be three commutative topological rings, $u : A \to B$, $v : B \to C$ two continuous homomorphisms, L a discrete C-module annihilated by an open ideal of C. There is a canonical exact sequence of A-modules, functorial in L,*

(20.3.7.1)
$$0 \longrightarrow \text{Dér. cont}_B(C, L) \xrightarrow{u^0} \text{Dér. cont}_A(C, L) \xrightarrow{v^0} \text{Dér. cont}_A(B, L) \xrightarrow{\partial}$$
$$\longrightarrow \mathrm{Exalcotop}_B(C, L) \xrightarrow{u^1} \mathrm{Exalcotop}_A(C, L) \xrightarrow{v^1} \mathrm{Exalcotop}_A(B, L).$$

Corollary (20.3.8). — *Under the hypotheses of* (20.3.5) (*resp.* (20.3.6)) *there is a canonical exact sequence, functorial in L*

(20.3.8.1)
$$0 \longrightarrow \text{Dér. cont}_B(C, L) \xrightarrow{u^0} \text{Dér. cont}_A(C, L) \xrightarrow{v^0} \text{Dér. cont}_A(B, L) \longrightarrow \mathrm{Exantop}_{B/A}(C, L) \longrightarrow 0$$

(*resp.*

(20.3.8.2)
$$0 \longrightarrow \text{Dér. cont}_B(C, L) \xrightarrow{u^0} \text{Dér. cont}_A(C, L) \xrightarrow{v^0} \text{Dér. cont}_A(B, L) \longrightarrow \mathrm{Exalcotop}_{B/A}(C, L) \longrightarrow 0).$$

We leave to the reader the task of writing the diagrams analogous to those of (20.2.5).

20.4. Principal parts and differentials In the remainder of this section and in the three following sections, all rings are assumed *commutative*. $\mathbf{0_{IV\text{-}1}}$ | 119

(20.4.1). Let A be a topological ring, B a topological A-algebra; the A-algebra $B \otimes_A B$ will be equipped with the *tensor product* topology, which in fact makes it into a topological A-algebra; we denote by p (or $p_{B/A}$) the surjective continuous A-homomorphism

(20.4.1.1)
$$p : B \otimes_A B \longrightarrow B$$

such that $p(b \otimes b') = bb'$; it is immediate that p is continuous. The *kernel* of p will be denoted $\mathfrak{J}_{B/A}$ (or simply $\mathfrak{J}$ if there is no risk of confusion). We denote by $j_1 : B \to B \otimes_A B$ and $j_2 : B \to B \otimes_A B$ the two canonical A-homomorphisms, such that

$$j_1(b) = b \otimes 1, \qquad j_2(b) = 1 \otimes b$$

which are continuous.

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Definition (20.4.2). — The *augmented B-algebra of principal parts of order* 1 of B over A is the topological A-algebra quotient

$$\mathrm{P}^1_{B/A} = (B \otimes_A B)/\mathfrak{J}^2 \tag{20.4.2.1}$$

equipped with the structure of B-algebra defined by the homomorphism $\bar{j}_1 : B \to \mathrm{P}^1_{B/A}$ (deduced from j_1 by composition with the canonical map $B \otimes_A B \to \mathrm{P}^1_{B/A}$), and with the augmentation of B-algebra $\varepsilon : \mathrm{P}^1_{B/A} \to B$ (also denoted $\varepsilon_{B/A}$) deduced from p by passage to the quotient. As $p(b \otimes 1) = b$ by definition, it is clear that ε is indeed an augmentation of B-algebra.

Definition (20.4.3). — The kernel of the augmentation $\varepsilon : \mathrm{P}^1_{B/A} \to B$,

$$\Omega^1_{B/A} = \mathfrak{J}_{B/A}/(\mathfrak{J}_{B/A})^2 \tag{20.4.3.1}$$

equipped with the topology induced by that of $\mathrm{P}^1_{B/A}$, which in fact makes it into a topological B-module, is called the B-module of 1-*differentials* (or simply *differentials*) of B over A.

We note that the topology of $\Omega^1_{B/A}$ is also the quotient topology of the topology induced on $\mathfrak{J}_{B/A}$ by that of $B \otimes_A B$ (Bourbaki, *Top. gén.*, chap. III, 3e éd., § 2, n° 7, prop. 20). If B is discrete it follows that so is $\Omega^1_{B/A}$. We denote $\widehat{\Omega}^1_{B/A}$ the *separated completion* of the topological B-module $\Omega^1_{B/A}$.

Every topological ring B can be considered as a $\mathbf{Z}$-algebra topological (where $\mathbf{Z}$ is equipped with the discrete topology), so one can define the topological B-module $\Omega^1_{B/\mathbf{Z}}$, which is sometimes also called the B-module of *absolute differentials* over B and which one denotes Ω^1_B. If B is a topological algebra over a prime field P (discrete), one has $B \otimes_P B = B \otimes_{\mathbf{Z}} B$, hence $\Omega^1_{B/P} = \Omega^1_B$.

Lemma (20.4.4). — *Let A be a ring, B an A-algebra. The ideal $\mathfrak{J}_{B/A}$ of $B \otimes_A B$ is generated by the elements $1 \otimes s - s \otimes 1$, where s runs through a set of generators of the A-algebra B.*

Proof. It is clear that for every $x \in B$, one has $x \otimes 1 - 1 \otimes x \in \mathfrak{J}$; on the other hand, for all x, y in B, one has $x \otimes y = xy \otimes 1 + (x \otimes 1)(1 \otimes y - y \otimes 1)$. If $\sum_i (x_i \otimes y_i) \in \mathfrak{J}$, one has by definition $\sum_i x_i y_i = 0$, hence

$$\sum_i (x_i \otimes y_i) = \sum_i (x_i \otimes 1)(1 \otimes y_i - y_i \otimes 1) \tag{20.4.4.1}$$

which proves that $\mathfrak{J}$ is generated by the elements $1 \otimes x - x \otimes 1$. Moreover, if $x = st$, one has

$$x \otimes 1 - 1 \otimes x = (s \otimes 1)(t \otimes 1 - 1 \otimes t) + (s \otimes 1 - 1 \otimes s)(1 \otimes t) \tag{20.4.4.2}$$

which immediately completes the proof by induction. □

Proposition (20.4.5). — *Let A be a topological ring, B a topological A-algebra. The topology of $\Omega^1_{B/A}$ is less fine than the topology deduced from that of B (19.0.2); if in B the square of every open ideal is open, these two topologies are identical.* $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 125

Proof. The first assertion is trivial, the tensor product topology on $B \otimes_A B$ being less fine than the topology deduced from that of B; *a fortiori* the topology induced on $\mathfrak{J} = \mathfrak{J}_{B/A}$ is less fine than the topology deduced from that of B. To prove the second assertion, write, by abuse of notation, $M \otimes N$ for the submodule $\mathrm{Im}(M \otimes_A N)$ for two sub-A-modules M, N of B. Using the relation

$$(xy) \otimes 1 - x \otimes (yz) = (x \otimes 1)(1 \otimes z)(y \otimes 1 - 1 \otimes y) + xz \cdot (y \otimes 1 - 1 \otimes y) - x \cdot (z \otimes 1 - 1 \otimes z)(y \otimes 1 - 1 \otimes y)$$

in the B-module $B \otimes_A B$ (defined by j_1), and taking into account (20.4.4), one sees immediately that, if $\mathfrak{K}$ is an ideal of B, one has

$$(\mathfrak{K}^2 \otimes B + B \otimes \mathfrak{K}^2) \cap \mathfrak{J} \subset \mathfrak{K}\mathfrak{J} + \mathfrak{J}^2. \tag{20.4.5.1}$$

Now, there is a fundamental system of neighbourhoods of 0 in $\mathfrak{J}$ (for the topology induced by that of $B \otimes_A B$) obtained by taking for neighbourhoods of 0 the sets $(\mathfrak{K} \otimes B + B \otimes \mathfrak{K}) \cap \mathfrak{J}$, where $\mathfrak{K}$ runs through the set of open ideals. As the topology of $\Omega^1_{B/A}$ deduced from $\mathfrak{J}$ is the quotient by $\mathfrak{J}^2$ of the topology of $\mathfrak{J}$, and the topology of B deduced from the open ideals of B and the relation (20.4.5.1) complete the proof of the second assertion. □

Definition (20.4.6). — Let p_1 and p_2 be the composed maps $B \xrightarrow{j_1} B \otimes_A B \to \mathrm{P}^1_{B/A}$, $B \xrightarrow{j_2} B \otimes_A B \to \mathrm{P}^1_{B/A}$, which are continuous A-homomorphisms such that $\varepsilon \circ p_1 = \varepsilon \circ p_2 = \mathrm{I}_B$. The continuous A-module homomorphism

(20.4.6.1) $$d_{B/A} = p_1 - p_2 : B \longrightarrow \Omega^1_{B/A}$$

is called the *exterior differential* of B relative to A; for every $x \in B$, $d_{B/A}(x)$ (also denoted dx) is called the *differential* of x (relative to A).

When $A = \mathbf{Z}$, one writes d_B instead of $d_{B/\mathbf{Z}}$; if B is an algebra over a prime field P, one has $d_B = d_{B/P}$.

Proposition (20.4.7). — *The B-module $\Omega^1_{B/A}$ is generated by the elements $d_{B/A}(x)$, where x runs through a system of generators of the A-algebra B.*

Proof. As $d_{B/A}(x)$ is the canonical image of $x \otimes 1 - 1 \otimes x$ in $\Omega^1_{B/A}$, the proposition is an immediate consequence of (20.4.4). □

Theorem (20.4.8). — *Let A be a topological ring, B a topological A-algebra topological.*

(i) *There exists a unique isomorphism of augmented topological B-algebras*

(20.4.8.1) $$\varphi : \mathrm{P}^1_{B/A} \xrightarrow{\sim} D_B(\Omega^1_{B/A})$$

which reduces to the identity on $\Omega^1_{B/A}$.

(ii) *The homomorphism $d_{B/A}$ is an A-derivation of B into $\Omega^1_{B/A}$, having the following universal property: for every topological B-module L, the map $u \mapsto u \circ d_{B/A}$ is an isomorphism of A-modules*

(20.4.8.2) $$\mathrm{Hom.\ cont}_B(\Omega^1_{B/A}, L) \xrightarrow{\sim} \mathrm{Dér.\ cont}_A(B, L).$$

Proof. (i) It is immediate that φ (using the notation of (20.4.6)) is necessarily the map $z \mapsto$ $\mathbf{0}_{\textbf{IV-1}}$ | 126
$(\varepsilon(z), z - p_1(\varepsilon(z)))$, and the inverse isomorphism the map $(b, x) \mapsto p_1(b) + x$; these two maps being continuous, this proves the first assertion. We note that the topology of B identifies (by p_1) with the *quotient* of the topology of $B \otimes_A B$ by $\mathfrak{J}_{B/A}$.

(ii) The fact that $d_{B/A}$ is an A-derivation of B follows from the definition (20.4.6) and from (20.1.1).

To prove the universal property, recall that Dér. $\mathrm{cont}_A(B, L)$ identifies canonically with the set of continuous B-algebra homomorphisms $u : B \to D_B(L)$ such that the composed $B \xrightarrow{u} D_B(L) \xrightarrow{q} B$ is the identity (where $q(b, 0) = b$ is the projection; cf. (20.1.6) and (20.3.2)). On the other hand, by virtue of the isomorphism φ, Hom. $\mathrm{cont}_B(\Omega^1_{B/A}, L)$ identifies canonically with the set of continuous B-algebra homomorphisms $v : \mathrm{P}^1_{B/A} \to D_B(L)$ such that the composed $\mathrm{P}^1_{B/A} \xrightarrow{v} D_B(L) \xrightarrow{q} B$ is the augmentation ε. As $p_2 = p_1 - d_{B/A}$ by definition, it suffices to prove that every $u \in G$ factors as

$$\begin{tikzcd} B \otimes_A B \arrow[r, "j_2"] & B \\ \arrow[u, hook, "j_1"'] & \\ B \arrow[r, "u"'] & D_B(L) \end{tikzcd}$$

where v is a continuous B-homomorphism. Now, one already has a continuous A-algebra homomorphism $j : b \mapsto (b, 0)$ from B into $D_B(L)$, which belongs to G; by definition of the topological tensor product of topological algebras ($\mathbf{0_I}$, 7.7.6), there therefore exists a continuous A-algebra homomorphism $w : B \otimes_A B \to D_B(L)$ making the diagram

$$\begin{tikzcd} B \otimes_A B & B \arrow[l, "j_1"'] \\ B \arrow[u, hook, "j_2"] \arrow[r, "u"'] & D_B(L) \end{tikzcd}$$

commutative. One therefore has by definition $w(b \otimes 1 - 1 \otimes b) = j(b) - u(b) \in L$, and by virtue of (20.4.4), this implies $w(\mathfrak{J}) \subset L$, hence $w(\mathfrak{J}^2) = 0$; w therefore factors as $B \otimes_A B \to \mathrm{P}^1_{B/A} \xrightarrow{v} D_B(L)$, where v is a continuous A-algebra homomorphism; moreover, as $v \circ p_1 = j$ is a homomorphism of B-algebras, v is a B-algebra homomorphism by the definition of the B-algebra structure of $\mathrm{P}^1_{B/A}$; as one has by definition $u = v \circ p_2$, this completes the proof. □

Theorem (20.4.9). — *Suppose that B is a topological A-algebra topological formally smooth. Then the topological B-module $\Omega^1_{B/A}$ is* formally projective.

Proof. Indeed, the hypothesis implies that $B \otimes_A B$ (equipped with the B-algebra structure defined by j_1) is a B-algebra topological formally smooth (19.3.5), (iii), and therefore also an A-algebra topological formally smooth (19.3.5), (ii); as B is topologically isomorphic to the A-algebra quotient $(B \otimes_A B)/\mathfrak{J}$ and is a formally smooth A-algebra, the conclusion follows from (19.5.3). □

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Corollary (20.4.10). — *Suppose moreover that in B the square of every open ideal is open; then, for every open ideal $\mathfrak{R}$ of B, $\Omega^1_{B/A} \otimes_B (B/\mathfrak{R}) = \Omega^1_{B/A}/\mathfrak{R}.\Omega^1_{B/A}$ is a projective $(B/\mathfrak{R})$-module.*

Proof. Indeed, the topology of $\Omega^1_{B/A}$ is then deduced from that of B (20.4.5), and it suffices to apply (19.2.4). □

Corollary (20.4.11). — *Let A, B be two local noetherian rings, $\rho : A \to B$ a local homomorphism, making B a topological A-algebra formally smooth (for the preadic topologies). Then, for every ideal of definition $\mathfrak{b}$ of B, $\Omega^1_{B/A} \otimes_B (B/\mathfrak{b})$ is a* free *$(B/\mathfrak{b})$-module.*

Proof. Indeed, it follows from (20.4.10) that this module is projective, and as $B/\mathfrak{b}$ is an artinian ring, every projective $(B/\mathfrak{b})$-module is free ($\mathbf{0}_{\mathbf{III}}$, 10.1.3). □

Proposition (20.4.12). — *Let A be a topological ring, B a topological A-algebra. If the structural homomorphism $A \to B$ is surjective, one has $\Omega^1_{B/A} = 0$.*

Proof. Indeed, one has $\mathrm{Der}_A(B, L) = 0$ for every B-module L by virtue of (20.1.1), and the proposition follows immediately from (20.4.8). □

Examples (20.4.13). — (i) Let A be a ring, $B = A[X_\alpha]_{\alpha \in I}$ a polynomial algebra over A, whose dX_α form a *basis* of the free B-module $\Omega^1_{B/A}$. On the other hand, if L is a free B-module with basis $(z_\alpha)_{\alpha \in I}$ whose index set I is the same, there exists an A-homomorphism u of B into $D_B(L)$ such that $u(X_\alpha) = (X_\alpha, z_\alpha)$ for every $\alpha \in I$, hence (20.1.5) a derivation D of B into L such that $D(X_\alpha) = z_\alpha$ for every α; by virtue of (20.4.8.1), this proves that the dX_α are linearly independent.

(ii) Let A be a ring, L an A-module, B the A-algebra $D_A(L)$; then the canonical homomorphism $x \mapsto d_{B/A}x$ of L into $\Omega^1_{B/A}$ is *bijective*, for it is immediate that the B-derivations of $B = D_A(L)$ into a B-module M are the maps of the form $(b, x) \mapsto u(x)$, where $u \in \mathrm{Hom}_A(L, M)$; one concludes by (20.4.8), (ii).

(iii) Suppose that B is the product of two topological A-algebras B_1, B_2 (identified with ideals of B). Then the images of $B_1 \otimes_A B_2$ and $B_2 \otimes_A B_1$ by the homomorphism p (20.4.1.1) are zero, whence it follows immediately that $\mathrm{P}^1_{B/A}$ identifies with the *product* (topological)

$\mathrm{P}^1_{B_1/A} \times \mathrm{P}^1_{B_2/A}$, and that the B-module $\Omega^1_{B/A}$ is the *direct sum* (topological) of the B-modules $\Omega^1_{B_1/A}$ and $\Omega^1_{B_2/A}$ (annihilated by B_2 and B_1 respectively).

(iv) Let A, B be two integral domains such that $A \subset B$, B integral over A, and the field of fractions of B a *separable* extension of that of A. Then the B-module $\Omega^1_{B/A}$ is a *torsion* module. Indeed, for every $x \in B$, the minimal polynomial $f(T)$ of x over the field of fractions K of A belongs to $A[T]$; as x is separable over K, one has $f'(x) \neq 0$ and on the other hand from the relation $f(x) = 0$ one deduces that $f'(x) d_{B/A} x = 0$, whence our assertion by virtue of (20.4.7).

Remarks (20.4.14). — (i) We note that the B-module $\Omega^1_{B/A}$, stripped of its topology, is independent of the topologies of A and of B.

(ii) We will introduce later the "algebra of principal parts of order n" of B over A, $\mathrm{P}^n_{B/A} = (B \otimes_A B)/(\mathfrak{J}_{B/A})^{n+1}$, which is the basis of the "differential calculus of order n".

20.5. Fundamental functorial properties of $\Omega^1_{B/A}$ $\mathbf{0_{IV\text{-}1}}$ | 128

(20.5.1). In this section and the following, unless expressly stated otherwise, all rings and modules considered are assumed equipped with the *discrete* topology.

(20.5.2). Let A be a ring, B, C two A-algebras, $u : B \to C$ an A-homomorphism; there is a commutative diagram

(20.5.2.1)
$$\begin{tikzcd} B \otimes_A B \arrow[r, "u\otimes u"] \arrow[d, "p_{B/A}"'] & C \otimes_A C \arrow[d, "p_{C/A}"] \\ B \arrow[r, "u"] & C \end{tikzcd}$$

whence by passage to quotients, an A-homomorphism of algebras

(20.5.2.2)
$$u' : \mathrm{P}^1_{B/A} \longrightarrow \mathrm{P}^1_{C/A}$$

such that the diagram

$$\begin{tikzcd} \mathrm{P}^1_{B/A} \arrow[r, "u'"] & \mathrm{P}^1_{C/A} \\ B \arrow[u, "p_1"] \arrow[r, "u"] & C \arrow[u, "p_1"'] \end{tikzcd}$$

is commutative; as $u \otimes u$ maps $\mathfrak{J}_{B/A}$ into $\mathfrak{J}_{C/A}$, one obtains, by restricting u' to $\Omega^1_{B/A}$, a map

(20.5.2.3)
$$u'' : \Omega^1_{B/A} \longrightarrow \Omega^1_{C/A}$$

such that the couple (u'', u) is a *di-homomorphism* for the B-module structure of $\Omega^1_{B/A}$ and the C-module structure of $\Omega^1_{C/A}$; this last fact allows one to deduce canonically a *homomorphism of C-modules*

(20.5.2.4)
$$u_{C/B/A} : \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A}.$$

Moreover, as the diagram $\mathbf{0_{IV\text{-}1}}$ | 129

(20.5.2.5)
$$\begin{tikzcd} \mathrm{P}^1_{B/A} \arrow[r, "u'"] & \mathrm{P}^1_{C/A} \\ B \arrow[u, "p_2"] \arrow[r, "u"] & C \arrow[u, "p_2"'] \end{tikzcd}$$

is also commutative, one deduces that the diagram

(20.5.2.6)
$$\begin{tikzcd} \Omega^1_{B/A} \arrow[r, "u''"] & \Omega^1_{C/A} \\ B \arrow[u, "d_{B/A}"] \arrow[r, "u"] & C \arrow[u, "d_{C/A}"'] \end{tikzcd}$$

is commutative. Finally, if $w : C \to D$ is a second homomorphism of A-algebras, one has the transitivity property

(20.5.2.7) $$(w \circ u)_{D/B/A} = w_{D/C/A} \circ (u_{C/B/A} \otimes \mathrm{I}_D)$$

as follows from the definition.

(20.5.3). Let now A, B be two rings, $v : A \to B$ a homomorphism of rings, C a B-algebra which becomes an A-algebra by means of v; then the canonical map $v_0 : C \otimes_A C \to C \otimes_B C$ is a surjective *di-homomorphism* of algebras (relative to $v : A \to B$) such that the diagram

(20.5.3.1) $$\begin{tikzcd} C \otimes_A C \arrow[r, "v_0"] \arrow[d, "p_{C/A}"'] & C \otimes_B C \arrow[d, "p_{C/B}"] \\ C \arrow[r, "\mathrm{I}_C"] & C \end{tikzcd}$$

is commutative; by passage to quotients, one deduces a di-homomorphism of algebras

(20.5.3.2) $$v' : \mathrm{P}^1_{C/A} \longrightarrow \mathrm{P}^1_{C/B}$$

As v_0 maps $\mathfrak{J}_{C/A}$ into $\mathfrak{J}_{C/B}$, one obtains, by restricting v' to $\Omega^1_{C/A}$, a map

(20.5.3.3) $$v_{C/B/A} : \Omega^1_{C/A} \longrightarrow \Omega^1_{C/B}$$

which is a *homomorphism of C-modules*. Moreover, the diagram **$0_{\text{IV-1}}$** | 130

(20.5.3.5) $$\begin{tikzcd} \Omega^1_{C/A} \arrow[r, "v_{C/B/A}"] & \Omega^1_{C/B} \\ C \arrow[u, "d_{C/A}"] \arrow[r, "\mathrm{I}_C"] & C \arrow[u, "d_{C/B}"'] \end{tikzcd}$$

is commutative. Finally, if $s : A' \to A$ is a second ring homomorphism, one has the transitivity property

(20.5.3.6) $$(v \circ s)_{C/B/A'} = v_{C/B/A} \circ s_{C/A/A'}.$$

(20.5.4). If now one has a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r, "v"] & B' \\ A \arrow[u, "f"] \arrow[r, "u"] & A' \arrow[u, "g"'] \end{tikzcd}$$

one deduces from (20.5.2.4) and (20.5.3.3), by composition, a homomorphism of B'-modules

(20.5.4.1) $$\Omega^1_{B/A} \otimes_B B' \xrightarrow{v_{B'/B/A}} \Omega^1_{B'/A} \xrightarrow{u_{B'/A'/A}} \Omega^1_{B'/A'}$$

such that the diagram of A'-homomorphisms

(20.5.4.2) $$\begin{tikzcd} \Omega^1_{B/A} \otimes_B B' \arrow[r] & \Omega^1_{B'/A'} \\ B' \arrow[u, "d_{B/A}\otimes 1"] \arrow[r, "\mathrm{I}_{B'}"] & B' \arrow[u, "d_{B'/A'}"'] \end{tikzcd}$$

is commutative. The homomorphism (20.5.4.1) corresponds moreover to a di-homomorphism of B-modules

(20.5.4.3) $$\Omega^1_{B/A} \longrightarrow \Omega^1_{B'/A'}.$$

Proposition (20.5.5). — *If A', B are two A-algebras and $B' = B \otimes_A A'$, the canonical homomorphism* (20.5.4.1)

(20.5.5.1) $$\Omega^1_{B/A} \otimes_B B' \longrightarrow \Omega^1_{B'/A'}$$

is bijective.

PROOF. The first member of (20.5.5.1) is none other than $\Omega^1_{B/A} \otimes_A A'$. One can write $B' \otimes_{A'} B' = (B \otimes_A B) \otimes_A A'$ up to canonical isomorphism, and $p_{B'/A'}$ then identifies with $p_{B/A} \otimes \mathrm{I}_{A'}$; hence (as $p_{B/A}$ is surjective) $\mathfrak{J}_{B'/A'} = \mathrm{Im}(\mathfrak{J}_{B/A} \otimes_A A')$ and $\mathfrak{J}^2_{B'/A'} = \mathrm{Im}(\mathfrak{J}^2_{B/A} \otimes_A A')$, whence $\mathrm{P}^1_{B'/A'} = \mathrm{P}^1_{B/A} \otimes_A A'$ up to canonical isomorphism, which transforms the augmentation ideals into each other; as the A-module $\mathrm{P}^1_{B/A}$ identifies canonically with the direct sum of B and $\Omega^1_{B/A}$, one has $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 131

(20.5.5.2) $$\Omega^1_{B/A} \otimes_A A' = \Omega^1_{B'/A'}$$

by the same isomorphism, and one verifies immediately that the composite of this isomorphism and the canonical isomorphism $\Omega^1_{B/A} \otimes_B B' \xrightarrow{\sim} \Omega^1_{B/A} \otimes_A A'$ is none other than (20.5.5.1). □

(20.5.6). The canonical homomorphisms (20.5.2.4) and (20.5.3.3) give, by functoriality, for every C-module L, canonical homomorphisms

(20.5.6.1) $$\mathrm{Hom}_C(\Omega^1_{C/A}, L) \longrightarrow \mathrm{Hom}_B(\Omega^1_{B/A} \otimes_B C, L) = \mathrm{Hom}_B(\Omega^1_{B/A}, L)$$

(20.5.6.2) $$\mathrm{Hom}_C(\Omega^1_{C/B}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/A}, L).$$

In view of the commutative diagrams (20.5.2.6) and (20.5.3.5), these homomorphisms are none other (up to canonical identification) than the homomorphisms (20.2.1.2) and (20.2.1.3) respectively.

Theorem (20.5.7). — *Let $u : A \to B$, $v : B \to C$ be two ring homomorphisms.*

(i) *The sequence of C-modules*

(20.5.7.1) $$\Omega^1_{B/A} \otimes_B C \xrightarrow{u_{C/B/A}} \Omega^1_{C/A} \xrightarrow{v_{C/B/A}} \Omega^1_{C/B} \longrightarrow 0$$

is exact.

(ii) *For $v_{C/B/A}$ to be left-invertible, it is necessary and sufficient that C be a B-algebra formally smooth relative to A (for the discrete topologies (cf. (19.9.1))); in particular, it suffices that C be a B-algebra formally smooth (for the discrete topologies).*

PROOF. (i) The exactness of the sequence (20.2.4.2) shows, taking into account (20.5.6), that the sequence

$$0 \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/B}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/A}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{B/A} \otimes_B C, L)$$

is exact for every C-module L. It is known that this implies the exactness of the sequence (20.5.7.1) (Bourbaki, *Alg.*, chap. II, 3e éd., § 2, n° 1, th. 1).

(ii) By virtue of the exactness of (20.5.7.1), to say that $v_{C/B/A}$ is left-invertible means that the sequence

(20.5.7.2) $$0 \longrightarrow \Omega^1_{B/A} \otimes_B C \xrightarrow{u_{C/B/A}} \Omega^1_{C/A} \xrightarrow{v_{C/B/A}} \Omega^1_{C/B} \longrightarrow 0$$

is *exact and split*; it is known (Bourbaki, *loc. cit.*, n° 1, prop. 1) that this is equivalent to saying that for every C-module L, the sequence

$$0 \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/B}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/A}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{B/A} \otimes_B C, L) \longrightarrow 0$$

is exact; taking into account (20.5.6) and (20.2.4.2), this condition is equivalent to $\mathrm{Exalcom}_{B/A}(C, L) = 0$ for every C-module L, and the conclusion follows from (19.9.8.1). □

We note moreover that if one has a commutative diagram of ring homomorphisms $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 132

$$\begin{tikzcd} A' \arrow[r] & B' \arrow[r] & C' \\ A \arrow[r] \arrow[u] & B \arrow[r] \arrow[u] & C \arrow[u] \end{tikzcd}$$

one deduces a commutative diagram

(20.5.7.3) $$\begin{tikzcd} \Omega^1_{B/A} \otimes_B C \arrow[r] \arrow[d] & \Omega^1_{C/A} \arrow[r] \arrow[d] & \Omega^1_{C/B} \arrow[r] \arrow[d] & 0 \\ \Omega^1_{B'/A'} \otimes_{B'} C' \arrow[r] & \Omega^1_{C'/A'} \arrow[r] & \Omega^1_{C'/B'} \arrow[r] & 0 \end{tikzcd}$$

where the vertical arrows come from the di-homomorphisms (20.5.4.3).

Corollary (20.5.8). — *Suppose that the homomorphism $v : B \to C$ makes C into a B-algebra formally étale (for the discrete topologies (19.10.2)); then the homomorphism (20.5.3.3)*

$$v_{C/B/A} : \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A}$$

is bijective.

Proof. Indeed, if C is a B-algebra formally unramified for the discrete topologies, it follows from (19.10.4), (20.4.8) and (20.1.1) that one has $\mathrm{Hom}_C(\Omega^1_{C/B}, L) = 0$ for *every* C-module L, hence $\Omega^1_{C/B} = 0$ (cf. (20.7.4)); on the other hand, if C is a B-algebra formally smooth for the discrete topologies, the sequence (20.5.7.2) is exact; whence the corollary. □

Corollary (20.5.9). — *Let A be a ring, B an A-algebra, S a multiplicative subset of B; then the canonical homomorphism*

$$S^{-1}\Omega^1_{B/A} \longrightarrow \Omega^1_{S^{-1}B/A} \tag{20.5.9.1}$$

is bijective.

Proof. It suffices to apply (20.5.8) to $C = S^{-1}B$, which is a B-algebra formally étale for the discrete topologies (19.10.3), (ii). □

Taking into account (20.5.5), one can therefore write

$$\Omega^1_{S^{-1}B/S^{-1}A} = S^{-1}\Omega^1_{B/A} = \Omega^1_{S^{-1}B/A}, \tag{20.5.9.2}$$

up to canonical isomorphisms.

Corollary (20.5.10). — *If k is a field and $K = k(X_\alpha)_{\alpha\in I}$ a pure transcendental extension of k, the dX_α form a basis of the K-vector space $\Omega^1_{K/k}$.*

Proof. As K is the field of fractions of the polynomial ring $k[X_\alpha]_{\alpha\in I}$, this follows from (20.4.13), (i) and (20.5.9). □

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(20.5.11). Let A be a ring, B an A-algebra, $\mathfrak{N}$ an ideal of B, C the A-algebra quotient $B/\mathfrak{N}$, and consider the composite homomorphism of A-modules

$$\delta : \mathfrak{N} \longrightarrow B \xrightarrow{d_{B/A}} \Omega^1_{B/A} \tag{20.5.11.1}$$

where the first arrow is the canonical injection; as $d(xy) = xdy + ydx$, we see that $\delta(\mathfrak{N}^2) \subset \mathfrak{N}.\Omega^1_{B/A}$, whence, by passing to quotients, a homomorphism of A-modules

$$\delta_{C/B/A} : \mathfrak{N}/\mathfrak{N}^2 \longrightarrow \Omega^1_{B/A} \otimes_B C = \Omega^1_{B/A}/(\mathfrak{N}.\Omega^1_{B/A}). \tag{20.5.11.2}$$

But in fact, $\delta_{C/B/A}$ is a homomorphism of C-*modules*, since for $x \in B$ and $y \in \mathfrak{N}$, we have $ydx \in \mathfrak{N}.\Omega^1_{B/A}$, hence $d(xy) \equiv xdy \pmod{\mathfrak{N}.\Omega^1_{B/A}}$, which proves first that (20.5.11.2) is a homomorphism of B-modules, and as $\mathfrak{N}$ annihilates both members, this establishes our assertion.

If B' is a second A-algebra, $u : B \to B'$ an A-homomorphism, $\mathfrak{N}'$ an ideal of B' such that $u(\mathfrak{N}) \subset \mathfrak{N}'$ and $C' = B'/\mathfrak{N}'$ the quotient algebra, one has a commutative diagram

$$\begin{tikzcd} \mathfrak{N}/\mathfrak{N}^2 \arrow[r, "\delta_{C/B/A}"] \arrow[d] & \Omega^1_{B/A} \otimes_B C \arrow[d] \\ \mathfrak{N}'/\mathfrak{N}'^2 \arrow[r, "\delta_{C'/B'/A}"] & \Omega^1_{B'/A} \otimes_{B'} C' \end{tikzcd} \tag{20.5.11.3}$$

where the vertical arrows come from u (20.5.2.4).

Theorem (20.5.12). — *Let A be a ring, B an A-algebra, C the A-algebra quotient $B/\mathfrak{N}$, $u : B \to C$ the canonical homomorphism.*

(i) *One has an exact sequence of C-modules*

(20.5.12.1) $$\mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{u_{C/B/A}} \Omega^1_{C/A} \longrightarrow 0$$

where $u_{C/B/A}$ and $\delta_{C/B/A}$ are defined by (20.5.2.4) *and* (20.5.11.2) *respectively.*

(ii) *If one sets $E = B/\mathfrak{N}^2$, the canonical homomorphism* (20.5.2.4)

$$\Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{E/A} \otimes_E C$$

is bijective.

(iii) *The three following conditions are equivalent:*

a) *$\delta_{C/B/A}$ is left-invertible.*

b) *Every A-extension of C by a C-module L, whose inverse image by $u : B \to C$ is A-trivial, is itself A-trivial.*

c) *The A-algebra $E = B/\mathfrak{N}^2$ is an A-trivial extension of C by $\mathfrak{N}/\mathfrak{N}^2$.*

(iv) *There is a canonical bijective correspondence between the left inverses of $\delta_{C/B/A}$ and the right inverses of the canonical homomorphism $E \to C$.*

PROOF. (i) As u is surjective, one has $\mathrm{Der}_B(C, L) = 0$ for every C-module L by virtue of (20.1.1). The exact sequence (20.2.3.1) therefore becomes $\mathbf{0}_{\textbf{IV-1}}$ | 134

(20.5.12.2)
$$0 \longrightarrow \mathrm{Der}_A(C, L) \xrightarrow{v^0} \mathrm{Der}_A(B, L) \xrightarrow{\partial} \mathrm{Exalcom}_B(C, L) \xrightarrow{v^1} \mathrm{Exalcom}_A(C, L) \xrightarrow{u^1} \mathrm{Exalcom}_A(B, L)$$

where v denotes the homomorphism $A \to B$. Recall on the other hand that $\mathrm{Exalcom}_B(C, L)$ is canonically identified with $\mathrm{Hom}_C(\mathfrak{N}/\mathfrak{N}^2, L)$ (18.3.8); one deduces therefore from (20.5.12.2) and (20.4.8) the exact sequence

(20.5.12.3)
$$0 \longrightarrow \mathrm{Hom}_C(\Omega^1_{C/A}, L) \longrightarrow \mathrm{Hom}_C(\Omega^1_{B/A} \otimes_B C, L) \xrightarrow{\varphi} \mathrm{Hom}_C(\mathfrak{N}/\mathfrak{N}^2, L) \xrightarrow{\psi} \mathrm{Ker}(u^1) \longrightarrow 0$$

with $\varphi = \partial \circ \eta^{-1}$ and $\psi = v^1 \circ \eta$. Returning to the definitions of ∂ (20.2.2) and of η (18.3.8), one verifies immediately that φ is precisely the homomorphism $\mathrm{Hom}(\delta_{C/B/A}, 1_L)$. The existence of the exact sequence formed by the first 4 terms of (20.5.12.3) shows therefore that the sequence (20.5.12.1) is exact (Bourbaki, *Alg.*, chap. II, 3[e] éd., § 2, n° 1, th. 1).

(ii) Apply to B and the ideal $\mathfrak{N}^2$ the exact sequence (20.5.12.1), which gives

(20.5.12.4) $$\mathfrak{N}^2/\mathfrak{N}^4 \longrightarrow \Omega^1_{B/A} \otimes_B E \longrightarrow \Omega^1_{E/A} \longrightarrow 0$$

whence, by tensoring with C (considered as an E-algebra), the exact sequence

$$\mathfrak{N}^2/\mathfrak{N}^4 \otimes_E C \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{E/A} \otimes_E C \longrightarrow 0.$$

But if x, y are two elements of $\mathfrak{N}$ and ζ the class of xy modulo $\mathfrak{N}^4$, the image $\delta'(\zeta \otimes 1)$ is $(x d_{B/A}(y) + y d_{B/A}(x)) \otimes 1$; but as the images of x and of y in C are zero, on has also $d_{B/A}(xy) \otimes 1 = 0$ in $\Omega^1_{B/A} \otimes_B C$, which proves our assertion.

(iii) To say that $\delta_{C/B/A}$ is left-invertible amounts, taking into account the exactness of (20.5.12.1), to saying that the sequence

(20.5.12.5) $$0 \longrightarrow \mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{u_{C/B/A}} \Omega^1_{C/A} \longrightarrow 0$$

is *exact and split*, and it is known (Bourbaki, *loc. cit.*) that to say that $\mathrm{Ker}(u^1) = 0$ in the exact sequence (20.5.12.3) for every L, which shows the equivalence of the conditions $a)$ and $b)$ (cf. (18.3.6.2)).

The fact that $b)$ implies $c)$ comes from the fact that the inverse image by $u : B \to C$ of the A-extension E of C by $\mathfrak{N}/\mathfrak{N}^2$ is A-trivial, u being composed of the canonical homomorphisms $B \to E \to C$ (18.1.6). Conversely, $c)$ implies $b)$, since every B-extension of C by L is B-equivalent to $E \otimes_{\mathfrak{N}/\mathfrak{N}^2} L$ (18.3.8), otherwise said its class is the image of the class of E (considered as B-extension) by the homomorphism $w_* : \mathrm{Exan}_B(C, \mathfrak{N}/\mathfrak{N}^2) \to \mathrm{Exan}_B(C, L)$ corresponding to a B-homomorphism $\mathbf{0}_{\textbf{IV-1}}$ | 135

$w : \mathfrak{N}/\mathfrak{N}^2 \to L$. The fact that $c)$ implies $b)$ then results from the commutativity of the diagram (18.3.6.5)

$$\begin{array}{ccc} \mathrm{Exan}_B(C, \mathfrak{N}/\mathfrak{N}^2) & \xrightarrow{w_*} & \mathrm{Exan}_B(C, L) \\ {\scriptstyle w^*}\downarrow & & \downarrow{\scriptstyle w^*} \\ \mathrm{Exan}_A(C, \mathfrak{N}/\mathfrak{N}^2) & \xrightarrow{w_*} & \mathrm{Exan}_A(C, L) \end{array}$$

(iv) We have seen (20.1.7) that the right inverses of $E \to C$ correspond bijectively, in a canonical fashion, to the set of elements $D \in \mathrm{Der}_A(E, \mathfrak{N}/\mathfrak{N}^2)$ such that $D(x) = x$ in $\mathfrak{N}/\mathfrak{N}^2$, hence also, by (20.4.8), to the set of E-homomorphisms $h : \Omega^1_{E/A} \to \mathfrak{N}/\mathfrak{N}^2$ such that the composite $\mathfrak{N}/\mathfrak{N}^2 \xrightarrow{d_{E/A}} \Omega^1_{E/A} \xrightarrow{h} \mathfrak{N}/\mathfrak{N}^2$ is the identity. By tensorisation with C, one deduces that the composite

$$\mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{h \otimes 1} \mathfrak{N}/\mathfrak{N}^2$$

is the identity; or, since $\mathfrak{N}/\mathfrak{N}^2$ is a C-module, $h \mapsto h \otimes 1$ is an isomorphism of the set $\mathrm{Hom}_E(\Omega^1_{E/A}, \mathfrak{N}/\mathfrak{N}^2)$ onto $\mathrm{Hom}_C(\Omega^1_{E/A} \otimes_E C, \mathfrak{N}/\mathfrak{N}^2)$; and by (ii) elsewhere one can canonically identify $\Omega^1_{E/A} \otimes_E C$ and $\Omega^1_{B/A} \otimes_B C$, $\delta_{C/E/A}$ being identified with $\delta_{C/B/A}$. □ □

Example (20.5.13). — Let $B = A[X_\alpha]_{\alpha \in I}$ be a polynomial algebra over A, $\mathfrak{N}$ an ideal of B and $C = B/\mathfrak{N}$; we know that $\Omega^1_{B/A}$ is a free B-module whose dX_α form a basis (20.4.13), (i), hence the dX_α also form a basis of the free C-module $\Omega^1_{B/A} \otimes_B C$. On the other hand, from the definition it results immediately that the image of $\mathfrak{N}/\mathfrak{N}^2$ by $\delta_{C/B/A}$ is the sub-C-module generated by the

$$dP_\lambda = \sum_\alpha \frac{\partial P_\lambda}{\partial X_\alpha} dX_\alpha.$$

One concludes that $\Omega^1_{C/A}$ is isomorphic to the quotient of the free C-module with basis the dX_α by the sub-C-module generated by the dP_λ, which gives a description of the module of differentials of an *arbitrary* A-algebra C that can always be obtained in the above manner.

Corollary (20.5.14). — *If C is an A-algebra formally smooth (for the discrete topologies), the sequence*

(20.5.14.1) $$0 \longrightarrow \mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{u_{C/B/A}} \Omega^1_{C/A} \longrightarrow 0$$

is exact and split.

PROOF. Indeed, every A-extension of C by a C-module is then trivial (19.4.1). □

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Remark (20.5.15). — Let $u : A \to B$ be a *surjective* homomorphism of rings; then, for every homomorphism of rings $v : B \to C$, the canonical homomorphism

(20.5.15.1) $$u_{C/B/A} : \Omega^1_{C/A} \longrightarrow \Omega^1_{C/B}$$

is *bijective*; this indeed results from the exact sequence (20.5.7.1), since $\Omega^1_{B/A} = 0$ (20.4.12).

20.6. Modules of imperfection and characteristic homomorphisms

Definition (20.6.1). — Given two ring homomorphisms $u : A \to B$, $v : B \to C$, one calls the *module of imperfection* of the B-algebra C relative to A, and one denotes by $\Upsilon_{C/B/A}$, the cokernel of the C-module homomorphism $v_{C/B/A} : \Omega^1_{B/A} \otimes_B C \to \Omega^1_{C/A}$. One therefore has by definition (cf. (20.5.7)) the exact sequence

$$0 \longrightarrow \Upsilon_{C/B/A} \longrightarrow \Omega^1_{B/A} \otimes_B C \xrightarrow{v_{C/B/A}} \Omega^1_{C/A} \xrightarrow{u_{C/B/A}} \Omega^1_{C/B} \longrightarrow 0. \tag{20.6.1.1}$$

When $A = \mathbf{Z}$ (so that $\Omega^1_{B/A}$ and $\Omega^1_{C/A}$ are the modules of "absolute" differentials), one writes $\Upsilon_{C/B}$ instead of $\Upsilon_{C/B/\mathbf{Z}}$; when B and C are algebras over a prime field P, one has $\Upsilon_{C/B/P} = \Upsilon_{C/B}$.

Let R, S be multiplicative subsets of B and C respectively, such that the image of R is contained in S. It then results from the exact sequence (20.6.1.1) and (20.5.9) that one has

$$\Upsilon_{S^{-1}C/R^{-1}B/A} = S^{-1}\Upsilon_{C/B/A}. \tag{20.6.1.2}$$

Proposition (20.6.2). — *If C is a B-algebra formally smooth relative to A (and in particular if C is a B-algebra formally smooth), one has $\Upsilon_{C/B/A} = 0$.*

PROOF. This results from (20.5.7), (ii). □

Proposition (20.6.3). — *Let k be a field, K an extension of k. For K to be a separable extension of k, it is necessary and sufficient that $\Upsilon_{K/k} = 0$ (or equivalently, the canonical homomorphism $\Omega^1_k \otimes_k K \to \Omega^1_K$ is injective, or again (20.4.8), every derivation of k in K extends to a derivation of K in K).*

PROOF. Indeed, it is equivalent to say that K is separable over k or that K is a k-algebra formally smooth (19.6.1); on the other hand, if P is the prime field of k, K is separable over P, hence also a P-algebra formally smooth, and it follows that K is a k-algebra formally smooth relative to P. Finally, to say that K is a k-algebra formally smooth relative to P amounts, by (20.5.7), (ii), to saying that the homomorphism $v_{K/k/P} : \Omega^1_k \otimes_k K \to \Omega^1_K$ is left-invertible; but as K is a field, this last condition is equivalent to saying that the kernel of $v_{K/k/P}$, that is to say $\Upsilon_{K/k/P}$, is zero. □

(20.6.4). Consider a commutative diagram of ring homomorphisms

$$\begin{tikzcd} A' \arrow[r, "u'"] & B' \arrow[r, "v'"] & C' \\ A \arrow[u, "f"] \arrow[r, "u"] & B \arrow[u, "g"] \arrow[r, "v"] & C \arrow[u, "h"] \end{tikzcd} \tag{20.6.4.1}$$

The commutativity of the diagram corresponding to (20.5.7.3) implies the existence of a unique C-homomorphism

$$\Upsilon_{C/B/A} \longrightarrow \Upsilon_{C'/B'/A'} \tag{20.6.4.2}$$

deduced canonically from (20.6.4.1) and rendering commutative the diagram

$$\begin{tikzcd} 0 \arrow[r] & \Upsilon_{C/B/A} \arrow[r] \arrow[d] & \Omega^1_{B/A} \otimes_B C \arrow[r, "v_{C/B/A}"] \arrow[d] & \Omega^1_{C/A} \arrow[r, "u_{C/B/A}"] \arrow[d] & \Omega^1_{C/B} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \Upsilon_{C'/B'/A'} \arrow[r] & \Omega^1_{B'/A'} \otimes_{B'} C' \arrow[r, "v_{C'/B'/A'}"] & \Omega^1_{C'/A'} \arrow[r, "u_{C'/B'/A'}"] & \Omega^1_{C'/B'} \arrow[r] & 0 \end{tikzcd} \tag{20.6.4.3}$$

where the vertical arrows come from the di-homomorphisms (20.5.4.3).

The definition of (20.6.4.2) is also equivalent to that of a C'-homomorphism

$$\Upsilon_{C/B/A} \otimes_C C' \longrightarrow \Upsilon_{C'/B'/A'} \tag{20.6.4.4}$$

which, composed with the canonical homomorphism $\Upsilon_{C/B/A} \to \Upsilon_{C/B/A} \otimes_C C'$, gives back (20.6.4.2). It is clear that (20.6.4.2) enjoys an evident transitivity property, allowing us to say that $\Upsilon_{C/B/A}$ is a *functor* in the triple (A, B, C).

(20.6.5). It will be convenient for us in what follows, under the conditions of (20.6.1), to introduce a *complex* (of chains) *of C-modules* $\mathrm{K}_\bullet(C/B/A)$ whose terms are zero except in degrees 0 and 1, where one takes

(20.6.5.1)
$$\begin{cases} \mathrm{K}_0(C/B/A) = \Omega^1_{C/A} \\ \mathrm{K}_1(C/B/A) = \Omega^1_{B/A} \otimes_B C \end{cases}$$

the differential operator being $v_{C/B/A} : \mathrm{K}_1 \to \mathrm{K}_0$. This allows writing (up to canonical isomorphisms) $\Omega^1_{C/B}$ and $\Upsilon_{C/B/A}$ as the *homology modules* of this complex

(20.6.5.2)
$$\mathrm{H}_0(\mathrm{K}_\bullet(C/B/A)) = \Omega^1_{C/B}, \qquad \mathrm{H}_1(\mathrm{K}_\bullet(C/B/A)) = \Upsilon_{C/B/A}.$$

Similarly:

Proposition (20.6.6). — *Under the hypotheses of (20.6.1), for every C-module L, one has canonical C-isomorphisms*

(20.6.6.1)
$$\mathrm{H}^0(\mathrm{K}_\bullet(C/B/A), L) \simeq \mathrm{Der}_B(C, L)$$

(20.6.6.2)
$$\mathrm{H}^1(\mathrm{K}_\bullet(C/B/A), L) \simeq \mathrm{Exalcom}_{B/A}(C, L).$$

PROOF. Indeed, the complex of cochains $\mathrm{Hom}_C(\mathrm{K}_\bullet(C/B/A), L)$ is none other, by virtue of (20.4.8) and (20.5.6), than the complex

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{Der}_A(C, L) \longrightarrow \mathrm{Der}_A(B, L) \longrightarrow 0 \longrightarrow \cdots$$

where the differential operator is v^0 (with the notations of (20.2.1)). The proposition results from the exact sequence (20.2.4.2) and from the definition of the cohomology modules **$0_{\text{IV-1}}$** | 138

(20.6.6.3)
$$\mathrm{H}^i(\mathrm{K}_\bullet, L) = \mathrm{H}^i(\mathrm{Hom}_C(\mathrm{K}_\bullet, L)).$$

□

If one has a commutative diagram of ring homomorphisms

$$\begin{array}{ccccc} A' & \longrightarrow & B' & \longrightarrow & C' \\ \uparrow & & \uparrow & & \uparrow \\ A & \longrightarrow & B & \longrightarrow & C \end{array}$$

the di-homomorphisms (20.5.4.3) define a di-homomorphism of complexes of modules

(20.6.6.4)
$$\mathrm{K}_\bullet(C/B/A) \longrightarrow \mathrm{K}_\bullet(C'/B'/A')$$

and the di-homomorphisms that one deduces for the homology or the cohomology are identified, by the formulas (20.6.5.2), (20.6.6.1) and (20.6.6.2), with the di-homomorphisms already defined in (20.5.4.3), (20.6.4.2), (20.2.1) and (18.3.7.2) (taking into account, for the last, the definition of the operator ∂ in the exact sequence (20.2.4.2)).

(20.6.7). One knows that for a complex $\mathrm{K}_\bullet$ of C-modules and a C-module L, one has canonical homomorphisms

$$\alpha : \mathrm{H}^i(\mathrm{K}_\bullet, L) \longrightarrow \mathrm{Hom}_C(\mathrm{H}_i(\mathrm{K}_\bullet), L)$$

(M, IV, 6). Here, the canonical homomorphism

$$\alpha_1 : \mathrm{H}^1(\mathrm{K}_\bullet(C/B/A), L) \longrightarrow \mathrm{Hom}_C(\mathrm{H}_1(\mathrm{K}_\bullet(C/B/A)), L)$$

is defined immediately as obtained by passing to the quotient by the image of $\mathrm{Hom}_C(\mathrm{K}_0, L)$ of the restriction homomorphism

$$\mathrm{Hom}_C(\mathrm{K}_1(C/B/A), L) \longrightarrow \mathrm{Hom}_C(\mathrm{H}_1(\mathrm{K}_\bullet(C/B/A)), L)$$

since $\mathrm{H}_1(\mathrm{K}_\bullet(C/B/A))$ is none other than the kernel of $\mathrm{K}_1 \to \mathrm{K}_0$; taking into account (20.6.6.2) and (20.6.5.2), one obtains therefore a canonical C-homomorphism

(20.6.7.1)
$$\mathrm{Exalcom}_{B/A}(C, L) \longrightarrow \mathrm{Hom}_C(\Upsilon_{C/B/A}, L)$$

which can be made explicit in the following way: by virtue of (20.2.4.2), every B-extension of C by L which is A-trivial comes from the datum of an A-derivation D of B in L, hence (20.4.8) of a

C-homomorphism f of $\Omega^1_{B/A} \otimes_B C$ in L; one makes correspond to this extension the *restriction* of f to $\Upsilon_{C/B/A}$, which in fact depends only on the class of this extension and not on the choice of D.

Definition (20.6.8). — Let $u : A \to B$, $v : B \to C$ be two ring homomorphisms. For every B-extension E of C by a C-module L, which is A-trivial (18.3.7), one calls the *characteristic homomorphism* of E, and one denotes by χ_E, the C-homomorphism $\Upsilon_{C/B/A} \to L$, image of the class of E by the canonical homomorphism (20.6.7.1).

One can define the homomorphism χ_E in another way: $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 139

Proposition (20.6.9). — *Let E be a B-extension of C by a C-module L, which is A-trivial* (18.3.7)*; then the diagram*

(20.6.9.1)
$$\begin{tikzcd}
0 \arrow[r] & \Upsilon_{C/B/A} \arrow[r] \arrow[d, "\chi_E"] & \Omega^1_{B/A} \otimes_B C \arrow[r, "v_{C/B/A}"] \arrow[d, "q_{E/B/A} \otimes 1_C"] & \Omega^1_{C/A} \arrow[d, "1^1"] & \\
0 \arrow[r] & L \arrow[r, "\delta_{C/E/A}"] & \Omega^1_{E/A} \otimes_E C \arrow[r, "p_{C/E/A}"] & \Omega^1_{C/A} \arrow[r] & 0
\end{tikzcd}$$

where $q : B \to E$ defines the structure of B-extension of E and $p : E \to C$ is the augmentation, is commutative and its rows are exact.

PROOF. The bottom row of the diagram is the sequence (20.5.12.5) relative to the two homomorphisms $A \xrightarrow{q \circ u} E$ and $p : E \to C = E/L$; as p is surjective and E is an A-trivial extension, this sequence is exact and split by virtue of (20.5.12), (iii). The commutativity of the right-hand square of (20.6.9.1) results from the relation $v = p \circ q$ (20.5.2.7); the image by $q_{E/B/A} \otimes 1_C$ of the kernel $\Upsilon_{C/B/A}$ of $v_{C/B/A}$ is therefore contained in the kernel L of $p_{C/E/A}$. On the other hand, let $h : C \to E$ be an A-homomorphism that is a right inverse of p, and let $j : L \to E$ be the canonical injection, so that one has $q(b) = h(v(b)) + j(D(b))$ for $b \in B$, where D is the A-derivation of B in L defining the B-extension E; one can write $D = f \circ d_{B/A}$, where $f : \Omega^1_{B/A} \to L$ is a B-homomorphism. By virtue of (20.5.2.6), on has $q_{E/B/A}(d_{B/A}(b)) = d_{E/A}(q(b)) = d_{E/A}(h(v(b))) + d_{E/A}(j(f(d_{B/A}(b))))$ for $b \in B$. Let then $z = \sum_i d_{B/A}(b_i) \otimes c_i$, where $b_i \in B$ and $c_i \in C$, be an element of $\Omega^1_{B/A} \otimes_B C$; one has

$$(q_{E/B/A} \otimes 1_C)(z) = \sum_i d_{E/A}(h(v(b_i))) \otimes c_i + \sum_i d_{E/A}(j(f(d_{B/A}(b_i)))) \otimes c_i.$$

In the first sum, one has $d_{E/A}(h(v(b_i))) = h_{E/C/A}(d_{C/A}(v(b_i)))$, hence this sum equals $(h_{E/C/A} \otimes 1_C)(v_{C/B/A}(z))$. So if one takes $z \in \Upsilon_{C/B/A}$, this sum is zero, and it remains, by definition of $\delta_{C/E/A}$,

$$(q_{E/B/A} \otimes 1_C)(z) = \sum_i \delta_{C/E/A}(c_i f(d_{B/A}(b_i))) = \delta_{C/E/A}(f(z))$$

which proves the commutativity of the left-hand square in (20.6.9.1). □

One notes that when one only assumes that $\delta_{C/B/A}$ is *injective* (and not necessarily left-invertible), this interpretation would still allow one to *define* χ_E as the restriction of $q_{E/B/A} \otimes 1_C$ to $\Upsilon_{C/B/A}$.

Corollary (20.6.10). — *If $\mathfrak{N}$ is an ideal of B, $C = B/\mathfrak{N}$, $E = B/\mathfrak{N}^2$, and if E is a B-extension A-trivial* (18.3.7) *of C by $\mathfrak{N}/\mathfrak{N}^2$, then the characteristic homomorphism $\chi_E : \Upsilon_{C/B/A} \to \mathfrak{N}/\mathfrak{N}^2$ is bijective.*

PROOF. Indeed, in the diagram (20.6.9.1), the two right-hand vertical arrows are bijective $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 140
(20.5.12), (ii). □

Theorem (20.6.11). — *Let $u : A \to B$, $v : B \to C$ be two ring homomorphisms, L a C-module. Suppose one of the following conditions verified:*

(i) *L is an injective C-module.*
(ii) *$\Upsilon_{C/B/A}$ is a direct factor of $\Omega^1_{B/A} \otimes_B C$ and $u_{C/B/A} : \Omega^1_{C/A} \to \Omega^1_{C/B}$ is right-invertible.*

Then the homomorphism canonical (20.6.7.1)

$$\mathrm{Exalcom}_{B/A}(C, L) \longrightarrow \mathrm{Hom}_C(\Upsilon_{C/B/A}, L)$$

is bijective.

In particular, if $\Omega^1_{C/B}$ and $\Omega^1_{C/A}$ are projective C-modules, the canonical homomorphism (20.6.7.1) *is bijective.*

PROOF. The fact that each of the conditions (i), (ii) implies that (20.6.7.1) is bijective results in both cases from the definition of α_1. We note moreover that condition (ii) is *necessary and sufficient* for the homomorphism (20.6.7.1) to be bijective for *every* C-module L (Bourbaki, *Alg.*, chap. II, 3e éd., §2, n° 1, prop. 1). If one assumes that $\Omega^1_{C/B}$ and $\Omega^1_{C/A}$ are projective C-modules, then, in the exact sequence (20.6.1.1), $\mathrm{Ker}(u_{C/B/A})$ is a projective C-module, since the sequence

$$0 \longrightarrow \mathrm{Ker}(u_{C/B/A}) \longrightarrow \Omega^1_{C/A} \longrightarrow \Omega^1_{C/B} \longrightarrow 0$$

is split, $\Omega^1_{C/B}$ being projective; as $\mathrm{Ker}(u_{C/B/A}) = \mathrm{Im}(v_{C/B/A})$, the sequence

$$0 \longrightarrow \Upsilon_{C/B/A} \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow \mathrm{Im}(v_{C/B/A}) \longrightarrow 0$$

is split. □

Corollary (20.6.12). — *Suppose that C is an A-algebra formally smooth. Then there exists a canonical homomorphism*

(20.6.12.1) $$\mathrm{Exalcom}_B(C, L) \longrightarrow \mathrm{Hom}_C(\Upsilon_{C/B/A}, L).$$

Moreover, this homomorphism is bijective if one of the conditions (i), (ii) *of* (20.6.11) *is verified.*

PROOF. Indeed, it results from the hypothesis on C that $\mathrm{Exalcom}_B(C, L) = \mathrm{Exalcom}_{B/A}(C, L)$, and the homomorphism (20.6.12.1) is none other than (20.6.7.1). □

Corollary (20.6.13). — *If C is an A-algebra formally smooth and if $\Omega^1_{C/B}$ is a projective C-module, the homomorphism* (20.6.12.1) *is bijective.*

PROOF. Indeed, one knows then that $\Omega^1_{C/A}$ is a projective C-module (20.4.9), the discrete topologies being considered. □

(20.6.14). The notations remaining the same, suppose now in addition that A, B, C are Λ-algebras and u, v are Λ-homomorphisms, which amounts to giving three ring homomorphisms $\Lambda \xrightarrow{s} A \xrightarrow{u} B \xrightarrow{v} C$.

One has therefore, apart from the module of imperfection $\Upsilon_{C/B/A}$, the modules of imperfection $\Upsilon_{B/A/\Lambda}$, $\Upsilon_{C/A/\Lambda}$ and $\Upsilon_{C/B/\Lambda}$, and one has already defined canonical C-module homomorphisms (20.6.4.2)

(20.6.14.1) $$u' : \Upsilon_{C/A/\Lambda} \longrightarrow \Upsilon_{C/B/\Lambda}$$

(20.6.14.2) $$s' : \Upsilon_{B/A/\Lambda} \longrightarrow \Upsilon_{C/B/A}.$$

As in the commutative diagram (20.5.7.3)

(20.6.14.3)
$$\begin{tikzcd}
\Omega^1_{A/\Lambda} \otimes_A B \arrow[r] \arrow[d] & \Omega^1_{B/\Lambda} \arrow[r] \arrow[d] & \Omega^1_{B/A} \arrow[r] \arrow[d] & 0 \\
\Omega^1_{A/\Lambda} \otimes_A C \arrow[r] & \Omega^1_{C/\Lambda} \arrow[r] & \Omega^1_{C/A} \arrow[r] & 0
\end{tikzcd}$$

the bottom row is formed of C-modules, one deduces by tensorisation a commutative diagram

(20.6.14.4)
$$\begin{tikzcd}
\Omega^1_{A/\Lambda} \otimes_A C \arrow[r] \arrow[d, equal] & \Omega^1_{B/\Lambda} \otimes_B C \arrow[r] \arrow[d] & \Omega^1_{B/A} \otimes_B C \arrow[r] \arrow[d] & 0 \\
\Omega^1_{A/\Lambda} \otimes_A C \arrow[r] & \Omega^1_{C/\Lambda} \arrow[r] & \Omega^1_{C/A} \arrow[r] & 0
\end{tikzcd}$$

where the first row is still exact and the left vertical arrow is the identity; if one puts

(20.6.14.5) $$\Upsilon^C_{B/A/\Lambda} = \mathrm{Ker}(u_{B/A/\Lambda} \otimes 1_C) = \mathrm{Ker}(\Omega^1_{A/\Lambda} \otimes_A C \longrightarrow \Omega^1_{B/\Lambda} \otimes_B C)$$

one sees, taking into account the definition of $\Upsilon_{C/A/\Lambda}$, that one has a unique C-homomorphism

(20.6.14.6) $$v' : \Upsilon^C_{B/A/\Lambda} \longrightarrow \Upsilon_{C/A/\Lambda}$$

rendering commutative the diagram

(20.6.14.7)
$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Y}^{C}_{B/A/\Lambda} \arrow[r] \arrow[d, "v'"] & \Omega^1_{A/\Lambda} \otimes_A C \arrow[r] \arrow[d, equal] & \Omega^1_{B/\Lambda} \otimes_B C \arrow[r] \arrow[d] & \Omega^1_{B/A} \otimes_B C \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{Y}_{C/A/\Lambda} \arrow[r] & \Omega^1_{A/\Lambda} \otimes_A C \arrow[r] & \Omega^1_{C/\Lambda} \arrow[r] & \Omega^1_{C/A} \arrow[r] & 0
\end{tikzcd}$$

whose rows are exact.

When $\Lambda = \mathbf{Z}$, one writes $\mathrm{Y}^{C}_{B/A}$ instead of $\mathrm{Y}^{C}_{B/A/\mathbf{Z}}$. If Λ is a prime field, one has $\mathrm{Y}^{C}_{B/A} = \mathrm{Y}^{C}_{B/A/\Lambda}$.

(20.6.15). To study the relations between the preceding modules, we introduce on the one hand the complex of B-modules $\mathrm{K}_\bullet(B/A/\Lambda)$ (20.6.5), and on the other hand the complexes of C-modules $\mathrm{K}_\bullet(C/A/\Lambda)$ and $\mathrm{K}_\bullet(C/B/\Lambda)$. We shall also set

(20.6.15.1)
$$\mathrm{K}^{C}_\bullet(B/A/\Lambda) = \mathrm{K}_\bullet(B/A/\Lambda) \otimes_B C.$$

On the other hand, we denote by $\mathrm{T}_\bullet(C/B/\Lambda)$ the complex of C-modules whose terms are zero except in degrees 0 and 1, and where

(20.6.15.2)
$$\mathrm{T}_0(C/B/\Lambda) = \mathrm{T}_1(C/B/\Lambda) = \Omega^1_{B/\Lambda} \otimes_B C$$

the differential operator being the identity, so that this complex is homotopic to 0; finally let us set

(20.6.15.3)
$$\mathrm{K}'_\bullet(C/A/\Lambda) = \mathrm{K}_\bullet(C/A/\Lambda) \oplus \mathrm{T}_\bullet(C/B/\Lambda).$$

By virtue of the trivial character of $\mathrm{T}_\bullet$, it is clear that one has

(20.6.15.4)
$$\mathrm{H}^i(\mathrm{K}'_\bullet, L) = \mathrm{H}^i(\mathrm{K}_\bullet, L) \quad \text{and} \quad \mathrm{H}_i(\mathrm{K}'_\bullet, L) = \mathrm{H}_i(\mathrm{K}_\bullet, L)$$

for every C-module L and every i.

(20.6.16). Let us now define an exact sequence of complexes, *split* in each degree

(20.6.16.1)
$$0 \longrightarrow \mathrm{K}^{C}_\bullet(B/A/\Lambda) \xrightarrow{j} \mathrm{K}'_\bullet(C/A/\Lambda) \xrightarrow{p} \mathrm{K}_\bullet(C/B/\Lambda) \longrightarrow 0$$

in the following way: let us denote temporarily by

$$f : \Omega^1_{A/\Lambda} \otimes_A C \longrightarrow \Omega^1_{B/\Lambda} \otimes_B C, \quad g : \Omega^1_{B/\Lambda} \otimes_B C \longrightarrow \Omega^1_{C/\Lambda}$$

the canonical homomorphisms $u_{B/A/\Lambda} \otimes 1_C$ and $v_{C/B/\Lambda}$ respectively, whose composite is $g \circ f = (v \circ u)_{C/A/\Lambda}$ (cf. (20.6.14.4)). One will take $j_1(x) = (x, f(x))$, $p_1(y,z) = z - f(y)$, $j_0(x) = (g(x), x)$, $p_0(y,z) = g(z) - y$ so that $\mathrm{Im}(j_1) = \mathrm{Ker}(p_1)$ is the graph of f, and $\mathrm{Im}(j_0) = \mathrm{Ker}(p_0)$ is the graph of g, complementary to $\{0\} \oplus \mathrm{T}_1$ and $\mathrm{K}_1(C/A/\Lambda) \oplus \{0\}$ respectively; the verification of the commutativity of the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{K}^{C}_1(B/A/\Lambda) \arrow[r, "j_1"] \arrow[d] & \mathrm{K}'_1(C/A/\Lambda) \arrow[r, "p_1"] \arrow[d] & \mathrm{K}_1(C/B/\Lambda) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{K}^{C}_0(B/A/\Lambda) \arrow[r, "j_0"] & \mathrm{K}'_0(C/A/\Lambda) \arrow[r, "p_0"] & \mathrm{K}_0(C/B/\Lambda) \arrow[r] & 0
\end{tikzcd}$$

where the vertical arrows are the differential operators, is immediate.

Theorem (20.6.17). — *One has an exact sequence of C-modules*

(20.6.17.1)
$$0 \longrightarrow \mathrm{Y}^{C}_{B/A/\Lambda} \xrightarrow{v'} \mathrm{Y}_{C/A/\Lambda} \xrightarrow{u'} \mathrm{Y}_{C/B/\Lambda} \xrightarrow{\partial} \Omega^1_{B/A} \otimes_B C \xrightarrow{v_{C/B/A}} \Omega^1_{C/A} \xrightarrow{u_{C/B/A}} \Omega^1_{C/B} \longrightarrow 0$$

where the boundary operator ∂ is the composite $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 143

(20.6.17.2)
$$\mathrm{Y}_{C/B/\Lambda} \xrightarrow{s'} \mathrm{Y}_{C/B/A} \longrightarrow \Omega^1_{B/A} \otimes_B C$$

the second arrow being the canonical injection.

PROOF. Writing the long exact homology sequence for the exact sequence of complexes (20.6.16.1), one obtains (20.6.17.1), the homology being zero in degrees distinct from 0 and 1; the fact that the homomorphisms of this exact sequence come from the functoriality of j and p is immediate. It remains to verify that ∂ is equal to (20.6.17.2); but an element $z \in \Upsilon_{C/B/\Lambda}$ is the image by p_1 of $(0, z)$, whence one deduces immediately that $\partial(z)$ is the image of z by the canonical homomorphism $s_{B/\Lambda} \otimes 1_C : \Omega^1_{B/\Lambda} \otimes_B C \to \Omega^1_{B/A} \otimes_B C$. Our assertion results from the commutativity of the diagram

(20.6.17.3)
$$\begin{tikzcd} \Upsilon_{C/B/\Lambda} \arrow[r, "s'"] & \Upsilon_{C/B/A} \\ \Omega^1_{B/\Lambda} \otimes_B C \arrow[u] \arrow[r] & \Omega^1_{B/A} \otimes_B C \arrow[u] \end{tikzcd}$$

(cf. (20.6.4.3)). □

Corollary (20.6.18). — (i) *The sequence of C-modules*

(20.6.18.1)
$$0 \longrightarrow \Upsilon^C_{B/A/\Lambda} \xrightarrow{s'} \Upsilon_{C/A/\Lambda} \xrightarrow{u'} \Upsilon_{C/B/\Lambda} \xrightarrow{s'} \Upsilon_{C/B/A} \longrightarrow 0$$

is exact.

(ii) *If B is an A-algebra formally smooth relative to Λ, one has $\Upsilon^C_{B/A/\Lambda} = 0$.*

PROOF. (i) In (20.6.17.1), the image of $\Upsilon_{C/B/A}$ by ∂ is the kernel of $v_{C/B/A}$, hence $\Upsilon_{C/B/A}$ by definition.

(ii) The hypothesis implies that the sequence

$$0 \longrightarrow \Omega^1_{A/\Lambda} \otimes_A B \longrightarrow \Omega^1_{B/\Lambda} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

is exact and split (20.5.7); by tensorisation with C, the sequence

$$0 \longrightarrow \Omega^1_{A/\Lambda} \otimes_A C \longrightarrow \Omega^1_{B/\Lambda} \otimes_B C \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow 0$$

remains therefore exact, whence our assertion. □

Corollary (20.6.19). — *Let K be a field, E, F two extensions of K such that $K \subset E \subset F$.*

(i) *If F is a separable extension of E, on has $\Upsilon_{F/E/K} = 0$ (or equivalently, the canonical homomorphism $\Omega^1_{E/K} \otimes_E F \to \Omega^1_{F/K}$ is injective).*

(ii) *Conversely, if F is a separable extension of K and if $\Upsilon_{F/E/K} = 0$, then F is a separable extension of E.*

PROOF. If P is the prime field of K, one has the exact sequence (20.6.18)

$$\Upsilon_{F/K/P} \longrightarrow \Upsilon_{F/E/P} \longrightarrow \Upsilon_{F/E/K} \longrightarrow 0.$$

If $\Upsilon_{F/E/P} = \Upsilon_{F/E} = 0$, one therefore has $\Upsilon_{F/E/K} = 0$, whence (i) by virtue of (20.6.3); conversely, if $\Upsilon_{F/E/K} = 0$ and $\Upsilon_{F/K/P} = \Upsilon_{F/K} = 0$, one has $\Upsilon_{F/E/P} = \Upsilon_{F/E} = 0$, whence (ii) by virtue of (20.6.3). □

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Corollary (20.6.20). — (i) *If K is an algebraic separable extension of k, one has $\Omega^1_{K/k} = 0$.*

(ii) *Let k be a field of characteristic 0, K an extension of k. For $\Omega^1_{K/k} = 0$, it is necessary and sufficient that K be an algebraic extension of k. In particular, for a field K of characteristic 0 to satisfy $\Omega^1_K = 0$, it is necessary and sufficient that K be an algebraic extension of* $\mathbf{Q}$ *(cf. (21.4.4) and (21.7.5)).*

PROOF. (i) For every $x \in K$, let f be the minimal polynomial of x over k; as $f'(x) \neq 0$ and $d_{K/k}(f(x)) = f'(x) d_{K/k}(x) = 0$, one has $d_{K/k}(x) = 0$ and our assertion results from (20.4.7).

(ii) There exists a pure extension L of k such that $k \subset L \subset K$ and K is an algebraic extension of L. As K is separable over L, it results from (20.6.19), (i) that the sequence

$$0 \longrightarrow \Omega^1_{L/k} \otimes_L K \longrightarrow \Omega^1_{K/k} \longrightarrow \Omega^1_{K/L} \longrightarrow 0$$

is exact, and from (i) that $\Omega^1_{K/L} = 0$. The relation $\Omega^1_{K/k} = 0$ is therefore equivalent to $\Omega^1_{L/k} = 0$, and as L is a pure extension of k, it results from (20.5.10) that the relation $\Omega^1_{L/k} = 0$ is equivalent to $L = k$. □

Remarks (20.6.21). — (i) As $\Upsilon_{B/A/\Lambda}$ is the kernel of $u_{B/A/\Lambda} : \Omega^1_{A/\Lambda} \otimes_A B \to \Omega^1_{B/\Lambda}$ and $\Upsilon^C_{B/A/\Lambda}$ the kernel of $u_{B/A/\Lambda} \otimes 1_C$, one has a canonical homomorphism

(20.6.21.1) $$\Upsilon_{B/A/\Lambda} \otimes_B C \longrightarrow \Upsilon^C_{B/A/\Lambda}.$$

This homomorphism is bijective when the sequence

(20.6.21.2) $$0 \longrightarrow \Upsilon_{B/A/\Lambda} \otimes_B C \longrightarrow \Omega^1_{A/\Lambda} \otimes_A C \longrightarrow \Omega^1_{B/\Lambda} \otimes_B C \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow 0$$

is exact, which occurs in the following cases:

1° C is a flat B-module.

2° The B-modules $\Omega^1_{B/\Lambda}$ and $\Omega^1_{B/A}$ are *flat*: indeed, the kernel of $\Omega^1_{B/\Lambda} \to \Omega^1_{B/A}$ is then also flat ($\mathbf{0_I}$, 6.1.2), and the sequence (20.6.21.2) is then exact by virtue of ($\mathbf{0_I}$, 6.1.2).

(ii) Consider a commutative diagram of ring homomorphisms

$$\begin{tikzcd}
\Lambda' \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \\
\Lambda \arrow[r] \arrow[u] & A \arrow[r] \arrow[u] & B \arrow[r] \arrow[u] & C \arrow[u]
\end{tikzcd}$$

Then the definitions of (20.6.16) show that one has a commutative diagram (where the vertical arrows come from (20.6.6.4))

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{K}^C_\bullet(B/A/\Lambda) \arrow[r] \arrow[d] & \mathrm{K}'_\bullet(C/A/\Lambda) \arrow[r] \arrow[d] & \mathrm{K}_\bullet(C/B/\Lambda) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{K}^{C'}_\bullet(B'/A'/\Lambda') \arrow[r] & \mathrm{K}'_\bullet(C'/A'/\Lambda') \arrow[r] & \mathrm{K}_\bullet(C'/B'/\Lambda') \arrow[r] & 0
\end{tikzcd}$$

whence, by passage to homology, a *commutative diagram* $\mathbf{0_{IV\text{-}1}}$ | 145

(20.6.21.3) $$\begin{tikzcd}
0 \arrow[r] & \Upsilon^C_{B/A/\Lambda} \arrow[r] \arrow[d] & \Upsilon_{C/A/\Lambda} \arrow[r] \arrow[d] & \Upsilon_{C/B/\Lambda} \arrow[r] \arrow[d] & \Omega^1_{B/A} \otimes_B C \arrow[r] \arrow[d] & \Omega^1_{C/A} \arrow[r] \arrow[d] & \Omega^1_{C/B} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Upsilon^{C'}_{B'/A'/\Lambda'} \arrow[r] & \Upsilon_{C'/A'/\Lambda'} \arrow[r] & \Upsilon_{C'/B'/\Lambda'} \arrow[r] & \Omega^1_{B'/A'} \otimes_{B'} C' \arrow[r] & \Omega^1_{C'/A'} \arrow[r] & \Omega^1_{C'/B'} \arrow[r] & 0
\end{tikzcd}$$

One has an analogous commutative diagram for (20.6.18.1).

Proposition (20.6.22). — *Let $s : \Lambda \to A$, $u : A \to B$ be two ring homomorphisms, $\mathfrak{R}$ an ideal of B, C the quotient ring $B/\mathfrak{R}$. Suppose that $E = B/\mathfrak{R}^2$ is a B-extension Λ-trivial of C by $\mathfrak{R}/\mathfrak{R}^2$. One then has an exact sequence*

(20.6.22.1) $$0 \longrightarrow \Upsilon^C_{B/A/\Lambda} \xrightarrow{v'} \Upsilon_{C/A/\Lambda} \xrightarrow{\chi_E} \mathfrak{R}/\mathfrak{R}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{v_{C/B/A}} \Omega^1_{C/A} \longrightarrow 0.$$

PROOF. Indeed, as $v : B \to C$ is surjective, one has $\Omega^1_{C/B} = 0$ (20.4.12). Moreover, it results from (20.6.10) that $\Upsilon_{C/B/\Lambda}$ is canonically identified with $\mathfrak{R}/\mathfrak{R}^2$. It suffices then to apply the exact sequences (20.6.17.1) and (20.6.18.1), noting that one has a commutative diagram

$$\begin{tikzcd}
 & \Upsilon_{C/A/\Lambda} \arrow[rrr] & & & \Omega^1_{A/\Lambda} \otimes_A C & \\
\Upsilon_{C/B/\Lambda} = & \Upsilon_{C/B/A} \arrow[u, "\chi_E"] = & \mathfrak{R}/\mathfrak{R}^2 \arrow[r, "\delta_{C/B/A}"] & \Omega^1_{B/A} \otimes_B C = & \Omega^1_{E/A} \otimes_E C \arrow[u] & \\
 & \mathfrak{R}/\mathfrak{R}^2 \arrow[u, "\ell"] \arrow[rrr, "\delta_{C/B/A}"] & & & \Omega^1_{B/A} \otimes_B C \arrow[u] &
\end{tikzcd}$$

and using the commutativity of the diagram (20.6.17.3). □

One has thus made precise the *kernel* of $\delta_{C/B/A}$ in the case where there exists a ring Λ such that A is a Λ-algebra and $B/\mathfrak{N}^2$ is Λ-trivial; one observes that this is the case in particular when C is a Λ-*algebra formally smooth* (for the discrete topology).

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Corollary (20.6.23). — *Under the hypotheses of (20.6.22), suppose in addition that B is an A-algebra formally smooth. Then one has an exact sequence*

(20.6.23.1)
$$0 \longrightarrow \Upsilon_{C/A/\Lambda} \xrightarrow{\chi_E} \mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A} \longrightarrow 0.$$

PROOF. This results from (20.6.18), (ii). □

(20.6.24). When the hypotheses of (20.6.22) are satisfied, one says also that the characteristic homomorphism χ_E is the *characteristic homomorphism of the A-algebra B, relative to Λ and to the ideal $\mathfrak{N}$* (suppressing these last precisions when there is no danger of confusion); one denotes it sometimes by χ_B or $\chi_{B/A}$.

Proposition (20.6.25). — *Let $s : \Lambda \to A$, $u : A \to B$, $v : B \to C$ be three ring homomorphisms, L a C-module. One then has an exact sequence*

(20.6.25.1)
$$\begin{aligned} &0 \longrightarrow \mathrm{Der}_B(C, L) \xrightarrow{u^1} \mathrm{Der}_A(C, L) \xrightarrow{v^1} \mathrm{Der}_A(B, L) \xrightarrow{\partial} \\ &\longrightarrow \mathrm{Exalcom}_{B/\Lambda}(C, L) \xrightarrow{u^1} \mathrm{Exalcom}_{A/\Lambda}(C, L) \xrightarrow{v^1} \mathrm{Exalcom}_{A/\Lambda}(B, L) \longrightarrow 0 \end{aligned}$$

where u^1, v^1 are homomorphisms defined in (18.3.6.4) and (18.3.6.2), and ∂ is defined as in (20.2.2).

PROOF. Indeed, as the exact sequence (20.6.16.1) is split, one deduces from it an exact sequence

$$0 \longrightarrow \mathrm{Hom}_C(\mathrm{K}_\bullet(C/B/\Lambda), L) \longrightarrow \mathrm{Hom}_C(\mathrm{K}'_\bullet(C/A/\Lambda), L) \longrightarrow \mathrm{Hom}_C(\mathrm{K}^C_\bullet(B/A/\Lambda), L) \longrightarrow 0.$$

If one applies to this complex the long exact cohomology sequence, taking into account (20.6.15.4) and (20.6.6), one obtains (20.6.25.1) since one has

$$\mathrm{Hom}_C(\mathrm{K}^C_\bullet(B/A/\Lambda), L) = \mathrm{Hom}_B(\mathrm{K}_\bullet(B/A/\Lambda), L)$$

by definition; the identification of u^1 and v^1 with the homomorphisms of (18.3.4.2) results from (20.6.6.4). □

Remark (20.6.26). — In this section, the complexes $\mathrm{K}_\bullet(C/B/A)$ have appeared as a technical device intended to simplify the exposition of certain functorial behaviours. In reality, these complexes, considered as objects of the category of complexes of C-modules "up to homotopy" (that is to say where the morphisms are the classes of homotopic homomorphisms) are finer invariants than the couple formed by $\Omega^1_{C/B}$ and $\Upsilon_{C/B/A}$. When k is a prime field, and C a k-algebra formally smooth (for example a regular local ring of finite type over an extension of k) (cf. (**IV**, 6.8.6)), one can show that the complex $\mathrm{K}_\bullet(C/A/k)$ can be described solely by making intervene C and A (to the exclusion of k): one expresses C as a quotient of a polynomial algebra B over A by an ideal $\mathfrak{Q}$, and one considers the complex $\mathrm{F}_\bullet(C/A)$ with $\geqslant 2$ nonzero terms

$$\cdots \longrightarrow 0 \longrightarrow \mathfrak{Q}/\mathfrak{Q}^2 \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow 0 \longrightarrow \cdots$$

(whose homology coincides with that of $\mathrm{K}_\bullet(C/A/k)$ by virtue of (20.6.23.1)). These complexes $\mathrm{F}_\bullet(C/A)$, from the point of view of homological algebra, will play the role of a conormal sheaf for $\mathrm{Spec}(C)$ over $\mathrm{Spec}(A)$, and will be of great importance in the chapters of this work devoted to the duality of coherent sheaves and to the theorem of Riemann-Roch.

20.7. Generalizations to topological rings

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(20.7.1). It results immediately from the definitions that if, in (20.5.2) and (20.5.3), the rings A, B, C are assumed to be topological rings and the ring homomorphisms u, v continuous, then the homomorphisms $u_{C/B/A}$ and $v_{C/B/A}$ are *continuous* (one must naturally take on $\Omega^1_{B/A} \otimes_B C$ the tensor product topology).

Moreover, $u_{C/B/A} : \Omega^1_{C/A} \to \Omega^1_{C/B}$ is a *strict surjective morphism* of topological C-modules; indeed, it is also so of the canonical homomorphism $C \otimes_A C \to C \otimes_B C$, taking into account the definition of the tensor product topology; by passing to quotients one deduces that the corresponding homomorphism $\mathrm{P}^1_{C/A} \to \mathrm{P}^1_{C/B}$ is a strict surjective morphism, and $u_{C/B/A}$ is the restriction of this last to $\Omega^1_{C/A}$.

In (20.5.5), if A' and B are topological A-algebras, and if $B' = \Omega^1_{B/A} \otimes_B B'$ are equipped with the tensor product topologies, the homomorphism (20.5.5.1) is a topological isomorphism, taking into account that $\mathrm{P}^1_{B/A}$ is the same as the direct topological sum of B and $\Omega^1_{B/A}$.

Proposition (20.7.2). — *Let $u : A \to B$, $v : B \to C$ be two continuous homomorphisms of topological rings. For the continuous homomorphism $v_{C/B/A} : \Omega^1_{B/A} \otimes_B C \to \Omega^1_{C/A}$ to be formally left-invertible* (*cf.* (19.1.5)), *it is necessary and sufficient that C be a B-algebra formally smooth relative to A* (19.9.1) (*and* a fortiori *it suffices that C be a B-algebra formally smooth*).

Proof. To say that $v_{C/B/A}$ is formally left-invertible means, given that the topologies of $\Omega^1_{C/A}$ and $\Omega^1_{B/A} \otimes_B C$ are finer than those deduced from the topology of C (20.4.5), that for every C-module discrete L, annihilated by an open ideal of C, the canonical homomorphism

$$\operatorname{Hom.cont}_C(\Omega^1_{B/A} \otimes_B C, L) \longrightarrow \operatorname{Hom.cont}_C(\Omega^1_{C/A}, L)$$

is *surjective* (19.1.5); as $\operatorname{Hom.cont}_C(\Omega^1_{B/A} \otimes_B C, L) = \operatorname{Hom.cont}_C(\Omega^1_{B/A}, L)$ by definition of the tensor product topology, it amounts to the same, by virtue of (20.4.8), to say that the canonical homomorphism $\mathbf{0_{IV\text{-}1}}$ | 148

$$\operatorname{Dér.\ cont}_A(C, L) \longrightarrow \operatorname{Dér.\ cont}_A(B, L)$$

is surjective. But the exact sequence (20.3.8.2) shows that this condition is equivalent to $\operatorname{Exalcotop}_{B/A}(C, L) = 0$, that is to say precisely that C is formally smooth relative to A (19.9.8). □

Corollary (20.7.3). — *Suppose that in B and in C, the square of every open ideal is also open. For C to be a B-algebra formally smooth relative to A, it is necessary and sufficient that, if one denotes by $(\mathfrak{R}_\lambda)$ a fundamental system of neighborhoods of 0 formed of ideals of C, then for every λ, the homomorphism*

(20.7.3.1) $$v_{C/B/A} \otimes 1_{C/\mathfrak{R}_\lambda} : \Omega^1_{B/A} \otimes_B (C/\mathfrak{R}_\lambda) \longrightarrow \Omega^1_{C/A} \otimes_C (C/\mathfrak{R}_\lambda)$$

is left-invertible.

Proof. Indeed, by virtue of (19.1.1), this homomorphism is surjective, and by injectivity in virtue of (20.7.2), it is also bijective; it results that the quotient of $\Omega^1_{C/A}$ by the closure of $\operatorname{Im}(v_{C/B/A})$ would be separated and $\neq 0$ and would therefore have a discrete quotient $L \neq 0$, contradicting what one has just seen (taking into account (20.4.8)). By virtue of $v_{C/B/A}$, which is a monomorphism in virtue of (20.7.2), it is also surjective in virtue of the image of $v_{C/B/A}$ being necessarily dense in $\Omega^1_{C/A}$, otherwise the quotient of $\Omega^1_{C/A}$ by the closure of $\operatorname{Im}(v_{C/B/A})$ would be separated and $\neq 0$ (taking into account (20.4.8)); by virtue of $v_{C/B/A}$, which is a formal monomorphism by virtue of (20.7.2), it is therefore a formal bimorphism (19.1.2). □

Proposition (20.7.4). — *Let A be a topological ring, B a topological A-algebra. For B to be formally non-ramified* (19.10.2), *it is necessary and sufficient that the separated completion $\widehat{\Omega}^1_{B/A}$ be zero.*

Proof. Indeed, it results immediately from (19.10.4) and (20.1.1) that, for B to be formally non-ramified, it is necessary and sufficient that for every open ideal $\mathfrak{R}$ of B, every open ideal $\mathfrak{H}$ of A such that $\mathfrak{H}.B \subset \mathfrak{R}$ and every $(B/\mathfrak{R})$-module L, one has $\operatorname{Der}_{A/\mathfrak{H}}(B/\mathfrak{R}, L) = 0$, that is to say $\operatorname{Dér.\ cont}_A(B, L) = 0$ for every B-module discrete L annihilated by an open ideal; by virtue of (20.4.8), this is equivalent to $\operatorname{Hom.cont}_B(\Omega^1_{B/A}, L) = 0$ for every such B-module, whence the proposition.

When B is discrete, the condition of (20.7.4) is equivalent to $\Omega^1_{B/A} = 0$. □

Corollary (20.7.5). — *Let A be a ring, $\mathfrak{m}$ an ideal of A, B a topological A-algebra; equip A of the topology deduced from that of A, and set $A_0 = A/\mathfrak{m}$, $B_0 = B/\mathfrak{m}B = B \otimes_A A_0$. Then, for B to be an A-algebra formally non-ramified, it is necessary and sufficient that $\Omega^1_{B_0/A_0} = 0$ (or equivalently that B_0 is an A_0-algebra formally non-ramified).*

Proof. Indeed, one has $\Omega^1_{B/A} \otimes_B B_0 = \Omega^1_{B/A}/\mathfrak{m}.\Omega^1_{B/A} = \Omega^1_{B_0/A_0}$ by virtue of (20.5.5); to write that $\Omega^1_{B/A}$ is equal to $\Omega^1_{B/A}$ is equivalent by (20.4.5) to writing that $\mathfrak{m}.\Omega^1_{B/A} = \Omega^1_{B/A}$, whence the conclusion. □

Proposition (20.7.6). — *Let $u : A \to B$, $v : B \to C$ be two continuous homomorphisms of topological rings, and suppose that v makes C into a B-algebra formally étale; then $v_{C/B/A} : \Omega^1_{B/A} \otimes_B C \to \Omega^1_{C/A}$ is a formal bimorphism (19.1.2).*

Proof. Indeed, for every C-module discrete L annihilated by an open ideal of C, one has then (20.7.4) $\text{Dér. cont}_B(C, L) = 0$ and by virtue of (20.3.6.1) the canonical homomorphism $\text{Dér. cont}_A(C, L) \to$ $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 149
$\text{Dér. cont}_A(B, L)$ is injective; it is moreover surjective by virtue of (20.7.2), hence it is bijective; it results that the image of $v_{C/B/A}$ is necessarily dense in $\Omega^1_{C/A}$, otherwise the quotient of $\Omega^1_{C/A}$ by the closure of $\text{Im}(v_{C/B/A})$ would be separated and $\neq 0$ and would therefore have a discrete quotient $L \neq 0$, contradicting what one has just seen (taking into account (20.4.8)). By virtue of $v_{C/B/A}$, which is a formal monomorphism by virtue of (20.7.2), it is therefore a formal bimorphism (19.1.2). □

Corollary (20.7.7). — *Suppose that in B and in C, the square of every open ideal is also open. If C is a B-algebra formally étale, then, for every open ideal $\mathfrak{R}_\lambda$ of C, the homomorphism (20.7.3.1) is bijective.*

Proof. Indeed, by virtue of (19.1.1), this homomorphism is surjective, and it is injective by (20.7.3). □

Proposition (20.7.8). — *Let $u : A \to B$ be a continuous homomorphism of topological rings, $\mathfrak{R}$ an ideal of B, C the topological quotient $B/\mathfrak{R}$, $v : B \to C$ the canonical homomorphism.*

(i) *In the exact sequence (20.5.12.1)*

$$\mathfrak{R}/\mathfrak{R}^2 \xrightarrow{\delta_{C/B/A}} \Omega^1_{B/A} \otimes_B C \xrightarrow{v_{C/B/A}} \Omega^1_{C/A} \longrightarrow 0$$

the homomorphism $\delta_{C/B/A}$ is continuous and the homomorphism $v_{C/B/A}$ is a strict morphism of topological C-modules.

(ii) *For $\delta_{C/B/A}$ to be formally left-invertible (19.1.5), it is necessary and sufficient that for every C-module discrete L annihilated by an open ideal of C, the canonical homomorphism*

(20.7.8.1) $$\text{Exalcotop}_A(C, L) \longrightarrow \text{Exalcotop}_A(B, L)$$

is injective.

Proof. (i) The first assertion is evident. On the other hand, the canonical homomorphism $v \otimes v : B \otimes_A B \to C \otimes_A C$ is a strict morphism by definition of the tensor product topology, and one deduces immediately that it is the same of $v_{C/B/A}$.

(ii) To say that $\delta_{C/B/A}$ is formally left-invertible means that for every C-module discrete L annihilated by an open ideal of C, the canonical homomorphism

$$\text{Hom.cont}_C(\Omega^1_{B/A} \otimes_B C, L) \longrightarrow \text{Hom.cont}_C(\mathfrak{R}/\mathfrak{R}^2, L)$$

is surjective. But, taking into account (18.4.3) and (20.4.8), this amounts to saying that the canonical homomorphism

$$\text{Dér. cont}_A(B, L) \longrightarrow \text{Exalcotop}_B(C, L)$$

is surjective, and the conclusion results from the exact sequence (20.3.6.1). □

Corollary (20.7.9). — *If the A-algebra topological $C = B/\mathfrak{R}$ is formally smooth, the canonical homomorphism $\delta_{C/B/A}$ is formally left-invertible.*

(20.7.10). In (20.6.1), when A, B, C are topological rings and u and v continuous homomorphisms, one equips $\Upsilon_{C/B/A}$ with the topology induced by that of $\Omega^1_{B/A} \otimes_B C$; the homomorphisms (20.6.4.2) and (20.6.4.4) are then continuous provided that, in (20.6.7), on supposes L a C-module discrete annihilated by an open ideal of C, one deduces, by passage to the inductive limit, the canonical C-homomorphism

$$\text{(20.7.10.1)} \qquad \mathrm{Exalcotop}_{B/A}(C,L) \longrightarrow \mathrm{Hom.cont}_C(\Upsilon_{C/B/A},L)$$

which can be made explicit in the following way (taking into account (18.5.3.1)): for every open ideal $\mathfrak{M}$ of A, every open ideal $\mathfrak{N}$ of B such that $\mathfrak{M}B \subset \mathfrak{N}$, every open ideal $\mathfrak{P}$ of C such that $\mathfrak{N}C \subset \mathfrak{P}$ and $\mathfrak{P}.L = 0$, every $(B/\mathfrak{N})$-extension E of $C/\mathfrak{P}$ by L which is $(A/\mathfrak{M})$-trivial comes from the datum of a continuous C-homomorphism f of $\Omega^1_{B/A} \otimes_B C$ in L, and the homomorphism (20.7.10.1) makes correspond to the class of E the restriction of f to $\Upsilon_{C/B/A}$, said *characteristic homomorphism of* E and again denoted χ_E.

Proposition (20.7.11). — *Suppose that the topology of C is such that the square of every open ideal is also open. If C is a topological A-algebra formally smooth and if $\Omega^1_{C/B}$ is a formally projective C-module, one has a canonical isomorphism*

$$\text{(20.7.11.1)} \qquad \mathrm{Exalcotop}_{B}(C,L) \xrightarrow{\sim} \mathrm{Hom.cont}_C(\Upsilon_{C/B/A},L)$$

for every C-module discrete L annihilated by an open ideal of C.

PROOF. One has in effect, in the exact sequence (20.3.7.1), $\mathrm{Exalcotop}_A(C,L) = 0$ (19.4.4), hence $\mathrm{Exalcotop}_B(C,L) = \mathrm{Exalcotop}_{B/A}(C,L)$ and the homomorphism (20.7.10.1) is none other than (20.7.8.1); the fact that it is bijective is deduced from (20.6.13) by passage to the inductive limit by (20.4.5) and (19.2.4). ☐

(20.7.12). In (20.6.14) one also equips $\Upsilon^C_{B/A/\Lambda}$ with the topology induced by that of $\Omega^1_{A/\Lambda} \otimes_A C$ and the homomorphism (20.6.14.6) is continuous, when the considered rings are topological and the homomorphisms of rings are continuous.

(20.7.13). If, in (20.6.23), one supposes that the rings Λ, A, B, C are topological, the homomorphisms s, u, v continuous and L a C-module discrete annihilated by an open ideal of C, one can pass to the inductive limit as in (20.3.6), and one obtains an exact sequence

$$\text{(20.7.13.1)} \qquad \begin{aligned} 0 \longrightarrow \mathrm{D\acute{e}r.\ cont}_B(C,L) \longrightarrow \mathrm{D\acute{e}r.\ cont}_A(C,L) \longrightarrow \mathrm{D\acute{e}r.\ cont}_A(B,L) \longrightarrow \\ \longrightarrow \mathrm{Exalcotop}_{B/\Lambda}(C,L) \longrightarrow \mathrm{Exalcotop}_{A/\Lambda}(C,L) \longrightarrow \mathrm{Exalcotop}_{A/\Lambda}(B,L) \longrightarrow 0. \end{aligned}$$

(20.7.14). Let A be a topological ring, B a topological A-algebra, $\mathfrak{M}'$ an open ideal of A, $\mathfrak{N}'$ an open ideal of B such that $\mathfrak{M}'B \subset \mathfrak{N}'$; set $A' = A/\mathfrak{M}'$, $B' = B/\mathfrak{N}'$; the kernel of the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ is $\mathfrak{U} = \mathrm{Im}(\mathfrak{N}' \otimes B + B \otimes \mathfrak{N}')$, whence it results immediately that the kernel of the homomorphism

$$\text{(20.7.14.1)} \qquad \varphi_{(\mathfrak{M}',\mathfrak{N}')} : \Omega^1_{B/A} \longrightarrow \Omega^1_{B'/A'}$$

is $((\mathfrak{J} \cap \mathfrak{U}) + \mathfrak{J}^2)/\mathfrak{J}^2$; moreover, the homomorphism (20.7.14.1) is *surjective*, as it results from (20.4.7). If $\mathfrak{M}''$ (resp. $\mathfrak{N}''$) is a second open ideal of A (resp. B) such that $\mathfrak{M}'' \subset \mathfrak{M}'$, $\mathfrak{N}'' \subset \mathfrak{N}'$ and $\mathfrak{M}''B \subset \mathfrak{N}''$, and if one sets $A'' = A/\mathfrak{M}''$, $B'' = B/\mathfrak{N}''$, one has a surjective homomorphism

$$\varphi_{(\mathfrak{M}'',\mathfrak{N}''),(\mathfrak{M}',\mathfrak{N}')} : \Omega^1_{B''/A''} \longrightarrow \Omega^1_{B'/A'}$$

and these homomorphisms form a *projective system*. If one notices that the $((\mathfrak{J} \cap \mathfrak{U}') + \mathfrak{J}^2)/\mathfrak{J}^2$ form a fundamental system of neighborhoods of 0 in $\Omega^1_{B/A}$, one sees therefore that the *separated completion* $\widehat{\Omega}^1_{B/A}$ of the topological B-module $\Omega^1_{B/A}$ is given, up to a canonical isomorphism, by

$$\text{(20.7.14.2)} \qquad \widehat{\Omega}^1_{B/A} = \varprojlim(\Omega^1_{B'/A'}).$$

Moreover, the canonical homomorphism $j : \Omega^1_{B/A} \to \widehat{\Omega}^1_{B/A}$ is the projective limit of the $\varphi_{(\mathfrak{M}',\mathfrak{N}')}$, hence $\varphi_{(\mathfrak{M}',\mathfrak{N}')}$ factors as $\Omega^1_{B/A} \xrightarrow{j} \widehat{\Omega}^1_{B/A} \to \Omega^1_{B'/A'}$, and as it is surjective one concludes that the canonical homomorphism

$$\text{(20.7.14.3)} \qquad \widehat{\Omega}^1_{B/A} \longrightarrow \Omega^1_{B'/A'}$$

is *surjective* for every couple $(\mathfrak{M}',\mathfrak{N}')$. One can moreover replace A' by A everywhere in the above.

Finally, if L is a separated and *complete* topological B-module, every continuous B-homomorphism of $\Omega^1_{B/A}$ in L extends in a unique fashion to a continuous $\widehat{B}$-homomorphism of $\widehat{\Omega}^1_{B/A}$ in L and reciprocally such a homomorphism gives by composition with $\Omega^1_{B/A} \to \widehat{\Omega}^1_{B/A}$ a continuous B-homomorphism, so that one has a canonical isomorphism

$$\mathrm{Hom.cont}_B(\Omega^1_{B/A}, L) \xrightarrow{\sim} \mathrm{Hom.cont}_{\widehat{B}}(\widehat{\Omega}^1_{B/A}, L).$$

More particularly, if L is a B-module discrete annihilated by an open ideal of B, one sees that the canonical isomorphism (20.4.8.2) can also be written

$$\mathrm{Hom.cont}_{\widehat{B}}(\widehat{\Omega}^1_{B/A}, L) \xrightarrow{\sim} \mathrm{Dér.\ cont}_A(B, L). \tag{20.7.14.4}$$

Proposition (20.7.15). — *Let A be a discrete ring, B a topological A-algebra, $\mathfrak{n}$-adique ($\mathbf{0_I}$, 7.1.9), $\mathfrak{n}$ an ideal of definition of B, $B_0 = B/\mathfrak{n}$. One supposes that $\Omega^1_{B/A}$ and $\mathfrak{n}/\mathfrak{n}^2$ are B_0-modules of finite type. Then $\widehat{\Omega}^1_{B/A}$ is a B-module of finite type.*

Proof. As the square of every open ideal of B is open, the topology of $\Omega^1_{B/A}$ is the $\mathfrak{n}$-preadic topology (20.4.5). Taking into account the hypothesis that B is an adique ring, it suffices, by virtue of ($\mathbf{0_I}$, 7.2.7 and 7.2.9), to prove that $\Omega^1_{B_0/A}$ is a B_0-module of finite type, which follows from (20.5.12.1) and the hypotheses. □

§21. Differentials in rings of characteristic p

The results of the present section, of a more specialized and technical nature than those of §§ 19, 20 and 22, will only serve occasionally in chapter IV. Their principal role here is in the proofs of three theorems of § 22 (22.3.3, 22.5.8 and 22.7.3), the first and last of which intervene in an essential way in the "fine" theory of noetherian local rings of chapter IV, § 7. $\mathbf{0_{IV\text{-}1}}$ | 153

21.1. Systems of p-generators and p-bases $\mathbf{0_{IV\text{-}1}}$ | 154

(21.1.1). Given a number p which is either 0, or a prime number, we say that a ring A is *of characteristic p* if there exists a ring homomorphism $P \to A$, where P is the *prime field* of characteristic p; we note that this homomorphism is then unique, the composite $\mathbf{Z} \to P \to A$ being the unique homomorphism of $\mathbf{Z}$ into A. If $A \neq 0$, this amounts to saying that A *contains a field of characteristic p*, the image of P being necessarily the only sub-field of A isomorphic to P (and moreover the only sub-field of A isomorphic to P).

(21.1.2). If $p > 0$, to say that A is of characteristic p is equivalent to saying that, in A, $p.1 = 0$, or equivalently $pA = 0$. If $p = 0$, to say that A is of characteristic p is equivalent to saying that for every integer $n \neq 0$, $n.1$ is *invertible* in A. If $A \neq 0$, there can exist only one p (prime or 0) such that A is of characteristic p; this follows from the identity of Bezout $ap + bq = 1$ for two distinct primes p, q. On the other hand, the reduced ring 0 is of characteristic p for *every* p.

(21.1.3). If A is of characteristic p, then so is every *algebra* over A. In particular, for every prime ideal $\mathfrak{p}$ of A, the *residue field* of A at $\mathfrak{p}$ is of characteristic p. Conversely, if $p = 0$ and if for every *maximal* ideal $\mathfrak{m}$ of A, the residue field of A at $\mathfrak{m}$ is of characteristic 0, then so is A, since for every integer $n \neq 0$, $n.1$ is then invertible in all the $A_{\mathfrak{m}}$, hence also in A. On the other hand, if $p > 0$, a local ring can have its residue field of characteristic p without being itself of characteristic p, as shown by the example of the *prime integer ring* $\mathbf{Z}_{p\mathbf{Z}}$ (19.8.3) or of the local artinian ring $\mathbf{Z}/p^n\mathbf{Z}$ for $n \geqslant 2$, which do not contain a field.

We note finally that for a ring (even reduced), the residue fields at its prime ideals can have different characteristics, as shown by the example of $\mathbf{Z}$.

(21.1.4). In all the rest of this section, we fix a *prime number* p and all rings will be assumed *of characteristic p*, unless express mention to the contrary. For such a ring A, the map $x \mapsto x^p$ is an *endomorphism* of A, which we denote F_A; if A is *reduced*, F_A is *injective*. We set $A^p = \mathrm{F}_A(A)$ (sub-ring of A formed by the x^p for $x \in A$); one can naturally consider A as an A^p-algebra.

One can also consider A as an A-algebra *by means of the homomorphism* $\mathrm{F}_A : x \mapsto x^p$ of A into A; in other words, this is the A-algebra $A_{[\mathrm{F}_A]}$, for which the product $\lambda.x$ of $x \in A$ by a scalar $\lambda \in A$ is the product $\lambda^p x$ in the ring A; we denote this A-algebra $A^{(p)}$. It is clear that for every ring homomorphism $u : A \to B$ the couple (u, u) is a *di-homomorphism* of A-algebras $A^{(p)} \to B^{(p)}$. For every A-module E, we set $E^{(p)} = E \otimes_A A^{(p)}$ where $A^{(p)}$ is considered as an (A, A)-bimodule, the structure of A-module on the left being that which one has just defined, and the structure of A-module on the right coming from the structure of A-module of $A^{(p)}$ provenant of the ring structure, so that for $x \in E$, α, β in A, one has

$$\alpha(x \otimes \beta) = x \otimes (\alpha\beta) \quad \text{and} \quad (\alpha x) \otimes \beta = x \otimes \alpha^p \beta.$$

Setting $x^{(p)} = x \otimes 1$, one therefore has $(\alpha x)^{(p)} = \alpha^p x^{(p)}$. When F_A is *injective*, one can identify $A^{(p)}$ as an algebra over its *sub-ring* A^p. $\mathbf{0_{IV\text{-}1}}$ | 155

Proposition (21.1.5). — *Let A, B be two rings, $u : A \to B$ a homomorphism.*

(i) *Every A-derivation of B in a B-module L is zero in $A[B^p]$ (hence is an $A[B^p]$-derivation).*

(ii) *For every sub-A-algebra A' of $A[B^p]$, if $j : A' \to B$ is the canonical injection, the canonical homomorphism*

$$j_{B/A'/A} : \Omega^1_{B/A} \longrightarrow \Omega^1_{B/A'}$$

is bijective.

(iii) *Suppose that there exists an integer $s \geqslant 0$ such that $u(A) \supset B^{p^s}$. Then, in the ring $B \otimes_A B$, $\mathfrak{J}_{B/A}$ is a nilideal.*

PROOF. (i) By induction from (20.1.1), one deduces, for every A-derivation $D : B \to L$, that $D(x^k) = kx^{k-1}D(x)$, whence in particular $D(x^p) = 0$.

(ii) In view of (20.4.8), the assertion (ii) is only a translation of (i), this last being written $\mathrm{Der}_{A[B^p]}(B, L) = \mathrm{Der}_A(B, L)$ for every B-module L.

(iii) For every $x \in B$, one has $(x \otimes 1 - 1 \otimes x)^{p^s} = x^{p^s} \otimes 1 - 1 \otimes x^{p^s} = 0$, since $x^{p^s} \in u(A)$. The conclusion results from the fact that the elements $1 \otimes x - x \otimes 1$ generate $\mathfrak{J}_{B/A}$ (20.4.4). □

It results immediately from (21.1.5) that one has, for every pair of ring homomorphisms $A \to B \to C$,

(21.1.5.1) $$\Upsilon_{C/B/A} = \Upsilon_{C/B/A'}$$

for every sub-A-algebra A' of $A[B^p]$.

On the other hand, (21.1.5) also shows in particular that for the modules of "absolute" differentials

(21.1.5.2) $$\Omega^1_A = \Omega^1_{A/A^p}$$

Corollary (21.1.6). — *Suppose that B is an A-algebra of finite type and that there exists an integer $s \geqslant 0$ such that $u(A) \supset B^{p^s}$. If $\Omega^1_{B/A} = 0$, u is surjective.*

PROOF. By (21.1.5), (iii), $\mathfrak{J}_{B/A}$ is a nilideal; moreover, by virtue of (20.4.4) and the hypothesis $\Omega^1_{B/A} = 0$, the relation $\Omega^1_{B/A} = 0$ signifies that $\mathfrak{J}_{B/A} = \mathfrak{J}^2_{B/A}$, whence one concludes that $\mathfrak{J}_{B/A} = 0$, or equivalently that $\pi : B \otimes_A B \to B$ is bijective. But every element of B has its p^s-th power in $u(A)$, B is *integral* over A, and being of finite type over A, B is thus reduced to proving the following lemma (where one *does not assume* that the rings considered are of characteristic p): □ $\mathbf{0_{IV\text{-}1}}$ | 156

Lemma (21.1.6.1). — *Let R be a ring, S an R-algebra finite; if the canonical homomorphism $\pi : S \otimes_R S \to S$ is bijective, then the structural homomorphism $u : R \to S$ is surjective.*

PROOF. It suffices to show that for every maximal ideal $\mathfrak{m}$ of R, if one sets $T = R - \mathfrak{m}$, the homomorphism $u_\mathfrak{m} : R_\mathfrak{m} \to T^{-1}S$ is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, th. 1); or the hypothesis implies that the homomorphism $(T^{-1}S) \otimes_{R_\mathfrak{m}} (S/\mathfrak{m}S) \to T^{-1}S$ is bijective, and as $T^{-1}S$ is an $R_\mathfrak{m}$-algebra finite, one sees that one can reduce to the case where R is a local ring. Denoting again by $\mathfrak{m}$ its maximal ideal, it suffices to prove that $u \otimes 1 : R/\mathfrak{m} \to S/\mathfrak{m}S$ is surjective, by virtue of the lemma of Nakayama; as the canonical homomorphism $(S/\mathfrak{m}S) \otimes_{R/\mathfrak{m}} (S/\mathfrak{m}S) \to S/\mathfrak{m}S$ is bijective and $S/\mathfrak{m}S$ is a finite $(R/\mathfrak{m})$-algebra, one is finally reduced to the case where R is a *field*; but then the ranks of $S \otimes_R S$ and of S over R must be equal, which is possible only if S is of rank 0 or 1, hence $u : R \to S$ is surjective. □

Proposition (21.1.7). — *Let A be a ring, B an A-algebra, $(x_\alpha)_{\alpha\in I}$ a family of elements of B. Consider the following properties:*

a) *$B = A[B^p, (x_\alpha)]$, or equivalently the $A[B^p]$-algebra B is generated by the family $(x_\alpha)_{\alpha\in I}$.*

b) *The $A[B^p]$-module B is generated by the monomials $\prod_\alpha x_\alpha^{n(\alpha)}$, where $(n(\alpha))_{\alpha\in I}$ is a family of integers with finite support such that $0 \leqslant n(\alpha) < p$ for all $\alpha \in I$.*

c) *The B-module $\Omega^1_{B/A}$ is generated by the $d_{B/A}(x_\alpha)$ $(\alpha \in I)$.*

Then the properties a) *and* b) *are equivalent and imply* c)*; if in addition B is an $A[B^p]$-algebra of finite type,* c) *is equivalent to* a) *and* b)*.*

Proof. It is clear that b) implies a), and conversely a) implies b), since every monomial $\prod_\alpha x_\alpha^{m(\alpha)}$, where the $m(\alpha)$ are integers $\geqslant 0$ (forming a family with finite support), can be written $(\prod_\alpha x_\alpha^{q(\alpha)})^p \prod_\alpha x_\alpha^{r(\alpha)}$, by dividing each $m(\alpha)$ by p, which gives $m(\alpha) = p.q(\alpha) + r(\alpha)$ with $0 \leqslant r(\alpha) < p$. The fact that a) implies c) results from (21.1.5), (ii) and (20.4.7).

Suppose conversely c) verified and that B is an $A[B^p]$-algebra of finite type; let B' be the sub-$A[B^p]$-algebra of B generated by the x_α; in the exact sequence (20.5.7.1)

$$\Omega^1_{B'/A[B^p]} \otimes_{B'} B \longrightarrow \Omega^1_{B/A[B^p]} \longrightarrow \Omega^1_{B/B'} \longrightarrow 0$$

the hypothesis implies that the left arrow is surjective (taking into account (21.1.5), (ii)); one therefore has $\Omega^1_{B/B'} = 0$, and as $B^p \subset B' \subset B$ and B is a B'-algebra of finite type, one necessarily has $B' = B$ by (21.1.6). □

Remark (21.1.8). — When B is a *field*, we will demonstrate the equivalence of properties a), b) and c) without hypothesis of finiteness (21.4.5).

Definition (21.1.9). — Let A be a ring (of characteristic p), B an A-algebra. One says that a family $(x_\alpha)_{\alpha\in I}$ of elements of B is *p-free over A* (resp. a *system of p-generators* of B over A, resp. a *p-base* of B over A) if the family of monomials $\prod_\alpha x_\alpha^{n(\alpha)}$ $(0 \leqslant n(\alpha) < p$, $(n(\alpha))_{\alpha\in I}$ with finite support) is a free family (resp. a system of generators, resp. a base) in the $A[B^p]$-module B.

One has corresponding definitions for a *subset* M of elements of B, by considering the family defined by the canonical injection $M \to B$. When one takes for A the *prime field* $\mathbf{F}_p$ (in which case $A[B^p] = B^p$), one omits the mention of A in the preceding definition (or says also that a family is "absolutely" p-free, resp. an absolute system of p-generators, resp. an absolute p-base).

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Lemma (21.1.10). — *Let C be a sub-A-algebra of B such that $B^p \subset C$, M a subset of C, N a subset of B.*

(i) *If M is a system of p-generators of C over A and N a system of p-generators of B over C, then $M \cup N$ is a system of p-generators of B over A.*

(ii) *Suppose that M is a p-base of C over A; then, for N to be p-free over C, it is necessary and sufficient that $M \cup N$ be p-free over A.*

Proof. One can reduce to the case where $B^p \subset A \subset C \subset B$. Then (i) is a particular case from the fact that if P (resp. Q) is a system of generators of the A-module C (resp. the C-module B), the set of products xy, $x \in P$ and $y \in Q$, is a system of generators of the A-module B. Preserving the same notations, if P is a base of the A-module C, to say that Q is a free family on C signifies that the relation $\sum_{\lambda,\mu} a_{\lambda\mu} x_\lambda y_\mu = 0$, where $a_{\lambda\mu} \in A$, $x_\lambda \in P$, $y_\mu \in Q$ (with x_λ (resp. y_μ) pairwise distinct) is equivalent to $a_{\lambda\mu} = 0$ for every couple (λ, μ), or equivalently $\sum_\lambda a_{\lambda\mu} x_\lambda = 0$ for every μ, whence the assertion (ii). □

21.2. p**-bases and formal smoothness**

Theorem (21.2.1). — *Let B be a ring, A a sub-ring of B such that $B^p \subset A \subset B$. Let E be an A-algebra, $u : A \to E$ the structural homomorphism, $q : E \to B$ an A-homomorphism surjective, $\mathfrak{N}$ its kernel, and suppose that $\mathfrak{N}^p = 0$. Then:*

(i) *For there to exist an A-homomorphism right inverse $v : B \to E$ of the homomorphism $q : E \to B$, it is necessary and sufficient that $u(q(z)^p) = z^p$ for every $z \in E$.*

(ii) *When the condition of* (i) *is satisfied, for every family $(z_\alpha)_{\alpha \in I}$ of elements of E such that $q(z_\alpha) = x_\alpha$ for all $\alpha \in I$, there exists an A-homomorphism $v : B \to E$ and* a unique one *such that $v(x_\alpha) = z_\alpha$ for all $\alpha \in I$, and v is right inverse to q.*

PROOF. If there exists an A-homomorphism $v : B \to E$, one must have $v(a) = u(a)$ for all $a \in A \subset B$, hence $v(q(z)^p) = u(q(z)^p)$ for all $z \in E$, since $B^p \subset A$. But one has $v(q(z)^p) = (v(q(z)))^p$ and by definition $v(q(z)) \equiv z \pmod{\mathfrak{N}}$ so $q \circ v = 1_B$, hence $(v(q(z)))^p = z^p$ since $\mathfrak{N}^p = 0$; whence the necessity of (i). The sufficiency of condition (i) will result from (ii). $\mathbf{0_{IV\text{-}1}}$ | 158

Under the hypotheses of (ii), the uniqueness of v is evident since the monomials $\prod_\alpha x_\alpha^{n(\alpha)}$ generate the A-module B; as these monomials also form a *base* of the A-module B, there exists an A-*linear* map v of B into E such that $v\big(\prod_\alpha x_\alpha^{n(\alpha)}\big) = \prod_\alpha z_\alpha^{n(\alpha)}$ for every family $(n(\alpha))_{\alpha \in I}$ with finite support with $0 \leqslant n(\alpha) < p$. It remains to see that v is a ring homomorphism. But, one can write $(\prod_\alpha x_\alpha^{m(\alpha)})(\prod_\alpha x_\alpha^{n(\alpha)}) = a.\prod_\alpha x_\alpha^{r(\alpha)}$, where $r(\alpha) = m(\alpha) + n(\alpha)$ if $m(\alpha) + n(\alpha) < p$, $r(\alpha) = m(\alpha) + n(\alpha) - p$ in the contrary case, and $a \in A$ is the product of the x_α^p for the $\alpha \in I$ such that $m(\alpha) + n(\alpha) \geqslant p$; it suffices to see that v is a ring homomorphism, and by construction, on has $f(q(z)^p) = z^p$ for every $z \in E$, the condition (i) of (21.2.1) is therefore satisfied by construction, and by the application of (21.2.1), one is therefore in the conditions of application of the theorem in this case.

For the general case, consider a fundamental system $(\mathfrak{J}_\lambda)$ of open ideals, and let $E_\lambda = E/\mathfrak{J}_\lambda$, $\mathfrak{N}_\lambda = (\mathfrak{N} + \mathfrak{J}_\lambda)/\mathfrak{J}_\lambda$, $C_\lambda = E_\lambda/\mathfrak{N}_\lambda = E/(\mathfrak{N} + \mathfrak{J}_\lambda)$; as $E = \varprojlim E_\lambda$ ($\mathbf{0_I}$, 7.2.2). For every couple (α, λ), denote by $z_{\alpha\lambda}$ the canonical image of z_α in E_λ, and by $p_\lambda : C_\lambda \to C_\lambda$ the canonical homomorphism and by $q_\lambda : E_\lambda \to C_\lambda$ the canonical homomorphism. As by hypothesis $\mathfrak{N}_\lambda$ is nilpotent, the first part of the proof shows the existence and uniqueness of an A-homomorphism $w_\lambda : B \to E_\lambda$ such that $p_\lambda \circ v = q_\lambda \circ w_\lambda$ and $w_\lambda(x_\alpha) = z_{\alpha\lambda}$ for all α. The uniqueness of the w_λ shows immediately that (w_λ) is a projective system of homomorphisms, and $w = \varprojlim w_\lambda$ answers the question. □

Corollary (21.2.2). — *If there exists a p-base of B over A, B is an A-algebra formally smooth relative to B^p* (*for the discrete topologies*).

PROOF. Indeed, let E be an A-extension of B by a B-module L, $q : E \to B$ the augmentation, $u : A \to E$ the structural homomorphism. To say that E is B^p-trivial means that there exists an A-homomorphism $v : B \to E$ such that $q(v(b)) = b$ and $v(b^p) = u(b^p)$ for all $b \in B$. But one has $u(q(z)^p) = v(q(z)^p) = (v(q(z)))^p$; but by the relation $L^2 = 0$, one also has $L^p = 0$, and as $v(q(z)) - z \in L$, on has $(v(q(z)))^p = z^p$; the condition of (21.2.1), (i) is therefore satisfied, and E is B-trivial. □

Corollary (21.2.3). — *Let B be a ring, A a sub-ring of B such that $B^p \subset A \subset B$, L a B-module. Then:*

(i) *For a derivation D of A in L to extend to a derivation of B in L, it is necessary and sufficient that D annihilate B^p.*

(ii) *If D annihilates B^p, then, for every family $(y_\alpha)_{\alpha \in I}$ of elements of L, there exists a unique derivation D' of B in L extending D and such that $D'(x_\alpha) = y_\alpha$ for all $\alpha \in I$.*

PROOF. Given a derivation D of A in L, consider the ring $E = \mathrm{D}_B(L)$ and the homomorphism $u : A \to E$ defined by $u(a) = (a, D(a))$; an A-homomorphism right inverse $v : B \to E$ is then of the form $x \mapsto (x, D'(x))$, where D' is a derivation of B in L extending D (20.1.5); as $u(q(z)^p) = (q(z)^p, D(q(z)^p))$ for $z \in E$, the corollary results immediately from (21.2.1) applied to E. □

It results from (21.2.3) that the sequence

(21.2.3.1) $$0 \longrightarrow \mathrm{Der}_A(B, L) \longrightarrow \mathrm{Der}_{B^p}(B, L) \longrightarrow \mathrm{Der}_{B^p}(A, L) \longrightarrow 0$$

(cf. (20.2.2.1)) is *exact*, and that the map

(21.2.3.2) $$D' \longmapsto (D'(x_\alpha))_{\alpha\in I}$$

is an isomorphism of the B-module $\mathrm{Der}_A(B,L)$ onto the B-module product L^I (by taking $D = 0$ in (21.2.3)).

Corollary (21.2.4). — *Under the hypotheses of (21.2.3), the sequence*

(21.2.4.1) $$0 \longrightarrow \Omega^1_{A/B^p} \otimes_A B \longrightarrow \Omega^1_{B/B^p} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

is exact and split, and the family $(d_{B/A}(x_\alpha))_{\alpha\in I}$ is a base of the B-module $\Omega^1_{B/A}$.

PROOF. This results immediately from (21.2.3) and from the formula $\mathrm{Der}_A(B,L) = \mathrm{Hom}_B(\Omega^1_{B/A}, L)$. $\square$

Corollary (21.2.5). — *Let A be a ring, B an A-algebra, $(x_\alpha)_{\alpha\in I}$ a p-base of B over A. Then $(d_{B/A}(x_\alpha))_{\alpha\in I}$ is a base of the B-module $\Omega^1_{B/A}$.*

PROOF. By using (21.1.5), (ii), one reduces to the case where $A = A[B^p]$ and it suffices then to apply (21.2.4). $\square$

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(21.2.6). Let A, B be two rings, $u : A \to B$ a ring homomorphism; one has (with the notations of (21.1.4)) a commutative diagram

(21.2.6.1) $$\begin{array}{ccc} A^{(p)} & \xrightarrow{u} & B^{(p)} \\ {\scriptstyle \mathrm{F}_A}\uparrow & & \uparrow{\scriptstyle \mathrm{F}_B} \\ A & \xrightarrow{u} & B \end{array}$$

One deduces therefore canonically a homomorphism of B-algebras

(21.2.6.2) $$u \otimes 1 : A^{(p)} \otimes_A B \longrightarrow B^{(p)}$$

whose image is the ring $A[B^p]$ (which is a B-algebra for the homomorphism $\mathrm{F}_B : B \to B^p$).

Theorem (21.2.7). — *Let A be a ring, B an A-algebra, $u : A \to B$ the structural homomorphism. One assumes the following conditions verified:*

(i) *The homomorphism (21.2.6.2) deduced from u is injective.*
(ii) *B admits a p-base $(x_\alpha)_{\alpha\in I}$ relative to A.*

Then B is an A-algebra formally smooth (for the discrete topologies). More precisely, let E be a topological A-algebra admissible ($\mathbf{0_I}$, 7.1.2), $\mathfrak{N}$ an ideal of definition of E, $C = E/\mathfrak{N}$, $v : B \to C$ an A-homomorphism, $q : E \to C$ the augmentation. Then, for every family $(z_\alpha)_{\alpha\in I}$ of elements of E such that $q(z_\alpha) = v(x_\alpha)$ for all $\alpha \in I$, there exists an A-homomorphism $w : B \to E$ and a unique one *such that $v = q \circ w$ and $w(x_\alpha) = z_\alpha$ for all $\alpha \in I$.*

PROOF. Consider first the case where E is *discrete*, hence $\mathfrak{N}$ is nilpotent; moreover, the reasoning $\mathbf{0_{IV\text{-}1}}$ | 160
of (19.4.3) allows one to suppose $\mathfrak{N}^2 = 0$ and by consequence $\mathfrak{N}^p = 0$. Finally, by considering the inverse image of the extension E of C by $\mathfrak{N}$, one can reduce to the case where $C = B$ and v is the identity (19.4.4). As the ring homomorphism $\mathrm{F}_B : z \mapsto z^p$ annihilates $\mathfrak{N}$, it factors as $E \xrightarrow{q} B \xrightarrow{\mathrm{F}'} E$, and F', considered as a homomorphism of B into $E^{(p)}$, is an *A-homomorphism* defining the structure of A-algebra on $E^{(p)}$ with the help of the composite homomorphism $A \xrightarrow{r} E \xrightarrow{\mathrm{F}_E} E^{(p)}$, where $r : A \to E$ is the structural homomorphism of the A-algebra E; moreover $r : A^{(p)} \to E^{(p)}$ is also an A-homomorphism. There is therefore a unique A-homomorphism $f : A^{(p)} \otimes_A B \to E^{(p)}$ such that the composites of f with the canonical homomorphisms $A^{(p)} \to A^{(p)} \otimes_A B$ and $B \to A^{(p)} \otimes_A B$ are respectively r and F'. But, by hypothesis, one can identify $A^{(p)} \otimes_A B$ with $A[B^p]$, the canonical homomorphisms $A^{(p)} \to A^{(p)} \otimes_A B$ and $B \to A^{(p)} \otimes_A B$ becoming identified respectively with u and F_B. One can therefore now consider E as an $A[B^p]$-algebra by means of the homomorphism f, and, by construction, one has $f(q(z)^p) = z^p$ for every $z \in E$; one is therefore in the conditions of application of (21.2.1), whence the theorem in this case.

For the general case, consider a fundamental system $(\mathfrak{J}_\lambda)$ of open ideals, and let $E_\lambda = E/\mathfrak{J}_\lambda$, $\mathfrak{N}_\lambda = (\mathfrak{N} + \mathfrak{J}_\lambda)/\mathfrak{J}_\lambda$, $C_\lambda = E_\lambda/\mathfrak{N}_\lambda$; as $E = \varprojlim E_\lambda$ ($\mathbf{0_I}$, 7.2.2). For every couple (α, λ), denote by $z_{\alpha\lambda}$ the canonical image of z_α in E_λ; the first part of the proof shows the existence and uniqueness of an A-homomorphism $w_\lambda : B \to E_\lambda$ such that $p_\lambda \circ v = q_\lambda \circ w_\lambda$ and $w_\lambda(x_\alpha) = z_{\alpha\lambda}$ for all α. The uniqueness of the w_λ shows immediately that (w_λ) is a projective system of homomorphisms, and $w = \varprojlim w_\lambda$ answers the question. □

Remark (21.2.8). — The verification of the existence and uniqueness of the homomorphism $w : B \to E$ such that $w(x_\alpha) = z_\alpha$ for all α can also be deduced directly from the fact that B is an A-algebra formally smooth and from the fact that the $d_{B/A}(x_\alpha)$ form a base of the B-module $\Omega^1_{B/A}$ (21.2.4), without invoking the existence of a p-base (so that the result is valid without supposing the rings of characteristic p). Indeed, one can reduce to the case where $\mathfrak{N}^2 = 0$; as $\mathrm{Der}_A(B, \mathfrak{N}) = \mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{N})$ by (20.4.8), there exists an A-derivation D of B in $\mathfrak{N}$ and a unique one such that $D(x_\alpha) = z'_\alpha$ for every family (z'_α) of elements of $\mathfrak{N}$; the conclusion results from (20.1.1).

21.3. p-bases and modules of imperfection

(21.3.1). Let A, B be two rings (of characteristic p), $i : A \to B$, $j : B \to A$ two ring homomorphisms such that one has

(21.3.1.1) $$j(i(a)) = a^p \quad \text{for all } a \in A,$$

(21.3.1.2) $$i(j(b)) = b^p \quad \text{for all } b \in B.$$

Most often, i will be *injective*, so that A is identified with a sub-ring of B; once this identification made, the existence of j implies that $B^p \subset A$, and j is then identified with F_B.

(21.3.2). If $h : A^p \to A$ is the canonical injection, one has, by (21.3.1.2) and (21.3.1.1), a commutative diagram

(21.3.2.1) $$\begin{tikzcd} B \arrow[r, "j"] & A \\ A \arrow[u, "i"] \arrow[r, "\mathrm{F}_A"] & A^p \arrow[u, "h"'] \end{tikzcd}$$

so that the couple (j, F_A) can be considered as a *di-homomorphism* of the A-algebra B (for i) in the A^p-algebra A (for h). One deduces a *canonical homomorphism of A-modules*

(21.3.2.2) $$\pi_{B/A} : \Omega^1_{B/A} \otimes_B A_{[j]} \longrightarrow \Omega^1_{A/A^p} = \Omega^1_A$$

(cf. (20.5.4); the identification of Ω^1_{A/A^p} and of the module of absolute differentials Ω^1_A comes from (21.1.5), (ii)).

One will set

(21.3.2.3) $$\Theta_{B/A} = \Omega^1_{B/A} \otimes_B A_{[j]}$$

(21.3.2.4) $$\Xi_{B/A} = \mathrm{Ker}(\pi_{B/A})$$

so that one has the exact sequence $\mathbf{0_{IV\text{-}1}}$ | 161

(21.3.2.5) $$0 \longrightarrow \Xi_{B/A} \longrightarrow \Theta_{B/A} \xrightarrow{\pi_{B/A}} \Omega^1_A$$

(one notes that these notations can be confusing since $\Theta_{B/A}$ and $\pi_{B/A}$ depend not only on A and B, but also on i and j).

(21.3.3). As by (21.3.1.2), one has $\mathrm{F}_B = i \circ j$, one can write for every B-module M (cf. (21.1.4))

(21.3.3.1) $$M^{(p)} = M \otimes_B B^{(p)} = (M \otimes_B A_{[j]}) \otimes_A B_{[i]}$$

whence in particular

(21.3.3.2) $$(\Omega^1_{B/A})^{(p)} = \Theta_{B/A} \otimes_A B_{[i]}$$

and one deduces therefore from (21.3.2.2) a *canonical homomorphism of B-modules*

(21.3.3.3) $$\pi_{B/A} \otimes 1_B : (\Omega^1_{B/A})^{(p)} \longrightarrow \Omega^1_A \otimes_A B_{[i]}.$$

Taking into account (20.5.4.2), the image of this homomorphism is the B-module generated by the elements $d_A(j(b)) \otimes 1$ for $b \in B$. Their image by the canonical homomorphism $i_{B/A} : \Omega^1_A \otimes_A B \to \Omega^1_B$ deduced from i (20.5.2) is therefore contained in the sub-B-module of Ω^1_B generated by the $d_B(i(j(b)))$ for $b \in B$; by virtue of (21.3.1.2), this image is *zero*. In other words, one has a sequence of homomorphisms

(21.3.3.4) $$0 \longrightarrow \Xi_{B/A} \otimes_A B_{[i]} \longrightarrow (\Omega^1_{B/A})^{(p)} \longrightarrow \Upsilon_{B/A}$$

which is not necessarily exact, but where the composite of two consecutive homomorphisms is zero.

Proposition (21.3.4). — *If B is a flat A-module* (*for i*), *the sequence* (21.3.3.4) *is exact.*

PROOF. This results from the definition of flatness. The proposition (21.3.4) applies in particular when $B^p \subset A \subset B$, i and j being respectively the canonical injection and F_B (so that B is then an A-module libre). But even in this case the kernel $\Xi_{B/A}$ is not necessarily zero. However: □

Proposition (21.3.5). — *Let A be a* reduced *ring, B a sub-ring of A such that $B^p \subset A \subset B$. Suppose that there exists a p-base $(x_\lambda)_{\lambda \in L}$ of B over A and a p-base $(y_\mu)_{\mu \in M}$ of A over B^p; then the canonical homomorphism* (21.3.3.4)

(21.3.5.1) $$(\Omega^1_{B/A})^{(p)} \longrightarrow \Upsilon_{B/A}$$

is bijective, and the elements $d_A(x^p_\lambda) \otimes 1$ form a base of the B-module $\Upsilon_{B/A}$.

PROOF. As B is reduced, $x \mapsto x^p$ is an isomorphism of structure of B onto B^p, and by transport $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 162
of structure via this isomorphism, one sees that (x^p_λ) and the y_μ form a p-base of B^p over A^p (21.1.10); by consequence (21.1.5), (ii) and (21.2.5) the $d_A(x^p_\lambda)$ and the $d_A(y_\mu)$ form a *base* of the A-module Ω^1_A. But, the image of $d_A(x_\lambda) \otimes 1$ by $i_{B/A}$ is $d_B(x^p_\lambda) = 0$; on the other hand, the image of $d_A(y_\mu)$ by $i_{B/A}$ is $d_B(i(y_\mu))$, and as the y_μ and the x_λ form a p-base of B over B^p (21.1.10), the $d_B(y_\mu)$ are part of a base of Ω^1_B (21.2.5), and are thus linearly independent over B. Hence the kernel of $i_{B/A}$ has a base formed by the $d_A(x^p_\lambda) \otimes 1$, and as these are the images by (21.3.5.1), these last form a base of the B-module $\Omega^1_{B/A} \otimes_B B^{(p)}$ (21.2.5), the homomorphism (21.3.5.1) of B-modules is bijective. □

Remarks (21.3.6). — (i) Let A', B' be two rings of characteristic p, and suppose one has two homomorphisms $i' : A' \to B'$, $j' : B' \to A'$ verifying (21.3.1.1) and (21.3.1.2), and ring homomorphisms $f : A \to A'$, $g : B \to B'$ rendering commutative the diagram

$$\begin{tikzcd} A' \arrow[r, "i'"] & B' \arrow[r, "j'"] & A' \\ A \arrow[u, "f"] \arrow[r, "i"] & B \arrow[u, "g"] \arrow[r, "j"] & A \arrow[u, "f"] \end{tikzcd}$$

Then the canonical homomorphism $\Omega^1_{B/A} \to \Omega^1_{B'/A'}$ (20.5.4.3) gives a di-homomorphism $\Theta_{B/A} \to \Theta_{B'/A'}$ rendering commutative the diagram

$$\begin{tikzcd} 0 \arrow[r] & \Xi_{B/A} \arrow[r] \arrow[d] & \Theta_{B/A} \arrow[r, "\pi_{B/A}"] \arrow[d] & \Omega^1_A \arrow[d] \\ 0 \arrow[r] & \Xi_{B'/A'} \arrow[r] & \Theta_{B'/A'} \arrow[r, "\pi_{B'/A'}"] & \Omega^1_{A'} \end{tikzcd}$$

(ii) Let A' be any A-algebra, and set $B' = B \otimes_A A'$; then $i' = i \otimes 1 : A' \to B'$ and $j' = j \otimes 1 : B' \to A'$ verify (21.3.1.1) and (21.3.1.2), and one has

$$\Omega^1_{B'/A'} = \Omega^1_{B/A} \otimes_A A' = \Omega^1_{B/A} \otimes_B B'$$

(20.5.5); one deduces an A'-isomorphism canonical

(21.3.6.1) $$\Theta_{B/A} \otimes_A A' \xrightarrow{\sim} \Theta_{B'/A'}$$

and also a B'-homomorphism canonical

(21.3.6.2) $$(\Omega^1_{B/A})^{(p)} \otimes_B B' \xrightarrow{\sim} (\Omega^1_{B'/A'})^{(p)}.$$

21.4. The case of field extensions

(21.4.0). Let K be a field of characteristic $p > 0$, k a sub-field of K; then the ring $k[K^p]$ is equal to the field $k(K^p)$ since k is algebraic over K^p. One can therefore apply the results of the preceding sections everywhere replacing A, B and $A[B^p]$ by k, K and $k(K^p)$.

Lemma (21.4.1). — *Let k be a field of characteristic $p > 0$, K an extension of k. For an element $x \in K$ to be p-free over k, it is necessary and sufficient that $x \notin k(K^p)$.*

PROOF. Indeed, x is a root of the polynomial $X^p - x^p$ of $k(K^p)[X]$, and one knows (Bourbaki, *Alg.*, chap. V, § 8, n° 1, prop. 1) that if $x \notin k(K^p)$ this polynomial is irreducible, so that the elements $1, x, \ldots, x^{p-1}$ form a base of the $k(K^p)$-module $k(K^p)(x)$. □

Theorem (21.4.2). — *Let k be a field of characteristic $p > 0$, K an extension of k, S a system of p-generators of K over k, $L \subset S$ a p-free subset over k. There exists then a p-base B of K over k such that $L \subset B \subset S$. In particular, every extension of k admits a p-base over k.*

PROOF. One can reduce to the case where $K^p \subset k$. By virtue of Zorn's theorem, there exists in K a subset B such that $L \subset B \subset S$, p-free over k and *maximal* amongst the subsets of S having these properties. It suffices to see that the sub-field K' of K generated by k and B is equal to K. In the contrary case, there would exist $x \in S$ not in K'; as $K^p \subset k$, one has $K' = K'(K^p)$; hence x would be p-free over K' (21.4.1), and hence $B \cup \{x\}$ would be p-free over k (21.1.10), contradicting the maximality of B. The last assertion of (21.4.2) is obtained by taking $L = \varnothing$, $S = K$. □ □

Corollary (21.4.3). — *Let k be a field of characteristic $p > 0$, K an extension of k. For a family (x_α) of elements of K to be p-free over k, it is necessary and sufficient that, for all α, x_α does not belong to the field K_α generated by $k(K^p)$ and the x_β of index $\beta \neq \alpha$.*

PROOF. The condition is necessary by virtue of (21.1.10). Conversely, suppose it is satisfied; one can reduce to the case where (x_α) is a system of p-generators of K. There exists then a sub-family of (x_α) which is a p-base of K (21.4.2), but this sub-family cannot be distinct from (x_α), since by hypothesis it would not be a family of p-generators of K. □

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Corollary (21.4.4). — *Let k be a field of characteristic $p > 0$, K an extension of k. For $\Omega^1_{K/k} = 0$, it is necessary and sufficient that $K = k(K^p)$. In particular, for $\Omega^1_K = 0$, it is necessary and sufficient that K be a* perfect field.

PROOF. Indeed, if (x_α) is a p-base of K over k, the $d_{K/k}(x_\alpha)$ form a base of the K-vector space $\Omega^1_{K/k}$ (21.2.5). □

Theorem (21.4.5). — *Let k be a field of characteristic $p > 0$, K an extension of k, (x_α) a family of elements of K. For (x_α) to be p-free over k (resp. a family of p-generators of K over k, resp. a p-base of K over k), it is necessary and sufficient that the family $(d_{K/k}(x_\alpha))$ be a free family (resp. a system of generators, resp. a base) in the K-vector space $\Omega^1_{K/k}$.*

PROOF. Let K' be the sub-field (equal to the sub-ring) of K generated by $k(K^p)$ and the x_α; taking into account (21.4.3), it suffices to see that, for each α, x_α does not belong to K_α. But, in the exact sequence $\mathbf{0_{IV\text{-}1}}$ | 164

$$\Omega^1_{K_\alpha/k(K^p)} \otimes_{K_\alpha} K \longrightarrow \Omega^1_{K/k(K^p)} \longrightarrow \Omega^1_{K/K_\alpha} \longrightarrow 0$$

the images by the left arrow of the $d_{K_\alpha/k}(x_\beta)$ for $\beta \neq \alpha$ are the $d_{K/k}(x_\beta)$, hence generate a sub-vector space of $\Omega^1_{K/k(K^p)}$ not containing $d_{K/k}(x_\alpha)$, and as this sub-vector space is the kernel of the right arrow, we see that $d_{K/K_\alpha}(x_\alpha) \neq 0$, hence $x_\alpha \notin K_\alpha$.

If now (x_α) is a p-free family, it is a sub-family of a p-base of K over k (21.4.2), hence the $d_{K/k}(x_\alpha)$ are part of a base of $\Omega^1_{K/k}$ (21.2.5), and are thus linearly independent over K. Let us prove conversely that if the $d_{K/k}(x_\alpha)$ are linearly independent over K, the family (x_α) is p-free over k. □

Corollary (21.4.6). — *Let k be a field of characteristic $p > 0$, K an extension of k, x an element of K. The three following conditions are equivalent:*

a) $x \notin k(K^p)$.
b) $d_{K/k}(x) \neq 0$.
c) *The element x is p-free over k.*

Proposition (21.4.7). — *Let L be a field of characteristic $p > 0$, K a sub-field of L such that $L^p \subset K \subset L$. Then the sequence of L-vector spaces*

$$0 \longrightarrow \Omega^1_{K/L^p} \otimes_K L \longrightarrow \Omega^1_{L/L^p} \longrightarrow \Omega^1_{L/K} \longrightarrow 0 \tag{21.4.7.1}$$

is exact; in other words, one has $\Upsilon_{L/K/L^p} = 0$.

PROOF. This is a particular case of (21.2.4), taking into account (21.4.2). □

Proposition (21.4.8). — *Under the hypotheses of (21.4.7), the canonical homomorphism*

$$(\Omega^1_{L/K})^{(p)} \longrightarrow \Upsilon_{L/K} \tag{21.4.8.1}$$

is bijective; if (x_λ) is a p-base of L over K, the elements $d_K(x^p_\lambda) \otimes 1$ form a base of the L-vector space $\Upsilon_{L/K}$.

PROOF. This is a particular case of (21.3.5). □

21.5. Application: criteria for separability In this and the following numbers, *we no longer assume* that the rings considered are of characteristic $p > 0$.

(21.5.1). Let us first note that the criterion (21.2.7) allows us to prove one part of the Cohen theorem on separable extensions (19.6.1), namely that if k is a field of characteristic $p > 0$ and if K is a separable extension of k, then K is a formally smooth k-algebra. Indeed, K admits a p-base over k (21.4.2), and on the other hand, it follows from the criterion of MacLane (Bourbaki, *Alg.*, chap. V, § 8, no. 2, prop. 3) that in an algebraic closure of K, $k^{p^{-1}}$ and K are linearly disjoint over k, and consequently the canonical homomorphism $k^{p^{-1}} \otimes_k K \to k^{p^{-1}}(K)$ is bijective, which is precisely the condition (i) of (21.2.7), after transport of structure by the isomorphism $x \mapsto x^p$.

Proposition (21.5.2). — (MacLane). *Let A, B be two discrete valuation rings, $\mathfrak{m}$, $\mathfrak{n}$ their respective maximal ideals, $K = A/\mathfrak{m}$ and $L = B/\mathfrak{n}$ their residue fields, $u : A \to B$ a homomorphism such that $Bu(\mathfrak{m}) = \mathfrak{n}$, $u_0 = u \otimes 1 : K \to L$ the corresponding homomorphism for the residue fields. Consider the following conditions:*

a) *L is a separable extension of K (for u_0).*
b) *For every complete discrete valuation ring B' with maximal ideal $\mathfrak{n}'$ and residue field L', every homomorphism $u' : A \to B'$ such that $B'u'(\mathfrak{m}) = \mathfrak{n}'$ and every K-isomorphism $\sigma : L \to L'$ (relative to u_0 and $u'_0 = u' \otimes 1 : K \to L'$), there exists an isomorphism $w : B \to B'$ such that $u' = w \circ u$ and that $w_0 = w \otimes 1 : L \to L'$ is equal to σ.*
b′) *For every homomorphism $u' : A \to B$ such that $Bu'(\mathfrak{m}) = \mathfrak{n}$ and such that $u'_0 = u' \otimes 1 : K \to L$ is equal to u_0, there exists an automorphism w of B such that $u' = w \circ u$ and that $w_0 = w \otimes 1 : L \to L$ is the identity.*
c) *Denoting by $u_1 : A/\mathfrak{m}^2 \to B/\mathfrak{n}^2$ the homomorphism deduced from u by passage to quotients, then for every local homomorphism $u'_1 : A/\mathfrak{m}^2 \to B/\mathfrak{n}^2$ such that $\mathrm{gr}_0(u'_1) = \mathrm{gr}_0(u_1)(= u_0)$, there exists an automorphism w_1 of $B/\mathfrak{n}^2$ such that $\mathrm{gr}_0(w_1)$ is the identity and that $u'_1 = w_1 \circ u_1$.*
c′) *For every local homomorphism $u'_1 : A/\mathfrak{m}^2 \to B/\mathfrak{n}^2$ such that $\mathrm{gr}_0(u'_1) = \mathrm{gr}_0(u_1)$ and $\mathrm{gr}_1(u'_1) = \mathrm{gr}_1(u_1)$ (homomorphism of $\mathfrak{m}/\mathfrak{m}^2$ into $\mathfrak{n}/\mathfrak{n}^2$), there exists an automorphism w_1 of $B/\mathfrak{n}^2$ such that $\mathrm{gr}_0(w_1)$ is the identity and that $u'_1 = w_1 \circ u_1$.*

Then one has the implications c) ⇔ c′) ⇔ a) ⇒ b) ⇒ b′).

If moreover A is a Cohen ring, the five conditions are equivalent.

PROOF. The implications $b) \Rightarrow b')$ and $c) \Rightarrow c')$ are trivial. Let us show that $a)$ implies $b)$. The homomorphism u makes B into a torsion-free A-module, since it transforms a uniformizer of A into a uniformizer of B; it follows that B is a *flat* A-module ($\mathbf{0_I}$, 6.3.4), hence a *Cohen A-algebra* (19.8.1); the fact that $a)$ implies $b)$ is then a consequence of (19.8.2), (i) applied to $C = B'$, $\mathfrak{J} = \mathfrak{n}'$, B' being considered as an A-algebra for u' and the homomorphism $B \to B'/\mathfrak{n}'$ being the composed $B \to B/\mathfrak{n} \xrightarrow{\sigma} B'/\mathfrak{n}'$, which is an A-homomorphism by virtue of the hypothesis $u'_0 = \sigma \circ u_0$. The same reasoning proves that a) implies c), taking $C = B/\mathfrak{n}^2$, $\mathfrak{J} = \mathfrak{n}/\mathfrak{n}^2$, C being considered as an A-algebra for u_1'.

Let us prove next that c') implies a). The two homomorphisms u_1 and u_1' are such that $u_1' = u_1 + D$, where D is a derivation of $A/\mathfrak{m}^2$ into $\mathfrak{n}/\mathfrak{n}^2$ (20.1.1); moreover, the hypothesis $\mathrm{gr}_1(u_1') = \mathrm{gr}_1(u_1)$ means that D is zero in $\mathfrak{m}/\mathfrak{m}^2$, and can thus be considered as a derivation of $K = A/\mathfrak{m}$ into $\mathfrak{n}/\mathfrak{n}^2$, and $\mathfrak{n}/\mathfrak{n}^2$ identifies (by choice of a uniformizer of B) with the K-module L (for u_0). On the other hand, the conditions imposed on w_1 imply that $\mathrm{gr}_1(w_1)$ is also the identity (since $u_1(\mathfrak{m}/\mathfrak{m}^2)$ generates $\mathfrak{n}/\mathfrak{n}^2$); w_1 is thus of the form $z \mapsto z + D'(z)$, where D' is this time a derivation of $B/\mathfrak{n}^2$ in the $(B/\mathfrak{n}^2)$-module $\mathfrak{n}/\mathfrak{n}^2$; as w_1 is the identity on $\mathfrak{n}/\mathfrak{n}^2$, D' can again (by the preceding identification of $\mathfrak{n}/\mathfrak{n}^2$ and of L) be considered as a derivation *of L into L*; finally, the relation $u_1' = w_1 \circ u_1$ means that D' *extends* D. Let us note on the other hand that *every* derivation D of K into L satisfying the conditions of c') (that is to say, every derivation of K into L extends to a derivation of L into itself, i.e. (20.6.5)) means that L is separable over K.

Let us prove finally that, when A is a Cohen ring, b') implies c'). Let us show first that under the hypotheses of c'), there exists a homomorphism $u' : A \to B$ satisfying the hypotheses of b') and that, by passage to quotients, gives $u_1' : A/\mathfrak{m}^2 \to B/\mathfrak{n}^2$. Indeed, this results from (19.8.6), (i) applied to the composed homomorphism $v' : A \to A/\mathfrak{m}^2 \xrightarrow{u_1'} B/\mathfrak{n}^2$; this last thus factors as $A \xrightarrow{u'} B \to B/\mathfrak{n}^2$ $\mathbf{0}_{\textbf{IV-1}}$ | 166
and the hypotheses on $\mathrm{gr}_0(u_1')$ and $\mathrm{gr}_1(u_1')$ imply $\mathrm{gr}_0(u') = \mathrm{gr}_0(u) = u_0$ and $\mathrm{gr}_1(u') = \mathrm{gr}_1(u)$, hence the image by u' of a uniformizer of A is a uniformizer of B, and therefore $Bu'(\mathfrak{m}) = \mathfrak{n}$. It then suffices, to obtain an automorphism w_1 satisfying the question, to take the automorphism deduced by passage to quotients from the automorphism w of B provided by the application of b') to the homomorphism u'. □

Remarks (21.5.3). — (i) The differential properties of fields allow us to solve the question of the *uniqueness* of the representative field in a complete local noetherian ring C (19.8.7), (ii). Let $\mathfrak{J}$ be the maximal ideal of C, $K = C/\mathfrak{J}$ the residue field of C; we can restrict to the case $\mathfrak{J} \neq 0$. Suppose there exists a homomorphism $u : K \to C$ that, composed with the augmentation $C \to K$, gives the identity; then, *for this homomorphism to be unique, it is necessary and sufficient that* $\Omega^1_K = 0$. Indeed, the condition $\Omega^1_K = 0$ implies that K is formally unramified on its prime field (20.7.4); if there existed a second homomorphism $v \neq u$ satisfying the question, there would exist a largest integer n such that $\mathfrak{J}^n$ contains the set of the $u(x) - v(x)$ for $x \in K$; by passage to the quotient, u and v would give two distinct homomorphisms u', v' of K into $C/\mathfrak{J}^{n+1}$, whose composites with $C/\mathfrak{J}^{n+1} \to C/\mathfrak{J}^n$ would be equal, which contradicts the definition (19.10.2). Inversely, suppose $\Omega^1_K \neq 0$; then there exists a derivation $D \neq 0$ of K into $\mathfrak{J}/\mathfrak{J}^2$ (20.4.8), hence a homomorphism $v_1 : K \to C/\mathfrak{J}^2$ such that, if $u_1 : K \to C/\mathfrak{J}^2$ is obtained by passage to quotients from u, one has $v_1 = u_1 + D$ (20.1.1). If k is the prime field of K, C is a k-algebra and K a formally smooth k-algebra (19.6.1), and the restrictions of u_1 and v_1 to k coincide, hence v_1 factors as $K \xrightarrow{v} C \to C/\mathfrak{J}^2$ and one has $v \neq u$.

Let us recall (20.6.20) and (21.4.4) that the condition $\Omega^1_K = 0$ means that K is *perfect* if it is of characteristic $\neq 0$, and an algebraic extension of $\mathbf{Q}$ if it is of characteristic 0.

(ii) In the same way, let W be a Cohen ring, C a complete local noetherian ring, $\mathfrak{J}$ a nonzero ideal of C contained in the maximal ideal; for the factorization $W \xrightarrow{v} C \to C/\mathfrak{J}$ in (19.8.6), (i) to be *unique*, it is necessary and sufficient that the residue field K of the Cohen ring W be such that $\Omega^1_K = 0$. Indeed, if $\Omega^1_K \neq 0$ and if $\mathfrak{m}$ designates the maximal ideal of W, it suffices to compose a derivation $D \neq 0$ of K into $\mathfrak{J}/\mathfrak{m}\mathfrak{J}$ to obtain a derivation $D' \neq 0$ of W into $\mathfrak{J}/\mathfrak{m}\mathfrak{J}$ and one finishes the reasoning as in (i) by forming the homomorphism $v_1 : W \to C/\mathfrak{m}\mathfrak{J}$ distinct from the homomorphism $u_1 : W \to C/\mathfrak{m}\mathfrak{J}$ deduced from u by passage to quotients; one finishes by invoking (19.8.6), (i). If on the contrary $\Omega^1_K = 0$, the uniqueness of v already results from (i) when W is a field of characteristic 0. In the contrary case, one has $\widehat{\Omega}^1_W = 0$; in fact, one then has $\mathfrak{m} = pW$, p being the characteristic of K (19.8.5), and the canonical homomorphism $\mathfrak{m}/\mathfrak{m}^2 \to \Omega^1_W/\mathfrak{m}\Omega^1_W$ (20.5.12.1) applied to W and to $K = W/\mathfrak{m}$ implies that $\Omega^1_W = \mathfrak{m}\Omega^1_W$, hence by Nakayama's lemma. But then (20.7.4) W is formally unramified (for its adic topology) over $\mathbf{Z}$, and the uniqueness of v is proved as in (i).

21.6. Admissible fields for an extension

(21.6.1). Given four fields $k_0 \subset k \subset K \subset L$, it results from (20.6.16) and (20.6.17) that one has an exact sequence

(21.6.1.1)
$$0 \longrightarrow \Upsilon_{K/k/k_0} \otimes_K L \xrightarrow{v} \Upsilon_{L/k/k_0} \xrightarrow{u} \Upsilon_{L/K/k_0} \xrightarrow{s} \Upsilon_{L/K/k} \longrightarrow 0.$$

When one fixes k_0, K, and L and one lets the intermediate field k between k_0 and K "vary", one evidently has $\Upsilon_{L/K/k_0} = \Upsilon_{L/K/k}$ when $k = k_0$. When the canonical homomorphism s of (21.6.1.1) is still *bijective*, one says that k is a *k_0-admissible field for the extension L of K*. The interest of the existence, under certain conditions, of such fields k, which are however "sufficiently close" to K (for example such that $[K : k]$ is finite) is that they allow us to replace in certain questions the modules of differentials Ω^1_{K/k_0} and Ω^1_{L/k_0} (which can be "too large", for example when k_0 is the prime field) by $\Omega^1_{K/k}$ and $\Omega^1_{L/k}$, which are more easily handled.

When k_0 is the *prime field*, one says "admissible field" instead of "k_0-admissible field".

We set

(21.6.1.2).
$$\Delta(L/K, k/k_0) = \mathrm{Coker}(\Upsilon_{K/k/k_0} \otimes_K L \longrightarrow \Upsilon_{L/k/k_0}) \simeq \mathrm{Ker}(\Upsilon_{L/K/k_0} \longrightarrow \Upsilon_{L/K/k})$$

(vector space over L); its rank will be denoted $d(L/K, k/k_0)$ and called the *defect of k_0-admissibility of k for the extension L of K* (it is evidently zero if and only if k is k_0-admissible for this extension). When k_0 is the prime field, one writes $\Delta(L/K, k)$ and $d(L/K, k)$ instead of $\Delta(L/K, k/k_0)$ and $d(L/K, k/k_0)$.

Proposition (21.6.2). — *Let $k_0 \subset k \subset K \subset L$ be four fields.*

(i) *The following conditions are equivalent:*
   a) *The field k is k_0-admissible for the extension L of K (in other words, the homomorphism $\Upsilon_{L/K/k_0} \to \Upsilon_{L/K/k}$ is injective, hence bijective).*
   b) *The canonical homomorphism $u : \Upsilon_{K/k/k_0} \to \Upsilon_{L/k/k_0}$ is zero.*
   c) *The canonical homomorphism $v : \Upsilon_{K/k/k_0} \otimes_K L \to \Upsilon_{L/k/k_0}$ is surjective (hence bijective).*
   d) *One has $d(L/K, k/k_0) = 0$ (or $\Delta(L/K, k/k_0) = 0$).*

(ii) *The equivalent conditions of* (i) *are satisfied when one is in one of the following cases: α) L is separable over k; β) L is separable over K; γ) one has $k \subset k_0(K^p)$, where p denotes the characteristic exponent of k_0.*

PROOF. (i) The assertions result trivially from the exactness of the sequence (21.6.1.1).

(ii) If L is separable over k, one has $\Upsilon_{L/K/k_0} = 0$ (20.6.19), hence condition b) of (i) is satisfied; if L is separable over K, one has $\Upsilon_{L/K/k_0} = 0$ (20.6.19), hence condition a) of (i) is satisfied; finally, if $k \subset k_0(K^p)$, it follows that $\Upsilon_{L/K/k_0} = \Upsilon_{L/K/k}$ by virtue of (21.1.5.1), and the condition a) of (i) is satisfied.

□

(21.6.3). Suppose that one has a commutative diagram of monomorphisms of fields

$$\begin{tikzcd} k'_0 \arrow[r] & k' \arrow[r] & K' \arrow[r] & L' \\ k_0 \arrow[r] \arrow[u] & k \arrow[r] \arrow[u] & K \arrow[r] \arrow[u] & L \arrow[u] \end{tikzcd}$$

It then results from (20.6.17), (ii) that one has a canonical homomorphism

(21.6.3.1)
$$\Delta(L/K, k/k_0) \longrightarrow \Delta(L'/K', k'/k'_0)$$

with an evident transitivity property, so that one can say that $\Delta(L/K, k/k_0)$ is a *functor* in the quadruplet (k_0, k, K, L).

Proposition (21.6.4). — (i) *Let $k \subset k' \subset k'' \subset K \subset L$ be five fields. One has an exact sequence of canonical homomorphisms*

(21.6.4.1) $$0 \longrightarrow \Delta(L/K, k'/k) \longrightarrow \Delta(L/K, k''/k) \longrightarrow \Delta(L/K, k''/k') \longrightarrow 0$$

and consequently the equality

(21.6.4.2) $$d(L/K, k''/k) = d(L/K, k''/k') + d(L/K, k'/k).$$

(ii) *Let $k_0 \subset k \subset K \subset L \subset M$ be five fields. One has an exact sequence of canonical homomorphisms*

(21.6.4.3) $$0 \longrightarrow \Delta(L/K, k/k_0) \otimes_L M \longrightarrow \Delta(M/K, k/k_0) \longrightarrow \Delta(M/L, k/k_0) \longrightarrow 0$$

and consequently the equality

(21.6.4.4) $$d(M/K, k/k_0) = d(M/L, k/k_0) + d(L/K, k/k_0).$$

PROOF. (i) Consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \Upsilon_{K/k'/k} \otimes_K L \arrow[r] \arrow[d] & \Upsilon_{K/k''/k} \otimes_K L \arrow[r] \arrow[d] & \Upsilon_{K/k''/k'} \otimes_K L \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Upsilon_{L/k'/k} \arrow[r] \arrow[d] & \Upsilon_{L/k''/k} \arrow[r] \arrow[d] & \Upsilon_{L/k''/k'} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Delta(L/K, k'/k) \arrow[r] & \Delta(L/K, k''/k) \arrow[r] & \Delta(L/K, k''/k') \arrow[r] & 0
\end{tikzcd}$$

where one can consider the three rows as *complexes* $T_1^\bullet$, $T_2^\bullet$, $T_3^\bullet$ respectively; the exact sequence (21.6.1.1) and the definition of Δ (21.6.1.2) show that one has an exact sequence of complexes $0 \to T_1^\bullet \to T_2^\bullet \to T_3^\bullet \to 0$; applying the exact cohomology sequence to it, and noting that, by virtue of the exactness of (21.6.1.1), the cohomology of $T_1^\bullet$ and that of $T_2^\bullet$ are zero except in a single degree, for which the cohomology modules are both equal to $\Upsilon_{k''/k'/k} \otimes_{k''} L$; as $T_1^\bullet$ and $T_2^\bullet$ have the same cohomology, that of $T_3^\bullet$ is necessarily zero, which proves (i).

(ii) Consider similarly the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \Delta(L/K, k/k_0) \otimes_L M \arrow[r] \arrow[d] & \Delta(M/K, k/k_0) \arrow[r] \arrow[d] & \Delta(M/L, k/k_0) \arrow[r] & 0 \\
0 \arrow[r] & \Upsilon_{L/K/k_0} \otimes_L M \arrow[r] \arrow[d] & \Upsilon_{M/K/k_0} \arrow[r] \arrow[d] & \Upsilon_{M/L/k_0} \arrow[r] & 0 \\
0 \arrow[r] & \Upsilon_{L/K/k} \otimes_L M \arrow[r] & \Upsilon_{M/K/k} \arrow[r] & \Upsilon_{M/L/k} \arrow[r] & 0
\end{tikzcd}$$

where again one considers the three rows as *complexes* $T_1'^\bullet$, $T_2'^\bullet$, and $T_3'^\bullet$; the exact sequence (21.6.1.1) and the definition of Δ (21.6.1.2) give here a short exact sequence of complexes $0 \to T_1'^\bullet \to T_2'^\bullet \to T_3'^\bullet \to 0$; applying the exact cohomology sequence to it again, this time, by virtue of the exactness of (21.6.1.1), the cohomology of $T_2'^\bullet$ and that of $T_3'^\bullet$ are zero except in a single degree, for which the cohomology modules are both equal to $\Upsilon_{M/L/K}$; one concludes therefore that the cohomology of $T_1'^\bullet$ is zero, which establishes (ii). □

Corollary (21.6.5). — (i) *Given five fields $k \subset k' \subset k'' \subset K \subset L$, for k'' to be k-admissible for the extension L of K, it is necessary and sufficient that k' be k-admissible and that k'' be k'-admissible for the extension L of K.*

(ii) *Given five fields $k_0 \subset k \subset K \subset L \subset M$, for k to be k_0-admissible for the extension M of K, it is necessary and sufficient that it be so for the extension L of K and for the extension M of L.*

This results immediately from the relations (21.6.4.2) and (21.6.4.4), the values of d being $\geqslant 0$.

Corollary (21.6.6). — *Let $k_0 \subset k \subset K \subset L$ be four fields, and suppose that k is k_0-admissible for the extension L of K. Then, if k_0', k', K', L' are four fields such that $k_0 \subset k_0' \subset k' \subset k \subset K \subset K' \subset L' \subset L$, k' is k_0'-admissible for the extension L' of K'.*

21.7. Cartier's equality The following result translates in terms of differentials a theorem of MacLane on derivations:

Theorem (21.7.1). — (Cartier). *Let K be a field, L a finite type extension of K. Then $\Omega^1_{L/K}$ and $\Upsilon_{L/K}$ are L-vector spaces of finite rank, and one has*

(21.7.1.1) $$\operatorname{rg}\Omega^1_{L/K} - \operatorname{rg}\Upsilon_{L/K} = \deg.\operatorname{tr}_K L.$$

PROOF. If L' is a field such that $K \subset L' \subset L$, one has the exact sequence (20.6.15.1) (applied to $\Lambda = P$, $A = K$, $B = L'$, $C = L$)

$$0 \longrightarrow \Upsilon_{L'/K} \otimes_{L'} L \longrightarrow \Upsilon_{L/K} \longrightarrow \Omega^1_{L'/K} \otimes_{L'} L \longrightarrow \Omega^1_{L/K} \longrightarrow \Omega^1_{L/L'} \longrightarrow 0$$

hence ($\mathbf{0_{III}}$, 11.10.2)

$$\operatorname{rg}_L \Omega^1_{L/K} - \operatorname{rg}_L \Upsilon_{L/K} = (\operatorname{rg}_{L'} \Omega^1_{L/L'} - \operatorname{rg}_{L'} \Upsilon_{L/L'}) + (\operatorname{rg}_{L'} \Omega^1_{L'/K} - \operatorname{rg}_{L'} \Upsilon_{L'/K}).$$

Since moreover $\deg.\operatorname{tr}_K L = \deg.\operatorname{tr}_K L' + \deg.\operatorname{tr}_{L'} L$, one sees, by recurrence on the number of generators of the extension L, that one is reduced to proving (21.7.1.1) when $L = K(x)$. We then distinguish three cases:

a) x is transcendent over K; as L is separable over K, one has $\Upsilon_{L/K} = 0$ (20.6.3); on the other hand, (20.5.9) and (20.4.13) show that $\Omega^1_{L/K}$ is of rank 1, hence (21.7.1.1) in this case.

b) L is a finite algebraic separable extension of K, so that one has again $\Upsilon_{L/K} = 0$ (20.6.3). On the other hand, one has $\Omega^1_{L/K} = 0$ by (20.6.20), hence again (21.7.1.1).

c) L is a finite algebraic inseparable extension of K; the reasoning at the beginning shows that one can restrict to the case where $L^p \subset K$; it then results from (21.4.8) that one has $\operatorname{rg}_L \Upsilon_{L/K} = \operatorname{rg}_L \Omega^1_{L/K}$, hence again (21.7.1.1). □

Corollary (21.7.2). — *Let K be a field, L a finite type extension of K, k a sub-field of K. Then $\Omega^1_{L/K}$ and $\Upsilon_{L/K/k}$ are vector spaces of finite rank over L, and one has*

(21.7.2.1) $$\operatorname{rg}\Omega^1_{L/K} - \operatorname{rg}\Upsilon_{L/K/k} = \deg.\operatorname{tr}_K L + d(L/K,k).$$

Moreover, one has the inequality

(21.7.2.2) $$\operatorname{rg}\Omega^1_{L/K} - \operatorname{rg}\Upsilon_{L/K/k} \geqslant \deg.\operatorname{tr}_K L.$$

Furthermore, for the two members of (21.7.2.2) to be equal, it is necessary and sufficient that k be an admissible field for the extension L of K.

PROOF. Indeed, as the homomorphism $\Upsilon_{L/K} \to \Upsilon_{L/K/k}$ is surjective, one has by definition (21.6.1.2) $\operatorname{rg}\Upsilon_{L/K} - \operatorname{rg}\Upsilon_{L/K/k} = d(L/K,k)$, and the corollary results immediately from (21.7.1) and from (21.6.2). □

Corollary (21.7.3). — *Let $k \subset K \subset L$ be three fields such that L is a finite type extension of k. Then one has*

(21.7.3.1) $$\operatorname{rg}_L \Omega^1_{L/k} - \operatorname{rg}_K \Omega^1_{K/k} = \deg.\operatorname{tr}_K L + d(L/K,k).$$

Moreover, one has the inequality $\mathbf{0_{IV\text{-}1}}$ | 171

(21.7.3.2) $$\operatorname{rg}_L \Omega^1_{L/k} - \operatorname{rg}_K \Omega^1_{K/k} \geqslant \deg.\operatorname{tr}_K L$$

the equality being attained if and only if k is an admissible field for the extension L of K.

PROOF. One has the exact sequence (20.6.1.1)

$$0 \longrightarrow \Upsilon_{L/K/k} \longrightarrow \Omega^1_{K/k} \otimes_K L \longrightarrow \Omega^1_{L/k} \longrightarrow \Omega^1_{L/K} \longrightarrow 0$$

hence the first member of (21.7.3.1) is equal to

$$\operatorname{rg}_L \Omega^1_{L/k} - \operatorname{rg}_L \Upsilon_{L/K/k}$$

and it suffices to apply (21.7.2). □

The interest of this last corollary is that it only involves modules of differentials, to the exclusion of modules of imperfection. We shall see moreover below (21.8.6) that for every finite type extension L of K, there exists a sub-field k of K such that $[K : k]$ is finite, and which is admissible for the extension L of K, so that the two members of (21.7.3.2) are then equal.

Corollary (21.7.4). — *Let K be a finite type extension of a field k. The following conditions are equivalent:*

a) *K is a finite separable extension of k.*
b) *K is a formally unramified k-algebra (19.10.2).*
c) *K is a formally étale k-algebra (19.10.2).*
d) *One has $\Omega^1_{K/k} = 0$.*

PROOF. Indeed, one knows (the topologies being discrete) that the conditions b) and d) are equivalent (20.7.4) and that a) implies that K is a formally smooth k-algebra (19.6.1), hence the conjunction of a) and b) is equivalent to c). The whole thing reduces to seeing that d) implies a). Now, by virtue of (21.7.1.1), the relation $\Omega^1_{K/k} = 0$ implies that K is algebraic (hence finite) over k and that $\Upsilon_{K/k} = 0$, that is to say (20.6.3) that K is separable over k. □

Remark (21.7.5). — By virtue of (20.6.3.2) and ($\mathbf{0_{III}}$, 11.10.2) the first member of (21.7.1.1) is none other than the *Euler-Poincaré characteristic* of the complex $\mathrm{K}_\bullet(C/B/A)$ introduced in (20.6.3), for $C = L$, $B = K$ and $A = P$ (prime field of K). In the chapter devoted to intersection theory and to the theorem of Riemann-Roch, an important role will be played equally by generalized Euler-Poincaré characteristics (with values in groups of classes of $\mathscr{O}_X$-Modules) of complexes generalizing the complexes $\mathrm{F}_\bullet(C/A)$ considered in (20.6.22).

21.8. Criteria for admissibility We return to our previous conventions and suppose therefore that all fields considered in this number are of characteristic $p > 0$.

Lemma (21.8.1). — *Let K be a field, k a sub-field of K, $(k_\alpha)_{\alpha\in I}$ a decreasing filtering family of sub-fields of K such that $k = \bigcap_\alpha k_\alpha$. Let V be a vector space over K, $(a_i)_{1\leqslant i\leqslant n}$ a finite family of vectors of V; if the family (a_i) is free over k, there exists an index γ such that it is also free over k_γ.*

PROOF. Let r be the rank of the family $(a_i)_{1\leqslant i\leqslant n}$ over K, and let us reason by recurrence on $n-r$; $\mathbf{0_{IV\text{-}1}}$ | 172
the proposition is evident for $n = r$, for the family $(a_i)_{1\leqslant i\leqslant r}$ is then free over K, hence over every sub-field of K. Suppose for example that $(a_i)_{1\leqslant i\leqslant r}$ is free over K, and write $a_{r+1} = \sum_{i=1}^r \lambda_i a_i$ with $\lambda_i \in K$; the family $(a_i)_{1\leqslant i\leqslant r+1}$ being free over k, the λ_i cannot all belong to k; suppose for example that $\lambda_{i_0} \notin k$. Then there is an index β such that $\lambda_{i_0} \notin k_\beta$; one concludes that the family $(a_i)_{1\leqslant i\leqslant r+1}$ is *free over* k_β; indeed, if the family $(a_i)_{1\leqslant i\leqslant r}$ were not free over k_β, a_{r+1} would be equal to a linear combination of the a_i with coefficients in k_β, and as these coefficients are necessarily the λ_i, one would arrive at a contradiction. It now suffices to apply the recurrence hypothesis by replacing K by k_β and the family $(k_\alpha)_{\alpha\in I}$ by the sub-family of the k_α contained in k_β. □

Lemma (21.8.2). — *Let K be a field, p its exponential characteristic, k_0 a sub-field of K, $(k_\alpha)_{\alpha\in I}$ a decreasing filtering family of sub-fields of K such that $\bigcap_\alpha k_\alpha(K^p) = k_0(K^p)$. If $(x_i)_{1\leqslant i\leqslant n}$ is a finite family of elements of K that is p-free over k_0, there exists an index α such that (x_i) is also p-free over k_α.*

PROOF. Indeed, to say that the family (x_i) is p-free over a sub-field k of K means that the finite family of monomials $\prod_{i=0}^n x_i^{m(i)}$ with $0 \leqslant m(i) < p$ is free over $k(K^p)$; it suffices therefore to apply the lemma (21.8.1) to this family of monomials in the vector space $V = K$, and to the sub-fields $k_\alpha(K^p)$ and $k_0(K^p)$ of K. □

Theorem (21.8.3). — *Let K be a field of characteristic $p > 0$, k_0 a sub-field of K, $(k_\alpha)_{\alpha\in I}$ a decreasing filtering family of sub-fields of K containing k_0. The following conditions are equivalent:*

a) *$\bigcap_\alpha k_\alpha(K^p) = k_0(K^p)$.*
b) *For every extension L of K such that $\Upsilon_{L/K/k_\alpha}$ is an L-vector space of finite rank* (*which holds in particular if L is a finite type extension in virtue of* (21.7.2)), *there exists $\alpha \in I$ such that k_α is k_0-admissible for the extension L of K.*
b′) *For every extension $L = K(x)$ of K, with $x^p \in K$, there exists $\alpha \in I$ such that k_α is k_0-admissible for the extension L of K.*
c) *The canonical application*

$$\Omega^1_{K/k_0} \longrightarrow \varprojlim_\alpha \Omega^1_{K/k_\alpha} \tag{21.8.3.1}$$

is injective.

PROOF. The canonical application (21.8.3.1) is understood as passage to the projective limit in the projective system of homomorphisms $\Omega^1_{K/k_0} \to \Omega^1_{K/k_\alpha}$ (20.5.3.3). We shall prove the logical scheme $c) \Rightarrow b) \Rightarrow b') \Rightarrow a) \Rightarrow c)$.

To say that k_α is k_0-admissible for the extension L of K means that the canonical homomorphism $\Upsilon_{L/K/k_0} \to \Upsilon_{L/K/k_\alpha}$ is injective; or, if N_α is the kernel of this homomorphism, the N_α form a decreasing filtering system of sub-vector spaces of $\Upsilon_{L/K/k_0}$, and as $\Upsilon_{L/K/k_0}$ is by hypothesis of *finite* rank, it amounts to the same to say that one of the N_α is 0 or that their *intersection* is 0. But this intersection is none other than the kernel of the projective limit homomorphism $\Upsilon_{L/K/k_0} \to \varprojlim_\alpha \Upsilon_{L/K/k_\alpha}$. Or, one has the commutative diagram

$$\begin{tikzcd}
\Upsilon_{L/K/k_0} \arrow[r] \arrow[d] & \varprojlim \Upsilon_{L/K/k_\alpha} \arrow[d] \\
\Omega^1_{K/k_0} \otimes_K L \arrow[r] & \varprojlim(\Omega^1_{K/k_\alpha} \otimes_K L)
\end{tikzcd}$$

where the left vertical arrow is injective by definition. To prove that $c)$ implies $b)$, it suffices therefore to show that the canonical application

$$\Omega^1_{K/k_0} \otimes_K V \longrightarrow \varprojlim_\alpha(\Omega^1_{K/k_\alpha} \otimes_K V) \tag{21.8.3.2}$$

is injective for every vector space V over K (and in particular for $V = L$). Now this is evident if $V = K^n$ since then $W \otimes_K V = W^n$ for every vector space W over K and projective limits and products commute. On the other hand, for every element z of the first member of (21.8.3.2) there exists a sub-space V' of V of finite rank containing z (identified canonically with a sub-space of $\Omega^1_{K/k_0} \otimes_K V'$); if $z \neq 0$, its image $\varprojlim(\Omega^1_{K/k_\alpha} \otimes_K V')$ is also non-zero; as the $\varprojlim$ functor is exact on the left, the image of z in the second member of (21.8.3.2) is therefore also $\neq 0$, which finishes the proof that $c)$ implies $b)$.

It is trivial that $b)$ implies $b')$; let us show that $b')$ implies $a)$. Using the notations of $b')$, on can suppose $L \neq K$. It then results from (21.4.7) that one has $\operatorname{rg} \Upsilon_{L/K} \leqslant 1$. If $\Upsilon_{L/K/k_\alpha} = 0$ there is nothing to prove by virtue of (21.6.1.1). Otherwise, $\Upsilon_{L/K/k_\alpha}$ is identified canonically with $\Upsilon_{L/K}$, and if one sets $a = x^p$, $\Upsilon_{L/K}$ is a one-dimensional space generated by the single element $d_K(a) \otimes 1$ (21.4.7); $\Upsilon_{L/K/k_\alpha}$ is thus a one-dimensional space generated by the single element $d_{K/k_\alpha}(a) \otimes 1$, i.e., to say that a sub-field k_α is k_0-admissible for the extension L of K means that one has $d_{K/k_\alpha}(a) \neq 0$, or equivalently (21.4.6) that $a \notin k_\alpha(K^p)$. Or, for every $a \notin k_0(K^p)$, one has $d_{K/k_0}(a) \neq 0$ and $K(a^{1/p}) \neq K$, hence one can apply $b')$, which proves the existence of an α such that $a \notin k_\alpha(K^p)$; as this holds for every $a \notin k_0(K^p)$, we have shown $b') \Rightarrow a)$.

It remains to show that $a)$ implies $c)$. Let $(x_\mu)_{\mu \in M}$ be a p-base of K over k_0; then the $d_{K/k_0}(x_\mu)$ form a base of the K-vector space Ω^1_{K/k_0} (21.4.5), and the condition $c)$ means that for every finite subset J of the set of indices M, there exists $\alpha \in I$ such that the $d_{K/k_\alpha}(x_\mu)$ for $\mu \in J$ are linearly independent in Ω^1_{K/k_α}; but this means also (21.4.5) that the x_μ for $\mu \in J$ form a p-free family *over* k_α, and the existence of an α having this property follows from (21.8.2) and from the hypothesis $a)$. □

Corollary (21.8.4). — *Let K be a field of characteristic $p > 0$, k_0 a sub-field of K, $(k_\alpha)_{\alpha \in I}$ a decreasing filtering family of sub-fields of K, containing k_0, such that*

$$\bigcap_\alpha k_\alpha(K^p) = k_0(K^p).$$

Let L be an extension of K such that $\Upsilon_{L/K/k_0}$ is an L-vector space of finite rank; then, for every field K' such that $K \subset K' \subset L$, there exists $\alpha \in I$ such that k_α is k_0-admissible for the extension L of K'.

PROOF. It suffices to apply (21.8.3) and (21.6.6). □

Corollary (21.8.5). — *Let K be a field of characteristic $p > 0$, k_0 a sub-field of K, $(k_\alpha)_{\alpha \in I}$ a decreasing filtering family of sub-fields of K containing k_0, such that*

$$\bigcap_\alpha k_\alpha(K^p) = k_0(K^p).$$

Then, for every extension L of K such that $\Upsilon_{L/K/k_0}$ is an L-vector space of finite rank, one has $\bigcap_\alpha k_\alpha(L^p) = k_0(L^p)$.

PROOF. Suppose indeed that M is an extension of L such that $\Upsilon_{M/L/k_0}$ is an M-vector space of finite rank. The exact sequence (21.6.1.1) **$0_{\text{IV-1}}$** | 174

$$0 \longrightarrow \Upsilon_{L/K/k_0} \otimes_L M \longrightarrow \Upsilon_{M/K/k_0} \longrightarrow \Upsilon_{M/L/k_0} \longrightarrow \Upsilon_{M/L/K} \longrightarrow 0$$

and the hypothesis show that $\Upsilon_{M/K/k_0}$ is also an M-vector space of finite rank. There exists therefore an index α such that k_α is k_0-admissible for the extension M of K (21.8.3); this holding also for the extension M of L (21.6.6); the equivalence of a) and b) in (21.8.3) gives the corollary. □

Corollary (21.8.6). — *Let K be a field of characteristic $p > 0$, k_0 a sub-field of K, $(k_\alpha)_{\alpha \in I}$ a decreasing filtering family of sub-fields of K containing k_0, such that*

$$\bigcap_\alpha k_\alpha(K^p) = k_0(K^p).$$

Let L be an extension of K such that $\Upsilon_{L/K/k_0}$ is an L-vector space of finite rank; then there exists a sub-field k of K, containing $k_0(K^p)$, such that $[K : k]$ is finite, and which is k_0-admissible for the extension L of K.

PROOF. It suffices indeed, by virtue of (21.8.3), to construct a decreasing filtering family $(k_\alpha)_{\alpha \in I}$ of sub-fields of K, containing K^p and k_0, for which $[K : k_\alpha] < +\infty$ and $\bigcap_\alpha k_\alpha = k_0(K^p)$. For this one considers a p-base $(x_\lambda)_{\lambda \in J}$ of K over k_0 and, for every finite subset H of J, one considers the sub-field k_H of K generated by $k_0(K^p)$ and the x_λ for $\lambda \in J - H$; it results from this definition that $(x_\lambda)_{\lambda \in H}$ is a p-base of K over k_H, and one concludes immediately that the k_H satisfy the required conditions. □

Remarks (21.8.7). —

(i) One has already seen (21.7.2) that if L is a *finite type* extension of K, $\Upsilon_{L/K/k_0}$ is of finite rank for every sub-field k_0 of K. It is the same if L is a *separable* extension of K, for in virtue of (20.6.19), one has $\Upsilon_{L/K/k_0} = 0$. Finally, if L is a *finite type extension of a separable extension L_0 of K*, the same reasoning as in (21.8.5) shows that $\Upsilon_{L/K/k_0}$ is still of finite rank (and in fact is isomorphic to a sub-space of $\Upsilon_{L/L_0/k_0}$).

(ii) In the statement of (21.8.5), and consequently in that of (21.8.3), b)), one can omit the hypothesis that $\Upsilon_{L/K/k_0}$ is of finite rank over L. Let us take for k_0 the prime field $\mathbf{F}_p$, for K a field such that $[K : K^p]$ is countably infinite (for example the field of rational fractions $\mathbf{F}_p(X_1, \ldots, X_n, \ldots)$ with a countable infinity of indeterminates); the construction procedure of (21.8.6) then shows that there exists an infinite strictly decreasing sequence (K_n) of sub-fields of K such that $K_0 = K$ and $\bigcap_n K_n = K^p$. We shall construct increasing finite extensions M_n of K, such that if M is the union of the M_n, the extension $L = M^{1/p}$ of K gives (21.8.5) a counterexample. Namely, take an element x of $K - K^p$, set $M_0 = K^p$, and $M_n = K^p(a_1 x, a_2 x, \ldots, a_n x)$ for $n \geqslant 1$, where the a_n are constructed by recurrence so that $a_n \in K_n$ and $a_{n+1} \notin M_n(x)$ for all n: this is possible, since $M_n(x)$ is of finite degree over K^p, while K_n is not of the same for K_n. One concludes also that $x \notin M = L^p$, but as $x = (a_n x) a_n^{-1}$, on the other hand, $a_n \in K_n(M) = K_n(L^p)$, one has $x \in K_n(L^p)$ for all n, hence $x \in \bigcap_n K_n(L^p)$, which proves our assertion.

Proposition (21.8.8). — *Let k be a field of characteristic $p > 0$, and let $A = k[[T_1, \ldots, T_r]]$ be the ring of formal power series in r indeterminates over k, $K = k((T_1, \ldots, T_r))$ its field of fractions. Then there exists a decreasing filtering family (A_α) of sub-rings noetherian of A such that A is an A_α-module* free of finite type *for all α and that, if K_α is the field of fractions of A_α, one has $\bigcap_\alpha K_\alpha = K^p$.*

PROOF. One can write $K^p = k^p((T_1^p, \ldots, T_r^p))$; as one has seen in the proof of (21.8.6), there exists a decreasing family (k_α) of sub-fields of k such that $[k : k_\alpha]$ is finite for all α and $\bigcap_\alpha k_\alpha = k^p$; it is clear that if one sets $A_\alpha = k_\alpha[[T_1^p, \ldots, T_r^p]]$, A is an A_α-module *free of finite type*; the whole thing therefore reduces to proving the relation

(21.8.8.1) $$\bigcap_\alpha k_\alpha((T_1^p, \ldots, T_r^p)) = k^p((T_1^p, \ldots, T_r^p)).$$

Since $\bigcap_\alpha k_\alpha = k^p$, this will result from the following two lemmas: □

Lemma (21.8.8.2). — *If k' is an extension of a field k, one has*

$$k'[[T_1,\ldots,T_r]]\cap k((T_1,\ldots,T_r))=k[[T_1,\ldots,T_r]].$$

PROOF. Indeed, set $C=k[[T_1,\ldots,T_r]]$, $D=k'[[T_1,\ldots,T_r]]$; as $k((T_1,\ldots,T_r))$ is the field of fractions of C, it will suffice to prove that D is a C-module *faithfully flat* (Bourbaki, *Alg. comm.*, chap. I, § 3, no. 5, prop. 10). Now, C and D are local rings noetherian, and if $\mathfrak{m}$ is the maximal ideal of C, one has $D/\mathfrak{m}^jD=(C/\mathfrak{m}^j)\otimes_k k'$, hence $D/\mathfrak{m}^jD$ is a flat $(C/\mathfrak{m}^j)$-module; it suffices to apply ($\mathbf{0_{III}}$, 10.2.1) and ($\mathbf{0_I}$, 6.6.2). □

Lemma (21.8.8.3). — *Let k be a field, (k_α) a decreasing filtering family of sub-fields of k, and set $k_0=\bigcap_\alpha k_\alpha$. Suppose there exists a power q of the characteristic exponent of k such that $k^q\subset k_0$. Then one has*

$$\bigcap_\alpha k_\alpha((T_1,\ldots,T_r))=k_0((T_1,\ldots,T_r)). \tag{21.8.8.4}$$

PROOF. It suffices to prove that a nonzero element f of the first member of (21.8.8.4) belongs to the second member. Let γ be an index, $g\in k_\gamma[[T_1,\ldots,T_r]]$ such that $gf\in k_\gamma[[T_1,\ldots,T_r]]$; replacing g by g^q, one can suppose that $g\in k_0[[T_1,\ldots,T_r]]$. Then, for all $\alpha\geqslant\gamma$, one has $\mathbf{0_{IV\text{-}1}}$ | 176

$$gf\in k_\gamma[[T_1,\ldots,T_r]]\cap k_\alpha((T_1,\ldots,T_r))=k_\alpha[[T_1,\ldots,T_r]]$$

by virtue of the lemma (21.8.8.2). But it is clear that the intersection of the rings $k_\alpha[[T_1,\ldots,T_r]]$ is $k_0[[T_1,\ldots,T_r]]$, and one has thus $gf\in k_0((T_1,\ldots,T_r))$. □

21.9. Completed modules of differentials in formal power series rings In this number, the fields are no longer necessarily supposed to be of characteristic >0.

Lemma (21.9.1). — *Let k be a field, A a local noetherian complete ring that is a k-algebra, K the residue field of A. If K is a finite type extension of k, the A-module $\widehat{\Omega}^1_{A/k}$ is of finite type.*

PROOF. By virtue of (20.7.15), it suffices to prove that $\Omega^1_{K/k}$ is a K-vector space of finite rank. Now, by hypothesis, K is the residue field of a k-algebra of finite type B; as $\Omega^1_{B/k}$ is a B-module of finite type by virtue of (20.4.7), the conclusion results from (20.5.9). □

Proposition (21.9.2). — *Let k_0 be a field, A a local noetherian complete ring that is a k_0-algebra formally smooth, $\mathfrak{m}$ its maximal ideal, $K=A/\mathfrak{m}$ its residue field* (*so that in particular A is isomorphic to a formal power series ring $K[[T_1,\ldots,T_r]]$* (19.6.5)). *If K is a finite type extension of k_0, then $\widehat{\Omega}^1_{A/k_0}$ is an A-module* free *of rank equal to* $\dim(A)+\deg.\mathrm{tr}_{k_0}K$. *Furthermore, for every sub-field k of k_0 such that $\Omega^1_{k_0/k}$ is of finite rank, $\widehat{\Omega}^1_{A/k}$ is an A-module free of rank equal to* $\dim(A)+\deg.\mathrm{tr}_{k_0}K+\mathrm{rg}_{k_0}(\Omega^1_{k_0/k})$.

PROOF. It is clear that, if P is the prime sub-field of k_0, K is a P-algebra formally smooth (19.6.1). Let us note on the other hand that since A is integral over A_0, A_0 integrally closed and E is separable over L_0, Ω^1_{A/A_0} is an A-module of *torsion* (20.4.13), (iv)); tensoring the exact sequence (21.9.2) by E, it comes an exact sequence

$$\widehat{\Omega}^1_{A_0/k}\otimes_{A_0}E\longrightarrow\widehat{\Omega}^1_{A/k}\otimes_A E\longrightarrow 0$$

hence $\mathrm{rg}_E(\widehat{\Omega}^1_{A/k}\otimes_A E)\leqslant r$. It remains to show that there exist, in $\widehat{\Omega}^1_{A/k}$, r elements linearly independent over E. Now, let $D_i=\partial/\partial T_i$ be the canonical derivations of A into P, which are restrictions of k-derivations continuous of A into A-modules topological of type fini (20.3.7.1); as E is a separable finite extension of L_0, they extend uniquely to derivations of A into E (still denoted D_i); by restriction to A, these derivations give derivations of A into E, and it is immediate that these derivations take their values in the same finite type sub-A-module of E; as they are continuous in A_0 and the topology of A is the $\mathfrak{m}_0$-adic topology, on has thus formed r *continuous* derivations of A into a topological A-module, which are evidently linearly independent, since $D_i(T_j)=\delta_{ij}$; this finishes the proof of the proposition. □ $\mathbf{0_{IV\text{-}1}}$ | 177

Corollary (21.9.3). — *Let k be a field, A the ring of formal power series $k[[T_1,\ldots,T_r]]$, equipped with its usual structure as a k-algebra. Then $\widehat{\Omega}^1_{A/k}$ is an A-module free of rank $r=\dim(A)$.*

PROOF. It suffices to note that A is a k-algebra formally smooth (19.3.4), and to apply (21.9.2) with $k_0=k=K$. □

Lemma (21.9.4). — *Let k be a field of characteristic $p > 0$, A a k-algebra that is a local noetherian complete ring, B a sub-k-algebra of A isomorphic to a topological algebra of the form $k[[T_1, \ldots, T_r]]$ and such that A is a B-algebra of finite type. If B_1 is the sub-algebra $k[[T_1^p, \ldots, T_r^p]]$ of B, $\widehat{\Omega}^1_{A/k}$ identifies canonically with Ω^1_{A/B_1}.*

PROOF. Every continuous derivation of B_1 into a topological A-module P, which is the restriction of a k-derivation continuous of A into P, is *zero*, for it is zero in the sub-ring $k[T_1^p, \ldots, T_r^p]$ of polynomials, and this last is dense in B_1. The exact sequence (20.3.7.1) then shows that the canonical homomorphism

$$\text{Dér. cont}_{B_1}(A, P) \longrightarrow \text{Dér. cont}_k(A, P)$$

is bijective; the conclusion results from (20.14.4), taking into account that Ω^1_{A/B_1} is an A-module of finite type (20.4.7), hence separated and complete since A is a local noetherian complete ring. □

Proposition (21.9.5). — *Let A be a local noetherian complete ring integral, A_0 a sub-ring of A isomorphic to a formal power series ring $k[[T_1, \ldots, T_r]]$ over a field k, such that A is an A_0-algebra finite and that the field of fractions E of A is a separable extension of the field of fractions L_0 of A_0. Then one has*

$$\text{(21.9.5.1)} \qquad \mathrm{rg}_A\, \widehat{\Omega}^1_{A/k} = \mathrm{rg}_E(\widehat{\Omega}^1_{A/k_0} \otimes_A E) = \dim(A).$$

PROOF. Let us note that if $\mathfrak{m}_0$ is the maximal ideal of A_0, the topology of A is the $\mathfrak{m}_0$-adic topology since A is an A_0-algebra finite (Bourbaki, *Alg. comm.*, chap. IV, § 2, no. 5, cor. 3 of prop. 9) and it induces on A_0 the $\mathfrak{m}_0$-adic topology (Bourbaki, *Alg. comm.*, chap. III, § 3, no. 4, th. 3). One knows (21.9.1) that $\widehat{\Omega}^1_{A/k}$ is an A-module of finite type and, moreover, Ω^1_{A/A_0} is an A-module of *torsion* (20.4.13), (iv)); since the exact sequence (20.5.2)

$$\text{(21.9.5.2)} \qquad \widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A \xrightarrow{u} \widehat{\Omega}^1_{A/k} \xrightarrow{v} \Omega^1_{A/A_0} \longrightarrow 0$$

is known to be exact by virtue of (20.7.17), and since Ω^1_{A/A_0} is a torsion module, we have $\mathrm{rg}_E(\widehat{\Omega}^1_{A/k} \otimes_A E) \leqslant r$.

It remains to show that there exist, in $\widehat{\Omega}^1_{A/k}$, r elements linearly independent over E. Now, let $D_i = \partial/\partial T_i$ be the canonical derivations of A_0 into itself; since E is a separable finite extension of L_0, they extend uniquely to derivations of E (still denoted D_i); by restriction to A, these derivations give derivations of A into E, and it is immediate that these derivations take their values in the same finite type sub-A-module of E; as they are continuous in A_0 and the topology of A is the $\mathfrak{m}_0$-adic topology, one has thus formed r continuous derivations of A into a topological A-module, which are evidently linearly independent, since $D_i(T_j) = \delta_{ij}$; this finishes the proof of the proposition. □

Proposition (21.9.6). — *Let $k_0 \subset k \subset K_0$ be three fields of characteristic $p > 0$, such that $[K_0 : k] < +\infty$; set*

$$L_0 = K_0((T_1, \ldots, T_r)), \quad M_0 = K_0((T_1^p, \ldots, T_r^p))$$

$$L = k((T_1, \ldots, T_r)), \quad M = k((T_1^p, \ldots, T_r^p)), \quad N = k((T_1^{p^2}, \ldots, T_r^{p^2})).$$

Let E be a finite type extension of L_0.

(i) *If Υ_{K_0/k_0} is of finite rank, one has*

$$\text{(21.9.6.1)} \qquad \mathrm{rg}_E\, \Omega^1_{E/M} - \mathrm{rg}_{K_0}\, \Omega^1_{K_0/k} = (r + \deg.\mathrm{tr}_{L_0} E) + d(E/K_0, k_0) + d(E/M_0, N/k_0)$$

(*notations of* (21.6.1)), *and in particular one has*

$$\text{(21.9.6.2)} \qquad \mathrm{rg}_E\, \Omega^1_{E/M} - \mathrm{rg}_{K_0}\, \Omega^1_{K_0/k} \geqslant r + \deg.\mathrm{tr}_{L_0} E + d(E/K_0, k_0).$$

Furthermore, if the two members of (21.9.6.2) *are equal, they remain so when one replaces k by a sub-field k' such that $k_0 \subset k' \subset k$.*

(ii) *Suppose moreover that $[k_0 : k_0^p] < +\infty$. Let (k_α) be a decreasing filtering family of sub-fields of K_0 containing k_0, such that $[K_0 : k_\alpha] < +\infty$ for all α and that $\bigcap_\alpha k_\alpha(K_0^p) = k_0(K_0^p)$; then there exists α such that, for $k = k_\alpha$, the two members of* (21.9.6.2) *are equal.*

PROOF. (i) One knows (21.1.5), (ii) that $\mathrm{rg}_E\,\Omega^1_{E/M}$ does not change when one replaces M by $M(L_0^p)$; as $L_0^p = K_0^p((T_1^p,\dots,T_r^p))$ and that $[k(K_0^p):k] < +\infty$, one has $M(L_0^p) = k(K_0^p)((T_1^p,\dots,T_r^p))$; similarly $\Omega^1_{K_0/k}$ does not change when one replaces k by $k(K_0^p)$, so that one can *suppose that* $K_0^p \subset k$, which we shall do in what follows. We introduce also the fields

$$k_0' = k_0(K_0^p), \quad N' = k_0'((T_1^{p^2},\dots,T_r^{p^2}))$$

so that one has the diagram of fields

$$\begin{array}{ccccccc} K_0 & \longrightarrow & M_0 & \longrightarrow & L_0 & \longrightarrow & E \\ \uparrow & & \uparrow & & \uparrow & & \\ k & \longrightarrow & N & \longrightarrow & M & & \\ \uparrow & & \uparrow & & \uparrow & & \\ k_0 & \longrightarrow & k_0' & \longrightarrow & N' & & \end{array}$$

We propose to evaluate the difference δ of the first and second members of (21.9.6.2). It results from (21.7.3) that one has

$$\mathrm{rg}_E\,\Omega^1_{E/M} - \mathrm{rg}_{M_0}\,\Omega^1_{M_0/M_0} = \deg.\mathrm{tr}_{M_0} E + d(E/M_0, M)$$

and as L_0 is a finite extension of M_0, one has $\deg.\mathrm{tr}_{L_0} E = \deg.\mathrm{tr}_{M_0} E$. On the other hand, a p-base of K_0 over $k = k(K_0^p)$ is also a p-base of M_0 over M, hence (21.4.5), one has

$$\mathrm{rg}_{M_0}\,\Omega^1_{M_0/M} = r + \mathrm{rg}_{K_0}\,\Omega^1_{K_0/k}$$

by virtue of (21.4.5) and (21.1.10), which proves (21.9.6.5).

On a again, by (21.6.4.2),

$$d(E/L_0, M) = d(E/L_0, M, k_0) + d(E/L_0, k_0, M/k_0)$$

and

$$d(L_0/M_0, N) = d(L_0/M_0, N/k) + d(L_0/M_0, k).$$

But as M_0 and L_0 are separable extensions of K_0 (21.9.6.4), one has $d(L_0/K_0, k) = 0$, and by (21.6.4.4)

$$d(L_0/M_0, k) = 0.$$

It follows therefore

$$\delta = d(E/L_0, M) + d(E/M_0, N/k_0) \geqslant 0$$

which proves (21.9.6.1).

On a again, by (21.6.4.2),

$$d(E/L_0, M, k_0) = d(E/L_0, M, k/k_0) + d(E/L_0, k/k_0, M/k_0).$$

Furthermore, when k is replaced by a sub-extension k' of k_0, N is replaced by a sub-extension N' of k_0, hence $d(E/M_0, N'/k_0) \leqslant d(E/M_0, N/k_0)$ (21.6.4.2), which proves the last assertion of (i).

(ii) For all α, set $N_\alpha = k_\alpha((T_1^{p^2},\dots,T_r^{p^2}))$. By virtue of (21.9.6.1) and of (21.8.3), it suffices to show (taking into account the fact that E is a finite type extension of M_0) that one has

$$\bigcap_\alpha N_\alpha = k_0(M_0^p).$$

Or, one has

(21.9.6.6)
$$\bigcap_\alpha N_\alpha = N' = (k_0(K_0^p))((T_1^{p^2},\dots,T_r^{p^2}))$$

by virtue of (21.8.8.3) and of the relation $\bigcap_\alpha k_\alpha = k_0(K_0^p)$ (21.8.6). On the other hand, one has $k_0(M_0^p) = k_0(K_0^p((T_1^{p^2},\dots,T_r^{p^2})))$. To finish the proof of (21.9.6), (ii), it remains therefore to prove

□

Lemma (21.9.6.4). — *For every field k, the field of series $K = k((T_1, \ldots, T_r))$ is a separable extension of k.*

PROOF. Let us recall rapidly the proof of this classical lemma. It suffices (Bourbaki, *Alg.*, chap. VIII, § 7, no. 3, proof of th. 1) to prove that for every *finite* extension k' of k, $K \otimes_k k'$ is without nilpotent element; but if one sets $A = k[[T_1, \ldots, T_r]]$ and $S = A - \{0\}$, $K \otimes_k k'$ is equal to $S^{-1}(A \otimes_k k')$, and $A \otimes_k k'$ identifies canonically with the integral ring $A' = k'[[T_1, \ldots, T_r]]$ and A to a sub-ring of A', hence the conclusion. □

Using this lemma and (21.6.2), (ii)), one has $d(L_0/K_0, k) = 0$, and by (21.6.4.4) also $d(L_0/M_0, k) = 0$ (21.6.2). It follows therefore

$$\delta = d(E/L_0, M) + d(E/M_0, N/k_0) \geqslant 0$$

which finishes the proof of (21.9.6), (i).

Lemma (21.9.6.7). — *Let k_0 be a field of characteristic $p > 0$ such that $[k_0 : k_0^p] < +\infty$, K_0 an extension of k_0. Then one has*

$$(k_0(K_0^p))((T_1, \ldots, T_r)) = k_0(K_0^p((T_1, \ldots, T_r))). \tag{21.9.6.8}$$

PROOF. As $k_0^p \subset K_0^p$, one has $k_0^p(K_0^p((T_1, \ldots, T_r))) = K_0^p((T_1, \ldots, T_r))$. If $(c_i)_{1 \leqslant i \leqslant n}$ is a base of k_0 over k_0^p, it is immediate that (c_i) is also a system of generators of both members of (21.9.6.8) over $K_0^p((T_1, \ldots, T_r))$. □

Remark (21.9.7). — The assertion of (21.9.6), (ii) *does not extend* to the case where $[k_0 : k_0^p]$ is infinite. Let us take for k_0 the field $\mathbf{F}_p(X_n)_{n \geqslant 0}$ of rational fractions with a countable infinity of indeterminates, so that the X_n^p form a p-base of k_0 over $\mathbf{F}_p$, and therefore $[k_0 : k_0^p] = +\infty$. In the statement of (21.9.6), take $K_0 = k_0$ (hence necessarily $k = k_0$), $L_0 = L = k_0((T))$, for E the extension $L_0(y)$, where y is a root of the polynomial $Y^p - \sum_{n=0}^{\infty} X_n T^{np}$. Then E is *separable* over k_0, without being an element of $k_0(L_0^p)$, and one sees immediately that this is not possible; one has therefore $d(E/K_0, k_0) = 0$, the formula (21.9.6.3) reduces to $\delta = d(E/M_0, M_0) - 1$, and it is enough to verify that this rank is 2. But a p-base of E over L_0 is formed of the single element y, and L_0 has evidently a p-base on M_0 formed of the single element T; hence, since E is an algebraic extension of M_0, our assertion results from (21.7.1) and (21.4.5).

Proposition (21.9.8). — *Let k_0 be a field (of arbitrary characteristic), K_0 an extension separable of k_0, A a local noetherian complete ring, which is a k_0-algebra and whose residue field is a finite extension of K_0. For every prime ideal $\mathfrak{p}$ of A, let $k(\mathfrak{p})$ be the residue field of A at $\mathfrak{p}$. Then, for every field k such that $k_0 \subset k \subset K_0$ and such that $[K_0 : k] < +\infty$, one has* $\mathbf{0_{IV\text{-}1}}$ | 182

$$\operatorname{rg}_{k(\mathfrak{p})}(\widehat{\Omega}^1_{A/(\mathfrak{p})/k} \otimes_{A/\mathfrak{p}} k(\mathfrak{p})) - \operatorname{rg}_{K_0} \Omega^1_{K_0/k} \geqslant \dim(A/\mathfrak{p}) + \operatorname{rg}_{k(\mathfrak{p})} \Upsilon_{k(\mathfrak{p})/k_0}. \tag{21.9.8.1}$$

Suppose moreover that $[k_0 : k_0^p] < +\infty$ (where p is the exponential characteristic of k_0). Then one can find a field k such that $k_0 \subset k \subset K_0$ and $[K_0 : k] < +\infty$, for which the two members of (21.9.8.1) are equal.

PROOF. As $A/\mathfrak{p}$ has the same residue field K that A and is complete, one can restrict to the case where A is integral and $\mathfrak{p} = 0$; one will then set $E = k(\mathfrak{p})$, field of fractions of A. As K_0 is formally smooth over k_0 (19.6.1), there exists a k_0-monomorphism $K_0 \to A$ making A into a K_0-algebra; one knows in particular that there exists a sub-K_0-algebra A_0 of A, K_0-isomorphic to a formal power series algebra $K_0[[T_1, \ldots, T_r]]$ and such that A is an A_0-algebra *finite* (19.8.9), which implies that E is a finite extension of the field of fractions $L_0 = K_0((T_1, \ldots, T_r))$ of A_0. One has $\Upsilon_{K_0/k_0} = 0$ since K_0 is separable over k_0; moreover, as A is an A_0-algebra finite, $\dim(A) = \dim(A_0)$ (16.1.5).

If $p = 1$, one has $\operatorname{rg}_E(\widehat{\Omega}^1_{A/k} \otimes_A E) = r$ by (21.9.5) and (17.1.4), (iii)), and $\Omega^1_{K_0/k} = 0$ since K_0 being a separable finite extension of k (21.7.4); on the other hand $\Upsilon_{E/k_0} = 0$, E being a separable extension of k_0; the two members of (21.9.8.1) are therefore equal in this case.

Suppose now $p > 1$; if $B = k[[T_1, \ldots, T_r]]$, A is a B-algebra, and setting $B_1 = k[[T_1^p, \ldots, T_r^p]]$ and M the field of fractions $k((T_1^p, \ldots, T_r^p))$, it results from (20.5.9) that $\widehat{\Omega}^1_{A/k} \otimes_A E$ identifies with $\Omega^1_{E/M}$ (21.9.4), and the inequality (21.9.8.1) is none other than (21.9.6.2), and the last assertion follows from (21.9.6), (ii). □

§22. Differential criteria for smoothness and regularity

The main results of this section are: $\mathbf{0}_{\textbf{IV-1}}$ | 182

a) The criteria (22.2.2) and (22.5.8), which give necessary and sufficient conditions for a complete Noetherian local ring A containing a field k to be formally smooth over k; the criterion (22.5.8) is stated without involving differential notions, and completes (19.6.6); the statement (22.2.2), of a slightly more technical nature, permits giving a *classification*, for fixed k, of the local rings A in question (22.2.5): they are determined by their dimension n and their residue field K, under the sole condition $n \geqslant \mathrm{rg}_{\mathrm{K}} \Upsilon_{\mathrm{K}/k}$.

b) The theorem (22.3.3), and the corollary (22.7.6) of the Jacobian criterion of Nagata (22.7.3), which give conditions allowing one to assert that a *localized* ring of a complete Noetherian local ring A containing a field k is geometrically regular over k; they will play an important role in the study of the "fine" properties of local rings carried out in chap. IV, § 7. We do not possess for the moment any proof of (22.7.6) independent of the Jacobian criterion of Nagata.

c) The Jacobian criterion of Zariski (22.6.7) and its variants, which are easy consequences of (20.7.8).

22.1. Lifting of formal smoothness $\mathbf{0}_{\textbf{IV-1}}$ | 183

Theorem (22.1.1). — *Let $s : \Lambda \to \mathrm{A}$, $u : \mathrm{A} \to \mathrm{B}$ be two ring homomorphisms, $\mathfrak{J}$ an ideal of* B, C *the quotient ring* $\mathrm{B}/\mathfrak{J}$. *Suppose that* C *is a Λ-algebra formally smooth for the discrete topology.*

(i) Suppose the following conditions are satisfied:
 α) $\mathfrak{J}/\mathfrak{J}^2$ is a projective C-module.
 β) The canonical homomorphism $\mathbf{S}^*_{\mathrm{C}}(\mathfrak{J}/\mathfrak{J}^2) \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ (19.5.2) is bijective.
 γ) The characteristic homomorphism $\chi : \Upsilon_{\mathrm{C}/\Lambda/\Lambda} \to \mathfrak{J}/\mathfrak{J}^2$ (20.6.20) is injective.
 δ) The C-module $\Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C}$ is projective.
 Then B, equipped with the $\mathfrak{J}$-preadic topology, is a formally smooth Λ-algebra.

(ii) Conversely, suppose that B is a Λ-algebra formally smooth for the $\mathfrak{J}$-preadic topology, and that A is a Λ-algebra formally smooth for the discrete topology. Then the conditions α), β), γ), δ) of (i) are satisfied.

Proof. By virtue of the exact sequence (20.6.22.1) (which is applicable, since $\mathrm{B}/\mathfrak{J}^2$ is a Λ-trivial extension of C by $\mathfrak{J}/\mathfrak{J}^2$, C being assumed to be a formally smooth Λ-algebra (19.4.4)), the condition γ) is equivalent to the following:

γ') *The homomorphism* $u_{\mathrm{B}/\Lambda/\Lambda} \otimes 1_{\mathrm{C}} : \Omega^1_{\Lambda/\Lambda} \otimes_{\mathrm{A}} \mathrm{C} \to \Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C}$ *is injective.*

(i) The conditions α) and β) imply that B is a Λ-algebra formally smooth for the $\mathfrak{J}$-preadic topology. It amounts to the same thing to say that B is a A-algebra formally smooth *relatively* to Λ, or a A-algebra formally smooth relatively to Λ. To see that B is a A-algebra formally smooth for the $\mathfrak{J}$-preadic topology, it suffices therefore, by virtue of (20.7.2), to prove that the continuous homomorphism of topological B-modules $u_{\mathrm{B}/\Lambda/\Lambda} : \Omega^1_{\Lambda/\Lambda} \otimes_{\mathrm{A}} \mathrm{B} \to \Omega^1_{\mathrm{B}/\Lambda}$ is *formally inversible on the left*. But since A is a Λ-algebra formally smooth, the topological B-module $\Omega^1_{\mathrm{B}/\Lambda}$ is formally projective (20.4.9), and its topology is deduced from that of B (20.4.5). By virtue of (19.1.9), it suffices therefore, for $u_{\mathrm{B}/\Lambda/\Lambda}$ to be formally inversible on the left, that $u_{\mathrm{B}/\Lambda/\Lambda} \otimes 1_{\mathrm{C}}$ be inversible on the left. But by virtue of γ'), this last application is injective, and on the other hand

$$0 \longrightarrow \Omega^1_{\Lambda/\Lambda} \otimes_{\mathrm{A}} \mathrm{C} \longrightarrow \Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C} \longrightarrow \Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C} \longrightarrow 0.$$

Finally, by virtue of δ), the C-module $\Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C}$ is projective, so the preceding exact sequence is split, which finishes proving (i).

(ii) If B is a Λ-algebra formally smooth for the $\mathfrak{J}$-preadic topology, B is a A-algebra formally $\mathbf{0}_{\textbf{IV-1}}$ | 184
smooth for the $\mathfrak{J}$-preadic topology as well, since A is a Λ-algebra formally smooth for the discrete topology; the conditions α) and β) result from (19.5.4). It amounts to the same thing to say that B is a A-algebra formally smooth relatively, or a A-algebra formally smooth relatively to Λ. To see that B is a Λ-algebra formally smooth, it suffices therefore, by virtue of (20.7.2), to prove that the continuous homomorphism of topological B-modules $u_{\mathrm{B}/\Lambda/\Lambda} : \Omega^1_{\Lambda/\Lambda} \otimes_{\mathrm{A}} \mathrm{B} \to \Omega^1_{\mathrm{B}/\Lambda}$ is *formally inversible on*

the left (19.1.7), and *a fortiori* injective. Finally, $\Omega^1_{\mathrm{B}/\Lambda}$ is a B-module formally projective (20.4.9), and consequently $\Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{C}$ is a projective C-module (19.2.4); the equivalence of *c*) results then from (19.1.9). □

Corollary (22.1.2). — *Let* $s : \Lambda \to \mathrm{A}$, $u : \mathrm{A} \to \mathrm{B}$ *be two ring homomorphisms,* $\mathfrak{m}$ *a maximal ideal of* B, $\mathrm{K} = \mathrm{B}/\mathfrak{m}$ *the quotient field. Suppose that the* Λ*-algebras* A *and* K *are formally smooth for the discrete topology. Then the following conditions are equivalent:*

a) B *is a* A*-algebra formally smooth for the* $\mathfrak{m}$*-preadic topology.*

b) The canonical homomorphism

(22.1.2.1)
$$\mathbf{S}^*_{\mathrm{K}}(\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \mathrm{gr}^*_{\mathfrak{m}}(\mathrm{B})$$

is bijective, and the characteristic homomorphism

(22.1.2.2)
$$\chi : \Upsilon_{\mathrm{K}/\Lambda/\Lambda} \longrightarrow \mathfrak{m}/\mathfrak{m}^2$$

is injective.

Proof. This follows immediately from (22.1.1), the conditions α) and δ) being trivially satisfied here, since K is a field. □

Remark (22.1.3). — Suppose the hypotheses of (22.1.2) are satisfied, and suppose in addition that $\mathfrak{m}/\mathfrak{m}^2$ is a K-vector space of *finite* rank. Then, for B to be a A-algebra formally smooth for the $\mathfrak{m}$-preadic topology, it suffices that the following conditions be satisfied:

(22.1.3.1). Given a polynomial algebra $\mathrm{E} = \mathrm{K}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ with indeterminates, and an ideal $\mathfrak{n}$ de E generated by the T_i $(1 \leqslant i \leqslant n)$, and the maximal ideal $\mathfrak{n}$ of E, every Λ-homomorphism $v : \mathrm{B} \to \mathrm{E}/\mathfrak{b}$, which, by passing to quotients, gives the identity $\mathrm{K} \to \mathrm{K}$, factors as $\mathrm{B} \xrightarrow{w} \mathrm{E}/\mathfrak{n}^k \to \mathrm{E}/\mathfrak{b}$, where w is a Λ-homomorphism.

(22.1.3.2). If F is the trivial extension $\mathrm{D}_{\mathrm{K}}(\mathrm{K})$ ("algebra of dual numbers over K"), $\rho : \mathrm{A} \to \mathrm{F}$ a ring homomorphism such that the composition $\mathrm{A} \xrightarrow{\rho} \mathrm{F} \to \mathrm{K}$ is the canonical homomorphism, then every Λ-homomorphism $v : \mathrm{B} \to \mathrm{K}$ which, by passing to quotients, gives the identity $\mathrm{K} \to \mathrm{K}$, factors as $\mathrm{B} \xrightarrow{w} \mathrm{F} \to \mathrm{K}$, where w is a Λ-homomorphism (for the A-algebra structure on F defined by ρ).

Indeed, it was seen (19.5.5) that the condition (22.1.3.1) implies that the canonical homomorphism (22.1.2.1) is bijective. By virtue of (22.1.2), it suffices then to see that $u_{\mathrm{B}/\Lambda/\Lambda} \otimes 1_{\mathrm{K}} : \Omega^1_{\Lambda/\Lambda} \otimes_{\mathrm{A}} \mathrm{K} \to \Omega^1_{\mathrm{B}/\Lambda} \otimes_{\mathrm{B}} \mathrm{K}$ is injective, or equivalently (since these are vector spaces over K) that for every K-vector space L, the canonical homomorphism $\mathrm{Der}_\Lambda(\mathrm{B}, \mathrm{L}) \to \mathrm{Der}_\Lambda(\mathrm{A}, \mathrm{L})$ *is surjective* (by virtue of (20.4.8)); it is clear that for this, it suffices that the canonical homomorphism $\mathrm{Der}_\Lambda(\mathrm{B}, \mathrm{K}) \to \mathrm{Der}_\Lambda(\mathrm{A}, \mathrm{K})$ be surjective, and our assertion follows, by virtue of (20.1.5).

One will note that the two conditions (22.1.3.1) and (22.1.3.2) are implied by the following:

(22.1.3.3). For every local artinian A-algebra E with residue field equal to K, and every nilpotent ideal $\mathfrak{r}$ of E, every Λ-homomorphism $\mathrm{B} \to \mathrm{E}/\mathfrak{r}$ which, by passing to quotients, gives the identity $\mathrm{K} \to \mathrm{K}$, factors as $\mathrm{B} \xrightarrow{u} \mathrm{E} \to \mathrm{E}/\mathfrak{r}$, where u is a Λ-homomorphism.

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Proposition (22.1.4). — *Let* $\varphi : \mathrm{A} \to \mathrm{B}$ *be a local homomorphism of local Noetherian rings,* k *(resp.* K*) the residue field of* A *(resp.* B*). For* B *to be a formally smooth* A*-algebra (for the preadic topologies), it is necessary and sufficient that for every local artinian ring* C*, with residue field isomorphic to* K*, every local homomorphism* $\mathrm{A} \to \mathrm{C}$ *(making* C *a topological* A*-algebra), and every nilpotent ideal* $\mathfrak{J}$ *of* C*, every local* A*-homomorphism* $\mathrm{B} \to \mathrm{C}/\mathfrak{J}$ *such that the homomorphism* $\mathrm{K} \to \mathrm{K}$ *obtained by passing to quotients is the identity, factors as* $\mathrm{B} \xrightarrow{f} \mathrm{C} \to \mathrm{C}/\mathfrak{J}$*, where* f *is a local* A*-homomorphism.*

Proof. The condition being necessary by definition (19.3.1), let us show that it is sufficient. Since the $\widehat{\mathrm{A}}$-local homomorphisms of $\widehat{\mathrm{B}}$ in this discrete $\widehat{\mathrm{A}}$-algebra come from, by extension, the A-local homomorphisms of B in this algebra, we may restrict ourselves to the case where A and B are *complete*, by virtue of (19.3.6). When A is a *field* k, the proposition follows from (22.1.3.3), where one takes Λ to be the prime subfield of k, and (19.6.1). In the general case, set $\mathrm{B}_0 = \mathrm{B}\otimes_{\mathrm{A}} k = \mathrm{B}/\mathfrak{m}\mathrm{B}$ ($\mathfrak{m}$ the maximal ideal of A), which is complete. If C_0 is a local artinian k-algebra with residue field isomorphic to K, $\mathfrak{J}_0$ a nilpotent ideal of C_0, every k-local homomorphism $g_0 : \mathrm{B}_0 \to \mathrm{C}_0/\mathfrak{J}_0$ giving the identity on K by passing to quotients, gives by composition a A-local homomorphism

$$g : \mathrm{B} \longrightarrow \mathrm{B}_0 \xrightarrow{g_0} \mathrm{C}_0/\mathfrak{J}_0$$

which by hypothesis factors as $\mathrm{B} \xrightarrow{f} \mathrm{C}_0 \to \mathrm{C}_0/\mathfrak{J}_0$, where f is a local A-homomorphism; but as C_0 is a k-algebra, f factors as $\mathrm{B} \to \mathrm{B}_0 \xrightarrow{f_0} \mathrm{C}_0$, where f_0 is a local k-homomorphism. We thus see that the hypotheses of the statement are still fulfilled when one substitutes B_0 and k for B and A respectively; therefore B_0 is a k-algebra *formally smooth* from what we have just seen. Applying (19.7.2) and (19.7.1), we see that there exists a local Noetherian complete ring B', a local homomorphism $\mathrm{A} \to \mathrm{B}'$ making B' a flat A-module and an A-algebra *formally smooth*, and a k-isomorphism $u'_0 : \mathrm{B}'_0 = \mathrm{B}'\otimes_{\mathrm{A}} k \xrightarrow{\sim} \mathrm{B}_0$. Let v'_0 be the inverse isomorphism of u'_0; we now propose to show that there exists a local A-homomorphism $v : \mathrm{B} \to \mathrm{B}'$ such that v'_0 comes from v by passing to quotients. For this, let us denote by $\mathfrak{n}$ and $\mathfrak{n}'$ the maximal ideals of B and B' respectively, and for every j, let $w_j : \mathrm{B}/(\mathfrak{n}^{j+1}+\mathfrak{m}\mathrm{B}) \to \mathrm{B}'/(\mathfrak{n}'^{j+1}+\mathfrak{m}\mathrm{B}')$ be the A-local homomorphism deduced from v'_0 by passing to quotients; it will suffice to form for each j a A-local homomorphism $v_j : \mathrm{B}/\mathfrak{n}^{j+1} \to \mathrm{B}'/\mathfrak{n}'^{j+1}$ such that the diagrams

$$\begin{tikzcd} \mathrm{B}/\mathfrak{n}^{j+1} \arrow[r, "v_j"] & \mathrm{B}'/\mathfrak{n}'^{j+1} \\ \mathrm{B}/\mathfrak{n}^{j} \arrow[u] \arrow[r, "v_{j-1}"] & \mathrm{B}'/\mathfrak{n}'^{j} \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} \mathrm{B}/\mathfrak{n}^{j+1} \arrow[r, "v_j"] & \mathrm{B}'/\mathfrak{n}'^{j+1} \\ \mathrm{B}/(\mathfrak{n}^{j+1}+\mathfrak{m}\mathrm{B}) \arrow[u] \arrow[r, "w_j"] & \mathrm{B}'/(\mathfrak{n}'^{j+1}+\mathfrak{m}\mathrm{B}') \arrow[u] \end{tikzcd}$$

are commutative; B and B' being complete, the homomorphism $v = \varprojlim v_j$ will give the answer. **$\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$** | 186
But the formation by recurrence of v_j results from the hypothesis on B and from the fact that $\mathfrak{n}'^j + (\mathfrak{n}'^{j+1}+\mathfrak{m}\mathrm{B}') = \mathfrak{n}'^j + \mathfrak{m}\mathrm{B}'$. It follows then from (19.7.1.4) that v is an A-*isomorphism*, for B' is a flat A-module, and B is complete for the $\mathfrak{m}$-preadic topology, the ideals $\mathfrak{m}^i\mathrm{B}$ being closed in B for the $\mathfrak{n}$-adic topology ($\mathbf{0}_{\mathrm{I}}$, 7.3.5). Q.E.D. □

22.2. Differential characterization of complete local algebras formally smooth over a field

(22.2.1). Let k be a field, P its prime subfield, A a k-algebra that is a *local* ring, $\mathfrak{m}$ its maximal ideal, $\mathrm{K} = \mathrm{A}/\mathfrak{m}$ its residue field. As K is separable over P, K is a P-algebra formally smooth for the discrete topology (19.6.1), and consequently the P-extension $\mathrm{A}/\mathfrak{m}^2$ of K by $\mathfrak{m}/\mathfrak{m}^2$ is P-*trivial*, and the characteristic homomorphism

$$\chi_{\mathrm{A}/k} : \mathrm{Y}_{\mathrm{K}/k} \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \tag{22.2.1.1}$$

is therefore defined (20.6.23).

Theorem (22.2.2). — *Under the conditions of (22.2.1), for A to be a k-algebra formally smooth for the $\mathfrak{m}$-preadic topology, it is necessary and sufficient that the following conditions be satisfied:*

(i) The canonical homomorphism $\mathbf{S}^*_{\mathrm{K}}(\mathfrak{m}/\mathfrak{m}^2) \to \mathrm{gr}^*_{\mathfrak{m}}(\mathrm{A})$ is bijective.
(ii) The characteristic homomorphism $\chi_{\mathrm{A}/k} : \mathrm{Y}_{\mathrm{K}/k} \to \mathfrak{m}/\mathfrak{m}^2$ is injective.

This is a particular case of (22.1.2), where one replaces Λ by P, A by k, and k and B by A; K and k, being separable over P, are indeed formally smooth P-algebras (19.6.1).

Remark (22.2.3). — The condition (ii) of (22.2.2) can be replaced by any one of the following:

a) The canonical homomorphism $\Omega^1_k \otimes_k \mathrm{K} \to \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$ is injective.
b) For every n, the canonical homomorphism $\Omega^1_k \otimes_k (\mathrm{A}/\mathfrak{m}^{n+1}) \to \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{m}^{n+1})$ is inversible on the left.
c) The canonical homomorphism $\Omega^1_k \widehat{\otimes}_k \mathrm{A} \to \widehat{\Omega}^1_{\mathrm{A}}$ is inversible on the left.
d) A is a k-algebra formally smooth (for the $\mathfrak{m}$-preadic topology) *relatively to the prime field*.

Indeed, it was already noted in (22.1.1) that the injectivity of $\chi_{\mathrm{A}/k}$ is equivalent to *a*), and that the conjunction of *a*) and condition (i) of (22.2.2) is equivalent to that of (i) and *d*). Finally, we have also seen in (22.1.1) that *b*) and *d*) are equivalent, and the equivalence of *b*) and *c*) results from (19.1.8).

(22.2.4). The main application of (22.2.2) concerns Noetherian local k-algebras that are *complete* (the case where $\mathfrak{m}/\mathfrak{m}^2$ is a K-vector space of *finite* rank). More precisely, let k be a field, K an extension of k, V a K-vector space of finite rank; we consider the triplets (A, α, β), where A is a complete Noetherian $\mathbf{0_{IV\text{-}1}}$ | 187
local k-algebra *formally smooth* with maximal ideal $\mathfrak{m}$, $\alpha : A/\mathfrak{m} \xrightarrow{\sim} K$ a k-algebra isomorphism, $\beta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\sim} V$ a di-isomorphism of vector spaces (for the isomorphism α). Given two such triplets (A, α, β), (A', α', β'), an *equivalence* from the first to the second is a k-isomorphism $u : A \xrightarrow{\sim} A'$ such that (if $\mathfrak{m}'$ is the maximal ideal of A'), the diagrams

$$\begin{tikzcd} A/\mathfrak{m} \arrow[r, "\mathrm{gr}^0(u)"] \arrow[d, "\alpha"] & A'/\mathfrak{m}' \arrow[d, "\alpha'"] \\ K \arrow[r, "1_K"] & K \end{tikzcd} \qquad \begin{tikzcd} \mathfrak{m}/\mathfrak{m}^2 \arrow[r, "\mathrm{gr}^1(u)"] \arrow[d, "\beta"] & \mathfrak{m}'/\mathfrak{m}'^2 \arrow[d, "\beta'"] \\ V \arrow[r, "1_V"] & V \end{tikzcd}$$

are commutative. It is immediate that the *equivalence classes* of triplets (A, α, β) form an *ensemble* that we shall denote $\mathrm{FL}(K/k, V)$. We now note that to every triplet (A, α, β) is associated the K-homomorphism composition of vector spaces $Y_{K/k} \xrightarrow{\chi_{A/k}} \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\beta} V$ (where $\chi_{A/k}$ is considered as a di-homomorphism relative to α^{-1}); the preceding definition shows (by (20.6.22.2)) that this K-homomorphism *depends only on the equivalence class* of (A, α, β) in $\mathrm{FL}(K/k, V)$; in other words we have thus defined a canonical application

$$\mathrm{FL}(K/k, V) \longrightarrow \mathrm{Hom}_K(Y_{K/k}, V). \tag{22.2.4.1}$$

Proposition (22.2.5). — *Let k be a field, K an extension of k, V a K-vector space of finite rank. The canonical application (22.2.4.1) is a bijection from $\mathrm{FL}(K/k, V)$ to the set of K-homomorphisms injective from $Y_{K/k}$ to V.*

Proof. (i) To show that (22.2.4.1) is injective, it must be proved that the following is true: let A, A' be two complete Noetherian local k-algebras formally smooth, $\mathfrak{m}$, $\mathfrak{m}'$ their respective maximal ideals, $\alpha : A/\mathfrak{m} \xrightarrow{\sim} K$, $\alpha' : A'/\mathfrak{m}' \xrightarrow{\sim} K$ two k-isomorphisms; let us be given two k-isomorphisms $v^0 : A/\mathfrak{m} \xrightarrow{\sim} A'/\mathfrak{m}'$, $v^1 : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\sim} \mathfrak{m}'/\mathfrak{m}'^2$ such that the diagrams

$$\begin{tikzcd} A/\mathfrak{m} \arrow[r, "v^0"] \arrow[d, "\alpha"] & A'/\mathfrak{m}' \arrow[dl, "\alpha'"] \\ K & \end{tikzcd} \qquad \begin{tikzcd} \mathfrak{m}/\mathfrak{m}^2 \arrow[r, "v^1"] \arrow[d, "\chi_A"] & \mathfrak{m}'/\mathfrak{m}'^2 \arrow[dl, "\chi_{A'}"] \\ Y_{K/k} & \end{tikzcd} \tag{22.2.5.1}$$

are commutative. One must prove that there exists a k-isomorphism $u : A \xrightarrow{\sim} A'$ such that $\mathrm{gr}^0(u) = v^0$ and $\mathrm{gr}^1(u) = v^1$. We note first that since K is a field, the canonical homomorphism $\mathbf{0_{IV\text{-}1}}$ | 188
(20.6.7.1) is *bijective* (20.6.11), so there already exists a k-isomorphism $v : A/\mathfrak{m}^2 \xrightarrow{\sim} A'/\mathfrak{m}'^2$ such that $\mathrm{gr}^0(v) = v^0$ and $\mathrm{gr}^1(v) = v^1$. We now note that since A' is metrizable and complete, and A a k-algebra formally smooth, the homomorphism composed $A \to A/\mathfrak{m}^2 \xrightarrow{v} A'/\mathfrak{m}'^2$ factors as $A \xrightarrow{u} A' \to A'/\mathfrak{m}'^2$, where u is a k-homomorphism (19.3.11), and one has thus already $\mathrm{gr}^0(u) = v^0$ and $\mathrm{gr}^1(u) = v^1$. But since v^0 and v^1 are bijective, and since A and A' are k-algebras formally smooth, one deduces from (22.2.2), (i)) that $\mathrm{gr}(u)$ is *bijective*; but since A and A' are separated and complete, this implies that u itself is *bijective* (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 of th. 1).

(ii) It remains to show that the image of (22.2.4.1) is the set of *injective* homomorphisms of $Y_{K/k}$ in V; one already knows by (22.2.2), (ii)) that it is contained in this set, so it remains to prove the □

Lemma (22.2.5.2). — *For every K-homomorphism $h : Y_{K/k} \to V$, there exists a k-algebra A that is a local Noetherian ring, regular and complete, with maximal ideal $\mathfrak{m}$, with residue field K, and a K-isomorphism $\beta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\sim} V$ such that $h = \beta \circ \chi_{A/k}$.*

Proof. Indeed, if this lemma is proved, the fact that A is regular implies that condition (i) of (22.2.2) is satisfied (17.1.1); if moreover h is injective, (22.2.2) will show that A is a k-algebra formally smooth whose equivalence class maps to h under (22.2.4.1) (while if h is not injective, the k-algebra A of which (22.2.5.2) proves the existence is not formally smooth).

To prove (22.2.5.2), let B be the K-algebra $\mathbf{S}^*_K(V)$, $\mathfrak{n}$ the augmentation ideal of the completion $\widehat{B}$ of B for the $\mathfrak{n}$-preadic topology (such that A is isomorphic to a formal power series algebra

$K[[T_1, \dots, T_n]]$ if $n = \mathrm{rg}_K V$). By virtue of (20.6.11) there exists on $A/\mathfrak{m}^2$ (where $\mathfrak{m} = \mathfrak{n}A$) a structure of k-extension of K by $\mathfrak{m}/\mathfrak{m}^2$ that is P-*trivial* (by definition, since the prime field P is separable), such that h is the characteristic homomorphism of this extension. If $f : k \xrightarrow{g} A \to A/\mathfrak{m}^2$ is the structural homomorphism, the fact that k is separable over the prime field permits (19.6.1) factoring f as $k \xrightarrow{g} A \to A/\mathfrak{m}^2$, and the k-algebra A thus defined satisfies (20.6.22.2). □

Theorem (22.2.6). — *Let k be a field, K an extension of k. For there to exist a k-algebra A, which is a local Noetherian ring, with maximal ideal $\mathfrak{m}$, such that $A/\mathfrak{m} = K$ and such that A be formally smooth (for the $\mathfrak{m}$-preadic topology), it is necessary and sufficient that $Y_{K/k}$ be a K-vector space of finite rank. The k-algebra A is then determined (up to k-isomorphism near, giving by passing to quotients the identity $K \to K$) by its dimension alone, which is subject to the sole condition of being $\geqslant \mathrm{rg}_K Y_{K/k}$.*

Proof. Since $\mathfrak{m}/\mathfrak{m}^2$ is of finite rank over K, the necessity of the condition results from (22.2.2), (ii)); conversely, (22.2.5) shows that the condition is sufficient and that one can take for $\mathfrak{m}/\mathfrak{m}^2$ a K-vector space of rank any $\geqslant \mathrm{rg}_K(Y_{K/k})$. □

In particular, the identity homomorphism of $Y_{K/k}$ associates *canonically* to K a complete Noe- $\mathbf{0_{IV\text{-}1}}$ | 189
therian k-algebra, formally smooth, for which $Y_{K/k}$ is isomorphic to $\mathfrak{m}/\mathfrak{m}^2$.

Remark (22.2.7). — (i) Let A be a local Noetherian ring, with residue field k; it follows from (19.7.1) and (19.7.2) that the determination of the Λ-algebras A that are complete Noetherian local rings and are formally smooth is *equivalent* to the same problem where Λ is replaced by k, and is thus in principle entirely solved by (22.2.5), which reduces the question to the structure of the *residue fields* of the k-algebras sought, as extensions of k.

(ii) The fact that $Y_{K/k}$ is of finite rank over K does *not imply* that K is of "finite radiciel multiplicity" over k (cf. (19.6.6)). For example, let k_0 be a perfect field of characteristic $p > 0$, $k = k_0(T)$ the field of rational fractions in one indeterminate, so that $K = k^{p^{-\infty}}$ is the smallest perfect extension of k. Then $\Omega^1_k = \Omega^1_{k/k} = 0$ (21.4.4), therefore $Y_{K/k} = \Omega^1_k \otimes_k K$, and since $k^p = k_0(T^p)$, T is a p-base of k over k^p, therefore (21.4.5), Ω^1_k is of rank 1 over k, and hence $Y_{K/k}$ is of rank 1 over K. But for every n the residue field of the local ring $K \otimes_k k^{p^{-n}}$ is isomorphic to K, which is therefore not separable over $k^{p^{-n}}$.

Proposition (22.2.8). — *Under the conditions of (22.2.1), suppose that the characteristic p of k is > 0; let (k_α) be a decreasing filtering family of subfields of k, such that $\bigcap k_\alpha(k^p) = k^p$. Then the condition (ii) of (22.2.2) can be replaced by any one of the following:*

a) There exists α_0 such that the canonical homomorphism

$$\Omega^1_{k/k_\alpha} \otimes_k K \longrightarrow \Omega^1_{A/k_\alpha} \otimes_A K$$

is injective for every $\alpha \geqslant \alpha_0$.

b) There exists α_0 such that A is a k-algebra formally smooth (for the $\mathfrak{m}$-preadic topology) relatively to k_α, for every $\alpha \geqslant \alpha_0$.

c) For every α and every integer $n \geqslant 0$, the canonical homomorphism

$$\Omega^1_{k/k_\alpha} \otimes_k (A/\mathfrak{m}^{n+1}) \longrightarrow \Omega^1_{A/k_\alpha} \otimes_A (A/\mathfrak{m}^{n+1})$$

is inversible on the left.

Proof. Indeed, if A is a formally smooth k-algebra, it is so *a fortiori* relatively to every subfield of k, which implies *c*) by virtue of (20.7.2) and (19.1.7); it is clear that *c*) implies *b*), and that *b*) implies *a*). Finally, taking into account (22.2.3), it remains to prove that *a*) implies that the canonical homomorphism $u : \Omega^1_k \otimes_k K \to \Omega^1_A \otimes_A K$ is *injective*. But for every α, one has a commutative $\mathbf{0_{IV\text{-}1}}$ | 190
diagram

$$\begin{tikzcd}
\Omega^1_k \otimes_k K \arrow[r, "u"] & \Omega^1_A \otimes_A K \\
\Omega^1_{k/k_\alpha} \otimes_k K \arrow[u, "v_\alpha"] \arrow[r, "u_\alpha"] & \Omega^1_{A/k_\alpha} \otimes_A K \arrow[u]
\end{tikzcd}$$

and the hypothesis that u_α is injective for $\alpha \geqslant \alpha_0$ implies that $\mathrm{Ker}(u)$ is contained in the intersection of the $\mathrm{Ker}(v_\alpha)$. But one has seen (21.8.3) that this intersection is zero by virtue of the hypothesis $\bigcap k_\alpha(k^p) = k^p$, which finishes the proof. □

Proposition (22.2.9). — *Under the hypotheses of (22.2.1), the following conditions are equivalent:*

a) $\mathrm{A}/\mathfrak{m}^2$ *is a k-extension k-trivial of* K *by* $\mathfrak{m}/\mathfrak{m}^2$.

b) $\chi_{\mathrm{A}/k} = 0$.

c) The canonical homomorphism (20.5.11.2)

$$\delta_{\mathrm{K}/\mathrm{A}/k} : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} \mathrm{K}$$

is injective (in other terms, the sequence

(22.2.9.1) $$0 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} \mathrm{K} \longrightarrow \Omega^1_{\mathrm{K}/k} \longrightarrow 0$$

is exact*)*.

PROOF. The equivalence of *b*) and *c*) results from the exact sequence (20.6.22.1); the equivalence of *c*) and *a*) results from (20.5.12), this sequence being split if it is exact, since these are vector spaces.

One will note that if K is *separable* over k, the conditions equivalent of (22.2.9) are satisfied, by virtue of the definition of $\chi_{\mathrm{A}/k}$ (22.2.1) and of (20.6.3). □

Corollary (22.2.10). — *Under the hypotheses of (22.2.1), consider a subfield k_0 of k. The following conditions are equivalent:*

a) $\mathrm{A}/\mathfrak{m}^2$ *is a k_0-extension k_0-trivial of* K *by* $\mathfrak{m}/\mathfrak{m}^2$.

b) The homomorphism composed $\mathrm{Y}_{\mathrm{K}/k_0} \to \mathrm{Y}_{\mathrm{K}/k} \xrightarrow{\chi_{\mathrm{A}/k_0}} \mathfrak{m}/\mathfrak{m}^2$ *is zero.*

c) The characteristic homomorphism $\chi_{\mathrm{A}/k} : \mathrm{Y}_{\mathrm{K}/k} \to \mathfrak{m}/\mathfrak{m}^2$ *factors through* $\mathrm{Y}_{\mathrm{K}/k} \to \mathrm{Y}_{\mathrm{K}/k/k_0} \xrightarrow{\chi'} \mathfrak{m}/\mathfrak{m}^2$.

Moreover, when these conditions are satisfied, the homomorphism χ' is determined in a unique and canonical way, and coincides with the characteristic homomorphism of the k_0-extension k_0-trivial $\mathrm{A}/\mathfrak{m}^2$ *of* K *by* $\mathfrak{m}/\mathfrak{m}^2$.

PROOF. Since the composition in *b*) is nothing other than χ_{A/k_0}, the equivalence of *a*) and *b*) results from (22.2.9); on the other hand, *b*) and *c*) are equivalent by virtue of the exact sequence (20.6.18.1); and since the canonical homomorphism $\mathrm{Y}_{\mathrm{K}/k} \to \mathrm{Y}_{\mathrm{K}/k/k_0}$ is surjective, this implies the uniqueness of χ'; the fact that χ' is the characteristic homomorphism of the k_0-extension $\mathrm{A}/\mathfrak{m}^2$ results from (20.6.22.2). □

Corollary (22.2.11). — *Under the hypotheses of (22.2.1), let p be the characteristic exponent of k, and let (k_α) be a decreasing filtering family of subfields of k such that $\bigcap k_\alpha(k^p) = k^p$. If $\mathrm{Y}_{\mathrm{K}/k}$ is of finite rank over* K, *there exists an index α such that* $\mathrm{A}/\mathfrak{m}^2$ *is a k_α-extension k_α-trivial of* K *by* $\mathfrak{m}/\mathfrak{m}^2$.

PROOF. It is known (21.8.3) that there exists then α such that the canonical homomorphism $\mathrm{Y}_{\mathrm{K}/k} \to \mathrm{Y}_{\mathrm{K}/k/k_\alpha}$ is bijective, so the condition *b*) of (22.2.10) (where k_α replaces k_0) is satisfied. □

22.3. Application to relations between certain local rings and their completions $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 191

Lemma (22.3.1). — *Let* A_0 *be a semi-local Noetherian ring,* A *a finite* A_0*-algebra (which is thus a semi-local Noetherian ring, cf. Bourbaki,* Alg. comm., *chap. IV, § 2, nº 5, cor. 3 of prop. 9). If* $\widehat{\mathrm{A}}_0$ *and* $\widehat{\mathrm{A}}$ *are the completions of* A_0 *and* A *for their preadic topologies respectively, then, when one equips* A_0, A *and* $\widehat{\mathrm{A}}$ *with discrete topologies,* $\widehat{\mathrm{A}}$ *is a* A*-algebra formally smooth relatively to* A_0 *(19.9.1).*

PROOF. It is known that $\widehat{\mathrm{A}} = \mathrm{A} \otimes_{\mathrm{A}_0} \widehat{\mathrm{A}}_0$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 3 of prop. 9 and chap. III, § 3, nº 4, th. 1), so the proposition results from (19.9.3). □

Proposition (22.3.2). — *Let* A *be a local Noetherian regular ring,* K *its field of fractions, p the characteristic of* K. *Suppose that one of the following hypotheses is satisfied:*

(i) $p = 0$.

(ii) $p > 0$ *and there exists a filtering decreasing family* (A_α) *of Noetherian subrings of* A, *such that* A *is a finite* A_α*-algebra for every α, such that* $\mathrm{A}^{p^{h(\alpha)}} \subset \mathrm{A}_\alpha$ *for a suitable integer $h(\alpha)$, and such that, if* K_α *denotes the field of fractions of* A_α, *we have* $\bigcap \mathrm{K}_\alpha(\mathrm{K}^p) = \mathrm{K}^p$.

Let then B *be a finite integral* A*-algebra containing* $\widehat{\mathrm{A}}$, $\mathfrak{n}$ *a prime ideal of* B, C *the local ring* $\mathrm{B}_{\mathfrak{n}}$, $\mathfrak{q}$ *a prime ideal of the completion* $\widehat{\mathrm{C}}$, *such that* $\mathfrak{q} \cap \mathrm{C} = 0$, *so that the local ring* $\widehat{\mathrm{C}}_{\mathfrak{q}}$ *is an algebra over the field of fractions* L *of* B. *Then* $\widehat{\mathrm{C}}_{\mathfrak{q}}$ *is an* L*-algebra formally smooth for its* $\mathfrak{q}$*-adic topology, and is therefore a geometrically regular ring over* L (19.6.5).

PROOF. We distinguish two cases.

I) Suppose that $\mathfrak{n}$ contains the maximal ideal $\mathfrak{m}$ of A, so that $\mathfrak{n} \cap \mathrm{A} = \mathfrak{m}$. Then $\mathfrak{n}$ is maximal. If $\mathfrak{r}$ is the radical of the semi-local ring B, $\widehat{\mathrm{C}}$ is the separated completion of B for the $\mathfrak{n}$-preadic topology, and a component of the semi-local ring $\widehat{\mathrm{B}}$, the completion of B for the $\mathfrak{r}$-preadic topology (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 4, prop. 8); one can thus write $\widehat{\mathrm{C}}_{\mathfrak{q}} = \widehat{\mathrm{B}}_{\mathfrak{q}'}$, where $\mathfrak{q}'$ is the prime ideal of $\widehat{\mathrm{B}}$ **$0_{\text{IV-1}}$** | 192
that is the inverse image of $\mathfrak{q}$. Let us now note that B is a subring of $\widehat{\mathrm{B}}$, and the hypothesis $\mathfrak{q} \cap \mathrm{C} = 0$ implies $\mathfrak{q}' \cap \mathrm{B} = 0$, so $\widehat{\mathrm{B}}_{\mathfrak{q}'}$ is also a localization of $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{L}$ at one of its prime ideals $\mathfrak{q}''$, of which $\mathfrak{q}'$ is the inverse image in $\widehat{\mathrm{B}}$; moreover we have $\widehat{\mathrm{B}} = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{B}$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. of prop. 9 and chap. III, § 3, n° 4, th. 1), so $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{L} = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{L}$; finally, if $\mathfrak{p}$ is the prime ideal of $\widehat{\mathrm{A}}$ that is the inverse image of $\mathfrak{q}''$, the fact that $\mathrm{A} \to \mathrm{B}$ is injective and the commutativity of the diagram

$$\begin{tikzcd} \mathrm{B} \arrow[r] & \widehat{\mathrm{B}} \\ \mathrm{A} \arrow[u] \arrow[r] & \widehat{\mathrm{A}} \arrow[u] \end{tikzcd}$$

imply that $\mathfrak{p} \cap \mathrm{A} = 0$, so $(\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{L})_{\mathfrak{q}''}$ is also a localization of the ring $\widehat{\mathrm{A}}_{\mathfrak{p}} \otimes_{\mathrm{A}} \mathrm{L} = (\widehat{\mathrm{A}}_{\mathfrak{p}} \otimes_{\mathrm{A}} \mathrm{K}) \otimes_{\mathrm{K}} \mathrm{L}$. To show that this local ring is an L-algebra formally smooth, it suffices therefore to prove that $\widehat{\mathrm{A}}_{\mathfrak{p}} \otimes_{\mathrm{A}} \mathrm{K}$ is a K-algebra *formally smooth*. First of all, as A is regular, so is $\widehat{\mathrm{A}}$ (17.3.2), and condition (i) of (22.2.2) is satisfied (17.1.1). It therefore suffices (when $p > 0$) to verify condition *b*) of (22.2.8). But (22.3.1) shows that, for every α, $\widehat{\mathrm{A}}$ is a A-algebra formally smooth relatively to A_{α} (for the discrete topologies); hence $\widehat{\mathrm{A}} \otimes_{\mathrm{A}_{\alpha}} \mathrm{K}_{\alpha} = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} (\mathrm{A} \otimes_{\mathrm{A}_{\alpha}} \mathrm{K}_{\alpha})$ is an $(\mathrm{A} \otimes_{\mathrm{A}_{\alpha}} \mathrm{K}_{\alpha})$-algebra formally smooth relatively to K_{α}, for the discrete topologies (19.9.2), (iii)). Since A is an A_{α}-algebra *finite*, we have $\mathrm{A} \otimes_{\mathrm{A}_{\alpha}} \mathrm{K}_{\alpha} = \mathrm{K}$; as $\widehat{\mathrm{A}}_{\mathfrak{p}}$ is a ring of fractions of $\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$, we deduce (19.9.2), (iv)) that $\widehat{\mathrm{A}}_{\mathfrak{p}}$ is a K-algebra formally smooth *relatively* to K_{α}, for the discrete topology, and *a fortiori* for its $\mathfrak{p}$-adic topology (19.3.8) and (19.9.5)). But this is precisely condition *b*) of (22.2.8) in this case. When $p = 0$, the residue field of $\widehat{\mathrm{A}}_{\mathfrak{p}}$ is separable over K, and the proposition is proved in this case by virtue of (19.6.4).

II) General case. Let $\mathfrak{s}$ be the prime ideal $\mathfrak{n} \cap \mathrm{A}$ of A, and set $\mathrm{S} = \mathrm{A} - \mathfrak{s}$, so that $\mathrm{S}^{-1}\mathrm{A} = \mathrm{A}_{\mathfrak{s}}$, and $\mathrm{C} = (\mathrm{S}^{-1}\mathrm{B})_{\mathfrak{s} \cdot \mathfrak{n}}$, where this time $\mathrm{S}^{-1}\mathfrak{n}$ contains the maximal ideal $\mathfrak{s}\mathrm{A}_{\mathfrak{s}}$ of $\mathrm{A}_{\mathfrak{s}}$. As $\mathrm{A}_{\mathfrak{s}}$ is regular (17.3.2) and $\mathrm{S}^{-1}\mathrm{B}$ a $\mathrm{A}_{\mathfrak{s}}$-algebra finite integral containing $\mathrm{A}_{\mathfrak{s}}$, it suffices therefore to verify condition (ii) for $\mathrm{A}_{\mathfrak{s}}$ when $p > 0$: but setting $\mathfrak{s}_{\alpha} = \mathfrak{s} \cap \mathrm{A}_{\alpha}$ and considering the local ring $(\mathrm{A}_{\alpha})_{\mathfrak{s}_{\alpha}}$. For every $t \notin \mathfrak{s}$, one has by hypothesis, setting $q = p^{h(\alpha)}$, $t^q \in \mathrm{A}_{\alpha}$ and $t^q \notin \mathfrak{s}_{\alpha}$; if one sets $\mathrm{S}_{\alpha} = \mathrm{A}_{\alpha} - \mathfrak{s}_{\alpha}$, one sees therefore that $\mathrm{A}_{\mathfrak{s}}$ is a $(\mathrm{A}_{\alpha})_{\mathfrak{s}_{\alpha}}$-algebra finite; moreover, K_{α} is also the field of fractions of $(\mathrm{A}_{\alpha})_{\mathfrak{s}_{\alpha}}$, which finishes the proof. □

Theorem (22.3.3). — *Let* A *be a local Noetherian complete ring,* $\mathfrak{p}$ *a prime ideal of* A, B *the localized ring* $\mathrm{A}_{\mathfrak{p}}$, $\mathfrak{q}$ *a prime ideal of the completion* $\widehat{\mathrm{B}}$, $\mathfrak{r} = \mathfrak{q} \cap \mathrm{B}$. *Then the localized ring* $\widehat{\mathrm{B}}_{\mathfrak{q}}$ *is a* $\mathrm{B}_{\mathfrak{r}}$*-algebra formally smooth for the preadic topologies.*

PROOF. The prime ideal $\mathfrak{r}$ of B is of the form $\mathfrak{n}_{\mathfrak{p}}$, where $\mathfrak{n}$ is a prime ideal of A, and $\mathrm{B}_{\mathfrak{r}} = \mathrm{A}_{\mathfrak{n}}$; let k be the residue field of $\mathrm{B}_{\mathfrak{r}}$, which is also the field of fractions of $\mathrm{A}/\mathfrak{n}$. As $\widehat{\mathrm{B}}$ is a $\mathrm{B}_{\mathfrak{r}}$-module **$0_{\text{IV-1}}$** | 193
flat ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), $\widehat{\mathrm{B}}_{\mathfrak{q}}$ is also a flat $\mathrm{B}_{\mathfrak{r}}$-module, and to prove that $\widehat{\mathrm{B}}_{\mathfrak{q}}$ is a $\mathrm{B}_{\mathfrak{r}}$-algebra formally smooth, it suffices to prove that $\widehat{\mathrm{B}}_{\mathfrak{q}} \otimes_{\mathrm{B}_{\mathfrak{r}}} k = \widehat{\mathrm{B}}_{\mathfrak{q}}/\mathfrak{r}\widehat{\mathrm{B}}_{\mathfrak{q}}$ is a k-algebra formally smooth. But $\widehat{\mathrm{B}}_{\mathfrak{q}}/\mathfrak{r}\widehat{\mathrm{B}}_{\mathfrak{q}} = (\widehat{\mathrm{B}}/\mathfrak{r}\widehat{\mathrm{B}})_{\mathfrak{q}'}$, where $\mathfrak{q}'$ is the canonical image of $\mathfrak{q}$ in $\widehat{\mathrm{B}}/\mathfrak{r}\widehat{\mathrm{B}}$; one has by definition $\mathfrak{q}' \cap (\mathrm{B}/\mathfrak{r}) = 0$ and $\mathrm{B}/\mathfrak{r} = \mathrm{A}_{\mathfrak{p}'}$, where $\mathfrak{p}' = \mathfrak{p}/\mathfrak{n}$; moreover $\widehat{\mathrm{B}}/\mathfrak{r}\widehat{\mathrm{B}}$ is the completion of $\mathrm{B}/\mathfrak{r}$.

It follows from (19.8.9) that there exists a subring A_0 of A that is a power series ring over a field of characteristic $p > 0$ or over a Cohen ring of characteristic 0; it is known that A is such that A is a finite A_0-algebra. Now, we know that the ring A_0 is *regular* (17.3.8), and it verifies one of the hypotheses (i), (ii) of (22.3.2), by virtue of (21.8.8); it suffices therefore to apply (22.3.2), replacing B by A and A by A_0. □

Corollary (22.3.4). — *Let* A *be a local Noetherian complete ring, integral,* K *its field of fractions,* $\mathfrak{p}$ *a prime ideal of* A, B *the localized ring* $\mathrm{A}_{\mathfrak{p}}$. *Then, for every prime ideal* $\mathfrak{q}$ *of the completion* $\widehat{\mathrm{B}}$, *such that* $\mathfrak{q} \cap \mathrm{B} = 0$, *the local ring* $\widehat{\mathrm{B}}_{\mathfrak{q}}$ *is a* K*-algebra formally smooth for its* $\mathfrak{q}$*-adic topology, and is therefore a geometrically regular ring over* K.

PROOF. Indeed, it follows from (19.8.9) that there exists a subring A_0 of A that is a power series ring over a field of characteristic $p > 0$ or over a Cohen ring of characteristic 0; it is known that the ring A_0 is *regular* (17.3.8), and it verifies one of hypotheses (i), (ii) of (22.3.2), by virtue of (21.8.8); it suffices therefore to apply (22.3.2), replacing B by A and A by A_0. □

22.4. Preliminary results on finite extensions of local rings whose maximal ideal has square zero

Proposition (22.4.1). — *Let* A *be a local ring whose maximal ideal* $\mathfrak{m}$ *is of square zero,* $\mathrm{K} = \mathrm{A}/\mathfrak{m}$ *its residue field; let us denote by* V *the ideal* $\mathfrak{m}$ *considered as a* K*-vector space. Let* T_i $(1 \leqslant i \leqslant r)$ *be indeterminates, and for each* i, *let* F_i *be a unitary polynomial of* $\mathrm{A}[\mathrm{T}_i]$, *of degree* d_i; *let* A' *be the quotient of the polynomial ring* $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ *by the ideal* $\mathfrak{b}$ *generated by the* r *polynomials* F_i. *Suppose that* A' *is a local ring, and let* $\mathfrak{m}'$ *and* $\mathrm{K}' = \mathrm{A}'/\mathfrak{m}'$ *be its maximal ideal and residue field. Then:*

(i) *If one sets* $d = \prod_{i=1}^{r} d_i$, A' *is a free* A*-module of rank* d, *and we have* $[\mathrm{K}' : \mathrm{K}] \leqslant d$.
(ii) *For* $[\mathrm{K}' : \mathrm{K}] = d$, *it is necessary and sufficient that* $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K} = \mathrm{A}'/\mathfrak{m}\mathrm{A}'$ *be a field (necessarily isomorphic to* K'*), in other words, that* $\mathfrak{m}' = \mathfrak{m}\mathrm{A}'$. *In this case,* $\mathfrak{m}'^2 = 0$, *and if* V' *denotes the ideal* $\mathfrak{m}'$ *considered as a* K'*-vector space, the canonical homomorphism* $\mathrm{V} \otimes_{\mathrm{K}} \mathrm{K}' \to \mathrm{V}'$ *is bijective.*

PROOF. Without assuming A' local, it is evident by Euclidean division and recurrence on r, that the monomials $\mathrm{M}_\nu = t_1^{\nu_1} t_2^{\nu_2} \cdots t_r^{\nu_r}$, where $0 \leqslant \nu_i \leqslant d_i - 1$, form a base of A' as an A-module. If A' is assumed local, K' is a quotient of $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$, so $[\mathrm{K}' : \mathrm{K}] \leqslant [\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K} : \mathrm{K}] = d$, and there cannot be equality unless $\mathfrak{m}' = \mathfrak{m}\mathrm{A}'$, which proves the first assertion of (ii). On the other hand, it is clear that when $\mathfrak{m}' = \mathfrak{m}\mathrm{A}'$, we have $\mathfrak{m}'^2 = 0$, and $\mathrm{V} \otimes_{\mathrm{K}} \mathrm{K}' = \mathfrak{m} \otimes_{\mathrm{A}} \mathrm{A}' = \mathfrak{m}\mathrm{A}' = \mathrm{V}'$ since A' is a free A-module. □

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Remark (22.4.2). — When A is artinian (that is to say $\mathrm{rg}_{\mathrm{K}}(\mathrm{V})$ is finite), the hypothesis that A' is local always implies $\mathrm{rg}_{\mathrm{K}'}(\mathfrak{m}'/\mathfrak{m}'^2) \geqslant \mathrm{rg}_{\mathrm{K}}(\mathrm{V})$, as we shall see further on (22.5.2). We will note that these two ranks can be equal without having $\mathfrak{m}' = \mathfrak{m}\mathrm{A}'$.

Proposition (22.4.3). — *With the hypotheses and notations of (22.4.1), suppose that* A *and the* K *be of characteristic* $p > 0$ *(in other words, that* $p.1 \in \mathfrak{m}$ *in* A*), and that the* F_i *be of the form*

$$\mathrm{F}_i(\mathrm{T}_i) = \mathrm{T}_i^{d_i} + \sum_{1 \leqslant k \leqslant d_i} c_{ik} \mathrm{T}_i^{d_i - k} \tag{22.4.3.1}$$

where $d_i \geqslant 2$ *for every* i, *and* $c_{ik} \in \mathfrak{m}$. *Let* A' *be the quotient of* $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ *by the ideal generated by the* r *polynomials* F_i. *Then:*

(i) A' *is a local ring, and its residue field* $\mathrm{K}' = \mathrm{A}'/\mathfrak{m}'$ *is isomorphic to* K *as a field over* A.
(ii) *For* A' *to be regular, it is necessary and sufficient that the classes* ξ_i *mod.* $\mathfrak{m}^2$ *of the elements* c_{i,d_i} $(1 \leqslant i \leqslant r)$ *be linearly independent over* K.

PROOF. (i) It is clear that $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K} = \mathrm{A}'/\mathfrak{m}\mathrm{A}'$ is isomorphic to $\mathrm{K}[\mathrm{T}_1, \ldots, \mathrm{T}_r]/\mathfrak{b}'$, where $\mathfrak{b}'$ is the ideal generated by the r polynomials $\mathrm{T}_i^{d_i}$; this is therefore a local ring of residue field K, and it is the same for A', from which (i) follows.

(ii) Let t_i $(1 \leqslant i \leqslant r)$ be the class of T_i mod. $\mathfrak{b}$; it follows from (i) that the maximal ideal $\mathfrak{m}'$ of A' is given by

$$\mathfrak{m}' = \mathfrak{m}\mathrm{A}' + \sum_i t_i \mathrm{A}'$$

from which, taking into account that $\mathfrak{m}^2 = 0$,

$$\mathfrak{m}'^2 = \sum_{i,j} t_i t_j \mathrm{A}' + \sum_i t_i \mathfrak{m}\mathrm{A}'.$$

One deduces that $A'/\mathfrak{m}'^2$ is isomorphic to the ring $A[T_1, \dots, T_r]/\mathfrak{c}$, where $\mathfrak{c}$ is the ideal generated by the elements

$$F_i \ (1 \leqslant i \leqslant r), \quad T_iT_j \ (1 \leqslant i \leqslant r,\ 1 \leqslant j \leqslant r), \quad xT_i \ (1 \leqslant i \leqslant r,\ x \in \mathfrak{m}).$$

Let $\mathfrak{c}_1 = \sum T_iT_jA[T_1, \dots, T_r]$, $\mathfrak{c}_2 = \mathfrak{c}_1 + \sum \mathfrak{m}T_iA[T_1, \dots, T_r]$, then $\mathfrak{c} = \mathfrak{c}_2 + \sum \xi_iA[T_1, \dots, T_r]$. It is clear that $A[T_1, \dots, T_r]/\mathfrak{c}_1$ is the A-module direct sum of A and of $At_i^{(1)}$, where $t_i^{(1)}$ is the class of T_i mod. $\mathfrak{c}_1$. One deduces that $A[T_1, \dots, T_r]/\mathfrak{c}_2$ is direct sum of A and of $Kt_i^{(2)}$, where $t_i^{(2)}$ is the class of T_i mod. $\mathfrak{c}_2$. Finally $A[T_1, \dots, T_r]/\mathfrak{c}$ is direct sum of $A/\mathfrak{n}$ and of Kt'_i, where t'_i is the class of T_i mod. $\mathfrak{c}$, and $\mathfrak{n}$ is the ideal of A generated by the ξ_i; in the identification of $A'/\mathfrak{m}'^2$ to $A[T_1, \dots, T_r]/\mathfrak{c}$, the image of V identifies with $\mathfrak{m}/\mathfrak{n}$, and this proves (ii). □

Proposition (22.4.4). — *With the hypotheses and notations of (22.4.1), suppose that* K *be of characteristic* $p > 0$ *(hence* $p.1 \in \mathfrak{m}$ *in* A*) and that the* F_i *be of the form*

$$F_i(T_i) = T_i^p + p\sum_{k=1}^{p-1} c_{ik}T_i^{p-k} - f_i \tag{22.4.4.1}$$

with $c_{ik} \in A$ *and* $f_i \in A$ *for every i and every k such that* $1 \leqslant k \leqslant p-1$*; suppose furthermore that if* a_i *is the class of* f_i *mod.* $\mathfrak{m}$*. Suppose that the family* $(a_i)_{1 \leqslant i \leqslant r}$ *is p-free over* K^p *("absolutely p-free"). Then:*

(i) A' *is a local ring, with maximal ideal* $\mathfrak{m}A'$*, whose residue field* K' *is isomorphic to the extension of* K *obtained by adjunction of the* $a_i^{1/p}$ $(1 \leqslant i \leqslant r)$*.*

(ii) The elements $d_{A'}f_i \otimes 1_{K'}$ form a base of the kernel $N = \Upsilon^{K'}_{A'/A}$ of the canonical homomorphism $\Omega^1_A \otimes_A K' \to \Omega^1_{A'} \otimes_{A'} K'$.

PROOF. (i) This time $A' \otimes_A K = A'/\mathfrak{m}A'$ is isomorphic to $K[T_1, \dots, T_r]/\mathfrak{b}'$, where $\mathfrak{b}'$ is the ideal generated by the polynomials $F_i = T_i^p - b_i$ $(1 \leqslant i \leqslant r)$; on deduces that A'_1 is isomorphic to the quotient of the polynomial ring $A_1[T_1, \dots, T_r]$ by the ideal generated by the F_i. The hypothesis made on the a_i implies that this quotient ring is a field (21.1.8), from which the assertions of (i) follow. $\mathbf{0_{IV\text{-}1}}$ | 195

Before proving (ii), we shall establish the

Lemma (22.4.4.2). — *Let* Λ *be a ring,* A *a* Λ*-algebra,* $\mathfrak{J}$ *an ideal of* A *such that the ring* $C = A/\mathfrak{J}$ *is of characteristic* $p > 0$ *(which is equivalent to saying that* $p.1 \in \mathfrak{J}$ *in* A*). Let* π *be the canonical image of* $p.1$ *in* $\mathfrak{J}/\mathfrak{J}^2$*. If* C *is a* $(\Lambda/p\Lambda)$*-algebra formally smooth for the discrete topologies, one has an exact sequence canonical*

$$0 \longrightarrow (\mathfrak{J}/\mathfrak{J}^2)/C.\pi \longrightarrow \Omega^1_{A/\Lambda} \otimes_A C \longrightarrow \Omega^1_{C/\Lambda} \longrightarrow 0. \tag{22.4.4.3}$$

PROOF OF THE LEMMA. Indeed, set $A' = A/pA$, $\mathfrak{J}' = \mathfrak{J}/pA$, so that $C = A'/\mathfrak{J}'$. As $A' = A \otimes_A (\Lambda/p\Lambda)$, we have $\Omega^1_{\Lambda/\Lambda} \otimes_A A' = \Omega^1_{A'/\Lambda'}$ where Λ' denotes $\Lambda/p\Lambda$ (20.5.5). Applying the exact sequence (20.5.14.1) to the Λ'-algebra formally smooth $C = A'/\mathfrak{J}'$, one obtains the exact sequence

$$0 \longrightarrow \mathfrak{J}'/\mathfrak{J}'^2 \longrightarrow \Omega^1_{A'/\Lambda'} \otimes_{A'} C \longrightarrow \Omega^1_{C/\Lambda} \longrightarrow 0$$

□

(22.4.4.4). (ii) As $\mathfrak{m}^2 = 0$, we shall also denote by V' the ideal $\mathfrak{m}'$ considered as a K'-vector space, and we shall set

$$V_0 = V/K.p = \mathfrak{m}/(\mathfrak{m}^2 + pA), \qquad V'_0 = V'/K'.p = \mathfrak{m}'/(\mathfrak{m}'^2 + pA').$$

Applying the exact sequence (22.4.4.3) in the case where $\Lambda = \mathbf{Z}$, to the $\mathbf{Z}$-algebra A (resp. A') and to its ideal $\mathfrak{m}$ (resp. $\mathfrak{m}'$), it comes, since K (resp. K') is separable over the prime field $\mathbf{Z}/p\mathbf{Z}$, hence a $(\mathbf{Z}/p\mathbf{Z})$-algebra formally smooth (19.6.1), the exact sequences

$$0 \longrightarrow V_0 \longrightarrow \Omega^1_A \otimes_A K \longrightarrow \Omega^1_K \longrightarrow 0, \qquad 0 \longrightarrow V'_0 \longrightarrow \Omega^1_{A'} \otimes_{A'} K' \longrightarrow \Omega^1_{K'} \longrightarrow 0. \tag{22.4.4.5}$$

Consider then the diagram (22.4.4.6) in which the two middle columns are the second exact sequence (22.4.4.5) and the first exact sequence (22.4.4.5) tensorized by K'; as the first exact sequence (22.4.4.5) is formed of K-vector spaces, the two middle columns of (22.4.4.6) are exact; moreover, the diagram formed by these columns and the horizontal arrows connecting them is commutative, by virtue of the proof of (22.4.4.2) and the commutativity of the diagram of (20.5.11.3). The last line $\mathbf{0_{IV\text{-}1}}$ | 196 of (22.4.4.6) is the exact sequence obtained from (20.6.1.1) by replacing A, B, C by $\mathbf{Z}$, K and K′; the middle line is obtained by tensorization with K′ from the K-module exact sequence deduced from (20.6.1.1) by replacing A, B, C by $\mathbf{Z}$, A, K respectively. The diagram formed by these two lines and the vertical arrows connecting them is commutative by virtue of the commutativity of (20.6.4.3). We note that one is under the conditions of application of (22.1.1), (ii)), so the upper line of the diagram (22.4.4.6) is *exact*; the extreme columns of the diagram

(22.4.4.6)
$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & & \\
0 \arrow[r] & \mathrm{V}_0 \otimes_{\mathrm{K}} \mathrm{K}' \arrow[r] \arrow[d] & \mathrm{V}'_0 \arrow[r] \arrow[d] & 0 & \\
0 \arrow[r] & \mathrm{N} \arrow[r] \arrow[d] & \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}' \arrow[r] \arrow[d] & \Omega^1_{\mathrm{A}'} \otimes_{\mathrm{A}'} \mathrm{K}' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{Y}_{\mathrm{K}'/\mathrm{K}} \arrow[r] \arrow[d] & \Omega^1_{\mathrm{K}} \otimes_{\mathrm{K}} \mathrm{K}' \arrow[r] \arrow[d] & \Omega^1_{\mathrm{K}'} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

are therefore formed respectively of the *kernels* and the *cokernels* of the horizontal arrows of the middle lines, which finishes proving that the diagram is *commutative*. Furthermore:

Lemma (22.4.4.7). — *The homomorphisms*

$$\mathrm{N} \longrightarrow \mathrm{Y}_{\mathrm{K}'/\mathrm{K}} \quad \textit{and} \quad \Omega^1_{\mathrm{A}'} \otimes_{\mathrm{A}'} \mathrm{K}' \longrightarrow \Omega^1_{\mathrm{K}'/\mathrm{K}}$$

of diagram (22.4.4.6) are bijective.

PROOF. Indeed, the snake diagram (Bourbaki, *Alg. comm.*, chap. I, § 1, n° 4, prop. 2) applied to the two middle columns of (22.4.4.6) gives an exact sequence

$$0 \longrightarrow \mathrm{N} \longrightarrow \mathrm{Y}_{\mathrm{K}'/\mathrm{K}} \longrightarrow 0 \longrightarrow \Omega^1_{\mathrm{A}'} \otimes_{\mathrm{A}'} \mathrm{K}' \longrightarrow \Omega^1_{\mathrm{K}'/\mathrm{K}} \longrightarrow 0.$$

But, if t_i is the canonical image of T_i in A′, the relation $\mathrm{F}_i(t_i) = 0$, together with the fact that $p.1 \in \mathfrak{m}\mathrm{A}' \subset \mathfrak{m}'$, shows immediately that $d_{\mathrm{A}'} f_i \in \mathfrak{m}'\Omega^1_{\mathrm{A}'}$, hence $d_{\mathrm{A}'} f_i \otimes 1_{\mathrm{K}'} = 0$; this proves that the $d_{\mathrm{A}} f_i \otimes 1_{\mathrm{K}}$ belong to N. Moreover, their images in $\mathrm{Y}_{\mathrm{K}'/\mathrm{K}}$ are the $d_{\mathrm{K}} a_i \otimes 1_{\mathrm{K}'}$, which form a *base* of $\mathrm{Y}_{\mathrm{K}'/\mathrm{K}}$ over K′ (21.4.7); taking into account (22.4.4.7), this finishes proving (22.4.4). □

□

(22.4.5). Now consider two rings A, B of *characteristic* $p > 0$, and two homomorphisms

$$\mathrm{A} \xrightarrow{i} \mathrm{B} \xrightarrow{j} \mathrm{A}$$

satisfying the conditions of (21.3.1). Let furthermore $\mathfrak{m}$ be an ideal *of square zero* in A; set **$0_{\text{IV-1}}$** | 197

$$\mathrm{K} = \mathrm{A}/\mathfrak{m}, \qquad \mathrm{B}_{(\mathrm{K})} = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{K} = \mathrm{B}/\mathrm{B}i(\mathfrak{m})$$

and let

$$\varphi : \mathrm{A} \longrightarrow \mathrm{K}, \qquad \psi : \mathrm{B} \longrightarrow \mathrm{B}_{(\mathrm{K})}$$

be the canonical homomorphisms. Since φ is surjective, $\Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}}$ identifies canonically with $\Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ (20.5.15). We note that, since $p \geqslant 2$, for every $x \in \mathfrak{m}$, one has $j(i(x)) = x^p = 0$, in other words $j(i(\mathfrak{m})) = 0$; by passing to quotients, one deduces from j a homomorphism

$$j' : \mathrm{B}_{(\mathrm{K})} \longrightarrow \mathrm{A}$$

such that, if $i' = \psi \circ i : \mathrm{A} \to \mathrm{B}_{(\mathrm{K})}$, we have

$$j'(i'(x)) = x^p \quad \text{and} \quad i'(j'(y)) = y^p \quad \text{for} \quad x \in \mathrm{A} \quad \text{and} \quad y \in \mathrm{B}_{(\mathrm{K})}.$$

Setting, following the notations of (21.3.2)

$$\Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} = \Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} \otimes_{\mathrm{B}_{(\mathrm{K})}} \mathrm{A}_{[j']} = \Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \otimes_{\mathrm{B}_{(\mathrm{K})}} \mathrm{A}_{[j']}$$

one has therefore (21.3.2.2) a canonical homomorphism

(22.4.5.1) $$\pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} : \Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} \longrightarrow \Omega^1_{\mathrm{A}}.$$

On the other hand, one deduces from i and j, by passing to quotients (taking into account that $j(i(\mathfrak{m})) = 0$) homomorphisms

$$\mathrm{K} \xrightarrow{i_0} \mathrm{B}_{(\mathrm{K})} \xrightarrow{j_0} \mathrm{K}$$

so that one has the commutative diagram

$$\begin{tikzcd} \mathrm{A} \arrow[r, "i"] & \mathrm{B} \arrow[r, "j"] \arrow[d, "\psi"] & \mathrm{A} \\ \mathrm{K} \arrow[r, "i_0"] & \mathrm{B}_{(\mathrm{K})} \arrow[r, "j_0"] \arrow[ur, "j'"] & \mathrm{K} \end{tikzcd}$$

It is clear furthermore that

$$j_0(i_0(s)) = s^p \quad \text{and} \quad i_0(j_0(t)) = t^p \quad \text{for} \quad s \in \mathrm{K} \quad \text{and} \quad t \in \mathrm{B}_{(\mathrm{K})}.$$

One therefore defines in the same way

$$\Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} = \Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \otimes_{\mathrm{B}_{(\mathrm{K})}} \mathrm{K}_{[j_0]} = \Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$$

hence, tensorizing (22.4.5.1) by K, one obtains a K-*homomorphism canonical*

(22.4.5.2) $$\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} = \pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{A}} \otimes 1_{\mathrm{K}} : \Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \longrightarrow \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}.$$

It follows immediately from the definitions that the diagram **$0_{\text{IV-1}}$** | 198

$$\begin{tikzcd} \Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \arrow[r, "\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}"] \arrow[dr, "\pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}"'] & \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \arrow[d, "\varphi_{\mathrm{K}/\mathrm{A}}"] \\ & \Omega^1_{\mathrm{K}} \end{tikzcd}$$

is commutative, $\pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ being the canonical homomorphism corresponding to i_0 and j_0 (21.3.2.2). By *restriction* to $\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} = \mathrm{Ker}(\pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}})$, the homomorphism $\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ therefore defines a homomorphism canonical

(22.4.5.3) $$\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \longrightarrow \mathrm{Ker}(\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \longrightarrow \Omega^1_{\mathrm{K}}) = \Upsilon_{\mathrm{K}/\mathrm{A}}.$$

(22.4.6). With the hypotheses and notations of (22.4.5), suppose in addition that $\mathfrak{m}$ be a *maximal* ideal of square zero, hence K a *field*, and let V denote the ideal $\mathfrak{m}$ considered as a K-vector space. As K is an algebra over its prime field $\mathbf{F}_p$ (so that A is an algebra over $\mathbf{F}_p$, in the sense that $\Omega^1_{\mathrm{A}} = \Omega^1_{\mathrm{A}/\mathbf{F}_p}$), one deduces from (20.5.14) that one has an *exact sequence*

(22.4.6.1) $$0 \longrightarrow \mathrm{V} \longrightarrow \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \longrightarrow \Omega^1_{\mathrm{K}} \longrightarrow 0.$$

One has therefore by (22.4.5.3) a homomorphism denoted

(22.4.6.2) $$\chi'_{\mathrm{B}/\mathrm{A}} : \Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \longrightarrow \mathrm{V}$$

that we shall call the *characteristic homomorphism* of the A-algebra B (relative to i, j and $\mathfrak{m}$). It follows from the definitions that for $\chi'_{\mathrm{B}/\mathrm{A}}$ to be *injective*, it is necessary and sufficient that $\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ be injective, the kernel of the latter being obviously contained in $\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$.

Proposition (22.4.7). — *Let* A *be a local artinian ring whose maximal ideal* $\mathfrak{m}$ *is of square zero,* $\mathrm{K} = \mathrm{A}/\mathfrak{m}$ *its residue field,* V *the ideal* $\mathfrak{m}$ *considered as a* K*-vector space; one supposes that* A *is of characteristic* $p > 0$*; let*

(22.4.7.1) $$\mathrm{F}_i(\mathrm{T}_i) = \mathrm{T}_i^p - f_i \qquad \textit{with} \quad f_i \in \mathrm{A} \quad (1 \leqslant i \leqslant r)$$

and let B *be the quotient ring of* $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ *by the ideal generated by the r polynomials* F_i*. Then:*

(i) B *is a local ring.*

(ii) *If* $\mathfrak{m}'$ *is the maximal ideal of* B, $\mathrm{K}' = \mathrm{B}/\mathfrak{m}'$ *its residue field,* V' *the* B*-module* $\mathfrak{m}'/\mathfrak{m}'^2$ *considered as a* K'*-vector space, then, for* $\mathrm{rg}_\mathrm{K}(\mathrm{V}) = \mathrm{rg}_{\mathrm{K}'}(\mathrm{V}')$*, it is necessary and sufficient that the characteristic homomorphism of* K*-vector spaces*

(22.4.7.2) $$\chi'_{\mathrm{B}/\mathrm{A}} : \Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \longrightarrow \mathrm{V}$$

(cf. (22.4.6.2)*) be injective.*

Proof. Let us again denote by a_i $(1 \leqslant i \leqslant r)$ the class of f_i mod. $\mathfrak{m}$; the a_i are not necessarily p-independent over K^p, but one can always suppose that the subfamily $(a_i)_{s+1 \leqslant i \leqslant r}$ is p-free over K^p and forms a p-base of K^p of the field $\mathrm{K}^p(a_1, \ldots, a_r)$ so that **$0_{\mathrm{IV}\text{-}1}$ | 199**

$$a_i \in \mathrm{K}^p(a_{s+1}, \ldots, a_r) \qquad \text{for} \quad 1 \leqslant i \leqslant s.$$

Let us denote by A'' the quotient ring of $\mathrm{A}[\mathrm{T}_{s+1}, \ldots, \mathrm{T}_r]$ by the ideal generated by the F_i of index $i \geqslant s+1$; then, by (22.4.4), A'' is a local ring whose maximal ideal is $\mathfrak{m}'' = \mathfrak{m}\mathrm{A}''$ and the residue field is $\mathrm{K}'' = \mathrm{K}(a_{s+1}^{1/p}, \ldots, a_r^{1/p})$. One has thus $a_i \in \mathrm{K}''^p$ for $1 \leqslant i \leqslant s$ and hence, in A'', the element f_i, for $1 \leqslant i \leqslant s$, can be written

(22.4.7.3) $$f_i = g_i^p + h_i$$

where $g_i \in \mathrm{A}''$ and $h_i \in \mathfrak{m}''$ (since $\mathfrak{m}''^2 = 0$ and $p\mathfrak{m}'' = 0$, it is immediate that h_i is uniquely determined by these conditions). Replacing T_i by $\mathrm{T}_i + g_i$, one sees thus that if one sets

$$\mathrm{G}_i(\mathrm{T}_i) = \mathrm{T}_i^p + \sum_{k=1}^{p-1} \binom{p}{k} g_i^{p-k}\mathrm{T}_i^k - h_i \qquad (1 \leqslant i \leqslant s)$$

B is the quotient ring of $\mathrm{A}''[\mathrm{T}_1, \ldots, \mathrm{T}_s]$ by the ideal generated by the G_i; but, all the coefficients of G_i except the leading coefficient belong to $\mathfrak{m}''$, we are in the situation of (22.4.3), which proves (i) (which indeed follows directly from the fact that $\mathrm{B}^p \subset \mathrm{A}$, and by consequence B has only a single ideal above $\mathfrak{m}$), and furthermore the residue field K' is *isomorphic* to K''.

Now let us note that for (22.4.7.2) to be injective, it is necessary and sufficient that **$0_{\mathrm{IV}\text{-}1}$ | 200**

$$\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} : \Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \otimes_{\mathrm{B}_{(\mathrm{K})}} \mathrm{K} \longrightarrow \Omega^1_\mathrm{A} \otimes_\mathrm{A} \mathrm{K}$$

be injective (by (22.4.6)). But $\mathrm{B}_{(\mathrm{K})} = \mathrm{B}/\mathfrak{m}\mathrm{B}$ is the quotient algebra $\mathrm{C} = \mathrm{K}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ by the ideal generated by the polynomials $\overline{\mathrm{F}}_i = \mathrm{T}_i^p - a_i$, and since $d_{\mathrm{C}/\mathrm{K}}\overline{\mathrm{F}}_i = 0$ it follows from (20.5.13) that $\Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ is a free $\mathrm{B}_{(\mathrm{K})}$-module with base $d_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}t_i$, where t_i is the canonical image of T_i. But by definition (22.4.5) the canonical image by $j' : \mathrm{B}_{(\mathrm{K})} \to \mathrm{A}$ of an element $x \in \mathrm{B}$ whose class mod. $\mathfrak{m}\mathrm{B}$ is the element $x^p \in \mathrm{A}$; one deduces immediately, since $t_i^p = f_i$, that the image of $d_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}t_i \otimes 1$ by $\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ is $d_\mathrm{A} f_i \otimes 1$ in $\Omega^1_\mathrm{A} \otimes_\mathrm{A} \mathrm{K}$; the condition that $\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ be injective is thus equivalent to the fact that the $d_\mathrm{A} f_i \otimes 1_\mathrm{K}$ are *linearly independent over* K, or equivalently that their images $d_\mathrm{A} f_i \otimes 1_{\mathrm{K}'}$ are *linearly independent over* K' in $\Omega^1_\mathrm{A} \otimes_\mathrm{A} \mathrm{K}'$.

Now, if one applies (22.4.4) to the A-algebra A'', one sees that the kernel of the canonical homomorphism $\Omega^1_\mathrm{A} \otimes_\mathrm{A} \mathrm{K}' \to \Omega^1_{\mathrm{A}''} \otimes_{\mathrm{A}''} \mathrm{K}'$ has for *base* the $d_\mathrm{A} f_i \otimes 1_{\mathrm{K}'}$ for $s+1 \leqslant i \leqslant r$. The preceding condition thus also amounts to saying that the $d_{\mathrm{A}''}h_i$ for $1 \leqslant i \leqslant s$ are linearly independent in $\Omega^1_{\mathrm{A}''} \otimes_{\mathrm{A}''} \mathrm{K}'$ (taking into account that $d_{\mathrm{A}''} f_i = d_{\mathrm{A}''}h_i$ by (22.4.7.3)); on the other hand, by (22.4.4), (ii), if V'' is the K'-vector space $\mathfrak{m}''/\mathfrak{m}''^2$, we have $\mathrm{rg}_{\mathrm{K}'}(\mathrm{V}'') = \mathrm{rg}_\mathrm{K}(\mathrm{V})$ (22.4.1), (ii)), and the condition $\mathrm{rg}_{\mathrm{K}'}(\mathrm{V}') = \mathrm{rg}_\mathrm{K}(\mathrm{V})$ is thus *equivalent* to $\mathrm{rg}_{\mathrm{K}'}(\mathrm{V}') = \mathrm{rg}_{\mathrm{K}'}(\mathrm{V}'')$. But it results from (22.4.3) that this last relation is *equivalent* to the fact that the classes h_i □

are linearly independent over K' in V''. But, in $\Omega^1_{\mathrm{A}''} \otimes_{\mathrm{A}''} \mathrm{K}'$, the images of the h_i by the injection $\mathrm{V}'' \to \Omega^1_{\mathrm{A}''} \otimes_{\mathrm{A}''} \mathrm{K}'$ (20.5.11.2) are the $d_{\mathrm{A}''}h_i \otimes 1_{\mathrm{K}'}$ (cf. (22.4.6.1)), and this finishes the proof of (22.4.7). **$0_{\mathrm{IV}\text{-}1}$ | 201**

Corollary (22.4.7.4). — *For* $\mathrm{rg}_{\mathrm{K}}(\mathrm{V}) = \mathrm{rg}_{\mathrm{K}'}(\mathrm{V}')$, *it is necessary and sufficient that the elements* $d_{\mathrm{A}} f_i \otimes 1_{\mathrm{K}}$ $(1 \leqslant i \leqslant r)$ *be linearly independent in the* K*-vector space* $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$.

Remark (22.4.8). — One can define the homomorphisms (22.4.5.2) and (22.4.6.2) in slightly different conditions. Let A, B be two rings, $\mathfrak{m}$ an ideal *of square zero* in A, $\mathrm{K} = \mathrm{A}/\mathfrak{m}$, p a prime number such that $p.1 \in \mathfrak{m}$ (in other words, K is of characteristic p, but not necessarily A). Let furthermore B be an A-algebra that is an A-module *faithfully flat* (so that $\mathrm{A} \subset \mathrm{B}$), and suppose that we have

$$y^p \in \mathrm{A} + p\mathrm{B} \subset \mathrm{A} + \mathfrak{m}\mathrm{B} \tag{22.4.8.1}$$

for every $y \in \mathrm{B}$. If $y^p = x + pz$ with $x \in \mathrm{A}$, $z \in \mathrm{B}$, the *class*$(A \cap pB)$ of x is well determined, and since $\mathrm{A} \cap p\mathrm{B} = p\mathrm{A}$ as B is a faithfully flat A-module, one has thus defined a canonical application $j : \mathrm{B} \to \mathrm{A}/p\mathrm{A}$, such that, for $x \in \mathrm{A}$, $j(x)$ is the class of x^p mod. $p\mathrm{A}$. Furthermore, if $y' \in \mathrm{B}$ and $y'^p = x' + pz'$ with $x' \in \mathrm{A}$, $z' \in \mathrm{B}$, it is immediate that the elements $(y + y')^p - x - x'$ and $(yy')^p - xx'$ belong to $p\mathrm{B}$ by virtue of the fact that the binomial coefficients $\binom{p}{i}$ are multiples of p for $1 \leqslant i \leqslant p - 1$. The application j is therefore a *ring homomorphism*. By composition, one deduces from j a homomorphism $j' : \mathrm{B} \xrightarrow{j} \mathrm{A}/p\mathrm{A} \to \mathrm{A}/\mathfrak{m} = \mathrm{K}$ and hence a homomorphism $j_0 : \mathrm{B}_{(\mathrm{K})} \to \mathrm{K}$; moreover, if $i_0 : \mathrm{K} \to \mathrm{B}_{(\mathrm{K})}$ is the canonical injection, i_0 and j_0 satisfy the conditions of (21.3.1) for the rings *of characteristic* p, K and $\mathrm{B}_{(\mathrm{K})}$. This being so, in order to define a K-homomorphism $\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} : \Omega^1_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \otimes_{\mathrm{B}_{(\mathrm{K})}} \mathrm{K}_{[j_0]} \to \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$, it suffices (20.4.8) to define a K-*derivation*

$$\mathrm{D}_0 : \mathrm{B}_{(\mathrm{K})} \longrightarrow \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \tag{22.4.8.2}$$

where the second member is considered as a $\mathrm{B}_{(\mathrm{K})}$-module by means of j_0. For this, it suffices to define an A-*derivation*

$$\mathrm{D} : \mathrm{B} \longrightarrow \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \tag{22.4.8.3}$$

that *vanishes on* $\mathfrak{m}\mathrm{B}$ (the second member being considered as a B-module by means of j'). Indeed, if $y^p = x + pz$ with $y \in \mathrm{B}$, $x \in \mathrm{A}$, $z \in \mathrm{B}$, the element $d_{\mathrm{A}} x \otimes 1_{\mathrm{K}}$ of $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$ depends only on the *class* $j(y)$ mod. $p\mathrm{A}$ of x (for since $d_{\mathrm{A}}(pt) = pd_{\mathrm{A}}(t)$, hence $d_{\mathrm{A}}(pt) \otimes 1 = 0$ in $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} = \Omega^1_{\mathrm{A}}/\mathfrak{m}\Omega^1_{\mathrm{A}}$, since $p.1 \in \mathfrak{m}$). One can therefore take

$$\mathrm{D}(y) = d_{\mathrm{A}} x \otimes 1_{\mathrm{K}}.$$

The fact that D is a derivation follows easily from the fact that j is a ring homomorphism; moreover, for $t \in \mathrm{A}$, one has $d_{\mathrm{A}}(t^p x) = t^p d_{\mathrm{A}}(x) + pt^{p-1} x d_{\mathrm{A}}(t)$ and one deduces that $\mathrm{D}(ty) = t\mathrm{D}(y)$, hence D is an A-derivation; finally, if in addition $t \in \mathfrak{m}$, $\mathrm{D}(ty) = t\mathrm{D}(y) = 0$ since $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$ is annihilated by $\mathfrak{m}$. We have thus defined the desired homomorphism (22.4.5.2), and it is immediate to verify that the composition

$$\Theta_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \xrightarrow{\pi'_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}} \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K} \longrightarrow \Omega^1_{\mathrm{K}}$$

is again the canonical homomorphism $\pi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}}$ analog to (22.4.5.3). The analog of (22.4.6.2) is also defined in the same way.

Now if K is a *field*, $\mathrm{A}/p\mathrm{A}$ is an $\mathbf{F}_p$-algebra, and one has the exact sequence (20.5.12.1)

$$p\mathrm{A} \longrightarrow \Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} (\mathrm{A}/p\mathrm{A}) \longrightarrow \Omega^1_{\mathrm{A}/p\mathrm{A}} \longrightarrow 0$$

which shows, tensorizing with K, that $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$ and $\Omega^1_{\mathrm{A}/p\mathrm{A}} \otimes_{\mathrm{A}/p\mathrm{A}} \mathrm{K}$ are isomorphic. The analog of (22.4.6.2) is therefore a homomorphism

$$\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \longrightarrow \mathrm{V}/p\mathrm{A} = \mathfrak{m}/(\mathfrak{m}^2 + p\mathrm{A}). \tag{22.4.8.4}$$

That said, the *criterion* (22.4.7), in which one replaces V and V′ by $\mathrm{V}/p\mathrm{A}$ and $\mathrm{V}'/p\mathrm{B}$ respectively, *remains valid supposing only that* K *be of characteristic* $p > 0$ (in other words, that $p.1 \in \mathfrak{m}$ in A), *and that the* $\mathrm{F}_i(\mathrm{T}_i)$ *be of the more general form* (22.4.4.1): indeed, B is a free A-module (hence faithfully flat), and it suffices to redo the argument of (22.4.7), replacing (22.4.6.2) by (22.4.8.4) and V″ by $\mathrm{V}''/p\mathrm{A}''$.

22.5. Geometrically regular algebras and formally smooth algebras

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Proposition (22.5.1). — *Let A be a regular local ring, A' a local ring containing A and being a finite A-algebra, $\mathfrak{m}$ (resp. $\mathfrak{m}'$) the maximal ideal of A (resp. A'), $\mathrm{K} = \mathrm{A}/\mathfrak{m}$ (resp. $\mathrm{K}' = \mathrm{A}'/\mathfrak{m}'$) its residue field. For A' to be a regular ring, it is necessary and sufficient that*

(22.5.1.1) $$\mathrm{rg}_{\mathrm{K}}(\mathfrak{m}/\mathfrak{m}^2) = \mathrm{rg}_{\mathrm{K}'}(\mathfrak{m}'/\mathfrak{m}'^2).$$

Proof. Indeed, the first member of (22.5.1.1) is $\dim(\mathrm{A})$ (17.1.1) and since $\mathrm{A}' \supset \mathrm{A}$ and A' is a finite A-algebra, $\dim(\mathrm{A}') = \dim(\mathrm{A})$ (16.1.5); the equality (22.5.1.1) is therefore necessary and sufficient for A' to be regular (17.1.1). □

Remark (22.5.2). — (i) Let $\mathrm{A}_1 = \mathrm{A}/\mathfrak{m}^2$, $\mathrm{A}'_1 = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_1 = \mathrm{A}'/\mathfrak{m}^2\mathrm{A}'$; as $\mathfrak{m}^2\mathrm{A}' \subset \mathfrak{m}'^2$, $\mathfrak{m}'_1 = \mathfrak{m}'/\mathfrak{m}^2\mathrm{A}'$ is the maximal ideal of A'_1 and $\mathfrak{m}'_1/\mathfrak{m}'^2_1$ is a K'-vector space isomorphic to $\mathfrak{m}'/\mathfrak{m}'^2$; the condition of regularity of A' depends therefore only on the structure of the A_1-*algebra* A'_1, which will allow us below to apply the preliminary results of (22.4) on algebras finite over artinian rings.

(ii) It results from the proof of (22.5.1) and from (16.2.6.1) that one has in all cases

(22.5.2.1) $$\mathrm{rg}_{\mathrm{K}}(\mathfrak{m}/\mathfrak{m}^2) \leqslant \mathrm{rg}_{\mathrm{K}'}(\mathfrak{m}'/\mathfrak{m}'^2).$$

(iii) Suppose that A and A' verify the general hypotheses of (22.4.1). It is known (19.8.10) that there exists a regular local ring B, with maximal ideal $\mathfrak{n}$, such that A is isomorphic to $\mathrm{B}/\mathfrak{n}^2$; let us denote by $\mathrm{G}_i \in \mathrm{B}[\mathrm{T}_i]$ a unitary polynomial whose canonical image in $\mathrm{A}[\mathrm{T}_i]$ is F_i $(1 \leqslant i \leqslant r)$; then, if B' is the quotient ring of $\mathrm{B}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ by the ideal generated by the G_i, it is clear that B' is a B-module of rank d and $\mathrm{A}' = \mathrm{B}' \otimes_{\mathrm{B}} \mathrm{A} = \mathrm{B}'/\mathfrak{n}^2\mathrm{B}'$; as A' is assumed to be a local ring, so is B', and so B' is isomorphic to K and $\mathfrak{n}'/\mathfrak{n}'^2$ isomorphic to $\mathfrak{m}'/\mathfrak{m}'^2$ as K'-vector spaces; the inequality (22.5.2.1) shows thus that we have, under the hypotheses of (22.4.1)

(22.5.2.2) $$\mathrm{rg}_{\mathrm{K}}(\mathrm{V}) \leqslant \mathrm{rg}_{\mathrm{K}'}(\mathfrak{m}'/\mathfrak{m}'^2).$$

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Corollary (22.5.3). — *Let A be a regular local ring, $\mathfrak{m}$ its maximal ideal, T_i $(1 \leqslant i \leqslant r)$ indeterminates, and for each i, let*

(22.5.3.1) $$\mathrm{F}_i(\mathrm{T}_i) = \mathrm{T}_i^{d_i} + \sum_{1 \leqslant k \leqslant d_i} c_{ik}\mathrm{T}_i^{d_i - k}$$

be a unitary polynomial of $\mathrm{A}[\mathrm{T}_i]$, with $d_i \geqslant 2$ for every i and $c_{ik} \in \mathfrak{m}$. Let A' be the quotient of the polynomial ring $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ by the ideal $\mathfrak{b}$ generated by the r polynomials F_i. Then:

(i) *A' is a local ring, and its residue field K' is isomorphic to the residue field K of A.*

(ii) *For A' to be regular, it is necessary and sufficient that the classes ξ_i mod. $\mathfrak{m}^2$ of the elements c_{i,d_i} $(1 \leqslant i \leqslant r)$ be linearly independent over K.*

Proof. Indeed, with the notations of (22.5.2), (i)), if $\overline{\mathrm{F}}_i$ denotes the polynomial of $\mathrm{A}_1[\mathrm{T}_i]$ obtained by applying to the coefficients of F_i the homomorphism $\mathrm{A} \to \mathrm{A}_1$, A'_1 is defined from A_1 and the polynomials $\overline{\mathrm{F}}_i$ as in (22.4.1), and the conclusion results from (22.5.1) and (22.4.3). □

Theorem (22.5.4). — *Let A be a regular local ring, K its residue field; suppose that K is of characteristic $p > 0$; let*

(22.5.4.1) $$\mathrm{F}_i(\mathrm{T}_i) = \mathrm{T}_i^p + p\sum_{k=1}^{p-1} c_{ik}\mathrm{T}_i^{p-k} - f_i$$

with $c_{ik} \in \mathrm{A}$ and $f_i \in \mathrm{A}$ $(1 \leqslant i \leqslant r)$; let B be the quotient ring of $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ by the ideal generated by the r polynomials F_i. Then B is a local ring, and the following conditions are equivalent:

a) *B is regular;*

b) *the elements $d_{\mathrm{A}}f_i \otimes 1_{\mathrm{K}}$ $(1 \leqslant i \leqslant r)$ are linearly independent in the K-vector space $\Omega^1_{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{K}$;*

c) *if $\mathrm{B}' = \mathrm{B}/\mathfrak{m}^2\mathrm{B}$ and if one sets $\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} = \Xi_{\mathrm{B}'_{(\mathrm{K})}/\mathrm{K}}$ (22.4.5), the characteristic homomorphism $\Xi_{\mathrm{B}_{(\mathrm{K})}/\mathrm{K}} \to \mathfrak{m}/\mathfrak{m}^2$ (22.4.6.2) is injective.*

PROOF. Indeed, B' is defined again from $A_1 = A/\mathfrak{m}^2$ and the polynomials $\overline{F}_i$ obtained by applying the homomorphism $A \to A_1$ to the coefficients. Taking into account that $\Omega^1_A \otimes_A K$ is canonically isomorphic to $\Omega^1_{A_1} \otimes_{A_1} K$ (20.5.12), (ii)), the theorem results from (22.5.1) and (22.4.7.4), using remark (22.4.8) when A_1 is not of characteristic p. □

(22.5.5). Let k be a field of characteristic $p > 0$, A a local ring, k' a *finite* radiciel extension of k such that $k'^p \subset k$, $\mathfrak{m}$ the maximal ideal of A, $K = A/\mathfrak{m}$ its residue field; recall that $A' = A \otimes_k k'$ is a local ring whose residue field is the same as that of $K \otimes_k k'$ (Bourbaki, *Alg. comm.*, chap. V, § 2, lemma 4). We note that one has $A'_{(K)} = A' \otimes_A K = K \otimes_k k'$, and hence (21.3.6) **$0_{IV-1}$** | 203

(22.5.5.1)
$$\Theta_{A'_{(K)}/K} = \Theta_{k'/k} \otimes_k K$$

to a canonical isomorphism. Furthermore, it is known (21.4.7) that one has $\Xi_{k'/k} = 0$, in other words said the canonical homomorphism (21.3.2.2)

$$\pi_{k'/k} : \Theta_{k'/k} \longrightarrow \Omega^1_k$$

is injective, so that one has a canonical *injection*

(22.5.5.2)
$$\Theta_{A'_{(K)}/K} \longrightarrow \Omega^1_k \otimes_k K$$

which makes explicit in the following way (taking into account (22.5.5.1)): for every $x' \in k'$, to the element $d_{k'/k}(x' \otimes 1_K) \otimes 1_K$ of $\Omega^1_{A'_{(K)}/K} \otimes_{A'_{(K)}} K$, one makes correspond the element $d_k(x'^p) \otimes 1_K$ of $\Omega^1_k \otimes_k K$.

Lemma (22.5.6). — *With the notations of* (22.5.5)*:*

(i) *The kernel of the canonical homomorphism* $\pi_{A'_{(K)}/K}$ (21.3.2.2) *identifies by* (22.5.5.2) *with* $(\Theta_{k'/k} \otimes_k K) \cap \Upsilon_{K/k}$.

(ii) *Let* $A_1 = A/\mathfrak{m}^2$, $A'_1 = A_1 \otimes_k k' = A'/\mathfrak{m}^2 A'$; *then the characteristic homomorphism* $\chi'_{A'_1/A_1}$ (22.4.6.2) *identifies with the restriction of the characteristic homomorphism* $\chi_{A/k} : \Upsilon_{K/k} \to V = \mathfrak{m}/\mathfrak{m}^2$ *to* $(\Theta_{k'/k} \otimes_k K) \cap \Upsilon_{K/k}$ (20.6.24).

PROOF. (i) By virtue of (20.6.9), applied replacing A by the prime field, C by K and E by $A_1 = A/\mathfrak{m}^2$ (extension of K by k, $V = \mathfrak{m}/\mathfrak{m}^2$), it suffices (cf. (22.5.5)) to verify that if $x' \in k'$ and if $z \in A_1$ is an element whose class mod. $\mathfrak{m}/\mathfrak{m}^2$ is $x'^p \in k$, then the image $d_k(x'^p) \otimes 1_K$ by the canonical homomorphism $\Omega^1_k \otimes_k K \to \Omega^1_{A_1} \otimes_{A_1} K$ is $d_{A_1} z \otimes 1_K$, which results from the definitions.

(ii) As $A'_{(K)} = K \otimes_k k' = (A'_1)_{(K)}$ (by virtue of the relation $K = A_1/(\mathfrak{m}/\mathfrak{m}^2)$), $\Theta_{A'_{(K)}/K}$ identifies with $\Theta_{(A'_1)_{(K)}/K}$, and the verification results from the same calculation as in (i) and from the definition of $\pi'_{A'_{(K)}/K}$. □

Proposition (22.5.7). — *Let k be a field of characteristic $p > 0$, A a k-algebra that is a regular local ring, $\mathfrak{m}$ its maximal ideal, $K = A/\mathfrak{m}$ its residue field; let $V = \mathfrak{m}/\mathfrak{m}^2$, considered as a K-vector space. Let k' be a finite extension of k such that $k'^p \subset k$, and $A' = A \otimes_k k'$; for A' to be a regular local ring, it is necessary and sufficient that the restriction of the characteristic homomorphism $\chi_{A/k} : \Upsilon_{K/k} \to V$* (20.6.24) *to $(\Theta_{k'/k} \otimes_k K) \cap \Upsilon_{K/k}$ be injective.*

PROOF. It is a matter of proving that the statement is equivalent to (22.5.1.1), by denoting by $\mathfrak{m}'$ the maximal ideal of A', with $K' = A'/\mathfrak{m}'$ its residue field. With the notations of (22.5.2), (i)), one has $A'_1 = A_1 \otimes_k k'$. Now, let $(a_i)_{1 \leqslant i \leqslant r}$ be a p-base of k'^p over k; the elements $d_k(x^p_\alpha)$ form a base of the k-vector space Ω^1_k (21.4.5), hence the elements $d_k(x^p_\alpha) \otimes 1_K$ form a base of the K-vector space $\Omega^1_k \otimes_k K$. One concludes (cf. (22.5.5)) that when k' ranges over the finite subextensions of $k^{1/p}$, the family of subspaces $\Theta_{k'/k} \otimes_k K$ of $\Omega^1_k \otimes_k K$ is filtering increasing and has as union $\Omega^1_k \otimes_k K$. It follows then from (22.5.7) that the condition b') implies that $\chi_{A/k}$ is injective, which is none other than condition (ii) of (22.2.2). **0_{IV-1}** | 204

One can apply the criterion (22.4.7), and the condition is equivalent to the fact that $\chi'_{A'_1/A_1}$ be injective; one concludes with the aid of (22.5.6). □

Theorem (22.5.8). — *Let k be a field, p its characteristic exponent, A a local Noetherian k-algebra. The following conditions are equivalent:*

a) A *is a k-algebra formally smooth (for its preadic topology).*
b) A *is geometrically regular over k.*
b′) For every finite extension k' of k such that $k'^p \subset k$, $\mathrm{A}' = \mathrm{A} \otimes_k k'$ is a regular ring.

Taking into account (19.6.6), *one can restrict oneself to the case $p > 1$.*

PROOF. The fact that a) implies b) is none other than (19.6.5), and b) trivially implies b'). Let us show that b') implies the conditions of (22.2.2) are satisfied; it is evident for the first (17.1.1) since b') implies first of all that A is regular. Next, let $(x_\alpha)_{\alpha\in\mathrm{I}}$ be a p-base of $k^{1/p}$ over k; we know that the elements $d_k(x_\alpha^p)$ form a base of the k-vector space Ω^1_k (21.4.5), hence the elements $d_k(x_\alpha^p) \otimes 1_\mathrm{K}$ form a base of the K-vector space $\Omega^1_k \otimes_k \mathrm{K}$. One concludes (cf. (22.5.5)) that when k' ranges over the finite subextensions of $k^{1/p}$, the family of subspaces $\Theta_{k'/k} \otimes_k \mathrm{K}$ is filtering increasing and has as union $\Omega^1_k \otimes_k \mathrm{K}$. It follows then from (22.5.7) that the condition b') implies that $\chi_{\mathrm{A}/k}$ is injective, which is none other than condition (ii) of (22.2.2). □

Corollary (22.5.9). — *Let k be a field,* A *a local Noetherian k-algebra that is formally smooth. Then, for every prime ideal $\mathfrak{p}$ of* A, *the local ring $\mathrm{A}_\mathfrak{p}$ is a k-algebra formally smooth (for the $\mathfrak{p}$-preadic topology).*

PROOF. Indeed, with the notations of (22.5.8), b')), it suffices to show that the local ring $\mathrm{A}_\mathfrak{p} \otimes_k k'$ is regular; but since this is a ring of fractions of the local ring $\mathrm{A} \otimes_k k'$ and the latter is regular by hypothesis, it is the same by (17.3.6). □

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Remark (22.5.10). — (i) The conclusion of (22.5.8) is in default when, in the condition b'), one restricts oneself to *monogene* extensions $k' = k(x)$ of k (with $x^p \in k$). One can indeed show that for all $x \in k^{1/p}$ such that $x \notin k$, one has $\Upsilon_{\mathrm{K}/k} \cap (\Theta_{k'/k} \otimes_k \mathrm{K}) = 0$, which implies that $\Upsilon_{\mathrm{K}/k} \neq 0$; this signifies that for every $x \in k^{1/p}$ such that $x \notin k$, one must have $d_\mathrm{K}(x^p) \otimes 1_\mathrm{K} \notin \Upsilon_{\mathrm{K}/k}$, or yet $d_\mathrm{K}(x^p) \neq 0$, that is to say $x^p \notin \mathrm{K}^p$; in other words, one must have $\mathrm{K}^p \cap k = k^p$, that is, K is an *inseparable* extension of k. Now, one constructs easily examples of such extensions: start from a perfect field k_0 of characteristic $p > 0$, let X, Y, Z be three indeterminates and set $k = k_0(\mathrm{X}, \mathrm{Y})$ and $\mathrm{K} = k(\mathrm{X}^{1/p} + \mathrm{Z}\mathrm{Y}^{1/p})$; one verifies easily that K satisfies the preceding conditions, hence $\Upsilon_{\mathrm{K}/k} \neq 0$. Applying now the lemma (22.2.5.2) taking $\mathrm{V} = \mathrm{K}$ and for h the zero homomorphism; A is then a valuation ring of discrete valuation having K as residue field, with $\chi_{\mathrm{A}/k} = 0$, and is not a k-algebra formally smooth by (22.2.2); however $\mathrm{A} \otimes_k k'$ is a regular ring for every *monogene* subextension $k' \subset k^{1/p}$ of k.

(ii) The corollary (22.5.9) leads to the following question: if A is a Noetherian k-algebra, under what conditions is the set of prime ideals $\mathfrak{p} \in \mathrm{Spec}(\mathrm{A})$ such that $\mathrm{A}_\mathfrak{p}$ is a k-algebra formally smooth an *open* set? We will address this later in certain particular cases.

22.6. Zariski's Jacobian criterion

Theorem (22.6.1). — (Jacobian criterion for formal smoothness.) *Let* A, B *be two topological rings, $u : \mathrm{A} \to \mathrm{B}$ a continuous homomorphism making* B *a formally smooth* A*-algebra, $\mathfrak{J}$ an ideal of* B *(not necessarily closed), $\mathrm{C} = \mathrm{B}/\mathfrak{J}$ the quotient topological algebra. For* C *to be a formally smooth* A*-algebra, it is necessary and sufficient that the canonical homomorphism (cf.* (20.5.11.2)*)*

(22.6.1.1)
$$\delta_{\mathrm{C}/\mathrm{B}/\mathrm{A}} : \mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_\mathrm{B} \mathrm{C}$$

be formally inversible on the left *(cf.* (19.1.5)*).*

PROOF. Indeed, saying that B (resp. C) is a formally smooth A-algebra signifies (19.4.4) that $\mathrm{Exalcotop}_\mathrm{A}(\mathrm{B}, \mathrm{L}) = 0$ (resp. $\mathrm{Exalcotop}_\mathrm{A}(\mathrm{C}, \mathrm{L}) = 0$) for every B-module (resp. C-module) discrete L annihilated by an open ideal of B (resp. C). By hypothesis $\mathrm{Exalcotop}_\mathrm{A}(\mathrm{B}, \mathrm{L}) = 0$ for every such discrete C-module L annihilated by an open ideal of C, saying that C is a formally smooth A-algebra thus signifies that the canonical homomorphism $\mathrm{Exalcotop}_\mathrm{A}(\mathrm{C}, \mathrm{L}) \to \mathrm{Exalcotop}_\mathrm{A}(\mathrm{B}, \mathrm{L})$ is *injective*, and the theorem results from (20.7.8). □

Recall (19.5.3) that when B and C are formally smooth A-algebras, $\mathfrak{J}/\mathfrak{J}^2$ is a C-module topologically *formally projective*; moreover, the canonical homomorphisms $\varphi_n : \mathbf{S}^n_{\mathrm{C}}(\mathfrak{J}/\mathfrak{J}^2) \to \mathfrak{J}^n/\mathfrak{J}^{n+1}$ are *formal bimorphisms* when B is assumed to be a preadmissible ring.

Corollary (22.6.2). — *With the hypotheses and notations of (22.6.1), suppose furthermore that the square of every open ideal of* B *is open, and that on* $\mathfrak{J}$ *the induced topology be identical to that deduced from that of* B (19.0) *(one will note that these two conditions are satisfied when* B *is* noetherian *and its topology preadic* ($\mathbf{0}_{\mathrm{I}}$, 7.3.2)*, or if* B *has the* $\mathfrak{J}$*-preadic topology). Let* $(\mathfrak{b}_\lambda)$ *be a fundamental system of open ideals of* B *and set* $\mathrm{B}_\lambda = \mathrm{B}/\mathfrak{b}_\lambda$ *for every* λ*. Then:*

(i) *The following conditions are equivalent:*

a) C *is a formally smooth* A*-algebra.*

b) For every λ*, the homomorphism of* B_λ*-modules*

$$(\mathfrak{J}/\mathfrak{J}^2) \otimes_{\mathrm{B}} \mathrm{B}_\lambda \longrightarrow \Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} \mathrm{B}_\lambda \tag{22.6.2.1}$$

deduced from $\delta_{\mathrm{C}/\mathrm{B}/\mathrm{A}}$ *by tensorization, is inversible on the left (in other words is an isomorphism onto a direct factor of* $\Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} \mathrm{B}_\lambda$*).*

(ii) *Suppose furthermore that the topological ring* C *be preadmissible* ($\mathbf{0}_{\mathrm{I}}$, 7.1.2) *and*

let $\mathfrak{L}$ be an ideal of definition of C; then the conditions *a*) and *b*) of (i) are also equivalent to: $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 206

c) The homomorphism of $(\mathrm{C}/\mathfrak{L})$-modules

$$(\mathfrak{J}/\mathfrak{J}^2) \otimes_{\mathrm{C}} (\mathrm{C}/\mathfrak{L}) \longrightarrow \Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} (\mathrm{C}/\mathfrak{L})$$

is inversible on the left.

Proof. The hypotheses imply that on $\Omega^1_{\mathrm{B}/\mathrm{A}}$ the topology is deduced from that of B (20.4.5), hence (i) results from (22.6.1) and (19.1.7). On the other hand, recall (20.4.9) that $\Omega^1_{\mathrm{B}/\mathrm{A}}$ is a B-module formally *projective*, hence $\Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} \mathrm{B}_\lambda$ is a B_λ-module projective for every λ (19.2.4); the equivalence of *c*) results then from (19.1.9). ☐

In particular:

Corollary (22.6.3). — *Let* A *be a ring,* B *a* A*-algebra,* $\mathfrak{J}$ *an ideal of* B, $\mathrm{C} = \mathrm{B}/\mathfrak{J}$*; one supposes* A, B, C *equipped with the discrete topologies and that* B *is a formally smooth* A*-algebra. For* C *to be a formally smooth* A*-algebra, it is necessary and sufficient that the canonical homomorphism (22.6.1.1) be inversible on the left.*

Proof. One will note that since *every* A-algebra C is a quotient of a polynomial algebra $\mathrm{A}[\mathrm{T}_\alpha]_{\alpha\in\mathrm{I}}$, and this last is formally smooth (19.3.3), the criterion (22.6.3) allows in principle to recognize if C is a formally smooth A-algebra. ☐

Proposition (22.6.4). — *Let* A *be a ring,* B *a formally smooth* A*-algebra (for the discrete topologies),* $\mathfrak{J}$ *an ideal of* B, $\mathrm{C} = \mathrm{B}/\mathfrak{J}$*; suppose that* $\mathfrak{J}/\mathfrak{J}^2$ *is a* C*-module of finite type. Let* $\mathfrak{p}$ *be a prime ideal of* C, $k(\mathfrak{p})$ *the residue field of* C *at* $\mathfrak{p}$, $\mathfrak{p}'$ *the inverse image of* $\mathfrak{p}$ *in* A*. The following conditions are equivalent:*

a) $\mathrm{C}_\mathfrak{p}$ *is a formally smooth* $\mathrm{A}_\mathfrak{q}$*-algebra (for the discrete topologies).*

a′) $\mathrm{C}_\mathfrak{p}$ *is a formally smooth* A*-algebra (for the discrete topologies).*

b) The canonical homomorphism

$$(\mathfrak{J}/\mathfrak{J}^2) \otimes_{\mathrm{C}} k(\mathfrak{p}) \longrightarrow \Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} k(\mathfrak{p}) \tag{22.6.4.1}$$

is injective.

c) There exists $f \in \mathrm{C} - \mathfrak{p}$ *such that* C_f *is a formally smooth* A*-algebra for the discrete topology.*

If furthermore B *is* noetherian*, the preceding conditions are also equivalent to:*

d) $\mathrm{C}_\mathfrak{p}$ *is a formally smooth* $\mathrm{A}_\mathfrak{q}$*-algebra for the* $\mathfrak{p}$*-preadic topology on* $\mathrm{C}_\mathfrak{p}$ *and the* $\mathfrak{q}$*-adic topology (or the discrete topology) on* $\mathrm{A}_\mathfrak{q}$*.*

Proof. As $\mathrm{A}_\mathfrak{q}$ is a formally smooth A-algebra (by (19.3.5), (iv) and (ii)), one already knows that $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 207
a) and *a′*) are equivalent (19.3.5), (ii)). On the other hand $\mathrm{C}_\mathfrak{p} = \mathrm{B}_{\mathfrak{p}'}/\mathfrak{J}\mathrm{B}_{\mathfrak{p}'}$, and $\mathrm{B}_{\mathfrak{p}'}$ is a A-algebra formally smooth for the discrete topology (19.3.5), (iv)); moreover $\Omega^1_{\mathrm{B}/\mathrm{A}}$ is a B-module *projective*, hence $\Omega^1_{\mathrm{B}/\mathrm{A}} \otimes_{\mathrm{B}} \mathrm{B}_\lambda$ is a B_λ-module projective for every λ (19.2.4); and $\mathfrak{J}/\mathfrak{J}^2$ a B-module of finite type, the equivalence of *a′*) and *b*) results then from the application of (22.6.3) to $\mathrm{B}_{\mathfrak{p}'}$ and $\mathrm{C}_\mathfrak{p}$, taking into

account the fact that $\Omega^1_{B/A}$ is a B-module projective (20.4.9) and $\mathfrak{J}/\mathfrak{J}^2$ a C-module of finite type. The application of (19.1.12) proves also the equivalence of *b*) and *c*). Finally, let us note that since $B_{\mathfrak{p}'}$ is a $A_{\mathfrak{q}}$-algebra formally smooth for the $\mathfrak{p}'$-preadic topology and its topology is preadic (19.3.8), and $\mathfrak{J}/\mathfrak{J}^2$ a C-module of finite type, and $C_{\mathfrak{p}}$ preadmissible et $\mathfrak{p}C_{\mathfrak{p}}$ is an ideal of definition for its topology, the equivalence of *c*) and *a*) in (i) results from (22.6.4), and the last assertion results from (22.6.2). □

Corollary (22.6.5). — *Under the general hypotheses of* (22.6.4)*, the set of* $\mathfrak{p} \in \operatorname{Spec}(C)$ *such that* $C_{\mathfrak{p}}$ *is a formally smooth* $A_{\mathfrak{q}}$*-algebra (or an* A*-algebra formally smooth), in designating by* $\mathfrak{q}$ *the inverse image of* $\mathfrak{p}$ *in* A, for the discrete topologies, *is open in* $\operatorname{Spec}(C)$.

PROOF. This results from form *c*) of (22.6.4). One will note that when B is noetherian, one can replace the discrete topologies by the preadic topologies. □

Corollary (22.6.6). — *Under the general hypotheses of* (22.6.4)*, the following conditions are equivalent:*

a) C *is a formally smooth* A*-algebra (for the discrete topologies).*

b) For every $\mathfrak{p} \in \operatorname{Spec}(C)$ *(or only for every* maximal *ideal* $\mathfrak{p}$ *of* C*),* $C_{\mathfrak{p}}$ *is a formally smooth* A*-algebra (or a formally smooth* $A_{\mathfrak{q}}$*-algebra, where* $\mathfrak{q}$ *designates the inverse image of* $\mathfrak{p}$ *in* A*)* for the discrete topologies.

Furthermore, when B *is noetherian, one can replace in b) the discrete topologies by the preadic topologies on* $A_{\mathfrak{q}}$ *and* $C_{\mathfrak{p}}$.

PROOF. The last assertion results from (22.6.4). By virtue of (22.6.3), the condition *a*) is equivalent to the fact that the homomorphism (22.6.1.1) be inversible on the left; the equivalence of *a*) and *b*) results from the application of (22.6.3) to $B_{\mathfrak{p}'}$ and $C_{\mathfrak{p}}$, taking into account the fact that $\Omega^1_{B/A}$ is a B-module projective and $\mathfrak{J}/\mathfrak{J}^2$ a C-module of finite type. □

Proposition (22.6.7). — *Let* k_0 *be a field,* k *a separable extension of* k_0*,* C *a* k*-algebra of finite type.*

(i) *Let* $\mathfrak{p}$ *be a prime ideal of* C*. The following conditions are equivalent:*

a) $C_{\mathfrak{p}}$ *is a* k_0*-algebra formally smooth for the discrete topology.*

b) $C_{\mathfrak{p}}$ *is a* k_0*-algebra formally smooth for the* $\mathfrak{p}$*-preadic topology.*

c) There exists $f \in C - \mathfrak{p}$ *such that* C_f *is a* k_0*-algebra formally smooth for the discrete topology.*

d) $C_{\mathfrak{p}}$ *is a geometrically regular ring over* k_0 (19.6.5).

Furthermore the set of $\mathfrak{p} \in \operatorname{Spec}(C)$ *having any of these properties is open in* $\operatorname{Spec}(C)$.

(ii) *The following conditions are equivalent:*

a) C *is a* k_0*-algebra formally smooth for the discrete topology.*

b) Every $\mathfrak{p} \in \operatorname{Spec}(C)$ *satisfies the equivalent conditions of (i).*

c) Every maximal ideal $\mathfrak{m}$ *of* C *satisfies the equivalent conditions of (i).*

d) For every extension k' *of* k_0*,* $C \otimes_{k_0} k'$ *is a regular ring* (17.3.6)*; one says also that* C *is a geometrically regular ring over* k_0*.*

PROOF. One knows that C is always isomorphic to a k-algebra of the form $B/\mathfrak{J}$. Since B is a k-algebra formally smooth (19.3.3) and k, being separable over k_0, is a k_0-algebra formally smooth (19.6.1), B is a k_0-algebra formally smooth (19.3.5), (ii)). The equivalence of conditions *a*), *b*), *c*) of (i) results therefore from (22.6.4), and the equivalence of conditions *a*) and *d*) in (i) results from (22.5.8); the equivalence of *d*) and *a*) in (ii) results from (22.6.6); the equivalence of *a*) and *d*) in (ii) results from the criterion of Zariski (22.6.4), and the last assertion results from (22.6.5). **$0_{\text{IV-1}}$** | 208

Finally, as Ω^1_{B/k_0} is a B-module projective, the equivalence of the criterion of Zariski *e*) and *b*) in (ii) results from the application of (19.1.12), taking into account the fact that $\Omega^1_{B/A}$ is a B-module projective (20.4.8), and using (22.6.3), taking into account (20.4.8). □

Corollary (22.6.8). — (Zariski.) *Let* C *be an algebra of finite type over a field. Then the set of* $\mathfrak{p} \in \operatorname{Spec}(C)$ *such that* $C_{\mathfrak{p}}$ *is a regular local ring is open in* $\operatorname{Spec}(C)$.

Remark (22.6.9). — (i) The statements of (22.6.7) are still valid if instead of supposing k separable over k_0, one supposes only that it is of *finite radiciel multiplicity*, that is to say (19.6.6) that there exists a finite radiciel extension k_0' of k_0 such that the residue field of the local artinian ring $k \otimes_{k_0} k_0'$ is a *separable* extension k' of k_0'. There exists then a k_0'-monomorphism $k' \to k \otimes_{k_0} k_0'$ which, composed with the canonical homomorphism

$k \otimes_{k_0} k'_0 \to k'$, gives the identity, since k' is a k'_0-algebra formally smooth (19.6.1); one concludes that $\mathrm{C} \otimes_{k_0} k'_0$ is equipped with the structure of a k'-algebra of finite type, and as k' is separable over k_0, one can apply (22.6.7) to this k'-algebra; as we shall not have occasion to use this generalization, we leave the detail to the reader. We do not know whether the results of (22.6.7) are valid without *any* hypothesis on the extension k of k_0.

(ii) Under the general hypotheses of (22.6.7), let (k_α) be a decreasing filtering family of subfields of k containing k_0 and such that $\bigcap k_\alpha(k^p) = k_0(k^p)$ (where p is the characteristic exponent of k_0). Using a dimension counting method due to Nagata, one can show that, in the Jacobian criterion (22.6.7), (iii)), one may restrict the k-derivations D_j to be k_α-derivations for a suitable α.

22.7. Nagata's Jacobian criterion

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(22.7.1). The Jacobian criterion of Nagata is the analog of the Jacobian criterion of Zariski, but for quotient rings of *formal power series* rings over a field. We shall give, as Nagata **[?]**, two versions, presented here as criteria of formal smoothness.

Proposition (22.7.2). — *Let k be a field, $\mathrm{A} = k[[\mathrm{T}_1, \dots, \mathrm{T}_r]]$ a formal power series ring, equipped with its usual k-algebra structure, $\mathfrak{q}$ an ideal of A, $\mathrm{B} = \mathrm{A}/\mathfrak{q}$. Let $\mathfrak{p}$ be a prime ideal of A containing $\mathfrak{q}$. Suppose that there exists a sub-k-algebra C' of $\mathrm{C} = \mathrm{A}/\mathfrak{p}$, isomorphic to a formal power series ring $k[[\mathrm{X}_1, \dots, \mathrm{X}_s]]$, such that C is finite over C' and that the field of fractions L of C is a separable extension of $\mathrm{L}' = k((\mathrm{X}_1, \dots, \mathrm{X}_s))$ (this hypothesis is always satisfied if k is of characteristic 0). Then the following conditions are equivalent:*

- *a) $\mathrm{B}_\mathfrak{p} = \mathrm{A}_\mathfrak{p}/\mathfrak{q}\mathrm{A}_\mathfrak{p}$ is a k-algebra formally smooth for the $\mathfrak{p}$-preadic topology.*
- *b) There exist k-derivations D_i of A into itself $(1 \leqslant i \leqslant m)$, and elements f_i of $\mathfrak{q}$ $(1 \leqslant i \leqslant m)$, such that the images of the f_i in $\mathfrak{q}\mathrm{A}_\mathfrak{p}$ generate this ideal of $\mathrm{A}_\mathfrak{p}$, and that $\det(\mathrm{D}_i f_j) \notin \mathfrak{p}$.*
- *b') There exists a sub-extension k' of K_0, containing K_0^p, such that $[\mathrm{K}_0 : k'] < +\infty$, k-derivations D_i of A into itself, and elements f_i of $\mathfrak{q}$ $(1 \leqslant i \leqslant m)$, such that the images of the f_i in $\mathfrak{q}\mathrm{A}_\mathfrak{p}$ generate this ideal of $\mathrm{A}_\mathfrak{p}$, and that $\det(\mathrm{D}_i f_j) \notin \mathfrak{p}$.*

PROOF. The residue field of $\mathrm{B}_\mathfrak{p}$ is $k(\mathfrak{p})$; as $k((\mathrm{X}_1, \dots, \mathrm{X}_s))$ is separable over k (21.9.6.4), $k(\mathfrak{p})$ is separable over k by hypothesis, and one already knows in these conditions that the properties *a*) and *c*) are equivalent (19.6.4).

The residue field of $\mathrm{B}_\mathfrak{p}$ is $k(\mathfrak{p})$; as $k((\mathrm{X}_1, \dots, \mathrm{X}_s))$ is separable over k, $k(\mathfrak{p})$ is separable over k by hypothesis. It results therefore from (22.6.2), (ii)) that the condition *a*) is equivalent to the fact that the homomorphism j_0 defined in (22.7.2.1) (where $\mathrm{L} = k(\mathfrak{p})$) be *injective*. $\mathbf{0_{IV\text{-}1}}$ | 210

Consider on the other hand the homomorphism composed

(22.7.2.1) $$j_0 : (\mathfrak{q}/\mathfrak{q}^2) \otimes_\mathrm{A} \mathrm{L} \longrightarrow \Omega^1_{\mathrm{A}/k} \otimes_\mathrm{A} \mathrm{L} = \Omega^1_{\mathrm{A}/k} \otimes_\mathrm{A} \mathrm{L}$$

be injective.

Consider further the composed homomorphism

(22.7.2.2) $$j : (\mathfrak{q}/\mathfrak{q}^2) \otimes_\mathrm{A} \mathrm{L} \xrightarrow{j_0} \Omega^1_{\mathrm{A}/k} \otimes_\mathrm{A} \mathrm{L} \longrightarrow \widehat{\Omega}^1_{\mathrm{A}/k} \otimes_\mathrm{A} \mathrm{L}$$

and let us show that condition *b*) is equivalent to saying that j is *injective*. We note indeed that $(\mathfrak{q}/\mathfrak{q}^2) \otimes_\mathrm{A} \mathrm{L} = (\mathfrak{q}\mathrm{A}_\mathfrak{p}/\mathfrak{q}^2\mathrm{A}_\mathfrak{p}) \otimes_{\mathrm{A}_\mathfrak{p}} \mathrm{L}$; the condition *b*) signifies that if $\mathrm{F}_i = j(f'_i \otimes 1)$, where f'_i is the image of f_i in $\mathfrak{q}\mathrm{A}_\mathfrak{p}/\mathfrak{q}^2\mathrm{A}_\mathfrak{p}$, the matrix $(\langle \mathrm{F}_i, \mathrm{D}_j \otimes 1 \rangle)$ is inversible; hence this implies that the F_i are linearly independent, and *a fortiori* the same is true of the $f'_i \otimes 1_\mathrm{K}$; but as the latter generate $(\mathfrak{q}\mathrm{A}_\mathfrak{p}/\mathfrak{q}^2\mathrm{A}_\mathfrak{p}) \otimes_{\mathrm{A}_\mathfrak{p}} \mathrm{L}$, one concludes that j is injective.

Conversely, suppose j injective, and take the f_i such that the $f'_i \otimes 1$ form a base of the image of j; but one knows (21.9.3) that $\widehat{\Omega}^1_{\mathrm{A}/k}$ is a free A-module of rank r and the k-derivations of A into itself generate its dual; it follows therefore that the existence of k-derivations D_i such that the matrix $(\langle \mathrm{F}_i, \mathrm{D}_j \otimes 1 \rangle)$ is inversible, in other words condition *b*).

Now, since $\mathrm{A}_\mathfrak{p}$ is regular (17.3.2), $\dim(\mathrm{A}_\mathfrak{p})$ is equal to the rank over L of $\mathfrak{p}\mathrm{A}_\mathfrak{p}/\mathfrak{p}^2\mathrm{A}_\mathfrak{p}$ (17.1.1), $\mathbf{0_{IV\text{-}1}}$ | 211 hence to the rank over L of $(\mathfrak{p}/\mathfrak{p}^2) \otimes_\mathrm{A} \mathrm{L}$, and similarly $\dim(\mathrm{A}/\mathfrak{p}) = r - \dim(\mathrm{A}_\mathfrak{p})$ by (16.5.11). Since $\mathrm{A}_\mathfrak{p}$ is regular (17.3.2), $\dim(\mathrm{A}_\mathfrak{p})$ is equal to the rank over L of $(\mathfrak{p}/\mathfrak{p}^2) \otimes_\mathrm{A} \mathrm{L}$, which finishes proving that i is injective. □

Lemma (22.7.2.4). — *Let* R *be a regular local ring,* $\mathfrak{m}$ *its maximal ideal,* $K = R/\mathfrak{m}$ *its residue field. If* $\mathfrak{n}$ *is an ideal of* R *such that* $R/\mathfrak{n}$ *is regular, then the canonical homomorphism*

$$\alpha : (\mathfrak{n}/\mathfrak{n}^2) \otimes_R K \longrightarrow (\mathfrak{m}/\mathfrak{m}^2) \otimes_R K \tag{22.7.2.5}$$

is injective.

PROOF. Indeed, one knows (17.1.6) that $\mathfrak{n}$ is generated by a subset $(x_i)_{1 \leqslant i \leqslant l}$ forming part of a regular system of parameters $(x_i)_{1 \leqslant i \leqslant n}$ of R; the images of the x_i for $1 \leqslant i \leqslant l$ form a base of $(\mathfrak{n}/\mathfrak{n}^2) \otimes_R K$, and their images by α form part of the base of $(\mathfrak{m}/\mathfrak{m}^2) \otimes_R K$ formed by the images of the x_i for $1 \leqslant i \leqslant n$, from which the lemma. □

Theorem (22.7.3). — *Let k be a field,* A *a local Noetherian complete ring with residue field* K*. Suppose that:*

1° $[k : k^p] < +\infty$ *(where p is the characteristic exponent of k);*
2° K *is a finite extension of a separable extension* K_0 *of k;*
3° A *is equipped with a structure of* K_0*-algebra formally smooth (for the preadic topology).*

Let $\mathfrak{q}$ *be an ideal of* A, $B = A/\mathfrak{q}$, $\mathfrak{p}$ *a prime ideal of* A *containing* $\mathfrak{q}$*. The following conditions are equivalent:*

a) $B_\mathfrak{p}$ *is a k-algebra formally smooth for the* $\mathfrak{p}$*-preadic topology.*
b) There exist k-derivations D_i *of* A *into itself* $(1 \leqslant i \leqslant m)$*, and elements* f_i $(1 \leqslant i \leqslant m)$ *of* $\mathfrak{q}$*, such that the images of the* f_i *in* $\mathfrak{q}A_\mathfrak{p}$ *generate this ideal of* $A_\mathfrak{p}$*, and that* $\det(D_i f_j) \notin \mathfrak{p}$*.*
b′) There exists a sub-extension k′ of K_0*, containing* K_0^p*, such that* $[K_0 : k'] < +\infty$*, k′-derivations* D_i *of* A *into itself, and elements* f_i *of* $\mathfrak{q}$ $(1 \leqslant i \leqslant m)$*, such that the images of the* f_i *in* $\mathfrak{q}A_\mathfrak{p}$ *generate this ideal of* $A_\mathfrak{p}$*, and that* $\det(D_i f_j) \notin \mathfrak{p}$*.*

PROOF. We distinguish two cases, according to whether $p = 1$ or $p > 1$.

A) k is of characteristic 0. Then K is a separable extension of K_0, so by (19.6.4) that A is K_0-isomorphic to a formal power series ring $K[[T_1, \ldots, T_r]]$ equipped with its usual K-algebra structure. But then, taking into account (19.8.8), (ii)) applied to $A/\mathfrak{p}$, replacing k by K, one sees that the general conditions of (22.7.2) are satisfied by replacing k by K. In this case one can immediately apply the conclusions of (22.7.2), and it is immediate that this proves the equivalence of *a*), *b*) (with $k' = K_0$ in *b′*)). **$0_{\text{IV-1}}$** | 212

B) k is of characteristic $p > 0$. As K_0 is separable over k, it results from (19.3.5), (ii)) and (19.6.1) that A is a k-algebra formally smooth; by virtue of (22.5.9), $A_\mathfrak{p}$ is also a k-algebra formally smooth for the preadic topology. It results therefore of (22.6.2), (ii)) that the condition *a*) is equivalent to the fact that the homomorphism j_0 defined in (22.7.2.1) (where $L = k(\mathfrak{p})$) be injective. Therefore (19.1.12), *c*)) shows already that *b*) implies *a*); as elsewhere *b′*) implies trivially *b*), all amounts to showing that the hypothesis that j_0 is injective implies *b′*).

Let us first denote by k' a sub-extension of K_0 such that $[K_0 : k'] < +\infty$. Note that $\widehat{\Omega}^1_{A/k'}$ is a free A-module of finite type. By the argument of (22.7.2) (for a suitable choice of k'), the composed homomorphism

$$j' : (\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \xrightarrow{j_0} \Omega^1_{A/k} \otimes_A L \longrightarrow \widehat{\Omega}^1_{A/k'} \otimes_A L \tag{22.7.3.1}$$

is injective. Consider again the commutative diagram

$$\begin{tikzcd}
(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \arrow[r, "j_0"] & \Omega^1_{A/k} \otimes_A L \arrow[r] & \widehat{\Omega}^1_{A/k'} \otimes_A L \\
(\mathfrak{p}/\mathfrak{p}^2) \otimes_A L \arrow[u, "\alpha"] \arrow[r, "i_0"] & \Omega^1_{A/k} \otimes_A L \arrow[u, equal] \arrow[r] & \widehat{\Omega}^1_{A/k'} \otimes_A L \arrow[u, equal]
\end{tikzcd} \tag{22.7.3.2}$$

Since j_0 is injective, we have $\operatorname{Ker}(i_0 \circ \alpha) = 0$. But as $A_\mathfrak{p}$ is regular, and the hypothesis *a*) implies that $B_\mathfrak{p} = A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$ is also regular (19.6.5), the lemma (22.7.2.4) shows that α is injective, from which $\alpha^{-1}(\operatorname{Ker}(i_0)) = 0$. If i' is the composed homomorphism of the second line of (22.7.3.2), one has similarly

$$\operatorname{Ker}(i') = \operatorname{Ker}(i). \tag{22.7.3.3}$$

But the same argument as in (22.7.2) shows that in (22.7.2), one has an exact sequence

$$(\mathfrak{p}/\mathfrak{p}^2) \otimes_{\mathrm{A}} \mathrm{L} \xrightarrow{i} \widehat{\Omega}^1_{\mathrm{A}/k'} \otimes_{\mathrm{A}} \mathrm{L} \longrightarrow \widehat{\Omega}^1_{(\mathrm{A}/\mathfrak{p})/k'} \otimes_{\mathrm{A}/\mathfrak{p}} \mathrm{L} \longrightarrow 0.$$

But, the hypothesis of separability made on L, together with (21.9.5), implies, by virtue of (21.9.5), that $\widehat{\Omega}^1_{(\mathrm{A}/\mathfrak{p})/k'} \otimes_{\mathrm{A}/\mathfrak{p}} \mathrm{L}$ is of rank r over L (where $r - s = \dim(\mathrm{A}) - \dim(\mathrm{A}/\mathfrak{p}) = \dim(\mathrm{A}_{\mathfrak{p}})$), from which, using (21.9.3),

(22.7.3.4) $$\mathrm{rg}_{\mathrm{L}}(\mathrm{Ker}(i')) = \mathrm{rg}_{\mathrm{L}}\,\Upsilon_{\mathrm{L}/k}.$$

On the other hand, the residue field of $\mathrm{A}_{\mathfrak{p}}$ being formally smooth over the prime field (20.6.22.1)

$$\Upsilon_{\mathrm{L}/k} \xrightarrow{\chi_{\mathrm{A}_{\mathfrak{p}}/k}} (\mathfrak{p}\mathrm{A}_{\mathfrak{p}})/(\mathfrak{p}\mathrm{A}_{\mathfrak{p}})^2 \xrightarrow{i} \Omega^1_{\mathrm{A}_{\mathfrak{p}}/k} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{L}$$

and as $\mathrm{A}_{\mathfrak{p}}$ is a k-algebra formally smooth for the $\mathfrak{p}$-preadic topology, it follows from (22.2.2) that $\chi_{\mathrm{A}_{\mathfrak{p}}/k}$ is an injective homomorphism, hence $\mathrm{Ker}(i)$ is isomorphic to $\Upsilon_{\mathrm{L}/k}$, and taking into account (22.7.3.4) and the fact that $\mathrm{Ker}(i) \subset \mathrm{Ker}(i')$, (22.7.3.3), this finishes proving the theorem. □

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Remark (22.7.4). — It was shown in fact (using (21.9.6)) that condition b') is even equivalent to the other conditions of (22.7.3) when one subjects k' to be one of the fields of a decreasing filtering family (k_α) of subfields of K_0, containing K_0^p, such that $[\mathrm{K}_0 : k_\alpha] < +\infty$ for every α and that $\bigcap k_\alpha = \mathrm{K}_0^p$.

Corollary (22.7.5). — *Under the general hypotheses of* (22.7.3), *the set of prime ideals* $\mathfrak{n} \in \mathrm{Spec}(\mathrm{B})$ *such that* $\mathrm{B}_{\mathfrak{n}}$ *is a* k*-algebra formally smooth (for the* $\mathfrak{n}$*-preadic topology) is open in* $\mathrm{Spec}(\mathrm{B})$.

PROOF. With the notations of (22.7.3), it suffices to see that if $\mathrm{B}_{\mathfrak{p}}$ is formally smooth, for every $\mathfrak{p}' \in \mathrm{V}(\mathfrak{q})$ sufficiently near to $\mathfrak{p}$ in $\mathrm{Spec}(\mathrm{A})$. But, if the k-derivations D_i and the elements f_i of $\mathfrak{q}$ verify the criterion b) of (22.7.3), one has also $f = \det(\mathrm{D}_i f_j) \notin \mathfrak{p}$ for every $\mathfrak{p}' \notin \mathrm{D}(f)$; by the other hand, there exists $g \notin \mathfrak{p}$ such that the images of f_i in $\mathfrak{q}\mathrm{A}_{\mathfrak{p}'}$ generate this ideal of $\mathrm{A}_{\mathfrak{p}'}$, from which this finishes the proof of the corollary. □

Corollary (22.7.6). — (Nagata.) *Let* B *be a local Noetherian complete ring containing a field. Then the set of* $\mathfrak{n} \in \mathrm{Spec}(\mathrm{B})$ *such that* $\mathrm{B}_{\mathfrak{n}}$ *is regular is open in* $\mathrm{Spec}(\mathrm{B})$.

PROOF. Indeed, B is an algebra on a *prime* field k, hence (19.8.8), (i)) from the proposition $\mathrm{A}/\mathfrak{q}$, where $\mathrm{A} = \mathrm{K}_0[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ is a formal power series ring and K_0 an extension of k; on the other hand, saying that $\mathrm{B}_{\mathfrak{n}}$ is regular is equivalent to saying that it is a k-algebra formally smooth (19.6.4). All the conditions of application of (22.7.5) are thus realized. □

Remark (22.7.7). — (i) We shall see further on, using (22.7.6), that the conclusion of (22.7.6) can be extended to the case of any complete Noetherian local ring (**IV**, 6.12.7).

(ii) The conclusion of the theorem (22.7.3) is no longer necessarily exact when one no longer assumes that $[k : k^p]$ is finite. Take for example $k = \mathbf{F}_p(\mathrm{X}_n)_{n \geqslant 0}$, $\mathrm{A} = k[[\mathrm{T}, \mathrm{U}]]$, $\mathfrak{q}$ the principal ideal $\mathrm{A}f$, where $f = \mathrm{U}^p - \sum_{n=0}^{\infty} \mathrm{X}_n \mathrm{T}^{np}$; A is a factorial ring in which f is an extremal element, for it is extremal in the polynomial ring $k((\mathrm{T}))[\mathrm{U}]$, hence also in the formal power series ring $k((\mathrm{T}))[[\mathrm{U}]]$ (Bourbaki, *Alg. comm.*, chap. VII, § 3), and *a fortiori* in A. Take $\mathfrak{p} = \mathfrak{q}$, so that $\mathrm{B}_{\mathfrak{p}}$ is the body of fractions designated by E in (21.9.7), which is separable over k (*loc. cit.*), hence a k-algebra formally smooth. But one has, with the notations of (22.7.3), $k = \mathrm{K}_0 = \mathrm{K}$, and the k-derivations of A into itself are combinations of $\partial/\partial\mathrm{T}$ and $\partial/\partial\mathrm{U}$; as one has $\partial f/\partial\mathrm{T} = \partial f/\partial\mathrm{U} = 0$, the criteria b) and b') of (22.7.3) are not satisfied.

§23. JAPANESE RINGS

The results of this section will be completed in **IV**, (**IV**, 7.6) and (**IV**, 7.7). $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 213

23.1. Japanese rings

Definition (23.1.1). — We say that an integral ring A is a *Japanese ring* if, for every finite extension K' of its field of fractions K, the integral closure A' of A in K' is an A-module of finite type (in other words, a finite A-algebra). We say that a ring A is *universally Japanese* if every integral A-algebra of finite type is a Japanese ring. **$0_{\text{IV-1}}$ | 214**

It is clear that if A is a Japanese ring, then every ring of fractions $S^{-1}A$ is also a Japanese ring, since (with the preceding notation) $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K', and is evidently an $S^{-1}A$-module of finite type.

A Noetherian integral ring, even a *discrete valuation ring*, is not necessarily a Japanese ring [**Nag62**].

Proposition (23.1.2). — *Let A be a Noetherian integral ring, K its field of fractions. If, for every finite radicial extension K' of K, the integral closure of A in K' is an A-module of finite type, then A is a Japanese ring.*

Proof. Since A is Noetherian, it suffices, to verify that A is a Japanese ring, to see that for every finite *quasi-Galois* extension L of K, the integral closure B of A in L is an A-module of finite type. Now L is a *Galois* extension of the largest *Galois* extension K' of K contained in L; and if A' is the integral closure of A in K', B is the integral closure of A' in L. But we know that in a *separable* extension, the integral closure of A' is an A'-module of finite type (Bourbaki, *Alg. comm.*, chap. V, §1, nº 6, cor. 1 of prop. 20), whence the proposition. □

It follows from (23.1.2) that when K is of *characteristic* 0, saying that A is a Japanese ring means that its *integral closure* A' is an A-module of finite type.

Theorem (23.1.3). — (Tate) *Let A be a Noetherian integral ring, $x \neq 0$ an element of A. Suppose the following conditions are satisfied:*

(i) *A is integrally closed.*
(ii) *$\mathfrak{p} = xA$ is prime and A is separated and complete for the $\mathfrak{p}$-preadic topology.*
(iii) *A/xA is a Japanese ring.*

Then A is a Japanese ring.

We will use the following lemma:

Lemma (23.1.3.1). — *Let A be a ring, x an element of A that is a non-zero-divisor in A and such that $xA = \mathfrak{p}$ is prime; then, for every integer $n > 0$, the inverse image of $x^n A_{\mathfrak{p}}$ in A is $x^n A$.*

Proof. Indeed, suppose that $b \in A$ is an element such that $b/1 = x^n a/s$ in $A_{\mathfrak{p}}$, where $a \in A$ and $s \notin \mathfrak{p}$; then there exists $s' \notin \mathfrak{p}$ such that $s'sb = x^n as'$, whence $s'sb \in \mathfrak{p}$, and since $s's \notin \mathfrak{p}$, this implies $b \in \mathfrak{p}$, that is to say $b = xb'$ with $b' \in A$; since x is a non-zero-divisor, we conclude that $s'sb' = x^{n-1}as$, and it suffices to argue by induction on n. □

Proof of Theorem (23.1.3). To prove the theorem, we can restrict to the case where the field of fractions K of A is of characteristic $p > 0$, since A is integrally closed. Let K' be a finite radicial extension of K, so that there exists a power $q = p^e$ such that $K'^q \subset K$; by replacing K' by a larger radicial extension, we may even suppose that there exists $y \in K'$ such that $y^q = x$. Let A' be the integral closure of A in K'; since A is integrally closed, A' is the set of $x' \in K'$ such that $x'^q \in A' \cap K = A$. Set $V = A_{\mathfrak{p}}$; the maximal ideal $\mathfrak{m} = xV$ of V being principal, we know that the local Noetherian ring V is a *discrete valuation ring* (17.1.4); since V is integrally closed, the same argument as above shows that the integral closure V' of V in K' is the set of $x' \in K'$ such that $x'^q \in V$; we know (Bourbaki, *Alg. comm.*, chap. VI, §8, nº 6, prop. 6, and chap. V, §2, nº 3, lemma 4) that V' is a *discrete valuation ring*, whose maximal ideal $\mathfrak{m}'$ is the set of $x' \in K'$ such that $x'^q \in \mathfrak{m}$, and moreover the residue field $V'/\mathfrak{m}'$ is a *finite* extension of $V/\mathfrak{m}$ (*loc. cit.*, chap. VI, §8, nº 1, lemma 2). Let us show that, for every integer $n > 0$, we have **$0_{\text{IV-1}}$ | 215**

(23.1.3.2) $$\mathfrak{m}'^n \cap A' = y^n A'.$$

Indeed, since $y^q = x$, we have $y \in \mathfrak{m}'$, hence $y^n A' \subset \mathfrak{m}'^n \cap A'$. Conversely, let $x' \in \mathfrak{m}'^n \cap A'$, and set $x' = z'y^n$ with $z' \in K'$. We have $x'^q \in A'$; on the other hand, x' is a sum of products $t'_1 \cdots t'_n$ with

$t'_i \in \mathfrak{m}'$, hence $t'^q_i \in \mathfrak{m}$, and we conclude that $x'^q \in \mathfrak{m}^n \cap A$. Now the lemma (23.1.3.1) proves that $\mathfrak{m}^n \cap A = x^n A$, hence we have $z'^q x^n \in x^n A$, or equivalently $z'^q \in A$ since $x \neq 0$, which establishes (23.1.3.2).

Let us show secondly that on A' the xA'-preadic topology is *separated*; this topology is also the yA'-preadic topology of A', and it follows from (23.1.3.2) that this topology is induced by the $\mathfrak{m}'$-*preadic* topology of V', which is separated since V' is a discrete valuation ring.

Let us then prove that A'/xA' is an A-module of *finite type*. Since $x = y^q$ and $y^k A'/y^{k+1}A'$ is isomorphic to A'/yA', it suffices to show that A'/yA' is an A-module of finite type; but the formula (23.1.3.2) applied for $n = 1$ shows that A'/yA' is a sub-ring of $V'/\mathfrak{m}'$, that is to say of the *residue field* of V', which is a *finite* extension of the residue field $V/\mathfrak{m}$. Now $V/\mathfrak{m}$ is the field of fractions of $A/\mathfrak{p}$, and since A' is integral over A, A'/yA' is integral over $A/\mathfrak{p}$, hence contained in the integral closure of $A/\mathfrak{p}$ in $V'/\mathfrak{m}'$; since by hypothesis $A/\mathfrak{p}$ is a Japanese Noetherian ring, A'/yA' is an $(A/\mathfrak{p})$-module of finite type, and *a fortiori* an A-module of finite type.

This being the case, the xA'-preadic topology of A' being separated, A' is a sub-ring of its completion $\widehat{A}'$ for this topology; but since A'/xA' is an A-module of finite type and A is separated and complete for the xA-preadic topology, $\widehat{A}'$ is an A-module of finite type by virtue of ($\mathbf{0}_{\mathbf{I}}$, 7.2.9), hence so is A'. □

Corollary (23.1.4). — *Let A be a Japanese Noetherian and integrally closed ring. Then every formal power series ring $A[[T_1, \dots, T_r]]$ is a Japanese ring.*

Proof. We know that for every integrally closed Noetherian ring B, the formal power series ring $B[[T]]$ is Noetherian and integrally closed (Bourbaki, *Alg. comm.*, chap. V, §1, nº 4, prop. 14); we can thus, by induction on n, restrict to proving that $A[[T]]$ is a Japanese ring. Now the element $x = T$ satisfies all the conditions of (23.1.3), whence the conclusion. □

Theorem (23.1.5). — (Nagata) *Every complete local Noetherian integral ring A is a Japanese ring.*

Proof. We know (19.8.8, ii) that there exists a sub-ring B of A that is a complete local ring and is in addition *regular*, such that A is a finite B-algebra; it clearly suffices to prove that B is a Japanese ring, that is to say, we can restrict to the case where A is in addition regular. Let us argue by induction on $n = \dim(A)$, the theorem being evident for $n = 0$. Suppose then $n > 0$, and, denoting by $\mathfrak{m}$ the maximal ideal of A, let x be an element of $\mathfrak{m} - \mathfrak{m}^2$; we know (17.1.8) that A/xA is a regular ring, and in addition ((17.1.7) and (16.3.4)) that $\dim(A/xA) = n - 1$; moreover, the ideal xA is closed in A ($\mathbf{0}_{\mathbf{I}}$, 7.3.5), hence A/xA is complete. By the induction hypothesis, A/xA is thus a Japanese ring, and it then follows from (23.1.3) that the same is true of A. □ $\mathbf{0}_{\mathbf{IV\text{-}1}}$ | 216

Corollary (23.1.6). — *Let A be a complete local Noetherian integral ring, K its field of fractions, K' a finite extension of K; then the integral closure A' of A in K' is a complete local ring.*

Proof. We already know from (23.1.5) that A' is a finite A-algebra, hence a semi-local Noetherian ring, and consequently a *direct product* of local rings; but since A' is integral, it is necessarily a local ring. □

Proposition (23.1.7). — *Let A be a local Noetherian integral ring, K its field of fractions, K' a finite extension of K, A' the integral closure of A in K'. Suppose that the completion $\widehat{A}$ is reduced, and let R denote its total ring of fractions.*

(i) *If the ring $R \otimes_K K'$ is* reduced *(which will be the case in particular when K' is a* separable *extension of K, R being a direct product of fields, hence extensions of K* (Bourbaki, *Alg.*, chap. VIII, §7, nº 3, cor. 1 of th. 1)*), then A' is an A-module of finite type.*

(ii) If in particular R is a separable K-algebra, then A is a Japanese ring.

Proof. (i). Set $A_1 = \widehat{A}$, $A'_1 = A' \otimes_A A_1$, $K'_1 = K' \otimes_A A_1$. Since A_1 is a flat A-module ($\mathbf{0}_{\mathbf{I}}$, 7.3.5), A'_1 is identified with a sub-ring of K'_1 and is evidently integral over A_1. On the other hand, if we set $K_1 = K \otimes_A A_1$, we can write $K'_1 = K' \otimes_K K_1$; since A is integral and A_1 is a flat A-module, every element $\neq 0$ of A is A_1-regular ($\mathbf{0}_{\mathbf{I}}$, 6.3.4); since $K = S^{-1}A$, where $S = A - \{0\}$, we have $K_1 = S^{-1}A_1$, and the preceding remark proves that K_1 is identified with a *sub-ring* of the total ring of fractions R of A_1. Since every K-module is flat, K'_1 is identified with a *sub-ring* of $K' \otimes_K R$, which by hypothesis is *reduced* and is a *finite* R-algebra. Denoting by $\mathfrak{q}_i$ ($1 \leqslant i \leqslant n$) the minimal prime ideals of A_1, by B_i the integral ring $A_1/\mathfrak{q}_i$, by L_i the field of fractions of B_i, we see that R is the direct product of the L_i, and $K' \otimes_K R$ the direct product of the $K' \otimes_K L_i$; these latter K-algebras, being reduced, are direct products of fields, *finite* extensions of the L_i. Since B_i is a complete local Noetherian integral ring, the theorem of Nagata (23.1.5) proves that the integral closure of B_i in $K' \otimes_K L_i$ is a B_i-module of finite type, and *a fortiori* an A_1-module of finite type; since every element of $K' \otimes_K L_i$ that is integral over A_1 is also integral over B_i (being annihilated by $\mathfrak{q}_i$), we conclude finally that the integral closure of A_1 in $K' \otimes_K R$ is an A_1-module of finite type. But since A'_1 is contained in this integral closure and A_1 is Noetherian, A'_1 is also an A_1-module of finite type. Finally, since A_1 is a faithfully flat A-module ($\mathbf{0}_\mathbf{I}$, 7.3.5), we conclude that A' is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. I, §3, n° 6, prop. 11).

(ii). The hypothesis implies that the L_i are separable extensions of K, hence $K' \otimes_K L_i$ is reduced (Bourbaki, *Alg.*, chap. VIII, §7, n° 3, th. 1) and we can apply (i) to *every* finite extension K' of K, which proves our assertion. □ $\mathbf{0}_{\mathbf{IV}\text{-}\mathbf{1}}$ | 217

23.2. Integral closure of a local Noetherian integral ring

(23.2.1). In what follows, for every ring A, we will write A_{red} for short for the quotient of A by its nilradical (so that if $X = \operatorname{Spec}(A)$, we have $X_{\mathrm{red}} = \operatorname{Spec}(A_{\mathrm{red}})$).

We will say that a *local* ring A is *unibranch* if A_{red} is integral and if the integral closure of A_{red} is a *local* ring, which generalizes the definition given in (**III**, 4.3.6). We will say that A is *geometrically unibranch* if it is unibranch and if the residue field of the local ring that is the integral closure of A_{red} is a *radicial* extension of that of A. It is clear that a local ring that is integrally closed is geometrically unibranch; it follows from (23.1.6) that a complete local Noetherian integral ring is unibranch.

Lemma (23.2.2). — *Let A be an integral ring, A' its integral closure; for the canonical morphism f : $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ to be radiciel, it is necessary and sufficient that for every $\mathfrak{p} \in \operatorname{Spec}(A)$, $A_\mathfrak{p}$ be geometrically unibranch; when this is so, f is a homeomorphism.*[15]

Proof. Indeed, for every $\mathfrak{p} \in \operatorname{Spec}(A)$, the integral closure of $A_\mathfrak{p}$ is $A'_\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, §1, n° 5, prop. 16), and all the prime ideals of $A'_\mathfrak{p}$ are maximal (*loc. cit.*, §2, n° 1, prop. 1). Saying that f is *injective* thus means that for every $\mathfrak{p} \in \operatorname{Spec}(A)$, $A'_\mathfrak{p}$ is a *local* ring, that is to say that the $A_\mathfrak{p}$ are unibranch. Saying that $A_\mathfrak{p}$ is geometrically unibranch then means that f is radiciel by virtue of (**I**, 3.5.8). When this is so, f is surjective and closed (**II**, 6.1.10), hence a homeomorphism. □

Lemma (23.2.3). — *Let A be an integral ring, K its field of fractions, B a Noetherian ring, $\phi : A \to B$ a ring homomorphism making B an A-module that is flat. Let K' be an extension of K; set $B_1 = B_{\mathrm{red}}$, and let R be the total ring of fractions of B_1. Then, for every sub-A-algebra A' of K', the canonical homomorphism*

(23.2.3.1) $$(A' \otimes_A B)_{\mathrm{red}} \longrightarrow (K' \otimes_A R)_{\mathrm{red}}$$

is injective.

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Proof. We can in fact consider the canonical homomorphism $h : A' \otimes_A B \to K' \otimes_A R$ as the composition of the following homomorphisms

$$A' \otimes_A B \xrightarrow{u} K' \otimes_A B \xrightarrow{v} K' \otimes_A B_1 \xrightarrow{w} K' \otimes_A R.$$

Since B is a flat A-module, u is injective; similarly, K' being a flat K-module and K a flat A-module, K' is a flat A-module, and w is therefore injective since B_1 is reduced, hence the homomorphism $B_1 \to R$ is injective. Finally, for the same reason, if $\mathfrak{N}$ is the nilradical of B, the kernel of v is $K' \otimes_A \mathfrak{N}$, hence it is contained in the nilradical of $K' \otimes_A B$; we immediately conclude that if the image by $h = w \circ v \circ u$ of an element $x \in A' \otimes_A B$ is nilpotent, then x is nilpotent, which proves the lemma. □

[15] In the rest of this section, we use notions and results from chap. IV, §§2 and 5; since the results of this section are not used before chap. IV, §6, this does not lead to circular reasoning.

Proposition (23.2.4). — *Let A be an integral ring, K its field of fractions, B a Noetherian ring, $\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ its minimal prime ideals, $\phi : A \to B$ a ring homomorphism. Suppose that the $B/\mathfrak{q}_i$ are Japanese rings, and that B is a flat A-module. Let K' be a finite extension of K, and let (C_α) be the increasing filtering family of sub-rings of K' that are finite A-algebras and admit K' as their field of fractions; the union A' of the C_α is thus the integral closure of A in K'. Then:*

(i) *There exists an index α such that for $\lambda \geqslant \alpha$, the canonical homomorphism*

$$(C_\alpha \otimes_A B)_{\mathrm{red}} \longrightarrow (C_\lambda \otimes_A B)_{\mathrm{red}} \tag{23.2.4.1}$$

is bijective.

(ii) *If in addition B is a faithfully flat A-module, the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(C_\alpha)$ *is* radiciel.

Proof. (i). Let $B_1 = B_{\mathrm{red}}$, and let R be the total ring of fractions of B_1; since A is integral and B is a flat A-module, every element $\neq 0$ of A is B-regular $(\mathbf{0_I}, 6.3.4)$, hence the composed homomorphism $A \to B \to B_{\mathrm{red}} = C$ is injective and extends to an injective homomorphism of the field of fractions K of A into the total ring of fractions R of C; and the integral closure B' of B in K' is a B-module of finite type, and the integral closure of B' of B in $(K' \otimes_A R)_{\mathrm{red}}$ is a finite B-module of finite type. By virtue of (23.2.3), the $(C_\alpha \otimes_A B)_{\mathrm{red}}$ are canonically identified with sub-rings of $(K' \otimes_A R)_{\mathrm{red}}$ which are *finite* B-algebras, hence contained in B'. Since B is Noetherian and B' a B-module of finite type, the filtering family of the $(C_\alpha \otimes_A B)_{\mathrm{red}}$ admits a *greatest element* $(C_\alpha \otimes_A B)_{\mathrm{red}}$, which proves (i).

(ii). Since B is a faithfully flat A-module, it suffices (**IV**, 2.6.1, v) to show that the morphism $\mathrm{Spec}(A' \otimes_A B) \to \mathrm{Spec}(C_\alpha \otimes_A B)$ is radiciel, or, what amounts to the same (**I**, 5.1.6), that the morphism $\mathrm{Spec}(A' \otimes_A B)_{\mathrm{red}} \to \mathrm{Spec}(C_\alpha \otimes_A B)_{\mathrm{red}}$ is radiciel; but we have $A' \otimes_A B = \varinjlim (C_\lambda \otimes_A B)$, hence (**IV**, 5.13.2) $(A' \otimes_A B)_{\mathrm{red}} = \varinjlim (C_\lambda \otimes_A B)_{\mathrm{red}}$, and the conclusion follows from (i). ☐

Corollary (23.2.5). — *Let A be a local Noetherian integral ring, K its field of fractions, K' a finite extension of K. Let (C_α) be the increasing filtering family of sub-rings of K' that are finite A-algebras (hence Noetherian semi-local rings* (Bourbaki, *Alg. comm.*, chap. IV, §2, nº 5, cor. 3 of prop. 9)*) and admit K' as their field of fractions. Then there exists α such that the homomorphism $(\widehat{C}_\alpha)_{\mathrm{red}} \to (\widehat{C}_\lambda)_{\mathrm{red}}$ is an isomorphism for $\lambda \geqslant \alpha$, and if A' is the integral closure of A in K', the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(C_\alpha)$ *is radiciel* (cf. (23.2.2)).

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Proof. We apply (23.2.4) taking $B = \widehat{A}$; since the $B/\mathfrak{q}_i$ are complete local Noetherian rings, they are Japanese rings (23.1.5) and B is a faithfully flat A-module $(\mathbf{0_I}, 7.3.5)$; moreover we have $C_\alpha \otimes_A B = \widehat{C}_\alpha$ (Bourbaki, *Alg. comm.*, chap. III, §3, nº 4, th. 3 and chap. IV, §2, nº 5, cor. 3 of prop. 9). ☐

Corollary (23.2.6). — *Under the hypotheses of (23.2.5), the integral closure A' of A in K' is a semi-local ring; if $\mathfrak{m}$ is the maximal ideal of A, then, for every maximal ideal $\mathfrak{m}'$ of A', $A'/\mathfrak{m}'$ is a finite extension of $A/\mathfrak{m}$.*

Proof. The fact that the homomorphism $(\widehat{C}_\alpha)_{\mathrm{red}} \to (\widehat{C}_\lambda)_{\mathrm{red}}$ is bijective for $\lambda \geqslant \alpha$ implies that the number of maximal ideals of C_α is *constant* for $\lambda \geqslant \alpha$, and that if $\mathfrak{m}_\alpha$ is the unique maximal ideal of C_α above $\mathfrak{m}_\alpha$, the fields $C_\alpha/\mathfrak{m}_\alpha$ and $C_\lambda/\mathfrak{m}_\lambda$ are canonically isomorphic. The conclusion follows from the fact that $\mathfrak{m}' = \varinjlim \mathfrak{m}_\alpha$ and $A'/\mathfrak{m}' = \varinjlim C_\alpha/\mathfrak{m}_\alpha$ $(\mathbf{0_{III}}, 10.3.1.3)$. ☐

Proposition (23.2.7). — (Y. Mori) *Let A be a local Noetherian integral ring, A' the integral closure of A. Then A' is a Krull ring* (Bourbaki, *Alg. comm.*, chap. VII, §1) *that is semi-local; in other words, there exists a family (V_λ) of discrete valuation rings having K as their field of fractions, such that $A' = \bigcap V_\lambda$ and that every $x \in K$ belongs to all but a finite number of the V_λ. Moreover, there exists a sub-algebra C of K which is a finite A-algebra, and such that the V_λ are the integral closures of the rings $C_{\mathfrak{p}_\lambda}$, where $(\mathfrak{p}_\lambda)$ is the family of prime ideals of height 1 of the local ring C.*

We will first prove the following lemma:

Lemma (23.2.7.1). — *Let A, B be two rings, $\phi : A \to B$ a homomorphism making B a faithfully flat A-module. Suppose that A is integral; then, if $C = B_{\mathrm{red}}$, the composed homomorphism $\psi : A \to B \to B_{\mathrm{red}} = C$ is injective and extends to an injective homomorphism of the field of fractions K of A into the total ring of fractions R of C; moreover, if C' is the integral closure of C in R, $C' \cap K$ is the integral closure of A.*

Proof. Since A is integral and ϕ is injective, the intersection of $\phi(A)$ and the nilradical $\mathfrak{N}$ of B is 0, and since $C = B/\mathfrak{N}$, ψ is injective. Every $a \neq 0$ in A being a non-zero-divisor does not become a zero-divisor in B by flatness ($\mathbf{0_I}$, 6.3.4); we deduce that a is not a non-zero-divisor in C either, since if $ax \in \mathfrak{N}$ for an $x \notin \mathfrak{N}$ in B, we would have $a^n x^n = 0$ for some integer n, which contradicts the above since $x^n \neq 0$. We can thus extend ψ to an injective homomorphism of K into R. To prove the last assertion, note that the integral closure A' of A is contained in $C' \cap K$. Conversely, let $x \in C' \cap K$; x is then integral over C, and *a fortiori* over B, that is to say $B[x]$ is a finite B-algebra; moreover, K is identified with a sub-ring of $B \otimes_A K$, and the sub-ring $B[x]$ of $B \otimes_A K$ is identified with $B \otimes_A A[x]$ by flatness; we conclude that $A[x]$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, §3, n$^{\text{o}}$ 6, prop. 11), hence $x \in A'$. □

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Proof of Proposition (23.2.7). This lemma being established, let us apply it to the canonical injection of A into its completion $\widehat{A}$, which is a faithfully flat A-module; $B = (\widehat{A})_{\text{red}} = \widehat{A}/\mathfrak{N}$ (where $\mathfrak{N}$ is the nilradical of $\widehat{A}$) is a complete local Noetherian reduced ring, whose total ring of fractions R is a direct product of a finite number of fields L_i; the canonical image B_i of B in L_i is a complete local Noetherian *integral* ring, whose field of fractions is L_i, and the integral closure B' of B in R is the *direct product* of the B'_i, where B'_i is the integral closure of B_i. But by virtue of (23.1.5), B'_i is a finite B_i-algebra, hence an integrally closed local Noetherian ring. We know (Bourbaki, *Alg. comm.*, chap. VII, §1, n$^{\text{o}}$ 3, cor. du th. 2) that B'_i is a Krull ring. Now, for every i, there is a homomorphism $\phi_i : K \to L_i$, which is injective, and so $\phi_i^{-1}(B'_i)$ is a Krull ring in K. But since $A' = B' \cap K$ by virtue of the lemma and $B' = \bigcap \phi_i^{-1}(B'_i)$, A' is an intersection of a finite number of Krull rings and is therefore a Krull ring. To prove the last assertion of the proposition, note that there exists a finite A-algebra $C \subset K$ such that the morphism $\operatorname{Spec}(A') \to \operatorname{Spec}(C)$ is radiciel and a homeomorphism (23.2.4); for every prime ideal $\mathfrak{p}' \in \operatorname{Spec}(A')$, $\mathfrak{p}'$ is thus the only prime ideal of A' above $\mathfrak{p} = \mathfrak{p}' \cap C$ and $A'_{\mathfrak{p}'}$ the integral closure of $C_{\mathfrak{p}}$; on the other hand the fact that $\operatorname{Spec}(A') \to \operatorname{Spec}(C)$ is a homeomorphism implies that the application $\mathfrak{p}' \mapsto \mathfrak{p}' \cap C$ is a *bijection* of the set of prime ideals of height 1 of A' onto the set of prime ideals of height 1 of C; whence the conclusion, by virtue of Bourbaki, *Alg. comm.*, chap. VII, §1, n$^{\text{o}}$ 6, th. 4. □

Remarks (23.2.8). —

(i) We should take care to note, in the application of (23.2.6) and (23.2.7), that the ring A' is *not necessarily Noetherian*, as an example of Nagata shows with $\dim(A) = 3$ [**Nag62**].

(ii) The conclusions of (23.2.7) remain valid if A' is the integral closure of A in a finite extension K' of K; it suffices in fact to consider a finite A-algebra B whose field of fractions is K'; B is a semi-local Noetherian ring, hence an intersection of a finite number of local Noetherian rings $B_{\mathfrak{m}_i}$ ($\mathfrak{m}_i$ the maximal ideals of B), and its integral closure (which is equal to A') is the intersection of the $A'_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. II, §3, n$^{\text{o}}$ 3, cor. 4 of th. 1) which are the integral closures of the $B_{\mathfrak{m}_i}$, hence Krull rings by (23.2.7); consequently A' is a Krull ring.

(iii) One can show ([**Nag62**], 3.3.10) that for every Noetherian integral ring A (not necessarily local), the integral closure of A is a Krull ring.

Corollary (23.2.9). — *Let A be a Noetherian integral ring, A' its integral closure. Suppose that for every ring C such that $A \subset C \subset A'$ and such that C is a finite A-algebra, and for every prime ideal $\mathfrak{q}$ of height 1 in C, $\mathfrak{q} \cap A$ is a prime ideal of height 1 in A. Let P be the set of prime ideals of height 1 in A; we then have*

$$A' = \bigcap_{\mathfrak{p} \in P} A'_{\mathfrak{p}}.$$

Proof. Since A' is an A-module without torsion, we have $A' = \bigcap A'_{\mathfrak{m}}$, where $\mathfrak{m}$ ranges over the set of maximal ideals of A (Bourbaki, *Alg. comm.*, chap. II, §3, n$^{\text{o}}$ 3, cor. 4 of th. 1); it will thus $\mathbf{0_{IV\text{-}1}}$ | 221
suffice to prove that $A'_{\mathfrak{m}}$ is the intersection of the $A'_{\mathfrak{p}}$ for the prime ideals $\mathfrak{p} \subset \mathfrak{m}$ of height 1, and since $A'_{\mathfrak{m}}$ is the integral closure of $A_{\mathfrak{m}}$, we see that it suffices to prove the corollary for $A_{\mathfrak{m}}$. Now there is, by virtue of (23.2.7), a finite $A_{\mathfrak{m}}$-algebra B contained in $A'_{\mathfrak{m}}$ such that $A'_{\mathfrak{m}}$ is the intersection of the integral closures of the rings $B_{\mathfrak{q}}$, where $\mathfrak{q}$ ranges over the set of prime ideals of height 1 of B. But B is of the form $C_{\mathfrak{m}}$, where C is a finite A-algebra; hence, for every prime ideal $\mathfrak{q}$ of height 1 in B,

$\mathfrak{q} \cap A$ is by hypothesis a prime ideal of height 1 in A; since the integral closure of $B_{\mathfrak{q}}$ contains that of $A_{\mathfrak{q}\cap A}$ and the latter contains $A'_{\mathfrak{m}}$, it follows that $A'_{\mathfrak{m}}$ is indeed the intersection of the $A'_{\mathfrak{p}}$ for $\mathfrak{p} \in P$ and $\mathfrak{p} \subset \mathfrak{m}$, which finishes the proof. □

Remarks (23.2.10). —

(i) We will see (**IV**, 5.6.3) and (**IV**, 5.6.10) that the hypothesis of (23.2.9) is satisfied when A is *universally catenary*, in particular (**IV**, 5.10.17) and (**IV**, 5.11.2) when A is a *quotient of a regular ring*; finally, it is evidently satisfied if $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a homeomorphism, in particular if this morphism is radiciel (23.2.2).

(ii) The conclusion of (23.2.9) is no longer necessarily exact when one omits the hypothesis on the finite sub-A-algebras of A'. Indeed, one can define a local Noetherian integral ring A, of dimension 2, whose integral closure A' is a finite A-algebra, but such that there is a prime ideal $\mathfrak{p}'$ of A' of height 1 above the maximal ideal $\mathfrak{m}$ (of height 2) of A, and such moreover that for every prime ideal $\mathfrak{p}$ of height 1 in A, $A_{\mathfrak{p}}$ is integrally closed (**IV**, 5.6.11); the intersection of these rings $A_{\mathfrak{p}}$ can then not be equal to A' since A' is a Krull ring (Bourbaki, *Alg. comm.*, chap. VII, §1, n° 5, prop. 9).

CHAPTER I

The language of schemes (EGA I)

SUMMARY

In §§1–8 we do little more than develop a language to be used in what follows. It should be noted, however, that, in accordance with the general spirit of this treatise, §§7–8 will be used less than the others, and in a less essential way; we speak of Chevalley schemes only for the purpose of relating to the language of Chevalley [**CC**] and Nagata [**Nag58a**]. Then, in §9, we give definitions and results concerning quasi-coherent sheaves, some of which are no longer simply a translation of known notions of commutative algebra into a "geometric" language, but are instead already of a global nature; they will be indispensable, in the following chapters, when it comes to the global study of morphisms. Finally, in §10, we introduce a generalization of the notion of a scheme, which will be used as an intermediary in Chapter III to conveniently formulate and prove the fundamental results of the cohomological study of proper morphisms; moreover, it should be noted that the notion of formal schemes seems indispensable in expressing certain facts about the "theory of modules" (classification problems of algebraic varieties). The results of §10 will not be used before §3 of Chapter III, and it is recommended to skip their reading until then. **I** | 79

§1. AFFINE SCHEMES

1.1. The prime spectrum of a ring

(1.1.1). *Notation.* Let A be a (commutative) ring, and M an A-module. In this chapter and the following, we will constantly use the following notation: **I** | 80

- $\operatorname{Spec}(A)$ = *set of prime ideals* of A, also called the *prime spectrum* of A; for $x \in X = \operatorname{Spec}(A)$, it will often be convenient to write $\mathfrak{j}_x$ instead of x. For $\operatorname{Spec}(A)$ to be *empty*, it is necessary and sufficient for the ring A to be 0.
- $A_x = A_{\mathfrak{j}_x}$ = *(local) ring of fractions* $S^{-1}A$, where $S = A - \mathfrak{j}_x$.
- $\mathfrak{m}_x = \mathfrak{j}_x A_{\mathfrak{j}_x}$ = *maximal ideal of* A_x.
- $k(x) = A_x/\mathfrak{m}_x$ = *residue field of* A_x, canonically isomorphic to the field of fractions of the integral ring $A/\mathfrak{j}_x$, with which we identify it.
- $f(x)$ = *class of* f *mod.* $\mathfrak{j}_x$ in $A/\mathfrak{j}_x \subset k(x)$, for $f \in A$ and $x \in X$. We also say that $f(x)$ is the *value* of f at a point $x \in \operatorname{Spec}(A)$; the equations $f(x) = 0$ and $f \in \mathfrak{j}_x$ are *equivalent*.
- $M_x = M \otimes_A A_x$ = *module of fractions with denominators in* $A - \mathfrak{j}_x$.
- $\mathfrak{r}(E)$ = *radical of the ideal of* A *generated by a subset* E *of* A.

- $V(E)$ = *set of* $x \in X$ *such that* $E \subset \mathfrak{j}_x$ (or the set of $x \in X$ such that $f(x) = 0$ for all $f \in E$), for $E \subset A$. So we have

$$\mathfrak{r}(E) = \bigcap_{x \in V(E)} \mathfrak{j}_x. \tag{1.1.1.1}$$

- $V(f) = V(\{f\})$ for $f \in A$.
- $D(f) = X - V(f)$ = *set of* $x \in X$ *where* $f(x) \neq 0$.

Proposition (1.1.2). — *We have the following properties:*

(i) $V(0) = X$, $V(1) = \varnothing$.
(ii) *The relation* $E \subset E'$ *implies* $V(E) \supset V(E')$.
(iii) *For each family* (E_λ) *of subsets of* A, $V(\bigcup_\lambda E_\lambda) = V(\sum_\lambda E_\lambda) = \bigcap_\lambda V(E_\lambda)$.
(iv) $V(EE') = V(E) \cup V(E')$.
(v) $V(E) = V(\mathfrak{r}(E))$.

PROOF. The properties (i), (ii), (iii) are trivial, and (v) follows from (ii) and from equation (1.1.1.1). It is evident that $V(EE') \supset V(E) \cap V(E')$; conversely, if $x \notin V(E)$ and $x \notin V(E')$, then there exists $f \in E$ and $f' \in E'$ such that $f(x) \neq 0$ and $f'(x) \neq 0$ in $k(x)$, hence $f(x)f'(x) \neq 0$, i.e., $x \notin V(EE')$, which proves (iv). □

Proposition (1.1.2) shows, among other things, that sets of the form $V(E)$ (where E varies over the subsets of A) are the *closed sets* of a topology on X, which we will call the *spectral topology* [1]; unless expressly stated otherwise, we always assume that $X = \operatorname{Spec}(A)$ is equipped with the spectral topology.

(1.1.3). For each subset Y of X, we denote by $\mathfrak{j}(Y)$ the set of $f \in A$ such that $f(y) = 0$ for all $y \in Y$; **I** | 81
equivalently, $\mathfrak{j}(Y)$ is the intersection of the prime ideals $\mathfrak{j}_y$ for $y \in Y$. It is clear that the relation $Y \subset Y'$ implies that $\mathfrak{j}(Y) \supset \mathfrak{j}(Y')$ and that we have

$$\mathfrak{j}\left(\bigcup_\lambda Y_\lambda\right) = \bigcap_\lambda \mathfrak{j}(Y_\lambda) \tag{1.1.3.1}$$

for each family (Y_λ) of subsets of X. Finally we have

$$\mathfrak{j}(\{x\}) = \mathfrak{j}_x. \tag{1.1.3.2}$$

Proposition (1.1.4). —

(i) *For each subset* E *of* A, *we have* $\mathfrak{j}(V(E)) = \mathfrak{r}(E)$.
(ii) *For each subset* Y *of* X, $V(\mathfrak{j}(Y)) = \overline{Y}$, *the closure of* Y *in* X.

PROOF. (i) is an immediate consequence of the definitions and (1.1.1.1); on the other hand, $V(\mathfrak{j}(Y))$ is closed and contains Y; conversely, if $Y \subset V(E)$, we have $f(y) = 0$ for $f \in E$ and all $y \in Y$, so $E \subset \mathfrak{j}(Y)$, $V(E) \supset V(\mathfrak{j}(Y))$, which proves (ii). □

Corollary (1.1.5). — *The closed subsets of* $X = \operatorname{Spec}(A)$ *and the ideals of* A *equal to their radicals (in other words, those that are the intersection of prime ideals) correspond bijectively by the inclusion-reversing maps* $Y \mapsto \mathfrak{j}(Y)$, $\mathfrak{a} \mapsto V(\mathfrak{a})$; *the union* $Y_1 \cup Y_2$ *of two closed subsets corresponds to* $\mathfrak{j}(Y_1) \cap \mathfrak{j}(Y_2)$, *and the intersection of any family* (Y_λ) *of closed subsets corresponds to the radical of the sum of the* $\mathfrak{j}(Y_\lambda)$.

Corollary (1.1.6). — *If* A *is a Noetherian ring,* $X = \operatorname{Spec}(A)$ *is a Noetherian space.*

Note that the converse of this corollary is false, as shown by any non-Noetherian integral ring with a single prime ideal $\neq \{0\}$ (for example a nondiscrete valuation ring of rank 1).

As an example of ring A whose spectrum is not a Noetherian space, one can consider the ring $\mathscr{C}(Y)$ of continuous real functions on an infinite compact space Y; we know that, as a set, Y corresponds to the set of maximal ideals of A, and it is easy to see that the topology induced on Y by that of $X = \operatorname{Spec}(A)$ is the original topology of Y. Since Y is not a Noetherian space, the same is true for X.

[1] The introduction of this topology in algebraic geometry is due to Zariski. So this topology is usually called the "Zariski topology" on X.

Corollary (1.1.7). — *For each $x \in X$, the closure of $\{x\}$ is the set of $y \in X$ such that $\mathfrak{j}_x \subset \mathfrak{j}_y$. For $\{x\}$ to be closed, it is necessary and sufficient that $\mathfrak{j}_x$ is maximal.*

Corollary (1.1.8). — *The space $X = \operatorname{Spec}(A)$ is a Kolmogoroff space.*

Proof. If x and y are two distinct points of X, we have either $\mathfrak{j}_x \not\subset \mathfrak{j}_y$ or $\mathfrak{j}_y \not\subset \mathfrak{j}_x$, so one of the points x, y does not belong to the closure of the other. □

(1.1.9). According to Proposition (1.1.2, iv), for two elements f, g of A, we have

$$D(fg) = D(f) \cap D(g). \tag{1.1.9.1}$$

Note also that the equality $D(f) = D(g)$ means, according to Proposition (1.1.4, i) and Proposition (1.1.2, v), that $\mathfrak{r}(f) = \mathfrak{r}(g)$, or that the minimal prime ideals containing (f) and (g) are the same; in particular, it is also the case when $f = ug$, where u is invertible.

Proposition (1.1.10). — I | 82

(i) *When f ranges over A, the sets $D(f)$ forms a basis for the topology of X.*
(ii) *For every $f \in A$, $D(f)$ is quasi-compact. In particular, $X = D(1)$ is quasi-compact.*

Proof.

(i) Let U be an open set in X; by definition, we have $U = X - V(E)$ where E is a subset of A, and $V(E) = \bigcap_{f \in E} V(f)$, hence $U = \bigcup_{f \in E} D(f)$.
(ii) By (i), it suffices to prove that, if $(f_\lambda)_{\lambda \in L}$ is a family of elements of A such that $D(f) \subset \bigcup_{\lambda \in L} D(f_\lambda)$, then there exists a finite subset J of L such that $D(f) \subset \bigcup_{\lambda \in J} D(f_\lambda)$. Let $\mathfrak{a}$ be the ideal of A generated by the f_λ; we have, by hypothesis, that $V(f) \supset V(\mathfrak{a})$, so $\mathfrak{r}(f) \subset \mathfrak{r}(\mathfrak{a})$; since $f \in \mathfrak{r}(f)$, there exists an integer $n \geqslant 0$ such that $f^n \in \mathfrak{a}$. But then f^n belongs to the ideal $\mathfrak{b}$ generated by the finite subfamily $(f_\lambda)_{\lambda \in J}$, and we have $V(f) = V(f^n) \supset V(\mathfrak{b}) = \bigcap_{\lambda \in J} V(f_\lambda)$, that is to say, $D(f) \supset \bigcup_{\lambda \in J} D(f_\lambda)$.

□

Proposition (1.1.11). — *For each ideal $\mathfrak{a}$ of A, $\operatorname{Spec}(A/\mathfrak{a})$ is canonically identified with the closed subspace $V(\mathfrak{a})$ of $\operatorname{Spec}(A)$.*

Proof. We know there is a canonical bijective correspondence (respecting the inclusion order structure) between ideals (resp. prime ideals) of $A/\mathfrak{a}$ and ideals (resp. prime ideals) of A containing $\mathfrak{a}$. □

Recall that the set $\mathfrak{N}$ of nilpotent elements of A (the *nilradical* of A) is an ideal equal to $\mathfrak{r}(0)$, the intersection of all the prime ideals of A (**0**, 1.1.1).

Corollary (1.1.12). — *The topological spaces $\operatorname{Spec}(A)$ and $\operatorname{Spec}(A/\mathfrak{N})$ are canonically homeomorphic.*

Proposition (1.1.13). — *For $X = \operatorname{Spec}(A)$ to be irreducible (**0**, 2.1.1), it is necessary and sufficient that the ring $A/\mathfrak{N}$ is integral (or, equivalently, that the ideal $\mathfrak{N}$ is prime).*

Proof. By virtue of Corollary (1.1.12), we can restrict to the case where $\mathfrak{N} = 0$. If X is reducible, then there exist two distinct closed subsets Y_1 and Y_2 of X such that $X = Y_1 \cup Y_2$, so $\mathfrak{j}(X) = \mathfrak{j}(Y_1) \cap \mathfrak{j}(Y_2) = 0$, since the ideals $\mathfrak{j}(Y_1)$ and $\mathfrak{j}(Y_2)$ are distinct from (0) (1.1.5); so A is not integral. Conversely, if there are elements $f \neq 0$, $g \neq 0$ of A such that $fg = 0$, we have $V(f) \neq X$, $V(g) \neq X$ (since the intersection of all the prime ideals of A is (0)), and $X = V(fg) = V(f) \cup V(g)$. □

Corollary (1.1.14). —

(i) *In the bijective correspondence between closed subsets of $X = \operatorname{Spec}(A)$ and ideals of A equal to their radicals, the irreducible closed subsets of X correspond to the prime ideals of A. In particular, the irreducible components of X correspond to the minimal prime ideals of A.*
(ii) *The map $x \mapsto \overline{\{x\}}$ establishes a bijective correspondence between X and the set of closed irreducible subsets of X (in other words, all closed irreducible subsets of X admit exactly one generic point).*

PROOF. (i) follows immediately from (1.1.13) and (1.1.11); and for proving (ii), we can, by (1.1.11), restrict to the case where X is irreducible; then, according to Proposition (1.1.13), there exists a smaller prime ideal $\mathfrak{N}$ in A, which corresponds to the generic point of X; in addition, X admits at most one generic point since it is a Kolmogoroff space ((1.1.8) and (**0**, 2.1.3)). ∎ I | 83

Proposition (1.1.15). — *If $\mathfrak{J}$ is an ideal in A containing the radical $\mathfrak{N}(A)$, the only neighbourhood of $V(\mathfrak{J})$ in $X = \operatorname{Spec}(A)$ is the whole space X.*

PROOF. Each maximal ideal of A belongs, by definition, to $V(\mathfrak{J})$. As each ideal $\mathfrak{a} \neq A$ of A is contained in a maximal ideal, we have $V(\mathfrak{a}) \cap V(\mathfrak{J}) \neq 0$, whence the proposition. ∎

1.2. Functorial properties of prime spectra of rings

(1.2.1). Let A, A' be two rings, and

$$\phi : A' \longrightarrow A$$

a homomorphism of rings. For each prime ideal $x = \mathfrak{j}_x \in \operatorname{Spec}(A) = X$, the ring $A'/\phi^{-1}(\mathfrak{j}_x)$ is canonically isomorphic to a subring of $A/\mathfrak{j}_x$, and so it is integral, or, in other words, $\phi^{-1}(\mathfrak{j}_x)$ is a prime ideal of A'; we denote it by ${}^a\phi(x)$, and we have thus defined a map

$$ {}^a\phi : X = \operatorname{Spec}(A) \longrightarrow X' = \operatorname{Spec}(A') $$

(also denoted $\operatorname{Spec}(\phi)$), that we call the map *associated* to the homomorphism ϕ. We denote by ϕ^x the injective homomorphism from $A'/\phi^{-1}(\mathfrak{j}_x)$ to $A/\mathfrak{j}_x$ induced by ϕ by passing to quotients, as well as its canonical extension to a monomorphism of fields

$$\phi^x : k({}^a\phi(x)) \longrightarrow k(x);$$

for each $f' \in A'$, we therefore have, by definition, Err$_{\text{II}}$

$$\phi^x(f'({}^a\phi(x))) = (\phi(f'))(x) \qquad (x \in X). \tag{1.2.1.1}$$

Proposition (1.2.2). —

(i) *For each subset E' of A', we have*

$$ {}^a\phi^{-1}(V(E')) = V(\phi(E')), \tag{1.2.2.1}$$

and in particular, for each $f' \in A'$,

$$ {}^a\phi^{-1}(D(f')) = D(\phi(f')). \tag{1.2.2.2}$$

(ii) *For each ideal $\mathfrak{a}$ of A, we have*

$$ \overline{{}^a\phi(V(\mathfrak{a}))} = V(\phi^{-1}(\mathfrak{a})). \tag{1.2.2.3}$$

PROOF. The relation ${}^a\phi(x) \in V(E')$ is, by definition, equivalent to $E' \subset \phi^{-1}(\mathfrak{j}_x)$, so $\phi(E') \subset \mathfrak{j}_x$, and finally $x \in V(\phi(E'))$, hence (i). To prove (ii), we can suppose that $\mathfrak{a}$ is equal to its radical, since $V(\mathfrak{r}(\mathfrak{a})) = V(\mathfrak{a})$ (1.1.2, (v)) and $\phi^{-1}(\mathfrak{r}(\mathfrak{a})) = \mathfrak{r}(\phi^{-1}(\mathfrak{a}))$; if we set $Y = V(\mathfrak{a})$, and $\mathfrak{a}' = \mathfrak{j}({}^a\phi(Y))$, then we have $\overline{{}^a(Y)} = V(\mathfrak{a}')$ ((1.1.4, (ii))) the relation $f' \in \mathfrak{a}'$ is, by definition, equivalent to $f'(x') = 0$ for each $x \in {}^a\phi(Y)$, so, by Equation (1.2.1.1), it is also equivalent to $\phi(f')(x) = 0$ for each $x \in Y$, or to $\phi(f') \in \mathfrak{j}(Y) = \mathfrak{a}$, since $\mathfrak{a}$ is equal to its radical; hence (ii). ∎

Corollary (1.2.3). — *The map ${}^a\phi$ is continuous.*

We remark that, if A'' is a third ring, and ϕ' a homomorphism $A'' \to A'$, then we have ${}^a(\phi' \circ \phi) = {}^a\phi \circ {}^a\phi'$; this result, with Corollary (1.2.3), says that $\operatorname{Spec}(A)$ is a *contravariant functor* in A, from the category of rings to that of topological spaces.

Corollary (1.2.4). — *Suppose that ϕ is such that every $f \in A$ can be written as $f = h\phi(f')$, where h is invertible in A (which, in particular, is the case when ϕ is surjective). Then ${}^a\phi$ is a homeomorphism from X to ${}^a\phi(X)$.* I | 84

PROOF. We show that for each subset $E \subset A$, there exists a subset E' of A' such that $V(E) = V(\phi(E'))$; according to the (T_0) axiom (1.1.8) and the formula (1.2.2.1), this implies first of all that ${}^a\phi$ is injective, and then, by (1.2.2.1), that ${}^a\phi$ is a homeomorphism. But it suffices, for each $f \in E$, to take $f' \in A'$ such that $h\phi(f') = f$ with h invertible in A; the set E' of these elements f' is exactly what we are searching for. ∎

(1.2.5). In particular, when ϕ is the canonical homomorphism from A to a ring quotient $A/\mathfrak{a}$, we again get (1.1.12), and ${}^a\phi$ is the *canonical injection* of $V(\mathfrak{a})$, identified with $\operatorname{Spec}(A/\mathfrak{a})$, into $X = \operatorname{Spec}(A)$.

Another particular case of (1.2.4):

Corollary (1.2.6). — *If S is a multiplicative subset of A, the spectrum $\operatorname{Spec}(S^{-1}A)$ is canonically identified (with its topology) with the subspace of $X = \operatorname{Spec}(A)$ consisting of the x such that $\mathfrak{j}_x \cap S = \varnothing$.*

PROOF. We know by (0, 1.2.6) that the prime ideals of $S^{-1}A$ are the ideals $S^{-1}\mathfrak{j}_x$ such that $\mathfrak{j}_x \cap S = \varnothing$, and that we have $\mathfrak{j}_x = (i_A^S)^{-1}(S^{-1}\mathfrak{j}_x)$. It then suffices to apply Corollary (1.2.4) to the i_A^S. □

Corollary (1.2.7). — *For ${}^a\phi(X)$ to be dense in X', it is necessary and sufficient for each element of the kernel $\operatorname{Ker}\phi$ to be nilpotent.*

PROOF. Applying Equation (1.2.2.3) to the ideal $\mathfrak{a} = (0)$, we have $\widetilde{{}^a\phi(X)} = V(\operatorname{Ker}\phi)$, and for $V(\operatorname{Ker}\phi) = X'$ to hold, it is necessary and sufficient for $\operatorname{Ker}\phi$ to be contained in all the prime ideals of A', or, equivalently, in the nilradical $\mathfrak{r}'$ of A'. □ **Err$_{\text{II}}$** **Err$_{\text{II}}$**

1.3. Sheaf associated to a module

(1.3.1). Let A be a commutative ring, M an A-module, f an element of A, and S_f the multiplicative set consisting of the f^n, where $n \geqslant 0$. Recall that we set $A_f = S_f^{-1}A$, $M_f = S_f^{-1}M$. If S'_f is the saturated multiplicative subset of A consisting of the $g \in A$ which divide an element of S_f, we know that A_f and M_f are canonically identified with ${S'_f}^{-1}A$ and ${S'_f}^{-1}M$ (**0**, 1.4.3).

Lemma (1.3.2). — *The following conditions are equivalent:*

(a) $g \in S'_f$;
(b) $S'_g \subset S'_f$;
(c) $f \in \mathfrak{r}(g)$;
(d) $\mathfrak{r}(f) \subset \mathfrak{r}(g)$;
(e) $V(g) \subset V(f)$;
(f) $D(f) \subset D(g)$.

PROOF. This follows immediately from the definitions and (1.1.5). □

(1.3.3). If $D(f) = D(g)$, then Lemma (1.3.2, b) shows that $M_f = M_g$. More generally, if $D(f) \supset D(g)$, then $S'_f \subset S'_g$, and we know (**0**, 1.4.1) that there exists a canonical functorial homomorphism

$$\rho_{g,f} : M_f \longrightarrow M_g,$$

and if $D(f) \supset D(g) \supset D(h)$, we have (**0**, 1.4.4)

$$\rho_{h,g} \circ \rho_{g,f} = \rho_{h,f}. \tag{1.3.3.1}$$

When f ranges over the elements of $A - \mathfrak{j}_x$ (for a given x in $X = \operatorname{Spec}(A)$), the sets S'_f constitute an increasing filtered set of subsets of $A - \mathfrak{j}_x$, since for elements f and g of $A - \mathfrak{j}_x$, S'_f and S'_g are contained in S'_{fg}; since the union of the S'_f over $f \in A - \mathfrak{j}_x$ is $A - \mathfrak{j}_x$, we conclude (**0**, 1.4.5) that the A_x-module M_x is canonically identified with the *inductive limit* $\varinjlim M_f$, relative to the family of homomorphisms $(\rho_{g,f})$. We denote by **I | 85**

$$\rho_x^f : M_f \longrightarrow M_x$$

the canonical homomorphism for $f \in A - \mathfrak{j}_x$ (or, equivalently, $x \in D(f)$).

Definition (1.3.4). — We define the structure sheaf of the prime spectrum $X = \operatorname{Spec}(A)$ (resp. the sheaf associated to the A-module M), denoted by $\widetilde{A}$ or $\mathscr{O}_X$ (resp. $\widetilde{M}$) as the sheaf of rings (resp. the $\widetilde{A}$-module) associated to the presheaf $D(f) \mapsto A_f$ (resp. $D(f) \mapsto M_f$), defined on the basis $\mathfrak{B}$ of X consisting of the $D(f)$ for $f \in A$ ((1.1.10), (**0**, 3.2.1), and (**0**, 3.5.6)).

We saw (**0**, 3.2.4) that the stalk $\widetilde{A}_x$ (resp. $\widetilde{M}_x$) *can be identified with the ring* A_x (resp. *the* A_x*-module* M_x); we denote by

$$\theta_f : A_f \longrightarrow \Gamma(D(f), \widetilde{A})$$

$$(\text{resp. } \theta_f : M_f \longrightarrow \Gamma(D(f), \widetilde{M})),$$

the canonical map, so that, for all $x \in D(f)$ and all $\xi \in M_f$, we have

$$(\theta_f(\xi))_x = \rho_x^f(\xi). \tag{1.3.4.1}$$

Proposition (1.3.5). — *$\widetilde{M}$ is an exact functor, covariant in M, from the category of A-modules to the category of $\widetilde{A}$-modules.*

Proof. Indeed, let M, N be two A-modules, and u a homomorphism $M \to N$; for each $f \in A$, u corresponds canonically to a homomorphism u_f from the A_f-module M_f to the A_f-module N_f, and the diagram (for $D(g) \subset D(f)$)

$$\begin{tikzcd} M_f \arrow[r, "u_f"] \arrow[d, "\rho_{g,f}"'] & N_f \arrow[d, "\rho_{g,f}"] \\ M_g \arrow[r, "u_g"] & N_g \end{tikzcd}$$

is commutative (1.4.1); these homomorphisms then define a homomorphism of $\widetilde{A}$-modules $\widetilde{u} : \widetilde{M} \to \widetilde{N}$ (**0**, 3.2.3). In addition, for each $x \in X$, $\widetilde{u}_x$ is the inductive limit of the u_f for $x \in D(f)$ ($f \in A$), and as a result (**0**, 1.4.5), if we canonically identify $\widetilde{M}_x$ and $\widetilde{N}_x$ with M_x and N_x respectively, then $\widetilde{u}_x$ is identified with the homomorphism u_x canonically induced by u. If P is a third A-module, v a homomorphism $N \to P$, and $w = v \circ u$, it is immediate that $w_x = v_x \circ u_x$, so $\widetilde{w} = \widetilde{v} \circ \widetilde{u}$. We have therefore clearly defined a *covariant (in M) functor* $\widetilde{M}$, from the category of A-modules to that of $\widetilde{A}$-modules. *This functor is exact*, since, for each $x \in X$, M_x is an exact functor in M (**0**, 1.3.2); in addition, we have $\operatorname{Supp}(M) = \operatorname{Supp}(\widetilde{M})$, by definition ((**0**, 1.7.1) and (**0**, 3.1.6)). □

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Proposition (1.3.6). — *For each $f \in A$, the open subset $D(f) \subset X$ is canonically identified with the prime spectrum* $\operatorname{Spec}(A_f)$, *and the sheaf $\widetilde{M_f}$ associated to the A_f-module M_f is canonically identified with the restriction $\widetilde{M}|D(f)$.*

Proof. The first assertion is a particular case of (1.2.6). In addition, if $g \in A$ is such that $D(g) \subset D(f)$, then M_g is canonically identified with the module of fractions of M_f whose denominators are the powers of the canonical image of g in A_f (**0**, 1.4.6). The canonical identification of $\widetilde{M_f}$ with $\widetilde{M}|D(f)$ then follows from the definitions. □

Theorem (1.3.7). — *For each A-module M and each $f \in A$, the homomorphism*

$$\theta_f : M_f \longrightarrow \Gamma(D(f), \widetilde{M})$$

is bijective (in other words, the presheaf $D(f) \mapsto M_f$ is a *sheaf*). *In particular, M can be identified with $\Gamma(X, \widetilde{M})$ via θ_1.*

Proof. We note that, if $M = A$, then θ_f is a homomorphism of structure rings; Theorem (1.3.7) then implies that, if we identify the rings A_f and $\Gamma(D(f), \widetilde{A})$ via θ_f, the homomorphism $\theta_f : M_f \to \Gamma(D(f), \widetilde{M})$ is an isomorphism of *modules*.

We show first that θ_f is *injective*. Indeed, if $\xi \in M_f$ is such that $\theta_f(\xi) = 0$, then, for each prime ideal $\mathfrak{p}$ of A_f, there exists $h \notin \mathfrak{p}$ such that $h\xi = 0$; as the annihilator of ξ is not contained in any prime ideal of A_f, each A_f integral, and so $\xi = 0$.

It remains to show that θ_f is *surjective*; we can restrict to the case where $f = 1$ (the general case then following by "localizing", using (1.3.6)). Now let s be a section of $\widetilde{M}$ over X; according to (1.3.4) and (1.1.10, ii), there exists a *finite* cover $(D(f_i))_{i \in I}$ of X ($f_i \in A$) such that, for each $i \in I$, the restriction $s_i = s|D(f_i)$ is of the form $\theta_{f_i}(\xi_i)$, where $\xi_i \in M_{f_i}$. If i, j are indices of I, and if the restrictions of s_i and s_j to $D(f_i) \cap D(f_j) = D(f_i f_j)$ are equal, then it follows, by definition of M, that

$$\rho_{f_i f_j, f_i}(\xi_i) = \rho_{f_i f_j, f_j}(\xi_j). \tag{1.3.7.1}$$

By definition, we can write, for each $i \in I$, $\xi_i = z_i/f_i^{n_i}$, where $z_i \in M$, and. since I is finite, by multiplying each z_i by a power of f_i, we can assume that all the n_i are equal to one single n. Then, by definition, (1.3.7.1) implies that there exists an integer $m_{ij} \geqslant 0$ such that $(f_i f_j)^{m_{ij}}(f_j^n z_i - f_i^n z_j) = 0$, and we can moreover suppose that the m_{ij} are equal to the one single m; then replacing the z_i by $f_i^m z_i$, it remains to prove the case where $m = 0$, in other words, the case where we have

$$f_j^n z_i = f_i^n z_j \tag{1.3.7.2}$$

for any i, j. We have $D(f_i^n) = D(f_i)$, and since the $D(f_i)$ form a cover of X, the ideal generated by the f_i^n is A; in other words, there exist elements $g_i \in A$ such that $\sum_i g_i f_i^n = 1$. Then consider the element $z = \sum_i g_i z_i$ of M; in (1.3.7.2), we have $f_i^n z = \sum_j g_j f_i^n z_j = (\sum_j g_j f_j^n) z_i = z_i$, where, by definition, $\xi_i = z/1$ in M_{f_i}. We conclude that the s_i are the restrictions to $D(f_i)$ of $\theta_1(z)$, which **I** | 87
proves that $s = \theta_1(z)$ and finishes the proof. □

Corollary (1.3.8). — *Let M and N be A-modules; the canonical homomorphism $u \mapsto \widetilde{u}$ from* $\operatorname{Hom}_A(M,N)$ *to* $\operatorname{Hom}_{\widetilde{A}}(\widetilde{M},\widetilde{N})$ *is bijective. In particular, the equations $M = 0$ and $\widetilde{M} = 0$ are equivalent.*

Proof. Consider the canonical homomorphism $v \mapsto \Gamma(v)$ from $\operatorname{Hom}_{\widetilde{A}}(\widetilde{M},\widetilde{N})$ to $\operatorname{Hom}_{\Gamma(\widetilde{A})}(\Gamma(\widetilde{M}),\Gamma(\widetilde{N}))$; the latter module is canonically identified with $\operatorname{Hom}_A(M,N)$, by Theorem (1.3.7). It remains to show that $u \mapsto \widetilde{u}$ and $v \mapsto \Gamma(v)$ are inverses of each other; it is evident that $\Gamma(\widetilde{u}) = u$ by definition of $\widetilde{u}$; on the other hand, if we let $u = \Gamma(v)$ for $v \in \operatorname{Hom}_{\widetilde{A}}(\widetilde{M},\widetilde{N})$, then the map $w : \Gamma(D(f),\widetilde{M}) \to \Gamma(D(f),\widetilde{N})$ canonically induced from v is such that the diagram

$$\begin{tikzcd} M \arrow[r, "u"] \arrow[d, "\rho_{f,1}"'] & N \arrow[d, "\rho_{f,1}"] \\ M_f \arrow[r, "w"] & N_f \end{tikzcd}$$

is commutative; so we necessarily have that $w = u_f$ for all $f \in A$ (**0**, 1.2.4), which shows that $\widetilde{\Gamma(v)} = v$. □

Corollary (1.3.9). —

(i) *Let u be a homomorphism from an A-module M to an A-module N; then the sheaves associated to* $\operatorname{Ker} u$, $\operatorname{Im} u$, *and* $\operatorname{Coker} u$, *are* $\operatorname{Ker}\widetilde{u}$, $\operatorname{Im}\widetilde{u}$, *and* $\operatorname{Coker}\widetilde{u}$ *(respectively). In particular, for $\widetilde{u}$ to be injective (resp. surjective, bijective), it is necessary and sufficient for u to be so too.*

(ii) *If M is an inductive limit (resp. direct sum) of a family of A-modules (M_λ), then $\widetilde{M}$ is the inductive limit (resp. direct sum) of the family $(\widetilde{M_\lambda})$, via a canonical isomorphism.*

Proof.

(i) It suffices to apply the fact that $\widetilde{M}$ is an exact functor in M (1.3.5) to the two exact sequences of A-modules

$$0 \longrightarrow \operatorname{Ker} u \longrightarrow M \longrightarrow \operatorname{Im} u \longrightarrow 0,$$
$$0 \longrightarrow \operatorname{Im} u \longrightarrow N \longrightarrow \operatorname{Coker} u \longrightarrow 0.$$

The second claim then follows from Theorem (1.3.7).

(ii) Let $(M_\lambda, g_{\mu\lambda})$ be an inductive system of A-modules, with inductive limit M, and let g_λ be the canonical homomorphism $M_\lambda \to M$. Since we have $\widetilde{g_{\nu\mu}} \circ \widetilde{g_{\mu\lambda}} = \widetilde{g_{\nu\lambda}}$ and $\widetilde{g_\lambda} = \widetilde{g_\mu} \circ \widetilde{g_{\mu\lambda}}$ for $\lambda \leqslant \mu \leqslant \nu$, it follows that $(\widetilde{M_\lambda}, \widetilde{g_{\mu\lambda}})$ is an inductive system of sheaves on X, and if we denote by h_λ the canonical homomorphism $\widetilde{M_\lambda} \to \varinjlim \widetilde{M_\lambda}$, then there is a unique homomorphism $v : \varinjlim \widetilde{M_\lambda} \to \widetilde{M}$ such that $v \circ h_\lambda = \widetilde{g_\lambda}$. To see that v is bijective, it suffices to check, for each $x \in X$, that v_x is a bijection from $(\varinjlim \widetilde{M_\lambda})_x$ to $\widetilde{M}_x$; but $\widetilde{M}_x = M_x$, and

$$(\varinjlim \widetilde{M_\lambda})_x = \varinjlim (\widetilde{M_\lambda})_x = \varinjlim (M_\lambda)_x = M_x \quad (\mathbf{0},\ 1.3.3).$$

Conversely, it follows from the definitions that $(\widetilde{g_\lambda})_x$ and (h_λ) are both equal to the canonical map from $(M_\lambda)_x$ to M_x; since $(\widetilde{g_\lambda})_x = v_x \circ (h_\lambda)_x$, v_x is the identity. Finally, if M is the direct sum of two A-modules N and P, it is immediate that $\widetilde{M} = \widetilde{N} \oplus \widetilde{P}$; each direct sum being the inductive limit of finite direct sums, the claims of (ii) are thus proved. I | 88

□

We note that Corollary (1.3.8) proves that the sheaves isomorphic to the associated sheaves of A-modules form an *abelian category* (T, I, 1.4).

We also note that Corollary (1.3.9) implies that, if M is an A-module *of finite type* (that is to say, there exists a surjective homomorphism $A^n \to M$) then there exists a surjective homomorphism $\widetilde{A^n} \to \widetilde{M}$, or, in other words, the $\widetilde{A}$-module $\widetilde{M}$ is *generated by a finite family of sections over* X (**0**, 5.1.1), and vice versa.

(1.3.10). If N is a submodule of an A-module M, the canonical injection $j : N \to M$ gives, by (1.3.9), an injective homomorphism $\widetilde{N} \to \widetilde{M}$, which allows us to canonically identify $\widetilde{N}$ with an $\widetilde{A}$-*submodule* of $\widetilde{M}$; we will always assume that we have made this identification. If N and P are submodules of M, then we have

(1.3.10.1) $$(N+P)^\sim = \widetilde{N} + \widetilde{P},$$

(1.3.10.2) $$(N\cap P)^\sim = \widetilde{N} \cap \widetilde{P},$$

since $N+P$ and $N \cap P$ are the image of the canonical homomorphism $N \oplus P \to M$ and the kernel of the canonical homomorphism $M \to (M/N) \oplus (M/P)$ (respectively), and it suffices to apply (1.3.9).

We conclude from (1.3.10.1) and (1.3.10.2) that, if $\widetilde{N} = \widetilde{P}$, then we have $N = P$.

Corollary (1.3.11). — *On the category of sheaves isomorphic to the associated sheaves of A-modules, the functor Γ is exact.*

Proof. Let $\widetilde{M} \xrightarrow{\widetilde{u}} \widetilde{N} \xrightarrow{\widetilde{v}} \widetilde{P}$ be an exact sequence corresponding to two homomorphisms $u : M \to N$ and $v : N \to P$ of A-modules. If $Q = \operatorname{Im} u$ and $R = \operatorname{Ker} v$, we have $\widetilde{Q} = \operatorname{Im} \widetilde{u} = \operatorname{Ker} \widetilde{v} = \widetilde{R}$ (Corollary (1.3.9)), hence $Q = R$. □

Corollary (1.3.12). — *Let M and N be A-modules.*

(i) *The sheaf associated to $M \otimes_A N$ is canonically identified with $\widetilde{M} \otimes_{\widetilde{A}} \widetilde{N}$.*
(ii) *If, in addition, M admits a finite presentation, then the sheaf associated to* $\operatorname{Hom}_A(M,N)$ *is canonically identified with $\mathscr{H}om_{\widetilde{A}}(\widetilde{M}, \widetilde{N})$.*

Proof.

(i) The sheaf $\mathscr{F} = \widetilde{M} \otimes_{\widetilde{A}} \widetilde{N}$ is associated to the presheaf
$$U \longmapsto \mathscr{F}(U) = \Gamma(U, \widetilde{M}) \otimes_{\Gamma(U,\widetilde{A})} \Gamma(U, \widetilde{N}),$$
with U varying over the basis (1.1.10, i) of X consisting of the $D(f)$, where $f \in A$. We know that $\mathscr{F}(D(f))$ is canonically identified with $M_f \otimes_{A_f} N_f$, by (1.3.7) and (1.3.6). Moreover, we know that the A_f-module $M_f \otimes_{A_f} N_f$ is canonically isomorphic to $(M \otimes_A N)_f$ (**0**, 1.3.4), which is itself canonically isomorphic to $\Gamma(D(f), (M \otimes_A N)^\sim)$ (Theorem (1.3.7) and Proposition (1.3.6)). In addition, we see immediately that the canonical isomorphisms
$$\mathscr{F}(D(f)) \simeq \Gamma(D(f), (M \otimes_A N)^\sim)$$
thus obtained satisfy the compatibility conditions with respect to the restriction operations (**0**, 1.4.2), so they define a canonical functorial isomorphism I | 89
$$\widetilde{M} \otimes_{\widetilde{A}} \widetilde{N} \simeq (M \otimes_A N)^\sim.$$

(ii) The sheaf $\mathscr{G} = \mathscr{H}om_{\widetilde{A}}(\widetilde{M}, \widetilde{N})$ is associated to the presheaf
$$U \longmapsto \mathscr{G}(U) = \operatorname{Hom}_{\widetilde{A}|U}(\widetilde{M}|U, \widetilde{N}|U),$$
with U varying over the basis of X consisting of the $D(f)$. We know that $\mathscr{G}(D(f))$ is canonically identified with $\operatorname{Hom}_{A_f}(M_f, N_f)$ (Proposition (1.3.6) and Corollary (1.3.8)),

which, according to the hypotheses on M, is canonically identified with $(\mathrm{Hom}_A(M,N))_f$ (**0**, 1.3.5). Finally, $(\mathrm{Hom}_A(M,N))_f$ is canonically identified with $\Gamma(D(f),(\mathrm{Hom}_A(M,N))^{\sim})$ (Proposition (1.3.6) and Theorem (1.3.7)), and the canonical isomorphisms $\mathscr{G}(D(f)) \simeq \Gamma(D(f),(\mathrm{Hom}_A(M,N))^{\sim})$ thus obtained are compatible with the restriction operations (**0**, 1.4.2); they thus define a canonical isomorphism $\mathscr{H}om_{\widetilde{A}}(\widetilde{M},\widetilde{N}) \simeq (\mathrm{Hom}_A(M,N))^{\sim}$.

□

(1.3.13). Now let B be a (commutative) A-algebra; this can be understood by saying that B is an A-module such that we have some given element $e \in B$ and an A-homomorphism $\phi : B \otimes_A B \to B$, so that *(a)* the diagrams

$$\begin{tikzcd} B \otimes_A B \otimes_A B \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes\phi"'] & B \otimes_A B \arrow[d, "\phi"] \\ B \otimes_A B \arrow[r, "\phi"] & B \end{tikzcd} \qquad \begin{tikzcd} B \otimes_A B \arrow[rr, "\sigma"] \arrow[dr, "\phi"'] & & B \otimes_A B \arrow[dl, "\phi"] \\ & B & \end{tikzcd}$$

(σ being the canonical symmetry map) are commutative; and *(b)* $\phi(e \otimes x) = \phi(x \otimes e) = x$. By Corollary (1.3.12), the homomorphism $\widetilde{\phi} : \widetilde{B} \otimes_{\widetilde{A}} \widetilde{B} \to \widetilde{B}$ of $\widetilde{A}$-modules satisfies the analogous conditions, and so it defines an $\widetilde{A}$-*algebra* structure on $\widetilde{B}$. In a similar way, the data of a B-module N is the same as the data of an A-module N and an A-homomorphism $\psi : B \otimes_A N \to N$ such that the diagram

$$\begin{tikzcd} B \otimes_A B \otimes_A N \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes\psi"'] & B \otimes_A N \arrow[d, "\psi"] \\ B \otimes_A N \arrow[r, "\psi"] & N \end{tikzcd}$$

is commutative and $\psi(e \otimes n) = n$; the homomorphism $\widetilde{\psi} : \widetilde{B} \otimes_{\widetilde{A}} \widetilde{N} \to \widetilde{N}$ satisfies the analogous condition, and so defines a $\widetilde{B}$-*module* structure on $\widetilde{N}$.

In a similar way, we see that if $u : B \to B'$ (resp. $v : N \to N'$) is a homomorphism of A-algebras (resp. of B-modules), then $\widetilde{u}$ (resp. $\widetilde{v}$) is a homomorphism of $\widetilde{A}$-algebras (resp. of $\widetilde{B}$-modules), and $\operatorname{Ker}\widetilde{u}$ is a $\widetilde{B}$-ideal (resp. $\operatorname{Ker}\widetilde{v}$, $\operatorname{Coker}\widetilde{v}$, and $\operatorname{Im}\widetilde{v}$ are $\widetilde{B}$-modules). If N is a B-module, then $\widetilde{N}$ is a $\widetilde{B}$-module of finite type if and only if N is a B-module of finite type (**0**, 5.2.3).

If M, N are B-modules, then the $\widetilde{B}$-module $\widetilde{M} \otimes_{\widetilde{B}} \widetilde{N}$ is canonically identified with $(M \otimes_B N)^{\sim}$; **I** | 90
similarly $\mathscr{H}om_{\widetilde{B}}(\widetilde{M},\widetilde{N})$ is canonically identified with $(\mathrm{Hom}_B(M,N))^{\sim}$ whenever M admits a finite presentation; the proofs are similar to those for Corollary (1.3.12).

If $\mathfrak{J}$ is an ideal of B, and N is a B-module, then we have $(\mathfrak{J}N)^{\sim} = \widetilde{\mathfrak{J}} \cdot \widetilde{N}$.

Finally, if B is an A-algebra *graded* by the A-submodules B_n ($n \in \mathbf{Z}$), then the $\widetilde{A}$-algebra $\widetilde{B}$, the direct sum of the $\widetilde{A}$-modules $\widetilde{B_n}$ (1.3.9), is graded by these $\widetilde{A}$-submodules, the axiom of graded algebras saying that the image of the homomorphism $B_m \otimes B_n \to B$ is contained in B_{m+n}. Similarly, if M is a B-module graded by the submodules M_n, then $\widetilde{M}$ is a $\widetilde{B}$-module graded by the $\widetilde{M_n}$.

(1.3.14). If B is an A-algebra, and M a submodule of B, then the $\widetilde{A}$-subalgebra of $\widetilde{B}$ generated by $\widetilde{M}$ (**0**, 4.1.3) is the $\widetilde{A}$-subalgebra $\widetilde{C}$, where we denote by C the subalgebra of B generated by M. Indeed, C is the direct sum of the submodules of B which are the images of the homomorphisms $\bigotimes^n M \to B$ ($n \geqslant 0$), so it suffices to apply (1.3.9) and (1.3.12).

1.4. Quasi-coherent sheaves over a prime spectrum

Theorem (1.4.1). — *Let X be the prime spectrum of a ring A, V a quasi-compact open subset of X, and $\mathscr{F}$ an $(\mathscr{O}_X|V)$-module. The following four conditions are equivalent.*

(a) *There exists an A-module M such that $\mathscr{F}$ is isomorphic to $\widetilde{M}|V$.*
(b) *There exists a finite open cover (V_i) of V by sets of the form $D(f_i)$ ($f_i \in A$) contained in V, such that, for each i, $\mathscr{F}|V_i$ is isomorphic to a sheaf of the form $\widetilde{M}_i$, where M_i is an A_{f_i}-module.*
(c) *The sheaf $\mathscr{F}$ is quasi-coherent* (**0**, 5.1.3).
(d) *The two following properties are satisfied:*
  (d1) *For each $f \in A$ such that $D(f) \subset V$, and for each section $s \in \Gamma(D(f), \mathscr{F})$, there exists an integer $n \geqslant 0$ such that $f^n s$ extends to a section of $\mathscr{F}$ over V.*
  (d2) *For each $f \in A$ such that $D(f) \subset V$ and for each section $t \in \Gamma(V, \mathscr{F})$ such that the restriction of t to $D(f)$ is 0, there exists an integer $n \geqslant 0$ such that $f^n t = 0$.*

(In the statement of the conditions (d1) and (d2), we have tacitly identified A and $\Gamma(\widetilde{A})$ using Theorem (1.3.7)).

PROOF. The fact that (a) implies (b) is an immediate consequence of Proposition (1.3.6) and the fact that the $D(f_i)$ form a basis for the topology of X (1.1.10). As each A-module is isomorphic to the cokernel of a homomorphism of the form $A^{(I)} \to A^{(J)}$, (1.3.6) implies that each sheaf associated to an A-module is quasi-coherent; so (b) implies (c). Conversely, if $\mathscr{F}$ is quasi-coherent, each $x \in V$ has a neighbourhood of the form $D(f) \subset V$ such that $\mathscr{F}|D(f)$ is isomorphic to the cokernel of a homomorphism $\widetilde{A_f}^{(I)} \to \widetilde{A_f}^{(J)}$, so also to the sheaf $\widetilde{N}$ associated to the module N, the cokernel of the corresponding homomorphism $A_f^{(I)} \to A_f^{(J)}$ (Corollaries (1.3.8) and (1.3.9)); since V is quasi-compact, it is clear that (c) implies (b).

To prove that (b) implies (d1) and (d2), we first assume that $V = D(g)$ for some $g \in A$, and that $\mathscr{F}$ is isomorphic to the sheaf $\widetilde{N}$ associated to an A_g-module N; by replacing X with V and A with A_g (1.3.6), we can reduce to the case where $g = 1$. Then $\Gamma(D(f), \widetilde{N})$ and N_f are canonically identified with one another (Proposition (1.3.6) and Theorem (1.3.7)), so a section $s \in \Gamma(D(f), \widetilde{N})$ is identified with an element of the form z/f^n, where $z \in N$; the section $f^n s$ is identified with the element $z/1$ of N_f and, as a result, is the restriction to $D(f)$ of the section of $\widetilde{N}$ over X that is identified with the element $z \in N$; hence (d1) in this case. Similarly, $t \in \Gamma(X, \widetilde{N})$ is identified with an element $z' \in N$, the restriction of t to $D(f)$ is identified with the image $z'/1$ of z' in N_f, and to say that this image is zero means that there exists some $n \geqslant 0$ such that $f^n z' = 0$ in N, or, equivalently, $f^n t = 0$. **I** | 91

To finish the proof, that (b) implies (d1) and (d2), it suffices to establish the following lemma.

Lemma (1.4.1.1). — *Suppose that V is the finite union of sets of the form $D(g_i)$, and that all of the sheaves $\mathscr{F}|D(g_i)$ and $\mathscr{F}|(D(g_i) \cap D(g_j)) = \mathscr{F}|D(g_i g_j)$ satisfy* (d1) *and* (d2)*; then $\mathscr{F}$ has the following two properties:*

(d′1) *For each $f \in A$ and for each section $s \in \Gamma(D(f) \cap V, \mathscr{F})$, there exists an integer $n \geqslant 0$ such that $f^n s$ extends to a section of $\mathscr{F}$ over V.*
(d′2) *For each $f \in A$ and for each section $t \in \Gamma(V, \mathscr{F})$ such that the restriction of t to $D(f) \cap V$ is 0, there exists an integer $n \geqslant 0$ such that $f^n t = 0$.*

We first prove (d′2): since $D(f) \cap D(g_i) = D(f g_i)$, there exists, for each i, an integer n_i such that the restriction of $(f g_i)^{n_i} t$ to $D(g_i)$ is zero: since the image of g_i in A_{g_i} is invertible, the restriction of $f^{n_i} t$ to $D(g_i)$ is also zero; taking n to be the largest of the n_i, we have proved (d′2).

To show (d′1), we apply (d1) to the sheaf $\mathscr{F}|D(g_i)$: there exists an integer $n_i \geqslant 0$ and a section s'_i of $\mathscr{F}$ over $D(g_i)$ extending the restriction of $(f g_i)^{n_i} s$ to $D(f g_i)$; since the image of g_i in A_{g_i} is invertible, there is a section s_i of $\mathscr{F}$ over $D(g_i)$ such that $s'_i = g_i^{n_i} s_i$, and s_i extends the restriction of $f^{n_i} s$ to $D(f g_i)$; in addition we can suppose that all the n_i are equal to a single integer n. By construction, the restriction of $s_i - s_j$ to $D(f) \cap D(g_i) \cap D(g_j) = D(f g_i g_j)$ is zero; by (d2) applied to the sheaf $\mathscr{F}|D(g_i g_j)$, there exists an integer $m_{ij} \geqslant 0$ such that the restriction to $D(g_i g_j)$ of $(f g_i g_j)^{m_{ij}}(s_i - s_j)$ is zero; since the image of $g_i g_j$ in $A_{g_i g_j}$ is invertible, the restriction of $f^{m_{ij}}(s_i - s_j)$ to $D(g_i g_j)$ is zero. We can then assume that all the m_{ij} are equal to a single integer m, and so there exists a section $s' \in \Gamma(V, \mathscr{F})$ extending the $f^m s_i$; as a result, this section extends $f^{n+m}s$, hence we have proved (d′1).

It remains to show that (d1) and (d2) imply (a). We first show that (d1) and (d2) imply that these conditions are satisfied for each sheaf $\mathscr{F}|D(g)$, where $g \in A$ is such that $D(g) \subset V$. It is evident for (d1); on the other hand, if $t \in \Gamma(D(g), \mathscr{F})$ is such that its restriction to $D(f) \subset D(g)$ is zero, there exists, by (d1), an integer $m \geqslant 0$ such that $g^m t$ extends to a section s of $\mathscr{F}$ over V; applying (d2), we see that there exists an integer $n \geqslant 0$ such that $f^n g^m t = 0$, and as the image of g in A_g is invertible, $f^n t = 0$. **I | 92**

That being so, since V is quasi-compact, Lemma (1.4.1.1) proves that the conditions (d′1) and (d′2) are satisfied. Consider then the A-module $M = \Gamma(V, \mathscr{F})$, and define a homomorphism of $\widetilde{A}$-modules $u : \widetilde{M} \to j_*(\mathscr{F})$, where j is the canonical injection $V \to X$. Since the $D(f)$ form a basis for the topology of X, it suffices, for each $f \in A$, to define a homomorphism $u_f : M_f \to \Gamma(D(f), j_*(\mathscr{F})) = \Gamma(D(f) \cap V, \mathscr{F})$, with the usual compatibility conditions (**0**, 3.2.5). Since the canonical image of f in A_f is invertible, the restriction homomorphism $M = \Gamma(V, \mathscr{F}) \to \Gamma(D(f) \cap V, \mathscr{F})$ factors as $M \to M_f \xrightarrow{u_f} \Gamma(D(f) \cap V, \mathscr{F})$ (**0**, 1.2.4), and the verification of these compatibility conditions for $D(g) \subset D(f)$ is immediate. This being so, we show that the condition (d′1) (resp. (d′2)) implies that each of the u_f are surjective (resp. injective), which proves that u is *bijective*, and as a result that $\mathscr{F}$ is the restriction to V of an $\widetilde{A}$-module isomorphic to $\widetilde{M}$. If $s \in \Gamma(D(f) \cap V, \mathscr{F})$, there exists, by (d′1), an integer $n \geqslant 0$ such that $f^n s$ extends to a section $z \in M$; we then have $u_f(z/f^n) = s$, so u_f is surjective. Similarly, if $z \in M$ is such that $u_f(z/1) = 0$, this means that the restriction to $D(f) \cap V$ of the section z is zero; according to (d′2), there exists an integer $n \geqslant 0$ such that $f^n z = 0$, hence $z/1 = 0$ in M_f, and so u_f is injective. □

Corollary (1.4.2). — *Each quasi-coherent sheaf over a quasi-compact open subset of X is induced by a quasi-coherent sheaf on X.*

Corollary (1.4.3). — *Every quasi-coherent $\mathscr{O}_X$-algebra over $X = \operatorname{Spec}(A)$ is isomorphic to an $\mathscr{O}_X$-algebra of the form $\widetilde{B}$, where B is an algebra over A; every quasi-coherent $\widetilde{B}$-module is isomorphic to a $\widetilde{B}$-module of the form $\widetilde{N}$, where N is a B-module.*

Proof. Indeed, a quasi-coherent $\mathscr{O}_X$-algebra is a quasi-coherent $\mathscr{O}_X$-module, and therefore of the form $\widetilde{B}$, where B is an A-module; the fact that B is an A-algebra follows from the characterization of the structure of an $\mathscr{O}_X$-algebra using the homomorphism $\widetilde{B} \otimes_{\widetilde{A}} \widetilde{B} \to \widetilde{B}$ of $\widetilde{A}$-modules, as well as Corollary (1.3.12). If $\mathscr{G}$ is a quasi-coherent $\widetilde{B}$-module, it suffices to show, in a similar way, that it is also a quasi-coherent $\widetilde{A}$-module to conclude the proof; since the question is local, we can, by restricting to an open subset of X of the form $D(f)$, assume that $\mathscr{G}$ is the cokernel of a homomorphism $\widetilde{B}^{(I)} \to \widetilde{B}^{(J)}$ of $\widetilde{B}$-modules (and *a fortiori* of $\widetilde{A}$-modules); the proposition then follows from Corollaries (1.3.8) and (1.3.9). □

1.5. Coherent sheaves over a prime spectrum

Theorem (1.5.1). — *Let A be a* Noetherian *ring, $X = \operatorname{Spec}(A)$ its prime spectrum, V an open subset of X, and $\mathscr{F}$ an $(\mathscr{O}_X|V)$-module. The following conditions are equivalent.*

(a) *$\mathscr{F}$ is coherent.*
(b) *$\mathscr{F}$ is of finite type and quasi-coherent.*
(c) *There exists an A-module M of finite type such that $\mathscr{F}$ is isomorphic to the sheaf $\widetilde{M}|V$.*

Proof. (a) trivially implies (b). To see that (b) implies (c), note that, since V is quasi-compact (**0**, 2.2.3), we have previously seen that $\mathscr{F}$ is isomorphic to a sheaf $\widetilde{N}|V$, where N is an A-module (1.4.1). We have $N = \varinjlim M_\lambda$, where M_λ run over the set of A-submodules of N of finite type, hence (1.3.9) $\mathscr{F} = \widetilde{N}|V = \varinjlim \widetilde{M_\lambda}|V$; but since $\mathscr{F}$ is of finite type, and V is quasi-compact, there exists an index λ such that $\mathscr{F} = \widetilde{M_\lambda}|V$ (**0**, 5.2.3). **I | 93**

Finally, we show that (c) implies (a). It is clear that $\mathscr{F}$ is then of finite type ((1.3.6) and (1.3.9)); in addition, since the questions is local, we can restrict to the case where $V = D(f)$, $f \in A$. Since A_f is Noetherian, we see that it suffices to prove that the kernel of a homomorphism $\widetilde{A^n} \to \widetilde{M}$, where M is an A-module, is of finite type. But such a homomorphism is of the form $\widetilde{u}$, where u is a homomorphism $A^n \to M$ (1.3.8), and if $P = \operatorname{Ker} u$ then we have $\widetilde{P} = \operatorname{Ker} \widetilde{u}$ (1.3.9). Since A is Noetherian, P is of finite type, which finishes the proof. ☐

Corollary (1.5.2). — *Under the hypotheses of (1.5.1), the sheaf $\mathscr{O}_X$ is a quasi-coherent sheaf of rings.*

Corollary (1.5.3). — *Under the hypotheses of (1.5.1), every coherent sheaf over an open subset of X is induced by a coherent sheaf on X.*

Corollary (1.5.4). — *Under the hypotheses of (1.5.1), every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is the inductive limit of the coherent $\mathscr{O}_X$-submodules of $\mathscr{F}$.*

Proof. Indeed, $\mathscr{F} = \widetilde{M}$, where M is an A-module, and M is the inductive limit of its submodules of finite type; we conclude the proof by appealing to (1.3.9) and (1.5.1). ☐

1.6. Functorial properties of quasi-coherent sheaves over a prime spectrum

(1.6.1). Let A, A' be rings,
$$\phi : A' \longrightarrow A$$
a homomorphism, and
$$^a\phi : X = \operatorname{Spec}(A) \longrightarrow X' = \operatorname{Spec}(A')$$
the continuous map associated to ϕ (1.2.1). We will define a *canonical homomorphism*
$$\widetilde{\phi} : \mathscr{O}_{X'} \longrightarrow {}^a\phi_*(\mathscr{O}_X)$$
of sheaves of rings. For each $f' \in A'$, we put $f = \phi(f')$; we have $^a\phi^{-1}(D(f')) = D(f)$ (1.2.2.2). The rings $\Gamma(D(f'), \widetilde{A'})$ and $\Gamma(D(f), \widetilde{A})$ are identified with $A'_{f'}$ and A_f (respectively) ((1.3.6) and (1.3.7)). The homomorphism ϕ canonically defines a homomorphism $\phi_{f'} : A'_{f'} \to A_f$ (**0**, 1.5.1), in other words, we have a homomorphism of rings
$$\Gamma(D(f), \widetilde{A'}) \longrightarrow \Gamma({}^a\phi^{-1}(D(f')), \widetilde{A}) = \Gamma(D(f'), {}^a\phi_*(\widetilde{A})).$$
In addition, these homomorphisms satisfy the usual compatibility conditions: for $D(f') \supset D(g')$, **I** | 94
the diagram
$$\begin{tikzcd}
\Gamma(D(f'), \widetilde{A'}) \arrow[r] \arrow[d] & \Gamma(D(f'), {}^a\phi_*(\widetilde{A})) \arrow[d] \\
\Gamma(D(g'), \widetilde{A'}) \arrow[r] & \Gamma(D(g'), {}^a\phi_*(\widetilde{A})
\end{tikzcd}$$
is commutative (**0**, 1.5.1); we have thus defined a homomorphism of $\mathscr{O}_{X'}$-algebras, as the $D(f')$ form a basis for the topology of X' (**0**, 3.2.3). The pair $\Phi = ({}^a\phi, \widetilde{\phi})$ is thus a *morphism* of ringed spaces
$$\Phi : (X, \mathscr{O}_X) \longrightarrow (X', \mathscr{O}_{X'}),$$
(**0**, 4.1.1).

We also note that, if we put $x' = {}^a\phi(x)$, then the homomorphism $\widetilde{\phi}_x^\sharp$ (**0**, 3.7.1) is exactly the homomorphism
$$\phi_x : A'_{x'} \longrightarrow A_x$$
canonically induced by $\phi : A' \to A$ (**0**, 1.5.1). Indeed, each $z' \in A'_{x'}$ can be written as g'/f', where f', g' are in A' and $f' \notin \mathfrak{j}_{x'}$; $D(f')$ is then a neighbourhood of x' in X', and the homomorphism $\Gamma(D(f'), \widetilde{A'}) \to \Gamma({}^a\phi^{-1}(D(f')), \widetilde{A})$ induced by $\widetilde{\phi}$ is exactly $\phi_{f'}$; by considering the section $s' \in \Gamma(D(f'), \widetilde{A'})$ corresponding to $g'/f' \in A'_{f'}$, we obtain $\widetilde{\phi}_x^\sharp(z') = \phi(g')/\phi(f')$ in A_x.

Example (1.6.2). — Let S be a multiplicative subset of A, and ϕ the canonical homomorphism $A \to S^{-1}A$; we have already seen (1.2.6) that $^a\phi$ is a *homeomorphism* from $Y = \operatorname{Spec}(S^{-1}A)$ to the subspace of $X = \operatorname{Spec}(A)$ consisting of the x such that $\mathfrak{j}_x \cap S = \varnothing$. In addition, for each x in this subspace, which is thus of the form $^a\phi(y)$ with $y \in Y$, the homomorphism $\widetilde{\phi}_y^\sharp : \mathscr{O}_x \to \mathscr{O}_y$ is *bijective* (**0**, 1.2.6); in other words, $\mathscr{O}_Y$ is identified with the sheaf on Y induced by $\mathscr{O}_X$.

Proposition (1.6.3). — *For every A-module M, there exists a canonical functorial isomorphism from the $\mathscr{O}_{X'}$-module $(M_{[\phi]})^\sim$ to the direct image $\Phi_*(\widetilde{M})$.*

Proof. For purposes of abbreviation, we write $M' = M_{[\phi]}$, and for each $f' \in A'$, we put $f = \phi(f')$. The modules of sections $\Gamma(D(f'), \widetilde{M'})$ and $\Gamma(D(f), \widetilde{M})$ are identified, respectively, with the modules $M'_{f'}$ and M_f (over $A'_{f'}$ and A_f, respectively); in addition, the $A'_{f'}$-module $(M_f)_{[\phi_{f'}]}$ is canonically isomorphic to $M'_{f'}$ (0, 1.5.2). We thus have a functorial isomorphism of $\Gamma(D(f'), \widetilde{A'})$-modules: $\Gamma(D(f'), \widetilde{M'}) \simeq \Gamma({}^a\phi^{-1}(D(f')), \widetilde{M})_{[\phi_{f'}]}$ and these isomorphisms satisfy the usual compatibility conditions with the restrictions (0, 1.5.6), thus defining the desired functorial isomorphism. We note that, in a precise way, if $u : M_1 \to M_2$ is a homomorphism of A-modules, it can be considered as a homomorphism $(M_1)_{[\phi]} \to (M_2)_{[\phi]}$ of A'-modules; if we denote this homomorphism by $u_{[\phi]}$, then $\Phi_*(\widetilde{u})$ is identified with $(u_{[\phi]})^{\sim}$. □

This proof also shows that, for each A-*algebra* B, the canonical functorial isomorphism $(B_{[\phi]})^{\sim} \simeq \Phi_*(\widetilde{B})$ is an isomorphism of $\mathscr{O}_{X'}$-*algebras*; if M is a B-module, the canonical functorial isomorphism $(M_{[\phi]})^{\sim} \simeq \Phi_*(\widetilde{M})$ is an isomorphism of $\Phi_*(\widetilde{B})$-modules. I | 95

Corollary (1.6.4). — *The direct image functor Φ_* is exact on the category of quasi-coherent $\mathscr{O}_X$-modules.*

Proof. Indeed, it is clear that $M_{[\phi]}$ is an exact functor in M and $\widetilde{M'}$ is an exact functor in M' (1.3.5). □

Proposition (1.6.5). — *Let N' be an A'-module, and N the A-module $N' \otimes_{A'} A_{[\phi]}$; then there exists a canonical functorial isomorphism from the $\mathscr{O}_X$-module $\Phi^*(\widetilde{N'})$ to $\widetilde{N}$.*

Proof. We first remark that $j : z' \mapsto z' \otimes 1$ is an A'-homomorphism from N' to $N_{[\phi]}$: indeed, by definition, for $f' \in A'$, we have $(f'z') \otimes 1 = z' \otimes \phi(f') = \phi(f')(z' \otimes 1)$. We have (1.3.8) a homomorphism $\widetilde{j} : \widetilde{N'} \to (N_{[\phi]})^{\sim}$ of $\mathscr{O}_{X'}$-modules, and, thanks to (1.6.3), we can consider $\widetilde{j}$ as mapping $\widetilde{N'}$ to $\Phi_*(\widetilde{N})$. There canonically corresponds to this homomorphism $\widetilde{j}$ a homomorphism $h = \widetilde{j}^\sharp$ from $\Phi^*(\widetilde{N'})$ to $\widetilde{N}$ (0, 4.4.3); we will see that, for each stalk, h_x is *bijective*. Put $x' = {}^a\phi(x)$ and let $f' \in A'$ be such that $x' \in D(f')$; let $f = \phi(f')$. The ring $\Gamma(D(f), \widetilde{A})$ is identified with A_f, the modules $\Gamma(D(f), \widetilde{N})$ and $\Gamma(D(f'), \widetilde{N'})$ with N_f and $N'_{f'}$ (respectively); let $s \in \Gamma(D(f'), \widetilde{N'})$, identified with n'/f'^p ($n' \in N'$), and s be its image under $\widetilde{j}$ in $\Gamma(D(f), \widetilde{N})$; s is identified with $(n' \otimes 1)/f^p$. On the other hand, let $t \in \Gamma(D(f), \widetilde{A})$, identified with g/f^q ($g \in A$); then, by definition, we have $h_x(s'_x \otimes t_x) = t_x \cdot s_x$ (0, 4.4.3). But we can canonically identify N_f with $N'_{f'} \otimes_{A'_{f'}} (A_f)_{[\phi_{f'}]}$ (0, 1.5.4); s then corresponds to the element $(n'/f'^p) \otimes 1$, and the section $y \mapsto t_y \cdot s_y$ with $(n'/f'^p) \otimes (g/f^q)$. The compatibility diagram of (0, 1.5.6) show that h_x is exactly the canonical isomorphism

$$N'_{x'} \otimes_{A'_{x'}} (A_x)_{[\phi_{x'}]} \simeq N_x = (N' \otimes_{A'} A_{[\phi]})_x. \tag{1.6.5.1}$$

In addition, let $v : N'_1 \to N'_2$ be a homomorphism of A'-modules; since $\widetilde{v}_{x'} = v_{x'}$ for each $x' \in X'$, it follows immediately from the above that $\Phi^*(\widetilde{v})$ is canonically identified with $(v \otimes 1)^{\sim}$, which finishes the proof of (1.6.5). □

If B' is an A'-algebra, the canonical isomorphism from $\Phi^*(\widetilde{B'})$ to $(B' \otimes_{A'} A_{[\phi]})^{\sim}$ is an isomorphism of $\mathscr{O}_X$-algebras; if, in addition, N' is a B'-module, then the canonical isomorphism from $\Phi^*(\widetilde{N'})$ to $(N' \otimes_{A'} A_{[\phi]})^{\sim}$ is an isomorphism of $\Phi^*(\widetilde{B'})$-modules.

Corollary (1.6.6). — *The sections of $\Phi^*(\widetilde{N'})$, the canonical images of the sections s', where s' varies over the A'-module $\Gamma(\widetilde{N'})$, generate the A-module $\Gamma(\Phi^*(N'))$.*

Proof. Indeed, these images are identified with the elements $z' \otimes 1$ of N, when we identify N' and N with $\Gamma(\widetilde{N'})$ and $\Gamma(\widetilde{N})$ (respectively) (1.3.7), and z' varies over N'. □

(1.6.7). In the proof of (1.6.5), we had proved in passing that the canonical map (**0**, 4.4.3.2) $\rho : \widetilde{N'} \to \Phi_*(\Phi^*(\widetilde{N'}))$ is exactly the homomorphism $\widetilde{j}$, where $j : N' \to N' \otimes_{A'} A_{[\phi]}$ is the homomorphism I | 96
$z' \mapsto z' \otimes 1$. Similarly, the canonical map (**0**, 4.4.3.3) $\sigma : \Phi^*(\Phi_*(\widetilde{M})) \to \widetilde{M}$ is exactly $\widetilde{p}$, where $p : M_{[\phi]} \otimes_{A'} A_{[\phi]} \to M$ is the canonical homomorphism, which sends each tensor product $z \otimes a$ ($z \in M$, $a \in A$) to $a \cdot z$; this follows immediately from the definitions ((0, 3.7.1), (0, 4.4.3), and (1.3.7)).

We conclude ((**0**, 4.4.3) and (**0**, 3.5.4.4)) that if $v : N' \to M_{[\phi]}$ is an A'-homomorphism, then $\widetilde{v}^\sharp = (v \otimes 1)^\sim$.

(1.6.8). Let N'_1 and N'_2 be A'-modules, and assume N'_1 admits a *finite presentation*; it then follows from (1.6.7) and (1.3.12, ii) that the canonical homomorphism (**0**, 4.4.6)

$$\Phi^*(\mathscr{H}om_{\widetilde{A'}}(\widetilde{N'_1}, \widetilde{N'_2})) \longrightarrow \mathscr{H}om_{\widetilde{A}}(\Phi^*(\widetilde{N'_1}), \Phi^*(\widetilde{N'_2}))$$

is exactly $\widetilde{\gamma}$, where γ denotes the canonical homomorphism of A-modules $\operatorname{Hom}_{A'}(N'_1, N'_2) \otimes_{A'} A \to \operatorname{Hom}_A(N'_1 \otimes_{A'} A, N'_2 \otimes_{A'} A)$.

(1.6.9). Let $\mathfrak{J}'$ be an ideal of A', and M an A-module; since, by definition, $\widetilde{\mathfrak{J}}'\widetilde{M}$ is the image of the canonical homomorphism $\Phi^*(\widetilde{\mathfrak{J}'}) \otimes_{\widetilde{A}} \widetilde{M} \to \widetilde{M}$, it follows from Proposition (1.6.5) and Corollary (1.3.12, i) that $\widetilde{\mathfrak{J}}'\widetilde{M}$ canonically identifies with $(\mathfrak{J}'M)^\sim$; in particular, $\Phi^*(\widetilde{\mathfrak{J}'})\widetilde{A}$ is identified with $(\mathfrak{J}'A)^\sim$, and, taking the right exactness of the functor Φ^* into account, the $\widetilde{A}$-algebra $\Phi^*((A'/\mathfrak{J}')^\sim)$ is identified with $(A/\mathfrak{J}'A)^\sim$.

(1.6.10). Let A'' be a third ring, ϕ' a homomorphism $A'' \to A'$, and write $\phi'' = \phi \circ \phi'$. It follows immediately from the definitions that ${}^a\phi'' = ({}^a\phi') \circ ({}^a\phi)$, and $\widetilde{\phi''} = \widetilde{\phi} \circ \widetilde{\phi'}$ (**0**, 1.5.7). We conclude that $\Phi'' = \Phi' \circ \Phi$; in other words, $(\operatorname{Spec}(A), \widetilde{A})$ is a *functor* from the category of rings to that of ringed spaces.

1.7. Characterization of morphisms of affine schemes

Definition (1.7.1). — We say that a ringed space $(X, \mathscr{O}_X)$ is an *affine scheme* if it is isomorphic to a ringed space of the form $(\operatorname{Spec}(A), \widetilde{A})$, where A is a ring; we then say that $\Gamma(X, \mathscr{O}_X)$, which is canonically identified with the ring A (1.3.7), is the ring of the affine scheme $(X, \mathscr{O}_X)$, and we denote it by $A(X)$ when there is no chance of confusion.

By abuse of language, when we speak of an *affine scheme* $\operatorname{Spec}(A)$; it will always be the ringed space $(\operatorname{Spec}(A), \widetilde{A})$.

(1.7.2). Let A and B be rings, and $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ the affine schemes corresponding to the prime spectra $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$. We have seen (1.6.1) that each ring homomorphism $\phi : B \to A$ corresponds to a morphism $\Phi = ({}^a\phi, \widetilde{\phi}) = \operatorname{Spec}(\phi) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$. We note that ϕ is entirely determined by Φ, since we have, by definition, $\phi = \Gamma(\widetilde{\phi}) : \Gamma(\widetilde{B}) \to \Gamma({}^a\phi_*(\widetilde{A}) = \Gamma(\widetilde{A})$.

Theorem (1.7.3). — [2] *Let $(X, \mathscr{O}_X)$ $(Y, \mathscr{O}_Y)$ be affine schemes. For a morphism of ringed spaces $(\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ to be of the form $({}^a\phi, \widetilde{\phi})$, where ϕ is a homomorphism of rings $A(Y) \to A(X)$, it is necessary and sufficient that, for each $x \in X$, $\theta_x^\sharp$ is a local homomorphism: $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$.*

PROOF. Let $A = A(X)$, $B = A(Y)$. The condition is necessary, since we saw (1.6.1) that $\widetilde{\phi}_x^\sharp$ is the I | 97
homomorphism from $B_{{}^a\phi(x)}$ to A_x canonically induced by ϕ, and, by definition, of ${}^a\phi(x) = \phi^{-1}(\mathfrak{j}_x)$, this homomorphism is local.

We now prove that the condition is sufficient. By definition, θ is a homomorphism $\mathscr{O}_Y \to \psi_*(\mathscr{O}_X)$, and we canonically obtain a ring homomorphism

$$\phi = \Gamma(\theta) : B = \Gamma(Y, \mathscr{O}_Y) \longrightarrow \Gamma(Y, \psi_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X) = A.$$

The hypotheses on $\theta_x^\sharp$ mean that this homomorphism induces, by passing to quotients, a monomorphism θ^x from the residue field $k(\psi(x))$ to the residue field $k(x)$, such that, for each section $f \in \Gamma(Y, \mathscr{O}_Y) = B$, we have $\theta^x(f(\psi(x))) = \phi(f)(x)$. The relation $f(\psi(x)) = 0$ is therefore equivalent to $\phi(f)(x) = 0$, which means that $\mathfrak{j}_{\psi(x)} = \mathfrak{j}_{{}^a\phi(x)}$, and we now write $\psi(x) = {}^a\phi(x)$ for each $x \in X$, or $\psi = {}^a\phi$. We also know that the diagram

$$\begin{tikzcd}
B = \Gamma(Y, \mathscr{O}_Y) \arrow[r, "\phi"] \arrow[d] & \Gamma(X, \mathscr{O}_X) = A \arrow[d] \\
B_{\psi(x)} \arrow[r, "\theta_x^\sharp"] & A_x
\end{tikzcd}$$

[2] *[Trans.] See (1.8) and the footnote there.*

is commutative (**0**, 3.7.2), which means that $\theta_x^\sharp$ is equal to the homomorphism $\phi_x : B_{\psi(x)} \to A_x$ canonically induced by ϕ (**0**, 1.5.1). As the data of the $\theta_x^\sharp$ completely characterize $\theta^\sharp$, and as a result θ (**0**, 3.7.1), we conclude that we have $\theta = \widetilde{\phi}$, by the definition of $\widetilde{\phi}$ (1.6.1). □

We say that a morphism (ψ, θ) of ringed spaces satisfying the condition of (1.7.3) is a *morphism of affine schemes*.

Corollary (1.7.4). — *If $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ are affine schemes, there exists a canonical isomorphism from the set of morphisms of affine schemes* $\operatorname{Hom}((X, \mathscr{O}_X), (Y, \mathscr{O}_Y))$ *to the set of ring homomorphisms from B to A, where $A = \Gamma(\mathscr{O}_X)$ and $B = \Gamma(\mathscr{O}_Y)$.*

Furthermore, we can say that the functors $(\operatorname{Spec}(A), \widetilde{A})$ in A and $\Gamma(X, \mathscr{O}_X)$ in $(X, \mathscr{O}_X)$ define an *equivalence* between the category of commutative rings and the opposite category of affine schemes (T, I, 1.2).

Corollary (1.7.5). — *If $\phi : B \to A$ is surjective, then the corresponding morphism $({}^a\phi, \widetilde{\phi})$ is a monomorphism of ringed spaces* (cf. (4.1.7)).

PROOF. Indeed, we know that ${}^a\phi$ is injective (1.2.5), and, since ϕ is surjective, for each $x \in X$, $\phi_x^\sharp : B_{{}^a\phi(x)} \to A_x$, which is induced by ϕ by passing to rings of fractions, is also surjective (**0**, 1.5.1); hence the conclusion (**0**, 4.1.1). □

1.8. Morphisms from locally ringed spaces to affine schemes Due to a remark by J. Tate, the **II** | 217
statements of Theorem (1.7.3) and Proposition (2.2.4) can be generalized as follows:[3]

Proposition (1.8.1). — *Let $(S, \mathscr{O}_S)$ be an affine scheme, and $(X, \mathscr{O}_X)$ a locally ringed space. Then there is a canonical bijection from the set of ring homomorphisms $\Gamma(S, \mathscr{O}_S) \to \Gamma(X, \mathscr{O}_X)$ to the set of morphisms* **II** | 218
of ringed spaces $(\psi, \theta) : (X, \mathscr{O}_X) \to (S, \mathscr{O}_S)$ such that, for each $x \in X$, $\theta_x^\sharp$ is a local homomorphism $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$.

PROOF. We note first that if $(X, \mathscr{O}_X)$ and $(S, \mathscr{O}_S)$ are any two ringed spaces, then a morphism (ψ, θ) from $(X, \mathscr{O}_X)$ to $(S, \mathscr{O}_S)$ canonically defines a ring homomorphism $\Gamma(\theta) : \Gamma(S, \mathscr{O}_S) \to \Gamma(X, \mathscr{O}_X)$, hence a first map

$$\rho : \operatorname{Hom}((X, \mathscr{O}_X), (S, \mathscr{O}_S)) \longrightarrow \operatorname{Hom}(\Gamma(S, \mathscr{O}_S), \Gamma(X, \mathscr{O}_X)). \tag{1.8.1.1}$$

Conversely, under the stated hypotheses, we set $A = \Gamma(S, \mathscr{O}_S)$, and consider a ring homomorphism $\phi : A \to \Gamma(X, \mathscr{O}_X)$. For each $x \in X$, it is clear that the set of the $f \in A$ such that $\phi(f)(x) = 0$ is a *prime ideal* of A, since $\mathscr{O}_x / \mathfrak{m}_x = k(x)$ is a field; it is therefore an element of $S = \operatorname{Spec}(A)$, which we denote by ${}^a\phi(x)$. In addition, for each $f \in A$, we have, by definition (**0**, 5.5.2), that ${}^a\phi^{-1}(D(f)) = X_f$, which proves that ${}^a\phi$ is a *continuous map* $X \to S$. We then define a homomorphism

$$\widetilde{\phi} : \mathscr{O}_S \longrightarrow {}^a\phi_*(\mathscr{O}_X)$$

of $\mathscr{O}_S$-modules; for each $f \in A$, we have $\Gamma(D(f), \mathscr{O}_S) = A_f$ (1.3.6); for each $s \in A$, we associate to $s/f \in A_f$ the element $(\phi(s)|X_f)(\phi(f)|X_f)^{-1}$ of $\Gamma(X_f, \mathscr{O}_X) = \Gamma(D(f), {}^a\phi_*(\mathscr{O}_X))$, and we immediately see (by passing from $D(f)$ to $D(fg)$) that this is a well-defined homomorphism of $\mathscr{O}_S$-modules, hence a morphism $({}^a\phi, \widetilde{\phi})$ of ringed spaces. In addition, with the same notation, and setting $y = {}^a\phi(x)$ for brevity, we immediately see (**0**, 3.7.1) that we have $\widetilde{\phi}_x^\sharp(s_y/f_y) = (\phi(s)_x)(\phi(f)_x)^{-1}$; since the relation $s_y \in \mathfrak{m}_y$ is, by definition, equivalent to $\phi(s)_x \in \mathfrak{m}_x$, we see that $\widetilde{\phi}_x^\sharp$ is a *local* homomorphism $\mathscr{O}_y \to \mathscr{O}_x$, and we have thus defined a second map $\sigma : \operatorname{Hom}(\Gamma(S, \mathscr{O}_S), \Gamma(X, \mathscr{O}_X)) \to \mathfrak{L}$, where $\mathfrak{L}$ is the set of the morphisms $(\psi, \theta) : (X, \mathscr{O}_X) \to (S, \mathscr{O}_S)$ such that $\theta_x^\sharp$ is local for each $x \in X$. It remains to prove that σ and ρ (restricted to $\mathfrak{L}$) are inverses of each other; the definition of $\widetilde{\phi}$ immediately shows that $\Gamma(\widetilde{\phi}) = \phi$, and, as a result, that $\rho \circ \sigma$ is the identity. To see that $\sigma \circ \rho$ is the identity, start with a morphism $(\psi, \theta) \in \mathfrak{L}$ and let $\phi = \Gamma(\theta)$; the hypotheses on $\theta_x^\sharp$ mean that this morphism induces, by passing to quotients, a monomorphism $\theta^x : k(\psi(x)) \to k(x)$ such that for each section

[3] *[Trans.] The following section (I.1.8) was added in the errata of EGA II, hence the temporary change in page numbers, which refer to EGA II.*

$f \in A = \Gamma(S, \mathscr{O}_S)$, we have $\theta^x(f(\psi(x))) = \phi(f)(x)$; the equation $f(\phi(x)) = 0$ is therefore equivalent to $\phi(f)(x) = 0$, which proves that ${}^a\phi = \psi$. On the other hand, the definitions imply that the diagram

$$\begin{tikzcd} A \arrow[r, "\phi"] \arrow[d] & \Gamma(X, \mathscr{O}_X) \arrow[d] \\ A_{\psi(x)} \arrow[r, "\theta_x^\sharp"] & \mathscr{O}_x \end{tikzcd}$$

is commutative, and it is the same for the analogous diagram where $\theta_x^\sharp$ is replaced by $\widetilde{\phi}_x^\sharp$, hence $\widetilde{\phi}_x^\sharp = \theta_x^\sharp$ (**0**, 1.2.4), and, as a result, $\widetilde{\phi} = \theta$. □

(1.8.2). When $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ are *locally* ringed spaces, we will consider the morphisms $(\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ such that, for each $x \in X$, $\theta_x^\sharp$ is a *local* homomorphism $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$. Henceforth when we speak of a *morphism of locally ringed spaces*, it will always be a morphism like the above; **II** | 219
with this definition of morphisms, it is clear that the locally ringed spaces form a *category*; for any two objects X and Y of this category, $\mathrm{Hom}(X, Y)$ thus denotes the set of morphisms of locally ringed spaces from X to Y (the set denoted $\mathfrak{L}$ in (1.8.1)); when we consider the set of *morphisms of ringed spaces* from X to Y, we will denote it by $\mathrm{Hom}_{\mathrm{rs}}(X, Y)$ to avoid any confusion. The map (1.8.1.1) is then written as

(1.8.2.1) $$\rho : \mathrm{Hom}_{\mathrm{rs}}(X, Y) \longrightarrow \mathrm{Hom}(\Gamma(Y, \mathscr{O}_Y), \Gamma(X, \mathscr{O}_X))$$

and its restriction

(1.8.2.2) $$\rho' : \mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(\Gamma(Y, \mathscr{O}_Y), \Gamma(X, \mathscr{O}_X))$$

is a *functorial* map in X and Y on the category of locally ringed spaces.

Corollary (1.8.3). — *Let $(Y, \mathscr{O}_Y)$ be a locally ringed space. For Y to be an affine scheme, it is necessary and sufficient that, for each locally ringed space $(X, \mathscr{O}_X)$, the map* (1.8.2.2) *be bijective.*

PROOF. Proposition (1.8.1) shows that the condition is necessary. Conversely, if we suppose that the condition is satisfied, and if we put $A = \Gamma(Y, \mathscr{O}_Y)$, then it follows from the hypotheses and from (1.8.1) that the functors $X \mapsto \mathrm{Hom}(X, Y)$ and $X \mapsto \mathrm{Hom}(X, \mathrm{Spec}(A))$, from the category of locally ringed spaces to that of sets, are *isomorphic*. We know that this implies the existence of a canonical isomorphism $X \to \mathrm{Spec}(A)$ (cf. **0**, 8). □

(1.8.4). Let $S = \mathrm{Spec}(A)$ be an affine scheme; denote by (S', A') the ringed space whose underlying space is *a point* and the structure sheaf A' is the (necessarily simple) sheaf on S' defined by the ring A. Let $\pi : S \to S'$ be the unique map from S to S'; on the other hand, we note that, for each open subset U of S, we have a canonical map $\Gamma(S', A') = \Gamma(S, \mathscr{O}_S) \to \Gamma(U, \mathscr{O}_S)$ which thus defines a *π-morphism* $\iota : A' \to \mathscr{O}_S$ of sheaves of rings. We have thus canonically defined a *morphism of ringed spaces* $i = (\pi, \iota) : (S, \mathscr{O}_S) \to (S', A')$. For each A-module M, we denote by M' the simple sheaf on S' defined by M, which is evidently an A'-module. It is clear that $i_*(\widetilde{M}) = M'$ (1.3.7).

Lemma (1.8.5). — *With the notation of* (1.8.4), *for each A-module M, the canonical functorial $\mathscr{O}_S$-homomorphism* (**0**, 4.4.3.3)

(1.8.5.1) $$i^*(i_*(\widetilde{M})) \longrightarrow \widetilde{M}$$

is an isomorphism.

PROOF. Indeed, the two parts of (1.8.5.1) are right exact (the functor $M \mapsto i_*(\widetilde{M})$ evidently being exact) and commute with direct sums; by considering M as the cokernel of a homomorphism $A^{(I)} \to A^{(J)}$, we can reduce to proving the lemma for the case where $M = A$, and it is evident in this case. □

Corollary (1.8.6). — *Let $(X, \mathscr{O}_X)$ be a ringed space, and $u : X \to S$ a morphism of ringed spaces. For each* **II** | 220
A-module M, we have (with the notation of (1.8.4)*) a canonical functorial isomorphism of $\mathscr{O}_X$-modules*

(1.8.6.1) $$u^*(\widetilde{M}) \simeq u^*(i^*(M')).$$

Corollary (1.8.7). — *Under the hypotheses of (1.8.6), we have, for each A-module M and each $\mathscr{O}_X$-module $\mathscr{F}$, a canonical isomorphism, functorial in M and $\mathscr{F}$,*

(1.8.7.1) $$\operatorname{Hom}_{\mathscr{O}_S}(\widetilde{M}, u_*(\mathscr{F})) \simeq \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F})).$$

PROOF. We have, according to (**0**, 4.4.3) and Lemma (1.8.5), a canonical isomorphism of bifunctors

$$\operatorname{Hom}_{\mathscr{O}_S}(\widetilde{M}, u_*(\mathscr{F})) \simeq \operatorname{Hom}_{A'}(M', i_*(u_*(\mathscr{F})))$$

and it is clear that the right hand side is exactly $\operatorname{Hom}_A(M, \Gamma(X, \mathscr{F}))$. We note that the canonical homomorphism (1.8.7.1) sends each $\mathscr{O}_S$-homomorphism $h : \widetilde{M} \to u_*(\mathscr{F})$ (in other words, each u-morphism $\widetilde{M} \to \mathscr{F}$) to the A-homomorphism $\Gamma(h) : M \to \Gamma(S, u_*(\mathscr{F})) = \Gamma(X, \mathscr{F})$. □

(1.8.8). With the notation of (1.8.4), it is clear (**0**, 4.1.1) that each morphism of ringed spaces $(\psi, \theta) : X \to S'$ is equivalent to the data of a ring homomorphism $A \to \Gamma(X, \mathscr{O}_X)$. We can thus interpret Proposition (1.8.1) as defining a canonical bijection $\operatorname{Hom}(X, S) \simeq \operatorname{Hom}(X, S')$ (where we understand that the right-hand side is the collection of morphisms of ringed spaces, since in general A is not a local ring). More generally, if X and Y are locally ringed spaces, and if (Y', A') is the ringed space whose underlying space is a point and whose sheaf of rings A' is the simple sheaf defined by the ring $\Gamma(Y, \mathscr{O}_Y)$, we can interpret (1.8.2.1) as a map

(1.8.8.1) $$\rho : \operatorname{Hom}_{\mathrm{rs}}(X, Y) \longrightarrow \operatorname{Hom}(X, Y').$$

The result of Corollary (1.8.3) is interpreted by saying that affine schemes are characterized among locally ringed spaces as those for which the restriction of ρ to $\operatorname{Hom}(X, Y)$:

(1.8.8.2) $$\rho' : \operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(X, Y')$$

is *bijective* for *every* locally ringed space X. In the following chapter, we generalize this definition, which allows us to associate to *any* ringed space Z (and not only to a ringed space whose underlying space is a point) a locally ringed space which we will call $\operatorname{Spec}(Z)$; this will be the starting point for a "relative" theory of preschemes over any ringed space, extending the results of Chapter I.

(1.8.9). We can consider the pairs $(X, \mathscr{F})$ consisting of a locally ringed space X and an $\mathscr{O}_X$-module $\mathscr{F}$ as forming a category, a *morphism* in this category being a pair (u, h) consisting of a morphism of locally ringed spaces $u : X \to Y$ and a u-morphism $h : \mathscr{G} \to \mathscr{F}$ of modules; these morphisms (for **II** | 221
$(X, \mathscr{F})$ and $(Y, \mathscr{G})$ fixed) form a set which we denote by $\operatorname{Hom}((X, \mathscr{F}), (Y, \mathscr{G}))$; the map $(u, h) \mapsto (\rho'(u), \Gamma(h))$ is a canonical map

(1.8.9.1) $$\operatorname{Hom}((X, \mathscr{F}), (Y, \mathscr{G})) \longrightarrow \operatorname{Hom}((\Gamma(Y, \mathscr{O}_Y), \Gamma(Y, \mathscr{G})), (\Gamma(X, \mathscr{O}_X), \Gamma(X, \mathscr{F})))$$

functorial in $(X, \mathscr{F})$ and $(Y, \mathscr{G})$, the right-hand side being the set of di-homomorphisms corresponding to the rings and modules considered (**0**, 1.0.2).

Corollary (1.8.10). — *Let Y be a locally ringed space, and $\mathscr{G}$ an $\mathscr{O}_Y$-module. For Y to be an affine scheme and $\mathscr{G}$ to be a quasi-coherent $\mathscr{O}_Y$-module, it is necessary and sufficient that, for each pair $(X, \mathscr{F})$ consisting of a locally ringed space X and an $\mathscr{O}_X$-module $\mathscr{F}$, the canonical map (1.8.9.1) be bijective.*

We leave the proof, which is modelled on that of (1.8.3), using (1.8.1) and (1.8.7), to the reader.

Remark (1.8.11). — The statements (1.7.3), (1.7.4), and (2.2.4) are particular cases of (1.8.1), as well as the definition in (1.6.1); similarly, (2.2.5) follows from (1.8.7). Corollary (1.8.7) also implies (1.6.3) (and, as a result, (1.6.4)) as a particular case, since if X is an affine scheme and $\Gamma(X, \mathscr{F}) = N$, then the functors $M \mapsto \operatorname{Hom}_{\mathscr{O}_S}(\widetilde{M}, u_*(\widetilde{N}))$ and $M \mapsto \operatorname{Hom}_{\mathscr{O}_S}(\widetilde{M}, (N_{[\phi]})^{\sim})$ (where $\phi : A \to \Gamma(X, \mathscr{O}_X)$ corresponds to u) are isomorphic, by Corollaries (1.8.7) and (1.3.8). Finally, (1.6.5) (and, as a result, (1.6.6)) follow from (1.8.6), and the fact that, for each $f \in A$, the A_f-modules $N' \otimes_{A'} A_f$ and $(N' \otimes_{A'} A)_f$ (with the notation of (1.6.5)) are canonically isomorphic.

§2. Preschemes and morphisms of preschemes

2.1. Definition of preschemes

(2.1.1). Given a ringed space $(X, \mathscr{O}_X)$, we say that an open subset V of X is an *affine open* subset if the ringed space $(V, \mathscr{O}_X|V)$ is an affine scheme (1.7.1).

Definition (2.1.2). — We define a prescheme to be a ringed space $(X, \mathscr{O}_X)$ such that every point of X admits an affine open neighbourhood.

Proposition (2.1.3). — *If $(X, \mathscr{O}_X)$ is a prescheme, then its affine open subsets form a basis for the topology of X.* **I** | 98

Proof. If V is an arbitrary open neighbourhood of $x \in X$, then there exists by hypothesis an open neighbourhood W of x such that $(W, \mathscr{O}_X|W)$ is an affine scheme; we write A to mean its ring. In the space W, $V \cap W$ is an open neighbourhood of x; so there exists some $f \in A$ such that $D(f)$ is an open neighbourhood of x contained inside $V \cap W$ (1.1.10, i). The ringed space $(D(f), \mathscr{O}_X|D(f))$ is thus an affine scheme, isomorphic to A_f (1.3.6), whence the proposition. ☐

Proposition (2.1.4). — *The underlying space of a prescheme is a Kolmogoroff space.*

Proof. If x and y are two distinct points of a prescheme X, then it is clear that there exists an open neighbourhood of one of these points that does not contain the other if x and y are not in the same affine open subset; and if they are in the same affine open subset, this is a result of (1.1.8). ☐

Proposition (2.1.5). — *If $(X, \mathscr{O}_X)$ is a prescheme, then every closed irreducible subset of X admits exactly one generic point, and the map $x \mapsto \overline{\{x\}}$ is thus a bijection of X onto its set of closed irreducible subsets.*

Proof. If Y is a closed irreducible subset of X and $y \in Y$, and if U is an affine open neighbourhood of y in X, then $U \cap Y$ is dense in Y, and also irreducible ((**0**, 2.1.1) and (**0**, 2.1.4)); thus, by Corollary (1.1.14), $U \cap Y$ is the closure in U of a point x, and so $Y \subset \overline{U}$ is the closure of x in X. The uniqueness of the generic point of X is a result of Proposition (2.1.4) and of (**0**, 2.1.3). ☐

(2.1.6). If Y is a closed irreducible subset of X, and y its generic point, then the local ring $\mathscr{O}_y$ (also written $\mathscr{O}_{X/Y}$) is called the *local ring of X along Y*, or the *local ring of Y in X*.

If X itself is irreducible and x its generic point then we say that $\mathscr{O}_x$ is the *ring of rational functions on X* (cf. §7).

Proposition (2.1.7). — *If $(X, \mathscr{O}_X)$ is a prescheme, then the ringed space $(U, \mathscr{O}_X|U)$ is a prescheme for every open subset U.*

Proof. This follows directly from Definition (2.1.2) and Proposition (2.1.3). ☐

We say that $(U, \mathscr{O}_X|U)$ is the prescheme *induced* on U by $(X, \mathscr{O}_X)$, or the *restriction* of $(X, \mathscr{O}_X)$ to U.

(2.1.8). We say that a prescheme $(X, \mathscr{O}_X)$ is *irreducible* (resp. *connected*) if the underlying space X is irreducible (resp. connected). We say that a prescheme is *integral* if it is *irreducible and reduced* (cf. (5.1.4)). We say that a prescheme $(X, \mathscr{O}_X)$ is *locally integral* if every $x \in X$ admits an open neighbourhood U such that the prescheme induced on U by $(X, \mathscr{O}_X)$ is integral.

2.2. Morphisms of preschemes

Definition (2.2.1). — Given two preschemes, $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$, we define a morphism (of preschemes) from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$ to be a morphism of ringed spaces (ψ, θ) such that, for all $x \in X$, $\theta^\sharp_x$ is a local homomorphism $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$.

By passing to quotients, the map $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$ gives us a monomorphism $\theta^x : k(\psi(x)) \to k(x)$, **I** | 99
which lets us consider $k(x)$ as an *extension* of the field $k(\psi(x))$.

(2.2.2). The composition (ψ'', θ'') of two morphisms (ψ, θ), (ψ', θ') of preschemes is also a morphism of preschemes, since it is given by the formula $\theta''^\sharp = \theta^\sharp \circ \psi^*(\theta'^\sharp)$ (**0**, 3.5.5). From this we conclude that preschemes form a *category*; using the usual notation, we will write $\operatorname{Hom}(X, Y)$ to mean the set of morphisms from a prescheme X to a prescheme Y.

Example (2.2.3). — If U is an open subset of X, then the canonical injection (**0**, 4.1.2) of the induced prescheme $(U, \mathscr{O}_X|U)$ into $(X, \mathscr{O}_X)$ is a morphism of preschemes; it is further a *monomorphism* of ringed spaces (and *a fortiori* a monomorphism of preschemes), which follows rapidly from (**0**, 4.1.1).

Proposition (2.2.4). — [4] *Let $(X, \mathscr{O}_X)$ be a prescheme, and $(S, \mathscr{O}_S)$ an affine scheme associated to a ring A. Then there exists a canonical bijective correspondence between morphisms of preschemes from $(X, \mathscr{O}_X)$ to $(S, \mathscr{O}_S)$ and ring homomorphisms from A to $\Gamma(X, \mathscr{O}_X)$.*

Proof. First note that, if $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ are two arbitrary ringed spaces, a morphism (ψ, θ) from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$ canonically defines a ring homomorphism $\Gamma(\theta) : \Gamma(Y, \mathscr{O}_Y) \to \Gamma(Y, \psi_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$. In the case that we consider, everything boils down to showing that any homomorphism $\phi : A \to \Gamma(X, \mathscr{O}_X)$ is of the form $\Gamma(\theta)$ for exactly one θ. Now, by hypothesis, there is a covering (V_α) of X by affine open subsets; by composing ϕ with the restriction homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(V_\alpha, \mathscr{O}_X|V_\alpha)$, we obtain a homomorphism $\phi_\alpha : A \to \Gamma(V_\alpha, \mathscr{O}_X|V_\alpha)$ that corresponds to a unique morphism $(\psi_\alpha, \theta_\alpha)$ from the prescheme $(V_\alpha, \mathscr{O}_X|V_\alpha)$ to $(S, \mathscr{O}_S)$, by Theorem (1.7.3). Furthermore, for each pair of indices (α, β), each point of $V_\alpha \cap V_\beta$ admits an affine open neighbourhood W contained inside $V_\alpha \cap V_\beta$ (2.1.3); it is clear that, by composing ϕ_α and ϕ_β with the restriction homomorphisms to W, we obtain the same homomorphism $\Gamma(S, \mathscr{O}_S) \to \Gamma(W, \mathscr{O}_X|W)$, so, with the equation $(\theta^\sharp_\alpha)_x = (\phi_\alpha)_x$ for all $x \in V_\alpha$ and all α (1.6.1), the restriction to W of the morphisms $(\psi_\alpha, \theta_\alpha)$ and $(\psi_\beta, \theta_\beta)$ coincide. From this we conclude that there is a morphism $(\psi, \theta) : (X, \mathscr{O}_X) \to (S, \mathscr{O}_S)$ of ringed spaces, and only one such that its restriction to each V_α is $(\psi_\alpha, \theta_\alpha)$, and it is clear that this morphism is a morphism of preschemes and such that $\Gamma(\theta) = \phi$.

Let $u : A \to \Gamma(X, \mathscr{O}_X)$ be a ring homomorphism, and $v = (\psi, \theta)$ the corresponding morphism $(X, \mathscr{O}_X) \to (S, \mathscr{O}_S)$. For each $f \in A$, we have that

$$\psi^{-1}(D(f)) = X_{u(f)} \tag{2.2.4.1}$$

with the notation of (**0**, 5.5.2) relative to the locally free sheaf $\mathscr{O}_X$. In fact, it suffices to verify this formula when X itself is affine, and then this is nothing but (1.2.2.2). ☐

Proposition (2.2.5). — *Under the hypotheses of Proposition (2.2.4), let $\phi : A \to \Gamma(X, \mathscr{O}_X)$ be a ring homomorphism, $f : (X, \mathscr{O}_X) \to (S, \mathscr{O}_S)$ the corresponding morphism of preschemes, $\mathscr{G}$ (resp. $\mathscr{F}$) an $\mathscr{O}_X$-module (resp. $\mathscr{O}_S$-module), and $M = \Gamma(S, \mathscr{F})$. Then there exists a canonical bijective correspondence between* **I** | 100
*f-morphisms $\mathscr{F} \to \mathscr{G}$ (**0**, 4.4.1) and A-homomorphisms $M \to (\Gamma(X, \mathscr{G}))_{[\phi]}$.*

Proof. Reasoning as in Proposition (2.2.4), we reduce to the case where X is affine, and the proposition then follows from Proposition (1.6.3) and from Corollary (1.3.8). ☐

(2.2.6). We say that a morphism of preschemes $(\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is *open* (resp. *closed*) if, for all open subsets U of X (resp. all closed subsets F of X), $\psi(U)$ is open (resp. $\psi(F)$ is closed) in Y. We say that (ψ, θ) is *dominant* if $\psi(X)$ is dense in Y, and *surjective* if ψ is surjective. We note that these conditions rely only on the continuous map ψ.

[4] *[Trans.] See (1.8) and the footnote there.*

Proposition (2.2.7). — *Let*

$$f = (\psi, \theta) : (X, \mathscr{O}_X) \longrightarrow (Y, \mathscr{O}_Y)$$

and

$$g = (\psi', \theta') : (Y, \mathscr{O}_Y) \longrightarrow (Z, \mathscr{O}_Z)$$

be morphisms of preschemes.

(i) *If f and g are both open (resp. closed, dominant, surjective), then so is $g \circ f$.*
(ii) *If f is surjective and $g \circ f$ closed, then g is closed.*
(iii) *If $g \circ f$ is surjective, then g is surjective.*

PROOF. Claims (i) and (iii) are evident. Write $g \circ f = (\psi'', \theta'')$. If F is closed in Y then $\psi^{-1}(F)$ is closed in X, so $\psi''(\psi^{-1}(F))$ is closed in Z; but since ψ is surjective, $\psi(\psi^{-1}(F)) = F$, so $\psi''(\psi^{-1}(F)) = \psi'(F)$, which proves (ii). □

Proposition (2.2.8). — *Let $f = (\psi, \theta)$ be a morphism $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$, and (U_α) an open cover of Y. For f to be open (resp. closed, surjective, dominant), it is necessary and sufficient for its restriction to each induced prescheme $(\psi^{-1}(U_\alpha), \mathscr{O}_X|\psi^{-1}(U_\alpha))$, considered as a morphism of preschemes from this induced prescheme to the induced prescheme $(U_\alpha, \mathscr{O}_Y|U_\alpha)$ to be open (resp. closed, surjective, dominant).*

PROOF. The proposition follows immediately from the definitions, taking into account the fact that a subset F of Y is closed (resp. open, dense) in Y if and only if each of the $F \cap U_\alpha$ are closed (resp. open, dense) in U_α. □

(2.2.9). Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two preschemes; suppose that X (resp. Y) has a finite number of irreducible components X_i (resp. Y_i) ($1 \leqslant i \leqslant n$); let ξ_i (resp. η_i) be the generic point of X_i (resp. Y_i) (2.1.5). We say that a morphism

$$f = (\psi, \theta) : (X, \mathscr{O}_X) \longrightarrow (Y, \mathscr{O}_Y)$$

is *birational* if, for all i, $\psi^{-1}(\eta_i) = \{\xi_i\}$ and $\theta^\sharp_{\xi_i} : \mathscr{O}_{\eta_i} \to \mathscr{O}_{\xi_i}$ is an *isomorphism*. It is clear that a birational morphism is dominant (**0**, 2.1.8), and thus it is surjective if it is also closed.

Notation (2.2.10). — In all that follows, when there is no risk of confusion, we *suppress* the structure sheaf (resp. the morphism of structure sheaves) from the notation of a prescheme (resp. morphism of preschemes). If U is an open subset of the underlying space X of a prescheme, then whenever we speak of U as a prescheme we always mean the induced prescheme on U.

2.3. Gluing preschemes

(2.3.1). It follows from Definition (2.1.2) that every ringed space obtained by *gluing* preschemes (**0**, 4.1.7) is again a prescheme. In particular, although every prescheme admits, by definition, a cover by affine open sets, we see that every prescheme can actually be obtained by *gluing affine schemes*. **I** | 101

Example (2.3.2). — Let K be a field, $B = K[s]$ and $C = K[t]$ polynomial rings in one indeterminate over K, and define $X_1 = \operatorname{Spec}(B)$ and $X_2 = \operatorname{Spec}(C)$, which are isomorphic affine schemes. In X_1 (resp. X_2), let U_{12} (resp. U_{21}) be the affine open $D(s)$ (resp. $D(t)$) where the ring B_s (resp. C_t) is formed of rational fractions of the form $f(s)/s^m$ (resp. $g(t)/t^n$) with $f \in B$ (resp. $g \in C$). Let u_{12} be the isomorphism of preschemes $U_{21} \to U_{12}$ corresponding (2.2.4) to the isomorphism from B_s to C_t that, to $f(s)/s^m$, associates the rational fraction $f(1/t)/(1/t^m)$. We can glue X_1 and X_2 along U_{12} and U_{21} by using u_{12}, because there is clearly no gluing condition. We later show that the prescheme X obtained in this manner is a particular case of a general method of construction (**II**, 2.4.3). Here we show only that X *is not an affine scheme*; this will follow from the fact that the ring $\Gamma(X, \mathscr{O}_X)$ is *isomorphic* to K, and so its spectrum reduces to a point. Indeed, a section of $\mathscr{O}_X$ over X has a restriction over X_1 (resp. X_2), identified with an affine open of X, that is a polynomial $f(s)$ (resp. $g(t)$), and it follows from the definitions that we should have $g(t) = f(1/t)$, which is only possible if $f = g \in K$.

2.4. Local schemes

(2.4.1). We say that an affine scheme is a *local scheme* if it is the affine scheme associated to a local ring A; there then exists, in $X = \operatorname{Spec}(A)$, a single *closed point* $a \in X$, and for all other $b \in X$ we have that $a \in \overline{\{b\}}$ (1.1.7).

For all preschemes Y and points $y \in Y$, the local scheme $\operatorname{Spec}(\mathscr{O}_y)$ is called the *local scheme of Y at the point y*. Let V be an affine open subset of Y containing y, and B the ring of the affine scheme V; then $\mathscr{O}_y$ is canonically identified with B_y (1.3.4), and the canonical homomorphism $B \to B_y$ thus corresponds (1.6.1) to a morphism of preschemes $\operatorname{Spec}(\mathscr{O}_y) \to V$. If we compose this morphism with the canonical injection $V \to Y$, then we obtain a morphism $\operatorname{Spec}(\mathscr{O}_y) \to Y$ which is *independent* of the affine open subset V (containing y) that we chose: indeed, if V' is some other affine open subset containing y, then there exists a third affine open subset W that contains y and is such that $W \subset V \cap V'$ (2.1.3); we can thus assume that $V \subset V'$, and then if B' is the ring of V', so everything relies on remarking that the diagram

$$\begin{tikzcd} B' \arrow[rr] \arrow[dr] & & B \arrow[dl] \\ & \mathscr{O}_y & \end{tikzcd}$$

is commutative (**0**, 1.5.1). The morphism

$$\operatorname{Spec}(\mathscr{O}_y) \longrightarrow Y$$

thus defined is said to be *canonical*.

Proposition (2.4.2). — *Let $(Y, \mathscr{O}_Y)$ be a prescheme; for all $y \in Y$, let (ψ, θ) be the canonical morphism* I | 102
$(\operatorname{Spec}(\mathscr{O}_y), \widetilde{\mathscr{O}_y}) \to (Y, \mathscr{O}_Y)$. Then ψ is a homeomorphism from $\operatorname{Spec}(\mathscr{O}_y)$ to the subspace S_y of Y given by the z such that $y \in \overline{\{z\}}$ (or, equivalently, the *generalizations* of y (**0**, 2.1.2)); *furthermore, if $z = \psi(\mathfrak{p})$, then $\theta_z^\sharp : \mathscr{O}_z \to (\mathscr{O}_y)_\mathfrak{p}$ is an isomorphism; (ψ, θ) is thus a monomorphism of ringed spaces.*

Proof. Since the unique closed point a of $\operatorname{Spec}(\mathscr{O}_y)$ is contained in the closure of any point of this space, and since $\psi(a) = y$, the image of $\operatorname{Spec}(\mathscr{O}_y)$ under the continuous map ψ is contained in S_y. Since S_y is contained in every affine open containing y, one can consider just the case where Y is an affine scheme; but then this proposition follows from (1.6.2). $\square$

We see (2.1.5) that there is a bijective correspondence between $\operatorname{Spec}(\mathscr{O}_y)$ and the set of closed irreducible subsets of Y containing y.

Corollary (2.4.3). — *For $y \in Y$ to be the generic point of an irreducible component of Y, it is necessary and sufficient for the only prime ideal of the local ring $\mathscr{O}_y$ to be its maximal ideal (*in other words, for $\mathscr{O}_y$ to be of *dimension zero).*

Proposition (2.4.4). — *Let $(X, \mathscr{O}_X)$ be a local scheme of some ring A, a its unique closed point, and $(Y, \mathscr{O}_Y)$ a prescheme. Every morphism $u = (\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ then factors uniquely as $X \to \operatorname{Spec}(\mathscr{O}_{\psi(a)}) \to Y$, where the second arrow denotes the canonical morphism, and the first corresponds to a local homomorphism $\mathscr{O}_{\psi(a)} \to A$. This establishes a canonical bijective correspondence between the set of morphisms $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ and the set of local homomorphisms $\mathscr{O}_y \to A$ for $(y \in Y)$.*

Indeed, for all $x \in X$, we have that $a \in \overline{\{x\}}$, so $\psi(a) \in \overline{\{\psi(x)\}}$, which shows that $\psi(X)$ is contained in every affine open subset that contains $\psi(a)$. So it suffices to consider the case where $(Y, \mathscr{O}_Y)$ is an affine scheme of ring B, and then we have that $u = ({}^a\phi, \widetilde{\phi})$, where $\phi \in \operatorname{Hom}(B, A)$ (1.7.3). Further, we have that $\phi^{-1}(\mathfrak{j}_a) = \mathfrak{j}_{\psi(a)}$, and hence that the image under ϕ of any element of $B - \mathfrak{j}_{\psi(a)}$ is invertible in the local ring A; the factorization in the result follows from the universal property of the ring of fractions (**0**, 1.2.4). Conversely, to each local homomorphism $\mathscr{O}_y \to A$ there is a unique corresponding morphism $(\psi, \theta) : X \to \operatorname{Spec}(\mathscr{O}_y)$ such that $\psi(a) = y$ (1.7.3), and, by composing with the canonical morphism $\operatorname{Spec}(\mathscr{O}_y) \to Y$, we obtain a morphism $X \to Y$, which proves the proposition.

(2.4.5). The affine schemes whose ring is a field K have an underlying space that is just a point. If A is a local ring with maximal ideal $\mathfrak{m}$, then each local homomorphism $A \to K$ has kernel equal to $\mathfrak{m}$, and so factors as $A \to A/\mathfrak{m} \to K$, where the second arrow is a monomorphism. The morphisms $\operatorname{Spec}(K) \to \operatorname{Spec}(A)$ thus correspond bijectively to monomorphisms of fields $A/\mathfrak{m} \to K$.

Let $(Y, \mathscr{O}_Y)$ be a prescheme; for each $y \in Y$ and each ideal $\mathfrak{a}_y$ of $\mathscr{O}_y$, the canonical homomorphism $\mathscr{O}_y \to \mathscr{O}_y/\mathfrak{a}_y$ defines a morphism $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to \operatorname{Spec}(\mathscr{O}_y)$; if we compose this with the canonical morphism $\operatorname{Spec}(\mathscr{O}_y) \to Y$, then we obtain a morphism $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to Y$, again said to be *canonical*. For $\mathfrak{a}_y = \mathfrak{m}_y$, this says that $\mathscr{O}_y/\mathfrak{a}_y = k(y)$, and so Proposition (2.4.4) says that:

Corollary (2.4.6). — *Let $(X, \mathscr{O}_X)$ be a local scheme whose ring K is a field, ξ the unique point of X, and $(Y, \mathscr{O}_Y)$ a prescheme. Then each morphism $u : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ factors uniquely as $X \to \operatorname{Spec}(k(\psi(\xi))) \to Y$, where the second arrow denotes the canonical morphism, and the first corresponds to a monomorphism $k(\psi(\xi)) \to K$. This establishes a canonical bijective correspondence between the set of morphisms $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ and the set of monomorphisms $k(y) \to K$ (for $y \in Y$).* I | 103

Corollary (2.4.7). — *For all $y \in Y$, every canonical morphism $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to Y$ is a monomorphism of ringed spaces.*

PROOF. We have already seen this when $\mathfrak{a}_y = 0$ (2.4.2), and it suffices to apply Corollary (1.7.5). □

Remark. — 2.4.8 Let X be a local scheme, and a its unique closed point. Since every affine open subset containing a is necessarily equal to the whole of X, every *invertible* $\mathscr{O}_X$-module (**0**, 5.4.1) is necessarily *isomorphic to* $\mathscr{O}_X$ (or, as we say, again, *trivial*). This property does not hold in general for an arbitrary affine scheme $\operatorname{Spec}(A)$; we will see in Chapter V that if A is a normal ring then this is true when A is a unique factorisation domain.

2.5. Preschemes over a prescheme

Definition (2.5.1). — Given a prescheme S, we say that the data of a prescheme X and a morphism of preschemes $\phi : X \to S$ defines a prescheme X *over the prescheme S*, or an *S-prescheme*; we say that S is the *base prescheme* of the S-prescheme X. The morphism ϕ is called the *structure morphism* of the S-prescheme X. When S is an affine scheme of ring A, we also say that X endowed with ϕ is a prescheme *over the ring A* (or an *A-prescheme*).

It follows from (2.2.4) that the data of a prescheme over a ring A is equivalent to the data of a prescheme $(X, \mathscr{O}_X)$ whose structure sheaf $\mathscr{O}_X$ is a sheaf of A-*algebras*. *An arbitrary prescheme can always be considered as a* $\mathbf{Z}$*-prescheme in a unique way*.

If $\phi : X \to S$ is the structure morphism of an S-prescheme X, we say that a point $x \in X$ is *over a point $s \in S$* if $\phi(x) = s$. We say that X *dominates* S if ϕ is a dominant morphism (2.2.6).

(2.5.2). Let X and Y be S-preschemes; we say that a morphism of preschemes $u : X \to Y$ is a *morphism of preschemes over S* (or an *S-morphism*) if the diagram

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

(where the diagonal arrows are the structure morphisms) is commutative: this ensures that, for all $s \in S$ and $x \in X$ over s, $u(x)$ also lies over s.

It follows immediately from this definition that the composition of any two S-morphisms is an S-morphism; S-preschemes thus form a *category*.

We denote by $\operatorname{Hom}_S(X, Y)$ the set of S-morphisms from an S-prescheme X to an S-prescheme Y; the identity morphism of an S-prescheme X is denoted by 1_X.

When S is an affine scheme of ring A, we will also say *A-morphism* instead of S-morphism.

(2.5.3). If X is an S-prescheme, and $v : X' \to X$ a morphism of preschemes, then the composition $X' \to X \to S$ endows X' with the structure of an S-prescheme; in particular, every prescheme induced by an open set U of X can be considered as an S-prescheme by the canonical injection. I | 104

If $u : X \to Y$ is an S-morphism of S-preschemes, then the restriction of u to any prescheme induced by an open subset U of X is also an S-morphism $U \to Y$. Conversely, let (U_α) be an open cover of X, and for each α let $u_\alpha : U_\alpha \to Y$ be an S-morphism; if, for all pairs of indices (α, β), the restrictions of u_α and u_β to $U_\alpha \cap U_\beta$ agree, then there exists an S-morphism $X \to Y$, and exactly one such that the restriction to each U_α is u_α.

If $u : X \to Y$ is an S-morphism such that $u(X) \subset V$, where V is an open subset of Y, then u, considered as a morphism from X to V, is also an S-morphism.

(2.5.4). Let $S' \to S$ be a morphism of preschemes; for all S'-preschemes, the composition $X \to S' \to S$ endows X with the structure of an S-prescheme. Conversely, suppose that S' is the induced prescheme of an open subset of S; let X be an S-prescheme and suppose that the structure morphism $f : X \to S$ is such that $f(X) \subset S'$; then we can consider X as an S'-prescheme. In this latter case, if Y is another S-prescheme whose structure morphism sends the underlying space to S', then every S-morphism from X to Y is also an S'-morphism.

(2.5.5). If X is an S-prescheme, with structure morphism $\phi : X \to S$, we define an *S-section of X* to be an S-morphism from S to X, that is to say a morphism of preschemes $\psi : S \to X$ such that $\phi \circ \psi$ is the identity on S. We denote by $\Gamma(X/S)$ the set of S-sections of X.

§3. Products of preschemes

3.1. Sums of preschemes Let (X_α) be any family of preschemes; let X be a topological space which is the *sum* of the underlying spaces X_α; X is then the union of pairwise disjoint open subspaces X'_α, and for each α there is a homomorphism ϕ_α from X_α to X'_α. If we equip each of the X'_α with the sheaf $(\phi_\alpha)_*(\mathscr{O}_{X_\alpha})$, it is clear that X becomes a prescheme, which we call the *sum* of the family of preschemes (X_α) and which we denote $\coprod_\alpha X_\alpha$. If Y is a prescheme, then the map $f \mapsto (f \circ \phi_\alpha)$ is a *bijection* from the set $\operatorname{Hom}(X, Y)$ to the product set $\prod_\alpha \operatorname{Hom}(X_\alpha, Y)$. In particular, if the X_α are S-preschemes, with structure morphisms ψ_α, then X is an S-prescheme by the unique morphism $\psi : X \to S$ such that $\psi \circ \phi_\alpha = \psi_\alpha$ for each α. The sum of two preschemes X and Y is denoted by $X \sqcup Y$. It is immediate that, if $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$, then $X \sqcup Y$ is canonically identified with $\operatorname{Spec}(A \times B)$.

3.2. Products of preschemes

Definition (3.2.1). — Given S-preschemes X and Y, we say that a triple (Z, p_1, p_2), consisting of an S-prescheme Z, and S-morphisms $p_1 : Z \to X$ and $p_2 : Z \to Y$, is a product of the S-preschemes **I** | 105
X and Y, if, for each S-prescheme T, the map $f \mapsto (p_1 \circ f, p_2 \circ f)$ is a bijection from the set of S-morphisms from T to Z, to the set of pairs consisting of an S-morphism $T \to X$ and an S-morphism $T \to Y$ (in other words, a bijection

$$\operatorname{Hom}_S(T, Z) \simeq \operatorname{Hom}_S(T, X) \times \operatorname{Hom}_S(T, Y)).$$

This is the general notion of a *product* of two objects in a category, applied to the category of S-preschemes (T, I, 1.1); in particular, a product of two S-preschemes is *unique* up to a unique S-isomorphism. Because of this uniqueness, most of the time we will denote a product of two S-preschemes X and Y by $X \times_S Y$ (or simply $X \times Y$, when there is no chance of confusion), with the morphisms p_1 and p_2 (the *canonical projections* of $X \times_S Y$ to X and to Y, respectively) being suppressed in the notation. If $g : T \to X$ and $h : T \to Y$ are S-morphisms, we denote by $(g, h)_S$, or simply (g, h), the S-morphism $f : T \to X \times_S Y$ such that $p_1 \circ f = g$, $p_2 \circ f = h$. If X' and Y' are two S-preschemes, p'_1 and p'_2 the canonical projections of $X' \times_S Y'$ (assumed to exist), and $u : X' \to X$ and $v : Y' \to Y$ S-morphisms, then we write $u \times_S v$ (or simply $u \times v$) for the S-morphism $(u \circ p'_1, v \circ p'_2)_S$ from $X' \times_S Y'$ to $X \times_S Y$.

When S is an affine scheme given by some ring A, we often replace S by A is the above notation.

Proposition (3.2.2). — *Let X, Y, and S be affine schemes, given by rings B, C, and A (respectively). Let $Z = \operatorname{Spec}(B \otimes_A C)$, and let p_1 and p_2 be the S-morphisms corresponding (2.2.4) to the canonical A-homomorphisms $u : b \mapsto b \otimes 1$ and $v : c \mapsto 1 \otimes c$ (respectively) from B and C to $B \otimes_A C$; then (Z, p_1, p_2) is a product of X and Y.*

PROOF. According to (2.2.4), it suffices to check that, if, to each A-homomorphism $f : B \otimes_A C \to L$ (where L is an A-algebra), we associate the pair $(f \circ u, f \circ v)$, then this defines a bijection $\mathrm{Hom}_A(B \otimes_A C, L) \simeq \mathrm{Hom}_A(B, L) \times \mathrm{Hom}_A(C, L)$,[5] which follows immediately from the definitions and the fact that $b \otimes c = (b \otimes 1)(1 \otimes c)$. □

Corollary (3.2.3). — *Let T be an affine scheme given by some ring D, and $\alpha = ({}^a\rho, \widetilde{\rho})$ (resp. $\beta = ({}^a\sigma, \widetilde{\sigma})$) an S-morphism $T \to X$ (resp. $T \to Y$), where ρ (resp. σ) is an A-homomorphism from B (resp. C) to D; then $(\alpha, \beta)_S = ({}^a\tau, \widetilde{\tau})$, where τ is the homomorphism $B \otimes_A C \to D$ such that $\tau(b \otimes c) = \rho(b)\sigma(c)$.*

Proposition (3.2.4). — *Let $f : S' \to S$ be a* monomorphism *of preschemes* (T, I, 1.1), *and let X and Y be S'-preschemes, also considered as S-preschemes via f. Every product of the S-preschemes X and Y is then a product of the S'-preschemes X and Y, and vice versa.*

PROOF. Let $\phi : X \to S'$ and $\psi : Y \to S'$ be the structure morphisms. If T is an S-prescheme, and $u : T \to X$ and $v : T \to Y$ are S-morphisms, then we have, by definition, that $f \circ \phi \circ u = f \circ \psi \circ v = \theta$, the structure morphism of T; the hypotheses on f imply that $\phi \circ u = \psi \circ v = \theta'$, and so we can consider T as an S'-prescheme with structure morphism θ', and u and v as S'-morphisms. The conclusion of the proposition follows immediately, taking (3.2.1) into account. □

Corollary (3.2.5). — *Let X and Y be S-preschemes, with structure morphisms $\phi : X \to S$ and $\psi : Y \to S$, and let S' be an open subset of S such that $\phi(X) \subset S'$ and $\psi(Y) \subset S'$. Every product of the S-preschemes X and Y is then also a product of the S'-preschemes X and Y, and conversely.*

It suffices to apply (3.2.4) to the canonical injection $S' \to S$. **I** | 106

Theorem (3.2.6). — *Given S-preschemes X and Y, there exists a product $X \times_S Y$.*

The proof proceeds in several steps.

Lemma (3.2.6.1). — *Let (Z, p, q) be a product of X and Y, and U and V open subsets of X and Y, respectively. If we let $W = p^{-1}(U) \cap q^{-1}(V)$, then the triple consisting of W and the restrictions of p and q to W* (considered as the morphisms $W \to U$ and $W \to V$, respectively) *is a product of U and V.*

Indeed, if T is an S-prescheme, then we can identify the S-morphisms $T \to W$ with the S-morphisms $T \to Z$ mapping T to W. Then, if $g : T \to U$ and $h : T \to V$ are any two S-morphisms, we can consider them as S-morphisms from T to X and to Y, respectively, and, by hypothesis, there is then a unique S-morphism $f : T \to Z$ such that $g = p \circ f$ and $h = q \circ f$. Since $p(f(Y)) \subset U$, $q(f(T)) \subset V$, we have

$$f(T) \subset p^{-1}(U) \cap q^{-1}(V) = W,$$

hence our claim.

Lemma (3.2.6.2). — *Let Z be an S-prescheme, $p : Z \to X$ and $q : Z \to Y$ both S-morphisms, (U_α) an open cover of X, and (V_λ) an open cover of Y. Suppose that, for each pair (α, λ), the S-prescheme $W_{\alpha\lambda} = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ and the restrictions of p and q to $W_{\alpha\lambda}$ form a product of U_α and V_λ. Then (Z, p, q) is a product of X and Y.*

We first show that, if f_1 and f_2 are S-morphisms $T \to Z$, then the equations $p \circ f_1 = p \circ f_2$ and $q \circ f_1 = q \circ f_2$ imply that $f_1 = f_2$. Indeed, Z is the union of the $W_{\alpha\lambda}$, so the $f_1^{-1}(W_{\alpha\lambda})$ form an open cover of T, and similarly for $f_2^{-1}(W_{\alpha\lambda})$. In addition, we have

$$f_1^{-1}(W_{\alpha\lambda}) = f_1^{-1}(p^{-1}(U_\alpha)) \cap f_1^{-1}(q^{-1}(V_\lambda)) = f_2^{-1}(p^{-1}(U_\alpha)) \cap f_2^{-1}(q^{-1}(V_\lambda)) = f_2^{-1}(W_{\alpha\lambda})$$

by hypothesis, and it thus reduces to noting that the the restrictions of f_1 and f_2 to $f_1^{-1}(W_{\alpha\lambda}) = f_2^{-1}(W_{\alpha\lambda})$ are identical for each pair of indices. But since these restrictions can be considered as S-morphisms from $f_1^{-1}(W_{\alpha\lambda})$ to $W_{\alpha\lambda}$, our claim follows from the hypotheses and Definition (3.2.1).

Suppose now that we are given S-morphisms $g : T \to X$ and $h : T \to Y$. Let $T_{\alpha\lambda} = g^{-1}(U_\alpha) \cap h^{-1}(V_\lambda)$; then the $T_{\alpha\lambda}$ form an open cover of T. By hypothesis, there exists an S-morphism $f_{\alpha\lambda}$ such that $p \circ f_{\alpha\lambda}$ and $q \circ f_{\alpha\lambda}$ are the restrictions of g and h to $T_{\alpha\lambda}$ (respectively). Now, we will show that

[5] The notation Hom_A denotes here the set of homomorphisms of *A-algebras*.

the restrictions of $f_{\alpha\lambda}$ and $f_{\beta\mu}$ to $T_{\alpha\lambda} \cap T_{\beta\mu}$ coincide, which will finish the proof of Lemma (3.2.6.2). The images of $T_{\alpha\lambda} \cap T_{\beta\mu}$ under $f_{\alpha\lambda}$ and $f_{\beta\mu}$ are contained in $W_{\alpha\lambda} \cap W_{\beta\mu}$ by definition. Since

$$W_{\alpha\lambda} \cap W_{\beta\mu} = p^{-1}(U_\alpha \cap U_\beta) \cap q^{-1}(V_\lambda \cap V_\mu),$$

it follows from Lemma (3.2.6.1) that $W_{\alpha\lambda} \cap W_{\beta\mu}$ and the restrictions to this prescheme of p and q form a *product* of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$. Since $p \circ f_{\alpha\lambda}$ and $p \circ f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$ and similarly for $q \circ f_{\alpha\lambda}$ and $q \circ f_{\beta\mu}$, we see that $f_{\alpha\lambda}$ and $f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$.

Lemma (3.2.6.3). — *Let (U_α) be an open cover of X, (V_λ) an open cover of Y, and suppose that, for each pair (α, λ), there exists a product of U_α and V_λ; then there exists a product of X and Y.* I | 107

Applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$, we see that there exists a product of S-preschemes induced, respectively, by X and Y on these open sets; in addition, the uniqueness of the product shows that, if we set $i = (\alpha, \lambda)$ and $j = (\beta, \mu)$, then there is a canonical isomorphism h_{ij} (resp. h_{ji}) from this product to an S-prescheme W_{ij} (resp. W_{ji}) induced by $U_\alpha \times_S V_\lambda$ (resp. $U_\beta \times_S V_\mu$) on an open set; then $f_{ij} = h_{ij} \circ h_{ji}^{-1}$ is an isomorphism from W_{ji} to W_{ij}. In addition, for a third pair $k = (\gamma, \nu)$, we have $f_{ik} = f_{ij} \circ f_{jk}$ on $W_{ki} \cap W_{kj}$, by applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta \cap U_\gamma$ and $V_\lambda \cap V_\mu \cap V_\nu$ in U_β and V_μ, respectively. It follows that we have a prescheme Z, an open cover (Z_i) of the underlying space of Z, and, for each i, an isomorphism g_i from the induced prescheme Z_i to the prescheme $U_\alpha \times_S V_\lambda$, so that, for each pair (i, j), we have $f_{ij} = g_i \circ g_j^{-1}$ (2.3.1); in addition, we have $g_i(Z_i \cap Z_j) = W_{ij}$. If p_i, q_i, and θ_i are the projections and the structure morphism of the S-prescheme $U_\alpha \times_S V_\lambda$ (respectively), we immediately see that $p_i \circ g_i = p_j \circ g_j$ on $Z_i \cap Z_j$, and similarly for the two other morphisms. We can thus define the morphisms of preschemes $p : Z \to X$ (resp. $q : Z \to Y$, $\theta : Z \to S$) by the condition that p (resp. q, θ) coincide with $p_i \circ g_i$ (resp. $q_i \circ g_i$, $\theta_i \circ g_i$) on each of the Z_i; Z, equipped with θ, is then an S-prescheme. We now show that $Z_i' = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ is equal to Z_i. For each index $j = (\beta, \mu)$, we have $Z_j \cap Z_i' = g_j^{-1}(p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda))$. We have, by Lemma (3.2.6.1),

$$p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = p_j^{-1}(U_\alpha \cap U_\beta) \cap q_j^{-1}(V_\lambda \cap V_\mu);$$

with the restrictions of p_j and q_j to $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda)$ defining, on this S-prescheme, the structure of a product of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$; but the uniqueness of the product then implies that $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = W_{ji}$. As a result, we have $Z_j \cap Z_i' = Z_j \cap Z_i$ for each j, hence $Z_i' = Z_i$. We then deduce from Lemma (3.2.6.2) that (Z, p, q) is a product of X and Y.

Lemma (3.2.6.4). — *Let $\phi : X \to S$ and $\psi : Y \to S$ be the structure morphisms of X and Y, (S_i) an open cover of S, and let $X_i = \phi^{-1}(S_i)$, $Y_i = \psi^{-1}(S_i)$. If each of the products $X_i \times_S Y_i$ exists, then $X \times_S Y$ exists.*

According to Lemma (3.2.6.3), everything follows from proving that the products $X_i \times_S Y_i$ exists for any i and j. Set $X_{ij} = X_i \cap X_j = \phi^{-1}(S_i \cap S_j)$, $Y_{ij} = Y_i \cap Y_j = \psi^{-1}(S_i \cap S_j)$; by Lemma (3.2.6.1), the product $Z_{ij} = X_{ij} \times_S Y_{ij}$ exists. We now note that, if T is an S-prescheme, and if $g : T \to X_i$ and $h : T \to Y_j$ are S-morphisms, then we necessarily have that $\phi(g(T)) = \psi(h(T)) \subset S_i \cap S_j$ by the definition of an S-morphism, and thus that $g(T) \subset X_{ij}$ and $h(T) \subset Y_{ij}$; it is then immediate that Z_{ij} is the product of X_i and Y_j.

(3.2.6.5). We can now complete the proof of Theorem (3.2.6). If S is an *affine scheme*, then there are covers (U_α) and (V_λ) of X and Y (respectively) consisting of affine open subsets; since $U_\alpha \times_S V_\lambda$ exists, by (3.2.2), $X \times_S Y$ exists similarly, by Lemma (3.2.6.3). If S is any prescheme, then there is a cover (S_i) of S consisting of affine open subsets. If $\phi : X \to S$ and $\psi : Y \to S$ are the structure morphisms, and if we set $X_i = \phi^{-1}(S_i)$ and $Y_i = \psi^{-1}(S_i)$, then the products $X_i \times_{S_i} Y_i$ exist, by the above; but then the products $X_i \times_S Y_i$ also exist (3.2.5), therefore $X \times_S Y$ exists similarly, by I | 108
Lemma (3.2.6.4). □

Corollary (3.2.7). — *Let $Z = X \times_S Y$ be the product of two S-preschemes, p and q the projections from Z to X and to Y (respectively),and ϕ (resp. ψ) the structure morphism of X (resp. Y). Let S' be an open subset of S, and U (resp. V) an open subset of X (resp. Y) contained in $\phi^{-1}(S')$ (resp. $\psi^{-1}(S')$). Then the product*

$U \times_{S'} V$ is canonically identified with the prescheme induced on Z by $p^{-1}(U) \cap q^{-1}(V)$ (considered as an S'-prescheme). In addition, if $f : T \to X$ and $g : T \to Y$ are S-morphisms such that $f(T) \subset U$ and $g(T) \subset V$, then the S'-morphism $(f, g)_{S'}$ can be identified with the restriction of $(f, g)_S$ to $p^{-1}(U) \cap q^{-1}(V)$.

PROOF. This follows from Corollary (3.2.5) and Lemma (3.2.6.1). □

(3.2.8). Let (X_α) and (Y_λ) be families of S-preschemes, and X (resp. Y) the sum of the family (X_α) (resp. (Y_λ)) (3.1). Then $X \times_S Y$ can be identified with the *sum* of the family $(X_\alpha \times_S Y_\lambda)$; this follows immediately from Lemma (3.2.6.3).

(3.2.9). [6] It follows from (1.8.1) that we can state (3.2.2) in the following manner: $Z = \operatorname{Spec}(B \otimes_A C)$ **II** | 221
is not only a product of $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(C)$ in the category of *S-preschemes*, but also in the category of *locally ringed spaces over S* (with a definition of S-morphisms modelled on that of (2.5.2)). The proof of (3.2.6) also proves that, for any two S-preschemes X and Y, the prescheme $X \times_S Y$ is not only the product of X and Y in the category of S-preschemes, but also in the category of locally ringed spaces over the prescheme S.

3.3. Formal properties of the product; change of the base prescheme

(3.3.1). The reader will notice that all the properties stated in this section, except (3.3.13) and (3.3.15), are true without modification in any category, whenever the products involved in the statements exist (since it is clear that the notions of an S-object and of an S-morphism can be defined exactly as in (2.5) for any object S of the category).

(3.3.2). First of all, $X \times_S Y$ is a *covariant bifunctor* in X and Y on the category of S-preschemes: it suffices in fact to note that the diagram

$$\begin{tikzcd} X \times Y \arrow[r, "f \times 1"] \arrow[d] & X' \times Y \arrow[r, "f' \times 1"] \arrow[d] & X'' \times Y \arrow[d] \\ X \arrow[r, "f"] & X' \arrow[r, "f'"] & X'' \end{tikzcd}$$

is commutative.

Proposition (3.3.3). — *For each S-prescheme X, the first (resp. second) projection from $X \times_S S$ (resp. $S \times_S X$) is a functorial isomorphism from $X \times_S S$ (resp. $S \times_S X$) to X, whose inverse isomorphism is $(1_X, \phi)_S$ (resp. $(\phi, 1_X)_S$), where we denote by ϕ the structure morphism $X \to S$; we can therefore write, up to a canonical isomorphism,*

$$X \times_S S = S \times_S X = X.$$

PROOF. It suffices to prove that the triple $(X, 1_X, \phi)$ is a product of X and S. If T is an S-prescheme, then the only S-morphism from T to S is necessarily the structure morphism $\psi : T \to S$. If f is an S-morphism from T to X, we necessarily have $\psi = \phi \circ f$, hence our claim. □

Corollary (3.3.4). — *Let X and Y be S-preschemes, with structure morphisms $\phi : X \to S$ and $\psi : Y \to S$. If we canonically identify X with $X \times_S S$, and Y with $S \times_S Y$, then the projections $X \times_S Y \to X$ and $X \times_S Y \to Y$ are identified with $1_X \times \psi$ and $\phi \times 1_Y$ (respectively).*

The proof is immediate and is left to the reader.

(3.3.5). We can define, in a manner similar to (3.2), the product of a finite number n of S-preschemes, **I** | 109
and the existence of these products follows from (3.2.6) by induction on n, and by noting that $(X_1 \times_S X_2 \times_S \cdots \times_S X_{n-1}) \times_S X_n$ satisfies the definition of a product. The uniqueness of the product implies, as in any category, its *commutativity* and *associativity* properties. If, for example, p_1, p_2, and p_3 denote the projections from $X_1 \times_S X_2 \times_S X_3$, and if we identify this prescheme with $(X_1 \times_S X_2) \times_S X_3$, then the projection to $X_1 \times_S X_2$ is identified with $(p_1, p_2)_S$.

[6] *[Trans.] (3.2.9) is from the errata of EGA II, on page 221, whence the change in page numbering.*

(3.3.6). Let S and S' be preschemes, and $\phi : S' \to S$ a morphism, which lets us consider S' as an S-prescheme. For each S-prescheme X, consider the product $X \times_S S'$, and let p and π' be the projections to X and to S' (respectively). Equipped with π', this product is an S'-prescheme; when we consider it as such, we denote it by $X_{(S')}$ or $X_{(\phi)}$, and we say that this is the prescheme obtained by *base change* (or *a change of base*) from S to S' by means of the morphism ϕ, or the *inverse image* of X by ϕ. We note that, if π is the structure morphism of X, and θ the structure morphism of $X \times_S S'$, considered as an S-prescheme, then the diagram

$$\begin{tikzcd} X \arrow[d, "\pi"'] & X_{(S')} \arrow[l, "p"'] \arrow[d, "\pi'"] \arrow[dl, "\theta"'] \\ S & S' \arrow[l, "\phi"] \end{tikzcd}$$

is commutative.

(3.3.7). With the notation of (3.3.6), for each S-morphism $f : X \to Y$, we denote by $f_{(S')}$ the S'-morphism $f \times_S 1 : X_{(S')} \to Y_{(S')}$, and we say that $f_{(S')}$ is the *base change* (or *inverse image*) of f by ϕ. Therefore $X_{(S')}$ is a *covariant functor* in X, from the category of S-preschemes to that of S'-preschemes.

(3.3.8). The prescheme $X_{(S')}$ can be considered as a solution to a *universal mapping problem*: each S'-prescheme T is also an S-prescheme via ϕ; each S-morphism $g : T \to X$ is then uniquely written as $g = p \circ f$, where f is an S'-morphism $T \to X_{(S')}$, as follows from the definition of the product applied to the S-morphisms f and $\psi : T \to S'$ (the structure morphism of T).

Proposition (3.3.9). — ("Transitivity of base change"). *Let S'' be a prescheme, and $\phi' : S'' \to S$ a morphism. For each S-prescheme X, there exists a canonical functorial isomorphism from the S''-prescheme $(X_{(\phi)})_{(\phi')}$ to the S''-prescheme $X_{(\phi\circ\phi')}$.*

Proof. Let T be a S''-prescheme, ψ its structure morphism, and g an S-morphism from T to X (T being considered as an S-prescheme with structure morphism $\phi \circ \phi' \circ \psi$). Since T is also an S'-prescheme with structure morphism $\phi' \circ \psi$, we can write $g = p \circ g'$, where g' is an S'-morphism $T \to X_{(\phi)}$, and then $g' = p' \circ g''$, where g'' is an S''-morphism $T \to (X_{(\phi)})_{(\phi')}$:

$$\begin{tikzcd} X \arrow[d, "\pi"'] & X_{(\phi)} \arrow[l, "p"'] \arrow[d, "\pi'"'] & (X_{(\phi)})_{(\phi')} \arrow[l, "p'"'] \arrow[d, "\pi''"] \\ S & S \arrow[l, "\phi"'] & S''. \arrow[l, "\phi'"'] \end{tikzcd}$$

So the result follows by the uniqueness of the solution to a universal mapping problem. □ **I** | 110

This result can be written as the equality (up to a canonical isomorphism) $(X_{(S')})_{(S'')} = X_{(S'')}$ (if there is no chance of confusion), or also as

$$(X \times_S S') \times_{S'} S'' = X \times_S S''; \tag{3.3.9.1}$$

the functorial nature of the isomorphism defined in (3.3.9) can similarly be expressed by the transitivity formula for base change morphisms

$$(f_{(S')})_{(S'')} = f_{(S'')} \tag{3.3.9.2}$$

for each S-morphism $f : X \to Y$.

Corollary (3.3.10). — *If X and Y are S-preschemes, then there exists a canonical functorial isomorphism from the S'-prescheme $X_{(S')} \times_{S'} Y_{(S')}$ to the S'-prescheme $(X \times_S Y)_{(S')}$.*

Proof. We have, up to canonical isomorphism,

$$(X \times_S S') \times_{S'} (Y \times_S S') = X \times_S (Y \times_S S') = (X \times_S Y) \times_S S'$$

according to (3.3.9.1) and the associativity of products of S-preschemes. □

The functorial nature of the isomorphism defined in Corollary (3.3.10) can be expressed by the formula

$$(u_{(S')}, v_{(S')})_{S'} = ((u,v)_S)_{(S')} \tag{3.3.10.1}$$

for each pair of S-morphisms $u : T \to X$, $v : T \to Y$.

In other words, the base change functor $X_{(S')}$ *commutes with products*; it also commutes with sums (3.2.8).

Corollary (3.3.11). — *Let Y be an S-prescheme, and $f : X \to Y$ a morphism which makes X a Y-prescheme (and, as a result, also an S-prescheme). The prescheme $X_{(S')}$ is then identified with the product $X \times_Y Y_{(S')}$, the projection $X \times_Y Y_{(S')} \to Y_{(S')}$ being identified with $f_{(S')}$.*

Proof. Let $\psi : Y \to S$ be the structure morphism of Y; we have the commutative diagram

$$\begin{tikzcd}
S' \arrow[d] & Y_{(S')} \arrow[l] \arrow[d] & X_{(S')} \arrow[l, "f_{(S')}"'] \arrow[d] \\
S & Y \arrow[l, "\psi"'] & X. \arrow[l, "f"']
\end{tikzcd}$$

We have that $Y_{(S')}$ is identified with $S'_{(\psi)}$, and $X_{(S')}$ with $S'_{(\psi\circ f)}$; taking (3.3.9) and (3.3.4) into account, we thus deduce the corollary. ☐

(3.3.12). Let $f : X \to X'$ and $g : Y \to Y'$ be S-morphisms which are *monomorphisms* of preschemes (T, I, 1.1); then $f \times_S g$ is a *monomorphism*. Indeed, if p and q are the projections of $X \times_S Y$, p' and q' the projections of $X' \times_S Y'$, and u and v both S-morphisms $T \to X \times_S Y$, then the equation $(f \times_S g) \circ u = (f \times_S g) \circ v$ implies that $p' \circ (f \times_S g) \circ u = p' \circ (f \times_S g) \circ v$, or, in other words, that $f \circ p \circ u = f \circ p \circ v$, and since f is a monomorphism, $p \circ u = p \circ v$; using the fact that g is a monomorphism, we similarly obtain $q \circ u = q \circ v$, hence $u = v$.

It follows that, for each base change $S' \to S$, **I** | 111

$$f_{(S')} : X_{(S')} \longrightarrow Y_{(S')}$$

is a monomorphism.

(3.3.13). Let S and S' be affine schemes of rings A and A' (respectively); a morphism $S' \to S$ then corresponds to a ring homomorphism $A \to A'$. If X is an S-prescheme, we denote by $X_{(A')}$ or $X \otimes_A A'$ the S'-prescheme $X_{(S')}$; when X is also affine of ring B, $X_{(A')}$ is affine of ring $B_{(A')} = B \otimes_A A'$ obtained by extension of scalars from the A-algebra B to A'.

(3.3.14). With the notation of (3.3.6), for each *S-morphism* $f : S' \to X$, we have that $f' = (f, 1_{S'})_S$ is an S'-morphism $S' \to X' = X_{(S')}$ such that $p \circ f' = f$, $\pi' \circ f' = 1_{S'}$, or, in other words, an *S'-section of of X'*; conversely, if f' is such an S'-section, then $f = p \circ f'$ is an S-morphism $S' \to X$. We thus define a canonical *bijective correspondence*

$$\operatorname{Hom}_S(S', X) \simeq \operatorname{Hom}_{S'}(S', X').$$

We say that f' is the *graph morphism* of f, and we denote it by Γ_f.

(3.3.15). Given a prescheme X, which we can always consider as a $\mathbf{Z}$-prescheme, it follows, in particular, from (3.3.14) that the *X-sections* of $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (where T is an indeterminate) correspond bijectively with *morphisms* $\mathbf{Z}[T] \to X$. We will show that these X-sections also correspond bijectively with *sections of the structure sheaf $\mathscr{O}_X$ over X*. Indeed, let (U_α) be a cover of X by affine open subsets; let $u : X \to X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ be an X-morphism, and let u_α be its restriction to U_α; if A_α is the ring of the affine scheme U_α, then $U_\alpha \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ is an affine scheme of ring $A_\alpha[T]$ (3.2.2), and u_α canonically corresponds to an A_α-homomorphism $A_\alpha[T] \to A_\alpha$ (1.7.3). Now, since such a homomorphism is completely determined by the data of the image of T in A_α, let $s_\alpha \in A_\alpha = \Gamma(U_\alpha, \mathscr{O}_X)$, and if we suppose that the restrictions of u_α and u_β to an affine open subset $V \subset U_\alpha \cap U_\beta$ coincide, then we see immediately that s_α and s_β coincide on V; thus the family (s_α) consists of the restrictions to U_α of a section s of $\mathscr{O}_X$ over X; conversely, it is clear that such a section defines a family (u_α) of morphisms which are the restrictions to U_α of an X-morphism $X \to X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$. This result is generalized in (**II**, 1.7.12).

3.4. Points of a prescheme with values in a prescheme; geometric points

(3.4.1). Let X be a prescheme; for each prescheme T, we then denote by $X(T)$ the set $\operatorname{Hom}(T, X)$ of morphisms $T \to X$, and the elements of this set are called *the points of X with values in T*. If we associate to each morphism $f : T \to T'$ the map $u' \mapsto u' \circ f$ from $X(T')$ to $X(T)$, we see, for fixed X, that $X(T)$ is a *contravariant functor in T*, from the category of preschemes to that of sets. In addition, each morphism of preschemes $g : X \to Y$ defines a functorial homomorphism $X(T) \to Y(T)$, which sends $v \in X(T)$ to $g \circ v$.

(3.4.2). Given sets P, Q, and R, and maps $\phi : P \to R$ and $\psi : Q \to R$, we define the *fibre product of P and Q over R* (with respect to ϕ and ψ) as the subset of the product set $P \times Q$ consisting of the pairs (p, q) such that $\phi(p) = \psi(q)$; we denote it by $P \times_R Q$. Definition (3.2.1) of the product of S-preschemes can be interpreted, with the notation of (3.4.1), via the formula **I | 112**

(3.4.2.1) $$(X \times_S Y)(T) = X(T) \times_{S(T)} Y(T).$$

the maps $X(T) \to S(T)$ and $Y(T) \to S(T)$ corresponding to the structure morphisms $X \to S$ and $Y \to S$.

(3.4.3). If we are given a prescheme S and we consider only the S-preschemes and S-morphisms, then we will denote by $X(T)_S$ the set $\operatorname{Hom}_S(T, X)$ of S-morphisms $T \to X$, and suppress the subscript S when there is no chance of confusion; we say that the elements of $X(T)_S$ are the *points* (or *S-points*, when there is a possibility of confusion) *of the S-prescheme X with values in the S-prescheme T*. In particular, an *S-section* of X is none other than a *point of X with values in S*. The formula (3.4.2.1) can then be written as

(3.4.3.1) $$(X \times_S Y)(T)_S = X(T)_S \times Y(T)_S;$$

more generally, if Z is an S-prescheme, and X, Y, and T are Z-preschemes (thus *ipso facto* S-preschemes), then we have

(3.4.3.2) $$(X \times_Z Y)(T)_S = X(T)_S \times_{Z(T)_S} Y(T)_S.$$

We note that, to show that a triple (W, r, s) consisting of an S-prescheme W and S-morphisms $r : W \to X$ and $s : W \to Y$ is a product of X and Y (over Z), it suffices, by definition, to check that, for *each S-prescheme T*, the diagram

$$\begin{tikzcd} W(T)_S \arrow[r, "r'"] \arrow[d, "s'"'] & X(T)_S \arrow[d, "\phi'"] \\ Y(T)_S \arrow[r, "\psi'"] & Z(T)_S \end{tikzcd}$$

makes $W(T)_S$ the fibre product of $X(T)_S$ and $Y(T)_S$ over $Z(T)_S$, where r' and s' correspond to r and s, and ϕ' and ψ' to the structure morphisms $\phi : X \to Z$ and $\psi : Y \to Z$.

(3.4.4). When T (resp. S) in the above is an affine scheme of ring B (resp. A), we replace T (resp. S) by B (resp. A) in the above notation, and we then call the elements of $X(B)$ the *points of X with values in the ring B*, and the elements of $X(B)_A$ the *points of the A-prescheme X with values in the A-algebra B*. We note that $X(B)$ and $X(B)_A$ are *covariant* functors in B. We similarly write $X(T)_A$ for the set of points of the A-prescheme X with values in the A-prescheme T.

(3.4.5). Consider, in particular, the case where T is of the form $\operatorname{Spec}(A)$, where A is a *local* ring; the elements of $X(A)$ then correspond bijectively to *local* homomorphisms $\mathcal{O}_x \to A$ for $x \in X$ (2.2.4); we say that the point x of the underlying space of X is the *location*[7] of the point of X with values in A to which it corresponds.

More specifically, we define the *geometric points* of a prescheme S to be the *points of X with values in a field K*: the data of such a point is equivalent to the data of its location x in the underlying subspace of X, and of an *extension* K of $k(x)$; K will be called the *field of values* of the corresponding geometric point, and we say that this geometric point is *located at x*. We also define a map $X(K) \to X$, sending a geometric point with values in K to its location. **I | 113**

[7] *[Trans.] We also say that* the geometric point lies over this x.

If $S' = \operatorname{Spec}(K)$ is an S-prescheme (in other words, if K is considered as an extension of the residue field $k(s)$, where $s \in S$), and if X is an S-prescheme, then an element of $X(K)_S$, or, as we say, a *geometric point of X lying over s with values in K*, consists of the data of a $k(s)$-monomorphism from the residue field $k(x)$ to K, where x is a point of X *lying over s* (therefore $k(x)$ is an extension of $k(s)$).

In particular, if $S = \operatorname{Spec}(K) = \{\xi\}$, *then the geometric points of X with values in K can be identified with the points $x \in X$ such that $k(x) = K$*; we say that these latter points are the *K-rational points of the K-prescheme X*; if K' is an extension of K, then the geometric points of X with values in K' bijectively correspond to the K'-rational points of $X' = X_{(K')}$ (3.3.14).

Lemma (3.4.6). — *Let X_i $(1 \leqslant i \leqslant n)$ be S-preschemes, s a point of S, and x_i $(1 \leqslant i \leqslant n)$ points of X_i lying over s. Then there exists an extension K of $k(s)$ and a geometric point of the product $Y = X_1 \times_S X_2 \times_S \cdots \times_S X_n$, with values in K, whose projections to the X_i are localized at the x_i.*

PROOF. There exist $k(s)$-monomorphisms $k(x_i) \to K$, all in the same extension K of $k(s)$ (Bourbaki, *Alg.*, chap. V, §4, prop. 2). The compositions $k(s) \to k(x_i) \to K$ are all identical, and so the morphisms $\operatorname{Spec}(K) \to X_i$ corresponding to the $k(x_i) \to K$ are all S-morphisms, and we thus conclude that they define a unique morphism $\operatorname{Spec}(K) \to Y$. If y is the corresponding point of Y, it is clear that its projection in each of the X_i is x_i. ☐

Proposition (3.4.7). — *Let X_i $(1 \leqslant i \leqslant n)$ be S-preschemes, and, for each index i, let x_i be a point of X_i. For there to exist a point y of $Y = X_1 \times_S X_2 \times \ldots \times_S X_n$ whose image is x_i under the ith projection for each $1 \leqslant i \leqslant n$, it is necessary and sufficient that the x_i all lie above the same point s of S.*

PROOF. The condition is evidently necessary; Lemma (3.4.6) proves that it is sufficient. ☐

In other words, if we denote by (X) the underlying set of X, we see that we have a canonical *surjective* function $(X \times_S Y) \to (X) \times_{(S)} (Y)$; we must point out that this function *is not injective* in general; in other words, *there can exist multiple distinct points z in $X \times_S Y$ that have the same projections $x \in X$ and $y \in Y$*; we have already seen this when S, X, and Y are prime spectra of fields k, K, and K' (respectively), since the tensor product $K \otimes_k K'$ has, in general, multiple distinct prime ideals (cf. (3.4.9)).

Corollary (3.4.8). — *Let $f : X \to Y$ be an S-morphism, and $f_{(S')} : X_{(S')} \to Y_{(S')}$ the S'-morphism induced by f by an extension $S' \to S$ of the base prescheme. Let p (resp. q) be the projection $X_{(S')} \to X$ (resp. $Y_{(S')} \to Y$); for every subset M of X, we have*

$$q^{-1}(f(M)) = f_{(S')}(p^{-1}(M)).$$

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PROOF. Indeed (3.3.11), $X_{(S')}$ can be identified with the product $X \times_Y Y_{(S')}$ thanks to the commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & X_{(S')} \arrow[l, "p"'] \arrow[d, "f_{(S')}"] \\ Y & Y_{(S')} \arrow[l, "q"'] \end{tikzcd}$$

By (3.4.7), the equation $q(y') = f(x)$ for $x \in M$ and $y' \in Y_{(S)}$ is equivalent to the existence of some $x' \in X_{(S')}$ such that $p(x') = x$ and $f_{(S')}(x') = y'$, whence the corollary. ☐

Lemma (3.4.6) can be made clearer in the following manner:

Proposition (3.4.9). — *Let X and Y be S-preschemes, x a point of X, and y a point of Y, with both x and y lying above the same point $s \in S$. The set of points of $X \times_S Y$ with projections x and y is in bijective correspondence with the set of types of extensions (?) composed of $k(x)$ and $k(y)$ considered as extensions of $k(s)$* (Bourbaki, *Alg.*, chap. VIII, §8, prop. 2)*.*

PROOF. Let p (resp. q) be the projection from $X \times_S Y$ to X (resp. Y), and E the subspace $p^{-1}(x) \cap q^{-1}(y)$ of the underlying space of $X \times_S Y$. First, note that the morphisms $\operatorname{Spec}(k(x)) \to S$

and $\mathrm{Spec}(k(y)) \to S$ factor as $\mathrm{Spec}(k(x)) \to \mathrm{Spec}(k(s)) \to S$ and $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(s)) \to S$; since $\mathrm{Spec}(k(s)) \to S$ is a monomorphism (2.4.7), it follows from (3.2.4) that we have

$$P = \mathrm{Spec}(k(x)) \times_S \mathrm{Spec}(k(y)) = \mathrm{Spec}(k(x)) \times_{\mathrm{Spec}(k(s))} \mathrm{Spec}(k(y)) = \mathrm{Spec}(k(x) \otimes_{k(s)} k(y)).$$

We will define two maps, $\alpha : P_0 \to E$ and $\beta : E \to P_0$, inverse to one another (where P_0 denotes the underlying set of the prescheme P). If $i : \mathrm{Spec}(k(x)) \to X$ and $j : \mathrm{Spec}(k(y)) \to Y$ are the canonical morphisms (2.4.5), we take α to be the map of underlying spaces corresponding to the morphism $i \times_S j$. On the other hand, every $z \in E$ defines, by hypothesis, two $k(s)$-monomorphisms, $k(x) \to k(z)$ and $k(y) \to k(z)$, and thus a $k(s)$-monomorphism $k(x) \otimes_{k(s)} k(y) \to k(z)$, and thus a morphism $\mathrm{Spec}(k(z)) \to P$; $\beta(z)$ will be the image of z in P_0 under this morphism. The verification of the fact that $\alpha \circ \beta$ and $\beta \circ \alpha$ are the identity maps follows from (2.4.5) and the definition of the product (3.2.1). Finally, we know that P_0 is in bijective correspondence with the set of types of extensions (?) composed of $k(x)$ and $k(y)$ (Bourbaki, *Alg.*, chap. VIII, §8, prop. 1). □

3.5. Surjections and injections

(3.5.1). In a general sense, consider a property **P** of morphisms of preschemes, and the following two propositions:

(i) If $f : X \to X'$ and $g : Y \to Y'$ are S-morphisms that have property **P**, then $f \times_S g$ also has property **P**.

(ii) If $f : X \to Y$ is an S-morphism that has property **P**, then every S'-morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$, induced by f by an extension of the base prescheme, also has property **P**.

Since $f_{(S')} = f \times_S 1_{S'}$, we see that, if, for every prescheme X, the *identity* 1_X has property **P**, then (i) implies (ii); on the other hand, since $f \times_S g$ is the composite morphism

$$X \times_S Y \xrightarrow{f \times 1_Y} X' \times_S Y \xrightarrow{1_{X'} \times g} X' \times_S Y',$$

we see that, if the *composition* of two morphisms has property **P**, then so does the product $f \times_S g$, and so (ii) implies (i).

A first application of this remark is

Proposition (3.5.2). —

(i) *If $f : X \to X'$ and $g : Y \to Y'$ are surjective S-morphisms, then $f \times_S g$ is surjective.*

(ii) *If $f : X \to Y$ is a surjective S-morphism, then $f_{(S')}$ is surjective for every extension S' of the base prescheme.*

Proof. The composition of any two surjections being a surjection, it suffices to prove (ii); but this proposition follows immediately from (3.4.8) applied to $M = X$. □

Proposition (3.5.3). — *For a morphism $f : X \to Y$ to be surjective, it is necessary and sufficient that, for every field K and every morphism $\mathrm{Spec}(K) \to Y$, there exist an extension K' of K and a morphism $\mathrm{Spec}(K') \to X$ that make the following diagram commute:*

$$\begin{tikzcd} X \arrow[d, "f"'] & \mathrm{Spec}(K') \arrow[l] \arrow[d] \\ Y & \mathrm{Spec}(K). \arrow[l] \end{tikzcd}$$

Proof. The condition is sufficient because, for all $y \in Y$, it suffices to apply it to a morphism $\mathrm{Spec}(K) \to Y$ corresponding to a monomorphism $k(y) \to K$, with K being an extension of $k(y)$ (2.4.6). Conversely, suppose that f is surjective, and let $y \in Y$ be the image of the unique point of $\mathrm{Spec}(K)$; there exists some $x \in X$ such that $f(x) = y$; we will consider the corresponding monomorphism $k(y) \to k(x)$ (2.2.1); it then suffices to take K' to be the extension of $k(y)$ such that there exist $k(y)$-monomorphisms from $k(x)$ and K to K' (Bourbaki, *Alg.*, chap. V, §4, prop. 2); the morphism $\mathrm{Spec}(K') \to X$ corresponding to $k(x) \to K'$ is exactly that for which we are searching. □

With the language introduced in (3.4.5), we can say that *every geometric point of Y with values in K comes from a geometric point of X with values in an extension of K.*

Definition (3.5.4). — We say that a morphism $f : X \to Y$ of preschemes is *universally injective*, or a *radicial morphism*, if, for every field K, the corresponding map $X(K) \to Y(K)$ is injective.

It follows also from the definitions that every *monomorphism of preschemes* (T, 1.1) is radicial.

(3.5.5). For a morphism $f : X \to Y$ to be radicial, it suffices that the condition of Definition (3.5.4) hold for every *algebraically closed* field. In fact, if K is an arbitrary field, and K' an algebraically-closed extension of K, then the diagram

$$\begin{tikzcd} X(K) \arrow[r, "\alpha"] \arrow[d, "\phi"'] & Y(K) \arrow[d, "\phi'"] \\ X(K') \arrow[r, "\alpha'"] & Y(K') \end{tikzcd}$$

commutes, where ϕ and ϕ' come from the morphism $\operatorname{Spec}(K') \to \operatorname{Spec}(K)$, and α and α' corresponding to f. However, ϕ is injective, and so too is α', by hypothesis; hence α is necessarily injective.

Proposition (3.5.6). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of preschemes.*

(i) *If f and g are radicial, then so is $g \circ f$.*
(ii) *Conversely, if $g \circ f$ is radicial, then so is f.*

PROOF. Taking into account Definition (3.5.4), the proposition reduces to the corresponding claims for the maps $X(K) \to Y(K) \to Z(K)$, and these claims are evident. □

Proposition (3.5.7). —

(i) *If the S-morphisms $f : X \to X'$ and $g : X \to X'$ are radicial, then so is $f \times_S g$.*
(ii) *If the S-morphism $f : X \to Y$ is radicial, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every extension $S' \to S$ of the base prescheme.*

PROOF. Given (3.5.1), it suffices to prove (i). We have seen (3.4.2.1) that

$$(X \times_S Y)(K) = X(K) \times_{S(K)} Y(K),$$
$$(X' \times_S Y')(K) = X'(K) \times_{S(K)} Y'(K),$$

with the map $(X \times_S Y)(K) \to (X' \times_S Y')(K)$ corresponding to $f \times_S g$ thus being identified with $(u, v) \to (f \circ u, g \circ v)$, and the proposition then follows. □

Proposition (3.5.8). — *For a morphism $f = (\psi, \theta) : X \to Y$ to be radicial, it is necessary and sufficient for ψ to be injective and for the monomorphism $\theta^x : k(\psi(x)) \to k(x)$ to make $k(x)$ a radicial extension of $k(\psi(x))$ for every $x \in X$.*

PROOF. We suppose that f is radicial and first show that the equation $\psi(x_1) = \psi(x_2) = y$ necessarily implies that $x_1 = x_2$. Indeed, there exists a field K, and an extension of $k(y)$, along with $k(y)$-monomorphisms $k(x_1) \to K$ and $k(x_2) \to K$ (Bourbaki, *Alg.*, chap. V, §4, prop. 2); the corresponding morphisms $u_1 : \operatorname{Spec}(K) \to X$ and $u_2 : \operatorname{Spec}(K) \to X$ are then such that $f \circ u_1 = f \circ u_2$, and so $u_1 = u_2$ by hypothesis, and this implies, in particular, that $x_1 = x_2$. We now consider $k(x)$ as the extension of $k(\psi(x))$ by means of θ^x: if $k(x)$ is not a radicial algebraically-closed extension, then there exist two distinct $k(\psi(x))$-monomorphisms from $k(x)$ to an algebraically-closed extension K of $k(\psi(x))$, and the two corresponding morphisms $\operatorname{Spec}(K) \to X$ would contradict the hypothesis. Conversely, taking (2.4.6) into account, it is immediate that the conditions stated are sufficient for f to be radicial. □

Corollary (3.5.9). — *If A is a ring, and S is a multiplicative set of A, then the canonical morphism* $\operatorname{Spec}(S^{-1}A) \to \operatorname{Spec}(A)$ *is radicial.*

PROOF. Indeed, this morphism is a monomorphism (1.6.2). □

Corollary (3.5.10). — *Let $f : X \to Y$ be a radicial morphism, $g : Y' \to Y$ a morphism, and $X' = X_{(Y')} = X \times_Y Y'$. Then the radicial morphism $f_{(Y')}$ (3.5.7, ii) is a bijection from the underlying space of X to $g^{-1}(f(X))$; further, for every field K, the set $X'(K)$ can be identified with the subset of $Y'(K)$ given by the inverse image of the map $Y'(K) \to Y(K)$ (corresponding to g) from the subset $X(K)$ of $Y(K)$.*

PROOF. The first claim follows from (3.5.8) and (3.4.8); the second, from the commutativity of the following diagram: I | 117

$$\begin{tikzcd} X'(K) \arrow[r] \arrow[d] & Y'(K) \arrow[d] \\ X(K) \arrow[r] & Y(K) \end{tikzcd}$$

□

Remark (3.5.11). — We say that a morphism $f = (\psi, \theta)$ of preschemes is *injective* if the map ψ is injective. For a morphism $f = (\psi, \theta) : X \to Y$ to be radicial, it is necessary and sufficient that, for every morphism $Y' \to Y$, the morphism $f_{(Y')} : X_{(Y')} \to Y'$ be injective (which justifies the terminology of a *universally injective* morphism). In fact, the condition is necessary by (3.5.7, ii) and (3.5.8). Conversely, the condition implies that ψ is injective; if, for some $x \in X$, the monomorphism $\theta^x : k(\psi(x)) \to k(x)$ were not radicial, then there would be an extension K of $k(\psi(x))$, and two distinct morphisms $\operatorname{Spec}(K) \to X$ corresponding to the same morphism $\operatorname{Spec}(K) \to Y$ (3.5.8). But then, setting $Y' = \operatorname{Spec}(K)$, there would be two distinct Y'-sections of $X_{(Y')}$ (3.3.14), which contradicts the hypothesis that $f_{(Y')}$ is injective.

3.6. Fibres

Proposition (3.6.1). — *Let $f : X \to Y$ be a morphism, y a point of Y, and $\mathfrak{a}_y$ an ideal of definition for $\mathscr{O}_y$ for the $\mathfrak{m}_y$-preadic topology. Then the projection $p : X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to X$ is a homeomorphism from the underlying space of the prescheme $X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ to the fibre $f^{-1}(y)$ equipped with the topology induced from that of the underlying space of X.*

PROOF. Since $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to Y$ is radicial ((3.5.4) and (2.4.7)), since $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ is a single point, and since the ideal $\mathfrak{m}_y/\mathfrak{a}_y$ is nilpotent by hypothesis (1.1.12), we already know ((3.5.10) and (3.3.4)) that p identifies, as *sets*, the underlying space of $X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ with $f^{-1}(y)$; everything reduces to proving that p is a homeomorphism. By (3.2.7), the question is local on X and Y, and so we can suppose that $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$, with B being an A-algebra. The morphism p then corresponds to the homomorphism $1 \otimes \phi : B \to B \otimes_A A'$, where $A' = A_y/\mathfrak{a}_y$ and ϕ is the canonical map from A to A'. Then every element of $B \otimes_A A'$ can be written as

$$\sum_i b_i \otimes \phi(a_i)/\phi(s) = \left(\sum_i (a_i b_i \otimes 1)\right)(1 \otimes \phi(s))^{-1},$$

where $s \notin \mathfrak{j}_y$, and Proposition (1.2.4) applies. □

(3.6.2). Throughout the rest of this treatise, whenever we consider a fibre $f^{-1}(y)$ of a morphism as having the structure of a $k(y)$-prescheme, *it will always be the prescheme obtained by transporting the structure of $X \times_Y \operatorname{Spec}(k(y))$ by the projection to X.* We will also write this (latter) product as $X \times_Y k(y)$, or $X \otimes_{\mathscr{O}_Y} k(y)$; more generally, if B is an $\mathscr{O}_y$-algebra, we will denote by $X \times_Y B$ or $X \otimes_{\mathscr{O}_Y} B$ the product $X \times_Y \operatorname{Spec}(B)$.

With the preceding convention, it follows from (3.5.10) that the points of X with values in an extension K of $k(y)$ are identified with the *points of $f^{-1}(y)$ with values in K.*

(3.6.3). Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and $h = g \circ f$ their composition; for all $z \in Z$, the fibre $h^{-1}(z)$ is a prescheme isomorphic to

$$X \times_Z \operatorname{Spec}(k(z)) = (X \times_Y Y) \times_Z \operatorname{Spec}(k(z)) = X \times_Y g^{-1}(z).$$

Inparticular, if U is an open subset of X, then the prescheme induced on $U \cap f^{-1}(y)$ by the prescheme I | 118
$f^{-1}(y)$ is isomorphic to $f_U^{-1}(y)$ (f_U being the restriction of f to U),

Proposition (3.6.4). — (Transitivity of fibres) *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms; let $X' = X \times_Y Y' = X_{(Y')}$ and $f' = f_{(Y')} : X' \to Y'$. For every $y' \in Y'$, if we let $y = g(y')$, then the prescheme $f'^{-1}(y')$ is isomorphic to $f^{-1}(y) \otimes_{k(y)} k(y')$.*

PROOF. Indeed, it suffices to remark that the two preschemes $(X \otimes_Y k(y)) \otimes_{k(y)} k(y')$ and $(X \times_Y Y') \otimes_{Y'} k(y')$ are both canonically isomorphic to $X \times_Y \operatorname{Spec}(k(y'))$ by (3.3.9.1). □

In particular, if V is an open neighbourhood of y in Y, and we denote by f_V the restriction of f to the induced prescheme on $f^{-1}(V)$, then the preschemes $f^{-1}(y)$ and $f_V^{-1}(y)$ are canonically identified.

Proposition (3.6.5). — *Let $f : X \to Y$ be a morphism, y a point of Y, Z the local prescheme* $\operatorname{Spec}(\mathcal{O}_y)$, *and $p = (\psi, \theta)$ the projection $X \times_Y Z \to X$; then p is a homeomorphism from the underlying space of $X \times_Y Z$ to the subspace $f^{-1}(Z)$ of X* (when the underlying space of Z is identified with a subspace of Y, cf. (2.4.2)*), and, for all $t \in X \times_Y Z$, letting $z = \psi(t)$, we have that $\theta_t^\sharp$ is an isomorphism from $\mathcal{O}_x$ to $\mathcal{O}_t$.*

PROOF. Since Z (identified as a subspace of Y) is contained inside every affine open containing y (2.4.2), we can, as in (3.6.1), reduce to the case where $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine schemes, with A being a B-algebra. Then $X \times_Y Z$ is the prime spectrum of $A \otimes_B B_y$, and this ring is canonically identified with $S^{-1}A$, where S is the image of $B - \mathfrak{j}_y$ in A (**0**, 1.5.2); since p then corresponds to the canonical homomorphism $A \to S^{-1}A$, the proposition follows from (1.6.2). □

3.7. Application: reduction of a prescheme mod. $\mathfrak{J}$ *This section, which makes use of notions and results from Chapter I and Chapter II, will not be used in what follows in this treatise, and is only intended for readers familiar with classical algebraic geometry.*

(3.7.1). Let A be a ring, X an A-prescheme, and $\mathfrak{J}$ an ideal of A; then $X_0 = X \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-prescheme, which we sometimes say is induced from X by *reduction* mod. $\mathfrak{J}$.

. This terminology is used foremost when A is a *local ring* and $\mathfrak{J}$ its maximal ideal, in such a way that X_0 is a prescheme over the residue field $k = A/\mathfrak{J}$ of A.

When A is also integral, with field of fractions K, we can consider the K-prescheme $X' = X \otimes_A K$. By an abuse of language which we will not use, it has been said, up until now, that X_0 is *induced by* X' by reduction mod. $\mathfrak{J}$. In the case where this language was used, A was a local ring of dimension 1 (most often a discrete valuation ring) and it was implied (be it more or less explicitly) that the given K-prescheme X' was a closed subprescheme of a K-prescheme P' (in fact, a projective space of the form (?) $\mathbf{P}_K^r$, cf. (**II**, 4.1.1)), itself of the form $P' = P \otimes_A K$, where P is a given A-prescheme (in fact, the A-scheme $\mathbf{P}_A^r$, with the notation of (**II**, 4.1.1)). In our language, the definition of X_0 in terms of X' is formulated as follows:

We consider the affine scheme $Y = \operatorname{Spec}(A)$, formed of two points, the unique closed point $y = \mathfrak{J}$ **I** | 119
and the generic point (0), the singleton set U of the generic point being thus an open $U = \operatorname{Spec}(K)$ in Y. If X is an A-prescheme (or, in other words, a Y-prescheme), then $X \otimes_A K = X'$ is exactly the prescheme induced by X on $\psi^{-1}(U)$, denoting by ψ the structure morphism $X \to Y$. In particular, if ϕ is the structure morphism $P \to Y$, then a closed subprescheme X' of $P' = \phi^{-1}(U)$ is a (locally closed) subprescheme of P. If P is Noetherian (for example, if A is Noetherian and P is of finite type over A), then there exists a smaller closed subprescheme $X = \overline{X'}$ of G that through which X' factors (9.5.10), and X' is the prescheme induced by X on the open $\phi^{-1}(U) \cap X$, and so is isomorphic to $X \otimes_A K$ (9.5.10). *The immersion of X' into $P' = P \otimes_A K$ thus lets us canonically consider X' as being of the form $X' = X \otimes_A K$, where X is an A-prescheme.* We can then consider the reduced mod. $\mathfrak{J}$ prescheme $X_0 = X \otimes_A k$, which is exactly the fibre $\psi^{-1}(y)$ of the closed point y. Up until now, lacking the adequate terminology, we had avoided explicitly introducing the A-prescheme X. One ought to, however, note that all the claims normally made about the "reduced mod. $\mathfrak{J}$" prescheme X_0 should be seen as consequences of more complicated claims about X itself, and cannot be satisfactorily formulated or understood except by interpreting them as such. It seems also that any hypotheses made on X_0 always reduce to hypotheses on X itself (independent of the prior data of an immersion of X' in $\mathbf{P}_K^r$), which lets us give more intrinsic statements.

(3.7.3). Lastly, we draw attention to a very particular fact, which has undoubtedly contributed to slowing the conceptual clarification of the situation envisaged here: if A is a discrete valuation ring, and if X is *proper* over A (which is indeed the case if X is a closed subprescheme of some $\mathbf{P}_A^r$, cf. (**II**, 5.5.4)), then the points of X with values in A and the points of X' with values in k are in bijective correspondence (**II**, 7.3.8). This is why we often believe that facts about X' have been proved, when in reality we have proved facts about X, and these remain valuable (in this form) whenever we no longer suppose that the base local ring is of dimension 1.

§4. Subpreschemes and immersion morphisms

4.1. Subpreschemes

(4.1.1). As the notion of a quasi-coherent sheaf (**0**, 5.1.3) is local, a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ over a prescheme X can be defined by the following condition: for each affine open V of X, $\mathscr{F}|V$ is isomorphic to the sheaf associated to a $\Gamma(V, \mathscr{O}_X)$-module (1.4.1). It is clear that, over a prescheme X, the structure sheaf $\mathscr{O}_X$ is quasi-coherent, and that the kernels, cokernels, and images of homomorphisms of quasi-coherent $\mathscr{O}_X$-modules, as well as inductive limits and direct sums of quasi-coherent $\mathscr{O}_X$-modules, are also quasi-coherent (Theorem (1.3.7) and Corollary (1.3.9)).

Proposition (4.1.2). — *Let X be a prescheme, and $\mathscr{I}$ a quasi-coherent sheaf of ideals of $\mathscr{O}_X$. Then the support Y of the sheaf $\mathscr{O}_X/\mathscr{I}$ is closed, and if we denote by $\mathscr{O}_Y$ the restriction of $\mathscr{O}_X/\mathscr{I}$ to Y, then $(Y, \mathscr{O}_Y)$ is a prescheme.*

Proof. It evidently suffices (2.1.3) to consider the case where X is an affine scheme, and to show that, in this case, Y is closed in X, and is an *affine scheme*. Indeed, if $X = \operatorname{Spec}(A)$, then we have $\mathscr{O}_X = \widetilde{A}$ and $\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is an ideal of A (1.4.1); Y is then equal to the closed subset $V(\mathfrak{I})$ of X and can be identified with the prime spectrum of the ring $B = A/\mathfrak{I}$ (1.1.11); in addition, if ϕ is the canonical homomorphism $A \to B = A/\mathfrak{I}$, then the direct image ${}^a\phi_*(\widetilde{B})$ is canonically identified with the sheaf $\widetilde{A}/\widetilde{\mathfrak{I}} = \mathscr{O}_X/\mathscr{I}$ (Proposition (1.6.3) and Corollary (1.3.9)), which finishes the proof. ☐ I | 120

We say that $(Y, \mathscr{O}_Y)$ is the *subprescheme* of $(X, \mathscr{O}_X)$ *defined by the sheaf of ideals* $\mathscr{I}$; this is a particular case of the more general notion of a *subprescheme*:

Definition (4.1.3). — We say that a ringed space $(Y, \mathscr{O}_Y)$ is a subprescheme of a prescheme $(X, \mathscr{O}_X)$ if:

1st. Y is a locally closed subspace of X;

2nd. if U denotes the largest open subset of X containing Y such that Y is closed in U (*equivalently*, the complement in X of the boundary of Y with respect to $\overline{Y}$), then $(Y, \mathscr{O}_Y)$ is a subprescheme of $(U, \mathscr{O}_X|U)$ defined by a quasi-coherent sheaf of ideals of $\mathscr{O}_X|U$.

We say that the subprescheme $(Y, \mathscr{O}_Y)$ of $(X, \mathscr{O}_X)$ is closed if Y is closed in X (in which case $U = X$).

It follows immediately from this definition and Proposition (4.1.2) that the closed subpreschemes of X are in canonical *bijective correspondence* with the quasi-coherent sheaves of ideals $\mathscr{J}$ of $\mathscr{O}_X$, since if two such sheaves $\mathscr{J}$ and $\mathscr{J}'$ have the same (closed) support Y, and if the restrictions of $\mathscr{O}_X/\mathscr{J}$ and $\mathscr{O}_X/\mathscr{J}'$ to Y are identical, then we have $\mathscr{J}' = \mathscr{J}$.

(4.1.4). Let $(Y, \mathscr{O}_Y)$ be a subprescheme of X, U the largest open subset of X such that Y is closed (and thus contained) in U, and V an open subset of X contained in U; then $V \cap Y$ is closed in V. In addition, if Y is defined by the quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X|U$, then $\mathscr{J}|V$ is a quasi-coherent sheaf of ideals of $\mathscr{O}_X|V$, and it is immediate that the prescheme induced by Y on $Y \cap V$ is the closed subprescheme of V defined by the sheaf of ideals $\mathscr{J}|V$. Conversely:

Proposition (4.1.5). — *Let $(Y, \mathscr{O}_Y)$ be a ringed space such that Y is a subspace of X, and there exists a cover (V_α) of Y by open subsets of X such that, for each α, $Y \cap V_\alpha$ is closed in V_α, and the ringed space $(Y \cap V_\alpha, \mathscr{O}_Y|(Y \cap V_\alpha))$ is a closed subprescheme of the prescheme induced on V_α by X. Then $(Y, \mathscr{O}_Y)$ is a subprescheme of X.*

Proof. The hypotheses imply that Y is locally closed in X and that the largest open U in which Y is closed (and thus contained) contains all the V_α; we can thus reduce to the case where $U = X$ and Y is closed in X. We then define a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ by taking $\mathscr{J}|V_\alpha$ to be the sheaf of ideals of $\mathscr{O}_X|V_\alpha$ which define the closed subprescheme $(Y \cap V_\alpha, \mathscr{O}_Y|(Y \cap V_\alpha))$, and, for each open subset W of X not intersecting Y, $\mathscr{J}|W = \mathscr{O}_X|W$. We see immediately, by Definition (4.1.3) and (4.1.4), that there exists a unique sheaf of ideals $\mathscr{J}$ satisfying these conditions, and that it defines the closed subprescheme $(Y, \mathscr{O}_Y)$. ☐

In particular, the *induced* (by X) prescheme on an *open subset* of X is a *subprescheme* of X.

Proposition (4.1.6). — *A subprescheme (resp. a closed subprescheme) of a subprescheme (resp. closed subprescheme) of X is canonically identified with a subprescheme (resp. closed subprescheme) of X.* I | 121

PROOF. Since a locally closed subset of a locally closed subspace of X is a locally closed subspace of X, it is clear (4.1.5) that the question is local and that we can thus suppose that X is affine; the proposition then follows from the canonical identification of $A/\mathfrak{J}'$ with $(A/\mathfrak{J})/(\mathfrak{J}'/\mathfrak{J})$, where A is some ring, and $\mathfrak{J}$ and $\mathfrak{J}'$ are ideals of A such that $\mathfrak{J} \subset \mathfrak{J}'$. □

In what follows, we will always make use of the above identification.

(4.1.7). Let Y be a subprescheme of a prescheme X, and denote by ψ the canonical injection $Y \to X$ of the *underlying subspaces*; we know that the inverse image $\psi^*(\mathscr{O}_X)$ is the restriction $\mathscr{O}_X|Y$ (**0**, 3.7.1). If, for each $y \in Y$, we denote by ω_y the canonical homomorphism $(\mathscr{O}_X)_y \to (\mathscr{O}_Y)_y$, then these homomorphisms are the restrictions to stalks of a *surjective* homomorphism ω of sheaves of rings $\mathscr{O}_X|Y \to \mathscr{O}_Y$: indeed, it suffices to check locally on Y, that is to say, we can suppose that X is affine and that the subprescheme Y is closed; if in this case $\mathscr{I}$ is the sheaf of ideals of $\mathscr{O}_X$ which defines Y, then the ω_y are none other than the restriction to stalks of the homomorphism $\mathscr{O}_X|Y \to (\mathscr{O}_X/\mathscr{I})|Y$. We have thus defined a *monomorphism of ringed spaces* (**0**, 4.1.1) $j = (\psi, \omega^\flat)$ which is evidently a morphism $Y \to X$ of preschemes (2.2.1), and we call this the *canonical injection morphism*.

If $f : X \to Z$ is a morphism, we then say that the composite morphism $Y \xrightarrow{j} X \xrightarrow{f} Z$ is the *restriction* of f to the subprescheme Y.

(4.1.8). Conforming to the general definitions (T, I, 1.1), we will say that a morphism of preschemes $f : Z \to X$ is *majoré*[8] by the injection morphism $j : Y \to X$ of a subprescheme Y of X if f factors as $Z \xrightarrow{g} Y \xrightarrow{j} X$, where g is a morphism of preschemes; further, g is necessarily *unique* since j is a monomorphism.

Proposition (4.1.9). — *For a morphism $f : Z \to X$ to be factor through an injection morphism $j : Y \to X$, it is necessary and sufficient that $f(Z) \subset Y$ and, for all $z \in Z$, letting $y = f(z)$, that the homomorphism $(\mathscr{O}_X)_y \to \mathscr{O}_z$ corresponding to f factor as $(\mathscr{O}_Z)_y \to (\mathscr{O}_Y)_y \to \mathscr{O}_z$* (or equivalently, *for the kernel of $(\mathscr{O}_X)_y \to \mathscr{O}_z$ to contain the kernel of $(\mathscr{O}_X)_y \to (\mathscr{O}_Y)_y$).*

PROOF. The conditions are evidently necessary. To see that they are sufficient, we can reduce to the case where Y is a *closed* subprescheme of X, by replacing X by an open U such that Y is closed in U (4.1.3) if necessary; Y is then defined by a quasi-coherent sheaf of ideals $\mathscr{I}$ of $\mathscr{O}_X$. Let $f = (\psi, \theta)$, and let $\mathscr{J}$ be the sheaf of ideals of $\psi^*(\mathscr{O}_X)$, kernel of $\theta^\sharp : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Z$; considering the properties of the functor ψ^* (**0**, 3.7.2), the hypotheses imply that, for each $z \in Z$, we have $(\psi^*(\mathscr{I}))_z \subset \mathscr{J}_z$, and, as a result, that $\psi^*(\mathscr{I}) \subset \mathscr{J}$. Thus $\theta^\sharp$ factors as

$$\psi^*(\mathscr{O}_X) \longrightarrow \psi^*(\mathscr{O}_X)/\psi^*(\mathscr{I}) = \psi^*(\mathscr{O}_X/\mathscr{I}) \xrightarrow{\omega} \mathscr{O}_Z,$$

the first arrow being the canonical homomorphism. Let ψ' be the continuous map $Z \to Y$ coinciding with ψ; it is clear that we have $\psi'^*(\mathscr{O}_Y) = \psi^*(\mathscr{O}_X/\mathscr{J})$; on the other hand, ω is evidently a local homomorphism, so $g = (\psi', \omega^\flat)$ is a morphism of preschemes $Z \to Y$ (2.2.1), which, according to I | 122
the above, is such that $f = j \circ g$, hence the proposition. □

Corollary (4.1.10). — *For an injection morphism $Z \to X$ to be factor through the injection morphism $Y \to X$, it is necessary and sufficient for Z to be a subprescheme of Y.*

We then write $Z \leqslant Y$, and this condition is evidently an *ordering* on the set of subpreschemes of X.

[8] *[Trans.] There doesn't seem to be an English equivalent of this, except for 'bounded above', which doesn't make much sense in this context. We would normally just say that 'f factors through j', but to avoid having to entirely restructure the often-lengthy sentences in the original, we sometimes (but as little as we can) use 'majoré'.*

4.2. Immersion morphisms

Definition (4.2.1). — We say that a morphism $f : Y \to X$ is an immersion (resp. a closed immersion, an open immersion) if it factors as $Y \xrightarrow{g} Z \xrightarrow{j} X$, where g is an isomorphism, Z is a subprescheme of X (resp. a closed subprescheme, a subprescheme induced by an open set), and j is the injection morphism.

The subprescheme Z and the isomorphism g are then determined in a *unique* way, since if Z' is a second subprescheme of X, j' the injection $Z' \to X$, and g' an isomorphism $Y \to Z'$ such that $j \circ g = j' \circ g'$, then we have $j' = j \circ g \circ g'^{-1}$, hence $Z' \leqslant Z$ (4.1.10), and we can similarly show that $Z \leqslant Z'$, hence $Z' = Z$, and, since j is a monomorphism of preschemes, $g' = g$.

We say that $f = j \circ g$ is the *canonical factorization* of the immersion f, and the subprescheme Z and the isomorphism g are those *associated to* f.

It is clear that an immersion is a *monomorphism* of preschemes (4.1.7) and *a fortiori* a *radicial* morphism (3.5.4).

Proposition (4.2.2). —

(a) *For a morphism $f = (\psi, \theta) : Y \to X$ to be an open immersion, it is necessary and sufficient for ψ to be a homeomorphism from Y to some open subset of X, and, for all $y \in Y$, that the homomorphism $\theta_y^\sharp : \mathscr{O}_{\psi(y)} \to \mathscr{O}_y$ be bijective.*

(b) *For a morphism $f = (\psi, \theta) : Y \to X$ to be an immersion (resp. a closed immersion), it is necessary and sufficient for ψ to be a homeomorphism from Y to some locally closed (resp. closed) subset of X, and, for all $y \in Y$, that the homomorphism $\theta_y^\sharp : \mathscr{O}_{\psi(y)} \to \mathscr{O}_y$ be surjective.*

PROOF.

(a) The conditions are evidently necessary. Conversely, if they are satisfied, then it is clear that $\theta^\sharp$ is an isomorphism from $\mathscr{O}_Y$ to $\psi^*(\mathscr{O}_X)$, and $\psi^*(\mathscr{O}_X)$ is the sheaf induced by "transport of structure" via ψ^{-1} from $\mathscr{O}_X|\psi(Y)$; hence the conclusion.

(b) The conditions are evidently necessary—we prove that they are sufficient. Consider first the particular case where we suppose that X is an affine scheme, and that $Z = \psi(Y)$ is *closed* in X. We then know (**0**, 3.4.6) that $\psi_*(\mathscr{O}_Y)$ has support equal to Z, and that, denoting its restriction to Z by $\mathscr{O}'_Z$, the ringed space $(Z, \mathscr{O}'_Z)$ is induced from $(Y, \mathscr{O}_Y)$ by transport of structure via the homeomorphism ψ considered as a map from Y to Z. Let us now show that $f_*(\mathscr{O}_Y) = \psi_*(\mathscr{O}_Y)$ is a *quasi-coherent* $\mathscr{O}_X$-module. Indeed, for all $x \notin Z$, $\psi_*(\mathscr{O}_Y)$ restricted to a suitable neighbourhood of x is zero. On the contrary, if $z \in Z$, then we have $x = \psi(y)$ for some well-defined $y \in Y$; let V be an affine open neighbourhood of y in Y; $\psi(V)$ is then open in Z, and so equal to the intersection of Z with an open subset U of X, and the restriction of U to $\psi_*(\mathscr{O}_Y)$ is identical to the restriction of U to the direct image $(\psi_V)_*(\mathscr{O}_Y|V)$, where ψ_V is the restriction of ψ to V. The restriction of the morphism (ψ, θ) **I** | 123
to $(V, \mathscr{O}_Y|V)$ is a morphism from this aforementioned prescheme to $(X, \mathscr{O}_X)$, and, as a result, is of the form $({}^a\phi, \widetilde{\phi})$, where ϕ is the homomorphism from the ring $A = \Gamma(X, \mathscr{O}_X)$ to the ring $\Gamma(V, \mathscr{O}_Y)$ (1.7.3); we conclude that $(\psi_V)_*(\mathscr{O}_Y|V)$ is a quasi-coherent $\mathscr{O}_X$-module (1.6.3), which proves our assertion, due to the local nature of quasi-coherent sheaves. In addition, the hypothesis that ψ is a homeomorphism implies (**0**, 3.4.5) that, for all $y \in Y$, ψ_y is an isomorphism $(\psi_*(\mathscr{O}_Y))_{\psi(y)} \to \mathscr{O}_y$; since the diagram

$$\begin{tikzcd}
\mathscr{O}_{\psi(y)} \arrow[r, "\theta_{\psi(y)}"] \arrow[d, "\psi_y \circ \rho_{\psi(y)}"'] & (\psi_*(\mathscr{O}_Y))_{\psi(y)} \arrow[d, "\psi_y"] \\
(\psi^*(\mathscr{O}_X))_y \arrow[r, "\theta_y^\sharp"] & \mathscr{O}_y
\end{tikzcd}$$

is commutative, and the vertical arrows are the isomorphisms (**0**, 3.7.2), the hypothesis that $\theta_y^\sharp$ is surjective implies that $\theta_{\psi(y)}$ is surjective as well. Since the support of $\psi_*(\mathscr{O}_Y)$

is $Z = \psi(Y)$, θ is a *surjective* homomorphism from $\mathscr{O}_X = \widetilde{A}$ to the quasi-coherent $\mathscr{O}_X$-module $f_*(\mathscr{O}_Y)$. As a result, there exists a unique isomorphism ω from a sheaf quotient $\widetilde{A}/\widetilde{\mathfrak{J}}$ (with $\mathfrak{J}$ being an ideal of A) to $f_*(\mathscr{O}_Y)$ which, when composed with the canonical homomorphism $\widetilde{A} \to \widetilde{A}/\widetilde{\mathfrak{J}}$, gives θ (1.3.8); if $\mathscr{O}_Z$ denotes the restriction of $\widetilde{A}/\widetilde{\mathfrak{J}}$ to Z, then $(Z, \mathscr{O}_Z)$ is a subprescheme of $(X, \mathscr{O}_X)$, and f factors through the canonical injection of this subprescheme into X and the isomorphism (ψ_0, ω_0), where ψ_0 is ψ considered as a map from Y to Z, and ω_0 the restriction of ω to $\mathscr{O}_Z$.

We now pass to the general case. Let U be an affine open subset of X such that $U \cap \psi(Y)$ is closed in U and nonempty. By restricting f to the prescheme induced by Y on the open subset $\psi^{-1}(U)$, and by considering it as a morphism from this prescheme to the prescheme induced by X on U, we reduce to the first case; the restriction of f to $\psi^{-1}(U)$ is thus a closed immersion $\psi^{-1}(U) \to U$, canonically factoring as $j_U \circ g_U$, where g_U is an isomorphism from the prescheme $\psi^{-1}(U)$ to a subprescheme Z_U of U, and j_U is the canonical injection $Z_U \to U$. Let V be a second affine open subset of X such that $V \subset U$; since the restriction Z'_V of Z_U to V is a subprescheme of the prescheme V, the restriction of f to $\psi^{-1}(V)$ factors as $j'_V \circ g'_V$, where j'_V is the canonical injection $Z'_V \to V$ and g'_V is an isomorphism from $\psi^{-1}(V)$ to Z'_V. By the uniqueness of the canonical factorization of an immersion (4.2.1), we necessarily have that $Z'_V = Z_V$ and $g'_V = g_V$. We conclude (4.1.5) that there is a subprescheme Z of X whose underlying space is $\psi(Y)$, and whose restriction to each $U \cap \psi(Y)$ is Z_U; the g_U are then the restrictions to $\psi^{-1}(U)$ of an isomorphism $g : Y \to Z$ such that $f = j \circ g$, where j is the canonical injection $Z \to X$.

□

Corollary (4.2.3). — *Let X be an affine scheme. For a morphism $f = (\psi, \theta) : Y \to X$ to be a closed immersion, it is necessary and sufficient for Y to be an affine scheme, and the homomorphism $\Gamma(\psi) : \Gamma(\mathscr{O}_X) \to \Gamma(\mathscr{O}_Y)$ to be surjective.*

Corollary (4.2.4). —

(a) *Let f be a morphism $Y \to X$, and (V_λ) a cover of $f(Y)$ by open subsets of X. For f to be an immersion (resp. an open immersion), it is necessary and sufficient for its restriction to each of the* I | 124 *induced preschemes $f^{-1}(V_\lambda)$ to be an immersion (resp. an open immersion) into V_λ.*

(b) *Let f be a morphism $Y \to X$, and (V_λ) an open cover of X. For f to be a closed immersion, it is necessary and sufficient for its restriction to each of the induced preschemes $f^{-1}(V_\lambda)$ to be a closed immersion into V_λ.*

PROOF. Let $f = (\psi, \theta)$; in the case (a), $\theta^\sharp_y$ is surjective (resp. bijective) for all $y \in Y$, and in the case (b), $\theta^\sharp_y$ is surjective for all $y \in Y$; it thus suffices to check, in case (a), that ψ is a homeomorphism from Y to a locally closed (resp. open) subset of X, and, in case (b), that ψ is a homeomorphism from Y to a closed subset of X. Now ψ is evidently injective, and sends each neighbourhood of y in Y to a neighbourhood of $\psi(y)$ is $\psi(Y)$ for all $y \in Y$, by virtue of the hypothesis; in case (a), $\psi(Y) \cap V_\lambda$ is locally closed (resp. open) in V_λ, so $\psi(Y)$ is locally closed (resp. open) in the union of the V_λ, and *a fortiori* in X; in case (b), $\psi(Y) \cap V_\lambda$ is closed in V_λ, so $\psi(Y)$ is closed in X since $X = \bigcup_\lambda V_\lambda$. □

Proposition (4.2.5). — *The composition of any two immersions (resp. of two open immersions, of two closed immersions) is an immersion (resp. an open immersion, a closed immersion).*

PROOF. This follows easily from (4.1.6). □

4.3. Products of immersions

Proposition (4.3.1). — *Let $\alpha : X' \to X$, $\beta : Y' \to Y$ be two S-morphisms; if α and β are immersions (resp. open immersions, closed immersions), then $\alpha \times_S \beta$ is an immersion (resp. an open immersion, a closed immersion). In addition, if α (resp. β) identifies X' (resp. Y') with a subprescheme X'' (resp. Y'') of X (resp. Y), then $\alpha \times_S \beta$ identifies the underlying space of $X' \times_S Y'$ with the subspace $p^{-1}(X'') \cap q^{-1}(Y'')$ of the underlying space of $X \times_S Y$, where p and q denote the projections from $X \times_S Y$ to X and Y respectively.*

PROOF. According to Definition (4.2.1), we can restrict to the case where X' and Y' are subpreschemes, and α and β the injection morphisms. The proposition has already been proven for the subpreschemes induced by open sets (3.2.7); since every subprescheme is a closed subprescheme of a prescheme induced by an open set (4.1.3), we can reduce to the case where X' and Y' are *closed* subpreschemes.

Let us first show that we can assume S to be *affine*. Let (S_λ) be a cover of S by affine open sets; if ϕ and ψ are the structure morphisms of X and Y, then let $X_\lambda = \phi^{-1}(S_\lambda)$ and $Y_\lambda = \psi^{-1}(S_\lambda)$. The restriction X'_λ (resp. Y'_λ) of X' (resp. Y') to $X_\lambda \cap X'$ (resp. $Y_\lambda \cap Y'$) is a closed subprescheme of X_λ (resp. Y_λ), the preschemes X_λ, Y_λ, X'_λ, and Y'_λ can be considered as S_λ-preschemes, and the products $X_\lambda \times_S Y_\lambda$ and $X_\lambda \times_{S_\lambda} Y_\lambda$ (resp. $X'_\lambda \times_S Y'_\lambda$ and $X'_\lambda \times_{S_\lambda} Y'_\lambda$) are identical (3.2.5). If the proposition is true when S is affine, then the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\lambda$ is thus an immersion (3.2.7). Since the product $X'_\lambda \times_S Y'_\mu$ (resp. $X_\lambda \times_S Y_\mu$) can be identified with $(X'_\lambda \cap X'_\mu) \times_S (Y'_\lambda \cap Y'_\mu)$ (resp. $(X_\lambda \cap X_\mu) \times_S (Y_\lambda \cap Y_\mu)$) (3.2.6.4), the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\mu$ is also an immersion; the same is true for $\alpha \times_S \beta$ by (4.2.4). **I** | 125

Next, we show that we can assume X and Y to be *affine*. Indeed, let (U_i) (resp. (V_j)) be a cover of X (resp. Y) by affine open sets, and let X'_i (resp. Y'_j) be the restriction of X' (resp. Y') to $X' \cap U_i$ (resp. $Y' \cap V_j$), which is a closed subprescheme of U_i (resp. V_j); $U_i \times_S V_j$ can be identified with the restriction of $X \times_S Y$ to $p^{-1}(U_i) \cap q^{-1}(V_j)$ (3.2.7); similarly, if p' and q' are the projections from $X' \times_S Y'$, then $X'_i \times_S Y'_j$ can be identified with the restriction of $X' \times_S Y'$ to $p'^{-1}(X'_i) \cap q'^{-1}(Y'_j)$. Set $\gamma = \alpha \times_S \beta$; we have, by definition, $p \circ \gamma = \alpha \circ p'$ and $q \circ \gamma = \beta \circ q'$; since $X'_i = \alpha^{-1}(U_i)$ and $Y'_j = \beta^{-1}(V_j)$, we also have $p'^{-1}(X'_i) = \gamma^{-1}(p^{-1}(U_i))$ and $q'^{-1}(Y'_j) = \gamma^{-1}(q^{-1}(V_j))$, hence

$$p'^{-1}(X'_i) \cap q'^{-1}(Y'_j) = \gamma^{-1}(p^{-1}(U_i) \cap q^{-1}(V_j)) = \gamma^{-1}(U_i \times_S V_j),$$

and we then conclude as in the previous part of the proof.

So suppose X, Y, and S are affine, and let B, C, and A be their respective rings. Then B and C are A-algebras, and X' and Y' are affine schemes whose rings are quotient algebras B' and C' of B and C respectively. In addition, we have $\alpha = ({}^a\rho, \widetilde{\rho})$ and $\beta = ({}^a\sigma, \widetilde{\sigma})$, where ρ and σ are (respectively) the canonical homomorphisms $B \to B'$ and $C \to C'$ (1.7.3). With that in mind, we know that $X \times_S Y$ (resp. $X' \times_S Y'$) is an affine scheme with ring $B \otimes_A C$ (resp. $B' \otimes_A C'$), and $\alpha \times_S \beta = ({}^a\tau, \widetilde{\tau})$, where τ is the homomorphism $\rho \otimes \sigma$ from $B \otimes_A C$ to $B' \otimes_A C'$ (Proposition (3.2.2) and Corollary (3.2.3)); since this homomorphism is surjective, $\alpha \times_S \beta$ is an immersion. In addition, if $\mathfrak{b}$ (resp. $\mathfrak{c}$) is the kernel of ρ (resp. σ), then the kernel of τ is $u(\mathfrak{b}) + v(\mathfrak{c})$, where u (resp. v) is the homomorphism $b \mapsto b \otimes 1$ (resp. $c \mapsto 1 \otimes c$). Since $p = ({}^a u, \widetilde{u})$ and $q = ({}^a v, \widetilde{v})$, this kernel corresponds, in the prime spectrum of $B \otimes_A C$, to the closed set $p^{-1}(X') \cap q^{-1}(Y')$ ((1.2.2.1) and Proposition (1.1.2, iii)), which finishes the proof. □

Corollary (4.3.2). — *If $f : X \to Y$ is an immersion (resp. an open immersion, a closed immersion) and an S-morphism, then $f_{(S')}$ is an immersion (resp. an open immersion, a closed immersion) for every extension $S' \to S$ of the base prescheme.*

4.4. Inverse images of a subprescheme

Proposition (4.4.1). — *Let $f : X \to Y$ be a morphism, Y' a subprescheme (resp. a closed subprescheme, a prescheme induced by an open set) of Y, and $j : Y' \to Y$ the injection morphism. Then the projection $p : X \times_Y Y' \to X$ is an immersion (resp. a closed immersion, an open immersion); the underlying space of the subprescheme of X associated to p is $f^{-1}(Y')$; in addition, if j' is the injection morphism of this subprescheme into X, then for a morphism $h : Z \to X$ to be such that $f \circ h : Z \to Y$ factors through j, it is necessary and sufficient for h to factor through j'.*

PROOF. Since $p = 1_X \times_Y j$ (3.3.4), the first claim follows from Proposition (4.3.1); the second is a particular case of Corollary (3.5.10) (after swapping the roles of X and Y'). Finally, if we have $f \circ h = j \circ h'$, where h' is a morphism $Z \to Y'$, then it follows from the definition of the product that we have $h = p \circ u$, where u is a morphism $Z \to X \times_Y Y'$, whence the last claim. □

We say that the subprescheme of X thus defined is the *inverse image* of the subprescheme Y' of Y under the morphism f, terminology which is consistent with that introduced more generally in (3.3.6). When we speak of $f^{-1}(Y')$ as a subprescheme of X, this will always be the subprescheme we mean. **I** | 126

When the preschemes $f^{-1}(Y')$ and X are identical, j' is the identity and each morphism $h : Z \to X$ thus factors through j', so *the morphism $f : X \to Y$ factors as $X \xrightarrow{g} Y' \xrightarrow{j} Y$.*

When y is a *closed* point of Y and $Y' = \operatorname{Spec}(k(y))$ is the smallest closed subprescheme of Y having $\{y\}$ as its underlying space (4.1.9), the closed subprescheme $f^{-1}(Y')$ is canonically isomorphic to the *fibre* $f^{-1}(y)$ defined in (3.6.2), and we will use this identification in all that follows.

Corollary (4.4.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms, and $h = g \circ f$ their composition. For each subprescheme Z' of Z, the subpreschemes $f^{-1}(g^{-1}(Z'))$ and $h^{-1}(Z')$ of X are identical.*

PROOF. This follows from the existence of the canonical isomorphism $X \times_Y (Y \times_Z Z') \simeq X \times_Z Z'$ (3.3.9.1). □

Corollary (4.4.3). — *Let X' and X'' be subpreschemes of X, and $j' : X' \to X$, and $j'' : X'' \to X$ their injection morphisms; then ${j'}^{-1}(X'')$ and ${j''}^{-1}(X')$ are both equal to the greatest lower bound $\inf(X', X'')$ of X' and X'' for the ordering $\leqslant$ on subpreschemes, and this is canonically isomorphic to $X' \times_X X''$.*

PROOF. This follows immediately from Proposition (4.4.1) and Corollary (4.1.10). □

Corollary (4.4.4). — *Let $f : X \to Y$ be a morphism, and Y' and Y'' subpreschemes of Y; then we have $f^{-1}(\inf(Y', Y'')) = \inf(f^{-1}(Y'), f^{-1}(Y''))$.*

PROOF. This follows from the existence of the canonical isomorphism between $(X \times_Y Y') \times_X (X \times_Y Y'')$ and $X \times_Y (Y' \times_Y Y'')$ (3.3.9.1). □

Proposition (4.4.5). — *Let $f : X \to Y$ be a morphism, and Y' a closed subprescheme of Y defined by a quasi-coherent sheaf of ideals $\mathscr{K}$ of $\mathscr{O}_Y$ (4.1.3); the closed subprescheme $f^{-1}(Y')$ of X is then defined by the quasi-coherent sheaf of ideals $f^*(\mathscr{K})\mathscr{O}_X$ of $\mathscr{O}_X$.*

PROOF. The statement is evidently local on X and Y; it thus suffices to note that if A is a B-algebra and $\mathfrak{K}$ an ideal of B, then we have $A \otimes_B (B/\mathfrak{K}) = A/\mathfrak{K}A$, and to then apply (1.6.9). □ **Err**$_{\mathbf{II}}$

Corollary (4.4.6). — *Let X' be a closed subprescheme of X defined by a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$, and i the injection $X' \to X$; for the restriction $f \circ i$ of f to X' to factor through the injection $j : Y' \to Y$* (in other words, for it to factor as $j \circ g$, with g a morphism $X' \to Y'$), *it is necessary and sufficient that $f^*(\mathscr{K}) \subset \mathscr{J}$.*

PROOF. It suffices to apply Proposition (4.4.1) to i, taking Proposition (4.4.5) into account. □

4.5. Local immersions and local isomorphisms

Definition (4.5.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is a local immersion at a point $x \in X$ if there exists an open neighbourhood U of x in X and an open neighbourhood V of $f(x)$ in Y such that the restriction of f to the induced prescheme U is a closed immersion of U into the induced prescheme V. We say that f is a local immersion if f is a local immersion at each point of X.

Definition (4.5.2). — We say that a morphism $f : X \to Y$ is a local isomorphism at a point $x \in X$ I | 127 if there exists an open neighbourhood U of x in X such that the restriction of f to the induced prescheme U is an open immersion of U into Y. We say that f is a local isomorphism if f is a local isomorphism at each point of X.

(4.5.3). An immersion (resp. a closed immersion) $f : X \to Y$ can be characterized as a local immersion such that f is a homeomorphism from the underlying space of X to a subset (resp. a closed subset) of Y. An open immersion f can be characterized as an *injective* local isomorphism.

Proposition (4.5.4). — *Let X be an irreducible prescheme, and $f : X \to Y$ a dominant injective morphism. If f is a local immersion, then f is an immersion, and $f(X)$ is open in Y.*

PROOF. Let $x \in X$, and let U be an open neighbourhood of x, and V an open neighbourhood of $f(x)$ in Y such that the restriction of f to U is a closed immersion into V; since U is dense in X, $f(U)$ is dense in Y by hypothesis, so $f(U) = V$, and f is a homeomorphism from U to V; the hypothesis that f is injective implies that $f^{-1}(V) = U$, hence the proposition. □

Proposition (4.5.5). —

(i) *The composition of any two local immersions (resp. of two local isomorphisms) is a local immersion (resp. a local isomorphism).*
(ii) *Let $f : X \to X'$ and $g : Y \to Y'$ be two S-morphisms. If f and g are local immersions (resp. local isomorphisms), then so too is $f \times_S g$.*
(iii) *If an S-morphism f is a local immersion (resp. a local isomorphism), then so too is $f_{(S')}$ for every extension $S' \to S$ of the base prescheme.*

PROOF. According to (3.5.1), it suffices to prove (i) and (ii).

(i) follows immediately from the transitivity of closed (resp. open) immersions (4.2.5) and from the fact that if f is a homeomorphism from X to a closed subset of Y, then for every open $U \subset X$, $f(U)$ is open in $f(X)$, so there exists an open subset V of Y such that $f(U) = V \cap f(X)$, and, as a result, $f(U)$ is closed in V.

To prove (ii), let p and q be the projections from $X \times_X Y$, and p' and q' the projections from $X' \times_S Y'$. There exists, by hypothesis, open neighbourhoods U, U', V, and V' of $x = p(z)$, $x' = p'(z')$, $y = q(z)$, and $y' = q'(z')$ (respectively), such that the restrictions of f and g to U and V (respectively) are closed (resp. open) immersions into U' and V' (respectively). Since the underlying spaces of $U \times_S V$ and $U' \times_S V'$ can be identified with the open neighbourhoods $p^{-1}(U) \cap q^{-1}(V)$ and $p'^{-1}(U') \cap q'^{-1}(V')$ of z and z' (respectively) (3.2.7), the proposition follows from Proposition (4.3.1). □

§5. REDUCED PRESCHEMES; THE SEPARATION CONDITION

5.1. Reduced preschemes

Proposition (5.1.1). — *Let $(X, \mathscr{O}_X)$ be a prescheme, and $\mathscr{B}$ a quasi-coherent $\mathscr{O}_X$-algebra. Then there exists a unique quasi-coherent $\mathscr{O}_X$-module $\mathscr{N}$ whose stalk $\mathscr{N}_x$ at any $x \in X$ is the nilradical of the ring $\mathscr{B}_x$. When X is affine, and, consequently, $\mathscr{B} = \widetilde{B}$, where B is an algebra over $A(X)$, then we have $\mathscr{N} = \widetilde{\mathfrak{N}}$, where $\mathfrak{N}$ is the nilradical of B.*

PROOF. The statement is local, so it suffices to show the latter claim. We know that $\widetilde{\mathfrak{N}}$ is a I | 128 quasi-coherent $\mathscr{O}_X$-module (1.4.1), and that its stalk at a point $x \in X$ is the ideal $\mathfrak{N}_x$ of the ring of fractions B_x; it remains to prove that the nilradical of B_x is contained in $\mathfrak{N}_x$, the converse inclusion being evident. Let z/s be an element of the nilradical of B_x, with $z \in B$, and $s \notin \mathfrak{j}_x$; by hypothesis, there exists an integer k such that $(z/s)^k = 0$, which implies that there exists some $t \notin \mathfrak{j}_x$ such that $tz^k = 0$. We conclude that $(tz)^k = 0$, and, as a result, that $z/s = (tz)/(ts) \in \mathfrak{N}_x$. □

We say that the quasi-coherent $\mathscr{O}_X$-module $\mathscr{N}$ thus defined is the *nilradical* of the $\mathscr{O}_X$-algebra $\mathscr{B}$; in particular, we denote by $\mathscr{N}_X$ the nilradical of $\mathscr{O}_X$.

Corollary (5.1.2). — *Let X be a prescheme; the closed subprescheme of X defined by the sheaf of ideals $\mathscr{N}_X$ is the only reduced subprescheme (**0**, 4.1.4) of X that has X as its underlying space; it is also the smallest subprescheme of X that has X as its underlying space.*

PROOF. Since the structure sheaf of the closed subprescheme of Y defined by $\mathscr{N}_X$ is $\mathscr{O}_X/\mathscr{N}_X$, it is immediate that Y is reduced and has X as its underlying space, because $\mathscr{N}_x \neq \mathscr{O}_x$ for any $x \in X$. To show the other claims, note that a subprescheme Z of X that has X as its underlying space is defined by a sheaf of ideals $\mathscr{I}$ (4.1.3) such that $\mathscr{I}_x \neq \mathscr{O}_x$ for any $x \in X$. We can restrict to the case where X is affine, say $X = \operatorname{Spec}(A)$ and $\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is an ideal of A; then, for every $x \in X$, we have $\mathfrak{I}_x \subset \mathfrak{j}_x$, and so $\mathfrak{I}$ is contained in every prime ideal of A, and so also in their intersection $\mathfrak{N}$, the nilradical of A. This proves that Y is the small subprescheme of X that has X as its underlying space (4.1.9); furthermore, if Z is distinct from Y, we necessarily have $\mathscr{I}_x \neq \mathscr{N}_x$ for at least one $x \in X$, and so (5.1.1) Z is not reduced. □

Definition (5.1.3). — We define the reduced prescheme associated to a prescheme X, denoted by X_{red}, to be the unique reduced subprescheme of X that has X as its underlying space.

Saying that a prescheme X is reduced thus implies that $X = X_{\text{red}}$.

Proposition (5.1.4). — *For the prime spectrum of a ring A to be a reduced (resp. integral) prescheme (2.1.7), it is necessary and sufficient for A to be a reduced (resp. integral) ring.*

PROOF. Indeed, it follows immediately from (5.1.1) that the condition $\mathscr{N} = (0)$ is necessary and sufficient for $X = \operatorname{Spec}(A)$ to be reduced; the claim corresponding to integral rings is then a consequence of (1.1.13). □

Since every ring of fractions $\neq \{0\}$ of an integral ring is integral, it follows from (5.1.4) that, for every *locally integral* prescheme X, $\mathscr{O}_x$ is an *integral* ring for every $x \in X$. The converse is true whenever the underlying space of X is *locally Noetherian*: indeed, X is then reduced, and if U is an affine open subset of X, which is a Noetherian space, then U has only a finite number of irreducible components, and so its ring A has only a finite number of minimal prime ideals (1.1.14). If two of the irreducible components of U had a common point x, then $\mathscr{O}_x$ would have at least two distinct minimal prime ideals, and would thus not be integral; the components of U are thus open subsets that are pairwise disjoint, and each of them is thus integral.

(5.1.5). Let $f = (\psi, \theta) : X \to Y$ be a morphism of preschemes; the homomorphism $\theta_x^\sharp : \mathscr{O}_{\psi(x)} \to \mathscr{O}_x$ **I** | 129
sends each nilpotent element of $\mathscr{O}_{\psi(x)}$ to a nilpotent element of $\mathscr{O}_x$; by passing to the quotients, $\theta^\sharp$ induces a homomorphism

$$\omega : \psi^*(\mathscr{O}_Y/\mathscr{N}_Y) \longrightarrow \mathscr{O}_X/\mathscr{N}_X.$$

It is clear that, for every $x \in X$, $\omega_x : \mathscr{O}_{\psi(x)}/\mathscr{N}_{\psi(x)} \to \mathscr{O}_x/\mathscr{N}_x$ is a local homomorphism, and so $(\psi, \omega^\flat)$ is a morphism of preschemes $X_{\text{red}} \to Y_{\text{red}}$, which we denote by f_{red}, and call the *reduced* morphism associated to f. It is immediate that, for morphisms $f : X \to Y$ and $g : Y \to Z$, we have $(g \circ f)_{\text{red}} = g_{\text{red}} \circ f_{\text{red}}$, and so we have defined X_{red} as a *functor*, *covariant* in X.

The preceding definition shows that the diagram

$$\begin{tikzcd} X_{\text{red}} \arrow[r, "f_{\text{red}}"] \arrow[d] & Y_{\text{red}} \arrow[d] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

is commutative, where the vertical arrows are the injection morphisms; in other words, $X_{\text{red}} \to X$ is a *functorial* morphism. We note in particular that, if X is reduced, then every morphism $f : X \to Y$ factors as $X \xrightarrow{f_{\text{red}}} Y_{\text{red}} \to Y$; in other words, f factors through the injection morphism $Y_{\text{red}} \to Y$.

Proposition (5.1.6). — *Let $f : X \to Y$ be a morphism; if f is surjective (resp. radicial, an immersion, a closed immersion, an open immersion, a local immersion, a local isomorphism), then so too is f_{red}. Conversely, if f_{red} is surjective (resp. radicial), then so too is f.*

PROOF. The proposition is trivial if f is surjective; if f is radicial, then the proposition follows from the fact that, for every $x \in X$, the field $k(x)$ is the same for the preschemes X and X_{red} (3.5.8). Finally, if $f = (\psi, \theta)$ is an immersion, a closed immersion, or a local immersion (resp. an open immersion, or a local isomorphism), then the proposition follows from the fact that, if $\theta_x^\sharp$ is surjective (resp. bijective), then so too is the homomorphism obtained by passing to the quotients by the nilradicals $\mathcal{O}_{\psi(x)}$ and $\mathcal{O}_x$ ((5.1.2) and (4.2.2)) (cf. (5.5.12)). □

Proposition (5.1.7). — *If X and Y are S-preschemes, then the preschemes $X_{\mathrm{red}} \times_{S_{\mathrm{red}}} Y_{\mathrm{red}}$ and $X_{\mathrm{red}} \times_S Y_{\mathrm{red}}$ are identical, and canonically identified with a subprescheme of $X \times_S Y$ that has the same underlying subspace as the two aforementioned products.*

PROOF. The canonical identification of $X_{\mathrm{red}} \times_S Y_{\mathrm{red}}$ with a subprescheme of $X \times_S Y$ that has the same underlying space follows from (4.3.1). Furthermore, if ϕ and ψ are the structure morphisms $X_{\mathrm{red}} \to S$ and $Y_{\mathrm{red}} \to S$ (respectively), then they factor through S_{red} (5.1.5), and since $S_{\mathrm{red}} \to S$ is a monomorphism, the first claim of the proposition follows from (3.2.4). □

Corollary (5.1.8). — *The preschemes $(X \times_S Y)_{\mathrm{red}}$ and $(X_{\mathrm{red}} \times_{S_{\mathrm{red}}} Y_{\mathrm{red}})_{\mathrm{red}}$ are canonically identified with one another.*

PROOF. This follows from (5.1.2) and (5.1.7). □

We note that, even if X and Y are reduced preschemes, $X \times_S Y$ might not be reduced, because the tensor product of two reduced algebras can have nilpotent elements.

Proposition (5.1.9). — *Let X be a prescheme, and $\mathscr{I}$ a quasi-coherent sheaf of ideals of $\mathcal{O}_X$ such that $\mathscr{I}^n = 0$ for some integer $n > 0$. Let X_0 be the closed subprescheme $(X, \mathcal{O}_X/\mathscr{I})$ of X; for X to be an affine scheme, it is necessary and sufficient for X_0 to be an affine scheme.* I | 130

The condition is clearly necessary, so we will show that it is sufficient. If we set $X_k = (X, \mathcal{O}_X/\mathscr{I}^{k+1})$, it is enough to prove by induction on k that X_k is affine, and so we are led to consider the base case, where $\mathscr{I}^2 = 0$. We set

$$A = \Gamma(X, \mathcal{O}_X)$$
$$A_0 = \Gamma(X_0, \mathcal{O}_{X_0}) = \Gamma(X, \mathcal{O}_X/\mathscr{I}).$$

The canonical homomorphism $\mathcal{O}_X \to \mathcal{O}_X/\mathscr{I}$ induces a homomorphism of rings $\phi : A \to A_0$. We will see below that ϕ is *surjective*, which implies that

(5.1.9.1) $$0 \longrightarrow \Gamma(X, \mathscr{I}) \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X, \mathcal{O}_X/\mathscr{I}) \longrightarrow 0$$

is an *exact* sequence. We now prove, assuming that this is true, the proposition. Note that $\mathfrak{K} = \Gamma(X, \mathscr{I})$ is an ideal whose square is zero in A, and thus a module over $A_0 = A/\mathfrak{K}$. By hypothesis, we have $X_0 = \operatorname{Spec}(A)$, and, since the underlying spaces of X_0 and X are identical, $\mathfrak{K} = \Gamma(X_0, \mathscr{I})$; Additionally, since $\mathscr{I}^2 = 0$, $\mathscr{I}$ is a quasi-coherent $(\mathcal{O}_X/\mathscr{I})$-module, so we have $\mathscr{I} \cong \widetilde{\mathfrak{K}}$ and $\mathfrak{K}_x = \mathscr{I}_x$ for all $x \in X_0$ (1.4.1). With this in mind, let $X' = \operatorname{Spec}(A)$, and consider the morphism $f = (\psi, \theta) : X \to X'$ of preschemes that corresponds to the identity map $A \to \Gamma(X, \mathcal{O}_X)$ (2.2.4). For every affine open subset V of X, the diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & \Gamma(V, \mathcal{O}_X|V) \arrow[d] \\ A_0 = A/\mathfrak{K} \arrow[r] & \Gamma(V, \mathcal{O}_{X_0}|V) \end{tikzcd}$$

commutes, whence the diagram

$$\begin{tikzcd} X' & X \arrow[l, "f"'] \\ X'_0 \arrow[u, "j'"] & X_0 \arrow[l, "f_0"'] \arrow[u, "j"'] \end{tikzcd}$$

also commutes (X'_0 being the closed subprescheme of X' defined by the quasi-coherent sheaf of ideals $\widetilde{R}$, and j and j' being the canonical injection morphisms). But since X_0 is affine, f_0 is an isomorphism, and since the underlying continuous maps of j and j' are identity maps, we see straight away that $\psi : X \to X'$ is a homeomorphism. Furthermore, the equation $\mathfrak{K}_x = \mathscr{I}_x$ shows that the restriction of $\theta^\sharp : \psi^*(\mathscr{O}_{X'}) \to \mathscr{O}_X$ is an *isomorphism* from $\psi^*(\widetilde{\mathfrak{K}})$ to $\mathscr{I}$; additionally, by passing to the quotients, $\theta^\sharp$ gives an *isomorphism* $\psi^*(\mathscr{O}_X/\widetilde{\mathfrak{K}}) \to \mathscr{O}_X/\mathscr{I}$, because f_0 is an isomorphism; we thus immediately conclude, by the 5 lemma (M, I, 1.1), that $\theta^\sharp$ is itself an isomorphism, and thus that f is an *isomorphism*, and thus that X is affine. So everything reduces to proving the exactitude of (5.1.9.1), which will follow from showing that $\mathrm{H}^1(X, \mathscr{I}) = 0$. But $\mathrm{H}^1(X, \mathscr{I}) = \mathrm{H}^1(X_0, \mathscr{I})$, and we have seen **I** | 131
that $\mathscr{I}$ is a quasi coherent $\mathscr{O}_{X_0}$-module. Our proof will thus follow from

Lemma (5.1.9.2). — *If Y is an affine scheme, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_Y$-module, then $\mathrm{H}^1(Y, \mathscr{F}) = 0$.*

PROOF. This lemma will be proven in Chapter III, §1, as a consequence of the more general theorem that $\mathrm{H}^i(Y, \mathscr{F}) = 0$ for all $i > 0$. To give an independent proof, note that $\mathrm{H}^1(Y, \mathscr{F})$ can be identified with the module $\mathrm{Ext}^1_{\mathscr{O}_Y}(Y; \mathscr{O}_Y, \mathscr{F})$ of extensions classes of the $\mathscr{O}_Y$-module $\mathscr{O}_Y$ by the $\mathscr{O}_Y$-module $\mathscr{F}$ (T, 4.2.3); so everything reduces to proving that such an extension $\mathscr{G}$ is trivial. But, for all $y \in Y$, there is a neighbourhood V of y in Y such that $\mathscr{G}|V$ is isomorphic to $\mathscr{F}|Y \oplus \mathscr{O}_Y|V$ (**0**, 5.4.9); from this we conclude that $\mathscr{G}$ is a *quasi-coherent* $\mathscr{O}_Y$-module. If A is the ring of Y, then we have $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$, where M and N are A-modules, and, by hypothesis, N is an extension of the A-module A by the A-module M (1.3.11). Since this extension is necessarily trivial, the lemma is proven, and thus so too is (5.1.9). □

Corollary (5.1.10). — *Let X be a prescheme such that $\mathscr{N}_X$ is nilpotent. For X to be an affine scheme, it is necessary and sufficient for X_{red} to be an affine scheme.*

5.2. Existence of a subprescheme with a given underlying space

Proposition (5.2.1). — *For every locally closed subspace Y of the underlying space of a prescheme X, there exists exactly one reduced subprescheme of X that has Y as its underlying space.*

PROOF. The uniqueness follows from (5.1.2), so it remains only to show the existence of the prescheme in question.

If X is affine, given by some ring A, and Y closed in X, then the proposition is immediate: $\mathfrak{j}(Y)$ is the largest ideal $\mathfrak{a} \subset A$ such that $V(\mathfrak{a}) = Y$, and it is equal to its radical (1.1.4, i), so $A/\mathfrak{j}(Y)$ is a reduced ring.

In the general case, for every affine open $U \subset X$ such that $U \cap Y$ is closed in U, consider the closed subprescheme Y_U of U defined by the sheaf of ideals associated to the ideal $\mathfrak{j}(U \cap Y)$ of $A(U)$, which is reduced. We can show that, if V is an affine open subset of X contained in U, then Y_V is *induced* by Y_U on $V \cap Y$; but this induced prescheme is a closed subprescheme (of V) which is reduced and has $V \cap Y$ as its underlying space; the uniqueness of Y_V thus implies our claim. □

Proposition (5.2.2). — *Let X be reduced, $f : X \to Y$ a morphism, and Z a sub-prescheme closed over Y such that $f(X) \subset Z$. Then, f factors as $X \xrightarrow{g} Z \xrightarrow{j} Y$, where j is an injective morphism.*

PROOF. It follows from the hypotheses that the closed subprescheme $f^{-1}(Z)$ of X has all of X as its underlying space (4.4.1); since X is reduced, this closed subprescheme agrees with X (5.1.2), and the proposition then follows from (4.4.1). □

Corollary (5.2.3). — *Let X be a reduced subprescheme of a prescheme Y; if Z is the closed reduced subprescheme of Y that has $\overline{X}$ as its underlying space, then X is a subprescheme induced on an open subset of Z.*

PROOF. There is indeed an open subset U of Y such that $X = U \cap \overline{X}$; since, by (5.2.2), X is a reduced subprescheme of Z, the subprescheme X is induced by Z on the open subspace X by uniqueness (5.2.1). ☐ I | 132

Corollary (5.2.4). — *Let $f : X \to Y$ be a morphism, and X' (resp. Y') a closed subprescheme of X (resp. Y) defined by a quasi-coherent sheaf of ideals $\mathscr{J}$ (resp. $\mathscr{K}$) of $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$). Suppose that X' is reduced, and that $f(X') \subset Y'$. Then $f^*(\mathscr{K})\mathscr{O}_X \subset \mathscr{J}$.*

PROOF. Since, by (5.2.2), the restriction of f to X' factors as $X' \to Y' \to Y$, it suffices to apply (4.4.6). ☐

5.3. Diagonal; graph of a morphism

(5.3.1). Let X be an S-prescheme; we define the *diagonal morphism* of X in $X \times_S X$, denoted by $\Delta_{X|S}$, or Δ_X, or even Δ if no confusion may arise, to be the S-morphism $(1_X, 1_X)_S$, or, in other words, the unique S-morphism Δ_X such that

(5.3.1.1) $$p_1 \circ \Delta_X = p_2 \circ \Delta_X = 1_X,$$

where p_1 and p_2 are the projections of $X \times_S X$ (Definition (3.2.1)). If $f : T \to X$ and $g : T \to Y$ are S-morphisms, we immediately have that

(5.3.1.2) $$(f, g)_S = (f \times_S g) \circ \Delta_{T|S}.$$

The reader will note that the preceding definition and the results stated in (5.3.1) to (5.3.8) are true in any category, *provided that the products used within exist in the category.*

Proposition (5.3.2). — *Let X and Y be S-preschemes; if we make the canonical identification between $(X \times Y) \times (X \times Y)$ and $(X \times X) \times (Y \times Y)$, then the morphism $\Delta_{X \times Y}$ is identified with $\Delta_X \times \Delta_Y$.*

PROOF. Indeed, if $p_1 : X \times X \to X$ and $q_1 : Y \times Y \to Y$ are the projections onto the first component, then the projection onto the first component $(X \times Y) \times (X \times Y) \to X \times Y$ is identified with $p_1 \times q_1$, and we have

$$(p_1 \times q_1) \circ (\Delta_X \times \Delta_Y) = (p_1 \circ \Delta_X) \times (q_1 \circ \Delta_Y) = 1_{X \times Y}$$

and we can argue similarly for the projections onto the second component. ☐

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Corollary (5.3.4). — *For every extension $S' \to S$ of the base prescheme, $\Delta_{X_{S'}}$ is canonically identified with $(\Delta_X)_{(S')}$.*

PROOF. It suffices to remark that $(X \times_S X)_{(S')}$ is canonically identified with $X_{(S')} \times_{S'} X_{(S')}$ (3.3.10). ☐

Proposition (5.3.5). — *Let X and Y be S-preschemes, and $\phi : S \to T$ a morphism of preschemes, which lets us consider every S-prescheme as a T-prescheme. Let $f : X \to S$ and $g : Y \to S$ be the structure morphisms, p and q the projections of $X \times_S Y$, and $\pi = f \circ p = g \circ q$ the structure morphism $X \times_S Y \to S$. Then the diagram*

(5.3.5.1) $$\begin{tikzcd} X \times_S Y \arrow[r, "(p,q)_T"] \arrow[d, "\pi"'] & X \times_T Y \arrow[d, "f \times_T g"] \\ S \arrow[r, "\Delta_{S|T}"] & S \times_T S \end{tikzcd}$$

commutes, and identifies $X \times_S Y$ with the product of the $(S \times_T S)$-preschemes S and $X \times_T Y$, and the projections with π and $(p, q)_T$. I | 133

PROOF. By (3.4.3), we are led to proving the corresponding proposition *in the category of sets*, replacing X, Y, and S by $X(Z)_T$, $Y(Z)_T$, and $S(Z)_T$ (respectively), with Z being an arbitrary T-prescheme. But, in the category of sets, the proof is immediate and left to the reader. ☐

Corollary (5.3.6). — *The morphism $(p, q)_T$ can be identified (letting $P = S \times_T S$) with $1_{X \times_T Y} \times_P \Delta_S$.*

PROOF. This follows from (5.3.5) and (3.3.4). □

Corollary (5.3.7). — *If $f : X \to Y$ is an S-morphism, then the diagram*

$$\begin{tikzcd} X \arrow[r, "(1_X,f)_S"] \arrow[d, "f"'] & X \times_S Y \arrow[d, "f\times_S 1_Y"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

commutes, and identifies X with the product of the $(Y \times_S Y)$-preschemes Y and $X \times_S Y$.

PROOF. It suffices to apply (5.3.5), replacing S by Y, and T by S, and noting that $X \times_Y Y = X$ (3.3.3). □

Proposition (5.3.8). — *For $f : X \to Y$ to be a monomorphism of preschemes, it is necessary and sufficient for $\Delta_{X|Y}$ to be an isomorphism from X to $X \times_Y X$.*

PROOF. Indeed, to say that f is a monomorphism implies that, for every Y-prescheme Z, the corresponding map $f' : X(Z)_Y \to Y(Z)_Y$ is an injection, and, since $Y(Z)_Y$ consists of a single element, this implies that $X(Z)_Y$ consists of a single element as well. But this can also be expressed by saying that $X(Z)_Y \times X(Z)_Y$ is canonically isomorphic to $X(Z)_Y$; the former is exactly the set $(X \times_Y X)(Z)_Y$ (3.4.3.1), which implies that $\Delta_{X|Y}$ is an isomorphism. □

Proposition (5.3.9). — *The diagonal morphism Δ_X is an immersion from X to $X \times_S X$.*

PROOF. Indeed, since the continuous maps p_1 and Δ_X from the underlying spaces are such that $p_1 \circ \Delta_X$ is the identity, Δ_X is a homeomorphism from X to $\Delta_X(X)$. Similarly, the composite homomorphism $\mathscr{O}_x \to \mathscr{O}_{\Delta_X(x)} \to \mathscr{O}_x$ (composed of the homomorphisms corresponding to p_1 and Δ_X) is the identity, which means that the homomorphism corresponding to Δ_X is surjective; the proposition thus follows from (4.2.2). □

We say that the subprescheme of $X \times_S X$ associated to the immersion Δ_X (4.2.1) is *the diagonal* of $X \times_S X$.

Corollary (5.3.10). — *Under the hypotheses of* (5.3.5), *$(p, q)_T$ is an immersion.*

PROOF. This follows from (5.3.6) and (4.3.1). □

We say (under the hypotheses of (5.3.5)) that $(p, q)_T$ is the *canonical immersion* of $X \times_S Y$ into $X \times_T Y$.

Corollary (5.3.11). — *Let X and Y be S-preschemes, and $f : X \to Y$ an S-morphism; then the graph morphism $\Gamma_f = (1_X, f)_S$ of f* (3.3.14) *is an immersion of X into $X \times_S Y$.*

PROOF. This is a particular case of Corollary (5.3.10), where we replace S by Y, and T by S (cf. (5.3.7)). □

The subprescheme of $X \times_S Y$ associated to the immersion Γ_f (4.2.1) is called *the graph* of the morphism f; the subpreschemes of $X \times_S Y$ that are graphs of morphisms $X \to Y$ are characterised by the property that the restriction to such a subprescheme G of the projection $p_1 : X \times_S Y \to X$ is an *isomorphism* g from G to X: G is the the graph of the morphism $p_2 \circ g^{-1}$, where p_2 is the projection $X \times_S Y \to Y$. **I | 134**

When we take, in particular, $X = S$, then the S-morphisms $S \to Y$ (which are exactly the *S-sections* of Y (2.5.5)) are equal to their graph morphisms; the subpreschemes of Y that are the graphs of S-sections (in other words, those that are isomorphic to S by the restriction of the structure morphism $Y \to S$) are then also called the *images* of these sections, or, by an abuse of language, the *S-sections* of Y.

Corollary (5.3.12). — *With the hypotheses and notation of* (5.3.11), *for every morphism $g : S' \to S$, let f' be the inverse image of f under g* (3.3.7); *then $\Gamma_{f'}$ is the inverse image of Γ_f under g.*

PROOF. This is a particular case of (3.3.10.1). □

Corollary (5.3.13). — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms; if $g \circ f$ is an immersion (resp. a local immersion), then so too is f.*

PROOF. Indeed, f factors as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$. Furthermore, p_2 can be identified with $(g \circ f) \times_Z 1_Y$ (3.3.4); if $g \circ f$ is an immersion (resp. a local immersion), then so too is p_2 ((4.3.1) and (4.5.5)), and since Γ_f is an immersion (5.3.11), we are done, by (4.2.4) (resp. (4.5.5)). ☐

Corollary (5.3.14). — *Let $j : X \to Y$ and $g : Z \to Z$ be S-morphisms. If j is an immersion (resp. a local immersion), then so too is $(j, g)_S$.*

PROOF. Indeed, if $p : Y \times_S Z \to Y$ is the projection onto the first component, then we have $j = p \circ (j, g)_S$, and it suffices to apply (5.3.13). ☐

Proposition (5.3.15). — *If $f : X \to Y$ is an S-morphism, then the diagram*

(5.3.15.1)
$$\begin{tikzcd} X \arrow[r, "\Delta_X"] \arrow[d, "f"'] & X \times_S X \arrow[d, "f \times_S f"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

commutes (in other words, Δ_X is a functorial morphism in the category of preschemes).

PROOF. The proof is immediate and left to the reader. ☐

Corollary (5.3.16). — *If X is a subprescheme of Y, then the diagonal $\Delta_X(X)$ can be identified with a subprescheme of $\Delta_Y(Y)$, and the underlying space can be identified with*

$$\Delta_Y(Y) \cap p_1^{-1}(X) = \Delta_Y \cap p_2^{-1}(X)$$

(p_1 and p_2 being the projections of $Y \times_S Y$).

PROOF. Applying (5.3.15) to the injection morphism $f : X \to Y$, we see that $f \times_S f$ is an immersion that identifies the underlying space of $X \times_S X$ with the subspace $p_1^{-1}(X) \cap p_2^{-1}(X)$ of $Y \times_S Y$ (4.3.1); further, if $z \in \Delta_Y(Y) \cap p_1^{-1}(X)$, then we have $z = \Delta_Y(y)$ and $y = p_1(z) \in X$, so **I** | 135
$y = f(y)$, and $z = \Delta_Y(f(y))$ belongs to $\Delta_X(X)$ by the commutativity of (5.3.15.1). ☐

Corollary (5.3.17). — *Let $f_1 : Y \to X$ and $f_2 : Y \to X$ be S-morphisms, and y a point of Y such that $f_1(y) = f_2(y) = x$, and such that the homomorphisms $k(x) \to k(y)$ corresponding to f_1 and f_2 are identical. Then, if $f = (f_1, f_2)_S$, the point $f(y)$ belongs to the diagonal $\Delta_{X|S}(X)$.*

PROOF. The two homomorphisms $k(x) \to k(y)$ corresponding to f_i $(i = 1, 2)$ define two S-morphisms $g_i : \operatorname{Spec}(k(y)) \to \operatorname{Spec}(k(x))$ such that the diagrams

$$\begin{tikzcd} \operatorname{Spec}(k(y)) \arrow[r, "g_i"] \arrow[d] & \operatorname{Spec}(k(x)) \arrow[d] \\ Y \arrow[r, "f_i"] & X \end{tikzcd}$$

commute. The diagram

$$\begin{tikzcd} \operatorname{Spec}(k(y)) \arrow[r, "(g_1, g_2)_S"] \arrow[d] & \operatorname{Spec}(k(x)) \times_S \operatorname{Spec}(k(x)) \arrow[d] \\ Y \arrow[r, "(f_1, f_2)_S"] & X \times_S X \end{tikzcd}$$

thus also commutes. But it follows from the equality $g_1 = g_2$ that the image under $(g_1, g_2)_S$ of the unique point of $\operatorname{Spec}(k(y))$ belongs to the diagonal of $\operatorname{Spec}(k(x)) \times_S \operatorname{Spec}(k(x))$; the conclusion then follows from (5.3.15). ☐

5.4. Separated morphisms and separated preschemes

Definition (5.4.1). — We say that a morphism of preschemes $f : X \to Y$ is *separated* if the diagonal morphism $X \to X \times_Y X$ is a *closed* immersion; we then also say that X is a *separated prescheme over* Y, or a Y-*scheme*. We say that a prescheme X is separated if it is separated over $\operatorname{Spec}(\mathbf{Z})$; we then also say that X is a *scheme*[9] *(cf.* (5.5.7)*)*.

By (5.3.9), for X to be separated over Y, it is necessary and sufficient for $\Delta_X(X)$ to be a *closed subspace* of the underlying space of $X \times_Y X$.

Proposition (5.4.2). — *Let $S \to T$ be a separated morphism. If X and Y are S-preschemes, then the canonical immersion $X \times_S Y \to X \times_T Y$* (5.3.10) *is closed.*

PROOF. Indeed, if we refer to the diagram in (5.3.5.1), we see that $(p,q)_T$ can be considered as being obtained from $\Delta_{S|T}$ by the extension $f \times_T g : X \times_T Y \to S \times_T S$ of the base prescheme $S \times_T S$; the proposition then follows from (4.3.2). □

Corollary (5.4.3). — *Let Y be an S-scheme, and $f : X \to Y$ an S-morphism. Then the graph morphism $\Gamma_f : X \to X \times_S Y$* (5.3.11) *is a closed immersion.*

PROOF. This is a particular case of (5.4.2), where we replace S by Y, and T by S. □

Corollary (5.4.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms, with g separated. If $g \circ f$ is a closed immersion, then so too is f.*

PROOF. The proof using (5.4.3) is the same as that of (5.3.13) using (5.3.11). □

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Corollary (5.4.5). — *Let Z be an S-scheme, and $j : X \to Y$ and $g : X \to Z$ S-morphisms. If j is a closed immersion, then so too is $(j,g)_S : X \to Y \times_S Z$.*

PROOF. The proof using (5.4.4) is the same as that of (5.3.14) using (5.3.13). □

Corollary (5.4.6). — *If X is an S-scheme, then every S-section of X* (2.5.5) *is a closed immersion.*

PROOF. If $\phi : X \to S$ is the structure morphism, and $\psi : S \to X$ an S-section of X, it suffices to apply (5.4.5) to $\phi \circ \psi = 1_S$. □

Corollary (5.4.7). — *Let X be an integral prescheme with generic point s, and X an S-scheme. If two S-sections f and g are such that $f(s) = g(s)$, then $f = g$.*

PROOF. Indeed, if $x = f(s) = g(s)$, then the homomorphisms $k(x) \to k(s)$ corresponding to f and g are necessarily identical. If $h = (f,g)_S$, we thus deduce (5.3.17) that $h(s)$ belongs to the diagonal $Z = \Delta_X(X)$; but since $S = \overline{\{s\}}$, and since Z is closed by hypothesis, we have $h(S) \subset Z$. It then follows from (5.2.2) that h factors as $S \to Z \to X \times_S X$, and we thus conclude that $f = g$, by definition of the diagonal. □

Remark (5.4.8). — If we suppose, conversely, that the conclusion of (5.4.3) is true when $f = 1_Y$, then we can conclude that Y is separated over S; similarly, if we suppose that the conclusion of (5.4.5) applies to the two morphisms $Y \xrightarrow{\Delta_Y} Y \times_Z Y \xrightarrow{p_1} Y$, then we can conclude that Δ_Y is a closed immersion, and thus that Y is separated over Z; finally, if we assume that the conclusion of (5.4.6) is true for the Y section Δ_Y of the Y-prescheme $Y \times_S Y \to Y$, then this implies that Y is separated over S.

[9] *[Trans.] We repeat here the warning given at the very start of this translation: the early versions of the EGA use* prescheme *to mean is now usually called a scheme, and* scheme *for what is now usually called a separated scheme.*. Grothendieck himself later said that the more modern terminology was preferable, but we have decided to keep this translation 'historically accurate' by using the older nomenclature.

5.5. Separation criteria

Proposition (5.5.1). —

(i) *Every monomorphism of preschemes (and, in particular, every immersion) is a separated morphism.*
(ii) *The composition of any two separated morphisms is separated.*
(iii) *If $f : X \to X'$ and $g : Y \to Y'$ are separated S-morphisms, then $f \times_S g$ is separated.*
(iv) *If $f : X \to Y$ is a separated S-morphism, then the S'-morphism $f_{(S')}$ is separated for every extension $S' \to S$ of the base prescheme.*
(v) *If the composition $g \circ f$ is separated, then f is separated.*
(vi) *For a morphism f to be separated, it is necessary and sufficient for f_{red} (5.1.5) to be separated.*

PROOF. Note that (i) is an immediate consequence of (5.3.8). If $f : X \to Y$ and $g : Y \to Z$ are morphisms, then the diagram

(5.5.1.1)
$$\begin{tikzcd} X \arrow[rr, "\Delta_{X|Z}"] \arrow[dr, "\Delta_{X|Y}"'] & & X \times_Z X \\ & X \times_Y X \arrow[ur, "j"'] & \end{tikzcd}$$

commutes (where j denotes the canonical immersion (5.3.10)), as can be immediately verified. If f and g are separated, then $\Delta_{X|Y}$ is a closed immersion by definition, and j is a closed immersion by (5.4.2), whence $\Delta_{X|Z}$ is a closed immersion by (4.2.4), which proves (ii). Given (i) and (ii), (iii) and (iv) **I** | 137
are equivalent (3.5.1), so it suffices to prove (iv). But $X_{(S')} \times_{Y_{(S')}} X_{(S')}$ is canonically identified with $(X \times_Y X) \times_Y Y_{(S')}$ by (3.3.11) and (3.3.9.1), and we immediately see that the diagonal morphism $\Delta_{X_{(S')}}$ can then be identified with $\Delta_X \times_Y 1_{Y_{(S')}}$; the proposition then follows from (4.3.1).

To prove (v), consider, as in (5.3.13), the factorisation $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ of f, noting that $p_2 = (g \circ f) \times_Z 1_Y$; the hypothesis (that $g \circ f$ is separated) implies that g_2 is separated, by (iii) and (i), and, since Γ_f is an immersion, Γ_f is separated, by (i), whence f is separated, by (ii). Finally, to prove (vi), recall that the preschemes $X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} X_{\mathrm{red}}$ and $X_{\mathrm{red}} \times_Y X_{\mathrm{red}}$ are canonically identified with one another (5.1.7); if we denote by j the injection $X_{\mathrm{red}} \to X$, then the diagram

$$\begin{tikzcd} X_{\mathrm{red}} \arrow[r, "\Delta_{X_{\mathrm{red}}}"] \arrow[d, "j"'] & X_{\mathrm{red}} \times_Y X_{\mathrm{red}} \arrow[d, "j \times_Y j"] \\ X \arrow[r, "\Delta_X"] & X \times_Y X \end{tikzcd}$$

commutes (5.3.15), and the proposition follows from the fact that the vertical arrows are homeomorphisms of the underlying spaces (4.3.1). □

Corollary (5.5.2). — *If $f : X \to Y$ is separated, then the restriction of f to any subprescheme of X is separated.*

PROOF. This follows from (5.5.1, i and ii). □

Corollary (5.5.3). — *If X and Y are S-preschemes such that Y is separated over S, then $X \times_S Y$ is separated over X.*

PROOF. This is a particular case of (5.5.1, iv). □

Proposition (5.5.4). — *Let X be a prescheme, and assume that its underlying space is a* finite *union of closed subsets X_k ($1 \leqslant k \leqslant n$); for each k, consider the reduced subprescheme of X that has X_k as its underlying space* (5.2.1)*, and denote this again by X_k. Let $f : X \to Y$ be a morphism, and, for each k, let Y_k be a closed subset of Y such that $f(X_k) \subset Y_k$; we again denote by Y_k the reduced subprescheme of Y that has Y_k as its underlying space, so that the restriction $X_k \to Y$ of f to X_k factors as $X_k \xrightarrow{f_k} Y_k \to Y$* (5.2.2)*. For f to be separated, it is necessary and sufficient for all the f_k to be separated.*

PROOF. The necessity follows from (5.5.1, i, ii, and v). Conversely, if the condition of the statement is satisfied, then each of the restrictions $X_k \to Y$ of f is separated (5.5.1, (i) and (ii)); if p_1 and p_2 are the projections of $X \times_Y X$, then the subspace $\Delta_{X_k}(X_k)$ can be identified with the subspace $\Delta_X(X) \cap p_1^{-1}(X_k)$ of the underlying space of $X \times_Y X$ (5.3.16); these subspaces are closed in $X \times_Y X$, and thus so too is their union $\Delta_X(X)$. □

Suppose, in particular, that the X_k are the *irreducible components* of X; then we can suppose that the Y_k are the irreducible components of Y (**0**, 2.1.5); Proposition (5.5.4) then, in this case, leads to the idea of separation in the case of *integral* preschemes (2.1.7).

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Proposition (5.5.5). — *Let (Y_λ) be an open cover of a prescheme Y; for a morphism $f : X \to Y$ to be separated, it is necessary and sufficient for each of its restrictions $f^{-1}(Y_\lambda) \to Y_\lambda$ to be separated.*

PROOF. If we set $X_\lambda = f^{-1}(Y_\lambda)$, everything reduces, by taking (4.2.4, b) and the identification of the products $X_\lambda \times_Y X_\lambda$ and $X_\lambda \times_{Y_\lambda} X_\lambda$ into account, to proving that the $X_\lambda \times_Y X_\lambda$ form a cover of $X \times_Y X$. But if we set $Y_{\lambda\mu} = Y_\lambda \cap Y_\mu$ and $X_{\lambda\mu} = X_\lambda \cap X_\mu = f^{-1}(Y_{\lambda\mu})$, then $X_\lambda \times_Y X_\mu$ can be identified with the product $X_{\lambda\mu} \times_{Y_{\lambda\mu}} X_{\lambda\mu}$ (3.2.6.4), and so also with $X_{\lambda\mu} \times_Y X_{\lambda\mu}$ (3.2.5), and finally with an open subset of $X_\lambda \times_Y X_\lambda$, which proves our claim (3.2.7). □

Proposition (5.5.4) allows us, by taking an affine open cover of Y, to restrict our study of separated morphisms to just those that take values in affine schemes.

Proposition (5.5.6). — *Let Y be an* affine *scheme, X a prescheme, and (U_α) a cover of X by affine open subsets. For a morphism $f : X \to Y$ to be separated, it is necessary and sufficient for $U_\alpha \cap U_\beta$ to be, for every pair of indices (α, β), an affine open subset,* and *for the ring $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X)$ to be generated by the union of the canonical images of the rings $\Gamma(U_\alpha, \mathscr{O}_X)$ and $\Gamma(U_\beta, \mathscr{O}_X)$.*

PROOF. The $U_\alpha \times_Y U_\beta$ form an open cover of $X \times_Y X$ (3.2.7); denoting the projections of $X \times_Y X$ by p and q, we have

$$\begin{aligned}\Delta_X^{-1}(U_\alpha \times_Y U_\beta) &= \Delta_X^{-1}(p^{-1}(U_\alpha) \cap q^{-1}(U_\beta)) \\ &= \Delta_X^{-1}(p^{-1}(U_\alpha)) \cap \Delta_X^{-1}(q^{-1}(U_\beta)) = U_\alpha \cap U_\beta;\end{aligned}$$

so everything reduces to proving that the restriction of Δ_X to $U_\alpha \cap U_\beta$ is a closed immersion into $U_\alpha \times_Y U_\beta$. But this restriction is exactly $(j_\alpha, j_\beta)_Y$, where j_α (resp. j_β) denotes the injection morphism from $U_\alpha \cap U_\beta$ to U_α (resp. U_β), as follows from the definitions. Since $U_\alpha \times_Y U_\beta$ is an affine scheme whose ring is canonically isomorphic to $\Gamma(U_\alpha, \mathscr{O}_X) \otimes_{\Gamma(Y, \mathscr{O}_Y)} \Gamma(U_\beta, \mathscr{O}_X)$ (3.2.2), we see that $U_\alpha \cap U_\beta$ must be an affine scheme, and that the map $h_\alpha \otimes h_\beta \mapsto h_\alpha h_\beta$ from the ring $A(U_\alpha \times_Y U_\beta)$ to $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X)$ must be surjective (4.2.3), which finishes the proof. □

Corollary (5.5.7). — *An affine scheme is separated* (and is thus a *scheme*, which justifies the terminology of (5.4.1)).

Corollary (5.5.8). — *Let Y be an affine scheme; for $f : X \to Y$ to be a separated morphism, it is necessary and sufficient for X to be separated* (in other words, for X to be a *scheme*).

PROOF. Indeed, we see that the criteria of (5.5.6) do not depend on f. □

Corollary (5.5.9). — *For a morphism $f : X \to Y$ to be separated, it is necessary and sufficient for the induced prescheme $f^{-1}(U)$ to be separated, for every open subset of U on which Y induces a separated prescheme, and it is sufficient for it to be the case for every affine open subset $U \subset Y$.*

PROOF. The necessity of the condition follows from (5.5.4) and (5.5.1, ii); the sufficiency follows from (5.5.4) and (5.5.8), taking into account the existence of affine open covers of Y. □

In particular, if X and Y are affine schemes, then *every* morphism $X \to Y$ is separated.

Proposition (5.5.10). — *Let Y be a* scheme, *and $f : X \to Y$ a morphism. For every affine open subset U of X, and every affine open subset V of Y, $U \cap f^{-1}(V)$ is affine.*

PROOF. Let p_1 and p_2 be the projections of $X \times_{\mathbf{Z}} Y$; the subspace $U \cap f^{-1}(V)$ is the image of $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ under p_1. But $p_1^{-1}(U) \cap p_2^{-1}(V)$ can be identified with the underlying space of the prescheme $U \times_{\mathbf{Z}} V$ (3.2.7), and is thus an affine scheme (3.2.2); since $\Gamma_f(X)$ is closed in $X \times_{\mathbf{Z}} Y$ (5.4.3), $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ is closed in $U \times_{\mathbf{Z}} V$, and so the prescheme induced by the subprescheme of $X \times_{\mathbf{Z}} Y$ associated to Γ_f (4.2.1) on the open subset $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ of its underlying space is a closed subprescheme of an affine scheme, and thus an affine scheme (4.2.3). The proposition then follows from the fact that Γ_f is an immersion. □ I | 139

Examples (5.5.11). — The prescheme from Example (2.3.2) ("the projective line over a field K") is *separated*, because, for the cover (X_1, X_2) of X by affine open subsets, $X_1 \cap X_2 = U_{12}$ is affine, and $\Gamma(U_{12}, \mathscr{O}_X)$, the ring of rational fractions of the form $f(s)/s^m$ with $f \in K[s]$, is generated by $K[s]$ and $1/s$, so the conditions of (5.5.6) are satisfied.

With the same choice of X_1, X_2, U_{12}, and U_{21} as in Example (2.3.2), now take u_{12} to be the isomorphism which sends $f(s)$ to $f(t)$; we now obtain, by gluing, a *non-separated integral* prescheme X, because the first condition of (5.5.6) is satisfied, but not the second. It is immediate here that $\Gamma(X, \mathscr{O}_X) \to \Gamma(X_1, \mathscr{O}_X) = K[s]$ is an isomorphism; the inverse isomorphism defines a morphism $f : X \to \operatorname{Spec}(K[s])$ that is surjective, and for every $y \in \operatorname{Spec}(K[s])$ such that $\mathfrak{j}_y \neq (0)$, $f^{-1}(y)$ consists of a single point, but for $\mathfrak{j}_y = (0)$, $f^{-1}(y)$ consists of *two* distinct points (we say that X is the "affine line over K with the point 0 doubled").

We can also give examples where *neither* of the two conditions of (5.5.6) are satisfied. First note that, in the prime spectrum Y of the ring $A = K[s,t]$ of polynomials in two indeterminates over a field K, the open subset U given by the union of $D(s)$ and $D(t)$ is *not an affine open subset*. Indeed, if z is a section of $\mathscr{O}_Y$ over U, there exist two integers $m, n \geqslant 0$ such that $s^m z$ and $t^n z$ are the restrictions of polynomials in s and t to U (1.4.1), which is clearly possible only if the section z extends to a section over the whole of Y, identified with a polynomial in s and t. If U were an affine open subset, then the injection morphism $U \to Y$ would be an isomorphism (1.7.3), which is a contradiction, since $U \neq Y$.

With the above in mind, take two affine schemes Y_1 and Y_2, prime spectra of the rings $A_1 = K[s_1, t_2]$ and $A_2 = K[s_2, t_2]$ (respectively); take $U_{12} = D(s_1) \cup D(t_1)$ and $U_{21} = D(s_2) \cup D(t_2)$, and take u_{12} to be the restriction of an isomorphism $Y_2 \to Y_1$ to U_{21} corresponding to the isomorphism of rings that sends $f(s_1, t_1)$ to $f(s_2, t_2)$; we then have an example where the conditions of (5.5.6) are not satisfied (the integral prescheme thus obtained is called "the affine plane over K with the point 0 doubled").

Remark (5.5.12). — Given some property **P** of morphisms of preschemes, consider the following propositions.

(i) *Every closed immersion has property* ***P***.
(ii) *The composition of any two morphisms that both have property* ***P*** *also has property* ***P***.
(iii) *If $f : X \to X'$ and $g : Y \to Y'$ are S-morphisms that have property* ***P***, *then $f \times_S g$ has property* ***P***. I | 140
(iv) *If $f : X \to Y$ is an S-morphism that has property* ***P***, *then every S'-morphism $f_{(S')}$ obtained by an extension $S' \to S$ of the base prescheme also has property* ***P***.
(v) *If the composition $g \circ f$ of two morphisms $f : X \to Y$ and $g : Y \to Z$ has property* ***P***, *and g is separated, then f has property* ***P***.
(vi) *If a morphism $f : X \to Y$ has property* ***P***, *then so too does f_{red} (5.1.5).*

If we suppose that (i) and (ii) are both true, then (iii) and (iv) are *equivalent*, and (v) and (vi) are *consequences* of (i), (ii), and (iii).

The first claim has already been shown (3.5.1). Consider the factorisation (5.3.13) $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ of f; the equation $p_2 = (g \circ f) \times_Z 1_Y$ shows that, if $g \circ f$ has property **P**, then so too does p_2, by (iii); if g is separated, then Γ_f is a closed immersion (5.4.3), and so also has property **P**, by (i); finally, by (ii), f has property **P**.

Finally, consider the commutative diagram

$$\begin{tikzcd} X_{\mathrm{red}} \arrow[r, "f_{\mathrm{red}}"] \arrow[d] & Y_{\mathrm{red}} \arrow[d] \\ X \arrow[r, "f"] & Y, \end{tikzcd}$$

where the vertical arrows are the closed immersions (5.1.5), and thus have property **P**, by (i). The hypothesis that f has property **P** implies, by (ii), that $X_{\mathrm{red}} \xrightarrow{f_{\mathrm{red}}} Y_{\mathrm{red}} \to Y$ has property **P**; finally, since a closed immersion is separated (5.5.1, i), f_{red} has property **P**, by (v).

Note that, if we consider the propositions

(i′) *Every immersion has property* ***P***;

(v′) *If $g \circ f$ has property* ***P***, *then so too does f*;

then the above arguments show that (v′) is a consequence of (i′), (ii), and (iii).

(5.5.13). Note that (v) and (vi) are again consequences of (i), (iii), and

(ii′) *If $j : X \to Y$ is a closed immersion, and $g : Y \to Z$ is a morphism that has property* ***P***, *then $g \circ j$ has property* ***P***.

Similarly, (v′) is a consequence of (i′), (iii), and

(ii″) *If $j : X \to Y$ is an immersion, and $g : Y \to Z$ is a morphism that has property* ***P***, *then $g \circ j$ has property* ***P***.

This follows immediately from the arguments of (5.5.12).

§6. Finiteness conditions

6.1. Noetherian and locally Noetherian preschemes

Definition (6.1.1). — We say that a prescheme X is Noetherian (resp. locally Noetherian) if it is a finite union (resp. union) of affine open V_α in such a way that the ring of the of the induced scheme on each of the V_α is Noetherian.

It follows immediately from (1.5.2) that, if X is locally Noetherian, then the structure sheaf $\mathscr{O}_X$ is a *coherent sheaf of rings*, since the questions is a local one. Every *quasi-coherent $\mathscr{O}_X$-submodule* (resp. quasi-coherent quotient $\mathscr{O}_X$-module) of a *coherent* $\mathscr{O}_X$-module $\mathscr{F}$ is *coherent*, as the question is once again a local one, and it suffices to apply (1.5.1), (1.4.1), and (1.3.10), combined with the fact that a submodule (resp. quotient module) of a module of finite type over a Noetherian ring is of finite type. In particular, every *quasi-coherent sheaf of ideals* of $\mathscr{O}_X$ is *coherent*. **I** | 141

If a prescheme X is a finite union (resp. union) of open subsets W_λ in such a way that the preschemes induced on the W_λ are Noetherian (resp. locally Noetherian), it is clear that X is Noetherian (resp. locally Noetherian).

Proposition (6.1.2). — *For a prescheme X to be Noetherian, it is necessary and sufficient for it to be locally Noetherian and have a quasi-compact underlying space. The underlying space itself is then also Noetherian.*

Proof. The first claim follows immediately from the definitions and (1.1.10, ii). The second follows from (1.1.6) and the fact that every space that is a finite union of Noetherian subspaces is itself Noetherian (**0**, 2.2.3). □

Proposition (6.1.3). — *Let X be an affine scheme given by a ring A. The following conditions are equivalent:* (a) *X is Noetherian;* (b) *X is locally Noetherian;* (c) *A is Noetherian.*

Proof. The equivalence between (a) and (b) follows from (6.1.2) ant the fact that the underlying space of every affine scheme is quasi-compact (1.1.10); it is furthermore clear that (c) implies (a). To see that (a) implies (c), we remark that there is a finite cover (V_i) of X by affine open subsets such that the ring A_i of the prescheme induced on V_i is Noetherian. So let $(\mathfrak{a}_n)$ be an increasing sequence of ideals of A; by a canonical bijective correspondence, there is a corresponding sequence $(\widetilde{\mathfrak{a}}_n)$ of sheaves of ideals in $\widetilde{A} = \mathscr{O}_X$; to see that the sequence $(\mathfrak{a}_n)$ is stable (?), it suffices to prove that the sequence $(\widetilde{\mathfrak{a}}_n)$ is. But the restriction $\widetilde{\mathfrak{a}}_n|V_i$ is a quasi-coherent sheaf of ideals of $\mathscr{O}_X|V_i$, being the inverse image of $\widetilde{\mathfrak{a}}_n$ under the canonical injection $V_i \to X$ (**0**, 5.1.4); $\widetilde{\mathfrak{a}}_n|V_i$ is thus of the form $\widetilde{\mathfrak{a}}_{ni}$, where $\mathfrak{a}_{ni}$ is an ideal of A_i (1.3.7). Since A_i is Noetherian, the sequence $(\mathfrak{a}_{ni})$ is stable for all i, whence the proposition. □

We note that the above argument proves also that *if X is a Noetherian prescheme, then every increasing sequence of coherent sheaves of ideals of $\mathscr{O}_X$ is stable (?).*

Proposition (6.1.4). — *Every subprescheme of a Noetherian (resp. locally Noetherian) prescheme is Noetherian (resp. locally Noetherian).*

PROOF. If suffices to give a proof for a Noetherian prescheme X; further, by definition (6.1.1), we can also restrict to the case where X is an affine scheme. Since every subprescheme of X is a closed subprescheme of a prescheme induced on an open subset (4.1.3), we can restrict to the case of a subprescheme Y, either closed or induced on an open subset of X. The proof in the case where Y is closed is immediate, since if A is the ring of X, we know that Y is an affine scheme given by the ring $A/\mathfrak{J}$, where $\mathfrak{J}$ is an ideal of A (4.2.3); since A is Noetherian (6.1.3), so too is $A/\mathfrak{J}$.

Now suppose that Y is open in X; the underlying space of Y is Noetherian (6.1.2), hence quasi-compact, and thus a finite union of open subsets $D(f_i)$ $(f_i \in A)$; everything reduces to showing the proposition in the case where $Y = D(f)$ with $f \in A$. But then Y is an affine scheme whose ring is isomorphic to A_f (1.3.6); since A is Noetherian (6.1.3), so too is A_f. □

(6.1.5). We note that the *product* of two Noetherian S-preschemes is not necessarily Noetherian, even if the preschemes are affine, since the tensor product of two Noetherian algebras in not necessarily a Noetherian ring (cf. (6.3.8)).

Proposition (6.1.6). — *If X is a Noetherian prescheme, the nilradical $\mathscr{N}_X$ of $\mathscr{O}_X$ is nilpotent.*

PROOF. We can in fact cover X with a finite number of affine open subsets U_i, and it suffices to prove that there exists whole numbers n_i such that $(\mathscr{N}_X|U_i)^{n_i} = 0$; if n is the largest of the n_i, then we will have $\mathscr{N}_X^n = 0$. We can thus restrict to the case where $X = \operatorname{Spec}(A)$ is affine, with A a Noetherian ring; by (5.1.1) and (1.3.13), it suffices to observe that the nilradical of A is nilpotent ([**Sam53b**, p. 127, cor. 4]). □

Corollary (6.1.7). — *Let X be a Noetherian prescheme; for X to be an affine scheme, it is necessary and sufficient that X_{red} be affine.*

PROOF. This follows from (6.1.6) and (5.1.10). □

Lemma (6.1.8). — *Let X be a topological space, x a point of X, and U an open neighbourhood of x having only a finite number of irreducible components. Then there exists a neighbourhood V of x such that every open neighbourhood of x contained in V is connected.*

PROOF. Let U_i $(1 \leqslant i \leqslant m)$ be the irreducible components of U not containing x; the complement (in U) of the union of the U_i is an open neighbourhood V of X inside U, and thus so too in X; it is also, incidentally, the complement (in X) of the union of the irreducible components of X that do not contain x (**0**, 2.1.6). So let W be an open neighbourhood of X contained in V. The irreducible components of W are the intersections of W with the irreducible components of U (**0**, 2.1.6), so these components contain x; since they are connected, so too is W. □

Corollary (6.1.9). — *A locally Noetherian topological space is locally connected (which implies, amongst other things, that its connected components are open).*

Proposition (6.1.10). — *Let X be a locally Noetherian topological space. The following conditions are equivalent.*

(a) *The irreducible components of X are open.*
(b) *The irreducible components of X are exactly its connected components.*
(c) *The connected components of X are irreducible.*
(d) *Two distinct irreducible components of X have an empty intersection.*

Finally, if X is a prescheme, then these conditions are also equivalent to

(e) *For every $x \in X$, $\operatorname{Spec}(\mathscr{O}_x)$ is irreducible (or, in other words, the nilradical of $\mathscr{O}_x$ is prime).*

PROOF. It is immediate that (a) implies (b), because an irreducible space is connected, and (a) implies that the irreducible components of X are the sets that are both open and closed. It is trivial that (b) implies (c); conversely, a closed set F containing a connected component C of X, with C distinct from F, cannot be irreducible, because not being connected means that F is the union of two disjoint nonempty sets that are both open and closed in F, and thus closed in X; as a result, (c) implies (b). We immediately conclude from this that (c) implies (d), since two distinct connected components have no points in common. **I** | 143

We have not yet used the fact that X is locally Noetherian. Suppose now that this is indeed the case, and we will show that (d) implies (a): by (**0**, 2.1.6), we can restrict ourselves to the case where the space X is Noetherian, and so has only a finite number of irreducible components. Since they are closed and pairwise disjoint, they are open.

Finally, the equivalence between (d) and (e) holds true even without the assumption that the underlying space of the prescheme X is locally Noetherian. We can in fact restrict ourselves to the case where $X = \operatorname{Spec}(A)$ is affine, by (**0**, 2.1.6); to say that x is contained in only one single irreducible component of X is to say that $\mathfrak{j}_x$ contains only one single minimal ideal of A (1.1.14), which is equivalent to saying that $\mathfrak{j}_x\mathscr{O}_x$ contains only one single minimal ideal of $\mathscr{O}_x$, whence the conclusion. □

Corollary (6.1.11). — *Let X be a locally Noetherian space. For X to be irreducible, it is necessary and sufficient that X be connected and nonempty, and that any two distinct irreducible components of X have an empty intersection. If X is a prescheme, this latter condition is equivalent to asking that $\operatorname{Spec}(\mathscr{O}_x)$ be irreducible for all $x \in X$.*

PROOF. The second claim has already been shown in (6.1.10); the only thing thus remaining to show is that the conditions in the first claim are sufficient. But by (6.1.10), these conditions imply that the irreducible components of X are exactly its connected components, and since X is connected and nonempty, it is irreducible. □

Corollary (6.1.12). — *Let X be a locally Noetherian prescheme. For X to be integral, it is necessary and sufficient that X be connected and that $\mathscr{O}_x$ be integral for all $x \in X$.*

Proposition (6.1.13). — *Let X be a locally Noetherian prescheme, and let $x \in X$ be a point such that the nilradical $\mathscr{N}_x$ of $\mathscr{O}_x$ is prime (resp. such that $\mathscr{O}_x$ is reduced, resp. integral); then there exists an open neighbourhood U of x that is irreducible (resp. reduced, resp. integral).*

PROOF. It suffices to consider two cases: where $\mathscr{N}_x$ is prime, and where $\mathscr{N}_x = 0$; the third hypotheses is a combination of the first two. If $\mathscr{N}_x$ is prime, then x belongs to only one single irreducible component Y of X (6.1.10); the union of the irreducible components of X that do not contain x is closed (the set of these components being locally finite), and the complement U of this union is thus open and contained in Y, and thus irreducible (**0**, 2.1.6) If $\mathscr{N}_x = 0$, we also have $\mathscr{N}_y = 0$ for any y in a neighbourhood of x, because $\mathscr{N}$ is quasi-coherent (5.1.1), and thus coherent, since X is locally Noetherian, and the conclusion then follows from (**0**, 5.2.2). □

6.2. Artinian preschemes

Definition (6.2.1). — We say that a prescheme is *Artinian* if it is affine, and given by an Artinian ring.

Proposition (6.2.2). — *Given a prescheme X, the following conditions are equivalent:* **I** | 144

(a) *X is an Artinian scheme;*
(b) *X is Noetherian and its underlying space is discrete;*
(c) *X is Noetherian and the points of its underlying space are closed* (the T_1 condition).

When any of the above hold, the underlying space of X is finite, and the ring A of X is the direct sum of local (Artinian) rings of points of X.

PROOF. We know that (a) implies the last claim ([**SZ60**, p. 205, th. 3]), so every prime ideal of A is thus maximal and is the inverse image of a maximal ideal of one of the local components of A, and so the space X is finite and discrete; (a) thus implies (b), and (b) clearly implies (c). To see that (c) implies (a), we first show that X is then finite; we can indeed restrict to the case where X is affine, and we know that a Noetherian ring whose prime ideals are all maximal is Artinian ([SZ60, p. 203]), whence our claim. The underlying space X is then discrete, the topological sum of a finite number of points x_i, and the local rings $\mathscr{O}_{x_i} = A_i$ are Artinian; it is clear that X is isomorphic to the prime spectrum affine scheme of the ring A (the direct sum of the A_i) (1.7.3). □

6.3. Morphisms of finite type

Definition (6.3.1). — We say that a morphism $f : X \to Y$ is *of finite type* if Y is the union of a family (V_α) of affine open subsets having the following property:

(P) $f^{-1}(V_\alpha)$ is a finite union of affine open subsets $U_{\alpha i}$ that are such that each ring $A(U_{\alpha i})$ is an algebra of finite type over $A(V_\alpha)$.

We then say that X is a prescheme of finite type over Y, or a Y-prescheme of finite type.

Proposition (6.3.2). — *If $f : X \to Y$ is a morphism of finite type, then every affine open subset W of Y satisfies property* (P) *of* (6.3.1).

We first show

Lemma (6.3.2.1). — *If $T \subset Y$ is an affine open subset, satisfying property* (P)*, then, for every $g \in A(T)$, $D(g)$ also satisfies property* (P)*.*

PROOF. By hypothesis, $f^{-1}(T)$ is a finite union of affine open subsets Z_j, that are such that $A(Z_j)$ is an algebra of finite type over $A(T)$; let $\phi_j : A(T) \to A(Z_j)$ be the homomorphism of rings corresponding to the restriction of f to Z_j (2.2.4), and set $g_j = \phi_j(g)$; we then have $f^{-1}(D(g)) \cap Z_j = D(g_j)$ (1.2.2.2). But $A(D(g_j)) = A(Z_j)_{g_j} = A(Z_j)[1/g_j]$ is of finite type over $A(Z_j)$, and *a fortiori* over $A(T)$ by the hypothesis, and so also over $A(D(g)) = A(T)[1/g]$, which proves the lemma. □

PROOF. With the above lemma, since W is quasi-compact (1.1.10), there exists a finite covering of W by sets of the form $D(g_i) \subset W$, where each g_i belongs to a ring $A(V_{\alpha(i)})$. Each $D(g_i)$, being $\mathbf{Err}_{\mathrm{II}}$ quasi-compact, is a finite union of sets $D(h_{ik})$, where $h_{ik} \in A(W)$; if $\phi_i : A(W) \to A(D(g_i))$ is the canonical map, then we have $D(h_{ik}) = D(\phi_i(h_{ik}))$ by (1.2.2.2). By (6.3.2.1), each of the $f^{-1}(D(h_{ik}))$ admits a finite covering by affine open subsets U_{ijk}, that are such that the $A(U_{ijk})$ are algebras of finite type over $A(D(h_{ik})) = A(W)[1/h_{ik}]$, whence the proposition. □

We can thus say that the notion of a prescheme of finite type over Y is *local on Y*. **I** | 145

Proposition (6.3.3). — *Let X and Y be affine schemes; for X to be of finite type over Y, it is necessary and sufficient that $A(X)$ be an algebra of finite type over $A(Y)$.*

PROOF. Since the condition clearly suffices, we show that it is necessary. Set $A = A(Y)$ and $B = A(X)$; by (6.3.2), there exists a finite affine open cover (V_i) of X such that each of the rings $A(V_i)$ is an A-algebra of finite type. Further, since the V_i are quasi-compact, we can cover each of them with a finite number of open subsets of the form $D(g_{ij}) \subset V_i$, where $g_{ij} \in B$; if ϕ_i is a homomorphism $B \to A(V_i)$ that corresponds to the canonical injection $V_i \to X$, then we have $B_{g_{ij}} = (A(V_i))_{\phi_i(g_{ij})} = A(V_i)[1/\phi_i(g_{ij})]$, so $B_{g_{ij}}$ is an A-algebra of finite type. We can thus restrict to the case where $V_i = D(g_i)$ with $g_i \in B$. By hypothesis, there exists a finite subset F_i of B and an integer $n_i \geqslant 0$ such that B_{g_i} is the algebra generated over A by the elements $b_i/g_i^{n_i}$, where the b_i run over all of F_i. Since there are only finitely many of the g_i, we can assume that all the n_i are equal to the same integer n. Further, since the $D(g_i)$ form a cover of X, the ideal generated in B by the g_i is equal to B, or, in other words, there exist $h_i \in B$ such that $\sum_i h_i g_i = 1$. So let F be the finite subset of B given by the union of the F_i, the set of the g_i, and the set of the h_i; we will show that the subring $B' = A[F]$ of B is equal to B. By hypothesis, for every $b \in B$ and every i, the canonical image of b in B_{g_i} is of the form $b'_i/g_i^{m_i}$, where $b'_i \in B'$; by multiplying the b'_i by suitable powers of the g_i, we can again assume that all the m_i are equal to the same integer m. By the definition of the ring of fractions, there is thus an integer N (dependant on b) such that $N \geqslant m$ and $g_i^N b \in B'$ for all i; but, *in the ring B'*, the g_i^N generate the ideal B', because the g_i do (and the h_i belong to B'); there are thus $c_i \in B'$ such that $\sum_i c_i g_i^N = 1$, whence $b = \sum_i c_i g_i^N b \in B'$, Q.E.D. □

Proposition (6.3.4). —

(i) *Every closed immersion is of finite type.*
(ii) *The composition of any two morphisms of finite type is of finite type.*
(iii) *If $f : X \to X'$ and $g : Y \to Y'$ are S-morphisms of finite type, then $f \times_S g$ is of finite type.*
(iv) *If $f : X \to Y$ is an S-morphism of finite type, then $f_{(S')}$ is of finite type for any extension $g : S' \to S$ of the base prescheme.*
(v) *If the composition $g \circ f$ of two morphisms is of finite type, with g separated, then f is of finite type.*
(vi) *If a morphism f is of finite type, then f_{red} is of finite type.*

PROOF. By (5.5.12), it suffices to prove (i), (ii), and (iv).

To show (i), we can restrict to the case of a canonical injection $X \to Y$, with X being a closed subprescheme of Y; further (6.3.2), we can assume that Y is affine, in which case X is also affine (4.2.3) and its ring is isomorphic to a quotient ring $A/\mathfrak{J}$, where A is the ring of Y and $\mathfrak{J}$ is an ideal of A; since $A/\mathfrak{J}$ is of finite type over A, the conclusion follows.

Now we show (ii). Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of finite type, and let U be an affine open subset of Z; g^{-1} admits a finite covering by affine open subsets V_i that are such that each $A(V_i)$ is an algebra of finite type over $A(U)$ (6.3.2); similarly. each of the f^{-1} admits a **I** | 146
finite cover by affine open subsets W_{ij} that are such that each $A(W_{ij})$ is an algebra of finite type over $A(V_i)$, and so also an algebra of finite type over $A(U)$, whence the conclusion.

Finally, to show (iv), we can restrict to the case where $S = Y$; then $f_{(S')}$ is also equal to $f_{Y_{(S')}}$, where we consider f as a Y-morphism, and the base extension is $Y_{(S')} \to Y$ (3.3.9). So let p and q be the projections $X_{(S')} \to X$ and $X_{(S')} \to S'$. Let V be an affine open subset of S; $f^{-1}(V)$ is a finite union of affine open subsets W_i, each of which is such that $A(W_i)$ is an algebra of finite type over $A(V)$ (6.3.2). Let V' be an affine open subset of S' contained in $g^{-1}(V)$; since $f \circ p = g \circ q$, $q^{-1}(V')$ is contained in the union of the $p^{-1}(W_i)$; on the other hand, the intersection $p^{-1}(W_i) \cap q^{-1}(V')$ can be identified with the product $W_i \times VV'$ (3.2.7), which is an affine scheme whose ring is isomorphic to $A(W_i) \otimes_{A(V)} A(V')$ (3.2.2); this ring is, by hypothesis, an algebra of finite type over $A(V')$, which proves the proposition. □

Corollary (6.3.5). — *Let $f : X \to Y$ be an immersion morphism. If the underlying space of Y (resp. X) is locally Noetherian (resp. Noetherian), then f is of finite type.*

PROOF. We can always assume that Y is affine (6.3.2); if the underlying space of Y is locally Noetherian, then we can further assume that it is Noetherian, and then the underlying space of X, which is a subspace, is also Noetherian. In other words, we can assume that Y is affine and that the underlying space of X is Noetherian; there then exists a covering of X by a finite number of affine open subsets $D(g_i) \subset Y$, where $g_i \in A(Y)$, that are such that the $X \cap D(g_i)$ are closed in $D(g_i)$ (and thus affine schemes (4.2.3)), because X is locally closed in Y (4.1.3). Then $A(X \cap D(g_i))$ is an algebra of finite type over $A(D(g_i))$, by (6.3.4, i) and (6.3.3), and $A(D(g_i)) = A(Y)_{g_i} = A(Y)[1/g_i]$ is of finite type over $A(Y)$, which finishes the proof. □

Corollary (6.3.6). — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms. If $g \circ f$ if of finite type, with either X Noetherian or $X \times_Z Y$ locally Noetherian, then f is of finite type.*

PROOF. This follows immediately from the proof of (5.5.12) and from (6.3.5) applied to the immersion morphism Γ_f. □

Proposition (6.3.7). — *Let $f : X \to Y$ be a morphism of finite type; if Y is Noetherian (resp. locally Noetherian), then X is Noetherian (resp. locally Noetherian).*

PROOF. We can restrict to proving the proposition for when Y is Noetherian. Then Y is a finite union of affine open subsets V_i that are such that the $A(V_i)$ are Noetherian rings. By (6.3.2), each of the $f^{-1}(V_i)$ is a finite union of affine open subsets W_{ij} that are such that the $A(W_{ij})$ are algebras of finite type over $A(V_i)$, and thus Noetherian rings; this proves that X is Noetherian. □

Corollary (6.3.8). — *Let X be a prescheme of finite type over S. For every base extension $S' \to S$ with S' Noetherian (resp. locally Noetherian), $X_{(S')}$ is Noetherian (resp. locally Noetherian).*

PROOF. This follows from (6.3.7), since $X_{(S')}$ is of finite type over S' by (6.3.4, iv). □

We can also says that, for a product $X \times_S Y$ of S-preschemes, if *one* of the factors X or Y is *of finite type* over S and *the other is Noetherian* (resp. *locally Noetherian*), then $X \times_S Y$ is *Noetherian* (resp. *locally Noetherian*). I | 147

Corollary (6.3.9). — *Let X be a prescheme of finite type over a locally Noetherian prescheme S. Then every S-morphism $f : X \to Y$ is of finite type.*

PROOF. In fact, we can assume that S is Noetherian; if $\phi : X \to S$ and $\psi : Y \to S$ are the structure morphisms, then we have $\phi = \psi \circ f$, and X is Noetherian by (6.3.7); f is thus of finite type by (6.3.6). □

Proposition (6.3.10). — *Let $f : X \to Y$ be a morphism of finite type. For f to be surjective, it is necessary and sufficient that, for every* algebraically closed *field Ω, the map $X(\Omega) \to Y(\Omega)$ that corresponds to f (3.4.1) be surjective.*

PROOF. The condition suffices, as we can see by considering, for all $y \in Y$, an algebraically closed extension Ω of $k(y)$, and the commutative diagram

$$\begin{tikzcd} X \arrow[dd, "f"'] & \\ & \mathrm{Spec}(\Omega) \arrow[ul] \arrow[dl] \\ Y & \end{tikzcd}$$

(cf. (3.5.3)). Conversely, suppose that f is surjective, and let $g : \{\xi\} = \mathrm{Spec}(\Omega) \to Y$ be a morphism, where Ω is an algebraically closed field. If we consider the diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & X_{(\Omega)} \arrow[l] \arrow[d, "f_{(\Omega)}"] \\ Y & \mathrm{Spec}(\Omega), \arrow[l] \end{tikzcd}$$

then it suffices to show that there exists a *rational point over* Ω in $X_{(\Omega)}$ ((3.3.14), (3.4.3), and (3.4.4)). Since f is surjective, $X_{(\Omega)}$ is nonempty (3.5.10), and since f is of finite type, so too is $f_{(\Omega)}$ (6.4.3, iv); thus $X_{(\Omega)}$ contains a nonempty affine open subset Z such that $A(Z)$ is an non-null algebra of finite type over Ω. By Hilbert's Nullstellensatz [**Zar47**], there exists an Ω-homomorphism $A(Z) \to \Omega$, and thus a section of $X_{(\Omega)}$ over $\mathrm{Spec}(\Omega)$, which proves the proposition. □

6.4. Algebraic preschemes

Definition (6.4.1). — Given a field K, we define an *algebraic K-prescheme* to be a prescheme X of finite type over K; K is called the base field of X. If in addition X is a scheme *(or if X is a* K-scheme, *which is equivalent* (5.5.8)*)*, we say that X is an *algebraic K-scheme*.

Every algebraic K-prescheme is *Noetherian* (6.3.7).

Proposition (6.4.2). — *Let X be an algebraic K-prescheme. For a point $x \in X$ to be closed, it is necessary and sufficient that $k(x)$ be an algebraic extension of K of finite degree.*

PROOF. We can assume that X is affine, with the ring A of X being a K-algebra of finite type. Indeed, the affine open subsets U of X such that $A(U)$ is a K-algebra of finite type form a finite cover of X (6.3.1). The closed points of X are thus the points such that $\mathfrak{j}_x$ is a maximal ideal of A, or in other words, such that $A/\mathfrak{j}_x$ is a field (necessarily equal to $k(x)$). Since $A/\mathfrak{j}_x$ is a K-algebra of finite type, we see that if x is closed, then $k(x)$ is a field that is an algebra of finite type over K, and so necessarily a K-algebra of *finite rank* [**Zar47**]. Conversely, if $k(x)$ is of finite rank over K, then so is $A/\mathfrak{j}_x \subset k(x)$, and since every integral ring that is also a K-algebra of finite rank is a field, we have that $A/\mathfrak{j}_x = k(x)$, and hence x is closed. □ I | 148

Corollary (6.4.3). — *Let K be an algebraically-closed field, and X an algebraic K-prescheme; the closed points of X are then the rational points over K (3.4.4) and can be canonically identified with the points of X with values in K.*

Proposition (6.4.4). — *Let X be an algebraic prescheme over a field K. The following properties are equivalent.*

(a) *X is Artinian.*
(b) *The underlying space of X is discrete.*
(c) *The underlying space of X has only a finite number of closed points.*
(c′) *The underlying space of X is finite.*
(d) *The points of X are closed.*
(e) *X is isomorphic to* $\mathrm{Spec}(A)$, *where A is a K-algebra of finite rank.*

PROOF. Since X is Noetherian, it follows from (6.2.2) that the conditions (a), (b), and (d) are equivalent, and imply (c) and (c′); it is also clear that (e) implies (a). It remains to see that (c) implies (d) and (e); we can restrict to the case where X is affine. Then $A(X)$ is a K-algebra of finite type (6.3.3), and thus a Jacobson ring ([**CC**, p. 3-11 and 3-12]), in which there are, by hypothesis, only a finite number of maximal ideals. Since a finite intersection of prime ideals can only be a prime ideal if it is equal to one of the prime ideals being intersected, every prime ideal of $A(X)$ is thus maximal, whence (d). Further, we then know (6.2.2) that $A(X)$ is an Artinian K-algebra of finite type, and so necessarily of *finite rank* [**Zar47**]. □

(6.4.5). When the conditions of (6.4.4) are satisfied, we say that X is a scheme *finite over K* (cf. (**II**, 6.1.1)), or a *finite K-scheme*, of *rank* $[A : K]$, which we also denote by $\mathrm{rg}_K(X)$; if X and Y are finite K-schemes, we have

(6.4.5.1) $$\mathrm{rg}_K(X \sqcup Y) = \mathrm{rg}_K(X) + \mathrm{rg}_K(Y),$$

(6.4.5.2) $$\mathrm{rg}_K(X \times_K Y) = \mathrm{rg}_K(X)\,\mathrm{rg}_K(Y),$$

as a result of (3.2.2).

Corollary (6.4.6). — *Let X be a finite K-scheme. For every extension K' of K, $X \otimes_K K'$ as a finite K'-scheme, and its rank over K' is equal to the rank of X over K.*

PROOF. If $A = A(X)$, then we have $[A \otimes_K K' : K'] = [A : K]$. □

Corollary (6.4.7). — *Let X be a scheme finite over a field K; we let $n = \sum_{x \in X} [k(X) : K]_S$* (we recall that if K' is an extension of K, then $[K' : K]_S$ is the *separable rank* of K' over k, the rank of the largest algebraic separable extension of K contained in K'); *then for every algebraically closed extension Ω of K,* **I** | 149
the underlying space of $X \otimes_K \Omega$ has exactly n points, which can be identified with the points of X with values in Ω.

PROOF. We can clearly restrict to the case where the ring $A = A(X)$ is *local* (6.2.2); let $\mathfrak{m}$ be its maximal ideal, and $L = A/\mathfrak{m}$ its residue field, an algebraic extension of K. The points of X with values in Ω then correspond, bijectively, to the Ω-sections of $X \otimes_K \Omega$ ((3.4.1) and (3.3.14)), and also to the K-homomorphisms from L to Ω (1.7.3), whence the proposition (Bourbaki, *Alg.*, chap. V, §7, nº 5, prop. 8), taking (6.4.3) into account. □

(6.4.8). The number n defined in (6.4.7) is called the *separable rank* of A (or of X) over K, or also the *geometric number of points* of X; it is equal to the number of elements of $X(\Omega)_K$. It follows immediately from this definition that, for every extension K' of K, $X \otimes_K K'$ has the same geometric number of points as X. If we denote this number by $n(X)$, it is clear that, if X and Y are two schemes, finite over K, then

(6.4.8.1) $$n(X \sqcup Y) = n(X) + n(Y).$$

Under the same hypotheses, we also have

(6.4.8.2) $$n(X \times_K Y) = n(X)n(Y)$$

because of the interpretation of $n(X)$ as the number of elements of $X(\Omega)_K$ and Equation (3.4.3.1).

Proposition (6.4.9). — *Let K be a field, X and Y algebraic K-preschemes, $f : X \to Y$ a K-morphism, and Ω an algebraically closed extension of K of infinite transcendence degree over K. For f to be surjective, it is necessary and sufficient that the map $X(\Omega)_K \to Y(\Omega)_K$ that corresponds to f (3.4.1) be surjective.*

PROOF. The necessity follows from (6.3.10), noting that f is necessarily of finite type (6.3.9). To see that the condition is sufficient, we argue as in (6.3.10), noting that, for every $y \in Y$, $k(y)$ is an extension of K of finite type, and so is K-isomorphic to a subfield of Ω. □

Remark (6.4.10). — We will see in chapter IV that the conclusion of (6.4.9) still holds without the hypothesis on the transcendence degree of Ω over K.

Proposition (6.4.11). — *If $f : X \to Y$ is a morphism of finite type, then, for every $y \in Y$, the fibre $f^{-1}(y)$ is an algebraic prescheme over the residue field $k(y)$, and for every $x \in f^{-1}(y)$, $k(x)$ is an extension of $k(x)$ of finite type.*

PROOF. Since $f^{-1}(y) = X \otimes_Y k(y)$ (6.3.6), the proposition follows from (6.3.4, iv) and (6.3.3). □

Proposition (6.4.12). — *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms; set $X' = X \times_Y Y'$, and let $f' = f_{(Y')} : X' \to Y'$. Let $y' \in Y'$ and set $y = g(y')$; if the fibre $f^{-1}(y)$ is a finite algebraic scheme over $k(y)$, then the fibre $f'^{-1}(y)$ is a finite algebraic scheme over $k(y')$, and has the same rank and geometric number of points as $f^{-1}(y)$ does.*

PROOF. Taking into account the transitivity of fibres (3.6.5), this follows immediately from (6.4.6) and (6.4.8). □

(6.4.13). Proposition (6.4.11) shows that the morphisms of finite type that correspond, intuitively, to the "algebraic families of algebraic varieties", with the points of Y playing the role of "parameters", **I** | 150
which gives these morphisms a "geometric" meaning. The morphisms which are not of finite type will show up in the following mostly in questions of "changing the base prescheme", by localisation or completion, for example.

6.5. Local determination of a morphism

Proposition (6.5.1). — *Let X and Y be S-preschemes, with Y of finite type over S; let $x \in X$ and $y \in Y$ lie over the same point $s \in S$.*

(i) *If two S-morphisms $f = (\psi, \theta)$ and $f' = (\psi', \theta')$ from X to Y are such that $\psi(x) = \psi'(x) = y$, and the (local) $\mathcal{O}_s$-homomorphisms $\theta_x^\sharp$ and $\theta'^\sharp_x$ from $\mathcal{O}_y$ to $\mathcal{O}_x$ are identical, then f and f' agree on an open neighbourhood of x.*
(ii) *Suppose further that S is locally Noetherian. For every local $\mathcal{O}_s$-homomorphism $\phi : \mathcal{O}_y \to \mathcal{O}_x$, there exists an open neighbourhood U of x in X, and an S-morphism $f = (\psi, \theta)$ from U to Y such that $\psi(x) = y$ and $\theta_x^\sharp = \phi$.*

PROOF.

(i) Since the question is local on S, X, and Y, we can assume that S, X, and Y are affine, given by rings A, B, and C (respectively), and with f and f' of the form $({}^a\phi, \widetilde{\phi})$ and $({}^a\phi', \widetilde{\phi}')$ (respectively), where ϕ and ϕ' are A-homomorphisms from C to B such that $\phi^{-1}(\mathfrak{j}_x) = \phi'^{-1}(\mathfrak{j}_x) = \mathfrak{j}_y$, and the homomorphisms ϕ_x and ϕ'_x from C_y to B_x, induced by ϕ and ϕ', are identical; we can further suppose that C is an A-algebra *of finite type*. Let c_i $(1 \leqslant i \leqslant n)$ be the generators of the A-algebra C, and set $b_i = \phi(c_i)$ and $b'_i = \phi'(c_i)$; by hypothesis, we have $b_i/1 = b'_i/1$ in the ring of fractions B_x $(1 \leqslant i \leqslant n)$. This implies that there exist elements $s_i \in B - \mathfrak{j}_x$ such that $s_i(b_i - b'_i) = 0$ for $1 \leqslant i \leqslant n$, and we can clearly assume that all the s_i are equal to a single element $g \in B - \mathfrak{j}_x$. From this, we conclude that we have $b_i/1 = b'_i/1$ for $1 \leqslant i \leqslant n$ in the ring of fractions B_g; if i_g is the canonical homomorphism $B \to B_g$, we then have $i_g \circ \phi = i_g \circ \phi'$; so the restrictions of f and f' to $D(g)$ are identical.
(ii) We can restrict to the situation as in (i), and further assume that the ring A is Noetherian. Let c_i $(1 \leqslant i \leqslant n)$ be the generators of the A-algebra C, and let $\alpha : A[X_1, \ldots, X_n] \to C$ be

the homomorphism of polynomial algebras that sends X_i to c_i for $1 \leqslant i \leqslant n$. Also let i_y be the canonical homomorphism $C \to C_y$, and consider the composite homomorphism

$$\beta : A[X_1, \ldots, X_n] \xrightarrow{\alpha} C \xrightarrow{i_y} C_y \xrightarrow{\phi} B_x.$$

We denote by $\mathfrak{a}$ the kernel of β; since A is Noetherian, so too is $A[X_1, \ldots, X_n]$, and so $\mathfrak{a}$ admits a finite system of generators $Q_j(X_1, \ldots, X_n)$ $(1 \leqslant j \leqslant m)$. Furthermore, each of the elements $\phi(i_y(c_i))$ can be written in the form b_i/s_i, where $b_i \in B$ and $s_i \notin \mathfrak{j}_x$; we can further **I** | 151
assume that all of the s_i are equal to a single element $g \in B - \mathfrak{j}_x$. With this, by hypothesis, we have $Q_j(b_1/g, \ldots, b_n/g) = 0$ in B_x; set

$$Q_j(X_1/T, \ldots, X_n/T) = P_j(X_1, \ldots, X_n, T)/T^{k_j}$$

where P_j is homogeneous of degree k_j. Then let $d_j = P_j(b_1, \ldots, b_n, g) \in B$. By hypothesis, we have $t_j d_j = 0$ for some $t_j \in B - \mathfrak{j}_x$ $(1 \leqslant j \leqslant m)$, and we can clearly assume that all the t_j are equal to a single element $h \in B - \mathfrak{j}_x$; from this we conclude that $P_j(hb_1, \ldots, hb_n, hg) = 0$ for $1 \leqslant j \leqslant m$. With this, consider the homomorphism ρ from $A[X_1, \ldots, X_n]$ to the ring of fractions B_{hg} which sends X_i to hb_i/hg $(1 \leqslant i \leqslant n)$; the image of $\mathfrak{a}$ under this homomorphism is 0, and is *a fortiori* the same as the image of the kernel $\alpha^{-1}(0)$ under ρ. So ρ factors as $A[X_1, \ldots, X_n] \xrightarrow{\alpha} C \xrightarrow{\gamma} B_{hg}$, with $\gamma(c_i) = hb_i/hg$, and it is clear that, if i_x is the canonical homomorphism $B_{hg} \to B_x$, then the diagram

(6.5.1.1)
$$\begin{tikzcd} C \arrow[r, "\gamma"] \arrow[d, "i_y"'] & B_{hg} \arrow[d, "i_x"] \\ C_y \arrow[r, "\phi"] & B_x \end{tikzcd}$$

is commutative; we thus have $\phi = \gamma_x$, and since ϕ is a local homomorphism, ${}^a\gamma(x) = y$; $f = ({}^a\gamma, \widetilde{\gamma})$ is thus an S-morphism from the neighbourhood $D(hg)$ of x to Y as claimed in the proposition.

□

Corollary (6.5.2). — *Under the hypotheses of (6.5.1, ii), if, further, X is of finite type over S, then we can assume that the morphism f is of finite type.*

PROOF. This follows from Corollary (6.3.6). □

Corollary (6.5.3). — *Suppose that the hypotheses of Proposition (6.5.1, ii), and suppose further that Y is integral, and that ϕ is an injective homomorphism. Then we can assume that $f = ({}^a\gamma, \widetilde{\gamma})$, where γ is injective.*

PROOF. Indeed, we can assume C to be integral (5.1.4), hence i_y injective; it then follows from the diagram (6.5.1.1) that γ is injective. □

Proposition (6.5.4). — *Let $f = (\psi, \theta) : X \to Y$ be a morphism of finite type, x a point of X, and $y = \psi(x)$.*

(i) *For f to be a local immersion at the point x (4.5.1), it is necessary and sufficient that $\theta_x^\sharp : \mathscr{O}_y \to \mathscr{O}_x$ be surjective.*
(ii) *Assume further that Y is locally Noetherian. For f to be a local isomorphism at the point x (4.5.2), it is necessary and sufficient that $\theta_x^\sharp$ be an isomorphism.*

PROOF.

(ii) By (6.5.1), there exists an open neighbourhood V of Y and a morphism $g : V \to X$ such that $g \circ f$ (resp. $f \circ g$) is defined and agrees with the identity on a neighbourhood of x (resp. y), whence we can easily see that f is a local isomorphism.
(i) Since the question is local on X and Y, we can assume that X and Y are affine, given by rings A and B (respectively); we have $f = ({}^a\phi, \widetilde{\phi})$, where ϕ is a homomorphism of rings $B \to A$ that makes A a B-algebra of finite type; we have $\phi^{-1}(\mathfrak{j}_x) = \mathfrak{j}_y$, and the homomorphism $\phi_x : B_y \to A_x$ induced by ϕ is *surjective*. Let (t_i) $(1 \leqslant i \leqslant n)$ be a system

of generators of the B-algebra A; the hypothesis on ϕ_x implies that there exist $b_i \in B$ and some $c \in B - \mathfrak{j}_x$ such that, in the ring of fractions A_x, we have $t_i/1 = \phi(b_i)/\phi(c)$ for $1 \leqslant i \leqslant n$. Then (1.3.3) there exists some $a \in A - \mathfrak{j}_x$ such that, if we let $g = a\phi(c)$, we also have $t_i/1 = a\phi(b_i)/g$ *in the ring of fractions* A_g. With this, there exists, by hypothesis, a polynomial $Q(X_1, \ldots, X_n)$, with coefficients in the ring $\phi(B)$, such that $a = Q(t_1, \ldots, t_n)$; let $Q(X_1/T, \ldots, X_n/T) = P(X_1, \ldots, X_n, T)/T^m$, where P is homogeneous of degree m. In **I** | 152
the ring A_g, we have

$$a/1 = a^m P(\phi(b_1), \ldots, \phi(b_n), \phi(c))/g^m = a^m \phi(d)/g^m$$

where $d \in B$. Since, in A_g, $g/1 = (a/1)(\phi(c)/1)$ is invertible by definition, so too are $a/1$ and $\phi(c)/1$, and we can thus write $a/1 = (\phi(d)/1)(\phi(c)/1)^{-m}$. From this we conclude that $\phi(d)/1$ is also invertible in A_g. So let $h = cd$; since $\phi(h)/1$ is invertible in A_g, the composite homomorphism $B \xrightarrow{\phi} A \to A_g$ factors as $B \to B_h \xrightarrow{\gamma} A_g$ (**0**, 1.2.4). We will show that γ is *surjective*; it suffices to show that the image of B_h in A_g contains the $t_i/1$ and $(g/1)^{-1}$. But we have $(g/1)^{-1} = (\phi(c)/1)^{m-1}(\phi(d)/1)^{-1} = \gamma(c^m/h)$, and $a/1 = \gamma(d^{m+1}/h^m)$, so $(a\phi(b_i))/1 = \gamma(b_i d^{m+1}/h^m)$, and since $t_i/1 = (a\phi(b_i)/1)(g/1)^{-1}$, our claim is proved. The choice of h implies that $\psi(D(g)) \subset D(h)$, and we also know that the restriction of f to $D(g)$ is equal to $({}^a\gamma, \widetilde{\gamma})$; since γ is surjective, this restriction is a closed immersion of $D(g)$ into $D(h)$ (4.2.3).

□

Corollary (6.5.5). — *Let $f = (\psi, \theta) : X \to Y$ be a morphism of finite type. Assume that X is irreducible, and denote by x its generic point, and let $y = \psi(x)$.*

(i) *For f to be a local immersion at any point of X, it is necessary and sufficient that $\theta_x^\sharp : \mathscr{O}_y \to \mathscr{O}_x$ be surjective.*

(ii) *Assume further that Y is irreducible and locally Noetherian. For f to be a local isomorphism at any point of X, it is necessary and sufficient that y be the generic point of Y* (or, equivalently (**0**, 2.1.4), that f be a *dominant* morphism) *and that $\theta_x^\sharp$ be an isomorphism* (in other words, that f be *birational* (2.2.9)).

Proof. It is clear that (i) follows from (6.5.4, i), taking into account the fact that every nonempty open subset of X contains x; similarly, (ii) follows from (6.5.4, ii). □

6.6. Quasi-compact morphisms and morphisms locally of finite type

Definition (6.6.1). — We say that a morphism $f : X \to Y$ is *quasi-compact* if the inverse image of any quasi-compact open subset of Y under f is quasi-compact.

Let $\mathfrak{B}$ be a base for the topology of Y consisting of quasi-compact open subsets (for example, affine open subsets); for f to be quasi-compact, it is necessary and sufficient that the inverse image of every set of $\mathfrak{B}$ under f be quasi-compact (or, equivalently, a *finite* union of affine open subsets), because every quasi-compact open subset of Y is a finite union of sets of $\mathfrak{B}$. For example, if X is *quasi-compact* and Y *affine*, then *every* morphism $f : X \to Y$ is quasi-compact: indeed, X is a finite union of affine open subsets U_i, and for every affine open subset V of Y, $U_i \cap f^{-1}(V)$ is affine (5.5.10), and so quasi-compact.

If $f : X \to Y$ is a quasi-compact morphism, it is clear that, for every open subset V of Y, the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \to V$. Conversely, if (U_α) is an open cover of Y, and $f : X \to Y$ a morphism such that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ are quasi-compact, then f is quasi-compact.

Definition (6.6.2). — We say that a morphism $f : X \to Y$ is *locally of finite type* if, for every $x \in X$, there exists an open neighbourhood U of x and an open neighbourhood $V \supset f(U)$ of y such that the restriction of f to U is a morphism of finite type from U to V. We then also say that X is a prescheme **I** | 153
locally of finite type over Y, or a Y-prescheme locally of finite type.

It follows immediately from (6.3.2) that, if f is locally of finite type, then, for every open subset W of Y, the restriction of f to $f^{-1}(W)$ is a morphism $f^{-1}(W) \to W$ that is locally of finite type.

If Y is locally Noetherian and X locally of finite type over Y, then X is locally Noetherian thanks to (6.3.7).

Proposition (6.6.3). — *For a morphism $f : X \to Y$ to be of finite type, it is necessary and sufficient that it be quasi-compact and locally of finite type.*

PROOF. The necessity of the conditions is immediate, given (6.3.1) and the remark following (6.6.1). Conversely, suppose that the conditions are satisfied, and let U be an affine open subset of Y, given by some ring A; for all $x \in f^{-1}(U)$, there is, by hypothesis, a neighbourhood $V(x) \subset f^{-1}(U)$ of x, and a neighbourhood $W(x) \subset U$ of $y = f(x)$ containing $f(V(x))$, and such that the restriction of f to $V(x)$ is a morphism $V(x) \to W(x)$ of finite type. Replacing $W(x)$ with a neighbourhood $W_1(x) \subset W(x)$ of x of the form $D(g)$ (with $g \in A$), and $V(x)$ with $V(x) \cap f^{-1}(W_1(x))$, we can assume that $W(x)$ is of the form $D(g)$, and thus of finite type over U (because its ring can be written as $A[1/g]$); so $V(x)$ is of finite type over U. Further, $f^{-1}(U)$ is quasi-compact by hypothesis, and so the finite union of open subsets $V(x_i)$, which finishes the proof. $\square$

Proposition (6.6.4). —

(i) *An immersion $X \to Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally Noetherian, or if the underlying space of X is Noetherian.*
(ii) *The composition of any two quasi-compact morphisms is quasi-compact.*
(iii) *If $f : X \to Y$ is a quasi-compact S-morphism, then so too is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any extension $g : S \to S'$ of the base prescheme.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two quasi-compact S-morphisms, then $f \times_S g$ is quasi-compact.*
(v) *If the composition of any two morphisms $f : X \to Y$ and $g : Y \to Z$ is quasi-compact, and if either g is separated or the underlying space of X is locally Noetherian, then f is quasi-compact.*
(vi) *For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be quasi-compact.*

PROOF. We note that (vi) is evident because the property of being quasi-compact, for a morphism, depends only on the corresponding continuous map of underlying spaces. We will similarly prove the part of (v) corresponding to the case where the underlying space of X is locally Noetherian. Set $h = g \circ f$, and let U be a quasi-compact open subset of Y; $g(U)$ is quasi-compact (but not necessarily open) in Z, and so contained in a finite union of quasi-compact open subsets V_j (2.1.3), and $f^{-1}(U)$ is thus contained in the union of the $h^{-1}(V_j)$, which are quasi-compact subspaces of X, and thus Noetherian subspaces. We thus conclude (**0**, 2.2.3) that $f^{-1}(U)$ is a Noetherian space, and *a fortiori* quasi-compact.

To prove the other claims, it suffices to prove (i), (ii), and (iii) (5.5.12). But (ii) is evident, and (i) follows from (6.3.5) whenever the space Y is locally Noetherian or the space X is Noetherian, and is evident for a closed immersion. To show (iii), we can restrict to the case where $S = Y$ (3.3.11); **I** | 154
let $f' = f_{(S')}$, and let U' be a quasi-compact open subset of S'. For every $s' \in U'$, let T be an affine open neighbourhood of $g(s')$ in S, and let W be an affine open neighbourhood of s' contained in $U' \cap g^{-1}(T)$; it will suffice to show that $f'^{-1}(W)$ is quasi-compact; in other words, we can restrict to showing that, when S and S' are *affine*, the underlying space of $X \times_S S'$ is quasi-compact. But since X is then, by hypothesis, a finite union of affine open subsets V_j, $X \times_S S'$ is a union of the underlying spaces of the affine schemes $V_j \times_S S'$ ((3.2.2) and (3.2.7)), which proves the proposition. $\square$

We note also that, if $X = X' \sqcup X''$ is the sum of two preschemes, a morphism $f : X \to Y$ is quasi-compact if and only if its restrictions to both X' and X'' are quasi-compact.

Proposition (6.6.5). — *Let $f : X \to Y$ be a quasi-compact morphism. For f to be dominant, it is necessary and sufficient that, for every generic point y of an irreducible component of Y, $f^{-1}(y)$ contain the generic point of an irreducible component of X.*

PROOF. It is immediate that the condition is sufficient (even without assuming that f is quasi-compact). To see that it is necessary, consider an affine open neighbourhood U of y; $f^{-1}(U)$ is quasi-compact, and so a *finite* union of affine open subsets V_i, and the hypothesis that f be dominant implies that y belongs to the closure *in* U of one of the $f(V_i)$. We can clearly assume X and Y to be reduced; since the closure in X of an irreducible component of V_i is an irreducible component on X

(**0**, 2.1.6), we can replace X by V_i, and Y by the closed reduced subprescheme of U that has $\overline{f(V_i)} \cap U$ as its underlying space (5.2.1), and we are thus led to proving the proposition when $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine and reduced. Since f is dominant, B is a subring of A (1.2.7), and the proposition then follows from the fact that every minimal prime ideal of B is the intersection of B with a minimal prime ideal of A (**0**, 1.5.8). □

Proposition (6.6.6). —

(i) *Every local immersion is locally of finite type.*

(ii) *If two morphisms $f : X \to Y$ and $g : Y \to Z$ are locally of finite type, then so too is $g \circ f$.*

(iii) *If $f : X \to Y$ is an S-morphism locally of finite type, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite type for any extension $S' \to S$ of the base prescheme.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are S-morphisms locally of finite type, then $f \times_S g$ is locally of finite type.*

(v) *If the composition $g \circ f$ of two morphisms is locally of finite type, then f is locally of finite type.*

(vi) *If a morphism f is locally of finite type, then so too is f_{red}.*

Proof. By (5.5.12), it suffices to prove (i), (ii), and (iii). If $j : X \to Y$ is a local immersion then, for every $x \in X$, there is an open neighbourhood V of $j(x)$ in Y and an open neighbourhood U of x in X such that the restriction of j to U is a closed immersion $U \to V$ (4.5.1), and so this restriction is of finite type. To prove (ii), consider a point $x \in X$; by hypothesis, there is an open neighbourhood W of $g(f(x))$ and an open neighbourhood V of $f(x)$ such that $g(V) \subset W$ and such that V is of of finite type over W; furthermore, $f^{-1}(V)$ is locally of finite type over V (6.6.2), so there is an open neighbourhood U of x that is contained in $f^{-1}(V)$ and of is finite type over V; thus we have I | 155
$g(f(U)) \subset W$, and that U is of finite type over W (6.3.4, ii). Finally, to prove (iii), we can restrict to the case where $Y = S$ (3.3.11); for every $x' \in X' = X_{(S')}$, let x be the image of x' in X, s the image of x in S, T an open neighbourhood of s, T' the inverse image of T in S', and U an open neighbourhood of x that is of finite type over T and whose image is contained in T; then $U \times_S T' = U \times_T T'$ is an open neighbourhood of x' (3.2.7) that is of finite type over T' (6.3.4, iv). □

Corollary (6.6.7). — *Let X and Y be S-preschemes that are locally of finite type over S. If S is locally Noetherian, then $X \times_S Y$ is locally Noetherian.*

Proof. Indeed, X being locally of finite type over S means that it is locally Noetherian, and that $X \times_S Y$ is locally of finite type over X, and so $X \times_S Y$ is also locally Noetherian. □

Remark (6.6.8). — Proposition (6.3.10) and its proof extend immediately to the case where we suppose only that the morphism f is locally of finite type. Similarly, propositions (6.4.2) and (6.4.9) hold true when we suppose only that the preschemes X and Y in the claim are locally of finite type over the field K.

§7. Rational maps

7.1. Rational maps and rational functions

(7.1.1). Let X and Y be preschemes, U and V dense open subsets of X, and f (resp. g) a morphism from U (resp. V) to Y; we say that f and g are *equivalent* if they agree on a dense open subset of $U \cap V$. Since a finite intersection of dense open subsets of X is a dense open subset of X, it is clear that this relation is an *equivalence relation*.

Definition (7.1.2). — Given preschemes X and Y, we define a rational map from X to Y to be an equivalence class of morphisms from a dense open subset of X to a dense open subset of Y, under the equivalence relation defined in (7.1.1). If X and Y are S-preschemes, we say that such a class is a rational S-map if there exists a representative of the class that is also an S-morphism. We define a rational S-section of X to be any rational S-map from S to X. We define a rational function on a prescheme X to be any rational X-section on the X-prescheme $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (where T is an indeterminate).

By an abuse of language, whenever we are discussing only S-preschemes, we will say "rational map" instead of "rational S-map" if no confusion may arise.

Let f be a rational map from X to Y, and U an open subset of X; if f_1 and f_2 are two morphisms belonging to the class of f, defined (respectively) on dense open subsets V and W of X, then the restrictions $f_1|(U \cap V)$ and $f_2|(U \cap W)$ agree on $U \cap V \cap W$, which is dense in U; the class f of morphisms thus defines a rational map from U to Y, called the *restriction of f to U*, and denoted by $f|U$.

If, to every S-morphism $f : X \to Y$, we take the corresponding rational S-map to which f belongs, we obtain a canonical map from $\mathrm{Hom}_S(X, Y)$ to the set of rational S-maps from X to Y. We denote by $\Gamma_{\mathrm{rat}}(X/Y)$ the set of rational Y-sections on X, and we thus have a canonical map $\Gamma(X/Y) \to \Gamma_{\mathrm{rat}}(X/Y)$. It is also clear that, if X and Y are S-preschemes, then the set of rational S-maps from X to Y is canonically identified with $\Gamma_{\mathrm{rat}}((X \times_S Y)/X)$ (3.3.14).

(7.1.3). It also follows from (7.1.2) and (3.3.14) that the rational functions on X are canonically identified with *equivalence classes of sections of the structure sheaf $\mathscr{O}_X$* over dense open subsets of X, where two such sections are equivalent if the agree on some dense open subset of X contained inside the intersection of the subsets on which they are defined. In particular, it follows that the rational functions on X form a *ring* $R(X)$.

(7.1.4). When X is an *irreducible* prescheme, every nonempty open subset of X is dense in X; so we can say that the nonempty open subsets of X are the *open neighbourhoods of the generic point x* of X. To say that two morphisms from nonempty open subsets of X to Y are equivalent thus means, in this case, that they have the *same germ* at the point x. In other words, the rational maps (resp. rational S-maps) $X \to Y$ are identified with the *germs of morphisms* (resp. *S-morphisms*) from nonempty open subsets of X to Y at the generic point x of X. In particular:

Proposition (7.1.5). — *If X is an irreducible prescheme, then the ring $R(X)$ of rational maps on X is canonically identified with the local ring $\mathscr{O}_x$ of the generic point x of X. It is a local ring of dimension 0, and thus a local Artinian ring when X is Noetherian; it is a field when X is integral, and, when X is further an affine scheme, it is identified with the field of fractions of $A(X)$.*

PROOF. Given the above, and the identification of rational functions with sections of $\mathscr{O}_X$ over a dense open subset of X, the first claim is nothing but the definition of the fibre of a sheaf above a point. For the other claims, we can reduce to the case where X is affine, given by some ring A; then $\mathfrak{j}_x$ is the nilradical of A, and $\mathscr{O}_x$ is thus of dimension 0; if A is integral, then $\mathfrak{j}_x = (0)$, and $\mathscr{O}_x$ is thus the field of fractions of A. Finally, if A is Noetherian, we know ([**Sam53b**, p. 127, cor. 4]) that $\mathfrak{j}_x$ is nilpotent, and $\mathscr{O}_x = A_x$ Artinian. □

If X is *integral*, then the ring $\mathscr{O}_z$ is integral for all $z \in X$; every affine open subset U containing z also contains x, and $R(U)$, being equal to the field of fractions of $A(U)$, is identified with $R(X)$; we thus conclude that $R(X)$ can also be identified with the *field of fractions of $\mathscr{O}_z$*: the canonical identification of $\mathscr{O}_z$ to a subring of $R(X)$ consists of associating, to every germ of a section $s \in \mathscr{O}_z$, the unique rational function on X, class of a section of $\mathscr{O}_X$, (necessarily defined on a dense open subset of X) having s as its germ at the point z.

(7.1.6). Now suppose that X has a *finite* number of irreducible components X_i ($1 \leqslant i \leqslant n$) (which will be the case whenever the underlying space of X is *Noetherian*); let X'_i be the open subset of X given by the complement of the $X_j \cap X_i$ for $j \neq i$ inside X_i; X'_i is irreducible, its generic point x_i is the generic point of X_i, and the X'_i are pairwise disjoint, with their union being dense in X (**0**, 2.1.6). For every dense open subset U of X, $U_i = U \cap X'_i$ is a nonempty dense open subset of X'_i, with the U_i being pairwise disjoint, and so $U' = \bigcup_i U'_i$ is dense in X. Giving a morphism from U' to Y consists of giving (arbitrarily) a morphism from each of the U_i to Y.

Thus:

Proposition (7.1.7). — *Let X and Y be two preschemes (resp. S-preschemes) such that X has a finite number of irreducible components X_i, with generic points x_i ($1 \leqslant i \leqslant n$). If R_i is the set of germs of morphisms (resp. S-morphisms) from open subsets of X to Y at the point x_i, then the set of rational maps (resp. rational S-maps) from X to Y can be identified with the product of the R_i ($1 \leqslant i \leqslant n$).*

Corollary (7.1.8). — *Let X be a Noetherian prescheme. The ring of rational functions on X is an Artinian ring, whose local components are the rings $\mathscr{O}_{x_i}$ of the generic points x_i of the irreducible components of X.*

Corollary (7.1.9). — *Let A be a Noetherian ring, and $X = \operatorname{Spec}(A)$. If Q is the complement of the union of the minimal prime ideals of A, then the ring of rational functions on X can be canonically identified with the ring of fractions $Q^{-1}A$.*

This will follow from the following lemma:

Lemma (7.1.9.1). — *For an element $f \in A$ to be such that $D(f)$ is dense in X, it is necessary and sufficient that $f \in Q$; every dense open subset of X contains an open subset of the form $D(f)$, where $f \in Q$.*

Proof. To show (7.1.9.1), we again denote by X_i ($1 \leqslant i \leqslant n$) the irreducible components of X; if $D(f)$ is dense in X then $D(f) \cap X_i \neq \varnothing$ for $1 \leqslant i \leqslant n$, and vice-versa; but this means that $f \notin \mathfrak{p}_i$ for $1 \leqslant i \leqslant n$, where we set $\mathfrak{p}_i = \mathfrak{j}(X_i)$, and since the $\mathfrak{p}_i$ are the minimal prime ideals of A (1.1.14), the conditions $f \notin \mathfrak{p}_i$ ($1 \leqslant i \leqslant n$) are equivalent to $f \in Q$, whence the first claim of the lemma. For the other claim, if U is a dense open subset of X, the complement of U is a set of the form $V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal which is not contained in any of the $\mathfrak{p}_i$; it is thus not contained in their union ([**Nor53**, p. 13]), and there thus exists some $f \in \mathfrak{a}$ belonging to Q; whence $D(f) \subset U$, which finishes the proof. □

(7.1.10). Suppose again that X is irreducible, with generic point x. Since every nonempty open subset U of X contains x, and thus also contains every $z \in X$ such that $x \in \overline{\{z\}}$, every morphism $U \to Y$ can be composed with the canonical morphism $\operatorname{Spec}(\mathscr{O}_x) \to X$ (2.4.1); and any two morphisms into Y from two nonempty open subsets of X which agree on a nonempty open subset of X give, by composition, the same morphism $\operatorname{Spec}(\mathscr{O}_x) \to Y$. In other words, to every rational map from X to Y there is a corresponding well-defined morphism $\operatorname{Spec}(\mathscr{O}_x) \to Y$.

Proposition (7.1.11). — *Let X and Y be two S-preschemes; suppose that X is irreducible with generic point x, and that Y is of finite type over S. Any two rational S-maps from X to Y that correspond to the same S-morphism $\operatorname{Spec}(\mathscr{O}_x) \to Y$ are then identical. If we further suppose S to be locally Noetherian, then every S-morphism from $\operatorname{Spec}(\mathscr{O}_x)$ to Y corresponds to exactly one rational S-map from X to Y.*

Proof. Taking into account that every nonempty subset of X is dense in X, this follows from (6.5.1). □

Corollary (7.1.12). — *Suppose that S is locally Noetherian, and that the other hypotheses of (7.1.11) are satisfied. The rational S-maps from X to Y can then be identified with points of the S-prescheme Y, with values in the S-prescheme $\operatorname{Spec}(\mathscr{O}_x)$.*

Proof. This is nothing but (7.1.11), with the terminology introduced in (3.4.1). □

Corollary (7.1.13). — *Suppose that the conditions of (7.1.12) are satisfied. Let s be the image of x in S. The data of a rational S-map from X to Y is equivalent to the data of a point y of Y over s along with a local $\mathscr{O}_s$-homomorphism $\mathscr{O}_y \to \mathscr{O}_x = R(X)$.*

Proof. This follows from (7.1.11) and (2.4.4). □

In particular:

Corollary (7.1.14). — *Under the conditions of (7.1.12), rational S-maps from X to Y depend only (for any given Y) on the S-prescheme $\operatorname{Spec}(\mathscr{O}_x)$, and, in particular, remain the same whenever X is replaced by $\operatorname{Spec}(\mathscr{O}_z)$, for any $z \in X$.*

Proof. Since $z \in \overline{\{x\}}$, x is the generic point of $Z = \operatorname{Spec}(\mathscr{O}_z)$, and $\mathscr{O}_{X,x} = \mathscr{O}_{Z,z}$. □

When X is integral, $R(X) = \mathscr{O}_x = k(x)$ is a field (7.1.5); the preceding corollaries then specialize to the following:

Corollary (7.1.15). — *Suppose that the conditions of (7.1.12) are satisfied, and further that X is integral. Let s be the image of x in S. Then rational S-maps from X to Y can be identified with the geometric points of $Y \otimes_S k(s)$ with values in the extension $R(X)$ of $k(s)$, or, in other words, every such map is equivalent to the data of a point $y \in Y$ above s along with a $k(s)$-monomorphism from $k(y)$ to $k(x) = R(X)$.*

Proof. The points of Y above s are identified with the points of $Y \otimes_S k(s)$ (3.6.3), and the local $\mathscr{O}_s$-homomorphisms $\mathscr{O}_y \to R(X)$ with the $k(s)$-monomorphisms $k(y) \to R(X)$. □

More precisely:

Corollary (7.1.16). — *Let k be a field, and X and Y two algebraic preschemes over k (6.4.1); suppose further that X is integral. Then the rational k-maps from X to Y can be identified with the geometric points of Y with values in the extension $R(X)$ of k (3.4.4).*

7.2. Domain of definition of a rational map

(7.2.1). Let X and Y be preschemes, and f rational map from X to Y. We say that f is *defined at a point* $x \in X$ if there exists a dense open subset U of X that contains x, and a morphism $U \to Y$ belonging to the equivalence class of f. The set of points $x \in X$ where f is defined is called the *domain of definition* of f; it is clear that it is an open dense subset of X.

Proposition (7.2.2). — *Let X and Y be S-preschemes such that X is reduced and Y is separated over S. Let f be a rational S-map from X to Y, with domain of definition U_0. Then there exists exactly one S-morphism $U_0 \to Y$ belonging to the class of f.* **I** | 159

Since, for every morphism $U \to Y$ belonging to the class of f, we necessarily have $U \subset U_0$, it is clear that the proposition will be a consequence of the following:

Lemma (7.2.2.1). — *Under the hypotheses of (7.2.2), let U_1 and U_2 be two dense open subsets of X, and $f_i : U_i \to Y$ $(i = 1, 2)$ two S-morphisms such that there exists an open subset $V \subset U_1 \cap U_2$, dense in X, and on which f_1 and f_2 agree. Then f_1 and f_2 agree on $U_1 \cap U_2$.*

PROOF. We can clearly restrict to the case where $X = U_1 = U_2$. Since X (and thus V) is reduced, X is the smallest closed subprescheme of X containing V (5.2.2). Let $g = (f_1, f_2)_S : X \to Y \times_S Y$; since, by hypothesis, the diagonal $T = \Delta_Y(Y)$ is a closed subprescheme of $Y \times_S Y$, $Z = g^{-1}(T)$ is a closed subprescheme of X (4.4.1). If $h : V \to Y$ is the common restriction of f_1 and f_2 to V, then the restriction of g to V is $g' = (h, h)_S$, which factors as $g' = \Delta_Y \circ h$; since $\Delta_Y^{-1}(T) = Y$, we have that $g'^{-1}(T) = V$, and so Z is a closed subprescheme of X inducing V, thus containing V, which implies that $Z = X$. From the equation $g^{-1}(T) = X$, we deduce (4.4.1) that g factors as $\Delta_Y \circ f$, where f is a morphism $X \to Y$, which implies, by the definition of the diagonal morphism, that $f_1 = f_2 = f$. □

It is clear that the morphism $U_0 \to Y$ defined in (7.2.2) is the unique morphism of the class f that *cannot be extended* to a morphism from an open subset of X that strictly contains U_0. *Under the hypotheses of* (7.2.2), we can thus *identify* the rational maps from X to Y with the *non-extendible* (to strictly larger open subsets) morphisms from dense open subsets of X to Y. With this identification, Proposition (7.2.2) implies:

Corollary (7.2.3). — *With the hypotheses from (7.2.2) on X and Y, let U be a dense open subset of X. Then there exists a canonical bijective correspondence between S-morphisms from U to Y and rational S-maps from X to Y that are defined at all points of U.*

PROOF. By (7.2.2), for every S-morphism f from U to Y, there exists exactly one rational S-map $\overline{f}$ from X to Y which extends f. □

Corollary (7.2.4). — *Let S be a scheme, X a reduced S-prescheme, Y an S-scheme, and $f : U \to Y$ an S-morphism from a dense open subset U of X to Y. If $\overline{f}$ is the rational* **Z***-map from X to Y that extends f, then $\overline{f}$ is an S-morphism (and thus the rational S-map from X to Y that extends f).*

PROOF. Indeed, if $\phi : X \to S$ and $\psi : Y \to S$ are the structure morphisms, U_0 the domain of definition of $\overline{f}$, and j the injection $U_0 \to X$, then it suffices to show that $\psi \circ \overline{f} = \phi \circ j$, but this follows from (7.2.2.1), since f is an S-morphism. □

Corollary (7.2.5). — *Let X and Y be two S-preschemes; suppose that X is reduced, and that X and Y are separated over S. Let $p : Y \to X$ be an S-morphism (making Y an X-prescheme), U a dense open subset of X, and f a U-section of Y; then the rational map $\overline{f}$ from X to Y extending f is a rational X-section of Y.*

PROOF. We have to show that $p \circ \overline{f}$ is the identity on the domain of definition of $\overline{f}$; since X is separated over S, this again follows from (7.2.2.1). □ **I** | 160

Corollary (7.2.6). — *Let X be a reduced prescheme, and U a dense open subset of X. Then there is a canonical bijective correspondence between sections of $\mathscr{O}_X$ over U and rational functions on X defined at every point of U.*

PROOF. Taking (7.2.3), (7.1.2), and (7.1.3) into account, it suffices to note that the X-prescheme $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ is separated over X (5.5.1, iv). □

Corollary (7.2.7). — *Let Y be a reduced prescheme, $f : X \to Y$ a separated morphism, U a dense open subset of Y, $g : U \to f^{-1}(U)$ a U-section of $f^{-1}(U)$, and Z the reduced subprescheme of X that has $\overline{g(U)}$ as its underlying space* (5.2.1)*. For g to be the restriction of a Y-section of X* (in other words (7.2.5), for the rational map from Y to X extending g to be defined everywhere)*, it is necessary and sufficient for the restriction of f to Z to be an isomorphism from Z to Y.*

PROOF. The restriction of f to $f^{-1}(U)$ is a separated morphism (5.5.1, i), so g is a closed immersion (5.4.6), and so $g(U) = Z \cap f^{-1}(U)$, and the subprescheme induced by Z on the open subset $g(U)$ of Z is identical to the closed subprescheme of $f^{-1}(U)$ associated to g (5.2.1). It is then clear that the stated condition is sufficient, because, if satisfied, and if $f_Z : Z \to Y$ is the restriction of f to Z, and $\overline{g} : Y \to Z$ is the inverse isomorphism, then $\overline{g}$ extends g. Conversely, if g is the restriction to U of a Y-section h of X, then h is a closed immersion (5.4.6), and so $h(Y)$ is closed, and, since it is contained in Z, is equal to Z, and it follows from (5.2.1) that h is necessarily an isomorphism from Y to the closed subprescheme Z of X. □

(7.2.8). Let X and Y be two S-preschemes, with X reduced, and Y separated over S. Let f be a rational S-map from X to Y, and let x be a point of X; we can compose f with the canonical S-morphism $\operatorname{Spec}(\mathscr{O}_x) \to X$ (2.4.1) provided that the intersection of $\operatorname{Spec}(\mathscr{O}_x)$ with the domain of definition of f is dense in $\operatorname{Spec}(\mathscr{O}_x)$ (identified with the set of $z \in X$ such that $x \in \overline{\{z\}}$ (2.4.2)). This will happen in the follow cases:

1st. X is *irreducible* (and thus *integral*), because then the generic point ξ of X is the generic point of $\operatorname{Spec}(\mathscr{O}_x)$; since the domain of definition U of f contains ξ, $U \cap \operatorname{Spec}(\mathscr{O}_x)$ contains ξ, and so is dense in $\operatorname{Spec}(\mathscr{O}_x)$.

2nd. X is *locally Noetherian*; our claim then follows from:

Lemma (7.2.8.1). — *Let X be a prescheme whose underlying space is locally Noetherian, and x a point of X. The irreducible components of* $\operatorname{Spec}(\mathscr{O}_x)$ *are the intersections of* $\operatorname{Spec}(\mathscr{O}_x)$ *with the irreducible components of X containing x. For an open subset $U \subset X$ to be such that $U \cap \operatorname{Spec}(\mathscr{O}_x)$ is dense* $\operatorname{Spec}(\mathscr{O}_x)$*, it is necessary and sufficient for it to have a nonempty intersection with the irreducible components of X that contain x* (which will be the case whenever U is *dense* in X).

PROOF. It suffices to show just the first claim, since the second then follows. Since $\operatorname{Spec}(\mathscr{O}_x)$ is contained in every affine open subset U that contains x, and since the irreducible components of U that contain x are the intersections of U with the irreducible components of X containing x (**0**, 2.1.6), we can suppose that X is affine, given by some ring A. Since the prime ideals of A_x correspond bijectively to the prime ideals of A that are contained in $\mathfrak{j}_x$ (2.1.6), the minimal prime ideals of A_x **I** | 161
correspond to the minimal prime ideals of A that are contained in $\mathfrak{j}_x$, hence the lemma. □

With this in mind, suppose that we are in one of the two cases mentioned in (7.2.8). If U is the domain of definition of the rational S-map f, then we denote by f' the rational map from $\operatorname{Spec}(\mathscr{O}_x)$ to Y which agrees (taking (2.4.2) into account) with f on $U \cap \operatorname{Spec}(\mathscr{O}_x)$; we say that this rational map is *induced by f*.

Proposition (7.2.9). — *Let S be a locally Noetherian prescheme, X a reduced S-prescheme, and Y an S-scheme of finite type. Suppose further that X is either irreducible or locally Noetherian. Then let f be a rational S-map from X to Y, and x a point of X. For f to be defined at a point x, it is necessary and sufficient for the rational map f' from* $\operatorname{Spec}(\mathscr{O}_x)$ *to Y, induced by f* (7.2.8)*, to be a morphism.*

PROOF. The condition clearly being necessary (since $\operatorname{Spec}(\mathscr{O}_x)$ is contained in every open subset containing x), we show that it is sufficient. By (6.5.1), there exists an open neighbourhood U of x in X, and an S-morphism g from U to Y that induces f' on $\operatorname{Spec}(\mathscr{O}_x)$. If X is irreducible, then U is dense in X, and, by (7.2.3), we can suppose that g is a rational S-map. Further, the generic point of

X belongs to both $\operatorname{Spec}(\mathscr{O}_x)$ and the domain of definition of f, and so s and g agree at this point, and thus on a nonempty open subset of X (6.5.1). But since f and g are rational S-maps, they are identical (7.2.3), and so f is defined at x.

If we now suppose that X is locally Noetherian, then we can suppose that U is Noetherian; then there are only a finite number of irreducible components X_i of X that contain x (7.2.8.1), and we can suppose that they are the only ones that have a nonempty intersection with U, by replacing, if needed, U with a smaller open subset (since there are only a finite number of irreducible components of X that have a nonempty intersection with U, because U is Noetherian). We then have, as above, that f and g agree on a nonempty open subset of each of the X_i. Taking into account the fact that each of the X_i is contained in $\overline{U}$, we consider the morphism f_1, defined on a dense open subset of $U \cup (X - \overline{U})$, equal to g on U, and to f on the intersection of $X - \overline{U}$ with the domain of definition of f. Since $U \cup (X - \overline{U})$ is dense in X, f_1 and f agree on a dense open subset of X, and since f is a rational map, f is an extension of f_1 (7.2.3), and is thus defined at the point x. □

7.3. Sheaf of rational functions

(7.3.1). Let X be a prescheme. For every open subset $U \subset X$, we denote by $R(U)$ the ring of rational functions on U (7.1.3); this is a $\Gamma(U, \mathscr{O}_X)$-algebra. Further, if $V \subset U$ is a second open subset of X, then every section of $\mathscr{O}_X$ over a dense (in X) open subset of V gives, by restriction to V, a section over a dense (in X) open subset of V, and if two sections agree on a dense (in X) open subset of U, then their restrictions to V agree on a dense (in X) open subset of V. We can thus define a di-homomorphism of algebras $\rho_{V,U} : R(U) \to R(V)$, and it is clear that, if $U \supset V \supset W$ are open subsets of X, then we have $\rho_{W,U} = \rho_{W,V} \circ \rho_{V,U}$; the $R(U)$ thus define a *presheaf* of algebras on X.

Definition (7.3.2). — We define the sheaf of rational functions on a prescheme X, denoted by $\mathscr{R}(X)$, to be the $\mathscr{O}_X$-algebra associated to the presheaf defined by the $R(U)$. **I** | 162

For every prescheme X and open subset $U \subset X$, it is clear that the induced sheaf $\mathscr{R}(X)|U$ is exactly $\mathscr{R}(U)$.

Proposition (7.3.3). — *Let X be a prescheme such that the family (X_λ) of its irreducible components is locally finite (which is the case whenever the underlying space of X is locally Noetherian). Then the $\mathscr{O}_X$-module $\mathscr{R}(X)$ is quasi-coherent, and for every open subset U of X that has a nonempty intersection with only finitely many of the components X_λ, $R(U)$ is equal to $\Gamma(U, \mathscr{R}(X))$, and can be canonically identified with the direct sum of the local rings of the generic points of the X_λ such that $U \cap X_\lambda \neq \varnothing$.*

PROOF. We can evidently restrict to the case where X has only a finite number of irreducible components X_i, with generic points x_i $(1 \leqslant i \leqslant n)$. The fact that $R(U)$ can be canonically identified with the direct sum of the $\mathscr{O}_{x_i} = R(X_i)$ such that $U \cap X_i \neq \varnothing$ then follows from (7.1.7). We will show that the presheaf $U \to R(U)$ satisfies the sheaf axioms, which will prove that $R(U) = \Gamma(U, \mathscr{R}(X))$. Indeed, it satisfies (F1) by what has already been discussed. To see that it satisfies (F2), consider a cover of an open subset U of X by open subsets $V_\alpha \subset U$; if the $s_\alpha \in R(V_\alpha)$ are such that the restrictions of s_α and s_β to $V_\alpha \cap V_\beta$ agree for every pair of indices, then we can conclude that, for every index i such that $U \cap X_i \neq \varnothing$, the components in $R(X_i)$ of all the s_α such that $V_\alpha \cap X_i \neq \varnothing$ are all the same; denoting this component by t_i, it is clear that the element of $R(U)$ that has the t_i as its components has s_α as its restriction to each V_α. Finally, to see that $\mathscr{R}(X)$ is quasi-coherent, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine; by taking U to be an affine open subset of the form $D(f)$, where $f \in A$, it follows from the above and from Definition (1.3.4) that we have $\mathscr{R}(X) = \widetilde{M}$, where M is the direct sum of the A-modules A_{x_i}. □

Corollary (7.3.4). — *Let X be a reduced prescheme that has only a finite number of irreducible components, and let X_i $(1 \leqslant i \leqslant n)$ be the closed reduced preschemes of X that have the irreducible components of X as their underlying spaces (5.2.1). If h_i is the canonical injection $X_i \to X$, then $\mathscr{R}(X)$ is the direct sum of the $\mathscr{O}_X$-algebras $(h_i)_*(\mathscr{R}(X_i))$.*

Corollary (7.3.5). — *If X is irreducible, then every quasi-coherent $\mathscr{R}(X)$-module $\mathscr{F}$ is a simple sheaf.*

PROOF. It suffices to show that every $x \in X$ admits a neighbourhood U such that $\mathscr{F}|U$ is a simple sheaf (**0**, 3.6.2); in other words, we are led to considering the case where X is affine; we can further suppose that $\mathscr{F}$ is the cokernel of a homomorphism $(\mathscr{R}(X))^I \to (\mathscr{R}(X))^J$ (**0**, 5.1.3), and everything then follows from showing that $\mathscr{R}(X)$ is a simple sheaf; but this is evident, because $\Gamma(U, \mathscr{R}(X)) = R(X)$ for every nonempty open subset U, where U contains the generic point of X. □

Corollary (7.3.6). — *If X is irreducible, then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a simple sheaf; if, further, X is reduced (and thus integral), then $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is isomorphic to a sheaf of the form $(\mathscr{R}(X))^{(I)}$.*

PROOF. The second claim follows from the fact that $R(X)$ is a field. □

Proposition (7.3.7). — *Suppose that the prescheme X is locally integral or locally Noetherian. Then $\mathscr{R}(X)$ is a quasi-coherent $\mathscr{O}_X$-algebra; if, further, X is reduced (which will be the case whenever X is locally integral), then the canonical homomorphism $\mathscr{O}_X \to \mathscr{R}(X)$ is injective.* **I** | 163

PROOF. Since the questions is local, the first claim follows from (7.3.3); the second follows from (7.2.3). □

(7.3.8). [10] Let X and Y be two *integral* preschemes, which implies that $\mathscr{R}(X)$ (resp. $\mathscr{R}(Y)$) is a quasi-coherent $\mathscr{O}_X$-module (resp. $\mathscr{O}_Y$-module) (7.3.3). Let $f : X \to Y$ be a *dominant* morphism; then there exists a canonical homomorphism of $\mathscr{O}_X$-modules

$$\tau : f^*(\mathscr{R}(Y)) \longrightarrow \mathscr{R}(X). \tag{7.3.8.1}$$

PROOF. Suppose first that $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine, given by integral rings A and B, with f thus corresponding to an injective homomorphism $B \to A$ which extends to a monomorphism $L \to K$ from the field of fractions L of B to the field of fractions K of A. The homomorphism (7.3.8.1) then corresponds to the canonical homomorphism $L \otimes_B A \to K$ (1.6.5). In the general case, for each pair of nonempty affine open sets $U \subset X$ and $V \subset Y$ such that $f(U) \subset V$, we define, as above, a homomorphism $\tau_{U,V}$ and we immediately have that, if $U' \subset U$, $V' \subset V$, $f(U') \subset V'$, then $\tau_{U,V}$ extends $\tau_{U',V'}$, and hence our assertion. If x and y are the generic points of X and Y respectively, then we have $f(x) = y$,

$$(f^*(\mathscr{R}(Y)))_x = \mathscr{O}_y \otimes_{\mathscr{O}_y} \mathscr{O}_x = \mathscr{O}_x$$

(**0**, 4.3.1) and τ_x is thus an *isomorphism*. □

7.4. Torsion sheaves and torsion-free sheaves

(7.4.1). Let X be an *integral* scheme. For every $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathscr{O}_X \to \mathscr{R}(X)$ defines, by tensoring, a homomorphism (again said to be *canonical*) $\mathscr{F} \to \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$, which, on each fibre, is exactly the homomorphism $z \to z \otimes 1$ from $\mathscr{F}_x$ to $\mathscr{F}_x \otimes_{\mathscr{O}_X} R(X)$. The *kernel* $\mathscr{T}$ of this homomorphism is an $\mathscr{O}_X$-submodule of $\mathscr{F}$, called the *torsion sheaf* of $\mathscr{F}$; it is quasi-coherent if $\mathscr{F}$ is quasi-coherent ((4.1.1) and (7.3.6)). We say that $\mathscr{F}$ is *torsion free* if $\mathscr{T} = 0$, and that $\mathscr{F}$ is a *torsion sheaf* if $\mathscr{T} = \mathscr{F}$. For every $\mathscr{O}_X$-module $\mathscr{F}$, $\mathscr{F}/\mathscr{T}$ is torsion free. We deduce from (7.3.5) that:

Proposition (7.4.2). — *If X is an integral prescheme, then every torsion-free quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is isomorphic to a subsheaf $\mathscr{G}$ of a simple sheaf of the form $(\mathscr{R}(X))^{(I)}$, generated (as a $\mathscr{R}(X)$-module) by $\mathscr{G}$.*

The cardinality of I is called the *rank* of $\mathscr{F}$; for every nonempty affine open subset U of X, the rank of $\mathscr{F}$ is equal to the rank of $\Gamma(U, \mathscr{F})$ as a $\Gamma(U, \mathscr{O}_X)$-module, as we see by considering the generic point of X, contained in U. In particular:

Corollary (7.4.3). — *On an integral prescheme X, every torsion-free quasi-coherent $\mathscr{O}_X$-module of rank* 1 *(in particular, every invertible $\mathscr{O}_X$-module) is isomorphic to an $\mathscr{O}_X$-submodule of $\mathscr{R}(X)$, and vice versa.*

Corollary (7.4.4). — *Let X be an integral prescheme, $\mathscr{L}$ and $\mathscr{L}'$ torsion-free $\mathscr{O}_X$-modules, and f (resp. f') a section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X. In order to have $f \otimes f' = 0$, it is necessary and sufficient for one of the sections f and f' to be zero.*

[10] *[Trans.] This paragraph was changed entirely in the Errata of EGA II.*

Proof. Let x be the generic point of X; we have, by hypothesis, that $(f \otimes f')_x = f_x \otimes f'_x = 0$. Since $\mathscr{L}_x$ and $\mathscr{L}'_x$ can be identified with $\mathscr{O}_X$-submodules of the field $\mathscr{O}_x$, the above equation leads to $f_x = 0$ or $f'_x = 0$, and thus $f = 0$ or $f' = 0$, since $\mathscr{L}$ and $\mathscr{L}'$ are torsion free (7.3.5). □

Proposition (7.4.5). — *Let X and Y be integral preschemes, and $f : X \to Y$ a dominant morphism. For every torsion-free quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $f_*(\mathscr{F})$ is a torsion-free $\mathscr{O}_Y$-module.*

Proof. Since f_* is left exact (**0**, 4.2.1), it suffices, by (7.4.2), to prove the proposition in the case where $\mathscr{F} = (\mathscr{R}(X))^{(I)}$. But every nonempty open subset U of Y contains the generic point of Y, so $f^{-1}(U)$ contains the generic point of X (**0**, 2.1.5), so we have that $\Gamma(U, f_*(\mathscr{F})) = \Gamma(f^{-1}(U), \mathscr{F}) = (R(X))^{(I)}$; in other words, $f_*(\mathscr{F})$ is the simple sheaf with fibre $(R(X))^{(I)}$, considered as a $\mathscr{R}(Y)$-module, and it is clearly torsion free. □ I | 164

Proposition (7.4.6). — *Let X be an integral prescheme, and x its generic point. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type, the following conditions are equivalent:* (a) *$\mathscr{F}$ is a torsion sheaf;* (b) *$\mathscr{F}_x = 0$;* (c) *$\operatorname{Supp}(\mathscr{F}) \neq X$.*

Proof. By (7.3.5) and (7.4.1), the equations $\mathscr{F}_x = 0$ and $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{R}(X) = 0$ are equivalent, so (a) and (b) are equivalent; then $\operatorname{Supp}(\mathscr{F})$ is closed in X (**0**, 5.2.2), and since every nonempty open subset of X contains x, (b) and (c) are equivalent. □

(7.4.7). We generalise (by an abuse of language) the definitions of (7.4.1) to the case where X is a *reduced* prescheme having only a *finite* number of irreducible components; it then follows from (7.3.4) that the equivalence between *a)* and *c)* in (7.4.6) still holds true for such a prescheme.

§8. Chevalley schemes

8.1. Allied local rings For each local ring A, we denote by $\mathfrak{m}(A)$ the maximal ideal of A.

Lemma (8.1.1). — *Let A and B be two local rings such that $A \subset B$; Then the following conditions are equivalent.*

(i) $\mathfrak{m}(B) \cap A = \mathfrak{m}(A)$.
(ii) $\mathfrak{m}(A) \subset \mathfrak{m}(B)$.
(iii) 1 *is not an element of the ideal of B generated by* $\mathfrak{m}(A)$.

Proof. It is evident that (i) implies (ii), and (ii) implies (iii); lastly, if (iii) is true, then $\mathfrak{m}(B) \cap A$ contains $\mathfrak{m}(A)$, and does not contain 1, and is thus equal to $\mathfrak{m}(A)$.

When the equivalent conditions of (8.1.1) are satisfied, we say that B *dominates* A; this is equivalent to saying that the injection $A \to B$ is a *local* homomorphism. It is clear that, in the set of local subrings of a ring R, the relation given by domination is an order. □

(8.1.2). Now consider a *field* R. For all subrings A of R, we denote by $L(A)$ the set of local rings $A_\mathfrak{p}$, where $\mathfrak{p}$ ranges over the prime spectrum of A; such local rings are identified with the subrings of R containing A. Since $\mathfrak{p} = (\mathfrak{p}A_\mathfrak{p}) \cap A$, the map $\mathfrak{p} \mapsto A_\mathfrak{p}$ from $\operatorname{Spec}(A)$ to $L(A)$ is bijective.

Lemma (8.1.3). — *Let R be a field, and A a subring of R. For a local subring M of R to dominate a ring $A_\mathfrak{p} \in L(A)$, it is necessary and sufficient that $A \subset M$; the local ring $A_\mathfrak{p}$ dominated by M is then unique, and corresponds to $\mathfrak{p} = \mathfrak{m}(M) \cap A$.*

Proof. If M dominates $A_\mathfrak{p}$, then $\mathfrak{m}(M) \cap A_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$, by (8.1.1), whence the uniqueness of $\mathfrak{p}$; on the other hand, if $A \subset M$, then $\mathfrak{m}M \cap A = \mathfrak{p}$ is prime in A, and since $A - \mathfrak{p} \subset M$, we have that $A_\mathfrak{p} \subset M$ and $\mathfrak{p}A_\mathfrak{p} \subset \mathfrak{m}(M)$, so M dominates $A_\mathfrak{p}$. □

Lemma (8.1.4). — *Let R be a field, M and N local subrings of R, and P the subring of R generated by $M \cup N$. Then the following conditions are equivalent.* I | 165

(i) *There exists a prime ideal $\mathfrak{p}$ of P such that $\mathfrak{m}(M) = \mathfrak{p} \cap M$ and $\mathfrak{m}(N) = \mathfrak{p} \cap N$.*
(ii) *The ideal $\mathfrak{a}$ generated in P by $\mathfrak{m}(M) \cup \mathfrak{m}(N)$ is distinct from P.*
(iii) *There exists a local subring Q of R simultaneously dominating both M and N.*

PROOF. It is clear that (i) implies (ii); conversely, if $\mathfrak{a} \neq P$, then $\mathfrak{a}$ is contained in a maximal ideal $\mathfrak{n}$ of P, and since $1 \notin \mathfrak{n}$, $\mathfrak{n} \cap M$ contains $\mathfrak{m}(M)$ and is distinct from M, so $\mathfrak{n} \cap M = \mathfrak{m}(M)$, and similarly $\mathfrak{n} \cap N = \mathfrak{m}(N)$. It is clear that, if Q dominates both M and N, then $P \subset Q$ and $\mathfrak{m}(M) = \mathfrak{m}(Q) \cap M = (\mathfrak{m}(Q) \cap P) \cap M$, and $\mathfrak{m}(N) = (\mathfrak{m}(Q) \cap P) \cap N$, so (iii) implies (i); the converse is evident when we take $Q = P_{\mathfrak{p}}$. □

When the conditions of (8.1.4) are satisfied, we say, with C. Chevalley, that the local rings M and N are *allied*.

Proposition (8.1.5). — *Let A and B be subrings of a field R, and C the subring of R generated by $A \cup B$. Then the following conditions are equivalent.*

(i) *For every local ring Q containing A and B, we have that $A_{\mathfrak{p}} = B_{\mathfrak{q}}$, where $\mathfrak{p} = \mathfrak{m}(Q) \cap A$ and $\mathfrak{q} = \mathfrak{m}(Q) \cap B$.*
(ii) *For all prime ideals $\mathfrak{r}$ of C, we have that $A_{\mathfrak{p}} = B_{\mathfrak{q}}$, where $\mathfrak{p} = \mathfrak{r} \cap A$ and $\mathfrak{q} = \mathfrak{r} \cap B$.*
(iii) *If $M \in L(A)$ and $N \in L(B)$ are allied, then they are identical.*
(iv) *$L(A) \cap L(B) = L(C)$.*

PROOF. Lemmas (8.1.3) and (8.1.4) prove that (i) and (iii) are equivalent; it is clear that (i) implies (ii) by taking $Q = C_{\mathfrak{r}}$; conversely, (ii) implies (i), because if Q contains $A \cup B$ then it contains C, and if $\mathfrak{r} = \mathfrak{m}(Q) \cap C$, then $\mathfrak{p} = \mathfrak{r} \cap A$ and $\mathfrak{q} = \mathfrak{r} \cap B$, by (8.1.3). It is immediate that (iv) implies (i), because if Q contains $A \cup B$ then it dominates a local ring $C_{\mathfrak{r}} \in L(C)$ by (8.1.3); by hypothesis we have that $C_{\mathfrak{r}} \in L(A) \cap L(B)$, and (8.1.1) and (8.1.3) prove that $C_{\mathfrak{r}} = A_{\mathfrak{p}} = B_{\mathfrak{q}}$. We prove finally that (iii) implies (iv). Let $Q \in L(C)$; Q dominates some $M \in L(A)$ and some $N \in L(B)$ (8.1.3), so M and N, being allied, are identical by hypothesis. As we then have that $C \subset M$, we know that M dominates some $Q' \in L(C)$ (8.1.3), so Q dominates Q', whence necessarily (8.1.3) $Q = Q' = M$, so $Q \in L(A) \cap L(B)$. Conversely, if $Q \in L(A) \cap L(B)$, then $C \subset Q$, so (8.1.3) Q dominates some $Q'' \in L(C) \subset L(A) \cap L(B)$; Q and Q'', being allied, are identical, so $Q'' = Q \in L(C)$, which completes the proof. □

8.2. Local rings of an integral scheme

(8.2.1). Let X be an *integral* prescheme, and R its field of rational functions, identical to the local ring of the generic point a of X; for all $x \in X$, we know that $\mathscr{O}_x$ can be canonically identified with a subring of R (7.1.5), and for every rational function $f \in R$, the domain of definition $\delta(f)$ of f is the open set of $x \in X$ such that $f \in \mathscr{O}_x$. It thus follows, from (7.2.6), that, for every open $U \subset X$, we have

$$\Gamma(U, \mathscr{O}_X) = \bigcap_{x \in U} \mathscr{O}_x. \tag{8.2.1.1}$$

Proposition (8.2.2). — *Let X be an integral prescheme, and R its field of rational fractions. For X to be a scheme, it is necessary and sufficient for the relation "$\mathscr{O}_x$ and $\mathscr{O}_y$ are allied" (8.1.4), for points x and y of X, to imply that $x = y$.* I | 166

PROOF. We suppose that this condition is satisfied, and aim to show that X is separated. Let U and V be two distinct affine open subsets of X, given by rings A and B (respectively), identified with subrings of R; U (resp. V) is thus identified (8.1.2) with $L(A)$ (resp. $L(B)$), and the hypotheses tell us (8.1.5) that C is the subring of R generated by $A \cup B$, and $W = U \cap V$ is identified with $L(A) \cap L(B) = L(C)$. Furthermore, we know ([**CC**], p. 5-03, 4 *bis*) that every subring E of R is equal to the intersection of the local rings belonging to $L(E)$; C is thus identified with the intersection of the rings $\mathscr{O}_z$ for $z \in W$, or, equivalently (8.2.1.1), with $\Gamma(W, \mathscr{O}_X)$. So consider the subprescheme induced by X on W; to the identity morphism $\phi : C \to \Gamma(W, \mathscr{O}_X)$ there corresponds (2.2.4) a morphism $\Phi = (\psi, \theta) : W \to \operatorname{Spec}(C)$; we will see that Φ is an *isomorphism* of preschemes, whence W is an *affine* open subset. The identification of W with $L(C) = \operatorname{Spec}(C)$ shows that ψ is *bijective*. On the other hand, for all $x \in W$, $\theta_x^\sharp$ is the injection $C_{\mathfrak{r}} \to \mathscr{O}_x$, where $\mathfrak{r} = \mathfrak{m}_x \cap C$, and, by definition, $C_{\mathfrak{r}}$ is identified with $\mathscr{O}_x$, so $\theta_x^\sharp$ is bijective. It thus remains to show that ψ is a *homeomorphism*, or, in other words, that for every closed subset $F \subset W$, $\psi(F)$ is closed in $\operatorname{Spec}(C)$. But F is the intersection of W with a closed subspace of U of the form $V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A; we will show that $\psi(F) = V(\mathfrak{a}C)$, which proves our claim. In fact, the prime ideals of C containing $\mathfrak{a}C$ are the prime ideals of C containing $\mathfrak{a}$, and so are the ideals of the form $\psi(x) = \mathfrak{m}_x \cap C$, where $\mathfrak{a} \subset \mathfrak{m}_x$ and $x \in W$; since $\mathfrak{a} \subset \mathfrak{m}_x$ is equivalent to $x \in V(\mathfrak{a}) = W \cap F$ for $x \in U$, we do indeed have that $\psi(F) = V(\mathfrak{a}C)$.

It follows that X is separated, because $U \cap V$ is affine and its ring C is generated by the union $A \cup B$ of the rings of U and V (5.5.6).

Conversely, suppose that X is separated, and let x and y be points of X such that $\mathscr{O}_x$ and $\mathscr{O}_y$ are allied. Let U (resp. V) be an affine open subset containing x (resp. y), of ring A (resp. B); we then know that $U \cap V$ is affine and that its ring C is generated by $A \cup B$ (5.5.6). If $\mathfrak{p} = \mathfrak{m}_x \cap A$ and $\mathfrak{q} = \mathfrak{m}_y \cap B$, then $A_\mathfrak{p} = \mathscr{O}_x$ and $B_\mathfrak{q} = \mathscr{O}_y$, and since $A_\mathfrak{p}$ and $B_\mathfrak{q}$ are allied, there exists a prime ideal $\mathfrak{r}$ of C such that $\mathfrak{p} = \mathfrak{r} \cap A$ and $\mathfrak{q} = \mathfrak{r} \cap B$ (8.1.4). But then there exists a point $z \in U \cap V$ such that $\mathfrak{r} = \mathfrak{m}_z \cap C$, since $U \cap V$ is affine, and so evidently $x = z$ and $y = z$, whence $x = y$. □

Corollary (8.2.3). — *Let X be an integral scheme, and x and y points of X. In order for $x \in \overline{\{y\}}$, it is necessary and sufficient for $\mathscr{O}_x \subset \mathscr{O}_y$, or, equivalently, for every rational function defined at x to also be defined at y.*

Proof. The condition is evidently necessary because the domain of definition $\delta(f)$ of a rational function $f \in R$ is open; we now show that it is sufficient. If $\mathscr{O}_x \subset \mathscr{O}_y$, then there exists a prime ideal $\mathfrak{p}$ of $\mathscr{O}_x$ such that $\mathscr{O}_y$ dominates $(\mathscr{O}_x)_\mathfrak{p}$ (8.1.3); but (2.4.2) there exists some $z \in X$ such that $x \in \overline{\{z\}}$ and $\mathscr{O}_z = (\mathscr{O}_x)_\mathfrak{p}$; since $\mathscr{O}_z$ and $\mathscr{O}_y$ are allied, we have that $z = y$ by (8.2.2), whence the corollary. □

Corollary (8.2.4). — *If X is an integral scheme then the map $x \to \mathscr{O}_x$ is injective; equivalently, if x and y are two distinct points of X, then there exists a rational function defined at one of these points but not the other.*

Proof. This follows from (8.2.3) and the axiom (T_0) (2.1.4). □ **I** | 167

Corollary (8.2.5). — *Let X be an integral scheme whose underlying space is Noetherian; letting f range over the field R of rational functions on X, the sets $\delta(f)$ generate the topology of X.*

In fact, every closed subset of X is thus a finite union of irreducible closed subsets, or, in other words, of the form $\overline{\{y\}}$ (2.1.5). But, if $x \notin \overline{\{y\}}$, then there exists a rational function f defined at x but not at y (8.2.3), or, equivalently, we have that $x \in \delta(f)$ and that $\delta(f)$ is not contained in $\overline{\{y\}}$. The complement of $\overline{\{y\}}$ is thus a union of sets of the form $\delta(f)$, and, by virtue of the first remark, every open subset of X is the union of finite intersections of open sets of the form $\delta(f)$.

(8.2.6). Corollary (8.2.5) shows that the topology of X is entirely characterised by the data of the local rings $(\mathscr{O}_x)_{x\in X}$ that have R as their field of fractions. It is equivalent to say that the closed subsets of X are defined in the following manner: given a finite subset $\{x_1, \ldots, x_n\}$ of X, consider the set of $y \in X$ such that $\mathscr{O}_y \subset \mathscr{O}_{x_i}$ for at least one index i, and these sets (over all choices of $\{x_1, \ldots, x_n\}$) are the closed subsets of X. Further, once the topology on X is known, the structure sheaf $\mathscr{O}_X$ is also determined by the family of the $\mathscr{O}_x$, since $\Gamma(U, \mathscr{O}_X) = \bigcap_{x\in U} \mathscr{O}_x$, by (8.2.1.1). The family $(\mathscr{O}_X)_{x\in X}$ thus completely determines the prescheme X when X is an integral scheme whose underlying space is Noetherian.

Proposition (8.2.7). — *Let X and Y be integral schemes, $f : X \to Y$ a dominant morphism (2.2.6), and K (resp.L) the field of rational functions on X (resp.Y). Then L can be identified with a subfield of K, and, for all $x \in X$, $\mathscr{O}_{f(x)}$ is the unique local ring of Y dominated by $\mathscr{O}_x$.*

Proof. If $f = (\psi, \theta)$ and a is the generic point of X, then $\psi(a)$ is the generic point of Y (**0**, 2.1.5); $\theta_a^\sharp$ is then a monomorphism of fields, from $L = \mathscr{O}_{\psi(a)}$ to $K = \mathscr{O}_a$. Since every nonempty affine open subset U of Y contains $\psi(a)$, it follows from (2.2.4) that the homomorphism $\Gamma(U, \mathscr{O}_Y) \to \Gamma(\psi^{-1}(U), \mathscr{O}_X)$ corresponding to f is the restriction of $\theta_a^\sharp$ to $\Gamma(U, \mathscr{O}_Y)$. So, for every $x \in X$, $\theta_x^\sharp$ is the restriction to $\mathscr{O}_{\psi(a)}$ of $\theta_a^\sharp$, and is thus a monomorphism. We also know that $\theta_x^\sharp$ is a local homomorphism, so, if we identify L with a subfield of K by $\theta_a^\sharp$, $\mathscr{O}_{\psi(x)}$ is dominated by $\mathscr{O}_x$ (8.1.1); it is also the only local ring of Y dominated by $\mathscr{O}_x$, since two local rings of Y that are allied are identical (8.2.2). □

Proposition (8.2.8). — *Let X be an* irreducible *prescheme, $f : X \to Y$ a local immersion (resp. local isomorphism), and suppose further that f is separated. Then f is an immersion (resp. an open immersion).*

PROOF. Let $f = (\psi, \theta)$; it suffices, in both cases, to prove that ψ is a *homeomorphism* from X to $\psi(X)$ (4.5.3). Replacing f by f_{red} ((5.1.6) and (5.5.1, vi)), we can assume that X and Y are *reduced*. If Y' is the closed reduced subprescheme of Y that has $\overline{\psi(X)}$ as its underlying space, then f factors as $X \xrightarrow{f'} Y' \xrightarrow{j} Y$, where j is the canonical injection (5.2.2). It follows from (5.5.1, v) that f' is again a separated morphism; further, f' is again a local immersion (resp. a local isomorphism), because, since the condition is local on X and Y, we can restrict to the case where f is a closed immersion (resp. open immersion), and our claim then follows immediately from (4.2.2). I | 168

We can thus suppose that f is a *dominant* morphism, which leads to the fact that Y is, itself, irreducible (**0**, 2.1.5), and so X and Y are both *integral*. Further, since the condition is local on Y, we can suppose that Y is an affine scheme; since f is separated, X is a scheme (5.5.1, ii), and we are finally at the hypotheses of Proposition (8.2.7). Then, for all $x \in X$, $\theta_x^\sharp$ is injective; but the hypothesis that f is a local immersion implies that $\theta_x^\sharp$ is surjective (4.2.2), so $\theta_x^\sharp$ is bijective, or, equivalently (with the identification of Proposition (8.2.7)) we have that $\mathscr{O}_{\psi(x)} = \mathscr{O}_x$. This implies, by Corollary (8.2.4), that ψ is an *injective* map, which already proves the proposition when f is a local isomorphism (4.5.3). When we suppose that f is only a local immersion, for all $x \in X$ there exists an open neighbourhood U of x in X and an open neighbourhood V of $\psi(x)$ in Y such that the restriction of ψ to U is a homeomorphism from U to a *closed* subset of V. But U is dense in X, so $\psi(U)$ is dense in Y and *a fortiori* in V, which proves that $\psi(U) = V$; since ψ is injective, $\psi^{-1}(V) = U$ and this proves that ψ is a homeomorphism from X to $\psi(X)$. □

8.3. Chevalley schemes

(8.3.1). Let X be a *Noetherian* integral scheme, and R its field of rational functions; we denote by X' the set of local subrings $\mathscr{O}_x \subset R$, where x ranges over all points of X. The set X' satisfies the following three conditions.

(Sch. 1) For all $M \in X'$, R is the field of fractions of M.
(Sch. 2) There exists a finite set of Noetherian subrings A_i of R such that $X' = \bigcup_i L(A_i)$, and, for all pairs of indices i, j, the subring A_{ij} of R generated by $A_i \cup A_j$ is an algebra of finite type over A_i.
(Sch. 3) Any two elements M and N of X' that are allied are identical.

We have seen in (8.2.1) that (Sch. 1) is satisfied, and (Sch. 3) follows from (8.2.2). To show (Sch. 2), it suffices to cover X by a finite number of affine open subsets U_i whose rings are Noetherian, and to take $A_i = \Gamma(U_i, \mathscr{O}_X)$; the hypothesis that X is a scheme implies that $U_i \cap U_j$ is affine, and also that $\Gamma(U_i \cap U_j, \mathscr{O}_X) = A_{ij}$ (5.5.6); further, since the space U_i is Noetherian, the immersion $U_i \cap U_j \to U_i$ is of finite type (6.3.5), so A_{ij} is an A_i-algebra of finite type (6.3.3).

(8.3.2). The structures whose axioms are (Sch. 1), (Sch. 2), and (Sch. 3) generalise "schemes", in the sense of C. Chevalley, who additionally supposes that R is an extension of finite type of a field K, and that the A_i are K-algebras of finite type (which renders a part of (Sch. 2) useless) [**CC**]. Conversely, if we have such a structure on a set X', then we can associate to it an integral scheme X by using the remarks from (8.2.6): the underlying space of X is equal to X' endowed with the topology defined in (8.2.6), and with the sheaf $\mathscr{O}_X$ such that $\Gamma(U, \mathscr{O}_X) = \bigcap_{x \in U} \mathscr{O}_x$ for all open $U \subset X$, with the evident definition of restriction homomorphisms. We leave to the reader the task of verifying that we thus obtain an integral scheme, whose local rings are the elements of X'; we will not use this result in what follows.

§9. Supplement on quasi-coherent sheaves

9.1. Tensor product of quasi-coherent sheaves

Proposition (9.1.1). — *Let X be a prescheme (resp. a locally Noetherian prescheme). Let $\mathscr{F}$ and $\mathscr{G}$ be quasi-coherent (resp. coherent) $\mathscr{O}_X$-modules; then $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}$ is quasi-coherent (resp. coherent); it is further of finite type if both $\mathscr{F}$ and $\mathscr{G}$ are of finite type. If $\mathscr{F}$ admits a finite presentation and if $\mathscr{G}$ is quasi-coherent (resp. coherent), then $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is quasi-coherent (resp. coherent).* I | 169

PROOF. Being a local proposition, we can suppose that X is affine (resp. Noetherian affine); further, if $\mathscr{F}$ is coherent, then we can assume that it is the cokernel of a homomorphism $\mathscr{O}_X^m \to \mathscr{O}_X^n$. The claims pertaining to quasi-coherent sheaves then follow from Corollaries (1.3.12) and (1.3.9); the claims pertaining to coherent sheaves follow from Theorem (1.5.1) and from the fact that if M and N are modules of finite type over a Noetherian ring A then $M \otimes_A N$ and $\operatorname{Hom}_A(M, N)$ are both A-modules of finite type. □

Definition (9.1.2). — Let X and Y be S-preschemes, p and q the projections of $X \times_S Y$, and $\mathscr{F}$ (resp.$\mathscr{G}$) a quasi-coherent $\mathscr{O}_X$-module (resp. quasi-coherent $\mathscr{O}_Y$-module). We define the tensor product of $\mathscr{F}$ and $\mathscr{G}$ over $\mathscr{O}_S$ (*or* over S), denoted by $\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{G}$ (*or* $\mathscr{F} \otimes_S \mathscr{G}$) to be the tensor product $p^*(\mathscr{F}) \otimes_{\mathscr{O}_{X\times_S Y}} q^*(\mathscr{G})$ over the prescheme $X \times_S Y$.

If X_i ($1 \leqslant i \leqslant n$) are S-preschemes, and $\mathscr{F}_i$ ($1 \leqslant i \leqslant n$) are quasi-coherent $\mathscr{O}_{X_i}$-modules, then we similarly define the tensor product $\mathscr{F}_1 \otimes_S \mathscr{F}_2 \otimes_S \cdots \otimes_S \mathscr{F}_n$ over the prescheme $Z = X_1 \times_S X_2 \times_S \cdots \times_S X_n$; it is a *quasi-coherent* $\mathscr{O}_Z$-module by virtue of (9.1.1) and (**0**, 5.1.4); it is further *coherent* if all the $\mathscr{F}_i$ are coherent *and* Z is *locally Noetherian*, by virtue of (9.1.1), (**0**, 5.3.11), and (6.1.1).

Note that, if we take $X = Y = S$, then Definition (9.1.2) gives us back the tensor product of $\mathscr{O}_S$-modules. Furthermore, since $q^*(\mathscr{O}_Y) = \mathscr{O}_{X\times_S Y}$ (**0**, 4.3.4), the product $\mathscr{F} \otimes_S \mathscr{O}_Y$ is canonically identified with $p^*(\mathscr{F})$, and, in the same way, $\mathscr{O}_X \otimes_S \mathscr{G}$ is canonically identified with $q^*(\mathscr{G})$. In particular, if we take $Y = S$ and denote by f the structure morphism $X \to Y$, then we have that $\mathscr{O}_X \otimes_Y \mathscr{G} = f^*(\mathscr{G})$: the ordinary tensor product and the inverse image thus appear as particular cases of the general tensor product.

Definition (9.1.2) leads immediately to the fact that, for fixed X and Y, $\mathscr{F} \otimes_S \mathscr{G}$ is a *right-exact additive covariant bifunctor* in $\mathscr{F}$ and $\mathscr{G}$.

Proposition (9.1.3). — *Let S, X, and Y be affine schemes of rings A, B, and C (respectively), with B and C being A-algebras. Let M (resp. N) be a B-module (resp. C-module), and $\mathscr{F} = \widetilde{M}$ (resp. $\mathscr{G} = \widetilde{N}$) the associated quasi-coherent sheaf; then $\mathscr{F} \otimes_S \mathscr{G}$ is canonically isomorphic to the sheaf associated to the $(B \otimes_A C)$-module $M \otimes_A N$.*

PROOF. According to Proposition (1.6.5), $\mathscr{F} \otimes_S \mathscr{G}$ is canonically isomorphic to the sheaf associated to the $(B \otimes_A C)$-module **I** | 170

$$\big(M \otimes_B (B \otimes_A C)\big) \otimes_{B\otimes_A C} \big((B \otimes_A C) \otimes_C N\big)$$

and, by the canonical isomorphisms between tensor products, this latter module is isomorphic to

$$M \otimes_B (B \otimes_A C) \otimes_C N = (M \otimes_B B) \otimes_A (C \otimes_C N) = M \otimes_A N.$$

□

Proposition (9.1.4). — *Let $f : T \to X$ and $g : T \to Y$ be S-morphisms, and $\mathscr{F}$ (resp. $\mathscr{G}$) a quasi-coherent $\mathscr{O}_X$-module (resp. quasi-coherent $\mathscr{O}_Y$-module). Then*

$$(f, g)_S^*(\mathscr{F} \otimes_S \mathscr{G}) = f^*(\mathscr{F}) \otimes_{\mathscr{O}_T} g^*(\mathscr{G}).$$

PROOF. If p, q are the projections of $X \times_S Y$, then the formula follows from the equalities $(f, g)_S^* \circ p^* = f^*$ and $(f, g)_S^* \circ q^* = g^*$ (**0**, 3.5.5), and the fact that the inverse image of a tensor product of algebraic sheaves is the tensor product of their inverse images (**0**, 4.3.3). □

Corollary (9.1.5). — *Let $f : X \to X'$ and $g : Y \to Y'$ be S-morphisms, and $\mathscr{F}'$ (resp. $\mathscr{G}'$) a quasi-coherent $\mathscr{O}_{X'}$-module (resp. quasi-coherent $\mathscr{O}_{Y'}$-module). Then*

$$(f, g)_S^*(\mathscr{F}' \otimes_S \mathscr{G}') = f^*(\mathscr{F}') \otimes_S g^*(\mathscr{G}')$$

PROOF. This follows from (9.1.4) and the fact that $f \times_S g = (f \circ p, g \circ q)_S$, where p and q are the projections of $X \times_S Y$. □

Corollary (9.1.6). — *Let X, Y, and Z be S-preschemes, and $\mathscr{F}$ (resp. $\mathscr{G}$, $\mathscr{H}$) a quasi-coherent $\mathscr{O}_X$-module (resp. quasi-coherent $\mathscr{O}_Y$-module, quasi-coherent $\mathscr{O}_Z$-module); then the sheaf $\mathscr{F} \otimes_S \mathscr{G} \otimes_S \mathscr{H}$ is the inverse image of $(\mathscr{F} \otimes_S \mathscr{G}) \otimes_S \mathscr{H}$ by the canonical isomorphism from $X \times_S Y \times_S Z$ to $(X \times_S Y) \times_S Z$.*

Proof. This isomorphism is given by $(p_1, p_2)_S \times_S p_3$, where p_1, p_2, and p_3 are the projections of $X \times_S Y \times_S Z$.

Similarly, the inverse image of $\mathscr{G} \otimes_S \mathscr{F}$ under the canonical isomorphism from $X \times_S Y$ to $Y \times_S X$ is $\mathscr{F} \otimes_S \mathscr{G}$. □

Corollary (9.1.7). — *If X is an S-prescheme, then every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is the inverse image of $\mathscr{F} \otimes_S \mathscr{O}_S$ by the canonical isomorphism from X to $X \times_S S$ (3.3.3).*

Proof. This isomorphism is $(1_X, \phi)_S$, where ϕ is the structure morphism $X \to S$, and the corollary follows from (9.1.4) and the fact that $\phi^*(\mathscr{O}_S) = \mathscr{O}_X$. □

(9.1.8). Let X be an S-prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module, and $\phi : S' \to S$ a morphism; we denote by $\mathscr{F}_{(\phi)}$ or $\mathscr{F}_{(S')}$ the quasi-coherent sheaf $\mathscr{F} \otimes_S \mathscr{O}_{S'}$ over $X \times_S S' = X_{(\phi)} = X_{(S')}$; so $\mathscr{F}_{(S')} = p^*(\mathscr{F})$, where p is the projection $X_{(S')} \to X$.

Proposition (9.1.9). — *Let $\phi'' : S'' \to S'$ be a morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ on the S-prescheme X, $(\mathscr{F}_{(\phi)})_{(\phi')}$ is the inverse image of $\mathscr{F}_{(\phi\circ\phi')}$ by the canonical isomorphism $(X_{(\phi)})_{(\phi')} \simeq X_{(\phi\circ\phi')}$ (3.3.9).*

Proof. This follows immediately from the definitions and from (3.3.9), and is written

$$(\mathscr{F} \otimes_S \mathscr{O}_{S'}) \otimes_{S'} \mathscr{O}_{S''} = \mathscr{F} \otimes_S \mathscr{O}_{S''}. \tag{9.1.9.1}$$

□

Proposition (9.1.10). — *Let Y be an S-prescheme, and $f : X \to Y$ an S-morphism. For every quasi-coherent $\mathscr{O}_Y$-module $\mathscr{G}$ and every morphism $S' \to S$, we have that $(f_{(S')})^*(\mathscr{G}_{(S')}) = (f^*(\mathscr{G}))_{(S')}$.*

Proof. This follows immediately from the commutativity of the diagram **I** | 171

$$\begin{tikzcd} X_{(S')} \arrow[r, "f_{(S')}"] \arrow[d] & Y_{(S')} \arrow[d] \\ X \arrow[r, "f"] & Y. \end{tikzcd}$$

□

Corollary (9.1.11). — *Let X and Y be S-preschemes, and $\mathscr{F}$ (resp. $\mathscr{G}$) a quasi-coherent $\mathscr{O}_X$-module (resp. quasi-coherent $\mathscr{O}_Y$-module). Then the inverse image of the sheaf $(\mathscr{F}_{(S')}) \otimes_{(S')} (\mathscr{G}_{(S')})$ by the canonical isomorphism $(X \times_S Y)_{(S')} \simeq (X_{(S')}) \times_{S'} (Y_{(S')})$ (3.3.10) is equal to $(\mathscr{F} \otimes_S \mathscr{G})_{(S')}$.*

Proof. If p and q are the projections of $X \times_S Y$, then the isomorphism in question is nothing but $(p_{(S')}, q_{(S')})_{S'}$; the corollary then follows from Propositions (9.1.4) and (9.1.10). □

Proposition (9.1.12). — *With the notation from Definition (9.1.2), let z be a point of $X \times_S Y$, and let $x = p(z)$, and $y = q(z)$; the stalk $(\mathscr{F} \otimes_S \mathscr{G})_z$ is isomorphic to $(\mathscr{F}_x \otimes_{\mathscr{O}_x} \mathscr{O}_z) \otimes_{\mathscr{O}_z} (\mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{O}_z) = \mathscr{F}_x \otimes_{\mathscr{O}_x} \mathscr{O}_z \otimes_{\mathscr{O}_y} \otimes \mathscr{G}_y$.*

Proof. Since we can reduce to the affine case, the proposition follows from Equation (1.6.5.1). □

Corollary (9.1.13). — *If $\mathscr{F}$ and $\mathscr{G}$ are of finite type, then*

$$\operatorname{Supp}(\mathscr{F} \otimes_S \mathscr{G}) = p^{-1}(\operatorname{Supp}(\mathscr{F})) \cap q^{-1}(\operatorname{Supp}(\mathscr{G})).$$

Proof. Since $p^*(\mathscr{F})$ and $q^*(\mathscr{G})$ are both of finite type over $\mathscr{O}_{X\times_S Y}$, we reduce, by Proposition (9.1.12) and by (**0**, 1.7.5), to the case where $\mathscr{G} = \mathscr{O}_Y$, that is, it remains to prove the following equation:

$$\operatorname{Supp}(p^*(\mathscr{F})) = p^{-1}(\operatorname{Supp}(\mathscr{F})). \tag{9.1.13.1}$$

The same reasoning as in (**0**, 1.7.5) leads us to prove that, for all $z \in X \times_S Y$, we have $\mathscr{O}_z/\mathfrak{m}_x\mathscr{O}_z \neq 0$ (with $x = p(z)$), which follows from the fact that the homomorphism $\mathscr{O}_x \to \mathscr{O}_z$ is *local*, by hypothesis. □

We leave it to the reader to extend the results in this section to the more general case of an arbitrary (but finite) number of factors, instead of just two.

9.2. Direct image of a quasi-coherent sheaf

Proposition (9.2.1). — *Let $f : X \to Y$ be a morphism of preschemes. We suppose that there exists a cover (Y_α) of Y by affine opens having the following property: every $f^{-1}(Y_\alpha)$ admits a* finite *cover $(X_{\alpha i})$ by affine opens that are contained in $f^{-1}(Y_\alpha)$ and that are such that every intersection $X_{\alpha i} \cap X_{\alpha j}$ is itself a* finite *union of affine opens. With these hypotheses, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $f_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_Y$-module.*

PROOF. Since this is a local condition on Y, we can assume that Y is equal to one of the Y_α, and thus omit the indices α.

(a) First, suppose that the $X_i \cap X_j$ are themselves *affine* opens. We set $\mathscr{F}_i = \mathscr{F}|X_i$ and $\mathscr{F}_{ij} = \mathscr{F}|(X_i \cap X_j)$, and let $\mathscr{F}'_i$ and $\mathscr{F}'_{ij}$ be the images of $\mathscr{F}_i$ and $\mathscr{F}_{ij}$ (respectively) by the restriction of f to X_i and to $X_i \cap X_j$ (respectively); we know that the $\mathscr{F}'_i$ and $\mathscr{F}'_{ij}$ are quasi-coherent (1.6.3). Set $\mathscr{G} = \bigoplus_i \mathscr{F}'_i$ and $\mathscr{H} = \bigoplus_{i,j} \mathscr{F}'_{ij}$; $\mathscr{G}$ and $\mathscr{H}$ are quasi-coherent $\mathscr{O}_Y$-modules; we will define a homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that $f_*(\mathscr{F})$ is the *kernel* of u; it will follow from this that $f_*(\mathscr{F})$ is quasi-coherent (1.3.9). It suffices to define u as a homomorphism of presheaves; taking into account the definitions of $\mathscr{G}$ and $\mathscr{H}$, it thus suffices, for every open subset $W \subset Y$, to define a homomorphism **I | 172**

$$u_W : \bigoplus_i \Gamma(f^{-1}(W) \cap X_i, \mathscr{F}) \longrightarrow \bigoplus_{i,j} \Gamma(f^{-1}(W) \cap X_i \cap X_j, \mathscr{F})$$

that satisfies the usual compatibility conditions when we let W vary. If, for every section $s_i \in \Gamma(f^{-1}(W) \cap X_i, \mathscr{F})$, we denote by $s_{i|j}$ its restriction to $f^{-1}(W) \cap X_i \cap X_j$, then we set

$$u_W\big((s_i)\big) = (s_{i|j} - s_{j|i})$$

and the compatibility conditions are clearly satisfied. To prove that the kernel $\mathscr{R}$ of u is $f_*(\mathscr{F})$, we define a homomorphism from $f_*(\mathscr{F})$ to $\mathscr{R}$ by sending each section $s \in \Gamma(f^{-1}(W), \mathscr{F})$ to the family (s_i), where s_i is the restriction of s to $f^{-1}(W) \cap X_i$; axioms (F1) and (F2) of sheaves (G, II, 1.1) tell us that this homomorphism is *bijective*, which finishes the proof in this case.

(b) In the general case, the same reasoning applies once we have established that the $\mathscr{F}_{ij}$ are quasi-coherent. But, by hypothesis, $X_i \cap X_j$ is a finite union of affine opens X_{ijk}; and since the X_{ijk} are affine opens *in a scheme*, the intersection of any two of them is again an affine open (5.5.6). We are thus led to the first case, and so we have proved Proposition (9.2.1).

□

Corollary (9.2.2). — *The conclusion of Proposition (9.2.1) holds true in each of the following cases:*

(a) *f is separated and quasi-compact;*
(b) *f is separated and of finite type;*
(c) *f is quasi-compact, and the underlying space of X is locally Noetherian.*

PROOF. In case (a), the $X_{\alpha i} \cap X_{\alpha j}$ are affine (5.5.6). Case (b) is a particular example of case (a) (6.6.3). Finally, in case (c), we can reduce to the case where Y is affine and the underlying space of X is Noetherian; then X admits a finite cover of affine opens (X_i), and the $X_i \cap X_j$, being quasi-compact, are finite unions of affine opens (2.1.3). □

9.3. Extension of sections of quasi-coherent sheaves

Theorem (9.3.1). — *Let X be a prescheme whose underlying space is Noetherian, or a scheme whose underlying space is quasi-compact. Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module* (**0**, 5.4.1), *f a section of $\mathscr{L}$ over X, X_f the open set of $x \in X$ such that $f(x) \neq 0$* (**0**, 5.5.1), *and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module.*

(i) *If $s \in \Gamma(X, \mathscr{F})$ is such that $s|X_f = 0$, then there exists an integer $n > 0$ such that $s \otimes f^{\otimes n} = 0$.*
(ii) *For every section $s \in \Gamma(X_f, \mathscr{F})$, there exists an integer $n > 0$ such that $s \otimes f^{\otimes n}$ extends to a section of $\mathscr{F} \otimes \mathscr{L}^{\otimes n}$ over X.*

Proof.

(i) Since the underlying space of X is quasi-compact, and thus the union of finitely-many affine opens U_i with $\mathscr{L}|U_i$ isomorphic to $\mathscr{O}_X|U_i$, we can reduce to the case where X is affine and $\mathscr{L} = \mathscr{O}_X$. In this case, f can be identified with an element of $A(X)$, and we have that $X_f = D(f)$; s can be identified with an element of an $A(X)$-module M, and $s|X_f$ to the corresponding element of M_f, and the result is then trivial, recalling the definition of a module of fractions. I | 173
(ii) Again, X is a finite union of affine opens U_i ($1 \leqslant i \leqslant r$) such that $\mathscr{L}|U_i \cong \mathscr{O}_X|U_i$, and, for every i, $(s \otimes f^{\otimes n})|(U_i \cap X_f)$ can be identified (by the aforementioned isomorphism) with $(f|(U_i \cap X_f))^n (s|(U_i \cap X_f))$. We then know (1.4.1) that there exists an integer $n > 0$ such that, for all i, $(s \otimes f^{\otimes n})|(U_i \cap X_f)$ extends to a section s_i of $\mathscr{F} \otimes \mathscr{L}^{\otimes n}$ over U_i. Let $s_{i|j}$ be the restriction of s_i to $U_i \cap U_j$; by definition we have that $s_{i|j} - s_{j|i} = 0$ on $X_f \cap U_i \cap U_j$. But, if X is a Noetherian space, then $U_i \cap U_j$ is quasi-compact; if X is a scheme, then $U_i \cap U_j$ is an affine open (5.5.6), and so again quasi-compact. By virtue of (i), there thus exists an integer m (independent of i and j) such that $(s_{i|j} - s_{j|i}) \otimes f^{\otimes m} = 0$. It immediately follows that there exists a section s' of $\mathscr{F} \otimes \mathscr{L}^{\otimes(n+m)}$ over X that restricts to $s_i \otimes f^{\otimes m}$ over each U_i, and restricts to $s \otimes f^{\otimes(n+m)}$ over X_f.

□

The following corollaries give an interpretation of Theorem (9.3.1) in a more algebraic language:

Corollary (9.3.2). — *With the hypotheses of (9.3.1), consider the graded ring $A_\bullet = \Gamma_\bullet(\mathscr{L})$ and the graded $A_\bullet$-module $M_* = \Gamma_\bullet(\mathscr{L}, \mathscr{F})$* (**0**, 5.4.6). *If $f \in A_n$, where $n \in \mathbf{Z}$, then there is a canonical isomorphism $\Gamma(X_f, \mathscr{F}) \simeq ((M_*)_f)_0$* (the subgroup of the module of fractions $(M_*)_f$ consisting of elements of degree 0).

Corollary (9.3.3). — *Suppose that the hypotheses of (9.3.1) are satisfied, and suppose further that $\mathscr{L} = \mathscr{O}_X$. Then, setting $A = \Gamma(X, \mathscr{O}_X)$ and $M = \Gamma(X, \mathscr{F})$, the A_f-module $\Gamma(X_f, \mathscr{F})$ is canonically isomorphic to M_f.*

Proposition (9.3.4). — *Let X be a Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-module, and $\mathscr{J}$ a coherent sheaf of ideals of $\mathscr{O}_X$, such that the support of $\mathscr{F}$ is contained in that of $\mathscr{O}_X|\mathscr{J}$. Then there exists an integer $n > 0$ such that $\mathscr{J}^n \mathscr{F} = 0$.*

Proof. Since X is a union of finitely-many affine opens whose rings are Noetherian, we can suppose that X is affine, given by some Noetherian ring A; then $\mathscr{F} = \widetilde{M}$, where $M = \Gamma(X, \mathscr{F})$ is an A-module of finite type, and $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J} = \Gamma(X, \mathscr{J})$ is an ideal of A ((1.4.1) and (1.5.1)). Since A is Noetherian, $\mathfrak{J}$ admits a finite system of generators f_i ($1 \leqslant i \leqslant m$). By hypothesis, every section of $\mathscr{F}$ over X is zero on each of the $D(f_i)$; if s_j ($1 \leqslant j \leqslant q$) are sections of $\mathscr{F}$ that generate M, then there exists an integer h, independent of i and j, such that $f_i^h s_j = 0$ (1.4.1), whence $f_i^h s = 0$ for all $s \in M$. We thus conclude that, if $n = mh$, then $\mathfrak{J}^n M = 0$, and so the corresponding $\mathscr{O}_X$-module $\mathscr{J}^n \mathscr{F} = \widetilde{\mathfrak{J}^n M}$ (1.3.13) is zero. □

Corollary (9.3.5). — *With the hypotheses of (9.3.4), there exists a closed subprescheme Y of X, whose underlying space is the support of $\mathscr{O}_X/\mathscr{J}$, such that, if $j : Y \to X$ is the canonical injection, then $\mathscr{F} = j_*(j^*(\mathscr{F}))$.*

PROOF. First, note that the supports of $\mathscr{O}_X/\mathscr{J}$ and $\mathscr{O}_X/\mathscr{J}^n$ are the same, since, if $\mathscr{J}_x = \mathscr{O}_x$, then $\mathscr{J}_x^n = \mathscr{O}_x$, and we also have that $\mathscr{J}_x^n \subset \mathscr{J}_x$ for all $x \in X$. We can, thanks to (9.3.4), thus suppose that $\mathscr{J}\mathscr{F} = 0$; we can then take Y to be the closed subprescheme of X defined by $\mathscr{J}$, and since $\mathscr{F}$ is then an $(\mathscr{O}_X/\mathscr{J})$-module, the conclusion follows immediately. ☐

9.4. Extension of quasi-coherent sheaves

(9.4.1). Let X be a topological space, $\mathscr{F}$ a sheaf of sets (resp. of groups, of rings) on X, U an open subset of X, $\psi : U \to X$ the canonical injection, and $\mathscr{G}$ a subsheaf of $\mathscr{F}|U = \psi^*(\mathscr{F})$. Since ψ_* is left exact, $\psi_*(\mathscr{G})$ is a subsheaf of $\psi_*(\psi^*(\mathscr{F}))$; we denote by ρ the canonical homomorphism $\mathscr{F} \to \psi_*(\psi^*(\mathscr{F}))$ (**0**, 3.5.3), and we denote by $\overline{\mathscr{G}}$ the subsheaf $\rho^{-1}(\psi_*(\mathscr{G}))$ of $\mathscr{F}$. It follows immediately from the definitions that, for every open subset V of X, $\Gamma(V, \overline{\mathscr{G}})$ consists of sections $s \in \Gamma(V, \mathscr{F})$ whose restriction to $V \cap U$ is a section of $\mathscr{G}$ over $V \cap U$. We thus have that $\overline{\mathscr{G}}|U = \psi^*(\overline{\mathscr{G}}) = \mathscr{G}$, and that $\overline{\mathscr{G}}$ is the *largest* subsheaf of $\mathscr{F}$ that restricts to $\mathscr{G}$ over U; we say that $\overline{\mathscr{G}}$ is the *canonical extension* of the subsheaf $\mathscr{G}$ of $\mathscr{F}|U$ to a subsheaf of $\mathscr{F}$. I | 174

Proposition (9.4.2). — *Let X be a prescheme, and U an open subset of X such that the canonical injection $j : U \to X$ is a quasi-compact morphism* (which will be the case for *all* U if the underlying space of X is *locally Noetherian* (6.6.4, i)). *Then:*

(i) *for every quasi-coherent $(\mathscr{O}_X|U)$-module $\mathscr{G}$, $j_*(\mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module, and $j_*(\mathscr{G})|U = j^*(j_*(\mathscr{G})) = \mathscr{G}$;*

(ii) *for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every quasi-coherent $(\mathscr{O}_X|U)$-submodule $\mathscr{G}$, the canonical extension $\overline{\mathscr{G}}$ of $\mathscr{G}$ (9.4.1) is a quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{F}$.*

PROOF. If $j = (\psi, \theta)$ (ψ being the injection $U \to X$ of underlying spaces), then, by definition, we have that $j_*(\mathscr{G}) = \psi_*(\mathscr{G})$ for every $(\mathscr{O}_X|U)$-module $\mathscr{G}$, and, further, that $j^*(\mathscr{H}) = \psi^*(\mathscr{H}) = \mathscr{H}|U$ for every $\mathscr{O}_X$-module $\mathscr{H}$, by definition of the prescheme induced over an open subset. So (i) is thus a particular case of ((9.2.2, a)); for the same reason, $j_*(j^*(\mathscr{F}))$ is quasi-coherent, and since $\overline{\mathscr{G}}$ is the inverse image of $j_*(\mathscr{G})$ by the homomorphism $\rho : \mathscr{F} \to j_*(j^*(\mathscr{F}))$, (ii) follows from (4.1.1). ☐

Note that the hypothesis that the morphism $j : U \to X$ is quasi-compact holds whenever the open subset U is *quasi-compact* and X is a *scheme*: indeed, U is then a union of finitely-many affine opens U_i, and, for every affine open V of X, $V \cap U_i$ is an affine open (5.5.6), and thus quasi-compact.

Corollary (9.4.3). — *Let X be a prescheme, and U a quasi-compact open subset of X such that the injection morphism $j : U \to X$ is quasi-compact. Suppose as well that every quasi-coherent $\mathscr{O}_X$-module is the inductive limit of its quasi-coherent $\mathscr{O}_X$-submodules of finite type* (which will be the case if X is an *affine scheme*). *Then let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module, and $\mathscr{G}$ a quasi-coherent $(\mathscr{O}_X|U)$-submodule of $\mathscr{F}|U$* of finite type. *Then there exists a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}'$ of $\mathscr{F}$* of finite type *such that $\mathscr{G}'|U = \mathscr{G}$.*

PROOF. We have $\mathscr{G} = \overline{\mathscr{G}}|U$, and $\overline{\mathscr{G}}$ is quasi-coherent, from (9.4.2), so the inductive limit of its quasi-coherent $\mathscr{O}_X$-submodules $\mathscr{H}_\lambda$ of finite type. It follows that $\mathscr{G}$ is the inductive limit of the $\mathscr{H}_\lambda|U$, and thus equal to one of the $\mathscr{H}_\lambda|U$, since it is of finite type (**0**, 5.2.3). ☐

Remark (9.4.4). — Suppose that for *every* affine open $U \subset X$, the injection morphism $U \to X$ is quasi-compact. Then, if the conclusion of (9.4.3) holds for every affine open U and for every quasi-coherent $(\mathscr{O}_X|U)$-submodule $\mathscr{G}$ of $\mathscr{F}|U$ of finite type, it follows that $\mathscr{F}$ is the inductive limit of its quasi-coherent $\mathscr{O}_X$-submodules of finite type. Indeed, for every affine open $U \subset X$, we have that $\mathscr{F}|U = \widetilde{M}$, where M is an $A(U)$-module, and since the latter is the inductive limit of its quasi-coherent submodules of finite type, $\mathscr{F}|U$ is the inductive limit of its $(\mathscr{O}_X|U)$-submodules of finite type (1.3.9). But, by hypothesis, each of these submodules is induced on U by a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}_{\lambda,U}$ of $\mathscr{F}$ of finite type. The finite sums of the $\mathscr{G}_{\lambda,U}$ are again quasi-coherent $\mathscr{O}_X$-modules of finite type, because this is a local property, and the case where X is affine was covered in (1.3.10); it is clear then that $\mathscr{F}$ is the inductive limit of these finite sums, whence our claim. I | 175

Corollary (9.4.5). — *Under the hypotheses of Corollary (9.4.3), for every quasi-coherent $(\mathscr{O}_X|U)$-module $\mathscr{G}$ of finite type, there exists a quasi-coherent $\mathscr{O}_X$-module $\mathscr{G}'$ of finite type such that $\mathscr{G}'|U = \mathscr{G}$.*

PROOF. Since $\mathscr{F} = j_*(\mathscr{G})$ is quasi-coherent (9.4.2) and $\mathscr{F}|U = \mathscr{G}$, it suffices to apply Corollary (9.4.3) to $\mathscr{F}$. ☐

Lemma (9.4.6). — *Let X be a prescheme, L a well-ordered set, $(V_\lambda)_{\lambda\in L}$ a cover of X by affine opens, and U an open subset of X; for all $\lambda \in L$, we set $W_\lambda = \bigcup_{\mu<\lambda} V_\mu$. Suppose that: (1) for every $\lambda \in L$, $V_\lambda \cap W_\lambda$ is quasi-compact; and (2) the immersion morphism $U \to X$ is quasi-compact. Then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every quasi-coherent $(\mathscr{O}_X|U)$-submodule $\mathscr{G}$ of $\mathscr{F}|U$* of finite type, *there exists a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}'$ of $\mathscr{F}$* of finite type *such that $\mathscr{G}'|U = \mathscr{G}$.*

Proof. Let $U_\lambda = U \cup W_\lambda$; we will define a family $(\mathscr{G}'_\lambda)$ by induction, where $\mathscr{G}'_\lambda$ is a quasi-coherent $(\mathscr{O}_X|U_\lambda)$-submodule of $\mathscr{F}|U_\lambda$ of finite type, such that $\mathscr{G}'_\lambda|U_\mu = \mathscr{G}'_\mu$ for $\mu < \lambda$ and $\mathscr{G}'_\lambda|U = \mathscr{G}$. The unique $\mathscr{O}_X$-submodule $\mathscr{G}'$ of $\mathscr{F}$ such that $\mathscr{G}'|U_\lambda = \mathscr{G}'$ for all $\lambda \in L$ (**0**, 3.3.1) will then give us what we want. So suppose that the $\mathscr{G}'_\mu$ are defined and have the preceding properties for $\mu < \lambda$; if λ does not have a predecessor then we take $\mathscr{G}'_\lambda$ to be the unique $(\mathscr{O}_X|U_\lambda)$-submodule of $\mathscr{F}|U_\lambda$ such that $\mathscr{G}'_\lambda|U_\mu = \mathscr{G}'_\mu$ for all $\mu < \lambda$, which is allowed since the U_μ with $\mu < \lambda$ then form a cover of U_λ. If, conversely, $\lambda = \mu + 1$, then $U_\lambda = U_\mu \cup V_\mu$, and it suffices to define a quasi-coherent $(\mathscr{O}_X|V_\mu)$-submodule $\mathscr{G}''_\mu$ of $\mathscr{F}|V_\mu$ of finite type such that

$$\mathscr{G}''_\mu|(U_\mu \cap V_\mu) = \mathscr{G}'_\mu|(U_\mu \cap V_\mu);$$

and then to take $\mathscr{G}'_\lambda$ to be the $(\mathscr{O}_X|U_\lambda)$-submodule of $\mathscr{F}|U_\lambda$ such that $\mathscr{G}'_\lambda|U_\mu = \mathscr{G}'_\mu$ and $\mathscr{G}'_\lambda|V_\mu = \mathscr{G}''_\mu$ (**0**, 3.3.1). But, since V_μ is affine, the existence of $\mathscr{G}''_\mu$ is guaranteed by (9.4.3) as soon as we show that $U_\mu \cap V_\mu$ is quasi-compact; but $U_\mu \cap V_\mu$ is the union of $U \cap V_\mu$ and $W_\mu \cap V_\mu$, which are both quasi-compact by virtue of the hypotheses. □

Theorem (9.4.7). — *Let X be a prescheme, and U an open subset of X. Suppose that one of the following conditions is verified:*

(a) *the underlying space of X is locally Noetherian;*
(b) *X is a quasi-compact scheme and U is a quasi-compact open.*

Then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every quasi-coherent $(\mathscr{O}_X|U)$-submodule $\mathscr{G}$ of $\mathscr{F}|U$ of finite type, *there exists a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}'$ of $\mathscr{F}$* of finite type *such that $\mathscr{G}'|U = \mathscr{G}$.*

Proof. Let $(V_\lambda)_{\lambda\in L}$ be a cover of X by affine opens, with L assumed to be finite in case (b); **I** | 176
since L is equipped with the structure of a well-ordered set, it suffices to check that the conditions of (9.4.6) are satisfied. It is clear in the case of (a), since the spaces V_λ are Noetherian. For case (b), the $V_\lambda \cap \lambda_\mu$ are affine (5.5.6), and thus quasi-compact, and, since L is finite, $V_\lambda \cap W_\lambda$ is quasi-compact. Whence the theorem. □

Corollary (9.4.8). — *Under the hypotheses of (9.4.7), for every quasi-coherent $(\mathscr{O}_X|U)$-module $\mathscr{G}$ of finite type, there exists a quasi-coherent $\mathscr{O}_X$-module $\mathscr{G}'$ of finite type such that $\mathscr{G}'|U = \mathscr{G}$.*

Proof. It suffices to apply (9.4.7) to $\mathscr{F} = j_*(\mathscr{G})$, which is quasi-coherent (9.4.2) and such that $\mathscr{F}|U = \mathscr{G}$. □

Corollary (9.4.9). — *Let X be a prescheme whose underlying space is locally Noetherian, or a quasi-compact scheme. Then every quasi-coherent $\mathscr{O}_X$-module is the inductive limit of its quasi-coherent $\mathscr{O}_X$-submodules of finite type.*

Proof. This follows from Theorem (9.4.7) and Remark (9.4.4). □

Corollary (9.4.10). — *Under the hypotheses of (9.4.9), if a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is such that every quasi-coherent $\mathscr{O}_X$-submodule of finite type of $\mathscr{F}$ is generated by its sections over X, then $\mathscr{F}$ is generated by its sections over X.*

Proof. Let U be an affine open neighbourhood of a point $x \in X$, and let s be a section of $\mathscr{F}$ over U; the $\mathscr{O}_X$-submodule $\mathscr{G}$ of $\mathscr{F}|U$ generated by s is quasi-coherent and of finite type, so there exists a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}'$ of $\mathscr{F}$ of finite type such that $\mathscr{G}'|U = \mathscr{G}$ (9.4.7). By hypothesis, there thus exists a finite number of sections t_i of $\mathscr{G}'$ over X and of sections a_i of $\mathscr{O}_X$ over a neighbourhood $V \subset U$ of x such that $s|V = \sum_i a_i(t_i|V)$, which proves the corollary. □

9.5. Closed image of a prescheme; closure of a subprescheme

Proposition (9.5.1). — *Let $f : X \to Y$ be a morphism of preschemes such that $f_*(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_Y$-module (which will be the case if f is quasi-compact* and, *in addition, either f is separated or X is locally Noetherian (9.2.2)). Then there exists a smaller subprescheme Y' of Y such that f factors through the canonical injection $j : Y' \to Y$ (*or, equivalently (4.4.1), such that the subprescheme $f^{-1}(Y')$ of X is *identical* to *X).*

More precisely:

Corollary (9.5.2). — *Under the conditions of (9.5.1), let $f = (\psi, \theta)$, and let $\mathscr{J}$ be the (quasi-coherent) kernel of the homomorphism $\theta : \mathscr{O}_Y \to f_*(\mathscr{O}_X)$. Then the closed subprescheme Y' of Y defined by $\mathscr{J}$ satisfies the conditions of (9.5.1).*

Proof. Since the functor ψ^* is exact, the canonical factorization $\theta : \mathscr{O}_Y \to \mathscr{O}_Y/\mathscr{J} \xrightarrow{\theta'} \psi_*(\mathscr{O}_X)$ gives (**0**, 3.5.4.3) a factorization $\theta^\sharp : \psi^*(\mathscr{O}_Y) \to \psi^*(\mathscr{O}_Y)/\psi^*(\mathscr{J}) \xrightarrow{\theta'^\sharp} \mathscr{O}_X$; since $\theta^\sharp_x$ is a local homomorphism for every $x \in X$, the same is true of ${\theta'}^\sharp_x$; if we denote by ψ_0 the continuous map ψ considered as a map from X to Y', and by θ_0 the restriction $\theta'|X' : (\mathscr{O}_Y/\mathscr{J})|Y' \to \psi_*(\mathscr{O}_X)|Y' = (\psi_0)_*(\mathscr{O}_X)$, **Err$_{\text{II}}$** then we see that $f_0 = (\psi_0, \theta_0)$ is a morphism of preschemes $X \to Y'$ (2.2.1) such that $f = j \circ f_0$. Now, if Y'' is a second closed subprescheme of Y, defined by a quasi-coherent sheaf of ideals $\mathscr{J}'$ **I | 177** of $\mathscr{O}_Y$, such that f factors through the injection $j' : Y'' \to Y$, then we should immediately have that $\psi(X) \subset Y''$, and thus that $Y' \subset Y''$, since Y'' is closed. Furthermore, for all $y \in Y''$, θ should factor as $\mathscr{O}_y \to \mathscr{O}_y/\mathscr{J}'_y \to (\psi_*(\mathscr{O}_X))_y$, which, by definition, leads to $\mathscr{J}'_y \subset \mathscr{J}_y$, and thus X' is a closed subprescheme of Y'' (4.1.10). ☐

Definition (9.5.3). — Whenever there exists a smaller subprescheme Y' of Y such that f factors through the canonical injection $j : Y' \to Y$, we say that Y' is the *closed image* prescheme of X under the morphism f.

Proposition (9.5.4). — *If $f_*(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_Y$-module, then the underlying space of the closed image of X under f is the closure $\overline{f(X)}$ in Y.*

Proof. As the support of $f_*(\mathscr{O}_X)$ is contained in $\overline{f(X)}$, we have (with the notation of (9.5.2)) $\mathscr{J}_y = \mathscr{O}_y$ for $y \notin \overline{f(X)}$, thus the support of $\mathscr{O}_Y/\mathscr{J}$ is contained in $\overline{f(X)}$. In addition, this support is closed and contains $f(X)$: indeed, if $y \in f(X)$, the unit element of the ring $(\psi_*(\mathscr{O}_X))_y$ is not zero, being the germ at y of the section

$$1 \in \Gamma(X, \mathscr{O}_X) = \Gamma(Y, \psi_*(\mathscr{O}_X));$$

since it is the image under θ of the unit element of $\mathscr{O}_y$, the latter does not belong to $\mathscr{J}_y$, hence $\mathscr{O}_y/\mathscr{J}_y \neq 0$; this finishes the proof. ☐

Proposition (9.5.5). — (Transitivity of closed images). *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of preschemes; we suppose that the closed image Y' of X under f exists, and that, if g' is the restriction of g to Y', then the closed image Z' of Y' under g' exists. Then the closed image of X under $g \circ f$ exists and is equal to Z'.*

Proof. It suffices (9.5.1) to show that Z' is the smallest closed subprescheme Z_1 of Z such that the closed subprescheme $(g \circ f)^{-1}(Z_1)$ of X (equal to $f^{-1}(g^{-1}(Z_1))$ by Corollary (4.4.2)) is equal to X; it is equivalent to say that Z' is the smallest closed subprescheme of Z such that f factors through the injection $g^{-1}(Z_1) \to Y$ (4.4.1). By virtue of the existence of the closed image Y', every Z_1 with this property is such that $g^{-1}(Z_1)$ factors through Y', which is equivalent to saying that $j^{-1}(g^{-1}(Z_1)) = g'^{-1}(Z_1) = Y'$, denoting by j the injection $Y' \to Y$. By the definition of Z', we indeed conclude that Z' is the smallest closed subprescheme of Z satisfying the preceding condition. ☐

Corollary (9.5.6). — *Let $f : X \to Y$ be an S-morphism such that Y is the closed image of X under f. Let Z be an S-scheme; if two S-morphisms g_1, g_2 from Y to Z are such that $g_1 \circ f = g_2 \circ f$, then $g_1 = g_2$.*

PROOF. Let $h = (g_1, g_2)_S : Y \to Z \times_S Z$; since the diagonal $T = \Delta_Z(Z)$ is a closed subprescheme of $Z \times_S Z$, $Y' = h^{-1}(T)$ is a closed subprescheme of Y (4.4.1). Let $u = g_1 \circ f = g_2 \circ f$; we then have, by definition of the product, $h' = h \circ f = (u, u)_S$, so $h \circ f = \Delta_Z \circ u$; since $\Delta_Z^{-1}(T) = Z$, we have $h'^{-1}(T) = u^{-1}(Z) = X$, so $f^{-1}(Y') = X$. From this, we conclude (4.4.1) that the f factors through the canonical injection $Y' \to Y$, so $Y' = Y$ by hypothesis; it then follows (4.4.1) that h factors as $\Delta_Z \circ v$, where v is a morphism $Y \to Z$, which implies that $g_1 = g_2 = v$. □

Remark (9.5.7). — If X and Y are S-schemes, Proposition (9.5.6) implies that, when Y is the closed image of X under f, f is an *epimorphism* in the category of *S-schemes* (T, 1.1). We will show in Chapter V that, conversely, if the closed image Y' of X under f exists and if f is an epimorphism of S-schemes, then we necessarily have that $Y' = Y$. I | 178

Proposition (9.5.8). — *Suppose that the hypotheses of* (9.5.1) *are satisfied, and let Y' be the closed image of X under f. For every open V of Y, let $f_V : f^{-1}(V) \to V$ be the restriction of f; then the closed image of $f^{-1}(V)$ under f_V in V exists and is equal to the prescheme induced by Y' on the open $V \cap Y'$ of Y'* (in other words, to the subprescheme $\inf(V, Y)$ of Y (4.4.3)).

PROOF. Let $X' = f^{-1}(V)$; since the direct image of $\mathscr{O}_{X'}$ by f_V is exactly the restriction of $f_*(\mathscr{O}_X)$ to V, it is clear that the kernel $\mathscr{J}'$ of the homomorphism $\mathscr{O}_V \to (f_V)_*(\mathscr{O}_{X'})$ is the restriction of $\mathscr{J}$ to V, from which the proposition quickly follows. □

We will see that this result can be understood as saying that taking the closed image commutes with an extension $Y_1 \to Y$ of the base prescheme, which is an *open immersion*. We will see in Chapter IV that it is the same for an extension $Y_1 \to Y$ which is a *flat* morphism, provided that f is separated and quasi-compact.

Proposition (9.5.9). — *Let $f : X \to Y$ be a morphism such that the closed image Y' of X under f exists.*

(i) *If X is reduced, then so too is Y'.*
(ii) *If the hypotheses of Proposition* (9.5.1) *are satisfied and X is irreducible (resp. integral), then so too is Y'.*

PROOF. By hypothesis, the morphism f factors as $X \xrightarrow{g} Y' \xrightarrow{j} Y$, where j is the canonical injection. Since X is reduced, g factors as $X \xrightarrow{h} Y'_{\text{red}} \xrightarrow{j'} Y'$, where j' is the canonical injection (5.2.2), and it then follows from the definition of Y' that $Y'_{\text{red}} = Y'$. If, moreover, the conditions of Proposition (9.5.1) are satisfied, then it follows from (9.5.4) that $f(X)$ is dense in Y'; if X is irreducible, then so is Y' (**0**, 2.1.5). The claim about integral preschemes follows from the conjunction of the two others. □

Proposition (9.5.10). — *Let Y be a subprescheme of a prescheme X, such that the canonical injection $i : Y \to X$ is a quasi-compact morphism. Then there exists a smaller closed subprescheme $\overline{Y}$ of X containing Y; its underlying space is the closure of that of Y; the latter is open in its closure, and the prescheme Y is induced on this open by $\overline{Y}$.*

PROOF. It suffices to apply Proposition (9.5.1) to the injection j, which is separated (5.5.1) and quasi-compact by hypothesis; (9.5.1) thus proves the existence of $\overline{Y}$, and (9.5.4) shows that its underlying space is the closure of Y in X; since Y is locally closed in X, it is open in $\overline{Y}$, and the last claim comes from (9.5.8) applied to an open subset V of X such that Y is closed in V. □

With the above notation, if the injection $V \to X$ is quasi-compact, and if $\mathscr{J}$ is the quasi-coherent sheaf of ideals of $\mathscr{O}_X|V$ defining the closed subprescheme Y of V, it follows from Proposition (9.5.1) that the quasi-coherent sheaf of ideals of $\mathscr{O}_X$ defining $\overline{Y}$ is the canonical extension (9.4.1) $\overline{\mathscr{J}}$ of $\mathscr{J}$, because it is clearly the largest quasi-coherent subsheaf of ideals of $\mathscr{O}_X$ inducing $\mathscr{J}$ on V.

Corollary (9.5.11). — *Under the hypotheses of Proposition* (9.5.10)*, every section of $\mathscr{O}_{\overline{Y}}$ over an open V of $\overline{Y}$ that is zero on $V \cap Y$ is zero.* I | 179

PROOF. By Proposition (9.5.8), we can reduce to the case where $V = \overline{Y}$. If we take into account that the sections of $\mathscr{O}_{\overline{Y}}$ over $\overline{Y}$ canonically correspond to the $\overline{Y}$-sections of $\overline{Y} \otimes_Z Z[T]$ (3.3.15) and that the latter is separated over $\overline{Y}$, then the corollary appears as a specific case of (9.5.6). □

When there exists a smaller closed subprescheme Y' of X containing a subprescheme Y of X, we say that Y' is the *closure* of Y in X, when there is little cause for confusion.

9.6. Quasi-coherent sheaves of algebras; change of structure sheaf

Proposition (9.6.1). — *Let X be a prescheme, and $\mathscr{B}$ a quasi-coherent $\mathscr{O}_X$-algebra (**0**, 5.1.3). For a $\mathscr{B}$-module $\mathscr{F}$ to be quasi-coherent (on the ringed space $(X, \mathscr{B})$) it is necessary and sufficient that $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module.*

PROOF. Since the question is local, we can assume X to be affine, given by some ring A, and thus $\mathscr{B} = \widetilde{B}$, where B is an A-algebra (1.4.3). If $\mathscr{F}$ is quasi-coherent on the ringed space $(X, \mathscr{B})$ then we can also assume that $\mathscr{F}$ is the cokernel of a $\mathscr{B}$-homomorphism $\mathscr{B}^{(I)} \to \mathscr{B}^{(J)}$; since this homomorphism is also an $\mathscr{O}_X$-homomorphism of $\mathscr{O}_X$-modules, and $\mathscr{B}^{(I)}$ and $\mathscr{B}^{(J)}$ are quasi-coherent $\mathscr{O}_X$-modules (1.3.9, ii), $\mathscr{F}$ is also a quasi-coherent $\mathscr{O}_X$-module (1.3.9, i).

Conversely, if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{F} = \widetilde{M}$, where M is a B-module (1.4.3); M is isomorphic to the cokernel of a B-homomorphism $B^{(I)} \to B^{(J)}$, so $\mathscr{F}$ is a $\mathscr{B}$-module isomorphic to the cokernel of the corresponding homomorphism $\mathscr{B}^{(I)} \to \mathscr{B}^{(J)}$ (1.3.13), which finishes the proof. □

In particular, if $\mathscr{F}$ and $\mathscr{G}$ are two quasi-coherent $\mathscr{B}$-modules, then $\mathscr{F} \otimes_{\mathscr{B}} \mathscr{G}$ is a quasi-coherent $\mathscr{B}$-module; similarly for $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ whenever we further suppose that $\mathscr{F}$ admits a finite presentation (1.3.13).

(9.6.2). Given a prescheme X, we say that a quasi-coherent $\mathscr{O}_X$-algebra $\mathscr{B}$ is of *finite type* if, for all $x \in X$, there exists an open *affine* neighbourhood U of x such that $\Gamma(U, \mathscr{B}) = B$ is an algebra of finite type over $\Gamma(U, \mathscr{O}_X) = A$. We then have that $\mathscr{B}|U = \widetilde{B}$ and, for all $f \in A$, the induced $(\mathscr{O}_X|D(f))$-algebra $\mathscr{B}|D(f)$ is of finite type, because it is isomorphic to $(B_f)^\sim$, and $B_f = B \otimes_A A_f$ is clearly an algebra of finite type over A_f. Since the $D(f)$ form a basis for the topology of U, we thus conclude that if $\mathscr{B}$ is a quasi-coherent $\mathscr{O}_X$-algebra of finite type then, for every open V of X, $\mathscr{B}|V$ is a quasi-coherent $(\mathscr{O}_X|V)$-algebra of finite type.

Proposition (9.6.3). — *Let X be a locally Noetherian prescheme. Then every quasi-coherent $\mathscr{O}_X$-algebra $\mathscr{B}$ of finite type is a coherent sheaf of rings (**0**, 5.3.7).*

PROOF. We can once again restrict to the case where X is an affine scheme given by a Noetherian ring A, and where $\mathscr{B} = \widetilde{B}$, with B being an A-algebra of finite type; B is then a Noetherian ring. With this, it remains to prove that the kernel $\mathscr{N}$ of a $\mathscr{B}$-homomorphism $\mathscr{B}^m \to \mathscr{B}$ is a $\mathscr{B}$-module of **I** | 180
finite type; but it is isomorphic (as a $\mathscr{B}$-module) to $\widetilde{N}$, where N is the kernel of the corresponding homomorphism of B-modules $B^m \to B$ (1.3.13). Since B is Noetherian, the submodule N of B^m is a B-module of finite type, so there exists a homomorphism $B^p \to B^m$ with image N; since the sequence $B^p \to B^m \to B$ is exact, so too is the corresponding sequence $\mathscr{B}^p \to \mathscr{B}^m \to \mathscr{B}$ (1.3.5), and since $\mathscr{N}$ is the image of $\mathscr{B}^p \to \mathscr{B}^m$ (1.3.9, i), this proves the proposition. □

Corollary (9.6.4). — *Under the hypotheses of (9.6.3), for a $\mathscr{B}$-module $\mathscr{F}$ to be coherent, it is necessary and sufficient that it be a quasi-coherent $\mathscr{O}_X$-module and a $\mathscr{B}$-module of finite type. If this is the case, and if $\mathscr{G}$ is a $\mathscr{B}$-submodule or a quotient module of $\mathscr{F}$, then in order for $\mathscr{G}$ to be a coherent $\mathscr{B}$-module, it is necessary and sufficient that it be a quasi-coherent $\mathscr{O}_X$-module.*

PROOF. Taking (9.6.1) into account, the conditions on $\mathscr{F}$ are clearly necessary; we will show that they are sufficient. We can restrict to the case where X is affine, given by some Noetherian ring A, where $\mathscr{B} = \widetilde{B}$, with B an A-algebra of finite type, where $\mathscr{F} = \widetilde{M}$, with M a B-module, and where there exists a surjective $\mathscr{B}$-homomorphism $\mathscr{B}^m \to \mathscr{F} \to 0$. We then have the corresponding exact sequence $B^m \to M \to 0$, so M is a B-module of finite type; further, the kernel P of the homomorphism $B^m \to M$ is then a B-module of finite type, since B is Noetherian. We thus conclude (1.3.13) that $\mathscr{F}$ is the cokernel of a $\mathscr{B}$-homomorphism $\mathscr{B}^m \to \mathscr{B}^n$, and is thus coherent, since $\mathscr{B}$ is a coherent sheaf of rings (**0**, 5.3.4). The same reasoning shows that a quasi-coherent $\mathscr{B}$-submodule (resp. a quotient $\mathscr{B}$-module) of $\mathscr{F}$ is of finite type, from whence the second part of the corollary. □

Proposition (9.6.5). — *Let X be a quasi-compact scheme, or a prescheme whose underlying space is Noetherian. For all quasi-compact $\mathscr{O}_X$-algebras $\mathscr{B}$ of finite type, there exists a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{E}$ of $\mathscr{B}$ of finite type such that $\mathscr{E}$ generates* (**0**, 4.1.3) *the $\mathscr{O}_X$-algebra $\mathscr{B}$.*

PROOF. In fact, by hypothesis, there exists a finite cover (U_i) of X consisting of affine opens such that $\Gamma(U_i, \mathscr{B}) = B_i$ is an algebra of finite type over $\Gamma(U_i, \mathscr{O}_X) = A_i$; let E_i be a A_i-submodule of B_i of finite type that generates the A_i-algebra B_i; thanks to (9.4.7), there exists an $\mathscr{O}_X$-submodule $\mathscr{E}_i$ of $\mathscr{B}$, quasi-coherent and of finite type, such that $\mathscr{E}_i|U_i = \widetilde{E}_i$. It is clear that the sum $\mathscr{E}$ of the $\mathscr{E}_i$ is the desired object. □

Proposition (9.6.6). — *Let X be a prescheme whose underlying space is locally Noetherian, or a quasi-compact scheme. Then every quasi-coherent $\mathscr{O}_X$-algebra $\mathscr{B}$ is the inductive limit of its quasi-coherent $\mathscr{O}_X$-subalgebras of finite type.*

PROOF. In fact, it follows from (9.4.9) that $\mathscr{B}$ is the inductive limit (as an $\mathscr{O}_X$-module) of its quasi-coherent $\mathscr{O}_X$-submodules of finite type; the latter generating quasi-coherent $\mathscr{O}_X$-*subalgebras* of $\mathscr{B}$ of finite type (1.3.14), and so $\mathscr{B}$ is *a fortiori* their inductive limit. □

§10. FORMAL SCHEMES

10.1. Formal affine schemes

(10.1.1). Let A be an *admissible* topological ring (**0**, 7.1.2); for each ideal of definition $\mathfrak{J}$ of A, $\operatorname{Spec}(A/\mathfrak{J})$ can be identified with the closed subspace $V(\mathfrak{J})$ of $\operatorname{Spec}(A)$ (1.1.11), the set of *open* prime ideals of A; this topological space does not depend on the ideal of definition $\mathfrak{J}$ considered; we denote this topological space by $\mathfrak{X}$. Let $(\mathfrak{J}_\lambda)$ be a fundamental system of neighbourhoods of 0 in A, consisting of ideals of definition, and for each λ, let $\mathscr{O}_\lambda$ be the structure sheaf of $\operatorname{Spec}(A/\mathfrak{J}_\lambda)$; this sheaf is induced on $\mathfrak{X}$ by $\widetilde{A}/\widetilde{\mathfrak{J}_\lambda}$ (which is zero outside of $\mathfrak{X}$). For $\mathfrak{J}_\mu \subset \mathfrak{J}_\lambda$, the canonical homomorphism $A/\mathfrak{J}_\mu \to A/\mathfrak{J}_\lambda$ thus defines a homomorphism $u_{\lambda\mu} : \mathscr{O}_\mu \to \mathscr{O}_\lambda$ of sheaves of rings (1.6.1), and $(\mathscr{O}_\lambda)$ is a *projective system of sheaves of rings* for these homomorphisms. Since the topology of $\mathfrak{X}$ admits a basis consisting of quasi-compact open subsets, we can associate to each $\mathscr{O}_\lambda$ a *pseudo-discrete sheaf of topological rings* (**0**, 3.8.1) which have $\mathscr{O}_\lambda$ as the underlying sheaf of rings (without topologies), and that we denote also by $\mathscr{O}_\lambda$; and the $\mathscr{O}_\lambda$ give again a *projective system of sheaves of topological rings* (**0**, 3.8.2). We denote by $\mathscr{O}_\mathfrak{X}$ the *sheaf of topological rings* on $\mathfrak{X}$, the projective limit of the system $(\mathscr{O}_\lambda)$; for each *quasi-compact* open subset U of $\mathfrak{X}$, $\Gamma(U, \mathscr{O}_\mathfrak{X})$ is a topological ring, the projective limit of the system of *discrete* rings $\Gamma(U, \mathscr{O}_\lambda)$ (**0**, 3.2.6). **I** | 181

Definition (10.1.2). — Given an admissible topological ring A, we define the formal spectrum of A, denoted by $\operatorname{Spf}(A)$, to be the closed subspace $\mathfrak{X}$ of $\operatorname{Spec}(A)$ consisting of the open prime ideals of A. We say that a topologically ringed space is a formal affine scheme if it is isomorphic to a formal spectrum $\operatorname{Spf}(A) = \mathfrak{X}$ equipped with a sheaf of topological rings $\mathscr{O}_\mathfrak{X}$ which is the projective limit of sheaves of pseudo-discrete topological rings $(\widetilde{A}/\widetilde{\mathfrak{J}_\lambda})|\mathfrak{X}$, where $\mathfrak{J}_\lambda$ varies over the filtered set of ideals of definition of A.

When we speak of a *formal spectrum* $\mathfrak{X} = \operatorname{Spf}(A)$ as a formal affine scheme, it will always be as the topologically ringed space $(\mathfrak{X}, \mathscr{O}_\mathfrak{X})$ where $\mathscr{O}_\mathfrak{X}$ is defined as above.

We note that every *affine scheme* $X = \operatorname{Spec}(A)$ can be considered as a formal affine scheme in only one way, by considering A as a discrete topological ring: the topological rings $\Gamma(U, \mathscr{O}_X)$ are then discrete whenever U is quasi-compact (but not, in general, when U is an arbitrary open subset of X).

Proposition (10.1.3). — *If $\mathfrak{X} = \operatorname{Spf}(A)$, where A is an admissible ring, then $\Gamma(\mathfrak{X}, \mathscr{O}_X)$ is topologically isomorphic to A.*

PROOF. Indeed, since $\mathfrak{X}$ is closed in $\operatorname{Spec}(A)$, it is quasi-compact, and so $\Gamma(\mathfrak{X}, \mathscr{O}_\mathfrak{X})$ is topologically isomorphic to the projective limit of the discrete rings $\Gamma(\mathfrak{X}, \mathscr{O}_\lambda)$; but $\Gamma(\mathfrak{X}, \mathscr{O}_\lambda)$ is isomorphic to $A/\mathfrak{J}_\lambda$ (1.3.7); since A is separated and complete, it is topologically isomorphic to $\varprojlim A/\mathfrak{J}_\lambda$ (**0**, 7.2.1), whence the proposition. □

Proposition (10.1.4). — *Let A be an admissible ring, $\mathfrak{X} = \operatorname{Spf}(A)$, and, for every $f \in A$, let $\mathfrak{D}(f) = D(f) \cap \mathfrak{X}$; then the topologically ringed space $(\mathfrak{D}(f), \mathscr{O}_{\mathfrak{X}}|\mathfrak{D}(f))$ is isomorphic to the formal affine spectrum* $\operatorname{Spf}(A_{\{f\}})$ *(**0**, 7.6.15).*

Proof. For every ideal of definition $\mathfrak{J}$ of A, the discrete ring $S_f^{-1}A/S_f^{-1}\mathfrak{J}$ is canonically identified with $A_{\{f\}}/\mathfrak{J}_{\{f\}}$ (**0**, 7.6.9), so, by (1.2.5) and (1.2.6), the topological space $\operatorname{Spf}(A_{\{f\}})$ is canonically identified with $\mathfrak{D}(f)$. Further, for every quasi-compact open subset U of $\mathfrak{X}$ contained in $\mathfrak{D}(f)$, $\Gamma(U, \mathscr{O}_\lambda)$ can be identified with the module of sections of the structure sheaf of $\operatorname{Spec}(S_f^{-1}A/S_f^{-1}\mathfrak{J}_\lambda)$ over U (1.3.6), so, setting $\mathfrak{Y} = \operatorname{Spf}(A_{\{f\}})$, $\Gamma(U, \mathscr{O}_{\mathfrak{X}})$ can be identified with the module of sections $\Gamma(U, \mathscr{O}_{\mathfrak{Y}})$, which proves the proposition. □

(10.1.5). As a sheaf of rings *without topology*, the structure sheaf $\mathscr{O}_{\mathfrak{X}}$ of $\operatorname{Spf}(A)$ admits, for every $x \in \mathfrak{X}$, a fibre which, by (10.1.4), can be identified with the inductive limit $\varinjlim A_{\{f\}}$ for the $f \notin \mathfrak{j}_x$. Then, by (**0**, 7.6.17) and (**0**, 7.6.18):

Proposition (10.1.6). — *For every $x \in \mathfrak{X} = \operatorname{Spf}(A)$, the fibre $\mathscr{O}_x$ is a local ring whose residue field is isomorphic to $k(x) = A_x/\mathfrak{j}_x A_x$. If, further, A is adic and Noetherian, then $\mathscr{O}_x$ is a Noetherian ring.*

Since $k(x)$ is not reduced at 0, we conclude from this that the *support* of the ring of sheaves $\mathscr{O}_{\mathfrak{X}}$ is *equal to* $\mathfrak{X}$.

10.2. Morphisms of formal affine schemes

(10.2.1). Let A, B be two admissible rings, and let $\phi : B \to A$ be a *continuous* morphism. The continuous map ${}^a\phi : \operatorname{Spec}(A) \to \operatorname{Spec}(B)$ (1.2.1) then maps $\mathfrak{X} = \operatorname{Spf}(A)$ to $\mathfrak{Y} = \operatorname{Spf}(B)$, since the inverse image under ϕ of an open prime ideal of A is an open prime ideal of B. Conversely, for all $g \in B$, ϕ defines a continuous homomorphism $\Gamma(\mathfrak{D}(g), \mathscr{O}_{\mathfrak{Y}}) \to \Gamma(\mathfrak{D}(\phi(g)), \mathscr{O}_{\mathfrak{X}})$ according to (10.1.4), (10.1.3), and (**0**, 7.6.7); since these homomorphisms satisfy the compatibility conditions for the restrictions corresponding to the change from g to a multiple of g, and since $\mathfrak{D}(\phi(g)) = {}^a\phi^{-1}(\mathfrak{D}(g))$, they define a *continuous* homomorphism of sheaves of topological rings $\mathscr{O}_{\mathfrak{Y}} \to {}^a\phi_*(\mathscr{O}_{\mathfrak{X}})$ (**0**, 3.2.5) that we denote by $\widetilde{\phi}$; we have thus defined a morphism $\Phi = ({}^a\phi, \widetilde{\phi})$ of topologically ringed spaces $\mathfrak{X} \to \mathfrak{Y}$. We note that, as a homomorphism of sheaves without topology, $\widetilde{\phi}$ defines a homomorphism $\widetilde{\phi}_x^\sharp : \mathscr{O}_{{}^a\phi(x)} \to \mathscr{O}_x$ on the stalks, for all $x \in \mathfrak{X}$.

Proposition (10.2.2). — *Let A and B be admissible topological rings, and let $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$. For a morphism $u = (\psi, \theta) : \mathfrak{X} \to \mathfrak{Y}$ of topologically ringed spaces to be of the form $({}^a\phi, \widetilde{\phi})$, where ϕ is a continuous ring homomorphism $B \to A$, it is necessary and sufficient that $\theta_x^\sharp$ be a local homomorphism $\mathscr{O}_{\phi(x)} \to \mathscr{O}_x$ for all $x \in \mathfrak{X}$.*

Proof. The condition is necessary: let $\mathfrak{p} = \mathfrak{j}_x \in \operatorname{Spf}(A)$, and let $\mathfrak{q} = \phi^{-1}(\mathfrak{j}_x)$; if $g \notin \mathfrak{q}$, then we have $\phi(g) \notin \mathfrak{p}$, and it is immediate that the homomorphism $B_{\{g\}} \to A_{\{\phi(g)\}}$ induced by ϕ (**0**, 7.6.7) sends $\mathfrak{q}_{\{g\}}$ to a subset of $\mathfrak{p}_{\{\phi(g)\}}$; by passing to the inductive limit, we see (taking (10.1.5) and (**0**, 7.6.17) into account) that $\widetilde{\phi}_x^\sharp$ is a local homomorphism.

Conversely, let (ψ, θ) be a morphism satisfying the condition of the proposition; by (10.1.3), θ defines a continuous ring homomorphism

$$\phi = \Gamma(\theta) : B = \Gamma(\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}) \longrightarrow \Gamma(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}) = A.$$

By virtue of the hypothesis on θ, for the section $\phi(g)$ of $\mathscr{O}_{\mathfrak{X}}$ over $\mathfrak{X}$ to be an invertible germ at the point x, it is necessary and sufficient that g be an invertible germ at the point $\psi(x)$. But, by (**0**, 7.6.17), the sections of $\mathscr{O}_{\mathfrak{X}}$ (resp. $\mathscr{O}_{\mathfrak{Y}}$) over $\mathfrak{X}$ (resp. $\mathfrak{Y}$) that have a non-invertible germ at the point x (resp. $\psi(x)$) are exactly the elements of $\mathfrak{j}_x$ (resp. $\mathfrak{j}_{\psi(x)}$); the above remark thus shows that ${}^a\phi = \psi$. Finally, **I** | 183
for all $g \in B$ the diagram

$$\begin{array}{ccc} B = \Gamma(\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}) & \xrightarrow{\phi} & \Gamma(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}) = A \\ \downarrow & & \downarrow \\ B_{\{g\}} = \Gamma(\mathfrak{D}(g), \mathscr{O}_{\mathfrak{Y}}) & \xrightarrow{\Gamma(\theta_{\mathfrak{D}(g)})} & \Gamma(\mathfrak{D}(\phi(g)), \mathscr{O}_{\mathfrak{X}}) = A_{\{\phi(g)\}} \end{array}$$

is commutative; by the universal property of completed rings of fractions (**0**, 7.6.6), $\theta_{\mathfrak{D}(g)}$ is equal to $\widetilde{\phi}_{\mathfrak{D}(g)}$ for all $g \in B$, and so (**0**, 3.2.5) we have $\theta = \widetilde{\phi}$. □

We say that a morphism (ψ, θ) of topologically ringed spaces satisfying the condition of Proposition (10.2.2) is a *morphism of formal affine schemes*. We can say that the functors $\operatorname{Spf}(A)$ in A and $\Gamma(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$ in $\mathfrak{X}$ define an *equivalence* between the category of admissible rings and the opposite category of formal affine schemes (T, I, 1.2).

(10.2.3). As a particular case of (10.2.2), note that, for $f \in A$, the canonical injection of the formal affine scheme induced by $\mathfrak{X}$ on $\mathfrak{D}(f)$ corresponds to the continuous canonical homomorphism $A \to A_{\{f\}}$. Under the hypotheses of Proposition (10.2.2), let h be an element of B, and g an element of A that is a multiple of $\phi(h)$; we then have $\psi(\mathfrak{D}(g)) \subset \mathfrak{D}(h)$; the restriction of u to $\mathfrak{D}(g)$, considered as a morphism from $\mathfrak{D}(g)$ to $\mathfrak{D}(h)$, is the unique morphism v making the diagram

$$\begin{tikzcd} \mathfrak{D}(g) \arrow[r, "v"] \arrow[d] & \mathfrak{D}(h) \arrow[d] \\ \mathfrak{X} \arrow[r, "u"] & \mathfrak{Y} \end{tikzcd}$$

commutate.

This morphism corresponds to the unique continuous homomorphism $\phi' : B_{\{h\}} \to A_{\{g\}}$ (**0**, 7.6.7) making the diagram

$$\begin{tikzcd} A \arrow[d] & B \arrow[l, "\phi"'] \arrow[d] \\ A_{\{g\}} & B_{\{h\}} \arrow[l, "\phi'"'] \end{tikzcd}$$

commutate.

10.3. Ideals of definition for a formal affine scheme

(10.3.1). Let A be an admissible ring, $\mathfrak{J}$ an open ideal of A, and $\mathfrak{X}$ the formal affine scheme $\operatorname{Spf}(A)$. Let $(\mathfrak{J}_\lambda)$ be the set of those ideals of definition for A that are contained in $\mathfrak{J}$; then $\widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}_\lambda$ is a sheaf of ideals of $\widetilde{A}/\widetilde{\mathfrak{J}}_\lambda$. Denote by $\mathfrak{J}^\Delta$ the projective limit of the sheaves on $\mathfrak{X}$ induced by $\widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}_\lambda$, which is identified with a *sheaf of ideals* of $\mathscr{O}_{\mathfrak{X}}$ (**0**, 3.2.6). For every $f \in A$, $\Gamma(\mathfrak{D}(f), \mathfrak{J}^\Delta)$ is the projective limit of the $S_f^{-1}\mathfrak{J}/S_f^{-1}\mathfrak{J}_\lambda$, or, in other words, can be identified with the open ideal $\mathfrak{J}_{\{f\}}$ of the ring $A_{\{f\}}$ (**0**, 7.6.9), and, in particular, $\Gamma(\mathfrak{X}, \mathfrak{J}^\Delta) = \mathfrak{J}$; we conclude (the $\mathfrak{D}(f)$ forming a basis for the topology of $\mathfrak{X}$) that

$$\mathfrak{J}^\Delta|\mathfrak{D}(f) = (\mathfrak{J}_{\{f\}})^\Delta. \tag{10.3.1.1}$$

(10.3.2). With the notation of (10.3.1), for all $f \in A$, the canonical map from $A_{\{f\}} = \Gamma(\mathfrak{D}(f), \mathscr{O}_{\mathfrak{X}})$ **I** | 184
to $\Gamma(\mathfrak{D}(f), (\widetilde{A}/\widetilde{\mathfrak{J}})|\mathfrak{X}) = S_f^{-1}A/S_f^{-1}\mathfrak{J}$ is *surjective* and has $\Gamma(\mathfrak{D}(f), \mathfrak{J}^\Delta) = \mathfrak{J}_{\{f\}}$ as its kernel (**0**, 7.6.9); these maps thus define a *surjective* continuous homomorphism, said to be *canonical*, from the sheaf of topological rings $\mathscr{O}_{\mathfrak{X}}$ to the sheaf of discrete rings $(\widetilde{A}/\widetilde{\mathfrak{J}})|\mathfrak{X}$, whose kernel is $\mathfrak{J}^\Delta$; this homomorphism is none other than $\widetilde{\phi}$ (10.2.1), where ϕ is the continuous homomorphism $A \to A/\mathfrak{J}$; the morphism $({}^a\phi, \widetilde{\phi}) : \operatorname{Spec}(A/\mathfrak{J}) \to \mathfrak{X}$ of formal affine schemes (where ${}^a\phi$ is the identity homeomorphism from $\mathfrak{X}$ to itself) is also said to be *canonical*. We thus have, according to the above, a *canonical isomorphism*

$$\mathscr{O}_{\mathfrak{X}}/\mathfrak{J}^\Delta \simeq (\widetilde{A}/\widetilde{\mathfrak{J}})|\mathfrak{X}. \tag{10.3.2.1}$$

It is clear (since $\Gamma(\mathfrak{X}, \mathfrak{J}^\Delta) = \mathfrak{J}$) that the map $\mathfrak{J} \to \mathfrak{J}^\Delta$ is *strictly increasing*; according to the above, for $\mathfrak{J} \subset \mathfrak{J}'$, the sheaf $\mathfrak{J}'^\Delta/\mathfrak{J}^\Delta$ is canonically isomorphic to $\widetilde{\mathfrak{J}}'/\widetilde{\mathfrak{J}} = (\mathfrak{J}'/\mathfrak{J})^\sim$.

(10.3.3). The hypotheses and notation being the same as in (10.3.1), we say that a sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_{\mathfrak{X}}$ is a *sheaf of ideals of definition* for $\mathfrak{X}$ (or an *ideal sheaf of definition* for $\mathfrak{X}$) if, for all $x \in \mathfrak{X}$, there exists an open neighbourhood of x of the form $\mathfrak{D}(f)$, where $f \in A$, such that $\mathscr{J}|\mathfrak{D}(f)$ is of the form $\mathfrak{H}^\Delta$, where $\mathfrak{H}$ is an ideal of definition for $A_{\{f\}}$.

Proposition (10.3.4). — *For all $f \in A$, each sheaf of ideals of definition for $\mathfrak{X}$ induces a sheaf of ideals of definition for $\mathfrak{D}(f)$.*

PROOF. This follows from (10.3.1.1). □

Proposition (10.3.5). — *If A is an admissible ring, then every sheaf of ideals of definition for $\mathfrak{X} = \operatorname{Spf}(A)$ is of the form $\mathfrak{J}^\Delta$, where $\mathfrak{J}$ is a uniquely determined ideal of definition for A.*

PROOF. Let $\mathscr{J}$ be a sheaf of ideals of definition of $\mathfrak{X}$; by hypothesis, and since $\mathfrak{X}$ is quasi-compact, there are finitely-many elements $f_i \in A$ such that the $\mathfrak{D}(f_i)$ cover $\mathfrak{X}$ and such that $\mathscr{J}|\mathfrak{D}(f_i) = \mathfrak{H}_i^\Delta$, where $\mathfrak{H}_i$ is an ideal of definition for $A_{\{f_i\}}$. For each i, there exists an open ideal $\mathfrak{K}_i$ of A such that $(\mathfrak{K}_i)_{\{f_i\}} = \mathfrak{H}_i$ (**0**, 7.6.9); let $\mathfrak{K}$ be an ideal of definition for A containing all the $\mathfrak{K}_i$. The canonical image of $\mathscr{J}/\mathfrak{K}^\Delta$ in the structure sheaf $(A/\mathfrak{K})^\sim$ of $\operatorname{Spec}(A/\mathfrak{K})$ (10.3.2) is thus such that its restriction to $\mathfrak{D}(f_i)$ is equal to its restriction to $(\mathfrak{K}_i/\mathfrak{K})^\sim$; we conclude that this canonical image is a *quasi-coherent* sheaf on $\operatorname{Spec}(A/\mathfrak{K})$, and so is of the form $(\mathfrak{J}/\mathfrak{K})^\sim$, where $\mathfrak{J}$ is an ideal of definition for A containing $\mathfrak{K}$ (1.4.1), whence $\mathscr{J} = \mathfrak{J}^\Delta$ (10.3.2); in addition, since for each i there exists an integer n_i such that $\mathfrak{H}_i^{n_i} \subset \mathfrak{K}_{\{f_i\}}$, we will have, by setting n to be the largest of the n_i, that $(\mathscr{J}/\mathfrak{K}^\Delta)^n = 0$, and, as a result (10.3.2), that $((\mathfrak{J}/\mathfrak{K})^\sim)^n = 0$, whence finally that $(\mathfrak{J}/\mathfrak{K})^n = 0$ (1.3.13), which proves that $\mathfrak{J}$ is an ideal of definition for A (**0**, 7.1.4). □

Proposition (10.3.6). — *Let A be an adic ring, and $\mathfrak{J}$ an ideal of definition for A such that $\mathfrak{J}/\mathfrak{J}^2$ is an $(A/\mathfrak{J})$-module of finite type. For any integer $n > 0$, we then have $(\mathfrak{J}^\Delta)^n = (\mathfrak{J}^n)^\Delta$.*

PROOF. For all $f \in A$, we have (since $\mathfrak{J}^n$ is an open ideal)

$$(\Gamma(\mathfrak{D}(f), \mathfrak{J}^\Delta))^n = (\mathfrak{J}_{\{f\}})^n = (\mathfrak{J}^n)_{\{f\}} = \Gamma(\mathfrak{D}(f^n), (\mathfrak{J}^n)^\Delta)$$

by (10.3.1.1) and (**0**, 7.6.12). The result then follows from the fact that $(\mathfrak{J}^\Delta)^n$ is associated to the presheaf $U \mapsto (\Gamma(U, \mathfrak{J}^\Delta))^n$ (**0**, 4.1.6), since the $\mathfrak{D}(f)$ form a basis for the topology of $\mathfrak{X}$. □ **I** | 185

(10.3.7). We say that a family $(\mathscr{J}_\lambda)$ of sheaves of ideals of definition for $\mathfrak{X}$ is a *fundamental system of sheaves of ideals of definition* if each sheaf of ideals of definition for $\mathfrak{X}$ contains one of the $\mathscr{J}_\lambda$; since $\mathscr{J}_\lambda = \mathfrak{J}_\lambda^\Delta$, it is equivalent to say that the $\mathfrak{J}_\lambda$ form a *fundamental system of neighbourhoods of* 0 in A. Let (f_α) be a family of elements of A such that the $\mathfrak{D}(f_\alpha)$ cover $\mathfrak{X}$. If $(\mathscr{J}_\lambda)$ is a filtered decreasing family of sheaves of ideals of $\mathscr{O}_\mathfrak{X}$ such that, for each α, the family $(\mathscr{J}_\lambda|\mathfrak{D}(f_\alpha))$ is a fundamental system of sheaves of ideals of definition for $\mathfrak{D}(f_\alpha)$, then $(\mathscr{J}_\lambda)$ is a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$. Indeed, for each sheaf of ideals of definition $\mathscr{J}$ for $\mathfrak{X}$, there is a finite cover of $\mathfrak{X}$ by $\mathfrak{D}(f_i)$ such that, for each i, $\mathscr{J}_{\lambda_i}|\mathfrak{D}(f_i)$ is a sheaf of ideals of definition for $\mathfrak{D}(f_i)$ that is contained in $\mathscr{J}|\mathfrak{D}(f_i)$. If μ is an index such that $\mathscr{J}_\mu \subset \mathscr{J}_{\lambda_i}$ for all i, then it follows from (10.3.3) that $\mathscr{J}_\mu$ is a sheaf of ideals of definition for $\mathfrak{X}$, evidently contained in $\mathscr{J}$, whence our claim.

10.4. Formal preschemes and morphisms of formal preschemes

(10.4.1). Given a topologically ringed space $\mathfrak{X}$, we say that an open $U \subset \mathfrak{X}$ is a *formal affine open* (resp. a *formal adic affine open*, resp. a *formal Noetherian affine open*) if the topologically ringed space induced on U by $\mathfrak{X}$ is a formal affine scheme (resp. a scheme whose ring is adic, resp. adic and Noetherian).

Definition (10.4.2). — A *formal prescheme* is a topologically ringed space $\mathfrak{X}$ which admits a formal affine open neighbourhood for each point. We say that the formal prescheme $\mathfrak{X}$ is adic (resp. locally Noetherian) if each point of $\mathfrak{X}$ admits a formal adic (resp. Noetherian) open neighbourhood. We say that $\mathfrak{X}$ is Noetherian if it is locally Noetherian and its underlying space is quasi-compact (and hence Noetherian).

Proposition (10.4.3). — *If $\mathfrak{X}$ is a formal prescheme (resp. a locally Noetherian formal prescheme), then the formal affine (resp. Noetherian affine) open sets form a basis for the topology of $\mathfrak{X}$.*

PROOF. This follows from Definition (10.4.2) and Proposition (10.1.4) by taking into account that, if A is an adic Noetherian ring, then so too is $A_{\{f\}}$ for all $f \in A$ (**0**, 7.6.11). □

Corollary (10.4.4). — *If $\mathfrak{X}$ is a formal prescheme (resp. a locally Noetherian formal prescheme, resp. a Noetherian formal prescheme), then the topologically ringed space induced on each open set of $\mathfrak{X}$ is also a formal prescheme (resp. a locally Noetherian formal prescheme, resp. a Noetherian formal prescheme).*

Definition (10.4.5). — Given two formal preschemes $\mathfrak{X}$ and $\mathfrak{Y}$, a morphism (of formal preschemes) from $\mathfrak{X}$ to $\mathfrak{Y}$ is a morphism (ψ, θ) of topologically ringed spaces such that, for all $x \in \mathfrak{X}$, $\theta_x^\sharp$ is a local homomorphism $\mathscr{O}_{\psi(x)} \to \mathscr{O}_x$.

It is immediate that the composition of any two morphisms of formal preschemes is again a morphism of formal preschemes; the formal preschemes thus form a *category*, and we denote by $\operatorname{Hom}(\mathfrak{X}, \mathfrak{Y})$ the set of morphisms from a formal prescheme $\mathfrak{X}$ to a formal prescheme $\mathfrak{Y}$.

If U is an open subset of $\mathfrak{X}$, then the canonical injection into $\mathfrak{X}$ of the formal prescheme induced I | 186
on U by $\mathfrak{X}$ is a morphism of formal preschemes (and in fact a *monomorphism* of topologically ringed spaces (**0**, 4.1.1)).

Proposition (10.4.6). — *Let $\mathfrak{X}$ be a formal prescheme, and $\mathfrak{S} = \operatorname{Spf}(A)$ a formal affine scheme. There exists a canonical bijective equivalence between the morphisms from the formal prescheme $\mathfrak{X}$ to the formal prescheme $\mathfrak{S}$ and the continuous homomorphisms from the ring A to the topological ring $\Gamma(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$.*

Proof. The proof is similar to that of (2.2.4), by replacing "homomorphism" with "continuous homomorphism", "affine open" with "formal affine open", and by using Proposition (10.2.2) instead of Theorem (1.7.3); we leave the details to the reader. □

(10.4.7). Given a formal prescheme $\mathfrak{S}$, we say that the data of a formal prescheme $\mathfrak{X}$ and a morphism $\phi : \mathfrak{X} \to \mathfrak{S}$ defines a formal prescheme $\mathfrak{X}$ *over* $\mathfrak{S}$ or an *formal $\mathfrak{S}$-prescheme*, ϕ being called the *structure morphism* of the $\mathfrak{S}$-prescheme $\mathfrak{X}$. If $\mathfrak{S} = \operatorname{Spf}(A)$, where A is an admissible ring, then we also say that the formal $\mathfrak{S}$-prescheme $\mathfrak{X}$ is a *formal A-prescheme* or a formal prescheme *over A*. An arbitrary formal prescheme can be considered as a formal prescheme over $\mathbf{Z}$ (equipped with the discrete topology).

If $\mathfrak{X}$ and $\mathfrak{Y}$ are formal $\mathfrak{S}$-preschemes, then we say that a morphism $u : \mathfrak{X} \to \mathfrak{Y}$ is a $\mathfrak{S}$-*morphism* if the diagram

$$\begin{tikzcd} \mathfrak{X} \arrow[rr, "u"] \arrow[dr] & & \mathfrak{Y} \arrow[dl] \\ & \mathfrak{S} & \end{tikzcd}$$

(where the downwards arrows are the structure morphisms) is commutative. With this definition, the formal $\mathfrak{S}$-preschemes (for some fixed $\mathfrak{S}$) form a *category*. We denote by $\operatorname{Hom}_{\mathfrak{S}}(\mathfrak{X}, \mathfrak{Y})$ the set of $\mathfrak{S}$-morphisms from a formal $\mathfrak{S}$-prescheme $\mathfrak{X}$ to a formal $\mathfrak{S}$-prescheme $\mathfrak{Y}$. When $\mathfrak{S} = \operatorname{Spf}(A)$, we sometimes say *$A$-morphism* instead of *$\mathfrak{S}$-morphism*.

(10.4.8). Since every affine scheme can be considered as a formal affine scheme (10.1.2), every (usual) prescheme can be considered as a formal prescheme. In addition, it follows from Definition (10.4.5) that, for the *usual* preschemes, the morphisms (resp. S-morphisms) of *formal* preschemes coincide with the morphisms (resp. S-morphisms) defined in §2.

10.5. Sheaves of ideals of definition for formal preschemes

(10.5.1). Let $\mathfrak{X}$ be a formal prescheme; we say that an $\mathscr{O}_{\mathfrak{X}}$-ideal $\mathscr{J}$ is a *sheaf of ideals of definition* (or an *ideal sheaf of definition*) for $\mathfrak{X}$ if every $x \in \mathfrak{X}$ has a formal affine open neighbourhood U such that $\mathscr{J}|U$ is a sheaf of ideals of definition for the formal affine scheme induced on U by $\mathfrak{X}$ (10.3.3); by (10.3.1.1) and Proposition (10.4.3), for each open $V \subset \mathfrak{X}$, $\mathscr{J}|V$ is then a sheaf of ideals of definition for the formal prescheme induced on V by $\mathfrak{X}$.

We say that a family $(\mathscr{J}_\lambda)$ of sheaves of ideals of definition for $\mathfrak{X}$ is a *fundamental system of* I | 187
sheaves of ideals of definition if there exists a cover (U_α) of $\mathfrak{X}$ by formal affine open sets such that, for each α, the family of the $\mathscr{J}_\lambda|U_\alpha$ is a fundamental system of sheaves of ideals of definition (10.3.6) for the formal affine scheme induced on U_α by $\mathfrak{X}$. It follows from the last remark of (10.3.7) that, when $\mathfrak{X}$ is a formal affine scheme, this definition coincides with the definition given in (10.3.7). For an open subset V of $\mathfrak{X}$, the restrictions $\mathscr{J}_\lambda|V$ then form a fundamental system of sheaves of ideals of definition for the formal prescheme induced on V, according to (10.3.1.1). If $\mathfrak{X}$ is a *locally Noetherian*

formal prescheme, and $\mathscr{J}$ is a sheaf of ideals of definition for $\mathfrak{X}$, then it follows from Proposition (10.3.6) that the powers $\mathscr{J}^n$ form a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$.

(10.5.2). Let $\mathfrak{X}$ be a formal prescheme, and $\mathscr{J}$ a sheaf of ideals of definition for $\mathfrak{X}$. Then the ringed space $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J})$ is a (usual) *prescheme*, which is affine (resp. locally Noetherian, resp. Noetherian) when $\mathfrak{X}$ is a formal affine scheme (resp. a locally Noetherian formal scheme, resp. a Noetherian formal scheme); we can reduce to the affine case, and then the proposition has already been proved in (10.3.2). In addition, if $\theta : \mathscr{O}_{\mathfrak{X}} \to \mathscr{O}_{\mathfrak{X}}/\mathscr{J}$ is the canonical homomorphism, then $u = (1_{\mathfrak{X}}, \theta)$ is a *morphism* (said to be *canonical*) of formal preschemes $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}) \to (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$, because, again, this was proved in the affine case (10.3.2), to which it is immediately reduced.

Proposition (10.5.3). — *Let $\mathfrak{X}$ be a formal prescheme, and $(\mathscr{J}_\lambda)$ a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$. Then the sheaf of topological rings $\mathscr{O}_{\mathfrak{X}}$ is the projective limit of the pseudo-discrete sheaves of rings (**0**, 3.8.1) $\mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda$.*

PROOF. Since the topology of $\mathfrak{X}$ admits a basis of formal quasi-compact affine open sets (10.4.3), we reduce to the affine case, where the proposition is a consequence of Proposition (10.3.5), (10.3.2), and Definition (10.1.1). □

It is not true that any formal prescheme admits a sheaf of ideals of definition. However:

Proposition (10.5.4). — *Let $\mathfrak{X}$ be a locally Noetherian formal prescheme. There exists a largest sheaf of ideals of definition $\mathscr{T}$ for $\mathfrak{X}$; this is the unique sheaf of ideals of definition $\mathscr{J}$ such that the prescheme $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J})$ is reduced. If $\mathscr{J}$ is a sheaf of ideals of definition for $\mathfrak{X}$, then $\mathscr{T}$ is the inverse image under $\mathscr{O}_{\mathfrak{X}} \to \mathscr{O}_{\mathfrak{X}}/\mathscr{J}$ of the nilradical of $\mathscr{O}_{\mathfrak{X}}/\mathscr{J}$.*

PROOF. Suppose first that $\mathfrak{X} = \operatorname{Spf}(A)$, where A is an adic Noetherian ring. The existence and the properties of $\mathscr{T}$ follow immediately from Propositions (10.3.5) and (5.1.1), taking into account the existence and the properties of the largest ideal of definition for A ((**0**, 7.1.6) and (**0**, 7.1.7)). **Err**$_{\text{II}}$

To prove the existence and the properties of $\mathscr{T}$ in the general case, it suffices to show that, if $U \supset V$ are two Noetherian formal affine open subsets of X, then the largest sheaf of ideals of definition $\mathscr{T}_U$ for U induces the largest sheaf of ideals of definition $\mathscr{T}_V$ for V; but as $(V, (\mathscr{O}_{\mathfrak{X}}|V)/(\mathscr{T}_U|V))$ is reduced, this follows from the above. □

We denote by $\mathfrak{X}_{\text{red}}$ the (usual) reduced prescheme $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{T})$.

Corollary (10.5.5). — *Let $\mathfrak{X}$ be a locally Noetherian formal prescheme, and $\mathscr{T}$ the largest sheaf of ideals of definition for $\mathfrak{X}$; for each open subset V of $\mathfrak{X}$, $\mathscr{T}|V$ is the largest sheaf of ideals of definition for the formal prescheme induced on V by $\mathfrak{X}$.*

Proposition (10.5.6). — *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be formal preschemes, $\mathscr{J}$ (resp. $\mathscr{K}$) be a sheaf of ideals of definition for $\mathfrak{X}$ (resp. $\mathfrak{Y}$), and $f : \mathfrak{X} \to \mathfrak{Y}$ a morphism of formal preschemes.*

(i) *If $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$, then there exists a unique morphism $f' : (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}) \to (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K})$ of usual preschemes making the diagram*

(10.5.6.1)
$$\begin{tikzcd} (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}) \arrow[r, "f"] & (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}) \\ (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}) \arrow[u] \arrow[r, "f'"] & (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K}) \arrow[u] \end{tikzcd}$$

commutate, where the vertical arrows are the canonical morphisms.

(ii) *Suppose that $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$ are formal affine schemes, $\mathscr{J} = \mathfrak{J}^\Delta$ and $\mathscr{K} = \mathfrak{K}^\Delta$, where $\mathfrak{J}$ (resp. $\mathfrak{K}$) is an ideal of definition for A (resp. B), and $f = ({}^a\phi, \widetilde{\phi})$, where $\phi : B \to A$ is a continuous homomorphism; for $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$ to hold, it is necessary and sufficient that $\phi(\mathfrak{K}) \subset \mathfrak{J}$, and, in this case, f' is then the morphism $({}^a\phi', \widetilde{\phi'})$, where $\phi' : B/\mathfrak{K} \to A/\mathfrak{J}$ is the homomorphism induced from ϕ by passing to quotients.*

PROOF.

(i) If $f = (\psi, \theta)$, then the hypotheses imply that the image under $\theta^\sharp : \psi^*(\mathscr{O}_{\mathfrak{Y}}) \to \mathscr{O}_{\mathfrak{X}}$ of the sheaf of ideals $\psi^*(\mathscr{K})$ of $\psi^*(\mathscr{O}_{\mathfrak{Y}})$ is contained in $\mathscr{J}$ (0, 4.3.5). By passing to quotients, we thus obtain from $\theta^\sharp$ a homomorphism of sheaves of rings

$$\omega : \psi^*(\mathscr{O}_{\mathfrak{Y}}/\mathscr{K}) = \psi^*(\mathscr{O}_{\mathfrak{Y}})/\psi^*(\mathscr{K}) \longrightarrow \mathscr{O}_{\mathfrak{X}}/\mathscr{J};$$

furthermore, since, for all $x \in \mathfrak{X}$, $\theta_x^\sharp$ is a *local* homomorphism, so too is ω_x. The morphism of ringed spaces $(\psi, \omega^\flat)$ is thus (2.2.1) the unique morphism f' of ringed spaces whose existence was claimed.

(ii) The canonical functorial correspondence between morphisms of formal affine schemes and continuous homomorphisms of rings (10.2.2) shows that, in the case considered, the relation $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathfrak{J}$ implies that we have $f' = ({}^a\phi', \widetilde{\phi'})$, where $\phi' : B/\mathfrak{K} \to A/\mathfrak{J}$ is the unique homomorphism making the diagram

(10.5.6.2)
$$\begin{tikzcd} B \arrow[r, "\phi"] \arrow[d] & A \arrow[d] \\ B/\mathfrak{K} \arrow[r, "\phi'"] & A/\mathfrak{J} \end{tikzcd}$$

commutate. The existence of ϕ' thus implies that $\phi(\mathfrak{K}) \subset \mathfrak{J}$. Conversely, if this condition is satisfied, then, denoting by ϕ' the unique homomorphism making the diagram (10.5.6.2) commutate and setting $f' = ({}^a\phi', \widetilde{\phi'})$, it is clear that the diagram (10.5.6.1) is commutative; considering the homomorphisms ${}^a\phi^*(\mathscr{O}_{\mathfrak{Y}}) \to \mathscr{O}_{\mathfrak{X}}$ and ${}^a\phi'^*(\mathscr{O}_{\mathfrak{Y}}/\mathscr{K}) \to \mathscr{O}_{\mathfrak{X}}/\mathscr{J}$ corresponding to f and f' respectively then leads to the fact that this implies the relation $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$.

□

It is clear that the correspondence $f \mapsto f'$ defined above is *functorial*.

10.6. Formal preschemes as inductive limits of preschemes

(10.6.1). Let $\mathfrak{X}$ be a formal prescheme, and $(\mathscr{J}_\lambda)$ a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$; for each λ, let f_λ be the canonical morphism $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda) \to \mathfrak{X}$ (10.5.2); for $\mathscr{J}_\mu \subset \mathscr{J}_\lambda$, the canonical morphism $\mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\mu \to \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda$ defines a canonical morphism $f_{\mu\lambda}$: **I | 189**
$(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda) \to (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\mu)$ of (usual) preschemes such that $f_\lambda = f_\mu \circ f_{\mu\lambda}$. The preschemes $X_\lambda = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda)$ and the morphisms $f_{\mu\lambda}$ thus form (by (10.4.8)) an *inductive system* in the category of formal preschemes.

Proposition (10.6.2). — *With the notation of (10.6.1), the formal prescheme $\mathfrak{X}$ and the morphisms f_λ form an inductive limit (T, I, 1.8) of the system $(X_\lambda, f_{\mu\lambda})$ in the category of formal preschemes.*

PROOF. Let $\mathfrak{Y}$ be a formal prescheme, and, for each index λ, let

$$g_\lambda = (\psi_\lambda, \theta_\lambda) : X_\lambda \longrightarrow \mathfrak{Y}$$

be a morphism such that $g_\lambda = g_\mu \circ f_{\mu\lambda}$ for $\mathscr{J}_\mu \subset \mathscr{J}_\lambda$. This latter condition and the definition of the X_λ imply first of all that the ψ_λ are identical to a single continuous map $\psi : \mathfrak{X} \to \mathfrak{Y}$ of the underlying spaces; in addition, the homomorphisms $\theta_\lambda^\sharp : \psi^*(\mathscr{O}_{\mathfrak{Y}}) \to \mathscr{O}_{X_i} = \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda$ form a *projective system* of homomorphisms of sheaves of rings. By passing to the projective limit, there is an induced homomorphism $\omega : \psi^*(\mathscr{O}_{\mathfrak{Y}}) \to \varprojlim \mathscr{O}_{\mathfrak{X}}/\mathscr{J}_\lambda = \mathscr{O}_{\mathfrak{X}}$, and it is clear that the morphism $g = (\psi, \omega^\flat)$ of *ringed spaces* is the *unique* morphism making the diagrams

(10.6.2.1)
$$\begin{tikzcd} X_\lambda \arrow[rr, "g_\lambda"] \arrow[dr, "f_\lambda"'] & & \mathfrak{Y} \\ & \mathfrak{X} \arrow[ur, "g"'] & \end{tikzcd}$$

commutative. It remains to prove that g is a morphism of *formal preschemes*; the question is local on $\mathfrak{X}$ and $\mathfrak{Y}$, so we can assume that $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$, with A and B admissible rings, and with $\mathscr{J}_\lambda = \mathfrak{J}_\lambda^\Delta$, where $(\mathfrak{J}_\lambda)$ is a fundamental system of ideals of definition for A (10.3.5);

since $A = \varprojlim A/\mathfrak{J}_\lambda$, the existence of a morphism g of formal affine schemes making the diagrams (10.6.2.1) commutate then follows from the bijective correspondence (10.2.2) between morphisms of formal affine schemes and continuous ring homomorphisms, and from the definition of the projective limit. But the uniqueness of g as a morphism of ringed spaces shows that it coincides with the morphism in the beginning of the proof. □

The following proposition establishes, under certain additional conditions, the existence of the inductive limit of a given inductive system of (usual) preschemes in the category of formal preschemes:

Proposition (10.6.3). — *Let $\mathfrak{X}$ be a topological space, and $(\mathscr{O}_i, u_{ji})$ a projective system of sheaves of rings on $\mathfrak{X}$, with* **N** *for its set of indices. Let $\mathscr{J}_i$ be the kernel of $u_{0i} : \mathscr{O}_i \to \mathscr{O}_0$. Suppose that:*

(a) *the ringed space $(\mathfrak{X}, \mathscr{O}_i)$ is a prescheme X_i;*
(b) *for all $x \in \mathfrak{X}$ and all i, there exists an open neighbourhood U_i of x in $\mathfrak{X}$ such that the restriction $\mathscr{J}_i|U_i$ is nilpotent; and* **Err$_{\text{II}}$**
(c) *the homomorphisms u_{ji} are surjective.*

Let $\mathscr{O}_{\mathfrak{X}}$ be the sheaf of topological rings given by the projective limit of the pseudo-discrete sheaves of rings $\mathscr{O}_i$, and let $u_i : \mathscr{O}_{\mathfrak{X}} \to \mathscr{O}_i$ be the canonical homomorphism. Then the topologically ringed space $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$ is a formal prescheme; the homomorphisms u_i are surjective; their kernels $\mathscr{J}^{(i)}$ form a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$, and $\mathscr{J}^{(0)}$ is the projective limit of the sheaves of ideals $\mathscr{J}_i$. **I | 190**

PROOF. We first note that, on each stalk, u_{ji} is a surjective homomorphism and *a fortiori* a local homomorphism; thus $v_{ij} = (1_{\mathfrak{X}}, u_{ji})$ is a morphism of preschemes $X_j \to X_i$ $(i \geqslant j)$ (2.2.1). Suppose first that each X_i is an affine scheme of ring A_i. There exists a *ring* homomorphism $\phi_{ji} : A_i \to A_j$ such that $u_{ji} = \widetilde{\phi_{ji}}$ (1.7.3); as a result (1.6.3), the sheaf $\mathscr{O}_j$ is a quasi-coherent $\mathscr{O}_i$-module over X_i (for the external law defined by u_{ji}), associated to A_j considered as an A_i-module by means of ϕ_{ji}. For all $f \in A_i$, let $f' = \phi_{ji}(f)$; by hypothesis, the open sets $D(f)$ and $D(f')$ are identical in $\mathfrak{X}$, and the homomorphism from $\Gamma(D(f), \mathscr{O}_i) = (A_i)_f$ to $\Gamma(D(f), \mathscr{O}_j) = (A_j)_{f'}$ corresponding to u_{ji} is exactly $(\phi_{ji})_f$ (1.6.1). But when we consider A_j as an A_i-module, $(A_j)_{f'}$ is the $(A_i)_f$-module $(A_j)_f$, so we also have $u_{ji} = \widetilde{\phi_{ji}}$, where ϕ_{ji} is now considered as a homomorphism of A_i-*modules*. Then, since u_{ji} is surjective, we conclude that ϕ_{ji} is also surjective (1.3.9), and if $\mathfrak{J}_{ji}$ is the kernel of ϕ_{ji}, then the kernel of u_{ji} is a quasi-coherent $\mathscr{O}_i$-module equal to $\widetilde{\mathfrak{J}_{ji}}$. In particular, we have $\mathscr{J}_i = \widetilde{\mathfrak{J}_i}$, where $\mathfrak{J}_i$ is the kernel of $\phi_{0i} : A_i \to A_0$. Hypothesis (b) implies that $\mathscr{J}_i$ is *nilpotent*: indeed, since $\mathfrak{X}$ is quasi-compact, we can cover $\mathfrak{X}$ by a finite number of open sets U_k such that $(\mathscr{J}_i|U_k)^{n_k} = 0$, and, by setting n to be the largest of the n_k, we have $\mathscr{J}_i^n = 0$. We thus conclude that $\mathfrak{J}_i$ is nilpotent (1.3.13). Then the ring $A = \varprojlim A_i$ is admissible (**0**, 7.2.2), the canonical homomorphism $\phi_i : A \to A_i$ is surjective, and its kernel $\mathfrak{J}^{(i)}$ is equal to the projective limit of the $\mathfrak{J}_{ik}$ for $k \geqslant i$; the $\mathfrak{J}^{(i)}$ form a fundamental system of neighbourhoods of 0 in A. The claims of Proposition (10.6.3) follow in this case from (10.1.1) and (10.3.2), with $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$ being $\operatorname{Spf}(A)$.

Again, in this particular case, we note that, if $f = (f_i)$ is an element of the projective limit $A = \varprojlim A_i$, then all the open sets $D(f_i)$ (which are affine open sets in X_i) can be identified with the open subset $\mathfrak{D}(f)$ of $\mathfrak{X}$, and the prescheme induced on $\mathfrak{D}(f)$ by X_i thus being identified with the affine scheme $\operatorname{Spec}((A_i)_{f_i})$.

In the general case, we remark first that, for every quasi-compact open subset U of $\mathfrak{X}$, each of the $\mathscr{J}_i|U$ is nilpotent, as shown by the above reasoning. We will show that, for every $x \in \mathfrak{X}$, there exists an open neighbourhood U of x in $\mathfrak{X}$ which is an *affine open set* for *all* the X_i. Indeed, we take U to be an affine open set for X_0, and observe that $\mathscr{O}_{X_0} = \mathscr{O}_{X_i}/\mathscr{J}_i$. Since, by the above, $\mathscr{J}_i|U$ is nilpotent, U is also an affine open set for each X_i, by Proposition (5.1.9). This being so, for each U satisfying the preceding conditions, the study of the affine case as above shows that $(U, \mathscr{O}_X|U)$ is a formal prescheme whose $\mathscr{J}^{(i)}|U$ form a fundamental system of sheaves of ideals of definition, and $\mathscr{J}^{(0)}|U$ is the projective limit of the $\mathscr{J}_i|U$; whence the conclusion. □

Corollary (10.6.4). — *Suppose that, for $i \geqslant j$, the kernel of u_{ji} is $\mathscr{J}_i^{j+1}$ and that $\mathscr{J}_1/\mathscr{J}_1^2$ is of finite type over $\mathscr{O}_0 = \mathscr{O}_1/\mathscr{J}_1$. Then $\mathfrak{X}$ is an adic formal prescheme, and if $\mathscr{J}^{(n)}$ is the kernel of $\mathscr{O}_{\mathfrak{X}} \to \mathscr{O}_n$,* **I | 191**

then $\mathscr{J}^{(n)} = \mathscr{J}^{n+1}$ and $\mathscr{J}/\mathscr{J}^2$ is isomorphic to $\mathscr{J}_1$. If, in addition, X_0 is locally Noetherian (resp. Noetherian), then $\mathfrak{X}$ is locally Noetherian (resp. Noetherian).

PROOF. Since the underlying spaces of $\mathfrak{X}$ and X_0 are the same, the question is local, and we can suppose that all the X_i are affine; taking into account the fact that $\mathscr{J}_{ij} = \widetilde{\mathfrak{J}_{ji}}$ (with the notation of Proposition (10.3.6)), we can immediately reduce the problem to the corresponding claims of Proposition (**0**, 7.2.7) and Corollary (**0**, 7.2.8), by noting that $\mathfrak{J}_1/\mathfrak{J}_1^2$ is then an A_0-module of finite type (1.3.9). □

In particular, *every locally Noetherian formal prescheme* $\mathfrak{X}$ is the inductive limit of a sequence (X_n) of locally Noetherian (usual) preschemes satisfying the conditions of Proposition (10.3.6) and Corollary (10.6.4): it suffices to consider a sheaf of ideals of definition $\mathscr{J}$ for $\mathfrak{X}$ (10.5.4) and to take $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$ ((10.5.1) and Proposition (10.6.2)).

Corollary (10.6.5). — *Let A be an admissible ring. For the formal affine scheme $\mathfrak{X} = \operatorname{Spf}(A)$ to be Noetherian, it is necessary and sufficient for A to be adic and Noetherian.*

PROOF. The condition is evidently sufficient. Conversely, suppose that $\mathfrak{X}$ is Noetherian, and let $\mathfrak{J}$ be an ideal of definition for A, and $\mathscr{J} = \mathfrak{J}^{\Delta}$ the corresponding sheaf of ideals of definition for $\mathfrak{X}$. The (usual) preschemes $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$ are then affine and Noetherian, so the rings $A_n = A/\mathfrak{J}^{n+1}$ are Noetherian (6.1.3), whence we conclude that $\mathfrak{J}/\mathfrak{J}^2$ is an $A/\mathfrak{J}$-module of finite type. Since the $\mathscr{J}^n$ form a fundamental system of sheaves of ideals of definition for $\mathfrak{X}$ (10.5.1), we have $\mathscr{O}_{\mathfrak{X}} = \varprojlim \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^n$ (10.5.3); we thus conclude (10.1.3) that A is topologically isomorphic to $\varprojlim A/\mathfrak{J}^n$, which is adic and Noetherian (**0**, 7.2.8). □

Remark (10.6.6). — With the notation of Proposition (10.6.3), let $\mathscr{F}_i$ be an $\mathscr{O}_i$-module, and suppose we are given, for $i \geqslant i$, a v_{ij}-morphism $\theta_{ji} : \mathscr{F}_i \to \mathscr{F}_j$ such that $\theta_{kj} \circ \theta_{ji} = \theta_{ki}$ for $k \leqslant j \leqslant i$. Since the underlying continuous map of v_{ij} is the identity, θ_{ji} is a homomorphism of sheaves of abelian groups to the space $\mathfrak{X}$; in addition, if $\mathscr{F}$ is the projective limit of the projective system $(\mathscr{F}_i)$ of sheaves of abelian groups, then the fact that the θ_{ji} are v_{ij}-morphisms lets us define an $\mathscr{O}_{\mathfrak{X}}$-module structure on $\mathscr{F}$ by passing to the projective limit; when equipped with this structure, we say that $\mathscr{F}$ is the *projective limit* (with respect to the θ_{ji}) of the system of $\mathscr{O}_i$-modules $(\mathscr{F}_i)$. In the particular case where $v_{ij}^*(\mathscr{F}_i) = \mathscr{F}_j$ and θ_{ji} is the *identity*, we say that $\mathscr{F}$ is the projective limit of a system $(\mathscr{F}_i)$ such that $v_{ij}^*(\mathscr{F}_i) = \mathscr{F}_j$ for $j \leqslant i$ (without mentioning the θ_{ji}).

(10.6.7). Let $\mathfrak{X}$ and $\mathfrak{Y}$ be formal preschemes, $\mathscr{J}$ (resp. $\mathscr{K}$) a sheaf of ideals of definition for $\mathfrak{X}$ (resp. $\mathfrak{Y}$), and $f : \mathfrak{X} \to \mathfrak{Y}$ a morphism such that $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$. We then have, for every integer $n > 0$, that $f^*(\mathscr{K}^n)\mathscr{O}_{\mathfrak{X}} = (f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}})^n \subset \mathscr{J}^n$; f thus induces (10.5.6) a morphism of (usual) preschemes $f_n : X_n \to Y_n$ by setting $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$ and $Y_n = (\mathscr{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K}^{n+1})$, and it immediately follows from the definitions that the diagrams

(10.6.7.1)
$$\begin{tikzcd} X_m \arrow[r, "f_m"] \arrow[d] & Y_m \arrow[d] \\ X_n \arrow[r, "f_n"] & Y_n \end{tikzcd}$$

commute for $m \leqslant n$; in other words, (f_n) is an *inductive system* of morphisms. **I** | 192

(10.6.8). Conversely, let (X_n) (resp. (Y_n)) be an inductive system of (usual) preschemes satisfying conditions (b) and (c) of Proposition (10.6.3), and let $\mathfrak{X}$ (resp. $\mathfrak{Y}$) be its inductive limit. By definition of the inductive limit, each sequence (f_n) of morphisms $X_n \to Y_n$ forms an inductive system that admits an *inductive limit* $f : \mathfrak{X} \to \mathfrak{Y}$, which is the unique morphism of formal preschemes that makes the diagrams

$$\begin{tikzcd} X_n \arrow[r, "f_n"] \arrow[d] & Y_n \arrow[d] \\ \mathfrak{X} \arrow[r, "f"] & \mathfrak{Y} \end{tikzcd}$$

commutate.

Proposition (10.6.9). — *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be locally Noetherian formal preschemes, and $\mathscr{J}$ (resp. $\mathscr{K}$) be a sheaf of ideals of definition for $\mathfrak{X}$ (resp. $\mathfrak{Y}$); the map $f \mapsto (f_n)$ defined in (10.6.7) is a bijection from the set of morphisms $f : \mathfrak{X} \to \mathfrak{Y}$ such that $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$ to the set of sequences (f_n) of morphisms that make the diagrams (10.6.7.1) commutate.*

PROOF. If f is the inductive limit of this sequence, then it remains to show that $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$. The statement, being local on $\mathfrak{X}$ and $\mathfrak{Y}$, can be reduced to the case where $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$ are affine, with A and B adic Noetherian rings, and with $\mathscr{J} = \mathfrak{J}^\Delta$ and $\mathscr{K} = \mathfrak{K}^\Delta$, where $\mathfrak{J}$ (resp. $\mathfrak{K}$) is an ideal of definition for A (resp. B). We then have that $X_n = \operatorname{Spec}(A_n)$ and $Y_n = \operatorname{Spec}(B_n)$, with $A_n = A/\mathfrak{J}^{n+1}$ and $B_n = B/\mathfrak{K}^{n+1}$, by Proposition (10.3.6) and (10.3.2); $f_n = ({}^a\phi_n, \widetilde{\phi_n})$, where the homomorphisms $\phi_n : B_n \to A_n$ forms a projective system, thus $f = ({}^a\phi, \widetilde{\phi})$, and so $f = ({}^a\phi, \widetilde{\phi})$, where $\phi = \varprojlim \phi_n$. The commutativity of the diagram (10.6.7.1) for $m = 0$ then gives the condition $\phi_n(\mathfrak{K}/\mathfrak{K}^{n+1}) \subset \mathfrak{J}/\mathfrak{J}^{n+1}$ for all n, so, by passing to the projective limit, we have $\phi(\mathfrak{K}) \subset \mathfrak{J}$, which implies that $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{J}$ (10.5.6, ii). □

Corollary (10.6.10). — *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be locally Noetherian formal preschemes, and $\mathscr{T}$ the largest sheaf of ideals of definition for $\mathfrak{X}$ (10.5.4).*

(i) *For every sheaf of ideals of definition $\mathscr{K}$ for $\mathfrak{Y}$ and every morphism $f : \mathfrak{X} \to \mathfrak{Y}$, we have $f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{T}$.*

(ii) *There is a canonical bijective correspondence between $\operatorname{Hom}(\mathfrak{X}, \mathfrak{Y})$ and the set of sequences (f_n) of morphisms making the diagrams (10.6.7.1) commute, where $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{T}^{n+1})$ and $Y_n = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K}^{n+1})$.*

PROOF. (ii) follows immediately from (i) and Proposition (10.6.9). To prove (i), we can reduce to the case where $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$, with A and B Noetherian, and with $\mathscr{T} = \mathfrak{T}^\Delta$ and $\mathscr{K} = \mathfrak{K}^\Delta$, where $\mathfrak{T}$ is the largest ideal of definition for A and $\mathfrak{K}$ is an ideal of definition for B. Let $f = ({}^a\phi, \widetilde{\phi})$, where $\phi : B \to A$ is a continuous homomorphism; since the elements of $\mathfrak{K}$ are topologically nilpotent (**0**, 7.1.4, ii), so too are those of $\phi(\mathfrak{K})$, and so $\phi(\mathfrak{K}) \subset \mathfrak{T}$, since $\mathfrak{T}$ is the set of topologically nilpotent elements of A (**0**, 7.1.6); hence, by Proposition (10.5.6, ii), we are done. □

Corollary (10.6.11). — *Let $\mathfrak{S}$, $\mathfrak{X}$, $\mathfrak{Y}$ be locally Noetherian formal preschemes, and $f : \mathfrak{X} \to \mathfrak{S}$ and $g : \mathfrak{Y} \to \mathfrak{S}$ the morphisms that make $\mathfrak{X}$ and $\mathfrak{Y}$ formal $\mathfrak{S}$-preschemes. Let $\mathscr{J}$ (resp. $\mathscr{K}$, $\mathscr{L}$) be a sheaf of ideals of definition for $\mathfrak{S}$ (resp. $\mathfrak{X}$, $\mathfrak{Y}$), and suppose that $f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{K}$ and $g^*(\mathscr{J})\mathscr{O}_{\mathfrak{Y}} = \mathscr{L}$; set $S_n = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{J}^{n+1})$, $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{K}^{n+1})$, and $Y_n = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{L}^{n+1})$. Then there exists a canonical bijective correspondence between $\operatorname{Hom}_{\mathfrak{S}}(\mathfrak{X}, \mathfrak{Y})$ and the set of sequences (u_n) of S_n-morphisms $u_n : X_n \to Y_n$ making the diagrams (10.6.7.1) commute.* **I** | 193

PROOF. For each $\mathfrak{S}$-morphism $u : \mathfrak{X} \to \mathfrak{Y}$, we have by definition that $f = g \circ u$, whence

$$u^*(\mathscr{L})\mathscr{O}_{\mathfrak{X}} = u^*(g^*(\mathscr{J}\mathscr{O}_{\mathfrak{Y}})\mathscr{O}_{\mathfrak{X}} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}} \subset \mathscr{K},$$

and the corollary then follows from Proposition (10.6.9). □

We note that, for $m \leqslant n$, the data of a morphism $f_n : X_n \to Y_n$ determines a unique morphism $f_m : X_m \to Y_m$ making the diagram (10.6.7.1) commutate, since we immediately see that we can reduce to the affine case; we have thus defined a map $\phi_{mn} : \operatorname{Hom}_{S_n}(X_n, Y_n) \to \operatorname{Hom}_{S_m}(X_m, Y_m)$, and the $\operatorname{Hom}_{S_n}(X_n, Y_n)$ form, with the ϕ_{mn}, a *projective system of sets*; Corollary (10.6.11) then says that there is a canonical bijection

$$\operatorname{Hom}_{\mathfrak{S}}(\mathfrak{X}, \mathfrak{Y}) \simeq \varprojlim_n \operatorname{Hom}_{S_n}(X_n, Y_n).$$

10.7. Products of formal preschemes

(10.7.1). Let $\mathfrak{S}$ be a formal prescheme; formal $\mathfrak{S}$-preschemes form a category, and we can define a notion of a *product* of formal $\mathfrak{S}$-preschemes.

Proposition (10.7.2). — *Let $\mathfrak{X} = \operatorname{Spf}(B)$ and $\mathfrak{Y} = \operatorname{Spf}(C)$ be formal affine schemes over a formal affine scheme $\mathfrak{S} = \operatorname{Spf}(A)$. Let $\mathfrak{Z} = \operatorname{Spf}(B\widehat{\otimes}_A C)$, and let p_1 and p_2 be the $\mathfrak{S}$-morphisms corresponding (10.2.2) to the canonical (continuous) A-homomorphisms $\rho : B \to B\widehat{\otimes}_A C$ and $\sigma : C \to B\widehat{\otimes}_A C$; then $(\mathfrak{Z}, p_1, p_2)$ is a product of the formal affine $\mathfrak{S}$-schemes $\mathfrak{X}$ and $\mathfrak{Y}$.*

Proof. By Proposition (10.4.6), it suffices to check that, if we associate, to each continuous A-homomorphism $\phi : B\widehat{\otimes}_A C \to D$ (where D is an admissible ring which is a topological A-algebra), the pair $(\phi \circ \rho, \phi \circ \sigma)$, then this defines a bijection

$$\operatorname{Hom}_A(B\widehat{\otimes}_A C, D) \simeq \operatorname{Hom}_A(B, D) \times \operatorname{Hom}_A(C, D),$$

which is exactly the universal property of the completed tensor product (**0**, 7.7.6). □

Proposition (10.7.3). — *Given formal $\mathfrak{S}$-preschemes $\mathfrak{X}$ and $\mathfrak{Y}$, their product $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ exists.*

Proof. The proof is similar to that of Theorem (3.2.6), replacing affine schemes (resp. affine open sets) by formal affine schemes (resp. formal affine open sets), and replacing Proposition (3.2.2) by Proposition (10.7.2). □

All the formal properties of the product of preschemes ((3.2.7) and (3.2.8), (3.3.1) and (3.3.12)) hold true without modification for the product of formal preschemes.

(10.7.4). Let $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$ be formal preschemes, and let $f : \mathfrak{X} \to \mathfrak{S}$ and $g : \mathfrak{Y} \to \mathfrak{S}$ be morphisms. Suppose that there exist, in $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$ respectively, three fundamental systems of sheaves of ideals of definitions $(\mathscr{J}_\lambda)$, $(\mathscr{K}_\lambda)$, and $(\mathscr{L}_\lambda)$, all having the same set I of indices, and such that $f^*(\mathscr{J}_\lambda)\mathscr{O}_{\mathfrak{X}} \subset \mathscr{K}_\lambda$ and $g^*(\mathscr{J}_\lambda)\mathscr{O}_{\mathfrak{Y}} \subset \mathscr{L}_\lambda$ for all λ. Set $S_\lambda = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{J}_\lambda)$, $X_\lambda = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{K}_\lambda)$, and $Y_\lambda = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{L}_\lambda)$; for $\mathscr{J}_\mu \subset \mathscr{J}_\lambda$, $\mathscr{K}_\mu \subset \mathscr{K}_\lambda$, and $\mathscr{L}_\mu \subset \mathscr{L}_\lambda$, note that S_λ (resp. X_λ, Y_λ) is a closed subprescheme of S_μ (resp. X_μ, Y_μ) that has the *same* underlying space (10.6.1). Since $S_\lambda \to S_\mu$ is a monomorphism of preschemes, we see that the products $X_\lambda \times_{S_\lambda} Y_\lambda$ and $X_\lambda \times_{S_\mu} Y_\lambda$ are identical (3.2.4), since $X_\lambda \times_{S_\mu} Y_\lambda$ can be identified with a closed subprescheme of $X_\mu \times_{S_\mu} Y_\mu$ that has the *same* underlying space (4.3.1). With this in mind, the product $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is the *inductive limit* of the usual preschemes $X_\lambda \times_{S_\lambda} Y_\lambda$: indeed, as we see in Proposition (10.6.2), we can reduce to the case where $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$ are formal affine schemes. Taking into account both Proposition (10.5.6, ii) and the hypotheses on the fundamental systems of sheaves of ideals of definition for $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$, we immediately see that our claim follows from the definition of the completed tensor product of two algebras (**0**, 7.7.1). **I** | 194

Furthermore, let $\mathfrak{Z}$ be a formal $\mathfrak{S}$-prescheme, $(\mathscr{M}_\lambda)$ a fundamental system of ideals of definition for $\mathfrak{Z}$ having I for its set of indices, and let $u : \mathfrak{Z} \to \mathfrak{X}$ and $v : \mathfrak{Z} \to \mathfrak{Y}$ be $\mathfrak{S}$-morphisms such that $u^*(\mathscr{K}_\lambda)\mathscr{O}_{\mathfrak{Z}} \subset \mathscr{M}_\lambda$ and $v^*(\mathscr{L}_\lambda)\mathscr{O}_{\mathfrak{Z}} \subset \mathscr{M}_\lambda$ for all λ. If we set $Z_\lambda = (\mathfrak{Z}, \mathscr{O}_{\mathfrak{Z}}/\mathscr{M}_\lambda)$, and if $u_\lambda : Z_\lambda \to X_\lambda$ and $v_\lambda : Z_\lambda \to Y_\lambda$ are the S_λ-morphisms corresponding to u and v (10.5.6), then we immediately have that $(u, v)_{\mathfrak{S}}$ is the inductive limit of the S_λ-morphisms $(u_\lambda, v_\lambda)_{S_\lambda}$.

The ideas of this section apply, in particular, to the case where $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$ are locally Noetherian, taking the systems consisting of the powers of a sheaf of ideals of definition (10.5.1) as the fundamental systems of sheaves of ideals of definition . However, we note that $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is not necessarily locally Noetherian (see however (10.13.5)).

10.8. Formal completion of a prescheme along a closed subset

(10.8.1). Let X be a *locally Noetherian* (usual) prescheme, and X' a closed subset of the underlying space of X; we denote by Φ the set of *coherent* sheaves of ideals $\mathscr{J}$ of $\mathscr{O}_X$ such that the support of $\mathscr{O}_X/\mathscr{J}$ is X'. The set Φ is nonempty ((5.2.1), (4.1.4), (6.1.1)); we order it by the relation $\supset$.

Lemma (10.8.2). — *The ordered set Φ is filtered; if X is Noetherian, then, for all $\mathscr{J}_0 \in \Phi$, the set of powers $\mathscr{J}_0^n$ ($n > 0$) is cofinal in Φ.*

Proof. If $\mathscr{J}_1$ and $\mathscr{J}_2$ are in Φ, and if we set $\mathscr{J} = \mathscr{J}_1 \cap \mathscr{J}_2$, then $\mathscr{J}$ is coherent since $\mathscr{O}_X$ is coherent ((6.1.1) and (**0**, 5.3.4)), and we have $\mathscr{J}_x = (\mathscr{J}_1)_x \cap (\mathscr{J}_2)_x$ for all $x \in X$, whence $\mathscr{J}_x = \mathscr{O}_x$ for $x \notin X'$, and $\mathscr{J}_x \neq \mathscr{O}_x$ for $x \in X'$, which proves that $\mathscr{J} \in \Phi$. On the other hand, if X is Noetherian and if $\mathscr{J}_0$ and $\mathscr{J}$ are in Φ, then there exists an integer $n > 0$ such that $\mathscr{J}_0^n(\mathscr{O}_X/\mathscr{J}) = 0$ (9.3.4), which implies that $\mathscr{J}_0^n \subset \mathscr{J}$. □

(10.8.3). Now let $\mathscr{F}$ be a *coherent* $\mathscr{O}_X$-module; for all $\mathscr{J} \in \Phi$, we have that $\mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J})$ is a coherent $\mathscr{O}_X$-module (9.1.1) with support contained in X', and we will usually identify it with its restriction to X'. When $\mathscr{J}$ varies over Φ, these sheaves form a *projective system* of sheaves of abelian groups.

Definition (10.8.4). — Given a closed subset X' of a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-module $\mathscr{F}$, we define the *completion of $\mathscr{F}$ along X'*, denoted by $\mathscr{F}_{/X'}$ (or $\widehat{\mathscr{F}}$ when there is little chance of confusion), to be the restriction to X' of the sheaf $\varprojlim_\Phi(\mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}))$; we say that its sections over X' are the *formal sections of $\mathscr{F}$ along X'*. I | 195

It is immediate that, for every open $U \subset X$, we have $(\mathscr{F}|U)_{/(U\cap X')} = (\mathscr{F}_{/X'})|(U \cap X')$.

By passing to the projective limit, it is clear that $(\mathscr{O}_X)_{/X'}$ is a sheaf of rings, and that $\mathscr{F}_{/X'}$ can be considered as an $(\mathscr{O}_X)_{/X'}$-module. In addition, since there exists a basis for the topology of X' consisting of quasi-compact open sets, we can consider $(\mathscr{O}_X)_{/X'}$ (resp. $\mathscr{F}_{/X'}$) as a *sheaf of topological rings* (resp. of *topological groups*), the projective limit of the *pseudo-discrete* sheaves of rings (resp. groups) $\mathscr{O}_X/\mathscr{F}$ (resp. $\mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{F}) = \mathscr{F}/\mathscr{J}\mathscr{F}$), and, by passing to the projective limit, $\mathscr{F}_{/X'}$ then becomes a *topological $(\mathscr{O}_X)_{/X'}$-module* ((**0**, 3.8.1) and (**0**, 3.8.2)); recall that, for every *quasi-compact* open $U \subset X$, $\Gamma(U \cap X', (\mathscr{O}_X)_{/X'})$ (resp. $\Gamma(U \cap X', \mathscr{F}_{/X'})$) is then the projective limit of the discrete rings (resp. groups) $\Gamma(U, \mathscr{O}_X/\mathscr{J})$ (resp. $\Gamma(U, \mathscr{F}/\mathscr{J}\mathscr{F})$).

Now, if $u : \mathscr{F} \to \mathscr{G}$ is a homomorphism of $\mathscr{O}_X$-modules, then there are canonically induced homomorphisms $u_{\mathscr{J}} : \mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}) \to \mathscr{G} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J})$ for all $\mathscr{J} \in \Phi$, and these homomorphisms form a projective system. By passing to the projective limit and restricting to X', these give a continuous $(\mathscr{O}_X)_{/X'}$-homomorphism $\mathscr{F}_{/X'} \to \mathscr{G}_{/X'}$, denoted $u_{/X'}$ or $\widehat{u}$, and called the *completion* of the homomorphism u along X'. It is clear that, if $v : \mathscr{G} \to \mathscr{H}$ is a second homomorphism of $\mathscr{O}_X$-modules, then we have $(v \circ u)_{/X'} = (v_{/X'}) \circ (u_{/X'})$, hence $\mathscr{F}_{/X'}$ is a *covariant additive functor* in $\mathscr{F}$ from the category of coherent $\mathscr{O}_X$-modules to the category of topological $(\mathscr{O}_X)_{/X'}$-modules.

Proposition (10.8.5). — *The support of $(\mathscr{O}_X)_{/X'}$ is X'; the topologically ringed space $(X', (\mathscr{O}_X)_{/X'})$ is a locally Noetherian formal prescheme, and, if $\mathscr{J} \in \Phi$, then $\mathscr{J}_{/X'}$ is a sheaf of ideals of definition for this formal prescheme. If $X = \operatorname{Spec}(A)$ is an affine scheme with Noetherian ring A, $\mathscr{J} = \widetilde{\mathfrak{J}}$ for some ideal $\mathfrak{J}$ of A, and $X' = V(\mathfrak{J})$, then $(X', (\mathscr{O}_X)_{/X'})$ is canonically identified with $\operatorname{Spf}(\widehat{A})$, where $\widehat{A}$ is the separated completion of A with respect to the $\mathfrak{J}$-preadic topology.*

Proof. We can evidently reduce to proving the latter claim. We know (**0**, 7.3.3) that the separated completion $\widehat{\mathfrak{J}}$ of $\mathfrak{J}$ with respect to the $\mathfrak{J}$-preadic topology can be identified with the ideal $\mathfrak{J}\widehat{A}$ of $\widehat{A}$, where $\widehat{A}$ is the Noetherian $\widehat{\mathfrak{J}}$-adic ring such that $\widehat{A}/\widehat{\mathfrak{J}}^n = A/\mathfrak{J}^n$ (**0**, 7.2.6). This latter equation shows that the open prime ideals of $\widehat{A}$ are the ideals $\widehat{\mathfrak{p}} = \mathfrak{p}\widehat{\mathfrak{J}}$, where $\mathfrak{p}$ is a prime ideal of A containing $\mathfrak{J}$, and that we have $\widehat{\mathfrak{p}} \cap A = \mathfrak{p}$, and hence $\operatorname{Spf}(\widehat{A}) = X'$. Since $\mathscr{O}_X/\mathscr{J}^n = (A/\mathfrak{J}^n)^\sim$, the proposition follows immediately from the definitions. □

We say that the formal prescheme defined above is the *completion of X along X'*, and we denote it by $X_{/X'}$ or $\widehat{X}$ when there is little chance of confusion. When we take $X' = X$, we can set $\mathscr{J} = 0$, and we thus have $X_{/X} = X$.

It is clear that, if U is a subprescheme induced on an open subset of X, then $U_{/(U\cap X')}$ is canonically identified with the formal subprescheme induced on $X_{/X'}$ by the open subset $U\cap X'$ of X'.

Corollary (10.8.6). — *The (usual) prescheme $\widehat{X}_{\mathrm{red}}$ is the unique reduced subprescheme of X having X' as its underlying space (5.2.1). For $\widehat{X}$ to be Noetherian, it is necessary and sufficient for $\widehat{X}_{\mathrm{red}}$ to be Noetherian, and it suffices that X be Noetherian.*

Proof. Since $\widehat{X}_{\mathrm{red}}$ is determined locally (10.5.4), we can assume that X is an affine scheme of some Noetherian ring A; with the notation of Proposition (10.8.5), the ideal $\mathfrak{T}$ of topologically nilpotent elements of $\widehat{A}$ is the inverse image under the canonical map $\widehat{A}\to\widehat{A}/\widehat{\mathfrak{J}} = A/\mathfrak{J}$ of the nilradical of $A/\mathfrak{J}$ (**0**, 7.1.3), so $\widehat{A}/\mathfrak{T}$ is isomorphic to the quotient of $A/\mathfrak{J}$ by its nilradical. The first claim then follows from Propositions (10.5.4) and (5.1.1). If $\widehat{X}_{\mathrm{red}}$ is Noetherian, then so too is its underlying space X', and so the $X'_n = \mathrm{Spec}(\mathscr{O}_X/\mathscr{J}^n)$ are Noetherian (6.1.2), and thus so too is $\widehat{X}$ (10.6.4); the converse is immediate, by Proposition (6.1.2). □

(10.8.7). The canonical homomorphisms $\mathscr{O}_X\to\mathscr{O}_X/\mathscr{J}$ (for $\mathscr{J}\in\Phi$) form a projective system, and give, by passing to the projective limit, a homomorphism of sheaves of rings $\theta:\mathscr{O}_X\to\psi_*((\mathscr{O}_X)_{/X'}) = \varprojlim_{\Phi}(\mathscr{O}_X/\mathscr{J})$, where ψ denotes the canonical injection $X'\to X$ of the underlying spaces. We denote by i (or i_X) the morphism (said to be *canonical*)

$$(\psi,\theta): X_{/X'}\longrightarrow X$$

of ringed spaces.

By taking tensor products, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphisms $\mathscr{O}_X\to\mathscr{O}_X/\mathscr{J}$ give homomorphisms $\mathscr{F}\to\mathscr{F}\otimes_{\mathscr{O}_X}(\mathscr{O}_X/\mathscr{J})$ of $\mathscr{O}_X$-modules which form a projective system, and thus give, by passing to the projective limit, a canonical functorial homomorphism $\gamma:\mathscr{F}\to\psi_*(\mathscr{F}_{/X'})$ of $\mathscr{O}_X$-modules.

Proposition (10.8.8). —

(i) *The functor $\mathscr{F}_{/X'}$ (in $\mathscr{F}$) is exact.*
(ii) *The functorial homomorphism $\gamma^\sharp: i^*(\mathscr{F})\to\mathscr{F}_{/X'}$ of $(\mathscr{O}_X)_{/X'}$-modules is an isomorphism.*

Proof.

(i) It suffices to prove that, if $0\to\mathscr{F}'\to\mathscr{F}\to\mathscr{F}''\to 0$ is an exact sequence of coherent $\mathscr{O}_X$-modules, and if U is an affine open subset of X with Noetherian ring A, then the sequence

$$0\longrightarrow\Gamma(U\cap X',\mathscr{F}'_{/X'})\longrightarrow\Gamma(U\cap X',\mathscr{F}_{/X'})\longrightarrow\Gamma(U\cap X',\mathscr{F}''_{/X'})\longrightarrow 0$$

is exact. We have that $\mathscr{F}|U=\widetilde{M}$, $\mathscr{F}'|U=\widetilde{M'}$, and $\mathscr{F}''|U=\widetilde{M''}$, where M, M', and M'' are three A-modules of finite type such that the sequence $0\to M'\to M\to M''\to 0$ is exact ((1.5.1) and (1.3.11)); let $\mathscr{J}\in\Phi$, and let $\mathfrak{J}$ be an ideal of A such that $\mathscr{J}|U=\widetilde{\mathfrak{J}}$. We then have

$$\Gamma(U\cap X',\mathscr{F}\otimes_{\mathscr{O}_X}\mathscr{O}_X/\mathscr{J}^n) = M\otimes_A(A/\mathfrak{J}^n)$$

(1.3.12); so, by definition of the projective limit, we have

$$\Gamma(U\cap X',\mathscr{F}_{/X'}) = \varprojlim_n(M\otimes_A(A/\mathfrak{J}^n)) = \widehat{M},$$

the separated completion of M with respect to the $\mathfrak{J}$-preadic topology, and similarly

$$\Gamma(U\cap X',\mathscr{F}'_{/X'}) = \widehat{M'},\ \Gamma(U\cap X',\mathscr{F}''_{/X'}) = \widehat{M''};$$

our claim then follows, since A is Noetherian, and since the functor $\widehat{M}$ in M is exact on the category of A-modules of finite type (**0**, 7.3.3). **I** | 197

(ii) The question is local, so we can assume that we have an exact sequence $\mathscr{O}_X^m \to \mathscr{O}_X^n \to \mathscr{F} \to 0$ (**0**, 5.3.2); since $\gamma^\sharp$ is functorial, and the functors $i^*(\mathscr{F})$ and $\mathscr{F}_{/X'}$ are right exact (by (i) and (**0**, 4.3.1)), we have the commutative diagram

$$\begin{tikzcd} i^*(\mathscr{O}_X^m) \arrow[r] \arrow[d, "\gamma^\sharp"'] & i^*(\mathscr{O}_X^n) \arrow[r] \arrow[d, "\gamma^\sharp"'] & i^*(\mathscr{F}) \arrow[r] \arrow[d, "\gamma^\sharp"'] & 0 \\ (\mathscr{O}_X^m)_{/X'} \arrow[r] & (\mathscr{O}_X^n)_{/X'} \arrow[r] & \mathscr{F}_{/X'} \arrow[r] & 0 \end{tikzcd} \tag{10.8.8.1}$$

whose rows are exact. Furthermore, the functors $i^*(\mathscr{F})$ and $\mathscr{F}_{/X'}$ commute with finite direct sums ((**0**, 3.2.6) and (**0**, 4.3.2)), and we thus reduce to proving our claim for $\mathscr{F} = \mathscr{O}_X$. We have $i^*(\mathscr{O}_X) = (\mathscr{O}_X)_{/X'} = \mathscr{O}_{\widehat{X}}$ (**0**, 4.3.4), and that $\gamma^\sharp$ is a homomorphism of $\mathscr{O}_{\widehat{X}}$*-modules*; so it suffices to check that $\gamma^\sharp$ sends the unit section of $\mathscr{O}_{\widehat{X}}$ over an open subset of X' to itself, which is immediate, and shows, in this case, that $\gamma^\sharp$ is the identity. □

Corollary (10.8.9). — *The morphism $i : X_{/X'} \to X$ of ringed spaces is flat.*

Proof. This follows from (**0**, 6.7.3) and Proposition (10.8.8, i). □

Corollary (10.8.10). — *If $\mathscr{F}$ and $\mathscr{G}$ are coherent $\mathscr{O}_X$-modules, then there exist canonical functorial (in $\mathscr{F}$ and $\mathscr{G}$) isomorphisms*

$$(\mathscr{F}_{/X'}) \otimes_{(\mathscr{O}_X)_{/X'}} (\mathscr{G}_{/X'}) \simeq (\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})_{/X'}, \tag{10.8.10.1}$$

$$(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G}))_{/X'} \simeq \mathscr{H}om_{(\mathscr{O}_X)_{/X'}}(\mathscr{F}_{/X'},\mathscr{G}_{/X'}). \tag{10.8.10.2}$$

Proof. This follows from the canonical identification of $i^*(\mathscr{F})$ with $\mathscr{F}_{/X'}$; the existence of the first isomorphism is then a result which holds for all morphisms of ringed spaces (**0**, 4.3.3.1), and the second is a result which holds for all flat morphisms (**0**, 6.7.6), by Corollary (10.8.9). □

Proposition (10.8.11). — *For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\Gamma(X,\mathscr{F}) \to \Gamma(X',\mathscr{F}_{/X'})$ induced by $\mathscr{F} \to \mathscr{F}_{/X'}$ has kernel consisting of the zero sections in some neighbourhood of X'.*

Proof. It follows from the definition of $\mathscr{F}_{/X'}$ that the canonical image of such a section is zero. Conversely, if $s \in \Gamma(X,\mathscr{F})$ has a zero image in $\Gamma(X',\mathscr{F}_{/X'})$, then it suffices to see that every $x \in X'$ admits a neighbourhood in X in which s is zero, and we can thus reduce to the case where $X = \operatorname{Spec}(A)$ is affine, A Noetherian, $X' = V(\mathfrak{J})$ for some ideal $\mathfrak{J}$ of A, and $\mathscr{F} = \widetilde{M}$ for some A-module M of finite type. Then $\Gamma(X',\mathscr{F}_{/X'})$ is the separated completion $\widehat{M}$ of M for the $\mathfrak{J}$-preadic topology, and the homomorphism $\Gamma(X,\mathscr{F}) \to \Gamma(X',\mathscr{F}_{/X'})$ is the canonical homomorphism $M \to \widehat{M}$. We know (**0**, 7.3.7) that the kernel of this homomorphism is the set of the $z \in M$ killed by an element of $1+\mathfrak{J}$. So we have $(1+f)s = 0$ for some $f \in \mathfrak{J}$; for every $x \in X'$ we have $(1_x + f_x)s_x = 0$, and, since $1_x + f_x$ is invertible in $\mathscr{O}_x$ ($\mathfrak{J}_x\mathscr{O}_x$ being contained in the maximal ideal of $\mathscr{O}_x$), we have $s_x = 0$, which proves the proposition. □

Corollary (10.8.12). — *The support of $\mathscr{F}_{/X'}$ is equal to* $\operatorname{Supp}(\mathscr{F}) \cap X'$.

Proof. It is clear that $\mathscr{F}_{/X'}$ is an $(\mathscr{O}_X)_{/X'}$-module of finite type ((10.8.8, ii) and (**0**, 5.2.4)), so **I** | 198
its support is closed (**0**, 5.2.2) and evidently contained in $\operatorname{Supp}(\mathscr{F}) \cap X'$. To show that it is equal to the latter set, we immediately reduce to proving that the equation $\Gamma(X',\mathscr{F}_{/X'}) = 0$ implies that $\operatorname{Supp}(\mathscr{F}) \cap X' = \varnothing$; this follows from Proposition (10.8.11) and Theorem (1.4.1). □

Corollary (10.8.13). — *Let $u : \mathscr{F} \to \mathscr{G}$ be a homomorphism of coherent $\mathscr{O}_X$-modules. For $u_{/X'} : \mathscr{F}_{/X'} \to \mathscr{G}_{/X'}$ to be zero, it is necessary and sufficient for u to be zero on a neighbourhood of X'.*

Proof. By Proposition (10.8.8, ii), $u_{/X'}$ can be identified with $i^*(u)$, so, if we consider u as a section over X of the sheaf $\mathscr{H} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$, then $u_{/X'}$ is the section over X' of $i^*(\mathscr{H}) = \mathscr{H}_{/X'}$ to which it canonically corresponds ((10.8.10.2) and (**0**, 4.4.6)). It thus suffices to apply Proposition (10.8.11) to the coherent $\mathscr{O}_X$-module $\mathscr{H}$. □

Corollary (10.8.14). — *Let $u : \mathscr{F} \to \mathscr{G}$ be a homomorphism of coherent $\mathscr{O}_X$-modules. For $u_{/X'}$ to be a monomorphism (resp. an epimorphism), it is necessary and sufficient for u to be a monomorphism (resp. an epimorphism) on a neighbourhood of X'.*

PROOF. Let $\mathscr{P}$ and $\mathscr{N}$ be the cokernel and kernel (respectively) of u, so that we have the exact sequence $0 \to \mathscr{N} \xrightarrow{v} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{w} \mathscr{P} \to 0$, hence (10.8.8, i) the exact sequence

$$0 \longrightarrow \mathscr{N}_{/X'} \xrightarrow{v_{/X'}} \mathscr{F}_{/X'} \xrightarrow{u_{/X'}} \mathscr{G}_{/X'} \xrightarrow{w_{/X'}} \mathscr{P}_{/X'} \longrightarrow 0.$$

If $u_{/X'}$ is a monomorphism (resp. an epimorphism), then we have $v_{/X'} = 0$ (resp. $w_{/X'} = 0$), so there exists a neighbourhood of X' on which $v = 0$ (resp. $w = 0$) by Corollary (10.8.13). □

10.9. Extension of morphisms to completions

(10.9.1). Let X and Y be locally Noetherian (usual) preschemes, $f : X \to Y$ a morphism, and X' (resp. Y') a closed subset of the underlying space X (resp. Y) such that $f(X') \subset Y'$. Let $\mathscr{J}$ (resp. $\mathscr{K}$) be a sheaf of ideals of $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$) such that the support of $\mathscr{O}_X/\mathscr{J}$ (resp. $\mathscr{O}_Y/\mathscr{K}$) is X' (resp. Y') and $f^*(\mathscr{K})\mathscr{O}_X \subset \mathscr{J}$; we note that there always exist such sheaves of ideals, since, for example, we can take $\mathscr{J}$ to be the largest sheaf of ideals of $\mathscr{O}_X$ defining a subprescheme of X with underlying space X' (5.2.1), and the hypothesis $f(X') \subset Y'$ implies that $f^*(\mathscr{K})\mathscr{O}_X \subset \mathscr{J}$ (5.2.4). For every integer $n > 0$ we have $f^*(\mathscr{K}^n)\mathscr{O}_X \subset \mathscr{J}^n$ (**0**, 4.3.5); as a result (4.4.6), if we set $X'_n = (X', \mathscr{O}_X/\mathscr{J}^{n+1})$ and $Y'_n = (Y', \mathscr{O}_Y/\mathscr{K}^{n+1})$, then f induces a morphism $f_n : X'_n \to Y'_n$, and it is immediate that the f_n form an inductive system. We denote its inductive limit (10.6.8) by $\widehat{f} : X_{/X'} \to Y_{/Y'}$, and we say (by abuse of language) that $\widehat{f}$ is the *extension of f to the completions of X and Y along X' and Y'*. It can be checked immediately that this morphism does not depend on the choice of sheaves of ideals $\mathscr{J}$ and $\mathscr{K}$ satisfying the above conditions. It suffices to consider the case where X and Y are Noetherian affine schemes with rings A and B (respectively); then $\mathscr{J} = \widetilde{\mathfrak{J}}$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$, where $\mathfrak{J}$ (resp. $\mathfrak{K}$) is an ideal of A (resp. B), f corresponds to a ring homomorphism $\phi : B \to A$ such that $\phi(\mathfrak{K}) \subset \mathfrak{J}$ ((4.4.6) and (1.7.4)); *$\widehat{f}$ is then the morphism corresponding* (10.2.2) *to the continuous homomorphism $\widehat{\phi} : \widehat{B} \to \widehat{A}$*, where $\widehat{A}$ (resp. $\widehat{B}$) is the separated completion of A (resp. B) with respect to the $\mathfrak{J}$-preadic (resp. $\mathfrak{K}$-preadic) topology (10.6.8); we know that, if we replace $\mathscr{J}$ by another sheaf of ideals $\mathscr{J}' = \widetilde{\mathfrak{J}'}$ **I** | 199
such that the support of $\mathscr{O}_X/\mathscr{J}'$ is X', then the $\mathfrak{J}$-preadic and $\mathfrak{J}'$-preadic topologies on A are the same (10.8.2).

We note that, by definition, the continuous map $X' \to Y'$ of the underlying spaces of $X_{/X'}$ and $Y_{/Y'}$ corresponding to $\widehat{f}$ is exactly the restriction to X' of f.

(10.9.2). It follows immediately from the above definition that the diagram of morphisms of ringed spaces

$$\begin{tikzcd} \widehat{X} \arrow[r, "\widehat{f}"] \arrow[d, "i_X"'] & \widehat{Y} \arrow[d, "i_Y"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

commutes, with the vertical arrows being the canonical morphisms (10.8.7).

(10.9.3). Let Z be a third prescheme, $g : Y \to Z$ a morphism, and Z' a closed subset of Z such that $g(Y') \subset Z'$. If $\widehat{g}$ denotes the completion of the morphism g along Y' and Z', then it immediately follows from (10.9.1) that we have $(g \circ f)^\wedge = \widehat{g} \circ \widehat{f}$.

Proposition (10.9.4). — *Let X and Y be locally Noetherian S-preschemes, with Y of finite type over S. Let f and g be S-morphisms from X to Y such that $f(X') \subset Y'$ and $g(X') \subset Y'$. For $\widehat{f} = \widehat{g}$ to hold, it is necessary and sufficient for f and g to coincide on a neighbourhood of X'.*

PROOF. The condition is evidently sufficient (even without the finiteness hypothesis on Y). To see that it is necessary, we remark first that the hypothesis $\widehat{f} = \widehat{g}$ implies that $f(x) = g(x)$ for all $x \in X'$. Also, since the questions is local, we can assume that X and Y are affine open neighbourhoods of x and $y = f(x) = g(x)$ respectively (with Noetherian rings), that S is affine, and that $\Gamma(Y, \mathscr{O}_Y)$ is a $\Gamma(S, \mathscr{O}_S)$-algebra of finite type (6.3.3). Then f and g correspond to $\Gamma(S, \mathscr{O}_S)$-homomorphisms ρ and σ (respectively) from $\Gamma(Y, \mathscr{O}_Y)$ to $\Gamma(X, \mathscr{O}_X)$ (1.7.3), and, by hypothesis, the extensions by continuity of these homomorphisms to the separated completion of $\Gamma(Y, \mathscr{O}_Y)$ are the same. We conclude from Proposition (10.8.11) that, for every section $s \in \Gamma(Y, \mathscr{O}_Y)$, the sections $\rho(s)$ and $\sigma(s)$ coincide on a neighbourhood (depending on s) of X'; since $\Gamma(Y, \mathscr{O}_Y)$ is an algebra of finite type over $\Gamma(S, \mathscr{O}_S)$, we have that there exists a neighbourhood V of X' such that $\rho(s)$ and $\sigma(s)$ coincide on V for *every* section $s \in \Gamma(Y, \mathscr{O}_Y)$. If $h \in \Gamma(X, \mathscr{O}_X)$ is such that $D(h)$ is a neighbourhood of x contained in V, then we conclude from the above and from Theorem (1.4.1, d) that f and g coincide on $D(h)$. □

Proposition (10.9.5). — *Under the hypotheses of (10.9.1), for every coherent $\mathscr{O}_Y$-module $\mathscr{G}$, there exists a canonical functorial isomorphism of $(\mathscr{O}_X)_{/X'}$-modules*

$$(f^*(\mathscr{G}))_{/X'} \simeq \widehat{f}^*(\mathscr{G}_{/Y'}).$$

PROOF. If we canonically identify $(f^*(\mathscr{G}))_{/X'}$ with $i_X^*(f^*(\mathscr{G}))$, and $\widehat{f}^*(\mathscr{G}_{/Y'})$ with $\widehat{f}^*(i_Y^*(\mathscr{G}))$ (10.8.8), then the proposition follows immediately from the commutativity of the diagram in (10.9.2). □

(10.9.6). Now let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, and let $\mathscr{G}$ be a coherent $\mathscr{O}_Y$-module. If $u : \mathscr{G} \to \mathscr{F}$ is an f-morphism, then it corresponds to an $\mathscr{O}_X$-homomorphism $u^\sharp : f^*(\mathscr{G}) \to \mathscr{F}$, thus, by completion, to a continuous $(\mathscr{O}_X)_{/X'}$-homomorphism $(u^\sharp)_{/X'} : (f^*(\mathscr{G}))_{/X'} \to \mathscr{F}_{/X'}$, and, by Proposition (10.9.5), there exists a unique $\widehat{f}$-morphism $v : \mathscr{G}_{/Y'} \to \mathscr{F}_{/X'}$ such that $v^\sharp = (u^\sharp)_{/X'}$. If we consider the triples $(\mathscr{F}, X, X')$ ($\mathscr{F}$ being a coherent $\mathscr{O}_X$-module, and X' a closed subset of X) as a *category*, with the morphisms $(\mathscr{F}, X, X') \to (\mathscr{G}, Y, Y')$ consisting of a morphism of preschemes $f : X \to Y$ such that $f(X') \subset Y'$ and an f-morphism $u : \mathscr{G} \to \mathscr{F}$, then we can say that $(X_{/X'}, \mathscr{F}_{/X'})$ is a *functor* in $(\mathscr{F}, X, X')$ with values in the category of pairs $(\mathfrak{Z}, \mathscr{H})$ consisting of a locally Noetherian formal prescheme $\mathfrak{Z}$ and an $\mathscr{O}_{\mathfrak{Z}}$-module $\mathscr{H}$, with the morphisms of the latter category being the pairs consisting of a morphism g of formal preschemes and a g-morphism. **I** | 200

Proposition (10.9.7). — *Let S, X, and Y be locally Noetherian preschemes, $g : X \to S$ and $h : Y \to S$ morphisms, S' a closed subset of S, and X' (resp. Y') a closed subset of X (resp. Y) such that $g(X') \subset S'$ (resp. $h(Y') \subset S'$); let $Z = X \times_S Y$; suppose that Z is locally Noetherian, and let $Z' = p^{-1}(X') \cap q^{-1}(Y')$, where p and q are the projections of $X \times_S Y$. With these conditions, the completion $Z_{/Z'}$ can be identified with the product $(X_{/X'}) \times_{S_{/S'}} (Y_{/Y'})$ of formal $S_{/S'}$-preschemes, where the structure morphisms are identified with $\widehat{g}$ and $\widehat{h}$, and the projections with $\widehat{p}$ and $\widehat{q}$.*

PROOF. It is immediate that the question is local for S, X, and Y, and we thus reduce to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(C)$, $S' = V(\mathfrak{J})$, $X' = V(\mathfrak{K})$, and $Y' = V(\mathfrak{L})$, with $\mathfrak{J}$, $\mathfrak{K}$, and $\mathfrak{L}$ ideals such that $\phi(\mathfrak{J}) \subset \mathfrak{K}$ and $\psi(\mathfrak{J}) \subset \mathfrak{L}$, where we denote by ϕ and ψ the homomorphisms $A \to B$ and $A \to C$ which correspond to g and h (respectively). We know that $Z = \operatorname{Spec}(B \otimes_A C)$ and that $Z' = V(\mathfrak{M})$, where $\mathfrak{M}$ is the ideal $\operatorname{Im}(\mathfrak{K} \otimes_A C) + \operatorname{Im}(B \otimes_A \mathfrak{L})$. The conclusion follows (10.7.2) from the fact that the completed tensor product $(\widehat{B} \otimes_{\widehat{A}} \widehat{C})^\wedge$ (where $\widehat{A}$, $\widehat{B}$, and $\widehat{C}$ are, respectively, the separated completions of A, B, and C with respect to the $\mathfrak{J}$-, $\mathfrak{K}$-, and $\mathfrak{L}$-preadic topologies) is the separated completion of the tensor product $B \otimes_A C$ with respect to the $\mathfrak{M}$-preadic topology (**0**, 7.7.2). □

In addition, we note that, if T is a locally Noetherian S-prescheme, $u : T \to X$ and $v : T \to Y$ both S-morphisms, and T' a closed subset of T such that $u(T') \subset X'$ and $v(T') \subset Y'$, then the extension to the completion $((u, v)_S)^\wedge$ can be identified with $(\widehat{u}, \widehat{v})_{S_{/S'}}$.

Corollary (10.9.8). — *Let X and Y be locally Noetherian S-preschemes such that $X \times_S Y$ is locally Noetherian; let S' be a closed subset of S, and X' (resp. Y') a closed subset of X (resp. Y) whose image in S is contained in S'. For every S-morphism $f : X \to Y$ such that $f(X') \subset Y'$, the graph morphism $\Gamma_{\widehat{f}}$ can be identified with the extension $(\Gamma_f)^\wedge$ of the graph morphism of f.*

Corollary (10.9.9). — *Let X and Y be locally Noetherian preschemes, $f : X \to Y$ a morphism, Y' a closed subset of Y, and $X' = f^{-1}(Y')$. Then the prescheme $X_{/X'}$ can be identified, by the commutative diagram*

$$\begin{tikzcd} X & X_{/X'} \arrow[l] \arrow[d, "\widehat{f}"] \\ Y & Y_{/Y'} \arrow[l] \arrow[from=1-1, to=2-1, "f"'] \end{tikzcd}$$

with the product $X \times_Y (Y_{/Y'})$ of formal preschemes.

Proof. It suffices to apply Proposition (10.9.7), replacing S and S' by Y, and X and X' by X. □

Remark (10.9.10). — If S is the sum $X_1 \sqcup X_2$ (3.1), X' the union $X_1' \cup X_2'$, where X_i' is a closed subset of X_i ($i = 1, 2$), then we have $X_{/X'} = X_{1/X_1'} \sqcup X_{2/X_2'}$. I | 201

10.10. Application to coherent sheaves on formal affine schemes

(10.10.1). In this paragraph, A denotes an *adic Noetherian ring*, and $\mathfrak{J}$ an ideal of definition for A. Let $X = \operatorname{Spec}(A)$, and $\mathfrak{X} = \operatorname{Spf}(A)$, which can be identified with the closed subset $V(\mathfrak{J})$ of X (10.1.2). In addition, Definitions (10.1.2) and (10.8.4) show that the *formal affine scheme* $\mathfrak{X}$ is identical the completion $X_{/\mathfrak{X}}$ of the affine scheme X along the closed subset $\mathfrak{X}$ of its underling space. To every coherent $\mathscr{O}_X$-module $\mathscr{F}$, corresponds an $\mathscr{O}_{\mathfrak{X}}$-module of finite type $\mathscr{F}_{/\mathfrak{X}}$, which is a sheaf of topological modules over the sheaf of topological rings $\mathscr{O}_{\mathfrak{X}}$. Every coherent $\mathscr{O}_X$-module $\mathscr{F}$ is of the form $\widetilde{M}$, where M is an A-module of finite type (1.5.1); we set $(\widetilde{M})_{/X} = M^\Delta$. In addition, if $u : M \to N$ is an A-homomorphism of A-modules of finite type, then it corresponds to a homomorphism $\widetilde{u} : \widetilde{M} \to \widetilde{N}$, and, as a result, to a continuous homomorphism $\widetilde{u}_{/X'} : (\widetilde{M})_{/X'} \to (\widetilde{N})_{/X'}$, which we denote by u^Δ. It is immediate that $(v \circ u)^\Delta = v^\Delta \circ u^\Delta$; we have thus defined a *covariant additive functor* M^Δ from the category of A-modules of finite type to the category of $\mathscr{O}_{\mathfrak{X}}$-modules of finite type. When A is a *discrete* ring, we have $M^\Delta = \widetilde{M}$.

Proposition (10.10.2). —

(i) *M^Δ is an exact functor in M, and there exists a canonical functorial isomorphism of A-modules $\Gamma(\mathfrak{X}, M^\Delta) \simeq M$.*

(ii) *If M and N are A-modules of finite type, then there exist canonical functorial isomorphisms*

(10.10.2.1) $$(M \otimes_A N)^\Delta \simeq M^\Delta \otimes_{\mathscr{O}_{\mathfrak{X}}} N^\Delta,$$

(10.10.2.2) $$(\operatorname{Hom}_A(M, N))^\Delta \simeq \mathscr{H}\!om_{\mathscr{O}_{\mathfrak{X}}}(M^\Delta, N^\Delta).$$

(iii) *The map $u \mapsto u^\Delta$ is a functorial isomorphism*

(10.10.2.3) $$\operatorname{Hom}_A(M, N) \simeq \operatorname{Hom}_{\mathscr{O}_X}(M^\Delta, N^\Delta).$$

Proof. The exactness of M^Δ follows from the exactness of the functors $\widetilde{M}$ (1.3.5) and $\mathscr{F}_{/X'}$ (10.8.8). By definition, $\Gamma(X, M^\Delta)$ is the separated completion of the A-module $\Gamma(X, \widetilde{M}) = M$ with respect to the $\mathfrak{J}$-preadic topology; but, since A is complete and M is of finite type, we know (**0**, 7.3.6) that M is separated and complete, which proves (i). The isomorphism (10.10.2.1) (resp. (10.10.2.2)) comes from the composition of the isomorphisms (1.3.12, i) and (10.8.10.1) (resp. (1.3.12, ii) and (10.8.10.2)). Finally, since $\operatorname{Hom}_A(M, N)$ is an A-module of finite type, we can apply (i), which identifies $\Gamma(\mathfrak{X}, (\operatorname{Hom}_A(M, N))^\Delta)$ with $\operatorname{Hom}_A(M, N)$, and we can use (10.10.2.2), which proves that the homomorphism (10.10.2.3) is an isomorphism. □

We deduce from Proposition (10.10.2) a series of results analogous to those of Theorem (1.3.7) and Corollary (1.3.12), whose formulation we leave to the reader.

We note that the exactness property of M^Δ, applied to the exact sequence $0 \to \mathfrak{J} \to A \to A/\mathfrak{J} \to 0$, shows that the sheaf of ideals of $\mathscr{O}_{\mathfrak{X}}$ denoted here by $\mathfrak{J}^\Delta$ coincides with the one denoted also by $\mathfrak{J}^\Delta$ in (10.3.1), by (10.3.2). I | 202

Proposition (10.10.3). — *Under the hypotheses of (10.10.1), $\mathscr{O}_{\mathfrak{X}}$ is a coherent sheaf of rings.*

PROOF. If $f \in A$, then we know that $A_{\{f\}}$ is an adic Noetherian ring (**0**, 7.6.11), and, since the question is local, we reduce (10.1.4) to proving that the kernel of the homomorphism $v : \mathscr{O}_{\mathfrak{X}}^n \to \mathscr{O}_{\mathfrak{X}}$ is an $\mathscr{O}_{\mathfrak{X}}$-module of finite type. We then have $v = u^\Delta$, where u is an A-homomorphism $A^n \to A$ (10.10.2); since A is Noetherian, the kernel of u is of finite type, or, equivalently, we have a homomorphism $A^m \xrightarrow{w} A^n$ such that the sequence $A^m \xrightarrow{w} A^n \xrightarrow{u} A$ is exact. We conclude (10.10.2) that the sequence $\mathscr{O}_{\mathfrak{X}}^m \xrightarrow{w^\Delta} \mathscr{O}_{\mathfrak{X}}^n \xrightarrow{v} \mathscr{O}_{\mathfrak{X}}$ is exact, which proves that the kernel of v is of finite type. □

(10.10.4). With the above notation, set $A_n = A/\mathfrak{J}^{n+1}$, and let S_n be the affine scheme $\operatorname{Spec}(A_n) = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$, with $\mathscr{J} = \mathfrak{J}^\Delta$ the sheaf of ideals of definition for $\mathscr{O}_{\mathfrak{X}}$ corresponding to the ideal $\mathfrak{J}$. Let u_{mn} be the morphism of preschemes $X_m \to X_n$ corresponding to the canonical homomorphism $A_n \to A_m$ for $m \leqslant n$; the formal scheme $\mathfrak{X}$ is the inductive limit of the X_n with respect to the u_{mn} (10.6.3).

Proposition (10.10.5). — *Under the hypothesis of (10.10.1), let $\mathscr{F}$ be an $\mathscr{O}_{\mathfrak{X}}$-module. The following conditions are equivalent:*

(a) *$\mathscr{F}$ is a coherent $\mathscr{O}_{\mathfrak{X}}$-module;*
(b) *$\mathscr{F}$ is isomorphic to the projective limit (10.6.6) of a sequence $(\mathscr{F}_n)$ of coherent $\mathscr{O}_{X_n}$-modules such that $u_{mn}^*(\mathscr{F}_n) = \mathscr{F}_m$; and*
(c) *there exists an A-module M of finite type (determined up to canonical isomorphism by Proposition (10.10.2, i)) such that $\mathscr{F}$ is isomorphic to M^Δ.*

PROOF. We first show that (b) implies (c). We have $\mathscr{F}_n = \widetilde{M_n}$, where M_n is an A_n-module of finite type, and the hypotheses imply that $M_m = M_n \otimes_{A_n} A_m$ for $m \leqslant n$ (1.6.5); the M_n thus form a projective system for the canonical di-homomorphisms $M_n \to M_m$ ($m \leqslant n$), and it follows immediately from the definition of the A_n that this projective system satisfies the conditions of (**0**, 7.2.9); as a result, its projective limit M is an A-module of finite type such that $M_n = M \otimes_A A_n$ for all n. We deduce that $\mathscr{F}_n$ is induced over X_n by $\widetilde{M} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\widetilde{\mathfrak{J}}^{n+1})$, and so $\mathscr{F} = M^\Delta$ by Definition (10.8.4).

Conversely, (c) implies (b); indeed, if u_n is the immersion morphism $X_n \to X$, then $u_n^*(\widetilde{M}) = (M \otimes_A A_n)^\sim$ is induced over X_n by $\widetilde{M} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\widetilde{\mathfrak{J}}^{n+1})$, and $M^\Delta = \varprojlim u_n^*(\widetilde{M})$ by Definition (10.8.4); since $u_m = u_n \circ u_{mn}$ for $m \leqslant n$, the $\mathscr{F}_n = u_n^*(\widetilde{M})$ satisfy the conditions of (b), whence our claim.

We now show that (c) implies (a): indeed, we have, by definition, that $\mathscr{O}_{\mathfrak{X}} = A^\Delta$; since M is the cokernel of a homomorphism $A^m \to A^n$, it follows from Proposition (10.10.2) that M^Δ is the cokernel of a homomorphism $\mathscr{O}_{\mathfrak{X}}^m \to \mathscr{O}_{\mathfrak{X}}^n$, and, since the sheaf of rings $\mathscr{O}_{\mathfrak{X}}$ is coherent (10.10.3), so too is M^Δ (**0**, 5.3.4).

Finally, (a) implies (b). Considered as an $\mathscr{O}_{\mathfrak{X}}$-module, we have that $\mathscr{O}_{X_n} = \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1} = A_n^\Delta$; **I | 203**
but $\mathscr{F}_n = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}$ is a coherent $\mathscr{O}_{\mathfrak{X}}$-module (**0**, 5.3.5), and, since it is also an $\mathscr{O}_{X_n}$-module, and $\mathscr{J}^{n+1}$ is coherent, we conclude that $\mathscr{F}_n$ is a coherent $\mathscr{O}_{X_n}$-module (**0**, 5.3.10), and it is immediate that $u_{mn}^*(\mathscr{F}_n) = \mathscr{F}_m$ for $m \leqslant n$ (recalling that the continuous map $X_m \to X_n$ of the underlying spaces is the identity on $\mathfrak{X}$). The sheaf $\mathscr{G} = \varprojlim \mathscr{F}_n$ is thus a coherent $\mathscr{O}_{\mathfrak{X}}$-module, since we have seen that (b) implies (a). The canonical homomorphisms $\mathscr{F} \to \mathscr{F}_n$ form a projective system, which, by passing to the limit, gives a canonical homomorphism $w : \mathscr{F} \to \mathscr{G}$, and it remains only to prove that w is bijective. The question is now *local*, so we can reduce to the case where $\mathscr{F}$ is the cokernel of a homomorphism $\mathscr{O}_{\mathfrak{X}}^p \to \mathscr{O}_{\mathfrak{X}}^q$; since this homomorphism is of the form v^Δ, where v is a homomorphism $A^m \to A^n$ (10.10.2), $\mathscr{F}$ is isomorphic to M^Δ, where $M = \operatorname{Coker} v$ (10.10.2). We then have, by Proposition (10.10.2), that $\mathscr{F}_n = M^\Delta \otimes_{\mathscr{O}_{\mathfrak{X}}} A_n^\Delta = (M \otimes_A A_n)^\Delta$, and, since the $\mathfrak{J}$-adic topology on $M \otimes_A A_n$ is discrete, we have $(M \otimes_A A_n)^\Delta = (M \otimes_A A_n)^\sim$ (as an $\mathscr{O}_{X_n}$-module); we have seen above that $M^\Delta = \varprojlim \mathscr{F}_n$, and w is thus the identity in this case. □

Corollary (10.10.6). — *If $\mathscr{F}$ satisfies condition* (b) *of Proposition (10.10.5), then the projective system $(\mathscr{F}_n)$ is isomorphic to the system of the $\mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}$.*

(10.10.7). Now let A and B be adic Noetherian rings, and $\phi : B \to A$ a continuous homomorphism; we denote by $\mathfrak{J}$ (resp. $\mathfrak{K}$) an ideal of definition for A (resp. B) such that $\phi(\mathfrak{K}) \subset \mathfrak{J}$, and we set $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, $\mathfrak{X} = \operatorname{Spf}(A)$, and $\mathfrak{Y} = \operatorname{Spf}(B)$. Let $f : X \to Y$ be the morphism of preschemes corresponding to ϕ (1.6.1), and $\widehat{f} : \mathfrak{X} \to \mathfrak{Y}$ its extension to the completions (10.9.1), which is also a morphism of formal preschemes that corresponds to ϕ (10.2.2).

Proposition (10.10.8). — *For every B-module N of finite type, there exists a canonical functorial isomorphism of $\mathscr{O}_{\mathfrak{X}}$-modules*

$$\widehat{f}^*(N^\Delta) \simeq (N \otimes_B A)^\Delta.$$

PROOF. Denoting by $i_X : \mathfrak{X} \to X$ and $i_Y : \mathfrak{Y} \to Y$ the canonical morphisms, we have (10.8.8), up to canonical functorial isomorphisms, $N^\Delta = i_Y^*(\widetilde{N})$ and

$$(N \otimes_B A)^\Delta = i_X^*((N \otimes_B A)^\sim) = i_X^*(f^*(\widetilde{N}))$$

(1.6.5); the proposition then follows from the commutativity of the diagram in (10.9.2). □

Corollary. — *For every ideal $\mathfrak{b}$ of B, we have $\widehat{f}^*(\mathfrak{b}^\Delta)\mathscr{O}_{\mathfrak{X}} = (\mathfrak{b}A)^\Delta$.*

PROOF. Let j be the canonical injection $\mathfrak{b} \to B$, to which corresponds the canonical injection $j^\Delta : \mathfrak{b}^\Delta \to \mathscr{O}_{\mathfrak{Y}}$ of sheaves of $\mathscr{O}_{\mathfrak{Y}}$-modules; by definition, $\widehat{f}^*(\mathfrak{b}^\Delta)\mathscr{O}_{\mathfrak{X}}$ is the image of the homomorphism $\widehat{f}^*(j^\Delta) : \widehat{f}^*(\mathfrak{b}^\Delta) \to \mathscr{O}_{\mathfrak{X}} = \widehat{f}^*(\mathscr{O}_{\mathfrak{Y}})$; but this homomorphism can be identified with $(j \otimes 1)^\Delta : (\mathfrak{b} \otimes_B A)^\Delta \to \mathscr{O}_{\mathfrak{X}} = (B \otimes_B A)^\Delta$ by Proposition (10.10.8). Since the image of $j \otimes 1$ is the ideal $\mathfrak{b}A$ of A, the image of $(j \otimes 1)^\Delta$ is thus $(\mathfrak{b}A)^\Delta$, by Proposition (10.10.2), whence the conclusion. □

10.11. Coherent sheaves on formal preschemes

Proposition (10.11.1). — *If $\mathfrak{X}$ is a locally Noetherian formal prescheme, then the sheaf of rings $\mathscr{O}_{\mathfrak{X}}$ is coherent, and every sheaf of ideals of definition for $\mathfrak{X}$ is coherent.* I | 204

PROOF. The question is local, so we can reduce to the case of a Noetherian affine formal scheme, and the proposition then follows from Propositions (10.10.3) and (10.10.5). □

(10.11.2). Let $\mathfrak{X}$ be a locally Noetherian formal prescheme, $\mathscr{J}$ a sheaf of ideals of definition for $\mathfrak{X}$, and X_n the locally Noetherian (usual) prescheme $(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$, so that $\mathfrak{X}$ is the *inductive limit* of the sequence (X_n) with respect to the canonical morphisms $u_{mn} : X_m \to X_n$ (10.6.3). With this notation:

Theorem (10.11.3). — *For an $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$ to be coherent, it is necessary and sufficient for it to be isomorphic to a projective limit of a sequence $(\mathscr{F}_n)$, where the $\mathscr{F}_n$ are coherent $\mathscr{O}_{X_n}$-modules such that $u_{mn}^*(\mathscr{F}_n) = \mathscr{F}_m$ for $m \leqslant n$ (10.6.6). The projective system $(\mathscr{F}_n)$ is then isomorphic to the system of the $u_n^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}$, where u_n is the canonical morphism $X_n \to \mathfrak{X}$.*

PROOF. The question is local, so we can reduce to the case where $\mathfrak{X}$ is a Noetherian affine formal scheme, and the theorem then is a consequence of Proposition (10.10.5) and Corollary (10.10.6). □

We can thus say that *the data of a coherent $\mathscr{O}_{\mathfrak{X}}$-module is equivalent to the data of a projective system $(\mathscr{F}_n)$ of coherent $\mathscr{O}_{X_n}$-modules such that $u_{mn}(\mathscr{F}_n) = \mathscr{F}_m$ for $m \leqslant n$.*

Corollary (10.11.4). — *If $\mathscr{F}$ and $\mathscr{G}$ are coherent $\mathscr{O}_{\mathfrak{X}}$-modules, then we can (with the notation of Theorem (10.11.3)) define a canonical functorial isomorphism*

$$\operatorname{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G}) \simeq \varprojlim_n \operatorname{Hom}_{\mathscr{O}_{X_n}}(\mathscr{F}_n, \mathscr{G}_n). \tag{10.11.4.1}$$

PROOF. The projective limit on the right-hand side is understood to be taken with respect to the maps $\theta_n \mapsto u_{mn}^*(\theta_n)$ ($m \leqslant n$) from $\operatorname{Hom}_{\mathscr{O}_{X_n}}(\mathscr{F}_n, \mathscr{G}_n)$ to $\operatorname{Hom}_{\mathscr{O}_{X_m}}(\mathscr{F}_m, \mathscr{F}_m)$. The homomorphism (10.11.4.1) sends an element $\theta \in \operatorname{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$ to the sequence $(u_n^*(\theta))$; we see that we can define an inverse homomorphism of the above by sending a projective system $(\theta_n) \in \varprojlim_n \operatorname{Hom}_{\mathscr{O}_{X_n}}(\mathscr{F}_n, \mathscr{G}_n)$ to its projective limit in $\operatorname{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$, taking into account Theorem (10.11.3). □

Corollary (10.11.5). — *For a homomorphism $\theta : \mathscr{F} \to \mathscr{G}$ to be surjective, it is necessary and sufficient for the corresponding homomorphism $\theta_0 = u_0^*(\theta) : \mathscr{F}_0 \to \mathscr{G}_0$ to be surjective.*

PROOF. The question is local, so we reduce to the case where $\mathfrak{X} = \mathrm{Spf}(A)$ with A an adic Noetherian ring, $\mathscr{F} = M^{\Delta}$, $\mathscr{G} = N^{\Delta}$, and $\theta = u^{\Delta}$, where M and N are A-modules of finite type, and u is a homomorphism $M \to N$; we then have that $\theta_0 = \widetilde{u_0}$, where u_0 is the homomorphism $u \otimes 1 : M \otimes_A A/\mathfrak{J} \to N \otimes_A A/\mathfrak{J}$; the conclusion follows from the fact θ and u (resp. θ_0 and u_0) are simultaneously surjective ((1.3.9) and (10.10.2)) and that u and u_0 are simultaneously surjective (**0**, 7.1.14). □

(10.11.6). Theorem (10.11.3) shows that we can consider every coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$ as a *topological $\mathscr{O}_{\mathfrak{X}}$-module*, considering it as a projective limit of *pseudo-discrete* sheaves of groups $\mathscr{F}_n$ (**0**, 3.8.1). It then follows from Corollary (10.11.4) that every homomorphism $u : \mathscr{F} \to \mathscr{G}$ of coherent $\mathscr{O}_{\mathfrak{X}}$-modules is automatically *continuous* (**0**, 3.8.2). Furthermore, if $\mathscr{H}$ is a coherent $\mathscr{O}_{\mathfrak{X}}$-submodule of a **I | 205**
coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, then, for every open $U \subset \mathfrak{X}$, $\Gamma(U, \mathscr{H})$ is a *closed* subgroup of the topological group $\Gamma(U, \mathscr{F})$, since the functor Γ is left exact, and $\Gamma(U, \mathscr{H})$ is the kernel of the homomorphism $\Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{F}/\mathscr{H})$, which is *continuous* by the above, since $\mathscr{F}/\mathscr{G}$ is coherent (**0**, 5.3.4); our claim follows from the fact that $\Gamma(U, \mathscr{F}/\mathscr{H})$ is a separated topological group.

Proposition (10.11.7). — *Let $\mathscr{F}$ and $\mathscr{G}$ be coherent $\mathscr{O}_{\mathfrak{X}}$-modules. We can define (with the notation of Theorem* (10.11.3)*) canonical functorial isomorphisms of topological $\mathscr{O}_{\mathfrak{X}}$-modules* (10.11.6)

$$\mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{G} \simeq \varprojlim_n (\mathscr{F}_n \otimes_{\mathscr{O}_{X_n}} \mathscr{G}_n), \tag{10.11.7.1}$$

$$\mathscr{H}om_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G}) \simeq \varprojlim_n \mathscr{H}om_{\mathscr{O}_{X_n}}(\mathscr{F}_n, \mathscr{G}_n). \tag{10.11.7.2}$$

PROOF. The existence of the isomorphism (10.11.7.1) follows from the formula

$$\mathscr{F}_n \otimes_{\mathscr{O}_{X_n}} \mathscr{G}_n = (\mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}) \otimes_{\mathscr{O}_{X_n}} (\mathscr{G} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}) = (\mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{G}) \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_n}$$

and from Theorem (10.11.3). The isomorphism (10.11.7.2), where both sides are considered as sheaves of modules without topology, follows from the definition of the sections of $\mathscr{H}om_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$ and $\mathscr{H}om_{\mathscr{O}_{X_n}}(\mathscr{F}_n, \mathscr{G}_n)$, and from the existence of the isomorphism (10.11.4.1), mapping a prescheme induced on an arbitrary Noetherian formal affine open set to $\mathfrak{X}$. It remains to prove that the isomorphism (10.11.7.2) is bicontinuous over a quasi-compact set, and we can thus reduce to the case where $\mathfrak{X} = \mathrm{Spf}(A)$ with A an adic Noetherian ring, and hence (10.10.5) to the case where $\mathscr{F} = M^{\Delta}$ and $\mathscr{G} = N^{\Delta}$, with M and N both A-modules of finite type; taking (10.10.2.1), (10.10.2.3), and Corollary (1.3.12, ii) into account, we reduce to showing that the canonical isomorphism $\mathrm{Hom}_A(M, N) \simeq \varprojlim_n \mathrm{Hom}_{A_n}(M_n, N_n)$ (with $M_n = M \otimes_A A_n$ and $N_n = N \otimes_A A_n$) is continuous, which has already been proved in (**0**, 7.8.2). □

(10.11.8). Since $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$ is the group of sections of the sheaf of topological groups $\mathscr{H}om_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$, it is equipped with a group topology. If $\mathfrak{X}$ is *Noetherian*, then it follows from (10.11.7.2) that the subgroups $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{J}^n\mathscr{G})$ (for arbitrary n) form a fundamental system of neighbourhoods of 0 in this group.

Proposition (10.11.9). — *Let $\mathfrak{X}$ be a Noetherian formal prescheme, and $\mathscr{F}$ and $\mathscr{G}$ coherent $\mathscr{O}_{\mathfrak{X}}$-modules. In the topological group $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$, the surjective (resp. injective, bijective) homomorphisms form an open set.*

PROOF. By Corollary (10.11.5), the set of surjective homomorphisms in $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$ is the inverse image under the continuous map $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}_{\mathscr{O}_{X_0}}(\mathscr{F}_0, \mathscr{G}_0)$ of a subset of the discrete group $\mathrm{Hom}_{\mathscr{O}_{X_0}}(\mathscr{F}_0, \mathscr{G}_0)$, whence the first claim. To show the second, we cover $\mathfrak{X}$ by a finite number of Noetherian formal affine subsets U_i. For $\theta \in \mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G})$ to be injective, it is necessary and sufficient for all of the images under the (continuous) restriction maps $\mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}_{\mathscr{O}_{\mathfrak{X}}|U_i)}(\mathscr{F}|U_i, \mathscr{G}|U_i)$ to be injective; we can thus reduce to the affine case, and then this has already been proved in (**0**, 7.8.3). □

10.12. Adic morphisms of formal preschemes

(10.12.1). Let $\mathfrak{X}$ and $\mathfrak{S}$ be *locally Noetherian* formal preschemes; we say that a morphism $f : \mathfrak{X} \to \mathfrak{S}$ is *adic* if there exists an ideal of definition $\mathscr{J}$ of $\mathfrak{S}$ such that $\mathscr{K} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}}$ is an ideal of definition of $\mathfrak{X}$; we then also say that $\mathfrak{X}$ is an *adic $\mathfrak{S}$-prescheme* (for f). Whenever this is the case, for *every* ideal of definition $\mathscr{J}_1$ of $\mathfrak{S}$, $\mathscr{K}_1 = f^*(\mathscr{J}_1)\mathscr{O}_{\mathfrak{X}}$ is an ideal of definition of $\mathfrak{X}$. Indeed, since the questions is local, we can assume that $\mathfrak{X}$ and $\mathfrak{S}$ are Noetherian and affine; there then exists a whole number n such that $\mathscr{J}^n \subset \mathscr{J}_1$ and $\mathscr{J}_1^n \subset \mathscr{J}$ ((10.3.6) and (**0**, 7.1.4)), whence $\mathscr{K}^n \subset \mathscr{K}_1$ and $\mathscr{K}_1^n \subset \mathscr{K}$. The first of these relations shows that $\mathscr{K}_1 = \mathfrak{K}_1^\Delta$, where $\mathfrak{K}_1$ is an open ideal of $A = \Gamma(\mathfrak{X}, \mathscr{O}_{\mathfrak{X}})$, and the second shows that $\mathfrak{K}_1$ is an ideal of definition of A (**0**, 7.1.4), whence our claim. I | 206

It follows immediately from the above that, if $\mathfrak{X}$ and $\mathfrak{Y}$ are adic $\mathfrak{S}$-preschemes, then *every $\mathfrak{S}$-morphism $u : \mathfrak{X} \to \mathfrak{Y}$ is adic*: indeed, if $f : \mathfrak{X} \to \mathfrak{S}$ and $g : \mathfrak{Y} \to \mathfrak{S}$ are the structure morphisms, and $\mathscr{J}$ is an ideal of definition of $\mathfrak{S}$, then we have $f = g \circ u$, and so $u^*(g^*(\mathscr{J})\mathscr{O}_{\mathfrak{Y}})\mathscr{O}_{\mathfrak{X}} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}}$ is an ideal of definition of $\mathfrak{X}$, and, by hypothesis, $g^*(\mathscr{J})\mathscr{O}_{\mathfrak{Y}}$ is an ideal of definition of $\mathfrak{Y}$.

(10.12.2). In what follows, we suppose that we have some fixed locally Noetherian formal prescheme $\mathfrak{S}$, and some ideal of definition $\mathscr{J}$ of $\mathfrak{S}$; we set $S_n = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{J}^{n+1})$. The (locally Noetherian) adic $\mathfrak{S}$-preschemes clearly form a *category*. We say that an inductive system (X_n) of locally Noetherian (usual) S_n-preschemes is an *adic inductive (S_n)-system* if the structure morphisms $f_n : X_n \to S_n$ are such that, for $m \leqslant n$, the diagrams

$$\begin{tikzcd} X_n \arrow[d, "f_n"'] & X_m \arrow[l] \arrow[d, "f_m"] \\ S_n & S_m \arrow[l] \end{tikzcd} \tag{10.12.2.1}$$

commute and *identify X_m with the product $X_n \times_{S_n} S_m = (X_n)_{(S_m)}$*. The adic inductive systems form a *category*: it suffices in fact to define a morphism $(X_n) \to (Y_n)$ of such systems to be an *inductive system of S_n-morphisms $u_n : X_n \to Y_n$* such that u_m is identified with $(u_n)_{(S_m)}$ for $m \leqslant n$. With this in mind:

Theorem (10.12.3). — *There is a canonical equivalence between the category of adic $\mathfrak{S}$-preschemes and the category of adic inductive (S_n)-systems.*

The equivalence in question is obtained in the following way: if $\mathfrak{X}$ is an adic $\mathfrak{S}$-prescheme, and $f : \mathfrak{X} \to \mathfrak{S}$ is the structure morphism, then $\mathscr{K} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}}$ is an ideal of definition of $\mathfrak{X}$, and we associate to $\mathfrak{X}$ the inductive system of the $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{K}^{n+1})$, with the structure morphism $f_n : X_n \to S_n$ corresponding to f (10.5.6). We first show that (X_n) is an *adic inductive system*: if $f = (\psi, \theta)$, then $\psi^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}} = \mathscr{K}$, so $\psi^*(\mathscr{J}^n)\mathscr{O}_{\mathfrak{X}} = \mathscr{K}^n$ for all n, and (by exactness of the functor ψ^*) $\mathscr{K}^{m+1}/\mathscr{K}^{n+1} = \psi^*(\mathscr{J}^{m+1}/\mathscr{J}^{n+1})(\mathscr{O}_{\mathfrak{X}}/\mathscr{K}^{n+1})$ for $m \leqslant n$; our conclusion thus follows from (4.4.5). Furthermore, it can be immediately verified that a $\mathfrak{S}$-morphism $u : \mathfrak{X} \to \mathfrak{Y}$ of adic $\mathfrak{S}$-preschemes corresponds (with the obvious notation) to an inductive system of S_n-morphisms I | 207
$u_n : X_n \to Y_n$ such that u_m is identified with $(u_n)_{(S_m)}$ for $m \leqslant n$.

The fact that this equivalence is well defined will follow from the more-precise following proposition.

Proposition (10.12.3.1). — *Let (X_n) be an inductive system of S_n-preschemes; suppose that the structure morphisms $f_n : X_n \to S_n$ are such that the diagrams in (10.12.2.1) commute and identify X_m with $X_n \times_{S_n} S_m$ for $m \leqslant n$. Then the inductive system (X_n) satisfies conditions* (b) *and* (c) *of (10.6.3); let $\mathfrak{X}$ be the inductive limit, and $f : \mathfrak{X} \to \mathfrak{S}$ the morphism given by the inductive limit of the inductive system (f_n). Then, if X_0 is locally Noetherian, $\mathfrak{X}$ is locally Noetherian, and f is an adic morphism.*

Proof. Since the sheaf of ideals of $\mathscr{O}_{S_n}$ that defines the subprescheme S_m of S_n is nilpotent, by (4.4.5), so too is the sheaf of ideals of $\mathscr{O}_{X_n}$ that defines the subprescheme X_m of X_n, and so the conditions of (10.6.3) are satisfied. Since the questions is local on $\mathfrak{X}$ and $\mathfrak{S}$, we can assume that $\mathfrak{S} = \operatorname{Spf}(A)$, $\mathscr{J} = \mathfrak{J}^\Delta$ (with A a Noetherian $\mathfrak{J}$-adic ring), and $X_n = \operatorname{Spec}(B_n)$; if $A_n = A/\mathfrak{J}^{n+1}$, then the hypothesis implies that B_0 is Noetherian, and if we set $\mathfrak{J}_n = \mathfrak{J}/\mathfrak{J}^{n+1}$, then $B_m = B_n/\mathfrak{J}_n^{m+1}B_n$. The kernel of $B_n \to B_0$ is thus $\mathfrak{K}_n = \mathfrak{J}_n B_n$, and the kernel of $B_n \to B_m$ is $\mathfrak{K}_n^{m+1}$ for $m \leqslant n$; further, since A_1 is Noetherian, $\mathfrak{J}_1$ is of finite type over A_1, and so $\mathfrak{K}_1 = \mathfrak{K}_1/\mathfrak{K}_1^2$ is of finite type over B_1, and *a fortiori* of finite type over $B_0 = B_1/\mathfrak{K}_1$; the fact that $\mathfrak{X}$ is Noetherian then follows from (10.6.4); if $B = \varprojlim B_n$, then we have $\mathfrak{X} = \operatorname{Spf}(B)$, and, if $\mathfrak{K}$ is the kernel of $B \to B_0$, then $B_n = B/\mathfrak{K}^{n+1}$. If $\rho_n : A/\mathfrak{J}^{n+1} \to B/\mathfrak{K}^{n+1}$ is the homomorphism corresponding to f_n, then we have that

$$\mathfrak{K}/\mathfrak{K}^{n+1} = (B/\mathfrak{K}^{n+1})\rho_n(\mathfrak{J}/\mathfrak{J}^{n+1})$$

since the homomorphism $\rho : A \to B$ corresponding to f is equal to $\varprojlim \rho_n$, and that the ideal $\mathfrak{J}B$ of B is dense in $\mathfrak{K}$, and, since every ideal of B is closed (**0**, 7.3.5), we also have that $\mathfrak{K} = \mathfrak{J}B$. If $\mathscr{K} = \mathfrak{K}^\Delta$, the equality $f^*(\mathscr{J})\mathscr{O}_\mathfrak{X} = \mathscr{K}$ then follows from (10.10.9), and finishes the proof. □

(10.12.3.2). The above equivalence gives, for adic $\mathfrak{S}$-preschemes $\mathfrak{X}$ and $\mathfrak{Y}$, a *canonical bijection*

$$\operatorname{Hom}_\mathfrak{S}(\mathfrak{X}, \mathfrak{Y}) \simeq \varprojlim_n \operatorname{Hom}_{S_n}(X_n, Y_n)$$

where the projective limit is relative to the maps $u_n \to (u_n)_{(S_m)}$ for $m \leqslant n$.

10.13. Morphisms of finite type

Proposition (10.13.1). — *Let $\mathfrak{Y}$ be a locally Noetherian formal prescheme, $\mathscr{K}$ an ideal of definition of $\mathfrak{Y}$, and $f : \mathfrak{X} \to \mathfrak{Y}$ a morphism of formal preschemes. Then the following conditions are equivalent.*

(a) *X is locally Noetherian, f is an adic morphism* (10.12.1), *and, if we set $\mathscr{J} = f^*(\mathscr{K})\mathscr{O}_\mathfrak{X}$, then the morphism $f_0 : (\mathfrak{X}, \mathscr{O}_\mathfrak{X}/\mathscr{J}) \to (\mathfrak{Y}, \mathscr{O}_\mathfrak{Y}/\mathscr{K})$ induced by f is of finite type.*

(b) *$\mathfrak{X}$ is locally Noetherian, and is the inductive limit of an adic inductive (Y_n)-system (X_n) such that the morphism $X_0 \to Y_0$ is of finite type.* I | 208

(1) *Every point of $\mathfrak{Y}$ has a Noetherian formal affine open neighbourhood V which satisfies the following property:*

  (**Q**) *$f^{-1}(V)$ is a finite union of Noetherian formal affine open subsets U_i such that the Noetherian adic ring $\Gamma(U_i, \mathscr{O}_\mathfrak{X})$ is topologically isomorphic to the quotient of a formal series algebra, restricted* (**0**, 7.5.1) *to $\Gamma(V, \mathscr{O}_\mathfrak{Y})$, by an ideal (which is necessarily closed).*

PROOF. It is immediate that (a) implies (b), by (10.12.3). To show that (b) implies (c), we can, since the question is local on $\mathfrak{Y}$, assume that $\mathfrak{Y} = \operatorname{Spf}(B)$, where B is Noetherian and adic; let $\mathscr{K} = \mathfrak{K}^\Delta$, with $\mathfrak{K}$ an ideal of definition of B. Since, by hypothesis, X_0 is of finite type over Y_0, X_0 is a finite union of affine open subsets U_i such that the ring A_{i0} of the affine scheme induced by X_0 on U_i is an algebra of finite type over the ring $B/\mathfrak{K}$ of Y_0 (6.3.2). By (5.1.9), U_i is also an affine open subset in each of the Noetherian preschemes X_n, and, if A_{in} is the ring of the affine scheme induced by X_n on U_i, then hypothesis (b) implies, for $m \leqslant n$, that A_{im} is isomorphic to $A_{in}/\mathfrak{K}^{m+1}A_{in}$. Consequently, the formal prescheme induced by $\mathfrak{X}$ on U_i is isomorphic to $\operatorname{Spf}(A_i)$, where $A_i = \varprojlim_n A_{in}$ (10.6.4); A_i is a $\mathfrak{K}A_i$-adic ring, and $A_i/\mathfrak{K}A_i$, being isomorphic to A_{i0}, is an algebra of finite type over $B/\mathfrak{K}$. We thus conclude (**0**, 7.5.5) that A_i is topologically isomorphic to a quotient of a formal series algebra restricted to B (by a necessarily closed ideal, because such an algebra is Noetherian (**0**, 7.5.4)).

To show that (c) implies (a), we can restrict to the case where $\mathfrak{X} = \operatorname{Spf}(A)$ is also affine, with A a Noetherian adic ring isomorphic to a quotient of a formal series algebra, restricted to B, by a closed ideal. Then (**0**, 7.5.5) $A/\mathfrak{K}A$ is an algebra of finite type over $B/\mathfrak{K}$, and $\mathfrak{K}A = \mathfrak{J}$ is an ideal of definition of A, and so, by (10.10.9), the conditions of (a) are satisfied. □

We note that, if the conditions of Proposition (10.13.1) are satisfied, then property (a) holds true for *any* ideal of definition $\mathscr{K}$ of $\mathfrak{Y}$ (by *(c)*), and so, in property (b), *all* the f_n are morphisms of finite type.

Corollary (10.13.2). — *If the conditions of* (10.13.1) *are satisfied, then every Noetherian formal affine open subset V of $\mathfrak{Y}$ has property* (**Q**)*, and, if $\mathfrak{Y}$ is Noetherian, then so too is $\mathfrak{X}$.*

PROOF. This follows immediately from (10.13.1) and (6.3.2). □

Definition (10.13.3). — When the equivalent properties (a), (b), and (c) of (10.13.1) are satisfied, we say that the morphism f is of finite type, or that $\mathfrak{X}$ is a formal $\mathfrak{Y}$-prescheme of finite type, or a formal prescheme of finite type over $\mathfrak{Y}$.

Corollary (10.13.4). — *Let $\mathfrak{X} = \operatorname{Spf}(A)$ and $\mathfrak{Y} = \operatorname{Spf}(B)$ be Noetherian formal affine schemes; for $\mathfrak{X}$ to be of finite type over $\mathfrak{Y}$, it is necessary and sufficient for the Noetherian adic ring A to be isomorphic to the quotient of a formal series algebra, restricted to B, by some closed ideal.*

Proof. With the notation of (10.13.1), if $\mathfrak{X}$ is of finite type over $\mathfrak{Y}$, then $A/\mathfrak{K}A$ is a $(B/\mathfrak{K})$-algebra of finite type by (6.3.3), and $\mathfrak{K}A$ is an ideal of definition of A (10.10.9). We are then done, by (**0**, 7.5.5). □

Proposition (10.13.5). —

(i) *The composition of any two morphisms (of formal preschemes) of finite type is again of finite type.* **I** | 209

(ii) *Let $\mathfrak{X}$, $\mathfrak{S}$, and $\mathfrak{S}'$ be locally Noetherian (resp. Noetherian) formal preschemes, and $f : \mathfrak{X} \to \mathfrak{S}$ and $\mathfrak{X} \to \mathfrak{S}'$ morphisms. If f is of finite type, then $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{S}'$ is locally Noetherian (resp. Noetherian) and of finite type over $\mathfrak{S}'$.*

(iii) *Let $\mathfrak{S}$ be a locally Noetherian formal prescheme, and $\mathfrak{X}'$ and $\mathfrak{Y}'$ formal $\mathfrak{S}$-preschemes such that $\mathfrak{X}' \times_{\mathfrak{S}} \mathfrak{Y}'$ is locally Noetherian. If $\mathfrak{X}$ and $\mathfrak{Y}$ are locally Noetherian formal $\mathfrak{S}$-preschemes, and $f : \mathfrak{X} \to \mathfrak{X}'$ and $g : \mathfrak{Y} \to \mathfrak{Y}'$ are $\mathfrak{S}$-morphisms of finite type, then $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is locally Noetherian, and $f \times_{\mathfrak{S}} g$ is a $\mathfrak{S}$-morphism of finite type.*

Proof. By the formal argument of (3.5.1), (iii) follows from (i) and (ii), so it suffices to prove (i) and (ii).

Let $\mathfrak{X}$, $\mathfrak{Y}$, and $\mathfrak{Z}$ be locally Noetherian formal preschemes, and $f : \mathfrak{X} \to \mathfrak{Y}$ and $g : \mathfrak{Y} \to \mathfrak{Z}$ morphisms of finite type. If $\mathscr{L}$ is an ideal of definition of $\mathfrak{Z}$, then $\mathscr{K} = g^*(\mathscr{L})\mathscr{O}_{\mathfrak{Y}}$ is an ideal of definition of $\mathfrak{Y}$, and $\mathscr{J} = f^*(g^*(\mathscr{L}))\mathscr{O}_{\mathfrak{X}}$ is an ideal of definition for $\mathfrak{X}$. Let $X_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J})$, $Y_0 = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K})$, and $Z_0 = (\mathfrak{Z}, \mathscr{O}_{\mathfrak{Z}}/\mathscr{L})$, and let $f_0 : X_0 \to Y_0$ and $g_0 : Y_0 \to Z_0$ be the morphisms corresponding to f and g (respectively). Since, by hypothesis, f_0 and g_0 are of finite type, so too is $g_0 \circ f_0$ (6.3.4), which corresponds to $g \circ f$; thus $g \circ f$ is of finite type, by (10.13.1).

Under the conditions of (ii), $\mathfrak{S}$ (resp. $\mathfrak{X}$, $\mathfrak{S}'$) is the inductive limit of a sequence (S_n) (resp. (X_n), (S'_n)) of locally Noetherian preschemes, and we can assume (10.13.1) that $X_m = X_n \times_{S_n} S_m$ for $m \leqslant n$. The formal prescheme $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{S}'$ is then the inductive limit of the preschemes $X_n \times_{S_n} S'_n$ (10.7.4), and we have that

$$X_m \times_{S_m} S'_m = (X_n \times_{S_n} S_m) \times_{S_m} S'_m = (X_n \times_{S_n} S'_n) \times_{S'_n} S'_m.$$

Furthermore, $X_0 \times_{S_0} S'_0$ is locally Noetherian, since X_0 is of finite type over S_0 (6.3.8). We thus conclude (10.12.3.1), first of all, that $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{S}'$ is locally Noetherian; then, since $X_0 \times_{S_0} S'_0$ is of finite type over S'_0 (6.3.8), it follows from (10.12.3.1) and (10.13.1) that $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{S}'$ is of finite type over $\mathfrak{S}'$, which proves (ii) (the claim about Noetherian preschemes being an immediate consequence of (6.3.8)). □

Corollary (10.13.6). — *Under the hypotheses of (10.9.9), if f is a morphism of finite type, then so too is its extension $\widehat{f}$ to the completions.*

10.14. Closed subpreschemes of formal preschemes

Proposition (10.14.1). — *Let $\mathfrak{X}$ be a locally Noetherian formal prescheme, and $\mathscr{A}$ a coherent sheaf of ideals of $\mathscr{O}_{\mathfrak{X}}$. If $\mathfrak{Y}$ if the (closed) support of $\mathscr{O}_{\mathfrak{X}}/\mathscr{A}$, then the topologically ringed space $(\mathfrak{Y}, (\mathscr{O}_{\mathfrak{X}}/\mathscr{A})|\mathfrak{Y})$ is a locally Noetherian formal prescheme that is Noetherian if $\mathfrak{X}$ is.*

Proof. Note that $\mathscr{O}_{\mathfrak{X}}/\mathfrak{A}$ is coherent by (10.10.3) and (**0**, 5.3.4), so its support $\mathfrak{Y}$ is closed (**0**, 5.2.2). Let $\mathscr{J}$ be an ideal of definition of $\mathfrak{X}$, and let $X_n = (\mathfrak{X}/\mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$; the sheaf of rings $\mathscr{O}_{\mathfrak{X}}/\mathscr{A}$ is the projective limit of the sheaves $\mathscr{O}_{\mathfrak{X}}/(\mathscr{A} + \mathscr{J}^{n+1}) = (\mathscr{O}_{\mathfrak{X}}/\mathscr{A}) \otimes_{\mathscr{O}_{\mathfrak{X}}} (\mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$ (10.11.3), all of which have support $\mathfrak{Y}$. The sheaf $(\mathscr{A} + \mathscr{J}^{n+1})/\mathscr{J}^{n+1}$ is a coherent $\mathscr{O}_{\mathfrak{X}}$-module, since $\mathscr{J}^{n+1}$ is coherent, and so $(\mathscr{A} + \mathscr{J}^{n+1})/\mathscr{J}^{n+1}$ is also a coherent $(\mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$-module (**0**, 5.3.10); if Y_n is the closed subprescheme of X_n defined by this sheaf of ideals, it is immediate that $(\mathfrak{Y}, (\mathscr{O}_{\mathfrak{X}}/\mathscr{A})|\mathfrak{Y})$ is **I** | 210 the formal prescheme given by the inductive limit of the Y_n, and, since the conditions of (10.6.4) are satisfied, this proves that this formal prescheme is locally Noetherian, and further Noetherian if $\mathfrak{X}$ is (since then Y_0 is, by (6.1.4)). □

Definition (10.14.2). — We define a closed subprescheme of a formal prescheme $\mathfrak{X}$ to be any formal prescheme of the form $(\mathfrak{Y}, (\mathscr{O}_{\mathfrak{X}}/\mathscr{A})|\mathfrak{Y})$ with $\mathscr{A}$ a coherent ideal of $\mathscr{O}_{\mathfrak{X}}$; we say that this prescheme is the subprescheme defined by $\mathscr{A}$. **Err$_{\text{II}}$**

It is clear that the correspondence thus defined between coherent ideals of $\mathscr{O}_{\mathfrak{X}}$ and closed subpreschemes of $\mathfrak{X}$ is bijective. **Err$_{\text{II}}$**

The morphism of topologically ringed spaces $j = (\psi, \theta) : \mathfrak{Y} \to \mathfrak{X}$, where ψ is the injection $\mathfrak{Y} \to \mathfrak{X}$ and $\theta^\sharp$ the canonical homomorphism $\mathscr{O}_{\mathfrak{X}} \to \mathscr{O}_{\mathfrak{X}}/\mathscr{A}$, is evidently (10.4.5) a morphism of formal preschemes, and we call it the *canonical injection* from $\mathfrak{Y}$ to $\mathfrak{X}$. Note that, if $\mathfrak{X} = \operatorname{Spf}(A)$, or if A is Noetherian and adic, then $\mathscr{A} = \mathfrak{a}^\Delta$, where $\mathfrak{a}$ is an ideal of A (10.10.5), and it then follows immediately from the above that $\mathfrak{Y} = \operatorname{Spf}(A/\mathfrak{a})$, up to isomorphism, and that j corresponds (10.2.2) to the canonical homomorphism $A \to A/\mathfrak{a}$.

We say that a morphism $f : \mathfrak{Z} \to \mathfrak{X}$ of locally Noetherian formal preschemes is a *closed immersion* if it factors as $\mathfrak{Z} \xrightarrow{g} \mathfrak{Y} \xrightarrow{j} \mathfrak{X}$, where g is an isomorphism from $\mathfrak{Z}$ to a closed subprescheme $\mathfrak{Y}$ of $\mathfrak{X}$, and j is the canonical injection. Since j is a monomorphism of ringed spaces, g and $\mathfrak{Y}$ are necessarily *unique*.

Proposition (10.14.3). — *A closed immersion is a morphism of finite type.*

Proof. We can immediately restrict to the case where $\mathfrak{X}$ is a formal affine scheme $\operatorname{Spf}(A)$, and $\mathfrak{Y} = \operatorname{Spf}(A/\mathfrak{a})$; the proposition then follows from Proposition (10.13.1, *c*). □

Lemma (10.14.4). — *Let $f : \mathfrak{Y} \to \mathfrak{X}$ be a morphism of locally Noetherian formal preschemes, and let (U_α) be a cover of $f(\mathfrak{Y})$ by Noetherian formal affine open subsets of $\mathfrak{X}$ such that the $f^{-1}(U_\alpha)$ are Noetherian formal affine open subsets of $\mathfrak{Y}$. For f to be a closed immersion, it is necessary and sufficient for $f(\mathfrak{Y})$ to be a closed subset of $\mathfrak{X}$ and, for all α, for the restriction of f to $f^{-1}(U_\alpha)$ to correspond* (10.4.6) *to a surjective homomorphism $\Gamma(U_\alpha, \mathscr{O}_{\mathfrak{X}}) \to \Gamma(f^{-1}(U_\alpha), \mathscr{O}_{\mathfrak{Y}})$.*

Proof. The conditions are clearly necessary. Conversely, if the conditions are satisfied, and if we denote by $\mathfrak{a}_\alpha$ the kernel of $\Gamma(U_\alpha, \mathscr{O}_{\mathfrak{X}}) \to \Gamma(f^{-1}(U_\alpha), \mathscr{O}_{\mathfrak{Y}})$, then we can define a coherent sheaf of ideals $\mathscr{A}$ of $\mathscr{O}_{\mathfrak{X}}$ by setting $\mathscr{A}|U_\alpha = \mathfrak{a}_\alpha^\Delta$ and taking $\mathscr{A}$ to be zero on the complement of the union of the U_α. Since $f(\mathfrak{Y})$ is closed, and since the support of $\mathfrak{a}_\alpha^\Delta$ is $U_\alpha \cap f(\mathfrak{Y})$, everything reduces to proving that $\mathfrak{a}_\alpha^\Delta$ and $\mathfrak{a}_\beta^\Delta$ induce the same sheaf on any Noetherian formal affine open subset $V \subset U_\alpha \cap U_\beta$. But the restriction to $f^{-1}(U_\alpha)$ of f is a closed immersion of this formal prescheme into U_α, $f^{-1}(V)$ is a Noetherian formal affine open subsets of $f^{-1}(U_\alpha)$, and the restriction of f to $f^{-1}(V)$ is a closed immersion; if $\mathfrak{b}$ is the kernel of the surjective homomorphism $\Gamma(V, \mathscr{O}_{\mathfrak{X}}) \to \Gamma(f^{-1}(V), \mathscr{O}_{\mathfrak{Y}})$ corresponding to this restriction, then it is immediate (10.10.2) that $\mathfrak{a}_\alpha^\Delta$ induces $\mathfrak{b}^\Delta$ on V. The sheaf of ideals $\mathscr{A}$ being thus defined, it is then clear that $f = g \circ j$, where $j : \mathfrak{Z} \to \mathfrak{X}$ is the canonical injection of the closed subprescheme $\mathfrak{Z}$ of $\mathfrak{X}$ defined by $\mathscr{A}$, and that g is an isomorphism from $\mathfrak{Y}$ to $\mathfrak{Z}$. □

Proposition (10.14.5). —

(i) *If $f : \mathfrak{Z} \to \mathfrak{Y}$ and $g : \mathfrak{Y} \to \mathfrak{X}$ are closed immersions of locally Noetherian formal preschemes, then $g \circ f$ is a closed immersion.* **I | 211**

(ii) *Let $\mathfrak{X}$, $\mathfrak{Y}$, and $\mathfrak{S}$ be locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{S}$ a closed immersion, and $g : \mathfrak{Y} \to \mathfrak{S}$ a morphism. Then the morphism $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y} \to \mathfrak{Y}$ is a closed immersion.*

(iii) *Let $\mathfrak{S}$ be a locally Noetherian formal prescheme, and $\mathfrak{X}'$ and $\mathfrak{Y}'$ locally Noetherian formal $\mathfrak{S}$-preschemes such that $\mathfrak{X}' \times_{\mathfrak{S}} \mathfrak{Y}'$ is locally Noetherian. If $\mathfrak{X}$ and $\mathfrak{Y}$ are locally Noetherian $\mathfrak{S}$-preschemes, and $f : \mathfrak{X} \to \mathfrak{X}'$ and $g : \mathfrak{Y} \to \mathfrak{Y}'$ are $\mathfrak{S}$-morphisms that are closed immersions, then $f \times_{\mathfrak{S}} g$ is a closed immersion.*

Proof. By (3.5.1), it again suffices to prove (i) and (ii).

To prove (i), we can assume that $\mathfrak{Y}$ (resp. $\mathfrak{Z}$) is a closed subprescheme of $\mathfrak{X}$ (resp. $\mathfrak{Y}$) defined by a coherent sheaf $\mathscr{J}$ (resp. $\mathscr{K}$) of ideals of $\mathscr{O}_{\mathfrak{X}}$ (resp. $\mathscr{O}_{\mathfrak{Y}}$); if ψ is the injection $\mathfrak{Y} \to \mathfrak{X}$ of underlying spaces, then $\psi_*(\mathscr{K})$ is a coherent sheaf of ideals of $\psi_*(\mathscr{O}_{\mathfrak{Y}}) = \mathscr{O}_{\mathfrak{X}}/\mathscr{J}$ (**0**, 5.3.12), and thus also a coherent $\mathscr{O}_{\mathfrak{X}}$-module (**0**, 5.3.10); the kernel $\mathscr{K}_1$ of $\mathscr{O}_{\mathfrak{X}} \to (\mathscr{O}_{\mathfrak{X}}/\mathscr{J})/\psi_*(\mathscr{K})$ is thus a coherent sheaf of ideals of $\mathscr{O}_{\mathfrak{X}}$ (**0**, 5.3.4), and $\mathscr{O}_{\mathfrak{X}}/\mathscr{K}_1$ is isomorphic to $\psi_*(\mathscr{O}_{\mathfrak{Y}}/\mathscr{K})$, which proves that $\mathfrak{Z}$ is an isomorphism to a closed subprescheme of $\mathfrak{X}$.

To prove (ii), we can immediately restrict to the case where $\mathfrak{S} = \mathrm{Spf}(A)$, $\mathfrak{X} = \mathrm{Spf}(B)$, and $\mathfrak{Y} = \mathrm{Spf}(C)$, with A a Noetherian $\mathfrak{J}$-adic ring, $B = A/\mathfrak{a}$ (where $\mathfrak{a}$ is an ideal of A), and C a Noetherian topological adic A-algebra. Everything then reduces to proving that the homomorphism $C \to C\widehat{\otimes}_A (A/\mathfrak{a})$ is *surjective*: but $A/\mathfrak{a}$ is an A-module of finite type, and its topology is the $\mathfrak{J}$-adic topology; it then follows from (**0**, 7.7.8) that $C\widehat{\otimes}_A (A/\mathfrak{a})$ can be identified with $C \otimes_A (A/\mathfrak{a}) = C/\mathfrak{a}C$, whence our claim. □

Corollary (10.14.6). — *Under the hypotheses of (10.14.5, ii), let $p : \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y} \to \mathfrak{X}$ and $q : \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y} \to \mathfrak{Y}$ be the projections, so that the diagram*

$$\begin{tikzcd} \mathfrak{X} \arrow[d, "f"'] & \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y} \arrow[l, "p"'] \arrow[d, "q"] \\ \mathfrak{S} & \mathfrak{Y} \arrow[l, "g"] \end{tikzcd}$$

commutes. For every coherent $\mathscr{O}_{\mathfrak{X}}$ module $\mathscr{F}$, we then have a canonical isomorphism of $\mathscr{O}_{\mathfrak{Y}}$-modules

$$u : g^* f_*(\mathscr{F}) \simeq q_* p^*(\mathscr{F}). \tag{10.14.6.1}$$

PROOF. We know that defining a homomorphism $g^* f_*(\mathscr{F}) \to q_* p^*(\mathscr{F})$ is equivalent to defining a homomorphism $f_*(\mathscr{F}) \to g_* q_* p^*(\mathscr{F}) = f_* p_* p^*(\mathscr{F})$ (**0**, 4.4.3): we take $u = f_*(\rho)$, where ρ is the canonical homomorphism $\mathscr{F} \to p_* p^*(\mathscr{F})$ (**0**, 4.4.3). To see that u is an isomorphism, we can immediately restrict to the case where $\mathfrak{S}$, $\mathfrak{X}$, and $\mathfrak{Y}$ are formal spectra of Noetherian adic rings A, B, and C (respectively), satisfying the conditions in (10.14.5, ii) above; we then have $\mathscr{F} = M^\Delta$, where M is an $(A/\mathfrak{a})$-module of finite type (10.10.5), and the two sides of (10.14.6.1) can then be identified, respectively, by (10.10.8), with $(C \otimes_A M)^\Delta$ and $((C/\mathfrak{a}C) \otimes_{A/\mathfrak{a}} M)^\Delta$, whence the corollary, since $(C/\mathfrak{a}C) \otimes_{A/\mathfrak{a}} M = (C \otimes_A (A/\mathfrak{a})) \otimes_{A/\mathfrak{a}} M$ is canonically identified with $C \otimes_A M$. □

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Corollary (10.14.7). — *Let X be a locally Noetherian usual prescheme, Y a closed subprescheme of X, j the canonical injection $Y \to X$, X' a closed subset of X, and $Y' = Y \cap X'$; then $\widehat{j} : Y_{/Y'} \to X_{/X'}$ is a closed immersion, and, for every coherent $\mathscr{O}_Y$-module $\mathscr{F}$, we have*

$$\widehat{j}_*(\mathscr{F}_{/Y'}) = (j_*(\mathscr{F}))_{/X'}.$$

PROOF. Since $Y' = j^{-1}(X')$, it suffices to use (10.9.9) and to apply (10.14.5) and (10.14.6). □

10.15. Separated formal preschemes

Definition (10.15.1). — Let $\mathfrak{S}$ be a formal prescheme, $\mathfrak{X}$ a formal $\mathfrak{S}$-prescheme, and $f : \mathfrak{X} \to \mathfrak{S}$ the structure morphism. We define the diagonal morphism $\Delta_{\mathfrak{X}|\mathfrak{S}} : \mathfrak{X} \to \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{X}$ (also denoted by $\Delta_{\mathfrak{X}}$) to be the morphism $(1_{\mathfrak{X}}, 1_{\mathfrak{X}})_{\mathfrak{S}}$. We say that $\mathfrak{X}$ is separated over $\mathfrak{S}$, or is a formal $\mathfrak{S}$-scheme, or that f is a separated morphism, if the image of the underlying space of $\mathfrak{X}$ under $\Delta_{\mathfrak{X}}$ is a closed subset of the underlying space of $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{X}$. We say that a formal prescheme $\mathfrak{X}$ is separated, or is a formal scheme, if it is separated over $\mathbf{Z}$.

Proposition (10.15.2). — *Suppose that the formal preschemes $\mathfrak{S}$ and $\mathfrak{X}$ are inductive limits of sequences (S_n) and (X_n) (respectively) of usual preschemes, and that the morphism $f : \mathfrak{X} \to \mathfrak{S}$ is the inductive limit of a sequence of morphisms $f_n : X_n \to S_n$. For f to be separated, it is necessary and sufficient for the morphism $f_0 : X_0 \to S_0$ to be separated.*

PROOF. Indeed, $\Delta_{\mathfrak{X}|\mathfrak{S}}$ is then the inductive limit of the sequence of morphisms $\Delta_{X_n|S_n}$ (10.7.4), and the image of the underlying space of $\mathfrak{X}$ (resp. of $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{X}$ under $\Delta_{\mathfrak{X}|\mathfrak{S}}$) is identical to the image of the underlying space of X_0 (resp. of $X_0 \times_{S_0} X_0$) under $\Delta_{X_0|S_0}$; whence the conclusion. □

Proposition (10.15.3). — *Suppose that all the formal preschemes (resp. morphisms of formal preschemes) in what follows are inductive limits of sequences of usual preschemes (resp. of morphisms of usual preschemes).*

(i) *The composition of any two separated morphisms is separated.*
(ii) *If $f : \mathfrak{X} \to \mathfrak{X}'$ and $g : \mathfrak{Y} \to \mathfrak{Y}'$ are separated $\mathfrak{S}$-morphisms, then $f \times_{\mathfrak{S}} g$ is separated.*
(iii) *If $f : \mathfrak{X} \to \mathfrak{Y}$ is a separated $\mathfrak{S}$-morphism, then the $\mathfrak{S}'$-morphism $f_{(\mathfrak{S}')}$ is separated for every extension $\mathfrak{S}' \to \mathfrak{S}$ of the base formal prescheme.*

(iv) *If the composition $g \circ f$ of two morphisms is separated, then f is separated.*

(In the above, it is implicit that if the same formal prescheme $\mathfrak{Z}$ is mentioned more than once in the same proposition, we consider it as the inductive limit of the *same* sequence (Z_n) of usual preschemes wherever it is mentioned, and the morphisms from $\mathfrak{Z}$ to another formal prescheme (resp. from a formal prescheme to $\mathfrak{Z}$) as inductive limits of morphisms from Z_n to some usual preschemes (resp. from some usual preschemes to Z_n)).

PROOF. With the notation of (10.15.2), we have, in fact, that $(g \circ f)_0 = g_0 \circ f_0$ and $(f \times_{\mathfrak{S}} g)_0 = f_0 \times_{S_0} g_0$; the claims of (10.15.3) are then immediate consequences of (10.15.2) and the corresponding claims in (5.5.1) for usual preschemes. □

We leave it to the reader to state, for the same type of formal preschemes and morphisms as in (10.15.3), the propositions corresponding to (5.5.5), (5.5.9), and (5.5.10) (by replacing "affine open subset" by "formal affine open subset satisfying condition (b) of (10.6.3)"). **I** | 213

A similar argument also shows that every *Noetherian* formal affine scheme is separated, which justifies the terminology.

Proposition (10.15.4). — *Let $\mathfrak{S}$ be a locally Noetherian formal prescheme, and $\mathfrak{X}$ and $\mathfrak{Y}$ locally Noetherian formal $\mathfrak{S}$-preschemes such that $\mathfrak{X}$ or $\mathfrak{Y}$ is of finite type over $\mathfrak{S}$ (so that $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is locally Noetherian) and such that $\mathfrak{Y}$ is separated over $\mathfrak{S}$. Let $f : \mathfrak{X} \to \mathfrak{Y}$ be an $\mathfrak{S}$-morphism; then the graph morphism $\Gamma_f(1_{\mathfrak{X}}, f)_{\mathfrak{S}} : \mathfrak{X} \to \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is a closed immersion.*

PROOF. We can assume that $\mathfrak{S}$ is the inductive limit of a sequence (S_n) of locally Noetherian preschemes, $\mathfrak{X}$ (resp. $\mathfrak{Y}$) the inductive limit of a sequence (X_n) (resp. (Y_n)) of S_n-preschemes, and f the inductive limit of a sequence $(f_n : X_n \to Y_n)$ of S_n-morphisms; then $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ is the inductive limit of the sequence $(X_n \times_{S_n} Y_n)$, and Γ_f the inductive limit of the sequence (Γ_{f_n}) (10.7.4); by hypothesis, Y_0 is separated over S_0 (10.15.2), so the space $\Gamma_{f_0}(X_0)$ is a closed subspace of $X_0 \times_{S_0} Y_0$; since the underlying spaces of $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$ (resp. $\Gamma_f(\mathfrak{X})$) and $X_0 \times_{S_0} Y_0$ (resp. $\Gamma_{f_0}(X_0)$) are identical, we already see that $\Gamma_f(\mathfrak{X})$ is a *closed* subspace of $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$. Now note that, when (U, V) runs over the set of pairs consisting of a Noetherian formal affine open subset U (resp. V) of $\mathfrak{X}$ (resp $\mathfrak{Y}$) such that $f(U) \subset V$, the open subsets $U \times_S V$ form a cover of $\Gamma_f(\mathfrak{X})$ in $\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y}$, and, if $f_U : U \to V$ is the restriction of f to U, then $\Gamma_{f_U} : U \to U \times_{\mathfrak{S}} V$ is the restriction of Γ_f to U. If we show that Γ_{f_U} is a closed immersion, then Γ_f will be a closed immersion (10.14.4), or, in other words, we are led to consider the case where $\mathfrak{S} = \mathrm{Spf}(A)$, $\mathfrak{X} = \mathrm{Spf}(B)$, and $\mathfrak{Y} = \mathrm{Spf}(C)$ are affine (with A, B, and C Noetherian adics), with f corresponding to a continuous A-homomorphism $\phi : C \to B$; then Γ_f corresponds to the unique continuous homomorphism $\omega : B \widehat{\otimes}_A C \to B$ which, when composed with the canonical homomorphisms $B \to B \widehat{\otimes}_A C$ and $C \to B \widehat{\otimes}_A C$, gives the identity and ϕ (respectively). But it is clear that ω is *surjective*, whence our claim. □

Corollary (10.15.5). — *Let $\mathfrak{S}$ be a locally Noetherian formal prescheme, and $\mathfrak{X}$ an $\mathfrak{S}$-prescheme of finite type; for $\mathfrak{X}$ to be separated over $\mathfrak{S}$, it is necessary and sufficient for the diagonal morphism $\mathfrak{X} \to \mathfrak{X} \times_{\mathfrak{S}} \mathfrak{X}$ to be a closed immersion.*

Proposition (10.15.6). — *A closed immersion $j : \mathfrak{Y} \to \mathfrak{X}$ of locally Noetherian formal preschemes is a separated morphism.*

PROOF. With the notation of (10.14.2), $j_0 : Y_0 \to X_0$ is a closed immersion, thus a separated morphism, and so it suffices to apply (10.15.2). □

Proposition (10.15.7). — *Let X be a locally Noetherian (usual) prescheme, X' a closed subset of X, and $\widehat{X} = X_{/X'}$. For $\widehat{X}$ to be separated, it is necessary and sufficient that $\widehat{X}_{\mathrm{red}}$ be separated, and it is sufficient that X be separated.*

PROOF. With the notation of (10.8.5), for $\widehat{X}$ to be separated, it is necessary and sufficient for X'_0 to be separated (10.15.2), and since $\widehat{X}_{\mathrm{red}} = (X'_0)_{\mathrm{red}}$, it is equivalent to ask for $\widehat{X}_{\mathrm{red}}$ to be separated (5.5.1, vi). □

CHAPTER II

Elementary global study of some classes of morphisms (EGA II)

Summary

§1. Affine morphisms.
§2. Homogeneous prime spectra.
§3. Homogeneous prime spectrum of a sheaf of graded algebras.
§4. Projective bundles; ample sheaves.
§5. Quasi-affine morphisms; quasi-projective morphisms; proper morphisms; projective morphisms.
§6. Integral morphisms and finite morphisms.
§7. Valuative criteria.
§8. Blowup schemes; projective cones; projective closure.

The various classes of morphisms studied in this chapter are used extensively in cohomological **II** | 5
methods; further study using these methods will be done in Chapter III, where we make particular use of §§2, 4, and 5 of Chapter II. On a first reading, §8 can be omitted: it supplements the formalism developed in §§1 and 3, reducing to easy applications of this formalism, and we will use it less consistently than the other results of this chapter.

§1. Affine morphisms

1.1. S-preschemes and $\mathscr{O}_S$-algebras

(1.1.1). Let S be a prescheme, X an S-prescheme, and $f : X \to S$ its structure morphism. We know (**0**, 4.2.4) that the direct image $f_*(\mathscr{O}_X)$ is an $\mathscr{O}_S$-algebra, which we denote $\mathscr{A}(X)$ when there is little **II** | 6
chance of confusion; if U is an open subset of S, then we have

$$\mathscr{A}(f^{-1}(U)) = \mathscr{A}(X)|U.$$

Similarly, for every $\mathscr{O}_X$-module $\mathscr{F}$ (resp. every $\mathscr{O}_X$-algebra $\mathscr{B}$), we write $\mathscr{A}(\mathscr{F})$ (resp. $\mathscr{A}(\mathscr{B})$) for the direct image $f_*(\mathscr{F})$ (resp. $f_*(\mathscr{B})$) which is an $\mathscr{A}(X)$-module (resp. an $\mathscr{A}(X)$-algebra) and not only an $\mathscr{O}_S$-module (resp. an $\mathscr{O}_S$-algebra).

(1.1.2). Let Y be a second S-prescheme, $g : Y \to S$ its structure morphism, and $h : X \to Y$ an S-morphism; we then have the commutative diagram

(1.1.2.1)
$$\begin{tikzcd} X \arrow[rr, "h"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S. & \end{tikzcd}$$

We have by definition $h = (\psi, \theta)$, where $\theta : \mathscr{O}_Y \to h_*(\mathscr{O}_X) = \psi_*(\mathscr{O}_X)$ is a homomorphism of sheaves of rings; we thus obtain (**0**, 4.2.2) a homomorphism of $\mathscr{O}_S$-algebras $g_*(\theta) : g_*(\mathscr{O}_Y) \to g_*(h_*(\mathscr{O}_X)) = f_*(\mathscr{O}_X)$, in other words, a homomorphism of $\mathscr{O}_S$-algebras $\mathscr{A}(Y) \to \mathscr{A}(X)$, which we denote by $\mathscr{A}(h)$. If $h' : Y \to Z$ is a second S-morphism, then it is immediate that $\mathscr{A}(h' \circ h) = \mathscr{A}(h) \circ \mathscr{A}(h')$. We have thus defined a *contravariant functor* $\mathscr{A}(X)$ from the category of S-preschemes to the category of $\mathscr{O}_S$-algebras.

Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module, $\mathscr{G}$ an $\mathscr{O}_Y$-module, and $u : \mathscr{G} \to \mathscr{F}$ an h-morphism, that is (**0**, 4.4.1) a homomorphism of $\mathscr{O}_Y$-modules $\mathscr{G} \to h_*(\mathscr{F})$. Then $g_*(u) : g_*(\mathscr{G}) \to g_*(h_*(\mathscr{F})) = f_*(\mathscr{F})$ is a homomorphism $\mathscr{A}(\mathscr{G}) \to \mathscr{A}(\mathscr{F})$ of $\mathscr{O}_S$-modules, which we denote by $\mathscr{A}(u)$; in addition, the pair $(\mathscr{A}(h), \mathscr{A}(u))$ form a *di-homomorphism* from the $\mathscr{A}(Y)$-module $\mathscr{A}(\mathscr{G})$ to the $\mathscr{A}(X)$-module $\mathscr{A}(\mathscr{F})$.

(1.1.3). If we fix the prescheme S, then we can consider the pairs $(X, \mathscr{F})$, where X is an S-prescheme and $\mathscr{F}$ is an $\mathscr{O}_X$-module, as forming a *category*, by defining a *morphism* $(X, \mathscr{F}) \to (Y, \mathscr{G})$ as a pair (h, u), where $h : X \to Y$ is an S-morphism and $u : \mathscr{G} \to \mathscr{F}$ is an h-morphism. We can then say that $(\mathscr{A}(X), \mathscr{A}(\mathscr{F}))$ is a *contravariant functor* with values in the category whose objects are pairs consisting of an $\mathscr{O}_S$-algebra and a module over that algebra, and the morphisms are the di-homomorphisms.

1.2. Affine preschemes over a prescheme

Definition (1.2.1). — Let X be an S-prescheme, and $f : X \to S$ its structure morphism. We say that X is *affine over* S if there exists a cover (S_α) of S by affine open sets such that for all α, the prescheme induced by X on the open set $f^{-1}(S_\alpha)$ is affine.

Example (1.2.2). — Every closed subprescheme of S is an affine S-prescheme over S ((**I**, 4.2.3) and (**I**, 4.2.4)).

Remark (1.2.3). — An affine prescheme X over S is not necessarily an affine scheme, as the example $X = S$ shows (1.2.2). On the other hand, if an affine scheme X is an S-prescheme, then X is not necessarily affine over S (see Example (1.3.3)). However, remember that if S is a *scheme*, then every **II | 7**
S-prescheme which is an affine scheme is affine over S (**I**, 5.5.10).

Proposition (1.2.4). — *Every S-prescheme which is affine over S is separated over S (in other words, it is an S-scheme).*

PROOF. This follows immediately from (**I**, 5.5.5) and (**I**, 5.5.8). □

Proposition (1.2.5). — *Let X be an S-scheme affine over S, and $f : X \to S$ its structure morphism. For every open $U \subset S$, $f^{-1}(U)$ is affine over U.*

PROOF. By Definition (1.2.1), we can reduce to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine; then $f = ({}^a\phi, \widetilde{\phi})$, where $\phi : A \to B$ is a homomorphism. As the $D(g)$ for $g \in A$ form a basis for S, we reduce to the case where $U = D(g)$; but we then know that $f^{-1}(U) = D(\phi(g))$ (**I**, 1.2.2.2), hence the proposition. □

Proposition (1.2.6). — *Let X be an S-scheme affine over S, and $f : X \to S$ its structure morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $f_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_S$-module.*

PROOF. Taking into account Proposition (1.2.4), this follows from (**I**, 9.2.2, a). □

In particular, the $\mathscr{O}_S$-algebra $\mathscr{A}(X) = f_*(\mathscr{O}_X)$ is *quasi-coherent*.

Proposition (1.2.7). — *Let X be an S-scheme affine over S. For every S-prescheme Y, the map $h \mapsto \mathscr{A}(h)$ from the set* $\operatorname{Hom}_S(Y, X)$ *to the set* $\operatorname{Hom}(\mathscr{A}(X), \mathscr{A}(Y))$ (1.1.2) *is bijective.*

PROOF. Let $f : X \to S$ and $g : Y \to S$ be the structure morphisms. First, suppose that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine; we must prove that for every homomorphism $\omega : f_*(\mathscr{O}_X) \to g_*(\mathscr{O}_Y)$ of $\mathscr{O}_S$-algebras, there exists a unique S-morphism $h : Y \to X$ such that $\mathscr{A}(h) = \omega$. By definition, for every open $U \subset S$, ω defines a homomorphism $\omega_U = \Gamma(U, \omega) : \Gamma(f^{-1}(U), \mathscr{O}_X) \to \Gamma(g^{-1}(U), \mathscr{O}_Y)$ of $\Gamma(U, \mathscr{O}_S)$-algebras. In particular, for $U = S$, this gives a homomorphism $\phi : \Gamma(X, \mathscr{O}_X) \to \Gamma(Y, \mathscr{O}_Y)$ of $\Gamma(S, \mathscr{O}_S)$-algebras, to which corresponds a well-defined S-morphism $h : Y \to X$, since X is affine (**I**, 2.2.4). It remains to prove that $\mathscr{A}(h) = \omega$, or, in other words, that, for every open set U of a basis for S, ω_U coincides with the homomorphism of algebras ϕ_U corresponding to the S-morphism $g^{-1}(U) \to f^{-1}(U)$, a restriction of h. We can reduce to the case where $U = D(\lambda)$, with $\lambda \in S$; then, if $f = ({}^a\rho, \widetilde{\rho})$, where $\rho : A \to B$ is a ring homomorphism, we have $f^{-1}(U) = D(\mu)$, where $\mu = \rho(\lambda)$, and $\Gamma(f^{-1}(U), \mathscr{O}_X)$ is the ring of fractions B_μ; the diagram

$$\begin{tikzcd} B \arrow[r, "\phi"] \arrow[d] & \Gamma(Y, \mathscr{O}_Y) \arrow[d] \\ B_\mu \arrow[r, "\phi_U"] & \Gamma(g^{-1}(U), \mathscr{O}_Y) \end{tikzcd}$$

is commutative, and so is the analogous diagram where ϕ_U is replaced by ω_U; the equality $\phi_U = \omega_U$ then follows from the universal property of rings of fractions (**0**, 1.2.4).

We now pass to the general case; let (S_α) be a cover of S by affine open sets such that the $f^{-1}(S_\alpha)$ are affine. Then every homomorphism $\omega : \mathscr{A}(X) \to \mathscr{A}(Y)$ of $\mathscr{O}_S$-algebras gives by restriction a family of homomorphisms **II | 8**

$$\omega_\alpha : \mathscr{A}(f^{-1}(S_\alpha)) \longrightarrow \mathscr{A}(g^{-1}(S_\alpha))$$

of $\mathscr{O}_{S_\alpha}$-algebras, hence a family of S_α-morphisms $h_\alpha : g^{-1}(S_\alpha) \to f^{-1}(S_\alpha)$ by the above. It remains to see that for every affine open set of a basis for $S_\alpha \cap S_\beta$, the restriction of h_α and h_β to $g^{-1}(U)$ coincide, which is evident since by the above, these restrictions both correspond to the homomorphism $\mathscr{A}(X)|U \to \mathscr{A}(Y)|U$, a restriction of ω. □

Corollary (1.2.8). — *Let X and Y be two S-schemes which are affine over S. For an S-morphism $h : Y \to X$ to be an isomorphism, it is necessary and sufficient for $\mathscr{A}(h) : \mathscr{A}(X) \to \mathscr{A}(Y)$ to be an isomorphism.*

PROOF. This follows immediately from Proposition (1.2.7) and from the functorial nature of $\mathscr{A}(X)$. □

1.3. Affine preschemes over S associated to an $\mathscr{O}_S$-algebra

Proposition (1.3.1). — *Let S be a prescheme. For every quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{B}$, there exists a prescheme X affine over S, defined up to unique S-isomorphism, such that $\mathscr{A}(X) = \mathscr{B}$.*

PROOF. The uniqueness follows from Corollary (1.2.8); we prove the existence of X. For every affine open $U \subset S$, let X_U be the prescheme $\operatorname{Spec}(\Gamma(U, \mathscr{B}))$; as $\Gamma(U, \mathscr{B})$ is a $\Gamma(U, \mathscr{O}_S)$-algebras, X_U is an S-prescheme (**I**, 1.6.1). In addition, as $\mathscr{B}$ is quasi-coherent, the $\mathscr{O}_S$-algebra $\mathscr{A}(X_U)$ canonically identifies with $\mathscr{B}|U$ ((**I**, 1.3.7), (**I**, 1.3.13), (**I**, 1.6.3)). Let V be a second affine open subset of S, and let $X_{U,V}$ be the prescheme induced by X_U on $f_U^{-1}(U \cap V)$, where f_U denotes the structure morphism $X_U \to S$; $X_{U,V}$ and $X_{V,U}$ are affine over $U \cap V$ (1.2.5), and by definition $\mathscr{A}(X_{U,V})$ and $\mathscr{A}(X_{V,U})$ canonically identify with $\mathscr{B}|(U \cap V)$. Hence there is (1.2.8) a canonical S-isomorphism $\theta_{U,V} : X_{U,V} \to X_{U,V}$; in addition, if W is a third affine open subset of S, and if $\theta'_{U,V}$, $\theta'_{V,W}$, and $\theta'_{U,W}$ are the restrictions of $\theta_{U,V}$, $\theta_{V,W}$, and $\theta_{U,W}$ to the inverse images of $U \cap V \cap W$ in X_V, X_W, and X_W respectively under the structure morphisms, then we have $\theta'_{U,V} \circ \theta'_{V,W} = \theta'_{U,W}$. As a result, there exists a prescheme X, a cover (T_U) of X by affine open sets, and for every U an isomorphism $\phi_U : X_U \to T_U$, such that ϕ_U maps $f_U^{-1}(U \cap V)$ to $T_U \cap T_V$, and we have $\theta_{U,V} = \phi_U^{-1} \circ \phi_V$ (**I**, 2.3.1). The morphism $g_U = f_U \circ \phi_U^{-1}$ makes T_U an S-prescheme, and the morphisms g_U and g_V coincide on $T_U \cap T_V$, hence X is an S-prescheme. In addition, it is clear by definition that X is affine over S and that $\mathscr{A}(T_U) = \mathscr{B}|U$, hence $\mathscr{A}(X) = \mathscr{B}$. □

We say that the S-scheme X defined in this way is *associated to the $\mathscr{O}_S$-algebra $\mathscr{B}$*, or is the *spectrum of $\mathscr{B}$*, and we denote it by $\operatorname{Spec}(\mathscr{B})$.

Corollary (1.3.2). — *Let X be a prescheme affine over S, $f : X \to S$ the structure morphism. For every affine open $U \subset S$, the induced prescheme on $f^{-1}(U)$ is the affine scheme with ring $\Gamma(U, \mathscr{A}(X))$.*

PROOF. As we can suppose that X is associated to an $\mathscr{O}_S$-algebra by Propositions (1.2.6) and **II | 9**
(1.3.1), the corollary follows from the construction of X described in Proposition (1.3.1). □

Example (1.3.3). — Let S be the affine plane over a field K, where the point 0 has been doubled (**I**, 5.5.11); with the notation of (**I**, 5.5.11), S is the union of two affine open sets Y_1 and Y_2; if f is the open immersion $Y_1 \to S$, then $f^{-1}(Y_2) = Y_1 \cap Y_2$ is not an affine open set in Y_1 (*loc. cit.*), hence we have an example of an affine scheme which is not affine over S.

Corollary (1.3.4). — *Let S be an affine scheme; for an S-prescheme X to be affine over S, it is necessary and sufficient for X to be an affine scheme.*

Corollary (1.3.5). — *Let X be a prescheme affine over a prescheme S, and let Y be an X-prescheme. For Y to be affine over S, it is necessary and sufficient for Y to be affine over X.*

PROOF. We immediately reduce to the case where S is an affine scheme, and then we can reduce to the case where X is an affine scheme (1.3.4); the two conditions of the statement then give that Y is an affine scheme (1.3.4). □

(1.3.6). Let X be a prescheme affine over S. To define a prescheme Y affine *over* X, it is equivalent, by Corollary (1.3.5), to give a prescheme Y affine *over* S, and an S-morphism $g : Y \to X$; in other words (Proposition (1.3.1) and (1.2.7)), it is equivalent to give a quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{B}$ and a homomorphism $\mathscr{A}(X) \to \mathscr{B}$ of $\mathscr{O}_S$-algebras (which can be considered as defining on $\mathscr{B}$ an $\mathscr{A}(X)$-algebra structure). If $f : X \to S$ is the structure morphism, then we have $\mathscr{B} = f_*(g_*(\mathscr{O}_Y))$.

Corollary (1.3.7). — *Let X be a prescheme affine over S; for X to be of finite type over S, it is necessary and sufficient for the quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{A}(X)$ to be of finite type* (**I**, 9.6.2).

PROOF. By definition (**I**, 9.6.2), we can reduce to the case where S is affine; then X is an affine scheme (1.3.4), and if $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, then $\mathscr{A}(X)$ is the $\mathscr{O}_S$-algebra $\widetilde{B}$; as $\Gamma(U, \widetilde{B}) = B$, the corollary follows from (**I**, 9.6.2) and (**I**, 6.3.3). □

Corollary (1.3.8). — *Let X be a prescheme affine over S; for X to be reduced, it is necessary and sufficient for the quasi-coherent $\mathscr{O}_X$-algebra $\mathscr{A}(X)$ to be reduced* (**0**, 4.1.4).

PROOF. The question is local on S; by Corollary (1.3.2), the corollary follows from (**I**, 5.1.1) and (**I**, 5.1.4). □

1.4. Quasi-coherent sheaves over an affine prescheme over S

Proposition (1.4.1). — *Let X be a prescheme affine over S, Y an S-prescheme, and $\mathscr{F}$ (resp. $\mathscr{G}$) a quasi-coherent $\mathscr{O}_X$-module (resp. an $\mathscr{O}_Y$-module). Then the map $(h, u) \mapsto (\mathscr{A}(h), \mathscr{A}(u))$ from the set of morphism $(Y, \mathscr{G}) \to (X, \mathscr{F})$ to the set of di-homomorphisms $(\mathscr{A}(X), \mathscr{A}(\mathscr{F})) \to (\mathscr{A}(Y), \mathscr{A}(\mathscr{G}))$ ((1.1.2) and (1.1.3)) is bijective.*

PROOF. The proof follows exactly as that of Proposition (1.2.7) by using (**I**, 2.2.5) and (**I**, 2.2.4), and the details are left to the reader. □

Corollary (1.4.2). — *If, in addition to the hypotheses of Proposition (1.4.1), we suppose that Y is affine over S, then for (h, u) to be an isomorphism, it is necessary and sufficient for $(\mathscr{A}(h), \mathscr{A}(u))$ to be a di-isomorphism.*

Proposition (1.4.3). — *For every pair $(\mathscr{B}, \mathscr{M})$ consisting of a quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{B}$ and a quasi-coherent $\mathscr{B}$-module $\mathscr{M}$* (considered as an $\mathscr{O}_S$-module or as a $\mathscr{B}$-module, which are equivalent (**I**, 9.6.1)), *there exists a pair $(X, \mathscr{F})$ consisting of a prescheme X affine over S and of a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, such that $\mathscr{A}(X) = \mathscr{B}$ and $\mathscr{A}(\mathscr{F}) = \mathscr{M}$; in addition, this couple is determined up to unique isomorphism.* **II** | 10

PROOF. The uniqueness follows from Proposition (1.4.1) and Corollary (1.4.2); the existence is proved as in Proposition (1.3.1), and we leave the details to the reader. □

We denote by $\widetilde{\mathscr{M}}$ the $\mathscr{O}_X$-module $\mathscr{F}$, and we say that it is *associated* to the quasi-coherent $\mathscr{B}$-module $\mathscr{M}$; for every affine open $U \subset S$, $\mathscr{M}|p^{-1}(U)$ (where p is the structure morphism $X \to S$) is canonically isomorphic to $(\Gamma(U, \mathscr{M}))^\sim$.

Corollary (1.4.4). — *On the category of quasi-coherent $\mathscr{B}$-modules, $\widetilde{\mathscr{M}}$ is an additive covariant exact functor in $\mathscr{M}$, which commutes with inductive limits and direct sums.*

PROOF. We immediately reduce to the case where S is affine, and the corollary then follows from (**I**, 1.3.5), (**I**, 1.3.9), and (**I**, 1.3.11). □

Corollary (1.4.5). — *Under the hypotheses of Proposition (1.4.3), for $\widetilde{\mathscr{M}}$ to be an $\mathscr{O}_X$-module of finite type, it is necessary and sufficient for $\mathscr{M}$ to be a $\mathscr{B}$-module of finite type.*

PROOF. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is an affine scheme. Then $\mathscr{B} = \widetilde{B}$, where B is an A-algebra of finite type (**I**, 9.6.2), and $\mathscr{M} = \widetilde{M}$, where M is a B-module (**I**, 1.3.13); *over the prescheme* X, $\mathscr{O}_X$ is associated to the ring B and $\widetilde{\mathscr{M}}$ to the B-module M; for $\widetilde{\mathscr{M}}$ to be of finite type, it is therefore necessary and sufficient for M to be of finite type (**I**, 1.3.13), hence our assertion. □

Proposition (1.4.6). — *Let Y be a prescheme affine over S, X and X' two preschemes affine over Y* (hence also over S (1.3.5)). *Let $\mathscr{B} = \mathscr{A}(Y)$, $\mathscr{A} = \mathscr{A}(X)$, and $\mathscr{A}' = \mathscr{A}(X')$. Then $X \times_Y X'$ is affine over Y* (thus also over S), *and $\mathscr{A}(X \times_Y X')$ canonically identifies with $\mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$.*

PROOF. By (I, 9.6.1), $\mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$ is a quasi-coherent $\mathscr{B}$-algebra, thus also a quasi-coherent $\mathscr{O}_S$-algebra (I, 9.6.1); let Z be the spectrum of $\mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$ (1.3.1). The canonical $\mathscr{B}$-homomorphisms $\mathscr{A} \to \mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$ and $\mathscr{A}' \to \mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$ correspond (1.2.7) to Y-morphisms $Z \to X$ and $p' : Z \to X'$. To see that the triple (Z, p, p') is a product $X \times_Y X'$, we can reduce to the case where S is an affine scheme with ring C (I, 3.2.6.4). But then Y, X, and X' are affine schemes (1.3.4) whose rings B, A, and A' are C-algebras such that $\mathscr{B} = \widetilde{B}$, $\mathscr{A} = \widetilde{A}$, and $\mathscr{A}' = \widetilde{A'}$. We then know (I, 1.3.13) that $\mathscr{A} \otimes_{\mathscr{B}} \mathscr{A}'$ canonically identifies with the $\mathscr{O}_S$-algebra $(A \otimes_B A')^\sim$, hence the ring $A(Z)$ identifies with $A \otimes_B A'$ and the morphisms p and p' correspond to the canonical homomorphisms $A \to A \otimes_B A'$ and $A' \to A \otimes_B A'$. The proposition then follows from (I, 3.2.2). □

Corollary (1.4.7). — *Let $\mathscr{F}$ (resp. $\mathscr{F}'$) be a quasi-coherent $\mathscr{O}_X$-module (resp. $\mathscr{O}_{X'}$-module); then $\mathscr{A}(\mathscr{F} \otimes_Y \mathscr{F}')$ canonically identifies with $\mathscr{A}(\mathscr{F}) \otimes_{\mathscr{A}(Y)} \mathscr{A}(\mathscr{F}')$.*

PROOF. We know that $\mathscr{F} \otimes_Y \mathscr{F}'$ is quasi-coherent over $X \times_Y X'$ (I, 9.1.2). Let $g : Y \to S$, $f : X \to Y$, and $f' : X' \to Y$ be the structure morphisms, such that the structure morphism $h : Z \to S$ is equal to $g \circ f \circ p$ and to $g \circ f' \circ p'$. We define a canonical homomorphism II | 11

$$\mathscr{A}(\mathscr{F}) \otimes_{\mathscr{A}(Y)} \mathscr{A}(\mathscr{F}') \longrightarrow \mathscr{A}(\mathscr{F} \otimes_Y \mathscr{F}')$$

in the following way: for every open $U \subset S$, we have canonical homomorphisms $\Gamma(f^{-1}(g^{-1}(U)), \mathscr{F}) \to \Gamma(h^{-1}(U), p^*(\mathscr{F}))$ and $\Gamma(f'^{-1}(g^{-1}(U)), \mathscr{F}') \to \Gamma(h^{-1}(U), p'^*(\mathscr{F}'))$ (0, 4.4.3), thus we obtain a canonical homomorphism

$$\Gamma(f^{-1}(g^{-1}(U)), \mathscr{F}) \otimes_{\Gamma(g^{-1}(U), \mathscr{O}_Y)} \Gamma(f'^{-1}(g^{-1}(U)), \mathscr{F}') \longrightarrow \Gamma(h^{-1}(U), p^*(\mathscr{F})) \otimes_{\Gamma(h^{-1}(U), \mathscr{O}_Z)} \Gamma(h^{-1}(U), p'^*(\mathscr{F}')).$$

To see that we have defined an isomorphism of $\mathscr{A}(Z)$-modules, we can reduce to the case where S (and as a result X, X', Y, and $X \times_Y X'$) are affine scheme, and (with the notation of Proposition (1.4.6)), $\mathscr{F} = \widetilde{M}$, $\mathscr{F}' = \widetilde{M'}$, where M (resp. M') is an A-module (resp. an A'-module). Then $\mathscr{F} \otimes_Y \mathscr{F}'$ identifies with the sheaf on $X \times_Y X'$ associated to the $(A \otimes_B A')$-module $M \otimes_B M'$ (I, 9.1.3), and the corollary follows from the canonical identification of the $\mathscr{O}_S$-modules $(M \otimes_B M')^\sim$ and $\widetilde{M} \otimes_{\widetilde{B}} \widetilde{M'}$ (where M, M', and B are considered as C-modules) ((I, 1.3.12) and (I, 1.6.3)). □

If we apply Corollary (1.4.7) in particular to the case where $X = Y$ and $\mathscr{F}' = \mathscr{O}_{X'}$, then we see that the $\mathscr{A}'$-module $\mathscr{A}(f'^*(\mathscr{F}))$ identifies with $\mathscr{A}(\mathscr{F}) \otimes_{\mathscr{B}} \mathscr{A}'$.

(1.4.8). In particular, when $X = X' = Y$ (X being affine over S), we see that if $\mathscr{F}$ and $\mathscr{G}$ are two quasi-coherent $\mathscr{O}_X$-modules, then we have

$$\mathscr{A}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) = \mathscr{A}(\mathscr{F}) \otimes_{\mathscr{A}(X)} \mathscr{A}(\mathscr{G}) \tag{1.4.8.1}$$

up to canonical functorial isomorphism. If in addition $\mathscr{F}$ admits a finite presentation, then it follows from (I, 1.6.3) and (I, 1.3.12) that

$$\mathscr{A}(\mathscr{H}om_X(\mathscr{F}, \mathscr{G})) = \mathscr{H}om_{\mathscr{A}(X)}(\mathscr{A}(\mathscr{F}), \mathscr{A}(\mathscr{G})) \tag{1.4.8.2}$$

up to canonical isomorphism.

Remark (1.4.9). — If X and X' are two preschemes affine over S, then the sum $X \sqcup X'$ is also affine over S, as the sum of two affine schemes is an affine scheme.

Proposition (1.4.10). — *Let S be a prescheme, $\mathscr{B}$ a quasi-coherent $\mathscr{O}_S$-algebra, and $X = \operatorname{Spec}(\mathscr{B})$. For a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{B}$, $\widetilde{\mathscr{J}}$ is quasi-coherent sheaf of ideals of $\mathscr{O}_X$, and the closed subprescheme Y of X defined by $\widetilde{\mathscr{J}}$ is canonically isomorphic to* $\operatorname{Spec}(\mathscr{B}/\mathscr{J})$.

PROOF. It follows immediately from (I, 4.2.3) that Y is affine over S; by Proposition (1.3.1), we reduce to the case where S is affine, and the proposition then follows immediately from (I, 4.1.2). □

We can also express the result of Proposition (1.4.10) by saying that if $h : \mathscr{B} \to \mathscr{B}'$ is a *surjective* homomorphism of quasi-coherent $\mathscr{O}_S$-algebras, $\mathscr{A}(h) : \operatorname{Spec}(\mathscr{B}') \to \operatorname{Spec}(\mathscr{B})$ is a *closed immersion*.

Proposition (1.4.11). — *Let S be a prescheme, $\mathscr{B}$ a quasi-coherent $\mathscr{O}_S$-algebra, and $X = \operatorname{Spec}(\mathscr{B})$. For every quasi-coherent sheaf of ideals $\mathscr{K}$ of $\mathscr{O}_S$, we have (denoting by f the structure morphism $X \to S$) $f^*(\mathscr{K})\mathscr{O}_X = (\mathscr{K}\mathscr{B})^\sim$ up to canonical isomorphism.* II | 12

PROOF. Since the questions is local on S, we can reduce to the case where $S = \mathrm{Spec}(A)$ is affine, and in this case the proposition is none other than (**I**, 1.6.9). □

1.5. Change of base prescheme

Proposition (1.5.1). — *Let X be a prescheme affine over S. For every extension $g : S' \to S$ of the base prescheme, $X' = X_{(S')} = X \times_S S'$ is affine over S'.*

PROOF. If f' is the projection $X' \to S'$, then it suffices to prove that $f'^{-1}(U')$ is an affine open set for every affine open subset U' of S' such that $g(U')$ is contained in an affine open subset U of S (1.2.1); we can thus reduce to the case where S and S' are affine, and it suffices to prove that X' is then an affine scheme (1.3.4). But then (1.3.4) X is an affine scheme, and if A, A', and B are the rings of S, S', and X respectively, then we know that X' is the affine scheme with ring $A' \otimes_A B$ (**I**, 3.2.2). □

Corollary (1.5.2). — *Under the hypotheses of Proposition (1.5.1), let $f : X \to S$ be the structure morphism, $f' : X' \to S'$ and $g' : X' \to X$ the projections, such that the diagram*

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "g"'] \end{tikzcd}$$

is commutative. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, there exists a canonical isomorphism of $\mathscr{O}_{S'}$-modules

$$u : g^*(f_*(\mathscr{F})) \simeq f'_*(g'^*(\mathscr{F})). \tag{1.5.2.1}$$

In particular, there exists a canonical isomorphism from $\mathscr{A}(X')$ to $g^*(\mathscr{A}(X))$.

PROOF. To define u, it suffices to define a homomorphism

$$v : f_*(\mathscr{F}) \longrightarrow g_*(f'_*(g'^*(\mathscr{F}))) = f_*(g'_*(g'^*(\mathscr{F})))$$

and to set $u = v^\sharp$ (**0**, 4.4.3). We take $v = f_*(\rho)$, where ρ is the canonical homomorphism $\mathscr{F} \to g'_*(g'^*(\mathscr{F}))$ (**0**, 4.4.3). To prove that u is an isomorphism, we can reduce to the case where S and S', hence X and X', are affine; with the notation of Proposition (1.5.1), we then have $\mathscr{F} = \widetilde{M}$, where M is a B-module. We then note immediately that $g^*(f_*(\mathscr{F}))$ and $f'_*(g'^*(\mathscr{F}))$ are both equal to the $\mathscr{O}_{S'}$-module associated to the A'-module $A' \otimes_A M$ (where M is considered as an A-module), and that u is the homomorphism associated to the identity ((**I**, 1.6.3), (**I**, 1.6.5), (**I**, 1.6.7)). □

Remark (1.5.3). — We do not have that Corollary (1.5.2) remains true when X is not assumed affine over S, even when $S' = \mathrm{Spec}(k(s))$ ($s \in S$) and $S' \to S$ is the canonical morphism (**I**, 2.4.5)—in which case X' is none other than the *fibre* $f^{-1}(s)$ (**I**, 3.6.2). In other words, when X is not affine over S, the operation "direct image of quasi-coherent sheaves" does not commute with the operation **II** | 13
of "passing to fibres". However, we will see in Chapter III (**III**, 4.2.4) a result in this sense, of an "asymptotic" nature, valid for *coherent* sheaves on X when f is proper (5.4) and S is Noetherian.

Corollary (1.5.4). — *For every prescheme X affine over S and every $s \in S$, the fibre $f^{-1}(s)$ (where f denoted the structure morphism $X \to S$) is an affine scheme.*

PROOF. It suffices to apply Proposition (1.5.1) with $S' = \mathrm{Spec}(k(s))$ and to use Corollary (1.3.4). □

Corollary (1.5.5). — *Let X be an S-prescheme, S' a prescheme affine over S; then $X' = X_{(S')}$ is a prescheme affine over X. In addition, if $f : X \to S$ is the structure morphism, then there is a canonical isomorphism of $\mathscr{O}_X$-algebras $\mathscr{A}(X') \simeq f^*(\mathscr{A}(S'))$, and for every quasi-coherent $\mathscr{A}(S')$-module $\mathscr{M}$, a canonical di-isomorphism $f^*(\mathscr{M}) \simeq \mathscr{A}(\widetilde{f'^*(\mathscr{M})})$, denoting by $f' = f_{(S')}$ the structure morphism $X' \to S'$.*

PROOF. It suffices to swap the roles of X and S' in (1.5.1) and (1.5.2). □

(1.5.6). Now let S, S' be two preschemes, $q : S' \to S$ a morphism, $\mathscr{B}$ (resp. $\mathscr{B}'$) a quasi-coherent $\mathscr{O}_S$-algebra (resp. $\mathscr{O}_{S'}$-algebra), $u : \mathscr{B} \to \mathscr{B}'$ a q-morphism (that is, a homomorphism $\mathscr{B} \to q_*(\mathscr{B}')$ of $\mathscr{O}_S$-algebras). If $X = \operatorname{Spec}(\mathscr{B})$, $X' = \operatorname{Spec}(\mathscr{B}')$, then we canonically obtain a morphism

$$v = \operatorname{Spec}(u) : X' \longrightarrow X$$

such that the diagram

(1.5.6.1)
$$\begin{tikzcd} X' \arrow[r, "v"] \arrow[d] & X \arrow[d] \\ S' \arrow[r, "q"] & S \end{tikzcd}$$

is commutative (the vertical arrows being the structure morphisms). Indeed, the data of u is equivalent to that of a homomorphism of quasi-coherent $\mathscr{O}_{S'}$-algebras $u^\sharp : q^*(\mathscr{B}) \to \mathscr{B}'$ (**0**, 4.4.3); this thus canonically defines an S'-morphism

$$w : \operatorname{Spec}(\mathscr{B}') \longrightarrow \operatorname{Spec}(q^*(\mathscr{B}))$$

such that $\mathscr{A}(w) = u^\sharp$ (1.2.7). On the other hand, it follows from (1.5.2) that $\operatorname{Spec}(q^*(\mathscr{B}))$ canonically identifies with $X \times_S S'$; the morphism v is the composition $X' \xrightarrow{w} X \times_S S' \xrightarrow{p_1} X$ of w with the first projection, and the commutativity of (1.5.6.1) follows from the definitions. Let U (resp. U') be an affine open of S (resp. S') such that $q(U') \subset U$, $A = \Gamma(U, \mathscr{O}_S)$, $A' = \Gamma(U', \mathscr{O}_{S'})$ their rings, $B = \Gamma(U, \mathscr{B})$, $B' = \Gamma(U', \mathscr{B}')$; the restriction of u to a $(q|U')$-morphism: $\mathscr{B}|U \to \mathscr{B}'|U'$ corresponds to a di-homomorphism of algebras $B \to B'$; if V, V' are the inverse images of U, U' in X, X' respectively, under the structure morphisms, then the morphism $V' \to V$, the restriction of v, corresponds (**I**, 1.7.3) to the above di-homomorphism.

(1.5.7). Under the same hypotheses as in (1.5.6), let $\mathscr{M}$ be a quasi-coherent $\mathscr{B}$-module; there is then a canonical isomorphism of $\mathscr{O}_{X'}$-modules

(1.5.7.1)
$$v^*(\widetilde{\mathscr{M}}) \simeq (q^*(\mathscr{M}) \otimes_{q^*(\mathscr{B})} \mathscr{B}')^\sim.$$

Indeed, the canonical isomorphism (1.5.2.1) gives a canonical isomorphism from $p_1^*(\widetilde{\mathscr{M}})$ to the sheaf on $\operatorname{Spec}(q^*(\mathscr{B}))$ associated to the $q^*(\mathscr{B})$-module $q^*(\mathscr{M})$, and it then suffices to apply (1.4.7). **II** | 14

1.6. Affine morphisms

(1.6.1). We say that a morphism $f : X \to Y$ of preschemes is *affine* if it defines X as a prescheme affine over Y. The properties of preschemes affine over another translates as follows in this language:

Proposition (1.6.2). —

(i) *A closed immersion is affine.*
(ii) *The composition of two affine morphisms is affine.*
(iii) *If $f : X \to Y$ is an affine S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(Y')}$ is affine for every base change $S' \to S$.*
(iv) *If $f : X \to Y$ and $f' : X' \to Y'$ are two affine S-morphisms, then*

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'$$

is affine.
(v) *If $f : X \to Y$ and $g : Y \to Z$ are two morphisms such that $g \circ f$ is affine and g is separated, then f is affine.*
(vi) *If f is affine, then so if f_{red}.*

Proof. By (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). But (i) is none other than Example (1.2.2), and (ii) is none other than Corollary (1.3.5); finally, (iii) follows from Proposition (1.5.1), since $X_{(S')}$ identifies with the product $X \times_Y Y_{(S')}$ (**I**, 3.3.11). □

Corollary (1.6.3). — *If X is an affine scheme and Y is a scheme, then every morphism $f : X \to Y$ is affine.*

Proposition (1.6.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type. For f to be affine, it is necessary and sufficient for f_{red} to be.*

PROOF. By (1.6.2, vi), we see only need to prove the sufficiency of the condition. It suffices to prove that if Y is affine and Noetherian, then X is affine; but Y_{red} is then affine, so the same is true for X_{red} by hypothesis. Now X is Noetherian, so the conclusion follows from (**I**, 6.1.7). □

1.7. Vector bundle associated to a sheaf of modules

(1.7.1). Let A be a ring, E an A-module. Recall that we call the *symmetric algebra* on E and denote by $\mathbf{S}(E)$ (or $\mathbf{S}_A(E)$) the quotient algebra of the tensor algebra $\mathbf{T}(E)$ by the two-sided ideal generated by the elements $x \otimes y - y \otimes x$, where x and y vary over E. The algebra $\mathbf{S}(E)$ is characterized by the following universal property: if σ is the canonical map $E \to \mathbf{S}(E)$ (obtained by composing $E \to \mathbf{T}(E)$ with the canonical map $\mathbf{T}(E) \to \mathbf{S}(E)$), then every A-linear map $E \to B$, where B is a *commutative* A-algebra, factors uniquely as $E \xrightarrow{\sigma} \mathbf{S}(E) \xrightarrow{g} B$, where g is an A-homomorphism *of algebras*. We immediately deduce from this characterization that for two A-modules E and F, we have

$$\mathbf{S}(E \oplus F) = \mathbf{S}(E) \otimes \mathbf{S}(F)$$

up to canonical isomorphism; in addition, $\mathbf{S}(E)$ is a covariant functor in E, from the category of A-modules to that of commutative A-algebras; finally, the above characterization also shows that if $E = \varinjlim E_\lambda$, then we have $\mathbf{S}(E) = \varinjlim \mathbf{S}(E_\lambda)$ up to canonical isomorphism. By abuse of language, a product $\sigma(x_1)\sigma(x_2)\cdots\sigma(x_n)$, where $x_i \in E$, is often denoted by $x_1 x_2 \cdots x_n$ if no confusion follows. The algebra $\mathbf{S}(E)$ is *graded*, $\mathbf{S}_n(E)$ being the set of linear combinations of n elements of E ($n \geqslant 0$); the algebra $\mathbf{S}(A)$ is canonically isomorphic to the polynomial algebra $A[T]$ is an indeterminate, and the algebra $\mathbf{S}(A^n)$ with the polynomial algebra in n indeterminates $A[T_1, \ldots, T_n]$. **II** | 15

(1.7.2). Let ϕ be a ring homomorphism $A \to B$. If F is a B-module, then the canonical map $F \to \mathbf{S}(F)$ gives a canonical map $F_{[\phi]} \to \mathbf{S}(F)_{[\phi]}$, which thus factors as $F_{[\phi]} \to \mathbf{S}(F_{[\phi]}) \to \mathbf{S}(F)_{[\phi]}$; the canonical homomorphism $\mathbf{S}(F_{[\phi]}) \to \mathbf{S}(F)_{[\phi]}$ is surjective, but not necessarily bijective. If E is an A-module, then every di-homomorphism $E \to F$ (that is to say, every A-homomorphism $E \to F_{[\phi]}$) thus canonically gives an A-homomorphism of algebras $\mathbf{S}(E) \to \mathbf{S}(F_{[\phi]}) \to \mathbf{S}(F)_{[\phi]}$, that is to say a di-homomorphism of algebras $\mathbf{S}(E) \to \mathbf{S}(F)$.

With the same notations, for every A-module E, $\mathbf{S}(E \otimes_A B)$ canonically identifies with the algebra $\mathbf{S}(E) \otimes_A B$; this follows immediately from the universal property of $\mathbf{S}(E)$ (1.7.1).

(1.7.3). Let R be a multiplicative subset of the ring A; apply (1.7.2) to the ring $B = R^{-1}A$, and remembering that $R^{-1}E = E \otimes_A R^{-1}A$, we see that we have $\mathbf{S}(R^{-1}E) = R^{-1}\mathbf{S}(E)$ up to canonical isomorphism. In addition, if $R' \supset R$ is a second multiplicative subset of A, then the diagram

$$\begin{array}{ccc} R^{-1}E & \longrightarrow & R'^{-1}E \\ \big\downarrow & & \big\downarrow \\ \mathbf{S}(R^{-1}E) & \longrightarrow & \mathbf{S}(R'^{-1}E) \end{array}$$

is commutative.

(1.7.4). Now let $(S, \mathscr{A})$ be a ringed space, and let $\mathscr{E}$ be a $\mathscr{A}$-module over S. If to any open $U \subset S$ we associate the $\Gamma(U, \mathscr{A})$-module $\mathbf{S}(\Gamma(U, \mathscr{E}))$, then we define (see the functorial nature of $\mathbf{S}(E)$ (1.7.2)) a presheaf of algebras; we say that the associated sheaf, which we denote by $\mathbf{S}(\mathscr{E})$ or $\mathbf{S}_{\mathscr{A}}(\mathscr{E})$ is the *symmetric $\mathscr{A}$-algebra* on the $\mathscr{A}$-module $\mathscr{E}$. It follows immediately from (1.7.1) that $\mathbf{S}(\mathscr{E})$ is a solution to a universal problem: every homomorphism of $\mathscr{A}$-modules $\mathscr{E} \to \mathscr{B}$, where $\mathscr{B}$ is an $\mathscr{A}$-algebra, factors uniquely as $\mathscr{E} \to \mathbf{S}(\mathscr{E}) \to \mathscr{B}$, the second arrow being a homomorphism of $\mathscr{A}$-algebras. There is thus a bijective correspondence between homomorphisms $\mathscr{E} \to \mathscr{B}$ of $\mathscr{A}$-modules and homomorphisms $\mathbf{S}(\mathscr{E}) \to \mathscr{B}$ of $\mathscr{A}$-algebras. In particular, every homomorphism $u : \mathscr{E} \to \mathscr{F}$ of $\mathscr{A}$-modules defines a homomorphism $\mathbf{S}(u) : \mathbf{S}(\mathscr{E}) \to \mathbf{S}(\mathscr{F})$ of $\mathscr{A}$-algebras, and $\mathbf{S}(\mathscr{E})$ is thus a covariant functor in $\mathscr{E}$.

By (1.7.2) and the commutativity of $\mathbf{S}$ with inductive limits, we have $(\mathbf{S}(\mathscr{E}))_x = \mathbf{S}(\mathscr{E}_x)$ for every point $x \in S$. If $\mathscr{E}, \mathscr{F}$ are two $\mathscr{A}$-modules, then $\mathbf{S}(\mathscr{E} \oplus \mathscr{F})$ canonically identifies with $\mathbf{S}(\mathscr{E}) \otimes_{\mathscr{A}} \mathbf{S}(\mathscr{F})$, as we see for the corresponding presheaves. **II** | 16

We also note that $\mathbf{S}(\mathscr{E})$ is a graded $\mathscr{A}$-algebra, the infinite direct sum of the $\mathbf{S}_n(\mathscr{E})$, where the $\mathscr{A}$-module $\mathbf{S}_n(\mathscr{E})$ is the sheaf associated to the presheaf $U \mapsto \mathbf{S}_n(\Gamma(U, \mathscr{E}))$. If we take in particular

$\mathscr{E} = \mathscr{A}$, then we see that $\mathbf{S}_{\mathscr{A}}(\mathscr{A})$ identifies with $\mathscr{A}[T] = \mathscr{A} \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (T indeterminate, $\mathbf{Z}$ being considered as a simple sheaf).

(1.7.5). Let $(T, \mathscr{B})$ be a second ringed space, f a morphism $(S, \mathscr{A}) \to (T, \mathscr{B})$. If $\mathscr{F}$ is a $\mathscr{B}$-module, then $\mathbf{S}(f^*(\mathscr{F}))$ canonically identifies with $f^*(\mathbf{S}(\mathscr{F}))$; indeed, if $f = (\psi, \theta)$, then by definition (**0**, 4.3.1),

$$\mathbf{S}(f^*(\mathscr{F})) = \mathbf{S}(\psi^*(\mathscr{F}) \otimes_{\psi^*(\mathscr{B})} \mathscr{A}) = \mathbf{S}(\psi^*(\mathscr{F})) \otimes_{\psi^*(\mathscr{B})} \mathscr{A}$$

(1.7.2); for every open U of S and every section h of $\mathbf{S}(\psi^*(\mathscr{F}))$ over U, h coincides, in a neighbourhood V of every point $s \in U$, with an element of $\mathbf{S}(\Gamma(V, \psi^*(\mathscr{F})))$; if we refer to the definition of $\psi^*(\mathscr{F})$ (**0**, 3.7.1) and take into account that every element of $\mathbf{S}(E)$ for a module E is a linear combination of a finite number of products of elements of E, then we see that there is a neighbourhood W of $\psi(s)$ in T, a section h' of $\mathbf{S}(\mathscr{F})$ over W, and a neighbourhood $V' \subset V \cap \psi^{-1}(W)$ of s such that h coincides with $t \mapsto h'(\psi(t))$ over V'; hence out assertion.

Proposition (1.7.6). — *Let A be a ring, $S = \operatorname{Spec}(A)$ its prime spectrum, $\mathscr{E} = \widetilde{M}$ the $\mathscr{O}_S$-module associated to an A-module M; then the $\mathscr{O}_S$-algebra $\mathbf{S}(\mathscr{E})$ is associated to the A-algebra $\mathbf{S}(M)$.*

Proof. For every $f \in A$, $\mathbf{S}(M_f) = (\mathbf{S}(M))_f$ (1.7.3), and the proposition thus follows from Definition (**I**, 1.3.4). □

Corollary (1.7.7). — *If S is a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_S$-module, then the $\mathscr{O}_S$-algebra $\mathbf{S}(\mathscr{E})$ is quasi-coherent. If in addition $\mathscr{E}$ is of finite type, then each of the $\mathscr{O}_S$-modules $\mathbf{S}_n(\mathscr{E})$ is of finite type.*

Proof. The first assertion is an immediate consequence of (1.7.6) and of (**I**, 1.4.1); the second follows from the fact that if E is an A-module of finite type, then $\mathbf{S}_n(E)$ is an A-module of finite type; we then apply (**I**, 1.3.13). □

Definition (1.7.8). — Let $\mathscr{E}$ be a quasi-coherent $\mathscr{O}_S$-module. We call the *vector bundle over S defined by $\mathscr{E}$* and denote by $\mathbf{V}(\mathscr{E})$ the spectrum (1.3.1) of the quasi-coherent $\mathscr{O}_S$-algebra $\mathbf{S}(\mathscr{E})$.

By (1.2.7), for every S-prescheme X, there is a canonical bijective correspondence between the S-morphisms $X \to \mathbf{V}(\mathscr{E})$ and the homomorphisms of $\mathscr{O}_S$-*algebras* $\mathbf{S}(\mathscr{E}) \to \mathscr{A}(X)$, thus also between these S-morphisms and the homomorphisms of $\mathscr{O}_S$-*modules* $\mathscr{E} \to \mathscr{A}(X) = f_*(\mathscr{O}_X)$ (where f is the structure morphism $X \to S$). In particular:

(1.7.9). Take for X a subprescheme induced by S on an *open* $U \subset S$. Then the S-morphisms $U \to \mathbf{V}(\mathscr{E})$ are none other than the U-sections (**I**, 2.5.5) of the U-prescheme induced by $\mathbf{V}(\mathscr{E})$ on the open $p^{-1}(U)$ (where p is the structure morphism $\mathbf{V}(\mathscr{E}) \to S$). From what we have just seen, these U-sections bijectively correspond to homomorphisms of $\mathscr{O}_S$-modules $\mathscr{E} \to j_*(\mathscr{O}_S|U)$ (where j is the canonical injection $U \to S$), or equivalently (**0**, 4.4.3) with the $(\mathscr{O}_S|U)$-homomorphisms $j^*(\mathscr{E}) = \mathscr{E}|U \to \mathscr{O}_S|U$. **II** | 17
In addition, it is immediate that the restriction to an open $U' \subset U$ of an S-morphism $U \to \mathbf{V}(\mathscr{E})$ corresponds to the restriction to U' of the corresponding homomorphism $\mathscr{E}|U \to \mathscr{O}_S|U$. We conclude that *the sheaf of germs of S-sections* of $\mathbf{V}(\mathscr{E})$ is canonically identified with the *dual* $\mathscr{E}^\vee$ of $\mathscr{E}$.

In particular, if we set $X = U = S$, then the *zero* homomorphism $\mathscr{E} \to \mathscr{O}_S$ corresponds to a canonical S-section of $\mathbf{V}(\mathscr{E})$, called the *zero S-section* (cf. (8.3.3)).

(1.7.10). Now take X to be the spectrum $\{\xi\}$ of a field K; the structure morphism $f : X \to S$ then corresponds to a monomorphism $k(s) \to K$, where $s = f(\xi)$ (**I**, 2.4.6); the S-morphisms $\{\xi\} \to \mathbf{V}(\mathscr{E})$ are none other than the *geometric points of* $\mathbf{V}(\mathscr{E})$ *with values in the extension K of* $k(s)$ (**I**, 3.4.5), points which are localized at the points of $p^{-1}(s)$. The set of these points, which we can call *the rational geometric fibre over K* of $\mathbf{V}(\mathscr{E})$ *over the point s*, is identified by (1.7.8) with the set of homomorphisms of $\mathscr{O}_S$-modules $\mathscr{E} \to f_*(\mathscr{O}_X)$, or, equivalently (**0**, 4.4.3) with the set of homomorphisms of $\mathscr{O}_X$-modules $f^*(\mathscr{E}) \to \mathscr{O}_X = K$. But we have by definition (**0**, 4.3.1) $f^*(\mathscr{E}) = \mathscr{E}_s \otimes_{\mathscr{O}_s} K = \mathscr{E}^s \otimes_{k(s)} K$, setting $\mathscr{E}^s = \mathscr{E}_s/\mathfrak{m}_s\mathscr{E}_s$; the geometric fibre of $\mathbf{V}(\mathscr{E})$ rational over K over s thus identifies with the *dual* of the *K-vector space* $\mathscr{E}^s \otimes_{k(s)} K$; if $\mathscr{E}^s$ or K is of finite dimension over $k(s)$, then this dual also identifies with $(\mathscr{E}^s)^\vee \otimes_{k(s)} K$, denoting by $(\mathscr{E}^s)^\vee$ the dual of the $k(s)$-vector space $\mathscr{E}^s$.

Proposition (1.7.11). —

(i) $\mathbf{V}(\mathscr{E})$ *is a contravariant functor in* $\mathscr{E}$ *from the category of quasi-coherent* $\mathscr{O}_S$*-modules to the category of affine* S*-schemes.*
(ii) *If* $\mathscr{E}$ *is an* $\mathscr{O}_S$*-module of finite type, then* $\mathbf{V}(\mathscr{E})$ *is of finite type over* S*.*
(iii) *If* $\mathscr{E}$ *and* $\mathscr{F}$ *are two quasi-coherent* $\mathscr{O}_S$*-modules, then* $\mathbf{V}(\mathscr{E} \oplus \mathscr{F})$ *canonically identifies with* $\mathbf{V}(\mathscr{E}) \times_S \mathbf{V}(\mathscr{F})$*.*
(iv) *Let* $g : S' \to S$ *be a morphism; for every quasi-coherent* $\mathscr{O}_S$*-module* $\mathscr{E}$*,* $\mathbf{V}(g^*(\mathscr{E}))$ *canonically identifies with* $\mathbf{V}(\mathscr{E})_{(S')} = \mathbf{V}(\mathscr{E}) \times_S S'$*.*
(v) *A surjective homomorphism* $\mathscr{E} \to \mathscr{F}$ *of quasi-coherent* $\mathscr{O}_S$*-modules corresponds to a closed immersion* $\mathbf{V}(\mathscr{F}) \to \mathbf{V}(\mathscr{E})$*.*

PROOF. (i) is an immediate consequence of (1.2.7), taking into account that every homomorphism of $\mathscr{O}_S$-modules $\mathscr{E} \to \mathscr{F}$ canonically defines a homomorphism of $\mathscr{O}_S$-algebras $\mathbf{S}(\mathscr{E}) \to \mathbf{S}(\mathscr{F})$. (ii) follows immediately from the definition (**I**, 6.3.1) and the fact that if E is an A-module of finite type, then $\mathbf{S}(E)$ is an A-algebra of finite type. To prove (iii), it suffices to start with the canonical isomorphism $\mathbf{S}(\mathscr{E} \oplus \mathscr{F}) \simeq \mathbf{S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}(\mathscr{F})$ (1.7.4) and to apply (1.4.6). Similarly, to prove (iv), it suffices to start with the canonical isomorphism $\mathbf{S}(g^*(\mathscr{E})) \simeq g^*(\mathbf{S}(\mathscr{E}))$ (1.7.5) and to apply (1.5.2). Finally, to establish (v), it suffices to remark that if the homomorphism $\mathscr{E} \to \mathscr{F}$ is surjective, then so is the corresponding homomorphism $\mathbf{S}(\mathscr{E}) \to \mathbf{S}(\mathscr{F})$ of $\mathscr{O}_S$-algebras, and the conclusion follows from (1.4.10). □

(1.7.12). Take in particular $\mathscr{E} = \mathscr{O}_S$; the prescheme $\mathbf{V}(\mathscr{O}_S)$ is the affine S-scheme, spectrum of the $\mathscr{O}_S$-algebra $\mathbf{S}(\mathscr{O}_S)$ which identifies with the $\mathscr{O}_S$-algebra $\mathscr{O}_S[T] = \mathscr{O}_S \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (T indeterminate); this **II** | 18
is evident when $S = \operatorname{Spec}(\mathbf{Z})$, by virtue of (1.7.6), and we pass from there to the general case by considering the structure morphism $S \to \operatorname{Spec}(\mathbf{Z})$ and using (1.7.11, iv). Because of this result, we set $\mathbf{V}(\mathscr{O}_S) = S[T]$, and we thus have the formula

$$S[T] = S \otimes_{\mathbf{Z}} \mathbf{Z}[T]. \tag{1.7.12.1}$$

The identification of the sheaf of germs of S-sections of $S[T]$ with $\mathscr{O}_S$, already seen in (**I**, 3.3.15), here in a more general context, as a special case of (1.7.9).

(1.7.13). For every S-prescheme X, we have seen (1.7.8) that $\operatorname{Hom}_S(X, S[T])$ canonically identifies with $\operatorname{Hom}_{\mathscr{O}_S}(\mathscr{O}_S, \mathscr{A}(X))$, which is canonically isomorphic to $\Gamma(S, \mathscr{A}(X))$, and as a result is equipped with the structure of a ring; in addition, to every S-morphism $h : X \to Y$ there corresponds a morphism $\Gamma(\mathscr{A}(h)) : \Gamma(S, \mathscr{A}(Y)) \to \Gamma(S, \mathscr{A}(X))$ for the ring structures (1.1.2). When we equip $\operatorname{Hom}_S(X, S[T])$ with a ring structure as defined, then we can see that $\operatorname{Hom}(X, S[T])$ can be considered as a *contravariant functor* in X, from the category of S-preschemes to that of rings. On the other hand, $\operatorname{Hom}_S(X, \mathbf{V}(\mathscr{E}))$ is likewise identified with $\operatorname{Hom}_{\mathscr{O}_S}(\mathscr{E}, \mathscr{A}(X))$ (where $\mathscr{A}(X)$ is considered as an $\mathscr{O}_S$*-module*); as a result, we can canonically equip it with a *module* structure over the ring $\operatorname{Hom}_S(X, S[T])$, and we see as above that the pair

$$(\operatorname{Hom}_S(X, S[T]), \operatorname{Hom}(X, \mathbf{V}(\mathscr{E})))$$

is a contravariant functor in X, with values in the category whose elements are the pairs (A, M) consisting of a ring A and an A-module M, the morphisms being di-homomorphisms.

We will interpret these facts by saying that $S[T]$ is an *S-scheme of rings* and that $\mathbf{V}(\mathscr{E})$ is an *S-scheme of modules* on the S-scheme of rings $S[T]$ (cf. Chapter 0, §8).

(1.7.14). We will see that the structure of an S-scheme of modules defined on the S-scheme $\mathbf{V}(\mathscr{E})$ allows us to reconstruct the $\mathscr{O}_S$-module $\mathscr{E}$ up to unique isomorphism: for this, we will show that $\mathscr{E}$ is canonically isomorphic to an $\mathscr{O}_S$-submodule of $\mathbf{S}(\mathscr{E}) = \mathscr{A}(\mathbf{V}(\mathscr{E}))$, defined by means of this structure. Indeed (1.7.4) the set $\operatorname{Hom}_{\mathscr{O}_S}(\mathbf{S}(\mathscr{E}), \mathscr{A}(X))$ of homomorphisms of $\mathscr{O}_S$*-algebras* is canonically identified with $\operatorname{Hom}_{\mathscr{O}_S}(\mathscr{E}, \mathscr{A}(X))$, the set of homomorphisms of $\mathscr{O}_S$*-modules*: if h and h' are two elements of this latter set, s_i ($1 \leqslant i \leqslant n$) sections of $\mathscr{E}$ over an open $U \subset S$, t a section of $\mathscr{A}(X)$ over U, then we have by definition

$$(h + h')(s_1 s_2 \cdots s_n) = \prod_{i=1}^{n} (h(s_i) + h'(s_i))$$

and

$$(t \cdot h)(s_1 s_2 \cdots s_n) = t^n \prod_{i=1}^{n} h(s_i).$$

This being so, if z is a section of $\mathbf{S}(\mathscr{E})$ over U, then $h \mapsto h(z)$ is a map from $\mathrm{Hom}_S(X, \mathbf{V}(\mathscr{E})) = \mathrm{Hom}_{\mathscr{O}_S}(\mathbf{S}(\mathscr{E}), \mathscr{A}(X))$ to $\Gamma(U, \mathscr{A}(X))$. We will show that $\mathscr{E}$ is identified with a submodule of $\mathbf{S}(\mathscr{E})$ **II** | 19 such that, *for every open $U \subset S$, every section z of this $\mathscr{O}_S$-submodule of U, and every S-prescheme X, the map $h \mapsto h(z)$ from $\mathrm{Hom}_{\mathscr{O}_S}(\mathbf{S}(\mathscr{E})|U, \mathscr{A}(X)|U)$ to $\Gamma(U, \mathscr{A}(X))$ is a homomorphism of $\Gamma(U, \mathscr{A}(X))$-modules.*

It is immediate that $\mathscr{E}$ has this property; to show the converse, we can reduce to proving that when $S = \mathrm{Spec}(A)$, $\mathscr{E} = \widetilde{M}$, a section z of $\mathbf{S}(\mathscr{E})$ over S that (for $U = S$) has the property stated above is necessarily a section of $\mathscr{E}$; we then have $z = \sum_{n=0}^{\infty} z_n$, where $z_n \in \mathbf{S}_n(M)$, and it is a question of proving that $z_n = 0$ for $n \neq 1$. Set $B = \mathbf{S}(M)$ and take for X the prescheme $\mathrm{Spec}(B[T])$, where T is an indeterminate. The set $\mathrm{Hom}_{\mathscr{O}_S}(\mathbf{S}(\mathscr{E}), \mathscr{A}(X))$ identifies with the set of ring homomorphisms $h : B \to B[T]$ (**I**, 1.3.13), and from what we saw above, we have $(T \cdot h)(z) = \sum_{n=0}^{\infty} T^n h(z_n)$: the hypothesis on z implies that we have $\sum_{n=0}^{\infty} T^n h(z_n) = T \cdot \sum_{n=0}^{\infty} h(z_n)$ for every homomorphism h. In in particular we take for h the canonical injection, then $\sum_{n=0}^{\infty} T^n z_n = T \cdot \sum_{n=0}^{\infty} z_n$, which implies the conclusion $z_n = 0$ for $n \neq 1$.

Proposition (1.7.15). — *Let Y be a prescheme whose underlying space is Noetherian, or a quasi-compact scheme. Every affine Y-scheme X of finite type over Y is Y-isomorphic to a closed Y-subscheme of a Y-scheme of the form $\mathbf{V}(\mathscr{E})$, where $\mathscr{E}$ is a quasi-coherent $\mathscr{O}_Y$-module of finite type.*

PROOF. The quasi-coherent $\mathscr{O}_Y$-algebra $\mathscr{A}(X)$ is of finite type (1.3.7). The hypotheses imply that $\mathscr{A}(X)$ is generated by a quasi-coherent $\mathscr{O}_Y$-submodule of finite type $\mathscr{E}$ (**I**, 9.6.5); by definition, this implies that the canonical homomorphism $\mathbf{S}(\mathscr{E}) \to \mathscr{A}(X)$ canonically extending the injection $\mathscr{E} \to \mathscr{A}(X)$ is *surjective*; the conclusion then follows from (1.4.10). □

§2. HOMOGENEOUS PRIME SPECTRA

2.1. Generalities on graded rings and modules

Notation (2.1.1). — Given a *positively graded* ring S, we denote by S_n the subset of S consisting of homogeneous elements of degree n ($n \geqslant 0$), by S_+ the (direct) sum of the S_n for $n > 0$; we have $1 \in S_0$, S_0 is a subring of S, S_+ is a graded ideal of S, and S is the direct sum of S_0 and S_+. If M is a *graded* module over S (with positive or negative degrees), we similarly denote by M_n the S_0-module consisting of homogeneous elements of M of degree n (with $n \in \mathbf{Z}$).

For every integer $d > 0$, we denote by $S^{(d)}$ the direct sum of the S_{nd}; by considering the elements of S_{nd} as homogeneous of degree n, the S_{nd} define on $S^{(d)}$ a graded ring structure.

For every integer k such that $0 \leqslant k \leqslant d-1$, we denote by $M^{(d,k)}$ the direct sum of the M_{nd+k} **II** | 20 ($n \in \mathbf{Z}$); this is a graded $S^{(d)}$-module when we consider the elements of M_{nd+k} as homogeneous of degree n. We write $M^{(d)}$ in place of $M^{(d,0)}$.

With the above notation, for every integer n (positive or negative), we denote by $M(n)$ the graded S-module defined by $(M(n))_k = M_{n+k}$ for every $k \in \mathbf{Z}$. In particular, $S(n)$ will be a graded S-module such that $(S(n))_k = S_{n+k}$, by agreeing to set $S_n = 0$ for $n < 0$. We say that a graded S-module M is *free* if it is isomorphic, considered as a *graded* module, to a direct sum of modules of the form $S(n)$; as $S(n)$ is a monogeneous S-module, generated by the element 1 of S considered as an element of degree $-n$, it is equivalent to say that M admits a *basis* over S consisting of *homogeneous* elements.

We say that a graded S-module M *admits a finite presentation* if there exists an exact sequence $P \to Q \to M \to 0$, where P and Q are finite direct sums of modules of the form $S(n)$ and the homomorphisms are of degree 0 (cf. (2.1.2)).

(2.1.2). Let M and N be two graded S-modules; we define on $M \otimes_S N$ a *graded* S-module structure in the following way. On the tensor product $M \otimes_{\mathbf{Z}} N$, we can define a graded $\mathbf{Z}$-module structure (where $\mathbf{Z}$ is graded by $\mathbf{Z}_0 = \mathbf{Z}$, $\mathbf{Z}_n = 0$ for $n \neq 0$) by setting $(M \otimes_{\mathbf{Z}} N)_q = \bigoplus_{m+n=q} M_m \otimes_{\mathbf{Z}} N_n$ (as M and N are respectively direct sums of the M_m and the N_n, we know that we can canonically identify $M \otimes_{\mathbf{Z}} N$ with the direct sum of all the $M_m \otimes_{\mathbf{Z}} N_n$). This being so, we have $M \otimes_S N = (M \otimes_{\mathbf{Z}} N)/P$, where P is the $\mathbf{Z}$-submodule of $M \otimes_{\mathbf{Z}} N$ generated by the elements $(xs) \otimes y - x \otimes (sy)$ for $x \in M$,

$y \in N$, $s \in S$; it is clear that P is a *graded* $\mathbf{Z}$-submodule of $M \otimes_{\mathbf{Z}} N$, and we see immediately that we obtain a graded S-module structure on $M \otimes_S N$ by passing to the quotient.

For two graded S-modules M and N, recall that a homomorphism $u : M \to N$ of S-modules is said to be *of degree* k if $u(M_j) \subset N_{j+k}$ for all $j \in \mathbf{Z}$. If H_n denotes the set of all the homomorphisms of degree n from M to N, then we denote by $\mathrm{Hom}_S(M, N)$ the (direct) *sum* of the H_n ($n \in \mathbf{Z}$) in the S-module H of all the homomorphisms (of S-modules) from M to N; in general, $\mathrm{Hom}_S(M, N)$ is not equal to the later. However, we have $H = \mathrm{Hom}_S(M, N)$ when M is *of finite type*; indeed, we can then suppose that M is generated by a finite number of homogeneous elements x_i ($1 \leqslant i \leqslant n$), and every homomorphism $u \in H$ can be written in a unique way as $\sum_{k \in \mathbf{Z}} u_k$, where for each k, $u_k(x_i)$ is equal to the homogeneous component of degree $k + \deg(x_i)$ of $u(x_i)$ ($1 \leqslant i \leqslant n$), which implies that $u_k = 0$ except for a finite number of indices; we have by definition that $u_k \in H_k$, hence the conclusion.

We say that the elements of degree 0 of $\mathrm{Hom}_S(M, N)$ are the *homomorphisms of graded S-modules*. It is clear that $S_m H_n \subset H_{m+n}$, so the H_n define on $\mathrm{Hom}_S(M, N)$ a graded S-module structure.

It follows immediately from these definitions that we have

(2.1.2.1) $$M(m) \otimes_S N(n) = (M \otimes_S N)(m + n),$$

(2.1.2.2) $$\mathrm{Hom}_S(M(m), N(n)) = (\mathrm{Hom}_S(M, N))(n - m),$$

for two graded S-modules M and N.

Let S and S' be two graded rings; a homomorphism of *graded rings* $\phi : S \to S'$ is a homomorphism of rings such that $\phi(S_n) \subset S'_n$ for all $n \in \mathbf{Z}$ (in other words, ϕ must be a homomorphism *of degree* 0 of graded $\mathbf{Z}$-modules). The data of such a homomorphism defines on S' a *graded* S'-module structure; equipped with this structure and its graded ring structure, we say that S' is a *graded* S'*-algebra*. **II | 21**

If M is also a graded S-module, then the tensor product $M \otimes_S S'$ of *graded* S-modules is equipped in a natural way with a *graded* S'-module structure, the grading being defined as above.

Lemma (2.1.3). — *Let S be a ring graded in positive degrees. For a subset E of S_+ consisting of homogeneous elements to generate S_+ as an S-module, it is necessary and sufficient for E to generate S an an S_0-algebra.*

Proof. The condition is evidently sufficient; we show that it is necessary. Let E_n (resp. E^n) be the set of elements of E equal to n (resp. $\leqslant n$); it suffices to show, by induction on $n > 0$, that S_n is the S_0-module generated by the elements of degree n which are products of elements of E^n. This is evident for $n = 1$ by virtue of the hypothesis; the latter also shows that $S_n = \sum_{p=0}^{n-1} S_p E_{n-p}$, and the induction argument is then immediate. □

Corollary (2.1.4). — *For S_+ to be an ideal of finite type, it is necessary and sufficient for S to be an S_0-algebra of finite type.*

Proof. We can always assume that a finite system of generators of the S_0-algebra S (resp. of the S-ideal S_+) consists of homogeneous elements, by replacing each of the generators considered by its homogeneous components. □

Corollary (2.1.5). — *For S to be Noetherian, it is necessary and sufficient for S_0 to be Noetherian and for S to be an S_0-algebra of finite type.*

Proof. The condition is evidently sufficient; it is necessary, since S_0 is isomorphic to S/S_+ and S_+ must be an ideal of finite type (2.1.4). □

Lemma (2.1.6). — *Let S be a ring graded in positive degrees, which is an S_0-algebra of finite type. Let M be a graded S-module of finite type. Then:*

(i) *The M_n are S_0-modules of finite type, and there exists an integer n_0 such that $M_n = 0$ for $n \leqslant n_0$.*
(ii) *There exists an integer n_1 and an integer $h > 0$ such that, for every integer $n \geqslant n_1$, we have $M_{n+h} = S_h M_n$.*
(iii) *For every pair of integers (d, k) such that $d > 0$, $0 \leqslant k \leqslant d - 1$, $M^{(d,k)}$ is an $S^{(d)}$-module of finite type.*
(iv) *For every integer $d > 0$, $S^{(d)}$ is an S_0-algebra of finite type.*

(v) *There exists an integer $h > 0$ such that $S_{mh} = (S_h)^m$ for all $m > 0$.*
(vi) *For every integer $n > 0$, there exists an integer m_0 such that $S_m \subset S_+^n$ for all $m \geqslant m_0$.*

PROOF. We can assume that S is generated (as an S_0-algebra) by homogeneous elements f_i, of degrees h_i ($1 \leqslant i \leqslant r$), and M is generated (as an S-module) by homogeneous elements x_j of degrees k_j ($1 \leqslant j \leqslant s$). It is clear that M_n is formed by linear combinations, with coefficients in S_0, of II | 22
elements $f_1^{\alpha_1} \cdots f_r^{\alpha_r} x_j$ such that the α_i are integers $\geqslant 0$ satisfying $k_j + \sum_i \alpha_i h_i = n$; for each j, there are only finitely many systems (α_i) satisfying this equation, since the h_i are > 0, hence the first assertion of (i); the second is evident. On the other hand, let h be the l.c.m. of the h_i and set $g_i = f_i^{h/h_i}$ ($1 \leqslant i \leqslant r$) such that all the g_i are of degree h; let z_μ be the elements of M of the form $f_1^{\alpha_1} \cdots f_r^{\alpha_r} x_j$ with $0 \leqslant \alpha_i < h/h_i$ for $1 \leqslant i \leqslant r$; there are finitely many of these elements, so let n_1 be the largest of their degrees. It is clear that for $n \geqslant n_1$, every element of M_{n+h} is a linear combination of the z_μ whose coefficients are monomials of degree > 0 with respect to the g_i, so we have $M_{n+h} = S_h M_n$, which establishes (ii). In a similar way, we see (for all $d > 0$) that an element of $M^{(d,k)}$ is a linear combinations, with coefficients in S_0, of elements of the form $g^d f_1^{\alpha_1} \cdots f_r^{\alpha_r} x_j$ with $0 \leqslant \alpha_i < d$, g being a homogeneous element of S; hence (iii); (iv) then follows from (iii) and from Lemma (2.1.3), by taking $M = S_+$, since $(S_+)^{(d)} = (S^{(d)})_+$. The assertion of (v) is deduced from (ii) by taking $M = S$. Finally, for a given n, there are finitely many systems (α_i) such that $\alpha_i \geqslant 0$ and $\sum_i \alpha_i < n$, so if m_0 is the largest value of the sum $\sum_i \alpha_i h_i$ of these systems, then we have $S_m \subset S_+^n$ for $m > m_0$, which proves (vi). □

Corollary (2.1.7). — *If S is Noetherian, then so is $S^{(d)}$ for every integer $d > 0$.*

PROOF. This follows from (2.1.5) and (2.1.6, iv). □

(2.1.8). Let $\mathfrak{p}$ be a *graded* prime ideal of the graded ring S; $\mathfrak{p}$ is thus a direct sum of the subgroups $\mathfrak{p}_n = \mathfrak{p} \cap S_n$. Suppose that $\mathfrak{p}$ *does not contain* S_+. Then if $f \in S_+$ is not in $\mathfrak{p}$, the relation $f^n x \in \mathfrak{p}$ is equivalent to $x \in \mathfrak{p}$; in particular, if $f \in S_d$ ($d > 0$), for all $x \in S_{m-nd}$, then the relation $f^n x \in \mathfrak{p}_m$ is equivalent to $x \in \mathfrak{p}_{m-nd}$.

Proposition (2.1.9). — *Let n_0 be an integer > 0; for all $n \geqslant n_0$, let $\mathfrak{p}_n$ be a subgroup of S_n. For there to exist a graded prime ideal $\mathfrak{p}$ of S not containing S_+ and such that $\mathfrak{p} \cap S_n = \mathfrak{p}_n$ for all $n \geqslant n_0$, it is necessary and sufficient for the following conditions to be satisfied:*

(1st) *$S_m \mathfrak{p}_n \subset \mathfrak{p}_{m+n}$ for all $m \geqslant 0$ and all $n \geqslant n_0$.*
(2nd) *For $m \geqslant n_0$, $n \geqslant n_0$, $f \in S_m$, $g \in S_n$, the relation $fg \in \mathfrak{p}_{m+n}$ implies $f \in \mathfrak{p}_m$ or $g \in \mathfrak{p}_n$.*
(3rd) *$\mathfrak{p}_n \neq S_n$ for at least one $n \geqslant n_0$.*

In addition, the graded prime ideal $\mathfrak{p}$ is then unique.

PROOF. It is evident that the conditions (1st) and (2nd) are necessary. In addition, if $\mathfrak{p} \not\supset S_+$, then there exists at least one $k > 0$ such that $\mathfrak{p} \cap S_k \neq S_k$; if $f \in S_k$ is not in $\mathfrak{p}$, the relation $\mathfrak{p} \cap S_n = S_n$ implies $\mathfrak{p} \cap S_{n-mk} = S_{n-mk}$ according to (2.1.8); therefore, if $\mathfrak{p} \cap S_n = S_n$ for a certain value of n, we would have $\mathfrak{p} \supset S_+$ contrary to the hypothesis, which proves that (3rd) is necessary. Conversely, suppose that the conditions (1st), (2nd), and (3rd) are satisfied. Note that if for an integer $d \geqslant n_0$, $f \in S_d$ is not in $\mathfrak{p}_d$, then, if $\mathfrak{p}$ exists, $\mathfrak{p}_m$, for $m < n_0$, is necessarily equal to the set of the $x \in S_m$ such that $f^r x \in \mathfrak{p}_{m+rd}$, except for a finite number of values of r. This already proves that if $\mathfrak{p}$ exists, then it is unique. It remains to show that if we define the $\mathfrak{p}_m$ for $m < n_0$ by the previous condition, then $\mathfrak{p} = \sum_{n=0}^\infty \mathfrak{p}_n$ is a prime ideal. First, note that by virtue of (2nd), for $m \geqslant n_0$, $\mathfrak{p}_m$ is also defined as the set of the $x \in S_m$ such that $f^r x \in \mathfrak{p}_{m+rd}$ except for a finite number of values of r. This being II | 23
so, if $g \in S_m$, $x \in \mathfrak{p}_n$, then we have $f^r g x \in \mathfrak{p}_{m+n+rd}$ except for a finite number of values of r, so $gx \in \mathfrak{p}_{m+n}$, which proves that $\mathfrak{p}$ is an ideal of S. To establish that this ideal is prime, in other words that the ring $S/\mathfrak{p}$, graded by the subgroups $S_n/\mathfrak{p}_n$, is an integral domain, it suffices (by considering the components of higher degree of two elements of $S/\mathfrak{p}$) to prove that if $x \in S_m$ and $y \in S_n$ are such that $x \notin \mathfrak{p}_m$ and $y \notin \mathfrak{p}_n$, then $xy \notin \mathfrak{p}_{m+n}$. If not, for r large enough, we would have $f^{2r} xy \in \mathfrak{p}_{m+n+2rd}$; but we have $f^r y \notin \mathfrak{p}_{n+rd}$ for all $r > 0$; it then follows from (2nd) that, except for a finite number of values of r, we have $f^r x \in \mathfrak{p}_{m+rd}$, and we conclude that $x \in \mathfrak{p}_m$ contrary to the hypothesis. □

(2.1.10). We say that a subset $\mathfrak{J}$ of S_+ is an *ideal of* S_+ if it is an ideal of S, and $\mathfrak{J}$ is a *graded prime ideal of* S_+ if it is the intersection of S_+ and a graded prime ideal of S *not containing* S_+ (this prime ideal is also unique according to Proposition (2.1.9)). If $\mathfrak{J}$ is an ideal of S_+, the *radical of* $\mathfrak{J}$ *in* S_+ is the set of elements of S_+ which have a power in $\mathfrak{J}$, in other words the set $\mathfrak{r}_+(\mathfrak{J}) = \mathfrak{r}(\mathfrak{J}) \cap S_+$; in particular, the radical of 0 in S_+ is then called the *nilradical* of S_+ and denoted by $\mathfrak{N}_+$: this is the set of nilpotent elements of S_+. If $\mathfrak{J}$ is an *graded* ideal of S_+, then its radical $\mathfrak{r}_+(\mathfrak{J})$ is a *graded* ideal: by passing to the quotient ring $S/\mathfrak{J}$, we can reduce to the case $\mathfrak{J} = 0$, and it remains to see that if $x = x_h + x_{h+1} + \cdots + x_k$ is nilpotent, then so are the $x_i \in S_i$ ($1 \leqslant h \leqslant i \leqslant k$); we can assume $x_k \neq 0$ and the component of highest degree of x^n is then x_k^n, hence x_k is nilpotent, and we then argue by induction on k. We say that the graded ring S is *essentially reduced* if $\mathfrak{N}_+ = 0$, in other words, if S_+ does not contain nilpotent elements $\neq 0$.

(2.1.11). We note that if, in the graded ring S, an element x is a zero-divisor, then so is its component of highest degree. We say that a ring S is *essentially integral* if the ring S_+ (*without the unit element*) does not contain a zero-divisor and is $\neq 0$; it suffices that a homogeneous element $\neq 0$ in S_+ is not a zero-divisor in this ring. It is clear that if $\mathfrak{p}$ is a graded prime ideal of S_+, then $S/\mathfrak{p}$ is essentially integral.

Let S be an essentially integral graded ring, and let $x_0 \in S_0$: if there then exists *a* homogeneous element $f \neq 0$ of S_+ such that $x_0 f = 0$, then we have $x_0 S_+ = 0$, since we have $(x_0 g)f = (x_0 f)g = 0$ for all $g \in S_+$, and the hypothesis thus implies $x_0 g = 0$. For S to be integral, it is necessary and sufficient for S_0 to be integral and the annihilator of S_+ in S_0 to be 0.

2.2. Rings of fractions of a graded ring

(2.2.1). Let S be a graded ring, in positive degrees, f a *homogeneous* element of S, of degree $d > 0$; then the ring of fractions $S' = S_f$ is graded, taking for S'_n the set of the x/f^k, where $x \in S_{n+kd}$ with $k \geqslant 0$ (we observe here that n can take arbitrary negative values); we denote the subring $S'_0 = (S_f)_0$ of S' consisting of elements *of degree* 0 by the notation $S_{(f)}$.

If $f \in S_d$, then the monomials $(f/1)^h$ in S_f (h a positive or negative integer) form a *free system* over the ring $S_{(f)}$, and the set of their linear combinations is none other than the ring $(S^{(d)})_f$, which II | 24
is thus *isomorphic to* $S_{(f)}[T, T^{-1}] = S_{(f)} \otimes_{\mathbf{Z}} \mathbf{Z}[T, T^{-1}]$ (where T is an indeterminate). Indeed, if we have a relation $\sum_{h=-a}^{b} z_h (f/1)^h = 0$ with $z_h = x_h/f^m$, where the x_h are in S_{md}, then this relation is equivalent by definition to the existence of a $k > -a$ such that $\sum_{h=-a}^{b} f^{h+k} x_h = 0$, and as the degrees of the terms of this sum are distinct, we have $f^{h+k} x_h = 0$ for all h, hence $z_h = 0$ for all h.

If M is a graded S-module, then $M' = M_f$ is a graded S_f-module, M'_n being the set of the z/f^k with $z \in M_{n+kd}$ ($k \geqslant 0$); we denote by $M_{(f)}$ the set of the homomogenous elements of degree 0 of M'; it is immediate that $M_{(f)}$ is an $S_{(f)}$-module and that we have $(M^{(d)})_f = M_{(f)} \otimes_{S_{(f)}} (S^{(d)})_f$.

Lemma (2.2.2). — *Let d and e be integers > 0, $f \in S_d$, $g \in S_e$. There exists a canonical ring isomorphism*

$$S_{(fg)} \simeq (S_{(f)})_{g^d/f^e};$$

if we canonically identify these two rings, then there exists a canonical module isomorphism

$$M_{(fg)} \simeq (M_{(f)})_{g^d/f^e}.$$

PROOF. Indeed, fg divides $f^e g^d$, and this latter element divides $(fg)^{de}$, so the graded rings S_{fg} and $S_{f^e g^d}$ are canonically identified; on the other hand, $S_{f^e g^d}$ also identifies with $(S_{f^e})_{g^d/1}$ (**0**, 1.4.6), and as $f^e/1$ is invertible in S_{f^e}, $S_{f^e g^d}$ also identifies with $(S_{f^e})_{g^d/f^e}$. The element g^d/f^e is of degree 0 in S_{f^e}; we immediately conclude that the subring of $(S_{f^e})_{g^d/f^e}$ consisting of elements of degree 0 is $(S_{(f^e)})_{g^d/f^e}$, and as we evidently have $S_{(f^e)} = S_{(f)}$, this proves the first part of the proposition; the second is established in a similar way. $\square$

(2.2.3). Under the hypotheses of (2.2.2), it is clear that the canonical homomorphism $S_f \to S_{fg}$ (**0**, 1.4.1), which sends x/f^k to $g^k x/(fg)^k$, is of degree 0, thus gives by restriction a *canonical homomorphism* $S_{(f)} \to S_{(fg)}$, such that the diagram

$$\begin{tikzcd} & S_{(f)} \arrow[dl] \arrow[dr] & \\ S_{(fg)} \arrow[rr, "\sim"] & & (S_{(f)})_{g^d/f^e} \end{tikzcd}$$

is commutative. We similarly define a canonical homomorphism $M_{(f)} \to M_{(fg)}$.

Lemma (2.2.4). — *If f and g are two homogeneous elements of S_+, then the ring $S_{(fg)}$ is generated by the union of the canonical images of $S_{(f)}$ and $S_{(g)}$.*

PROOF. By virtue of Lemma (2.2.2), it suffices to see that $1/(g^d/f^e) = f^{d+e}/(fg)^d$ belongs to the canonical image of $S_{(g)}$ in $S_{(fg)}$, which is evident by definition. □

Proposition (2.2.5). — *Let d be an integer > 0 and let $f \in S_d$. Then there exists a canonical ring isomorphisms $S_{(f)} \simeq S^{(d)}/(f-1)S^{(d)}$; if we identify these two rings by this isomorphism, then there exists a canonical module isomorphism $M_{(f)} \simeq M^{(d)}/(f-1)M^{(d)}$.*

PROOF. The first of these isomorphisms is defined by sending x/f^n, where $x \in S_{nd}$, to the element $\overline{x}$, the class of x mod. $(f-1)S^{(d)}$; this map is well-defined, because we have the congruence $f^h x \equiv x\,(\text{mod.}\,(f-1)S^{(d)})$ for all $x \in S^{(d)}$, so if $f^h x = 0$ for an $h > 0$, then we have $\overline{x} = 0$. On the **II** | 25
other hand, if $x \in S_{nd}$ is such that $x = (f-1)y$ with $y = y_{hd} + y_{(h+1)d} + \cdots + y_{kd}$ with $y_{jd} \in S_{jd}$ and $y_{hd} \neq 0$, then we necessarily have $h = n$ and $x = -y_{hd}$, as well as the relations $y_{(j+1)d} = f y_{jd}$ for $h \leqslant j \leqslant k-1$, $f y_{kd} = 0$, which ultimately gives $f^{k-n}x = 0$; we send every class $\overline{x}$ mod. $(f-1)S^{(d)}$ of an element $x \in S_{nd}$ to the element x/f^n of $S_{(f)}$, since the preceding remark shows that this map is well-defined. It is immediate that these two maps thus defined are ring homomorphisms, each the reciprocal of the other. We proceed exactly the same way for M. □

Corollary (2.2.6). — *If S is Noetherian, then so is $S_{(f)}$ for f homogeneous of degree > 0.*

PROOF. This follows immediately from Corollary (2.1.7) and Proposition (2.2.5). □

(2.2.7). Let T be a multiplicative subset of S_+ consisting of *homogeneous* elements; $T_0 = T \cup \{1\}$ is then a multiplicative subset of S; as the elements of T_0 are homogeneous, the ring $T_0^{-1}S$ is still graded in the evident way; we denote by $S_{(T)}$ the subring of $T_0^{-1}S$ consisting of elements of order 0, that is to say, the elements of the form x/h, where $h \in T$ and x is homogeneous of degree equal to that of h. We know (**0**, 1.4.5) that $T_0^{-1}S$ is canonically identified with the inductive limit of the rings S_f, where f varies over T (with respect to the canonical homomorphisms $S_f \to S_{fg}$); as this identification respects the degrees, it identifies $S_{(T)}$ with the *inductive limit* of the $S_{(f)}$ for $f \in T$. For every graded S-module M, we similarly define the module $M_{(T)}$ (over the ring $S_{(T)}$) consisting of elements of degree 0 of $T_0^{-1}M$, and we see that this module is the inductive limit of the $M_{(f)}$ for $f \in T$.

If $\mathfrak{p}$ is a graded prime ideal of S_+, then we denote by $S_{(\mathfrak{p})}$ and $M_{(\mathfrak{p})}$ the ring $S_{(T)}$ and the module $M_{(T)}$ respectively, where T is the set of *homogeneous* elements of S_+ which do not belong to $\mathfrak{p}$.

2.3. Homogeneous prime spectrum of a graded ring

(2.3.1). Given a graded ring S, in positive degrees, we call the *homogeneous prime spectrum* of S and denote it by $\mathrm{Proj}(S)$ the set of graded prime ideals of S_+ (2.1.10), or equivalently the set of graded prime ideals of S *not containing* S_+; we will define a *scheme* structure having $\mathrm{Proj}(S)$ as the underlying set.

(2.3.2). For every subset E of S, let $V_+(E)$ be the set of graded prime ideals of S containing E and not containing S_+; this is thus the subset $V(E) \cap \mathrm{Proj}(S)$ of $\mathrm{Spec}(S)$. From (**I**, 1.1.2) we deduce:

$$V_+(0) = \mathrm{Proj}(S),\ V_+(S) = V_+(S_+) = \varnothing, \tag{2.3.2.1}$$

$$V_+\big(\textstyle\bigcup_\lambda E_\lambda\big) = \bigcap_\lambda V_+(E_\lambda), \tag{2.3.2.2}$$

$$V_+(EE') = V_+(E) \cup V_+(E'). \tag{2.3.2.3}$$

We do not change $V_+(E)$ by replacing E with the graded ideal generated by E; in addition, if $\mathfrak{J}$ is a graded ideal of S, then we have

$$V_+(\mathfrak{J}) = V_+\big(\textstyle\bigcup_{q\geqslant n}(\mathfrak{J} \cap S_q)\big) \tag{2.3.2.4}$$

for all $n > 0$: indeed, if $\mathfrak{p} \in \mathrm{Proj}(S)$ contains the homogeneous elements of $\mathfrak{J}$ of degree $\geqslant n$, then as **II | 26**
by hypothesis there exists a homogeneous element $f \in S_d$ not contained in $\mathfrak{p}$, for every $m \geqslant 0$ and every $x \in S_m \cap \mathfrak{J}$, we have $f^r x \in \mathfrak{J} \cap S_{m+rd}$ for all but finitely many values of r, so $f^r x \in \mathfrak{p} \cap S_{m+rd}$, which implies that $x \in \mathfrak{p} \cap S_m$ (2.1.9).

Finally, we have, for every graded ideal $\mathfrak{J}$ of S,

$$V_+(\mathfrak{J}) = V_+(\mathfrak{r}_+(\mathfrak{J})). \tag{2.3.2.5}$$

(2.3.3). By definition, the $V_+(E)$ are the closed subsets of $X = \mathrm{Proj}(S)$ for the topology induced by the spectral topology of $\mathrm{Spec}(S)$, which we also call the *spectral topology* on X. For all $f \in S$, we set

$$D_+(f) = D(f) \cap \mathrm{Proj}(S) = \mathrm{Proj}(S) - V_+(f) \tag{2.3.3.1}$$

and so, for any two elements f and g of S (**I**, 1.1.9.1),

$$D_+(fg) = D_+(f) \cap D_+(g). \tag{2.3.3.2}$$

Proposition (2.3.4). — *The $D_+(f)$, as f runs over the set of homogeneous elements of S_+, form a base for the topology of $X = \mathrm{Proj}(S)$.*

PROOF. It follows from (2.3.2.2) and (2.3.2.4) that every closed subset of X is the intersection of sets of the form $V_+(f)$, where f is homogeneous of degree > 0. ☐

(2.3.5). Let f be a *homogeneous* element of S_+, of degree $d > 0$; for every graded prime ideal $\mathfrak{p}$ of S that does not contain f, we know that the set of the x/f^n, where $x \in \mathfrak{p}$ and $n \geqslant 0$, is a prime ideal of the ring of fractions S_f (**0**, 1.2.6); its intersection with $S_{(f)}$ is thus a prime ideal of $S_{(f)}$, which we denote by $\psi_f(\mathfrak{p})$: it is the set of the x/f^n for $n \geqslant 0$ and $x \in \mathfrak{p} \cap S_{nd}$. We have thus defined a map

$$\psi_f : D_+(f) \longrightarrow \mathrm{Spec}(S_{(f)});$$

furthermore, if $g \in S_e$ is another homogeneous element of S_+, then we have a commutative diagram

$$\begin{tikzcd} D_+(f) \arrow[r, "\psi_f"] & \mathrm{Spec}(S_{(f)}) \\ D_+(fg) \arrow[u] \arrow[r, "\psi_{fg}"'] & \mathrm{Spec}(S_{(fg)}) \arrow[u] \end{tikzcd} \tag{2.3.5.1}$$

where the vertical arrow on the left is the inclusion, and the vertical arrow on the right is the map ${}^a\omega_{fg,f}$ induced by the canonical homomorphism $\omega = \omega_{fg,f} : S_{(f)} \to S_{(fg)}$ (**I**, 1.2.1). Indeed, if $x/f^n \in \omega^{-1}(\psi_{fg}(\mathfrak{p}))$, with $fg \notin \mathfrak{p}$, then, by definition, $g^n x/(fg)^n \in \psi_{fg}(\mathfrak{p})$, so $g^n x \in \mathfrak{p}$, and so $x \in \mathfrak{p}$; the converse is evident.

Proposition (2.3.6). — *The map ψ_f is a homeomorphism from $D_+(f)$ to $\mathrm{Spec}(S_{(f)})$.*

PROOF. Firstly, ψ_f is continuous; this is since, if $h \in S_{nd}$ is such that $h/f^n \in \psi_f(\mathfrak{p})$, then, by definition, $h \in \mathfrak{p}$, and conversely, and so $\psi_f^{-1}(D(h/f^n)) = D_+(hf)$, and our claim then follows from (2.3.3.2). Furthermore, the $D_+(hf)$, where h runs over the sets S_{nd}, form a topology of $D_+(f)$, by (2.3.4) and (2.3.3.2); the above thus proves, taking into account the (T_0) axiom, which holds in $D_+(f)$ **II | 27**
and in $\operatorname{Spec}(S_{(f)})$, that ψ_f is injective and that the inverse map $\psi_f(D_+(f)) \to D_+(f)$ is continuous. Finally, to see that ψ_f is surjective, we note that, if $\mathfrak{q}_0$ is a prime ideal of $S_{(f)}$, and if, for all $n > 0$, we denote by $\mathfrak{p}_n$ the set of $x \in S_n$ such that $x^d/f^n \in \mathfrak{q}_0$, then the $\mathfrak{p}_n$ satisfy the conditions of (2.1.9): if $x, y \in S_n$ are such that $x^d/f^n, y^d/f^n \in \mathfrak{q}_0$, then $(x+y)^{2d}/f^{2n} \in \mathfrak{q}_0$, whence $(x+y)^d/f^n \in \mathfrak{q}_0$, since $\mathfrak{q}_0$ is prime; this proves that the $\mathfrak{p}_n$ are subgroups of the S_n, and the verification of the other conditions of (2.1.9) is immediate, taking into account the fact that $\mathfrak{q}_0$ is prime. If $\mathfrak{p}$ is the graded prime ideal of S thus defined, then indeed $\psi_f(\mathfrak{p}) = \mathfrak{q}_0$, since, if $x \in S_{nd}$, then having $x/f^n \in \mathfrak{q}_0$ and $x^d/f^{nd} \in \mathfrak{q}_0$ is equivalent to $\mathfrak{q}_0$ being prime. □

Corollary (2.3.7). — *To have $D_+(f) = \varnothing$, it is necessary and sufficient for f to be nilpotent.*

PROOF. To have $\operatorname{Spec}(S_{(f)}) = \varnothing$, it is necessary and sufficient to have $S_{(f)} = 0$, or indeed to have $1 = 0$ in S_f, which means, by definition, that f is nilpotent. □

Corollary (2.3.8). — *Let E be a subset of S_+. Then the following conditions are equivalent:*

(a) $V_+(E) = X = \operatorname{Proj}(S)$.
(b) *Every element of E is nilpotent.*
(c) *The homogeneous components of every element of E are nilpotent.*

PROOF. It is clear that (c) implies (b), and that (b) implies (a). If $\mathfrak{J}$ is the graded ideal of S generated by E, then condition (a) is equivalent to requiring that $V_+(\mathfrak{J}) = X$; *a fortiori*, (a) implies that every homogeneous element $f \in \mathfrak{J}$ is such that $V_+(f) = X$, and so f is nilpotent by (2.3.7). □

Corollary (2.3.9). — *If $\mathfrak{J}$ is a graded ideal of S_+, then $\mathfrak{r}_+(\mathfrak{J})$ is the intersection of the graded prime ideals of S_+ that contain $\mathfrak{J}$.*

PROOF. By considering the graded ring $S/\mathfrak{J}$, we can reduce to the case where $\mathfrak{J} = 0$. We need to prove that, if $f \in S_+$ is not nilpotent, then there exists a graded prime ideal of S that does not contain f; but at least one of the homogeneous components of f is not nilpotent, and we can thus suppose f to be homogeneous; the claim then follows from (2.3.7). □

(2.3.10). For every subset Y of $X = \operatorname{Proj}(S)$, let $\mathfrak{j}_+(Y)$ be the set of $f \in S_+$ such that $Y \subset V_+(f)$; this is equivalent to saying that $\mathfrak{j}_+(Y) = \mathfrak{j}(Y) \cap S_+$; then $\mathfrak{j}_+(Y)$ is an ideal of S_+ that is equal to its radical in S_+.

Proposition (2.3.11). —

(i) *For every subset E of S_+, $\mathfrak{j}_+(V_+(E))$ is the radical in S_+ of the graded ideal of S_+ generated by E.*
(ii) *For every subset Y of X, $V_+(\mathfrak{j}_+(Y)) = \overline{Y}$, where $\overline{Y}$ is the closure of Y in X.*

PROOF.

(i) If $\mathfrak{J}$ is the graded ideal of S_+ generated by E, then $V_+(E) = V_+(\mathfrak{J})$, and the claim then follows from (2.3.9).
(ii) Since $V_+(\mathfrak{J}) = \bigcap_{f \in \mathfrak{J}} V_+(f)$, having $Y \subset V_+(\mathfrak{J})$ implies that $Y \subset V_+(f)$ for every $f \in \mathfrak{J}$, and thus $\mathfrak{j}_+(Y) \supset \mathfrak{J}$, whence $V_+(\mathfrak{j}_+(Y)) \subset V_+(\mathfrak{J})$, which proves (ii) by the definition of the closed subsets.

□

Corollary (2.3.12). — *The closed subsets Y of $X = \operatorname{Proj}(S)$ are in bijective correspondence with the graded ideals of S_+ that are equal to their radical in S_+, via the inclusion-reversing maps $Y \mapsto \mathfrak{j}_+(Y)$ and $\mathfrak{J} \mapsto V_+(\mathfrak{J})$; the union $Y_1 \cup Y_2$ of two closed subsets of X corresponds to $\mathfrak{j}_+(Y_1) \cap \mathfrak{j}_+(Y_2)$, and the* **II | 28**
intersection of an arbitrary family (Y_λ) of closed subsets corresponds to the radical in S_+ of the sum of the $\mathfrak{j}_+(Y_\lambda)$.

Corollary (2.3.13). — *Let $\mathfrak{J}$ be a graded ideal of S_+; to have $V_+(\mathfrak{J}) = \varnothing$, it is necessary and sufficient for every element f of S_+ to have a power f^n in $\mathfrak{J}$.*

This above corollary can also be expressed in one of the following equivalent forms:

Corollary (2.3.14). — *Let (f_α) be a family of homogeneous elements of S_+. For the $D_+(f_\alpha)$ to form a cover of $X = \operatorname{Proj}(S)$, it is necessary and sufficient for every element of S_+ to have a power in the ideal generated by the f_α.*

Corollary (2.3.15). — *Let (f_α) be a family of homogeneous elements of S_+, and f an element of S_+. Then the following are equivalent:*

(a) $D_+(f) \subset \bigcup_\alpha D_+(f_\alpha)$;
(b) $V_+(f) \supset \bigcap_\alpha V_+(f_\alpha)$;
(c) *f has a power in the ideal generated by the f_α.*

Corollary (2.3.16). — *For $X = \operatorname{Proj}(S)$ to be empty, it is necessary and sufficient for every element of S_+ to be nilpotent.*

Corollary (2.3.17). — *In the bijective correspondence described in (2.3.12), the* irreducible *closed subsets of X correspond to the graded* prime *ideals of S_+.*

PROOF. If $Y = Y_1 \cup Y_2$, where Y_1 and Y_2 are distinct closed subsets of Y, then

$$\mathfrak{j}_+(Y) = \mathfrak{j}_+(Y_1) \cap \mathfrak{j}_+(Y_2)$$

with the ideals $\mathfrak{j}_+(Y_1)$ and $\mathfrak{j}_+(Y_2)$ being distinct from $\mathfrak{j}_+(Y)$, and so $\mathfrak{j}_+(Y)$ is not prime. Conversely, if $\mathfrak{J}$ is a graded ideal of S_+ that is not prime, then there exists elements $f, g \in S_+$ such that $f \notin \mathfrak{J}$ and $g \notin \mathfrak{J}$, but $fg \in \mathfrak{J}$; then $V_+(f) \supset V_+(\mathfrak{J})$ and $V_+(g) \supset V_+(\mathfrak{J})$, but $V_+(\mathfrak{J}) \subset V_+(f) \cup V_+(g)$, by (2.3.2.3); we thus conclude that $V_+(\mathfrak{J})$ is the union of the closed subsets $V_+(f) \cap V_+(\mathfrak{J})$ and $V_+(g) \cap V_+(\mathfrak{J})$, which are distinct from $V_+(\mathfrak{J})$. □

2.4. The scheme structure on $\operatorname{Proj}(S)$

(2.4.1). Let f and g be homogeneous elements of S_+; consider the affine schemes $Y_f = \operatorname{Spec}(S_{(f)})$, $Y_g = \operatorname{Spec}(S_{(g)})$, and $Y_{fg} = \operatorname{Spec}(S_{(fg)})$. By (2.2.2), the morphism $w_{fg,f} = ({}^a\omega_{fg,f}, \widetilde{\omega}_{fg,f})$ from Y_{fg} to Y_f, corresponding to the canonical homomorphism $\omega_{fg,f} : S_{(f)} \to S_{(fg)}$, is an *open immersion* (**I**, 1.3.6). Using the inverse homeomorphism of $\psi_f : D_+(f) \to Y_f$ (2.3.6), we can transport the affine scheme structure of Y_f to $D_+(f)$; by the commutativity of diagram (2.3.5.1), the affine scheme $D_+(fg)$ can thus be identified with the induced scheme on the open subset $D_+(fg)$ of the underlying space of the affine scheme $D_+(f)$. It is then clear (taking (2.3.4) into account) that $X = \operatorname{Proj}(S)$ is endowed with a unique *prescheme* structure, whose restriction to each $D_+(f)$ is the affine scheme that we have just defined. Furthermore:

Proposition (2.4.2). — *The prescheme* $\operatorname{Proj}(S)$ *is a scheme.*

PROOF. It suffices (**I**, 5.5.6) to show, for any homogeneous f and g in S_+, that $D_+(f) \cap D_+(g) = D_+(fg)$ is affine, and that its ring is generated by the canonical images of the rings of $D_+(f)$ and $D_+(g)$; the first point is evident by definition, and the second follows from (2.2.4). □

Whenever we speak of the homogeneous prime spectrum $\operatorname{Proj}(S)$ as a *scheme*, it will always mean with respect to the structure that we have just defined. **II** | 29

Example (2.4.3). — Let $S = K[T_1, T_2]$, where K is a field, T_1 and T_2 are indeterminates, and S is graded by total degree. It follows from (2.3.14) that $\operatorname{Proj}(S)$ is the union of $D_+(T_1)$ and $D_+(T_2)$; we immediately see that these affine schemes are isomorphic to $K[T]$, and that $\operatorname{Proj}(S)$ is obtained by the gluing of these two affine schemes as described in (**I**, 2.3.2) (cf. (7.4.14)).

Proposition (2.4.4). — *Let S be a positively graded ring, and let X be the scheme* $\operatorname{Proj}(S)$.

(i) *If $\mathfrak{N}_+$ is the nilradical of S_+ (2.1.10), then the scheme X_{red} is canonically isomorphic to* $\operatorname{Proj}(S/\mathfrak{N}_+)$; *in particular, if S is essentially reduced, then* $\operatorname{Proj}(S)$ *is reduced.*
(ii) *If S is essentially reduced, then, for X to be integral, it is necessary and sufficient for S to be essentially integral.*

PROOF.

(i) Let $\overline{S}$ be the graded ring $S/\mathfrak{N}_+$, and denote by $x \mapsto \overline{x}$ the canonical homomorphism $S \to \overline{S}$ of degree 0. For all $f \in S_d$ ($d > 0$), the canonical homomorphism $S_f \to \overline{S}$ (**0**, 1.5.1) is surjective and of degree 0, and thus gives, by restriction, a surjective homomorphism $S_{(f)} \to \overline{S}_{(\overline{f})}$; if we suppose that $f \notin \mathfrak{N}_+$, then we immediately see that $\overline{S}_{(\overline{f})}$ is reduced, and that the kernel of the above homomorphism is the nilradical of $S_{(f)}$, or, in other words, that $\overline{S}_{(\overline{f})} = (S_{(f)})_{\mathrm{red}}$. So to this homomorphism corresponds a closed immersion $D_+(\overline{f}) \to D_+(f)$ that identifies $D_+(\overline{f})$ with $(D_+(f))_{\mathrm{red}}$ (**I**, 5.1.2), and which is, in particular, a homeomorphism of the underlying spaces of these two affine schemes. Furthermore, if $g \notin \mathfrak{N}_+$ is another homogeneous element of S_+, then the diagram

$$\begin{tikzcd} S_{(f)} \arrow[r] \arrow[d] & \overline{S}_{(\overline{f})} \arrow[d] \\ S_{(fg)} \arrow[r] & \overline{S}_{(\overline{fg})} \end{tikzcd}$$

is commutative; since, further, the $D_+(f)$, for f homogeneous, of degree > 0, and $f \notin \mathfrak{N}_+$, form a cover of $X = \mathrm{Proj}(S)$ (2.3.7), we see that the morphisms $D_+(\overline{f}) \to D_+(f)$ are the restrictions of a closed immersion $\mathrm{Proj}(\overline{S}) \to \mathrm{Proj}(S)$ which is a homeomorphism of the underlying spaces; whence the conclusion (**I**, 5.1.2).

(ii) Suppose that S is essentially integral, or, in other words, that (0) is a graded prime ideal of S_+ that is distinct from S_+; then X is reduced, by (i), and irreducible, by (2.3.17). Conversely, suppose that S is essentially reduced and that X is integral; then, for homogeneous $f \neq 0$ in S_+, we have that $D_+(f) \neq \varnothing$ (2.3.7); the hypothesis that X is irreducible implies that $D_+(f) \cap D_+(g) \neq \varnothing$ for homogeneous $f, g \neq 0$ in S_+; thus $fg \neq 0$, by (2.3.3.2), and we thus conclude that S_+ has no zero divisors, whence the first claim.

□

(2.4.5). Given a commutative ring A, recall that we say that a graded ring S is a *graded A-algebra* if it is endowed with the structure of an A-algebra such that each of its subgroups S_n is an A-module; for this, it suffices for S_0 to be an A-algebra, or, in other words, we define the structure of a graded **II** | 30
A-algebra on S by defining the structure of an A-algebra on S_0 and setting $\alpha \cdot x = (\alpha \cdot 1)x$ for $\alpha \in A$ and $x \in S_n$.

Proposition (2.4.6). — *Suppose that S is a graded A-algebra. Then, on $X = \mathrm{Proj}(S)$, the structure sheaf $\mathscr{O}_X$ is an A-algebra (where A is considered as a simple sheaf on X); in other words, X is a scheme over* $\mathrm{Spec}(A)$.

PROOF. It suffices to note that, for every homogeneous f in S_+, $S_{(f)}$ is an algebra over A, and that the diagram

$$\begin{tikzcd} S_{(f)} \arrow[rr] & & S_{(fg)} \\ & A \arrow[ul] \arrow[ur] & \end{tikzcd}$$

is commutative, for homogeneous f, g in S_+. □

Proposition (2.4.7). — *Let S be a positively graded ring.*

(i) *For every integer $d > 0$, there exists a canonical isomorphism from the scheme* $\mathrm{Proj}(S)$ *to the scheme* $\mathrm{Proj}(S^{(d)})$.

(ii) *Let S' be the graded ring such that $S_0 = \mathbf{Z}$ and $S'_n = S_n$ (considered as a $\mathbf{Z}$-module) for $n > 0$. Then there exists a canonical isomorphism from the scheme* $\mathrm{Proj}(S)$ *to the scheme* $\mathrm{Proj}(S')$.

PROOF.

(i) We first show that the map $\mathfrak{p} \mapsto \mathfrak{p} \cap S^{(d)}$ is a bijection from the set $\mathrm{Proj}(S)$ to the set $\mathrm{Proj}(S^{(d)})$. Indeed, suppose that we have a graded prime ideal $\mathfrak{p}' \in \mathrm{Proj}(S^{(d)})$, and let $\mathfrak{p}_{nd} = \mathfrak{p}' \cap S_{nd}$ ($n \geqslant 0$). For all $n > 0$ that are not multiples of d, define $\mathfrak{p}_n$ as the set of

$x \in S_n$ such that $x^d \in \mathfrak{p}_{nd}$; if $x, y \in \mathfrak{p}_n$, then $(x+y)^{2d} \in \mathfrak{p}_{2nd}$, and so $(x+y)^d \in \mathfrak{p}_{nd}$, since $\mathfrak{p}'$ is prime; it is immediate that the $\mathfrak{p}_n$ thus defined, for $n \geqslant 0$, satisfy the conditions of (2.1.9), and so there exists a unique prime ideal $\mathfrak{p} \in \mathrm{Proj}(S)$ such that $\mathfrak{p} \cap S^{(d)} = \mathfrak{p}'$. Since, for every homogeneous f in S_+, we have that $V_+(f) = V_+(f^d)$ (2.3.2.3), we see that the above bijection is a *homeomorphism* of topological spaces. Finally, with the same notation, $S_{(f)}$ and $S_{(f^d)}$ are canonically identified (2.2.2), and so $\mathrm{Proj}(S)$ and $\mathrm{Proj}(S^{(d)})$ are canonically identified as *schemes*.

(ii) If, to each $\mathfrak{p} \in \mathrm{Proj}(S)$, we associate the unique prime ideal $\mathfrak{p}' \in \mathrm{Proj}(S')$ such that $\mathfrak{p}' \cap S_n = \mathfrak{p} \cap S_n$ for every $n > 0$ (2.1.9), then it is clear that we have defined a canonical homeomorphism $\mathrm{Proj}(S) \xrightarrow{\sim} \mathrm{Proj}(S')$ of the underlying spaces, since $V_+(f)$ is the same set for S and S' when f is a homogeneous element of S_+. Since, further, $S_{(f)} = S'_{(f)}$, $\mathrm{Proj}(S)$ and $\mathrm{Proj}(S')$ can be identified as *schemes*.

□

Corollary (2.4.8). — *If S is a graded A-algebra, and S'_A the graded A-algebra such that $(S'_A)_0 = A$ and $(S'_A)_n = S_n$ for $n > 0$, then there exists a canonical isomorphism from $\mathrm{Proj}(S)$ to $\mathrm{Proj}(S'_A)$.*

PROOF. In fact, these two schemes are both canonically isomorphic to $\mathrm{Proj}(S')$, using the notation of (2.4.7, (ii)). □

2.5. The sheaf associated to a graded module

(2.5.1). Let M be a *graded* S-module. Then, for every homogeneous f in S_+, $M_{(f)}$ is an $S_{(f)}$-module, and thus has a corresponding quasi-coherent associated sheaf $(M_{(f)})^\sim$ on the affine scheme $\mathrm{Spec}(S_{(f)})$, identified with $D_+(f)$ (**I**, 1.3.4).

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Proposition (2.5.2). — *There exists on $X = \mathrm{Proj}(S)$ exactly one quasi-coherent $\mathscr{O}_X$-module $\widetilde{M}$ such that, for f homogeneous in S_+, we have $\Gamma(D_+(f), \widetilde{M}) = M_{(f)}$, with the restriction homomorphism $\Gamma(D_+(f), \widetilde{M}) \to \Gamma(D_+(fg), \widetilde{M})$, for f and g homogeneous in S_+, being the canonical homomorphism $M_{(f)} \to M_{(fg)}$* (2.2.3).

PROOF. Suppose that $f \in S_d$ and $g \in S_e$. Since $D_+(fg)$ can be identified with the prime spectrum of $(S_{(f)})_{g^d/f^e}$ by (2.2.2), the restriction to $D_+(fg)$ of the sheaf $(M_{(f)})^\sim$ on $D_+(f)$ is canonically identified with the sheaf associated to the module $(M_{(f)})_{g^d/f^e}$ (**I**, 1.3.6), and thus also with $(M_{(fg)})^\sim$ (2.2.2); we thus conclude that there exists a canonical isomorphism

$$\theta_{g,f} : (M_{(f)})^\sim | D_+(fg) \xrightarrow{\sim} (M_{(g)})^\sim | D_+(fg)$$

such that, if h is a third homogeneous element of S_+, then $\theta_{f,h} = \theta_{f,g} \circ \theta_{g,h}$ in $D_+(fgh)$. Consequently (**0**, 3.3.1) there exists a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ on X, and, for every homogeneous f in S_+, an isomorphism η_f from $\mathscr{F}|D_+(f)$ to $(M_f)^\sim$ such that $\theta_{g,f} = \eta_g \circ \eta_f^{-1}$. If we then consider the sheaf $\mathscr{G}$ associated to the presheaf (on the base for the topology of X given by the $D_+(f)$) defined by $D_+(f) \mapsto M_{(f)}$, with the canonical homomorphisms $M_{(f)} \to M_{(fg)}$ as restriction homomorphisms, then the above proves that $\mathscr{F}$ and $\mathscr{G}$ are isomorphic (taking (**I**, 1.3.7) into account); the sheaf $\mathscr{G}$ is denoted by $\widetilde{M}$, and indeed satisfies the conditions of the statement. We have, in particular, $\widetilde{S} = \mathscr{O}_X$. □

Definition (2.5.3). — We say that the quasi-coherent $\mathscr{O}_X$-module $\widetilde{M}$ defined in (2.5.2) is *associated* to the graded S-module M.

Recall that the graded S-modules form a category when we restrict from arbitrary homomorphisms of graded modules to homomorphisms *of degree* 0. With this convention:

Proposition (2.5.4). — *The functor $M \mapsto \widetilde{M}$ is an exact additive covariant functor from the category of graded S-modules to the category of quasi-coherent $\mathscr{O}_X$-modules, and it commutes with inductive limits and direct sums.*

PROOF. Indeed, since these properties are local, it suffices to show that they are satisfied for the sheaves of the form $\widetilde{M}|D_+(f) = (M_{(f)})^\sim$; but the functors $M \mapsto M_f$, $N \mapsto N_0$ (to the category of graded S_f-modules), and $P \mapsto \widetilde{P}$ (to the category of $S_{(f)}$-modules) all have the three properties of exactness and of commutativity with inductive limits and direct sums ((**I**, 1.3.5) and (**I**, 1.3.9)); whence the proposition. □

We denote by $\widetilde{u}$ the homomorphism $\widetilde{M} \to \widetilde{N}$ corresponding to a homomorphism $u : M \to N$ of degree 0. We immediately deduce from (2.5.4) that the results of (**I**, 1.3.9) and (**I**, 1.3.10) still hold for graded S-modules and homomorphisms of degree 0 (with the sense given here to $\widetilde{M}$), with the proofs being purely formal.

Proposition (2.5.5). — *For all $\mathfrak{p} \in X = \operatorname{Proj}(S)$, we have $\widetilde{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}$.*

PROOF. By definition, $\widetilde{M}_{\mathfrak{p}} = \varinjlim \Gamma(D_+(f), \widetilde{M})$, where f runs over the set of homogeneous elements of S_+ such that $f \notin \mathfrak{p}$; since $\Gamma(D_+(f), \widetilde{M}) = M_{(f)}$, the proposition follows from the definition of $M_{(\mathfrak{p})}$ (2.2.7) □

In particular, the *local ring* $(\mathscr{O}_X)_{\mathfrak{p}}$ is exactly the ring $S_{(\mathfrak{p})}$, the set of fractions x/f with f homogeneous in S_+ and not belonging to $\mathfrak{p}$, and with x homogeneous of the same degree as f. **II** | 32

Even more particularly, if S is *essential integral*, so that $\operatorname{Proj}(S) = X$ is *integral* (2.4.4), and if $\xi = (0)$ is the generic point of X, then the *field of rational functions* $R(X) = \mathscr{O}_\xi$ is field consisting of elements fg^{-1} with f and g homogeneous of the same degree in S_+, and with $g \neq 0$.

Proposition (2.5.6). — *If, for all $z \in M$ and all homogeneous f in S_+, there exists a power of f that annihilates z, then $\widetilde{M} = 0$. This sufficient condition is also necessary if the S_0-algebra S is generated by the set S_1 of homogeneous elements of degree* 1.

PROOF. The condition $\widetilde{M} = 0$ is equivalent to $M_{(f)} = 0$ for all homogeneous f in S_+. On the other hand, if $f \in S_d$, to say that $M_{(f)} = 0$ implies that, for all homogeneous $z \in M$ whose degree is some multiple of d, there exists a power f^n such that $f^n z = 0$. If $M_{(f)} = 0$ for all $f \in S_1$, then the condition of the statement is thus satisfied for all $f \in S_1$; the condition is *a fortiori* satisfied for all homogeneous f in S_+ if S_1 generates S, since every homogeneous element of S_+ is then a linear combination of products of elements of S_1. □

Proposition (2.5.7). — *Let $d > 0$ be an integer, and let $f \in S_d$. Then, for all $n \in \mathbf{Z}$, the $(\mathscr{O}_X|D_+(f))$-module $(S(nd))^\sim|D_+(f)$ is canonically isomorphic to $\mathscr{O}_X|D_+(f)$.*

PROOF. Indeed, multiplication by the invertible element $(f/1)^n$ of S_f gives a bijection from $S_{(f)} = (S_f)_0$ to $(S_f)_{nd} = (S_f(nd))_0 = (S(nd)_f)_0 = S(nd)_{(f)}$; in other words, the $S_{(f)}$-modules $S_{(f)}$ and $S(nd)_{(f)}$ are canonically isomorphic, whence the proposition. □

Corollary (2.5.8). — *On the open subset $U = \bigcup_{f \in S_d} D_+(f)$, the restriction of the $\mathscr{O}_X$-module $(S(nd))^\sim$ is an invertible $(\mathscr{O}_X|U)$-module* (**0**, 5.4.1).

Corollary (2.5.9). — *If the ideal S_+ of S is generated by the set S_1 of homogeneous elements of degree* 1, *then the $\mathscr{O}_X$-module $(S(n))^\sim$ is invertible for all $n \in \mathbf{Z}$.*

PROOF. It suffices to remark that $X = \bigcup_{f \in S_1} D_+(f)$, by the hypothesis (2.3.14) and to apply (2.5.8) with $U = X$. □

(2.5.10). We set, for the rest of this section,

$$\mathscr{O}_X(n) = (S(n))^\sim \tag{2.5.10.1}$$

for all $n \in \mathbf{Z}$, and, for every open subset U of X, and every $(\mathscr{O}_X|U)$-module $\mathscr{F}$,

$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X|U} (\mathscr{O}_X(n)|U) \tag{2.5.10.2}$$

for all $n \in \mathbf{Z}$. If the ideal S_+ is generated by S_1, then the functor $\mathscr{F}(n)$ is *exact* in $\mathscr{F}$ for all $n \in \mathbf{Z}$, since $\mathscr{O}_X(n)$ is then an *invertible* $\mathscr{O}_X$-module.

(2.5.11). Let M and N be graded S-modules. For all $f \in S_d$ ($d > 0$), we define a canonical functorial homomorphism of $S_{(f)}$-modules by

(2.5.11.1) $$\lambda_f : M_{(f)} \otimes_{S_{(f)}} N_{(f)} \longrightarrow (M \otimes_S N)_{(f)}$$

by composing the homomorphism $M_{(f)} \otimes_{S_{(f)}} N_{(f)} \to M_f \otimes_{S_f} N_f$ (coming from the injections $M_{(f)} \to M_f$, $N_{(f)} \to N_f$, and $S_{(f)} \to S_f$) with the canonical isomorphism $M_f \otimes_{S_f} N_f \xrightarrow{\sim} (M \otimes_S N)_f$ (**0**, 1.3.4), and by noting that, by the definition of the tensor product of two graded modules, this latter isomorphism preserves degrees; for $x \in M_{md}$ and $y \in N_{nd}$ ($m, n \geqslant 0$), we thus have II | 33

$$\lambda_f((x/f^m) \otimes (y/f^n)) = (x \otimes y)/f^{m+n}.$$

It immediately follows from this definition that, if $g \in S_e$ ($e > 0$), then the diagram

$$\begin{tikzcd}
M_{(f)} \otimes_{S_{(f)}} N_{(f)} \arrow[r, "\lambda_f"] \arrow[d] & (M \otimes_S N)_{(f)} \arrow[d] \\
M_{(fg)} \otimes_{S_{(fg)}} N_{(fg)} \arrow[r, "\lambda_{fg}"'] & (M \otimes_S N)_{(fg)}
\end{tikzcd}$$

(where the vertical arrow on the right is the canonical homomorphism, and the one on the left comes from the canonical homomorphisms) commutes. Thus λ induces a canonical functorial homomorphism of $\mathscr{O}_X$-modules

(2.5.11.2) $$\lambda : \widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N} \longrightarrow (M \otimes_S N)^{\sim}.$$

Consider, in particular, graded ideals $\mathfrak{J}$ and $\mathfrak{K}$ of S; since $\widetilde{\mathfrak{J}}$ and $\widetilde{\mathfrak{K}}$ are sheaves of ideals of $\mathscr{O}_X$, we have a canonical homomorphism $\widetilde{\mathfrak{J}} \otimes_{\mathscr{O}_X} \widetilde{\mathfrak{K}} \to \mathscr{O}_X$; the diagram

(2.5.11.3) $$\begin{tikzcd}
\widetilde{\mathfrak{J}} \otimes_{\mathscr{O}_X} \widetilde{\mathfrak{K}} \arrow[rr, "\lambda"] \arrow[dr] & & (\mathfrak{J} \otimes_S \mathfrak{K})^{\sim} \arrow[dl] \\
& \mathscr{O}_X &
\end{tikzcd}$$

then commutes. Indeed, we can reduce to verifying this on each open subset $D_+(f)$ (for f homogeneous in S_+), and this follows immediately from the definition (2.5.11.1) of λ_f and from (**I**, 1.3.13).

Finally, note that, if M, N, and P are graded S-modules, then the diagram

(2.5.11.4) $$\begin{tikzcd}
\widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N} \otimes_{\mathscr{O}_X} \widetilde{P} \arrow[r, "\lambda\otimes 1"] \arrow[d, "1\otimes\lambda"'] & (M \otimes_S N)^{\sim} \otimes_{\mathscr{O}_X} \widetilde{P} \arrow[d, "\lambda"] \\
\widetilde{M} \otimes_{\mathscr{O}_X} (N \otimes_S P)^{\sim} \arrow[r, "\lambda"'] & (M \otimes_S N \otimes_S P)^{\sim}
\end{tikzcd}$$

commutes. It again suffices to verify this on each open subset $D_+(f)$, and this follows immediately from the definitions and from (**I**, 1.3.13). II | 34

(2.5.12). Under the hypotheses of (2.5.11), we define a functorial canonical homomorphism of $S_{(f)}$-modules

(2.5.12.1) $$\mu_f : (\mathrm{Hom}_S(M, N))_{(f)} \longrightarrow \mathrm{Hom}_{S_{(f)}}(M_{(f)}, N_{(f)})$$

by sending u/f^n, where u is a homomorphism of degree nd, to the homomorphism $M_{(f)} \to N_{(f)}$ that sends x/f^m ($x \in M_{md}$) to $u(x)/f^{m+n}$. For $g \in S_e$ ($e > 0$), we again have a commutative diagram:

$$\begin{tikzcd}
(\mathrm{Hom}_S(M,N))_{(f)} \arrow[r, "\mu_f"] \arrow[d] & \mathrm{Hom}_{S_{(f)}}(M_{(f)}, N_{(f)}) \arrow[d] \\
(\mathrm{Hom}_S(M,N))_{(fg)} \arrow[r, "\mu_{fg}"'] & \mathrm{Hom}_{S_{(fg)}}(M_{(fg)}, N_{(fg)})
\end{tikzcd}$$

(where the vertical arrow on the left is the canonical homomorphism, and the one on the right comes from the canonical homomorphisms). We thus again conclude (taking (**I**, 1.3.8) into account) that the μ_f define a functorial canonical homomorphism of $\mathscr{O}_X$-modules

(2.5.12.2) $$\mu : (\mathrm{Hom}_S(M,N))^\sim \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(\widetilde{M}, \widetilde{N})$$

Proposition (2.5.13). — *Suppose that the ideal S_+ is generated by S_1. Then the homomorphism λ (2.5.11.2) is an isomorphism; so too is the homomorphism μ (2.5.12.2) if the graded S-module M admits a finite presentation (2.1.1)*

PROOF. Since X is the union of the $D_+(f)$ for $f \in S_1$ (2.3.14), we are led to proving that λ_f and μ_f are isomorphisms, under the given hypotheses, whenever f is homogeneous and *of degree* 1. But we can then define a $\mathbf{Z}$-bilinear map $M_m \times N_n \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$ by sending (x,y) to the element $(x/f^m) \otimes (y/f^n)$ (if $m < 0$, we write x/f^m to mean $f^{-m}x/1$); these maps define a $\mathbf{Z}$-linear map $M \otimes_{\mathbf{Z}} N \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$, and, if $s \in S_q$, this map sends $(sx) \otimes y$ to $(s/f^q)((x/f^m) \otimes (y/f^n))$ (for $x \in M_m$ and $y \in N_n$). We thus obtain a di-homomorphism of modules $\gamma_f : M \otimes_S N \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$, with respect to the canonical homomorphism $S \to S_{(f)}$ (sending $s \in S_q$ to s/f^q). Suppose furthermore that, for an element $\sum_i (x_i \otimes y_i)$ of $M \otimes_S N$ (with x_i and y_i homogeneous of degree m_i and n_i, respectively), we have that $f^r \sum_i (x_i \otimes y_i) = 0$, or, in other words, that $\sum_i (f^r x_i \otimes y_i) = 0$. We thus deduce, by (**0**, 1.3.4), that $\sum_i (f^r x_i / f^{m_i + r}) \otimes (y_i / f^{n_i}) = 0$, i.e. $\gamma_f(\sum_i (x_i \otimes y_i)) = 0$. Then γ_f factors as $M \otimes_S N \to (M \otimes_S N)_f \xrightarrow{\gamma'_f} M_{(f)} \otimes_{S_{(f)}} N_{(f)}$; if λ'_f is the restriction of γ' to $(M \otimes_S N)_{(f)}$, **II** | 35
then we can immediately show that λ_f and λ'_f are inverse $S_{(f)}$-homomorphisms, whence the first part of the proposition.

To prove the second part, suppose that M is the cokernel of a homomorphism $P \to Q$ of graded S-modules, with P and Q being direct sums of a finite number of modules of the form $S(n)$; using the left-exactness of $\mathrm{Hom}_S(L,N)$ in L, and the exactness of $M_{(f)}$ in M, we can immediately reduce to proving that μ_f is an isomorphism whenever $M = S(n)$. But, for any homogeneous z in N, let u_z be the homomorphism from $S(n)$ to N such that $u_z(1) = z$; we immediately see that $\eta : z \to u_z$ is an isomorphism of degree 0 from $N(-n)$ to $\mathrm{Hom}_S(S(n), N)$. There is a corresponding isomorphism

$$\eta_f : (N(-n))_{(f)} \longrightarrow (\mathrm{Hom}_S(S(n), N))_{(f)}.$$

Now let η'_f be the isomorphism $N_{(f)} \to \mathrm{Hom}_{S_{(f)}}(S(n)_{(f)}, N_{(f)})$ that, to any $z' \in N_{(f)}$, associates the homomorphism $v_{z'}$ that is such that $v_{z'}(s/f^k) = sz'/f^{n+k}$ (for $s \in S_{n+k} = (S(n))_k$). We easily note that the composed map

$$(N(-n))_{(f)} \xrightarrow{\eta_f} (\mathrm{Hom}_S(S(n), N))_{(f)} \xrightarrow{\mu_f} \mathrm{Hom}_{S_{(f)}}(S(n)_{(f)}, N_{(f)}) \xrightarrow{\eta'^{-1}_f} N_{(f)}$$

is the isomorphism $z/f^h \mapsto z/f^{h-n}$ from $(N(-n))_{(f)}$ to $N_{(f)}$, and thus μ_f is an isomorphism. □

If the ideal S_+ is generated by S_1, then we deduce from (2.5.13) that, for every graded ideal $\mathfrak{J}$ of S, and for every graded S-module M, we have

(2.5.13.1) $$\widetilde{\mathfrak{J}} \cdot \widetilde{M} = (\mathfrak{J} \cdot M)^\sim$$

up to canonical isomorphism; this follows from the commutativity of the diagram

$$\begin{tikzcd}
\widetilde{\mathfrak{J}} \otimes_{\mathscr{O}_X} \widetilde{M} \arrow[rr, "\lambda"] \arrow[dr] & & (\mathfrak{J} \otimes_S M)^\sim \arrow[dl] \\
& \widetilde{M} &
\end{tikzcd}$$

which we can verify as we did for (2.5.11.3).

Corollary (2.5.14). — *Suppose that S is generated by S_1. For any $m, n \in \mathbf{Z}$, we then have:*

$$\mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{O}_X(n) = \mathscr{O}_X(m+n) \tag{2.5.14.1}$$

$$\mathscr{O}_X(n) = (\mathscr{O}_X(1))^{\otimes n} \tag{2.5.14.2}$$

up to canonical isomorphism.

Proof. The first equation follows from (2.5.13) and from the existence of the canonical isomorphism $S(m) \otimes_S S(n) \xrightarrow{\sim} S(m+n)$ of degree 0 that sends the element $1 \otimes 1$ (where the first 1 is in $(S(m))_{-m}$ and the second is in $(S(n))_{-n}$) to the element $1 \in (S(m+n))_{-(m+n)}$. It then suffices to prove the second equation for $n = -1$, and, by (2.5.13), this reduces to seeing that $\operatorname{Hom}_S(S(1), S)$ is canonically isomorphic to $S(-1)$, which can be immediately proven by going back to the definitions (2.1.2) and by remembering that $S(1)$ is a monogeneous S-module. □

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Corollary (2.5.15). — *Suppose that S is generated by S_1. Then, for every graded S-module M, and for every $n \in \mathbf{Z}$, we have*

$$(M(n))^{\sim} = \widetilde{M}(n) \tag{2.5.15.1}$$

up to canonical isomorphism.

Proof. This follows from definitions (2.5.10.2) and (2.5.10.1), from Proposition (2.5.13), and from the existence of a canonical isomorphism $M(n) \xrightarrow{\sim} M \otimes_S S(n)$ of degree 0 that, to every $z \in (M(n))_h = M_{n+h}$, associates $z \otimes 1 \in M_{n+h} \otimes (S(n))_{-n} \subset (M \otimes_S S(n))_h$. □

(2.5.16). We denote by S' the graded ring such that $S'_0 = \mathbf{Z}$, and $S'_n = S_n$ for $n > 0$. Then, if $f \in S_d$ ($d > 0$), we have that $(S(n))_{(f)} = (S'(n))_{(f)}$ for all $n \in \mathbf{Z}$, since an element of $(S'(n))_{(f)}$ is of the form x/f^k, with $x \in S'_{n+kd}$ ($k > 0$), and we can always take k to be such that $n + kd \neq 0$. Since $X = \operatorname{Proj}(S)$ and $X' = \operatorname{Proj}(S')$ are canonically identified (2.4.7, (ii)), we see that, for all $n \in \mathbf{Z}$, $\mathscr{O}_X(n)$ and $\mathscr{O}_{X'}(n)$ are the images of one another under the above identification.

Note also that, for all $d > 0$ and all $n \in \mathbf{Z}$, we have

$$(S^{(d)}(n))_h = S_{(n+h)d} = (S(nd))_{hd}$$

for $f \in S_d$, and thus $(S^{(d)}(n))_{(f)} = (S(nd))_{(f)}$. We know that the schemes $X = \operatorname{Proj}(S)$ and $X^{(d)} = \operatorname{Proj}(S^{(d)})$ are canonically identified (2.4.7, (ii)); the above shows that, if the S_0-algebra $S^{(d)}$ is generated by S_d, then $\mathscr{O}_X(nd)$ and $\mathscr{O}_{X^{(d)}}(n)$ are the images of one another under this identification, for all $n \in \mathbf{Z}$.

Proposition (2.5.17). — *Let $d > 0$ be an integer, and let $U = \bigcup_{f \in S_d} D_+(f)$. Then the restriction to U of the canonical homomorphism $\mathscr{O}_X(nd) \otimes_{\mathscr{O}_X} \mathscr{O}_X(-nd) \to \mathscr{O}_X$ is an isomorphism for every integer n.*

Proof. By (2.5.16), we can restrict to the case where $d = 1$, and the conclusion then follows from the proof of (2.5.13). □

2.6. The graded S-module associated to a sheaf on $\mathrm{Proj}(S)$ *We suppose all throughout this section that the ideal S_+ of S is generated by the set S_1 of homogeneous elements of degree* 1.

(2.6.1). The $\mathscr{O}_X$-module $\mathscr{O}_X(1)$ is *invertible* (2.5.9); we thus define, for every $\mathscr{O}_X$-module $\mathscr{F}$ (**0**, 5.4.6),

$$\Gamma_\bullet(\mathscr{F}) = \Gamma_\bullet(\mathscr{O}_X(1), \mathscr{F}) = \bigoplus_{n\in\mathbf{Z}} \Gamma(X, \mathscr{F}(n)) \tag{2.6.1.1}$$

taking (2.5.14.2) into account. Recall (**0**, 5.4.6) that $\Gamma_\bullet(\mathscr{O}_X)$ is endowed with the structure of a *graded ring*, and $\Gamma_\bullet(\mathscr{F})$ with the structure of a *graded* $\Gamma_\bullet(\mathscr{O}_X)$*-module*.

Since $\mathscr{O}_X(n)$ is locally free, $\Gamma_\bullet(\mathscr{F})$ is a *left exact* additive covariant functor in $\mathscr{F}$; in particular, if $\mathscr{J}$ is a sheaf of ideals of $\mathscr{O}_X$, then $\Gamma_\bullet(\mathscr{J})$ is canonically identified with a *graded ideal* of $\Gamma_\bullet(\mathscr{O}_X)$.

(2.6.2). Let M be a graded S-module; for every $f \in S_d$ $(d > 0)$, $x \mapsto x/1$ is a homomorphism of abelian groups $M_0 \to M_{(f)}$, and, since $M_{(f)}$ is canonically identified with $\Gamma(D_+(f), \widetilde{M})$, we thus obtain a homomorphism of abelian groups $\alpha_0^f : M_0 \to \Gamma(D_+(f), \widetilde{M})$. It is clear that, for every $g \in S_e$ $(e > 0)$, the diagram **II | 37**

$$\begin{tikzcd} & \Gamma(D_+(f), \widetilde{M}) \arrow[d] \\ M_0 \arrow[ur, "\alpha_0^f"] \arrow[dr, "\alpha_0^{fg}"'] & \\ & \Gamma(D_+(fg), \widetilde{M}) \end{tikzcd}$$

commutes; this implies that, for all $x \in M_0$, the sections $\alpha_0^f(x)$ and $\alpha_0^g(x)$ of M agree on $D_+(f) \cap D_+(g)$, and thus there exists a unique section $\alpha_0(x) \in \Gamma(X, \widetilde{M})$ whose restriction to each $D_+(f)$ is $\alpha_0^f(x)$. We have thus defined (without using the hypothesis that S be generated by S_1) a homomorphism of abelian groups

$$\alpha_0 : M_0 \longrightarrow \Gamma(X, \widetilde{M}). \tag{2.6.2.1}$$

Applying this result to the graded S-module $M(n)$ (for each $n \in \mathbf{Z}$), we obtain, for each $n \in \mathbf{Z}$, a homomorphism of abelian groups

$$\alpha_n : M_n = (M(n))_0 \longrightarrow \Gamma(X, \widetilde{M}(n)) \tag{2.6.2.2}$$

(taking (2.5.15)); whence we obtain a functorial homomorphism (of degree 0) of graded abelian groups

$$\alpha : M \longrightarrow \Gamma_\bullet(\widetilde{M}) \tag{2.6.2.3}$$

(also denoted by α_M) which, on each M_n, agrees with α_n.

If we take, in particular, $M = S$, then we immediately see (taking into account the definition (**0**, 5.4.6) of multiplication in $\Gamma_\bullet(\mathscr{O}_X)$) that $\alpha : S \to \Gamma_\bullet(\mathscr{O}_X)$ is a homomorphism of graded rings, and that, for every graded S-module M, (2.6.2.3) is a di-homomorphism of graded modules.

Proposition (2.6.3). — *For every $f \in S_d$ $(d > 0)$, $D_+(f)$ is identical to the set of $\mathfrak{p} \in X$ on which the section $\alpha_d(f)$ of $\mathscr{O}_X(d)$ does not vanish* (**0**, 5.5.2).

Proof. Since $X = \bigcup_{g\in S_1} D_+(g)$ by hypothesis, it suffices to show that, for all $g \in S_1$, the set of $\mathfrak{p} \in D_+(g)$ on which $\alpha_d(f)$ does not vanish is identical to $D_+(fg)$. But the restriction of $\alpha_d(f)$ to $D_+(g)$ is, by definition, the section corresponding to the element $f/1$ of $(S(d))_{(g)}$; under the canonical isomorphism $(S(d))_{(g)} \xrightarrow{\sim} S_{(g)}$ (2.5.7), this section of $\mathscr{O}_X(d)$ over $D_+(g)$ is identified with the section of $\mathscr{O}_X$ over $D_+(g)$ that corresponds to the element f/g^d of $S_{(g)}$; to say that this section vanishes at $\mathfrak{p} \in D_+(g)$ implies that $f/g^d \in \mathfrak{q}$, where $\mathfrak{q}$ is the prime ideal of $S_{(g)}$ corresponding to $\mathfrak{p}$ (2.3.6); by definition, this implies that $f \in \mathfrak{p}$, whence the proposition. □

(2.6.4). Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module, and set $M = \Gamma_\bullet(\mathscr{F})$; by the existence of the homomorphism of graded rings $\alpha : S \to \Gamma_\bullet(\mathscr{O}_X)$, we can consider M as a graded S-module. For every $f \in S_d$ $(d > 0)$, it follows from (2.6.3) that the restriction to $D_+(f)$ of the section $\alpha_d(f)$ of $\mathscr{O}_X(d)$ is invertible; thus so too is the restriction to $D_+(f)$ of the section $\alpha_d(f^n)$ of $\mathscr{O}_X(nd)$, for all $n > 0$. So let $z \in M_{nd} = \Gamma(X, \mathscr{F}(nd))$ $(n > 0)$; if there exists an integer $k \geqslant 0$ such that the restriction to $D_+(f)$ of $f^k z$, i.e. the section $(z|D_+(f))(\alpha_d(f^k)|D_+(f))$ of $\mathscr{F}((n+k)d)$, is zero, then, by the above remark, we also have that $z|D_+(f) = 0$. This shows that we have defined an $S_{(f)}$-homomorphism $\beta_f : M_{(f)} \to \Gamma(D_+(f), \mathscr{F})$ by sending the element z/f^n to the section $(z|D_+(f))(\alpha_d(f^n)|D_+(f))^{-1}$ of $\mathscr{F}$ over $D_+(f)$. We can further immediately show that the diagram **II | 38**

(2.6.4.1)
$$\begin{tikzcd} M_{(f)} \arrow[r, "\beta_f"] \arrow[d] & \Gamma(D_+(f), \mathscr{F}) \arrow[d] \\ M_{(fg)} \arrow[r, "\beta_{fg}"'] & \Gamma(D_+(fg), \mathscr{F}) \end{tikzcd}$$

commutes for $g \in S_e$ $(e > 0)$. If we recall that $M_{(f)}$ is canonically identified with $\Gamma(D_+(f), \widetilde{M})$, and that the $D_+(f)$ form a base for the topology of X (2.3.4), then we see that the β_f come from a unique canonical homomorphism of $\mathscr{O}_X$-modules

(2.6.4.2)
$$\beta : (\Gamma_\bullet(\mathscr{F}))^\sim \longrightarrow \mathscr{F}$$

(also denoted by $\beta_{\mathscr{F}}$) which is evidently functorial.

Proposition (2.6.5). — *Let M be a graded S-module, and $\mathscr{F}$ an $\mathscr{O}_X$-module; then the composite homomorphisms*

(2.6.5.1)
$$\widetilde{M} \xrightarrow{\tilde{\alpha}} (\Gamma_\bullet(\widetilde{M}))^\sim \xrightarrow{\beta} \widetilde{M}$$

(2.6.5.2)
$$\Gamma_\bullet(\mathscr{F}) \xrightarrow{\alpha} \Gamma_\bullet((\Gamma_\bullet(\mathscr{F}))^\sim) \xrightarrow{\Gamma_\bullet(\beta)} \Gamma_\bullet(\mathscr{F})$$

are the identity isomorphisms.

Proof. The proof for (2.6.5.1) is local: in an open subset $D_+(f)$, it follows immediately from the definitions, along with the fact that β, applied to quasi-coherent sheaves, is determined by its action on the sections over $D_+(f)$ (I, 1.3.8). The proof for (2.6.5.2) is done for each degree separately: if we set $M = \Gamma_\bullet(\mathscr{F})$, then $M_n = \Gamma(X, \mathscr{F}(n))$, and $(\Gamma_\bullet(\widetilde{M}))_n = \Gamma(X, \widetilde{M}(n)) = \Gamma(X, (M(n))^\sim)$. But if $f \in S_1$ and $z \in M_n$, then $\alpha_n^f(z)$ is the element $z/1$ of $(M(n))_{(f)}$, equal to $(f/1)^n(z/f^n)$; it corresponds, via β_f, to the section

$$\Big((\alpha_1(f))^n|D_+(f)\Big)\Big((z|D_+(f))\,((\alpha_1(f))^n|D_+(f))^{-1}\Big)$$

over $D_+(f)$, i.e. the restriction of z to $D_+(f)$, which finishes the proof for (2.6.5.2). □

2.7. Finiteness conditions

Proposition (2.7.1). —
(i) *If S is a graded Noetherian ring, then $X = \mathrm{Proj}(S)$ is a Noetherian scheme.*
(ii) *If S is a graded A-algebra of finite type, then $X = \mathrm{Proj}(S)$ is a scheme of finite type over $Y = \mathrm{Spec}(A)$.*

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Proof.

(i) If S is Noetherian, then the ideal S_+ admits a finite system of homogeneous generators f_i $(1 \leqslant i \leqslant p)$, thus (2.3.14) the underlying space X is the union of the $D_+(f_i) = \mathrm{Spec}(S_{(f_i)})$, and everything then reduces to showing that each of the $S_{(f_i)}$ is Noetherian, which follows from (2.2.6).

(ii) The hypothesis implies that S_0 is an A-algebra of finite type, and that S is an S_0-algebra of finite type, and so S_+ is an ideal of finite type (2.1.4). We are thus reduced, as in (i), to showing that $S_{(f)}$ is an A-algebra of finite type for all $f \in S_d$. By (2.2.5), it suffices to show that $S^{(d)}$ is an A-algebra of finite type, which follows from (2.1.6).

□

(2.7.2). In what follows, we consider the following finiteness conditions for a graded S-module M:

(TF) There exists an integer n such that the submodule $\bigoplus_{k\geqslant n} M_k$ is an S-module of finite type.
(TN) There exists an integer n such that $M_k = 0$ for $k \geqslant n$.

If M satisfies (TN), then $M_{(f)} = 0$ for all homogeneous f in S_+, and thus $\widetilde{M} = 0$.

Let M and N be graded S-modules; we say that a homomorphism $u : M \to N$ of degree 0 is (TN)-*injective* (resp. (TN)-*surjective*, (TN)-*bijective*) if there exists an integer n such that $u_k : M_k \to N_k$ is injective (resp. surjective, bijective) for $k \geqslant n$. Saying that u is (TN)-injective (resp. (TN)-surjective) thus reduces to saying that $\operatorname{Ker} u$ (resp. $\operatorname{Coker} u$) satisfies (TN). By (2.5.4), if u is (TN)-injective (resp. (TN)-surjective, (TN)-bijective), then $\tilde{u}$ is injective (resp. surjective, bijective); if u is (TN)-bijective, then we also say that u is a (TN)-*isomorphism*.

Proposition (2.7.3). — *Let S be a graded ring such that the ideal S_+ is of finite type, and let M be a graded S-module.*

(i) *If M satisfies condition (TF) then the $\mathscr{O}_X$-module $\widetilde{M}$ is of finite type.*
(ii) *Suppose that M satisfies (TF); for $\widetilde{M} = 0$, it is necessary and sufficient for M to satisfy (TN).*

Proof. We have just seen that condition (TN) implies that $\widetilde{M} = 0$. If M satisfies (TF), then the graded submodule $M' = \bigoplus_{k\geqslant n} M_k$, which is, by hypothesis, of finite type, is such that M/M' satisfies (TN); thus $(M/M')^{\sim}$, and the exactness of the functor $\widetilde{M}$ (2.5.4) implies that $\widetilde{M} = \widetilde{M'}$; to prove that $\widetilde{M}$ is of finite type, we can thus reduce to the case where M is *of finite type* . Since the question is local, it suffices to prove that $M_{(f)}$ is an $S_{(f)}$-module of finite type (**I**, 1.3.9); but $M^{(d)}$ is an $S^{(d)}$-module of finite type (2.1.6, iii), and our claim then follows from (2.2.5).

Now suppose that M satisfies (TF) and that $\widetilde{M} = 0$; then, with the same notation as above, we have that $\widetilde{M'} = 0$, and condition (TN) for M' is equivalent to condition (TN) for M, so to prove that $\widetilde{M} = 0$ implies that M satisfies (TN), we can again restrict to the case where M is generated by a finite number of homogeneous elements x_i $(1 \leqslant i \leqslant p)$; let $(f_j)_{1\leqslant j\leqslant q}$ be a system of homogeneous generators of the ideal S_+. By hypothesis, $M_{(f_j)} = 0$ for all j, and so there exists an integer n such that $f_j^n x_i = 0$ for any i and j. Let n_j be the degree of f_j, and let m be the largest value of $\sum_j r_j n_j$ for the system of finitely many integers (r_j) such that $\sum_j r_j \leqslant nq$; it is then clear that, if $k > m$, then $S_k x_i = 0$ **II** | 40
for all i; if h is the largest of the degrees of the x_i, then we conclude that $M_k = 0$ for $k > h + m$, which finishes the proof. □

Corollary (2.7.4). — *Let S be a graded ring such that the ideal S_+ is of finite type; for $X = \operatorname{Proj}(S) = \varnothing$, it is necessary and sufficient for there to exist n such that $S_k = 0$ for $k \geqslant n$.*

Proof. The condition $X = \varnothing$ is equivalent to $\mathscr{O}_X = \widetilde{S} = 0$, and S is a monogeneous S-module. □

Theorem (2.7.5). — *Suppose that the ideal S_+ is generated by a finite number of homogeneous elements of degree 1; let $X = \operatorname{Proj}(S)$. Then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\beta : (\Gamma_\bullet(\mathscr{F}))^{\sim} \to \mathscr{F}$ (2.6.4) is an isomorphism.*

Proof. If S_+ is generated by a finite number of elements $f_i \in S_1$, then X is the union of the subspaces $\operatorname{Spec}(S_{(f_i)})$ (2.3.6), which are quasi-compact, and so X is quasi-compact; furthermore, X is a scheme (2.4.2); by (**I**, 9.3.2), (2.5.14.2), and (2.6.3), we have, for all $f \in S_d$ $(d > 0)$, a canonical isomorphism $(\Gamma_\bullet(\mathscr{F}))_{(\alpha_d(f))} \xrightarrow{\sim} \Gamma(D_+(f), \mathscr{F})$; also, by definition, $(\Gamma_\bullet(\mathscr{F}))_{(\alpha_d(f))}$ (where $\Gamma_\bullet(\mathscr{F})$ is considered as a $\Gamma_\bullet(\mathscr{O}_X)$-module) is exactly $(\Gamma_\bullet(\mathscr{F}))_{(f)}$ (where $\Gamma_\bullet(\mathscr{F})$ is considered as an S-module); if we refer to the definition (**I**, 9.3.1) of the above canonical isomorphism, then we see that it agrees with the homomorphism β_f, whence the theorem. □

Remark (2.7.6). — If we suppose that the graded ring S is *Noetherian*, then the condition of (2.7.5) is satisfied *ipso facto* as soon as we suppose that the ideal S_+ is generated by the set S_1 of homogeneous elements of degree 1.

Corollary (2.7.7). — *Under the hypotheses of (2.7.5), every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is isomorphic to an $\mathscr{O}_X$-module of the form $\widetilde{M}$, where M is a graded S-module.*

Corollary (2.7.8). — *Under the hypotheses of (2.7.5), every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type is isomorphic to an $\mathscr{O}_X$-module of the form $\widetilde{N}$, where N is a graded S-module of finite type.*

Proof. We can suppose that $\mathscr{F} = \widetilde{M}$, where M is a graded S-module (2.7.7). Let $(f_\lambda)_{\lambda \in L}$ be a system of homogeneous generators of M; for every finite subset H of L, let M_H be the graded submodule of M generated by the f_λ such that $\lambda \in H$; it is clear that M is the inductive limit of its submodules M_H, and so $\mathscr{F}$ is the inductive limit of its sub-$\mathscr{O}_X$-modules $\widetilde{M_H}$ (2.5.4). But, since $\mathscr{F}$ is of finite type, and since the underlying space of X is quasi-compact, it follows from (**0**, 5.2.3) that $\mathscr{F} = \widetilde{M_H}$ for some finite subset H of L. □

Corollary (2.7.9). — *Under the hypotheses of (2.7.5), let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type. Then there exists an integer n_0 such that, for all $n \geqslant n_0$, $\mathscr{F}(n)$ is isomorphic to a quotient of an $\mathscr{O}_X$-module of the form $\mathscr{O}_X^k$ (for some $k > 0$ depending on n), and is thus generated by a finite number of its sections over X (**0**, 5.1.1).*

Proof. By (2.7.8), we can suppose that $\mathscr{F} = \widetilde{M}$, where M is a quotient of a finite direct sum of S-modules of the form $S(m_i)$; by (2.5.4), we can thus restrict to the case where $M = S(m)$, and so $\mathscr{F}(n) = (S(m+n))^\sim = \mathscr{O}_X(m+n)$ (2.5.15). It thus suffices to prove 41

Lemma (2.7.9.1). — *Under the hypotheses of (2.7.5), for all $n \geqslant 0$, there exists an integer k (depending on n) and a surjective homomorphism $\mathscr{O}_X^k \to \mathscr{O}_X(n)$.*

It suffices (2.7.2) to show that, for suitable k, there is a (TN)-surjective homomorphism u *of degree* 0 from the graded product S-module S^k to $S(n)$. But $(S(n))_0 = S_n$, and, by hypothesis, $S_h = S_1^h$ for all $h > 0$, and so $SS_n = S_n + S_{n+1} + \ldots$. Since S_n is an S_0-module of finite type ((2.1.5) and (2.1.6, i)), consider a system of generators $(a_i)_{1 \leqslant i \leqslant k}$ of this module; consider the homomorphism u that sends a_i to the i-th element of the canonical basis of S^k $(1 \leqslant i \leqslant k)$; since $\operatorname{Coker} u$ can then be identified with $(S(n))_{-n} + \ldots + (S(n))_{-1}$, u is indeed the desired homomorphism. □

Corollary (2.7.10). — *Under the hypotheses of (2.7.5), let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type. Then there exists an integer n_0 such that, for all $n \geqslant n_0$, $\mathscr{F}$ is isomorphic to a quotient of an $\mathscr{O}_X$-module of the form $(\mathscr{O}_X(-n))^k$ (for some k depending on n).*

Proposition (2.7.11). — *Suppose that the hypotheses of (2.7.5) are satisfied, and let M be a graded S-module. Then:*

- (i) *The canonical homomorphism $\widetilde{\alpha} : \widetilde{M} \to (\Gamma_\bullet(\widetilde{M}))^\sim$ is an isomorphism.*
- (ii) *Let $\mathscr{G}$ be a quasi-coherent sub-$\mathscr{O}_X$-module of $\widetilde{M}$, and let N be the graded sub-S-module of M given by the inverse image of $\Gamma_\bullet(\mathscr{G})$ under α. Then $\widetilde{N} = \mathscr{G}$ (where $\widetilde{N}$ is identified, by (2.5.4), with a sub-$\mathscr{O}_X$-module of $\widetilde{M}$).*

Proof. Since $\beta : (\Gamma_\bullet(\widetilde{M}))^\sim \to \widetilde{M}$ is an isomorphism (by (2.7.5)), $\widetilde{\alpha}$ is the inverse isomorphism (by (2.6.5.1)), whence (i). Let P be the graded submodule $\alpha(M)$ of $\Gamma_\bullet(\widetilde{M})$; since $\widetilde{M}$ is an exact functor (2.5.4), the image of $\widetilde{M}$ under $\widetilde{\alpha}$ is equal to $\widetilde{P}$, and so, by (i), $\widetilde{P} = (\Gamma_\bullet(\widetilde{M}))^\sim$. Set $Q = \Gamma_\bullet(\mathscr{G}) \cap P$; by the above, and by (2.5.4), we have that $\widetilde{Q} = (\Gamma_\bullet(\mathscr{G}))^\sim$, and so the restriction of β to $\widetilde{Q}$ is an *isomorphism* from this $\mathscr{O}_X$-module to G by (2.7.5). But, by the definition of N, and by (2.5.4), the restriction of the isomorphism $\widetilde{\alpha}$ to $\widetilde{N}$ is an isomorphism from $\widetilde{N}$ to $\widetilde{Q}$, whence the conclusion, by (2.6.5.1). □

2.8. Functorial behaviour

(2.8.1). Let S and S' be positively graded rings, and $\phi : S' \to S$ a homomorphism of graded rings. We denote by $G(\phi)$ the open subset of $X = \operatorname{Proj}(S)$ given by the complement of $V_+(\phi(S'_+))$, or, equivalently, the union of the $D_+(\phi(f'))$ where f' runs over the set of homogeneous elements of S'_+. The restriction to $G(\phi)$ of the continuous map ${}^a\phi$ from $\operatorname{Spec}(S)$ to $\operatorname{Spec}(S')$ (I, 1.2.1) is thus a continuous map from $G(\phi)$ to $\operatorname{Proj}(S')$, which we again denote, with an abuse of language, by ${}^a\phi$. If $f' \in S'_+$ is homogeneous, then

$$ {}^a\phi^{-1}(D_+(f')) = D_+(\phi(f')) \tag{2.8.1.1} $$

taking into account the fact that ${}^a\phi$ sends $G(\phi)$ to $\operatorname{Proj}(S')$, as well as (I, 1.2.2.2). The homomorphism ϕ also canonically defines (with the same notation) a homomorphism of graded rings $S'_{f'} \to S_f$, whence, by restriction to the degree 0 elements, a homomorphism $S'_{(f')} \to S_{(f)}$, which we denote II | 42
by $\phi_{(f)}$; there is a corresponding (I, 1.6.1) morphism of affine schemes $({}^a\phi_{(f)}, \widetilde{\phi}_{(f)}) : \operatorname{Spec}(S_{(f)}) \to \operatorname{Spec}(S'_{(f')})$. If we canonically identify $\operatorname{Spec}(S_{(f)})$ with the scheme induced by $\operatorname{Proj}(S)$ on $D_+(f)$ (2.3.6), then we have defined a morphism $\Phi_f : D_+(f) \to D_+(f')$, and ${}^a\phi_{(f)}$ is identified with the restriction of ${}^a\phi$ to $D_+(f)$. It is also immediate that, if g' is another homogeneous element of S'_+, and $g = \phi(g')$, then the diagram

$$\begin{tikzcd} D_+(f) \arrow[r, "\Phi_f"] & D_+(f') \\ D_+(fg) \arrow[u] \arrow[r, "\Phi_{fg}"'] & D_+(f'g') \arrow[u] \end{tikzcd}$$

commutes, by the fact that the diagram

$$\begin{tikzcd} S'_{(f')} \arrow[r, "\phi_{(f)}"] \arrow[d, "\omega_{f'g',f'}"'] & S_{(f)} \arrow[d, "\omega_{fg,f}"] \\ S'_{(f'g')} \arrow[r, "\phi_{(fg)}"'] & S_{(fg)} \end{tikzcd}$$

commutes. Taking the definition of $G(\phi)$, along with (2.3.3.2), we thus see that:

Proposition (2.8.2). — *Given a homomorphism of graded rings $\phi : S' \to S$, there exists exactly one morphism $({}^a\phi, \widetilde{\phi})$ from the induced prescheme $G(\phi)$ to $\operatorname{Proj}(S')$ (said to be* associated to ϕ, *and denoted by* $\operatorname{Proj}(\phi)$*) such that, for every homogeneous element $f' \in S'_+$, the restriction of this morphism to $D_+(\phi(f'))$ agrees with the morphism associated to the homomorphism $S'_{(f')} \to S_{(\phi(f'))}$ corresponding to ϕ.*

PROOF. With the above notation, if $f' \in S'_d$, then the diagram

$$\begin{tikzcd} S'_{(f')} \arrow[r, "\phi_{(f)}"] \arrow[d, "\sim"'] & S_{(f)} \arrow[d, "\sim"] \\ S'^{(d)}/(f'-1)S'^{(d)} \arrow[r] & S^{(d)}/(f-1)S^{(d)} \end{tikzcd} \tag{2.8.2.1}$$

commutes (the vertical arrows being the isomorphisms (2.2.5)). □

Corollary (2.8.3). — (i) *The morphism* $\operatorname{Proj}(\phi)$ *is affine.*

(ii) *If* $\operatorname{Ker}(\phi)$ *is nilpotent (and, in particular, if ϕ is injective), then the morphism* $\operatorname{Proj}(\phi)$ *is dominant.*

PROOF. Claim (i) is an immediate consequence of (2.8.2) and (2.8.1.1). Claim (ii) follows since, if $\operatorname{Ker}(\phi)$ is nilpotent, then, for every homogeneous f' in S'_+, we immediately see that $\operatorname{Ker}(\phi_f)$ (with $f = \phi(f')$) is nilpotent, and thus so too is $\operatorname{Ker}(\phi_{(f)})$; we then apply (2.8.2) and (I, 1.2.7) □

We note that there are, in general, morphisms from $\operatorname{Proj}(S)$ to $\operatorname{Proj}(S')$ that are not affine, and that thus do not come from homomorphisms of graded rings $S' \to S$; an example is the structure morphism $\operatorname{Proj}(S) \to \operatorname{Spec}(A)$ when A is a field ($\operatorname{Spec}(A)$ thus being identified with $\operatorname{Proj}(A[T])$, cf. (3.1.7)); indeed, this follows from (**I**, 2.3.2). **II** | 43

(2.8.4). Let S'' be another positively graded ring, and $\phi' : S'' \to S'$ a homomorphism of graded rings, and set $\phi'' = \phi \circ \phi'$. By (2.8.1.1) and the formula ${}^a\phi'' = ({}^a\phi') \circ ({}^a\phi)$, we immediately see that $G(\phi'') \subset G(\phi)$, and that, if Φ, Φ', and Φ'' are the morphisms associated to ϕ, ϕ', and ϕ'' (respectively), then $\Phi'' = \Phi' \circ (\Phi|G(\phi''))$.

(2.8.5). Suppose that S (resp. S') is a graded A-algebra (resp. a graded A'-algebra), and let $\psi : A' \to A$ be a ring homomorphism such that the diagram

(2.8.5.1)
$$\begin{tikzcd} A' \arrow[r, "\psi"] \arrow[d] & A \arrow[d] \\ S' \arrow[r, "\phi"'] & S \end{tikzcd}$$

commutes. We can then consider $G(\phi)$ (resp. $\operatorname{Proj}(S')$) as a scheme over $\operatorname{Spec}(A)$ (*resp.* $\operatorname{Spec}(A')$); if Φ (resp. Ψ) is the morphism associated to ϕ (resp. ψ), then the diagram

$$\begin{tikzcd} G(\phi) \arrow[r, "\Phi"] \arrow[d] & \operatorname{Proj}(S') \arrow[d] \\ \operatorname{Spec}(A) \arrow[r, "\Psi"'] & \operatorname{Spec}(A') \end{tikzcd}$$

commutes: it suffices to prove this for the restriction of Φ to $D_+(f)$, where $f = \phi(f')$, with f' homogeneous in S'_+; this then follows from the fact that the diagram

$$\begin{tikzcd} A' \arrow[r, "\psi"] \arrow[d] & A \arrow[d] \\ S'_{(f')} \arrow[r, "\phi_{(f)}"'] & S_{(f)} \end{tikzcd}$$

commutes.

(2.8.6). Now let M be a graded S-module, and consider the S'-module $M_{[\phi]}$, which is evidently graded. Let f' be homogeneous in S'_+, and let $f = \phi(f')$; we know (**0**, 1.5.2) that there is a canonical isomorphism $(M_{[\phi]})_{f'} \xrightarrow{\sim} (M_f)_{[\phi_f]}$, and it is immediate that this isomorphism preserves degree, whence a canonical isomorphism $(M_{[\phi]})_{(f')} \xrightarrow{\sim} (M_{(f)})_{[\phi_{(f)}]}$. To this isomorphism, there canonically corresponds an isomorphism of sheaves $(M_{[\phi]})^{\sim}|D_+(f') \xrightarrow{\sim} (\Phi_f)_*(\widetilde{(M)}|D_+(f))$ ((2.5.2) and (**I**, 1.6.3)). Furthermore, if g' is another homogeneous element of S'_+, and $g = \phi(g')$, then the diagram | 43

$$\begin{tikzcd} (M_{[\phi]})_{(f')} \arrow[r, "\sim"] \arrow[d] & (M_{(f)})_{[\phi_{(f)}]} \arrow[d] \\ (M_{[\phi]})_{(f'g')} \arrow[r, "\sim"] & (M_{(fg)})_{[\phi_{(fg)}]} \end{tikzcd}$$

commutes, whence we immediately conclude that the isomorphism

$$(M_{[\phi]})^{\sim}|D_+(f'g') \xrightarrow{\sim} (\Phi_{fg})_*(\widetilde{M}|D_+(fg))$$

is the restriction to $D_+(f'g')$ of the isomorphism $(M_{[\phi]})^{\sim}|D_+(f') \xrightarrow{\sim} (\Phi_f)_*(\widetilde{M}|D_+(f))$. Since Φ_f is the restriction to $D_+(f)$ of the morphism Φ, we see that, taking (2.8.1.1) into account, and setting $X' = \operatorname{Proj}(S)'$:

Proposition (2.8.7). — *There exists a canonical functorial isomorphism from the $\mathscr{O}_{X'}$-module $(M_{[\phi]})^{\sim}$ to the $\mathscr{O}_{X'}$-module $\Phi_*(\widetilde{M}|G(\phi))$.*

We thus immediately deduce a canonical functorial map from the set of ϕ-morphisms $M' \to M$ from a graded S'-module to the graded S-module M, to the set of Φ-morphisms $\widetilde{M'} \to \widetilde{M}|G(\phi)$. With the notation of (2.8.4), if M'' is a graded S''-module, then, to the composition of a ϕ-morphism $M' \to M$ and a ϕ'-morphism $M'' \to M'$, canonically corresponds the composition of $\widetilde{M'}G(\phi') \to \widetilde{M}|G(\phi'')$ and $\widetilde{M''} \to \widetilde{M'}|G(\phi')$.

Proposition (2.8.8). — *Under the hypotheses of (2.8.1), let M' be a graded S'-module. Then there exists a canonical functorial homomorphism ν from the $(\mathscr{O}_X|G(\phi))$-module $\Phi^*(\widetilde{M'})$ to the $(\mathscr{O}_X|G(\phi))$-module $(M' \otimes_{S'} S)^{\sim}|G(\phi)$. If the ideal S'_+ is generated by S'_1, then ν is an isomorphism.*

PROOF. Indeed, for $f' \in S'_d$ $(d > 0)$, we define a canonical functorial homomorphism of $S_{(f)}$-modules (where $f = \phi(f')$)

(2.8.8.1) $$\nu_f : M'_{(f')} \otimes_{S'_{(f')}} S_{(f)} \longrightarrow (M' \otimes_{S'} S)_{(f)}$$

by composing the homomorphism $M'_{(f')} \otimes_{S'_{(f')}} S_{(f)} \to M'_{f'} \otimes_{S'_{f'}} S_f$ and the canonical isomorphism $M'_{f'} \otimes_{S'_{f'}} S_f \xrightarrow{\sim} (M' \otimes_{S'} S)_f$ (**0**, 1.5.4), and noting that the latter preserves degrees. We can immediately verify the compatibility of ν_f with the restriction operators from $D_+(f)$ to $D_+(fg)$ (for any $g' \in S'_+$ and $g = \phi(g')$), whence the definition of the homomorphism

$$\nu : \Phi^*(\widetilde{M'}) \longrightarrow (M' \otimes_{S'} S)^{\sim}|G(\phi)$$

taking (**I**, 1.6.5) into account. To prove the second claim, it suffices to show that ν_f is an isomorphism for all $f' \in S_1$, since $G(\phi)$ is then a union of the $D_+(\phi(f'))$. We first define a $\mathbf{Z}$-bilinear $M'_m \times S_n \to M'_{(f')} \otimes_{S'_{(f')}} S_{(f)}$ by sending (x', s) to the element $(x'/f'^m) \otimes (s/f^n)$ (with the convention that x'/f'^m is $f'^{-m}x'/1$ when $m < 0$). We claim that, in the proof of (2.5.13), this map gives rise to a **II** | 45
di-homomorphism of modules

$$\eta_f : M' \otimes_{S'} S \longrightarrow M'_{(f')} \otimes_{S_{(f')}} S_{(f)}.$$

Furthermore, if, for $r > 0$, we have $f^r \sum_i (x'_i \otimes s_i) = 0$, then this can also be written as $\sum_i (f'^r x'_i \otimes s_i) = 0$, whence, by (**0**, 1.5.4), $\sum_i (f'^r x_i / f'^{m_i+r}) \otimes (s_i/f^{n_i}) = 0$, i.e. $\eta_f(\sum_i x_i \otimes y_i)0 =$, which proves that η_f factors as $M' \otimes_{S'} S \to (M' \otimes_{S'} S)_f \xrightarrow{\eta'_f} M'_{(f')} \otimes_{S'_{(f')}} S_{(f)}$; we finally can prove that η'_f and ν_f are inverse isomorphisms to one another.

In particular, it follows from (2.1.2.1) that we have a canonical homomorphism

(2.8.8.2) $$\Phi^*(\mathscr{O}_{X'}(n)) \xrightarrow{\sim} \mathscr{O}_X(n)|G(\phi)$$

for all $n \in \mathbf{Z}$. □

(2.8.9). Let A and A' be rings, and $\psi : A' \to A$ a ring homomorphism, defining a morphism $\Psi : \operatorname{Spec}(A) \to \operatorname{Spec}(A')$. Let S' be a positively graded A'-algebra, and set $S = S' \otimes_{A'} A$, which is evidently an A-algebra graded by the $S'_n \otimes_{A'} A$; the map $\phi : s' \to s' \otimes 1$ is then a graded ring homomorphism that makes the diagram (2.8.5.1) commute. Since S_+ is here the A-module generated by $\phi(S'_+)$, we have $G(\phi) = \operatorname{Proj}(S) = X$; whence, setting $X' = \operatorname{Proj}(S')$, we have the commutative diagram

(2.8.9.1)
$$\begin{tikzcd} X \arrow[r, "\Phi"] \arrow[d, "p"'] & X' \arrow[d] \\ Y \arrow[r, "\Psi"'] & Y' \end{tikzcd}$$

Now let M' be a graded S'-module, and set $M = M' \otimes_{A'} A = M' \otimes_{S'} S$. Under these conditions:

Proposition (2.8.10). — *The diagram (2.8.9.1) identifies the scheme X with the product $X' \times_{Y'} Y$; furthermore, the canonical homomorphism $\nu : \Phi^*(\widetilde{M'}) \to \widetilde{M}$ (2.8.8) is an isomorphism.*

PROOF. The first claim will be proven if we show that, for every homogenous f' in S'_+, setting $f = \phi(f')$, the restrictions of Φ and p to $D_+(f)$ identify this scheme with the product $D_+(f') \times_{Y'} Y$ (**I**, 3.2.6.2); in other words, it suffices (**I**, 3.2.2) to prove that $S_{(f)}$ is canonically identified with $S_f \xrightarrow{\sim} S'_{f'} \otimes_{A'} A$, which is immediate by the existence of the canonical isomorphism $S_f \xrightarrow{\sim} S'_{f'} \otimes_{A'} A$ that preserves degrees (**0**, 1.5.4). The second claim then follows from the fact that $M'_{(f')} \otimes_{S'_{(f')}} S_{(f)}$ can be identified, by the above, with $M'_{(f')} \otimes_{A'} A$, and this can be identified with $M_{(f)}$, since M_f is canonically identified with $M'_{f'} \otimes_{A'} A$ by an isomorphism that preserves degrees. □

Corollary (2.8.11). — *For every integer $n \in \mathbf{Z}$, $\widetilde{M}(n)$ can be identified with $\Phi^*(\widetilde{M'}(n)) = \widetilde{M'}(n) \otimes_{Y'} \mathscr{O}_Y$; in particular, $\mathscr{O}_X(n) = \Phi^*(\mathscr{O}_{X'}(n)) = \mathscr{O}_{X'}(n) \otimes_{Y'} \mathscr{O}_Y$.*

PROOF. This follows from (2.8.10) and (2.5.15). □

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(2.8.12). Under the hypotheses of (2.8.9), for $f' \in S'_d$ ($d > 0$) and $f = \phi(f')$, the diagram

(2.8.12.1)
$$\begin{tikzcd} M'_{(f')} \arrow[r, "\sim"] \arrow[d] & M'^{(d)}/(f'-1)M'^{(d)} \arrow[d] \\ M_{(f)} \arrow[r, "\sim"] & M^{(d)}/(f-1)M^{(d)} \end{tikzcd}$$

(cf. (2.2.5)) commutes.

(2.8.13). Keep the notation and hypotheses of (2.8.9), and let $\mathscr{F}'$ be an $\mathscr{O}_{X'}$-module; if we set $\mathscr{F} = \Phi^*(\mathscr{F}')$, then, for all $n \in \mathbf{Z}$, we have $\mathscr{F}(n) = \Phi^*(\mathscr{F}'(n))$, by (2.8.11) and (**0**, 4.3.3). Then (**0**, 3.7.1) we have a canonical homomorphism

$$\Gamma(\rho) : \Gamma(X', \mathscr{F}'(n)) \longrightarrow \Gamma(X, \mathscr{F}(n))$$

which gives a canonical di-homomorphism of graded modules

$$\Gamma_\bullet(\mathscr{F}') \longrightarrow \Gamma_\bullet(\mathscr{F}).$$

Suppose that the ideal S_+ is generated by S_1, and that $\mathscr{F}' = \widetilde{M'}$, thus $\mathscr{F} = \widetilde{M}$ with $M = M' \otimes_{A'} A$. If f' is homogeneous in S'_+, and $f = \phi(f')$, then we have seen that $M_{(f)} = M'_{(f')} \otimes_{A'} A$, and the diagram

$$\begin{tikzcd} M'_0 \arrow[r] \arrow[d] & M'_{(f')} \arrow[d] & = \Gamma(D_+(f'), \widetilde{M'}) \\ M_0 \arrow[r] & M_{(f)} & = \Gamma(D_+(f), \widetilde{M}) \end{tikzcd}$$

thus commutes; we immediately conclude from this remark, and from the definition of the homomorphism α (2.6.2), that the diagram

(2.8.13.1)
$$\begin{tikzcd} M' \arrow[r, "\alpha_{M'}"] \arrow[d] & \Gamma_\bullet(\widetilde{M'}) \arrow[d] \\ M \arrow[r, "\alpha_M"'] & \Gamma_\bullet(\widetilde{M}) \end{tikzcd}$$

commutes. Similarly, the diagram

(2.8.13.2)
$$\begin{tikzcd} (\Gamma_\bullet(\mathscr{F}'))^\sim \arrow[r, "\beta_{\mathscr{F}'}"] \arrow[d] & \mathscr{F}' \arrow[d] \\ (\Gamma_\bullet(\mathscr{F}))^\sim \arrow[r, "\beta_{\mathscr{F}}"'] & \mathscr{F} \end{tikzcd}$$

commutes (the vertical arrow on the right being the canonical Φ-morphism $\mathscr{F}' \to \Phi^*(\mathscr{F}') = \mathscr{F}$).

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(2.8.14). Still keeping the notation and hypotheses of (2.8.9), let N' be another graded S'-module, and let $N = N' \otimes_{A'} A$. It is immediate that the canonical di-homomorphisms $M' \to M$ and $N' \to N$ give a di-homomorphism $M' \otimes_{S'} N' \to M \otimes_S N$ (with respect to the canonical ring homomorphism $S' \to S$), and thus also an S-homomorphism $(M' \otimes_{S'} N') \otimes_{A'} A \to M \otimes_S N$ of degree 0, to which corresponds (taking (2.8.10) into account) a homomorphism of $\mathscr{O}_X$-modules

$$\Phi^*((M' \otimes_{S'} N')^\sim) \longrightarrow (M \otimes_S N)^\sim. \tag{2.8.14.1}$$

Furthermore, we can immediately verify that the diagram

$$\begin{tikzcd} \Phi^*(\widetilde{M'} \otimes_{\mathscr{O}_{X'}} \widetilde{N'}) \arrow[r, "\sim"] \arrow[d, "\Phi^*(\lambda)"'] & \widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N} \arrow[d, "\lambda"] & = \Phi^*(\widetilde{M'}) \otimes_{\mathscr{O}_X} \Phi^*(\widetilde{N'}) \\ \Phi^*((M' \otimes_{S'} N')^\sim) \arrow[r] & (M \otimes_S N)^\sim & \end{tikzcd} \tag{2.8.14.2}$$

commutes, with the first line being the canonical isomorphism (**0**, 4.3.3). If the ideal S'_+ is generated by S'_1, then it is clear that S_+ is generated by S_1, and the two vertical arrows of (2.8.14.2) are then isomorphisms (2.5.13); it is thus also the case for (2.8.14.1).

We similarly have a canonical di-homomorphism $\mathrm{Hom}_{S'}(M', N') \to \mathrm{Hom}_S(M, N)$ by sending a homomorphism u' of degree k to the homomorphism $u' \otimes 1$, which is also of degree k; from this, we again deduce a S-homomorphism of degree 0

$$(\mathrm{Hom}_{S'}(M', N')) \otimes_{A'} A \longrightarrow \mathrm{Hom}_S(M, N)$$

whence a homomorphism of $\mathscr{O}_X$-modules

$$\Phi^*((\mathrm{Hom}_{S'}(M', N'))^\sim) \longrightarrow (\mathrm{Hom}_S(M, N))^\sim. \tag{2.8.14.3}$$

Furthermore, the diagram

$$\begin{tikzcd} \Phi^*((\mathrm{Hom}_{S'}(M', N'))^\sim) \arrow[r] \arrow[d, "\Phi^*(\mu)"'] & (\mathrm{Hom}_S(M, N))^\sim \arrow[d, "\mu"] \\ \Phi^*(\mathscr{H}om_{\mathscr{O}_{X'}}(\widetilde{M'}, \widetilde{N'})) \arrow[r] & \mathscr{H}om_{\mathscr{O}_X}(\widetilde{M}, \widetilde{N}) \end{tikzcd}$$

commutes (the second horizontal line being the canonical homomorphism (**0**, 4.4.6)).

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(2.8.15). With the notation and hypotheses of (2.8.1), it follows from (2.4.7) that we do not change the morphism Φ, up to isomorphism, when we replace S_0 and S'_0 by $\mathbf{Z}$, and ϕ_0 by the identity map, and thus when we replace S and S' by $S^{(d)}$ and $S'^{(d)}$ (respectively) ($d > 0$), and ϕ by its restriction $\phi^{(d)}$ to $S^{(d)}$.

2.9. Closed subschemes of a scheme $\mathrm{Proj}(S)$

(2.9.1). If $\phi : S \to S'$ is a homomorphism of graded rings, then we say that ϕ is (TN)-*surjective* (resp. (TN)-*injective*, (TN)-*bijective*) if there exists an integer n such that, for $k \geqslant n$, $\phi_k : S_k \to S'_k$ is *surjective* (resp. *injective*, *bijective*). Instead of saying that ϕ is (TN)-bijective, we sometimes say that it is a (TN)-*isomorphism*.

Proposition (2.9.2). — *Let S be a positively graded ring, and let $X = \mathrm{Proj}(S)$.*

(i) *If $\phi : S \to S'$ is a (TN)-surjective homomorphism of graded rings, then the corresponding morphism Φ (2.8.1) is defined on the whole of $\mathrm{Proj}(S')$, and is a closed immersion of $\mathrm{Proj}(S')$ into X. If $\mathfrak{J}$ is the kernel of ϕ, then the closed subscheme of X associated to Φ is defined by the quasi-coherent sheaf of ideals $\widetilde{\mathfrak{J}}$ of $\mathscr{O}_X$.*

(ii) *Suppose further that the ideal S_+ is generated by a finite number of homogeneous elements of degree* 1*. Let X' be a closed subscheme of X defined by a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$. Let $\mathfrak{J}$ be the graded ideal of S given by the inverse image of $\Gamma_\bullet(\mathscr{J})$ under the canonical homomorphism $\alpha : S \to \Gamma_\bullet(\mathscr{O}_X)$ (2.6.2), and set $S' = S/\mathfrak{J}$. Then X' is the subscheme associated to the closed immersion* $\operatorname{Proj}(S') \to X$ *corresponding to the canonical homomorphism of graded rings $S \to S'$.*

PROOF.

(i) We can suppose that ϕ is surjective (2.9.1). Since, by hypothesis, $\phi(S_+)$ generates S'_+, we have $G(\phi) = \operatorname{Proj}(S')$. Now, the second claim can be checked locally on X; so let f be a homogeneous element of S_+, and set $f' = \phi(f)$. Since ϕ is a surjective homomorphism of graded rings, we immediately see that $\phi_{(f')} : S_{(f)} \to S'_{(f')}$ is surjective, and that its kernel is $\mathfrak{J}_{(f)}$, which proves (i) (**I**, 4.2.3).

(ii) By (i), we are led to proving that the homomorphism $\widetilde{j} : \widetilde{\mathfrak{J}} \to \mathscr{O}_X$ induced by the canonical injection $j : \mathfrak{J} \to S$ is an isomorphism from $\widetilde{\mathfrak{J}}$ to $\mathscr{J}$, which follows from (2.7.11).

□

We note that $\mathfrak{J}$ is the *largest* of the graded ideals $\mathfrak{J}'$ of S such that $\widetilde{j}(\widetilde{\mathfrak{J}}') = \mathscr{J}$, since we can immediately show, using the definitions (2.6.2), that this equation implies that $\alpha(\mathfrak{J}') \subset \Gamma_\bullet(\mathscr{J})$.

Corollary (2.9.3). — *Suppose that the hypotheses of (2.9.2, (i)) are satisfied, and further that the ideal S_+ is generated by S_1; then $\Phi^*((S(n))^\sim)$ is canonically isomorphic to $(S'(n))^\sim$ for all $n \in \mathbf{Z}$, and so $\Phi^*(\mathscr{F}(n))$ is canonically isomorphic to $\Phi^*(\mathscr{F})(n)$ for every $\mathscr{O}_X$-module $\mathscr{F}$.*

PROOF. This is a particular case of (2.8.8), taking (2.5.10.2) into account. □

Corollary (2.9.4). — *Suppose that the hypotheses of (2.9.2, (ii)) are satisfied. For the closed sub-prescheme X' of X to be integral, it is necessary and sufficient for the graded ideal $\mathfrak{J}$ to be prime in S.*

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PROOF. Since X' is isomorphic to $\operatorname{Proj}(S/\mathfrak{J})$, the condition is sufficient by (2.4.4). To see that it is necessary, consider the exact sequence $0 \to \mathscr{J} \to \mathscr{O}_X \to \mathscr{O}_X/\mathscr{J}$, which gives the exact sequence

$$0 \longrightarrow \Gamma_\bullet(\mathscr{J}) \longrightarrow \Gamma_\bullet(\mathscr{O}_X) \longrightarrow \Gamma_\bullet(\mathscr{O}_X/\mathscr{J}).$$

It suffices to prove that, if $f \in S_m$ and $g \in S_n$ are such that the image in $\Gamma_\bullet(\mathscr{O}_X/\mathscr{J})$ of $\alpha_{n+m}(fg)$ is zero, then the image of either $\alpha_m(f)$ or $\alpha_n(g)$ is zero. But, by definition, these images are sections of invertible $(\mathscr{O}_X/\mathscr{J})$-modules $\mathscr{L} = (\mathscr{O}_X/\mathscr{J})(m)$ and $\mathscr{L}' = (\mathscr{O}_X/\mathscr{J})(n)$ over the integral scheme X'; the hypothesis implies that the product of these two sections is zero in $\mathscr{L} \otimes \mathscr{L}'$ ((2.9.3) and (2.5.14.1)), and so one of them is zero by (**I**, 7.4.4). □

Corollary (2.9.5). — *Let A be a ring, M an A-module, S a graded A-algebra generated by the set S_1 of homogeneous elements of degree* 1*, $u : M \to S_1$ a surjective homomorphism of A-modules, and $\overline{u} : \mathbf{S}(M) \to S$ the homomorphism (of A-algebras) from the symmetric algebra $\mathbf{S}(M)$ of M to S that extends u. Then the morphism corresponding to $\overline{u}$ is a closed immersion of* $\operatorname{Proj}(S)$ *into* $\operatorname{Proj}(\mathbf{S}(M))$.

PROOF. Indeed, $\overline{u}$ is surjective by hypothesis, and so it suffices to apply (2.9.2) □

§3. HOMOGENEOUS SPECTRUM OF A SHEAF OF GRADED ALGEBRAS

3.1. Homogeneous spectrum of a quasi-coherent graded $\mathscr{O}_Y$-algebra

(3.1.1). Let Y be a prescheme, $\mathscr{S}$ a graded $\mathscr{O}_Y$-algebra, and $\mathscr{M}$ a graded $\mathscr{S}$-module. If $\mathscr{S}$ is *quasi-coherent*, then each of its homogenous components $\mathscr{S}_n$ is a *quasi-coherent* $\mathscr{O}_Y$-module, since they are the images of $\mathscr{S}$ under a homomorphism from $\mathscr{S}$ to itself ((**I**, 1.3.8) and (**I**, 1.3.9)); similarly, if $\mathscr{M}$ is quasi-coherent as an $\mathscr{O}_Y$-module, then its homogenous components $\mathscr{M}_n$ are also quasi-coherent, and the converse is also true. For an integer $d > 0$, we denote by $\mathscr{S}^{(d)}$ the direct sum of the $\mathscr{O}_Y$-modules $\mathscr{S}_{nd}$ (for $n \in \mathbf{Z}$), which is quasi-coherent if $\mathscr{S}$ is (**I**, 1.3.9); for every integer k such that $0 \leqslant k \leqslant d-1$, we denote by $\mathscr{M}^{(d,k)}$ (or $\mathscr{M}^{(d)}$, for $k = 0$) the direct sum of the $\mathscr{M}_{nd+k}$ (for $n \in \mathbf{Z}$), which is a graded $\mathscr{S}^{(d)}$-module, and quasi-coherent if both $\mathscr{S}$ and $\mathscr{M}$ are quasi-coherent (**I**, 9.6.1). We denote by $\mathscr{M}(n)$

the graded $\mathscr{S}$-module such that $(\mathscr{M}(n))_k = \mathscr{M}_{n+k}$ for all $k \in \mathbf{Z}$; if $\mathscr{S}$ and $\mathscr{M}$ are quasi-coherent, then $\mathscr{M}(n)$ is a quasi-coherent graded $\mathscr{S}$-module (**I**, 9.6.1).

We say that $\mathscr{M}$ is a graded $\mathscr{S}$-module *of finite type* (resp. admitting a *finite presentation*) if, for all $y \in Y$, there exists an open neighbourhood U of y, along with integers n_i (resp. integers m_i and n_j) such that there is a surjective degree 0 homomorphism $\bigoplus_{i=1}^r (\mathscr{S}(n_i)|U) \to \mathscr{M}|U$ (resp. such that $\mathscr{M}|U$ is isomorphic to the cokernel of a degree 0 homomorphism $\bigoplus_{i=1}^r (\mathscr{S}(m_i)|U) \to \bigoplus_{j=1}^s (\mathscr{S}(n_j)|U)$).

Let U be an affine open of Y, with ring $A = \Gamma(U, \mathscr{O}_Y)$; by hypothesis, the graded $(\mathscr{O}_Y|U)$-algebra $\mathscr{S}|U$ is isomorphic to $\widetilde{S}$, where $S = \Gamma(U, \mathscr{S})$ is a graded A-algebra (**I**, 1.4.3); set $X_U = \operatorname{Proj}(\Gamma(U, \mathscr{S}))$. **II** | 50
Let $U' \subset U$ be another affine open of Y, with ring A', and let $j : U' \to U$ be the canonical injection, which corresponds to the restriction homomorphism $A \to A'$; we have that $\mathscr{S}|U' = j^*(\mathscr{S}|U)$, and so $S' = \Gamma(U', \mathscr{S})$ is canonically identified with $S \otimes_A A'$ (**I**, 1.6.5). We conclude (2.8.10) that $X_{U'}$ is canonically identified with $X_U \times_U U'$, and thus also with $f_U^{-1}(U')$, where we denote by f_U the structure morphism $X_U \to U$ (**I**, 4.4.1). We denote by $\sigma_{U',U}$ the canonical isomorphism $f_U^{-1}(U') \xrightarrow{\sim} X_{U'}$ thus defined, and by $\rho_{U',U}$ the open immersion $X_{U'} \to X_U$ obtained by composing $\sigma_{U',U}^{-1}$ with the canonical injection $f_U^{-1}(U') \to X_U$. It is immediate that, if $U'' \subset U'$ is another affine open of Y, then $\rho_{U'',U} = \rho_{U'',U'} \circ \rho_{U',U}$.

Proposition (3.1.2). — *Let Y be a prescheme. For every quasi-coherent positively graded $\mathscr{O}_Y$-algebra, there exists exactly one (up to Y-isomorphism) prescheme X over Y satisfying the following property: if $f : X \to Y$ is the structure morphism, then, for every affine open U of Y, there exists an* isomorphism *η_U from the induced prescheme $f^{-1}(U)$ to $X_U = \operatorname{Proj}(\Gamma(U, \mathscr{S}))$ such that, if V is another affine open of Y that is contained in U, then the diagram*

(3.1.2.1)
$$\begin{tikzcd} f^{-1}(V) \arrow[r, "\eta_V"] \arrow[d] & X_V \arrow[d, "\rho_{V,U}"] \\ f^{-1}(U) \arrow[r, "\eta_V"'] & X_U \end{tikzcd}$$

commutes.

Proof. Given affine opens U and V of Y, let $X_{U,V}$ be the prescheme induced on $f_U^{-1}(U \cap V)$ by X_U; we are going to define a Y-isomorphism $\theta_{U,V} : X_{V,U} \xrightarrow{\sim} X_{U,V}$. For this, we consider an affine open $W \subset U \cap V$: by composing the isomorphisms

$$f_U^{-1}(W) \xrightarrow{\sigma_{W,U}} X_W \xrightarrow{\sigma_{W,V}^{-1}} f_V^{-1}(W),$$

we obtain an isomorphism τ_W, and we immediately see that, if $W' \subset W$ is an affine open, then $\tau_{W'}$ is the restriction of τ_W to $f_U^{-1}(W')$; the τ_W are thus indeed the restrictions of a Y-isomorphism $\theta_{V,U}$. Further, if U, V, and W are affine open subsets of Y, and $\theta'_{U,V}$, $\theta'_{V,W}$, and $\theta'_{U,W}$ the restrictions of $\theta_{U,V}$, $\theta_{V,W}$, and $\theta_{U,W}$ (respectively) to the inverse images of $U \cap V \cap W$ in X_V, X_W, and X_W (respectively), then it follows from the above definitions that we have $\theta'_{U,V} \circ \theta'_{V,W} = \theta'_{U,W}$. The existence of some X satisfying the properties in the statement thus follows from (**I**, 2.3.1); its uniqueness up to Y-isomorphism is trivial, taking (3.1.2.1) into account. □

(3.1.3). We say that the prescheme X defined in (3.1.2) is the *homogeneous spectrum* of the quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$, and we denote it by $\operatorname{Proj}(\mathscr{S})$. It is immediate that $\operatorname{Proj}(\mathscr{S})$ is *separated over Y* ((2.4.2) and (**I**, 5.5.5)); if $\mathscr{S}$ is an $\mathscr{O}_Y$-algebra *of finite type* (**I**, 9.6.2), then $\operatorname{Proj}(\mathscr{S})$ is *of finite type* over Y ((2.7.1, (ii)) and (**I**, 6.3.1)).

If f is the structure morphism $X \to Y$, then it is immediate that, for every prescheme induced by Y on an open subset U of Y, $f^{-1}(U)$ can be identified with the homogeneous spectrum $\operatorname{Proj}(\mathscr{S}|U)$.

Proposition (3.1.4). — *Let $f \in \Gamma(Y, \mathscr{S}_d)$ for $d > 0$. Then there exists an open subset X_f of the underlying space of $X = \operatorname{Proj}(\mathscr{S})$ that satisfies the following property: for every affine open U of Y, we have $X_f \cap \phi^{-1}(U) = D_+(f|U)$ in $\phi^{-1}(U)$ identified with $X_U = \operatorname{Proj}(\Gamma(U, \mathscr{S}))$, where ϕ denotes the structure morphism $X \to Y$. Furthermore, the prescheme induced on X_f is affine over Y, and is canonically isomorphic* **II** | 51
to $\operatorname{Spec}(\mathscr{S}^{(d)}/(f-1)\mathscr{S}^{[d]})$ (1.3.1).

PROOF. We have $f|U \in \Gamma(U, \mathscr{S}_d) = (\Gamma(U, \mathscr{S}))_d$. If U and U' are affine opens of Y such that $U' \subset U$, then $f|U'$ is the image of $f|U$ under the restriction homomorphism

$$\Gamma(U, \mathscr{S}) \longrightarrow \Gamma(U', \mathscr{S})$$

and so $D_+(f|U')$ is equal (with the notation of (3.1.1)) to the prescheme induced on the inverse image $\rho_{U',U}^{-1}(D_+(f|U))$ in $X_{U'}$ (2.8.1); whence the first claim. Furthermore, the prescheme induced on $D_+(f|U)$ by X_U is canonically identified with $\operatorname{Spec}((\Gamma(U, \mathscr{S}))_{f|U})$, with these identifications being compatible with the restriction homomorphisms (2.8.1); the second claim then follows from (2.2.5) and from the commutativity of the diagram (2.8.2.1). □

We also say that X_f (as an open subset of the underlying space X) is the set of $x \in X$ where f *does not vanish*.

Corollary (3.1.5). — *If $f \in \Gamma(Y, \mathscr{S}_d)$ and $g \in \Gamma(Y, \mathscr{S}_e)$, then*

$$X_{fg} = X_f \cap X_g. \tag{3.1.5.1}$$

PROOF. It suffices to consider the intersection of the two sets with a set $\phi^{-1}(U)$, where U is an affine open in Y, and to then apply formula (2.3.3.2). □

Corollary (3.1.6). — *Let (f_α) be a family of sections of $\mathscr{S}$ over Y such that $f_\alpha \in \Gamma(Y, \mathscr{S}_{d_\alpha})$; if the sheaf of ideals of $\mathscr{S}$ generated by this family* (**0**, 5.1.1) *contains all the $\mathscr{S}_n$ starting from a certain rank, then the underlying space X is the union of the X_{f_α}.*

PROOF. For every affine open U of Y, $\phi^{-1}(U)$ is the union of the $X_{f_\alpha} \cap \phi^{-1}(U)$ (2.3.14). □

Corollary (3.1.7). — *Let $\mathscr{A}$ be a quasi-coherent $\mathscr{O}_Y$-algebra; set*

$$\mathscr{S} = \mathscr{A}[T] = \mathscr{A} \otimes_{\mathbf{Z}} \mathbf{Z}[T]$$

where T is an indeterminate (and $\mathbf{Z}$ and $\mathbf{Z}[T]$ are considered as simple sheaves over Y). Then $X = \operatorname{Proj}(\mathscr{S})$ is canonically identified with $\operatorname{Spec}(\mathscr{A})$. In particular, $\operatorname{Proj}(\mathscr{O}_Y[T])$ is identified with Y.

PROOF. By applying (3.1.6) to the unique section $f \in \Gamma(Y, \mathscr{S})$ that is equal to T at each point of Y we see that $X_f = X$. Further, here we have $d = 1$, and $\mathscr{S}^{(1)}/(f-1)\mathscr{S}^{(1)} = \mathscr{S}/(f-1)\mathscr{S}$ is canonically isomorphic to $\mathscr{A}$, whence the corollary (1.2.2). □

Let $g \in \Gamma(Y, \mathscr{O}_Y)$; if we take $\mathscr{S} = \mathscr{O}_Y[T]$, then $g \in \Gamma(Y, \mathscr{S}_0)$; let

$$h = gT \in \Gamma(Y, \mathscr{S}_1).$$

If $X = \operatorname{Proj}(\mathscr{S})$, then the canonical identification defined in (3.1.7) identifies X_h with the open subset Y_g of Y (in the sense of (**0**, 5.5.2)): indeed, we can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine, and everything then reduces (taking (2.2.5) into account) to the fact that the ring of fractions A_g is canonically identified with $A[T]/(gT-1)A[T]$ (**0**, 1.2.3).

Proposition (3.1.8). — *Let $\mathscr{S}$ be a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra. Then*

(i) *For all $d > 0$, there exists a canonical Y-isomorphism from $\operatorname{Proj}(\mathscr{S})$ to $\operatorname{Proj}(\mathscr{S}^{(d)})$.* II | 52
(ii) *Let $\mathscr{S}'$ be the graded $\mathscr{O}_Y$-algebra given by the direct sum of $\mathscr{O}_Y$ with the $\mathscr{S}_n$ (for $n \geqslant 0$); then $\operatorname{Proj}(\mathscr{S}')$ and $\operatorname{Proj}(\mathscr{S})$ are canonically Y-isomorphic.*
(iii) *Let $\mathscr{L}$ be an invertible $\mathscr{O}_Y$-module* (**0**, 5.4.1)*, and let $\mathscr{S}_{(\mathscr{L})}$ be the graded $\mathscr{O}_Y$-algebra given by the direct sum of the $\mathscr{S}_d \otimes \mathscr{L}^{\otimes d}$ (for $d \geqslant 0$); then $\operatorname{Proj}(\mathscr{S})$ and $\operatorname{Proj}(\mathscr{S}_{(\mathscr{L})})$ are canonically Y-isomorphic.*

PROOF. In each of the three cases, it suffices to define the isomorphism locally on Y, since the verification of compatibility with the restriction operations from one open subset to a smaller one is trivial. We can thus suppose that Y is affine, and then (i) follows from (2.4.7, (i)), and (ii) follows from (2.4.8). As for (iii), if we further suppose that $\mathscr{L}$ is isomorphic to $\mathscr{O}_Y$ (which we are allowed to do, since the question is local on Y), then the isomorphism between $\operatorname{Proj}(\mathscr{S})$ and $\operatorname{Proj}(\mathscr{S}_{(\mathscr{L})})$ is evident; to define a *canonical* isomorphism, let $Y = \operatorname{Spec}(A)$ and $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra, and let c be a generator of the free A-module L such that $\mathscr{L} = \widetilde{L}$; then, for all $n > 0$, $x_n \mapsto x_n \otimes c^{\otimes n}$ is an A-isomorphism from S_n to $S_n \otimes L^{\otimes n}$, and these A-isomorphisms define an A-isomorphism of graded algebras $\phi_c : S \to S_{(L)} = \bigoplus_{n\geqslant 0} S_n \otimes L^{\otimes n}$. So let $f \in S_+$ be homogeneous of degree d; for all $x \in S_{nd}$, we have that $(x \otimes c^{nd})/(f \otimes c^d)^n = (x \otimes (\varepsilon c)^{nd})/(f \otimes (\varepsilon c)^d)^n$ for every invertible element $\varepsilon \in A$, which shows that the isomorphism $S_{(f)} \to (S_{(L)})_{(f\otimes c^d)}$ induced from ϕ_c is *independent* of the generator c of L considered, and thus finishes the proof. □

(3.1.9). Recall ((**0**, 4.1.3) and (**I**, 1.3.14)) that, for the quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ to be *generated by the $\mathscr{O}_Y$-module $\mathscr{S}_1$*, it is necessary and sufficient for there to exist an affine open cover (U_α) of Y such that the graded algebra $\Gamma(U_\alpha, \mathscr{S})$ over $\Gamma(U_\alpha, \mathscr{S}_0)$ is generated by the set $\Gamma(U_\alpha, \mathscr{S}_1)$ of its homogeneous elements of degree 1. For every open V of Y, $\mathscr{S}|V$ is then generated by the $(\mathscr{O}_Y|V)$-module $\mathscr{S}_1|V$.

Proposition (3.1.10). — *Suppose that there exists a finite affine open cover (U_i) of Y such that each graded algebra $\Gamma(U_i, \mathscr{S})$ is of finite type over $\Gamma(U_i, \mathscr{O}_Y)$. Then there exists $d > 0$ such that $\mathscr{S}^{(d)}$ is generated by $\mathscr{S}_d$, with $\mathscr{S}_d$ an $\mathscr{O}_Y$-module of finite type.*

PROOF. Indeed, it follows from (2.1.6, (v)) that, for each i, there exists an integer m_i such that $\Gamma(U_i, \mathscr{S}_{nm_i}) = (\Gamma(U_i, \mathscr{S}_{m_i}))^n$ for all $n > 0$; it suffices to take d to be a common multiple of all the m_i, taking (2.1.6, (i)) into account. □

Corollary (3.1.11). — *Under the hypotheses of* (3.1.10), $\operatorname{Proj}(\mathscr{S})$ *is Y-isomorphic to a homogeneous spectrum* $\operatorname{Proj}(\mathscr{S}')$, *where $\mathscr{S}'$ is a graded $\mathscr{O}_Y$-algebra generated by $\mathscr{S}'_1$, with $\mathscr{S}'_1$ an $\mathscr{O}_Y$-module of finite type.*

PROOF. It suffices to take $\mathscr{S}' = \mathscr{S}^{(d)}$, where d satisfies the property of (3.1.10), and to then apply (3.1.8, (i)) □

(3.1.12). If $\mathscr{S}$ is a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra, we know (**I**, 5.1.1) that its *nilradical* $\mathscr{N}$ is a quasi-coherent $\mathscr{O}_Y$-module; we say that $\mathscr{N}_+ = \mathscr{N} \cap \mathscr{S}_+$ is the *nilradical of $\mathscr{S}_+$*; this is a quasi-coherent graded $\mathscr{S}_0$-module, since we can immediately reduce to the case where Y is affine, and the proposition then follows from (2.1.10). For all $y \in Y$, $(\mathscr{N}_+)_y$ is then the nilradical of $(\mathscr{S}_+)_y = (\mathscr{S}_y)_+$ (**I**, 5.1.1). We say that the graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ is *essentially reduced* if $\mathscr{N}_+ = 0$, which is equivalent to saying that $\mathscr{S}_y$ is an essentially reduced graded $\mathscr{O}_y$-algebra for all $y \in Y$. For every graded II | 53
$\mathscr{O}_Y$-algebra $\mathscr{S}$, $\mathscr{S}/\mathscr{N}_+$ is essentially reduced.

We say that $\mathscr{S}$ is *integral* if, for all $y \in Y$, $\mathscr{S}_y$ is an integral ring and, furthermore, $(\mathscr{S}_y)_+ = (\mathscr{S}_+)_y \neq 0$.

Proposition (3.1.13). — *Let $\mathscr{S}$ be a positively-graded $\mathscr{O}_Y$-algebra. If $X = \operatorname{Proj}(\mathscr{S})$, then the Y-scheme X_{red} is canonically isomorphic to* $\operatorname{Proj}(\mathscr{S}/\mathscr{N}_+)$; *in particular, if $\mathscr{S}$ is essentially reduced, then X is reduced.*

PROOF. The fact that $X' = \operatorname{Proj}(\mathscr{S}/\mathscr{N}_+)$ is reduced follows immediately from (2.4.4, (i)), since the property is local; further, for every affine open $U \subset Y$, $\phi'^{-1}(U)$ is equal to $(\phi^{-1}(U))_{\mathrm{red}}$ (where we denote by ϕ and ϕ' the structure morphisms $X \to Y$ and $X' \to Y$, respectively); we immediately see that the canonical U-morphisms $\phi'^{-1}(U) \to \phi^{-1}(U)$ are compatible with the restriction operations, and thus define a closed immersion $X' \to X$ that is a homeomorphism of the underlying spaces; whence the conclusion (**I**, 5.1.2). □

Proposition (3.1.14). — *Let Y be an integral prescheme, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra such that $\mathscr{S}_0 = \mathscr{O}_Y$.*

- *(i) If $\mathscr{S}$ is integral* (3.1.12), *then $X = \operatorname{Proj}(\mathscr{S})$ is integral, and the structure morphism $\phi : X \to Y$ is dominant.*
- *(ii) Suppose furthermore that $\mathscr{S}$ is essentially reduced. Then, conversely, if X is integral and ϕ is dominant, then $\mathscr{S}$ is integral.*

PROOF.

- (i) If (U_α) is a base of Y consisting of non-empty affine opens, then it suffices to prove the proposition in the case where Y is replaced by one of the U_α, and $\mathscr{S}$ by $\mathscr{S}|U_\alpha$: indeed, one one hand it will follow that the underlying space $\phi^{-1}(U_\alpha)$ are irreducible opens (and thus non-empty) of X such that $\phi^{-1}(U_\alpha) \cap \phi^{-1}(U_\beta) \neq \varnothing$ for any pair of indices (since $U_\alpha \cap U_\beta$ contains some U_γ), and so X is irreducible (**0**, 2.1.4); on the other hand, X will be reduced, since this is a local property, and so X will indeed be integral, with $\phi(X)$ dense in Y.

So suppose that $Y = \operatorname{Spec}(A)$, where A is integral (**I**, 5.1.4), and that $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra; the hypothesis is that, for every $y \in Y$, $\widetilde{S}_y = S_y$ is an integral graded ring such that $(S_y)_+ \neq 0$. It suffices to prove that S is an *integral* ring, since then we will have that $S_+ \neq 0$, and we can then apply (2.4.4, (ii)). But let $f, g \neq 0$ be elements of S, and suppose that $fg = 0$; for all $y \in Y$ we then have that $(f/1)(g/1) = 0$ in S_y, and so $f/1 = 0$ or $g/1 = 0$ by hypothesis. Suppose, for example, that $f/1 = 0$ in S_y; this implies that there exists $a \in A$ such that $a \notin \mathfrak{j}_y$ and $af = 0$; we then have, for *all* $z \in Y$, that $(a/1)(f/1) = 0$ in the *integral* ring S_z, and since $a/1 \neq 0$ (since A is integral), $f/1 = 0$, which implies that $f = 0$.

(ii) Since the question is local on Y, we can again suppose that $Y = \operatorname{Spec}(A)$, with A integral, and that $\mathscr{S} = \widetilde{S}$. By hypothesis, for all $y \in Y$, $(S_y)_+$ does not contain any non-zero nilpotent elements, and the same is true of $(S_0)_y = A_y$ by hypothesis; so S_y is a reduced ring for all $y \in Y$, and we thus conclude first of all that S itself is reduced (**I**, 5.1.1). The hypothesis that X is integral implies that S is essentially integral (2.4.4, (ii)), and everything then reduces to showing that the annihilator $\mathfrak{J}$ of S_+ in $A = S_0$ is just 0 (2.1.11). If this were not the case, we would have that $(S_h)_+ = 0$ for some $h \neq 0$ in $\mathfrak{J}$, and thus (3.1.1) that $\phi^{-1}(D(h)) = \varnothing$, and $\phi(X)$ would not be dense in Y, contradicting the hypothesis (since $D(h) \neq \varnothing$, since h is not nilpotent). **II** | 54

□

3.2. Sheaf on $\operatorname{Proj}(\mathscr{S})$ associated to a graded $\mathscr{S}$-module

(3.2.1). Let Y be a prescheme, $\mathscr{S}$ a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra, and $\mathscr{M}$ a quasi-coherent graded $\mathscr{S}$-module (on $(Y, \mathscr{O}_Y)$, or, equivalently (**I**, 9.6.1), on the ringed space $(Y, \mathscr{S})$). With the notation of (3.1.1), we denote by $\widetilde{\mathscr{M}_U}$ the quasi-coherent $\mathscr{O}_{X_U}$-module $(\Gamma(U, \mathscr{M}))^\sim$; for $U' \subset U$, $\Gamma(U', \mathscr{M})$ is canonically identified with $\Gamma(U, \mathscr{M}) \otimes_A A'$ (**I**, 1.6.4); thus we have $\widetilde{\mathscr{M}_{U'}} = \rho^*_{U',U}(\widetilde{\mathscr{M}_U})$ (2.8.11).

Proposition (3.2.2). — *There exists on* $\operatorname{Proj}(\mathscr{S}) = X$ *exactly one quasi-coherent* $\mathscr{O}_X$*-module* $\widetilde{\mathscr{M}}$ *such that, for every affine open* U *of* Y*, we have* $\eta_U^*((\Gamma(U, \mathscr{M}))^\sim) = \widetilde{M}|f^{-1}(U)$ *(denoting by* η_U *the isomorphism* $f^{-1}(U) \xrightarrow{\sim} \operatorname{Proj}(\Gamma(U, \mathscr{S}))$*), where* f *is the structure morphism* $X \to Y$*.*

Proof. Since $\rho_{U',U}$ is identified with the injection morphism $f^{-1}(U') \to f^{-1}(U)$ (3.1.2.1), the proposition follows immediately from the relation $\widetilde{\mathscr{M}_{U'}} = \rho^*_{U',U}(\widetilde{\mathscr{M}_U})$ and from the gluing principle for sheaves (**0**, 3.3.1). □

We say that $\widetilde{\mathscr{M}}$ is the $\mathscr{O}_X$-module *associated to* the quasi-coherent graded $\mathscr{S}$-module $\mathscr{M}$.

Proposition (3.2.3). — *Let* $\mathscr{M}$ *be a quasi-coherent graded* $\mathscr{S}$*-module, and let* $f \in \Gamma(Y, \mathscr{S}_d)$ *(for* $d > 0$*). If* ξ_f *is the canonical isomorphism from* X_f *to the* Y*-prescheme* $Z_f = \operatorname{Spec}(\mathscr{S}^{(d)}/(f-1)\mathscr{S}^{(d)})$ (3.1.4)*, then* $(\xi_f)_*(\widetilde{\mathscr{M}}|X_f)$ *is the* $\mathscr{O}_{Z_f}$*-module* $(\mathscr{M}^{(l)}/(f-1)\mathscr{M}^{(d)})$ (1.4.3)*.*

Proof. Since the question is local on Y, we can immediately reduce to (2.2.5), taking into account the commutativity of the diagram in (2.8.12.1). □

Proposition (3.2.4). — *The* $\mathscr{O}_X$*-module* $\widetilde{\mathscr{M}}$ *is an exact additive covariant functor in* $\mathscr{M}$*, from the category of quasi-coherent graded* $\mathscr{S}$*-modules to the category of quasi-coherent* $\mathscr{O}_X$*-modules, that commutes with inductive limits and direct sums.*

Proof. Since the question is local on Y, we can reduce to (**I**, 1.3.11), (**I**, 1.3.9), and (2.5.4). □

In particular, if $\mathscr{N}$ is a quasi-coherent graded sub-$\mathscr{S}$-module of $\mathscr{M}$, then $\widetilde{\mathscr{N}}$ is canonically identified with with a quasi-coherent sub-$\mathscr{O}_X$-module of $\widetilde{\mathscr{M}}$; in particular, for every quasi-coherent graded sheaf $\mathscr{J}$ of ideals of $\mathscr{S}$, $\widetilde{\mathscr{J}}$ is a quasi-coherent sheaf of ideals of $\mathscr{O}_X$.

If $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module, and $\mathscr{I}$ a quasi-coherent sheaf of ideals of $\mathscr{O}_Y$, then $\mathscr{I}\mathscr{M}$ is a quasi-coherent graded sub-$\mathscr{S}$-module of $\mathscr{M}$, and we have

$$(\mathscr{I}\mathscr{M})^\sim = \mathscr{I} \cdot \widetilde{\mathscr{M}} \tag{3.2.4.1}$$

(where the right-hand side is defined as in (**0**, 4.3.5)). It suffices to verify this formula in the case where $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, with S a graded A-algebra, $\mathscr{M} = \widetilde{M}$, with M a graded S-module, and $\mathscr{I} = \mathfrak{I}$, with $\mathfrak{I}$ an ideal of A. For every homogeneous element f of S_+, the restriction to $D_+(f) = \operatorname{Spec}(S_{(f)})$ of the left-hand side of (3.2.4.1) can be associated with $(\mathfrak{I}M)_{(f)} = \mathfrak{I} \cdot M_{(f)}$, and the same is true of the restriction of the right-hand side, given (**I**, 1.3.13) and (**i**, 1.6.9). II | 55

Proposition (3.2.5). — *Let $f \in \Gamma(Y, \mathscr{S}_d)$ (for $d > 0$). On the open subset X_f, the $(\mathscr{O}_X|X_f)$-module $(\mathscr{S}(nd))^\sim X_f$ is canonically isomorphic to $\mathscr{O}_X|X_f$ for all $n \in \mathbf{Z}$. In particular, if the $\mathscr{O}_Y$-algebra $\mathscr{S}$ is generated by $\mathscr{S}_1$ (3.1.9), then the $\mathscr{O}_X$-modules $(\mathscr{S}(n))^\sim$ are invertible for all $n \in \mathbf{Z}$.*

PROOF. Indeed, for every affine open U of Y, we defined in (2.5.7) a canonical isomorphism from $(\mathscr{S}(nd))^\sim|(X_f \cap \phi^{-1}(U))$ to $\mathscr{O}_X|(X_f \cap \phi^{-1}(U))$, taking (3.1.4) into account (where ϕ is the structure morphism $X \to Y$); it is immediate that these isomorphisms are compatible with the restriction from U to an affine open $U' \subset U$, whence the first claim. To prove the second, it suffices to note that, if $\mathscr{S}$ is generated by $\mathscr{S}_1$, then there is a cover (U_α) of Y by affine opens such that $\Gamma(U_\alpha, \mathscr{S})$ is generated by $\Gamma(U_\alpha, \mathscr{S})_1 = \Gamma(U_\alpha, \mathscr{S}_1)$; we can then apply the result of (2.5.9), since the property of being invertible is local. □

We again set, for all $n \in \mathbf{Z}$,

(3.2.5.1) $$\mathscr{O}_X(n) = (\mathscr{S}(n))^\sim$$

and, for all $\mathscr{O}_X$-modules $\mathscr{F}$,

(3.2.5.2) $$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X(n).$$

It follows immediately from these definitions that, for every open subset U of Y, we have

$$((\mathscr{S}|U)(n))^\sim = \mathscr{O}_X(n)|f^{-1}(U)$$

where f is the structure morphism $X \to Y$.

Proposition (3.2.6). — *Let $\mathscr{M}$ and $\mathscr{N}$ be quasi-coherent graded $\mathscr{S}$-modules. Then there exists a canonical functorial (in $\mathscr{M}$ and $\mathscr{N}$) homomorphism*

(3.2.6.1) $$\lambda : \widetilde{\mathscr{M}} \otimes_{\mathscr{O}_X} \widetilde{\mathscr{N}} \longrightarrow (\mathscr{M} \otimes_{\mathscr{S}} \mathscr{N})^\sim$$

and a canonical functorial (in $\mathscr{M}$ and $\mathscr{N}$) homomorphism

(3.2.6.2) $$\mu : (\mathscr{H}om_{\mathscr{S}}(\mathscr{M}, \mathscr{N}))^\sim \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(\widetilde{\mathscr{M}}, \widetilde{\mathscr{N}}).$$

Furthermore, if $\mathscr{S}$ is generated by $\mathscr{S}_1$ (3.1.9), then λ is an isomorphism; if, further, $\mathscr{M}$ admits a finite presentation (3.1.1), then μ is an isomorphism.

PROOF. The isomorphisms λ and μ were defined in (2.5.11.2) and (2.5.12.2) in the case where Y is affine; since these definitions are local, they transfer immediately to the general case considered here, taking (2.8.14) into account. □

Corollary (3.2.7). — *If $\mathscr{S}$ is generated by $\mathscr{S}_1$, then, for any $m, n \in \mathbf{Z}$,*

(3.2.7.1) $$\mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{O}_X(n) = \mathscr{O}_X(m+n)$$

(3.2.7.2) $$\mathscr{O}_X(n) = (\mathscr{O}_X(1))^{\otimes n}$$

up to canonical isomorphism.

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Corollary (3.2.8). — *If $\mathscr{S}$ is generated by $\mathscr{S}_1$, then, for any graded $\mathscr{S}$-module $\mathscr{M}$ and any $n \in \mathbf{Z}$,*

(3.2.8.1) $$(\mathscr{M}(n))^\sim = \widetilde{\mathscr{M}}(n)$$

up to canonical isomorphism.

PROOF. This follows from the corresponding properties in the case where Y is affine ((2.5.14) and (2.5.15)), along with (2.8.11). □

Remarks (3.2.9). —

(1) If $\mathscr{S} = \mathscr{A}[T]$, with $\mathscr{A}$ a quasi-coherent $\mathscr{O}_Y$-algebra (3.1.7), then we immediately see that all the invertible $\mathscr{O}$-modules $\mathscr{O}(n)$ are canonically isomorphic to $\mathscr{O}_X$.

Furthermore, let $\mathscr{N}$ be a quasi-coherent $\mathscr{A}$-module, and set $\mathscr{M} = \mathscr{N} \otimes_{\mathscr{A}} \mathscr{A}[T]$. It then follows from (3.2.3) and (3.1.7) that, under the canonical identification of $X = \operatorname{Proj}(\mathscr{A}[T])$ with $X' = \operatorname{Spec}(\mathscr{A})$, the $\mathscr{O}_X$-module $\widetilde{\mathscr{M}}$ is identified with the $\mathscr{O}_{X'}$-module $\widetilde{\mathscr{N}}$ associated to $\mathscr{N}$ (in the sense of (1.4.3)).

(2) Let $\mathscr{S}$ be an arbitrary graded $\mathscr{O}_Y$-algebra, and $\mathscr{S}'$ the graded $\mathscr{O}_Y$-algebra such that $\mathscr{S}'_0 = \mathscr{O}_Y$ and $\mathscr{S}'_n = \mathscr{S}_n$ for all $n > 0$; then the canonical isomorphism from $X = \operatorname{Proj}(\mathscr{S})$ to $X' = \operatorname{Proj}(\mathscr{S}')$ (3.1.8, (ii)) identifies $\mathscr{O}_X(n)$ with $\mathscr{O}_{X'}(n)$ for all $n \in \mathbf{Z}$. This follows from the same proposition for the affine case (2.5.16) and from the fact that the identifications, for the affine opens of Y, commute with the restriction operations. Similarly, let $X^{(d)} = \operatorname{Proj}(\mathscr{S}^{(d)})$; then the canonical isomorphism from X to $X^{(d)}$ (3.1.8, (i)) identifies $\mathscr{O}_X(nd)$ with $\mathscr{O}_{X^{(d)}}(n)$ for all $n \in \mathbf{Z}$.

Proposition (3.2.10). — *Let $\mathscr{L}$ be an invertible $\mathscr{O}_Y$-module, and g the canonical isomorphism from $X_{(\mathscr{L})} = \operatorname{Proj}(\mathscr{S}_{(\mathscr{L})})$ to $X = \operatorname{Proj}(\mathscr{S})$ (3.1.8, (iii)). Then, for any $n \in \mathbf{Z}$, $g_*(\mathscr{O}_{X_{(\mathscr{L})}}(n))$ is canonically isomorphic to $\mathscr{O}_X(n) \otimes_Y \mathscr{L}^{\otimes n}$.*

PROOF. Suppose first of all that Y is affine, of ring A, and that $\mathscr{L} = \widetilde{L}$, where L is a free monogenous A-module. With the notation from the proof of (3.1.8, (iii)), we define, for $f \in S_d$, an isomorphism from $(S(n))_{(f)} \otimes_A L^{\otimes n}$ to $(S_{(L)}(n))_{(f \otimes c^d)}$ by sending $(x/f^k) \otimes c^n$, where $x \in S_{kd+n}$, to the element $(x \otimes c^{n+kd})/(f \otimes c^d)^k$; it is immediate that this isomorphism is independent of the chosen generator c of L; further, the isomorphisms thus defined for each $f \in S_+$ are compatible with the restriction operators $D_+(f) \to D_+(fg)$. Finally, in the general case, we easily see, from the definitions (3.1.1), that the isomorphisms thus defined for each affine open U of Y are compatible with passing from U to an affine open $U' \subset U$. □

3.3. Graded $\mathscr{S}$-module associated to a sheaf on $\operatorname{Proj}(\mathscr{S})$ *Throughout this entire section we suppose that the graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ is generated by $\mathscr{S}_1$* (3.1.9). Recall that, by (3.1.8, (i)), this restrictive assumption is not essential, thanks to finiteness conditions (3.1.10).

(3.3.1). Let p be the structure morphism $X = \operatorname{Proj}(\mathscr{S}) \to Y$. For every $\mathscr{O}_X$-module $\mathscr{F}$, set

(3.3.1)
$$\mathbf{\Gamma}_*(\mathscr{F}) = \bigoplus_{n \in \mathbf{Z}} p_*(\mathscr{F}(n))$$

and, in particular, II | 57

(3.3.1.2)
$$\mathbf{\Gamma}_*(\mathscr{O}_X) = \bigoplus_{n \in \mathbf{Z}} p_*(\mathscr{O}_X(n)).$$

We know (**0**, 4.2.2) that there exists a canonical homomorphism

$$p_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} p_*(\mathscr{G}) \longrightarrow p_*(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})$$

for $\mathscr{O}_X$-modules $\mathscr{F}$ and $\mathscr{G}$; we thus deduce from (3.2.7.1) that $\mathbf{\Gamma}_*(\mathscr{O}_X)$ is endowed with the structure of a *graded $\mathscr{O}_Y$-algebra*, and (3.2.5.2) defines the structure of a *graded $\mathbf{\Gamma}_*(\mathscr{O}_X)$-module* on $\mathbf{\Gamma}_*(\mathscr{F})$.

By (3.2.5), and by left exactness of the functor p_* (**0**, 4.2.1), $\mathbf{\Gamma}_*(\mathscr{F})$ is an additive and *left exact* covariant functor in $\mathscr{F}$ from the category of $\mathscr{O}_X$-modules to the category of graded $\mathscr{O}_Y$-modules (where the morphisms are the homomorphisms of degree 0). In particular, if $\mathscr{J}$ is a sheaf of ideals of $\mathscr{O}_X$, then $\mathbf{\Gamma}_*(\mathscr{J})$ can be identified with a *graded sheaf of ideals* in $\mathbf{\Gamma}_*(\mathscr{O}_X)$.

(3.3.2). Let $\mathscr{M}$ be a quasi-coherent graded $\mathscr{S}$-module. For every affine open U of Y, we defined in (2.6.2) a homomorphism of abelian groups

$$\alpha_{0,U} : \Gamma(U, \mathscr{M}_0) \longrightarrow \Gamma(p^{-1}(U), \widetilde{\mathscr{M}}).$$

It is immediate that these homomorphisms commute with the restriction operations (2.8.13.1) and thus define (without using the hypothesis that $\mathscr{S}$ is generated by $\mathscr{S}_1$) a homomorphism of sheaves of abelian groups

(3.3.2.1)
$$\alpha_0 : \mathscr{M}_0 \longrightarrow p_*(\widetilde{\mathscr{M}}).$$

Applying this result to each of the $\mathscr{M}_n = (\mathscr{M}(n))_0$, and taking (3.2.8.1) into account, we can define a homomorphism of sheaves of abelian groups

$$\alpha_n : \mathscr{M}_n \longrightarrow p_*(\widetilde{\mathscr{M}}(n)) \tag{3.3.2.2}$$

for all $n \in \mathbf{Z}$, whence a functorial homomorphism (of degree 0) of graded sheaves of abelian groups

$$\alpha : \mathscr{M} \longrightarrow \mathbf{\Gamma}_*(\widetilde{\mathscr{M}}) \tag{3.3.2.3}$$

(also denoted $\alpha_{\mathscr{M}}$).

By taking $\mathscr{M} = \mathscr{S}$ in particular, we see that $\alpha : \mathscr{S} \to \mathbf{\Gamma}_*(\mathscr{O}_X)$ is a homomorphism of graded $\mathscr{O}_Y$-algebra, and that (3.3.2.3) is a di-homomorphism of graded modules, with respect to this homomorphism of graded algebras.

We again note that to each of the α_n there corresponds (**0**, 4.4.3) a canonical homomorphism of $\mathscr{O}_X$-modules

$$\alpha_n^\sharp : p^*(\mathscr{M}_n) \longrightarrow \widetilde{\mathscr{M}}(n). \tag{3.3.2.4}$$

We can easily verify that this homomorphism is exactly the one which corresponds functorially (3.2.4) to the canonical homomorphism (of degree 0) of *graded* $\mathscr{O}_Y$-modules

$$\mathscr{M}_n \otimes_{\mathscr{O}_Y} \mathscr{S} \longrightarrow \mathscr{M}(n) \tag{3.3.2.5}$$

where the grading of the right-hand side comes naturally from that of $\mathscr{S}$. We can restrict to the case **II** | 57 where $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{M} = \widetilde{M}$, and $\mathscr{S} = \widetilde{S}$, with the graded A-algebra S being generated by S_1, so that, as f runs over S_1, the $D_+(f)$ form a cover of X. By the definitions (2.6.2), we see then see, taking (**I**, 1.6.7) into account, that the restriction to $D_+(f)$ of the homomorphism (3.3.2.4) corresponds (**I**, 1.3.8) to the homomorphism of $S_{(f)}$-modules $M_n \otimes_A S_{(f)} \to (S(n))_{(f)}$ that sends $x \otimes 1$ (where $x \in M_n$) to $x/1$; this proves the claim.

Proposition (3.3.3). — *For every section $f \in \Gamma(Y, \mathscr{S}_d)$ (where $d > 0$), X_f is identical to the set of points of X where $\alpha_d(f)$ (considered as a section of $\mathscr{O}_X(d)$) does not vanish* (**0**, 5.5.2).

Proof. (Note that $\alpha_d(f)$ is a section of $p_*(\mathscr{O}_X(d))$ over Y, but by definition such a section is also a section of $\mathscr{O}(d)$ over X (**0**, 4.2.1)). The definition of X_f (3.1.4) lets us reduce to the case where Y is affine, which has already been dealt with in (2.6.3). □

(3.3.4). From now on, we suppose, in addition to the hypothesis at the start of this section, that, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $p_*(\mathscr{F}(n))$ are *quasi-coherent* on Y, so that $\mathbf{\Gamma}_*(\mathscr{F}) = \bigoplus_{n\in\mathbf{Z}} p_*(\mathscr{F}(n))$ is also a quasi-coherent $\mathscr{O}_Y$-module ((**I**, 1.4.1) and (**I**, 1.3.9)); this will always be the case if X is *of finite type* over Y (**I**, 9.2.2). We thus conclude that $(\mathbf{\Gamma}_*(\mathscr{F}))^\sim$ is defined, and is a quasi-coherent $\mathscr{O}_X$-module. For every affine open U of Y< we have ((**I**, 1.3.9) and (**I**, 2.5.4))

$$\begin{aligned}\Big(\Gamma(U, \bigoplus_{n\in\mathbf{Z}} p_*(\mathscr{F}(n)))\Big)^\sim &= \bigoplus_{n\in\mathbf{Z}} \Big(\Gamma(U, p_*(\mathscr{F}(n)))\Big)^\sim \\ &= \bigoplus_{n\in\mathbf{Z}} \Big(\Gamma(p^{-1}(U), \mathscr{F}(n))\Big)^\sim \\ &= \Big(\bigoplus_{n\in\mathbf{Z}} \Gamma(p^{-1}(U), \mathscr{F}(n))\Big)^\sim \\ &= (\mathbf{\Gamma}_*(\mathscr{F}|p^{-1}(U)))^\sim\end{aligned}$$

and so (2.6.4) we have a canonical homomorphism

$$\beta_U : \Big(\Gamma(U, \bigoplus_{n\in\mathbf{Z}}) p_*(\mathscr{F}(n))\Big)^\sim \longrightarrow \mathscr{F}|p^{-1}(U).$$

Furthermore, the commutativity of (2.8.13.2) shows that these homomorphism commute with the restriction operations on Y; we thus obtain a canonical functorial homomorphism

$$\beta : (\mathbf{\Gamma}_*(\mathscr{F}))^\sim \longrightarrow \mathscr{F} \tag{3.3.4.1}$$

(also denoted $\beta_{\mathscr{F}}$) for quasi-coherent $\mathscr{O}_X$-modules.

Proposition (3.3.5). — *Let $\mathscr{M}$ be a quasi-coherent graded $\mathscr{S}$-module, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module; then the composite homomorphisms*

(3.3.5.1) $$\widetilde{\mathscr{M}} \xrightarrow{\widetilde{\alpha}} (\mathbf{\Gamma}_*(\widetilde{\mathscr{M}}))^\sim \xrightarrow{\beta} \widetilde{\mathscr{M}}$$

(3.3.5.2) $$\mathbf{\Gamma}_*(\mathscr{F}) \xrightarrow{\alpha} \mathbf{\Gamma}_*((\mathbf{\Gamma}_*(\mathscr{F}))^\sim) \xrightarrow{\mathbf{\Gamma}_*(\beta)} \mathbf{\Gamma}_*(\mathscr{F})$$

are the identity isomorphisms.

Proof. The question is local on Y, so we can reduce to (2.6.5). □

3.4. Finiteness conditions

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Proposition (3.4.1). — *Let Y be a prescheme, and $\mathscr{S}$ a quasi-coherent $\mathscr{O}_Y$-algebra generated by $\mathscr{S}_1$ (3.1.9); suppose further that $\mathscr{S}_1$ is of finite type. Then $X = \operatorname{Proj}(\mathscr{S})$ is of finite type over Y.*

Proof. Since the question is local on Y, we can suppose that Y is affine of ring A; then $\mathscr{S} = \widetilde{S}$, where $S = \Gamma(Y, \mathscr{S})$, and by hypothesis S is an A-algebra generated by $S_1 = \Gamma(Y, \mathscr{S}_1)$, where we can further suppose that S_1 is an A-module of finite type ((**I**, 1.3.9) and (**I**, 1.3.12)). Then S is a graded A-algebra of finite type, and we can reduce to (2.7.1, (ii)). □

(3.4.2). Let $\mathscr{S}$ be a quasi-coherent graded $\mathscr{O}_Y$-algebra; for a quasi-coherent graded $\mathscr{S}$-module $\mathscr{M}$, consider the following finiteness conditions:

(**TF**) There exists an integer n such that the $\mathscr{S}$-module $\bigoplus_{k\geqslant n} \mathscr{M}_k$ is of finite type.
(**TN**) There exists an integer n such that $\mathscr{M}_k = 0$ for $k \geqslant n$.

If $\mathscr{M}$ satisfies (**TN**), then $\widetilde{\mathscr{M}} = 0$, since this is a local property on Y (2.7.2).

Let $\mathscr{M}$ and $\mathscr{N}$ be quasi-coherent graded $\mathscr{S}$-modules; we say that a homomorphism $u : \mathscr{M} \to \mathscr{N}$ of degree 0 is *(**TN**)-injective* (resp. *(**TN**)-surjective*, *(**TN**)-bijective*) if there exists an integer n such that $u_k : \mathscr{M}_k \to \mathscr{N}_k$ is injective (resp. surjective, bijective) for $k \geqslant n$; then $\widetilde{u} : \widetilde{\mathscr{M}} \to \widetilde{\mathscr{N}}$ is injective (resp. surjective, bijective) by (2.7.2), since this is a local property on Y, and taking (**I**, 1.3.9) into account; if u is (**TN**)-bijective, then we also say that u is a *(**TN**)-isomorphism*.

Proposition (3.4.3). — *Let Y be a prescheme, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra generated by $\mathscr{S}_1$, with S_1 assumed to be of finite type. Let $\mathscr{M}$ be a quasi-coherent graded $\mathscr{S}$-module.*

*(i) If $\mathscr{M}$ satisfies (**TF**), then $\widetilde{\mathscr{M}}$ is of finite type.*
*(ii) Suppose that $\mathscr{M}$ satisfies (**TF**); for $\widetilde{\mathscr{M}} = 0$, it is necessary and sufficient for $\mathscr{M}$ to satisfy (**TN**).*

Proof. Since the questions are local on Y, we can reduce to the case where Y is affine of ring A, $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra such that the ideal S_+ is of finite type, and $\mathscr{M} = \widetilde{M}$, where M is a graded S-module; the proposition then follows from (2.7.3). □

Theorem (3.4.4). — *Let Y be a prescheme, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra generated by $\mathscr{S}_1$, where $\mathscr{S}_1$ is assumed to be of finite type; let $X = \operatorname{Proj}(\mathscr{S})$. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism β (3.3.4) is an isomorphism.*

Proof. Note first of all that β is defined, by (3.4.1). To see that β is an isomorphism, we can reduce to the case where Y is affine of ring A, $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra generated by S_1, and S_1 is an A-module of finite type. It then suffices to apply (2.7.5). □

Corollary (3.4.5). — *Under the hypotheses of (3.4.4), every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is isomorphic to an $\mathscr{O}_X$-module of the form $\widetilde{\mathscr{M}}$, where $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module. If, further, $\mathscr{F}$ is of finite type, and if we assume that Y is a quasi-compact scheme, or that the underlying space of Y is Noetherian, then we can take $\mathscr{M}$ to be of finite type.*

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Proof. The first claim follows immediately from (3.4.4) by taking $\mathscr{M} = \mathbf{\Gamma}_*(\mathscr{F})$. To prove the second, it suffices to show that $\mathscr{M}$ is the inductive limit of its *graded* sub-$\mathscr{S}$-modules of finite type $\mathscr{N}_\lambda$: indeed, it will follow from this that $\widetilde{\mathscr{M}}$ is the inductive limit of the $\widetilde{\mathscr{N}_\lambda}$ (3.2.4), and so $\mathscr{F}$ is the inductive limit of the $\beta(\widetilde{\mathscr{N}_\lambda})$; since X is quasi-compact ((3.4.1) and (I, 6.3.1)) and $\mathscr{F}$ is of finite type, $\mathscr{F}$ will necessarily be equal to one of the $\beta(\widetilde{\mathscr{N}_\lambda})$ (0, 5.2.3).

To define the $\mathscr{N}_\lambda$ having $\mathscr{M}$ as their inductive limit, it suffices to consider, for each $n \in \mathbf{Z}$, the quasi-coherent $\mathscr{O}_Y$-module $\mathscr{M}_n$, which is the inductive limit of its sub-$\mathscr{O}_Y$-modules $\mathscr{M}_n^{(\mu_n)}$ of finite type, by the hypotheses on Y (I, 9.4.9); it is immediate that $\mathscr{P}_{\mu_n} = \mathscr{S} \cdot \mathscr{M}_n^{(\mu_n)}$ is a graded $\mathscr{S}$-module of finite type, and we immediately see that taking $\mathscr{N}_\lambda$ to be the finite sums of the $\mathscr{S}$-modules of the form $\mathscr{P}_{\mu_n}$ gives the desired objects. □

Corollary (3.4.6). — *Suppose that the hypotheses of (3.4.4) are satisfied, and further that the underlying space of Y is quasi-compact; let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type. Then there exists n_0 such that, for all $n \geqslant n_0$, the canonical homomorphism $\sigma : p^*(p_*(\mathscr{F}(n))) \to \mathscr{F}(n)$ (0, 4.4.3) is surjective.*

PROOF. For all $y \in Y$, let U be an affine open neighbourhood of y in Y. There exists an integer $n_0(U)$ such that, for all $n \geqslant n_0(U)$, $\mathscr{F}(n)|p^{-1}(U)$ is generated by a finite number of its sections over $p^{-1}(U)$ (2.7.9); but these sections are the canonical images of sections of $p^*(p_*(\mathscr{F}(n)))$ over $p^{-1}(U)$ ((0, 3.7.1) and (0, 4.4.3)), so $\mathscr{F}(n)|p^{-1}(U)$ is equal to the canonical image of $p^*(p_*(\mathscr{F}(n)))|p^{-1}(U)$. Finally, since Y is quasi-compact, there exists a finite cover of Y by affine opens U_i, and taking n_0 to be the largest of the $n_0(U_i)$ finishes the proof. □

Remarks (3.4.7). — If $p = (\psi, \theta) : X \to Y$ is a morphism of ringed spaces, and $\mathscr{F}$ an $\mathscr{O}_X$-module, the fact that the canonical homomorphism $\sigma : p^*(p_*(\mathscr{F})) \to \mathscr{F}$ is surjective can be explained in the following way (0, 4.4.1): for all $x \in X$, and every section s of $\mathscr{F}$ over an open neighbourhood V of x, there exists an open neighbourhood U of $p(x)$ in Y, a finite number of sections t_i (for $1 \leqslant i \leqslant m$) of $\mathscr{F}$ over $p^{-1}(U)$, a neighbourhood $W \subset V \cap p^{-1}(U)$ of x, and sections a_i (for $1 \leqslant i \leqslant m$) of $\mathscr{O}_X$ over W such that

$$s|W = \sum_{i=1}^{m} a_i \cdot (t_i|W).$$

If Y is an *affine scheme* and $p_*(\mathscr{F})$ is *quasi-coherent*, this condition is equivalent to $\mathscr{F}$ being *generated by its sections over* X (0, 5.5.1): indeed, if $Y = \operatorname{Spec}(A)$, we can suppose that $U = D(f)$ with $f \in A$; then there exists an integer $n > 0$ and sections s_i of $\mathscr{F}$ over X such that t_i is the restriction to $p^{-1}(U)$ of $s_i g^n$, where $g = \theta(f)$ (by applying (I, 1.4.1) to $p_*(\mathscr{F})$); since g is invertible over $p^{-1}(U)$, we thus have

$$s|W = \sum_i b_i \cdot (s_i|W)$$

where $b_i = a_i(g|W)^{-n}$, whence the claim. If Y is affine, then the corollary (3.4.6) recovers (2.7.9).

We thus conclude that, when Y is an arbitrary prescheme, the following three conditions, for a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ such that $p_*(\mathscr{F})$ is *quasi-coherent*, are equivalent: **II** | 61

a) *The canonical homomorphism $\sigma : p^*(p_*(\mathscr{F})) \to \mathscr{F}$ is surjective.*
b) *There exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{G}$ and a surjective homomorphism $p^*(\mathscr{G}) \to \mathscr{F}$.*
c) *For every affine open U of Y, $\mathscr{F}|p^{-1}(U)$ is generated by its sections over $p^{-1}(U)$.*

Indeed, we have just proven the equivalence between *a)* and *c)*. It is also clear that *a)* implies *b)*, since $p_*(\mathscr{F})$ is quasi-coherent by hypothesis. Conversely, every homomorphism $u : p^*(\mathscr{G}) \to \mathscr{F}$ factors as $p^*(\mathscr{G}) \to p^*(p_*(\mathscr{F})) \xrightarrow{\sigma} \mathscr{F}$ (0, 3.5.4.4), so if u is surjective then so too is σ, which proves that *b)* implies *a)*.

Corollary (3.4.8). — *Suppose that the hypotheses of (3.4.4) are satisfied, and suppose further that Y is a quasi-compact scheme, or that the underlying space of Y is Noetherian. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type; then there exists an integer n_0 such that, for all $n \geqslant n_0$, $\mathscr{F}$ is isomorphic to a quotient of an $\mathscr{O}_X$-module of the form $(p^*(\mathscr{G}))(-n)$, where $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_Y$-module of finite type (that depends on n).*

PROOF. Since the structure morphism $X \to Y$ is separated and of finite type, $p_*(\mathscr{F}(n))$ is quasi-coherent (I, 9.2.2, b)), and thus the inductive limit of its quasi-coherent sub-$\mathscr{O}_Y$-modules of finite type, by the hypotheses on Y (I, 9.4.9). We thus deduce, by (3.4.6), (0, 4.3.2), and (0, 5.2.3), that $\mathscr{F}(n)$ is the canonical image of an $\mathscr{O}_X$-module of the form $p^*(\mathscr{G})$, where $\mathscr{G}$ is a quasi-coherent sub-$\mathscr{O}_Y$-module of $p_*(\mathscr{F}(n))$ of finite type; the corollary then follows from (3.2.5.2) and (3.2.7.1). □

3.5. Functorial behaviour

(3.5.1). Let Y be a prescheme, and $\mathscr{S}$ and $\mathscr{S}'$ quasi-coherent positively-graded $\mathscr{O}_Y$-algebras; let $X = \operatorname{Proj}(\mathscr{S})$ and $X' = \operatorname{Proj}(\mathscr{S}')$, and let p and p' be the structure morphisms from X and X' to Y. Let $\phi : \mathscr{S}' \to \mathscr{S}$ be an $\mathscr{O}_Y$-homomorphism of graded algebras. For every affine open U of Y, let $S_U = \Gamma(U, \mathscr{S})$ and $S'_U = \Gamma(U, \mathscr{S}')$; the homomorphism ϕ defines a homomorphism $\phi_U : S'_U \to S_U$ of graded A_U-algebras, where $A_u = \Gamma(U, \mathscr{O}_Y)$. There is a corresponding open subset $G(\phi_U)$ in $p^{-1}(U)$ and morphism $\Phi_U : G(\phi_U) \to p'^{-1}(U)$ (2.8.1). Furthermore, if $V \subset U$ is an affine open, then the diagram

$$\begin{tikzcd} S'_U \arrow[r, "\phi_U"] \arrow[d] & S_U \arrow[d] \\ S'_V \arrow[r, "\phi_V"'] & S_V \end{tikzcd} \tag{3.5.1.1}$$

commutes, and we immediately see, by definition (2.8.1), that we have

$$G(\phi_V) = G(\phi_U) \cap p^{-1}(V)$$

and that Φ_V is the restriction to $G(\phi_V)$ of Φ_U. We have thus defined an open subset $G(\phi)$ of X such that $G(\phi) \cap p^{-1}(U) = G(\phi_U)$ for every affine open $U \subset Y$, and an *affine* Y-morphism $\Phi : G(\phi) \to X'$, which we say is *associated* to ϕ, and which we denote by $\operatorname{Proj}(\phi)$. If, for all $y \in Y$, there is an affine **II | 62**
neighbourhood U of y such that the $\Gamma(U, \mathscr{O}_Y)$-module $\Gamma(U, \mathscr{S}_+)$ is generated by $\phi(\Gamma(U, \mathscr{S}'_+))$, then $G(\phi_U) = p^{-1}(U)$, and so $G(\phi) = X$.

Proposition (3.5.2). —

(i) If $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module, then there exists a canonical functorial isomorphism from the $\mathscr{O}_{X'}$-module $(\mathscr{M}_{[\phi]})^\sim$ to the $\mathscr{O}_{X'}$-module $\Phi_*(\widetilde{\mathscr{M}}|G(\phi))$.

(ii) If $\mathscr{M}'$ is a quasi-coherent graded $\mathscr{S}'$-module, then there exists a canonical functorial isomorphism ν from the $(\mathscr{O}_X|G(\phi))$-module $\Phi^*(\widetilde{\mathscr{M}'})$ to the $(\mathscr{O}_X|G(\phi))$-module $(\mathscr{M}' \otimes_{\mathscr{S}'} \mathscr{S})^\sim|G(\phi)$. If $\mathscr{S}'$ is generated by $\mathscr{S}'_1$, then ν is an isomorphism.

PROOF. The homomorphisms in question are indeed already defined if Y is affine ((2.8.7) and (2.8.8)), and in the general case it suffices to check that they are compatible with the restriction of an affine open of Y to a smaller open, which follows immediately from the commutativity of (3.5.1.1). □

In particular, for all $n \in \mathbf{Z}$, we have a canonical homomorphism

$$\Phi^*(\mathscr{O}_{X'}(n)) \longrightarrow \mathscr{O}_X(n)|G(\phi). \tag{3.5.2.1}$$

Proposition (3.5.3). — *Let Y and Y' be preschemes, $\psi : Y' \to Y$ a morphism, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra; set $\mathscr{S}' = \psi^*(\mathscr{S})$. Then the Y'-scheme $X' = \operatorname{Proj}(\mathscr{S}')$ is canonically identified with $\operatorname{Proj}(\mathscr{S}) \times_Y Y'$. Furthermore, if $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module, then the $\mathscr{O}_{X'}$-module $(\psi^*(\mathscr{M}))^\sim$ can be identified with $\widetilde{\mathscr{M}} \otimes_Y \mathscr{O}_{Y'}$.*

PROOF. Note first of all that $\psi^*(\mathscr{S})$ and $\psi^*(\mathscr{M})$ are quasi-coherent $\mathscr{O}_{Y'}$-modules, as are their homogenous components (**0**, 5.1.4). Let U be an affine open of Y, $U' \subset \psi^{-1}(U)$ an affine open of Y', and A and A' the rings of U and U', respectively; then $\mathscr{S}|U = \widetilde{S}$, where S is a graded A-algebra, and $\mathscr{S}'|U'$ can be identified with $(S \otimes_A A')^\sim$ (**I**, 1.6.5); the first claim then follows from (2.8.10) and (**I**, 3.2.6.2), since we immediately see that the projection $\operatorname{Proj}(\mathscr{S}'|U') \to \operatorname{Proj}(\mathscr{S}|U)$ defined by the above identification is compatible with the restriction operations on U and U', and thus indeed defines a morphism $\operatorname{Proj}(\mathscr{S}') \to \operatorname{Proj}(\mathscr{S})$. Now let

$$q : \operatorname{Proj}(\mathscr{S}) \longrightarrow Y$$
$$q' : \operatorname{Proj}(\mathscr{S}') \longrightarrow Y'$$

be the structure morphisms; $q'^{-1}(U')$ can then be identified with $q^{-1}(U) \times_U U'$, and the two sheaves $(\psi^*(\mathscr{M}))^\sim|q'^{-1}(U')$ and $(\widetilde{\mathscr{M}} \otimes_Y \mathscr{O}_{Y'})|q'^{-1}(U')$ are then both canonically identified with $(M \otimes_A A')^\sim$, where we set $M = \Gamma(U, \mathscr{M})$, by (2.8.10) and (I, 1.6.5); whence the second claim, since we can again immediately see the compatibility of the above identifications with the restriction operations. $\square$

Corollary (3.5.4). — *With the notation of (3.5.3), $\mathscr{O}_{X'}(n)$ is canonically identified with $\mathscr{O}_X(n) \otimes_Y \mathscr{O}_{Y'}$ for all $n \in \mathbf{Z}$ (where $X = \mathrm{Proj}(\mathscr{S})$).*

PROOF. Indeed, with the notation of (3.5.3), it is clear that $\psi^*(\mathscr{S}(n)) = \mathscr{S}'(n)$ for all $n \in \mathbf{Z}$. $\square$

(3.5.5). Keeping the above notation, denote by Ψ the canonical projection $X' \to X$, and set $\mathscr{M}' = \psi^*(\mathscr{M})$; we further suppose that $\mathscr{S}$ is generated by $\mathscr{S}_1$, and that X is of finite type over Y; it then **II | 63**
follows that $\mathscr{S}'$ is generated by $\mathscr{S}'_1$ (as can be seen by reducing to the case where Y and Y' are affine), and that X' is of finite type over Y' (I, 6.3.4). Let $\mathscr{F}$ be an $\mathscr{O}_X$-module, and set $\mathscr{F}' = \Psi^*(\mathscr{F})$; it then follows from (3.5.4) and (0, 4.3.3) that $\mathscr{F}'(n) = \Psi^*(\mathscr{F}(n))$ for all $n \in \mathbf{Z}$. We further define a canonical Ψ-homomorphism $\theta_n : q_*(\mathscr{F}(n)) \to q'_*(\mathscr{F}'(n))$ in the following way: given the commutativity of the diagram

$$\begin{tikzcd} X \arrow[d, "q"'] & X' \arrow[l, "\Psi"'] \arrow[d, "q'"] \\ Y & Y' \arrow[l, "\psi"] \end{tikzcd}$$

it is enough to define a homomorphism $q_*(\mathscr{F}(n)) \to \psi_*(q'_*(\Psi^*(\mathscr{F}(n)))) = q_*(\Psi_*(\Psi^*(\mathscr{F}(n))))$, and it suffices to take the homomorphism $\theta_n = q_*(\rho_n)$, where ρ_n is the canonical homomorphism $\mathscr{F}(n) \to \Psi_*(\Psi^*(\mathscr{F}(n)))$ (0, 4.4.3). It is immediate that, for every affine open U of Y and every affine open U' of Y' such that $U' \subset \psi^{-1}(U)$, the homomorphism θ_n gives, on sections, the canonical homomorphism (0, 3.7.2) $\Gamma(q^{-1}(U), \mathscr{F}(n)) \to \Gamma(q'^{-1}(U'), \mathscr{F}'(n))$. The commutativity of (2.8.13.2) then shows that, if $\mathscr{F}$ is quasi-coherent, the diagram

$$\begin{tikzcd} \mathscr{F} \arrow[r, "\rho"] & \mathscr{F}' \\ (\mathbf{\Gamma}_*(\mathscr{F}))^{\sim} \arrow[u, "\beta_{\mathscr{F}}"] \arrow[r, "\widetilde{\theta}"'] & (\mathbf{\Gamma}_*(\mathscr{F}'))^{\sim} \arrow[u, "\beta_{\mathscr{F}'}"'] \end{tikzcd}$$

commutes (the top horizontal arrow being the canonical Ψ-morphism $\mathscr{F} \to \Psi^*(\mathscr{F})$).

Similarly, the commutativity of (2.8.13.1) shows that the diagram

$$\begin{tikzcd} \mathbf{\Gamma}_*(\widetilde{\mathscr{M}}) \arrow[r, "\theta"] & \mathbf{\Gamma}_*(\widetilde{\mathscr{M}'}) \\ \mathscr{M} \arrow[u, "\alpha_{\mathscr{M}}"] \arrow[r, "\rho"'] & \mathscr{M}' \arrow[u, "\alpha_{\mathscr{M}'}"'] \end{tikzcd}$$

commutes (the bottom horizontal arrow being the canonical ψ-morphism $\mathscr{M} \to \psi^*(\mathscr{M})$).

(3.5.6). Now consider preschemes Y and Y', a morphism $g : Y' \to Y$, a quasi-coherent graded $\mathscr{O}_Y$-algebra (resp. quasi-coherent graded $\mathscr{O}_{Y'}$-algebra) $\mathscr{S}$ (resp. $\mathscr{S}'$), and a g-morphism of graded algebras $u : \mathscr{S} \to \mathscr{S}'$, i.e. a $\mathscr{O}_Y$-homomorphism of graded algebras $\mathscr{S} \to g_*(\mathscr{S}')$; we already know that this is equivalent to giving a $\mathscr{O}_{Y'}$-homomorphism of graded algebras $u^\sharp : g^*(\mathscr{S}) \to \mathscr{S}'$. We thus canonically obtain from $u^\sharp$ a Y'-morphism $W = \mathrm{Proj}(u^\sharp) : G(u^\sharp) \to \mathrm{Proj}(g^*(\mathscr{S}))$, where $G(u^\sharp)$ is an open of $X' = \mathrm{Proj}(\mathscr{S}')$ (3.5.1). We also know that $X'' = \mathrm{Proj}(g^*(\mathscr{S}))$ is canonically identified with $X \times_Y Y'$, by taking $X = \mathrm{Proj}(\mathscr{S})$ (3.5.3); composing the first projection $p : X \times_Y Y' \to X$ with **II | 64**
$\mathrm{Proj}(u^\sharp)$, we thus obtain a morphism $v : G(u^\sharp) \to X$, which we denote by $\mathrm{Proj}(u)$, and which is such that the diagram

$$\begin{tikzcd} G(u^\sharp) \arrow[r, "v"] \arrow[d] & X \arrow[d] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

commutes.

Furthermore, for every quasi-coherent graded $\mathscr{O}_Y$-module $\mathscr{M}$, we have a canonical v-morphism

$$\nu : \widetilde{\mathscr{M}} \longrightarrow (g^*(\mathscr{M}) \otimes_{g^*(\mathscr{S})} \mathscr{S}')^{\sim}|G(u^\sharp). \tag{3.5.6.1}$$

Indeed, $\nu^\sharp$ is given by composing the homomorphisms

$$v^*(\widetilde{\mathscr{M}}) = w^*(p^*(\widetilde{\mathscr{M}})) \longrightarrow w^*((g^*(\mathscr{M}))^{\sim}) \longrightarrow (g^*(\mathscr{M}) \otimes_{g^*(\mathscr{S})} \mathscr{S}')^{\sim}|G(u^\sharp)$$

where the first arrow comes from the isomorphism (3.5.3) and the second is the homomorphism (3.5.2, (i)); if $\mathscr{S}$ is generated by $\mathscr{S}_1$, then it follows from (3.5.2) that $\nu^\sharp$ is an *isomorphism*.

As a particular case of (3.5.6.1), we have, for all $n \in \mathbf{Z}$, a canonical v-morphism

$$\nu : \mathscr{O}_X(n) \longrightarrow \mathscr{O}_{X'}(n)|G(u^\sharp). \tag{3.5.6.2}$$

3.6. Closed subpreschemes of a prescheme $\operatorname{Proj}(\mathscr{S})$

(3.6.1). Let Y be a prescheme, and $\phi : \mathscr{S} \to \mathscr{S}'$ a degree 0 homomorphism of quasi-coherent graded $\mathscr{O}_Y$-algebras. We say that ϕ is *(**TN**)-surjective* (resp. *(**TN**)-injective*, *(**TN**)-bijective*) if there exists n such that, for all $k \geqslant n$, $\phi_k : \mathscr{S}_k \to \mathscr{S}'_k$ is surjective (resp. injective, bijective). If this is the case, then we can reduce the study of the corresponding morphism $\Phi : \operatorname{Proj}(\mathscr{S}') \to \operatorname{Proj}(\mathscr{S})$ to the case where ϕ is *surjective* (resp. *injective*, *bijective*). We prove this as in (2.9.1) (which is the particular case where Y is affine) by using (3.1.8). Instead of saying that ϕ is (**TN**)-bijective, we also say that it is a *(**TN**)-isomorphism*.

Proposition (3.6.2). — *Let Y be a prescheme, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra; set $X = \operatorname{Proj}(\mathscr{S})$.*

- *(i) If $\phi : \mathscr{S} \to \mathscr{S}'$ is a (**TN**)-surjective homomorphism of graded $\mathscr{O}_Y$-algebras, then the corresponding morphism $\Phi = \operatorname{Proj}(\phi)$ (3.5.1) is defined on all of $\operatorname{Proj}(\mathscr{S}')$ and is a closed immersion of $\operatorname{Proj}(\mathscr{S}')$ into X. If $\mathscr{J}$ is the kernel of ϕ, then the closed subprescheme of X associated to Φ is defined by the quasi-coherent sheaf of ideals $\widetilde{\mathscr{J}}$ in $\mathscr{O}_X$.*
- *(ii) Suppose further that $\mathscr{S}_0 = \mathscr{O}_Y$, that $\mathscr{S}$ is generated by $\mathscr{S}_1$, and that $\mathscr{S}_1$ is of finite type. Let X' be a closed subprescheme of $X = \operatorname{Proj}(\mathscr{S})$, defined by a quasi-coherent sheaf of ideals $\mathscr{I}$ in $\mathscr{O}_X$. Let $\mathscr{J}$ be the quasi-coherent graded sheaf of ideals of $\mathscr{S}$, given by the inverse image of $\mathbf{\Gamma}_*(\mathscr{I})$ by the canonical homomorphism $\alpha : \mathscr{S} \to \mathbf{\Gamma}_*(\mathscr{O}_X)$ (3.3.2), and set $\mathscr{S}' = \mathscr{S}|\mathscr{J}$. Then X' is the subprescheme associated (**I**, 4.2.1) to the closed immersion $\operatorname{Proj}(\mathscr{S}') \to X$ corresponding to the canonical homomorphism $\mathscr{S} \to \mathscr{S}'$ of graded $\mathscr{O}_Y$-algebras.* II | 65

PROOF.

- (i) We can assume that ϕ is surjective (3.6.1). Then, for every affine open U of Y, $\Gamma(U, \mathscr{S}) \to \Gamma(U, \mathscr{S}')$ is surjective (**I**, 1.3.9), so (3.5.1) $G(\phi) = X$. We can then immediately reduce to proving the proposition in the case where Y is affine, and this follows from (2.9.2, (i)).
- (ii) We can reduce to proving that the homomorphism $\widetilde{\mathscr{J}} \to \mathscr{O}_X$ induced by the canonical injection $\mathscr{J} \to \mathscr{S}$ is an isomorphism from $\widetilde{\mathscr{J}}$ to $\mathscr{I}$; since the question is local on Y, we can take Y to be affine of ring A, which implies that $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra generated by S_1, with S_1 of finite type over A. It then suffices to apply (2.9.2, (ii)).

□

Corollary (3.6.3). — *Under the conditions of (3.6.2, (i)), suppose further that $\mathscr{S}$ is generated by $\mathscr{S}_1$. Then $\Phi^*(\mathscr{O}_X(n))$ is canonically identified with $\mathscr{O}_{X'}(n)$ for all $n \in \mathbf{Z}$.*

PROOF. We have defined such a canonical isomorphism when Y is affine (2.9.3); in the general case, it suffices to show that the isomorphisms thus defined for each affine open U of Y are compatible with the passage from U to an affine open $U' \subset U$, which is immediate. □

Corollary (3.6.4). — *Let Y be a prescheme, $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra generated by $\mathscr{S}_1$, $\mathscr{M}$ a quasi-coherent $\mathscr{O}_Y$-module, u a surjective $\mathscr{O}_Y$-homomorphism $\mathscr{M} \to \mathscr{S}_1$, and $\overline{u} : \mathbf{S}_{\mathscr{O}_Y}(\mathscr{M}) \to \mathscr{S}$ the homomorphism of graded $\mathscr{O}_Y$-algebras that extends u (1.7.4). Then the morphism corresponding to $\overline{u}$ is a closed immersion of $\operatorname{Proj}(\mathscr{S})$ into $\operatorname{Proj}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{M}))$.*

PROOF. Indeed, $\overline{u}$ is surjective by hypothesis, and we apply (3.6.1, (i)). □

3.7. Morphisms from a prescheme to a homogeneous spectrum

(3.7.1). Let $q : X \to Y$ be a morphism of preschemes, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and $\mathscr{S}$ a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra; then $q^*(\mathscr{S})$ is a quasi-coherent positively-graded $\mathscr{O}_X$-algebra. Consider the quasi-coherent graded $\mathscr{O}_X$-algebra $\mathscr{S}' = \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$; suppose that we have some $\mathscr{O}_X$-homomorphism of graded algebras

$$\psi : q^*(\mathscr{S}) \longrightarrow \mathscr{S}' = \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$$

which is equivalent to having a q-morphism of graded algebras

$$\psi^\flat : \mathscr{S} \longrightarrow q_*(\mathscr{S}').$$

We know that $\mathrm{Proj}(\mathscr{S}')$ is canonically identified with X ((3.1.7) and (3.1.8, (iii))); we thus canonically obtain from ψ an open subset $G(\psi)$ of X, and a Y-morphism

(3.7.1.1) $$r_{\mathscr{L},\psi} : G(\psi) \longrightarrow \mathrm{Proj}(\mathscr{S}) = P$$

which we call the morphism *associated to $\mathscr{L}$ and ψ*; recall (3.5.6) that this morphism is obtained by composing the Y-morphism **II** | 66

$$\tau = \mathrm{Proj}(\psi) : G(\psi) \longrightarrow \mathrm{Proj}(q^*(\mathscr{S}))$$

with the first projection $\pi : \mathrm{Proj}(q^*(\mathscr{S})) = P \times_Y X \to P$.

(3.7.2). We will explicitly describe $r = r_{\mathscr{L},\psi}$ in the case where $Y = \mathrm{Spec}(A)$ is affine, so that $\mathscr{S} = \widetilde{S}$, where S is a positively-graded A-algebra. Suppose first of all that $X = \mathrm{Spec}(B)$ is also affine, and that we have $\mathscr{L} = \widetilde{L}$, where L is a free B-module of rank 1. Then $q^*(\mathscr{S}) = (S \otimes_A B)^\sim$ (**I**, 1.6.5); if c is a generator of L, then $\psi_n : q^*(\mathscr{S}_n) \to \mathscr{L}^{\otimes n}$ corresponds to a homomorphism $w_n : s \otimes b \mapsto bv_n(s)c^{\otimes n}$ from $S_n \otimes A_B$to $L^{\otimes n}$, where $v_n : S_n \to B$ is a homomorphism of A-modules, constituting a homomorphism of algebras $S \to B$. Let $f \in S_d$ (where $d > 0$), and set $g = v_d(f)$; we have $\pi^{-1}(D_+(f)) = D_+(f \otimes 1)$ by (2.8.10) and the identification of $D_+(f)$ with $\mathrm{Spec}(S_{(f)})$ (2.3.6); we also know, from (2.8.1.1) (taking into account the canonical identification of X with $\mathrm{Proj}(\mathscr{S}')$), that

$$\tau^{-1}(D_+(f \otimes 1)) = D(g)$$

whence

(3.7.2.1) $$r^{-1}(D_+(f)) = D(g).$$

Furthermore, the morphism $\tau = \mathrm{Proj}(\psi)$, restricted to $D(g)$, corresponds to the homomorphism that sends $(s \otimes 1)/(f \otimes 1)^n$ (where $s \in S_{nd}$) to $v_{nd}(s)/g^n$ (2.8.1), and the projection π, restricted to $D_+(f \otimes 1)$, corresponds t the homomorphism $s/f^n \mapsto (s \otimes 1)/(f \otimes 1)^n$; we thus conclude that r, restricted to $D(g)$, corresponds to the homomorphism of A-algebras $\omega : S_{(f)} \to B_g$ such that $\omega(s/f^n) = v_{nd}(s)/g^n$ (where $s \in S_{nd}$ and $n > 0$). Passing to the case where X is arbitrary (but Y still affine), we thus obtain, taking (2.8.1) into account, the following:

Proposition (3.7.3). — *If $Y = \mathrm{Spec}(A)$ is affine and $\mathscr{S} = \widetilde{S}$, where S is a graded A-algebra, then, for all $f \in S_d = \Gamma(Y, \mathscr{S}_d)$, we have*

(3.7.3.1) $$r^{-1}_{\mathscr{L},\psi}(D_+(f)) = X_{\psi^\flat(f)}$$

(where $\psi^\flat(f) \in \Gamma(X, \mathscr{L}^{\otimes d})$) and the restriction $X_{\psi^\flat(f)} \to D_+(f) = \mathrm{Spec}(S_{(f)})$ of $r_{\mathscr{L},\psi}$ corresponds (**I**, 2.2.4) *to the homomorphism of algebras*

(3.7.3.2) $$\psi^\flat_{(f)} : S_{(f)} \longrightarrow \Gamma(X_{\psi^\flat(f)}, \mathscr{O}_X)$$

such that, for $s \in S_{nd} = \Gamma(Y, \mathscr{S}_{nd})$,

(3.7.3.3) $$\psi^\flat_{(f)}(s/f^n) = (\psi^\flat(s)|X_{\psi^\flat(f)})(\psi^\flat(f)|X_{\psi^\flat(f)})^{-n}.$$

We say that $r_{\mathscr{L},\psi}$ is *everywhere defined* if $G(\psi) = X$. For this to be the case, it is evidently necessary and sufficient that $G(\psi) \cap q^{-1}(U) = q^{-1}(U)$ for every affine open $U \subset Y$; in other words, the question is *local* on Y. If Y is affine, then $G(\psi)$ is the union of the $r^{-1}(D_+(f))$ over f homogeneous in S_+ (2.8.1); by (3.7.3.1), the $X_{\psi^\flat(f)}$ must then form a *cover* of X; in other words:

Corollary (3.7.4). — *Under the hypotheses of (3.7.3), for $r_{\mathscr{L},\psi}$ to be everywhere defined, it is necessary and sufficient that, for every $x \in X$, there exist an integer $n > 0$ and a section s of $\mathscr{S}_n$ over Y such that, if we set $t(= \psi^\flat(s) \in \Gamma(X, \mathscr{L}^{\otimes n})$, then $t(x) \neq 0$.*

Note that this condition is always satisfied if ψ is *(**TN**)-surjective.* **II** | 67

Similarly, the question of if $r_{\mathscr{L},\psi}$ is *dominant* is local on Y, and we have:

Corollary (3.7.5). — *Under the hypotheses of (3.7.3), for $r_{\mathscr{L},\psi}$ to be dominant, it is necessary and sufficient that, for every integer $n > 0$, every section $s \in S_n$ such that $\psi^\flat(s) \in \Gamma(X, \mathscr{L}^{\otimes n})$ is locally nilpotent is itself nilpotent.*

Proof. We have to show that $r^{-1}_{\mathscr{L},\psi}(D_+(s))$ is not empty if $D_+(s)$ is empty, and the corollary thus follows from (3.7.3.1) and (2.3.7). □

Proposition (3.7.6). — *Let $q : X \to Y$ be a morphism, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, $\mathscr{S}$ and $\mathscr{S}'$ quasi-coherent graded $\mathscr{O}_Y$-algebras, $u : \mathscr{S}' \to \mathscr{S}$ a homomorphism of graded algebras, and $\psi : q^*(\mathscr{S}) \to \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$ a homomorphism of graded algebras; let $\psi' = \psi \circ q^*(u)$ be the composite homomorphism. If $r_{\mathscr{L},\psi'}$ is everywhere defined, then so too is $r_{\mathscr{L},\psi}$; if u is (**TN**)-surjective, then if $r_{\mathscr{L},\psi'}$ is dominant, so too is $r_{\mathscr{L},\psi}$; conversely, if u is (**TN**)-injective, then if $r_{\mathscr{L},\psi}$ is dominant, so too is $r_{\mathscr{L},\psi'}$.*

Proof. We know that $G(\psi') \subset G(\psi)$ (2.8.4), whence the first claim; if u is (**TN**)-surjective, then $\mathrm{Proj}(u) : \mathrm{Proj}(S) \to \mathrm{Proj}(S')$ is everywhere defined and a closed immersion; since $r_{\mathscr{L},\psi'}$ is the composition of $\mathrm{Proj}(u)$ and the restriction to $G(\psi')$ of $r_{\mathscr{L},\psi}$, we thus conclude that, if $r_{\mathscr{L},\psi'}$ is dominant, then so too is $r_{\mathscr{L},\psi}$. Finally, if u is (**TN**)-injective, then we know that $\mathrm{Proj}(u)$ is a dominant morphism from $G(u)$ to $\mathrm{Proj}(S')$ (2.8.3); since $G(\psi')$ is the inverse image of $G(u)$ under $r_{\mathscr{L},\psi}$, we see that, if $r_{\mathscr{L},\psi}$ is dominant, then so too is $r_{\mathscr{L},\psi'}$. □

Proposition (3.7.7). — *Let Y be a quasi-compact prescheme, $q : X \to Y$ a quasi-compact morphism, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra given by the filtered inductive limit of an inductive system $(\mathscr{S}^\lambda)$ of quasi-coherent $\mathscr{O}_Y$-algebras. Let $\phi_\lambda : \mathscr{S}^\lambda \to \mathscr{S}$ be the canonical homomorphism, $\psi : q^*(\mathscr{S}) \to \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$ a homomorphism of graded rings, and set $\psi_\lambda = \psi \circ q^*(\phi_\lambda)$. For $r_{\mathscr{L},\psi}$ to be everywhere defined, it is necessary and sufficient that there exist some λ such that $r_{\mathscr{L},\psi_\lambda}$ be everywhere defined; $r_{\mathscr{L},\psi_\mu}$ is then everywhere defined for $\mu \geqslant \lambda$.*

Proof. The condition is sufficient by (3.7.6). Conversely, suppose that $r_{\mathscr{L},\psi}$ is everywhere defined; we can reduce to the case where Y is affine, since, if, for every affine open $U \subset Y$ there exists $\lambda(U)$ such that the restriction of $r_{\mathscr{L},\psi_{\lambda(U)}}$ to $q^{-1}(U)$ is everywhere defined, then it will suffice (Y being quasi-compact) to cover Y by a finite number of affine opens U_i, and to take $\lambda \geqslant \lambda(U_i)$ for all indices i, by (3.7.6). If Y is affine, then the hypothesis implies that, for all $x \in X$, there exists a section $s^{(x)}$ of S_n such that, if we set $t^{(x)} = \psi^\flat(s^{(x)})$, then $t^{(x)}(x) \neq 0$ (where $t^{(x)}$ is considered as a section of $\mathscr{L}^{\otimes n}$ over X), which implies that $t^{(x)}(z) \neq 0$ for any z in a neighbourhood $V(x)$ of x. Cover X by a finite number of $V(x_i)$, and let $s^{(i)}$ be the corresponding sections of S; then there exists some λ such that the $s^{(i)}$ are all of the form $\phi_\lambda({s'}_\lambda^{(i)})$, with ${s'}_\lambda^{(i)} \in S^\lambda$ for all i; it then follows from (3.7.4) that $r_{\mathscr{L},\psi_\lambda}$ is everywhere defined. The final claim is a trivial consequence of (3.7.6). □

Corollary (3.7.8). — *Under the hypotheses of (3.7.7), if the $r_{\mathscr{L},\psi_\lambda}$ are dominant, then so too is $r_{\mathscr{L},\psi}$; the converse is true if the ϕ_λ are all injective.*

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Proof. The second claim is a particular case of (3.7.6); to show that $r_{\mathscr{L},\psi}$ is dominant, we can restrict to the case where Y is affine; if $s \in S$ is such that $\psi^\flat(s)$ is locally nilpotent, since we can write $s = \phi_\lambda(s_\lambda)$ for at least one λ, we conclude from the hypotheses and from (3.7.5) that s_λ is nilpotent, and thus so too is s, and the criteria of (3.7.5) then apply. □

Remarks (3.7.9). —

(i) With the notation of (3.7.1), and taking (3.2.10) into account, we have, for all $n \in \mathbf{Z}$, a canonical homomorphism

(3.7.9.1) $$\theta : r^*_{\mathscr{L},\psi}(\mathscr{O}_P(n)) \longrightarrow \mathscr{L}^{\otimes n}$$

defined in a general way in (3.5.6.2). We immediately see that, under the conditions of (3.7.3), the restriction of this homomorphism to $X_{\psi^\flat(f)}$ sends s/f^k (where $s \in S_{n+kd}$) to the element $(\psi^\flat(s)|X_{\psi^\flat(f)})(\psi^\flat(f)|X_{\psi^\flat(f)})^{-k}$.

(ii) Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module, and suppose that q is quasi-compact and separated, so that, for all $n \geqslant 0$, $q_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n})$ is a quasi-coherent $\mathscr{O}_Y$-module (**I**, 9.2.2). Let $\mathscr{M}' = \bigoplus_{n\geqslant 0} \mathscr{F} \otimes \mathscr{L}^{\otimes n}$, which is a quasi-coherent graded $\mathscr{S}'$-module, and consider its image $\mathscr{M} = q_*(\mathscr{M}') = \bigoplus_{n\geqslant 0}(\mathscr{F} \otimes \mathscr{L}^{\otimes n})$ (which is a quasi-coherent $\mathscr{S}$-module, via the homomorphism $\psi^\flat$). We are going to show the existence of a canonical homomorphism of $\mathscr{O}_X$-modules

(3.7.9.2) $$\xi : r^*_{\mathscr{L},\psi}(\widetilde{\mathscr{M}}) \longrightarrow \mathscr{F}|G(\psi).$$

Indeed, we have already defined (3.5.6.1) a canonical homomorphism

(3.7.9.3) $$r^*_{\mathscr{L},\psi}(\widetilde{\mathscr{M}}) \longrightarrow (q^*(\mathscr{M}) \otimes_{q^*(\mathscr{S})} \mathscr{S}')^\sim|G(\psi)$$

where the right-hand side is considered as a quasi-coherent sheaf on $\mathrm{Proj}(\mathscr{S}')$. We also have a canonical homomorphism

(3.7.9.4) $$q^*(q_*(\mathscr{M}')) \otimes_{q^*(\mathscr{S})} \mathscr{S}' \longrightarrow \mathscr{M}'$$

for any quasi-coherent graded $\mathscr{S}'$-module $\mathscr{M}'$: for any open U of X, any section t' of $q^*(q_*(\mathscr{M}'_h))$ over U, and any section b' of $\mathscr{S}'_k$ over U, we send $t' \otimes b'$ to the section $b'\sigma(t')$ of $\mathscr{M}'_{h+k}$, where $\sigma(t')$ is the section of $\mathscr{M}'_h$ over U that corresponds canonically (**0**, 4.4.3) to t'. We thus obtain a canonical homomorphism

(3.7.9.5) $$(q^*(q_*(\mathscr{M}')) \otimes_{q^*(\mathscr{S})} \mathscr{S}')^\sim|G(\psi) \longrightarrow \widetilde{\mathscr{M}'}|G(\psi)$$

and, finally, since $\widetilde{\mathscr{M}'}$ is canonically identified with $\mathscr{F}$ (3.2.9, (i)), we indeed obtain the desired canonical homomorphism.

Under the conditions of (3.7.3), the restriction of this homomorphism to $X_{\psi^\flat(f)}$ acts as follows: given a section t_{nd} of $\mathscr{F} \otimes \mathscr{L}^{\otimes nd}$ over X, if t'_{nd} is the section t_{nd} considered as a section of $q_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n})$ over Y, then it sends the element t'_{nd}/f^n to the section $(t_{nd}|X_{\psi^\flat(f)})(\psi^\flat(f)|X_{\psi^\flat(f)})^{-n}$ of $\mathscr{F}$ over $X_{\psi^\flat(f)}$.

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(3.8.1). With the notation of (3.7.1), the question of if $r_{\mathscr{L},\psi}$ is an immersion (resp. an open immersion, a closed immersion) is clearly *local on* Y.

Proposition (3.8.2). — *Under the hypotheses of (3.7.3), for $r_{\mathscr{L},\psi}$ to be everywhere defined and an immersion, it is necessary and sufficient that there exist a family of sections $s_\alpha \in S_{n_\alpha}$ (where $n_\alpha > 0$) such that, if we set $f_\alpha = \psi^\flat(s_\alpha)$, the following conditions are satisfied:*

(i) The X_{f_α} form a cover of X.

(ii) The X_{f_α} are affine opens.

(iii) For every α and every $t \in \Gamma(X_{f_\alpha}, \mathscr{O}_X)$, there exists an integer $m > 0$ and some $s \in S_{mn_\alpha}$ such that $t = (\psi^\flat(s)|X_{f_\alpha})(f_\alpha|X_{f_\alpha})^{-m}$.

For $r_{\mathscr{L},\psi}$ to be everywhere defined and an open immersion, it is necessary and sufficient that there exist a family (s_α) satisfying conditions (i), (ii), and (iii) above, as well as:

(iv) For all $n > 0$ and every $s \in S_{nn\alpha}$ such that $\psi^\flat(s)|X_{f_\alpha} = 0$, there exists an integer $k > 0$ such that $s^k_\alpha = 0$.

For $r_{\mathscr{L},\psi}$ to be everywhere defined and a closed immersion, it is necessary and sufficient that there exist a family (s_α) satisfying conditions (i), (ii), and (iii) above, as well as:

(v) The $D_+(s_\alpha)$ form a cover of $P = \mathrm{Proj}(S)$.

Proof. For r to be an immersion (resp. a closed immersion), it is necessary and sufficient that there exist a cover of $r(G(\psi))$ (resp. of P) by the $D_+(s_\alpha)$, such that, if we set $V_\alpha = r^{-1}(D_+(s_\alpha))$, then the restriction of r to U_α is a *closed* immersion of V_α into $D_+(s_\alpha)$ (**I**, 4.2.4). Condition (i) says that r is everywhere defined and also that the $D_+(s_\alpha)$ cover $r(X)$, by (3.7.3.1); since $D_+(s_\alpha)$ is affine, conditions (ii) and (iii) say that the restriction of r to X_{f_α} is a closed immersion into $D_+(s_\alpha)$, by (**I**, 4.2.3); finally, since conditions (iii) and (iv) say that $\psi^\flat_{(s_\alpha)}$ is bijective (using the notation of (3.7.3.2)), conditions (ii), (iii), and (iv) say that the restriction of r to X_{f_α} is an isomorphism to $D_+(s_\alpha)$ for all α, and so conditions (i), (ii), (iii), and (iv) together say that r is an open immersion. □

Corollary (3.8.3). — *Under the hypotheses of (3.7.6), if $r_{\mathscr{L},\psi'}$ is everywhere defined and an immersion, then so too is $r_{\mathscr{L},\psi}$. If we further suppose that u is (**TN**)-surjective, then if $r_{\mathscr{L},\psi'}$ is an open (resp. closed) immersion, so too is $r_{\mathscr{L},\psi}$.*

Proof. By (3.8.2), there exists a family $s'_\alpha \in S'_{n_\alpha}$ such that, if we set $f_\alpha = \psi'^\flat(s'_\alpha)$, conditions (i), (ii), and (iii) are satisfied. But if we set $s_\alpha = u(s'_\alpha)$, then we also have that $f_\alpha = \psi^\flat(s_\alpha)$, and if $t = (\psi'^\flat(s')|X_{f_\alpha})(f_\alpha|X_{f_\alpha})^{-m}$, then also $t = (\psi^\flat(s)|X_{f_\alpha})(f_\alpha|X_{f_\alpha})^{-m}$ by setting $s = u(s')$, whence the first claim. The second claim follows immediately from the fact that $\operatorname{Proj}(u)$ is a closed immersion. □

Proposition (3.8.4). — *Suppose that the hypotheses of (3.7.7) are satisfied, and further that $q : X \to Y$ is a morphism of finite type. Then, for $r_{\mathscr{L},\psi}$ to be everywhere defined and an immersion, it is necessary and sufficient that there exist some λ such that $r_{\mathscr{L},\psi_\lambda}$ is everywhere defined and an immersion; in this case, $r_{\mathscr{L},\psi_\mu}$ is also everywhere defined and an immersion for all $\mu \geqslant \lambda$.*

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Proof. Taking (3.8.3) into account, it suffices to show that, if $r_{\mathscr{L},\psi}$ is everywhere defined and an immersion, then so too is $r_{\mathscr{L},\psi_\lambda}$ for at least one λ. By the same argument as in (3.7.7), using the fact that Y is quasi-compact, (3.8.2) shows the existence of a *finite* family $(s_i \in S_{n_i})$ of elements of S satisfying conditions (i), (ii), and (iii). The morphism $X_{f_i} \to Y$ (where $f_i = \psi^\flat(s_i)$) is of finite type: it is a morphism of affine schemes, and is thus quasi-compact (**I**, 6.6.1), and also locally of finite type, since q is of finite type (**I**, 6.3.2), and the conclusion then follows from (**I**, 6.6.3). The ring B_i of X_{f_i} is thus an A-algebra of finite type (**I**, 6.6.3); let (t_{ij}) be a family of generators of this algebra. By hypothesis, there exist elements $s'_{ij} \in S_{m_{ij}n_i}$ such that

$$t_{ij} = (\psi^\flat(s'_{ij})|X_{f_i})(\psi^\flat(s_i)|X_{f_i})^{-m_{ij}}.$$

Also by hypothesis, there exists some λ and elements $s_{i\lambda} \in S^\lambda_{n_i}$ and $s'_{ij\lambda} \in S^\lambda_{m_{ij}n_i}$ whose images under ϕ_λ are s_i and s'_{ij}, respectively; it is clear that $r_{\mathscr{L},\psi_\lambda}$ satisfies conditions (i), (ii), and (iii) of (3.8.2). □

Proposition (3.8.5). — *Let Y be a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, $q : X \to Y$ a morphism* of finite type, *$\mathscr{L}$ an invertible $\mathscr{O}_X$-module, $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra, and $\psi : q^*(\mathscr{S}) \to \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$ a homomorphism of graded algebras. For $r_{\mathscr{L},\psi}$ to be everywhere defined and an immersion, it is necessary and sufficient that there exist an integer $n > 0$ and a quasi-coherent sub-$\mathscr{O}_Y$-module* of finite type *$\mathscr{E}$ of $\mathscr{S}_n$ such that*

a) the homomorphism $\psi_n \circ q^*(j_n) : q^*(\mathscr{E}) \to \mathscr{L}^{\otimes n}$ (where $j_n : \mathscr{E} \to \mathscr{S}$ is the canonical injection) is surjective; and
b) if we denote by $\mathscr{S}'$ the (graded) sub-$\mathscr{O}_Y$-algebra of $\mathscr{S}$ generated by $\mathscr{E}$, and by ψ' the homomorphism $\psi \circ q^*(j')$ (where j' is the injection of $\mathscr{S}'$ into $\mathscr{S}$), $r_{\mathscr{L},\psi'}$ is everywhere defined and an immersion.

If this is the case, then every quasi-coherent sub-$\mathscr{O}_Y$-module $\mathscr{E}'$ of $\mathscr{S}_n$ that contains $\mathscr{E}$ also possesses the same property, as does the image of $\otimes^k\mathscr{E}$ in $\mathscr{S}_{kn}$ for all $k > 0$.

Proof. The fact that the condition is sufficient, and the last two claims, are particular cases of (3.8.3), taking into account the canonical isomorphism between $\operatorname{Proj}(\mathscr{S})$ and $\operatorname{Proj}(\mathscr{S}^{(k)})$ (3.1.8).

Let (U_i) be a finite cover of Y by affine opens, and set $A_i = A(U_i)$. Since $q^{-1}(U_i)$ is compact, the hypothesis that $r_{\mathscr{L},\psi}$ be everywhere defined and an immersion implies, along with (3.8.2), the existence of a finite family $(s_{ij} \in S^{(i)}_{n_{ij}})$ of elements of $S^{(i)} = \Gamma(U_i, \mathscr{S})$ satisfying conditions (i), (ii), and

(iii). We see, as in the proof of (3.8.4), that the morphism $X_{f_{ij}} \to U_i$ (where $f_{ij} = \psi^\flat(s_{ij})$) is of finite type, and that the ring B_{ij} of $X_{f_{ij}}$ is thus an A_i-algebra of finite type, having a system of generators of the form $(\psi^\flat(t_{ijk})X_{f_{ij}})(f_{ij}|X_{f_{ij}})^{-m_{ijk}}$, where $t_{ijk} \in S^{(i)}_{m_{ijk}n_{ij}}$. Let n be a common multiple of the $m_{ijk}n_{ij}$; replacing (for each (i,j,k)) s_{ij} by some power s_{ij}^ρ such that $\rho m_{ijk}n_{ij} = n$, and multiplying t_{ijk} by $s_{ij}^{\rho-m_{ijk}}$, we can assume that, for each i, the s_{ij} and t_{ijk} belong to $S_n^{(i)}$ and that $m_{ijk} = 1$. Let E_i be the sub-A_i-module of $S^{(i)}$ generated by these elements (for fixed i). Then there exists a coherent sub-$\mathscr{O}_Y$-module $\mathscr{E}_i$ of $\mathscr{S}_n$ of finite type such that $\mathscr{E}_i|U_i = (E_i)^\sim$ (**I**, 9.4.7). It is clear that the sub- **II** | 71
$\mathscr{O}_Y$-module $\mathscr{E}$ of $\mathscr{S}_n$ given by the sum of the $\mathscr{E}_i$ is the desired object (since each section f_{ij} is such that, for all $x \in X_{f_{ij}}$, there exists an affine neighbourhood $V \subset X_{f_{ij}}$ of x such that $f|V$ is a basis for $\Gamma(V, \mathscr{L}^{\otimes n})$). □

§4. Projective bundles; ample sheaves

4.1. Definition of projective bundles

Definition (4.1.1). — Let Y be a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-module, and $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E})$ the symmetric $\mathscr{O}_Y$-algebra of $\mathscr{E}$ (1.7.4), which is quasi-coherent (1.7.7). We define the *projective bundle on Y defined by $\mathscr{E}$*, denoted $\mathbf{P}(\mathscr{E})$, to be the Y-scheme $P = \operatorname{Proj}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))$. The $\mathscr{O}_P$-module $\mathscr{O}_P(1)$ is called the *fundamental sheaf on P*.

When Y is affine of ring A, then we have $\mathscr{E} = \widetilde{E}$ for some A-module E, and we then write $\mathbf{P}(E)$ instead of $\mathbf{P}(\widetilde{E})$.

When we take $\mathscr{E} = \mathscr{O}_Y^n$, we write $\mathbf{P}_Y^{n-1}$ instead of $\mathbf{P}(\mathscr{E})$; if, further, Y is affine of ring A, then we also write $\mathbf{P}_A^{n-1}$ instead of $\mathbf{P}_Y^{n-1}$. Since $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{O}_Y)$ is canonically identified with $\mathscr{O}_Y[T]$ (1.7.4), $\mathbf{P}_Y^0$ is canonically identified with Y (3.1.7); Example (2.4.3) is then exactly $\mathbf{P}_K^1$.

(4.1.2). Let $\mathscr{E}$ and $\mathscr{F}$ be quasi-coherent $\mathscr{O}_Y$-modules; let $u : \mathscr{E} \to \mathscr{F}$ be an $\mathscr{O}_Y$-homomorphism; there is a canonically corresponding homomorphism $\mathbf{S}(u) : \mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}) \to \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ of graded $\mathscr{O}_Y$-algebras (1.7.4). If u is *surjective*, then so too is $\mathbf{S}(u)$, and thus (3.6.2, (i)) $\operatorname{Proj}(\mathbf{S}(u))$ is a *closed immersion* $\mathbf{P}(\mathscr{F}) \to \mathbf{P}(\mathscr{E})$, which we denote by $\mathbf{P}(u)$. We can thus say that $\mathbf{P}(\mathscr{E})$ is a *contravariant functor* in $\mathscr{E}$, with the condition that we only consider *surjective* morphisms of quasi-coherent $\mathscr{O}_Y$-modules.

Still supposing that u is surjective, and letting $P = \mathbf{P}(\mathscr{E})$, $Q = \mathbf{P}(\mathscr{F})$, and $j = \mathbf{P}(u)$, we have, up to isomorphism, that

(4.1.2.1) $$j^*(\mathscr{O}_P(n)) = \mathscr{O}_Q(n) \qquad \text{for all } n \in \mathbf{Z}$$

by (3.6.3).

(4.1.3). Now let $\psi : Y' \to Y$ be a morphism, and let $\mathscr{E}' = \psi^*(\mathscr{E})$; then $\mathbf{S}_{\mathscr{O}_{Y'}}(\mathscr{E}') = \psi^*(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))$ (1.7.5); thus (3.5.3)

(4.1.3.1) $$\mathbf{P}(\psi^*(\mathscr{E})) = \mathbf{P}(\mathscr{E}) \times_Y Y'$$

up to canonical isomorphism; furthermore, we clearly have that

$$\psi^*((\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))(n)) = (\mathbf{S}_{\mathscr{O}_{Y'}}(\mathscr{E}'))(n)$$

for all $n \in \mathbf{Z}$, whence, letting $P = \mathbf{P}(\mathscr{E})$ and $P' = \mathbf{P}(\mathscr{E}')$, we have (3.5.4), up to isomorphism, that

(4.1.3.2) $$\mathscr{O}_{P'}(n) = \mathscr{O}_p(n) \otimes_Y \mathscr{O}_{Y'} \qquad \text{for all } n \in \mathbf{Z}.$$

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Proposition (4.1.4). — *Let $\mathscr{L}$ be an invertible $\mathscr{O}_Y$-module. For every quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$, there exists a canonical Y-isomorphism $i_{\mathscr{L}} : \mathbf{P}(\mathscr{E}) \xrightarrow{\sim} \mathbf{P}(\mathscr{E} \otimes \mathscr{L})$; furthermore, if we let $P = \mathbf{P}(\mathscr{E})$ and $Q = \mathbf{P}(\mathscr{E} \otimes \mathscr{L})$, then $i_{\mathscr{L}}^*(\mathscr{O}_Q(n))$ is canonically isomorphic to $\mathscr{O}_P(n) \otimes_Y \mathscr{L}^{\otimes n}$ for all $n \in \mathbf{Z}$.*

Proof. Note first of all that, if A is a ring, E an A-module, and L a *free monogenous* A-module, then we can canonically define a homomorphism of A-modules

$$\mathbf{S}_n(E \otimes L) \longrightarrow \mathbf{S}_n(E) \otimes L^{\otimes n}$$

by sending $(x_1 \otimes y_1) \ldots (x_n \otimes y_n)$ to the element

$$(x_1 x_2 \ldots x_n) \otimes (y_1 \otimes y_2 \otimes \ldots \otimes y_n) \qquad (x_i \in E, y_i \in L, \text{for } i \leqslant i \leqslant n);$$

we can immediately see (by restricting to the case where $L = A$) that this homomorphism is in fact an isomorphism. We thus obtain a canonical isomorphism of graded A-algebras $\mathbf{S}_A(E \otimes L) \xrightarrow{\sim} \bigoplus_{n \geqslant 0} \mathbf{S}_n(E) \otimes L^{\otimes n}$. By returning to the conditions of (4.1.4), the above remarks allow us to define a canonical isomorphism of graded $\mathscr{O}_Y$-algebras

$$\text{(4.1.4.1)} \qquad \mathbf{S}_{\mathscr{O}_Y}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{L}) \xrightarrow{\sim} \bigoplus_{n \geqslant 0} \mathbf{S}_n(\mathscr{E}) \otimes_{\mathscr{O}_Y} \mathscr{L}^{\otimes n}$$

by defining this isomorphism as an isomorphism of presheaves, and taking into account (1.7.4), (**I**, 1.3.9), and (**I**, 1.3.12). The proposition then follows from (3.1.8, (iii)) and (3.2.10). □

(4.1.5). With the hypotheses of (4.1.1), let $P = \mathbf{P}(\mathscr{E})$, and denote by p the structure morphism $P \to Y$. Since, by definition, $\mathscr{E} = (\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))_1$, we have a canonical homomorphism $\alpha_1 : \mathscr{E} \to p_*(\mathscr{O}_P(1))$ (3.3.2.2), and thus (**0**, 4.4.3) also a canonical homomorphism

$$\text{(4.1.5.1)} \qquad \alpha_1^\sharp : p^*(\mathscr{E}) \longrightarrow \mathscr{O}_P(1).$$

Proposition (4.1.6). — *The canonical homomorphism* (4.1.5.1) *is surjective.*

Proof. We have seen, in (3.3.2), that $\alpha_1^\sharp$ corresponds functorially to the canonical homomorphism $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}) \to (\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))(1)$; since, by definition, $\mathscr{E}$ generates $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E})$, this homomorphism is surjective, whence the conclusion, by (3.2.4) □

4.2. Morphisms from a prescheme to a projective bundle

(4.2.1). Keeping the notation of (4.1.5), let X be a Y-prescheme, $q : X \to Y$ the structure morphism, and let $r : X \to P$ be a Y-*morphism* such that the following diagram commutes:

$$\begin{tikzcd} P \arrow[d, "p"'] & X \arrow[l, "r"'] \arrow[dl, "q"] \\ Y & \end{tikzcd}$$

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Since the functor r^* is right exact (**0**, 4.3.1), we obtain, from the surjective homomorphism in (4.1.5.1), a surjective homomorphism

$$r^*(\alpha_1^\sharp) : r^*(p^*(\mathscr{E})) \longrightarrow r^*(\mathscr{O}_P(1)).$$

But $r^*(p^*(\mathscr{E})) = q^*(\mathscr{E})$, and $r^*(\mathscr{O}_P(1))$ is locally isomorphic to $r^*(\mathscr{O}_P) = \mathscr{O}_X$, or, in other words, the latter is an *invertible* sheaf $\mathscr{L}_r$ on $\mathscr{O}_X$, and so we have defined, given r, a canonical surjective $\mathscr{O}_X$-homomorphism

$$\text{(4.2.1.1)} \qquad \phi_r : q^*(\mathscr{E}) \longrightarrow \mathscr{L}_r.$$

When $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{E} = \widetilde{E}$, we can further clarify this homomorphism in the following way: given $f \in E$, it follows from (2.6.3) that

$$\text{(4.2.1.2)} \qquad r^{-1}(D_+(f)) = X_{\phi_r^\flat(f)}.$$

Now let V be an affine open subset of X that is contained inside $r^{-1}(D_+(f))$, and let B be its ring, which is an A-algebra; let $S = \mathbf{S}_A(E)$; the restriction of r to V corresponds to an A-homomorphism $\omega : \mathbf{S}_f \to B$, and we have that $q^*(\mathscr{E})|V = (E \otimes_A B)^\sim$ and $\mathscr{L}_r|V = \widetilde{L}_r$, whence $L_r = (S(1))_{(f)} \otimes_{S_{(f)}} B_{[\omega]}$ (**I**, 1.6.5). The restriction of ϕ_r to $q^*(\mathscr{E})|V$ corresponds to the B-homomorphism $u : E \otimes_A B \to L_r$, which sends $x \otimes 1$ to $(x/1) \otimes f = (f/1) \otimes \omega(x/f)$. The canonical extension of ϕ_r to a homomorphism of $\mathscr{O}_X$-algebras

$$\psi_r : q^*(\mathbf{S}(\mathscr{E})) = \mathbf{S}(q^*(\mathscr{E})) \longrightarrow \mathbf{S}(\mathscr{L}_r) = \bigoplus_{n \geqslant 0} \mathscr{L}_r^{\otimes n}$$

is thus such that the restriction of ψ_r to $q^*(\mathbf{S}_n(\mathscr{E}))|V$ corresponds to the homomorphism $\mathbf{S}_n(\mathscr{E} \otimes_A B) = \mathbf{S}_n(E) \otimes_A B \to L_r^{\otimes n}$ that sends $s \otimes 1$ to $(f/1)^{\otimes n} \otimes \omega(s/f^n)$.

(4.2.2). Conversely, suppose that we are given a morphism $q : X \to Y$, an invertible $\mathscr{O}_X$-module $\mathscr{L}$, and a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$; to each homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$ there canonically corresponding homomorphism of quasi-coherent $\mathscr{O}_X$-algebras

$$\psi : \mathbf{S}(q^*(\mathscr{E})) = q^*(\mathbf{S}(\mathscr{E})) \longrightarrow \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$$

and thus (3.7.1) a Y-morphism $r_{\mathscr{L},\psi} : G(\psi) \to \operatorname{Proj}(\mathbf{S}(\mathscr{E})) = \mathbf{P}(\mathscr{E})$, which we denote by $r_{\mathscr{L},\phi}$. If ϕ is *surjective*, then so too is ψ, and thus (3.7.4) $r_{\mathscr{L},\phi}$ is *everywhere defined*. Furthermore, with the notation of (4.2.1) and (4.2.2):

Proposition (4.2.3). — *Given a morphism $q : X \to Y$ and a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$, maps $r \to (\mathscr{L}_r, \phi_r)$ and $(\mathscr{L}, \phi) \to r_{\mathscr{L},\phi}$ give a bijective correspondence between the set of Y-morphisms $r : X \to \mathbf{P}(\mathscr{E})$ and the set of equivalence classes of pairs $(\mathscr{L}, \phi)$ of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ and a surjective homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$, where such pairs $(\mathscr{L}, \phi)$ and $(\mathscr{L}', \phi')$ are defined to be equivalent if there exists an $\mathscr{O}_X$-isomorphism $\tau : \mathscr{L} \xrightarrow{\sim} \mathscr{L}'$ such that $\phi' = \tau \circ \phi$.*

Proof. Start first with a Y-morphism $r : X \to \mathbf{P}(\mathscr{E})$, and construct $\mathscr{L}_r$ and ϕ_r (4.2.1), and let $r' = r_{\mathscr{L}_r,\phi_r}$; it follows immediately from (4.2.1) and (3.7.2) that the morphisms r and r' are identical (by taking the generator of $\mathscr{L}_r$ to be the element $(f/1) \otimes 1$ to define the homomorphisms v_n of (3.7.2)). Conversely, take a pair $(\mathscr{L}, \phi)$ and construct $r'' = r_{\mathscr{L},\phi}$, and then $\mathscr{L}_{r''}$ and $\phi_{r''}$; we will show that there exists a canonical isomorphism $\tau : \mathscr{L}_{r''} \xrightarrow{\sim} \mathscr{L}$ such that $\phi = \tau \circ \phi_{r''}$; to define it, we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and (with the notation of (4.2.1) and (3.7.2)) associate to each element $(x/1) \otimes 1$ of $L_{r''}$ (where $x \in E$) the element $v_1(x)c$ of L. We immediately see that τ does not depend on the chosen generator c of L; since v_1 is surjective by hypothesis, to prove that τ is an isomorphism it suffices to to show that, if $x/1 = 0$ in $(S(1))_{(f)}$, then $v_1(x)/1 = 0$ in B_g; but the first equality implies that $f^n x = 0$ in $\mathbf{S}_{n+1}(E)$ for some n, and this implies that $v_{n+1}(f^n x) = g^n v_1(x) = 0$ in B, whence the conclusion. Finally, it is immediate that, for any two equivalent pairs $(\mathscr{L}, \phi)$ and $(\mathscr{L}', \phi')$, we have $r_{\mathscr{L},\phi} = r_{\mathscr{L}',\phi'}$. ∎ II | 74

In particular, for $X = Y$:

Theorem (4.2.4). — *The set of Y-sections of $\mathbf{P}(\mathscr{E})$ is in canonical bijective correspondence with the set of quasi-coherent sub-$\mathscr{O}_Y$-modules $\mathscr{F}$ of $\mathscr{E}$ such that $\mathscr{E}/\mathscr{F}$ is invertible.*

We note that this property corresponds to the classical definition of "projective space" as the set of hyperplanes of a vector space (the classical case corresponding to $Y = \operatorname{Spec}(K)$, where K is a field, and $\mathscr{E} = \widetilde{E}$, where E is a finite-dimensional K-vector space; the $\mathscr{F}$ having the property described in (4.2.4) then correspond to the hyperplanes of E, and we already know that the Y-sections of $\mathbf{P}(\mathscr{E})$ are then the *K-rational points of* $\mathbf{P}(\mathscr{E})$ (**I**, 3.4.5)).

Remark (4.2.5). — Since there is a canonical bijective correspondence between Y-morphisms from X to P and their graph morphisms, X-sections of $P \times_Y X$ (**I**, 3.3.14), we see that, conversely, (4.2.3) can be deduced from (4.2.4). Denote by $\operatorname{Hyp}_Y(X, \mathscr{E})$ the set of quasi-coherent sub-$\mathscr{O}_X$-modules $\mathscr{F}$ of $\mathscr{E} \otimes_Y \mathscr{O}_X = q^*(\mathscr{E})$ such that $q^*(\mathscr{E})/\mathscr{F}$ is an invertible $\mathscr{O}_X$-module. If $g : X' \to X$ is a Y-morphism, then it follows from the fact that g^* is right exact that $g^*q^*(\mathscr{E})/\mathscr{F}) = g^*q^*(\mathscr{E}))/g^*(\mathscr{F})$, and so the latter sheaf is invertible, and thus $\operatorname{Hyp}_Y(X, \mathscr{E})$ is a *contravariant functor* into the category of Y-preschemes. We can thus interpret the theorem (4.2.4) as defining a *canonical isomorphism* of functors $\operatorname{Hom}_Y(X, \mathbf{P}(\mathscr{E}))$ and $\operatorname{Hyp}_Y(X, \mathscr{E})$, where both functors are contravariant in the variable X and map into the category of Y-preschemes. This also gives a characterisation of the projective bundle $P = \mathbf{P}(\mathscr{E})$ by the following *universal property*, which is much closer to the geometric intuition than the constructions from §§2–3: for every morphism $q : X \to Y$ and every invertible $\mathscr{O}_X$-module $\mathscr{L}$ that is a quotient of $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{O}_X$, there exists a unique Y-morphism $r : X \to P$ such that $\mathscr{L} = r^*(\mathscr{O}_P(1))$.

We will see later that we can similarly define, amongst other things, "Grassmannian" schemes.

Corollary (4.2.6). — *Suppose that every invertible $\mathscr{O}_Y$-module is trivial* (**I**, 2.4.8)*. Let V be the group $\operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{E}, \mathscr{O}_Y)$, considered as a module over the ring $A = \Gamma(Y, \mathscr{O}_Y)$, and let V^* be the subset of V consisting of surjective homomorphisms. Then the set of Y-sections of $\mathbf{P}(\mathscr{E})$ is canonically identified with V^*/A^*, where A^* is the group of units of A.*

In particular: **II** | 75

(1) The corollary (4.2.6) applies whenever Y is a *local scheme* (**I**, 2.4.8). Let Y be an arbitrary prescheme, y a point of Y, and $Y' = \operatorname{Spec}(k(y))$; then the fibre $p^{-1}(y)$ of $\mathbf{P}(\mathscr{E})$ can, by (4.1.3.1), be identified with $\mathbf{P}(\mathscr{E}^y)$, where $\mathscr{E}^y = \mathscr{E}_y \otimes_{\mathscr{O}_y} k(y) = \mathscr{E}_y/\mathfrak{m}_y\mathscr{E}_y$ is considered as a vector space over $k(y)$. More generally, if K is an extension of $k(y)$, then $p^{-1}(y) \otimes_{k(y)} K$ can be identified with $\mathbf{P}(\mathscr{E}^y \otimes_{k(y)} K)$. The corollary (4.2.6) then shows that the set of *geometric points of* $\mathbf{P}(\mathscr{E})$ *with values in the extension* K *of* $k(y)$ (**I**, 3.4.5), which we can also call the *rational geometric fibre over* K *of* $\mathbf{P}(\mathscr{E})$ *over the point* y, can be identified with the *projective space* associated to the *dual* of the K-vector space $\mathscr{E}^y \otimes_{k(y)} K$.

(2) Suppose that Y is affine of ring A, and, further, that every invertible $\mathscr{O}_Y$-module is trivial; further, take $\mathscr{E} = \mathscr{O}_Y^n$; then, in (4.2.6), V can be identified with A^n (**I**, 1.3.8), and V^* with the sets of systems $(f_i)_{1\leqslant i\leqslant n}$ of elements of A that generate the ideal A; any two such systems define the same Y-section of $\mathbf{P}_Y^{n-1} = \mathbf{P}_A^{n-1}$, or, in other words, *the same point of* $\mathbf{P}_A^{n-1}$ *with values in* A, if and only if one of them can be obtained from the other by multiplication by an invertible element of A.

These properties justify the terminology "projective bundle" for $\mathbf{P}(\mathscr{E})$. We note that the definitions that we will similarly obtain for "projective space" is in fact *dual* to the classical definition; this is imposed upon us by the necessity of being able to define $\mathbf{P}(\mathscr{E})$ for *arbitrary* quasi-coherent $\mathscr{O}_Y$-modules $\mathscr{E}$, and not just locally free ones.

Remark (4.2.7). — We will see, in Chapter V, that, if Y is connected and locally Noetherian, and if $\mathscr{E}$ is locally free, then, letting $P = \mathbf{P}(\mathscr{E})$, every invertible $\mathscr{O}_P$-module is isomorphic to an $\mathscr{O}_P$-module of the form $\mathscr{L}' \otimes_{\mathscr{O}_Y} \mathscr{O}_P(m)$, with $\mathscr{L}'$ some invertible $\mathscr{O}_Y$-module, well defined up to isomorphism, and m some well defined integer. In other words, $\mathrm{H}^1(P, \mathscr{O}_P^*)$ is isomorphic to $\mathbf{Z} \times \mathrm{H}^1(Y, \mathscr{O}_Y^*)$ (**0**, 5.4.7). We will also see ((**III**, 2.1.14), taking (**0**, 5.4.10) into account) that $p_*(\mathscr{L}^{\otimes m}) = 0$ if $m < 0$, and $p_*(\mathscr{L}^{\otimes m})$ is isomorphic to $\mathscr{L}' \otimes_{\mathscr{O}_Y} (\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))_m$ if $m \geqslant 0$. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_Y$-module, then every Y-morphism $\mathbf{P}(\mathscr{E}) \to \mathbf{P}(\mathscr{F})$ is determined by the data of an invertible $\mathscr{O}_Y$-module, an integer $m \geqslant 0$, and an $\mathscr{O}_Y$-homomorphism $\psi : \mathscr{F} \to \mathscr{L}' \otimes_{\mathscr{O}_Y} (\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}))_m$ such that the corresponding homomorphism $\psi^\sharp$ of $\mathscr{O}_{\mathbf{P}(\mathscr{F})}$-modules is surjective. We will also see that, if the Y-morphism in question is an isomorphism, then $m = 1$ and $\mathscr{F}$ is isomorphic to $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{L}'$ (the converse of (4.1.4)). This will allow us to determine the sheaf of germs of automorphisms of $\mathbf{P}(\mathscr{E})$ as the quotient of the sheaf of groups $\mathscr{A}ut(\mathscr{E})$ (which is locally isomorphic to $\mathbf{GL}(n, \mathscr{O}_Y)$ is $\mathscr{E}$ is of rank n) by $\mathscr{O}_Y^*$.

(4.2.8). Keeping the notation of (4.2.1), let $u : X' \to X$ be a morphism; if the Y-morphism $r : X \to P$ corresponds to the homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$, then the Y-morphism $r \circ u$ corresponds to $u^*(\phi) : u^*(q^*(\mathscr{E})) \to u^*(\mathscr{L})$, as follows immediately from the definitions.

(4.2.9). Let $\mathscr{E}$ and $\mathscr{F}$ be quasi-coherent $\mathscr{O}_Y$-modules, $v : \mathscr{E} \to \mathscr{F}$ a surjective homomorphism, and $j = \mathbf{P}(v)$ the corresponding closed immersion $\mathbf{P}(\mathscr{F}) \to \mathbf{P}(\mathscr{E})$ (4.1.2). If the Y-morphism $r : X \to \mathbf{P}(\mathscr{F})$ corresponds to the homomorphism $\phi : q^*(\mathscr{F}) \to \mathscr{L}$, then the Y-morphism $j \circ r$ **II** | 76
corresponds to $q^*(\mathscr{E}) \xrightarrow{q^*(v)} q^*(\mathscr{F}) \xrightarrow{\phi} \mathscr{L}$; this again follows from the definition given in (4.2.1).

(4.2.10). Let $\psi : Y' \to Y$ be a morphism, and let $\mathscr{E}' = \psi^*(\mathscr{E})$. If the Y-morphism $r : X \to P$ corresponds to the homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$, then the Y'-morphism

$$r_{(Y')} : X_{(Y')} \longrightarrow P' = \mathbf{P}(\mathscr{E}')$$

corresponds to $\phi_{(Y')} : q^*_{(Y')}(\mathscr{E}') = q^*(\mathscr{E}) \otimes_Y \mathscr{O}_{Y'} \to \mathscr{L} \otimes_Y \mathscr{O}_{Y'}$. Indeed, by (4.1.3.1), we have the commutative diagram

$$\begin{tikzcd}
Y' \arrow[d] & P' = P_{(Y')} \arrow[l, "p_{(Y')}"'] \arrow[d, "u"] & X_{(Y')} \arrow[l, "r_{(Y')}"'] \arrow[d, "v"] \\
Y & P \arrow[l, "p"] & X \arrow[l, "r"]
\end{tikzcd}$$

From (4.1.3.1), we have

$$(r_{(Y')})^*(\mathscr{O}_{P'}(1)) = (r_{(Y')})^*(u^*(\mathscr{O}_P(1))) = v^*(r^*(\mathscr{O}_P(1))) = v^*(\mathscr{L}) = \mathscr{L} \otimes_Y \mathscr{O}_{Y'};$$

we also know that $u^*(\alpha_1^\sharp)$ is exactly the canonical homomorphism $\alpha_1^\sharp : (p_{(Y')})^*(\mathscr{E}') \to \mathscr{O}_{P'}(1)$; we can see this by explicitly calculating the canonical homomorphisms $\alpha_1^\sharp$ to P and P' as in (4.1.6). Whence our claim.

4.3. The Segre morphism

(4.3.1). Let Y be a prescheme, and $\mathscr{E}$ and $\mathscr{F}$ quasi-coherent $\mathscr{O}_Y$-modules; let $P_1 = \mathbf{P}(\mathscr{E})$ and $P_2 = \mathbf{P}(\mathscr{F})$, and denote the structure morphisms by $p_1 : P_1 \to Y$ and $p_2 : P_2 \to Y$. Let $Q = P_1 \times_Y P_2$, and let $q_1 : Q \to P_1$ and $q_2 : Q \to P_2$ be the canonical projections; then the $\mathscr{O}_Q$-module $\mathscr{L} = \mathscr{O}_{P_1}(1) \otimes_Y \mathscr{O}_{P_2}(1) = q_1^*(\mathscr{O}_{P_1}(1)) \otimes_{\mathscr{O}_Q} q_2^*(\mathscr{O}_{P_2}(1))$ is invertible, since it is the tensor product of of two invertible $\mathscr{O}_Q$-modules (**0**, 5.4.4). Also, if $r = p_1 \circ q_1 = p_2 \circ q_2$ is the structure morphism $Q \to Y$, then $r^*(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) = q_1^*(p_1^*(\mathscr{E})) \otimes_{\mathscr{O}_Q} q_2^*(p_2^*(\mathscr{F}))$ (**0**, 4.3.3); the canonical surjective homomorphisms (4.1.5.1) $p_1^*(\mathscr{E}) \to \mathscr{O}_{P_1}(1)$ and $p_2^*(\mathscr{F}) \to \mathscr{O}_{P_2}(1)$ thus give, by taking the tensor product, a canonical homomorphism

$$s : r^*(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) \longrightarrow \mathscr{L} \tag{4.3.1.1}$$

which is evidently surjective; from this we obtain (4.2.2) a canonical morphism, called the *Segre morphism*:

$$\varsigma : \mathbf{P}(\mathscr{E}) \times_Y \mathbf{P}(\mathscr{F}) \longrightarrow \mathbf{P}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}). \tag{4.3.1.2}$$

We can study the morphism ς more explicitly in the case where $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{E} = \widetilde{E}$ and $\mathscr{F} = \widetilde{F}$, where E and F are A-modules, whence $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F} = (E \otimes_A F)^\sim$ (**I**, 1.3.12); let $R = \mathbf{S}_A(E)$, $S = \mathbf{S}_A(F)$, and $T = \mathbf{S}_A(E \otimes_A F)$; let $f \in E$ and $g \in F$, and consider the affine open

$$D_+(f) \times_Y D_+(g) = \operatorname{Spec}(B)$$

of Q, where $B = R_{(f)} \otimes_A S_{(g)}$; the restriction of $\mathscr{L}$ to this affine open is $\widetilde{L}$, where **II | 77**

$$L = (R(1))_{(f)} \otimes_A (S(1))_{(g)}$$

and the element $c = (f/1) \otimes (g/1)$ is a generator of L considered as a free B-module (2.5.7). The homomorphism (4.3.1.1) corresponds to the homomorphism

$$(x \otimes y) \otimes b \longmapsto b((x/1) \otimes (y/1))$$

from $(E \otimes_A F) \otimes_A B$ to L. With the notation of (3.7.2), we thus have that $v_1(x \otimes y) = (x/f) \otimes (y/g)$; the restriction of ς to $D_+(f) \times_Y D_+(g)$ is a morphism from this affine scheme to $D_+(f \otimes g)$, corresponding to the ring homomorphism $\omega : T_{(f\otimes g)} \to R_{(f)} \otimes_A S_{(g)}$ defined by

$$\omega((x \otimes y)/(f \otimes g)) = (x/f) \otimes (y/g) \tag{4.3.1.3}$$

for $x \in E$ and $y \in F$.

(4.3.2). It follows from (4.2.3) that we have a canonical isomorphism

$$\tau : \varsigma^*(\mathscr{O}_P(1)) \xrightarrow{\sim} \mathscr{O}_{P_1}(1) \otimes_Y \mathscr{O}_{P_2}(1) \tag{4.3.2.1}$$

where we let $P = \mathbf{P}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$. We will show that, for $x \in \Gamma(Y, \mathscr{E})$ and $y \in \Gamma(Y, \mathscr{F})$, we have

$$\tau(\alpha_1(x \otimes y)) = \alpha_1(x) \otimes \alpha_1(y). \tag{4.3.2.2}$$

Indeed, we can restrict to the case where Y is affine, and we then have, with the notation of (4.3.1) and (2.6.2), that $\alpha_1^{f\otimes g}(x \otimes y) = (x \otimes y)/1$ in $(T(1))_{(f\otimes g)}$, that $\alpha_1^f(x) = x/1$ in $(R(1))_{(f)}$, and that $\alpha_1^g(y) = y/1$ in $(S(1))_{(g)}$. The definition of τ given in (4.2.3) and the calculation of v_1 done in (4.3.1) then immediately prove the claim (4.3.2.2). From this we obtain the equation

$$\varsigma^{-1}(P_{x\otimes y}) = (P_1)_x \times_Y (P_2)_y \tag{4.3.2.3}$$

with the notation of (3.1.4). Indeed, taking (3.3.3) into account, the equation (4.3.2.2) (by restricting to the affine case, with the help of (**I**, 3.2.7) and (**I**, 3.2.3)) leaves us only to prove the following lemma:

Lemma (4.3.2.4). — *Let B and B' be A-algebras, and let $Y = \operatorname{Spec}(A)$, $Z = \operatorname{Spec}(B)$, and $Z' = \operatorname{Spec}(B')$; then $D(t \otimes t') = D(t) \times_Y D(t')$ for any $t \in B$, $t' \in B'$.*

PROOF. Indeed, if $p : Z \times_Y Z' \to Z$ and $p' : Z \times_Y Z' \to Z'$ are the canonical projections, then it follows from (I, 1.2.2.2) that $p^{-1}(D(t)) = D(t \otimes 1)$ and $p'^{-1}(D(t')) = D(1 \otimes t')$; the conclusion follows from (I, 3.2.7) and (I, 1.1.9.1), since $(t \otimes 1)(1 \otimes t') = t \otimes t'$. □

Proposition (4.3.3). — *The Segre morphism is a closed immersion.*

PROOF. Since the question is local on Y, we can restrict to the case where Y is affine. With the notation of (4.3.1) and (4.3.1), the $D_+(f \otimes g)$ then form a basis for the topology of P, since the $f \otimes g$ generate T when f runs over E and g runs over F. By (4.3.2.3), we also know that $\varsigma^{-1}(D_+(f \otimes g)) = D_+(f) \times_Y D_+(g)$. It thus suffices (I, 4.2.4) to prove that the restriction of ς to $D_+(f) \times_Y D_+(g)$ is a closed immersion into $D_+(f \otimes g)$. But, with the same notation, the equation (4.3.1.3) shows that ω is *surjective*, which completes the proof. □

(4.3.4). The Segre morphism is *functorial* in $\mathscr{E}$ and $\mathscr{F}$, if we consider only *surjective* homomorphisms of quasi-coherent $\mathscr{O}_Y$-modules. Indeed, we must then show that, if $\mathscr{E} \to \mathscr{E}'$ is a surjective $\mathscr{O}_Y$-homomorphism, then the diagram II | 78

$$\begin{tikzcd} \mathbf{P}(\mathscr{E}') \times \mathbf{P}(\mathscr{F}) \arrow[r, "j\times 1"] \arrow[d, "\varsigma"'] & \mathbf{P}(\mathscr{E}) \times \mathbf{P}(\mathscr{F}) \arrow[d, "\varsigma"] \\ \mathbf{P}(\mathscr{E}' \otimes \mathscr{F}) \arrow[r] & \mathbf{P}(\mathscr{E} \otimes \mathscr{F}) \end{tikzcd}$$

commutes, where j denotes the canonical closed immersion $\mathbf{P}(\mathscr{E}') \to \mathbf{P}(\mathscr{E})$. Let $P_1' = \mathbf{P}(\mathscr{E}')$ and keep the notation from (4.3.1); then $j \times 1$ is a closed immersion (I, 4.3.1) and, up to isomorphism,

$$(j \times 1)^*(\mathscr{O}_{P_1}(1) \otimes \mathscr{O}_{P_2}(1)) = j^*(\mathscr{O}_{P_1}(1)) \otimes \mathscr{O}_{P_2}(1) = \mathscr{O}_{P_1'}(1) \otimes \mathscr{O}_{P_2}(1)$$

by (4.1.2.1) and (I, 9.1.5); our claim then follows from (4.2.8) and (4.2.9).

(4.3.5). With the notation of (4.3.1), let $\psi : Y' \to Y$ be a morphism, and let $\mathscr{E}' = \psi^*(\mathscr{E})$ and $\mathscr{F}' = \psi^*(\mathscr{F})$; then the Segre morphism $\mathbf{P}(\mathscr{E}') \times \mathbf{P}(\mathscr{F}') \to \mathbf{P}(\mathscr{E}' \otimes \mathscr{F}')$ can be identified with $\varsigma_{(Y')}$. Indeed, keeping the notation of (4.3.1), let $P_1' = \mathbf{P}(\mathscr{E}')$ and $P_2 = \mathbf{P}(\mathscr{F}')$; we know (4.1.3.1) that P_i' can be identified with $(P_i)_{(Y')}$ $(i = 1, 2)$, and so the structure morphism $P_1' \times P_2' \to Y'$ can be identified with $r_{(Y')}$. Also $\mathscr{E}' \otimes \mathscr{F}'$ can be identified with $\psi^*(\mathscr{E} \otimes \mathscr{F})$, and so $\mathbf{P}(\mathscr{E}' \otimes \mathscr{F}')$ can be identified with $(\mathbf{P}(\mathscr{E} \otimes \mathscr{F}))_{(Y')}$ (4.1.3.1). Finally, $\mathscr{O}_{P_1'}(1) \otimes_Y \mathscr{O}_{P_2'}(1) = \mathscr{L}'$ can be identified with $\mathscr{L} \otimes_Y \mathscr{O}_{Y'}$, by (4.1.3.1) and (I, 9.1.11). The canonical homomorphism $(r_{(Y)})^*(\mathscr{E}' \otimes \mathscr{F}') \to \mathscr{L}'$ can then be identified with $s_{(Y')}$, and our claim follows from (4.2.10).

Remark (4.3.6). — The prescheme given by the *sum* of $\mathbf{P}(\mathscr{E})$ and $\mathbf{P}(\mathscr{F})$ is even canonically isomorphic to a *closed subprescheme of* $\mathbf{P}(\mathscr{E} \oplus \mathscr{F})$. Indeed, the surjective homomorphisms $\mathscr{E} \oplus \mathscr{F} \to \mathscr{E}$ and $\mathscr{E} \oplus \mathscr{F} \to \mathscr{F}$ correspond to closed immersions $\mathbf{P}(\mathscr{E}) \to \mathbf{P}(\mathscr{E} \oplus \mathscr{F})$ and $\mathbf{P}(\mathscr{F}) \to \mathbf{P}(\mathscr{E} \oplus \mathscr{F})$; everything then reduces to showing that the underlying spaces of the closed subpreschemes of $\mathbf{P}(\mathscr{E} \oplus \mathscr{F})$ obtained in this way have empty intersection. Since the question is local on Y, we can adopt the notation of (4.3.1); but $\mathbf{S}_n(E)$ and $\mathbf{S}_n(F)$ can be identified with submodules of $\mathbf{S}_n(E \oplus F)$ with intersection consisting only of 0; if $\mathfrak{p}$ is a graded prime ideal of $\mathbf{S}(E)$ such that $\mathfrak{p} \cap \mathbf{S}_n(E) \neq \mathbf{S}_n(E)$ for any $n \geqslant 0$, then there exists a corresponding graded prime ideal of $\mathbf{S}(E \oplus F)$ whose intersection with $\mathbf{S}_n(E)$ is $\mathfrak{p} \cap \mathbf{S}_n(E)$, but who also *contains* $\mathbf{S}_+(F)$, as we immediately see; thus no point in $\mathrm{Proj}(\mathbf{S}(E))$ can have the same image in $\mathrm{Proj}(\mathbf{S}(E \oplus F))$ as any point in $\mathrm{Proj}(\mathbf{S}(F))$.

4.4. Immersions into projective bundles; very ample sheaves

Proposition (4.4.1). — *Let Y be a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, $q : X \to Y$ a morphism* of finite type, *and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module.*

(i) *Let $\mathscr{S}$ be a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra, and $\psi : q^*(\mathscr{S}) \to \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$ a homomorphism of graded algebras. For $r_{\mathscr{L},\psi}$ to be everywhere defined and an immersion, it is necessary and sufficient for there to exist an integer $n \geqslant 0$ and a quasi-coherent sub-$\mathscr{O}_Y$-module* of finite type *$\mathscr{E}$ of $\mathscr{S}_n$ such that the homomorphism $\psi' = \psi_n \circ q^*(j) : q^*(\mathscr{E}) \to \mathscr{L}^{\otimes n} = \mathscr{L}'$ (where j is the injection $\mathscr{E} \to \mathscr{S}_n$) is surjective and such that the morphism $r_{\mathscr{L}',\psi'} : X \to \mathbf{P}(\mathscr{E})$ is an immersion.* II | 79

(ii) *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_Y$-module, and $\phi : q^*(\mathscr{F}) \to \mathscr{L}$ a surjective homomorphism. For the morphism $r_{\mathscr{L},\phi}$ to be an immersion $X \to \mathbf{P}(\mathscr{F})$, it is necessary and sufficient for there to exist a quasi-coherent sub-$\mathscr{O}_Y$-module* of finite type *$\mathscr{E}$ of $\mathscr{F}$ such that the homomorphism $\phi' = \phi \circ q(j) : q^*(\mathscr{E}) \to \mathscr{L}$ (where j is the canonical injection $\mathscr{E} \to \mathscr{F}$) is surjective and such that the morphism $r_{\mathscr{L},\phi'} : X \to \mathbf{P}(\mathscr{E})$ is an immersion.*

PROOF.

(i) The fact that $r_{\mathscr{L},\phi}$ is everywhere defined and is an immersion is equivalent, by (3.8.5), to the existence of some $n \geqslant 0$ and $\mathscr{E}$ such that, if $\mathscr{S}'$ is the subalgebra of $\mathscr{S}$ generated by $\mathscr{E}$, the homomorphism $q^*(\mathscr{E}) \to \mathscr{L}^{\otimes n}$ is surjective and the morphism $r_{\mathscr{L},\psi'} : X \to \mathrm{Proj}(\mathscr{S}')$ is everywhere defined and is an immersion. We already have a canonical surjective homomorphism $\mathbf{S}(\mathscr{E}) \to \mathscr{S}'$ to which there exists a corresponding closed immersion $\mathrm{Proj}(\mathscr{S}') \to \mathbf{P}(\mathscr{E})$ (3.6.2); whence the conclusion.

(ii) Since $\mathscr{F}$ is the inductive limit of its quasi-coherent submodules of finite type $\mathscr{E}_\lambda$ (**I**, 9.4.9), $\mathbf{S}(\mathscr{F})$ is the inductive limit of the $\mathbf{S}(\mathscr{E}_\lambda)$; the conclusion then follows from (3.8.4), by observing that we can take all the n_i in the proof of (3.8.4) to be equal to 1: indeed, supposing that Y is affine, if $r = r_{\mathscr{L},\phi}$ is an immersion, then $r(X)$ is a quasi-compact subspace of $\mathbf{P}(\mathscr{F})$ that we can cover by finitely many open subsets of $\mathbf{P}(\mathscr{F})$ of the form $D_+(f)$, with $f \in F$, such that $D_+(f) \cap r(X)$ is closed.

□

Definition (4.4.2). — Let Y be a prescheme, and $q : X \to Y$ a morphism. We say that an invertible $\mathscr{O}_X$-module $\mathscr{L}$ is *very ample for q*, or *relative to q* (or *very ample for* (or *relative to*) Y, or simply *very ample*, if q is clear from the context) if there exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ and a Y-immersion i from X to $P = \mathbf{P}(\mathscr{E})$ such that $\mathscr{L}$ is isomorphic to $i^*(\mathscr{O}_P(1))$.

It is equivalent (4.2.3) to say that there exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ and a *surjective* homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$ such that $r_{\mathscr{L},\phi} : X \to \mathbf{P}(\mathscr{E})$ is an *immersion*.

We note that the existence of a very ample (for Y) $\mathscr{O}_X$-module implies that q is *separated* ((3.1.3) and (**I**, 5.5.1, (i) and (ii))).

Corollary (4.4.3). — *Suppose that there exists a quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$, generated by $\mathscr{S}_1$, and a Y-immersion $i : X \to P = \mathrm{Proj}(\mathscr{S})$ such that $\mathscr{L}$ is isomorphic to $i^*(\mathscr{O}_P(1))$; then $\mathscr{L}$ is very ample relative to q.*

PROOF. If $\mathscr{F} = \mathscr{S}_1$, then the canonical homomorphism $\mathbf{S}(\mathscr{F}) \to \mathscr{S}$ is surjective, and so, by compositing with the corresponding closed immersion $\mathrm{Proj}(\mathscr{S}) \to \mathbf{P}(\mathscr{F})$ (3.6.2) and the immersion i, we obtain an immersion $j : X \to \mathbf{P}(\mathscr{F}) = P'$ such that $\mathscr{L}$ is isomorphic to $j^*(\mathscr{O}_{P'}(1))$ (3.6.3). □

Proposition (4.4.4). — *Let $q : X \to Y$ be a quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. Then the following properties are equivalent:*

(a) *$\mathscr{L}$ is very ample relative to q.*

(b) *$q_*(\mathscr{L})$ is quasi-coherent, the canonical homomorphism $\sigma : q^*(q_*(\mathscr{L})) \to \mathscr{L}$ is surjective, and the morphism $r_{\mathscr{L},\sigma} : X \to \mathbf{P}(q_*(\mathscr{L}))$ is an immersion.*

PROOF. Since q is quasi-compact, we know that $q_*(\mathscr{L})$ is quasi-coherent if q is separated (**I**, 9.2.2).

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We know (3.4.7) that the existence of a surjective homomorphism $\phi : q^*(\mathscr{E}) \to \mathscr{L}$ (with $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-module) implies that σ is surjective; furthermore, given the factorisation $q^*(\mathscr{E}) \to q^*(q_*(\mathscr{L})) \xrightarrow{\sigma} \mathscr{L}$ of ϕ, there is a canonically corresponding factorisation

$$q^*(\mathbf{S}(\mathscr{E})) \longrightarrow q^*(\mathbf{S}(q_*(\mathscr{L}))) \longrightarrow \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$$

and so (3.8.3) the hypothesis that $r_{\mathscr{L},\phi}$ is an immersion implies that so too is $j = r_{\mathscr{L},\sigma}$; furthermore (4.2.4), $\mathscr{L}$ is isomorphic to $j^*(\mathscr{O}_{P'}(1))$, where $P' = \mathbf{P}(q_*(\mathscr{L}))$. We thus see that (a) and (b) are equivalent. □

Corollary (4.4.5). — *Suppose that q is quasi-compact. For $\mathscr{L}$ to be very ample relative to Y, it is necessary and sufficient for there to exist an open cover (U_α) of Y such that $\mathscr{L}|q^{-1}(U_\alpha)$ is very ample relative to U_α for every α.*

PROOF. Indeed, condition (b) of (4.4.4) is local on Y. □

Proposition (4.4.6). — *Let Y be a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, $q : X \to Y$ a morphism* of finite type*, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. Then conditions (a) and (b) of (4.4.4) are equivalent to the following:*

(a′) *There exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$* of finite type *and a surjective homomorphism ϕ : $q^*(\mathscr{E}) \to \mathscr{L}$ such that $r_{\mathscr{L},\phi}$ is an immersion.*

(b′) *There exists a coherent sub-$\mathscr{O}_Y$-module $\mathscr{E}$ of $q_*(\mathscr{L})$* of finite type *that has the properties stated in condition (a′).*

PROOF. It is clear that (a′) or (b′) imply (a); also (a) implies (a′), by (4.4.1), and similarly (b) implies (b′). □

Corollary (4.4.7). — *Suppose that Y is a quasi-compact scheme, or a Noetherian prescheme. If $\mathscr{L}$ is very ample for q, then there exists a quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ such that $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$, and also a* dominant open *Y-immersion $i : X \to P = \operatorname{Proj}(\mathscr{S})$ such that $\mathscr{L}$ is isomorphic to $i^*(\mathscr{O}_P(1))$.*

PROOF. Indeed, condition (b) of (4.4.6) is satisfied; the structure morphism $p : \mathbf{P}(\mathscr{E}) = P' \to Y$ is then separated and of finite type (3.1.3), and so P' is a quasi-compact scheme (resp. a Noetherian prescheme) if Y is a quasi-compact scheme (resp. a Noetherian prescheme). Let Z be the closure (**I**, 9.5.11) of the subprescheme X' of P' associated to the immersion $j = r_{\mathscr{L},\phi}$ from X into P'; it is clear that j factors as a dominant open immersion $i : X \to Z$ followed by the canonical injection $Z \to P'$. But Z can be identified with a prescheme $\operatorname{Proj}(\mathscr{S})$, where $\mathscr{S}$ is a graded $\mathscr{O}_Y$-algebra equal to the quotient of $\mathbf{S}(\mathscr{E})$ by a quasi-coherent graded sheaf of ideals (3.6.2), and it is clear that $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$; furthermore, $\mathscr{O}_Z(1)$ is the inverse image of $\mathscr{O}_{P'}(1)$ by the canonical injection (3.6.3), and so $\mathscr{L} = i^*(\mathscr{O}_Z(1))$. □

Proposition (4.4.8). — *Let $q : X \to Y$ be a morphism, $\mathscr{L}$ a very ample (relative to q) $\mathscr{O}_X$-module, and $\mathscr{L}'$ an invertible $\mathscr{O}_X$-module, such that there exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}'$ and a surjective homomorphism $q^*(\mathscr{E}') \to \mathscr{L}'$. Then $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}'$ is very ample relative to q.*

PROOF. The hypothesis implies the existence of a Y-morphism $r' : X \to P' = \mathbf{P}(\mathscr{E}')$ such that $\mathscr{L}' = r'^*(\mathscr{O}_{P'}(1))$ (4.2.1). There is, by hypothesis, a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ and a Y-immersion **II** | 81
$r : X \to P = \mathbf{P}(\mathscr{E})$ such that $\mathscr{L} = r^*(\mathscr{O}_P(1))$. Let $Q = \mathbf{P}(\mathscr{E} \otimes \mathscr{E}')$, and consider the Segre morphism $\varsigma : P \times_Y P' \to Q$ (4.3.1). Since r is an immersion, so too is $(r, r')_Y : X \to P \times_Y P'$ (**I**, 5.3.14); but since ς is an immersion (4.3.3), so too is $r'' : X \xrightarrow{(r,r')} P \times_Y P' \xrightarrow{\varsigma} Q$. But also (4.3.2.1) $\varsigma(\mathscr{O}_Q(1))$ is isomorphic to $\mathscr{O}_P(1) \otimes_Y \mathscr{O}_{P'}(1)$, and so (**I**, 9.1.4) $r''^*(\mathscr{O}_Q(1))$ is isomorphic to $\mathscr{L} \otimes \mathscr{L}'$, which proves the proposition. □

Corollary (4.4.9). — *Let $q : X \to Y$ be a morphism.*

(1) *Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module, and $\mathscr{K}$ an invertible $\mathscr{O}_Y$-module. For $\mathscr{L}$ to be very ample relative to q, it is necessary and sufficient for $\mathscr{L} \otimes q^*(\mathscr{K})$ to be so.*

(2) *If $\mathscr{L}$ and $\mathscr{L}'$ are very ample (relative to q) $\mathscr{O}_X$-modules, then so too is $\mathscr{L} \otimes \mathscr{L}'$; in particular, $\mathscr{L}^{\otimes n}$ is very ample relative to q for all $n > 0$.*

PROOF. Claim (ii) is an immediate consequence of (4.4.8), as well as the necessity of condition (i); conversely, if $\mathscr{L} \otimes q^*(\mathscr{K})$ is very ample, then so too is $(\mathscr{L} \otimes q^*(\mathscr{K})) \otimes q^*(\mathscr{K}^{-1})$, by the above, and the latter $\mathscr{O}_X$-module is isomorphic to $\mathscr{L}$ ((**0**, 5.4.3) and (**0**, 5.4.5)). □

Proposition (4.4.10). —

(i) *For every prescheme Y, every invertible $\mathscr{O}_Y$-module $\mathscr{L}$ is very ample relative to the identity morphism 1_Y.*

(i *bis*) *Let $f : X \to Y$ be a morphism, and $j : X' \to X$ an immersion. If $\mathscr{L}$ is a very ample (relative to f) $\mathscr{O}_X$-module, then $j^*(\mathscr{L})$ is very ample relative to $f \circ j$.*

(ii) *Let Z be a quasi-compact prescheme, $f : X \to Y$ a morphism of finite type, $g : Y \to Z$ a quasi-compact morphism, $\mathscr{L}$ a very ample (relative to f) $\mathscr{O}_X$-module, and $\mathscr{K}$ a very ample (relative to g) $\mathscr{O}_Y$-module. Then there exists some integer $n_0 > 0$ such that $\mathscr{L} \otimes f^*(\mathscr{K}^{\otimes n})$ is very ample relative to $g \circ f$ for all $n \geqslant n_0$.*

(iii) *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms, and let $X' = X_{(Y')}$. If $\mathscr{L}$ is a very ample (relative to f) $\mathscr{O}_X$-module, then $\mathscr{L}' = \mathscr{L} \otimes_Y \mathscr{O}_{Y'}$ is a very ample (relative to $f_{(Y')}$) $\mathscr{O}_X$-module.*

(iv) *Let $f_i : X_i \to Y_i$ $(i = 1, 2)$ be S-morphism. If $\mathscr{L}_i$ is a very ample (relative to f_i) $\mathscr{O}_{X_i}$-module $(i = 1, 2)$, then $\mathscr{L}_1 \otimes_S \mathscr{L}_2$ is very ample relative to $f_1 \times_S f_2$.*

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms. If an $\mathscr{O}_X$-module $\mathscr{L}$ is very ample relative to $g \circ f$, then it is also very ample relative to f.*

(vi) *Let $f : X \to Y$ be a morphism, and j the canonical injection $X_{\text{red}} \to X$. If an $\mathscr{O}_X$-module $\mathscr{L}$ is very ample relative to f, then $j^*(\mathscr{L})$ is very ample relative to f_{red}.*

PROOF. Property (i *bis*) follows immediately from the definition (4.4.2), and it is immediate that (vi) follows formally from (i *bis*) and (v), by an argument copied from the proof of (**I**, 5.5.12), which we leave to the reader. To prove (v), we consider, as in (**I**, 5.5.12), the factorisation $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$, where $p_2 = (g \circ f) \times 1_Y$. It follows from the hypothesis and from (i) and (iv) that $\mathscr{L} \otimes_{\mathscr{O}_Z} \mathscr{O}_Y$ is very ample for p_2; but also $\mathscr{L} = \Gamma_f^*(\mathscr{L} \otimes_{\mathscr{O}_Z} \mathscr{O}_Y)$ (**I**, 9.1.4), and Γ_f is an immersion (**I**, 5.3.11); we can thus apply (i *bis*).

To prove (i), we apply the definition (4.4.2) with $\mathscr{E} = \mathscr{L}$, and note that then $\mathbf{P}(\mathscr{E})$ can be identified with Y (4.1.4). **II** | 82

Now we prove (iii). There exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ and a Y-immersion $i : X \to \mathbf{P}(\mathscr{E}) = P$ such that $\mathscr{L} = i^*(\mathscr{O}_P(1))$; if we let $\mathscr{E}' = g^*(\mathscr{E})$, then $\mathscr{E}'$ is a quasi-coherent $\mathscr{O}_{Y'}$-module, and we have that $P' = \mathbf{P}(\mathscr{E}') = P_{(Y')}$ (4.1.3.1), that $i_{(Y')}$ is an immersion from $X_{(Y')}$ into P' (**I**, 4.3.2), and that $\mathscr{L}'$ is isomorphic to $(i_{(Y')})^*(\mathscr{O}_{P'}(1))$ (4.2.10).

To prove (iv), note that there is, by hypothesis, a Y_i-immersion $r_i : X_i \to P_i = \mathbf{P}(\mathscr{E}_i)$, where $\mathscr{E}_i$ is a quasi-coherent $\mathscr{O}_{Y_i}$-module, and $\mathscr{L}_i = r_i^*(\mathscr{O}_{P_i}(1))$ $(i = 1, 2)$; $r_1 \times_S r_2$ is an S-immersion of $X_1 \times_S X_2$ into $P_1 \times_S P_2$ (**I**, 4.3.1), and the inverse image of $\mathscr{O}_{P_1}(1) \otimes_S \mathscr{O}_{P_2}(1)$ under this immersion is $\mathscr{L}_1 \otimes_S \mathscr{L}_2$. Now let $T = Y_1 \times_S Y_2$, and let p_1 and p_2 be the projections from T to Y_1 and Y_2, respectively. If we let $P_i' = \mathbf{P}(p_i^*(\mathscr{E}_i))$ $(i = 1, 2)$, then $P_i' = P_i \times_{Y_i} T$, by (4.1.3.1), and so

$$P_1' \times_T P_2' = (P_1 \times_{Y_1} T) \times_T (P_2 \times_{Y_2} T) = P_1 \times_{Y_1} (T \times_{Y_2} P_2) = P_1 \times_{Y_1} (Y_1 \times_S P_2) = P_1 \times_S P_2$$

up to canonical isomorphism. Similarly, $\mathscr{O}_{P_i'}(1) = \mathscr{O}_{P_i}(1) \otimes_{Y_i} \mathscr{O}_T$ (4.1.3.2), and an analogous calculation (based in particular on (**I**, 9.1.9.1) and (**I**, 9.1.2)) shows that, in the above identification, $\mathscr{O}_{P_1'}(1) \otimes_T \mathscr{O}_{P_2'}(1)$ can be identified with $\mathscr{O}_{P_1} \otimes_S \mathscr{O}_{P_2}(1)$. We can thus consider $r_1 \times_S r_2$ as a T-immersion from $X_1 \times_S X_2$ into $P_1' \times_T P_2'$, with the inverse image of $\mathscr{O}_{P_1'}(1) \otimes_T \mathscr{O}_{P_2'}(1)$ under this immersion being $\mathscr{L}_1 \otimes_S \mathscr{L}_2$. We then finish the argument as in (4.4.8) by using the Segre morphism.

It remains only to prove (ii). We can first of all restrict to the case where Z is an affine scheme, since, in general, there exists a finite cover (U_i) of Z by affine opens; if the proposition were proven for $\mathscr{K}|g^{-1}(U_i)$, $\mathscr{L}|f^{-1}(g^{-1}(U))$, and an integer n_i, then it would suffice to take n_0 to be the largest of the n_i to prove the proposition for $\mathscr{K}$ and $\mathscr{L}$ (4.4.5). The hypothesis implies that f and g are separated morphisms, and so X and Y are quasi-compact *schemes*.

There is an immersion $r : X \to P = \mathbf{P}(\mathscr{E})$, where $\mathscr{E}$ is a quasi-coherent $\mathscr{O}_Y$-module *of finite type*, and $\mathscr{L} = r^*(\mathscr{O}_P(1))$, by (4.4.6). We will see that there exists a very ample (relative to the composed morphism $P \xrightarrow{h} Y \xrightarrow{g} Z$) $\mathscr{O}_P$-module $\mathscr{M}$ such that $\mathscr{O}_P(1)$ is isomorphic to $\mathscr{M} \otimes_Y \mathscr{K}^{\otimes(-m)}$ for some integer m. For $n \geqslant m+1$, $\mathscr{O}_P(1) \otimes_Y \mathscr{K}^{\otimes n}$ will then be very ample for Z, by hypothesis and by (iv) applied to the morphisms $h : P \to Y$ and 1_Y; since r is an immersion and $\mathscr{L} \otimes f^*(\mathscr{K}^{\otimes n}) = r^*(\mathscr{O}_P(1) \otimes_Y \mathscr{K}^{\otimes n})$, the conclusion will then follow from (i *bis*). To prove our claim concerning $\mathscr{O}_P(1)$, we will use the following lemma:

Lemma (4.4.10.1). — *Let Z be a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, and let $g : Y \to Z$ be a quasi-compact morphism, $\mathscr{K}$ a very ample (with respect to g) invertible $\mathscr{O}_Y$-module, and $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-module of finite type. Then there exists an integer m_0 such that, for all $m \geqslant m_0$, $\mathscr{E}$ is isomorphic to a quotient of an $\mathscr{O}_Y$-module of the form $g^*(\mathscr{F}) \otimes \mathscr{K}^{\otimes(-m)}$, where $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_Z$-module of finite type (depending on m).*

This lemma will be proven in (4.5.10.1); the reader can verify that (4.4.10) is not used anywhere in (4.5).

Assuming this lemma, there exists a closed immersion j_1 from P to II | 83

$$P_1 = \mathbf{P}(g^*(\mathscr{F}) \otimes \mathscr{K}^{\otimes(-m)})$$

such that $\mathscr{O}_P(1)$ is isomorphic to $j_1^*(\mathscr{O}_{P_1}(1))$ (4.1.2). Now, there exists an isomorphism from P_1 to $P_2 = \mathbf{P}(g^*(\mathscr{F}))$, sending $\mathscr{O}_{P_2}(1) \otimes_Y \mathscr{K}^{\otimes(-m)}$ to $\mathscr{O}_{P_1}(1)$ (4.1.4); we thus have a closed immersion $j_2 : P \to P_2$ such that $\mathscr{O}_P(1)$ is isomorphic to $j_2^*(\mathscr{O}_{P_2}(1)) \otimes_Y \mathscr{K}^{\otimes(-m)}$. Finally, P_2 can be identified with $P_3 \times_Z Y$, where $P_3 = \mathbf{P}(\mathscr{F})$, and $\mathscr{O}_{P_2}(1)$ with $\mathscr{O}_{P_3}(1) \otimes_Z \mathscr{O}_Y$ (4.1.3). By definition, $\mathscr{O}_{P_3}(1)$ is very ample for Z; since so too is $\mathscr{K}$, we conclude, from (iv), that $\mathscr{O}_{P_2}(1) \otimes_Y \mathscr{K}$ is very ample for Z; so too is $\mathscr{M} = j_2^*(\mathscr{O}_{P_2}(1) \otimes_Y \mathscr{K})$ by (i *bis*), and $\mathscr{O}_P(1)$ is isomorphic to $\mathscr{M} \otimes_Y \mathscr{K}^{\otimes(-m-1)}$, which finishes the proof. □

Proposition (4.4.11). — *Let $f : X \to Y$ and $f' : X' \to Y$ be morphisms, X'' the sum prescheme $X \sqcup X'$, and f'' the morphism $X'' \to Y$ that agrees with f (resp. f') on X (resp. X'). Let $\mathscr{L}$ (resp. $\mathscr{L}'$) be an invertible $\mathscr{O}_X$-module (resp. invertible $\mathscr{O}_{X'}$-module), and let $\mathscr{L}''$ be the invertible $\mathscr{O}_{X''}$-module that agrees with $\mathscr{L}$ (resp. $\mathscr{L}'$) on X (resp. X'). For $\mathscr{L}''$ to be very ample relative to f'', it is necessary and sufficient for $\mathscr{L}$ to be very ample relative to f and for $\mathscr{L}'$ to be very ample relative to f'.*

Proof. We can immediately restrict to the case where Y is affine. If $\mathscr{L}''$ is very ample then so too are $\mathscr{L}$ and $\mathscr{L}'$, by (4.4.10, (i *bis*)). Conversely, if $\mathscr{L}$ and $\mathscr{L}'$ are very ample, then it follows immediately from the definition (4.4.2) and from (4.3.6) that $\mathscr{L}''$ is very ample. □

4.5. Ample sheaves

(4.5.1). Given a prescheme X and an invertible $\mathscr{O}_X$-module $\mathscr{L}$, we define, for every $\mathscr{O}_X$-module $\mathscr{F}$ (when there will be no confusion possible over $\mathscr{L}$) $\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}$ ($n \in \mathbf{Z}$); we also define $S = \bigoplus_{n \geqslant 0} \Gamma(X, \mathscr{L}^{\otimes n})$ (a graded subring of the ring $\Gamma_\bullet(\mathscr{L})$ defined in (**0**, 5.4.6)). If we consider X as a $\mathbf{Z}$-prescheme, and we denote by p the structure morphism $X \to \operatorname{Spec}(\mathbf{Z})$, then there is a bijective correspondence between homomorphisms $p^*(\widetilde{S}) \to \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$ of graded $\mathscr{O}_X$-algebras and endomorphisms of the graded ring S (**I**, 2.2.5); the homomorphism $\varepsilon : p^*(\widetilde{S}) \to \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$ that then corresponds to the *identity* automorphism of S is said to be *canonical*. There is a corresponding (3.7.1) morphism $G(\varepsilon) \to \operatorname{Proj}(S)$ that is also said to be *canonical*.

Theorem (4.5.2). — *Let X be a quasi-compact scheme or a prescheme whose underlying space is Noetherian, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and S the graded ring $\bigoplus_{n \geqslant 0} \Gamma(X, \mathscr{L}^{\otimes n})$. Then the following conditions are equivalent:*

- (a) *When f runs over the set of homogeneous elements of S_+, the X_f form a base for the topology of X.*
- (a′) *When f runs over the set of homogeneous elements of S_+, the X_f that are affine form a cover of X.*
- (b) *The canonical morphism $G(\varepsilon) \to \operatorname{Proj}(S)$ (4.5.1) is everywhere defined and is a dominant open immersion.* II | 84
- (b′) *The canonical morphism $G(\varepsilon) \to \operatorname{Proj}(S)$ is everywhere defined and is a homeomorphism from the underlying space of X to a subspace of $\operatorname{Proj}(S)$.*
- (c) *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, if we denote by $\mathscr{F}_n$ the sub-$\mathscr{O}_X$-module of $\mathscr{F}(n)$ generated by the sections of $\mathscr{F}(n)$ over X, then $\mathscr{F}$ is the sum of the sub-$\mathscr{O}_X$-modules $\mathscr{F}_n(-n)$ over the integers $n > 0$.*
- (c′) *Property* (c) holds for every quasi-coherent sheaf of ideals of $\mathscr{O}_X$.

Furthermore, if (f_α) is a family of homogeneous elements of S_+ such that the X_{f_α} are affine, then the restriction to $\bigcup_\alpha X_{f_\alpha}$ of the canonical morphism $X \to \operatorname{Proj}(S)$ is an isomorphism from $\bigcup_\alpha X_{f_\alpha}$ to $\bigcup_\alpha (\operatorname{Proj}(S))_{f_\alpha}$.

PROOF. It is clear that (b) implies (b′), and (b′) implies (a) by (3.7.3.1) (taking into account the fact that $\varepsilon^\flat$ is the identity). Condition (a) implies (a′), since every $x \in X$ has an affine neighbourhood U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_X|U$; if $f \in \Gamma(X, \mathscr{L}^{\otimes n})$ is such that $x \in X_f \subset U$, then X_f is also the set of $x' \in U$ such that $(f|U)(x') \neq 0$, and it is thus an affine open subset (**I**, 1.3.6). To prove that (a′) implies (b), it suffices to prove the last claim of the theorem, and to further prove that, if $X = \bigcup_\alpha X_{f_\alpha}$, then condition (iv) of (3.8.2) is satisfied. This latter point follows immediately from (**I**, 9.3.1, (i)). As for the last claim of (4.5.2), since X_{f_α} is the inverse image of $(\operatorname{Proj}(S))_{f_\alpha}$ under $G(\varepsilon) \to \operatorname{Proj}(S)$, it suffices to apply (**I**, 9.3.2). Thus (a), (a′), (b), and (b′) are all equivalent.

To show that (a′) implies (c), note that, if X_f is affine (with $f \in S_k$), then $\mathscr{F}|X_f$ is generated by its sections over X_f (**I**, 1.3.9); on the other hand (**I**, 9.3.1, (ii)), such a section s is of the form $(t|X_f) \otimes (f|X_f)^{-m}$, where $t \in \Gamma(X, \mathscr{F}(km))$; by definition, t is also a section of $\mathscr{F}_{km}$, so s is indeed a section of $\mathscr{F}_{km}(-km)$ over X_f, which proves (c). It is clear that (c) implies (c′), so it remains only to show that (c′) implies (a). But let U be an open neighbourhood of $x \in X$, and let $\mathscr{J}$ be a quasi-coherent sheaf of ideals of $\mathscr{O}_X$ defining a closed subprescheme of X that has $X - U$ as its underlying space (**I**, 5.2.1). Hypothesis (c′) implies that there exists an integer $n > 0$ and a section f of $\mathscr{J}(n)$ over X such that $f(x) \neq 0$. But we clearly have $f \in S_n$, and $x \in X_f \subset U$, which proves (a). □

When X is a prescheme whose underlying space is Noetherian, the equivalent conditions of (4.5.2) imply that X is a *scheme*, since it is isomorphic to a subprescheme of the scheme $S = \operatorname{Proj}(A)$, by (4.5.2, (b)).

Definition (4.5.3). — We say that an invertible $\mathscr{O}_X$-module $\mathscr{L}$ is *ample* if X is a quasi-compact scheme and if the equivalent conditions of (4.5.2) are satisfied.

It evidently follows from criterion (a) of (4.5.2) that, if $\mathscr{L}$ is an ample $\mathscr{O}_X$-module, then, for every open subset U of X, $\mathscr{L}|U$ is an ample $(\mathscr{O}_X|U)$-module.

It follows from the proof of (4.5.2) that the *affine* X_f form a base for the topology of X. Furthermore:

Corollary (4.5.4). — *Let $\mathscr{L}$ be an ample $\mathscr{O}_X$-module. For every finite subspace Z of X and every neighbourhood U of Z, there exists an integer n and some $f \in \Gamma(X, \mathscr{L}^{\otimes n})$ such that X_f is an affine neighbourhood of Z contained in U.*

PROOF. By (4.5.2, (b)), it suffices to prove that, for every finite subset Z' of $\operatorname{Proj}(S)$ and every open neighbourhood U of Z', there exists a homogeneous element $f \in S_+$ such that $Z \subset (\operatorname{Proj}(S))_f \subset U$ (2.4.1). But, by definition, the closed set Y, complement of U in $\operatorname{Proj}(S)$, is of the form $V_+(\mathfrak{I})$, where $\mathfrak{I}$ is a graded ideal of S that does not contain S_+ (2.3.2); also, the points of Z' are, by definition, graded ideals $\mathfrak{p}_i$ of S_+ that do not contain $\mathscr{J}$ (2.3.1). There thus exists an element $f \in \mathfrak{I}$ that does not belong to any of the $\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, no. 1, prop. 2), and, since the $\mathfrak{p}_i$ are graded, the argument made *loc. cit.* shows that we can even take f to be homogeneous; this element then satisfies the claim. □ **II** | 85

Proposition (4.5.5). — *Suppose that X is a quasi-compact scheme or a prescheme whose underlying space is Noetherian. Then conditions (a) to (c′) of (4.5.2) are equivalent to the following:*

(d) *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type, there exists an integer n_0 such that, for all $n \geqslant n_0$, $\mathscr{F}(n)$ is generated by its sections over X.*

(d′) *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type, there exist integers $n > 0$ and $k > 0$ such that $\mathscr{F}$ is isomorphic to a quotient of the $\mathscr{O}_X$-module $\mathscr{L}^{\otimes(-n)} \otimes \mathscr{O}_X^k$.*

(d″) *Property (d′) holds for every quasi-coherent sheaf of ideals of $\mathscr{O}_X$ of finite type.*

PROOF. Since X is quasi-compact, if a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type is such that $\mathscr{F}(n)$ (which is of finite type) is generated by its sections over X, then $\mathscr{F}(n)$ is generated by a *finite* number of these sections (**0**, 5.2.3), and so (d) implies (d′), and it is clear that (d′) implies (d″).

Since every quasi-coherent $\mathscr{O}_X$-module $\mathscr{G}$ is the inductive limits of its sub-$\mathscr{O}_X$-modules of finite type (**I**, 9.4.9), to satisfy condition (c') of (4.5.2), it suffices to do so for a quasi-coherent sheaf of ideals of $\mathscr{O}_X$ that is of finite type, and (d'') thus implies (c'). It remains only to show that, if $\mathscr{L}$ is ample, then property (d) is satisfied. Consider a finite cover of X by X_{f_i} ($f_i \in S_{n_i}$), that we can assume to be affine; by replacing the f_i with suitable powers (which does not alter the X_{f_i}), we can assume that all the n_i are equal to one single integer m. The sheaf $\mathscr{F}|X_{f_i}$, being of finite type, by hypothesis, is generated by a finite number of its sections h_{ij} over X_{f_i} (**I**, 1.3.13); so there exists an integer k_0 such that the section $h_{ij} \otimes f_i^{\otimes k_0}$ extends to a section of $\mathscr{F}(k_0 m)$ over X for every pair (i,j) (**I**, 9.3.1). *A fortiori*, the $h_{ij} \otimes f_i^{\otimes k_0}$ extend to sections of $\mathscr{F}(km)$ over X for every $k \geqslant k_0$, and, for these values of k, $\mathscr{F}(km)$ is thus generated by its sections over X. For every p such that $0 < p < m$, $\mathscr{F}(p)$ is also of finite type, and so there exists an integer k_p such that $\mathscr{F}(p)(km) = \mathscr{F}(p+km)$ is generated by its sections over X for all $k \geqslant k_p$. Taking n_0 to be the largest of the $k_p m$, we thus conclude that $\mathscr{F}(n)$ is generated by its sections over X for all $n \geqslant n_0$, since such an n is of the form $n = km + p$, with $k \geqslant k_p$ and $0 \leqslant p < m$. □

Proposition (4.5.6). — *Let X be a quasi-compact scheme, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module.*

(i) *Let $n > 0$ be an integer. For $\mathscr{L}$ to be ample, it is necessary and sufficient for $\mathscr{L}^{\otimes n}$ to be ample.*
(ii) *Let $\mathscr{L}'$ be an invertible $\mathscr{O}_X$-module such that, for all $x \in X$, there exists an integer $n > 0$ and a section s' of $\mathscr{L}'^{\otimes n}$ over X such that $s'(x) \neq 0$. Then, if $\mathscr{L}$ is ample, so too is $\mathscr{L} \otimes \mathscr{L}'$.* II | 86

Proof. Property (i) is an evident consequence of criterion (a) of (4.5.2), since $X_{f^{\otimes n}} = X_f$. On the other hand, if $\mathscr{L}$ is ample, then, for every $x \in X$ and every neighbourhood U of x, there exists some $m > 0$ and $f \in \Gamma(X, \mathscr{L}^{\otimes m})$ such that $x \in X_f \subset U$ (4.5.2, (a)); if $f' \in \Gamma(X, \mathscr{L}'^{\otimes n})$ is such that $f'(x) \neq 0$, then $s(x) \neq 0$ for $s = f^{\otimes n} \otimes f'^{\otimes m} \in \Gamma(X, (\mathscr{L} \otimes \mathscr{L}')^{\otimes mn})$, and so $x \in X_s \subset X_f \subset U$, which proves that $\mathscr{L} \otimes \mathscr{L}'$ is ample (4.5.2, (a)). □

Corollary (4.5.7). — *The tensor product of two ample $\mathscr{O}_X$-modules is ample.*

Corollary (4.5.8). — *Let $\mathscr{L}$ be an ample $\mathscr{O}_X$-module, and $\mathscr{L}'$ an invertible $\mathscr{O}_X$-module; then there exists an integer $n_0 > 0$ such that $\mathscr{L}^{\otimes n} \otimes \mathscr{L}'$ is ample and generated by its sections over X for $n \geqslant n_0$.*

Proof. It follows from (4.5.5) that there exists an integer m_0 such that $\mathscr{L}^{\otimes m} \otimes \mathscr{L}'$ is generated by its sections over X for all $m \geqslant m_0$; by (4.5.6), we can then take $n_0 = m_0 + 1$. □

Remark (4.5.9). — Let $P = \mathrm{H}^1(X, \mathscr{O}_X^\times)$ be the group of classes of invertible $\mathscr{O}_X$-modules (**0**, 5.4.7), and let P^+ be the subset of P consisting of classes of ample sheaves. Suppose that P^+ is *non-empty*. Then it follows from (4.5.7) and (4.5.8) that

$$P^+ + P^+ \subset P^+ \quad \text{and} \quad P^+ - P^+ = P$$

or, in other words, $P^+ \cup \{0\}$ is the set of *positive* elements in P for a *preorder* structure on P that is compatible with its group structure, and is even *archimedean*, by (4.5.8). This is why we sometimes say "positive sheaf" instead of ample sheaf, and "negative sheaf" for the inverse of an ample sheaf (but we will not use this terminology).

Proposition (4.5.10). — *Let Y be an affine scheme, $q : X \to Y$ a quasi-compact separated morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module.*

(i) *If $\mathscr{L}$ is very ample for q, then $\mathscr{L}$ is ample.*
(ii) *Suppose further that the morphism q is* of finite type. *Then, for $\mathscr{L}$ to be ample, it is necessary and sufficient for it to posses one of the following properties:*
  (e) *There exists $n_0 > 0$ such that, for every integer $n \geqslant n_0$, $\mathscr{L}^{\otimes n}$ is very ample for q.*
  (e') *There exists $n > 0$ such that $\mathscr{L}^{\otimes n}$ is very ample for q.*

Proof. The first claim follows from the definition (4.4.2) of a very ample $\mathscr{O}_X$-module: if A is the ring of Y, then there exists an A-module E and a surjective homomorphism

$$\psi : q^*((\mathbf{S}(E))^\sim) \longrightarrow \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$$

such that $i = r_{\mathscr{L},\psi}$ is an everywhere-defined immersion $X \to P = \mathbf{P}(\widetilde{E})$ and such that $\mathscr{L} = i^*(\mathscr{O}_P(1))$; since the $D_+(f)$ for f homogeneous in $(\mathbf{S}(E))_+$ form a base for the topology of P, and since $i^{-1}(D_+(f)) = X_{\psi^\flat(f)}$, by (3.7.3.1), we see that condition (a) of (4.5.2) is satisfied, and so $\mathscr{L}$ is ample.

Now to prove that, if q is of finite type and $\mathscr{L}$ is ample, then condition (e) is satisfied. Firstly, it follows from criterion (b) of (4.5.2) and from (4.4.1, (i)) that there exists an integer k_0 such that $\mathscr{L}^{\otimes k_0}$ **II** | 87
is very ample relative to q. Also, by (4.5.5), there exists an integer m_0 such that, for all $m \geqslant m_0$, $\mathscr{L}^{\otimes m}$ is generated by its sections over X. Let $n_0 = k_0 + m_0$; if $n \geqslant n_0$, then we can write $n = k_0 + m$ with $m \geqslant m_0$, whence $\mathscr{L}^{\otimes n} = \mathscr{L}^{\otimes k_0} \otimes \mathscr{L}^{\otimes m}$. Since $\mathscr{L}^{\otimes m}$ is generated by its sections over X, it follows from (4.4.8) and (3.4.7) that $\mathscr{L}^{\otimes n}$ is very ample relative to q. Finally, it is clear that (e) implies (e'), and (e') implies that $\mathscr{L}$ is ample by (i) and by (4.5.6, (i))

(4.5.10.1). *[Proof of Lemma (4.4.10.1)].* Let $\mathscr{E}(n) = \mathscr{E} \otimes \mathscr{K}^{\otimes n}$; since g is separated (4.4.2), the proof of (3.4.8) applies, and shows that the canonical homomorphism $g^*(g_*(\mathscr{E}(n))) \to \mathscr{E}(n)$ is surjective for n large enough. Furthermore, since Z is quasi-compact, the proof of (3.4.6) means that if suffices to prove the claim in the case where Z is affine. But $\mathscr{K}$ is then ample, by (4.5.10, (i)), and the conclusion follows from (4.5.5, (d')).

□

Corollary (4.5.11). — *Let Y be an affine scheme, $q : X \to Y$ a separated morphism of finite type, $\mathscr{L}$ an ample $\mathscr{O}_X$-module, and $\mathscr{L}'$ an invertible $\mathscr{O}_X$-module. Then there exists an integer n_0 such that, for all $n \geqslant n_0$, $\mathscr{L}^{\otimes n} \otimes \mathscr{L}'$ is very ample relative to q.*

Proof. There exists m_0 such that, for $m \geqslant m_0$, $\mathscr{L}^{\otimes m} \otimes \mathscr{L}'$ is generated by its sections over X (4.5.8); there also exists k_0 such that $\mathscr{L}^{\otimes k}$ is very ample relative to q for $k \geqslant k_0$. Then $\mathscr{L}^{\otimes (k+m_0)} \otimes \mathscr{L}'$ is very ample if $k \geqslant k_0$ ((4.4.8) and (3.4.7)). □

Remark (4.5.12). — We do not know if the hypothesis that an $\mathscr{O}_X$-module $\mathscr{L}$ is such that $\mathscr{L}^{\otimes n}$ is very ample (relative to q) implies the same conclusion for $\mathscr{L}^{\otimes (n+1)}$.

Proposition (4.5.13). — *Let X be a quasi-compact prescheme, Z a closed prescheme of X defined by a* nilpotent *quasi-coherent sheaf $\mathscr{J}$ of ideals of $\mathscr{O}_X$, and j the canonical injection $Z \to X$. For an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample, it is necessary and sufficient for $\mathscr{L}' = j^*(\mathscr{L})$ to be an ample $\mathscr{O}_Z$-module.*

Proof. The condition is *necessary*. Indeed, for every section f of $\mathscr{L}^{\otimes n}$ over X, let f' be its canonical image $f \otimes 1$, which is a section of $\mathscr{L}'^{\otimes n} = \mathscr{L}^{\otimes n} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J})$ over the space Z (which is identical to X); it is clear that $X_f = Z_{f'}$, and so the criterion (a) of (4.5.2) shows that $\mathscr{L}'$ is ample.

To see that the condition is *sufficient*, note first of all that we can restrict to the case where $\mathscr{J}^2 = 0$, by considering the (finite) sequence of preschemes $X_k = (X, \mathscr{O}_X/\mathscr{J}^{k+1})$ with each prescheme being a closed subprescheme of the next, defined by a square-zero sheaf of ideals. But X is a scheme, since X_{red} is a scheme by hypothesis ((4.5.3) and (**I**, 5.5.1)). Criterion (a) of (4.5.2) shows that it suffices to prove

Lemma (4.5.13.1). — *Under the hypotheses of (4.5.13), suppose further that $\mathscr{J}$ is square-zero; with $\mathscr{L}$ being an invertible $\mathscr{O}_X$-module, let g be a section of $\mathscr{L}'^{\otimes n}$ over Z such that Z_g is affine. Then there exists an integer $m > 0$ such that $g^{\otimes m}$ is the canonical image of a section f of $\mathscr{L}^{\otimes nm}$ over X.*

Proof. We have the exact sequence of $\mathscr{O}_X$-modules

$$0 \longrightarrow \mathscr{J}(n) \longrightarrow \mathscr{O}_X(n) = \mathscr{L}^{\otimes n} \longrightarrow \mathscr{O}_Z(n) = \mathscr{L}'^{\otimes n} \longrightarrow 0$$

since $\mathscr{F}(n)$ is an exact functor in $\mathscr{F}$; from this, we have the exact sequence of cohomology **II** | 88

$$0 \longrightarrow \Gamma(X, \mathscr{J}(n)) \longrightarrow \Gamma(X, \mathscr{L}^{\otimes n}) \longrightarrow \Gamma(X, \mathscr{L}'^{\otimes n}) \xrightarrow{\partial} \mathrm{H}^1(X, \mathscr{J}(n))$$

that sends, in particular, g to an element $\partial g \in \mathrm{H}^1(X, \mathscr{J}(n))$. □

Note that, since $\mathscr{J}^2 = 0$, $\mathscr{J}$ can be considered as a quasi-coherent $\mathscr{O}_Z$-module, and, for all k, $\mathscr{L}'^{\otimes k} \otimes_{\mathscr{O}_Z} \mathscr{J}(n) = \mathscr{J}(n+k)$; for every section $s \in \Gamma(X, \mathscr{L}'^{\otimes k})$, tensor multiplication with s is thus a homomorphism $\mathscr{J}(n) \to \mathscr{J}(n+k)$ of $\mathscr{O}_Z$-modules, which then gives a homomorphism $\mathrm{H}^i(X, \mathscr{J}(n)) \xrightarrow{s} \mathrm{H}^i(X, \mathscr{J}(n+k))$ of cohomology groups.

With this, we will see that

$$g^{\otimes m} \otimes \partial g = 0 \tag{4.5.13.2}$$

for $m > 0$ large enough. In fact, Z_g is an *affine open* subset of Z, and so $\mathrm{H}^1(Z_g, \mathscr{J}(n)) = 0$ when $\mathscr{J}(n)$ is considered as an $\mathscr{O}_Z$*-module* (**I**, 5.1.9.2). In particular, if we set $g' = g|Z_g$, and if we consider its image under the map $\partial : \Gamma(Z_g, \mathscr{L}'^{\otimes n}) \to \mathrm{H}^1(Z_g, \mathscr{J}(n))$, then $\partial g' = 0$. To better explain this equation, we note that, in dimension 1, the cohomology of a sheaf of abelian groups is the same as its Čech cohomology (G, II, 5.9); to calculate ∂g, we must thus consider a fine-enough open cover (U_α) of X, that we can suppose to be *finite* and consisting of affine opens, and take, for each α, a section $g_\alpha \in \Gamma(U_\alpha, \mathscr{L}^{\otimes n})$ whose canonical image in $\Gamma(U_\alpha, \mathscr{L}'^{\otimes n})$ is $g|U_\alpha$, and to consider the cocycle class $(g_{\alpha|\beta} - g_{\beta|\alpha})$, with $g_{\alpha|\beta}$ being the restriction of g_α to $U_\alpha \cap U_\beta$ (with this cocycle taking values in $\mathscr{J}(n)$). We can further suppose that $\partial g'$ is calculated in the same way, by means of a cover given by the $U_\alpha \cap Z_g$, and restrictions $g_\alpha|(U_\alpha \cap Z_g)$ (by replacing, if necessary, (U_α) by a finer cover); the equation $\partial g' = 0$ then implies that there exists, for each α, a section $h_\alpha \in \Gamma(U_\alpha \cap Z_g, \mathscr{J}(n))$ such that $(g_{\alpha|\beta} - g_{\beta|\alpha})|(U_\alpha \cap U_\beta \cap Z_g) = h_{\alpha|\beta} - h_{\beta|\alpha}$, where $h_{\alpha|\beta}$ denotes the restriction of h_α to $U_\alpha \cap U_\beta \cap Z_g$ (G, II, 5.11). Then there exists an integer $m > 0$ such that $g^{\otimes m} \otimes h_\alpha$ is the restriction to $U_\alpha \cap Z_g$ of a section $t_\alpha \in \Gamma(U_\alpha, \mathscr{J}(n+nm))$ for all α (**I**, 9.3.1); thus $g^{\otimes m} \otimes (g_{\alpha|\beta} - g_{\beta|\alpha}) = t_{\alpha|\beta} - t_{\beta|\alpha}$ for every pair of indices, which proves (4.5.13.2).

Now note that, if $s \in \Gamma(X, \mathscr{O}_Z(p))$ and $t \in \Gamma(X, \mathscr{O}_Z(q))$, then, in the group $\mathrm{H}^1(X, \mathscr{J}(p+q))$,

$$\partial(s \otimes t) = (\partial s) \otimes t + s \otimes (\partial t). \tag{4.5.13.3}$$

Indeed, we can again calculate the two members by considering an open cover (U_α) of X, and, for each α, a section $s_\alpha \in \Gamma(U_\alpha, \mathscr{O}_X(p))$ (resp. $t_\alpha \in \Gamma(U_\alpha, \mathscr{O}_X(q))$) whose canonical image in $\Gamma(U_\alpha, \mathscr{O}_Z(p)$ (resp. $\Gamma(U_\alpha, \mathscr{O}_Z(q))$) is $s|U_\alpha$ (resp. $t|U_\alpha$); equation (4.5.13.3) then follows from the equations

$$(s_{\alpha|\beta} \otimes t_{\alpha|\beta}) - (s_{\beta|\alpha} \otimes t_{\beta|\alpha}) = (s_{\alpha|\beta} - s_{\beta|\alpha}) \otimes t_{\alpha|\beta} + s_{\beta|\alpha} \otimes (t_{\alpha|\beta} - t_{\beta|\alpha})$$

with the same notation as above. By induction on k, we thus have

$$\partial(g^{\otimes k}) = (kg^{\otimes(k-1)}) \otimes (\partial g) \tag{4.5.13.4}$$

and we thus conclude from (4.5.13.2) and (4.5.13.4) that $\partial(g^{\otimes(m+1)} = 0$; thus $g^{\otimes(m+1)}$ is the canonical image of a section f of $\mathscr{L}^{\otimes n(m+1)}$ over X, which proves (4.5.13). ∎ **II** | 89

Corollary (4.5.14). — *Let X be a Noetherian scheme, and j the canonical injection $X_{\mathrm{red}} \to X$. For an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample, it is necessary and sufficient for $j^*(\mathscr{L})$ to be an ample $\mathscr{O}_{X_{\mathrm{red}}}$-module.*

PROOF. This follows from (**I**, 6.1.6) ∎

4.6. Relatively ample sheaves

Definition (4.6.1). — Let $f : X \to Y$ be a quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. We say that $\mathscr{L}$ is *ample relative to* f, or *relative to* Y, or f*-ample*, or Y*-ample* (or even simply *ample* if no confusion may arise with the notion defined in (4.5.3)) if there exists an affine open cover (U_α) of Y such that, if we set $X_\alpha = f^{-1}(U_\alpha)$, then $\mathscr{L}|X_\alpha$ is an ample $\mathscr{O}_{X_\alpha}$-module for all α.

The existence of an f-ample $\mathscr{O}_X$-module implies that f is necessarily *separated* ((4.5.3) and (**I**, 5.5.5)).

Proposition (4.6.2). — *Let $f : X \to Y$ be a quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. If $\mathscr{L}$ is very ample relative to f, then it is ample relative to f.*

PROOF. This follows from the local (on Y) character of the notion of a very ample sheaf (4.4.5), from the definition (4.6.1), and from criterion (4.5.10, (i)). ∎

Proposition (4.6.3). — *Let $f : X \to Y$ be a quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and let $\mathscr{S}$ be the graded $\mathscr{O}_Y$-algebra $\bigoplus_{n\geqslant 0} f_*(\mathscr{L}^{\otimes n})$. Then the following conditions are equivalent:*

(a) *$\mathscr{L}$ is f-ample.*

(b) *$\mathscr{S}$ is quasi-coherent, and the canonical homomorphism $\sigma : f^*(\mathscr{S}) \to \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n}$* (**0**, 4.4.3) *is such that the Y-morphism $r_{\mathscr{L},\sigma} : G(\sigma) \to \operatorname{Proj}(\mathscr{S}) = P$ is everywhere defined and is an dominant open immersion.*

(b′) *The morphism f is separated, and the Y-morphism $r_{\mathscr{L},\sigma}$ is everywhere defined and is a homeomorphism from the underlying space of X to a subspace of* $\operatorname{Proj}(\mathscr{S})$.

Furthermore, if any of the above are satisfied, then, for all $n \in \mathbf{Z}$, the canonical homomorphism

$$r^*_{\mathscr{L},\sigma}(\mathscr{O}_P(n)) \longrightarrow \mathscr{L}^{\otimes n} \tag{4.6.3.1}$$

defined in (3.7.9.1) *is an isomorphism.*

Finally, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, if we set $\mathscr{M} = \bigoplus_{n\geqslant 0} f_*(\mathscr{F}\otimes\mathscr{L}^{\otimes n})$, then the canonical homomorphism

$$r^*_{\mathscr{L},\sigma}(\widetilde{\mathscr{M}}) \longrightarrow \mathscr{F} \tag{4.6.3.2}$$

defined in (3.7.9.2) *is an isomorphism.*

PROOF. We note that (a) implies that f is separated, and thus that $\mathscr{S}$ is quasi-coherent (**I**, 9.2.2, (a)). Since $r_{\mathscr{L},\sigma}$ being an everywhere defined immersion is of a local (on Y) character, to prove that (a) implies (b) we can suppose that Y is affine and $\mathscr{L}$ ample; the claim then follows from (4.5.2, (b)). It is clear that (b) implies (b′); finally, to prove that (b′) implies (a), it suffices to consider an affine open cover (U_α) of Y and to apply the criterion (4.5.2, (b′)) to each sheaf $\mathscr{L}|f^{-1}(U_\alpha)$. **II** | 90

For the final two claims, we use the fact that $\sigma^\flat$ is here the identity, and the clarification of the homomorphisms (3.7.9.1) and (3.7.9.2); from this, it immediately follows that (4.6.3.1) is an isomorphism. As for (4.6.3.2), we can restrict to the case where Y is affine, and thus $\mathscr{L}$ ample; it is clear that the homomorphism (4.6.3.2) is injective, and criterion (4.5.2, (c)) shows that it is surjective, whence the conclusion. □

Corollary (4.6.4). — *Let (U_α) be an open cover of Y. For $\mathscr{L}$ to be ample relative to Y, it is necessary and sufficient for $\mathscr{L}|f^{-1}(U_\alpha)$ to be ample relative to U_α for all α.*

PROOF. Condition (b) is in fact local on Y. □

Corollary (4.6.5). — *Let $\mathscr{K}$ be an invertible $\mathscr{O}_Y$-module. For $\mathscr{L}$ to be Y-ample, it is necessary and sufficient for $\mathscr{L}\otimes f^*(\mathscr{K})$ to be Y-ample.*

PROOF. This is an evident consequence of (4.6.4), by taking the U_α to be such that $\mathscr{K}|U_\alpha$ is isomorphic to $\mathscr{O}_Y|U_\alpha$ for all α. □

Corollary (4.6.6). — *Suppose that Y is affine; for $\mathscr{L}$ to be Y-ample, it is necessary and sufficient for $\mathscr{L}$ to be ample.*

PROOF. This is an immediate consequence of the definition (4.6.1) and the criteria (4.6.3, (b)) and (4.5.2, (b)), since here $\operatorname{Proj}(\mathscr{S}) = \operatorname{Proj}(\Gamma(Y,\mathscr{S}))$ by definition. □

Corollary (4.6.7). — *Let $f : X \to Y$ be a quasi-compact morphism. Suppose that there exists a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$, and a Y-morphism $g : X \to P = \mathbf{P}(\mathscr{E})$ that is a homeomorphism from the underlying space of X to a subspace of $\mathbf{P}$; then $\mathscr{L} = g^*(\mathscr{O}_P(1))$ is Y-ample.*

PROOF. We can assume Y to be affine; the corollary then follows from the criterion (4.5.2, (a)), from equation (3.7.3.1), and from (4.2.3). □

Proposition (4.6.8). — *Let X be a quasi-compact scheme or a prescheme whose underlying space is Noetherian, and let $f : X \to Y$ be a quasi-compact separated morphism. For an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be f-ample, it is necessary and sufficient for one of the following equivalent conditions to be satisfied:*

(c) *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ of finite type, there exists an integer $n_0 > 0$ such that, for all $n \geqslant n_0$, the canonical homomorphism $\sigma : f^*(f_*(\mathscr{F}\otimes\mathscr{L}^{\otimes n})) \to \mathscr{F}\otimes\mathscr{L}^{\otimes n}$ is surjective.*

(c′) *For every quasi-coherent sheaf $\mathscr{J}$ of ideals of $\mathscr{O}_X$ of finite type, there exists an integer $n > 0$ such that the canonical homomorphism $\sigma : f^*(f_*(\mathscr{J}\otimes\mathscr{L}^{\otimes n})) \to \mathscr{J}\otimes\mathscr{L}^{\otimes n}$ is surjective.*

PROOF. Since X is quasi-compact, so too is $f(X)$, and so there exists a finite cover (U_i) of $f(X)$ consisting of affine open subsets U_i of Y. To prove condition (c) when $\mathscr{L}$ is f-ample, we can replace Y by the U_i, and X by the $f^{-1}(U_i)$, since, if we obtain, for each i, an integer n_i such that (c) holds true (for U_i, $f^{-1}(U_i)$, and $\mathscr{L}|f^{-1}(U_i)$) for all $n \geqslant n_i$, then it suffices to take n_0 to be the largest of the n_i in order to obtain (c) for Y, X, and $\mathscr{L}$. But if Y is affine, condition (c) follows from (4.5.5, (d)), taking (4.6.6) into account. It is trivial that (c) implies (c′). Finally, to prove that (c′) implies that $\mathscr{L}$ is f-ample, we can again restrict to the case where Y is affine: in fact, every quasi-coherent sheaf $\mathscr{J}_i$ of ideals of $\mathscr{O}_X|f^{-1}(U_i)$ of finite type is the restriction of a coherent sheaf of ideals of $\mathscr{O}_X$ of finite type (I, 9.4.7), and hypothesis (c′) implies that $\mathscr{J}_i \otimes (\mathscr{L}^{\otimes n}|f^{-1}(U_i))$ is generated by its sections (taking (I, 9.2.2) and (3.4.7) into account); it thus suffices to apply criterion (4.5.5, (d″)). □ II | 91

Proposition (4.6.9). — *Let $f : X \to Y$ be a quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module.*

(i) *Let $n > 0$ be an integer. For $\mathscr{L}$ to be f-ample, it is necessary and sufficient for $\mathscr{L}^{\otimes n}$ to be f-ample.*

(ii) *Let $\mathscr{L}'$ be an invertible $\mathscr{O}_X$-module, and suppose that there exists an integer $n > 0$ such that the canonical homomorphism $\sigma : f^*(f_*(\mathscr{L}'^{\otimes n})) \to \mathscr{L}'^{\otimes n}$ is surjective. Then, if $\mathscr{L}$ if f-ample, so too is $\mathscr{L} \otimes \mathscr{L}'$.*

PROOF. We can in fact immediately restrict to the case where Y is affine, and the proposition is then an immediate consequence of (4.5.6). □

Corollary (4.6.10). — *The tensor product of two f-ample $\mathscr{O}_X$-modules is f-ample.*

Proposition (4.6.11). — *Let Y be a quasi-compact prescheme, $f : X \to Y$ a morphism of finite type, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. For $\mathscr{L}$ to be f-ample, it is necessary and sufficient for it to posses one of the following equivalent properties:*

(d) *There exists some $n_0 > 0$ such that, for every integer $n \geqslant n_0$, $\mathscr{L}^{\otimes n}$ is very ample relative to f.*

(d′) *There exists some $n > 0$ such that $\mathscr{L}^{\otimes n}$ is very ample relative to f.*

PROOF. If $\mathscr{L}$ is ample relative to f, then there exists a *finite* cover (U_i) of Y by affine open subsets such that the $\mathscr{L}|f^{-1}(U_i)$ are ample. We thus conclude (4.5.10) that there exists an integer n_0 such that $\mathscr{L}^{\otimes n}|f^{-1}(U_i)$ is very ample relative to $f^{-1}(U_i) \to U_i$ for all $n \geqslant n_0$ and every i, and so $\mathscr{L}^{\otimes n}$ is very ample relative to f (4.5.5). Conversely, (d′) already implies that $\mathscr{L}^{\otimes n}$ is f-ample (4.6.2), and thus so too is $\mathscr{L}$ (4.6.9, (i)). □

Corollary (4.6.12). — *Let Y be a quasi-compact prescheme, $f : X \to Y$ a morphism of finite type, and $\mathscr{L}$ and $\mathscr{L}'$ invertible $\mathscr{O}_X$-modules. If $\mathscr{L}$ is f ample, then there exists some n_0 such that $\mathscr{L}^{\otimes n} \otimes \mathscr{L}'$ is very ample relative to f for all $n \geqslant n_0$.*

PROOF. We argue as in (4.6.11), by using a finite affine open cover of Y and (4.5.11). □

Proposition (4.6.13). — (i) *For every prescheme Y, every invertible $\mathscr{O}_Y$-module $\mathscr{L}$ is ample relative to the identity morphism 1_Y.*

(i *bis*) *Let $f : X \to Y$ be a quasi-compact morphism, and $j : X' \to X$ a quasi-compact morphism that is a homeomorphism from the underlying space of X' to a subspace of X. If $\mathscr{L}$ is an $\mathscr{O}_X$-module that is ample relative to f, then $j^*(\mathscr{L})$ is ample relative to $f \circ j$.*

(ii) *Let Z be a quasi-compact prescheme, $f : X \to Y$ and $g : Y \to Z$ quasi-compact morphisms, $\mathscr{L}$ an $\mathscr{O}_X$-module that is ample relative to f, and $\mathscr{K}$ an $\mathscr{O}_Y$-module that is ample relative to g. Then there exists an integer $n_0 > 0$ such that $\mathscr{L} \otimes f^*(\mathscr{K}^{\otimes n})$ is ample relative to $g \circ f$ for all $n \geqslant n_0$.*

(iii) *Let $f : X \to Y$ be a quasi-compact morphism, and $g : Y' \to Y$ a morphism, and let $X' = X_{(Y')}$. If $\mathscr{L}$ is an $\mathscr{O}_X$-module that is ample relative to f, then $\mathscr{L}' = \mathscr{L} \otimes_Y \mathscr{O}_{Y'}$ is an $\mathscr{O}_{X'}$-module that is ample relative to $f_{(Y')}$.*

(iv) *Let $f_i : X_i \to Y_i$ ($i = 1, 2$) be quasi-compact S-morphisms. If $\mathscr{L}_i$ is an $\mathscr{O}_{X_i}$-modules that is ample relative to f_i ($i = 1, 2$), then $\mathscr{L}_1 \otimes_S \mathscr{L}_2$ is ample relative to $f_1 \times_S f_2$.*

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms such that $g \circ f$ is quasi-compact. If an $\mathscr{O}_X$-module $\mathscr{L}$ is ample relative to $g \circ f$, and if g is separated or the underlying space of X is locally Noetherian, then $\mathscr{L}$ is ample relative to f.* II | 92

(vi) *Let $f : X \to Y$ be a quasi-compact morphism, and j the canonical injection $X_{\mathrm{red}} \to X$. If $\mathscr{L}$ is an $\mathscr{O}_X$-module that is ample relative to f, then $j^*(\mathscr{L})$ is ample relative to f_{red}.*

PROOF. Note first of all that (v) and (vi) follow from (i), (i *bis*), and (iv) by the same argument as in (4.4.10), by using (4.6.4) instead of (4.4.5); we leave the details of the argument to the reader. Claim (i) is trivially a consequence of (4.4.10, (i)) and (4.6.2). To prove (i *bis*), (iii), and (iv), we will use the following lemma:

Lemma (4.6.13.1). — (i) *Let $u : Z \to S$ be a morphism, $\mathscr{L}$ an invertible $\mathscr{O}_S$-module, and s a section of $\mathscr{L}$ over S, and let s' be the canonically corresponding section $u^*(\mathscr{L}) = \mathscr{L}'$ over Z. Then $Z_{s'} = u^{-1}(S_s)$.*

(ii) *Let Z and Z' be S-preschemes, p and p' the projections of $T = Z \times_S Z'$, $\mathscr{L}$ (resp. $\mathscr{L}'$) an invertible $\mathscr{O}_Z$-module (resp. invertible $\mathscr{O}_{Z'}$-module), and let t (resp. t') be a section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over Z (resp. Z'), and s (resp. s') the corresponding section of $p^*(\mathscr{L})$ (resp. $p'^*(\mathscr{L}')$) over $Z \times_S Z'$. Then $T_{s\otimes s'} = Z_t \times_S Z'_{t'}$.*

PROOF. It follows from the definitions that we can restrict to the case where all the preschemes in question are affine. Furthermore, in (i), we can suppose that $\mathscr{L} = \mathscr{O}_B$; claim (i) then follows immediately from (**I**, 1.2.2.2). Similarly, in (ii), we can restrict to the case where $\mathscr{L} = \mathscr{O}_Z$ and $\mathscr{L}' = \mathscr{O}_{Z'}$, and then the claim reduces to Lemma (4.3.2.4). □

We now prove (i *bis*). We can assume that Y is affine (4.6.4), and thus that $\mathscr{L}$ is ample (4.6.6); if s runs over the set of the union of the $\Gamma(X, \mathscr{L}^{\otimes n})$ $(n > 0)$, then the X_s form a base for the topology of X (4.5.2, (a)), and so, by hypothesis, the $j^{-1}(X_s)$ form a base for the topology of X'; we thus conclude, by Lemma (4.6.13.1, (i)) and (4.5.2, (a)), that $j^*(\mathscr{L})$ is ample.

Next we prove (iii). We can again suppose that Y and Y' are affine (4.6.4), whence it follows that the projection $h : X' \to X$ is affine (1.5.5). Since $\mathscr{L}$ is ample (4.6.6), if s runs over the set of sections of the $\mathscr{L}^{\otimes n}$ $(n > 0)$ over X such that X_s is affine, then the X_s cover X (4.5.2, (a')), and so the $h^{-1}(X_s)$ are affine (1.2.5) and cover X'; it thus follows again from Lemma (4.6.13.1, (i)) and from (4.5.2, (a')) that $\mathscr{L}'$ is ample, since the morphism $f_{(Y')}$ is quasi-compact (**I**, 6.6.4, (iii)).

To prove (iv), note first of all that $f_1 \times_S f_2$ is quasi-compact (**I**, 6.6.4, (iv)). We can further suppose that S, Y_1, and Y_2 are affine ((4.6.4) and (**I**, 3.2.7)), and thus that $\mathscr{L}_i$ is ample $(i = 1, 2)$ (4.6.6). The open subsets $(X_1)_{s_1} \times_S (X_2)_{s_2}$ form a cover of $X_1 \times_S X_2$, where s_i runs over the sections of $\mathscr{L}_i^{\otimes n_i}$ such that $(X_i)_{s_i}$ is affine (4.5.2, (a')). Then, replacing s_1 and s_2 with suitable powers, which does not change the $(X_i)_{s_i}$, we can assume that $n_1 = n_2$. We thus deduce, from (4.6.13.1, (ii)) and (4.5.2, (a')), that $\mathscr{L}_1 \otimes_S \mathscr{L}_2$ is ample, whence the claim, since $Y_1 \times_S Y_2$ is affine (4.6.6).

It remains only to prove (ii). By the same argument as in (4.4.10), but here using (4.6.4), we can restrict to the case where Z is affine. Since $\mathscr{K}$ is then ample, and Y quasi-compact, there exists a finite number of sections $s_i \in \Gamma(Y, \mathscr{K}^{\otimes k_i})$ such that the Y_{s_i} are *affine* and cover Y (4.5.2, (a')); **II** | 93
replacing the s_i with suitable powers, we can further suppose that all the k_i are equal to one single integer k. Let s'_i be the sections of $f^*(\mathscr{K}^{\otimes k})$ over X that canonically correspond to the s_i, so that the $X_{s'_i} = f^{-1}(Y_{s_i})$ (4.6.13.1, (i)) cover X. Since $\mathscr{L}|X_{s'_i}$ is ample ((4.6.4) and (4.6.6)), there exists, for each i, a finite number of sections $t_{ij} \in \Gamma(X, \mathscr{L}^{\otimes n_{ij}})$ such that the X_{ij} are affine, contained in the $X_{s'_i}$, and cover $X_{s'_i}$ (4.5.2, (a')); we can also suppose that all the n_{ij} are equal to one single integer n. With this in mind, X is separated and quasi-compact, and so there exists an integer $m > 0$, and, for every (i, j), a section

$$u_{ij} \in \Gamma(X, \mathscr{L}^{\otimes n} \otimes_X f^*(\mathscr{K}^{\otimes mk}))$$

such that $t_{ij} \otimes s_i'^{\otimes m}$ is the restriction to $X_{s'_i}$ of u_{ij} (**I**, 9.3.1); furthermore, $X_{u_{ij}} = X_{t_{ij}}$, and so the $X_{u_{ij}}$ are affine and cover X. We can also suppose that m is of the form nr; if we set $n_0 = rk$, then we see (4.5.2, (a')) that $\mathscr{L} \otimes_{\mathscr{O}_X} f^*(\mathscr{K}^{\otimes n_0})$ is ample. Furthermore, there exists $h_0 > 0$ such that $\mathscr{K}^{\otimes h}$ is generated by its sections over Y for all $h \geqslant h_0$ (4.5.5); *a fortiori*, $f^*(\mathscr{K}^{\otimes h})$ is generated by its sections over X for all $h \geqslant h_0$, by definition of the inverse images ((**0**, 3.7.1) and (4.4.1)). We thus conclude that $\mathscr{L} \otimes f^*(\mathscr{K}^{\otimes n_0 + h})$ is ample for all $h \geqslant h_0$ (4.5.6), which finishes the proof. □

Remark (4.6.14). — Under the conditions of (ii), we refrain from believing that $\mathscr{L} \otimes f^*(\mathscr{K})$ is ample for $g \circ f$; in fact, since $\mathscr{L} \otimes f^*(\mathscr{K}^{-1})$ is also ample for f (4.6.5), we would conclude that $\mathscr{L}$ is ample for $g \circ f$; taking, in particular, g to be the identity morphism, *every* invertible $\mathscr{O}_X$-module would be ample for f, which is not the case in general (see (5.1.6), (5.3.4, (i)), and (5.3.1)).

Proposition (4.6.15). — *Let $f : X \to Y$ be a quasi-compact morphism, $\mathscr{J}$ a quasi-coherent locally-nilpotent sheaf of ideals of $\mathscr{O}_X$, Z the closed subprescheme of X defined by $\mathscr{J}$, and $j : Z \to X$ the canonical injection. For an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample for f, it is necessary and sufficient for $j^*(\mathscr{L})$ to be ample for $f \circ j$.*

PROOF. Since the question is local on Y (4.6.4), we can suppose Y to be affine; since X is then quasi-compact, we can suppose $\mathscr{J}$ to be nilpotent. Taking (4.6.6) into account, the proposition is then exactly the same as (4.5.13). □

Corollary (4.6.16). — *Let X be a locally Noetherian prescheme, and $f : X \to Y$ a quasi-compact morphism. For an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample for f, it is necessary and sufficient for its inverse image $\mathscr{L}'$ under the canonical injection $X_{\text{red}} \to X$ to be ample for f_{red}.*

PROOF. We have already seen that the condition is necessary (4.6.13, (vi)); conversely, if it is satisfied, then we can restrict, to prove that $\mathscr{L}$ is ample for f, to the case where Y is affine (4.6.4); then Y_{red} is also affine, and so $\mathscr{L}'$ is ample (4.6.6), and so too is $\mathscr{L}$ by (4.5.13), since X is then Noetherian and X_{red} a closed subprescheme of X defined by a quasi-coherent nilpotent sheaf of ideals (**I**, 6.1.6). □

Proposition (4.6.17). — *With the notation and hypotheses of (4.4.11), for $\mathscr{L}''$ to be ample relative to f'', it is necessary and sufficient for $\mathscr{L}$ to be ample relative to f and $\mathscr{L}'$ ample relative to f'.*

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PROOF. The necessity of the condition follows from (4.6.13, (i *bis*)),. To see that the condition is sufficient, we can restrict to the case where Y is affine, and then the fact that $\mathscr{L}''$ is ample follows from criterion (4.5.2, (a)) applied to $\mathscr{L}$, $\mathscr{L}'$, and $\mathscr{L}''$, noting that a section of $\mathscr{L}$ over X can be extended (by 0) to a section of $\mathscr{L}''$ over X''. □

Proposition (4.6.18). — *Let Y be a quasi-compact prescheme, $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra of finite type, and $X = \operatorname{Proj}(\mathscr{S})$, and let $f : X \to Y$ the structure morphism. Then f is of finite type, and there exists an integer $d > 0$ such that $\mathscr{O}_X(d)$ is invertible and f-ample.*

PROOF. By (3.1.10), there exists an integer $d > 0$ such that $\mathscr{S}^{(d)}$ is generated by $\mathscr{S}_d$. We know that, under the canonical isomorphism between X and $X^{(d)} = \operatorname{Proj}(\mathscr{S}^{(d)})$, $\mathscr{O}_X(d)$ is identified with $\mathscr{O}_{X^{(d)}}(1)$ (3.2.9, (ii)). We thus see that we can restrict to the case where $\mathscr{S}$ is generated by $\mathscr{S}_1$; the proposition then follows from (4.4.3) and (4.6.2) (taking into account the fact that f is a morphism of finite type (3.4.1)). □

§5. QUASI-AFFINE MORPHISMS; QUASI-PROJECTIVE MORPHISMS; PROPER MORPHISMS; PROJECTIVE MORPHISMS

5.1. Quasi-affine morphisms

Definition (5.1.1). — We define a quasi-affine scheme to be a scheme isomorphic to some subscheme induced on some quasi-compact open subset of an affine scheme. We say that a morphism $f : X \to Y$ is quasi-affine, or that X (considered as a Y-prescheme via f) is a quasi-affine Y-scheme, if there exists a cover (U_α) of Y by affine open subsets such that the $f^{-1}(U_\alpha)$ are quasi-affine schemes.

It is clear that a quasi-affine morphism is *separated* ((**I**, 5.5.5) and (**I**, 5.5.8)) and *quasi-compact* (**I**, 6.6.1); every affine morphisms is evidently quasi-affine.

Recall that, for any prescheme X, setting $A = \Gamma(X, \mathscr{O}_X)$, the identity homomorphism $A \to A = \Gamma(X, \mathscr{O}_X)$ defines a morphism $X \to \operatorname{Spec}(A)$, said to be *canonical* (**I**, 2.2.4); this is nothing but the canonical morphism defined in (4.5.1) for the specific case where $\mathscr{L} = \mathscr{O}_X$, if we remember that $\operatorname{Proj}(A[T])$ is canonically identified with $\operatorname{Spec}(A)$ (3.1.7).

Proposition (5.1.2). — *Let X be a quasi-compact scheme or a prescheme whose underlying space is Noetherian, and A the ring $\Gamma(X, \mathscr{O}_X)$. The following conditions are equivalent.*

(a) *X is a quasi-affine scheme.*
(b) *The canonical morphism $u : X \to \operatorname{Spec}(A)$ is an open immersion.*

(b′) *The canonical morphism $u : X \to \operatorname{Spec}(A)$ is a homeomorphism from X to some subspace of the underlying space of* $\operatorname{Spec}(A)$.

(c) *The $\mathscr{O}_X$-module $\mathscr{O}_X$ is very ample relative to u (4.4.2).*

(c′) *The $\mathscr{O}_X$-module $\mathscr{O}_X$ is ample (4.5.1).*

(d) *When f ranges over A, the X_f form a basis for the topology of X.*

(d′) *When f varies over A, the X_f that are affine form a cover of X.* **II** | 95

(e) *Every quasi-coherent $\mathscr{O}_X$-module is generated by its sections over X.*

(e′) *Every quasi-coherent sheaf of ideals of $\mathscr{O}_X$ of finite type is generated by its sections over X.*

Proof. It is clear that (b) implies (a), and (a) implies (c) by (4.4.4, b) applied to the identity morphism (taking into account the remark preceding this proposition); Furthermore, (c) implies (c′) (4.5.10, i), and (c′), (b), and (b′) are all equivalent by (4.5.2, b) and (4.5.2, b′). Finally, (c′) is the same as each of (d), (d′), (e), and (e′) by (4.5.2, a), (4.5.2, a′), (4.5.2, c), and (4.5.5, d″). ☐

We further observe that, with the previous notation, the X_f that are affine form a *basis* for the topology of X, and that the canonical morphism u is *dominant* (4.5.2).

Corollary (5.1.3). — *Let X be a quasi-compact prescheme. If there exists a morphism $v : X \to Y$ from X to some affine scheme Y (which would be a homeomorphism from X to some open subspace of Y), then X is quasi-affine.*

Proof. There exists a family (g_α) of sections of $\mathscr{O}_Y$ over Y such that the $D(g_\alpha)$ form a basis for the topology of $v(X)$; if $v = (\psi, \theta)$ and we set $f_\alpha = \theta(g_\alpha)$, then we have $X_{f_\alpha} = \psi^{-1}(D(g_\alpha))$ (**I**, 2.2.4.1), so the X_{f_α} form a basis for the topology of X, and the criterion (5.1.2, d) is satisfied. ☐

Corollary (5.1.4). — *If X is a quasi-affine scheme, then* every *invertible $\mathscr{O}_X$-module is very ample (relative to the canonical morphism), and* a fortiori *ample.*

Proof. Such a module $\mathscr{L}$ is generated by its sections over X (5.1.2, e), so $\mathscr{L} \otimes \mathscr{O}_X = \mathscr{L}$ is very ample (4.4.8). ☐

Corollary (5.1.5). — *Let X be a quasi-compact prescheme. If there exists an invertible $\mathscr{O}_X$-module $\mathscr{L}$ such that $\mathscr{L}$ and $\mathscr{L}^{-1}$ are ample, then X is a quasi-affine scheme.*

Proof. Indeed, $\mathscr{O}_X = \mathscr{L} \otimes \mathscr{L}^{-1}$ is then ample (4.5.7). ☐

Proposition (5.1.6). — *Let $f : X \to Y$ be a quasi-compact morphism. Then the following conditions are equivalent.*

(a) *The morphism f is quasi-affine.*

(b) *The $\mathscr{O}_Y$-algebra $f_*(\mathscr{O}_X) = \mathscr{A}(X)$ is quasi-coherent, and the canonical morphism $X \to \operatorname{Spec}(\mathscr{A}(X))$ corresponding to the identity morphism $\mathscr{A}(X) \to \mathscr{A}(X)$ (1.2.7) is an open immersion.*

(b′) *The $\mathscr{O}_Y$-algebra $\mathscr{A}(X)$ is quasi-coherent, and the canonical morphism $X \to \operatorname{Spec}(\mathscr{A}(X))$ is a homeomorphism from X to some subspace of* $\operatorname{Spec}(\mathscr{A}(X))$.

(c) *The $\mathscr{O}_X$-module $\mathscr{O}_X$ is very ample for f.*

(c′) *The $\mathscr{O}_X$-module $\mathscr{O}_X$ is ample for f.*

(d) *The morphism f is separated, and, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\sigma : f^*(f_*(\mathscr{F})) \to \mathscr{F}$ (**0**, 4.4.3) is surjective.*

Furthermore, whenever f is quasi-affine, every invertible $\mathscr{O}_X$-module $\mathscr{L}$ is very ample relative to f.

Proof. The equivalence between (a) and (c′) follows from the local (on Y) character of the f-ampleness (4.6.4), Definition (5.1.1), and the criterion (5.1.2, c′). The other properties are local on Y and thus follow immediately from (5.1.2) and (5.1.4), taking into account the fact that $f_*(\mathscr{F})$ is **II** | 96
quasi-coherent whenever f is separated (**I**, 9.2.2, a). ☐

Corollary (5.1.7). — *Let $f : X \to Y$ be a quasi-affine morphism. For every open subset U of Y, the restriction $f^{-1}(U) \to U$ of f is quasi-affine.*

Corollary (5.1.8). — *Let Y be an affine scheme, and $f : X \to Y$ a quasi-compact morphism. For f to be quasi-affine, it is necessary and sufficient for X to be a quasi-affine scheme.*

Proof. This is an immediate consequence of (5.1.6) and (4.6.6). ☐

Corollary (5.1.9). — *Let Y be a quasi-compact scheme or a prescheme whose underlying space is Noetherian, and $f : X \to Y$ a morphism of* finite type. *If f is quasi-affine, then there exists a quasi-coherent $\mathscr{O}_Y$-subalgebra $\mathscr{B}$ of $\mathscr{A}(X) = f_*(\mathscr{O}_X)$ of* finite type (**I**, 9.6.2) *such that the morphism $X \to \operatorname{Spec}(\mathscr{B})$ corresponding to the canonical injection $\mathscr{B} \to \mathscr{A}(X)$ is an immersion. Further, every quasi-coherent $\mathscr{O}_Y$-subalgebra $\mathscr{B}'$ of finite type over $\mathscr{A}(X)$ containing $\mathscr{B}$ has the same property.*

PROOF. Indeed, $\mathscr{A}(X)$ is the inductive limit of its quasi-coherent $\mathscr{O}_Y$-subalgebras of finite type (**I**, 9.6.5); the result is then a particular case of (3.8.4), taking into account the identification of $\operatorname{Spec}(\mathscr{A}(X))$ with $\operatorname{Proj}(\mathscr{A}(X)[T])$ (3.1.7). □

Proposition (5.1.10). —

(i) *A quasi-compact morphism $X \to Y$ that is a homeomorphism from the underlying space of X to some subspace of the underlying space of Y (so, in particular, any closed immersion) is quasi-affine.*
(ii) *The composition of any two quasi-affine morphisms is quasi-affine.*
(iii) *If $f : X \to Y$ is a quasi-affine S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is a quasi-affine morphism for any extension $S' \to S$ of the base prescheme.*
(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are quasi-affine S-morphisms, then $f \times_S g$ is quasi-affine.*
(v) *If $f : X \to Y$ and $g : Y \to Z$ are morphisms such that $g \circ f$ is quasi-affine, and if g is separated or the underlying space of X is locally Noetherian, then f is quasi-affine.*
(vi) *If f is a quasi-affine morphism, then so is f_{red}.*

PROOF. Taking into account the criterion (5.1.6, c′), all of (i), (iii), (iv), (v), and (vi) follow immediately from (4.6.13, i *bis*), (4.6.13, iii), (4.6.13, iv), (4.6.13, v), and (4.6.13, vi) (respectively). To prove (ii), we can restrict to the case where Z is affine, and then the claim follows directly from applying (4.6.13, ii) to $\mathscr{L} = \mathscr{O}_X$ and $\mathscr{K} = \mathscr{O}_Y$. □

Remark (5.1.11). — Let $f : X \to Y$ and $g : Y \to Z$ be morphisms such that $X \times_Z Y$ is locally Noetherian. Then the graph immersion $\Gamma_f : X \to X \times_Z Y$ is quasi-affine, since it is quasi-compact (**I**, 6.3.5), and since (**I**, 5.5.12) shows that, in (v), the conclusion still holds true if we remove the hypothesis that g is separated.

Proposition (5.1.12). — *Let $f : X \to Y$ be a quasi-compact morphism, and $g : X' \to X$ a quasi-affine morphism. If $\mathscr{L}$ is an ample (for f) $\mathscr{O}_X$-module, then $g^*(\mathscr{L})$ is an ample (for $f \circ g$) $\mathscr{O}_{X'}$-module.*

PROOF. Since $\mathscr{O}_{X'}$ is very ample for g, and the question is local on Y (4.6.4), it follows from (4.6.13, ii) that there exists (for Y affine) an integer n such that

$$g^*(\mathscr{L}^{\otimes n}) = (g^*(\mathscr{L}))^{\otimes n}$$

is ample for $f \circ g$, and so $g^*(\mathscr{L})$ is ample for $f \circ g$ (4.6.9) □

5.2. Serre's criterion

Theorem (5.2.1). — (Serre's criterion). *Let X be a quasi-compact scheme or a prescheme whose underlying space is Noetherian. The following conditions are equivalent.*

(a) *X is an affine scheme.*
(b) *There exists a family of elements $f_\alpha \in A = \Gamma(X, \mathscr{O}_X)$ such that the X_{f_α} are affine, and such that the ideal generated by the f_α in A is equal to A itself.*
(c) *The functor $\Gamma(X, \mathscr{F})$ is exact in $\mathscr{F}$ on the category of quasi-coherent $\mathscr{O}_X$-modules, or, in other words, if*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0 \tag{*}$$

is an exact sequence of quasi-coherent $\mathscr{O}_X$-modules, then the sequence

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is also exact.

(c′) *Condition* (c) *holds for every exact sequence $(*)$ of quasi-coherent $\mathscr{O}_X$-modules such that $\mathscr{F}$ is isomorphic to an $\mathscr{O}_X$-submodule of $\mathscr{O}_X^n$ for some finite n.*
(d) *$\mathrm{H}^1(X, \mathscr{F}) = 0$ for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$.*
(d′) *$\mathrm{H}^1(X, \mathscr{J}) = 0$ for every quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$.*

PROOF. It is evident that (a) implies (b); furthermore, (b) implies that the X_{f_α} cover X, because, by hypothesis, the section 1 is a linear combination of the f_α, and the $D(f_\alpha)$ thus cover $\operatorname{Spec}(A)$. The final claim of (4.5.2) thus implies that $X \to \operatorname{Spec}(A)$ is an isomorphism.

We know that (a) implies (c) (I, 1.3.11), and (c) trivially implies (c′). We now prove that (c′) implies (b). First of all, (c′) implies that, for every *closed* point $x \in X$ and every open neighbourhood U of x, there exists some $f \in A$ such that $x \in X_f \subset X - U$. Let $\mathscr{J}$ (resp. $\mathscr{J}'$) be the quasi-coherent sheaf of ideals of $\mathscr{O}_X$ defining the closed reduced subprescheme of X that has $X - U$ (resp. $(X-U) \cup \{x\}$) as its underlying space (I, 5.2.1); it is clear that we have $\mathscr{J}' \subset \mathscr{J}$, and that $\mathscr{J}'' = \mathscr{J}/\mathscr{J}'$ is a quasi-coherent $\mathscr{O}_X$ module that has support equal to $\{x\}$, and such that $\mathscr{J}''_x = k(x)$. Hypothesis (c′) applied to the exact sequence $0 \to \mathscr{J}' \to \mathscr{J} \to \mathscr{J}'' \to 0$ shows that $\Gamma(X, \mathscr{J}) \to \Gamma(X, \mathscr{J}'')$ is surjective. The section of $\mathscr{J}''$ whose germ at x is 1_x is thus the image of some section $f \in \Gamma(X, \mathscr{J}) \subset \Gamma(X, \mathscr{O}_X)$, and we have, by definition, that $f(x) = 1_x$ and $f(y) = 0$ in $X - U$, which establishes our claim. Now, if U is affine, then so is X_f (I, 1.3.6), so the union of the X_f that are affine ($f \in A$) is an open set Z that contains *all the closed points* of X; since X is a quasi-compact Kolmogoroff space, we necessarily have $Z = X$ (0, 2.1.3). Because X is quasi-compact, there are a *finite* number of elements $f_i \in A$ ($1 \leqslant i \leqslant n$) such that the X_{f_i} are affine and cover X. So consider the homomorphism $\mathscr{O}_X^n \to \mathscr{O}_X$ defined by the sections f_i (0, 5.1.1); since, for all $x \in X$, at least one of the $(f_i)_x$ is invertible, this homomorphism is *surjective*, and we thus have an exact sequence $0 \to \mathscr{R} \to \mathscr{O}_X^n \to \mathscr{O}_X \to 0$, where $\mathscr{R}$ is a quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{O}_X$. It then follows from (c′) that the corresponding homomorphism $\Gamma(X, \mathscr{O}_X^n) \to \Gamma(X, \mathscr{O}_X)$ is surjective, which proves (b). **II** | 98

Finally, (a) implies (d) (I, 5.1.9.2), and (d) trivially implies (d′). It remains to show that (d′) implies (c′). But if $\mathscr{F}'$ is a quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{O}_X^n$, then the filtration $0 \subset \mathscr{O}_X \subset \mathscr{O}_X^2 \subset \ldots \subset \mathscr{O}_X^n$ defines a filtration of $\mathscr{F}'$ given by the $\mathscr{F}'_k = \mathscr{F} \cap \mathscr{O}_X^k$ ($0 \leqslant k \leqslant n$), which are quasi-coherent $\mathscr{O}_X$-modules (I, 4.1.1), and $\mathscr{F}'_{k+1}/\mathscr{F}'_k$ is isomorphic to a quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{O}_X^{k+1}/\mathscr{O}_X^k = \mathscr{O}_X$, which is to say, a quasi-coherent sheaf of ideals of $\mathscr{O}_X$. Hypothesis (d′) thus implies that $\mathrm{H}^1(X, \mathscr{F}'_{k+1}/\mathscr{F}'_k) = 0$; the exact cohomology sequence $\mathrm{H}^1(X, \mathscr{F}'_k) \to \mathrm{H}^1(X, \mathscr{F}'_{k+1}) \to \mathrm{H}^1(X, \mathscr{F}'_{k+1}/\mathscr{F}'_k) = 0$ then lets us prove by induction on k that $H^1(X, \mathscr{F}'_k) = 0$ for all k. □

Remark (5.2.1.1). — When X is a *Noetherian* prescheme, we can replace "quasi-coherent" by "coherent" in the statements of (c′) and (d′). Indeed, in the proof of the fact that (c′) implies (b), $\mathscr{J}$ and $\mathscr{J}'$ are then *coherent* sheaves of ideals, and, furthermore, every quasi-coherent submodule of a coherent module is coherent (I, 6.1.1); whence the conclusion.

Corollary (5.2.2). — *Let $f : X \to Y$ be a separated quasi-compact morphism. The following conditions are equivalent.*

(a) *The morphism f is an affine morphism.*
(b) *The functor f_* is exact on the category of quasi-coherent $\mathscr{O}_X$-modules.*
(c) *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $\mathrm{R}^1 f_*(\mathscr{F}) = 0$.*
(c′) *for every quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$, we have $\mathrm{R}^1 f_*(\mathscr{J}) = 0$.*

PROOF. All these conditions are local on Y, by definition of the functor $\mathrm{R}^1 f_*$ (T, 3.7.3), and so we can assume that Y is affine. If f is affine, then X is affine, and property (b) is nothing more than (I, 1.6.4). Conversely, we now show that (b) implies (a): for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have that $f_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_Y$-module (I, 9.2.2, a). By hypothesis, the functor $f_*(\mathscr{F})$ is exact in $\mathscr{F}$, and the functor $\Gamma(Y, \mathscr{G})$ is exact in $\mathscr{G}$ (in the category of quasi-coherent $\mathscr{O}_Y$-modules) because Y is affine (I, 1.3.11); so $\Gamma(Y, f_*(\mathscr{F})) = \Gamma(X, \mathscr{F})$ is exact in $\mathscr{F}$, which proves our claim, by (5.2.1, c).

If f is affine, then $f^{-1}(U)$ is affine for every affine open subset U of Y (1.3.2), and so $\mathrm{H}^1(f^{-1}(U), \mathscr{F}) = 0$ (5.2.1, d), which, by definition, implies that $\mathrm{R}^1 f_*(\mathscr{F}) = 0$. Finally, suppose that condition (c′) is satisfied; the exact sequence of terms of low degree in the Leray spectral sequence (G, II, 4.17.1 and I, 4.5.1) give, in particular, the exact sequence

$$0 \longrightarrow \mathrm{H}^1(Y, f_*(\mathscr{J})) \longrightarrow \mathrm{H}^1(X, \mathscr{J}) \longrightarrow \mathrm{H}^0(Y, \mathrm{R}^1 f_*(\mathscr{J})).$$

Since Y is affine, and $f_*(\mathscr{J})$ quasi-coherent (**I**, 9.2.2, a), we have that $\mathrm{H}^1(Y, f_*(\mathscr{J})) = 0$ (5.2.1); hypothesis (c′) thus implies that $\mathrm{H}^1(X, \mathscr{J}) = 0$, and we conclude, by (5.2.1), that X is an affine scheme. □

Corollary (5.2.3). — *If $f : X \to Y$ is an affine morphism, then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{H}^1(Y, f_*(\mathscr{F})) \to \mathrm{H}^1(X, \mathscr{F})$ is bijective.*

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PROOF. We have the exact sequence

$$0 \longrightarrow \mathrm{H}^1(Y, f_*(\mathscr{F})) \longrightarrow \mathrm{H}^1(X, \mathscr{F}) \longrightarrow \mathrm{H}^0(Y, \mathrm{R}^1 f_*(\mathscr{F}))$$

of terms of low degree in the Leray spectral sequence, and the conclusion follows from (5.2.2). □

Remark (5.2.4). — In Chapter III, §1, we prove that, if X is affine, then we have $\mathrm{H}^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent $\mathscr{O}_X$-modules $\mathscr{F}$.

5.3. Quasi-projective morphisms

Definition (5.3.1). — We say that a morphism $f : X \to Y$ is *quasi-projective*, or that X (considered as a Y-prescheme via f) is *quasi-projective over Y*, or that X is a *quasi-projective Y-scheme*, if f is of finite type and there exists an invertible f-ample $\mathscr{O}_X$-module.

We note that this notion *is not local on Y*: the counterexamples of Nagata [**Nag58b**] and Hironaka show that, even if X and Y are non-singular algebraic schemes over an algebraically closed field, every point of Y can have an affine neighbourhood U such that $f^{-1}(U)$ is quasi-projective over U, without f being quasi-projective.

We note that a quasi-projective morphism is necessarily *separated* (4.6.1). When Y is quasi-compact, it is equivalent to say either that f is quasi-projective, or that f is of finite type and there exists a *very ample* (relative to f) $\mathscr{O}_X$-module ((4.6.2) and (4.6.11)). Further:

Proposition (5.3.2). — *Let Y be a quasi-compact scheme or a prescheme whose underlying space is Noetherian, and let X be a Y-prescheme. The following conditions are equivalent.*

(a) *X is a quasi-projective Y-scheme.*
(b) *X is of finite type over Y, and there exists some quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ of finite type such that X is Y-isomorphic to a subprescheme of $\mathbf{P}(\mathscr{E})$.*
(c) *X is of finite type over Y, and there exists some quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ such that $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$, and such that X is Y-isomorphic to a induced subprescheme on some everywhere-dense open subset of $\mathrm{Proj}(\mathscr{S})$.*

PROOF. This follows immediately from the previous remark and from (4.4.3), (4.4.6), and (4.4.7). □

We note that, whenever Y is a *Noetherian* prescheme, we can, in conditions (b) and (c) of (5.3.2), remove the hypothesis that X is of finite type over Y, since this automatically satisfied (**I**, 6.3.5).

Corollary (5.3.3). — *Let Y be a quasi-compact scheme such that there exists an ample $\mathscr{O}_Y$-module $\mathscr{L}$ (4.5.3). For a Y-scheme X to be quasi-projective, it is necessary and sufficient for it to be of finite type over Y and also isomorphic to a Y-subscheme of a projective bundle of the form $\mathbf{P}^r_Y$.*

PROOF. If $\mathscr{E}$ is a quasi-coherent $\mathscr{O}_Y$-module of finite type, then $\mathscr{E}$ is isomorphic to a quotient of an $\mathscr{O}_Y$-module $\mathscr{L}^{\otimes(-n)} \otimes_{\mathscr{O}_Y} \mathscr{O}_Y^k$ (4.5.5), and so $\mathbf{P}(\mathscr{E})$ is isomorphic to a closed subscheme of $\mathbf{P}_Y^{k-1}$ ((4.1.2) and (4.1.4)). □

Proposition (5.3.4). —

(i) *A quasi-affine morphism of finite type (and, in particular, a quasi-compact immersion, or an affine morphism of finite type) is quasi-projective.*
(ii) *If $f : X \to Y$ and $g : Y \to Z$ are quasi-projective, and if Z is quasi-compact, then $g \circ f$ is quasi-projective.* **II** | 100
(iii) *If $f : X \to Y$ is a quasi-projective S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is quasi-projective for every extension $S' \to S$ of the base prescheme.*

(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are quasi-projective S-morphisms, then $f \times_S g$ is quasi-projective.*
(v) *If $f : X \to Y$ and $g : Y \to Z$ are morphisms such that $g \circ f$ is quasi-projective, and if g is separated or X locally Noetherian, then f is quasi-projective.*
(vi) *If f is a quasi-projective morphism, then so is f_{red}.*

PROOF. (i) follows from (5.1.6) and (5.1.10, i). The other claims are immediate consequences of Definition (5.3.1), of the properties of morphisms of finite type (**I**, 6.3.4), and of (4.6.13). □

Remark (5.3.5). — We note that we can have f_{red} being quasi-projective without f being quasi-projective, even if we assume that Y is the spectrum of an algebra of finite rank over $\mathbf{C}$ and that f is proper.

Corollary (5.3.6). — *If X and X' are quasi-projective Y-schemes, then $X \sqcup X'$ is a quasi-projective Y-scheme.*

PROOF. This follows from (4.6.18). □

5.4. Proper morphisms and universally closed morphisms

Definition (5.4.1). — We say that a morphism of preschemes $f : X \to Y$ is *proper* if it satisfies the following two conditions:

(a) f is separated and of finite type; and
(b) for every prescheme Y' and every morphism $Y' \to Y$, the projection $f_{(Y')} : X \times_Y Y' \to Y'$ is a closed morphism (**I**, 2.2.6).

When this is the case, we also say that X (considered as a Y-prescheme with structure morphism f) is proper over Y.

It is immediate that conditions (a) and (b) are *local* on Y. To show that the image of a closed subset Z of $X \times_Y Y'$ under the projection $q : X \times_Y Y' \to Y'$ is closed in Y, it suffices to see that $q(Z) \cap U'$ is closed in U' for every affine open subset U' of Y'; since $q(Z) \cap U' = q(Z \cap q^{-1}(U'))$, and since $q^{-1}(U')$ can be identified with $X \times_Y U'$ (**I**, 4.4.1), we see that to satisfy condition (b) of Definition (5.4.1), we can restrict to the case where Y is an *affine* scheme. We further see (5.3.6) that, if Y is locally Noetherian, then we can even restrict to proving (b) in the case where Y' is of finite type over Y.

It is clear that every proper morphism is *closed*.

Proposition (5.4.2). —

(i) *A closed immersion is a proper morphism.*
(ii) *The composition of two proper morphisms is proper.*
(iii) *If X and Y are S-preschemes, and $f : X \to Y$ a proper S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is proper for every extension $S' \to S$ of the base prescheme.*
(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are proper S-morphisms, then $f \times_S g : X \times_S Y \to X' \times_S Y'$ is a proper S-morphism.*

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PROOF. It suffices to prove (i), (ii), and (iii) (**I**, 3.5.1). In each of the three cases, verifying condition (a) of (5.4.1) follows from previous results ((**I**, 5.5.1) and (6.4.3)); it remains to verify condition (b). It is immediate in case (i), because if $X \to Y$ is a closed immersion, then so is $X \times_Y Y' \to Y \times_Y Y' = Y'$ ((**I**, 4.3.2) and (3.3.3)). To prove (ii), consider two proper morphisms $X \to Y$ and $Y \to Z$, and a morphism $Z' \to Z$. We can write $X \times_Z Z' = X \times_Y (Y \times_Z Z')$ (**I**, 3.3.9.1), and so the projection $X \times_Z Z' \to Z'$ factors as $X \times_Y (Y \times_Z Z') \to Y \times_Z Z' \to Z'$. Taking the initial remark into account, (ii) follows from the fact that the composition of two closed morphisms is closed. Finally, for every morphism $S' \to S$, we can identify $X_{(S')}$ with $X \times_Y Y_{(S')}$ (**I**, 3.3.11); for every morphism $Z \to Y_{(S')}$, we can write

$$X_{(S')} \times_{Y_{(S')}} Z = (X \times_Y Y_{(S')}) \times_{Y_{(S')}} Z = X \times_Y Z;$$

since by hypothesis $X \times_Y Z \to Z$ is closed, this proves (iii). □

Corollary (5.4.3). — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms such that $g \circ f$ is proper.*

(i) *If g is separated, then f is proper.*
(ii) *If g is separated and of finite type, and if f is surjective, then g is proper.*

PROOF. (i) follows from (5.4.2) by the general procedure (**I**, 5.5.12). To prove (ii), we need only verify that condition (b) of Definition (5.4.1) is satisfied. For every morphism $Z' \to Z$, the diagram

$$\begin{tikzcd} X \times_Z Z' \arrow[r, "f \times 1_{Z'}"] \arrow[dr, "p"'] & Y \times_Z Z' \arrow[d, "p'"] \\ & Z' \end{tikzcd}$$

(where p and p' are the projections) commutes (**I**, 3.2.1); furthermore, $f \times 1_{Z'}$ is surjective because f is surjective (**I**, 3.5.2), and p is a closed morphism by hypothesis. Every closed subset F of $Y \times_Z Z'$ is thus the image under $f \times 1_{Z'}$ of some closed subset E of $X \times_Z Z'$, so $p'(F) = p(E)$ is closed in Z' by hypothesis, whence the corollary. □

Corollary (5.4.4). — *If X is a proper prescheme over Y, and $\mathscr{S}$ a quasi-coherent $\mathscr{O}_Y$-algebra, then every Y-morphism $f : X \to \operatorname{Proj}(\mathscr{S})$ is proper (and* a fortiori *closed).*

PROOF. The structure morphism $p : \operatorname{Proj}(\mathscr{S}) \to Y$ is separated, and $p \circ f$ is proper by hypothesis. □

Corollary (5.4.5). — *Let $f : X \to Y$ be a separated morphism of finite type. Let $(X_i)_{1 \leqslant i \leqslant n}$ (resp. $(Y_i)_{1 \leqslant i \leqslant n}$) be a finite family of closed subpreschemes of X (resp. Y), and j_i (resp. h_i) the canonical injection $X_i \to X$ (resp. $Y_i \to Y$). Suppose that the underlying space of X is the union of the X_i, and that, for all i, there is a morphism $f_i : X_i \to Y_i$, such that the diagram*

$$\begin{tikzcd} X_i \arrow[r, "f_i"] \arrow[d, "j_i"'] & Y_i \arrow[d, "h_i"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

commutes. Then, for f to be proper, it is necessary and sufficient for all of the f_i to be proper.

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PROOF. If f is proper, then so is $f \circ j_i$, because j_i is a closed immersion (5.4.2); since h_i is a closed immersion, and thus a separated morphism, f_i is proper, by (5.4.3). Conversely, suppose that all of the f_i are proper, and consider the prescheme Z given by the *sum* of the X_i; let u be the morphism $Z \to X$ which reduces to j_i on each X_i. The restriction of $f \circ u$ to each X_i is equal to $f \circ j_i = h_i \circ f_i$, and is thus proper, because both the h_i and the f_i are (5.4.2); it then follows immediately from Definition (5.4.1) that u is proper. But since by hypothesis u is surjective, we conclude that f is proper by (5.4.3). □

Corollary (5.4.6). — *Let $f : X \to Y$ be a separated morphism of finite type; for f to be proper, it is necessary and sufficient for $f_{\text{red}} : X_{\text{red}} \to Y_{\text{red}}$ to be proper.*

PROOF. This is a particular case of (5.4.5), with $n = 1$, $X_1 = X_{\text{red}}$, and $Y_1 = Y_{\text{red}}$ (**I**, 5.1.5). □

(5.4.7). If X and Y are Noetherian preschemes, and $f : X \to Y$ a separated morphism of finite type, then we can, to show that f is proper, restrict to the the case of *dominant* morphisms and *integral* preschemes. Indeed, let X_i ($1 \leqslant i \leqslant n$) be the (finitely many) irreducible components of X, and consider, for each i, the unique closed reduced subprescheme of X that has X_i as its underlying space, which we again denote by X_i (**I**, 5.2.1). Let Y_i be the unique closed reduced subprescheme of Y that has $\overline{f(X_i)}$ as its underlying space. If g_i (resp. h_i) is the injection morphism $X_i \to X$ (resp. $Y_i \to Y$), then we conclude that $f \circ g_i = h_i \circ f_i$, where f_i is a dominant morphism $X_i \to Y_i$ (**I**, 5.2.2); we are then under the right conditions to apply (5.4.5), and for f to be proper, it is necessary and sufficient for all the f_i to be proper.

Corollary (5.4.8). — *Let X and Y be separated S-preschemes of finite type over S, and $f : X \to Y$ an S-morphism. For f to be proper, it is necessary and sufficient that, for every S-prescheme S', the morphism $f \times_S 1_{S'} : X \times_S S' \to Y \times_S S'$ be closed.*

PROOF. First note that, if $g : X \to S$ and $h : Y \to S$ are the structure morphisms, then we have, by definition, $g = h \circ f$, and so f is separated and of finite type ((**I**, 5.5.1) and (6.3.4)). If f is proper, then so is $f \times_S 1_{S'}$ (5.4.2); *a fortiori*, $f \times_S 1_{S'}$ is closed. Conversely, suppose that the conditions of the statement are satisfied, and let Y' be a Y-prescheme; Y' can also be considered as an S-prescheme, and since $Y \to S$ is separated, $X \times_Y Y'$ can be identified with a closed subprescheme of $X \times_S Y'$ (**I**, 5.4.2). In the commutative diagram

$$\begin{array}{ccc} X \times_Y Y' & \xrightarrow{f \times 1_{Y'}} & Y \times_Y Y' = Y' \\ \downarrow & & \downarrow \\ X \times_S Y' & \xrightarrow{f \times 1_{S'}} & Y \times_S Y', \end{array}$$

the vertical arrows are closed immersions; it thus immediately follows that if $f \times 1_{S'}$ is a closed morphism, then so is $f \times 1_{Y'}$ □

Remark (5.4.9). — We say that a morphism $f : X \to Y$ is *universally closed* if it satisfies condition (b) of Definition (5.4.1). The reader will observe that, in (5.4.2) to (5.4.8), we can replace every occurrence of "proper" with "universally closed" without changing the validity of the results (and in the hypotheses of (5.4.3), (5.4.5), (5.4.6), and (5.4.8), we can omit the finiteness conditions). **II** | 103

(5.4.10). Let $f : X \to Y$ be a morphism of finite type. We say that a closed subset Z of X is *proper on Y* (or *Y-proper*, or *proper for f*) if the restriction of f to a closed subprescheme of X, with underlying space Z (**I**, 5.2.1), is *proper*. Since this restriction is then separated, it follows from (5.4.6) and (**I**, 5.5.1, vi) that the preceding property *does not depend* on the closed subprescheme of X that has Z as its underlying space. If $g : X' \to X$ is a *proper* morphism, then $g^{-1}(Z)$ is a *proper* subset of X': if T is a subprescheme of X that has Z as its underlying space, it suffices to note that the restriction of g to the closed subprescheme $g^{-1}(T)$ of X' is a proper morphism $g^{-1}(T) \to T$, by (5.4.2, iii), and to then apply (5.4.2, ii). Further, if X'' is a *Y-scheme* of finite type, and $u : X \to X''$ a Y-morphism, then $u(Z)$ is a *proper* subset of X''; indeed, let us take T to be the closed reduced subprescheme of X having Z as its underlying space; then the restriction of f to T is proper, and thus so is the restriction of u to T (5.4.3, i), thus $u(Z)$ is closed in X''; let T'' be a closed subprescheme of X'' having $u(Z)$ as its underlying space (**I**, 5.2.1), such that $u|T$ factors as $T \xrightarrow{v} T'' \xrightarrow{j} X''$, where j is the canonical injection (**I**, 5.2.2), and v is thus proper and surjective (5.4.5); if g is the restriction to T'' of the structure morphism $X'' \to Y$, then g is separated and of finite type, and we have that $f|T = g \circ v$; it thus follows from (5.4.3, ii) that g is proper, whence our assertion.

It follows, in particular, from these remarks that, if Z is a Y-proper subset of X, then

(1) for every closed subprescheme X' of X, $Z \cap X'$ is a Y-proper subset of X'; and
(2) if X is a subprescheme of a Y-scheme of finite type X'', then Z is also a Y-proper subset of X'' (and so, in particular, is *closed in X''*).

5.5. Projective morphisms

Proposition (5.5.1). — *Let X be a Y-prescheme. The following conditions are equivalent.*

(a) *X is Y-isomorphic to a* closed *subprescheme of a projective bundle $\mathbf{P}(\mathscr{E})$, where $\mathscr{E}$ is a quasi-coherent $\mathscr{O}_Y$-module of finite type.*

(b) *There exists a quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S}$ such that $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$, and such that X is Y-isomorphic to* $\operatorname{Proj}(\mathscr{S})$.

PROOF. Condition (a) implies (b), by (3.6.2, ii): if $\mathscr{J}$ is a quasi-coherent graded sheaf of ideals of $\mathbf{S}(\mathscr{E})$, then the quasi-coherent graded $\mathscr{O}_Y$-algebra $\mathscr{S} = \mathbf{S}(\mathscr{E})/\mathscr{J}$ is generated by $\mathscr{S}_1$, and $\mathscr{S}_1$, the canonical image of $\mathscr{E}$, is an $\mathscr{O}_Y$-module of finite type. Condition (b) implies (a) by (3.6.2) applied to the case where $\mathscr{M} \to \mathscr{S}_1$ is the identity map. □

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Definition (5.5.2). — We say that a Y-prescheme X is *projective* on Y, or is a projective Y-scheme, if it satisfies either of the (equivalent) conditions (a) and (b) of (5.5.1). We say that a morphism $f : X \to Y$ is projective if it makes X a projective Y-scheme.

It is clear that if $f : X \to Y$ is projective, then there exists a *very ample* (relative to f) $\mathscr{O}_X$-module (4.4.2).

Theorem (5.5.3). —

(i) *Every projective morphism is quasi-projective and proper.*

(ii) *Conversely, let Y be a quasi-compact scheme or a prescheme whose underlying space is Noetherian; then every morphism $f : X \to Y$ that is quasi-projective and proper is projective.*

PROOF.

(i) It is clear that if $f : X \to Y$ is projective, then it is of finite type and quasi-projective (thus, in particular, separated); furthermore, it follows immediately from (5.5.1, b) and (3.5.3) that if f is projective, then so is $f \times_Y 1_{Y'} : X \times_Y Y' \to Y'$ for every morphism $Y' \to Y$. To show that f is universally closed, it is thus enough to show that a projective morphism f is *closed*. Since the question is local on Y, we can suppose that $Y = \operatorname{Spec}(A)$, thus (5.5.1) $X = \operatorname{Proj}(S)$, where S is a graded A-algebra generated by a finite number of elements of S_1. For all $y \in Y$, the fibre $f^{-1}(y)$ can be identified with $\operatorname{Proj}(S) \times_Y \operatorname{Spec}(k(y))$ (**I**, 3.6.1), and so also with $\operatorname{Proj}(S \otimes_A k(y))$ (2.8.10); so $f^{-1}(y)$ is empty if and only if $S \otimes_A k(y)$ satisfies condition (TN) (2.7.4), or, in other words, if $S_n \otimes_A k(y) = 0$ for sufficiently large n. But since $(S_n)_y$ is an $\mathscr{O}_y$-module of finite type, the preceding condition implies that $(S_n)_y = 0$ for sufficiently large N, by Nakayama's lemma. If $\mathfrak{a}_n$ is the annihilator in A of the A-module S_n, then the preceding condition also implies that $\mathfrak{a}_n \subset \mathfrak{j}_n$ for sufficiently large n (**0**, 1.7.4). But since $S_n S_1 = S_{n+1}$, by hypothesis, we have that $\mathfrak{a}_n \subset \mathfrak{a}_{n+1}$, and if $\mathfrak{a}$ is the union of the $\mathfrak{a}_n$, then we see that $f(X) = V(\mathfrak{a})$, which proves that $f(X)$ is closed in Y. If now X' is an arbitrary closed subset of X, then there exists a closed subprescheme of X that has X' as its underlying space (**I**, 5.2.1), and it is clear (5.5.1, a) that the morphism $X' \to X \xrightarrow{f} Y$ is projective, and so $f(X')$ is closed in Y.

(ii) The hypothesis on Y and the fact that f is quasi-projective implies the existence of a quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E}$ of finite type, as well as a Y-immersion $j : X \to \mathbf{P}(\mathscr{E})$ (5.3.2). But since f is proper, j is *closed*, by (5.4.4), and so f is projective.

□

Remark (5.5.4). —

(i) Let $f : X \to Y$ be a morphism such that f is proper, such that there exists a *very ample* (relative to f) $\mathscr{O}_X$-module $\mathscr{L}$, and such that the quasi-coherent $\mathscr{O}_Y$-module $\mathscr{E} = f_*(\mathscr{L})$ is *of finite type*. Then f is a *projective* morphism: indeed (4.4.4), there is then a Y-immersion $r : X \to \mathbf{P}(\mathscr{E})$, and, since f is proper, r is a *closed* immersion (5.4.4). We will see in Chapter III, §3, that when Y is *locally Noetherian*, the third condition above ($\mathscr{E}$ being of finite type) is a consequence of the first two, and so the first two conditions *characterise*, in this case, the projective morphisms, and if Y is quasi-compact, then we can replace the second condition (the existence of a very ample (relative to f) $\mathscr{O}_X$-module $\mathscr{L}$) by the hypothesis that there exists an *ample* (relative to f) $\mathscr{O}_X$-module (4.6.11).

(ii) Let Y be a quasi-compact scheme such that there exists an ample $\mathscr{O}_Y$-module. For a Y-scheme X to be *projective*, it is necessary and sufficient for it to be Y-isomorphic to a *closed* Y-subscheme of a projective bundle of the form $\mathbf{P}^r_Y$. The condition is clearly sufficient. Conversely, if X is projective over Y, then it is quasi-projective, and so there exists a **II** | 105
Y-immersion j of X into some $\mathbf{P}^r_Y$ (5.3.3) that is *closed*, by (5.4.4) and (5.5.3).

(iii) The argument of (5.5.3) shows that, for every prescheme Y and every integer $r \geqslant 0$, the structure morphism $\mathbf{P}^r_Y \to Y$ is *surjective*, because if we set $\mathscr{S} = \mathbf{S}_{\mathscr{O}_Y}(\mathscr{O}_Y^{r+1})$, then we evidently have $\mathscr{S}_y = \mathbf{S}_{k(y)}(k(y)^{r+1})$ (1.7.3), and so $(\mathscr{S}_n)_y \neq 0$ for any $y \in Y$ or any $n \geqslant 0$.

(iv) It follows from the examples of Nagata [**Nag58b**] that there exist proper morphisms that are not quasi-projective.

Proposition (5.5.5). —

(i) *A closed immersion is a projective morphism.*

(ii) *If $f : X \to Y$ and $g : Y \to Z$ are projective morphisms, and if Z is a quasi-compact scheme or a prescheme whose underlying space is Noetherian, then $g \circ f$ is projective.*

(iii) *If $f : X \to Y$ is a projective S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is projective for every extension $S' \to S$ of the base prescheme.*

(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are projective S-morphisms, then so is $f \times_S g$.*

(v) *If $g \circ f$ is a projective morphism, and if g is separated, then f is projective.*

(vi) *If f is projective, then so is f_{red}.*

Proof. (i) follows immediately from (3.1.7). We have to show (iii) and (iv) separately, because of the restriction introduced on Z in (ii) (cf. (**I**, 3.5.1)). To show (iii), we restrict to the case where $S = Y$ (**I**, 3.3.11), and the claim then immediately follows from (5.5.1, b) and (3.5.3). To show (iv), we are immediately led to the case where $X = \mathbf{P}(\mathscr{E})$ and $X = \mathbf{P}(\mathscr{E}')$, where $\mathscr{E}$ (resp. $\mathscr{E}'$) is a quasi-coherent $\mathscr{O}_Y$-module (resp. quasi-coherent $\mathscr{O}_{Y'}$-module) of finite type. Let p and p' be the canonical projections of $T = Y \times_S Y'$ to Y and Y' (respectively); by (4.1.3.1), we have $\mathbf{P}(p^*(\mathscr{E})) = \mathbf{P}(\mathscr{E}) \times_Y T$ and $\mathbf{P}(p'^*(\mathscr{E}')) = \mathbf{P}(\mathscr{E}') \times_{Y'} T$; whence

$$\begin{aligned}\mathbf{P}(p^*(\mathscr{E})) \times_T \mathbf{P}(p'^*(\mathscr{E}')) &= (\mathbf{P}(\mathscr{E}) \times_Y T) \times_T (T \times_{Y'} \mathbf{P}(\mathscr{E}')) \\ &= \mathbf{P}(\mathscr{E}) \times_Y (T \times_{Y'} \mathbf{P}(\mathscr{E}')) = \mathbf{P}(\mathscr{E}) \times_S \mathbf{P}(\mathscr{E}')\end{aligned}$$

by replacing T with $Y \times_S Y'$, and using (**I**, 3.3.9.1). But $p^*(\mathscr{E})$ and $p'^*(\mathscr{E}')$ are of finite type over T (**0**, 5.2.4), and thus so is $p^*(\mathscr{E}) \otimes_{\mathscr{O}_T} p'^*(\mathscr{E}')$; since $\mathbf{P}(p^*(\mathscr{E})) \times_T \mathbf{P}(p'^*(\mathscr{E}'))$ can be identified with a closed subprescheme of $p^*(\mathscr{E}) \otimes_{\mathscr{O}_T} p'^*(\mathscr{E}')$ (4.3.3), this proves (iv). To show (v) and (vi), we can apply (**I**, 5.5.13), because every closed subprescheme of a projective Y-scheme is a projective Y-scheme, by (5.5.1, a).

It remains to prove (ii); by the hypothesis on Z, this follows from (5.5.3), (5.3.4, ii), and (5.4.2, ii). □

Proposition (5.5.6). — *If X and X' are projective Y-schemes, then $X \sqcup X'$ is a projective Y-scheme.*

Proof. This is an evident consequence of (5.5.2) and (4.3.6). □

Proposition (5.5.7). — *Let X be a projective Y-scheme, and $\mathscr{L}$ a Y-ample $\mathscr{O}_X$-module; then, for every section f of $\mathscr{L}$ over X, X_f is affine over Y.*

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Proof. Since the question is local on Y, we can assume that $Y = \operatorname{Spec}(A)$; furthermore, $X_{f^{\otimes n}} = X_f$, so by replacing $\mathscr{L}$ with some suitable $\mathscr{L}^{\otimes n}$, we can assume that $\mathscr{L}$ is very ample relative to the structure morphism $q : X \to Y$ (4.6.11). The canonical homomorphism $\sigma : q^*(q_*(\mathscr{L})) \to \mathscr{L}$ is thus surjective, and the corresponding morphism

$$r = r_{\mathscr{L},\sigma} : X \longrightarrow P = \mathbf{P}(q_*(\mathscr{L}))$$

is an immersion such that $\mathscr{L} = r^*(\mathscr{O}_P(1))$ (4.4.4); furthermore, since X is proper over Y, the immersion r is closed (5.4.4). But by definition, $f \in \Gamma(Y, q_*(\mathscr{L}))$, and $\sigma^\flat$ is the identity of $q_*(\mathscr{L})$; it then follows from Equation (3.7.3.1) that we have $X_f = r^{-1}(D_+(f))$; so X_f is a closed subprescheme of the affine scheme $D_+(f)$, and is thus also an affine scheme. □

In the particular case where $Y = X$, we obtain (taking (4.6.13, i) into account) the following corollary, whose direct proof is immediate anyway:

Corollary (5.5.8). — *Let X be a prescheme, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. For every section f of $\mathscr{L}$ over X, X_f is affine over X (and thus also an affine scheme whenever X is an affine scheme).*

5.6. Chow's lemma

Theorem (5.6.1). — (Chow's lemma). *Let S be a prescheme, and X an S-scheme of finite type. Suppose that the following conditions are satisfied:*

(a) *S is Noetherian;*
(b) *S is a quasi-compact scheme, and X has a finite number of irreducible components.*

Under these hypotheses,

(i) *there exists a* quasi-projective *S-scheme X', and an S-morphism $f : X' \to X$ that is both* projective *and* surjective;
(ii) *we can take X' and f to be such that there exists an open subset $U \subset X$ for which $U' = f^{-1}(U)$ is dense in X', and for which the restriction of f to U' is an isomorphism $U' \simeq U$; and*
(iii) *if X is reduced (resp. irreducible, integral), then we can assume that X' is reduced (resp. irreducible, integral).*

Proof. The proof proceeds in multiple steps.

(A) We can first restrict to the case where X is *irreducible*. Indeed, in hypothesis (a), X is Noetherian, and so, in the two hypotheses, the irreducible components X_i of X are finite in number. If the theorem is shown to be true for each of the closed reduced preschemes of X having the X_i as their underlying spaces, and if X'_i and $f_i : X'_i \to X_i$ are the prescheme and the morphism corresponding to X_i (respectively), then the prescheme X' given by the *sum* of the X'_i, and the morphism $f : X' \to X$ whose restriction to each X'_i is $j_i \circ f_i$ (where j_i is the canonical injection $X_i \to X$) satisfy the conclusion of the theorem. It is immediate that X' is reduced if all of the X'_i are; furthermore, we can satisfy (ii) by taking U to be the union of the sets $U_i \cap \complement\left(\bigcup_{j\neq i} X_j\right)$. Finally, since the X'_i are quasi-projective over S, so is X' (5.3.6); similarly, the morphisms $X'_i \to X$ are projective by (5.5.5, i) and (5.5.5, ii), and so f **II** | 107
is projective (5.5.6), and is clearly surjective, by definition.

(B) Now suppose that X is *irreducible*. Since the structure morphism $r : X \to S$ is of finite type, there exists a finite cover (S_i) of S by affine open subsets, and for each i there is a finite cover (T_{ij}) of $r^{-1}(S_i)$ by affine open subsets, and the morphisms $T_{ij} \to S_i$ are of finite type, and so quasi-projective (5.3.4, i); since in both hypotheses (a) and (b) the immersion $S_i \to S$ is quasi-compact, it is also quasi-projective (5.3.4, i), and so the restriction of r to T_{ij} is a quasi-projective morphism (5.3.4, ii). Denote the T_{ij} by U_k ($1 \leqslant k \leqslant n$). There exists, for each index k, an open immersion $\phi_k : U_k \to P_k$, where P_k is projective over S ((5.3.2) and (5.5.2)). Let $U = \bigcap_k U_k$; since X is irreducible, and the U_k nonempty, U is nonempty, and thus dense in X; the restrictions of the ϕ_k to U define a morphism

$$\phi : U \longrightarrow P = P_1 \times_S P_2 \times_S \cdots \times_S P_n$$

such that the diagrams

(5.6.1.1)
$$\begin{tikzcd} U \arrow[r, "\phi"] \arrow[d, "j_k"'] & P \arrow[d, "p_k"] \\ U_k \arrow[r, "\phi_k"] & P_k \end{tikzcd}$$

commute, where j_k is the canonical injection $U \to U_k$, and p_k the canonical projection $P \to P_k$. If j is the canonical injection $U \to X$, then the morphism $\psi = (j, \phi)_S : U \to X \times_S P$

is an *immersion* (**I**, 5.3.14). In hypothesis (a), $X \times_S P$ is locally Noetherian ((3.4.1), (**I**, 6.3.7), and (**I**, 6.3.8)); in hypothesis (b), $X \times_S P$ is a quasi-compact scheme ((**I**, 5.5.1) and (**I**, 6.6.4)); in both cases, the *closure* X' in $X \times_S P$ of the subprescheme Z associated to ψ (and so with underlying space $\psi(U)$) exists, and ψ factors as

$$\psi : U \xrightarrow{\psi'} X' \xrightarrow{h} X \times_S P \tag{5.6.1.2}$$

where ψ' is an *open immersion* and h a *closed immersion* (**I**, 9.5.10). Let $q_1 : X \times_S P \to X$ and $q_2 : X \times_S P \to P$ be the canonical projections; we set

$$f : X' \xrightarrow{h} X \times_S P \xrightarrow{q_1} X, \tag{5.6.1.3}$$

$$g : X' \xrightarrow{h} X \times_S P \xrightarrow{q_2} P. \tag{5.6.1.4}$$

We will see that X' and f satisfy the conclusion of the theorem.

(C) First we show that f is *projective* and *surjective*, and that the restriction of f to $U' = f^{-1}(U)$ is an *isomorphism* from U' to U. Since the P_k are projective over S, so is P (5.5.5, iv), and so $X \times_S P$ is projective over X (5.5.5, iii), and thus so is X', which is a closed subprescheme of $X \times_S P$. Furthermore, we have $f \circ \psi' = q_1 \circ (h \circ \psi') = q_1 \circ \psi = j$, so $f(X')$ contains the open everywhere-dense subset U of X; but f is a *closed* morphism (5.5.3), so $f(X') = X$. Now note that $q_1^{-1}(U) = U \times_S P$ is induced on an open subset of $X \times_S P$, and, by definition, the prescheme $U' = h^{-1}(U \times_S P)$ is induced by X' on the open subset U'; it is thus the closure *relative to* $U \times_S P$ of the prescheme Z (**I**, 9.5.8). But the immersion ψ factors as **II** | 108
$U \xrightarrow{\Gamma_\phi} U \times_S P \xrightarrow{j \times 1} X \times_S P$, and since P is separated over S, the graph morphism Γ_ϕ is a closed immersion (**I**, 5.4.3), and so Z is a *closed* subprescheme of $U \times_S P$, whence $U' = Z$. Since ψ is an immersion, the restriction of f to U' is an isomorphism onto U, and the inverse of ψ'; finally, by the definition of X', U' is dense in X'.

(D) We now show that g is an *immersion*, which will imply that X' is *quasi-projective* over S, because P is projective over S. Set

$$\begin{aligned} V_k &= \phi_k(U_k) \quad \text{(open subset of } P_k) \\ W_k &= p_k^{-1}(V_k) \quad \text{(open subset of } P) \\ U'_k &= f^{-1}(U_k) \quad \text{(open subset of } X') \\ U''_k &= g^{-1}(W_k) \quad \text{(open subset of } X'). \end{aligned}$$

It is clear that the U'_k form an open cover of X'; we will first see that the U''_k also form an open cover of X', by showing that $U'_k \subset U''_k$. For this, it will suffice to show that the diagram

$$\begin{tikzcd} U'_k \arrow[r, "g|U'_k"] \arrow[d, "f|U'_k"'] & P \arrow[d, "p_k"] \\ U_k \arrow[r, "\phi_k"] & P_k \end{tikzcd} \tag{5.6.1.5}$$

commutes. But the prescheme $U'_k = h^{-1}(U_k \times_S P)$ is induced by X' on the open subset U'_k, and is thus the closure of $Z = U' \subset U'_k$ relative to U'_k (**I**, 9.5.8). To show the commutativity of (5.6.1.5), it thus suffices (since P_k is an S-scheme) to show that composing the diagram with the canonical injection $U' \to U'_k$ (or, equivalently, thanks to the isomorphism from U' to U, with ψ) gives us a commutative diagram (**I**, 9.5.6). But, by definition, the diagram thus obtains is exactly (5.6.1.1), whence our claim.

The W_k thus form an open cover of $g(X')$; to show that g is an immersion, it suffices to show that each of the restrictions $g|U''_k$ is an immersion into W_k (**I**, 4.2.4). For this, consider the morphism $u_k : W_k \xrightarrow{p_k} V_k \xrightarrow{\phi_k^{-1}} U_k \to X$; since X is separated over S, the graph morphism $\Gamma_{u_k} : W_k \to X \times_S W_k$ is a closed immersion (**I**, 5.4.3), and so the graph $T_k = \Gamma_{u_k}(W_k)$ is a closed subprescheme of $X \times_S W$; if we show that $U' \to X \times_S W_k$ factors

through this subprescheme, then the map from the subprescheme induced by X' on the open subset X''_k of X' to $X \times_S W_k$ will also factor through this graph, by (I, 9.5.8). Since the restriction of q_2 to T_k is an isomorphism onto W_k, the restriction of g to X''_k will be an immersion into W_k, and our claim will be proven. Let v_k be the canonical injection $U' \to X \times_S W_k$; we have to show that there exists a morphism $w_k : U' \to W_k$ such that $v_k = \Gamma_{u_k} \circ w_k$. By the definition of the product, it suffices to prove that $q_1 \circ v_k = u_k \circ q_2 \circ v_k$ (I, 3.2.1), or, by composing on the right with the isomorphism $\psi' : U \to U'$, that $q_1 \circ \psi =$ II | 109
$u_k \circ q_2 \circ \psi$. But since $q_1 \circ \psi = j$ and $q_2 \circ \psi = \phi$, our claim follows from the commutativity of (5.6.1.1), taking into account the definition of u_k.

(E) It is clear that since U, and thus U', is irreducible, so is the X' from the preceding construction, and the morphism f is thus *birational* (I, 2.2.9). If in addition X is reduced, then so is U', and hence X' is also reduced (I, 9.5.9). This finishes the proof.

□

Corollary (5.6.2). — *Suppose that one of the hypotheses,* (a) *and* (b), *of* (5.6.1) *is satisfied. For X to be proper over S, it is necessary and sufficient for there to exist a projective scheme X' over S, and a surjective S−morphism $f : X' \to X$ (which is thus projective, by* (5.5.5, v)*). Whenever this is the case, we can further choose f to be such that there exists a dense open subset U of X for which the restriction of f to $f^{-1}(U)$ is an isomorphism $f^{-1}(U) \simeq U$, and for which $f^{-1}(U)$ is dense in X'. If in addition X is irreducible (resp. reduced), then we can assume that X' is also irreducible (resp. reduced); when X and X' are irreducible, f is a birational morphism.*

PROOF. The condition is sufficient, by (5.5.3) and (5.4.3, ii). It is necessary because, with the notation of (5.6.1), if X is proper over S, then X' is proper over S, because it is projective over X, and thus proper over X (5.5.3), and our claim follows from (5.4.2, ii); furthermore, since X' is quasi-projective over S, it is projective over S, by (5.5.3). □

Corollary (5.6.3). — *Let S be a locally Noetherian prescheme, and X an S-scheme of finite type over S, with structure morphism $f_0 : X \to S$. For X to be proper over S, it is necessary and sufficient that, for every morphism* of finite type *$S' \to S$, $(f_0)_{(S')} : X_{(S')} \to S'$ be a closed morphism. It even suffices for this condition to be verified only for every S-prescheme of the form $S' = S \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_n]$ (where the T_i are indeterminates).*

PROOF. The condition being clearly necessary, we now show that it is sufficient. Since the question is local on S and S' (5.4.1), we can suppose that S and S' are affine and Noetherian. By Chow's lemma, there exists a projective S-scheme P, an immersion $j : X' \to P$, and a surjective projective morphism $f : X' \to X$, such that the diagram

$$\begin{tikzcd} X \arrow[d, "f_0"'] & X' \arrow[l, "f"'] \arrow[d, "j"] \\ S & P \arrow[l, "r"] \end{tikzcd}$$

commutes. Since P is of finite type over S, the first hypothesis implies that the projection $q_2 : X \times_S P \to P$ is a *closed* morphism. But the immersion j is the composition of q_2 and the morphism $f \times 1$ from $X' \times_S P$ to $X \times_S P$; but f, being projective, is proper (5.5.3), and so $f \times 1$ is closed. We thus conclude that j is a closed immersion, and thus proper (5.4.2, i). Furthermore, the structure morphism $r : P \to S$ is projective, and thus proper (5.5.3), so $f_0 \circ f = r \circ j$ is proper (5.4.2, ii); finally, since f is surjective, f_0 is proper, by (5.4.3).

To prove the proposition using only the second, weaker hypothesis (where S' is of the form $S \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_n]$), it suffices to show that it implies the first. But, if S' is affine and of finite type over $S = \operatorname{Spec}(A)$, then we have $S' = \operatorname{Spec}(A[c_1, \ldots, c_n])$ (I, 6.3.3), and S' is thus isomorphic to a closed II | 110
subprescheme of $S'' = \operatorname{Spec}(A[T_1, \ldots, T_n])$ (where the T_i are indeterminates). In the commutative

diagram

$$\begin{array}{ccc} X \times_S S' & \xrightarrow{1_X \times j} & X \times_S S'' \\ {\scriptstyle (f_0)_{(S')}} \downarrow & & \downarrow {\scriptstyle (f_0)_{(S'')}} \\ S' & \xrightarrow{j} & S'' \end{array}$$

both j and $1_X \times j$ are closed immersions (**I**, 4.3.1), and $(f_0)_{(S')}$ is closed by hypothesis; thus $(f_0)_{(S'')}$ is also closed. □

§6. INTEGRAL MORPHISMS AND FINITE MORPHISMS

6.1. Preschemes integral over another prescheme

Definition (6.1.1). — Let X be an S-prescheme, with structure morphism $f : X \to S$. We say that X is *integral over* S, or that f is an *integral morphism*, if there exists a cover (S_α) of S by affine opens such that, for all α, the induced prescheme $f^{-1}(S_\alpha)$ is an affine scheme whose ring B_α is an integral algebra over the ring A_α of S_α. We say that X is *finite over* S, or that f is a *finite morphism* if X is integral and of finite type over S.

If S is affine of ring A, then we also say "integral (resp. finite) over A" instead of "integral (resp. finite) over S".

(6.1.2). It is clear that, if X is integral over S, then it is *affine* over S. For an affine prescheme X over S to be integral (resp. finite) over S it is necessary and sufficient that the associated quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{A}(X)$ be such that there exist a cover (S_α) of S by affine opens having the property that, for all α, $\Gamma(S_\alpha, \mathscr{A}(X))$ is an integral (resp. integral and of finite type) algebra over $\Gamma(S_\alpha, \mathscr{O}_S)$. A quasi-coherent $\mathscr{O}_S$-algebra with this property is said to be *integral* (resp. *finite*) over $\mathscr{O}_S$. Giving an integral (resp. finite) prescheme over S is thus (1.3.1) the same as giving a quasi-coherent $\mathscr{O}_S$-algebra that is integral (resp. finite) over $\mathscr{O}_S$. Note that a quasi-coherent $\mathscr{O}_S$-algebra $\mathscr{B}$ is finite if and only if it is an *$\mathscr{O}_S$-module of finite type* (**I**, 1.3.9); it is equivalent to say that $\mathscr{B}$ is an *integral* $\mathscr{O}_S$-algebra *of finite type*, since an algebra that is integral and of finite type over a ring A is an A-module of finite type.

Proposition (6.1.3). — *Let S be a locally Noetherian prescheme. For an affine prescheme X over S to be finite over S, it is necessary and sufficient that the $\mathscr{O}_S$-algebra $\mathscr{A}(X)$ be coherent.*

PROOF. Taking the preceding remark into account, this reduces to noting that, if S is locally Noetherian, then the quasi-coherent $\mathscr{O}_S$-modules of finite type are exactly the coherent $\mathscr{O}_S$-modules (**I**, 1.5.1). □

Proposition (6.1.4). — *Let X be an integral (resp. finite) prescheme over S, with structure morphism $f : X \to S$. Then, for every affine open $U \subset S$ of ring A, $f^{-1}(U)$ is an affine scheme whose ring B is an integral (resp. finite) algebra over A.*

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PROOF. We first prove the following lemma:

Lemma (6.1.4.1). — *Let A be a ring, M an A-module, and $(g_i)_{1 \leqslant i \leqslant m}$ a finite system of elements of A such that the $D(g_i)$ (for $1 \leqslant i \leqslant m$) cover* $\operatorname{Spec}(A)$. *If, for all i, M_{g_i} is an A_{g_i}-module of finite type, then M is an A-module of finite type.*

PROOF. We can assume that M_{g_i} admits a finite system of generators (m_{ij}/g_i^n) with $m_{ij} \in M$, with n the same for all indices i. We will show that the m_{ij} for a system of generators of M. By hypothesis, for each i, there exist $a_{ij} \in A$ and some integer p (independent of i) such that, in M_{g_i}, $m/1 = (\sum_i a_{ij} m_{ij})/g_i^p$; this implies that there exists an integer $r \geqslant p$ such that, for all i, we have $g_i^r m \in M'$. But, since the $D(g_i^r) = D(g_i)$ cover $\operatorname{Spec}(A)$, the ideal of A generated by the g_i^r is equal to A, or, in other words, there exist elements $a_i \in A$ such that $\sum_i a_i g_i^r$; then $m = (\sum_i a_i g_i^r) m \in M'$, whence the lemma. □

Now we already know (1.3.2) that $f^{-1}(U)$ is affine. If ϕ is the homomorphism $A \to B$ corresponding to f, then there exists a finite cover of U by opens $D(g_i)$ (where $g_i \in A$) such that, if $h_i = \phi(g_i)$, B_{h_i} is an integral (resp. integral and finite) algebra over A_{g_i}. Indeed, there exists a cover of U by affine opens $V_\alpha \subset U$ such that, if $A_\alpha = A(V_\alpha)$, $B_\alpha = A(f^{-1}(V_\alpha))$ is an integral (resp. finite) algebra over A_α. Every $x \in U$ belongs to some V_α, so there exists $g \in A$ such that $x \in D(g) \subset V_\alpha$; if g_α is the image of g in A_α, then $A(D(g)) = A_g = (A_\alpha)_{g_\alpha}$; let $h = \phi(g)$, and let h_α be the image of g_α in B_α; we have

$$A(D(h)) = B_h = (B_\alpha)_{h_\alpha}$$

and, since B_α is integral (resp. finite) over A_α, $(B_\alpha)_{h_\alpha}$ is integral (resp. finite) over $(A_\alpha)_{g_\alpha}$. It now suffices to use the fact that U is quasi-compact to obtain the desired cover.

If we suppose first of all that the B_{h_i} are integral and finite over the A_{g_i}, then since B_{h_i} can also be written as B_{g_i} as an A_{g_i}-module, Lemma (6.1.4.1) shows that, in this case, B is an A-module of finite type.

Now suppose only that each B_{h_i} is integral over A_{g_i}; let $b \in B$, and let C be the sub-A-algebra of B generated by b. For all i, C_{h_i} is the algebra over A_{g_i} generated by $b/1$ in B_{h_i}; it follows from the hypothesis that each C_{h_i} is an A_{g_i}-module of finite type, and so (6.1.4.1) C is an A-module of finite type, which proves that B is integral over A. □

Proposition (6.1.5). —

(i) *A closed immersion is finite (and* a fortiori *integral).*
(ii) *The composition of two finite (resp. integral) morphisms is finite (resp. integral).*
(iii) *If $f : X \to Y$ is a finite (resp. integral) S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is finite (resp. integral) for any base extension $S' \to S$.*
(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are finite (resp. integral) S-morphisms, then $f \times_S g : X \times_S Y \to X' \times_S Y'$ is finite (resp. integral).*
(v) *If $f : X \to Y$ and $g : Y \to Z$ are morphisms such that $g \circ f$ is finite (resp. integral), if g is separated, then f is finite (resp. integral).*
(vi) *If $f : X \to Y$ is a finite (resp. integral) morphism, then f_{red} is finite (resp. integral).*

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PROOF. By (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). To prove that a closed immersion $X \to S$ is finite, we can restrict to the case where $S = \operatorname{Spec}(A)$, and everything then follows from noting that a quotient ring $A/\mathfrak{J}$ is a monogeneous A-module. To prove that the composition of two finite (resp. integral) morphism $X \to Y$, $Y \to Z$ is finite (resp. integral), we can again assume that Z (and thus X and Y (1.3.4)) is affine, and then the claim is equivalent to saying that, if B is a finite (resp. integral) A-algebra and C a finite (resp. integral) B-algebra, then C is a finite (resp. integral) A-algebra, which is immediate. Finally, to prove (iii), we can restrict to the case where $S = Y$, since $X_{(S')}$ can be identified with $X \times_Y Y_{(S')}$ (**I**, 3.3.11); we can further suppose that $S = \operatorname{Spec}(A)$ and $S' = \operatorname{Spec}(A')$; then X is affine of ring B (1.3.4), and $X_{(S')}$ affine of ring $A' \otimes_A B$, and it suffices to note that, if B is a finite (resp. integral) A-algebra, then $A' \otimes_A B$ is a finite (resp. integral) A'-algebra. □

We also note that, if X and Y are S-preschemes that are finite (resp. integral) over S, then their *sum* $X \sqcup Y$ is a finite (resp. integral) prescheme over S, since this reduces to showing that, if B and C are finite (resp. integral) A-algebras over A, then so too is $B \times C$.

Corollary (6.1.6). — *If X is an integral (resp. finite) prescheme over S, then, for every open $U \subset S$, $f^{-1}(U)$ is integral (resp. finite) over U.*

PROOF. This is a particular case of (6.1.5, (iii)). □

Corollary (6.1.7). — *Let $f : X \to Y$ be a finite morphism. Then, for all $y \in Y$, the fibre $f^{-1}(y)$ is a finite algebraic scheme over $k(y)$, and* a fortiori *its underlying space is discrete and finite.*

PROOF. Indeed, as a $k(y)$-prescheme, $f^{-1}(y)$ can be identified with $X \times_Y \operatorname{Spec}(k(y))$ (**I**, 3.6.1), which is finite over $\operatorname{Spec}(k(y))$ (6.1.5, (iii)); it is thus an affine scheme whose ring is an algebra of finite rank over $k(y)$ (6.1.4). The corollary then follows from (**I**, 6.4.4). □

Corollary (6.1.8). — *Let X and S be integral preschemes, and $f : X \to S$ a* dominant *morphism. If f is integral (resp. finite) then the field $R(X)$ of rational functions on X is algebraic (resp. algebraic of finite degree) over the field $R(S)$ of rational functions on S.*

PROOF. Let s be the generic point of S; the $k(s)$-prescheme $f^{-1}(s)$ is integral (resp. finite) over $\operatorname{Spec}(k(s))$ (6.1.5, (iii)) and contains, by hypothesis, the generic point x of X; since the local ring of x in $f^{-1}(s)$ is equal to $k(x)$ (**I**, 3.6.5), and is thus a local ring of an integral (resp. finite) algebra over $k(s)$ (6.1.4), whence the corollary. □

Remark (6.1.9). — The hypothesis that g be *separated* is essential for the validity of (6.1.5, (v)): if Y is not separated over S, then the identity 1_Y is the composite morphism $Y \xrightarrow{\Delta_Y} Y \times_Z Y \xrightarrow{p_1} Y$, but Δ_Y is not an integral morphism, as follows from (6.1.10):

Proposition (6.1.10). — *Every integral morphism is universally closed.*

PROOF. Let $f : X \to Y$ be an integral morphism; by (6.1.5, (iii)), it suffices to show that f is *closed*. Let Z be a closed subset of X; then there exists a subprescheme of X whose underlying space is Z (**I**, 5.4.1), and it thus follows from (6.1.5, (i) and (ii)) that it suffices to prove that $f(X)$ is *closed* in Y. By (6.1.5, (vi)), we can suppose that X and Y are *reduced*; further, if T is the closed reduced subprescheme of Y whose underlying space is $\overline{f(X)}$ (**I**, 5.2.1) then we know that f factors as $X \to T \xrightarrow{j} Y$, where j is the injection morphism (**I**, 5.2.2), and since j is separated (**I**, 5.5.1, (i)), it follows from (6.1.5, (v)) that g is an integral morphism. We can thus suppose that $f(X)$ is *dense* in Y. Finally, since the question is local on Y, we can restrict to the case where $Y = \operatorname{Spec}(A)$. Then $X = \operatorname{Spec}(B)$, where B is an A-algebra that is integral over A (6.1.4); furthermore, A is reduced (**I**, 5.1.4), and the hypothesis that $f(X)$ be dense in Y implies that the homomorphism $\phi : A \to B$ corresponding to f is *injective* (**I**, 1.2.7). Under these conditions, saying that $f(X) = Y$ implies that every prime ideal of A is the intersection with A of a prime ideal of B, which is exactly the first theorem of Cohen–Seidenberg ([**?**, t. I, p. 257, th. 3]). □ II | 113

Corollary (6.1.11). — *Every finite morphism $f : X \to Y$ is projective.*

PROOF. Since f is affine, $\mathscr{O}_X$ is a *very ample* $\mathscr{O}_X$-module with respect to f (5.1.2); furthermore, $f_*(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_Y$-module *of finite type* (6.1.2); finally, f is separated, of finite type, and universally closed (6.1.10), and thus satisfies the conditions of criterion (5.5.4, (i)). □

Proposition (6.1.12). — *Let $f : X' \to X$ be a finite morphism, and let $\mathscr{B} = f_*(\mathscr{O}_{X'})$ (which is a quasi-coherent $\mathscr{O}_X$-algebra, and a $\mathscr{O}_X$-module of finite type). Let $\mathscr{F}'$ be a quasi-coherent $\mathscr{O}_{X'}$-module; for $\mathscr{F}'$ to be locally free of rank r, it is necessary and sufficient that $f_*(\mathscr{F}')$ be a locally free $\mathscr{B}$-module of rank r.*

PROOF. It is clear that, if $f_*(\mathscr{F}')|U$ is isomorphic to $\mathscr{B}^r|U$ (where $U \subset X$ is open), then $\mathscr{F}'|f^{-1}(U)$ is isomorphic to $\mathscr{O}_{X'}^r|f^{-1}(U)$ (1.4.2). Conversely, suppose that $\mathscr{F}'$ is locally free of rank r; we will show that $f_*(\mathscr{F}')$ is locally isomorphic to $\mathscr{B}^r$ as a $\mathscr{B}$-module. Let x be a point of X; as U runs over a fundamental system of affine neighbourhoods of x, $f^{-1}(U)$ runs over a fundamental system of affine neighbourhoods (1.2.5) of the finite set $f^{-1}(x)$, since f is closed (6.1.10). The proposition then follows from the following lemma: □

Lemma (6.1.12.1). — *Let Y be a prescheme, $\mathscr{E}$ a locally free $\mathscr{O}_Y$-module of rank r, and Z a finite subset of Y contained inside some affine open V. Then there exists a neighbourhood $U \subset V$ of Z such that $\mathscr{E}|U$ is isomorphic to $\mathscr{O}_Y^r|U$.*

PROOF. We can evidently assume that Y is affine; for all $z_i \in Z$, there exists in the closure $\overline{\{z_i\}}$ at least one closed point z_i' (**0**, 2.1.3); if Z' is the set of the z_i' then every neighbourhood of Z' is a neighbourhood of Z, and we can thus assume that Z is discrete and closed in Y. Consider the closed reduced subprescheme of Y that has Z has its underlying space (**I**, 5.2.1) and let $j : Z \to Y$ be the canonical injection; $j^*(\mathscr{E}) = \mathscr{E} \otimes_Y \mathscr{O}_Z$ is locally free of rank r on the discrete scheme Z, and is thus isomorphic to $\mathscr{O}_Z^r$; in other words, there exist r sections s_i (for $1 \leqslant i \leqslant r$) of $\mathscr{E} \otimes_Y \mathscr{O}_Z$ over Z such that the homomorphism $\mathscr{O}_Z^r \to \mathscr{E} \otimes_Y \mathscr{O}_Z$ defined by these sections is bijective. But $Y = \operatorname{Spec}(A)$ is affine, Z is defined by an ideal $\mathfrak{J}$ of A, and we have $\mathscr{E} = \widetilde{M}$, where M is an A-module; the s_i are elements of $M \otimes_A (A/\mathfrak{J})$ and are thus images of r elements $t_i \in M = \Gamma(Y, \mathscr{E})$. For all $z_j \in Z$ there is thus a II | 114 neighbourhood V_j of z_j such that the restrictions of the t_i to V_j defined an isomorphism $\mathscr{O}_Y^r|V_j \to \mathscr{E}|V_j$ (**0**, 5.5.4); the neighbourhood U given by the union of the V_j is the desired neighbourhood. ☐

Proposition (6.1.13). — *Let $g : X' \to X$ be an integral morphism of preschemes, Y a normal locally integral prescheme, and f a rational map from Y to X' such that $g \circ f$ is an everywhere defined rational map* (**I**, 7.2.1). *Then f is everywhere defined.*

Proof. If f_1 and f_2 are morphisms (from dense open subsets of Y to X') in the class f, then it is clear that $g \circ f_1$ and $g \circ f_2$ are equivalent morphisms, which justifies the notation $g \circ f$ for their equivalence class. We recall also that, if we further suppose Y to be *locally Noetherian*, then the hypothesis that Y is normal already implies that Y is locally integral (**I**, 6.1.13).

To prove the proposition, note first of all that the question is local on Y, and so we can suppose that there exists in the class $g \circ f$ a *morphism* $h : Y \to X$. Consider the inverse image $Y' = X'_{(h)} = X'_{(Y)}$, and note that the morphism $g' = g_{(Y)} : Y' \to Y$ is *integral* (6.1.5, (iii)). Given the correspondence between rational maps from Y to X' and rational Y-sections of Y' (**I**, 7.1.2), we see that it suffices to prove the specific case of (6.1.13) where $X = Y$; in other words, the following: ☐

Corollary (6.1.14). — *Let X be a normal locally integral prescheme, $g : X' \to X$ an* integral *morphism, and f a rational X-section of X'. Then f is everywhere defined.*

Proof. Since the question is local on X, we can assume that X is integral, and then f is identified with a morphism from an open U of X to X' (**I**, 7.2.2) that is a U-section of $g^{-1}(U)$. Since g is separated, f is a closed immersion from U into $g^{-1}(U)$ (**I**, 5.4.6); let Z be the closed subprescheme of $g^{-1}(U)$ associated to f (**I**, 4.2.1), which is isomorphic to U, and thus integral; let X_1 be the reduced subprescheme of X' whose underlying space is the closure $\overline{Z}$ of Z in X' (**I**, 5.2.1); then Z is an induced subprescheme on an open of X_1 (**I**, 5.2.3), and, since it is irreducible, so too is X_1, which is thus integral. The morphism f can then be considered as a rational X-section of X_1; since the restriction of g to X_1 is an integral morphism (6.1.5, (i) and (ii)), we can finally reduce to proving (6.1.14) in the specific case where $X' = X_1$; in other words, the following: ☐

Corollary (6.1.15). — *Let X be a normal integral prescheme, X' an integral prescheme, and $g : X' \to X$ an* integral *morphism. If there exists a rational X-section f of X', then g is an isomorphism.*

Proof. Since the question is local on X, we can assume that X is affine of integral ring A, and then X' is affine of ring A' with A' integral over A (6.1.4) and integral; furthermore, the argument of (6.1.14) shows that there exists a dense open of X that is isomorphic to a dense open of X', and so A and A' have the same field of fractions. Also, by (**I**, 8.2.1.1), and the hypothesis that the $\mathscr{O}_x$ are integrally closed, the ring A is integrally closed, and so $A' = A$, which finishes the proof of (6.1.13) ☐

6.2. Quasi-finite morphisms

Proposition (6.2.1). — *Let $f : X \to Y$ be morphism locally of finite type, and x a point of X. Then the following conditions are equivalent:* **II** | 115

a) The point x is isolated in its fibre $f^{-1}(f(x))$.
b) The ring $\mathscr{O}_x$ is a quasi-finite $\mathscr{O}_{f(x)}$-module (**0**, 7.4.1).

Proof. Since the question is clearly local on X and on Y, we can assume that $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine, with A a B-algebra of finite type (**I**, 6.3.3). Furthermore, we can replace X by $X \times_Y \operatorname{Spec}(\mathscr{O}_{f(x)})$ without changing either the fibre $f^{-1}(f(x))$ or the local ring $\mathscr{O}_x$ (**I**, 3.6.5); we can thus assume that B is a local ring (equal to $\mathscr{O}_{f(x)}$); if $\mathfrak{n}$ is the maximal ideal of B, then $f^{-1}(f(x))$ is an affine scheme of ring $A/\mathfrak{n}A$, of finite type over $k(f(x)) = B/\mathfrak{n}$ (**I**, 6.4.11). With this, if (a) is satisfied then we can further suppose that $f^{-1}(f(x))$ consists of the single point x; thus $A/\mathfrak{n}A$ is of finite rank over $B/\mathfrak{n}$ (**I**, 6.4.4), or, in other words, A is a quasi-finite B-module. Conversely, if A is a quasi-finite B-module, then $f^{-1}(f(x))$ is an Artinian affine scheme, and thus discrete (**I**, 6.4.4); so x is isolated in its fibre, and this shows that (b) implies (a). ☐

Corollary (6.2.2). — *Let $f : X \to Y$ be a morphism of finite type. Then the following conditions are equivalent:*

a) *Every point $x \in X$ is isolated in its fibre $f^{-1}(f(x))$ (or, in other words, the subspace $f^{-1}(f(x))$ is* discrete*).*
b) *For all $x \in X$, the prescheme $f^{-1}(f(x))$ is a finite $k(f(x))$-prescheme.*
c) *For all $x \in X$, the ring $\mathscr{O}_x$ is a quasi-finite $\mathscr{O}_{f(x)}$-module.*

PROOF. The equivalence of (a) and (c) follows from (6.2.1). Since $f^{-1}(f(x))$ is an algebraic $k(f(x))$-prescheme (**I**, 6.4.11), the equivalence of (a) and (b) follows from (**I**, 6.4.4). □

Definition (6.2.3). — If $f : X \to Y$ is a morphism of finite type satisfying the equivalent conditions of (6.2.2), we say that f is *quasi-finite*, or that X is *quasi-finite over Y*.

It is clear that every *finite* morphism is quasi-finite (6.1.8).

Proposition (6.2.4). —

(i) *An immersion $X \to Y$ that is closed, or such that X is Noetherian, is a quasi-finite morphism.*
(ii) *If $f : X \to Y$ and $g : Y \to Z$ are quasi-finite morphisms, then $g \circ f$ is quasi-finite.*
(iii) *If X and Y are S-preschemes, and $f : X \to Y$ a quasi-finite S-morphism, then $f_{(S')} : X_{(S')} \to Y_{(S')}$ is quasi-finite for any base extension $g : S' \to S$.*
(iv) *If $f : X \to Y$ and $g : X' \to Y'$ are quasi-finite S-morphisms, then*
$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$
is quasi-finite.
(v) *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms such that $g \circ f$ is quasi-finite; if, further, g is separated, or X is Noetherian, or $X \times_Z Y$ is locally Noetherian, then f is quasi-finite.*
(vi) *If f is quasi-finite, then f_{red} is quasi-finite.*

PROOF. If $f : X \to Y$ is an immersion, then every fibre consists of a single point, and claim (i) then follows from ((**I**, 6.3.4, (i)) and (**I**, 6.3.5)). To prove (ii), note first of all that $h = g \circ f$ is of finite type (**I**, 6.3.4, (ii)); furthermore, if $z = h(x)$ and $y = f(x)$, then y is isolated in $g^{-1}(z)$, and so there exists an open neighbourhood V of y in Y that does not meet any point of $g^{-1}(z)$ apart from y; thus $f^{-1}(V)$ is an open neighbourhood of x that does not meet any of the $f^{-1}(y')$, where $y' \neq y$ is in $g^{-1}(z)$; since x is isolated in $f^{-1}(y)$, it is isolated in $h^{-1}(z) = f^{-1}(g^{-1}(z))$. To prove (iii), we can **II** | 116
reduce to the case where $Y = S$ (**I**, 3.3.11); we again note first of all that $f' = f_{(S')}$ is of finite type (**I**, 6.3.4, (iii)); also, if $x' \in X' = X_{(S')}$, and if we set $y' = f'(x')$ and $y = g(y')$, then $f'^{-1}(y')$ can be identified with $f^{-1}(y) \otimes_{k(y)} k(y')$ (**I**, 3.6.5); since $f^{-1}(y)$ is of finite rank over $k(y)$ by hypothesis, $f'^{-1}(y')$ is of finite rank over $k(y')$, and thus discrete. Claims (iv), (v), and (vi) follows from (i), (ii), and (iii) by the general method (**I**, 5.5.12), except for when the hypotheses in (v) are not "g is separated"; in these cases, we remark first of all that, if x is isolated in $f^{-1}(g^{-1}(g(f(x))))$, then it is *a fortiori* isolated in $f^{-1}(f(x))$; the fact that f is of finite type then follows from (**I**, 6.3.6). □

Proposition (6.2.5). — *Let A be a* complete *local Noetherian ring, X an A-scheme locally of finite type, and x a point of X over the closed point y of $Y = \operatorname{Spec}(A)$. Suppose that x is isolated in its fibre $f^{-1}(y)$ (where f is the structure morphism $X \to Y$). Then $\mathscr{O}_x$ is an A-module of finite type, and X is Y-isomorphic to the sum* (**I**, 3.1) *of $X' = \operatorname{Spec}(\mathscr{O}_x)$ (which is a finite Y-scheme) and an A-scheme X''.*

PROOF. It follows from (6.2.1) that $\mathscr{O}_x$ is a quasi-finite A-module. Since $\mathscr{O}_x$ is Noetherian (**I**, 6.3.7), and the homomorphism $A \to \mathscr{O}_x$ is local, the hypothesis that A is *complete* implies that $\mathscr{O}_x$ is an A-module *of finite type* (**0**, 7.4.3). Let $X' = \operatorname{Spec}(\mathscr{O}_x)$ be the local scheme of X at the point x (**I**, 2.4.1), and let $g : X' \to X$ be the canonical morphism. Since the composite $X' \xrightarrow{g} X \xrightarrow{f} Y$ is finite (6.1.1), and since f is separated, g is finite (6.1.5, (v)), and so $g(X')$ is *closed* in X (6.1.10); also, since g is of finite type, g is a local isomorphism at the closed point x' of X' by the definition of g and (**I**, 6.5.4); but since X' is the only open neighbourhood of x', this implies that g is an open immersion, and so $g(X')$ is also *open* in X, which finishes the proof. □

Corollary (6.2.6). — *Let A be a* complete *local Noetherian ring, $Y = \operatorname{Spec}(A)$, and $f : X \to Y$ a separated quasi-finite morphism. Then X is Y-isomorphic to a sum $X' \sqcup X''$, where X' is a finite Y-scheme, and X'' is a quasi-finite Y-scheme such that, if y is the closed point of Y, then $X'' \cap f^{-1}(y) = \varnothing$.*

PROOF. Indeed, the fibre $f^{-1}(y)$ is finite and discrete by hypothesis, and the corollary then follows, by induction on the number of points in this fibre, from (6.2.5). □

Remark (6.2.7). — In Chapter V, we will see that, if Y is *locally Noetherian*, then a separated quasi-finite morphism $X \to Y$ is necessarily *quasi-affine*.

6.3. Integral closure of a prescheme

Proposition (6.3.1). — *Let $(X, \mathscr{A})$ be a ringed space, $\mathscr{B}$ a (commutative) $\mathscr{A}$-algebra, and f a section of $\mathscr{B}$ over X. Then the following properties are equivalent:*

a) *The sub-$\mathscr{A}$-module of $\mathscr{B}$ generated by the f^n for $n \geqslant$ (**0**, 5.1.1) is of finite type.*
b) *There exists a sub-$\mathscr{A}$-algebra $\mathscr{C}$ of $\mathscr{B}$ that is an $\mathscr{A}$-module of finite type and such that $f \in \Gamma(X, \mathscr{C})$.*
c) *For all $x \in X$, f_x is integral over the fibre $\mathscr{A}_x$.*

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PROOF. Since the sub-$\mathscr{A}$-module of $\mathscr{B}$ generated by the f^n is an $\mathscr{A}$-algebra, it is clear that (a) implies (b). On the other hand, (b) implies that, for all $x \in X$, the $\mathscr{A}_x$-module $\mathscr{C}_x$ is of finite type, which implies that every element of the algebra $\mathscr{C}_x$, and, in particular, f_x, is integral over $\mathscr{A}_x$. Finally, if we have, for some $x \in X$, a relation of the form

$$f_x^n + (a_1)_x f_x^{n-1} + \ldots + (a_n)_x = 0$$

where the a_i (for $1 \leqslant i \leqslant n$) are sections of $\mathscr{A}$ over an open neighbourhood U of x, then the section $f^n|U + a_1 \cdot f^{n-1}|U + \ldots + a_n$ is zero over a neighbourhood $V \subset U$ of x, from which it immediately follows that the $f^k|V$ (for $k \geqslant 0$) are linear combinations of the $f^j|V$ (for $0 \leqslant j \leqslant n-1$) with coefficients in $\Gamma(V, \mathscr{A})$; we thus conclude that (c) implies (a). □

If the equivalent conditions of (6.3.1) are satisfied then we say that the section f is *integral over $\mathscr{A}$*.

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), there exists a (unique) sub-$\mathscr{A}$-module $\mathscr{A}'$ of $\mathscr{B}$ such that, for all $x \in X$, $\mathscr{A}'_x$ is the set of germs $f_x \in \mathscr{B}_x$ that are integral over $\mathscr{A}_x$. For every open $U \subset X$, the sections of $\mathscr{A}'$ over U are the sections $f \in \Gamma(U, \mathscr{B})$ that are integral over $\mathscr{A}|U$.*

PROOF. The existence of $\mathscr{A}'$ is immediate, by taking $\Gamma(U, \mathscr{A}')$ to be the set of $f \in \Gamma(U, \mathscr{B})$ such that f_x is integral over $\mathscr{A}_x$ for all $x \in U$. The second claim follows immediately from (6.3.1). □

It is clear that $\mathscr{A}'$ is a sub-$\mathscr{A}$-algebra of $\mathscr{B}$; we say that it is the *integral closure of $\mathscr{A}$ in $\mathscr{B}$*.

(6.3.3). Let $(X, \mathscr{A})$ and $(Y, \mathscr{B})$ be ringed spaces, and let

$$g = (\psi, \theta) : X \longrightarrow Y$$

be a morphism. Let $\mathscr{C}$ (resp. $\mathscr{D}$) be an $\mathscr{A}$-algebra (resp. $\mathscr{B}$-algebra), and let

$$u : \mathscr{D} \longrightarrow \mathscr{C}$$

be a g-morphism (**0**, 4.4.1). Then, if $\mathscr{A}'$ (resp. $\mathscr{B}'$) is the integral closure of $\mathscr{A}$ (resp. $\mathscr{B}$) in $\mathscr{C}$ (resp. $\mathscr{D}$), the *restriction* of u to $\mathscr{B}'$ is a g-morphism

$$u' : \mathscr{B}' \longrightarrow \mathscr{A}'.$$

Indeed, if j is the canonical injection $\mathscr{B}' \to \mathscr{D}$, then it suffices to show that

$$v = u^\sharp \circ g^*(j) : j^*(\mathscr{B}') \longrightarrow \mathscr{C}'$$

sends $g^*(\mathscr{B}')$ to $\mathscr{A}'$. But an element of $g^*(\mathscr{B}')_x = \mathscr{B}'_{\psi(x)} \otimes_{\mathscr{B}_{\psi(x)}} \mathscr{A}_x$ is integral over $\mathscr{A}_x$ by the definition of $\mathscr{B}'$, and thus so too is its images under v_x, which proves the claim.

Proposition (6.3.4). — *Let X be a prescheme, and $\mathscr{A}$ a quasi-coherent $\mathscr{O}_X$-algebra. Then the integral closure $\mathscr{O}'_X$ of $\mathscr{O}_X$ in $\mathscr{A}$ is a quasi-coherent $\mathscr{O}_X$-algebra, and, for every affine open U of X, $\Gamma(U, \mathscr{O}'_X)$ is the integral closure of $\Gamma(U, \mathscr{O}_X)$ in $\Gamma(U, \mathscr{A})$.*

PROOF. We can restrict to the case where $X = \operatorname{Spec}(B)$ is affine, and $\mathscr{A} = \widetilde{A}$, where A is a B-algebra; let B' be the integral closure of B in A. Everything then reduces to proving that, for all $x \in X$, an element of A_x which is integral over B_x necessarily belongs to B'_x, which follows from the fact that, for a commutative ring C, the operations of taking the integral closure in a C-algebra and passing to a ring of fractions (with respect to a multiplicative subset of C) commute [SZ60, t. I, pp. 261 and 257]. □ II | 118

The X-scheme $X' = \operatorname{Spec}(\mathscr{O}'_X)$ is then called the *integral closure of X with respect to $\mathscr{A}$* (or *with respect to* $\operatorname{Spec}(\mathscr{A})$); it is clear that X' is *integral* over X (6.1.2).

We immediately deduce from (6.1.4) that, if $f : X' \to X$ is the structure morphism, then, for every open U of X, $f^{-1}(U)$ is the *integral closure of the prescheme induced by X on U, with respect to $\mathscr{A}|U$.*

(6.3.5). Let X and Y be preschemes, $f : X \to Y$ a morphism, $\mathscr{A}$ (resp. $\mathscr{B}$) a quasi-coherent $\mathscr{O}_X$-algebra (resp. quasi-coherent $\mathscr{O}_Y$-algebra), and let $u : \mathscr{B} \to \mathscr{A}$ be an f-morphism. We have seen (6.3.3) that we obtain an f-morphism $u' : \mathscr{O}'_Y \to \mathscr{O}'_X$, where $\mathscr{O}'_X$ (resp. $\mathscr{O}'_Y$) is the integral closure of $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$) in $\mathscr{A}$ (resp. $\mathscr{B}$). Then, if X' (resp. Y') is the integral closure of X (resp. Y) with respect to $\mathscr{A}$ (resp. $\mathscr{B}$), we canonically obtain from u a morphism $f' = \operatorname{Spec}(u') : X' \to Y'$ (1.5.6) such that the diagram

(6.3.5.1)
$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d] & Y' \arrow[d] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

commutes.

(6.3.6). Suppose that X has only a *finite* number of irreducible components X_i (for $1 \leqslant i \leqslant r$), with generic points ξ_i, and consider, in particular, the integral closure of X with respect to some quasi-coherent $\mathscr{R}(X)$-*algebra* $\mathscr{A}$ (quasi-coherent as either an $\mathscr{O}_X$-algebra or as an $\mathscr{R}(X)$-algebra, since the two are equivalent). We know (**I**, 7.3.5) that $\mathscr{A}$ is a direct sum of r quasi-coherent $\mathscr{O}_X$-algebras $\mathscr{A}_i$, where the support of $\mathscr{A}_i$ is contained inside X_i, and the sheaf induced by $\mathscr{A}_i$ on X_i is a simple sheaf whose fibre A_i is an algebra over $\mathscr{O}_{\xi_i}$. It is then clear (6.3.4) that the integral closure $\mathscr{O}'_X$ of $\mathscr{O}_X$ in $\mathscr{A}$ is the *direct sum* of the integral closures $\mathscr{O}_X^{(i)}$ of $\mathscr{O}_X$ in each of the $\mathscr{A}$, and thus the integral closure $X' = \operatorname{Spec}(\mathscr{O}'_X)$ of X with respect to $\mathscr{A}$ is the X-scheme given by the *sum* of the $\operatorname{Spec}(\mathscr{O}_X^{(i)}) = X'_i$ (for $1 \leqslant i \leqslant r$).

Suppose further that the $\mathscr{O}_X$-algebra $\mathscr{A}$ is *reduced*, or, equivalently, that each of the algebras A_i are reduced, and can thus be considered as an algebra over the *field* $k(\xi_i)$ (equal to the field of rational functions of the reduced subprescheme of X that has X_i as its underlying space); then (1.3.8) each of the X'_i is a *reduced* X-scheme, and X' is also the integral closure of X_{red}. Suppose further that each of the algebras A_i is a *direct sum of a finite number of fields* K_{ij} (for $1 \leqslant j \leqslant s_i$); if $\mathscr{K}_{ij}$ is the sub-algebra of $\mathscr{A}_i$ corresponding to K_{ij}, then it is clear that $\mathscr{O}_X^{(i)}$ is the *direct sum* of the integral closures $\mathscr{O}_X^{(ij)}$ of $\mathscr{O}_X$ in each of the $\mathscr{K}_{ij}$. Then X'_i is the *sum* of the X-schemes $X'_{ij} = \operatorname{Spec}(\mathscr{O}_X^{(ij)})$ (for $1 \leqslant j \leqslant s_i$). Furthermore, under these hypotheses, and with this notation, we have:

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Proposition (6.3.7). — *Each of the X'_{ij} is an integral normal X-prescheme, and its field of rational functions $R(X'_{ij})$ is canonically identified with the algebraic closure K'_{ij} of $k(\xi_i)$ in K_{ij}.*

PROOF. By the above, we can suppose that X is integral, and so $r = 1$ and $s_1 = 1$, so that the unique algebra A_1 is a field K; let ξ be the generic point of X, and let $f : X \to X'$ be the structure morphism. For every non-empty affine open U of X, $f^{-1}(U)$ can be identified with the spectrum of the integral closure B'_U in the field K of the integral ring $B_U = \Gamma(Y, \mathscr{O}_X)$ (6.3.4); since the ring B'_U is integral and integrally closed, so too are the local rings of the points of its spectrum, and thus $f^{-1}(U)$ is by definition an *integral* and *normal* scheme ((**0**, 4.1.4) and (**I**, 5.1.4)). Furthermore, since

(0) is the only prime ideal of B'_U over the prime ideal (0) of B_U [**SZ60**, t. 1, p. 259], $f^{-1}(\xi)$ consists of a single point ξ', and $k(\xi')$ is the field of fractions K' of B'_U, which is exactly the algebraic closure of $k(\xi)$ in K. Finally, X' is irreducible, since, if U runs over the set of non-empty affine opens of X, then the $f^{-1}(U)$ give an open cover of X' by irreducible opens; furthermore, the intersection $f^{-1}(U \cap V)$ of any two of these opens contains ξ', and is thus non-empty, and we conclude by (**0**, 2.1.4). $\square$

Corollary (6.3.8). — *Let X be a reduced prescheme that has only a finite number of irreducible components X_i (where $1 \leqslant i \leqslant r$), and let ξ_i be the generic point of X_i. Then the integral closure X' of X with respect to $\mathscr{R}(X)$ is the sum of the r X-schemes X'_i, which are integral and normal. If $f : X' \to X$ is the structure morphism, then $f^{-1}(\xi_i)$ consists only of the generic point ξ'_i of X'_i, and we have $k(\xi'_i) = k(\xi_i)$, or, in other words, that f is birational.*

In this case, we say that X' is the *normalisation* of the reduced prescheme X; we note that f, since it is birational and integral, is *surjective* (6.1.10). Then in order to have $X' = X$, it is necessary and sufficient that X be *normal*. If X is an integral prescheme, then it follows from (6.3.8) that its normalisation X' is integral.

(6.3.9). Let X and Y be integral preschemes, $f : X \to Y$ a dominant morphism, and $L = R(X)$ (resp. $K = R(Y)$) the field of rational functions of X (resp. Y); there is a canonical injection $K \to L$ corresponding to f, and if we identify K (resp. L) with the simple sheaf $\mathscr{R}(Y)$ (resp. $\mathscr{R}(X)$), this injection is an f-morphism. Let K_1 (resp. L_1) be an extension of K (resp. L), and suppose we have a morphism $K_1 \to L_1$ such that the diagram

$$\begin{tikzcd} K_1 \arrow[r] & L_1 \\ K \arrow[u] \arrow[r] & L \arrow[u] \end{tikzcd}$$

commutes; if K_1 (resp. L_1) is considered as a simple sheaf on Y (resp. X), and thus as an $\mathscr{R}(Y)$-algebra (resp. $\mathscr{R}(X)$-algebra), this implies that $K_1 \to L_1$ is an f-morphism. With this, if X' (resp. Y') is the integral closure of X (resp. Y) with respect to L_1 (resp. K_1), then X' (resp. Y') is a normal integral prescheme (6.3.6) whose field of rational functions is canonically identified with the algebraic closure L' (resp. K') of L (resp. K) in L_1 (resp K_1), and there exists a (necessarily dominant) canonical morphism $f' : X' \to Y'$ that makes the diagram (6.3.5.1) commute. **II** | 120

The most important case is when we take $L_1 = L$, with K_1 then being an extension of K contained in L, and when we suppose X to be integral and *normal*, so that $X' = X$. The above then shows that, since X is normal, if Y' is the integral closure of Y with respect to a field $K_1 \subset L = R(X)$, then every dominant morphism $f : X \to Y$ factors as

$$f : X \xrightarrow{f'} Y' \longrightarrow Y$$

where f' is dominant; also, since the monomorphism $K_1 \to L$ is given, f' is necessarily unique, as we see by reducing to the case where X and Y are affine. We thus see that, given Y, L, and a K-monomorphism $K_1 \to L$, the integral closure Y' of Y with respect to K_1 is the solution of a *universal problem*.

Remark (6.3.10). — Consider the hypothesis of (6.3.6), and suppose further that each of the algebras A_i is *of finite rank* over $k(\xi_i)$ (which implies that A_i is a direct sum of a finite number of fields); we can then show, in certain cases, that the structure morphism $X' \to X$ is not just integral, but even *finite*. Restrict to the case where X is *reduced*; since the question is local on X, we can further suppose that X is affine of ring C, and that C has only a finite number of minimal ideals $\mathfrak{p}_i$ (for $1 \leqslant i \leqslant r$) such the that $C_i = C/\mathfrak{p}_i$ are integral; then X' is finite over X if the integral closure of each C_i in finite-dgree extension of its field of fractions is a *C-module of finite type* (6.3.4). We know that this condition is always satisfied if C is an *algebra of finite type over a field* [**SZ60**, t. I, p. 267, th. 9], or *over* **Z** [**Nag58a**, I,p. 93, th. 3], or *over a complete Noetherian local ring* [**Nag55**, p. 298]. We thus conclude that $X' \to X$ will be a finite morphism whenever X is a scheme *of finite type* over a field, or over **Z**, or over a complete Noetherian local ring.

6.4. Determinant of an endomorphism of $\mathscr{O}_X$-modules

(6.4.1). Let A be a (commutative) ring, E a free A-module of rank n, and u an endomorphism of E; recall that, to define the *characteristic polynomial* of u, we consider the endomorphism $u \otimes 1$ of the free $A[T]$-module of rank n, $E \otimes_A A[T]$ (where T is an indeterminate), and we set

(6.4.1.1) $$P(u,T) = \det(T \cdot I - (u \otimes 1))$$

(where I is the identity automorphism of $E \otimes_A A[T]$). We have

(6.4.1.2) $$P(u,T) = T^n - \sigma_1(u)T^{n-1} + \ldots + (-1)^n \sigma_n(u)$$

where $\sigma_i(u)$ is an element of A, equal to a homogeneous polynomial of degree i (with integer coefficients) with respect to the entries of the matrix of u in some arbitrary basis of E; we say that the $\sigma_i(u)$ are the *elementary symmetric functions* of u; in particular, we have that $\sigma_1(u) = \operatorname{Tr} u$ and $\sigma_n(u) = \det u$. Recall that, by the Hamilton–Cayley theorem, we have

(6.4.1.3) $$P(u,u) = u^n - \sigma_1(u)u^{n-1} + \ldots + (-1)^n \sigma_n(u) = 0$$

which can also be written as **II** | 121

(6.4.1.4) $$(\det u) \cdot I_E = uQ(u)$$

(where I_E is the identity automorphism of E), where

(6.4.1.5) $$Q(u) = (-1)^{n+1}(u^{n-1} - \sigma_1(u)u^{n-2} + \ldots + (-1)^{n-1}\sigma_{n-1}(u)).$$

Let $\phi : A \to B$ be a ring homomorphism, which makes B into an A-algebra; consider the B-module $E_{(B)} = E \otimes_A B$, which is free of rank n, and the extension $u \otimes 1$ of u to an endomorphism of $E_{(B)}$; it is immediate that we have $\sigma_i(u \otimes 1) = \phi(\sigma_i(u))$ for all indices i.

(6.4.2). Now suppose that A is an *integral* ring, with field of fractions K, and let E be an A-module *of finite type* (but not necessarily free). Let n be the *rank* of E, i.e. the rank of the free K-module $E \otimes_A K$; to every endomorphism u of E there canonically corresponds the endomorphism $u \otimes 1$ of $E \otimes_A K$; by an abuse of language, we also define the *characteristic polynomial* of u, denoted by $P(u,T)$, to be the polynomial $P(u \otimes 1, T)$, whose coefficients $\sigma_i(u \otimes 1)$ (which belong to K) are also denoted by $\sigma_i(u)$ and called the *elementary symmetric functions of* u; in particular, $\det u = \det(u \otimes 1)$ by definition. With this notation, Equations (6.4.1.3) to (6.4.1.5) make sense and still hold true, if we interpret u^j as the composite homomorphism $E \to E \otimes_A K$ of the endomorphism $u^j \otimes 1 = (u \otimes 1)^j$ of $E \otimes_A K$ and the canonical homomorphism $x \mapsto x \otimes 1$.

If F is the torsion submodule of E, and if $E_0 = E/F$, then $u(F) \subset F$, and so, by passing to quotients, u gives an endomorphism u_0 of E_0; furthermore, $E \otimes_A K$ can be identified with $E_0 \otimes_A K$, and $u \otimes 1$ with $u_0 \otimes 1$, and so $\sigma_i(u) = \sigma_i(u_0)$ for $1 \leqslant i \leqslant n$.

If E is *torsion free*, then E is canonically identified with a sub-A-module of $E \otimes_A K$, and the relation $u \otimes 1 = 0$ is equivalent to $u = 0$. If A is a free A-module, then the two definitions of $\sigma_i(u)$ given in (6.4.1) and (6.4.2) coincide, by the above, which justifies the notation used.

We also note that, if E is a *torsion module*, or, in other words, if $E_0 = \{0\}$, then the exterior algebra of E_0 consists of K, and the determinant of the unique endomorphism u_0 of E_0 is *equal to* 1.

Proposition (6.4.3). — *Let A be an integral ring, E an A-module of finite type, and u an endomorphism of E; then the elementary symmetric functions $\sigma_i(u)$ of u (and, in particular,* $\det u$*) are elements of K that are integral over A.*

Proof. Let A' be the integral closure of A; since $A'[T]$ is an integrally closed ring [**Jaf60**, p. 99], it is the integral closure of $A[T]$ in its field of fractions $K(T)$. Replacing u by $T \cdot I - u \otimes 1$, and A by $A[T]$, we see that we can reduce to proving that u is integral over A. If n is the rank of E, then $\det(u) = \det(\vee^n u)$ and $(\vee^n u \otimes 1) = \vee^n(u \otimes 1)$, so we can suppose that $n = 1$. But then the map $u \mapsto \det u$ is a homomorphism from the A-module $\operatorname{Hom}_A(E,E)$ to K; since E is of finite type, $\operatorname{Hom}_A(E,E)$ is isomorphic to a sub-A-module of the A-module of finite type E^n (if E admits a system of n generators), and so the elements $\det u$ belong to a sub-A-module of K of finite type, and are thus integral over A. □

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Corollary (6.4.4). — *Under the hypotheses of (6.4.3), if we further suppose A to be normal, then the $\sigma_i(u)$ (and, in particular, $\operatorname{Tr} u$ and $\det u$) belong to A.*

Proposition (6.4.5). — *Let A be an integral ring, E an A-module of finite type of rank n, and u an endomorphism of E such that the $\sigma_i(u)$ belong to A. For u to be an automorphism of E, it is necessary that $\det u$ be invertible in A; this condition is sufficient if E is torsion free.*

Proof. The condition is *sufficient* when E is torsion free, since the hypothesis and equations (6.4.1.4) and (6.4.1.5) (which hold in E, and not just in $E \otimes_A K$, since E is torsion free) prove that $(\det u)^{-1}Q(u)$ is the inverse of u.

The condition is *necessary*, since, if u is invertible, then it follows from (6.4.3) that $\det(u^{-1})$ belongs to the integral closure A' of A in its field of fractions K, and is evidently the inverse of $\det u$ *in* A'. Our claim then follows from: □

Lemma (6.4.5.1). — *Let A be a subring of a ring A' such that A' is* integral *over A. If an element $x \in A$ is invertible in A', then it is also invertible in A.*

Proof. In the contrary case, x would belong to a maximal ideal $\mathfrak{m}$ of A, and it follows from the first theorem of Cohen–Seidenberg [**SZ60**, t. I, p. 257, th. 3] that there would be a maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m} = \mathfrak{m}' \cap A$; we would then have that $x \in \mathfrak{m}'$, which is a contradiction. □

Corollary (6.4.6). — *Let A be an integral and integrally closed ring, E a torsion-free A-module of finite type, and u an endomorphism of E> For u to be an automorphism of E, it is necessary and sufficient that $\det u$ be invertible in A.*

Proof. This follows from (6.4.4) and (6.4.5). □

Remark (6.4.7). — We will later need a generalisation of the above results. Consider a *reduced* Noetherian ring A; let $\mathfrak{p}_\alpha$ (for $1 \leqslant \alpha \leqslant r$) be its minimal ideals, K_α the field of fractions of the integral ring $A/\mathfrak{p}_\alpha$, and K the total ring of fractions of A, which is the *direct sum* of the fields K_α. Let E be an A-module of finite type, and *suppose* that $E \otimes_A K$ is a *free* K-module of rank n (which here is merely a consequence of the other hypotheses); equivalently, we can ask that all the K_α-vector spaces $E \otimes_A K_\alpha = E_\alpha$ have *the same dimension* n; if then u is an endomorphism of E, we again set $P(u,T) = P(u \otimes 1, T)$ and $\sigma_j(u) = \sigma_j(u \otimes 1)$, and, in particular, $\det u = \det(u \otimes 1)$; the $\sigma_j(u)$ are thus elements of K. It is immediate that $E \otimes_A K$ is the direct sum of the E_α, and that the latter are stable under $u \otimes 1$; the restriction of $u \otimes 1$ to E_α is exactly the extension u_α of u to this K_α-vector space; we thus conclude that $\sigma_j(u)$ is the element of K who components in the K_α are the $\sigma_j(u_\alpha)$. Since the integral closure of A in K is the *direct sum* of the integral closures of A in the K_α, the $\sigma_j(u)$ are *integral* over A.

Lemma (6.4.7.1). — *The sub-A-algebra of K generated by all the elements $\sigma_j(u)$ (for $1 \leqslant j \leqslant n$), where u runs over $\operatorname{Hom}_A(E,E)$, is an A-module of finite type.*

Proof. It suffices to prove that the sub-$A[T]$-algebra of $K[T]$ generated by the $P(u,T)$ is an $A[T]$-module of finite type, since if the $F_i(T)$ (for $1 \leqslant i \leqslant m$) form a system of generators for this $A[T]$-module, then the coefficients of the $F_i(T)$ are integral over A, and thus generate an A-algebra that is an A-module of finite type [**SZ60**, t. I, p. 255, th. 1]. We can thus replace A by $A[T]$ (which **II** | 123
is Noetherian), and E by $E \otimes_A A[T] = E'$, which is such that $E' \otimes_{A[T]} K[T] = E \otimes_A K[T]$ is a free $K[T]$-module of rank n. Using the initial notation, we thus see that it suffices to prove that the A-module generated by the elements $\det u$, where u runs over $\operatorname{Hom}_A(E,E)$, is of finite type; *a fortiori* (since every submodule of an A-module of finite type is of finite type) it suffices to prove that, as v runs over the set of endomorphisms of $\wedge^n E$, the A-module generated by the $\det v$ is of finite type; in other words, we can again reduce to the case where $n = 1$. But then the proposition follows from the fact that $\operatorname{Hom}_A(E,E)$ is an A-module of finite type, and that $v \mapsto \det v$ is a homomorphism from this A-module into K. □

Let F be the kernel of the canonical homomorphism $E \to E \otimes_A K$, and let $E_0 = E/F$; we see, as above, that $E \otimes_A K$ can be identified with $E_0 \otimes_A K$, that $u(F) \subset F$, and that, if u_0 is the endomorphism of E_0 induced from u by passing to the quotient, then $u \otimes 1$ can be identified with $u_0 \otimes 1$, and $\sigma_j(u) = \sigma_j(u_0)$ for all j. *If we have $F = 0$*, then equations (6.4.1.3) and (6.4.1.5) still make sense and hold true whenever E is identified with a submodule of $E \otimes_A K$, and the u^j with homomorphisms $E \to E \otimes_A K$; then Proposition (6.4.5) extends also to this case, with the same proof.

(6.4.8). Let $(X, \mathscr{A})$ be a ringed space, $\mathscr{E}$ a *locally free* $\mathscr{A}$-module (of finite rank), and u an endomorphism of $\mathscr{E}$. By hypothesis, there exists a base $\mathfrak{B}$ for the topology of X such that, for all $V \in \mathfrak{B}$, $\mathscr{E}|V$ is isomorphic to some $\mathscr{A}^n|V$ (where n may vary with V). Let u be an endomorphism of $\mathscr{E}$; for all $V \in \mathfrak{B}$, u_V is thus an endomorphism of the $\Gamma(V, \mathscr{A})$-module $\Gamma(V, \mathscr{E})$, which is free by hypothesis; the determinant $\det u_V$ is thus defined and belongs to $\Gamma(V, \mathscr{A})$. Furthermore, if $e_1, \ldots, e_n$ form a basis of $\Gamma(V, \mathscr{E})$, then their restrictions to every open $W \subset V$ form a basis of $\Gamma(W, \mathscr{E})$ over $\Gamma(W, \mathscr{A})$, and so $\det u_W$ is the restriction of $\det u_V$ to W. There thus exists exactly one section of $\mathscr{A}$ over X, which we denote by $\det u$, and call the *determinant* of u, such that the restriction of $\det u$ to each $V \in \mathfrak{B}$ is $\det u_V$. It is clear that, for all $x \in X$, we have $(\det u)_x = \det u_x$; for endomorphisms u and v of $\mathscr{E}$, we have

$$\det(u \circ v) = (\det u)(\det v) \tag{6.4.8.1}$$

as well as

$$\det(1_{\mathscr{E}}) = 1_{\mathscr{A}} \tag{6.4.8.2}$$

and, if the rank of $\mathscr{E}$ is constant (which will be the case (**0**, 5.4.1) if X is connected) and equal to n,

$$\det(s \cdot u) = s^n \det u \tag{6.4.8.3}$$

for all $s \in \Gamma(X, \mathscr{A})$ (note that $\det(0) = 0_{\mathscr{A}}$ if $n \geqslant 1$, but $\det(0) = 1_{\mathscr{A}}$ for $n = 0$). Furthermore, for u to be an *automorphism* of $\mathscr{E}$, it is necessary and sufficient that u be *invertible* in $\Gamma(X, \mathscr{A})$.

If the rank of $\mathscr{E}$ is constant, then we can similarly define the elementary symmetric functions $\sigma_i(u)$, which are elements of $\Gamma(X, \mathscr{A})$; we again have the relations in (6.4.1.3) and (6.4.1.5).

We have thus defined a homomorphism $\det : \mathrm{Hom}_{\mathscr{A}}(\mathscr{E}, \mathscr{E}) \to \Gamma(X, \mathscr{A})$ of multiplicative monoids; if we note that $\mathrm{Hom}_{\mathscr{A}}(\mathscr{E}, \mathscr{E}) = \Gamma(X, \mathscr{H}om_{\mathscr{A}}(\mathscr{E}, \mathscr{E}))$ by definition, then we see that we can replace X in this definition by an arbitrary open U of X, which immediately defines a homomorphism $\det : \mathscr{H}om_{\mathscr{A}}(\mathscr{E}, \mathscr{E}) \to \mathscr{A}$ of sheaves of multiplicative monoids. If $\mathscr{E}$ has constant rank, then we similarly define homomorphisms $\sigma_i : \mathscr{H}om_{\mathscr{A}}(\mathscr{E}, \mathscr{E}) \to \mathscr{A}$ of sheaves of sets; for $i = 1$, the homomorphism $\sigma_1 = \mathrm{Tr}$ is a homomorphism of $\mathscr{A}$-modules. **II** | 124

Let $(Y, \mathscr{B})$ be another ringed space, and let $f : (X, \mathscr{A}) \to (Y, \mathscr{B})$ be a morphism of ringed spaces; if $\mathscr{F}$ is a locally free $\mathscr{B}$-module, then $f^*(\mathscr{F})$ is a locally free $\mathscr{A}$-module (which is of the same rank as $\mathscr{F}$ if the latter is of constant rank) (**0**, 5.4.5). For every endomorphism v of $\mathscr{F}$, $f^*(v)$ is an endomorphism of $f^*(\mathscr{F})$, and it follows immediately from the definitions that $\det f^*(v)$ is the section of $\mathscr{A} = f^*(\mathscr{B})$ over X that canonically corresponds to $\det v \in \Gamma(Y, \mathscr{B})$. We can further say that the homomorphism $f^*(\det) : f^*(\mathscr{H}om_{\mathscr{B}}(\mathscr{F}, \mathscr{F})) \to f^*(\mathscr{B}) = \mathscr{A}$ is the composition

$$f^*(\mathscr{H}om_{\mathscr{B}}(\mathscr{F}, \mathscr{F})) \xrightarrow{\gamma^\sharp} \mathscr{H}om_{\mathscr{A}}(f^*(\mathscr{F}), f^*(\mathscr{F})) \xrightarrow{\det} \mathscr{A}$$

(**0**, 4.4.6). We have analogous results for the σ_i.

(6.4.9). Now suppose that X is a *locally integral prescheme*, so that the sheaf $\mathscr{R}(X)$ of rational functions on X is a locally simple sheaf of fields (**I**, 7.4.3), and quasi-coherent as an $\mathscr{O}_X$-module. If $\mathscr{E}$ is a quasi-coherent $\mathscr{O}_X$-module *of finite type*, then $\mathscr{E}' = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module (**I**, 7.3.6); for every endomorphism u of $\mathscr{E}$, $u \otimes 1_{\mathscr{R}(X)}$ is then an endomorphism of $\mathscr{E}'$, and $\det(u \otimes 1)$ is a section of $\mathscr{R}(X)$ over X, which we also call the *determinant* of u, and also denote by $\det u$. It follows from (6.4.3) that $\det u$ is a section of the *integral closure* of $\mathscr{O}_X$ in $\mathscr{R}(X)$ (6.3.2); if, further, X is *normal*, then $\det u$ is a *section of* $\mathscr{O}_X$ over X, and if we further suppose that $\mathscr{E}$ is *torsion free*, then for u to be an automorphism of $\mathscr{E}$ it is necessary and sufficient that $\det u$ be *invertible*, by (6.4.6). Equations (6.4.8.1) to (6.4.8.3) still hold true; from the homomorphism $u \mapsto \det u$, applied to the modules of sections of $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{E})$, we obtain a homomorphism of sheaves $\det : \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{E}) \to \mathscr{R}(X)$, which takes values in $\mathscr{O}_X$ if X is normal. We have analogous definitions and results for the other elementary symmetric functions $\sigma_j(u)$, if $\mathscr{E}'$ is of *constant rank*; if, further, X is *normal*, then the $\sigma_j(u)$ are *sections of* $\mathscr{O}_X$ over X.

Finally, let X and Y be integral preschemes, and $f : X \to Y$ a *dominant* morphism. We know that there exists a canonical homomorphism $f^*(\mathscr{R}(Y)) \to \mathscr{R}(X)$ (**I**, 7.3.8), whence we obtain, for every quasi-coherent $\mathscr{O}_Y$-module $\mathscr{F}$ of finite type, a canonical homomorphism $\theta : f^*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{R}(Y)) \to f^*(\mathscr{F}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$. If v is an endomorphism of $\mathscr{F}$, then $f^*(v \otimes 1_{\mathscr{R}(Y)})$ is an endomorphism of **II** | 125
$f^*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{R}(Y))$, and we have a commutative diagram

$$\begin{array}{ccc} f^*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{R}(Y)) & \xrightarrow{f^*(v\otimes 1)} & f^*(\mathscr{F} \otimes \mathscr{O}_Y \mathscr{R}(Y)) \\ {\scriptstyle\theta}\downarrow & & \downarrow{\scriptstyle\theta} \\ f^*(\mathscr{F}) \otimes_{\mathscr{O}_X} \mathscr{R}(X) & \xrightarrow[f^*(v)\otimes 1]{} & f^*(\mathscr{F}) \otimes_{\mathscr{O}_X} \mathscr{R}(X) \end{array}$$

We thus easily conclude that $\det f^*(v)$ is the canonical image, under the homomorphism $f^*(\mathscr{R}(Y)) \to \mathscr{R}(X)$, of the section $\det v$ of $\mathscr{R}(Y)$; indeed, we can immediately reduce to the case where $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine, with A and B integral rings with fields of fractions K and L (resp.), with the homomorphism $B \to A$ an injection and thus extending to a monomorphism $L \to K$; if $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type, then the rank of $M \otimes_B L$ over L is *equal* to that of $(M \otimes_B A) \otimes_A K$ over K, and $\det((u \otimes 1) \otimes 1)$ is the image in K of $\det(u \otimes 1)$ for any endomorphism u of M, whence the conclusion.

(6.4.10). Finally, suppose that X is a *locally Noetherian reduced prescheme*, so that the sheaf $\mathscr{R}(X)$ of rational functions on X is again a quasi-coherent $\mathscr{O}_X$-module (**I**, 7.3.4); let $\mathscr{E}$ be a *coherent* $\mathscr{O}_X$-module such that $\mathscr{E}' = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is *locally free* (of finite rank). By (6.4.7), if $\mathscr{E}'$ is of constant rank, then we can, for any endomorphism u of $\mathscr{E}$, define the $\sigma_j(u)$, which are sections of $\mathscr{R}(X)$ over X. If we do not suppose $\mathscr{E}'$ to be of constant rank, we can still define the homomorphism $\det : \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{E}) \to \mathscr{R}(X)$.

6.5. Norm of an invertible sheaf

(6.5.1). Let $(X, \mathscr{A})$ be a ringed space and $\mathscr{B}$ a (commutative) $\mathscr{A}$-algebra. The $\mathscr{A}$-module $\mathscr{B}$ is canonically identified with a sub-$\mathscr{A}$-module of $\mathscr{H}om_{\mathscr{A}}(\mathscr{B}, \mathscr{B})$, and a section f of $\mathscr{B}$ over an open U of X is identified with multiplication by this section. If $(X, \mathscr{A})$ and $\mathscr{B}$ satisfy one of the conditions given in (6.4.8), (6.4.9), or (6.4.10), then we can define $\det(f)$ (and, in certain cases, the $\sigma_j(f)$) as sections of $\mathscr{A}$ or $\mathscr{R}(X)$ over U, that we call the *norm* of f (resp. the *elementary symmetric functions* of f), and denote by $N_{\mathscr{B}/\mathscr{A}}(f)$. We will suppose that *one* of the following conditions is satisfied:

(I) *$\mathscr{B}$ is a locally free $\mathscr{A}$-module (of finite rank).*
(II) *$(X, \mathscr{A})$ is a locally Noetherian reduced prescheme, $\mathscr{B}$ is a coherent $\mathscr{A}$-module such that $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module, and, for every section $f \in \Gamma(U, \mathscr{B})$ over an open U of X, the norm $N_{\mathscr{B}/\mathscr{A}}(f)$ is a section of $\mathscr{A}$ over U.*

Note that the latter condition is automatically satisfied whenever the locally Noetherian prescheme **II** | 126
X is *normal* (6.4.9).

The hypothesis that $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is locally free can also be expressed in the following way: denote by X_α the closed reduced subpreschemes of X whose underlying spaces are the irreducible components of X (**I**, 5.2.1), which are thus locally Noetherian integral preschemes. Every $x \in X$ belongs to a finite number of the subspaces X_α; on the other hand, $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X_\alpha)$ is a locally free $\mathscr{R}(X_\alpha)$-module of constant rank k_α (**I**, 7.3.6); to say that $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module implies that, for all $x \in X$, *the ranks k_α that correspond to the indices such that $x \in X_\alpha$ are equal*. This is a local statement, and we can reduce to the case $X = \operatorname{Spec}(C)$, where C is a reduced Noetherian ring, and $\mathscr{B} = \widetilde{D}$, where D is a C-algebra that is a C-module of finite type; if $\mathfrak{p}_i$ (for $1 \leqslant i \leqslant m$) are the minimal prime ideals of C, then the total ring of fractions L of C is the direct sum of the fields of fractions K_i of the integral rings $C/\mathfrak{p}_i$, and $D \otimes_C L$ is the direct sum of the $D \otimes_C K_i$, whence the conclusion.

It is clear that, under hypothesis (I) or (II), we have thus defined a *homomorphism of sheaves of multiplicative monoids* $N_{\mathscr{B}/\mathscr{A}} : \mathscr{B} \to \mathscr{A}$, also denoted by N if no confusion may arise, and we call this homomorphism the *norm*. For sections f and g of $\mathscr{B}$ over the same open U, we thus have

$$N_{\mathscr{B}/\mathscr{A}}(fg) = N\mathscr{B}/\mathscr{A}(f)\mathscr{B}/\mathscr{A}(g) \tag{6.5.1.1}$$

for the corresponding sections of $\mathscr{A}$ over U;

$$N\mathscr{B}/\mathscr{A}(1_{\mathscr{B}}) = 1_{\mathscr{A}}; \tag{6.5.1.2}$$

finally, for any section s of $\mathscr{A}$ over U,

$$N\mathscr{B}/\mathscr{A}(s \cdot 1_{\mathscr{B}}) = s^n \tag{6.5.1.3}$$

if the rank of $\mathscr{B}$ is constant and equal to n (for $s = 0_{\mathscr{A}}$, this formula gives $N(0_{\mathscr{B}}) = 0_{\mathscr{A}}$ if $n \geqslant 1$, and $N(0_{\mathscr{B}}) = N(1_{\mathscr{B}}) = 1_{\mathscr{A}}$ if $n = 0$).

In hypothesis (I), for $f \in \Gamma(U, \mathscr{B})$ to be invertible it is necessary and sufficient that $N(f) \in \Gamma(U, \mathscr{A})$ be invertible. In hypothesis (II), this condition is necessary; it is sufficient (by (6.4.7)) if we suppose that $\mathscr{B} \to \mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is *injective* and that the following more restrictive hypothesis is satisfied:

(II *bis*) *$(X, \mathscr{A})$ is a locally Noetherian reduced prescheme, $\mathscr{B}$ is a coherent $\mathscr{A}$-module such that $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module, and, for any section $f \in \Gamma(U, \mathscr{B})$ over an open U such that $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)|U$ is of constant rank n on $\mathscr{R}(X)|U$, the $\sigma_j(f)$ (for $1 \leqslant j \leqslant n$) are sections of $\mathscr{A}$ over U.*

(We again note that this condition is satisfied if X is *normal*.)

(6.5.2). Suppose that one of the hypotheses (I) or (II) of (6.5.1) are satisfied, and let $\mathscr{L}'$ be an *invertible* $\mathscr{B}$-module. We will canonically associate (up to unique isomorphism) an *invertible* $\mathscr{A}$-module in the following way. Denote by $\mathscr{A}^*$ (resp. $\mathscr{B}^*$) the subsheaf of $\mathscr{A}$ (resp. of $\mathscr{B}$) such that $\Gamma(U, \mathscr{A}^*)$ (resp. $\Gamma(U, \mathscr{B}^*)$) is the set of invertible elements of $\Gamma(U, \mathscr{A})$ (resp. of $\Gamma(U, \mathscr{B})$) for every open $U \subset X$; these II | 127
are sheaves of multiplicative groups, and $N_{\mathscr{B}/\mathscr{A}}$ restricted to $\mathscr{B}^*$ is a *homomorphism* $\mathscr{B}^* \to \mathscr{A}^*$ of sheaves of groups (6.5.1). Let $\mathfrak{L}$ be the set of pairs (U_λ, ν_λ) that have the following property: U_λ is an open of X, and ν_λ is an isomorphism $\mathscr{L}'|U_\lambda \xrightarrow{\sim} \mathscr{B}|U_\lambda$ of $(B|U_\lambda)$-modules. By hypothesis, the U_λ form a cover of X; for arbitrary indices λ and μ, set $\omega_{\lambda\mu} = (\eta_\lambda|U_\lambda \cap U_\mu) \circ (\nu_\mu|U_\lambda \cap U_\mu)^{-1}$, which is an automorphism of $\mathscr{B}|U_\lambda \cap U_\mu$ which is canonically identified with a section of $\mathscr{B}^*$ over $U_\lambda \cap U_\mu$, and $(\omega_{\lambda\mu})$ is a 1-cocycle for the cover $\mathfrak{U} = (U_\lambda)$ with values in $\mathscr{B}^*$ (**0**, 5.4.7). The fact that $N_{\mathscr{B}/\mathscr{A}} : \mathscr{B}^* \to \mathscr{A}^*$ is a homomorphism implies that $(N_{\mathscr{B}/\mathscr{A}}\omega_{\lambda\mu})$ is a 1-cocycle of $\mathfrak{U}$ with values in $\mathscr{A}^*$, and thus corresponds (up to unique isomorphism) to an invertible $\mathscr{A}$-module $\mathscr{L}$ (**0**, 5.4.7) which we denote by $N_{\mathscr{B}/\mathscr{A}}(\mathscr{L}')$, and call the *norm* of the invertible $\mathscr{B}$-module $\mathscr{L}'$.

Let $\mathfrak{M}$ be a subset of $\mathfrak{L}$ such that the corresponding U_λ still form a cover of X, and let $\mathfrak{V}$ be the resulting cover; the restriction of the cocycle $(\omega_{\lambda\mu})$ to $\mathfrak{V}$ defines a 1-cocycle $(N_{\mathscr{B}/\mathscr{A}}\omega_{\lambda\mu})$ of $\mathfrak{V}$ with values in $\mathscr{A}^*$, the restriction of the 1-cocycle $(N_{\mathscr{B}/\mathscr{A}}\omega_{\lambda\mu})$ of $\mathfrak{U}$; it is clear that there is a canonical isomorphism from the invertible $\mathscr{A}$-module defined by this 1-cocycle of $\mathfrak{V}$ to $N_{\mathscr{B}/\mathscr{A}}(\mathscr{L}')$, which allows us to define the invertible $\mathscr{A}$-module by an arbitrary sub-cover of $\mathfrak{U}$. This possibility immediately shows that, if $\mathscr{L}'_1$ and $\mathscr{L}'_2$ are invertible $\mathscr{B}$-modules, then, by (6.5.1.1) and (6.5.1.2),

$$N(\mathscr{L}'_1 \otimes_{\mathscr{B}} \mathscr{L}'_2) = N(\mathscr{L}'_1) \otimes_{\mathscr{A}} N(\mathscr{L}'_2) \tag{6.5.2.1}$$

and

$$N_{\mathscr{B}/\mathscr{A}}(\mathscr{B}) = \mathscr{A} \tag{6.5.2.2}$$

as well as

$$N(\mathscr{L}'^{-1}) = (N(\mathscr{L}'))^{-1} \tag{6.5.2.3}$$

up to canonical isomorphisms. Furthermore, it follows from (6.5.1.3) that, if $\mathscr{L}$ is an invertible $\mathscr{A}$-module = and if $\mathscr{B}$ is of constant rank n on $\mathscr{A}$ in case (I) (resp. $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is of constant rank n on $\mathscr{R}(X)$ in case (II)), we have, up to canonical isomorphism,

$$N_{\mathscr{B}/\mathscr{A}}(\mathscr{L} \otimes_{\mathscr{A}} \mathscr{B}) = \mathscr{L}^{\otimes n}. \tag{6.5.2.4}$$

(6.5.3). We now show that $N_{\mathscr{B}/\mathscr{A}}$ is a covariant *functor* in the category of invertible $\mathscr{B}$-modules. Let $h' : \mathscr{L}'_1 \to \mathscr{L}'_2$ be a homomorphism of invertible $\mathscr{B}$-modules, and let $\mathfrak{V} = (U_\lambda)$ be an open cover of X such that, for all λ, we have $(\mathscr{B}|U_\lambda)$-isomorphisms $\eta_\lambda^{(1)} : \mathscr{L}'_1|U_\lambda \xrightarrow{\sim} \mathscr{B}|U_\lambda$ and $\eta_\lambda^{(2)} : \mathscr{L}'_2|U_\lambda \xrightarrow{\sim} \mathscr{B}|U_\lambda$; there is thus, for each λ, an endomorphism h'_λ of $\mathscr{B}|U_\lambda$ such that $h'_\lambda \circ \eta_\lambda^{(1)} = \eta_\lambda^{(2)} \circ (h'|U_\lambda)$, and we can evidently identify h'_λ with a section of $\mathscr{B}$ over U_λ (**0**, 5.1.1). So, for every pair (λ, μ)

of indices, the restrictions to $U_\lambda \cap U_\mu$ of $(\eta_\lambda^{(2)})^{-1} \circ h'_\lambda \circ \eta_\lambda^{(1)}$ and $(\eta_\mu^{(2)})^{-1} \circ h'_\mu \circ \eta_\mu^{(1)}$ agree. We thus obtain, for 1-cocycles $(\omega_{\lambda\mu}^{(1)})$ and $(\omega_{\lambda\mu}^{(2)})$ with values in $\mathscr{B}^*$ corresponding to $\mathscr{L}'_1$ and $\mathscr{L}'_2$, the relation

$$\omega_{\lambda\mu}^{(2)} h'_\mu = h'_\lambda \omega_{\lambda\mu}^{(1)}.$$

If we set $h_\lambda = N(h'_\lambda)$, we thus have the analogous relations **II** | 128

$$N(\omega_{\lambda\mu}^{(2)}) h_\mu = h_\lambda N(\omega_{\lambda\mu}^{(1)})$$

and so the h_λ define a homomorphism $N(\mathscr{L}'_1) \to N(\mathscr{L}'_2)$ which we denote by $N_{\mathscr{B}/\mathscr{A}}(h')$, or $N(h')$. In hypothesis (I), for h' to be an *isomorphism*, it is necessary and sufficient (since it is a local question) for $N_{\mathscr{B}/\mathscr{A}}(h')$ to be an isomorphism. In hypothesis (II), this condition is again necessary; it is sufficient if hypothesis (II *bis*) is satisfied and $\mathscr{B} \to \mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is injective.

Take, in particular, $\mathscr{L}'_1 = \mathscr{B}$; the homomorphisms $\mathscr{B} \to \mathscr{L}'$ can then be identified (**0**, 5.1.1) with the sections of $\mathscr{L}'$ over X, whence a canonical map

$$N_{\mathscr{B}/\mathscr{A}} : \Gamma(X, \mathscr{L}') \longrightarrow \Gamma(X, N_{\mathscr{B}/\mathscr{A}}(\mathscr{L}')).$$

It again follows from (6.5.1.1) that, if $f'_1 \in \Gamma(X, \mathscr{L}'_1)$ and $f'_2 \in \Gamma(X, \mathscr{L}'_2)$, then

$$N(f'_1 \otimes f'_2) = N(f'_1) \otimes N(f'_2). \tag{6.5.3.1}$$

For every invertible $\mathscr{A}$-module $\mathscr{L}$ and every section $f \in \Gamma(X, \mathscr{L})$, we have, taking (6.5.2.4) into account, that

$$N_{\mathscr{B}/\mathscr{A}}(f \otimes 1_{\mathscr{B}}) = f^{\otimes n} \tag{6.5.3.2}$$

whenever $\mathscr{B}$ is of constant rank n in hypothesis (I) (resp. whenever $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is of constant rank n in hypothesis (II)). Finally, for the homomorphism $\mathscr{B} \to \mathscr{L}'$ corresponding to a section f' of $\mathscr{L}'$ over X to be an isomorphism, it is necessary and sufficient for f'_x to be a basis for $\mathscr{L}'_x$ for all $x \in X$; in hypothesis (I), this condition is thus equivalent to saying that $(N(f'))_x$ is a basis for $(N(\mathscr{L}'))_x$ for all x; in hypothesis (II), this condition is again necessary, and it is sufficient whenever $\mathscr{B}$ satisfies hypothesis (II *bis*) and $\mathscr{B} \to \mathscr{B} \otimes_{\mathscr{A}} \mathscr{R}(X)$ is injective.

(6.5.4). Let $(X, \mathscr{A})$ and $(X', \mathscr{A}')$ be ringed spaces, $f : X' \to X$ a morphism, $\mathscr{B}$ an $\mathscr{A}$-algebra, and $\mathscr{B}' = f^*(\mathscr{B})$ the inverse image $\mathscr{A}'$-algebra. Suppose that one of the following hypotheses is satisfied:

(1) $\mathscr{B}$ satisfies hypothesis (I) of (6.5.1).

(2) $(X, \mathscr{A})$ and $\mathscr{B}$ satisfy hypothesis (II) of (6.5.1), $(X', \mathscr{A}')$ is a locally Noetherian reduced prescheme, and, if we denote by X_α and X'_β the closed reduced subpreschemes of X and X' (respectively) that have the irreducible components of these spaces as their underlying spaces, then the restriction of f to each X'_β is a *dominant* morphism from X'_β to X_α.

Under these conditions, $\mathscr{B}'$ satisfies hypothesis (I) or hypothesis (II) (respectively) of (6.5.1). The first claim is immediate; to establish the second, it suffices to show that, for all $x' \in X'$, the ranks of the $\mathscr{B}' \otimes_{\mathscr{O}_{X'}} \mathscr{R}(X'_\beta)$ are *the same* for the β such that $x' \in X'_\beta$. But if the restriction of f to X'_β is a dominant morphism to X_α, then the rank of $\mathscr{B}' \otimes_{\mathscr{O}_{X'}} \mathscr{R}(X'_\beta)$ is equal to that of $\mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{R}(X_\alpha)$ (as we immediately see by reducing to the affine case, as in (6.4.9)), whence our claim, by hypothesis (II) and (6.5.1).

With this in mind, it follows from (6.4.8), (6.4.9), and (6.4.10) that, if s is a section of $\mathscr{B}$ over an open $U \subset X$, and s' the corresponding section of $\mathscr{B}'$ over $f^{-1}(U)$, then $N_{\mathscr{B}'/\mathscr{A}'}(s')$ is the section of $\mathscr{A}'$ over $f^{-1}(U)$ that corresponds to the section $N_{\mathscr{B}/\mathscr{A}}(s)$ of $\mathscr{A}$ over U. **II** | 129

If $\mathscr{M}$ is an invertible $\mathscr{B}$-module, then we deduce from the above that, if $\mathscr{M}' = f^*(\mathscr{M})$ (which is an invertible $\mathscr{B}'$-module), we have $N_{\mathscr{B}'/\mathscr{A}'}(\mathscr{M}') = f^*(N_{\mathscr{B}/\mathscr{A}}(\mathscr{M}))$ up to canonical isomorphism.

(6.5.5). Suppose from now on that $(X, \mathscr{A})$ is a *prescheme*. The data of a quasi-coherent $\mathscr{A}$-algebra $\mathscr{B}$, which is an *$\mathscr{A}$-module of finite type*, is then equivalent, as we know, to that of a *finite* morphism $g : X' \to X$ such that $g_*(\mathscr{O}_{X'}) = \mathscr{B}$, defined up to X-isomorphism ((6.1.2) and (1.3.1)). Furthermore, the data of a quasi-coherent $\mathscr{O}_{X'}$-module $\mathscr{F}'$ is equivalent to that of a quasi-coherent $\mathscr{B}$-module $\mathscr{F}$ such that $g_*(\mathscr{F}) = \mathscr{F}$ (1.4.3), and for $\mathscr{F}'$ to be invertible it is necessary and sufficient that $\mathscr{F}$ be invertible (6.1.12). To translate the above results in terms of finite morphisms g, it will be necessary to suppose either that $g_*(\mathscr{O}_{X'})$ is a *locally free* $\mathscr{O}_X$-module (of finite type) or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy hypothesis (II). For every invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$, we thus set

$$N_{X'/X}(\mathscr{L}') = N_{g_*(\mathscr{O}_{X'})/\mathscr{O}_X}(g_*(\mathscr{L}')) \tag{6.5.5.1}$$

and we call this the *norm* (with respect to g) of $\mathscr{L}'$. Similarly, if $h' : \mathscr{L}'_1 \to \mathscr{L}'_2$ is a homomorphism of invertible $\mathscr{O}_{X'}$-modules, then we set

$$N_{X'/X}(h') = N_{g_*(\mathscr{O}_{X'})/\mathscr{O}_X}(g_*(h')) : N_{X'/X}(\mathscr{L}'_1) \longrightarrow N_{X'/X}(\mathscr{L}'_2). \tag{6.5.5.2}$$

In particular, for $\mathscr{L}'_1 = \mathscr{O}_{X'}$, we thus obtain a canonical map

$$N_{X'/X} : \Gamma(X', \mathscr{L}') \longrightarrow \Gamma(X, N_{X'/X}(\mathscr{L}')). \tag{6.5.5.3}$$

We leave to the reader the majority of these translations, and we restrict ourselves to spelling out the details of the following:

Proposition (6.5.6). — *Let $g : X' \to X$ be a finite morphism, and suppose that either $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy (II* bis*) (which will be the case, in particular, if X is locally Noetherian and normal). For a homomorphism $h' : \mathscr{L}'_1 \to \mathscr{L}'_2$ of invertible $\mathscr{O}_{X'}$-modules to be an isomorphism, it is necessary and sufficient, in the first hypothesis, that $N_{X'/X}(h')$ be an isomorphism; in the second hypothesis, this condition is again necessary, and is sufficient if the homomorphism $g_*(\mathscr{O}_{X'}) \to g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is injective.*

PROOF. Note that we use here the fact that, for $g_*(h')$ to be an isomorphism, it is necessary and sufficient for h' to be an isomorphism (1.4.2). □

Corollary (6.5.7). — *Let $g : X' \to X$ be a finite morphism, and suppose that either $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy (II* bis*) and $g_*(\mathscr{O}_{X'}) \to g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is injective. Let $\mathscr{L}'$ be an invertible $\mathscr{O}_{X'}$-module, f' a section of $\mathscr{L}'$ over X', and $f = N_{X'/X}(f')$ the corresponding section of $\mathscr{L} = N_{X'/X}(\mathscr{L}')$ over X (6.5.5.1). Then $g(X' \setminus X'_{f'}) = X \setminus X_f$, and X_f is the largest open subset* **II** | 130
U of X such that $g^{-1}(U) \subset X'_{f'}$.

PROOF. Indeed, $g(X' \setminus X'_{f'})$ is closed in X (6.1.10); it thus suffices to prove the last claim. But the relation $U \subset X_f$ is equivalent to the fact that the homomorphism $\mathscr{O}_X|U \to \mathscr{L}|U$ defined by $f|U$ is an isomorphism. By (6.5.6), this is equivalent to saying that the homomorphism $\mathscr{O}_{X'}|g^{-1}(U) \to \mathscr{L}'|g^{-1}(U)$ defined by $f'|g^{-1}(U)$ is an isomorphism, which is equivalent to the relation $g^{-1}(U) \subset X'_{f'}$. □

Proposition (6.5.8). — *Let $g : X' \to X$ be a finite morphism, and $f : Y \to X$ a morphism; let $Y' = X'_{(Y)}$, $g' = g_{(Y)}$, and $f' = f_{(X')}$, so that the diagram*

$$\begin{tikzcd} X' \arrow[d, "g"'] & Y' \arrow[l, "f'"'] \arrow[d, "g'"] \\ X & Y \arrow[l, "f"] \end{tikzcd}$$

commutes.

Suppose that either $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy (II). Suppose further that Y is a locally Noetherian reduced prescheme, and that the restriction of f to any irreducible component of Y is a dominant morphism to an irreducible component of X. Then, for every invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$, we have

$$N_{Y'/Y}(f'^*(\mathscr{L}')) = f^*(N_{X'/X}(\mathscr{L}'))$$

up to canonical isomorphism.

PROOF. Note that we have $f^*(g_*(\mathscr{L}')) = g'_*(f'^*(\mathscr{L}'))$, by (1.5.2), and, in particular, $g'_*(\mathscr{O}_{Y'}) = f^*(g_*(\mathscr{O}_{X'}))$; if $g_*(\mathscr{O}_{X'})$ is locally free, then so too is $g'_*(\mathscr{O}_{Y'})$. The conclusion then follows from the definitions and (6.5.4). □

Remark (6.5.9). — We later generalise the notion of norm developed above, by placing it relation to the notion of direct image of a divisor.

6.6. Application: criteria for ampleness

Proposition (6.6.1). — *Let Y be a prescheme, $f : X \to Y$ a quasi-compact morphism, and $g : X' \to X$ a finite and surjective morphism. Suppose that either $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy (II* bis*). Then, for every invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$ that is ample for $f \circ g$, $\mathscr{L} = N_{X'/X}(\mathscr{L}')$ is ample for f.*

PROOF. We can suppose Y to be affine (4.6.4), and then, by (4.6.6), the statement is equivalent to: □

Corollary (6.6.2). — *Let X be a quasi-compact prescheme, $g : X' \to X$ a finite surjective morphism, such that either $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module or that $(X, \mathscr{O}_X)$ and $g_*(\mathscr{O}_{X'})$ satisfy (II* bis*). Then, for every ample $\mathscr{O}_{X'}$-module $\mathscr{L}'$, $\mathscr{L} = N_{X'/X}(\mathscr{L}')$ is ample.*

PROOF. In the second hypothesis, we can further suppose that the canonical homomorphism $g_*(\mathscr{O}_{X'}) \to g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is *injective*. Indeed, if not, let $\mathscr{T}$ be the kernel of this homomorphism, **II** | 131
which is a coherent ideal of $\mathscr{B} = g_*(\mathscr{O}_{X'})$ (**I**, 6.1.1), and set $X'' = \operatorname{Spec}(\mathscr{B}/\mathscr{T})$; we thus have a commutative diagram

$$\begin{tikzcd} X'' \arrow[rr, "h"] \arrow[dr, "g'"'] & & X' \arrow[dl, "g"] \\ & X & \end{tikzcd}$$

where h is a closed immersion (1.4.10). Furthermore, we know that the support of $\mathscr{T}$ is a closed set (**0**, 5.2.2) that is rare in X (**I**, 7.4.6), whence we conclude that, for the generic point x of an irreducible component of X, there is an affine open neighbourhood U of x such that $\mathscr{B}|U = (\mathscr{B}/\mathscr{T})|U$. Since g is, by hypothesis, surjective, we thus conclude that $x \in g'(X'')$; g' is thus dominant, and, since it is a finite morphism, it is *surjective* (6.1.10); by definition,

$$g'_*(\mathscr{O}_{X''}) \otimes_{\mathscr{O}_X} \mathscr{R}(X) = (\mathscr{B}/\mathscr{T}) \otimes_{\mathscr{O}_X} \mathscr{R}(X) = g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$$

thus $(X, \mathscr{O}_X)$ and $g'_*(\mathscr{O}_{X''})$ satisfy (II *bis*), and furthermore $g'_*(\mathscr{O}_{X''}) \to g'_*(\mathscr{O}_{X''}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is injective. Finally, $h^*(\mathscr{L}') = \mathscr{L}''$ is an ample $\mathscr{O}_{X''}$-module (4.6.13, (i *bis*)), and we have that $N_{X''/X}(\mathscr{L}'') = N_{X'/X}(\mathscr{L}')$. Indeed, to define these two invertible $\mathscr{O}_X$-modules, we can use the same affine open cover (U_λ) of X such that the restrictions of $g_*(\mathscr{L}')$ and $g'_*(\mathscr{L}'')$ to U_λ are isomorphic to $\mathscr{B}|U_\lambda$ and $(\mathscr{B}/\mathscr{T})|U_\lambda$ (respectively). We immediately see that, for every isomorphism $\eta_\lambda : g_*(\mathscr{L}')|U_\lambda \to \mathscr{B}|U_\lambda$, there is a canonically corresponding isomorphism

$$\eta'_\lambda : g'_*(\mathscr{L}'')|U_\lambda \longrightarrow (\mathscr{B}/\mathscr{T})|U_\lambda$$

so that, if $(\omega_{\lambda\mu})$ and $(\omega'_{\lambda\mu})$ are the 1-cocycles corresponding to the systems of isomorphisms (η_λ) and (η'_λ) (6.5.2), $\omega'_{\lambda\mu}$ is the canonical image in $\Gamma(U_\lambda \cap U_\mu, \mathscr{B}/\mathscr{T})$ of $\omega_{\lambda\mu} \in \Gamma(U_\lambda \cap U_\mu, \mathscr{B})$. By the definition of $\mathscr{T}$, we thus conclude that

$$N_{\mathscr{B}/\mathscr{A}}(\omega_{\lambda\mu}) = N_{(\mathscr{B}/\mathscr{T})/\mathscr{A}}(\omega'_{\lambda\mu})$$

(where $\mathscr{A} = \mathscr{O}_X$), whence the claimed equality.

So suppose that the homomorphism $g_*(\mathscr{O}_{X'}) \to g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is injective when we are in hypothesis (II *bis*). It suffices to prove that, as f runs over the sections of $\mathscr{L}^{\otimes n}$ (for $n > 0$) over X, the X_f form a base for the topology of X (4.5.2). But let $x \in X$, and let U be an arbitrary neighbourhood of x; since $g^{-1}(x)$ is finite (6.1.7) and $\mathscr{L}'$ is ample, there exists an integer $n > 0$ and a section f' of $\mathscr{L}'^{\otimes n}$ over X' such that $X'_{f'}$ is a neighbourhood of $g^{-1}(x)$ contained inside $g^{-1}(U)$ (4.5.4). Since

$$\mathscr{L}^{\otimes n} = N_{X'/X}(\mathscr{L}'^{\otimes n})$$

it suffices to take $f = N_{X'/X}(f')$; indeed, then $X \setminus X_f = g(X' \setminus X'_{f'})$ (6.5.7), and so $x \in X_f \subset U$. □

Corollary (6.6.3). — *Under the hypotheses of (6.6.1), for an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample for f, it is necessary and sufficient that $\mathscr{L}' = g^*(\mathscr{L})$ be ample for $f \circ g$.*

PROOF. The condition is necessary, since g is affine (5.1.12). To prove that the condition is sufficient, we can suppose that Y is affine (4.6.4), and so X and X' are quasi-compact and $\mathscr{L}'$ is ample (4.6.6), and we need to show that $\mathscr{L}$ is ample. But the set of points $x \in X$ that admit a neighbourhood where $g_*(\mathscr{O}_{X'})$ (resp. $g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$) has a given rank n in the first (resp. second) hypothesis is simultaneously open and closed in X, and so X is the prescheme given by the sum of a finite number of these opens, and we can thus suppose that it is equal to one of them (4.6.17). But then $N_{X'/X}(\mathscr{L}') = \mathscr{L}^{\otimes n}$, and so $\mathscr{L}^{\otimes n}$ is ample by (6.6.2), and thus so too is $\mathscr{L}$ (4.5.6). □ **II** | 132

Corollary (6.6.4). — *Suppose that the hypotheses of (6.6.1) are satisfied, and suppose further that $f : X \to Y$ is of finite type. Then, for f to be quasi-projective, it is necessary and sufficient that $f \circ g$ be quasi-projective. If we further suppose that Y is a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, then for f to be projective, it is necessary and sufficient that $f \circ g$ be projective.*

PROOF. The hypothesis implies that $f \circ g$ is of finite type. Taking into account the definition of quasi-projective morphisms (5.3.1), the first claim follows from (6.6.1) and (6.6.3). Taking into account this result, along with (5.5.3, (ii)), it remains to show that, if f is quasi-projective, then for f to be proper, it is necessary and sufficient that $f \circ g$ be proper. But f is then separated (5.3.1) and of finite type; since g is surjective, our claim follows from (5.4.2, (ii)) and (5.4.3, (ii)). □

In particular:

Corollary (6.6.5). — *Let X be a prescheme of finite type over a field K, and K' a finite-degree extension of K. For X to be projective (resp. quasi-projective) over K, it is necessary and sufficient that $X' = X \otimes_K K'$ be projective (resp. quasi-projective) over K'.*

PROOF. The condition is necessary ((5.3.4, (iii)) and (5.5.5, (iii))). Conversely, suppose that it is satisfied, and let $g : X' \to X$ be the canonical projection. It is clear that g is a finite morphism (6.1.5, (iii)) and surjective (**I**, 3.5.2, (ii)). Furthermore, $g_*(\mathscr{O}_{X'})$ is a locally free $\mathscr{O}_X$-module, since it is isomorphic to $\mathscr{O}_X \otimes_K K'$ (1.5.2). It then follows from the hypothesis and from (6.1.11) and (5.5.5, (ii)) that X' is projective (resp. quasi-projective) *over* K; we then deduce from (6.6.4) that X is projective (resp. quasi-projective) over K. □

In Chapter V, we will show that the statement of (6.6.5) remains true when K' is an *arbitrary* extension of K.

The rest of this section is dedicated to the proof of the criterion in (6.6.11), which is a rather technical refinement of (6.6.1); it can be omitted on a first reading.

Lemma (6.6.6). — *Let X be a reduced Noetherian prescheme, and $\mathscr{E}$ a coherent $\mathscr{O}_X$-module such that $\mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module of constant rank n. Then there exists a reduced Noetherian prescheme Z and a birational finite morphism $h : Z \to X$ that has the following property: the morphisms of sheaves of sets $\sigma_i : \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E},\mathscr{E}) \to \mathscr{R}(X)$ (for $1 \leqslant i \leqslant n$) (cf. (6.4.10)) send $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{E},\mathscr{E})$ to the coherent $\mathscr{O}_X$-algebra $h_*(\mathscr{O}_Z)$.*

PROOF. Consider an affine open U of X, of ring $A(U) = A$; let $E = \Gamma(U,\mathscr{E})$, and let C_U be the subalgebra of $R(U)$ generated by the $\sigma_i(u)$ where u runs over $\operatorname{Hom}_A(E,E)$; we have seen (6.4.7.1) that this A-algebra is of *finite rank*. Furthermore, it is clear that forming the algebras C_U commutes with the restriction operations of an affine open U to an affine open $U' \subset U$. We have thus defined a *finite* sub-$\mathscr{O}_X$-algebra $\mathscr{C}$ of $\mathscr{R}(X)$ such that $\Gamma(U,\mathscr{C}) = C_U$ for every affine open U of X. We take $Z = \operatorname{Spec}(\mathscr{C})$, and take h to be the structure morphism, which is thus finite (6.1.2); since $\mathscr{C}$ is reduced, Z is a reduced Noetherian prescheme (1.3.8). Finally, the total ring of fractions of C_U is $R(U)$, by definition, and since C_U is contained inside the integral closure of $A(U)$ in $R(U)$, there is a bijective correspondence between minimal prime ideals of $A(U)$ and minimal prime ideals of C_U [**SZ60**, t. I, p. 259], which proves that h is birational and finishes the proof. □

Corollary (6.6.7). — *Under the hypotheses of (6.6.6), let W be an open of X such that, for all $x \in W$, either X is normal at the point x, or $\mathscr{E}_x$ is a free $\mathscr{O}_x$-module. Then we can suppose h to be defined such that the restriction of h to $h^{-1}(W)$ is an isomorphism from $h^{-1}(W)$ to W.*

PROOF. Either hypothesis implies that, if $U \subset W$ is an affine open, then, with the notation of (6.6.6), $(\sigma_i(u))_x \in A_x$ for all $x \in U$ (6.4.3), and so $\sigma_i(u) \in A$, and the conclusion follows from the definition of h given in (6.6.6). □

(6.6.8). Let X be a reduced Noetherian prescheme, and $g : X' \to X$ a finite surjective morphism, so that $\mathscr{B} = g_*(\mathscr{O}_{X'})$ is a *coherent* $\mathscr{O}_X$-algebra; suppose further that $\mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a *locally free* $\mathscr{R}(X)$-module *of constant rank* n. We can then apply Lemma (6.6.6), taking $\mathscr{E} = \mathscr{B}$, whence, with the notation of (6.6.6), we obtain a homomorphism of sheaves of multiplicative monoids $\sigma_n : \mathscr{H}om_{\mathscr{O}_X}(\mathscr{B}, \mathscr{B}) \to h_*(\mathscr{O}_Z)$, and by composing this homomorphism with the canonical homomorphism $\mathscr{B} \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{B}, \mathscr{B})$ (6.5.1), we thus obtain a homomorphism of sheaves of multiplicative monoids:

$$N' : \mathscr{B} = g_*(\mathscr{O}_{X'}) \longrightarrow h_*(\mathscr{O}_Z) = \mathscr{C}. \tag{6.6.8.1}$$

With this in mind, for every invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$, $g_*(\mathscr{L}')$ is an invertible $\mathscr{B}$-module (6.1.12), and the method of (6.5.2) allows us to functorially associate to $\mathscr{L}'$ an invertible $\mathscr{C}$-module, which we denote by $N'(g_*(\mathscr{L}'))$.

Lemma (6.6.9). — *Let X be a reduced Noetherian prescheme, and $g : X' \to X$ a finite surjective morphism such that $g_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a locally free $\mathscr{R}(X)$-module of constant rank. Then there exists a reduced Noetherian prescheme Z and a finite birational morphism $h : Z \to X$ that has the following property: for every ample $\mathscr{O}_{X'}$-module $\mathscr{L}'$, the invertible $\mathscr{O}_Z$-module $\mathscr{M}$ such that $h_*(\mathscr{M}) = N'(g_*(\mathscr{L}'))$ (using the notation of (6.6.8)) is ample.*

PROOF. Suppose first of all that the homomorphism $\mathscr{B} \to \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is *injective*. Define Z and h as in (6.6.6) (with $\mathscr{E} = g_*(\mathscr{O}_{X'})$). Let $z \in Z$; we have to show that there exists an integer $m > 0$ and a section t of $\mathscr{M}^{\otimes m}$ over Z such that Z_t is an affine open that contains z (4.5.2). Let $x = h(z)$, and let U be an affine open of X that contains x; then $h^{-1}(U)$ is an affine open neighbourhood of z, and it suffices to find t such that $z \in Z_t \subset h^{-1}(U)$, since Z_t will then necessarily be affine (5.5.8). There exists, by hypothesis, an integer $n > 0$ and a section s' of $\mathscr{L}'^{\otimes n}$ over X' such that

$$g^{-1}(x) \subset X'_{s'} \subset g^{-1}(U) \tag{6.6.9.1}$$

by (4.5.4). By definition, s' is also a section of $g_*(\mathscr{L}')$ over X, and it corresponds, as in (6.5.2), to a section $s = N'(s')$ of $N'(g_*(\mathscr{L}'))$ over X. We will show that, if t is the section s considered as a section of $\mathscr{M}$ over Z, then t is the desired section. Set **II** | 134

$$V = X \setminus g(X' \setminus X'_{s'}) \tag{6.6.9.2}$$

which is an open of X that contains x and is contained in U, by (6.6.9.1) and (6.1.10). We will show that

$$h^{-1}(V) \subset Z_t \subset h^{-1}(U) \tag{6.6.9.3}$$

which will finish the proof. It is equivalent to say that the set T of $y \in X$ such that s_y is invertible contains V and is contained in U. For this, consider first of all an affine open W contained in V; then $g^{-1}(W)$ is an affine open in X', and by (6.6.9.2) $s'_{y'}$ is invertible for all $y' \in g^{-1}(W)$; by the hypotheses on X and $\mathscr{B}$, we can apply the results of (6.4.7), and we see that, if $y = g(y')$, then s_y is invertible; in other words, $V \subset T$. On the other hand, it also follows from (6.4.7) that, conversely, if s_y is invertible, then so too is $s'_{y'}$, which implies that $y' \in g^{-1}(U)$ by (6.6.9.1), and so $y \in U$, whence $T \subset U$ in this case.

We pass from this to the general case by the same argument as in (6.6.2), replacing X' by X'' such that $g'_*(\mathscr{O}_{X''}) \to g'_*(\mathscr{O}_{X''}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is injective, and $\mathscr{L}'$ by an ample $\mathscr{O}_{X''}$-module $\mathscr{L}''$ such that $N'(g_*(\mathscr{L}')) = N'(g'_*(\mathscr{L}''))$. Lemma (6.6.9) is then proven in all cases (with a suitable choice of h). □

Corollary (6.6.10). — *Suppose that the hypotheses of (6.6.9) are satisfied; for every invertible $\mathscr{O}_X$-module $\mathscr{L}$ such that $g^*(\mathscr{L})$ is ample, $h^*(\mathscr{L})$ is ample.*

PROOF. If we set $\mathscr{L}' = g^*(\mathscr{L})$, then $g_*(\mathscr{L}') = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{B}$ (**0**, 5.4.10), so

$$N'(g_*(\mathscr{L}')) = (\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{C})^{\otimes n}$$

(by the same argument as for (6.5.2.4)). We thus conclude that $\mathscr{M} = (h^*(\mathscr{L}))^{\otimes n}$, and since $\mathscr{M}$ is ample, so too is $h^*(\mathscr{L})$ (4.5.6). $\square$

Proposition (6.6.11). — *Let Y be an affine scheme, X an reduced Noetherian prescheme, $f : X \to Y$ a quasi-compact morphism, and $g : X' \to X$ a finite surjective morphism. Let W be an open subset of X such that, for all $x \in W$, either X is normal at the point x or there exists an open neighbourhood $T \subset W$ of x such that $(g_*(\mathscr{O}_{X'}))|T$ is a locally free $(\mathscr{O}_X|T)$-module. Then there exists a reduced Y-prescheme Z and a* finite birational *Y-morphism $h : Z \to X$ such that the restriction of h to $h^{-1}(W)$ is an isomorphism from $h^{-1}(W)$ to W and has the following property: for every invertible $\mathscr{O}_X$-module $\mathscr{L}$ such that $g^*(\mathscr{L})$ is ample for $f \circ g$, $h^*(\mathscr{L})$ is ample for $f \circ h$.*

Proof. Since Y is affine, $g^*(\mathscr{L})$ is ample, and the problem thus reduces to proving that, for a suitable choice of h, $h^*(\mathscr{L})$ is ample (4.6.6). We will show that we can replace g by a finite surjective morphism $g' : X'' \to X$ such that $g'^*(\mathscr{L})$ is ample and $g'_*(\mathscr{O}_{X''}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is a *locally free* $\mathscr{R}(X)$-module *of constant rank*; we will then have reduced to the conditions of (6.6.10), and the proposition will thus be proven.

For this, let $\mathscr{B} = g_*(\mathscr{O}_{X'})$; denote by X_i (for $1 \leqslant i \leqslant n$) the closed reduced subpreschemes of X that have the irreducible components of X as their underlying space (**I**, 5.2.1); they are integral by hypothesis. Let X'_i be the closed subprescheme $g^{-1}(X_i)$ of X', and $g_i : X'_i \to X_i$ the morphism **II** | 135
g restricted to X'_i, which is finite (6.1.5, (iii)) and surjective; let k_i be the *rank* of the $\mathscr{O}_{X_i}$-algebra $\mathscr{B}_i = (g_i)_*(\mathscr{O}_{X'_i})$. Since $\mathscr{B}_i \otimes_{\mathscr{O}_{X_i}} \mathscr{R}(X_i)$ is a constant presheaf (**I**, 7.3.5), the rank k_i is also the rank of the $(\mathscr{O}_X|U)$-algebra $\mathscr{B}|U$ for every open U of X that does not meet the *only* irreducible component X_i. If T is an open subset of X such that $\mathscr{B}|T$ is isomorphic to $\mathscr{O}_X^m|T$, then it follows from the above remark that the numbers k_i are equal to m for all the indices i such that $T \cap X_i \neq \varnothing$. So let U be the open subset of X given by the points that admit a neighbourhood on which $\mathscr{B}$ is a locally free $\mathscr{O}_X$-module, and let U_j (for $1 \leqslant j \leqslant s$) be its connected components, which are open in X and finitely many (since U is Noetherian); denote by V_j the closed subprescheme of X' given by the *closure* of the subprescheme induced on the open $g^{-1}(U_j)$ (**I**, 9.5.11). By the above, for all indices i such that $X_i \cap U_j \neq \varnothing$, the ranks k_i are all equal to a single integer m_j; note also that a single X_i cannot meet two U_j with different indices j. Let i_λ be the indices i such that $X_i \cap U = \varnothing$. Consider the product k of all the k_i, set $n_i = k/k_i$, and let X'' be the prescheme defined as follows. For each j (for $1 \leqslant j \leqslant s$), consider k/m_j preschemes isomorphic to V_j, and for each λ, k/k_{i_λ} preschemes isomorphic to X'_{i_λ}; then X'' is the *sum* of all these preschemes. We define a morphism $g'' : X'' \to X'$ that reduces to the canonical injection on each of the summands of X''; it is clear that g'' is a finite dominant morphism, and thus surjective (since a finite morphism is closed (6.1.10)); set $g' = g \circ g''$, which is a finite surjective morphism $X'' \to X$; we have $g'^*(\mathscr{L}) = g''^*(g^*(\mathscr{L}))$, and so $g'^*(\mathscr{L})$ is an ample $\mathscr{O}_{X''}$-module (5.1.12). It is then clear that, for this new prescheme X'', the ranks defined as the k_i for X' are all *equal to* k; taking (**I**, 7.3.3) into account, we immediately conclude that, for every affine open T of X, $(g'_*(\mathscr{O}_{X''}) \otimes_{\mathscr{O}_X} \mathscr{R}(X))|T$ is an $(\mathscr{R}(X)|T)$-module isomorphic to $(\mathscr{R}(X)|T)^k$. $\square$

Corollary (6.6.12). — *If, in the statement of (6.6.11), we have $W = X$, then for an invertible $\mathscr{O}_X$-module $\mathscr{L}$ to be ample for f, it is necessary and sufficient that $g^*(\mathscr{L})$ be ample for $f \circ g$.*

Remark (6.6.13). — In Chapter III, we will see that, if Y is Noetherian, and f of finite type, and if the restriction of f to the closed reduced subprescheme of X that has $X \setminus W$ as its underlying space is *proper*, then the conclusion of (6.6.12) still holds true. But we will give, in Chapter V, examples of algebraic schemes X over a field K (with the structure morphism $X \to \operatorname{Spec}(K)$ not proper) whose normalisation X' is quasi-affine, but which are not quasi-affine (in that $\mathscr{O}_X$ is not ample, even though $\mathscr{O}_{X'}$ is (5.1.12) and the morphism $X' \to X$ is finite and surjective (6.3.10)). We will see in the next section that this circumstance cannot happen when we replace "quasi-affine" by "affine".

6.7. Chevalley's theorem We are going to prove (with the help of Serre's criterion (5.2.1)) the following theorem, which was proven by C. Chevalley by other methods, in the case of *algebraic* schemes.

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Theorem (6.7.1). — *Let X be an affine scheme, Y a Noetherian prescheme, and $f : X \to Y$ a finite surjective morphism. Then Y is an affine scheme.*

PROOF. It is clear that $f_{\text{red}} : X_{\text{red}} \to Y_{\text{red}}$ is finite (6.1.5, (vi)); since X_{red} is an affine scheme, and since saying that Y is affine is equivalent to saying that Y_{red} is affine (since Y is Noetherian (**I**, 6.1.7)), we see that we can suppose Y to be *reduced*. For every closed subset Y' of Y, there exists exactly one reduced subprescheme of Y that has Y' as its underlying space (**I**, 5.1.2); its inverse image $f^{-1}(Y')$, which is canonically isomorphic to $X \times_Y Y'$ (**I**, 4.4.1), is affine as a closed subprescheme of X, and the restriction of f to $f^{-1}(Y')$, which can be identified with $f \times_Y 1_{Y'}$, is a finite surjective morphism (6.1.5, (iii)). By the principal of Noetherian induction (**0**, 2.2.2), we can thus (taking (**I**, 6.1.7) into account) reduce to proving the theorem under the hypothesis that for *every* closed subset $Y' \neq Y$, every closed subprescheme of Y that has Y' as its underlying space is affine. We thus conclude that, *for every coherent $\mathscr{O}_Y$-module $\mathscr{F}$ whose (closed) support Z is distinct from Y, we have* $\mathrm{H}^1(Y, \mathscr{F}) = 0$. Indeed, there exists a closed subprescheme Y' of Y that has Z as its underlying space and is such that, if $j : Y \to Y$ is the canonical injection, then $\mathscr{F} = j_*(j^*(\mathscr{F}))$ (**I**, 9.3.5); then (5.2.3) $\mathrm{H}^1(Y, \mathscr{F}) = \mathrm{H}^1(Y', j^*(\mathscr{F})) = 0$, by (**I**, 5.1.9.2).

Suppose first of all that Y is not irreducible, and let Y' be an irreducible component of Y, and $Y'' = Y \setminus Y'$; we again denote by Y' the closed reduced subprescheme of Y that has Y' as its underlying space, and by j the canonical injection $Y' \to Y$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_Y$-module, and consider the canonical homomorphism

$$\rho : \mathscr{F} \longrightarrow \mathscr{F}' = j_*(j^*(\mathscr{F}))$$

(**0**, 4.4.3); $\mathscr{F}'$ is a coherent $\mathscr{O}_Y$-module by (**0**, 5.3.10) and (**0**, 5.3.12), since $j_*(\mathscr{O}_{Y'}) = \mathscr{O}_Y / \mathscr{J}$, where we denote by $\mathscr{J}$ the sheaf of ideals of $\mathscr{O}_Y$ that defines the subprescheme Y'. Then $\mathscr{G} = \operatorname{Ker} \rho$ and $\mathscr{K} = \operatorname{Im} \rho$ are also coherent $\mathscr{O}_Y$-modules (**0**, 5.3.4); but, by definition, the fibre $\mathscr{F}'_y$ of $\mathscr{F}'$ at the generic point y of Y' is equal to $\mathscr{F}_y$, since y is interior to Y' and thus $\mathscr{J}_y = 0$, since Y is reduced. We thus conclude that y is not contained in the (closed) support of $\mathscr{G}$; also, the support of $\mathscr{F}'$ (and *a fortiori* that of $\mathscr{K}$) is contained in Y'; in other words, the supports of $\mathscr{G}$ and $\mathscr{K}$ are *distinct from* Y. We thus deduce that $\mathrm{H}^1(Y, \mathscr{G}) = \mathrm{H}^1(Y, \mathscr{K}) = 0$, and the exact sequence of cohomology applied to the exact sequence $0 \to \mathscr{G} \to \mathscr{F} \to \mathscr{K} \to 0$ implies that $\mathrm{H}^1(Y, \mathscr{F}) = 0$. We thus conclude by Serre's criterion (5.2.1).

So suppose that Y is irreducible, and thus *integral*. We can also suppose that X is *integral*: if we denote by X_i the closed reduced subpreschemes of X that have the irreducible components of X as their underlying space (**I**, 5.2.1), and by g_i the restriction of g to X_i, then at least one of the g_i is dominant, and since it is a finite morphism (6.1.5), it is surjective (6.1.10); since X_i is also an affine scheme, we see that we can replace X by X_i in the statement.

Lemma (6.7.1.1). — *Let X and Y be integral Noetherian preschemes, x (resp. y) the generic point of X (resp. Y), and $f : X \to Y$ a finite surjective morphism. Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module such that there exists an affine open neighbourhood U of y and a section $g \in \Gamma(X, \mathscr{L})$ such that $x \in X_g \subset f^{-1}(U)$. Then there exist* **II** | 137 *integers $m, n > 0$, a homomorphism $u : \mathscr{O}_Y^m \to f_*(\mathscr{L}^{\otimes n})$, and an open neighbourhood V of y such that the restriction $u|V$ is an isomorphism from $\mathscr{O}_Y^m|V$ to $f_*(\mathscr{L}^{\otimes n})|V$.*

PROOF. Let C be the (integral) ring of U, and $k = \mathscr{O}_y$ its field of fractions: since f is finite, $U' = f^{-1}(U)$ is affine (1.3.2); let D be its (integral) ring with field of fractions $K = \mathscr{O}_x$; by hypothesis, D is a C-module of finite type (6.1.4), and so K is an extension of finite rank of k. The fibre $f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y)) = X \times_Y \operatorname{Spec}(\mathscr{O}_y)$ can be identified with $\operatorname{Spec}(K)$ (**I**, 3.6.5); let s_i (for $1 \leqslant i \leqslant m$) be elements of D that form a basis of K over k. There exists $n > 0$ such that the sections $(s_i|X_g)g^{\otimes n}$ of $\mathscr{L}^{\otimes n}$ over X_g extend to sections b_i (for $1 \leqslant i \leqslant m$) of $\mathscr{L}^{\otimes n}$ over X (**I**, 9.3.1). The b_i are also, by definition, sections of $f_*(\mathscr{L}^{\otimes n})$ over Y, and thus define a homomorphism $u : \mathscr{O}_Y^m \to f_*(\mathscr{L}^{\otimes n})$ (**0**, 5.1.1); we will show that u is the desired homomorphism. We have that

$\mathscr{L}^{\otimes n}|U' = \widetilde{M}$, where M is a D-module of finite type, so if ϕ is the injection $C \to D$ corresponding to the morphism $f^{-1}(U) \to U$ given by the restriction of f, then $M_{[\phi]}$ is a C-module of finite type; then

$$f_*(\mathscr{L}^{\otimes n})|U = (M_{[\phi]})^\sim$$

(**I**, 1.6.3) is coherent, and since U is an arbitrary affine open of Y, $f_*(\mathscr{L}^{\otimes n})$ is coherent; furthermore, $u|U = \widetilde{\theta}$, where θ is a C-homomorphism $C^m \to M_{[\phi]}$, and $u_y = \theta_y$ is the homomorphism $\theta \otimes 1 : K^m = C^m \otimes K \to M_{[\phi]} \otimes K$; but the latter is, by definition, an *isomorphism*, since the $(b_i)_x$ form a basis of $M_{[\phi]} \otimes K$ over k, with g_x being, by hypothesis, a generator of $(\mathscr{L}^{\otimes n})_x$. We thus conclude that the supports of the $\mathscr{O}_Y$-modules $\operatorname{Ker} u$ and $\operatorname{Coker} u$ do not contain y; since these $\mathscr{O}_Y$-modules are coherent (**0**, 5.3.3), their supports are closed (**0**, 5.2.2), whence the lemma. □

With this, the hypotheses of Lemma (6.7.1.1) are satisfied in the case that we are considering, by taking $\mathscr{L} = \mathscr{O}_X$, since X is affine (**I**, 1.1.10); set $\mathscr{A} = \mathscr{O}_Y$ and $\mathscr{B} = f_*(\mathscr{O}_X)$. By Serre's criterion (5.2.1.1), it suffices to prove that, for every coherent $\mathscr{O}_Y$-module $\mathscr{F}$, we have that $\mathrm{H}^1(Y, \mathscr{F}) = 0$; it even suffices to prove this in the case where $\mathscr{F} \subset \mathscr{O}_Y$, which implies that $\mathscr{F}$ is torsion free, since Y is integral; in fact, we will show that $\mathrm{H}^1(Y, \mathscr{O}_Y) = 0$ for *every* coherent *torsion-free* $\mathscr{O}_Y$-module $\mathscr{F}$. But the homomorphism $u : \mathscr{A}^m \to \mathscr{B}$ defines a homomorphism

$$v : \mathscr{G} = \mathscr{H}om_{\mathscr{A}}(\mathscr{B}, \mathscr{F}) \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{A}^m, \mathscr{F}) = \mathscr{F}^m.$$

We will first show that v is *injective*: by hypothesis, $\mathscr{T} = \operatorname{Coker} u$ has a support that does not meet V, and is thus a *torsion* $\mathscr{O}_Y$-module (**I**, 7.4.6); the exact sequence

$$\mathscr{A}^m \longrightarrow \mathscr{B} \longrightarrow \mathscr{T} \longrightarrow 0$$

gives, by left exactness of the functor $\mathscr{H}om_{\mathscr{A}}$, the exact sequence

$$0 \longrightarrow \mathscr{H}om_{\mathscr{A}}(\mathscr{T}, \mathscr{F}) \longrightarrow \mathscr{G} \xrightarrow{v} \mathscr{F}^m.$$

But since $\mathscr{F}$ is torsion free, we have that $\mathscr{H}om_{\mathscr{A}}(\mathscr{T}, \mathscr{F}) = 0$ (**0**, 5.2.6), whence our claim. We thus have the exact sequence

$$0 \longrightarrow \mathscr{G} \longrightarrow \mathscr{F}^m \longrightarrow \operatorname{Coker} v \longrightarrow 0$$

where $\mathscr{G}$ and $\operatorname{Coker} v$ are coherent $\mathscr{O}_Y$-modules ((**0**, 5.3.4) and (**0**, 5.3.5)); by the exact sequence **II** | 138
of cohomology, it would suffice to show that $\mathrm{H}^1(Y, \mathscr{G}) = \mathrm{H}^1(Y, \operatorname{Coker} v) = 0$ in order to deduce that $\mathrm{H}^1(Y, \mathscr{F}^m) = (\mathrm{H}^1(Y, \mathscr{F}))^m = 0$, and thus that $\mathrm{H}^1(Y, \mathscr{F}) = 0$. But the restriction $v|V$ is an isomorphism, and so the support of $\operatorname{Coker} v$ is distinct from Y, whence $\mathrm{H}^1(Y, \operatorname{Coker} v) = 0$ by the hypothesis at the start. Now, $\mathscr{G}$ is a coherent *$\mathscr{B}$-module* (**I**, 9.6.4); since X is affine over Y, there exists a quasi-coherent $\mathscr{O}_X$-module $\mathscr{K}$ such that $\mathscr{G}$ is isomorphic to $f_*(\mathscr{K})$ (1.4.3); since X is affine, we have that $\mathrm{H}^1(X, \mathscr{K}) = 0$ (**I**, 5.1.9.2), and so $\mathrm{H}^1(Y, \mathscr{G}) = 0$ by (5.2.3), which finishes the proof of Theorem (6.7.1). □

Corollary (6.7.2). — *Let X be a Noetherian prescheme, $(X_i)_{1 \leqslant i \leqslant n}$ a finite cover of the space X consisting of closed subsets. For X to be affine, it is necessary and sufficient that, for each i, there exist a closed subprescheme of X that has X_i as its underlying space and is affine.*

Proof. Indeed, if this is the case, then let X' be the scheme given by the *sum* of the X_i; it is clear that X' is affine, and we define a surjective morphism $f : X' \to X$ by taking the restriction of f to X_i to be the canonical injection. Everything reduces to showing that f is *finite*, by (6.7.1), and this we have already seen in (6.1.5). □

Corollary (6.7.3). — *For a Noetherian prescheme X to be affine, it is necessary and sufficient that the closed reduced subpreschemes whose underlying spaces are the irreducible components of X be affine.*

§7. Valuative criteria

In this section, we give valuative criteria for separation and properness for a given morphism, that is, criteria which introduce a variable auxiliary scheme of the form $\operatorname{Spec}(A)$, where A is a valuation ring. Under certain suitable "Noetherian" hypotheses, we can refine our criteria and restrict to the case where A is a *discrete* valuation ring. This will be the only case that we need to concern ourselves with in all that follows, and we introduce arbitrary valuation rings, in the general case, only to discuss the links with the classical study of such objects.

7.1. Reminder on valuation rings

(7.1.1). Amongst the many diverse equivalent properties that characterise valuation rings, we will use the following: a ring A is said to be a *valuation ring* if it is an integral ring which is not a field, and A is *maximal* in the set of local rings strictly contained in the field of fractions K of A under the domination relation (**I**, 8.1.1). Recall that a valuation ring is *integrally closed*. If A is a valuation ring, then so too is $A_{\mathfrak{p}}$ for any prime ideal $\mathfrak{p} \neq 0$ of A.

(7.1.2). Let K be a field, and A a local subring of K that is not a field; then there exists a valuation ring that both dominates A and has K as its field of fractions ([**CC**, p. 1-07, lemma 2]). **II** | 139

Now let B be a valuation ring, k its residue field, K its field of fractions, and L an extension of k. Then there exists a *complete* valuation ring C that dominates B and whose residue field is L. Indeed, L is the algebraic extension of a pure transcendental extension $L' = k(T_\mu)_{\mu \in M}$; we know that we can extend the valuation of B corresponding to B to a valuation of $K' = K(T_\mu)_{\mu \in M}$ in such a way that L' is the residue field of this valuation ([**Jaf60**, p. 98]); replacing B by the completion of the ring of this extended valuation, we see that that we can restrict to the case where B is complete and L is an algebraic closure of k. If $\overline{K}$ is an algebraic closure of K, we can then extend the valuation that defines B to $\overline{K}$, and the corresponding residue field is an algebraic closure of k, as we can see by lifting to $\overline{K}$ the coefficients of a unitary polynomial of $k[T]$. We are thus finally led to the case where $L = k$, and it then suffices to take C to be the completion of B in order to satisfy our claim.

(7.1.3). Let K be a field, and A a subring of K; the integral closure A' of A in K is the intersection of the valuation rings that contain A and have K as their field of fractions ([**Sam53b**, p. 51, th. 2]). Proposition (7.1.2) can then be expressed geometrically in an equivalent form:

Proposition (7.1.4). — *Let Y be a prescheme, $p : X \to Y$ a morphism, x a point of X, $y = p(x)$, and $y' \neq y$ a specialisation* (**0**, 2.1.2) *of y. Then there exists a local scheme Y' which is the spectrum of some valuation ring, and a separated morphism $f : Y' \to Y$ such that, denoting the unique closed point of Y' by a and the generic point of Y' by b, we have $f(a) = y'$ and $f(b) = y$. We can furthermore suppose that one of the two additional following properties are satisfied:*

(i) *Y' is the spectrum of a complete valuation ring whose residue field is algebraically closed, and there exists a $k(y)$-homomorphism $k(x) \to k(b)$.*
(ii) *There exists a $k(y)$-isomorphism $k(x) \xrightarrow{\sim} k(b)$.*

PROOF. Let Y_1 be the closed reduced subprescheme of Y that has $\overline{\{y\}}$ as its underlying space (**I**, 5.2.1), and let X_1 be the closed subprescheme given by the inverse image $p^{-1}(Y_1)$; since $y' \in \overline{\{y\}}$ by hypothesis, and since $k(x)$ is the same in X and in X_1, we can assume that Y is *integral*, with generic point y; $\mathscr{O}_{y'}$ is then an integral local ring that is not a field, and whose field of fractions is $\mathscr{O}_y = k(y)$, and $k(x)$ is then an extension of $k(y)$. To satisfy the conditions $f(a) = y'$ and $f(b) = y$ as well as the additional condition (i) (resp. (ii)), we take $Y' = \operatorname{Spec}(A')$, where A' is a valuation ring that dominates $\mathscr{O}_{y'}$ (resp. a valuation ring that dominates $\mathscr{O}_{y'}$ and whose field of fractions is $k(x)$); the existence such an of A' is guaranteed by (7.1.2). □

(7.1.5). Recall that a local ring A is said to be *of dimension 1* if there exists a prime ideal distinct from the maximal ideal $\mathfrak{m}$, and if every prime ideal of A distinct from $\mathfrak{m}$ is a *minimal* prime ideal; when A is *integral*, it is equivalent to ask that $\mathfrak{m}$ and (0) be the only prime ideals, with $\mathfrak{m} \neq (0)$; in other words, $Y = \operatorname{Spec}(A)$ consists of two points a and b: a is the unique *closed* point, we have $\mathfrak{j}_a = \mathfrak{m}$, and $k(a) = k$ is the *residue field* $k = A/\mathfrak{m}$; b is the *generic point* of Y, $\mathfrak{j}_b = (0)$, with the set $\{b\}$ being the unique open subset of Y distinct from both $\varnothing$ and Y (an open subset which is thus *everywhere dense*), and $k(b) = K$ is the *field of fractions* of A. **II** | 140

(7.1.6). For a local ring A, Noetherian and of dimension 1, we know ([**CC**, pp. 2-08 and 17-01]) that the following conditions are equivalent:

(a) A is normal;
(b) A is regular;

(c) A is a valuation ring;

furthermore, A is then a *discrete valuation ring*. Propositions (7.1.2) and (7.1.3) then have the following analogues for discrete valuation rings:

Proposition (7.1.7). — *Let A be an integral local Noetherian ring that is not a field, K its field of fractions, and L an extension of finite type of K; then there exists a discrete valuation ring that dominates A and has L as its field of fractions.*

PROOF. Suppose first of all that $L = K$. Let $\mathfrak{m}$ be the maximal ideal of A, $(x_1, \ldots, x_n)$ a system of non-null generators of $\mathfrak{m}$, and B the subring $A[x_2/x_1, \ldots, x_n/x_1]$ of K, which is Noetherian. It is immediate that the ideal $\mathfrak{m}B$ of B is identical to the principal ideal $x_1 B$; if $\mathfrak{p}$ is a minimal prime ideal of $x_1 B$, then $\mathfrak{p}$ is of rank 1 ([**SZ60**, t. I, p. 277]); in other words, $B_{\mathfrak{p}}$ is a local Noetherian ring *of dimension 1*; it is clear that $\mathfrak{p}B_{\mathfrak{p}} \cap A$ is an ideal of A that contains $\mathfrak{m}$ and that does not contain 1, and is thus equal to $\mathfrak{m}$, and so $B_{\mathfrak{p}}$ *dominates* A (**I**, 8.1.1). It follows from the Krull-Akizuki Theorem ([**Nag55**, p. 293]) that the integral closure C of $B_{\mathfrak{p}}$ is a Noetherian ring (even though C is not necessarily a $B_{\mathfrak{p}}$-module of finite type); if $\mathfrak{n}$ is a maximal ideal of C, then $C_{\mathfrak{n}}$ is a normal local Noetherian ring of dimension 1 ([**Nag55**, p. 295]), and thus a discrete valuation ring that dominates $B_{\mathfrak{p}}$ and *a fortiori* A.

Now, if L is an extension of finite type of K, we can, by the above, restrict to the case where A is already a discrete valuation ring. Let w be a valuation of K associated to A; there exists a discrete valuation w' of L that *extends* w: we can restrict, by induction on the number of generators of L, to the case where $L = K(\alpha)$, and then the proposition is classical ([**Jaf60**, p. 106]). □

Corollary (7.1.8). — *Let A be a Noetherian integral ring, K its field of fractions, and L an extension of finite type of K. Then the integral closure of A in L is the intersection of the discrete valuation rings that have L as their field of fractions and that contain A.*

PROOF. Indeed, such a discrete valuation ring, being normal, contains *a fortiori* every element of L that is integral over A. It thus suffices to prove that, if $x \in L$ is not integral over A, then there exists a discrete valuation ring C that has L as its field of fractions, contains A, and does not contain x. The hypothesis on x implies that $x \notin B = A[1/x]$, or, in other words, that $1/x$ is not invertible in the Noetherian ring B. There is thus a prime ideal $\mathfrak{p}$ of B that contains $1/x$. The integral local ring $B_{\mathfrak{p}}$ is Noetherian and contained in L, which is an extension of finite type of the field of fractions of $B_{\mathfrak{p}}$ (with the latter containing K). By (7.1.7), there thus exists a discrete valuation ring C that dominates $B_{\mathfrak{p}}$ and has L as its field of fractions; since $1/x \in \mathfrak{p}B_{\mathfrak{p}}$ belongs to the maximal ideal of C, we have that $x \notin C$, which concludes the proof. □

The geometric form of (7.1.7) is the following:

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Proposition (7.1.9). — *Let Y be a locally Noetherian prescheme, $p : X \to Y$ a morphism locally of finite type, x a point of X, $y = p(x)$, and $y' \neq y$ a specialisation of y. Then there exists a local scheme Y', spectrum of a discrete valuation ring, a separated morphism $f : Y' \to Y$, and a rational Y-map g from Y' to X, such that, denoting the closed point of Y' by a, and the generic point of Y' by b, we have $f(a) = y'$, $f(b) = y$, $g(b) = x$, and such that, in the commutative diagram*

(7.1.9.1)
$$\begin{tikzcd} & k(x) \arrow[dl, "\gamma"'] \\ k(b) & k(y) \arrow[l, "\phi"] \arrow[u, "\pi"'] \end{tikzcd}$$

(where π, ϕ, and γ are the homomorphisms corresponding to p, f, and g, respectively) the morphism γ is a bijection.

PROOF. As in (7.1.4), we can restrict to the case where Y is integral with generic point y (taking (**I**, 6.4.3, iv) into account), and, since the question is local on X and Y, we can assume that p is of finite type; we are then in the situation of (7.1.4), with the additional property that $k(x)$ is an extension *of finite type* of $k(y)$ (**I**, 6.4.11) and that $\mathscr{O}_{y'}$ is Noetherian; this lets us apply (7.1.7) and take $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring that dominates $\mathscr{O}_{y'}$ and whose field of fractions is $k(x)$. We have thus defined a commutative diagram (7.1.9.1) where γ is a bijection, with π and ϕ corresponding to the morphisms p and f. Furthermore, since X and Y are locally Noetherian (**I**, 6.6.2) and since Y' is integral, there exists exactly one rational Y-map g from Y' to X to which corresponds the isomorphism γ (**I**, 7.1.15), which finishes the proof. □

7.2. Valuative criterion for separatedness

Proposition (7.2.1). — *Let X and Y be preschemes, and $f : X \to Y$ a quasi-compact morphism. The following two conditions are equivalent:*

(a) *The morphism f is closed.*
(b) *For all $x \in X$, and every specialisation $y' \neq y$ of $y = f(x)$, there exists a specialisation x' of x such that $f(x') = x$.*

Proof. Condition (b) implies that $f(\overline{\{x\}}) = \overline{\{y\}}$, and is thus a consequence of (a). To show that (b) implies (a), consider a closed subset X' of the underlying space X; let $Y' = \overline{f(X')}$, and show that $Y' = f(X')$ as follows. Consider the closed reduced subpreschemes of X and Y whose underlying spaces are X' and Y' (respectively) (**I**, 5.2.1); there then exists a morphism $f' : X' \to Y'$ such that the diagram

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d] & Y' \arrow[d] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

commutes (**I**, 5.2.2), and, since f is quasi-compact, so too is f'. We are thus led to proving that, if f is a quasi-compact and *dominant* morphism, then condition (b) implies that $f(X) = Y$. But let y' be a point of Y, and let y be the generic point of an irreducible component of Y that contains y'; by (b), it suffices to show that $f^{-1}(y)$ is not empty. But we know that this property is a consequence of the fact that f is quasi-compact and dominant (**I**, 6.6.5). □ **II** | 142

Corollary (7.2.2). — *Let $f : X \to Y$ be a quasi-compact immersion. For the underlying space X to be closed in Y, it is necessary and sufficient for it to contain every specialisation (in Y) of all of its points.*

Proposition (7.2.3). — *Let Y be a prescheme (resp. a locally Noetherian prescheme), and $f : X \to Y$ a morphism (resp. a morphism locally of finite type). The following conditions are equivalent:*

(a) *f is separated.*
(b) *The diagonal morphism $X \to X \times_Y X$ is quasi-compact, and, for every Y-prescheme of the form $Y' = \operatorname{Spec}(A)$, with A some valuation ring (resp. some discrete valuation ring), any two Y-morphisms from Y' to X that agree on the generic point of Y' are equal.*
(c) *The diagonal morphism $X \to X \times_Y X$ is quasi-compact, and, for every Y-prescheme of the form $Y' = \operatorname{Spec}(A)$, with A some valuation ring (resp. some discrete valuation ring), any two Y'-sections of $X' = X_{(Y')}$ that agree on the generic point of Y' are equal.*

Proof. The equivalence of (b) and (c) follows from the bijective correspondence between Y-morphisms from Y' to X and Y'-sections of X' (**I**, 3.3.14). If X is separated over Y, condition (b) is satisfied, by (**I**, 7.2.2.1), since Y' is integral. It remains to show that (b) implies that the diagonal morphism $\Delta : X \to X \times_Y X$ is closed, and it is equivalent to show that it satisfies the criteria of (7.2.2). But let z be a point of the diagonal $\Delta(X)$, and $z' \neq z$ a specialisation of z in $X \times_Y X$. There then exists (7.1.4) a valuation ring A and a morphism f from $Y' = \operatorname{Spec}(A)$ to $X \times_Y X$ such that f sends the closed point a of Y' to z', and the generic point b of Y' to z; this morphism makes Y' an $(X \times_Y X)$-prescheme, and *a fortiori* a Y-prescheme. If we compose the two projections of $X \times_Y X$ with f, then we obtain two Y-morphisms, g_1 and g_2, from Y' to X, which, by hypothesis, agree on the point b; they are thus equal to one single morphism g, which implies (**I**, 5.3.1) that f factors as $f = \Delta \circ g$, and thus $z' \in \Delta(X)$. If we suppose that Y is locally Noetherian and f is locally of finite type, then $X \times_Y X$ is locally Noetherian (**I**, 6.6.7); we can thus follow the same argument as before by supposing that A is a discrete valuation ring, by (7.1.9). □

Remark (7.2.4). — (i) The hypothesis that the morphism Δ is quasi-compact is always satisfied whenever Y is locally Noetherian and f is locally of finite type, because $X \times_Y X$ is then locally Noetherian (**I**, 6.6.4, i). In the general case, this also implies that, for every cover (U_α) of X by affine opens, the sets $U_\alpha \cap U_\beta$ are *quasi-compact*.

(ii) For f to be separated, it is *sufficient* for condition (b) or (c) to be satisfied for some valuation ring A that is *complete* and whose residue field is *algebraically closed*; this follows from the proofs of (7.2.3) and (7.2.4).

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7.3. Valuative criterion for properness

Proposition (7.3.1). — *Let A be a valuation ring, $Y = \operatorname{Spec}(A)$, b the generic point of Y, X an integral* scheme, *and $f : X \to Y$ a* closed *morphism such that $f^{-1}(b)$ consists of a single point x and such that the corresponding homomorphism $k(b) \to k(x)$ is bijective. Then f is an isomorphism.*

Proof. Since f if closed and dominant, we have that $f(X) = Y$; it suffices (**I**, 4.2.2) to prove that, for all $y' \neq b$ in Y, there exists *exactly one* point x' such that $f(x') = y'$, and that the corresponding homomorphism $\mathscr{O}_{y'} \to \mathscr{O}_{x'}$ is bijective, since then f will be a homeomorphism. But if $f(x') = y'$ then $\mathscr{O}_{x'}$ is a local ring contained in $K = k(x) = k(b)$ and dominating $\mathscr{O}_{y'}$; the latter is the local ring $A_{y'}$, and is thus a valuation ring (7.1.1) that has K as its field of fractions. Also, $\mathscr{O}_{x'} \neq K$, since x' is not the generic point of X (**0**, 2.1.3); we thus conclude that $\mathscr{O}_{x'} = \mathscr{O}_{y'}$. Since X is an integral scheme, the fact that $\mathscr{O}_{x'} = \mathscr{O}_{x''}$ implies that $x' = x''$ (**I**, 8.2.2), which finishes the proof. □

(7.3.2). Let A be a valuation ring, K its field of fractions, $Y = \operatorname{Spec}(A)$, and b the generic point of Y, such that $\mathscr{O}_b = k(b)$ is equal to K; let $f : X \to Y$ be a morphism. We know (**I**, 7.1.4) that the *rational Y-sections* of X are in bijective correspondence with the *germs* of Y-sections (defined in a neighbourhood of b) at the point b, whence we have a canonical map

$$\Gamma_{\mathrm{rat}}(X/Y) \longrightarrow \Gamma(f^{-1}(b)/\operatorname{Spec}(K)) \tag{7.3.2.1}$$

with the elements of $\Gamma(f^{-1}(b)/\operatorname{Spec}(K))$ being identified, by definition (**I**, 3.4.5), with the *points of $f^{-1}(b) = X \otimes_A K$ that are rational over K*. When f is *separated*, it follows from (**I**, 5.4.7) that the map (7.3.2.1) is *injective*, since Y is an integral scheme.

Composing (7.3.2.1) with the canonical map $\Gamma(X/Y) \to \Gamma_{\mathrm{rat}}(X/Y)$ (**I**, 7.1.2), we obtain a canonical map

$$\Gamma(X/Y) \longrightarrow \Gamma(f^{-1}(b)/\operatorname{Spec}(K)). \tag{7.3.2.2}$$

When f is *separated*, this map is again *injective* (**I**, 5.4.7).

Proposition (7.3.3). — *Let A be a valuation ring with field of fractions K, $Y = \operatorname{Spec}(A)$, b the generic point of Y, and $f : X \to Y$ a* separated *and* closed *morphism. Then the canonical map* (7.3.2.2) *is bijective (which is equivalent to saying that it is* surjective, *and implies that the rational Y-sections of X are* everywhere defined*).*

Proof. So let x be a point of $f^{-1}(b)$ that is *rational* over K. Since f is separated, so too is the morphism $f^{-1}(b) \to \operatorname{Spec}(K)$ corresponding to f (**I**, 5.5.1, iv), and, since every section of $f^{-1}(b)$ is a closed immersion (**I**, 5.4.6), $\{x\}$ is closed *in* $f^{-1}(b)$. Consider the closed reduced subprescheme X' of X that has the closure $\overline{\{x\}}$ of $\{x\}$ *in* X as its underlying space. It is clear that the restriction of f to X' satisfies the hypotheses of (7.3.1), and is thus an *isomorphism* from X' to Y, whose inverse isomorphism is the desired Y-section of X. □

(7.3.4). To state the two following results, we use a terminology that will be justified and discussed in chapter IV: if F is a subset of a prescheme Y, we define the *codimension* of F in Y, denoted $\operatorname{codim}_Y F$, **II** | 144
to be the lower bound of the integers $\dim(\mathscr{O}_z)$ over all z in F.

Corollary (7.3.5). — *Let Y be a locally Noetherian reduced prescheme, and N the set of points $y \in Y$ where Y is not regular* (**0**, 4.1.4)*; suppose that* $\operatorname{codim}_Y N \geqslant 2$. *Let $f : X \to Y$ be a morphism of finite type, both* separated *and* closed, *and let g be a rational Y-section of X; if Y' is the set of points of Y where g is not defined (a set which is* closed (**I**, 7.2.1)*), then* $\operatorname{codim}_Y Y' \geqslant 2$.

Proof. It suffices to prove that g is defined at every point $z \in Y$ such that $\dim \mathscr{O}_z \leqslant 1$. If $\dim \mathscr{O}_z = 0$, then z is the generic point of an irreducible component of Y (**I**, 1.1.14), and so belongs to every everywhere-dense open subset of Y, and, in particular, to the domain of definition of g. So suppose that $\dim \mathscr{O}_z = 1$; by hypothesis, $\mathscr{O}_z$ is then a regular Noetherian local ring, and thus (7.1.6) a *discrete valuation ring*. Let $Z = \operatorname{Spec}(\mathscr{O}_z)$; since $U = Y - Y'$ is everywhere dense, $U \cap Z$ is nonempty (**I**, 2.4.2); let g' be the rational map from Z to X induced by g (**I**, 7.2.8); it suffices to show that g' is a *morphism* (**I**, 7.2.9). But g' can be considered as a rational Z-section of the Z-prescheme $f^{-1}(Z) = X \times_Y Z$; it is clear that the morphism $f^{-1}(Z) \to Z$ corresponding to f is closed, and it follows from (**I**, 5.5.1, i) that it is separated; we thus conclude from (7.3.3) that g' is everywhere defined; since Z is reduced, and X is separated over Y, g' is a morphism (**I**, 7.2.2). □

Corollary (7.3.6). — *Let S be a locally Noetherian prescheme, and X and Y both S-preschemes; suppose that Y is reduced, and further that the set N of points $y \in Y$ where Y is not regular is such that $\operatorname{codim}_Y N \geqslant 2$; suppose finally that the structure morphism $X \to S$ is proper. Let f be a rational S-map from Y to X, and let Y' be the points of Y where f is not defined; then $\operatorname{codim}_Y Y' \geqslant 2$.*

Proof. We know (**I**, 7.1.2) that we can identify the rational S-maps from Y to X with the rational Y-sections of $X \times_S Y$; since the structure morphism $X \times_S Y \to Y$ is closed (5.4.1), we can apply (7.3.5), whence the corollary. □

Remark (7.3.7). — The hypotheses on Y in (7.3.5) and (7.3.6) will be satisfied in particular when Y is *normal* (**0**, 4.1.4), by (7.1.6).

We can characterise the universally closed morphisms (resp. proper morphisms) by a converse of (7.3.3):

Theorem (7.3.8). — *Let Y be a prescheme (resp. a locally Noetherian prescheme), and $f : X \to Y$ a quasi-compact separated morphism (resp. a morphism of finite type). The following conditions are equivalent:*

(a) *f is universally closed (resp. proper).*
(b) *For every Y-scheme of the form $Y' = \operatorname{Spec}(A)$, where A is a valuation ring (resp. a discrete valuation ring) with field of fractions K, the canonical map*

$$\operatorname{Hom}_Y(Y', X) \longrightarrow \operatorname{Hom}_Y(\operatorname{Spec}(K), X)$$

corresponding to the canonical injection $A \to K$ is surjective (resp. bijective). **II** | 145
(c) *For every Y-scheme of the form $Y' = \operatorname{Spec}(A)$, where A is a valuation ring (resp. a discrete valuation ring), the canonical map (7.3.2.2) with respect to the Y'-prescheme $X_{(Y')}$ is surjective (resp. bijective).*

Proof. The equivalence of (b) and (c) follows immediately from (**I**, 3.3.14); (a) implies (b), since (a) implies, in either case, that $f_(Y')$ is separated (**I**, 5.5.1, iv) and closed, and it suffices to apply (7.3.3). It remains to prove that (b) implies (a). We first consider the case where Y is arbitrary, and f is separated and quasi-compact. If condition (b) is satisfied by f, then it is also satisfied by $f_{(Y'')} : X_{(Y'')} \to Y''$, where Y'' is an arbitrary Y-prescheme, thanks to the equivalence between (b) and (c), and the fact that $X_{(Y'')} \times_{Y''} Y' = X \times_Y Y'$ for every morphism $Y' \to Y''$ (**I**, 3.3.9.1); since, further, $f_{(Y'')}$ is separated and quasi-compact whenever f is ((**I**, 5.5.1, iv) and (**I**, 6.6.4, iii)), we are led to proving that (b) implies that f is *closed*. For this, it suffices to verify condition (b) of (7.2.1). So let $x \in X$, and y' be a specialisation of $y = f(x)$, distinct from y; by (7.1.4), there exists a scheme Y', the spectrum of some valuation ring, and a separated morphism $g : Y' \to Y$ such that, letting a denote the closed point and b the generic point of Y', we have that $g(a) = y'$, $g(b) = y$, and that there exists a $k(y)$-homomorphism $k(x) \to k(b)$. The latter corresponds canonically to a Y-morphism $\operatorname{Spec}(k(b)) \to X$ (**I**, 2.4.6), and it thus follows from (b) that there exists a Y-morphism $h : Y' \to X$ to which the previous morphism corresponds. We then have that $h(b) = x$; if we set $h(a) = x'$, then x' is a specialisation of x, and we have that $f(x') = f(h(a)) = g(a) = y'$.

If now Y is locally Noetherian and f of finite type, then hypothesis (b) implies, first of all, that f is *separated*, by (7.2.3), with the diagonal morphism $X \to X \times_Y X$ being quasi-compact (7.2.4). Further, to show that f is proper, it suffices to show that $f_{(Y'')} : X_{(Y'')} \to Y''$ is *closed* for every Y-prescheme Y'' *of finite type*, taking (5.6.3) into account. Since Y'' is then locally Noetherian, we can follow the same reasoning as in the first case by taking Y' to be the spectrum of a discrete valuation ring, and applying (7.1.9) instead of (7.1.4). ☐

Remarks (7.3.9). — (i) Whenever Y is an arbitrary prescheme and f a separated morphism, for f to be universally closed, it *suffices* that condition (b) or (c) be satisfied for the *complete* valuation rings A whose residue field is *algebraically closed*; this follows from the above proof and from (7.1.4).

(ii) From criterion (c) of (7.3.8) we obtain a new proof of the fact that a projective morphism $X \to Y$ is closed (5.5.3), and it is closer to the classical approach. We can indeed assume that Y is affine, and thus that X can be identified with a closed subprescheme of a projective bundle $\mathbf{P}^n_Y$ (5.3.3); to prove that $X \to Y$ is closed, it suffices to verify that the structure morphism $\mathbf{P}^n_Y \to Y$ is closed, and criteria (c) of (7.3.8), combined with (4.1.3.1), tells us that we can reduce to proving the following fact: *if Y is the spectrum of a valuation ring A, with field of fractions K, then every point of $\mathbf{P}^n_Y$ with values in K comes from (by restriction to the generic point of Y) a point of $\mathbf{P}^n_Y$ with values in A.* But every invertible $\mathscr{O}_Y$-module is trivial (**I**, 2.4.8); so it follows from (4.2.6) that a point of $\mathbf{P}^n_Y$ with values in K can be identified with a class of elements $(\zeta c_0, \zeta c_1, \ldots, \zeta c_n)$ of K, where $\zeta \neq 0$ and the c are elements of K that are not all zero. However, by multiplying the c_i by an element of A of suitable valuation, we can **II** | 146
suppose that the c_i all belong to A, and that at least one of them is invertible. But then (4.2.6) the system $(c_0, \ldots, c_n)$ also defines a point of $\mathbf{P}^n_Y$ with values in A, which proves our claim.

(iii) Criteria (7.2.3) and (7.3.8) are particularly simple when we consider the data of a Y-prescheme X as being equivalent to the data of the functor

$$X(Y') = \mathrm{Hom}_Y(Y', X)$$

for Y-preschemes Y'; these criteria allow us, for example, to prove that, under certain conditions, the "Picard schemes" are proper.

Corollary (7.3.10). — *Let Y be an integral scheme (resp. a locally Noetherian integral scheme), X an integral scheme, and $f : X \to Y$ a dominant morphism.*

(i) *If f is quasi-compact and universally closed, then every valuation ring whose field of fractions is the field $R(X)$ of rational functions on X, and which is dominated by a local ring Y, also dominates by a local ring of X.*

(ii) *Conversely, suppose that f is of finite type, and that the property described in (i) is verified by every valuation ring (resp. every discrete valuation ring) that has $R(X)$ as its field of fractions. Then f is proper.*

PROOF. Note first of all that the hypotheses imply, in any case, that f is separated (**I**, 5.5.9).

(i) Let $K = R(Y)$, $L = R(X)$, y a point of Y, and A a valuation ring that dominates $\mathscr{O}_y$ and has L as its field of fractions; the injection $\mathscr{O}_y \to A$ then defines a morphism h from $Y' = \mathrm{Spec}(A)$ to Y (**I**, 2.4.4) such that $h(a) = y$, where we write a to denote the closed point of Y'; furthermore, if η is the generic point of Y, which is also the generic point of $\mathrm{Spec}(\mathscr{O}_y)$, then we have $h(b) = \eta$, writing b to denote the generic point of Y' (since $K \subset L$ by hypothesis). If ξ is the generic point of X, then $k(\xi) = k(b) = L$ by hypothesis, whence we have a Y-morphism $g : \mathrm{Spec}(L) \to X$ such that $g(b) = \xi$; by (7.3.8), g comes from a Y-morphism $g' : Y' \to X$. If $x = g'(a)$, it is clear that A dominates $\mathscr{O}_x$.

(ii) Since the questions is local on Y, we can always suppose that Y is affine (resp. affine and Noetherian). Since f is of finite type, we can apply, in either case, Chow's lemma (5.6.1). There is thus a projective morphism $p : P \to Y$, an immersion morphism $j : X' \to P$, and a projective morphism $g : X' \to X$ that is both surjective and birational, with X integral, such that the diagram

$$\begin{tikzcd} P \arrow[d, "p"'] & X' \arrow[l, "j"'] \arrow[d, "g"] \\ Y & X \arrow[l, "f"] \end{tikzcd}$$

commutes. It suffices to prove that j is a *closed* immersion, since then $f \circ g = p \circ j$ will be a projective morphism, and thus closed, and, since g is surjective, f will also be proper (5.4.3). Let Z be the closed reduced subprescheme of P that has $\overline{j(X')}$ as its underlying space (**I**, 5.2.1); since X' is integral, j factors as $i \circ h$, where $i : Z \to P$ is the canonical injection, $h : X' \to Z$ a dominant open immersion (**I**, 5.2.3), and Z is integral; furthermore, **II** | 147
Z is projective over Y, and we see that we can restrict to the case where P is *integral* and j is *dominant* and *birational*, and everything then reduces to showing that j is *surjective*. But let $z \in P$; then $\mathscr{O}_z$ is an integral (resp. integral and Noetherian) local ring whose field of fractions is

$$L = R(P) = R(X') = R(X).$$

We can restrict to the case where z is not the generic point of P. There then exists ((7.1.2) and (7.1.7)) a valuation ring (resp. a discrete valuation ring) A which dominates $\mathscr{O}_z$ and has L as its field of fractions. *A fortiori*, A dominates $\mathscr{O}_y$, where $y = p(z)$, and, by hypothesis, there thus exists some $x \in X$ such that A dominates $\mathscr{O}_x$. Since g is proper, the first part of the proof shows that A also dominates $\mathscr{O}_{x'}$ for some $x' \in X'$; it then follows that $\mathscr{O}_z$ and $\mathscr{O}_{j(x')} = \mathscr{O}_{x'}$ are allied (**I**, 8.1.4), and, since P is a scheme, this implies that $z = j(x')$ (**I**, 8.2.2) and finishes the proof.

□

Corollary (7.3.11). — *Let X and Y be integral schemes, and $f : X \to Y$ a dominant, quasi-compact, and universally closed morphism. Suppose further that Y is affine of (integral) ring B. Then $\Gamma(X, \mathscr{O}_X)$ is canonically isomorphic to a subring of the integral closure of B in $R(X)$.*

Proof. Indeed (**I**, 8.2.1.1), $\Gamma(X, \mathscr{O}_X)$ can be identified with the intersection of the $\mathscr{O}_x$ over $x \in X$; by (7.3.10), (7.1.2), and (7.1.3), $\Gamma(X, \mathscr{O}_X)$ is then contained in the intersection of the valuation rings that contain B and that have $R(X)$ as their field of fractions; the conclusion then follows from (7.1.3). □

Remarks (7.3.12). — Under the hypotheses of (7.3.11), and when we suppose that $R(X)$ is an extension of finite type of $R(Y)$, we can, in many cases, conclude that $\Gamma(X, \mathscr{O}_X)$ is a *module of finite type* over the ring $B = \Gamma(Y, \mathscr{O}_X)$. For example, this will be the case whenever B is an *algebra of finite type over a field*, since we then know that the integral closure of B in an extension of finite type of its field of fractions is a B-module of finite type ([**SZ60**, t. I, p. 267, th. 9]); the conclusion then follows from (7.3.11) and the fact that B is Noetherian.

In particular, *a proper affine scheme X over a field K is finite*. Indeed, by (1.6.4), (5.4.6), and (**I**, 6.4.4, (c)), we can restrict to the case where X is *reduced*. Furthermore, it suffices to prove that each of the closed subpreschemes of X that have an irreducible component of X as their underlying space (of which there are finitely many) is finite over K, which means (taking (5.4.5) into account) that we are finally reduced to the case where X is *integral*. But then the result follows from the above remarks.

In chapter III, we will again prove this above proposition by other methods, and as a consequence of more general results, by showing that, if $f : X \to Y$ is proper and Y is locally Noetherian, $f_*(\mathscr{F})$ is coherent for any coherent $\mathscr{O}_X$-module $\mathscr{F}$ (**III**, 4.4.2).

Finally, note that criterion (7.3.10) is taken as the *definition* of proper morphisms in classical algebraic geometry. We only mention this here as a remark, since criterion (7.3.8) seems more manageable in all the applications with which we are familiar.

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7.4. Algebraic curves and function fields of dimension 1 The aim of this section is to show how to formulate the classical notion of algebraic curves (as introduced, by example, in the book of C. Chevalley [**Che51**]) in the language of schemes. All throughout this section, *we write k to mean a field, all the schemes in question are k-schemes of finite type, and all the morphisms are k-morphisms*.

Proposition (7.4.1). — *Let X be a prescheme of finite type over k (and thus Noetherian); let x_i ($1 \leqslant i \leqslant n$) be the generic points of the irreducible components X_i of X, and let $K_i = k(x_i)$ ($1 \leqslant i \leqslant n$). Then the following conditions are equivalent:*

(a) *Each of the K_i is an extension of k with transcendence degree equal to* 1.
(b) *For every closed point x of X, the local ring $\mathscr{O}_x$ is of dimension* 1 (7.1.5).
(c) *The closed irreducible subsets of X that are distinct from the X_i are exactly the closed points of X.*

PROOF. Since X is quasi-compact, every closed irreducible subset F of X contains a closed point (**0**, 2.1.3). By (**I**, 2.4.2), there is a bijective correspondence between the prime ideals of $\mathscr{O}_x$ and the closed irreducible subsets of X that contain x (**I**, 1.1.14); the equivalence between (b) and (c) follows immediately from this. Now, if $\mathfrak{p}_\alpha$ ($1 \leqslant \alpha \leqslant r$) are the minimal prime ideals of the local Noetherian ring $\mathscr{O}_x$, then the local rings $\mathscr{O}_x/\mathfrak{p}_\alpha$ are integral, and have the K_i such that $x \in X_i$ as their fields of fractions. Furthermore, we know ([**CC**, p. 4-06, th. 2]) that the dimension of a local integral k-algebra of finite type is equal to the transcendence degree over k of its field of fractions. Finally, the dimension of $\mathscr{O}_x$ is bounded above by the dimensions of the $\mathscr{O}_x/\mathfrak{p}_\alpha$; but condition (a) implies that these dimensions are equal to 1, and so (a) implies (b); conversely, if $\mathscr{O}_x$ is of dimension 1, then none of the $\mathfrak{p}_\alpha$ can be equal to the maximal ideal of $\mathscr{O}_x$, otherwise $\mathscr{O}_x$ would be of dimension 0; thus each of the $\mathscr{O}_x/\mathfrak{p}_\alpha$ are of dimension 1, which shows that (b) implies (a). ☐

We note that, under the conditions of (7.4.1), the set X is either *empty* or *infinite*, as an immediate result of (**I**, 6.4.4).

Definition (7.4.2). — We define an *algebraic curve over k* to be a non-empty algebraic *scheme* over k that satisfies the conditions of (7.4.1).

In the language of dimensions, which will be introduced in Chapter IV, this can be expressed by saying that an algebraic curve over k is a non-empty algebraic k-scheme *whose irreducible components are all of dimension* 1.

We note that, if X is an algebraic curve over k, then the closed reduced subpreschemes X_i ($1 \leqslant i \leqslant n$) of X that have the irreducible components of X as their underlying space are also algebraic curves over k.

Corollary (7.4.3). — *Let X be an irreducible algebraic curve. The only non-closed point of X is its generic point. The closed subsets of X that are distinct from X are the finite sets of closed points; these are also the only subsets of X that are not everywhere dense.*

PROOF. If a point $x \in X$ is not closed, then its closure in X is an irreducible closed subset of X, and thus necessarily the whole of X, by (7.4.1), and thus x is the generic point of X. A closed subset F of X that is distinct from X cannot contain the generic point of X, and so all its points are closed **II** | 149
(in X, and *a fortiori* in F); by considering the closed reduced subpreschemes of X that have F as their underlying space (**I**, 5.2.1), it thus follows from (**I**, 6.2.2) that F is finite and discrete. The closure in X of any infinite subset of X is thus necessarily equal to X itself. ☐

If X is an arbitrary algebraic curve, by applying (7.4.3) to the irreducible components of X, we see that the only non-closed points of X are the generic points of these components.

Corollary (7.4.4). — *Let X and Y be irreducible algebraic curves over k, and $f : X \to Y$ a k-morphism. For f to be dominant, it is necessary and sufficient for $f^{-1}(y)$ to be finite for all $y \in Y$.*

PROOF. Indeed, if f is not dominant, then $f(X)$ is necessarily a *finite* subset of Y, by (7.4.3), and so it is not possible for $f^{-1}(y)$ to be finite for every point of Y, since otherwise X would be finite, which is a contradiction (7.4.1). Conversely, if f is dominant, then for any $y \in Y$ distinct from the generic point η of Y, we have that $f^{-1}(y)$ is closed in X, since $\{y\}$ is closed in Y (7.4.3); also, by hypothesis, $f^{-1}(y)$ does not contain the generic point ξ of X, and is thus finite, by (7.4.3). Finally, to see that, when f is dominant, $f^{-1}(\eta)$ is finite, we note that the fibre $f^{-1}(\eta)$ is an irreducible scheme of finite type over $k(\eta)$, and with generic point ξ ((**I**, 6.3.9) and (**I**, 6.4.11)). Since $k(\xi)$ and $k(\eta)$ are extensions of finite type of k, both of transcendence degree 1, we have that $k(\xi)$ is necessarily an extension of finite degree of $k(\eta)$, and so ξ is closed in $f^{-1}(\eta)$ (**I**, 6.4.2), and $f^{-1}(\eta)$ thus consists of a single point ξ. □

We will see, in Chapter III, that, if $f : X \to Y$ is a *proper* morphism of Noetherian preschemes such that $f^{-1}(y)$ is finite for all $y \in Y$, then f is necessarily *finite*; it will thus follow from (7.4.4) that a proper dominant morphism from an irreducible algebraic curve to an algebraic curve is *finite*.

Corollary (7.4.5). — *Let X be an algebraic curve over k. For X to be regular, it is necessary and sufficient for X to be normal, or for the local rings of its closed points to be discrete valuation rings.*

PROOF. This follows immediately from (7.4.1, (b)) and (7.1.6). □

Corollary (7.4.6). — *Let X be a reduced algebraic curve, and $\mathscr{A}$ a reduced coherent $\mathscr{R}(X)$-algebra; then the integral closure X' of X with respect to $\mathscr{A}$ (6.3.4) is a normal algebraic curve, and the canonical morphism $X' \to X$ is finite.*

PROOF. The fact that $X' \to X$ is finite follows from (6.3.10); X' is thus an algebraic k-scheme; furthermore, if x_i $(1 \leqslant i \leqslant n)$ are the generic points of the irreducible components of X, and x'_j $(1 \leqslant j \leqslant m)$ the generic points of the irreducible components of X', then each of the $k(x'_j)$ is a finite algebraic extension of one of the $k(x_i)$ (6.3.6), and thus of transcendence degree 1 over k. So X' is indeed an algebraic curve over k, and, furthermore, we know that X' is a finite sum of normal integral schemes ((6.3.6) and (6.3.7)). □

(7.4.7). We say that an algebraic curve X over k is *complete* if it is *proper* over k.

Corollary (7.4.8). — *For a reduced algebraic curve X over k to be complete, it is necessary and sufficient for its normalisation X' to be complete.*

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PROOF. The canonical morphism $f : X' \to X$ is finite (7.4.6), and thus proper (6.1.11) and surjective (6.3.8); if $g : X \to \operatorname{Spec}(k)$ is the structure morphism, then g and $\circ f$ are both proper, by (5.4.2, (ii)) and (5.4.3, (ii)), since g is separated by hypothesis. □

Proposition (7.4.9). — *Let X be a* normal *algebraic curve over k, and Y a* proper *algebraic k-scheme over k. Then every rational k-map from X to Y is everywhere defined, or, in other words, is a morphism.*

PROOF. It follows from (7.3.7) that, at the points $x \in X$ where such a map is not defined, the dimension of $\mathscr{O}_x$ must be $\geqslant 2$, and so the set of such points is empty; the final claim follows from (**I**, 7.2.3). □

Corollary (7.4.10). — *A normal algebraic curve over k is quasi-projective over k.*

PROOF. Since X is a finite sum of normal integral algebraic curves (6.3.8), we can restrict to the case where X is integral (5.3.6). Since X is quasi-compact, it is covered by a finite number of affine open subsets U_i $(1 \leqslant i \leqslant n)$, and, since each of these U_i is of finite type over k, for each i there exists some integer n_i along with a k-immersion $f_i : U_i \to \mathbf{P}_k^{n_i}$ ((5.3.3) and (5.3.4, (i))). Since U_i is dense in X, it follows from (7.4.9) that f_i can be extended to a k-morphism $g_i : X \to \mathbf{P}_k^{n_i}$, whence we obtain a k-morphism $g = (g_1, \ldots, g_n)_k$ from X to the product P of the $\mathbf{P}_k^{n_i}$ over k. Furthermore, for each i, since the restriction of g_i to U_i is an immersion, so too is the restriction of g to U_i (**I**, 5.3.14). Since the U_i cover X, and since g is separated (**I**, 5.5.1, (v)), g is an immersion from X into P (**I**, 8.2.8). Since the Segre morphism (4.3.3) gives an immersion of P into $\mathbf{P}_k^N$, this proves that X is quasi-projective. □

Corollary (7.4.11). — *Any normal algebraic curve X is isomorphic to the scheme induced by some complete normal algebraic curve $\widehat{X}$ on some everywhere dense open subset, and this $\widehat{X}$ is unique up to unique isomorphism.*

Proof. If X_1 and X_2 are complete normal curves, then it follows from (7.4.9) that every isomorphism from any dense open U_1 in X_1 to any dense open U_2 in X_2 can be uniquely extended to an isomorphism from X_1 to X_2; whence the uniqueness claim. To prove the existence of $\widehat{X}$, it suffices to note that we can consider X as a subscheme of a projective bundle $\mathbf{P}^n_k$ (7.4.10). Let $\overline{X}$ be the closure of X in $\mathbf{P}^n_k$ (**I**, 9.5.11); since X is induced by $\overline{X}$ on a dense open subset of $\overline{X}$ (**I**, 9.5.10), the generic points x_i of the irreducible components of X are also the generic points of the irreducible components of $\overline{X}$, and the $k(x_i)$ are the same for both of these schemes, and so (7.4.1) $\overline{X}$ is an algebraic curve over k that is reduced (**I**, 9.5.9) and projective over k (5.5.1), whence complete (5.5.3). So we take for $\widehat{X}$ the *normalisation* of $\overline{X}$, which is again complete (7.4.8); furthermore, if $h : \widehat{X} \to \overline{X}$ is the canonical morphism, then the restriction of h to $h^{-1}(X)$ is an isomorphism to X, since X is normal (6.3.4), and since $h^{-1}(X)$ contains the generic points of the irreducible components of $\widehat{X}$ (6.3.8), it is dense in $\widehat{X}$, which finishes the proof. □

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Remark (7.4.12). — We will show, in Chapter V, that the conclusion of (7.4.10) still holds true without the assumption that the curve is normal (or even reduced); we will also show that, for an algebraic curve (reduced or not) to be affine, it is necessary and sufficient for its (reduced) irreducible components to *not* be complete.

Corollary (7.4.13). — *Let X be a* normal *irreducible curve over the field $K = R(X)$, and Y a* complete *integral curve over the field $L = R(Y)$. Then there is a canonical bijective correspondence between* dominant *k-morphisms $X \to Y$ and k-monomorphisms $L \to K$.*

Proof. By (7.4.9), rational k-map from X to Y can be identified with k-morphisms $u : X \to Y$. Since the dominant morphisms $u : X \to Y$ are characterised by being those such that $u(x) = y$ (writing x and y to denote the generic points of X and Y, respectively), the corollary follows from these remarks and from (**I**, 7.1.13). □

(7.4.14). We can refine the result of (7.4.13) in the case where Y is the *projective line* $\mathbf{P}^1_k = \mathrm{Proj}(k[T_0, T])$, where T_0 and T are indeterminates. Then Y is an integral scheme (2.4.4), and the scheme induced on the open subset $D_+(T_0)$ of Y is isomorphic to $\mathrm{Spec}(k[T])$ (2.3.6), and so the generic point of Y is the ideal (0) of $k[T]$, and the field of rational functions of Y is $k(T)$, which proves that Y is a complete algebraic curve over k. Furthermore, the only graded prime ideal of $S = k[T_0, T]$ that contains T_0 and is distinct from S_+ is the principal ideal (T_0), and so the complement of $D_+(T_0)$ in $Y = \mathbf{P}^1_k$ consists of *one closed point*, called the "point at infinity", which we denote by ∞ (for a general study of the links between vector bundles and projective bundles, see (8.4)). With these notations:

Corollary (7.4.15). — *Let X be a* normal *irreducible curve over the field $K = R(X)$. Then there exists a canonical bijective correspondence between the set K and the set of morphisms u from X to $\mathbf{P}^1_k$ that are distinct from the constant morphism with value ∞. For such a rational map to be dominant, it is necessary and sufficient for the corresponding element of K to be transcendent over k.*

Proof. This claim follows immediately from (7.4.9) and the following:

Lemma. — *Let X be an integral prescheme over k, and let $K = R(X)$ be its field of rational functions. Then there exists a canonical bijective correspondence between the set K and the set of rational maps u from X to $\mathbf{P}^1_k$ that are distinct from the constant morphism with value ∞. For such a rational map to be dominant, it is necessary and sufficient for the corresponding element of K to be transcendent over k.*

First of all, rational maps from X to $\mathbf{P}^1_k$ correspond bijectively to points of $\mathbf{P}^1_k$ with values in the extension K of k (**I**, 7.1.12). If such a point is located (**I**, 3.4.5) at the generic point of $\mathbf{P}^1_k$, then the corresponding rational map is clearly dominant. In the converse case, since every point of $\mathbf{P}^1_k$ that is distinct from the generic point is closed (7.4.3), the image of the domain of definition U of u by the unique morphism $U \to \mathbf{P}^1_k$ of the class u (**I**, 7.2.2) consists of one closed point y of $\mathbf{P}^1_k$, and this morphism (which is not necessarily everywhere defined on X) is thus not dominant; as an abuse of language, we thus say that the rational map u is "constant, of value y". It remains to place in bijective correspondence the points of $\mathbf{P}^1_k$ with value in K that are located (**I**, 3.4.5) *not* at ∞, and the elements of K, and then to verify that the location of such a point is the generic point of $\mathbf{P}^1_k$ if and only if it corresponds to an element that is transcendental over k. But this is immediate (4.2.6, example 1). □ **II** | 152

Corollary (7.4.16). — *Let X and Y be algebraic curves over k that are* normal, complete, *and irreducible; let $K = R(X)$ and $L = R(Y)$ be their fields. Then there exists a canonical bijective correspondence between the set of k-isomorphisms $X \xrightarrow{\sim} Y$ and the set of k-isomorphisms $L \xrightarrow{\sim} K$.*

Proof. This is an evident consequence of (7.4.13). □

(7.4.17). This corollary (7.4.16) shows that an algebraic curve over k that is normal, complete, and irreducible, is *determined by its field of rational functions K up to unique isomorphism*; by definition, K is an extension of finite type of k, of transcendence degree 1 (we classically call this a *field of algebraic functions of one variable*). Furthermore:

Proposition (7.4.18). — *For every extension K of k of finite type and of transcendence degree* 1*, there exists an algebraic curve X (determined up to unique isomorphism) that is normal, complete, and irreducible, and such that $R(X) = K$. The set of local rings of X can be identified* (**I**, 8.2.1) *with the set consisting of the elements of K and the elements of the valuation rings that contain k and have K as their field of fractions.*

Proof. Indeed, K is an extension of finite degree of a pure transcendental extension $k(T)$ of k, which can be identified, as we have seen, with the field of rational functions of the projective line $Y = \mathbf{P}^1_k$. Let X be the *integral closure* of Y with respect to K (6.3.4); then X is a normal algebraic curve over the field K (6.3.7), and it is complete, since the morphism $X \to Y$ is finite (7.4.6). The local rings $\mathscr{O}_x$ of X are either the field K, when x is the generic point, or discrete valuation rings that contain k and have K as their field of fractions, when x is distinct from the generic point (7.4.5). Conversely, let A be such a ring; since the morphism $X \to \operatorname{Spec}(k)$ is proper, the fact that A dominates k implies that A also dominates a local ring $\mathscr{O}_x$ of X (7.3.10); since the latter is a valuation ring that has K as a field of fractions, it is necessarily equal to A. □

Remarks (7.4.19). — It follows from (7.4.16) and (7.4.18) that *the data of an algebraic curve over k that is normal, complete, and irreducible, is essentially equivalent to the data of an extension K of k that is of finite type and of transcendence degree* 1. We note that, if k' is an extension of the base field k, then $X \otimes_k k'$ will again be a complete algebraic curve over k' (5.4.2, (iii)), but, in general, it will be neither reduced nor irreducible. It will, however, be both reduced and irreducible if K is a separable extension of k, and k is algebraically closed in K (this can be expressed, in classical terminology, which we will not use, by saying that K is a "regular extension" of k). But even in this case, it is possible for $X \otimes_k k'$ to not be normal. The reader will find details on these questions in Chapter IV.

§8. Blowup schemes; based cones; projective closure

8.1. Blowup preschemes

(8.1.1). Let Y be a prescheme, and, for every integer $n \geqslant 0$, let $\mathscr{I}_n$ be a quasi-coherent sheaf of ideals of $\mathscr{O}_Y$; suppose that the following conditions are satisfied:

$$\mathscr{I}_0 = \mathscr{O}_Y,\ \mathscr{I}_n \subset \mathscr{I}_m \text{ for } m \leqslant n, \tag{8.1.1.1}$$

$$\mathscr{I}_m \mathscr{I}_n \subset \mathscr{I}_{m+n} \text{ for any } m, n. \tag{8.1.1.2}$$

We note that these hypotheses imply **II** | 153

$$\mathscr{I}_1^n \subset \mathscr{I}_n. \tag{8.1.1.3}$$

Set

$$\mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{I}_n. \tag{8.1.1.4}$$

It follows from (8.1.1.1) and (8.1.1.2) that $\mathscr{S}$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra, and thus defines a Y-scheme $X = \operatorname{Proj}(\mathscr{S})$. If $\mathscr{J}$ is an *invertible* sheaf of ideals of $\mathscr{O}_Y$, then $\mathscr{I}_n \otimes_{\mathscr{O}_Y} \mathscr{J}^{\otimes n}$ is canonically identified with $\mathscr{I}_n \mathscr{J}^n$. If we then replace the $\mathscr{I}_n$ by the $\mathscr{I}_n \mathscr{J}^n$, and, in doing so, replace $\mathscr{S}$ by a quasi-coherent $\mathscr{O}_Y$-algebra $\mathscr{S}_{(\mathscr{J})}$, then $X_{(\mathscr{J})} = \operatorname{Proj}(\mathscr{S}_{(\mathscr{J})})$ is canonically isomorphic to X (3.1.8).

(8.1.2). Suppose that Y is *locally integral*, so that the sheaf $\mathscr{R}(Y)$ of rational functions is a quasi-coherent $\mathscr{O}_Y$-algebra (**I**, 7.3.7). We say that an $\mathscr{O}_Y$-submodule $\mathscr{I}$ of $\mathscr{R}(Y)$ is a *fractional ideal* of $\mathscr{R}(Y)$ if it is of *finite type* (**0**, 5.2.1). Suppose we have, for all $n \geqslant 0$, a quasi-coherent fractional ideal $\mathscr{I}_n$ of $\mathscr{R}(Y)$, such that $\mathscr{I}_0 = \mathscr{O}_Y$, and such that condition (8.1.1.2) (but not necessarily the second condition (8.1.1.1)) is satisfied; we can then again define a quasi-coherent graded $\mathscr{O}_Y$-algebra by Equation (8.1.1.4), and the corresponding Y-scheme $X = \operatorname{Proj}(\mathscr{S})$; we will again have a canonical isomorphism from X to $X_{\mathscr{J}}$ for every *invertible* fractional ideal $\mathscr{J}$ of $\mathscr{R}(Y)$.

Definition (8.1.3). — Let Y be a prescheme (resp. a locally integral prescheme), and $\mathscr{I}$ a quasi-coherent ideal of $\mathscr{O}_Y$ (resp. a quasi-coherent fractional ideal of $\mathscr{R}(Y)$). We say that the Y-scheme $X = \operatorname{Proj}(\bigoplus_{n\geqslant 0} \mathscr{I}^n)$ is obtained by blowing up the ideal $\mathscr{I}$, or is the blow-up prescheme of Y relative to $\mathscr{I}$. When $\mathscr{I}$ is a quasi-coherent ideal of $\mathscr{O}_Y$, and Y' is the closed subprescheme of Y defined by $\mathscr{I}$, we also say that X is the Y-scheme obtained by blowing up Y'.

By definition, $\mathscr{S} = \bigoplus_{n\geqslant 0} \mathscr{I}^n$ is then generated by $\mathscr{S}_1 = \mathscr{I}$; if $\mathscr{I}$ is an $\mathscr{O}_Y$-module of *finite type*, then X is *projective* over Y (5.5.2). Without any hypotheses on $\mathscr{I}$, the $\mathscr{O}_X$-module $\mathscr{O}_X(1)$ is *invertible* (3.2.5) and *very ample*, by (4.4.3) applied to the structure morphism $X \to Y$.

We note that, if $j : X \to Y$ is the structure morphism, then the restriction of f to $f^{-1}(Y - Y')$ is an *isomorphism* to $Y - Y'$ whenever $\mathscr{I}$ is an *ideal of* $\mathscr{O}_Y$ and Y' is the closed subprescheme that it defines: indeed, since the questions is local on Y, it suffices to assume that $\mathscr{I} = \mathscr{O}_Y$, and our claim then follows from (3.1.7).

If we replace $\mathscr{I}$ by $\mathscr{I}^d$ ($d > 0$), then the blow-up Y-scheme X is replaced by a canonically isomorphic Y-scheme X' (8.1.1); similarly, for every *invertible* ideal (resp. *invertible* fractional ideal) $\mathscr{J}$, the blow-up prescheme $X_{(\mathscr{J})}$ relative to the ideal $\mathscr{I}\mathscr{J}$ is canonically isomorphic to X (8.1.1).

In particular, whenever $\mathscr{I}$ is an *invertible* ideal (resp. *invertible* fractional ideal), the Y-scheme obtained by blowing up $\mathscr{I}$ is *isomorphic to* Y (3.1.7).

Proposition (8.1.3). — *Let Y be an integral prescheme.*

(i) *For every sequence $(\mathscr{I}_n)$ of quasi-coherent fractional ideals of $\mathscr{R}(Y)$ that satisfies (8.1.1.2) and such that $\mathscr{I}_0 = \mathscr{O}_Y$, the Y-scheme $X = \operatorname{Proj}(\bigoplus_{n\geqslant 0} \mathscr{I}^n)$ is integral, and the structure morphism $f : X \to Y$ is dominant.* **II | 154**

(ii) *Let $\mathscr{I}$ be a quasi-coherent fractional ideal of $\mathscr{R}(Y)$, and let X be the Y-scheme given by the blow up of Y relative to $\mathscr{I}$. If $\mathscr{I} \neq 0$, then the structure morphism $f : X \to Y$ is then birational and surjective.*

PROOF.

(i) This follows from the fact that $\mathscr{S} = \bigoplus_{n\geqslant 0} \mathscr{I}_n$ is an *integral* $\mathscr{O}_Y$-algebra ((3.1.12) and (3.1.14)), since, for all $y \in Y$, $\mathscr{O}_y$ is an integral ring (**I**, 5.1.4).

(ii) By (i), X is integral; if, furthermore, x and y are the generic points of X and Y (respectively), then we have $f(x) = y$, and it remains to show that $k(x)$ is of rank 1 over $k(y)$. But x is also the generic point of the fibre $f^{-1}(y)$; if ψ is the canonical morphism $Z \to Y$, where $Z = \operatorname{Spec}(k(y))$, then the prescheme $f^{-1}(y)$ can be identified with $\operatorname{Proj}(\mathscr{S}')$, where $\mathscr{S}' = \psi^*(\mathscr{S})$ (3.5.3). But it is clear that $\mathscr{S}' = \bigoplus_{n\geqslant 0}(\mathscr{I}_y)^n$, and, since $\mathscr{I}$ is a quasi-coherent fractional ideal of $\mathscr{R}(Y)$ that is not zero, $\mathscr{I}_y \neq 0$ (**I**, 7.3.6), whence $\mathscr{I}_y = k(y)$; then $\operatorname{Proj}(\mathscr{S}')$ can be identified with $\operatorname{Spec}(k(y))$ (3.1.7), whence the conclusion.

□

We show a *converse* of (8.1.4) in (**III**, 2.3.8).

(8.1.5). We return to the setting and notation of (8.1.1). By definition, the injection homomorphisms $\mathscr{I}_{n+1} \to \mathscr{I}_n$ (8.1.1.1) define, for every $k \in \mathbf{Z}$, an injective homomorphism of degree zero of graded $\mathscr{S}$-modules

$$u_k : \mathscr{S}_+(k+1) \longrightarrow \mathscr{S}(k); \tag{8.1.5.1}$$

since $\mathscr{S}_+(k+1)$ and $\mathscr{S}(k+1)$ are canonically **(TN)**-isomorphic, they give a canonical correspondence between u_k and an injective homomorphism of $\mathscr{O}_X$-modules (3.4.2):

$$\widetilde{u}_k : \mathscr{O}_X(k+1) \longrightarrow \mathscr{O}_X(k). \tag{8.1.5.2}$$

Recall as well (3.2.6) that we have defined canonical homomorphisms

$$\lambda : \mathscr{O}_X(h) \otimes_{\mathscr{O}_X} \mathscr{O}_X(k) \longrightarrow \mathscr{O}_X(h+k) \tag{8.1.5.3}$$

and, since the diagram

$$\begin{tikzcd} \mathscr{S}(h) \otimes_{\mathscr{S}} \mathscr{S}(k) \otimes_{\mathscr{S}} \mathscr{S}(l) \arrow[r] \arrow[d] & \mathscr{S}(h+k) \otimes_{\mathscr{S}} \mathscr{S}(l) \arrow[d] \\ \mathscr{S}(h) \otimes_{\mathscr{S}} \mathscr{S}(k+l) \arrow[r] & \mathscr{S}(h+k+l) \end{tikzcd}$$

commutes, it follows from the functoriality of the λ (3.2.6) that the homomorphisms (8.1.5.3) define the structure of a *quasi-coherent graded $\mathscr{O}_X$-algebra* on

$$\mathscr{S}_X = \bigoplus_{n\in\mathbf{Z}} \mathscr{O}_X(n). \tag{8.1.5.4}$$

Furthermore, the diagram

$$\begin{tikzcd} \mathscr{S}(h) \otimes_{\mathscr{S}} \mathscr{S}(k+1) \arrow[r] \arrow[d, "1\otimes u_k"'] & \mathscr{S}(h+k+1) \arrow[d, "u_{k+h}"] \\ \mathscr{S}(h) \otimes_{\mathscr{S}} \mathscr{S}(k) \arrow[r] & \mathscr{S}(h+k) \end{tikzcd}$$

commutes; the functoriality of the λ then implies that we have a commutative diagram

$$\begin{tikzcd} \mathscr{O}_X(h) \otimes_{\mathscr{O}_X} \mathscr{O}_X(k+1) \arrow[r, "\lambda"] \arrow[d, "1\otimes\widetilde{u}_k"'] & \mathscr{O}_X(h+k+1) \arrow[d, "\widetilde{u}_{k+h}"] \\ \mathscr{O}_X(h) \otimes_{\mathscr{O}_X} \mathscr{O}_X(k) \arrow[r, "\lambda"] & \mathscr{O}_X(h+k) \end{tikzcd} \tag{8.1.5.5}$$

where the horizontal arrows are the canonical homomorphisms. We can thus say that the $\widetilde{u}_k$ define an *injective homomorphism* (of degree zero) *of graded $\mathscr{S}_X$-modules*

$$\widetilde{u} : \mathscr{S}_X(1) \longrightarrow \mathscr{S}_X. \tag{8.1.5.6}$$

(8.1.6). Keeping the notation from (8.1.5), we now note that, for $n \geqslant 0$, the composite homomorphism $\widetilde{v}_n = \widetilde{u}_{n-1} \circ \widetilde{u}_{n-2} \circ \ldots \circ \widetilde{u}_0$ is an *injective* homomorphism $\mathscr{O}_X(n) \to \mathscr{O}_X$; we denote by $\mathscr{I}_{n,X}$ its image, which is thus a quasi-coherent ideal of $\mathscr{O}_X$, *isomorphic* to $\mathscr{O}_X(n)$. Furthermore, the diagram

$$\begin{tikzcd} \mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{O}_X(n) \arrow[r, "\lambda"] \arrow[d, "\widetilde{v}_m\otimes\widetilde{v}_n"'] & \mathscr{O}_X(m+n) \arrow[d, "\widetilde{v}_{m+n}"] \\ \mathscr{O}_X \arrow[r, "\mathrm{id}"'] & \mathscr{O}_X \end{tikzcd}$$

commutes for $m \geqslant 0$, $n \geqslant 0$. We thus deduce the following inclusions:

$$\mathscr{I}_{0,X} = \mathscr{O}_X, \quad \mathscr{I}_{n,X} \subset \mathscr{I}_{m,X} \qquad \text{for } 0 \leqslant m \leqslant n; \tag{8.1.6.1}$$

$$\mathscr{I}_{m,X}\mathscr{I}_{n,X} \subset \mathscr{I}_{m+n,X} \qquad \text{for } m \geqslant 0,\ n \geqslant 0. \tag{8.1.6.2}$$

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Proposition (8.1.7). — *Let Y be a prescheme, $\mathscr{I}$ a quasi-coherent ideal of $\mathscr{O}_Y$, and $X = \mathrm{Proj}(\bigoplus_{n\geqslant 0} \mathscr{I}^n)$ the Y-scheme given by blowing up $\mathscr{I}$. We then have, for all $n > 0$, a canonical isomorphism*

$$\mathscr{O}_X(n) \xrightarrow{\sim} \mathscr{I}^n \mathscr{O}_X = \mathscr{I}_{n,X} \tag{8.1.7.1}$$

*(cf. (**0**, 4.3.5)), and thus that $\mathscr{I}^n\mathscr{O}_X$ is a very-ample invertible $\mathscr{O}_X$-module if $n > 0$.*

PROOF. The last claim is immediate, since $\mathscr{O}_X(1)$ is invertible (3.2.5) and very ample for Y by definition ((4.4.3) and (4.4.9)). Also by definition, the image of v_n is exactly $\mathscr{I}^n\mathscr{S}$, and (8.1.7.1) then follows from the exactness of the functor $\widetilde{\mathscr{M}}$ (3.2.4) and from Equation (3.2.4.1). □

Corollary (8.1.8). — *Under the hypotheses of (8.1.7), if $f : X \to Y$ is the structure morphism, and Y' the closed subprescheme of Y defined by $\mathscr{I}$, then the closed subprescheme $X' = f^{-1}(Y')$ of X is defined by $\mathscr{I}\mathscr{O}_X$ (which is canonically isomorphic to $\mathscr{O}_X(1)$), from which we obtain a canonical short exact sequence*

$$0 \longrightarrow \mathscr{O}_X(1) \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_{X'} \longrightarrow 0. \tag{8.1.8.1}$$

PROOF. This follows from (8.1.7.1) and from (**I**, 4.4.5). □

(8.1.9). Under the hypotheses of (8.1.7), we can be more precise about the structure of the $\mathscr{I}_{n,X}$. Note that the homomorphism

$$\widetilde{u}_{-1} : \mathscr{O}_X \longrightarrow \mathscr{O}_X(-1)$$

canonically corresponds to a section s of $\mathscr{O}_X(-1)$ over X, which we call the *canonical section* (relative to $\mathscr{I}$) (**0**, 5.1.1). In the diagram in (8.1.5.5), the horizontal arrows are isomorphisms (3.2.7); by replacing h with k, and k with -1 in this diagram, we obtain that $\widetilde{u}_k = 1_k \otimes \widetilde{u}_{-1}$ (where 1_k denotes the identity on $\mathscr{O}_X(h)$), or, equivalently, that the homomorphism $\widetilde{u}_k$ is given exactly by *tensoring with the canonical section s* (for all $k \in \mathbf{Z}$). The homomorphism $\widetilde{u}$ (8.1.5.6) can then be understood in the same way.

Thus, for all $n \geqslant 0$, the homomorphism $\widetilde{v}_n : \mathscr{O}_X(n) \to \mathscr{O}_X$ is given exactly by tensoring with $s^{\otimes n}$; we thus deduce:

Corollary (8.1.10). — *With the notation of (8.1.8), the underlying space of X' is the set of $x \in X$ such that $s(x) = 0$, where s denotes the canonical section of $\mathscr{O}_X(-1)$.*

PROOF. Indeed, if c_x is a generator of the fibre $(\mathscr{O}_X(1))_x$ at a point x, then $s_x \otimes c_x$ is canonically identified with a generator of the fibre of $\mathscr{I}_{1,X}$ at the point x, and is thus invertible if and only if $s_x \notin \mathfrak{m}_x(\mathscr{O}_x(-1))_x$, or, equivalently, if and only if $s(x) \neq 0$. □

Proposition. — *Let Y be an integral prescheme, $\mathscr{I}$ a quasi-coherent fractional ideal of $\mathscr{R}(Y)$, and X the Y-scheme given by blowing up $\mathscr{I}$. Then $\mathscr{I}\mathscr{O}_X$ is an invertible $\mathscr{O}_X$-module that is very ample for Y.*

PROOF. Since the questions is local on Y (4.4.5), we can reduce to the case where $Y = \operatorname{Spec}(A)$, with A some integral ring of ring of fractions K, and $\mathscr{I} = \widetilde{\mathfrak{I}}$, with $\mathfrak{I}$ some fractional ideal of K; there then exists an element $a \neq 0$ of A such that $a\mathfrak{I} \subset A$. Let $S = \bigoplus_{n\geqslant 0}\mathfrak{I}^n$; the map $x \mapsto ax$ is an A-isomorphism from $\mathfrak{I}^{n+1} = (S(1))_n$ to $a\mathfrak{I}^{n+1} = a\mathfrak{I}S_n \subset \mathfrak{I}^n = S_n$, and thus defines a (TN)-isomorphism of degree zero of graded S-modules $S_+(1) \to a\mathfrak{I}S$. On the other hand, $x \mapsto a^{-1}x$ is an isomorphism of degree zero of graded S-modules $a\mathfrak{I}S \xrightarrow{\sim} \mathfrak{I}S$. We thus obtain, by composition (3.2.4), an isomorphism of $\mathscr{O}_X$-modules $\mathscr{O}_X(1) \xrightarrow{\sim} \mathscr{I}\mathscr{O}_X$, and, since S is generated by $S_1 = \mathfrak{I}$, $\mathscr{O}_X(1)$ is invertible (3.2.5) and very ample ((4.4.3) and (4.4.9)), whence our claim. □ II | 157

8.2. Preliminary results on the localisation of graded rings

(8.2.1). Let S be a graded ring, but not assumed (for the moment) to be only in positive degree. We define

$$S^{\geqslant} = \bigoplus_{n\geqslant 0} S_n, \qquad S^{\leqslant} = \bigoplus_{n\leqslant 0} S_n \tag{8.2.1.1}$$

which are both graded subrings of S, in only positive and negative degrees (respectively). If f is a homogeneous elements of degree d (positive or negative) of S, then the ring of fraction $S_f = S'$ is again endowed with the structure of a graded ring, by taking S'_n ($n \in \mathbf{Z}$) to be the set of the x/f^k for $x \in S_{n+kd}$ ($k \geqslant 0$); we define $S_{(f)} = S'_0$, and will write $S_f^{\geqslant}$ and $S_f^{\leqslant}$ for $S'^{\geqslant}$ and $S'^{\leqslant}$ (respectively). If $d > 0$, then

$$(S^{\geqslant})_f = S_f \tag{8.2.1.2}$$

since, if $x \in S_{n+kd}$ with $n + kd < 0$, then we can write $x/f^k = xf^h/f^{h+k}$, and we also have that $n + (h+k)d > 0$ for h sufficiently large and > 0. We thus conclude, by definition, that

$$(S^{\geqslant})_{(f)} = (S_f^{\geqslant})_0 = S_{(f)}. \tag{8.2.1.3}$$

If M is a graded S-module, then we similarly define

(8.2.1.4) $$M^{\geqslant} = \bigoplus_{n\geqslant 0} M_n, \qquad M^{\leqslant} = \bigoplus_{n\leqslant 0} M_n$$

which are (respectively) a graded $S^{\geqslant}$-module and a graded $S^{\leqslant}$-module, and their intersection is the S_0 module M_0. If $f \in S_d$, then we define M_f to be the graded S_f-module whose elements of degree n are the z/f^k for $z \in M_{n+kd}$ ($k \geqslant 0$); we denote by $M_{(f)}$ the set of elements of degree zero of M_f, and this is an $S_{(f)}$-module, and we will write $M_f^{\geqslant}$ and $M_f^{\leqslant}$ to mean $(M_f)^{\geqslant}$ and $(M_f)^{\leqslant}$ (respectively). If $d > 0$, then we see, as above, that

(8.2.1.5) $$(M^{\geqslant})_f = M_f$$

and

(8.2.1.6) $$(M^{\geqslant})_{(f)} = (M_f^{\geqslant})_0 = M_{(f)}.$$

(8.2.2). Let $\mathbf{z}$ be an indeterminate, we we will call the *homogenisation variable*. If S is a graded ring (in positive or negative degrees), then the polynomial algebra[1]

(8.2.2.1) $$\widehat{S} = S[\mathbf{z}]$$

is a graded S-algebra, where we define the degree of $f\mathbf{z}^n$ ($n \geqslant 0$), with f homogeneous, as **II** | 158

(8.2.2.2) $$\deg(f\mathbf{z}^n) = n + \deg f.$$

Lemma (8.2.3). — (i) *There are canonical isomorphisms of (non-graded) rings*

(8.2.3.1) $$\widehat{S}_{(\mathbf{z})} \xrightarrow{\sim} \widehat{S}/(\mathbf{z}-1)\widehat{S} \xrightarrow{\sim} S.$$

(ii) *There is a canonical isomorphism of (non-graded) rings*

(8.2.3.2) $$\widehat{S}_{(f)} \xrightarrow{\sim} S_f^{\leqslant}$$

for all $f \in S_d$ with $d > 0$.

Proof. The first of the isomorphisms in (8.2.3.1) was defined in (2.2.5), and the second is trivial; the isomorphism $\widehat{S}_{(\mathbf{z})} \xrightarrow{\sim} S$ thus defined thus gives a correspondence between $x\mathbf{z}^n/\mathbf{z}^{n+k}$ (where $\deg(x) = k$ for $k \geqslant -n$) and the element x. The homomorphism (8.2.3.2) gives a correspondence between $x\mathbf{z}^n/f^k$ (where $\deg(x) = kd - n$) and the element x/f^k of degree $-n$ in $S_f^{\leqslant}$, and it is again clear that this does indeed give an isomorphism. □

(8.2.4). Let M be a graded S-module. It is clear that the S-module

(8.2.4.1) $$\widehat{M} = M \otimes_S \widehat{S} = M \otimes_S S[\mathbf{z}]$$

is the direct sum of the S-modules $M \otimes S\mathbf{z}^n$, and thus of the abelian groups $M_k \otimes S\mathbf{z}^n$ ($k \in \mathbf{Z}$, $n \geqslant 0$); we define on $\widehat{M}$ the structure of a graded $\widehat{S}$-module by setting

(8.2.4.2) $$\deg(x \otimes \mathbf{z}^n) = n + \deg x$$

for all homogeneous x in M. We leave it to the reader to prove the analogue of (8.2.3):

Lemma (8.2.5). — (i) *There is a canonical di-isomorphism of (non-graded) modules*

(8.2.5.1) $$\widehat{M}_{(\mathbf{z})} \xrightarrow{\sim} M.$$

(ii) *For all $f \in S_d$ ($d > 0$), there is a di-isomorphism of (non-graded) modules*

(8.2.5.2) $$\widehat{M}_{(f)} \xrightarrow{\sim} M_f^{\leqslant}.$$

[1] This should not be confused with the use of the notation $\widehat{S}$ to denote the completed separation of a ring.

(8.2.6). Let S be a *positively*-graded ring, and consider the decreasing sequence of graded ideals of S

(8.2.6.1) $$S_{[n]} = \bigoplus_{m \geqslant n} S_m \qquad (n \geqslant 0)$$

(so, in particular, we have $S_{[0]} = S$ and $S_{[1]} = S_+$). Since it is evident that $S_{[m]}S_{[n]} \subset S_{[m+n]}$, we can define a *graded ring* $S^\natural$ by setting

(8.2.6.2) $$S^\natural = \bigoplus_{n \geqslant 0} S^\natural_n \quad \text{with} \quad S^\natural_n = S_{[n]}.$$

$S^\natural_0$ is then the ring S considered as a *non-graded* ring, and $S^\natural$ is thus an $S^\natural_0$-algebra. For every homogeneous element $f \in S_d$ $(d > 0)$, we denote by $f^\natural$ the element f considered as belonging to $S_{[d]} = S^\natural_d$. With this notation:

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Lemma (8.2.7). — *Let S be a positively-graded ring, and f a homogeneous element of S_d $(d > 0)$. There are canonical ring isomorphisms*

(8.2.7.1) $$S_f \xrightarrow{\sim} \bigoplus_{n \in \mathbf{Z}} S(n)_{(f)}$$

(8.2.7.2) $$(S^{\geqslant}_f)_{f/1} \xrightarrow{\sim} S_f$$

(8.2.7.3) $$S^\natural_{(f^\natural)} \xrightarrow{\sim} S^{\geqslant}_f$$

where the first two are isomorphisms of graded rings.

Proof. It is immediate, by definition, that we have $(S_f)_n = (S(n)_f)_0$, whence the isomorphism in (8.2.7.1), which is exactly the identity. Next, since $f/1$ is invertible in S_f, there is a canonical isomorphism $S_f \xrightarrow{\sim} (S^{\geqslant}_f)_{f/1} = (S_f)_{f/1}$, by (8.2.1.2) applied to S_f; the inverse isomorphism is, by definition, the isomorphism in (8.2.7.2). Finally, if $x = \sum_{m \geqslant n} y_m$ is an element of $S_{[n]}$ with $n = kd$, then the element $x/(f^\natural)^k$ corresponds to the element $\sum_m y_m/f^k$ of $S^{\geqslant}_f$, and we can quickly verify that this defines an isomorphism (8.2.7.3). □

(8.2.8). If M is a graded S-module, then we similarly define, for all $n \in \mathbf{Z}$,

(8.2.8.1) $$M_{[n]} = \bigoplus_{m \geqslant n} M_m$$

and, since $S_{[m]}M_{[n]} \subset M_{[m+n]}$ $(m \geqslant 0)$, we can define a graded $S^\natural$-module $M^\natural$ by setting

(8.2.8.2) $$M^\natural = \bigoplus_{n \in \mathbf{Z}} \quad \text{with} \quad M^\natural_n = M_{[n]}.$$

We leave to the reader the proof of:

Lemma (8.2.9). — *With the notation of (8.2.7) and (8.2.8), there are canonical di-isomorphisms of modules*

(8.2.9.1) $$M_f \xrightarrow{\sim} \bigoplus_{n \in \mathbf{Z}} M(n)_{(f)}$$

(8.2.9.2) $$(M^{\geqslant}_f)_{f/1} \xrightarrow{\sim} M_f$$

(8.2.9.3) $$M^\natural_{(f^\natural)} \xrightarrow{\sim} M^{\geqslant}_f$$

where the first two are di-isomorphisms of graded modules.

Lemma (8.2.10). — *Let S be a positively-graded ring.*

(i) *For $S^\natural$ to be an $S^\natural_0$-algebra of finite type (resp. a Noetherian $S^\natural_0$-algebra), it is necessary and sufficient for S to be an $S^\natural_0$-algebra of finite type (resp. a Noetherian $S^\natural_0$-algebra).*

(ii) *For $S^\natural_{n+1} = S^\natural_1 S^\natural_n$ $(n \geqslant n_0)$, it is necessary and sufficient for $S_{n+1} = S_1 S_n$ $(n \geqslant n_0)$.*

(iii) *For $S_n^\natural = S_1^\natural$ ($n \geqslant n_0$), it is necessary and sufficient for $S_n = S_1^n$ ($n \geqslant n_0$).*
(iv) *If (f_α) is a set of homogeneous elements of S_+ such that S_+ is the radical in S_+ of the ideal of S_+ generated by the f_α, then $S_+^\natural$ is the radical in $S_+^\natural$ of the ideal of $S_+^\natural$ generated by the $f_\alpha^\natural$.*

PROOF.

(i) If $S^\natural$ is an $S_0^\natural$-algebra of finite type, then $S_+ = S_1^\natural$ is a module of finite type over $S = S_0^\natural$, by (2.1.6, i), and so S is an S_0-algebra of finite type (2.1.4); if $S^\natural$ is a Noetherian ring, then so too is $S_0^\natural = S$ (2.1.5). Conversely, if S is an S_0-algebra of finite type, then we know (2.1.6, ii) **II** | 160
that there exist $h > 0$ and $m_0 > 0$ such that $S_{n+h} = S_h S_n$ for $n \geqslant m_0$; we can clearly assume that $m_0 \geqslant h$. Furthermore, the S_m are S_0-modules of finite type (2.1.6, i). So, if $n \geqslant m_0 + h$, then $S_n^\natural = S_h S_{n-h}^\natural = S_h^\natural S_{n-h}^\natural$; and if $m < m_0 + h$ then, letting $E = S_{m_0} + \ldots + S_{m_0+h-1}$, we have that

$$S_m^\natural = S_m + \ldots + S_{m_0+h-1} + S_h E + S_h^2 E + \ldots.$$

For $1 \leqslant m \leqslant m_0$, let G_m be the union of the finite systems of generators of the S_0-modules S_i for $m \leqslant i \leqslant m_0 + h - 1$, considered as a subset of $S_{[m]}$. For $m_0 + 1 \leqslant m \leqslant m_0 + h - 1$, let G_m be the union of the finite system of generators of the S_0-modules S_i for $m \leqslant i \leqslant m_0 + h - 1$ and of $S_h E$, considered as a subset of $S_{[m]}$. It is clear that $S_m^\natural = S_0^\natural G_m$ for $1 \leqslant m \leqslant m_0 + h - 1$, and thus the union G of the G_m for $1 \leqslant m \leqslant m_0 + h - 1$ is a system of generators of the $S_0^\natural$-algebra $S^\natural$. We thus conclude that, if $S = S_0^\natural$ is a Noetherian ring, then so too is $S^\natural$.

(ii) It is clear that, if $S_{n+1} = S_1 S_n$ for $n \geqslant n_0$, then $S_{n+1}^\natural = S_1 S_n^\natural$, and *a fortiori* $S_{n+1}^\natural = S_1^\natural S_n^\natural$ for $n \geqslant n_0$. Conversely, this last equality can be written as

$$S_{n+1} + S_{n+2} + \ldots = (S_1 + S_2 + \ldots)(S_n + S_{n+1} + \ldots)$$

and comparing terms of degree $n+1$ (in S) on both sides gives that $S_{n+1} = S_1 S_n$.

(iii) If $S_n = S_1^n$ for $n \geqslant n_0$, then $S_n^\natural = S_1^n + S_1^{n+1} + \ldots$; since $S_1^\natural$ contains $S_1 + S_1^2 + \ldots$, we have that $S_n^\natural \subset S_1^{\natural n}$, and thus $S_n^\natural = S_1^{\natural n}$ for $n \geqslant n_0$. Conversely, the only terms of $S_1^{\natural n} = (S_1 + S_2 + \ldots)^n$ that are of degree n in S are those of S_1^n; the equality $S_n^\natural = S_1^{\natural n}$ thus implies that $S_n = S_1^n$.

(iv) It suffices to show that, if an element $g \in S_{k+h}$ is considered as an element of $S_k^\natural$ ($k > 0$, $h \geqslant 0$), then there exists an integer $n > 0$ such that g^n is a linear combination (in $S_{kn}^\natural$) of the $f_\alpha^\natural$ with coefficients in $S^\natural$. By hypothesis, there exists an integer m_0 such that, for $m \geqslant m_0$, we have, *in* S, that $g^m = \sum_\alpha c_{\alpha m} f_\alpha$, where the indices α here are *independent of* m; furthermore, we can clearly assume that the $c_{\alpha m}$ are homogeneous, with

$$\deg(c_{\alpha m}) = m(k+h) - \deg f_\alpha$$

in S. So take m_0 sufficiently large enough to ensure that $km_0 > \deg f_\alpha$ for all the f_α that appear in g^{m_0}; for all α, let $c'_{\alpha m}$ be the element $c_{\alpha m}$ considered as having degree $km - \deg f_\alpha$ *in* $S^\natural$; we then have, in $S^\natural$, that $g^m = \sum_\alpha c'_{\alpha m} f_\alpha^\natural$, which finishes the proof.

□

(8.2.11). Consider the graded S_0-algebra

(8.2.11.1) $$S^\natural \otimes_S S_0 = S^\natural / S_+ S^\natural = \bigoplus_{n \geqslant 0} S_{[n]} / S_+ S_{[n]}.$$

Since S_n is a quotient S_0-module of $S_{[n]} / S_+ S_{[n]}$, there is a canonical homomorphism of graded S_0-algebras

(8.2.11.2) $$S^\natural \otimes_S S_0 \longrightarrow S$$

which is clearly *surjective*, and thus corresponds (2.9.2) to a canonical *closed immersion*

(8.2.11.3) $$\mathrm{Proj}(S) \longrightarrow \mathrm{Proj}(S^\natural \otimes_S S_0).$$

Proposition (8.2.12). — *The canonical morphism (8.2.11.3) is bijective. For the homomorphism (8.2.11.2) to be (TN)-bijective, it is necessary and sufficient for there to exist some n_0 such that $S_{n+1} = S_1 S_n$ for $n \geqslant n_0$. If this latter condition is satisfied, then (8.2.11.3) is an isomorphism; the converse is true whenever S is Noetherian.*

PROOF. To prove the first claim, it suffices (2.8.3) to show that the kernel $\mathfrak{I}$ of the homomorphism (8.2.11.2) consists of *nilpotent* elements. But if $f \in S_{[n]}$ is an element whose class modulo $S_+ S_{[n]}$ belongs to this kernel, then this implies that $f \in S_{[n+1]}$; then f^{n+1}, considered as an element of $S_{[n(n+1)]}$, is also an element of $S_+ S_{[n(n+1)]}$, since it can be written as $f \cdot f^n$; so the class of f^{n+1} modulo $S_+ S_{[n(n+1)]}$ is zero, which proves our claim. Since the hypothesis that $S_{n+1} = S_1 S_n$ for $n \geqslant n_0$ is equivalent to $S^\natural_{n+1} = S^\natural_1 S^\natural_n$ for $n \geqslant n_0$ (8.2.10, ii), this hypothesis is equivalent, by definition, to the fact that (8.2.11.2) is (TN)-injective, and thus (TN)-bijective, and so (8.2.11.3) is an isomorphism, by (2.9.1). Conversely, if (8.2.11.3) is an isomorphism, then the sheaf $\widetilde{\mathfrak{I}}$ on $\operatorname{Proj}(S^\natural \otimes_S S_0)$ is zero (2.9.2, i); since $S^\natural \otimes_S S_0$ is Noetherian, as a quotient of $S^\natural$ (8.2.10, i), we conclude from (2.7.3) that $\mathfrak{I}$ satisfies condition (TN), and so $S^\natural_{n+1} = S^\natural_1 S^\natural_n$ for $n \geqslant n_0$, and this finishes the proof, by (8.2.10, ii). □

(8.2.13). Consider now the canonical injections $(S_+)^n \to S_{[n]}$, which define an injective homomorphism of degree zero of graded rings

$$\bigoplus_{n \geqslant 0} (S_+)^n \longrightarrow S^\natural. \tag{8.2.13.1}$$

Proposition (8.2.14). — *For the homomorphism (8.2.13.1) to be a (TN)-isomorphism, it is necessary and sufficient for there to exist some n_0 such that $S_n = S_1^n$ for all $n \geqslant n_0$. Whenever this is the case, the morphisms corresponding to (8.2.13.1) is everywhere defined and also an isomorphism*

$$\operatorname{Proj}(S^\natural) \xrightarrow{\sim} \operatorname{Proj}\Big(\bigoplus_{n \geqslant 0} (S_+)^n\Big);$$

the converse is true whenever S is Noetherian.

PROOF. The first two claims are evident, given (8.2.10, iii) and (2.9.1). The third will follow from (8.2.10, i and iii) and the following lemma:

Lemma (8.2.14.1). — *Let T be a positively-graded ring that is also a T_0-algebra of finite type. If the morphism corresponding to the injective homomorphism $\bigoplus_{n \geqslant 0} T_1^n \to T$ is everywhere defined and also an isomorphism $\operatorname{Proj}(T) \to \operatorname{Proj}(\bigoplus_{n \geqslant 0} T_1^n)$, then there exists some n_0 such that $T_n = T_1^n$ for $n \geqslant n_0$.*

Let g_i ($1 \leqslant i \leqslant r$) be generators of the T_0-module T_1. The hypothesis implies first of all that the $D_+(g_i)$ cover $\operatorname{Proj}(T)$ (2.8.1). Let $(h_j)_{1 \leqslant j \leqslant s}$ be a system of homogeneous elements of T_+, with $\deg(h_j) = n_j$, that form, with the g_i, a system of generators of the ideal T_+, or, equivalently (2.1.3), a system of generators of T as a T_0-algebra; if we set $T' = \bigoplus_{n \geqslant 0} T_1^n$, then the element $h_j / g_i^{n_j}$ of the ring $T_{(g_i)}$ must, by hypothesis, belong to the subring $T'_{(g_i)}$, and so there exists some integer k such that $T_1^k h_j \subset T_1^{k+n_j}$ for all j. We thus conclude, by induction on r, that $T_1^k h_j^r \subset T'$ for all $r \geqslant 1$, and, by definition of the h_j, we thus have that $T_1^k T \subset T'$. Also, there exists, for all j, an integer m_j such that $h_j^{m_j}$ belongs to the ideal of T generated by the g_i (2.3.14), so $h_j^{m_j} \in T_1 T$, and $h_j^{m_j k} \in T_1^k T \subset T'$. **II** | 162
There is thus an integer $m_0 \geqslant k$ such that $h_j^m \in T_1^{mn}$ for $m \geqslant m_0$. So, if q is the largest of the integers n_j, then $n_0 = qsm_0 + k$ is the required number. Indeed, an element of S_n, for $n \geqslant n_0$, is the sum of monomials belonging to $T_1^\alpha u$, where u is a product of powers of the h_j; if $\alpha \geqslant k$, then it follows from the above that $T_1^\alpha u \subset T_1^n$; in the other case, one of the exponents of the h_j is $\geqslant m_0$, so $u \in T_1^\beta v$, where $\beta \geqslant k$ and v is again a product of powers of the h_j; we can then reduce to the previous case, and so we conclude that $T_1^\alpha u \subset T_1^n$ in all cases. □

Remark (8.2.15). — The condition $S_n = S_1^n$ for $n \geqslant n_0$ clearly implies that $S_{n+1} = S_1 S_n$ for $n \geqslant n_0$, but the converse is not necessarily true, even if we assume that S is Noetherian. For example, let K be a field, $A = K[\mathbf{x}]$, and $B = K[\mathbf{y}]/\mathbf{y}^2 K[\mathbf{y}]$, where $\mathbf{x}$ and $\mathbf{y}$ are indeterminates, with $\mathbf{x}$ taken to have degree 1 and $\mathbf{y}$ to have degree 2, and let $S = A \otimes_K B$, so that S is a graded algebra over K that has a basis given by the elements 1, $\mathbf{x}^n$ ($n \geqslant 1$), and $\mathbf{x}^n\mathbf{y}$ ($n \geqslant 0$). It is immediate that $S_{n+1} = S_1 S_n$ for $n \geqslant 2$, but $S_1^n = K\mathbf{x}^n$ while $S_n = K\mathbf{x}^n + K\mathbf{x}^n\mathbf{y}$ for $n \geqslant 2$.

8.3. Based cones

(8.3.1). Let Y be a prescheme; in all of this section, we will consider only *Y-preschemes* and *Y-morphisms*. Let $\mathscr{S}$ be a quasi-coherent *positively*-graded $\mathscr{O}_Y$-algebra; *we further assume that* $\mathscr{S}_0 = \mathscr{O}_Y$. Following the notation introduced in (8.2.2), we let

$$\widehat{\mathscr{S}} = \mathscr{S}[\mathbf{z}] = \mathscr{S} \otimes_{\mathscr{O}_Y} \mathscr{O}_Y[\mathbf{z}] \tag{8.3.1.1}$$

which we consider as a positively-graded $\mathscr{O}_Y$-algebra by defining the degrees as in (8.2.2.2), so that, for every affine open subset U of Y, we have

$$\Gamma(U, \widehat{\mathscr{S}}) = (\Gamma(U, \mathscr{S}))[\mathbf{z}].$$

In what follows, we write

$$X = \operatorname{Proj}(\mathscr{S}), \quad C = \operatorname{Spec}(\mathscr{S}), \quad \widehat{C} = \operatorname{Proj}(\widehat{\mathscr{S}}) \tag{8.3.1.2}$$

(where, in the definition of C, we consider $\mathscr{S}$ as a non-graded $\mathscr{O}_Y$-algebra), and we say that C (resp. $\widehat{C}$) is the *affine cone* (resp. *projective cone*) defined by $\mathscr{S}$; we will sometimes say "cone" instead of "affine cone". By an abuse of language, we also say that C (resp. $\widehat{C}$) is the *affine cone based at X (?)* (resp. the *projective cone based at X (?)*)[2], with the implicit understanding that the prescheme X is given in the form $\operatorname{Proj}(\mathscr{S})$; finally, we say that $\widehat{C}$ is the *projective closure* of C (with the data of $\mathscr{S}$ being implicit in the structure of C).

Proposition (8.3.2). — *There exist canonical Y-morphisms*

$$Y \xrightarrow{\varepsilon} C \xrightarrow{i} \widehat{C} \tag{8.3.2.1}$$

$$X \xrightarrow{j} \widehat{C} \tag{8.3.2.2}$$

such that ε and j are closed immersions, and i is an affine morphism, which is a dominant open immersion, for which

$$i(C) = \widehat{C} - j(X); \tag{8.3.2.3}$$

furthermore, $\widehat{C}$ is the smallest closed subprescheme of $\widehat{C}$ containing $i(C)$.

PROOF. To define i, consider the open subset of $\widehat{C}$ given by

$$\widehat{C}_{\mathbf{z}} = \operatorname{Spec}(\widehat{\mathscr{S}}/(\mathbf{z}-1)\widehat{\mathscr{S}}) \tag{8.3.2.4}$$

(3.1.4), where $\mathbf{z}$ is canonically identified with a section of $\mathscr{S}$ over Y. The isomorphism $i : C \xrightarrow{\sim} \widehat{C}_{\mathbf{z}}$ then corresponds to the canonical isomorphism (8.2.3.1)

$$\widehat{\mathscr{S}}/(\mathbf{z}-1)\widehat{\mathscr{S}} \xrightarrow{\sim} \mathscr{S}.$$

The morphism ε corresponds to the augmentation homomorphism $\mathscr{S} \to \mathscr{S}_0 = \mathscr{O}_Y$, which has kernel $\mathscr{S}_+$ (1.2.7), and, since the latter is surjective, ε is a closed immersion (1.4.10). Finally, j corresponds (3.5.1) to the surjective homomorphism of degree zero $\widehat{\mathscr{S}} \to \mathscr{S}$, which restricts to the identity on $\mathscr{S}$ and is zero on $\mathbf{z}\widehat{\mathscr{S}}$, which is its kernel; j is everywhere defined, and is a closed immersion, by (3.6.2).

To prove the other claims of (8.3.2), we can clearly restrict to the case where $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{S} = \widetilde{S}$, with S a graded A-algebra, whence $\widehat{\mathscr{S}} = (\widehat{S})^\sim$; the homogeneous elements f of S_+ can then be identified with sections of $\widehat{\mathscr{S}}$ over Y, and the open subset of $\widehat{C}$, denoted $D_+(f)$ in (2.3.3), can then be written as $\widehat{C}_f$ (3.1.4); similarly, the open subset of C denoted $D(f)$ in (**I**, 1.1.1) can be written as C_f (**0**, 5.5.2). With this in mind, it follows from (2.3.14) and from the definition of $\widehat{S}$

[2] *[Trans.] A more literal translation of the French (*côtne projetant (affine/projectif)*) would be the* projecting (affine/projective) cone*, but it seems that this terminology already exists to mean something else.*

that, in this case, the open subsets $\widehat{C}_{\mathbf{z}} = i(C)$ and $\widehat{C}_f$ (with f homogeneous in S_+) form a *cover* of $\widehat{C}$. Furthermore, with this notation,

$$i^{-1}(\widehat{C}_f) = C_f; \tag{8.3.2.5}$$

indeed, $\widehat{C}_f \cap i(C) = \widehat{C}_f \cap \widehat{C}_{\mathbf{z}} = \widehat{C}_{f\mathbf{z}} = \operatorname{Spec}(\widehat{S}_{(f\mathbf{z})})$. But, if $d = \deg(f)$, then $\widehat{S}_{(f\mathbf{z})}$ is canonically isomorphic to $(\widehat{S}_{(\mathbf{z})})_{f/\mathbf{z}^d}$ (2.2.2), and it follows from the definition of the isomorphism in (8.2.3.1) that the image of $(\widehat{S}_{(\mathbf{z})})_{f/\mathbf{z}^d}$ under the corresponding isomorphism of rings of fractions is exactly S_f. Since $C_f = \operatorname{Spec}(S_f)$, this proves (8.3.2.5) and shows, at the same time, that the morphism i is affine; furthermore, the restriction of i to C_f, considered as a morphism to $\widehat{C}_f$, corresponds (**I**, 1.7.3) to the canonical homomorphism $\widehat{S}_{(f)} \to \widehat{S}_{(f\mathbf{z})}$, and, by the above and (8.2.3.2), we can claim the following result:

(8.3.2.6). If $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{S} = \widetilde{S}$, then, for every homogeneous f in S_+, $\widehat{C}_f$ is canonically identified with $\operatorname{Spec}(S_f^{\leqslant})$, and the morphism $C_f \to \widehat{C}_f$ given by restricting i then corresponds to the canonical injection $S_f^{\leqslant} \to S_f$.

Now note that (for Y affine) the complement of $\widehat{C}_{\mathbf{z}}$ in $\widehat{C} = \operatorname{Proj}(\widehat{S})$ is, by definition, the set of graded prime ideals of $\widehat{S}$ containing $\mathbf{z}$, which is exactly $j(X)$, by definition of j, which proves (8.3.2.3). **II** | 164

Finally, to prove the last claim of (8.3.2), we can assume that Y is affine. With the above notation, note that, in the ring $\widehat{S}$, $\mathbf{z}$ is not a zero divisor; since $i(C) = \widehat{C}$, it suffices to prove the following lemma:

Lemma (8.3.2.7). — *Let T be a positively-graded ring, $Z = \operatorname{Proj}(T)$, and g a homogeneous element of T of degree $d > 0$. If g is not a zero divisor in T, then Z is the smallest closed subprescheme of Z that contains $Z_g = D_+(g)$.*

By (**I**, 4.1.9), the question is local on Z; for every homogeneous element $h \in T_e$ ($e > 0$), it thus suffices to prove that Z_h is the smallest closed subprescheme of Z_h that contains Z_{gh}; it follows from the definitions and from (**I**, 4.3.2) that this condition is equivalent to asking for the canonical homomorphism $T_{(h)} \to T_{(gh)}$ to be *injective*. But this homomorphism can be identified with the canonical homomorphism $T_{(h)} \to (T_{(h)})_{g^e/h^d}$ (2.2.3). But since g^e is not a zero divisor in T, g^e/h^d is not a zero divisor in T_h (nor *a fortiori* in $T_{(h)}$), since the fact that $(g^e/h^d)(t/h^m) = 0$ (for $t \in T$ and $m > 0$) implies the existence of some $n > 0$ such that $h^n g^e t = 0$, whence $h^n t = 0$, and thus $t/h^m = 0$ in T_h. This thus finishes the proof (**0**, 1.2.2). □

(8.3.3). We will often identify the affine cone C with the subprescheme induced by the projective cone $\widehat{C}$ on the open subset $i(C)$ by means of the open immersion i. The closed subprescheme of C associated to the closed immersion ε is called the *vertex prescheme* (?) of C; we also say that ε, which is a Y-section of C, is the *vertex section* (?), or the *null section*, or C; we can identify Y with the vertex prescheme (?) of C by means of ε. Also, $i \circ \varepsilon$ is a Y-section of $\widehat{C}$, and thus also a closed immersion (**I**, 5.4.6), corresponding to the canonical surjective homomorphism of degree zero $\widehat{\mathscr{S}} = \mathscr{S}[\mathbf{z}] \to \mathscr{O}_Y[\mathbf{z}]$ (3.1.7), whose kernel is $\mathscr{S}_+[\mathbf{z}] = \mathscr{S}_+\widehat{\mathscr{S}}$; the subprescheme of $\widehat{C}$ associated to this closed immersion is also called the *vertex prescheme* (?) of $\widehat{C}$, and $i \circ \varepsilon$ the *vertex section* (?) of $\widehat{C}$; it can be identified with Y by means of $i \circ \varepsilon$. Finally, the closed subprescheme of $\widehat{C}$ associated to j is called the *part at infinity* of $\widehat{C}$, and can be identified with X by means of j.

(8.3.4). The subpreschemes of C (resp. $\widehat{C}$) induced on the *open* subsets

$$E = C - \varepsilon(Y), \qquad \widehat{E} = \widehat{C} - i(\varepsilon(Y)) \tag{8.3.4.1}$$

are called (by an abuse of language) the *pointed affine cone* and the *pointed projective cone* (respectively) defined by $\mathscr{S}$; we note that, despite this nomenclature, *E is not necessarily affine* over Y, nor $\widehat{E}$ projective over Y (8.4.3). When we identify C with $i(C)$, we thus have the underlying spaces

$$C \cup \widehat{E} = \widehat{C}, \qquad C \cap \widehat{E} = E \tag{8.3.4.2}$$

so that $\widehat{C}$ can be considered as being obtained by *gluing* the open subpreschemes C and $\widehat{E}$; furthermore, by (8.3.2.3),

$$E = \widehat{E} - j(X). \tag{8.3.4.3}$$

If $Y = \operatorname{Spec}(A)$ is affine, then, with the notation of (8.3.2),

$$E = \bigcup C_f, \qquad \widehat{E} = \bigcup \widehat{C}_f, \qquad C_f = C \cap \widehat{C}_f \tag{8.3.4.4}$$

where f runs over the set of homogeneous elements of S_+ (or only a subset M of this set, with M generating an ideal of S_+ whose radical in S_+ is S_+ itself, or, equivalently, such that the X_f for $f \in M$ cover X (2.3.14)). The gluing of C and $\widehat{C}_f$ along C_f is thus determined by the injection morphisms $C_f \to C$ and $C_f \to \widehat{C}_f$, which, as we have seen (8.3.2.6), correspond (respectively) to the canonical homomorphisms $S \to S_f$ and $S_f^{\leqslant} \to S_f$.

Proposition (8.3.5). — *With the notation of (8.3.1) and (8.3.4), the morphism associated (3.5.1) to the canonical injection $\phi : \mathscr{S} \to \widehat{\mathscr{S}} = \mathscr{S}[\mathbf{z}]$ is a surjective affine morphism (called the canonical retraction)*

$$p : \widehat{E} \longrightarrow X \tag{8.3.5.1}$$

such that

$$p \circ j = 1_X. \tag{8.3.5.2}$$

Proof. To prove the proposition, we can restrict to the case where Y is affine. Taking into account the expression in (8.3.4.4) for $\widehat{E}$, the fact that the domain of definition $G(\phi)$ of p is equal to $\widehat{E}$ will follow from the first of the following claims:

(8.3.5.3). If $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{S} = \widetilde{S}$, then, for all homogeneous $f \in S_+$,

$$p^{-1}(X_f) = \widehat{C}_f \tag{8.3.5.4}$$

and the restriction of p to $\widehat{C}_f = \operatorname{Spec}(S_f^{\leqslant})$, considered as a morphism from $\widehat{C}_f$ to X_f, corresponds to the canonical injection $S_{(f)} \to S_f^{\leqslant}$. If, further, $f \in S_1$, then $\widehat{C}_f$ is isomorphic to $X_f \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (where T is an indeterminate).

Indeed, Equation (8.3.5.4) is exactly a particular case of (2.8.1.1), and the second claim is exactly the definition of $\operatorname{Proj}(\phi)$ whenever Y is affine (2.8.1). Then Equation (8.3.5.2) and the fact that p is surjective show that the composition $\mathscr{S} \to \widehat{\mathscr{S}} \to \mathscr{S}$ of the canonical homomorphisms is the identity on $\mathscr{S}$. Finally, the last claim of (8.3.5.3) follows from the fact that $S_f^{\leqslant}$ is isomorphic to $S_{(f)}[T]$ whenever $f \in S_1$ (2.2.1). □

Corollary (8.3.6). — *The restriction*

$$\pi : E \longrightarrow X \tag{8.3.6.1}$$

of p to E is a surjective affine morphism. If Y is affine and f homogeneous in S_+, then

$$\pi^{-1}(X_f) = C_f \tag{8.3.6.2}$$

and the restriction of π to C_f corresponds to the canonical injection $S_{(f)} \to S_f$. If, further, $f \in S_1$, then C_f is isomorphic to $X_f \otimes_{\mathbf{Z}} \mathbf{Z}[T, T^{-1}]$ (where T is an indeterminate).

Proof. Equation (8.3.6.2) follows immediately from (8.3.5.3) and (8.3.2.5), and shows the surjectivity of π; we have already seen that the immersion i, restricted to C_f, corresponds to the injection $S_f^{\leqslant} \to S_f$ (8.3.2). Finally, the last claim is a consequence of the fact that, for $f \in S_1$, S_f is isomorphic to $S_{(f)}[T, T^{-1}]$ (2.2.1). □ **II | 166**

Remark (8.3.7). — Whenever Y is affine, the elements of the underlying space of E are the (not-necessarily-graded) prime ideals $\mathfrak{p}$ of S not containing S_+, by definition of the immersion ε (8.3.2). For such an ideal $\mathfrak{p}$, the $\mathfrak{p} \cap S_n$ clearly satisfy the conditions of (2.1.9), and so there exists exactly one *graded* prime ideal $\mathfrak{q}$ of S such that $\mathfrak{q} \cap S_n = \mathfrak{p} \cap S_n$ for all n; the map $\pi : E \to X$ of underlying spaces can then be understood via the equation

$$\pi(\mathfrak{p}) = \mathfrak{q}. \tag{8.3.7.1}$$

Indeed, to prove this equation, it suffices to consider some homogeneous f in S_+ such that $\mathfrak{p} \in D(f)$, and to note that $\mathfrak{q}_{(f)}$ is the inverse image of $\mathfrak{p}_f$ under the injection $S_{(f)} \to S_f$.

Corollary (8.3.8). — *If $\mathscr{S}$ is generated by $\mathscr{S}_1$, then the morphisms p and π are of finite type; for all $x \in X$, the fibre $p^{-1}(x)$ is isomorphic to* $\operatorname{Spec}(k(x)[T])$*, and the fibre π^{-1} isomorphic to* $\operatorname{Spec}(k(x)[T, T^{-1}])$

PROOF. This follows immediately from (8.3.5) and (8.3.6) by noting that, whenever Y is affine and S is generated by S_1, the X_f, for $f \in S_1$, form a cover of X (2.3.14). □

Remark (8.3.9). — The pointed affine cone corresponding to the graded $\mathscr{O}_Y$-algebra $\mathscr{O}_Y[T]$ (where T is an indeterminate) can be identified with $G_m = \operatorname{Spec}(\mathscr{O}_Y[T, T^{-1}])$, since it is exactly C_T, as we have seen in (8.3.2) (see (8.4.4) for a more general result). This prescheme is canonical endowed with the structure of a "*Y-scheme in commutative groups*". This idea will be explained in detail later on, but, for now, can be quickly summarised as follows. A Y-scheme in groups is a Y-scheme G endowed with two Y-morphisms, $p : G \times_Y G \to G$ and $s : G \to G$, that satisfy conditions formally analogous to the axioms of the composition law and the symmetry law of a group: the diagram

$$\begin{tikzcd} G \times G \times G \arrow[r, "p\times 1"] \arrow[d, "1\times p"'] & G \times G \arrow[d, "p"] \\ G \times G \arrow[r, "p"'] & G \end{tikzcd}$$

should commute ("associativity"), and there should be a condition which corresponds to the fact that, for groups, the maps

$$(x,y) \longmapsto (x, x^{-1}, y) \longmapsto (x, x^{-1}y) \longmapsto x(x^{-1}y)$$

and

$$(x,y) \longmapsto (x, x^{-1}, y) \longmapsto (x, yx^{-1}) \longmapsto (yx^{-1})x$$

should both reduce to $(x,y) \mapsto y$; the sequence of morphisms corresponding, for example, to the first composite map is

$$G \times G \xrightarrow{(1,s)\times 1} G \times G \times G \xrightarrow{1\times p} G \times G \xrightarrow{p} G$$

and the reader should write down the second sequence.

It is immediate (**I**, 3.4.3) that the data of a structure of a Y-scheme in groups on a Y-scheme G is equivalent to the data, for *every* Y-prescheme Z, of a *group* structure on the set $\operatorname{Hom}_Y(Z, G)$, where these structures should be such that, for every Y-morphism $Z \to Z'$, the corresponding map $\operatorname{Hom}_Y(Z', G) \to \operatorname{Hom}_Y(Z, G)$ is a group homomorphism. In the particular case of G_m that we consider here, $\operatorname{Hom}_Y(Z, G)$ can be identified with the set of Z-sections of $Z \times_Y G_m$ (**I**, 3.3.14), and thus with the set of Z-sections of $\operatorname{Spec}(\mathscr{O}_Z[T, T^{-1}])$; finally, the same reasoning as in (**I**, 3.3.15) shows that this set is canonically identified with the set of *invertible* elements of the ring $\Gamma(Z, \mathscr{O}_Z)$, and the group structure on this set is the structure coming from the multiplication in the ring $\Gamma(Z, \mathscr{O}_Z)$. The reader can verify that the morphisms p and s from above are obtained in the following way: they correspond, by (1.2.7) and (1.4.6), to the homomorphisms of $\mathscr{O}_Y$-algebras **II** | 167

$$\pi : \mathscr{O}_Y[T, T^{-1}] \longrightarrow \mathscr{O}_Y[T, T^{-1}, T', T'^{-1}]$$
$$\sigma : \mathscr{O}_Y[T, T^{-1}] \longrightarrow \mathscr{O}_Y[T, T^{-1}]$$

and are entirely defined by the data of $\pi(T) = TT'$ and $\sigma(T) = T^{-1}$.

With this in mind, G_m can be considered as a "*universal domain of operators*" for every *affine cone* $C = \operatorname{Spec}(\mathscr{S})$, where $\mathscr{S}$ is a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra. This means that we can canonically define a Y-morphism $G_m \times_Y C \to C$ which has the formal properties of an external law of a set endowed with a group of operators; or, again, as above for schemes in groups, we can give, for every Y-prescheme Z, an external law on $\operatorname{Hom}_Y(Z, C)$, having the group $\operatorname{Hom}_Y(Z, G_m)$ as its set of operators, with the usual axioms of sets endowed with a group of operators, and a compatibility condition with respect to the Y-morphisms $Z \to Z'$. In the current case, the morphism $G_m \times_Y C \to C$ is defined by the data of a homomorphism of $\mathscr{O}_Y$-algebras $\mathscr{S} \to \mathscr{S} \otimes_{\mathscr{O}_Y} \mathscr{O}_Y[T, T^{-1}] = \mathscr{S}[T, T^{-1}]$, which associates, to each section $s_n \in \Gamma(U, \mathscr{S}_n)$ (where U is an open subset of Y), the section $s_n T^n \in \Gamma(U, \mathscr{S} \otimes_{\mathscr{O}_Y} \mathscr{O}_Y[T, T^{-1}])$.

Conversely, suppose that we are given a quasi-coherent, *a priori non-graded*, $\mathscr{O}_Y$-algebra, and, on $C = \operatorname{Spec}(\mathscr{S})$, a structure of a "*$Y$-scheme in sets endowed with a group of operators*" that has the Y-scheme in groups G_m as its domain of operators; then we canonically obtain a *grading* of $\mathscr{O}_Y$-algebras on $\mathscr{S}$. Indeed, the data of a Y-morphism $G_m \times_Y C \to C$ is equivalent to that of a homomorphism of $\mathscr{O}_Y$-algebras $\psi : \mathscr{S} \to \mathscr{S}[T, T^{-1}]$, which can be written as $\psi = \sum_{n \in \mathbf{Z}} \psi_n T^n$, where the $\psi_n : \mathscr{S} \to \mathscr{S}$ are homomorphisms of $\mathscr{O}_Y$-modules (with $\psi_n(s) = 0$ except for finitely many n for every section $s \in \Gamma(U, \mathscr{S})$, for any open subset U of Y). We can then prove that the axioms of sets endowed with a group of operators imply that the $\psi_n(\mathscr{S}) = \mathscr{S}_n$ define a grading (in positive or negative degree) of $\mathscr{O}_Y$-algebras on $\mathscr{S}$, with the ψ_n being the corresponding projectors. We also have the notation of a structure of an "*affine cone*" on every affine Y-scheme, defined in a "geometric" way without any reference to any prior grading. We will not further develop this point of view here, and we leave the work of precisely formulating the definitions and results corresponding to the information given above to the reader. **II** | 168

8.4. Projective closure of a vector bundle

(8.4.1). Let Y be a prescheme, and $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-module. If we take $\mathscr{S}$ to be the graded $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E})$, then Definition (8.3.1.1) shows that $\widehat{\mathscr{S}}$ can be identified with $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E} \oplus \mathscr{O}_Y)$. With the affine cone $\operatorname{Spec}(\mathscr{S})$ defined by $\mathscr{S}$ being, by definition, $\mathbf{V}(\mathscr{E})$, and $\operatorname{Proj}(\mathscr{S})$ being, by definition, $\mathbf{P}(\mathscr{E})$, we see that:

Proposition (8.4.2). — *The projective closure of a vector bundle $\mathbf{V}(\mathscr{E})$ on Y is canonically isomorphic to $\mathbf{P}(\mathscr{E} \oplus \mathscr{O}_Y)$, and the part at infinity of the latter is canonically isomorphic to $\mathbf{P}(\mathscr{E})$.*

Remark (8.4.3). — Take, for example, $\mathscr{E} = \mathscr{O}_Y^r$ with $r \geqslant 2$; then the pointed cones E and $\widehat{E}$ defined by $\mathscr{S}$ are nether affine nor projective on Y if $Y \neq \varnothing$. The second claim is immediate, because $\widehat{C} = \mathbf{P}(\mathscr{O}_Y^{r+1})$ is projective on Y, and the underlying spaces of E and $\widehat{E}$ are non-closed open subsets of $\widehat{C}$, and so the canonical immersions $E \to \widehat{C}$ and $\widehat{E} \to \widehat{C}$ are not projective (5.5.3), and we conclude by appealing to (5.5.5, v). Now, supposing, for example, that $Y = \operatorname{Spec}(A)$ is affine, and $r = 2$, then $C = \operatorname{Spec}(A[T_1, T_2])$, and E is then the prescheme induced by C on the open subset $D(T_1) \cup D(T_2)$; but we have already seen that the latter is not affine (**I**, 5.5.11); *a fortiori* $\widehat{E}$ cannot be affine, since E is the open subset where the section $\mathbf{z}$ over $\widehat{E}$ does not vanish (8.3.2).

However:

Proposition (8.4.4). — *If $\mathscr{L}$ is an invertible $\mathscr{O}_Y$-module, then there are canonical isomorphisms for both the pointed cones E and $\widehat{E}$ corresponding to $C = \mathbf{V}(\mathscr{L})$:*

$$\operatorname{Spec}\left(\bigoplus_{n \in \mathbf{Z}} \mathscr{L}^{\otimes n}\right) \xrightarrow{\sim} E \tag{8.4.4.1}$$

$$\mathbf{V}(\mathscr{L}^{-1}) \xrightarrow{\sim} \widehat{E}. \tag{8.4.4.2}$$

Furthermore, there exists a canonical isomorphism from the projective closure of $\mathbf{V}(\mathscr{L})$ to the projective closure of $\mathbf{V}(\mathscr{L}^{-1})$ that sends the null section (resp. the part at infinity) of the former to the part at infinity (resp. the null section) of the second.

Proof. We have here that $\mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$; the canonical injection

$$\mathscr{S} \longrightarrow \bigoplus_{n \in \mathbf{Z}} \mathscr{L}^{\otimes n}$$

defines a canonical dominant morphism

$$\operatorname{Spec}\left(\bigoplus_{n \in \mathbf{Z}} \mathscr{L}^{\otimes n}\right) \longrightarrow \mathbf{V}(\mathscr{L}) = \operatorname{Spec}\left(\bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}\right) \tag{8.4.4.3}$$

and it suffices to prove that this morphism is an isomorphism from the scheme $\operatorname{Spec}(\bigoplus_{n \in \mathbf{Z}} \mathscr{L}^{\otimes n})$ to E. Since the questions is local on Y, we can assume that $Y = \operatorname{Spec}(A)$ is affine and that $\mathscr{L} = \mathscr{O}_Y$, **II** | 169 and so $\mathscr{S} = (A[T])^{\sim}$ and $\bigoplus_{n\in\mathbf{Z}} \mathscr{L}^{\times n} = (A[T,T^{-1}])^{\sim}$. But $A[T,T^{-1}]$ is the ring of fractions $A[T]_T$ of $A[T]$, and thus (8.4.4.3) identifies $\bigoplus_{n\in\mathbf{Z}} \mathscr{L}^{\otimes n}$ (?) with the prescheme induced by $C = \mathbf{V}(\mathscr{L})$ on the open subset $D(T)$; the complement $V(T)$ of this open subset in C is the underlying space of the closed subprescheme of C defined by the ideal $TA[T]$, which is exactly the null section of C, and so $E = D(T)$.

The isomorphism in (8.4.4.2) will be a consequence of the last claim, since $\mathbf{V}(\mathscr{L}^{-1})$ is the complement of the part at infinity of its projective closure, and $\widehat{E}$ is the complement of the null section of the projective closure $C = \mathbf{V}(\mathscr{L})$. But these projective closures are $\mathbf{P}(\mathscr{L}^{-1} \oplus \mathscr{O}_Y)$ and $\mathbf{P}(\mathscr{L} \oplus \mathscr{O}_Y)$ (respectively); but we can write $\mathscr{L} \oplus \mathscr{O}_Y = \mathscr{L} \otimes (\mathscr{L}^{-1} \oplus \mathscr{O}_Y)$. The existence of the desired canonical isomorphism then follows from (4.1.4), and everything reduces to showing that this isomorphism swaps the null sections and the parts at infinity. For this, we can reduce to the case where $Y = \operatorname{Spec}(A)$ is affine, $L = Ac$, and $L^{-1} = Ac'$, with the canonical isomorphism $L \otimes L^{-1} \to A$ sending $c \otimes c'$ to the element 1 of A. Then $\mathbf{S}(L \oplus A)$ is the tensor product of $A[\mathbf{z}]$ with $\bigoplus_{n\geqslant 0} Ac^{\otimes n}$, and $\mathbf{S}(L^{-1} \oplus A)$ is the tensor product of $A[\mathbf{z}]$ with $\bigoplus_{n\geqslant 0} Ac'^{\otimes n}$, and the isomorphism defined in (4.1.4) sends $\mathbf{z}^h \otimes c'^{\otimes(n-h)}$ to the element $\mathbf{z}^{n-h} \otimes c^{\otimes h}$. But, in $\mathbf{P}(\mathscr{L}^{-1} \oplus \mathscr{O}_Y)$, the part at infinity is the set of points where the section $\mathbf{z}$ vanishes, and the null section is the section of points where the section c' vanishes; since we have analogous definitions for $\mathbf{P}(\mathscr{L} \oplus \mathscr{O}_Y)$, the conclusion follows immediately from the above explanation. □

8.5. Functorial behaviour

(8.5.1). Let Y and Y' be prescheme, $q : Y' \to Y$ a morphism, and $\mathscr{S}$ (resp. $\mathscr{S}'$) a quasi-coherent *positively*-graded $\mathscr{O}_Y$-algebra (resp. quasi-coherent *positively*-graded $\mathscr{O}_{Y'}$-algebra). Consider a q-morphism of graded algebras

(8.5.1.1) $$\phi : \mathscr{S} \longrightarrow \mathscr{S}'.$$

We know (1.5.6) that this corresponds, canonically, to a morphism

$$\Phi = \operatorname{Spec}(\phi) : \operatorname{Spec}(\mathscr{S}') \longrightarrow \operatorname{Spec}(\mathscr{S})$$

such that the diagram

(8.5.1.2)
$$\begin{tikzcd} C' \arrow[r, "\Phi"] \arrow[d] & C \arrow[d] \\ Y' \arrow[r, "q"'] & Y \end{tikzcd}$$

commutes, where we write $C = \operatorname{Spec}(\mathscr{S})$ and $C' = \operatorname{Spec}(\mathscr{S}')$. *Suppose, further, that* $\mathscr{S}_0 = \mathscr{O}_Y$ *and* $\mathscr{S}'_0 = \mathscr{O}_{Y'}$; let $\varepsilon : Y \to C$ and $\varepsilon : Y' \to C'$ be the canonical immersions (8.3.2); we then have a commutative diagram

(8.5.1.3)
$$\begin{tikzcd} Y' \arrow[r, "q"] \arrow[d, "\varepsilon'"'] & Y \arrow[d, "\varepsilon"] \\ C' \arrow[r, "\Phi"'] & C \end{tikzcd}$$

which corresponds to the diagram **II** | 170

$$\begin{tikzcd} \mathscr{S} \arrow[r, "\phi"] \arrow[d] & \mathscr{S}' \arrow[d] \\ \mathscr{O}_Y \arrow[r] & \mathscr{O}_{Y'} \end{tikzcd}$$

where the vertical arrows are the augmentation homomorphisms, and so the commutativity follows from the hypothesis that ϕ is assumed to be a homomorphism of *graded* algebras.

Proposition (8.5.2). — *If E (resp. E') is the pointed affine cone defined by $\mathscr{S}$ (resp. $\mathscr{S}'$), then $\Phi^{-1}(E) \subset E'$; if, further,* $\operatorname{Proj}(\phi) : G(\phi) \to \operatorname{Proj}(\mathscr{S})$ *is everywhere defined (or, equivalently, if $G(\phi) = \operatorname{Proj}(\mathscr{S}')$), then $\Phi^{-1}(E) = E'$, and conversely.*

PROOF. The first claim follows from the commutativity of (8.5.1.3). To prove the second, we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $Y' = \operatorname{Spec}(A')$ are affine, and $\mathscr{S} = \widetilde{S}$ and $\mathscr{S}' = \widetilde{S}'$. For every homogeneous f in S_+, writing $f' = \phi(f)$, we have that $\Phi^{-1}(C_f) = C'_{f'}$ (I, 2.2.4.1); saying that $G(\phi) = \operatorname{Proj}(S')$ implies that the radical (in S'_+) of the ideal generated by the $f' = \phi(f)$ is S'_+ itself ((2.8.1) and (2.3.14)), and this is equivalent to saying that the $C'_{f'}$ cover E' (8.3.4.4). □

(8.5.3). The q-morphism ϕ canonically extends to a q-morphism of graded algebras

(8.5.3.1)
$$\widehat{\phi} : \widehat{\mathscr{S}} \longrightarrow \widehat{\mathscr{S}'}$$

by letting $\widehat{\phi}(\mathbf{z}) = \mathbf{z}$. This induces a morphism

$$\widehat{\Phi} = \operatorname{Proj}(\widehat{\phi}) : G(\widehat{\phi}) \longrightarrow \widehat{C} = \operatorname{Proj}(\widehat{\mathscr{S}})$$

such that the diagram

$$\begin{tikzcd} G(\widehat{\phi}) \arrow[r, "\widehat{\Phi}"] \arrow[d] & \widehat{C} \arrow[d] \\ Y' \arrow[r, "q"'] & Y \end{tikzcd}$$

commutes (3.5.6). It follows immediately from the definitions that, if we write $i : C \to \widehat{C}$ and $i' : C' \to \widehat{C}'$ to mean the canonical open immersions (8.3.2), then $i'(C') \subset G(\widehat{\phi})$, and the diagram

(8.5.3.2)
$$\begin{tikzcd} C' \arrow[r, "\Phi"] \arrow[d, "i"'] & C \arrow[d, "i'"] \\ G(\widehat{\phi}) \arrow[r, "\widehat{\Phi}"'] & \widehat{C} \end{tikzcd}$$

commutes. Finally, if we let $X = \operatorname{Proj}(\mathscr{S})$ and $X' = \operatorname{Proj}(\mathscr{S}')$, and if $j : X \to \widehat{C}$ and $j' : X' \to \widehat{C}'$ are the canonical closed immersions (8.3.2), then it follows from the definition of these immersions that $j'(G(\phi)) \subset G(\widehat{\phi})$, and that the diagram **II** | 171

(8.5.3.3)
$$\begin{tikzcd} G(\phi) \arrow[r, "\operatorname{Proj}(\phi)"] \arrow[d, "j'"'] & X \arrow[d, "j"] \\ G(\widehat{\phi}) \arrow[r, "\widehat{\Phi}"'] & \widehat{C} \end{tikzcd}$$

commutes.

Proposition (8.5.4). — *If $\widehat{E}$ (resp. $\widehat{E}'$) is the pointed projective cone defined by $\mathscr{S}$ (resp. by $\mathscr{S}'$), then $\widehat{\Phi}^{-1}(\widehat{E}) \subset \widehat{E}'$; furthermore, if $p : \widehat{E} \to X$ and $p' : \widehat{E}' \to X'$ are the canonical retractions, then $p'(\widehat{\Phi}^{-1}(\widehat{E})) \subset G(\widehat{\phi})$, and the diagram*

(8.5.4.1)
$$\begin{tikzcd} \widehat{\Phi}^{-1}(\widehat{E}) \arrow[r, "\widehat{\Phi}"] \arrow[d, "p'"'] & \widehat{E} \arrow[d, "p"] \\ G(\phi) \arrow[r, "\operatorname{Proj}(\phi)"'] & X \end{tikzcd}$$

commutes. If $\operatorname{Proj}(\phi)$ *is everywhere defined, then so too is $\widehat{\Phi}$, and we have that $\widehat{\Phi}^{-1}(\widehat{E}) = \widehat{E}'$*

PROOF. The first claim follows from the commutativity of Diagrams (8.5.1.3) and (8.5.3.2), and the two following claims from the definition of the canonical retractions (8.3.5) and the definition of $\widehat{\phi}$. To see that $\widehat{\Phi}$ is everywhere defined whenever $\operatorname{Proj}(\phi)$ is, we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $Y' = \operatorname{Spec}(A')$ are affine, and where $\mathscr{S} = \widetilde{S}$ and $\mathscr{S}' = \widetilde{S}'$; the hypothesis is that, when f runs over the set of homogeneous elements of S_+, the radical in S'_+ of the ideal generated in S'_+ by the $\phi(f)$ is S'_+ itself; we thus immediately conclude that the radical in $(S'[\mathbf{z}])_+$ of the ideal generated by $\mathbf{z}$ and the $\phi(f)$ is $(S'[\mathbf{z}])_+$ itself, whence our claim; this also shows that $\widehat{E}'$ is the union of the $\widehat{C}'_{\phi(f)}$, and hence equal to $\widehat{\Phi}^{-1}(\widehat{E})$. □

Corollary (8.5.5). — *Whenever* $\operatorname{Proj}(\phi)$ *is everywhere defined, the inverse image under* $\widehat{\Phi}$ *of the underlying space of the part at infinity (resp. of the vertex prescheme) of* $\widehat{C}'$ *is the underlying space of the part at infinity (resp. of the vertex prescheme) of* $\widehat{C}$.

PROOF. This follows immediately from (8.5.4) and (8.5.2), taking into account the equalities (8.3.4.1) and (8.3.4.2). □

8.6. A canonical isomorphism for pointed cones

(8.6.1). Let Y be a prescheme, $\mathscr{S}$ a quasi-coherent positively-graded $\mathscr{O}_Y$-algebra *such that* $\mathscr{S}_0 = \mathscr{O}_Y$, and let X be the Y-scheme $\operatorname{Proj}(\mathscr{S})$. We are going to apply the results of (8.5) to the case where $Y' = X$, and $q : X \to Y$ is the structure morphism; let

(8.6.1.1) $$\mathscr{S}_X = \bigoplus_{n \in \mathbf{Z}} \mathscr{O}_X(n)$$

which is a quasi-coherent graded $\mathscr{O}_X$-algebra, with multiplication defined by means of the canonical homomorphisms (3.2.6.1) II | 172

$$\mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{O}_X(n) \longrightarrow \mathscr{O}_X(m+n)$$

whose associativity is ensured by the commutative diagram in (2.5.11.4). Let $\mathscr{S}'$ be the quasi-coherent positively-graded $\mathscr{O}_X$-subalgebra $\mathscr{S}_X^{\geqslant} = \bigoplus_{n \geqslant 0} \mathscr{O}_X(n)$ of $\mathscr{S}_X$.

Finally, consider the canonical q-morphism

(8.6.1.2) $$\alpha : \mathscr{S} \longrightarrow \mathscr{S}_X^{\geqslant}$$

defined in (3.3.2.3) as a homomorphism $\mathscr{S} \to q_*(\mathscr{S}_X)$, but which clearly sends $\mathscr{S}$ to $q_*(\mathscr{S}_X^{\geqslant})$. Write

(8.6.1.3) $$C_X = \operatorname{Spec}(\mathscr{S}_X^{\geqslant}), \quad \widehat{C}_X = \operatorname{Proj}(\mathscr{S}_X^{\geqslant}[\mathbf{z}]), \quad X' = \operatorname{Proj}(\mathscr{S}_X^{\geqslant})$$

and denote by E_X and $\widehat{E}_X$ the corresponding pointed affine and pointed projective cones (respectively); denote the canonical morphisms defined in (8.3) by $\varepsilon_X : X \to C_X$, $i_X : C \to \widehat{C}_X$, $j_X : X' \to \widehat{C}_X$, $p_X : \widehat{E}_X \to X'$, and $\pi_X : E_X \to X'$.

Proposition (8.6.2). — *The structure morphism* $u : X' \to X$ *is an* isomorphism, *and the morphism* $\operatorname{Proj}(\alpha)$ *is everywhere defined and identical to* u. *The morphism* $\operatorname{Proj}(\widehat{\alpha}) : \widehat{C}_X \to \widehat{C}$ *is everywhere defined, and its restrictions to* $\widehat{E}_X$ *and* E_X *are* isomorphisms *to* $\widehat{E}$ *and* E *(respectively). Finally, if we identify* X' *with* X *via* u, *then the morphisms* p_X *and* π_X *are identified with the structure morphisms of the* X*-preschemes* $\widehat{E}_X$ *and* E_X.

PROOF. We can clearly restrict to the case where $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{S} = \widetilde{S}$; then X is the union of affine open subsets X_f, where f runs over the set of homogeneous elements of S_+, with the ring of each X_f being $S_{(f)}$. It follows from (8.2.7.1) that

(8.6.2.1) $$\Gamma(X_f, \mathscr{S}_X^{\geqslant}) = S_f^{\geqslant}.$$

So $u^{-1}(X_f) = \operatorname{Proj}(S_f^{\geqslant})$. But if $f \in S_d$ ($d > 0$), then $\operatorname{Proj}(S_f^{\geqslant})$ is canonically isomorphic to $\operatorname{Proj}((S_f^{\geqslant})^{(d)})$ (2.4.7), and we also know that $(S_f^{\geqslant})^{(d)} = (S^{(d)})_f^{\geqslant}$ can be identified with $S_{(f)}[T]$ (2.2.1) by the map $T \mapsto f/1$; we thus conclude (3.1.7) that the structure morphism $u^{-1}(X_f) \to X_f$ is an isomorphism, whence the first claim. To prove the second, note that the restriction $u^{-1}(X_f) \cap G(\alpha) \to X = \operatorname{Proj}(S)$ of $\operatorname{Proj}(\alpha)$ corresponds to the canonical map $x \mapsto x/1$ from S to $S_f^{\geqslant}$ (2.6.2); we thus deduce, first of all, that $G(\alpha) = X'$, and then, taking into account the fact that $u^{-1}(X_f) = (u^{-1}(X_f))_{f/1}$, that it follows from (2.8.1.1) that the image of $u^{-1}(X_f)$ under $\operatorname{Proj}(\alpha)$ is contained in X_f, and the restriction of $\operatorname{Proj}(\alpha)$ to $u^{-1}(X_f)$, considered as a morphism to $X_f = \operatorname{Spec}(S_{(f)})$, is indeed identical to that of u. Finally, applying (8.3.5.4) to p_X instead of p, we see that $p_X^{-1}(u^{-1}(X_f)) = \operatorname{Spec}((S_f^{\geqslant})^{\leqslant}_{f/1})$, and this open subset is, by (8.5.4.1), the inverse image under $\operatorname{Proj}(\widehat{\alpha})$ of $p^{-1}(X_f) = \operatorname{Spec}(S_f^{\leqslant})$ (8.3.5.3). Taking (8.2.3.2) into account, the restriction of $\operatorname{Proj}(\widehat{\alpha})$ to

$p_X^{-1}(u^{-1}(X_f))$ corresponds to the isomorphism inverse to (8.2.7.2), restricted to $S_f^{\leqslant}$, whence the third claim; the last claim is evident by definition.

We note also that it follows from the commutative diagram in (8.5.3.2) that *the restriction to* C_X *of* $\operatorname{Proj}(\widehat{\alpha})$ *is exactly the morphism* $\operatorname{Spec}(\alpha)$. ☐ II | 173

Corollary (8.6.3). — *Considered as X-schemes,* $\widehat{E}_X$ *is canonically isomorphic to* $\operatorname{Spec}(\mathscr{S}_X^{\leqslant})$*, and* E_X *to* $\operatorname{Spec}(\mathscr{S}_X)$*.*

Proof. Since we know that the morphisms p_X and π_X are affine ((8.3.5) and (8.3.6)), it suffices (given (1.3.1)) to prove the corollary in the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathscr{S} = \widetilde{S}$. The first claim follows from the existence of the canonical isomorphisms (8.2.7.2) $(S_f^{\geqslant})_{f/1}^{\leqslant} \xrightarrow{\sim} S_f^{\leqslant}$ and from the fact that these isomorphisms are compatible with the map sending f to fg (where f and g are homogeneous in S_+). Similarly, applying (8.3.6.2) to π_X instead of π, we see that $\pi_X^{-1}(u^{-1}(X_f)) = \operatorname{Spec}((S_f^{\geqslant})_{f/1})$ for f homogeneous in S_+, and the second claim then follows from the existence of the canonical isomorphisms (8.2.7.2) $(S_f^{\geqslant})_{f/1} \xrightarrow{\sim} S_f$.

We can then say that $\widehat{C}_X$, considered as an X-scheme, is given by *gluing* the affine X-schemes $C_X = \operatorname{Spec}(\mathscr{S}_X^{\geqslant})$ and $\widehat{E}_X = \operatorname{Spec}(\mathscr{S}_X^{\leqslant})$ over X, where the intersection of the two affine X-schemes is the open subset $E_X = \operatorname{Spec}(\mathscr{S}_X)$. ☐

Corollary (8.6.4). — *Assume that* $\mathscr{O}_X(1)$ *is an invertible* $\mathscr{O}_X$*-module, and that* $\mathscr{S}_X$ *is isomorphic to* $\bigoplus_{n\in\mathbf{Z}}(\mathscr{O}_X(1))^{\otimes n}$ *(which will be the case, in particular, whenever* $\mathscr{S}$ *is generated by* $\mathscr{S}_1$ *((3.2.5) and (3.2.7))). Then the pointed projective cone* $\widehat{E}$ *can be identified with the rank-1 vector bundle* $\mathbf{V}(\mathscr{O}_X(-1))$ *on* X*, and the pointed affine cone* E *with the subprescheme of this vector bundle induced on the complement of the null section. With this identification, the canonical retraction* $\widehat{E} \to X$ *is identified with the structure morphism of the* X*-scheme* $\mathbf{V}(\mathscr{O}_X(-1))$*. Finally, there exists a canonical* Y*-morphism* $\mathbf{V}(\mathscr{O}_X(1)) \to C$*, whose restriction to the complement of the null section of* $\mathbf{V}(\mathscr{O}_X(1))$ *is an isomorphism from this complement to the pointed affine cone* E*.*

Proof. If we write $\mathscr{L} = \mathscr{O}_X(1)$, then $\mathscr{S}_X^{\geqslant}$ is identical to $\mathbf{S}_{\mathscr{O}_X}(\mathscr{L})$, and so $\widehat{E}_X$ is canonically identified with $\mathbf{V}(\mathscr{L}^{-1})$, by (8.6.3), and C_X with $\mathbf{V}(\mathscr{L})$. The morphism $\mathbf{V}(\mathscr{L}) \to C$ is the restriction of $\operatorname{Proj}(\widehat{\alpha})$, and the claims of the corollary are then particular cases of (8.6.2). ☐

We note that the inverse image under the morphism $\mathbf{V}(\mathscr{O}_X(1)) \to C$ of the underlying space of the vertex prescheme of C is the underlying space of the null section of $\mathbf{V}(\mathscr{O}_X(1))$ (8.5.5); but, in general, the corresponding subpreschemes of C and of $\mathbf{V}(\mathscr{O}_X(1))$ are not isomorphic. This problem will be studied below.

8.7. Blowing up based cones

(8.7.1). Under the conditions of (8.6.1), we have, writing $r = \operatorname{Proj}(\widehat{\alpha})$, a commutative diagram

$$\begin{tikzcd} X \arrow[r, "i_X \circ \varepsilon_X"] \arrow[d, "q"'] & \widehat{C}_X \arrow[d, "r"] \\ Y \arrow[r, "i \circ \varepsilon"] & \widehat{C} \end{tikzcd} \tag{8.7.1.1}$$

by (8.5.1.3) and (8.5.3.2); furthermore, the restriction of r to the complement $\widehat{C}_X - i_X(\varepsilon_X(X))$ of the null section is an *isomorphism* to the complement $\widehat{C} - i(\varepsilon(Y))$ of the null section, by (8.6.2). If we suppose, to simplify things, that Y is affine, that $\mathscr{S}$ is of finite type and generated by $\mathscr{S}_1$, and that X is projective over Y and $\widehat{C}_X$ projective over X (5.5.1), then $\widehat{C}_X$ is projective over Y (5.5.5, ii), and *a fortiori* over $\widehat{C}$ (5.5.5, v). We then have a projective Y-morphism $r : \widehat{C}_X \to \widehat{C}$ (whose restriction to C_X is a projective Y-morphism $C_X \to C$) that *contracts* X *to* Y (?) and that induces an *isomorphism* when we restrict to the *complements of* X *and* Y. We thus have a connection between C_X and C, analogous to that which exists between a blow-up prescheme and the original prescheme (8.1.3). We will effectively show that C_X can be identified with the homogeneous spectrum of a graded $\mathscr{O}_C$-algebra. II | 174

(8.7.2). Keeping the notation of (8.6.1), consider, for all $n \geqslant 0$, the quasi-coherent ideal

(8.7.2.1) $$\mathscr{S}_{[n]} = \bigoplus_{m \geqslant n} \mathscr{S}_m$$

of the graded $\mathscr{O}_Y$-algebra $\mathscr{S}$. It is clear that

(8.7.2.2) $$\mathscr{S}_{[0]} = \mathscr{S}, \qquad \mathscr{S}_{[n]} \subset \mathscr{S}_{[m]} \qquad \text{for } m \leqslant n$$

(8.7.2.3) $$\mathscr{S}_n \mathscr{S}_{[m]} \subset \mathscr{S}_{[m+n]}.$$

Consider the $\mathscr{O}_C$-module associated to $\mathscr{S}_{[n]}$, which is a quasi-coherent ideal of $\mathscr{O}_C = \widetilde{\mathscr{S}}$ (1.4.4)

(8.7.2.4) $$\mathscr{I}_n = (\mathscr{S}_{[n]})^{\sim}.$$

We thus deduce, from (8.7.2.2) and (8.7.2.3), using (1.4.4) and (1.4.8.1), the analogous formulas

(8.7.2.5) $$\mathscr{I}_{[0]} = \mathscr{O}_C, \qquad \mathscr{I}_{[n]} \subset \mathscr{I}_{[m]} \qquad \text{for } m \leqslant n$$

(8.7.2.6) $$\mathscr{I}_n \mathscr{I}_{[m]} \subset \mathscr{I}_{[m+n]}.$$

We are thus in the setting of (8.1.1), which leads us to introduce the quasi-coherent graded $\mathscr{O}_C$-algebra

(8.7.2.7) $$\mathscr{S}^{\natural} = \bigoplus_{n \geqslant 0} \mathscr{I}_n = \left(\bigoplus_{n \geqslant 0} \mathscr{S}_{[n]} \right)^{\sim}.$$

Proposition (8.7.3). — *There is a canonical C-isomorphism*

(8.7.3.1) $$h : C_X \xrightarrow{\sim} \operatorname{Proj}(\mathscr{S}^{\natural}).$$

PROOF. Suppose first of all that $Y = \operatorname{Spec}(A)$ is affine, so that $\mathscr{S} = \widetilde{S}$, with S a positively-graded A-algebra, and $C = \operatorname{Spec}(S)$. Definition (8.7.2.4) then shows, with the notation of (8.2.6), that $\mathscr{S}^{\natural} = (S^{\natural})^{\sim}$. To define (8.7.3.1), consider a homogeneous element $f \in S_d$ ($d > 0$) and the corresponding element $f^{\natural} \in S^{\natural}$ (8.2.6); the S-isomorphism in (8.2.7.3) then defines a C-isomorphism

(8.7.3.2) $$\operatorname{Spec}(S_f^{\geqslant}) \xrightarrow{\sim} \operatorname{Spec}(S^{\natural}_{(f^{\natural})}).$$

But with the notation of (8.6.2), if $v : C_X \to X$ is the structure morphism, then it follows from (8.6.2.1) that $v^{-1}(X_f) = \operatorname{Spec}(S_f^{\geqslant})$. We also have that $\operatorname{Spec}(S^{\natural}_{(f^{\natural})}) = D_+(f^{\natural})$, which means that (8.7.3.2) defines an isomorphism $v^{-1}(X_f) \to D_+(f^{\natural})$. Furthermore, if $g \in S_e$ ($e > 0$), then the diagram **II | 175**

$$\begin{tikzcd} v^{-1}(X_{fg}) \arrow[r, "\sim"] \arrow[d] & D_+(f^{\natural}g^{\natural}) \arrow[d] \\ v^{-1}(X_f) \arrow[r, "\sim"] & D_+(f^{\natural}) \end{tikzcd}$$

commutes, by definition of the isomorphism in (8.2.7.3). Finally, by definition, S_+ is generated by the homogeneous f, and so it follows from (8.2.10, iv) and from (2.3.14) that the $D_+(f^{\natural})$ form a cover of $\operatorname{Proj}(S^{\natural})$, and that the $v^{-1}(X_f)$ form a cover of C_X, since the X_f form a cover of X; in this case, we have thus defined the isomorphism (8.7.3.1).

To prove (8.7.3) in the general case, it suffices to show that, if U and U' are affine open subsets of Y, given by rings A and A' (respectively), and such that $U' \subset U$, then, setting $\mathscr{S}|U = \widetilde{S}$ and $\mathscr{S}|U' = \widetilde{S}'$, the diagram

(8.7.3.3) $$\begin{tikzcd} C_{U'} \arrow[r] \arrow[d] & \operatorname{Proj}(S'^{\natural}) \arrow[d] \\ C_U \arrow[r] & \operatorname{Proj}(S^{\natural}) \end{tikzcd}$$

commutes. But S is canonically identified with $S \otimes_A A'$, and so $S'^{\natural}$ is canonically identified with

$$S^{\natural} \otimes_S S' = S^{\natural} \otimes_A A';$$

thus $\mathrm{Proj}(S'^{\natural}) = \mathrm{Proj}(S^{\natural}) \times_U U'$ (2.8.10); similarly, if $X = \mathrm{Proj}(S)$ and $X' = \mathrm{Proj}(S')$, then $X' = X \times_U U'$ and $\mathscr{S}_{X'} = \mathscr{S}_X \otimes_{\mathscr{O}_U} U'$ (3.5.4), or, equivalently, $\mathscr{S}_{X'} = j^*(\mathscr{S}_X)$, where j is the projection $X' \to X$. We then (1.5.2) have that $C_{U'} = C_U \times_X X' = C_U \times_U U'$, and the commutativity of (8.7.3.3) is then immediate. □

Remark (8.7.4). — (i) The end of the proof of (8.7.3) can be immediately generalised in the following way. Let $g : Y' \to Y$ be a morphism, $\mathscr{S}' = g^*(\mathscr{S})$, and $X' = \mathrm{Proj}(\mathscr{S}')$; then we have a commutative diagram

(8.7.4.1)
$$\begin{tikzcd} C_{X'} \arrow[r] \arrow[d] & \mathrm{Proj}(\mathscr{S}'^{\natural}) \arrow[d] \\ C_X \arrow[r] & \mathrm{Proj}(\mathscr{S}^{\natural}) \end{tikzcd}$$

Now let $\phi : \mathscr{S}'' \to \mathscr{S}$ be a homomorphism of graded $\mathscr{O}_Y$-algebras such that, if we write $X'' = \mathrm{Proj}(\mathscr{S}'')$, then $u = \mathrm{Proj}(\phi) : X \to X''$ is everywhere defined; we also have a Y-morphism $v : C \to C''$ (with $C'' = \mathrm{Spec}(\mathscr{S}'')$) such that $\mathscr{A}(v) = \phi$, and, since ϕ is a homomorphism of graded algebras, ϕ induces a v-morphism of graded algebras $\psi : \mathscr{S}''^{\natural} \to \mathscr{S}^{\natural}$ (1.4.1). Furthermore, it follows from (8.2.10, iv) and from the hypothesis on ϕ that $\mathrm{Proj}(\psi)$ is everywhere defined. Finally, taking (3.5.6.1) into account, there is a canonical u-morphism $\mathscr{S}_{X''} \to \mathscr{S}_X$, whence (1.5.6) a morphism $w : C_{X''} \to C_X$. With this in mind, the diagram II | 176

(8.7.4.2)
$$\begin{tikzcd} C_{X''} \arrow[r, "\sim"] \arrow[d, "w"'] & \mathrm{Proj}(\mathscr{S}''^{\natural}) \arrow[d, "\mathrm{Proj}(\psi)"] \\ C_X \arrow[r, "\sim"] & \mathrm{Proj}(\mathscr{S}^{\natural}) \end{tikzcd}$$

is commutative, as we can immediately verify by restricting to the case where Y is affine.

(ii) Note that, by (8.7.2.5) and (8.7.2.6), we have $\mathscr{I}_1^m \subset \mathscr{I}_m \subset \mathscr{I}_1$ for all $m > 0$. But, by definition, $\mathscr{I}_1 = (\mathscr{S}_+)^\sim$, and so $\mathscr{I}_1$ defines the closed subprescheme $\varepsilon(Y)$ in C ((1.4.10) and (8.3.2)); we thus conclude that, for all $m > 0$, *the support of $\mathscr{O}_C/\mathscr{I}_m$ is contained in the underlying space of the vertex prescheme $\varepsilon(Y)$*; on the inverse image of the pointed affine cone E, the structure morphism $\mathrm{Proj}(\mathscr{S}^{\natural}) \to C$ thus restricts to an *isomorphism* (by (8.7.3) and (8.7.1)). Furthermore, by canonically identifying C with an open subset of $\widehat{C}$ (8.3.3), we can clearly extend the ideals $\mathscr{I}_m$ of $\mathscr{O}_C$ to ideals $\mathscr{J}_m$ of $\mathscr{O}_{\widehat{C}}$, by asking for it to agree with $\mathscr{O}_{\widehat{C}}$ on the open subset $\widehat{E}$ of $\widehat{C}$. If we define $\mathscr{T} = \bigoplus_{n \geqslant 0} \mathscr{J}_m$, which is a quasi-coherent graded $\mathscr{O}_{\widehat{C}}$-algebra, we can extend the isomorphism (8.7.3.1) to a $\widehat{C}$-isomorphism

(8.7.4.3)
$$\widehat{C}_X \xrightarrow{\sim} \mathrm{Proj}(\mathscr{T}).$$

Indeed, over $\widehat{E}$, it follows from the above that $\mathrm{Proj}(\mathscr{T})$ is canonically identified with $\widehat{E}$, and we thus define the isomorphism (8.7.4.3) over $\widehat{E}$ by asking for it to agree with the canonical isomorphism $\widehat{E}_X \to \widehat{E}$ (8.6.2); it is clear that this isomorphism and (8.7.3.1) then agree over $\widehat{E}$.

Corollary (8.7.5). — *Suppose that there exists some $n_0 > 0$ such that*

(8.7.5.1)
$$\mathscr{S}_{n+1} = \mathscr{S}_1 \mathscr{S}_n \qquad \textit{for } n \geqslant n_0.$$

Then the vertex subprescheme (?) of C_X (isomorphic to X) is the inverse image under the canonical morphism $r : C_X \to C$ of the vertex subprescheme of C (isomorphic to Y). Conversely, if this property is true, and if we further assume that Y is Noetherian and that $\mathscr{S}$ is of finite type, then there exists some $n_0 > 0$ such that (8.7.5.1) holds true.

PROOF. Since the first claim is local on Y, we can assume that $Y = \mathrm{Spec}(A)$ is affine, so that $\mathscr{S} = \widetilde{S}$, with S a positively-graded A-algebra. The claim then follows from (8.2.12), since $\mathrm{Proj}(S^{\natural} \otimes_S S_0) = C_X \times_C \varepsilon(Y)$ (by the identification in (8.7.3.1)), or, in other words, since this prescheme is the inverse image of $\varepsilon(Y)$ in C_X (**I**, 4.4.1). The converse also follows from (8.2.12) whenever Y is

Noetherian affine and S is of finite type. If Y is Noetherian (but not necessarily affine) and $\mathscr{S}$ is of finite type, then there exists a finite cover of Y by Noetherian affine open subsets U_i, and we then deduce from the above that, for all i, there exists an integer n_i such that $\mathscr{S}_{n+1}|U_i = (\mathscr{S}_1|U_i)(\mathscr{S}_n|U_i)$ for $n \geqslant n_i$; the largest of the n_i then ensures that (8.7.5.1) holds true. □ II | 177

(8.7.6). Now consider the C-prescheme Z given by *blowing up* the *vertex subprescheme* $\varepsilon(Y)$ in the affine cone C; by Definition (8.1.3), it is exactly the prescheme $\operatorname{Proj}(\bigoplus_{n\geqslant 0}\mathscr{S}_+^n)$; the canonical injection

(8.7.6.1) $$\iota : \bigoplus_{n\geqslant 0}\mathscr{S}_+^n \longrightarrow \mathscr{S}^\natural$$

defines (by the identification in (8.7.3)) a canonical dominant C-morphism

(8.7.6.2) $$G(\iota) \longrightarrow Z$$

where $G(\iota)$ is an open subset of C_X (3.5.1); note that it could be the case that $G(\iota) \neq C_X$, as shown by the example where $Y = \operatorname{Spec}(K)$, with K a field, and $\mathscr{S} = \widetilde{S}$, with $S = K[\mathbf{y}]$, where $\mathbf{y}$ is an indeterminate *of degree* 2; if R_n denotes the set $(S_+)^n$, considered as a subset of $S_{[n]} = S_n^\natural$, then $S_+^\natural$ is not the radical in $S_+^\natural$ of the ideal generated by the union of the R_n (cf. (2.3.14)).

Corollary (8.7.7). — *Assume that there exists some $n_0 > 0$ such that*

(8.7.7.1) $$\mathscr{S}_n = \mathscr{S}_1^n \quad \textit{for } n \geqslant n_0.$$

Then the canonical morphism (8.7.6.2) is everywhere defined, and is an isomorphism $C_X \xrightarrow{\sim} Z$. Conversely, if this property is true, and if we further assume that Y is Noetherian and that $\mathscr{S}$ is of finite type, then there exists some n_0 such that (8.7.7.1) holds true.

PROOF. The first claim is local on Y, and thus follows from (8.2.14); the converse follows similarly, arguing as in (8.7.5). □

Remark (8.7.8). — Since condition (8.7.7.1) implies (8.7.5.1), we see that, whenever it holds true, not only can C_X be identified with the prescheme given by blowing up the vertex (identified with Y) of the affine cone C, but also the vertex (identified with X) of C_X can be identified with the closed subprescheme given by the inverse image of the vertex Y of C. Furthermore, hypothesis (8.7.7.1) implies that, on $X = \operatorname{Proj}(\mathscr{S})$, the $\mathscr{O}_X$-modules $\mathscr{O}_X(n)$ are invertible ((3.2.5) and (3.2.9)), and that $\mathscr{O}_X(n) = \mathscr{L}^{\otimes n}$ with $\mathscr{L} = \mathscr{O}_X(1)$ ((3.2.7) and (3.2.9)); by Definition (8.6.1.1), C_X is thus the *vector bundle* $\mathbf{V}(\mathscr{L})$ on X, and its vertex is the *null section* of this vector bundle.

8.8. Ample sheaves and contractions

(8.8.1). Let Y be a prescheme, $f : X \to Y$ a *separated* and *quasi-compact* morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module that is *ample relative to* f. Consider the positively-graded $\mathscr{O}_Y$-algebra

(8.8.1.1) $$\mathscr{S} = \mathscr{O}_Y \oplus \bigoplus_{n\geqslant 1} f_*(\mathscr{L}^{\otimes n})$$

which is quasi-coherent (**I**, 9.2.2, a). There is a canonical homomorphisms of graded $\mathscr{O}_X$-algebras II | 178

(8.8.1.2) $$\tau : f^*(\mathscr{S}) \longrightarrow \bigoplus_{n\geqslant 0}\mathscr{L}^{\otimes n}$$

which, in degrees $\geqslant 1$, agrees with the canonical homomorphism $\sigma : f^*(f_*(\mathscr{L}^{\otimes n})) \to \mathscr{L}^{\otimes n}$ (**0**, 4.4.3), and is the identity in degree 0. The hypothesis that $\mathscr{L}$ is f-ample then implies ((4.6.3) and (3.6.1)) that the corresponding Y-morphism

(8.8.1.3) $$r = r_{\mathscr{L},\tau} : X \longrightarrow P = \operatorname{Proj}(\mathscr{S})$$

is everywhere defined and is a *dominant open immersion*, and that

(8.8.1.4) $$r^*(\mathscr{O}_P(n)) = \mathscr{L}^{\otimes n} \qquad \text{for all } n \in \mathbf{Z}.$$

Proposition (8.8.2). — *Let $C = \operatorname{Spec}(\mathscr{S})$ be the affine cone defined by $\mathscr{S}$; if $\mathscr{L}$ is f-ample, then there exists a canonical Y-morphism*

(8.8.2.1) $$g : V = \mathbf{V}(\mathscr{L}) \longrightarrow C$$

such that the diagram

(8.8.2.2)
$$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "f"'] & \mathbf{V}(\mathscr{L}) \arrow[r, "\pi"] \arrow[d, "g"] & X \arrow[d, "f"] \\ Y \arrow[r, "\varepsilon"] & C \arrow[r, "\psi"] & Y \end{tikzcd}$$

commutes, where ψ and π are the structure morphisms, and j and ε the canonical immersions sending X and Y (respectively) to the null section of $\mathbf{V}(\mathscr{L})$ and the vertex prescheme of C (respectively). Furthermore, the restriction of g to $\mathbf{V}(\mathscr{L}) - j(X)$ is an open immersion

(8.8.2.3) $$\mathbf{V}(\mathscr{L}) - j(X) \longrightarrow E = C - \varepsilon(Y)$$

into the pointed affine cone E corresponding to $\mathscr{S}$.

Proof. With the notation of (8.8.1), let $\mathscr{S}_P^{\geqslant} = \bigoplus_{n \geqslant 0} \mathscr{O}_P(n)$ and $C_p = \operatorname{Spec}(\mathscr{S}_P^{\geqslant})$. We know (8.6.2) that there is a canonical morphism $h = \operatorname{Spec}(\alpha) : C_p \to C$ such that the diagram

(8.8.2.4)
$$\begin{tikzcd} C_P \arrow[r] \arrow[d, "h"'] & P \arrow[d, "p"] \\ C \arrow[r, "\psi"] & Y \end{tikzcd}$$

commutes; furthermore, if $\varepsilon_P : P \to C_P$ is the canonical immersion, then the diagram

(8.8.2.5)
$$\begin{tikzcd} P \arrow[r, "p"] \arrow[d, "\varepsilon_P"'] & C_P \arrow[d, "h"] \\ Y \arrow[r, "\varepsilon"] & C \end{tikzcd}$$

commutes (8.7.1.1), and, finally, the restriction of H to the pointed affine cone E_P is an *isomorphism* $E_P \xrightarrow{\sim} E$ (8.6.2). It follows from (8.8.1.4) that

$$r^*(\mathscr{S}_P^{\geqslant}) = \mathbf{S}_{\mathscr{O}_X}(\mathscr{L})$$

and so we have a canonical P-morphism $q : \mathbf{V}(\mathscr{L}) \to C_P$, with the commutative diagram **II** | 179

(8.8.2.6)
$$\begin{tikzcd} \mathbf{V}(\mathscr{L}) \arrow[r, "\pi"] \arrow[d, "q"'] & X \arrow[d, "r"] \\ C_P \arrow[r] & P \end{tikzcd}$$

identifying $\mathbf{V}(\mathscr{L})$ with the product $C_P \times_P X$ (1.5.2); since r is an open immersion, so too is q (I, 4.3.2). Furthermore, the restriction of q to $\mathbf{V}(\mathscr{L}) - j(X)$ sends this prescheme to E_p, by (8.5.2), and the diagram

(8.8.2.7)
$$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "r"'] & \mathbf{V}(\mathscr{L}) \arrow[d, "q"] \\ P \arrow[r, "\varepsilon_P"] & C_P \end{tikzcd}$$

is commutative (since it is a particular case of (8.5.1.3)). The claims of (8.8.2) immediately follow from these facts, by taking g to be the composite morphism $h \circ q$. $\square$

Remark (8.8.3). — Assume further that Y is a *Noetherian* prescheme, and that f is a *proper* morphism. Since r is then *proper*(5.4.4), and thus closed, and since it is also a dominant open immersion, r is necessarily an *isomorphism* $X \xrightarrow{\sim} P$. Furthermore, we will see, in Chapter III (**III**, 2.3.5.1), that $\mathscr{S}$ is then necessarily an $\mathscr{O}_Y$-algebra *of finite type*. It then follows that $\mathscr{S}^\natural$ is an $\mathscr{S}_0^\natural$-algebra *of finite type*

((8.2.10, i) and (8.7.2.7)); since C_P is C-isomorphic to $\operatorname{Proj}(\mathscr{S}^\natural)$ (8.7.3), we see that the morphism $h : C_P \to C$ is *projective*; since the morphism r is an isomorphism, so too is $q : \mathbf{V}(\mathscr{L}) \to C_P$, and we thus conclude that the morphism $g : \mathbf{V}(\mathscr{L}) \to C$ is *projective*. Furthermore, since the restriction of h to E_P is an isomorphism to E, and since q is an isomorphism, the restriction (8.8.2.3) of g is an isomorphism $\mathbf{V}(\mathscr{L}) - j(X) \xrightarrow{\sim} E$.

If we further assume that L is *very ample* for f, then, as we will also see in Chapter III (**III**, 2.3.5.1), there exists some integer $n_0 > 0$ such that $\mathscr{S}_n = \mathscr{S}_1^n$ for $n \geqslant n_0$. We then conclude, by (8.7.7), that $\mathbf{V}(\mathscr{L})$ can be identified with the prescheme Z given by *blowing up the vertex prescheme* (identified with Y) *in the affine cone* C, and that the *null section* of $\mathbf{V}(\mathscr{L})$ (identified with Y) is the *inverse image* of the vertex subprescheme Y of C.

Some of the above results can in fact be proven even without the Noetherian hypothesis:

Corollary (8.8.4). — *Let Y be a prescheme (resp. a quasi-compact scheme), $f : X \to Y$ a proper morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module that is ample relative to f. Then the morphism in* (8.8.2.1) *is proper (resp. projective), and its restriction* (8.8.2.3) *is an isomorphism.*

PROOF. To prove that g is proper, we can restrict to the case where Y is affine, and it then suffices to consider the case where Y is a quasi-compact scheme. The same arguments as in (8.8.3) first of all show that r is an *isomorphism* $X \xrightarrow{\sim} P$; then q is also an isomorphism, and, since the restriction of h to E_P is an isomorphism $E_P \xrightarrow{\sim} E$, we have already seen that (8.8.2.3) is an isomorphism. It remains only to prove that g is *projective*.

Since f is of finite type, by hypothesis, we can apply (3.8.5) to the homomorphism τ from **II** | 180
(8.8.1.2): there is an integer $d > 0$ and a quasi-coherent $\mathscr{O}_Y$-submodule $\mathscr{E}$ of finite type of $\mathscr{S}_d$ such that, if $\mathscr{S}'$ is the $\mathscr{O}_Y$-subalgebra of $\mathscr{S}$ generated by $\mathscr{E}$, and $\tau' = \tau \circ q^*(\phi)$ (where ϕ is the canonical injection $\mathscr{S}' \to \mathscr{S}$), then $r' = r_{\mathscr{L},\tau'}$ is an immersion

$$X \longrightarrow P' = \operatorname{Proj}(\mathscr{S}').$$

Furthermore, since ϕ is injective, r' is also a *dominant immersion* (3.7.6); the same argument as for r then shows that r' is a *surjective closed immersion*; since r' factors as $X \xrightarrow{r} \operatorname{Proj}(\mathscr{S}) \xrightarrow{\Phi} \operatorname{Proj}(\mathscr{S}')$, where $\Phi = \operatorname{Proj}(\phi)$, we thus conclude that Φ is also a *surjective closed immersion*. But this implies that Φ is an *isomorphism*; we can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine, and $\mathscr{S} = \widetilde{S}$ and $\mathscr{S}' = \widetilde{S}'$, with S a graded A-algebra and S' a graded subalgebra of S. For every homogeneous element $t \in S'$, we have that $S'_{(t)}$ is a subring of $S_{(t)}$; if we return to the definition of $\operatorname{Proj}(\phi)$ (2.8.1), we see that it suffices to prove that, if B' is a subring of a ring B, and if the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(B')$ corresponding to the canonical injection $B' \to B$ is a closed immersion, then this morphism is necessarily an *isomorphism*; but this follows from (**I**, 4.2.3). Furthermore, $\Phi^*(\mathscr{O}_{P'}(n)) = \mathscr{O}_P(n)$ ((3.5.2, ii) and (3.5.4)), and so $r'^*(\mathscr{O}_{P'}(n))$ is isomorphic to $\mathscr{L}^{\otimes n}$ (4.6.3). Let $\mathscr{S}'' = \mathscr{S}'^{(d)}$, so that (3.1.8, i) X is canonically identified with $P'' = \operatorname{Proj}(\mathscr{S}'')$, and $\mathscr{L}'' = \mathscr{L}^{\otimes d}$ with $\mathscr{O}_{P''}(1)$ (3.2.9, ii).

Now, if $C'' = \operatorname{Spec}(\mathscr{S}'')$, then $\mathscr{S}_{P''}^{\geqslant} = \bigoplus_{n \geqslant 0} \mathscr{O}_{P''}(n)$ can be identified with $\bigoplus_{n \geqslant 0} \mathscr{L}''^{\otimes n}$, and thus $C_{P''} = \operatorname{Spec}(\mathscr{S}_{P''}^{\geqslant})$ with $\mathbf{V}(\mathscr{L}'')$; we also know (8.7.3) that $C_{P''}$ is C''-isomorphic to $\operatorname{Proj}(\mathscr{S}''^\natural)$; by the definition of $\mathscr{S}''$, we know that $\mathscr{S}''^\natural$ is generated by $\mathscr{S}_1''^\natural$, and that $\mathscr{S}_1''^\natural$ is of finite type over $\mathscr{S}_0''^\natural = \mathscr{S}''$ ((8.2.10, i and iii)), and so $\operatorname{Proj}(\mathscr{S}''^\natural)$ is *projective* over C'' (5.5.1). Consider the diagram

$$\begin{tikzcd}
\mathbf{V}(\mathscr{L}) \arrow[r, "g"] \arrow[d, "u"'] & \operatorname{Spec}(\mathscr{S}) = C \arrow[d, "v"] \\
\mathbf{V}(\mathscr{L}'') \arrow[r, "g''"] & \operatorname{Spec}(\mathscr{S}'') = C''
\end{tikzcd} \tag{8.8.4.1}$$

where g and g'' correspond, by (1.5.6), to the canonical j-morphisms

$$\mathscr{S} \longrightarrow \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n} \quad \text{and} \quad \mathscr{S}'' \longrightarrow \bigoplus_{n \geqslant 0} \mathscr{L}''^{\otimes n}$$

(3.3.2.3) (see (8.8.5) below), and v and u to the inclusion morphisms $\mathscr{S}'' \to \mathscr{S}$ and $\bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes nd} \to \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$ (respectively); it is immediate (3.3.2) that this diagram is commutative. We have just seen that g'' is a projective morphism; we also know that u is a *finite* morphism. Since the question is local

on X, we can assume that X is affine of ring A, and that $\mathscr{L} = \mathscr{O}_X$; everything then reduces to noting that the ring $A[T]$ is a module of finite type over its subring $A[T^d]$ (with T an indeterminate). Since Y is a quasi-compact scheme, and since C'' is affine over Y, we know that C'' is also a quasi-compact scheme, and so $g'' \circ u$ is a projective morphism (5.5.5, ii); by commutativity of (8.8.4.1), $v \circ g$ is also **II** | 181
projective, and, since v is affine, thus separated, we finally conclude that g is projective (5.5.5, v). □

(8.8.5). Consider again the situation in (8.8.1). We will see that the morphism $g : \mathbf{V}(\mathscr{L}) \to C$ can be also be defined in a way that works for any invertible (but not necessarily ample) $\mathscr{O}_X$-module $\mathscr{L}$. For this, consider the f-morphism

$$\tau^\flat : \mathscr{S} \longrightarrow \bigoplus_{n\geqslant 0} \mathscr{L}^{\otimes n} \tag{8.8.5.1}$$

corresponding to the morphism τ from (8.8.1.2). This induces (1.5.6) a morphism $g' : V \to C$ such that, if $\pi : V \to X$ and $\psi : C \to Y$ are the structure morphisms, the diagrams

$$\begin{tikzcd} X \arrow[d, "f"'] & V \arrow[l, "\pi"'] \arrow[d, "g'"] \\ Y & C \arrow[l, "\psi"'] \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "j"] \arrow[d, "f"'] & V \arrow[d, "g'"] \\ Y \arrow[r, "\varepsilon"] & C \end{tikzcd} \tag{8.8.5.2}$$

commute ((8.5.1.2) and (8.5.1.3)). We will show that (if we assume that $\mathscr{L}$ is f-ample) *the morphisms g and g' are identical.*

Since the questions is local on Y, we can assume that $Y = \operatorname{Spec}(A)$ is affine, and (by (8.8.1.3)) identify X with an open subset of $P = \operatorname{Proj}(S)$, where $S = A \oplus \bigoplus_{n\geqslant 0} \Gamma(X, \mathscr{L}^{\otimes n})$; we then deduce, by (8.8.1.4), that $\Gamma(X, \mathscr{O}_P(n)) = \Gamma(X, \mathscr{L}^{\otimes n})$ for all $n \in \mathbf{Z}$. Taking into account the definition of $h = \operatorname{Spec}(\alpha)$, where α is the canonical p-morphism $\widetilde{S} \to \mathscr{S}_P^{\geqslant}$ (8.6.1.2), we have to show that the restriction to X of $\alpha^\sharp : p^*(\widetilde{S}) \to \mathscr{S}_P^{\geqslant}$ is identical to τ. Taking (**0**, 4.4.3) into account, it suffices to show that, if we compose the canonical homomorphism $\alpha_n : S_n \to \Gamma(P, \mathscr{O}_P(n))$ with the restriction homomorphism $\Gamma(P, \mathscr{O}_P(n)) \to \Gamma(X, \mathscr{O}_P(n)) = \Gamma(X, \mathscr{L}^{\otimes n})$, then we obtain the identity, for all $n > 0$; but this follows immediately from the definition of the algebra S and of α_n (2.6.2).

Proposition (8.8.6). — *Assume (with the notation of (8.8.5)) that, if we write $f = (f_0, \lambda)$, then the homomorphism $\lambda : \mathscr{O}_Y \to j_*(\mathscr{O}_X)$ is bijective; then:*

(i) *if we write $g = (g_0, \mu)$, then $\mu : \mathscr{O}_C \to g_*(\mathscr{O}_V)$ is an isomorphism; and*
(ii) *if X is integral (resp. locally integral and normal), then C is integral (resp. normal).*

Proof. Indeed, the f-morphism $\tau^\flat$ is then an *isomorphism*

$$\tau^\flat : \mathscr{S} = \psi_*(\mathscr{O}_C) \longrightarrow f_*(\pi_*(\mathscr{O}_V)) = \psi_*(g_*(\mathscr{O}_V))$$

and the Y-morphism g can be considered as that for which the homomorphism $\mathscr{A}(g)$ (1.1.2) is equal to $\tau^\flat$. To see that μ is an isomorphism of $\mathscr{O}_C$-modules, it suffices (1.4.2) to see that $\mathscr{A}(\mu) : \psi_*(\mathscr{O}_C) \to \psi_*(g_*(\mathscr{O}_V))$ is an isomorphism. But, by Definition (1.1.2), we have that $\mathscr{A}(\mu) = \mathscr{A}(g)$, whence the conclusion of (i).

To prove (ii), we can restrict to the case where Y is affine, and so $\mathscr{S} = \widetilde{S}$, with $S = \bigoplus_{n\geqslant 0} \Gamma(X, \mathscr{L}^{\otimes n})$; **II** | 182
the hypothesis that X is integral implies that the ring S is integral (**I**, 7.4.4), and thus so too is C (**I**, 5.1.4). To show that C is normal, we will use the following lemma:

Lemma (8.8.6.1). — *Let Z be a normal integral prescheme. Then the ring $\Gamma(Z, \mathscr{O}_Z)$ is integral and integrally closed.*

Proof. It follows from (**I**, 8.2.1.1) that $\Gamma(Z, \mathscr{O}_Z)$ is the intersection, in the field of rational functions $R(Z)$, of the integrally closed rings $\mathscr{O}_z$ over all $z \in Z$. □

With this in mind, we first show that V is *locally integral* and *normal*; for this, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, with ring A integral and integrally closed (6.3.8), and where $\mathscr{L} = \mathscr{O}_X$. Since then $V = \operatorname{Spec}(A[T])$, and $A[T]$ is integral and integrally closed [**Jaf60**, p. 99], this proves our claim. For every affine open subset U of C, $g^{-1}(U)$ is quasi-compact, since the morphism g is quasi-compact; since V is locally integral, the connected components of $g^{-1}(U)$ are open integral preschemes in $g^{-1}(U)$, and thus finite in number, and, since V is normal, these preschemes are also normal (6.3.8). Then $\Gamma(U, \mathscr{O}_C)$, which is equal to $\Gamma(g^{-1}(U), \mathscr{O}_V)$, by (i), is the direct sum (?) of finitely-many integral and integrally closed rings (8.8.6.1), which proves that C is normal (6.3.4). □

8.9. Grauert's ampleness criterion: statement We intend to show that the properties proven in (8.8.2) *characterise* f-ample $\mathscr{O}_X$-modules, and, more precisely, to prove the following criterion:

Theorem (8.9.1). — (Grauert's criterion). *Let Y be a prescheme, $p : X \to Y$ a separated and quasi-compact morphism, and $\mathscr{L}$ an invertible $\mathscr{O}_X$-module. For $\mathscr{L}$ to be ample relative to p, it is necessary and sufficient for there to exist a Y-prescheme C, a Y-section $\varepsilon : Y \to C$ of C, and a Y-morphism $q : \mathbf{V}(\mathscr{L}) \to C$, satisfying the following properties:*

(i) *the diagram*

(8.9.1.1)
$$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "p"'] & \mathbf{V}(\mathscr{L}) \arrow[d, "q"] \\ Y \arrow[r, "\varepsilon"] & C \end{tikzcd}$$

commutes, where j is the null section of the vector bundle $\mathbf{V}(\mathscr{L})$; and

(ii) *the restriction of q to $\mathbf{V}(\mathscr{L}) - j(X)$ is a quasi-compact open immersion*
$$\mathbf{V}(\mathscr{L}) - j(X) \longrightarrow X$$
whose image does not intersect $\varepsilon(Y)$.

Note that, if C is *separated* over Y, we can, in condition (ii), remove the hypothesis that the open immersion is quasi-compact; to see that this property (of quasi-compactness) is in fact a consequence of the other conditions, we can restrict to the case where Y is affine, and the claim then follows from (**I**, 5.5.1)i and (**I**, 5.5.10). We can also remove the same hypothesis if we assume that X is Noetherian, **II** | 183
since then V is also Noetherian, and the claim follows from (**I**, 6.3.5).

Corollary (8.9.2). — *If the morphism $p : X \to Y$ is proper, then we can, in the statement of Theorem (8.9.1), assume that q is proper, and replace "open immersion" by "isomorphism".*

In a more suggestive manner, we can say (whenever $p : X \to Y$ is proper) that *$\mathscr{L}$ is ample relative to p if and only if we can* "contract" *the null section of the vector bundle $\mathbf{V}(\mathscr{L})$ to the base prescheme Y.* An important particular case is that where Y is the spectrum of a field, and where the operation of "contraction" consists of contract the null section $\mathbf{V}(\mathscr{L})$ *to a single point*.

(8.9.3). The necessity of the conditions in Theorem (8.9.1) and Corollary (8.9.2) follow immediately from (8.8.2) and (8.8.4).

To show that the conditions of (8.9.1) suffices, consider a slightly more general situation. For this, let (with the notation of (8.8.2))
$$\mathscr{S}' = \bigoplus_{n \geqslant 0} \mathscr{L}^{\otimes n}$$
and
$$V = \mathbf{V}(\mathscr{L}) = \operatorname{Spec}(\mathscr{S}').$$

The closed subprescheme $j(X)$, null section of $\mathbf{V}(\mathscr{L})$, is defined by the quasi-coherent sheaf of ideals $\mathscr{J} = (\mathscr{S}'_+)^\sim$ of $\mathscr{O}_V$ (1.4.10). This $\mathscr{O}_V$-module is *invertible*, since this property is local on X, and this reduces to remarking that the ideal $TA[T]$ in a ring of polynomials $A[T]$ is a free cyclic $A[T]$-module. Furthermore, it is immediate (again, because the question is local on X) that
$$\mathscr{L} = j^*(\mathscr{J})$$
and
$$j_*(\mathscr{L}) = \mathscr{J} / \mathscr{J}^2.$$

Now, if
$$\pi : \mathbf{V}(\mathscr{L}) \longrightarrow X$$
is the structure morphism, then $\pi_*(\mathscr{J}) = \mathscr{S}'_+$ and $\pi_*(\mathscr{J}/\mathscr{J}^2) = \mathscr{L}$; there are thus canonical homomorphisms $\mathscr{L} \to \pi_*(\mathscr{J}) \to \mathscr{L}$, the first being the canonical injection $\mathscr{L} \to \mathscr{S}'_+$, and the second

the canonical projection from $\mathscr{S}'_+$ to $\mathscr{S}'_1 = \mathscr{L}$, and their composition being the identity. We can also canonically embed $\pi_*(\mathscr{J}) = \mathscr{S}'_+ = \bigoplus_{n\geqslant 1}\mathscr{L}^{\otimes n}$ into the *product* $\prod_{n\geqslant 1}\mathscr{L}^{\otimes n} = \varprojlim_n \pi_*(\mathscr{J}/\mathscr{J}^{n+1})$ (since $\pi_*(\mathscr{J}/\mathscr{J}^{n+1}) = \mathscr{L}\oplus\mathscr{L}^{\otimes 2}\oplus\ldots\oplus\mathscr{L}^{\otimes n}$), and we thus have canonical homomorphisms

(8.9.3.1) $$\mathscr{L} \longrightarrow \varprojlim \pi_*(\mathscr{J}/\mathscr{J}^{n+1}) \longrightarrow \mathscr{L}$$

whose composition is the identity.

With this in mind, the generalisation of (8.9.1) that we are going to prove is the following:

Proposition (8.9.4). — *Let Y be a prescheme, V a Y-prescheme, and X a closed subprescheme of V defined by an ideal $\mathscr{J}$ of $\mathscr{O}_V$, which is an* invertible *$\mathscr{O}_V$-module; if $j : X \to V$ is the canonical injection, then let* **II | 184** *$\mathscr{L} = j^*(\mathscr{J}) = \mathscr{J}\otimes_{\mathscr{O}_V}\mathscr{O}_X$, so that $j_*(\mathscr{L}) = \mathscr{J}/\mathscr{J}^2$. Assume that the structure morphism $p : X \to Y$ is separated and quasi-compact, and that the following conditions are satisfied:*

(i) *there exists a Y-morphism $\pi : V \to X$ of finite type such that $\pi\circ j = 1_X$, and so $\pi_*(\mathscr{J}/\mathscr{J}^2) = \mathscr{L}$;*
(ii) *there exists a homomorphism of $\mathscr{O}_X$-modules $\phi : \mathscr{L} \to \varprojlim \pi_*(\mathscr{J}/\mathscr{J}^{n+1})$ such that the composition*
$$\mathscr{L} \xrightarrow{\phi} \varprojlim \pi_*(\mathscr{J}/\mathscr{J}^{n+1}) \xrightarrow{\alpha} \pi_*(\mathscr{J}/\mathscr{J}^2) = \mathscr{L}$$
(where α is the canonical homomorphism) is the identity;
(iii) *there exists a Y-prescheme C, a Y-section ε of C, and a Y-morphism $q : V \to C$ such that the diagram*

(8.9.4.1) $$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "p"'] & V \arrow[d, "q"] \\ Y \arrow[r, "\varepsilon"'] & C \end{tikzcd}$$

commutes; and
(iv) *the restriction of q to $W = V - j(X)$ is a quasi-compact open immersion into C, whose image does not intersect $\varepsilon(Y)$.*

Then $\mathscr{L}$ is ample relative to p.

8.10. Grauert's ampleness criterion: proof

Lemma (8.10.1). — *Let $\pi : V \to X$ be a morphism, $j : X \to V$ an X-section of V that is also a closed immersion, and $\mathscr{J}$ a quasi-coherent ideal of $\mathscr{O}_V$ that defines the closed subprescheme of V associated to j. Then the following all hold true.*

(i) *For all $n\geqslant 0$, $\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$ and $\pi_*(\mathscr{J}/\mathscr{J}^{n+1})$ are quasi-coherent $\mathscr{O}_X$-modules, and $\pi_*(\mathscr{O}_V/\mathscr{J}) = \mathscr{O}_X$ and $\pi_*(\mathscr{J}/\mathscr{J}^2) = j^*(\mathscr{J})$.*
(ii) *If $X = \{\xi\} = \operatorname{Spec}(k)$, where k is a field, then $\varprojlim \pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$ is isomorphic to the separated completion of the local ring $\mathscr{O}_{j(\xi)}$ for the $\mathfrak{m}_{j(\xi)}$-preadic topology.*
(iii) *Assume that $\mathscr{J}$ is an invertible $\mathscr{O}_V$-module (which implies that*
$$\mathscr{L} = j^*(\mathscr{J}) = \pi_*(\mathscr{J}/\mathscr{J}^2)$$
is an invertible $\mathscr{O}_X$-module), and that there exists a homomorphism $\phi : \mathscr{L} \to \varprojlim \pi_*(\mathscr{J}/\mathscr{J}^{n+1})$ such that the composition $\mathscr{L} \xrightarrow{\phi} \varprojlim \pi_*(\mathscr{J}/\mathscr{J}^{n+1}) \xrightarrow{\alpha} \pi_*(\mathscr{J}/\mathscr{J}^2)$ (where α is the canonical homomorphism) is the identity. If we write $\mathscr{S} = \bigoplus_{n\geqslant 0}\mathscr{L}^{\otimes n}$, then ϕ canonically induces an isomorphism of $\mathscr{O}_X$-algebras from the completion $\widehat{\mathscr{S}}$ of $\mathscr{S}$ relative to its canonical filtration (the completion being isomorphic to the product $\prod_{n\geqslant 0}\mathscr{L}^{\otimes n}$) to $\varprojlim \pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$.

PROOF. Note first of all that the support of the $\mathscr{O}_V$-module $\mathscr{O}_V/\mathscr{J}^{n+1}$ is $j(X)$, and the support of $\mathscr{J}/\mathscr{J}^{n+1}$ is contained in $j(X)$. In the case of (ii), $j(X)$ is a closed point $j(\xi)$ of V, and, by definition, **II | 185** $\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$ is the fibre of $\mathscr{O}_V/\mathscr{J}^{n+1}$ at the point $j(\xi)$, or, equivalently, setting $C = \mathscr{O}_{j(\xi)}$, and denoting by $\mathfrak{m}$ the maximal ideal of C, the C-module $C/\mathfrak{m}^{n+1}$; claim (ii) is then evident.

To prove (i), note that the question is local on X; we can thus restrict to the case where X is affine. Let U be an affine open subset of V; then $j(X)\cap U$ is an affine open subset of $j(X)$, so $U_0 = \pi(j(X) \cap U)$, which is isomorphic to it, is an affine open subset of X; for every affine open subset $W_0 \subset U_0$ in X, $W = \pi^{-1}(W_0) \cap U$ is an affine open subset of V, since X is a scheme (**I**, 5.5.10); in particular, $U' = U \cap \pi^{-1}(U_0)$ is an affine open subset of V, and clearly $\pi(U') = U_0$ and $j(U_0) = j(X) \cap U$. Then, by definition, $\Gamma(W_0, \pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})) = \Gamma(\pi^{-1}(W_0), \mathscr{O}_V/\mathscr{J}^{n+1})$; but since every point of $\pi^{-1}(W_0)$ not belonging to $j(W_0)$ has an open neighbourhood in $\pi^{-1}(W_0)$ not intersecting $j(X)$, and in which $\mathscr{O}_V/\mathscr{J}^{n+1}$ is thus zero, it is clear that the sections of $\mathscr{O}_V/\mathscr{J}^{n+1}$ over $\pi^{-1}(W_0)$ and over W are in bijective correspondence. In other words, if π' is the restriction of π to U', then the $(\mathscr{O}_X|U_0)$-modules $\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})|U_0$ and $\pi'_*((\mathscr{O}_V/\mathscr{J}^{n+1})|U')$ are identical. Since U' and U_0 are affine, and since the U_0 cover X, we thus conclude (**I**, 1.6.3) that $\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$ is quasi-coherent, and the proof is identical for $\pi_*(\mathscr{J}/\mathscr{J}^{n+1})$.

Finally, to prove (iii), note that $\mathscr{S}$ is exactly $\mathbf{S}_{\mathscr{O}_X}(\mathscr{L})$; so ϕ canonically induces a homomorphism of $\mathscr{O}_X$-algebras $\psi : \mathscr{S} \to \varprojlim \pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$ (1.7.4); furthermore, this homomorphism sends $\mathscr{L}^{\otimes n}$ to $\varprojlim_m \pi_*(\mathscr{J}^n/\mathscr{J}^{n+1})$, and is thus continuous for the topologies considered, and indeed then extends to a homomorphism $\widehat{\psi} : \widehat{\mathscr{S}} \to \varprojlim \pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$. To see that this is indeed an isomorphism, we can, as in the proof of (i), restrict to the case where $X = \operatorname{Spec}(A)$ and $V = \operatorname{Spec}(B)$ are affine, with $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of B; there is an injection $A \to B$ corresponding to π that identifies A with a subring of B that is *complementary* to B, and $\mathscr{L}$ (resp. $\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1})$) is the quasi-coherent $\mathscr{O}_X$-module associated to the A-module $L = \mathfrak{J}/\mathfrak{J}^2$ (resp. $B/\mathfrak{J}^{n+1}$). Since $\mathscr{J}$ is an *invertible* $\mathscr{O}_V$-module, we can further assume that $\mathfrak{J} = Bt$, where t is not a zero divisor in B. From the fact that $B = A \oplus Bt$, we deduce that, for all $n > 0$,

$$B = A \oplus At \oplus At^2 \oplus \ldots \oplus At^n \oplus Bt^{n+1}$$

and so there exists a canonical A-isomorphism from the ring of formal series $A[[T]]$ to $C = \varprojlim B/\mathfrak{J}^{n+1}$ that sends T to t. We also have that $L = A\bar{t}$, where $\bar{t}$ is the class of t modulo Bt^2, and the homomorphism ϕ sends, by hypothesis, $\bar{t}$ to an element $t' \in C$ that is congruent to t modulo Ct^2. We thus deduce, by induction on n, that

$$A \oplus At' \oplus \ldots \oplus At'^n \oplus Ct^{n+1} = A \oplus At \oplus \ldots \oplus At^n \oplus Ct^{n+1}$$

which proves that the homomorphism $\widehat{\psi}$ does indeed correspond to an isomorphism from $\prod_{n \geqslant 0} L^{\otimes n}$ to C. □

Lemma (8.10.2). — *Under the hypotheses of Lemma (8.10.1), let $g : X' \to X$ be a morphism, write $V' = V \times_X X'$, and let $\pi' : V' \to X'$ and $g : V' \to V$ be the canonical projections, so that we have the commutative diagram* II | 186

$$\begin{tikzcd} V & V' \arrow[l, "g'"'] \arrow[d, "\pi'"] \\ X \arrow[u, leftarrow, "\pi"'] & X' \arrow[l, "g"] \end{tikzcd}$$

Then $j' = j \times 1_{X'}$ is an X'-section of V' that is also a closed immersion, and $\mathscr{J}' = g'^*(\mathscr{J})\mathscr{O}_{V'}$ is the quasi-coherent ideal of $\mathscr{O}_{V'}$ that defines the closed subprescheme of V' associated to j'. Furthermore, $\pi'_*(\mathscr{O}_{V'}/\mathscr{J}'n+1) = g^*(\pi_*(\mathscr{O}_V/\mathscr{J}^{n+1}))$. Finally, $\mathscr{J}'$ is an $\mathscr{O}_{V'}$-module that is canonically isomorphic to $g'^*(\mathscr{J})$, and is, in particular, invertible if $\mathscr{J}$ is an invertible $\mathscr{O}_V$-module.

Proof. The fact that j' is a closed immersion follows from (**I**, 4.3.1), and it is an X'-section of V' by functoriality of extension of the base prescheme. Furthermore, if Z (resp. Z') is the closed subprescheme of V (resp. V') associated to j (resp. j'), then $Z' = g'^{-1}(Z)$ (**I**, 4.3.1), and the second claim then follows from (**I**, 4.4.5). To prove the other claims, we see, as in (8.10.1), that we can restrict to the case where X, V, and X' (and thus also V') are affine; we keep the notation from the proof of (8.10.1), and let $X' = \operatorname{Spec}(A')$. Then $V' = \operatorname{Spec}(B')$, where $B' = B \otimes_A A'$, and $\mathscr{J}' = \widetilde{\mathfrak{J}''}$, where $\mathfrak{J}' = \operatorname{Im}(\mathfrak{J} \otimes_A A')$. Then $B'/\mathfrak{J}'^{n+1} = (B/\mathfrak{J}^{n+1}) \otimes_A A'$; furthermore, since $\mathfrak{J}$ is a direct factor (as an A-module) of B, $\mathfrak{J} \otimes_A A'$ is a direct factor (as an A'-module) of B', and is thus canonically identified with $\mathfrak{J}'$. □

Corollary (8.10.3). — *Assume that the hypotheses of Lemma (8.10.1) are satisfied, and assume further that π is of finite type, and that $\mathscr{J}$ is an invertible $\mathscr{O}_V$-module. Then, for all $x \in X$, the local ring at the point $j(x)$ of the fibre $\pi^{-1}(x)$ is a regular (thus integral) ring of dimension 1, whose completion is isomorphic to the formal series ring $k(x)[[T]]$ (where T is an indeterminate); furthermore, there exists exactly one irreducible component of $\pi^{-1}(x)$ that contains $j(x)$.*

PROOF. Since $\pi^{-1}(x) = V \times_X \operatorname{Spec}(k(x))$, we are led, by (8.10.2), to the case where X is the spectrum of a field K. Since π is of finite type (**I**, 6.4.3, iv), $\mathscr{O}_{j(x)}$ is a Noetherian local ring, and thus separated for the $\mathfrak{m}_{j(x)}$-preadic topology (**0**, 7.3.5); it follows from (8.10.1, ii and iii) that the completion of this ring is isomorphic to $K[[T]]$, and so $\mathscr{O}_{j(x)}$ is regular and of dimension 1 ([**CC**, p. 17-01, th. 1]); finally, since $\mathscr{O}_{j(x)}$ is integral, $j(x)$ belongs to exactly one of the (finitely many) irreducible components of V (**I**, 5.1.4). □

Corollary (8.10.4). — *Suppose that the hypotheses of Lemma (8.10.1) are satisfied, and assume further that $\mathscr{J}$ is an invertible $\mathscr{O}_V$-module. Let $W = V - j(X)$; for every quasi-coherent ideal $\mathscr{K}$ of $\mathscr{O}_X$, let $\mathscr{K}_V = \pi^*(\mathscr{K})\mathscr{O}_V$ and $\mathscr{K}_W = \mathscr{K}_V|W$. Then $\mathscr{K}_V$ is the largest quasi-coherent ideal of $\mathscr{O}_V$ whose restriction to W is $\mathscr{K}_W$.*

PROOF. Indeed, we see as in (8.10.1) that the question is local on X and V; we can thus reuse the notation from the proof of (8.10.1), with $\mathfrak{J} = Bt$, where t is not a zero divisor in B. Furthermore, we have $W = \operatorname{Spec}(B_t)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$, where $\mathfrak{K}$ is an ideal of A; whence $\pi^*(\mathscr{K})\mathscr{O}_V = (\mathfrak{K}.B)^\sim$ (**I**, 1.6.9), $\mathscr{K}_W = (\mathfrak{K}.B_t)^\sim$, and the largest ideal of B whose canonical image in B_t is $\mathfrak{K}.B_t$ is the inverse image of $\mathfrak{K}.B_t$, that is, the set of $s \in B$ such that, for some integer $n > 0$, we have $t^n s \in \mathfrak{K}.B$. We have to show that this last relation implies that $s \in \mathfrak{K}.B$, or again that the canonical image of t is not a zero divisor in $B/\mathfrak{K}B = (A/\mathfrak{K}) \otimes_A B$, which follows from (8.10.2) applied to $X' = \operatorname{Spec}(A/\mathfrak{K})$. □ **II** | 187

Corollary (8.10.5). — *Suppose that the hypotheses of (8.10.3) are satisfied; let $W = V - j(X)$, x be a point of X, $\mathscr{K}$ a quasi-coherent ideal of $\mathscr{O}_X$, and z the generic point of the irreducible component of $\pi^{-1}(x)$ that contains $j(x)$ (8.10.3).*

(i) *Let g be a section of $\mathscr{O}_V$ over V such that $g|W$ is a section of $\mathscr{K}_W$ over W (using the notation from (8.10.4)). Then g is a section of $\mathscr{K}_V$; if further $g(z) \neq 0$, and if, for every integer $m > 0$, we denote by g_m^x the germ at the point x of the canonical image g_m of g in $\Gamma(X, \pi_*(\mathscr{O}_V/\mathscr{J}^{m+1}))$, then there exists an integer $m > 0$ such that the image of g_m^x in*

$$(\pi_*(\mathscr{O}_V/\mathscr{J}^{m+1}))_x \otimes_{\mathscr{O}_x} k(x)$$

is $\neq 0$.

(ii) *Suppose further that the conditions of (8.10.1, iii) are fulfilled. Then, if there exists a section g of $\mathscr{K}_V$ over V such that $g(z) \neq 0$, then there exists an integer $n \geqslant 0$ and a section f of $\mathscr{K}.\mathscr{L}^{\otimes n} = \mathscr{K} \otimes \mathscr{L}^{\otimes n} \subset \mathscr{L}^{\otimes n}$ such that $f(x) \neq 0$. If g is a section of $\mathscr{J}$, we can take $n > 0$.*

PROOF.

(i) Since the ideal of $\mathscr{O}_W$ generated by $g|W$ is contained in $\mathscr{K}_W$ by hypothesis, the ideal of $\mathscr{O}_V$ generated by g is contained in $\mathscr{K}_V$ by (8.10.4), or, in other words, g is a section of $\mathscr{K}_V$. To prove the second claim of (i), we can again assume that X and V are affine, and reuse the notation from (8.10.1); the fibre $\pi^{-1}(x)$ is then affine of ring $B' = B \otimes_A k(x)$, and there exists in B' an element t' which is not a zero divisor and is such that $B' = k(x) \oplus B't'$. Since $j(x)$ is a specialisation of z and since $g(z) \neq 0$, we necessarily have that $g_{(j)x} \neq 0$. But $\mathscr{O}_{j(x)}$ is a separated local ring (8.10.3), and thus embeds into its completion, and the image of g in this completion is thus not null. But this completion is isomorphic to $\varprojlim_n (B'/B't'^{n+1})$ (8.10.3); if $g' = g \otimes 1 \in B'$, there then exists an integer m such that $g' \notin B't'^{m+1}$, or, again, the image g'_m of g' in $B'/B't'^{m+1}$ is not null. But since g'_m is exactly the image of g_m^x, our claim is proved.

(ii) By (8.10.1, iii), $\pi_*(\mathscr{O}_V/\mathscr{J}^{m+1})$ is isomorphic to the direct sum of the $\mathscr{L}^{\otimes k}$ for $0 \leqslant k \leqslant m$; we denote by f_k the section of $\mathscr{L}^{\otimes k}$ over X that is the component of the element of $\bigoplus_{k=0}^m \Gamma(X, \mathscr{L}^{\otimes k})$ which corresponds to g_m by this isomorphism. Choosing m as in (i), there is thus an index k such that $f_k(x) \neq 0$, by (i). To see that f_k is a section of $\mathscr{K}\mathscr{L}^{\otimes k}$, it suffices to consider, as above, the case where X and V are affine, and this follows immediately from the fact that $g \in \mathfrak{K}.B$ (with the notation from (8.10.4)). The final claim follows from the fact that the hypothesis $g \in \Gamma(V, \mathscr{J})$ implies that $f_0 = 0$.

□

(8.10.6). *Proof of* (8.9.4). The question is local on Y (4.6.4); since ε is a Y-section, we can thus replace C by an affine open neighbourhood U of a point of $\varepsilon(Y)$ such that $\varepsilon(Y) \cap U$ is closed in U. In other words, we can assume that C is affine, and that Y is a closed subprescheme of C (and thus also affine) defined by a quasi-coherent sheaf $\mathscr{I}$ of ideals of $\mathscr{O}_C$. Since p is separated and quasi-compact, **II** | 188
X is thus a quasi-compact scheme, and we are reduced to proving that $\mathscr{L}$ is *ample* (4.6.4). By criterion (4.5.2, a)), we must thus prove the following: for every quasi-coherent ideal $\mathscr{K}$ of $\mathscr{O}_X$ and every point $x \in X$ not belonging to the support of $\mathscr{O}_X/\mathscr{K}$, there exists an integer $n > 0$ and a section f of $\mathscr{K} \otimes \mathscr{L}^{\otimes n}$ over X such that $f(x) \neq 0$.

For this, set

$$\mathscr{K}_V = \pi^*(\mathscr{K})\mathscr{O}_V$$
$$\mathscr{K}_W = \mathscr{K}_V|W$$

where $W = V - j(X)$; since the restriction of q to W is a quasi-compact immersion to C, it follows from (**I**, 9.4.2) that $\mathscr{K}_W$ is the restriction to W of a quasi-coherent ideal $\mathscr{K}'_V$ of $\mathscr{O}_V$ of the form

$$\mathscr{K}'_V = q^*(\mathscr{K}_C)\mathscr{O}_V$$

where $\mathscr{K}_C$ is a quasi-coherent ideal of $\mathscr{O}_C$. Furthermore, since, by hypotheses, $q^{-1}(Y) \subset j(X)$, and since Y is defined by the ideal $\mathscr{I}$, the restriction to W of $q^*(\mathscr{I})\mathscr{O}_V$ is identical to that of $\mathscr{O}_V$, and so $\mathscr{K}_W$ is also the restriction to W of $q^*(\mathscr{I}\mathscr{K}_C)\mathscr{O}_V$, and we can thus suppose that $\mathscr{K}_C \subset \mathscr{I}$, whence

$$\mathscr{K}'_V \subset q^*(\mathscr{I})\mathscr{O}_V \subset \mathscr{J} \tag{8.10.6.1}$$

taking into account (**I**, 4.4.6) and the commutativity of (8.9.4.1). Furthermore, we deduce from (8.10.4) that

$$\mathscr{K}'_V \subset \mathscr{K}_V. \tag{8.10.6.2}$$

With this in mind, it follows from (8.10.3) that $j(x)$ belongs to exactly one irreducible component of $\pi^{-1}(x)$; let z be the generic point of this component, and let $z' = q(z)$. By (8.10.5), the proof will be finished (taking (8.10.6.1) and (8.10.6.2) into account) if we show the existence of a section g of $\mathscr{K}'_V$ over V such that $g(z) \neq 0$. But, by hypothesis, $\mathscr{K}$ has a restriction equal to that of $\mathscr{O}_X$ in an open neighbourhood of x; also, it follows from (8.10.3) that $z \neq j(x)$, and so $z \in W$, and thus $(\mathscr{K}_W)_z = \mathscr{O}_{V,z}$, whence, by definition, $(\mathscr{K}_C)_{z'} = \mathscr{O}_{C,z}$. Since C is affine, there is thus a section g' of $\mathscr{K}_C$ over C such that $g'(z') \neq 0$, and by taking g to be the section of $\mathscr{K}'_V$ corresponding canonically to g', we indeed have $g(z) \neq 0$, which finishes the proof.

Remark (8.10.7). — We ignore the question of whether or not condition (ii) in (8.9.4) is superfluous or not. In any case, the conclusion does not hold if we do not assume the existence of a Y-morphism $\pi : V \to X$ such that $\pi \circ j = 1_X$; we briefly point out how we can indeed construct a counterexample, whose details will not be developed until later on. We take $Y = \operatorname{Spec}(k)$, where k is a field, and $C = \operatorname{Spec}(A)$, where $A = k[T_1, T_2]$, and the Y-section ε corresponding to the augmentation homomorphism $A \to k$. We denote by C' the scheme induced by C by blowing up the closed point $a = \varepsilon(Y)$ of C; if D is the inverse image of a in C', we consider in D a closed point b, and we denote **II** | 189
by V the scheme induced by C' by blowing up b; X is the closed subprescheme of V given by the inverse image of a by the structure morphism $q : V \to C$. We now show that X is the union of two irreducible components, X_1 and X_2, where X_1 is the inverse image of b in V. It is immediate that the ideal $\mathscr{J}$ of $\mathscr{O}_V$ that defines X is again invertible, we we can show that $j^*(\mathscr{J}) = \mathscr{L}$ (where j is the canonical injection $X \to V$) is not ample, by considering the "degree" of the inverse image of $\mathscr{L}$ in X_1, which would be > 0 if $\mathscr{L}$ were ample, but we can show (by an elementary intersection calculation) that it is in fact equal to 0.

8.11. Uniqueness of contractions

Lemma (8.11.1). — *Let U and V be preschemes, and $h = (h_0, \lambda) : U \to V$ a surjective morphism. Suppose that*

(1) $\lambda : \mathscr{O}_V \to h_*(\mathscr{O}_U) = (h_0)_*(\mathscr{O}_U)$ is an isomorphism;
*(2) the underlying space of V can be identified with the quotient of the underlying space of U by the relation $h_0(x) = h_0(y)$ (*a condition which always holds whenever the morphism h is *open* or *closed*, or, *a fortiori* when h is *proper*.)

Then, for every prescheme W, the map

(8.11.1.1) $$\operatorname{Hom}(V, W) \longrightarrow \operatorname{Hom}(U, W)$$

that, to each morphism $v = (v_0, \nu)$ from V to W, associates the morphism $u = v \circ h = (u_0, \mu)$, is a bijection from $\operatorname{Hom}(V, W)$ *to the set of u such that u_0 is constant on every fibre $h_0^{-1}(x)$.*

Proof. It is clear that, if $u = v \circ h$, so that $u_0 = v_0 \circ h_0$, then u_0 is constant on every set $h_0^{-1}(x)$. Conversely, if u has this property, we will show that there exists exactly one $v \in \operatorname{Hom}(V, W)$ such that $u = v \circ h$. The existence and uniqueness of the continuous map $v_0 : V \to W$ such that $u_0 = v_0 \circ h_0$ follows from the hypotheses, since h_0 can be identified with the canonical map from U to U/R. We can also, replacing V by some isomorphic prescheme if necessary, suppose that λ is the identity; by hypothesis, μ is then a homomorphism $\mu : \mathscr{O}_W \to (u_0)_*(\mathscr{O}_U) = (v_0)_*((h_0)_*(\mathscr{O}_U))$ such that the corresponding homomorphism $\mu^\sharp : u_0^*(\mathscr{O}_W) \to \mathscr{O}_U$ is *local* on every fibre. Since $(v_0)_*((h_0)_*(\mathscr{O}_U)) = (v_0)_*(\mathscr{O}_V)$, we necessarily have that $\nu = u$, and everything then reduces to showing that the corresponding homomorphism $\nu^\sharp : v_0^*(\mathscr{O}_W) \to \mathscr{O}_V$ is local on every fibre. But every $y \in V$ is of the form $h_0(x)$ for some $x \in U$; let $z = v_0(y) = u_0(x)$. Then (**0**, 3.5.5) the homomorphism $\mu_x^\sharp$ factors as

$$\mu_x^\sharp : \mathscr{O}_z \xrightarrow{\nu_y^\sharp} \mathscr{O}_y \xrightarrow{\lambda_x^\sharp} \mathscr{O}_x.$$

By hypothesis, $\lambda_x^\sharp$ and $\mu_x^\sharp$ are local homomorphisms; thus $\lambda_x^\sharp$ sends every invertible element of $\mathscr{O}_y$ to an invertible element of $\mathscr{O}_x$; if $\nu_y^\sharp$ sent a non-invertible element of $\mathscr{O}_z$ to an invertible element of $\mathscr{O}_y$, then $\mu_x^\sharp$ would send this element of $\mathscr{O}_z$ to an invertible element of $\mathscr{O}_x$, contradicting the hypothesis, whence the lemma. □

Corollary (8.11.2). — *Let U be an integral prescheme, and V a normal prescheme; then every morphism $h : U \to V$ that is universally closed, birational, and radicial, is also an isomorphism.*

Proof. If $h = (h_0, \lambda)$, then it follows from the hypotheses that h_0 is injective and closed, and that $h_0(U)$ is dense in V, and so h_0 is a *homeomorphism* from U to V. To prove the corollary, it will suffice **II** | 190
to show that $\lambda : \mathscr{O}_V \to (h_0)_*(\mathscr{O}_U)$ is an isomorphism: we can then apply (8.11.1), which proves that the map (8.11.1.1) is bijective (the fibres $h_0^{-1}(x)$ each consisting of a single point); thus h will be an isomorphism. The question clearly being local on V, we can suppose that $V = \operatorname{Spec}(A)$ is affine, of an integral and integrally closed ring (8.8.6.1); h then corresponds (**I**, 2.2.4) to a homomorphism $\phi : A \to \Gamma(U, \mathscr{O}_U)$, and everything reduces to showing that ϕ is an isomorphism. But, if K is the field of fractions of A, then $\Gamma(U, \mathscr{O}_U)$ has, by hypothesis, K as its field of fractions, and A is a subring of $\Gamma(U, \mathscr{O}_U)$, with ϕ being the canonical injection (**I**, 8.2.7). Since the morphism h satisfies the hypotheses of (7.3.11), $\Gamma(U, \mathscr{O}_U)$ is a subring of the integral closure of A in K, and is thus identical to A by hypothesis. □

Remark (8.11.3). — We will see in chapter III (**III**, 4.4.11) that, whenever V is a *locally Noetherian* prescheme, every morphism $h : U \to V$ that is proper and quasi-finite (in particular, every morphism satisfying the hypotheses of (8.11.2)) is necessarily *finite*. The conclusion of (8.11.2) then follows in this case from (6.1.15).

(8.11.4). We will now see that, in Grauert's criterion, we can often prove that the prescheme C and the "contraction" q are determined in an *essentially unique* manner.

Lemma (8.11.5). — *Let Y be a prescheme, $p : X \to Y$ a proper morphism, $\mathscr{L}$ a p-ample invertible $\mathscr{O}_X$-module, C a Y-prescheme, $\varepsilon : Y \to C$ a Y-section, and $q : V = \mathbf{V}(\mathscr{L}) \to C$ a Y-morphism, all such* *that the diagram in (8.9.1.1) commutes. Suppose further that, if $p = (p_0, \theta)$, then $\theta : \mathscr{O}_Y \to p_*(\mathscr{O}_X)$ is an isomorphism. Let $\mathscr{S}' = \bigoplus_{n \geqslant 0} p_*(\mathscr{L}^{\otimes n})$ and $C' = \operatorname{Spec}(\mathscr{S}')$, and let $q' : \mathbf{V}(\mathscr{L}) \to C'$ be the canonical Y-morphism (8.8.5). Then there exists exactly one Y-morphism $u : C' \to C$ such that $q = u \circ q'$.*

Proof. The hypothesis on θ implies, in particular, that p is surjective; since, by (8.8.4), the restriction of q' to $\mathbf{V}(\mathscr{L}) - j(X)$ is an *isomorphism* to $C' - \varepsilon'(Y)$ (where ε is the vertex section of C'), it follows from (8.8.4) that q' is *proper* and *surjective*; furthermore, by (8.8.6), if we let $q' = (q'_0, \tau)$, then $\tau : \mathscr{O}_{C'} \to q'_*(\mathscr{O}_V)$ is an isomorphism. We are thus in a situation where we can apply (8.11.1), and we will have proven the lemma if we show that q is constant on every fibre $q'^{-1}(z')$, where $z' \in C'$. But this condition is trivially satisfied for $z' \notin \varepsilon'(Y)$. If $z' \in \varepsilon'(Y)$, then there exists exactly one $y \in Y$ such that $z' = \varepsilon'(y)$, and, by commutativity of (8.8.5.2) and the fact that q' sends $\mathbf{V}(\mathscr{L}) - j(X)$ to $C' - \varepsilon'(Y)$, $q'^{-1}(z') = j(p^{-1}(y))$; the commutativity of the diagram in (8.9.1.1) then proves our claim. □

Corollary (8.11.6). — *Under the hypotheses of (8.11.5), suppose further that q is proper, and that the restriction of q to $\mathbf{V}(\mathscr{L}) - j(X)$ is an isomorphism to $C - \varepsilon(Y)$. Then the morphism u is universally closed, surjective, and radicial, and its restriction to $C' - \varepsilon'(Y)$ is an isomorphism to $C - \varepsilon(Y)$.*

Proof. Since q' is an isomorphism from $\mathbf{V}(\mathscr{L}) - j(X)$ to $C' - \varepsilon'(Y)$ (8.8.4), the last claim follows immediately from the fact that $q = u \circ q'$. Furthermore, the commutativity of the diagrams in (8.8.5.2) **II** | 191
and (8.9.1.1) shows that the restriction of u to the closed subprescheme $\varepsilon'(Y)$ of C' is an isomorphism to the closed subprescheme $\varepsilon(Y)$ of C, from which we immediately deduce that, for all $z' \in \varepsilon'(Y)$, if $z = u(z')$, then u defines an isomorphism from $k(z)$ to $k(z')$. These remarks prove that u is bijective and radicial; furthermore, if $\psi : C \to Y$ and $\psi' : C' \to Y$ are the structure morphisms, then $\psi' = \psi \circ u$, and, since ψ' is separated (1.2.4), so too is u (**I**, 5.5.1, v). We have already seen, in the proof of (8.11.5), that q' is surjective; since $q = u \circ q'$ is proper, we finally conclude, from (5.4.3) and (5.4.9), that u is universally closed. □

Proposition (8.11.7). — *Let Y be a prescheme, X an* integral *prescheme, $p : X \to Y$ a proper morphism, $\mathscr{L}$ a p-ample invertible $\mathscr{O}_X$-module, C a* normal *Y-prescheme, $\varepsilon : Y \to C$ a Y-section, and $q : V = \mathbf{V}(\mathscr{L}) \to C$ a Y morphism, all such that the diagram in (8.9.1.1) commutes. Suppose further that, if $p = (p_0, \theta)$, then $\theta : \mathscr{O}_Y \to p_*(\mathscr{O}_X)$ is an isomorphism. Let $\mathscr{S}' = \bigoplus_{n \geqslant 0} p_*(\mathscr{L}^{\otimes n})$ and $C' = \operatorname{Spec}(\mathscr{S}')$, and let $q' : \mathbf{V}(\mathscr{L}) \to C'$ be the canonical Y-morphism (8.8.5). Then the unique Y-morphism $u : C' \to C$ such that $q = u \circ q'$ is an* isomorphism.

Proof. It follows from (8.8.6) that C' is integral; since u is a homeomorphism of the underlying subspaces $C' \to C$ (u being bijective and closed, by (8.11.6)), C is irreducible, thus integral, and, since the restriction of u to a non-empty open subset of C' is an isomorphism to an open subset of C, u is birational. Since C is assumed to be normal, it suffices to apply (8.11.2) to obtain the conclusion. □

Remark (8.11.8). — (i) The hypothesis that C is normal implies that X is also normal. Indeed, $C' = \operatorname{Spec}(\mathscr{S}')$ is then normal, being isomorphic to C, and integral, by (8.8.6); we thus conclude that $\operatorname{Proj}(\mathscr{S}')$ is *normal*. Indeed, the question is local on Y; if Y is affine, with $\mathscr{S}' = \widetilde{S}'$, then the ring $S' = \Gamma(C', \mathscr{S}')$ is integral and integrally closed (8.8.6.1), and so, for every homogeneous element $f \in S'_+$, the graded ring S'_f is integral and integrally closed [**SZ60**, t. I, p. 257 and 261], and thus so too is the ring $S'_{(f)}$ of its degree-zero terms, because the intersection of S'_f with the field of fractions of $S'_{(f)}$ is equal to $S'_{(f)}$; this proves our claim (6.3.4). Finally, since X is isomorphic to an open subprescheme of $\operatorname{Proj}(\mathscr{S}')$ (8.8.1), X is indeed normal. We can thus express (8.11.7) in the following form: *If X is integral and normal, and $p = (p_0, \theta) : X \to Y$ is a proper morphism such that $\theta : \mathscr{O}_Y \to p_*(\mathscr{O}_X)$ is an isomorphism, then, for every p-ample $\mathscr{O}_X$-module $\mathscr{L}$, there exists exactly one way of contracting the null section of $V = \mathbf{V}(\mathscr{L})$ to obtain a normal Y-scheme C and a proper Y-morphism $q : V \to C$.*

(ii) When p is proper, the hypothesis $p_*(\mathscr{O}_X) = \mathscr{O}_Y$ can be considered as an auxiliary hypothesis, not really restricting the generality of the result. Indeed, if it is not satisfied, then it suffices to replace Y with the Y-scheme $Y' = \operatorname{Spec}(p_*(\mathscr{O}_X))$, and to consider X as a Y'-scheme. We will return to this general method in chapter III, § 4.

8.12. Quasi-coherent sheaves on based cones

(8.12.1). Let us use the hypotheses and notation of (8.3.1). Let $\mathscr{M}$ be a *quasi-coherent graded $\mathscr{S}$-module*; to avoid any confusion, we denote by $\widetilde{\mathscr{M}}$ the quasi-coherent $\mathscr{O}_C$-module associated to $\mathscr{M}$ (1.4.3) when $\mathscr{M}$ is considered as a *non-graded* $\mathscr{S}$-module, and by $\mathscr{P}roj_0(\mathscr{M})$ the quasi-coherent $\mathscr{O}_X$-module associated to $\mathscr{M}$, $\mathscr{M}$ being considered this time as a graded $\mathscr{S}$-module (in other words, the $\mathscr{O}_X$-module denoted by $\widetilde{\mathscr{M}}$ in (3.2.2)). In addition, we set II | 192

(8.12.1.1) $$\mathscr{M}_X = \mathscr{P}roj_0(\mathscr{M}) = \bigoplus_{n\in\mathbf{Z}} \mathscr{P}roj_0(\mathscr{M}(n));$$

the quasi-coherent graded $\mathscr{O}_X$-algebra $\mathscr{S}_X$ being defined by (8.6.1.1), $\mathscr{P}roj(\mathscr{M})$ is equipped with a structure of a *(quasi-coherent) graded $\mathscr{S}_X$-module*, by means of the canonical homomorphisms (3.2.6.1)

(8.12.1.2) $$\mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{P}roj_0(\mathscr{M}(n)) \longrightarrow \mathscr{P}roj_0(\mathscr{S}(m) \otimes_{\mathscr{S}} \mathscr{M}(n)) \longrightarrow \mathscr{P}roj_0(\mathscr{M}(m+n)),$$

the verification of the axioms of sheaves of modules being done using the commutative diagram in (2.5.11.4).

If $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, and $\mathscr{M} = \widetilde{M}$, where S is a graded A-algebra and M is a graded S-module, then, for every homogeneous element $f \in S_+$, we have

(8.12.1.3) $$\Gamma(X_f, \mathscr{P}roj(\widetilde{M})) = M_f$$

by the definitions and (8.2.9.1).

Now consider the quasi-coherent graded $\widehat{\mathscr{S}}$-module

(8.12.1.4) $$\widehat{\mathscr{M}} = \mathscr{M} \otimes_{\mathscr{S}} \widehat{\mathscr{S}}$$

($\widehat{\mathscr{S}}$ being defined by (8.3.1.1)); this induces a quasi-coherent graded $\mathscr{O}_{\widehat{C}}$-module $\mathscr{P}roj_0(\widehat{\mathscr{M}})$, which we will also denote by

(8.12.1.5) $$\mathscr{M}^{\square} = \mathscr{P}roj_0(\widehat{\mathscr{M}}).$$

It is clear (3.2.4) that $\mathscr{M}^{\square}$ is an additive functor which is *exact* in $\mathscr{M}$, commuting with direct sums and with inductive limits.

Proposition (8.12.2). — *With the notation of (8.3.2), we have canonical functorial isomorphisms*

(8.12.2.1) $$i^*(\mathscr{M}^{\square}) \xrightarrow{\sim} \widetilde{\mathscr{M}}, \quad j^*(\mathscr{M}^{\square}) \xrightarrow{\sim} \mathscr{P}roj_0(\mathscr{M}).$$

Indeed, $i^*(\mathscr{M}^{\square})$ is canonically identified with $(\widehat{\mathscr{M}}/(\mathbf{z}-1)\widehat{\mathscr{M}})^{\sim}$ on $\operatorname{Spec}(\widehat{\mathscr{S}}/(\mathbf{z}-1)\widehat{\mathscr{S}})$ by (3.2.3); the first of the canonical isomorphisms (8.12.2.1) is then immediately induced (1.4.1) by the canonical isomorphism $\widehat{\mathscr{M}}/(\mathbf{z}-1)\widehat{\mathscr{M}} \xrightarrow{\sim} \mathscr{M}$. The canonical immersion $j : X \to C$ corresponds to the canonical homomorphism $\widehat{\mathscr{S}} \to \mathscr{S}$ with kernel $\mathbf{z}\widehat{\mathscr{S}}$ (8.3.2); the second homomorphism (8.12.2.1) is the particular case of the canonical homomorphism (3.5.2, ii), since here we have $\widehat{\mathscr{M}} \otimes_{\widehat{\mathscr{S}}} \mathscr{S} = \mathscr{M}$; to verify that this is an isomorphism, we can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, and $\mathscr{M} = \widetilde{M}$; by appealing to (2.8.8), the proof that, for all homogeneous f in S_+, the preceding homomorphism, restricted to X_f, restricts to an isomorphism, is then immediate.

By an abuse of language, we again say, thanks to the existence of the first isomorphism (8.12.2.1), that $\mathscr{M}^{\square}$ is the *projective closure* of the $\mathscr{O}_X$-module $\widetilde{\mathscr{M}}$ (it being implicit that the data of the $\mathscr{O}_C$-module $\widetilde{\mathscr{M}}$ includes the grading of the $\mathscr{S}$-module $\mathscr{M}$). II | 193

(8.12.3). With the notation of (8.3.5), we have a canonical functorial homomorphism

(8.12.3.1) $$p^*(\mathscr{P}roj(\mathscr{M})) \longrightarrow \mathscr{M}^{\square}|\widehat{E}.$$

Indeed, this is a particular case of the homomorphism $\nu^\sharp$ defined more generally in (3.5.6). If $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, and $\mathscr{M} = \widetilde{M}$, then, by appealing to (2.8.8), the restriction of (8.12.3.1) to $p^{-1}(X_f) = \widehat{C}_f$ (for some homogeneous f in S_+) corresponds to the canonical homomorphism

(8.12.3.2) $$M_{(f)} \otimes_{S_{(f)}} S_f^{\leqslant} \longrightarrow M_f^{\leqslant}$$

taking into account (8.2.3.2) and (8.2.5.2).

(8.12.4). Let us place ourselves in the settings of (8.5.1), and assume its hypotheses and keep its notation. It follows from (1.5.6) that, for every quasi-coherent graded $\mathscr{S}$-module $\mathscr{S}$, we have, on one hand, a canonical isomorphism

$$\Phi^*(\widetilde{\mathscr{M}}) \xrightarrow{\sim} (q^*(\mathscr{M}) \otimes_{q^*(\mathscr{S})} \mathscr{S}')^\sim \tag{8.12.4.1}$$

of $\mathscr{O}_{C'}$-modules; on the other hand, (3.5.6) implies the existence of a canonical $\mathrm{Proj}(\phi)$-morphism

$$\mathscr{P}\!roj_0 \mathscr{M} \longrightarrow (\mathscr{P}\!roj_0(q^*(\mathscr{M})) \otimes_{q^*(\mathscr{S})} \mathscr{S}')|G(\phi) \tag{8.12.4.2}$$

and also of a canonical $\widehat{\Phi}$-morphism

$$\mathscr{P}\!roj_0 \widehat{\mathscr{M}} \longrightarrow (\mathscr{P}\!roj_0(q^*(\widehat{\mathscr{M}})) \otimes_{q^*(\widehat{\mathscr{S}})} \widehat{\mathscr{S}'})|G(\widehat{\phi}). \tag{8.12.4.3}$$

(8.12.5). Consider now the setting of (8.6.1), with the same notation; we thus take $Y' = X$, the morphism $q : X \to Y$ being the structure morphism, and ϕ the canonical q-morphism (8.6.1.2). We then have a canonical isomorphism

$$q^*(\mathscr{M}) \otimes_{q^*(\mathscr{S})} \mathscr{S}_X^{\geqslant} \xrightarrow{\sim} \mathscr{M}_X^{\geqslant} \tag{8.12.5.1}$$

by setting $\mathscr{M}_X^{\geqslant} = \bigoplus_{n\geqslant 0} \mathscr{P}\!roj_0(\mathscr{M}(n))$. We can indeed restrict to the case where $Y = \mathrm{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, and $\mathscr{M} = \widetilde{M}$, and define the isomorphism (8.12.5.1) on each of the affine open subsets X_f (where f is homogeneous in S_+), by verifying the compatibility with taking a homogeneous multiple of f. But the restriction to X_f of the left-hand side of (8.12.5.1) is $\widetilde{M}' = ((M \otimes_A S_{(f)}) \otimes_{S\otimes_A S_{(f)}} S_f^{\geqslant})^\sim$ by (8.6.2.1); since we have a canonical isomorphism from $M \otimes_A S_{(f)}$ to $M \otimes_S (S \otimes_A S_{(f)})$, we have an induced isomorphism from $\widetilde{M}'$ to $(M \otimes_S S_f^{\geqslant})^\sim$, and the latter is canonically isomorphic, by (8.2.9.1), to the restriction to X_f of the right-hand side of (8.12.5.1), and satisfies the required compatibility conditions. **II** | 194

Replacing $\mathscr{M}$ by $\widehat{\mathscr{M}}$, $\mathscr{S}$ by $\widehat{\mathscr{S}}$, and $\mathscr{S}_X$ by $(\mathscr{S}_X^{\geqslant})^\wedge$ in the previous argument, we similarly have a canonical isomorphism

$$q^*(\widehat{\mathscr{M}}) \otimes_{q^*(\widehat{\mathscr{S}})} (\mathscr{S}_X^{\geqslant})^\wedge \xrightarrow{\sim} (\mathscr{M}_X^{\geqslant})^\wedge. \tag{8.12.5.2}$$

If we recall (8.6.2) that the structure morphism $u : \mathrm{Proj}(\mathscr{S}_X^{\geqslant}) \to X$ is an isomorphism, then we deduce, first of all, from the above, that we have a canonical u-isomorphism

$$\mathscr{P}\!roj_0 \mathscr{M} \xrightarrow{\sim} \mathscr{P}\!roj_0(\mathscr{M}_X^{\geqslant}) \tag{8.12.5.3}$$

as a particular case of (8.12.4.2). We note that, with the notation from the proof of (8.6.2), this reduces to seeing that the canonical homomorphism $M_{(f)} \otimes_{S_{(f)}} (S_f^{\geqslant})^{(d)} \to (M_f^{\geqslant})^{(d)}$ is an isomorphism whenever $f \in S_d$, which is immediate.

Secondly, the isomorphism (8.12.5.2) gives us, by this time applying (8.12.4.3) to the canonical morphism $r = \mathrm{Proj}(\widehat{\alpha}) : \widehat{C}_X \to \widehat{C}$, a canonical r-morphism

$$\mathscr{M}^\square \longrightarrow (\mathscr{M}_X^{\geqslant})^\square. \tag{8.12.5.4}$$

Recall now (8.6.2) that the restrictions of r to the pointed cones $\widehat{E}_X$ and E_X are *isomorphisms* to $\widehat{E}$ and E (respectively). Furthermore:

Proposition (8.12.6). — *The restrictions to $\widehat{E}_X$ and E_X of the canonical r-morphism* (8.12.5.4) *are isomorphisms*

$$\mathscr{M}^\square|\widehat{E} \xrightarrow{\sim} (\mathscr{M}_X^{\geqslant})^\square|\widehat{E}_X \tag{8.12.6.1}$$

$$\mathscr{M}^\sim|\widehat{E} \xrightarrow{\sim} (\mathscr{M}_X^{\geqslant})^\sim|\widehat{E}_X. \tag{8.12.6.2}$$

PROOF. We restrict to the case where Y is affine, as in the proof of (8.6.2) (whose notation we adopt); by reducing to definitions (2.8.8), we have to show that the canonical homomorphism

$$\widehat{M}_{(f)} \otimes_{\widehat{S}_{(f)}} (S_f^{\geqslant})^\wedge_{(f/1)} \longrightarrow (M \otimes_S S_f^{\geqslant})^\wedge_{(f/1)}$$

is an isomorphism; but, by (8.2.3.2) and (8.2.5.2), the left-hand side is canonically identified with $M_f^{\leqslant} \otimes_{S_f^{\leqslant}} (S_f^{\geqslant})_{f/1}^{\leqslant}$, and thus with $M_f^{\leqslant}$, by (8.2.7.2), and the right-hand side with $(M_f^{\geqslant})_{f/1}^{\leqslant}$, and thus also with $M_f^{\leqslant}$, by (8.2.9.2), whence the conclusion concerning (8.12.6.1); (8.12.6.2) then follows from (8.12.6.1) and (8.12.2.1). □

Corollary (8.12.7). — *With the identifications of (8.6.3), the restriction of $(\mathscr{M}_X^{\geqslant})^{\square}$ to $\widehat{E}_X$ can be identified with $(\mathscr{M}_X^{\leqslant})^{\sim}$, and the restriction of $(\mathscr{M}_X^{\geqslant})^{\square}$ to E_x with $\widetilde{\mathscr{M}}_X$.*

Proof. We can restrict to the affine case, and this follows from the identification of $(M_f^{\geqslant})_{f/1}^{\leqslant}$ with $M_f^{\leqslant}$, and of $(M_f^{\geqslant})_{f/1}$ with M_f (8.2.9.2). □

Proposition (8.12.8). — *Under the hypotheses of (8.6.4), the canonical homomorphism (8.12.3.1) is an isomorphism.*

Proof. Taking into account the fact that $\operatorname{Proj}(\mathscr{S}_X^{\geqslant}) \to X$ is an isomorphism (8.6.2), and the isomorphisms (8.12.5.4) and (8.12.6.1), we are led to proving the corresponding proposition for the canonical homomorphism $p_X^*(\mathscr{P}roj_0(\mathscr{M}_X^{\geqslant})) \to (\mathscr{M}_X^{\geqslant})^{\square}|E_X$, or, in other words, we can restrict to the case where $\mathscr{S}_1$ is an invertible $\mathscr{O}_Y$-module, and where $\mathscr{S}$ is generated by $\mathscr{S}_1$. With the notation of (8.12.3), we then have, for some $f \in S_1$, that $S_f^{\leqslant} = S_{(f)}[1/f]$, and the canonical homomorphism $M_{(f)} \otimes_{S_{(f)}} S_f^{\leqslant} \to M_f^{\leqslant}$ is an isomorphism, by the definition of $M_f^{\leqslant}$. □ **II | 195**

(8.12.9). Consider now the quasi-coherent $\mathscr{S}$-modules

$$\mathscr{M}_{[n]} = \bigoplus_{m \geqslant n} \mathscr{M}_m$$

and (with the notation of (8.7.2)) the quasi-coherent graded $\mathscr{S}^{\natural}$-module

$$\mathscr{M}^{\natural} = \left(\bigoplus_{n \geqslant 0} \mathscr{M}_{[n]} \right)^{\sim}. \tag{8.12.9.1}$$

We have seen (8.7.3) that there exists a canonical C-isomorphism $h : C_X \xrightarrow{\sim} \operatorname{Proj}(\mathscr{S}^{\natural})$. Furthermore:

Proposition (8.12.10). — *There exists a canonical h-isomorphism*

$$\mathscr{P}roj_0(\mathscr{M}^{\natural}) \xrightarrow{\sim} \widetilde{\mathscr{M}}_X. \tag{8.12.10.1}$$

Proof. We argue as in (8.7.3), this time using the existence of the di-isomorphism (8.2.9.3) instead of (8.2.7.3). We leave the details to the reader. □

8.13. Projective closures of subsheaves and closed subschemes

(8.13.1). With hypotheses and notation as in (8.12.1), consider a *not-necessarily graded* quasi-coherent sub-$\mathscr{S}$-module $\mathscr{N}$ of $\mathscr{M}$. We can then consider the quasi-coherent $\mathscr{O}_C$-module $\widetilde{\mathscr{N}}$ associated to $\mathscr{N}$, which is a sub-$\mathscr{O}_C$-module of $\widetilde{\mathscr{M}}$. We have seen elsewhere (8.12.2.1) that $\widetilde{\mathscr{M}}$ can be identified with the restriction of $\mathscr{M}^{\square}$ to C. Since the canonical injection $i : C \to \widehat{C}$ is an affine morphism (8.3.2), and *a fortiori* quasi-compact, the *canonical extension* $(\widetilde{\mathscr{N}})^{-}$, the largest sub-$\mathscr{O}_{\widehat{C}}$-module contained in $\mathscr{M}^{\square}$ and inducing $\widetilde{\mathscr{N}}$ on C, is a *quasi-coherent* $\mathscr{O}_{\widehat{C}}$-module (**I**, 9.4.2). We will give a more explicit description by using a graded $\widehat{\mathscr{S}}$-module.

(8.13.2). For this, consider, for every integer $n \geqslant 0$, the homomorphism $\bigoplus_{i \leqslant n} \mathscr{M}_i \to \mathscr{M}$ which, for every open U of Y, sends the family

$$(s_i) \in \bigoplus_{i \leqslant n} \Gamma(U, \mathscr{M}_i)$$

to the section $\sum_i s_i \in \Gamma(U, \mathscr{M})$. Denote by $\mathscr{N}_n'$ the inverse image of $\mathscr{N}$ by this homomorphism, which is a quasi-coherent sub-$\mathscr{S}$-module of $\bigoplus_{i \leqslant n} \mathscr{M}_i$. Now consider the homomorphism $\bigoplus_{i \leqslant n} \mathscr{M}_i \to \widehat{\mathscr{M}} = \mathscr{M}[\mathbf{z}]$ which sends (s_i) to the section $\sum_{i \leqslant n} s_i \mathbf{z}^{n-i} \in \Gamma(U, \widehat{\mathscr{M}}_n)$, and let $\mathscr{N}_n$ be the image of

$\mathscr{N}_n'$ under this homomorphism; we immediately have that $\overline{\mathscr{N}} = \bigoplus_{n\geqslant 0}\mathscr{N}_n$ is a (quasi-coherent) sub-$\widehat{\mathscr{S}}$-module of $\widehat{\mathscr{M}}$; we say that $\overline{\mathscr{N}}$ is induced from $\mathscr{N}$ by *homogenisation*, via the "homogenising variable" $\mathbf{z}$. We note that, if $\mathscr{N}$ is already a *graded* sub-$\mathscr{S}$-module of $\mathscr{M}$, then $\overline{\mathscr{N}}$ can be identified with the direct sum of the components $\widehat{\mathscr{N}_n}$ of degree $n \geqslant 0$ in $\widehat{\mathscr{N}} = \mathscr{N}[\mathbf{z}]$. **II** | 196

Proposition (8.13.3). — *The $\mathscr{O}_{\widehat{C}}$-module $\mathscr{P}roj_0(\overline{\mathscr{N}})$ is the canonical extension $(\widetilde{\mathscr{N}})^-$ of $\widetilde{\mathscr{N}}$ to $\widehat{C}$.*

PROOF. The question is local on Y and $\widehat{C}$ by the definition of the canonical extension (**I**, 9.4.1). We can thus already suppose that $Y = \operatorname{Spec}(A)$ is affine, with $\mathscr{S} = \widetilde{S}$, $\mathscr{M} = \widetilde{M}$, and $\mathscr{N} = \widetilde{N}$, where N is a non-necessarily-graded sub-S-module of M. Furthermore (8.3.2.6), $\widehat{C}$ is a union of affine opens $\widehat{C}_z = C$ and $\widehat{C}_f = \operatorname{Spec}(S_f^{\leqslant})$ (with f homogeneous in S_+). It thus suffices to show that: (1) the restriction of $\mathscr{P}roj_0(\overline{\mathscr{N}})$ to C is $\widetilde{\mathscr{N}}$; (2) the restriction of $\mathscr{P}roj_0(\overline{\mathscr{N}})$ to each $\widehat{C}_f$ is the canonical extension of the restriction of $\mathscr{N}$ to $C \cap \widehat{C}_f = \operatorname{Spec}(S_f)$ (8.3.2.6). For the first point, note that $\mathscr{P}roj_0(\overline{\mathscr{N}})|C$ can be identified with $(\overline{N}_{(\mathbf{z})})^\sim$ (8.3.2.4); but $\overline{N}_{(\mathbf{z})}$ is canonically identified (2.2.5) with the image of $\overline{N}$ in $\widehat{M}/(\mathbf{z}-1)\widehat{M}$, and by the canonical isomorphism of the latter with M (8.2.5), this image can be identified with N, by the definition of $\overline{N}$ given in (8.13.2).

To prove the second point, note that the injection $i : C \cap \widehat{C}_f \to \widehat{C}$ corresponds to the canonical injection $S_f^{\leqslant} \to S_f$ (8.3.2.6); we also have that $\Gamma(\widehat{C}_f, \mathscr{M}^\square) = M_f^{\leqslant}$, that $\Gamma(\widehat{C}_f, i_*(\widetilde{\mathscr{N}})) = N$, and, by (8.12.2.1), that $\Gamma(\widehat{C}_f, i_*(i^*(\mathscr{M}^\square))) = M_f$. Taking (**I**, 9.4.2) into account, we are thus led to showing that $\overline{N}_{(f)} \subset \widehat{M}_{(f)} = M_f^{\leqslant}$ is canonically identified with the inverse image of N_f under the canonical injection $M_f^{\leqslant} \to M_f$. Indeed, let $d = \deg(f) > 0$, and suppose that an element $(\sum_{k\leqslant md} x_k)/f^m$ of M_f (with $x_k \in M_k$) is of the form y/f^m with $y \in N$. By multiplying y and the x_k by one single suitable f^h, we can already assume that $\sum_{k\leqslant md} x_k = y$. But in the identification of (8.2.5.2), $(\sum_{k\leqslant md} x_k)/f^m$ corresponds to $\sum_{k\leqslant md} x_k\mathbf{z}^{md-k}/f^m$, and this is indeed an element of $\overline{N}_{(f)}$, since $\sum_{k\leqslant md} x_k \in N$; the converse is evident. □

Remark (8.13.4). — (i) The most important case of application of (8.13.3) is that where $\mathscr{M} = \mathscr{S}$, with $\widetilde{\mathscr{N}}$ then being an *arbitrary* quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_C$ (1.4.3), corresponding bijectively to a *closed subprescheme* Z of C. Then the canonical extension $\overline{\mathscr{J}}$ of $\mathscr{J}$ is the quasi-coherent sheaf of ideals of $\mathscr{O}_{\widehat{C}}$ that defines the *closure* $\overline{Z}$ of Z in $\widehat{C}$ (**I**, 9.5.10); Proposition (8.13.3) gives a canonical way of defining $\overline{Z}$ by using a graded ideal in $\widehat{\mathscr{S}} = \mathscr{S}[\mathbf{z}]$.

(ii) Suppose, to simplify things, that Y is affine, and adopt the notation from the proof of (8.13.3). For every non-zero $x \in N$, let $d(x)$ be the largest degree of the homogeneous components x_i of x in M; by definition, $\overline{N}$ is the submodule of $\widehat{M}$ consisting of 0 and elements of the form $h(x,k) = \mathbf{z}^k \sum_{i\leqslant d(x)} x_i\mathbf{z}^{d(x)-i}$ (for integral $k \geqslant 0$); it is thus generated, as a module over $\widehat{S} = S[\mathbf{z}]$, by the elements of the form

$$h(x,0) = \sum_{i\leqslant d(x)} x_i\mathbf{z}^{d(x)-i}.$$

We say that $h(x,0)$ is induced from x by *homogenisation* via the "homogenising variable" $\mathbf{z}$. **II** | 197
But since $h(x,0)$ does not depend additively on x (nor *a fortiori* S-linearly), *we will refrain from believing* (even when $M = S$) that the $h(x,0)$ form a *system of generators* of the graded $\widehat{S}$-module $\overline{N}$ when we let x run over a *system of generators* of the S-module N. This is, however, the case (considered only in elementary algebraic geometry) when N is a *free cyclic* S-module, since, if t is a basis of N, then $h(t,0)$ generates the $\widehat{S}$-module $\overline{N}$.

8.14. Supplement on sheaves associated to graded $\mathscr{S}$-modules

(8.14.1). Let Y be a prescheme, $\mathscr{S}$ a *positively-graded* quasi-coherent $\mathscr{O}_Y$-algebra, $X = \operatorname{Proj}(\mathscr{S})$, and $q : X \to Y$ the structure morphism (which is separated, by (3.1.3)). Using the notation of (8.12.1), we have defined a functor $\mathscr{M}_X = \mathscr{P}roj(\mathscr{M})$ in $\mathscr{M}$, from the category of quasi-coherent graded $\mathscr{S}$-modules to the category of quasi-coherent graded $\mathscr{S}_X$-modules; it is further clear (3.2.4) that this is an *additive* and *exact* functor, commuting with inductive limits.

Note, furthermore, that it follows immediately from the definition (8.12.1.1) that we have

$$\mathscr{P}roj(\mathscr{M}(n)) = (\mathscr{P}roj(\mathscr{M}))(n) \quad \text{for all } n \in \mathbf{Z}. \tag{8.14.1.1}$$

(8.14.2). We will first extend the canonical homomorphisms λ and μ, defined in (3.2.6), to $\mathscr{S}_X$-modules of the form $\mathscr{P}roj(\mathscr{M})$. For this, note that, for any $m \in \mathbf{Z}$ and $n \in \mathbf{Z}$, we have, by (2.1.2.1), a canonical homomorphism of $\mathscr{O}_X$-modules

$$\lambda_{mn} : \mathscr{P}roj_0((\mathscr{H}om_{\mathscr{S}}(\mathscr{M},\mathscr{N}))(n-m)) \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}roj_0(\mathscr{M}(m)), \mathscr{P}roj_0(\mathscr{N}(n))) \tag{8.14.2.1}$$

for any quasi-coherent graded $\mathscr{S}$-modules $\mathscr{M}$ and $\mathscr{N}$. This induces a homomorphism

$$\mu_k : \mathscr{P}roj_0((\mathscr{H}om_{\mathscr{S}}(\mathscr{M},\mathscr{N}))(k)) \longrightarrow (\mathscr{H}om_{\mathscr{S}_X}(\mathscr{P}roj(\mathscr{M}), \mathscr{P}roj(\mathscr{N})))_k \tag{8.14.2.2}$$

given by sending every $u \in \Gamma(U, \mathscr{P}roj_0((\mathscr{H}om_{\mathscr{S}}(\mathscr{M},\mathscr{N}))(k)))$ to the homomorphism $\mu_k(u)$, of degree k, of graded $\mathbf{Z}$-modules $\Gamma(U, \mathscr{P}roj(\mathscr{M})) \to \Gamma(U, \mathscr{P}roj(\mathscr{N}))$ (where U is open in X) which, in each $\Gamma(U, \mathscr{P}roj_0(\mathscr{M}(m)))$, agrees with $\mu_{m,m+k}(u)$; furthermore, by returning to the definition of the μ_{mn} (2.5.12.1), we immediately see that $\mu_k(u)$ is in fact a homomorphism of degree k of graded $\Gamma(U, \mathscr{S}_X)$-modules, and, furthermore, that the μ_k define a homomorphism of *graded $\mathscr{S}_X$-modules*

$$\mathscr{P}roj(\mathscr{H}om_{\mathscr{S}}(\mathscr{M},\mathscr{N})) \longrightarrow \mathscr{H}om_{\mathscr{S}_X}(\mathscr{P}roj(\mathscr{M}), \mathscr{P}roj(\mathscr{N})). \tag{8.14.2.3}$$

Similarly, taking the associativity diagram (2.5.11.4) into account, the homomorphisms (8.14.2.1) give a homomorphism of *graded $\mathscr{S}_X$-modules*

$$\lambda : \mathscr{P}roj(\mathscr{M}) \otimes_{\mathscr{S}_X} \mathscr{P}roj(\mathscr{N}) \longrightarrow \mathscr{P}roj(\mathscr{M} \otimes_{\mathscr{S}} \mathscr{N}). \tag{8.14.2.4}$$

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Proposition (8.14.3). — *The homomorphism (8.14.2.4) is bijective; so too is (8.14.2.3) whenever the graded $\mathscr{S}$-module $\mathscr{M}$ admits a finite presentation (3.1.1).*

Proof. The question is clearly local on X and Y; we can thus suppose that $Y = \operatorname{Spec}(A)$ is affine, with $\mathscr{S} = \widetilde{S}$, $\mathscr{M} = \widetilde{M}$, and $\mathscr{N} = \widetilde{N}$, where S is a positively-graded A-algebra, and M and N are graded S-modules. If f is a homogeneous element of S_+, then the homomorphisms (8.14.2.1) and (8.14.2.2), restricted to the affine open $D_+(f)$, correspond to the canonical homomorphisms (2.5.11.1) and (2.5.12.1):

$$M(m)_{(f)} \otimes_{S_{(f)}} N(n)_{(f)} \longrightarrow (M \otimes_S N)(m+n)_{(f)}$$
$$(\operatorname{Hom}_S(M,N))(n-m)_{(f)} \longrightarrow \operatorname{Hom}_{S_{(f)}}(M(m)_{(f)}, N(n)_{(f)}).$$

If we refer to the definitions of these homomorphisms, we thus see (taking (8.2.9.1) into account) that the restriction of (8.14.2.4) to $D_+(f)$ corresponds to the canonical homomorphism

$$M_f \otimes_{S_f} N_f \longrightarrow (M \otimes_S N)_f$$

defined in (**0**, 1.3.4), and we know that this latter homomorphism is an isomorphism. Similarly, the restriction of (8.14.2.3) to $D_+(f)$ corresponds to the canonical homomorphism (**0**, 1.3.5)

$$(\operatorname{Hom}_S(M,N))_f \longrightarrow \operatorname{Hom}_{S_f}(M_f, N_f)$$

taking into account the fact that, since M is of finite type, the module $\operatorname{Hom}_S(M,N)$, the direct sum of the subgroups consisting of *homogeneous* homomorphisms of S-modules (2.1.2), agrees with the set of *all* homomorphisms $M \to N$ of S-modules. The hypothesis that M admits a finite presentation then implies (**0**, 1.3.5) that the canonical homomorphism in question is indeed an isomorphism. □

Proposition (8.14.4). — *If U is a quasi-compact open of X, then there exists an integer d such that, for every integer n that is a multiple of d, $\mathscr{O}_X(n)|U$ is invertible, with its inverse being $\mathscr{O}_X(-n)|U$.*

PROOF. Since $q(U)$ is quasi-compact, it is covered by a finite number of affine opens V_i, and so every $x \in U$ is contained in some affine open of the form $D_+(f)$, where f is a homogeneous element of degree > 0 of one of the rings $\Gamma(V_i, \mathscr{S})$. Since U is quasi-compact, we can cover it by a finite number of such opens $D_+(f_j)$; let d be a common multiple of the degrees of the f_j. This d satisfies the desired property, by (2.5.17). □

(8.14.5). With the hypotheses and notation of (8.14.1), we defined, in (3.3.2), canonical homomorphisms of $\mathscr{O}_Y$-modules

$$\alpha_n : \mathscr{M}_n \longrightarrow q_*(\mathscr{P}roj_0(\mathscr{M}(n))) \qquad (n \in \mathbf{Z}). \tag{8.14.5.1}$$

Generalising the notation of (3.3.1), we set, for every *graded* $\mathscr{S}_X$*-module* $\mathscr{F}$,

$$\mathbf{\Gamma}_*(\mathscr{F}) = \bigoplus_{n \in \mathbf{Z}} q_*(\mathscr{F}_n). \tag{8.14.5.2}$$

In particular, $\mathbf{\Gamma}(\mathscr{S}_X) = \bigoplus_{n\in\mathbf{Z}} q_*(\mathscr{O}_X(n))$ is the graded $\mathscr{O}_Y$-algebra denoted by $\mathbf{\Gamma}_*(\mathscr{O}_X)$ in (3.3.1.2); it is clear that $\mathbf{\Gamma}(\mathscr{F})$ is a *graded* $\mathbf{\Gamma}_*(\mathscr{S}_X)$*-algebra* (**0**, 4.2.2). When we take $\mathscr{M} = \mathscr{S}$ in the homomorphisms (8.14.5.1), we obtain the homomorphism of graded $\mathscr{O}_Y$-algebras **II | 199**

$$\alpha : \mathscr{S} \longrightarrow \mathbf{\Gamma}(\mathscr{S}_X) \tag{8.14.5.3}$$

previously defined in (3.3.2), and which makes $\mathbf{\Gamma}_*(\mathscr{F})$ a *graded* $\mathscr{S}$*-module*; the homomorphisms (8.14.5.1) then define a homomorphism (of degree 0) of *graded* $\mathscr{S}$*-modules*

$$\alpha : \mathscr{M} \longrightarrow \mathbf{\Gamma}_*(\mathscr{P}roj(\mathscr{M})). \tag{8.14.5.4}$$

(8.14.6). In general, for a quasi-coherent graded $\mathscr{S}_X$-module $\mathscr{F}$, it is not certain that the graded $\mathscr{S}$-module $\mathbf{\Gamma}_*(\mathscr{F})$ will necessarily be quasi-coherent. Consider an open X' of X such that the restriction $q' : X' \to Y$ of q to X' is a *quasi-compact* morphism. Since q' is further separated, $q'_*(\mathscr{F}')$ is then a quasi-coherent $\mathscr{O}_Y$-module for every quasi-coherent $\mathscr{O}_{X'}$ module $\mathscr{F}'$ (**I**, 9.2.2, b). We set

$$\mathscr{S}_{X'} = \mathscr{S}_X|X' = \bigoplus_{n\in\mathbf{Z}} \mathscr{O}_X(n)|X' \tag{8.14.6.1}$$

and, for every graded $\mathscr{S}_{X'}$-module $\mathscr{F}'$,

$$\mathbf{\Gamma}'_*(\mathscr{F}') = \bigoplus_{n\in\mathbf{Z}} q'_*(\mathscr{F}'_n). \tag{8.14.6.2}$$

The previous remark then shows that, if $\mathscr{F}'$ is a quasi-coherent $\mathscr{S}_{X'}$-module, then $\mathbf{\Gamma}'_*(\mathscr{F}')$ is a graded *quasi-coherent* $\mathscr{S}$-module (**I**, 9.6.1).

We note also that the canonical injection $j : X' \to X$ is *quasi-compact*, because $q' = q \circ j$ is quasi-compact and q is separated (**I**, 6.6.4, v). Then $\mathscr{F} = j_*(\mathscr{F}')$ is a quasi-coherent graded $\mathscr{S}_X$-module for every quasi-coherent graded $\mathscr{S}_{X'}$-module $\mathscr{F}'$, and it follows from the previous definitions that

$$\mathbf{\Gamma}'_*(\mathscr{F}') = \mathbf{\Gamma}_*(\mathscr{F}). \tag{8.14.6.3}$$

With the same hypotheses on X', for every quasi-coherent graded $\mathscr{S}$-module $\mathscr{M}$, we set

$$\mathscr{P}roj'(\mathscr{M}) = \mathscr{P}roj(\mathscr{M})|X' \tag{8.14.6.4}$$

which is a quasi-coherent graded $\mathscr{S}_{X'}$-module. The canonical homomorphism

$$\mathscr{P}roj(\mathscr{M}) \longrightarrow j_*(\mathscr{P}roj'(\mathscr{M}))$$

(**0**, 4.4.3) thus gives a canonical homomorphism $\mathbf{\Gamma}_*(\mathscr{P}roj(\mathscr{M})) \to \mathbf{\Gamma}'_*(\mathscr{P}roj'(\mathscr{M}))$ of graded $\mathscr{S}$-modules, and, by composition with (8.14.5.4), we obtain a functorial canonical homomorphism (of degree 0) of quasi-coherent graded $\mathscr{S}$-modules

$$\alpha' : \mathscr{M} \longrightarrow \mathbf{\Gamma}'_*(\mathscr{P}roj'(\mathscr{M})). \tag{8.14.6.5}$$

(8.14.7). Keeping the hypotheses on X' from (8.14.6), let $\mathscr{F}'$ be a *quasi-coherent graded* $\mathscr{S}_{X'}$*-module* such that $\mathscr{P}roj'(\mathbf{\Gamma}'_*(\mathscr{F}'))$ is also a graded *quasi-coherent* $\mathscr{S}_{X'}$-module. We will define a functorial canonical homomorphism (of degree 0) of graded $\mathscr{S}_{X'}$-modules **II | 200**

$$\beta' : \mathscr{P}roj'(\mathbf{\Gamma}'_*(\mathscr{F}')) \longrightarrow \mathscr{F}'. \tag{8.14.7.1}$$

Suppose first of all that $Y = \mathrm{Spec}(A)$ is affine, and that $\mathscr{S} = \widetilde{S}$, where S is a positively-graded A-algebra; then $\mathbf{\Gamma}'_*(\mathscr{F}') = \widetilde{M}$, where $M = \bigoplus n \in \mathbf{Z}\Gamma(X', \mathscr{F}'_n)$ is a graded S-module. Let $f \in S_d$ be such that $D_+(f) \subset X'$; by definition (2.6.2), $\alpha_d(f)$ restricted to $D_+(f)$ is the section of $\mathscr{O}_X(d)$ over $D_+(f)$ corresponding to the element $f/1$ of $(S(d))_{(f)}$, and is thus invertible; thus so too is $\alpha_d(f^n)$ for every $n > 0$. From this, we immediately conclude that we have defined an S_f-homomorphism (of degree 0) of graded modules $\beta_f : M_f \to \Gamma(D_+(f), \mathscr{F}')$ by sending each element $z/f^n \in M_f$ (where $z \in M$) to the section $(z|D_+(f))(\alpha_d(f^n)|D_+(f))^{-1}$ of $\mathscr{F}'$ over $D_+(f)$. Furthermore, we have a commutative diagram corresponding to (2.6.4.1), whence the definition of β' in this case. To pass to the general case, we must consider an A-algebra A', the graded A'-algebra $S' = S \otimes_A A'$, and use the commutative diagram analogous to (2.8.13.2); we leave the details to the reader.

Proposition (8.14.8). — *If X' is an open of $X = \mathrm{Proj}(\mathscr{S})$ such that $q' : X' \to Y$ is quasi-compact, then the homomorphism β' defined in (8.14.7) is bijective.*

PROOF. We can clearly restrict to the case where Y is affine, and everything then reduces to proving (with the notation of (8.14.7)) that the homomorphism $\beta_f : M_f \to \Gamma(D_+(f), \mathscr{F}')$ is an isomorphism. But replacing f by one of its powers changes neither $D_+(f)$ nor β_f; since X' is *quasi-compact* by hypothesis, we can always assume, by (8.14.4), that the sheaf $\mathscr{O}_X(d)$ is *invertible*. Since X' is a scheme (because q' is separated), the proposition is then exactly (**I**, 9.3.1). □

Corollary (8.14.9). — *Under the hypotheses of (8.14.8), every quasi-coherent graded $\mathscr{S}_{X'}$-module is isomorphic to a graded $\mathscr{S}_{X'}$-module of the form $\mathscr{P}roj'(\mathscr{M})$, where $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module. Further, if $\mathscr{F}'$ is of finite type, and if we assume that Y is a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, then we can assume that $\mathscr{M}$ is of finite type.*

PROOF. The proof starting from (8.14.8) follows exactly the same route as the proof of (3.4.5) starting from (3.4.4), and we leave the details to the reader. □

Proposition (8.14.10). — *Under the hypotheses of (8.14.7), let $\mathscr{M}$ be a quasi-coherent graded $\mathscr{S}$-module, and $\mathscr{F}'$ a quasi-coherent graded $\mathscr{S}_{X'}$-module; the composite homomorphisms*

(8.14.10.1) $$\mathscr{P}roj'(\mathscr{M}) \xrightarrow{\mathscr{P}roj'(\alpha')} \mathscr{P}roj'(\mathbf{\Gamma}'_*(\mathscr{P}roj'(\mathscr{M}))) \xrightarrow{\beta'} \mathscr{P}roj'(\mathscr{M})$$

(8.14.10.2) $$\mathbf{\Gamma}'_*(\mathscr{F}') \xrightarrow{\alpha'} \mathbf{\Gamma}'_*(\mathscr{P}roj'(\mathbf{\Gamma}'_*(\mathscr{F}'))) \xrightarrow{\mathbf{\Gamma}'_*(\beta')} \mathbf{\Gamma}'_*(\mathscr{F}')$$

are the identity isomorphisms.

PROOF. The question is local on Y, and the proof follows as in (2.6.5); we leave the details to the reader. □

Remark (8.14.11). — In chapter III (**III**, 2.3.1), we will see that, when Y is *locally Noetherian*, and $\mathscr{S}$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra *of finite type* (in which case we can take $X' = X$), then **II | 201**
the homomorphism α (8.14.5.4) is **(TN)**-*bijective* for every quasi-coherent graded $\mathscr{S}$-module $\mathscr{M}$ satisfying condition **(TF)**.

Remark (8.14.12). — The situation described in (8.14.4) is a particular case of the following. Let X be a ringed space, and $\mathscr{S}$ a (positively- and negatively-) graded $\mathscr{O}_X$-algebra; suppose that there exists an integer $d > 0$ such that $\mathscr{S}_d$ and $\mathscr{S}_{-d}$ are *invertible*, with the canonical homomorphism

(8.14.12.1) $$\mathscr{S}_d \otimes_{\mathscr{O}_X} \mathscr{S}_{-d} \longrightarrow \mathscr{O}_X$$

being an *isomorphism* (such that $\mathscr{S}_{-d}$ is identified with $\mathscr{S}_d^{-1}$). We then say that the graded $\mathscr{O}_X$-algebra $\mathscr{S}$ is *periodic, of period d*. This nomenclature stems from the following property: *under the preceding hypotheses, for every graded $\mathscr{S}$-module $\mathscr{F}$, the canonical homomorphism*

(8.14.12.1) $$\mathscr{S}_d \otimes \mathscr{F}_n \longrightarrow \mathscr{F}_{n+d}$$

is an isomorphism for all $n \in \mathbf{Z}$. Indeed, the question is local on X, and we can assume that $\mathscr{S}_d$ has an *invertible* section s over X, with its inverse s' being a section of $\mathscr{S}_{-d}$. The homomorphism $\mathscr{F}_{n+d} \to \mathscr{S}_d \otimes \mathscr{F}_n$, which sends each section $z \in \Gamma(U, \mathscr{F}_{n+d})$ to the section $(s|U) \otimes (s'|U)z$ of

$\mathscr{S}_d \otimes \mathscr{F}_n$ over U, is then the inverse of (8.14.12.2), whence our claim. This induces, for all $k \in \mathbf{Z}$, a canonical isomorphism

$$(\mathscr{S}_d)^{\otimes k} \otimes \mathscr{F}_n \xrightarrow{\sim} \mathscr{F}_{n+kd}.$$

Then *the data of a graded $\mathscr{S}$-module $\mathscr{F}$ is equivalent to the data of $\mathscr{S}_0$-modules $\mathscr{F}_i$ ($0 \leqslant i \leqslant d-1$) and canonical homomorphisms*

$$\mathscr{S}_i \otimes \mathscr{F}_j \longrightarrow \mathscr{F}_{i+j} \qquad \text{for } 0 \leqslant i, j \leqslant d-1$$

(setting $\mathscr{F}_{i+j} = \mathscr{S}_d \otimes_{\mathscr{S}_0} \mathscr{F}_{i+j-d}$ whenever $i+j \geqslant d$). Of course, for theses homomorphisms to give a well-defined $\mathscr{S}$-module structure on the direct sum of the $(\mathscr{S}_d)^{\otimes k} \otimes \mathscr{F}_i$ ($k \in \mathbf{Z}$, $0 \leqslant i \leqslant d-1$), they should satisfy some associativity conditions that we will not explain.

In the case where $d = 1$ (which is the one considered in (3.3)), we can thus say that the category of graded $\mathscr{S}$-modules (resp. quasi-coherent $\mathscr{S}$-modules if X is a prescheme and $\mathscr{S}$ is quasi-coherent) is *equivalent* to the category of arbitrary $\mathscr{S}_0$-modules (resp. quasi-coherent $\mathscr{S}_0$-modules); it is in this way that we can think of the results of this paragraph as generalising those of §3. Furthermore, we see that, under suitable finiteness conditions, the results of this paragraph (along with (8.14.11)) reduces, in some sense, the study of quasi-coherent graded algebras on a prescheme, and graded modules "modulo **(TN)**" on such algebras, to the study of the particular case where the algebras in question are *periodic* (and where condition **(TN)** for $\mathscr{M}$ (3.4.2) thus implies that $\mathscr{M} = 0$).

Remark (8.14.13). — Under the hypotheses of (8.14.1), let d be an integer > 0; we have defined a canonical Y-isomorphism h from X to $X^{(d)} = \operatorname{Proj}(\mathscr{S}^{(d)})$ (3.1.8). For every quasi-coherent graded **II | 202**
$\mathscr{S}$-module $\mathscr{M}$ and every integer k such that $0 \leqslant k \leqslant d-1$, we also have (with the notation of (3.1.1)) a canonical h-isomorphism

(8.14.13.1)
$$(\mathscr{P}\mathit{roj}(\mathscr{M}))^{(d,k)} \xleftarrow{\sim} \mathscr{P}\mathit{roj}(\mathscr{M}^{(d,k)}).$$

Suppose, first of all, that $Y = \operatorname{Spec}(A)$ is affine, $\mathscr{S} = \widetilde{S}$, and $\mathscr{M} = \widetilde{M}$, where S is a positively-graded A-algebra, and M a graded S-module. We know, for every $f \in S_e$ ($e > 0$), that h sends $D_+(f)$ to $D_+(f^d)$, and corresponds to the canonical isomorphism $S_{(f^d)} \to S_{(f)}$ (2.2.2). The restriction of (8.14.13.1) to $D_+(f^d)$ then corresponds to the canonical di-isomorphism $M_{f^d} \to M_f$ restricted to the elements of M_{f^d} whose degree is congruent to k (modulo d). We leave to the reader the task of showing that these isomorphisms are compatible with passing from f to some homogeneous multiple fg, and then that there is an analogous compatibility with passing from S to a graded A'-algebra $S' = S \otimes_A A'$, where A' is some A-algebra. In particular, this gives us an h-isomorphism

(8.14.13.2)
$$(\mathscr{S}^{(d)})_{X^{(d)}} \xrightarrow{\sim} (\mathscr{S}_X)^{(d)}$$

that respects the multiplicative structures of both the source and the target, and that, thanks to (8.14.13.1), becomes an h-di-isomorphism from a graded $(\mathscr{S}^{(d)})_{X^{(d)}}$-module to a graded $(\mathscr{S}_X)^{(d)}$-module. Similarly, we have an h-isomorphism

(8.14.13.3)
$$\mathscr{P}\mathit{roj}_0(\mathscr{S}^{(d,k)}(n)) \xrightarrow{\sim} \mathscr{O}_X(nd+k),$$

which completes the result of (3.2.9, ii).

The isomorphism in (8.14.13.1) immediately induces an isomorphism of graded $\mathscr{S}^{(d)}$-modules

(8.14.13.4)
$$\mathbf{\Gamma}_*^{(d)}(\mathscr{P}\mathit{roj}(\mathscr{M}^{(d,k)})) \xrightarrow{\sim} \mathbf{\Gamma}_*((\mathscr{P}\mathit{roj}(\mathscr{M}))^{(d,k)})$$

where $\mathbf{\Gamma}_*^{(d)}$ corresponds to the structure morphism $q^{(d)} : X^{(d)} \to Y$; it can be immediately verified that the canonical homomorphism α (8.14.5.4), and the analogous homomorphism $\alpha^{(d)}$ for $X^{(d)}$, make the following diagram commute:

(8.14.13.5)
$$\begin{tikzcd} & \mathscr{M}^{(d,k)} \arrow[dl, "\alpha^{(d)}"'] \arrow[dr, "\alpha"] & \\ \mathbf{\Gamma}_*^{(d)}(\mathscr{P}\mathit{roj}(\mathscr{M}^{(d,k)})) \arrow[rr, "\sim"] & & \mathbf{\Gamma}_*((\mathscr{P}\mathit{roj}(\mathscr{M}))^{(d,k)}) \end{tikzcd}$$

where we proceed by supposing that Y is affine and then calculating the restrictions of the images under $\alpha^{(d)}$ and α of some single element of $M^{(d,k)}$ to the open subsets $D_+(f^d)$ and $D_+(f)$ (using the same notation as above).

Proposition (8.14.14). — *Let Y be a quasi-compact prescheme, $\mathscr{S}$ a quasi-coherent graded $\mathscr{O}_Y$-algebra of finite type, and $\mathscr{M}$ a quasi-coherent graded $\mathscr{S}$-module satisfying condition* **(TF)***; let $X = \operatorname{Proj}(\mathscr{S})$. Then $\mathscr{S}_X$ is a periodic graded $\mathscr{O}_X$-algebra (8.14.12), and there exists some period d of $\mathscr{S}_X$ such that the $(\mathscr{P}roj(\mathscr{M}))^{(d,k)}$ $(0 \leqslant k \leqslant d-1)$ are $(\mathscr{S}_X)^{(d)}$-modules of finite type.* II | 203

Proof. Indeed, (3.1.10) proves that there exists some d such that $\mathscr{S}^{(d)}$ is generated by $\mathscr{S}_d = (\mathscr{S}^{(d)})_1$, with the latter being an $\mathscr{S}_0$-module of finite type. To prove the first claim, we can thus, by (8.14.13.2), restrict to the case where $d = 1$, and the proposition then follows from (3.2.7). Furthermore, taking (8.14.13.1) into account, the second claim is a consequence of (2.1.6, iii) and (3.4.3). ☐

CHAPTER III

Cohomological study of coherent sheaves (EGA III)

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SUMMARY

This chapter gives the fundamental theorems concerning the cohomology of coherent algebraic sheaves, with the exception of theorems explaining the theory of residues (duality theorems), which will be the subject of a later chapter. Amongst all those included here, there are essentially six fundamental theorems, and each one is the subject of one of the first six chapters. These results will prove to be essential tools in all that follows, even in questions which are not truly cohomological in their nature; the reader will see the first such examples starting from §4. §7 gives some more technical results, but ones which are constantly used in applications. Finally, in §§8–11, we will develop certain results, related to the duality of coherent sheaves, that are particularly important for applications, and which can be explained even before the introduction of the full general theory of residues.

The content of §§1 and 2 is due to J.-P. Serre, and the reader will observe that we have had only to follow (FAC). §8 and 9 are equally inspired by (FAC) (the changes necessitated by the different contexts, however, being less evident). Finally, as we said in the Introduction, §4 should be considered as the formalisation, in modern language, of the fundamental "invariance theorem" of Zariski's "theory of holomorphic functions".

We draw attention to the fact that the results of n°3.4 (and the preliminary propositions of (**0**, 13.4 to 13.7)) will not be used in what follows Chapter III, and can thus be skipped in a first reading.

§1. COHOMOLOGY OF AFFINE SCHEMES

1.1. Review of the exterior algebra complex

(1.1.1). Let A be a ring, $\mathbf{f} = (f_i)_{1\leqslant i\leqslant r}$ a system of r elements of A. The *exterior algebra complex* $K_\bullet(\mathbf{f})$ corresponding to $\mathbf{f}$ is a chain complex (G, I, 2.2) defined in the following way: the graded A-module $K_\bullet(\mathbf{f})$ is equal to the *exterior algebra* $\wedge(A^r)$, graded in the usual way, and the boundary map is the *interior multiplication* $i_\mathbf{f}$ by $\mathbf{f}$ considered as an element of the dual $(A^r)^\vee$; we recall that $i_\mathbf{f}$ is an

[1]EGA IV does not depend on §§8–11, and will probably be published before these chapters. *[Trans.] These last four chapters were never published.*

antiderivation of degree -1 of $\wedge(A^r)$, and if $(\mathbf{e}_i)_{1\leqslant i\leqslant r}$ is the canonical basis of A^r, then we have $i_{\mathbf{f}}(\mathbf{e}_i) = f_i$; the verification of the condition $i_{\mathbf{f}} \circ i_{\mathbf{f}} = 0$ is immediate.

An equivalent definition is the following: for each i, we consider a chain complex $K_\bullet(f_i)$ defined as follows: $K_0(f_i) = K_1(f_i) = A$, $K_n(f_i) = 0$ for $n \neq 0, 1$: the boundary map is defined by the condition that $d_1 : A \to A$ is *multiplication by* f_i. We then take $K_\bullet(\mathbf{f})$ to be the *tensor product* $K_\bullet(f_1) \otimes K_\bullet(f_2) \otimes \cdots \otimes K_\bullet(f_r)$ (G, I, 2.7) with its total degree; the verification of the isomorphism from this complex to the complex defined above is immediate.

(1.1.2). For every A-module M, we define the *chain complex*

$$K_\bullet(\mathbf{f}, M) = K_\bullet(\mathbf{f}) \otimes_A M \tag{1.1.2.1}$$

and the *cochain complex* (G, I, 2.2)

$$K^\bullet(\mathbf{f}, M) = \mathrm{Hom}_A(K_\bullet(\mathbf{f}, M). \tag{1.1.2.2}$$

If g is a k-cochain of this latter complex, and if we set

$$g(i_1, \ldots, i_k) = g(\mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_k}),$$

then g identifies with an *alternating* map from $[1, r]^k$ to M, and it follows from the above definitions that we have

$$d^k g(i_1, i_2, \ldots, i_{k+1}) = \sum_{h=1}^{k+1} (-1)^{h-1} f_{i_h} g(i_1, \ldots, \widehat{i_h}, \ldots, i_{k+1}). \tag{1.1.2.3}$$

(1.1.3). From the above complexes, we deduce as usual the *homology and cohomology A-modules* (G, I, 2.2) **III** | 83

$$\mathrm{H}_\bullet(\mathbf{f}, M) = \mathrm{H}_\bullet(K_\bullet(\mathbf{f}, M)), \tag{1.1.3.1}$$

$$\mathrm{H}^\bullet(\mathbf{f}, M) = \mathrm{H}^\bullet(K^\bullet(\mathbf{f}, M)). \tag{1.1.3.2}$$

We define an *A-isomorphism* $K_\bullet(\mathbf{f}, M) \simeq K^\bullet(\mathbf{f}, M)$ by sending each chain $z = \sum(\mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_k}) \otimes z_{i_1,\ldots,i_k}$ to the cochain g_z such that $g_z(j_1, \ldots, j_{r-k}) = \varepsilon z_{i_1,\ldots,i_k}$, where $(j_h)_{1\leqslant h\leqslant r-k}$ is the strictly increasing sequence complementary to the strictly increasing sequence $(i_h)_{1\leqslant h\leqslant k}$ in $[1, r]$ and $\varepsilon = (-1)^\nu$, where ν is the number of inversions of the permutation $i_1, \ldots, i_k, j_1, \ldots, j_{r-k}$ of $[1, r]$. We verify that $g_{dz} = d(g_z)$, which gives an isomorphism

$$\mathrm{H}^i(\mathbf{f}, M) \simeq \mathrm{H}_{r-i}(\mathbf{f}, M) \text{ for } 0 \leqslant i \leqslant r. \tag{1.1.3.3}$$

In this chapter, we will especially consider the cohomology modules $\mathrm{H}^\bullet(\mathbf{f}, M)$.

For a given $\mathbf{f}$, it is immediate (G, I, 2.1) that $M \mapsto \mathrm{H}^\bullet(\mathbf{f}, M)$ is a *cohomological functor* (T, II, 2.1) from the category of A-modules to the category of graded A-modules, zero in degrees < 0 and $> r$. In addition, we have

$$\mathrm{H}^0(\mathbf{f}, M) = \mathrm{Hom}_A(A/(\mathbf{f}), M), \tag{1.1.3.4}$$

denoting by $(\mathbf{f})$ the ideal of A generated by $f_1, \ldots, f_r$; this follows immediately from (1.1.2.3), and it is clear that $\mathrm{H}^0(\mathbf{f}, M)$ identifies with the submodule of M *killed by* $(\mathbf{f})$. Similarly, we have by (1.1.2.3) that

$$\mathrm{H}^r(\mathbf{f}, M) = M/\left(\sum_{i=1}^r f_i M\right) = (A/(\mathbf{f})) \otimes_A M. \tag{1.1.3.5}$$

We will use the following known result, which we will recall a proof of to be complete:

Proposition (1.1.4). — *Let A be a ring, $\mathbf{f} = (f_i)_{1\leqslant i\leqslant r}$ a finite family of elements of A, and M an A-module. If, for $1 \leqslant i \leqslant r$, the scaling $z \mapsto f_i \cdot z$ on $M_{i-1} = M/(f_1 M + \cdots + f_{i-1} M)$ is injective, then we have $\mathrm{H}^i(\mathbf{f}, M) = 0$ for $i \neq r$.*

It suffices to prove that $\mathrm{H}_i(\mathbf{f}, M) = 0$ for all $i > 0$ according to (1.1.3.3). We argue by induction on r, the case $r = 0$ being trivial. Set $\mathbf{f}' = (f_i)_{1\leqslant i\leqslant r-1}$; this family satisfies the conditions in the statement, so if we set $L_\bullet = K_\bullet(\mathbf{f}', M)$, then we have $\mathrm{H}_i(L_\bullet) = 0$ for $i > 0$ by hypothesis, and $\mathrm{H}_0(L_\bullet) = M_{r-1}$ by virtue of (1.1.3.3) and (1.1.3.5). To abbreviate, set $K_\bullet = K_\bullet(f_r) = K_0 \oplus K_1$, with $K_0 = K_1 = A$, $d_1 : K_1 \to K_0$ multiplication by f_r; we have by definition (1.1.1) that $K_\bullet(\mathbf{f}, M) = K_\bullet \otimes_A L_\bullet$. We have the following lemma:

Lemma (1.1.4.1). — *Let $K_\bullet$ be a chain complex of free A-modules, zero except in dimensions* 0 *and* 1*. For every chain complex $L_\bullet$ of A-modules, we have an exact sequence*

$$0 \longrightarrow \mathrm{H}_0(K_\bullet \otimes \mathrm{H}_p(L_\bullet)) \longrightarrow \mathrm{H}_p(K_\bullet \otimes L_\bullet) \longrightarrow \mathrm{H}_1(K_\bullet \otimes \mathrm{H}_{p-1}(L_\bullet)) \longrightarrow 0$$

for every index p.

This is a particular case of an exact sequence of low-order terms of the Künneth spectral sequence (M, XVII, 5.2 (a) and G, I, 5.5.2); it can be proved directly as follows. Consider K_0 and K_1 as chain complexes (zero in dimensions $\neq 0$ and $\neq 1$ respectively); we then have an exact sequence of complexes **III | 84**

$$0 \longrightarrow K_0 \otimes L_\bullet \longrightarrow K_\bullet \otimes L_\bullet \longrightarrow K_1 \otimes L_\bullet \longrightarrow 0,$$

to which we can apply the exact sequence in homology

$$\cdots \longrightarrow \mathrm{H}_{p+1}(K_1 \otimes L_\bullet) \xrightarrow{\partial} \mathrm{H}_p(K_0 \otimes L_\bullet) \longrightarrow \mathrm{H}_p(K_\bullet \otimes L_\bullet) \longrightarrow \mathrm{H}_p(K_1 \otimes L_\bullet) \xrightarrow{\partial} \mathrm{H}_{p-1}(K_0 \otimes L_\bullet) \longrightarrow \cdots .$$

But it is evident that $\mathrm{H}_p(K_0 \otimes L_\bullet) = K_0 \otimes \mathrm{H}_p(L_\bullet)$ and $\mathrm{H}_p(K_1 \otimes L_\bullet) = K_1 \otimes \mathrm{H}_{p-1}(L_\bullet)$ for all p; in addition, we verify immediately that the operator $\partial : K_1 \otimes \mathrm{H}_p(L_\bullet) \to K_0 \otimes \mathrm{H}_p(L_\bullet)$ is none other than $d_1 \otimes 1$; the lemma thus follows from the above exact sequence and the definition of $\mathrm{H}_0(K_\bullet \otimes \mathrm{H}_p(L_\bullet))$ and $\mathrm{H}_1(K_\bullet \otimes \mathrm{H}_{p-1}(L_\bullet))$.

The lemma having been established, the end of the proof of Proposition (1.1.4) is immediate: the induction hypothesis of Lemma (1.1.4.1) gives $\mathrm{H}_p(K_\bullet \otimes L_\bullet) = 0$ for $p \geqslant 2$; in addition if we show that $\mathrm{H}_1(K_\bullet, \mathrm{H}_0(L_\bullet)) = 0$, then we also deduce from Lemma (1.1.4.1) that $\mathrm{H}_1(K_\bullet \otimes L_\bullet) = 0$; but by definition, $\mathrm{H}_1(K_\bullet, \mathrm{H}_0(L_\bullet))$ is none other than the kernel of the scaling $z \mapsto f_r \cdot z$ on M_{r-1}, and as by hypothesis this kernel is zero, this finishes the proof.

(1.1.5). Let $\mathbf{g} = (g_i)_{1 \leqslant i \leqslant r}$ be a second sequence of r elements of A, and set $\mathbf{fg} = (f_i g_i)_{1 \leqslant i \leqslant r}$. We can define a canonical homomorphism of complexes

$$\phi_{\mathbf{g}} : K_\bullet(\mathbf{fg}) \longrightarrow K_\bullet(\mathbf{f}) \tag{1.1.5.1}$$

as the canonical extension to the exterior algebra $\wedge(A^r)$ of the A-linear map $(x_1, \ldots, x_r) \mapsto (g_1 x_1, \ldots, g_r x_r)$ from A^r to itself. To see that we have a homomorphism of complexes, it suffices to note, in general, that if $u : E \to F$ is an A-linear map, and if $\mathbf{x} \in F^\vee$ and $\mathbf{y} = {}^t u(\mathbf{x}) \in E^\vee$, then we have the formula

$$(\wedge u) \circ i_{\mathbf{y}} = i_{\mathbf{x}} \circ (\wedge u); \tag{1.1.5.2}$$

indeed, the two elements are antiderivations of $\wedge F$, and it suffices to check that they coincide on F, which follows immediately from the definitions.

When we identify $K_\bullet(\mathbf{f})$ with the tensor product of the $K_\bullet(f_i)$ (1.1.1), $\phi_{\mathbf{g}}$ is the tensor product of the ϕ_{g_i}, where ϕ_{g_i} is the identity in degree 0 and multiplication by g_i in degree 1.

(1.1.6). In particular, for every pair of integers m and n such that $0 \leqslant n \leqslant m$, we have homomorphisms of complexes

$$\phi_{\mathbf{f}^{m-n}} : K_\bullet(\mathbf{f}^m) \longrightarrow K_\bullet(\mathbf{f}^n) \tag{1.1.6.1}$$

and as a result, homomorphisms

$$\phi_{\mathbf{f}^{m-n}} : K^\bullet(\mathbf{f}^n, M) \longrightarrow K^\bullet(\mathbf{f}^m, M), \tag{1.1.6.2}$$

$$\phi_{\mathbf{f}^{m-n}} : \mathrm{H}^\bullet(\mathbf{f}^n, M) \longrightarrow \mathrm{H}^\bullet(\mathbf{f}^m, M). \tag{1.1.6.3}$$

The latter homomorphisms evidently satisfy the transitivity condition $\phi_{\mathbf{f}^{m-p}} = \phi_{\mathbf{f}^{m-n}} \circ \phi_{\mathbf{f}^{n-p}}$ for $p \leqslant n \leqslant m$; they therefore define two *inductive systems* of A-modules; we set **III | 85**

$$\mathrm{C}^\bullet((\mathbf{f}), M) = \varinjlim_n K^\bullet(\mathbf{f}^n, M), \tag{1.1.6.4}$$

$$\mathrm{H}^\bullet((\mathbf{f}), M) = \mathrm{H}^\bullet(\mathrm{C}^\bullet((\mathbf{f}), M)) = \varinjlim_n \mathrm{H}^\bullet(\mathbf{f}^n, M), \tag{1.1.6.5}$$

the last equality following from the fact that passing to the inductive limit commutes with the functor $\mathrm{H}^\bullet$ (G, I, 2.1). We will later see (1.4.3) that $\mathrm{H}^\bullet((\mathbf{f}), M)$ does not depend on the *ideal* $(\mathbf{f})$ of A (and similarly on the $(\mathbf{f})$-pre-adic topology on A), which justifies the notations.

It is clear that $M \mapsto C^\bullet((\mathbf{f}), M)$ is an exact A-linear functor, and $M \mapsto \mathrm{H}^\bullet((\mathbf{f}), M)$ is a cohomological functor.

(1.1.7). Set $\mathbf{f} = (f_i) \in A^r$ and $\mathbf{g} = (g_i) \in A^r$; denote by $e_\mathbf{g}$ the left multiplication by the vector $\mathbf{g} \in A^r$ on the exterior algebra $\wedge(A^r)$; we know that we have the *homotopy formula*

$$i_\mathbf{f} e_\mathbf{g} + e_\mathbf{g} i_\mathbf{f} = \langle \mathbf{g}, \mathbf{f} \rangle 1 \tag{1.1.7.1}$$

in the A-module A^r (1 denotes the identity automorphism of A^r); this relation also implies that *in the complex $K_\bullet(\mathbf{f})$* we have

$$d e_\mathbf{g} + e_\mathbf{g} d = \langle \mathbf{g}, \mathbf{f} \rangle 1. \tag{1.1.7.2}$$

If the ideal $(\mathbf{f})$ is equal to A, then there exists a $\mathbf{g} \in A^r$ such that $\langle \mathbf{g}, \mathbf{f} \rangle = \sum_{i=1}^r g_i f_i = 1$. As a result (G, I, 2.4):

Proposition (1.1.8). — *Suppose that the ideal $(\mathbf{f})$ generated by the f_i is equal to A. Then the complex $K_\bullet(\mathbf{f})$ is homotopically trivial, and so are the complexes $K_\bullet(\mathbf{f}, M)$ and $K^\bullet(\mathbf{f}, M)$ for every A-module M.*

Corollary (1.1.9). — *If $(\mathbf{f}) = A$, then we have $\mathrm{H}^\bullet(\mathbf{f}, M) = 0$ and $\mathrm{H}^\bullet((\mathbf{f}), M) = 0$ for every A-module M.*

PROOF. Indeed, we then have $(\mathbf{f}^n) = A$ for all n. ☐

Remark (1.1.10). — With the same notations as above, set $X = \mathrm{Spec}(A)$ and Y the closed subprescheme of X defined by the ideal $(\mathbf{f})$. We will prove in §9 that $\mathrm{H}^\bullet((\mathbf{f}), M)$ is isomorphic to the cohomology $\mathrm{H}^\bullet_Y(X, \widetilde{M})$ corresponding to the antifilter Φ of closed subsets of Y (T, 3.2). We will also show that Proposition (1.2.3) applied to X and to $\mathscr{F} = \widetilde{M}$ is a particular case of an exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^p_Y(X, \mathscr{F}) \longrightarrow \mathrm{H}^p(X, \mathscr{F}) \longrightarrow \mathrm{H}^p(X - Y, \mathscr{F}) \longrightarrow \mathrm{H}^{p+1}_Y(X, \mathscr{F}) \longrightarrow \cdots.$$

1.2. Čech cohomology of an open cover

Notation (1.2.1). — In this section, we denote:

(1) X a prescheme;
(2) $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module; **III** | 86
(3) $A = \Gamma(X, \mathscr{O}_X)$, $M = \Gamma(X, \mathscr{F})$;
(4) $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant r}$ a finite system of elements of A;
(5) $U_i = X_{f_i}$, the open set (**0**, 5.5.2) of the $x \in X$ such that $f_i(x) \neq 0$;
(6) $U = \bigcup_{i=1}^r U_i$;
(7) $\mathfrak{U}$ the cover $(U_i)_{1 \leqslant i \leqslant r}$ of U.

(1.2.2). Suppose that X is either a prescheme whose underlying space is *Noetherian* or a *scheme* whose underlying space is *quasi-compact*. We then know (**I**, 9.3.3) that we have $\Gamma(U_i, \mathscr{F}) = M_{f_i}$. We set

$$U_{i_0 i_1 \cdots i_p} = \bigcap_{k=0}^{p} U_{i_k} = X_{f_{i_0} f_{i_1} \cdots f_{i_p}}$$

(**0**, 5.5.3); so we also have

$$\Gamma(U_{i_0 i_1 \cdots i_p}, \mathscr{F}) = M_{f_{i_0} f_{i_1} \cdots f_{i_p}}. \tag{1.2.2.1}$$

We have (**0**, 1.6.1) that $M_{f_{i_0} f_{i_1} \cdots f_{i_p}}$ identifies with the inductive limit $\varinjlim_n M^{(n)}_{i_0 i_1 \cdots i_p}$, where the inductive system is formed by the $M^{(n)}_{i_0 i_1 \cdots i_p} = M$, the homomorphisms $\phi_{nm} : M^{(m)}_{i_0 i_1 \cdots i_p} \to M^{(n)}_{i_0 i_1 \cdots i_p}$ being multiplication by $(f_{i_0} f_{i_1} \cdots f_{i_p})^{n-m}$ for $m \leqslant n$. We denote by $C^p_n(M)$ the set of *alternating* maps from $[1, r]^{p+1}$ to M (for all n); these A-modules also form an inductive system with respect to the ϕ_{nm}. If $C^p(\mathfrak{U}, \mathscr{F})$ is the group of *alternating* Čech p-cochains relative to the cover $\mathfrak{U}$, with coefficients in $\mathscr{F}$ (G, II, 5.1), then it follows from the above that we can write

$$C^p(\mathfrak{U}, \mathscr{F}) = \varinjlim_n C^p_n(M). \tag{1.2.2.2}$$

With the notations of (1.1.2), $C_n^p(M)$ identifies with $K^{p+1}(\mathbf{f}^n, M)$, and the map ϕ_{nm} identifies with the map $\phi_{\mathbf{f}^{n-m}}$ defined in (1.1.6). We thus have, for every $p \geqslant 0$, a canonical functorial isomorphism

(1.2.2.3) $$C^p(\mathfrak{U}, \mathscr{F}) \simeq C^{p+1}((\mathbf{f}), M).$$

In addition, the formula (1.1.2.3) and the definition of the cohomology of a cover (G, II, 5.1) shows that the isomorphisms (1.2.2.3) are compatible with the coboundary maps.

Proposition (1.2.3). — *If X is a prescheme whose underlying space in Noetherian or a scheme whose underlying space is quasi-compact, then there exists a canonical functorial isomorphism in $\mathscr{F}$*

(1.2.3.1) $$\mathrm{H}^p(\mathfrak{U}, \mathscr{F}) \simeq \mathrm{H}^{p+1}((\mathbf{f}), M) \text{ for } p \geqslant 1.$$

In addition, we have a functorial exact sequence in $\mathscr{F}$

(1.2.3.2) $$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(\mathfrak{U}, \mathscr{F}) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0.$$

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Proof. The isomorphisms (1.2.3.1) are immediate consequences of what we saw in (1.2.2). On the other hand, we have $C^0(\mathfrak{U}, \mathscr{F}) = C^1((\mathbf{f}), M)$; as a result, $\mathrm{H}^0(\mathfrak{U}, \mathscr{F})$ identifies with the subgroup of 1-cocycles of $C^1((\mathbf{f}), M)$; as $M = C^0((\mathbf{f}), M)$, the exact sequence (1.2.3.2) is none other than the one given by the definition of the cohomology groups $\mathrm{H}^0((\mathbf{f}), M)$ and $\mathrm{H}^1((\mathbf{f}), M)$. □

Corollary (1.2.4). — *Suppose that the X_{f_i} are quasi-compact and that there exists $g_i \in \Gamma(U, \mathscr{F})$ such that $\sum_i g_i(f_i|U) = 1|U$. Then for every quasi-coherent $(\mathscr{O}_X|U)$-module $\mathscr{G}$, we have $\mathrm{H}^p(\mathfrak{U}, \mathscr{G}) = 0$ for $p > 0$; if in addition $U = X$, then the canonical homomorphism (1.2.3.2) $M \to \mathrm{H}^0(\mathfrak{U}, \mathscr{F})$ is bijective.*

Proof. As by hypothesis the $U_i = X_{f_i}$ are quasi-compact, so is U, and we can reduce to the case where $U = X$; the hypothesis then implies that $\mathrm{H}^p((\mathbf{f}), M) = 0$ for all $p \geqslant 0$ (1.1.9). The corollary then follows immediately from (1.2.3.1) and (1.2.3.2). □

We note that since $\mathrm{H}^0(\mathfrak{U}, \mathscr{F}) = \mathrm{H}^0(U, \mathscr{F})$ (G, II, 5.2.2), we have again proved (**I**, 1.3.7) as a special case.

Remark (1.2.5). — Suppose that X is an *affine scheme*; then the $U_i = X_{f_i} = D(f_i)$ are affine open sets, as well as the $U_{i_0 i_1 \cdots i_p}$ (but U is not necessarily affine). In this case, the functors $\Gamma(X, \mathscr{F})$ and $\Gamma(U_{i_0 i_1 \cdots i_p}, \mathscr{F})$ are exact in $\mathscr{F}$ (**I**, 1.3.11). If we have an exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of quasi-coherent $\mathscr{O}_X$-modules, then the sequence of complexes

$$0 \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{F}') \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{F}) \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{F}'') \longrightarrow 0$$

is exact, and thus gives an exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^p(\mathfrak{U}, \mathscr{F}) \longrightarrow \mathrm{H}^p(\mathfrak{U}, \mathscr{F}) \longrightarrow \mathrm{H}^p(\mathfrak{U}, \mathscr{F}'') \xrightarrow{\partial} \mathrm{H}^{p+1}(\mathfrak{U}, \mathscr{F}') \longrightarrow \cdots.$$

On the other hand, if we set $M' = \Gamma(X, \mathscr{F}')$ and $M'' = \Gamma(X, \mathscr{F}'')$, then the sequence $0 \to M' \to M \to M'' \to 0$ is exact; as $C^\bullet((\mathbf{f}), M)$ is an exact functor in M, we also have the exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^p((\mathbf{f}), M') \longrightarrow \mathrm{H}^p((\mathbf{f}), M) \longrightarrow \mathrm{H}^p((\mathbf{f}), M'') \xrightarrow{\partial} \mathrm{H}^{p+1}((\mathbf{f}), M') \longrightarrow \cdots.$$

This being so, as the diagram

$$\begin{tikzcd}
0 \arrow[r] & C^\bullet(\mathfrak{U}, \mathscr{F}') \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U}, \mathscr{F}) \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U}, \mathscr{F}'') \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & C^\bullet((\mathbf{f}), M') \arrow[r] & C^\bullet((\mathbf{f}), M) \arrow[r] & C^\bullet((\mathbf{f}), M'') \arrow[r] & 0
\end{tikzcd}$$

is commutative, we conclude that the diagrams **III | 88**

(1.2.5.1) $$\begin{tikzcd}
\mathrm{H}^p(\mathfrak{U}, \mathscr{F}'') \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^{p+1}(\mathfrak{U}, \mathscr{F}') \arrow[d] \\
\mathrm{H}^{p+1}((\mathbf{f}), M'') \arrow[r, "\partial"] & \mathrm{H}^{p+2}((\mathbf{f}), M')
\end{tikzcd}$$

are commutative for all p (G, I, 2.1.1).

1.3. Cohomology of an affine scheme

Theorem (1.3.1). — *Let X be an affine scheme. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $\mathrm{H}^p(X, \mathscr{F}) = 0$ for all $p > 0$.*

PROOF. Let $\mathfrak{U}$ be a finite cover of X by the affine open sets $X_{f_i} = D(f_i)$ $(1 \leqslant i \leqslant r)$; we then know that the ideal of $A = \Gamma(X, \mathscr{O}_X)$ generated by the f_i is equal to A. We thus conclude from Corollary (1.2.4) that we have $\mathrm{H}^p(\mathfrak{U}, \mathscr{F}) = 0$ for $p > 0$. As there are finite covers of X by affine open sets which are arbitrarily fine (I, 1.1.10), the definition of Čech cohomology (G, II, 5.8) shows that we also have $\check{\mathrm{H}}^p(X, \mathscr{F}) = 0$ for $p > 0$. But this also applies to every prescheme X_f for $f \in A$ (I, 1.3.6), hence $\check{\mathrm{H}}^p(X_f, \mathscr{F}) = 0$ for $p > 0$. As we have $X_f \cap X_g = X_{fg}$, we deduce that we also have $\mathrm{H}^p(X, \mathscr{F}) = 0$ for all $p > 0$, by virtue of (G, II, 5.9.2). □

Corollary (1.3.2). — *Let Y be a prescheme, $f : X \to Y$ an affine morphism (II, 1.6.1). For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $\mathrm{R}^q f_*(\mathscr{F}) = 0$ for $q > 0$.*

PROOF. By definition $\mathrm{R}^q f_*(\mathscr{F})$ is the $\mathscr{O}_Y$-module associated to the presheaf $U \mapsto \mathrm{H}^q(f^{-1}(U), \mathscr{F})$, where U varies over the open subsets of Y. But the affine open sets form a basis for Y, and for such an open set U, $f^{-1}(U)$ is affine (II, 1.3.2), hence $\mathrm{H}^q(f^{-1}(U), \mathscr{F}) = 0$ by Theorem (1.3.1), which proves the corollary. □

Corollary (1.3.3). — *Let Y be a prescheme, $f : X \to Y$ an affine morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{H}^p(Y, f_*(\mathscr{F})) \to \mathrm{H}^p(X, \mathscr{F})$ (**0**, 12.1.3.1) is bijective for all p.*

PROOF. It suffices (by (**0**, 12.1.7)) to show that the edge homomorphisms $''E_2^{p0} = \mathrm{H}^p(Y, f_*(\mathscr{F})) \to \mathrm{H}^p(X, \mathscr{F})$ of the second spectral sequence of the composite functor Γf_* are bijective. But the E_2-term of this spectral sequence is given by $''E_2^{pq} = \mathrm{H}^p(Y, \mathrm{R}^q f_*(\mathscr{F}))$ (G, II, 4.17.1), so it follows from Corollary (1.3.2) that $''E_2^{pq} = 0$ for $q > 0$, and the spectral sequence degenerates; hence our assertion (**0**, 11.1.6). □

Corollary (1.3.4). — *Let $f : X \to Y$ be an affine morphism, $g : Y \to Z$ a morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{R}^p g_*(f_*(\mathscr{F})) \to \mathrm{R}^p (g \circ f)_*(\mathscr{F})$ (**0**, 12.2.5.1) is bijective for all p.* **III | 89**

PROOF. It suffices to note that, according to Corollary (1.3.3), for every affine open subset W of Z, the canonical homomorphism $\mathrm{H}^p(g^{-1}(W), f_*(\mathscr{F})) \to \mathrm{H}^p(f^{-1}(g^{-1}(W)), \mathscr{F})$ is bijective; this proves that the homomorphism of presheaves defining the canonical homomorphism $\mathrm{R}^p g_*(f_*(\mathscr{F})) \to \mathrm{R}^p(g \circ f)_*(\mathscr{F})$ is bijective (**0**, 12.2.5). □

1.4. Application to the cohomology of arbitrary preschemes

Proposition (1.4.1). — *Let X be a scheme, $\mathfrak{U} = (U_\alpha)$ be a cover of X by affine open sets. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the cohomology modules $\mathrm{H}^\bullet(X, \mathscr{F})$ and $\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F})$ (over $\Gamma(X, \mathscr{O}_X)$) are canonically isomorphic.*

PROOF. As X is a scheme, every finite intersection V of open sets in the cover $\mathfrak{U}$ is affine (I, 5.5.6), so $\mathrm{H}^q(V, \mathscr{F}) = 0$ for $g \geqslant 1$ by Theorem (1.3.1). The proposition then follows from a theorem of Leray (G, II, 5.4.1). □

Remark (1.4.2). — We note that the result of Proposition (1.4.1) is still true when the finite intersections of the sets U_α are affine, even when we do not necessarily assume that X is a scheme.

Corollary (1.4.3). — *Let X be a scheme with quasi-compact underlying space, $A = \Gamma(X, \mathscr{O}_X)$, and $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant r}$ a finite sequence of elements of A such that the X_{f_i} (notation of (1.2.1)) are affine. Then (with the notations of (1.2.1)), for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have a canonical isomorphism which is functorial in $\mathscr{F}$*

(1.4.3.1) $$\mathrm{H}^q(U, \mathscr{F}) \simeq \mathrm{H}^{q+1}((\mathbf{f}), M) \text{ for } q \geqslant 1,$$

and an exact sequence which is functorial in $\mathscr{F}$

(1.4.3.2) $$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(U, \mathscr{F}) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0.$$

PROOF. This follows immediately from Propositions (1.4.1) and (1.2.3). □

(1.4.4). If X is an *affine scheme*, then it follows from Remark (1.2.5) and Proposition (1.4.1) that for all $q \geqslant 0$, the diagrams

$$\begin{tikzcd} \mathrm{H}^q(\mathfrak{U},\mathscr{F}'') \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^{q+1}(\mathfrak{U},\mathscr{F}') \arrow[d] \\ \mathrm{H}^{q+1}((\mathbf{f}),M'') \arrow[r, "\partial"] & \mathrm{H}^{q+2}((\mathbf{f}),M') \end{tikzcd} \tag{1.4.4.1}$$

corresponding to an exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of quasi-coherent $\mathscr{O}_X$-modules (with the notations of Remark (1.2.5)) are commutative.

Proposition (1.4.5). — *Let X be a quasi-compact scheme, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and consider the graded ring $A_\bullet = \Gamma_\bullet(\mathscr{L})$ (0, 5.4.6); then $\mathrm{H}^\bullet(\mathscr{F},\mathscr{L}) = \bigoplus_{n\in\mathbf{Z}} \mathrm{H}^\bullet(X,\mathscr{F}\otimes\mathscr{L}^{\otimes n})$ is a graded $A_\bullet$-module, and for all $f \in A_n$, we have a canonical isomorphism* III | 90

$$\mathrm{H}^\bullet(X_f,\mathscr{F}) \simeq (\mathrm{H}^\bullet(\mathscr{F},\mathscr{L}))_{(f)} \tag{1.4.5.1}$$

of $(A_\bullet)_{(f)}$-modules.

PROOF. As X is a quasi-compact scheme, we can calculate the cohomology of all the $\mathscr{O}_X$-modules $\mathscr{F}\otimes\mathscr{L}^{\otimes n}$ using the same finite cover $\mathfrak{U} = (U_i)$ consisting of the affine open sets such that the restriction $\mathscr{L}|U_i$ is isomorphic to $\mathscr{O}_X|U_i$ for each i (1.4.1). It is then immediate that the $U_i \cap X_f$ are affine open sets (I, 1.3.6), and we can thus calculate the cohomology $\mathrm{H}^\bullet(X_f,\mathscr{F}\otimes\mathscr{L}^{\otimes n})$ using the cover $\mathfrak{U}|X_f = (U_i \cap X_f)$ (1.4.1). It is immediate that for all $f \in A_n$, multiplication by f defines a homomorphism $\mathrm{C}^\bullet(\mathfrak{U},\mathscr{F}\otimes\mathscr{L}^m) \to \mathrm{C}^\bullet(\mathfrak{U},\mathscr{F}\otimes\mathscr{L}^{\otimes(m+n)})$, hence a homomorphism $\mathrm{H}^\bullet(\mathfrak{U},\mathscr{F}\otimes\mathscr{L}^{\otimes m}) \to \mathrm{H}^\bullet(\mathfrak{U},\mathscr{F}\otimes\mathscr{L}^{\otimes(m+n)})$, which establishes the first assertion. On the other hand, for a given $f \in A_n$, it follows from (I, 9.3.2) that we have an isomorphism of complexes of $(A_\bullet)_{(f)}$-modules

$$\mathrm{C}^\bullet(\mathfrak{U}|X_f,\mathscr{F}) \simeq \left(\mathrm{C}^\bullet\left(\mathfrak{U},\bigoplus_{n\in\mathbf{Z}}\mathscr{F}\otimes\mathscr{L}^{\otimes n}\right)\right)_{(f)},$$

taking into account (I, 1.3.9, ii). Passing to the cohomology of these complexes, we induce the isomorphism (1.4.5.1), recalling that the functor $M \mapsto M_{(f)}$ is exact on the category of graded $A_\bullet$-modules. □

Corollary (1.4.6). — *Suppose that the hypotheses of Proposition (1.4.5) are satisfied, and in addition suppose that $\mathscr{L} = \mathscr{O}_X$. If we set $A = \Gamma(X,\mathscr{O}_X)$, then for all $f \in A$, we have a canonical isomorphism $\mathrm{H}^\bullet(X_f,\mathscr{F}) \simeq (\mathrm{H}^\bullet(X,\mathscr{F}))_f$ of A_f-modules.*

Corollary (1.4.7). — *Let X be a quasi-compact scheme, f an element of $\Gamma(X,\mathscr{O}_X)$.*

(i) *Suppose that the open set X_f is affine. Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, every $i > 0$, and every $\xi \in \mathrm{H}^i(X,\mathscr{F})$, there exists an integer $n > 0$ such that $f^n\xi = 0$.*
(ii) *Conversely, suppose that X_f is quasi-compact and that for every quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ and every $\zeta \in \mathrm{H}^1(X,\mathscr{J})$, there exists an $n > 0$ such that $f^n\zeta = 0$. Then X_f is affine.*

PROOF.

(i) If X_f is affine, then we have $\mathrm{H}^i(X_f,\mathscr{F}) = 0$ for all $i > 0$ (1.3.1), so the assertion follows directly from Corollary (1.4.6).
(ii) By virtue of Serre's criterion (II, 5.2.1), it suffices to prove that for every quasi-coherent sheaf of ideals $\mathscr{K}$ of $\mathscr{O}_X|X_f$, we have $\mathrm{H}^1(X_f,\mathscr{K}) = 0$. As X_f is a quasi-compact open set in a quasi-compact scheme X, there exists a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ such that $\mathscr{K} = \mathscr{J}|X_f$ (I, 9.4.2). According to Corollary (1.4.6), we have $\mathrm{H}^1(X_f,\mathscr{K}) = (\mathrm{H}^1(X,\mathscr{J}))_f$, and the hypothesis implies that the right hand side is zero, hence the assertion.

□

Remark (1.4.8). — We note that Corollary (1.4.7, i) gives a simpler proof of the relation (**II**, 4.5.13.2).

Lemma (1.4.9). — *Let X be a quasi-compact scheme, $\mathfrak{U} = (U_i)_{1 \leqslant i \leqslant n}$ a finite cover of X by affine open sets, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. The complex of sheaves $\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F})$ defined by the cover $\mathfrak{U}$ (G, II, 5.2) is then a quasi-coherent $\mathscr{O}_X$-module.* III | 91

PROOF. It follows from the definitions (G, II, 5.2) that $\mathscr{C}^p(\mathfrak{U}, \mathscr{F})$ is the direct sum of the direct image sheaves of the $\mathscr{F}|U_{i_0 \cdots i_p}$ under the canonical injection $U_{i_0 \cdots i_p} \to X$. The hypothesis that X is a scheme implies that these injections are affine morphisms (I, 5.5.6), hence the $\mathscr{C}^p(\mathfrak{U}, \mathscr{F})$ are quasi-coherent (II, 1.2.6). □

Proposition (1.4.10). — *Let $u : X \to Y$ be a separated and quasi-compact morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{R}^q u_*(\mathscr{F})$ are quasi-coherent $\mathscr{O}_Y$-modules.*

PROOF. The question is local on Y, so we can suppose that Y is affine. Then X is a finite union of affine open sets U_i $(1 \leqslant i \leqslant n)$; let $\mathfrak{U}$ be the cover (U_i). In addition, as Y is a scheme, it follows from (I, 5.5.10) that for every affine open $V \subset Y$, the canonical injection $u^{-1}(V) \to X$ is an affine morphism; we conclude (Proposition (1.4.1) and (G, II, 5.2)) that we have a canonical isomorphism

(1.4.10.1) $$\mathrm{H}^\bullet(u^{-1}(V), \mathscr{F}) \simeq \mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)),$$

where we set $\mathscr{K}^\bullet = u_*(\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F}))$. According to Lemma (1.4.1) and (I, 9.2.2), $\mathscr{K}^\bullet$ is a quasi-coherent $\mathscr{O}_Y$-module; moreover, it constitutes a *complex of sheaves* since so is $\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F})$. It then follows from the definition of the cohomology $\mathscr{H}^\bullet(\mathscr{K}^\bullet)$ (G, II, 4.1) that the latter consists of quasi-coherent $\mathscr{O}_Y$-modules (I, 4.1.1). As (for V affine in Y) the functor $\Gamma(V, \mathscr{G})$ is exact in $\mathscr{G}$ on the category of quasi-coherent $\mathscr{O}_Y$-modules, we have (G, II, 4.1)

(1.4.10.2) $$\mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)) = \Gamma(V, \mathscr{H}^\bullet(\mathscr{K}^\bullet)).$$

Finally, we note that it follows from the definition of the canonical homomorphism

$$\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F}) \longrightarrow \mathrm{H}^\bullet(X, \mathscr{F}),$$

given in (G, II, 5.2), that if $V' \subset V$ is a second affine open subset of Y, then the diagram

$$\begin{tikzcd} \mathrm{H}^\bullet(u^{-1}(V), \mathscr{F}) \arrow[r, "\sim"] \arrow[d] & \mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)) \arrow[d] \\ \mathrm{H}^\bullet(u^{-1}(V'), \mathscr{F}) \arrow[r, "\sim"] & \mathrm{H}^\bullet(\Gamma(V', \mathscr{K}^\bullet)) \end{tikzcd}$$

is commutative. We thus conclude from the above that the isomorphisms (1.4.10.1) define an isomorphism of $\mathscr{O}_Y$-modules

(1.4.10.3) $$\mathrm{R}^\bullet u_*(\mathscr{F}) \simeq \mathscr{H}^\bullet(\mathscr{K}^\bullet),$$

and as a result, $\mathrm{R}^\bullet u_*(\mathscr{F})$ is quasi-coherent. □ III | 92

In addition, it follows from (1.4.10.3), (1.4.10.2), and (1.4.10.1) that:

Corollary (1.4.11). — *Under the hypotheses of Proposition (1.4.10), for every affine open set V of Y, the canonical homomorphism*

(1.4.11.1) $$\mathrm{H}^q(u^{-1}(V), \mathscr{F}) \longrightarrow \Gamma(V, \mathrm{R}^1 u_*(\mathscr{F}))$$

is an isomorphism for all $q \geqslant 0$.

Corollary (1.4.12). — *Suppose that the hypotheses of Proposition (1.4.10) are satisfied, and in addition suppose that Y is quasi-compact. Then there exists an integer $r > 0$ such that for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every integer $q > r$, we have $\mathrm{R}^q u_*(\mathscr{F}) = 0$. If Y is affine, then we can take for r an integer such that there exists a cover of X consisting of r affine open sets.*

PROOF. As we can cover Y by a finite number of affine open sets, we can reduce to proving the second assertion, by virtue of Corollary (1.4.11). If $\mathfrak{U}$ is a cover of X by r affine open sets, then we have $\mathrm{H}^q(\mathfrak{U}, \mathscr{F}) = 0$ for $q > r$, since the cochains of $C^q(\mathfrak{U}, \mathscr{F})$ are alternating; the assertion thus follows from Proposition (1.4.1). □

Corollary (1.4.13). — *Suppose that the hypotheses of Proposition (1.4.10) are satisfied, and in addition suppose that $Y = \mathrm{Spec}(A)$ is affine. Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every $f \in A$, we have*

$$\Gamma(Y_f, \mathrm{R}^q u_*(\mathscr{F})) = (\Gamma(Y, \mathrm{R}^q u_*(\mathscr{F}))_f$$

up to canonical isomorphism.

PROOF. This follows from the fact that $\mathrm{R}^q u_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_Y$-module (**I**, 1.3.7). □

Proposition (1.4.14). — *Let $f : X \to Y$ be a separated and quasi-compact morphism, $g : Y \to Z$ an affine morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{R}^p(g \circ f)_*(\mathscr{F}) \to g_*(\mathrm{R}^p f_*(\mathscr{F}))$ (**0**, 12.2.5.2) is bijective for all p.*

PROOF. For every affine open subset W of Z, $g^{-1}(W)$ is an affine open subset of Y. The homomorphism of presheaves defining the canonical homomorphism

$$\mathrm{R}^p(g \circ f)_*(\mathscr{F}) \longrightarrow g_*(\mathrm{R}^p f_*(\mathscr{F}))$$

(**0**, 12.2.5) is thus bijective by Corollary (1.4.11). □

Proposition (1.4.15). — *Let $u : X \to Y$ be a separated morphism of finite type, $v : Y' \to Y$ a flat morphism of preschemes (**0**, 6.7.1); let $u' = u_{(Y')}$, such that we have the commutative diagram*

(1.4.15.1)
$$\begin{tikzcd} X \arrow[d, "u"'] & X' = X_{(Y')} \arrow[l, "v'"'] \arrow[d, "u'"] \\ Y & Y'. \arrow[l, "v"'] \end{tikzcd}$$

Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $\mathrm{R}^q u'_*(\mathscr{F}')$ is canonically isomorphic to $\mathrm{R}^q u_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} = v^*(\mathrm{R}^q u_*(\mathscr{F}))$ for all $q \geqslant 0$, where $\mathscr{F}' = v'^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$.

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PROOF. The canonical homomorphism $\rho : \mathscr{F} \to v'_*(v'^*(\mathscr{F}))$ (**0**, 4.4.3.2) defines by functoriality a homomorphism

(1.4.15.2)
$$\mathrm{R}^q u_*(\mathscr{F}) \longrightarrow \mathrm{R}^q u_*(v'_*(\mathscr{F}')).$$

On the other hand, we have, by setting $w = u \circ v' = v \circ u'$, the canonical homomorphisms (**0** ,12.2.5.1 and 12.2.5.2)

(1.4.15.3)
$$\mathrm{R}^q u_*(v'_*(\mathscr{F}')) \longrightarrow \mathrm{R}^q w_*(\mathscr{F}') \longrightarrow v_*(\mathrm{R}^q u'_*(\mathscr{F}')).$$

Composing (1.4.15.3) and (1.4.15.2), we have a homomorphism

$$\psi : \mathrm{R}^q u_*(\mathscr{F}) \longrightarrow v_*(\mathrm{R}^q u'_*(\mathscr{F}')),$$

and finally we obtain a canonical homomorphism (whose definition does not make *any assumptions* on v)

(1.4.15.4)
$$\psi^\sharp : v^*(\mathrm{R}^q u_*(\mathscr{F})) \longrightarrow \mathrm{R}^q u'_*(\mathscr{F}'),$$

and it is necessary to prove that it is an isomorphism when v is *flat*. It is clear that the question is local on Y and Y', and we can therefore suppose that $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(B)$; we will also use the following lemma:

Lemma (1.4.15.5). — *Let $\phi : A \to B$ be a ring homomorphism, $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $f : X \to Y$ the morphism corresponding to ϕ, and M a B-module. For the $\mathscr{O}_X$-module $\widetilde{M}$ to be f-flat (**0**, 6.7.1), it is necessary and sufficient for M to be a flat A-module. In particular, for the morphism f to be flat, it is necessary and sufficient for B to be a flat A-module.*

This follows from the definition (**0**, 6.7.1) and from (**0**, 6.3.3), taking into account (**I**, 1.3.4).

This being so, it follows from (1.4.11.1) and the definitions of the homomorphisms (1.4.15.3) (cf. (**0**, 12.2.5)) that ψ then corresponds to the composite morphism

$$\mathrm{H}^q(X, \mathscr{F}) \xrightarrow{\rho_q} \mathrm{H}^q(X, v'_*(v'^*(\mathscr{F}))) \xrightarrow{\theta_q} \mathrm{H}^q(X', v'^*(v'_*(v'^*(\mathscr{F})))) \xrightarrow{\sigma_q} \mathrm{H}^q(X', v'^*(\mathscr{F})),$$

where ρ_q and σ_q are the homomorphisms in cohomology corresponding to the canonical morphisms ρ and $\sigma : v'^*(v'_*(\mathscr{G}')) \to \mathscr{G}'$, and θ_q is the ϕ-morphism (**0**, 12.1.3.1) relative to the $\mathscr{O}_X$-module $v'_*(v'^*(\mathscr{F}))$. But by the functoriality of θ_q, we have the commutative diagram

$$\begin{tikzcd} \mathrm{H}^q(X,\mathscr{F}) \arrow[r, "\rho_q"] \arrow[d, "\theta_q"'] & \mathrm{H}^q(X, v'_*(v'^*(\mathscr{F}))) \arrow[d, "\theta_q"] \\ \mathrm{H}^q(X', v'^*(\mathscr{F})) \arrow[r, "v'^*(\rho_q)"] & \mathrm{H}^q(X', v'^*(v'_*(v'^*(\mathscr{F})))), \end{tikzcd}$$

and as by definition (**0**, 4.4.3) $v'^*(\rho)$ is the inverse of σ, we see that the composite morphism considered above is finally none other than θ_q; as a result, $\psi^\sharp$ is the associated B-homomorphism $\mathrm{H}^q(X,\mathscr{F}) \otimes_A B \to \mathrm{H}^q(X',\mathscr{F}')$. As u is of finite type, X is a finite union of affine open sets U_i ($1 \leqslant i \leqslant r$); let $\mathfrak{U}$ be the cover (U_i). As v is an affine morphism, so is v' (**II**, 1.6.2, iii), and as a result the $U'_i = v'^{-1}(U_i)$ form an affine open cover $\mathfrak{U}'$ of X'. We then know (**0**, 12.1.4.2) that the diagram **III** | 94

$$\begin{tikzcd} \mathrm{H}^q(\mathfrak{U},\mathscr{F}) \arrow[r, "\theta_q"] \arrow[d] & \mathrm{H}^q(\mathfrak{U}',\mathscr{F}') \arrow[d] \\ \mathrm{H}^q(X,\mathscr{F}) \arrow[r, "\theta_q"] & \mathrm{H}^q(X',\mathscr{F}') \end{tikzcd}$$

is commutative, and the vertical arrows are isomorphisms since X and X' are schemes (**I**, 1.4.1). As a result, it suffices to prove that the canonical ϕ-morphism $\theta_q : \mathrm{H}^q(\mathfrak{U},\mathscr{F}) \to \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$ is such that the associated B-homomorphism

$$\mathrm{H}^q(\mathfrak{U},\mathscr{F}) \otimes_A B \longrightarrow \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$$

is an isomorphism. For every sequence $\mathbf{s} = (i_k)_{0 \leqslant k \leqslant p}$ of $p+1$ indices of $[1,r]$, set $U_{\mathbf{s}} = \bigcap_{k=0}^p U_{i_k}$, $U'_{\mathbf{s}} = \bigcap_{k=0}^p U'_{i_k} = v'^{-1}(U_{\mathbf{s}})$, $M_{\mathbf{s}} = \Gamma(U_{\mathbf{s}},\mathscr{F})$, and $M'_{\mathbf{s}} = \Gamma(U'_{\mathbf{s}},\mathscr{F}')$. The canonical map $M_{\mathbf{s}} \otimes_A B \to M'_{\mathbf{s}}$ is an isomorphism (**I**, 1.6.5), hence the canonical map $C^p(\mathfrak{U},\mathscr{F}) \otimes_A B \to C^p(\mathfrak{U}',\mathscr{F}')$ is an isomorphism, by which $d \otimes 1$ identifies with the coboundary map $C^p(\mathfrak{U}',\mathscr{F}') \to C^{p+1}(\mathfrak{U}',\mathscr{F}')$. As B is a *flat* A-module, it follows from the definition of the cohomology modules that the canonical map $\mathrm{H}^q(\mathfrak{U},\mathscr{F}) \otimes_A B \to \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$ is an isomorphism (**0**, 6.1.1). This result will later be generalized in §6. □

Corollary (1.4.16). — *Let A be a ring, X an A-scheme of finite type, and B an A-algebra which is faithfully flat over A. For X to be affine, it is necessary and sufficient for $X \otimes_A B$ to be.*

Proof. The condition is evidently necessary (**I**, 3.2.2); we show that it is sufficient. As X is separated over A and the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is flat, it follows from Proposition (1.4.1) that we have

(1.4.16.1) $$\mathrm{H}^i(X \otimes_A B, \mathscr{F} \otimes_A B) = \mathrm{H}^i(X,\mathscr{F}) \otimes_A B$$

for every $i \geqslant 0$ and every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$. If $X \otimes_A B$ is affine, the left hand side of (1.4.16.1) is zero for $i = 1$, hence so is $\mathrm{H}^1(X,\mathscr{F})$ since B is a faithfully flat A-module. As X is a quasi-compact scheme, we finish the proof by Serre's criterion (**II**, 5.2.1). □

Proposition (1.4.17). — *Let X be a prescheme, $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H} \to 0$ an exact sequence of $\mathscr{O}_X$-modules. If $\mathscr{F}$ and $\mathscr{H}$ are quasi-coherent, then so is $\mathscr{G}$.*

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Proof. The question is local on X, so we can suppose that $X = \mathrm{Spec}(A)$ is affine, and it then suffices to prove that $\mathscr{G}$ satisfies the conditions (d1) and (d2) of (**I**, 1.4.1) (with $V = X$). The verification of (d2) is immediate, because if $t \in \Gamma(X,\mathscr{G})$ is zero when restricted to $D(f)$, then so is its image $v(t) \in \Gamma(X,\mathscr{H})$; therefore there exists an $m > 0$ such that $f^m v(t) = v(f^m t) = 0$ (**I**, 1.4.1), and as Γ is left exact, $f^m t = u(s)$, where $s \in \Gamma(X,\mathscr{F})$; as u is injective, the restriction of s to $D(f)$

is zero, hence (I, 1.4.1) there exists an integer $n > 0$ such that $f^n s = 0$; we finally deduce that $f^{m+n} t = u(f^n s) = 0$.

We now check (d1); let $t' \in \Gamma(D(f), \mathscr{G})$; as $\mathscr{H}$ is quasi-coherent, there exists an integer m such that $f^m v(t') = v(f^m t')$ extends to a section $z \in \Gamma(X, \mathscr{H})$ (I, 1.4.1). But in virtue of Theorem (1.3.1) (or (I, 5.1.9.2)) applied to the quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the sequence $\Gamma(X, \mathscr{G}) \to \Gamma(X, \mathscr{H}) \to 0$ is exact, so there exists $t \in \Gamma(X, \mathscr{G})$ such that $z = v(t)$; we thus see that $v(f^m t' - t'') = 0$, denoting by t'' the restriction of t to $D(f)$; thus we have $f^m t' - t'' = u(s')$, where $s' \in \Gamma(D(f), \mathscr{F})$. But as $\mathscr{F}$ is quasi-coherent, there exists an integer $n > 0$ such that $f^n s'$ extends to a section $s \in \Gamma(X, \mathscr{F})$; as $f^{m+n} t' - f^n t'' = u(f^n s')$, we see that $f^{m+n} t'$ is the restriction to $D(f)$ of a section $f^n t + u(f^n s) \in \Gamma(X, \mathscr{G})$, which finishes the proof. □

§2. Cohomological study of projective morphisms

2.1. Explicit calculations of certain cohomology groups

§3. Finiteness theorem for proper morphisms

3.1. The dévissage lemma

Definition (3.1.1). — Let $\mathtt{C}$ be an abelian category. We say that a subset $\mathtt{C}'$ of the set of objects of $\mathtt{C}$ is *exact* if $0 \in \mathtt{C}'$ and if, for every exact sequence $0 \to A' \to A \to A'' \to 0$ in $\mathtt{C}$ such that two of the objects A, A', A'' are in $\mathtt{C}'$, then the third is also in $\mathtt{C}'$.

Theorem (3.1.2). — *Let X be a Noetherian prescheme; we denote by $\mathtt{C}$ the abelian category of coherent $\mathscr{O}_X$-modules. Let $\mathtt{C}'$ be an exact subset of $\mathtt{C}$, X' a closed subset of the underlying space of X. Suppose that for every closed irreducible subset Y of X', with generic point y, there exists an $\mathscr{O}_X$-module $\mathscr{G} \in \mathtt{C}'$ such that $\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1. Then every coherent $\mathscr{O}_X$-module with support contained in X' is in $\mathtt{C}'$ (and in particular, if $X' = X$, then we have $\mathtt{C}' = \mathtt{C}$).*

Proof. Consider the following property $\mathbf{P}(Y)$ of a closed subset Y of X': every coherent $\mathscr{O}_X$-module with support contained in Y is in $\mathtt{C}'$. By virtue of the principle of Noetherian induction (0, 2.2.2), we see that we can reduce to showing that *if Y is a closed subset of X' such that the property $\mathbf{P}(Y')$ is true for every closed subset Y' of Y, distinct from Y, then $\mathbf{P}(Y)$ is true.*

Therefore, let $\mathscr{F} \in \mathtt{C}$ have support contained in Y, and we show that $\mathscr{F} \in \mathtt{C}'$. Denote also by Y the closed reduced subprescheme of X having Y for its underlying space (I, 5.2.1); it is defined by a coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$. We know (I, 9.3.4) that there exists an integer $n > 0$ such that $\mathscr{J}^n \mathscr{F} = 0$; for $1 \leqslant k \leqslant n$, we thus have an exact sequence

$$0 \longrightarrow \mathscr{J}^{k-1}\mathscr{F} / \mathscr{J}^k \mathscr{F} \longrightarrow \mathscr{F} / \mathscr{J}^k \mathscr{F} \longrightarrow \mathscr{F} / \mathscr{J}^{k-1} \mathscr{F} \longrightarrow 0$$

of coherent $\mathscr{O}_X$-modules ((I, 5.3.6) and (I, 5.3.3)); as $\mathtt{C}'$ is exact, we see, by induction on k, that it suffices to show that each of the $\mathscr{F}_k = \mathscr{J}^{k-1}\mathscr{F} / \mathscr{J}^k \mathscr{F}$ is in $\mathtt{C}'$. We thus reduce to proving that $\mathscr{F} \in \mathtt{C}'$ under the additional hypothesis that $\mathscr{J}\mathscr{F} = 0$; it is equivalent to say that $\mathscr{F} = j_*(j^*(\mathscr{F}))$, where j is the canonical injection $Y \to X$. Let us now consider two cases:

(a) Y is *reducible*. Let $Y = Y' \cap Y''$, where Y' and Y'' are closed subsets of Y, distinct from Y; denote also by Y' and Y'' the closed reduced subpreschemes of X having Y and Y'' for their respective underlying spaces, which are defined respectively by sheaves of ideals $\mathscr{J}'$ and $\mathscr{J}''$ of $\mathscr{O}_X$. Set $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X / \mathscr{J}')$ and $\mathscr{F}'' = \mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X / \mathscr{J}'')$. The canonical homomorphisms $\mathscr{F} \to \mathscr{F}'$ and $\mathscr{F} \to \mathscr{F}''$ thus define a homomorphism $u : \mathscr{F} \to \mathscr{F}' \oplus \mathscr{F}''$. We show that for every $z \notin Y' \cap Y''$, the homomorphism $u_z : \mathscr{F}_z \to \mathscr{F}'_z \oplus \mathscr{F}''_z$ is *bijective*. Indeed, we have $\mathscr{J}' \cap \mathscr{J}'' = \mathscr{J}$, since the question is local and the above equality follows from ((I, 5.2.1) and (I, 1.1.5)); if $z \notin Y''$, then we have $\mathscr{J}'_z = \mathscr{J}_z$, hence $\mathscr{F}'_z = \mathscr{F}_z$ and $\mathscr{F}''_z = 0$, which establishes our assertion in this case; we reason similarly for $z \notin Y'$. As a result, the kernel and cokernel of u, which are in $\mathtt{C}$ (0, 5.3.4), have their support in $Y' \cap Y''$, and thus is in $\mathtt{C}'$ by hypothesis; for the same reason, $\mathscr{F}'$ and $\mathscr{F}''$ are in $\mathtt{C}'$, hence also $\mathscr{F}' \oplus \mathscr{F}''$, as $\mathtt{C}'$ is exact. The conclusion then follows from the consideration of the two exact sequences III | 116

$$0 \longrightarrow \operatorname{Im} u \longrightarrow \mathscr{F}' \oplus \mathscr{F}'' \longrightarrow \operatorname{Coker} u \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{Ker} u \longrightarrow \mathscr{F} \longrightarrow \operatorname{Im} u \longrightarrow 0,$$

and the hypothesis that $\mathtt{C}'$ is exact.

(b) Y is irreducible, and as a result, the subprescheme Y of X is *integral*. If y is its generic point, then we have $(\mathscr{O}_Y)_y = k(y)$, and as $j^*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module, $\mathscr{F}_y = (j^*(\mathscr{F}))_y$ is a $k(y)$-vector space of finite dimension m. By hypothesis, there is a coherent $\mathscr{O}_X$-module $\mathscr{G} \in \mathtt{C}'$ (necessarily of support Y) such that $\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1. As a result, there is a $k(y)$-isomorphism $(\mathscr{G}_y)^m \simeq \mathscr{F}_y$, which is also an $\mathscr{O}_Y$-isomorphism, and as $\mathscr{G}^m$ and $\mathscr{F}$ are coherent, there exists an open neighbourhood W of y in X and an isomorphism $\mathscr{G}^m|W \simeq \mathscr{F}|W$ (**0**, 5.2.7). Let $\mathscr{H}$ be the graph of this isomorphism, which is a coherent $(\mathscr{O}_X|W)$-submodule of $(\mathscr{G}^m \oplus \mathscr{F})|W$, canonically isomorphic to $\mathscr{G}^m|W$ and to $\mathscr{F}|W$; there thus exists a coherent $\mathscr{O}_X$-submodule $\mathscr{H}_0$ of $\mathscr{G}^m \oplus \mathscr{F}$, inducing $\mathscr{H}$ on W and 0 on $X - Y$, since $\mathscr{G}^m$ and $\mathscr{F}$ have Y for their support (**I**, 9.4.7). The restrictions $v : \mathscr{H}_0 \to \mathscr{G}^m$ and $w : \mathscr{H}_0 \to \mathscr{F}$ of the canonical projections of $\mathscr{G}^m \oplus \mathscr{F}$ are then homomorphisms of coherent $\mathscr{O}_X$-modules, which, on W and on $X - Y$, reduce to isomorphisms; in other words, the kernels and cokernels of v and w have their support in the closed set $Y - (Y \cap W)$, distinct from Y. They are in $\mathtt{C}'$; on the other hand, we have $\mathscr{G}^m \in \mathtt{C}'$ since $\mathscr{G} \in \mathtt{C}'$ and since $\mathtt{C}'$ is exact. We conclude successively, by the exactness of $\mathtt{C}'$, that $\mathscr{H}_0 \in \mathtt{C}'$, then $\mathscr{F} \in \mathtt{C}'$. Q.E.D.

□

Corollary (3.1.3). — *Suppose that the exact subset $\mathtt{C}'$ of $\mathtt{C}$ has in addition the property that any coherent direct factor of a coherent $\mathscr{O}_X$-module $\mathscr{M} \in \mathtt{C}'$ is also in $\mathtt{C}'$. In this case, the conclusion of Theorem (3.1.2) is still valid when the condition "$\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1" is replaced by $\mathscr{G}_y \neq 0$ (this is equivalent to* $\operatorname{Supp}(\mathscr{G}) = Y$*).*

Proof. The reasoning of Theorem (3.1.2) must be modified only in the case (b); now $\mathscr{G}_y$ is a $k(y)$-vector space of dimension $q > 0$, and as a result, we have an $\mathscr{O}_Y$-isomorphism $(\mathscr{G}_y)^m \simeq (\mathscr{F}_y)^q$; the end of the reasoning in Theorem (3.1.2) then proves that $\mathscr{F}^q \in \mathtt{C}'$, and the additional hypothesis on $\mathtt{C}'$ implies that $\mathscr{F} \in \mathtt{C}'$. □

3.2. The finiteness theorem: the case of usual schemes

Theorem (3.2.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism. For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathscr{O}_Y$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent for $q \geqslant 0$.*

Proof. Since the questions is local on Y, we can suppose Y Noetherian, thus X Noetherian (**I**, 6.3.7). The coherent $\mathscr{O}_X$-modules $\mathscr{F}$ for which the conclusion of Theorem (3.2.1) is true forms an *exact* subset $\mathtt{C}'$ of the category $\mathtt{C}$ of coherent $\mathscr{O}_X$-modules. Indeed, let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is an exact sequence of coherent $\mathscr{O}_X$-modules; suppose for example that $\mathscr{F}'$ and $\mathscr{F}''$ belong to $\mathtt{C}'$; we have the long exact sequence in cohomology **III | 117**

$$\mathrm{R}^{q-1} f_*(\mathscr{F}'') \xrightarrow{\partial} \mathrm{R}^q f_*(\mathscr{F}') \longrightarrow \mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \mathrm{R}^q f_*(\mathscr{F}'') \xrightarrow{\partial} \mathrm{R}^{q+1} f_*(\mathscr{F}'),$$

in which by hypothesis the outer four terms are coherent; it is the same for the middle term $\mathrm{R}^q f_*(\mathscr{F})$ by ((**0**, 5.3.4) and (**0**, 5.3.3)). We show in the same way that when $\mathscr{F}$ and $\mathscr{F}'$ (resp. $\mathscr{F}$ and $\mathscr{F}''$) are in $\mathtt{C}'$, then so is $\mathscr{F}''$ (resp. $\mathscr{F}'$). In addition, every coherent *direct factor* $\mathscr{F}'$ of an $\mathscr{O}_X$-module $\mathscr{F} \in \mathtt{C}'$ belongs to $\mathtt{C}'$: indeed, $\mathrm{R}^q f_*(\mathscr{F}')$ is then a direct factor of $\mathrm{R}^q f_*(\mathscr{F})$ (G, II, 4.4.4), therefore it is of finite type, and as it is quasi-coherent (1.4.10), it is coherent, as Y is Noetherian. By virtue of Corollary (3.1.3), we reduce to proving that when X is *irreducible* with generic point x, there exists *one* coherent $\mathscr{O}_X$-module $\mathscr{F}$ belonging to $\mathtt{C}'$, such that $\mathscr{F}_x \neq 0$: indeed, if this point is established, then it can be applied to any irreducible closed subprescheme Y of X, since if $j : Y \to X$ is the canonical injection, then $f \circ j$ is proper (**II**, 5.4.2), and if $\mathscr{G}$ is a coherent $\mathscr{O}_Y$-module with support Y, then $j_*(\mathscr{G})$ is a coherent $\mathscr{O}_X$-module such that $\mathrm{R}^q(f \circ j)_*(\mathscr{G}) = \mathrm{R}^q f_*(j_*(\mathscr{G}))$ (G, II, 4.9.1), therefore we can apply Corollary (3.1.3).

By virtue of Chow's lemma (**II**, 5.6.2), there exists an irreducible prescheme X' an a *projective* and surjective morphism $g : X' \to X$ such that $f \circ g : X' \to Y$ is *projective*. There exists an ample $\mathscr{O}_X$-module $\mathscr{L}$ for g (**II**, 5.3.1); we apply the fundamental theorem of projective morphisms (2.2.1) to

$g : X' \to X$ and with $\mathscr{L}$: there thus exists an integer n such that $\mathscr{F} = g_*(\mathscr{O}_{X'}(n))$ is a coherent $\mathscr{O}_X$-module and $\mathrm{R}^q g_*(\mathscr{O}_{X'}(n)) = 0$ for all $q > 0$; in addition, as $g^*(g_*(\mathscr{O}_{X'}(n))) \to \mathscr{O}_{X'}(n)$ is surjective for n large enough (2.2.1), we see that we can suppose, at the generic point x of X, that we have $\mathscr{F}_x \neq 0$ (**II**, 3.4.7). On the other hand, as $f \circ g$ is projective as Y is Noetherian, the $\mathrm{R}^q(f \circ g)_*(\mathscr{O}_{X'}(n))$ are *coherent* (2.2.1). This being so, $\mathrm{R}^\bullet(f \circ g)_*(\mathscr{O}_{X'}(n))$ is the abutment of a Leray spectral sequence, whose E_2-term is given by $E_2^{pq} = \mathrm{R}^p f_*(\mathrm{R}^q g_*(\mathscr{O}_{X'}(n)))$; the above shows that this spectral sequence degenerates, and we then know (**0**, 11.1.6) that $E_2^{p0} = \mathrm{R}^p f_*(\mathscr{F})$ is isomorphic to $\mathrm{R}^p(f \circ g)_*(\mathscr{O}_{X'}(n))$, which finishes the proof. □

Corollary (3.2.2). — *Let Y be a locally Noetherian prescheme. For every proper morphism $f : X \to Y$, the direct image under f of any coherent $\mathscr{O}_X$-module is a coherent $\mathscr{O}_Y$-module.*

Corollary (3.2.3). — *Let A be a Noetherian ring, X a proper scheme over A; for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{H}^p(X, \mathscr{F})$ are A-modules of finite type, and there exists an integer $r > 0$ such that for every coherent $\mathscr{O}_X$-module $\mathscr{F}$ and all $p > r$, $\mathrm{H}^p(X, \mathscr{F}) = 0$.*

PROOF. The second assertion has already been proved (1.4.12); the first follows from the finiteness theorem (3.2.1), taking into account Corollary (1.4.11). □

In particular, if X is a *proper algebraic scheme* over a field k, then, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{H}^p(X, \mathscr{F})$ are *finite-dimensional* k-vector spaces.

Corollary (3.2.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type. For every coherent $\mathscr{O}_X$-module $\mathscr{F}$ whose support in proper over Y* (**II**, 5.4.10), *the $\mathscr{O}_Y$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent.*

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PROOF. Since the questions is local on Y, we can suppose Y Noetherian, and it is the same for X (**I**, 6.3.7). By hypothesis, every closed subprescheme Z of X whose underlying space is $\mathrm{Supp}(\mathscr{F})$ is proper over Y, in other words, if $j : Z \to X$ is the canonical injection, then $f \circ j : Z \to Y$ is proper. We can suppose that Z is such that $\mathscr{F} = j_*(\mathscr{G})$, where $\mathscr{G} = j^*(\mathscr{F})$ is a coherent $\mathscr{O}_Z$-module (**I**, 9.3.5); as we have $\mathrm{R}^q f_*(\mathscr{F}) = \mathrm{R}^q(f \circ j)_*(\mathscr{G})$ by Corollary (1.3.4), the conclusion follows immediately from Theorem (3.2.1). □

3.3. Generalization of the finiteness theorem (usual schemes)

Proposition (3.3.1). — *Let Y be a Noetherian prescheme, $\mathscr{S}$ a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, graded in positive degrees, $Y' = \mathrm{Proj}(\mathscr{S})$, and $g : Y' \to Y$ the structure morphism. Let $f : X \to Y$ be a proper morphism, $\mathscr{S}' = f^*(\mathscr{S})$, $\mathscr{M} = \bigoplus_{k \in \mathbf{Z}} \mathscr{M}_k$ a quasi-coherent graded $\mathscr{S}'$-module of finite type. Then the $\mathrm{R}^p f_*(\mathscr{M}) = \bigoplus_{k \in \mathbf{Z}} \mathrm{R}^p f_*(\mathscr{M}_k)$ are graded $\mathscr{S}$-modules of finite type for all p. Suppose in addition that the $\mathscr{S}$ are generated by $\mathscr{S}_1$; then, for every $p \in \mathbf{Z}$, there exists an integer k_p such that for all $k \geqslant k_p$ and all $r > 0$, we have*

$$\mathrm{R}^p f_*(\mathscr{M}_{k+r}) = \mathscr{S}_r \mathrm{R}^p f_*(\mathscr{M}_k). \tag{3.3.1.1}$$

PROOF. The first assertion is identical to the statement of Theorem (2.4.1, i), where we have simply replaced "projective morphism" by "proper morphism". In the proof of Theorem (2.4.1, i), the hypothesis on f was only used to show (with the notation of this proof) that $\mathrm{R}^p f'_*(\widetilde{\mathscr{M}})$ is a coherent $\mathscr{O}_{Y'}$-module. With the hypothesis of Proposition (3.3.1), f' is proper (**II**, 5.4.2, iii), so we can resume without change in the proof of Theorem (2.4.1, i), thanks to the finiteness theorem (3.2.1).

As for the second assertion, it suffices to remark that there is a finite affine open cover (U_i) of Y such that the restrictions to the U_i of the two sides of (3.3.1.1) are equal for all $k \geqslant k_{p,i}$ (**II**, 2.1.6, ii); it suffices to take for k_p the largest of the $k_{p,i}$. □

Corollary (3.3.2). — *Let A be a Noetherian ring, $\mathfrak{m}$ an ideal of A, X a proper A-scheme, and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. Then, for all $p \geqslant 0$, the direct sum $\bigoplus_{k \geqslant 0} \mathrm{H}^p(X, \mathfrak{m}^k \mathscr{F})$ is a module of finite type over the ring $S = \bigoplus_{k \geqslant 0} \mathfrak{m}^k$; in particular, there exists an integer $k_p \geqslant 0$ such that for all $k \geqslant k_p$ and all $r > 0$, we have*

$$\mathrm{H}^p(X, \mathfrak{m}^{k+r} \mathscr{F}) = \mathfrak{m}^r \mathrm{H}^p(X, \mathfrak{m}^k \mathscr{F}). \tag{3.3.2.1}$$

PROOF. It suffices to apply Proposition (3.3.1) with $Y = \operatorname{Spec}(A)$, $\mathscr{S} = \widetilde{S}$, $\mathscr{M}_k = \mathfrak{m}^k\mathscr{F}$, taking into account Corollary (1.4.11). $\square$

It should be remembered that the S-module structure on $\bigoplus_{k\geqslant 0} \mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F})$ is obtained by considering, for every $a \in \mathfrak{m}^r$, the map $\mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F}) \to \mathrm{H}^p(X, \mathfrak{m}^{k+r}\mathscr{F})$, which comes from the passage to cohomology of the multiplication map $\mathfrak{m}^r\mathscr{F} \to \mathfrak{m}^{k+r}\mathscr{F}$ defined by a (2.4.1).

3.4. Finiteness theorem: the case of formal schemes The results of this section (except the definition (3.4.1)) will not be used in the rest of this chapter. **III** | 119

(3.4.1). Let $\mathfrak{X}$ and $\mathfrak{S}$ be two locally Noetherian formal preschemes (**I**, 10.4.2), $f : \mathfrak{X} \to \mathfrak{S}$ a morphism of formal preschemes. We say that f is a *proper* morphism if it satisfies the following conditions:

1st. *f is a morphism of finite type* (**I**, 10.13.3).

2nd. *If $\mathscr{K}$ is a sheaf of ideals of definition for $\mathfrak{S}$ and if we set $\mathscr{J} = f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}}$, $X_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J})$, $S_0 = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K})$, then the morphism $f_0 : X_0 \to S_0$ induced by f* (**I**, 10.5.6) *is proper*.

It is immediate that this definition does not depend on the sheaf of ideals of definition $\mathscr{K}$ for $\mathscr{S}$ considered; indeed, if $\mathscr{K}'$ is a second sheaf of ideals of definition such that $\mathscr{K}' \subset \mathscr{K}$, and if we set $\mathscr{J}' = f^*(\mathscr{K}')\mathscr{O}_{\mathfrak{X}}$, $X'_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}')$, $S'_0 = (\mathscr{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K}')$, then the morphism $f'_0 : X'_0 \to S'_0$ induced by f is such that the diagram

$$\begin{tikzcd} X_0 \arrow[r, "f_0"] \arrow[d, "i"'] & S_0 \arrow[d, "j"] \\ X'_0 \arrow[r, "f'_0"] & S'_0 \end{tikzcd}$$

is commutative, i and j being surjective immersions; it is equivalent to say that f_0 or f'_0 is proper, by virtue of (**II**, 5.4.5).

We note that, for all $n \geqslant 0$, if we set $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$, $S_n = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K}^{n+1})$, then the morphism $f_n : X_n \to S_n$ induced by f (**I**, 10.5.6) is proper for all n whenever it is for $n = 0$ (**II**, 5.4.6).

If $g : Y \to Z$ is a proper morphism of locally Noetherian usual preschemes, Z' a closed subset of Z, Y' a closed subset of Y such that $g(Y') \subset Z'$, then the extension $\widehat{g} : Y_{/Y'} \to Z_{/Z'}$ of g to the completions (**I**, 10.9.1) is a proper morphism of formal preschemes, as it follows from the definition and from (**II**, 5.4.5).

Let $\mathfrak{X}$ and $\mathfrak{S}$ be two locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{S}$ a morphism *of finite type* (**I**, 10.13.3); the notation being the same as above, we say that a subset Z of the underlying space of $\mathfrak{X}$ is *proper* over $\mathfrak{S}$ (or proper for f) if, considered as a subset of X_0, Z is *proper over* S_0 (**II**, 5.4.10). All the properties of proper subsets of usual preschemes stated in (**II**, 5.4.10) are still true for the proper subsets of formal preschemes, as it follows immediately from the definitions.

Theorem (3.4.2). — *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{Y}$ a proper morphism. For every coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, the $\mathscr{O}_{\mathfrak{Y}}$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent for all $q \geqslant 0$.*

Let $\mathscr{J}$ be a sheaf of ideals of definition for $\mathfrak{Y}$, $\mathscr{K} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}}$, and consider the $\mathscr{O}_{\mathfrak{X}}$-modules

(3.4.2.1) $$\mathscr{F}_k = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{Y}}} (\mathscr{O}_{\mathfrak{Y}}/\mathscr{J}^{k+1}) = \mathscr{F}/\mathscr{K}^{k+1}\mathscr{F} \quad (k \geqslant 0)$$

which evidently form a *projective system* of topological $\mathscr{O}_{\mathfrak{X}}$-modules, such that $\mathscr{F} = \varprojlim_k \mathscr{F}_k$ (**I**, 10.11.3). **III** | 120
On the other hand, it follows from Theorem (3.4.2) that each of the $\mathrm{R}^q f_*(\mathscr{F})$, being coherent, is naturally equipped with a topological $\mathscr{O}_{\mathfrak{Y}}$-module structure (**I**, 10.11.6), and so are the $\mathrm{R}^q f_*(\mathscr{F}_k)$. The canonical homomorphisms $\mathscr{F} \to \mathscr{F}_k = \mathscr{F}/\mathscr{K}^{k+1}\mathscr{F}$ canonically correspond to homomorphisms

$$\mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \mathrm{R}^q f_*(\mathscr{F}_k),$$

which are necessarily continuous for the topological $\mathscr{O}_{\mathfrak{Y}}$-module structures above (**I**, 10.11.6), and form a projective system, giving the limit a canonical functorial homomorphism

(3.4.2.2) $$\mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \varprojlim_k \mathrm{R}^q f_*(\mathscr{F}_k),$$

which will be a continuous homomorphism of topological $\mathscr{O}_{\mathfrak{Y}}$-modules. We will prove along with Theorem (3.4.2) the

Corollary (3.4.3). — *Each of the homomorphisms (3.4.2.2) is a topological isomorphism. In addition, if $\mathfrak{Y}$ is Noetherian, then the projective system $(\mathrm{R}^q f_*(\mathscr{F}/\mathscr{K}^{k+1}\mathscr{F}))_{k\geqslant 0}$ satisfies the* (ML)*-condition* (**0**, 13.1.1).

We will begin by establishing Theorem (3.4.2) and Corollary (3.4.3) when Y is a Noetherian formal affine scheme (**I**, 10.4.1):

Corollary (3.4.4). — *Under the hypotheses of Theorem (3.4.2), suppose in addition that $\mathfrak{Y} = \mathrm{Spf}(A)$, where A is an adic Noetherian ring. Let $\mathfrak{J}$ be an ideal of definition for A, and set $\mathscr{F}_k = \mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}$ for $k \geqslant 0$. Then the $\mathrm{H}^n(\mathfrak{X},\mathscr{F})$ are A-modules of finite type; the projective system $(\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))_{k\geqslant 0}$ satisfies the* (ML)*-condition for all n; if we set*

$$N_{n,k} = \mathrm{Ker}\left(\mathrm{H}^n(\mathfrak{X},\mathscr{F}) \longrightarrow \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)\right) \tag{3.4.4.1}$$

(also equal to $\mathrm{Im}(\mathrm{H}^n(\mathfrak{X},\mathfrak{J}^{k+1}\mathscr{F}) \to \mathrm{H}^n(\mathfrak{X},\mathscr{F}))$ by the exact sequence in cohomology), then the $N_{n,k}$ define on $\mathrm{H}^n(\mathfrak{X},\mathscr{F})$ a $\mathfrak{J}$-good filtration (**0**, 13.7.7)*; finally, the canonical homomorphism*

$$\mathrm{H}^n(\mathfrak{X},\mathscr{F}) \longrightarrow \varprojlim_k \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k) \tag{3.4.4.2}$$

is a topological isomorphism for all n (the left hand side being equipped with the $\mathfrak{J}$-adic topology, the $\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$ with the discrete topology).

Set

$$S = \mathrm{gr}(A) = \bigoplus_{k\geqslant 0} \mathfrak{J}^k/\mathfrak{J}^{k+1},\ \mathscr{M} = \mathrm{gr}(\mathscr{F}) = \bigoplus_{k\geqslant 0} \mathfrak{J}^k\mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}.$$

We know that $\mathfrak{J}^\Delta$ is a sheaf of ideals of definition for $\mathfrak{Y}$ (**I**, 10.3.1); let $\mathscr{K} = f^*(\mathfrak{J}^\Delta)\mathscr{O}_{\mathfrak{X}}$, $X_0 = (\mathfrak{X},\mathscr{O}_{\mathfrak{X}}/\mathscr{K})$, $Y_0 = (\mathfrak{Y},\mathscr{O}_{\mathfrak{Y}}/\mathfrak{J}^\Delta) = \mathrm{Spec}(A_0)$, with $A_0 = A/\mathfrak{J}$. It is clear that the $\mathscr{M}_k = \mathfrak{J}^k\mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}$ are coherent $\mathscr{O}_{X_0}$-modules (**I**, 10.11.3). Consider on the other hand the quasi-coherent graded $\mathscr{O}_{X_0}$-algebra

$$\mathscr{S} = \mathscr{O}_{X_0} \otimes_{A_0} S = \mathrm{gr}(\mathscr{O}_{\mathfrak{X}}) = \bigoplus_{k\geqslant 0} \mathscr{K}^k/\mathscr{K}^{k+1}. \tag{3.4.4.4}$$

The hypothesis that $\mathscr{F}$ is an $\mathscr{O}_{\mathfrak{X}}$-module of finite type implies first that $\mathscr{M}$ is a graded $\mathscr{S}$-module *of finite type*. Indeed, the question is local on $\mathfrak{X}$, and we can thus suppose that $\mathfrak{X} = \mathrm{Spf}(B)$, where B is an adic Noetherian ring, and $\mathscr{F} = N^\Delta$, where N is a B-module of finite type (**I**, 10.10.5); we have in addition $X_0 = \mathrm{Spec}(B_0)$, where $B_0 = B/\mathfrak{J}B$, and the quasi-coherent $\mathscr{O}_{X_0}$-modules $\mathscr{S}$ and $\mathscr{M}$ are respectively equal to $\widetilde{S'}$ and $\widetilde{M'}$, where $S' = \bigoplus_{k\geqslant 0}((\mathfrak{J}^k/\mathfrak{J}^{k+1}) \otimes_{A_0} B_0)$ and $M' = \bigoplus_{k\geqslant 0}((\mathfrak{J}^k/\mathfrak{J}^{k+1}) \otimes_{A_0} N_0)$, with $N_0 = N/\mathfrak{J}N$; we then evidently have $M' = S' \otimes_{B_0} N_0$, and as N_0 is a B_0-module of finite type, M' is a S'-module of finite type, hence our assertion (**I**, 1.3.13). **III | 121**

As the morphism $f_0 : X_0 \to Y_0$ is *proper* by hypothesis, we can apply Corollary (3.3.2) to $\mathscr{S}$, $\mathscr{M}$, and the morphism f_0: taking into account Corollary (1.4.11), we conclude that for *all* $n \geqslant 0$, $\bigoplus_{k\geqslant 0} \mathrm{H}^n(X_0,\mathscr{M}_k)$ is a graded S-module *of finite type*. This proves that the condition (F_n) of (**0**, 13.7.7) is satisfied for *all* $n \geqslant 0$, when we consider the strictly projective system $(\mathscr{F}/\mathfrak{J}^k\mathscr{F})_{k\geqslant 0}$ of sheaves of abelian groups on X_0, each equipped with its natural "filtered A-module" structure. We can thus apply (**0**, 13.7.7), which proves that:

1st. The projective system $(\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))_{k\geqslant 0}$ satisfies the (ML)-condition.
2nd. If $\mathrm{H}'^n = \varprojlim_k \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$, then H'^n is an A-module of finite type.
3rd. The filtration defined on H'^n by the kernels of the canonical homomorphisms $\mathrm{H}'^n \to \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$ is $\mathfrak{J}$-good.

Note that on the other hand, if we set $X_k = (\mathfrak{X},\mathscr{O}_{\mathfrak{X}}/\mathscr{K}^{k+1})$, then $\mathscr{F}_k$ is a coherent $\mathscr{O}_{X_k}$-module (**I**, 10.11.3), and if U is an affine open set in X_0, then U is also an affine open set in each of the X_k (**I**, 5.1.9), so $\mathrm{H}^n(U,\mathscr{F}_k) = 0$ for all $n > 0$ and all k (1.3.1) and $\mathrm{H}^0(U,\mathscr{F}_k) \to \mathrm{H}^0(U,\mathscr{F}_h)$ is surjective for $h \leqslant k$ (**I**, 1.3.9). We are thus in the conditions of (**0**, 13.3.2) and applying (**0**, 13.3.1) proves that H'^n canonically identifies with $\mathrm{H}^n(\mathfrak{X},\varprojlim_k \mathscr{F}_k) = \mathrm{H}^n(\mathfrak{X},\mathscr{F})$; this finishes the proof of Corollary (3.4.4).

(3.4.5). We return to the proof of (3.4.2) and (3.4.3). We first prove the propositions for the case $\mathfrak{Y} = \mathrm{Spf}(A)$ envisaged in (3.4.4); for this, for all $g \in A$, apply (3.4.4) to the Noetherian affine formal scheme induced on the open set $\mathfrak{Y}_g = \mathfrak{D}(g)$ of $\mathfrak{Y}$, which is equal to $\mathrm{Spf}(A_{\{g\}})$, and to the formal prescheme induced by $\mathfrak{X}$ on $f^{-1}(\mathfrak{Y}_g)$; note that $\mathfrak{Y}_g$ is also an affine open set in the prescheme $Y_k = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/(\mathfrak{J}^\Delta)^{k+1})$, and as $\mathscr{F}_k$ is a coherent $\mathscr{O}_{X_k}$-module, we have

$$\mathrm{H}^n(f^{-1}(\mathfrak{Y}_g), \mathscr{F}_k) = \Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F}_k))$$

for all $k \geqslant 0$ by virtue of Corollary (1.4.11). The canonical homomorphism

$$\mathrm{H}^n(f^{-1}(\mathfrak{Y}_g), \mathscr{F}) \longrightarrow \varprojlim_k \Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F}_k))$$

is an isomorphism; but we have (**0**, 3.2.6)

$$\varprojlim_k \Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F}_k)) = \Gamma(\mathfrak{Y}_g, \varprojlim_k \mathrm{R}^n f_*(\mathscr{F}_k)),$$

and as the sheaf $\mathrm{R}^n f_*(\mathscr{F})$ is the sheaf associated to the presheaf $\mathfrak{Y}_g \mapsto \mathrm{H}^n(f^{-1}(\mathfrak{Y}_g), \mathscr{F})$ on the $\mathfrak{Y}_g$ (**0**, 3.2.1), we have shown that the homomorphism (3.4.2.2) is *bijective*. Let us now prove that $\mathrm{R}^n f_*(\mathscr{F})$ is a coherent $\mathscr{O}_{\mathfrak{Y}}$-module, and more precisely that we have **III | 122**

(3.4.5.1) $$\mathrm{R}^n f_*(\mathscr{F}) = \big(\mathrm{H}^n(\mathfrak{X}, \mathscr{F})\big)^\Delta.$$

With the above notation, we have, since $\mathscr{F}_k$ is a coherent $\mathscr{O}_{X_k}$-module (1.4.13),

$$\Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F}_k)) = (\Gamma(\mathfrak{Y}, \mathrm{R}^n f_*(\mathscr{F}_k)))_g = (\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k))_g.$$

Now the $\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k)$ form a projective system satisfying (ML), and their projective limit $\mathrm{H}^n(\mathfrak{X}, \mathscr{F})$ is an A-module of finite type. We conclude (**0**, 13.7.8) that we have

$$\varprojlim_k \big((\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k))_g\big) = \mathrm{H}^n(\mathfrak{X}, \mathscr{F}) \otimes_A A_{\{g\}} = \Gamma(\mathfrak{Y}_g, (\mathrm{H}^n(\mathfrak{X}, \mathscr{F}))^\Delta),$$

taking into account (**I**, 10.10.8) applied to A and $A_{\{g\}}$; this proves (3.4.5.1) since $\Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F})) = \varprojlim_k \Gamma(\mathfrak{Y}_g, \mathrm{R}^n f_*(\mathscr{F}_k))$.

As (3.4.2.2) is then an isomorphism of coherent $\mathscr{O}_{\mathfrak{Y}}$-modules, it is necessarily a *topological* isomorphism (**I**, 10.11.6). Finally, it follows from the relations $\mathrm{R}^n f_*(\mathscr{F}_k) = (\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k))^\Delta$ that the projective system $(\mathrm{R}^n f_*(\mathscr{F}_k))_{k \geqslant 0}$ satisfies (ML) (**I**, 10.10.2).

Once (3.4.2) and (3.4.3) are proved in the case where the formal prescheme $\mathfrak{Y}$ is affine Noetherian, it is immediate to pass to the general case for (3.4.2) and the first assertion of (3.4.3), which are local on $\mathfrak{Y}$. As for the second assertion of (3.4.3), it suffices, $\mathfrak{Y}$ being Noetherian, to cover it by a finite number of Noetherian affine open sets U_i and to note that the restrictions of the projective system $(\mathrm{R}^q f_*(\mathscr{F}_k))$ to each of the U_i satisfies (ML).

Along the way, we have in addition proved:

Corollary (3.4.6). — *Under the hypotheses of Corollary (3.4.4), the canonical homomorphism*

(3.4.6.1) $$\mathrm{H}^q(\mathfrak{X}, \mathscr{F}) \longrightarrow \Gamma(\mathfrak{Y}, \mathrm{R}^q f_*(\mathscr{F}))$$

is bijective.

§4. The fundamental theorem of proper morphisms. Applications

4.1. The fundamental theorem

§5. An existence theorem for coherent algebraic sheaves

5.1. Statement of the theorem

§6. Local and global Tor functors; Künneth formula

6.1. Introduction

§7. Base change for homological functors of sheaves of modules

7.1. Functors of A-modules

CHAPTER IV

Local study of schemes and their morphisms (EGA IV)

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SUMMARY

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The subjects discussed in the chapter call for the following remarks. **IV-1** | 223

(a) The common property of all the subjects discussed is that they all related to *local* properties of preschemes or morphisms, i.e. considered at a point, or the points of a fibre, or on a (non-specified) neighbourhood of a point or of a fibre. These properties are generally of a *topological*, *differential*, or *dimensional* nature (i.e. bringing the ideas of *dimension* and *depth* into play), and are linked to the properties of the *local rings* at the points considered. One type of problem is the relating, for a given morphisms $f : X \to Y$ and point $x \in X$, of the properties of X at x with those of Y at $y = f(x)$ and those of the fibre $X_y = f^{-1}(y)$ at x. Another is the determining of the topological nature (for example, the constructibility, or the fact of being open or closed) of the set of points $x \in X$ at which X has a certain property, or for which the fibre $X_{f(x)}$ passing through x has a certain property at x. Similarly, we are interested in the topological nature of the set of points $y \in Y$ such that X has a certain property at all the points of the fibre X_y, or those such that this fibre itself has a certain property.

[1]The order and content of §§11–21 are given only as an indication of what the titles will be, and will possibly be modified before their publication. *[Trans.] This was indeed the case: many of §§11–21 ended up having entirely different titles or content. See here.*

(b) The most important idea for the following chapters is that of *flat morphisms of finite presentation*, as well as the particular cases of *smooth morphisms* and *étale morphisms*. Their detailed study (as well as that of connected questions) really starts in §11.

(c) Sections §§1–10 can be considered as being preliminary in nature, and as developing three types of techniques, used, not only in the other sections of the chapter, but also, of course, in the follow chapters:

 (c1) Sections §§1–4 are envisaged as treating the diverse aspects of the idea of *change of base*, above all in relation with the conditions of *finiteness* or *flatness*; we there initiate the technique of *descent*, with its most elementary aspects (the questions of "effectiveness" linked to this technique will be studied in Chapter V).

 (c2) Sections §§5–7 are focused on what we may call *Noetherian* techniques, since the preschemes considered are always locally Noetherian, whereas, on the contrary, there is generally no finiteness condition imposed on the *morphisms*; this is essentially due to the fact that the ideas of dimension and depth are hardly manageable except in the case of Noetherian local rings. Recall that §7 constitutes a "delicate (?)" theory of Noetherian local rings, not much used in what follows in the chapter.

 (c3) Sections §§8–10 describe, amongst other things, the means of *eliminating the Noetherian hypotheses* on the preschemes considered, by substituting such hypotheses for suitable ones of *finiteness* ("finite presentation") on the *morphisms* considered: the advantage of this substitution is that the latter such hypotheses (those of finiteness on the morphisms) are *stable under base change*, which is not the case for the Noetherian hypotheses on the preschemes. The technique permitting this substitution relies, in some part, on the use of the idea of the *projective limit* of preschemes, thanks to which we can reduce a question to the same question with *Noetherian* hypotheses; on the other hand, it relies on the systematic use of *constructible sets*, which have the double interest of being preserved under taking inverse images (of arbitrary morphisms) and by direct images **IV-1** | 224
 (of morphisms of finite presentation), and having manageable topological properties in locally Noetherian preschemes. The same techniques often even allow to restrict to the case of more specific Noetherian rings, for example the **Z**-*algebras of finite type*, and it is here that the properties of "excellent" rings (studied in §7) intervene in a decisive manner. Independently of the question of elimination of Noetherian hypotheses, the techniques of §§8–10, elementary in nature, find constant use in nearly all applications.

§1. Relative finiteness conditions. Constructible sets in preschemes

In this section, we will resume the exposé of "finiteness conditions" for a morphism of preschemes $f : X \to Y$ given in (**I**, 6.3 and 6.6). There are essentially two notions of "finiteness" of a *global* nature on X, that of *quasi-compact* morphism (defined in (**I**, 6.6.1)) and that of a *quasi-separated* morphism; on the other hand, there are two notions of "finiteness" of a *local* nature on X, that of a morphism *locally of finite type* (defined in (**I**, 6.6.2)) and that of a morphism *locally of finite presentation*. By combining these local notions with the preceding global notions, we obtain the notion of a morphism *of finite type* (defined in (**I**, 6.3.1)) and of a morphism *of finite presentation*. For the convenience of the reader, we will give again in this section the properties stated in (**I**, 6.3 and 6.6), referring to their labels in Chapter I for their proofs.

In n^{os} 1.8 and 1.9, we complete, in the context of preschemes, and making use of the previous notions of finiteness, the results on constructible sets given in ($\mathbf{0}_{\mathrm{III}}$, §9).

1.1. Quasi-compact morphisms

Definition (1.1.1). — We say that a morphism of preschemes $f : X \to Y$ is *quasi-compact* if the continuous map f from the topological space X to the topological space Y is quasi-compact (**0**, 9.1.1), in other words, if the inverse image $f^{-1}(U)$ of every quasi-compact open subset U of Y is quasi-compact (cf. (**I**, 6.6.1)).

If $\mathfrak{B}$ is a basis for the topology of Y consisting of affine open sets, then for f to be quasi-compact, it is necessary and sufficient that for all $V \in \mathfrak{B}$, $f^{-1}(V)$ is a *finite union of affine open sets*. For example, if Y is affine and X is quasi-compact, *every* morphism $f : X \to Y$ is quasi-compact (**I**, 6.6.1).

If $f : X \to Y$ is a quasi-compact morphism, then it is clear that for every open subset V of Y, the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \to V$. Conversely, if (U_α) is an open cover of Y and $f : X \to Y$ is a morphism such that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ are quasi-compact, then f is quasi-compact. As a result, if $f : X \to Y$ is an S-morphism of S-preschemes, and if there **IV-1** | 225 exists an open cover (S_λ) of S such that the restrictions $g^{-1}(S_\lambda) \to h^{-1}(S_\lambda)$ of f (where g and h are the structure morphisms) are quasi-compact, then f is quasi-compact.

Proposition (1.1.2). — (i) *An immersion $X \to Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally noetherian, or if the underlying space of X is noetherian.*

(i) *The composite of two quasi-compact morphisms is quasi-compact.*

(i) *If $f : X \to Y$ is an S-morphism quasi-compact, then the same is true of $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every base extension $g : S' \to S$ of the base prescheme.*

(i) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms quasi-compact,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is quasi-compact.

(i) *If the composite $g \circ f$ of two morphisms $f : X \to Y$, $g : Y \to Z$ is quasi-compact, and if g is separated, or X locally noetherian, f is quasi-compact.*

(i) *For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be so.*

For the proof, see (**I**, 6.6.4). We note that (vi) is also a consequence of the following more general proposition:

Proposition (1.1.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If $g \circ f$ is quasi-compact and f surjective, g is quasi-compact.*

Indeed, if V is a quasi-compact open subset in Z, $f^{-1}(g^{-1}(V))$ is quasi-compact by hypothesis, and since $g^{-1}(V) = f(f^{-1}(g^{-1}(V)))$, since f is surjective, $g^{-1}(V)$ is quasi-compact.

Corollary (1.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; suppose g quasi-compact and surjective* (*resp. dominant*)*. Then, for f to be quasi-compact* (*resp. dominant*)*, it suffices that f' be so.*

If $g' : X' \to X$ is the canonical projection, g' is a surjective morphism (**I**, 3.5.2), and so $f \circ g' = g \circ f'$; if f' is quasi-compact (resp. dominant), the same is true of $g \circ f'$ since g is quasi-compact (1.1.2) (resp. surjective); as $f \circ g'$ is quasi-compact (resp. dominant) and g' is surjective, we deduce that f is quasi-compact (1.1.3) (resp. dominant).

To abbreviate, we will call a *maximal point* of a prescheme X any generic point of an irreducible component of X; if $X = \operatorname{Spec}(A)$ is affine, the maximal points of X are the *minimal prime ideals* of A.

Proposition (1.1.5). — *Let $f : X \to Y$ be a quasi-compact morphism. The following conditions are equivalent:*

a) *f is dominant;*

a) *for every maximal point y of Y, $f^{-1}(y) \neq 0$;*

a) *for every maximal point y of Y, $f^{-1}(y)$ contains a maximal point of X.*

The equivalence of a) and c) has been proved in (**I**, 6.6.5). It is clear that c) implies b); on the other hand, if X' is an irreducible component of X meeting $f^{-1}(y)$, $f(X')$ is contained in the **IV-1** | 226 irreducible component $Y' = \overline{\{y\}}$ of Y ($\mathbf{0}_{\mathbf{I}}$, 2.1.6); if x is the generic point of X', we necessarily have $f(x) = y$ ($\mathbf{0}_{\mathbf{I}}$, 2.1.5); thus b) implies c).

Proposition (1.1.6). — *Let $f' : X' \to Y$, $f'' : X'' \to Y$ be two morphisms, X the prescheme sum $X' \coprod X''$, $f : X \to Y$ the morphism which coincides with f' in X' and with f'' in X''. For f to be quasi-compact, it is necessary and sufficient that f' and f'' be so.*

This follows immediately from the definition.

1.2. Quasi-separated morphisms

Definition (1.2.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *quasi-separated* (or that X is *quasi-separated over* Y) if the diagonal morphism $\Delta_f = (1_X, 1_X)_Y : X \to X \times_Y X$ is quasi-compact. We say that a prescheme X is *quasi-separated* if it is quasi-separated over $\operatorname{Spec}(\mathbf{Z})$.

By definition, every separated morphism is quasi-separated, Δ_f being a closed immersion (1.1.2), (i)); a *scheme* X is a prescheme quasi-separated, being separated over $\operatorname{Spec}(\mathbf{Z})$ (**I**, 5.4.1).

Proposition (1.2.2). — (i) *Every monomorphism of preschemes $f : X \to Y$ (in particular every immersion) is quasi-separated.*

(i) *If $f : X \to Y$ and $g : Y \to Z$ are two quasi-separated morphisms, $g \circ f : X \to Z$ is quasi-separated.*

(i) *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism quasi-separated. Then, for every base change $g : S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is quasi-separated.*

(i) *If $f : X \to Y$, $f' : X' \to Y'$ are two S-morphisms quasi-separated, then*

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'$$

is quasi-separated.

(i) *If $f : X \to Y$, $g : Y \to Z$ are two morphisms such that $g \circ f$ is quasi-separated, then f is quasi-separated.*

(i) *If $f : X \to Y$ is a quasi-separated morphism, $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is quasi-separated.*

By virtue of (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertion (i) is trivial, every monomorphism being separated (**I**, 5.5.1). To prove (iii), we immediately reduce to the case where $Y = S$ (**I**, 3.3.11) and the assertion follows from the fact that $\Delta_{f_{(S')}} = (\Delta_f)_{(S')}$ (**I**, 5.3.4) and from (1.1.2), (iii). To prove (ii), consider the projections p and q of $X \times_Y X$ onto X; if $i = (p, q)_Z$, we know that the diagram

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_Z Y \arrow[d, "1 \times_Z f"] \\ Y \arrow[r, "\Delta_g"] & Y \times_Z Y \end{tikzcd}$$

is commutative and identifies $X \times_Y X$ as the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.5). If g is quasi-separated, Δ_g is a quasi-compact morphism, and the same is true of Γ_f (1.1.2), (iii)); the same is true of $i \circ \Delta_f$ (1.1.2), (ii)) which equals $\Delta_{g \circ f}$. **IV-1** | 227

Corollary (1.2.3). — (i) *Let X be a quasi-separated prescheme. Then every morphism $f : X \to Y$ is quasi-separated.*

(i) *Let Y be a quasi-separated prescheme. For a morphism $f : X \to Y$ to be quasi-separated, it suffices that the prescheme X be quasi-separated.*

This follows immediately from (1.2.2), (ii) and (v)) applied to X, Y and $\operatorname{Spec}(\mathbf{Z})$.

Proposition (1.2.4). — *Let $f : X \to Y$ be a morphism, $g : Y \to Z$ a morphism quasi-separated. If $g \circ f$ is quasi-compact, then the same is true of f.*

We know that f is the composite morphism $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{\mathrm{pr}_2} Y$ (**I**, 5.3.13); on the other hand, pr_2 is identified with $(g \circ f) \times_Z 1_Y$ (**I**, 3.3.4) and if $g \circ f$ is quasi-compact, the same is true of pr_2 (1.1.2), (iv)). Finally, we have the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_Z Y \arrow[d, "f \times 1_Y"] \\ Y \arrow[r, "\Delta_g"] & Y \times_Z Y \end{tikzcd}$$

(1.2.4.1)

which identifies X as the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z Y$ (**I**, 5.3.7). Since by hypothesis Δ_g is a quasi-compact morphism, the same is true of Γ_f (1.1.2), (iii)); the conclusion follows from (1.1.2), (ii)).

Proposition (1.2.5). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If g is quasi-compact and surjective and f' quasi-separated, then f is quasi-separated.*

Indeed $(X' \times_{Y'} X') = (X \times_Y X)_{(Y')}$ and $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4); since the projection $X' \times_{Y'} X' \to X \times_Y X$ is quasi-compact (1.1.2), (iii)) and surjective (**I**, 3.5.2), one can apply (1.1.4), which shows that since $\Delta_{f'}$ is a quasi-compact morphism, the same is true of Δ_f.

Proposition (1.2.6). — *Let $f : X \to Y$ be a morphism, (U_α) an open cover of Y such that the preschemes induced U_α are quasi-separated. For f to be quasi-separated, it suffices that each of the preschemes induced by X on the open sets $f^{-1}(U_\alpha)$ be quasi-separated.*

The inverse image $X_\alpha \times_Y X_\alpha$ of U_α is $X_\alpha \times_{U_\alpha} X_\alpha$, with $X_\alpha = f^{-1}(U_\alpha)$, and the restriction $X_\alpha \times_{U_\alpha} X_\alpha$ to X_α of Δ_f is none other than Δ_{f_α}, where f_α denotes the restriction $X_\alpha \to U_\alpha$ of f; by virtue of the definition (1.2.1) and of the local character of the notion of quasi-compact morphism (1.1.1), for f to be quasi-separated, it suffices that each f_α be quasi-separated. But as by hypothesis the morphism $U_\alpha \to \operatorname{Spec}(\mathbf{Z})$ is quasi-separated, it amounts to the same to say that f_α is quasi-separated or that the composite $X_\alpha \to U_\alpha \to \operatorname{Spec}(\mathbf{Z})$ is (1.2.2), (ii) and (v)), whence the proposition.

To verify that a morphism is quasi-separated, one is thus led to give criteria for a prescheme to be quasi-separated:

Proposition (1.2.7). — *Let X be a prescheme, (U_α) an open cover of X consisting of quasi-compact open sets. The following properties are equivalent:*

a) *X is quasi-separated.*
a) *For every quasi-compact open subset U of X, the canonical immersion $U \to X$ is quasi-compact (in other words, U is* retrocompact *in X).*
b') *The intersection of two quasi-compact open subsets of X is quasi-compact.*
a) *For every pair of indices (α, β), the intersection $U_\alpha \cap U_\beta$ is quasi-compact.*

The properties b) and b') are trivially equivalent by definition of a quasi-compact morphism. Since a quasi-compact open is a finite union of affine open sets, for two quasi-compact open sets U, V of X, $U \times_{\mathbf{Z}} V$ is a quasi-compact open subset of $X \times_Z X$ (**I**, 3.2.7), whose inverse image by Δ_X is $U \cap V$, so a) implies b'). It is trivial that b') implies c); finally, if c) is satisfied, the $U_\alpha \times_{\mathbf{Z}} U_\beta$ form a cover of $X \times_{\mathbf{Z}} X$ by quasi-compact open sets, and we know that for the morphism $\Delta_X : X \to X \times_{\mathbf{Z}} X$ to be quasi-compact, it suffices that the inverse images of $U_\alpha \times_{\mathbf{Z}} U_\beta$ by Δ_X be quasi-compact (1.1.1); hence c) implies a). **IV-1** | 228

Corollary (1.2.8). — *Every prescheme X whose underlying space is locally noetherian* (*for example a locally noetherian prescheme*) *is quasi-separated; every morphism $f : X \to Y$ is then quasi-separated.*

Proposition (1.2.9). — *Let X', X'' be two preschemes, $X = X' \coprod X''$ their sum, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism which coincides with f' in X' and with f'' in X''. For f to be quasi-separated, it is necessary and sufficient that f' and f'' be so.*

Indeed, $X \times_Y X$ is the sum of the four preschemes $X' \times_Y X'$, $X' \times_Y X''$, $X'' \times_Y X'$ and $X'' \times_Y X''$, and Δ_f is the morphism which coincides with $\Delta_{f'}$ in X' and with $\Delta_{f''}$ in X''; the proposition therefore follows immediately from the definitions.

1.3. Morphisms locally of finite type

(1.3.1). Recall that if A is a ring, a A-algebra (commutative) B is said to be *of finite type* if it is generated by a finite number of elements of B, or, what amounts to the same, if it is isomorphic to a quotient of a polynomial algebra $A[T_1, \dots, T_n]$ by an ideal of this algebra. It is clear that for every A-algebra (commutative) A', $B \otimes_A A'$ is then an A'-algebra of finite type. If B is an A-algebra of finite type and C a B-algebra of finite type, C is an A-algebra of finite type, for C is a quotient of $B[T_1, \dots, T_n]$ and B a quotient of $A[S_1, \dots, S_m]$, $B[T_1, \dots, T_n] = B \otimes_A A[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]$ and is a quotient of $A[S_1, \dots, S_m, T_1, \dots, T_n]$, so the same is true of C.

Definition (1.3.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite type in x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $A(U)$ is an $A(V)$-algebra of finite type. We say that f is *locally of finite type* if it is of finite type at every point of X.

Proposition (1.3.3). — *If Y is a locally noetherian prescheme and $f : X \to Y$ a morphism locally of finite type, then X is locally noetherian.*

For the proof, see (**I**, 6.3.7).

Proposition (1.3.4). — (i) *Every local immersion is locally of finite type.*
(i) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite type, the same is true of $g \circ f$.*
(i) *If $f : X \to Y$ is an S-morphism locally of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite type for every base extension $S' \to S$ of the base prescheme.*
(i) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite type, $f \times_S g$ is locally of finite type.*
(i) *If the composite $g \circ f$ of two morphisms is locally of finite type, f is locally of finite type.*
(i) *If a morphism f is locally of finite type, the same is true of f_{red}.*

For the proof, see (**I**, 6.3.6). **IV-1** | 229

Corollary (1.3.5). — *Let $f : X \to Y$ be a morphism locally of finite type. For every morphism $Y' \to Y$ such that Y' is locally noetherian, $X \times_Y Y'$ is locally noetherian.*

Indeed, $f_{(Y')} : X \times_Y Y' \to Y'$ is locally of finite type by (1.3.4), (iii)), and it suffices to apply (1.3.3).

Proposition (1.3.6). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ to be locally of finite type, it is necessary and sufficient that B be an A-algebra of finite type.*

For the proof, see (**I**, 6.3.3), taking into account that f is quasi-compact and (**I**, 6.6.3).

Proposition (1.3.7). — *For a morphism locally of finite type $f : X \to Y$ to be surjective, it is necessary and sufficient that, for every algebraically closed field Ω, the map $X(\Omega) \to Y(\Omega)$ corresponding to f (**I**, 3.4.1) be surjective.*

For the proof, see (**I**, 6.3.10).

(1.3.8). Let A be a ring. We say that an A-algebra B is *essentially of finite type* if B is A-isomorphic to an A-algebra of the form $S^{-1}C$, where C is an A-algebra of finite type and S a multiplicative subset of C.

Proposition (1.3.9). — (i) *If B is an A-algebra essentially of finite type and C a B-algebra essentially of finite type, then C is an A-algebra essentially of finite type.*
(i) *Let B be an A-algebra essentially of finite type, A' an A-algebra; then $B' = B \otimes_A A'$ is an A'-algebra essentially of finite type.*

(i) Let $B = S'^{-1}B'$ and $C = T'^{-1}C'$, where B' (resp. C') is an A-algebra (resp. a B-algebra) of finite type and S' (resp. T') a multiplicative subset of B' (resp. C'); then C' is also the form $S'^{-1}C''$, where C'' is a B'-algebra of finite type, so C is also a ring of fractions of C''; as C'' is an A-algebra of finite type, this proves our assertion.

(ii) If B is a ring of fractions of an A-algebra of finite type C, B' is a ring of fractions of the A'-algebra of finite type $C \otimes_A A'$, whence (ii).

(1.3.10). If B is an A-algebra *locally* essentially of finite type, B is of the form $C_{\mathfrak{q}}$, where C is an A-algebra of finite type and $\mathfrak{q}$ a prime ideal of C (Bourbaki, *Alg. comm.*, chap. II, §5, prop. 11). Let $\mathfrak{p}$ be the inverse image in A of the ideal $\mathfrak{q}$; if we set $S = A - \mathfrak{p}$, $C_{\mathfrak{q}}$ is also a local ring at a prime ideal of $S^{-1}C$; as $S^{-1}C$ is an algebra of finite type over $A_{\mathfrak{p}} = S^{-1}A$, we see that B is also an $A_{\mathfrak{p}}$-algebra essentially of finite type, and furthermore the homomorphism $A_{\mathfrak{p}} \to B$ is *local*.

Proposition (1.3.11). — *If B is an A-algebra locally essentially of finite type, B is A-isomorphic to an A-algebra quotient of a localisation $C_{\mathfrak{q}}$, where C is a polynomial algebra $A[T_1, \ldots, T_n]$ and $\mathfrak{q}$ a prime ideal of C.*

Indeed, by definition, B is isomorphic to $C'_{\mathfrak{q}'}$, where C' is an A-algebra of finite type; but $C' = C/\mathfrak{b}$, where $C = A[T_1, \ldots, T_n]$ and $\mathfrak{b}$ is an ideal of C; so $\mathfrak{q}' = \mathfrak{q}/\mathfrak{b}$, where $\mathfrak{q}$ is a prime ideal of C, and $C'_{\mathfrak{q}'}$ is isomorphic to $C_{\mathfrak{q}}/\mathfrak{b}C_{\mathfrak{q}}$. **IV-1** | 230

1.4. Morphisms locally of finite presentation

(1.4.1). Given a ring A, we say that an A-algebra B (commutative) is *of finite presentation* if it is isomorphic to $A[T_1, \ldots, T_n]/\mathfrak{a}$ for a *finitely generated* ideal $\mathfrak{a}$ of $A[T_1, \ldots, T_n]$. For every A-algebra (commutative) A', $B \otimes_A A'$ is then an A'-algebra of finite presentation, being isomorphic to the quotient of $A'[T_1, \ldots, T_n]$ by the image of $\mathfrak{a}$ via the canonical map $A[T_1, \ldots, T_n] \to A'[T_1, \ldots, T_n]$, which is then an ideal of finite type. If B is an A-algebra of finite presentation and C a B-algebra of finite presentation, then C is an A-algebra of finite presentation. Indeed, let $B = A[S_1, \ldots, S_m]/\mathfrak{a}$ and $C = B[T_1, \ldots, T_n]/\mathfrak{b}$, where $\mathfrak{a}$ (resp. $\mathfrak{b}$) is a finitely generated ideal of $A[S_1, \ldots, S_m]$ (resp. $B[T_1, \ldots, T_n]$); the ring $B[T_1, \ldots, T_n]$ is isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{a}'$, where $\mathfrak{a}'$ is the image of $\mathfrak{a}$ via the canonical map of $A[S_1, \ldots, S_m]$ to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$, and thus an ideal of finite type; the ideal $\mathfrak{b}$ of $B[T_1, \ldots, T_n]$ is of the form $\mathfrak{b}'/\mathfrak{a}'$, where $\mathfrak{b}'$ is an ideal of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$; as $\mathfrak{a}'$ and $\mathfrak{b}'/\mathfrak{a}'$ are finitely generated modules, the same is true of $\mathfrak{b}'$, and C, isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{b}'$, is indeed an A-algebra of finite presentation. Note finally that if A is *noetherian*, the same is true of $A[T_1, \ldots, T_n]$, so every A-algebra of finite type is then of finite presentation.

Definition (1.4.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite presentation in* x if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $A(U)$ is an $A(V)$-algebra of finite presentation. We say that f is *locally of finite presentation* if it is of finite presentation at every point of X.

If Y is locally noetherian, it amounts to the same to say that f is locally of finite type or locally of finite presentation.

Proposition (1.4.3). — (i) *Every local isomorphism is locally of finite presentation.*

(i) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite presentation, the same is true of $g \circ f$.*

(i) *If $f : X \to Y$ is an S-morphism locally of finite presentation, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite presentation for every base extension $S' \to S$ of the base prescheme.*

(i) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite presentation, $f \times_S g$ is locally of finite presentation.*

(i) *If the composite $g \circ f$ of two morphisms is locally of finite presentation and if g is locally of finite type, then f is locally of finite presentation.*

The assertion (i) is trivial. To prove (iii), one can restrict to the case where $S = Y$ (**I**, 3.3.11); let **IV-1** | 231
x' be a point of $X_{(S')}$, s' and x its projections onto S' and X respectively, s the common projection of s' and x onto S. By hypothesis, there exists an affine open neighbourhood V (resp. V') of s (resp. of x) such that the image of U be contained in V and such that $A(U)$ be an $A(V)$-algebra of finite presentation; let V' be an affine open neighbourhood of s' whose image in S is contained in V, and $U' = \mathrm{Spec}(A \otimes_{A(V)} A(V'))$ an affine open neighbourhood of x' whose image in S' is contained in V' and whose ring $A(U) \otimes_{A(V)} A(V')$ is an $A(V')$-algebra of finite presentation.

One knows that (iv) follows from (i), (ii), and (iii) (**I**, 3.5.1). To prove (v), let us consider a point $x \in X$, and let $y = f(x)$, $z = g(f(x))$. There exists an affine open neighbourhood $W = \mathrm{Spec}(C)$ (resp.

$V = \mathrm{Spec}(B)$) of z (resp. of y) such that $g(V) \subset W$ and such that B is a C-algebra of finite type. On the other hand, there exists an affine open neighbourhood $V' = \mathrm{Spec}(B')$ (resp. $U = \mathrm{Spec}(A)$) of y (resp. of x) such that $f(U) \subset V'$ and such that A is a B'-algebra of finite presentation. Let $s \in B$ be such that the open set $\mathrm{D}(B_s)$ in V is a neighbourhood of y contained in V', and let $U' = f^{-1}(\mathrm{D}(s)) \subset U$; we have $U' = \mathrm{Spec}(A \otimes_{B'} B_s)$, and $A \otimes_{B'} B_s$ is a B_s-algebra of finite presentation (1.4.1); on the other hand $B_s = B[T]/(1 - sT)$ is a B-algebra of finite presentation by definition; so by (1.4.1), $A \otimes_{B'} B_s$ is a C-algebra of finite presentation. One sees therefore that on the other hand f is also locally of finite presentation.

On the other hand, consider the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_Z Y \arrow[d, "f \times 1"] \\ Y \arrow[r, "\Delta_g"] & Y \times_Z Y \end{tikzcd}$$

which identifies X as the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z Y$ (**I**, 5.3.7). If Δ_g is locally of finite presentation, the same follows for Γ_f by (iii); on the other hand, we have the factorisation of f as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ (**I**, 5.3.13), where the projection p_2 equals $(g \circ f) \times_Z 1_Y$; as $g \circ f$ is assumed locally of finite presentation, the same is true of p_2 by (iv), and it will follow from (ii) that f is also locally of finite presentation. One sees therefore that we are led to proving the

Corollary (1.4.3.1). — *Let $g : Y \to Z$ be a morphism locally of finite type; then the diagonal morphism $\Delta_g : Y \to Y \times_Z Y$ is locally of finite presentation.*

One can restrict to the case where $Z = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, B being an A-algebra of finite **IV-1** | 232
type; the morphism Δ_g corresponds then to the *augmentation homomorphism* $\delta : B \otimes_A B \to B$ such that $\delta(x \otimes y) = xy$. In view of the definition (1.4.2), it suffices to show that the kernel $\mathfrak{J}$ of δ is an ideal of finite type, which follows from ($\mathbf{0_{IV}}$, 20.4.4).

Proposition (1.4.4). — *Let A be a ring, B an A-algebra, B' a polynomial algebra $A[T_1, \dots, T_n]$, $u : B' \to B$ a surjective homomorphism of A-algebras. For B to be an A-algebra of finite presentation, it is necessary and sufficient that the kernel $\mathfrak{J}$ of u be a finitely generated ideal of B'.*

The condition is sufficient by definition (1.4.1). To see that it is necessary, note that the morphism $g : \mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is locally of finite type; if B is an A-algebra of finite presentation, the morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(B')$ corresponding to u is locally of finite presentation, so $g \circ f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is locally of finite presentation. But, to prove that the ideal $\mathfrak{J}$ is of finite type, it suffices to prove that $\widetilde{\mathfrak{J}}$ is a $\widetilde{B}'$-Module of finite type (**II**, 6.1.4.1). Returning to the definition (1.4.2), we are finally led to proving the

Lemma (1.4.4.1). — *Let $\mathfrak{a}$ be an ideal of a ring C, s an element of C, such that $C_s/\mathfrak{a}_s$ is a C-algebra of finite presentation; then $\mathfrak{a}_s$ is a finitely generated ideal of C_s.*

By hypothesis, there exists a C-algebra of polynomials $C' = C[T_1, \dots, T_n]$ and a C-homomorphism surjective $v : C' \to C_s/\mathfrak{a}_s$ whose kernel $\mathfrak{b}$ is of finite type. Let $\rho : C_s \to C_s/\mathfrak{a}_s$ be the canonical homomorphism; for each i, there exists $t_i \in C$ such that $\rho(t_i/s^k) = v(T_i)$ (one may suppose that the exponent k is the same for all indices i). Consider then the C_s-algebra $D = C_s[T_1, \dots, T_n]$, and the homomorphism of C_s-algebras $w : D \to C_s$ such that $w(T_i) = t_i/s^k$; w is evidently surjective, and by the same construction $v' = \rho \circ w : D \to C_s/\mathfrak{a}_s$. On the other hand, every polynomial of D can be written P/s^m, where $P \in C' = C[T_1, \dots, T_n]$, and we have $v'(P/s^m) = (1/s^m)v(P)$; as $1/s$ is invertible in C_s, the relation $v'(P/s^m) = 0$ in $C_s/\mathfrak{a}_s$ is equivalent to $v(P) = 0$, and by the same the kernel $\mathfrak{b}'$ of v' is generated by $\mathfrak{b}$ in the ring D. But as $\mathfrak{b}'$ is the image of $\mathfrak{b}$ by the canonical map of $\mathfrak{b}$ to D, and the kernel of w is the ideal generated by the $F_i(T_i)$ in D; this proves that B' is an A-algebra of finite presentation.

This lemma being proved, keep the same notation, and let $w = v \circ u : B'' \to B$. If $\mathfrak{b}$ (resp. $\mathfrak{a}$) is the kernel of w (resp. u), we have $\mathfrak{a} = v(\mathfrak{b})$ since v is surjective. But (1.4.4), if B is an A-algebra of finite presentation, $\mathfrak{b}$ is an ideal of finite type of B'', so $\mathfrak{a}$ is a finitely generated A-module of finite type since B is an A-algebra of finite type; as B' is an A-module of finite presentation, $\mathfrak{a}$ is *a fortiori* a finitely generated ideal of B'; by the same, B is a B'-algebra of finite presentation, and as B' is an A-algebra of finite presentation, B is an A-algebra of finite presentation.

1.5. Morphisms of finite type

Proposition (1.5.1). — *Let $f : X \to Y$ be a morphism, (U_α) an open cover of Y consisting of affine open sets. The following conditions are equivalent:*

a) *f is locally of finite type and quasi-compact.*
a) *For each α, $f^{-1}(U_\alpha)$ is a finite union of affine open sets $V_{\alpha i}$ such that each ring $A(V_{\alpha i})$ is an $A(U_\alpha)$-algebra of finite type.*
a) *For every affine open subset U of Y, $f^{-1}(U)$ is a finite union of affine open sets V_j such that the rings $A(V_j)$ are $A(U)$-algebras of finite type.*

For the proof, see (**I**, 6.3.2) and (**I**, 6.6.3).

Definition (1.5.2). — We say that a morphism f is *of finite type* if it satisfies the equivalent conditions of the proposition (1.5.1).

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Proposition (1.5.3). — *Let $f : X \to Y$ be a morphism of finite type; if Y is noetherian, the same is true of X.*

For the proof, see (**I**, 6.3.7).

Proposition (1.5.4). — (i) *Let $j : X \to Y$ be a morphism of immersion. If j is quasi-compact* (*which is the case if j is a closed immersion, or if the underlying space of X is noetherian, or if the underlying space of Y is locally noetherian*), *j is of finite type.*
(i) *The composite of two morphisms of finite type is of finite type.*
(i) *If $f : X \to Y$ is an S-morphism of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite type for every base extension $S' \to S$ of the base prescheme.*
(i) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms of finite type, $f \times_S g$ is of finite type.*
(i) *If the composite $g \circ f$ of two morphisms is of finite type and if g is quasi-separated or if X is noetherian, f is of finite type.*
(i) *If a morphism f is of finite type, the same is true of f_{red}.*

This follows immediately (except the case (v) where X is noetherian, proved in (**I**, 6.3.6)) from the definitions and from (1.1.2), (1.2.4) and (1.3.4).

Corollary (1.5.5). — *Let $f : X \to Y$ be a morphism of finite type. For every morphism $Y' \to Y$ such that Y' is noetherian, $X \times_Y Y'$ is noetherian.*

This follows from (1.5.4), (iii)) and from (1.5.3).

Corollary (1.5.6). — *Let X be a prescheme of finite type over a locally noetherian prescheme S. Then every S-morphism $f : X \to Y$ is of finite type.*

For the proof, see (**I**, 6.3.9).

Proposition (1.5.7). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be of finite type, it is necessary and sufficient that φ make B an A-algebra of finite type.*

For the proof, see (**I**, 6.3.3).

1.6. Morphisms of finite presentation

Definition (1.6.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *of finite presentation* if it satisfies the following three conditions:

(i) f is locally of finite presentation;
(i) f is quasi-compact (which, together with (i), is equivalent to saying that f is of finite type (1.5.2));
(i) f is quasi-separated.

We then also say that X is *of finite presentation over* Y, or a Y*-prescheme of finite presentation*.

The condition (iii) is automatically satisfied if f is separated, or if X is locally noetherian (1.2.8); when Y is locally noetherian, it amounts to the same to say that f is *of finite presentation* or that f is *of finite type* (the latter condition implying that X is locally noetherian (1.3.3)). **IV-1** | 235

Proposition (1.6.2). — (i) *Every quasi-compact immersion* (*in particular every quasi-compact open immersion*) *is of finite presentation.*

(i) *The composite of two morphisms of finite presentation is of finite presentation.*

(i) *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism of finite presentation. Then, for every base change $S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite presentation.*

(i) *Let $f : X \to Y$, $f' : X' \to Y'$ be two S-morphisms of finite presentation; then $f \times_S f' : X \times_S X' \to Y \times_S Y'$ is of finite presentation.*

(i) *If the composite $g \circ f$ of two morphisms is of finite presentation and if g is quasi-separated and locally of finite type, then f is of finite presentation.*

This follows immediately from (1.1.2), (1.2.2), (1.2.4) and (1.4.3).

It follows in particular from (1.6.2), (iii)) that if f is a morphism of finite presentation and U an open subset of Y, the restriction $f^{-1}(U) \to U$ of f is a morphism of finite presentation. Conversely, let (U_α) be a cover of Y by affine open sets, and suppose that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ of f are morphisms of finite presentation; then f is evidently locally of finite presentation, quasi-compact by virtue of (1.1.1) and quasi-separated by virtue of (1.2.6). One can therefore say that the property for a morphism $f : X \to Y$ of being of finite presentation is *local on* Y.

If X is a quasi-separated prescheme, every morphism $f : X \to Y$ is quasi-separated (1.2.3). So, if f is quasi-compact and locally of finite presentation, f is of finite presentation.

In particular:

Corollary (1.6.3). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ to be of finite presentation, it is necessary and sufficient that φ make B an A-algebra of finite presentation.*

The necessity of the condition follows indeed from (1.4.6).

Remark (1.6.4). — In the definition (1.6.1), the condition (iii) is not a consequence of the other two. Take for example Y an affine scheme whose underlying space is not noetherian, and let U be an open non quasi-compact subset in Y. Let X be the prescheme obtained by gluing two preschemes Y_1, Y_2 isomorphic to Y along the open sets U_1, U_2 corresponding to U ($\mathbf{0_I}$, 4.1.7), so that X is the union of two open subsets isomorphic to Y_1 and Y_2 (which we identify with the latter), with $Y_1 \cap Y_2 = U$. Furthermore, there is a morphism $f : X \to Y$ which coincides in Y_i with the isomorphism $Y_i \to Y$ given. It is clear that f satisfies condition (i) of (1.6.1), being a local isomorphism; it also satisfies (ii), as the inverse image of an open quasi-compact set is quasi-compact; but as $Y_1 \cap Y_2$ is not quasi-separated (1.2.7), f is not quasi-separated.

Proposition (1.6.5). — *Let X', X'' be two preschemes, $X = X' \coprod X''$ their sum, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism which coincides with f' in X' and with f'' in X''. For f to be of finite presentation, it is necessary and sufficient that f' and f'' be so.*

It suffices to show that for f to possess one of the three properties of the definition (1.6.1), it is necessary and sufficient that f' and f'' possess it; this is evident for the property of being locally of finite presentation, which is local on X; for the property of being quasi-compact, this was seen in (1.1.6) and for the property of being quasi-separated, in (1.2.9). **IV-1** | 236

1.7. Improvements of earlier results We will give in this number a list of propositions proved in earlier chapters, whose statement can be improved with the help of the new finiteness conditions introduced above.

(1.7.1). In the statements of (**I**, 6.4.2), (**I**, 6.4.3) and (**I**, 6.4.9), one can, instead of supposing that X and Y are of finite type over the field K, suppose only that they are *locally of finite type* over K; for (**I**, 6.4.2), it suffices to observe that every point of a prescheme X that is closed in every closed subset of X is a closed subset in X, and an affine scheme locally of finite type over K is *ipso facto* of finite type (1.5.1). For (**I**, 6.4.9), one uses (1.3.7) and (1.3.4), (v)).

(1.7.2). In (**I**, 6.5.1), (ii)), one can, instead of supposing that S is locally noetherian and Y of finite type over S, suppose only that Y is *locally of finite presentation over* S, as one sees immediately from the proof (by applying the definition (1.4.2)). Similarly in (**I**, 6.5.4), (ii)) and (**I**, 6.5.5), (ii)) it suffices to suppose that f is an S-morphism, Y being *locally of finite presentation* over S.

(1.7.3). In (**I**, 7.1.11), for the second assertion, instead of supposing S locally noetherian and Y of finite type over S, one can suppose only Y locally of finite presentation over S (the proof depending on (**I**, 6.5.1), (ii)); similarly in (**I**, 7.1.12) to (**I**, 7.1.15).

(1.7.4). In (**I**, 9.2.2), one can replace the three hypotheses by the single hypothesis (implied by each of the others) that f is *quasi-compact and quasi-separated*, by virtue of (1.2.6) and (1.2.7). We note that in (**I**, 9.2.1), the hypothesis means exactly that f is quasi-compact and quasi-separated.

(1.7.5). In (**I**, 9.3.1), (**I**, 9.3.2) and (**I**, 9.3.3), one can replace the two hypotheses on X by the (weaker) hypothesis that X is a *quasi-compact and quasi-separated* prescheme, the proof using exactly these two properties (by virtue of (1.2.7)).

(1.7.6). In (**I**, 9.3.4) and (**I**, 9.3.5), it suffices to suppose X quasi-compact, $\mathscr{F}$ and $\mathscr{J}$ quasi-coherent and of finite type.

(1.7.7). In (**I**, 9.4.7), one can weaken the hypothesis $b)$ by supposing only X *quasi-separated*, since it suffices that the canonical immersion $U \to X$ be quasi-compact (1.2.7). One can make the same modification in (**I**, 9.4.8), (**I**, 9.4.9) and (**I**, 9.4.10).

(1.7.8). In (**I**, 9.5.1), the conditions in parentheses in the statement can be replaced by the condition that f be *quasi-compact and quasi-separated*; the current proof is indeed insufficient (with the hypotheses made), because of the fact that for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\mathrm{R}^1 f_*(\mathscr{F}) = 0$, it does not necessarily follow, for an affine open subset U of Y, that the same property holds for the $(\mathscr{O}_X|f^{-1}(U))$-Modules quasi-coherent, if one does not know whether such a Module is the *restriction* of an $\mathscr{O}_X$-Module quasi-coherent (the same remark applying to conditions $b)$ and $c')$); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (**I**, 9.4.7), modified above (1.7.7).

(1.7.9). In (**II**, 5.3.2), it suffices to suppose Y *quasi-compact and quasi-separated* (this being what intervenes in (**II**, 4.4.6) and (**II**, 4.4.7)). Similarly in (**II**, 5.5.3), (ii)).

(1.7.10). In (**II**, 1.7.15), one can suppose only that Y is quasi-compact and quasi-separated, the proof using only (**I**, 9.6.5).

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(1.7.11). In the second assertion of (**II**, 3.4.5), one can substitute for the two hypotheses on Y the weaker hypothesis that Y is quasi-compact and quasi-separated. Similarly in (**II**, 3.4.8).

(1.7.12). In (**II**, 3.8.5), one can similarly suppose only that Y is quasi-compact and quasi-separated, this hypothesis being that which intervenes in (**I**, 9.4.7).

(1.7.13). In (**II**, 4.4.1), (**II**, 4.4.6) and (**II**, 4.4.7), it suffices to suppose that Y is *quasi-compact and quasi-separated*, taking into account (1.2.3) in the proof of (**II**, 4.4.7). Similarly, in (**II**, 4.4.10.1), it suffices to suppose Z quasi-compact and quasi-separated.

(1.7.14). In (**II**, 4.5.2) and (**II**, 4.5.5), it suffices to suppose X *quasi-compact and quasi-separated*, this hypothesis being that which intervenes in (**I**, 9.3.1) and (**I**, 9.3.2).

(1.7.15). In (**II**, 4.6.8), it suffices again to suppose X quasi-compact and quasi-separated, this hypothesis being that which intervenes in (**I**, 9.4.7).

(1.7.16). In (**II**, 5.1.2), it suffices to suppose that X is quasi-compact and quasi-separated (the hypothesis intervening in (**II**, 4.5.2) and (**II**, 4.5.5)). Similarly in (**II**, 5.1.9), where one uses (**I**, 9.6.5).

(1.7.17). In (**II**, 5.2.1), it suffices always to suppose X quasi-compact and quasi-separated (the hypothesis intervening in (**II**, 4.5.2)).

(1.7.18). In (**II**, 5.2.2), one must suppose f *quasi-compact* and X *quasi-separated*; the current proof is indeed insufficient (with the hypotheses made), because of the fact that for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\mathrm{R}^1 f_*(\mathscr{F}) = 0$, it does not necessarily follow, for an affine open subset U of Y, that the same property holds for the $(\mathscr{O}_X|f^{-1}(U))$-Modules quasi-coherent, if one does not know whether such a Module is the restriction of an $\mathscr{O}_X$-Module quasi-coherent (the same remark applying to conditions b) and c')); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (**I**, 9.4.7), modified above (1.7.7).

(1.7.19). In (**II**, 5.3.2), it suffices to suppose Y *quasi-compact and quasi-separated* (this being what intervenes in (**II**, 4.4.6) and (**II**, 4.4.7)). Similarly in (**II**, 5.5.3), (ii)).

(1.7.20). In (**III**, 1.2.2, 1.2.3 and 1.2.4), one can replace the two hypotheses on X by the weaker hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using only (**I**, 9.3.3).

(1.7.21). In (**III**, 1.4.10 to 1.4.14), one can replace "separated" by "quasi-separated", this hypothesis serving only to be able to apply (**I**, 9.2.2) (see above (1.7.4)). Similarly, in (**III**, 1.4.15), it suffices to suppose that the morphism u is *quasi-separated and quasi-compact*.

(1.7.22). In (**III**, 2.1.3), one can again replace the hypotheses by the hypothesis that X is a prescheme quasi-compact and quasi-separated, the reasoning being the same as in (**III**, 1.2.2).

1.8. Morphisms of finite presentation and constructible sets **IV-1** | 238

(1.8.1). Let X be a prescheme; one knows that every quasi-compact open subset in X is a finite union of affine open sets, and conversely. For an open subset U of X to be *retrocompact* in X ($\mathbf{0_{III}}$, 9.1.1), it is necessary and sufficient that for every *affine* open subset V of X, $U \cap V$ be quasi-compact. When X is *quasi-separated* (1.2.1), every quasi-compact open subset in X is retrocompact in X (1.2.7), so every locally constructible subset of X is retrocompact in X ($\mathbf{0_{III}}$, 9.1.13); if *in addition* X is *quasi-compact*, there is identity between constructible subsets and locally constructible subsets in X ($\mathbf{0_{III}}$, 9.1.12), and between constructible open subsets and quasi-compact open subsets ($\mathbf{0_{III}}$, 9.1.5).

Proposition (1.8.2). — *Let X, Y be two preschemes, $f : X \to Y$ a morphism. For every constructible* (*resp. locally constructible*) *subset Z of Y, $f^{-1}(Z)$ is a constructible* (*resp. locally constructible*) *subset of X.*

Suppose Z locally constructible; for every $x \in X$, there will be an affine open neighbourhood V of $f(x)$ such that $Z \cap V$ be constructible in V; as the proposition is proved for constructible subsets, $f^{-1}(Z) \cap f^{-1}(V)$ will be constructible in $f^{-1}(V)$, so $f^{-1}(Z)$ will be locally constructible. It suffices therefore to consider the case where Z is constructible, and taking into account ($\mathbf{0_{III}}$, 9.1.3), we are reduced to the case where Z is open and retrocompact in Y, in other words such that the canonical injection $Z \to Y$ is a quasi-compact morphism; it suffices then to see that $f^{-1}(Z)$ is *retrocompact in* X, in other words that the injection $f^{-1}(Z) \to X$ is a quasi-compact morphism; but this follows from (1.1.2), (iii)), since $f^{-1}(Z) = Z \times_Y X$ (**I**, 4.4.1).

Lemma (1.8.3). — *Let X be a quasi-compact and quasi-separated prescheme, Z a constructible subset of X. There exist then an affine scheme X' and a morphism of finite presentation $f : X' \to X$ such that $f(X') = Z$.*

Since X is quasi-compact, it is a finite union of affine open sets X_i ($i \in I$), and since X is quasi-separated, it follows from (1.2.7) that each of the canonical immersions $g_i : X_i \to X$ is a morphism of finite presentation; one concludes that if $f_i : X'_i \to X_i$ is a morphism of finite presentation, the same is true of $g_i \circ f_i : X'_i \to X$ (1.6.2), and if X' is the prescheme sum of the X'_i, the morphism $h : X' \to X$ which coincides with $g_i \circ f_i$ in each X'_i is also of finite presentation (1.6.5). Since $Z \cap X_i$ is constructible in X_i ($\mathbf{0_{III}}$, 9.1.8, (i)), one sees that one is led to proving the lemma when X_i is *affine*, that is a ring A. As Z is then a finite union of subsets of the form $U \cap \complement V$, where U and V are quasi-compact open subsets ($\mathbf{0}_{\mathbf{III}}$, 9.1.3), one sees, by considering a suitable sum of preschemes, that one can even reduce to the case where $Z = U \cap \complement V$; as U is a finite union of open sets of the form X_{f_i}, where $f_i \in A$; let $\mathfrak{J}$ be the ideal of A generated by the f_i, and let Z' be the closed affine sub-scheme of X which it defines (and which is by construction of finite presentation on X); we have by definition $V = \complement Z'$, so $U \cap \complement V = U \cap Z'$. Consider the affine scheme $X' = Z' \times_X U$ and let $f : X' \to X$ be the structural morphism; one sees that the canonical immersion $U \to X$, which is of finite presentation, and the same is true of the open immersion $U \to X$, which is quasi-compact since U and X are affine; we conclude therefore from (1.6.2), (iv)) that f is of finite presentation, and we have $f(X') = Z' \cap U$ (**I**, 3.4.8), which finishes the proof. **IV-1** | 239

Theorem (1.8.4). — *(Chevalley). — Let $f : X \to Y$ be a morphism of finite presentation. For every locally constructible subset Z of X, $f(Z)$ is locally constructible in Y.*

Let $y \in Y$ and V an affine open neighbourhood of y; as f is quasi-compact and quasi-separated, the same is true of its restriction $f^{-1}(V) \to V$, so $f^{-1}(V)$ is a prescheme quasi-compact and quasi-separated (1.2.3); since $Z \cap f^{-1}(V)$ (1.8.1) is constructible in $f^{-1}(V)$, one sees that it suffices to consider the case where Y is affine and Z constructible; but then it will suffice to prove the

Lemma (1.8.4.1). — *Let Y be an affine scheme, $f : X \to Y$ a morphism quasi-compact and locally of finite presentation; then $f(X)$ is a constructible subset of Y.*

Since X is quasi-compact, it is a finite union of affine open sets; one can therefore reduce to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, B being an A-algebra of finite presentation. As Y is quasi-compact and separated, it is also quasi-separated, whence (1.8.3), there exists a morphism of finite presentation $g : X' \to X$ such that $Z = g(X')$; as $f \circ g$ is of finite presentation (1.6.2), and $f(Z) = f(g(X'))$, we have seen that the question is reduced to proving that $f(X)$ is constructible; since f is of finite presentation (1.6.2), (iv)), it suffices to apply (1.8.4.1).

Lemma (1.8.4.2). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; if B is an A-algebra of finite presentation, there exist a λ and an A_λ-algebra of finite presentation B_λ such that B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$.*

By hypothesis, B is isomorphic to an algebra of the form $A[T_1, \ldots, T_n]/\mathfrak{J}$, where the T_i are indeterminates and $\mathfrak{J}$ an ideal of finite type; let (F_j) ($1 \leqslant j \leqslant m$) be a system of generators of $\mathfrak{J}$. Let $\varphi_\lambda : A_\lambda \to A$ be the canonical homomorphism. Since L is filtering, there exists λ such that each of the coefficients of the polynomials F_j is the image by φ_λ of an element of A_λ. In other words, there exist a system of polynomials $(F_{j\lambda})_{1 \leqslant j \leqslant m}$ of $A_\lambda[T_1, \ldots, T_n]$ such that $F_j = \varphi_\lambda(F_{j\lambda})$ for $1 \leqslant j \leqslant m$. If $\mathfrak{J}_\lambda$ is the ideal of $A_\lambda[T_1, \ldots, T_n]$ generated by the $F_{j\lambda}$, $\mathfrak{J}$ is the image of $\mathfrak{J}_\lambda \otimes_{A_\lambda} A$ in $A[T_1, \ldots, T_n] = A_\lambda[T_1, \ldots, T_n] \otimes_{A_\lambda} A$; the ring $B_\lambda = A_\lambda[T_1, \ldots, T_n]/\mathfrak{J}_\lambda$ answers the question.

We will apply this lemma by considering A as an inductive limit of its sub-$\mathbf{Z}$-algebras of *finite type*. One sees therefore that there exist such a sub-$\mathbf{Z}$-algebra A_0 of A, and an A_0-algebra of finite presentation B_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$; setting $X_0 = \operatorname{Spec}(B_0)$, $Y_0 = \operatorname{Spec}(A_0)$, so that $X = X_0 \times_{Y_0} Y$, f being the projection $X \to Y$; it follows from (**I**, 3.4.8) that $f(X) = g_0^{-1}(f_0(X_0))$; taking into account (1.8.2), it suffices therefore to show that $f_0(X_0)$ is constructible. In other words, we are finally reduced to proving the **IV-1** | 240

Corollary (1.8.5). — *Let Y be a noetherian prescheme, $f : X \to Y$ a morphism of finite type. For every constructible subset Z of X, $f(Z)$ is a constructible subset of Y.*

One reduces as above to proving that $f(X)$ is constructible. Applying the criterion ($\mathbf{0}_{\mathbf{III}}$, 9.2.3): one must prove that for every irreducible closed subset T of Y, $T \cap f(f^{-1}(T))$ is rare in T or contains a non-empty open subset of T; by taking a sub-prescheme of Y having T as underlying space (**I**, 5.2.1) and considering its inverse image by f, one is finally (taking into account (1.5.4), (iii))) reduced to proving that if Y is irreducible and f dominant, $f(X)$ contains a non-empty open subset of Y. One can in addition suppose Y affine; then X is a finite union of affine open sets V_i, and as $f(X)$ is dense in Y, the same is true of at least one of the $f(V_i)$. One can therefore also suppose X affine; if (X_j) is the family (finite) of the irreducible components of X, one sees as above that at least one of the $f(X_j)$ is dense in Y; one can therefore suppose X irreducible. Finally, replacing f by f_{red} (1.5.4), (vi)) one can suppose X and Y *integral*. Then $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$, where A is a noetherian integral ring and B an integral A-algebra of finite type containing A (1.2.7). It suffices to show that there exists $g \in A$ such that, for every $y \in D(g)$ (that is to say $g \notin \mathfrak{j}_y$) the ideal $\mathfrak{j}_y$ the ideal of A such that y is the intersection of A and of a prime ideal of B, it will be shown that $D(g) \subset f(X)$. Finally it suffices to prove the

Lemma (1.8.5.1). — *Let A be an integral ring, B an integral A-algebra of finite type. There exists $g \in A$ such that every homomorphism of A into an algebraically closed field Ω, non-zero on g, extends to a homomorphism of B into Ω.*

But this is a classical result of commutative algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 3 of th. 1).

Corollary (1.8.6). — *Let Y be an irreducible prescheme, η its generic point, $f : X \to Y$ a morphism locally of finite type. If $f^{-1}(\eta)$ is not empty, there exists an open neighbourhood U of η in Y such that $f^{-1}(y)$ is not empty for every $y \in U$. If $\mathscr{F}$ is an $\mathscr{O}_X$-Module quasi-coherent of finite type, and if $\mathscr{F}_\eta = \mathscr{F} \otimes_{\mathscr{O}_Y} k(\eta)$ is not zero, then there exists an open neighbourhood U' of η in Y such that $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is not zero for every $y \in U'$.*

If p_y is the canonical projection of the fibre $f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y))$ into X, we have $\mathscr{F}_y = p_y^*(\mathscr{F})$, so $\operatorname{Supp}(\mathscr{F}_y) = p_y^{-1}(\operatorname{Supp}(\mathscr{F})) = \operatorname{Supp}(\mathscr{F}) \cap f^{-1}(y)$ by virtue of (**I**, 9.13.1) and (**I**, 3.6.1), since $\mathscr{F}$ is of finite type; moreover $\operatorname{Supp}(\mathscr{F})$ is closed in X ($\mathbf{0_I}$, 5.2.2), and if Z is a closed sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space and j the canonical immersion $Z \to X$, $f \circ j$ is locally of finite type (1.3.4), (vi)); this proves that the first assertion implies the second. To prove the first assertion, remark first that one can suppose X and Y *reduced* (1.3.4), (vi)), in other words Y *integral*. Furthermore, if Z is the closed sub-prescheme reduced of X having $\overline{\{\xi\}}$ as underlying space (for a point ξ of $f^{-1}(\eta)$); one can, as above, replace X by Z, in other words suppose B *integral*. Since the morphism f is then dominant, the corresponding homomorphism $\varphi : A \to B$ is injective (**I**, 1.2.7); the corollary therefore follows finally from the lemma (1.8.5.1).

Proposition (1.8.7). — *Let S be a prescheme, $g : X \to S$, $h : Y \to S$ two morphisms of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \operatorname{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \operatorname{Spec}(k(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set of $s \in S$ such that f_s is surjective* (*resp. radiciel*) *is locally constructible.*

Let E be the set of s such that f_s is surjective; the set $S - E$ is equal to $h(Y - f(X))$; but f is of finite presentation (1.6.2), (v)) so $f(X)$ is locally constructible in Y (1.8.4); as h is of finite presentation, $h(Y - f(X))$ is also locally constructible in S, hence also E. **IV-1 | 241**

To show that the set E' of s such that f_s is radiciel is locally constructible, we will use the

Lemma (1.8.7.1). — *Let U, V be two preschemes; for a morphism $f : U \to V$ to be radiciel, it suffices that the diagonal morphism $\Delta_f : U \to U \times_V U$ be surjective. Moreover, every radiciel morphism is separated.*

Indeed, Δ_f being an immersion (**I**, 5.3.9), is a morphism locally of finite type, and $\Delta_{f_s} : X_s \to X_s \times_{Y_s} X_s$ is deduced from $\Delta_f : X \to X \times_Y X$ by the base change $\operatorname{Spec}(k(s)) \to S$. As f is of finite presentation, Δ_f is also of finite presentation (1.6.2), (v)); the same is true of Δ_{f_s} (1.6.2), (iv)), so it suffices to apply the first part of the proposition to Δ_f, using the lemma (1.8.7.1).

1.9. Pro-constructible and ind-constructible sets

Lemma (1.9.1). — *Let $(A_\alpha)_{\alpha\in I}$ be an inductive system of rings whose index set is right filtering, and let $A = \varinjlim A_\alpha$. For $A = 0$, it is necessary and sufficient that there exist an index γ such that $A_\gamma = 0$ (and then $A_\beta = 0$ for all $\beta \geqslant \gamma$).*

Indeed, for every $\alpha \in I$, the canonical homomorphism $\varphi_\alpha : A_\alpha \to A$ transforms the unit element into the unit element; to say that $A = 0$ means that $\varphi_\alpha(1) = \varphi_\alpha(0)$, or equivalently that $\varphi_\alpha(1) = 0$; as the set is filtering, this is equivalent to saying that there exists an index $\gamma \geqslant \alpha$ such that $\varphi_{\gamma\alpha} : A_\alpha \to A_\gamma$ satisfies $\varphi_{\gamma\alpha}(1) = \varphi_{\gamma\alpha}(0)$, and this last relation is equivalent to $A_\gamma = 0$, whence the lemma.

Proposition (1.9.2). — *Let B be a ring, $(A_\alpha)_{\alpha\in I}$ an inductive system of B-algebras whose index set I is right filtering, and let $A = \varinjlim A_\alpha$; set $Y = \operatorname{Spec}(B)$, $X_\alpha = \operatorname{Spec}(A_\alpha)$, $X = \operatorname{Spec}(A)$, and let $u : X \to Y$, $u_\alpha : X_\alpha \to Y$ be the morphisms corresponding to the structural homomorphisms $B \to A$, $B \to A_\alpha$. Then:*

(i) *For $X = \varnothing$, it is necessary and sufficient that there exist $\gamma \in I$ such that $X_\gamma = \varnothing$, and then $X_\alpha = \varnothing$ for $\alpha \geqslant \gamma$.*

(i) *We have*

$$u(X) = \bigcap_{\alpha\in I} u_\alpha(X_\alpha). \tag{1.9.2.1}$$

The assertion (i) is none other than the translation of (1.9.1). To prove (ii), note that, since u factorises as $X \xrightarrow{u_\alpha} X_\alpha \to Y$, the first member is contained in the second. Conversely, let $y \in Y - u(X)$; on a $u^{-1}(y) = \varnothing$, and $u^{-1}(y)$ is the $k(y)$-prescheme $\operatorname{Spec}(A \otimes_B k(y))$; as in the category of B-modules the functor $\varinjlim$ commutes with the tensor product, $A \otimes_B k(y)$ is the inductive limit of the system $(A_\alpha \otimes_B k(y))$; the lemma (1.9.1) shows that there exists α such that $A_\alpha \otimes_B k(y) = 0$, so $u_\alpha^{-1}(y) = \varnothing$, and hence $y \notin u_\alpha(X_\alpha)$. Q.E.D.

Proposition (1.9.3). — *Let X be a quasi-compact and quasi-separated prescheme, $(X_\alpha)_{\alpha\in I}$ an open cover of X, and for every $\alpha \in I$, set $E_\alpha = E \cap X_\alpha$. The following conditions are equivalent:*

a) *There exists a quasi-compact morphism $f : X' \to X$ such that $E = f(X')$.*

a′) *For every α, there exists a quasi-compact morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $E_\alpha = f_\alpha(X'_\alpha)$.*

a″) *For every α, there exists an affine scheme X'_α and a morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $f_\alpha(X'_\alpha) = E_\alpha$.*

a) *E is an intersection of constructible subsets of X.*

b′) *$F = X - E$ is a union of constructible subsets of X.*

It is clear that $b)$ and $b')$ are equivalent. The condition $a)$ implies $a')$ by taking for X'_α the prescheme induced on $f^{-1}(X_\alpha)$ by X' and for f_α the restriction of f. To see that $a')$ implies $a'')$, it suffices to remark that X'_α is a finite union of affine open sets; let X''_α be the prescheme *sum* of the $X''_{\alpha\lambda}$ ($\lambda \in L_\alpha$), which is an affine scheme; if $g_\alpha : X''_\alpha \to X'_\alpha$ is the morphism coinciding with the identity in each of the $X''_{\alpha\lambda}$, it is clear that $f'_\alpha = f_\alpha \circ g_\alpha$ on a $E_\alpha = f'_\alpha(X''_\alpha)$ since g_α is surjective. To show that $a'')$ implies $a)$, note that since X_α covers X for $\alpha \in J$, and let $f : X' \to X$ be the morphism which coincides with $g_\alpha \circ f_\alpha$ in each X'_α, it is clear that $E = f(X')$ and the hypothesis that X is quasi-separated implies that j_α is quasi-compact (1.2.7), hence f is quasi-compact (1.1.6) and finality of f (1.1.2).

It remains to prove the equivalence of $a'')$ and $b)$. Suppose first that $a'')$ holds; for every $\alpha \in J$, where $X_\alpha = \operatorname{Spec}(A)$ is affine, and $E = f(X')$ where $X' = \operatorname{Spec}(A')$ is affine, $f : X' \to X$ a morphism corresponding to a homomorphism of rings $A \to A'$. Note now that A' is the inductive limit of the family (A'_λ) of its sub-A-algebras of *finite type*, ordered by inclusion; if $X'_\lambda = \operatorname{Spec}(A'_\lambda)$, f factorises as $X' \to X'_\lambda \xrightarrow{f_\lambda} X$ and it follows from (1.9.2), (ii)) that $f(X') = \bigcap_\lambda f_\lambda(X'_\lambda)$; if one shows that $f_\lambda(X'_\lambda)$ is an intersection of constructible subsets of X, the same will be true of $f(X')$. One can therefore restrict to the case where A' is an A-algebra of *finite type*. But then, by the theorem of Chevalley (1.8.5), $f(X')$ is constructible in X, whence $b)$.

Let us show finally that $b)$ implies $a'')$. If E is an intersection of a family $(G_\lambda)_{\lambda\in L}$ of constructible subsets of X, each E_α is an intersection of constructible subsets $G_\lambda \cap X_\alpha$, which are constructible in X_α ($\mathbf{0}_{\mathbf{III}}$, 9.1.8, (i)), hence we are reduced to the case where $X = \operatorname{Spec}(A)$ is affine. We know then (1.8.3) that for each $\lambda \in L$ there exists a morphism $f_\lambda : X'_\lambda \to X$, where $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ is affine, such that $f_\lambda(X'_\lambda) = G_\lambda$. It suffices then to take for B' the product tensor $A'_J = \bigotimes_{\lambda\in J} A'_\lambda$ (product tensor of A-algebras), $X'_J = \mathrm{Spec}(A'_J)$, and let $f_J : X'_J \to X$ be the structural morphism. If $A' = \varinjlim A'_J$, J ranging over the filtering set of finite subsets of L, $X' = \mathrm{Spec}(A')$, and if $f : X' \to X$ is the structural morphism, we have $f(X') = \bigcap_{\lambda\in L} f_\lambda(X'_\lambda)$.

Lemma (1.9.3.1). — *Let A be a ring, A' an A-algebra of finite type. Then A' is a limit $\varinjlim A''_\mu$ where the A''_μ are A-algebras of finite presentation (1.4.1).*

Indeed, one can write $A' = B/\mathfrak{J}$, with $B = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ an ideal of B. But $\mathfrak{J}$ is the inductive limit of the *finitely generated* ideals $\mathfrak{J}'_\mu$ of B, contained in $\mathfrak{J}$, ordered by inclusion; as in the category of B-modules the functor $\varinjlim$ is exact, we conclude that $A' = \varinjlim B/\mathfrak{J}'_\mu$ up to A-isomorphism.

Lemma (1.9.3.2). — *Let $(A'_\lambda)_{\lambda\in L}$ be a family of A-algebras, $X = \mathrm{Spec}(A)$, $X'_\lambda = \mathrm{Spec}(A'_\lambda)$, and let $f_\lambda : X'_\lambda \to X$ be the structural morphism. For every finite subset J of L, set $A'_J = \bigotimes_{\lambda\in J} A'_\lambda$ (tensor product of A-algebras), $X'_J = \mathrm{Spec}(A'_J)$, and let $f_J : X'_J \to X$ be the structural morphism. If $A' = \varinjlim A'_J$, J ranging over the filtering set of finite subsets of L, $X' = \mathrm{Spec}(A')$, and if $f : X' \to X$ is the structural morphism, we have $f_J(X'_J) = \bigcap_{\lambda\in J} f_\lambda(X'_\lambda)$. If $A' = \varinjlim A'_J$, $X' = \mathrm{Spec}(A')$, and $f : X' \to X$ the structural morphism, we have $f(X') = \bigcap_{\lambda\in L} f_\lambda(X'_\lambda)$.*

The first assertion follows from (**I**, 3.4.7); the second follows from (1.9.2), (ii)).

Definition (1.9.4). — Let X be a topological space. We say that a subset E of X is *pro-constructible* (resp. *ind-constructible*) in X if, for every $x \in X$, there exists an open neighbourhood U of x such that $E \cap U$ is an intersection (resp. union) of locally constructible subsets of U.

Taking into account (1.2.7) and ($\mathbf{0_{III}}$, 9.1.8 and 9.1.12), the equivalent conditions of the proposition (1.9.3) are expressed by saying that E is *pro-constructible* in X, the equivalent conditions on the locally constructible subsets in X being identical to the constructible subsets when X is quasi-compact and quasi-separated ($\mathbf{0_{III}}$, 9.1.5).

Proposition (1.9.5). — *Let X be a prescheme.*

(i) *For a subset E of X to be pro-constructible, it is necessary and sufficient that $X - E$ be ind-constructible.*
(i) *Every finite union and every finite intersection of pro-constructible* (*resp. ind-constructible*) *subsets of X are pro-constructible* (*resp. ind-constructible*).
(i) *Every intersection* (*resp. union*) *of locally constructible subsets of X is pro-constructible* (*resp. ind-constructible*). *Conversely, if X is quasi-compact and quasi-separated, every pro-constructible* (*resp. ind-constructible*) *subset of X is an intersection* (*resp. union*) *of constructible subsets.*
(i) *Let (U_α) be an open cover of X. For a subset E of X to be pro-constructible* (*resp. ind-constructible*) *in X, it is necessary and sufficient that for every α, $E \cap U_\alpha$ be pro-constructible* (*resp. ind-constructible*) *in U_α.*
(i) *Every pro-constructible closed subset E of X is retrocompact in X.*
(i) *Let $f : X' \to X$ be a morphism of preschemes; for every pro-constructible* (*resp. ind-constructible*) *subset E of X, $f^{-1}(E)$ is pro-constructible* (*resp. ind-constructible*) *in X'.*
(i) *Let $f : X' \to X$ be a quasi-compact morphism; for every pro-constructible subset E' of X', $f(E')$ is pro-constructible in X; in particular $f(X')$ is pro-constructible in X.*
(i) *Let $f : X' \to X$ be a morphism locally of finite presentation; for every ind-constructible subset E' of X', $f(E')$ is ind-constructible in X; in particular $f(X')$ is ind-constructible in X.*
(i) *Suppose X quasi-compact and quasi-separated; then, for a subset E of X to be pro-constructible, it is necessary and sufficient that there exist an affine scheme X' and a* (*necessarily quasi-compact*) *morphism $f : X' \to X$ such that $E = f(X')$.*

The properties (i), (ii), (iv), and (iii) (first assertion) follow from the definition (1.9.4) and are valid for any topological space, using the distributivity of intersection and union, and the fact that if T is locally constructible in X, $T \cap U$ is locally constructible in U for every open U of X. If X is quasi-compact and quasi-separated and E is pro-constructible in X, there is a finite open cover (U_i) of X by affine open sets such that $E \cap U_i$ is intersection of constructible subsets in X_i for every i; E

being a union of the $E \cap U_i$, is intersection of finite unions $\bigcup_i F_{i,\lambda(i)}$, for all choices of $\lambda(i) \in L_i$; these unions are constructible in X; this proves the second assertion of (iii) and also (ix) by (1.9.3).

To prove (v), one can restrict to the case where X is affine, and prove that E is quasi-compact for the induced topology of X. The properties (vi) and (viii) translate respectively (1.9.5), (vi) and (vii).

Similarly, to prove (vii), one can restrict to the case where X is affine; then X' is a finite union of affine open sets X'_i and $f(E')$ is a union of the $f(E' \cap X'_i)$, so one can even reduce to X' affine, in which case X' is a union of affine open sets, and $f(E')$ is a union of the $f(E' \cap X'_i)$; hence $f(E')$ is pro-constructible by an application continued.

To prove (viii), one can also restrict to the case where X is affine; then X' is a finite union of affine open sets X'_i and $f(E')$ is a union of the $f(E' \cap X'_i)$, so from (ii) one can restrict to the case where X' is affine; this then follows from the theorem of Chevalley (1.8.4) and (iii).

Finally, to prove (viii), one can still suppose X affine and, by (iv), also restrict to the case where E' is ind-constructible in X'; the conclusion then follows from (iii) and (1.8.4). **IV-1** | 245

Examples (1.9.6). — Every *finite* subset of a prescheme X is pro-constructible: indeed, it suffices to consider the singleton $\{x\}$ reduced to a single point (1.9.5), (ii)) and $\{x\}$ is the image of $\operatorname{Spec}(k(x))$ by the canonical morphism $\operatorname{Spec}(k(x)) \to X$, which is quasi-compact; whence the conclusion by (1.9.5), (vii)). On the other hand, a subset $\{x\}$ reduced to a point is not necessarily ind-constructible; for example, let A be a noetherian integral ring having infinitely many maximal ideals, and let x be the generic point of $X = \operatorname{Spec}(A)$; if $\{x\}$ were ind-constructible in X, it would be constructible by virtue of (1.9.5), (iii)) and would therefore contain a non-empty open subset of X ($\mathbf{0_{III}}$, 9.2.2); but this contradicts the hypothesis that A has infinitely many maximal ideals, by virtue of the theorem of Artin–Tate ($\mathbf{0_{I}}$, 16.3.3).

Every *closed* subset Y of a prescheme X is pro-constructible, by (1.9.5), (v)). Every open subset of X is therefore ind-constructible, but an open subset U of X is not pro-constructible in general unless it is *retrocompact*, by virtue of (1.9.5), (v)).

Finally, if X is a *noetherian* prescheme, hence quasi-separated (1.2.8), it follows from (1.9.5), (iii)) and ($\mathbf{0_{III}}$, 9.1.7) that the ind-constructible subsets of X are the *unions of locally closed subsets* of X.

Theorem (1.9.7). — *Let X be a quasi-compact prescheme, E an ind-constructible subset, $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible subsets of X such that $\bigcap_{\lambda \in L} F_\lambda \subset E$; then there exists a finite subset J of L such that $\bigcap_{\lambda \in J} F_\lambda \subset E$.*

Note that the sets $F = X - E$ and $F \cap F_\lambda$ are pro-constructible, so we are reduced to the

Corollary (1.9.8). — *Let X be a quasi-compact prescheme, $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible subsets of X whose intersection is empty; then there exists a finite sub-family $(F_\lambda)_{\lambda \in J}$ whose intersection is empty.*

Covering X by a finite number of affine open sets, and using (1.9.5), (iv)), we are reduced to the case where X is affine; by virtue of (1.9.5), (ix)), we have $F_\lambda = f_\lambda(X'_\lambda)$, where $X'_\lambda = \operatorname{Spec}(A'_\lambda)$ **IV-1** | 246
is affine, and f_λ is a morphism $X'_\lambda \to X$; then, with the notation of (1.9.3.2), on a by hypothesis $f(X') = \varnothing$, so $X' = \varnothing$; but this implies by (1.9.2), (i)) that there exists a finite subset J of L such that $X'_J = \varnothing$, hence $f_J(X'_J) = \bigcap_{\lambda \in J} F_\lambda = \varnothing$.

Corollary (1.9.9). — *Let X be a quasi-compact prescheme, F a pro-constructible subset, $(E_\lambda)_{\lambda \in L}$ a family of ind-constructible subsets of X such that $F \subset \bigcup_{\lambda \in L} E_\lambda$; then there exists a finite subset J of L such that $F \subset \bigcup_{\lambda \in J} E_\lambda$. In particular, from every cover of X by ind-constructible subsets, one can extract a finite cover of X.*

This follows from (1.9.7) by passing to complements.

Proposition (1.9.10). — *Let X be a prescheme. For a subset E of X to be ind-constructible, it is necessary that, for every $x \in X$, the intersection $E \cap \overline{\{x\}}$ be a neighbourhood of x in $\overline{\{x\}}$. If X is locally noetherian, this condition is sufficient.*

Suppose E ind-constructible, and let Y be the intersection of $\overline{\{x\}}$ and an affine open subset in X, containing x; there is then a sub-prescheme of X having Y as underlying space (**I**, 5.2.1), and if $j : Y \to X$ is the canonical injection, $E \cap Y = j^{-1}(E)$ is ind-constructible in Y (1.9.5), (vi)). Since Y is quasi-compact and separated, $E \cap Y$ is therefore a union of constructible subsets of Y (1.9.5), (iii));

furthermore, there are two open subsets U, V of Y such that $x \in U \cap \complement V \subset E \cap Y$ ($\mathbf{0_{III}}$, 9.1.3). But as x is the generic point of the irreducible space Y, V is necessarily empty and we have $U \subset E \cap Y$.

Now suppose X locally noetherian, and let E be a subset of X satisfying the condition of the statement. By virtue of the definition (1.9.4), one can restrict to the case where X is *noetherian*. Then, for every $x \in E$, $E \cap \overline{\{x\}}$ contains a non-empty open subset of the form $U \cap \overline{\{x\}}$, where U is open in X; this shows that E is a union of locally closed subsets, hence is ind-constructible (1.9.6).

Proposition (1.9.11). — *Let X be a prescheme, E a subset of X. The following properties are equivalent:*

a) *E is locally constructible.*
a) *E is ind-constructible and pro-constructible.*
a) *E and $X - E$ are pro-constructible.*
a) *E and $X - E$ are ind-constructible.*

It suffices evidently to prove that d) implies a), and one can restrict to the case where X is affine. Then (1.9.5), (iii)), E (resp. $X - E$) is a union of a family (E_λ) (resp. (E'_μ)) of constructible subsets of X; as the E_λ and the E'_μ form a cover of X, by (1.9.9) it follows that there are finitely many λ_i, μ_j such that the E_{λ_i} and the E'_{μ_j} form a cover of X; this implies that E is the union of the E_{λ_i}, hence is constructible.

Corollary (1.9.12). — *Let $f : X \to Y$ be a surjective morphism, either quasi-compact, or locally of finite presentation. For a subset E of Y to be locally constructible* (*resp. pro-constructible, resp. ind-constructible*), *it is necessary and sufficient that $f^{-1}(E)$ be so in X.*

One knows that the condition is necessary (1.8.2) and (1.9.5), (vi))); to show that it is sufficient, **IV-1** | 247
one can restrict (by virtue of (1.9.5), (iv))) to the case where X is affine, or the case where $f^{-1}(E)$ is pro-constructible, *or* the case where $f^{-1}(E)$ is ind-constructible. If f is surjective and quasi-compact, and $f^{-1}(E)$ pro-constructible, it follows from (1.9.5), (vii)) that $E = f(f^{-1}(E))$ is pro-constructible. If f is surjective and locally of finite presentation, and $f^{-1}(E)$ ind-constructible, $E = f(f^{-1}(E))$ is ind-constructible by (1.9.5), (viii)), which finishes the proof.

(1.9.13). Let X be a prescheme, $\mathscr{T}$ its topology; it follows from (1.9.5), (i) and (ii)) that the ind-constructible (resp. pro-constructible) subsets of X are the open subsets (resp. closed subsets) for a topology on X, called the *constructible topology* and which we will denote by $\mathscr{T}^{\mathrm{cons}}$; we will denote by X^{cons} the set X equipped with the topology $\mathscr{T}^{\mathrm{cons}}$.

Proposition (1.9.14). — *Let X be a prescheme, $\mathscr{T}$ its topology, $\mathscr{T}^{\mathrm{cons}}$ the constructible topology on X.*

(i) *The topology $\mathscr{T}^{\mathrm{cons}}$ is finer than $\mathscr{T}$.*
(i) *The locally constructible subsets of X are identical to the subsets that are both open and closed in X^{cons}.*
(i) *For every morphism $f : X \to Y$, the underlying map of X^{cons} to Y^{cons} is continuous; we denote it f^{cons}.*
(i) *If the morphism $f : X \to Y$ is quasi-compact, f^{cons} is a closed map; in particular, if f is quasi-compact and bijective, f^{cons} is a homeomorphism.*
(i) *If the morphism $f : X \to Y$ is locally of finite presentation, f^{cons} is an open map; in particular, if f is bijective and locally of finite presentation, f^{cons} is a homeomorphism.*
(i) *For every open subset U of X, the topology induced by $\mathscr{T}^{\mathrm{cons}}$ on U is identical to the topology of U^{cons}.*

Indeed, (i) follows from the fact that every open set for $\mathscr{T}$ is an open set for $\mathscr{T}^{\mathrm{cons}}$ (1.9.6), and (ii) is a translation of (1.9.11); (iii), (iv) and (v) translate respectively (1.9.5), (vi), (vii) and (viii). Finally, to prove (vi), it suffices to remark that the canonical injection $j : U \to X$ is locally of finite presentation (1.4.3), so $j^{\mathrm{cons}} : U^{\mathrm{cons}} \to X^{\mathrm{cons}}$ is a continuous open map.

Proposition (1.9.15). — *Let X be a prescheme.*

(i) *For X^{cons} to be quasi-compact, it is necessary and sufficient that X be quasi-compact.*
(i) *For X^{cons} to be separated, it is necessary and sufficient that X be quasi-separated, and then X^{cons} is locally compact and totally disconnected.*
(i) *For X^{cons} to be compact, it is necessary and sufficient that X be quasi-compact and quasi-separated.*
(i) *Every point of X admits, for the topology $\mathscr{T}^{\mathrm{cons}}$, an open and compact neighbourhood.*
(i) *For a quasi-compact morphism $f : X \to Y$ to be quasi-compact, it is necessary and sufficient that the continuous map $f^{\mathrm{cons}} : X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ be proper.*

(1.9.16). We will show later how one can, for every prescheme X, equip the space X^{cons} with a sheaf of rings which in fact makes it a prescheme whose local rings are all fields, identical to the residue fields at the points of X. Such preschemes arise naturally for example in the translation, in the language of schemes, of Néron's constructions in his theory of the reduction of abelian varieties.

1.10. Application to open morphisms IV-1 | 249

Theorem (1.10.1). — *Let X be a prescheme, E an ind-constructible subset of X (1.9.4), x a point of X. For x to be interior to E, it is necessary and sufficient that every generalisation ($\mathbf{0_I}$, 2.1.2) x' of x belong to E, in other words that* $\operatorname{Spec}(\mathscr{O}_{X,x}) \subset E$.

The condition is evidently necessary, every neighbourhood of x containing the generalisations of x. To see that it is sufficient, one can evidently restrict to the case where $X = \operatorname{Spec}(A)$ is affine, x being a prime ideal $\mathfrak{p}$ of A. One knows then (1.9.5), (ix)) that there exist an affine scheme $X' = \operatorname{Spec}(A')$ and a morphism $f : X' \to X$ such that $f(X') = X - E$. Set $Y = \operatorname{Spec}(\mathscr{O}_{X,x})$ and $Y' = X' \times_X Y$; the hypothesis means (by virtue of (**I**, 3.6.5)) that $Y' = \varnothing$, in other words $A' \otimes_A A_{\mathfrak{p}} = 0$. As $A_{\mathfrak{p}} = \varinjlim A_t$, where t ranges over $A - \mathfrak{p}$ ($\mathbf{0_I}$, 1.4.5), on a $\varinjlim A'_t = 0$, the functor $\varinjlim$ commuting with the tensor product. By (1.9.1), there exists $t \in A - \mathfrak{p}$ such that $A'_t = 0$, and $\mathrm{D}(t)$ is then an open neighbourhood of x contained in E.

Corollary (1.10.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, and let $y \in Y$. For y to be interior to $f(X)$ it is necessary and sufficient that every generalisation y' of y belong to $f(X)$.*

One knows indeed (1.9.5), (viii)) that $f(X)$ is ind-constructible in Y and it suffices to apply (1.10.1).

We will say that a continuous map $f : X \to Y$ is *open at a point* $x \in X$ if the image by f of *every* neighbourhood of x in X is a neighbourhood of $f(x)$ in Y.

Proposition (1.10.3). — *Let $f : X \to Y$ be a morphism, x a point of X, $y = f(x)$. Consider the following conditions:*

a) *f is open at the point x.*
a) *For every generalisation y' of y, there exists a generalisation x' of x such that $f(x') = y'$.*
b′) *The image by f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,y})$.
a) *For every irreducible closed subset Y' of Y containing y, there exists an irreducible component of $X' = f^{-1}(Y')$ containing x and dominating Y'.*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) $\Rightarrow$ c). If in addition f is locally of finite presentation, the four conditions are equivalent.

By definition of a generalisation, the image by f of $\operatorname{Spec}(\mathscr{O}_{X,x})$ is always contained in $\operatorname{Spec}(\mathscr{O}_{Y,y})$, so *b*) and *b*′) are equivalent. It is immediate that *c*) implies *b*), for if y' is a generalisation of y, $Y' = \overline{\{y'\}}$ is an irreducible closed subset of Y containing y; if x' is the generic point of an irreducible component of $f^{-1}(Y')$ containing x and dominating Y', we have $f(x') = y'$ ($\mathbf{0_I}$, 2.1.5) and x' is a generalisation of x. Conversely, *b*) implies *c*): indeed, let y' be the generic point of Y' and let x' be a generalisation of x such that $f(x') = y'$; let Z' be an irreducible component of $f^{-1}(Y')$ containing x' (hence also x); as $f(x')$ is already dense in Y', it is thus *a fortiori* so for $f(Z')$.

To see that *a*) implies *b*′), one can evidently restrict to the case where $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, x being a prime ideal $\mathfrak{q}$ of B, so that $\mathscr{O}_{X,x} = B_{\mathfrak{q}} = \varinjlim B_t$, where t ranges over $B - \mathfrak{q}$. We have, by (1.9.2), $f(\operatorname{Spec}(\mathscr{O}_{X,x})) = \bigcap_t f(\operatorname{Spec}(B_t)) = \bigcap_t f(\mathrm{D}(t))$, and by hypothesis, for every $t \in B - \mathfrak{q}$, y is interior to $f(\mathrm{D}(t))$, so $f(\mathrm{D}(t))$ contains $\operatorname{Spec}(\mathscr{O}_{Y,y})$; whence $\operatorname{Spec}(\mathscr{O}_{Y,y}) \subset f(\operatorname{Spec}(\mathscr{O}_{X,x}))$, which establishes our assertion. Finally, when f is locally of finite presentation, *b*) implies *a*) by virtue of (1.10.2) applied to the restriction $U \to Y$ of f to an arbitrary open neighbourhood U of x. IV-1 | 250

Corollary (1.10.4). — *Let $f : X \to Y$ be a morphism. Consider the following conditions:*

a) *f is open.*
a) *For every $x \in X$ and every generalisation y' of $y = f(x)$, there exists a generalisation x' of x such that $f(x') = y'$.*
b′) *For every $x \in X$, the image by f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,f(x)})$.

a) *For every irreducible closed subset Y' of Y, every irreducible component of $f^{-1}(Y')$ dominates Y'.*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b') $\Rightarrow$ c). If in addition f is locally of finite presentation, the four conditions are equivalent.

To say that f is open means that f is open at every point $x \in X$, hence the implications $a) \Rightarrow b) \Leftrightarrow b')$ follow from the analogous implications in (1.10.3), as well as the implication $b) \Rightarrow a)$ when f is locally of finite presentation. The implication $c) \Rightarrow b)$ also follows from the analogous implication in (1.10.3); let us prove finally that $b)$ implies $c)$. Indeed, let x' be the generic point of an irreducible component of $f^{-1}(Y')$, and let us show that $y' = f(x')$ is the generic point of Y'. Let y'' be a generalisation of y'; it exists by hypothesis a generalisation x'' of x' such that $f(x'') = y''$, and as $x'' \in f^{-1}(Y')$, we necessarily have $x'' = x'$, hence $y'' = y'$, which finishes the proof.

§2. Base change and flatness

This section (contrary to §6) only exceptionally uses Noetherian techniques. Nos. 1 and 2 are scarcely more than translations of elementary properties of flatness in commutative algebra (cf. Bourbaki, *Alg. comm.*, chap. I), and are included here for convenience of reference. The following numbers are primarily devoted to properties of "descent" by flat or faithfully flat morphisms: if $g : Y' \to Y$ is such a morphism, the goal is to be able to assert that a part of Y, or an $\mathscr{O}_Y$-Module, or a morphism $X \to Y$, has a certain property, when one knows that its inverse image by g possesses this property. We restrict ourselves here to properties that do not make use of the general technique of "descent", which will be developed in Chapter V. **IV-2** | 5

2.1. Flat modules on preschemes

(2.1.1). Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module; recall ($\mathbf{0_I}$, 6.7.1) that $\mathscr{F}$ is said to be *f-flat* (or *Y-flat*) *at a point* $x \in X$ if $\mathscr{F}_x$ is a flat $\mathscr{O}_{f(x)}$-module, *f-flat* (or *Y-flat*) if it is f-flat at every $x \in X$, and finally that the morphism f is said to be *flat at the point* $x \in X$ (resp. *flat*) if $\mathscr{O}_X$ is f-flat at the point x (resp. f-flat). When $f = 1_X$, we will simply say that an $\mathscr{O}_X$-Module $\mathscr{F}$ is *flat at the point* x (resp. *flat*) if it is X-flat at this point (resp. at every point $x \in X$), that is to say if $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module (resp. if this holds for every $x \in X$). Recall that we have shown (**III**, 1.4.15.1) the following property:

Proposition (2.1.2). — *Let A, B be two rings, $\varphi : A \to B$ a ring homomorphism, $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$, $f : X \to Y$ the morphism corresponding to φ, and M a B-module. For $\mathscr{F} = \widetilde{M}$ to be f-flat, it is necessary and sufficient that M be a flat A-module.*

Proposition (2.1.3). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:* **IV-2** | 6

a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules, is exact.*

a') *Condition* a) *is satisfied for all the canonical morphisms $g : \operatorname{Spec}(\mathscr{O}_y) \to Y$* (**I**, 2.4.1), *where y ranges over Y.*

b) *$\mathscr{F}$ is f-flat.*

Proof. The questions being local on X and Y, we can reduce to the case where $Y = \operatorname{Spec}(A)$, $\mathscr{F} = \widetilde{M}$, where M is a B-module. It is clear that a) implies a'); condition a') implies that for every $x \in X$, the functor $N \mapsto M_{\mathfrak{n}} \otimes_{B_{\mathfrak{n}}} N$ is exact in the category of $B_{\mathfrak{n}}$-modules, $\mathfrak{n}$ being the ideal $\mathfrak{j}_{f(x)}$ of B; this means that $M_{\mathfrak{n}}$ is a flat $B_{\mathfrak{n}}$-module, and it follows from ($\mathbf{0_I}$, 6.3.3) and from (2.1.2) that $\mathscr{F}$ is f-flat. Finally, to see that b) implies a), we can also reduce to the case where $Y' = \operatorname{Spec}(A')$ and $\mathscr{G}' = \widetilde{N'}$, where N' is an A'-module; the conclusion then follows again from (2.1.2) and the definition of flatness, since $(M \otimes_A N')^\sim = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$. $\square$

Proposition (2.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, and let g' be the canonical projection $X' \to X$. Let x' be a point of X', $x = g'(x')$, $y' = f'(x')$, $y = g(y') = f(x)$. If $\mathscr{F}$ is f-flat at the point x, then $\mathscr{F}'$ is f'-flat at the point x'; in particular if $\mathscr{F}$ is f-flat, $\mathscr{F}'$ is f'-flat; if f is flat, f' is flat.*

PROOF. It suffices to prove the first assertion; applying (**I**, 3.6.5) three times, as well as (**I**, 2.4.4), we can reduce to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$, $X = \operatorname{Spec}(\mathscr{O}_x)$, $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$, $\mathscr{F} = \widetilde{M}$, where $M = \mathscr{F}_x$; the hypothesis and (2.1.2) then imply that $\mathscr{F}$ is f-flat; we are thus reduced to proving a particular case of the second assertion, which follows immediately from (2.1.3). □

Proposition (2.1.5). — *Consider a commutative diagram of morphisms of preschemes*

$$\begin{tikzcd} X & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ Y \arrow[from=u, "f"'] & Y' \arrow[l, "g"] \arrow[d, "h"] \\ & Z \end{tikzcd}$$

where $X' = X \times_Y Y'$ and $f' = f_{(Y')}$. Let x' be a point of X', and set $x = g'(x')$, $y' = f'(x')$, $y = f(x) = g(y')$, $z = h(y')$. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module f-flat at the point x (resp. f-flat), and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_{Y'}$-Module h-flat at the point y' (resp. h-flat); then $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is a quasi-coherent $\mathscr{O}_{X'}$-Module $(h \circ f')$-flat at the point x' (resp. $(h \circ f')$-flat).

PROOF. As in (2.1.4), we can reduce to the case where $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$ and $Z = \operatorname{Spec}(\mathscr{O}_z)$, and **IV-2 | 7**
it then suffices to prove that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is $(h \circ f')$-flat. Taking into account (2.1.2), the proposition follows from Bourbaki, *Alg. comm.*, chap. I, §2, no. 7, prop. 8. □

Corollary (2.1.6). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module. If $\mathscr{F}$ is f-flat at the point $x \in X$ and if g is flat at the point $f(x)$, then $\mathscr{F}$ is $(g \circ f)$-flat at the point x. In particular, if f and g are flat morphisms, the same is true of $g \circ f$.*

PROOF. This is the particular case of (2.1.5) with $Y' = Y$, $\mathscr{G} = \mathscr{O}_Y$. □

Corollary (2.1.7). — *If $f : X \to X'$, $g : Y \to Y'$ are two flat S-morphisms, then*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is a flat morphism.

PROOF. This follows from (2.1.4) and (2.1.6) (cf. (**I**, 3.5.1)). □

Proposition (2.1.8). — *Let $f : X \to Y$ be a morphism of preschemes,*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules, such that $\mathscr{F}''$ is Y-flat.

(i) *For every $g : Y' \to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the sequence*

$$0 \longrightarrow \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{F}'' \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow 0$$

of $\mathscr{O}_{X'}$-Modules (where $X' = X \times_Y Y'$) is exact.

(i) *For $\mathscr{F}$ to be Y-flat, it is necessary and sufficient that $\mathscr{F}'$ be so.*

PROOF. We can evidently suppose X, Y, Y' affine; the conclusion then follows from (2.1.2) and ($\mathbf{0_I}$, 6.1.2). □

Corollary (2.1.9). — *Let $\mathscr{L}^\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules, i an index such that, denoting by $d^i : \mathscr{L}^i \to \mathscr{L}^{i+1}$ the derivation operator, $\mathscr{B}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Im}(d^i)$ and $\mathscr{Z}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Coker}(d^i)$ are Y-flat. Then, with the notations of (2.1.8), the canonical homomorphism*

$$\mathscr{H}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$$

is bijective.

PROOF. Since the tensor product is right exact, we have

$$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathrm{Coker}(d^i \otimes 1) = \mathscr{Z}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$$

and $\mathscr{Z}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{Z}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$. Furthermore, in the exact sequence

$$0 \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \longrightarrow \mathscr{L}^{i+1} \longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet) \longrightarrow 0$$

$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, so it follows from (2.1.8) (i) that we have the exact sequence

$$0 \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{L}^{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow 0$$

whence $\mathscr{B}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathrm{Im}(d^i \otimes 1) = \mathscr{B}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$. Then, since in the exact sequence

$$0 \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet) \longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet) \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \longrightarrow 0$$

$\mathscr{B}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, it follows from (2.1.8) (i) and from what precedes that the sequence

$$0 \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow 0$$

is exact, which proves the corollary. □

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Corollary (2.1.10). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module that is Y-flat, $\mathscr{L}_\bullet = (\mathscr{L}_i)$ a left resolution of $\mathscr{F}$ by quasi-coherent $\mathscr{O}_X$-Modules that are Y-flat. Then, for every morphism $g : Y' \to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}' = (\mathscr{L}_i \otimes_{\mathscr{O}_Y} \mathscr{G}')_{i \geqslant 0}$ is a left resolution of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$.*

Furthermore, if $\mathscr{Z}_i(\mathscr{L}_\bullet) = \mathrm{Ker}(\mathscr{L}_i \to \mathscr{L}_{i-1})$, the $\mathscr{Z}_i(\mathscr{L}_\bullet)$ are Y-flat, and we have $\mathscr{Z}_i(\mathscr{L}_\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{Z}_i(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') = \mathrm{Ker}(\mathscr{L}_i \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{L}_{i-1} \otimes_{\mathscr{O}_Y} \mathscr{G}')$.

PROOF. Set $\mathscr{R}_i = \mathrm{Im}(\mathscr{L}_{i+1} \to \mathscr{L}_i) = \mathscr{Z}_i(\mathscr{L}_\bullet)$; we then have the exact sequences

$$0 \leftarrow \mathscr{F} \leftarrow \mathscr{L}_0 \leftarrow \mathscr{R}_0 \leftarrow 0$$
$$0 \leftarrow \mathscr{R}_i \leftarrow \mathscr{L}_{i+1} \leftarrow \mathscr{R}_{i+1} \leftarrow 0$$

and since $\mathscr{F}$ and the $\mathscr{L}_i$ are Y-flat, we deduce from (2.1.8) (ii) by induction that all the $\mathscr{R}_i$ are also Y-flat; using (2.1.8) (i), we therefore have the exact sequences

$$0 \leftarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{L}_0 \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{R}_0 \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow 0$$
$$0 \leftarrow \mathscr{R}_i \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{L}_{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{R}_{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow 0 \quad (i \geqslant 0)$$

which prove the corollary. □

Proposition (2.1.11). — *Let $f : X \to Y$ be a flat morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_Y$-Module of finite presentation. If $\mathscr{J}$ is the ideal of $\mathscr{O}_Y$ that annihilates $\mathscr{F}$, then $f^*(\mathscr{J})$ is the ideal of $\mathscr{O}_X$ that annihilates $f^*(\mathscr{F})$.*

PROOF. Indeed, by definition we have an exact sequence ($\mathbf{0_I}$, 5.3.7)

$$0 \longrightarrow \mathscr{J} \longrightarrow \mathscr{O}_Y \longrightarrow \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F})$$

whence, since f is flat, an exact sequence

$$0 \longrightarrow f^*(\mathscr{J}) \longrightarrow \mathscr{O}_X \longrightarrow f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F}))$$

and since by hypothesis $\mathscr{F}$ is a finitely presented $\mathscr{O}_Y$-Module, $f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F}))$ is canonically identified with $\mathscr{H}om_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{F}))$ ($\mathbf{0_I}$, 6.7.6), whence the conclusion. □

Proposition (2.1.12). — *Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation, x a point of X. The following conditions are equivalent:*

a) *$\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module.*
a) *There exists an open neighbourhood U of x such that $\mathscr{F}|U$ is a locally free $(\mathscr{O}_X|U)$-Module.*

PROOF. Indeed, $\mathscr{F}_x$ is a finitely presented $\mathscr{O}_x$-module and $\mathscr{O}_x$ a local ring; it therefore amounts to the same to say that $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module or a free $\mathscr{O}_x$-module (Bourbaki, *Alg. comm.*, chap. II, §3, no. 2, cor. 2 of prop. 5); whence the conclusion, taking into account ($\mathbf{0_I}$, 5.2.7). We note that the proposition is valid for an arbitrary locally ringed space. □

Proposition (2.1.13). — *Let $f : X \to Y$ be a morphism of preschemes. If f is flat at a point $x \in X$ and if the ring $\mathscr{O}_x$ is reduced (resp. integral and integrally closed), then the ring $\mathscr{O}_{f(x)}$ is reduced (resp. integral and integrally closed). If f is faithfully flat and if X is reduced (resp. normal), then Y is reduced (resp. normal).* **IV-2** | 9

PROOF. Set $\mathscr{O}_{f(x)} = A$, $\mathscr{O}_x = B$. If B is a flat A-module, it is also a faithfully flat A-module ($\mathbf{0_I}$, 6.6.2), so A is identified with a subring of B; if B is reduced, the same is therefore true of A. Suppose now that B is integral and integrally closed, and let L be its field of fractions; denote by $K \subset L$ the field of fractions of A. The hypothesis implies that $B \cap K = A$ (Bourbaki, *Alg. comm.*, chap. I, §3, no. 5, prop. 10). If then $t \in K$ is integral over A, it is also integral over B, and so belongs to B by hypothesis, and therefore $t \in A$, which proves that A is integrally closed. □

Proposition (2.1.14). — *Let $f : X \to Y$ be a faithfully flat morphism. If X is integral and if the underlying space of Y is locally noetherian, then Y is locally integral.*

PROOF. Indeed, every $y \in Y$ is of the form $f(x)$ for some $x \in X$ and by hypothesis $\mathscr{O}_y$ is identified with a subring of $\mathscr{O}_x$ ($\mathbf{0_I}$, 6.6.1); since $\mathscr{O}_x$ is integral, the same is true of $\mathscr{O}_y$, and this proves the proposition (**I**, 5.1.4). □

2.2. Faithfully flat modules on preschemes

Proposition (2.2.1). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$, from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules, is exact and faithful.*

a′) *Condition* a) *is satisfied for all the canonical morphisms $g : \mathrm{Spec}(\mathscr{O}_y) \to Y$* (**I**, 2.4.1)*, where y ranges over Y.*

a″) *Condition* a) *is satisfied for all the canonical immersions $Y' \to Y$, where Y' ranges over the set of affine open subschemes of Y.*

a) *$\mathscr{F}$ is Y-flat and, for every $y \in Y$, denoting by X_y the fibre $f^{-1}(y)$, the $\mathscr{O}_{X_y}$-Module $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is $\neq 0$.*

PROOF. It is clear that a) implies a′) and a″); condition a′) implies first that $\mathscr{F}$ is Y-flat (2.1.3); on the other hand, it implies that for every $y \in Y$, the functor $N \mapsto \mathscr{F} \otimes_{\mathscr{O}_y} \widetilde{N}$ is faithful in the category of $\mathscr{O}_y$-modules; taking in particular $N = k(y)$, we obtain the second assertion of b). For the second assertion of b) to imply a), on the other hand, it suffices to restrict to the case where Y is affine, whence the question is local on Y, and one can also assume X affine, and the condition then reduces to showing that the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is exact and faithful, which proves that b′) implies a). For showing that b) implies a″), we can restrict to the case where Y is affine; then f is faithfully flat in the category of $\mathscr{O}_Y$-modules, and so the functor $\mathscr{G}' \hookrightarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is exact and faithful, which shows that b′) implies c) and concludes the proof. □ **IV-2** | 10

Proposition (2.2.2). — *Let $Y = \mathrm{Spec}(A)$ be an affine scheme, $f : X \to Y$ a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. Condition* a) *of* (2.2.1) *is equivalent to each of the following:*

b′) *$\mathscr{F}$ is Y-flat and, for every closed point y of Y, $\mathscr{F}_y \neq 0$.*

c) *The functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ from the category of quasi-coherent $\mathscr{O}_Y$-Modules to that of quasi-coherent $\mathscr{O}_X$-Modules, is exact and faithful.*

PROOF. If b′) is satisfied, there is at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x \neq 0$; let $U = \mathrm{Spec}(B)$ be an affine neighbourhood of x, so that $\mathscr{F}|U = \widetilde{M}$, where M is a B-module. Then b′) implies that $M/\mathfrak{j}_y M \neq 0$ (since M is an A-module that is flat (2.1.2)), which implies that $M \otimes_A A_y \neq 0$ for every closed point $y \in Y$, that $M \otimes_A A_y \widetilde{N}$ is exact and faithful ($\mathbf{0_I}$, 6.4.5). The relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$ implies $\mathscr{G} = 0$, otherwise said the functor $\mathscr{G} \hookrightarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is faithful; since it is also exact (2.1.3), this shows that b′) implies c), and it is clear that c) implies a). □

Suppose now that c) holds and that $g : Y' \to Y$ is also an affine morphism; moreover, the functor $\mathscr{H}' \mapsto g'_*(\mathscr{H}')$ is then exact in the category of quasi-coherent $\mathscr{O}_{X'}$-Modules (**I**, 1.6.4), and if $g'_*(\mathscr{H}') = \mathscr{H}'$, on has $\mathscr{H}' = \widetilde{\mathscr{H}}$; the preceding functor is also faithful; to see that $\mathscr{G}' \hookrightarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$

is exact and faithful, it suffices therefore to see that the functor $\mathscr{G}' \mapsto g'_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}')$ is so. Or, if $f' = f_{(Y')} : X' \to Y'$, the fact that g is affine implies that we have a canonical isomorphism

$$\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(g'^*(\mathscr{F}) \otimes_{\mathscr{O}_{X'}} f'^*(\mathscr{G}')). \tag{2}$$

Indeed, we know (**II**, 1.5.2) that there is a canonical isomorphism $f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(f'^*(\mathscr{G}'))$, and on the other hand a canonical homomorphism $\mathscr{F} \to g'_*(g'^*(\mathscr{F}))$ ($\mathbf{0_I}$, 4.4.3.2); composing the homomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \to g'_*(g'^*(\mathscr{F})) \otimes_{\mathscr{O}_X} g'_*(f'^*(\mathscr{G}'))$ with the canonical morphism ($\mathbf{0_I}$, 4.2.2.1), we deduce the homomorphism (2.2.2.1); the verification that it is an isomorphism is immediate, reducing to the case where X is affine. This being the case, the functor $\mathscr{G}' \mapsto g_*(\mathscr{G}')$ is exact and faithful and by hypothesis the same is true of $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$; their composite is therefore exact and faithful (2.2.1) and (2.2.2).

Corollary (2.2.3). — *Let $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ be two affine schemes, $f : X \to Y$ a morphism, $\mathscr{F} = \widetilde{M}$ a quasi-coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to satisfy the equivalent conditions of* (2.2.1) (*or* (2.2.2)), *it is necessary and sufficient that the A-module M be faithfully flat.*

Proof. Indeed, condition c) of (2.2.2) means that the functor $N \mapsto M \otimes_A N$ from the category of A-modules to that of B-modules is exact and faithful, and the conclusion follows from ($\mathbf{0_I}$, 6.4.1). □

Definition (2.2.4). — When the equivalent conditions of (2.2.1) are satisfied, we say that the quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is *faithfully flat relative to f* (or relative to Y).

We note that this notion is local on Y, but *not on X*; in particular one can have $\mathscr{F}_x = 0$ for certain **IV-2** | 11
$x \in X$, in other words $\operatorname{Supp}(\mathscr{F})$ is not necessarily equal to X. However, it follows immediately from the criterion (2.2.1) b) that for every $y \in Y$, there exists at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x = \mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \neq 0$, and *a fortiori* $\mathscr{F}_x \neq 0$; in other words:

Corollary (2.2.5). — *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module, faithfully flat relative to f, then $f(\operatorname{Supp}(\mathscr{F})) = Y$, and* a fortiori *$f$ is a surjective morphism.*

This result admits a partial converse:

Corollary (2.2.6). — *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module of finite type. For $\mathscr{F}$ to be faithfully flat relative to f, it is necessary and sufficient that $\mathscr{F}$ be f-flat and that $f(\operatorname{Supp}(\mathscr{F})) = Y$.*

Proof. Indeed (**I**, 9.1.13) and (**I**, 3.6.1), we have $\operatorname{Supp}(\mathscr{F}_y) = f^{-1}(y) \cap \operatorname{Supp}(\mathscr{F})$, so in this case the criterion (2.2.1) b) is none other than the condition of the corollary. □

In particular, the $\mathscr{O}_X$-Module $\mathscr{O}_X$ is faithfully flat relative to f if and only if it is f-flat and f is surjective, in other words if and only if the morphism f is *faithfully flat* ($\mathbf{0_I}$, 6.7.8).

Let us make explicit the properties intervening in Definition (2.2.4):

Proposition (2.2.7). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to f. Then, for a sequence $\mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$ of quasi-coherent $\mathscr{O}_Y$-Modules to be exact, it is necessary and sufficient that the corresponding sequence $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ be so. In particular, for a homomorphism $u : \mathscr{G} \to \mathscr{G}'$ of quasi-coherent $\mathscr{O}_Y$-Modules to be injective* (*resp. surjective, bijective, zero*)*, it is necessary and sufficient that $1_{\mathscr{F}} \otimes f^*(u) : \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be so. For a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$ to be zero, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$. For every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$, the map $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the set of quasi-coherent sub-$\mathscr{O}_Y$-Modules of $\mathscr{G}$ to the set of quasi-coherent sub-$\mathscr{O}_X$-Modules of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is injective.*

Proof. To prove the last assertion, that is to say that for two quasi-coherent sub-$\mathscr{O}_Y$-Modules $\mathscr{G}'$, $\mathscr{G}''$ of $\mathscr{G}$, the relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ implies $\mathscr{G}' = \mathscr{G}''$, we can reduce to the case where $\mathscr{G}' \subset \mathscr{G}''$, and it then suffices to apply the second assertion of the statement to the injection $u : \mathscr{G}' \to \mathscr{G}''$. □

Corollary (2.2.8). — *Let $f : X \to Y$ be a faithfully flat morphism. For every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$, the canonical map*

$$\Gamma(Y, \mathscr{G}) \longrightarrow \Gamma(X, f^*(\mathscr{G})) \tag{3}$$

is injective.

PROOF. Indeed, $\Gamma(Y, \mathscr{G})$ is canonically identified with $\mathrm{Hom}_{\mathscr{O}_Y}(\mathscr{O}_Y, \mathscr{G})$ ($\mathbf{0_I}$, 5.1.1) and similarly $\Gamma(X, f^*(\mathscr{G}))$ with $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, f^*(\mathscr{G}))$. By virtue of (2.2.1) and (2.2.4), the hypothesis that the functor $\mathscr{G} \mapsto f^*(\mathscr{G})$ is exact and faithful in the category of $\mathscr{O}_Y$-Modules implies that a homomorphism $u : \mathscr{O}_Y \to \mathscr{G}$ is zero if and only if $f^*(u) : f^*(\mathscr{O}_Y) = \mathscr{O}_X \to f^*(\mathscr{G})$ is zero. □

Remark (2.2.9). — The results of (2.2.7) and (2.2.8) are still true when the $\mathscr{O}_Y$-Modules $\mathscr{G}', \mathscr{G}, \mathscr{G}''$ that appear there are arbitrary (not necessarily quasi-coherent). Indeed, for every $y \in Y$, there exists $x \in f^{-1}(y)$ such that $\mathscr{F}_x$ is a faithfully flat $\mathscr{O}_y$-module and consequently the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ **IV-2** | 12
is faithful; since moreover for every $x \in f^{-1}(y)$ the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ is exact, we immediately deduce the conclusion.

Proposition (2.2.10). — *Let $f : X \to Y$, $g : Y' \to Y$, $h : Y' \to Z$ be three morphisms of preschemes, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_{Y'}$-Module faithfully flat relative to Z. For $\mathscr{G}$ to be plat* (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be a flat $\mathscr{O}_{X'}$-Module* (*resp. faithfully flat*) *relative to Z.*

PROOF. We already know that if $\mathscr{G}$ is Z-flat, the same is true of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ (2.1.5). Consider an arbitrary base change $Z'' \to Z$ and set $X'' = X' \times_Z Z''$; if $\mathscr{G}$ is faithfully flat relative to Z, the functor

$$\mathscr{H}'' \longmapsto \mathscr{H}'' \otimes_{\mathscr{O}_Z} \mathscr{G}' \longrightarrow (\mathscr{H}'' \otimes_{\mathscr{O}_Z} \mathscr{G}') \otimes_{\mathscr{O}_Y} \mathscr{F} = \mathscr{H}'' \otimes_{\mathscr{O}_Z} (\mathscr{G}' \otimes_{\mathscr{O}_Y} \mathscr{F}) \tag{4}$$

from the category of $\mathscr{O}_{X''}$-Modules quasi-coherent to that of $\mathscr{O}_{X''}$-Modules quasi-coherent is composed of two exact and faithful functors, so is exact and faithful. Conversely, if this composed functor is exact (resp. exact and faithful), the same is true of $\mathscr{H}'' \mapsto \mathscr{H}'' \otimes_{\mathscr{O}_Z} \mathscr{G}'$, since the functor $\mathscr{H}' \mapsto \mathscr{H}' \otimes_{\mathscr{O}_Y} \mathscr{F}$ (from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules) is exact and faithful by hypothesis. □

Corollary (2.2.11). — (i) *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to Y, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$ is faithfully flat relative to Y'.*

(i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y. For g to be a faithfully flat morphism, it is necessary and sufficient that $\mathscr{F}$ be faithfully flat relative to Z.*

(i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-Module. Suppose the morphism f is faithfully flat. For $\mathscr{G}$ to be flat* (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $f^*(\mathscr{G})$ be flat* (*resp. faithfully flat*) *relative to Z.*

(i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, x a point of X. Suppose f is flat at the point x. For $g \circ f$ to be flat at the point x, it is necessary and sufficient that g be flat at the point $f(x)$.*

PROOF. To show (i), apply (2.2.10) replacing Z by Y' and $\mathscr{G}'$ by $\mathscr{O}_{Y'}$. To show (ii), apply (2.2.10) taking $Y' \to Y$ the identity and replacing $\mathscr{G}'$ by $\mathscr{O}_Y$. To show (iii), apply (2.2.10) by replacing $Y' \to Y$ by the identity, $\mathscr{F}$ by $\mathscr{O}_X$ and $\mathscr{G}'$ by $\mathscr{G}$. Finally, (iv) follows from (ii) by replacing $\mathscr{F}$ by $\mathscr{O}_X$ and $\mathscr{G}$ by $\mathscr{O}_Y$, replacing X by $\mathrm{Spec}(\mathscr{O}_x)$, $\mathscr{F}$ by $\mathscr{O}_X$, Y by $\mathrm{Spec}(\mathscr{O}_{f(x)})$ and Z by $\mathrm{Spec}(\mathscr{O}_{g(f(x))})$, and taking into account ($\mathbf{0_I}$, 6.6.2). □

Corollary (2.2.12). — *Let Y be an affine scheme, $f : X \to Y$ a quasi-compact morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to f, there exists an affine scheme X' and a surjective local isomorphism $g : X' \to X$ such that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$.*

PROOF. Since X is quasi-compact, and therefore a finite union of affine opens X_i; it suffices to take for X' the affine scheme *sum* of the preschemes X_i, f being the canonical morphism (**I**, 3.1), the **IV-2** | 13
hypothesis implies that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$, by virtue of (2.2.11) (iii). □

Corollary (2.2.13). — (i) *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If f is a faithfully flat morphism, the same is true of f'.*

(i) *If $f : X \to X'$, $g : Y \to Y'$ are two faithfully flat S-morphisms,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is faithfully flat.

(i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms such that f is faithfully flat (resp. faithfully flat). For g to be a flat (resp. faithfully flat) morphism, it is necessary and sufficient that $g \circ f$ be a flat (resp. faithfully flat) morphism.*

Proposition (2.2.14). — *Let $f : X \to Y$ be a faithfully flat and quasi-compact morphism. If X is locally noetherian, the same is true of Y.*

PROOF. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine; since f is quasi-compact, we can also restrict to the case where $X = \operatorname{Spec}(B)$ is affine. Then B is a noetherian ring by hypothesis, and a faithfully flat A-module (2.2.3); therefore A is noetherian ($\mathbf{0_I}$, 6.5.2). □

Proposition (2.2.15). — *Let $f : X \to Y$ be a faithfully flat morphism. If $\mathfrak{S}(X)$ and $\mathfrak{S}(Y)$ denote respectively the sets of sub-preschemes of X and of Y, the map $Z \mapsto f^{-1}(Z)$ from $\mathfrak{S}(Y)$ to $\mathfrak{S}(X)$ is injective.*

PROOF. Since f is surjective, we have $f(f^{-1}(Z)) = Z$. On the other hand, if U is an open of Y containing Z and in which Z is closed, $f^{-1}(Z)$ is closed in $f^{-1}(U)$, and the restriction $f^{-1}(U) \to U$ of f is faithfully flat. We can therefore restrict to considering only closed sub-preschemes of Y. Now, if Z is a sub-prescheme of Y corresponding to a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_Y$, we know that $f^{-1}(Z)$ corresponds to the quasi-coherent ideal $f^*(\mathscr{J})\mathscr{O}_X$ of $\mathscr{O}_X$ (**I**, 4.4.5), and since f is flat, $f^*(\mathscr{J})\mathscr{O}_X$ is identified with $f^*(\mathscr{J})$. But the map $\mathscr{J} \hookrightarrow f^*(\mathscr{J})$ from the set of quasi-coherent ideals of $\mathscr{O}_Y$ to the set of quasi-coherent ideals of $\mathscr{O}_X$ is injective (2.2.7), whence the conclusion. □

Corollary (2.2.16). — *Let X, Y be two S-preschemes; if $S' \to S$ is a faithfully flat morphism, the map $f \mapsto f_{(S')}$ from $\operatorname{Hom}_S(X,Y)$ to $\operatorname{Hom}_{S'}(X_{(S')}, Y_{(S')})$ is injective.*

PROOF. Since $X_{(S')} \times_{S'} Y_{(S')} = (X \times_S Y)_{(S')}$ (**I**, 3.3.10), the morphism projection $X_{(S')} \times_{S'} Y_{(S')} \to X \times_S Y$ is faithfully flat (2.2.13). The elements of $\operatorname{Hom}_S(X,Y)$ correspond bijectively to the sub-preschemes of $X \times_S Y$ which are the *graphs* of these S-morphisms (**I**, 5.3.11), and if Z_f is the graph of $f \in \operatorname{Hom}_S(X,Y)$, then $Z_{f(S')} = g^{-1}(Z_f)$ (**I**, 5.3.12). It therefore suffices to apply g to Proposition (2.2.15). □

Proposition (2.2.17). — *Let A be a ring, B an A-algebra such that B is a faithfully flat A-module of finite presentation. Then the structural homomorphism $\varphi : A \to B$ is an isomorphism of the A-module A onto a direct summand of the A-module B. If A is a local ring, B is a free A-module of finite type, and there exists a basis of this module containing the unit element.*

PROOF. By virtue of Bourbaki, *Alg. comm.*, chap. II, §3, no. 3, prop. 12, it suffices to prove the proposition when A is a local ring; one then has (*loc. cit.*, no. 2, cor. 2 of prop. 5) that B is a free A-module of finite type, and the conclusion follows from *loc. cit.*, prop. 5. □ **IV-2** | 14

2.3. Topological properties of flat morphisms

Lemma (2.3.1). — *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, $g : Y' \to Y$ a flat morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the canonical homomorphism*

$$g^*(f_*(\mathscr{F})) \longrightarrow f'_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}) \tag{5}$$

is bijective.

PROOF. This is a particular case of (**III**, 1.4.15) (improved in (1.7.21)). □

Proposition (2.3.2). — *Let S be a prescheme, $f : X \to Y$ an S-morphism quasi-compact and quasi-separated, $j : Z \to Y$ the closed image of X by f (**I**, 9.5.3) and (1.7.8), and let $g : X \to Z$ be a morphism such that $f = j \circ g$, where $g : X \to Z$ is a morphism (loc. cit.). Let $h : S' \to S$ be a flat morphism, and set $f' = f_{(S')} : X_{(S')} \to Y_{(S')}$; then $j' = j_{(S')} : Z_{(S')} \to Y_{(S')}$, the closed image of $X_{(S')}$ by f'.*

PROOF. As the morphism $Y_{(S')} \to Y$ is flat (2.1.4), we can restrict to the case where $S = Y$ (**I**, 3.3.11); we can then set $f' = (\psi, \theta)$, on sees easily (for example by reducing to the case where Y and Y' are affine and using (**I**, 2.2.4)) that $\theta' : \mathscr{O}_{S'} \to f'_*(\mathscr{O}_{X'})$ is composed of the canonical homomorphism (2.3.1) $h^*(f_*(\mathscr{O}_X)) \to f'_*(\mathscr{O}_{X'})$ and of $h^*(\theta) : h^*(\mathscr{O}_Y) \to h^*(f_*(\mathscr{O}_X))$; the conclusion thus follows from (2.3.1) and from the fact that $\operatorname{Ker}(\theta)$, being an $\mathscr{O}_Y$-Module quasi-coherent, commutes with flat base change ($\mathbf{0_I}$, 6.7.8). □

(2.3.3). We will say that a morphism $f : X \to Y$ is *quasi-flat* if there exists a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ of finite type which is f-flat and whose support is equal to X. We will say that f is *quasi-faithfully flat* if it is quasi-flat and surjective. Every flat morphism plat (resp. faithfully flat) morphism is quasi-flat (resp. quasi-faithfully flat), since then $\mathscr{F} = \mathscr{O}_X$ satisfies the preceding conditions.

It follows immediately from (2.1.4) and from (**I**, 9.1.13), that if f is quasi-flat, then for every morphism $Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is quasi-flat. Similarly (taking into account (**I**, 3.5.2)), if f is quasi-faithfully flat, the same is true of $f_{(Y')}$.

Proposition (2.3.4). — *Let $f : X \to Y$ be a quasi-flat morphism (2.3.3). Then f has the following properties (which are equivalent by virtue of (1.10.4)):*

(i) *For every $x \in X$ and every generisation y' of $y = f(x)$, there exists a generisation x' of x such that $f(x') = y'$.*

(i) *For every $x \in X$, the image by f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,y})$. **IV-2** | 15

(i) *For every closed irreducible part Y' of Y, every irreducible component of $f^{-1}(Y')$ dominates Y'.*

PROOF. It suffices, for example, to prove (ii). By hypothesis, there is a quasi-coherent $\mathscr{O}_X$-Module of finite type $\mathscr{F}$, which is f-flat and such that $\operatorname{Supp}(\mathscr{F}) = X$. For every $x \in X$, $\mathscr{F}_x$ is then a finitely generated $\mathscr{O}_x$-module, non-reduced to 0, and moreover $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module *flat*, and consequently the homomorphism $\rho : \mathscr{O}_y \to \mathscr{O}_x$. Since the latter is local and $\mathscr{F}_x \neq 0$, the Nakayama lemma shows that $\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \neq 0$, so $\mathscr{F}_x$ is a faithfully flat $\mathscr{O}_y$-module ($\mathbf{0_I}$, 6.4.1). It follows that for every prime ideal $\mathfrak{q}$ of $\mathscr{O}_y$, there exists a prime ideal $\mathfrak{p}$ of $\mathscr{O}_x$ such that $\mathfrak{q} = \rho^{-1}(\mathfrak{p})$ ($\mathbf{0_I}$, 6.5.1), which proves (ii). □

Corollary (2.3.5). — *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of (2.3.4) (in particular a quasi-flat or open morphism (1.10.4)).*

(i) *Let Z, Z' be two closed irreducible parts of Y such that $Z \subset Z'$, and let T be an irreducible component of $f^{-1}(Z)$; then there exists an irreducible component T' of $f^{-1}(Z')$ containing T (and dominating Z').*

(i) *For every irreducible component Y' of Y, $\overline{f(T)}$ is an irreducible component of Y.*

(i) *Suppose Y irreducible, and, if y is its generic point, suppose that $f^{-1}(y)$ is irreducible. Then X is irreducible.*

PROOF. (i) It suffices to apply (2.3.4) (i) by taking for x the generic point of T (the generic point of Z) and for y' the generic point of Z'.

(ii) It is clear that $\overline{f(T)} = Z$ is irreducible, and by virtue of (i), Z cannot be strictly contained in a closed irreducible component Z' of Y.

(iii) By virtue of (ii), every irreducible component of X dominates Y, so every such component meets $f^{-1}(y)$; the conclusion then follows from ($\mathbf{0_I}$, 2.1.8). □

Proposition (2.3.6). — *Let Y be a prescheme whose set of irreducible components is locally finite (which is the case if the underlying space of Y is locally noetherian (cf. (**I**, 6.1.9))).*

(i) *Every stable closed part W of Y by generisation ($\mathbf{0_I}$, 2.1.2) is open. In particular, every connected component of Y is open.*

(i) *Let $f : X \to Y$ be a closed morphism satisfying moreover the equivalent conditions* (i), (ii), (iii) *of (2.3.4) (which will be the case if f is quasi-flat or open). Then the image by f of every connected component C of X is a connected component of Y.*

(i) *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions* (i), (ii), (iii) *of (2.3.4) and such that moreover for every $y \in Y$, the set $f^{-1}(y)$ is finite (which will be the case if f is quasi-finite or radiciel). Then the set of irreducible components of X is locally finite.*

PROOF. (i) If $y \in W$, the generic points η_i ($1 \leqslant i \leqslant m$) of the irreducible components Y_i of Y **IV-2** | 16
containing y belong by hypothesis to W, so, since W is closed, these components are themselves contained in W; since there is by hypothesis a neighbourhood U of y such that U is a union of the $U \cap Y_i$ ($\mathbf{0_I}$, 2.1.6), then $U \subset W$, so W is open. Since for every $y \in Y$, $\overline{\{y\}}$ is connected, a connected component of Y is stable by generisation, so the second assertion follows immediately from the first.

(ii) Since C is closed in X, $f(C)$ is closed in Y by hypothesis. Moreover, as C is stable by generisation, the hypothesis on f implies that $f(C)$ is stable by generisation; therefore, by virtue of (i), $f(C)$ is open and closed, and since it is connected, it is a connected component of Y.

(iii) Let $x \in X$; by hypothesis, there is a neighbourhood U of $y = f(x)$ which intersects only a finite number of irreducible components Y_i of Y $(1 \leqslant i \leqslant n)$; let y_i be the generic point of Y_i (2.3.4). Since each of these sets is finite by hypothesis, this proves our assertion. □

Proposition (2.3.7). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a quasi-flat morphism, $f' = f_{(Y')} : X' \to Y'$.*

(i) *If f is quasi-compact and dominant, the same is true of f'.*
(i) *If every irreducible component of X dominates an irreducible component of Y, then every irreducible component of X' dominates an irreducible component of Y'.*

PROOF. Denote by $g' : X' \to X$ the canonical projection, which is a quasi-flat morphism (2.3.3).

(i) We already know (1.1.2) that f' is quasi-compact; moreover, if y' is a maximal point (1.1.4) of Y', $y = g(y')$ is a maximal point of Y, as it follows from (2.3.4) (iii). By hypothesis (1.1.5), there exists $x \in X$ such that $f(x) = y$; therefore (**I**, 3.4.7), there exists $x' \in X'$ such that $f'(x') = y'$.

(ii) Let x' be a maximal point of X', and let $x = g'(x')$; it follows from (2.3.4) (ii) that x is a maximal point of X, and by hypothesis $y = f(x)$ is a maximal point of Y. Set $y' = f'(x')$, so that $g(y') = y$; it suffices to show that y' is a maximal point of Y'. By virtue of (**I**, 5.1.7) and (2.3.3), we can restrict to the case where X and Y are *reduced*, so $\mathscr{O}_x$ and $\mathscr{O}_y$ are fields, and by virtue of (**I**, 2.4.4) applied twice and of (**I**, 3.6.5), we can restrict to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$ and $Y' = \operatorname{Spec}(\mathscr{O}_y)$. Then f is a flat morphism since $\mathscr{O}_y$ is a field (2.1.2), and it follows likewise from (2.1.4) that $y' = f'(x')$ is a maximal point of Y'; therefore it follows from (2.3.4) (ii) that $y' = f'(x')$ is a maximal point. □

Corollary (2.3.8). — (Zariski). — *Let A, B be two noetherian local rings, $\mathfrak{m}$, $\mathfrak{n}$ their maximal ideals respectively, $\varphi : A \to B$ a local homomorphism. Suppose the following hypotheses are satisfied:*

1° *B is an A-algebra essentially of finite type (1.3.8).*
2° *The completion $\widehat{A}$ of A for the $\mathfrak{m}$-adic topology is integral.*
3° *φ is injective.*

Then the $\mathfrak{m}$-adic topology on A is induced by the $\mathfrak{n}$-adic topology of B.

PROOF. Set $B' = B \otimes_A \widehat{A}$, where C is an A-algebra of finite type and S is a multiplicative part **IV-2** | 17
of C, so B' is an A-algebra of finite type. Since A is identified with a subring of $\widehat{A}$ ($\mathbf{0_I}$, 7.3.5), A is integral by 2°. Hypothesis 3° implies that there exists a prime ideal $\mathfrak{a}$ of B inducing the ideal $\mathfrak{o}$ of A ($\mathbf{0_I}$, 1.5.8), and therefore the local homomorphism $A \to B/\mathfrak{a}$ is injective. We can therefore restrict to proving (2.3.8) by adding the hypothesis that B is a local *integral* ring. Let us apply (2.3.7) (ii) to $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(\widehat{A})$ and $X' = \operatorname{Spec}(B')$; since the morphism $Y' \to Y$ is flat and X dominates Y, every irreducible component of X' dominates Y'; since by hypothesis Y' is integral, X' is also integral. Identifying A and $\widehat{A}$ as subrings of $\mathscr{O}_{x'}$, and denoting by $\mathfrak{r}$ the maximal ideal of $\mathscr{O}_{x'}$, the intersection of the ideals $\mathfrak{r}^k \cap \widehat{A}$ are open in $\widehat{A}$ ($\mathbf{0_I}$, 7.3.5). Moreover, the $\mathfrak{n}$-preadic topology of B induces on A a topology that is *finer* than the $\mathfrak{m}$-preadic topology of A ($\mathbf{0_I}$, 7.3.5). On the other hand, $\mathfrak{n}^k \cap \mathfrak{r}^k \cap A$, so the $\mathfrak{n}$-preadic topology of B induces on A the $\mathfrak{m}$-preadic topology; but $\mathfrak{m}^k \subset \mathfrak{n}^k \cap A$, so these two topologies are *identical*. □

Remark (2.3.9). — We will see further on (**IV**, 7.8.3) (vii), that for the most common noetherian local rings A in algebraic geometry, the hypothesis that A is integral and integrally closed implies that $\widehat{A}$ is integral. This is why, in algebraic geometry over a base field, one generally states (2.3.8) under the hypothesis that A is *integral and integrally closed*.

Theorem (2.3.10). — *Let $f : X \to Y$ be a quasi-flat morphism (2.3.3). Then, for every pro-constructible part (1.9.4) Z of Y, we have $f^{-1}(\overline{Z}) = \overline{f^{-1}(Z)}$.*

PROOF. Since f is continuous, we have $f^{-1}(\overline{Z}) \supset \overline{f^{-1}(Z)}$ and it suffices to prove that for every $x \in X$ such that $f(x) \in \overline{Z}$, x is adherent to $f^{-1}(Z)$; it is clear that the question is local on Y, so we can suppose Y affine. By virtue of the hypothesis, there exists an affine scheme Y' and a morphism $g : Y' \to Y$ such that $g(Y') = Z$ (1.9.5) (ix). Let Y_1 be the image by f of Y' (**I**, 9.5.3), and let X_1 be the closed sub-prescheme $f^{-1}(Y_1)$ of X; if $f_1 : X_1 \to Y_1$ is the restriction of f to X_1, we know that f_1 is quasi-flat (2.3.3); we can therefore replace X, Y by X_1, Y_1 respectively, in other words suppose that g

is *dominant*. Set then $X' = X \times_Y Y'$, and let f' and g' be the projections of X' to Y' and X respectively, so that we have the commutative diagram

$$\begin{tikzcd} X & X' \arrow[l, "g'"'] \\ Y \arrow[u, "f"] & Y' \arrow[l, "g"] \arrow[u, "f'"'] \end{tikzcd}$$

Since f is quasi-flat, g quasi-compact and dominant, it follows from (2.3.7) (where the roles of f and g are interchanged) that g' is a dominant morphism, which proves the theorem. □

Corollary (2.3.11). — *Let f be a quasi-flat and quasi-compact morphism, F a closed part of X such that $F = f^{-1}(f(F))$; then $F = f^{-1}(\overline{f(F)})$.*

Proof. Let Y' be the reduced sub-prescheme of X having F as underlying space (**I**, 5.2.1) **IV-2** | 18
and let $j : Y' \to X$ be the canonical injection; then $f \circ j$ is quasi-compact (1.1.2), so $Z = f(F)$ is pro-constructible in Y (1.9.5) (vii), and the corollary follows from the fact that F is closed. □

We can also write the result of (2.3.11) in the form $F = f^{-1}(f(X) \cap \overline{f(F)})$, in other words, the closed subsets of the subspace $f(X)$ of Y are the parts $Z \subset f(X)$ such that $f^{-1}(Z)$ is closed in X; this also means that *the topology induced by Y on $f(X)$ is the quotient of that of X by the equivalence relation defined by f*. In particular:

Corollary (2.3.12). — *Let $f : X \to Y$ be a quasi-faithfully flat morphism* (2.3.3) *and quasi-compact. Then the topology of Y is the quotient of that of X by the equivalence relation defined by f (in other words, for $Z \subset Y$ to be open (resp. closed) in Y, it is necessary and sufficient that $f^{-1}(Z)$ be open (resp. closed) in X).*

Proof. Indeed, we then have $f(X) = Y$. □

Corollary (2.3.13). — *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism faithfully flat and quasi-compact. For Y to be separated over S, it is necessary and sufficient that the canonical immersion $j : X \times_Y X \to X \times_S X$* (**I**, 5.3.10) *be a closed immersion.*

Proof. Note for this that we have the commutative diagram (**I**, 5.3.5)

$$\begin{tikzcd} X \times_Y X \arrow[r, "j"] \arrow[d, "\pi"'] & X \times_S X \arrow[d, "f \times_S f"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

identifying $X \times_Y X$ with the product of the $(Y \times_S Y)$-preschemes Y and $X \times_S X$. Since f is surjective, the same is true of π and of $f \times_S f$, so the diagonal $\Delta_Y(Y)$ has as inverse image by $f \times_S f$ the image $j(X \times_Y X)$ (**I**, 3.4.8). Since $f \times_S f$ is faithfully flat and quasi-compact (1.1.2) and (2.2.13), it suffices to apply (2.3.12) to this morphism. □

Corollary (2.3.14). — *Let $f : X \to Y$ be a quasi-faithfully flat and quasi-compact morphism, and let Z be a part of Y. For Z to be a locally closed pro-constructible part of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a locally closed pro-constructible part of X.*

Proof. We already know (1.9.12) that, for Z to be a pro-constructible part of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a pro-constructible part of X. The condition is evidently necessary. To prove that it is sufficient, consider the closed reduced sub-prescheme Y_1 of Y having $\overline{Z}$ as underlying space and, let X_1 be its inverse image $f^{-1}(Y_1)$, which has $\overline{f^{-1}(Z)}$ as underlying space by virtue of (2.3.10). Since $f^{-1}(Z)$ is locally closed in X, it is open in $\overline{f^{-1}(Z)}$, and therefore in X_1; or, the morphism $f_1 : X_1 \to Y_1$ induced by f by restriction is quasi-faithfully flat and quasi-compact (1.1.2) and (2.3.3); we deduce from (2.3.12) that Z is open in $\overline{Z}$, which shows that Z is locally closed in Y. □

Remark (2.3.15). — It does not suffice, in (2.3.12), to suppose only that f is *faithfully flat*. For example, take for Y a reduced irreducible algebraic curve (**II**, 7.4.2), for X the prescheme *sum* of the schemes $\mathrm{Spec}(\mathscr{O}_y)$, where y ranges over Y (**I**, 3.1), and for f the canonical morphism, which is faithfully flat (**I**, 2.4.2); if η denotes the generic point of Y, $Z = \{\eta\}$ is not open in Y (**II**, 7.4.3), but $f^{-1}(Z) = \mathrm{Spec}(\mathscr{O}_\eta)$ since $\mathscr{O}_\eta$ is a field, and $f^{-1}(Z)$ is open in X. **IV-2** | 19

2.4. Universally open morphisms and flat morphisms

(2.4.1). We have already defined (**II**, 5.4.9) the notion of a *universally closed* morphism; in the same manner, we introduce the following definitions:

Definition (2.4.2). — We say that a morphism of preschemes $f : X \to Y$ is *universally open* (resp. *universally bicontinuous*, resp. a *universal homeomorphism*) if, for every morphism $g : Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is open (resp. a homeomorphism onto its image, resp. a homeomorphism onto Y').

We will see further on (**IV**, 14.3.2) that when Y is locally noetherian, the definition of universally open morphism given here is equivalent to Definition (**III**, 4.3.9) for morphisms of finite type; the reader will be able to verify that we do not use this last definition before §14.

Proposition (2.4.3). — (i) *An immersion (resp. an open immersion, resp. a closed immersion) is universally bicontinuous (resp. universally open, resp. universally closed).*

(i) *The composite of two universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms) is so as well.*

(i) *If $f : X \to Y$ is an S-morphism universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every base change $S' \to S$.*

(i) *If $f : X \to X'$, $g : Y \to Y'$ are two S-morphisms universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms), the same is true of $f \times_S g : X \times_S Y \to X' \times_S Y'$.*

(i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms such that f is surjective; if $g \circ f$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of g.*

(i) *For a morphism f to be universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), it is necessary and sufficient that f_{red} be so.*

(i) *Let (U_α) be an open cover of Y. For $f : X \to Y$ to be universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), it is necessary and sufficient that for every α, its restriction $f^{-1}(U_\alpha) \to U_\alpha$ be so.*

Proof. Assertion (i) follows from (**I**, 4.3.2). Assertion (ii) also follows from the definitions, and it is the same for (iii) by reducing to the case where $Y = S$, $Y' = S'$, which one can do using (**I**, 3.3.11); one knows that (iv) follows from (ii) and (iii) (**I**, 3.5.1). To prove (v), note that for every morphism $Z' \to Z$, $f_{(Z')} : X_{(Z')} \to Y_{(Z')}$ is surjective (**I**, 3.5.2), and it is then a purely topological question. For the case where $g \circ f$ is open (resp. closed), the fact that g is then open (resp. closed) follows from Bourbaki, *Top. gén.*, chap. I, 3^e^ éd., §5, no. 1, prop. 1; for the two other cases, we can reduce to supposing that $g(f(X)) = g(Y) = Z$, in other words that $g \circ f$ is a homeomorphism from X to Z; since f is surjective, g is necessarily bijective, and since g is a continuous open map by what precedes, g is indeed a homeomorphism from Y to Z. **IV-2** | 20

To prove (vi), note that to say that a morphism g is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism bijective) is equivalent to saying that g_{red} has the same property. On the other hand (**I**, 5.1.8), for every morphism $Y' \to Y$, one has $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y'_{\mathrm{red}})_{\mathrm{red}} = (X \times_Y Y')_{\mathrm{red}}$, so the previous remark shows that if f_{red} is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of f. The converse is proved in the same way, noting here that for every morphism $Y'' \to Y_{\mathrm{red}}$, one has $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y'')_{\mathrm{red}} = (X \times_Y Y'')_{\mathrm{red}}$ (**I**, 5.1.8).

Finally, the necessity of (vii) follows immediately from (iii). Conversely, suppose (vii) is fulfilled, and let $g : Y' \to Y$ be a morphism; then the $g^{-1}(U_\alpha) = U'_\alpha$ form an open cover of Y', and if we denote f_α the restriction $f^{-1}(U_\alpha) \to U_\alpha$ of f, and f' the morphism $f_{(Y')}$, the restriction $f'^{-1}(U'_\alpha) \to U'_\alpha$ is none other than $(f_\alpha)_{(U'_\alpha)}$. We can therefore restrict to proving that f is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism onto Y), which is immediate. □

Proposition (2.4.4). — *A universally bicontinuous morphism $f : X \to Y$ is radiciel (hence separated* (1.7.7.1)).

PROOF. Indeed, f is universally injective by hypothesis (**I**, 3.5.11). □

Proposition (2.4.5). — (i) *A morphism $f : X \to Y$ which is* integral, surjective *and* radiciel *is a universal homeomorphism.*

(i) *Inversely, suppose Y locally noetherian. Then, if a morphism of finite type $f : X \to Y$ is a universal homeomorphism, it is finite, surjective and radiciel.*

PROOF. (i) It suffices to observe that the three properties of f are conserved by base change (**I**, 3.5.2), (**I**, 3.5.7) and (**II**, 6.1.5), and as an integral morphism is closed (**II**, 6.1.10), it is clear that f is a universal homeomorphism from X to Y.

(ii) Since f is of finite type and closed by hypothesis and separated by (2.4.4), it is proper (**II**, 5.4.4), and for every $y \in Y$, $f^{-1}(y)$ is reduced to a single element; therefore (**III**, 4.4.2) f is finite; it is clear that f is surjective, and it is radiciel since it is universally injective (**I**, 3.5.11). □

Theorem (2.4.6). — *Let $f : X \to Y$ be a quasi-flat morphism* (2.3.3) *and locally of finite presentation. Then f is universally open. In particular, a flat morphism locally of finite presentation is universally open.*

PROOF. We know that for every base change $Y' \to Y$, $f_{(Y')}$ is quasi-flat (2.3.3) and locally of finite presentation (1.4.3) (iii). It therefore suffices to prove that f is an open morphism. But this follows from the criterion (1.10.4) for morphisms locally of finite presentation, the conditions b), b′), and c) of (1.10.4) being none other than the conditions (i), (ii), (iii) of (2.3.4). □ **IV-2** | 21

Corollary (2.4.7). — *For every prescheme Y, the structural morphism $\mathbf{V}^n_Y \to Y$ (where $\mathbf{V}^n_Y = Y \otimes_{\mathbf{Z}} \operatorname{Spec}(\mathbf{Z}[T_1, \ldots, T_n])$, also written $Y[T_1, \ldots, T_n]$) is universally open.*

PROOF. Indeed, for $Y = \operatorname{Spec}(A)$, $Y[T_1, \ldots, T_n] = \operatorname{Spec}(A[T_1, \ldots, T_n])$, and $A[T_1, \ldots, T_n]$ is a free A-algebra of finite presentation. □

Remarks (2.4.8). — (i) Note that a morphism $f : X \to Y$ faithfully flat and quasi-compact *need not be universally open*, even when X and Y are noetherian. Take for example for Y a reduced irreducible algebraic curve (**II**, 7.4.2) of generic point y, and let X be the prescheme *sum* $Y \amalg \operatorname{Spec}(k(y))$, f being the canonical morphism; it is clear that f is flat and surjective, hence faithfully flat, and quasi-compact, but the image by f of the open part $\operatorname{Spec}(k(y))$ of X is the singleton $\{y\}$, which is not open in Y (**II**, 7.4.3).

(ii) For every prescheme X, the canonical morphism $f : X_{\text{red}} \to X$ is a closed immersion and a *universal homeomorphism* (2.4.3) (vi); but when X is locally noetherian, f is not *flat* unless X is reduced, that is $f = 1_X$ (2.2.17).

Proposition (2.4.9). — *Let Y be a discrete prescheme. Then every morphism $f : X \to Y$ is universally open.*

PROOF. The question being local on Y (2.4.3) (vii), we can reduce to the case where the underlying space of Y is a point; replacing f by f_{red} (2.4.3) (vi), we can suppose moreover that Y is the spectrum of a field k; on the other hand, for every base change $Y' \to Y$, the open sets of $X' = X \times_Y Y'$ are the inverse images of open affine sets of X', so we can suppose $X = \operatorname{Spec}(B)$ affine, B being a k-algebra. It suffices then to prove that for every k-algebra A', if we set $B' = B \otimes_k A'$, the image by $f' : \operatorname{Spec}(B') \to \operatorname{Spec}(A')$ of every open set of the form $U = D(t)$ $(t \in B')$ is open in $\operatorname{Spec}(A')$. Now, B is the inductive limit of the family of its sub-k-algebras of finite type B_α, so $B' = \varinjlim(B_\alpha \otimes_k A')$; there exists an index α such that t is the image in B' of an element $t_\alpha \in B'_\alpha = B_\alpha \otimes_k A'$. But since k is a field, B_α is a k-algebra of finite presentation and a flat k-module, hence B'_α is an A'-algebra of finite presentation and a flat A'-module; we conclude from (2.4.6) that the image by $f'_\alpha : \operatorname{Spec}(B'_\alpha) \to \operatorname{Spec}(A')$ is *open* in $\operatorname{Spec}(A')$. It suffices to check that for every point $y' \in f'_\alpha(U_\alpha)$, the intersection $V = U \cap f'^{-1}(y')$ is non-empty. But by definition of y', $V_\alpha = U_\alpha \cap f'^{-1}_\alpha(y')$ is non-empty; that is, V_α is a non-empty open subset of $\operatorname{Spec}(B'_\alpha \otimes_k k(y'))$ and V is its inverse image by the morphism $v : \operatorname{Spec}(B \otimes_k k(y')) \to \operatorname{Spec}(B_\alpha \otimes_k k(y'))$. As B_α is a sub-algebra of B and k is a field, the homomorphism $B_\alpha \otimes_k k(y') \to B \otimes_k k(y')$ is injective, so the morphism v is dominant (**I**, 1.2.7), which concludes the proof. □ **IV-2** | 22

Corollary (2.4.10). — *Let k be a field, X, Y two k-preschemes; then the projection morphism $X \times_k Y \to X$ is universally open. In particular, for every extension K of k, the projection morphism $X_{(K)} \to X$ is universally open.*

PROOF. It suffices to apply (2.4.9) to the structural morphism $Y \to \mathrm{Spec}(k)$. □

Remark (2.4.11). — If $f : X \to Y$ is an *open* morphism, we know that, for every part E of Y, one has $f^{-1}(\overline{E}) = \overline{f^{-1}(E)}$ (Bourbaki, *Top. gén.*, chap. I, 3ᵉ éd., §5, no. 4, prop. 7). This remark applies for example when f is a flat morphism *locally of finite presentation* (2.4.6), or a morphism projection $X \times_k Y \to X$ where X, Y are preschemes over a field k (2.4.10), and generalises then (2.3.10).

2.5. Permanence of properties of Modules by faithfully flat descent

Proposition (2.5.1). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, $g : Y' \to Y$ a faithfully flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$. For $\mathscr{F}$ to be flat* (*resp. faithfully flat*) *relative to f, it is necessary and sufficient that $\mathscr{F}'$ be flat* (*resp. faithfully flat*) *relative to f'.*

PROOF. It suffices to apply (2.2.10) replacing $X, Y, Z, \mathscr{F}, \mathscr{G}'$ by $Y', X, Y, \mathscr{O}_{Y'}, \mathscr{F}$ respectively; we conclude that for $\mathscr{F}$ to be flat (resp. faithfully flat) relative to f, it is necessary and sufficient that $\mathscr{F}'$ be flat (resp. faithfully flat) relative to $g \circ f'$. But since $g' : X' \to X$ is the canonical projection, g' is faithfully flat (2.2.13) and we have $g \circ f' = f \circ g'$; therefore for $\mathscr{F}'$ to be flat (resp. faithfully flat) relative to $f \circ g'$, it is necessary and sufficient that $\mathscr{F}$ be so relative to f (2.2.11) (iii). □

Proposition (2.5.2). — *Let $f : X' \to X$ be a faithfully flat and quasi-compact morphism, $\mathscr{F}' = f^*(\mathscr{F})$. Consider, for a quasi-coherent Module, the property of being:*

(i) *of finite type;*
(i) *of finite presentation;*
(i) *locally free of finite type;*
(i) *locally free of rank n.*

Then, if **P** *denotes one of the preceding properties, for $\mathscr{F}$ to have property* **P**, *it is necessary and sufficient that $\mathscr{F}'$ have it.*

PROOF. Note that the question is local on X, and we can therefore suppose X affine; we then know (2.2.12) that there is an affine scheme X'' and a faithfully flat morphism $g : X'' \to X'$ which is a surjective local isomorphism; by hypothesis, the question is local on X, and taking into account (2.2.3) and (**II**, 6.1.4.1), it therefore suffices to prove the **IV-2** | 23

Lemma (2.5.3). — *Let A be a ring, A' a faithfully flat A-algebra, M an A-module; $M' = M \otimes_A A'$. For M to be of finite type* (*resp. of finite presentation*), *it is necessary and sufficient that M' be so.*

The proof follows from Bourbaki, *Alg. comm.*, chap. I, §3, no. 6, prop. 11. □

Remark (2.5.4). — The assertions of (2.5.2) for properties (i) and (ii) are still valid when one supposes only f *quasi-faithfully flat* (2.3.3) and quasi-compact. This is because one is reduced to Bourbaki, *loc. cit.* in this way.

Proposition (2.5.5). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite type and f-flat, $\mathscr{E}$ an $\mathscr{O}_Y$-Module quasi-coherent of finite type; for every $y \in Y$, set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} k(y)$. For a point $x \in X$ to be a maximal point of* $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, *it is necessary and sufficient that $y = f(x)$ be a maximal point of* $\mathrm{Supp}(\mathscr{E})$, *and that x be a maximal point of* $\mathrm{Supp}(\mathscr{F}_y)$ *in $f^{-1}(y)$. If this is so, then we have*

$$\mathrm{long}((\mathscr{E} \otimes_Y \mathscr{F})_x) = \mathrm{long}(\mathscr{E}_y) \cdot \mathrm{long}((\mathscr{F}_y)_x). \tag{6}$$

2.6. Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent

Proposition (2.6.1). — *Let $f : X \to Y$ be an S-morphism of S-preschemes, $g : S' \to S$ a surjective morphism, $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \to Y'$. Consider, for a morphism, the property of being:*

(i) *surjective;*
(i) *injective;*
(i) *with finite fibres (as sets);*
(i) *bijective;*
(i) *radiciel.*

Then, if **P** *denotes one of the preceding properties and if f' has property* **P**, *the same is true of f.*

Proposition (2.6.2). — *With the notations being those of (2.6.1), suppose the morphism $g : S' \to S$ faithfully flat and quasi-compact. Consider for a morphism the property of being:*

(i) *open;*
(i) *closed;*
(i) *quasi-compact and universally bicontinuous onto its image;*
(i) *a universal homeomorphism;*

Then, if **P** *denotes one of the preceding properties and if f' has property* **P**, *the same is true of f.*

Proof. Since the morphism $Y' \to Y$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), **IV-2** | 28
one can, by virtue of (**I**, 3.3.11), reduce to the case where $Y = S$, $Y' = S'$. If g' is the projection $X' \to X$, we know that for every part M of X, $g^{-1}(f(M)) = f'(g'^{-1}(M))$ (**I**, 3.4.8). If f' is an open (resp. closed) morphism, then, for every open (resp. closed) part M of X, $f'(g'^{-1}(M))$ is open (resp. closed) in Y', and since g is faithfully flat and quasi-compact, we conclude by virtue of (2.3.12) that $f(M)$ is open (resp. closed) in Y. This proves the proposition for cases (i) and (ii). Let us prove it in case (iii) (which will imply the conclusion in case (iv), taking into account (2.6.1) (iv)); by virtue of (2.6.1) (ii), f is injective, and it remains to prove that f, considered as a map from X onto $f(X)$, is a quasi-compact and *open* map. Since f' is quasi-compact, the same is true of f (1.1.4). It then suffices for every closed set Z of X to have $Z = f^{-1}(\overline{f(Z)})$; since g' is surjective, this relation is equivalent to $g'^{-1}(Z) = g'^{-1}(f^{-1}(\overline{f(Z)}))$ or to $g'^{-1}(Z) = f'^{-1}(g^{-1}(\overline{f(Z)}))$. Now, since f is quasi-compact, its composite with the canonical injection $Z \to X$ (where Z is the reduced sub-prescheme of X having Z as underlying space) is also quasi-compact, and one concludes that $f(Z)$ is pro-constructible. Applying (2.3.10) to $f(Z)$ in Y (image of the morphism $f|Z$), we obtain $g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}$; the formula to prove is therefore equivalent to: $Z' = f'^{-1}(\overline{f'(Z')})$, where $Z' = g'^{-1}(Z)$; but this last formula follows from the hypothesis that f' is a homeomorphism onto $f'(X')$. □

Remark (2.6.3). — In cases (i) and (ii), the conclusions of (2.6.2) are still valid when one supposes only g *quasi-faithfully flat* (2.3.3) and *quasi-compact*; indeed, by virtue of (2.1.4), (**I**, 3.5.2) and (**I**, 9.1.13.1), one can still reduce to the case where $Y = S$, $Y' = S'$; the conclusion then follows from (2.3.12). In cases (iii) and (iv), the conclusions remain valid when one supposes only g *quasi-faithfully flat*; indeed, one then uses (2.3.10) and the fact that g is surjective. Finally, if g is faithfully flat and locally of finite presentation, or if g is surjective and S discrete, the conclusion of (2.6.2) is valid when **P** is the property:

(iii bis) *of being a homeomorphism onto a sub-space image*;

this follows indeed from the proof given in (2.6.2) and from Remark (2.4.11).

Corollary (2.6.4). — *With the notations being those of (2.6.1), suppose the morphism $g : S' \to S$ faithfully flat and quasi-compact. Consider for a morphism the property of being:*

(i) *universally open;*
(i) *universally closed;*
(i) *quasi-compact and universally bicontinuous;*
(i) *a universal homeomorphism;*
(i) *quasi-compact;*
(i) *quasi-compact and dominant.*

Then, if **P** *denotes one of the preceding properties, for f to have property* **P**, *it is necessary and sufficient that f' have it.*

PROOF. Properties (v) and (vi) are included for the record, being consequences of (1.1.4), (1.1.6) and (2.3.7). For the others, the condition is necessary by virtue of (2.4.3). Conversely, suppose for example that f' is universally open, and let $Y_1 \to Y$ be an arbitrary morphism; set $X_1 = X \times_Y Y_1$ and $f_1 = f_{(Y_1)}$. On the other hand, set $Y_1' = Y_1 \times_Y Y'$, $X_1' = X' \times_{Y'} Y_1' = X_1 \times_{Y_1} Y_1'$, $f_1' = f'_{(Y_1')} = (f_1)_{(Y_1')} : X_1' \to Y_1'$; since $g' = g_{(Y_1')} : Y_1' \to Y'$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), and that f' is universally open, f_1' is open, and it follows from (2.6.2) that f_1 is universally open. The same reasoning applies for the other cases. □ **IV-2 | 29**

2.7. Permanence of various properties of morphisms by faithfully flat descent

Proposition (2.7.1). — *Let $f : X \to Y$ be an S-morphism of S-preschemes, $g : S' \to S$ a faithfully flat and quasi-compact morphism, $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \to Y'$. Consider, for a morphism, the property of being:*

(i) *separated;*
(i) *quasi-separated;*
(i) *locally of finite type;*
(i) *locally of finite presentation;*
(i) *of finite type;*
(i) *of finite presentation;*
(i) *proper;*
(i) *an isomorphism;*
(i) *a monomorphism;*
(i) *an open immersion;*
(i) *a quasi-compact immersion;*
(i) *a closed immersion;*
(i) *affine;*
(i) *quasi-affine;*
(i) *finite;*
(i) *quasi-finite;*
(i) *integral.*

Then, if **P** *denotes one of the preceding properties, for f to have property* **P**, *it is necessary and sufficient that f' have it.*

PROOF. It has been shown in Chapters I, II and in Chapter IV, §1, that if f has one of the properties **P** above, the same is true of f' (without hypothesis on the morphism $g : S' \to S$). It therefore remains to prove the converse; since the projection $Y' \to Y$ is a faithfully flat and quasi-compact morphism (2.2.13) and (1.1.2), one can reduce to the case where $S = Y$, $S' = Y'$ by virtue of (**I**, 3.3.11). **IV-2 | 30**

(i) To say that f is separated means that the diagonal morphism $\Delta_f : X \to X \times_Y X$ is *closed*; since $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4), if $\Delta_{f'}$ is closed, the same is true of Δ_f by virtue of (2.6.2).

(ii) has already been proved under weaker hypotheses (1.2.5).

(iii) and (iv): The question is evidently local on X and Y, and, taking into account (2.2.12), it therefore suffices to prove the

Lemma (2.7.1.1). — *Let A be a ring, B an A-algebra, A' a faithfully flat A-algebra and $B' = B \otimes_A A'$. For B to be an A-algebra of finite type* (*resp. of finite presentation*), *it is necessary and sufficient that B' be an A'-algebra of finite type* (*resp. of finite presentation*).

We already know that the condition is necessary without hypothesis on A' (1.3.4), (1.3.6), (1.4.3) and (1.4.6). Suppose that B' is an A'-algebra of finite type; let $(B_\alpha)_{\alpha \in I}$ be the increasing filtering family of A-sub-algebras of B of finite type, so that $B = \varinjlim B_\alpha$, and consequently $B' = \varinjlim (B_\alpha \otimes_A A')$, the tensor product commuting with inductive limits; if (x_i') is a finite system of generators of the A'-algebra B', there exists an index α such that all the x_i' belong to the sub-algebra $B_\alpha \otimes_A A'$ of B', whence $B' = B_\alpha \otimes_A A'$, and since A' is faithfully flat, $B = B_\alpha$ ($\mathbf{0_I}$, 6.4.1).

Suppose now that B' is an A'-algebra of finite presentation; we already know from the preceding that B is an A-algebra of finite type, so there exists a surjective homomorphism $C = A[T_1, \ldots, T_n] \to B$ of A-algebras and an A-homomorphism $C \to B$; let $\mathfrak{J}$ be the kernel of this homomorphism, so that one has an exact sequence $0 \to \mathfrak{J} \to C \to B' \to 0$ (since A' is A-flat), $\mathfrak{J}' = \mathfrak{J} \otimes_A A'$ (identified with an ideal of C'). Since B' is an A'-algebra of finite presentation, $\mathfrak{J}'$ is a C'-module of finite type (1.4.4); but $\mathfrak{J} = \mathfrak{J} \otimes_C C'$ and C' is a faithfully flat C-module (2.2.13) and (2.2.3); we know then that $\mathfrak{J}$ is a C-module of finite type (Bourbaki, *Alg. comm.*, chap. I, §3, no. 6, prop. 11), hence B is an A-algebra of finite presentation.

(v) follows from (iii) and from (2.6.2) (v) by virtue of (1.5.2).

(vi) follows similarly from (iv), (v) and (ii) by virtue of (1.6.1).

(vii) follows from (i), (v) and (2.6.4) (ii) (**II**, 5.4.1).

(viii) First note that since f' is an isomorphism, it is a universal homeomorphism, hence the same is true of f (2.6.4); we conclude already that f is quasi-compact and separated (2.4.4); it remains to prove that $\theta : \mathscr{O}_Y \to f_*(\mathscr{O}_X)$ is an isomorphism of $\mathscr{O}_Y$-Modules. Now, if we set $f' = (\psi', \theta')$, $\theta' : \mathscr{O}_{Y'} \to f'_*(\mathscr{O}_{X'})$ is composed of the canonical homomorphism $g^*(f_*(\mathscr{O}_X)) \to f'_*(\mathscr{O}_{X'})$ and of $g^*(\theta)$ (2.3.2); but the first of these two homomorphisms is bijective by virtue of the hypothesis on g (2.3.1), so if θ' is bijective, θ is bijective (2.2.7), which proves (viii). **IV-2** | 31

(ix) The proposition follows from (viii), from (**I**, 5.3.4) and from (**I**, 5.3.8) which reduces the monomorphisms to isomorphisms.

(x) If f' is an open immersion, $f'(X')$ is open in Y', and we have $f'(X') = g^{-1}(f(X))$ (**I**, 3.4.8); it follows from (2.3.12) that $f(X)$ is open. One can then replace Y (resp. Y') by the sub-prescheme induced on the open $f(X)$ (resp. $f'(X')$), taking into account (1.1.2) and (2.2.13); then f' becomes an isomorphism, so the same is true of f by (viii), and this establishes (x).

(xi) If f' is a quasi-compact immersion, f' is a quasi-compact morphism and quasi-separated (**II**, 5.1.1), hence the same is true of f by (ii) and (2.6.2) (v). Let Z be the closed sub-prescheme of Y closed image of X by f (1.7.8), and set $f = j \circ g'$ with $j' = j_{(Y')}$, $g' = g_{(Y')}$, and we know that j' is the canonical closed immersion $Z' \to Y'$ of the sub-prescheme Z' of Y', closed image of X' by f' (2.3.2). The hypothesis on f' means that g' is an open immersion; it follows by (x) that g is also an open immersion, and this shows that f is an immersion.

(xii) To say that f (resp. f') is a closed immersion means that f (resp. f') is a quasi-compact immersion and a closed morphism; one sees therefore that (xii) follows from (xi) and from (2.6.2) (ii).

(xiii) and (xiv) Suppose f' affine (resp. quasi-affine); note then that f' is quasi-compact and quasi-separated (**II**, 5.1.1), hence the same is true of f by (ii) and (2.6.2) (v). Set $\mathscr{A} = f_*(\mathscr{O}_X)$, $\mathscr{A}' = f'_*(\mathscr{O}_{X'})$; by virtue of (2.3.1), the canonical homomorphism of $\mathscr{O}_{Y'}$-Algebras $g^*(\mathscr{A}) \to \mathscr{A}'$ is bijective; consequently, if $h : Z = \operatorname{Spec}(\mathscr{A}) \to Y$ is the structural morphism, the structural morphism $h' : Z' = \operatorname{Spec}(\mathscr{A}') \to Y'$ is identified with $h_{(Y')}$ (**II**, 1.5.2). Let then $u : X \to Z$ (resp. $u' : X' \to Z'$) be the Y-morphism (resp. Y'-morphism) canonically corresponding to the identical homomorphism of $\mathscr{A}$ (resp. $\mathscr{A}'$) (**II**, 1.2.7); since we have the commutative diagram

$$\begin{tikzcd} X & X' \arrow[l] \\ Z \arrow[u, "u"] & Z' \arrow[l] \arrow[u, "u'"'] \\ Y \arrow[u, "h"] & Y' \arrow[l] \arrow[u, "h'"'] \end{tikzcd}$$

and that $h' \circ u' = f'$, it follows from (**II**, 1.2.7) that $u' = u_{(Z')}$. Moreover, g' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). This being the case, the hypothesis on f' means that u' is an isomorphism (resp. an open immersion) (**II**, 5.1.6); it follows then from (viii) (resp. (x)) that u is an isomorphism (resp. an open immersion), whence (xiii) (resp. (xiv)).

(xv) If f' is finite, it is affine, hence so is f by (xiii); moreover, with the notations of the proof of (xiii), $\mathscr{A}'$ is an $\mathscr{O}_{Y'}$-Module of finite type, and $\mathscr{A}'$ is isomorphic to $g^*(\mathscr{A})$; it follows from (2.5.2) that $\mathscr{A}$ is an $\mathscr{O}_Y$-Module of finite type, hence f is a finite morphism.

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(xvi) To say that f is quasi-finite means that f is a morphism of finite type and that for every $y \in Y$, $f^{-1}(y)$ is finite (**II**, 6.2.2) and (**I**, 6.4.4); the conclusion follows therefore from (v) and (1.5.2).

(xvii) One sees as in (xv) that f is affine. We can reduce to the case where $Y = \operatorname{Spec}(A)$, $Y' = \operatorname{Spec}(A')$, and then $X = \operatorname{Spec}(B)$, $X' = \operatorname{Spec}(B')$, where $B' = B \otimes_A A'$; B is equal to the inductive limit of its sub-A-algebras of finite type B_α, hence $B' = \varinjlim B'_\alpha$, where $B'_\alpha = B_\alpha \otimes_A A'$, and B'_α is an A'-algebra of finite type. But by hypothesis B' is integral over A', hence B'_α is an A'-module of finite type, and B_α is therefore an A-module of finite type (2.5.2). □

Corollary (2.7.2). — *With the hypotheses and notations being those of (2.7.1), suppose f quasi-compact; let $\mathscr{L}$ be a quasi-coherent $\mathscr{O}_X$-Module inversible, $\mathscr{L}' = \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, its inverse image. For $\mathscr{L}$ to be ample (resp. very ample) for f, it is necessary and sufficient that $\mathscr{L}'$ be ample (resp. very ample) for f'.*

Remarks (2.7.3). — (i) It follows from (2.6.1), (2.6.4) and (2.5.4.1) that the conclusions of (2.7.1) are valid in cases (i), (iii), (v), (vii) and (xvi) when one supposes only that g is quasi-compact and *quasi-faithfully flat* (2.3.3); we have already remarked that (2.7.1) is valid in case (ii) only assuming g surjective and quasi-compact.

(ii) With the notations and hypotheses of (2.7.1), it can happen that f is *proper* and f' *projective* without f being *quasi-projective*. Indeed, Hironaka [**?**] has given an example of a proper non-projective morphism $f : X \to Y$, where X and Y are two regular algebraic schemes ($\mathbf{0_I}$, 4.1.4) over the same field k, Y being projective over k; moreover Y is a union of two affine opens Y_i ($i = 1, 2$) such that $f_i : X \times_Y Y_i \to Y_i$ is projective for $i = 1, 2$. Let then $Y' = Y_1 \amalg Y_2$ be the sum prescheme; it is clear that the canonical morphism $g : Y' \to Y$ coinciding with the canonical injections into Y_1 and Y_2, is faithfully flat and quasi-compact by virtue of (**I**, 5.5.10); however, although $f' : X \times_Y Y' \to Y'$ (coinciding with f_i on each of the Y_i) is projective (**II**, 5.5.6), it is not of the same type as f.

(iii) Under the hypotheses of (2.7.1), it can happen that f' is a *local isomorphism* without f being a local immersion.

2.8. Preschemes over a regular base of dimension 1; closure of a closed subprescheme of the generic fibre

Proposition (2.8.1). — *Let Y be a locally Noetherian, regular, irreducible prescheme of dimension 1, with generic point η, $f : X \to Y$ a morphism, $X_\eta = f^{-1}(\eta)$ the fibre at the generic point, $i : X_\eta \to X$ the canonical morphism. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, $\mathscr{F}_\eta = i^*(\mathscr{F})$, $\mathscr{G}'$ a quotient $\mathscr{O}_{X_\eta}$-Module of $\mathscr{F}_\eta$, and let $\overline{\mathscr{G}'}$ be the $\mathscr{O}_X$-Module image of $\mathscr{F}$ by the composed homomorphism ($\mathbf{0_I}$, 4.4.3.2)*

$$\mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} i_*(i^*(\mathscr{F})) \longrightarrow i_*(\mathscr{G}').$$

Then $\overline{\mathscr{G}'}$ is a quotient $\mathscr{O}_X$-Module of $\mathscr{F}$, quasi-coherent and f-flat, such that $i^*(\overline{\mathscr{G}'}) = \mathscr{G}'$, and it is the only quotient $\mathscr{O}_X$-Module of $\mathscr{F}$ having these properties.

Proof. Since i_* is quasi-compact and quasi-separated (1.1.2) and (1.2.2), it follows from (1.7.4) that for every quasi-coherent $\mathscr{O}_{X_\eta}$-Module $\mathscr{H}'$, $i_*(\mathscr{H}')$ is a quasi-coherent $\mathscr{O}_X$-Module; moreover, for every open subset U of X, we have $(i_*(\mathscr{H}'))|U = (i|U \cap X_\eta)_*(\mathscr{H}'|U \cap X_\eta)$ by definition ($\mathbf{0_I}$, 3.4.1). If we prove the proposition when X and Y are affine, it will follow by gluing in the general case, given the uniqueness assertion valid in the affine case. In other words, we are reduced to proving the following: □ **IV-2** | 34

Lemma (2.8.1.1). — *Let A be a regular Noetherian ring ($\mathbf{0_{IV}}$, 17.3.6), integral of dimension 1, K its field of fractions, M an A-module, N' a quotient K-module of $M_{(K)} = M \otimes_A K$ by a sub-K-module P', $\overline{N'}$ the image of M by the composed homomorphism $M \to M_{(K)} \to N'$. Then $\overline{N'}$ is a flat A-module, and is the unique quotient module N of M which is a flat A-module and such that the kernel of the surjective homomorphism $M_{(K)} \to N_{(K)}$ is equal to P'.*

Proof. Since for every maximal ideal $\mathfrak{m}$ of A, $A_{\mathfrak{m}}$ is a regular local ring of dimension 1, hence a discrete valuation ring, it amounts to the same to say that an A-module N is flat or that it is without torsion ($\mathbf{0_I}$, 6.3.4). Since N' is a K-vector space, it is an A-module without torsion, hence so is $\overline{N'}$, being a submodule of N'; moreover, it is immediate to verify that $\overline{N'}_{(K)}$ identifies with N'. Conversely, if N is a quotient A-module of M having the properties stated, the fact that N is a flat

A-module implies that the canonical homomorphism $N \to N_{(K)} = N \otimes_A K$ is *injective*. Since $N_{(K)}$ identifies with N', the conclusion follows from the commutativity of the diagram

$$\begin{tikzcd} M \arrow[r] \arrow[d] & N \arrow[d] \\ M_{(K)} \arrow[r] & N_{(K)} \end{tikzcd}$$

□

Corollary (2.8.2). — *Under the conditions of (2.8.1), for $\mathscr{F}$ to be f-flat, it is necessary and sufficient that the canonical homomorphism $\mathscr{F} \to i_*(\mathscr{F}_\eta) = i_*(i^*(\mathscr{F}))$ be injective.*

(2.8.3). The formation of the $\mathscr{O}_X$-Module $\overline{\mathscr{G}'}$ is functorial in $\mathscr{F}$ and $\mathscr{G}'$: more precisely, if $\mathscr{F}_1$, $\mathscr{F}_2$ are two quasi-coherent $\mathscr{O}_X$-Modules, $u : \mathscr{F}_1 \to \mathscr{F}_2$ an $\mathscr{O}_X$-homomorphism, $\mathscr{G}'_i$ a quotient $\mathscr{O}_{X_\eta}$-Module of $(\mathscr{F}_i)_\eta$ $(i = 1, 2)$ and $v : \mathscr{G}'_1 \to \mathscr{G}'_2$ a homomorphism making the diagram

$$\begin{tikzcd} (\mathscr{F}_1)_\eta \arrow[r, "i^*(u)"] \arrow[d] & (\mathscr{F}_2)_\eta \arrow[d] \\ \mathscr{G}'_1 \arrow[r, "v"] & \mathscr{G}'_2 \end{tikzcd}$$

commutative (homomorphism determined in a unique way (when it exists) by this property), then the diagram

$$\begin{tikzcd} \mathscr{F}_1 \arrow[r, "u"] \arrow[d] & \mathscr{F}_2 \arrow[d] \\ \overline{\mathscr{G}'_1} \arrow[r, "w"] & \overline{\mathscr{G}'_2} \end{tikzcd}$$

is commutative, and there is therefore a unique homomorphism $w : \overline{\mathscr{G}'_1} \to \overline{\mathscr{G}'_2}$ making the diagram **IV-2** | 35

$$\begin{tikzcd} \mathscr{F}_1 \arrow[r, "u"] \arrow[d] & \mathscr{F}_2 \arrow[d] \\ \overline{\mathscr{G}'_1} \arrow[r, "w"] & \overline{\mathscr{G}'_2} \end{tikzcd}$$

commutative.

Proposition (2.8.4). — *The hypotheses on Y being those of (2.8.1), let X_1, X_2 be two Y-preschemes, $\mathscr{F}_i$ a quasi-coherent $\mathscr{O}_{X_i}$-Module, $\mathscr{G}'_i$ a quotient $\mathscr{O}_{(X_i)_\eta}$-Module of $(\mathscr{F}_i)_\eta$ $(i = 1, 2)$. Then we have*

(2.8.4.1) $$\overline{\mathscr{G}'_1} \otimes_{\mathscr{O}_Y} \overline{\mathscr{G}'_2} = \overline{\mathscr{G}'_1 \otimes_{k(\eta)} \mathscr{G}'_2}.$$

Proof. Indeed, setting $X = X_1 \times_Y X_2$, the first member of (2.8.4.1) is a quasi-coherent $\mathscr{O}_X$-Module which is Y-flat ($\mathbf{0_I}$, 6.2.1), whose inverse image in X_η is $\mathscr{G}'_1 \otimes_{k(\eta)} \mathscr{G}'_2$ (**I**, 9.1.5) and which is a quotient of $\mathscr{F}_1 \otimes_{\mathscr{O}_Y} \mathscr{F}_2$; the conclusion therefore follows from the uniqueness property of (2.8.1). □

Proposition (2.8.5). — *The hypotheses on X and Y being those of (2.8.1), let Z' be a closed subprescheme of X_η. There then exists a unique closed subprescheme $\overline{Z'}$ of X which is Y-flat and such that $i^{-1}(\overline{Z'}) = Z'$.*

Proof. If $\mathscr{J}'$ is the quasi-coherent Ideal of $\mathscr{O}_{X_\eta}$ defining Z', it suffices to apply (2.8.1) to the case where $\mathscr{F} = \mathscr{O}_X$ and $\mathscr{G}' = \mathscr{O}_{X_\eta}/\mathscr{J}'$; if $\overline{\mathscr{G}'} = \mathscr{O}_X/\mathscr{J}$, we have indeed $\mathscr{J}' = i^*(\mathscr{J})\mathscr{O}_{X_\eta}$, hence $i^{-1}(\overline{Z'}) = Z'$ (**I**, 4.4.5). □

We note that the prescheme $\overline{Z'}$ is the *closed image* of Z' by the composed morphism $Z' \to X_\eta \xrightarrow{i} X$, where the first arrow is the canonical injection (**I**, 9.5.3); its underlying space is the *closure* in X of Z' (**I**, 9.5.4), which justifies the notation adopted. We also say that $\overline{Z'}$ is the subprescheme of X which is the *closure* of Z'.

Corollary (2.8.6). — *Let X_1, X_2 be two Y-preschemes, Z'_i a closed subprescheme of $(X_i)_\eta$ ($i = 1, 2$). Then we have*

(2.8.6.1) $$\overline{Z'_1 \times_{k(\eta)} Z'_2} = \overline{Z'_1} \times_Y \overline{Z'_2}.$$

PROOF. This follows from (2.8.4) and (2.8.5). □

§3. ASSOCIATED PRIME CYCLES AND PRIMARY DECOMPOSITION

We give in this section primarily the translation of results on modules presented in Bourbaki, *Alg. comm.*, chap. IV, which we follow very closely. The notions that follow seem scarcely to be of interest except in the case of locally Noetherian preschemes. **IV-2** | 36

3.1. Associated prime cycles of a Module

Definition (3.1.1). — Let X be a prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. We say that a point $x \in X$ is *associated* to $\mathscr{F}$ if the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$ is associated to the $\mathscr{O}_x$-Module $\mathscr{F}_x$ (in other words, if $\mathfrak{m}_x$ is the annihilator of an element of $\mathscr{F}_x$). We denote by $\mathrm{Ass}(\mathscr{F})$ the set of $x \in X$ associated to $\mathscr{F}$. We say that an irreducible closed subset Z of X is an *associated prime cycle* of $\mathscr{F}$ if its generic point is associated to $\mathscr{F}$. When $\mathscr{F} = \mathscr{O}_X$, we also say that the points (resp. prime cycles) associated to $\mathscr{F}$ are *associated to the prescheme X*.

We say that an associated prime cycle of $\mathscr{F}$ (resp. X) is *immersed* if it is contained in another associated prime cycle of $\mathscr{F}$ (resp. X) (in other words, if it is not *maximal* among these cycles).

If X is locally Noetherian, the *irreducible components* of X are none other than the maximal (or *non-immersed*) prime cycles associated to X.

It is clear that if $x \in \mathrm{Ass}(\mathscr{F})$, then $\mathscr{F}_x \neq 0$, in other words

(3.1.1.1) $$\mathrm{Ass}(\mathscr{F}) \subset \mathrm{Supp}(\mathscr{F}).$$

If $x \in X$ is associated to $\mathscr{F}$, it is obviously associated to $\mathscr{F}|U$ for every open neighbourhood U of x, and conversely, if it is associated to $\mathscr{F}|U$ for one of these neighbourhoods, it is associated to $\mathscr{F}$.

We note finally that for a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the existence of immersed associated prime cycles is a *local* question, since if y and z are two points of $\mathrm{Ass}(\mathscr{F})$ such that $y \in \overline{\{z\}}$, every neighbourhood of y contains z.

Proposition (3.1.2). — *Let A be a Noetherian ring, M an A-module, $X = \mathrm{Spec}(A)$, $\mathscr{F} = \widetilde{M}$; for a point $x \in X$ to be associated to $\mathscr{F}$, it is necessary and sufficient that the prime ideal $\mathfrak{j}_x$ of A be associated to the module M (in other words, be the annihilator of an element $f \in M$).*

PROOF. This follows from the definition (3.1.1) and from Bourbaki, *loc. cit.*, §1, no. 2, cor. of prop. 5, applied to $S = A - \mathfrak{j}_x$. □

Proposition (3.1.3). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, x a point of X, Z the reduced closed subprescheme of X having $\overline{\{x\}}$ as underlying space (**I**, 5.2.1). The following conditions are equivalent:*

a) *$x \in \mathrm{Ass}(\mathscr{F})$.*

b) *There exists an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is equal to $\mathrm{Supp}(\mathscr{O}_U \cdot f)$.*

b') *There exists an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is an irreducible component of $\mathrm{Supp}(\mathscr{O}_U \cdot f)$.*

c) *There exists an open neighbourhood U of x and a sub-Module of $\mathscr{F}|U$ isomorphic to $\mathscr{O}_Z|U$ ($\mathscr{O}_Z$ being identified with a quotient of $\mathscr{O}_X$).*

c') *There exists an open neighbourhood U of x and a coherent sub-Module $\mathscr{G}$ of $\mathscr{F}|U$ such that $U \cap Z$ is an irreducible component of $\mathrm{Supp}(\mathscr{G})$.*

PROOF. It is clear that c) implies b), for it suffices to take for f the element of $\Gamma(U, \mathscr{F})$ corresponding to the unit section of $\mathscr{O}_Z|U$. Since $U \cap Z$ is irreducible ($\mathbf{0_I}$, 2.1.6), b) implies b') and b') implies c') since $\mathscr{O}_X$ is coherent ($\mathbf{0_I}$, 5.3.4). To see that c') implies a), we can restrict to the case $U = X = \mathrm{Spec}(A)$ is affine, A being therefore Noetherian, and where $\mathscr{F} = \widetilde{M}$, where M is an

A-module, and $\mathscr{G} = \widetilde{N}$, where N is a submodule of M, of finite type. We know then that the minimal elements of $\mathrm{Supp}(\mathscr{G})$ are the maximal points of $V(\mathrm{Ann}(N))$ ($\mathbf{0_I}$, 1.7.4), and these are also the minimal elements of $\mathrm{Ass}(N)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 3, cor. 1 of prop. 7); since $\mathrm{Ass}(N) \subset \mathrm{Ass}(M) = \mathrm{Ass}(\mathscr{F})$, we see that c') implies a). Finally a) implies c) by virtue of (3.1.2), taking again X affine, $\mathscr{F} = \widetilde{M}$, and Z defined by the ideal fA (with the notations of (3.1.2)). □

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Corollary (3.1.4). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then the maximal associated prime cycles of $\mathscr{F}$ are the irreducible components of* $\mathrm{Supp}(\mathscr{F})$*, and the generic points of these components are the points $x \in X$ such that $\mathscr{F}_x$ is an $\mathscr{O}_x$-Module of finite length and $\neq 0$.*

PROOF. Indeed, if x is the generic point of one of the irreducible components Z of $\mathrm{Supp}(\mathscr{F})$, it follows from the equivalence of a) and c') in (3.1.3) that x belongs to $\mathrm{Ass}(\mathscr{F})$, and Z is an associated prime cycle of $\mathscr{F}$, necessarily maximal by virtue of (3.1.1.1); the converse follows trivially from (3.1.1.1); finally, the last assertion being obviously local follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 2 of prop. 7. □

Corollary (3.1.5). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. For $\mathscr{F} = 0$, it is necessary and sufficient that* $\mathrm{Ass}(\mathscr{F}) = \varnothing$.

PROOF. The question being local, we are reduced to the case where X is affine, and the conclusion follows immediately from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 1 of prop. 2. □

Proposition (3.1.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module; then* $\mathrm{Ass}(\mathscr{F})$ *is locally finite* (*that is to say, every point of X admits a neighbourhood whose intersection with* $\mathrm{Ass}(\mathscr{F})$ *is finite*).

PROOF. It suffices to consider the case where X is affine, hence Noetherian, and then the proposition follows from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, cor. of th. 2. □

Proposition (3.1.7). — *Let X be a prescheme.*

(i) *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules. Then* $\mathrm{Ass}(\mathscr{F}') \subset \mathrm{Ass}(\mathscr{F}) \subset \mathrm{Ass}(\mathscr{F}') \cup \mathrm{Ass}(\mathscr{F}'')$.

(ii) *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, $(\mathscr{F}_\alpha)$ a family of quasi-coherent sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$ such that $\mathscr{F}$ is the union of the $\mathscr{F}_\alpha$. Then* $\mathrm{Ass}(\mathscr{F}) = \bigcup_\alpha \mathrm{Ass}(\mathscr{F}_\alpha)$.

(iii) *For every family $(\mathscr{F}_\alpha)$ of quasi-coherent $\mathscr{O}_X$-Modules, we have* $\mathrm{Ass}(\bigoplus \mathscr{F}_\alpha) = \bigcup \mathrm{Ass}(\mathscr{F}_\alpha)$.

PROOF. We are immediately reduced to the corresponding propositions for modules (Bourbaki, *loc. cit.*, §1, no. 1, formula (1), prop. 3 and cor. 1 of prop. 3). □ **IV-2 | 38**

Proposition (3.1.8). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, U an open subset of X, $\mathscr{J}$ a coherent Ideal of $\mathscr{O}_X$ defining a closed subprescheme of X having $X - U$ as underlying space. The following conditions are equivalent:*

a) $\mathrm{Ass}(\mathscr{F}) \subset U$.

b) For every affine open V of X, every section of $\mathscr{F}$ over V whose restriction to $V \cap U$ is zero, is equal to 0.

c) The canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{F})$ is injective.

PROOF. The question being local, we can suppose $X = \mathrm{Spec}(A)$ affine, A being a Noetherian ring, $\mathscr{F} = \widetilde{M}$, where M is an A-module, and $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A. The homomorphism $M \to \mathrm{Hom}_A(\mathfrak{J}, M)$ sends $m \in M$ to the homomorphism $x \mapsto xm$ from $\mathfrak{J}$ to M; to say that it is not injective means that there exists an $m \neq 0$ in M such that $\mathfrak{J}m = 0$.

Let us first show that c) implies a): indeed, if $\mathrm{Ass}(\mathscr{F})$ met $X - U$, there would exist a prime ideal $\mathfrak{p} \in \mathrm{Ass}(M)$ containing $\mathfrak{J}$, hence an element $m \neq 0$ of M such that $\mathfrak{J}m = 0$. Secondly, b) implies c): indeed, if $\mathfrak{J}m = 0$ for some $m \neq 0$ in M, for every prime ideal $\mathfrak{q} \not\supset \mathfrak{J}$, there exists an $a \in \mathfrak{J}$ not in $\mathfrak{q}$, hence the relation $am = 0$ implies that the canonical image of m in $M_\mathfrak{q}$ is 0, in other words m would be a section $\neq 0$ of $\mathscr{F}$ over X whose restriction to U would be zero. Finally a) implies b). To see this, note that the canonical homomorphism $M \to \prod_{\mathfrak{p} \in \mathrm{Ass}(M)} M_\mathfrak{p}$ is *injective*: indeed, if N is the kernel of this homomorphism, we have $\mathrm{Ass}(N) \subset \mathrm{Ass}(M)$; if there existed a $\mathfrak{p} \in \mathrm{Ass}(N)$, there would be an element $n \in N$ whose $\mathfrak{p}$ would be the annihilator; but by definition of N, there is an element $s \notin \mathfrak{p}$ such that $sn = 0$, which is absurd; we conclude that $\operatorname{Ass}(N) = \varnothing$, whence $N = 0$. Now condition a) implies that if $m \in M$ is a section of $\mathscr{F}$ whose restriction to U is zero, the canonical image of m in $M_{\mathfrak{p}}$ is zero for every $\mathfrak{p} \in \operatorname{Ass}(M)$, hence $m = 0$. □

Corollary (3.1.9). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, f a section of $\mathscr{O}_X$ over X. For f to be $\mathscr{F}$-regular ($\mathbf{0_{IV}}$, 15.2.2), it is necessary and sufficient that $\operatorname{Ass}(\mathscr{F}) \subset X_f$.*

PROOF. Indeed, it is immediate that to say that f is $\mathscr{F}$-regular means that the canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(f\mathscr{O}_X, \mathscr{F})$ is injective, and it suffices to apply (3.1.8) to the Ideal $\mathscr{J} = f\mathscr{O}_X$. □

Proposition (3.1.10). — *Let X, Y be locally Noetherian preschemes, $f : X \to Y$ an integral morphism. Then, for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we have $f(\operatorname{Ass}(\mathscr{F})) = \operatorname{Ass}(f_*(\mathscr{F}))$.* **IV-2 | 39**

PROOF. The question being local on Y and the morphism f being affine, we are immediately reduced to the case where Y is affine, in other words to the following: □

Lemma (3.1.10.1). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a ring homomorphism making B an integral A-algebra, M a B-module. Then the prime ideals $\mathfrak{p} \in \operatorname{Ass}(M_{[\rho]})$ are the inverse images by ρ of the prime ideals $\mathfrak{q} \in \operatorname{Ass}(M)$.*

PROOF. Indeed, if $\mathfrak{q} \in \operatorname{Ass}(M)$, $\mathfrak{q}$ is the annihilator in B of an element $x \in M$, hence $\rho^{-1}(\mathfrak{q})$ is the annihilator in A of x. Conversely, let $\mathfrak{p} \in \operatorname{Ass}(M_{[\rho]})$, so that $\mathfrak{p}$ is the inverse image by ρ of the annihilator $\mathfrak{b}$ in B of an element $x \in M$; it follows from the first theorem of Cohen-Seidenberg that there exists a prime ideal $\mathfrak{q}$ of B containing $\mathfrak{b}$ and whose inverse image is $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, cor. 2 of th. 1); by considering one of the minimal prime ideals among those contained in $\mathfrak{q}$ and containing $\mathfrak{b}$, we can obviously suppose that $\mathfrak{q}$ itself is one of these minimal ideals. But since $B \cdot x \subset M$ is isomorphic to $B/\mathfrak{b}$, we know that $\mathfrak{q} \in \operatorname{Ass}(B/\mathfrak{b}) \subset \operatorname{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, th. 2). □

Corollary (3.1.11). — *Under the hypotheses of (3.1.10), for $\mathscr{F}$ to be without immersed associated prime cycle, it suffices that the same be true of $f_*(\mathscr{F})$.*

PROOF. Suppose indeed that $f_*(\mathscr{F})$ has no immersed associated prime cycle. Note that if A is an integral algebra over a field k, all the prime ideals of A are maximal (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, prop. 1); it therefore follows from (**I**, 6.2.2) that the fibres of f are *discrete* spaces; if x, x' are two distinct points of $\operatorname{Ass}(\mathscr{F})$, neither of them can be adherent to the other if $f(x) = f(x')$; and if $f(x) \neq f(x')$, (3.1.10) and the hypothesis imply that neither of the two points $f(x)$, $f(x')$ can be adherent to the other, hence the same is true of x and x'. □

Remark (3.1.12). — Under the hypotheses of (3.1.10), it can on the contrary happen that $\mathscr{F}$ be without immersed associated prime cycle, but not $f_*(\mathscr{F})$. Take for example $Y = \operatorname{Spec}(k[\mathrm{T}])$ where k is a field ("affine line"), and X the sum of $X_1 = Y$ and $X_2 = \operatorname{Spec}(k)$, the morphism $X_2 \to Y$ corresponding to the canonical homomorphism $k[\mathrm{T}] \to k[\mathrm{T}]/\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal (T). It is clear that the morphism $f : X \to Y$ is finite; if one takes $\mathscr{F} = \mathscr{O}_X$, $\mathscr{F}$ is without immersed associated prime cycle, but $f_*(\mathscr{F}) = \widetilde{M}$, where M is the $k[\mathrm{T}]$-module direct sum of $k[\mathrm{T}]$ and k, so $\operatorname{Ass}(M)$ is formed by the generic point (0) of Y and the point $\mathfrak{m}$.

Proposition (3.1.13). — *Let X be a locally Noetherian prescheme, U an open subset of X, $i : U \to X$ the canonical injection. For every quasi-coherent $\mathscr{O}_U$-Module $\mathscr{F}$, we have $\operatorname{Ass}(i_*(\mathscr{F})) = \operatorname{Ass}(\mathscr{F})$.*

PROOF. Recall that $i_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-Module (1.2.2) and (1.7.4); since $i_*(\mathscr{F})|U = \mathscr{F}$, we have $(\operatorname{Ass}(i_*(\mathscr{F}))) \cap U = \operatorname{Ass}(\mathscr{F})$, and it therefore remains to prove that $\operatorname{Ass}(i_*(\mathscr{F})) \subset U$. But by (3.1.8), this relation means that for every affine open V of X, every section of $i_*(\mathscr{F})$ over V which is zero in $U \cap V$, is zero, a condition trivially satisfied since $\Gamma(V, i_*(\mathscr{F})) = \Gamma(U \cap V, \mathscr{F}) = \Gamma(U \cap V, i_*(\mathscr{F}))$. □

3.2. Irredundant decompositions

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Proposition (3.2.1). — *Let X be a locally Noetherian prescheme, U an open dense subset of X. The following conditions are equivalent:*

a) X is reduced.

b) The sub-prescheme induced on U is reduced and X is without immersed prime cycle.

c) X is without immersed prime cycle and for every generic point x of an irreducible component of X, we have $\operatorname{long}(\mathscr{O}_x) = 1$.

The prime cycles associated to X are then identical to the irreducible components of X.

Proof. It is clear that if X is reduced it is the same for the sub-prescheme induced on U. Moreover, the existence of immersed prime cycles being a local question, we can restrict to the case $X = \operatorname{Spec}(A)$ is affine, A being Noetherian. If A is reduced, we know that the minimal prime ideals of A form a reduced primary decomposition of (0) (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, prop. 10) and are the elements of $\operatorname{Ass}(A)$, so there are no immersed associated prime ideals of A, which shows that a) implies b). It is immediate that b) implies c), since a generic point x of an irreducible component of X belongs to U, hence $\mathscr{O}_x$ is a field. Finally, c) implies a): it suffices to remark that if $\mathscr{N}$ is the Nilradical of $\mathscr{O}_X$, which is a coherent Ideal, $\operatorname{Supp}(\mathscr{N})$ cannot contain any of the generic points of the irreducible components of X by hypothesis; if $\operatorname{Supp}(\mathscr{N})$ were not empty and x was one of the maximal points of this closed set, the criterion (3.1.3), c′) would show that $x \in \operatorname{Ass}(\mathscr{O}_X)$, and $\{x\}$ would be an immersed prime cycle of X, contradicting the hypothesis; hence $\mathscr{N} = 0$. □

We say that a locally Noetherian prescheme X is *irredundant* if $\mathscr{O}_X$ is irredundant (which implies that X is irreducible); for X to be *integral*, it is necessary and sufficient that the $\mathscr{O}_X$-Module $\mathscr{O}_X$ be integral (**I**, 2.1.8) and (3.2.1). If $\mathscr{O}_X$ is integral, $\mathscr{O}_X$ is integral at every point x, that is to say $\mathscr{O}_x$ is an integral ring. We say that a closed subprescheme Y of X is *primary* in X if the Ideal $\mathscr{J}$ of $\mathscr{O}_X$ defining Y is primary in $\mathscr{O}_X$.

Definition (3.2.2). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. We say that $\mathscr{F}$ is *reduced* if it satisfies the two following conditions: 1° $\mathscr{F}$ is without immersed associated prime cycle; 2° for every maximal point x of $\operatorname{Supp}(\mathscr{F})$, we have $\operatorname{long}(\mathscr{F}_x) = 1$.

Condition 1° means that the associated prime cycles of $\mathscr{F}$ are the irreducible components of $\operatorname{Supp}(\mathscr{F})$ (3.1.4), and condition 2° means that for every generic point x of such a component we have $\operatorname{long}(\mathscr{F}_x) = 1$.

For an affine scheme X, this definition gives in particular the notion of *reduced module* over a Noetherian ring A; an A-module M of finite type is said *reduced* if it has no immersed associated prime ideals and if, for every $\mathfrak{p} \in \operatorname{Ass}(M)$, $\operatorname{long}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we say that $\mathscr{F}$ is *reduced at a point* $x \in X$ if $\mathscr{F}_x$ is a reduced $\mathscr{O}_x$-module; that means that, on the local scheme $\operatorname{Spec}(\mathscr{O}_x)$, $\widetilde{\mathscr{F}_x}$ is reduced; it amounts therefore to saying that x does not belong to any immersed associated prime cycle of $\mathscr{F}$ and that $\operatorname{long}(\mathscr{F}_z) = 1$ for all maximal points z of $\operatorname{Supp}(\mathscr{F})$ such that $x \in \overline{\{z\}}$. It is clear that if $\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module that is reduced, it is reduced at every point of X; conversely, if $\mathscr{F}$ is *reduced at a point* x, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is a reduced $\mathscr{O}_U$-Module: it suffices in fact to take U not meeting any immersed associated prime cycle of $\mathscr{F}$ (such a neighbourhood exists since these cycles form a locally finite set of closed subsets of X). To say that $\mathscr{O}_X$ is reduced at a point x is equivalent to saying that X is *reduced* at the point x.

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Proposition (3.2.3). — *Let X be a locally Noetherian prescheme, U an open subset of X, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module such that $U \cap \operatorname{Supp}(\mathscr{F})$ is dense in* $\operatorname{Supp}(\mathscr{F})$. *The following conditions are equivalent:*

a) $\mathscr{F}$ is reduced.

b) $\mathscr{F}|U$ is reduced, and $\mathscr{F}$ is without immersed associated prime cycle.

c) There exists a reduced closed subprescheme X' of X and a coherent $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ without torsion and of rank 1 on every irreducible component of X', such that if $j : X' \to X$ is the canonical injection, we have $j_*(\mathscr{F}') = \mathscr{F}$.

Moreover, when this holds, the subprescheme X' is defined by the Ideal $\mathscr{J}$ of $\mathscr{O}_X$ which is the annihilator of $\mathscr{F}$.

PROOF. To see that $c)$ implies $a)$, we can, by virtue of (3.1.3), limit to the case where $X' = X$ is integral, with generic point x and one can further suppose $X = \operatorname{Spec}(A)$ affine and $\mathscr{F} = \widetilde{M}$, where A is integral and M is an A-module without torsion of rank 1; the annihilator of every element of M being then reduced to 0, we have $\operatorname{Ass}(\mathscr{F}) = \{x\}$ and $\mathscr{F}_x$ is isomorphic to the field of fractions of A, so the conditions of the definition (3.2.2) are satisfied. It is clear that if $\mathscr{F}$ has no immersed associated prime cycles and if $U \cap \operatorname{Supp}(\mathscr{F})$ is dense in $\operatorname{Supp}(\mathscr{F})$, then $\operatorname{Ass}(\mathscr{F}) = \operatorname{Ass}(\mathscr{F}|U)$, hence $a)$ and $b)$ are equivalent. If $a)$ is satisfied, take for X' the reduced closed subprescheme of X whose $\operatorname{Supp}(\mathscr{F})$ is the underlying space (**I**, 5.2.1), and let $\mathscr{F}' = j^*(\mathscr{F})$; a point x of $\operatorname{Ass}(\mathscr{F})$ is necessarily a maximal point of X', and as $\operatorname{long}(\mathscr{F}_x) = 1$, $\mathscr{F}'_x$ is isomorphic to $\mathscr{F}_x$, hence to the field $k(x) = \mathscr{O}_{X',x}$, which proves that $\mathscr{F}'$ is without torsion and of rank 1 (**I**, 7.4.6) and (**I**, 7.4.2). Finally, the last assertion is trivial, since for every $y \in X'$, the annihilator of the $\mathscr{O}_{X',y}$-Module $\mathscr{F}'_y$ is zero. □

Definition (3.2.4). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. We say that $\mathscr{F}$ is *irredundant* if $\operatorname{Ass}(\mathscr{F})$ is reduced to a single element; if $\mathscr{F}$ is of finite type, we say that $\mathscr{F}$ is *integral* if moreover $\mathscr{F}$ is reduced (in other words if $\operatorname{long}(\mathscr{F}_x) = 1$). We say that a quasi-coherent sub-$\mathscr{O}_X$-Module $\mathscr{G}$ of a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is *primary* in $\mathscr{F}$ if $\mathscr{F}/\mathscr{G}$ is irredundant.

For an affine scheme X, this definition gives in particular the notion of *integral module* over a Noetherian ring A; an A-module M of finite type is said *integral* if $\operatorname{Ass}(M)$ is reduced to a single prime ideal $\mathfrak{p}$, and if moreover $\operatorname{long}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we say that $\mathscr{F}$ is *integral at a point $x \in X$* if $\mathscr{F}_x$ is an *integral $\mathscr{O}_x$-module*: that means that x belongs to only *a single* associated prime cycle (necessarily non immersed) of $\mathscr{F}$ and that $\operatorname{long}(\mathscr{F}_z) = 1$ at its generic point z. It is clear that if $\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module that is integral, it is integral at every point of X; conversely, if $\mathscr{F}$ is *integral at a point x*, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is an integral $\mathscr{O}_U$-Module: it suffices in fact to take U meeting no immersed associated prime cycle of $\mathscr{F}$ (such a neighbourhood exists since these cycles form a locally finite set of closed subsets of X).

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Definition (3.2.5). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. A *decomposition* of $\mathscr{F}$ is called *irredundant* if it is a family $(\mathscr{F}_\alpha)_{\alpha \in I}$ of quotient $\mathscr{O}_X$-Modules of $\mathscr{F}$ such that the $\mathscr{F}_\alpha$ are irredundant, the family $(\operatorname{Supp}(\mathscr{F}_\alpha))$ is locally finite, and the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is injective. Such a decomposition is said to be *reduced* if the sets $\operatorname{Ass}(\mathscr{F}_\alpha)$ are pairwise distinct, and if there exists no subset $J \neq I$ such that the subfamily $(\mathscr{F}_\alpha)_{\alpha \in J}$ is an irredundant decomposition of $\mathscr{F}$.

If $(\mathscr{F}_\alpha)_{\alpha \in I}$ is an irredundant decomposition (resp. reduced irredundant decomposition) of $\mathscr{F}$, and if one sets $\mathscr{F}_\alpha = \mathscr{F}/\mathscr{G}_\alpha$, we say also that the family $(\mathscr{G}_\alpha)_{\alpha \in I}$ of sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$ is a *primary decomposition* (resp. a *reduced primary decomposition*) of 0 in $\mathscr{F}$; one observes that the injectivity of the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is equivalent to the condition $\bigcap_{\alpha \in I} \mathscr{G}_\alpha = 0$.

Proposition (3.2.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then there exists a reduced irredundant decomposition $(\mathscr{F}^{(x)})_{x \in \operatorname{Ass}(\mathscr{F})}$, formed of coherent $\mathscr{O}_X$-Modules such that for every $x \in \operatorname{Ass}(\mathscr{F})$, we have $\operatorname{Ass}(\mathscr{F}^{(x)}) = \{x\}$. For every $x \in \operatorname{Ass}(\mathscr{F})$ such that $\{x\}$ is not immersed, $\mathscr{F}^{(x)}$ is uniquely determined as the image of the canonical homomorphism $\mathscr{F} \to i_*(i^*(\mathscr{F}))$, where i is the canonical morphism* $\operatorname{Spec}(k(x)) \to X$.

PROOF. For every $x \in \operatorname{Ass}(\mathscr{F})$, let U be an affine open neighbourhood of x, with ring A, and **IV-2 | 43**
let $\mathscr{F}|U = \widetilde{M}$, where M is an A-module of finite type. We know (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 3, prop. 4) that there exists a submodule N of M such that, if we set $P = M/N$, we have $\operatorname{Ass}(P) = \{x\}$ and $\operatorname{Ass}(N) = \operatorname{Ass}(M) - \{x\}$. Let $\mathscr{G} = \widetilde{P}$, which is a quasi-coherent $\mathscr{O}_U$-Module; let $u : j^*(\mathscr{F}) \to \mathscr{G}$ be the surjective homomorphism corresponding to $M \to P$; from this we derive by composition a homomorphism $j_*(u) : j_*(j^*(\mathscr{F})) \to j_*(\mathscr{G})$, from which by composition a homomorphism

$$v : \mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} j_*(j^*(\mathscr{F})) \xrightarrow{j_*(u)} j_*(\mathscr{G})$$

whose u is the restriction to U; we denote by $\mathscr{F}^{(x)}$ the image of $\mathscr{F}$ by this homomorphism, which is a coherent $\mathscr{O}_X$-Module (**I**, 6.1.1). We have $\mathrm{Ass}(j_*(\mathscr{G})) = \{x\}$ by virtue of (3.1.13), and *a fortiori* (3.1.7) $\mathrm{Ass}(\mathscr{F}^{(x)}) = \{x\}$, since $\mathscr{F}^{(x)} \neq 0$. Moreover, if $\mathscr{H}$ is the kernel of v, $\mathrm{Ass}(\mathscr{H})$ is contained in the intersection of the $\mathrm{Ass}(\mathscr{N}^{(x)})$, which is empty; by (3.1.5), $\mathscr{H} = 0$. In view of the fact that $\mathrm{Ass}(\mathscr{F})$ is locally finite (3.1.6), it is clear that $(\mathscr{F}^{(x)})_{x \in \mathrm{Ass}(\mathscr{F})}$ is a reduced irredundant decomposition of $\mathscr{F}$ satisfying the conditions of the statement. The characterisation of $\mathscr{F}^{(x)}$ when $\{x\}$ is not immersed follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 3, prop. 5, the question being local, and taking into account (**I**, 1.6.7). □

Corollary (3.2.7). — *Under the hypotheses of (3.2.6), if $\mathscr{F}$ has no immersed associated prime cycle, there exists only one reduced irredundant decomposition of $\mathscr{F}$.*

Corollary (3.2.8). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. There exists a finite filtration $(\mathscr{F}_i)_{0 \leqslant i \leqslant n}$ of $\mathscr{F}$ such that $\mathscr{F}_0 = \mathscr{F}$, $\mathscr{F}_n = 0$, formed of coherent $\mathscr{O}_X$-Modules and such that the quotients $\mathscr{F}_i / \mathscr{F}_{i+1}$ are zero or irredundant, and $\mathrm{Ass}(\mathscr{F}_i/\mathscr{F}_{i+1}) \subset \mathrm{Ass}(\mathscr{F})$.*

Proof. Indeed, $\mathscr{F}$ is isomorphic to a sub-$\mathscr{O}_X$-Module of a finite direct sum $\bigoplus_{j=1}^{n} \mathscr{G}_j$, where the $\mathscr{G}_j$ are irredundant and coherent (3.2.6); since every quasi-coherent sub-$\mathscr{O}_X$-Module of $\mathscr{G}_j$ is either zero or irredundant (3.1.7), the $\mathscr{F}_i = \mathscr{F} \cap (\bigoplus_{j=1}^{n-i} \mathscr{G}_j)$ answer the question, $\mathscr{F}_i/\mathscr{F}_{i+1}$ being isomorphic to a sub-$\mathscr{O}_X$-Module coherent of $\mathscr{G}_{n-i}$. □

3.3. Relations with flatness

Proposition (3.3.1). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module which is f-flat, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-Module. If, for every $y \in Y$, we set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$, then*

$$\mathrm{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) \supset \bigcup_{y \in \mathrm{Ass}(\mathscr{E})} \mathrm{Ass}(\mathscr{F}_y) \tag{3.3.1.1}$$

and the two members are equal if Y is locally Noetherian.

Proof. (It is understood that $\mathscr{F}_y$ is a sheaf on the fibre $f^{-1}(y)$, and we identify this fibre with a subspace of X (**I**, 3.6.1).) The question being local on X and on Y, we are reduced to the case where X and Y are affine, and the proposition is then proved in Bourbaki, *Alg. comm.*, chap. IV, §2, no. 6, th. 2. □

Corollary (3.3.2). — *Let Y be a locally Noetherian prescheme without immersed associated prime cycles, $f : X \to Y$ a morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module which is f-flat. Then, for every $x \in \mathrm{Ass}(\mathscr{F})$, $f(x)$ is a maximal point of Y.*

Proof. It suffices to apply (3.3.1) with $\mathscr{E} = \mathscr{O}_Y$, since $\mathrm{Ass}(\mathscr{O}_Y)$ is by hypothesis the set of maximal points of Y. □

Corollary (3.3.3). — *Under the hypotheses of (3.3.1), suppose moreover that X and Y are locally Noetherian, $\mathscr{E}$ and $\mathscr{F}$ coherent. Then the following conditions are equivalent:*

a) $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ is without immersed associated prime cycle.

b) For every point $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ is without immersed associated prime cycle.

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Proof. Suppose a) verified. The hypotheses imply that $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ is a coherent $\mathscr{O}_X$-Module ($\mathbf{0_I}$, 5.3.11 and (**I**, 5.3.5)); its associated prime cycles are therefore the irreducible components of $\mathrm{Supp}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$ (3.1.4) and for every maximal point x of $\mathrm{Supp}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$, $f(x) = y$ is a maximal point of $\mathrm{Supp}(\mathscr{E})$ and x is a maximal point of $\mathrm{Supp}(\mathscr{F}_y)$ (2.5.5). Since, by virtue of (3.3.1) and of the hypothesis, $y \in f(\mathrm{Supp}(\mathscr{F}))$, we see that $\mathscr{F}_y \neq 0$ (**I**, 9.1.13), every point of $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$ is the image by f of a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$, whence condition b) is verified.

Conversely, suppose b) verified, and show that if z, z' are two distinct points of $\mathrm{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$, neither of them can be adherent to the other. Firstly, if $f(z) = f(z') = y$, we have $y \in \mathrm{Ass}(\mathscr{E})$ and z and z' belong to $\mathrm{Ass}(\mathscr{F}_y)$ by (3.3.1), hence by hypothesis none of the two points z, z' is adherent to the other *in* $f^{-1}(y)$, so none of them can be adherent to the other *in* X. If $y = f(z)$ and $y' = f(z')$ are distinct, they belong to $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, hence neither of them can be adherent to the other in Y; it follows from the continuity of f that neither of the two points z, z' can be adherent to the other in X. □

Proposition (3.3.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module which is f-flat. Then the following conditions are equivalent:*

a) $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ is reduced (3.2.2).

b) For every point $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$, $\mathrm{long}(\mathscr{E}_y) = 1$ and $\mathscr{F}_y$ is reduced.

PROOF. Suppose *a*) verified. On already knows from (3.3.3) that for every $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ is without immersed associated prime cycle. Moreover (2.5.5), for every $x \in \mathrm{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) \cap f^{-1}(y)$, we have $1 = \mathrm{long}((\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})_x) = \mathrm{long}(\mathscr{E}_y) \cdot \mathrm{long}((\mathscr{F}_y)_x)$, hence $\mathrm{long}(\mathscr{E}_y) = 1$, which proves *b*).

Conversely, suppose *b*) verified; one already knows that every $x \in \mathrm{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$ is a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})$, that $y = f(x)$ is a maximal point of $\mathrm{Supp}(\mathscr{E})$ and x a maximal point of $\mathrm{Supp}(\mathscr{F}_y)$ (3.3.1) and (3.3.3); moreover it follows from the hypothesis and (2.5.5) that $\mathrm{long}((\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F})_x) = 1$, which proves *a*). □

Corollary (3.3.5). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a* flat *morphism; if Y is reduced at the points of $f(X)$ and if $f^{-1}(y)$ is a reduced $k(y)$-prescheme for every $y \in f(X)$, then X is reduced.*

PROOF. Since the Nilradical $\mathscr{N}_Y$ is coherent, the set of points where Y is reduced is open ($\mathbf{0_I}$, 5.2.2), and one can restrict to the case where Y is reduced. It suffices then to apply (3.3.4) with $\mathscr{E} = \mathscr{O}_Y$ and $\mathscr{F} = \mathscr{O}_X$. □

Proposition (3.3.6). — *Let $f : X \to S$, $g : Y \to S$ be two morphisms, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, $\mathscr{G}$* **IV-2 | 45**
a quasi-coherent $\mathscr{O}_Y$-Module. Suppose that: 1° *$\mathscr{G}$ is g-flat;* 2° *X is locally Noetherian, and for every $s \in S$, $g^{-1}(s)$ is locally Noetherian (which will hold if Y is also locally Noetherian). Let $Z = X \times_S Y$; for every couple (x, y) such that $x \in X$, $y \in Y$ and $f(x) = g(y) = s$, let $\mathrm{T}_{x,y}$ be the prescheme* $\mathrm{Spec}(k(x) \otimes_{k(s)} k(y))$*, and let $\mathrm{I}_{x,y}$ be the image of* $\mathrm{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}})$ *by the canonical monomorphism $\mathrm{T}_{x,y} \to Z$ (**I**, 3.4.9). Then*

$$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \Big(\bigcup_{y \in \mathrm{Ass}(\mathscr{G}_{f(x)})} \mathrm{I}_{x,y} \Big) \tag{3.3.6.1}$$

where for every $s \in S$, $\mathscr{G}_s = \mathscr{G} \otimes_{\mathscr{O}_S} k(s)$.

PROOF. Let $p : Z \to X$, $q : Z \to Y$ be the canonical projections, so that we have the commutative diagram

$$\begin{tikzcd}
X & Z \arrow[l, "p"'] \\
S \arrow[u, "f"] & Y \arrow[l, "g"] \arrow[u, "q"']
\end{tikzcd}$$

Setting $\mathscr{G}' = q^*(\mathscr{G})$, so that $\mathscr{F} \otimes_S \mathscr{G} = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}'$; since $\mathscr{G}'$ is p-flat (2.1.4), it follows from (3.3.1) that

$$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \mathrm{Ass}(\mathscr{G}'_x) \tag{3.3.6.2}$$

with $\mathscr{G}'_x = \mathscr{G}' \otimes_{\mathscr{O}_X} k(x)$. If $s = f(x)$, we have $\mathscr{G}'_x = \mathscr{G}_s \otimes_{k(s)} k(x)$, and $p^{-1}(x) = g^{-1}(s) \otimes_{k(s)} k(x)$; moreover, since the field $k(x)$ is a flat $k(s)$-module, the morphism $p^{-1}(x) \to g^{-1}(s)$ is flat (2.1.4); applying (3.3.1) to this morphism, it follows that

$$\mathrm{Ass}(\mathscr{G}'_x) = \bigcup_{y \in \mathrm{Ass}(\mathscr{G}_s)} \mathrm{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}}) \tag{3.3.6.3}$$

from which the proposition follows. □

We note that if, in the statement, we drop hypothesis 2°, one can still deduce, by virtue of (3.3.1), the relation

(3.3.6.4) $$\mathrm{Ass}(\mathscr{F}\otimes_{\mathrm{S}}\mathscr{G}) \supset \bigcup_{x\in\mathrm{Ass}(\mathscr{F})}\Big(\bigcup_{y\in\mathrm{Ass}(\mathscr{G}_{f(x)})}\mathrm{I}_{x,y}\Big).$$

Corollary (3.3.7). — *Under the hypotheses of (3.3.6), suppose moreover that* S *is locally Noetherian and that* $f(\mathrm{Ass}(\mathscr{F}))\subset\mathrm{Ass}(\mathscr{O}_{\mathrm{S}})$. *Then*

(3.3.7.1) $$\mathrm{Ass}(\mathscr{F}\otimes_{\mathrm{S}}\mathscr{G}) = \bigcup_{(x,y)\in\mathrm{C}}\mathrm{I}_{x,y}$$

where C *is the set of couples* (x,y) *such that* $x\in\mathrm{Ass}(\mathscr{F})$, $y\in\mathrm{Ass}(\mathscr{G})$ *and* $f(x)=g(y)$.

PROOF. Since $\mathscr{G}$ is g-flat, it follows from (3.3.1) that the relation "$s\in\mathrm{Ass}(\mathscr{G}_s)$" is equivalent to $y\in\mathrm{Ass}(\mathscr{G})$: the conclusion follows from (3.3.6.1). □

Remarks (3.3.8). — We will see later (4.2.2) that under the hypotheses of (3.3.6), $\mathrm{T}_{x,y}$ is a prescheme *without immersed associated prime cycle*; it will follow that if $\mathscr{F}$ and the $\mathscr{G}_s$ are without immersed associated prime cycle, the same is true of $\mathscr{F}\otimes_{\mathrm{S}}\mathscr{G}$.

Corollary (3.3.9). — *Under the conditions of (3.3.7), we have*

(3.3.9.1) $$q(\mathrm{Ass}(\mathscr{F}\otimes_{\mathrm{S}}\mathscr{G}))\subset\mathrm{Ass}(\mathscr{G})$$

(*where* $q: X\times_{\mathrm{S}} Y\to Y$ *is the canonical projection*).

PROOF. Indeed, if $(x,y)\in Z$, we have $q(\mathrm{I}_{x,y})=\{y\}\subset\mathrm{Ass}(\mathscr{G})$. □

3.4. Properties of the sheaves $\mathscr{F}/t\mathscr{F}$ IV-2 | 46

Proposition (3.4.1). — *Let X be a locally Noetherian prescheme, t a section of* $\mathscr{O}_X$ *over X, Y the closed subprescheme of X defined by the Ideal* $t\mathscr{O}_X$ *of* $\mathscr{O}_X$. *Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_X$*-Module,* S *the closed subprescheme having* $\mathrm{Supp}(\mathscr{F})$ *as underlying space,* (S_i) *the family of the reduced closed subpreschemes of X having as underlying spaces the irreducible components of* S; *we denote by* s_i *the generic point of* S_i. *Finally, let Z be an irreducible component of* $\mathrm{Supp}(\mathscr{F}/t\mathscr{F})=\mathrm{S}\cap Y$, *z its generic point.*

(i) *For every i such that* $Z\subset\mathrm{S}_i$, *Z is an irreducible component of* $\mathrm{S}_i\cap Y$.

(ii) *If Z is not equal to any of the* S_i, *we have*

(3.4.1.1) $$\mathrm{long}((\mathscr{F}/t\mathscr{F})_z)\geqslant\sum_i\mathrm{long}(\mathscr{F}_{s_i})$$

where the sum of the second member is extended to all i such that $Z\subset\mathrm{S}_i$.

(iii) *Suppose that Z is not equal to any* S_i. *For the two members of (3.4.1.1) to be equal, it is necessary and sufficient that the two following conditions be satisfied:*
- α) t_z *is* $\mathscr{F}_z$*-regular* ($\mathbf{0_{IV}}$, *15.1.4*).
- α) *For every i such that* $M_{\mathfrak{p}_i}\neq 0$, *the image of the germ* t_z *in* $\mathscr{O}_{\mathrm{S}_i,z}$ *generates the maximal ideal of this ring* (*which implies that* $\mathscr{O}_{\mathrm{S}_i,z}$ *is a discrete valuation ring whose* t_z *is a uniformiser*).

PROOF. Suppose that Z is not equal to any S_i. As in the demonstration of (3.4.1), we can reduce to the case where $X=\mathrm{Spec}(\mathscr{O}_z)$; the proposition is then (taking into account (3.1.2)) a consequence of ($\mathbf{0_{IV}}$, 16.4.6.3).

(i) If $j: Y\to X$ is the canonical injection, we have $\mathscr{F}/t\mathscr{F}=\mathscr{F}\otimes_{\mathscr{O}_X}\mathscr{O}_Y=j^*(\mathscr{F})$, hence $\mathrm{Supp}(\mathscr{F}/t\mathscr{F})=j^{-1}(\mathrm{S})=\mathrm{S}\cap Y$ (**I**, 9.1.13), whence the assertion.

(ii) and (iii): Suppose that Z is not equal to any of the S_i. We can, to prove *(ii)* and *(iii)*, replace X by $\mathrm{Spec}(\mathscr{O}_z)$; and if $M=\mathscr{F}_z$, we can suppose $\mathscr{F}=\widetilde{M}$, denoting by $\mathfrak{p}_i$ the minimal ideals of $\mathscr{O}_z$. Moreover, since M is an $\mathscr{O}_z$-module of finite type, on a $\mathrm{S}=\mathrm{S}'_{\mathrm{red}}$, with $\mathrm{S}'=\mathrm{Spec}(\mathscr{O}_z/\mathfrak{a})$, where $\mathfrak{a}$ is the annihilator of M in $\mathscr{O}_z$ ($\mathbf{0_I}$, 1.7.4), and the two members of (3.4.1.1) have the same values whether one considers M as an $\mathscr{O}_z$-module or as an $(\mathscr{O}_z/\mathfrak{a})$-module; one can therefore finally replace X by $\mathrm{S}'=\mathrm{Spec}(A)$, where A is a local Noetherian ring, M being a faithful A-module; since Z is closed in S, the hypothesis that $Z\neq\mathrm{S}_i$ for all i means that $s_i\notin Z$, hence $\dim(A)>0$; finally, to say that z is the generic point of Z, an irreducible component of $\mathrm{S}_i\cap V(t)$, means that A/tA is an Artinian local ring. We are therefore reduced to proving the following statement: □

Lemma (3.4.1.2). — *Let A be a local Noetherian ring of dimension > 0, $\mathfrak{p}_i$ the minimal prime ideals of A, $\mathfrak{m}$ its maximal ideal, t an element of $\mathfrak{m}$ such that A/tA is Artinian. Then, for every A-module M of finite type, we have*

(3.4.1.3) $$\operatorname{long}(M/tM) \geqslant \sum_i \operatorname{long}(M_{\mathfrak{p}_i})\,;$$

moreover, for the two members of (3.4.1.3) to be equal, it is necessary and sufficient that the following conditions be satisfied:

α) t is M-regular;

α) for every i such that $M_{\mathfrak{p}_i} \neq 0$, the image of t in $A/\mathfrak{p}_i$ generates the maximal ideal of this ring (which implies that $A/\mathfrak{p}_i$ is a discrete valuation ring).

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Proof. Since A is not of dimension 0 and A/tA is Artinian, we necessarily have $\dim(A) = 1$ ($\mathbf{0_{IV}}$, 16.3.4) and $t \notin \mathfrak{p}_i$ for every i: the principal ideal (t) is therefore an ideal of definition, and contains by consequence a power of its maximal ideal $\mathfrak{m}$. Let N be the submodule of elements of M annihilated by a power of t (or by a power of $\mathfrak{m}$, which amounts to the same since one has just seen this); if one sets $P = M/N$, t is P-regular, since for $x \in M$ the relation $tx \in N$ implies $t^k(tx) = 0$ for an integer k, hence $x \in N$. This being so, we have the

Lemma (3.4.1.4). — *Let A be a ring,*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

an exact sequence of A-modules. If $t \in A$ is M''-regular, the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact.

Proof. Since $M/tM = M \otimes_A (A/tA)$, it suffices to prove the exactness at M'/tM'; now, if the image x' of an element x' of M' is such that $x' = ty$ with $y \in M$, we deduce, for the images x'', y'' in M'', $x'' = ty''$; but as $x'' = 0$, the hypothesis implies $y'' = 0$, hence y is the image of an element $y' \in M'$, and the relation $x' = ty'$ proves that $M' \to M$ is injective. □

This lemma being established, we derive the relation

(3.4.1.5) $$\operatorname{long}(M/tM) = \operatorname{long}(N/tN) + \operatorname{long}(P/tP).$$

On the other hand, for every i, we have $N_{\mathfrak{p}_i} = 0$ since $t \notin \mathfrak{p}_i$; hence $M_{\mathfrak{p}_i} = P_{\mathfrak{p}_i}$; to prove (3.4.1.3), it suffices to do so replacing M by P; moreover, if the two members of (3.4.1.3) are equal, it follows from the same inequality for P and from (3.4.1.5) that necessarily $\operatorname{long}(N/tN) = 0$, hence $N/tN = 0$ and finally $N = 0$, by the lemma of Nakayama, N being of finite type; now, $N = 0$ means that t is M-regular. $\operatorname{Ass}(M) \subset \bigcup\{\mathfrak{p}_i\}$.

We proceed then by induction on $n = \sum \operatorname{long}(M_{\mathfrak{p}_i})$. If $n = 0$, we have $M_{\mathfrak{p}_i} = 0$ for every i, hence $M = 0$ since none of the $\mathfrak{p}_i$ belongs to $\operatorname{Ass}(M)$; the two members of (3.4.1.3) are then zero, and the assertion β) of (3.4.1.2) is trivially satisfied. If $n > 0$, the reasoning from the beginning of the proof of (3.4.1) allows us moreover to suppose that the A-module M is *faithful*: this implies $M_{\mathfrak{p}_i} \neq 0$ for every i (Bourbaki, *Alg. comm.*, chap. II, §2, no. 2, cor. 2 of prop. 4), and therefore $\operatorname{Ass}(M) = \bigcup\{\mathfrak{p}_i\}$.

Suppose first $n = 1$; there is then only a single minimal prime ideal $\mathfrak{p}$ of A, and to say that $M_{\mathfrak{p}}$ is of length 1 means that $M_{\mathfrak{p}}$ is isomorphic to the residue field $k = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, hence $\mathfrak{p}$ is the annihilator of M as an $A_{\mathfrak{p}}$-module. By consequence $M_{\mathfrak{p}}$ is annihilated by $\mathfrak{p}A_{\mathfrak{p}}$, hence $\mathfrak{p}$ is the annihilator of M since M is supposed faithful. This being so, the hypothesis $M \neq 0$ implies $M/tM \neq 0$ by the lemma of Nakayama, hence $\operatorname{long}(M/tM) \geqslant 1$, which proves (3.4.1.3) in this case. Moreover, if $\operatorname{long}(M/tM) = 1$, M is necessarily *monogenic* (Bourbaki, *Alg. comm.*, chap. II, §3, no. 2, cor. 2 of prop. 4) hence isomorphic to a quotient $A/\mathfrak{b}$; since it is faithful, we necessarily have $\mathfrak{b} = 0$ and M is isomorphic to A; since $\operatorname{long}(A/tA) = 1$, tA is necessarily equal to the maximal ideal $\mathfrak{m}$, and since A is a local integral Noetherian ring, this proves that A is a discrete valuation ring (Bourbaki, *Alg. comm.*, chap. VI, §3, no. 6, prop. 9), whose t is the uniformiser. Conversely, if A is a discrete **IV-2 | 48** valuation ring, t its uniformiser, M is without torsion, hence isomorphic to a submodule of A (M being of finite type), and therefore isomorphic to A itself, whence $\operatorname{long}(M/tA) = \operatorname{long}(A/tA) = 1$.

Suppose now $n \geqslant 2$; there exists then an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with $M' \neq 0$, $M'' \neq 0$ and $\operatorname{Ass}(M) = \operatorname{Ass}(M') \cup \operatorname{Ass}(M'')$; indeed, if $\operatorname{Ass}(M)$ is not reduced to a single element, this follows from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, prop. 4; if on the contrary $\operatorname{Ass}(M)$ is reduced to a single prime ideal, the latter is necessarily the unique minimal prime ideal $\mathfrak{p}$ of A; the hypothesis implies $\operatorname{long}(M_{\mathfrak{p}}) \geqslant 2$ and it suffices to take for M' the inverse image of a submodule of $M_{\mathfrak{p}}$ distinct from 0 and from $M_{\mathfrak{p}}$. Since t is M-regular, t does not belong to any of the prime ideals of $\operatorname{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 2 of prop. 2), hence, for the same reason, t is M'-regular and M''-regular. This last property implies by (3.4.1.4) that the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact; as moreover it is the same for the sequence

$$0 \longrightarrow M'_{\mathfrak{p}_i} \longrightarrow M_{\mathfrak{p}_i} \longrightarrow M''_{\mathfrak{p}_i} \longrightarrow 0$$

for every i, we have

$$\operatorname{long}(M/tM) = \operatorname{long}(M'/tM') + \operatorname{long}(M''/tM'')$$
$$\operatorname{long}(M_{\mathfrak{p}_i}) = \operatorname{long}(M'_{\mathfrak{p}_i}) + \operatorname{long}(M''_{\mathfrak{p}_i})$$

and the induction hypothesis implies the inequality (3.4.1.3). Moreover, the two members can only be equal if the analogous inequalities for M' and M'' are also equalities. By virtue of the induction hypothesis, this is equivalent to the property β) for the $\mathfrak{p}_i$ such that $M'_{\mathfrak{p}_i} \neq 0$ or $M''_{\mathfrak{p}_i} \neq 0$; but these ideals are precisely those for which $M_{\mathfrak{p}_i} \neq 0$. □

Corollary (3.4.2). — *Under the general hypotheses of (3.4.1), suppose that Z is not equal to any of the S_i and that $\operatorname{long}((\mathscr{F}/t\mathscr{F})_z) = 1$. Then there exists a single S_i containing Z, and for this value of i, we have $\operatorname{long}(\mathscr{F}_{s_i}) = 1$; moreover $\mathscr{O}_{\mathrm{S}_i,z}$ is a discrete valuation ring whose t_z is a uniformiser, and t_z is $\mathscr{F}_z$-regular.*

Proof. This follows from (3.4.1), the two members of (3.4.1.1) being then equal. □

Proposition (3.4.3). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, Y the closed subprescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, T an associated prime cycle of $\mathscr{F}$, T' an irreducible component of $\mathrm{T} \cap Y$, x the generic point of T'. Suppose that t_x is $\mathscr{F}_x$-regular; then $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$.* **IV-2 | 49**

Proof. As in the proof of (3.4.1), we can reduce to the case where $X = \operatorname{Spec}(\mathscr{O}_x)$; the proposition is then (taking into account (3.1.2)) a consequence of ($\mathbf{0_{IV}}$, 16.4.6.3). □

Proposition (3.4.4). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, Y the closed subprescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, (S_i) the family of the irreducible components of $\operatorname{Supp}(\mathscr{F})$. Let y be a point of Y such that t_y is $\mathscr{F}_y$-regular and such that none of the associated prime cycles immersed of $\mathscr{F}/t\mathscr{F}$ contains y. Then the irreducible components of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing y are exactly the irreducible components of the $\mathrm{S}_i \cap Y$ containing y, and the prime cycles associated to $\mathscr{F}$ containing y are non-immersed.*

Proof. Let us first prove the last assertion. Let $\mathrm{T} \supset \mathrm{T}_1$ be two associated prime cycles of $\mathscr{F}$ containing y; if x is the generic point of an irreducible component of $\mathrm{T} \cap Y$ containing y, x is a generalisation of y, hence contained in every neighbourhood of y, and the hypothesis that t_y is $\mathscr{F}_y$-regular implies that t_x is $\mathscr{F}_x$-regular ($\mathbf{0_{IV}}$, 15.2.4), hence, by virtue of (3.4.3), $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$. Let x_1 be the generic point of an irreducible component of $\mathrm{T}_1 \cap Y$ containing y, and let x be the generic point of an irreducible component of $\mathrm{T} \cap Y$ containing x_1; it follows from what precedes that x_1 and x both belong to $\operatorname{Ass}(\mathscr{F}/t\mathscr{F})$, and since $x_1 \in \overline{\{x\}}$, the hypothesis implies that $x_1 = x$. Let us still designate by T and T_1 the integral closed subpreschemes of X having T and T_1 as underlying spaces respectively, and set $A = \mathscr{O}_{\mathrm{T},x}$, $A_1 = \mathscr{O}_{\mathrm{T}_1,x}$; then $A_1 = A/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A. By definition of x and x_1, A/tA and A_1/tA_1 are Artinian rings; on the other hand, it was seen above that t_x is $\mathscr{F}_x$-regular, hence x cannot belong to $\operatorname{Ass}(\mathscr{F})$, and consequently A_1 is not Artinian.

We therefore have $\dim A = \dim A_1 = 1$; but this implies $\mathfrak{p} = 0$ and $A = A_1$ ($\mathbf{0_{IV}}$, 16.1.2.2); since $\operatorname{Spec}(A)$ and $\operatorname{Spec}(A_1)$ are respectively dense in T and T_1, we have indeed $\mathrm{T} = \mathrm{T}_1$.

The S_i containing y are therefore all the associated prime cycles of $\mathscr{F}$ containing y; if x is the generic point of an irreducible component of $\mathrm{S}_i \cap Y$ containing y, we deduce also from ($\mathbf{0_{IV}}$, 15.2.4) that t_z is $\mathscr{F}_z$-regular, hence, by (3.4.3), $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$; this demonstrates the first assertion of (3.4.4). □

Proposition (3.4.5). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, Y the closed subprescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, y a point of Y; suppose that t_y is $\mathscr{F}_y$-regular and that $\mathscr{F}/t\mathscr{F}$ is integral at the point y (3.2.4). Then $\mathscr{F}$ is integral at the point y.*

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Proof. In view of (3.4.4), it suffices to prove that y is contained in a *single* irreducible component of $\operatorname{Supp}(\mathscr{F})$, and that if s is the generic point of this component, we have $\operatorname{long}(\mathscr{F}_s) = 1$; the conclusion then follows from (3.4.2). □

Proposition (3.4.6). — *The hypotheses being those of (3.4.1), let x be a point of Y. Suppose that Y contains none of the S_i containing x, and that $\mathscr{F}/t\mathscr{F}$ is reduced at the point x (3.2.2). Then t_z is $\mathscr{F}_z$-regular and $\mathscr{F}$ is reduced at the point x. Moreover, if z_j is the generic point of an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, z_j is contained in only one of the S_i, and $\mathscr{O}_{\mathrm{S}_i,z_j}$ is a discrete valuation ring whose t_{z_j} is a uniformiser.*

Proof. The fact that t_z is $\mathscr{F}_z$-regular follows from the following lemma applied to the ring $\mathscr{O}_z$:

Lemma (3.4.6.1). — *Let A be a Noetherian ring, M an A-module of finite type, $\mathfrak{p}_i$ the minimal elements of $\operatorname{Supp}(M)$, t an element of A. Suppose that t does not belong to any of the $\mathfrak{p}_i$ and that M/tM is a reduced A-module (3.2.2). Then t is M-regular.*

Every prime ideal $\mathfrak{p} \in \operatorname{Supp}(M)$ contains one of the $\mathfrak{p}_i$; since t does not belong to any of the $\mathfrak{p}_i$, the homothety of ratio t in $M_\mathfrak{p}$ is not nilpotent (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, cor. of prop. 9). Denote by N the submodule of M formed by the elements annihilated by a power of t, and set $P = M/N$; we shall show that $N = 0$. Since t is P-regular, on a an exact sequence (3.4.1.4)

$$0 \longrightarrow N/tN \longrightarrow M/tM \longrightarrow P/tP \longrightarrow 0.$$

Since N is of finite type, it is annihilated by a power of t, and it suffices therefore to prove that $N/tN = 0$ for every $\mathfrak{p} \in \operatorname{Ass}(M/tM)$, or equivalently that the homomorphism $u_\mathfrak{p} : (M/tM)_\mathfrak{p} \to (P/tP)_\mathfrak{p}$ is bijective for every $\mathfrak{p} \in \operatorname{Ass}(M/tM)$. Now, we have $(P/tP)_\mathfrak{p} \neq 0$; in effect, since $\mathfrak{p} \in \operatorname{Supp}(M/tM) = \operatorname{Supp}(M) \cap V(t)$ ($\mathbf{0_I}$, 1.7.5), the image of t in $A_\mathfrak{p}$ is contained in the maximal ideal $\mathfrak{p}A_\mathfrak{p}$, hence the hypothesis $P_\mathfrak{p} = tP_\mathfrak{p}$ would imply $P_\mathfrak{p} = 0$ by the lemma of Nakayama; one would then have $M_\mathfrak{p} = N_\mathfrak{p}$ and the homothety of ratio t in $M_\mathfrak{p}$ would be nilpotent, since $\mathfrak{p} \in \operatorname{Supp}(M)$. This being so, the hypothesis that M/tM is reduced implies $\operatorname{long}((M/tM)_\mathfrak{p}) = 1$, and since $(P/tP)_\mathfrak{p} \neq 0$, $u_\mathfrak{p}$ is necessarily bijective, which proves the lemma.

By hypothesis, none of the associated prime cycles immersed of $\mathscr{F}/t\mathscr{F}$ contains x, hence none of the associated prime cycles immersed of $\mathscr{F}$ contains x, by virtue of (3.4.4). On the other hand, applying (3.4.2) to an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, we see that $\operatorname{long}(\mathscr{F}_{s_i}) = 1$ for every S_i containing x, which completes the proof that $\mathscr{F}_x$ is reduced; the last assertions are also consequences of (3.4.2). □

Corollary (3.4.7). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, M an A-module of finite type, $(x_i)_{1 \leqslant i \leqslant k}$ a family of elements of $\mathfrak{m}$ forming part of a system of parameters for M ($\mathbf{0_{IV}}$, 16.3.6). If the A-module $N = M/(\sum_{i=1}^k x_i M)$ is integral (3.2.4), then M is integral and the sequence $(x_i)_{1 \leqslant i \leqslant k}$ is M-regular.* **IV-2** | 51

Proof. By induction on k, one is immediately reduced to the case $k = 1$; we write x instead of x_1; the hypothesis that x forms part of a system of parameters for M implies $\dim(M) = \dim(M) - 1$ ($\mathbf{0_{IV}}$, 16.3.7). Setting $n = \dim(N)$; there is therefore an element $\mathfrak{p}$ of $\operatorname{Supp}(M)$ such that $\dim(M/\mathfrak{p}M) = n + 1$ ($\mathbf{0_{IV}}$, 16.3.5), and moreover x forms part of a system of parameters for $M/\mathfrak{p}M$ ($\mathbf{0_{IV}}$, 16.3.5); in addition, the support of N', being isomorphic to N, is integral; moreover, the support of N' (equal to the intersection of $\operatorname{Supp}(M)$ and $V(x)$) cannot contain $\operatorname{Supp}(M')$, and this last set is irreducible by construction. The hypothesis implies then that x is M'-regular by virtue of (3.4.6). One concludes that the kernel of the homothety $z \mapsto xz$ in M is contained in $\mathfrak{p}^j M \subset \mathfrak{m}^j M$ for every integer j, and this kernel is therefore reduced to 0 ($\mathbf{0_{IV}}$, 7.3.5), which proves that x is M-regular. One can then apply (3.4.5). ☐

Remark (3.4.8). — An analogous proposition to (3.4.7), where one replaces "integral" by "reduced", is no longer necessarily exact. Consider for example the ring of polynomials $C = K[X, Y, Z]$ over a field K, the quotient ring $B = C/\mathfrak{pq}$, where $\mathfrak{p} = CZ$, $\mathfrak{q} = CX^2 + CY$; let A be the local ring of B corresponding to the maximal image of $CX + CY + CZ$ in B. If x, z are the canonical images of X, Z in A, it is clear that $xz \neq 0$ but $x^2z^2 = 0$; on the other hand, since A/xA is isomorphic to $K[Y, Z]/(YZ)$, we have $\dim(A/xA) = 1$ while $\dim(A) = 2$, hence x forms part of a system of parameters of A ($\mathbf{0_{IV}}$, 16.3.4), A/xA is reduced, but A is not.

Proposition (3.4.9). — *Let A be a Noetherian ring, M an A-module of finite type, f an M-regular element such that M/fM has no immersed associated prime ideal. If $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) are the prime ideals associated to M/fM, then, for every integer $n > 0$, $f^n M$ is the intersection of the inverse images of $f^n M_{\mathfrak{p}_i}$ by the canonical homomorphisms $M \to M_{\mathfrak{p}_i}$ ($1 \leqslant i \leqslant m$).*

Proof. It amounts to showing that the $\mathfrak{p}_i$ are also the prime ideals associated to $M/f^n M$, for then the saturations of $f^n M$ for the $\mathfrak{p}_i$ are the submodules of the reduced primary decomposition (necessarily unique) of $f^n M$ in M. Now, we have

$$\operatorname{Ass}(f^{n-1}M/f^n M) \subset \operatorname{Ass}(M/f^n M) \subset \operatorname{Ass}(M/f^{n-1}M) \cup \operatorname{Ass}(f^{n-1}M/f^n M)$$

by (3.1.7), and since f is M-regular, $f^{n-1}M/f^n M$ is isomorphic to M/fM; it therefore suffices to reason by induction on n. ☐

§4. Change of base field for algebraic preschemes

We will develop in §5 the general theory of the dimension of preschemes; but the theory of the dimension of *algebraic* preschemes can be developed in a more elementary way, and since it moreover presents numerous special features, we will give here a rapid exposition, independent of the general theory. **IV-2** | 52

If K is a field and L an extension of K, we will denote by $\deg.\mathrm{tr}_K L$ the *transcendence degree* of L over K.

4.1. Dimension of algebraic preschemes

Definition (4.1.1). — Let k be a field, X a prescheme locally of finite type over k. The *dimension* of X is defined to be the number

(4.1.1.1) $$\dim X = \sup_x \deg.\mathrm{tr}_k\, k(x)$$

where x ranges over the set of maximal points of X.

We will see in §5 (5.2.2) that the definition (4.1.1) depends only in appearance on the base field k over which X is locally of finite type, and that the number $\dim(X)$ thus defined coincides with the dimension of the *underlying space* X (defined in ($\mathbf{0_{IV}}$, 14.1.2)). We obviously have $\dim X = \dim X_{\mathrm{red}}$.

We note that each $k(x)$ is a finitely generated extension of k (**I**, 6.3.3) and therefore has a finite transcendence degree over k. If X is of finite type over k and non-empty, it is Noetherian (**I**, 6.3.7), hence has only a finite number of irreducible components, and therefore $\dim(X)$ is finite and $\geqslant 0$; the definition (4.1.1) gives

$$\dim(\varnothing) = -\infty.$$

It is clear that, if we denote by (X_α) the family of reduced closed subpreschemes of X having as underlying spaces the irreducible components of X (**I**, 5.2.1), then

(4.1.1.2) $$\dim(X) = \sup_\alpha \dim(X_\alpha).$$

This reduces the calculation of the dimension to the case of *integral* preschemes (locally of finite type over k). Finally, we obviously have

(4.1.1.3) $$\dim(X) = \dim(U)$$

for every sub-prescheme induced on an everywhere dense open subset U of X; this finally reduces the notion of dimension to the case of an *affine scheme of finite type* over k.

Theorem (4.1.2). — *Let X, Y be two preschemes locally of finite type over a field k, $f : X \to Y$ a k-morphism.*

(i) *If f is quasi-compact and dominant, then* $\dim(Y) \leqslant \dim(X)$.
(ii) *If f is quasi-finite, then* $\dim(X) \leqslant \dim(Y)$.
(iii) *Suppose that X is of finite type over k. For* $\dim(X) \geqslant n$ (*resp.* $\dim(X) \leqslant n$, $\dim(X) = n$), *it is necessary and sufficient that there exist an open dense subset U of X, and a k-morphism $g : U \to \mathbf{V}(k^n)$ $(= \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n]))$, which we will also denote $\mathbf{V}_k^n$) which is surjective* (*resp. finite, resp. finite and surjective*). *If X is an affine scheme, one can take $U = X$.*

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Proof. (i) Let y be a maximal point of Y; we know that $f^{-1}(y)$ contains a maximal point x of X (1.1.5), hence $k(x)$ is an extension of $k(y)$, whence the inequality $\operatorname{deg.tr}_k k(y) \leqslant \operatorname{deg.tr}_k k(x) \leqslant \dim(X)$, which proves (i).

(ii) If x is a maximal point of X, $k(x)$ is a finite extension of $k(f(x))$ (**II**, 6.2.2), hence of the same transcendence degree over k. If one considers the reduced closed subprescheme of Y having $\overline{\{f(x)\}}$ as underlying space, one sees that one is reduced to proving the

Corollary (4.1.2.1). — *Let Y be a k-prescheme locally of finite type; for every subprescheme Z of Y,* $\dim(Z) \leqslant \dim(Y)$. *Suppose moreover that the irreducible components of Y all have the same dimension; then, for Z to be* rare *in Y, it is necessary and sufficient that* $\dim(Z) < \dim(Y)$.

Indeed, let z be a maximal point of Z; by considering one of the reduced closed subpreschemes of Y having as underlying space one of the irreducible components of Y containing z, one reduces to the case where Y is integral with generic point y, then, in considering within Y an affine open containing z, to the case where Y is affine and of finite type over k, say $Y = \operatorname{Spec}(A)$, and Z closed in Y, say $Z = \operatorname{Spec}(A/\mathfrak{a})$ where $\mathfrak{a}$ is an integral ideal of A. By virtue of the Noether normalisation lemma (Bourbaki, *Alg. comm.*, chap. V, §3, no. 1, th. 1) there exists a finite sequence $(x_i)_{1\leqslant i\leqslant n}$ of elements of A, algebraically independent over k, such that A is integral over the ring $B = k[x_1, \dots, x_n]$ and that $\mathfrak{a} \cap B$ is generated by a subfamily $(x_i)_{1\leqslant i\leqslant p}$ (possibly empty) of A. Let $g : Y \to \operatorname{Spec}(B) = \mathbf{V}_k^n$ be the finite dominant morphism corresponding to the injection canonical $B \to A$; since $C = B/(\mathfrak{a} \cap B)$ is isomorphic to $k[x_{p+1}, \dots, x_n]$, g induces on Z a dominant morphism $Z \to \mathbf{V}_k^{n-p}$; we have therefore $\operatorname{deg.tr}_k k(y) = n$ and $\operatorname{deg.tr}_k k(z) = n - p$, hence $\operatorname{deg.tr}_k k(z) \leqslant \operatorname{deg.tr}_k k(y) = \dim(Y)$, which demonstrates (i). Moreover, if $p = 0$, Z contains an open non-empty subset of Y, hence Z is not rare; if on the contrary $p > 0$, we have $\overline{\{z\}}$ is rare in $\overline{\{y\}}$, which finishes proving (4.1.2.1).

(iii) The proof that the conditions stated are sufficient follows from (i) and (ii). To prove that they are necessary, one can consider an open dense subset U of X, union of affine opens U_i pairwise disjoint and irreducible ($1 \leqslant i \leqslant m$), each containing one of the maximal points of X ($\mathbf{0_I}$, 2.1.6); one can moreover suppose X reduced and if X is affine, one can of course take $U = X$. The same reasoning shows that if $\dim(U_i) = n_i$, there exists a k-morphism $g_i : U_i \to \mathbf{V}_k^{n_i}$ which is finite and dominant, hence surjective (**II**, 6.1.10); for each n_i there is then a morphism $h_i : \mathbf{V}_k^{n_i} \to \mathbf{V}_k^n$ which is a closed immersion corresponding to the canonical homomorphism $k[\mathrm{T}_1, \dots, \mathrm{T}_n] \to k[\mathrm{T}_1, \dots, \mathrm{T}_{n_i}]$, and which is the identity for $n_i = n$. Since U is the sum of the U_i, one takes for g the morphism which IV-2 | 54
on each U_i coincides with $h_i \circ g_i$; it is obviously finite and surjective. When $n' \geqslant n$, one composes with the canonical closed immersion $\mathbf{V}_k^n \to \mathbf{V}_k^{n'}$; when $n' \leqslant n$, one composes with g the canonical morphism $p : \mathbf{V}_k^n \to \mathbf{V}_k^{n'}$ corresponding to the canonical injection $k[\mathrm{T}_1, \dots, \mathrm{T}_{n'}] \to k[\mathrm{T}_1, \dots, \mathrm{T}_n]$, noting that p is faithfully flat, hence surjective (**I**, 3.5.2) and (**II**, 6.1.5), whence the corollary. □

Remark (4.1.3). — The corollary (4.1.2.1) shows that in the formula (4.1.1.1), one can suppose that x ranges over the set of *all* points of X.

Corollary (4.1.4). — *Let X be a prescheme locally of finite type over a field k, K an extension of k; then* $\dim(X \otimes_k K) = \dim(X)$.

PROOF. One can restrict to the case where X is of finite type over k; then the morphism $u : X \otimes_k K \to X$ is faithfully flat (2.2.13), (i), hence if U is an open dense subset in X, $u^{-1}(U) = U \otimes_k K$ is dense in $X \otimes_k K$ (2.3.10); if $g : U \to \mathbf{V}_k^n$ is finite and surjective, it is the same for $g_{(K)} : U \otimes_k K \to \mathbf{V}_k^n$ (**I**, 3.5.2) and (**II**, 6.1.5), whence the corollary. ☐

Corollary (4.1.5). — *Let X and Y be two preschemes locally of finite type over a field k; then* $\dim(X \times_k Y) = \dim(X) + \dim(Y)$.

PROOF. It suffices to prove that if U (resp. V) is an affine neighbourhood of a point x (resp. y) of X (resp. Y) such that $\deg.\mathrm{tr}_k\, k(x) = m = \dim U$ and $\deg.\mathrm{tr}_k\, k(y) = n = \dim V$, then $\dim(U \times_k V) = m + n$, in other words, by virtue of (4.1.2) that there exist finite surjective k-morphisms $f : X \to \mathbf{V}_k^m$, $g : Y \to \mathbf{V}_k^n$; then $f \times g : X \times_k Y \to \mathbf{V}_k^{m+n}$ is finite and surjective (**I**, 3.5.2) and (**II**, 6.1.5), whence the corollary. ☐

4.2. Associated prime cycles on algebraic preschemes

Proposition (4.2.1). — *Let K and L be two extensions of a field k, such that $K \otimes_k L$ is Noetherian. Then the prime ideals associated to $K \otimes_k L$ are minimal, and if E is the residue field of the local ring of such an ideal, we have*

$$\deg.\mathrm{tr}_K E = \deg.\mathrm{tr}_k L, \qquad \deg.\mathrm{tr}_L E = \deg.\mathrm{tr}_k K \tag{4.2.1.1}$$

whence

$$\deg.\mathrm{tr}_k E = \deg.\mathrm{tr}_k K + \deg.\mathrm{tr}_k L. \tag{4.2.1.2}$$

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PROOF. We know that K is an algebraic extension of a purely transcendental extension $K' = k(\mathbf{t})$, where $\mathbf{t} = (t_\alpha)_{\alpha \in I}$ is a family of indeterminates; the ring $k[\mathbf{t}] \otimes_k L = L[\mathbf{t}]$ is integral, hence the same is true of $K' \otimes_k L$, which is a ring of fractions, and the field of fractions of $K' \otimes_k L$ is $L(\mathbf{t})$; we have the commutative diagram of canonical homomorphisms

$$\begin{tikzcd}
K \arrow[r] & K \otimes_k L \arrow[r] & K \otimes_{K'} L(\mathbf{t}) \\
K' \arrow[u] \arrow[r] & K' \otimes_k L \arrow[u] \arrow[r] & L(\mathbf{t}) \arrow[u] \\
& k \arrow[ul] \arrow[u] & L \arrow[ul] \arrow[u]
\end{tikzcd}$$

Since K is faithfully flat over K', $K \otimes_k L = K \otimes_{K'} (K' \otimes_k L)$ is faithfully flat over $K' \otimes_k L$ ($\mathbf{0_I}$, 6.5.2); moreover $K' \otimes_k L$ identifies with a subring of $K \otimes_k L$; the trace on $K' \otimes_k L$ of a prime ideal $\mathfrak{p}$ associated to $K \otimes_k L$ is the ideal 0, an element $\neq 0$ of $K' \otimes_k L$ not being a zero divisor in $K \otimes_k L$ ($\mathbf{0_I}$, 6.3.4 and Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 3 of prop. 2). Since moreover K is algebraic over K', $K \otimes_k L$ is an integral algebra over $K' \otimes_k L$, and the prime ideals of $K \otimes_k L$ inducing 0 on $K' \otimes_k L$ are necessarily without mutual inclusion relation (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, cor. 1 of prop. 1); this proves the first assertion (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 3, cor. 1 of prop. 7). Furthermore, the residue field E of $\mathfrak{p}$ is algebraic over the residue field of the ideal (0) of $K' \otimes_k L$, that is to say $L(\mathbf{t})$; hence $\deg.\mathrm{tr}_k E = \mathrm{Card}(I) = \deg.\mathrm{tr}_k K$, in other words we have the first relation (4.2.1.1); exchanging the roles of K and L, we have the second relation (4.2.1.1), whence (4.2.1.2). ☐

Corollary (4.2.2). — *Under the hypotheses of (3.3.6), if the preschemes $\mathrm{T}_{x,y}$ are locally Noetherian, they have no immersed associated prime cycle.*

Corollary (4.2.3). — *Under the hypotheses of (3.3.6) (resp. (3.3.7)), if the $\mathrm{T}_{x,y}$ are locally Noetherian and if $\mathscr{F}$ and the $\mathscr{G}_s$ (for $s \in \mathrm{S}$) (resp. $\mathscr{F}$ and $\mathscr{G}$) are without immersed associated prime cycle, the same is true of $\mathscr{F} \otimes_{\mathrm{S}} \mathscr{G}$.*

PROOF. This follows from (4.2.2), (3.3.2) and the proof of (3.3.6). In particular, since every prescheme over a field k is *flat* over k, we have thus proved assertion (i) of the following: □

Proposition (4.2.4). — *Let k be a field, X and Y two k-preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Suppose moreover X and Y integral. Then:*

(i) *$X \times_k Y$ is without immersed associated prime cycle; each of the irreducible components of $X \times_k Y$ dominates X and Y, and the set of these components is in bijective correspondence with the set of the irreducible components of $\mathrm{Spec}(\mathrm{R}(X) \otimes_k \mathrm{R}(Y))$ (in other words the set of the minimal prime ideals of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$), where $\mathrm{R}(X)$ and $\mathrm{R}(Y)$ are the fields of rational functions of X and Y respectively.*
(ii) *If a maximal point z of $X \times_k Y$ is identified with a minimal prime ideal $\mathfrak{p}$ of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, the local ring $\mathscr{O}_{X \times_k Y, z}$ is isomorphic to the ring of fractions $(\mathrm{R}(X) \otimes_k \mathrm{R}(Y))_{\mathfrak{p}}$. In particular, if $\mathrm{R}(X)$ or $\mathrm{R}(Y)$ is separable over k, $X \times_k Y$ is reduced.*
(iii) *If moreover X and Y are locally of finite type over k, every irreducible component of $X \times_k Y$ is of dimension $\dim(X) + \dim(Y)$.*

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PROOF. Assertion (iii) follows from (4.1.2), taking into account (i) and (ii). To prove (ii), one can obviously restrict to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$ are affine, A and B being integral rings with fields of fractions $K = \mathrm{R}(X)$ and $L = \mathrm{R}(Y)$; assertion (i) shows that every minimal prime ideal $\mathfrak{q}$ of $A \otimes_k B$ is the trace on $A \otimes_k B$ of a minimal prime ideal $\mathfrak{p}$ of $K \otimes_k L$; since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$, the isomorphism of the rings of fractions $(A \otimes_k B)_{\mathfrak{q}}$ and $(K \otimes_k L)_{\mathfrak{p}}$ follows from ($\mathbf{0_I}$, 1.2.5). Finally, if for example $\mathrm{R}(Y)$ is separable over k, we know that the ring $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$ is reduced (Bourbaki, *Alg.*, chap. VIII, §7, no. 3, th. 1); so the same is true of the local rings at the maximal points of $X \times_k Y$. One deduces that $X \times_k Y$ is reduced (3.2.1). □

Proposition (4.2.5). — *Let k be a field, X and Y two k-preschemes locally Noetherian, $\mathscr{F}$ (resp. $\mathscr{G}$) a quasi-coherent $\mathscr{O}_X$-Module (resp. $\mathscr{O}_Y$-Module). Let (Z'_λ) (resp. (Z''_μ)) be the family of the associated prime cycles of $\mathscr{F}$ (resp. $\mathscr{G}$), and denote also by Z'_λ (resp. Z''_μ) the reduced subprescheme of X (resp. Y) having Z'_λ (resp. Z''_μ) as underlying space. Then, if $Z'_\lambda \times_k Z''_\mu$ is locally Noetherian, the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of the distinct associated prime cycles of $\mathscr{F} \otimes_k \mathscr{G}$.*

PROOF. It suffices to apply (4.2.4) to the product $\mathrm{Spec}(k(x)) \times_k \mathrm{Spec}(k(y))$.
In particular: □

Corollary (4.2.6). — *Let k be a field, X, Y two k-preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Let (Z'_λ) (resp. (Z''_μ)) be the family of reduced subpreschemes having as underlying spaces the irreducible components of X (resp. Y). Then the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of the irreducible components of $X \times_k Y$.*

PROOF. Indeed, one can restrict to the case where X and Y are reduced (**I**, 5.1.8); the irreducible components of X (resp. Y) are then the associated prime cycles of $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$) (3.2.1). Applying (4.2.5) to $\mathscr{F} = \mathscr{O}_X$ and $\mathscr{G} = \mathscr{O}_Y$, and noting that by definition $\mathscr{O}_X \otimes_k \mathscr{O}_Y = \mathscr{O}_{X \times_k Y}$, the corollary follows from the fact that $\mathscr{O}_X$ and $\mathscr{O}_Y$ have no immersed associated prime cycles by hypothesis (4.2.3). □

Let us apply the preceding results to the case where Y is the spectrum of an extension K of k:

Proposition (4.2.7). — *Let k be a field, X a k-prescheme, K an extension of k such that $X \otimes_k K$ is locally Noetherian, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, x' a point of $X \otimes_k K$, x its image in X.*

(i) *Let (Z_λ) be the family of reduced subpreschemes of X having as underlying spaces the associated prime cycles of $\mathscr{F}$; then the irreducible components $Z_{\lambda\mu}$ of the $Z_\lambda \otimes_k K$ are the cycles premiers associated to $\mathscr{F} \otimes_k K$, and $Z_{\lambda\mu}$ dominates Z_λ; moreover, for $Z_{\lambda\mu}$ to be immersed, it is necessary and sufficient that Z_λ be so.*
(ii) *For x to belong to an immersed associated prime cycle of $\mathscr{F}$, it is necessary and sufficient that x' belong to an immersed associated prime cycle of $\mathscr{F} \otimes_k K$; for $\mathscr{F}$ to be without immersed associated prime cycle, it is necessary and sufficient that the same hold for $\mathscr{F} \otimes_k K$.*

(iii) *The indices λ such that $x \in Z_\lambda$ are the same as those for which there exists an index μ such that $x' \in Z_{\lambda\mu}$. In particular, if x' belongs to only one associated prime cycle of $\mathscr{F} \otimes_k K$, x belongs to only one associated prime cycle of $\mathscr{F}$.*

(iv) *If X is locally of finite type over k, we have $\dim(Z_{\lambda\mu}) = \dim(Z_\lambda)$.*

PROOF. One will note that the hypothesis implies that X itself is locally Noetherian (2.2.13) and (2.2.14); assertion (i) results from (4.2.5) and the proof of (4.2.3), for $\mathscr{G} = \mathscr{O}_Y$, with $Y = \operatorname{Spec}(K)$; (ii) and (iii) result from (i) and (2.3.5); finally, (iv) is a particular case of (4.2.4), (iii). □

Corollary (4.2.8). — *If X is locally of finite type over k, the set of dimensions of associated prime cycles is the same for $\mathscr{F}$ and $\mathscr{F} \otimes_k K$; the set of dimensions of the irreducible components is the same for X and $X \otimes_k K$.*

Proposition (4.2.9). — *Suppose the hypotheses of (4.2.5) are satisfied, and suppose moreover that $\mathscr{F}$ and $\mathscr{G}$ are coherent. Let $(\mathscr{F}_\lambda)_{\lambda \in \mathrm{L}}$ and $(\mathscr{G}_\mu)_{\mu \in \mathrm{M}}$ be irredundant decompositions of $\mathscr{F}$ and $\mathscr{G}$ respectively; for every couple $(\lambda, \mu) \in \mathrm{L} \times \mathrm{M}$, let $(\mathscr{H}_{\lambda\mu\nu})_{\nu \in \mathrm{S}(\lambda,\mu)} = \operatorname{Ass}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$ (3.2.5). Then $(\mathscr{H}_{\lambda\mu\nu})$ (for all triples (λ, μ, ν)) is an irredundant decomposition of $\mathscr{F} \otimes_k \mathscr{G}$; it is reduced if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are.*

PROOF. Note that $\mathscr{F} \otimes_k \mathscr{G}$ and each of the $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ are coherent, and each of the $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ identifies with a quotient of $\mathscr{F} \otimes_k \mathscr{G}$; each of the $\mathscr{H}_{\lambda\mu\nu}$ identifies by definition with a coherent quotient of $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$, hence also with a coherent quotient of $\mathscr{F} \otimes_k \mathscr{G}$. The family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite, for $\operatorname{Supp}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$ is contained in the underlying space of $\operatorname{Supp}(\mathscr{F}_\lambda) \times_k \operatorname{Supp}(\mathscr{G}_\mu)$ (where $\operatorname{Supp}(\mathscr{F}_\lambda)$ and $\operatorname{Supp}(\mathscr{G}_\mu)$ denote the corresponding reduced closed subpreschemes), and for λ and μ given, the family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite by hypothesis; our assertion therefore results from the fact that the families $(\operatorname{Supp}(\mathscr{F}_\lambda))$ and $(\operatorname{Supp}(\mathscr{G}_\mu))$ are locally finite. To show that $(\mathscr{H}_{\lambda\mu\nu})$ is an irredundant decomposition of $\mathscr{F} \otimes_k \mathscr{G}$, it suffices therefore to prove that the canonical homomorphism $\mathscr{F} \otimes_k \mathscr{G} \to \bigoplus_{\lambda,\mu,\nu} \mathscr{H}_{\lambda\mu\nu}$ is injective; but it is the composite of the homomorphisms $\mathscr{F} \otimes_k \mathscr{G} \to \bigoplus_{\lambda,\mu} (\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu) \to \bigoplus_{\lambda,\mu,\nu} \mathscr{H}_{\lambda\mu\nu}$; the second of these homomorphisms is injective by definition, and the same holds for the first, which is none other than the tensor product of the canonical injective homomorphisms $\mathscr{F} \to \bigoplus_\lambda \mathscr{F}_\lambda$, $\mathscr{G} \to \bigoplus_\mu \mathscr{G}_\mu$ (recall that $\mathscr{F}$ and $\mathscr{G}$ are flat over k). Finally, if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are reduced, one can suppose that $\mathrm{M} = \operatorname{Ass}(\mathscr{F})$ and $\mathrm{M} = \operatorname{Ass}(\mathscr{G})$; the fact that the $\mathscr{H}_{\lambda\mu\nu}$ then form a partition of $\operatorname{Ass}(\mathscr{F} \otimes \mathscr{G})$ follows from (4.2.2) (cf. (3.2.5)). □

Corollary (4.2.10). — *Under the hypotheses of (4.2.7), suppose moreover $\mathscr{F}$ coherent, and let $(\mathscr{F}_\lambda)_{\lambda \in \mathrm{L}}$ be an irredundant decomposition of $\mathscr{F}$; for every $\lambda \in \mathrm{L}$, let $(\mathscr{F}_{\lambda\mu})_{\mu \in \operatorname{Ass}(\mathscr{F}_\lambda \otimes_k K)}$ be a reduced irredundant decomposition of $\mathscr{F}_\lambda \otimes_k K$. Then $(\mathscr{F}_{\lambda\mu})$ is an irredundant decomposition of $\mathscr{F} \otimes_k K$; it is reduced if $(\mathscr{F}_\lambda)$ is.*

PROOF. One applies (4.2.9) with $Y = \operatorname{Spec}(K)$, $\mathscr{G} = \mathscr{O}_Y = K$, M reduced to a single element. □

4.3. Reminders on tensor products of fields For the convenience of the reader, we recall here certain properties of tensor products of fields which we will use in the following sections; for proofs, we refer to [**?**]. **IV-2** | 58

(4.3.1). Recall that an extension L of a field k is called *primary* if the largest algebraic separable extension of k contained in L is k itself.

Proposition (4.3.2). — *Let K, L be two extensions of a field k. If L is a primary extension of k, then $\operatorname{Spec}(L \otimes_k K)$ is irreducible, and if ξ is its generic point, $k(\xi)$ is a primary extension of K. Conversely, if for every finite separable extension K of k, $\operatorname{Spec}(L \otimes_k K)$ is irreducible, L is a primary extension of k.*

PROOF. See the proof in [**?**], p. 14-03 to 14-06. □

Corollary (4.3.3). — *If k is separably closed (i.e. if its algebraic closure is purely inseparable over k), then, for any two extensions K, L of k, $\operatorname{Spec}(L \otimes_k K)$ is irreducible, and conversely.*

PROOF. This follows immediately from (4.3.2). □

Corollary (4.3.4). — *Let L be an extension of a field k, L_s the largest algebraic separable extension of k contained in L (sometimes called the* separable algebraic closure *of k in L); suppose L_s of finite degree over k, and let K be a Galois extension of k containing L_s. Then $\operatorname{Spec}(L \otimes_k K)$ has $[L_s : k]$ irreducible components, the sum of whose residue fields at the generic points are the primary extensions of K.*

PROOF. Indeed, we have $L \otimes_k K = L \otimes_{L_s} (L_s \otimes_k K)$ and one knows that $L_s \otimes_k K$ is isomorphic to a product of $[L_s : k]$ fields isomorphic to K (Bourbaki, *Alg.*, chap. VIII, §7, no. 3, cor. 2 of th. 1); hence $L \otimes_k K$ is isomorphic to a product of $[L_s : k]$ rings isomorphic to $L \otimes_{L_s} K$; since L is a primary extension of L_s, the conclusion follows from (4.3.2). □

Proposition (4.3.5). — *Let K, L be two extensions of a field k. If L is a separable extension of k, the ring $L \otimes_k K$ is reduced. Conversely, if for every finite purely inseparable extension K of k, the ring $L \otimes_k K$ is reduced, L is a separable extension of k.*

PROOF. For the proof, see Bourbaki, *Alg.*, chap. VIII, §7, no. 3, th. 1. □

Corollary (4.3.6). — *If k is perfect, then for any two extensions K, L of k, $K \otimes_k L$ is reduced, and conversely.*

PROOF. This follows immediately from (4.3.5). □

Corollary (4.3.7). — *Let L be a separable extension of k and let K be an extension of k such that K or L is finite over k; then the residue fields of the semi-local ring $L \otimes_k K$ are separable extensions of K.*

PROOF. One knows indeed that $L \otimes_k K$ is a direct product of these residue fields E_i; for every extension K' of K, $L \otimes_k K'$ is therefore also isomorphic to the direct product of the rings $E_i \otimes_K K'$; since $L \otimes_k K'$ is reduced, the E_i are separable over K by virtue of (4.3.5). □

Corollary (4.3.8). — *If K and L are two finite separable extensions of k, $K \otimes_k L$ is a product of a finite number of separable extensions of k.*

Proposition (4.3.9). — *If k is algebraically closed, then, for any two extensions K, L of k, $K \otimes_k L$ is integral, and conversely.*

PROOF. This is a consequence of (4.3.3) and (4.3.6), a perfect and separably closed field being algebraically closed. □

4.4. Irreducible and connected preschemes over an algebraically closed field IV-2 | 59

(4.4.1). Let k be a field, X a k-prescheme, K an extension of k. Since the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ is faithfully flat, quasi-compact, and universally open (2.4.9), the same holds for the projection morphism $p : X \otimes_k K \to X$ (2.2.13). It therefore follows from (2.3.5) that every irreducible component of $X \otimes_k K$ *dominates* an irreducible component of X, and (since p is surjective) every irreducible component of X is dominated by an irreducible component of $X \otimes_k K$; one thus deduces from p a *surjective* application from the set of irreducible components of $X \otimes_k K$ to the set of irreducible components of X. In the same way, it follows from the fact that p is continuous and surjective that every connected component of X is the image by p of a union of connected components of $X \otimes_k K$, whence a surjective application from the set of connected components of $X \otimes_k K$ to the set of connected components of X. One concludes that every irreducible (resp. connected) component Z' of $X \otimes_k K$ is contained in a *unique* set of the form $p^{-1}(Z)$, where Z is an irreducible (resp. connected) component of X; for every subprescheme Z of X having Z as underlying space (and still denoted Z), Z' is an irreducible (resp. connected) component of $Z \otimes_k K$. In particular, if $X \otimes_k K$ has a finite number n (resp. n') of irreducible (resp. connected) components (which is the case when $X \otimes_k K$ is of *finite type* over K, for then $X \otimes_k K$ is of finite type over K, hence Noetherian), the number of irreducible (resp. connected) components of X is $\leqslant n$ (resp. $\leqslant n'$); to say that it is equal to n (resp. n') means that for every reduced subprescheme Z of X having as underlying space an irreducible (resp. connected) component of X, $Z \otimes_k K$ is irreducible (resp. connected) (**I**, 5.1.8); the number of irreducible (resp. connected) components of $X \otimes_k K$ is therefore independent of K if and only if for every irreducible (resp. connected) component Z of X, $Z \otimes_k K$ is irreducible (resp. connected) for *every* extension K of k.

In particular, if $X \otimes_k K$ is irreducible (resp. connected), the same holds for X (which already follows from the fact that p is surjective). In the same way, if $X \otimes_k K$ is *reduced* (resp. *integral*), the same holds for X since p is faithfully flat (2.1.13).

In this section and the following, we shall examine more closely what one can say about the irreducible (resp. connected) components of $X \otimes_k K$ when K varies; what precedes already shows that the number of these components is a *increasing* function of K.

Lemma (4.4.2). — *Let X', X be two topological spaces, $f : X' \to X$ a continuous map. Suppose the following conditions are satisfied:*

(i) *f is surjective and open (resp. such that the topological space X identifies canonically with the quotient of X' by the equivalence relation defined by f).*

(ii) *For every $x \in X$, $f^{-1}(x)$ is irreducible (resp. connected).*

Then for X' to be irreducible (resp. connected), it is necessary and sufficient that X be so.

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PROOF. The assertion relative to connectedness is none other than Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., §11, no. 3, prop. 7. Let us prove the assertion relative to irreducibility; the condition being trivially necessary since f is surjective, let us show it is sufficient. Let X'_1, X'_2 be two closed parts of X' such that $X' = X'_1 \cup X'_2$, and let X_i $(i = 1, 2)$ denote the set of $x \in X$ such that $f^{-1}(x) \subset X'_i$; since $X_i = X - f(X' - X'_i)$, and since f is open, we see that X_i is a *closed* part of X $(i = 1, 2)$. On the other hand, for every $x \in X$, $f^{-1}(x)$ is irreducible by hypothesis, and is the union of the two closed parts $X'_1 \cap f^{-1}(x)$ and $X'_2 \cap f^{-1}(x)$; one of these two parts must therefore be equal to $f^{-1}(x)$, which means that $x \in X_1$ or $x \in X_2$; hence $X = X_1 \cup X_2$, and since X is irreducible, we deduce $X = X_1$ or $X = X_2$, and therefore $X' = X'_1$ or $X' = X'_2$. □

Remark (4.4.3). — If the topology of X is not the quotient of that of X' by the equivalence relation defined by f, it can happen that X and all the fibres $f^{-1}(x)$ are irreducible, without X' being connected: one obtains an example by taking for X an irreducible scheme (for example the affine line $\mathrm{Spec}(k[\mathrm{T}])$) over a field k algebraically closed, considering a closed point x_0 of X and taking for X' the space that is the *sum* of $\{x_0\}$ and of the open subspace $X - \{x_0\}$ of X. The same if one replaces the hypothesis "f open" by "f closed" (and even "f proper") (which still implies that the topology of X is the quotient of that of X' by the relation defined by f), it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being so. Let us take for X the affine line over k, denote by S the product $X \times_k \mathbf{P}$, where $\mathbf{P}$ is the projective line over k; if x_0 is a closed point of X, t_0 a closed point of $\mathbf{P}$, $p : \mathrm{S} \to X$, $q : \mathrm{S} \to \mathbf{P}$ the projections, we denote by X' the reduced subprescheme having as underlying space the closed set $p^{-1}(x_0) \cup q^{-1}(t_0)$, and we take for f the restriction of p to X'; f is proper but X' is not irreducible.

Theorem (4.4.4). — *Let k be an algebraically closed field, X a k-prescheme. If X is irreducible (resp. connected), then $X \otimes_k K$ is also irreducible (resp. connected) for every extension K of k.*

PROOF. One saw in (4.4.1) that the morphism $p : X \otimes_k K \to X$ is faithfully flat and open; to be able to apply the lemma (4.4.2), it suffices therefore to verify that the fibres $p^{-1}(x)$ are irreducible for every $x \in X$. But the $k(x)$-prescheme $p^{-1}(x)$ is isomorphic to $\mathrm{Spec}(k(x) \otimes_k K)$ (**I**, 3.6.2), hence is integral by virtue of the hypothesis on k and of (4.3.9). □

Corollary (4.4.5). — *Let k be an algebraically closed field, X a k-prescheme, K an extension of k, $p : X \otimes_k K \to X$ the canonical projection. If Z is an irreducible (resp. connected) part of X, $p^{-1}(Z)$ is irreducible (resp. connected); in particular, if X_0 is an irreducible (resp. connected) component of X containing Z, $p^{-1}(X_0)$ is an irreducible (resp. connected) component of $X \otimes_k K$ containing $p^{-1}(Z)$.*

PROOF. The second assertion results from (4.4.1) and the first applied by replacing Z by X_0; to prove the first, let X' denote the sub-prescheme of $X \otimes_k K$ having X_0 as underlying space, $Z' = p^{-1}(Z)$; the equivalence relation R defined by p on X' is open, hence the same holds for the one it induces on the saturated part Z', and Z identifies with the quotient space $Z'/\mathrm{R}_{Z'}$ (Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., §5, no. 2, prop. 4); one can therefore apply the lemma (4.4.2) to Z and Z', whence the first assertion from (4.4.4). □

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Corollary (4.4.6). — *Under the hypotheses and with the notation of (4.4.5), the map $Z \mapsto p^{-1}(Z)$ is a bijection from the set of irreducible (resp. connected) components of X to the set of irreducible (resp. connected) components of $X \otimes_k K$, whose inverse bijection is $Z' \mapsto p(Z')$.*

4.5. Geometrically irreducible and geometrically connected preschemes

Proposition (4.5.1). — *Let k be a field, X a k-prescheme, Ω an algebraically closed extension of k. Let n (resp. n') be the number of irreducible (resp. connected) components of $X \otimes_k \Omega$; this number is independent of the algebraically closed extension Ω chosen. Moreover, for every extension K of k, the number of irreducible (resp. connected) components of $X \otimes_k K$ is $\leqslant n$ (resp. $\leqslant n'$).*

Proof. Indeed, two algebraically closed extensions Ω, Ω' of k can always be considered as sub-extensions of a same extension E of k; it suffices then to apply (4.4.6) to $X \otimes_k \Omega$ and $X \otimes_k \Omega'$ and to their common extension E of Ω and Ω' to show the first assertion. The second is obtained by taking for Ω an algebraically closed extension of K and using (4.4.1). □

Definition (4.5.2). — The number n (resp. n') of (4.5.1) is called the *geometric number of irreducible* (resp. *connected*) *components* of X (relative to k). If $n = 1$ (resp. $n' \leqslant 1$), one says that X is a k-prescheme *geometrically irreducible* (resp. *geometrically connected*).

One thus recovers the definition given for a particular case in (**III**, 4.3.4). By virtue of (4.5.1), the following properties are equivalent:

a) X is geometrically irreducible (resp. geometrically connected).
b) For an algebraically closed extension Ω of k, $X \otimes_k \Omega$ is irreducible (resp. connected).
c) For every extension K of k, $X \otimes_k K$ is irreducible (resp. connected).

(4.5.3). Let X be a k-prescheme, Z a locally closed part of X. If we still denote by Z any of the subpreschemes of X having Z as underlying space, it results from (**I**, 5.1.8) that for an extension K of k, the number of irreducible (resp. connected) components of $Z \otimes_k K$ is independent of the subprescheme with underlying space Z that one has chosen; one thus also defines the geometric number of irreducible (resp. connected) components of Z as that of any subprescheme of X having Z as underlying space.

Proposition (4.5.4). — *Let X, Y be two k-preschemes, $f : X \to Y$ a k-morphism surjective irreducible (resp. geometrically connected); it is the same for Y.*

Proof. Indeed, for every extension K of k, $f_{(K)} : X_{(K)} \to Y_{(K)}$ is surjective (**I**, 3.5.2), and $X_{(K)}$ being irreducible (resp. connected) by hypothesis, the same holds for $Y_{(K)}$. □

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Definition (4.5.5). — One says that a morphism $f : X \to Y$ of preschemes is *irreducible* (resp. *connected*), if, for every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is geometrically irreducible (resp. geometrically connected).

Proposition (4.5.6). — (i) *Let X be a k-prescheme, K an extension of k. Then the geometric number of irreducible (resp. connected) components of $X_{(K)}$ relative to K is equal to the geometric number of irreducible (resp. connected) components of X relative to k. In particular, for the K-prescheme $X_{(K)}$ to be geometrically irreducible (resp. geometrically connected), it is necessary and sufficient that the k-prescheme X be so.*

(ii) *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, and set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If f is irreducible (resp. connected), the same holds for f'. The converse is true if g is surjective.*

Proof. (i) If Ω is an algebraically closed extension of K, we have $X \otimes_k \Omega = X_{(K)} \otimes_K \Omega$, whence the assertion.

(ii) For every $y' \in Y'$, if $y = g(y')$, we have $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (**I**, 3.6.4), hence the two assertions result from (i). □

Proposition (4.5.7). — *Let $f : X \to Y$ be a surjective irreducible (resp. connected) morphism, $g : Y' \to Y$ a morphism, and set $X' = X \times_Y Y'$. If moreover Y' is irreducible (resp. connected), and f universally open (resp. flat and quasi-compact, or universally open, or universally closed), then X' is irreducible (resp. connected).*

Proof. The properties of f that one considers being all stable by base change, one can reduce to the case where $Y' = Y$; it then suffices to apply (4.4.2) (taking into account (2.3.12)). □

Corollary (4.5.8). — *Let X, Y be two k-preschemes.*

(i) *If X is geometrically irreducible (resp. geometrically connected) and Y is irreducible (resp. connected), then $X \times_k Y$ is irreducible (resp. connected).*
(ii) *If X and Y are geometrically irreducible (resp. geometrically connected), the same holds for $X \times_k Y$.*

PROOF. (i) The structural morphism $X \to \operatorname{Spec}(k)$ is surjective and universally open (2.4.9) and in addition irreducible (resp. connected); it suffices therefore to apply (4.5.7).

(ii) If Ω is an algebraically closed extension of k, we have $(X \times_k Y)_{(\Omega)} = X_{(\Omega)} \times_\Omega Y_{(\Omega)}$ (**III**, 3.3.10); by hypothesis $X_{(\Omega)}$ and $Y_{(\Omega)}$ are geometrically irreducible (resp. geometrically connected) Ω-preschemes (4.5.6), and it suffices therefore to apply (i). □

Proposition (4.5.9). — *Let X be a k-prescheme. The following conditions are equivalent:*

a) *X is geometrically irreducible (that is to say, for every extension K of k, $X \otimes_k K$ is irreducible).*
b) *For every finite separable extension K of k, $X \otimes_k K$ is irreducible.*
c) *X is irreducible, and if x is its generic point, $k(x)$ is a primary extension of k.*

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PROOF. It is clear that a) implies b). To see that b) implies c), consider a finite separable extension K of k; if $p : X \otimes_k K \to X$ is the canonical projection, the maximal points of $X \otimes_k K$ are those of the fibre $p^{-1}(x) = \operatorname{Spec}(k(x) \otimes_k K)$ (2.3.4) and ($\mathbf{0_I}$, 2.1.8); to say that this fibre is irreducible for every finite separable extension K of k amounts to saying that $k(y)$ is a primary extension of k, by virtue of (4.3.2). Conversely, if c) is satisfied, the same reasoning (taking into account (4.3.2)) shows that for every *extension* K of k, $X \otimes_k K$ is irreducible, hence c) implies a). □

Corollary (4.5.10). — *Let X be a k-prescheme irreducible, x its generic point, k' the algebraic separable closure of k in $k(x)$, k'' a Galois extension of k (of finite degree or not) containing k'. Suppose k' of finite degree over k; then the irreducible components of $X \otimes_k k''$ are geometrically irreducible; their number, equal to $[k' : k]$ and is also the geometric number of irreducible components of X.*

PROOF. As in (4.5.9), one is reduced to considering the maximal points of the fibre $\operatorname{Spec}(k(x) \otimes_k k'')$; by virtue of (4.3.4), they are equal in number to $[k' : k]$ and the residue fields at these points are primary extensions of k'', which, taking into account (4.5.9), proves the corollary. □

Corollary (4.5.11). — *Let X be a k-prescheme. Suppose that X has only a finite number of maximal points x_i ($1 \leqslant i \leqslant r$) and that for each i, the algebraic separable closure k'_i of k in $k(x_i)$ is a finite extension of k. Then there exists a finite separable extension L of k such that the irreducible components of $X \otimes_k L$ are geometrically irreducible; their number, equal to $\sum_i [k'_i : k]$, is the geometric number of irreducible components of X.*

PROOF. The maximal points of $X \otimes_k L$ are those of the various fibres $\operatorname{Spec}(k(x_i) \otimes_k L)$ (2.3.4) and ($\mathbf{0_I}$, 2.1.8), applied to the irreducible components of X and to their inverse images in $X \otimes_k L$. It suffices therefore to take for L a Galois extension finite of k containing all the k'_i and to apply (4.5.10). □

One will note that (4.5.11) applies in particular to every k-prescheme X of *finite type* over k, for X is then Noetherian, hence has only a finite number of irreducible components, and each of the $k(x_i)$ is a finite type extension of k, hence the same holds for the algebraic separable closure of k in $k(x_i)$ ([**?**], p. 6-06, lemma 5).

Remarks (4.5.12). — (i) The notions defined in (4.5.2) depend on the base field k. For short, and whenever it is necessary to mention the base field, one says "k-irreducible" (resp. "k-connected") instead of "geometrically irreducible (resp. connected) relative to k". One will note that this terminology, quite natural in the context of schemes, is exactly the *opposite* of that of Weil: for this author, "k-irreducible" (resp. "k-connected", "k-normal", etc.) denotes an *intrinsic* property of a prescheme X, independent of the choice of the field k taken as "base field" (in other words, of the morphism $X \to \operatorname{Spec}(k)$ chosen); we express these properties by saying simply that X is irreducible (resp. connected, normal, etc.). Weil says on the other hand "absolutely irreducible", "absolutely connected", "absolutely normal", etc., for the corresponding notions relative to the prescheme $X \otimes_k \Omega$, where Ω is an algebraically closed suitable extension of k. We will avoid in what follows using the qualifier "absolute", which may lend to confusion, and whenever we use the abbreviated terminology introduced above (in conflict with the accepted terminology, see above), we will refer to (4.5.12) to avoid all ambiguity.

(ii) The proposition (4.5.9) gives a "birational" criterion (that is, depending only on the residue field at the generic point) for a k-prescheme X to be geometrically irreducible. There is no analogous criterion for X to be geometrically connected.

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Proposition (4.5.13). — *Let X, Y be two k-preschemes, $f : Y \to X$ a k-morphism. Suppose Y geometrically connected and non-empty, and X connected. Then X is geometrically connected.*

PROOF. Let Ω be an algebraic closure of k, $X' = X_{(\Omega)}$, $Y' = Y_{(\Omega)}$, $f' = f_{(\Omega)} : Y' \to X'$; note that the morphisms $p : X' \to X$ and $q : Y' \to Y$ are *open* (2.4.10) and *closed* since Ω is an algebraic extension of k (**II**, 6.1.10). Hence $U = p(\mathrm{U}')$ is open and closed, non-empty in X, and hence is equal to X. Since Y is non-empty, one concludes (**I**, 3.4.7) that $\mathrm{V}' = f'^{-1}(\mathrm{U}')$ is non-empty; furthermore, it is an open and closed part of Y', and since this last space is connected by hypothesis, $\mathrm{V}' = Y'$. We deduce that $\mathrm{U}' = X'$, without which the open and closed set $X' - \mathrm{U}'$ would show, applying the same reasoning to $X' - \mathrm{U}'$, that $f'^{-1}(X' - \mathrm{U}') = Y'$, which is absurd since Y' is non-empty. □

Corollary (4.5.13.1). — (i) *Let X, Y be two k-preschemes, $f : Y \to X$ a k-morphism. If Y is geometrically connected and non-empty, the connected component X_0 of X containing $f(Y)$ is geometrically connected.*

(ii) *Let X be a k-prescheme. If Y is an irreducible component of X which is geometrically irreducible, the connected component X_0 of X containing Y is geometrically connected.*

PROOF. (i) One can reduce to the case where Y is reduced, so that f factorizes as $f : Y \xrightarrow{g} X_0 \xrightarrow{j} X$, denoting X_0 as the reduced subprescheme having X_0 as underlying space (**I**, 5.2.2); it suffices to apply (4.5.13) to g.

(ii) Considering a prescheme having Y as underlying space, and remarking that Y is *a fortiori* geometrically connected, it suffices to apply (i) to the canonical injection $Y \to X$. □

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Corollary (4.5.14). — *Let X be a k-prescheme. If x is a point of X such that $k(x)$ is a primary extension of k (in particular if x is a rational point of X), the connected component of x in X is geometrically connected.*

PROOF. It suffices to apply (4.5.13.1) to $Y = \mathrm{Spec}(k(x))$ and to the canonical morphism $Y \to X$, taking into account that Y is geometrically irreducible by (4.5.9). □

Proposition (4.5.15). — *Let X be a k-prescheme, x a point of X, k' the algebraic separable closure of k in $k(x)$. Suppose:*

(i) *X is connected.*
(ii) *k' is a finite extension of k.*

Then the geometric number of connected components of X is $\leqslant [k' : k]$, and if k'' is a Galois extension finite of k containing k', the connected components of $X \otimes_k k''$ are geometrically connected.

PROOF. There exist finite Galois extensions k'' of k containing k' by virtue of condition (ii). Set $X'' = X \otimes_k k''$; the projection morphism $p : X'' \to X$ is faithfully flat and finite, hence closed (**II**, 6.1.10) and it follows (2.3.6), (ii)) the image by p of every connected component X''_α of X'' is equal to X; X''_α therefore contains a point $x''_\alpha \in p^{-1}(x)$. But $p^{-1}(x)$ has a number of points equal to the number of irreducible components of $\mathrm{Spec}(k(x) \otimes_k k'')$ (**I**, 3.4.9), that is $[k' : k]$ (4.3.4), and *a fortiori* the number of connected components of X'' is $\leqslant [k' : k]$. For every $x''_\alpha \in p^{-1}(x)$, $k(x''_\alpha)$ is a primary extension of k'' by (4.3.5) and (**I**, 3.4.9). One can therefore apply the corollary (4.5.14) to x''_α and X'', which proves that all the connected components of X'' are geometrically connected. □

Corollary (4.5.16). — *Let X be a k-prescheme, (X_α) the family of its connected components, and for every α, let $x_\alpha \in X_\alpha$. Suppose:*

(i) *The family (X_α) is finite.*
(ii) *For every α, the algebraic separable closure k'_α of k in $k(x_\alpha)$ is a finite extension of k.*

Then the geometric number of connected components of X is at most $\sum_\alpha [k'_\alpha : k]$, and there exists a finite separable extension k'' of k such that all the connected components of $X \otimes_k k''$ are geometrically connected.

PROOF. Noting that the connected components of X are open by virtue of condition (i), it suffices to apply to each of the subpreschemes of X induced on the open sets X_α the result of (4.5.15); for an extension k'' answering the question, it suffices to take a Galois extension finite of k containing all the k'_α. □

Corollary (4.5.17). — *Suppose that the k-prescheme X contains a point x such that the algebraic separable closure of k in $k(x)$ is finite over k. For X to be geometrically connected, it is necessary and sufficient that for every finite separable extension K of k, $X \otimes_k K$ be connected.*

Remark (4.5.18). — We shall see in §8 (4.8.5) that the conclusion of (4.5.17) is still valid if, instead of supposing verified the condition of the statement, one supposes that X is *quasi-compact*.

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Proposition (4.5.19). — *Let X be a k-prescheme, k' an algebraically closed extension of k, $X' = X \otimes_k k'$, $p : X' \to X$ the canonical projection. Suppose $p^{-1}(Z)$ is contained in a single irreducible component X'_0 of X'. Then $a = X_0 = p(X'_0)$ is an irreducible component of X containing Z, and moreover X_0 is geometrically irreducible. If moreover X'_0 is the only irreducible component of X' that meets Z', then X_0 is the only irreducible component of X that meets Z.*

PROOF. To show that $X'_0 = p^{-1}(X_0)$, we will apply the following: □

Lemma (4.5.19.1). — *Let T, T' be two preschemes, $p : \mathrm{T}' \to \mathrm{T}$ a morphism; consider the prescheme $\mathrm{T}'' = \mathrm{T}' \times_{\mathrm{T}} \mathrm{T}'$, and let $p_1 : \mathrm{T}'' \to \mathrm{T}'$, $p_2 : \mathrm{T}'' \to \mathrm{T}'$ be the canonical projections. For an open subset U' of T' to be of the form $p^{-1}(\mathrm{U})$, where $\mathrm{U} \subset \mathrm{T}$, it is necessary and sufficient that $p_1^{-1}(\mathrm{U}') = p_2^{-1}(\mathrm{U}')$.*

PROOF. The structural morphism $q : \mathrm{T}'' \to \mathrm{T}$ is equal to $p \circ p_1$ and to $p \circ p_2$ by definition, hence the relation is necessary. Conversely, suppose it is verified; if u' is a point of U', t a point of $p^{-1}(p(u'))$; since $p(t') = p(u')$, there exists a point $t'' \in \mathrm{T}''$ such that $p_1(t'') = u'$, $p_2(t'') = t'$ (I, 3.4.7); hence $t'' \in p_1^{-1}(\mathrm{U}') = p_2^{-1}(\mathrm{U}')$ by hypothesis, hence $t' = p_2(t'') \in \mathrm{U}'$, which proves the lemma. □

This lemma being established, to show that $X'_0 = p^{-1}(X_0)$, let us form the product $X'' = X' \times_X X'$ (relative to the morphism $p : X' \to X$); if p_1 and p_2 are the canonical projections, it suffices to show that $p_1^{-1}(X'_0) = p_2^{-1}(X'_0)$. Now, if we set $\mathrm{S} = \mathrm{Spec}(k)$, $\mathrm{S}' = \mathrm{Spec}(k')$, observe that if $\mathrm{S}'' = \mathrm{S}' \times_{\mathrm{S}} \mathrm{S}' = \mathrm{Spec}(k' \otimes_k k')$, we can also write $X'' = X' \times_{\mathrm{S}'} \mathrm{S}''$. If $\varphi'' : \mathrm{X}'' \to \mathrm{S}''$ is the structural morphism, it suffices to prove that for every $s'' \in \mathrm{S}''$, $p_1^{-1}(X'_0) \cap \varphi''^{-1}(s'') = p_2^{-1}(X'_0) \cap \varphi''^{-1}(s'')$. If one sets $\mathrm{T} = \mathrm{Spec}(k(s''))$, $\mathrm{U} = X'' \times_{\mathrm{S}''} \mathrm{T}$, and if $v : U \to X''$ is the canonical projection, and since we have the diagram

$$\begin{tikzcd}
X & X' \arrow[l, "p_1"'] & X'' \arrow[l, "v"'] & \mathrm{U} \arrow[l] \\
\mathrm{S} \arrow[u] & \mathrm{S}' \arrow[l] \arrow[u] & \mathrm{S}'' \arrow[l] \arrow[u] & \mathrm{T} \arrow[l] \arrow[u]
\end{tikzcd}$$

By construction, U_1 and U_2 contain both the set $w^{-1}(Z)$, where $w = p \circ p_1 \circ v$, and the two irreducible components of X' containing Z'; they are therefore all equal to X'_0 by (4.4.5), and finally $\mathrm{U}_1 = \mathrm{U}_2$.

Since p is surjective and X'_0 dominates an irreducible component of X, $X_0 = p(X'_0)$ is an irreducible component of X containing Z, and since $p^{-1}(X_0)$ is an irreducible component of $X \otimes_k K$, X_0 is geometrically irreducible.

To prove the last assertion, let X_1 be a second irreducible component of X meeting Z. Taking $z \in Z \cap X_1$, one sees that there exists an irreducible component of X' meeting Z' and dominating X_1 (2.3.5); but since X'_0 is by hypothesis the only irreducible component of X' meeting Z', we have necessarily $X_1 = X_0$.

Remark (4.5.20). — The fact that Z' is contained in a single irreducible component X'_0 of X' does *not imply* that $X_0 = p(X'_0)$ is the only irreducible component of X containing Z.

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Proposition (4.5.21). — *Let k be a separably closed field, and let k' be an algebraic closure of k. For every k-prescheme X, the canonical projection $X \otimes_k k' \to X$ is a universal homeomorphism. In particular the irreducible (resp. connected) components of X are geometrically irreducible (resp. geometrically connected).*

PROOF. By definition, k' is a purely inseparable extension of k; the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is integral, surjective, and radical, hence the first assertion results from (2.4.5), (i)); the second follows from (4.5.1). □

4.6. Geometrically reduced algebraic preschemes

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Proposition (4.6.1). — *Let k be a field, X a k-prescheme, Ω a perfect extension of k. The following conditions are equivalent:*

a) *For every reduced k-prescheme S, $X \times_k \mathrm{S}$ is reduced.*
b) *For every extension K of k, $X \otimes_k K$ is reduced.*
c) *$X \otimes_k \Omega$ is reduced.*
d) *For every finite purely inseparable extension k' of k, $X \otimes_k k'$ is reduced.*
e) *X is reduced, and for every irreducible component X_α of X, with generic point x_α, $k(x_\alpha)$ is a separable extension of k.*

PROOF. It is trivial that $a) \Rightarrow b) \Rightarrow c)$; since k' can be considered as a sub-extension of Ω, we saw (4.4.1) that $c)$ implies $d)$. Let us show that $d)$ implies $e)$. Taking $k' = k$, one sees first that X is reduced; to prove that $k(x_\alpha)$ is separable over k, it suffices to show that $k(x_\alpha) \otimes_k k'$ is a reduced ring for every finite purely inseparable extension k' of k (4.3.5). One can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, A being a reduced k-algebra; then by hypothesis the ring $A \otimes_k k'$ is reduced (**I**, 5.1.4) and $k(x_\alpha) \otimes_k k'$ is a ring of fractions of this ring, hence is reduced ($\mathbf{0_I}$, 1.2.8). Finally, to prove that $e)$ implies $a)$, one can restrict to the case where $X = \mathrm{Spec}(A)$, $\mathrm{S} = \mathrm{Spec}(B)$ are affine, A and B being k-algebras. The $k(x_\alpha)$ are the fields of fractions of the quotient rings $A/\mathfrak{p}_\alpha$, where $\mathfrak{p}_\alpha = \mathfrak{i}_{x_\alpha}$ are the minimal prime ideals of A; since by hypothesis $\bigcap_\alpha \mathfrak{p}_\alpha = 0$, A is contained in the product $\prod A/\mathfrak{p}_\alpha$, hence in $\prod k(x_\alpha)$; one can similarly consider B as a subalgebra of a product $\prod L_\beta$ of extensions of k. It follows that $A \otimes_k B$ identifies with a subalgebra of $(\prod_\alpha k(x_\alpha)) \otimes_k (\prod_\beta L_\beta)$, and this tensor product itself identifies with a subalgebra of $\prod_{\alpha,\beta}(k(x_\alpha) \otimes_k L_\beta)$ (Bourbaki, *Alg.*, chap. II, 3e éd., §7, no. 7, prop. 15). Now by hypothesis the $k(x_\alpha) \otimes_k L_\beta$ are reduced (4.3.5), hence the same holds for $A \otimes_k B$. □

Definition (4.6.2). — When the equivalent conditions a) to e) of the proposition (4.6.1) are satisfied, one says that X is *geometrically reduced* (or *separable*, or *universally reduced*) over k. One says that a k-prescheme X is *geometrically integral* over k if, for every extension K of k, $X \otimes_k K$ is integral; this amounts (by virtue of (4.6.1)) to saying that X is separable and geometrically irreducible over k.

One says that a k-algebra A (commutative) is *separable* if $\mathrm{Spec}(A)$ is separable over k; this means therefore that for every extension K of k, the ring $A_{(K)} = A \otimes_k K$ is reduced. One will note that this definition coincides with that of Bourbaki, *Alg.*, chap. VIII, §7, no. 5, def. 1, when A is a ring of finite rank over k (loc. cit., cor. of prop. 7), but not in general, a k-algebra being able to have zero radical $\neq 0$ even if it is integral.

Corollary (4.6.3). — *Let X be a k-prescheme integral; for X to be geometrically reduced (resp. geometrically integral) over k, it is necessary and sufficient that its field of rational functions $\mathrm{R}(X)$ be a separable (resp. separable and primary) extension of k.*

PROOF. This follows immediately from (4.5.9) and (4.6.1). □

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Corollary (4.6.4). — *Let X be a k-prescheme reduced. Then, for every separable extension k' of k, $X' = X \otimes_k k'$ is reduced.*

PROOF. It suffices to apply the equivalence of $a)$ and $e)$ in (4.6.1), replacing X by $\mathrm{Spec}(k')$ and S by X. □

Proposition (4.6.5). — (i) *Let X be a k-prescheme, K an extension of k. For X to be geometrically reduced (resp. geometrically integral) over k, it is necessary and sufficient that $X \otimes_k K$ be geometrically reduced (resp. geometrically integral) over K.*

(ii) *Let X, Y be two k-preschemes. If X and Y are geometrically reduced (resp. geometrically integral) over k, the same holds for $X \times_k Y$.*

PROOF. Assertion (i) is a trivial consequence of the definitions and of (4.4.1). For the part of assertion (ii) concerning separability, observe that if Ω is an algebraically closed extension of k, $(X \times_k Y) \otimes_k \Omega = X \times_k (Y \otimes_k \Omega)$; since $Y \otimes_k \Omega$ is reduced, the same holds for $X \times_k (Y \otimes_k \Omega)$ by virtue of (4.6.1), a)). The rest of assertion (ii) results from what precedes and from (4.5.8), (ii). □

Proposition (4.6.6). — *Let X be a k-prescheme of finite type. There exists a finite purely inseparable extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k'.*

PROOF. Since X can be covered by a finite number of open affines U_i, one can restrict to showing that for each i, there is a finite purely inseparable extension k_i of k such that $(U_i \otimes_k k_i)_{\mathrm{red}}$ is geometrically reduced over k_i; one can then suppose that the k_i are contained in a same finite purely inseparable extension k' of k; $(U_i \otimes_k k')_{\mathrm{red}}$ identifies with $(U_i \otimes_k k_i)_{\mathrm{red}} \otimes_{k_i} k'$ by virtue of (**I**, 5.1.8), hence is geometrically reduced over k' by hypothesis, and the same holds for $(X \otimes_k k')_{\mathrm{red}}$. Suppose therefore $X = \operatorname{Spec}(A)$, where A is a k-algebra of finite type, and let Ω be the algebraic closure of k. Denote by $\mathfrak{N}'$ the nilradical of $A \otimes_k \Omega$; since $A \otimes_k \Omega$ is a Noetherian ring, $\mathfrak{N}'$ is an ideal generated by a finite number of elements of the form $y_i = \sum_j x_{ij} \otimes \xi_{ij}$ where $x_{ij} \in A$, $\xi_{ij} \in \Omega$. Let K be a finite sub-extension of Ω containing the ξ_{ij}, and $\mathfrak{N}$ the ideal of $A \otimes_k K$ generated by the y_i ($A \otimes_k K$ being identified with a subring of $A \otimes_k \Omega$); it is clear that the y_i are nilpotent in $A \otimes_k K$; on the other hand, since $\mathfrak{N} \otimes_K \Omega = \mathfrak{N}'$ (and hence $\mathfrak{N} \cap (A \otimes_k K) = \mathfrak{N}$), $\mathfrak{N}$ contains the nilradical of $A \otimes_k K$, hence it is equal to this nilradical. Since $(A \otimes_k \Omega)/\mathfrak{N}' = ((A \otimes_k K)/\mathfrak{N}) \otimes_K \Omega$, one sees that $(X \otimes_k K)_{\mathrm{red}} \otimes_K \Omega$ is reduced, and by (4.6.1), $(X_{(K)})_{\mathrm{red}}$ is geometrically reduced over K. Replacing K by the quasi-Galois extension over k that it generates in Ω, one sees by (**I**, 5.1.8) that one can suppose moreover K quasi-Galois over k, hence a separable extension of a finite purely inseparable extension k' of k (Bourbaki, *Alg.*, chap. V, §10, no. 9, prop. 14). By virtue of (**I**, 5.1.8), $(X_{(k')})_{\mathrm{red}} \otimes_{k'} \overline{k}$ is isomorphic to $(X_{(K)})_{\mathrm{red}}$, hence is geometrically reduced; the same holds for $(X_{(k')})_{\mathrm{red}} \otimes_{k'} \overline{k} \cdot K$ by (4.6.5), (i); replacing K again we are done. □

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Corollary (4.6.7). — *If K is an extension of finite type of k, there exists a finite purely inseparable extension k' of k such that the residue fields of the semi-local ring $K \otimes_k k'$ are separable over k'.*

PROOF. It suffices to apply (4.6.6) to $X = \operatorname{Spec}(A)$, where A is a k-algebra of finite type whose field of fractions is K. □

Corollary (4.6.8). — *Let X be a k-prescheme of finite type. There exists a finite extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' and the irreducible components of $X_{(k')}$ are geometrically irreducible and the connected components of $X_{(k')}$ are geometrically connected.*

PROOF. This follows immediately from (4.6.6), (4.5.10) and (4.5.15). □

Definition (4.6.9). — Let k be a field, X a k-prescheme, x a point of X. One says that X is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at the point x over k, if, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, X' is reduced (resp. integral) at x' (i.e., $\mathscr{O}_{X',x'}$ is reduced (resp. integral)). One says that X is *geometrically punctually integral* if X is geometrically punctually integral at all its points, that is, if for every extension k' of k, all the local rings of $X \otimes_k k'$ are integral (in which case one says also that $X \otimes_k k'$ is *punctually integral*).

Note that, for X to be geometrically reduced over k (4.6.2), it is necessary and sufficient that it be geometrically reduced over k at every point $x \in X$.

Proposition (4.6.10). — *Let k be a field, X a k-prescheme, k' an extension of k, x' a point of $X' = X \otimes_k k'$, x its image in X. For X to be geometrically punctually reduced (resp. geometrically punctually integral) at x over k, it is necessary and sufficient that X' be so at x' over k'. In particular, for X to be geometrically* *punctually reduced* (*resp. geometrically punctually integral*) *over* k, *it is necessary and sufficient that* X' *be so over* k'.

PROOF. It suffices to prove the first assertion. The condition being obviously necessary, let us prove it is sufficient. Suppose therefore X' geometrically punctually reduced (resp. geometrically punctually integral) at x' over k', and let us prove that X is so at x over k, that is, for every extension k'' of k and every point x'' of $X'' = X \otimes_k k''$ above x, $\mathscr{O}_{X'',x''}$ is reduced (resp. integral). For this, note that there exists a point z of $X' \times_X X''$ projecting to x' and x'' (**I**, 3.4.7). Let s be the point of $\operatorname{Spec}(k' \otimes_k k'')$ image of z (**I**, 3.4.9); set $K = k(s)$, $Z = X \otimes_k K$, so that K can be considered as a composite extension of k' and k'', and z as a point of Z whose image in X' (resp. X'') is x' (resp. x''). Since X' is geometrically punctually reduced (resp. geometrically punctually integral) at x' over k', the ring $\mathscr{O}_{Z,z}$ is reduced (resp. integral), and since $\mathscr{O}_{Z,z}$ is an $\mathscr{O}_{X'',x''}$-module faithfully flat, it follows that $\mathscr{O}_{X'',x''}$ is reduced (resp. integral) ($\mathbf{0_I}$, 6.5.1). ☐

Corollary (4.6.11). — *Suppose* k *perfect* (*resp. algebraically closed*). *For* X *to be geometrically punctually reduced* (*resp. geometrically punctually integral*) *at the point* x *over* k, *it is necessary and sufficient that* $\mathscr{O}_{X,x}$ *be reduced* (*resp. integral*).

PROOF. This is obviously necessary. Conversely, if this condition is satisfied, then, for every extension k' of k, $\operatorname{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$ is reduced (4.6.4) (resp. integral (4.4.4)); hence, for every point x' of $X' = X \otimes_k k'$ above x, $\mathscr{O}_{X',x'}$, which is isomorphic to a local ring of $\operatorname{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$, is reduced (resp. integral). ☐

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Proposition (4.6.12). — *Let* k *be a field,* X *a* k*-prescheme,* x *a point of* X, Ω *a perfect extension of* k. *The following conditions are equivalent:*

a) *X is geometrically reduced at x over k, that is, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, $\mathscr{O}_{X',x'}$ is reduced.*
b) *The prescheme $X \otimes_k \Omega$ is reduced at every point above x.*
c) *For every finite purely inseparable extension k' of k, $X' = X \otimes_k k'$ is reduced at the unique point above x.*
d) $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is geometrically reduced over k.*
e) *$\mathscr{O}_{X,x}$ is reduced, and for every irreducible component Z of X containing x, with generic point z, $k(z)$ is a separable extension of k.*

PROOF. The implications $d) \Rightarrow a) \Rightarrow b) \Rightarrow c)$ are immediate, taking into account for the last that every purely inseparable extension of k is isomorphic to a sub-extension of Ω. Let us prove that $c)$ implies $d)$. It suffices to show that for every finite purely inseparable extension k' of k, $\mathscr{O}_{X,x} \otimes_k k'$ is reduced (4.6.1); now this ring is the local ring of $X' = X \otimes_k k'$ at the unique point x' of X' above x (**I**, 3.5.8) and (3.5.7); it is therefore reduced by hypothesis $c)$. Finally, the equivalence of $d)$ and $e)$ results from the equivalence of $b)$ and $e)$ in (4.6.1), since the irreducible components of X containing x correspond bijectively to the irreducible components of $\operatorname{Spec}(\mathscr{O}_{X,x})$ (**I**, 2.4.2). ☐

Corollary (4.6.13). — *Under the hypotheses of* (4.6.12), *suppose moreover that X is locally Noetherian. Then conditions a) to e) of* (4.6.12) *are also equivalent to the following:*

f) *There exists a neighbourhood U of x that is geometrically reduced over k.*

PROOF. Indeed, this condition trivially implies that X is geometrically reduced at x over k. Conversely, if this is so, since $\mathscr{O}_{X,x}$ is reduced, there exists a reduced open neighbourhood U of x in X (**I**, 6.1.13); taking U small enough, we can moreover suppose that U meets no irreducible components of X not containing x. Then the criterion (4.6.12), $e)$) proves that U is geometrically reduced over k, taking into account the criterion (4.6.1), $e)$). ☐

It follows from (4.6.13) that when X is locally Noetherian, the set of $x \in X$ at which X is geometrically reduced over k is *open*, and it is the largest open subprescheme of X that is geometrically reduced over k.

(4.6.14). It results from (4.6.10) and (4.6.11) that if Ω is an algebraically closed extension of k, it amounts to the same to say that X is geometrically punctually integral at x, or that $X \otimes_k \Omega$ is integral at every point above x.

For X to be geometrically punctually integral at x, it is necessary that X be integral at x, and that if z is the generic point of the unique irreducible component of X containing x, $k(z)$ be a separable extension of k; this follows from (4.6.10). But these conditions are *not sufficient*, as shown by the example in (4.5.12), (ii)). A sufficient but non-necessary condition is that X be geometrically reduced at x and that x belong to only a single irreducible component of X, and that this component be moreover geometrically irreducible; this follows from (4.6.1) and (4.5.9).

Recall that if a prescheme *locally Noetherian* is punctually integral, it is *locally integral* (**I**, 6.1.13). If a k-prescheme X, locally of finite type over k, is geometrically punctually integral over k, it follows from this remark that, for every extension k' of k, $X \otimes_k k'$ is *locally integral* (one says in this case that X is *geometrically locally integral*). **IV-2** | 72

Proposition (4.6.15). — (i) *Let k be a field, X a k-prescheme of finite type. For X to be geometrically punctually integral, it is necessary and sufficient that it be geometrically reduced and that the geometric number of its irreducible components be equal to the geometric number of its connected components.*

(ii) *Let X be a prescheme locally of finite type over k. If X is geometrically punctually integral at a point x, there exists an open neighbourhood U of x that is geometrically punctually integral. In other words, the set of $x \in X$ at which X is geometrically punctually integral is open.*

PROOF. (i) Given an algebraically closed extension Ω of k, $X_{(\Omega)}$ is punctually integral (or locally integral, which amounts to the same since $X_{(\Omega)}$ is Noetherian) if and only if it is reduced and the number of its connected components is equal to the number of its irreducible components (**I**, 6.1.10). It then suffices to apply (4.5.1), (4.6.1) and the first remark of (4.6.14).

(ii) The question being local on X, one can restrict to the case where X is a prescheme of finite type over k. Moreover, one knows that there is an open neighbourhood of x in X which is geometrically reduced (4.6.13), so one can suppose X geometrically reduced. There then exists a finite extension k' of k such that $X' = X \otimes_k k'$ is reduced and its irreducible components are geometrically irreducible (4.6.8). Let $p : X' \to X$ be the canonical projection, which is a finite surjective morphism, and let x'_j ($1 \leqslant j \leqslant r$) be the points of $p^{-1}(x)$; by hypothesis, X' is integral at each x'_j, hence each x'_j has an open integral neighbourhood V'_j in X'. Since p is closed (**II**, 6.1.10), each of the $p(V'_j)$ is a neighbourhood of x, hence there exists an open neighbourhood U of x such that $p^{-1}(U)$ is contained in the union of the V'_j, and is therefore integral at each of its points, hence locally integral. The irreducible components of $p^{-1}(U)$ are therefore open and pairwise disjoint, and since they are geometrically irreducible (4.5.9), $p^{-1}(U) = U \otimes_k k'$ is geometrically punctually integral by (i), and the same holds for U by definition. ☐

Proposition (4.6.16). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

a) *For every extension of finite type k' of k, setting $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$, $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module* (3.2.2).

b) *For every finite purely inseparable extension k' of k, $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module.*

c) *$\mathscr{F}$ is reduced, and if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed subprescheme defined by $\mathscr{J}$ is geometrically reduced over k.*

Moreover, if X is locally of finite type over k, these conditions are also equivalent to:

d) *For every extension k' of k (or for an algebraically closed extension k' of k), $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module.*

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PROOF. It is clear that $a)$ implies $b)$. Condition $b)$ implies first that $\mathscr{F}$ is reduced: this means (3.2.3) that if Y is the closed subprescheme of X defined by the coherent Ideal $\mathscr{J}$ of $\mathscr{O}_X$, annihilator of $\mathscr{F}$, Y is reduced, and there is an $\mathscr{O}_Y$-Module coherent without torsion $\mathscr{G}$ of rank 1 on every irreducible component of Y, such that $j_*(\mathscr{G}) = \mathscr{F}$, $j : Y \to X$ being the canonical injection. Now, for *every* extension k' of k, the annihilator of $\mathscr{F}'$ is $\mathscr{J}' = \mathscr{J} \otimes_k k'$, the morphism projection $X' \to X$ being flat (2.1.11); the closed subprescheme of X' defined by $\mathscr{J}'$ is $Y' = Y \otimes_k k'$, and if $j' : Y' \to X'$ is the canonical injection, and $\mathscr{G}' = \mathscr{G} \otimes_k k'$, we have $\mathscr{F}' = j'_*(\mathscr{G}')$. Note also that, for every extension

k' of k for which X' is locally Noetherian, $\mathscr{F}'$ is without immersed associated prime cycle (4.2.7). Moreover, every maximal point y' of Y' is above a maximal point y of Y; since $\mathscr{G}_{y'}$ is isomorphic to $\mathscr{O}_{Y,y}$, $\mathscr{G}'_{y'}$ is isomorphic to $\mathscr{O}_{Y',y'}$. One sees therefore (by (3.2.3)) that, whether $\mathscr{F}'$ is reduced or the prescheme Y' is reduced, it amounts to the same thing. The fact that b) implies c) and that c) implies a) are therefore consequences of (4.6.1), d) and b)); in a), the hypothesis that k' is an extension of finite type of k is only used to ensure that X' is locally Noetherian. When X is locally of finite type over k, X' is locally Noetherian for every extension k' of k, which proves in this case the equivalence of d) and the other conditions. □

Definition (4.6.17). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.16), one says that $\mathscr{F}$ is *geometrically reduced* (or *separable*) over k.

Proposition (4.6.18). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

a) *For every extension of finite type k' of k, setting $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$, $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module* (3.2.4).
b) *For every finite extension k' of k, $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module.*
c) *$\mathscr{F}$ is reduced (or integral), and if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed subprescheme defined by $\mathscr{J}$ is geometrically integral over k.*

Moreover, if X is locally of finite type over k, these conditions are also equivalent to:

d) *For every extension k' of k (or for an algebraically closed extension k' of k), $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module.*

PROOF. Conditions a), b) and c) imply that $\mathscr{F}$ is integral, hence $\mathrm{Ass}(\mathscr{F})$ reduces to a single point x, which is the generic point of $\mathrm{Supp}(\mathscr{F})$. For *every* extension k' of k for which X' is locally Noetherian, the associated prime cycles of $\mathscr{F}'$ are therefore non-immersed, and are the maximal points of $\mathrm{Supp}(\mathscr{F}')$, which are all above x. On the other hand, by virtue of (4.6.16), each of the conditions a), b), c) implies that $\mathscr{F}$ is geometrically reduced; it is trivial that a) implies b), and b) implies that $\mathrm{Supp}(\mathscr{F})$ is geometrically irreducible (4.5.9), hence c), since the subprescheme Y defined by the annihilator $\mathscr{J}$ of $\mathscr{F}$ is then geometrically reduced and geometrically irreducible, hence geometrically integral; this shows that c) implies a), and c) implies d) when X is locally of finite type over k. □

Definition (4.6.19). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.18), one says that $\mathscr{F}$ is *geometrically integral* over k.

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Proposition (4.6.20). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. Then, if $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) over k, $\mathscr{F} \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K.*

PROOF. Indeed, for every finite extension K' of K, $(X \otimes_k K) \otimes_K K' = X \otimes_k K'$ is locally Noetherian, and one saw in the proofs of (4.6.16) and (4.6.18) that the hypothesis on $\mathscr{F}$ implies that $\mathscr{F} \otimes_k K'$ is reduced (resp. integral) over K, whence the conclusion. □

Proposition (4.6.21). — *Let k be a field, X, Y two k-preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, $\mathscr{G}$ a coherent $\mathscr{O}_Y$-Module.*

(i) *If $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) and $\mathscr{G}$ reduced (resp. integral), then $\mathscr{F} \otimes_k \mathscr{G}$ is reduced (resp. integral).*
(ii) *If $\mathscr{F}$ and $\mathscr{G}$ are geometrically reduced (resp. geometrically integral), the same holds for $\mathscr{F} \otimes_k \mathscr{G}$.*

PROOF. (i) Using (3.2.3), (4.6.16) and (4.6.18), one can restrict to the case where $\mathrm{Supp}(\mathscr{F}) = X$, $\mathrm{Supp}(\mathscr{G}) = Y$, X being geometrically reduced (resp. geometrically integral), Y reduced (resp. integral), $\mathscr{F}$ and $\mathscr{G}$ without immersed associated prime cycles, and $\mathscr{F}_x$ isomorphic to $\mathscr{O}_x$ at every maximal point x of X, $\mathscr{G}_y$ isomorphic to $\mathscr{O}_y$ at every maximal point y of Y. One knows then (4.2.3) that $\mathscr{F} \otimes_k \mathscr{G}$ is without immersed associated prime cycle, and that the associated prime cycles of $\mathscr{F} \otimes_k \mathscr{G}$ are exactly the irreducible components of $X \times_k Y$ (4.2.5). One can therefore restrict to the case where X and Y are irreducible; then, if X is geometrically reduced and Y reduced, one knows (4.2.4), (ii)) that $X \times_k Y$ is reduced; if X is geometrically integral and Y integral, one knows that $X \times_k Y$ is also irreducible (4.5.8), (i)). Finally, every maximal point z of $X \times_k Y$ being above x and y, the hypotheses on $\mathscr{F}$ and $\mathscr{G}$ imply, by virtue of (**I**, 9.1.12) that $(\mathscr{F} \otimes_k \mathscr{G})_z$ is isomorphic to $\mathscr{O}_z$, which finishes the proof in this case.

(ii) The reasoning is analogous, but here (after reduction to the case where $X = \operatorname{Supp}(\mathscr{F})$, $Y = \operatorname{Supp}(\mathscr{G})$), $X \times_k Y$ is geometrically reduced (resp. geometrically integral) by virtue of (4.6.5), (ii). □

The definitions (4.6.18) and (4.6.19) localize:

Definition (4.6.22). — Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. One says that $\mathscr{F}$ is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at a point $x \in X$ if, for every finite purely inseparable (resp. finite) extension k' of k, $\mathscr{F} \otimes_k k'$ is reduced (resp. integral) at each of the points x' of $X \otimes_k k'$ above x (cf. (3.2.2) and (3.2.4)).

If $\mathscr{F}$ is reduced at x, to say that $\mathscr{F}$ is geometrically reduced at x is equivalent to saying that $\mathscr{F}$ is reduced at x and that the closed subprescheme Y defined by the annihilator $\mathscr{J}$ of $\mathscr{F}$ is geometrically reduced at x; it follows that if X is locally of finite type over k, saying that $\mathscr{F}$ is geometrically punctually integral at x is equivalent to saying that $\mathscr{F}$ is reduced at x and that the subprescheme Y is *geometrically punctually integral at* x. **IV-2** | 75

4.7. Multiplicities in the primary decomposition on an algebraic prescheme

Lemma (4.7.1). — *Let A, B be two local rings, $\mathfrak{m}$ the maximal ideal of A, $\rho : A \to B$ a local homomorphism. Suppose that B is a flat A-module, and $B/\mathfrak{m}B$ a B-module of finite length. Then, for an A-module M of finite length, it is necessary and sufficient that $M_{(B)}$ be a B-module of finite length, and we have*

$$\operatorname{long}_B(M_{(B)}) = \operatorname{long}_A(M) \cdot \operatorname{long}_B(B/\mathfrak{m}B). \tag{4.7.1.1}$$

PROOF. This is a particular case of (2.5.5.2). □

Corollary (4.7.2). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a ring homomorphism. Let $\mathfrak{q}$ be a minimal prime ideal of B, $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$. If B is a flat A-module, then $\mathfrak{p}$ is a minimal prime ideal of A, $M_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module of finite length, and we have*

$$\operatorname{long}_{B_{\mathfrak{q}}}(M_{(B)})_{\mathfrak{q}} = (\operatorname{long}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}))(\operatorname{long}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})). \tag{4.7.2.1}$$

PROOF. Indeed, $B_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$-module ($\mathbf{0_I}$, 6.3.2) and the homomorphism $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ is $A_{\mathfrak{p}}$-flat, hence faithfully flat ($\mathbf{0_I}$, 6.6.2); the fact that $\mathfrak{p}$ is minimal follows from (2.3.5). Moreover, $A_{\mathfrak{p}}$ is then an Artinian ring (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, prop. 9) and $M_{\mathfrak{p}}$ is therefore an $A_{\mathfrak{p}}$-module of finite length; the formula (4.7.2.1) is therefore a particular case of (4.7.1.1). □

Proposition (4.7.3). — *Let k be a field of characteristic exponent p, X an integral prescheme locally Noetherian, $K = \mathrm{R}(X)$ the field of rational functions of X. Let k' be an extension of k, (X'_α) the family of irreducible components of $X' = X_{(k')}$, x'_α the generic point of X'_α.*

(i) *Suppose that X' is locally Noetherian (which occurs if X is of finite type over k, or if k' is an extension of finite type of k). Then for k'' an extension of k', (X''_β) the family of irreducible components of X'', x''_β the generic point of X''_β. If x''_β is above x'_α, we have*

$$\operatorname{long}(\mathscr{O}_{x''_\beta}) = \operatorname{long}(\mathscr{O}_{x'_\alpha}) \cdot \operatorname{long}(\mathscr{O}_{Z'',x''_\beta}) \tag{4.7.3.1}$$

where we have set $Z'' = (X'_{\mathrm{red}}) \otimes_k k''$. In particular, if k'' is a separable extension of k' such that Z'' is locally Noetherian, we have $\operatorname{long}(\mathscr{O}_{x''_\beta}) = \operatorname{long}(\mathscr{O}_{x'_\alpha})$.

(ii) *If X is locally of finite type over k, or if k' is an extension of finite type of k, the numbers $\operatorname{long}(\mathscr{O}_{x'_\alpha})$ are powers of p.*

(iii) *Suppose X is locally of finite type over k. Then, if k' is perfect, all the numbers $\operatorname{long}(\mathscr{O}_{x'_\alpha})$ are equal and do not depend on the particular perfect extension K of k considered. Moreover, there exists a finite purely inseparable extension k_1 of k such that, if x_1 is the generic point of $X_1 = X \otimes_k k_1$ (which is irreducible), we have $\operatorname{long}(\mathscr{O}_{x'_\alpha}) = \operatorname{long}(\mathscr{O}_{x_1})$ for every α.*

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Proof. (i) Recall (4.2.4) that the irreducible components X'_α, which are finite in number, correspond bijectively to the minimal prime ideals $\mathfrak{p}_\alpha$ of $K \otimes_k k'$, and that $\mathscr{O}_{x'_\alpha}$ is an Artinian ring, isomorphic to $(K \otimes_k k')_{\mathfrak{p}_\alpha}$; these rings therefore do not depend on the extensions K and k' of k (and are therefore the same, k' being fixed, for all k-preschemes integral having the same fields of rational functions). The formula (4.7.3.1) results from (4.7.2.1) applied to the Noetherian ring A of an affine neighbourhood of x'_α in X', and to the ring B of an affine neighbourhood small enough of x''_β in X'', $\mathfrak{p}$ and $\mathfrak{q}$ being the minimal prime ideals corresponding to x'_α and x''_β respectively, and taking $M = A$: one notes that $A/\mathfrak{p}$ is the ring of an affine neighbourhood of x'_α in X'_{red}, and that $B = A \otimes_{k'} k''$, and that $\mathscr{O}_{Z'',x''_\beta} = (B/\mathfrak{p}B)_\mathfrak{q} = B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}$, whence (4.7.3.1). The last assertion of (i) results from the fact that in this case Z'' is reduced (4.2.4), hence $\mathfrak{p}B_\mathfrak{q} = \mathfrak{q}B_\mathfrak{q}$.

(ii) If k' is an extension of finite type of k, there is a purely transcendental extension k'_0 of k such that k' is a finite extension of k'_0; since k'_0 is separable over k, it results from (i) applied with k' replaced by k and k'' by k'_0 respectively, that we can replace X by $X \otimes_k k'_0$, in other words suppose k' is an extension *finite* of k. Then there is a finite quasi-Galois extension k'' of k containing k', and the formula (4.7.3.1) shows that it suffices to prove the assertion for k''. But k'' is a separable extension of a finite purely inseparable extension k_1 of k, hence, by virtue of (i), one is reduced to proving the assertion when $k' = k_1$ is *purely inseparable*. One knows that $K \otimes_k k_1$ is an Artinian ring having only a single minimal prime ideal (4.3.2); if $\overline{K}$ is the algebraic closure of K, the length of $K \otimes_k k_1$ divides that of the Artinian ring $\overline{K} \otimes_k k_1$ by virtue of (4.7.1.1). But $\overline{K} \otimes_k k_1$ is a $\overline{K}$-algebra having only a single prime ideal (necessarily maximal), and its quotient by this ideal is necessarily a finite extension of $\overline{K}$, hence necessarily equal to $\overline{K}$; the length of $\overline{K} \otimes_k k_1$ is therefore equal to its rank over $\overline{K}$, that is, $[k_1 : k]$, which is a power of p.

Suppose next that X is locally of finite type over k; then, X'' is locally Noetherian for every extension k'' of k'. Taking for k'' an algebraically closed extension, one is reduced, by virtue of (4.7.3.1), to proving the assertion when k' is the *algebraic closure* $\overline{k}$. Now, one knows (replacing X by an open affine of finite type over k) that there is a finite purely inseparable extension k_1 of k such that $(X \otimes_k k_1)_{\mathrm{red}}$ is separable over k_1 (4.6.6); hence $(X \otimes_k k_1)_{\mathrm{red}} \otimes_{k_1} \overline{k}$ is reduced (4.6.4), and the same holds for one and we conclude as above.

(iii) Note that two extensions k', k'' of k can always be contained in a same extension E of k; it suffices then to compare the numbers for k' and for an algebraically closed extension E common to Ω and Ω' to show the first assertion. The second is obtained by taking for Ω an algebraically closed extension $\overline{k}$ and using the fact that the extension k_1 provided by (ii) determines as unique generic point of $X_1 = X \otimes_k k_1$, and the relation $\mathrm{long}(\mathscr{O}_{x'_\alpha}) = \mathrm{long}(\mathscr{O}_{x_1})$ for every α, follows from (4.7.3.1). □

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Definition (4.7.4). — Let k be a field, X a k-prescheme integral, K the field of rational functions of X. The *purely inseparable multiplicity* of K (or of X) over k is defined to be the upper bound of the lengths of the Artinian rings $K \otimes_k k_1$, when k_1 ranges over the set of finite purely inseparable extensions of k. The *separable multiplicity* of K (or of X) over k is defined to be the geometric number of irreducible components of X if this number is finite, and $+\infty$ otherwise. The *total multiplicity* of K (or of X) over k is the product of the purely inseparable multiplicity and the separable multiplicity.

One will note that if p is the characteristic exponent of k and if the purely inseparable multiplicity of K over k is finite, it is a power $q = p^f$ of p, as follows from (4.7.3), (ii)); when $p \neq 1$, f is well-determined and is called the *exponent of inseparability* of K over k; when $p = 1$, the purely inseparable multiplicity is always equal to 1.

To say that the purely inseparable multiplicity (resp. separable, resp. total) of X over k is 1 means that X is geometrically reduced over k (resp. geometrically irreducible, resp. geometrically integral) over k.

When X is locally of finite type over k, the purely inseparable multiplicity of X over k is also the common length of the local rings A_i at the minimal prime ideals of $A = K \otimes_k k'$, for every perfect extension k' of k; the separable multiplicity of X over k is the number of minimal prime ideals of

$A = K \otimes_k \Omega$ for every algebraically closed extension Ω of k, and finally the total multiplicity is the sum of the lengths of the local rings A_i of $A = K \otimes_k \Omega$ at its minimal prime ideals; these numbers are all three *finite* in this case.

Definition (4.7.5). — Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, x a point of X such that $\mathscr{F}_x$ is an $\mathscr{O}_x$-module of finite length. The *geometric length* of $\mathscr{F}$ at x (relative to k), or the *purely inseparable multiplicity of x for $\mathscr{F}$*, is the product $\lambda_x(\mathscr{F}) = (\mathrm{long}_{\mathscr{O}_x}(\mathscr{F}_x))\mu_i(k(x) \mid k)$, where $\mu_i(k(x) \mid k)$ denotes the purely inseparable multiplicity of $k(x)$ over k.

When X is *integral*, and x the generic point of X, $\mathrm{long}(\mathscr{O}_x) = 1$, hence the purely inseparable multiplicity of x for $\mathscr{O}_X$ is none other than the purely inseparable multiplicity of X over k, defined in (4.7.4).

Proposition (4.7.6). — *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of coherent $\mathscr{O}_X$-Modules. If at a point $x \in X$ the geometric length $\lambda_x(\mathscr{F})$ of $\mathscr{F}$ at this point is defined, the same holds for $\lambda_x(\mathscr{F}')$ and $\lambda_x(\mathscr{F}'')$, of $\mathscr{F}'$ and $\mathscr{F}''$ at the point x, and conversely; moreover, we have*

(4.7.6.1) $$\lambda_x(\mathscr{F}) = \lambda_x(\mathscr{F}') + \lambda_x(\mathscr{F}'').$$

Proof. The proposition results immediately from the definition, for $\mathrm{long}(\mathscr{F}_x)$ is finite if and only if $\mathrm{long}(\mathscr{F}'_x)$ and $\mathrm{long}(\mathscr{F}''_x)$ are finite, and we have $\mathrm{long}(\mathscr{F}_x) = \mathrm{long}(\mathscr{F}'_x) + \mathrm{long}(\mathscr{F}''_x)$. □

(4.7.7). Under the hypotheses of (4.7.5), the only points $x \in X$ such that $\mathscr{F}_x$ is a non-zero $\mathscr{O}_x$-module of finite length are (by virtue of (3.1.2)) the *maximal points* of $\mathrm{Supp}(\mathscr{F})$, as follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 2 of prop. 7, the question being local on X. One knows that there exists a closed subprescheme Y having $\mathrm{Supp}(\mathscr{F})$ as underlying space, and a coherent $\mathscr{O}_Y$-Module $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, $j : Y \to X$ being the canonical injection. If x is a maximal point of Y, we see that the geometric lengths of $\mathscr{F}$ and of $\mathscr{G}$ at x are equal, $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,x}$ having the same residue fields.

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Proposition (4.7.8). — *Under the hypotheses of (4.7.5), let x be a maximal point of $\mathrm{Supp}(\mathscr{F})$. Let k' be a perfect extension of k; set $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$; then the geometric length $\lambda_x(\mathscr{F})$ of $\mathscr{F}$ at x is equal to the length of the $\mathscr{O}_{x'}$-module $\mathscr{F}'_{x'}$ at every maximal point x' of $\mathrm{Supp}(\mathscr{F}')$ above x. Moreover, there exists a finite purely inseparable extension k_1 of k such that, setting $X_1 = X \otimes_k k_1$, $\mathscr{F}_1 = \mathscr{F} \otimes_k k_1$, $\lambda_x(\mathscr{F})$ is also equal to the length of the $\mathscr{O}_{x_1}$-module $(\mathscr{F}_1)_{x_1}$ at every maximal point x_1 of $\mathrm{Supp}(\mathscr{F}_1)$ above x.*

Proof. The remark of (4.7.7) shows that one can restrict to the case where x is a maximal point of X and one can therefore suppose $X = \mathrm{Spec}(A)$ affine Noetherian, $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type; let $\mathfrak{p}$ (resp. $\mathfrak{q}$) be the minimal prime ideal of A (resp. B) corresponding to x (resp. x'). We have then, applying the formula (4.7.2.1):

$$\mathrm{long}_{\mathscr{O}_{x'}}(\mathscr{F}'_{x'}) = \mathrm{long}_{\mathscr{O}_x}(\mathscr{F}_x) \cdot \mathrm{long}_{B_\mathfrak{q}}(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}).$$

But since $k(x) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$, the same reasoning as in (4.7.3) shows that $\mathrm{long}_{B_\mathfrak{q}}(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q})$ is equal to the length of the local ring $(B/\mathfrak{p}B)_{\mathfrak{q}'}$, where $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$, and this length is by definition $\mu_i(k(x) \mid k)$, whence the first assertion. The second is proved similarly, replacing k' by k_1 and using (4.7.3), (iii). □

One says, in the case where x is a maximal point of $\mathrm{Supp}(\mathscr{F})$, that the geometric length of $\mathscr{F}$ at x is the *purely inseparable multiplicity of the maximal prime cycle $\{x\}$ associated to $\mathscr{F}$*.

Corollary (4.7.9). — *Under the hypotheses of (4.7.5), let k' be an extension of k, $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$; let Z be an irreducible component of $\mathrm{Supp}(\mathscr{F})$, and Z' an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ that dominates Z (4.2.7). Then the purely inseparable multiplicities of Z and of Z' with respect to $\mathscr{F}$ and $\mathscr{F}'$ respectively are the same.*

Proof. It suffices to consider a perfect extension k'' of k' and, setting $X'' = X \otimes_k k''$, $\mathscr{F}'' = \mathscr{F} \otimes_k k''$, a maximal point x'' of $\mathrm{Supp}(\mathscr{F}'')$ above the generic point of Z'; then by (4.7.8) $\mathrm{long}(\mathscr{F}''_{x''})$ is equal to the purely inseparable multiplicities of Z and of Z'. □

Proposition (4.7.10). — *Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, x a point of X, z_i the generic points of the irreducible components of* $\mathrm{Supp}(\mathscr{F})$ *containing x* (4.6.22). *For $\mathscr{F}$ to be geometrically reduced at x over k, it is necessary and sufficient that x not belong to any immersed associated prime cycle of $\mathscr{F}$ and that, for every i, the geometric length $\lambda_{z_i}(\mathscr{F})$ of $\mathscr{F}$ at the point z_i be equal to* 1*.*

Proof. Let us first show that the conditions of the statement are sufficient. Indeed, if k' is a finite extension of k, x' a point of $X' = X \otimes_k k'$ above x, and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, x' does not belong to any immersed associated prime cycle of $\mathscr{F}'$ by (4.2.7); moreover, by (4.7.8), the geometric length of $\mathscr{F}'$ is equal to 1 at every generic point z'_j of an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ containing x', such a point being above one of the z_i by (2.3.4); *a fortiori* $\mathrm{long}(\mathscr{F}'_{z'_j}) = 1$, which proves that $\mathscr{F}'$ is reduced at x'.

Conversely, it results first from (4.2.7) that if, for *one* finite extension k' of k, $\mathscr{F}'$ is reduced at all points x' of X' above x, then x does not belong to any immersed associated prime cycle of $\mathscr{F}$. Moreover, taking for k' the field k_1 from the statement of (4.7.8) (where x is replaced by one of the z_i), the hypothesis that $\mathscr{F}$ is geometrically reduced at x implies $\lambda_{z_i}(\mathscr{F}) = 1$. □

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Proposition (4.7.11). — *Let k be a field, X a prescheme locally of finite type over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$, x a point of X, x' a point of X' above x. For $\mathscr{F}$ to be geometrically punctually reduced* (*resp. geometrically punctually integral*) *at x, it is necessary and sufficient that $\mathscr{F}'$ be geometrically punctually reduced* (*resp. geometrically punctually integral*) *at x'.*

Proof. The irreducible components of $\mathrm{Supp}(\mathscr{F}')$ containing x' each dominate an irreducible component of $\mathrm{Supp}(\mathscr{F})$ containing x, and conversely each of these latter is dominated by an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ containing x' ((4.2.7), (i)) and (2.3.4)). The assertion concerning separability results therefore from (4.2.7), (ii), (4.7.9) and (4.7.10). Let Y be the closed subprescheme of X defined by the Ideal $\mathscr{J}$ annihilator of $\mathscr{F}$, so that $Y' = Y \otimes_k k'$ is the closed subprescheme of X' defined by the Ideal $\mathscr{J}' = \mathscr{J} \otimes_k k'$ annihilator of $\mathscr{F}'$; to say that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically punctually reduced (resp. integral) at x (resp. x') is the same as saying that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically punctually reduced at x (resp. x') and that Y (resp. Y') is geometrically punctually integral at x (resp. x'). By virtue of the assertion of the statement concerning separability, the hypothesis that $\mathscr{F}$ is geometrically punctually integral at x, or the hypothesis that $\mathscr{F}'$ is geometrically punctually integral at x', imply both that there exists in Y a neighbourhood of x that is separable, and the conclusion results from (4.6.10). □

Definition (4.7.12). — Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, x a maximal point of the support of $\mathscr{F}$. The *total multiplicity* of x (or of the prime cycle $\overline{\{x\}}$) for $\mathscr{F}$ (relative to k) is the product of the purely inseparable multiplicity of x for $\mathscr{F}$ by the separable multiplicity of $k(x)$ over k.

It amounts to the same to say that the total multiplicity of x for $\mathscr{F}$ is the product of the *length* $\mathrm{long}_{\mathscr{O}_x}(\mathscr{F}_x)$ by the total multiplicity of $k(x)$ over k (4.7.4). With the notation of (4.7.11), the total multiplicity of x for $\mathscr{F}$ is equal to the *sum* of the total multiplicities for $\mathscr{F}'$ of the maximal points x'_i of $\mathrm{Supp}(\mathscr{F}')$ that are above x; moreover, there exists a finite extension k' of k such that the total multiplicity of x for $\mathscr{F}$ be *equal to* $\sum_i \mathrm{long}(\mathscr{F}'_{x'_i})$. **IV-2 | 80**

Proposition (4.7.13). — *Let the hypotheses and notation be those of* (4.7.10)*. For $\mathscr{F}$ to be geometrically punctually integral at the point x, it suffices that x belong to only one associated prime cycle* (*necessarily non-immersed*) *of $\mathscr{F}$ and that the total multiplicity for $\mathscr{F}$ of the generic point z of this cycle be equal to* 1*.*

Proof. Indeed, if k' is a finite extension of k, x' a point of $X' = X \otimes_k k'$ above x, and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, x' can belong to only one associated prime cycle of $\mathscr{F}'$, since there is only one maximal point of $\mathrm{Supp}(\mathscr{F}')$ above z by hypothesis; moreover, $\mathscr{F}$ is geometrically reduced by (4.7.10), whence the conclusion. □

§5. Dimension, depth, and regularity in locally Noetherian preschemes

This section merely restates, in geometric language, and with various complements of a technical nature, the notions and results of commutative Algebra exposed in chapter 0, §§16 and 17. **IV-2** | 86

5.1. Dimension of preschemes

(5.1.1). Let A be a ring, $\mathfrak{J}$ an ideal of A. Recall $(\mathbf{0}, 16.1)$ that the definitions of the dimension of a ring $(\mathbf{0}, 16.1.1)$ and those of the theory of combinatorial dimension of topological spaces $(\mathbf{0}, 14.1.2)$ give the following relations:

(5.1.1.1) $$\dim(\operatorname{Spec}(A)) = \dim(A)$$

(5.1.1.2) $$\dim(\mathrm{V}(\mathfrak{J})) = \dim(A/\mathfrak{J})$$

(5.1.1.3) $$\operatorname{codim}(\mathrm{V}(\mathfrak{J}), \operatorname{Spec}(A)) = \operatorname{ht}(\mathfrak{J})$$

(5.1.1.4) $\operatorname{Spec}(A)$ is a catenary space $\Leftrightarrow$ A is a catenary ring.
(5.1.1.5) $\operatorname{Spec}(A)$ is equidimensional $\Leftrightarrow$ A is equidimensional $\Leftrightarrow$ the minimal prime ideals $\mathfrak{p}_\alpha$ of A are such that the dimensions of the rings $A/\mathfrak{p}_\alpha$ are all equal.
(5.1.1.6) $\operatorname{Spec}(A)$ is equicodimensional $\Leftrightarrow$ A is equicodimensional $\Leftrightarrow$ all the maximal ideals have the same height.

Recall that a Noetherian ring A is called *biequidimensional* if $\operatorname{Spec}(A)$ is biequidimensional, that is if $\operatorname{Spec}(A)$ is Noetherian and A is equidimensional, catenary and of finite dimension.

Proposition (5.1.2). — *Let X be a prescheme, Y a closed irreducible part of X, y the generic point of Y. We have*

(5.1.2.1) $$\operatorname{codim}(Y, X) = \dim(\mathscr{O}_{X,y}).$$

Proof. Indeed (**I**, 2.4.2) the closed irreducible parts of X containing y are canonically in bijective correspondence with the closed irreducible parts of $\operatorname{Spec}(\mathscr{O}_{X,y})$, hence in bijective correspondence with the prime ideals of $\mathscr{O}_{X,y}$, and (5.1.2.1) results from the definitions. □

In particular, one recovers the fact that the generic points of the irreducible components of X are the *only* points $x \in X$ such that $\dim(\mathscr{O}_x) = 0$ (**I**, 1.1.14).

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Corollary (5.1.3). — *For every closed part Y of a prescheme X, we have*

(5.1.3.1) $$\operatorname{codim}(Y, X) = \inf_{y \in Y} \dim(\mathscr{O}_{X,y}).$$

Moreover, if X is locally Noetherian, then for every $x \in Y$

(5.1.3.2) $$\operatorname{codim}_x(Y, X) = \inf_{y \in Y,\, x \in \overline{\{y\}}} \dim(\mathscr{O}_{X,y}).$$

Proof. The relation (5.1.3.2) follows from (5.1.3.1) and $(\mathbf{0}, 14.2.6)$. □

This corollary permits us to *define*, for an *arbitrary* part Y of a prescheme X, the *codimension* $\operatorname{codim}(Y, X)$ of Y in X as equal to the second member of (5.1.3.1).

Proposition (5.1.4). — *For every prescheme X, we have*

(5.1.4.1) $$\dim(X) = \sup_{x \in X}(\dim(\mathscr{O}_x)).$$

If every closed irreducible part of X contains a closed point, we also have

(5.1.4.2) $$\dim(X) = \sup_{x \in \mathrm{F}}(\dim(\mathscr{O}_x))$$

where F *is the set of* closed *points of X.*

Proof. It suffices to remark (by virtue of (**I**, 2.4.2)) that the chains of closed irreducible parts of X correspond bijectively to the chains of closed irreducible parts of a local scheme of X; when every closed irreducible part of X contains a closed point, one can obviously restrict to the local schemes at the closed points of X. □

One will note that every closed irreducible part of X contains a closed point when X is *quasi-compact* ($\mathbf{0_I}$, 2.1.3); we shall see a little further (5.1.11) that this also holds when X is *locally Noetherian*.

Corollary (5.1.5). — *For a prescheme X to be catenary, it is necessary and sufficient that, for every $x \in X$, the local ring $\mathscr{O}_x$ be catenary. If moreover every closed irreducible part of X contains a closed point, for X to be catenary, it suffices, for every* closed *point x of X, that $\mathscr{O}_x$ be catenary.*

Proof. The reasoning is the same as in (5.1.4), since it involves comparing chains of closed irreducible parts having the same endpoints. □

(5.1.6). We shall now examine properties more special linked to Noetherian hypotheses.

Recall ($\mathbf{0}$, 16.2.3) that a *local Noetherian* ring $A \neq 0$ is of *finite* dimension, equal to the minimum number of generators of an ideal of definition of A; for every prime ideal $\mathfrak{p}$ of A, the height of $\mathfrak{p}$, equal to $\dim(A_\mathfrak{p})$, is therefore also finite. These properties, combined with (5.1.2), (5.1.3) and (5.1.4), show that:

Proposition (5.1.7). — *For every non-empty closed part Y of a locally Noetherian prescheme X,* $\operatorname{codim}(Y, X)$ *is finite. If X is Noetherian and affine and Y a closed irreducible part of X,* $\operatorname{codim}(Y, X)$ *is equal to the minimum number of sections s_i of $\mathscr{O}_X$ above X such that Y is an irreducible component of the set of $x \in X$ such that $s_i(x) = 0$ for every i.*

Corollary (5.1.8). — *Let X be a locally Noetherian prescheme, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible, f a section of $\mathscr{L}$ above X. Then every irreducible component of the set Z of $x \in X$ such that $f(x) = 0$ is of codimension $\leqslant 1$ in X; it is of codimension 1 if Z contains no irreducible component of X.*

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Proof. One can restrict to the case where $\mathscr{L} = \mathscr{O}_X$. If y is the generic point of an irreducible component of Z, the ideal (f_y) of $\mathscr{O}_{X,y}$ must be such that $\mathscr{O}_{X,y}/(f_y)$ has only a single prime ideal, which means that f_y generates an ideal of definition of the local Noetherian ring $\mathscr{O}_{X,y}$; we therefore have $\operatorname{codim}(Y, X) \leqslant 1$ (5.1.7); if Z contains no irreducible component of X, we cannot have $\operatorname{codim}(Y, X) = 0$ by virtue of ($\mathbf{0}$, 14.2.1). □

Proposition (5.1.9). — *Let X be a locally Noetherian prescheme, Y a closed part of X. If $x \in Y$ is such that $\mathscr{O}_{X,x}$ is a catenary ring, we have*

$$\operatorname{codim}_x(Y, X) = \dim(\mathscr{O}_{X,x}) - \operatorname{codim}(\overline{\{x\}}, Y) = \dim(\mathscr{O}_{X,x}) - \dim(\mathscr{O}_{Y,x}). \tag{5.1.9.1}$$

Proof. Let Y_i ($1 \leqslant i \leqslant n$) be the irreducible components of Y containing x (which are finite in number since Y is locally Noetherian), and let y_i be the generic point of Y_i. Setting $A = \mathscr{O}_{X,x}$, and $\mathfrak{p}_i$ being the prime ideal of A corresponding to $Y_i \cap \operatorname{Spec}(A)$, the closed irreducible parts of Y containing x correspond bijectively to the prime ideals of A containing one of the $\mathfrak{p}_i$, hence $\dim(\mathscr{O}_{Y,x}) = \sup_i \dim(A/\mathfrak{p}_i)$; on the other hand, $\mathscr{O}_{X,y_i} = A_{\mathfrak{p}_i}$, and the hypothesis on A implies $\dim(A) = \dim(A/\mathfrak{p}_i) + \dim(A_{\mathfrak{p}_i})$ ($\mathbf{0}$, 16.1.4); the conclusion results therefore from the relation $\operatorname{codim}_x(Y, X) = \inf_i(\dim(\mathscr{O}_{X,y_i}))$ ((5.1.2) and ($\mathbf{0}$, 14.2.6)). □

Proposition (5.1.10). — (i) *In every prescheme X, every locally constructible non-empty part E contains a point x such that $\{x\}$ is a locally closed part of X (or, which amounts to the same, such that x is isolated in $\overline{\{x\}}$).*

(ii) *Let X be a locally Noetherian prescheme, x a point of X such that $\{x\}$ is locally closed in X; then $\dim(\overline{\{x\}}) \leqslant 1$; every point $y \neq x$ of $\overline{\{x\}}$ is therefore a closed point in X.*

Proof. (i) The result is a particular case of the following lemma: □

Lemma (5.1.10.1). — *Let X be a topological space having the following property: for every locally closed non-empty part $Z \neq \varnothing$ of X, there exists a part Z' of Z, locally closed in X (or in Z, which amounts to the same), and a point $x \in Z'$, closed in Z'. Then every locally closed non-empty part $Z \neq \varnothing$ of X contains a point x isolated in $\overline{\{x\}}$.*

Proof. Indeed, let $Z' \subset Z$ be a locally closed part containing a point x such that x be closed *in* Z'. There is a neighbourhood U open of x in X such that $Z' \cap U$ is closed in U, hence x is also closed in U; this means that $U \cap \overline{\{x\}} = \{x\}$, in other words x is isolated in $\overline{\{x\}}$. □

The lemma applies to every space underlying a prescheme X, for Z is then also an underlying space of a prescheme (**I**, 5.2.1) and it suffices to take for Z' an open affine *in* Z, which is a Kolmogorov space quasi-compact, hence contains a closed point ($\mathbf{0_I}$, 2.1.3).

(ii) Let Z be the reduced subprescheme of X having $\overline{\{x\}}$ as underlying space; the hypothesis implies that $\{x\}$ is open *in* Z; hence, for every $z \in Z$, the generic point x of $\operatorname{Spec}(\mathscr{O}_{Z,z})$ is *isolated* in $\operatorname{Spec}(\mathscr{O}_{Z,z})$; but $A = \mathscr{O}_{Z,z}$ is a local Noetherian integral ring, and the hypothesis implies that there exists an $f \in A$ such that A_f is a field; one knows (**0**, 16.3.3) that this implies $\dim(A) \leqslant 1$. Since $\dim(\mathscr{O}_{Z,z}) \leqslant 1$ for every $z \in Z$, we have $\dim(Z) \leqslant 1$. **IV-2** | 89

Corollary (5.1.11). — *If X is a locally Noetherian prescheme non-empty and containing a closed point, every closed part of X contains a closed point.*

PROOF. Indeed, every closed part of X is constructible ($\mathbf{0_{III}}$, 9.1.1 and 9.1.5) and it suffices to apply (5.1.10). □

(5.1.12). Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent *of finite type*, $\mathrm{S} = \operatorname{Supp}(\mathscr{F})$ its support, which is *closed* in X ($\mathbf{0_I}$, 5.2.2). If, for every $x \in X$, one considers $\operatorname{Supp}(\mathscr{F}_x)$ as a closed part of the local scheme $\operatorname{Spec}(\mathscr{O}_x)$, by definition (**0**, 16.1.7) $\dim(\mathscr{F}_x) = \dim(\operatorname{Supp}(\mathscr{F}_x))$; but we have

$$\operatorname{Supp}(\mathscr{F}_x) = \mathrm{S} \cap \operatorname{Spec}(\mathscr{O}_{X,x}) = \operatorname{Spec}(\mathscr{O}_{\mathrm{S},x})$$

where, in the last term, S denotes any closed subprescheme of X having S as underlying space. Remarking that the irreducible components of $\operatorname{Spec}(\mathscr{O}_{X,x})$ and the irreducible components of $\operatorname{Spec}(\mathscr{O}_{\mathrm{S},x})$ are the intersections of the irreducible components of S containing x, and correspond bijectively to these latter, one sees that

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{\mathrm{S},x}) \tag{5.1.12.1}$$

by virtue of (5.1.1.1); it also results from (5.1.2) that

$$\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, \mathrm{S}) \tag{5.1.12.2}$$

when X is locally Noetherian.

One says that $\mathscr{F}$ is *equidimensional at* $x \in X$ if $\mathscr{F}_x$ is an equidimensional $\mathscr{O}_{X,x}$-module (**0**, 16.1.7), that is if $\operatorname{Supp}(\mathscr{F}_x)$ is equidimensional as a closed part of $\operatorname{Spec}(\mathscr{O}_x)$; this amounts to saying that the ring $\mathscr{O}_{\mathrm{S},x}$ is *equidimensional*. The *dimension* of $\mathscr{F}$ is denoted $\dim(\mathscr{F})$ and means the dimension of the *support* $\operatorname{Supp}(\mathscr{F})$; it results from (5.1.4) and (5.1.12.1) that

$$\dim(\mathscr{F}) = \sup_{x \in X} \dim(\mathscr{F}_x). \tag{5.1.12.3}$$

If $X = \operatorname{Spec}(A)$ is an affine scheme and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type, one has $\dim(\mathscr{F}) = \dim(M)$ by virtue of (**0**, 16.1.7) and (5.1.4).

Proposition (5.1.13). — *Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite type, x a point of X, x' a generisation of x in X; then*

$$\dim(\mathscr{F}_{x'}) \leqslant \dim(\mathscr{F}_x). \tag{5.1.13.1}$$

PROOF. This follows immediately from (5.1.12.1) and the definitions. □

5.2. Dimension of an algebraic prescheme IV-2 | 90

Proposition (5.2.1). — *Let k be a field, X an irreducible prescheme locally of finite type over k, ξ its generic point. Then X is biequidimensional and we have* $\dim(X) = \deg.\mathrm{tr}_k\, k(\xi)$.

Proof. Every local ring $\mathscr{O}_x$ is the local ring of a k-algebra of finite type, hence a quotient of a regular local ring (**0**, 17.3.9), and one knows (**0**, 16.5.12) that these rings are catenary; since X is irreducible, (5.1.5) shows that X is catenary, and it suffices to show that for every *closed point* $x \in X$, we have

$$\dim(\mathscr{O}_x) = \deg.\mathrm{tr}_k\, k(\xi). \tag{5.2.1.1}$$

One can obviously suppose X reduced and affine, hence integral, with ring of fractions A, $k(\xi) = K$ the field of fractions of A. One knows (Bourbaki, *Alg. comm.*, chap. V, §3, no. 1, th. 1) that there exists a sub-k-algebra $B = k[t_1, \ldots, t_n]$ of A, where the t_i are algebraically independent over k, such that A is a finite B-algebra. Let $\mathfrak{m} = \mathfrak{i}_x$, which by hypothesis is a maximal ideal of A; $\mathfrak{n} = B \cap \mathfrak{m}$ is therefore also a maximal ideal of B (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, prop. 1), and $A_{\mathfrak{m}}$ is a local ring of the $B_{\mathfrak{n}}$-algebra *finite* $\mathrm{S}^{-1}A$, where $\mathrm{S} = B - \mathfrak{n}$; since $B_{\mathfrak{n}}$ is integrally closed and $\mathrm{S}^{-1}A$ integral, $\dim(A_{\mathfrak{m}}) = \dim(B_{\mathfrak{n}})$ (**0**, 16.1.6). One can therefore restrict to the case where $A = k[t_1, \ldots, t_n]$; one knows then that $k' = k(x)$ is a *finite* extension of k (**I**, 6.4.2); let $A' = k'[t_1, \ldots, t_n]$; taking into account (**I**, 2.4.6) and (3.3.14), there is a maximal ideal $\mathfrak{m}'$ of A' above $\mathfrak{m}$ and such that k' is the residue field of $A'_{\mathfrak{m}'}$. Since A' is integral and is a finite A-algebra, the same reasoning as above shows $\dim(A'_{\mathfrak{m}'}) = \dim(A_{\mathfrak{m}})$, so one is reduced to the case where $A/\mathfrak{m} = k$. Now in this case $\mathfrak{m}$ is generated by the polynomials $t_i - a_i$ (where $a_i \in k$, $1 \leqslant i \leqslant n$, the a_i being the images of the t_i in $A/\mathfrak{m}$). Replacing t_i by $t_i - a_i$, one sees that $\mathfrak{m} = \sum_{i=1}^{n} At_i$. The completion of the local ring $A_{\mathfrak{m}}$ is then the ring of formal power series $k[[t_1, \ldots, t_n]]$; one knows that it has the same dimension as $A_{\mathfrak{m}}$, and on the other hand, the dimension of $k[[t_1, \ldots, t_n]]$ is equal to n (**0**, 17.1.4); whence the conclusion. □

Corollary (5.2.2). — *For a prescheme X locally of finite type over a field k,* $\dim(X)$ *coincides with the number defined in* (4.1.1).

Proof. Indeed, if X_α are the reduced subpreschemes having as underlying spaces the irreducible components of X, we have $\dim(X) = \sup_\alpha(\dim(X_\alpha))$ (**0**, 14.1.2.1) and it suffices to apply (5.2.1) to the X_α. □

Corollary (5.2.3). — *Let X be a prescheme locally of finite type over a field k, x a point of X. We have*

$$\dim_x(X) = \dim(\mathscr{O}_x) + \deg.\mathrm{tr}_k\, k(x). \tag{5.2.3.1}$$

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Proof. One knows that there exists an open neighbourhood U of x in X such that $\dim_x(X) = \dim(U)$ (**0**, 14.1.4.1), and one can suppose that the irreducible components of U are the $X_i \cap U$, where the X_i are the irreducible components of X containing x; since $U \cap X_i$ is dense in X_i, it follows from (4.1.1.3) that $\dim(X_i) = \dim(U \cap X_i)$, hence $\dim_x(X) = \sup_i(\dim(X_i))$. Moreover, the minimal prime ideals of $\mathscr{O}_x$ correspond to the generic points of the X_i, hence (**0**, 16.1.1.1), $\dim(\mathscr{O}_x) = \sup_i(\dim(\mathscr{O}_{X_i,x}))$. One is therefore reduced to the case where X is irreducible; by (5.2.1), X is biequidimensional, hence $\dim(X) = \dim(\overline{\{x\}}) + \mathrm{codim}(\overline{\{x\}}, X)$, and one knows that $\dim(\overline{\{x\}}) = \deg.\mathrm{tr}_k\, k(x)$ by (5.1.2) and $\mathrm{codim}(\overline{\{x\}}, X) = \dim(\mathscr{O}_x)$ by (5.1.2). □

Corollary (5.2.4). — *Let X be a prescheme locally of finite type over a field k, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible, f a section of $\mathscr{L}$ above X, Y the set of $x \in X$ such that $f(x) = 0$ and that is rare in X. Then* $\dim(Y) \leqslant \dim(X) - 1$; *the two members of this inequality are equal if Y meets every irreducible component of maximal dimension of X.*

If (X_α) is the family of irreducible components of X, we have

$$\dim(Y) = \sup_\alpha(\dim(Y \cap X_\alpha))$$

and one is therefore reduced to the case where X is irreducible ($Y \cap X_\alpha$ being rare in X_α since X_α has non-empty interior in X). One can restrict to the case $Y \neq \varnothing$; then, for every maximal point x of Y, we have *(since X is biequidimensional)* $\dim(\overline{\{x\}}) = \dim(X) - \operatorname{codim}(\overline{\{x\}}, X)$, *and since Y is rare in X, we have* $\operatorname{codim}(\overline{\{x\}}, X) = 1$ *by (5.1.8); whence the corollary.*

Remarks (5.2.5). — (i) Contrary to what occurs for algebraic k-preschemes, a locally Noetherian prescheme X is not necessarily catenary (cf. (5.6.11)); it is however if all its local rings $\mathscr{O}_x$ are quotients of regular rings (**0**, 16.5.12) and in particular if X is *regular* (**I**, 4.1.4). But even an integral scheme $X = \operatorname{Spec}(A)$, where A is a regular ring, is not necessarily *biequidimensional*, in other words the codimensions in X of the various closed points of X are not necessarily the same.

(ii) When X is a prescheme locally of finite type over a field k, one saw in (4.1.1.3) that $\dim(U) = \dim(X)$ for every open everywhere dense subset U of X. This result does *not extend* to Noetherian preschemes, even if they are biequidimensional.

5.3. Dimension of the support of a Module and Hilbert polynomial This section uses results from chapter III; it will not be used later in this chapter. **IV-2 | 92**

Proposition (5.3.1). — *Let A be a local Artinian ring, X a projective scheme over* $\operatorname{Spec}(A)$, *$\mathscr{L}$ an $\mathscr{O}_X$-Module invertible very ample relative to A, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module $\neq 0$; if we set $\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}$ for $n \in \mathbf{Z}$, the degree of the Hilbert polynomial $\mathrm{P}(n) = \chi_A(\mathscr{F}(n))$ of $\mathscr{F}$ with respect to A (**III**, 2.5.3) is equal to the dimension of* $\operatorname{Supp}(\mathscr{F})$.

Proof. We reason by induction on $d = \dim(\operatorname{Supp}(\mathscr{F}))$. One knows that there exists a closed subprescheme Y of X whose $\operatorname{Supp}(\mathscr{F})$ is the underlying space, and an $\mathscr{O}_Y$-Module coherent $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, where $j : Y \to X$ is the canonical injection ($\mathbf{Err}_{\mathbf{III}}$, 30). It is immediate that the Hilbert polynomials of $\mathscr{F}$ and of $\mathscr{G}$ are the same, so one can restrict to the case where $X = \operatorname{Supp}(\mathscr{F})$. Suppose first $d = 0$; all the points of X are closed, X is an Artinian scheme (**I**, 6.2.2), hence $\mathscr{F}(n) = \mathscr{F}$ for every integer n, and one has by (**III**, 2.5.3)

$$\chi_A(\mathscr{F}(n)) = \operatorname{long}_A(\Gamma(X, \mathscr{F}))$$

for n large enough, and the degree of the polynomial is therefore 0. Suppose now $d > 0$, and set $Z = \operatorname{Ass}(\mathscr{F})$, which is a finite set (3.1.6); there exists an integer $m > 0$ and a section $f \in \Gamma(X, \mathscr{L}^{\otimes m})$ such that X_f is a neighbourhood of Z (**II**, 4.5.4). Multiplication by f is a homomorphism

$$\mu_f : \mathscr{F} \longrightarrow \mathscr{F}(m)$$

which is injective (3.1.8); one therefore has an exact sequence

$$0 \longrightarrow \mathscr{F} \xrightarrow{\mu_f} \mathscr{F}(m) \longrightarrow \mathscr{G} \longrightarrow 0 \tag{5.3.1.1}$$

where $\mathscr{G}$ is coherent. By virtue of the Nakayama lemma, the points $x \in \operatorname{Supp}(\mathscr{G})$ are exactly those for which $f(x) = 0$. We shall now deduce that

$$\dim(\operatorname{Supp}(\mathscr{G})) = d - 1. \tag{5.3.1.2}$$

This will result from the following lemma: □

Lemma (5.3.1.3). — *Let A be a local Artinian ring, X a projective scheme over* $\operatorname{Spec}(A)$, *$\mathscr{L}$ an $\mathscr{O}_X$-Module invertible ample relative to A; then, for every section g of $\mathscr{L}$ above X such that $g(x) = 0$ meets every irreducible component of X of dimension $\neq 0$.*

Proof. Indeed (**II**, 5.5.7), the set X_g is an open affine of X, and if it contains an irreducible component X' of X, the reduced closed subprescheme of X having X' as underlying space is both projective and affine over A (**III**, 4.4.2), and is therefore finite over A, hence an Artinian scheme, therefore of dimension 0. □

This lemma being established, note that since Z contains the maximal points of X, X_f is dense, hence $\operatorname{Supp}(\mathscr{G})$ is rare in X, and the relation (5.3.1.2) results from the lemma and from (5.2.4).

From the exact sequence (5.3.1.1), we deduce, for every n, the exact sequence

$$0 \longrightarrow \mathscr{F}(n) \longrightarrow \mathscr{F}(n+m) \longrightarrow \mathscr{G}(n) \longrightarrow 0$$

and for n large enough, one therefore has the exact sequence (**III**, 2.2.3)

$$0 \longrightarrow \Gamma(X, \mathscr{F}(n)) \longrightarrow \Gamma(X, \mathscr{F}(n+m)) \longrightarrow \Gamma(X, \mathscr{G}(n)) \longrightarrow 0$$

whence, taking into account (**III**, 2.5.3), for n large enough,

$$\chi_A(\mathscr{G}(n)) = \chi_A(\mathscr{F}(n+m)) - \chi_A(\mathscr{F}(n)).$$

Since by virtue of (5.3.1.2) and the induction hypothesis, the degree of the polynomial $\chi_A(\mathscr{G}(n))$ is $d-1$, the preceding relation implies that the degree of $\chi_A(\mathscr{F}(n))$ is d, Q.E.D.

5.4. Dimension of the image of a morphism

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Proposition (5.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism.*

(i) *If f is quasi-finite, we have* $\dim(X) \leqslant \dim(\overline{f(X)}) \leqslant \dim(Y)$.

(ii) *If f is surjective and open* (*resp. surjective and closed*), *we have* $\dim(X) \geqslant \dim(Y)$.

Proof. (i) One can replace f by f_{red} (**II**, 6.2.4), hence suppose X and Y reduced; if Z is the closed reduced subprescheme of Y having $\overline{f(X)}$ as underlying space, we have $f = j \circ g$, where $j : Z \to Y$ is the canonical injection and $g : X \to Z$ is a quasi-finite morphism (**I**, 5.2.2) and (**II**, 6.2.4). One can therefore restrict to the case where $f(X)$ is dense in Y. For every $x \in X$, $\mathscr{O}_x$ is then an $\mathscr{O}_{f(x)}$-module quasi-finite (**II**, 6.2.2), and it follows that $\mathfrak{m}_{f(x)}\mathscr{O}_x$ is an ideal of definition of $\mathscr{O}_x$ ($\mathbf{0}_{\mathbf{I}}$, 7.4.4); but one knows (**0**, 16.3.10) that if $A \to B$ is a local homomorphism of local Noetherian rings, such that, if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}B$ is an ideal of definition of B, then $\dim(B) \leqslant \dim(A)$; this finishes proving (i) by virtue of (5.1.4).

(ii) The definition of dimension (**0**, 14.1.2) shows that it suffices to prove that for every suite $(y_i)_{0\leqslant i\leqslant n}$ of distinct elements of Y such that $y_i \in \overline{\{y_{i+1}\}}$ for $0 \leqslant i \leqslant n-1$, there exists a suite $(x_i)_{0\leqslant i\leqslant n}$ of points of X such that $x_i \in \overline{\{x_{i+1}\}}$ for $0 \leqslant i \leqslant n-1$ and $f(x_i) = y_i$ for every i. Suppose first f surjective and *open*, and let us show the existence of x_i by induction on i; the existence of $x_0 \in X$ such that $f(x_0) = y_0$ results from the fact that f is surjective. If the x_i have been determined for $i \leqslant m$ in a way satisfying $f(x_i) = y_i$ for $i \leqslant m$ and $x_i \in \overline{\{x_{i+1}\}}$ for $i \leqslant m$, we note that since f is surjective and y_{m+1} a generisation of y_m (1.10.3), there exists $x_{m+1} \in f^{-1}(y_{m+1})$ which is a generisation of x_m and the induction can continue.

Suppose now f surjective and *closed*, and let us show the existence of x_i by *descending* induction on i; the existence of x_n such that $f(x_n) = y_n$ results again from the fact that f is surjective. Suppose now f surjective and closed and let us show the existence of the x_i by descending induction on i: if the x_i have been determined satisfying the conditions sought for $i > m$, we note $f(\overline{\{x_{m+1}\}}) = \overline{\{y_{m+1}\}}$ since f is closed (Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., §5, no. 4, prop. 9); there exists therefore $x_m \in \overline{\{x_{m+1}\}}$ such that $f(x_m) = y_m$ and the descending induction can continue. □

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Corollary (5.4.2). — *If X, Y are two locally Noetherian preschemes, $f : X \to Y$ a finite morphism* (*hence closed*), *we have* $\dim(X) = \dim(\overline{f(X)})$. *If moreover f is surjective,* $\dim(X) = \dim(Y)$.

Remarks (5.4.3). — (i) One saw in (4.1.2) that if X and Y are two preschemes locally of finite type over a field k and f a k-morphism, the inequality $\dim(Y) \leqslant \dim(X)$ is already verified when f is *quasi-compact and dominant*. On the contrary, one can have $\dim(Y) > \dim(X)$ even when f is of *finite type, bijective* and a *local immersion*, if one only supposes X and Y locally Noetherian.

(ii) If A and B are two Noetherian rings such that $A \subset B$ and B is a finite A-algebra, the corollary (5.4.2) shows again that $\dim(B) = \dim(A)$ (**0**, 16.1.5). Suppose moreover that A is a *local* Noetherian ring; then B is a Noetherian *semi-local* ring (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 3 of prop. 9); if $\mathfrak{n}_i$ ($1 \leqslant i \leqslant r$) are the maximal ideals of B, we have therefore

(5.4.3.1) $$\dim(A) = \sup_i \dim(B_{\mathfrak{n}_i}).$$

One will observe that, under these conditions, one does not necessarily have $\dim(B_{\mathfrak{n}_i}) = \dim(A)$ for *every* i.

5.5. Dimension formula for a morphism of finite type

(5.5.1). Recall (**0**, 16.3.9) that if A, B are two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, we have

(5.5.1.1) $$\dim(B) \leqslant \dim(A) + \dim(B \otimes_A k).$$

We deduce:

Proposition (5.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, x a point of X, $y = f(x)$. Then*

(5.5.2.1) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)).$$

In particular, if x is a maximal point of the fibre $f^{-1}(y)$, we have **IV-2** | 95

(5.5.2.2) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y)$$

since $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$ is the local ring of x in the prescheme $f^{-1}(y)$, and is of dimension 0 by hypothesis.

We shall now obtain a more precise formula than (5.5.2.1) when one supposes that f is a morphism *of finite type*.

Proposition (5.5.3). — *Let A be a Noetherian ring,* T *an indeterminate, $\mathfrak{p}$ a prime ideal of A; then $\mathfrak{p}' = \mathfrak{p}A[\mathrm{T}]$ is a prime ideal of $B = A[\mathrm{T}]$ and $\mathfrak{p}' \cap A = \mathfrak{p}$. There exists an infinity of prime ideals of B distinct from $\mathfrak{p}'$, whose intersection with A is $\mathfrak{p}$; these ideals are pairwise without inclusion relation. Moreover, if $\mathfrak{q}$ is such an ideal, we have*

(5.5.3.1) $$\dim(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{p}'}) + 1 = \dim(A_{\mathfrak{p}}) + r$$

PROOF. In the first assertions, we immediately reduce, by replacing A by $A/\mathfrak{p}$ and observing that $A[\mathrm{T}]/\mathfrak{p}A[\mathrm{T}] = (A/\mathfrak{p})[\mathrm{T}]$, to the case $\mathfrak{p} = 0$; they then result from the fact that the prime ideals of $B = A[\mathrm{T}]$ whose intersection with A reduces to 0 are exactly those which do not meet the multiplicative subset $S = A - \{0\}$ of the integral ring A; now there is an increasing bijection from the set of these prime ideals onto the set of prime ideals of $S^{-1}A[\mathrm{T}] = K[\mathrm{T}]$, where K is the field of fractions of A *(Bourbaki, Alg. comm., chap. II, §2, no. 5, prop. 11)*. Moreover, by (5.5.1.2), $\dim(B_{\mathfrak{q}}) \leqslant \dim(A_{\mathfrak{p}}) + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})$, and if k is the field of fractions of $A/\mathfrak{p}$, $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ identifies canonically with $(k[\mathrm{T}])_{\bar{\mathfrak{q}}}$, hence is a discrete valuation ring, hence of dimension 1. Finally, if $\mathfrak{p} = \mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_m$ is a chain of prime ideals of A of maximal length, the ideals $\mathfrak{p}_j B$ $(0 \leqslant j \leqslant m)$ are prime, pairwise distinct in B, and contained in $\mathfrak{q}$; so $\dim(B_{\mathfrak{q}}) \geqslant m + 1 = \dim(A_{\mathfrak{p}}) + 1$ and hence $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1$. This relation also writes $\mathrm{ht}(\mathfrak{q}) = \mathrm{ht}(\mathfrak{p}) + 1$; since $\mathfrak{q} \neq \mathfrak{p}'$, we also have $\mathrm{ht}(\mathfrak{q}) \geqslant \mathrm{ht}(\mathfrak{p}') + 1 \geqslant \mathrm{ht}(\mathfrak{p}) + 1$ by definition of the height of a prime ideal; this finishes proving (5.5.3.1). □

Corollary (5.5.4). — *For every Noetherian ring A, we have*

(5.5.4.1) $$\dim(A[\mathrm{T}_1, \ldots, \mathrm{T}_r]) = \dim(A) + r$$

(T_i *indeterminates*).

Corollary (5.5.5). — *For every locally Noetherian prescheme X, the dimension of $X[\mathrm{T}_1, \ldots, \mathrm{T}_r] = X \otimes_{\mathbf{Z}} \mathbf{Z}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ (T_i indeterminates) is $\dim(X) + r$.*

PROOF. This results from (5.5.4) and (**0**, 14.1.7). □

Corollary (5.5.6). — *Under the hypotheses of (5.5.3), let $\mathfrak{q}$ be a prime ideal of $B = A[\mathrm{T}]$ such that $\mathfrak{q} \cap A = \mathfrak{p}$; if k and k' are the residue fields of $A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}$ respectively, we have*

(5.5.6.1) $$\dim(A_{\mathfrak{p}}) + 1 = \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_k\, k'.$$

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PROOF. If $\mathfrak{q} = \mathfrak{p}B$, by (5.5.3.1), $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}})$, and in this case $k' = k(\mathrm{T})$, so formula (5.5.6.1) is indeed verified. In the contrary case, $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1$, and since $\mathfrak{q}$ corresponds to a nonzero prime ideal $\mathfrak{q}' \neq 0$ of $k[\mathrm{T}]$, k' is an algebraic extension of k, so we again have formula (5.5.6.1). □

Lemma (5.5.7). — *Let A be a local integral Noetherian ring, $\mathfrak{m}$ its maximal ideal, k its residue field.*

label=(i) *For every integral ring B containing A, such that $B = A[x]$ (for some $x \in B$) and every prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{m}$, we have*

$$\dim(A) + \deg.\mathrm{tr}_A B \geqslant \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_k k' \tag{5.5.7.1}$$

where k' denotes the residue field of $B_\mathfrak{q}$ and $\deg.\mathrm{tr}_A B$ the degree of transcendence of the field of fractions of B over that of A.

lbbel=(ii) *Suppose that for every maximal ideal $\mathfrak{n}$ of $A[\mathrm{T}]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, the ring $(A[\mathrm{T}])_\mathfrak{n}$ is catenary; then the two members of (5.5.7.1) are equal.*

Proof. *(i)* If x is transcendental over the field of fractions of A, then $\deg.\mathrm{tr}_A B = 1$ and the two members of (5.5.7.1) are equal by virtue of (5.5.6). In the contrary case, $B = A[\mathrm{T}]/\mathfrak{p}$, where $\mathfrak{p}$ is a nonzero prime ideal of $A[\mathrm{T}]$ such that $\mathfrak{p} \cap A = 0$ since B contains A; so $\mathrm{ht}(\mathfrak{p}) = 1$ by virtue of (5.5.3). The ideal $\mathfrak{q}$ of B is of the form $\mathfrak{n}/\mathfrak{p}$, where $\mathfrak{n} \supset \mathfrak{p}$ is a prime ideal of $A[\mathrm{T}]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, and $B_\mathfrak{q} = (A[\mathrm{T}])_\mathfrak{n}/\mathfrak{p}(A[\mathrm{T}])_\mathfrak{n}$; formula (**0**, 16.1.4.1), applied to $X = \mathrm{Spec}((A[\mathrm{T}])_\mathfrak{n})$ and $Y = \mathrm{Spec}(B_\mathfrak{q})$, gives

$$\dim((A[\mathrm{T}])_\mathfrak{n}) \geqslant \mathrm{ht}(\mathfrak{p}(A[\mathrm{T}])_\mathfrak{n}) + \dim(B_\mathfrak{q}) \tag{5.5.7.2}$$

since $\mathrm{ht}(\mathfrak{p}(A[\mathrm{T}])_\mathfrak{n}) = \mathrm{ht}(\mathfrak{p}) = 1$ by virtue of the bijective correspondence between prime ideals of $A[\mathrm{T}]$ contained in $\mathfrak{n}$ and prime ideals of $(A[\mathrm{T}])_\mathfrak{n}$. Finally, formula (5.5.6.1) gives

$$\dim((A[\mathrm{T}])_\mathfrak{n}) = \dim(A) + 1 - \deg.\mathrm{tr}_k k' \tag{5.5.7.3}$$

since the residue fields of $B_\mathfrak{q}$ and of $(A[\mathrm{T}])_\mathfrak{n}$ are the same; moreover, $\deg.\mathrm{tr}_A B = 0$, which finishes proving (5.5.7.1).

(ii) If $(A[\mathrm{T}])_\mathfrak{n}$ is catenary, the two members of (5.5.7.2) are also equal (**0**, 16.1.4), hence so are the two members of (5.5.7.1). □

Theorem (5.5.8). — (Dimension formula). *Let A be a local Noetherian integral ring, B an integral ring containing A and which is an A-algebra of finite type, $\mathfrak{q}$ a prime ideal of B such that $\mathfrak{q} \cap A$ is the maximal ideal $\mathfrak{m}$ of A, k and k' the residue fields of A and $B_\mathfrak{q}$ respectively. We then have the inequality*

$$\dim(A) + \deg.\mathrm{tr}_A B \geqslant \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_k k'. \tag{5.5.8.1}$$

Moreover, the two members are equal if, for every sub-A-algebra A' of finite type of $B[\mathrm{T}]$ and every maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m}' \cap A = \mathfrak{m}$, $A'_{\mathfrak{m}'}$ is catenary.

Proof. Let $B = A[x_1, \ldots, x_n]$ and argue by induction on n. Set $C = A[x_1, \ldots, x_{n-1}]$ and $\mathfrak{r} = \mathfrak{q} \cap C$; $C_\mathfrak{r}$ is a local integral Noetherian ring, and if we set $S = C - \mathfrak{r}$, $B' = S^{-1}B$, $B' = C_\mathfrak{r}[x_n]$, and $\mathfrak{q}' = S^{-1}\mathfrak{q}$, we have $B_\mathfrak{q} = B'_{\mathfrak{q}'}$ and $\mathfrak{q}' \cap C_\mathfrak{r} = \mathfrak{r}C_\mathfrak{r}$; moreover the fields of fractions of B' and $C_\mathfrak{r}$ are those of B and C respectively. By (5.5.7.1), we thus have

$$\dim(C_\mathfrak{r}) + \deg.\mathrm{tr}_C B \geqslant \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_{k_1} k'. \tag{5.5.8.2}$$

On the other hand, the induction hypothesis gives **IV-2** | 97

$$\dim(A) + \deg.\mathrm{tr}_A C \geqslant \dim(C_\mathfrak{r}) + \deg.\mathrm{tr}_k k_1 \tag{5.5.8.3}$$

whence (5.5.8.1) by adding (5.5.8.2) and (5.5.8.3) member by member. To prove the second assertion, note first that every sub-A-algebra of finite type of $C[\mathrm{T}]$ is also a sub-A-algebra of finite type of $B[\mathrm{T}]$; so by induction on n, we can suppose that the two members of (5.5.8.3) are equal. On the other hand, to see that the two members of (5.5.8.2) are equal, it suffices, by virtue of (5.5.7), to verify that if $\mathfrak{n}$ is a maximal ideal of $C_\mathfrak{r}[\mathrm{T}]$ such that $\mathfrak{n} \cap C_\mathfrak{r} = \mathfrak{r}C_\mathfrak{r}$, then $(C_\mathfrak{r}[\mathrm{T}])_\mathfrak{n}$ is catenary; but $C_\mathfrak{r}[\mathrm{T}] = S'^{-1}C[\mathrm{T}]$ where $S' = C - \mathfrak{r}$; the ideal $\mathfrak{n}$ is thus of the form $S'^{-1}\mathfrak{n}'$, where $\mathfrak{n}'$ is a prime ideal of $C[\mathrm{T}]$ such that $\mathfrak{n}' \cap C = \mathfrak{r}$, whence $(C_\mathfrak{r}[\mathrm{T}])_\mathfrak{n} = (C[\mathrm{T}])_{\mathfrak{n}'}$; if $\mathfrak{m}'$ is a maximal ideal of $C[\mathrm{T}]$ containing $\mathfrak{n}'$, $(C[\mathrm{T}])_{\mathfrak{m}'}$ is thus a local ring of $(C[\mathrm{T}])_{\mathfrak{n}'}$, and since by hypothesis this last ring is catenary, so is the former by (**0**, 16.1.4). □

5.6. Dimension formula and universally catenary rings

Proposition (5.6.1). — *Let A be a Noetherian ring. The following properties are equivalent:*

label=a) *Every polynomial ring $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ is catenary.*
lbbel=b) *Every A-algebra of finite type is catenary.*
lcbel=c) *A is catenary, and for every local integral A-algebra, essentially of finite type (1.3.8), B', designating by $\mathfrak{p}$ the inverse image in A of the maximal ideal $\mathfrak{q}$ of B', by A' the image of $A_\mathfrak{p}$ in B', by k and k' the residue fields of A' and B' respectively,*

$$\dim(A') + \deg.\mathrm{tr}_{A'}(B') = \dim(B') + \deg.\mathrm{tr}_k k'. \tag{5.6.1.1}$$

Proof. The fact that *b)* implies *c)* results from (5.5.8); indeed, B' is an A'-algebra essentially of finite type (1.3.10), hence of the form $B''_{\mathfrak{q}''}$, where B'' is a sub-A'-algebra of finite type of B', $\mathfrak{q}''$ a prime ideal of B'' above the maximal ideal $\mathfrak{p}'$ of A'; moreover the fields of fractions of B' and B'' are the same, hence $\deg.\mathrm{tr}_{A'}(B') = \deg.\mathrm{tr}_{A'}(B'')$. To prove (5.6.1.1), it suffices to show that (under hypothesis *b)*) every sub-A'-algebra A_1 of finite type of $B'[\mathrm{T}]$ is catenary; indeed the two members of (5.5.8.1), where A, B, and $\mathfrak{q}$ are replaced by A', B'', and $\mathfrak{q}''$, will then be equal, whence the equality (5.6.1.1). But hypothesis *b)* implies that every A-algebra essentially of finite type is catenary (**0**, 16.1.4), and A_1 is such an A-algebra (1.3.9).

It is trivial that *b)* implies *a)*; conversely, *a)* implies *b)*, since every A-algebra of finite type is a quotient of a polynomial algebra (**0**, 16.1.4).

It remains to prove that *c)* implies *b)*. Since every quotient by a prime ideal of an A-algebra of finite type is an integral A-algebra of finite type, if B is an integral A-algebra of finite type, $\mathfrak{q}$ and $\mathfrak{q}'$ two prime ideals of B such that $\mathfrak{q}' \subset \mathfrak{q}$, by (**0**, 16.1.4.2) **IV-2 | 98**

$$\dim(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) + \dim(B_{\mathfrak{q}'}) = \dim(B_\mathfrak{q}). \tag{5.6.1.2}$$

Let $\mathfrak{p}$, $\mathfrak{p}'$ be the inverse images of $\mathfrak{q}$, $\mathfrak{q}'$ respectively in A, $\mathfrak{n}$ the kernel of the homomorphism $A_\mathfrak{p} \to B_\mathfrak{q}$. The image A' of $A_\mathfrak{p}$ in $B_\mathfrak{q}$ being isomorphic to $A_\mathfrak{p}/\mathfrak{n}$, formula (5.6.1.1) applied to A' and $B' = B_\mathfrak{q}$ writes

$$\dim(A_\mathfrak{p}/\mathfrak{n}) + \deg.\mathrm{tr}_{A_\mathfrak{p}/\mathfrak{n}}(B_\mathfrak{q}) = \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_{k(\mathfrak{p})} k(\mathfrak{q}). \tag{5.6.1.3}$$

On the other hand, the kernel of $A_{\mathfrak{p}'} \to B_{\mathfrak{q}'}$ is $\mathfrak{n}A_{\mathfrak{p}'}$, so, applying formula (5.6.1.1) where A' is replaced by $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and B' by $B_{\mathfrak{q}'}$, we have

$$\dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) + \deg.\mathrm{tr}_{A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}}(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}'}) + \deg.\mathrm{tr}_{k(\mathfrak{p}')} k(\mathfrak{q}'). \tag{5.6.1.4}$$

Finally, $B/\mathfrak{q}'$ is an integral A-algebra of finite type, the inverse image of $\mathfrak{q}/\mathfrak{q}'$ in A is $\mathfrak{p}$, and the kernel of the homomorphism $A_\mathfrak{p} \to B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}$ is $\mathfrak{p}'A_\mathfrak{p}$, so, applying (5.6.1.1) where A' is replaced by $A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}$ and B' by $B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}$, we have

$$\dim(A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}) + \deg.\mathrm{tr}_{A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}}(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) = \dim(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) + \deg.\mathrm{tr}_{k(\mathfrak{p})} k(\mathfrak{q}). \tag{5.6.1.5}$$

Adding (5.6.1.4) and (5.6.1.5) member by member, and noting that $k(\mathfrak{p}')$ (resp. $k(\mathfrak{q}')$) is the field of fractions of $A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}$ (resp. $B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}$); moreover $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and $A_\mathfrak{p}/\mathfrak{n}$ have the same field of fractions, and $B_{\mathfrak{q}'}$ and $B_\mathfrak{q}$ have the same field of fractions; finally, since A is supposed catenary, by (**0**, 16.1.4.2)

$$\dim(A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}) + \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) = \dim(A_\mathfrak{p}/\mathfrak{n}).$$

Taking these remarks into account and (5.6.1.3), we obtain relation (5.6.1.2). □

Definition (5.6.2). — A Noetherian ring A is said to be *universally catenary* if it satisfies the equivalent conditions *a)*, *b)*, *c)* of (5.6.1).

Remarks (5.6.3). — *(i)* If A is universally catenary, then so is $S^{-1}A$ for every multiplicative subset S of A, as follows immediately from ((5.6.1), *a)*) and the fact that every ring of fractions of a catenary ring is catenary. Conversely, if for every prime ideal $\mathfrak{p}$ of A, the ring $A_\mathfrak{p}$ is universally catenary, then A is universally catenary: this follows from ((5.6.1), *c)*), noting that by setting $S = A - \mathfrak{p}$, $S^{-1}B$ is an $A_\mathfrak{p}$-algebra of finite type, and $B_\mathfrak{q} = (S^{-1}B)_{S^{-1}\mathfrak{q}}$.

(ii) A locally Noetherian prescheme X is said to be *universally catenary* if, for every $x \in X$, the ring $\mathscr{O}_x$ is universally catenary. It follows from (i) that for A to be a universally catenary ring, it is necessary and sufficient that the scheme $\operatorname{Spec}(A)$ be universally catenary.

(iii) The criterion ((5.6.1), *c)*) only involves the images of A in integral A-algebras of finite type, hence *integral quotient rings* of A. We conclude that if $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ are the minimal prime ideals of A, it amounts to the same to say that A is universally catenary or that each of the $A/\mathfrak{p}_i$ is. This also shows that a locally Noetherian prescheme X is universally catenary, or that X_{red} is.

(iv) The criterion ((5.6.1), *b)*) and remark (i) show that, if A is a universally catenary ring, then so is every A-algebra *essentially of finite type* (1.3.8). **IV-2** | 99

Proposition (5.6.4). — *Every quotient of a regular ring is universally catenary.*

PROOF. It suffices to see that every regular ring A is universally catenary ((5.6.3), (iv)); or, we know that $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ is regular (**0**, 17.3.7), hence catenary (**0**, 16.5.12), and we conclude by ((5.6.1), *a)*). □

In particular, it follows from the Cohen structure theorem (**0**, 19.8.8) that every *complete local Noetherian ring* is universally catenary. Similarly, every algebra of finite type over a field, every prescheme being a quotient of a regular prescheme, and every prescheme *locally of finite type over a field* is universally catenary.

We will see further on (6.3.7) that in (5.6.4), one can replace "regular ring" by "Cohen-Macaulay ring".

Proposition (5.6.5). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a dominant morphism locally of finite type. Let ξ (resp. η) be the generic point of X (resp. Y), and let $e = \dim(f^{-1}(\eta)) = \deg.\mathrm{tr}_{k(\eta)}\, k(\xi)$* ("dimension of the generic fibre", cf. ($\mathbf{0_I}$, 2.1.8) and (4.1.1)). *For every $x \in X$, setting $y = f(x)$, we have*

(5.6.5.1) $$e + \dim(\mathscr{O}_y) \geqslant \deg.\mathrm{tr}_{k(y)}\, k(x) + \dim(\mathscr{O}_x)$$

(5.6.5.2) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) - \delta(x)$$

where $\delta(x) = \dim_x(f^{-1}(y)) - e$.

Moreover, if Y is universally catenary, the two members of (5.6.5.1) (*resp.* (5.6.5.2)) *are equal. If furthermore x is closed in $f^{-1}(y)$, we have*

(5.6.5.3) $$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + e.$$

PROOF. We can evidently restrict to the case where X and Y are affine (hence f of finite type) and reduced (1.5.4), hence integral. Since f is dominant, $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_x$, and the fields of fractions of $\mathscr{O}_y$ and of $\mathscr{O}_x$ are $k(\eta)$ and $k(\xi)$ respectively (**I**, 8.2.7); moreover, $\mathscr{O}_x$ is a local ring of an $\mathscr{O}_y$-algebra integral of finite type B at a prime ideal $\mathfrak{q}$ (**I**, 3.6.5); applying (5.5.8.1) with $A = \mathscr{O}_y$, B, and $\mathfrak{q}$, we obtain (5.6.5.1). Moreover, $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y) = \mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$ is none other than the local ring of the fibre $f^{-1}(y)$ at the point x (**I**, 3.6.1); since $f^{-1}(y)$ is a prescheme of finite type over $k(y)$, it follows from (5.2.3) that

$$\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) = \dim_x(f^{-1}(y)) - \deg.\mathrm{tr}_{k(y)}\, k(x)\,;$$

so the inequality (5.6.5.2) is just another form of (5.6.5.1).

The fact that the two members of (5.6.5.1) are equal when Y is universally catenary is none other than the equality (5.6.1.1), applied to $A = \mathscr{O}_y$ and $B' = \mathscr{O}_x$. To show that moreover (5.6.5.3) holds when x is closed in $f^{-1}(y)$, it suffices to note that, in general, $\deg.\mathrm{tr}_{k(y)}\, k(x)$ is the dimension of $\overline{\{x\}} \cap f^{-1}(y)$, as follows from (5.2.1) applied to the closed reduced sub-prescheme of $f^{-1}(y)$ having this subspace as underlying space. **IV-2** | 100

We will show further on (13.1.1) that, under the conditions of (5.6.5), we *always* have $\delta(x) \geqslant 0$, and (5.6.5.2) makes precise in this case (5.2.1). □

Corollary (5.6.6). — *Under the general hypotheses of (5.6.5), we have*

$$\dim(X) \leqslant \dim(Y) + e. \tag{5.6.6.1}$$

If we furthermore suppose Y universally catenary, then, for the two members of (5.6.6.1) to be equal, it is necessary and sufficient that

$$\dim(Y) = \sup_{y \in f(X)} \dim(\mathscr{O}_y). \tag{5.6.6.2}$$

In particular, the two members of (5.6.6.1) are equal when Y is locally of finite type over a field.

PROOF. The inequality (5.6.6.1) results from (5.6.5.1) applied to a closed point x of X, taking into account (5.1.4.2) and (5.1.11). On the other hand, every nonempty fibre $f^{-1}(y)$ contains a closed point x in this fibre (5.1.11), and if we suppose Y universally catenary, at this point we have the relation (5.6.5.3). Taking the upper bounds of the two members of (5.6.5.3) when x runs through X, and noting that every closed point x of X is *a fortiori* closed in $f^{-1}(f(x))$, the second assertion of the corollary follows from (5.1.4.1) and (5.1.4.2).

We note that condition (5.6.6.2) is trivially satisfied when f is surjective. □

Corollary (5.6.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, n an integer. If, for every $y \in Y$, $\dim(f^{-1}(y)) \leqslant n$, then*

$$\dim(X) \leqslant \dim(Y) + n. \tag{5.6.7.1}$$

PROOF. By virtue of $(\mathbf{0}, 14.1.7)$ and of (1.5.4), we can restrict to the case where X and Y are affine, hence f of finite type, X and Y reduced; let X_i $(1 \leqslant i \leqslant m)$ be the sub-preschemes (integral) of X having for underlying spaces the irreducible components of X; we have $\dim(X) = \sup(\dim(X_i))$. If Z_i is the closed reduced sub-prescheme of Y having for underlying space $\overline{f(X_i)}$, Z_i is integral $(\mathbf{0_I}, 2.1.5)$; the restriction $X_i \to Y$ of f factors as $X_i \xrightarrow{g_i} Z_i \xrightarrow{j_i} Y$, where j_i is the canonical injection (**I**, 5.2.2), and g_i is dominant and of finite type (1.5.4). We have $\dim(Z_i) \leqslant \dim(Y)$ for every i; on the other hand, if z_i is the generic point of Z_i, $\dim(g_i^{-1}(z_i)) \leqslant n$ for every i by hypothesis; so to prove (5.6.7.1), it suffices to show that $\dim(X_i) \leqslant \dim(Z_i) + n$ for every i, which follows from (5.6.6.1). □ **IV-2 | 101**

Corollary (5.6.8). — *Let Y be a locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a morphism locally of finite type, n an integer $\geqslant 0$. Suppose that Y is universally catenary, and that for every $y \in Y$, $\dim(f^{-1}(y)) \geqslant n$ (resp. $\dim(f^{-1}(y)) = n$). Then*

$$\dim(X) \geqslant \dim(Y) + n \tag{5.6.8.1}$$

(*resp.*

$$\dim(X) = \dim(Y) + n \tag{5.6.8.2}$$

).

PROOF. Since $n \geqslant 0$, the hypothesis implies that f is *surjective*, hence Y irreducible; moreover, if η is the generic point of Y, $\dim(f^{-1}(\eta)) \geqslant n$ (resp. $\dim(f^{-1}(\eta)) = n$). The conclusion follows from (5.6.6). □

Remarks (5.6.9). — *(i)* Even if Y is regular, X irreducible, $f : X \to Y$ dominant and of finite type, the two members of (5.6.6.1) are not necessarily equal, as shown by the example where $Y = \operatorname{Spec}(A)$, A a discrete valuation ring, $X = \operatorname{Spec}(K)$ where K is the field of fractions of A, f being the canonical morphism.

(ii) Example ((5.4.3), (i)) shows that in (5.6.6) and (5.6.8), one cannot suppress the hypothesis that X is irreducible, the other hypotheses being satisfied. However, we will see (10.6.1) that with additional hypotheses on Y, satisfied for example when Y is a prescheme locally of finite type over a field, or over a Dedekind ring having an infinity of prime ideals, such phenomena cannot occur.

Proposition (5.6.10). — *Let A be a local integral Noetherian universally catenary ring, B an integral ring containing A and which is an A-algebra of finite type. Then, for every maximal ideal $\mathfrak{n}$ of B, $\dim(B_{\mathfrak{n}}) = \dim(A)$.*

PROOF. Indeed, $\deg.\mathrm{tr}_A B = 0$ and the residue field k' of $B_{\mathfrak{n}}$ is an algebraic extension of the field of fractions of A. The conclusion follows from formula (5.6.1.1), every maximal ideal being above that of A. ☐

Example (5.6.11). — Let A be a local Noetherian integral ring and *integrally closed*; then, as we have seen ($\mathbf{0}$, 16.1.6), the conclusion of (5.6.10) is valid for every integral finite A-algebra B containing A. On the other hand, we will construct an example of a local Noetherian integral *catenary* ring A for which the conclusion of (5.6.10) will fail. We use the construction of ((5.2.5), (i)). Let k_0 be a field, k a pure transcendental extension $k_0(S_i)_{i\in\mathbf{N}}$ of infinite transcendence degree, V the discrete valuation ring $k[\mathrm{S}]_{(\mathrm{S})}$, local ring of the polynomial ring $k[\mathrm{S}]$ at the prime ideal (S); finally let E be the polynomial ring $V[\mathrm{T}]$; we have seen that in E the maximal ideal $\mathfrak{m} = (\mathrm{S}) + (\mathrm{T})$ is of height 2 and the maximal ideal $\mathfrak{m}' = (\mathrm{ST} - 1)$ of height 1; the residue fields are $k(\mathfrak{m}) = k$ and $k(\mathfrak{m}') = k(\mathrm{S})$, fields of fractions of V; by choice of k, these fields are isomorphic. Let ε and ε' be the canonical homomorphisms of E onto $E/\mathfrak{m} = k(\mathfrak{m})$ and $E/\mathfrak{m}' = k(\mathfrak{m}')$ respectively; let σ be an isomorphism of $k(\mathfrak{m})$ onto $k(\mathfrak{m}')$, and consider the subring C of E formed by $x \in E$ such that $\varepsilon'(x) = \sigma(\varepsilon(x))$ (this construction is a particular case of "gluing procedures" which will be systematically studied in chap. V). It is immediate that $\mathfrak{n} = \mathfrak{m}\mathfrak{m}' = \mathfrak{m} \cap \mathfrak{m}'$ is a maximal ideal of C, $C/\mathfrak{m}\mathfrak{m}'$ identifying with the subfield of $(E/\mathfrak{m}) \times (E/\mathfrak{m}')$ formed by the pairs $(z, \sigma(z))$. We have $E = C + C(\mathrm{ST} - 1)$, hence E is a C-algebra of finite type; we will now see that C is Noetherian: this will result from the following lemma:

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Lemma (5.6.11.1). — *Let R be a ring, S a subring, $\mathfrak{R} = \mathrm{Ann}_S(R/S)$ the conductor of S in R (largest ideal of S which is also an ideal of R).*

label=(i) For every ideal $\mathfrak{J} \subset \mathfrak{R}$ of R, there exists a strictly increasing map from the set of ideals $\mathfrak{L}$ of S such that $R \cdot \mathfrak{L} = \mathfrak{J}$ onto the set of sub-$(S/\mathfrak{R})$-modules of $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$.

lbbel=(ii) If $S/\mathfrak{R}$ and R are Noetherian and if R is an S-module of finite type, then every increasing sequence of ideals of S contained in $\mathfrak{R}$ is stationary.

PROOF. *(i)* This is clear since $\mathfrak{L}$ is an ideal of S and $R \cdot \mathfrak{L} = \mathfrak{J}$ ($\mathfrak{L}$ being an ideal of S), whence the conclusion.

(ii) Let $(\mathfrak{L}_n)$ be an increasing sequence of ideals of S contained in $\mathfrak{R}$. The sequence of ideals $R \cdot \mathfrak{L}_n$ of R is stationary since R is Noetherian; we can thus suppose that all the ideals $R \cdot \mathfrak{L}_n$ are equal to the same ideal $\mathfrak{J}$ of R. Since R is Noetherian, $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$ is also a finitely generated S-module, hence also an $(S/\mathfrak{R})$-module of finite type. But since $S/\mathfrak{R}$ is Noetherian, $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$ is an $(S/\mathfrak{R})$-Noetherian module, and the conclusion follows from (i). ☐

We apply this lemma with $R = E$, $S = C$; it is clear that $\mathfrak{n}$ is the conductor of C in E; moreover, for every ideal $\mathfrak{a}$ of C, $\mathfrak{a}/(\mathfrak{n} \cap \mathfrak{a})$ is isomorphic to $(\mathfrak{a} + \mathfrak{n})/\mathfrak{n}$, hence a sub-$C$-module of $C/\mathfrak{n}$, which is a simple C-module; it therefore suffices to show that every ideal $\mathfrak{a} \cap \mathfrak{n}$ of C is of finite type, and this follows from (5.6.11.1). It follows from ($\mathbf{0}$, 16.1.5) that $\dim(C) = \dim(E) = 2$. Let $A = C_{\mathfrak{n}}$; if $U = C - \mathfrak{n}$, the ring of fractions $B = U^{-1}E$ is thus the integral closure of A, and it is an A-module of finite type; moreover, since $\mathfrak{m}$ and $\mathfrak{m}'$ are the only prime ideals of E containing $\mathfrak{n}$, B is a semi-local ring whose local rings at the two maximal ideals are isomorphic to $E_{\mathfrak{m}}$ and $E_{\mathfrak{m}'}$ respectively, hence are of dimension 2 and 1 respectively. This shows that $\dim(B) = 2$, hence also $\dim(A) = 2$ ($\mathbf{0}$, 16.1.5). Since A is a local integral ring, it is necessarily catenary (every prime ideal distinct from 0 and from the maximal ideal being necessarily of height 1); but it does not satisfy the conclusion of (5.6.10), and *a fortiori* is not universally catenary.

Let us also note that $\mathfrak{n}$ is a prime ideal of height 2 in C, and that for every prime ideal $\mathfrak{p}$ of height 1 IV-2 | 103
in C, there exists a *single* prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, necessarily of height 1, and such that $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$. Indeed, there is at least one prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, and it follows from the Cohen–Seidenberg theorem that such an ideal is necessarily of height 1; moreover, since $\mathfrak{p} \not\supset \mathfrak{n}$, $C_{\mathfrak{p}}$ is integrally closed *(Bourbaki, Alg. comm., chap. V, §1, no. 5, cor. 5 of prop. 16)*, hence $E_{\mathfrak{p}} = C_{\mathfrak{p}}$, which proves our assertions.

It would be interesting to know whether every local integral Noetherian ring satisfying the conclusion of (5.6.10) is universally catenary; this is what Nagata [33] has affirmed, but his proof does not appear to be complete.

5.7. Depth and property (S_k)

Definition (5.7.1). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. We call *depth* (resp. *codepth*) of $\mathscr{F}$ at a point $x \in X$ the number $\mathrm{prof}(\mathscr{F}_x)$ (resp. $\mathrm{coprof}(\mathscr{F}_x)$) (**0**, 16.4.5 and 16.4.9). We call *codepth* of $\mathscr{F}$ the number

$$\mathrm{coprof}(\mathscr{F}) = \sup_{x \in X} \mathrm{coprof}(\mathscr{F}_x). \tag{5.7.1.1}$$

We say that $\mathscr{F}$ is an *$\mathscr{O}_X$-Module de Cohen-Macaulay* at a point x if $\mathscr{F}_x$ is an $\mathscr{O}_x$-module de Cohen-Macaulay, that is to say (**0**, 16.5.1) if $\mathrm{coprof}(\mathscr{F}_x) = 0$. We say that $\mathscr{F}$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay if it is so at every point, that is if $\mathrm{coprof}(\mathscr{F}) = 0$. A point $x \in X$ such that $\mathscr{O}_x$ is a Cohen-Macaulay ring is also called a *Cohen-Macaulay point* of X.

We call *codepth* of X and write $\mathrm{coprof}(X)$ the number $\mathrm{coprof}(\mathscr{O}_X)$. We say that X is a *Cohen-Macaulay prescheme* if $\mathscr{O}_X$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay, that is if $\mathrm{coprof}(X) = 0$. Every locally Noetherian prescheme of dimension 0 is evidently a Cohen-Macaulay prescheme. To say that $\mathrm{Spec}(A)$ is a Cohen-Macaulay scheme means that A is a Cohen-Macaulay ring (**0**, 16.5.13).

Definition (5.7.2). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, k a positive or negative integer. We say that $\mathscr{F}$ possesses *property* (S_k) if, for every $x \in X$, we have

$$\mathrm{prof}(\mathscr{F}_x) \geqslant \inf(k, \dim(\mathscr{F}_x)). \tag{5.7.2.1}$$

We say that $\mathscr{F}$ possesses property (S_k) *at a point* $x \in X$ if, for every generisation x' of x in X, we have

$$\mathrm{prof}(\mathscr{F}_{x'}) \geqslant \inf(k, \dim(\mathscr{F}_{x'})). \tag{5.7.2.2}$$

We say that X satisfies property (S_k) at a point x (resp. satisfies (S_k)) if $\mathscr{O}_X$ satisfies (S_k) at the point x (resp. satisfies (S_k)).

For $\mathscr{F}$ to satisfy (S_k), it is necessary and sufficient that it satisfy it at every point. If U is an open subset of X and if $\mathscr{F}$ satisfies (S_k), then so does $\mathscr{F}|U$; reciprocally, if (U_α) is an open covering of X and if $\mathscr{F}|U_\alpha$ satisfies (S_k) for every α, then $\mathscr{F}$ satisfies (S_k).

Remarks (5.7.3). — *(i)* Recall that one always has $\mathrm{prof}(\mathscr{F}_x) \leqslant \dim(\mathscr{F}_x)$ (**0**, 16.4.5.1) if $\mathscr{F}_x \neq 0$. To **IV-2** | 104
say that $\mathscr{F}$ possesses property (S_k) thus means that one has $\mathrm{prof}(\mathscr{F}_x) \geqslant k$ *except* at points $x \in X$ such that $\dim(\mathscr{F}_x) < k$, and that at these latter points one has $\dim(\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_x)$, that is to say (**0**, 16.5.1) that $\mathscr{F}$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay at these points; we note that only at points where $\dim(\mathscr{F}_x) = k$ does one have $\mathrm{prof}(\mathscr{F}_x) = k$, hence $\mathscr{F}$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay at these points. To say that $\mathscr{F}$ possesses (S_k) *for all k* signifies that $\mathscr{F}$ is an *$\mathscr{O}_X$-Module de Cohen-Macaulay*. It is clear that for $k' \geqslant k$, property $(\mathrm{S}_{k'})$ implies (S_k); for $k \leqslant 0$, *every* $\mathscr{O}_X$-Module coherent satisfies (S_k).

(ii) To verify condition (5.7.2.1), one can restrict to the case $x \in \mathrm{Supp}(\mathscr{F})$; in the contrary case indeed $\dim(\mathscr{F}_x) = -\infty$ (**0**, 14.1.2).

(iii) If $X = \mathrm{Spec}(A)$, where A is a Noetherian ring, and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type, one says that M possesses property (S_k) if $\mathscr{F}$ possesses this property. For a locally Noetherian prescheme X, to say that $\mathscr{F}$ possesses property (S_k) *at a point* $x \in X$ signifies that if one sets $Y = \mathrm{Spec}(\mathscr{O}_x)$, the $\mathscr{O}_Y$-Module $\widetilde{\mathscr{F}_x}$ possesses property (S_k); one notes that condition (5.7.2.1) in general *does not entail* (5.7.2.2) for every generisation x' of x, since there is no inequality relation between $\mathrm{prof}(\mathscr{F}_x)$ and $\mathrm{prof}(\mathscr{F}_{x'})$ (**0**, 16.4.6).

(iv) It follows immediately from the definition that if $\mathscr{F}$ satisfies (S_k) at a point x, it also satisfies (S_k) at every point x' which is a generisation of x.

(v) Property (S_k) is especially important for $k = 1$ and $k = 2$; it was introduced for $k = 2$ by Serre, to express his criterion of normality (cf. (5.8.5)).

(vi) Let X be a locally Noetherian prescheme, $j : Y \to X$ the canonical closed immersion, $\mathscr{G}$ an $\mathscr{O}_Y$-Module coherent. It is clear that for every $x \in Y$, $\dim(\mathscr{G}_x) = \dim((j_*(\mathscr{G}))_x)$ and $\mathrm{prof}(\mathscr{G}_x) =$

$\mathrm{prof}((j_*(\mathscr{G}))_x)$, whence $\mathrm{coprof}(\mathscr{G}_x) = \mathrm{coprof}((j_*(\mathscr{G}))_x)$. For $\mathscr{G}$ to satisfy (S_k), it suffices that $j_*(\mathscr{G})$ satisfy (S_k).

Proposition (5.7.4). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Set $S = \mathrm{Supp}(\mathscr{F})$, and for every $n \geqslant 0$, denote by Z_n the set of $z \in X$ such that $\mathrm{coprof}(\mathscr{F}_z) > n$, so that $Z_n \subset S$.*

label=(i) For $\mathscr{F}$ to satisfy property (S_k), it is necessary and sufficient that, for every $n \geqslant 0$, we have

(5.7.4.1) $$\mathrm{codim}(Z_n, S) > n + k.$$

lbbel=(ii) Suppose furthermore that the Z_n are closed in X. Then, for $\mathscr{F}$ to satisfy property (S_k) at a point $x \in S$, it is necessary and sufficient that, for every $n \geqslant 0$, we have

(5.7.4.2) $$\mathrm{codim}_x(Z_n, S) > n + k.$$

PROOF. *(i)* By definition (5.1.3), $\mathrm{codim}(Z_n, S) = \inf_{z \in Z_n}(\dim(\mathscr{O}_{S,z}))$ and the inequality (5.7.4.1) means that, for every $z \in X$, and every $n \geqslant 0$, the relation

$$\dim(\mathscr{F}_z) - \mathrm{prof}(\mathscr{F}_z) \geqslant n + 1$$

implies the relation

$$\dim(\mathscr{F}_z) \geqslant n + k + 1.$$

But if one sets $a = \dim(\mathscr{F}_z)$, $b = \mathrm{prof}(\mathscr{F}_z)$, we have $b \leqslant a$, and to say that for every $n \geqslant 0$, $b \leqslant a - n - 1$ implies $k \leqslant a - n - 1$ is equivalent to saying $b \geqslant \inf(k, a)$, whence the proposition.

(ii) The argument is the same as in (i), except that one restricts to $z \in X$ which are generisations of x, and takes account of (5.1.3.2). □

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Proposition (5.7.5). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. For $\mathscr{F}$ to satisfy (S_k) $(k \geqslant 1)$, it is necessary and sufficient that for every integer r such that $0 \leqslant r < k$, every open subset U of X, and every $(\mathscr{F}|U)$-regular suite $(f_i)_{1 \leqslant i \leqslant r}$ of sections of $\mathscr{O}_X$ above U, $(\mathscr{F}|U)/(\sum_{i=1}^{r} f_i(\mathscr{F}|U))$ be without embedded associated prime cycle.*

PROOF. Let us first prove the proposition for $k = 1$; it then says that *for $\mathscr{F}$ to satisfy* (S_1), *it is necessary and sufficient that $\mathscr{F}$ be without embedded associated prime cycle*. Indeed, to say that $\mathscr{F}$ satisfies (S_1) means that at the points $x \in \mathrm{Supp}(\mathscr{F})$ such that $\dim(\mathscr{F}_x) > 0$, one has $\mathrm{prof}(\mathscr{F}_x) \geqslant 1$ (since at the other points of $\mathrm{Supp}(\mathscr{F})$ one has $\dim(\mathscr{F}_x) = 0$, hence $\mathrm{prof}(\mathscr{F}_x) = 0$); but to say that $\mathrm{prof}(\mathscr{F}_x) \geqslant 1$ means that $\mathfrak{m}_x$ *is not associated* to $\mathscr{F}$ (**0**, 16.4.6, (i)), or equivalently that x is not associated to $\mathscr{F}$ (3.1.1); on the other hand, if S is a sub-prescheme of X having for underlying space $\mathrm{Supp}(\mathscr{F})$, by (5.1.12.1) $\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{S,x})$; to say that $\dim(\mathscr{F}_x) > 0$ means that x is not a maximal point of $\mathrm{Supp}(\mathscr{F})$ (5.1.2), whence the conclusion.

In the second place, let us show that, for k arbitrary > 1, the condition of the statement is *necessary*. One can restrict to considering the case $U = X$, and our assertion will be proved (by induction on k and by virtue of the first part of the argument) if we show that when $\mathscr{F}$ satisfies (S_k) $(k > 1)$ and f is an $\mathscr{F}$-regular section of $\mathscr{O}_X$ above X, then $\mathscr{F}/f\mathscr{F}$ satisfies (S_{k-1}) (i.e. satisfies (5.7.2.1) where k is replaced by $k-1$). Indeed, for every $x \in X$, f_x is an $\mathscr{F}_x$-regular element; if it is invertible, $\mathscr{F}_x/f_x\mathscr{F}_x = 0$ and the conclusion is trivial. If on the contrary $f_x \in \mathfrak{m}_x$, one knows that $\mathrm{prof}(\mathscr{F}_x/f_x\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_x) - 1$ (**0**, 16.4.6, (i)), and $\dim(\mathscr{F}_x/f_x\mathscr{F}_x) = \dim(\mathscr{F}_x) - 1$ (**0**, 16.3.4), and our assertion follows.

Let us finally prove that for $k > 1$, the condition of the statement is *sufficient*. We proceed by induction on k; let x be a point of X, and suppose first that $\dim(\mathscr{F}_x) \geqslant k$. The induction hypothesis implies that $\mathscr{F}$ satisfies (S_{k-1}), hence $\mathrm{prof}(\mathscr{F}_x) \geqslant k-1$; taking account of (**0**, 15.2.4), there is thus an open neighbourhood U of x and an $(\mathscr{F}|U)$-regular suite $(f_i)_{1 \leqslant i \leqslant k-1}$ of sections of $\mathscr{O}_X$ above U, such that $(f_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant k-1$. The hypothesis implies that $\mathscr{G} = (\mathscr{F}|U)/(\sum_{i=1}^{k-1} f_i(\mathscr{F}|U))$ is without embedded associated prime cycle; but $\dim(\mathscr{G}_x) = \dim(\mathscr{F}_x) - (k-1) \geqslant 1$ (**0**, 16.3.4), hence x is not associated to $\mathscr{G}$; so $\mathrm{prof}(\mathscr{G}_x) \geqslant 1$ (**0**, 16.4.6), and since $\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(\mathscr{F}_x) - (k-1)$ (**0**, 16.4.6), we have $\mathrm{prof}(\mathscr{F}_x) \geqslant k$. Suppose in the second place that $\dim(\mathscr{F}_x) = r < k$; since $\mathscr{F}$ satisfies (S_{k-1}), we have $\mathrm{prof}(\mathscr{F}_x) \geqslant \inf(k-1, r) = r$, and this finishes the proof. □

Corollary (5.7.6). — *Suppose that $\mathscr{F}$ satisfies* (S_k) $(k \geqslant 1)$*; if* (f_i) $(1 \leqslant i \leqslant r)$ *is an $\mathscr{F}$-regular suite of sections of $\mathscr{O}_X$ above X and if $r < k$, $\mathscr{F}/(\sum_{i=1}^r f_i\mathscr{F})$ satisfies* (S_{k-r}).

PROOF. This results immediately from (5.7.5). □

Corollary (5.7.7). — *For $\mathscr{F}$ to satisfy* (S_2)*, it is necessary and sufficient that $\mathscr{F}$ be without embedded associated prime cycle and that for every open subset U of X, every $(\mathscr{F}|U)$-regular section f of $\mathscr{O}_X$ above U, $(\mathscr{F}|U)/f(\mathscr{F}|U)$ be without embedded associated prime cycle.*

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Remarks (5.7.8). — Let X be a locally Noetherian prescheme *of dimension* 1; it then amounts to the same to say that X is a Cohen-Macaulay prescheme, or that it satisfies (S_1), or that it satisfies one of the properties (S_n) for $n \geqslant 1$, by virtue of definitions (5.7.1) and (5.7.2). By virtue of (5.7.5), it also amounts to the same, for preschemes of dimension 1, to say that X is a Cohen-Macaulay prescheme or that it has no embedded associated prime cycle. For example, a *reduced* locally Noetherian prescheme of dimension 1 is a Cohen-Macaulay prescheme.

Proposition (5.7.9). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a homomorphism of rings, M a B-module such that $M_{[\mathfrak{p}]}$ is an A-module of finite type. Let $\mathfrak{p}$ be a prime ideal of A; the prime ideals of B above $\mathfrak{p}$ and belonging to* $\mathrm{Supp}(M)$ *are finite in number, and if $(\mathfrak{q}_i)_{1 \leqslant i \leqslant n}$ is the family of these ideals, we have*

$$\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \sup_i \dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) \tag{5.7.9.1}$$

$$\mathrm{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \inf_i \mathrm{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}). \tag{5.7.9.2}$$

PROOF. If $\mathfrak{b}$ is the annihilator of M, $B/\mathfrak{b}$ identifies with a sub-A-module of $\mathrm{End}_A(M_{[\mathfrak{p}]})$, hence is of finite type since A is Noetherian; so there are only a finite number of prime ideals of B above $\mathfrak{p}$ and containing $\mathfrak{b}$, and these are precisely those belonging to $\mathrm{Supp}(M)$. Replacing B by $B/\mathfrak{b}$, which does not change the second members of (5.7.9.1) and (5.7.9.2) (by (**0**, 16.1.9) and (**0**, 16.4.8)), one can thus suppose that B is a finite A-algebra. Set $S = A - \mathfrak{p}$ and $B' = S^{-1}B$; B' is a semi-local Noetherian ring whose maximal ideals are $S^{-1}\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ and since $M_\mathfrak{p}$ is an $A_\mathfrak{p}$-module of finite type, $\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \dim_{B'}(M_\mathfrak{p})$ (**0**, 16.1.9). This being so, relation (5.7.9.1) becomes a particular case of (**0**, 16.1.7.4). To prove relation (5.7.9.2), we immediately reduce as in (**0**, 16.4.8) to the case $\mathrm{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 0$, and the same argument as in (**0**, 16.4.8) shows that $M_\mathfrak{p}$ contains a sub-B'-module of *finite length*, and hence also a *simple* sub-B'-module; but such a sub-module is necessarily isomorphic to the residue field of one of the $B_{\mathfrak{q}_i}$, hence there is at least one index i such that $\mathrm{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) = 0$ (**0**, 16.4.6), which finishes the proof. □

Corollary (5.7.10). — *Suppose verified the hypotheses of* (5.7.9)*, and suppose in addition that A is a* local *ring; then, for M to be an A-module de Cohen-Macaulay, it is necessary and sufficient that, for all prime ideals $\mathfrak{q}_i$ of B above the maximal ideal of A, $M_{\mathfrak{q}_i}$ be a $B_{\mathfrak{q}_i}$-module de Cohen-Macaulay, and moreover that all the $\dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$ be equal.*

PROOF. It follows from (5.7.9.1) and (5.7.9.2) that these conditions are equivalent to $\dim_A(M) = \mathrm{prof}_A(M)$. □

Corollary (5.7.11). — *Suppose verified the hypotheses of* (5.7.9).

label=(i) *If $M_{[\mathfrak{p}]}$ possesses property* (S_k)*, then so does M.*

lbbel=(ii) *Suppose that for every pair of prime ideals $\mathfrak{q}$, $\mathfrak{q}'$ of B such that $\rho^{-1}(\mathfrak{q}) = \rho^{-1}(\mathfrak{q}')$, $\dim_{B_\mathfrak{q}}(M_\mathfrak{q}) = \dim_{B_{\mathfrak{q}'}}(M_{\mathfrak{q}'})$. Then, if M possesses property* (S_k)*, so does $M_{[\mathfrak{p}]}$.*

PROOF. This results immediately from (5.7.9.1) and (5.7.9.2) and the definition of property (S_k). □

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(5.7.12). Conforming to definitions (5.7.1), given a Noetherian ring A and an A-module M of finite type, one defines $\mathrm{coprof}_A(M)$ as $\sup_{x\in X}(\mathrm{coprof}_{A_x}(M_x))$, where $X = \mathrm{Spec}(A)$; we will see further on (6.11.5) that this definition *coincides* with that of (**0**, 16.4.9) when A is a local Noetherian ring.

Corollary (5.7.13). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a homomorphism of rings, M a B-module such that $M_{[\mathfrak{p}]}$ is an A-module of finite type. Then*

(5.7.13.1) $$\operatorname{coprof}_A(M_{[\mathfrak{p}]}) \geqslant \operatorname{coprof}_B(M).$$

Proof. This results from the preceding definition and from (5.7.9.1) and (5.7.9.2). □

5.8. Regular preschemes and property (R_k). Serre's normality criterion

(5.8.1). Recall $(\mathbf{0_I}, 4.1.4)$ that a ringed space $(X, \mathscr{O}_X)$ is called *regular* at a point $x \in X$ if $\mathscr{O}_x$ is a *regular local ring*. When dealing with preschemes, we will only use this terminology when X is *locally Noetherian*.

Definition (5.8.2). — A locally Noetherian prescheme X is said to be *regular in codimension* $\leqslant k$, or *possesses property* (R_k) if the set of points where X is *not* regular is of codimension $> k$ (that is to say (5.1.3), for every $x \in X$, the relation $\dim(\mathscr{O}_x) \leqslant k$ implies that $\mathscr{O}_x$ is regular).

To say that X is regular signifies that X possesses property (R_k) for every k. If $X = \operatorname{Spec}(A)$, where A is a Noetherian ring, one says that A possesses property (R_k) if X possesses this property; to say that X is regular signifies that the ring A is *regular* $(\mathbf{0}, 17.3.6)$. For a locally Noetherian prescheme X, we say that X possesses property (R_k) *at a point* $x \in X$ if the local ring $\mathscr{O}_x$ possesses property (R_k); this means that for every generisation x' of x in X, the relation $\dim(\mathscr{O}_{x'}) \leqslant k$ implies that $\mathscr{O}_{x'}$ is regular. To say that X is regular at a point x is equivalent to saying that X satisfies (R_n) for *every* $n \geqslant 0$ at the point x, by virtue of $(\mathbf{0}, 17.3.6)$.

Proposition (5.8.3). — *If k is a field, X a prescheme locally of finite type over k, then, for every $x \in X$, there exists an open neighbourhood of x in X isomorphic to a sub-scheme of a regular k-scheme.*

Proof. Indeed, there is an affine open neighbourhood U of x isomorphic to a k-scheme of the form $\operatorname{Spec}(A)$, where A is a k-algebra of finite type; A is therefore isomorphic to a quotient of a polynomial algebra $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, and it is known that this last algebra is a regular ring $(\mathbf{0}, 17.3.7)$. □

(5.8.4). By virtue of (5.8.2), to say that X possesses property (R_0) signifies that for every maximal point x of X, $\mathscr{O}_x$ is a *field* $(\mathbf{0}, 17.1.4)$, or equivalently that X is *reduced* at this point. Since the set U of $x \in X$ where X is reduced is *open* (the nilradical of X being an $\mathscr{O}_X$-Module coherent (**I**, 6.1.1) and $(\mathbf{0_I}$, 5.2.2)), it amounts to the same to say that X possesses property (R_0) or that the set U is *everywhere dense*. Consequently:

Proposition (5.8.5). — *For a locally Noetherian prescheme X to be reduced, it is necessary and sufficient that it satisfy properties* (S_1) *and* (R_0).

Proof. Taking into account (5.7.5), this results from the preceding remark and from (3.2.1). □

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Theorem (5.8.6). — (Serre's criterion). *Let X be a locally Noetherian prescheme. For X to be* normal, *it is necessary and sufficient that X satisfy properties* (S_2) *and* (R_1), *that is to say that, for every $x \in X$, the following properties hold:*

label=(i) *If* $\dim(\mathscr{O}_x) \leqslant 1$, $\mathscr{O}_x$ *is regular* (*that is, is a field or a discrete valuation ring*).
lbbel=(ii) *If* $\dim(\mathscr{O}_x) \geqslant 2$, *then* $\operatorname{prof}(\mathscr{O}_x) \geqslant 2$.

Proof. The conditions are *necessary*. Indeed, to say that X is normal signifies that for every $x \in X$, $\mathscr{O}_x$ is a local Noetherian integrally closed ring. If $\dim(\mathscr{O}_x) = 0$ (resp. $\dim(\mathscr{O}_x) = 1$), we conclude that $\mathscr{O}_x$ is a field since $\mathscr{O}_x$ is integral (resp. that $\mathscr{O}_x$ is a discrete valuation ring, by virtue of (**II**, 7.1.6)). On the other hand, for every nonzero element $f_x \neq 0$ of $\mathscr{O}_x$, one knows (Bourbaki, *Alg. comm.*, chap. VII, §1, no. 4, prop. 8) that the prime ideals associated to $\mathscr{O}_x/f_x\mathscr{O}_x$ are non-immersed, hence $\mathscr{O}_x$ satisfies (S_2) (5.7.7).

The conditions are *sufficient*. The question being local, one can moreover suppose that $X = \operatorname{Spec}(A)$, where A is a local Noetherian ring *reduced* (**I**, 5.1.4); if R is the total ring of fractions of A, R is a direct product of a finite number of fields, and (taking account of (**II**, 6.3.6)), it suffices to prove that A is *integrally closed* in R. So let $h = f/g$ be an element of R integral over A, g and f being elements of A such that g is a non-zero-divisor of 0. We have a relation of the form

(5.8.6.1) $$f^n + \sum_{i=1}^{n} a_i f^{n-i} g^i = 0 \qquad \text{with } a_i \in A \text{ for } 1 \leqslant i \leqslant n.$$

Let $\mathfrak{p}$ be a prime ideal of A such that $\dim(A_\mathfrak{p}) = 1$; if $f_\mathfrak{p}$ and $g_\mathfrak{p}$ are the images of f and g in $A_\mathfrak{p}$, it follows from (5.8.6.1) that $f_\mathfrak{p}/g_\mathfrak{p}$ (which belongs to the total ring of fractions of $A_\mathfrak{p}$, since $g_\mathfrak{p}$ is a non-zero-divisor of 0 in $A_\mathfrak{p}$ by flatness ($\mathbf{0_I}$, 5.3.1)) is integral over $A_\mathfrak{p}$; but since $A_\mathfrak{p}$ is regular, hence integrally closed, we have $f_\mathfrak{p}/g_\mathfrak{p} \in A_\mathfrak{p}$. But hypothesis (S_2) implies (5.7.7) that g is a non-zero-divisor and that A/gA has no non-immersed associated prime ideals $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$; or, gA is the intersection of the primary ideals $\mathfrak{q}_i$ corresponding to the $\mathfrak{p}_i$, and from what we have just seen, the $\mathfrak{q}_i$ are the inverse images in A, by the homomorphisms $A \to A_{\mathfrak{p}_i}$, of the ideals $(gA)_{\mathfrak{p}_i}$ *(Bourbaki, Alg. comm., chap. IV, §2, no. 3, prop. 5)*. But by the Hauptidealsatz ($\mathbf{0}$, 16.3.2) $\dim(A_{\mathfrak{p}_i}) = 1$ for $1 \leqslant i \leqslant n$, hence $(fA)_{\mathfrak{p}_i} \subset (gA)_{\mathfrak{p}_i}$ for every i by what precedes; since fA is contained in the intersection of the inverse images of $(fA)_{\mathfrak{p}_i}$ $(1 \leqslant i \leqslant n)$, we have $fA \subset gA$, that is to say $f/g \in A$. □

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5.9. ***Z*-pure and *Z*-closed Modules** Some of the notions and results of this section and the following are particular cases of notions and results developed in chapter III in the theory of local cohomology. For the convenience of the reader, we give here an independent treatment.

(5.9.1). Let X be a locally Noetherian prescheme, Z a subset of X *stable by specialisation*: this means that for every finite subset M of Z, the closure of M is contained in Z, and hence Z is the union of an increasing filtering family (Z_α) of closed subsets of X; conversely, it is clear that such a union is stable by specialisation.

Set $U_\alpha = X - Z_\alpha$, so that $X - Z$ is the intersection of the filtering decreasing family of opens U_α; let $i_\alpha : U_\alpha \to X$ be the canonical injection, and for $U_\alpha \supset U_\beta$, let $i_{\alpha\beta} : U_\beta \to U_\alpha$ be the canonical injection, so that $i_\beta = i_\alpha \circ i_{\alpha\beta}$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module (not necessarily quasi-coherent); from the canonical homomorphism ($\mathbf{0_I}$, 4.4.3.2)

$$\mathscr{F}|U_\alpha \longrightarrow (i_{\alpha\beta})_*(\mathscr{F}|U_\beta)$$

we deduce, by application of the functor $(i_\alpha)_*$, a homomorphism

$$\rho_{\beta\alpha} : (i_\alpha)_*(\mathscr{F}|U_\alpha) \longrightarrow (i_\beta)_*(\mathscr{F}|U_\beta)$$

and one verifies immediately that $\rho_{\gamma\alpha} = \rho_{\gamma\beta} \circ \rho_{\beta\alpha}$ for $U_\alpha \supset U_\beta \supset U_\gamma$; in other words, the $\mathscr{O}_X$-Modules $(i_\alpha)_*(\mathscr{F}|U_\alpha)$ form an *inductive system* for the homomorphisms $\rho_{\beta\alpha}$. We set

(5.9.1.1) $$\mathscr{H}^0_{X/Z}(\mathscr{F}) = \varinjlim_\alpha (i_\alpha)_*(\mathscr{F}|U_\alpha).$$

This $\mathscr{O}_X$-Module does not depend on the increasing family (Z_α) of closed sets whose union is Z: indeed, let V be a Noetherian open subset of X; one knows (G, II, 3.10.1) that in the category of $\mathscr{O}_V$-Modules, the functor $\mathscr{G} \mapsto \Gamma(V, \mathscr{G})$ commutes with inductive limits; so by virtue of (5.9.1.1)

(5.9.1.2) $$\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_\alpha \Gamma(V \cap U_\alpha, \mathscr{F}).$$

Let then (Z'_λ) be a second increasing filtering family of closed sets of X, with union Z; $V \cap Z_\alpha$ is therefore the union of the $V \cap Z_\alpha \cap Z'_\lambda$; but $V \cap Z_\alpha$ is locally closed in X, hence every closed irreducible subset of $V \cap Z_\alpha$ admits a generic point; since the $V \cap Z_\alpha \cap Z'_\lambda$ are closed in $V \cap Z_\alpha$ and form (for α fixed) an increasing filtering family, there exists an index λ such that $V \cap Z_\alpha \cap Z'_\lambda = V \cap Z_\alpha$ ($\mathbf{0_{III}}$, 9.2.4), in other words $V \cap Z_\alpha \subset V \cap Z'_\lambda$. This proves that the filtering decreasing families $V \cap U_\alpha$, $V \cap U'_\lambda$ (where $U'_\lambda = X - Z'_\lambda$) are cofinal in each other, whence our assertion, by virtue of (5.9.1.2). IV-2 | 110

We note that the set Z is not necessarily constructible: one has an example of this by taking $X = \mathrm{Spec}(A)$, where A is an integral Noetherian ring having an infinity of maximal ideals, and Z the complement of the generic point of X.

If Z is closed and if $i : X - Z \to X$ is the canonical injection, we have

$$\mathscr{H}^0_{X/Z}(\mathscr{F}) = i_*(\mathscr{F}|X - Z) \tag{5.9.1.3}$$

and in particular, for $Z = \varnothing$, $\mathscr{H}^0_{X/\varnothing}(\mathscr{F}) = \mathscr{F}$.

Proposition (5.9.2). — *label=(i) The functor $\mathscr{F} \mapsto \mathscr{H}^0_{X/Z}(\mathscr{F})$ is exact on the left.*
lbbel=(ii) If $\mathscr{F}$ is quasi-coherent, then so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$.

Proof. Assertion (i) results from the definition (5.9.1.1), from the fact that $(i_\alpha)_*$ is a left exact functor and from the fact that the inductive limit preserves exactness in the category of $\mathscr{O}_X$-Modules. Assertion (ii) results from (**I**, 9.2.2) and the fact that an inductive limit of quasi-coherent $\mathscr{O}_X$-Modules is quasi-coherent (**I**, 1.3.9). □

Remark (5.9.3). — If $\mathscr{F}$ is an $\mathscr{O}_X$-Algebra, then so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$ ($\mathbf{0_I}$, 4.2.4); in particular $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_X$-Algebra, and for every $\mathscr{O}_X$-Module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$-Module, which is quasi-coherent if $\mathscr{F}$ is quasi-coherent (**I**, 9.6.1). More particularly, suppose that $X = \operatorname{Spec}(A)$, where A is integral and Noetherian; then $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is the $\mathscr{O}_X$-Algebra $\widetilde{B}$, where

$$B = \bigcap_{\mathfrak{p} \in X - Z} A_\mathfrak{p}. \tag{5.9.3.1}$$

This results indeed from (5.9.1.2) and (**I**, 8.2.1.1).

Proposition (5.9.4). — *Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, X' a locally Noetherian prescheme, $f : X' \to X$ a flat morphism. Then $Z' = f^{-1}(Z)$ is stable by specialisation and for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we have a canonical isomorphism*

$$f^*(\mathscr{H}^0_{X/Z}(\mathscr{F})) \xrightarrow{\sim} \mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F})). \tag{5.9.4.1}$$

Proof. Indeed, with the notations of (5.9.1), $Z'_\alpha = f^{-1}(Z_\alpha)$ is closed in X' and Z' is the union of the Z'_α; moreover, $(i_\alpha)_{(X')}$ is the canonical injection $i'_\alpha : U'_\alpha \to X'$, where $U'_\alpha = X' - Z'_\alpha = f^{-1}(U_\alpha)$. Since f is flat, one knows (2.3.1) that the canonical homomorphism $f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \to (i'_\alpha)_*(f^*(\mathscr{F})|U'_\alpha)$ is bijective; since, for $\alpha \leqslant \beta$, the diagram IV-2 | 111

$$\begin{tikzcd} f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \arrow[r, "\sim"] \arrow[d] & (i'_\alpha)_*(f^*(\mathscr{F})|U'_\alpha) \arrow[d] \\ f^*((i_\beta)_*(\mathscr{F}|U_\beta)) \arrow[r, "\sim"] & (i'_\beta)_*(f^*(\mathscr{F})|U'_\beta) \end{tikzcd}$$

is commutative, passing to the limit one obtains a canonical isomorphism $\varinjlim_\alpha (f^*((i_\alpha)_*(\mathscr{F}|U_\alpha))) \xrightarrow{\sim} \mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$. But since the functor f^* commutes with inductive limits ($\mathbf{0_I}$, 4.3.2), this gives by definition the isomorphism (5.9.4.1) sought. □

Corollary (5.9.5). — *Under the hypotheses of (5.9.4), if $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent, then so is $\mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$. The converse is true when f is a faithfully flat and quasi-compact morphism.*

Proof. The first assertion results from (5.9.4.1) and ($\mathbf{0_I}$, 5.3.11); the second amounts to saying that if $\mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$ is of finite type, then so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$; this results from (5.9.4.1) and (2.5.2). □

Corollary (5.9.6). — *Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. For every $x \in X$, setting $X_x = \operatorname{Spec}(\mathscr{O}_x)$, $Z_x = Z \cap X_x$, we have a canonical functorial isomorphism*

$$(\mathscr{H}^0_{X/Z}(\mathscr{F}))_x \xrightarrow{\sim} \mathscr{H}^0_{X_x/Z_x}(\widetilde{\mathscr{F}_x}). \tag{5.9.6.1}$$

Proof. It suffices to apply (5.9.4) to the canonical morphism $X_x \to X$, which is flat, and (**I**, 1.6.5). □

(5.9.7). With the notations of (5.9.1), for every α one has a canonical functorial homomorphism $\mathscr{F} \to (i_\alpha)^*(\mathscr{F}|U_\alpha)$ ($\mathbf{0_I}$, 4.4.3.2), and these homomorphisms form an inductive system; passing to the inductive limit, one deduces a canonical functorial homomorphism

$$\rho_{X/Z} : \mathscr{F} \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F}). \tag{5.9.7.1}$$

Proposition (5.9.8). — *Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, $\mathscr{F}$ an $\mathscr{O}_X$-Module. The following properties are equivalent:*

label=a) *The homomorphism $\rho_{X/Z}$ (5.9.7.1) is injective (resp. bijective).*

lbbel=b) *For every Noetherian open subset V of X, the homomorphism*

$$(\rho_{X/Z})_V : \Gamma(V, \mathscr{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_\alpha, \mathscr{F})$$

is injective (resp. bijective).

a′) *For every closed subset $T \subset Z$ of X, the canonical homomorphism ($\mathbf{0_I}$, 4.4.3.2)*

$$\mathscr{F} \longrightarrow i_*(\mathscr{F}|X - T)$$

(where $i : X - T \to X$ is the canonical injection) is injective (resp. bijective).

b′) *For every closed subset $T \subset Z$ of X and every Noetherian open subset V of X, the restriction homomorphism*

$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F})$$

is injective (resp. bijective).

PROOF. Taking account of (5.9.1.2), the equivalence of *a)* and *b)* (resp. *a′)* and *b′)*) results from the **IV-2 | 112**
definition of the functor Γ and from the fact that it is left exact. Since the homomorphism $(\rho_{X/Z})_V$ is the composite

$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_\alpha, \mathscr{F}) \tag{5.9.8.1}$$

for every closed subset $T \subset Z$, if $(\rho_{X/Z})_V$ is injective, then so is $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap (X - T), \mathscr{F})$; on the other hand, the fact that *b′)* implies *b)* results from the definition of an inductive limit. It remains to show that if $\rho_{X/Z}$ is bijective, then so is $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap (X - T), \mathscr{F})$, and for this it suffices, by virtue of (5.9.8.1), to see that if $U' \subset U$ are two open sets contained in V and containing $V \cap Z$, the restriction homomorphism $\Gamma(U, \mathscr{F}) \to \Gamma(U', \mathscr{F})$ is injective; but this results from the fact that $\rho_{X/Z}$ is injective, by replacing in what precedes V by U and $V \cap (X - T)$ by U'. The equivalence of *a)* and *a′)* follows from (5.9.1.3) and the fact that the filtering families (U_α) and $(X - T)_{T \subset Z, T \text{ closed}}$ are cofinal. □

Definition (5.9.9). — Under the hypotheses of (5.9.8), we say that $\mathscr{F}$ is *Z-pure* (resp. *Z-closed*) if the homomorphism $\rho_{X/Z}$ is injective (resp. bijective).

If $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, where M is an A-module, we say that M is Z-pure (resp. Z-closed) when $\mathscr{F}$ is Z-pure (resp. Z-closed).

We say that $\mathscr{F}$ is *Z-pure* (resp. *Z-closed*) *at a point* $x \in X$ if (with the notations of (5.9.6)) $\mathscr{F}_x$ is Z_x-pure (resp. Z_x-closed); this amounts to the same, by virtue of (5.9.6), as saying that the canonical homomorphism $\mathscr{F}_x \to (\mathscr{H}^0_{X/Z}(\mathscr{F}))_x$ is injective (resp. bijective).

We note that for every $x \in X - Z$, $\mathscr{F}$ is Z-closed at the point x, by virtue of (5.9.8).

Corollary (5.9.10). — label=(i) *Let (V_λ) be an open covering of X. For $\mathscr{F}$ to be Z-pure (resp. Z-closed), it is necessary and sufficient that for every λ, $\mathscr{F}|V_\lambda$ be $(Z \cap V_\lambda)$-pure (resp. $(Z \cap V_\lambda)$-closed).*

lbbel=(ii) *Let Z' be a subset of Z stable under specialisation. If $\mathscr{F}$ is Z-pure (resp. Z-closed), it is Z'-pure (resp. Z'-closed).*

PROOF. This follows immediately from (5.9.8), *b′)*. □

Proposition (5.9.11). — *Under the hypotheses of (5.9.8), the $\mathscr{O}_X$-Modules $\operatorname{Ker}(\rho_{X/Z})$ and $\operatorname{Coker}(\rho_{X/Z})$ have their support contained in Z, and the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed. Furthermore, if $u : \mathscr{F} \to \mathscr{F}'$ is a homomorphism of $\mathscr{O}_X$-Modules such that $\mathscr{F}'$ is Z-closed, u factors in a unique way as $\mathscr{F} \xrightarrow{\rho_{X/Z}} \mathscr{H}^0_{X/Z}(\mathscr{F}) \xrightarrow{v} \mathscr{F}'$. If in addition the supports of $\operatorname{Ker}(u)$ and of $\operatorname{Coker}(u)$ are contained in Z, v is an isomorphism.*

PROOF. The first assertion means that, for every $x \in X - Z$, we have

$$\operatorname{Ker}(\rho_{X/Z})_x = \operatorname{Coker}(\rho_{X/Z})_x = 0,$$

i.e. that $(\rho_{X/Z})_x$ is bijective, or in other words that $\mathscr{F}$ is Z-pure at x, as has been noted above (5.9.9).

To show that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, it suffices to see that for every Noetherian open subset V of X, $\Gamma(V\cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F}))$ is equal to $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$; but by definition

$$\Gamma(V\cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_{\alpha} \Gamma(V\cap U_\alpha \cap U_\beta, \mathscr{F}).$$

Now, the double family $(U_\alpha \cap U_\beta)$ is a decreasing filtering family, and our assertion results from (5.9.1) and the theorem of the double inductive limit.

Let us pass to the second part of the proposition. The existence and uniqueness of v result from the fact that for every α, $(i_\alpha)_*(u)$ is the unique homomorphism making the diagram **IV-2 | 113**

$$\begin{tikzcd} \mathscr{F} \arrow[r, "u"] \arrow[d] & \mathscr{F}' \arrow[d] \\ (i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[r, "(i_\alpha)_*(u)"] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \end{tikzcd}$$

commutative, from which there is a unique homomorphism w making all the diagrams

$$\begin{tikzcd} (i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[r, "(i_\alpha)_*(u)"] \arrow[d] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \arrow[d] \\ \mathscr{H}^0_{X/Z}(\mathscr{F}) \arrow[r, "w"] & \mathscr{H}^0_{X/Z}(\mathscr{F}') \end{tikzcd}$$

commutative, and finally that $\mathscr{F}'$ and $\mathscr{H}^0_{X/Z}(\mathscr{F}')$ identify canonically by hypothesis.

It remains to see that if the supports of $\operatorname{Ker}(u)$ and of $\operatorname{Coker}(u)$ are contained in Z, v is an isomorphism. It suffices to see that for every Noetherian open subset V, the corresponding homomorphism $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) \to \Gamma(V, \mathscr{F}')$ is then an isomorphism. Now, if a section $t \in \Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$ has image 0 in $\Gamma(V, \mathscr{F}')$, note that for some index α, we have $t \in \Gamma(V\cap U_\alpha, \mathscr{F})$, and by virtue of the hypothesis on u, we have $t_y = 0$ for every $y \in V \cap (X-Z)$; there is thus an open set containing $V\cap(X-Z)$ such that the restriction of t to this open set is zero, hence by definition t is the element 0 of $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$. Let us now prove that every section $s' \in \Gamma(V, \mathscr{F}')$ is the image of a section of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above V. By hypothesis, for every $x \in V\cap(X-Z)$ there exists a section $s^{(x)}$ of $\mathscr{F}$ above a neighbourhood $W^{(x)}$ of x in X, whose image by u is $s'|W^{(x)}$; $s'|W^{(x)}$ is thus also the image by v of the section $t^{(x)}$ of $\mathscr{H}^0_{X/Z}(\mathscr{F})$, the canonical image of $s^{(x)}$. Furthermore, since as we have seen v is injective, the restrictions of $t^{(x)}$ and $t^{(x')}$ to $W^{(x)}\cap W^{(x')}$ are identical for any two points x, x' of $V\cap(X-Z)$; the $t^{(x)}$ are therefore restrictions of the same section t of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above an open neighbourhood U of $(X-Z)\cap V$. But since $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, t extends in a unique way to a section of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above V, whose image by v has the same restriction as s' to U, and coincides therefore with s' for the same reason.

We say that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is the Z-*closure* of $\mathscr{F}$. □

Remarks (5.9.12). — **IV-2 | 114**

label=*(i)* Let $\mathsf{C}(X)$ be the category of $\mathscr{O}_X$-Modules, and let $\mathsf{C}_Z(X)$ be the subcategory of $\mathsf{C}(X)$ formed by the $\mathscr{O}_X$-Modules with support contained in Z; this subcategory is localising in the sense of Gabriel, and the functor $\mathscr{H}^0_{X/Z}$ is none other than the localisation functor of Gabriel (cf. [**?**]); this would furnish another proof of (5.9.11)). When Z is closed, the functor $i^* : \mathsf{C}(X) \to \mathsf{C}(X-Z)$ (where $i : X-Z \to X$ is the canonical injection) defines an *equivalence* of categories $\mathsf{C}(X)/\mathsf{C}_Z(X) \approx \mathsf{C}(X-Z)$.

lbbel=*(ii)* It results from (5.9.11) that the condition $\mathscr{H}^0_{X/Z}(\mathscr{F}) = 0$ is *equivalent* to $\operatorname{Supp}(\mathscr{F}) \subset Z$. It entails indeed this last condition since the kernel of $\rho_{X/Z}$ is then equal to $\mathscr{F}$. Conversely, if $\operatorname{Supp}(\mathscr{F}) \subset Z$, it suffices to apply the second part of (5.9.11) to the unique homomorphism $u : \mathscr{F} \to 0$ to conclude that the corresponding morphism $v : \mathscr{H}^0_{X/Z}(\mathscr{F}) \to 0$ is an isomorphism.

lcbel=*(iii)* The above developments hold in their entirety for any ringed space which is locally Noetherian, whose every irreducible closed subset admits exactly one generic point.

In particular they apply, on a locally Noetherian prescheme, to *sheaves of abelian groups* (considered as Modules over the simple sheaf associated to the constant presheaf **Z**). We have moreover for every $x \in X$ the canonical isomorphism (5.9.6.1), in which $\widetilde{\mathscr{F}_x}$ denotes the sheaf induced on the subspace X_x of X by the sheaf $\mathscr{F}$; the direct proof results immediately from the definition (5.9.1.2) and the theorem of the double inductive limit.

5.10. Property (S_2) and Z-closure

(5.10.1). Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module; for every subset T of X, we set

$$\operatorname{prof}_T(\mathscr{F}) = \inf_{x\in T} \operatorname{prof}(\mathscr{F}_x). \tag{5.10.1.1}$$

Proposition (5.10.2). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

label=a) $\mathscr{F}$ *is Z-pure.*
lbbel=b) $\operatorname{Ass}(\mathscr{F})$ *does not meet Z.*

If furthermore $\mathscr{F}$ is coherent, these conditions are equivalent to the following:

resume,label=a) $\operatorname{prof}_Z(\mathscr{F}) \geqslant 1$.

PROOF. To say that $\mathscr{F}$ is Z-pure means that for every Noetherian open subset V of X, and every open set $U \supset X - Z$, the restriction homomorphism $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap U, \mathscr{F})$ is injective (5.9.8); but by (3.1.8) this is equivalent to $V \cap \operatorname{Ass}(\mathscr{F}) \subset U$, whence the equivalence of *a)* and *b)*. Furthermore, to say that $x \in \operatorname{Ass}(\mathscr{F})$ means that no element of $\mathfrak{m}_x$ is $\mathscr{F}_x$-regular (3.1.2), hence, when $\mathscr{F}$ is coherent, this can be written $\operatorname{prof}(\mathscr{F}_x) = 0$; from this we immediately deduce in this case the equivalence of *b)* and *c)*. □

Corollary (5.10.3). — *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules. If $\mathscr{F}$ is Z-pure, so is $\mathscr{F}'$; conversely, if $\mathscr{F}'$ and $\mathscr{F}''$ are Z-pure, so is $\mathscr{F}$.*

PROOF. This results from the form (5.10.2), *b)* of the condition for a quasi-coherent $\mathscr{O}_X$-Module to be Z-pure, and from (3.1.7). □

Corollary (5.10.4). — *Suppose $\mathscr{F}$ coherent. For $\mathscr{F}$ to be Z-pure at a point $x \in X$, it is necessary and sufficient that* $\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 1$ (*with the notations of* (5.9.6)).

PROOF. This results immediately from (5.10.2) and (5.9.6). □

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Theorem (5.10.5). — *Let X be a locally Noetherian prescheme, Z a subset stable under specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to be Z-closed, it is necessary and sufficient that* $\operatorname{prof}_Z(\mathscr{F}) \geqslant 2$.

PROOF. By virtue of (5.10.2), we can limit ourselves to the case where $\mathscr{F}$ is Z-pure and $\operatorname{prof}_Z(\mathscr{F}) \geqslant 1$. Furthermore, to say that $\operatorname{prof}_Z(\mathscr{F}) \geqslant 2$ is equivalent to saying that for every closed subset Z_α of Z, $\operatorname{prof}_{Z_\alpha}(\mathscr{F}) \geqslant 2$; and similarly, it results from (5.9.8) that to say that $\mathscr{F}$ is Z-closed for every α is equivalent to saying that $\mathscr{F}$ is Z-closed in the case where Z is *closed*. The question being local, it suffices, for every $x \in Z$, to prove the theorem for $\mathscr{F}|U$, where U is an affine neighbourhood of x, and one can therefore reduce to the case where $X = U$ is affine. We know then that $\operatorname{Ass}(\mathscr{F})$ is finite (3.1.6), and since $\operatorname{Ass}(\mathscr{F}) \subset X - Z$, there is a section f of $\mathscr{O}_X$ above X such that $\operatorname{Ass}(\mathscr{F}) \subset X_f \subset X - Z$ (**II**, 4.5.4); we deduce from this that f is $\mathscr{F}$-regular (3.1.9) and that for every $y \in Z$, we have $f_y \in \mathfrak{m}_y$, hence $\operatorname{prof}(\mathscr{F}_y) = 1 + \operatorname{prof}(\mathscr{F}_y/f_y\mathscr{F}_y)$ (**0**, 16.4.6). The condition $\operatorname{prof}_Z(\mathscr{F}) \geqslant 2$ is thus equivalent to $\operatorname{prof}_Z(\mathscr{F}/f\mathscr{F}) \geqslant 1$, or equivalently (5.10.2) to the fact that $\mathscr{F}/f\mathscr{F}$ is Z-*pure*, and it suffices to see that this last property is equivalent to the fact that $\mathscr{F}$ is Z-closed.

Consider the exact sequence $0 \to \mathscr{F} \xrightarrow{f} \mathscr{F} \to \mathscr{F}/f\mathscr{F} \to 0$ (the homothety of ratio $f : \mathscr{F} \xrightarrow{f} \mathscr{F}$ being injective by hypothesis); setting $W = X - Z$, we have the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(X,\mathscr{F}) \arrow[r, "f"] \arrow[d] & \Gamma(X,\mathscr{F}) \arrow[r] \arrow[d] & \Gamma(X,\mathscr{F}/f\mathscr{F}) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \Gamma(W,\mathscr{F}) \arrow[r, "f"] & \Gamma(W,\mathscr{F}) \arrow[r] & \Gamma(W,\mathscr{F}/f\mathscr{F}) & \end{tikzcd}$$

where the two rows are exact (X being affine). If the restriction homomorphism $\Gamma(X,\mathscr{F}) \to \Gamma(W,\mathscr{F})$ is bijective, we deduce from this diagram that

$$\Gamma(X,\mathscr{F}/f\mathscr{F}) \longrightarrow \Gamma(W,\mathscr{F}/f\mathscr{F})$$

is injective, and this shows (5.9.8) that if $\mathscr{F}$ is Z-closed, $\mathscr{F}/f\mathscr{F}$ is Z-pure. Conversely, suppose that $\mathscr{F}/f\mathscr{F}$ is Z-pure, and let s be a section of $\mathscr{F}$ above W; since $X_f \subset W$, as $f^n(s|X_f)$ extends to a section t of $\mathscr{F}$ above X (**I**, 1.4.1); moreover, the restrictions of t and of $f^n s$ to X_f being the same, it follows that the restriction of t to W is equal to $f^n s$ by virtue of the relation $\mathrm{Ass}(\mathscr{F}) \subset X_f$ (5.10.2); since f is $\mathscr{F}$-regular, it suffices to see that t is of the form $f^n t'$, where $t' \in \Gamma(X,\mathscr{F})$ to show that the homomorphism $\Gamma(X,\mathscr{F}) \to \Gamma(W,\mathscr{F})$ is surjective, hence bijective. Now, to say that $t = f^n t'$ means that the image of t in $\Gamma(X,\mathscr{F}/f^n\mathscr{F})$ is zero. But since $f^k\mathscr{F}/f^{k+1}\mathscr{F}$ is isomorphic to $\mathscr{F}/f\mathscr{F}$, hence Z-pure by hypothesis, we deduce from (5.10.3) that $\mathscr{F}/f^n\mathscr{F}$ is Z-pure. But by definition the image of $t|W = f^n s$ in $\Gamma(W,\mathscr{F}/f^n\mathscr{F})$ is zero, whence the conclusion. ☐ **IV-2** | 116

Corollary (5.10.6). — *Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to be Z-closed at a point x, it is necessary and sufficient that* $\mathrm{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2$.

Proof. This results from (5.9.6) and (5.10.5). ☐

Theorem (5.10.7). — (Hartshorne). *Let X be a locally Noetherian prescheme, Y a closed subset of X. Suppose that for every $y \in Y$, we have* $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$*; then for every connected component C of X, $C - (C \cap Y)$ is connected.*

Proof. We can limit ourselves to the case where X is connected; it results then from (5.10.5) that the canonical homomorphism $\mathscr{O}_X \to i_*(\mathscr{O}_X|X - Y)$ (where $i : X - Y \to X$ is the canonical injection) is bijective. It follows that the restriction homomorphism $\Gamma(X,\mathscr{O}_X) \to \Gamma(X - Y,\mathscr{O}_X)$ is also bijective. It now suffices to apply the lemma (**III**, 7.8.6.1). ☐

Corollary (5.10.8). — *Let X be a locally Noetherian prescheme, d an integer such that for every $x \in X$, the relation* $\dim(\mathscr{O}_x) \geqslant d$ *implies* $\mathrm{prof}(\mathscr{O}_x) \geqslant 2$*. Suppose X connected; then, if X', X'' are two distinct irreducible components of X, there exists a suite $(X_i)_{0 \leqslant i \leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and that, for $1 \leqslant i \leqslant n$,* $\mathrm{codim}(X_{i-1} \cap X_i, X) \leqslant d - 1$ *(we say then that X is* connected in codimension $\leqslant d - 1$*).*

If Y is a closed subset of X such that $\mathrm{codim}(Y, X) \geqslant d$*, we have* $\dim(\mathscr{O}_{X,y}) \geqslant d$ *for every $y \in Y$* (5.1.3)*, hence* $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$ *for every $y \in Y$, and it results from* (5.10.7) *that $X - Y$ is connected. On the other hand, for* $\mathrm{codim}(Y, X) \geqslant d$*, it is necessary and sufficient that* $\mathrm{codim}(Y \cap V, V) \geqslant d$ *for every open neighbourhood V of y in X (**0**, 14.2.3). We note finally that if $\mathfrak{Y}$ denotes the set of closed subsets Y of X of codimension $\geqslant d$, the union of two sets of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$ (**0**, 14.2.5), and every closed set contained in a set of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$; this is also expressed by saying that $\mathfrak{Y}$ is an* antifilter *of closed subsets of X. The corollary then results from the following topological lemma:*

Lemma (5.10.8.1). — *Let X be a connected locally Noetherian topological space, $\mathfrak{Y}$ an antifilter of closed subsets of X. Suppose that if Y is a closed subset of X such that for every $y \in Y$, there is a neighbourhood V of y in X and a $Y_y \in \mathfrak{Y}$ such that $V \cap Y = V \cap Y_y$, then $Y \in \mathfrak{Y}$. The following conditions are then equivalent:*

label=a) *For every $Y \in \mathfrak{Y}$, $X - Y$ is connected.*

lbbel=b) *If X' and X'' are two distinct irreducible components of X, there exists a suite $(X_i)_{0 \leqslant i \leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$ and that, for $1 \leqslant i \leqslant n$, $X_{i-1} \cap X_i \notin \mathfrak{Y}$.*

PROOF. Suppose *b)* verified and let us prove that $U = X - Y$ is connected for every $Y \in \mathfrak{Y}$. If U' **IV-2 | 117**
and U'' are two distinct irreducible components of U, there exist two irreducible components X', X'' of X such that $X' \cap U = U'$, $X'' \cap U = U''$ ($\mathbf{0_I}$, 2.1.6); let us form for these two components a suite (X_i) having the property stated in *b)* and set $U_i = X_i \cap U$ for $1 \leqslant i \leqslant n$; then U_i is an irreducible component of U ($\mathbf{0_I}$, 2.1.6), and moreover $U_i \cap U_{i-1} \neq \varnothing$ for $1 \leqslant i \leqslant n$; if we had $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i with $1 \leqslant i \leqslant n$, we would deduce $X_{i-1} \cap X_i \subset Y$ by definition of Y, whence $U_{i-1} \cap U_i = \varnothing$, contradicting the hypothesis. This completes the proof that U is connected.

Let us now show that *a)* implies *b)*. Denote by Y the union of the family $(X_\alpha \cap X_\beta)$, where (X_α, X_β) ranges over the set of couples of *distinct* irreducible components of X such that $X_\alpha \cap X_\beta \in \mathfrak{Y}$. For every point $y \in Y$ in a neighbourhood V of y in X meeting only a finite number of irreducible components of X; it follows on the one hand that Y is closed and on the other that $V \cap Y$ is the intersection of V and of a set of $\mathfrak{Y}$; by virtue of the hypothesis on $\mathfrak{Y}$, we have $Y \in \mathfrak{Y}$, hence $U = X - Y$ is connected; moreover Y is rare in X. Let X', X'' be two distinct irreducible components of X, U', U'' their respective traces on U; these are distinct irreducible components of U ($\mathbf{0_I}$, 2.1.6). Now the union of the irreducible components W of U such that there exist irreducible components $(U_i)_{0 \leqslant i \leqslant n}$ of U for which $U_0 = U'$, $U_{i-1} \neq U_i$ and $U_{i-1} \cap U_i \neq \varnothing$ for $1 \leqslant i \leqslant n$ and $U_n = W$ is an *open and closed* set in U, since U is locally Noetherian and its irreducible components form a locally finite family of closed sets. There is thus such a suite (U_i) for which $U_n = U''$; let X_i $(0 \leqslant i \leqslant n)$ be the irreducible component of X such that $X_i \cap U = U_i$ ($\mathbf{0_I}$, 2.1.6); since $U_{i-1} \neq U_i$, we have $X_{i-1} \neq X_i$ for $1 \leqslant i \leqslant n$; if we had $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i with $1 \leqslant i \leqslant n$, we would deduce $X_{i-1} \cap X_i \subset Y$ by definition of Y, whence $U_{i-1} \cap U_i = \varnothing$, contradicting the hypothesis. This completes the proof of the lemma.

We note that the hypothesis on X in (5.10.8) is verified when X is a *Cohen-Macaulay prescheme* and $d \geqslant 2$. □

Corollary (5.10.9). — *If a Noetherian local ring A satisfies* (S_2) *and is catenary, it is equidimensional.*

PROOF. The hypothesis of (5.10.8) is then verified by $X = \operatorname{Spec}(A)$, with $d = 2$. To show that all the irreducible components of X have the same dimension, it suffices then, by virtue of (5.10.8), to show that two such components X', X'' have the same dimension when one further supposes that $\operatorname{codim}(X' \cap X'', X) = 1$. There is then an irreducible component Z of $X' \cap X''$ such that $\operatorname{codim}(Z, X) = 1$, hence $\operatorname{codim}(Z, X') = 1$, since $\operatorname{codim}(Z, X') \geqslant 1$; similarly $\operatorname{codim}(Z, X'') = 1$, and as X is catenary, this implies $\dim(X') = \dim(X'')$. □

Proposition (5.10.10). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and suppose that the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent. Then:*

label=(i) *We have* $\operatorname{prof}_Z(\mathscr{H}^0_{X/Z}(\mathscr{F})) \geqslant 2$.

lbbel=(ii) *For every point $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$, we have* $\operatorname{codim}(Z \cap \overline{\{x\}}, \overline{\{x\}}) \geqslant 2$.

lcbel=(iii) *The set U of the $x \in X$ such that $\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2$ (notations of (5.9.6)) is open in X; we have $X - U \subset Z$, and U is the largest open subset of X such that $\mathscr{F}|U$ is $(Z \cap U)$-closed.*

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PROOF. Set $\mathscr{F}' = \mathscr{H}^0_{X/Z}(\mathscr{F})$ for short. We know (5.9.11) that $\mathscr{F}'$ is Z-closed, and assertion (i) thus results from (5.10.5) applied to $\mathscr{F}'$. Let $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$; since the restrictions of $\mathscr{F}$ and $\mathscr{F}'$ to $X - Z$ are canonically isomorphic (5.9.11), we have $x \in \operatorname{Ass}(\mathscr{F}')$. Consider a point $y \in Z \cap \overline{\{x\}}$; the prime ideal $\mathfrak{p}$ of $\mathscr{O}_y$ corresponding to x is associated to the $\mathscr{O}_y$-module $\mathscr{F}'_y$, hence we have, by virtue of (i) and (**0**, 16.4.6.2), $2 \leqslant \operatorname{prof}(\mathscr{F}'_y) \leqslant \dim(\mathscr{O}_y/\mathfrak{p}) = \operatorname{codim}(\{y\}, \overline{\{x\}})$, whence (ii). To prove (iii), note that U is the set of the $x \in X$ such that $\widetilde{\mathscr{F}_x}$ is Z_x-closed (5.10.5), or equivalently, by virtue of (5.9.6), the set of points where the canonical homomorphism $\widetilde{\mathscr{F}_x} \to \widetilde{\mathscr{F}'_x}$ is bijective; it is therefore the complement of the union of the supports of $\operatorname{Ker}(\rho_{X/Z})$ and $\operatorname{Coker}(\rho_{X/Z})$, and these last are coherent $\mathscr{O}_X$-Modules by virtue of the hypothesis ($\mathbf{0_I}$, 5.3.4), hence have closed support ($\mathbf{0_I}$, 5.2.2); this shows that U is open, and U is clearly the largest open set such that $\mathscr{F}|U$ is $(Z \cap U)$-closed; finally the inclusion $X - U \subset Z$ results from (5.9.11).

We will see further on (5.11.1) that in the most important cases assertion (ii) conversely implies that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent. □

(5.10.11). Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation; we have seen that $\mathscr{A} = \mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_X$-Algebra (5.9.3); the X-scheme $X' = \operatorname{Spec}(\mathscr{H}^0_{X/Z}(\mathscr{O}_X))$ (**II**, 1.3.1) is called the *Z-closure* of X. Furthermore for every $\mathscr{O}_X$-Module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is an $\mathscr{A}$-Module, which is quasi-coherent; in this last case, there is thus a unique $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that, denoting by $g : X' \to X$ the structural morphism (**II**, 1.4.3)

(5.10.11.1) $$\mathscr{H}^0_{X/Z}(\mathscr{F}) = g_*(\mathscr{F}').$$

Proposition (5.10.12). — *The notations being those of (5.10.11):*

label=(i) *Let x be a point of X. For the morphism $X'_x \to X_x$ $(= \operatorname{Spec}(\mathscr{O}_{X,x}))$ deduced from g (notations of (5.10.12)) to be an isomorphism, it is necessary and sufficient that $\mathscr{O}_X$ be Z-closed at the point x (which holds for every $x \in X - Z$).*

lbbel=(ii) *Suppose furthermore that g be a finite morphism (see in (5.11.2) sufficient conditions for this to hold). Then the set U of the points $x \in X$ where X satisfies (S_2) is open in X ((5.7.2)) and* $\operatorname{codim}(X - U, X) \geqslant 2$.

lcbel=(iii) *Under the same hypotheses as in* (ii), *X' satisfies (S_2) and for every $x' \in X'$ such that $\operatorname{codim}(\overline{\{x'\}}, X') \leqslant 1$, the point $x = g(x')$ is such that $\operatorname{codim}(\overline{\{x\}}, X) = \operatorname{codim}(\overline{\{x'\}}, X')$.*

(5.10.13). We will now apply the results above to the case where Z is one of the sets $Z^{(n)}(X)$ (or simply $Z^{(n)}$), defined as the set of the $x \in X$ such that $\dim(\mathscr{O}_x) \geqslant n$; it is clear that $Z^{(n)}$ is stable under specialisation; for a closed subset T of X to be contained in $Z^{(n)}$, it is necessary and sufficient that $\operatorname{codim}(T, X) \geqslant n$. We shall be interested here in the case $n = 2$. **IV-2** | 119

Proposition (5.10.14). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module with support equal to X.*

label=(i) *For $\mathscr{F}$ to satisfy (S_1), it is necessary and sufficient that it be $Z^{(1)}$-pure.*

lbbel=(ii) *For $\mathscr{F}$ to satisfy (S_2), it is necessary and sufficient that it be $Z^{(2)}$-closed and $Z^{(1)}$-pure, or equivalently that it be $Z^{(2)}$-closed and have no associated prime cycle of codimension* 1.

Proof. label=*(i)* To say that $\mathscr{F}$ has the property (S_1) means that $\mathscr{F}$ has no immersed associated prime cycle (5.7.5), or equivalently that for every $x \in \operatorname{Ass}(\mathscr{F})$, we have $\dim(\mathscr{F}_x) = 0$ (3.1.4), in other words $\dim(\mathscr{O}_x) = 0$ (5.1.12.1); but this is equivalent to saying that $\operatorname{Ass}(\mathscr{F})$ does not meet $Z^{(1)}$, and the conclusion results from (5.10.2).

lbbel=*(ii)* To say that $\mathscr{F}$ is $Z^{(2)}$-closed means that $\operatorname{prof}_{Z^{(2)}}(\mathscr{F}) \geqslant 2$ (5.10.5), or equivalently that, for every $x \in X$, $\dim(\mathscr{F}_x) \geqslant 2$ implies $\operatorname{prof}(\mathscr{F}_x) \geqslant 2$; this shows that property (S_2) implies this condition; it implies in addition that $\mathscr{F}$ satisfies (S_1), hence has no immersed associated prime cycle. Conversely, suppose that $\mathscr{F}$ is $Z^{(2)}$-closed and has no associated prime cycle of codimension 1; to verify (S_2), it remains to show that if $x \in X$ is such that $\dim(\mathscr{F}_x) = 1$ (or, what amounts to the same, $\dim(\mathscr{O}_x) = 1$), then $\operatorname{prof}(\mathscr{F}_x) = 1$. But the hypothesis $\dim(\mathscr{O}_x) = 1$ implies $x \notin \operatorname{Ass}(\mathscr{F})$, and this last relation is equivalent to $\operatorname{prof}(\mathscr{F}_x) \neq 0$, i.e. $\operatorname{prof}(\mathscr{F}_x) \geqslant 1$. If $\mathscr{F}$ is $Z^{(1)}$-pure, it is noted above that it satisfies (S_1), hence all the associated prime cycles of $\mathscr{F}$ are of codimension 0, hence the preceding applies.

We note that it can happen that $\mathscr{F}$ is $Z^{(2)}$-closed and does not satisfy (S_1): this is the case for example when X is of dimension 1 (for then $Z^{(2)} = \varnothing$, and every $\mathscr{O}_X$-Module is $Z^{(2)}$-closed) and has immersed associated prime cycles.

Recall that in Chapter III, in the study of local cohomology, a cohomological characterisation of property (S_n) for all $n \geqslant 1$, generalising (5.10.14), is given. □

Corollary (5.10.15). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module with support X. Suppose that $\mathscr{F}$ has no associated prime cycles of codimension 1 and that $\mathscr{F}' = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent. Then:* **IV-2** | 120

label=(i) *$\mathscr{F}'$ satisfies the property (S_2).*

lbbel=(ii) *The set U of the $x \in X$ such that $\mathscr{F}$ satisfies (S_2) at the point x (5.7.2) is open in X and* $\operatorname{codim}(X - U, X) \geqslant 2$.

PROOF. label=*(i)* We know (5.9.11) that $\mathscr{F}'$ is $Z^{(2)}$-closed, and furthermore $\mathrm{Supp}(\mathscr{F}') = X$, since the maximal points of X belong to $X - Z^{(2)}$ and at these points $\mathscr{F}'_x = \mathscr{F}_x \neq 0$, hence the support of $\mathscr{F}'$ is dense in X, and as $\mathscr{F}'$ is coherent, $\mathrm{Supp}(\mathscr{F}')$ is closed, hence $\mathrm{Supp}(\mathscr{F}') = X$. It remains to see that $\mathscr{F}'$ has no associated prime cycles of codimension 1. But if $x \in \mathrm{Ass}(\mathscr{F}')$ and $\dim(\mathscr{F}'_x) = \dim(\mathscr{O}_x) = 1$, we have $x \in X - Z^{(2)}$, hence since $\mathscr{F}'_x = \mathscr{F}_x$ we would have $x \in \mathrm{Ass}(\mathscr{F})$, contrary to the hypothesis, which completes the proof of (i).

lbbel=*(ii)* We have $Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$, with the notations of (5.9.6), taking into account (**I**, 2.4.2); on the other hand, the hypothesis that $\mathscr{F}$ has no associated prime cycles of codimension 1 implies the same hypothesis for $\widetilde{\mathscr{F}_x}$; for $\widetilde{\mathscr{F}_x}$ to satisfy (S_2), it is therefore necessary and sufficient, by virtue of (5.10.14), that $\widetilde{\mathscr{F}_x}$ be $Z^{(2)}(X_x)$-closed; assertion (ii) therefore results from (5.10.6) and (5.10.10), (iii).

□

Proposition (5.10.16). — *Let X be a locally Noetherian prescheme,*

$$X' = \mathrm{Spec}(\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{O}_X))$$

its $Z^{(2)}$-closure, $g : X' \to X$ the structural morphism. Suppose that X has no associated prime cycle of codimension 1.

label=(i) *For X to satisfy (S_2) at a point $x \in X$, it is necessary and sufficient that the morphism $X'_x \to X_x$ deduced from g (notations of* (5.10.12)*) be an isomorphism. This condition is always satisfied if* $\mathrm{codim}(\overline{\{x\}}, X) \leqslant 1$.

lbbel=(ii) *Suppose furthermore that g be a finite morphism (see in* (5.11.2) *sufficient conditions for this to hold). Then the set U of the points where X satisfies (S_2) is open and* $\mathrm{codim}(X - U, X) \geqslant 2$; *furthermore U is the largest open subset of X such that the restriction $g^{-1}(U) \to U$ of g is an isomorphism.*

lcbel=(iii) *Under the same hypotheses as in* (ii), *X' satisfies (S_2) and for every $x' \in X'$ such that* $\mathrm{codim}(\overline{\{x'\}}, X') \leqslant 1$, *the point $x = g(x')$ is such that* $\mathrm{codim}(\overline{\{x\}}, X) = \mathrm{codim}(\overline{\{x'\}}, X')$.

ldbel=(iv) *The hypotheses being those of* (ii), *let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module with support X, such that $\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent; then the $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that $g_*(\mathscr{F}') = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent and satisfies (S_2), and its support is a union of irreducible components of X'.*

PROOF. Assertions (i) and (ii) are included for reference, having already been proved in substance: (i) results in fact from (5.10.12), (i) and (5.10.14), and (ii) is a particular case of (5.10.15), (ii). **IV-2** | 121

Let us prove (iii); set $x = g(x')$; since g is finite, it is of the same for the morphism $X'_x \to X_x$, hence $\dim(\mathscr{O}_{X',x'}) \leqslant \dim(X_x) \leqslant \dim(\mathscr{O}_{X,x})$ by virtue of (5.4.1). Suppose first that $\dim(\mathscr{O}_{X',x'}) \leqslant 1$ and let us show that then $\dim(\mathscr{O}_{X,x}) \leqslant 1$. Otherwise, we would have $\dim(\mathscr{O}_{X,x}) \geqslant 2$, i.e. $x \in Z^{(2)}$; then we would have $\mathrm{prof}(\mathscr{O}_{X',x'}) \geqslant 2$ by (5.10.12), which is absurd. On the other hand, $x \in X - Z^{(2)}$, and hence $\mathscr{O}_{X',x'}$ is isomorphic to $\mathscr{O}_{X,x}$ (5.10.12), (i), whence $\dim(\mathscr{O}_{X',x'}) = \dim(\mathscr{O}_{X,x})$. Furthermore, since X has no associated prime cycles of codimension 1, the hypothesis $\dim(\mathscr{O}_{X,x}) = 1$ implies $x \notin \mathrm{Ass}(\mathscr{O}_X)$, hence $\mathrm{prof}(\mathscr{O}_{X,x}) \neq 0$, and hence also $\mathrm{prof}(\mathscr{O}_{X',x'}) = 1$. Suppose now $\dim(\mathscr{O}_{X',x'}) \geqslant 2$, hence $\dim(\mathscr{O}_{X,x}) \geqslant 2$, i.e. $x \in Z^{(2)}$; then we deduce from (5.10.12) that $\mathrm{prof}(\mathscr{O}_{X',x'}) \geqslant 2$ by (5.10.12), which establishes the assertions of (iii).

To prove (iv) it suffices to replace $\mathscr{O}_X$ by $\mathscr{F}$ in the preceding reasoning, which establishes that $\mathscr{F}'$ satisfies (S_2) and that if $\dim(\mathscr{F}'_{x'}) \leqslant 1$, $\mathscr{F}_x$ and $\mathscr{F}'_{x'}$ are isomorphic; in particular if $\dim(\mathscr{F}'_{x'}) = 0$, we have $\dim(\mathscr{O}_{X,x}) = 0$, hence $\dim(\mathscr{F}_x) = 0$ since $\mathscr{F}$ has support X, and finally $\dim(\mathscr{O}_{X',x'}) = 0$; every irreducible component of $\mathrm{Supp}(\mathscr{F}')$ is therefore an irreducible component of X', since $\mathscr{F}'$ is coherent, hence $\mathrm{Supp}(\mathscr{F}')$ is closed, hence $\mathrm{Supp}(\mathscr{F}')$ is a union of irreducible components of X'. □

Proposition (5.10.17). — *Let A be an integral Noetherian ring, and denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height* 1. *Suppose that $A^{(1)}$ is a finite A-algebra. Then:*

label=(i) *The ring $A^{(1)}$ satisfies the condition* (S_2).

lbbel=(ii) *The set U of the $\mathfrak{p} \in \operatorname{Spec}(A)$ such that the canonical homomorphism $A_{\mathfrak{p}} \to (A^{(1)})_{\mathfrak{p}}$ is bijective is equal to the set of the $\mathfrak{p}$ such that $A_{\mathfrak{p}}$ satisfies* (S_2); *U is open in $X = \operatorname{Spec}(A)$ and* $\operatorname{codim}(X - U, X) \geqslant 2$.

lcbel=(iii) *For every multiplicative subset S of A, $(S^{-1}A)^{(1)}$ is a $(S^{-1}A)$-algebra that is finite.*

ldbel=(iv) *Let B be a finite A-algebra, integral and containing A. Then $B^{(1)}$ is a finite B-algebra. Furthermore, for every prime ideal $\mathfrak{q}$ of B of height* 1, *the prime ideal $\mathfrak{q} \cap A$ of A is of height* 1.

PROOF. Taking into account the formula (5.9.3.1), we see that $X' = \operatorname{Spec}(A^{(1)})$ is the $\mathrm{Z}^{(2)}$-closure of $X = \operatorname{Spec}(A)$; as A has no immersed prime ideals, properties (i) and (ii) are particular cases of (5.10.16), (i), (ii) and (iii). To prove (iii), it suffices to remark that $(S^{-1}A)^{(1)} = S^{-1}A^{(1)}$, which is a particular case of (5.9.4): indeed $S^{-1}A$ is a flat A-module, the prime ideals of $S^{-1}A$ are the $S^{-1}\mathfrak{p}$, where $\mathfrak{p} \in \operatorname{Spec}(A)$ does not meet S, and $\operatorname{ht}(S^{-1}\mathfrak{p}) = \operatorname{ht}(\mathfrak{p})$, whence (iii). As $A^{(1)}$ is a finite A-algebra, $S^{-1}A^{(1)}$ is a finite $S^{-1}A$-algebra, whence (iii). To prove (iv), set $Y = \operatorname{Spec}(B)$, and let $f : Y \to X$ be the structural morphism; since it is finite, it results from (5.4.1) that for every $y \in Y$, $\dim(\mathscr{O}_{Y,y}) \leqslant \dim(\mathscr{O}_{X,f(y)})$; hence, if $T = f^{-1}(\mathrm{Z}^{(2)}(X))$, we have $T \supset \mathrm{Z}^{(2)}(Y)$. Let us show that $\mathscr{G} = \mathscr{H}^0_{Y/\mathrm{Z}^{(2)}(Y)}(f_*(\mathscr{O}_Y))$ is coherent; indeed $f_*(\mathscr{O}_Y) = \widetilde{B}$, B being considered as an A-module; but since B is an integral finite A-algebra, its field of fractions is finite over the field of fractions of A, and therefore B is contained in a free A-module of finite type, and by (5.9.2), (i) $\mathscr{G}$ is a quasi-coherent sub-$\mathscr{O}_X$-Module of $(\mathscr{H}^0_{X/\mathrm{Z}^{(2)}(X)}(\mathscr{O}_X))^n$ for a suitable n; this last being coherent by hypothesis, so is $\mathscr{G}$. Now, it results from the definition (5.9.1.2) that $\mathscr{G}$ is isomorphic to $f_*(\mathscr{H}^0_{Y/T}(\mathscr{O}_Y))$; this proves *a fortiori* that $\mathscr{H}^0_{Y/T}(\mathscr{O}_Y)$ is a coherent $\mathscr{O}_Y$-Module. It results then from (5.10.10), (ii), applied to $\mathscr{O}_Y$ and to codimension 2, that $\mathrm{Z}^{(2)}(Y) \subset T$, and finally $T = \mathrm{Z}^{(2)}(Y)$. This proves the two assertions of (iv). □

5.11. Coherence criteria for the Modules $\mathscr{H}^0_{X/Z}(\mathscr{F})$ **IV-2** | 122

Proposition (5.11.1). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Denote by (x_α) the family of points of* $\operatorname{Ass}(\mathscr{F}) \cap (X - Z)$ *and, for every α, let Y_α be the reduced closed subprescheme of X with underlying space $\overline{\{x_\alpha\}}$, and $Z_\alpha = Z \cap \overline{\{x_\alpha\}}$. Then the following two conditions are equivalent:*

label=a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-Module.*

lbbel=b) *For every α, $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$ is a coherent $\mathscr{O}_{Y_\alpha}$-Module.*

These two conditions imply the following:

resume,label=a) *For every α, we have* $\operatorname{codim}(Z_\alpha, Y_\alpha) \geqslant 2$.

Moreover, the three conditions a), b), c) *are equivalent when in addition one of the following properties is satisfied:*

label=(i) *Every point of X admits an open neighbourhood isomorphic to a subscheme of a regular scheme* (*in which case we also say that X is* locally immersible in a regular scheme).

lbbel=(ii) *For every α, Y_α is universally catenary* (5.6.2) *and its normalisation* (**II**, 6.3.8) *Y'_α is finite over Y_α.*

PROOF. All the properties under consideration are local on X, so we can limit ourselves to the case where $X = \operatorname{Spec}(A)$ is affine, and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type. Then, for every α, if h_α is the canonical injection $Y_\alpha \to X$, $(h_\alpha)_*(\mathscr{O}_{Y_\alpha}) = \mathscr{G}_\alpha$ is the $\mathscr{O}_X$-Module corresponding to the A-module quotient $A/\mathfrak{J}_{x_\alpha}$, and, by definition of $\operatorname{Ass}(\mathscr{F})$, this A-module is isomorphic to a sub-A-module of M. Since $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$ is a quasi-coherent sub-$\mathscr{O}_X$-Module of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ (5.9.2), the hypothesis that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent implies that so is $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$. On the other hand, it results from the definition (5.9.1.2) that $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$ is isomorphic to $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$; this proves that *a)* implies *b)*. To see that *b)* implies *a)*, it suffices to show that there is a finite filtration $(\mathscr{F}_i)_{0 \leqslant i \leqslant n}$ of $\mathscr{F}$ formed by coherent $\mathscr{O}_X$-Modules, with $\mathscr{F}_0 = \mathscr{F}$, $\mathscr{F}_n = 0$, and such that $\mathscr{H}^0_{X/Z}(\mathscr{F}_i/\mathscr{F}_{i+1})$ is coherent for every i: this results, by descending induction on i, from the exact sequences

$$0 \longrightarrow \mathscr{F}_{i+1} \longrightarrow \mathscr{F}_i \longrightarrow \mathscr{F}_i/\mathscr{F}_{i+1}$$

from the fact that $\mathscr{H}^0_{X/Z}$ is a left exact functor (5.9.2), and finally from ($\mathbf{0_I}$, 5.3.3) and (**I**, 6.1.1). By virtue of (3.2.8), it therefore suffices to prove that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent when $\mathscr{F}$ is irredundant; in other words, $\mathrm{Ass}(M) = \{\mathfrak{p}\}$ is reduced to a single element. We now note the

Lemma (5.11.1.1). — *Let A be a Noetherian ring, M an A-module of finite type, such that $\mathrm{Ass}(M) = \{\mathfrak{p}\}$. There exists a finite filtration $(M_h)_{0\leqslant h\leqslant m}$ of M such that $M_0 = M$, $M_m = 0$ and that M_h/M_{h+1} is isomorphic to a submodule of $A/\mathfrak{p}$.*

PROOF. First note that the canonical homomorphism $M \to M_{\mathfrak{p}} = N$ is injective (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 2, prop. 6). Set $B = A_{\mathfrak{p}}$, $\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}$, the maximal ideal of B; we have $\mathrm{Ass}(N) = \{\mathfrak{m}\}$ (*loc. cit.*, prop. 5), and since N is a B-module of finite type, there is an integer r such that $\mathfrak{m}^r N = 0$; if we set $N'_j = \mathfrak{m}^j N$ for $0 \leqslant j \leqslant r$, N'_j/N'_{j+1} is an $(A/\mathfrak{p})$-module of finite type, hence a direct sum of a finite number of submodules isomorphic to $B/\mathfrak{m}$; in other words, there is a finite filtration $(N_h)_{0\leqslant h\leqslant m}$ of N such that $N_0 = N$, $N_m = 0$ and that N_h/N_{h+1} is isomorphic to $B/\mathfrak{m}$, i.e. to the field of fractions of $A/\mathfrak{p}$; the filtration of the $M_h = M \cap N_h$ answers the question, since M_h/M_{h+1} is isomorphic to a sub-$(A/\mathfrak{p})$-module of finite type of $N_h/N_{h+1} = B/\mathfrak{m}$; but one knows that such a submodule is isomorphic to a submodule of $A/\mathfrak{p}$. □ **IV-2 | 123**

The existence of the filtration (M_h) then shows, by the same reasoning as above, that to prove (granting *b)*) that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent, we can limit ourselves to showing that $\mathscr{H}^0_{X/Z}(\mathscr{P})$ is coherent, where $\mathscr{P} = (A/\mathfrak{q})^{\sim}$, $\mathfrak{q}$ being an ideal associated to M. But if $\mathfrak{q} = \mathfrak{J}_y$ with $y \in Z$, the support of $\mathscr{P}$ is a closed set contained in Z, since Z is stable under specialisation; the definition (5.9.1.2) then shows that $\mathscr{H}^0_{X/Z}(\mathscr{P}) = 0$. If on the contrary $\mathfrak{q} = \mathfrak{J}_{x_\alpha}$ for some α, we have $\mathscr{P} = (h_\alpha)_*(\mathscr{O}_{Y_\alpha})$ by definition, and we have seen above that $\mathscr{H}^0_{X/Z}(\mathscr{P})$ is isomorphic to $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$, hence is coherent by virtue of *b)*.

We have already seen (5.10.10), (ii) that *a)* implies *c)*. It remains to prove that, under one or the other of the hypotheses (i), (ii), *c)* implies *b)*. Note that if X satisfies (i), so does each Y_α. Similarly (5.11.1); it therefore suffices to prove (5.9.3.1) the corollary below. □

Corollary (5.11.2). — *Let A be a Noetherian integral ring, satisfying one of the two following hypotheses:*

label=(i) *A is a quotient of a regular Noetherian ring.*

lbbel=(ii) *A is universally catenary, and its integral closure A' is a finite A-algebra.*

Then the ring $A^{(1)}$, intersection of the $A_{\mathfrak{p}}$ where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1, is a finite A-algebra.

PROOF. (i) Set $X = \mathrm{Spec}(A)$. The set U of the points $x \in X$ where X satisfies (S_2) is *open* in X by (6.11.2)[2]. Furthermore, if one sets $Z = X - U$, we have $\mathrm{codim}(Z, X) \geqslant 2$: indeed, for every $x \in X$ such that $\dim(A_{x'}) \leqslant 1$, we have $\dim(A_{x'}) \leqslant 1$ for every generalisation x' of x, hence $\mathrm{prof}(A_{x'}) \geqslant 1$, hence X satisfies (S_2) at the point x. We have thus $Z \subset Z^{(2)}$, and $\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{O}_X) = j_*(\mathscr{H}^0_{U/Z^{(2)}}(\mathscr{O}_U))$, where $j : U \to X$ is the canonical injection (5.9.1.2). But since the prescheme U satisfies (S_2), it results from (5.10.14) that $\mathscr{H}^0_{U/Z^{(2)}}(\mathscr{O}_U)$ is isomorphic to $\mathscr{O}_U$; on the other hand, since $\mathrm{codim}(Z, X) \geqslant 2$, one knows, from Chapter III, § 9, that $j_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_X$-Module, which proves the proposition in case (i).

(ii) The ring A' is, by virtue of hypothesis (ii), a Noetherian integral ring integrally closed, hence (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4) the intersection of its local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A'. Now, for such a prime ideal $\mathfrak{p}'$, if one sets $\mathfrak{p} = \mathfrak{p}' \cap A$ and $S = A - \mathfrak{p}$, $A'_{\mathfrak{p}'}$ is a local ring at a prime ideal $S^{-1}\mathfrak{p}'$ of $S^{-1}A'$; since $S^{-1}\mathfrak{p}'$ lies above the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$, it is a maximal ideal of $S^{-1}A'$; and $S^{-1}A'$ is by hypothesis a finite $A_{\mathfrak{p}}$-algebra; since $S^{-1}\mathfrak{p}'$ lies above the maximal ideal of $A_{\mathfrak{p}}$, $A_{\mathfrak{p}}$ is universally catenary. We deduce from (5.6.10) that $\dim(A_{\mathfrak{p}}) = \dim(A'_{\mathfrak{p}'}) = 1$. It results from this that $A^{(1)} \subset A'$, and as A' is a finite module of A by hypothesis, so is $A^{(1)}$ since A is Noetherian; the proposition is therefore proved in case (ii). □ **IV-2 | 124**

Remark (5.11.3). — The fact that $A^{(1)}$ is a finite A-algebra is not strictly necessary in hypothesis (ii) of (5.11.2): one only supposes that A is *catenary*. An example is given by the catenary local ring A

[2] The reader may verify that (5.11.2) is not used in the proof of (6.11.2).

constructed in (5.6.11), whose integral closure A' (denoted B in (5.6.11)) is a finite A-algebra; if $A^{(1)}$ were a finite A-algebra, it would be contained in the field of fractions of A, hence contained in A'. But on the other hand, with the notations of (5.6.11), every prime ideal $\mathfrak{r}$ of height 1 in A is of the form $\mathfrak{p}A$, where $\mathfrak{p}$ is a prime ideal of height 1 in C, and $A_{\mathfrak{p}\mathfrak{A}} = C_{\mathfrak{p}}$; we know (5.6.11) that $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$, where $\mathfrak{p}'$ is the unique prime ideal of E above $\mathfrak{p}$, hence $A_{\mathfrak{p}\mathfrak{A}}$ is integrally closed and contains A'; by definition, $A^{(1)} \subset A'$, and the hypothesis that $A^{(1)}$ is a finite A-algebra would imply finally $A^{(1)} = A'$. But this conclusion is absurd, since one of the two prime ideals of $\mathfrak{n}A$ of A is of height 1, while $\mathfrak{n}A$ is of height 2, which would contradict (5.10.17), (iv).

Corollary (5.11.4). — *Let X be a locally Noetherian prescheme, Z a closed subset of X, $U = X - Z$, $i : U \to X$ the canonical injection, $\mathscr{F}$ an $\mathscr{O}_U$-Module coherent. For $i_*(\mathscr{F})$ to be a coherent $\mathscr{O}_X$-Module, it is necessary that for every $x \in \operatorname{Ass}(\mathscr{F})$, $\operatorname{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$. This condition is sufficient in each of the two following cases:*

label=(i) *The prescheme X is locally immersible in a regular scheme.*

lbbel=(ii) *The prescheme X is universally catenary and universally Japanese (which means, by definition, that every point $x \in X$ admits an affine open neighbourhood whose ring is universally Japanese (**0**, 23.1.1)).*

Proof. One knows (**I**, 9.4.7) that there exists a quasi-coherent sub-$\mathscr{O}_X$-Module $\mathscr{G}$ of $i_*(\mathscr{F})$ such that $\mathscr{G}|U = \mathscr{F}$. We have clearly $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{G}) \subset \operatorname{Ass}(i_*(\mathscr{F}))$, and since $\operatorname{Ass}(i_*(\mathscr{F})) = \operatorname{Ass}(\mathscr{F})$ (3.1.13), on has $\operatorname{Ass}(\mathscr{G}) = \operatorname{Ass}(\mathscr{F})$; it suffices then to apply (5.11.1) to the coherent $\mathscr{O}_X$-Module $\mathscr{G}$, noting that $i_*(\mathscr{F}) = \mathscr{H}^0_{X/Z}(\mathscr{G})$ and that, when X is universally catenary and universally Japanese, hypothesis (ii) of (5.11.1) is satisfied by definition. □

Corollary (5.11.5). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module such that $\operatorname{Ass}(\mathscr{F}) \subset X - Z$. Then the condition:*

label=a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-Module*

implies the following:

ldbel=d) *For every closed subset T of Z in X (or only for a* filtering family

of closed subsets T of Z, with union Z), $i_*(\mathscr{F}|X - T)$ (where $i : X - T \to X$ is the canonical injection) is a coherent $\mathscr{O}_X$-Module. **IV-2** | 125

Furthermore, the conditions a) *and* d) *are satisfied in each of the two following cases:*

label=(i) *A is a quotient of a regular Noetherian ring.*

lbbel=(ii) *A is universally catenary and universally Japanese.*

We then have the implications

$$a) \Leftrightarrow b) \Rightarrow c) \Leftrightarrow d)$$

Proof. When X satisfies one of the hypotheses (i), (ii) of (5.11.1), *a)* and *d)* are equivalent. Note that $\operatorname{Ass}(i_*(\mathscr{F}|X - T)) = \operatorname{Ass}(\mathscr{F})$ by virtue of the hypothesis and of (3.1.13); it results therefore from (5.10.2) that the canonical applications

$$i_*(\mathscr{F}|X - T) \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F})$$

are *injective*; the fact that *a)* implies *d)* is therefore a consequence of this remark. Conversely, the condition *d)* implies, by virtue of (5.11.1) that $\operatorname{codim}(T \cap Y_\alpha, Y_\alpha) \geqslant 2$ with the notations of (5.11.1); hence $\operatorname{codim}(Z \cap Y_\alpha, Y_\alpha) \geqslant 2$ since Z is a union of its closed subsets in X, and the last assertion of the corollary follows from (5.11.1). □

Corollary (5.11.6). — *Let A be a Noetherian integral ring, $X = \operatorname{Spec}(A)$. Consider the following properties:*

label=a) *For every integral quotient ring B of A, the ring $B^{(1)}$ (notation of (5.11.2)) is a finite B-algebra.*

lbbel=b) *For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$ and every closed subset Z of X, stable under specialisation, and such that for every $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$ we have $\operatorname{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$, the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.*

lcbel=c) *For every closed subset T of X and every coherent $\mathscr{O}_U$-Module $\mathscr{G}$ (where $U = X - T$) such that for every $x \in \operatorname{Ass}(\mathscr{G})$ we have $\operatorname{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geqslant 2$, $i_*(\mathscr{G})$ (where $i : U \to X$ is the canonical injection) is a coherent $\mathscr{O}_X$-Module.*

ldbel=d) *For every integral quotient ring B of A and every ideal $\mathfrak{J}$ of height $\geqslant 2$ in B, the ring $\bigcap_{\mathfrak{q}\not\supset\mathfrak{J}} B_{\mathfrak{q}}$ is a finite B-algebra.*

We then have the implications

$$a) \Leftrightarrow b) \Rightarrow c) \Leftrightarrow d)$$

Furthermore, the conditions a), b), c) *and* d) *are satisfied in each of the two following cases:*

label=(i) *A is a quotient of a regular Noetherian ring.*

lbbel=(ii) *A is universally catenary and universally Japanese.*

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Proof. Let $B = A/\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of A, so that $Y = \operatorname{Spec}(B)$ is the closed subset $V(\mathfrak{q})$ of X; set $Z = Z^{(2)}(Y)$, which is a subset of X stable under specialisation. If condition *b)* is satisfied, one can apply it to the $\mathscr{O}_X$-Module coherent $\mathscr{F} = (A/\mathfrak{q})^{\sim}$ and to $Z = V(\mathfrak{J})$; inversely, one proves that *d)* implies *c)* by using the equivalence of *a)* and *b)* in (5.11.1). It is clear that *c)* is a particular case of *b)*. Finally, the fact that *a)* (and hence each of the other conditions) is satisfied when A satisfies one of the hypotheses (i), (ii) results from (5.11.2), by the definition of universally catenary rings and of universally Japanese rings. □

Remarks (5.11.7). — label=*(i)* If, in (5.11.5), condition *d)* implies *a)* without supplementary hypothesis on X; we shall see further on (7.2.4) that this is indeed the case also when X is a *local* scheme. Similarly, we shall prove that when A is a local Noetherian ring the four properties *a)*, *b)*, *c)*, *d)* of (5.11.6) are equivalent (7.2.4). If A satisfies property *a)* of (5.11.6), it is of the same for every *ring of fractions* $S^{-1}A$ (5.10.17), (iii) and (iv); in particular, it extends to all Noetherian rings.

lbbel=*(ii)* If A satisfies property *a)* of (5.11.6), it is of the same for every *integral finite A-algebra C*. Indeed every integral quotient ring of $S^{-1}A$ is of the form $S^{-1}(A/\mathfrak{q})$, where $\mathfrak{q}$ is a prime ideal of A not meeting S; and on the other hand, if $\mathfrak{r}$ is a prime ideal of C, $\mathfrak{p}$ its inverse image in A, $C_{\mathfrak{r}}$ is an $(A/\mathfrak{p})$-algebra that is finite integral with the field of fractions finite over that of $A/\mathfrak{p}$; our assertions are therefore consequences of (5.10.17), (iii) and (iv).

5.12. Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA We have already seen in (3.4) relations between the properties of A and of A/tA concerning the associated prime ideals, as well as the properties of being integral or reduced. In this section we give other relations between the properties of these rings, related to the notions of dimension and of depth.

Proposition (5.12.1). — *Let A be a Noetherian local ring, $X = \operatorname{Spec}(A)$, t an element of A belonging to a system of parameters* (**0**, *16.3.6*), *X_0 the closed subspace $V(t)$ of X, X_1 the complementary open subset $X - X_0$. Let Z be a subset of X such that every specialisation of a point of Z belongs to Z. Suppose that X is biequidimensional* (*which will be the case if A is equidimensional and a quotient of a regular local ring* (**0**, *16.5.12*)). *Then if we set $Z_0 = Z \cap X_0$, $Z_1 = Z \cap X_1$, we have*

(5.12.1.1) $$\operatorname{codim}(Z_0, X_0) \leqslant \operatorname{codim}(Z, X) \leqslant \operatorname{codim}(Z_1, X_1).$$

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Proof. The second inequality results from the definition (5.1.3); to prove the first, we can limit ourselves to proving it when Z is *closed*: indeed, by hypothesis Z is a union of closed subsets Z_α, and if $\operatorname{codim}(X_0 \cap Z_\alpha, X_0) \leqslant \operatorname{codim}(Z_\alpha, X)$ for every α, we will have also $\operatorname{codim}(Z_0, X_0) = \inf_\alpha(\operatorname{codim}(X_0 \cap Z_\alpha, X_0)) \leqslant \inf_\alpha(\operatorname{codim}(Z_\alpha, X)) = \operatorname{codim}(Z, X)$. Suppose therefore Z closed, so that

$$\operatorname{codim}(Z, X) = \dim(X) - \dim(Z).$$

On the other hand, X_0 is clearly catenary, equidimensional and of dimension $\dim(X) - 1$ by virtue of (**0**, 16.3.4); we thus also have

$$\operatorname{codim}(Z_0, X_0) = \dim(X_0) - \dim(Z_0) = \dim(X) - 1 - \dim(Z_0).$$

But we have $\dim(Z_0) \geqslant \dim(Z) - 1$ (**0**, 16.3.4), which completes the proof of the first inequality (5.12.1.1). □

Proposition (5.12.2). — *Let A be a Noetherian local ring, M an A-module of finite type, t an M-regular element belonging to the maximal ideal $\mathfrak{m}$ of A, k an integer $\geqslant 1$. Suppose that A is a catenary ring. Then, if M/tM is equidimensional and satisfies (S_k), the same is true of M.*

PROOF. Taking into account ($\mathbf{Err}_{\mathbf{III}}$, 30) and (5.7.3), (vi), we can limit ourselves to the case where $\mathrm{Supp}(M) = \mathrm{Spec}(A) = X$. Set $X_0 = V(t) = \mathrm{Spec}(A/tA)$; since M/tM satisfies (S_1) by hypothesis, it has no immersed associated prime ideals (5.7.5). Applying (3.4.4) to the closed point of X (and of every closed subprescheme of X), we see that the irreducible components of $\mathrm{Supp}(M/tM)$ are exactly those of $Y_i \cap X_0$, where Y_i $(1 \leqslant i \leqslant r)$ are the irreducible components of X. Since t is by hypothesis M-regular, $V(t)$ contains no maximal point of X (3.1.8); each of the irreducible components of $Y_i \cap X_0$ is therefore of codimension 1 in Y_i (5.1.8); since X is catenary and since all the irreducible components of $\mathrm{Supp}(M/tM)$ have by hypothesis the same dimension, we conclude that the Y_i all have the same dimension, in other words X is equidimensional, hence *biequidimensional* since A is local and catenary. To see that M satisfies (S_k), let us apply the criterion (5.7.4) (with the notations of (5.7.4)): we must verify that, for every integer $n \geqslant 0$, $\mathrm{codim}(Z_n, X) > n + k$. But the hypothesis on M/tM and the criterion (5.7.4) show that $\mathrm{codim}(Z_n \cap X_0, X_0) > n + k$ for every integer $n \geqslant 0$. But every specialisation of a point of Z_n still belongs to Z_n, as we shall see in (6.11.5)[3]; since X is biequidimensional and t belongs to a system of parameters for M ($\mathbf{0}$, 16.4.1), and since t is a non-zero-divisor for A (the annihilator of M being nilpotent by virtue of the hypothesis $\mathrm{Supp}(M) = \mathrm{Spec}(A)$), the conclusion results from (5.12.1). □

Remark (5.12.3). — If, in the statement of (5.12.2), one only supposes that M/tM is equidimensional, the conclusion does not necessarily follow that M is equidimensional.

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Corollary (5.12.4). — *Let A be a Noetherian local catenary ring, t a regular element belonging to the maximal ideal of A, k an integer $\geqslant 1$. If A/tA satisfies (S_k), the same is true of A.*

PROOF. If $k = 1$, the corollary results from (3.4.4), in view of the interpretation of condition (S_1) (5.7.5). If $k \geqslant 2$, it results from the criterion of Hartshorne (5.10.9) that A/tA is equidimensional, and one can apply (5.12.2). □

Proposition (5.12.5). — *Let A be a Noetherian local catenary ring, t an element of A belonging to the maximal ideal of A, k an integer $\geqslant 0$. If the ring A/tA is reduced, equidimensional and satisfies the property (R_k), the ring A also has these three properties.*

PROOF. It is already known (3.4.6) that A is reduced, and it results from (5.12.2), applied for $k = 1$, that A is equidimensional. Set $X = \mathrm{Spec}(A)$, $X_0 = V(t) = \mathrm{Spec}(A/tA)$, and let Z be the set of points $x \in X$ where X is not regular; by virtue of ($\mathbf{0}$, 17.3.2), every specialisation of a point of Z also belongs to Z. It results on the other hand from ($\mathbf{0}$, 17.1.8) that for every point $x \in X_0$ where X is regular, X is also regular, hence the set Z_0' of the points where X_0 is not regular contains $Z \cap X_0$; the hypothesis therefore implies that

$$k \leqslant \mathrm{codim}(Z_0', X_0) \leqslant \mathrm{codim}(Z_0, X_0).$$

But, since A is equidimensional and catenary, X is biequidimensional, hence from (5.12.1)

$$\mathrm{codim}(Z_0, X_0) \leqslant \mathrm{codim}(Z, X)$$

which proves that X satisfies (R_k). □

Remark (5.12.6). — If, in the statement of (5.12.5), one only supposes that A/tA is reduced and possesses property (R_k), it does not necessarily follow that A possesses property (R_k).

Corollary (5.12.7). — *Let A be a Noetherian local catenary ring, t a regular element belonging to the maximal ideal of A. If A/tA is integral and integrally closed, the same is true of A.* **IV-2 | 129**

PROOF. By virtue of Serre's criterion (5.8.6) and of the fact that A is local, it suffices to show that $\mathrm{Spec}(A)$ satisfies (S_2) and (R_1). But the hypothesis on A/tA implies that A satisfies (R_1) by (5.12.5), and that A satisfies (S_2) by (5.12.4). □

[3]The reader may verify that (5.12.2) is not used in the proof of (6.11.5).

Proposition (5.12.8). — (Hironaka's lemma). *Let A be a Noetherian local ring, reduced, equidimensional and catenary. Suppose furthermore that for every minimal prime ideal $\mathfrak{q}_i$ of A, the ring $B_i = A/\mathfrak{q}_i$ is such that $B_i^{(1)}$ (notation of (5.10.17)) is a finite B_i-algebra (which occurs for instance when A possesses one of the properties (i) or (ii) of (5.11.6)). Let t be an element of A, non-zero-divisor in A, and such that:*

*label=(i) The A-module A/tA has only one non-immersed associated prime ideal $\mathfrak{p}$ (necessarily of height 1 by the Hauptidealsatz (**0**, 16.3.2)), but one does not suppose that this is the only prime ideal associated to A/tA.*

lbbel=(ii) The ideal $\mathfrak{p}A_\mathfrak{p}$ of $A_\mathfrak{p}$ is generated by $t/1$.

lcbel=(iii) The ring $A/\mathfrak{p}$ is integrally closed.

Under these conditions, the ring A is integral and integrally closed, and we have $\mathfrak{p} = tA$.

PROOF. Since $Z = V(tA)$ is an irreducible closed subset of $X = \mathrm{Spec}(A)$, whose generic point z **IV-2** | 130
is such that $\mathfrak{J}_z = \mathfrak{p}$, hypothesis (i) shows that $\mathfrak{p}$ cannot contain more than *one* minimal prime ideal $\mathfrak{q}$ of A, in other words $\mathfrak{p}$ belongs to a *single* irreducible component of X; moreover $\mathfrak{q}$ is the inverse image of (0) by the canonical homomorphism $A \to A_\mathfrak{p}$, and $A_\mathfrak{p}$ is therefore isomorphic to $(A/\mathfrak{q})_{\mathfrak{p}\mathfrak{q}}$, which proves that the integral quotient $A/\mathfrak{q}$ is *integrally closed* as an intersection of integrally closed rings (**I**, 8.2.1.1). If A is *Noetherian and normal*, A is composed of a finite direct product of integral domains that are integrally closed (**I**, 5.1.4). Let us now note that since $A^{(1)}$ is a finite A-algebra, it is a semi-local Noetherian ring, and every non-immersed prime ideal associated to $A^{(1)}/tA^{(1)}$ is of height 1 by the Hauptidealsatz (**0**, 16.3.2); now, for such a prime ideal $\mathfrak{r}$, $\mathfrak{r} \cap A$ is of height 1 by virtue of (5.10.17), (iv), and as it contains tA, it cannot be $\mathfrak{p}$ by virtue of hypothesis (i). On the other hand, $A^{(1)}$ is contained in the integral closure A' of A; if one sets $S = A - \mathfrak{p}$, the integral closure of $A_\mathfrak{p}$ is $S^{-1}A'$ (Bourbaki, *Alg. comm.*, chap. V, § 1, no 5, prop. 17), hence $S^{-1}A' = A_\mathfrak{p}$ since $A_\mathfrak{p}$ is a discrete valuation ring; it follows from this that there exists only *one* prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$, and *a fortiori* one prime ideal $\mathfrak{p}^{(1)}$ of $A^{(1)}$ above $\mathfrak{p}$, and we have $A_\mathfrak{p} = A'_{\mathfrak{p}'} = A^{(1)}_{\mathfrak{p}^{(1)}}$. Note now that $A^{(1)}$ satisfies (S_2) (5.10.17), (i) and since t is a non-zero-divisor in $A^{(1)}$, $A^{(1)}/tA^{(1)}$ has no immersed associated prime ideals (5.7.7). The prime ideal $\mathfrak{p}^{(1)}$ is therefore the unique associated prime ideal **IV-2** | 131
of $A^{(1)}/tA^{(1)}$, in other words $tA^{(1)}$ is a $\mathfrak{p}^{(1)}$-primary ideal of $A^{(1)}$; this means also that the inverse image of $tA^{(1)}$ by the canonical projection $A^{(1)} \to A^{(1)}/\mathfrak{p}^{(1)}$ is the maximal ideal of $A_\mathfrak{p}$, from (ii); hence $tA^{(1)} = \mathfrak{p}^{(1)}$ is prime in $A^{(1)}$. On the other hand, $A^{(1)}/\mathfrak{p}^{(1)}$ is finite over $A/\mathfrak{p}$ and has the same field of fractions (since the residue field of $A^{(1)}_{\mathfrak{p}^{(1)}} = A_\mathfrak{p}$), hence it is identical to $A/\mathfrak{p}$ by virtue of hypothesis (iii). One can therefore write $A^{(1)} = A + \mathfrak{p}^{(1)} = A + tA^{(1)}$, and since t is contained in the radical of the semi-local Noetherian ring $A^{(1)}$, the lemma of Nakayama implies $A^{(1)} = A$, whence $\mathfrak{p}^{(1)} = \mathfrak{p} = tA$. But since A is catenary and A/tA is integral and integrally closed by virtue of hypothesis (iii), we conclude from (5.12.7) that A is integrally closed. □

Corollary (5.12.10). — *Let A be an integral Noetherian local ring, satisfying one of the two following hypotheses:*

label=a) A is universally catenary and universally Japanese.

lbbel=b) A is a quotient of a regular ring.

Let $(x_i)_{1 \leqslant i \leqslant n}$ be a suite of n elements of A; set $\mathfrak{J} = x_1A + \ldots + x_nA$, and suppose the following conditions satisfied:

label=(i) There exists only one non-immersed prime ideal $\mathfrak{p}$ associated to $A/\mathfrak{J}$ and $\mathfrak{p}$ is of height n.

lbbel=(ii) The maximal ideal $\mathfrak{p}A_\mathfrak{p}$ of $A_\mathfrak{p}$ is generated by the $x_i/1$ (hence $A_\mathfrak{p}$ is a regular local ring of dimension n).

lcbel=(iii) The ring $A/\mathfrak{p}$ is integrally closed.

Under these conditions, for every integer i such that $0 \leqslant i \leqslant n$, the quotient ring $A/(x_1A + \ldots + x_iA)$ is integrally closed and of dimension $\dim(A) - i$. In particular, $\mathfrak{J}$ is equal to $\mathfrak{p}$, A is integrally closed and $(x_i)_{1 \leqslant i \leqslant n}$ is an A-regular sequence.

PROOF. We proceed by induction on n, the proposition being trivial for $n = 0$ and reducing to Hironaka's lemma (5.12.8) for $n = 1$. We may therefore suppose $n \geqslant 2$. Set $\mathfrak{J}' = x_1A + \ldots + x_{n-1}A$, and let $\mathfrak{q}$ be a minimal prime ideal among those containing $\mathfrak{J}'$. If $\mathfrak{q}$ is minimal among the prime ideals of A containing $\mathfrak{q} + x_n A$, $\mathfrak{p}'/\mathfrak{q}$ is of height 1 by the Hauptidealsatz (**0**, 16.3.1); let us show that $\mathfrak{q} \subset \mathfrak{p}$. Indeed, a prime ideal $\mathfrak{n}$ of B minimal among those containing $\mathfrak{q} + x_n A$ is of height $\leqslant n-1$ by the Hauptidealsatz, and since $\mathfrak{q}$ is minimal, there is at most one such ideal. Since it contains $\mathfrak{J}$, it is necessarily equal to $\mathfrak{p}$, whence $\mathfrak{q} \subset \mathfrak{p}$.

Consider now the integral ring $B = A/\mathfrak{q}$, and let t be the image of x_n in B. It is immediate (cf. (5.6.3), (iv)) that if A satisfies hypothesis *a)* (resp. *b)*), then so does B, and B and t satisfy the hypotheses of Hironaka's lemma. Application of (5.12.8) proves that $B = A/\mathfrak{q}$ is integrally closed. Let us now use the induction hypothesis, which shows that $A/(x_1 A + \ldots + x_i A)$ is integrally closed and of dimension $\dim(A) - i$ for $0 \leqslant i \leqslant n-1$ and that $\mathfrak{q} = \mathfrak{J}'$, whence $\mathfrak{p} = \mathfrak{J}' + x_n A = \mathfrak{J}$. We conclude that $A/\mathfrak{J} = A/\mathfrak{p}$ is integrally closed by hypothesis (iii), and that $\dim(A/\mathfrak{p}) = n$ and that $\dim(A/\mathfrak{p}) = \dim(A) - n$ since A is biequidimensional, which completes the proof. □

5.13. Permanence properties by passage to the inductive limit Throughout this section, $(A_\alpha, \varphi_{\beta\alpha})$ denotes an inductive system of rings whose index set is a pre-ordered filtering set; we set $A = \varinjlim A_\alpha$ and denote by φ_α the canonical homomorphism $A_\alpha \to A$.

Proposition (5.13.1). — label=(i) *For every index α, let $\mathfrak{a}_\alpha$ be an ideal of A_α such that, for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}(\mathfrak{a}_\alpha) \subset \mathfrak{a}_\beta$. Then the $\mathfrak{a}_\alpha$ form an inductive system for the restrictions of the $\varphi_{\beta\alpha}$, $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$ identifies canonically with an ideal of A, $A/\mathfrak{a}$ with $\varinjlim(A_\alpha/\mathfrak{a}_\alpha)$. If furthermore $\mathfrak{a}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{a}_\beta)$ for $\alpha \leqslant \beta$, then $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α.*

lbbel=(ii) *Conversely, for every ideal $\mathfrak{a}$ of A, if we set $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α, the $\mathfrak{a}_\alpha$ form an inductive system such that $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$.*

PROOF. It is clear that (ii) is a particular case of (i). In (i), the fact that $\mathfrak{a}$ identifies canonically with an ideal of A and $A/\mathfrak{a}$ with $\varinjlim(A_\alpha/\mathfrak{a}_\alpha)$ comes from the exactness of the functor $\varinjlim$ in the category of abelian groups. Furthermore, $\mathfrak{a}$ is the union of the increasing family of the $\varphi_\alpha(\mathfrak{a}_\alpha)$, hence, if $x_\alpha \in A_\alpha$ is such that $\varphi_\alpha(x_\alpha) \in \mathfrak{a}$, there exists $\beta \geqslant \alpha$ such that $\varphi_\alpha(x_\alpha) = \varphi_\beta(y_\beta)$ for some $y_\beta \in \mathfrak{a}_\beta$; but this implies the existence of a $\gamma \geqslant \beta$ such that $\varphi_{\gamma\alpha}(x_\alpha) = \varphi_{\gamma\beta}(y_\beta) \in \mathfrak{a}_\gamma$, and hence $x_\alpha \in \mathfrak{a}_\alpha$ if $\mathfrak{a}_\alpha = \varphi_{\gamma\alpha}^{-1}(\mathfrak{a}_\gamma)$. □

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Corollary (5.13.2). — *If $\mathfrak{N}_\alpha$ is the nilradical of A_α, the nilradical of A identifies with $\varinjlim \mathfrak{N}_\alpha$, and A_{red} with $\varinjlim (A_\alpha)_{\mathrm{red}}$. In particular, if the A_α are reduced, so is A.*

PROOF. It is clear that $\varphi_{\beta\alpha}(\mathfrak{N}_\alpha) \subset \mathfrak{N}_\beta$ for $\alpha \leqslant \beta$, and that $\varinjlim \mathfrak{N}_\alpha$ is a nilpotent ideal of A, hence contained in the nilradical $\mathfrak{N}$ of A. Conversely, if $x \in \mathfrak{N}$, we have $x^h = 0$ for some integer h; one can write $x = \varphi_\alpha(x_\alpha)$ for some index α and some $x_\alpha \in A_\alpha$; the relation $\varphi_\alpha(x_\alpha^h) = 0$ implies the existence of a $\beta \geqslant \alpha$ such that $\varphi_{\beta\alpha}(x_\alpha^h) = 0$, or equivalently $x_\beta^h = 0$ on setting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$; hence $x_\beta \in \mathfrak{N}_\beta$ and since $x = \varphi_\beta(x_\beta)$ this completes the proof, taking account of (5.13.1). □

Proposition (5.13.3). — label=(i) *For every index α, let $\mathfrak{p}_\alpha$ be a prime ideal of A_α such that, for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}(\mathfrak{p}_\alpha) \subset \mathfrak{p}_\beta$. Then $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ is a prime ideal of A. In particular, if the A_α are integral, so is A.*

lbbel=(ii) *If furthermore $\mathfrak{p}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{p}_\beta)$ for $\alpha \leqslant \beta$, then $A_\mathfrak{p}$ identifies canonically with $\varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$.*

lcbel=(iii) *If A is a local ring, $\mathfrak{p}$ its maximal ideal, and if the homomorphisms φ_α are injective, then, setting $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, the homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ are injective.*

PROOF. The two assertions of (i) are in fact equivalent, since $A/\mathfrak{p} = \varinjlim(A_\alpha/\mathfrak{p}_\alpha)$. Suppose therefore all the A_α integral, and let x, y be two elements of A such that $xy = 0$. There exists thus an index α such that $x = \varphi_\alpha(x_\alpha)$, $y = \varphi_\alpha(y_\alpha)$ with x_α, y_α in A_α, and $\varphi_\alpha(x_\alpha y_\alpha) = 0$; but this implies the existence of an index $\beta \geqslant \alpha$ such that, setting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, $y_\beta = \varphi_{\beta\alpha}(y_\alpha)$, $x_\beta y_\beta = 0$; since A_β is integral, we have $x_\beta = 0$ or $y_\beta = 0$, which completes the proof that A is integral.

Let us now prove (ii). The hypothesis implies that the $A_\alpha - \mathfrak{p}_\alpha$ form an inductive system of multiplicative subsets of the A_α; hence the local rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ form an inductive system (where the transition homomorphisms are *local*), and the canonical homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \to A_\mathfrak{p}$ an inductive system of homomorphisms, whose inductive limit $\psi : \varinjlim((A_\alpha)_{\mathfrak{p}_\alpha}) \to A_\mathfrak{p}$ is a surjective homomorphism. It remains to see that ψ is injective. Now, if $x_\alpha \in A_\alpha$ and $s_\alpha \in A_\alpha - \mathfrak{p}_\alpha$ are such that $\varphi_\alpha(x_\alpha)/\varphi_\alpha(s_\alpha) = 0$ in $A_\mathfrak{p}$, there exists $s' \in A - \mathfrak{p}$ such that $s'\varphi_\alpha(x_\alpha) = 0$, and, replacing α by a larger index, we can suppose $s' = \varphi_\alpha(s'_\alpha)$ with $s'_\alpha \in A_\alpha - \mathfrak{p}_\alpha$, whence $\varphi_\alpha(s'_\alpha x_\alpha) = 0$; there is thus a $\beta \geqslant \alpha$ such that $\varphi_{\beta\alpha}(s'_\alpha x_\alpha) = 0$ in $(A_\beta)_{\mathfrak{p}_\beta}$, which proves that the canonical image of x_α/s_α in $\varinjlim((A_\alpha)_{\mathfrak{p}_\alpha})$ is zero.

Finally, the first assertion of (iii) results from (ii) and from the fact that the elements of $A_\alpha - \mathfrak{p}_\alpha$ are invertible in A, and that $A_{\mathfrak{p}_\alpha} = A$, and the fact that $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ is injective follows by flatness from the hypothesis that φ_α is injective. ☐

Corollary (5.13.4). — *Suppose that the A_α are all integral, and that the $\varphi_{\beta\alpha}$ are injective. Then, if A'_α is the integral closure of A_α, $A' = \varinjlim A'_\alpha$ is the integral closure of A. In particular, if A_α is integrally closed for every α, so is A. If the A_α are local rings that are unibranch* (*resp. geometrically unibranch*) (**0**, *23.2.1) and the $\varphi_{\beta\alpha}$ are injective local homomorphisms, A is unibranch* (*resp. geometrically unibranch*).

Proof. If K_α is the field of fractions of A_α, it results from (5.13.3), (ii) applied to $\mathfrak{p}_\alpha = 0$, that $K = \varinjlim K_\alpha$ is the field of fractions of A, and $A' = \varinjlim A'_\alpha$ identifies with a subring of K, clearly integral over A. Conversely, let $z \in K$ be an element satisfying an integral dependence equation $z^n + \sum_{i=1}^n c_i z^{n-i} = 0$, where the $c_i \in A$. There exists an index α, a $z_\alpha \in K_\alpha$ and $c_{i\alpha} \in A_\alpha$ such that z (resp. the c_i) are the images of z_α (resp. the $c_{i\alpha}$), which proves that $z \in A'$. It follows immediately that if the A_α are local and the $\varphi_{\beta\alpha}$ injective local homomorphisms, A' is local and the $\varphi_{\beta\alpha}$ local, hence if the A'_α are local, A' is a single extension of k_α for every α, $k' = \varinjlim k'_\alpha$, which is the residue field of A', is a purely inseparable (resp. radicial) extension of k, and A is therefore geometrically unibranch. ☐ **IV-2** | 133

(5.13.5). We shall say that a ring A is *normal*, equivalently, if for every prime ideal $\mathfrak{p}$ of A, $A_\mathfrak{p}$ is *integral* and *integrally closed*. We note that this implies that A is *reduced* and that $\mathfrak{p}$ can contain only a *single* minimal prime ideal of A, in other words $\mathfrak{p}$ belongs to a single irreducible component of $\operatorname{Spec}(A)$; moreover $\mathfrak{q}$ is the inverse image of (0) by the canonical homomorphism $A \to A_\mathfrak{p}$, and $A_\mathfrak{p}$ is therefore isomorphic to $(A/\mathfrak{q})_{\mathfrak{p}\mathfrak{q}}$, which proves that the integral quotient $A/\mathfrak{q}$ is *integrally closed* as an intersection of integrally closed rings (**I**, 8.2.1.1). If A is *Noetherian* and *normal*, A is a composed direct product of a finite number of integral domains that are integrally closed (**I**, 5.1.4).

Corollary (5.13.6). — *Suppose that the A_β are normal and that for $\alpha \leqslant \beta$, every irreducible component of* $\operatorname{Spec}(A_\beta)$ *dominates an irreducible component of* $\operatorname{Spec}(A_\alpha)$. *Then $A = \varinjlim A_\alpha$ is normal.*

Proof. Indeed, let $\mathfrak{p}$ be a prime ideal of A and for every α, set $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$. By hypothesis, for every α, $\mathfrak{p}_\alpha$ contains a single minimal ideal $\mathfrak{q}_\alpha$ of A_α, and for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}^{-1}(\mathfrak{q}_\beta)$ is a minimal ideal of A_α contained in $\mathfrak{p}_\alpha$, hence equal to $\mathfrak{q}_\alpha$. Setting $B_\alpha = A_\alpha/\mathfrak{q}_\alpha$, the integral rings B_α therefore form an inductive system for the homomorphisms $\varphi'_{\beta\alpha} : B_\alpha \to B_\beta$ deduced from the $\varphi_{\beta\alpha}$ by passage to quotients, which are injective; furthermore the B_α are integrally closed since they are the same by definition; as furthermore $(B_\alpha)_{\mathfrak{p}'_\alpha} = (A_\alpha)_{\mathfrak{p}_\alpha}$; since $\varinjlim(B_\alpha)_{\mathfrak{p}'_\alpha} = \varinjlim(A_\alpha)_{\mathfrak{p}_\alpha} = A_\mathfrak{p}$ is an integral ring integrally closed (5.13.3), this proves the corollary. ☐

Proposition (5.13.7). — *Suppose that the A_α are regular and that, for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}$ makes A_β a flat A_α-module; suppose furthermore that $A = \varinjlim A_\alpha$ is Noetherian. Then A is regular.*

Proof. By definition, it suffices to prove that for every prime ideal $\mathfrak{p}$ of A, the local Noetherian ring $A_\mathfrak{p}$ is regular. Setting $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, we have $A_\mathfrak{p} = \varinjlim((A_\alpha)_{\mathfrak{p}_\alpha})$ by (5.13.3); since by hypothesis the $(A_\alpha)_{\mathfrak{p}_\alpha}$ are regular and $(A_\beta)_{\mathfrak{p}_\beta}$ is a flat $(A_\alpha)_{\mathfrak{p}_\alpha}$-module for $\alpha \leqslant \beta$ ($\mathbf{0}_\mathbf{I}$, 6.3.2), we see that we are reduced to the case where the A_α and A are *local* rings. It then suffices to show that, if M, N are two A-modules of finite type, $\operatorname{Tor}_i^A(M, N) = 0$ for i sufficiently large (**0**, 17.3.1 and 17.2.6). But we have the following lemma (where the A_α are again arbitrary): **IV-2** | 134

Lemma (5.13.7.1). — *For every A-module M of finite presentation, there exists an index α and an A_α-module of finite presentation M_α such that M is isomorphic to $M_\alpha \otimes_{A_\alpha} A$.*

Proof. Indeed, one can suppose that $M = A^r/P$, where P is a sub-A-module of finite type. Since $A' = \varinjlim(A_\alpha^r)$ and since P is of finite type, there exists an α and a sub-A_α-module of finite type

P_α of A_α^r such that P is the image canonique of $P_\alpha \otimes_{A_\alpha} A$; the exactness to the right of the tensor product then shows that M is isomorphic to $(A_\alpha^r/P_\alpha) \otimes_{A_\alpha} A$. □

This lemma being established, M and N are of finite presentation, since A is Noetherian; write therefore $M = M_\alpha \otimes_{A_\alpha} A$, $N = N_\alpha \otimes_{A_\alpha} A$; by virtue of the hypothesis, A is a flat A_α-module, hence $\operatorname{Tor}_i^A(M,N) = \operatorname{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \otimes_{A_\alpha} A$ up to isomorphism (**III**, 6.3.9); since by hypothesis A_α is regular, there exists i_0 such that $\operatorname{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) = 0$ for $i \geqslant i_0$. □

Corollary (5.13.8). — *Suppose that the A_α are Noetherian and satisfy property (R_k) for an integer $k \geqslant 0$. Suppose furthermore that for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}$ makes A_β a flat A_α-module, that $A = \varinjlim A_\alpha$ is Noetherian and that, for every α, ${}^a\varphi_\alpha$ is a homeomorphism of $\operatorname{Spec}(A)$ onto $\operatorname{Spec}(A_\alpha)$. Then A satisfies (R_k).*

Proof. Indeed let $\mathfrak{p}$ be a prime ideal of A and, for every α, set $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, so that $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ and $A_\mathfrak{p} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$ (5.13.3). It results from (**I**, 2.4.2) and from the hypothesis that the morphism $\operatorname{Spec}(A_\mathfrak{p}) \to \operatorname{Spec}((A_\alpha)_{\mathfrak{p}_\alpha})$ is a surjective homeomorphism for every α; we thus have $\dim(A_\mathfrak{p}) = \dim((A_\alpha)_{\mathfrak{p}_\alpha})$ for every α. Suppose then that $\dim(A_\mathfrak{p}) \leqslant k$; we have thus $\dim((A_\alpha)_{\mathfrak{p}_\alpha}) \leqslant k$, and the hypothesis implies that $(A_\alpha)_{\mathfrak{p}_\alpha}$ is a regular ring; since furthermore, for $\alpha \leqslant \beta$, $(A_\beta)_{\mathfrak{p}_\beta}$ is a flat $(A_\alpha)_{\mathfrak{p}_\alpha}$-module ($\mathbf{0_I}$, 6.3.2), we conclude from (5.13.7) that $A_\mathfrak{p}$ is regular, which proves the corollary. □

§6. Flat morphisms of locally Noetherian preschemes

Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism. For every $y \in Y$, the fibre $f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y))$ is also a locally Noetherian prescheme, since it suffices to see this when $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are spectra of Noetherian rings, and then $B \otimes_A k(y)$ is a ring of fractions of the quotient ring $B/\mathfrak{i}_y B$, and hence is Noetherian. We first propose in this section to relate the properties of X, of Y, and of the fibres $f^{-1}(y)$ (where y ranges over Y), under the hypothesis that the morphism f is *flat*. The questions treated will reduce to the study of the relations between the local Noetherian rings $\mathscr{O}_y$, $\mathscr{O}_x$, and $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$, where the homomorphism $\mathscr{O}_y \to \mathscr{O}_x$ is local and makes $\mathscr{O}_x$ a flat $\mathscr{O}_y$-module. In (6.11) to (6.13) (which are fairly separate from the rest of the section, by virtue of their "absolute" rather than "relative" nature), we apply some of the preceding results to find criteria allowing us to assert that the singular locus (or certain analogous sets) of certain preschemes are closed sets; these criteria will play an important role in §7. **IV-2** | 134

6.1. Flatness and dimension **IV-2** | 135

Proposition (6.1.1). — *Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A, $k = A/\mathfrak{m}$ its residue field, $\varphi : A \to B$ a local homomorphism. Suppose that for every prime ideal $\mathfrak{p}$ of A, distinct from $\mathfrak{m}$, and for every minimal element $\mathfrak{q}$ of the set of prime ideals of B containing $\mathfrak{p}B$, $\varphi^{-1}(\mathfrak{q})$ is distinct from $\mathfrak{m}$ (in other words, that no irreducible component of $\operatorname{Spec}(B/\mathfrak{p}B)$ is contained in the inverse image of the closed point $\mathfrak{m}$ of $\operatorname{Spec}(A)$ in $\operatorname{Spec}(B)$). Then we have*

$$\dim(B) = \dim(A) + \dim(B \otimes_A k). \tag{6.1.1.1}$$

Proof. We argue by induction on $n = \dim(A)$; the assertion is trivial for $n = 0$, since $\mathfrak{m}B$ is then contained in the nilradical of B because $\mathfrak{m}$ is the nilradical of A. We can thus suppose $n > 0$. Let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant m)$ be the minimal prime ideals of B, and set $\mathfrak{p}_i = \varphi^{-1}(\mathfrak{q}_i)$; for every i, we have $\mathfrak{p}_i \neq \mathfrak{m}$; otherwise there would exist (since $n > 0$) a prime ideal $\mathfrak{p} \neq \mathfrak{m}$ contained in $\mathfrak{p}_i$ and since $\mathfrak{q}_i$ is minimal among the prime ideals of B containing $\mathfrak{p}B$, one would arrive at a contradiction with the hypothesis. It follows that $\mathfrak{m}$ is distinct from the union of the $\mathfrak{p}_i$ and the minimal prime ideals $\mathfrak{p}'_j$ $(1 \leqslant j \leqslant r)$ of A (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2) and there exists an $a \in \mathfrak{m}$ belonging to none of the $\mathfrak{p}_i$ nor the $\mathfrak{p}'_j$. Set $A' = A/aA$, $B' = B/aB$; we have (**0**, 16.3.4)

$$\dim(A') = \dim(A) - 1, \quad \dim(B') = \dim(B) - 1$$

by construction of a, and on the other hand $B' \otimes_{A'} k = B \otimes_A k = B/\mathfrak{m}B$, hence

$$\dim(B' \otimes_{A'} k) = \dim(B \otimes_A k)$$

and it therefore suffices to prove (6.1.1.1) for A' and B'; but, by virtue of the bijective correspondence between ideals of A (resp. B) containing a and ideals of A' (resp. B'), the hypotheses of the statement are also valid for A' and B'; one can thus apply the induction hypothesis, which completes the proof. □

Corollary (6.1.2). — *Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, $M \neq 0$ a finite type A-module, $N \neq 0$ a finite type B-module. If N is a flat A-module (resp. if B is a flat A-module), then we have*

(6.1.2.1) $$\dim_B(M \otimes_A N) = \dim_A(M) + \dim_{B \otimes_A k}(N \otimes_A k)$$

(resp. (6.1.1.1)).

PROOF. It suffices to prove the assertion concerning N; on the other hand, if $\mathfrak{b}$ is the annihilator of N, one can replace B by $B/\mathfrak{b}$, and hence suppose that $\operatorname{Supp}(N) = \operatorname{Spec}(B)$; the hypothesis then means that the morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ corresponding to φ is quasi-flat (2.3.3); we conclude (2.3.4) that for every prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of A, every irreducible component of $f^{-1}(\mathrm{V}(\mathfrak{p}))$ dominates $\mathrm{V}(\mathfrak{p})$, and hence the condition of (6.1.1) is satisfied, whence the conclusion. □ **IV-2** | 136

Corollary (6.1.3). — *Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A, k its residue field, $\varphi : A \to B$ a local homomorphism. Suppose that* $\dim(B \otimes_A k) = 0$ *(that is to say* (**0**, *16.2.1) that $\mathfrak{m}B$ is an ideal of definition of B). Then* $\dim(B) \leqslant \dim(A)$. *If in addition there exists a finite type B-module N that is a flat A-module and whose support equals* $\operatorname{Spec}(B)$ *(which is the case in particular if B is a flat A-module), then* $\dim(B) = \dim(A)$.

PROOF. The first assertion follows from (5.5.1) and the second from (6.1.2). □

Corollary (6.1.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a surjective morphism. For every closed subset Z of Y, we then have*

(6.1.4.1) $$\operatorname{codim}(f^{-1}(Z), X) \leqslant \operatorname{codim}(Z, Y).$$

Moreover, if f is quasi-flat (2.3.3)*, the two sides of* (6.1.4.1) *are equal.*

PROOF. Indeed, if y is a maximal point of Z and x a maximal point of the fibre $f^{-1}(y)$, we have $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y})$ by (5.5.2); the inequality (6.1.4.1) then follows from (5.1.2.1) and (5.1.3.1) and the fact that f is surjective. If in addition f is quasi-flat, we know (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates an irreducible component of Z; the maximal points of $f^{-1}(Z)$ are thus the maximal points of the fibres $f^{-1}(y)$, where y ranges over the set of maximal points of Z ($\mathbf{0_I}$, 2.1.8), and at each of these maximal points x we have $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) = 0$; since in addition $\mathscr{O}_x$ is a flat $\mathscr{O}_y$-module, we have $\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y)$ by (6.1.1), whence the equality in (6.1.4.1), taking into account that f is surjective. □

Proposition (6.1.1) admits the following partial converse:

Proposition (6.1.5). — *Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, M a finite type B-module. Suppose that:*

1° *A is a regular ring.*
2° *M is a Cohen-Macaulay B-module.*
3° *We have* $\dim_B(M) = \dim(A) + \dim_{B \otimes_A k}(M \otimes_A k)$.

Then M is a flat A-module.

PROOF. We proceed by induction on $n = \dim(A)$. If $n = 0$, A is a field since it is regular and the assertion is trivial. Suppose $n > 0$; let $\mathfrak{m}$ be the maximal ideal of A, and let $x \in \mathfrak{m}$ be an element not belonging to $\mathfrak{m}^2$; we then know (**0**, 17.1.8) that $A' = A/xA$ is regular and $\dim(A') = \dim(A) - 1$. Set

$$B' = B/xB, \quad M' = M/xM = M \otimes_A A';$$

we thus have

$$B' \otimes_{A'} k = B \otimes_A k, \quad M' \otimes_{A'} k = M \otimes_A k$$

and by (5.5.1.2) we have

$$\dim_{B'}(M') \leqslant \dim(A') + \dim_{B' \otimes_{A'} k}(M' \otimes_{A'} k);$$

we thus conclude from $(\mathbf{0}, 16.3.4)$ that **IV-2 | 137**

$$\dim_B(M) \leqslant \dim_{B'}(M') + 1 \leqslant \big(\dim(A') + \dim_{B\otimes_A k}(M \otimes_A k)\big) + 1 = \dim(A) + \dim_{B\otimes_A k}(M \otimes_A k).$$

Since by hypothesis the extreme members of these inequalities are equal, we necessarily have: (i) $\dim_{B'}(M') = \dim_B(M) - 1$, and since M is by hypothesis a Cohen-Macaulay B-module, this means that x is M-regular $(\mathbf{0}$, 16.1.9 and 16.5.5); (ii) $\dim_{B'}(M') = \dim(A') + \dim_{B'\otimes_{A'}k}(M' \otimes_{A'} k)$; since M' is a Cohen-Macaulay B'-module by (i) and $(\mathbf{0}$, 16.1.9 and 16.5.5) and A' is regular, the induction hypothesis shows that M' is a flat A'-module; we then deduce from $(\mathbf{0}_{\mathbf{III}}$, 10.2.7) that M is a flat A-module, since x is M-regular by (i). □

6.2. Flatness and projective dimension

Proposition (6.2.1). — (i) *Let A, B be two rings, $\varphi : A \to B$ a homomorphism such that B is a flat A-module. Then, for every A-module M, we have*

(6.2.1.1) $$\operatorname{dim.proj}_B(M \otimes_A B) \leqslant \operatorname{dim.proj}_A(M).$$

(ii) *Suppose in addition that A is a Noetherian ring, B a faithfully flat A-module, and M a finite type A-module; then*

(6.2.1.2) $$\operatorname{dim.proj}_B(M \otimes_A B) = \operatorname{dim.proj}_A(M).$$

PROOF. (i) We can restrict to the case where $n = \operatorname{dim.proj}_A(M)$ is finite; there thus exists a left resolution

$$0 \longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \ldots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

of M by projective A-modules; since B is a flat A-module, the sequence

$$0 \longrightarrow P_n \otimes_A B \longrightarrow P_{n-1} \otimes_A B \longrightarrow \ldots \longrightarrow P_0 \otimes_A B \longrightarrow M \otimes_A B \longrightarrow 0$$

is exact; on the other hand, the $P_i \otimes_A B$ are projective B-modules; whence the conclusion.

(ii) Suppose that $\operatorname{dim.proj}_B(M \otimes_A B) = m$, and consider an exact sequence

$$0 \longrightarrow R \longrightarrow P_{m-1} \longrightarrow P_{m-2} \longrightarrow \ldots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where the P_i $(0 \leqslant i \leqslant m-1)$ are projective A-modules of finite type; since A is Noetherian, R is also a finite type A-module. Since B is a flat A-module, we also have an exact sequence

$$0 \longrightarrow R \otimes_A B \longrightarrow P_{m-1} \otimes_A B \longrightarrow \ldots \longrightarrow P_0 \otimes_A B \longrightarrow M \otimes_A B \longrightarrow 0$$

and the hypothesis on $M \otimes_A B$ implies that $R \otimes_A B$ is a projective B-module $(\mathbf{0}$, 17.2.1). Since B is a faithfully flat A-module, we conclude that R is a projective A-module (Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, §3, n$^{\text{o}}$ 6, prop. 12); hence $\operatorname{dim.proj}_A(M) \leqslant m$, which completes the proof.

□

Corollary (6.2.2). — *Let $f : X \to Y$ be a flat morphism of preschemes, $\mathscr{E}$ an $\mathscr{O}_Y$-Module quasi-coherent. If* $\operatorname{dim.proj}(\mathscr{E}) \leqslant n$ *$(\mathbf{0}$, 17.2.14), then* $\operatorname{dim.proj}(f^*(\mathscr{E})) \leqslant n$.

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Proposition (6.2.3). — *Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, N a finite type B-module. Suppose that B and N are flat A-modules. Then we have*

(6.2.3.1) $$\operatorname{dim.proj}_B(N) = \operatorname{dim.proj}_{B\otimes_A k}(N \otimes_A k).$$

PROOF. Consider a left resolution of N by free B-modules of finite type

$$\ldots \longrightarrow L_i \longrightarrow L_{i-1} \longrightarrow \ldots \longrightarrow L_0 \longrightarrow N \longrightarrow 0.$$

Since the L_i and N are flat A-modules (B being a flat A-module) it follows from (2.1.10) that the $\mathrm{Z}_i(L_\cdot)$ are flat A-modules, that $L_\cdot \otimes_A k$ is a left resolution of the $(B \otimes_A k)$-module $N \otimes_A k$, and that $\mathrm{Z}_i(L_\cdot \otimes_A k) = \mathrm{Z}_i(L_\cdot) \otimes_A k$ for every i. Note that since B is Noetherian, the $\mathrm{Z}_i(L_\cdot)$ are B-modules of finite type, hence the $\mathrm{Z}_i(L_\cdot \otimes_A k)$ are $(B \otimes_A k)$-modules of finite type. Now, to say that a B-module (resp. a $(B \otimes_A k)$-module) of finite type is flat is equivalent to saying that it is *free* or equivalently *projective* $(\mathbf{0}_{\mathbf{III}}$, 10.1.3). Since the $\mathrm{Z}_i(L_\cdot)$ are flat A-modules, it follows moreover from $(\mathbf{0}_{\mathbf{III}}$, 10.2.5) (taking $C = B$) that, for $\mathrm{Z}_i(L_.)$ to be a flat B-module, it is necessary and sufficient that $\mathrm{Z}_i(L_. \otimes_A k)$ be a flat $(B \otimes_A k)$-module. The smallest integer n such that $\mathrm{Z}_{n-1}(L_.)$ is a free B-module is thus also the smallest integer such that $\mathrm{Z}_{n-1}(L_. \otimes_A k)$ is a free $(B \otimes_A k)$-module, which proves the proposition (**0**, 17.2.1). □

6.3. Flatness and depth

Proposition (6.3.1). — *Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, M a finite type A-module, N a finite type B-module. If N is a flat A-module $\neq 0$, then we have*

$$\operatorname{prof}_B(M \otimes_A N) = \operatorname{prof}_A(M) + \operatorname{prof}_{B \otimes_A k}(N \otimes_A k). \tag{6.3.1.1}$$

Proof. We can restrict to the case where $M \neq 0$, since otherwise the two sides of (6.3.1.1) are both equal to $+\infty$. We proceed by induction on the integer n equal to the second member of (6.3.1.1) (which is finite by the hypotheses, (**0**, 16.4.6.2) and the Nakayama lemma applied to N). Suppose first that $n = 0$; then, if $\mathfrak{m}$ and $\mathfrak{n}$ denote the maximal ideals of A and B respectively, $\mathfrak{m}$ is an ideal premier associated to M and $\mathfrak{n}/\mathfrak{m}B$ an ideal premier associated to $N \otimes_A k$ (**0**, 16.4.6, (i)). We then conclude from (3.3.1) that $\mathfrak{n}$ is a prime ideal of B associated to $M \otimes_A N$, hence (**0**, 16.4.6, (i)) $\operatorname{prof}_B(M \otimes_A N) = 0$. Suppose now $n > 0$, and distinguish two cases:

a) Suppose $\operatorname{prof}_A(M) > 0$. Let $x \in \mathfrak{m}$ be an M-regular element, and set

$$A' = A/xA, \quad B' = B/xB, \quad M' = M/xM, \quad N' = N/xN;$$

we have **IV-2 | 139**

$$B' \otimes_{A'} k = B \otimes_A k, \quad N' \otimes_{A'} k = N \otimes_A k$$

and, since N is a flat A-module, the hypothesis that x is M-regular implies that x is also $(M \otimes_A N)$-regular ($\mathbf{0_I}$, 6.1.1). We have by (**0**, 16.4.6, (ii) and 16.4.8)

$$\operatorname{prof}_{A'}(M') = \operatorname{prof}_A(M) - 1, \quad \operatorname{prof}_{B'}(M' \otimes_{A'} N') = \operatorname{prof}_B(M \otimes_A N) - 1$$

and

$$\operatorname{prof}_{B' \otimes_{A'} k}(N' \otimes_{A'} k) = \operatorname{prof}_{B \otimes_A k}(N \otimes_A k).$$

The equality (6.3.1.1) is thus a consequence of the same relation for A', B', M', and N'; but since N is a flat A-module, $N' = N \otimes_A A'$ is a flat A'-module ($\mathbf{0_I}$, 6.2.1); one can therefore apply the induction, which proves (6.3.1.1) in this case.

b) Suppose $\operatorname{prof}_{B \otimes_A k}(N \otimes_A k) > 0$. Consider a $y \in \mathfrak{n}$ that is $(N \otimes_A k)$-regular; it follows from ($\mathbf{0_{III}}$, 10.2.4) that y is N-regular and that

$$N' = N/yN$$

is a flat A-module, since N is assumed to be a flat A-module. Applying ($\mathbf{0_I}$, 6.1.2) to the exact sequence of A-modules

$$0 \longrightarrow N \xrightarrow{y} N \longrightarrow N' \longrightarrow 0$$

we obtain isomorphisms

$$(M \otimes_A N)/y(M \otimes_A N) \xrightarrow{\sim} M \otimes_A N' \quad \text{and} \quad N' \otimes_A k \xrightarrow{\sim} (N \otimes_A k)/y(N \otimes_A k)$$

and moreover y is $(M \otimes_A N)$-regular. We thus have

$$\operatorname{prof}_{B \otimes_A k}(N' \otimes_A k) = \operatorname{prof}_{B \otimes_A k}(N \otimes_A k) - 1$$

and

$$\operatorname{prof}_B(M \otimes_A N') = \operatorname{prof}_B(M \otimes_A N) - 1;$$

the induction hypothesis shows that the relation (6.3.1.1) holds for A, B, M, and N', and from what precedes, we deduce (6.3.1.1) for A, B, M, and N. □

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), suppose in addition that $M \neq 0$ and $N \neq 0$; then we have*

$$\operatorname{coprof}_B(M \otimes_A N) = \operatorname{coprof}_A(M) + \operatorname{coprof}_{B \otimes_A k}(N \otimes_A k). \tag{6.3.2.1}$$

PROOF. This follows immediately from (6.3.1.1) and (6.1.2) and from the definition of coprofondeur (**0**, 16.4.9). ☐

Corollary (6.3.3). — *Under the hypotheses of (6.3.1), suppose in addition that $M \neq 0$ and $N \neq 0$; for $M \otimes_A N$ to be a Cohen-Macaulay B-module, it is necessary and sufficient that M be a Cohen-Macaulay A-module and that $N \otimes_A k$ be a $(B \otimes_A k)$-module de Cohen-Macaulay.*

PROOF. This follows from the corollary (6.3.2) and the definition of Cohen-Macaulay modules, by the fact that their coprofondeur is zero (**0**, 16.5.1), taking into account that the condition $N \neq 0$ is equivalent to $N \otimes_A k \neq 0$ by the Nakayama lemma. ☐

Corollary (6.3.4). — *Under the hypotheses of (6.3.1), suppose in addition that $N \otimes_A k$ is a B-module of finite length; then we have*

(6.3.4.1) $$\operatorname{prof}_B(M \otimes_A N) = \operatorname{prof}_A(M).$$

If in addition $M \neq 0$ and $N \neq 0$, then we have **IV-2** | 140

(6.3.4.2) $$\operatorname{coprof}_B(M \otimes_A N) = \operatorname{coprof}_A(M).$$

PROOF. It amounts to saying that $N \otimes_A k$ is a B-module of finite length, or a $(B \otimes_A k)$-module of finite length, and we know (**0**, 16.2.3) that modules of finite length $\neq 0$ are of dimension 0 and depth 0. The corollary (6.3.4) is applicable in particular when $B \otimes_A k$ is a ring of finite length, that is to say when $\mathfrak{m}B$ is an *ideal of definition* of the ring B. ☐

Corollary (6.3.5). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) *If, at a point $x \in X$, $\operatorname{coprof}(\mathscr{O}_x) \leqslant n$, then, setting $y = f(x)$, $\operatorname{coprof}(\mathscr{O}_y) \leqslant n$ and $\operatorname{coprof}(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) \leqslant n$; in particular, if $\mathscr{O}_x$ is a Cohen-Macaulay ring, the same is true of $\mathscr{O}_y$ and of $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$.*

(ii) *Suppose conversely that $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$ is a Cohen-Macaulay ring. Then, if $\mathscr{O}_y$ is a Cohen-Macaulay ring, $\operatorname{coprof}(\mathscr{O}_x) \leqslant n$ and $\operatorname{coprof}(\mathscr{O}_y) \leqslant n$ (resp. $\mathscr{O}_x$ is a Cohen-Macaulay ring).*

PROOF. It suffices to apply (6.3.2) for $M = A = \mathscr{O}_y$ and $N = B = \mathscr{O}_x$. ☐

Corollary (6.3.6). — *Let A be a Cohen-Macaulay ring. Then $A' = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ (T_i indeterminates) is a Cohen-Macaulay ring.*

PROOF. Indeed, if we set $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(A')$ and $f : X \to Y$ is the canonical morphism, f is *flat* (since A' is a free A-module (2.1.2)) and for every $y \in Y$, $k(y)[\mathrm{T}_1, \dots, \mathrm{T}_n]$ is a regular ring (**0**, 17.3.7), hence *a fortiori* a Cohen-Macaulay ring; it therefore suffices to apply (6.3.5). ☐

Corollary (6.3.7). — *Every quotient of a Cohen-Macaulay ring is universally catenary.*

PROOF. This follows from (6.3.6), (5.6.1), and (**0**, 16.5.12). ☐

Proposition (6.3.8). — *Let A be a local Noetherian ring; suppose that there exists a Cohen-Macaulay finite type A-module M whose support is equal to $\operatorname{Spec}(A)$. Let B be a quotient ring of A, and let $f : \operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ be the canonical morphism; then, for every $x \in \operatorname{Spec}(B)$, $f^{-1}(x)$ is a Cohen-Macaulay prescheme.*

PROOF. Since B is a finite type A-module, on a $\widehat{B} = B \otimes_A \widehat{A}$ ($\mathbf{0_I}$, 7.3.3); hence f is flat and finite, **IV-2** | 141
and the fibres of the two morphisms $\operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ and $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$ at the same point are the same. One is therefore reduced to proving the proposition when $B = A$. Let $\mathfrak{p} = \mathfrak{i}_x$ be a prime ideal; since $\mathfrak{p} \in \operatorname{Supp}(M)$, we know (**0**, 16.5.6 and 16.5.10) that $M_{\mathfrak{p}}$ is a Cohen-Macaulay $A_{\mathfrak{p}}$-module. Taking into account (**I**, 3.6.5), we see that if we set $A'' = A' \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}$ and $M'' = M' \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}$, M'' is an A''-module de Cohen-Macaulay; on the other hand, $M_{\mathfrak{p}}$ is by hypothesis a Cohen-Macaulay $A_{\mathfrak{p}}$-module and $A''_{\mathfrak{p}''}$ is a flat $A_{\mathfrak{p}}$-module ($\mathbf{0_I}$, 7.3.3 and 6.3.2). We conclude from (6.3.3) that, if k is the residue field of $A_{\mathfrak{p}}$, $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. But since $A_{\mathfrak{p}}$ is of dimension 0, the prime ideals of A'' correspond bijectively to those of $k \otimes_{A_{\mathfrak{p}}} A''$ (**I**, 3.5.7), and if $\mathfrak{q}''$ is the prime ideal of $k \otimes_{A_{\mathfrak{p}}} A''$ corresponding to $\mathfrak{p}''$, the local rings $(k \otimes_{A_{\mathfrak{p}}} A'')_{\mathfrak{q}''}$ and $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ are isomorphic. It follows that $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. Q.E.D. ☐

6.4. Flatness and property (S_k)

Proposition (6.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module that is f-flat.*

(i) *Let $x \in \mathrm{Supp}(\mathscr{F})$ be such that $\mathscr{E} \otimes_Y \mathscr{F}$ has property (S_k) at the point x; then $\mathscr{E}$ has property (S_k) at the point $y = f(x)$.*

(ii) *Suppose that for every $y \in f(\mathrm{Supp}(\mathscr{F}))$, $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} k(y)$ has property (S_k); if for a point $y \in f(\mathrm{Supp}(\mathscr{F}))$, $\mathscr{E}$ has property (S_k) at the point y, then $\mathscr{E} \otimes_Y \mathscr{F}$ has property (S_k) at every point of $f^{-1}(y)$.*

Proof. (i) Setting $y = f(x)$ and let y' be a generisation of y; as in the proof of (6.4.1) there exists a generic point x' of an irreducible component of $f^{-1}(y')$ that is a generisation of x; hence on a $\dim(\mathscr{O}_{x'} \otimes_{\mathscr{O}_{y'}} k(y')) = 0$ and by the hypothesis and (6.1.2), $\dim(\mathscr{O}_{x'}) = \dim(\mathscr{O}_{y'})$. The hypothesis implies, either that $\dim(\mathscr{O}_{x'}) > k$, in which case $\dim(\mathscr{O}_{y'}) > k$, or that $\mathscr{O}_{x'}$ is a regular ring, and then $\mathscr{O}_{y'}$ is a regular ring by (6.5.1).

(ii) For every generisation x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generisation of y, we can restrict to verifying that if $x \in \mathrm{Supp}(\mathscr{F})$ and $f(x) = y$, we have $\mathrm{prof}(\mathscr{G}_x) \geqslant \inf(k, \dim(\mathscr{G}_x))$; it follows from (6.1.2) and (6.3.1) that

$$\dim(\mathscr{G}_x) = \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y))$$
$$\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(\mathscr{E}_y) + \mathrm{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y)).$$

By hypothesis, we have

$$\mathrm{prof}(\mathscr{E}_y) \geqslant \inf(k, \dim(\mathscr{E}_y))$$
$$\mathrm{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y)) \geqslant \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y)))$$

whence, by adding member by member,

$$\mathrm{prof}(\mathscr{G}_x) \geqslant \inf(k, \dim(\mathscr{E}_y)) + \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y))) \geqslant$$
$$\geqslant \inf(k, \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y))) = \inf(k, \dim(\mathscr{G}_x))$$

which proves (ii).

□

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Corollary (6.4.2). — *Let X, Y be two locally Noetherian preschemes, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module, $f : X \to Y$ a flat morphism. Suppose that for every $y \in Y$, the prescheme $f^{-1}(y)$ verifies (S_k) at every point x; then, for $f^*(\mathscr{E})$ to verify (S_k) at a point x, it is necessary and sufficient that $\mathscr{E}$ verify (S_k) at the point $f(x)$.*

Remarks (6.4.3). — Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, M a finite type B-module that is a flat A-module, and suppose in addition that the A-module A and the $(B \otimes_A k)$-module $M \otimes_A k$ both have property (S_n); can one then conclude that the B-module M has property (S_n)? We do not know the answer even when $n = 1$, $M = B$ and $B = \widehat{A}$, in other words, we do not know whether in general the *completion* of a local Noetherian ring satisfying (S_n) also satisfies (S_n), even for $n = 1$. Setting $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $\mathscr{F} = \widetilde{M}$, it would suffice, by (6.4.1), (ii), to show that for *every* $y \in Y$, $\mathscr{F}_y$ has property (S_k) (and not only when y is the closed point of Y); it would also suffice to show that the set U of $x \in X$ such that $\mathscr{F}_x \otimes_{\mathscr{O}_{f(x)}} k(f(x))$ satisfies (S_k) is *open* in X (since it contains the closed point of X by hypothesis). We shall show later (12.1.4) that U is open when B is a local ring of a finite type A-algebra, and that all the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes (otherwise said satisfy (S_n) for *all* n) when the hypotheses of (6.3.8) are satisfied. On the other hand, for $B = \widehat{A}$ and $M = B$, we have seen in (6.3.8) that the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes for all n, for which it suffices that A satisfies (S_n), *it is necessary and sufficient that its completion $\widehat{A}$ satisfies* (S_n). It remains to see whether this property holds without restriction on A.

6.5. Flatness and property (R_k)

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Proposition (6.5.1). — *Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, for which B is a flat A-module. Then:*

(i) *If B is regular, A is regular.*
(ii) *If A and $B \otimes_A k$ are regular, B is regular.*

This proposition is given for the record, having already been proved in $(\mathbf{0}, 17.3.3)$.

Corollary (6.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) *If X is regular at a point x, Y is regular at the point $f(x)$.*
(ii) *If for some $y \in f(X)$, Y is regular at the point y and if $f^{-1}(y)$ is a regular prescheme at a point x, then X is regular at the point x.*

Proposition (6.5.3). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) *If X has property (R_k) at a point x (resp. if X has property (R_k)), Y has property (R_k) at the point $f(x)$ (resp. Y has property (R_k) at every point of $f(X)$).*
(ii) *Suppose that for every $y \in f(X)$, the prescheme $f^{-1}(y)$ has property (R_k); then, if for a point $y \in f(X)$, Y has property (R_k) at the point y, X has property (R_k) at every point of $f^{-1}(y)$.*

PROOF. (i) Set $y = f(x)$ and let y' be a generisation of y; as in the proof of (6.4.1), there exists a generic point x' of an irreducible component of $f^{-1}(y')$ that is a generisation of x; hence $\dim(\mathscr{O}_{x'} \otimes_{\mathscr{O}_{y'}} k(y')) = 0$ and by the hypothesis and (6.1.2), $\dim(\mathscr{O}_{x'}) = \dim(\mathscr{O}_{y'})$. The hypothesis implies, either that $\dim(\mathscr{O}_{x'}) > k$, in which case $\dim(\mathscr{O}_{y'}) > k$, or that $\mathscr{O}_{x'}$ is a regular ring, and then $\mathscr{O}_{y'}$ is a regular ring by (6.5.1).

(ii) For every generisation x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generisation of y; we can restrict to verifying that if $\dim(\mathscr{O}_x) \leqslant k$, then $\mathscr{O}_x$ is a regular ring. Now, by (6.1.2)

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y))$$

hence if $\dim(\mathscr{O}_x) \leqslant k$, we have *a fortiori* $\dim(\mathscr{O}_y) \leqslant k$ and $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) \leqslant k$ and the hypothesis implies that $\mathscr{O}_y$ and $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$ are regular rings. We then conclude from (6.5.1) that $\mathscr{O}_x$ is a regular ring. □

Corollary (6.5.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) *If X is normal at a point x, Y is normal at the point $f(x)$.*
(ii) *If, for every $y \in f(X)$, $f^{-1}(y)$ is a normal prescheme and if Y is normal at a point $y \in f(X)$, then X is normal at every point of $f^{-1}(y)$.*

PROOF. This follows immediately from the Serre normality criterion (5.8.6) and from (6.4.1) applied for $k = 2$ and (6.5.3) applied for $k = 1$. □

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Remarks (6.5.5). — (i) Let A, B be two local Noetherian rings, $\varphi : A \to B$ a local homomorphism such that B is a flat A-module. Let k be the residue field of A, and suppose that the two rings A and $B \otimes_A k$ satisfy property (R_i) (5.8.2); it does *not* then necessarily follow that B satisfies (R_i), even in the particular case where $i = 0$ or $i = 1$ and where B is the completion $\widehat{A}$ of A. Nagata has indeed given an example where A is normal (hence satisfies (R_1)) but where $\widehat{A}$ is not even reduced (hence does not satisfy (R_0)) *[30]*. One cannot apply the proposition (6.5.3) in this case because the fibres $f^{-1}(y)$ do not necessarily satisfy property (R_i) at the points distinct from the closed point of $Y = \mathrm{Spec}(A)$. We shall show however later (7.8.3), (v) that such phenomena do not occur for the local Noetherian rings most commonly encountered in applications.

(ii) The property for a prescheme of being *integral* does not at all behave like the properties we have just examined in this number and the preceding ones: it can happen that $f : X \to Y$ is a flat morphism of finite type, that all the fibres $f^{-1}(y)$ are regular (and even geometrically regular (6.7.6)) and that Y is integral, without X even being locally integral. For example, let k be an algebraically closed field of characteristic 0, and let A be the integral ring $k[\mathrm{U},\mathrm{V}]/(\mathrm{UV}-(\mathrm{U}+\mathrm{V})^3)$ (whose spectrum is therefore a "cubic with a double point"); A is not integrally closed, and if u, v are the canonical images of U, V in A, the integral closure of A is $C = A[t]$, where $t = (u-v)/(u+v)$, which satisfies the equation $t^2 = 1+u+v$, whence $u = \frac{1}{2}(t^3+t^2-t)$, $v = \frac{1}{2}(-t^3+t^2+t)$ and hence $C = k[t]$, isomorphic to the polynomial ring in one indeterminate over k. If $\mathfrak{m} = Au + Av$, the maximal ideal of A (corresponding to the "double point" of the cubic), there are two maximal ideals $\mathfrak{n}_1$, $\mathfrak{n}_2$ of C, generated by $t-1$ and $t+1$ respectively. Then B is the subring of the product $C \times C$ consisting of the pairs of polynomials (f,g) such that $f(1) = g(-1)$ and $f(-1) = g(1)$ (Spec(B) is the scheme obtained by "gluing" two copies of Spec(C), the point $\mathfrak{n}_1$ (resp. $\mathfrak{n}_2$) of one being "glued" to the point $\mathfrak{n}_2$ (resp. $\mathfrak{n}_1$) of the other; cf. chap. V, where this operation will be discussed in a general fashion). There are thus two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of B above the maximal ideal $\mathfrak{m}$ of A. Moreover, since the "gluing" process commutes with localisation and completion, one easily verifies that the canonical homomorphisms $\widehat{A}_{\mathfrak{m}} \to \widehat{B}_{\mathfrak{r}_1}$ and $\widehat{A}_{\mathfrak{m}} \to \widehat{B}_{\mathfrak{r}_2}$ are bijective, and hence $B_{\mathfrak{r}_1}$ and $B_{\mathfrak{r}_2}$ are flat $A_{\mathfrak{m}}$-modules (Bourbaki, *Alg. comm.*, chap. III, §3, n° 5, prop. 10) having moreover the same residue field as $A_{\mathfrak{m}}$. For every other maximal ideal $\mathfrak{p}$ of A, it is immediate that there are two maximal ideals $\mathfrak{q}_1$, $\mathfrak{q}_2$ of B above $\mathfrak{p}$, and that the homomorphisms $A_{\mathfrak{p}} \to B_{\mathfrak{q}_1}$ and $A_{\mathfrak{p}} \to B_{\mathfrak{q}_2}$ are bijective. One thus sees that the morphism Spec$(B) \to$ Spec(A) is flat and finite, and that all its fibres are geometrically regular (it is even *étale*, as we shall see later (17.6.3)); however it is immediate that B is not integral.

6.6. Transitivity properties

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Proposition (6.6.1). — *For a locally Noetherian prescheme Z, let us denote by $\mathbf{P}(Z)$ any one of the following properties:*

a) *Z is a Cohen-Macaulay prescheme.*
b) *Z satisfies* (S_n).
c) *Z is regular.*
d) *Z satisfies* (R_n).
e) *Z is normal.*
f) *Z is reduced.*

Let X, Y, Z be three locally Noetherian preschemes, $f : X \to Y$, $g : Y \to Z$ two morphisms.

(i) *Suppose that f and g are flat and that for every $y \in f(X)$ (resp. every $z \in g(Y)$), the prescheme $f^{-1}(y)$ (resp. $g^{-1}(z)$) satisfies* $\mathbf{P}$. *Then $h = g \circ f$ is flat and for every $z \in h(X)$, $h^{-1}(z)$ satisfies* $\mathbf{P}$.

(ii) *Suppose the following conditions are satisfied: f is faithfully flat, $h = g \circ f$ is flat, for every $z \in h(X)$, the prescheme $f^{-1}(z)$ satisfies* $\mathbf{P}$, *and for every $z \in g(Y)$, the prescheme $h^{-1}(z)$ satisfies* $\mathbf{P}$. *Then g is flat and for every $z \in g(Y)$, $g^{-1}(z)$ satisfies* $\mathbf{P}$.

PROOF. (i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f is flat (resp. faithfully flat, g is flat (2.1.6) and (2.2.13)). Furthermore, for every $z \in Z$, $f_z = f \otimes 1_{k(z)} : h^{-1}(z) \to g^{-1}(z)$ is flat (2.1.4), and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4). The conclusion follows respectively from (6.3.5), (ii), (6.4.2), (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii), and (3.3.5), (ii).

(ii) We already know that g is flat (2.2.13); moreover, for every $z \in Z$, f_z is faithfully flat (2.2.13). The conclusion follows respectively from (6.3.5), (i), (6.4.2), (i), (6.5.2), (i), (6.5.3), (i), (6.5.4), (i), and (2.1.13).

□

Remark (6.6.2). — Suppose h flat, f faithfully flat, and suppose that for every $z \in Z$, $h^{-1}(z)$ is of *coprofondeur* $\leqslant n$ (5.7.1); then it follows from the reasoning of (6.6.1), (ii), and from (6.3.2) that $g^{-1}(z)$ is of coprofondeur $\leqslant n$ for every $z \in g(Y)$ and that for every $y \in Y$, $f^{-1}(y)$ is of coprofondeur $\leqslant n$.

6.7. Application to base changes in algebraic preschemes

Proposition (6.7.1). — *Let k be a field, X a locally Noetherian k-prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Let k' be an extension of k; set $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$ and let $p : X' \to X$ be the canonical projection. Suppose, either that X is locally of finite type over k, or that k' is a finite extension of k, so that in both cases X' is locally Noetherian. Let x' be a point of X', $x = p(x')$. Then:*

(i) *We have $\operatorname{coprof}(\mathscr{F}_x) = \operatorname{coprof}(\mathscr{F}'_{x'})$; in particular, for $\mathscr{F}_x$ to be a Cohen-Macaulay $\mathscr{O}_x$-module, it is necessary and sufficient that $\mathscr{F}'_{x'}$ be a Cohen-Macaulay $\mathscr{O}_{x'}$-module.*

(ii) *For $\mathscr{F}$ to satisfy property (S_n) at the point x, it is necessary and sufficient that $\mathscr{F}'$ satisfy (S_n) at the point x'.*

Proof. We know that p is faithfully flat (2.2.13); the two assertions are therefore consequences respectively of (6.3.2) and (6.4.2), provided that we prove that $p^{-1}(x) = \operatorname{Spec}(k(x) \otimes_k k')$ is a Cohen-Macaulay prescheme; since one of the two fields $k(x)$, k' is a finite extension of k by hypothesis, we are reduced to proving the following: □

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Lemma (6.7.1.1). — *Let K, L be two extensions of a field k, one of which is of finite type (so that $K \otimes_k L$ is a Noetherian ring). Then $K \otimes_k L$ is a Cohen-Macaulay ring.*

Proof. Suppose for example that L is a finite type extension of k, so that L is a finite extension of a pure transcendental extension $k(\mathbf{t})$ of k ($\mathbf{t}$ being a finite system $(t_i)_{1 \leqslant i \leqslant m}$ of indeterminates). Then $A = K \otimes_k k(\mathbf{t})$, $B = K \otimes_k L$; B is a finite type A-module, and B is an A-module flat ($\mathbf{0_I}$, 6.2.1) and of finite type; the morphism $h : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is therefore finite, and since every Artinian prescheme is Cohen-Macaulay (6.3.5), or if S is the set of elements $\neq 0$ of $k[\mathbf{t}]$, we have $A' = S^{-1}A'$, where $A' = K \otimes_k k[\mathbf{t}] = K[\mathbf{t}]$; by ($\mathbf{0}$, 16.5.13), it suffices to prove that A' is a Cohen-Macaulay ring, and this follows from the fact that A' is regular ($\mathbf{0}$, 17.3.7). □

Corollary (6.7.2). — *For $\mathscr{O}_x$ to be a Cohen-Macaulay ring (resp. for X to satisfy (S_n) at the point x), it is necessary and sufficient that $\mathscr{O}_{x'}$ be a Cohen-Macaulay ring (resp. that X' satisfy (S_n) at the point x').*

Corollary (6.7.3). — *Let X, Y be two k-preschemes locally Noetherian, at least one of which is locally of finite type over k. Let $\mathscr{F}$ (resp. $\mathscr{G}$) be a coherent $\mathscr{O}_Y$-Module (resp. a coherent $\mathscr{O}_X$-Module). If $\mathscr{F}$ and $\mathscr{G}$ satisfy property (S_n), then so does $\mathscr{F} \otimes_k \mathscr{G}$.*

Proof. The hypothesis implies that $X \times_k Y$ is locally Noetherian; let $p : X \times_k Y \to X$ and $q : X \times_k Y \to Y$ be the canonical projections, which are flat morphisms. Suppose for example that X is locally of finite type over k; on can write $\mathscr{F} \otimes_k \mathscr{G} = p^*(\mathscr{F}) \otimes_k \mathscr{G}$, and since $\mathscr{F}$ is flat relative to the structural morphism $X \to \operatorname{Spec}(k)$, $p^*(\mathscr{F})$ is q-flat (2.1.4); it suffices to see that for every $y \in Y$, $p^*(\mathscr{F}) \otimes_{\mathscr{O}_Y} k(y)$ has property (S_n); but $p^*(\mathscr{F}) \otimes_{\mathscr{O}_Y} k(y) = \mathscr{F} \otimes_k k(y)$, and since X is locally of finite type over k, the conclusion follows from (6.7.1), (ii). □

Proposition (6.7.4). — *For a locally Noetherian prescheme Z, and a point $z \in Z$, let $\mathbf{P}(Z, z)$ be one of the following properties:*

a) *Z is regular at the point z.*
b) *Z satisfies (R_n) at the point z.*
c) *Z is normal at the point z.*
d) *Z is reduced at the point z.*

This being so, under the hypotheses and with the notations of (6.7.1), if $\mathbf{P}(X', x')$ is true, then so is $\mathbf{P}(X, x)$; the converse is true if k' is a separable extension of k.

Proof. The case where $\mathbf{P}$ is property d) has already been treated for the record, having already been treated (2.1.13) and (4.6.1). The same reasoning as at the beginning of the proof of (6.7.1) shows that the first assertion follows respectively from (6.5.2), (i), (6.5.3), (i), (6.5.4), (i); of the same kind, the second assertion follows from (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii). □

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Lemma (6.7.4.1). — *Let K, L be two extensions of a field k. If L is a separable extension of k, then $K \otimes_k L$ is a regular ring.*

PROOF. We distinguish two cases:

A) L is a *finite type* extension of k. Then, with the notations of (6.7.1.1), we can suppose that L is a *finite separable* extension of $k(\mathbf{t})$. For every $s \in \operatorname{Spec}(A)$, $k(s) \otimes_{k(t)} L$ is composed of a direct sum of a finite number of fields, finite separable extensions of $k(s)$, hence a regular ring; it follows from (6.5.2), (ii) that it suffices to prove that A is a regular ring; as $A = S^{-1}A'$, it suffices to prove that A' is regular ($\mathbf{0}$, 17.3.6), but one has seen that this was indeed the case in the proof of (6.7.1.1).

B) Let (L_λ) be the filtered family of sub-extensions of L that are of finite type over k; by A), each of the rings $C_\lambda = K \otimes_k L_\lambda$ is regular; on the other hand, for $L_\lambda \subset L_\mu$, L_μ is an L_λ-module plat, hence C_μ is a C_λ-module plat ($\mathbf{0_I}$, 6.2.1). Since $C = K \otimes_k L = \varinjlim (K \otimes_k L_\lambda)$ is Noetherian, we can apply (5.13.7), and C is therefore regular. □

Remarks (6.7.5). — (i) In the proof, the fact that $\mathbf{P}(X, x)$ holds implies $\mathbf{P}(X', x')$, without the hypothesis that k' is a *separable* extension of k (4.3.1); but even when X is regular it can happen that X' is not regular without k' separable (6.7.6). We have already seen (4.6.1) that when $\mathbf{P}$ is property d), it can happen that X is regular without X' being regular; it is interesting to give an example of this in which k is *algebraically closed* in the field of rational functions K of X.

(ii) As recalled in (i), it follows from (4.6.1) that when X is a prescheme that is algebraic over k which is not geometrically reduced, there are finite radiciel extensions k' of k such that $X' = X \otimes_k k'$ is not reduced.

Definition (6.7.6). — Let k be a field, X a k-prescheme locally Noetherian. We say that X is *geometrically regular* at a point x (resp. has property (R_n) geometric, resp. is *geometrically normal* at a point x) if, for every finite extension k' of k, $X' = X_{(k')} = X \otimes_k k'$ is regular (resp. has property (R_n), resp. is normal) at every point x' whose projection in X is x. We say that X is *geometrically regular* (or equivalently *geometrically regular in codimension* $\leqslant n$), resp. *geometrically normal*) if X is geometrically regular (resp. has the geometric property (R_n), resp. is geometrically normal) at every point.

We say that an algebra A over k is a *geometrically regular* (resp. *geometrically normal*, resp. *geometrically reduced*, resp. *having geometric property* (R_n)) ring if $\operatorname{Spec}(A)$ has the same property.

If $X = \operatorname{Spec}(K)$, where K is an extension of k, it amounts to the same thing to say that X is geometrically regular, or geometrically normal, or geometrically reduced, or that K is a *separable* extension of k: this follows from (4.6.1) and from the fact that if K is separable from k and k' a finite extension, $K \otimes_k k'$ is composed of a direct sum of a finite number of fields (Bourbaki, *Alg.*, chap. VIII, §7, n° 3, cor. 1 of th. 1).

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Proposition (6.7.7). — *Let k be a field, X a locally Noetherian k-prescheme, x a point of X; let us denote by $\mathbf{Q}(k')$ the relation "k' is an extension of k, and $\mathbf{P}(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ whose projection is x", where $\mathbf{P}(Z, z)$ is one of the properties a), b), c) of (6.7.4). Then the following properties are equivalent:*

a) *$\mathbf{Q}(k')$ is true for every finite extension k' of k.*
b) *$\mathbf{Q}(k')$ is true for every finite radiciel extension k' of k.*
c) *$\mathbf{Q}(k')$ is true for every finite type extension k' of k.*

Suppose in addition that X is locally of finite type over k; then the three preceding properties are also equivalent to the following:

d) *$\mathbf{Q}(k')$ is true for every extension k' of k.*
e) *$\mathbf{Q}(k')$ is true for every perfect extension k' of k.*
f) *$\mathbf{Q}(k')$ is true for every extension $k' = k^{p^{-s}}$, where p is the characteristic exponent of k, and $s > 0$.*

PROOF. To prove the equivalence of a), b), and c), it clearly suffices to establish that b) implies c). Let K be a finite type extension of k, which is therefore the field of fractions of a k-algebra of finite type A; set $Y = \operatorname{Spec}(A)$. We know (4.6.6) that there exists a finite radiciel extension k' of k such that $Y' = (Y \otimes_k k')_{\mathrm{red}}$ is separable over k'; if η' is the generic point of Y', $K' = k(\eta')$ is a separable extension of k' by (4.6.1). Since, by hypothesis, $\mathbf{P}(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ above x, it follows from (6.7.4) that $\mathbf{P}(X_{(K')}, x'')$ is true for every point x'' of $X_{(K')}$ above x, since $X_{(K')} = (X_{(k')})_{(K')}$, and x'' is above a point x' of $X_{(k')}$ above x, and K' is separable over k'. But we also have $X_{(K')} = (X_{(K)})_{(K')}$ and for every $x_0 \in X_{(K)}$ above x, there exists $x'' \in X_{(K')}$ above x_0; it follows from (6.7.4) that $\mathbf{P}(X_{(K)}, x_0)$ is true.

Suppose now X locally of finite type over k, so that for *every* extension K of k, $X_{(K)}$ is locally Noetherian. Since a radiciel extension of k is isomorphic to a sub-extension of a perfect extension of k, it follows immediately from (6.7.4) that $e)$ implies $f)$; similarly, $f)$ implies $b)$; $d)$ implies trivially $e)$, and finally $e)$ implies $d)$. We therefore need to prove that $b)$ implies $e)$. We take $k' = k^{p^{-\infty}}$ in $e)$. Since the question is local on X, one can moreover restrict to the case where X is affine and of finite type over k, by replacing X by a neighbourhood of x; let thus $X = \operatorname{Spec}(B)$, where B is a k-algebra of finite type; moreover, by definition of property $\mathbf{P}$ in all cases considered, one can also replace X by the local scheme of X at the point x, hence suppose that B is a *local* Noetherian ring. Set $B' = B \otimes_k k'$; k' is the inductive limit of its *finite* sub-extensions k'_λ and if we set $B'_\lambda = B \otimes_k k'_\lambda$, the morphism $f_\lambda : \operatorname{Spec}(B') \to \operatorname{Spec}(B'_\lambda)$ is a *homeomorphism* of $\operatorname{Spec}(B')$ onto $\operatorname{Spec}(B'_\lambda)$ for every λ; indeed f_λ is **IV-2** | 150 bijective by (**I**, 3.5.2), (**I**, 3.5.7), and (**I**, 3.5.8); on the other hand, it is closed by (**II**, 6.1.10). One can now apply (5.13.6) which proves (by hypothesis $b)$) that $b)$ implies $e)$ when $\mathbf{P}(Z, z)$ is property (R_n) at the point z. The case where $\mathbf{P}(Z, z)$ is the property of being regular (i.e. of having property (R_n) for all n) follows trivially. Finally, taking into account the Serre normality criterion (5.8.6), $b)$ implies $e)$ when $\mathbf{P}(Z, z)$ is the property of being normal at the point z, since being normal at z is equivalent to having both properties (S_2) and (R_1): one then applies what precedes for $n = 1$, and (6.7.2), (ii) for $n = 2$. □

Corollary (6.7.8). — *Let k be a field; for a k-prescheme locally Noetherian X and a point $x \in X$, let us denote by $\mathbf{P}(X, x)$ one of the following properties:*

(i) $\operatorname{coprof}(\mathscr{O}_x) \leqslant n$.
(ii) *$\mathscr{O}_x$ is a Cohen-Macaulay ring.*
(iii) *X satisfies property (S_n) at the point x.*
(iv) *X is geometrically regular at the point x.*
(v) *X has property (R_n) geometric at the point x.*
(vi) *X is geometrically normal at the point x.*
(vii) *X is geometrically reduced (i.e.* separable*) at the point x.*

Let k' be an extension of k; suppose, either that k' is of finite type, or that X is locally of finite type over k, so that $X' = X \otimes_k k'$ is locally Noetherian. Let x' be a point of X' whose projection in X is x. Then, for $\mathbf{P}(X, x)$ to hold, it is necessary and sufficient that $\mathbf{P}(X', x')$ hold.

Proof. This has already been seen for property (vii) (4.6.11), and for properties (i), (ii), and (iii) (6.7.1). For (iv), (v), and (vi), this follows from (6.7.7): indeed, the fact that the condition is necessary follows from the equivalence of the criteria $a)$ and $c)$, and from the equivalence of $c)$ and $d)$ when X is locally of finite type over k. To see that the condition is sufficient, either k'' is a finite radiciel extension of k; one can always consider k' and k'' as sub-extensions of an extension K of k; set $X'' = X \otimes_k k''$, $X_0 = X \otimes_k K$, and note that since k'' is a radiciel extension of k, there is only *one* point x'' of X'' above x (**I**, 3.5.7) and (**I**, 3.5.8). Let x_0 be any point of X_0 above x; if $\mathbf{P}(X', x')$ is true, then so is $\mathbf{P}(X_0, x_0)$ by (6.7.7), $c)$ and $d)$) (since k' is a finite type extension of k and K is an extension of k', one can suppose that k' is of finite type, and if X is locally of finite type over k, X' is locally of finite type over k'). We deduce from (6.7.4) that the property $\mathbf{Q}(k'')$ is true (with the notations of (6.7.7)), hence $\mathbf{P}(X, x)$ is true by (6.7.7), $b)$). □

6.8. Regular, normal, reduced, smooth morphisms

Definition (6.8.1). — Let X, Y be two preschemes, $f : X \to Y$ a morphism such that the fibre $f^{-1}(y)$ is a locally Noetherian prescheme for every $y \in Y$, x a point of X. We say respectively that f is a morphism:

(i) of coprofondeur $\leqslant n$ at the point x;
(ii) of Cohen-Macaulay at the point x;
(iii) (S_n) at the point x;
(iv) regular at the point x;
(v) (R_n) at the point x;
(vi) normal at the point x;
(vii) reduced at the point x;

if f is flat at the point x, and if in addition the corresponding property $\mathbf{P}(f^{-1}(f(x)), x)$ (notation of (6.7.8)) is true.

We say that f is *smooth* at the point x if it is locally of finite presentation in a neighbourhood of x in X, and if it is regular at the point x. We say that f is a morphism of coprofondeur $\leqslant n$, Cohen-Macaulay, (S_n), regular, (R_n), normal, reduced, smooth, if it has the corresponding property at every point of X.

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Proposition (6.8.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, x a point of X. Let us denote by $\mathbf{M}(f, x)$ one of the properties (i) to (vii) of the definition (6.8.1), or the property for f to be smooth at the point x. Let Y' be a locally Noetherian prescheme, $g : Y' \to Y$ a morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Suppose that f or g is locally of finite type. Then, for every $x' \in X'$ above x, the property $\mathbf{M}(f, x)$ implies $\mathbf{M}(f', x')$.*

Proof. Set $y = f(x)$, $y' = f'(x')$; by transitivity of fibres (I, 3.6.4), on a $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; since $f^{-1}(y)$ is locally of finite type over $k(y)$ or $k(y')$ is of finite type over $k(y)$ (I, 6.4.11), it follows from (6.7.8) that the properties $\mathbf{P}(f^{-1}(y), x)$ and $\mathbf{P}(f'^{-1}(y'), x')$ are equivalent; moreover, if f is flat at the point x, f' is flat at the point x' (2.1.4), which proves the proposition, taking into account (1.4.3), (iii) and (2.7.1), (iv). □

Proposition (6.8.3). — *For a morphism f of locally Noetherian preschemes, let $\mathbf{M}(f)$ be one of the following properties: to be Cohen-Macaulay, (S_n), regular, (R_n), normal, reduced.*

(i) *Let X, Y, Z be three locally Noetherian preschemes, $f : X \to Y$, $g : Y \to Z$ two morphisms. If $\mathbf{M}(f)$ and $\mathbf{M}(g)$ are true, $\mathbf{M}(g \circ f)$ is true.*
(ii) *Conversely, if f is surjective and if $\mathbf{M}(f)$ and $\mathbf{M}(g \circ f)$ are true, then $\mathbf{M}(g)$ is true.*
(iii) *Let X, Y, Y' be three locally Noetherian preschemes, $f : X \to Y$, $h : Y' \to Y$ two morphisms; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Suppose that f or h is locally of finite type. Then, if $\mathbf{M}(f)$ is true, so is $\mathbf{M}(f')$; the converse is true when h is faithfully flat.*

The conclusions of (i) and (iii) are still true when $\mathbf{M}$ is the property of being smooth and (in (iii)) h is quasi-compact.

Proof. (i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f is flat and f surjective, g is flat (2.1.6) and (2.2.13). Moreover, for every $z \in Z$, $f_z = f \otimes 1_{k(z)} : u^{-1}(z) \to g^{-1}(z)$ is flat (resp. faithfully flat if f is) and the morphism $f_z^{-1}(y) \otimes_{k(y)} k(y')$ is isomorphic to $f^{-1}(y)$ dans $g^{-1}(z)$ (I, 3.6.4). When $\mathbf{M}$ is the property of being regular, (R_n), normal or reduced, the hypothesis implies that $\mathbf{M}(f)$ and $\mathbf{M}(g)$ hold, and each of the fibres $f_z^{-1}(y')$ has for $y' \in Y'_z$ the corresponding property $c)$, $d)$, $e)$, $f)$ of (6.6.1); we deduce from (6.6.1), (i) that X'_z has the same property, hence $\mathbf{M}(g \circ f)$ is true.

(ii) Conversely, the hypothesis that $\mathbf{M}(g \circ f)$ and $\mathbf{M}(f)$ are true and that f is surjective implies that Y'_z has the corresponding property for every z, by (6.6.1), (ii), f_z being surjective for every z.

(iii) The first assertion follows from (6.8.2). On the other hand, if h is faithfully flat, f is flat (2.4.1); moreover, with the notations of (6.8.2), the properties $\mathbf{P}(f^{-1}(y), x)$ and $\mathbf{P}(f'^{-1}(y'), x')$ are equivalent (6.7.8), hence $\mathbf{M}(f)$ and $\mathbf{M}(f')$ are equivalent. Finally, the last assertion of the proposition follows from (1.4.3), (iii) and (2.7.1), (iv).

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Remarks (6.8.4). — (i) If f is faithfully flat, g flat, and $g \circ f$ is of coprofondeur $\leqslant n$, then g is of coprofondeur $\leqslant n$, as follows from (6.6.2).

(ii) When, in (6.8.1), one takes Y to be the spectrum of a field k, the notions (iv), (v), and (vi) reduce to those defined in (6.7.5). It is clear that these latter are the ones relative to the base field k. The definition (6.8.1) leads then to saying that a prescheme X is "regular (resp. (R_n), resp. normal) *over* k" instead of saying that it is "geometrically regular (resp. (R_n), resp. normal)"; one will avoid confusing this notion with the property of being regular (resp. (R_n), resp. normal) which is independent of k. One can make the same remarks in this regard as in (4.5.12).

Proposition (6.8.5). — *Let k be a field, X, Y two k-preschemes locally Noetherian, at least one of which is locally of finite type over k. For a k-prescheme Z, and a point $z \in Z$, let us denote by $\mathbf{P}(Z)$ one of the properties c), d), e) (resp. f)) of (6.6.1); the property "$\mathbf{P}(Z)$ geometric" is then defined in (6.7.6) (resp. (4.6.1)). Then:*

(i) *If X has property $\mathbf{P}$ and Y has property $\mathbf{P}$ geometric, $X \times_k Y$ has the geometric property $\mathbf{P}$.*
(ii) *If X and Y have property $\mathbf{P}$ geometric, so does $X \times_k Y$.*

PROOF. Let $f : X \times_k Y \to X$, $g : X \to \operatorname{Spec}(k)$ be the structural morphisms, which are faithfully flat (2.2.13). The hypothesis that Y has the geometric property $\mathbf{P}$ implies by (6.8.2) that $\mathbf{M}(f)$ is true, $\mathbf{M}$ being the property corresponding to $\mathbf{P}$; under hypothesis (ii), $\mathbf{M}(g)$ is also true, hence assertion (ii) follows from (6.8.3), (i). As for assertion (i), it follows directly from (6.6.1), (i). □

Theorem (6.8.6). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. The following properties are equivalent:*

a) *f is smooth at the point x.*
b) *f is regular at the point x.*
c) *$\mathscr{O}_x$ is a formally smooth $\mathscr{O}_y$-algebra $(\mathbf{0}$, 19.3.1) for the preadic topologies on $\mathscr{O}_x$ and $\mathscr{O}_y$.*
c′) *$\mathscr{O}_x$ is a formally smooth $\mathscr{O}_y$-algebra $(\mathbf{0}$, 19.3.1) for the discrete topologies on $\mathscr{O}_x$ and $\mathscr{O}_y$.*

The equivalence of a) and b) follows from the definitions (6.8.1) and from the fact that for locally Noetherian preschemes, morphisms locally of finite type are locally of finite presentation (1.4.2). Secondly, for $\mathscr{O}_x$ to be an $\mathscr{O}_y$-algebra smooth for the preadic topologies, it is necessary and sufficient that $\mathscr{O}_x$ be a flat $\mathscr{O}_y$-module and that $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$ be a $k(y)$-algebra formally smooth for its preadic topology $(\mathbf{0}$, 19.7.1); but for $\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)$ to be a $k(y)$-algebra formally smooth for its preadic topology, it is necessary and sufficient that it be a $k(y)$-algebra geometrically regular $(\mathbf{0}$, 19.6.6); this shows the equivalence of b) and c). Finally, to prove the equivalence of c) and c′), we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(C)$ are affine, A being Noetherian and C a finite type A-algebra, and one can therefore write C in the form $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]/\mathfrak{J}$. Since here $\mathfrak{J}/\mathfrak{J}^2$ is a C-module of finite presentation, the equivalence of c) and c′) follows from $(\mathbf{0}$, 22.6.4) applied to A, $B = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ and $C = B/\mathfrak{J}$. **IV-2 | 153**

Corollary (6.8.7). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism locally of finite type. Then the set of points $x \in X$ where f is smooth (or regular) is open in X.*

PROOF. It follows from $(\mathbf{0}$, 22.6.5) that the set of $x \in X$ satisfying condition c') of (6.8.6) is open in X, and the conclusion follows from (6.8.6). □

Remark (6.8.8). — In (17.5.1), we shall show the equivalence of b) and c') in (6.8.6), as well as the corollary (6.8.7), remain valid without the Noetherian hypothesis on X and Y, provided one restricts to morphisms *locally of finite presentation*.

6.9. The theorem of generic flatness

Theorem (6.9.1). — *Let Y be a locally Noetherian prescheme, integral, $u : X \to Y$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. There then exists a non-empty open subset U of Y such that $\mathscr{F}|u^{-1}(U)$ is flat over U.*

PROOF. One can clearly restrict to the case where $Y = \operatorname{Spec}(A)$ is affine; then X is a finite union of open affines X_i of finite type over Y; if, for each i, there is an open non-empty U_i in Y such that $\mathscr{F}|(X_i \cap u^{-1}(U_i))$ is flat over U_i, it is clear that taking U as the intersection of the U_i answers the question; one can therefore suppose that $X = \operatorname{Spec}(B)$, where B is a finite type A-algebra. We then have $\mathscr{F} = \widetilde{M}$, where M is a finite type B-module; the theorem will therefore follow from the following: □

Lemma (6.9.2). — *Let A be an integral Noetherian ring, B a finite type A-algebra, M a finite type B-module. Then there exists $f \neq 0$ in A such that M_f is a free A_f-module.*

PROOF. Let K be the field of fractions of A; then $B \otimes_A K$ is a finite type algebra over K and $M \otimes_A K$ a $(B \otimes_A K)$-module of finite type. We argue by induction on the *dimension* n of the support of $M \otimes_A K$, which is $-\infty$ or finite and $\geqslant 0$. Suppose first $n = -\infty$, that is to say $M \otimes_A K = 0$; if $(m_i)_{1 \leqslant i \leqslant r}$ is a generating system of the B-module M, there exists $f \neq 0$ in A such that $fm_i = 0$ for $1 \leqslant i \leqslant r$; hence $M_f = 0$ and the lemma is true in this case. Suppose now $n \geqslant 0$. We know that there exists a composition series $M = M_1 \supset M_2 \supset \ldots \supset M_q = 0$ of the B-module M such that each of the quotients $N_i = M_i/M_{i+1}$ is isomorphic to a B-module of the form $B/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of B (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, th. 1). If the theorem is true for each of the N_i, there is for each i an $f_i \neq 0$ in A such that $(N_i)_{f_i}$ is a free A_{f_i}-module; setting $f = f_1 f_2 \cdots f_{q-1}$, it follows that $(N_i)_f$ is a free A_f-module for $1 \leqslant i \leqslant q-1$. But $(N_i)_f = (M_i)_f/(M_{i+1})_f$ ($\mathbf{0_I}$, 1.3.2) and since an extension of free modules is free, we deduce that M_f is a free A_f-module. Replacing B by $B/\mathfrak{p}$ ($\mathfrak{p}$ a prime ideal of B), which is still of finite type over A, one can thus restrict to the case where $M = B$ and B is *integral*. We then know (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 1 of th. 1) that there exists an element $g \neq 0$ in A and elements t_i $(1 \leqslant i \leqslant m)$ of B, algebraically independent over A and such that B_g is entire over $A_g[t_1, \ldots, t_m]$. One can replace A by A_g, B by B_g and hence assume that B is entire over $C = A[t_1, \ldots, t_m]$, hence a C-module of finite type and *without torsion*. We know moreover (4.1.2) that the dimension of $\operatorname{Spec}(B \otimes_A K)$ is equal to m, hence on a $m = n$. **IV-2** | 154

This being so, if h is the rank of the torsion-free C-module B, there exists an exact sequence of C-modules

$$0 \longrightarrow C^h \longrightarrow B \longrightarrow M' \longrightarrow 0$$

where M' is a C-module of *torsion* of finite type; the support of M' therefore does not contain the generic point of $\operatorname{Spec}(C)$ (**I**, 7.4.6) and hence the support of $M' \otimes_A K$ does not contain the generic point of $\operatorname{Spec}(C \otimes_A K)$ (**I**, 9.1.13.1); we thus conclude (4.1.2.1) that its dimension is $< n$. By the induction hypothesis, there exists an $f \neq 0$ in A such that M'_f is a free A_f-module; on the other hand C^h_f is an A_f-module free; hence it follows that M_f is a free A_f-module. Q.E.D. □

Corollary (6.9.3). — *Let S be a Noetherian prescheme, $u : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then there is a partition $(\mathrm{S}_i)_{1 \leqslant i \leqslant n}$ of S in a finite number of locally closed subsets of S such that, if one denotes by S_i the reduced sub-prescheme having space S_i as underlying space, and if one sets $X_i = X \times_S S_i$, $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_i}$, the $\mathscr{O}_{X_i}$-Module $\mathscr{F}_i$ is flat over S_i.*

PROOF. We proceed by Noetherian induction ($\mathbf{0_I}$, 2.2.2) on the set of closed subsets T of S such that the reduced sub-prescheme having T for underlying space satisfies the conclusion of (6.9.3). One can thus restrict to proving the corollary for S having for underlying space a closed subset $\neq S$. Since the morphism $S_{\mathrm{red}} \to S$ is of finite type and surjective, one can replace S by S_{red}, hence suppose S reduced and non-empty. Since S is Noetherian, the interior T of an irreducible component of S is non-empty, and the prescheme induced on T is integral; there is therefore by (6.9.1) a non-empty open subset $U \subset T$ such that $\mathscr{F}|u^{-1}(U)$ is flat over U. If Y is the sub-prescheme reduced by S having for underlying space $S - U$, it is clear by hypothesis that there is a partition (Y_i) of Y into locally closed sets in S, such that the Y_i and U form a partition answering the question. □ **IV-2** | 155

6.10. Dimension and depth of a Module normally flat along a closed sub-prescheme

(6.10.1). Let X be a locally Noetherian prescheme, $\mathscr{J}$ a quasi-coherent Ideal of $\mathscr{O}_X$, Y the closed sub-prescheme of X defined by $\mathscr{J}$, $j : Y \to X$ the canonical injection. For every integer $k \geqslant 0$, the $\mathscr{O}_X$-Module $\mathscr{J}^k\mathscr{F} / \mathscr{J}^{k+1}\mathscr{F}$ is annihilated by $\mathscr{J}$, hence is of the form $j_*(\mathscr{G}_k)$, where $\mathscr{G}_k = j^*(\mathscr{J}^k\mathscr{F} / \mathscr{J}^{k+1}\mathscr{F}) = j^*(\mathscr{J}^k\mathscr{F})$ is a coherent $\mathscr{O}_Y$-Module. By abuse of language, we denote $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ the graded $\mathscr{O}_Y$-Module equal to the direct sum

$$\bigoplus_{k=0}^{\infty} \mathscr{G}_k = j^*\left(\bigoplus_{k=0}^{\infty} \mathscr{J}^k\mathscr{F} / \mathscr{J}^{k+1}\mathscr{F}\right);$$

in particular, we have $\mathrm{gr}^0_{\mathscr{J}}(\mathscr{F}) = \mathscr{G}_0 = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_Y = j^*(\mathscr{F})$. We shall say (following Hironaka) that $\mathscr{F}$ is *normally flat along* Y if $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is a flat $\mathscr{O}_Y$-Module. This is the same as saying ((2.1.12) and ($\mathbf{0_I}$, 6.1.2)) that each of the $\mathscr{O}_Y$-Modules $\mathscr{G}_k$ is *locally free*.

Proposition (6.10.2). — *Let X be a locally Noetherian prescheme, Y a closed integral sub-prescheme of X. For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, there exists an open subset U of X such that $Y \cap U \neq \varnothing$ and that $\mathscr{F}|U$ is normally flat along $Y \cap U$.*

Proof. Let $\mathscr{J}$ be the coherent Ideal of $\mathscr{O}_X$ defining Y; the $\mathscr{O}_Y$-Algebra $\mathscr{B} = \mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$, an image of $\mathscr{J} / \mathscr{J}^2$, is quasi-coherent and *of finite type* since it is generated by $\mathrm{gr}^1_{\mathscr{J}}(\mathscr{O}_X)$, image of $\mathscr{J} / \mathscr{J}^2$; as $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is a graded $\mathscr{B}$-module quasi-coherent generated by $\mathrm{gr}^1_{\mathscr{J}}(\mathscr{F})$, it is a finite type $\mathscr{B}$-Module. If we set $Z = \mathrm{Spec}(\mathscr{B})$, the structural morphism $u : Z \to Y$ is then of finite type, and if $\mathscr{H}$ is the coherent $\mathscr{O}_Z$-Module such that $u_*(\mathscr{H}) = \mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$, it suffices to apply u and to $\mathscr{H}$ the theorem of generic flatness (6.9.1) to prove the proposition. □

Proposition (6.10.3). — *With the notations of (6.10.1), suppose that $\mathscr{F}$ is normally flat along Y. Then:*

(i) *$Y \cap \mathrm{Supp}(\mathscr{F})$ is a part that is both open and closed in Y (in other words, a union of connected components of Y).*

(ii) *If $\mathscr{J}$ is locally nilpotent, $\mathrm{Supp}(\mathscr{F})$ is an open and closed part of X, and for every $x \in \mathrm{Supp}(\mathscr{F})$, we have*

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{Y,x}) \tag{6.10.3.1}$$

$$\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{O}_{Y,x}) \tag{6.10.3.2}$$

$$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{O}_{Y,x}). \tag{6.10.3.3}$$

Corollary (6.10.4). — *Let $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ (resp. $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$) be the set of $x \in X$ such that $\mathscr{F}$ satisfies (S_n) at the point x (resp. such that $\mathrm{coprof}(\mathscr{F}_x) \leqslant n$). If $\mathscr{F}$ is normally flat along Y, then $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F}) = \mathrm{U}_{\mathrm{S}_n}(\mathscr{O}_Y)$ and $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F}) = \mathrm{U}_{\mathrm{C}_n}(\mathscr{O}_Y)$.*

Proposition (6.10.5). — *With the notations of (6.10.1), suppose that Y is connected and non-empty, and that $\mathscr{F}$ is normally flat along Y. For every integer $n \geqslant 0$, we set*

$$r(n) = \mathrm{rg}_{\mathscr{O}_Y}(\mathrm{gr}^n_{\mathscr{J}}(\mathscr{F})) \tag{6.10.5.1}$$

(the $\mathscr{O}_Y$-Module locally free $\mathrm{gr}^n_{\mathscr{J}}(\mathscr{F})$ being necessarily of constant rank). Then:

(i) *There exists a polynomial $P \in \mathbf{Q}[\mathrm{T}]$ such that $P(n) = r(n)$ for n sufficiently large.*

(ii) *We have $Y \cap \mathrm{Supp}(\mathscr{F}) = \varnothing$, or $Y \cap \mathrm{Supp}(\mathscr{F}) = Y$ (in other words $Y \subset \mathrm{Supp}(\mathscr{F})$).*

In the second case, let $d-1$ denote the degree of P; for every maximal point y of Y, we have

$$\dim(\mathscr{F}_y) = d \tag{6.10.5.2}$$

and in particular

$$\mathrm{codim}(Y, \mathrm{Supp}(\mathscr{F})) = d. \tag{6.10.5.3}$$

(iii) *Suppose $Y \subset \mathrm{Supp}(\mathscr{F})$. For every $x \in Y$, we have*

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{Y,x}) + d. \tag{6.10.5.4}$$

Proposition (6.10.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, Y a closed irreducible sub-prescheme of X, of generic point $y \in \operatorname{Supp}(\mathscr{F})$. There then exists a neighbourhood U of y in X such that, for every $x \in U \cap Y$, we have* IV-2 | 157

(6.10.6.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{F}_y) + \dim(\mathscr{O}_{Y,x})$$

(6.10.6.2) $$\operatorname{prof}(\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_y) + \operatorname{prof}(\mathscr{O}_{Y,x})$$

(6.10.6.3) $$\operatorname{coprof}(\mathscr{F}_x) = \operatorname{coprof}(\mathscr{F}_y) + \operatorname{coprof}(\mathscr{O}_{Y,x}).$$

6.11. Criteria for the sets $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ or $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ to be open IV-2 | 158

Lemma (6.11.1). — *Let X be a locally Noetherian prescheme, locally immersible in a regular scheme (5.11.1); let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. Then:*

(i) *(M. Auslander) The function $x \mapsto \operatorname{coprof}(\mathscr{F}_x)$ is semi-continuous superiorly in X (in other words, for every integer n, the set $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ of $x \in X$ such that $\operatorname{coprof}(\mathscr{F}_x) \leqslant n$ is open).*

(ii) *For every integer n, the set $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ of $x \in X$ such that $\mathscr{F}$ has property (S_n) at the point x is open.*

Proposition (6.11.2). — *Let X be a locally Noetherian prescheme, regular and locally immersible in a regular scheme (5.11.1); let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. Then:*

(i) *(M. Auslander) The function $x \mapsto \operatorname{coprof}(\mathscr{F}_x)$ is semi-continuous superiorly in X (in other words, for every integer n, the set $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ of $x \in X$ such that $\operatorname{coprof}(\mathscr{F}_x) \leqslant n$ is open).*

(ii) *For every integer n, the set $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ of $x \in X$ such that $\mathscr{F}$ has property (S_n) at the point x is open.*

Corollary (6.11.3). — *Under the hypotheses of (6.11.2), the set $\mathrm{CM}(\mathscr{F})$ of points $x \in X$ such that $\mathscr{F}_x$ is a Cohen-Macaulay module is open in X.*

Proof. Indeed, it is the set $\mathrm{U}_{\mathrm{C}_0}(\mathscr{F})$. □

Remarks (6.11.4). — (i) The reasoning of (6.11.2), (ii) shows that (without hypothesis on X) when $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ is open for every integer n, then $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ is also open.

(ii) We have $\mathrm{CM}(\mathscr{F}) = \bigcap_n \mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$. If every point $x \in X$ admits a neighbourhood V that is open such that V is of *finite* dimension, the decreasing sequence of intersections $V \cap \mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ is *stationary* since $\dim(\mathscr{F}_x)$ is finite for every x; hence, if the $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ are open, $\mathrm{CM}(\mathscr{F})$ is then also open in X.

(iii) We write $\mathrm{U}_{\mathrm{S}_n}(X)$, $\mathrm{U}_{\mathrm{C}_n}(X)$, $\mathrm{CM}(X)$ instead of $\mathrm{U}_{\mathrm{S}_n}(\mathscr{O}_X)$, $\mathrm{U}_{\mathrm{C}_n}(\mathscr{O}_X)$, $\mathrm{CM}(\mathscr{O}_X)$.

Proposition (6.11.5). — *Let A be a local Noetherian ring, M a finite type A-module. For every prime ideal $\mathfrak{p}$ of A, we have*

(6.11.5.1) $$\operatorname{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leqslant \operatorname{coprof}_A(M).$$

Proof. We can restrict to the case where $M_{\mathfrak{p}} \neq 0$. Since $\widehat{A}$ is a faithfully flat A-module $(\mathbf{0_I}, 6.5.1)$, there exists a prime ideal $\mathfrak{q}$ of $\widehat{A}$ above $\mathfrak{p}$ $(\mathbf{0_I}, 6.5.1)$; since $\widehat{M}_{\mathfrak{q}} = (M \otimes_A \widehat{A})_{\mathfrak{q}} = M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \widehat{A}_{\mathfrak{q}}$ and $\widehat{A}_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$-module, hence a flat A-module flat $(\mathbf{0_I}, 6.2.1)$, it follows from (6.3.2) that $\operatorname{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leqslant \operatorname{coprof}_{\widehat{A}_{\mathfrak{q}}}(\widehat{M}_{\mathfrak{q}})$. On the other hand, $X = \operatorname{Spec}(\widehat{A})$ is a sub-scheme of a regular scheme by the Cohen structure theorem $(\mathbf{0}, 19.8.8, (i))$; on deduces from (6.11.2) that $\operatorname{coprof}_{\widehat{A}_{\mathfrak{q}}}(\widehat{M}_{\mathfrak{q}}) \leqslant \operatorname{coprof}_{\widehat{A}}(\widehat{M})$; finally, we know that $\operatorname{coprof}_{\widehat{A}}(\widehat{M}) = \operatorname{coprof}_A(M)$ $(\mathbf{0}, 16.4.10)$, which completes the proof of (6.11.5.1). □

Proposition (6.11.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, n an integer > 0. Suppose that for every closed integral sub-prescheme Y of X, there exists a non-empty open subset W of Y such that the sub-prescheme of Y induced on the open W satisfies (S_n). Then the set $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ is open in X.* IV-2 | 160

Corollary (6.11.7). — *With the notations of (6.11.6):*

(i) *The set $\mathrm{U}_{\mathrm{S}_1}(\mathscr{F})$ is open in X.*

(ii) *For $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ to be open, it suffices that every maximal point x of $\operatorname{Supp}(\mathscr{F})$, belonging to $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$, be interior to $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$.*

Proposition (6.11.8). — *Let X be a locally Noetherian prescheme satisfying the following property:*

(CMU) *Every closed integral sub-prescheme Y of X contains a non-empty open W such that the prescheme induced by X on W is a Cohen-Macaulay prescheme.*

Then, for every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the function $x \mapsto \operatorname{coprof}(\mathscr{F}_x)$ is locally constructible and semi-continuous superiorly; the sets $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ and $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ are open in X.

PROOF. Indeed, let Y be a closed integral sub-prescheme of X, of generic point y; by (6.10.6), there is a neighbourhood V of y in X such that, for every $x \in V \cap Y$, we have

(6.11.8.1) $$\operatorname{coprof}(\mathscr{F}_x) = \operatorname{coprof}(\mathscr{F}_y) + \operatorname{coprof}(\mathscr{O}_{Y,x}).$$

But by hypothesis there exists a non-empty open W of Y such that, for $x \in V \cap W$ we have $\operatorname{coprof}(\mathscr{O}_{Y,x}) = 0$, hence $\operatorname{coprof}(\mathscr{F}_x)$ is constant in a neighbourhood of y in Y, which proves that the function $x \mapsto \operatorname{coprof}(\mathscr{F}_x)$ is locally constructible ($\mathbf{0_{III}}$, 9.3.2); moreover it follows from (6.11.5) and from ($\mathbf{0_{III}}$, 9.3.4) that this function is semi-continuous superiorly. The last assertion follows from (6.11.4), (i), or equivalently from (6.11.6). □

Remarks (6.11.9). — (i) If X satisfies the hypothesis (CMU) of (6.11.8), then, for every morphism locally of finite type $u : X' \to X$, X' also satisfies (CMU).

(ii) Suppose that there exists a coherent $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\operatorname{Supp}(\mathscr{F}) = X$ and that $\mathscr{F}$ is a Cohen-Macaulay $\mathscr{O}_X$-Module. Then X satisfies condition (CMU).

6.12. Nagata's criteria for $\operatorname{Reg}(X)$ to be open

(6.12.1). Given a locally Noetherian prescheme X, the *singular locus* of X is defined to be the set of points $x \in X$ such that X is not regular at the point x (in other words, such that the local ring $\mathscr{O}_x$ is not regular); it is denoted $\operatorname{Sing}(X)$; the complementary set $X - \operatorname{Sing}(X)$, consisting of the $x \in X$ where X is regular, is denoted $\operatorname{Reg}(X)$. We propose in this number to give conditions under which $\operatorname{Sing}(X)$ is *closed* (i.e. $\operatorname{Reg}(X)$ is *open*).

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Proposition (6.12.2). — *Let X be a locally Noetherian prescheme. The following conditions are equivalent:*

a) $\operatorname{Reg}(X)$ *is open in X.*
b) *For every $x \in \operatorname{Reg}(X)$, there exists a non-empty open subset of $\overline{\{x\}}$ contained in* $\operatorname{Reg}(X)$.

Moreover, these conditions are implied by the following:

c) *For every $x \in \operatorname{Reg}(X)$, denoting by Y the closed reduced sub-prescheme of X having $\overline{\{x\}}$ as underlying space,* $\operatorname{Reg}(Y)$ *is a neighbourhood of x in Y.*

PROOF. The equivalence of a) and b) follows from the fact that $\operatorname{Reg}(X)$ is stable by generisation ($\mathbf{0}$, 17.3.2) and from ($\mathbf{0_{III}}$, 9.2.6). To prove that c) implies b), consider a closed integral sub-prescheme Z of Y of generic point z; if Y' is the sub-prescheme integral of X having as underlying space $Y' = \overline{\{z\}}$ of z in X, Z is open in Y' and the sub-prescheme Z of Y', being reduced, is induced by Y' on the open Z of the underlying space of Y'. The hypothesis implies that $\operatorname{Reg}(Y')$ is a neighbourhood of z in Y'; it therefore suffices to apply (6.12.2) by replacing X by Y and Y by Z. □

Corollary (6.12.3). — *Let X be a locally Noetherian prescheme. The following conditions are equivalent:*

a) *For every sub-prescheme Y of X,* $\operatorname{Reg}(Y)$ *is open in Y.*
b) *For every closed integral sub-prescheme Y of X,* $\operatorname{Reg}(Y)$ *contains a non-empty open part of Y.*

PROOF. It is clear that a) implies b), since if Y is integral and y is its generic point, $\mathscr{O}_{Y,y}$ is a field, hence $y \in \operatorname{Reg}(Y)$. Conversely, to see that b) implies a), consider a closed integral sub-prescheme Z of Y of generic point z; if Y' is the sub-prescheme integral of X having $\overline{\{z\}}$ as underlying space, $Y' = \overline{\{z\}}$ in X, and the sub-prescheme Z of Y', being reduced, is induced by Y' on the open part of its underlying space. The hypothesis implies that $\operatorname{Reg}(Y')$ is a neighbourhood of z in Y'; it suffices then to apply (6.12.2) by replacing X and Y by Z. □

Theorem (6.12.4). — (Nagata). — *Let A be a Noetherian ring, $X = \operatorname{Spec}(A)$. The following conditions are equivalent:*

a) *For every prescheme X' locally of finite type over X,* $\operatorname{Reg}(X')$ *is open in X'.*
b) *For every finite integral A-algebra A', there exists a non-empty open of $X' = \operatorname{Spec}(A')$ contained in* $\operatorname{Reg}(X')$.

c) *For every prime ideal $\mathfrak{p}$ of A and every finite radiciel extension K' of the field of fractions K of $A/\mathfrak{p}$, finite on A, having K' for field of fractions, and such that in $X' = \operatorname{Spec}(A')$, a non-empty open subset is contained in* $\operatorname{Reg}(X')$.

Corollary (6.12.5). — (Zariski). — *Let k be a field; for every k-prescheme X locally of finite type over k, the set of $x \in X$ where X is regular (resp.* geometrically regular *over k) is open in X.*

Corollary (6.12.6). — *Let A be a ring having one of the following properties:*

(i) *A is a Dedekind ring and its field of fractions K is of characteristic* 0.
(ii) *A is a semi-local Noetherian ring of dimension $\leqslant 1$.*

Then, for every prescheme X locally of finite type over A, $\operatorname{Reg}(X)$ *is open in X.*

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Theorem (6.12.7). — (Nagata). — *Let A be a local Noetherian* complete *ring, $X = \operatorname{Spec}(A)$. Then* $\operatorname{Reg}(X)$ *is open in X.*

Proof. By (6.12.2), we are reduced to the case where A is moreover *integral*, and we must show that $\operatorname{Reg}(X)$ contains a non-empty open subset of X. We distinguish two cases:

I) *The field of fractions K of A is of characteristic* 0. — We know then (**0**, 19.8.8) that there exists a complete discrete valuation ring C and a sub-ring B of A such that A is a *finite* B-algebra and such that B is isomorphic to a formal power series ring $C[[T_1, \dots, T_r]]$. As C is regular (**II**, 7.1.6), so is B (**0**, 17.3.8); moreover, the field of fractions L of B being of characteristic 0, K is a *separable* finite extension of L, so we can apply (6.12.4), I) to B, which proves the proposition in this case.

II) *The field of fractions K of A is of characteristic $p > 0$.* — Then A contains the prime field $\mathbf{F}_p$, and the theorem has already been proved in this case (**0**, 22.7.6). □

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Corollary (6.12.8). — *Let A be a local Noetherian complete ring. Then, for every prescheme X locally of finite type over A,* $\operatorname{Reg}(X)$ *is open in X.*

Proof. We verify condition c) of (6.12.4). If $\mathfrak{p}$ is prime in A, $A/\mathfrak{p}$ is again a local Noetherian complete ring; if K' is a finite extension of the field of fractions K of $A/\mathfrak{p}$, K' is the field of fractions of a sub-A-algebra A' of K', generated by a finite system of generators of K' over K, integral over A; A' is a semi-local integral ring of dimension $\leqslant 1$ (**0**, 16.1.5), and therefore, in $X' = \operatorname{Spec}(A')$, the set reduced to the generic point is open and evidently contained in $\operatorname{Reg}(X')$, which proves condition c) of (6.12.4) in this case. This corollary applies in particular when $A = \mathbf{Z}$. □

Proposition (6.12.9). — *Let X be a locally Noetherian prescheme such that* $\operatorname{Reg}(X)$ *is open; then, for every $n \geqslant 0$, the set $\mathrm{U}_{\mathrm{R}_n}(X)$ of the points $x \in X$ where X satisfies (R_n) is open.*

Proof. Indeed (**0**, 8.2), $\mathrm{U}_{\mathrm{R}_n}(X)$ is the union of $\operatorname{Reg}(X)$ and the set of $x \in \operatorname{Sing}(X)$ such that the generic points z_i of the irreducible components of $\operatorname{Sing}(X)$ containing x satisfy the relation $\dim(\mathscr{O}_{X,z_i}) \geqslant n+1$ (since we have $z_i \notin \operatorname{Reg}(X)$); in other words, these points $x \in \operatorname{Sing}(X)$ are those for which $\operatorname{codim}_x(\operatorname{Sing}(X), X) \geqslant n+1$ (**0**, 1.2); as the function $x \mapsto \operatorname{codim}_x(\operatorname{Sing}(X), X)$ is semi-continuous inferiorly (**0**, 14.2.6), the set of these points is open in $\operatorname{Sing}(X)$, which completes the proof. □

We also note the following elementary result:

Proposition (6.12.10). — *Let X be a locally Noetherian prescheme. The set $\mathrm{U}_{\mathrm{R}_1}(X)$ is open in X. For $\mathrm{U}_{\mathrm{R}_0}(X)$ to be open in X, it is necessary and sufficient that every maximal point $x \in X$ such that $x \in \mathrm{U}_{\mathrm{R}_1}(X)$ (which means that X is reduced at x) be interior to $\mathrm{U}_{\mathrm{R}_1}(X)$.*

Proof. By definition (**0**, 8.2), $\mathrm{U}_{\mathrm{R}_1}(X)$ is the set $X - \bigcup X'_\alpha$, where X'_α ranges over the set of irreducible components of X such that X is not reduced at their generic point; as the set of irreducible components of X is locally finite, $\mathrm{U}_{\mathrm{R}_1}(X)$ is open. Regarding $\mathrm{U}_{\mathrm{R}_0}(X)$, the condition of the statement being trivially necessary, let us prove that it is sufficient. Let X''_β be the irreducible components of X such that at their generic point x''_β the prescheme X is reduced; by hypothesis there exists an open

$U \subset \mathrm{U}_{\mathrm{R}_1}(X)$ containing all the x''_β. Denote then by Z_λ the irreducible components of the closed set $Z = X - U$ whose generic point z_λ is such that $\dim(\mathscr{O}_{X,z_\lambda}) \leqslant 1$ and that $\mathscr{O}_{X,z_\lambda}$ is non-regular. No point belonging to one of the Z_λ can belong to $\mathrm{U}_{\mathrm{R}_1}(X)$; but conversely, if $x \in Z$ does not belong to any of the Z_λ, then, for every generisation x' of x, either x' belongs to U, or $\dim(\mathscr{O}_{X,x'}) \geqslant 2$, or $\dim(\mathscr{O}_{X,x'}) = 1$ and, as x' does not belong to any of the Z_λ, $\overline{\{x'\}}$ is necessarily an irreducible component of Z, of codimension 1 in X, distinct from the Z_λ, hence $\mathscr{O}_{X,x'}$ is regular by definition. We conclude that $\mathrm{U}_{\mathrm{R}_1}(X) = X - \bigcup Z_\lambda$; as the set of the Z_λ is locally finite in Z, $\bigcup Z_\lambda$ is closed, which finishes proving that $\mathrm{U}_{\mathrm{R}_1}(X)$ is open in X. ☐ **IV-2** | 168

6.13. Criteria for $\mathrm{Nor}(X)$ to be open

(6.13.1). Given a locally Noetherian prescheme X, we denote by $\mathrm{Nor}(X)$ the set of $x \in X$ where X is *normal*; this set contains $\mathrm{Reg}(X)$ and is contained in the (open) set of points x such that $\mathscr{O}_{X,x}$ is integral (i.e. the set of points x where X is reduced, and which belong to only a single irreducible component of X).

Proposition (6.13.2). — *Let X be a locally Noetherian prescheme, normalisé (**II**, 6.3.8), X' its normalisation. If the canonical morphism $f : X' \to X$ is finite,* $\mathrm{Nor}(X)$ *is open in X.*

Proof. Indeed, on a $X' = \mathrm{Spec}(f_*(\mathscr{O}_{X'}))$ and $f_*(\mathscr{O}_{X'})$ is a coherent $\mathscr{O}_X$-Algebra by hypothesis (**II**, 6.1.3). To say that X is normal at a point x means that the canonical homomorphism $\mathscr{O}_x \to (f_*(\mathscr{O}_{X'}))_x$ is bijective (**II**, 6.3.4); but the set of such points is open since $\mathscr{O}_X$ and $f_*(\mathscr{O}_{X'})$ are coherent ($\mathbf{0_I}$, 5.2.7). ☐

Corollary (6.13.3). — *If A is an integral Noetherian* Japanese *ring,* $\mathrm{Nor}(\mathrm{Spec}(A))$ *is open in* $\mathrm{Spec}(A)$.

Proposition (6.13.4). — *Let X be a locally Noetherian prescheme. For* $\mathrm{Nor}(X)$ *to be open, it is necessary and sufficient that every maximal point x of X such that X is reduced at the point x (which means that $x \in \mathrm{Nor}(X)$) be interior to* $\mathrm{Nor}(X)$.

Proof. It suffices to prove the sufficiency of the condition. By virtue of the remark made in (6.13.1), we can restrict (by replacing X by an open subset of X in which X is reduced) to the case where X is *reduced*, i.e. suppose the maximal points of X belong to $\mathrm{U}_{\mathrm{S}_1}(X)$ and to $\mathrm{U}_{\mathrm{R}_1}(X)$; as $\mathrm{Nor}(X) = \mathrm{U}_{\mathrm{S}_1}(X) \cap \mathrm{U}_{\mathrm{R}_1}(X)$ by Serre's criterion (**0**, 8.6), we deduce from (6.11.7) and (6.12.10) that these two sets are open; hence $\mathrm{Nor}(X)$ is open in X. ☐

Corollary (6.13.5). — *If X is such that* $\mathrm{Reg}(X)$ *is open in X, then* $\mathrm{Nor}(X)$ *is open in X.*

Proof. Indeed, every maximal point x is reduced (hence $\mathscr{O}_{X,x}$ is a field), hence belongs to $\mathrm{Reg}(X)$, hence is interior to $\mathrm{Reg}(X)$ by hypothesis, and *a fortiori* is interior to $\mathrm{Nor}(X)$. ☐

Proposition (6.13.6). — *Let A be an integral Noetherian ring, K its field of fractions, K' a finite extension of K, A' the integral closure of A in K'. For A' to be a finite A-algebra, it is necessary and sufficient that the two following conditions be satisfied:* **IV-2** | 169

(i) *There exists $f \neq 0$ in A such that the integral closure A'_f of the ring of fractions A_f is a finite A_f-algebra.*
(ii) *For every prime ideal $\mathfrak{p} \in \mathrm{Spec}(A)$, the integral closure $A'_\mathfrak{p}$ of the local ring $A_\mathfrak{p}$ is a finite $A_\mathfrak{p}$-algebra.*

Proof. The conditions are necessary, for if S is any multiplicative subset of A, $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K' and is by hypothesis a $S^{-1}A$-module of finite type. To see that the conditions are sufficient, consider the increasing filtrant family (B_λ) of sub-A-algebras of K' which are finite A-algebras and have K' as field of fractions. Set $Y = \mathrm{Spec}(A)$, $X_\lambda = \mathrm{Spec}(B_\lambda)$, and denote by u_λ the morphism $X_\lambda \to Y$, and let $S_\lambda = X_\lambda - \mathrm{Nor}(X_\lambda)$, $T_\lambda = u_\lambda(S_\lambda)$. We can restrict to those B_λ which contain a finite system of generators of the A-module A'_f; this ensures that $(B_\lambda)_f = A'_f$ for all λ, or equivalently that $u_\lambda^{-1}(\mathrm{D}(f))$ is contained in $\mathrm{Nor}(X_\lambda)$. By (6.13.2), S_λ is therefore *closed* in X_λ, and as u_λ is a finite morphism, T_λ is *closed* in Y. But for every $\mathfrak{p} \in \mathrm{Spec}(A)$, there exists by hypothesis (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and hence all the points of X_λ above $\mathfrak{p}$ belong to $\mathrm{Nor}(X_\lambda)$; in other words on a $\bigcap_\lambda T_\lambda = \varnothing$; as Y is Noetherian and the T_λ are closed, there exists λ_0 such that $T_{\lambda_0} = \varnothing$, hence $B_{\lambda_0} = A'$. Q.E.D. ☐

Proposition (6.13.7). — *Let A be a Noetherian ring, $X = \operatorname{Spec}(A)$. The following conditions are equivalent:*

a) *For every prescheme X' locally of finite type over X, $\operatorname{Nor}(X')$ is open in X'.*
b) *For every finite integral A-algebra A', $\operatorname{Nor}(\operatorname{Spec}(A'))$ is open in $\operatorname{Spec}(A')$.*
c) *For every prime ideal $\mathfrak{p}$ of A and every finite radiciel extension K' of the field of fractions K of $A/\mathfrak{p}$, there exists a sub-A-algebra A' of K', finite over A, having K' as field of fractions and such that $\operatorname{Nor}(\operatorname{Spec}(A'))$ is open in $\operatorname{Spec}(A')$.*

PROOF. The fact that a) implies b) and that b) implies c) is proved as in (6.12.4). To show that c) implies a), by (6.13.2), we are reduced as in (6.12.4) to proving that if A' is an integral A-algebra of finite type, the generic point of $\operatorname{Spec}(A')$ is interior to $\operatorname{Nor}(\operatorname{Spec}(A'))$. We proceed as in (6.12.4) by first proving the following: □

Lemma (6.13.7.1). — *Let A be an integral Noetherian ring, A' an integral A-algebra of finite type, containing A and such that the field of fractions K' of A' is a separable extension of the field of fractions K of A. If $\operatorname{Nor}(\operatorname{Spec}(A))$ is open in $\operatorname{Spec}(A)$, then $\operatorname{Nor}(\operatorname{Spec}(A'))$ is open in $\operatorname{Spec}(A')$.*

PROOF. We must prove (by (6.13.2)) that the generic point of $\operatorname{Spec}(A')$ is interior to $\operatorname{Nor}(\operatorname{Spec}(A'))$. The proof follows the same approach as that of (6.12.4), I), and we preserve the notations. The proof of (6.12.4), I) shows (by replacing A by an A_f with $f \neq 0$) that the fibres $g^{-1}(z)$ of the morphism $g : X' \to X$ are regular and *a fortiori* normal. Moreover g is flat and X is normal, hence (**0**, 4.2), (ii)) X' is normal. This lemma being established, we pass to the general case as in (6.12.4), II), whose notations we still preserve; applying hypothesis c), we see each time that $\operatorname{Nor}(X'_1)$ is open and we are reduced as before to the case where X' is normal and $g : X' \to X'$ flat and surjective; we conclude each time that X' is normal with the aid of (**0**, 4.2), (i)). □

6.14. Base change and integral closure

Proposition (6.14.1). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \to Y$ a* normal *morphism (6.8.1). Then, for every normal Y-prescheme X, the prescheme $X' = X \times_Y Y'$ is normal.*

PROOF SKETCH. We note that in this statement we do not suppose X locally Noetherian. The proof proceeds in several stages. □

Lemma (6.14.1.1). — *Let R be a ring that is a finite direct product of a finite number of fields.*

(i) *For a sub-ring A of R having R as its total ring of fractions to be normal, it is necessary and sufficient that it be integrally closed in R.*
(ii) *Let (A_λ) be a family of normal sub-rings of R; if $A = \bigcap_\lambda A_\lambda$ admits R as total ring of fractions, A is normal.*

PROOF. (i) As A is a sub-ring of R, $A_\mathfrak{p}$ is a sub-ring of $R_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A; moreover $R_\mathfrak{p}$ is a ring of fractions of $A_\mathfrak{p}$, and is necessarily a finite direct product of fields, hence every element which is not a zero divisor of 0 in $R_\mathfrak{p}$ is invertible; this proves that $R_\mathfrak{p}$ is the total ring of fractions of $A_\mathfrak{p}$. If A is integrally closed in R, $A_\mathfrak{p}$ is hence integrally closed in $R_\mathfrak{p}$; but if $R_\mathfrak{p}$ is a finite direct product of 2 fields at least, the integral closure of $A_\mathfrak{p}$ in $R_\mathfrak{p}$ is a direct product of 2 rings at least which are not reduced to 0, which is absurd since $A_\mathfrak{p}$ is a local ring; hence $R_\mathfrak{p}$ is necessarily a field and $A_\mathfrak{p}$ is integral and integrally closed, which by definition means that A is normal. Conversely, if A is normal, $R_\mathfrak{p}$, total ring of fractions of an integral ring $A_\mathfrak{p}$, is a field and $A_\mathfrak{p}$ is integrally closed in $R_\mathfrak{p}$; if $x \in R$ is an integer over A, its image in each $R_\mathfrak{p}$ belongs to $A_\mathfrak{p}$, hence $x \in A$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 1, cor. 1 du th. 1), and hence A is integrally closed in R.

(ii) As $A \subset A_\lambda \subset R$ for each λ, R is also the total ring of fractions of each A_λ. In view of the characterisation of normal rings having R as total ring of fractions given in (i), assertion (ii) follows from Bourbaki, *Alg. comm.*, chap. V, § 1, n° 3, prop. 12. □

The proof of (6.14.1) proceeds in several stages:

I) *Reduction to the case where $Y = \operatorname{Spec}(A)$, $Y' = \operatorname{Spec}(A')$, $X = \operatorname{Spec}(B)$, A, A' are local Noetherian rings, A integral, B the integral closure of A.* — We must prove that for every $x' \in X'$, the local ring $\mathscr{O}_{X',x'}$ is integral and integrally closed; let x, y, y' be the images of x' in X, Y, Y' respectively; if we set $A = \mathscr{O}_{Y,y}$, $A' = \mathscr{O}_{Y',y'}$, $B = \mathscr{O}_{X,x}$, the rings $\mathscr{O}_{X',x'}$, for x, y, y' fixed, are the localisations of

$B' = B \otimes_A A'$ at the prime ideals of B' (**I**, 3.6.5), and it suffices to prove that B' is a *normal* ring. Note moreover that $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a normal morphism by definition (6.8.1) and **I**, 3.6.5); one is hence reduced to the case of local schemes. Denote by (B_α) the family of *integral closures* of the sub-A-algebras *of finite type* of B; it is clear that B is the union of the increasing filtrant family of the B_α. As $\varinjlim$ commutes with the tensor product, B' is hence isomorphic to $\varinjlim B'_\alpha$, where we have set $B'_\alpha = B_\alpha \otimes_A A'$. If we prove that C'_α is the integral closure of B'_α in C', then it will follow that C' is the integral closure of A' in B'. We can hence restrict to the case where B is a A-algebra of finite type; as B is supposed *reduced*, it is sufficient to prove the claim when B is *integral* and is the *integral closure* of A.

II)-V) The remaining reductions and the completion of the proof are technically involved. In II), one reduces to the case where A is a local integral ring of dimension 1, B a discrete valuation ring, and the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *radiciel*. In III), one establishes a key lemma:

Lemma (6.14.1.2). — *Let A be a local Noetherian integral ring of dimension 1, A' a local Noetherian ring, $A \to A'$ a local homomorphism such that the corresponding morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is* normal. *Let K be the field of fractions of A, $\mathfrak{m}$ the maximal ideal of A, $k = A/\mathfrak{m}$ its residue field.*

(i) *If $\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$ are the minimal ideals of A', $K' = K \otimes_A A'$ is a finite direct product of integral and integrally closed rings $K'_j \supset A'/\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$, K_j having for field of fractions the field of fractions L'_j of $A'/\mathfrak{q}'_j$, so that the total ring of fractions L' of A' identifies to the direct product of the L'_j and A' to a sub-ring of K'.*

(ii) *The ideal $\mathfrak{p}' = \mathfrak{m}A'$ of A' is prime; the ring $A'_{\mathfrak{p}'}$ is of dimension 1 and identifies to a sub-product of the fields L'_j for the indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$.*

(iii) *If $\overline{A}'_{\mathfrak{p}'}$ denotes the sub-ring of L', product of $A'_{\mathfrak{p}'}$ and of the L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$, one has*

$$A' = K' \cap \overline{A}'_{\mathfrak{p}'}.$$

Lemma (6.14.1.4). — *Let R be a ring, $\mathfrak{a}$ and $\mathfrak{b}$ two ideals of R; if $R/\mathfrak{a}$ and $R/\mathfrak{b}$ are Noetherian, so is $R/(\mathfrak{a} \cap \mathfrak{b})$.*

PROOF. Indeed, let $\mathfrak{c}$ be an ideal of R such that $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{c}$; we have therefore $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{c} \subset \mathfrak{c}$; but $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{c})$ is an R-module isomorphic to $(\mathfrak{a} + \mathfrak{c})/\mathfrak{a}$, hence an ideal of $R/\mathfrak{a}$, and is therefore of finite type; on the other hand $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is a sub-R-module of $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$, and this last is isomorphic to $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$, hence is of finite type as ideal of $R/\mathfrak{b}$; hence $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is of finite type, and it is so finally for $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{b})$. □

The proof of (6.14.1) is completed using these lemmas along with the normality criteria of (6.8.2) and the results of (5.13.6). **IV-2** | 173

Corollary (6.14.2). — *Let k be a field, X a normal k-prescheme (not necessarily locally Noetherian). Then, for every separable extension k' of k, $X_{(k')} = X \otimes_k k'$ is normal.*

PROOF. We know indeed by (6.7.6) that the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is then normal. □

Corollary (6.14.3). — *Let k be a field, X, Y two integral k-preschemes, normal, whose fields of rational functions $K = R(X)$, $L = R(Y)$ are separable extensions of k. Then the prescheme $X \times_k Y$ is normal.*

PROOF. The question being local on X and Y, we can suppose that $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, where A and B are two integral and integrally closed rings, of fields of fractions K and L respectively; suppose first that A is a k-algebra of finite type, hence K a finite type extension of k. By flatness, $A \otimes_k B$ is a sub-k-algebra of $K \otimes_k L$; but $K \otimes_k L$ is a Noetherian normal ring (by (6.7.4), I) or (6.14.2)), hence a finite direct product of integral rings C_i, so that if E_i is the field of fractions of C_i, the direct product E of the E_i is the total ring of fractions of $K \otimes_k L$ and also of $A \otimes_k B$. This shows that $A \otimes_k L$ and $K \otimes_k B$ are normal (as they are intersections of the C_i with appropriate direct factors), and it follows from (6.14.2) that $A \otimes_k B = (A \otimes_k L) \cap (K \otimes_k B)$, hence $A \otimes_k B$ is normal by (6.14.1.1).

In the general case, A is the union of the increasing filtrant family of its sub-k-algebras of finite type A_α, and $A \otimes_k B$ is the limit of the $A_\alpha \otimes_k B$. It follows from (**0**, 13.6) that to prove that every irreducible component of $\mathrm{Spec}(A_\beta \otimes_k B)$ dominates an irreducible component of $\mathrm{Spec}(A'_\alpha)$ for $\alpha \leqslant \beta$, it suffices to use (**I**, 3.7), (ii) and the fact that A_α and A_β are integral rings. □

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Proposition (6.14.4). — *Let A be a Noetherian ring, A' a Noetherian A-algebra such that the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ *is* normal. *Let B be an A-algebra, C the integral closure of A in B; set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, C' identifying to a sub-ring of B'; then C' is the integral closure of A' in B'.*

PROOF. The proof proceeds in several steps:

I) *Reduction to the case where B is reduced.* — Set $B_0 = B_{\mathrm{red}} = B/\mathfrak{N}$, where $\mathfrak{N}$ is the nilradical of B, and let C_0 be the integral closure of A in B_0. By (6.14.4.1) below, the inverse image C of C_0 by the canonical homomorphism $\varphi : B \to B_0$ is the integral closure of A in B.

II) *Reduction to the case where B is an integral A-algebra of finite type containing A.* — Let (B_α) be the increasing filtrant family of sub-A-algebras of finite type of B, and let C_α be the integral closure of A in B_α. It follows from the definition of integral closure that C is the union of the C_α; if we set $B'_\alpha = B_\alpha \otimes_A A'$, $C'_\alpha = C_\alpha \otimes_A A'$, C'_α is contained in B'_α and it suffices to show that C'_α is the integral closure of A' in B'_α.

III-V) The remaining cases reduce eventually to the case where A is a field and B is algebraically closed over A, where the conclusion follows from (6.14.1). □

Lemma (6.14.4.1). — *Let A be a ring, B an A-algebra, $B_0 = B/\mathfrak{n}$ the quotient of B by a nilideal. If C_0 is the integral closure of A in B_0, the inverse image C of C_0 by the canonical homomorphism $\varphi : B \to B_0$ is the integral closure of A in B.*

PROOF. Indeed, if $x \in B$ is such that $\varphi(x)$ satisfies a monic integral dependence equation with coefficients in A, we deduce that x satisfies a relation of the form $x^n + a_1 x^{n-1} + \ldots + a_n \in \mathfrak{n}$ with $a_i \in A$, whence, raising to a sufficiently large power, we obtain a monic integral dependence equation for x with coefficients in A. We have therefore an exact sequence of A-modules $0 \to C \to B \to B_0/C_0$, hence, by flatness, the exact sequence $0 \to C' \to B' \to B'_0/C'_0$, where we have set $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$. If we prove that C'_0 is the integral closure of A' in B'_0, the lemma (6.14.4.1) will show that C' is the integral closure of A' in B'. □

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Corollary (6.14.5). — *Let A be a Noetherian ring, A' a Noetherian A-algebra such that the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ *is* normal. *Let B, C be two A-algebras, $\varphi : B \to C$ an A-homomorphism making C a B-algebra, D the integral closure of B in C; if we set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, $D' = D \otimes_A A'$, D' is the integral closure of B' in C'.*

PROOF. Let (B_λ) be the increasing filtrant family of sub-A-algebras of finite type of B, and set $B'_\lambda = B'_\lambda \otimes_A A'$, so that B' is the union of the filtrant increasing family of the sub-A'-algebras of finite type B'_λ. Let D_λ be the integral closure of B_λ in C, and set $D'_\lambda = D_\lambda \otimes_A A'$; if we prove that D'_λ is the integral closure of B'_λ in C', it will follow that D' is the integral closure of B' in C'. But B_λ and B'_λ are Noetherian and the morphism $\mathrm{Spec}(B'_\lambda) \to \mathrm{Spec}(B_\lambda)$ is normal (6.8.2); we can hence apply (6.14.4) by replacing A, A' and B by B_λ, B'_λ and C respectively, hence the corollary. □

One can for example apply (6.14.5) when A is a local ring and A' its completion $\widehat{A}$, a *local excellent ring*, since in this case $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a regular morphism (7.8.2).

6.15. Geometrically unibranch preschemes

(6.15.1). We say that a prescheme X is *unibranch* (resp. *geometrically unibranch*) at a point x, or that the point x is *unibranch* (resp. *geometrically unibranch*) in X if the local ring $\mathcal{O}_{X,x}$ is unibranch (resp. geometrically unibranch) (**0**, 23.2.1); we say that X is *unibranch* (resp. *geometrically unibranch*) if it is so at every point. It follows from this definition that, for X to be unibranch (resp. geometrically unibranch) at a point, it is necessary and sufficient that X_{red} be so at this point.

(6.15.2). It should be noted that with the definitions of (6.15.1), a *local* ring A can be unibranch (resp. geometrically unibranch) *without* $\operatorname{Spec}(A)$ being so; in other words, it can happen that there exist prime ideals $\mathfrak{p}$ of A such that $A_{\mathfrak{p}}$ is not unibranch; this also amounts to saying that the notion of a geometrically unibranch point on a prescheme *is not stable by generisation*. We shall see this in the following example.

(6.15.3). We say, to abbreviate, that a morphism $f : X \to Y$ is *radiciel at a point* $y \in Y$, if $f^{-1}(y)$ is empty or reduced to a single point x and if $k(x)$ is a radiciel extension of $k(y)$. It amounts to the same to say that f is radiciel at all the points of Y, or that f is radiciel (**I**, 3.5.8). If $g : f^{-1}(y) \to \operatorname{Spec}(k(y))$ is the morphism obtained from f by the base change $\operatorname{Spec}(k(y)) \to Y$, it amounts to the same to say that f is radiciel at the point $y \in Y$, or that g is radiciel.

Lemma (6.15.3.1). — (i) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If f is radiciel at a point y and g radiciel at the point $z = g(y)$, then $g \circ f$ is radiciel at the point $g(y)$. The converse is true if f is surjective.*

(ii) *Let $f : X \to Y$, $h : Y' \to Y$ be two morphisms, and let $f' = f_{(Y')} : X \times_Y Y' \to Y'$. Let $y' \in Y'$ and $y = h(y')$; then for f to be radiciel at the point y, it is necessary and sufficient that f' be radiciel at the point y'.*

Proof. (i) If f is radiciel at y and g radiciel at $z = g(y)$, $g^{-1}(z)$ is reduced to y and $f^{-1}(g^{-1}(z)) = f^{-1}(y)$ is empty or reduced to a single point x; moreover $k(x)$ is radiciel extension of $k(z)$, hence $k(x)$ is radiciel extension of $k(z)$. Conversely, if f is surjective and $g \circ f$ is radiciel at $z = g(y)$, $g^{-1}(z)$ is reduced to at most two points; but $f^{-1}(g^{-1}(z))$ would have at least two distinct points, a contradiction; furthermore, $f^{-1}(y) = f^{-1}(g^{-1}(z))$ is reduced to a single point x, and by hypothesis $k(z) \subset k(y) \subset k(x)$ with $k(x)$ radiciel over $k(z)$, hence $k(y)$ radiciel over $k(z)$ and $k(x)$ radiciel over $k(y)$.

(ii) Let $g : f^{-1}(y) \to \operatorname{Spec}(k(y))$, $g' : f'^{-1}(y') \to \operatorname{Spec}(k(y'))$ be the fibre morphisms of f and f' respectively; as $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (**I**, 3.6.4), g' is the base change of g, and the assertion follows from (**I**, 6.1), (v)). ◻

(6.15.4). Generalising the definition given in (**I**, 2.2.9), for two reduced preschemes X, Y, we say that a morphism $f : X \to Y$ is *birational* if the restriction of f to the set of maximal points of X is a bijection onto the set of maximal points of Y, and if, for every maximal point y of Y, the morphism $X \times_Y \operatorname{Spec}(\mathcal{O}_y) \to \operatorname{Spec}(\mathcal{O}_y)$ induced by f is an isomorphism (in other words, if the fibre $f^{-1}(y)$ is reduced to a single point x (maximal in X) and if the homomorphism $k(y) \to k(x)$ induced by f is bijective); this notion reduces to that of (**I**, 2.2.9) when X and Y have only a *finite* number of irreducible components.

Lemma (6.15.4.1). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a* flat *morphism. If f_{red} is birational, so is f'_{red}.*

Proof. Indeed, if y' is a maximal point of Y', we know (**I**, 3.4), (ii)) that $y = g(y')$ is a maximal point of Y; there exists a unique point x of X in $f^{-1}(y)$ by hypothesis; since moreover x is maximal in X and $k(x) = k(y)$. It follows therefore by (**I**, 3.4.9) that $f'^{-1}(y')$ is reduced to a single point x' and that $k(x') = k(y')$. We conclude by the remark that, by (**I**, 3.7), (ii)), every irreducible component of X' dominates one irreducible component of Y'. ◻

Proposition (6.15.5). — *Let $f : X \to Y$ be a morphism such that f_{red} is entire and birational (6.15.4), hence surjective.*

(i) *For Y to be geometrically unibranch at a point y, it is necessary and sufficient that f be radiciel at the point y and that X be geometrically unibranch at the unique point x of $f^{-1}(y)$.*

(ii) *For Y to be geometrically unibranch, it is necessary and sufficient that X be geometrically unibranch and that f be radiciel.*

PROOF. We can evidently restrict to the case where X and Y are *reduced*; the fact that f is surjective results from f being a closed morphism (**II**, 6.1.10) and from $f(X)$ containing all the maximal points of Y. This immediately shows that (i) implies (ii). Using (**I**, 3.6.5) and (**II**, 6.1.5), we can suppose, to prove (i), that $Y = \mathrm{Spec}(\mathscr{O}_{X,y})$, and similarly $X = \mathrm{Spec}(B)$. If A is geometrically unibranch, it is *integral*, and hence B is also integral (since f is birational). Conversely, if f is radiciel at y and X is geometrically unibranch at x, B has a single maximal ideal (since every ideal of A lies above the maximal of A and A is local), hence B is a local ring, *integral*. As f is dominant by hypothesis and A is reduced (**I**, 1.2.7), we conclude that A is also *integral*. Thus, to prove (i), it amounts to showing that A is integral and geometrically unibranch au point x means B is integral and B is the integral closure, i.e. the morphism $g : Y \to X$ is radiciel at the point x (6.15.3). □

Proposition (6.15.6). — *Let k be a field, X a k-prescheme. If X is normal, then, for every extension k' of k, $X' = X \otimes_k k'$ is geometrically unibranch.*

PROOF. We know that k' is the algebraic extension of a pure extension k_0 of k, and if k'' is the largest separable extension of k_0 contained in k', k' is a radiciel extension of k''. We know (6.14.2) that $X \otimes_k k''$ is normal; as $X'' \otimes_{k''} k' = X'' \otimes_{k''} k'$, we see that we can restrict to the case where the extension k' of k is *radiciel*. Moreover (**I**, 3.6.5), we can suppose that $X = \mathrm{Spec}(A)$, where A is a local integral and integrally closed ring (since X is normal). The projection morphism $f : X' \to X$ is a universal homeomorphism, since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism (**I**, 4.5); as $X' = \mathrm{Spec}(A')$ where $A' = A \otimes_k k'$, we see that A' is a local ring whose nilradical $\mathfrak{N}'$ is the unique minimal prime ideal, hence $X'_{\mathrm{red}} = \mathrm{Spec}(A'_0)$, where $A'_0 = A'/\mathfrak{N}'$ is a local integral ring; moreover, if K is the field of fractions of A, the field of fractions K'_0 of A'_0 is radiciel over K, since the morphism f is radiciel. As A'_0 is entire over A, its integral closure B is also the integral closure of A in K'_0; moreover, every element of K'_0 whose p^m-th power (for some m) belongs to A belongs to K (for p the characteristic exponent of k); in particular B is an integral closure of A in K'_0. But as A is integrally closed and K'_0 is radiciel over K, every element of K'_0 integral over A belongs to A (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 3, lemme 3), and *a fortiori* A'_0 is geometrically unibranch, and it is so also of X'. □

Proposition (6.15.7). — *Let k be a field, X a k-prescheme, k' an extension of k, $X' = X \otimes_k k'$. Let x' be a point of X', x its projection in X. For X to be geometrically unibranch at the point x, it is necessary and sufficient that X' be geometrically unibranch at the point x'. For X to be geometrically unibranch, it is necessary and sufficient that X' be so.*

PROOF. The second assertion follows from the first and from the fact that the projection $f : X' \to X$ is surjective. To demonstrate the first assertion, we can, by (**I**, 5.1.8), restrict to the case where X is reduced, and by (**I**, 3.6.5), to the case where $X = \mathrm{Spec}(A)$ (with $A = \mathscr{O}_{X,x}$) is a local scheme. Note that $A' = \mathscr{O}_{X',x'}$ is by hypothesis an A-module faithfully flat, hence containing A, and hence the hypothesis that X is geometrically unibranch at x, or the hypothesis that X' is geometrically unibranch at x', both imply that A is a local integral ring (since A, being reduced, is also isomorphic to a sub-ring of A'_{red}). Let K be the field of fractions of A, B the integral closure of A, and set $Y = \mathrm{Spec}(B)$; it amounts to saying that X is geometrically unibranch at the point x, or that the morphism $g : Y \to X$ is radiciel at the point x (6.15.3). Let $Y' = Y \otimes_k k' = Y \times_X X'$, so that we have the commutative diagram:

$$\begin{tikzcd} Y & Y' \arrow[l, "h"'] \\ X \arrow[u, "g"] & X' \arrow[l, "f"'] \arrow[u, "g'"'] \end{tikzcd}$$

Note that f is a flat morphism; hence by (6.15.4.1) the entire morphism g'_{red} is birational. Moreover, as Y is normal, Y' is geometrically unibranch by (6.15.6). As f is entire and birational, f is radiciel at this point (6.15.5); on the other hand, x is the projection of x' in X and it is equivalent to say that g is radiciel at the point x or that g' is radiciel at the point x' (6.15.3.1); finally, for X to be geometrically unibranch at x, it suffices that g be radiciel at this point, hence the proposition. □

Lemma (6.15.8). — *Let k be a* separably closed *field (in other words, such that the algebraic closure of k is* radiciel *over k), X a k-prescheme locally of finite type over k, x a closed point of X. If X is unibranch at the point x, it is geometrically unibranch at this point.*

PROOF. Indeed, we know (**I**, 6.4.2) that $k(x)$ is an algebraic extension of k; as the integral closure of $(\mathscr{O}_x)_{\mathrm{red}}$ (integral closure which is by hypothesis a local ring) is an extension *algebraic* of $k(x)$, it is therefore an extension *radiciel* of $k(x)$, hence also of $k(x)$ by hypothesis, hence also of $k(x)$. □

Corollary (6.15.9). — *Let k be a field, X a k-prescheme locally of finite type over k, k' a separably closed extension of k (in other words, such that the algebraic closure of k' is radiciel over k'); then, for X to be geometrically unibranch, it is necessary and sufficient that $X' = X \otimes_k k'$ be unibranch.*

PROOF. By (6.15.7), we are reduced to proving that if X is unibranch and k separably closed, then X is geometrically unibranch. But X is geometrically unibranch at its closed points (6.15.8); we conclude from (10.4.9) that X is geometrically unibranch at all its points, relying on the fact (noted at the end of (6.15.2)) that the set of points where X is geometrically unibranch is constructible (the corollary (6.15.9) will of course not be used up to this point). □

Proposition (6.15.10). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \to Y$ a normal morphism (6.8.1), $f : X \to Y$ a morphism; set $X' = X \times_Y Y'$ and let $p : X' \to X$ be the canonical projection, x' a point of X'. Then, if X is reduced (resp. geometrically unibranch) at the point $x = p(x')$, X' is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point x'.*

PROOF. By (**I**, 3.6.5), we can restrict to the case where $Y = \mathrm{Spec}(A)$, $Y' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, where A, A' and B are local Noetherian rings, the homomorphisms $A \to A'$ and $A \to B$ being local, and x being the closed point of X; it suffices to prove that $B' = B \otimes_A A'$ is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch at its points). Suppose first B reduced only; B is the union of the increasing filtrant family of its sub-A-algebras of finite type B_λ, and by flatness, B' is the union of the filtrant increasing family of the sub-A'-algebras of finite type $B'_\lambda = B_\lambda \otimes_A A'$. But, the morphism $\mathrm{Spec}(B'_\lambda) \to \mathrm{Spec}(B_\lambda)$ is normal (6.8.2), and *a fortiori* reduced; the fact that B'_λ is reduced follows then; one concludes that B' is itself reduced by (**0**, 13.2).

Suppose now B geometrically unibranch; by (**I**, 5.1.8), we can moreover suppose B reduced, hence integral. Let C be its integral closure, which is by hypothesis a local ring; set $Z = \mathrm{Spec}(C)$, $C' = C \otimes_A A'$, $Z' = \mathrm{Spec}(C') = Z \times_X X'$, so that we have the commutative diagram:

$$\begin{tikzcd} Z & Z' \arrow[l, "h"'] \\ X \arrow[u, "l"] & X' \arrow[l, "p"'] \arrow[u, "l'"'] \end{tikzcd}$$

By the first part of the argument, X' is *reduced*; moreover, as $Z' = Z \times_Y Y'$, it follows from (6.14.1) that Z' is *normal* (and *a fortiori* geometrically unibranch). As f is entire and birational, so is f' by (6.15.4.1), since p is flat. Finally, the hypothesis that X is geometrically unibranch at x implies that f is radiciel at this point (6.15.5); we conclude by (6.15.3.1) that f' is radiciel at every point $x' \in p^{-1}(x)$, and it follows from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. □

Remarks (6.15.11). —
(i) In the proof of (6.15.10), one cannot avoid appealing to (6.14.1), even when $X = Y$, as one makes use of the integral closure of the ring B, which is not necessarily a Noetherian ring even if B is.
(ii) The example (**0**, 5.2), (ii)) shows that in (6.15.10), one cannot replace "geometrically unibranch" by "integral" in the statement, even if the residue field $k(x)$ is algebraically closed and the morphism g is étale.

§7. Relations between a local Noetherian ring and its completion. Excellent rings

7.1. Formal equidimensionality and formally catenary rings

Definition (7.1.1). — We say that a local Noetherian ring A is *formally equidimensional* if its completion $\widehat{A}$ is equidimensional ($\mathbf{0}$, 16.1.4).

Proposition (7.1.2). — *Let A be a local Noetherian ring, $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ its minimal prime ideals. For A to be formally equidimensional, it is necessary and sufficient that the $A/\mathfrak{p}_i$ be equidimensional (in other words that the $A/\mathfrak{p}_i$ have the same dimension).*

PROOF. Indeed, this follows from the fact that for each of the $\mathfrak{p}_i$, $\mathfrak{p}' \cap A$ contains one of the $\mathfrak{p}_i$, hence $\mathfrak{p}'$ contains one of the $\mathfrak{p}_i\widehat{A}$, and $\widehat{A}/\mathfrak{p}_i\widehat{A} = (A/\mathfrak{p}_i) \otimes_A \widehat{A}$ is the completion of the local ring $A/\mathfrak{p}_i$ ($\mathbf{0_I}$, 7.3.3), hence a maximal chain of prime ideals of $\widehat{A}$ identifies therefore canonically to a chain of maximal prime ideals of one of the $\widehat{A}/\mathfrak{p}_i\widehat{A}$, and reciprocally, hence the conclusion. □

Proposition (7.1.3). — *Let A, A' be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A, $\varphi : A \to A'$ a local homomorphism; suppose that A' is a flat A-module, and that A' is equidimensional and catenary. Then:*

(i) *A is equidimensional and catenary.*

(ii) *Suppose moreover that $\mathfrak{m}A'$ is an ideal of definition of A'. Then, if $\mathfrak{a}$ is an ideal of A, for $A'/\mathfrak{a}A'$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be so. In particular, for every prime ideal $\mathfrak{p}$ of A, $A'/\mathfrak{p}A'$ is formally equidimensional.*

PROOF. Let us first prove the second assertion of (ii). Set $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(A')$, $Y = V(\mathfrak{p}) = \operatorname{Spec}(A/\mathfrak{p})$, $Y' = V(\mathfrak{p}A') = \operatorname{Spec}(A'/\mathfrak{p}A')$; if $f : X' \to X$ is the morphism corresponding to φ, Y' is also the closed sub-prescheme $f^{-1}(Y)$ of X'. As $\mathfrak{m}A'$ is an ideal of definition of A', one has

$$\dim(X) = \dim(X') \tag{7.1.3.1}$$

by (6.1.3); similarly $\mathfrak{m}A'/\mathfrak{p}A'$ is an ideal of definition of $A'/\mathfrak{p}A'$ and Y' is flat over Y (**I**, 1.4), hence (6.1.3)

$$\dim(Y) = \dim(Y'). \tag{7.1.3.2}$$

Let y be the generic point of Y, Y'_i $(1 \leqslant i \leqslant r)$ the irreducible components of Y', y'_i the generic point of Y'_i $(1 \leqslant i \leqslant r)$; one has $f(y'_i) = y$ for all i (**I**, 3.4). On the other hand y'_i is a maximal point of the fibre $f^{-1}(y)$ ($\mathbf{0_I}$, 2.1.8), hence $\mathscr{O}_{X',y'_i} \otimes_{\mathscr{O}_{X,y}} k(y) = \mathscr{O}_{X',y'_i}/\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is of dimension 0, in other words $\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is an ideal of definition of $\mathscr{O}_{X',y'_i}$; one can again apply (6.1.3), which gives

$$\dim(\mathscr{O}_{X',y'_i}) = \dim(\mathscr{O}_{X,y}) \tag{7.1.3.3}$$

that is to say ($\mathbf{0}$, 1.2)

$$\operatorname{codim}(Y'_i, X') = \operatorname{codim}(Y, X) \tag{7.1.3.4}$$

for every i; by (7.1.3.1), (7.1.3.2) and (7.1.3.4) and the inequality $\dim(Y'_i) \leqslant \dim(Y')$, one deduces **IV-2** | 184

$$\dim(X) \leqslant \dim(Y) + \operatorname{codim}(Y, X) \tag{7.1.3.6}$$

hence the two sides of this inequality are equal by ($\mathbf{0}$, 14.2.2.2); moreover

$$\dim(Y'_i) = \dim(Y'_i) = \dim(Y) \tag{7.1.3.7}$$

for all i, which proves the second assertion of (ii).

Let us now demonstrate the first assertion of (ii). Let $\mathfrak{p}_j$ $(1 \leqslant j \leqslant n)$ be the minimal prime ideals of A among those containing $\mathfrak{a}$, and let $\mathfrak{p}'_{jh}$ $(1 \leqslant h \leqslant m_j)$ be the minimal prime ideals of A' among those containing $\mathfrak{p}_jA'$; we know then (**I**, 3.4) that the $\mathfrak{p}'_{jh}$ $(1 \leqslant j \leqslant n$, $1 \leqslant h \leqslant m_j$ for every $j)$ are also the minimal prime ideals of A' among those containing $\mathfrak{a}A'$. From what we have seen above, for every j, the dimensions of the rings $A'/\mathfrak{p}'_{jh}$ $(1 \leqslant h \leqslant m_j)$ are all equal and are equal to $\dim(A'/\mathfrak{p}_jA')$, which itself is equal to $\dim(A/\mathfrak{p}_j)$ by (7.1.3.2). To say that $A'/\mathfrak{a}A'$ is equidimensional means therefore that the dimensions of the $A/\mathfrak{p}_j$ are all equal, that is to say that $A/\mathfrak{a}$ is equidimensional.

Let us now prove (i). Let $\mathfrak{r}'$ be a minimal prime ideal of A' among those containing $\mathfrak{m}A'$, and let $A'' = A'_{\mathfrak{r}'}$; as A'' is a flat A'-module, it is also a flat A-module; moreover A'' is equidimensional and catenary ($\mathbf{0}$, 16.1.4), and $\mathfrak{m}A''$ is an ideal of definition of A''. One can hence restrict to the case where $\mathfrak{m}A'$ is an ideal of definition of A'. If then $\mathfrak{p}$ and $\mathfrak{q} \subset \mathfrak{p}$ are two prime ideals of A, one can apply (ii) to $A/\mathfrak{q}$ and to $A'/\mathfrak{q}A'$, which is equidimensional and flat over $A/\mathfrak{q}$; if $Z = \operatorname{Spec}(A/\mathfrak{q})$ and

$Y = \operatorname{Spec}(A/\mathfrak{p})$, one can apply (7.1.3.6), hence $\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z)$; moreover one can apply (7.1.3.6); as X is equicodimensional, these relations show that it is biequidimensional by (**0**, 14.3.3). □

Corollary (7.1.4). — *Let A be a local Noetherian ring formally equidimensional. Then:*

(i) *A is equidimensional and catenary (in other words,* biequidimensional*).*
(ii) *For every ideal $\mathfrak{a}$ of A, for $A/\mathfrak{a}$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be so. In particular, for every prime ideal $\mathfrak{p}$ of A, $A/\mathfrak{p}$ is formally equidimensional.*

PROOF. If $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}\widehat{A}$ is an ideal of definition of $\widehat{A}$. By hypothesis $\widehat{A}$ is equidimensional, and we know that it is catenary (**0**, 6.4); it therefore suffices to apply (7.1.3) to $A' = \widehat{A}$. □

Corollary (7.1.5). — *Let A be a local Noetherian ring such that there exists an A-module of finite type M which is an A-module of Cohen-Macaulay and for which* $\operatorname{Supp}(M) = \operatorname{Spec}(A)$ *(this is the case for example if A is a Cohen-Macaulay ring). Then A is formally equidimensional, hence (7.1.4) for every quotient ring B of A to be formally equidimensional, it is necessary and sufficient that it be equidimensional.*

PROOF. Indeed, $\widehat{M} = M \otimes_A \widehat{A}$ is a $\widehat{A}$-module of Cohen-Macaulay (**0**, 16.5.2), of support equal to $\operatorname{Spec}(\widehat{A})$; hence (**0**, 16.5.4), $\widehat{A}$ is equidimensional. □ **IV-2** | 185

Remark (7.1.6). — We have seen (6.3.8) that under the hypotheses of (7.1.5), for every quotient ring B of A, the fibres of the canonical morphism $\operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ are Cohen-Macaulay schemes; hence ((6.4.3) and (**0**, 7.5)), *for B to have no immersed associated prime ideals, it is necessary and sufficient that it be so for $\widehat{B}$.*

Lemma (7.1.7). — *Let A be a local Noetherian ring, $\mathfrak{p}_i$ $(1 \leqslant i \leqslant r)$ its minimal prime ideals. Suppose that each of the rings $A/\mathfrak{p}_i$ is formally equidimensional. Then, for every prime ideal $\mathfrak{q}$ of A and every i such that $\mathfrak{p}_i \subset \mathfrak{q}$, the ring $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ is formally equidimensional.*

PROOF. As $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ is the local ring of $A/\mathfrak{p}_i$ at the prime ideal $\mathfrak{q}/\mathfrak{p}_i$, we can restrict to the case where A is integral and formally equidimensional. Set $A' = \widehat{A}$, and let $\mathfrak{q}'$ be one of the minimal prime ideals of A' among those containing $\mathfrak{q}A'$; set $B = A_\mathfrak{q}$, $B' = A'_{\mathfrak{q}'}$. Set $C = \widehat{B}$, $C' = \widehat{B}'$; as B' is a B-module flat ($\mathbf{0_I}$, 6.3.2), we know that C' is a C-module flat (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 4); as B' is catenary (**0**, 6.4) and equidimensional by hypothesis (by (7.1.3), (i)), it is the same for $B' = A'_{\mathfrak{q}'}$ (**0**, 16.1.4); moreover, as A' is isomorphic to a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), it is the same of B' (**0**, 17.3.9); hence we conclude from (7.1.5) that C' is equidimensional. On the other hand C' is catenary (**0**, 6.4), hence C is equidimensional by (7.1.3), (i)). □

Theorem (7.1.8). — *Let A be a local Noetherian ring. The following conditions are equivalent:*

a) *Every integral quotient ring of A is formally equidimensional.*
b) *The quotient rings of A by its minimal prime ideals are formally equidimensional.*
c) *Every local ring B which is an A-algebra essentially of finite type* (**0**, 3.8) *and which is equidimensional, is formally equidimensional.*

PROOF. It is trivial that c) implies a) and that a) implies b). On the other hand, as every integral prime ideal of A contains a minimal prime ideal of A, b) implies a) by (7.1.4), (ii). It remains to see that a) implies c).

It suffices to prove that the quotients of B by its minimal prime ideals are formally equidimensional (7.1.2); as every quotient ring of B is still an A-algebra essentially of finite type (**0**, 3.9), one can in the first instance suppose B *integral*. If $\mathfrak{q}$ is the inverse image in A of the maximal ideal of B, we know that B is also an $A_\mathfrak{q}$-algebra essentially of finite type (**0**, 3.10), and it follows from a) and from (7.1.7) that every integral quotient of $A_\mathfrak{q}$ is formally equidimensional; one can hence suppose that the homomorphism $A \to B$ is a *local* homomorphism. The kernel $\mathfrak{r}$ of this homomorphism is then a prime ideal of A, and by a), every integral quotient of $A/\mathfrak{r}$ is formally equidimensional; as B is an $(A/\mathfrak{r})$-algebra essentially of finite type, one sees that one can also suppose that A is *integral* and is a sub-ring of B.

Set $A' = \widehat{A}$, and $C' = A'[T_1, \dots, T_n] = C \otimes_A A'$; there is a unique prime ideal $\mathfrak{p}'$ of C' above the maximal ideal $\mathfrak{m}A'$ of A', hence above $\mathfrak{p}$; set $B' = C'_{\mathfrak{p}'}$; the homomorphism $B \to B'$ is local and makes B' a B-module flat. We know then that $\widehat{B}'$ is *equidimensional*, from which we will deduce that $\widehat{B}$ is the same, which will complete the proof. **IV-2 | 186**

Indeed, A' is a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), hence it is the same for B' (**0**, 17.3.9); by (7.1.5), in order to show that B' is *equidimensional*, it suffices to show that C' is *equidimensional*, since C' is a quotient of a regular ring. But the minimal prime ideals of C' are the ideals $\mathfrak{q}'C'$, where $\mathfrak{q}'$ ranges over the minimal prime ideals of A' (**0**, 5.3), and one has $C'/\mathfrak{q}'C' = (A'/\mathfrak{q}')[T_1, \dots, T_n]$. As A' is equidimensional by hypothesis (since A is integral), it is the same of C' by (**0**, 5.4). Q.E.D. □

Definition (7.1.9). — When the equivalent conditions of (7.1.8) are satisfied, we say that A is a *formally catenary* ring. For a local Noetherian ring, to be formally equidimensional or formally catenary and equidimensional is therefore the same, by (7.1.2).

Proposition (7.1.10). — *Every quotient local ring of a local Cohen-Macaulay ring is formally catenary.*

PROOF. This follows immediately from (7.1.8) and from (7.1.5) applied to the integral quotient rings of A. □

Proposition (7.1.11). — (i) *A local Noetherian ring that is formally catenary is universally catenary.*

(ii) *If A is a local Noetherian ring formally catenary, every local ring B which is an A-algebra essentially of finite type is formally catenary.*

PROOF. Assertion (ii) follows immediately from condition c) of (7.1.8) and from the fact that if B is a local ring which is an A-algebra essentially of finite type, C is also an A-algebra essentially of finite type (**0**, 3.9). To prove (i), note first that, by (7.1.8), b), a local Noetherian ring formally catenary A is catenary, since the quotients of A by its minimal prime ideals are catenary (7.1.4). If now E is an A-algebra of finite type, it follows from what precedes and from (ii) that every local ring of E at a prime ideal is catenary, hence E is catenary. Hence A is universally catenary. □

Remarks (7.1.12). — (i) It is not known whether, conversely, a local Noetherian ring which is universally catenary is always formally catenary.

(ii) A local Noetherian ring A of *dimension* 1 is necessarily formally equidimensional, its maximal ideal not being able to be minimal (for in this case A would be of dimension 0). As $\dim(\widehat{A}) = 1$, this shows that A is even *formally equidimensional*, and *a fortiori* formally catenary (hence *universally catenary*). On the other hand, the local ring of dimension 2 defined in (**0**, 6.11), which is catenary but non-universally catenary, is *a fortiori* not formally catenary.

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Corollary (7.1.13). — *Let Y be a locally Noetherian prescheme irreducible of dimension* 1*, X an irreducible prescheme, $f : X \to Y$ a dominant morphism of finite type, ξ, η the generic points of X and Y respectively. Then, for every $y \in f(X)$, the dimensions of the irreducible components of $f^{-1}(y)$ are all equal to* $\deg.\mathrm{tr}_{k(\eta)} k(\xi)$.

PROOF. By hypothesis, $f^{-1}(\eta)$ is irreducible of generic point ξ and of dimension $\deg.\mathrm{tr}_{k(\eta)} k(\xi)$ (**0**, 2.1). As Y is irreducible and of dimension 1, one has $\dim(\mathscr{O}_y) = 1$ for every $y \neq \eta$, and $\mathscr{O}_y$ is universally catenary by (7.1.12), (ii). If $y \in f(X)$ and if z is the generic point of an irreducible component Z of $f^{-1}(y)$, one has, by (**0**, 6.5)

$$\dim(Z) = 1 + \dim(f^{-1}(\eta)) - \dim(\mathscr{O}_{X,z}).$$

But one has $\dim(\mathscr{O}_{X,z}) \leqslant \dim(\mathscr{O}_y)$ (**0**, 16.3.9), and on the other hand, as z is not the generic point of X, $\dim(\mathscr{O}_{X,z}) > 0$; hence $\dim(\mathscr{O}_{X,z}) = 1$ and $\dim(Z) = \dim(f^{-1}(\eta))$. □

7.2. Strictly formally catenary rings

(7.2.1). *Notations* (7.2.1). — For an *integral* Noetherian ring A, we denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A *of height* 1 ($\mathbf{0}$, 10.17). If A is a local Noetherian integral ring, we denote by $A^{(\omega)}$ the intersection of the $A_{\mathfrak{p}}$ where this time $\mathfrak{p}$ ranges over the set of all prime ideals of A *distinct from the maximal ideal*.

We say that a local Noetherian ring A is *strictly equidimensional* if, for every associated prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$, one has $\dim(A/\mathfrak{p}) = \dim(A)$; this amounts to saying that A is *equidimensional* and *without immersed associated prime ideals*.

(7.2.1.1). *Example* (7.2.1.1). — Let A be a local Noetherian ring of *dimension* 1. Then the following conditions are equivalent:

a) A is without immersed associated prime ideals.
a′) $\widehat{A}$ is without immersed associated prime ideals.
b) A is strictly equidimensional.
b′) $\widehat{A}$ is strictly equidimensional.
c) A is a Cohen-Macaulay ring.

Indeed a) signifies that the maximal ideal $\mathfrak{m}$ of A is not associated to A. Since a) implies b). Inversely, b) implies that every associated prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$ is distinct from $\mathfrak{m}$, hence minimal, and hence b) implies a). We already know that a) and c) are equivalent for a local ring of dimension 1 ($\mathbf{0}$, 7.8), and that a') and b') are equivalent by the same reasoning. As $\dim(\widehat{A}) = 1$, one sees also that a') and b') are equivalent to c).

Proposition (7.2.2). — *Let A be a local Noetherian integral ring; set $A' = \widehat{A}$, $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$ and denote by $f : X' \to X$ the canonical morphism. Let a be the closed point of X, $j : X - \{a\} \to X$ the canonical injection. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, $\mathscr{F}' = f^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$; then the following conditions are equivalent:*

a) *The $\mathscr{O}_X$-Module $j_*(\mathscr{F}|X - \{a\})$ is coherent.*
b) *For every $x' \in \mathrm{Ass}(\mathscr{F}')$, one has $\dim(\overline{\{x'\}}) \geqslant 2$.*

Proof. Let a' be the unique point of the fibre $f^{-1}(a)$, and let $j' : X' - \{a'\} \to X'$ be the canonical injection. As the morphism f is faithfully flat and quasi-compact, it is equivalent to say that $j_*(\mathscr{F}|X - \{a\})$ is coherent and that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent (5.9.5). On the other hand, as A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen ($\mathbf{0}$, 19.8.8), it follows from (5.11.4) that condition b) of the statement is equivalent to the fact that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent. Hence the proposition. □

Proposition (7.2.3). — *Let A be a local Noetherian integral ring. The notations being those of (7.2.2), the following conditions are equivalent:*

a) *$A^{(1)}$ is a finite A-algebra.*
b) *For every closed part T of X of codimension $\geqslant 2$, if one denotes by $i : X - T \to X$ the canonical injection, $i_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_X$-Module.*
c) *For every $x' \in \mathrm{Ass}(\mathscr{O}_{X'})$ and every closed part T of X of codimension $\geqslant 2$, one has $\mathrm{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$.*

Proof. Set $Z = Z^{(2)}(X)$ ($\mathbf{0}$, 10.13) and $Z' = f^{-1}(Z)$, which are stable by specialisation; the conditions a) and b) are equivalent respectively to the following properties: a_1) $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is coherent; b_1) $\mathscr{H}^0_{X/T}(\mathscr{O}_X)$ is coherent for every closed part T of X of codimension $\geqslant 2$. By (5.9.5), these two last properties are equivalent respectively to: a'_1) $\mathscr{H}^0_{X'/Z'}(\mathscr{O}_{X'})$ is coherent; b'_1) posing $T' = f^{-1}(T)$, $\mathscr{H}^0_{X'/T'}(\mathscr{O}_{X'})$ is coherent for every closed part T of X of codimension $\geqslant 2$. Since every point of $\mathrm{Ass}(\mathscr{O}_{X'})$ projects to the generic point of X since f is a flat morphism ($\mathbf{I}$, 3.2); since by definition this point does not belong to Z, we see that $\mathrm{Ass}(\mathscr{O}_{X'})$ does not meet Z'; the equivalence of a'_1) and b'_1) follows from (5.11.4), and this gives the equivalence of a), b) and c). □

Proposition (7.2.4). — *Let A be a local Noetherian ring and set $X = \mathrm{Spec}(A)$. The following conditions are equivalent:*

a) *For every integral quotient ring B of A, the ring $B^{(1)}$ is a finite B-algebra.*
b) *For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$ and every part Z, stable by specialisation, and such that for every $x \in \mathrm{Ass}(\mathscr{F}) \cap (X - Z)$, one has $\mathrm{codim}(\overline{\{x\}} \cap Z, \{x\}) \geqslant 2$, the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.*
c) *For every coherent $\mathscr{O}_U$-Module $\mathscr{G}$ (where $U = X - T$) such that, for every $x \in \mathrm{Ass}(\mathscr{G})$, one has $\mathrm{codim}(\overline{\{x\}} \cap T, \{x\}) \geqslant 2$, $i_*(\mathscr{G})$ (where $i : U \to X$ is the canonical injection) is a coherent $\mathscr{O}_X$-Module.*
d) *For every integral quotient ring B of A and every ideal $\mathfrak{J}$ of height $\geqslant 2$ in B, the ring $\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$ is a finite B-algebra.*
e) *For every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geqslant 2$, the ring $C^{(\omega)}$ is a finite C-algebra (or, what amounts to the same, if $Y = \mathrm{Spec}(C)$, and $j : U \to Y$ is the complement in Y of the closed point of C and $j : U \to Y$ the canonical injection, $j_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_Y$-Module).*
f) *For every integral quotient ring B of A and every local ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geqslant 2$, one has $\dim(\widehat{C}/\mathfrak{r}) \geqslant 2$ for every $\mathfrak{r} \in \mathrm{Ass}(\widehat{C})$.*

Theorem (7.2.5). — *Let A be a local Noetherian ring. The following conditions are equivalent:*

a) *For every integral quotient ring B of A, the completion $\widehat{B}$ is strictly equidimensional (7.2.1).*
b) *A is formally catenary (7.1.9) and the fibres of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ satisfy (S_1) (in other words,* have no immersed associated prime cycle*).*
c) *A is universally catenary* (**0**, 6.2) *and for every integral quotient ring B of A, the ring $B^{(1)}$ is a finite B-algebra (cf. (7.2.4)).*
d) *A is universally catenary and for every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geqslant 2$, the ring $C^{(\omega)}$ is a finite C-algebra.*
e) *A is universally catenary and for every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geqslant 2$, the completion $\widehat{C}$ has no associated prime cycle of dimension* 1.

Moreover, when these conditions are satisfied, then, for every quotient ring B of A which is strictly equidimensional, the completion $\widehat{B}$ is strictly equidimensional.

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Definition (7.2.6). — When a local Noetherian ring A satisfies the equivalent conditions of (7.2.5), we say that it is *strictly formally catenary*.

Corollary (7.2.7). — *Let A be a local Noetherian ring such that there exists an A-module of finite type M which is a Cohen-Macaulay A-module and for which $\mathrm{Supp}(M) = \mathrm{Spec}(A)$; then A is strictly formally catenary.*

PROOF. Indeed, A is formally catenary (7.1.5) and (7.1.9)), and on the other hand the fibres of $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ are Cohen-Macaulay preschemes (6.3.8), hence *a fortiori* satisfy (S_1). □

Corollary (7.2.8). — *If A is a local Noetherian ring strictly formally catenary, every local ring B which is an A-algebra essentially of finite type is strictly formally catenary.*

PROOF. Indeed, B is formally catenary (7.1.11) and it follows from (7.4.4)[4] that the fibres of $\mathrm{Spec}(\widehat{B}) \to \mathrm{Spec}(B)$ satisfy (S_1), hence the conclusion. □

Corollary (7.2.9). — *Every local Noetherian ring of dimension* 1 *is strictly formally catenary.*

PROOF. Indeed, every integral quotient ring of such a ring A is of dimension 0 or 1, hence its completion is strictly equidimensional (7.2.1.1). □

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Remarks (7.2.10). — (i) It follows from (7.2.7) and (7.2.8) that every quotient of a Cohen-Macaulay local ring is strictly formally catenary. A local Noetherian ring of dimension 2 satisfying (S_2) is strictly formally catenary, since it is then a Cohen-Macaulay ring.

[4] The corollary (7.2.8) is not used to demonstrate (7.4.4).

Recall however that there are local Noetherian integral rings of dimension 2 which are not universally catenary, nor *a fortiori* strictly formally catenary (**0**, 6.11).

(ii) It is not known whether a formally catenary ring (7.1.9) is strictly formally catenary; this is related to whether one knows whether for a local Noetherian integral ring, $\widehat{B}$ has no immersed associated prime ideals (6.4.3). We also do not know whether, in the equivalent conditions $c)$, $d)$, $e)$ of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is so when A is henselian (18.9.6).

7.3. Formal fibres of local Noetherian rings

(7.3.1). We shall consider in this number and the next two properties $\mathbf{P}(Z,k)$ of the following form:

> "Z is a locally Noetherian prescheme on a field k, and for every $z \in Z$, one has $Q(\mathscr{O}_z, k)$"

where $Q(A,k)$ is a property of a local Noetherian ring A which is a k-algebra; one supposes that if k' is a field isomorphic to k, and if A' is a local Noetherian ring which is a k'-algebra di-isomorphic to A, the properties $Q(A,k)$ and $Q(A',k')$ are equivalent.

If X, Y are two locally Noetherian preschemes, we say that a morphism $f : X \to Y$ is a **P**-*morphism* if:

1° f is *flat*;

2° for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), k(y))$ is true.

Lemma (7.3.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism. The following properties are equivalent:*

a) *f is a* **P**-*morphism.*

b) *For every $x \in X$, if one sets $y = f(x)$, then $\mathscr{O}_x$ is an $\mathscr{O}_y$-module flat and the property $Q(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y), k(y))$ is true.*

c) *For every $x \in X$, if one sets $y = f(x)$, the morphism* $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ *corresponding to $\mathscr{O}_y \to \mathscr{O}_x$ is a* **P**-*morphism.*

c') *For every closed point $x \in X$, the morphism* $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ *is a* **P**-*morphism.*

PROOF. The equivalence of a) and b) follows from the definitions; it is the same for c), by (**I**, 2.4.2); finally the equivalence of b) and c') follows from the fact that for every $x \in X$, $\overline{\{x\}}$ contains a closed point (**0**, 1.11). □

Corollary (7.3.3). — *Let A, B be two Noetherian rings, $\varphi : A \to B$ a homomorphism such that the corresponding morphism* $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ *is a* **P**-*morphism. Then, for every multiplicative subset S of A, the morphism* $\operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$ *is a* **P**-*morphism.*

PROOF. This follows immediately from (7.3.2) and from (**I**, 1.6.2). □

(7.3.4). We shall always suppose in what follows that the property Q is such that the three following conditions are satisfied:

($\mathrm{P_I}$) *(transitivity).* — If $f : X \to Y$ is a regular morphism (6.8.1) and $g : Y \to Z$ a **P**-morphism, then $g \circ f$ is a **P**-morphism.

($\mathrm{P_{II}}$) *(descent).* — If $f : X \to Y$ and $g : Y \to Z$ are morphisms of locally Noetherian preschemes such that f is faithfully flat and that $g \circ f$ is a **P**-morphism, then g is a **P**-morphism.

($\mathrm{P_{III}}$) — For every field k, the property $\mathbf{P}(\operatorname{Spec}(k), k)$ is true.

Remarks (7.3.5). — (i) Conditions ($\mathrm{P_I}$) and ($\mathrm{P_{III}}$) imply that every *regular morphism* is a **P**-morphism.

(ii) Note that the hypotheses of ($\mathrm{P_I}$) (resp. ($\mathrm{P_{II}}$)) imply that $h = g \circ f$ (resp. g) is flat (**I**, 2.13); the hypotheses of ($\mathrm{P_I}$) or of ($\mathrm{P_{II}}$) also imply that for every $z \in Z$, $f_z : h^{-1}(z) \to g^{-1}(z)$ is flat (**I**, 1.4); as for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4), this shows that it suffices, to verify conditions ($\mathrm{P_I}$) and ($\mathrm{P_{II}}$), to do so only when Z is the *spectrum of a field*.

(iii) In certain cases, the property Q will be such that the following condition is also satisfied:

$(\mathrm{P}'_{\mathrm{I}})$ — If $f : X \to Y$ and $g : Y \to Z$ are two **P**-morphisms, then $g \circ f$ is a **P**-morphism.

(7.3.6). We shall say that the property **P** is *geometric* if it satisfies (in addition to $(\mathrm{P}_{\mathrm{I}})$, $(\mathrm{P}_{\mathrm{II}})$ and $(\mathrm{P}_{\mathrm{III}})$) the condition:

$(\mathrm{P}_{\mathrm{IV}})$ *(finite extension of base field).* — If $\mathbf{P}(Z, k)$ is true, then for every extension k' of finite type of k, $\mathbf{P}(Z \otimes_k k', k')$ is true.

Lemma (7.3.7). — *Let $f : X \to Y$ be a **P**-morphism of locally Noetherian preschemes, $g : Y' \to Y$ a morphism locally of finite type. Then, if **P** satisfies condition $(\mathrm{P}_{\mathrm{IV}})$, the morphism $f' = f_{(Y')} : X \times_Y Y' \to Y'$ is a **P**-morphism.*

PROOF. Indeed, for every $y' \in Y'$, if one sets $y = g(y')$, $k(y')$ is an extension of finite type of $k(y)$ (**I**, 6.4.11) and $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (**I**, 3.6.4); it therefore suffices to apply $(\mathrm{P}_{\mathrm{IV}})$. □

(7.3.8). *Examples* (7.3.8). — The following properties $\mathbf{P}(Z, k)$ satisfy conditions $(\mathrm{P}_{\mathrm{I}})$, $(\mathrm{P}_{\mathrm{II}})$ and $(\mathrm{P}_{\mathrm{III}})$:

(i) (also denoted (i_n)) Z is of coprofondeur $\leqslant n$.
(ii) Z is a Cohen-Macaulay prescheme.
(iii) (also denoted (iii_n)) Z satisfies (S_n).
(iv) Z is regular.
(v) (also denoted (v_n)) Z satisfies (R_n).
(vi) Z is reduced.
(vii) Z is normal.

For properties (ii) to (vii), this follows from (6.6.1) which, in fact, demonstrates the stronger condition $(\mathrm{P}'_{\mathrm{I}})$. For (i), property $(\mathrm{P}_{\mathrm{I}})$ follows from (6.6.2); property $(\mathrm{P}_{\mathrm{II}})$ follows from (6.3.2) and from the fact that a regular prescheme is of coprofondeur 0.

Moreover, it follows from (6.7.1) that properties (i), (ii) and (iii) are *geometric*; and by (6.7.8), it is the same for the following:

(iv′) Z is geometrically regular.
(v′) (also denoted (v'_n)) Z possesses the geometric property (R_n).
(vi′) Z is geometrically reduced.
(vii′) Z is geometrically normal.

Remarks (7.3.9). — (i) Each of properties (iv′), (v'_n), (vi′), (vii′) implies the corresponding property (iv), (v_n), (vi), (vii). Property (iv) implies all of (i) through (vii), and property (iv′) implies all the properties enumerated in (7.3.8). Let us also recall that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and (v'_0)) is equivalent to (vi) (resp. (vi′)) (**0**, 8.5); finally the conjunction of (iii_2) and (v_1) (resp. (iii_2) and (v'_1)) is equivalent to (vii) (resp. (vii′)) (**0**, 8.6).

(ii) In all the examples of (7.3.8), the property $Q(\mathscr{O}_z, k)$ which serves to define **P** is such that, for every generisation z' of z in Z, $Q(\mathscr{O}_z, k)$ *implies* $Q(\mathscr{O}_{z'}, k)$: by (**I**, 3.4), it suffices to verify this for properties (i) to (vii) (and even (i) to (v) by the remark (7.3.9), (i)); this follows from (6.3.9) for (i), and finally from (**0**, 16.5.10 and 17.3.2) for (ii) and (iv).

(7.3.10). We shall say that a property $\mathbf{P}(Z, k)$ is *of the first type* if the property $Q(A, k)$ which serves to define **P** is a property $\mathbf{R}(A)$ *not involving* k and is true when A is a field. It is clear that in the examples of (7.3.8), properties (i) to (vii) are of the first type, properties (iv′) to (vii′) of the second type.

We shall say that $\mathbf{P}(Z, k)$ is *of the second type* if the property $Q(A, k)$ is of the form

"For every extension k' of finite type of k and every closed point z' of $\mathrm{Spec}(A \otimes_k k')$ above the closed point of $\mathrm{Spec}(A)$, one has $\mathbf{R}(\mathscr{O}_{z'})$"

$\mathbf{R}(A)$ being assumed true when A is a field.

(7.3.13). Given a semi-local Noetherian ring A, we call *formal fibres* of A the fibres of the canonical morphism $f : \mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$; for every prime ideal $\mathfrak{p}$ of A, the formal fibre at $\mathfrak{p}$ is then the prescheme $\mathrm{Spec}(\widehat{A} \otimes_A k(\mathfrak{p}))$; as the completion of the local ring $A/\mathfrak{p}$ is $\widehat{A}/\mathfrak{p}\widehat{A} = (A/\mathfrak{p}) \otimes_A \widehat{A}$, one sees that the formal fibre of $A/\mathfrak{p}$ at the generic point (o) of $\mathrm{Spec}(A/\mathfrak{p})$ is also the formal fibre of A at $\mathfrak{p}$.

The property $\mathbf{P}$ being defined as in (7.3.1), we shall say that *the formal fibres of A have property $\mathbf{P}$*, or that A is a *$\mathbf{P}$-ring*, if, for every $x \in \operatorname{Spec}(A)$, $\mathbf{P}(f^{-1}(x), k(x))$ is true. As f is *flat*, this amounts to saying that f is a $\mathbf{P}$-morphism.

Proposition (7.3.14). — *Let A be a semi-local Noetherian ring, $\mathfrak{m}_i$ $(1 \leqslant i \leqslant r)$ its maximal ideals, and set $A_i = A_{\mathfrak{m}_i}$; for A to be a $\mathbf{P}$-ring, it is necessary and sufficient that each of the A_i be so.*

Proof. Indeed, $\widehat{A}$ is the direct product of the $\widehat{A}_i$, hence the formal fibre of A at a point $x \in \operatorname{Spec}(A)$ is the sum of the formal fibres at x of those A_i such that $x \in \operatorname{Spec}(A_i)$. □

Proposition (7.3.15). — *Let A be a semi-local Noetherian ring.*

(i) *If A is a $\mathbf{P}$-ring, every quotient ring of A is a $\mathbf{P}$-ring.*
(ii) *If $\mathbf{P}$ satisfies moreover condition $(\mathrm{P_{IV}})$ (7.3.6), then every finite A-algebra is a $\mathbf{P}$-ring.*

Proof. For every ideal $\mathfrak{a}$ of A, $\widehat{A} \otimes_A (A/\mathfrak{a})$ is the completion of $A/\mathfrak{a}$, and the formal fibres of $A/\mathfrak{a}$ are the formal fibres of A at the points of $\mathrm{V}(\mathfrak{a})$, hence (i). On the other hand, if B is a finite A-algebra, $\widehat{B} = B \otimes_A \widehat{A}$, and (ii) follows from (7.3.7). □

Proposition (7.3.16). — *Suppose that the property $\mathbf{P}$ is of the first (resp. second) type, defined from a property $\mathbf{R}$ (7.3.10). When $\mathbf{P}$ is of the second type, suppose in addition that the relation*

"for every finite extension k' of k, and every $z' \in Z \otimes_k k'$, $\mathbf{R}(\mathscr{O}_{z'})$ is true"

implies $\mathbf{P}(Z, k)$ for the examples (iv') to (vii') of (7.3.8), by (6.7.7) and (4.6.1)).

Let A be a semi-local Noetherian ring. If the property $\mathbf{R}$ satisfies conditions $(\mathrm{R_I})$ and $(\mathrm{R_{II}})$, the following properties are equivalent:

a) *A is a $\mathbf{P}$-ring.*
b) *For every integral quotient ring B of A (resp. every finite integral A-algebra B) and every prime ideal $\mathfrak{q}$ of $\widehat{B}$ whose inverse image in B is 0, $\mathbf{R}(\widehat{B}_\mathfrak{q})$ is true.*

If moreover $\mathbf{R}$ satisfies condition $(\mathrm{R'_I})$, properties a) and b) are also equivalent to:

c) *For every integral quotient ring B of A (resp. every finite A-algebra B) if one sets $Y = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(\widehat{B})$, and if $g : Y' \to Y$ is the canonical morphism, one has (with the notations of* (7.3.12)*)*

$$\mathrm{U}_\mathbf{R}(Y') = g^{-1}(\mathrm{U}_\mathbf{R}(Y)). \tag{7.3.16.1}$$

Proposition (7.3.17). — *Suppose that the property $\mathbf{R}$ satisfies conditions $(\mathrm{R'_I})$ and $(\mathrm{R_{II}})$ and that $\mathbf{P}$ is a property of the first (resp. second) type defined from $\mathbf{R}$ (7.3.10). Then, if A is a $\mathbf{P}$-ring, $\mathbf{R}(A)$ and $\mathbf{R}(\widehat{A})$ are equivalent.*

Proof. This follows from (7.3.12.1) applied to $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(\widehat{A})$. □

Proposition (7.3.18). — *Suppose that $\mathbf{P}$ is a property of the first (resp. second) type defined from a property $\mathbf{R}$ satisfying conditions $(\mathrm{R'_I})$ and $(\mathrm{R_{II}})$ as well as the following condition:*

$(\mathrm{R_{III}})$ *For every local Noetherian complete ring C, if one sets $Z = \operatorname{Spec}(C)$, the set $\mathrm{U}_\mathbf{R}(Z)$* (7.3.12) *is open in Z.*

Then, if A is a $\mathbf{P}$-ring and if one sets $X = \operatorname{Spec}(A)$, the set $\mathrm{U}_\mathbf{R}(X)$ is open in X.

Proof. Indeed, if $X' = \operatorname{Spec}(\widehat{A})$ and if $f : X' \to X$ is the canonical morphism, on a $\mathrm{U}_\mathbf{R}(X') = f^{-1}(\mathrm{U}_\mathbf{R}(X))$ (7.3.12). As f is faithfully flat and quasi-compact and by hypothesis $\mathrm{U}_\mathbf{R}(X')$ is open in X', the conclusion follows from (**I**, 3.12). □

Remarks (7.3.19). — (i) The property $\mathbf{R}$ which serves to define $\mathbf{P}$ satisfies condition $(\mathrm{R_{III}})$ in all the examples enumerated in (7.3.8). For (i), (ii) and (iii), this follows from (6.11.2) and from the theorem of Cohen (**0**, 19.8.8); when $\mathbf{P}$ is one of the properties (iv), (iv'), (v), (v'), this follows from (6.12.7) and (6.12.9), and when $\mathbf{P}$ is one of the properties (vii), (vii'), from (6.12.7) and (6.13.4); finally, for (vi) and (vi') the assertion of (7.3.18) is trivial, being true for every locally Noetherian prescheme.

(ii) We have already noted (6.4.3) that when **P** is one of the properties (ii) or (iii_n) of (7.3.8), we do not know whether every local Noetherian ring is a **P**-ring; recall however that when A is a *quotient of a Cohen-Macaulay ring*, the formal fibres of A are Cohen-Macaulay preschemes (6.3.8).

(iii) The most interesting case of the notion of **P**-ring is that which corresponds to the strongest property (iv′) of (7.3.8), that is to say the rings whose *formal fibres are geometrically regular*. Fields, and more generally local Noetherian complete rings, trivially satisfy this property.

(iv) Let A be a local Noetherian ring of *dimension* 1; $\mathrm{Spec}(A)$ is then formed of the closed point a, corresponding to the maximal ideal $\mathfrak{m}$, and of the maximal points b_i $(1 \leqslant i \leqslant r)$ corresponding to the minimal prime ideals of A. One has $\dim(\widehat{A}) = 1$ and the maximal ideal of $\widehat{A}$ is $\mathfrak{m}\widehat{A}$; the formal fibre at a is then $\mathrm{Spec}(k)$, where $k = A/\mathfrak{m}$ is the residue field; the formal fibre at each b_i is the spectrum of an Artinian ring whose residue fields are the residue fields L_{ij} of $\mathrm{Spec}(\widehat{A})$ at the maximal points b_{ij} above b_i. As an Artinian ring is a Cohen-Macaulay ring, one sees that A is a **P**-ring when **P** is property (ii) of (7.3.8). Moreover, as a reduced Artinian ring is a direct sum of fields, the following properties are equivalent (6.7.7):

a) the formal fibres of A are geometrically reduced;
b) the formal fibres of A are geometrically normal;
c) the formal fibres of A are geometrically regular;

moreover, when A is reduced, they are also equivalent to:

d) $\widehat{A}$ is reduced and L_{ij} is a separable extension of K_i for every couple (i, j) (4.6.1).

In particular, if A is a *discrete valuation ring*, K its field of fractions, and if $\widehat{K}$ is the completion of K for a valuation corresponding to A (field of fractions of $\widehat{A}$), for the formal fibres of A to be geometrically regular, it is necessary and sufficient that $\widehat{K}$ be a *separable extension* of K. This is always the case when K is of *characteristic* 0.

7.4. Permanence of properties of formal fibres

(7.4.1). In the remainder of this number, we suppose that the property **P** is of the form defined in (7.3.1), and *satisfies conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ and $(\mathrm{P_{III}})$ of (7.3.4). Moreover, we suppose that the property Q which serves to define **P** is such that, for every generisation z' of z in Z, $Q(\mathscr{O}_z, k)$ *implies* $Q(\mathscr{O}_{z'}, k)$.

Lemma (7.4.2). — *Let A, A' be local Noetherian rings, $\varphi : A \to A'$ a local homomorphism such that $f = {}^a\varphi : \mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a* **P***-morphism. Then, if the formal fibres of A' are geometrically regular, A is a* **P***-ring.*

Proof. Consider the completed homomorphism $\widehat{\varphi} : \widehat{A} \to \widehat{A}'$ and the corresponding morphism $\widehat{f} = {}^a\widehat{\varphi}$; one has the commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(\widehat{A}) & \xleftarrow{\widehat{f}} & \mathrm{Spec}(\widehat{A}') \\ {\scriptstyle g}\uparrow & & \uparrow{\scriptstyle g'} \\ \mathrm{Spec}(A) & \xleftarrow{f} & \mathrm{Spec}(A') \end{array}$$

where g and g' are the canonical morphisms. As by hypothesis f is a **P**-morphism and g' a regular morphism, it follows from $(\mathrm{P_I})$ that $f \circ g' = g \circ \widehat{f}$ is a **P**-morphism. On the other hand, the hypothesis that f is a **P**-morphism implies that f is flat, hence it is the same for $\widehat{f}$ (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, cor. de la prop. 3), which is moreover a local homomorphism, hence faithfully flat ($\mathbf{0_I}$, 6.6.2); it follows then from $(\mathrm{P_{II}})$ that g is a **P**-morphism. □

Corollary (7.4.3). — (i) *Let A be a* **P***-ring local Noetherian, $A' = \widehat{A}$ its completion. Suppose that for every prime ideal $\mathfrak{p}'$ of A', the formal fibres of $A'_{\mathfrak{p}'}$ are geometrically regular; then, for every prime ideal $\mathfrak{p}$ of A, $A_{\mathfrak{p}}$ is a* **P***-ring.*

(ii) *Suppose that* **P** *satisfies condition* $(\mathrm{P_{IV}})$ (7.3.6). *Let A be a* **P***-ring local Noetherian, B a local A-algebra essentially of finite type, and such that the homomorphism $A \to B$ is local. Set $A' = \widehat{A}$, and let $\mathfrak{n}'$ be the unique prime ideal of $B' = B \otimes_A A'$ above the maximal ideal of B and the maximal ideal of A'. If the formal fibres of $B'_{\mathfrak{n}'}$ are geometrically regular, then B is a* **P***-ring.*

Theorem (7.4.4). — *The hypotheses on* $\mathbf{P}$ *being those of (7.4.1), let* A *be a* $\mathbf{P}$*-ring local Noetherian.*

(i) *For every prime ideal* $\mathfrak{p}$ *of* A*,* $A_{\mathfrak{p}}$ *is a* $\mathbf{P}$*-ring.*
(ii) *Suppose moreover that* $\mathbf{P}$ *satisfies condition* $(\mathrm{P_{IV}})$*. Then every local ring which is an* A*-algebra essentially of finite type is a* $\mathbf{P}$*-ring.*

Corollary (7.4.5). — *Suppose that the property* $\mathbf{P}$ *satisfies conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$, $(\mathrm{P_{III}})$*. The following conditions are equivalent:*

a) *For every prime ideal* $\mathfrak{p}$ *of* A*,* $A_{\mathfrak{p}}$ *is a* $\mathbf{P}$*-ring.*
b) *For every maximal ideal* $\mathfrak{m}$ *of* A*,* $A_{\mathfrak{m}}$ *is a* $\mathbf{P}$*-ring.*

If moreover $\mathbf{P}$ *satisfies* $(\mathrm{P_{IV}})$*, properties a) and b) are also equivalent to:*

c) *For every* A*-algebra of finite type* B *and every prime ideal* $\mathfrak{q}$ *of* B*,* $B_{\mathfrak{q}}$ *is a* $\mathbf{P}$*-ring.*

The equivalence of a) and b) follows from (7.4.4), (i)), and that of a) and c) from (7.4.4), (ii). When condition a) of (7.4.5) is satisfied, for a semi-local Noetherian ring, this definition coincides with that of (7.3.13). The ring $\mathbf{Z}$ *is a* $\mathbf{P}$*-ring when conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ *and* $(\mathrm{P_{III}})$ *are satisfied. The ring* $\mathbf{Z}$ *is a* $\mathbf{P}$*-ring (7.3.19), (iv)); every local Noetherian complete ring is a* $\mathbf{P}$*-ring.*

Proposition (7.4.6). — *Suppose that the property* $\mathbf{P}$ *satisfies conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ *and* $(\mathrm{P_{III}})$*. Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A*,* $\widehat{A}$ *the separated completion of* A *for the* $\mathfrak{J}$*-preadic topology. Then, if* A *is a* $\mathbf{P}$*-ring (7.4.5), the canonical morphism* $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ *is a* $\mathbf{P}$*-morphism.*

Corollary (7.4.7). — *Suppose that* $\mathbf{P}$ *satisfies conditions* $(\mathrm{P_I})$, $(\mathrm{P'_I})$, $(\mathrm{P_{II}})$, $(\mathrm{P_{III}})$ *and* $(\mathrm{P_{IV}})$*. Then, if* A *is a* $\mathbf{P}$*-ring (7.4.5), the canonical morphism* $\mathrm{Spec}(A[[T_1, \dots, T_r]]) \to \mathrm{Spec}(A)$ *is a* $\mathbf{P}$*-morphism. In particular, if moreover* A *is integral,* K *its field of fractions, and if* $\mathfrak{p}$ *is a prime ideal of* $B = A[[T_1, \dots, T_r]]$ *such that* $\mathfrak{p} \cap A = 0$*, then the property* $\mathbf{P}(B_{\mathfrak{p}}, K)$ *is true.*

Proof. Indeed, the canonical morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ factorises as

$$\mathrm{Spec}(A[[T_1, \dots, T_r]]) \xrightarrow{f} \mathrm{Spec}(A[T_1, \dots, T_r]) \xrightarrow{g} \mathrm{Spec}(A).$$

It is clear that the morphism g is regular ($\mathbf{0}$, 17.3.7); by (7.4.5), $A[T_1, \dots, T_r]$ is a $\mathbf{P}$-ring, hence it follows from (7.4.6) that f is a $\mathbf{P}$-morphism; as g is regular, hence also a $\mathbf{P}$-morphism (7.3.5), (i)), it is the same for $g \circ f$ by $(\mathrm{P'_I})$. □

Remark (7.4.8). — The preceding results pose the following problems:

A) Let A be a Zariski ring *complete*, and let $\mathfrak{J}$ be an ideal of definition of A; if $A/\mathfrak{J}$ is a $\mathbf{P}$-ring, is it the same for A?
B) Let k be a complete non-discrete valued field; we call the ring of *restricted formal power series* $k\{T_1, \dots, T_n\}$ the sub-ring of $k[[T_1, \dots, T_n]]$ formed by the series whose coefficients *tend to* 0. Is such a ring a $\mathbf{P}$-ring?
C) If A is a linearly topologised $\mathbf{P}$-ring, S a multiplicative subset of A, are the rings $A\{S^{-1}\}$ ($\mathbf{0_I}$, 7.6.1) and $A_{\{S\}}$ ($\mathbf{0_I}$, 7.6.15) also $\mathbf{P}$-rings?

7.5. A criterion for P-morphisms

(7.5.0). This number will not be used in the sequel of chapter IV, and can therefore be omitted on a first reading. We shall see later (7.9.8) that the results of this number can be considerably improved when one has the "resolution of singularities" at one's disposal. **IV-2 | 203**

In the remainder of this number, we shall consider a property $\mathbf{R}(A)$, and we denote by $\mathbf{P}(Z, k)$ the property

"Z is a locally Noetherian prescheme on a field k, and, for every *finite* extension k' of k, if one sets $Z' = Z \otimes_k k'$, the property $\mathbf{R}(\mathscr{O}_{z'})$ is true for every $z' \in Z'$."

Theorem (7.5.1). — *Let* A*,* B *be two local Noetherian complete rings,* $\mathfrak{m}$ *the maximal ideal of* A*,* $k = A/\mathfrak{m}$ *its residue field,* $\varphi : A \to B$ *a local homomorphism such that:*

(i) *The residue field of* B *is a finite extension of* k*.*
(ii) B *is a flat* A*-module.*

Let moreover $\mathbf{R}(C)$ *be a property satisfying condition* $(\mathrm{R_{III}})$ *(7.3.18) and the following condition:*

($\mathrm{R_{IV}}$) *For every local ring C in a local Noetherian complete ring and every C-regular element t above the maximal ideal of C, the property $\mathbf{R}(C/tC)$ implies $\mathbf{R}(C)$.*

Then, for every ideal $\mathfrak{p}$ of A, $\mathbf{R}(\widehat{B}_{\mathfrak{q}})$ is true for every prime ideal $\mathfrak{q}$ of $\widehat{B}$ above $\mathfrak{p}$, if and only if $\mathbf{R}(\widehat{A}_{\mathfrak{p}'})$ is true for every prime ideal $\mathfrak{p}'$ of $\widehat{A}$ above $\mathfrak{p}$ and the fibre of the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ over $\mathfrak{p}$ satisfies $\mathbf{P}$.

7.6. Two applications to the finiteness of integral closure

(7.6.1). Let A be a local Noetherian integral ring. We shall say (following Nagata) that A is a N-*1 ring* (or a (N_1)-*ring*) if its integral closure A' is a *finite* A-module. We shall say that A is a N-*2 ring* (or a (N_2)-*ring*, or a *Japanese ring*) if, for every extension K' of finite degree of the field of fractions K of A, the integral closure A' of A in K' is a finite A-module.

Proposition (7.6.2). — *Let A be a local Noetherian integral ring whose formal fibres are geometrically reduced. Then A is a (N_1)-ring.*

Proposition (7.6.3). — *Let A be a local Noetherian integral ring whose formal fibres are geometrically normal. Then A is an (N_2)-ring.*

Corollary (7.6.4). — *Every local Noetherian* complete *integral ring is an (N_2)-ring (i.e.* Japanese*).*

PROOF. Indeed, the formal fibres of a complete local ring are geometrically regular (7.3.19), (iii)), hence *a fortiori* geometrically normal. ☐

Corollary (7.6.5). — *Let A be a Noetherian ring whose formal fibres are geometrically normal (in the sense of (7.4.5)). Then A is* universally Japanese *(**0**, 23.1.1).*

Proposition (7.6.6). — *Let A be a reduced local Noetherian ring, A' its normalisation, $f : X' = \mathrm{Spec}(A') \to X = \mathrm{Spec}(A)$ the corresponding morphism. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, U the complement of the closed point a in X, $U' = f^{-1}(U)$, $\mathscr{F}' = f^*(\mathscr{F})$; suppose that, for every point $x \in U'$, the $\mathscr{O}_x$-module $\mathscr{F}'_x = \mathscr{F}_x \otimes A'_x$ is* reflexive *(i.e. that $\mathscr{F}'$ is reflexive on U'). Then the canonical homomorphism $\mathscr{F} \to f_*(f^*(\mathscr{F}))$ is bijective.*

Corollary (7.6.7). — *Under the hypotheses of (7.6.6), the canonical homomorphism $A \to \Gamma(U', \mathscr{O}_{X'})$ is bijective. Moreover, if A is integral and A' is finite over A, the normalisation A' of A identifies to $\Gamma(U, \mathscr{O}_X)$.*

7.7. Nagata rings

Definition (7.7.1). — We say that a local Noetherian ring A is a *Nagata ring* if the following two conditions are satisfied:

(i) A is universally Japanese (**0**, 23.1.1).

(ii) For every prime ideal $\mathfrak{p}$ of A and every finite extension of the field of fractions of $A/\mathfrak{p}$, the integral closure of $A/\mathfrak{p}$ in this extension has an open regular locus.

A Noetherian ring A is said to be a *Nagata ring* if, for every prime ideal $\mathfrak{p}$ of A, $A_{\mathfrak{p}}$ is a Nagata ring in the above sense.

Proposition (7.7.2). — *A local Noetherian ring A whose formal fibres are geometrically regular is a Nagata ring. In particular, every complete local Noetherian ring is a Nagata ring.*

Proposition (7.7.3). — *Let A be a Nagata ring. Then every A-algebra of finite type is a Nagata ring.*

7.8. Excellent rings

Definition (7.8.1). — A Noetherian ring A is said to be *quasi-excellent* if it satisfies the following conditions:

(i) For every prime ideal $\mathfrak{p}$ of A, the formal fibres of $A_{\mathfrak{p}}$ are geometrically regular.
(ii) For every A-algebra of finite type B, $\operatorname{Reg}(B)$ is open in $\operatorname{Spec}(B)$.

A Noetherian ring A is said to be *excellent* if it is quasi-excellent and universally catenary.

Proposition (7.8.2). — *A Noetherian ring A is quasi-excellent if and only if it satisfies the condition (i) of (7.8.1) and the following condition:*

(ii') *For every integral quotient ring B of A, there exists a non-empty open subset of* $\operatorname{Spec}(B)$ *which is regular.*

Theorem (7.8.3). — (i) *Every quotient ring of a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).*

(ii) *Every algebra of finite type over a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).*
(iii) *Every ring of fractions of a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).*
(iv) *If A is a quasi-excellent (resp. excellent) local ring, its completion $\widehat{A}$ is quasi-excellent (resp. excellent).*

Corollary (7.8.4). — *Every complete Noetherian local ring is excellent. The ring* $\mathbf{Z}$ *is excellent. Every field is excellent. Every Dedekind ring whose field of fractions is of characteristic* 0 *is excellent.*

Corollary (7.8.5). — *Every algebra of finite type over an excellent ring is excellent. In particular, every algebra of finite type over* $\mathbf{Z}$ *or over a field is excellent.*

Corollary (7.8.6). — *Let A be an excellent ring. Then A is a Nagata ring* (7.7.1). *In particular, A is universally Japanese.*

Proposition (7.8.7). — *Let A be a quasi-excellent ring. Then, for every ideal $\mathfrak{J}$ of A, the $\mathfrak{J}$-adic completion $\widehat{A}$ is quasi-excellent.*

Remark (7.8.8). — Let A be a quasi-excellent local ring, $\widehat{A}$ its completion. Then the canonical morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$ is a *regular* morphism (6.8.1). This is one of the key properties of excellent rings.

7.9. Chains of prime ideals in polynomial rings

(7.9.0). In this final number of § 7, we give some applications of the theory of excellent rings to the study of chains of prime ideals in polynomial rings over Noetherian rings.

Proposition (7.9.1). — *Let A be a Noetherian ring. Then the polynomial ring $A[T_1, \ldots, T_n]$ is catenary if and only if A is catenary.*

Proposition (7.9.2). — *Let A be an excellent Noetherian ring. Then, for every A-algebra of finite type B, B is universally catenary.*

Remark (7.9.8). — When one has the "resolution of singularities" at one's disposal, the results of § 7.5 can be considerably improved. We shall not enter into these questions here.

§8. Projective limits of preschemes

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. 8.1.1 The notion of a projective limit of preschemes was defined in (II, 2.4.1); if the ordered set $\mathbf{L}$ of indices is filtered and increasing, and if the morphisms of the projective system are *affine*, then the projective limit exists, and is a prescheme. The results of the present paragraph will essentially show that the algebraic geometry of $\mathscr{O}_y$-preschemes of finite presentation (and the theory of Modules of finite presentation on these preschemes) is essentially equivalent to the algebraic geometry of preschemes of finite presentation over "sufficiently small" open neighborhoods of y. A particularly important case, and to a certain extent a classical one, is the one where Y is integral and where y is its generic point, so that $\mathscr{O}_y$ is nothing other than *the field* $\mathrm{R}(Y) = K$ *of rational functions on* Y. The results of the present paragraph thus allow reinterpreting algebraic geometry over K in terms of algebraic geometry over non-empty open subsets "sufficiently small" of Y, that is to say, intuitively, in terms of "families" of geometric objects indexed by the points of such an open subset. This point of view has moreover been commonly used for a long time, not only in Algebraic Geometry over algebraically closed fields, but also in the study of arithmetic of varieties defined over a *number field* K (a finite extension of $\mathbf{Q}$), by considering the latter as the field of fractions of its ring of integers A ("the theory of *reduction modulo* $\mathfrak{p}$" ($\mathfrak{p}$ a prime ideal of A); cf. (**I**, 3.7)). The results of §§ 8 and 9 thus provide, among other things, some of the foundations of the language of "reduction modulo $\mathfrak{p}$" in arithmetic.

We note that in the example considered here, the morphisms $S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) are the *canonical open immersions* $U_\mu \to U_\lambda$, and *a fortiori* are *flat* morphisms (but not faithfully flat in general), which explains the interest of the statements that do not make such a restriction.

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. 8.1.2 *b)* Suppose that the A_λ are *fields*, so that $A = \varinjlim A_\lambda$ is also a field. This case arises generally when one starts from geometric data over a field K of any kind (for example k the prime subfield of K). There is then interest in considering K as an inductive limit of its sub-extensions that are *of finite type over* k, which allows in many cases reducing questions to the case where K is a finite type extension of k. Using also the method sketched in *a)*, one can then reduce more generally to the case of a base ring A which is an *algebra of finite type over* k.

We note that in this example, the morphisms $S_\mu \to S_\lambda$ are *faithfully flat*.

c) Suppose one is interested in properties, local over Y, of preschemes of finite presentation Y over any prescheme Y, which one can then assume to be an affine scheme of ring A. There is then interest in considering A as an inductive limit of its sub-rings which are $\mathbf{Z}$-*algebras of finite type*, which allows reducing many questions to the case where Y is the spectrum of such an algebra. This is the basis for the "Kroneckerian point of view" according to which Algebraic Geometry reduces to the Algebraic Geometry of preschemes of finite type over $\mathbf{Z}$ (which one sometimes qualifies as

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. 8.1.2.cont "Absolute Algebraic Geometry"). This example shows us in particular that for most "relative" questions on a prescheme of base Y, one can reduce to the case where Y is *noetherian*.

We note that in this example, in contrast to the preceding ones, the morphisms $S_\mu \to S_\lambda$ have in general no particular regularity property.

. 8.1.3 In example *a)* of (8.1.2), we saw that if Y is an integral prescheme of generic point y, $f : X \to Y$ a morphism of finite presentation, then, if the generic fibre $f^{-1}(y)$ is proper over $k(y)$, there is an open neighborhood U of y in Y such that $f^{-1}(U)$ is proper over U; it follows *a fortiori* that for all $s \in U$, $f^{-1}(s)$ is proper over $k(s)$. One arrives at a need for a converse, ensuring that if $f^{-1}(s)$ is proper over $k(s)$ for "sufficiently many" points $s \in U$, then $f^{-1}(y)$ is proper over $k(y)$. For example, suppose that X and Y are preschemes algebraic over a field k algebraically closed (one can take for k the field $\mathbf{C}$ of complex numbers, to fix ideas); one sometimes needs to know that if, for all $s \in Y$, *rational over* k, the fibre $f^{-1}(s)$ is proper over $k(s)$, then $f^{-1}(y)$ is proper over $k(y)$, and consequently $f^{-1}(U)$ is proper over U for a neighborhood U of y. This statement will follow easily from the following: the set E of points $s \in Y$ such that $f^{-1}(s)$ is proper over $k(s)$ is *constructible* (and hence equal to all of Y if it contains the closed points of Y, by the Hilbert Nullstellensatz ((10.4.8))); this amounts to saying that if $f^{-1}(y)$ *is not proper* over $k(y)$, then there exists an open neighborhood U of y such that $f^{-1}(s)$ is not proper over $k(s)$ for all $s \in U$ (cf. ((9.6.1), (iv))). This example illustrates the interest of systematically developing *constructibility criteria* for the most important notions: this is what will be done in § 9.

8.1. Projective limits of preschemes

. 8.2.1 Let S_0 be a ringed space, $\mathbf{L}$ a preordered filtered increasing set, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ an inductive system of $\mathscr{O}_{S_0}$-Algebras (not necessarily commutative), with $\mathbf{L}$ as index set. We know that, considered as an inductive system of $\mathscr{O}_{S_0}$-Modules, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ admits an inductive limit $\mathscr{A}$; denote by $\varphi_\lambda : \mathscr{A}_\lambda \to \mathscr{A}$ the canonical homomorphism. Let $m_\lambda : \mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \to \mathscr{A}_\lambda$ be the homomorphism of $\mathscr{O}_{S_0}$-Modules that defines the multiplication in the $\mathscr{O}_{S_0}$-Algebra $\mathscr{A}_\lambda$; the hypothesis on the $\varphi_{\mu\lambda}$ implies that the m_λ form an inductive system of homomorphisms, and as the functor $\varinjlim$ commutes with the tensor product, **IV-3** | 8
$m = \varinjlim m_\lambda$ is a homomorphism $\mathscr{A} \otimes \mathscr{A} \to \mathscr{A}$ of $\mathscr{O}_{S_0}$-Modules; by passage to the limit on the commutative diagrams expressing the associativity of m_λ and the existence of the identity section in $\mathscr{A}_\lambda$, one sees that m defines on $\mathscr{A}$ a structure of $\mathscr{O}_{S_0}$-Algebra and that φ_λ is a homomorphism of $\mathscr{O}_{S_0}$-Algebras for all $\lambda \in \mathbf{L}$. Furthermore $\mathscr{A}$ is the inductive limit of the system $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ *in the category of $\mathscr{O}_{S_0}$-Algebras*, in other words, for every $\mathscr{O}_{S_0}$-Algebra $\mathscr{B}$, the canonical map

(8.2.1.1)
$$\mathrm{Hom}_{S_0\text{-Alg.}}(\mathscr{A}, \mathscr{B}) \longrightarrow \varprojlim_{\lambda} \mathrm{Hom}_{S_0\text{-Alg.}}(\mathscr{A}_\lambda, \mathscr{B})$$

which to every homomorphism $f : \mathscr{A} \to \mathscr{B}$ of $\mathscr{O}_{S_0}$-Algebras associates the family $(f \circ \varphi_\lambda)$, is *bijective*. Indeed, one already knows that it is injective and identifies $\mathrm{Hom}_{S_0\text{-Alg.}}(\mathscr{A}, \mathscr{B})$ with a part of $\varprojlim \mathrm{Hom}_{S_0\text{-Mod.}}(\mathscr{A}_\lambda, \mathscr{B})$; it suffices to see that if (f_λ) is an inductive system of homomorphisms of $\mathscr{O}_{S_0}$-Algebras, $f_\lambda : \mathscr{A}_\lambda \to \mathscr{B}$, its inductive limit $f : \mathscr{A} \to \mathscr{B}$, which is by definition a homomorphism of $\mathscr{O}_{S_0}$-*Modules*, is also a homomorphism of $\mathscr{O}_{S_0}$-*Algebras*; but this follows from passage to the inductive limit in the commutative diagram

$$\begin{tikzcd}
\mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \arrow[r, "f_\lambda \otimes f_\lambda"] & \mathscr{B} \otimes \mathscr{B} \\
\mathscr{A}_\lambda \arrow[u, "m_\lambda"] \arrow[r, "f_\lambda"'] & \mathscr{B} \arrow[u]
\end{tikzcd}$$

and from the fact that the functor $\varinjlim$ commutes with tensor products. We note finally that if the $\mathscr{A}_\lambda$ are commutative $\mathscr{O}_{S_0}$-Algebras, so is $\mathscr{A}$.

. 8.2.2 Suppose now that S_0 is a *prescheme*, and that the $\mathscr{A}_\lambda$ are *quasi-coherent* $\mathscr{O}_{S_0}$-Algebras (commutative); we know then that $\mathscr{A} = \varinjlim \mathscr{A}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_0}$-Algebra (**I**, 4.1.1). Denote by S_λ (resp. $\mathscr{A}$) the spectrum of the $\mathscr{O}_{S_0}$-Algebra $\mathscr{A}_\lambda$ (resp. $\mathscr{A}$) (**II**, 1.3.1), and let $u_{\lambda\mu} : S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) and $u_\lambda : S \to S_\lambda$ be the S_0-morphisms corresponding to the homomorphisms $\varphi_{\mu\lambda}$ and φ_λ respectively (**II**, 1.2.7); it is clear that $(S_\lambda, u_{\lambda\mu})$ is a *projective system* in the category of S_0-preschemes. We note that the $u_{\lambda\mu}$ and u_λ are affine morphisms (**II**, 1.6.2), hence *quasi-compact* and *separated*.

Proposition (8.2.3). — *With the notations of* (8.2.2), *the morphisms $u_\lambda : S \to S_\lambda$ make S a projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of preschemes. Furthermore, if $h : S_0 \to T$ is a morphism, making every S_0-prescheme into a T-prescheme, S is also a projective limit of the system $(S_\lambda, u_{\lambda\mu})$ in the category of T-preschemes.*

PROOF. It suffices to show that if X is any S_0-prescheme, the canonical map **IV-3** | 9

(8.2.3.1)
$$\mathrm{Hom}_{S_0}(X, S) \longrightarrow \varprojlim_{\lambda} \mathrm{Hom}_{S_0}(X, S_\lambda)$$

which to every S_0-morphism $v : X \to S$ associates the family $(u_\lambda \circ v)$, is *bijective*. Now, if $g : X \to S_0$ is the structural morphism and if one sets $\mathscr{B} = g_*(\mathscr{O}_X)$, which is an $\mathscr{O}_{S_0}$-Algebra, the map (8.2.3.1) identifies canonically with (8.2.1.1) (**II**, 1.2.7), and the conclusion follows from what we saw in (8.2.1.1).

The other assertions of (8.1) are consequences of the following general lemma: □

Lemma (8.2.4). — *Let $\mathscr{C}$ be a category, T an object of $\mathscr{C}$, $\mathscr{C}_T$ the subcategory of objects of $\mathscr{C}$ above T. Let $(S_\lambda, u_{\lambda\mu})$ be a projective system in $\mathscr{C}_T$; then every projective limit of this system in $\mathscr{C}_T$ is also a projective limit in $\mathscr{C}$, and conversely.*

PROOF. Let $f_\lambda : S_\lambda \to T$ be the structural morphism. Suppose that S is the projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathscr{C}$, and denote by $u_\lambda : S \to S_\lambda$ the canonical corresponding morphisms. Consider then a projective system of T-morphisms $w_\lambda : Y \to S_\lambda$, where $Y \in \mathscr{C}_T$. By hypothesis there exists a unique morphism (in $\mathscr{C}$) $w : Y \to S$ such that $w_\lambda = u_\lambda \circ w$ for all λ. The hypothesis that the u_λ are T-morphisms implies that the morphisms $f_\lambda \circ u_\lambda : S \to T$ are all the same, and this morphism f makes S into a T-object. If $g : Y \to T$ is the structural morphism of Y, one has $f \circ w = f_\lambda \circ u_\lambda \circ w = f_\lambda \circ w_\lambda = g$ for all λ, which proves that w is a T-morphism. Conversely, suppose (with the same notations) that S is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathscr{C}_T$, and consider now a projective system of morphisms (of $\mathscr{C}$) $w_\lambda : Y \to S_\lambda$. The composed morphisms $f_\lambda \circ w_\lambda : Y \to T$ are then all the same: indeed, for any two indices λ, μ, there is an index ν such that $\lambda \leqslant \nu$ and $\mu \leqslant \nu$, hence $f_\nu = f_\lambda \circ u_{\lambda\nu} = f_\mu \circ u_{\mu\nu}$; as $w_\lambda = u_{\lambda\nu} \circ w_\nu$ and $w_\mu = u_{\mu\nu} \circ w_\nu$, one has $f_\lambda \circ w_\lambda = f_\lambda \circ u_{\lambda\nu} \circ w_\nu = f_\nu \circ w_\nu$ and similarly $f_\mu \circ w_\mu = f_\nu \circ w_\nu$. If $g : Y \to T$ is the unique morphism thus defined, g makes Y into a T-object, and the w_λ are T-morphisms; they thus have a projective limit $w : Y \to S$ which is a T-morphism, and *a fortiori* a morphism of $\mathscr{C}$; moreover the first part of the reasoning shows that every projective limit w' (in $\mathscr{C}$) of the projective system (w_λ) is necessarily also a T-morphism, hence equal to w, which finishes proving the lemma. □

Proposition (8.2.5). — *Under the conditions of (8.2.2), let α be an element of $\mathbf{L}$, X_α an S_α-prescheme. For all $\lambda \geqslant \alpha$, set $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, and for $\alpha \leqslant \lambda \leqslant \mu$, set $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, so that $(X_\lambda, v_{\lambda\mu})$ is a projective system of X_α-preschemes, whose index set is formed of the $\lambda \geqslant \alpha$ in $\mathbf{L}$. Set also $X = X_\alpha \times_{S_\alpha} S$ and $v_\lambda = 1_{X_\alpha} \times u_\lambda$. Then the X_α-morphisms $v_\lambda : X \to X_\lambda$ make X a projective limit of the projective system $(X_\lambda, v_{\lambda\mu})$ in the category of X_α-preschemes, or in the category of all preschemes.*

Proof. This will follow from the following general lemma: □

Lemma (8.2.6). — *Let $\mathscr{C}$ be a category in which fibered products exist, $q : T' \to T$ a morphism of $\mathscr{C}$, $\mathscr{C}_T$ (resp. $\mathscr{C}_{T'}$) the category of objects of $\mathscr{C}$ above T (resp. T'). Let $(S_\lambda, u_{\lambda\mu})$ be a projective system (not necessarily filtered) in $\mathscr{C}_T$, and set $S'_\lambda = S_\lambda \times_T T'$, $u'_{\lambda\mu} = u_{\lambda\mu} \times 1_{T'}$, so that $(S'_\lambda, u'_{\lambda\mu})$ is a projective system in $\mathscr{C}_{T'}$. Then, if (S, u_λ) is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathscr{C}_T$, $(S \times_T T', u_\lambda \times 1_{T'})$ is a projective limit of $(S'_\lambda, u'_{\lambda\mu})$ in $\mathscr{C}_{T'}$.* **IV-3 | 10**

Proof. By hypothesis, for all λ, one has a commutative diagram

$$\begin{tikzcd}
S' \arrow[r, "u'_\lambda"] \arrow[d, "p"'] & S'_\lambda \arrow[r, "h'_\lambda"] & T' \\
S \arrow[r, "u_\lambda"'] & S_\lambda \arrow[u, "p_\lambda"] \arrow[r, "h_\lambda"'] & T \arrow[u, "q"']
\end{tikzcd}$$

where we have set $S' = S \times_T T'$, $u'_\lambda = u_\lambda \times 1_{T'}$, $h'_\lambda = h_\lambda \times 1_{T'}$. Let Y be a T'-object, and consider a projective system of T'-morphisms $w'_\lambda : Y \to S'_\lambda$, where $w'_\lambda = p_\lambda \circ w'_\lambda$ are T-morphisms from Y to S_λ. The hypothesis that the u_λ are T-morphisms implies that the morphisms $f_\lambda \circ u_\lambda : S \to T$ are all the same, and this morphism f makes S into a T-object. By definition of the fibered product, there exists therefore by hypothesis a unique T-morphism $w : Y \to S$ such that $u_\lambda \circ w = w_\lambda$ for all λ. By definition of the fibered product, there is a unique T'-morphism $w' : Y \to S'$ such that $p \circ w' = w$. One has then $u'_\lambda \circ w' = p_\lambda \circ w'_\lambda$ for each λ; on the other hand, writing that w' is a T'-morphism and w'_λ a T'-morphism, one sees that $h'_\lambda \circ u'_\lambda \circ w' = g' = h'_\lambda \circ w'_\lambda$. The definition of S'_λ as fibered product $S_\lambda \times_T T'$, and the fact that w' is a T'-morphism, imply that w' is the unique T'-morphism verifying these relations, whence the lemma. □

Remark (8.2.7). — Given any ringed space S, the inductive limits with respect to any preordered (not necessarily filtered) set $\mathbf{L}$ of (not necessarily commutative) $\mathscr{O}_S$-Algebras exist by (8.2.1), and on the other hand, for two homomorphisms of $\mathscr{O}_S$-Algebras $\mathscr{A} \to \mathscr{B}$, $\mathscr{A} \to \mathscr{C}$, the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is the "amalgamated sum" in this category.

When S is a prescheme, the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is a quasi-coherent $\mathscr{O}_S$-Algebra whenever $\mathscr{A}$, $\mathscr{B}$, and $\mathscr{C}$ are (**I**, 1.3.13); one concludes that, in the category of quasi-coherent $\mathscr{O}_S$-Algebras, the inductive limits for any preordered index set always exist. This allows generalizing the definition of a projective limit of preschemes and the propositions (8.2.3) and (8.2.5) to the case where the preordered set $\mathbf{L}$ is not necessarily filtered.

. 8.2.8 With the notations of (8.2.2), set $u_{\lambda\mu} = (\psi_{\lambda\mu}, \theta_{\lambda\mu})$ and $u_\lambda = (\psi_\lambda, \theta_\lambda)$; $(S_\lambda, \psi_{\lambda\mu})$ is then a projective system of topological spaces and (ψ_λ) a projective system of continuous maps from the spaces underlying S and S_λ respectively.

Proposition (8.2.9). — *With the notations of (8.2.8), the projective limit of the projective system (ψ_λ) of continuous maps is a homeomorphism from the space underlying S to the projective limit of the projective system $(S_\lambda, \psi_{\lambda\mu})$ of topological spaces.*

PROOF. Let T be the topological limit space of the system $(S_\lambda, \psi_{\lambda\mu})$ and set $\psi = \varprojlim \psi_\lambda : S \to T$. **IV-3** | 11
One can reduce to the case where $S_0 = S_\alpha$ for an $\alpha \in \mathbf{L}$, and $\lambda \geqslant \alpha$.

(i) Let us show first that the topology of S is the inverse image by ψ of the topology of T; in other words, if $\pi_\lambda : T \to S_\lambda$ is the canonical map, the map $\psi^{-1}(\pi_\lambda^{-1}(U_\lambda))$, where U_λ runs through the open subsets of S_λ. Now every open subset of S is by definition a union of open subsets obtained as follows: one considers an affine open U_0 of S_0, of ring A_0, so that $\psi_\alpha^{-1}(U_0)$ is the open affine of S of ring $A = \Gamma(U_0, \mathscr{A})$, and one takes an element $f \in A$ and one considers in $\operatorname{Spec}(A)$, identified with $\psi_\alpha^{-1}(U_0)$, the open subset $\mathrm{D}(f)$. These open subsets $\mathrm{D}(f)$ form a base of the topology of S (**II**, 1.3.1). Now, if one sets $A_\lambda = \Gamma(U_0, \mathscr{A}_\lambda)$, one has $A = \varinjlim A_\lambda$ (**I**, 1.3.9), hence there exists an index λ such that f is the canonical image of an element $f_\lambda \in A_\lambda$; one has then $\mathrm{D}(f) = \psi_\lambda^{-1}(\mathrm{D}(f_\lambda))$ (**I**, 1.2.2), and as $\psi_\lambda = \pi_\lambda \circ \psi$, our assertion is shown.

(ii) Let us show now that ψ is *bijective*, which will finish the proof. As S is a Kolmogoroff space, it follows already from (i) that ψ is injective, and it remains to show that ψ is surjective. One can evidently replace T by an open $\pi_\alpha^{-1}(U_0)$, where U_0 is open affine in $S_\alpha = S_0$, hence one can reduce to the case where the S_λ and S are affine, in other words $\mathscr{A}_\lambda$ is the sheaf associated to an A_0-algebra A_λ, and $A = \varinjlim A_\lambda$ the canonical homomorphisms. By definition, an element of T is a family $(\mathfrak{p}_\lambda)_{\lambda \in \mathbf{L}}$ where $\mathfrak{p}_\lambda$ is a prime ideal of A_λ and where $\mathfrak{p}_\lambda = \varphi_{\mu\lambda}^{-1}(\mathfrak{p}_\mu)$ for $\lambda \leqslant \mu$. One knows then ((5.13.3) and (5.13.1)) that there is a prime ideal $\mathfrak{p}$ of A such that $\mathfrak{p}_\lambda = \varphi_\lambda^{-1}(\mathfrak{p})$ for all $\lambda \in \mathbf{L}$, which finishes the proof. □

In particular, one has thus shown the

Corollary (8.2.10). — *Let $(A_\lambda)_{\lambda \in \mathbf{L}}$ be a filtered inductive system of rings, and let $A = \varinjlim A_\lambda$, $\varphi_\lambda : A_\lambda \to A$ the canonical homomorphisms. The canonical map $\mathfrak{p} \mapsto (\varphi_\lambda^{-1}(\mathfrak{p}))$ is a homeomorphism from* $\operatorname{Spec}(A)$ *to the topological space* $\varprojlim \operatorname{Spec}(A_\lambda)$.

Corollary (8.2.11). — *With the notations of (8.2.8), for every quasi-compact open subset U of S, there exists an index λ and a quasi-compact open subset U_λ of S_λ such that $U = \psi_\lambda^{-1}(U_\lambda)$.*

PROOF. This follows from the fact that, by definition of the projective limit topology, the $\psi_\lambda^{-1}(U_\lambda)$ (U_λ open quasi-compact of S_λ) form a base of the topology of S, and since the index set $\mathbf{L}$ is filtered. □

Corollary (8.2.12). — *With the notations of (8.2.8), the inductive limit of the system of homomorphisms $\theta_\lambda^\# : \psi_\lambda^*(\mathscr{O}_{S_\lambda}) \to \mathscr{O}_S$ of sheaves of rings on S is an isomorphism*

(8.2.12.1) $$\varinjlim_\lambda \psi_\lambda^*(\mathscr{O}_{S_\lambda}) \xrightarrow{\sim} \mathscr{O}_S.$$

PROOF. One can evidently suppose the S_λ affine; with the notations of the proof of (8.2.9), it suffices to see that the inductive limit of the system of canonical maps $(A_\lambda)_{\mathfrak{p}_\lambda} \to A_{\mathfrak{p}}$ is an isomorphism, which is nothing other than $(0_I, 5.13.3, (ii))$. □

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Proposition (8.2.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions, so that S_μ identifies with an open subset of S_λ for $\lambda \leqslant \mu$. Then, for all $\alpha \in \mathbf{L}$, the underlying space of the prescheme S identifies with the subspace of S_α which is the intersection of the S_λ for $\lambda \geqslant \alpha$, and the structural sheaf $\mathscr{O}_S$ with the sheaf induced by $\mathscr{O}_{S_\alpha}$ on this intersection; more generally, for every $\mathscr{O}_{S_\alpha}$-Module $\mathscr{F}_\alpha$, $u_\alpha^*(\mathscr{F}_\alpha)$ identifies with the $\mathscr{O}_S$-Module induced by $\mathscr{F}_\alpha$ on S.*

PROOF. The first assertion follows from (8.2.9), the definition of a projective limit of topological spaces; moreover the $\psi_\lambda^*(\mathscr{O}_{S_\lambda})$ are all equal to the sheaf induced by $\mathscr{O}_{S_\alpha}$ on S; with the notations of the proof of (8.2.9), the homomorphisms $(A_\lambda)_{\mathfrak{p}_\lambda} \to (A_\mu)_{\mathfrak{p}_\mu}$ are bijective for a system $(\mathfrak{p}_\lambda)$ of prime ideals corresponding to a single point of S; the assertion relative to $\mathscr{O}_S$ follows from (8.2.12). The last assertion then follows from the definition of $u_\alpha^*(\mathscr{F}_\alpha)$ $(0_I, 4.3.1)$. ☐

Remark (8.2.14). — The results of (8.2.9) and (8.2.12) show that S is the projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of *all* ringed spaces (or of all locally ringed spaces). Indeed, let Y be a ringed space, and consider a projective system of morphisms of ringed spaces $w_\lambda : Y \to S_\lambda$. If one sets $w_\lambda = (\rho_\lambda, \omega_\lambda)$, the ρ_λ form a projective system of continuous maps, and the ω_λ form a system of homomorphisms of sheaves of rings; in view of (8.2.9), their limits projective give respectively a continuous map $\rho : Y \to S$ and, in view of (8.2.12), a homomorphism of sheaves of rings $\omega^\# : \rho^*(\mathscr{O}_S) \to \mathscr{O}_Y$ such that $\omega_\lambda^\# = \omega^\# \circ \rho^*(\theta_\lambda^\#)$, which shows our assertion.

8.2. Constructible parts in a projective limit of preschemes

. 8.3.1 Throughout this paragraph, we assume the conditions of (8.2.2) are satisfied, and we keep the same notations.

Theorem (8.3.2). — *For all λ, let E_λ, F_λ be two subsets of S_λ. Set*

$$E = \bigcap_\lambda u_\lambda^{-1}(E_\lambda), \qquad F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda). \tag{8.3.2.1}$$

Suppose the following conditions are satisfied:

(i) *For all λ, E_λ is pro-constructible and F_λ is ind-constructible (1.9.4).*

(ii) *For $\lambda \leqslant \mu$, we have*

$$E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda), \qquad F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda).$$

(iii) *There exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact (which implies that S_λ is quasi-compact for $\lambda \geqslant \alpha$).*

Then the following conditions are equivalent:

a) *$E \subset F$.*

b) *There exists $\lambda \geqslant \alpha$ such that $u_\lambda^{-1}(E_\lambda) \subset u_\lambda^{-1}(F_\lambda)$ (and then $u_\mu^{-1}(E_\mu) \subset u_\mu^{-1}(F_\mu)$ for $\mu \geqslant \lambda$).*

c) *There exists $\lambda \geqslant \alpha$ such that $E_\lambda \subset F_\lambda$ (and then $E_\mu \subset F_\mu$ for $\mu \geqslant \lambda$).*

The remarks in parentheses in b) *and* c) *follow from (ii). Set* **IV-3 | 13**

$$G_\lambda = E_\lambda \cap (S_\lambda - F_\lambda), \qquad G = E \cap (S - F).$$

Then G_λ is a pro-constructible part of S_λ (1.9.5, (i)), and by (8.3.2.1) and (ii), one has

$$G_\mu \subset u_{\lambda\mu}^{-1}(G_\lambda) \quad \text{for } \lambda \leqslant \mu$$

$$G = \bigcap_\lambda u_\lambda^{-1}(G_\lambda).$$

One is thus reduced to proving the particular case of (8.3.2) corresponding to $F_\lambda = \varnothing$ for all λ:

Corollary (8.3.3). — *For all λ, let E_λ be a pro-constructible part of S_λ such that, for $\lambda \leqslant \mu$, $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$. Suppose there exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact. Then the following conditions are equivalent:*

a) *$E = \bigcap_\lambda u_\lambda^{-1}(E_\lambda) = \varnothing$.*

b) *There exists λ such that $u_\lambda^{-1}(E_\lambda) = \varnothing$ (and then $u_\mu^{-1}(E_\mu) = \varnothing$ for $\mu \geqslant \lambda$).*

c) *There exists λ such that $E_\lambda = \varnothing$ (and then $E_\mu = \varnothing$ for $\mu \geqslant \lambda$).*

PROOF. It is clear that *c)* implies *b)*: since S_λ is quasi-compact, so is S (8.2.9); E_λ being pro-constructible, so is $u_\lambda^{-1}(E_\lambda)$ (1.9.5, (vi)); then the filtering decreasing family of pro-constructible sets $u_\lambda^{-1}(E_\lambda)$ has empty intersection, hence (1.9.9) one of them is empty.

Finally, let us show that *b)* implies *c)*. As S_α is quasi-compact and $\mathbf{L}$ is filtered, one can replace S_α by an open affine, hence suppose (8.2.2) that S and the S_λ for $\lambda \geqslant \alpha$ are affine; one has then (1.9.2.1), for $\lambda \geqslant \alpha$,

$$u_\lambda(S) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu).$$

Hence $E_\lambda \cap u_\lambda(S) = \bigcap_{\mu \geqslant \lambda}(E_\lambda \cap u_{\lambda\mu}(S_\mu))$. Now, as u_λ and the $u_{\lambda\mu}$ are quasi-compact, $u_\lambda(S)$ and $u_{\lambda\mu}(S_\mu)$ are pro-constructible in S_λ (1.9.5, (vii)), hence the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ for $\mu \geqslant \lambda$ form a filtering decreasing family of pro-constructible parts of S_λ. As S_λ is quasi-compact, the hypothesis *b)* implies that the intersection of this family is empty, hence (1.9.9) one of the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ is empty, hence $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ is empty. Q.E.D. □

Corollary (8.3.4). — *For all λ, let F_λ be an ind-constructible part of S_λ such that for $\lambda \leqslant \mu$, $u_{\lambda\mu}^{-1}(F_\lambda) \subset F_\mu$. Suppose there exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact. Then the following conditions are equivalent:*

a) *The set $F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda)$ is equal to S.*
b) *There exists λ such that $u_\lambda^{-1}(F_\lambda) = S$ (and then $u_\mu^{-1}(F_\mu) = S$ for $\mu \geqslant \lambda$).*

This follows immediately from (8.3.3) by passage to complements.

Corollary (8.3.5). — *For all λ, let E_λ, F_λ be two constructible parts of S_λ such that, for $\mu \geqslant \lambda$, $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ and $F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda)$. Suppose there exists an index α such that S_α is quasi-compact. Then, for $E \subset F$ (resp. $E = F$), it is necessary and sufficient that there exists λ such that $E_\lambda \subset F_\lambda$ (resp. $E_\lambda = F_\lambda$), in which case we also have $E_\mu \subset F_\mu$ (resp. $E_\mu = F_\mu$) for $\mu \geqslant \lambda$. This is a particular case of (8.3.2).* **IV-3** | 14

Corollary (8.3.6). — *Suppose there exists α such that S_α is quasi-compact. For $S = \varnothing$, it is necessary and sufficient that there exists λ such that $S_\lambda = \varnothing$.*

Corollary (8.3.7). — *On a, for all λ,*

$$u_\lambda(S) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu). \tag{8.3.7.1}$$

PROOF. It is clear that the first member of (8.3.7.1) is contained in the second. Let s be a point of S_λ and set $X_\lambda = \operatorname{Spec}(k(s))$; consider the projective system $(X_\mu, v_{\mu\nu})$ where $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ and $v_{\mu\nu} = 1 \times u_{\mu\nu}$ for $\lambda \leqslant \mu \leqslant \nu$; its projective limit is $X = X_\lambda \times_{S_\lambda} S$ (8.2.5). If $s \in \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu)$, this implies that $X_\mu \neq \varnothing$ for all $\mu \geqslant \lambda$ (**I**, 3.4.8); it follows from (8.3.6) that $X \neq \varnothing$, hence $s \in u_\lambda(S)$, hence (**I**, 3.4.8). □

Proposition (8.3.8). — (i) *For the morphism $u_\lambda : S \to S_\lambda$ to be dominant (resp. surjective), it is necessary and sufficient that for $\mu \geqslant \lambda$ the morphism $u_{\lambda\mu} : S_\mu \to S_\lambda$ be dominant (resp. surjective).*

(ii) *If, for an index λ, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat (resp. faithfully flat) for $\mu \geqslant \lambda$, then the morphism $u_\lambda : S \to S_\lambda$ is flat (resp. faithfully flat).*

(iii) *Suppose that the morphisms $u_\mu : S \to S_\mu$ are surjective for $\mu \geqslant \lambda$. For u_λ to be an open (resp. universally open) morphism, it is necessary and sufficient that, for $\mu \geqslant \lambda$, $u_{\lambda\mu}$ be an open (resp. universally open) morphism.*

PROOF. (i) As $u_\lambda(S) \subset u_{\lambda\mu}(S_\mu)$ for $\mu \geqslant \lambda$, the necessity of the conditions is trivial, and it follows immediately from (8.3.7.1) that if the $u_{\lambda\mu}$ are surjective, so is u_λ. Suppose now the $u_{\lambda\mu}$ dominant for $\mu \geqslant \lambda$, and consider in S_λ an open quasi-compact non-empty subset U_λ; then $u_{\lambda\mu}(S_\mu) \cap U_\lambda$ is non-empty for all μ, hence by (8.3.7.1), $u_\lambda(S) \cap U_\lambda$ is non-empty, hence u_λ is dominant. **IV-3** | 15

We note that it can happen that all the $u_{\lambda\mu}$ are open without it being so for u_λ if the u_μ are not surjective.

(ii) By (i), it suffices to consider the case where the $u_{\lambda\mu}$ are flat; one can then reduce to the case where S_λ is affine, hence the S_μ for $\mu \geqslant \lambda$ and S are also affine, and the assertion follows from $(0_I, 6.2.3)$.

(iii) By (8.2.5) and (**I**, 3.5.2), it suffices to treat the case of open morphisms. As $u_\lambda = u_{\lambda\mu} \circ u_\mu$ and u_μ is surjective, one knows that if u_λ is open it is the same of $u_{\lambda\mu}$ (Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., § 5, n^o 1, prop. 1). □

Conversely, to show that u_λ is open when all the $u_{\lambda\mu}$ are for $\mu \geqslant \lambda$, it suffices to see that for every open quasi-compact V of S, $u_\lambda(V)$ is open in S_λ; but there exists then $\mu \geqslant \lambda$ such that $V = u_\mu^{-1}(V_\mu)$, where V_μ is open in S_μ (8.2.11); as u_μ is surjective, one has $V_\mu = u_\mu(V)$ and $u_\lambda(V) = u_{\lambda\mu}(u_\mu(V))$ is therefore open by hypothesis.

. 8.3.9 For any prescheme X, we will denote as usual by $\mathfrak{P}(X)$ the power set of the space underlying X, by $\mathfrak{C}(X)$ (resp. $\mathfrak{Oc}(X)$, $\mathfrak{Fc}(X)$, $\mathfrak{LFc}(X)$) the set of constructible parts (resp. constructible and open, constructible and closed, locally constructible and closed) of X. It is clear that $(\mathfrak{P}(S_\lambda), u_{\lambda\mu}^{-1})$ is an *inductive system of sets* and that the maps $u_\lambda^{-1} : \mathfrak{P}(S_\lambda) \to \mathfrak{P}(S)$ form an inductive system of maps, whence, passing to the inductive limit, a canonical map

$$u_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \longrightarrow \mathfrak{P}(S). \tag{8.3.9.1}$$

Furthermore, it follows from (1.8.2) that $u_{\lambda\mu}^{-1}$ sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Oc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{LFc}(S_\lambda)$) into $\mathfrak{C}(S_\mu)$ (resp. $\mathfrak{Oc}(S_\mu)$, $\mathfrak{Fc}(S_\mu)$, $\mathfrak{LFc}(S_\mu)$) and that u_λ^{-1} sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Oc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{LFc}(S_\lambda)$) into $\mathfrak{C}(S)$ (resp. $\mathfrak{Oc}(S)$, $\mathfrak{Fc}(S)$, $\mathfrak{LFc}(S)$). Hence by restriction of (8.3.9.1), canonical maps

$$\varinjlim \mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{C}(S) \tag{8.3.9.2}$$

$$\varinjlim \mathfrak{Oc}(S_\lambda) \longrightarrow \mathfrak{Oc}(S) \tag{8.3.9.3}$$

$$\varinjlim \mathfrak{Fc}(S_\lambda) \longrightarrow \mathfrak{Fc}(S) \tag{8.3.9.4}$$

$$\varinjlim \mathfrak{LFc}(S_\lambda) \longrightarrow \mathfrak{LFc}(S). \tag{8.3.9.5}$$

. 8.3.10 Let $g_\alpha : X_\alpha \to S_\alpha$ be a morphism; with the notations of (8.2.5) one has a canonical map $v_{\mathfrak{P}} : \varinjlim \mathfrak{P}(X_\lambda) \to \mathfrak{P}(X)$; on the other hand, for all $\lambda \geqslant \alpha$ and a morphism of projection $g_\lambda : X_\lambda \to S_\lambda$, it is clear that $g_\lambda^{-1} : \mathfrak{P}(S_\lambda) \to \mathfrak{P}(X_\lambda)$ form an inductive system of maps, and that the diagrams

$$\begin{tikzcd} \varinjlim \mathfrak{P}(S_\lambda) \arrow[r, "u_{\mathfrak{P}}"] \arrow[d, "\varinjlim g_\lambda^{-1}"'] & \mathfrak{P}(S) \arrow[d, "g^{-1}"] \\ \varinjlim \mathfrak{P}(X_\lambda) \arrow[r, "v_{\mathfrak{P}}"'] & \mathfrak{P}(X) \end{tikzcd} \tag{8.3.10.1}$$

are commutative, and one has analogous diagrams replacing $\mathfrak{P}$ by $\mathfrak{C}$, $\mathfrak{Oc}$, $\mathfrak{Fc}$, or $\mathfrak{LFc}$.

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Theorem (8.3.11). — *Suppose there exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact and quasi-separated. Then the canonical maps* (8.3.9.2), (8.3.9.3), (8.3.9.4), *and* (8.3.9.5) *are* bijective*.*

PROOF. By the preceding remark, it follows from (8.3.5) that under the hypothesis that for an $\alpha \in \mathbf{L}$, S_α is *quasi-compact*, the map (8.3.9.2) is *injective*. Furthermore:

It follows that these maps are *surjective*; as every constructible part of S is a finite union of sets of the form $U \cap V^c$, where U and V are open and constructible, it will suffice to prove that (8.3.9.3) is surjective (and also (8.3.9.4), by passage to complements). Now, as the morphisms $S_\lambda \to S_\alpha$ and $S \to S_\alpha$ are separated, the S_λ for $\lambda \geqslant \alpha$ and S are quasi-compact and quasi-separated, and one knows that the constructible open parts in such a prescheme are the quasi-compact open subsets (1.8.1). The conclusion follows hence from (8.2.11), except for the map (8.3.9.5). To prove that the latter is surjective, consider a locally closed and constructible part Z in S; Z is then quasi-compact (0_{III}, 9.1.8). As every point $x \in Z$ admits by hypothesis a quasi-compact open neighborhood V_x in S such that $Z \cap V_x$ is closed in V_x, one can cover Z by a finite number of the V_x, hence there is a quasi-compact open U containing Z such that Z is closed in U; as Z is also constructible in S, it is also constructible in U (0_{III}, 9.1.8). One knows (8.2.11) that there is an index λ and a quasi-compact open U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$. Applying to U (which is a projective limit of the $U_\mu = u_{\lambda\mu}^{-1}(U_\lambda)$ for $\mu \geqslant \lambda$) the fact that (8.3.9.4) is surjective, one sees that there exists $\mu \geqslant \lambda$ and a closed constructible set Z_μ *in* U_μ such that $Z = u_\mu^{-1}(Z_\mu)$. But as the canonical immersion $U_\mu \to S_\mu$ is quasi-compact by hypothesis (1.2.7), each U_μ is open quasi-compact in S_μ, so that Z_μ is a constructible part of S_μ; each of the $Z_{\mu\lambda}$ is of finite presentation on S_μ (1.6.2). This finishes the proof. □

Corollary (8.3.12). — *Suppose S_0 quasi-compact, and let, for all λ, Z_λ be a constructible part of S_λ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ for $\mu \geqslant \lambda$. For Z to be open (resp. closed, resp. locally closed) in S, it is necessary and sufficient that there exists α such that Z_λ is so in S_λ.* **IV-3 | 17**

PROOF. Cover S_α by a finite number of affine open subsets $U_\alpha^{(i)}$; then the $U^{(i)} = u_\alpha^{-1}(U_\alpha^{(i)})$ form a finite open affine covering of S, and for Z to be open (resp. closed) in S, it is necessary and sufficient that each $Z \cap U^{(i)}$ be open (resp. closed) in $U^{(i)}$. As $\mathbf{L}$ is filtered and S_α is affine, hence quasi-compact and quasi-separated, the result follows from (8.3.11). □

Proposition (8.3.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat for $\lambda \leqslant \mu$, and that there exists α such that S_α is quasi-compact. For all λ, let Z'_λ, Z''_λ be two pro-constructible parts of S_λ, such that $Z'_\mu = u_{\lambda\mu}^{-1}(Z'_\lambda)$ and $Z''_\mu = u_{\lambda\mu}^{-1}(Z''_\lambda)$ for $\mu \geqslant \lambda$; suppose in addition that $\overline{Z''_\alpha}$ is constructible in S_α. Let $Z' = u_\lambda^{-1}(Z'_\lambda)$, $Z'' = u_\lambda^{-1}(Z''_\lambda)$; for $Z'' \subset \overline{Z'}$, it is necessary and sufficient that there exists $\lambda \geqslant \alpha$ such that $Z''_\lambda \subset \overline{Z'_\lambda}$.*

PROOF. Indeed, one knows that $u_\lambda : S \to S_\lambda$ is also a flat morphism for all λ (8.3.8); as Z'_λ is pro-constructible, the closure of Z'_μ for $\mu \geqslant \lambda$ (resp. of Z') in S_μ (resp. S) is equal to $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ (resp. $u_\lambda^{-1}(\overline{Z'_\lambda})$) (2.3.10). As the $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ and $u_\lambda^{-1}(\overline{Z'_\lambda})$ are constructible (1.8.2), the conclusion follows from (8.3.2). □

8.3. Criteria for irreducibility and connectedness for projective limits of preschemes

Proposition (8.4.1). — *Suppose there exists an index α such that S_α is quasi-compact.*

(i) *If S is not irreducible and if moreover the underlying space of S is noetherian and S_α is quasi-separated, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not irreducible.*

(ii) *If S is not connected, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not connected.*

PROOF. Suppose that S is the union of two distinct closed sets S', S'' (resp. disjoint non-empty closed sets S', S'') in S. In case (i), S' and S'' are constructible since the space S is noetherian. By (8.3.11), there exist $\lambda \geqslant \alpha$ and two closed constructible sets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$; as $S = S' \cup S''$, it follows from (8.3.11) that one can suppose $S_\lambda = S'_\lambda \cup S''_\lambda$; as S'_λ and S''_λ are distinct from S_λ, this proves that S_λ is not irreducible.

In case (ii), S' and S'' are open quasi-compact, hence, by (8.2.11), there exist $\lambda \geqslant \alpha$ and two open quasi-compact subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$. Moreover, S' and S'' are at once pro-constructible and ind-constructible, hence constructible (1.9.11), and it follows from (8.3.5) that one can take λ such that $S_\lambda = S'_\lambda \cup S''_\lambda$ and $S'_\lambda \cap S''_\lambda = \varnothing$; this shows that S_λ is not connected. □

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Proposition (8.4.2). — *Suppose that the underlying space of S is noetherian and that one of the two following conditions is satisfied:*

a) *For $\lambda \leqslant \mu$, $u_{\lambda\mu} : S_\mu \to S_\lambda$ is dominant, and there exists α such that S_α is quasi-compact.*

b) *There exists α such that the underlying space of S_α is noetherian, and for $\mu \geqslant \lambda$, $u_{\lambda\mu}$ is a homeomorphism from S_μ to a subspace of S_λ.*

Under these conditions, there exists λ such that, for $\mu \geqslant \lambda$:

(i) *For every irreducible component Y_i of S $(1 \leqslant i \leqslant m)$, $\overline{u_\mu(Y_i)}$ is an irreducible component of S_μ, and the map $Y_i \mapsto \overline{u_\mu(Y_i)}$ is a bijection from the set of irreducible components of S to the set of irreducible components of S_μ.*

(ii) *For every connected component C_j of S $(1 \leqslant j \leqslant n)$, $\overline{u_\mu(C_j)}$ is a connected component of S_μ, and the map $C_j \mapsto \overline{u_\mu(C_j)}$ is a bijection from the set of connected components of S to the set of connected components of S_μ.*

PROOF. We first establish the

Lemma (8.4.2.1). — *Under condition* a) *or* b) *of (8.4.2), there exists λ such that, for $\mu \geqslant \lambda$, $u_\mu : S \to S_\mu$ is dominant.*

In case *a)*, this has already been proved without supposing S noetherian (8.3.8). In case *b)*, set $Z_\alpha = \overline{u_\alpha(S)}$; as a closed part of the noetherian S_α, Z_α is constructible, and as $u_\alpha^{-1}(Z_\alpha) = S$, it follows from (8.3.5) that there exists $\lambda \geqslant \alpha$ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda) = S_\mu$. But as $u_{\lambda\mu}$ is a homeomorphism from S_μ to a subspace and the composed map $S \to Z_\mu \to Z_\alpha$ is dominant, it is the same of $S \to Z_\mu = S_\mu$.

The lemma being established, we can suppose that for all λ, u_λ is dominant.

(i) Each S_λ is a union of the $\overline{u_\lambda(Y_i)}$, which are irreducible. On the other hand, if U_i is the complement of S of the union of the Y_j of index $j \neq i$ $(1 \leqslant i \leqslant m)$, the U_i are pairwise disjoint and $Y_i = \overline{U_i}$ (0_I, 2.1.6). As the space S is noetherian, the U_i are quasi-compact, hence there exists an index λ and open subsets $U_{i\lambda}$ of S_λ such that $U_i = u_\lambda^{-1}(U_{i\lambda})$ for $1 \leqslant i \leqslant m$ (8.2.11). One concludes that if one sets $U_{i\mu} = u_{\lambda\mu}^{-1}(U_{i\lambda})$ for $\lambda \leqslant \mu$, the $U_{i\mu}$ are pairwise disjoint, the $U_i = u_\mu^{-1}(U_{i\mu})$ are the same and u_μ is dominant. Consequently, no $\overline{U_{i\mu}}$ is contained in another; hence $\overline{U_{i\mu}} = u_\mu(Y_i)$, which proves that the $\overline{U_{i\mu}}$ are the irreducible components of S_μ, and finishes the proof.

(ii) As the space S is noetherian, the C_j are open and closed in S and quasi-compact; the same reasoning as in (i) shows that there exist λ and open subsets $V_{j\lambda}$ of S_λ such that $C_j = u_\lambda^{-1}(V_{j\lambda})$ for $1 \leqslant j \leqslant n$. One sees also, as in (i), that if one sets $V_{j\mu} = u_{\lambda\mu}^{-1}(V_{j\lambda})$ for $\mu \geqslant \lambda$, the $V_{j\mu}$ are pairwise disjoint, and $u_\mu(C_j)$ is dense in $V_{j\mu}$. Furthermore, it follows from (8.3.4) that for μ large enough, the union of the $V_{j\mu}$ is S_μ, since every open subset in a prescheme is ind-constructible (1.9.6). The $V_{j\mu}$ are thus the connected components of S_μ. ☐

Corollary (8.4.3). — *Suppose one of the conditions* a), b) *of (8.4.2) is satisfied, the underlying space of S being noetherian; then, for S to be irreducible, it is necessary and sufficient that there exists λ such that S_μ is irreducible for all $\mu \geqslant \lambda$.*

Proposition (8.4.4). — *Suppose there exists α such that S_α is quasi-compact and that, for $\lambda \leqslant \mu$, $u_{\lambda\mu} : S_\mu \to S_\lambda$ is dominant. Then, for S to be connected, it is necessary and sufficient that there exists λ_0 such that S_λ is connected for all $\lambda \geqslant \lambda_0$.* **IV-3 | 19**

PROOF. The condition is sufficient by (8.4.1); on the other hand, by (8.3.8), (i), we have seen that $u_\lambda : S \to S_\lambda$ is dominant for all λ sufficiently large, so if S is connected, since $u_\lambda(S)$ is dense in S_λ and connected, S_λ is connected. ☐

Corollary (8.4.5). — *Let k be a field, X a k-prescheme quasi-compact and connected. For X to be geometrically connected (4.5.2), it is necessary and sufficient that, for every finite and separable extension K of k, $X \otimes_k K$ be connected.*

PROOF. The condition is trivially necessary. To see that it is sufficient, we note that if Ω is an algebraic closure of k, $X \otimes_k \Omega$ is connected (4.5.1) and, for $k \subset k' \subset k'' \subset \Omega$, the morphism $X \otimes_k k'' \to X \otimes_k k'$ is surjective. One is thus reduced, by (8.4.4), to proving that $X \otimes_k k'$ is connected for every finite extension k' of k. But if K is the largest separable extension contained in k', the morphism $X \otimes_k k' \to X \otimes_k K$ is finite, surjective and radiciel, hence (2.4.5) a homeomorphism, and as $X \otimes_k K$ is connected by hypothesis, so is $X \otimes_k k'$. ☐

Remarks (8.4.6). —

(i) The proof of (8.4.2) shows that the conclusion of the proposition is valid if one only supposes that the underlying space of S is noetherian, that there exists α such that S_α is quasi-compact, and finally that the $u_\lambda : S \to S_\lambda$ are dominant.

(ii) On the other hand, the conclusion of (8.4.2) can fail when the u_λ are not dominant for λ sufficiently large, even when the S_λ and S are noetherian, as the following example shows, taking $\mathbf{N}$ for the set of indices, all the S_n equal to $\mathrm{Spec}(A \times K) = \mathrm{Spec}(A) \amalg \mathrm{Spec}(K)$, where K is a field, A a K-algebra of any kind, and all the morphisms $u_{n,n+1}$ equal to the same morphism corresponding to the homomorphism $(x, y) \mapsto (j(y), y)$ of $A \times K$ into itself, where $j : K \to A$ is the canonical homomorphism. One easily verifies that the inductive limit of this system of rings is K, the canonical homomorphism u_n corresponding to the second projection $A \times K \to K$. One sees then that $S = \mathrm{Spec}(K)$ is irreducible even though none of the S_n are connected.

8.4. Modules of finite presentation on a projective limit of preschemes

. 8.5.1 We continue to use the notations and hypotheses of (8.2.2); we will moreover limit ourselves in this section to the case where S_0 is one of the S_λ, which one can always reduce to.

When, in this section, we consider a family $(\mathscr{F}_\lambda)$, where, for all $\lambda \in \mathbf{L}$, $\mathscr{F}_\lambda$ is an $\mathscr{O}_{S_\lambda}$-Module, it will be understood that this family satisfies the condition

(8.5.1.1) $$\mathscr{F}_\mu = u^*_{\lambda\mu}(\mathscr{F}_\lambda) \quad \text{for } \lambda \leqslant \mu.$$

We will then set **IV-3** | 20

(8.5.1.2) $$\mathscr{F} = u^*_\lambda(\mathscr{F}_\lambda)$$

which is an $\mathscr{O}_S$-Module not depending on the index λ, by virtue of (8.5.1.1).

Let now $(\mathscr{F}_\lambda)$, $(\mathscr{G}_\lambda)$ be two such families of $\mathscr{O}_{S_\lambda}$-Modules. It is clear that the maps $f_\lambda \mapsto u^*_{\lambda\mu}(f_\lambda)$ from $\mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda)$ into $\mathrm{Hom}_{\mathscr{O}_{S_\mu}}(\mathscr{F}_\mu, \mathscr{G}_\mu)$ define an inductive system of abelian groups $(\mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda), u^*_{\lambda\mu})$, and that the maps $f_\lambda \mapsto u^*_\lambda(f_\lambda)$ form an *inductive system* of homomorphisms of abelian groups

$$u^*_\lambda : \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{F}, \mathscr{G});$$

whence, passing to the inductive limit, a canonical homomorphism of abelian groups

(8.5.1.3) $$u_{\mathscr{F},\mathscr{G}} : \varinjlim \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{F}, \mathscr{G}).$$

We note that when $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$, the condition (8.5.1.1) is satisfied, and $\mathscr{F} = \mathscr{O}_S$; the homomorphism (8.5.1.3) gives then (0_I, 5.1.1) a canonical homomorphism of abelian groups

(8.5.1.4) $$u_{\mathscr{G}} : \varinjlim \Gamma(S_\lambda, \mathscr{G}_\lambda) \longrightarrow \Gamma(S, \mathscr{G}).$$

Theorem (8.5.2). — (i) *Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated) and that, for an $\lambda \in \mathbf{L}$, $\mathscr{F}_\lambda$ is quasi-coherent and of finite type (resp. of presentation finie) and $\mathscr{G}_\lambda$ quasi-coherent. Then the homomorphism $u_{\mathscr{F},\mathscr{G}}$ is* injective *(resp.* bijective*).*

(ii) *Suppose S_0 quasi-compact and quasi-separated. For every $\mathscr{O}_S$-Module quasi-coherent $\mathscr{F}$ of finite presentation, there exist $\lambda \in \mathbf{L}$ and an $\mathscr{O}_{S_\lambda}$-Module quasi-coherent $\mathscr{F}_\lambda$ of finite presentation such that $\mathscr{F}$ is isomorphic to $u^*_\lambda(\mathscr{F}_\lambda)$.*

PROOF. (i) One can evidently reduce to the case where $S_0 = S_\lambda$ and the morphisms $u_{0\lambda} : S_\lambda \to$ **IV-3** | 21
S_0 are affine. Consider first the case where $S_0 = \mathrm{Spec}(A_0)$ is *affine*. Then the assertion (i) is equivalent to the

Lemma (8.5.2.1). — *Let A_0 be a ring, $(A_\lambda)_{\lambda\in\mathbf{L}}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; let M_0, N_0 be two A_0-modules, and set $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$, $M = M_0 \otimes_{A_0} A = \varinjlim M_\lambda$, $N = N_0 \otimes_{A_0} A = \varinjlim N_\lambda$. If M_0 is of finite type (resp. of finite presentation), the canonical homomorphism*

(8.5.2.2) $$\varinjlim \mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \longrightarrow \mathrm{Hom}_A(M, N)$$

is injective (resp. bijective).

One knows that there are canonical functorial isomorphisms

$$\mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \simeq \mathrm{Hom}_{A_0}(M_0, N_\lambda), \qquad \mathrm{Hom}_A(M, N) \simeq \mathrm{Hom}_{A_0}(M_0, N)$$

so that (8.5.2.2) is nothing other, up to canonical isomorphisms, than the canonical homomorphism

(8.5.2.3) $$\varinjlim \mathrm{Hom}_{A_0}(M_0, N_\lambda) \longrightarrow \mathrm{Hom}_{A_0}(M_0, \varinjlim N_\lambda)$$

which to every inductive system of A_0-module homomorphisms $\theta_\lambda : M_0 \to N_\lambda$ associates its inductive limit. Now, if M_0 is of finite type (resp. of finite presentation), one has an exact sequence $A_0^n \to M_0 \to 0$ (resp. $A_0^p \to A_0^n \to M_0 \to 0$); as it is clear that (8.5.2.3) is bijective when $M_0 = A_0^n$, it suffices to use the left exactness of the $\mathrm{Hom}_{A_0}(M_0, P)$ functor and the exactness of the functor $\varinjlim$ (in the category of abelian groups) to conclude.

Let us now pass to the case where S_0 is quasi-compact, and let (U_i) be a finite open affine covering of S_0; for all λ, the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form a finite open affine covering of S_λ, and the $V_i = u_0^{-1}(U_i)$ a finite open affine covering of S. To see that $u_{\mathscr{F},\mathscr{G}}$ is injective, it suffices, if $f = u^*_\lambda(f_\lambda)$

is such that $f = 0$, to show that there exists $\mu \geqslant \lambda$ such that $f_\mu = u^*_{\lambda\mu}(f_\lambda) = 0$. By the lemma (8.5.2.1), for each i there exists $\mu \geqslant \lambda$ such that $f_\mu | U_{i,\mu} = 0$; it thus suffices to take μ larger than all the λ_i.

Suppose moreover S_0 quasi-separated and $\mathscr{F}_0$ of finite presentation, and let $f : \mathscr{F} \to \mathscr{G}$ be an $\mathscr{O}_S$-homomorphism. By the lemma (8.5.2.1), for each i, there exist λ_i and a homomorphism $f^{(i)}_{\lambda_i} : \mathscr{F}_{\lambda_i} | U_{i,\lambda_i} \to \mathscr{G}_{\lambda_i} | U_{i,\lambda_i}$ such that $u^*_{\lambda_i}(f^{(i)}_{\lambda_i}) = f | V_i$ (with the notations of the lemma). As **L** is filtered, one can suppose all the λ_i equal to a same λ. Note now that S_λ is quasi-separated (0_I, 5.2.5), and $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_\lambda}$-Module of finite presentation; as, for every pair of indices i, j, $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.2.5) that for every pair (i, j), there exists an index $\lambda_{ij} \geqslant \lambda$ such that $u^*_{\lambda\mu}(f^{(i)}_\lambda | U_{ij\lambda}) = u^*_{\lambda\mu}(f^{(j)}_\lambda | U_{ij\lambda})$ for $\mu \geqslant \lambda_{ij}$; taking μ larger than all the λ_{ij}, the $u^*_{\lambda\mu}(f^{(i)}_\lambda) | U_{ij\mu}$ coincide in $U_{i\mu} \cap U_{j\mu}$ for all pairs (i, j), and therefore define an S_μ-morphism $f_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ such that $f = u^*_\mu(f_\mu)$.

(ii) Consider first the case where $S_0 = \operatorname{Spec}(A_0)$ is *affine*. Then the assertion is equivalent to the lemma (5.13.7.1). In the general case, starting from a finite open affine covering (U_i) of S_0, one deduces (8.5.2.1) that for each i, there exist $\lambda(i)$ and an $\mathscr{O}_{U_{i,\lambda(i)}}$-Module quasi-coherent of finite presentation $\mathscr{F}_i$ such that $u^*_{\lambda(i)}(\mathscr{F}_i) = \mathscr{F} | V_i$ (with the notations of the proof of (i)). As **L** is filtered, one can suppose all the $\lambda(i)$ equal to a same λ. Note now that S_λ is quasi-separated (1.2.7), that $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_\lambda}$-Module of finite presentation; as, for every pair (i, j), $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.1) that for every pair (i, j), there exists an isomorphism $\theta_{ij} : \mathscr{F}_i | U_{ij\lambda} \xrightarrow{\sim} u^*_{\lambda,\lambda_{ij}}(\mathscr{F}_j) | U_{ij\lambda}$ such that $u^*_\lambda(\theta_{ij})$ is the identity automorphism of $\mathscr{F} | (V_i \cap V_j)$; one can further suppose all the λ_{ij} equal to λ, so that $\theta_{ij} : \mathscr{F}_i | U_{ij\lambda} \xrightarrow{\sim} \mathscr{F}_j | U_{ij\lambda}$. Finally, for all triples i, j, k, if one sets $U_{ijk,\lambda} = U_{i\lambda} \cap U_{j\lambda} \cap U_{k\lambda}$, $U_{ijk,\lambda}$ is quasi-compact, and if θ'_{ij}, θ'_{jk} and θ'_{ik} denote the restrictions of θ_{ij}, θ_{jk} and θ_{ik} to $U_{ijk,\lambda}$, one has $u^*_\lambda(\theta'_{ik}) = u^*_\lambda(\theta'_{ij} \circ \theta'_{jk})$. There is thus, by (i), an index $\mu \geqslant \lambda$ such that $u^*_{\lambda\mu}(\theta'_{ik}) = u^*_{\lambda\mu}(\theta'_{ij} \circ \theta'_{jk})$, hence the isomorphisms $u^*_{\lambda\mu}(\theta_{ij}) : u^*_{\lambda\mu}(\mathscr{F}_i) | U_{ij\mu} \xrightarrow{\sim} u^*_{\lambda\mu}(\mathscr{F}_j) | U_{ij\mu}$ satisfy the gluing condition, and define on S_μ an $\mathscr{O}_{S_\mu}$-Module quasi-coherent $\mathscr{F}_\mu$ of finite presentation (0_I, 3.3.1) such that $\mathscr{F}$ is isomorphic to $u^*_\mu(\mathscr{F}_\mu)$. □ IV-3 | 22

Scholium (8.5.3). — *The result of (8.5.2) can be expressed by saying that if S_0 is quasi-compact and quasi-separated, the category of $\mathscr{O}_S$-Modules quasi-coherent of finite presentation is determined up to equivalence by the data of the categories of $\mathscr{O}_{S_\lambda}$-Modules quasi-coherent of finite presentation, the functors $u^*_{\lambda\mu}$ between these categories, and the transitivity isomorphisms $u_{\mu\lambda} \circ u_{\lambda\mu} \xrightarrow{\sim} u_{\mu\lambda}$. In an image-like fashion, one can say that the data of an $\mathscr{O}_S$-Module quasi-coherent of finite presentation $\mathscr{F}$ is "functorially" equivalent to the data of an $\mathscr{O}_{S_\lambda}$-Module quasi-coherent of finite presentation $\mathscr{F}_\lambda$ for λ sufficiently large, and that, if an $\mathscr{O}_{S_\mu}$-Module quasi-coherent of finite presentation $\mathscr{F}''_\mu$ also has $\mathscr{F}$ as its inverse image, then $\mathscr{F}'_\lambda$ and $\mathscr{F}''_\mu$ have the same inverse image in an S_ν for ν large enough ($\nu \geqslant \lambda, \nu \geqslant \mu$).*

Corollary (8.5.4). — *Suppose S_0 quasi-compact and quasi-separated; then, for every quasi-coherent $\mathscr{O}_{S_\lambda}$-Module $\mathscr{G}_\lambda$, the canonical homomorphism* (8.5.1.4) *is bijective.* IV-3 | 23

PROOF. Indeed, it suffices to apply (8.5.2), (i) to the case where $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$, which is of finite presentation. □

Proposition (8.5.5). — *Suppose S_0 quasi-compact, and suppose that $\mathscr{F}_\lambda$ is an $\mathscr{O}_{S_\lambda}$-Module quasi-coherent of finite presentation. For $\mathscr{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $\mathscr{F}_\mu$ is locally free (resp. locally free of rank n).*

PROOF. The condition being trivially sufficient, let us prove it is necessary. If $\mathscr{F}$ is locally free (resp. locally free of rank n), there exists a finite open affine covering (V_i) of S such that $\mathscr{F} | V_i$ is isomorphic to $\mathscr{O}^n_S | V_i$ (resp. $\mathscr{O}^n_S | V_i$) for all i. By (8.2.11) there exists $\nu \geqslant \lambda$ and for each i a quasi-compact open $U_{i\nu}$ of S_ν such that $V_i = u^{-1}_\nu(U_{i\nu})$. As S_ν is quasi-compact, each $U_{i\nu}$ is a finite union of open affines; one is thus reduced to the case where S_0 is affine and $\mathscr{F} = \mathscr{O}^n_S$; one knows then (8.2.5) that there exists $\mu \geqslant \lambda$ such that $\mathscr{F}_\mu$ is isomorphic to $\mathscr{O}^n_{S_\mu}$ (8.5.2). □

Proposition (8.5.6). — *Suppose S_0 quasi-compact, and consider an exact sequence*

$$\mathscr{F}_\lambda \longrightarrow \mathscr{G}_\lambda \longrightarrow \mathscr{H}_\lambda \longrightarrow 0$$

of homomorphisms of $\mathscr{O}_{S_\lambda}$-Modules quasi-coherent, where $\mathscr{F}_\lambda$ and $\mathscr{G}_\lambda$ are of finite type and $\mathscr{H}_\lambda$ of finite presentation. For the corresponding sequence $\mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ to be exact, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that the sequence $\mathscr{F}_\mu \to \mathscr{G}_\mu \to \mathscr{H}_\mu \to 0$ is exact (in which case it is exact for $\nu \geqslant \mu$).

PROOF. The fact that the condition is sufficient and the last assertion follow from the fact that the functor u_λ^* (resp. $u_{\lambda\mu}^*$) is right exact. To show that the condition is necessary, note that by ((8.5.2), (i)) there exists $\nu \geqslant \lambda$ such that the composed $\mathscr{F}_\nu \to \mathscr{G}_\nu \to \mathscr{H}_\nu$ is zero. Changing notations, one can therefore already suppose that $g_\lambda \circ f_\lambda = 0$. If one sets $\mathscr{H}'_\lambda = \operatorname{Coker}(\mathscr{F}_\lambda \to \mathscr{G}_\lambda)$, one has then a homomorphism $f_\lambda : \mathscr{H}'_\lambda \to \mathscr{H}_\lambda$; by hypothesis, $u_\lambda^*(f_\lambda)$ is an isomorphism, and it follows from (8.5.2.4) that there exists $\mu \geqslant \lambda$ such that $u_{\lambda\mu}^*(f_\lambda)$ is an isomorphism, which finishes the proof. □

Corollary (8.5.7). — *Suppose S_0 quasi-compact, $\mathscr{F}_\lambda$ quasi-coherent of finite type, $\mathscr{G}_\lambda$ quasi-coherent. Let $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ be a homomorphism. For $f = u_\lambda^*(f_\lambda) : \mathscr{F} \to \mathscr{G}$ to be surjective, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ is surjective.*

PROOF. This is the particular case of (8.5.6) applied to the sequence $0 \to \mathscr{H}'_\lambda \to 0 \to 0$, where $\mathscr{H}_\lambda = \operatorname{Coker}(f_\lambda)$, which is quasi-coherent and of finite type (taking into account that $\operatorname{Im} f_\lambda$ and $\operatorname{Ker} g_\lambda$ are the reciprocal images of $\operatorname{Im} f_\lambda$ and $\operatorname{Ker} g_\lambda$ for $\mu \geqslant \lambda$), and in view of exactness to the right of the functors u_λ^* and $u_{\lambda\mu}^*$. □

Corollary (8.5.8). — *Suppose S_0 quasi-compact and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ flat. Then:* **IV-3** | 24

(i) *Let $\mathscr{F}_\lambda \xrightarrow{f_\lambda} \mathscr{G}_\lambda \xrightarrow{g_\lambda} \mathscr{H}_\lambda$ be a sequence of homomorphisms of $\mathscr{O}_{S_\lambda}$-Modules quasi-coherent, such that* $\operatorname{Im} f_\lambda$ *and* $\operatorname{Ker} g_\lambda$ *are of finite type. For the corresponding sequence $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ to be exact, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that the sequence $\mathscr{F}_\mu \xrightarrow{f_\mu} \mathscr{G}_\mu \xrightarrow{g_\mu} \mathscr{H}_\mu$ is exact.*

(ii) *Let $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ be a homomorphism of $\mathscr{O}_{S_\lambda}$-Modules quasi-coherent such that* $\operatorname{Ker} f_\lambda$ *is of finite type. For $f = u_\lambda^*(f_\lambda) : \mathscr{F} \to \mathscr{G}$ to be injective, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ is injective.*

PROOF. (i) Taking into account (8.3.8), (ii), note that, by flatness, $\operatorname{Im} f$ and $\operatorname{Ker} g$ (resp. $\operatorname{Im} f_\mu$ and $\operatorname{Ker} g_\mu$) are the reciprocal images of $\operatorname{Im} f_\lambda$ and $\operatorname{Ker} g_\lambda$ for $\mu \geqslant \lambda$ (0_I, 6.7.2). Suppose the sequence $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ is exact. Since $\operatorname{Im} f_\lambda$ is of finite type, there exists $\mu \geqslant \lambda$ such that the composed $\mathscr{F}_\mu \to \mathscr{G}_\mu \to \mathscr{H}_\mu$ is zero, by (8.5.2), (i). Changing notations, one can therefore suppose that $g_\lambda \circ f_\lambda = 0$. Then, as $\operatorname{Ker} g_\lambda$ is of finite type, it follows from (8.5.7) that there exists $\mu \geqslant \lambda$ such that the homomorphism $\mathscr{F}_\mu \to \operatorname{Ker} g_\mu$ is surjective, which finishes proving (i).

(ii) This is the particular case of (i) applied to the sequence $0 \to \mathscr{F}_\lambda \to \mathscr{G}_\lambda$. □

Lemma (8.5.9). — *Suppose S_0 quasi-compact and quasi-separated, $\mathscr{F}_\lambda$ quasi-coherent of finite type; let $\mathscr{G}'_\lambda$, $\mathscr{G}''_\lambda$ be two quasi-coherent quotients of $\mathscr{F}_\lambda$, $\mathscr{G}'_\lambda$ being in addition supposed of finite presentation. For $\mathscr{G}''$ to be a quotient of $\mathscr{G}'$, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $\mathscr{G}''_\mu$ is a quotient of $\mathscr{G}'_\mu$.*

PROOF. By hypothesis, there are two surjective homomorphisms $p'_\lambda : \mathscr{F}_\lambda \to \mathscr{G}'_\lambda$, $p''_\lambda : \mathscr{F}_\lambda \to \mathscr{G}''_\lambda$; by exactness to the right of u_λ^* and $u_{\lambda\mu}^*$, $p' = u_\lambda^*(p'_\lambda)$, $p'' = u_\lambda^*(p''_\lambda)$, $p'_\mu = u_{\lambda\mu}^*(p'_\lambda)$, $p''_\mu = u_{\lambda\mu}^*(p''_\lambda)$ are also surjective; moreover, if there exists a homomorphism $f : \mathscr{G}' \to \mathscr{G}''$ (resp. $f_\mu : \mathscr{G}'_\mu \to \mathscr{G}''_\mu$) such that $p'' = f \circ p'$ (resp. $p''_\mu = f_\mu \circ p'_\mu$), this homomorphism is necessarily unique, which shows that the question is *local* on S_λ, and that one can therefore (as S_0 is quasi-compact and **L** filtered) suppose S_λ affine, hence *quasi-separated*. It is clear that the condition in the statement is sufficient. Conversely, if there exists a homomorphism $f : \mathscr{G}' \to \mathscr{G}''$ such that $p'' = f \circ p'$, it follows from (8.5.2), (i) that there exists $\mu \geqslant \lambda$ and a homomorphism $f_\mu : \mathscr{G}'_\mu \to \mathscr{G}''_\mu$ such that $f = u_\mu^*(f_\mu)$. Furthermore, by (8.5.7), one can suppose μ taken large enough for p_μ to be surjective, which finishes the proof. □

. 8.5.10 To finish this section, for every quasi-coherent Module $\mathscr{F}$ on a prescheme, denote by $\mathfrak{Q}(\mathscr{F})$ the set of Modules which are quotients of $\mathscr{F}$ and are *of finite presentation*. If $\mathscr{F}_\lambda$ is quasi-coherent and $\mathscr{G}_\lambda \in \mathfrak{Q}(\mathscr{F}_\lambda)$, it follows from the fact that $u_{\lambda\mu}^*$ and u_λ^* are exact to the right that $\mathscr{G}_\mu \in \mathfrak{Q}(\mathscr{F}_\mu)$

for $\mu \geqslant \lambda$ and $\mathscr{G} \in \mathfrak{Q}(\mathscr{F})$; it is clear that $(\mathfrak{Q}(\mathscr{F}_\lambda), u^*_{\lambda\mu})$ is an inductive system of sets, and that the $u^*_\lambda : \mathfrak{Q}(\mathscr{F}_\lambda) \to \mathfrak{Q}(\mathscr{F})$ form an inductive system of maps, whence, passing to the inductive limit, a canonical map

(8.5.10.1) $$u_{\mathfrak{Q}} : \varinjlim \mathfrak{Q}(\mathscr{F}_\lambda) \longrightarrow \mathfrak{Q}(\mathscr{F}).$$

Furthermore, if $(\mathscr{F}'_\lambda)$ is a second family of quasi-coherent $\mathscr{O}_{S_\lambda}$-Modules and if, for all λ, $\mathscr{F}'_\lambda$ is a quotient of $\mathscr{F}_\lambda$, then $\mathscr{F}'$ is a quotient of $\mathscr{F}$ and one has a commutative diagram

(8.5.10.2) $$\begin{array}{ccc} \varinjlim \mathfrak{Q}(\mathscr{F}_\lambda) & \xrightarrow{u_{\mathfrak{Q}}} & \mathfrak{Q}(\mathscr{F}) \\ \uparrow & & \uparrow \\ \varinjlim \mathfrak{Q}(\mathscr{F}'_\lambda) & \xrightarrow[u_{\mathfrak{Q}}]{} & \mathfrak{Q}(\mathscr{F}') \end{array}$$

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Proposition (8.5.11). — *Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Suppose that $\mathscr{F}_\lambda$ is quasi-coherent of finite type (resp. of finite presentation) for all λ; then the canonical map* (8.5.10.1) *is* injective *(resp.* bijective*).*

PROOF. The first assertion follows more precisely from the lemma (8.5.9). For the second, consider an $\mathscr{O}_S$-Module quotient $\mathscr{G}$ of $\mathscr{F}$ which is of finite presentation. It follows from (8.5.2), (ii) that there exist $\lambda' \in \mathbf{L}$ and an $\mathscr{O}_{S_{\lambda'}}$-Module quasi-coherent of finite presentation $\mathscr{G}_{\lambda'}$ such that $\mathscr{G} = u^*_{\lambda'}(\mathscr{G}_{\lambda'})$; as $\mathbf{L}$ is filtered, one can replace λ and λ' by an index majoring them and suppose $\lambda' = \lambda$. Consider then the canonical homomorphism $p : \mathscr{F} \to \mathscr{G}$; it follows from (8.5.2), (i) that there exists $\mu \geqslant \lambda$ and a homomorphism $p_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ such that $p = u^*_\mu(p_\mu)$. Furthermore, by (8.5.7), one can suppose μ taken large enough for p_μ to be surjective, which finishes the proof. ☐

8.5. Sub-preschemes of finite presentation of a projective limit of preschemes

. 8.6.1 Given a prescheme Y, we will denote in this section by $\mathfrak{Spr}(Y)$ the ordered set (**I**, 4.1.10) of sub-preschemes of Y which are *of finite presentation* on Y (1.6.1), by $\mathfrak{Spro}(Y)$ (resp. $\mathfrak{Sprf}(Y)$) the part of $\mathfrak{Spr}(Y)$ formed of sub-preschemes induced on open subsets (resp. closed sub-preschemes) of Y, of finite presentation on Y; recall that saying a sub-prescheme Z of Y belongs to $\mathfrak{Spro}(Y)$ means it is induced on an open subset and that the underlying space Z is *retrocompact* in Y (resp. that it is closed and that the ideal $\mathscr{J}$ of $\mathscr{O}_Y$ which defines it is *of finite type*, which also means that $j_*(\mathscr{O}_Z) \in \mathfrak{Q}(\mathscr{O}_Y)$ if $j : Z \to Y$ is the canonical injection) (1.6.1 and 1.4.5).

. 8.6.2 We keep the notations of (8.2.2) and suppose that S_0 is one of the S_λ. Let Y_λ be a sub-prescheme of S_λ; then $Y_\mu = u^{-1}_{\lambda\mu}(Y_\lambda)$ (resp. $Y = u^{-1}_\lambda(Y_\lambda)$) is a sub-prescheme of S_μ for $\mu \geqslant \lambda$ (resp. of S); it is induced on an open subset (resp. it is closed) if Y_λ is so, and is of finite presentation on S_μ (resp. S) if Y_λ is of finite presentation on S_λ (1.6.2, (iii)). Hence $(\mathfrak{Spr}(S_\lambda), u^{-1}_{\lambda\mu})$ (resp. $(\mathfrak{Spro}(S_\lambda), u^{-1}_{\lambda\mu})$, $(\mathfrak{Sprf}(S_\lambda), u^{-1}_{\lambda\mu})$) is an inductive system and the maps $u^{-1}_\lambda : \mathfrak{Spr}(S_\lambda) \to \mathfrak{Spr}(S)$ (resp. $\mathfrak{Spro}(S_\lambda) \to \mathfrak{Spro}(S)$, $\mathfrak{Sprf}(S_\lambda) \to \mathfrak{Sprf}(S)$) form an inductive system of maps; whence, passing to the inductive limit, canonical maps

(8.6.2.1) $$\varinjlim \mathfrak{Spr}(S_\lambda) \longrightarrow \mathfrak{Spr}(S)$$

(8.6.2.2) $$\varinjlim \mathfrak{Spro}(S_\lambda) \longrightarrow \mathfrak{Spro}(S)$$

(8.6.2.3) $$\varinjlim \mathfrak{Sprf}(S_\lambda) \longrightarrow \mathfrak{Sprf}(S).$$

Recall (**I**, 4.1.10) that $\mathfrak{Spr}(X)$, for any prescheme X, is an *ordered* set by the relation "Z is a sub-prescheme of the sub-prescheme Y" which we write $Z \leqslant Y$. The maps $u^{-1}_{\lambda\mu}$ and u^{-1}_λ are *increasing* for the corresponding order relations in $\mathfrak{Spr}(S_\lambda)$, $\mathfrak{Spr}(S_\mu)$, $\mathfrak{Spr}(S)$. Furthermore, one defines an order relation on the set $\varinjlim \mathfrak{Spr}(S_\lambda)$ by writing $\eta \leqslant \eta'$ for two elements of this set whenever there exist λ and two elements Y_λ, Y'_λ of $\mathfrak{Spr}(S_\lambda)$, whose η and η' are the canonical images, and such that $Y_\lambda \leqslant Y'_\lambda$; one verifies easily that this does not depend on the choice of representatives, and that one obtains a *relation of order*. This being the case, the fact that the u^{-1}_λ are increasing implies immediately that the canonical map (8.6.2.1) is *increasing*; it is the same evidently for (8.6.2.2) and (8.6.2.3). **IV-3 | 26**

Proposition (8.6.3). — *Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Then the maps* (8.6.2.1), (8.6.2.2), (8.6.2.3) *are* injective *(resp.* bijective*).*

PROOF. Taking into account the remarks of (8.6.1), the assertions relative to (8.6.2.3) follow from (8.5.11) applied to $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$; similarly, the assertions relative to (8.6.2.2) are particular cases of (8.3.5) and (8.3.11), taking into account that the S_λ and S are quasi-compact. It remains to consider the map (8.6.2.1). Let us first prove that it is surjective when S_0 is quasi-compact and quasi-separated. Let Z be a sub-prescheme of S, of finite presentation on S; as S is quasi-compact, it is the same of Z, hence there exists a quasi-compact open U of S such that Z is a sub-prescheme closed of U, of finite presentation on U. There exists then an index λ and a quasi-compact open U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$ (8.2.11); as S_λ is quasi-separated, each U_λ is of the same type of U_λ (1.2.7), and by the same argument as in the case of (8.6.2.3), it is surjective.

Finally, to see that (8.6.2.1) is injective when S_0 is quasi-compact, it suffices to prove the following more precise result: □

Corollary (8.6.3.1). — *Suppose S_0 quasi-compact, and let Z'_λ, Z''_λ be two sub-preschemes of S_λ, of finite presentation on S_λ. For $Z' = u_\lambda^{-1}(Z'_\lambda)$ to be majorized by $Z'' = u_\lambda^{-1}(Z''_\lambda)$* (**I**, 4.1.10)*, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $Z'_\mu = u_{\lambda\mu}^{-1}(Z'_\lambda)$ is majorized by $Z''_\mu = u_{\lambda\mu}^{-1}(Z''_\lambda)$.*

Corollary (8.6.4). — *Suppose S_0 quasi-compact, and let Z_λ be a sub-prescheme of S_λ, of finite presentation on S_λ. For $Z = u_\lambda^{-1}(Z_\lambda)$ to be a sub-prescheme induced on an open subset (resp. a closed sub-prescheme) of S, it is necessary and sufficient that there exists $\mu \geqslant \lambda$ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ is induced on an open subset (resp. a closed sub-prescheme) of S_μ.* **IV-3 | 27**

PROOF. Let $(U_\lambda^{(i)})$ be a finite open affine covering of S_λ, and set $U_\mu^{(i)} = u_{\lambda\mu}^{-1}(U_\lambda^{(i)})$ for $\mu \geqslant \lambda$ and $U^{(i)} = u_\lambda^{-1}(U_\lambda^{(i)})$. If Z is open (resp. closed) in S, $Z \cap U^{(i)}$ is open (resp. closed) in $U^{(i)}$, and conversely. Since **L** is filtered, it suffices to show the corollary when S_λ is affine, hence quasi-separated; but then the result follows from the bijectivity of the maps (8.6.2.1), (8.6.2.2), and (8.6.2.3). □

8.6. Criteria for a projective limit of preschemes to be reduced (resp. integral)

Proposition (8.7.1). — *Suppose that S is not reduced. Then there exists λ_0 such that for $\lambda \geqslant \lambda_0$, S_λ is not reduced.*

PROOF. The question being local on S_0, one can suppose $S_0 = \operatorname{Spec}(A_0)$ affine, whence $S = \operatorname{Spec}(A)$, where $A = \varinjlim A_\lambda$ is the inductive limit of an inductive system of A_0-algebras (A_λ). One knows then (5.13.2) that the nilradical of A is the inductive limit of the nilradicals of the A_λ; so if it is not zero, one of the A_λ contains a nilpotent element $a_\lambda \neq 0$ whose image in A is non-zero, and the image of a_λ in the A_μ for $\mu \geqslant \lambda$ is therefore a nilpotent element $\neq 0$. □

Proposition (8.7.2). — *Suppose one of the following hypotheses is satisfied:*

a) *S_0 is quasi-compact, the nilradical $\mathscr{N}_0$ of $\mathscr{O}_{S_0}$ is an ideal of finite type (which will be the case when S_0 is noetherian), and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Under these conditions, for S to be reduced, it is necessary and sufficient that there exists λ_0 such that S_λ is reduced for $\lambda \geqslant \lambda_0$.

Furthermore, in case b)*, if S is reduced, all the S_λ are reduced.*

PROOF. The last assertion follows from the fact that the morphism $S \to S_\lambda$ is then faithfully flat for all λ (8.3.8) and from (2.1.13). On the other hand, (8.7.1) proves that the condition is necessary in hypothesis *a)*; then (8.2.13), the underlying space of S identifies with the intersection of the underlying spaces of the S_λ (the S_λ being identified with sub-preschemes induced on open subsets of S_0), and the structural sheaf $\mathscr{O}_S$ identifies with the sheaf induced by all the $\mathscr{O}_{S_\lambda}$; in particular for all $s \in S$ and for all S_λ, the local ring $\mathscr{O}_s$ is the same for S and for all S_λ. If $\mathscr{N}_\lambda$ is the Nilradical of $\mathscr{O}_{S_\lambda}$, the Nilradical $\mathscr{N}$ of $\mathscr{O}_S$ has at each point of S the same fibre $\mathscr{N}_s$ (nilradical of $\mathscr{O}_s$) (induced on S); the hypothesis that S is reduced implies by $u_\lambda^*(\mathscr{N}_\lambda) = 0$; as $\mathscr{N}_0$ is supposed of finite type, it is the same of $\mathscr{N}_\lambda$, and the conclusion follows from (8.5.7). □ **IV-3 | 28**

Corollary (8.7.3). — *Suppose one of the following hypotheses is satisfied:*

a) *S_0 is a noetherian prescheme and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Then, for S to be integral, it is necessary and sufficient that there exists λ_0 such that S_λ is integral for $\lambda \geqslant \lambda_0$.

PROOF. Recall that saying a prescheme is integral means it is at once reduced and irreducible; the corollary thus follows from (8.7.2) and (8.4.3). ☐

Remark (8.7.4). — If one does not make hypotheses on the $u_{\lambda\mu}$, it can happen that S is integral even though all the S_λ are not reduced and not connected, as the example (8.4.4) shows, where one takes A non-reduced.

8.7. Preschemes of finite presentation on a projective limit of preschemes

. 8.8.1 Keeping always the notations and hypotheses of (8.2.2), we will assume given in this section two S_α-preschemes X_α, Y_α, which defines (8.2.5) by setting $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, $Y_\lambda = Y_\alpha \times_{S_\alpha} S_\lambda$, $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, $w_{\lambda\mu} = 1_{Y_\alpha} \times u_{\lambda\mu}$ (for $\alpha \leqslant \lambda \leqslant \mu$), whose projective limits are respectively $X = X_\alpha \times_{S_\alpha} S$, $Y = Y_\alpha \times_{S_\alpha} S$, the canonical morphisms $v_\lambda : X \to X_\lambda$ and $w_\lambda : Y \to Y_\lambda$ being respectively equal to $1_{X_\alpha} \times u_\lambda$ and $1_{Y_\alpha} \times u_\lambda$. For $\alpha \leqslant \lambda \leqslant \mu$, one has a canonical map $e_{\mu\lambda}$: $\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu)$, which to every S_λ-morphism $f_\lambda : X_\lambda \to Y_\lambda$ associates the morphism $f_\mu = f_\lambda \times 1_{S_\mu} : X_\lambda \times_{S_\lambda} S_\mu \to Y_\lambda \times_{S_\lambda} S_\mu$, and it is clear that $(\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda), e_{\lambda\mu})$ is an *inductive system* of sets. Similarly, one has a canonical map $e_\lambda : \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_S(X, Y)$ which to f_λ associates $f = f_\lambda \times 1_S : X_\lambda \times_{S_\lambda} S \to Y_\lambda \times_{S_\lambda} S$ and (e_λ) is an inductive system of maps; whence, passing to the inductive limit, a canonical map, functorial in S_α, X_α and Y_α: **IV-3 | 28**

$$e : \varinjlim \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Hom}_S(X, Y). \tag{8.8.1.1}$$

Theorem (8.8.2). — (i) *Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and Y_α locally of finite type (resp. locally of finite presentation) on S_α. Then the map* (8.8.1.1) *is* injective *(resp.* bijective*).*

(ii) *Suppose S_0 quasi-compact and quasi-separated. For every prescheme X of finite presentation on S, there exist $\lambda \in \mathbf{L}$, a prescheme X_λ of finite presentation on S_λ, and an S-isomorphism $X \xrightarrow{\sim} X_\lambda \times_{S_\lambda} S$.*

PROOF. (i) Consider first the case where $S_0 = \mathrm{Spec}(A_0)$, $X_\alpha = \mathrm{Spec}(B_\alpha)$ and $Y_\alpha = \mathrm{Spec}(C_\alpha)$ are affine; then the $S_\lambda = \mathrm{Spec}(A_\lambda)$ and $S = \mathrm{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$, and the assertions of (i) are equivalent to the **IV-3 | 29**

Lemma (8.8.2.1). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in \mathbf{L}}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; let B_α be an A_α-algebra, C_α an A_α-algebra of finite type (resp. of finite presentation). Then the canonical homomorphism*

$$\varinjlim \mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \longrightarrow \mathrm{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \tag{8.8.2.2}$$

is injective (resp. bijective).

Lemma (8.8.2.3). — *Let E be a ring, G an E-algebra of finite type (resp. of finite presentation), (F_λ) an inductive system of E-algebras. Then the canonical homomorphism*

$$\varinjlim \mathrm{Hom}_{E\text{-alg.}}(G, F_\lambda) \longrightarrow \mathrm{Hom}_{E\text{-alg.}}(G, \varinjlim F_\lambda)$$

which to every inductive system of homomorphisms $\theta_\lambda : G \to F_\lambda$ of E-algebras associates its inductive limit, is injective (resp. bijective).

These lemmas are proved by reduction to the case of polynomial algebras and quotients thereof, using the universal properties of polynomial algebras and the finite presentation hypothesis.

(ii) follows from (i) and the gluing arguments of (8.5.2). ☐ **IV-3 | 32**

Corollary (8.8.2.4). — *Suppose S_0 quasi-compact and quasi-separated, X_α of finite presentation on S_α, Y_α of finite type on S_α and quasi-separated on S_α (which will be the case for example if Y_α is also of finite presentation on S_α). Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. For $f : X \to Y$ to be an isomorphism, it is necessary and sufficient that there exists $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to Y_\lambda$ is an isomorphism.*

Corollary (8.8.2.5). — *Suppose S_0 quasi-compact and quasi-separated, X_α and Y_α of finite presentation on S_α. For X and Y to be S-isomorphic, it is necessary and sufficient that there exists $\lambda \geqslant \alpha$ such that X_λ and Y_λ are S_λ-isomorphic. Moreover, for every S-isomorphism $f : X \xrightarrow{\sim} Y$, there exists $\mu \geqslant \lambda$ and an S_μ-isomorphism $f_\mu : X_\mu \xrightarrow{\sim} Y_\mu$ such that $f = f_\mu \times 1_S$.*

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Scholium (8.8.3). — *The essential content of (8.8.2) can be expressed by saying that if S_0 is quasi-compact and quasi-separated, the category of S-preschemes of finite presentation is determined up to equivalence by the data of the categories of S_λ-preschemes of finite presentation, the functors $\rho_{\mu\lambda} : X_\lambda \mapsto X_\lambda \times_{S_\lambda} S_\mu$ (for $\lambda \leqslant \mu$) between these categories, and the transitivity isomorphisms $\rho_{\nu\mu} \circ \rho_{\mu\lambda} \xrightarrow{\sim} \rho_{\nu\lambda}$. In an image-like fashion, one can say that the data of an S-prescheme of finite presentation X is "functorially" equivalent to the data of an S_λ-prescheme of finite presentation X'_λ for λ sufficiently large; if an S_μ-prescheme of finite presentation X''_μ is such that X is S-isomorphic to $X''_\mu \times_{S_\mu} S$, there exists ν such that $\lambda \leqslant \nu$, $\mu \leqslant \nu$ and such that $X'_\lambda \times_{S_\lambda} S_\nu$ and $X''_\mu \times_{S_\mu} S_\nu$ are S_ν-isomorphic. The fact that $\mathbf{L}$ is filtered implies moreover that if one gives a* finite *family $(X^{(i)})_{i\in I}$ of S-preschemes of finite presentation, and a* finite *family $(f^{(j)})_{j\in J}$ of S-morphisms between these preschemes (thus $f^{(j)}$ being of the form $X^{(\sigma(j))} \to X^{(\tau(j))}$ where σ and τ are two maps of J into I), there is then an index $\lambda \in \mathbf{L}$, a family $(X_\lambda^{(i)})_{i\in I}$ of S_λ-preschemes of finite presentation and a family $(f_\lambda^{(j)})_{j\in J}$ of S_λ-morphisms such that $X^{(i)}$ identifies with $X_\lambda^{(i)} \times_{S_\lambda} S$ for all i and j; moreover, if one has a relation $f^{(j)} = f^{(k)}$, one can suppose $f_\lambda^{(j)} = f_\lambda^{(k)}$.*

8.8. First applications to the elimination of noetherian hypotheses IV-3 | 34

Proposition (8.9.1). — *Let A be a ring, X an A-prescheme.*

(i) *The following conditions are equivalent:*
   a) *X is of finite presentation on A.*
   b) *There exists a noetherian ring A_0, a prescheme X_0 of finite type on A_0, a ring homomorphism $A_0 \to A$, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
   c) *There exists a sub-ring A_0 of A, which is a $\mathbf{Z}$-algebra of finite type, a prescheme X_0 of finite type on A_0, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*

(ii) *If $\mathscr{F}$ is an $\mathscr{O}_X$-Module quasi-coherent of finite presentation, one can suppose the sub-ring A_0 of A is such that there exists an $\mathscr{O}_{X_0}$-Module coherent $\mathscr{F}_0$ such that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{A_0} A$;* $\mathrm{Supp}(\mathscr{F})$ *is constructible and closed in X, and there is a closed sub-prescheme Z of X, having* $\mathrm{Supp}(\mathscr{F})$ *as underlying space, such that the canonical immersion $Z \to X$ is of finite presentation.*

(iii) *If Y is a second A-prescheme of finite presentation, and $f : X \to Y$ an A-morphism, one can suppose the sub-ring A_0 of A such that there exists a prescheme Y_0 of finite type on A_0 and an A_0-morphism $f_0 : X_0 \to Y_0$ (necessarily of finite type) such that f identifies with $f_0 \otimes 1_A$.*

PROOF. (i) As A is the inductive limit of its sub-rings that are of finite type over $\mathbf{Z}$, *a)* implies *c)* by (8.8.2), (ii); *c)* implies *b)* since a $\mathbf{Z}$-algebra of finite type is a noetherian ring; finally, if A_0 is noetherian, an A_0-prescheme of finite type is of finite presentation (1.6.1), hence *b)* implies *a)* by (1.6.2, (iii)).

The assertion (ii) follows immediately from (8.5.2), (ii), (8.3.11) and (1.8.2), and the assertion (iii) from (8.8.2), (i). □

. 8.9.2 Proposition (8.9.1) and the results of (8.5.2) and (8.8.2) allow extending in many cases to morphisms of finite presentation $X \to Y$ results proved by noetherian techniques assuming Y is *locally noetherian*. We will see many examples in the sequel and will limit ourselves here to giving a few results of this type.

Proposition (8.9.3). — *Let A be a ring, M an A-module of finite presentation. Then every surjective A-endomorphism of M is bijective.* IV-3 | 35

PROOF. Indeed, consider A as the inductive limit of its sub-$\mathbf{Z}$-algebras of finite type. It follows from (8.5.2.6) that there is one of these sub-algebras A_0 and an A_0-module M_0 of finite type such that M is A-isomorphic to $M_0 \otimes_{A_0} A$; moreover, if $f : M \to M$ is a surjective A-endomorphism, one can suppose (8.5.2), (i) that there exists an A_0-endomorphism $f_0 : M_0 \to M_0$ such that $f = f_0 \otimes 1_A$;

finally (8.5.7) one can suppose that f_0 is *surjective*. But as A_0 is noetherian and M_0 an A_0-module of finite type (Bourbaki, *Alg.*, chap. VIII, § 2, n° 2, lemma 3) f_0 is bijective, and it is the same of f. □

Proposition (8.9.4). — *("generic flatness theorem"). Let Y be an integral prescheme, $u : X \to Y$ a morphism of finite type and locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists a non-empty open subset U of Y such that $\mathscr{F}|u^{-1}(U)$ is flat over U.*

PROOF. The reasoning from the beginning of (6.9.1) reduces to proving the

Lemma (8.9.4.1). — *Let A be an integral ring, B an A-algebra of finite presentation, M a B-module of finite presentation. Then there exists $f \neq 0$ in A such that M_f is an A_f-module libre.*

Indeed, by (8.9.1) there is a sub-**Z**-algebra of finite type A_0 of A, a A_0-algebra of finite type B_0 and a B_0-module of finite type M_0 such that B is A-isomorphic to $B_0 \otimes_{A_0} A$ and M is B-isomorphic to $M_0 \otimes_{A_0} A$; by (6.9.2), there exists $f_0 \neq 0$ in A_0 such that $(M_0)_{f_0}$ is an $(A_0)_{f_0}$-module libre. As $M_{f_0} = (M_0)_{f_0} \otimes_{(A_0)_{f_0}} A_{f_0}$ and $A_{f_0} = (A_0)_{f_0} \otimes_{A_0} A$, M_{f_0} is an A_{f_0}-module libre. □

Corollary (8.9.5). — *Let S be a prescheme quasi-compact and quasi-separated, $u : X \to S$ a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists a partition $(S_i)_{1 \leqslant i \leqslant n}$ of S into a finite number of locally closed sets in S, having for each $1 \leqslant i \leqslant n$ a sub-prescheme S'_i of S of finite presentation, having S_i as underlying space, such that if one sets $X_i = X \times_S S'_i$, the $\mathscr{O}_{X_i}$-Module $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'_i}$ is flat over S'_i.*

8.9. Permanence of properties of morphisms by passage to the projective limit IV-3 | 36

. 8.10.0 In this section we keep the general hypotheses and notations of (8.8.1).

Proposition (8.10.1). — *If there exists λ such that, for $\mu \geqslant \lambda$, f_μ is an open (resp. universally open) morphism, then f is an open (resp. universally open) morphism.*

PROOF. As $X = X_\lambda \times_{Y_\lambda} Y$, the assertion relative to universally open morphisms is a particular case of (2.4.3, (iii)). Suppose f_μ open for $\mu \geqslant \lambda$; it suffices to see that for every open *quasi-compact* U of X, $f(U)$ is open in Y. Now, it exists $\mu \geqslant \lambda$ such that $U = v_\mu^{-1}(U_\mu)$, where U_μ is a quasi-compact open in X_μ (8.2.11); one has then $f(v_\mu^{-1}(U_\mu)) = w_\mu^{-1}(f_\mu(U_\mu))$ (**I**, 3.4.8), hence $f(U)$ is open in Y. □

Corollary (8.10.2). — *Let $f : X \to Y$ be a morphism. For f to be universally open, it suffices that, for every integer $n > 0$, if one sets $Y_n = Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_n] (= \mathbf{V}_Y^n)$, and $X_n = X \times_Y Y_n$, the projection $f_n = f \times 1_{Y_n} : X_n \to Y_n$ is an open morphism.*

Proposition (8.10.3). — *Suppose there exists α such that: 1° S_α is quasi-compact; 2° the morphisms $X_\alpha \to S_\alpha$, $Y_\alpha \to S_\alpha$ are quasi-compact and the morphism $Y_\alpha \to S_\alpha$ is quasi-separated; 3° for $\alpha \leqslant \lambda \leqslant \mu$, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat; 4° $f_\alpha(X_\alpha)$ is constructible in Y_α. Then, for f to be dominant, it is necessary and sufficient that there exists $\lambda \geqslant \alpha$ such that f_λ is dominant.*

Proposition (8.10.4). — *Suppose there exists α such that Y_α is quasi-compact and of finite type on S_α. For the morphism f to be separated, it is necessary and sufficient that there exists $\lambda \geqslant \alpha$ such that f_λ is separated.* IV-3 | 37

Theorem (8.10.5). — *Suppose S_0 quasi-compact, X_α and Y_α of finite presentation on S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Consider, for a morphism, the property of being:*

(i) *an isomorphism;*
(i bis) *a monomorphism;*
(ii) *an immersion;*
(iii) *an open immersion;*
(iv) *a closed immersion;*
(v) *separated;*
(vi) *surjective;*
(vii) *radiciel;*
(viii) *affine;*
(ix) *quasi-affine;*
(x) *finite;*
(xi) *quasi-finite;*

(xii) *proper.*

Then, if **P** *denotes one of the preceding properties, for* f *to possess* **P**, *it is necessary and sufficient that there exists* $\lambda \geqslant \alpha$ *such that* f_λ *possesses property* **P** *(in which case* f_μ *also possesses it for* $\mu \geqslant \lambda$*).*

If S_0 *is moreover supposed quasi-separated, the same conclusion is valid when* **P** *is the property of being:* **IV-3** | 38

(xiii) *projective;*

(xiv) *quasi-projective.*

PROOF *(sketch)*. The case where **P** is one of properties (i) or (v) was not inserted in the statement since, for those cases, the theorem follows from what was already shown in (8.8.2.4) and (8.10.4) respectively. Moreover, taking into account (I, 5.4.1) and (5.3.4), (v) follows also from (iv). The case (i bis) follows immediately from (i), using (I, 5.3.8) and noting that the diagonal Δ_{f_λ} of $X_\lambda \times_{Y_\lambda} X_\lambda$ is deduced from Δ_f by base change $S \to S_\lambda$.

We note on the other hand that (vi), (vii) and (xi) are in fact conditions on the *fibres* $f^{-1}(y)$ of the morphisms considered, taking into account the transitivity of fibres by base change (I, 3.6.4): condition (vi) means that all fibres are non-empty; condition (vii) that they must be radiciel (I, 3.5.8); and condition (xi) that they must be finite (II, 6.2.2) and (I, 6.4.4), taking into account that f and the f_λ are morphisms of finite type by (1.5.4, (v)). The theorem in these three cases will thus be a consequence of a general result on a type of properties concerning only the fibres, which will be established in (9.3.3); we postpone the proof of the theorem in case (xi) until then, and note that (8.11) and (8.12) will not use the theorem up to (9.3.3) and that (8.11) and (8.12) are not used before (9.3.3).

For the remaining cases, one can reduce to proving that the condition in the statement is necessary, all considered properties **P** being invariant by base change (see chap. I$^{\text{er}}$ and II in the sections concerning each of these properties). One can moreover suppose $S_\alpha = S_0$ and $Y_\alpha = S_\alpha$, hence $Y_\lambda = S_\lambda$ for all $\lambda \geqslant \alpha$. Finally, properties (i) through (xii) are local on S_0, hence one can further suppose $S_0 = \operatorname{Spec}(A_0)$ affine (and quasi-separated). We designate by A_λ the ring of S_λ (resp. S).

Cases (ii), (iii), (iv): these follow from (8.6.3) applied to the sub-prescheme X' associated to f.

Cases (viii) *and* (ix): follow from (8.10.5), applied to the morphism $j : X \to \mathbf{V}_S^r = \operatorname{Spec}(S[T_1, \ldots, T_r])$ of S-preschemes of finite presentation.

Case (x): By hypothesis, $X = \operatorname{Spec}(B)$, where B is an A-algebra which is an A-module of finite presentation (1.4.7); it follows from (8.5.2), (ii) that there is a λ and an A_λ-module of finite presentation B_λ such that the A-module B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$. The A-algebra structure on B is defined by an A-homomorphism $m : B \otimes_A B \to B$; by (8.5.2), (i) there exists $\mu \geqslant \lambda$ an A_μ-homomorphism $m_\mu : B_\mu \otimes_{A_\mu} B_\mu \to B_\mu$ such that $m = m_\mu \otimes 1$. Considering the usual diagrams expressing the associativity and commutativity of m, one sees, applying (8.5.2), (i) again, that one can define on B_ν a multiplication making it into an A_ν-algebra for ν taken large enough, so that $X'_\nu = \operatorname{Spec}(B_\nu)$ is S_ν-isomorphic to $X_\nu \times_{S_\nu} S$, by (i), hence there exist $\rho \geqslant \nu$ such that X_ρ and X'_ρ are S_ρ-isomorphic, which finishes the proof in this case.

Case (xii): Apply to the morphism $f_0 : X_0 \to S_0$ the proposition (8.10.5.1) (Chow's lemma for morphisms of finite presentation).

Cases (xiii) *and* (xiv): By (xii) and (II, 5.5.3) (which is applicable since the S_λ are quasi-compact and quasi-separated, taking into account (1.7.19)), it suffices to consider the case where f is quasi-projective. □

Proposition (8.10.5.1). — *(Chow's lemma for morphisms of finite presentation). Let* A *be a ring,* X, Y *two* **IV-3** | 39
A-preschemes of finite presentation, $f : X \to Y$ *a separated* A*-morphism. Then there exist two* A*-preschemes* X', P *of finite presentation and* A*-morphisms* $p : P \to Y$, $j : X' \to P$, $g : X' \to X$, *such that the diagram*

$$\begin{tikzcd} X' \arrow[r, "j"] & P \\ X \arrow[u, "g"] \arrow[r, "f"'] & Y \arrow[u, "p"'] \end{tikzcd}$$

is commutative, and such that: 1° p *is projective; 2°* g *is projective and surjective; 3°* j *is an open immersion.*

8.10. Application to quasi-finite morphisms We propose to show in this section the two following theorems: **IV-3 | 41**

Theorem (8.11.1). — *Let $f : X \to Y$ be a morphism which is proper, locally of finite presentation, and quasi-finite. Then the morphism f is finite.*

Theorem (8.11.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, quasi-finite and separated. Then the morphism f is quasi-affine, and* a fortiori *quasi-projective.*

Remark (8.11.3). — We will see below that the proofs of (8.11.1) and (8.11.2), in the case where Y is locally noetherian, yield another proof of Chevalley's theorem (**III**, 4.4.2).

. 8.11.4 The hypotheses and conclusions of (8.11.1) and (8.11.2) are all local on Y (1.6.1, 1.2.6, (**II**, 5.1.1), (**II**, 5.4.1) and (**II**, 6.2.2)), hence one can suppose $Y = \operatorname{Spec}(A)$ affine. One knows that there exists then a sub-ring A_0 of A, which is a **Z**-algebra of finite type, and a morphism of finite type $f_0 : X_0 \to \operatorname{Spec}(A_0)$ such that X identifies with $X_0 \otimes_{A_0} A$ and f with $f_0 \times 1$ (8.9.1). Furthermore, A can be considered as an inductive limit of its sub-rings containing A_0 and which are **Z**-algebras of finite type; using (8.1.2, *c*)) and (8.10.5), (v), (xi) and (xii), one sees that it suffices to prove theorems (8.11.1) and (8.11.2) for f_0; using now the method of (8.1.2, *a*)) and (8.10.5), (v), (ix), (x), (xi) and (xii), one can replace Y by $\operatorname{Spec}(\mathscr{O}_y)$, where y is any point of Y.

Proposition (8.11.5). — *Let $f : X \to Y$ be a morphism of finite presentation. The following properties are equivalent:*

a) *f is a closed immersion.*
b) *f is a proper monomorphism.*
c) *f is proper and for all $y \in Y$, $f^{-1}(y)$ is radiciel and geometrically reduced over $k(y)$ (that is to say either empty or $k(y)$-isomorphic to* $\operatorname{Spec}(k(y))$*).*

PROOF. It is clear that *a)* implies *b)*. To see that *b)* implies *c)*, note (**I**, 3.3.12) that for all $y \in Y$, the morphism $f^{-1}(y) \to \operatorname{Spec}(k(y))$ deduced from f by base extension is a monomorphism, hence is injective, and in any case affine. Moreover, if A is the ring of $f^{-1}(y)$, the homomorphism canonical $A \otimes_k A \to A$ is bijective (**I**, 5.3.8). This evidently implies that $A = k$, unless $A = 0$.

It remains to prove that *c)* implies *a)*. It follows first from (8.11.1) that f is finite, then a fortiori $X = \operatorname{Spec}(\mathscr{A})$, where $\mathscr{A}$ is an $\mathscr{O}_Y$-Algebra finite. It suffices then to prove that the canonical homomorphism $\mathscr{O}_Y \to \mathscr{A}_y$ is surjective (**II**, 1.4.10), or equivalently that for all $y \in Y$, the homomorphism $\mathscr{O}_y \to \mathscr{A}_y$ is surjective. But by hypothesis $f^{-1}(y) = \operatorname{Spec}(\mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y)$ (**II**, 1.5.5) is such that the corresponding homomorphism $k(y) = \mathscr{O}_y/\mathfrak{m}_y \to \mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y$ is bijective; as $\mathscr{A}_y$ is an $\mathscr{O}_y$-module of finite type, the lemma of Nakayama shows that $\mathscr{O}_y \to \mathscr{A}_y$ is surjective, which finishes the proof. □

Proposition (8.11.6). — *If a morphism $f : X \to Y$ of finite presentation is a universal homeomorphism, it is finite, surjective, and radiciel (the converse being true by (2.4.5, (i))).*

PROOF. Indeed, f being of finite type, universally closed, and separated by (2.4.4), is proper by definition (**II**, 5.4.1). As it is evidently quasi-finite (**II**, 6.2.3), it is finite by (8.11.1). One knows moreover that it is surjective (2.4.2), and evidently radiciel. □

8.11. New proof and generalization of Zariski's "Main Theorem" **IV-3 | 43**

Lemma (8.12.1). — *Let $f : X \to Y$ be a morphism quasi-compact and quasi-separated, $\mathscr{C}$ a quasi-coherent $\mathscr{O}_Y$-Algebra, $Z = \operatorname{Spec}(\mathscr{C})$, which is a Y-prescheme affine above Y. Let $g : X \to Z$ be a Y-morphism, $\varphi = \mathscr{A}(g) : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism of $\mathscr{O}_Y$-Algebras* (**II**, 1.2.7)*. Suppose that g is an immersion. Then, for the closed image of X by g to be equal to Z, it is necessary and sufficient that φ be injective; g is then an open immersion.*

Lemma (8.12.2). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a morphism quasi-separated and of finite type, $\mathscr{C}$ a quasi-coherent $\mathscr{O}_Y$-Algebra, $Z = \operatorname{Spec}(\mathscr{C})$, $\varphi : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism. Let $(\mathscr{C}_\lambda)$ be the filtering increasing family of quasi-coherent sub-$\mathscr{O}_Y$-Algebras of finite type of $\mathscr{C}$ (whose $\mathscr{C}$ is the union* (**I**, 9.6.6) *and (1.7.9)); set $Z_\lambda = \operatorname{Spec}(\mathscr{C}_\lambda)$ and let g_λ be the composed morphism $X \xrightarrow{g} Z \to Z_\lambda$. Then the following conditions are equivalent:*

a) *The morphism $g : X \to Z$ is an immersion.*

b) *There exists λ such that $g_\lambda : X \to Z_\lambda$ is an immersion.*
c) *There exists λ such that $g_\lambda : X \to Z_\lambda$ is an immersion.*

Moreover, when g_λ is an immersion, it is the same of g_μ for $\mu \geqslant \lambda$, and there exists λ such that, for $\mu \geqslant \lambda$, g_μ is an open immersion.

Proposition (8.12.3). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a morphism separated and of finite type. Let $\mathscr{B} = f_*(\mathscr{O}_X)$, which is a quasi-coherent $\mathscr{O}_Y$-Algebra (**I**, 9.2.2); let $\mathscr{C}$ be the integral closure of $\mathscr{O}_Y$ in $\mathscr{B}$, which is a quasi-coherent $\mathscr{O}_Y$-Algebra (**II**, 6.3.4). Set $Z = \operatorname{Spec}(\mathscr{C})$, and let $g : X \to Z$ be the Y-morphism corresponding to the canonical injection $\mathscr{C}_\lambda \to \mathscr{B} = f_*(\mathscr{O}_X)$ (**II**, 1.2.7). Let $(\mathscr{C}_\lambda)$ be the filtering increasing family of sub-$\mathscr{O}_Y$-Algebras quasi-coherent of finite type of $\mathscr{C}$ (whose $\mathscr{C}$ is the union (**I**, 9.6.6) and (1.7.9)), and, for all λ, $g_\lambda : X \to Z_\lambda$ the Y-morphism corresponding to the injection $\mathscr{C}_\lambda \to \mathscr{B} = f_*(\mathscr{O}_X)$. Then the following conditions are equivalent:* IV-3 | 44

a) *There exists a factorization of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

a′) *There exists a factorization $X \xrightarrow{f'} Y' \xrightarrow{u} Y$ of f, where f' is an open immersion and u a finite morphism.*
b) *The morphism $g : X \to Z$ is an immersion.*
c) *There exists λ such that $g_\lambda : X \to Z_\lambda$ is an immersion.*

Moreover, when g_λ is an immersion, so is g_μ for $\mu \geqslant \lambda$, and there exists λ such that, for $\mu \geqslant \lambda$, g_μ is an open immersion.

We will say that a morphism $f : X \to Y$, where Y is quasi-compact and quasi-separated, is *pseudo-finite*, if it is of finite type and satisfies condition *a)* of (8.12.3) (in which case it is necessarily separated).

Corollary (8.12.4). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a morphism.* IV-3 | 45

(i) *Suppose f pseudo-finite. Then, for every morphism $Y' \to Y$, where Y' is quasi-compact and quasi-separated, $f_{(Y')} : X' = X_{(Y')} \to Y'$ is pseudo-finite.*
(ii) *Let (U_λ) be a covering of Y by open quasi-compact subsets. For f to be pseudo-finite, it is necessary and sufficient that for all λ, the restriction $f_\lambda : f^{-1}(U_\lambda) \to U_\lambda$ of f is a pseudo-finite morphism.*
(iii) *Suppose moreover Y noetherian, and f of finite type. Then, for f to be pseudo-finite, it is necessary and sufficient that, for every $y \in Y$, the morphism $f_y = f \times 1 : X_y = X \times_Y \operatorname{Spec}(\mathscr{O}_y) \to \operatorname{Spec}(\mathscr{O}_y)$ be pseudo-finite.*

. 8.12.5 We can now give a proof of Zariski's "Main theorem" (**III**, 4.4.3) not using the "global" cohomological results of chap. III, but appealing instead to finer properties of local noetherian rings; we will moreover generalize the statement of the theorem by eliminating the noetherian hypotheses:

Theorem (8.12.6). — *("Main theorem" of Zariski). Let Y be a prescheme quasi-compact and quasi-separated. If a morphism $f : X \to Y$ is quasi-finite, separated and of finite presentation, there exists a factorization of f*

(8.12.6.1) $$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

§9. Constructible properties

9.1. The principle of finite extension

Proposition (9.1.1). — *(Principle of finite extension). Let k be a field, $\mathfrak{C}$ a set of extensions of k. Suppose the following conditions are satisfied:* IV-3 | 54

label=(i) *If $K \in \mathfrak{C}$ and if there exists a k-homomorphism $K \to K'$ (where K' belongs to the universe under consideration), then $K' \in \mathfrak{C}$.*
lbbel=(ii) *If $K \in \mathfrak{C}$, there exists a sub-extension K' of K, of finite type over k, such that $K' \in \mathfrak{C}$.*
lcbel=(iii) *If $K \in \mathfrak{C}$ is the field of fractions of an integral k-algebra A of finite type, there exists $f \in A - \{0\}$ such that for every maximal ideal $\mathfrak{m}$ of A we have $A_f/\mathfrak{m} \in \mathfrak{C}$.*

Let Ω be an algebraically closed extension of k (in the universe under consideration). The following conditions are equivalent: IV-3 | 55

label=(a) *$\mathfrak{C}$ is non-empty.*
lbbel=(b) *There exists $K' \in \mathfrak{C}$ which is a finite extension of k.*
lcbel=(c) *We have $\Omega \in \mathfrak{C}$.*

PROOF. The condition (i) implies that *b)* entails *c)*, and *c)* trivially entails *a)*; let us prove that *a)* entails *b)*. By virtue of (ii) and (iii) there exists an extension $K' \in \mathfrak{C}$ which is of the form $A/\mathfrak{m}$, where A is a k-algebra of finite type on k and $\mathfrak{m}$ a maximal ideal of A. By the Hilbert Nullstellensatz (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 2 of th. 1), K' is a finite extension of k. □

Corollary (9.1.2). — *Under the hypotheses* (i), (ii), (iii) *of* (9.1.1)*, if k is algebraically closed and if $\mathfrak{C}$ is non-empty, $\mathfrak{C}$ contains all the extensions of k in the universe under consideration.*

PROOF. Indeed, as *a)* entails *c)*, we have $k \in \mathfrak{C}$, and the conclusion follows from hypothesis (i). □

Remark (9.1.3). — In practice, one will verify condition (ii) of (9.1.1) by noting that K is the inductive limit of its sub-extensions of finite type K_α, and by applying the results of § 8, taking into account the fact that for $K_\alpha \subset K_\beta$, K_β is faithfully flat over K_α. Frequently the set $\mathfrak{C}$ is formed by *fields* belonging to a set $\mathfrak{C}'$ of k-algebras satisfying the following condition:

(i bis) If $A \in \mathfrak{C}'$ and if there exists a homomorphism of k-algebras $A \to A'$ (where A' belongs to the universe under consideration), then $A' \in \mathfrak{C}'$.

When this is the case, condition (i) is trivially satisfied, and one will verify generally condition (iii) of (9.1.1) by noting that the field of fractions K of A is the inductive limit of the algebras A_f (for the relation $\mathrm{D}(f) \supset \mathrm{D}(g)$), and by applying the results of § 8, taking into account eventually the fact that the morphisms $\mathrm{D}(g) \to \mathrm{D}(f)$ are open immersions.

Proposition (9.1.4). — *Let k be a field, Ω an algebraically closed extension of k, X and Y two preschemes of finite type over k. The following conditions are equivalent:*

label=(a) *There exists an Ω-morphism $X_{(\Omega)} \to Y_{(\Omega)}$ (resp. an Ω-morphism having one (determined) of the properties* (i) *to* (xiv) *of* (8.10.5)*).*
lbbel=(b) *There exists a finite extension k' of k and a k'-morphism $X_{(k')} \to Y_{(k')}$ (resp. a k'-morphism having the considered property).*
lcbel=(c) *There exists an extension K of k and a K-morphism $X_{(K)} \to Y_{(K)}$ (resp. a K-morphism having the considered property).*

PROOF. We apply the remark (9.1.3), taking for $\mathfrak{C}'$ the set of all k-algebras A (of the universe under consideration) such that there exists an A-morphism $X \otimes_k A \to Y \otimes_k A$ (resp. a morphism having one of the properties of (8.10.5)). The condition (i bis) of (9.1.3) is then verified because the properties considered in (8.10.5) are all stable under base change. The condition (iii) of (9.1.1) is therefore satisfied by virtue of (8.8.2, , (i)) (resp. (8.10.5)), since Spec(k) is quasi-compact and quasi-separated. It remains to verify condition (ii) of (9.1.1) which also follows from (8.8.2, , (i)) and (8.10.5), by considering K as an inductive limit of its sub-extensions of finite type. We conclude therefore by (9.1.1). In particular, if there exists an extension K of k and a K-isomorphism $X_{(K)} \xrightarrow{\sim} Y_{(K)}$, we say that X and Y are *geometrically isomorphic*. □ IV-3 | 56

The following corollary generalizes (**II**, 6.6.5):

Corollary (9.1.5). — *Let k be a field, X a k-prescheme. If there exists an extension K of k such that $X_{(K)}$ is projective (resp. quasi-projective) over K, then X is projective (resp. quasi-projective) over k.*

PROOF. The morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ being faithfully flat and quasi-compact, and X being of finite type over k already by (**II**, 2.7.1, , (v)), the hypothesis means that there exists a closed immersion (resp. an immersion) $X_{(K)} \to \mathbf{P}^r_K = \mathbf{P}^r_k \otimes_k K$ ((**II**, 5.4.2), (ii) and (**II**, 5.5.3)); applying property (iv) (resp. (ii)) of (8.10.5), we deduce that there exists a finite extension k' of k and a closed immersion (resp. an immersion) $X_{(k')} \to \mathbf{P}^r_{k'}$, in other words $X_{(k')}$ is projective (resp. quasi-projective) over k'. We conclude by (**II**, 6.6.5). □

Proposition (9.1.6). — *Let k be a field, Ω an algebraically closed extension of k, X a prescheme of finite type over k, $\mathscr{F}$, $\mathscr{G}$ two coherent $\mathscr{O}_X$-Modules. Suppose that there exists an isomorphism $\mathscr{F} \otimes_k \Omega \xrightarrow{\sim} \mathscr{G} \otimes_k \Omega$. Then there exists a finite extension k' of k and an isomorphism $\mathscr{F} \otimes_k k' \xrightarrow{\sim} \mathscr{G} \otimes_k k'$.*

PROOF. The reasoning is the same as in (9.1.4), applying (8.5.2, , (i)) (one uses in the proof the property (iii) of (9.1.1)), the fact that the morphisms $\mathrm{D}(g) \to \mathrm{D}(f)$ (with the notations of (9.1.3)) are open immersions, and $a\ fortiori$ the morphisms are flat. □

9.2. Constructible and ind-constructible properties

Definition (9.2.1). — Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation. We say that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) *property of algebraic preschemes* if the two following conditions are satisfied: **IV-3 | 56**

label=(i) If k is a field, X an algebraic prescheme over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, for $\mathbf{P}(X, \mathscr{F}, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, \mathscr{F} \otimes_k k', k')$ be true.

lbbel=(ii) Let S be an integral noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $s \in S$, set $X_s = u^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_S} k(s)$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, k(s))$ is true. Then one of the two sets E, $S - E$ contains an open non-empty set (and consequently is a neighborhood of η, or E is a neighborhood of η if it contains η).

Remarks (9.2.2). — *(i)* It is understood here a convention of language and not a mathematical definition in the strict sense. We have analogous "definitions" for relations between k, one or several k-preschemes, coherent Modules on these preschemes, or homomorphisms between these Modules. **IV-3 | 57**

(ii) We shall also have to consider relations involving *constructible parts* of preschemes. For example, let $\mathbf{P}(X, Z, k)$ be a relation; we say (by abuse of language) that $\mathbf{P}$ is a constructible (resp. ind-constructible) property of the constructible part Z of X if the two following conditions are satisfied:

1° If k is a field, X an algebraic prescheme over k, Z a *constructible part* of X, k' an extension of k, then, for $\mathbf{P}(X, Z, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, p^{-1}(Z), k')$ be true ($p : X_{(k')} \to X$ being the projection).

2° Let S be an integral noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, Z a constructible part of X. For every $s \in S$, set $X_s = u^{-1}(s)$, $Z_s = Z \cap X_s$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, Z_s, k(s))$ is true. Then one of the two sets E, $S - E$ contains an open non-empty set (resp. contains an open non-empty set if it contains η).

We note that in condition 2° one must assume that Z is a constructible part of X, and *not merely* a constructible part of X_s for every s; the conjunction of these two properties entails the second (**0**, 1.8.2), but not conversely.

(iii) If $\mathbf{P}$ is a constructible property, it is clear that the same is true of "non $\mathbf{P}$". If $\mathbf{P}$, $\mathbf{Q}$ are two constructible (resp. ind-constructible) properties, then so are the properties "$\mathbf{P}$ or $\mathbf{Q}$" and "$\mathbf{P}$ and $\mathbf{Q}$"; indeed, if, under the hypotheses of (9.2.1, , (ii)), E, E' are two parts of S and if E contains an open non-empty set, then so does $E \cup E'$, and if $S - E$ and $S - E'$ each contain an open non-empty set, then so does $(S - E) \cap (S - E') = (S - (E \cup E'))$.

(iv) Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation satisfying condition (9.2.1, , (i)); let S be a prescheme, $u : X \to S$ a morphism of type fini, with the notations of (9.2.1, , (ii)), and let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, k(s))$ is true. Let moreover $g : S' \to S$ be any morphism, and set $X' = X_{(S')}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$; it follows from the transitivity of the fibres (**I**, 3.6.4) and condition (9.2.1, , (i)) that the set of $s' \in S'$ such that $\mathbf{P}(X'_{s'}, \mathscr{F}'_{s'}, k(s'))$ is true is equal to $g^{-1}(E)$. This extends immediately to the case where $\mathbf{P}$ involves several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules.

(v) As we shall see in the sequel of this paragraph and in the rest of chap. IV, most of the properties $\mathbf{P}$ satisfying condition (9.2.1, , (i)) also satisfy (9.2.1, , (ii)). As possible exceptions, note the property of being *projective*, or *affine*, or *quasi-affine* on the base field (for an algebraic prescheme); we shall see (9.6.2) that these properties are *ind-constructible*, but we shall give later an example where S is an open non-empty part of $\mathrm{Spec}(\mathbf{Z})$ (or an open part of an elliptic curve over a finite field) and

where all the fibres X_s except the generic fibre X_η are projective over $k(s)$ (all the preschemes X_s being of dimension 2). **IV-3** | 58

Proposition (9.2.3). — *Let* **P** *be a constructible (resp. ind-constructible) property of algebraic preschemes, S a prescheme, X a prescheme of finite presentation over S, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set E of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, k(s))$ is true is locally constructible (resp. ind-constructible). Moreover, if S is irreducible of generic point η, one of the two sets E, $S - E$ is a neighborhood of η in S (resp. E is a neighborhood of η if it contains this point).*

Proof. To prove these assertions, one can reduce to the case where $S = \operatorname{Spec}(A)$ is affine. One knows then that there exists a sub-ring A_0 of A which is a $\mathbf{Z}$-algebra of finite type, an A_0-prescheme of finite type X_0 and a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that X is isomorphic to $X_0 \otimes_{A_0} A$ and $\mathscr{F}$ to $\mathscr{F}_0 \otimes_{A_0} A$ (8.9.1). Let $p : S \to S_0 = \operatorname{Spec}(A_0)$ be the morphism corresponding to the injection $A_0 \to A$, and let E_0 be the set of $s_0 \in S_0$ such that

$$\mathbf{P}((X_0)_{s_0}, (\mathscr{F}_0)_{s_0}, k(s_0))$$

is true; then, by virtue of (9.2.2, , (iv)), we have $E = p^{-1}(E_0)$; one can therefore reduce to the case where S is the spectrum of a $\mathbf{Z}$-algebra of finite type, hence a noetherian scheme. We use the criterion of constructibility ($\mathbf{0}_{\text{III}}$, 9.2.3) (resp. the criterion of ind-constructibility (**I**, 1.9.10)); we are then led, using as in (9.2.2, , (iv)) above and replacing S by a closed integral *subscheme* of S, to the case where S is noetherian and integral, and where it suffices to prove that E is rare in S or contains an open non-empty set of S (resp. that E contains an open non-empty set if it contains the generic point); but this is guaranteed by virtue of condition (9.2.1, , (ii)). ☐

Corollary (9.2.4). — *Let* **P** *be a constructible (resp. ind-constructible) property of algebraic preschemes, X, Y two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, set $X_s = X \times_S \operatorname{Spec}(k(s))$, $Y_s = Y \times_S \operatorname{Spec}(k(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set E of $s \in S$ such that the property $\mathbf{P}(f_s^{-1}(y), k(y))$ is true for every $y \in Y_s$ is locally constructible (resp. ind-constructible).*

Proof. Indeed, let Z be the set of $y \in Y$ such that $\mathbf{P}(f^{-1}(y), k(y))$ is true. As the fibres $f^{-1}(y)$ and $f_s^{-1}(y)$ are isomorphic, we see that E is the set of $s \in S$ such that $Y_s \subset Z$; if $g : Y \to S$ is the structural morphism, we therefore have $E = S - g(Y - Z)$. Since f is of finite presentation (**I**, 1.6.2, , (v)), it follows from (9.2.3) that Z is locally constructible (resp. ind-constructible) in Y. As g is of finite presentation, $g(Y - Z)$ is locally constructible (resp. pro-constructible) in S, by virtue of the theorem of Chevalley (**I**, 1.8.4) (resp. of (**I**, 1.9.5, , (vii))); hence E is locally constructible (resp. ind-constructible) in S. ☐ **IV-3** | 59

Remark (9.2.5). — We note that if **P** is a property of algebraic preschemes for which Proposition (9.2.3) is true, then **P** also satisfies the condition (9.2.1, , (ii)): this follows from the fact that in an irreducible noetherian space, a constructible set is either rare or contains an open non-empty set ($\mathbf{0}_{\text{III}}$, 9.2.3).

Proposition (9.2.6). — *Denote by* **P** *one of the following properties of a k-prescheme algebraic X:*

label=(i) X is empty.
lbbel=(ii) X is finite over k.
lcbel=(iii) X is radiciel over k.
ldbel=(iv) $\dim(X)$ belongs to a given part Φ of the set $\overline{\mathbf{Z}} = \mathbf{Z} \cup \{-\infty\}$.

Then **P** *is constructible.*

Proof. It is clear that (i) and (ii) are particular cases of (iv), by taking Φ to be the set $\{-\infty\}$ and the set $\{-\infty, 0\}$ respectively. It thus suffices to prove (iii) and (iv). In each of the two cases, condition (i) of (9.2.1) is satisfied by virtue of (2.7.1, , (xv)) and (4.1.4). On the other hand, in case (iii), property **P** satisfies the conclusion of (9.2.3) by virtue of (1.8.7); it remains to see that the same holds in case (iv). This will follow from the following more precise proposition: ☐

Proposition (9.2.6.1). — *If $f : X \to S$ is a morphism of finite presentation, the function $s \mapsto \dim(f^{-1}(s))$ is locally constructible.*

PROOF. The question is local on S, so one can assume that $S = \operatorname{Spec}(A)$ is affine and prove that for every n, the set E of $s \in S$ such that $\dim(X_s) = n$ is constructible. The same reasoning as in (9.2.3) reduces us to the case where A is noetherian and integral. □

Corollary (9.2.6.2). — *Let S be an integral noetherian prescheme of generic point η, $f : X \to S$ a morphism of type fini. Then there exists a neighborhood U of η in S such that the function $s \mapsto \dim(X_s)$ is constant in this neighborhood.*

9.3. Constructible properties of morphisms of algebraic preschemes

Proposition (9.3.1). — *Let $\mathbf{P}(X, k)$ be a constructible (resp. ind-constructible) property of algebraic preschemes. Denote by $\mathbf{P}'(f, X, Y, k)$ the following relation: $f : X \to Y$ is a k-morphism of k-preschemes algebraic such that for every $y \in Y$, we have the property $\mathbf{P}(f^{-1}(y), k(y))$. Then $\mathbf{P}'$ is constructible (resp. ind-constructible).* **IV-3 | 60**

PROOF. Indeed, as $\mathbf{P}$ satisfies condition (9.2.1, , (i)), so does $\mathbf{P}'$ by virtue of the transitivity of the fibres (**I**, 3.6.4); on the other hand, condition (9.2.1, , (ii)) follows from (9.2.4), by virtue of the remark (9.2.5). □

Proposition (9.3.2). — *Denote by $\mathbf{P}$ one of the following properties of a k-morphism of k-preschemes algebraic $f : X \to Y$:*

label=(i) f is surjective.
lbbel=(ii) f is quasi-finite.
lcbel=(iii) f is radiciel.
ldbel=(iv) For every $y \in Y$, $\dim(f^{-1}(y))$ belongs to Φ (notation of (9.2.6)).

Then $\mathbf{P}$ is constructible.

PROOF. This follows immediately from (9.3.1) and (9.2.6) if one takes into account that f is of finite type (**I**, 1.5.4, , (v)), the characterization of radiciel morphisms (**I**, 3.5.8), and that of quasi-finite morphisms (**II**, 6.2.2). □

Proposition (9.3.3). — *Suppose verified the hypotheses of (8.8.1), and let us moreover retain the notations; suppose moreover that S_α is quasi-compact, X_α and Y_α are of finite presentation over S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Let $\mathbf{P}$ be an ind-constructible property of morphisms of algebraic preschemes. For every $s \in S$ (resp. $s_\alpha \in S_\alpha$) set $X_s = X \times_S \operatorname{Spec}(k(s))$, $Y_s = Y \times_S \operatorname{Spec}(k(s))$, $f_s = f_\alpha \times 1 : X_s \to Y_s$ (resp. $X_{\lambda,s_\alpha} = X_\lambda \times_{S_\lambda} \operatorname{Spec}(k(s_\lambda))$, $Y_{\lambda,s_\alpha} = Y_\lambda \times_{S_\lambda} \operatorname{Spec}(k(s_\lambda))$, $f_{\lambda,s_\alpha} = f_\lambda \times 1 : X_{\lambda,s_\alpha} \to Y_{\lambda,s_\alpha}$). Then, in order that for every $s \in S$ we have $\mathbf{P}(f_s)$, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that for every $s_\lambda \in S_\lambda$ we have $\mathbf{P}(f_{\lambda,s_\lambda})$.*

PROOF. Indeed, let E (resp. E_λ) be the set of $s \in S$ (resp. $s_\lambda \in S_\lambda$) such that property $\mathbf{P}(f_s)$ (resp. $\mathbf{P}(f_{\lambda,s_\lambda})$) is true; it follows from (9.2.2, , (iv)) that $E_\mu = u_{\lambda\mu}^{-1}(E_\lambda)$ for $\lambda \leqslant \mu$, and $E = u_\lambda^{-1}(E_\lambda)$; by virtue of (9.2.3), E (resp. E_λ) is ind-constructible in S (resp. S_λ); the proposition then follows from (8.3.4) applied to the ind-constructible part E of S. □

Remark (9.3.4). — The combination of (9.3.3) and (9.3.2, , (ii)) proves the assertion (8.10.5, , (xi)).

Proposition (9.3.5). — *Let $\mathbf{P}(X, Y, k)$ be the property: "X and Y are two preschemes of finite type over the field k, and there exists an extension k' of k and a k'-morphism $g : X_{(k')} \to Y_{(k')}$ satisfying $\mathbf{Q}(g)$", where $\mathbf{Q}$ is one of the properties* (i) *to* (xiv) *of (8.10.5). Then $\mathbf{P}$ is an ind-constructible property.* **IV-3 | 61**

PROOF. The definition of $\mathbf{P}$ shows that condition (9.2.1, , (i)) is satisfied, taking into account that the property $\mathbf{Q}(g)$ is stable under change of the base field, and that two extensions of k can always be considered as sub-extensions of a third extension. To verify (9.2.1, , (ii)), one can reduce to the case where $S = \operatorname{Spec}(A)$ is affine; if $K = k(\eta)$, the field of fractions of A, and a finite extension K' of K and a K'-morphism $g' : (X_\eta)_{(K')} \to (Y_\eta)_{(K')}$, satisfying $\mathbf{Q}(g')$, and K' is evidently the field of fractions of an integral A-algebra finite A'. If one sets $S' = \operatorname{Spec}(A')$, it follows from (8.10.5) that there is a neighborhood U' of the generic point η' of S' such that, if one sets $X' = X \otimes_A A'$, $Y' = Y \otimes_A A'$, there exists for every $s' \in U'$, a morphism $X'_{s'} \to Y'_{s'}$ having property $\mathbf{Q}$. But the morphism $h : S' \to S$ is finite, hence closed, and since $h^{-1}(\eta) = \{\eta'\}$, $h(U')$ contains a neighborhood U of η

in S; for every $s \in U$, there is therefore an $s' \in U'$ such that $h(s') = s$, and as $X'_{s'} = X_s \otimes_{k(s)} k(s')$, $Y'_{s'} = Y_s \otimes_{k(s)} k(s')$, property $\mathbf{P}(X_s, Y_s, k(s))$ is true for every $s \in U$. □

Example (9.3.6). — Take for example $\mathbf{Q}$ the property of being an isomorphism. Then, combining (9.3.5) and (9.3.3), one obtains the following property: with the notations and hypotheses being those of (8.8.1), S_α being quasi-compact, X_α and Y_α of finite presentation on S_α, in order that, for every $s \in S$, X_s and Y_s be *geometrically isomorphic* (9.1.4), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that, for every $s_\lambda \in S_\lambda$, X_{λ,s_λ} and Y_{λ,s_λ} be geometrically isomorphic.

An analogous result holds when the preschemes considered are endowed with "composition laws" of a certain kind, for example "preschemes in groups", "preschemes in rings", etc. ($\mathbf{0}_{\mathrm{III}}$, 8.2.1). Then the preceding statement is also valid when one replaces "isomorphism" by "isomorphisms" which are also *homomorphisms* for the considered composition laws ($\mathbf{0}_{\mathrm{III}}$, 8.2.2); it suffices here to use not only (8.10.5) but also (8.8.2, , (i)) by noting that the notion of homomorphism for a composition law is expressed by writing that diagrams of morphisms of preschemes are commutative.

One can also, instead of considering morphisms of preschemes as in (9.3.5), consider homomorphisms of Modules, using (9.1.6) instead of (9.1.5).

9.4. Constructibility of certain properties of modules

(9.4.1). Notations. In this number and the following ones until the end of § 9, we shall systematically use the following notations: given a morphism $f : X \to S$, we set, for every $s \in S$, $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(k(s))$; for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, $\mathscr{F}_s$ will denote the $\mathscr{O}_{X_s}$-Module $\mathscr{F} \otimes_{\mathscr{O}_S} k(s)$, and for every homomorphism $u : \mathscr{F} \to \mathscr{G}$ of $\mathscr{O}_X$-Modules quasi-coherent, $u_s : \mathscr{F}_s \to \mathscr{G}_s$ will be the morphism $p^*(u)$, p denoting the canonical projection $X_s \to X$. For every section g of $\mathscr{F}$ over X, we will denote by g_s the image of g by the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_s, \mathscr{F}_s)$. For every closed part Z of X, we will denote by Z_s the preimage $p^{-1}(Z) = Z \cap X_s$ (**I**, 3.6.1). Finally, if Y is a second S-prescheme and $h : X \to Y$ an S-morphism, we will denote h_s the morphism $h \times 1 : X_s \to Y_s$. **IV-3** | 62

Proposition (9.4.2). — *Let S be an integral noetherian prescheme of generic point η, $f : X \to S$ a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ three quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to \mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules, and suppose that the sequence*

$$\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$$

is exact. Then there exists a neighborhood U of η in S such that, for every $s \in U$, the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact.

Proof. With the general notations of (9.2.1), it is here the relation $\mathbf{P}(X, \mathscr{F}, \mathscr{G}, \mathscr{H}, u, v, k)$: "$X$ is an algebraic prescheme over the field k, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is an exact sequence of coherent $\mathscr{O}_X$-Modules". As for every extension k' of k, the projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.13), the condition (9.2.1, , (i)) is satisfied (2.2.7). By virtue of (9.2.3), it suffices to reduce to the case where S is affine, noetherian, and integral, and where $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ are coherent $\mathscr{O}_X$-Modules, and the hypothesis implies that there exists a neighborhood U of η in S such that the sequence is exact on $f^{-1}(U)$, applying the general method of (8.1.2, , *a*)) and the fact that it suffices to prove that $\operatorname{Ker}(v_s) = (\operatorname{Ker} v)_s$ and $\operatorname{Im}(u_s) = (\operatorname{Im} u)_s$ for s close to η in S; by the flatness results ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), one is led to prove the proposition in the particular case where the sequence $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H} \to 0$ is exact. But then there is an open U in S containing η and such that $\mathscr{H}|f^{-1}(U)$ is flat on U (6.9.1); it follows from (2.1.8) that for every $s \in U$, the sequence $0 \to \mathscr{F}_s \to \mathscr{G}_s \to \mathscr{H}_s \to 0$ is exact, which completes the proof. □

Corollary (9.4.3). — *Let S be an integral noetherian prescheme, of generic point η, $f : X \to S$ a morphism of finite presentation. Let $\mathscr{L}^\bullet = (\mathscr{L}^i)_{i \in \mathbf{Z}}$ be a complex of $\mathscr{O}_X$-Modules quasi-coherent of finite presentation. For every i, there exists a neighborhood U of η in S such that the canonical homomorphisms*

$$(\mathscr{H}^i(\mathscr{L}^\bullet))_s \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet_s)$$

are bijective for every $s \in U$.

PROOF. One can evidently reduce to a complex of three terms of degrees -1, 0, $+1$: $\mathscr{M} \xrightarrow{u} \mathscr{N} \xrightarrow{v} \mathscr{P}$, and to $i = 0$; the homomorphism to consider is then the canonical homomorphism $(\operatorname{Ker} v / \operatorname{Im} u)_s \to \operatorname{Ker}(v_s)/\operatorname{Im}(u_s)$. Using (8.9.1) and (8.5.2, , (i)), one sees that one can reduce to the case where S (hence also X) is noetherian, and hence $\mathscr{M}$, $\mathscr{N}$, $\mathscr{P}$ are $\mathscr{O}_X$-Modules *coherent*; then $\operatorname{Im}(u)$ and $\operatorname{Ker}(v)$ are also coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4) and there exists moreover a neighborhood U of η such that for $s \in U$, we have $\operatorname{Ker}(v_s) = (\operatorname{Ker}(v))_s$ and $\operatorname{Im}(u_s) = (\operatorname{Im}(u))_s$; the conclusion then follows from (9.4.2) applied to the suite exact $0 \to (\operatorname{Im} u)_\eta \to (\operatorname{Ker} v)_\eta \to (\operatorname{Ker} v/\operatorname{Im} u)_\eta \to 0$, taking into account the fact that $\mathscr{O}_\eta = k(\eta)$ (since S is integral) and by the flatness of the suite exact on the right. □ IV-3 | 63

Proposition (9.4.4). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ three $\mathscr{O}_X$-Modules quasi-coherent of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to \mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules. Then the set E of $s \in S$ such that the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact is locally constructible.*

PROOF. Taking into account (9.2.3), it suffices to establish that the property $\mathbf{P}$ considered in (9.4.2) is locally constructible. We have already remarked in the proof of (9.4.2) that condition (9.2.1, , (i)) is satisfied, and it remains to verify condition (9.2.1, , (ii)). Suppose therefore S integral noetherian of generic point η, and let us prove that E or $S - E$ is a neighborhood of η. If $\eta \in E$, our assertion follows from (9.4.2), and one can therefore assume $\eta \notin E$, that is, the sequence $\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$ is not exact. We distinguish two cases:

1º Set $w = v \circ u$, and suppose first that $w_\eta = v_\eta \circ u_\eta \neq 0$. As $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ are coherent, so is $\mathscr{N} = \operatorname{Ker}(w)$ ($\mathbf{0}_{\mathrm{I}}$, 5.3.4); it follows from (9.4.2) that there is a neighborhood U of η in S such that, for $s \in S$, $\operatorname{Ker}(w_s) = \mathscr{N}_s$; restricting S, one can therefore assume this relation is verified for every $s \in S$. Let j the canonical injection $\mathscr{N} \to \mathscr{F}$, and set $\mathscr{M} = \operatorname{Coker}(j)$; the exactness on the right of the functor $\mathscr{F}_s$ entails that $\mathrm{M}_s = \operatorname{Coker}(j_s)$ for every $s \in S$; as M is coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), the hypothesis $w_\eta \neq 0$ means that $M_\eta \neq 0$; it follows from (1.8.6) that there is a neighborhood U of η in S such that $M_s \neq 0$ for $s \in U$, hence $w_s \neq 0$ for $s \in U$, and therefore $\mathscr{F}_s$ is not g_s-flat, i.e., $S - E$ is a neighborhood of η.

2º Suppose that $w_\eta = 0$; by virtue of (8.5.2, , (i)), applied following the general method of (8.1.2, , *a*)), there exists a neighborhood U of η such that $w|f^{-1}(U) = 0$; replacing S by U, one can already suppose $w = 0$. Then $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is a complex of three terms $\mathscr{L}^\bullet$, to which one can apply (9.4.3); by hypothesis we have $(\mathscr{H}^0(\mathscr{L}^\bullet))_\eta \neq 0$, and $\mathscr{H}^0(\mathscr{L}^\bullet)$ is coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.3 and 5.3.4), so it follows from (1.8.6) that there is a neighborhood U of η such that $(\mathscr{H}^0(\mathscr{L}^\bullet))_s \neq 0$ for every $s \in U$; but as one can assume that $\mathscr{H}^0(\mathscr{L}^\bullet_s) = (\mathscr{H}^0(\mathscr{L}^\bullet))_s$ for $s \in U$ by (9.4.3), one sees that $S - E$ is again a neighborhood of η in S. □

Corollary (9.4.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$ two $\mathscr{O}_X$-Modules quasi-coherent of finite presentation, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\mathscr{O}_X$-Modules. Then the set of $s \in S$ such that u_s is injective (resp. surjective, bijective, zero) is locally constructible.* IV-3 | 64

PROOF. It suffices to apply (9.4.4) to the sequences $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G}$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{1_{\mathscr{G}}} \mathscr{G}$. □

Corollary (9.4.6). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Let h be a section of $\mathscr{F}$ over X; for every $s \in S$, let h_s be the corresponding section of $\mathscr{F}_s$ over X_s (for the projection morphism $X_s \to X$). Then, the set of $s \in S$ such that $h_s = 0$ is locally constructible.*

PROOF. It suffices to remark that h corresponds to a homomorphism $u : \mathscr{O}_X \to \mathscr{F}$ ($\mathbf{0}_{\mathrm{I}}$, 5.1.1) and h_s to the homomorphism u_s. □

Proposition (9.4.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. The set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module locally free (resp. locally free of rank n) is locally constructible.*

PROOF. If X is an algebraic prescheme over a field k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, then, for $\mathscr{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that

$\mathscr{F} \otimes_k k'$ also be, since the projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.7). In other words, the condition (9.2.1, , (i)) is verified for the properties considered. The constructibility then follows, with the structure of the proof considering four separate cases depending on the generic fiber. □

Lemma (9.4.7.1). — *Let S be an integral noetherian prescheme of generic point η, X, Y two S-preschemes of type fini, $g : X \to Y$ an S-morphism, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. If $\mathscr{F}_\eta$ is not g_η-flat, then there exists a neighborhood U of η in S such that for every $s \in U$, $\mathscr{F}_s$ is not g_s-flat.*

Proof. Taking into account (2.1.2) and Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, § 2, n$^{\text{o}}$ 3, Remark 1, the hypothesis means that there exists an open non-empty V of Y_η, and an $\mathscr{O}_\eta$-Module coherent $\mathscr{G}$ and an $\mathscr{O}_Z$-homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that $(1 \otimes u)_s$ is not injective for $s \in U$. Using (8.5.2, , (i) and (ii)) and the method of (8.1.2, , *a*)), one can assume that these data are defined over a neighborhood of η; the result follows from (9.4.5). □

Remark (9.4.7.2). — The lemma (9.4.7.1) will later be generalized and freed from noetherian hypotheses (11.2.8).

Proposition (9.4.8). — *Let S be a prescheme locally noetherian, $f : X \to S$ a morphism of type fini, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that for every $s \in S$, X_s is a locally integral prescheme. Then the set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module of torsion (resp. a torsion-free $\mathscr{O}_{X_s}$-Module) is locally constructible.* **IV-3** | 65

Proof. One can evidently assume $S = \operatorname{Spec}(A)$ affine and noetherian and prove E (resp. E') is constructible; replacing S by the reduced closed subprescheme having as underlying space a closed part of S, and noting (**I**, 3.6.4) that for $s \in Y$, the fibre $(X_{(Y)})_s$ is canonically identified to X_s and the sheaf $(\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_Y)_s$ to $\mathscr{F}_s$, one sees that one is brought to the case where S is integral of generic point η, and has to prove that E or $S - E$ (resp. E' or $S - E'$) is a neighborhood of η. Note moreover that X is a finite union of affine opens U_i, each of the $(U_i)_s$ is induced on an open of X_s, hence locally integral; moreover, if $(U_i)_\eta$ is empty (resp. non-empty), one knows (9.2.6) that it is also true (resp. non-empty) in a neighborhood of η. One can therefore assume that all the $(U_i)_\eta$ are non-empty and integral; saying that $\mathscr{F}_s$ is of torsion (resp. torsion-free) is equivalent to saying that each $\mathscr{F}_s|(U_i)_s$ is of torsion (resp. torsion-free). One is therefore brought to the case where $X = \operatorname{Spec}(B)$ is affine, and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type; one will set $B_s = B \otimes_A k(s)$, $M_s = M \otimes_A k(s)$ and one can assume B_s is integral. We have four cases to consider:

1$^{\text{o}}$ $\eta \in E$; M_η is then a B_η-module of torsion of finite type, and there is therefore an element $h \neq 0$ in B_η such that $hM_\eta = 0$; by virtue of (8.5.2, , (i)), applied following the method of (8.1.2, , *a*)), one can (replacing S by a neighborhood of η) assume that $h \in \Gamma(X, \mathscr{O}_X)$ is of the form g_η, where $g \in \Gamma(X, \mathscr{O}_X)$; let $u : \mathscr{F} \to \mathscr{F}$ the endomorphism of $\mathscr{F}$ defined by g; by hypothesis, $u_\eta = 0$, hence (9.4.5) the endomorphism $u_s : \mathscr{F}_s \to \mathscr{F}_s$, which is the multiplication by g_s, is zero in a neighborhood of η. On the other hand, let $v : \mathscr{O}_X \to \mathscr{O}_X$ the endomorphism defined by the multiplication by g; as B_η is integral and $h = g_\eta \neq 0$, v_η is injective, and it follows hence from (9.4.5) that v_s is an injective endomorphism of $\mathscr{O}_{X_s}$ for s close to η, i.e., g_s is an $\mathscr{O}_{X_s}$-regular element for these values of s; hence $\mathscr{F}_s$ is of torsion in a neighborhood of η.

2$^{\text{o}}$ $\eta \in S - E$. Say that a B_η-module of finite type M_η is not a module of torsion means that its quotient M_η/T by its torsion submodule is $\neq 0$, and as it is a torsion-free module of finite type, it is isomorphic to a submodule of B_η^n; there exists therefore an injective homomorphism $w : M_\eta \to B_\eta^n$ which is $\neq 0$. Applying (8.5.2, , (i)) and (9.4.5) as in 1$^{\text{o}}$, there exists a homomorphism $v : \mathscr{F} \to \mathscr{O}_X^n$ such that $v_\eta = \widetilde{w}$, and such that for s close to η, $v_s : \mathscr{F}_s \to \mathscr{O}_{X_s}^n$ is injective; for these values of s, $\mathscr{F}_s$ is therefore torsion-free.

3$^{\text{o}}$ As M_η is a B_η-module of finite type with torsion, i.e., it is not locally free, following from the analysis above there exists a homomorphism $v : \mathscr{F} \to \mathscr{O}_X^n$ such that v_η is injective, and such that for s in a neighborhood of η, $v_s : \mathscr{F}_s \to \mathscr{O}_{X_s}^n$ is injective; for these values of s, $\mathscr{F}_s$ is torsion-free.

4$^{\text{o}}$ $\eta \in S - E'$. Let T the torsion submodule of M_η; by hypothesis $T \neq 0$, and T is of finite type since M_η is noetherian. Using this time (8.5.2, , (i) and (ii)) a coherent $\mathscr{O}_X$-Module $\mathscr{G}$ and an injective homomorphism $u : \mathscr{G} \to \mathscr{F}$ such that $\mathscr{G}_\eta = \widetilde{T}$ and that u_η is the canonical injection $\widetilde{T} \to \mathscr{F}_\eta$. It follows from 1$^{\text{o}}$ and from (1.8.6) that in a neighborhood of η, $\mathscr{G}_s$ is a torsion $\mathscr{O}_{X_s}$-Module $\neq 0$, and on the other hand it follows from (9.4.5) that in a neighborhood of η, u_s is injective. We conclude that in a neighborhood of η, the torsion submodule of $\mathscr{F}_s$ is not zero. ☐

Remark (9.4.9). — The property "X is a k-prescheme algebraic locally integral" does not satisfy condition (9.2.1, , (i)), and it is therefore not certain that the statement (9.4.8) remains valid when one makes no hypothesis on S and one only assumes that f is a morphism of finite presentation and $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation. Let us consider however the following particular case: S_0 being a locally noetherian prescheme, let $f_0 : X_0 \to S_0$ a morphism of type fini, such that the fibres $(X_0)_{s_0}$ are locally integral (for every $s_0 \in S_0$), and $\mathscr{F}_0$ a coherent $\mathscr{O}_{X_0}$-Module; let $g : S \to S_0$ be any morphism, set $X = X_0 \times_{S_0} S$, $F = \mathscr{F}_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_S$, $f = f_0 \times 1_S : X \to S$, and *suppose* that for every $s \in S$, the fibre X_s is again locally integral. Then the set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is of torsion (resp. torsion-free) is again *locally constructible*. **IV-3** | 66

9.5. Constructibility of topological properties

Proposition (9.5.1). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. Then the set E of $s \in S$ such that $Z_s \neq \varnothing$ is locally constructible.* **IV-3** | 67

PROOF. Indeed, $E = f(Z)$, and it suffices to apply the theorem of Chevalley (**I**, 1.8.4). ☐

Corollary (9.5.2). — *If Z', Z'' are two locally constructible parts of X, the set of $s \in S$ such that $Z'_s \subset Z''_s$ (resp. $Z'_s = Z''_s$) is locally constructible.*

PROOF. Indeed, the relation $Z'_s \subset Z''_s$ is equivalent to $(Z' \cap (\complement Z''))_s = \varnothing$ and $Z' \cap \complement Z''$ is locally constructible. ☐

Proposition (9.5.3). — *Let $f : X \to S$ be a morphism of finite presentation, Z, Z' two locally constructible parts of X such that $Z \subset Z'$. Then the set E of $s \in S$ such that Z_s is dense in Z'_s is locally constructible in S.*

PROOF. We must verify the two conditions of (9.2.2, , (ii)). For the first condition, concerning an algebraic prescheme X over a field k, two constructible parts $Z \subset Z'$, and an extension k' of k, the projection $p : X_{(k')} \to X$ is faithfully flat and quasi-compact, hence $p^{-1}(\overline{Z}) = \overline{p^{-1}(Z)}$ and $p^{-1}(\overline{Z'}) = \overline{p^{-1}(Z')}$ by virtue of (2.3.10); as p is surjective, the relation $\overline{Z} = \overline{Z'}$ is equivalent to $\overline{p^{-1}(Z)} = \overline{p^{-1}(Z')}$.

Let us now verify the second condition, supposing S affine, noetherian, integral, and of generic point η. We distinguish two cases:

1° $\eta \in S - E$, hence Z_η is not dense in Z'_η. Montrons first that one can assume Z' closed. We have $\overline{Z'_\eta} = \overline{Z_\eta}$ (closure in X_η); set $V_\eta = X_\eta - \overline{Z_\eta}$, which is open in X_η and does not meet Z'_η; one can assume V_η of the form $V \cap X_\eta$, where V is open (hence constructible) in X, and the hypothesis $V_\eta \cap Z'_\eta = \varnothing$ entails then by (9.5.1) that $V_s \cap Z'_s = \varnothing$ for $s \in U$; hence Z_s is not dense in Z'_s for $s \in U$, otherwise $U \subset S - E$ is a neighborhood of η.

2° $\eta \in E$, hence Z_η is dense in Z'_η. It suffices to show that one can assume $Z' = Z$, and this follows from the method of reducing to irreducible closed parts. ☐

Corollary (9.5.4). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. The set E of $s \in S$ such that Z_s is open (resp. closed, resp. locally closed) in X_s is locally constructible in S.*

PROOF. Saying that Z_s is open in X_s means that $(X - Z)_s = X_s - Z_s$ is closed in X_s, and as $X - Z$ is locally constructible, one can reduce to considering the set of $s \in S$ such that Z_s is closed, and the set of s such that Z_s is locally closed. We verify in each case the two conditions of (9.2.2, , (ii)). The first condition is a consequence of the fact that $p : X_{(k')} \to X$ is faithfully flat and quasi-compact, and of (2.3.12). We then verify the second condition, assuming S affine, noetherian, and integral. ☐

Proposition (9.5.5). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. For every $s \in S$, let $\mathrm{D}(s) \subset \mathbf{Z} \cup \{-\infty\}$ be the set of dimensions of the irreducible components of Z_s. Then the function $s \mapsto \mathrm{D}(s)$ is locally constructible in S.* **IV-3** | 69

Corollary (9.5.6). — *Let S be a prescheme noetherian and irreducible of generic point η, X a prescheme irreducible, $f : X \to S$ a morphism dominant of type fini. Then there exists a neighborhood U of η in S such that, for $s \in U$, all the irreducible components of X_s are of dimension $n = \dim(X_\eta)$.*

Remark (9.5.7). — One must be careful that under the hypotheses of (9.5.6), it can happen that X_s is *not irreducible for any* $s \neq \eta$ in a neighborhood of η; in other words, the property "X is an irreducible k-prescheme algebraic" is *not constructible*. **IV-3** | 70

9.6. Constructibility of certain properties of morphisms

Proposition (9.6.1). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. Let E be the set of $s \in S$ for which f_s has one of the following properties:* **IV-3** | 71

label=(i) being surjective;
lbbel=(ii) being dominant;
lcbel=(iii) being separated;
ldbel=(iv) being proper;
lebel=(v) being radiciel;
lfbel=(vi) being finite;
lgbel=(vii) being quasi-finite;
lhbel=(viii) being an immersion;
libel=(ix) being a closed immersion;
ljbel=(x) being an open immersion;
lkbel=(xi) being an isomorphism;
llbel=(xii) being a monomorphism.

Then E is locally constructible in S.

Proof. The assertions for (i), (v) and (vii) are inserted here only as a reminder, having already been established in (9.3.2).

(ii): Note first that f is of finite presentation (**I**, 1.6.2, , (v)), hence, by the theorem of Chevalley (**I**, 1.8.4), $f(X)$ is locally constructible in Y. On the other hand, $f_s(X_s) = (f(X))_s$ (**I**, 3.6.1); the set of s such that f_s is dominant is the set of s such that $(f(X))_s$ is dense in Y_s; the conclusion follows from (9.5.3).

(iii): As $f : X \to Y$ is of finite presentation, the diagonal immersion $\Delta_f : X \to X \times_Y X$ is of finite presentation (**I**, 1.6.2, , (iv) and (v)); it follows hence from (1.8.4.1) that $\Delta_f(X) = \Delta_X$ is a locally constructible part of $X \times_Y X$. Saying that f_s is separated means (taking into account (**I**, 5.3.4)) that $(\Delta_X)_s$ is closed in $X_s \times_{Y_s} X_s = (X \times_Y X)_s$; the conclusion follows from (9.5.4).

(iv): We show that the property "X and Y are algebraic preschemes over a field k, and $f : X \to Y$ a k-morphism proper" is constructible. Condition (9.2.1, , (i)) is verified by virtue of (2.7.1, , (vii)). It thus remains to verify (9.2.1, , (ii)), hence one can assume S is affine, noetherian, and integral of generic point η, and one must prove that E or $S - E$ is a neighborhood of η. The proof considers separately the cases where $\eta \in E$ and $\eta \notin E$.

(vi): The property is the conjunction of the properties (iv) and (vii) (8.11.1), hence the proposition follows from what has already been proved.

(ix): Verification of condition (9.2.1, , (i)) follows from (2.7.1, , (xi)). One can reduce to the case where S is affine, noetherian, and integral of generic point η, and prove that E or $S - E$ is a neighborhood of η. **IV-3** | 72

(viii): The verification of condition (9.2.1, , (i)) is done as in (ix), using (2.7.1, , (x)) and the fact that every immersion of one prescheme noetherian into another is quasi-compact.

(x): Using this time (2.7.1, , (ix)) and (8.10.5, , (iii)), we are brought to the case where S is affine, noetherian, integral of generic point η, and where $\eta \in S - E$. We distinguish three cases: 1° $f_\eta(X_\eta)$ is not a locally closed part of Y_η. 2° $f_\eta(X_\eta)$ is locally closed in Y_η but f_η is not an immersion. 3° $f_\eta(X_\eta)$ is open in Y_η and f_η is an immersion.

(xi): The property is the conjunction of (i) and (x) and the conclusion follows from what has been proved.

(xii): By virtue of (**I**, 5.3.8), saying that f_s is a monomorphism means that $\Delta_{f_s} = (\Delta_f)_s : X_s \to X_s \times_{Y_s} X_s = (X \times_Y X)_s$ is an isomorphism, and as $X \times_Y X$ is an S-prescheme of finite presentation (**I**, 1.6.2, , (iv)), the conclusion follows from (xi). ☐

Proposition (9.6.2). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism.* **IV-3** | 73

label=I) Let E be the set of $s \in S$ for which f_s has one of the following properties:

label=(i) *being affine;*
lbbel=(ii) *being quasi-affine;*
lcbel=(iii) *being projective;*
ldbel=(iv) *being quasi-projective.*
Then E is ind-constructible in S.

label=II) *Let $\mathscr{L}$ be an $\mathscr{O}_X$-Module inversible. Then the set E' of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to f_s is ind-constructible in S.*

PROOF. I) We verify conditions (i) and (ii) of (9.2.1). For the condition (9.2.1, , (i)), it follows, for properties (i) and (ii), from (2.7.1, , (xiii) and (xiv)); for the properties (iii) and (iv), it follows from (9.1.5). We then verify (9.2.1, , (ii)), assuming S noetherian, integral of generic point η, and that $f_\eta : X_\eta \to Y_\eta$ has one of the properties (i) to (iv) of the statement. Applying (8.10.5, , (viii), (ix), (xiii) and (xiv)), one sees that there exists a neighborhood U of η such that, if V and W are the preimages of U in X and Y respectively by the structural morphisms, the restriction $V \to W$ of f has the same property. The conclusion follows from the stability of these properties under base change (**II**, 4.6.13) and (4.4.10).

II) One proceeds similarly. For condition (9.2.1, , (i)) it follows this time from (2.7.2). For the condition (9.2.1, , (ii)), with the same notations as in I), it follows from (8.10.5, , (vi) and (x)), applied following the method of (8.1.2, , *a*))), that there exists a neighborhood U of η in S such that the restriction $\mathscr{L}|V$ is ample (resp. very ample) relatively to the restriction $V \to W$ of f. The conclusion follows from the stability of the properties considered under base change (**II**, 4.6.13) and the stability of amplitude by base change (**III**, 4.7.1) and (**II**, 4.4.10). □

Proposition (9.6.3). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism proper, $\mathscr{L}$ an $\mathscr{O}_X$-Module inversible. Then the set E (resp. E') of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to $f_s : X_s \to Y_s$ is locally constructible in S.* **IV-3** | 74

Corollary (9.6.4). — *Let $f : X \to S$ be a morphism proper of finite presentation, $\mathscr{L}$ an $\mathscr{O}_X$-Module inversible. Then the set E of $s \in S$ such that $\mathscr{L}_s$ is ample (relatively to f_s) is open in S, and $\mathscr{L}|f^{-1}(E)$ is ample relatively to the restriction $f^{-1}(E) \to E$ of f.*

Corollary (9.6.5). — *Under the hypotheses of (9.6.4), for $\mathscr{L}$ to be ample relatively to f, it is necessary and sufficient that, for every $s \in S$, $\mathscr{L}_s$ be ample relatively to f_s.*

Remarks (9.6.7). — *(i)* For each of the properties **P** considered in (9.6.1) the proposition (9.3.3) is applicable, and these properties (for the morphisms f_s) are therefore "stable" under passage to the projective limit of a projective system of essentially affine (8.13.4) S-preschemes X_λ to a suitable one of them. **IV-3** | 75

(ii) Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is *open* in X_s, and denote by Z_s the sub-prescheme of X_s induced on the open $Z \cap X_s$. Then, in the propositions (9.6.1) and (9.6.2, , (I)), one can everywhere replace f_s by its restriction $f_s|Z_s : Z_s \to Y_s$ without changing the conclusions.

9.7. Constructibility of properties of separability, geometric irreducibility, and geometric connectedness

Lemma (9.7.1). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η has n irreducible components (resp. n' connected components), there exists a neighborhood U of η in S such that for every $s \in U$, X_s has at least n irreducible components (resp. at least n' connected components).* **IV-3** | 76

PROOF. One can reduce to the case where S is affine, hence X quasi-compact and quasi-separated. Let V_i ($1 \leqslant i \leqslant n$) be the interiors of the irreducible components of X_η, which are two by two disjoint and quasi-compact, and whose reunion is dense in X_η; by virtue of (8.2.11), applied following the method of (8.1.2, , *a*))), there exist for each i open quasi-compact W_i in X such that $W_i \cap X_\eta = V_i$; as X is quasi-separated, the intersections $W_i \cap W_j$ are quasi-compact (**I**, 1.2.7), hence, replacing S by a neighborhood of η, one can assume $W_i \cap W_j = \varnothing$ for $i \neq j$ by virtue of (8.3.3). Moreover, as the W_i are constructible, there is a neighborhood U of η in S such that for $s \in U$ the

$(W_i)_s$ are non-empty (9.5.1) and (9.2.3) and that the reunion of the $(W_i)_s$ is dense (9.5.3) and (9.2.3). The rest of the proof follows by considering the components of X_s.

The case of connected components is treated similarly, replacing the V_i by the connected components C_j $(1 \leqslant j \leqslant n')$ of X_η. □

Lemma (9.7.2). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η is not reduced, there exists a neighborhood U of η in S such that, for every $s \in U$, X_s is not reduced.* IV-3 | 77

PROOF. Indeed, let $\mathscr{N}$ the Nilradical of $\mathscr{O}_X$; it follows from (8.2.13), applied following the method of (8.1.2, , *a*))) that $\mathscr{N}_\eta$ is the nilradical of $\mathscr{O}_{X_\eta}$, and the hypothesis is that $\mathscr{N}_\eta \neq 0$. We conclude from (9.4.5) and (9.2.3) that there is a neighborhood U of η such that, for every $s \in U$, $\mathscr{N}_s \neq 0$; and as $\mathscr{N}_s$ is evidently contained in the Nilradical of $\mathscr{O}_{X_s}$, one sees that X_s is not reduced for $s \in U$. □

Lemma (9.7.4). — *Let k be a field, Ω an algebraically closed extension of k, F a non-constant polynomial of $k[T_1, \dots, T_n]$. The following conditions are equivalent:*

label=(a) $X = \operatorname{Spec}(k[T_1, \dots, T_n]/(F))$ *is geometrically integral.*
lbbel=(b) $F_{(K)}$ *is irreducible for every extension K of k.*
lcbel=(c) $F_{(\Omega)}$ *is irreducible.*

We say in this case that F is geometrically irreducible.

Lemma (9.7.5). — *Let A be an integral ring, K its field of fractions, F a non-constant polynomial of $A[T_1, \dots, T_n]$ of degree d such that $F_{(K)}$ is geometrically irreducible. Then there exists $f \neq 0$ in A such that for every $x \in \mathrm{D}(f)$, $F_{(k(x))}$ is geometrically irreducible.* IV-3 | 78

Lemma (9.7.5.1). — *Let A be an integral ring, $S = \operatorname{Spec}(A)$, η the generic point of S, B the polynomial ring $A[T_1, \dots, T_m]$, $(P_i)_{i \in I}$ a finite family of elements of B, $\mathfrak{a}$ the ideal of B generated by the P_i. For every $s \in S$, let $\mathfrak{a}_s$ the ideal of $k(s)[T_1, \dots, T_m]$ generated by the polynomials $(P_i)_{(k(s))}$ $(i \in I)$. Then, if $V(\mathfrak{a}_\eta) = \varnothing$, there exists a neighborhood U of η in S such that $V(\mathfrak{a}_s) = \varnothing$ for every $s \in U$.*

PROOF. Indeed, let $Y = \operatorname{Spec}(B)$, and let Z the closed part $V(\mathfrak{a})$ of Y; as $\mathfrak{a}$ is an ideal of finite type, Z is constructible in Y ($\mathbf{0}_{\mathrm{III}}$, 9.1.5) and with the notations introduced in (9.4.1), $V(\mathfrak{a}_s) = Z_s$ for every $s \in S$; as the structural morphism $Y \to S$ is of finite presentation, the conclusion follows from (9.5.1) and (9.2.3). □

Proposition (9.7.6). — *Let S be a noetherian prescheme integral of generic point η, $f : X \to S$ a morphism of type fini, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. If $\mathscr{F}_\eta$ is without first associated immersed cycle, then there exists a neighborhood U of η in S such that, for every $s \in U$, $\mathscr{F}_s$ is without first associated immersed cycle.*

Theorem (9.7.7). — *Let $f : X \to S$ be a morphism of finite presentation, and let E be the set of $s \in S$ for which X_s has one of the following properties:* IV-3 | 79

label=(i) being geometrically irreducible;
lbbel=(ii) being geometrically connected;
lcbel=(iii) being geometrically reduced;
ldbel=(iv) being geometrically integral.

Then E is locally constructible in S.

PROOF. To see that the properties considered are constructible, remark first that they trivially verify condition (9.2.1, , (i)), by virtue of (4.5.6, , (i)) and (4.6.5, , (i)). It thus remains to verify (9.2.1, , (ii)), hence one can assume that $S = \operatorname{Spec}(A)$ is affine noetherian and integral of generic point η. By virtue of (4.6.8), there exists a finite extension K' of $K = k(\eta)$ such that $(X_\eta)_{(K')}$ has its connected components (resp. irreducible components) geometrically irreducible (resp. geometrically connected) and that $((X_\eta)_{(K')})_{\mathrm{red}}$ is geometrically reduced.

Suppose first that $\eta \in S - E$ for one of the properties (i) to (iv). The proof proceeds by showing that X_s is not geometrically irreducible (resp. connected, reduced, integral) in a neighborhood of η, using the lemmas (9.7.1) and (9.7.2). IV-3 | 80

Now suppose that $\eta \in E$ and examine each of the four properties separately.

1° Suppose X_η geometrically integral. The proof uses the function field and the lemma on geometrically irreducible polynomials (9.7.4) and (9.7.5).

2° Suppose X_η geometrically irreducible. One replaces X by X_{red} (**I**, 1.5.8) and concludes from 1°.

3° Suppose X_η geometrically connected. The proof uses the surjection (5.10.8.1) applied to **IV-3 | 81**
$\mathfrak{F} = \{\varnothing\}$ and the characterization of geometric connectedness (4.5.13.1).

4° Suppose X_η geometrically reduced. The proof uses the results on geometrically irreducible components and (9.7.6). $\square$

Proposition (9.7.8). — *Let S be an irreducible prescheme of generic point η, and $f : X \to S$ a morphism of finite presentation. Let n (resp. n') be the geometric number of irreducible components (resp. connected components) of X_η (4.5.2). Then there exists a neighborhood U of η in S such that for every $s \in U$, the geometric number of irreducible components (resp. connected components) of X_s is equal to n (resp. n').* **IV-3 | 82**

Corollary (9.7.9). — *Let $f : X \to S$ be a morphism of finite presentation. For every $s \in S$, let $n(s)$ (resp. $n'(s)$) be the geometric number of irreducible components (resp. connected components) of $f^{-1}(s)$ (4.5.2). Then the function $s \mapsto n(s)$ (resp. $s \mapsto n'(s)$) is locally constructible in S.*

Corollary (9.7.10). — *Let X be a locally noetherian prescheme such that, if X' is the normalization of X_{red}, the canonical morphism $f : X' \to X$ is finite. Then the set of points $x \in X$ such that X is geometrically unibranch at the point x is locally constructible in X.*

PROOF. Indeed, this set is by definition the set of points $x \in X$ such that the number of geometric points of $f^{-1}(x)$ is equal to 1. But as f is finite, this number is also the number of geometric irreducible components of the discrete space $f^{-1}(x)$ (**II**, 6.3.8) and of (4.5.11); the conclusion follows from (9.7.9). $\square$

Remark (9.7.11). — Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is open in X_s. Then, in theorem (9.7.7), one can everywhere replace X_s by Z_s without changing the conclusion: one sees it by repeating the reasoning made in (9.6.7, , (ii)).

Proposition (9.7.12). — *Let $f : X \to S$ be a morphism of finite presentation, and $g : S \to X$ a S-section of X (**I**, 2.5.5). For every $s \in S$, let X^0_s be the connected component of $f^{-1}(s)$ which contains $g(s)$; then, the reunion X^0 of the X^0_s for $s \in S$ is a locally constructible part of X.*

PROOF. We first show that one can always reduce to the case where $S = \mathrm{Spec}(A)$ is affine; with the notations of (9.2.3), we have $f = (f_0)_{(S)}$, where $f_0 : X_0 \to S_0$ is a morphism of type fini, and one can moreover assume that there exists a S_0-section $g_0 : S_0 \to X_0$ such that $g = (g_0)_{(S)}$ (8.9.1). Let x **IV-3 | 83**
be a point of X, Y the sub-prescheme reduced of S having $\overline{\{f(x)\}}$ as underlying space, and Y' the sub-prescheme reduced of S having $\overline{f(Z)}$ as underlying space; as the restriction of f to Z factorizes in $Z \to Y \to S$ (**I**, 5.2.2), one can assume, replacing S by Y, that $f(x) = \eta$, the generic point of the integral prescheme S. By hypothesis, X_η is the sum of two sub-preschemes X^0_η, X^1_η, induced on open complementary sets of X_η. As $g : S \to X$ is continuous and injective, one of these two opens is empty, and as $g(\eta) \in X^0_\eta$ by definition, we have $g^{-1}(X^1) = \varnothing$. In other words, g is a S-section of X^0; on the other hand, as $(X^0)_\eta$ is geometrically connected (4.5.13), it follows that for s close to η by (9.7.7), $(X^0)_s = X^0_s$. $\square$

9.8. Primary decomposition in a neighborhood of a generic fibre

Proposition (9.8.1). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-* **IV-3** | 83
coherent of finite presentation. Then the set E of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module without first associated immersed cycle is locally constructible.

Proof. One knows that for quasi-coherent Modules on algebraic preschemes, the property of being without first associated immersed cycle is invariant by extension of the base field (4.2.7); one can therefore reduce to the case where S is affine, noetherian, and integral of generic point η, and has to prove that E or $S - E$ is a neighborhood of η. One has seen in (9.7.6) that if $\eta \in E$, E **IV-3** | 84
is a neighborhood of η; it remains to consider the case where $\mathscr{F}_\eta$ has associated immersed cycles. One can reduce to the case where X is affine, and where there is a sub-$\mathscr{O}_X$-Module coherent $\mathscr{H}$ of $\mathscr{F}_\eta$ whose support Z' is non-empty and *rare* with respect to the support Z of $\mathscr{F}$; then, by (8.5.2) and (8.5.8), applied following the method of (8.1.2, , *a*))), there exist quotients $\mathscr{F}_i$ of $\mathscr{F}$ ($i \in I$) such that $\mathscr{G}_i = (\mathscr{F}_i)_\eta$ for every i, and a homomorphism $g : \mathscr{F} \to \bigoplus \mathscr{F}_i$ such that $h = g_\eta$. Moreover (**I**, 9.3.5), there exists an integer m such that $(\mathscr{J}_i^m \mathscr{F}_i)_\eta = (\mathscr{J}_i)_\eta^m \mathscr{G}_i = 0$ and, restricting S again, one can therefore assume that $\mathscr{J}_i^m \mathscr{F}_i = 0$ (8.5.2.5), hence that the support of $\mathscr{F}_i$ is contained in Z_i. □

Proposition (9.8.3). — *Under the conditions of* (9.8.2), *for every $s \in S$ and every $i \in I$, denote by $x_{i_{s\alpha}}$ ($\alpha \in J_{s,i}$) the maximal points of $(Z_i)_s$. There exists a neighborhood U of η in S such that, for every $s \in U$, the $x_{i_{s\alpha}}$ (for $i \in I$ and $\alpha \in J_{s,i}$) are two by two distinct and that* $\mathrm{Ass}(\mathscr{F}_s)$ *is the set $\{x_{i_{s\alpha}}\}$ (in other words, the first associated cycles of $\mathscr{F}_s$ are the irreducible components of $(Z_i)_s$). Moreover, one can shrink U such that the closure of $\{x_{i_{s\alpha}}\}$ in X_s is a first associated cycle of $\mathscr{F}_s$, and it is necessary and sufficient that $(Z_i)_\eta$ be a maximal first associated cycle of $\mathscr{F}_\eta$.*

Corollary (9.8.4). — *With the notations and hypotheses of* (9.8.2), *there exists a neighborhood U of η in S such that, for every $s \in U$, each of the $(\mathscr{F}_i)_s$ is without first associated immersed cycle, and the family $(\mathscr{F}_{i_{s\alpha}})_{\alpha \in J_{s,i}}$ is the unique irredundant reduced decomposition of $(\mathscr{F}_i)_s$, while the family $(\mathscr{F}_{i_{s\alpha}})_{i \in I, \alpha \in J_{s,i}}$ is an irredundant reduced decomposition of $\mathscr{F}_s$.*

Proposition (9.8.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-* **IV-3** | 85
coherent of finite presentation. For every $s \in S$, let $E(s)$ (resp. $E'(s)$) be the set (finite part of $\mathbf{Z} \cup \{-\infty\}$*) of dimensions of the first associated (resp. maximal first associated, i.e., the irreducible components of* $\mathrm{Supp}(\mathscr{F}_s)$*) cycles of $\mathscr{F}_s$. Then the functions $s \mapsto E(s)$ and $s \mapsto E'(s)$ are locally constructible in S.*

Proposition (9.8.6). — *With the hypotheses and notations of* (9.8.2), *let $i \in I$ be such that $(Z_i)_\eta$ is maximal* **IV-3** | 86
(in other words, an irreducible component of $\mathrm{Supp}(\mathscr{F}_\eta)$*). Then there exists a neighborhood U of η in S such that for every $s \in U$ and every $\alpha \in J_{s,i}$, the geometric length of $\mathscr{F}_s$ in $x_{i_{s\alpha}}$ (relative to $k(s)$)* (4.7.5) *is equal to the geometric length of $\mathscr{F}_\eta$ in x_i (relative to $k(\eta)$).*

Corollary (9.8.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent* **IV-3** | 87
of finite presentation. For every $s \in S$, let $M(s)$ be the set of couples (d, m) such that there exists an irreducible component of $\mathrm{Supp}(\mathscr{F}_s)$ *of dimension d and of radiciel multiplicity m for $\mathscr{F}_s$* (4.7.8). *Then the function $s \mapsto M(s)$ is locally constructible in S.*

Proposition (9.8.8). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. For every $s \in S$, let $l(s)$ be the sum of the total multiplicities for $\mathscr{F}_s$ (relative to $k(s)$) of the generic points of the irreducible components of* $\mathrm{Supp}(\mathscr{F}_s)$ (4.7.12). *Then the function $s \mapsto l(s)$ is locally constructible in S.*

Remark (9.8.9). — One can make more precise the preceding propositions in various ways; let us limit ourselves to one statement, given as an example. Say that a finite part P of a k-prescheme algebraic X is *saturated* if, for every couple of points x, y of P, the generic points of the irreducible components of $\overline{\{x\}} \cap \overline{\{y\}}$ also belong to P; for every finite part Q of X, there exists a smallest finite part P of X containing Q and saturated; we say that P is the *saturate* of Q. For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we call *primary skeleton* of $\mathscr{F}$ the system (P, Q, ω, d, m) where $Q = \mathrm{Ass}(\mathscr{F})$, P is the saturate of Q, ω is the order relation $\{x\} \subset \overline{\{y\}}$ on P, d the function $x \mapsto \dim \overline{\{x\}}$ on P, m the application of the set of maximal elements of Q for the relation ω. We define in an evident manner the notion of *isomorphism* of two virtual skeletons, and finally, with the previous notations, we call *geometric primary type* of $\mathscr{F}$ the class (for the relation of isomorphism of virtual skeletons) of the primary skeleton of $\mathscr{F}$.

(9.8.9.1). Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. For every $s \in S$, let $T(s)$ be the geometric primary type of $\mathscr{F}_s$. Then the function $s \mapsto T(s)$ is locally constructible in S. **IV-3** | 88

9.9. Constructibility of local properties of fibres

Proposition (9.9.1). — *Let $f : X \to S$ be a morphism locally of finite presentation, Z a locally constructible part of X such that for every $s \in S$, Z_s is closed in X_s, Φ a finite part of $\mathbf{Z} \cup \{\pm\infty\}$. The following parts of X are locally constructible:* **IV-3** | 88

label=(i) *The set of $x \in X$ such that $\dim_x(Z_{f(x)})$ belongs to Φ.*
lbbel=(ii) *The set of $x \in X$ such that $\operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ belongs to Φ.*
lcbel=(iii) *The set of $x \in X$ such that the local ring $\mathscr{O}_{X_{f(x)},x}$ is equidimensional.*

We note that properties (i) and (ii) express respectively that the functions $x \mapsto \dim_x(Z_{f(x)})$ and $x \mapsto \operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ are locally constructible in X ($\mathbf{0}_{\mathrm{III}}$, 9.3.1).

Proposition (9.9.2). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation, Φ a finite part of $\mathbf{Z} \cup \{\pm\infty\}$. The following parts of X are locally constructible:* **IV-3** | 90

label=(i) *The set of points $x \in X$ such that $\mathscr{F}_{f(x)}$ is equidimensional at the point x (5.1.12).*
lbbel=(ii) *The set of points $x \in X$ such that the geometric lengths of $\mathscr{F}_{f(x)}$, relative to $k(f(x))$, of the generic points of the irreducible components of $\operatorname{Supp}(\mathscr{F}_{f(x)})$ containing x (4.7.5), belong to Φ.*
lcbel=(iii) *The set of points $x \in X$ such that the dimensions of the first associated cycles of $\mathscr{F}_{f(x)}$ and containing x belong to Φ.*
ldbel=(iv) *The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically reduced at the point x (4.6.17).*
lebel=(v) *The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically integral at the point x (4.6.22).*
lfbel=(vi) *The set of $x \in X$ such that $\dim.\operatorname{proj}((\mathscr{F}_{f(x)})_x) \in \Phi$.*
lgbel=(vii) *The set of $x \in X$ such that $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) \in \Phi$.*
lhbel=(viii) *The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ possesses the property (S_k) at the point x (5.7.2).*
libel=(ix) *The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is a Cohen–Macaulay $\mathscr{O}_{X_{f(x)}}$-Module at the point x (5.7.1).*

Corollary (9.9.3). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation, **P** one of the properties* (i) *to* (ix) *of (9.9.2). Then the set of $s \in S$ such that the property **P** is true for all points $x \in X_s$ is locally constructible in S.* **IV-3** | 93

Proof. Indeed, its complement is the image by f of the complement of E, the set of points where **P** holds. As E is locally constructible in X, $X - E$ is also, hence $f(X - E)$ is locally constructible in S, by virtue of the theorem of Chevalley (**I**, 1.8.4). □

Proposition (9.9.4). — *Let $f : X \to S$ be a morphism locally of finite presentation. The set of $x \in X$ such that $X_{f(x)}$ has at the point x one (determined) of the following properties:*

label=(i) *being geometrically regular;*
lbbel=(ii) *possessing the geometric property (R_k);*
lcbel=(iii) *being geometrically normal;*
ldbel=(iv) *being geometrically reduced (i.e. separable);*
lebel=(v) *being geometrically punctually integral;*

is a locally constructible part of X.

Proof. For properties (iv) and (v), this follows from (9.9.2, , (iv) and (v)) applied to $\mathscr{F} = \mathscr{O}_X$. For the other properties, taking into account (6.7.8), one reduces, as at the beginning of (9.9.2), to the case where S is noetherian and integral of generic point $\eta = f(x)$. Moreover, by the theorem of generic flatness (6.9.1), one can, by replacing S by a neighborhood of η, assume that f is a *flat* morphism. Then, saying that $X_{f(x)}$ is geometrically regular at the point y means that f is *regular* at the point y (6.8.1), and one knows that the set L of these points is *open* in X (6.8.7), which proves the proposition in case (i). □ **IV-3** | 94

Corollary (9.9.5). — *Let $f : X \to S$ be a morphism of finite presentation, and denote by* **P** *one of the properties* (i) *to* (v) *of* (9.9.4)*. Then the set of $s \in S$ such that the property* **P** *is true for all points $x \in X_s$ is locally constructible in S.*

PROOF. The proof is the same as that of (9.9.3) starting from (9.9.4). □

Proposition (9.9.6). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{L}_\bullet$ a complex formed of $\mathscr{O}_X$-Modules quasi-coherent of finite presentation; for every $s \in S$, let $(\mathscr{L}_\bullet)_s$ the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} k(s)$ of $\mathscr{O}_{X_s}$-Modules coherent of finite type. For a given integer n, the set of $x \in X$ such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is locally constructible in X.*

PROOF. One can reduce to the case where $\mathscr{L}_i = 0$ except for $i = 0$, 1 or 2, and where $n = 1$. Moreover, the question being local on X, one can reduce to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, B being a finitely presented A-algebra, and the complex $\mathscr{L}_\bullet$ is a complex of $\mathscr{O}_X$-Modules of finite presentation, zero except in degrees 0, 1 and 2. There exists then a sub-ring A_0 of A which is a **Z**-algebra of finite type, an A_0-prescheme of finite type X_0 and a complex $\mathscr{L}_\bullet^{(0)}$ of $\mathscr{O}_{X_0}$-Modules coherent, zero except in degrees 0, 1 and 2, such that X is isomorphic to $X_0 \otimes_{A_0} A$ and $\mathscr{L}_\bullet \simeq \mathscr{L}_\bullet^{(0)} \otimes_{A_0} A$. For every $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, the fibres of X and X_0 at s and s_0 are the same, and $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = (\mathscr{H}_n((\mathscr{L}_\bullet^{(0)})_{f_0(x)}))_x$, hence one can reduce to the case where A is noetherian. Then, by (1.8.2), one is led to show that, for every $x \in X$, there is a neighborhood V of x in $\overline{\{x\}}$ such that, for every $x' \in V$, the set of dimensions of the first cycles associated to $\mathscr{F}_{f(x')}$ containing x' is the same. But this follows from (9.5.1) applied to the support Z of $\mathscr{H}_n(\mathscr{L}_\bullet)$, and $Z \cap \overline{\{x\}}$ is closed in the noetherian space X; hence $Z_s \cap \overline{\{x\}} = \varnothing$ (resp. $\neq \varnothing$) for s in a neighborhood of η by (9.5.1). The set $V = f^{-1}(U) \cap \overline{\{x\}}$ is then the desired neighborhood. □

§10. JACOBSON PRESCHEMES

We have already had occasion to observe (5.2.5) that even *excellent* preschemes (7.8.5) do not always behave like the "varieties" of classical Algebraic Geometry, in particular concerning dimension questions; thus if X is the spectrum of a complete discrete valuation ring, the set of closed points of X (reduced to a single element) is not everywhere dense in X, and its complement (also reduced to a single element) is an open set everywhere dense in X whose dimension is zero, hence $< \dim(X)$. We examine in this paragraph general conditions under which such phenomena do not occur; it turns out that, under such conditions, the behaviour is much more satisfactory in many respects for the relations between dimension, codimension, depth, and coprofundity in these preschemes ((10.6) and (10.8)). Furthermore, in the preschemes considered, the fact that the set of closed points is everywhere dense (and even "very dense" (10.1.3)) allows many demonstrations to be simplified; one thus recovers the classical viewpoint of "algebraic varieties" which, from our point of view, are the sets of closed points of *algebraic preschemes* over a field, and one makes the link between the language of schemes and that of "varieties" or "algebraic spaces" of Serre ((10.9) and (10.10)). **IV-3 | 95**

10.1. Very dense subsets of a topological space

(10.1.1). We say that a subset of a topological space X is *quasi-constructible* if it is a finite union of locally closed parts of X. We say that a subset T of X is *locally quasi-constructible* if for every $x \in X$, there exists an open neighbourhood V of x such that $T \cap V$ is quasi-constructible in V. It is clear that every quasi-constructible part of X is locally quasi-constructible: the converse is true if X is quasi-compact. The reasoning of $(\mathbf{0_{III}}, 9.1.3)$, where one omits the word "retrocompact", shows that the set of quasi-constructible (resp. locally quasi-constructible) parts of X, which we will denote by $\mathfrak{Qc}(X)$ (resp. $\mathfrak{Qqc}(X)$), is stable under finite union, finite intersection, and passage to complements. If $f : X \to Y$ is a continuous map, it follows immediately from these definitions that, for every quasi-constructible (resp. locally quasi-constructible) part Z of Y, $f^{-1}(Z)$ is quasi-constructible (resp. locally quasi-constructible) in X.

The constructible (resp. locally constructible) parts of X are evidently quasi-constructible (resp. locally quasi-constructible); the converse is true when X is Noetherian (resp. locally Noetherian).

In what follows, we will denote respectively by $\mathfrak{O}(X)$, $\mathfrak{F}(X)$, $\mathfrak{Qf}(X)$, $\mathfrak{C}(X)$, $\mathfrak{Lc}(X)$ the set of subsets of X that are respectively open, closed, locally closed, constructible, locally constructible.

Proposition (10.1.2). — *Let X be a topological space, X_0 a subset of X. The following conditions are equivalent:* IV-3 | 96

label=a) *For every non-empty locally closed part Z of X, we have $Z \cap X_0 \neq \varnothing$.*

a′) *For every closed part Z of X, we have $Z = \overline{Z \cap X_0}$.*

lbbel=b) *For every non-empty locally quasi-constructible part Z of X, we have $\overline{Z} \cap X_0 \neq \varnothing$.*

b′) *For every locally quasi-constructible part Z of X, we have $Z \subset \overline{Z \cap X_0}$ (equivalently, $Z \cap X_0$ is dense in Z).*

lcbel=c) *The map $U \mapsto U \cap X_0$ from $\mathfrak{O}(X)$ to $\mathfrak{O}(X_0)$ is injective (hence bijective).*

c′) *The map $F \mapsto F \cap X_0$ from $\mathfrak{F}(X)$ to $\mathfrak{F}(X_0)$ is injective (hence bijective).*

c″) *The map $Z \mapsto Z \cap X_0$ from $\mathfrak{Lc}(X)$ to $\mathfrak{Lc}(X_0)$ is injective (hence bijective).*

c‴) *The map $Z \mapsto Z \cap X_0$ from $\mathfrak{Lqc}(X)$ to $\mathfrak{Lqc}(X_0)$ is injective.*

Furthermore, when these conditions are satisfied, the map $Z \mapsto Z \cap X_0$ from $\mathfrak{Lc}(X)$ to $\mathfrak{Lqc}(X_0)$ is bijective.

Proof. Note that the surjectivity assertions in *c)* and *c′)* are trivial; they imply that every locally closed part of X_0 is the trace on X_0 of a locally closed part of X, and hence the map $\mathfrak{Lc}(X) \to \mathfrak{Lc}(X_0)$ defined in *c″)* is also surjective.

We will prove the implications

$$c''') \Rightarrow c'') \Rightarrow c') \Rightarrow c) \Rightarrow b) \Rightarrow b') \Rightarrow a') \Rightarrow a) \Rightarrow c''').$$

The first three are trivial. To see that *c)* implies *b)*, note first that *c)* implies that X_0 is *dense* in X. Replacing X and X_0 by a suitable open set U of X and $U \cap X_0$ respectively, we can reduce to the case where Z is locally closed in X, hence $Z = V \cap \complement W$, where V and W are open in X; the hypothesis $Z \neq \varnothing$ means that $V \not\subset W$, equivalently $V \cup W \neq W$. By virtue of *c)*, we have $(V \cup W) \cap X_0 \neq W \cap X_0$, hence $V \cap X_0 \not\subset W$, and therefore $(V \cap \complement W) \cap X_0 \neq \varnothing$.

To see that *b)* implies *b′)*, it suffices to apply *b)* to $Z \cap U$, where U is an arbitrary open neighbourhood of a point of Z. As $\overline{Z} \cap X_0 \subset \overline{Z}$, it is trivial that *b′)* implies *a′)*. To show that *a′)* implies *a)*, note that if Z is locally closed in X, we can write $Z = F - F'$ where F and F' are closed in X and $F' \subset F$; hence $Z \cap X_0 = (F \cap X_0) - (F' \cap X_0)$. If $Z \cap X_0 = \varnothing$, we would deduce that $F \cap X_0 = F' \cap X_0$, hence $F = F'$ by *a′)*, that is to say $Z = \varnothing$.

To see that *a)* implies *c‴)*, it suffices to show that if $Z \neq \varnothing$ is locally quasi-constructible, the relation $Z' \cap X_0 = Z'' \cap X_0$ is equivalent to $((Z' \cup Z'') - (Z' \cap Z'')) \cap X_0 = \varnothing$. In other words, it suffices to prove that *a)* implies *b)*; moreover, replacing X and X_0 by an open set U of X and $U \cap X_0$ respectively, we reduce to the case where Z is locally closed in X, whence the conclusion.

It remains to show that the map $\mathfrak{Lqc}(X) \to \mathfrak{Lqc}(X_0)$ is surjective. Let Z_0 be a locally quasi-constructible part of X_0: there is a covering (V_α) of X_0 by open sets of X_0 such that $Z_0 \cap V_\alpha$ is quasi-constructible in V_α (and hence also in X_0). By virtue of *c)*, for each α there is a single open set U_α of X such that $X_0 \cap U_\alpha = V_\alpha$, and by virtue of *c″)* a single quasi-constructible set Z_α in X such that $Z_0 \cap V_\alpha = X_0 \cap Z_\alpha$. If α and β are any two indices, we have $Z_\beta \cap U_\alpha \cap Z_0 \cap V_\beta = V_\alpha \cap Z_0 \cap V_\beta = Z_\alpha \cap X_0 \cap V_\beta \subset Z_\alpha \cap X_0$; as $Z_\beta \cap U_\alpha$ and Z_α are quasi-constructible in X, it follows from *c″)* that $Z_\beta \cap U_\alpha \subset Z_\alpha$. Setting $Z = \bigcup Z_\alpha$, since the V_α cover X_0, it follows from *c)* that the U_α cover X, and we see that Z is locally quasi-constructible in X and $Z_0 = Z \cap X_0$. ☐ IV-3 | 97

Definition (10.1.3). — When a subset X_0 of a topological space satisfies the equivalent conditions of (10.1.2), we say that X_0 is *very dense* in X.

We have already seen in the course of the proof of (10.1.2) that X_0 is then *dense* in X.

Corollary (10.1.4). — *If X_0 is very dense in X and U an open subset of X, $U \cap X_0$ is very dense in U. Conversely, if (U_α) is an open covering of X such that $U_\alpha \cap X_0$ is very dense in U_α for every α, then X_0 is very dense in X.*

Proof. As every locally closed part of U is locally closed in X, the first assertion follows from criterion *a)* of (10.1.2); similarly for the second, since if $Z \neq \varnothing$ is locally closed in X, $Z \cap U_\alpha$ is locally closed in U_α for every α, and $Z \cap U_\alpha \neq \varnothing$ for at least one α. ☐

10.2. Quasi-homeomorphisms

Proposition (10.2.1). — *Let X_0, X be two topological spaces, $f : X_0 \to X$ a continuous map. The following conditions are equivalent:*

label=a) *The map $U \mapsto f^{-1}(U)$ from $\mathfrak{D}(X)$ to $\mathfrak{D}(X_0)$ is bijective.*
a') *The map $F \mapsto f^{-1}(F)$ from $\mathfrak{F}(X)$ to $\mathfrak{F}(X_0)$ is bijective.*
lbbel=b) *The topology of X_0 is the inverse image by f of that of X, and $f(X_0)$ is very dense (10.1.3) in X.*
lcbel=c) *The functor $\mathscr{F} \mapsto f^*(\mathscr{F})$ from the category of sheaves of sets (resp. sheaves of abelian groups) on X to the category of sheaves of sets (resp. sheaves of abelian groups) on X_0 is an equivalence of categories.*

PROOF. It is clear that *a)* and *a')* are equivalent, and *a)* implies that the topology of X_0 is the inverse image by f of that of X. On the other hand, if $f(X_0)$ is not very dense in X, there exist two distinct open sets U_1, U_2 of X such that $U_1 \cap f(X_0) = U_2 \cap f(X_0)$, and hence $f^{-1}(U_1) = f^{-1}(U_2)$, which shows that *a)* implies *b)*. Conversely, condition *b)* implies that the maps $U \mapsto U \cap f(X_0)$ from $\mathfrak{D}(X)$ to $\mathfrak{D}(f(X_0))$ and $V \mapsto f^{-1}(V)$ from $\mathfrak{D}(f(X_0))$ to $\mathfrak{D}(X_0)$ are bijective, hence so is their composite $U \mapsto f^{-1}(U)$. **IV-3** | 98

To see that *a)* implies *c)*, it suffices to apply the definition of $f^*(\mathscr{F})$ ($\mathbf{0_I}$, 3.7.1) and the axioms of sheaves: *a)* implies that for every open set U of X, the canonical map $\Gamma(U, \mathscr{F}) \to \Gamma(f^{-1}(U), f^*(\mathscr{F}))$ is a functorial bijection in $\mathscr{F}$, whence *c)*. It remains to show that *c)* implies *b)*.

Suppose first that $f(X_0)$ is not very dense in X; there then exist two distinct closed parts $Y \supset Y'$ of X such that $Y \cap f(X_0) = Y' \cap f(X_0)$. Let $\mathscr{F}$ (resp. $\mathscr{F}'$) be the sheaf of abelian groups on X that is the direct image by the canonical injection $Y \to X$ (resp. $Y' \to X$) of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y (resp. Y'); the definition of f^* ($\mathbf{0_I}$, 3.7.1) shows that $f^*(\mathscr{F})$ is isomorphic to $f^*(\mathscr{F}')$ but $\mathscr{F}$ is not isomorphic to $\mathscr{F}'$, so condition *c)* is not satisfied. (One notes that the functor f^* is not even *faithful*, for if u and v are the identity automorphism and the zero endomorphism of $\mathscr{F}|\mathscr{F}'$, $f^*(u)$ and $f^*(v)$ are equal.)

Let us now show that *c)* implies that the topology of X_0 is the inverse image of that of X by f. Note that if condition *c)* is satisfied for the category of sheaves of sets, it is also satisfied for the category of sheaves of abelian groups, since it follows immediately from the definitions ($\mathbf{0_{III}}$, 8.2.5) that the latter is none other than the category of *group objects* in the category of sheaves of sets. It suffices therefore to prove our assertion assuming *c)* verified for the categories of sheaves of abelian groups on X and X_0. Now, if $\mathbf{Z}_X$ denotes the simple sheaf on X associated to the constant presheaf equal to $\mathbf{Z}$, we have a canonical isomorphism $\Gamma(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{Hom}(\mathbf{Z}_X, \mathscr{F})$ functorial in $\mathscr{F}$ (by the same reasoning as in ($\mathbf{0_I}$, 5.1.1)); as it is clear by definition ($\mathbf{0_I}$, 3.7.1) that $f^*(\mathbf{Z}_X) = \mathbf{Z}_{X_0}$, the hypothesis that f^* is an equivalence implies that the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_0, f^*(\mathscr{F}))$ is *bijective* for every sheaf of abelian groups $\mathscr{F}$ on X. Now, let Y_0 be a closed part of X_0; let $\mathscr{F}_0$ be the sheaf that is the direct image by the canonical injection $Y_0 \to X_0$ of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y_0. As f^* is an equivalence, $\mathscr{F}_0$ is isomorphic to a sheaf of the form $f^*(\mathscr{F})$, where $\mathscr{F}$ is a sheaf of abelian groups on X. Consider then the section s_0 of $\mathscr{F}_0$ over X_0 such that $(s_0)_{x_0} = 1_{x_0}$ if $x_0 \in Y_0$, $(s_0)_{x_0} = 0_{x_0}$ if $x_0 \notin Y_0$. The preceding remarks show that there exists a section s of $\mathscr{F}$ over X such that $(s_0)_{x_0} = s_{f(x_0)}$ for every $x_0 \in X_0$ (the fibres $(\mathscr{F}_0)_{x_0}$ and $\mathscr{F}_{f(x_0)}$ being canonically identified ($\mathbf{0_I}$, 3.7.1)); this implies that Y_0 is the inverse image by f of the set of $x \in X$ such that $s_x \neq 0_x$, and y is closed in X. Q.E.D. □

Definition (10.2.2). — When a continuous map $f : X_0 \to X$ satisfies the equivalent conditions of (10.2.1), we say that f is a *quasi-homeomorphism* of X_0 into X.

By virtue of (10.2.1, b)), to say that a subset X_0 of a topological space X is very dense in X means therefore that the canonical injection $X_0 \to X$ is a quasi-homeomorphism.

Corollary (10.2.3). — *The composite of two quasi-homeomorphisms is a quasi-homeomorphism.*

PROOF. This follows immediately from (10.2.1, a)). □

Corollary (10.2.4). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible part of Y, $X' = f^{-1}(Y')$; then the restriction $f' = f|X' : X' \to Y'$ is a quasi-homeomorphism.*

PROOF. It is clear by virtue of (10.2.1, b)) that the topology induced on X' by that of X is the inverse image of the topology induced on Y' by that of Y. Let $Z \neq \varnothing$ be a closed part *in* Y', and $z \in Z$; there is an open neighbourhood U of z in Y such that $U \cap Y'$ is the union of a finite number of parts Y'_i closed in U; if j is an index such that $z \in Y'_j$, $Z \cap Y'_j$ is closed in U. Since $f(X) \cap U$ is very dense in U (10.1.4), $Z \cap Y'_j \cap f(X)$ is non-empty (10.1.2, a)), and *a fortiori* $Z \cap f(X)$; but as $Z \subset Y'$, we have $Z \cap f(X) = Z \cap f'(X')$; the criterion (10.1.2, a)) shows that $f'(X')$ is very dense in Y', and we conclude with the help of (10.2.1, b)). ☐ IV-3 | 99

Corollary (10.2.5). — *Let $f : X \to Y$ be a continuous map, (V_α) an open covering of Y. If, for every α, the restriction $f^{-1}(V_\alpha) \to V_\alpha$ of f is a quasi-homeomorphism, then f is a quasi-homeomorphism.*

PROOF. This follows immediately from the criterion (10.2.1, b)) and (10.1.4). ☐

Corollary (10.2.6). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible part of Y, $X' = f^{-1}(Y')$. For Y' to be quasi-compact (resp. Noetherian, resp. retrocompact in Y), it is necessary and sufficient that X' be quasi-compact (resp. Noetherian, resp. retrocompact in X).*

PROOF. Let us prove the first two assertions; by virtue of (10.2.4), we can restrict to the case where $Y' = Y$. To say that X is quasi-compact (resp. Noetherian) means that for every increasing filtered family (U_α) in $\mathfrak{O}(X)$ having X as its greatest element (resp. for every increasing filtered family (U_α) in $\mathfrak{O}(X)$), there exists γ such that $U_\alpha = U_\gamma$ for $\alpha \geqslant \gamma$. As $U \mapsto f^{-1}(U)$ is a bijection from $\mathfrak{O}(Y)$ to $\mathfrak{O}(X)$, our assertion follows immediately from the preceding remark.

The quasi-compact open sets of X are therefore the sets of the form $f^{-1}(U)$ where U is a quasi-compact open set in Y, by virtue of (10.2.1, a)) and of what precedes. For X' to be retrocompact in X, it is necessary and sufficient that for every quasi-compact open set U in Y, $f^{-1}(U) \cap X' = f^{-1}(U \cap Y')$ be quasi-compact ($\mathbf{0}_{\mathbf{III}}$, 9.1.1); the first part of the proof shows that this is equivalent to saying that $U \cap Y'$ is quasi-compact for every quasi-compact open set U, that is to say that Y' is retrocompact in Y. ☐

Proposition (10.2.7). — *Let $f : X \to Y$ be a quasi-homeomorphism. Then the map $Z \mapsto f^{-1}(Z)$ from $\mathfrak{P}(Y)$ to $\mathfrak{P}(X)$ defines by restriction the following bijections (cf. (10.1.1) for the notation):*

$$\mathfrak{O}(Y) \xrightarrow{\sim} \mathfrak{O}(X)$$
$$\mathfrak{F}(Y) \xrightarrow{\sim} \mathfrak{F}(X)$$
$$\mathfrak{Qf}(Y) \xrightarrow{\sim} \mathfrak{Qf}(X)$$
$$\mathfrak{Qc}(Y) \xrightarrow{\sim} \mathfrak{Qc}(X)$$
$$\mathfrak{Qqc}(Y) \xrightarrow{\sim} \mathfrak{Qqc}(X)$$
$$\mathfrak{C}(Y) \xrightarrow{\sim} \mathfrak{C}(X)$$
$$\mathfrak{Lc}(Y) \xrightarrow{\sim} \mathfrak{Lc}(X).$$

PROOF. For the first two, this is the definition (10.2.2); as the topology of X is the inverse image by f of that of $f(X)$, the five remaining maps, where one replaces Y by $f(X)$, are bijective. One can therefore (by (10.2.1, b))) restrict to the case where X is a very dense subspace of Y, and the fact that the maps $\mathfrak{Qf}(Y) \to \mathfrak{Qf}(X)$, $\mathfrak{Qc}(Y) \to \mathfrak{Qc}(X)$, $\mathfrak{Qqc}(Y) \to \mathfrak{Qqc}(X)$ are bijective has already been proved (10.1.2). Every locally constructible part being locally quasi-constructible, the maps $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ and $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$ are injective; moreover, for every open set $U \subset Y$, the restriction $f^{-1}(U) \to U$ of f is a quasi-homeomorphism (10.2.4), hence if one shows that $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ is surjective, the same will be true of $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$. But by virtue of (10.2.6), every retrocompact open part Z in X is of the form $f^{-1}(T)$, where T is a retrocompact open set in Y; this clearly proves the surjectivity of $\mathfrak{C}(Y) \to \mathfrak{C}(X)$. ☐ IV-3 | 100

Remarks (10.2.8). — (i) One has seen in the proof of (10.2.1) that if f is a quasi-homeomorphism, the canonical map

(10.2.8.1) $$\Gamma(Y, \mathscr{F}) \longrightarrow \Gamma(X, f^*(\mathscr{F}))$$

is an isomorphism of abelian groups, functorial in $\mathscr{F}$, in the category of sheaves of abelian groups on X. As f^* is exact in this category ($\mathbf{0_I}$, 3.7.2), this implies that the canonical homomorphisms

$$\mathrm{H}^i(Y,\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, f^*(\mathscr{F}))$$

are *bijective* for all i.

(ii) If $f : X \to Y$ is a continuous map and $\mathscr{F}, \mathscr{G}$ two sheaves of abelian groups on Y, we have $f^*(\mathscr{F} \otimes_{\mathbf{Z}_Y} \mathscr{G}) = f^*(\mathscr{F}) \otimes_{\mathbf{Z}_X} f^*(\mathscr{G})$ up to canonical isomorphism ($\mathbf{0_I}$, 4.3.3). We conclude that if f is a quasi-homeomorphism, f^* is again an equivalence of the category of sheaves of rings on Y and the category of sheaves of rings on X; the datum of a structure of ringed space on Y is therefore equivalent to that of a structure of ringed space on X.

Given two ringed spaces $(X, \mathscr{O}_X)$, $(Y, \mathscr{O}_Y)$, we say that a morphism of ringed spaces $f = (\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a *quasi-isomorphism* if ψ is a quasi-homeomorphism of X into Y and $\theta^\# : \psi^*(\mathscr{O}_Y) \to \mathscr{O}_X$ is an isomorphism of sheaves of rings; in this case, the ringed space $(X, \mathscr{O}_X)$ is entirely determined, up to isomorphism, by $(Y, \mathscr{O}_Y)$, the space X, and the quasi-homeomorphism ψ (which one can take arbitrarily). When f is a quasi-isomorphism of ringed spaces, the functor

$$\mathscr{F} \longmapsto f^*(\mathscr{F})$$

is an equivalence of the category of $\mathscr{O}_Y$-Modules and that of $\mathscr{O}_X$-Modules, since $f^*(\mathscr{F})$ is canonically identified here with $\psi^*(\mathscr{F})$. We conclude for example with isomorphisms of bi-δ-functors

$$\mathrm{Ext}^i_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{G}) \xrightarrow{\sim} \mathrm{Ext}^i_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{G})).$$

In general, the usual constructions of the theory of sheaves and homological algebra, carried out on the ringed space Y or the ringed space X, are equivalent.

10.3. Jacobson spaces IV-3 | 101

Definition (10.3.1). — We say that a topological space X is a *Jacobson space* if the set X_0 of *closed* points of X is very dense in X (equivalently, if the canonical injection $X_0 \to X$ is a *quasi-homeomorphism*).

This means therefore (10.1.2) that every non-empty locally closed (or even merely locally quasi-constructible) part Z of X contains a closed point of X, or equivalently that *every closed part of X is the closure of the set of its closed points*.

Proposition (10.3.2). — *Let X be a Jacobson space, Z a locally quasi-constructible part of X; then the subspace Z of X is a Jacobson space, and for a point of Z to be closed in Z, it is necessary and sufficient that it be closed in X.*

Proof. If X_0 is the set of closed points of X, $Z \cap X_0$ is very dense in Z by virtue of (10.2.4) applied to the injection $i : X_0 \to X$; as the set Z_0 of closed points of Z evidently contains $Z \cap X_0$, it is *a fortiori* very dense in Z, hence Z is a Jacobson space. Let us now show that in fact $Z_0 = Z \cap X_0$; let $x \in Z$ be a closed point of Z; let $\overline{\{x\}}$ be its closure in X; $\overline{\{x\}} \cap Z = \{x\}$ is a locally quasi-constructible part, and as its intersection with X_0 is non-empty (10.1.2), we have $x \in X_0$. □

Proposition (10.3.3). — *Let X be a topological space, (U_α) an open covering of X. For X to be a Jacobson space, it is necessary and sufficient that each of the subspaces U_α be a Jacobson space.*

Proof. The condition is necessary by virtue of (10.3.2). Conversely, let us show that the hypothesis that the U_α are Jacobson spaces implies that for a closed point $x \in X$ to be closed in X, it suffices that it be closed in U_α. It suffices indeed to see that x is also closed in each of the U_β that contain it; but $U_\alpha \cap U_\beta$ is open in U_α, hence x is closed in $U_\alpha \cap U_\beta$, and by virtue of (10.3.2), x is also closed in U_β, which completes the proof. □

10.4. Jacobson preschemes and Jacobson rings

Definition (10.4.1). — We say that a prescheme X is a *Jacobson prescheme* if the underlying topological space of X is a Jacobson space. We say that a ring A is a *Jacobson ring* if $\operatorname{Spec}(A)$ is a Jacobson prescheme.

Every closed part of $X = \operatorname{Spec}(A)$ is of the form $Z = \mathrm{V}(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal equal to its radical, and the set X_0 of closed points of X is the set of maximal ideals of A; to say that $Z \cap X_0$ is dense in Z therefore means that $\mathfrak{a}$ is an *intersection of maximal ideals* (**I**, 1.1.4); as $\mathfrak{a}$ is an intersection of prime ideals, this is equivalent to saying that every prime ideal of A is an intersection of maximal ideals; by virtue of (10.3.1) and (10.1.2), the usual definition of Jacobson rings (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, déf. 1) therefore coincides with the definition (10.4.1).

Proposition (10.4.2). — *Let X be a prescheme, (U_α) a covering of X by affine open sets. For X to be a Jacobson prescheme, it is necessary and sufficient that the rings $\Gamma(U_\alpha, \mathscr{O}_X)$ be Jacobson rings.* **IV-3 | 102**

PROOF. This follows from (10.3.3) and the definition (10.4.1). □

(10.4.3). A discrete prescheme is a Jacobson prescheme; an Artinian ring is therefore a Jacobson ring. Every principal ideal domain having an infinity of maximal ideals (for example $\mathbf{Z}$) is a Jacobson ring; a local Noetherian ring A is a Jacobson ring if and only if its maximal ideal is its only prime ideal, that is to say if A is Artinian. Every sub-prescheme of a Jacobson prescheme is a Jacobson prescheme by virtue of (10.3.2).

Proposition (10.4.4). — *Let B be an integral ring. The following conditions are equivalent:*

label=a) *There exists $f \neq 0$ in B such that B_f is a field.*
lbbel=b) *The field of fractions of B is a B-algebra of finite type.*
lcbel=c) *There exists a field K containing B which is a B-algebra of finite type.*
ldbel=d) *The generic point of* $\operatorname{Spec}(B)$ *is isolated in* $\operatorname{Spec}(B)$.

PROOF. It is clear that *d)* is equivalent to *a)*, since *d)* means that there exists $f \neq 0$ in B such that $\mathrm{D}(f)$ is reduced to the generic point of $\operatorname{Spec}(B)$. It is trivial that *a)* implies *b)* and *b)* implies *c)*. Finally, *c)* implies *a)*, by virtue of Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 2 of th. 1. □

Proposition (10.4.5). — *Let A be a ring. The following conditions are equivalent:*

label=a) *A is a Jacobson ring.*
lbbel=b) *For every non-maximal prime ideal $\mathfrak{p}$ of A and every $f \neq 0$ in $B = A/\mathfrak{p}$, B_f is not a field.*
b′) *Every A-algebra of finite type K that is a field is a finite A-algebra.*

PROOF. We know that *a)* implies *b′)* (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, cor. 3 of th. 3). Moreover, the kernel of the homomorphism $A \to K$ is then a *maximal* ideal $\mathfrak{m}$ of A, and K is a *finite* extension of $A/\mathfrak{m}$ (*loc. cit.*). It is trivial that *b′)* implies *b)*, since $A/\mathfrak{p}$ is not a field, every B-algebra of finite type that is a field is an A-algebra of finite type, and B_f is a B-algebra of finite type. It remains to see that *b)* implies *a)*. We will use the following lemma: □

Lemma (10.4.5.1). — *Let X be a topological space having the following property: for every non-empty locally closed part Z of X, there exists a part Z' of Z, locally closed in X (or in Z, which amounts to the same), and a point $x \in Z'$, closed in Z'. Then, for X to be a Jacobson space, it is necessary and sufficient that no non-closed point x of X be isolated in $\overline{\{x\}}$.*

PROOF. If X is a Jacobson space, a non-closed point x of X cannot be isolated in $\overline{\{x\}}$, for this would mean that there exists an open set U of X such that $U \cap \overline{\{x\}} = \{x\}$; but $U \cap \overline{\{x\}}$ is locally closed in X and would contain no closed point of X, which is contrary to the hypothesis. Conversely, suppose the condition of the statement is satisfied; then the set of closed points X_0 of X is identical to the set of $x \in X$ that are isolated in $\overline{\{x\}}$; but it follows from (5.1.10.1) that this set is *very dense* in X, hence X is a Jacobson space by definition. **IV-3 | 103**

Recall (5.1.10) that the hypothesis made on X in (10.4.5.1) is always satisfied when the underlying space of X is a *prescheme*.

Returning to the proof of (10.4.5), condition *b)* implies that for every non-closed point x of $\operatorname{Spec}(A)$, the generic point x of $\operatorname{Spec}(A/\mathfrak{j}_x) = \mathrm{V}(\mathfrak{j}_x) = \overline{\{x\}}$ is not isolated in $\overline{\{x\}}$, hence *b)* implies *a)* by virtue of the lemma (10.4.5.1) and (10.4.4). □

Corollary (10.4.6). — *Every algebra of finite type B over a Jacobson ring A is a Jacobson ring, and the inverse image in A of every maximal ideal of B is a maximal ideal of A. In particular, every algebra of finite type over a field or over $\mathbf{Z}$ is a Jacobson ring.*

PROOF. A B-algebra of finite type K which is a field is also an A-algebra of finite type, hence an A-algebra of finite type and *a fortiori* a B-module of finite type, whence the first assertion; the second was demonstrated in the course of the proof of (10.4.5), applied to $K = B/\mathfrak{m}$. □

Corollary (10.4.7). — *If X is a Jacobson prescheme, $f : Y \to X$ a morphism locally of finite type, then Y is a Jacobson prescheme and the image by f of every closed point in Y is a closed point in X.*

PROOF. The question being local on X and on Y, we reduce to the case where X and Y are affine, and the corollary then follows from (10.4.6). □

Corollary (10.4.8). — *If k is an algebraically closed field, X a k-prescheme locally of finite type, the set of rational points of X over k is very dense in X.*

PROOF. Indeed, X is a Jacobson prescheme (10.4.7) and the closed points of X are exactly the rational points over k (**I**, 6.4.2). □

(10.4.9). The fact that preschemes locally of finite type over a field or over $\mathbf{Z}$ are Jacobson preschemes is particularly important, because of the possibility of reducing to this case in many questions of Algebraic Geometry (8.1.2, c)). We will give two examples:

(10.4.10). *Applications I: Proof of* (6.15.9). Let k be a separably closed field, X a k-prescheme locally of finite type over k and *unibranch*. We know that the integral closure of a k-algebra of finite type A in a finite extension of its field of fractions is a finite A-algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, nº 2, th. 2), hence every k-algebra of finite type is a universally Japanese ring; we conclude that the set of points $x \in X$ where X is geometrically unibranch is locally constructible (9.7.10). But the hypothesis and the lemma (6.15.8) imply that this set contains all the *closed* points of X. The conclusion follows therefore from (10.4.6), (10.3.1), and the bijectivity of the map $\mathfrak{Lc}(X) \to \mathfrak{Lc}(X_0)$ (where X_0 is the set of closed points of X) (10.2.7).

(10.4.11). *Applications II:*

Proposition (10.4.11). — *Let S be a prescheme, X an S-prescheme of finite type. Every S-endomorphism of X which is radicial is surjective (hence bijective).*

PROOF. Let $f : X \to S$ be the structural morphism, $g : X \to X$ the S-endomorphism considered, **IV-3** | 104
and, for every $s \in S$, let g_s be the morphism deduced from g by the base change $\operatorname{Spec}(k(s)) \to S$, which is a $k(s)$-endomorphism of the fibre $f^{-1}(s) = X \times_S \operatorname{Spec}(k(s))$.

To prove that g is surjective, it suffices to prove that g_s is surjective for every $s \in S$, hence one can (by virtue of (**I**, 3.5.7)) restrict to the case of a field k, in which case f is a morphism of finite presentation, since S is Noetherian. Applying (8.9.1) and (8.10.5, vii)), we reduce to the case where $S = \operatorname{Spec}(A)$, where A is a sub-$\mathbf{Z}$-algebra of finite type of k. But X is then a Jacobson prescheme (10.4.7), and $g(X)$ is constructible in X (1.8.5), hence, to prove that $g(X) = X$, it suffices to show that $g(X)$ contains all the closed points of X (10.3.1). □

Lemma (10.4.11.1). — *Let Y be a $\mathbf{Z}$-prescheme of finite type.*

label=(i) For a point $y \in Y$ to be closed in Y, it is necessary and sufficient that $k(y)$ be a finite field.

lbbel=(ii) For every prime number p and every integer $d \geqslant 1$, the set of points $y \in Y$ such that $k(y)$ is an extension of $\mathbf{F}_p$ whose degree divides d is finite.

PROOF. Assertion (i) follows from the fact that the image of a closed point $y \in Y$ in $\operatorname{Spec}(\mathbf{Z})$ is a closed point (10.4.7), hence a prime number, and from (**I**, 6.4.2). On the other hand, as Y is a finite union of affine open sets of finite type over $\mathbf{Z}$, we can restrict to proving (ii) in the case where $Y = \operatorname{Spec}(C)$, where C is a $\mathbf{Z}$-algebra of finite type. Now, the points $y \in Y$ such that the degree of $k(y)$ over $\mathbf{F}_p$ divides d correspond bijectively to the homomorphisms $C \to \mathbf{F}_{p^d}$; but if (t_i) is a system of generators of the $\mathbf{Z}$-algebra C, every homomorphism of C is determined by its values on the elements t_i, and hence there are at most a finite number of homomorphisms of C into a finite field. □

Since X is a $\mathbf{Z}$-prescheme of finite type, let $T_{p,d}$ be the set of closed points $z \in X$ such that $k(z)$ is an extension of $\mathbf{F}_p$ of degree dividing d; it follows from (10.4.11.1) that the set $T_{p,d}$ is finite and that the set of closed points of X is the union of the $T_{p,d}$. Moreover, if $z \in T_{p,d}$ and if h is any endomorphism of X, $h(z) \in T_{p,d}$, that is to say $T_{p,d}$ is *stable* under every endomorphism of X. As g is by hypothesis injective, its restriction to $T_{p,d}$ is *bijective* since $T_{p,d}$ is finite, which completes the proof of the proposition.

We will see later (17.9.7) that when one furthermore supposes that X is an S-prescheme of finite presentation, and moreover that g is a monomorphism, then one can affirm that g is an *automorphism* of X.

10.5. Noetherian Jacobson preschemes

Proposition (10.5.1). — *Let B be an integral Noetherian ring. The conditions equivalent* a) *to* d) *of* (10.4.4) *are then also equivalent to the following:* **IV-3** | 105

lebel=e) Spec(B) *is finite.*
lfbel=f) *B is a semi-local ring of dimension $\leqslant 1$.*

Proof. It follows from the Artin–Tate theorem $(\mathbf{0}, 16.3.3)$ that conditions *a)* and *f)* are equivalent. Condition *f)* implies that Spec(B) is the union of the finite set of its closed points and its generic point, hence *f)* implies *e)* without supposing A Noetherian. Finally *e)* implies *d)* without supposing A Noetherian, for the generic point x of X is the complement of the union of the closures $\overline{\{y\}}$, where y runs through the set of points $y \neq x$, and as these points are finite in number, $\{x\}$ is the complement of a closed set in X. □

Corollary (10.5.2). — *Let A be a Noetherian ring. For A to be a Jacobson ring, it is necessary and sufficient that there exist no prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is a semi-local ring of dimension* 1.

Proof. This follows immediately from (10.5.1) and condition *b)* of (10.4.5), the prime ideals $\mathfrak{p}$ such that $A/\mathfrak{p}$ is semi-local of dimension 0 being the maximal ideals of A. □

Corollary (10.5.3). — *Let X be a Noetherian irreducible prescheme. The following conditions are equivalent:*

label=a) *The generic point of X is isolated.*
lbbel=b) *X is finite.*
lcbel=c) *X is of dimension $\leqslant 1$ and the set of its closed points is finite.*

Proof. There exists a finite number of irreducible affine open sets U_i $(1 \leqslant i \leqslant n)$ covering X, each of which contains the generic point of X; it suffices to prove the equivalence of *a)*, *b)*, and *c)* for each of the $(U_i)_{\mathrm{red}}$ (taking into account $(\mathbf{0}, 14.1.7)$). But this equivalence follows from (10.5.1). □

Remark (10.5.4). — A Noetherian prescheme X satisfying the equivalent conditions of (10.5.3) is not necessarily an affine scheme; in fact it can even be *non-separated*. One obtains an example by replacing, in the example (**I**, 5.5.11) of the "affine line with doubled point", X_1 and X_2 by the spectra of the valuation ring of a discrete valuation, U_{12} and U_{21} by the open sets reduced to the generic point in X_1 and X_2 respectively; the non-separated prescheme X thus obtained has exactly 3 points.

Proposition (10.5.5). — *Let X be a Noetherian prescheme.*

label=(i) The set X_0 of points $x \in X$ such that $\overline{\{x\}}$ is finite is very dense in X.
lbbel=(ii) For X to be a Jacobson prescheme, it is necessary and sufficient that there exist no sub-prescheme of X isomorphic to the spectrum of an integral semi-local ring of dimension 1.

Proof. The condition that $\overline{\{x\}}$ is finite is equivalent to the fact that x is isolated in $\overline{\{x\}}$ (10.5.3), hence $\overline{\{x\}}$, being the underlying space of a sub-prescheme (Noetherian) of X, and assertion (i) follows from (5.1.10.1); taking into account (10.4.5.1), for a non-closed point x of X to belong to X_0, it is necessary and sufficient that the sub-prescheme closed and integral Y of X having $\overline{\{x\}}$ as underlying space be of dimension 1 and finite, hence a finite union of sub-preschemes that are spectra of integral semi-local rings of dimension 1. Conversely, if there exists a sub-prescheme Z of X which is the spectrum of an integral semi-local ring of dimension 1, Z is not a Jacobson prescheme (10.5.2), hence neither is X (10.3.2). □

Remark (10.5.6). — Assertion (ii) of (10.5.5) is still valid when X is *locally Noetherian*: indeed, if (V_α) is a covering of X by affine Noetherian open sets, every sub-prescheme of one of the V_α is a sub-prescheme of X; conversely, if a sub-prescheme Z of X is isomorphic to the spectrum of an integral semi-local ring of dimension 1, there is an α such that $V_\alpha \cap Z$ contains an affine open set U of Z not reduced to the generic point of Z, which is therefore also the spectrum of an integral semi-local ring of dimension 1. We conclude with the help of (10.3.3). **IV-3 | 106**

Proposition (10.5.7). — *Let X be a locally Noetherian prescheme, Y a closed part of X such that every non-empty closed part of X meets Y. Then the prescheme induced on the open set $X - Y$ is a Jacobson prescheme.*

PROOF. Let us apply the criterion (10.5.5, ii)), and suppose that there is a sub-prescheme Z of $X - Y$ which is the spectrum of an integral semi-local ring of dimension 1, with generic z of Z being therefore isolated in Z (or in the closure $\overline{Z}$ of Z in X), and Z being distinct from $\{z\}$. Let $y \neq z$ be a point of Z; as it does not belong to Y, it is not a closed point of X, and its closure $\overline{\{y\}}$ in X meets Y at a point $x \neq y$, which is therefore not in Z. The existence of the chain $\{x\} \subset \overline{\{y\}} \subset \overline{\{z\}}$ shows that the dimension of $\overline{\{z\}}$ would be $\geqslant 2$, and it would be the same for the dimension of $U \cap \overline{\{z\}}$, where U is an affine open neighbourhood (hence Noetherian) of x in X. But this contradicts the fact that the generic point z of $U \cap \overline{\{z\}}$ is isolated in $U \cap \overline{\{z\}}$ (10.5.3). □

Corollary (10.5.8). — *Let A be a Noetherian ring; for every element f of the radical $\mathfrak{R}$ of A, the ring A_f is a Jacobson ring, and the open set* $\operatorname{Spec}(A) - \operatorname{V}(\mathfrak{R})$ *is a Jacobson scheme.*

PROOF. Setting $X = \operatorname{Spec}(A)$, $Y = \operatorname{V}(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of A, to say that every non-empty closed part of X meets Y is equivalent ($\mathbf{0_I}$, 2.1.3) to saying that Y contains all the *closed* points of X, or equivalently that $\mathfrak{J}$ is *contained in the radical* $\mathfrak{R}$ of A. If $f \in \mathfrak{R}$, the open set $\operatorname{D}(f)$ does not meet $\operatorname{V}(\mathfrak{R})$, and is a Jacobson space by virtue of (10.5.7). □

Corollary (10.5.9). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, $X = \operatorname{Spec}(A)$, $Y = X - \{\mathfrak{m}\}$; then Y is a Jacobson scheme, whose closed points are the prime ideals $\mathfrak{p} \in \operatorname{Spec}(A)$ such that $\dim(A/\mathfrak{p}) = 1$.*

PROOF. The first assertion is a particular case of (10.5.8); on the other hand, the closed points of Y are the prime ideals $\mathfrak{p}$ of A which are maximal elements of the set of prime ideals $\neq \mathfrak{m}$, which, by definition of the dimension, means that $\dim(A/\mathfrak{p}) = 1$. □

Proposition (10.5.10). — *Let A be a local Noetherian ring, reduced and complete, and which is not a field. Then the finite intersections of the kernels of the local homomorphisms of A into discrete valuation rings V, which make V a finite A-algebra, form a filter base tending towards 0 for the adic topology of A.*

PROOF. It suffices (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8) to prove that the intersection of the kernels in question is reduced to 0. Suppose first that A is integral and of dimension 1; by virtue of the theorem of Nagata ($\mathbf{0}$, 23.1.5 and 23.1.6), the integral closure A' of A is a complete local integral ring and a finite A-algebra; it is therefore a discrete valuation ring, and the proposition follows immediately in this case. **IV-3 | 107**

Let us pass to the general case; let x be the closed point of $X = \operatorname{Spec}(A)$, and set $Y = X - \{x\}$; we know (10.5.9) that Y is a Jacobson prescheme. Let us show that the intersection of the prime ideals $\mathfrak{p}$ of A such that $\dim(A/\mathfrak{p}) = 1$ is reduced to 0; the proposition will follow since, for each such prime $\mathfrak{p}$, the intersection of the kernels of the local homomorphisms of $A/\mathfrak{p}$ into discrete valuation rings V, making V an $(A/\mathfrak{p})$-algebra of finite type, is reduced to 0. But to say that the intersection of these prime ideals is reduced to 0 means that the set of these ideals is dense in X, or equivalently that Y is dense in X, and this also follows from (10.5.9). □

10.6. Dimension in Jacobson preschemes The results of this subsection refine in certain cases and generalize results of § 5.

Proposition (10.6.1). — *Let S be a locally Noetherian prescheme, satisfying moreover the following conditions:* 1^o *S is a Jacobson prescheme;* 2^o *for every $s \in S$, $\mathscr{O}_{S,s}$ is universally catenary* (5.6.2); 3^o *every irreducible component S' of S is equicodimensional (equivalently, for every closed point s of S' and every sub-prescheme of S having S' as underlying space, we have $\dim(S') = \dim(\mathscr{O}_{S',s})$). We then have the following properties:*

label=(i) For every morphism locally of finite type $g : X \to S$, X satisfies the conditions 1^o, 2^o, *and* 3^o *above. In particular, if X is equidimensional (for example if X is irreducible), X is catenary and for every closed point x of X, we have $\dim(\mathscr{O}_{X,x}) = \dim(X)$ (**0**, 14.3.3).*

lbbel=(ii) Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism; suppose X irreducible and f dominant. If ξ (resp. η) is the generic point of X (resp. Y) and $e = \dim(f^{-1}(\eta)) = \deg.\mathrm{tr}_{k(\eta)} k(\xi)$, we have

$$\dim(X) = \dim(Y) + e. \tag{10.6.1.1}$$

lcbel=(iii) Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism, n an integer $\geqslant 0$ such that $\dim(f^{-1}(y)) \geqslant n$ (resp. $\dim(f^{-1}(y)) \leqslant n$) for every $y \in Y$. Then we have

$$\dim(X) \geqslant \dim(Y) + n \tag{10.6.1.2}$$

(resp.

$$\dim(X) \leqslant \dim(Y) + n). \tag{10.6.1.3}$$

Proof. (i) Property 1^o for X follows from (10.4.7). For every $x \in X$, $\mathscr{O}_{X,x}$ is a local ring of an $\mathscr{O}_{S,g(x)}$-algebra of finite type and a prime ideal, and the homomorphism $\mathscr{O}_{S,g(x)} \to \mathscr{O}_{X,x}$ is local; hence (5.6.3, iv)) $\mathscr{O}_{X,x}$ is universally catenary. To show that X satisfies condition 3^o, we consider several cases: **IV-3 | 108**

a) X is a closed irreducible sub-prescheme of S; let S' be an irreducible component of S containing X, ξ the generic point of X, x a closed point of X; for every $s \in S'$, $\mathscr{O}_{S',s}$, quotient of $\mathscr{O}_{S,s}$, is catenary (5.6.1), hence the conditions 2^o and 3^o imply that S' is biequidimensional (**0**, 14.3.3). By virtue of (5.1.2) and (**0**, 14.3.3.2), we have

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S',x}) - \dim(\mathscr{O}_{S',\xi}) = \dim(S') - \dim(\mathscr{O}_{S',\xi})$$

which proves that $\dim(\mathscr{O}_{X,x})$ does not depend on the closed point x considered, whence the assertion in this case (5.1.4).

b) X is irreducible and g dominant. Then, for every closed point x of X, $g(x)$ is a closed point of S by (10.4.7); as $\mathscr{O}_{S,g(x)}$ is universally catenary, it follows from (5.6.5.3) that

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S,g(x)}) + e$$

where $e = \dim(g^{-1}(\zeta))$, ζ being the generic point of S. As $\dim(S) = \dim(\mathscr{O}_{S,g(x)})$ by virtue of condition 3^o for S, we have $\dim(\mathscr{O}_{X,x}) = \dim(S) + e$ for every closed point $x \in X$; this proves condition 3^o for X (5.1.4), and at the same time the formula

$$\dim(X) = \dim(S) + e. \tag{10.6.1.4}$$

c) General case. Considering a reduced sub-prescheme X' of X having as underlying space an irreducible component of X, we reduce to the case where X is integral; using (**I**, 5.2.2) and the case *a)* proved above, one can then replace S by the sub-prescheme reduced having $g(X)$ as underlying space; we are then reduced to case *b)*, and this completes the proof of (i).

(ii) The morphism f being locally of finite type (**I**, 1.3.4), we can apply the results of (i) by replacing S by Y; moreover, as X is irreducible and f dominant, (10.6.1.4) gives (10.6.1.1).

(iii) The assertion relative to the case $\dim(f^{-1}(y)) \leqslant n$ for every $y \in Y$ has already been proved under more general hypotheses in (5.6.7). Suppose that $\dim(f^{-1}(y)) \geqslant n$ for every $y \in Y$, and consider a generic point η of an irreducible component Y' of Y; there exists at least one irreducible component Z of $f^{-1}(\eta)$ of dimension $\geqslant n$; if ξ is the generic point of Z, ξ is also the generic point of an irreducible component X' of X such that $f(X')$ is dense in Y' ($\mathbf{0_I}$, 2.1.8). Considering the reduced sub-preschemes of X, Y having as underlying spaces X', Y' respectively, and the restriction $X' \to Y'$

of f (**I**, 5.2.2); it follows from (ii) that $\dim(X') \geqslant \dim(Y') + n$; this being true for every irreducible component Y' of Y, we conclude the inequality (10.6.1.2). □

Corollary (10.6.2). — *Suppose that X satisfies conditions 1^o, 2^o, and 3^o of (10.6.1). Then, for every dense open set U in X, we have $\dim(U) = \dim(X)$.* **IV-3** | 109

PROOF. One knows that $\dim(U) \leqslant \dim(X)$ (**0**, 14.1.4); moreover, as X is a Jacobson prescheme, U contains a closed point of every irreducible component of X, hence $\dim(U) = \dim(X)$ by virtue of (10.6.1) and (**0**, 14.1.2.1). □

Proposition (10.6.3). — *Suppose that X satisfies conditions 1^o, 2^o, and 3^o of (10.6.1), and let Y be a closed part of X. Then, for every $x \in Y$, and every open neighbourhood U of x in X not meeting the irreducible components of Y that do not contain x, we have*

(10.6.3.1) $$\dim_x(Y) = \dim(U \cap Y) = \dim \overline{\{x\}} + \operatorname{codim}(\overline{\{x\}}, Y) = \sup_i \dim(Y_i)$$

where Y_i ($1 \leqslant i \leqslant m$) are the irreducible components of Y containing x.

PROOF. Considering a closed sub-prescheme of X having Y as underlying space, one can restrict, by virtue of (10.6.1, i)), to the case where $Y = X$. By virtue of the choice of U, U is the union of the $U \cap X_i$, hence $\dim(U) = \sup \dim(U \cap X_i)$; but by (10.6.2) applied to a closed sub-prescheme of X having X_i as underlying space, one has (taking into account (10.6.1, i))), $\dim(U \cap X_i) = \dim(X_i)$; this proves that the second and fourth terms of (10.6.3.1) are equal. As $U \cap X_i$ is biequidimensional (10.6.1, i)), we have

(10.6.3.2) $$\dim(X_i) = \dim(U \cap X_i) = \dim(U \cap \overline{\{x\}}) + \operatorname{codim}(U \cap \overline{\{x\}}, U \cap X_i)$$

applied to a closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, one has $\dim(U \cap \overline{\{x\}}) = \dim(\overline{\{x\}})$; as one also has, for the same reasons,

(10.6.3.3) $$\dim(X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X_i)$$

we obtain

$$\dim(U) = \dim(\overline{\{x\}}) + \sup_i \operatorname{codim}(\overline{\{x\}}, X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by definition of the codimension (**0**, 14.2.1). This shows that $\dim(U)$ of the open neighbourhood U of x satisfying the conditions of the statement is *independent* of the choice of U, hence is equal to $\dim_x(X)$, by (**0**, 14.1.4.1). □

Corollary (10.6.4). — *Under the hypotheses of (10.6.3), let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, Y its support. For every $x \in Y$, we have*

(10.6.4.1) $$\dim(\overline{\{x\}}) + \dim(\mathscr{F}_x) = \dim_x Y.$$

PROOF. This follows from (10.6.3.1) and the formula $\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, Y)$ (5.1.12.2). □

10.7. Examples and counter-examples

(10.7.1). Let S be a locally Noetherian prescheme of *dimension* $\leqslant 1$ and suppose that S is a Jacobson prescheme; when S is Noetherian, this amounts to saying that the irreducible components of S of dimension 1 are infinite, since every $x \in S$ which is not a closed point is the generic point of such a component (10.4.5) and (10.5.4). Then S also satisfies conditions 2° and 3° of (10.6.1): indeed, every local ring $\mathscr{O}_{S,s}$ is of dimension 0 or 1, and is hence necessarily catenary, and by (5.6.4), universally catenary (7.2.9); on the other hand, an irreducible component S' of S, either is reduced to a point, or is of dimension 1, and for every closed point $s \in S'$, $\mathscr{O}_{S',s}$ is necessarily of dimension 1. We deduce from these remarks and from (10.6.1) that every prescheme locally of finite type over S satisfies also the properties 1°, 2°, and 3° of (10.6.1): this includes in particular preschemes *locally of finite type over a field or over* $\mathbf{Z}$.

(10.7.2). Let A be a local Noetherian ring *universally catenary* and let S be the complement of the closed point in $X = \operatorname{Spec}(A)$. Then S satisfies conditions 1°, 2°, and 3° of (10.6.1): indeed, we have already seen that S is a Jacobson prescheme (10.5.9); similarly, for the same reason as for the local rings $A_{\mathfrak{p}}$, it is a Jacobson prescheme, and moreover the local rings $\mathscr{O}_{S,s} = \mathscr{O}_{X,s}$ are universally catenary. On the other hand, an irreducible component X' of X is the complement of $\mathfrak{a}$ in X; for every closed point x of S, the adherence of x is $\{x, \mathfrak{a}\}$, and X' is by hypothesis biequidimensional ($\mathbf{0}$, 14.3.3), hence $\dim(\mathscr{O}_{X',x}) = \dim(X') - 1$ ($\mathbf{0}$, 14.3.2), which proves that S' is equicodimensional (5.1.4). **IV-3** | 110

(10.7.3). Let A be a discrete valuation ring; let us show that in the ring $B = A[T_1, \ldots, T_n]$ of polynomials in n indeterminates, there exist two maximal ideals $\mathfrak{m}$, $\mathfrak{n}$ of heights n and $n+1$ respectively. We have already seen in (5.2.5, i)) (for $n = 1$; the proof is by induction on n). Let $A[T_1, \ldots, T_{n+1}]$ be a faithfully flat free $A[T_1, \ldots, T_n]$-module; hence there are in $A[T_1, \ldots, T_{n+1}]$ two maximal ideals $\mathfrak{m}'$, $\mathfrak{n}'$ of heights $n+1$ and $n+2$ respectively (5.5.1); moreover, by (5.5.3), these ideals are necessarily of heights $n+1$ and $n+2$. Suppose $n \geqslant 2$; let $R = 1 + 3$ which is a multiplicative part of B; setting $B' = R^{-1}B$, the ideal $\mathfrak{J}' = R^{-1}\mathfrak{J}$ is contained in the *radical* of B' (Bourbaki, *Alg. comm.*, chap. III, §3, n° 5, prop. 12); so the topological subspace $X' = \operatorname{Spec}(B')$ is identified with a subspace of X, and the local rings $\mathscr{O}_{S,x}$ and $\mathscr{O}_{X',x}$ are the same by (**I**, 1.6.2). Considering then in X' the closed set $Y' = \mathrm{V}(\mathfrak{J}')$, and setting $S = X' - Y'$; we know (10.5.7) that S is a Jacobson prescheme, evidently irreducible and Noetherian; moreover the local rings $\mathscr{O}_{S,s} = \mathscr{O}_{X',s}$ are universally catenary, being the same as the local rings of A.

However, there exist two closed points a, b of S such that $\mathscr{O}_{S,a}$ and $\mathscr{O}_{S,b}$ *do not have the same dimension*, that is to say S does not satisfy condition 3° of (10.6.1). To see this, consider the two maximal ideals $\mathfrak{m}' = R^{-1}\mathfrak{m}$ and $\mathfrak{n}' = R^{-1}\mathfrak{n}$, of heights n and $n+1$ respectively; the corresponding points a', b' are the only closed points of X', corresponding to $\mathfrak{a}$ and $\mathfrak{b}$. Since $B_{\mathfrak{m}'}$ is biequidimensional, $\mathfrak{p}'$ is then a prime ideal of height $n-1$. Similarly $\mathfrak{n}'$ corresponds to a point b of X' such that the adherence of b in X' is $\{b, b'\}$. Since a and b are closed points of S, and are in S, this proves our claim.

10.8. Rectified depth

Definition (10.8.1). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $x \in X$, we call the *rectified depth* of $\mathscr{F}$ at the point x and denote by $\operatorname{prof}^*_x(\mathscr{F})$ the number (an integer $\geqslant 0$ or $+\infty$) equal to

$$\operatorname{prof}^*_x(\mathscr{F}) = \operatorname{prof}(\mathscr{F}_x) + \dim(\overline{\{x\}}) \tag{10.8.1.1}$$

where $\overline{\{x\}}$ is the closure of the point x in X. For any part Z of X, we call the *rectified depth* of $\mathscr{F}$ along Z and denote by $\operatorname{prof}^*_Z(\mathscr{F})$ the number

$$\operatorname{prof}^*_Z(\mathscr{F}) = \inf_{x \in Z} \operatorname{prof}^*_x(\mathscr{F}). \tag{10.8.1.2}$$

In other terms, for every integer n, the relation $\operatorname{prof}^*_Z(\mathscr{F}) \geqslant n$ is equivalent to $\operatorname{prof}^*_x(\mathscr{F}) \geqslant n$ for every $x \in Z$. If $Z = X$, one writes $\operatorname{prof}^*(\mathscr{F})$ instead of $\operatorname{prof}^*_X(\mathscr{F})$.

Remarks (10.8.2). — (i) At every closed point $x \in X$, the rectified depth is equal to the depth. **IV-3** | 111

(ii) Let Y be a closed sub-prescheme of X, $j : Y \to X$ the canonical injection, $\mathscr{G}$ a coherent $\mathscr{O}_Y$-Module. We know (5.7.3, vi)) that $\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(j_*(\mathscr{G})_x)$ for every $x \in Y$; we deduce that $\mathrm{prof}^*_x(\mathscr{G}) = \mathrm{prof}^*_{j_*(x)}(j_*(\mathscr{G}))$ for every $x \in Y$.

(iii) The notion of rectified depth is only of interest when it is of *local* character, that is to say when it does not change upon replacing X by an arbitrary open neighbourhood U of x. This requires evidently that x not be isolated in $\overline{\{x\}}$, hence x is not a closed point and hence X must be a *Jacobson prescheme* (10.4.5.1); most often, it will also be necessary to know that $\dim(U) = \dim(X)$ for every dense open U in X, and one will therefore need to suppose that X also satisfies conditions 2° and 3° of (10.6.1).

Lemma (10.8.3). — *Let X be a regular and biequidimensional prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then, for every $x \in X$,*

(10.8.3.1) $$\mathrm{prof}^*_x(\mathscr{F}) = \dim(X) - \dim.\mathrm{proj}(\mathscr{F}_x).$$

PROOF. Indeed, as X is biequidimensional, we have (**0**, 14.3.5.1)

$$\dim(\overline{\{x\}}) = \dim(X) - \mathrm{codim}(\overline{\{x\}}, X) = \dim(X) - \dim(\mathscr{O}_{X,x})$$

by virtue of (5.1.2). On the other hand, as X is regular, we have by (**0**, 17.3.4)

$$\mathrm{prof}(\mathscr{F}_x) = \dim(\mathscr{O}_{X,x}) - \dim.\mathrm{proj}(\mathscr{F}_x)$$

whence the lemma. □

Corollary (10.8.4). — *Under the hypotheses of (10.8.3), the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is lower semi-continuous.*

PROOF. This follows from (10.8.3.1), since $x \mapsto \dim.\mathrm{proj}(\mathscr{F}_x)$ is upper semi-continuous (6.11.1). □

Proposition (10.8.5). — *Let S be a locally Noetherian prescheme, X a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that S satisfies the following conditions: 1° S is a Jacobson prescheme; 2° S is regular; 3° the irreducible components of S are equicodimensional. Then the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is lower semi-continuous in X; in other words, for every integer n, the set U_n of $x \in X$ such that $\mathrm{prof}^*_x(\mathscr{F}) \geqslant n$ is open.*

PROOF. As the local rings $\mathscr{O}_{S,s}$ of S are regular, they are universally catenary (**0**, 16.5.12), hence S satisfies conditions 1°, 2°, and 3° of (10.6.1), and X does too (10.6.1, i)). The notion of rectified depth being of local character (10.8.2, iii)), we can restrict to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, A being a regular ring and B an A-algebra of finite type, hence a quotient of a polynomial ring $C = A[T_1, \ldots, T_n]$, and the latter is regular (**0**, 17.3.7). One can therefore suppose that X is a closed sub-prescheme of a regular prescheme Y also satisfying conditions 1°, 2°, and 3° of (10.6.1); taking into account the remark (10.8.2, ii)), we are thus reduced to the case where X is moreover regular and Noetherian. But then, as the local rings of X are integral, the irreducible components of X are also irreducible. Then, as the local rings of X are catenary (**0**, 16.5.12), l'hypothesis 3° of (10.6.1) implies that X is biequidimensional ((5.1.5) and (**0**, 14.3.3)); it therefore suffices to apply (10.8.4). □ **IV-3 | 112**

Corollary (10.8.6). — *Under the hypotheses of (10.8.5), for every $x \in X$, the number $\mathrm{prof}^*_x(\mathscr{F})$ is the unique integer n having the following property: there exists an open neighbourhood U of x in X such that for every point $x' \in U \cap \overline{\{x\}}$, closed in U, we have $\mathrm{prof}(\mathscr{F}_{x'}) = n$. In particular, for $\mathrm{prof}^*_x(\mathscr{F}) \geqslant m$ (resp. $\mathrm{prof}^*_x(\mathscr{F}) \leqslant m$), it is necessary and sufficient that there exist an open neighbourhood V of x such that, for every point $x' \in V \cap \overline{\{x\}}$, closed in V, we have $\mathrm{prof}(\mathscr{F}_{x'}) \geqslant m$ (resp. $\mathrm{prof}(\mathscr{F}_{x'}) \leqslant m$).*

PROOF. Indeed, one can restrict to the case where x is not a closed point; if $\mathrm{prof}^*_y(\mathscr{F}) \leqslant n$, the set of $y \in X$ such that $\mathrm{prof}^*_y(\mathscr{F}) \leqslant n$ is closed by virtue of (10.8.5), hence contains $\overline{\{x\}}$, and by virtue of the lower semi-continuity of $y \mapsto \mathrm{prof}^*_y(\mathscr{F})$, there exists an open neighbourhood U of x such that $\mathrm{prof}(\mathscr{F}_{x'}) = n$ for every $y \in U \cap \overline{\{x\}}$, hence $\mathrm{prof}(\mathscr{F}_{x'}) = n$ if $x' \in U \cap \overline{\{x\}}$ is *closed* in U (since the rectified depth is local).

For preschemes satisfying the hypotheses of (10.8.5), the notion of rectified depth can therefore be defined with the aid of the depth at the *closed points* only (these forming a set that is very dense in every closed part of X). □

Proposition (10.8.7). — *Let S be a locally Noetherian prescheme satisfying conditions 1^o, 2^o, and 3^o of (10.6.1). Let X be a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, $Y = \operatorname{Supp}(\mathscr{F})$. Then, for every $x \in Y$, we have*

$$\operatorname{prof}^*_x(\mathscr{F}) = \dim_x(Y) - \operatorname{coprof}(\mathscr{F}_x). \tag{10.8.7.1}$$

Proof. Indeed, by definition, $\operatorname{coprof}(\mathscr{F}_x) = \dim(\mathscr{F}_x) - \operatorname{prof}(\mathscr{F}_x)$, and it follows from (10.6.4) that $\dim_x(Y) = \dim(\overline{\{x\}}) + \dim(\mathscr{F}_x)$; whence (10.8.7.1) by the definition of $\operatorname{prof}^*_x(\mathscr{F})$. □

Corollary (10.8.8). — *Under the hypotheses on f and $\mathscr{F}$ of (9.9.1), the function $x \mapsto \operatorname{prof}^*_x(\mathscr{F}_{f(x)})$ is constructible.*

Proof. Note that for every $s \in S$, the fibre X_s, being a prescheme locally of finite type over $k(s)$, is a Jacobson prescheme (10.4.7); as moreover $\operatorname{Spec}(k(s))$ satisfies the conditions of (10.6.1), we have $\operatorname{prof}^*_x(\mathscr{F}_{f(x)}) = \dim_x(\operatorname{Supp}(\mathscr{F}_{f(x)})) - \operatorname{coprof}((\mathscr{F}_{f(x)})_x)$ (10.8.7). Now, if $Z = \operatorname{Supp}(\mathscr{F})$, we have $Z_{f(x)} = \operatorname{Supp}(\mathscr{F}_{f(x)})$ (**I**, 9.1.13) and Z is locally constructible (8.9.1); hence the functions $x \mapsto \dim_x(\operatorname{Supp}(\mathscr{F}_{f(x)}))$ and $x \mapsto \operatorname{coprof}((\mathscr{F}_{f(x)})_x)$ are locally constructible ((9.9.1) and (9.9.3)), which proves the proposition. □

10.9. Maximal spectra and ultrapreschemes

(10.9.1). Let X be a Jacobson prescheme, and let $S(X)$ be the *ringed space* whose underlying space is the subspace of closed points of X, and whose sheaf of rings is the sheaf induced on this subspace by $\mathscr{O}_X$, in other words the sheaf $\psi^*(\mathscr{O}_X)$, with $\psi : S(X) \to X$ denoting the canonical injection. As ψ is a quasi-homeomorphism, one has seen in (10.2.8, ii)) that if $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_{S(X)})$ is the homomorphism of sheaves of rings such that $\theta^\#$ is the identity, then $j_X = (\psi, \theta)$ is a *quasi-isomorphism* of ringed spaces, and $\mathscr{F} \mapsto j^*_X(\mathscr{F})$ is an equivalence of the category of $\mathscr{O}_X$-Modules locally free (resp. coherent) and that of $\mathscr{O}_{S(X)}$-Modules locally free (resp. coherent); moreover, if $\mathscr{L}$ is an $\mathscr{O}_X$-Module locally free and if U is an open set of X, such that $\mathscr{L}|U$ is isomorphic to $(\mathscr{O}_X|U)^n$, $j^*_X(\mathscr{L})|(U \cap S(X))$ is isomorphic to $(\mathscr{O}_{S(X)}|(U \cap S(X)))^n$.

(10.9.2). Let X, Y be two Jacobson preschemes and $f = (\rho, \lambda) : X \to Y$ a morphism *locally of finite type*. We have seen in (10.4.7) that $\rho(S(X)) \subset S(Y)$ and that ρ restricted to $S(X)$ gives a continuous map $S(X) \to S(Y)$. On the other hand, for every open set V of Y, one defines by composition a homomorphism of rings

$$\Gamma(V \cap S(Y), \mathscr{O}_{S(Y)}) \longrightarrow \Gamma(f^{-1}(V), \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(V) \cap S(X), \mathscr{O}_{S(X)})$$

where the two extreme isomorphisms have been defined in (10.9.1); it is clear that this defines a homomorphism of sheaves of rings $S(\lambda) : \mathscr{O}_{S(Y)} \to \rho_*(\mathscr{O}_{S(X)})$ (recalling that the open sets of $S(Y)$ correspond bijectively to those of Y (resp. $S(X)$) (10.2.1)); we thus obtain a morphism of ringed spaces $S(f) = (S(\rho), S(\lambda)) : (S(X), \mathscr{O}_{S(X)}) \to (S(Y), \mathscr{O}_{S(Y)})$ such that the diagram

$$\begin{tikzcd} S(X) \arrow[r, "S(f)"] \arrow[d, "j_X"'] & S(Y) \arrow[d, "j_Y"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

is commutative; moreover, if Z is a third Jacobson prescheme and $g : Y \to Z$ a morphism locally of finite type, it is clear that $S(g \circ f) = S(g) \circ S(f)$. We have thus defined a *covariant functor* $S : C \to C'$, where C' is the category of *ringed spaces with local rings* and C the category whose objects are the *Jacobson preschemes* and whose morphisms are the morphisms *locally of finite type* between Jacobson preschemes.

(10.9.3). Let us determine the sub-category C'' of C' consisting of the ringed spaces isomorphic to $S(X)$ and whose morphisms come from $S(f)$. Suppose first that $X = \operatorname{Spec}(A)$, where A is a **IV-3 | 113**

Jacobson ring; then $S(X)$ is the set of *maximal ideals* of A, equipped with: 1° the topology induced by that of X, so that a base of this topology is formed by the $\mathrm{D}^m(h) = \mathrm{D}(h) \cap S(X)$, the set of maximal ideals $\mathfrak{m}$ such that $h \notin \mathfrak{m}$, where h runs through A; 2° the sheaf of rings $\mathscr{O}_{S(X)}$ such that $\Gamma(\mathrm{D}^m(h), \mathscr{O}_{S(X)}) = A_f$. We call this ringed space the *maximal spectrum* of A and denote it $\mathrm{Spm}(A)$.

One notes that if $j : \mathrm{D}(h) \to X$ is the canonical injection, the ringed space induced by $S(X)$ on $S(\mathrm{D}(h))$ is $S(\mathrm{D}(h))$.

Let B be a second Jacobson ring, $Y = \mathrm{Spec}(B)$, $\varphi : B \to A$ a homomorphism of rings making A a B-algebra of finite type, $f = ({}^a\varphi, \widetilde{\varphi}) : X \to Y$ the corresponding morphism of preschemes, and $S(f) : \mathrm{Spm}(A) \to \mathrm{Spm}(B)$ the morphism of ringed spaces corresponding to f. It is clear that $S(f) = (\psi, \theta)$ is a morphism of *ringed spaces with local rings*, that is ($\mathrm{Err}_{\mathrm{II}}$, 1.8.2) for every $x \in \mathrm{Spm}(A)$, $\theta_x^\#$ is a local homomorphism. Reciprocally:

Proposition (10.9.4). — *Let A, B be two Jacobson rings. If $u = (\psi, \theta) : \mathrm{Spm}(A) \to \mathrm{Spm}(B)$ is a morphism of ringed spaces with local rings such that $\Gamma(\theta) : B \to A$ makes A a B-algebra of finite type, there exists one and only one morphism of preschemes $f : \mathrm{Spec}(A) \to \mathrm{Spec}(B)$ such that $u = S(f)$.*

PROOF. The uniqueness of f is evident, since if $f = ({}^a\varphi, \widetilde{\varphi})$, one must have $\varphi = \Gamma(\theta)$; it remains to see that $S(f)$ is defined and equal to $u = S(f)$. Now, the first assertion follows from the fact that φ is supposed to make A a B-algebra of finite type, and the fact that $\theta_x^\#$ is a local homomorphism for every $x \in \mathrm{Spm}(A)$ allows us to repeat the reasoning of (I, 1.7.3) restricting to the maximal points of $\mathrm{Spm}(A)$: one shows thus successively that $\psi(x) = {}^a\varphi(x)$ for every $x \in \mathrm{Spm}(A)$, then that $\theta_x^\# = \varphi_x : B_{\psi(x)} \to A_x$, which completes the proof that $u = S(f)$. □

(10.9.5). Let us now consider a ringed space $(X, \mathscr{O}_X)$; we will say that an open part U of X is *ultra-affine* if the ringed space $(U, \mathscr{O}_X|U)$ is isomorphic to a maximal spectrum $\mathrm{Spm}(A)$, where A is a Jacobson ring. We will say that X is an *ultra-prescheme* if every point of X admits an ultra-affine open neighbourhood. One shows, as in (I, 2.1.3) and (I, 2.1.4), that the ultra-affine open sets form a *base* of the topology of X and that X is a Kolmogoroff space. If Y is a second ultra-prescheme, we will say that a morphism of ringed spaces $f : X \to Y$ is a *morphism of ultra-preschemes* if it satisfies the following conditions: 1° f is a morphism of *ringed spaces with local rings*; 2° for every $x \in X$, there is an ultra-affine open neighbourhood V of $f(x)$ in Y and an ultra-affine open neighbourhood U of x in X such that $f(U) \subset V$ and the homomorphism of rings $\Gamma(V, \mathscr{O}_Y) \to \Gamma(U, \mathscr{O}_X)$ corresponding to f makes $\Gamma(U, \mathscr{O}_X)$ a $\Gamma(V, \mathscr{O}_Y)$-algebra of finite type. **IV-3** | 114

It is immediate that one thus composes morphisms well, the composite of two morphisms of ultra-preschemes being an ultra-prescheme morphism thanks to the final remark of (10.9.3). It is clear that the category C_0'' thus defined is a sub-category of C' that contains C''; we now propose to show that $C'' = C_0''$, in other words:

Proposition (10.9.6). — *The functor $X \mapsto S(X)$ from C to C_0'' is* fully faithful, *that is for X, Y in C, the canonical map*

$$\mathrm{Hom}_C(X, Y) \longrightarrow \mathrm{Hom}_{C_0''}(S(X), S(Y))$$

is bijective. It is an equivalence of categories.

PROOF. Let us first show that it is *injective*: let f, g be two morphisms locally of finite type from X to Y and suppose that $S(f) = S(g)$. This implies first that for every open set V of Y, $f^{-1}(V) \cap S(X) = g^{-1}(V) \cap S(X)$, hence (10.3.1) and (10.2.7) $f^{-1}(V) = g^{-1}(V)$; it therefore suffices to prove that for every affine open V of Y, f and g coincide on $f^{-1}(V) = g^{-1}(V)$, which reduces to the case where X and Y are affine. It suffices (I, 2.2.4) to prove that the two homomorphisms of rings $B \to \Gamma(X, \mathscr{O}_X)$ corresponding to f and g are the same; now, for every $s \in B$, the images of s by these homomorphisms are two sections of $\mathscr{O}_X$ over X which, by hypothesis, induce the *same* section over $S(X)$; we say therefore (10.2.8) that these sections are identical, whence our assertion.

Let us show in the second place that every morphism $h : S(X) \to S(Y)$ (in the category C_0'') is of the form $S(f)$, where $f : X \to Y$ is a morphism locally of finite type. By hypothesis, there is an ultra-affine open covering (U') of $S(X)$ (resp. (V'_μ)) and for each λ, it corresponds under h to an ultra-affine open of $S(Y)$. By (10.9.4) and the remark following, we can determine the unique morphism $f : X \to Y$ and conclude that $h = S(f)$.

2º It remains to prove that every ultra-prescheme X' is of the form $S(X)$ for some Jacobson prescheme X (which will necessarily be unique up to isomorphism, by 1º). There are ultra-affine open sets covering X', each isomorphic to $S(U_\alpha)$ by open sets of X', U_α being the spectrum of a Jacobson ring; for every pair of indices α, β, consider the unique open $U_{\alpha\beta}$ of U_α whose trace on $S(U_\alpha)$ is $S(U_\alpha) \cap S(U_\beta)$; by virtue of 1º, the identity of $U_\alpha^\beta \cap U_\beta^\alpha$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \to U_{\beta\alpha}$ is an isomorphism of preschemes. One checks immediately (by virtue of 1º) that the family $(\theta_{\alpha\beta})$ satisfies the recollement condition ($\mathbf{0_I}$, 4.1.7), and that this family defines a prescheme X, such that the U_α are identified with affine open sets of X; it is clear that $X' = S(X)$, which completes the proof. □

10.10. Serre's algebraic spaces

(10.10.1). The language introduced by Serre in (FAC) is sometimes convenient, in particular in the questions where the principal interest is attached to the rational points over a base field k (algebraically closed by hypothesis) of the "algebraic varieties" over k that one considers. We will briefly outline here this language, to allow the reader to translate the results of Serre into the language of schemes by connecting them to the preceding considerations. It is moreover possible to develop the language of Serre equally for preschemes over a non-algebraically-closed field (and even over an Artinian ring); but this introduces considerable technical complications, and moreover, over an arbitrary base field, the advantages (mainly psychological) of the Serre viewpoint disappear; thus we will confine ourselves to the framework fixed by Serre. The present subsection, like the preceding one, will not be used in the rest of this Treatise, and we will therefore limit ourselves to brief indications. **IV-3** | 115

(10.10.2). Given a fixed ring R, one can naturally (as in every category) define the notion of R-ultra-prescheme. Consider in particular an *algebraically closed* field k; $\mathrm{Spm}(k)$ is then identical to $\mathrm{Spec}(k)$; we say that an $\mathrm{Spm}(k)$-ultra-prescheme X' is a *prealgebraic space* over k: it is therefore a k-ringed space with local rings whose every point has an open neighbourhood isomorphic to the maximal spectrum of a k-algebra of finite type; it amounts to the same to say (by (10.9.6)) that $X' = S(X)$, where X is a k-prescheme locally of finite type. If X, Y, Z are three k-preschemes locally of finite type over k, $X \times_k Y$ is also a k-prescheme locally of finite type (**I**, 1.3.4), hence by virtue of (10.9.6), the notion of product exists in the category of prealgebraic spaces over k (the "fibre product" $X' \times_{k'} Y'$, where X', Y', Z' are three prealgebraic spaces over k). One can also define the diagonal morphism $\Delta_{X'} : X' \to X' \times_{k'} X'$ (which is none other than $S(\Delta_X)$); we say that X' is an *algebraic space* over k if X is a scheme, and this amounts to the same as saying that the image of $\Delta_{X'}$ is a *closed part* of $X' \times_{k'} X'$.

(10.10.3). The simplifications which follow from the hypothesis that k is algebraically closed stem from the fact that, for a k-prescheme X locally of finite type, there is a bijective correspondence between *closed points* of X, *points of X with values in k* (**I**, 3.4.4), and *rational points of X* (**I**, 3.4.5), in virtue of (**I**, 6.4.2). This shows in particular (in virtue of (**I**, 4.3.1)) that for two k-prealgebraic spaces X', Y', the underlying set of the product $X' \times_{k'} Y'$ is *identical to the set-theoretic product $X' \times Y'$* (although of course the topology of the underlying space of $X' \times_{k'} Y'$ is *not* the product topology of the topologies of the underlying spaces of X' and Y'; it is in general strictly finer than the latter).

On the other hand, the local rings $\mathscr{O}_{x'}$ at the points of a prealgebraic space X' over k are the k-algebras whose residue field is *isomorphic* to k; if A and B are two such local k-algebras, every k-homomorphism $\varphi : A \to B$ is necessarily *local*: indeed, if an element x of the maximal ideal of A were such that $\varphi(x)$ is invertible, there would exist $\lambda \in k$ non-zero such that $\varphi(x - \lambda \cdot 1)$ belongs to the maximal ideal of B, which is absurd since $x - \lambda \cdot 1$ is invertible in A. We conclude immediately that if $X' = S(X)$, $Y' = S(Y)$ are two k-prealgebraic spaces, every morphism $X' \to Y'$ of *k-ringed spaces* is also a morphism of *ringed spaces with local rings* (**I**, 1.8.2); moreover, if A and B are two k-algebras of finite type, φ makes A a B-algebra of finite type; hence *every* morphism of k-prealgebraic spaces over k is, by virtue of (10.9.6), of the form $S(f)$, where $f : X \to Y$ is a *morphism of k-preschemes*.

(10.10.4). We say that a k-prealgebraic space $X' = S(X)$ is *reduced* if X is reduced; as the set of $x \in X$ where X is reduced is open ($\mathbf{0_I}$, 5.2.2), its complement contains at least one closed point if it is non-empty (5.1.11), and it amounts to the same to say that X' is reduced or that each of its local rings $\mathscr{O}_{X'}$ is reduced. We have just seen in (10.10.3) that one can define $\mathscr{O}_{X'}$ as a *subsheaf* of $\mathscr{A}(X')$ (in (FAC), Serre limits himself in fact to *reduced* prealgebraic spaces). One notes that if X' and Y' are two reduced k-prealgebraic spaces, the same is true of $X' \times_{k'} Y'$: indeed, it suffices to see that if A and B are two reduced k-algebras of finite type, the same is true of $A \otimes_k B$; but we have already seen that the radicals of A and B are reduced to 0, and as k is algebraically closed, A and B are "separable" algebras in the sense of Bourbaki (*Alg.*, chap. VIII, §7, n° 5, prop. 5); hence $A \otimes_k B$ is without radical (*loc. cit.*, n° 6, cor. 3 of th. 3). However, if Z' is a third prealgebraic space over k, the "fibre product" $X' \times_{k'} Y'$ of two reduced prealgebraic spaces over Z' is not in general reduced, which implies that the category of these spaces is insufficient for many questions (notably in the theory of algebraic groups). But, as one has just seen above, one can keep the language of Serre without limiting oneself, as in (FAC), to considering only *quasi-compact* prealgebraic spaces, to the case of reduced prealgebraic spaces.

(10.10.5). Finally, one can also consider ultra-preschemes over an *arbitrary* field k while preserving a language close to that of Serre, and by introducing, as Weil did, an algebraically closed extension K of k (chosen large enough, for example of transcendence degree infinite over k, to have sufficiently many "generic points" in the sense of Weil). For every prescheme X locally of finite type over k, one associates the set $S_K(X) = S(X \otimes_k K)$ of points of X with values in K; one has a canonical map $j : S_K(X) \to X$ which one shows to be a quasi-homeomorphism when one equips $S_K(X)$ with the topology that is the inverse image of that of X by j; one equips $S_K(X)$ with the sheaf of k-algebras $j^*(\mathscr{O}_X)$, and one thus obtains a sub-category of the category of (k, K)-prealgebraic spaces. One can show that one can again define products and generalize the results of (10.10.3) and (10.10.4) ($\mathscr{A}(X')$ being replaced here by the sheaf $\mathscr{A}_K(X')$ of germs of maps from X' *into* K). However, this point of view makes an algebraically closed sub-field of k play an artificial role, and we note it only to set it aside. **IV-3** | 116

§11. Topological properties of finitely presented flat morphisms. Flatness criteria

§12. Study of fibres of finitely presented flat morphisms

§13. Equidimensional morphisms

§14. Universally open morphisms

§15. Study of fibres of a universally open morphism

§16. Differential invariants. Differentially smooth morphisms

In this paragraph we will present, in global form, some notions of differential calculus particularly useful in algebraic geometry. We will ignore many classic developments in differential geometry (connections, infinitesimal transformations associated to vector fields, jets, etc.), although these notions are translated in a particularly natural way for schemes. We will similarly ignore phenomena exclusive to characteristic $p > 0$ (some of which are seen, in the affine case, in (**0**, 21). For certain complements to the differential formalism for preschemes the reader may consult Exposés II and VII of [**eAG64**] as well as subsequent chapters of this treatise. **IV-4** | 5

16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathscr{O}_X), (Y, \mathscr{O}_Y)$ be two ringed spaces and $f = (\psi, \theta) : Y \to X$ a morphism of ringed spaces (**0**, 4.1.1) such that the homomorphism

$$\theta^\# : \psi^*(\mathscr{O}_X) \longrightarrow \mathscr{O}_Y$$

is surjective, so that $\mathscr{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$. We can then endow $\psi^*(\mathscr{O}_X)$ with the $\mathscr{I}_f$-preadic filtration.

Definition (16.1.2). — The $\mathscr{O}_Y$-augmented sheaf of rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ is called the n'th *normal invariant* of f; the ringed space $(Y, \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1})$ is called the n'th *infinitesimal neighbourhood* of Y along f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathscr{O}_X)$

(16.1.2.1) $$\mathscr{G}r_\bullet(f) = \bigoplus_{n \geqslant 0} (\mathscr{I}_f^n / \mathscr{I}_f^{n+1})$$

is called the sheaf of graded rings *associated to* f. The sheaf $\mathscr{G}r_1(f) = \mathscr{I}_f/\mathscr{I}_f^2$ is called the *conormal sheaf* of f (that will be denoted by $\mathscr{N}_{Y/X}$ when there is no risk of confusion).

It is clear that the $\mathscr{O}_{Y^{(n)}} = \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ (that we also denote $\mathscr{O}_{Y_f^{(n)}}$) form a projective system of sheaves of rings on Y, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ for $n \leqslant m$ identifies $\mathscr{O}_{Y^{(n)}}$ with the quotient of $\mathscr{O}_{Y^{(m)}}$ by the power $(\mathscr{I}_f/\mathscr{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathscr{O}_{Y^{(n)}}$, kernel of $\phi_{0n} : \mathscr{O}_{Y^{(n)}} \to \mathscr{O}_Y$. The $Y^{(n)}$ therefore form a inductive system of ringed spaces, all having underlying space Y, and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \to X$ equal to (ψ, θ_n), where $\theta_n^\#$ is the canonical morphism $\psi^*(\mathscr{O}_X) \to \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$. It is clear that the sheaf $\mathscr{G}r_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathscr{O}_Y = \mathscr{G}r_0(f)$ and the $\mathscr{G}r_k(f)$ of $\mathscr{O}_Y$-modules. **IV-4 | 6**

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathscr{O}_Y$-algebras

(16.1.2.2) $$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_\bullet(f)$$

which coincide in degrees 0 and 1 with the identities.

Examples (16.1.3). —

(i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathscr{O}_y$) and that, if $x = \psi(y)$, $\theta^\# : \mathscr{O}_x \to \mathscr{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$. So the $\mathscr{O}_{Y^{(n)}}$ are identified with the rings $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$ and $\mathscr{G}r_\bullet(f)$ with the graded ring associated with the local ring $\mathscr{O}_x$ endowed with the $\mathfrak{m}_x$-preadic filtration.

(ii) Suppose that Y is a closed subset of an open subspace U of X and that the $\mathscr{O}_Y$ is induced on Y by a quotient sheaf $\mathscr{O}_U/\mathscr{I}$, where $\mathscr{I}$ is an ideal of $\mathscr{O}_U$ such that $\mathscr{I}_x = \mathscr{O}_x$ for every $x \notin Y$; if X is a locally ringed space we also suppose that $\mathscr{I}_x \neq \mathscr{O}_x$ for $y \in Y$ so that $(Y, \mathscr{O}_Y)$ is a locally ringed space.

Let $\psi_0 : Y \to U$ be the canonical injection and denote by $\theta_0 : \mathscr{O}_U \to (\psi_0)_*(\mathscr{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathscr{O}_U) = \mathscr{O}_U|Y \to (\mathscr{O}_U/\mathscr{I})|Y$, so that $j_0 = (\psi_0, \theta_0) : Y \to U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \to X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \to X$ is the canonical injection and $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathscr{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathscr{O}_U/\mathscr{I}^{n+1})$, and we have $(\psi_0)_*(\mathscr{O}_{Y^{(n)}}) = \mathscr{O}_U/\mathscr{I}^{n+1}$, and $\mathscr{G}r_n(j) = \mathscr{G}r_n(j_0) = \psi_0^*(\mathscr{I}^n/\mathscr{I}^{n+1}) = j_0^*(\mathscr{I}^n/\mathscr{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathscr{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathscr{O}_Y$-module*, or *a fortiori* with a structure of an $\mathscr{O}_Y$-algebra. The data of such structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n}; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \to Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \to Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \to X$ be an immersion of preschemes. Then:*

(i) *$\mathscr{G}r_\bullet(f)$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra.* IV-4 | 7

(ii) *The $Y^{(n)}$ are preschemes, canonically isomorphic to subpreschemes of X.*

(iii) *Every homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n}, makes the $\mathscr{O}_{Y^{(n)}}$ and $\mathscr{O}_{Y^{(k)}}$ for $k \leqslant n$ quasi-coherent $\mathscr{O}_Y$-algebras; the $\mathscr{O}_Y$-module structures induced from the above structures on the $\mathscr{G}r_k(f)$ for $k \leqslant n$ coincide with the ones defined in (16.1.2).*

PROOF. (i) Since the question is local on X and Y, we can reduce to the case where Y is a closed subpreschemes of X defined by an quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$; since $\mathscr{O}_Y$ is the restriction to Y of $\mathscr{O}_X/\mathscr{I}$ the assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathscr{I}^{n+1}$ of $\mathscr{O}_X$. Finally, to prove (iii) we notice that the data of λ_n makes the ideal $\mathscr{I}/\mathscr{I}^n$ of the augmentation ϕ_{0n} and their quotients $\mathscr{I}/\mathscr{I}^{k+1} (1 \leqslant k \leqslant n)$ $\mathscr{O}_Y$-modules, and it suffices to prove by induction on k that the $\mathscr{I}/\mathscr{I}^{k+1}$ are quasi-coherent $\mathscr{O}_Y$-modules and the structure of quotient $\mathscr{O}_Y$-module induced on $\mathscr{I}^k/\mathscr{I}^{k+1}$ is the same as defined on (16.1.2). The second assertion is immediate, $\mathscr{I}^k/\mathscr{I}^{k+1}$ being killed by $\mathscr{I}/\mathscr{I}^{n+1}$; the first result, by induction on k, is trivial for $k = 1$ and for $\mathscr{I}/\mathscr{I}^{k+1}$ being an extension of $\mathscr{I}/\mathscr{I}^k$ by $\mathscr{I}^k/\mathscr{I}^{k+1}$ (1.4.17). □

Corollary (16.1.6). — *Under the general hypotheses of (16.1.5), if the immersion f is locally of finite presentation then the $\mathscr{G}r_n(f)$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

PROOF. Indeed, with the notation from the proof of (16.1.5), $\mathscr{I}$ is an ideal of finite type of $\mathscr{O}_X$ (1.4.7), therefore the $\mathscr{I}^n/\mathscr{I}^{n+1}$ are $\mathscr{O}_Y$-modules of finite type, hence the conclusion. □

Corollary (16.1.7). — *Under the general hypotheses of (16.1.5), let $g : X \to Y$ be a morphism of preschemes, left inverse to f. Therefore, for every n, the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n}, making $\mathscr{O}_{Y^{(n)}}$ a quasi-coherent $\mathscr{O}_Y$-algebra; via these homomorphisms, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ ($n \leqslant m$) are homomorphisms of $\mathscr{O}_Y$-algebras. Also, if g is locally of finite type, then the $\mathscr{O}_{Y^{(n)}}$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

PROOF. The first assertion is an immediate result from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathscr{G}r_n(f)$ being then quasi-coherent $\mathscr{O}_Y$-modules of finite type by (16.1.6), the same goes for the $\mathscr{O}_Y$-modules $\mathscr{I}/\mathscr{I}^{n+1}$, being extensions of a finite number of the $\mathscr{G}r_k(f)$ (**III**, 1.4.17). □

Proposition (16.1.8). — *Let X be a locally Noetherian prescheme, $j : Y \to X$ an immersion; Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathscr{G}r_n(j)$ are coherent $\mathscr{O}_Y$-modules and the $\mathscr{G}r_\bullet(j)$ is a coherent sheaf of rings over the space Y.*

PROOF. Everything is local on X and Y, so we reduce to the case where X is affine and j is a closed immersion and therefore all the assertions are evident except for the last, which follows from the fact that if A is a Noetherian ring and $\mathfrak{J}$ is an ideal of A, then $\mathrm{gr}^\bullet_{\mathfrak{J}}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (**0**, 5.3.7). □

Proposition (16.1.9). — *Let X be a prescheme, $j : Y \to X$ an immersion locally of finite presentation, y a point of Y. The following conditions are equivalent:* IV-4 | 8

(a) *There exists an open neighbourhood U of y in Y such that $j|U$ is a homeomorphism of U onto an open set of X.*

(b) *There is an integer $n > 0$ such that the canonical homomorphism*

$$(\phi_{n-1,n})_y : \mathscr{O}_{Y^{(n)},y} \longrightarrow \mathscr{O}_{Y^{(n-1)},y}$$

is bijective.

(c) *There is an integer $n > 0$ such that $(\mathscr{G}r_n(j))_y = 0$.*

In addition, if the integer n satisfies (b) *or* (c), *then there is a neighbourhood V of y in Y such that $\mathscr{G}r_m(j)|V = 0$ for $m \geqslant n$ and that $\phi_{nm}|V : \mathscr{O}_{Y^{(m)}}|V \to \mathscr{O}_{Y^{(n)}}|V$ is bijective for $m \geqslant n$.*

PROOF. Since the questions is local on Y, we can restrict ourselves to the case where j is a closed immersion, Y being defined by a quasi-coherent ideal *of finite type* $\mathfrak{I}$ of $\mathscr{O}_X$. The equivalence of (b) and (c), for a given n, is immediate; also, since $\mathscr{I}^n/\mathscr{I}^{n+1}$ is an $\mathscr{O}_X$-module of finite type, there is an open neighbourhood U of y in X such that $\mathscr{I}^n|U = \mathscr{I}^{n+1}|U$ (**0**, 5.2.2), so we also have $\mathscr{I}^n|U = \mathscr{I}^m|U$ for $m \geqslant n$ proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the cases where the underlying space of Y is equal to the underlying space of X and where $\mathscr{I}$ is generated by a finite number of sections over X: since $\mathscr{I}$ is contained in the nilradical $\mathscr{N}$ of $\mathscr{O}_X$ (**I**, 5.1.2), it is now nilpotent which proves b). Finally, to prove that (b) implies (a), we can restrict ourselves to the case where $\mathscr{I}^n = \mathscr{I}^m$; therefore, for every $y \in Y$, since $\mathscr{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathscr{O}_{X,y}$, we must have $\mathscr{I}_y^n = 0$ because of Nakayama's lemma, since $\mathscr{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathscr{I}_x^n = 0$ is an open U of X contained in Y (**0**, 5.2.2); since on the other hand $\mathscr{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. □

Corollary (16.1.10). — *For a restriction of the immersion j to an open neighbourhood of y in Y to be an open immersion (in other words, for j to be a* local isomorphism *on the point y), it is necessary and sufficient that $(\mathscr{G}r_1(j))_y = (\mathscr{N}_{Y/X})_y = 0$.*

PROOF. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. □

Remark (16.1.11). —

(i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathscr{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings over Y is called the *normal invariant of infinite order* of f, and sometimes denoted by $\mathscr{O}_{Y^{(\infty)}}$. When X is a locally noetherian prescheme, $j : Y \to X$ a closed immersion, Y then is a closed subprescheme of X defined by a coherent ideal $\mathscr{I}$ and $\mathscr{O}_{Y^{(\infty)}}$ is exactly the *formal completion* of $\mathscr{O}_X$ along Y (**I**, 10.8.4), and $Y^{(\infty)} = (Y, \mathscr{O}_{Y^{(\infty)}})$ is the formal prescheme that is the *completion* of X along Y (**I**, 10.8.5). In all cases, we could say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (via the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n.

(ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \to X$, it can happen that the homomorphism $\theta^{\#} : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Y$ is surjective without f being a local immersion and **IV-4** | 9
without f being injective. We have an example by taking Y to be a sum of preschemes Y_λ all isomorphic to $\operatorname{Spec}(\mathscr{O}_x)$, where $x \in X$, ad taking f to be the morphism equal to the canonical morphism in each of the Y_λ.

16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \to X$ and $f' = (\psi', \theta') : Y' \to X'$ by two morphisms of ringed spaces such that $\theta^{\#}$ and $\theta'^{\#}$ are surjective; consider a commutative diagram of morphisms of ringed spaces

(16.2.1.1)
$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \end{tikzcd}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X))$ and as a result a commutative diagram of homomorphisms of sheaves of rings over Y'

$$\begin{tikzcd} \rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X)) \arrow[r, "\psi'^*(\mu^{\#})"] \arrow[d, "\rho^*(\theta^{\#})"'] & \psi'^*(\mathscr{O}_{X'}) \arrow[d, "\theta'^{\#}"] \\ \rho^*(\mathscr{O}_Y) \arrow[r, "\lambda^{\#}"'] & \mathscr{O}_{Y'} \end{tikzcd}$$

from which we conclude, if $\mathscr{I}$ and $\mathscr{I}'$ are the kernels of $\theta^{\#}$ and $\theta'^{\#}$, that we have $\psi'^*(\mu^{\#})(\rho^*(\mathscr{I})) \subset \mathscr{I}'$, having in mind the exactness of the functor ρ^*. We deduce that, for every integer n, $\psi'^*(\mu^{\#})(\rho^*(\mathscr{I}^n)) \subset$

$\mathscr{I}'^{n}$, which shows that $\psi'^*(\mu^\#)$ defines, passing to the quotients, a homomorphism of sheaves of rings

(16.2.1.2) $$\nu_n : \rho^*(\psi^*(\mathscr{O}_X)/\mathscr{I}^{n+1}) \longrightarrow \psi'^*(\mathscr{O}_{X'})/\mathscr{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, \nu_n) : Y'^{(n)} \to Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{tikzcd} Y^{(n)} \arrow[r, "h_{mn}"] & Y^{(m)} \arrow[r, "h_m"] & X \\ Y'^{(n)} \arrow[u, "w_n"] \arrow[r, "h'_{mn}"'] & Y'^{(m)} \arrow[u, "w_m"] \arrow[r, "h'_m"'] & X' \arrow[u, "v"'] \end{tikzcd} \qquad (n \leqslant m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness **IV-4** | 10
of the functor ρ^*, we obtain a di-homomorphism of graded algebras (relative to the morphism $\lambda^\# : \rho^*(\mathscr{O}_Y) \to \mathscr{O}_{Y'}$)

(16.2.1.3) $$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f')$$

(or, if you like, a ρ-morphism (**0**, 3.5.1) $\mathscr{G}r_\bullet(f) \to \mathscr{G}r_\bullet(f')$), and in particular a di-homomorphism of conormal sheaves

$$\mathrm{gr}_1(u) : \rho^*(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_1(f').$$

It is also immediate that these homomorpisms give rise to a commutative diagram

(16.2.1.4) $$\begin{tikzcd} \rho^*(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f))) \arrow[r] \arrow[d, "\mathbf{S}(\mathrm{gr}_1(u))"'] & \rho^*(\mathscr{G}r_\bullet(f)) \arrow[d, "\mathrm{gr}(u)"] \\ \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f')) \arrow[r] & \mathscr{G}r_\bullet(f') \end{tikzcd}$$

where the horizontal arrow are the canonical morphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \\ Y'' \arrow[u, "u'"] \arrow[r, "f''"'] & X'' \arrow[u, "v'"'] \end{tikzcd}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^\#$ is surjective, and if w'_n and w''_n are defined from u', v' for one and $u'' = u \circ u'$, $v'' = v \circ v'$ for the other, we have $w''_n = w_n \circ w'_n$, which follows immediately from the definitions and from (**0**, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^*(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathscr{G}r_\bullet(f)$ *depend functorially* on f.

Proposition (16.2.2). — *With the notation and hypotheses of* (16.2.1), *suppose also that f, f', u, and v are morphisms of preschemes. We have:*

(i) *The morphisms $w_n : Y'^{(n)} \to Y^{(n)}$ are morphisms of preschemes.*

(ii) *If $Y' = Y \times_X X'$, u and f' the canonical projections, and if f is an immersion or if v is flat, we have $Y'^{(n)} = Y^{(n)} \times_X X'$.*

(iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes I : \mathscr{G}r_\bullet(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_\bullet(f')$$

is bijective (resp. surjective).

PROOF.

(i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho^*_{y'}(\theta^\#_{\psi'(y')})$ is a *local* homomorphism (I, 1.6.2), so w_n is a morphism of preschemes (I, 2.2.1). **IV-4** | 11

(ii) and (iii) If f is an immersion, we can restrict ourselves to the case where f is a closed immersion, Y being defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$ and $Y^{(n)}$ by the ideal $\mathscr{I}^{n+1}$; the assertions follows from (I, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, $X' = \operatorname{Spec}(A')$ are affines, A' being a flat A-module; so $Y' = \operatorname{Spec}(B')$ where $B' = B \otimes_A A'$; in addition, if $\mathfrak{I}$ is the kernel of the homomorphism $A \to B$, the kernel $\mathfrak{I}'$ of $A' \to B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and $\mathscr{I}'^n/\mathscr{I}'^{n+1}$ is equal to

$$\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim) \otimes_{\sigma^*(\mathscr{O}_X)} \mathscr{O}_{X'}) =$$
$$\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1}))^\sim) \otimes_{\psi'^*(\sigma^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'}) = \rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

and in particular for $n = 0$, we have

$$\mathscr{O}_{Y'} = \rho^*(\mathscr{O}_Y) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

from which we have canonical isomorphism of $\mathscr{I}'^n/\mathscr{I}'^{n+1}$ with

$$\rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\mathscr{O}_Y)} \mathscr{O}_{Y'} = (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

which proves (iii). Let now $C_n = \Gamma(Y, \mathscr{O}_{Y^{(n)}}), C'_n = \Gamma(Y', \mathscr{O}_{Y'^{(n)}})$. As $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \to C_{n-1}$ (resp. $C'_n \to C'_{n-1}$) is $\Gamma(Y, \mathscr{I}^n/\mathscr{I}^{n+1})$ (resp. $\Gamma(Y, \mathscr{I}'^n/\mathscr{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{K}_n \otimes_A A' \arrow[r] \arrow[d, "r"] & C_n \otimes_A A' \arrow[r] \arrow[d, "s_n"] & C_{n-1} \otimes_A A' \arrow[r] \arrow[d, "s_{n-1}"] & 0 \\
0 \arrow[r] & \mathfrak{K}'_n \arrow[r] & C'_n \arrow[r] & C'_{n-1} \arrow[r] & 0
\end{tikzcd}$$

where the vertical arrow of the left is bijective and the two lines are exact (A' being a flat A-module). We deduce by induction that s_n is bijective for every n, because it is true by hypothesis for $n = 0$, and is deduced by application of the five lemma for all n. That proves the second assertion of (ii).

□

Corollary (16.2.3). — *Let $g : X \to Y$, $u : Y' \to Y$ be two morphisms of preschemes, $X' = X \times_Y Y'$, $g' : X' \to Y'$ and $v : X' \to X$ by the canonical projections. Let $f : Y \to X$ by a Y-section of X (and therefore an immersion), $f' = f_{(Y')} : Y' \to X'$ the Y'-section of X' deduced from f by the base change u. We have:*

(i) *The morphism $w_n : Y'^{(n)}_{f'} \to Y^{(n)}_f$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : Y'^{(n)}_{f'} \to X'$ identifies $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_X X'$.*

(ii) *If we endow $\mathscr{O}_{Y^{(n)}_f}$ (resp. $\mathscr{O}_{Y'^{(n)}_{f'}}$) with the structure of an $\mathscr{O}_Y$-algebra defined by g (resp. with the structure of an $\mathscr{O}_{Y'}$-algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathscr{O}_{Y'}$-algebras*

(16.2.3.1) $$\rho^*(\mathscr{O}_{Y^{(n)}_f}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{O}_{Y'^{(n)}_{f'}}$$

induced by the homomorphism v_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathscr{O}_{Y'}$-modules **IV-4** | 12

(16.2.3.2) $$\operatorname{Gr}_1(u) : \mathscr{G}r_1(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_1(f')$$

is bijective.

PROOF.

(i) Let us first note that $f' : Y' \to X'$ and $u : Y' \to Y$ identifies Y' with the product $Y \times_X X'$ (via the structure morphisms $f : Y \to X$ and $v : X' \to X$) (14.5.12.1). The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.

(ii) The commutative diagram

$$\begin{tikzcd} Y_f^{(n)} & Y'^{(n)}_{f'} \arrow[l, "w_n"'] \arrow[d, "h'_n"] \\ X \arrow[u, leftarrow, "h_n"] \arrow[d, "g"] & X' \arrow[l, "v"'] \arrow[d, "g'"] \\ Y & Y' \arrow[l, "u"'] \end{tikzcd}$$

identifies $Y'^{(n)}_{f'}$ with the product $Y_f^{(n)} \times_X X'$, so (**I**, 3.3.9) it identifies (via the morphisms $g' \circ h'_n$ and w_n) $Y'^{(n)}_{f'}$ to the product $Y_f^{(n)} \times_Y Y'$. Since $Y_f^{(n)}$ (resp. $Y'^{(n)}_{f'}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathscr{O}_Y$-algebran $\mathscr{O}_{Y_f^{(n)}}$ (resp. to the $\mathscr{O}_{Y'}$-algebran $\mathscr{O}_{Y'^{(n)}_{f'}}$), the fact that the canonical homomorphism (16.2.3.1) is bijective follows from (**II**, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathscr{O}_{Y_f^{(n)}} \to \mathscr{O}_Y$ and $\mathscr{O}_{Y'^{(n)}_{f'}} \to \mathscr{O}_{Y'}$; since $\mathscr{O}_{Y_f^{(n)}}$ is a direct sum (as an $\mathscr{O}_Y$-module) of $\mathscr{O}_Y$ and the augmentation ideal $\mathscr{I}/\mathscr{I}^{n+1}$, we can therefore see that the canonical homomorphism (16.2.3.1), restricted to $\mathscr{I}/\mathscr{I}^{n+1} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, is a bijection of the latter onto $\mathscr{I}'/\mathscr{I}'^{n+1}$. For $n = 1$ this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geqslant 2$. However:

Corollary (16.2.4). — *Under the hypotheses of* (16.2.3), *suppose that $u : Y' \to Y$ is a flat morphism (resp. that the $\mathscr{G}r_n(f)$ are flat $\mathscr{O}_Y$-modules for $n \leqslant m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathscr{G}r_n(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_n(f')$$

is bijective for all n (resp. for $n \leqslant m$).

Proof. If u is flat, then we deduce by base change that the same is true for $v : X' \to X$, and we already know in this case that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathscr{G}r_n(f)$ are flat for $n \leqslant m$, then we first see by induction on n that the same holds for $\mathscr{I}/\mathscr{I}^{n+1}$ for $n \leqslant m$, because of the exact sequences

$$0 \longrightarrow \mathscr{I}^n/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^n \longrightarrow 0$$

(**0**, 6.1.2); in addition, we have the commutative diagram **IV-4** | 13

$$\begin{tikzcd} 0 \arrow[r] & (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^n) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \mathscr{I}'^n/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^n \arrow[r] & 0 \end{tikzcd}$$

in which the lines are exact (the first by flatness (**0**, 6.1.2)) and the two last vertical arrows are bijective by virtue of (16.2.2, (ii)); hence the conclusion. □

Remarks (16.2.5). —

(i) The reasoning of (16.2.2, (i)) still applies to (16.2.1.1) when these are morphisms of *locally ringed spaces* (**I**, 1.8.2).

(ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{I} = 0$ but the kernel $\mathfrak{I}'$ of $A' \to B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n, but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty}(\mathbf{Z}/m^h\mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the particular case of the diagram (16.2.1.1) where $X' = X$, v is the identity, X a prescheme, Y a subprescheme of X, Y' a subprescheme of Y, f, u, and $f' = f \circ u$ the canonical injections; the di-homomorphism (16.2.1.3) gives us, by tensoring with $\mathscr{O}_{Y'}$ over $\rho^*(\mathscr{O}_Y)$, a homomorphism of graded $\mathscr{O}_{Y'}$-algebras

$$u^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f'). \tag{16.2.6.1}$$

On the other hand, we identify $\mathscr{O}_Y$ to $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$ and $\mathscr{O}_{Y'}$ to $\rho^*(\mathscr{O}_Y)/\mathscr{I}_u$; since ρ^* is an exact functor, we have $\rho^*(\mathscr{O}_Y) = \rho^*(\psi^*(\mathscr{O}_X))/\rho^*(\mathscr{I}_f) = \psi'^*(\mathscr{O}_X)/\rho^*(\mathscr{I}_f)$, and since $\mathscr{O}_{Y'}$ is moreover identified with $\psi'^*\mathscr{O}_X/\mathscr{I}_{f'}$, we see that $\mathscr{I}_u = \mathscr{I}_{f'}/\rho^*(\mathscr{I}_f)$. We deduce that for every integer n there is a canonical homomorphism $\mathscr{I}_{f'}^n/\mathscr{I}_{f'}^{n+1} \to \mathscr{I}_u^n/\mathscr{I}_u^{n+1}$, from which we have a canonical morphism of graded $\mathscr{O}_{Y'}$-algebras

$$\mathscr{G}r_\bullet(f') \longrightarrow \mathscr{G}r_\bullet(u). \tag{16.2.6.2}$$

Proposition (16.2.7). — *Let X be a prescheme, Y a subprescheme of X, Y' a subprescheme of Y, $j : Y' \to Y$ the canonical injection. We then have an exact sequence of conormal sheaves ($\mathscr{O}_{Y'}$-modules)*

$$j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0 \tag{16.2.7.1}$$

where the arrows are the degree 1 *components of the canonical homomorphisms* (16.2.6.1) *and* (16.2.6.2).

Proof. The problem being local, we can restrict to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(A/\mathfrak{J})$ and $Y' = \operatorname{Spec}(A/\mathfrak{K})$, $\mathfrak{J}$ and $\mathfrak{K}$ being ideals of A such that $\mathfrak{J} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \to \mathfrak{K}/\mathfrak{K}^2 \to (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \to 0$ is exact, which is **IV-4** | 14
immediate given that the image of $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{J}+\mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2$ is identified with $\mathfrak{K}/(\mathfrak{J}+\mathfrak{K}^2)$. □

It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{J} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{J}+\mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \to S$ be a morphism of preschemes, $\Delta_f : X \to X \times_S X$ the corresponding diagonal morphism, which is an immersion (**I**, 5.3.9). We denote by $\mathscr{P}_f^n$ or $\mathscr{P}_{X/S}^n$, and call the *sheaf of principal parts of order n of the S-prescheme X*, the $\mathscr{O}_X$-augmented sheaf of rings, n-th normal invariant of Δ_f (16.1.2). We will also write $\mathscr{P}_f^\infty = \mathscr{P}_{X/S}^\infty = \varprojlim_n \mathscr{P}_{X/S}^n$, $\mathscr{G}r_n(\mathscr{P}_f) = \mathscr{G}r_n(\mathscr{P}_{X/S}) = \mathscr{G}r_n(\Delta_f)$ (16.1.2); the $\mathscr{O}_X$-module $\mathscr{G}r_1(\Delta_f)$, augmentation sheaf of ideals of $\mathscr{P}_{X/S}^1$, is denoted by Ω_f^1 or $\Omega_{X/S}^1$, and is called the $\mathscr{O}_X$-module of 1-*differentials* of f, or of X with respect to S, or of the S-prescheme X.

It follows from this definition that $\mathscr{P}_{X/S}^0$ is canonically identified with $\mathscr{O}_X$ (16.1.2).

We have (16.1.2.2) a canonical surjective morphism of graded $\mathscr{O}_X$-algebras

$$\mathbf{S}_{\mathscr{O}_X}^\bullet(\Omega_{X/S}^1) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.3.1.1}$$

And it follows from Definition (16.3.1) that for every open U of X we have $\mathscr{P}_{f|U}^n = \mathscr{P}_f^n|U$, $\mathscr{P}_{f|U}^\infty = \mathscr{P}_f^\infty|U$, $\mathscr{G}r_n(\mathscr{P}_{f|U}) = \mathscr{G}r_n(\mathscr{P}_f)|U$, $\Omega_{f|U}^1 = \Omega_f^1|U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Denote by p_1, p_2 the two canonical projections of the product $X \times_S X$; since Δ_f is an X-section of $X \times_S X$ for both p_1 and p_2, *each* of these morphisms define, for all n, a homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}_{X/S}^n$, right inverse of the augmentation $\mathscr{P}_{X/S}^n \to \mathscr{O}_X$ (16.1.7); we can also say that we thus define on $\mathscr{P}_{X/S}^n$ *two quasi-coherent augmented $\mathscr{O}_X$-algebra* structures; the corresponding $\mathscr{O}_X$-module structures on on $\mathscr{G}r_n(\mathscr{P}_{X/S}^n)$ are the same. We also have, by passing to the limit, two $\mathscr{O}_X$-algebra structures on $\mathscr{P}_{X/S}^\infty$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \to X \times_S X$ is an *involutive automorphism* of $X \times_S X$, called the *canonical symmetry*, such that

(16.3.3.1) $$p_1 \circ s = p_2, \qquad p_2 \circ s = p_1, \qquad s \circ \Delta_f = \Delta_f.$$

If we put $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ $(i = 1, 2)$, $\Delta_f = (\delta, \nu)$, $\lambda^\#$ is then an isomorphism of $\rho^*(\pi_1^*(\mathscr{O}_X))$ onto $\pi_2^*(\mathscr{O}_X)$, and $\delta^*(\lambda^\#)$ fixes $\delta^*(\mathscr{O}_{X\times_S X})$ and the kernel $\mathscr{I}$ of the homomorphism $\nu^\# : \delta^*(\mathscr{O}_{X\times_S X}) \to \mathscr{O}_X$. Therefore:

Proposition (16.3.4). — *The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the* canonical symmetry*) is an involutive automorphism of the projective system $(\mathscr{P}^n_{X/S})$ of $\mathscr{O}_X$-augmented sheaves of rings, and as a result also of the projective limit $\mathscr{P}^\infty_{X/S}$. This automorphism permutes the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$.* **IV-4** | 15

(16.3.5). In what follows, the two $\mathscr{O}_X$-algebra structures defined on the $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$ will play very different roles: *we will now agree, unless said otherwise, that when $\mathscr{P}^n_{X/S}$ or $\mathscr{P}^\infty_{X/S}$ is considered as an $\mathscr{O}_X$-algebra, it is the algebra structure induced by p_1.*

For every open U of X and every section $t \in \Gamma(U, \mathscr{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^n_{X/S})$ (resp. $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^\infty_{X/S})$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d^n_f, or $d^n_{X/S}$ (resp. d^∞_f, or $d^\infty_{X/S}$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ (resp. $\mathscr{O}_X \to \mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S}$) induced by p_2. For every open U of X, and every $t \in \Gamma(U, \mathscr{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t. We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega^1_{X/S})$, also denoted $d_{X/S}(t)$).

It follows immediately [5] from this definition that we have

(16.3.6.1) $$d(t_1 t_2) = t_1 dt_2 + t_2 dt_1$$

for every t_1, t_2 in $\Gamma(U\mathscr{O}_X)$, that is, d is a *derivation* of the ring $\Gamma(U, \mathscr{O}_X)$ in the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

In all notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \operatorname{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine schemes, B then being an A-algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \to B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{I} = \mathfrak{I}_{B/A}$ (**0**, 20.4.1); $\mathscr{P}^n_f$ is the structure sheaf of the prescheme $\operatorname{Spec}(P^n_{B/A})$, where

$$P^n_{B/A} = (B \otimes_A B)/\mathfrak{I}^{n+1};$$

$\mathscr{G}r_\bullet(\mathscr{P}_f)$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the graded B-module

$$\operatorname{gr}^\bullet_{\mathfrak{I}}(B \otimes_A B) = \bigoplus_{n \geqslant 0} (\mathfrak{I}^n / \mathfrak{I}^{n+1});$$

in particular $\Omega^1_f = \Omega^1_{X/S}$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the B-module of 1-differentials of B over A, $\Omega^1_{B/A}$ (**0**, 20.4.3). The projection morphisms $p_1 : X \times_S X \to X$, $p_2 : X \times_S X \to X$ corresponding to the two homomorphisms of rings $j_1 : B \to B \otimes_A B$, $j_2 : B \to B \otimes_A B$ such that $j_1(b) = b \otimes 1$, $j_2(b) = 1 \otimes b$, so that (by the convention of (16.3.5)), $P^n_{B/A}$ is always considered as a B-algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \to P^n_{B/A}$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \to P^n_{B/A}$ is denoted by $d^n_{B/A}$ and corresponds to $d^n_{X/S}$ acting on $\Gamma(X, \mathscr{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A} t$, defined in (**0**, 20.4.6).

If $\pi_n : B \otimes_A B \to P^n_{B/A}$ is the canonical homomorphism, so we have, in light of the preceding definitions,

(16.3.7.1) $$\pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d^n_{B/A}(b') \quad \text{for } b \in B, b' \in B.$$

[5][Trans.] This is, locally we have (**0**, 20.1.1).

Proposition (16.3.8). — *The image of the canonical homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$.* **IV-4 | 16**

PROOF. We immediately reduce to the case where $X = \operatorname{Spec}(B)$ and $S = \operatorname{Spec}(A)$ are affine and the proposition follows from (16.3.7.1) since π_n is surjective. We note that in general $d^n_{X/S}$ *is not surjective* (even for $n = 1$). ☐

Proposition (16.3.9). — *Suppose that $f : X \to S$ is a morphism locally of finite type. Then the $\mathscr{P}^n_f$ and the $\mathscr{G}r_n(\mathscr{P}_f)$ are quasi-coherent $\mathscr{O}_X$-modules of finite type.*

PROOF. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (**I**, 4.3.1). ☐

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "w"] \end{tikzcd}$$

We deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & & X' \arrow[ll, "u"'] \arrow[d, "\Delta_{f'}"] \\ X \times_S X & & X' \times_{S'} X' \arrow[ll, "v"] \end{tikzcd}$$

where v is the composite homomorphism (**I**, 5.3.5) and (**I**, 5.3.15).

(16.4.1.2)
$$X' \times_{S'} X' \xrightarrow{(p'_1, p'_2)_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

So we induce from u and v, as explained in (16.2.1), homomorphisms of augmented sheaves of rings

(16.4.1.3)
$$\nu_n : \rho^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{X'/S'}$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system, and therefore give at the limit a homomorphism of sheaves of graded rings

(16.4.1.4)
$$\nu_\infty : \rho^*(\mathscr{P}^\infty_{X/S}) \longrightarrow \mathscr{P}^\infty_{X'/S'};$$

on the other hand, by passing to the quotient, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

(16.4.1.5)
$$\operatorname{gr}(u) : \rho^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram **IV-4 | 17**

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] & X'' \arrow[l, "u'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "w"] & S'' \arrow[l, "w'"] \end{tikzcd}$$

we deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & & X' \arrow[ll, "u"'] \arrow[d, "\Delta_{f'}"] & & X'' \arrow[ll, "u'"'] \arrow[d, "\Delta_{f''}"] \\ X \times_S X & & X' \times_{S'} X' \arrow[ll, "v"] & & X'' \times_{S''} X'' \arrow[ll, "v'"] \end{tikzcd}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f'. We verify immediately that if $u'' = u \circ u'$, $w'' = w \circ w'$, then the composite homomorphism $v \circ v'$ is equal to the homomorphism v'' deduced from u'', v'', f, f'' as v is from u, w, f, f'. If we put $u' = (\rho', \lambda')$, $u'' = (\rho'', \lambda'')$ it follows (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathscr{P}^n_{X/S}) \to \mathscr{P}^n_{X''/S''}$ is equal to the composite

$$\rho'^*(\rho^*(\mathscr{P}^n_{X/S})) \xrightarrow{\rho'^*(\nu^\#_n)} \rho'^*(\mathscr{P}^n_{X'/S'}) \xrightarrow{\nu'_n} \mathscr{P}^n_{X''/S''},$$

and we have analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), which lets us say that the $\mathscr{P}^n_{X/S}$, $\mathscr{P}^\infty_{X/S}$ and $\mathscr{G}r_\bullet(\mathscr{P}_{X/S})$ *depend functorially on* f.

(16.4.3). We verify immediately (for example, by restricting ourselves to the affine case with help of (16.3.7)) that with the notation of (16.4.1), the diagram

(16.4.3.1)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d] & \mathscr{O}_{X'} \arrow[d] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

where the vertical arrows are the ones defining the algebra structure chosen in (16.3.5) (that is to say, the ones coming from the first projections) is commutative; the same goes for the diagram

(16.4.3.2)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d, "\rho^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

the vertical arrows defining here the algebra structure from the second projection; besides, if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have **IV-4** | 18

$$\nu_n \circ \rho^*(\sigma) = \sigma' \circ \nu_n$$

which switches one diagram with the other. We deduce from (16.4.3.1) a canonical homomorphism of *augmented* $\mathscr{O}_{X'}$*-algebras*

(16.4.3.3)
$$P^n(u) : u^*(\mathscr{P}^n_{X/S}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{P}^n_{X'/S'}$$

and it follows from (16.4.3.2) that the diagram

(16.4.3.4)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "u^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ u^*(\mathscr{P}^n_{X/S}) \arrow[r, "P^n(u)"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative. We deduce a homomorphism of *graded* $\mathscr{O}_{X'}$*-algebras*

(16.4.3.5)
$$\mathrm{Gr}_\bullet(u) : u^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'})$$

and in particular a homomorphism of $\mathscr{O}_{X'}$-modules

(16.4.3.6)
$$\mathrm{Gr}_1(u) : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/S'}$$

giving rise to a commutative diagram

(16.4.3.7)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "d_{X/S} \otimes 1"'] & \mathscr{O}_{X'} \arrow[d, "d_{X'/S'}"] \\ \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] & \Omega^1_{X'/S'} \end{tikzcd}$$

(16.4.4). When $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[u] \arrow[r] & A' \arrow[u] \end{tikzcd}$$

the image of $\mathfrak{I}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathfrak{I}_{B'/A'}$, and the homomorphism ν_n corresponds to the homomorphism of rings $P^n_{B/A} \to P^n_{B'/A'}$ induced from the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (**0**, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (**0**, 20.5.4.2).

Proposition (16.4.5). — *Suppose that $X' = X \times_S S'$, f' and u the canonical projections. Then the canonical homomorphisms $P^n(u)$ (16.4.3.3) and $\mathrm{Gr}_1(u)$ (16.4.3.6) are bijective.* **IV-4** | 19

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)) replacing g by the first $p_1 : X \times_S X \to X$ and f by the diagonal Δ_f. □

We note that under the hypotheses of (16.4.5) the homomorphism $\mathrm{Gr}_\bullet(u)$ (16.4.3.5) is *surjective*, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — *Under the hypotheses of (16.4.5), suppose in addition that $w : S \to S'$ is flat (resp. that $\mathscr{G}r_n(\mathscr{P}^n_{X/S})$ are flat $\mathscr{O}_X$-modules for $n \leqslant m$); then the homomorphism*

$$\mathrm{Gr}_n(u) : u^*(\mathscr{G}r(\mathscr{P}^n_{X/S})) \longrightarrow \mathscr{G}r(\mathscr{P}^n_{X'/S'})$$

is bijective for each n (resp. for $n \leqslant m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \to X \times_S X$, so the conclusion follows from (16.2.4). □

(16.4.7). Let S be a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_S$-Module, and set $X = \mathbf{V}(\mathscr{E})$ (**II**, 1.7.8), the vector bundle associated to $\mathscr{E}$, equal to $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$. Let $f : X \to S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathscr{E})$, t is identified with a section of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ over U; let t' be its image in $\Gamma(f^{-1}(U), \mathscr{O}_X) = \Gamma(U, f_*(\mathscr{O}_X)) = \Gamma(U, \mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$, and set

$$\delta(t) = d^n_{X/S}(t') - t' \in \Gamma(f^{-1}(U), \mathscr{P}^n_{X/S}); \tag{16.4.7.1}$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathscr{O}_S) \to \Gamma(f^{-1}(U), \mathscr{O}_X)$) of $\Gamma(U, \mathscr{E})$ into $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$, and therefore the image belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$. We deduce (by varying U) a canonical homomorphism of $\mathscr{O}_X$-algebras

$$f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{P}^n_{X/S} \tag{16.4.7.2}$$

and in view of the above remark, if $\mathscr{K}$ is the ideal kernel of augmentation $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \to \mathscr{O}_S$, the image of $\mathscr{K}^{n+1}$ by (16.4.7.2) is zero, so that by factoring by $\mathscr{K}^{n+1}$, we finally have a canonical homomorphism

$$\delta_n : f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})/\mathscr{K}^{n+1}) \longrightarrow \mathscr{P}^n_{X/S}. \tag{16.4.7.3}$$

Proposition (16.4.8). — *Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathscr{O}_S$-algebras*

$$f^*(\mathbf{S}^\bullet_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.4.8.1}$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, since both filtrations involved in (16.4.7.3) are finite (Bourbaki, *Alg. comm.*, chap. III, §2, no 8, cor. 3 of th. 1). To do this, consider the split exact sequence of $\mathscr{O}_S$-modules **IV-4** | 20

$$0 \longrightarrow \mathscr{E} \xrightarrow{u} \mathscr{E} \oplus \mathscr{E} \xrightarrow{v} \mathscr{E} \longrightarrow 0 \tag{16.4.8.2}$$

where, for every pair of sections s, t of $\mathscr{E}$ over an open U of S, we take $u(s) = (-s, s)$ and $v(s,t) = s + t$. We have

$$X \times_S X = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}))$$

((**II**, 1.4.6) and (**II**, 1.7.11)), and the diagonal morphism $X \to X \times_S X$ corresponds (**II**, 1.2.7) to the homomorphism of $\mathscr{O}_X$-algebras $\mathbf{S}(v) : \mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}) \to \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ (**II**, 1.7.4), such that if $\mathscr{I}$ is the kernel of this homomorphism, then we have

$$\mathscr{P}^n_{X/S} = f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E})/\mathscr{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: □

Lemma (16.4.8.3). — *Let Y be a ringed space, $0 \to \mathscr{F}' \xrightarrow{u} \mathscr{F} \xrightarrow{v} \mathscr{F}'' \to 0$ an exact sequence of $\mathscr{O}_Y$ modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \to \mathscr{F}'|V \to \mathscr{F}|V \to \mathscr{F}''|V \to 0$ is split. Let $\mathscr{I}$ be the kernel ideal of $\mathbf{S}(v)$:*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}) \longrightarrow \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}''),$$

and let $\operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$ be the graded $\mathscr{O}_Y$-algebra associated to the $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ endowed with the $\mathscr{I}$-preadic filtration. *Then the homomorphism of graded $\mathscr{O}_Y$-algebras*

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'') \longrightarrow \operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})) \tag{16.4.8.4}$$

(where the first member is the graded tensor product of symmetric $\mathscr{O}_Y$-algebras endowed with the canonical gradation (**II**, 1.7.4) *and* (**II**, 2.1.2)*), induced by the canonical injection*

$$\mathscr{F}' \longrightarrow \mathscr{I} = \operatorname{gr}^1_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})),$$

is bijective.

Proof. The injection $\mathscr{F}' \to \mathscr{I}$ indeed canonically gives a homomorphism of graded $\mathscr{O}_Y$-algebras $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \to \operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}))$, and since the second member is by definition a graded $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$-algebra, we induce the canonical homomorphism (16.4.8.4) by tensoring the above with $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F})$ is canonically identified with the graded tensor product $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$ (**II**, 1.7.4), and it is immediate that $\mathscr{I}$ is therefore the ideal $\mathscr{I}' \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where $\mathscr{I}'$ is the augmentation ideal of $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}')$, that is to say the (direct) sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant 1$. We conclude that $\mathscr{I}^n = \mathscr{I}'^n \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where this time $\mathscr{I}'^n$ is the direct sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant n$; we have therefore $\mathscr{I}^n/\mathscr{I}^{n+1} = \mathbf{S}^n_{\mathscr{O}_Y}(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, which proves that (16.4.8.4) is bijective. □

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is the image by f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); we can easily see that it follows from the definition of u (16.4.8.2) and of δ (16.4.7.1), given the definition of the $\mathscr{O}_X$-algebra structures of $\mathscr{P}^n_{X/S}$ and of the $d^n_{X/S}$ (16.3.5) and (16.4.3.6). **IV-4 | 21**

In particular:

Corollary (16.4.9). — *Under the conditions of* (16.4.7)*, we have a canonical isomorphism*

$$\operatorname{gr}_1(\delta) : f^*(\mathscr{E}) \simeq \Omega^1_{X/S}. \tag{16.4.9.1}$$

Corollary (16.4.10). — *If $S = \operatorname{Spec}(A)$, $\mathscr{E} = \mathscr{O}^m_S$, so that*

$$X = \operatorname{Spec}(A[T_1, \ldots, T_m]),$$

then $\mathscr{P}^n_{X/S}$ is canonically identified with the $\mathscr{O}_X$-algebra corresponding to the quotient $A[T_1, \ldots, T_m]$-algebra $A[T_1, \ldots, T_m, U_1, \ldots, U_m]/\mathfrak{K}^{n+1}$, where the U_i ($1 \leqslant i \leqslant m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \ldots, U_m$.

We thus recover in particular the structure of $\Omega^1_{X/S}$ in this case (**0**, 20.5.13).

In addition, note that the $d^n_{X/S}$ then corresponds to a polynomial $F(T_1, \ldots, T_m)$, the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \ldots, T_m + U_m)$, which follows from the definition (16.4.7.1).

Proposition (16.4.11). — *Let $f : X \to S$ be a morphism, $g : S \to X$ a S-section of X, $S^{(n)}$ the n-th infinitesimal neighbourhood of S by the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathscr{O}_S$-algebras*

(16.4.11.1) $$\varpi_n : g^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{O}_{S^{(n)}_g}$$

(via the $\mathscr{O}_S$-algebra structure on $\mathscr{O}_{S^{(n)}_g}$ defined by f (16.1.7)), making the diagram

(16.4.11.2)
$$\begin{tikzcd} \mathscr{O}_S = g^*(\mathscr{O}_X) \arrow[rr, "\lambda_n"] \arrow[dr, "g^*(d^n_{X/S})"'] & & \mathscr{O}_{S^{(n)}_g} \\ & g^*(\mathscr{P}^n_{X/S}) \arrow[ur, "\varpi_n"'] & \end{tikzcd}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (I, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g, the diagrams

(16.4.11.3)
$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(g\circ f, 1_X)_S"'] & X \times_S X \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(1_X, g\circ f)_S"'] & X \times_S X \end{tikzcd}$$

identifies S with the product of the $(X \times_S X)$-preschemes X and X by the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams **IV-4 | 22**

(16.4.11.4)
$$\begin{tikzcd} X \arrow[r, "(g\circ f, 1_X)_S"] \arrow[d, "f"'] & X \arrow[d, "p_1"] \\ S \arrow[r, "g"'] & X \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "(1_X, g\circ f)_S"] \arrow[d, "f"'] & X \arrow[d, "p_2"] \\ S \arrow[r, "g"'] & X \end{tikzcd}$$

identify X to the product of X-preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (particular case of the associativity formula (I, 3.3.9.1)). We can say that Δ_f, considered as an X-section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S-sections of X: each of these sections g in fact are deduced by *base change* $(g \circ f, 1_X)_S : X \to X \times_S X$. The definition of the homomorphism $\bar{\omega}_n$ and the fact that it is bijective follows from the remarks of (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of the first diagram (16.4.11.4) follows also from (16.2.3, (ii)) this time applied to the second diagram (16.11.4). To explain ϖ_n, we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X, and it is clear that $g|W$ is a W-section of the morphism $U \cap f^{-1}(W)$. We can then suppose that S is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{K}$. Then the preceding definitions show that if W is an open of S, t is a section of $\mathscr{O}_X$ over $f^{-1}(W)$, $\varpi_n(d^n t|W)$ is equal to the canonical image of t in $\Gamma(W, (\mathscr{O}_X/\mathscr{K}^{n+1})|W)$. The uniqueness of ϖ_n then follows since the image of $\mathscr{O}_X$ under $d^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$ (16.3.8). ☐

Corollary (16.4.12). — *Let k be a field, X a k-prescheme, x a point of X* rational over k. *Then $(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$-algebra) to $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k-section g of X such that $g(\mathrm{Spec}(k)) = \{x\}$. ☐

Corollary (16.4.13). — *Let $f : X \to S$ be a morphism, s a point of S, $X_s = X \times_S \mathrm{Spec}(k(s))$ the fibre of f in s. If $x \in X_s$ is* rational over $k(s)$, *$(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_s} k(s)$ is canonically isomorphic to $\mathscr{O}_{X_s,x}/\mathfrak{m}'^{\,n+1}_x$, is the maximal ideal of $\mathscr{O}_{X_s,x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$* (where t is a section of $\mathscr{O}_X$ over an open neighbourhood of x in X) *to the class of $t_x \otimes 1$ modulo $\mathfrak{m}'^{\,n+1}_x$.*

PROOF. This follows from (16.4.5) and (16.4.12). □

The preceding corollaries justify the terminology "sheaf of principal parts of order n".

Proposition (16.4.14). — *Let $\rho : A \to B$ be a morphism of rings, S a multiplicative subset of B. Then the canonical homomorphisms*

(16.4.14.1) $$S^{-1}P^n_{B/A} \longrightarrow P^n_{S^{-1}B/A}$$

deduced from the canonical homomorphisms $P^n_{B/A} \to P^n_{S^{-1}B/A}$ (16.4.4), form a projective system and are bijective. **IV-4** | 23

PROOF. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{J}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{J})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (**I**, 1.3.4). □

Corollary (16.4.15). — *The notation being that of (16.4.14), let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

(16.4.15.1) $$S^{-1}P^n_{B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

forming a projective system.

PROOF. It evidently suffices to define canonical isomorphisms

(16.4.15.2) $$P^n_{S^{-1}B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

that is to say that we reduce to the case there $\rho(R)$ is consists of *invertible* elements of B. But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \to B \otimes_{R^{-1}A} B$ by passing to quotients (**I**, 1.5.3). □

Corollary (16.4.16). — *Let $f : X \to S$ be a morphism of preschemes, x a point of X, $s = f(x)$. Then we have canonical isomorphisms*

(14.4.16.1) $$(\mathscr{P}^n_{X/S})_x \simeq P^n_{\mathscr{O}_x/\mathscr{O}_s}$$

forming a projective system.

We deduce from these isomorphisms of the associated graded rings, and in particular a canonical isomorphism

(16.4.16.2) $$(\Omega^1_{X/S})_x \simeq \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}.$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \ldots, T_r)$. Then, for every integer n, the homomorphism of $K[U_1, \ldots, U_r]$ (U_i indeterminates) into $P^n_{K/k}$ which sends U_i to $d^n T_i - T_i.1$ is surjective and defines an isomorphism from the quotient $K[U_1, \ldots, U_n]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P^n_{K/k}$.*

PROOF. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \ldots, T_r]$ and $S = B - \{0\}$. □

We thus recover the fact that the dT_i form a basis of the K-vector space $\Omega^1_{K/k}$ (**0**, 20.5.10).

(16.4.18). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, and consider the canonical homomorphism of augmented $\mathscr{O}_X$-algebras (16.4.3.3)

(16.4.18.1) $$g_{X/Y/Z} : \mathscr{P}^n_{X/Z} \longrightarrow \mathscr{P}^n_{X/Y}$$

(16.4.18.2) $$f_{X/Y/Z} : f^*(\mathscr{P}^n_{Y/Z}) \longrightarrow \mathscr{P}^n_{X/Z}.$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathscr{P}^n_{X/Z})$.

PROOF. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and **IV-4** | 24
$S = Z$, $u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively.

We have a commutative diagram (**I**, 3.5)

(16.4.18.3)
$$\begin{tikzcd} X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "j"] \arrow[d, "p"] & X \times_Z X \arrow[d, "f \times_Z f"] \\ & Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g \circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathscr{O}_X$ and $\mathscr{O}_{X \times_Y X}$ are identified respectively with $\mathscr{O}_{X \times_Z X}/\mathscr{I}$ and $\mathscr{O}_{X \times_Z X}/\mathscr{L}$, where $\mathscr{L} \supset \mathscr{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g \circ f}$ and j. The $\mathscr{O}_X$-algebra $\mathscr{P}^n_{X/Z}$ is identified with $\mathscr{O}_{X \times_Z X}/\mathscr{I}^{n+1}$, and $\mathscr{P}^n_{X/Y}$ is identified with $\mathscr{O}_{X \times_Y X}/(\mathscr{I}/\mathscr{L})^{n+1}$, which is to say with $\mathscr{O}_{X \times_Z X}/(\mathscr{I}^{n+1} + \mathscr{L})$, and therefore with the quotient of $\mathscr{P}^n_{X/Z}$ by $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$. But we know (*loc. cit*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$, so if $\mathscr{O}_Y$ is identified to $\mathscr{O}_{Y \times_Z Y}/\mathscr{K}$, where $\mathscr{K}$ is the ideal corresponding to Δ_g, $\mathscr{L}$ is equal to $(f \times_Z f)^*(\mathscr{K}).\mathscr{O}_{X \times_Z X}$ (**I**, 4.4.5). Since $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$ is the ideal of $\mathscr{P}^n_{X/Z}$ generated by the image of $\mathscr{L}$, we deduce the proposition. □

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.19.1)
$$f^*(\Omega^1_{Y/Z}) \xrightarrow{f_{X/Y/Z}} \Omega^1_{X/Z} \xrightarrow{g_{X/Y/Z}} \Omega^1_{X/Y} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ a closed immersion, $\mathscr{K}$ the quasi-coherent sheaf of ideals of $\mathscr{O}_Y$ corresponding to j. It follows that $\mathscr{P}^n_{X/Y} = \mathscr{O}_X = \mathscr{O}_Y/\mathscr{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathscr{P}^n_{Y/Z}) \to \mathscr{P}^n_{X/Z}$ is surjective, and its kernel is the ideal of $j^*(\mathscr{P}^n_{Y/Z})$ generated by $j^*(\mathscr{O}_Y \cdot d^n_{Y/Z}(\mathscr{K}))$ (it should be noted that $d^n_{Y/Z}(\mathscr{K})$ is a subsheaf of abelian groups of $\mathscr{P}^n_{X/Z}$, but not an $\mathscr{O}_Y$-module in general).*

PROOF. We know (**I**, 5.3.8) that the diagonal $\Delta_j : X \to X \times_Y X$ is an isomorphism, from which the first assertion follows. If ϖ_1 and ϖ_2 are the two canonical homomorphisms of algebras $\mathscr{O}_Y \to \mathscr{P}^n_{Y/Z}$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \to Y$, recall that by definition ((16.3.5) and (16.3.6)) ϖ_1 is the structure homomorphism of the $\mathscr{O}_Y$-algebra $\mathscr{P}^n_{Y/Z}$ and $\varpi_2 = d^n_{Y/Z}$. The $\mathscr{O}_X$-algebra $j^*(\mathscr{P}^n_{Y/Z})$ is therefore identified with $\mathscr{P}^n_{Y/Z}/\varpi_1(\mathscr{K})\mathscr{P}^n_{Y/Z}$ and its quotient by the ideal generated by $j^*(d^n_{Y/Z}(\mathscr{K}))$ to $\mathscr{P}^n_{Y/Z}/(\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K}))\mathscr{P}^n_{Y/Z}$. Now note that we have a commutative diagram **IV-4** | 25

$$\begin{tikzcd} Y \arrow[d, "\Delta_f"'] & X \arrow[l, "j"'] \arrow[d, "\Delta_{f \circ j}"] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

identifying X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta^n_{Y/Z}$ and $\Delta^n_{X/Z}$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f \circ j}$ respectively, then we have a diagram

$$\begin{tikzcd} \Delta^n_{Y/Z} \arrow[d] & \Delta^n_{X/Z} \arrow[l] \arrow[d] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

making $\Delta^n_{X/Z}$ the product of the $(Y \times_Z Y)$-preschemes $\Delta^n_{Y/Z}$ and $X \times_Z X$. We can also say that $\mathscr{P}^n_{X/Z}$ is identified with the sheaf of rings $\mathscr{P}^n_{Y/Z} \otimes_{\mathscr{O}_{Y\times_Z Y}} \mathscr{O}_{X\times_Z X}$. But we see immediately that (for example, by restricting to the affine case) that $\mathscr{O}_{X\times_Z X} = \mathscr{O}_{Y\times_Z Y}/(p_1^*(\mathscr{K}) + p_2^*(\mathscr{K}))\mathscr{O}_{Y\times_Z Y}$. Therefore $\mathscr{P}^n_{X/Z}$ is identified with the quotient of $\mathscr{P}^n_{Y/Z}$ by the ideal generated by the image in $\mathscr{P}^n_{Y/Z}$ of $p_1^*(\mathscr{K}) + p_2^*(\mathscr{K})$. But by definition this ideal is generated by $\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K})$. ☐

Corollary (16.4.21). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ an immersion. We have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.21.1) $$\mathscr{N}_{X/Y} \longrightarrow j^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{P}^n_{X/S}$ and $\Omega^1_{X/S}$ are quasi-coherent $\mathscr{O}_X$-modules of finite presentation.*

PROOF. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is affine, $X = \operatorname{Spec}(B)$, where $B = A[T_1, \ldots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \ldots, T_r]$. Applying (16.4.20) where $Z = S$, $Y = \operatorname{Spec}(C)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$. Then $j^*(\mathscr{P}^n_{Y/Z})$ is a free $\mathscr{O}_X$-module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathscr{O}_Y.d^n_{Y/Z}(\mathscr{K}))$ generates a quasi-coherent $\mathscr{O}_X$-module of finite type; hence the conclusion. ☐

Proposition (16.4.23). — *Let X, Y be two S-preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ the canonical projections. Then the canonical homomorphism*

(16.4.23.1) $$p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega^1_{X/S}) \oplus q^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{(X\times_S Y)/S}$$

is bijective.

PROOF. The commutative diagram **IV-4 | 26**

$$\begin{tikzcd} Y \arrow[d, "g"'] & X \times_S Y \arrow[l, "q"'] \arrow[d, "h"] & X \times_S Y \arrow[l, "\mathrm{id}"'] \arrow[d, "p"] \\ S & S \arrow[l, "\mathrm{id}"] & X \arrow[l, "f"] \end{tikzcd}$$

gives us a factorization of the canonical *isomorphism* $P^n(p)$ (16.4.5)

$$p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}$$

and similarly, switching X with Y, we have a factorization of the *isomorphism* $P^n(q)$

$$q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \quad (\text{resp. } q_{Z/X/S} : q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S})$$

is *injective*, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y} \quad (\text{resp. } \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X})$$

is direct summand of the image $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, this kernel is, by virtue of (16.4.18), generated by the image by $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathscr{P}^n_{Y/S})$ (resp. $p^*(\mathscr{P}^n_{X/S})$). We conclude the proposition by considering the case $n = 1$. ☐

We immediately generalize (16.4.23) to the case of a product of any finite number of S-preschemes.

Remarks (16.4.24). —

(i) We will see (17.2.3) that when the morphism $f : X \to Y$ in (16.4.18) is *smooth*, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally *left invertible* and in particular injective. Similarly, when the morphism $f \circ j : X \to Z$ of (16.4.20) is *smooth*, the homomorphism on the left in (16.4.21.1) is locally *left invertible* and *a fortiori* injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the "imperfection modules" studied in (**0**, 20.6), and the exact sequences where they occur.

(ii) Let X be a topological space, $\mathscr{A}$ a sheaf of rings over X and $\mathscr{B}$ a $\mathscr{A}$-algebra over X. Then it is clear that

$$U \longmapsto P^n_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{A})} \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U,\mathscr{B})$-algebras, and therefore the associated sheaf $\mathscr{P}^n_{\mathscr{B}/\mathscr{A}}$ is an augmented $\mathscr{B}$-algebra. In the particular case where X is a prescheme, $f = (\psi, \theta) : X \to S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathscr{P}^n_{X/S}$ is canonically isomorphic to $\mathscr{P}^n_{\mathscr{O}_X/\psi^*(\mathscr{O}_S)}$. It follows that the formalism developed in the present paragraph could be considered as a particular case of a differential formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of "varieties", the "global" constructions of the $\mathscr{P}^n$ analogous to those we have used here are also better suited for applications. **IV-4** | 27

16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi, \theta) : X \to S$ be a morphism of ringed spaces. For every $\mathscr{O}_X$-module $\mathscr{F}$, we say *S-derivation* (or *(X/S)-derivation*, or *f-derivation*) *of $\mathscr{O}_X$ to $\mathscr{F}$* for every homomorphism of *sheaves of additive groups $D : \mathscr{O}_X \to \mathscr{F}$* satisfying the following conditions:

(a) for every open V of X, and all pair of sections (t_1, t_2) of $\mathscr{O}_X$ over V, we have

$$D(t_1 t_2) = t_1 D(t_2) + D(t_1) t_2; \tag{16.5.1.1}$$

(b) for every open V of X, every section t of $\mathscr{O}_X$ over V, and every section s of $\mathscr{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

$$D((s|V)t) = (s|V) D(t). \tag{16.5.1.2}$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathscr{O}_x \to \mathscr{F}_x$ is an *$\mathscr{O}_{f(x)}$-derivation*.

Another interpretation consists of considering the $\mathscr{O}_X$-algebra $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$ as equal to $\mathscr{O}_X \oplus \mathscr{F}$, the algebra structure being defined by the condition that for every open V of X, the product of two sections of $\mathscr{O}_X$ (resp. of a section of $\mathscr{O}_X$ and a section of $\mathscr{F}$) over V is defined by the ring structure of $\Gamma(V, \mathscr{O}_X)$ (resp. the $\Gamma(V, \mathscr{O}_X)$-module structure on $\Gamma(V, \mathscr{F})$), and the product of two sections of $\mathscr{F}$ over V is chosen to be zero; then $\mathscr{F}$ is an ideal of $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, the kernel of the canonical augmentation $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F}) \to \mathscr{O}_X$, and to say that D is an S-derivation of $\mathscr{O}_X$ to $\mathscr{F}$ means that $1_{\mathscr{O}_X} + D$ is an *$\mathscr{O}_S$-homomorphism of algebras* from $\mathscr{O}_X$ to $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, which, composed with the augmentation, gives $1_{\mathscr{O}_X}$.

The S-derivations of $\mathscr{O}_X$ to $\mathscr{F}$ clearly form a $\Gamma(X, \mathscr{O}_X)$*-module* $\mathrm{Der}_{\mathscr{O}_S}(\mathscr{O}_X, \mathscr{F})$.

When $\mathscr{F} = \mathscr{O}_X$, an S-derivation of $\mathscr{O}_X$ to itself is simply called an *S-derivation of $\mathscr{O}_X$*.

Proposition (16.5.2). — *Let A be a ring, B an A-algebra, L a B-module; let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\mathscr{F} = \widetilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S-derivation D of $\mathscr{O}_X$ to $\mathscr{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ of B to L, is an isomorphism of B-modules from $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{F})$ to $\mathrm{Der}_A(B, L)$ (cf. (**0**, 20.1.2)).*

PROOF. This follows immediately from the given interpretation of S-derivations in terms of homomorphisms of algebras, analogous to the interpretation given in (**0**, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathscr{O}_X$-algebras and homomorphisms of B-algebras ((**I**, 1.3.13) and (**I**, 1.3.8)). ☐ **IV-4** | 28

Proposition (16.5.3). — *Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes.*

(i) *The differential $d_{X/S} : \mathscr{O}_X \to \Omega^1_{X/S}$ (16.3.6) is an S-derivation.*

(ii) *For every $\mathscr{O}_X$-module $\mathscr{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathscr{O}_X)$-modules*

$$\mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \simeq \mathrm{Der}_S(\mathscr{O}_X, \mathscr{F}). \tag{16.5.3.1}$$

PROOF. The assertion (i) has already been written (16.3.6). On the other hand, it is immediate (in light of (**0**, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, considering the restrictions to a fibre $\mathscr{O}_x$ of the two sides and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S-derivation $D : \mathscr{O}_X \to \mathscr{F}$; for every affine open $V = \mathrm{Spec}(B)$ of X, such that $f(V)$ is contained in an affine open $U = \mathrm{Spec}(A)$ of S, $D_V : B \to \Gamma(V, \mathscr{F})$ is an A-derivation, and therefore there exists a *unique* B-homomorphism $u_V : \Omega^1_{B/A} \to \Gamma(V, \mathscr{F})$ such that $D_V = u_V \circ d_{B/A}$ (**0**, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V|W$, and therefore the u_V define a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{O}_X \to \mathscr{F}$ answering the question. □

(16.5.4). With the notation of (16.5.1), for every open U of X, $\mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a $\Gamma(U, \mathscr{O}_X)$-module and it is clear that the map $U \mapsto \mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathscr{O}_X$-module), in light of the pointwise characterization of S-derivations, seen in (16.5.1). This $\mathscr{O}_X$-module is denoted by $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ and is called the *sheaf of S-derivations of $\mathscr{O}_X$ in $\mathscr{F}$*, and what we have seen is further expressed in the following corollary:

Corollary (16.5.5). — *For every $\mathscr{O}_X$-module $\mathscr{F}$, the homomorphism of $\mathscr{O}_X$-modules induced by $u \mapsto u \circ d_{X/S}$*

(16.5.5.1) $$\mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \longrightarrow \mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$$

is bijective.

Corollary (16.5.6). —
(i) *If the morphism $f : X \to S$ is locally of finite presentation and if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module.*
(ii) *If in addition S is locally Noetherian and if $\mathscr{F}$ is coherent, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*

PROOF. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (**I**, 1.3.12); the assertion (ii) follows from (**0**, 5.3.5). □

(16.5.7). We set

(16.5.7.1) $$\mathfrak{G}_{X/S} = \mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \mathscr{D}er_S(\mathscr{O}_X, \mathscr{O}_X),$$

and say that it is the *sheaf of S-derivations of $\mathscr{O}_X$*, or even the *tangent sheaf of X relative to S*: it is therefore the *dual* of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6). **IV-4** | 29

(16.5.8). Suppose in particular that $\Omega^1_{X/S}$ is a *locally free* $\mathscr{O}_X$-module (of finite rank) (which will be the case then f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is locally free $\mathscr{O}_X$-module of the same rank as $\Omega^1_{X/S}$ at each point. More specifically, suppose that $\Omega^1_{X/S}$ is of rank n at a point x; then there are n sections s_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$ form a basis if this $k(x)$-vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathscr{O}_x$-module $(\Omega^1_{X/S})_x$, and therefore, by restricting U, we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$. So the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leqslant i \leqslant n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leqslant i \leqslant n}$ the dual basis of $(ds_i)_{1 \leqslant i \leqslant n}$, so that, by (16.5.3), we have

(16.5.8.1) $$D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad \text{(Kronecker's symbol).}$$

Every $\Gamma(S, \mathscr{O}_S)$-derivation of the $\Gamma(S, \mathscr{O}_S)$-algebra $\Gamma(U, \mathscr{O}_X)$ is therefore written in an unique way as

$$D = \sum_{i=1}^n a_i D_i = \sum_{i=1}^n a_i \frac{\partial}{\partial s_i},$$

where the a_i $(1 \leqslant i \leqslant n)$ are sections of $\mathscr{O}_X$ over U. For every section $g \in \Gamma(U, \mathscr{O}_X)$, if we put $dg = \sum_{i=1}^n c_i ds_i$, then we have $c_i = \langle D_i, dg \rangle = D_i g$ by virtue of (16.5.8.1), in other words,

(16.5.8.2) $$dg = \sum_{i=1}^n (D_i g) ds_i = \sum_{i=1}^n \frac{\partial g}{\partial s_i} ds_i.$$

(16.5.9). Let D_1, D_2 be two S-derivations of $\mathscr{O}_X$. For every open U of X, if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathscr{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation in this ring, and therefore the $\psi^*(\mathscr{O}_S)$-endomorphism of $\mathscr{O}_X$

(16.5.9.1) $$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$$

is also an S-derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{O}_X)$ a $\Gamma(S, \mathscr{O}_S)$-*Lie algebra* structure. Since the definition of this structure commutes with the restriction to an open of X, we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathscr{O}_S)$-*Lie algebra* structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is *not* $\Gamma(X, \mathscr{O}_X)$-*bilinear*.

(16.5.10). For every base change $g : S' \to S$, if we set $X' = X \times_S S'$, then we see (16.4.5) that we have a canonical isomorphism

(16.5.10.1) $$\Omega^1_{X/S} \otimes_S S' \simeq \Omega^1_{X'/S'},$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3) **IV-4** | 30

(16.5.10.2) $$\mathfrak{G}_{X/S} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'},$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) *If $g : S' \to S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism (16.5.10.2) is injective (resp. bijective).*

(ii) *If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of finite type, then the homomorphism (16.5.10.2) is bijective.*

PROOF. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of finite type (resp. of finite presentation) ((16.3.9) (16.4.22)). ☐

(16.5.12). Since $\Omega^1_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module, we can consider the vector bundle over X defined by $\Omega^1_{X/S}$ (**II**, 1.7.8)

(16.5.12.1) $$T_{X/S} = \mathbf{V}(\Omega^1_{X/S})$$

which is called the *tangent bundle of X relative to S*. We have therefore a canonical bijection (**II**, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S-sections* of the tangent bundle of X relative to S. If $f : X \to Y$ is an S-morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega^1_{Y/S}) \to \Omega^1_{X/S}$, which, having in mind that

$$\mathbf{V}(f^*(\Omega^1_{Y/S})) = \mathbf{V}(\Omega^1_{Y/S}) \times_Y X \quad (\textbf{II}, 1.7.11),$$

gives us an X-morphism $T_{X/S}(f) : T_{X/S} \to T_{Y/S} \times_Y X$. If $g : Y \to Z$ is a second S-morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (**0**, 20.5.4.1).

It follows from (16.5.10.1) and from (**II**, 1.7.11) that for every base change $g : S' \to S$ we have a canonical isomorphism

(16.5.12.2) $$T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'.$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \mathrm{Spec}(k(x))$ that are *rational over* $k(x)$, that is, the set

(16.5.13.1) $$T_{X/S}(x) = \mathrm{Hom}_{k(x)}(\Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x), k(x)),$$

which is the *dual* of the $k(x)$-vector space $\Omega^1_{\mathscr{O}_x/\mathscr{O}_s} / \mathfrak{m}_x \cdot \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}$. When $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of *finite type*, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g : S \to S'$, **IV-4** | 31
and every point $x' \in X' = X \times_S S'$ over x, we have a canonical isomorphism

(16.5.13.2) $$T_{X'/S'}(x') \simeq T_{X/S} \otimes_{k(x)} k(x').$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \to k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

(16.5.13.3)
$$T_{X/S}(x) = T_{X_s/k(s)}(x) = \mathrm{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}'^2_x, k(x)),$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$. In the case where S is the spectrum of a field k, we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k, as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S-prescheme and let $g : Y \to X$ be an S-morphism; then we have a canonical homomorphism of $\mathscr{O}_Y$-modules (16.4.19)

(16.5.13.4)
$$g_{Y/X/S} : g^*(\Omega^1_{X/S}) \longrightarrow \Omega^1_{Y/S}.$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y) = (\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module *of finite type*, then we can identify

$$\mathrm{Hom}_{k(y)}(g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of $k(y)$-*vector spaces*

(16.5.13.5)
$$T_y(g) : T_{Y/S}(y) \longrightarrow T_{X/S}(x) \otimes_{k(x)} k(y)$$

called the *linear map tangent to g at the point y*. When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$-vector spaces $T_{Y/S}(y) \to T_{X/S}(x)$; also note that in this case, $g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y)$ is identified with $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega^1_{X/S}$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y.

(16.5.14). The interpretation of derivations of an A-algebra B to a B-module L, given in (**0**, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \to S$, $g : Y \to S$, and a closed subprescheme Y_0 of Y defined by a *zero-square* ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (so that Y and Y_0 have the *same underlying subspace*). Suppose we are given an S-morphism $u_0 : Y_0 \to X$, so that we have a commutative diagram

(16.5.14.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & Y_0 \arrow[l, "u_0"'] \arrow[d, "j"] \\ S & Y \arrow[l, "g"] \arrow[ul, dashed, "u"'] \end{tikzcd}$$

and we suggest looking for an *S-morphism* $u : Y \to X$ such that $u_0 = u \circ j$ (in other words, if it is **IV-4** | 32
possible to complete the diagram above by the dotted arrow u, keeping it *commutative*).

For that, consider an affine open $U = \mathrm{Spec}(C)$ of Y; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \mathrm{Spec}(C/\mathfrak{L})$, where $\mathfrak{L} = \Gamma(U, \mathscr{J})$, a *zero-square* ideal in C; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \mathrm{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \mathrm{Spec}(A)$ of S, so that B and C are A-algebras and $u_0|U_0$ corresponds to an A-homomorphism ψ from B to $C/\mathfrak{L}$; Let $P(U_0)$ be the set of restrictions $u|U$ of the sought homomorphisms, which corresponds canonically to A-homomorphisms of algebras $\phi : B \to C$ such that the *composite* $B \xrightarrow{\phi} C \to C/\mathfrak{L}$ *is equal to* ψ. So we know (**0**, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \mathrm{Der}_A(B, \mathfrak{L})$; when $P(U_0)$ is not empty, the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ (or even a *principal homogeneous space* (or *torsor*) *under* $\mathrm{Der}_A(B, \mathfrak{L})$).

Now notice that, since $\mathfrak{L}$ is equipped with a B-module structure via ψ, we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{L})$ onto $\mathrm{Der}_A(B, \mathfrak{L})$ (**0**, 20.4.8). Besides, as $\mathfrak{L}$ is square-zero, therefore a $(C/\mathfrak{L})$-module, every B-homomorphism $v : \Omega^1_{B/A} \to \mathfrak{L}$ can be considered as a $(C/\mathfrak{L})$-homomorphism $\Omega^1_{B/A} \otimes_B (C/\mathfrak{L}) \to \mathfrak{L}$. As $\mathscr{J}$ is square-zero, it can be considered as a quasi-coherent

$\mathscr{O}_{Y_0}$-module; let's introduce the $\mathscr{O}_{Y_0}$-module

(16.5.14.2) $$\mathscr{G} = \mathscr{H}om(u_0^*(\Omega^1_{X/S}), \mathscr{I});$$

it follows from the fact that $\Omega^1_{B/A} = \Gamma(V, \Omega^1_{X/S})$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathscr{G})$.

As $P(U_0)$ is defined as a set of S-morphisms $U \to X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathscr{P}$ on Y_0. We can use this fact to prove that the map $h : \Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y, U'_0 its inverse image in Y_0, then the diagram

(16.5.14.3) $$\begin{tikzcd} \Gamma(U_0, \mathscr{G}) \times P(U_0) \arrow[r, "h"] \arrow[d] & P(U_0) \arrow[d] \\ \Gamma(U'_0, \mathscr{G}) \times P(U'_0) \arrow[r, "h'"'] & P(U'_0) \end{tikzcd}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V, W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h, this **IV-4** | 33
follows from the commutativity of the diagram (**0**, 20.5.4.2).

The mapping $\Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ therefore define a homomorphism of *sheaf of sets* $m : \mathscr{G} \times \mathscr{P} \times \mathscr{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathscr{P}) \neq \varnothing$, $m_{U_0} : \Gamma(U_0, \mathscr{G}) \times \Gamma(U_0, \mathscr{P}) \to \Gamma(U_0, \mathscr{P})$ is an external law defining in $\Gamma(U_0, \mathscr{P})$ a torsor structure for the group $\Gamma(U_0, \mathscr{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathscr{P}$ over a topological space Z, a sheaf of groups $\mathscr{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathscr{P}) \neq \varnothing$, $m_U : \Gamma(U, \mathscr{G}) \times \Gamma(U, \mathscr{P}) \to \Gamma(U, \mathscr{P})$ makes $\Gamma(U, \mathscr{P})$ a *torsor* under the group $\Gamma(U, \mathscr{G})$, then we say that $\mathscr{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathscr{G}$. We say that $\mathscr{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathscr{G}$ [6] if in addition $\Gamma(U, \mathscr{P}) \neq \varnothing$ for every open $U \neq \varnothing$ in a suitable basis for the topology of Z.

For the general theory of torsors, we refer to [**eAG64**]; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathscr{G}$) and elements from the cohomology set $\mathrm{H}^1(Z, \mathscr{G})$. Consider a torsor $\mathscr{P}$ under $\mathscr{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathscr{P}) \neq \varnothing$ for every λ; denote by p_λ an element of $\Gamma(U_\lambda, \mathscr{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \varnothing$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathscr{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu | U_\lambda \cap U_\mu) = p_\lambda | U_\lambda \cap U_\mu$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \varnothing$, then the restrictions $\gamma'_{\lambda\mu}$, $\gamma'_{\mu\nu}$, $\gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}$, $\gamma_{\mu\nu}$, $\gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu}\gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-*cocycle* of the cover (U_λ) with values in $\mathscr{G}$. If, for every λ, p'_λ is a second element of $\Gamma(U_\lambda, \mathscr{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathscr{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ), an automorphism $\theta_{\lambda\mu}$ of the sheaf *of sets* $\mathscr{G} | U_\lambda \cap U_\mu$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathscr{G} | U_\lambda$ via the automorphisms $\theta_{\lambda\mu}$ (**0**, 3.3.1); we thus obtain a torsor under $\mathscr{G}$, denoted $\mathscr{P}$, and if we take for p_λ the unit section over U_λ, then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathscr{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-*coboundary*, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda \beta_\mu^{-1}$, then the torsor $\mathscr{P}$ obtained is *isomorphic to* $\mathscr{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathscr{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (**III**, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Z$-module; then every torsor over $\mathscr{G}$ is trivial.*

[6] [Trans.] This is nowadays more commonly called a $\mathscr{G}$-torsor rather then a torsor *under* $\mathscr{G}$.

Returning to the problem considered in (16.5.13), we thus obtain: **IV-4 | 34**

Proposition (16.5.17). — *Let X, Y be two S-preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_Y$ such that $\mathscr{I}^2 = 0$, $j : Y_0 \to Y$ the canonical injection. Let $u_0 : Y_0 \to X$ be an S-morphism, and $\mathscr{P}$ the sheaf of sets on Y such that, for every open U of Y, $\Gamma(U, \mathscr{P})$ is the set of S-morphisms $u : U \to X$ such that $u_0|U_0 = u \circ (j|U_0)$, where $U_0 = j^{-1}(U)$. Then there exists on $\mathscr{P}$ the structure of a pseudo-torsor over the $\mathscr{O}_{Y_0}$-module $\mathscr{G} = \mathscr{H}om_{\mathscr{O}_{Y_0}}(u_0^*(\Omega^1_{X/S}), \mathscr{I})$.*

In particular:

Corollary (16.5.18). — *With the notation of (16.5.16), suppose that Y is affine and $\Omega^1_{X/S}$ is of finite presentation; if there is a open cover (U_α) of Y, and, for every index α, an S-morphism $v_\alpha : U_\alpha \to X$ such that, if $U_\alpha^0 = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|U_\alpha^0) = u_0|U_\alpha^0$, then there is an S-morphism $u : Y \to X$ such that $u \circ j = u_0$.*

Proof. Indeed, $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_{Y_0}$-module (**I**, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathscr{P}$, which is by hypothesis a torsor over $\mathscr{G}$, and not only a pseudo-torsor, is *trivial*; but if w is an isomorphism from $\mathscr{G}$ to $\mathscr{P}$ (as it is a torsor over $\mathscr{G}$), the image under w of the zero section of $\mathscr{G}$ is the S-morphism we want. □

16.6. Sheaf of p-differentials and exterior differentials.

(16.6.1). Let $f : X \to S$ be a morphism of preschemes. We define the *sheaf of p-differentials of X relative to S* (p integer) to be the *p'th exterior power* (**0**, 4.1.5) of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$, denoted by

$$\Omega^p_{X/S} = \bigwedge^p(\Omega^1_{X/S}). \tag{16.6.1.1}$$

So we have $\Omega^0_{X/S} = \mathscr{O}_X$, and $\Omega^p_{X/S} = 0$ for $p < 0$; the $\Omega^p_{X/S}$ are the homogeneous components of the exterior algebra of $\Omega^1_{X/S}$

$$\Omega^\bullet_{X/S} = \bigwedge(\Omega^1_{X/S}) = \bigoplus_{p\in\mathbf{Z}} \bigwedge^p(\Omega^1_{X/S}), \tag{16.6.1.2}$$

which is therefore a quasi-coherent graded anti-commutative $\mathscr{O}_X$-algebra whose elements of degree 1 are square-zero. For every affine U of X, we have $\Gamma(U, \Omega^\bullet_{X/S}) = \bigwedge(\Gamma(U, \Omega^1_{X/S}))$, where $\Gamma(U, \Omega^1_{X/S})$ is considered as a $\Gamma(U, \mathscr{O}_X)$-module.

When $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines, B being then an A-algebra, we have (**0**, 4.1.5) $\Omega^p_{X/S} = (\Omega^p_{B/A})^\sim$, by putting $\Omega^p_{B/A} = \bigwedge^p \Omega^1_{B/A}$.

Theorem (16.6.2). — *There is one and only one endomorphism d of the sheaf of additive groups $\Omega^\bullet_{X/S}$ with the following properties:*

(i) $d \circ d = 0$.
(ii) *For every open set U of X and every section $f \in \Gamma(U, \mathscr{O}_X)$ we have $df = d_{X/S}f$.* **IV-4 | 35**
(iii) *For every open set U of X, every pair of integers p, q and every pair of sections $\omega'_p \in \Gamma(U, \Omega^p_{X/S})$, $\omega''_q \in \Gamma(U, \Omega^q_{X/S})$, we have*

$$d(\omega'_p \wedge \omega''_q) = (d\omega'_p) \wedge \omega''_q + (-1)^p \omega'_p \wedge d\omega''_q. \tag{16.6.2.1}$$

Also, d is an endomorphism of graded $\psi^*(\mathscr{O}_X)$-modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d. For every affine open U of X, every section of $\Omega^p_{X/S}$ over U is (because of (ii)) a linear combination if a finite number of elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathscr{O}_X$ over U (**0**, 20.4.7). The conditions (i) and (iii) then show, by induction on p, that we necessarily have

$$d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p. \tag{16.6.2.2}$$

This therefore proves the *uniqueness* of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d, we can restrict ourselves to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rd ed., §10) to define an

A-derivation D of degree $+1$ of an exterior algebra $\bigwedge(M)$ (where M is a B-module and an A-algebra), such derivation taking its values in a *graded anti-commutative* A-algebra $C = \bigoplus_{n=0}^{\infty} C_n$, whose elements of degree 1 are square-zero, it suffices to give arbitrarily an A-derivation D_0 of B in C_1 and an A-homomorphism D_1 of M in C_2; then it exists one and only one A-anti-derivation D of $\Lambda(M)$ in C coinciding with D_0 in B and D_1 in M.

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A-homomorphism u of $\Omega^1_{B/A}$ in $\Omega^2_{B/A}$ such that

(16.6.2.3) $$u(g.df) = dg \wedge df$$

for whichever f, g in A; it suffices to show that there is an A-homomorphism $v : B \otimes_A \Omega^1_{B/A} \to \Omega^2_{B/A}$ such that

(16.6.2.4) $$v(g.\omega) = dg \wedge \omega$$

for $g \in B$ and $\omega \in \Omega^1_{B/A}$. Finally, since $\Omega^1_{B/A} = \mathfrak{I}/\mathfrak{I}^2$ (where $\mathfrak{I} = \mathfrak{I}_{B/A}$ is the kernel of the canonical homomorphism $B \times_A B \to B$) and that $\Omega^1_{B/A}$ is generated by elements of the form $g.df$, it is enough to define an A-homomorphism $w : B \otimes_A (B \otimes_A B) \to \Omega^2_{B/A}$ such that

(16.6.2.5) $$w(g' \otimes g \otimes f) = dg' \wedge (g.df)$$

and such that w is zero on the image of $B \otimes A\mathfrak{I}^2$. Or, since the second member of (16.6.2.5) is A-trilinear in g', g and f, the existence of w verifying (16.6.2.5) is immediate. Since, on the other hand, $\mathfrak{I}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Or, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes$
$y - y \otimes x$, the formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, **IV-4** | 36
which is to say that d is a derivation.

It remains to be shown that d verifies the condition (i). Now, the square of an anti-derivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega^\bullet_{B/A}$ is generated by $\Omega^1_{B/A}$ as a B-algebra, it is enough to verify that $d(dz) = 0$ for $z \in B$ and $x \in \Omega^1_{B/A}$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B, and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The anti-derivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — *For every base change $g : S' \to S$, if we put $X' = X \times_S S'$, the canonical morphism*

(16.6.4.1) $$\Omega^\bullet_{X/S} \otimes_S S' \longrightarrow \Omega^\bullet_{X'/S'}$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega^\bullet_{X/S}$ over an open set U of X, $s \otimes 1$ its inverse image, section of $\Omega^\bullet_{X'/S'}$ over the inverse image U' of U in X', we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

PROOF. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathscr{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega^1_{X/S}$ is an locally free $\mathscr{O}_X$-module of rank n in a point x, so that we have n sections $s_i \in \Gamma(U, \mathscr{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$ (16.5.8). Then, for every integer $p \geqslant 1$, the p-differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 \leqslant i_2 \leqslant \ldots \leqslant i_p$ elements of $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega^p_{X/S})$ over $\Gamma(U, \mathscr{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathscr{O}_X)$, we have

(16.6.5.1) $$d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p}$$

where, in the second member, k varies in the set of the $n - p$ indexes different from the i_h, i_r being the biggest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathscr{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathscr{P}^n_{X/S}(\mathscr{F})$.

(16.7.1). Let $f : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-module. We denote by $X^{(n)}_{\Delta_f}$ the *n'th infinitesimal neighbourhood* of X via the diagonal morphism $\Delta_f : X \to X \times_S X$, by $h_n : X^{(n)}_{\Delta_f} \to X \times_S X$ the canonical morphism (16.1.2), and consider the two composite morphisms IV-4 | 37

$$p_1^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p_2^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p_1^{(n)}$ corresponds to the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ which we have chosen to define the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ (16.3.5), and $p_2^{(n)}$ to the homomorphism of sheaves of rings $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (16.3.6). Since $X^{(n)}_{\Delta_f}$ and X have the same underlying subspace, we can write

(16.7.1.1)
$$\mathscr{P}^n_{X/S} = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{O}_X)).$$

More generally, we define

(16.7.1.2)
$$\mathscr{P}^n_{X/S}(\mathscr{F}) = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{F})).$$

so that $\mathscr{P}^n_{X/S} = \mathscr{P}^n_{X/S}(\mathscr{O}_X)$; by definition, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is an $\mathscr{O}_X$-module.

(16.7.2). If we come back to the definition of the inverse image of modules on ringed spaces (**0**, 4.3.1) and having in mind that $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we see that we can write the definition (16.7.1.2) in the form

(16.7.2.1)
$$\mathscr{P}^n_{X/S}(\mathscr{F}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F},$$

but where you have to be careful that, in the interpretation of the symbol $\otimes$, $\mathscr{P}^n_{X/S}$ is endowed with the structure of $\mathscr{O}_X$-module defined by *the homomorphism of sheaves of rings* $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathscr{P}^n_{X/S}(\mathscr{F})$ is canonically equipped with a $\mathscr{P}^n_{X/S}$-module structure.

Proposition (16.7.3). — (i) *The functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$ from the category of $\mathscr{O}_X$-modules to the category of $\mathscr{P}^n_{X/S}$-modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathscr{P}^n_{X/S}$ is flat.*

(ii) *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module (resp. of finite type, resp. of finite presentation), then $\mathscr{P}^n_{X/S}(\mathscr{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).*

PROOF. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathscr{P}^n_{X/S}$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$. □

(16.7.4). The two structures of $\mathscr{O}_X$-module on $\mathscr{P}^n_{X/S}$ define in $\mathscr{P}^n_{X/S}(\mathscr{F})$ two structures of $\mathscr{O}_X$-modules, which happen to be permutable, and therefore an $\mathscr{O}_X$-bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (chosen in (16.3.5)) on the *left* and the one coming from the homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ on the *right*. On other words, for every open U of X, and every triplet $a \in \Gamma(U, \mathscr{O}_X)$, $b \in \Gamma(U, \mathscr{P}^n_{X/S})$, $t \in \Gamma(U, \mathscr{F})$, we have by definition

(16.7.4.1)
$$a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b.d^n a) \otimes t = b \otimes (at) = (d^n a).(b \otimes t).$$

The $\mathscr{O}_X$-module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the *left* $\mathscr{O}_X$-module structure. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then the same is true for $\mathscr{P}^n_{X/S}(\mathscr{F})$ for any one of its $\mathscr{O}_X$-module structures. If also $\mathscr{F}$ is of finite type (resp. of finite presentation) and $f : X \to S$ is locally of finite type (resp. locally of finite presentation), $\mathscr{P}^n_{X/S}(\mathscr{F})$ is (for any one of its IV-4 | 38

$\mathscr{O}_X$-module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

$$d^n_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad \text{(also denoted } d^n_{X/S}) \tag{16.7.5.1}$$

such that, in the notations of (16.7.4), we have

$$d^n_{X/S,\mathscr{F}}(t) = 1 \otimes t \tag{16.7.5.2}$$

and consequently, because of (16.7.4.1)

$$d^n_{X/S,\mathscr{F}}(at) = (1 \otimes t)a = (d^n_{X/S,\mathscr{F}}(t)).a \tag{16.7.5.3}$$

$$d^n_{X/S,\mathscr{F}}(at) = (d^n_{X/S,\mathscr{F}}(a)).(1 \otimes t) = (d^n_{X/S,\mathscr{F}}(a))(d^n_{X/S,\mathscr{F}}(t)). \tag{IV.16.7.5.4}$$

Therefore it is $\mathscr{O}_X$-linear for the structure of *right* $\mathscr{O}_X$-module on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and *semilinear* (relative to the homomorphism σ (16.3.4)) for the *left* $\mathscr{O}_X$-module structure.

Proposition (16.7.6). — *The right $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}(\mathscr{F})$ is generated by the image of $\mathscr{F}$ by the homomorphism $d_{X/S,\mathscr{F}}$.*

PROOF. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathscr{F} = \mathscr{O}_X$ (16.3.8). □

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathscr{P}^m_{X/S} \longrightarrow \mathscr{P}^n_{X/S}$$

for $n \leqslant m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (n \leqslant m)$$

which are homomorphisms of $\mathscr{O}_X$-bimodules in light of (16.1.6) and (7.4.1); also we have commutative diagrams

$$\begin{tikzcd} \mathscr{P}^m_{X/S}(\mathscr{F}) \arrow[rr] & & \mathscr{P}^n_{X/S}(\mathscr{F}) \\ & \mathscr{F} \arrow[ul, "d^n_{X/S,\mathscr{F}}"] \arrow[ur, "d^n_{X/S,\mathscr{F}}"'] & \end{tikzcd}$$

We have therefore a projective system of $\mathscr{O}_X$-bimodules $(\mathscr{P}^n_{X/S}(\mathscr{F}))$, and we define

$$\mathscr{P}^\infty_{X/S}(\mathscr{F}) = \varprojlim \mathscr{P}^n_{X/S}(\mathscr{F}). \tag{16.7.7.1}$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

$$d^\infty_{X/S,\mathscr{F}} : \mathscr{F} \to \mathscr{P}^\infty_{X/S}(\mathscr{F}). \tag{16.7.7.2}$$

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(16.7.8). Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) \simeq \mathscr{P}^n_{X/S}(\mathscr{F}) \otimes_{\mathscr{P}^n_{X/S}} \mathscr{P}^n_{X/S}(\mathscr{G}) \tag{16.7.8.1}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n.1, prop. 3).

We conclude in particular (or we see directly from the definition (16.7.2.1)) that if $\mathscr{F}$ has an $\mathscr{O}_X$-algebra structure (not necessarily associative), $\mathscr{P}^n_{X/S}(\mathscr{F})$ has a canonical $\mathscr{O}_X$-algebra structure; the latter is associative (resp. commutative, res. unital, resp. a Lie algebra) if $\mathscr{F}$ is so. Also the canonical homomorphisms $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ for $n \leqslant m$ (16.7.7) are then algebra di-homomorphisms; similarly, (16.7.5.1) is then an $\mathscr{O}_X$-algebra homomorphisms when $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the $\mathscr{O}_X$-algebra structure from its structure of *right* $\mathscr{O}_X$-module.

With the same notations, we equally have a canonical homomorphisms of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{P}^n_{X/S}}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{P}^n_{X/S}(\mathscr{G})) \tag{16.7.8.2}$$

(Bourbaki, *Alg.*, chap. II, 3[rd]ed., §5, n.3), which is bijective when $\mathscr{P}^n_{X/S}$ is *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathscr{O}_{X'}$-bimodules

(16.7.9.1) $$u^*(\mathscr{P}^n_{X/S}(\mathscr{F})) \longrightarrow \mathscr{P}^n_{X'/S'}(u^*(\mathscr{F})).$$

We leave it to the reader to extend the properties seen in (16.4) in the case $\mathscr{F} = \mathscr{O}_X$.

Remark (16.7.10). — The definition of $\mathscr{P}^n_{X/S}(\mathscr{F})$ in the form (16.7.1.2) still makes sense when $\mathscr{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (**0**, 3.7.1)); a variant of this definition allows us to define the "jet schemes" (relatively to S) for any prescheme X.

16.8. Differential operators. [7]

Definition (16.8.1). — Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-modules, n an integer $\geqslant 0$. We say that a morphism $D : \mathscr{F} \to \mathscr{G}$ of sheaves of additive groups is a differential operator of order $\leqslant n$ (relative to S) if there is a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G}$ (where $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the structure of left $\mathscr{O}_X$-modules (16.7.4)) such that we have $D = u \circ d^n_{X/S,\mathscr{F}}$.

It is clear, because of the existence of canonical morphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F})$$

for $n \leqslant m$ (16.7.7), that a differential operator of order $\leqslant n$ is a differential operator of order $\leqslant m$ **IV-4** | 40
for $n \leqslant m$. If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, then, for every open set U of X, $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$.

We say that a homomorphism $D : \mathscr{F} \to \mathscr{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geqslant 0$ such that $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$. The *order* of a differential operator is the least upper bound of all integers n so that D is a differential operator of order $\leqslant n$ (and therefore $+\infty$ if there is no such integer); such order is always finite if X is *quasi-compact*. The differential operator of order 0 are exactly the homomorphisms of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G}$; the operators of order < 0 are *zero* by convention. For $n \geqslant 0$ a differential operator of order n is not in general a homomorphism of $\mathscr{O}_X$-modules but always a homomorphism of $\psi^*(\mathscr{O}_S)$-modules.

When $\mathscr{F} = \mathscr{O}_X$, a differential operator of order $\leqslant 1$ of $\mathscr{O}_X$ to $\mathscr{G}$ can be put in the form of $v + D$, where $v : \mathscr{O}_X \to \mathscr{G}$ is an $\mathscr{O}_X$-homomorphism, and D is an *S-derivation* (16.5.1) of $\mathscr{O}_X$ to $\mathscr{G}$: this results from the structure of $P_{B/A}$ (**0**, 20.4.8).

(16.8.2). To describe in a more precise manner a differential operator of order $\leqslant n$, $D : \mathscr{F} \to \mathscr{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V, to characterize the homomorphism $D = D_U : \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G})$. If we put $\Gamma(V, \mathscr{O}_S) = A$, $\Gamma(U, \mathscr{O}_X) = B$, so that B is an A-algebra, we have $\Gamma(U, \mathscr{P}^n_{X/S}(\mathscr{F})) = (B \otimes_A B)/\mathfrak{I}^{n+1}$, where we abbreviate $\mathfrak{I} = \mathfrak{I}_{B/A}$. Also put $M = \Gamma(U, \mathscr{F})$, $N = \Gamma(U, \mathscr{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A-homomorphism $D : M \to N$ factors through

$$M \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical morphism $t \mapsto 1 \otimes t$, and v is a B-homomorphism, the structure of B-module coming from the first factor (whereas we recall that in the formation of the tensor product over B, the structure of B-module on $(B \otimes_A B)/\mathfrak{I}^{n+1}$ comes from the second factor B). Note also that the B-module $((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M)/\mathfrak{I}^{n+1}(B \otimes_A M)$, where $(B \otimes_A M)$ is considered as a $(B \otimes_A B)$ module and its structure of B-module comes from $t \mapsto 1 \otimes t$ of B in $B \otimes_A B$. Let then D' be the B-homomorphism of $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; then condition of factorization of D is to say that D' must be zero on the B-module $\mathfrak{I}^{n+1}(B \otimes_A M)$.

[7]For a more general formalism, see the exposé VII of [**eAG64**] (due to P. Gabriel).

(16.8.3). It is clear that the set of differential operators of order $\leqslant n$ from $\mathscr{F}$ to $\mathscr{G}$ forms an additive group, denoted by $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$; when $\mathscr{F}=\mathscr{G}=\mathscr{O}_X$, we also write $\mathrm{Diff}^n_{X/S}$ instead of $\mathrm{Diff}^n_{X/S}(\mathscr{O}_X,\mathscr{O}_X)$.

We have seen (16.8.1), that given two open sets $U\supset V$ of X, we have a restriction homomorphism

$$\mathrm{Diff}^n_{X/S}(\mathscr{F}|U,\mathscr{G}|U)\longrightarrow\mathrm{Diff}^n_{X/S}(\mathscr{F}|V,\mathscr{G}|V)$$

from which we deduce that $U\mapsto\mathrm{Diff}^n_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a presheaf of additive groups; in fact, it is actually a *sheaf*, since for an open set U varying in X, the homomorphisms $u\mapsto u\circ d^n_{U/S,\mathscr{F}|U}$ are *isomorphisms* of sheaves of additive groups **IV-4** | 41

(16.8.3.1) $$\mathrm{Hom}_{\mathscr{O}_U}(\mathscr{P}^n_{U/S}(\mathscr{F}|U),\mathscr{G}|U)\simeq\mathrm{Diff}^n_{U/S}(\mathscr{F}|U,\mathscr{G}|U),$$

because of the fact that the image of $\mathscr{F}$ by $d^n_{X/S,\mathscr{F}}$ generates $\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.6). We denote this sheaf by $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms (16.8.3.1) define an isomorphism of sheaves of additive groups*

(18.4.1) $$\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G})\simeq\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}).$$

When $\mathscr{F}=\mathscr{G}=\mathscr{O}_X$, we also write $\mathscr{D}\!\mathit{iff}^n_{X/S}$ instead of $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{O}_X,\mathscr{O}_X)$; it results from (16.6.4) that $\mathscr{D}\!\mathit{iff}^n_{X/S}$ is the *dual* of $\mathscr{P}^n_{X/S}$; so we also write $\langle t,D\rangle$ instead of $u(t)$ if t is a section of $\mathscr{P}^n_{X/S}$ over an open set and if u is a homomorphism from $\mathscr{P}^n_{X/S}$ to $\mathscr{O}_X$ corresponding to D.

(16.8.5). When $\mathscr{P}^n_{X/S}(\mathscr{F})$ has an $\mathscr{O}_X$-bimodule structure (16.7.4), we deduce canonicallly an $\mathscr{O}_X$-bimodule structure on $\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G})$, and therefore on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ because of (16.8.4.1). More precisely, to the left $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$ corresponds, because of (16.8.1), the left $\mathscr{O}_X$-module structure on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ explained as follows: for every open set U of X, every section $a\in\Gamma(U,\mathscr{O}_X)$ and every differential operator $D:\mathscr{F}|U\to\mathscr{G}|U$, aD is the differential operator which, for every section $t\in\Gamma(U,\mathscr{F})$, makes correspond the section

(16.8.5.1) $$(aD)(t)=a(D(t))$$

of $\Gamma(U,\mathscr{G})$. Similarly, the right $\mathscr{O}_X$-module structure on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t\in\Gamma(U,\mathscr{F})$, makes correspond the section

(16.8.5.2) $$(Da)(t)=D(at).$$

Proposition (16.8.6). — *If $f:X\to S$ is a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module for any of the structures defined in (16.8.5).*

PROOF. The proposition follows from the fact that, under these hypothesis, $\mathscr{P}^n_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module of finite presentation (16.7.4) and of (**I**, 1.3.12) □

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$; we also see as in (16.8.3) that $U\mapsto\mathrm{Diff}_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$. It is immediate that $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$ is the reunion of the increasing filtered family of its subsheaves $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$ is similarly **IV-4** | 42
the union of its subgroups $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$ (16.8.1). The $\mathscr{O}_X$-bimodule structure on the $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ induce therefore an $\mathscr{O}_X$-bimodule structure on $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$, further explained in (16.8.5.1) and (16.8.5.2).

Note that, for $n\leqslant m$, we have a commutative diagram

(16.8.7.1) $$\begin{tikzcd}\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] \arrow[d] & \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}) \arrow[d] \\ \mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^m_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] & \mathscr{D}\!\mathit{iff}^m_{X/S}(\mathscr{F},\mathscr{G})\end{tikzcd}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the horizontal arrow on the left comes from the canonical morphism $\mathscr{P}^m_{X/S}(\mathscr{F})\to\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.7). For every open set U of X, we then endow $\Gamma(U, \mathscr{P}^\infty_{X/S}(\mathscr{F})) = \varprojlim \Gamma(U, \mathscr{P}^n_{X/S}(\mathscr{F}))$ of the projective limit topology of the discrete topologies on $\Gamma(U, \mathscr{P}^n_{X/S}(\mathscr{F}))$, which defines on $\Gamma(U, \mathscr{P}^\infty_{X/S}(\mathscr{F}))$ a topological $\Gamma(U, \mathscr{O}_X)$-bimodule structure, so that $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ shows itself as a sheaf valued in the category of topological commutative groups (**0**, 3.2.6). So [**God58**, II, 1.11], the limit of the inductive system of sheaves of commutative groups $(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}))$ is precisely the sheaf of *continuous* germs of homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$ (the latter equipped with the discrete topology): the continuous homomorphisms $\Gamma(U, \mathscr{P}^\infty_{X/S}(\mathscr{F}))$ into the discrete group $\mathscr{G}$ indeed correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U, \mathscr{P}^n_{X/S}(\mathscr{F})) \to \Gamma(U, \mathscr{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathscr{H}om.\mathrm{cont}_{\mathscr{O}_X}(\mathscr{P}^\infty_{X/S}(\mathscr{F}), \mathscr{G}) \simeq \mathscr{D}iff_{X/S}(\mathscr{F}, \mathscr{G})$$

where the first member denotes the sheaf of germs of continuous homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$.

Proposition (16.8.8). — *Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules, $D : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\psi^*(\mathscr{O}_S)$-modules, n an integer $\geqslant 0$. The following conditions are equivalent:*

(a) D is a differential operator of order $\leqslant n$.

(b) For all sections a of $\mathscr{O}_X$ over an open set U, the homomorphism $D_a : \mathscr{F}|U \to \mathscr{G}|U$ such that, for every section t of $\mathscr{F}$ over an open set $V \subset U$, we have

(16.8.8.1) $$D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leqslant n - 1$.

(c) For every open set U of X, every family $(a_i)_{1 \leqslant i \leqslant n+1}$ of $n + 1$ sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, we have the identity

(16.8.8.2) $$\sum_{H \subset I_{n+1}} (-1)^{\mathrm{Card}(H)} \Big(\prod_{i \in H} a_i\Big) D\Big(\big(\prod_{i \notin H} a_i\big)t\Big) = 0$$

(where I_{n+1} is the interval $1 \leqslant i \leqslant n + 1$ of $\mathbb{N}$).

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PROOF. Lets prove first the equivalence of (a) and (b). By definition, to prove that D is a differential operator of order $\leqslant n$, it suffices to prove so for the restriction $D|U : \mathscr{F}|U \to \mathscr{G}|U$ to any affine open set U of X, and on the other hand the property (c) is valid for every open set U if it is so on every affine open set. We can therefore restrict ourselves to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affines. Because of (16.8.2) (where we use the same notations), the condition (a) means that the A-homomorphism $D' : B \otimes_A M \to N$ such that $D'(b \otimes t) = bD(t)$ is zero on $\mathfrak{I}^{n+1}(B \otimes_A M)$, which is, because of (**0**, 20.4.4), equivalent to saying that D' is annihilates every element of the form

$$\Big(\prod_{i=1}^{n+1} (a_i \otimes 1 - 1 \otimes a_i)\Big).(1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now, this element can be written as $\sum_{H \subset I_{n+1}} (\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i)t)$, and the image under D' of this element is the first member of (16.8.8.2), which proves the equivalence of (a) and (c).

Let us prove now the equivalence of (b) and (c). Let us reason by induction on n, the statement being trivial for $n = 0$. Writing a_{n+1} instead of a in the condition (b), we see, by the induction hypothesis, that condition (b) means that for every family $(a_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, that

$$\sum_{H' \subset I_{n+1}} (-1)^{\mathrm{Card}(H')} \Big(\prod_{i \in H'} a_i\Big) D_{a_{n+1}}\Big(\big(\prod_{i \notin H'} a_i\big)t\Big) = 0.$$

But if we replace on this relation $D_{a_{n+1}}$ by the definition (16.8.8.1), we check immediately that we have, up to sign, the first member of (16.8.8.2); from which we conclude. □

Proposition (16.8.9). — *If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, and $D' : \mathscr{G} \to \mathscr{H}$ a differential operator of order $\leqslant n'$, then $D' \circ D : \mathscr{F} \to \mathscr{H}$ is a differential operator of order $\leqslant n + n'$.*

PROOF. By hypothesis, we can write $D = u \circ d^n_{X/S,\mathscr{F}}$ and $D' = v \circ d^n_{X/S,\mathscr{G}}$, where $u : \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{G}$ and $v : \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ are $\mathscr{O}_X$-homomorphisms. It all comes down to showing that the composite homomorphism of sheaves of additive groups

$$\mathscr{F} \xrightarrow{d^n_{X/S,\mathscr{F}}} \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{d^{n'}_{X/S,\mathscr{G}}} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

factors as

$$\mathscr{F} \xrightarrow{d^{n+n'}_{X/S,\mathscr{F}}} \mathscr{P}^{n+n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{w} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

where w is an $\mathscr{O}_X$-homomorphism. It suffices to prove the

Lemma (16.8.9.1). — *There is one and only one $\mathscr{O}_X$-homomorphism*

(16.8.9.2) $$\delta : \mathscr{P}^{n+n'}_{X/S} \longrightarrow \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) = \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{P}^n_{X/S}$$

making the following diagram commute **IV-4** | 44

(16.8.9.3)
$$\begin{tikzcd}
\mathscr{O}_X \arrow[r, "d^{n+n'}_{X/S}"] \arrow[d, "d^n_{X/S}"'] & \mathscr{P}^{n+n'}_{X/S} \arrow[d, "\delta"] \\
\mathscr{P}^n_{X/S} \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S})
\end{tikzcd}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathscr{F}$

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "d^{n+n'}_{X/S,\mathscr{F}}"] \arrow[d, "d^n_{X/S,\mathscr{F}}"'] & \mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \arrow[d, "\delta\otimes 1"] \\
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}((\mathscr{F})))
\end{tikzcd}$$

and on the other hand, we verify immediately the from definition (16.7.5) that the diagram

$$\begin{tikzcd}
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "u"] \arrow[d, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{G} \arrow[d, "d^{n'}_{X/S,\mathscr{G}}"] \\
\mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}((\mathscr{F}))) \arrow[r, "1\otimes u"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{G})
\end{tikzcd}$$

is commutative. We finish the proof by taking w to be the composite $\mathscr{O}_X$-homomorphism

$$\mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \xrightarrow{\delta\otimes 1} \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \xrightarrow{1\otimes u} \mathscr{P}^{n'}_{X/S}(\mathscr{G}).$$

□

PROOF (16.8.9.3). It remains to prove the lemma (16.8.9.3). Considering (16.7.6), which proves the uniqueness of δ, we are brought back to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines; letting $\mathfrak{I} = \mathfrak{I}_{B/A}$, it suffices to define a canonical homomorphism of B-modules

$$\phi : (B \otimes_A B)/\mathfrak{I}^{n+n'+1} \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{I}^{n+1})$$

the B-module structure of the two members coming from the first B factor; recall that on tensor product of the second member, $(B \otimes_A B)/\mathfrak{I}^{n'+1}$ must be considered as a right B-module by its **IV-4** | 45
second B factor , and $(B \otimes_A B)/\mathfrak{I}^{n+1}$ as a left B-module by its first B factor (16.7.2). It is the same to define a homomorphism of B-modules

$$\phi_0 : B \otimes_A B \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{I}^{n+1})$$

and prove it is zero on $\mathfrak{I}^{n+n'+1}$. Now, we immediately define a homomorphism by the condition that

$$\phi_0(b \otimes b') = \pi_{n'}(b \otimes 1) \otimes \pi_n(1 \otimes b') \quad \text{for } b, b' \text{ in } B$$

under the notations of (16.3.7). Also, it is immediate that ϕ_0 is a homomorphism of *rings*. Now, we can write

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) - \pi_{n'}(1\otimes 1)\otimes \pi_n(1\otimes b)$$

and we have

$$\pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)b\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes b\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1)$$

from which, finally

(16.8.9.4) $\quad \phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1 - 1\otimes b).$

A product of $n + n' + 1$ of terms of the form (16.8.9.4) is therefore necessarily zero, because the same is true for the product of $n+1$ terms of the form $\pi_n(b\otimes 1 - 1\otimes b)$ and of $n'+1$ terms of the form $\pi_{n'}(b\otimes 1 - 1\otimes b)$. The conclusion therefore results from (**0**, 20.4.4). ☐

Corollary (16.8.10). — *The sheaf $\mathscr{D}\!iff_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$ (also denoted $\mathscr{D}\!iff_{X/S}$) is canonically endowed with the structure of sheaf of rings, and the $\mathscr{D}\!iff^n_{X/S}$ form an increasing filtration compatible with such structure.*

In particular, $\mathscr{D}\!iff^0_{X/S}$ is a sheaf of subrings of $\mathscr{D}\!iff_{X/S}$, which is canonically identified with $\mathscr{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathscr{O}_X$-bimodule of $\mathscr{D}\!iff_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathscr{O}_X$ considered as a sheaf of subrings of $\mathscr{D}\!iff_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathscr{F} = \oplus_{\lambda\in L}\mathscr{F}_\lambda$; then it is clear (16.7.2.1) that $\mathscr{P}^n_{X/S}(\mathscr{F}) = \oplus_{\lambda\in L}\mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; since the functor $U \mapsto \Gamma(U,\mathscr{F})$ commutes with the formation of arbitrary direct sums, $d^n_{X/S,\mathscr{F}}$ is the homomorphism whose restriction to each $\mathscr{F}_\lambda$ is $d^n_{X/S,\mathscr{F}_\lambda} : \mathscr{F}_\lambda \to \mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; then we conclude immediately that we have

$$\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\mathrm{Diff}^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}),$$

and therefore also (**0**, 3.2.6)

$$\mathscr{D}\!iff^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\mathscr{D}\!iff^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}).$$

Moreover, if $\mathscr{G} = \prod_{\mu\in M}\mathscr{G}_\mu$ (**0**, 3.2.6), we have

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) = \prod_{\mu\in M}\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}_\mu),$$

every homomorphism u from $\mathscr{P}^n_{X/S}(\mathscr{F})$ to $\mathscr{G}$ corresponds bijectively to the family of its **IV-4** | 46
composites $u_\mu : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G} \to \mathscr{G}_\mu$. We have therefore

$$\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\mu\in M}\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}_\mu),$$

and consequently also

$$\mathscr{D}\!iff^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\mu\in M}\mathscr{D}\!iff^n_{X/S}(\mathscr{F},\mathscr{G}_\mu).$$

(ii) So far, we have hardly encountered differential operators $\mathscr{F} \to \mathscr{G}$ where $\mathscr{F}$ and $\mathscr{G}$ are not locally free of finite rank, in which case the structure is reduced locally, because of (i), to the case of the sheaf $\mathscr{D}\!iff_{X/S}$; the latter will be studied later (16.11) in a particular case.

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathscr{I}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$, there is an open neighbourhood of x in X and a regular sequence (0, 15.2.2) (resp. quasi-regular (**0**, 15.2.2)) of elements of $\Gamma(U, \mathscr{O}_X)$ which generates $\mathscr{I}|U$.

We say that a regular (resp. quasi-regular) sequence of sections of $\mathscr{O}_X$ over U which generates $\mathscr{I}|U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathscr{I}|U$.

Definition (16.9.2). — Let $j : Y \to X$ be an immersion of preschemes and let U be an open set such that $j(Y) \subset U$ and that j is a closed immersion of Y in U. We say that j is regular (resp. quasi-regular) if the closed subprescheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathscr{O}_U$ (condition independent of the choice of U).

We say that a subprescheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \to X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subprescheme and $\mathscr{I}$ is the ideal of $\mathscr{O}_X$ that defines Y, it is the same as asking $\mathscr{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an *integral* ring, f and element $\neq 0$ of A, the closed subprescheme $V(f)$ of $\operatorname{Spec}(A)$ (isomorphic to $\operatorname{Spec}(A/(f))$) is *regularly immersed* in $\operatorname{Spec}(A)$.

Every regular ideal is quasi-regular (**0**, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — *Let X be a ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$, $(f_i)_{1 \leqslant i \leqslant m}$ a finite sequence of sections of $\mathscr{O}_X$ over X generating $\mathscr{I}$. For the (f_i) to be a quasi-regular sequence* (**0**, 15.2.2)*, it is necessary and sufficient that the following conditions are verified:*

(i) *The canonical images of f_i in $\mathscr{I}/\mathscr{I}^2$ form a basis of this $\mathscr{O}_X/\mathscr{I}$-module.*
(ii) *The canonical surjective homomorphism* (16.1.2.2)

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Also, if this is true, then every sequence $(f'_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{I}$ over X which generates $\mathscr{I}$ is quasi-regular.

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PROOF. The two conditions stated above only translate the ones in (**0**, 15.2.2), given the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). □

Corollary (16.9.4). — *Let X be a locally ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$. For $\mathscr{I}$ to be quasi-regular, it is necessary and sufficient that it verifies the following conditions:*

(i) *$\mathscr{I}$ is of finite type.*
(ii) *$\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module.*
(iii) *The canonical homomorphism*

(16.9.4.1)
$$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

PROOF. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough to show, because of (16.9.3), that if, in a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$, there is a neighbourhood U of x in X and n-sections f_i $(1 \leqslant i \leqslant n)$ of $\mathscr{I}$ over U such that the image in $\mathscr{I}/\mathscr{I}^2$ form a basis of $(\mathscr{I}/\mathscr{I}^2)|U$ over $(\mathscr{O}_X/\mathscr{I})|U$, so there will be a neighbourhood $V \subset U$ of x such that the $f_i|V$ generate $\mathscr{I}|V$. Now, by hypothesis, we have $\mathscr{I}_x \neq \mathscr{O}_x$, so that $\mathscr{I}_x$ is contained in the maximal ideal of $\mathscr{O}_X$; since $\mathscr{I}_x$ is an $\mathscr{O}_x$ module of finite type and the classes of $(f_i)_x$ in $\mathscr{I}_x/\mathscr{I}_x^2$ generate this $(\mathscr{O}_x/\mathscr{I}_x)$-module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathscr{I}_x$. Since $\mathscr{I}$ is of finite type, we conclude by (**0**, 5.2.2). □

Corollary (16.9.5). — *Let X be a locally ringed space, $\mathscr{I}$ a quasi-regular ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{I}$ over X, x a point of $\operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$. The following conditions are equivalent:*

(a) There is a neighbourhood of x in X such that $f_i|U$ form a quasi-regular sequence of elements $\Gamma(U,\mathscr{O}_X)$ generating $\mathscr{I}|U$.
(b) The $(f_i)_x$ form a system of generators of $\mathscr{I}_x$ whose size is as small as possible.
(b′) The $(f_i)_x$ form a minimal set of generators of $\mathscr{I}_x$.
(c) If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X,\mathscr{I}/\mathscr{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathscr{O}_x/\mathscr{I}_x)$-module $\mathscr{I}_x/\mathscr{I}_x^2$.

PROOF. By hypothesis, $\mathscr{O}_x$ is a local ring, $\mathscr{I}_x$ an ideal of finite type of $\mathscr{O}_x$ contained in the maximal ideal of $\mathscr{O}_x$; the equivalence of (b), (b′) and (c) results from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (**0**, 5.2.2) it follows that if condition (c) is verified (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathscr{I}/\mathscr{I}^2)|U$ has constant rank equal to n, and that the $f_i|U$ generate $\mathscr{I}|U$; it suffices now to apply the last assertion of (16.9.3) to U. □

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathscr{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathscr{O}_y/\mathscr{I}_y)$-module $(\mathscr{I}_y/\mathscr{I}_y^2)$ for all $y\in X$. We have an example by taking $X=\operatorname{Spec}(A)$, where A is a Dedekind ring, **IV-4** | 48
and $\mathscr{I}=\widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is a *non-principal* ideal of A; then indeed $\mathscr{I}_y/\mathscr{I}_y^2=0$ in every point y different from $x\in X$ corresponding to $\mathfrak{I}$, and $\mathscr{I}_x/\mathscr{I}_x^2$ has rank 1 over the field $\mathscr{O}_x/\mathscr{I}_x$; also, $\mathscr{I}$ is evidently a regular ideal.

(ii) In (16.9.5), one cannot replace "quasi-regular" by "regular", even when X is a prescheme (cf. (16.9.12)). Indeed, denote by B the ring of germs of infinitely differentiable functions on the point 0 on $\mathbb{R}$; it has a maximal ideal $\mathfrak{m}$ generated by the germ of t of the identity mapping to 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k>0$ is not reduced to 0. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1,f_2) is *regular* in A: indeed, f_1 is not a zero divisor in A, because the relation $tP[T]\in\mathfrak{n}TB[T]$, for a polynomial $P\in B[T]$, implies that the products of t by the coefficients of P belongs to the ideal $\mathfrak{n}$, and it results immediately that the coefficients are the same in $\mathfrak{n}$, so $P[T]\in\mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the ring of polynomials $\mathbb{R}[T]$, therefore integral, and the image of f_2 in A/f_1A, being equal to T, is not a zero divisor, so that our claim is true. However, f_2 is a zero divisor in A, since for every non-zero element $x\in\mathfrak{n}$, the image of x in A is $\neq 0$, but the image of xT is zero. We conclude that the sequence (f_2,f_1) is *not regular* in A; on the other hand, the ideal $\mathfrak{I}=f_1A+f_2A$ is distinct from A, so the conditions (b), (b′) and (c) of (16.9.5) do not imply the condition (a) when we replace "quasi-regular" by "regular".

(16.9.7). If $X=\operatorname{Spec}(A)$ is an affine scheme, we'll say that the ideal $\mathfrak{I}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathscr{J}=\widetilde{\mathfrak{I}}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular); we note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming in A a regular (resp. quasi-regular) sequence as the example (16.9.5) shows; however this is true if A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — *Let $j:Y\to X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:*

(i) j is an immersion locally of finite presentation.
(ii) The conormal sheaf $\mathscr{G}r^1(j)=\mathscr{N}_{Y/X}$ (16.1.2) is a locally free $\mathscr{O}_Y$-module.
(iii) The canonical homomorphism

$$\mathbf{S}^{\bullet}_{\mathscr{O}_Y}(\mathscr{G}r^1(j))\longrightarrow\mathscr{G}r^{\bullet}(j)$$

(16.1.2.2) *is bijective.*

PROOF. The problem being local on Y, we can restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X, so the translation of (16.9.4) into (16.9.8) results from the description of $\mathscr{Gr}^1(j)$ and $\mathscr{Gr}^\bullet(j)$ in terms of the ideal $\mathscr{I}$ of $\mathscr{O}_X$ defining the subprescheme Y (16.1.3, (ii)). □

Corollary (16.9.9). — *Let Y be a prescheme, X an Y-prescheme, $j : Y \to X$ a Y-section of X, so that the n'th normal invariant $\mathscr{A}^{(n)}$ of j (16.1.2) is an augmented $\mathscr{O}_Y$-algebra (16.1.7); take $\mathscr{A}^{(\infty)} = \varprojlim \mathscr{A}^{(n)}$. For j to be a quasi-regular immersion it is necessary and sufficient that j is locally of finite presentation, and that every $y \in Y$ admits an affine open neighbourhood U of ring C such that $\mathscr{A}^{(\infty)}|U$ is isomorphic as an augmented $\mathscr{O}_U$-algebra to $\mathscr{O}_U[[T_1, \ldots, T_n]]$.*

Proof. We can reduce to the case where j is a closed immersion by restricting to a small neighbourhood of y (see the argument on (16.4.11)), and therefore $\mathscr{O}_Y$ is identified with the quotient algebran $\mathscr{O}_X/\mathscr{I}$ and the canonical surjective homomorphism $\mathscr{O}_X \to \mathscr{O}_Y$ admits a right inverse **IV-4 | 49**
(16.1.7). We can therefore suppose $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ are affines, B being then an augmented A-algebra, and the augmentation ideal $\mathfrak{I}$ being of finite type. Since $\mathscr{A}^{(n)}$ is identified with $(B/\mathfrak{I}^{n+1})^\sim$, the corollary follows from the equivalence of (b) and (c) in (**0**, 19.5.4) when $B/\mathfrak{I} = A$. □

We note that, on the affine case considered, the fact that j is an quasi-regular immersion is further equivalent, because of (**0**, 19.5.4), to saying that B is a *formally smooth* A-algebra for the $\mathfrak{I}$-preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \to Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, $j : Y \to X$ the canonical injection, y a point of Y.*

- (i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \to U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathscr{I}_y$ of the surjective homomorphism $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$.*
- (ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a *coherent* ideal $\mathscr{I}$ of $\mathscr{O}_X$. The condition is clearly necessary. Conversely, if $\mathscr{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathscr{I}$ over an open neighbourhood U of y in X, we can suppose that the s_i generate $\mathscr{I}|U$ (**0**, 5.5.2) and form a regular sequence (**0**, 15.2.4), from which the claim follows.

(ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences of $\mathscr{O}_{X,y}$, formed from elements of the maximal ideal (**0**, 15.1.11)

□

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathscr{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and therefore *a fortiori* neither are quasi-regular morphisms in the sense of (6.8.1).

(ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (**0**, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\operatorname{Spec}(k(x)) \to X$ to be a *regular* immersion.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, Y' a subprescheme of Y, such that the canonical injection $j : Y' \to Y$ is regular. Then the sequence of $\mathscr{O}_{Y'}$-modules*

$$0 \longrightarrow j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0 \qquad (16.9.13.1)$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U of the homomorphisms of (16.9.13.1) form a split exact sequence. **IV-4 | 50**

Let us first prove the following lemma:

Lemma (16.9.13.2). — *Let A be a ring, $\mathfrak{I}$ an ideal of A, $A' = A/\mathfrak{I}$, $(f_i)_{1\leqslant i\leqslant r}$ a sequence of elements of A which is A'-regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geqslant n$, we have the relation*

$$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n + \mathfrak{I}\mathfrak{K}^N. \tag{16.9.13.3}$$

Proof. It is clearly sufficient to prove that every element of the first is contained in the second, and, by induction on n, we reduce to the case $N = n+1$. An element of the first member of (16.9.13.3), being in $\mathfrak{K}^n$, is written as $P(f_1, \ldots, f_r)$, where $P \in A[T_1, \ldots, T_r]$ is homogeneous of degree n. If f'_i is the canonical image of f_i in A', the hypothesis $P(f_1, \ldots, f_n) \in \mathfrak{I}$ means that $P(f'_1, \ldots, f'_n) = 0$. But $P(f'_1, \ldots, f'_n) \in \mathfrak{K}'^n$, so the canonical image of $P(f'_1, \ldots, f'_n)$ in $\mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A'-regular implies that the canonical homomorphism $\mathbf{S}^n_C(\mathfrak{K}'/\mathfrak{K}'^2) \to \mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is bijective (**0**, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$. It follows immediately that $P(f_1, \ldots, f_n) \in \mathfrak{I}\mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \ldots, f_n) \in \mathfrak{I}$, we finally have $P(f_1, \ldots, f_n) \in \mathfrak{I}\mathfrak{K}^n + \mathfrak{I}\mathfrak{K}^{n+1}$ which proves the lemma. □

By taking the quotient of both side of (16.9.13.3) by $\mathfrak{I}\mathfrak{K}^n$, we see that the relations (16.9.13.3) for $N \geqslant n$ entail

$$(\mathfrak{I} \cap \mathfrak{K}^n)/\mathfrak{I}\mathfrak{K}^n \subset \bigcap_{N \geqslant n} \mathfrak{K}^N.(A/(\mathfrak{I}\mathfrak{K}^n)). \tag{16.9.13.4}$$

We deduce the

Corollary (16.9.13.5). — *Suppose that the hypothesis of (16.9.13.2) are verified and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A. Then for every integer $n > 0$,*

$$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n. \tag{16.9.13.6}$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{I}\mathfrak{K}^n$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). □

Taking in particular $n = 2$ in (16.9.13.6), and remarking that we have $\mathfrak{L}^2 = \mathfrak{I}^2 + \mathfrak{I}\mathfrak{K} + \mathfrak{K}^2 = \mathfrak{I}\mathfrak{L} + \mathfrak{K}^2$; since $\mathfrak{I}\mathfrak{L} \subset \mathfrak{L}^2$, we deduce that

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L} + (\mathfrak{I} \cap \mathfrak{K}^2) = \mathfrak{I}\mathfrak{L} + \mathfrak{I}\mathfrak{K}^2 = \mathfrak{I}\mathfrak{L},$$

in other words,

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L}, \tag{16.9.13.7}$$

which we can also express in saying that the canonical homomorphism

$$\mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow (\mathfrak{I} + \mathfrak{J}^2)/\mathfrak{J}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that the sequence of stalks of the sheaves appearing in (16.9.13.1), in a **IV-4** | 51
point $x \in Y'$, is exact. Now, if we take $A = \mathscr{O}_{X,x}$, we can write $\mathscr{O}_{Y,x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A, then $\mathscr{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an *A'-regular* sequence of elements of A', which themselves are images of elements from a A'-regular sequence of elements of A belonging to the maximal ideal of A. If $\mathfrak{K}$ is the ideal generated by the latter and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, we have $\mathscr{O}_{Y',x} = A/\mathfrak{L}$, and since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I}/\mathfrak{I}\mathfrak{L} \to (\mathfrak{I} + \mathfrak{J}^2)/\mathfrak{J}^2$ is bijective. But this shows that the sequence

$$0 \longrightarrow \mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow \mathfrak{L}/\mathfrak{L}^2 \longrightarrow (\mathfrak{I}/\mathfrak{L})/(\mathfrak{I}/\mathfrak{L})^2 \longrightarrow 0$$

is exact (see the demonstration of (16.2.7)), and the modules making up this sequence are precisely the stalks in x of the sheaves of (16.9.13.1). The second claim follows from the fact that $\mathscr{N}_{X/Y}$ is a locally free $\mathscr{O}_{Y'}$-module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rd ed., §1, n°11, prop. 21. □

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \to S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

(i) $\Omega^1_{X/S}$ is a locally projective $\mathscr{O}_X$-module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega^1_{X/S})$ is a projective $\Gamma(U, \mathscr{O}_X)$-module (not necessarily of finite type).

(ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_{\bullet}(\mathscr{P}_{X/S})$$

is bijective.

In particular, if $\Omega^1_{X/S}$ is locally free of finite rank, the $\mathscr{P}^n_{X/S}$ are locally free of finite rank $\mathscr{O}_X$-modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \to S$ is differentially smooth, since $\Omega^1_{X/S} = 0$ because of (**I**, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Lets limit ourselves to proving the following proposition:

Proposition (16.10.2). — *Let A be a ring, B a formally smooth A-algebra for the discrete topology* (**0**, 19.3.1)*. Then* Spec(B) *is differentially smooth over* Spec(A)*.*

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B-algebra (by on or the other canonical homomorphisms $b \mapsto b \otimes 1$, $b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (**0**, 19.3.5, (iii)); **IV-4** | 52
therefore $B \otimes_A B$ is a formally smooth A-algebra with the discrete topology (**0**, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A-algebra for the $\mathfrak{I}$-preadic topology (**0**, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A-algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (**0**, 19.5.4). □

Proposition (16.10.3). — *For a morphism $f : X \to S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x, of ring A, such that $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is an augmented topological A-algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A-module and B being endowed with the B^+-preadic topology (where B^+ is the augmentation ideal). If $\Omega^1_{X/S}$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \dots, T_n]]$.*

Proof. The notion of a differentially smooth morphism is clearly local on X, so we reduce to the case where $S = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(C)$. Consider $C \otimes_B C$ as a C-algebra (by the first factor); take $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow C with the $\mathfrak{I}$-preadic topology; we can apply to the topological C algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (**0**, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C-algebra for the discrete topologies. The topology of $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is clearly the projective limit topology of this ring (16.1.11). □

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega^1_{X/S}$ in the point x. We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fiber $f^{-1}(f(x))$.

Proposition (16.10.4). — *Let $f : X \to S$, $g : S' \to S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \to S'$.*

(i) *If f is differentially smooth, the same is true for f'.*

(ii) *Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega^1_{X'/S'}$ is a finite type $\mathscr{O}_{X'}$-module, f is differentially smooth and $\Omega^1_{X/S}$ is a finite type $\mathscr{O}_X$-module.*

Proof. Indeed, if f is differentially smooth, the $\mathscr{G}r_n(\mathscr{P}^n_{X/S})$ are *flat* $\mathscr{O}_X$-modules; therefore by (16.4.6), the homomorphism $\mathscr{G}r_n(\mathscr{P}^n_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}^n_{X'/S'})$ are bijective for every n, because of the commutative of the diagram (16.2.1.3), it follows from the definition (16.10.1) that f' is differentially smooth. On the other hand, if g is faithfully flat and quasi-compact, it follows also from (16.4.6) that $\mathscr{G}r_n(\mathscr{P}^n_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}^n_{X'/S'})$ is bijective for every n. Suppose also that f' is differentially smooth and $\Omega^1_{X'/S'}$ is of finite rank. Since the canonical projection $X' \to X$ is a faithfully flat and quasi-compact morphism, it results first from (2.5.2) that $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. □

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \to S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \to X \times_S X$ to be quasi-regular.*

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PROOF. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subprescheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (**I**, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathscr{I}$ of finite type, and $\Omega^1_{X/S} = \mathscr{I}/\mathscr{I}^2$ is an $\mathscr{O}_X$-module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). □

Remark (16.10.6). — Let $f : X \to S$ be a morphism such that the $\mathscr{O}_X$-module $\Omega^1_{X/S}$ is locally free of finite rank. It results from (**0**, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda \in L}$ of sections of $\mathscr{O}_X$ over U for which $(dz_\lambda)_{\lambda\in L}$ forms a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

16.11. Differential operators on a differentially smooth S-prescheme

(16.11.1). Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the dz_λ form system of generators of $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. Let m be an integer or the symbol ∞, and take, for every λ

(16.11.1.1) $$\zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathscr{P}^m_{X/S}).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

(16.11.1.2) $$|\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p} = \prod_\lambda (p_\lambda!)$$

(16.11.1.3) $$\binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}!/(\mathbf{q}!(\mathbf{p}-\mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leqslant \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \not\leqslant \mathbf{p}$,

(16.11.1.4) $$\mathbf{z}^{\mathbf{p}} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^{\mathbf{p}} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

(16.11.1.5) $$d^m(\mathbf{z}^{\mathbf{p}}) = (d^m(\mathbf{z}))^{\mathbf{p}} = (\boldsymbol{\zeta} + \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^{\mathbf{q}}$$

(16.11.1.6) $$\boldsymbol{\zeta}^{\mathbf{p}} = (d^m \mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} d^m(\mathbf{z}^{\mathbf{q}}).$$

Since the dz_λ generate $\Omega^1_{X/S}$, and are the images of δz_λ, and as the canonical morphism (16.3.1.1) is surjective, we conclude that for finite m, the δz_λ generate the $\mathscr{O}_U$-algebra $\mathscr{P}^n_{U/S}$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 du th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leqslant m$) generate the $\mathscr{O}_U$-module $\mathscr{P}^n_{U/S}$. A differential operator $D \in \mathrm{Diff}^m_{U/S}$ is consequently entirely determined by the values of $\langle \boldsymbol{\zeta}^{\mathbf{p}}, D\rangle$for $|\mathbf{p}| \leqslant m$, or, which amounts to the same by (16.11.1.5) and (16.11.1.6), by the values of the **IV-4** | 54
$\langle d^m(\mathbf{z}), D\rangle = D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leqslant m$; more precisely, it follows from (16.11.5) that we have

(16.11.1.7) $$D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \langle \boldsymbol{\zeta}^{\mathbf{q}}, D\rangle \mathbf{z}^{\mathbf{p}-\mathbf{q}}.$$

Theorem (16.11.2). — *Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the family $(dz_\lambda)_{\lambda\in L}$ generates $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. The following conditions are equivalent:*

(a) *$f|U$ is differentially smooth and (dz_λ) is a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$.*

(b) *There is a family $(D_{\mathbf{p}})_{\mathbf{p}\in\mathbb{N}^{(L)}}$ of differential operators of $\mathscr{O}_U$ to itself verifying the conditions*

$$D_{\mathbf{p}}(\mathbf{z}^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}}\mathbf{z}^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p},\mathbf{q} \text{ in } \mathbb{N}^{(L)}). \tag{16.11.2.1}$$

Also, when these conditions are verified, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

$$D_{\mathbf{q}}\circ D_{\mathbf{p}} = D_{\mathbf{p}}\circ D_{\mathbf{q}} = \frac{(\mathbf{p}+\mathbf{q})!}{\mathbf{p}!\mathbf{q}!}D_{\mathbf{p}+\mathbf{q}} \quad (\mathbf{p},\mathbf{q} \text{ in } \mathbb{N}^{(L)}). \tag{16.11.2.2}$$

Finally, if L is finite, for every integer m, the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\!\mathit{iff}^m_{U/S}$, in other words, every differential operator of order $\leqslant m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}|\leqslant m} a_{\mathbf{p}}D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathscr{O}_X$ over U.

PROOF. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

$$\langle\boldsymbol{\zeta}^{\mathbf{p}}, D_{\mathbf{q}}\rangle = \delta_{\mathbf{pq}} \quad \text{(Kronecker's symbol)}. \tag{16.11.2.3}$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$. Then, for every integer $m \geqslant 1$, we deduce similarly from (16.11.2.3) that the $\boldsymbol{\zeta}^{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from the definition (16.10.1), the fact that $\boldsymbol{\zeta}^{\mathbf{p}}$ form a basis of $\mathscr{P}^n_{U/S}$ for $|\mathbf{p}| \leqslant m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathscr{P}^n_{U/S} \to \mathscr{O}_U(|\mathbf{q}| \leqslant m)$ such that $\langle\zeta^{\mathbf{p}}, u_{\mathbf{q},m}\rangle = \delta\mathbf{p},\mathbf{q}$ for $|\mathbf{p}| \leqslant m$, $|\mathbf{q}| \leqslant m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geqslant |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined, and that, if L is finite, for $|\mathbf{p}| \leqslant m$, the $D_{\mathbf{p}}$ form a basis of the dual $\mathscr{D}\!\mathit{iff}^m_{U/S}$ of $\mathscr{P}^n_{U/S}$. Finally, the relations (16.11.2.2) follows immediately from the expression of the values of the three operators considered on the $\mathbf{z}^{\mathbf{r}}$, and of the fact that the $\boldsymbol{\zeta}^{\mathbf{r}}$ for $|\mathbf{r}| \leqslant m$ generate $\mathscr{P}^n_{U/S}$. ☐

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Remarks (16.11.3). — (i) The fact that, because of (16.11.2.2), the $D_{\mathbf{p}}$ are pairwise permutable naturally do not imply that the $\mathscr{O}_U$-algebra $\mathscr{D}\!\mathit{iff}_{U/S}$ is commutative, the $D_{\mathbf{p}}$ do not commute with the sections of $\mathscr{O}_U$ unless $n = 0$.

(ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\boldsymbol{\epsilon}_\lambda = (\epsilon_{\lambda\mu})_{\mu\in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators $D_{\boldsymbol{\epsilon}_\lambda}$ are exactly the S-derivations D_i introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).

(iii) For every integer $r \geqslant 1$, we can define the notion of *differentially smooth up to order r* by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}^m_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}\!\mathit{r}_m(\mathscr{P}^n_{X/S})$$

are bijective *for every $m \leqslant r$*. The argument of (16.11.2) proves also that if, in the condition (a), we replace "differentially smooth" by "differentially smooth up to order r", this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leqslant r$, $|\mathbf{q}| \leqslant r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is *of characteristic* p (p equal to zero or a prime number) if, for every affine open set U of X, the ring $\Gamma(U, \mathscr{O}_X)$ is of characteristic p (**0**, 21.1.1). It follows from (**0**, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every *closed* point x of X, the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$-prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a scheme of characteristic 0, $f : X \to S$ a morphism. If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X, we can suppose there is a family (z_λ) of sections of $\mathscr{O}_X$ over X such that the (dz_λ) is a basis for the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_{\lambda} D_\lambda^{p_\lambda}$$

(where the D_λ are the coordinate forms corresponding to the basis (dz_λ)) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_λ are derivations. ☐

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \mathrm{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \mathrm{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega^1_{X/S}$ has rank 1, and that the morphism $X \to S$ **IV-4 | 56**
has rank 1, and that the morphism $X \to S$ is differentially smooth up to order $p-1$ (16.11.3, (iii)), but not of order p. However, the proof of (16.12.2) proves that if $\Omega^1_{X/S}$ is locally free, and if $n!1_{\mathscr{O}_X}$ is inversible in $\Gamma(X, \mathscr{O}_X)$, then X is differentially smooth over S up to order n.

§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.

In this paragraph, we revisit the concepts studied in ($\mathbf{0}_{\mathrm{III}}$, 9), expressed in the geometric language of schemes from a global point of view, for preschemes locally of finite presentation over a given base.

Most of the results (except 17.7, 17.8, 17.9, 17.13, and 17.16) are reduced to various properties already encountered in ($\mathbf{0}_{\mathrm{III}}$, 9).

For more specific results on étale morphisms, the reader should consult §18.

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y', all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathscr{J}$ of $\mathscr{O}_{Y'}$, and every morphism $Y' \to Y$, the map

$$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X) \tag{17.1.1.1}$$

induced by the canonical map $Y'_0 \to Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y.

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

(i) Suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \to B$. According to (**0**, 19.3.1) and (**0**, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ, B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A-algebra, for the *discrete* topologies on A and B.

(ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathscr{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathscr{J}^2 = 0$, and if we have $\mathscr{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals $\mathscr{J}^{j+1}$ for $0 \leqslant j \leqslant n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leqslant j \leqslant n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). **IV** | 57

(iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* ($\mathbf{0}_{\mathbf{III}}$, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y-preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

(iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an *arbitrary* Y-prescheme Z and a closed subprescheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Z$. Then the map

(17.1.2.1) $$\mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \to Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathscr{J}|U_\alpha$ are nilpotent, and for each α, let U_α^0 be the inverse image of U_α in Z_0, which is the closed subprescheme of U_α defined by $\mathscr{J}|U_\alpha$. Let $f_0 : Z_0 \to X$ by a Y-morphism; by hypothesis, for each α, there is at most one (resp. one and only one) Y-morphism $f_\alpha : U_\alpha \to X$ whose restriction to Z_0 coincides with $f_0|U_\alpha$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|V = f_\beta|V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y-morphism $f : Z \to X$ whose restriction to Z_0 coincides with f_0.

Proposition (17.1.3). —

(i) *A monomorphism of preschemes is formally unramified; an open immersion is formally étale.*

(ii) *The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).*

(iii) *If $f : X \to Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f.*

(vi) *If $f : X \to Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$.*

PROOF. According to (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \to Y$, $g : Y \to Z$, an affine scheme Z', a closed subscheme Z'_0 of Z defined by a nilpotent ideal and a morphism $Z' \to Z$. Suppose that f and g formally smooth, and consider a Z-morphism $u_0 : Z'_0 \to X$; the hypothesis on g implies that there exists a Z-morphism $v : Z' \to Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \to Z$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \to X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \to Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified. **IV** | 58

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'', a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \to Y'$ making Y'' a Y'-prescheme; we then know by (**I**, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with

$\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). □

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f.*

PROOF. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \to Y$ a morphism, $j : Y'_0 \to Y'$ the canonical injection, $u_0 : Y'_0 \to Y$ a Y-morphism, such that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \to X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z-morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y-morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. □

Corollary (17.1.5). — *Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.*

PROOF. This follows from (17.1.4) and (17.1.3, (ii) and (iv)). □

Proposition (17.1.6). — *Let $f : X \to Y$ be a morphism of preschemes.*

(i) *Let (U_α) be an open covering of X and, for each α, let $i_\alpha : U_\alpha \to X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.*

(ii) *Let (V_λ) be an open covering of Y. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \to V_\lambda$ of f is.*

PROOF. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \to Y$ and $i_\lambda : f^{-1}(V_\lambda) \to X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \to V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if the $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f. **IV** | 59

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathscr{J}$, which we may assume to satisfy $\mathscr{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \to Y$ be a morphism. Suppose we are given a Y-morphism $u_0 : Y'_0 \to X$; denote by W_α (resp. W^0_α) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the *same underlying topological space*). Let us first suppose that the $f \circ i_\alpha$ are *formally unramified*, and show that, if u' and u'' are two Y-morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α, we have $u'|W_\alpha = u''|W_\alpha$, since the restrictions of both Y-morphisms to W^0_α coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are *formally smooth* and prove the existence of a Y-morphism $u : Y' \to X$ whose restriction to Y'_0 is u_0. Now, since Y' is an *affine scheme*, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u. □

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y, which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \to Y$ to be formally unramified, it is necessary and sufficient that $\Omega^1_f = 0$ (what we still write $\Omega^1_{X/Y} = 0$ (16.3.1)).*

PROOF. Taking into account (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the conclusion then follows from (**0**, 20.7.4) and the interpretation of $\Omega^1_{X/Y}$ in this case (16.3.7). □

Corollary (17.2.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms. For f being formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is surjective.

PROOF. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). □

Proposition (17.2.3). — *Let $f : X \to Y$ be a formally smooth morphism.*

(i) *The $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega^1_{X/Y}$ is locally free and of finite type.*

(ii) *For all morphisms $g : Y \to Z$, the sequence (16.4.19) of $\mathscr{O}_X$-modules*

(17.2.3.1)
$$0 \longrightarrow f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split *exact sequence.*

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PROOF.

(i) We know (16.3.9) that if f is locally of finite type, then Ω^1_f is an $\mathscr{O}_X$-module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (**0**, 20.4.9) and (**0**, 19.2.1).

(ii) Again, we can restrict to the case where X, Y, and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (**0**, 20.5.7).

□

Corollary (17.2.4). — *If $f : X \to Y$ is formally étale, then, for all morphisms $g : Y \to Z$, the canonical homomorphism of $\mathscr{O}_X$-modules*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is bijective.

PROOF. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega^1_{X/Y} = 0$ (17.2.1). □

Proposition (17.2.5). — *Let $f : X \to Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathscr{O}_X$-modules (16.4.21)*

(17.2.5.1)
$$0 \longrightarrow \mathscr{N}_{X'/X} \longrightarrow \Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) form a split *exact sequence.*

PROOF. By virtue of (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and $X' = \operatorname{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B. The conormal sheaf $\mathscr{N}_{X'/X}$ then corresponds to the B-module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (**0**, 20.5.14). □

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

(a) *f is a monomorphism.*
(b) *f is radicial and formally unramified.*
(c) *For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$-isomorphic to* $\mathrm{Spec}(k(y))$ *(in other words, it is reduced to a single point z such that $k(y) \to \mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ is an isomorphism).*

PROOF. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega^1_{X/Y} = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is quasi-coherent of finite type (16.3.9). It follows from (**I**, 9.1.13.1) that, for $(\Omega^1_{X/Y})_x = 0$, it is necessary and sufficient that if we set $Y_1 = \mathrm{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega^1_{X_1/Y_1})_x = 0$; but as the morphism $f_1 : X_1 \to Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \to X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega^1_{X/Y}$ is by definition the conormal sheaf $\mathscr{G}r_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathscr{G}r_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation (1.4.3.1); therefore the hypothesis $\mathscr{G}r_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (**I**, 5.3.8). ☐ **IV | 61**

17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \to Y$ is *smooth* (resp. *unramified*, or *net* [8] resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over* Y.

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remark (17.3.2). —

(i) Note that definition (17.3.1) can be phrased using only the functor
$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$
considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

(ii) Let A be a ring and B an A-algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A-algebra if the corresponding morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to say that B is an A-algebra *of finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.

(iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y*.

Proposition (17.3.3). —

(i) *An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient to it be locally of finite presentation.*

(ii) *The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).*

(iii) *If $f : X \to Y$ is a smooth (resp. unramified, resp. étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are smooth (resp. unramified, resp. étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.* **IV | 62**

[8] The words "net" and "formally net" seem more preferable to the terminology used in "unramified" (resp. formally unramified") and will be used almost exclusively in Chapter V. In this chapter, we have kept the old terminology so as not to conflict with **0**, 19.10.

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.*

PROOF. This follows from (1.4.3) and (17.1.3). □

Proposition (17.3.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f.*

PROOF. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). □

Corollary (17.3.5). — *Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.*

PROOF. This follows from (17.3.4) and (17.3.3, (ii)). □

Proposition (17.3.6). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism* (16.4.19)

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective.

PROOF. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). □

Definition (17.3.7). — Let $f : X \to Y$ be a morphism. We say that f is *smooth* (resp. *unramifed*, resp. *étale*) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y.

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over Y at the point x*.

Taking into account remark (17.3.2, (iii)), it is equivalent to say that f is smooth (resp. unramified, resp. étale) and to say that f is smooth (resp. unramified, resp. étale) at all points of X.

It is clear that the set of points of X at which the morphism $f : X \to Y$ is smooth (resp. unramified, resp. étale) is *open* in X.

Proposition (17.3.8). — *For all preschemes Y and all locally free $\mathscr{O}_Y$-modules $\mathscr{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathscr{E})$ (**II**, 1.7.8) associated to $\mathscr{E}$ is a smooth Y-prescheme.*

PROOF. Indeed (17.3.2, (iii)), we can restrict ourselves to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{V}(\mathscr{E}) = \operatorname{Spec}(A[T_1, \ldots, T_r])$; as $A[T_1, \ldots, T_r]$ is a formally smooth A-algebra for the discrete topologies (**0**, 19.3.2), and of finite presentation, this proves the proposition (17.3.2, (ii)) □

Corollary (17.3.9). — *Under the hypotheses of (17.3.8), the projective prescheme $\mathbf{P}(\mathscr{E})$ (**II**, 4.1.1) is a smooth Y-prescheme.*

PROOF. We can still restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{P}(\mathscr{E}) = \mathbf{P}^r_Y$. We then know (**II**, 2.3.14) that we have a finite open cover of $\mathbf{P}^r_A$ by the $D_+(T_i)$ $(0 \leqslant i \leqslant r)$ respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \ldots, T_r]$ and f for T_i; but it follows immediately from the definition of $S_{(f)}$ (**II**, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \ldots, T_{i-1}, T_{i+1}, \ldots, T_r]$; hence the corollary follows by (17.3.8). □

17.4. Characterizations of unramified morphisms. **IV** | 63

§18. Supplement on étale morphisms. Henselian local rings and strictly local rings

§19. Regular immersions and normal flatness

§20. Meromorphic functions and pseudo-morphisms

20.0. Introduction Most of the concepts and results of §§20 and 21 directly relate to Chapter I, and hardly depend on Chapters I and IV, except for the occasional usage of the notion of depth and of a local regular ring (in (20.6), (21.11), (21.13), and (21.15)), of Zariski's "Main theorem" in (20.4) and (21.12), and the properties of transversely regular immersions in (20.6) and (21.15).

In §20, we introduce several variants of the concept of a rational map, already studied in (I, 7) **IV-4** | 226 from a point of view still fairly close to the classical point of view, and for this reason quite ill-suited to the case of not necessarily reduced preschemes. The notions and results of §20 are used in §21 (n^{os}(21.1) and (21.7)) to develop the general notion of a divisor and its most elementary properties. This notion is especially convenient when the local rings of the preschemes considered are Noetherian and integrally closed, and especially when they are also *factorial* ((21.6) and (21.7)), because of their identification in the latter case with the notion of a 1-*codimensional cycle* (a linear combination of irreducible subpreschemes of codimension 1). In (21.9), we determine the divisors on a Noetherian prescheme of dimension 1 but not necessarily normal, which is useful for various applications. The (21.11) and (21.12) give two important theorems, due respectively to Auslander–Buchsbaum and Van der Waerden, and relate the notion of a factorial ring (the n^{os}(21.9), (21.11), and (21.12) are independent of each other). In the n^{os}(21.13) and (21.14), also independent of the previous three, we study a useful variant of the notion of a local factorial ring, that of a local *parafactorial* ring, which is introduce in particular **[?]** in the development of comparison theorem of the Picard group of a projective prescheme X over a field k and a "hyperplane section". We will see in (21.14.1) (Ramanujam–Samuel theorem) that the local parafactorial rings are much more numerous than one would have expected *a priori*.

In (20.5), (20.6), and (21.15), we review the previous notions but from a point of view "relative" to a fixed base prescheme. For the moment these notions are used only relatively rarely; in particular, the concept of a relative divisor is hardly used except when it comes to positive divisors, and in this case it is explained advantageously without the help of the notion of a relative meromorphic functions, using the notion of a transversely regular immersion of codimension 1. It will therefore be advantageous to omit these sections on a first reading.

20.1. Meromorphic functions

(20.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space, and let $\mathscr{S}$ be a subsheaf *of sets* of $\mathscr{O}_X$. For every open U of X, consider the *ring of fractions* $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1). It is immediate that the map $U \mapsto \Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ is a *presheaf of rings* ((**0**, 1.5.1) and (**0**, 1.5.7)). We denote by $\mathscr{O}_X[\mathscr{S}^{-1}]$ the *sheaf of rings* associated to this presheaf and we say that this is the *sheaf of rings of fractions of $\mathscr{O}_X$ with denominators in $\mathscr{S}$*; this is a *flat* $\mathscr{O}_X$-module. It is immediate that for every $x \in X$, we have a canonical isomorphism

(20.1.1.1) $$(\mathscr{O}_X[\mathscr{S}^{-1}])_x \simeq \mathscr{O}_x[\mathscr{S}_x^{-1}],$$

since the reasoning of (**0**, 1.4.5) generalizes immediately in the case where we have an inductive system $(A_\alpha, \phi_{\beta\alpha})$ of rings, and for each index α a subset S_α of A_α such that $\phi_{\beta\alpha}(S_\alpha) \subset S_\beta$ for $\alpha \leqslant \beta$; **IV-4** | 227 we then take for S the inductive limit in $A = \varinjlim A_\alpha$ of the inductive limit of the subsets (S_α).

(20.1.2). Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module. We then set

(20.1.2.1) $$\mathscr{F}[\mathscr{S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X[\mathscr{S}^{-1}],$$

and we say that this is the *sheaf of modules of fractions of $\mathscr{F}$ with denominators in $\mathscr{S}$*; it is immediate that it is associated to the presheaf of modules $U \mapsto \Gamma(U, \mathscr{F})[\Gamma(U, \mathscr{S})^{-1}]$, and that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{F}[\mathscr{S}^{-1}])_x \simeq \mathscr{F}_x[\mathscr{S}_x^{-1}]. \tag{20.1.2.2}$$

(20.1.3). We will focus here on the case where $\mathscr{S}$ is the subsheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{O}_X$ such that for every open U, $\Gamma(U, \mathscr{S})$ is the *set of regular elements* of the ring $\Gamma(U, \mathscr{O}_X)$; it is immediate that it is a sheaf (and not only a presheaf), the regularity of a section of $\mathscr{O}_X$ over U being verified "fibre by fibre" (i.e. meaning that the germ of the section in x is regular in $\mathscr{O}_X$ for all $x \in U$), in other words $\mathscr{S}(\mathscr{O}_X)_x$ is none other then the set of regular elements of $\mathscr{O}_{X,x}$. The corresponding sheaf of rings

$$\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$$

is called the *sheaf of germs of meromorphic functions on* X, and the sections of $\mathscr{M}_X$ over X are called the *meromorphic functions on* X; they form a ring which we denote by $M(X)$. For every $\mathscr{O}_X$*-module* $\mathscr{F}$,

$$\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{F}[\mathscr{S}^{-1}]$$

is also denoted $\mathscr{M}_X(\mathscr{F})$ and called the *sheaf of germs of meromorphic sections of* $\mathscr{F}$; its sections over X form an $M(X)$-module denoted by $M(X, \mathscr{F})$, whose elements are called *meromorphic sections of* $\mathscr{F}$ *over* X. These definitions imply that for every open U of X, we have a canonical isomorphism $\mathscr{M}_X(\mathscr{F})|U \simeq \mathscr{M}_U(\mathscr{F}|U)$, in particular $\mathscr{M}_X|U \simeq \mathscr{M}_U$.

(20.1.3.1). If X is a *reduced prescheme*, then we note that if an element $s \in \Gamma(U, \mathscr{O}_X)$ is such that $s_\xi \neq 0$ for every maximal point ξ of U, then s is *regular*. Indeed, if $st = 0$ for a $t \in \Gamma(U, \mathscr{O}_X)$, then we have $s_\xi t_\xi = 0$, so $t_\xi = 0$ since $\mathscr{O}_{X,\xi}$ is a field, and say that $t_\xi = 0$ for every maximal point ξ of X means that $t = 0$: we are immediately reduced to the case where U is affine, and an element of a reduced ring which belongs to all the minimal prime ideals is zero by definition. The converse is true if the set of irreducible components of X is *locally finite*. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine; if $\mathfrak{p}_i$ ($1 \leqslant i \leqslant n$) are the minimal prime ideals of A and if $s \in \mathfrak{p}_i$ for an index i, then there exists $t \in A$ such that $t \in \mathfrak{p}_j$ for $j \neq i$ and $t \notin \mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, n°1, Prop. 1); so we have $st \in \mathfrak{p}_i$ for all i, and as a result $st = 0$ since A is reduced; s is therefore nonregular.

(20.1.4). For every open U of X, the homomorphism $t \mapsto t/1$ from $\Gamma(U, \mathscr{O}_X)$ to $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (which is none other than the *total ring of fractions* of $\Gamma(U, \mathscr{O}_X)$) is injective; these homomorphisms **IV-4** | 228
thus define a *canonical injective homomorphism*

$$i : \mathscr{O}_X \longrightarrow \mathscr{M}_X \tag{20.1.4.1}$$

which allows us to identify $\mathscr{O}_X$ with a subsheaf of $\mathscr{M}_X$. Given a meromorphic function $\phi \in M(X)$, we say that ϕ is *defined* on an open U of X if $\phi|U$ is a *section of* $\mathscr{O}_U$ over U; the axioms of sheaves show that there is, for a given section ϕ, a *largest* open on which ϕ is defined; we call this the *domain of definition* of ϕ and denote it by $\operatorname{dom}(\phi)$.

(20.1.5). For every $\mathscr{O}_X$-module $\mathscr{F}$, we obtain from (20.1.4.1) a di-homomorphism consisting of i and the homomorphism of sheaves of additive groups

$$1_{\mathscr{F}} \otimes i : \mathscr{F} \longrightarrow \mathscr{M}_X(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X. \tag{20.1.5.1}$$

We note that the latter is not injective in general; when it is injective, we say that $\mathscr{F}$ is *strictly torsion-free*: this means that for every open U of X and every section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element in this ring the homothety $z \mapsto sz$ of $\Gamma(U, \mathscr{F})$ is injective; this condition is evidently satisfied if $\mathscr{F}$ is *locally free*.

Proposition (20.1.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. For $\mathscr{F}$ to be strictly torsion-free, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{O}_X)$.

PROOF. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, and we know that the elements s of A belonging to an ideal of $\operatorname{Ass}(M)$ are exactly those for which the homothety $z \mapsto sz$ is not injective (Bourbaki, *Alg. comm.*, chap. IV, §1, n°1, cor. 2 of prop. 2). □

(20.1.7). If u is a section of $\mathscr{M}_X(\mathscr{F})$ over X, then we say that u is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X such that $u|V$ is the image of a section of $\mathscr{F}$ over V under the di-homomorphism (20.1.5.1). We say that u is *defined* on an open U of X if it is defined at every point in U; there is still a larger open in which u is defined, called the *domain of definition* of u and denoted

$\operatorname{dom}(u)$. When $\mathscr{F}$ is strictly torsion-free, such that $\mathscr{F}$ is identified by (20.1.5.1) with a subsheaf of $\mathscr{M}_X(\mathscr{F})$, then saying u is defined on U means that $u|V$ is a *section of $\mathscr{F}$ over U*.

(20.1.8). In accordance with the general notation of ($\mathbf{0_I}$, 5.4.7), we denote by $\mathscr{M}_X^*$ the sheaf of multiplicative groups such that $\Gamma(U, \mathscr{M}_X^*)$ is (for every open U of X) the group of *invertible elements* of $\Gamma(U, \mathscr{M}_X)$. This sheaf is none other than the sheaf $\mathscr{S}(\mathscr{M}_X)$ defined in (20.1.3): indeed, if $s \in \Gamma(U, \mathscr{S}(\mathscr{M}_X))$, then for every $x \in U$, there exists an open neighbourhood $V \subset U$ of x such that $s|V$ is a regular element in the *total ring of fractions* of $\Gamma(V, \mathscr{O}_X)$, and we know that such an element is necessarily invertible in this ring of fractions. We say that the sections of $\mathscr{M}_X^*$ over X are the *regular meromorphic functions* (note that we are deviating here from the terminology followed by certain authors, who call "regular" meromorphic functions those which are *sections of* $\mathscr{O}_X$, identified with a subsheaf of $\mathscr{M}_X$).

Let $\mathscr{L}$ be an *invertible $\mathscr{O}_X$-module* ($\mathbf{0_I}$, 5.4.1); then it is clear that $\mathscr{M}_X(\mathscr{L}) = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}_X$ is an *invertible $\mathscr{M}_X$-module*. Let U be an open such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$; as every automorphism of $\mathscr{M}_U$ is multiplication by an invertible element of $\Gamma(U, \mathscr{M}_X)$ ($\mathbf{0_I}$, 5.4.7), it is equivalent to say that a section $s \in \Gamma(U, \mathscr{M}_X(\mathscr{L}))$ has an invertible image in $\Gamma(U, \mathscr{M}_X)$ under *an* isomorphism or by *any* isomorphism on $\Gamma(U, \mathscr{M}_X)$; we say in this case that s is a *regular meromorphic section of $\mathscr{L}$* over U; a section s of $\mathscr{L}$ over X is called a *regular meromorphic section of $\mathscr{L}$* if, for every open U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $s|U$ is a regular meromorphic section of $\mathscr{L}$ over U. We denote by $(\mathscr{M}_X(\mathscr{L}))^*$ the subsheaf of $\mathscr{M}_X(\mathscr{L})$ such that for every open U, $\Gamma(U, (\mathscr{M}_X(\mathscr{L}))^*)$ is the set of regular meromorphic sections of $\mathscr{L}$ over U. Let s be a meromorphic section of $\mathscr{L}$ over X (i.e. a section of $\mathscr{M}_X(\mathscr{L})$); it defines a homomorphism $h_s : \mathscr{M}_X \to \mathscr{M}_X(\mathscr{L})$ which sends every section t of $\mathscr{M}_X$ over an open U to $(s|U)t$. It follows immediately from the above that for s to be *regular*, it is necessary and sufficient for h_s to be *injective*, and in fact h_s is then a *bijective* homomorphism from $\mathscr{M}_X$ to $\mathscr{M}_X(\mathscr{L})$, and its restriction to $\mathscr{M}_X^*$ is a bijection to $(\mathscr{M}_X(\mathscr{L}))^*$. We conclude that the homothety $t \mapsto ts$ is an isomorphism from $M(X)$ to $M(X, \mathscr{L})$. **IV-4** | 229

(20.1.9). Let s be a regular meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; then for every $\mathscr{O}_X$-module $\mathscr{F}$, s similarly defines a homomorphism $h_s \otimes 1_{\mathscr{F}} : \mathscr{M}_X(\mathscr{F}) \to \mathscr{M}_X(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L})$, which is again *bijective*.

(20.1.10). Let s be a meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; for s to be regular, it is necessary and sufficient for there to exists a meromorphic section s' of $\mathscr{L}^{-1}$ over X such that the canonical image of $s \otimes s'$ in $\mathscr{M}_X$ ($\mathbf{0_I}$, 5.4.3) is the unit section, and this section s' is then unique: indeed, the necessity of the local existence of such a section is evident, and its local uniqueness implies its global (and unique) existence; moreover, the existence of s' is trivially sufficient for s to be regular. We will take $s' = s^{-1}$.

Finally, if $\mathscr{L}'$ is a second invertible $\mathscr{O}_X$-module, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, then $s \otimes s'$ is evidently a regular meromorphic section of $\mathscr{L} \otimes \mathscr{L}'$ over X.

(20.1.11). If $f : X' \to X$ is a morphism of ringed spaces, then there is in general no natural map sending a meromorphic function on X to a meromorphic function on X'. For example, if X is the spectrum of a local integral domain A, X' its residue field k, then there is no natural homomorphism from the field of fractions K of A to k, and we can only send an element of K to an element of k if it is already in A.

In general, if $f = (\psi, \theta)$, then for every open U of X, denote by $\mathscr{S}_f(U)$ the set of *regular* sections $s \in \Gamma(U, \mathscr{O}_X)$ such that the image of s under

$$\Gamma(\theta^\#) : \Gamma(U, \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathscr{O}_{X'})$$

is a *regular* section. It is immediate that $U \mapsto \mathscr{S}_f(U)$ is a *subsheaf* of the sheaf of sets $\mathscr{S}(\mathscr{O}_X)$, which we denote by $\mathscr{S}_f$. We set $\mathscr{M}_f = \mathscr{O}_X[\mathscr{S}_f^{-1}]$; this is a subsheaf of rings of $\mathscr{M}_X$, and we canonically obtain from $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ extending $\theta^\#$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1, prop. 2); hence, recalling that $f^*(\mathscr{M}_f) = \psi^*(\mathscr{M}_f) \otimes_{\psi^*(\mathscr{O}_X)} \mathscr{O}_{X'}$, we get a canonical homomorphism of $\mathscr{O}_{X'}$-algebras **IV-4** | 230

(20.1.11.1) $$f^*(\mathscr{M}_f) \longrightarrow \mathscr{M}_{X'}.$$

For every meromorphic function ϕ on X that *is a section of* $\mathscr{M}_f$, $\Gamma(\theta'^{\#})(\phi)$ is a meromorphic function on X', called the *inverse image of* ϕ *under* f, and denoted by $\phi \circ f$ is there is no cause for confusion.

Similarly, if $\mathscr{F}$ is an $\mathscr{O}_X$-module, then we set $\mathscr{M}_f(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_f$, and we immediately obtain from $\theta'^{\#}$ a canonical homomorphism (which is also written as $u \mapsto u \circ f$)

$$\Gamma(X, \mathscr{M}_f(\mathscr{F})) \longrightarrow \Gamma(X', \mathscr{M}_{X'}(f^*(\mathscr{F}))).$$

In addition, if $u \in \Gamma(X, \mathscr{M}_f(\mathscr{F}))$ is defined (20.1.7) at a point x, then u coincides, on a neighbourhood U of x, with a section of the form $\sum_i h_i \otimes (t_i/s_i)$, where the h_i belong to $\Gamma(U, \mathscr{F})$, the t_i to $\Gamma(U, \mathscr{O}_X)$, and the s_i to $\Gamma(U, \mathscr{S}_f)$. As by hypothesis the images of the s_i in $\Gamma(f^{-1}(U), \mathscr{O}_{X'})$ are regular, we see that $u \circ f$ is defined at every point of $f^{-1}(U)$; in other words, we have

(20.1.11.2) $$f^{-1}(\operatorname{dom}(u)) \subset \operatorname{dom}(u \circ f).$$

We will see later (20.6.5, (i)) examples (with $\mathscr{F} = \mathscr{O}_X$) where the two sides of (20.1.11.2) can be different.

Consider in particular the case where $\mathscr{M}_f = \mathscr{M}_X$; then, if $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module, the image in $\mathscr{M}_{X'}(f^*(\mathscr{L}))$, under $\Gamma(\theta'^{\#})$, of a *regular* meromorphic section of $\mathscr{L}$ over X (20.1.8) is a *regular* meromorphic section of $f^*(\mathscr{L})$ over X', as it follows immediately from the definition of its sections, and from the fact that a homomorphism of rings sends an invertible element to an invertible element.

Let $f' : X'' \to X'$ be a second morphism of ringed spaces, and suppose that $\mathscr{M}_f = \mathscr{M}_X$ and $\mathscr{M}_{f'} = \mathscr{M}_{X'}$; then, if we set $f'' = f \circ f'$, we also have $\mathscr{M}_{f''} = \mathscr{M}_X$, and we immediately see that for every meromorphic section u of $\mathscr{F}$ over X, we have $u \circ f'' = (u \circ f) \circ f'$.

Proposition (20.1.12). — *If the morphism $f : X' \to X$ is flat* ($\mathbf{0_I}$, 6.7.1), *then we have $\mathscr{M}_f = \mathscr{M}_X$, and the homomorphism $\phi \mapsto \phi \circ f$ is defined on all of $M(X)$. In addition, if f is a (flat) morphism of locally ringed spaces, then we have* $\operatorname{dom}(\phi \circ f) = f^{-1}(\operatorname{dom}(\phi))$; *if in addition f is surjective (thus faithfully flat), then the homomorphism $\phi \mapsto \phi \circ f$ is injective.*

PROOF. The first assertion follows from the fact that, if B is an A-algebra which is a flat A-module, then every element of A not a divisor of 0 in A is not a divisor of 0 in B ($\mathbf{0_I}$, 6.3.4). To prove the other assertions, note that, for every $x' \in X'$, if $x = f(x')$, then $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, and as the homorpism $\mathscr{O}_{X,x} \to \mathscr{O}_{X',x'}$ is *local* by hypothesis, it is injective (($\mathbf{0_I}$, 6.5.1) and ($\mathbf{0_I}$, 6.6.2)); if we set $A = \mathscr{O}_{X,x}$, $B = \mathscr{O}_{X',x'}$, such that A identifies with a subring of B, then $(f^*(\mathscr{M}_X))_{x'}$ is equal to $S^{-1}A \otimes_A B = S^{-1}B$, where S is the set of regular elements of A, $(\mathscr{M}_{X'})_{x'}$ is equal to $T^{-1}B$, where T is the set of regular elements of B, and as we have seen that $S \subset T$, the homomorphism $S^{-1}B \to T^{-1}B$ **IV-4** | 231
is injective; in other words, this proves that the homomorphism (20.1.11.1) $f^*(\mathscr{M}_X) \to \mathscr{M}_{X'}$ is *injective* (hence the last assertion of the statement). The quotient $f^*(\mathscr{M}_X)/\mathscr{O}_{X'}$ identifies with an $\mathscr{O}_{X'}$-submodule of $\mathscr{M}_{X'}/\mathscr{O}_{X'}$, and $(f^*(\mathscr{M}_X)/\mathscr{O}_{X'})_{x'}$ identifies with $(\mathscr{M}_X/\mathscr{O}_X)_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{X',x'}$. Then suppose that $x \notin \operatorname{dom}(\phi)$; the image of ϕ_x in $(\mathscr{M}_X/\mathscr{O}_X)_x$ is therefore $\neq 0$; by faithful flatness, we deduce that that it is the same for the image of $(\phi \circ f)_{x'}$, so $x' \notin \operatorname{dom}(\phi \circ f)$, which finishes the proof. □

Remark (20.1.13). — Let X be a *reduced* complex analytic space; then the notion of a meromorphic function on X defined above coincides with the usual notion. Consider on the other hand a prescheme Y, locally of finite type over the field $\mathbf{C}$; we then know that we can associate to Y an analytic space Y^{an} having the same underlying topological space, and the canonical morphism $f : Y^{\mathrm{an}} \to Y$ is *flat* [**?**]; by virtue of (20.1.12), the canonical homomorphism $u \mapsto u \circ f$ from $M(Y)$ to $M(Y^{\mathrm{an}})$ is therefore always defined and is injective; but it is not *surjective* in general. For example, when $Y = \mathbf{V}_0^r$ ($\mathbf{Err_{III}}$, 14) is the affine space of dimension r over $\mathbf{C}$, $M(Y)$ canonically identifies with the field $R(Y)$ of rational functions on Y (20.2.13, (i)), while $M(Y^{\mathrm{an}})$ is the field of usual meromorphic functions on $\mathbf{C}^r$. Because of this fact, it is often preferable, in algebraic geometry, to abstain from the terminology introduced in this section, and to use the equivalent terminology of "pseudo-function" which will be defined below.

20.2. Pseudo-morphisms and pseudo-functions

§21. DIVISORS

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